The Hodge Project

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Chapter 1

Introduction

1.1 What this project is

The Hodge Project is a cumulative reference whose goal is to develop, with complete proofs, all the mathematics needed to state, understand and work with the Hodge conjecture, together with the theory that surrounds it.

Informally, the Hodge conjecture says the following: on a smooth complex projective variety X, every rational cohomology class of type (p, p) is a rational linear combination of the classes of algebraic subvarieties of X. A precise statement is given in Chapter 68. Each word of the informal statement hides a substantial theory: rational cohomology classes come from algebraic topology (Chapter 22), the type (p, p) comes from the Hodge decomposition of a compact Kähler manifold (Chapter 48), which rests on the analysis of elliptic operators (Chapter 42), and the classes of subvarieties come from algebraic geometry (Chapters 53 and 63). This project builds all of it, starting from logic and set theory.

The organisation is modelled on the Stacks project: the text is written in a strict logical order, every result has a permanent tag, and the whole project is a single document that can be read as a book or browsed as a website.

1.2 How the project is organised

The chapters are grouped into parts.

- Part I, *Foundations*: logic, set theory, the number systems and category theory.
- Part II, *Algebra*: from groups to derived categories, including a chapter of elementary number theory.
- Part III, *Topology*: general topology, homotopy theory, singular homology and cohomology, Poincaré duality.
- Part IV, *Sheaves*: sheaves and sheaf cohomology.
- Part V, *Analysis*: real, complex and functional analysis, several complex variables and Sobolev spaces.
- Part VI, *Differential Geometry*: from smooth manifolds to the Hodge theorem for harmonic forms.
- Part VII, *Complex Geometry*: complex and Kähler manifolds, the Hodge decomposition, the Lefschetz theorems, the Kodaira vanishing and embedding theorems, deformations of complex structures and moduli, and Calabi–Yau manifolds.

- Part VIII, *Algebraic Geometry*: varieties, schemes, coherent cohomology, GAGA, curves, algebraic groups, abelian varieties, algebraic cycles, the cycle class map and étale cohomology.

- Part IX, *Hodge Theory*: Hodge structures and Mumford–Tate groups, the Hodge conjecture, the evidence for it, its failed variants, mixed Hodge structures, absolute Hodge classes, and the Tate and standard conjectures around it.

The order of the chapters is the logical order of the project, not a recommended reading order. A reader interested mainly in the Hodge conjecture can start from Part IX and follow references backwards as needed.

1.3 A path to the Hodge conjecture

The following chapters form the spine of the project; everything else supports them or grows out of them.

1. *Cohomology.* Singular cohomology and Poincaré duality (Chapters 22 and 23), sheaf cohomology (Chapter 26) and de Rham's theorem (Chapter 37) give the rational cohomology of a manifold and its description by differential forms.

2. *Harmonic theory.* Elliptic operators (Chapter 42) give the Hodge theorem: every de Rham class has a unique harmonic representative (Chapter 43).

3. *Kähler geometry.* On a compact Kähler manifold (Chapter 47) the Laplacian respects the bidegree of forms, which gives the Hodge decomposition $H^k = \bigoplus_{p+q=k} H^{p,q}$ (Chapter 48), hard Lefschetz and the Hodge–Riemann relations (Chapter 49).

4. *Algebraic geometry.* Projective varieties are Kähler, and GAGA identifies analytic and algebraic objects on them (Chapter 58). Subvarieties have cohomology classes, which are rational and of type (p, p) (Chapter 63).

5. *The conjecture.* Hodge structures (Chapter 66) isolate the linear algebra, and the Hodge conjecture asserts the converse of the last statement (Chapter 68). It is a theorem for divisors (Chapter 69), it is known in further cases (Chapter 72), and its natural strengthenings are false (Chapter 73).

6. *The surrounding theory.* Variations of Hodge structure and normal functions (Chapters 70 and 71), mixed Hodge structures (Chapter 74), absolute Hodge classes (Chapter 75), and the Tate and standard conjectures (Chapters 76 and 77) place the conjecture in its wider landscape.

Remark 1.3.1. The book also contains material that is not needed on the way to the Hodge conjecture, and is there because it belongs to the same circle of ideas or because it is beautiful. Chapters and sections of this kind say so at their beginning, in a paragraph "On a first reading". For instance, Chapter 5 is not used by any other chapter, Chapter 2 is needed only for the foundations of set theory, and most of Chapter 40 (geodesics beyond the basics, surfaces, minimal surfaces, space forms) develops geometry for its own sake. The **Prerequisites** line of each chapter lists the chapters it actually depends on.

1.4 Sources and status

The mathematics in this project is standard, and each chapter lists the references it follows; the bibliography collects them. The chapters of Part IX are based in part on the expository text *The Hodge Conjecture* (2026), whose claims were checked against the original literature and corrected where necessary. Where a theorem is quoted rather than proved, the text says so and gives a reference (Remark 1.5.3). All chapters are written; none has yet been checked by an independent reader, which is the condition for the status *checked* (Remark 1.5.6).

1.5 Conventions of the text

Notation 1.5.1. All numbered items of a section share a single counter: Lemma 5.2.3 is the third numbered item of Section 5.2, and it is followed by Example 5.2.4. Chapters are numbered consecutively through all the parts.

Notation 1.5.2. The text uses the following kinds of numbered items.

- **Definition**: introduces a notion.
- **Theorem**, **Proposition**, **Lemma**, **Corollary**: proved statements, in roughly decreasing order of importance.
- **Conjecture**: a statement that is not known to be true.
- **Example**: an instance of a definition or a result.
- **Counterexample**: shows that a hypothesis cannot be dropped, or that a plausible statement is false.
- **Remark**: context, motivation or intuition. Remarks are never used in proofs.
- **Notation**: fixes notation or a convention.
- **Exercise**: a statement for the reader to prove, sometimes followed by a solution.

Remark 1.5.3. A result followed by *Proof* has a complete proof that uses only earlier material. A result followed by *Sketch of proof* has an argument whose main ideas are given but whose omitted verifications are named. A result followed by no proof is quoted from the literature, with a reference. Sketches and quotations are temporary: the aim of the project is to replace all of them with complete proofs.

Remark 1.5.4. The project has no forward references: a proof may only use definitions and results that appear before it. Examples, counterexamples, remarks and exercises may mention notions that are defined later (for instance, the category of topological spaces appears as an example in Chapter 6), but such mentions are never used in a proof. The only exception is this introduction, which refers forward freely.

Remark 1.5.5. Every labelled item (chapter, section, definition, result, example, and so on) carries a permanent four-character *tag*, shown next to it on the website. The number of an item changes whenever material is inserted before it; its tag never changes. To refer to an item from outside the project, cite its tag.

Remark 1.5.6. Each chapter has a status. *Planned*: only an outline exists. *Draft*: the chapter is being written, and its proofs have not been checked. *Complete*: all planned material is written, and the proofs await independent checking. *Checked*: every proof in the chapter has been checked by a reader other than its author.

1.6 Mathematical conventions

Notation 1.6.1. We work in ZFC set theory together with the assumption that a Grothendieck universe exists, and we fix such a universe $\mathcal{U}$ once and for all. A set is *small* if it is an element of $\mathcal{U}$. Chapter 3 develops these notions; Chapter 6 explains why they are needed.

Notation 1.6.2. The natural numbers include zero: $\mathbb{N} = \{0, 1, 2, \ldots\}$. This is the convention of set theory and logic, where $\mathbb{N}$ is built starting from $0 = \emptyset$ and each natural number $n = \{0, 1, \ldots, n-1\}$ is a set with n elements (Remark 3.4.3). Many texts, especially in analysis and number theory, start from 1 instead; for the positive integers we write $\mathbb{N}_{>0} = \{1, 2, 3, \ldots\}$, and more generally $\mathbb{N}_{\geq k} = \{k, k+1, \ldots\}$. A real number x is *positive* if $x > 0$ and *non-negative* if $x \geq 0$.

Notation 1.6.3. The composition of maps $f\colon X \to Y$ and $g\colon Y \to Z$ is written $g \circ f$ or gf. The identity map of X is written id_X.

Notation 1.6.4. A ring is associative and has an identity element 1, and ring homomorphisms preserve identity elements. Rings are not assumed to be commutative, except from Chapter 14 onwards, where "ring" means "commutative ring" unless stated otherwise.

Part I

Foundations

Chapter 2

Mathematical Logic

2.1 Introduction

Mathematical logic studies the language in which mathematics is written: what a formula is, what it means for a formula to be true in a structure, and what it means to prove it. The main result of this chapter is Gödel's completeness theorem (Theorem 2.6.1 and Corollary 2.6.2): a statement follows from a set of axioms in every structure if and only if it can be formally deduced from them. Its consequences, the compactness and Löwenheim–Skolem theorems, are the first tools of model theory. We end with the statements of Gödel's incompleteness theorems, which limit what any axiomatic foundation, including the set theory of Chapter 3, can prove about itself.

We follow the approach of [End01]; other references are [Sho67] and [vD13], and [CK90] for model theory.

Prerequisites. None.

On a first reading. This chapter is needed only for the formal foundation of set theory (Chapter 3) and for model theory (Chapter 5). A reader who is comfortable with the usual informal use of sets and logic can skip it and come back to it later.

Remark 2.1.1. This chapter studies formal languages *from the outside*. The reasoning we use about formulas, structures and deductions (induction on the natural numbers, finite sequences, sets, countability) is ordinary informal mathematics, sometimes called the *metatheory*. In Chapter 3, set theory is itself formulated as a first-order theory in the sense of this chapter. There is no vicious circle: informal mathematics is used to describe formal systems, and one of these formal systems is then used as the official foundation of the rest of the project. The proofs in this chapter use only elementary facts about finite and countable sets, and in particular never the axiom of choice.

2.2 Propositional logic

Definition 2.2.1. Let P be a set, whose elements we call *propositional variables*, and let $\neg$, $\to$, (and) be four symbols not in P. A *string* is a finite sequence of symbols. The set of *propositional formulas* over P is the smallest set of strings such that

1. every $p \in P$, viewed as a string of length one, is a formula;
2. if φ is a formula, then $\neg\varphi$ is a formula;
3. if φ and ψ are formulas, then $(\varphi \to \psi)$ is a formula.

Here $\neg\varphi$ denotes the string consisting of $\neg$ followed by the symbols of φ, and similarly for $(\varphi \to \psi)$.

Remark 2.2.2. Every formula is obtained from propositional variables by finitely many applications of the two operations, since the set of such strings contains P and is closed under the operations. Hence statements about all formulas can be proved by *induction on formulas*: it suffices to prove them for propositional variables and to show that they are preserved by the two operations.

Notation 2.2.3. We use the abbreviations $(\varphi \wedge \psi) := \neg(\varphi \to \neg\psi)$, $(\varphi \vee \psi) := (\neg\varphi \to \psi)$ and $(\varphi \leftrightarrow \psi) := ((\varphi \to \psi) \wedge (\psi \to \varphi))$, and we omit outer parentheses when no confusion is possible.

Lemma 2.2.4. *Let φ be a propositional formula. Then φ contains as many symbols (as symbols). Moreover, every proper initial segment of φ either consists only of symbols $\neg$ (possibly none), or contains strictly more symbols (than symbols).*

Proof. For a string u let $b(u)$ be the number of symbols (in u minus the number of symbols) in u. We argue by induction on φ.

If $\varphi = p$ is a variable, then $b(p) = 0$ and the only proper initial segment is empty.

If $\varphi = \neg\psi$, then $b(\varphi) = b(\psi) = 0$. A proper initial segment of φ is empty or of the form $\neg u$ with u a proper initial segment of ψ. By induction, u consists of symbols $\neg$ or $b(u) > 0$, and the same holds for $\neg u$.

If $\varphi = (\psi \to \chi)$, then $b(\varphi) = 1 + b(\psi) + b(\chi) - 1 = 0$. A nonempty proper initial segment of φ has one of the forms (u with u an initial segment of ψ, or ($\psi \to u$ with u an initial segment of χ (possibly empty or equal to χ). By induction every initial segment u of ψ or χ satisfies $b(u) \geq 0$, so these segments have $b \geq 1$. □

Theorem 2.2.5 (Unique readability)**.** *Every propositional formula φ satisfies exactly one of the following: φ is a propositional variable; $\varphi = \neg\psi$ for a formula ψ; $\varphi = (\psi \to \chi)$ for formulas ψ and χ. In the second case ψ is uniquely determined, and in the third case ψ and χ are uniquely determined.*

Proof. The three cases are distinguished by the first symbol of φ, and in the second case ψ is φ with its first symbol removed. Suppose $(\psi \to \chi) = (\psi' \to \chi')$ with $\psi \neq \psi'$. After the first symbol the two strings agree, so one of ψ, ψ' is a proper initial segment of the other, say ψ of ψ'. By Lemma 2.2.4, $b(\psi) = 0$, so ψ must consist only of symbols $\neg$. But every formula contains at least one propositional variable (by induction on formulas), a contradiction. Hence $\psi = \psi'$, and then $\chi = \chi'$. □

Definition 2.2.6. A *truth assignment* is a map $v\colon P \to \{0,1\}$. By Theorem 2.2.5, there is a unique map $\bar{v}$ from the set of formulas to $\{0,1\}$, defined by recursion on formulas, such that $\bar{v}(p) = v(p)$ for $p \in P$, $\bar{v}(\neg\varphi) = 1 - \bar{v}(\varphi)$, and $\bar{v}(\varphi \to \psi) = 0$ if and only if $\bar{v}(\varphi) = 1$ and $\bar{v}(\psi) = 0$. We say that v *satisfies* φ if $\bar{v}(\varphi) = 1$, and that v satisfies a set of formulas Σ if it satisfies every element of Σ.

Definition 2.2.7. A formula φ is a *tautology* if every truth assignment satisfies it. A set Σ of formulas *tautologically implies* φ, written $\Sigma \models \varphi$, if every truth assignment satisfying Σ satisfies φ. A set of formulas is *satisfiable* if some truth assignment satisfies it.

Lemma 2.2.8. *For every truth assignment v: $\bar{v}(\varphi \wedge \psi) = 1$ if and only if $\bar{v}(\varphi) = \bar{v}(\psi) = 1$; $\bar{v}(\varphi \vee \psi) = 1$ if and only if $\bar{v}(\varphi) = 1$ or $\bar{v}(\psi) = 1$; and $\bar{v}(\varphi \leftrightarrow \psi) = 1$ if and only if $\bar{v}(\varphi) = \bar{v}(\psi)$.*

Proof. $\bar{v}(\neg(\varphi \to \neg\psi)) = 1$ if and only if $\bar{v}(\varphi \to \neg\psi) = 0$, that is, $\bar{v}(\varphi) = 1$ and $\bar{v}(\neg\psi) = 0$. Next, $\bar{v}(\neg\varphi \to \psi) = 0$ if and only if $\bar{v}(\varphi) = 0$ and $\bar{v}(\psi) = 0$. The last statement follows from the first and the definition of $\bar{v}$ on implications. □

Example 2.2.9. The following are tautologies, as one checks by computing $\bar{v}$ for the finitely many possible values of v on the variables involved: $\varphi \vee \neg\varphi$ (excluded middle); $\neg\neg\varphi \leftrightarrow \varphi$; $((\varphi \to \psi) \to \varphi) \to \varphi$ (Peirce's law); $\neg(\varphi \wedge \psi) \leftrightarrow (\neg\varphi \vee \neg\psi)$ (De Morgan); $(\varphi \to (\psi \to \chi)) \to ((\varphi \to \psi) \to (\varphi \to \chi))$. By contrast, $(\varphi \to \psi) \to (\psi \to \varphi)$ is not a tautology when φ and ψ are distinct variables.

Proposition 2.2.10. *Let $p_1, \dots, p_n$ be distinct propositional variables, $n \geq 1$, and let $F\colon \{0,1\}^n \to \{0,1\}$ be any function. There is a formula φ whose variables are among $p_1, \dots, p_n$ such that $\bar{v}(\varphi) = F(v(p_1), \dots, v(p_n))$ for every truth assignment v.*

Proof. For $\varepsilon \in \{0,1\}^n$ let $\gamma_\varepsilon = \ell_1 \wedge \cdots \wedge \ell_n$, where $\ell_i = p_i$ if $\varepsilon_i = 1$ and $\ell_i = \neg p_i$ if $\varepsilon_i = 0$ (with parentheses grouped to the left). By Lemma 2.2.8, $\bar{v}(\gamma_\varepsilon) = 1$ if and only if $v(p_i) = \varepsilon_i$ for all i. If F is identically 0, take $\varphi = p_1 \wedge \neg p_1$. Otherwise let φ be the disjunction of the γ_ε over all ε with $F(\varepsilon) = 1$. Then $\bar{v}(\varphi) = 1$ if and only if $(v(p_1), \dots, v(p_n))$ is one of these ε, that is, if and only if $F(v(p_1), \dots, v(p_n)) = 1$. □

Counterexample 2.2.11. Proposition 2.2.10 fails if only $\wedge$ and $\vee$ may be used, taken as primitive connectives with the truth tables of Lemma 2.2.8. Indeed, by induction on such formulas, if $v(p) \leq w(p)$ for every variable p then $\bar{v}(\varphi) \leq \bar{w}(\varphi)$. So no such formula computes the function $F(x) = 1 - x$.

Theorem 2.2.12 (Compactness, propositional case)**.** *Let P be countable and let Σ be a set of formulas over P. If every finite subset of Σ is satisfiable, then Σ is satisfiable.*

Proof. Call a set of formulas *finitely satisfiable* if all its finite subsets are satisfiable. The formulas are finite strings over a countable set of symbols, so they can be enumerated as $\varphi_0, \varphi_1, \dots$. Put $\Delta_0 = \Sigma$, and let $\Delta_{n+1} = \Delta_n \cup \{\varphi_n\}$ if this set is finitely satisfiable, and $\Delta_{n+1} = \Delta_n \cup \{\neg\varphi_n\}$ otherwise.

Each Δ_n is finitely satisfiable. Suppose Δ_n is, but neither $\Delta_n \cup \{\varphi_n\}$ nor $\Delta_n \cup \{\neg\varphi_n\}$ is. Then there are finite $\Delta, \Delta' \subseteq \Delta_n$ such that $\Delta \cup \{\varphi_n\}$ and $\Delta' \cup \{\neg\varphi_n\}$ are not satisfiable. A truth assignment v satisfying $\Delta \cup \Delta'$ satisfies φ_n or $\neg\varphi_n$, hence one of the two sets; contradiction.

Let $\Delta = \bigcup_n \Delta_n$. Every finite subset of Δ lies in some Δ_n, so Δ is finitely satisfiable; in particular Δ never contains both ψ and $\neg\psi$. For every formula ψ, exactly one of $\psi, \neg\psi$ lies in Δ. Define $v(p) = 1$ if and only if $p \in \Delta$. We show by induction on ψ that $\bar{v}(\psi) = 1$ if and only if $\psi \in \Delta$. For variables this is the definition. For $\neg\psi$: $\bar{v}(\neg\psi) = 1$ iff $\psi \notin \Delta$ iff $\neg\psi \in \Delta$. For $\psi \to \chi$: if $(\psi \to \chi) \in \Delta$, $\psi \in \Delta$ and $\chi \notin \Delta$, then $\neg\chi \in \Delta$ and the finite set $\{\psi \to \chi, \psi, \neg\chi\}$ is not satisfiable. If $(\psi \to \chi) \notin \Delta$, then $\neg(\psi \to \chi) \in \Delta$; if moreover $\psi \notin \Delta$ or $\chi \in \Delta$, then Δ contains $\{\neg(\psi \to \chi), \neg\psi\}$ or $\{\neg(\psi \to \chi), \chi\}$, neither of which is satisfiable. So $(\psi \to \chi) \in \Delta$ if and only if $\psi \notin \Delta$ or $\chi \in \Delta$, which by induction is equivalent to $\bar{v}(\psi \to \chi) = 1$. Hence v satisfies $\Delta \supseteq \Sigma$. □

Remark 2.2.13. The theorem holds for arbitrary sets P, but the proof then needs a form of the axiom of choice (for instance Zorn's lemma, Chapter 3) to replace the enumeration of all formulas.

2.3 First-order languages

Definition 2.3.1. A *first-order language* $\mathcal{L}$ consists of a set of *constant symbols*, and for each $n \geq 1$ a set of n-ary *function symbols* and a set of n-ary *relation symbols*, all these sets being pairwise disjoint. The language is *countable* if the union of these sets is countable. Every language also has the *logical symbols*: a countably infinite set of *variables* $v_0, v_1, v_2, \dots$, disjoint from the other symbols, and the symbols $=$, $\neg$, $\to$, $\forall$.

Definition 2.3.2. Let $\mathcal{L}$ be a language. The set of *terms* is the smallest set that contains the variables and the constant symbols and contains the tuple $(f, t_1, \dots, t_n)$ whenever f is an n-ary function symbol and $t_1, \dots, t_n$ are terms. We write $f(t_1, \dots, t_n)$ for this tuple.

The *atomic formulas* are the tuples $(=, t_1, t_2)$ and $(R, t_1, \dots, t_n)$, where the t_i are terms and R is an n-ary relation symbol; we write $t_1 = t_2$ and $R(t_1, \dots, t_n)$. The set of *formulas* is the smallest set that contains the atomic formulas and contains $(\neg, \varphi)$, $(\to, \varphi, \psi)$ and $(\forall, x, \varphi)$ whenever φ, ψ are formulas and x is a variable. We write $\neg\varphi$, $(\varphi \to \psi)$ and $\forall x\, \varphi$ for these tuples.

Remark 2.3.3. We assume, as we may, that no symbol is itself a finite tuple of the above shapes. Since two tuples are equal only if they have the same length and the same entries, every term and every formula decomposes *uniquely* into its immediate constituents. This is the analogue of Theorem 2.2.5, obtained here for free: we have defined formulas as trees rather than strings. As in Remark 2.2.2, we can prove statements by induction on terms and formulas, and define functions on them by recursion. The *complexity* $c(\varphi)$ of a formula is the number of occurrences of $\neg$, $\to$ and $\forall$ in it: $c = 0$ on atomic formulas, $c(\neg\varphi) = c(\varphi) + 1$, $c(\varphi \to \psi) = c(\varphi) + c(\psi) + 1$ and $c(\forall x\, \varphi) = c(\varphi) + 1$.

In practice we write formulas as usual, for instance $x \cdot y = y \cdot x$ instead of $(=, (\cdot, x, y), (\cdot, y, x))$, and we use variables named $x, y, z, \dots$ for arbitrary distinct variables.

Notation 2.3.4. Besides the abbreviations of Notation 2.2.3, we write $\exists x\, \varphi$ for $\neg\forall x\, \neg\varphi$ and $t \neq s$ for $\neg(t = s)$.

Example 2.3.5.

1. The language of groups has a constant symbol e, a unary function symbol $^{-1}$ and a binary function symbol $\cdot$.
2. The language of rings has constant symbols $0, 1$ and binary function symbols $+, \cdot$ and a unary function symbol $-$.
3. The language of orders has a single binary relation symbol $<$.
4. The language of set theory has a single binary relation symbol $\in$.
5. The language of arithmetic $\mathcal{L}_A$ has a constant symbol 0, a unary function symbol S and binary function symbols $+$ and $\cdot$.

All these languages are countable.

Definition 2.3.6. The set $\mathrm{Var}(t)$ of variables of a term t is defined as usual. The set $\mathrm{FV}(\varphi)$ of *free variables* of a formula is defined by recursion: $\mathrm{FV}(t_1 = t_2) = \mathrm{Var}(t_1) \cup \mathrm{Var}(t_2)$, $\mathrm{FV}(R(t_1, \dots, t_n)) = \bigcup_i \mathrm{Var}(t_i)$, $\mathrm{FV}(\neg\varphi) = \mathrm{FV}(\varphi)$, $\mathrm{FV}(\varphi \to \psi) = \mathrm{FV}(\varphi) \cup \mathrm{FV}(\psi)$ and $\mathrm{FV}(\forall x\, \varphi) = \mathrm{FV}(\varphi) \setminus \{x\}$. A *sentence* is a formula without free variables. A variable *occurs* in φ if it appears anywhere in the tree φ, free or not.

Definition 2.3.7. Let x be a variable and t a term. For a term u, the term $u[t/x]$ is obtained by replacing every occurrence of x in u by t. For formulas, $\varphi[t/x]$ is defined by recursion: $(t_1 = t_2)[t/x] = (t_1[t/x] = t_2[t/x])$ and similarly for $R(t_1, \dots, t_n)$; $(\neg\varphi)[t/x] = \neg(\varphi[t/x])$; $(\varphi \to \psi)[t/x] = (\varphi[t/x] \to \psi[t/x])$; $(\forall y\, \varphi)[t/x] = \forall y\, \varphi$ if $y = x$, and $(\forall y\, \varphi)[t/x] = \forall y\, (\varphi[t/x])$ if $y \neq x$.

The term t is *substitutable* for x in φ if: φ is atomic; or $\varphi = \neg\psi$ and t is substitutable for x in ψ; or $\varphi = (\psi \to \chi)$ and t is substitutable for x in ψ and in χ; or $\varphi = \forall y\, \psi$ and either $x \notin \mathrm{FV}(\varphi)$, or y does not occur in t and t is substitutable for x in ψ.

Remark 2.3.8. Two facts follow at once by induction: if $x \notin \mathrm{FV}(\varphi)$ then $\varphi[t/x] = \varphi$; and substitution does not change the complexity, $c(\varphi[t/x]) = c(\varphi)$. A constant symbol, or more generally a term without variables, is substitutable for every variable in every formula.

Lemma 2.3.9. *Let x and y be variables, and suppose y does not occur in the formula φ. Then y is substitutable for x in φ, x is substitutable for y in $\varphi[y/x]$, and $\varphi[y/x][x/y] = \varphi$.*

Proof. For terms u not containing y, clearly $u[y/x][x/y] = u$. This gives the claims for atomic formulas, and the cases $\neg$ and $\rightarrow$ follow by induction. Let $\varphi = \forall z\, \psi$; then $z \neq y$ and y does not occur in ψ. The variable z does not occur in the term y, so by induction y is substitutable for x in φ. If $z = x$, then $\varphi[y/x] = \varphi$, in which y is not free, so x is substitutable for y in it and $\varphi[x/y] = \varphi$. If $z \neq x$, then $\varphi[y/x] = \forall z\, \psi[y/x]$; since z does not occur in the term x, the induction hypothesis shows that x is substitutable for y in $\varphi[y/x]$, and $\varphi[y/x][x/y] = \forall z\, \psi[y/x][x/y] = \forall z\, \psi$. □

2.4 Structures and satisfaction

Definition 2.4.1. A *structure* $\mathcal{A}$ for a language $\mathcal{L}$ consists of a nonempty set A, its *universe*, together with an element $c^{\mathcal{A}} \in A$ for every constant symbol c, a map $f^{\mathcal{A}} \colon A^n \to A$ for every n-ary function symbol f, and a subset $R^{\mathcal{A}} \subseteq A^n$ for every n-ary relation symbol R. An *assignment* in $\mathcal{A}$ is a map s from the set of variables to A. For an assignment s, a variable x and $a \in A$, we write $s[x \mapsto a]$ for the assignment that agrees with s except that it sends x to a.

Definition 2.4.2. Let $\mathcal{A}$ be a structure and s an assignment. The *value* $t^{\mathcal{A}}[s] \in A$ of a term is defined by $x^{\mathcal{A}}[s] = s(x)$, $c^{\mathcal{A}}[s] = c^{\mathcal{A}}$ and $f(t_1, \ldots, t_n)^{\mathcal{A}}[s] = f^{\mathcal{A}}(t_1^{\mathcal{A}}[s], \ldots, t_n^{\mathcal{A}}[s])$. The relation $\mathcal{A} \models \varphi[s]$, read "$\mathcal{A}$ satisfies φ under s", is defined by recursion on φ:

1. $\mathcal{A} \models (t_1 = t_2)[s]$ if and only if $t_1^{\mathcal{A}}[s] = t_2^{\mathcal{A}}[s]$;
2. $\mathcal{A} \models R(t_1, \ldots, t_n)[s]$ if and only if $(t_1^{\mathcal{A}}[s], \ldots, t_n^{\mathcal{A}}[s]) \in R^{\mathcal{A}}$;
3. $\mathcal{A} \models \neg\varphi[s]$ if and only if $\mathcal{A} \not\models \varphi[s]$;
4. $\mathcal{A} \models (\varphi \rightarrow \psi)[s]$ if and only if $\mathcal{A} \not\models \varphi[s]$ or $\mathcal{A} \models \psi[s]$;
5. $\mathcal{A} \models \forall x\, \varphi[s]$ if and only if $\mathcal{A} \models \varphi[s[x \mapsto a]]$ for every $a \in A$.

Remark 2.4.3. With the abbreviations of Notation 2.3.4, $\mathcal{A} \models \exists x\, \varphi[s]$ if and only if $\mathcal{A} \models \varphi[s[x \mapsto a]]$ for some $a \in A$, and the connectives $\wedge$, $\vee$, $\leftrightarrow$ have their usual meaning, by the same computation as in Lemma 2.2.8.

Lemma 2.4.4 (Coincidence lemma)**.** *Let s and s' be assignments in $\mathcal{A}$. If they agree on* $\mathrm{Var}(t)$*, then $t^{\mathcal{A}}[s] = t^{\mathcal{A}}[s']$. If they agree on* $\mathrm{FV}(\varphi)$*, then $\mathcal{A} \models \varphi[s]$ if and only if $\mathcal{A} \models \varphi[s']$.*

Proof. The statement for terms follows by induction on terms, and implies the statement for atomic formulas; the cases $\neg$ and $\rightarrow$ follow by induction. If $\varphi = \forall x\, \psi$ and s, s' agree on $\mathrm{FV}(\varphi) = \mathrm{FV}(\psi) \setminus \{x\}$, then for each $a \in A$ the assignments $s[x \mapsto a]$ and $s'[x \mapsto a]$ agree on $\mathrm{FV}(\psi)$, and the claim follows from the induction hypothesis. □

Notation 2.4.5. By Lemma 2.4.4, whether $\mathcal{A} \models \sigma[s]$ holds for a sentence σ does not depend on s; we then write $\mathcal{A} \models \sigma$ and say that σ is *true* in $\mathcal{A}$.

Lemma 2.4.6 (Substitution lemma)**.** *Let t be substitutable for x in φ, let s be an assignment in $\mathcal{A}$, and put $a = t^{\mathcal{A}}[s]$. Then $\mathcal{A} \models \varphi[t/x][s]$ if and only if $\mathcal{A} \models \varphi[s[x \mapsto a]]$. Also $u[t/x]^{\mathcal{A}}[s] = u^{\mathcal{A}}[s[x \mapsto a]]$ for every term u.*

Proof. The statement for terms follows by induction on u and gives the atomic case; the cases $\neg$ and $\to$ follow by induction. Let $\varphi = \forall y\, \psi$.

If $x \notin \mathrm{FV}(\varphi)$, then $\varphi[t/x] = \varphi$ (Remark 2.3.8), and s and $s[x \mapsto a]$ agree on $\mathrm{FV}(\varphi)$, so the claim follows from Lemma 2.4.4.

Otherwise $y \neq x$, y does not occur in t, and t is substitutable for x in ψ. Then $\varphi[t/x] = \forall y\, \psi[t/x]$. For $b \in A$, since y does not occur in t, we have $t^{\mathcal{A}}[s[y \mapsto b]] = a$, so by induction $\mathcal{A} \models \psi[t/x][s[y \mapsto b]]$ if and only if $\mathcal{A} \models \psi[s[y \mapsto b][x \mapsto a]]$. Since $x \neq y$, the assignment $s[y \mapsto b][x \mapsto a]$ equals $s[x \mapsto a][y \mapsto b]$. Quantifying over all $b \in A$ gives the claim. □

Counterexample 2.4.7. The hypothesis that t is substitutable cannot be dropped. Let φ be $\exists y\, (x \neq y)$ and $t = y$. In a structure with at least two elements, φ is true under every assignment, but $\varphi[y/x] = \exists y\, (y \neq y)$ is false: the variable y of t has been "captured" by the quantifier $\exists y$.

Definition 2.4.8. Let Γ be a set of formulas. A structure $\mathcal{A}$ with an assignment s *satisfies* Γ if $\mathcal{A} \models \gamma[s]$ for all $\gamma \in \Gamma$. The set Γ *logically implies* φ, written $\Gamma \models \varphi$, if every $(\mathcal{A}, s)$ satisfying Γ satisfies φ; φ is *valid* if $\emptyset \models \varphi$. A *theory* is a set of sentences. A *model* of a theory T is a structure in which all sentences of T are true. For a structure $\mathcal{A}$, the *theory of* $\mathcal{A}$, written $\mathrm{Th}(\mathcal{A})$, is the set of sentences true in $\mathcal{A}$.

Example 2.4.9. Let T_{grp} consist of the sentences $\forall x \forall y \forall z\, ((x \cdot y) \cdot z = x \cdot (y \cdot z))$, $\forall x\, (x \cdot e = x \wedge e \cdot x = x)$ and $\forall x\, (x \cdot x^{-1} = e \wedge x^{-1} \cdot x = e)$ in the language of groups. Its models are exactly the groups (with the inverse map and the identity element as interpretations of $^{-1}$ and e). Adding $\forall x \forall y\, (x \cdot y = y \cdot x)$ gives a theory whose models are the abelian groups.

Definition 2.4.10. An *isomorphism* of $\mathcal{L}$-structures $h \colon \mathcal{A} \to \mathcal{B}$ is a bijection $h \colon A \to B$ with $h(c^{\mathcal{A}}) = c^{\mathcal{B}}$, $h(f^{\mathcal{A}}(a_1, \ldots, a_n)) = f^{\mathcal{B}}(h(a_1), \ldots, h(a_n))$ and $(a_1, \ldots, a_n) \in R^{\mathcal{A}}$ if and only if $(h(a_1), \ldots, h(a_n)) \in R^{\mathcal{B}}$, for all symbols and all $a_i \in A$.

Proposition 2.4.11. *Let $h \colon \mathcal{A} \to \mathcal{B}$ be an isomorphism and s an assignment in $\mathcal{A}$. Then $\mathcal{A} \models \varphi[s]$ if and only if $\mathcal{B} \models \varphi[h \circ s]$. In particular, isomorphic structures have the same theory.*

Proof. By induction on terms, $h(t^{\mathcal{A}}[s]) = t^{\mathcal{B}}[h \circ s]$. Since h is injective and preserves relations in both directions, the atomic case follows; the cases $\neg$, $\to$ follow by induction. For $\forall x\, \psi$, note that $h \circ (s[x \mapsto a]) = (h \circ s)[x \mapsto h(a)]$ and that $a \mapsto h(a)$ is a bijection $A \to B$, so quantifying over $a \in A$ is the same as quantifying over $b \in B$. □

2.5 Formal deductions

We now define formal proofs. The system below, a Hilbert-style system with modus ponens as its only rule, is the one of [End01].

Definition 2.5.1. A formula is *prime* if it is atomic or of the form $\forall x\, \varphi$. Every formula is built from prime formulas using $\neg$ and $\to$ alone. A *truth assignment* for $\mathcal{L}$ is a map v from the set of prime formulas to $\{0, 1\}$; it extends to $\bar{v}$ on all formulas as in Definition 2.2.6. A formula is a *tautology* if $\bar{v}(\varphi) = 1$ for all v, and a set Σ *tautologically implies* φ if $\bar{v}(\varphi) = 1$ for all v with $\bar{v}(\Sigma) = \{1\}$.

Definition 2.5.2. A *generalization* of a formula α is a formula $\forall x_1 \cdots \forall x_n\, \alpha$ with $n \geq 0$. The *logical axioms* are all generalizations of formulas of the following forms, where x, y, z are variables, t is a term and α, β are formulas:

(A1) tautologies;

(A2) $\forall x\, \alpha \to \alpha[t/x]$, where t is substitutable for x in α;

(A3) $\forall x\, (\alpha \to \beta) \to (\forall x\, \alpha \to \forall x\, \beta)$;

(A4) $\alpha \to \forall x\, \alpha$, where $x \notin \mathrm{FV}(\alpha)$;

(A5) $x = x$;

(A6) $x = y \to (\alpha[x/z] \to \alpha[y/z])$, where α is atomic.

Definition 2.5.3. Let Γ be a set of formulas. A *deduction* from Γ is a finite sequence $(\alpha_0, \ldots, \alpha_n)$ of formulas such that each α_i is a logical axiom, or belongs to Γ, or is obtained by *modus ponens* from earlier formulas, that is, there are $j, k < i$ with $\alpha_k = (\alpha_j \to \alpha_i)$. We write $\Gamma \vdash \varphi$, and say that φ is a *theorem* of Γ, if there is a deduction from Γ whose last formula is φ. The set Γ is *inconsistent* if there is a formula β with $\Gamma \vdash \beta$ and $\Gamma \vdash \neg\beta$, and *consistent* otherwise.

Remark 2.5.4. A deduction is finite, so it uses only finitely many formulas of Γ. Hence $\Gamma \vdash \varphi$ if and only if $\Gamma_0 \vdash \varphi$ for some finite $\Gamma_0 \subseteq \Gamma$, and Γ is consistent if and only if all its finite subsets are. Whether $\Gamma \vdash \varphi$ holds could depend on the language in which deductions are carried out; see Lemma 2.5.12 below.

Lemma 2.5.5 (Rule T)**.** *If $\Gamma \vdash \alpha_1, \ldots, \Gamma \vdash \alpha_n$ and $\{\alpha_1, \ldots, \alpha_n\}$ tautologically implies β, then $\Gamma \vdash \beta$. In particular, if Γ is inconsistent, then $\Gamma \vdash \beta$ for every formula β.*

Proof. The formula $\alpha_1 \to (\alpha_2 \to \cdots(\alpha_n \to \beta)\cdots)$ is a tautology, hence a logical axiom. Applying modus ponens n times gives $\Gamma \vdash \beta$. For the second claim, $\{\gamma, \neg\gamma\}$ tautologically implies every formula. □

Theorem 2.5.6 (Soundness)**.** *If $\Gamma \vdash \varphi$, then $\Gamma \models \varphi$. In particular, a set of formulas that is satisfied by some structure and assignment is consistent.*

Proof. We first show that every logical axiom is valid. If α is valid, so is every generalization of α, by the definition of satisfaction. Fix $\mathcal{A}$ and s. (A1) Let $v(\pi) = 1$ if and only if $\mathcal{A} \models \pi[s]$, for prime π. By induction, $\bar{v}(\alpha) = 1$ if and only if $\mathcal{A} \models \alpha[s]$; so tautologies are satisfied. (A2) If $\mathcal{A} \models \forall x\, \alpha[s]$ then $\mathcal{A} \models \alpha[s[x \mapsto t^{\mathcal{A}}[s]]]$, which by Lemma 2.4.6 means $\mathcal{A} \models \alpha[t/x][s]$. (A3) is immediate from the definitions. (A4) If $x \notin \mathrm{FV}(\alpha)$, then s and $s[x \mapsto a]$ agree on $\mathrm{FV}(\alpha)$, so this follows from Lemma 2.4.4. (A5) is clear. (A6) If $s(x) = s(y) = a$, then by Lemma 2.4.6 both $\alpha[x/z]$ and $\alpha[y/z]$ are satisfied under s if and only if α is satisfied under $s[z \mapsto a]$.

Now let $(\alpha_0, \ldots, \alpha_n)$ be a deduction from Γ and let $(\mathcal{A}, s)$ satisfy Γ. By induction on i, $\mathcal{A} \models \alpha_i[s]$: this holds for axioms and elements of Γ, and if $\mathcal{A} \models \alpha_j[s]$ and $\mathcal{A} \models (\alpha_j \to \alpha_i)[s]$ then $\mathcal{A} \models \alpha_i[s]$. For the last claim, no $(\mathcal{A}, s)$ satisfies both β and $\neg\beta$. □

Theorem 2.5.7 (Generalization theorem)**.** *If $\Gamma \vdash \varphi$ and x is not free in any formula of Γ, then $\Gamma \vdash \forall x\, \varphi$.*

Proof. We show by induction on i that $\Gamma \vdash \forall x\, \alpha_i$ for every formula α_i of a deduction $(\alpha_0, \ldots, \alpha_n)$ from Γ. If α_i is a logical axiom, so is its generalization $\forall x\, \alpha_i$. If $\alpha_i \in \Gamma$, then $x \notin \mathrm{FV}(\alpha_i)$, so $\alpha_i \to \forall x\, \alpha_i$ is an instance of (A4), and modus ponens gives $\Gamma \vdash \forall x\, \alpha_i$. If α_i is obtained from α_j and $\alpha_j \to \alpha_i$ by modus ponens, then by induction $\Gamma \vdash \forall x\, \alpha_j$ and $\Gamma \vdash \forall x\, (\alpha_j \to \alpha_i)$; using the instance $\forall x\, (\alpha_j \to \alpha_i) \to (\forall x\, \alpha_j \to \forall x\, \alpha_i)$ of (A3) and modus ponens twice, $\Gamma \vdash \forall x\, \alpha_i$. □

Theorem 2.5.8 (Deduction theorem)**.** *$\Gamma \cup \{\gamma\} \vdash \varphi$ if and only if $\Gamma \vdash \gamma \to \varphi$.*

Proof. If $\Gamma \vdash \gamma \to \varphi$, then $\Gamma \cup \{\gamma\}$ proves both $\gamma \to \varphi$ and γ, hence φ. Conversely, let $(\alpha_0, \ldots, \alpha_n)$ be a deduction from $\Gamma \cup \{\gamma\}$; we show $\Gamma \vdash \gamma \to \alpha_i$ by induction on i. If $\alpha_i = \gamma$, then $\gamma \to \gamma$ is a tautology. If α_i is a logical axiom or lies in Γ, then $\Gamma \vdash \alpha_i$, and α_i tautologically implies $\gamma \to \alpha_i$; apply Lemma 2.5.5. If α_i comes from α_j and $\alpha_j \to \alpha_i$ by modus ponens, then by induction Γ proves $\gamma \to \alpha_j$ and $\gamma \to (\alpha_j \to \alpha_i)$, and these two formulas tautologically imply $\gamma \to \alpha_i$. □

Corollary 2.5.9 (Reductio ad absurdum)**.** *If $\Gamma \cup \{\neg\varphi\}$ is inconsistent, then $\Gamma \vdash \varphi$. If $\Gamma \cup \{\varphi\}$ is inconsistent, then $\Gamma \vdash \neg\varphi$.*

Proof. If $\Gamma \cup \{\neg\varphi\}$ proves β and $\neg\beta$, then by Theorem 2.5.8 Γ proves $\neg\varphi \to \beta$ and $\neg\varphi \to \neg\beta$, which tautologically imply φ. The second statement is proved in the same way, using that $\{\varphi \to \beta, \varphi \to \neg\beta\}$ tautologically implies $\neg\varphi$. □

For a constant symbol c and a variable y, let $\alpha[y/c]$ denote the result of replacing every occurrence of c in α by y (defined by recursion like substitution, but also inside the scope of quantifiers).

Lemma 2.5.10. *Let $(\alpha_0, \ldots, \alpha_n)$ be a deduction from Γ, let c be a constant symbol that does not occur in any formula of Γ, and let y be a variable that does not occur in any α_i. Then $(\alpha_0[y/c], \ldots, \alpha_n[y/c])$ is a deduction from Γ.*

Proof. Elements of Γ are unchanged, and replacement commutes with $\to$, so applications of modus ponens remain applications of modus ponens. It remains to see that the replacement of a logical axiom α is a logical axiom. Replacement commutes with generalization, so we may assume α is of one of the forms (A1)–(A6). Note that y does not occur in α.

(A1) Let w be a truth assignment, and define $v(\pi) = w(\pi[y/c])$ for prime π; replacement sends prime formulas to prime formulas and commutes with $\neg$ and $\to$, so $\bar{v}(\alpha) = \bar{w}(\alpha[y/c])$. Hence if α is a tautology so is $\alpha[y/c]$. (A2) By induction on formulas, $\beta[t/x][y/c] = \beta[y/c][t[y/c]/x]$. The term $t[y/c]$ is substitutable for x in $\beta[y/c]$: the only variable it may contain besides those of t is y, and y is not quantified anywhere in β. (A3), (A5) are clear, and (A4) is preserved because the variable x occurs in α, so $x \neq y$, and replacing a constant by y does not make x free. (A6) Write the axiom as $x = x' \to (\beta[x/z] \to \beta[x'/z])$ with β atomic; renaming z if necessary we may assume $z \neq y$. Then its replacement is $x = x' \to (\beta[y/c][x/z] \to \beta[y/c][x'/z])$, again an instance of (A6). □

Corollary 2.5.11 (Generalization on constants)**.** *Suppose $\Gamma \vdash \varphi[c/x]$, where the constant symbol c occurs neither in Γ nor in φ. Then $\Gamma \vdash \forall x\, \varphi$.*

Proof. Choose a deduction of $\varphi[c/x]$ from Γ, and a variable $y \neq x$ that occurs neither in this deduction nor in φ. Since c does not occur in φ, we have $\varphi[c/x][y/c] = \varphi[y/x]$. By Lemma 2.5.10, there is a deduction of $\varphi[y/x]$ from a finite subset $\Gamma_0 \subseteq \Gamma$ in whose formulas y does not occur. By Theorem 2.5.7, $\Gamma_0 \vdash \forall y\, \varphi[y/x]$.

By Lemma 2.3.9, x is substitutable for y in $\varphi[y/x]$ and $\varphi[y/x][x/y] = \varphi$, so $\forall y\, \varphi[y/x] \to \varphi$ is an instance of (A2). Hence $\{\forall y\, \varphi[y/x]\} \vdash \varphi$. The variable x is not free in $\forall y\, \varphi[y/x]$, so Theorem 2.5.7 and Theorem 2.5.8 give $\vdash \forall y\, \varphi[y/x] \to \forall x\, \varphi$. By modus ponens, $\Gamma \vdash \forall x\, \varphi$. □

Lemma 2.5.12. *Let $\mathcal{L}'$ be obtained from $\mathcal{L}$ by adding new constant symbols. If $\Gamma \cup \{\varphi\}$ is a set of $\mathcal{L}$-formulas and $\Gamma \vdash \varphi$ by a deduction in $\mathcal{L}'$, then $\Gamma \vdash \varphi$ by a deduction in $\mathcal{L}$. In particular, a set of $\mathcal{L}$-formulas that is consistent in $\mathcal{L}$ is consistent in $\mathcal{L}'$.*

Proof. A deduction in $\mathcal{L}'$ contains finitely many new constants. Replace them one at a time by distinct variables not occurring in the deduction. By Lemma 2.5.10 (applied in $\mathcal{L}'$), the result is a deduction from Γ of φ, which is unchanged because it contains no new constants.

All its formulas are $\mathcal{L}$-formulas, and an $\mathcal{L}$-formula that is a logical axiom of $\mathcal{L}'$ is a logical axiom of $\mathcal{L}$ (for (A1), every prime $\mathcal{L}$-formula is a prime $\mathcal{L}'$-formula). The last statement follows by applying the first to $\varphi = (x = x)$ and $\varphi = \neg(x = x)$, using Lemma 2.5.5. □

Lemma 2.5.13 (Equality)**.** *For all terms $t, t', t'', t_1, \dots, t_n, s_1, \dots, s_n$:*

1. $\vdash t = t$;
2. $\{t = t'\} \vdash t' = t$;
3. $\{t = t', t' = t''\} \vdash t = t''$;
4. $\{t_1 = s_1, \dots, t_n = s_n\} \vdash f(t_1, \dots, t_n) = f(s_1, \dots, s_n)$ *for every n-ary function symbol f;*
5. $\{t_1 = s_1, \dots, t_n = s_n, R(t_1, \dots, t_n)\} \vdash R(s_1, \dots, s_n)$ *for every n-ary relation symbol R.*

Proof. We use the following observation. If $\vdash \beta$ where β has no quantifiers, and $x_1, \dots, x_k$ are distinct variables not occurring in the terms $u_1, \dots, u_k$, then $\vdash \beta[u_1/x_1] \cdots [u_k/x_k]$. Indeed, by Theorem 2.5.7, $\vdash \forall x_1 \cdots \forall x_k \, \beta$, and we remove the quantifiers one at a time with (A2): each term u_i is substitutable because it contains none of the quantified variables and the remaining formula has no other quantifiers. In each part below, the variables used are chosen distinct and not occurring in the terms of the statement.

(1) Apply the observation to the axiom $x = x$. (2) Let α be $z = x$. The instance $x = y \to (x = x \to y = x)$ of (A6), with $x := t$ and $y := t'$, together with (1) gives the claim. (3) Let α be $z = w$. The instance $x = y \to (x = w \to y = w)$ of (A6), with $x := t'$, $y := t$, $w := t''$, gives $\vdash t' = t \to (t' = t'' \to t = t'')$; combine with (2). (4) For $0 \le k \le n$ let $a_k = f(y_1, \dots, y_k, x_{k+1}, \dots, x_n)$. With α the atomic formula $f(x_1, \dots, x_n) = f(y_1, \dots, y_{k-1}, z, x_{k+1}, \dots, x_n)$, axiom (A6) gives $\vdash x_k = y_k \to (f(x_1, \dots, x_n) = a_{k-1} \to f(x_1, \dots, x_n) = a_k)$. Starting from $f(x_1, \dots, x_n) = a_0$, which holds by (1), we obtain $\{x_1 = y_1, \dots, x_n = y_n\} \vdash f(x_1, \dots, x_n) = a_n$. By the deduction theorem this becomes a deducible quantifier-free implication, and the observation with $x_i := t_i$, $y_i := s_i$ gives the claim. (5) is proved in the same way as (4), with α the atomic formula $R(y_1, \dots, y_{k-1}, z, x_{k+1}, \dots, x_n)$. □

Lemma 2.5.14 (Alphabetic variants)**.** *Let φ be a formula, x a variable and t a term. There is a formula φ' with $c(\varphi') = c(\varphi)$ such that t is substitutable for x in φ', $\vdash \varphi \to \varphi'$ and $\vdash \varphi' \to \varphi$.*

Proof. By induction on φ. For atomic φ take $\varphi' = \varphi$. For $\neg\psi$ and $\psi \to \chi$, take $\neg\psi'$ and $\psi' \to \chi'$ and use Lemma 2.5.5. Let $\varphi = \forall y \, \psi$, and let ψ' be given by induction. Choose a variable z that does not occur in ψ' or in t and is different from x, and put $\varphi' = \forall z \, \psi'[z/y]$.

The term t is substitutable for x in φ'. If $x = y$, then x is not free in φ'. If $x \neq y$, then z does not occur in t, and t is substitutable for x in $\psi'[z/y]$ because it is substitutable in ψ' and $\psi'[z/y]$ has the same quantifiers as ψ' in the same places.

From $\vdash \psi \to \psi'$, Theorem 2.5.7 and (A3) give $\vdash \forall y \, \psi \to \forall y \, \psi'$, and similarly $\vdash \forall y \, \psi' \to \forall y \, \psi$. By Lemma 2.3.9, $\forall y \, \psi' \to \psi'[z/y]$ and $\forall z \, \psi'[z/y] \to \psi'$ are instances of (A2). The variable z is not free in $\forall y \, \psi'$, and y is not free in $\forall z \, \psi'[z/y]$; so by Theorems 2.5.7 and 2.5.8, $\vdash \forall y \, \psi' \to \forall z \, \psi'[z/y]$ and $\vdash \forall z \, \psi'[z/y] \to \forall y \, \psi'$. Combining these implications with Lemma 2.5.5 gives the claim. Finally $c(\varphi') = c(\psi') + 1 = c(\varphi)$. □

Example 2.5.15. Let us check that $\vdash \forall x \forall y \, \varphi \to \forall y \forall x \, \varphi$. By (A2) twice (a variable is substitutable for itself), $\{\forall x \forall y \, \varphi\} \vdash \varphi$. Neither x nor y is free in $\forall x \forall y \, \varphi$, so the generalization theorem gives $\{\forall x \forall y \, \varphi\} \vdash \forall y \forall x \, \varphi$, and the deduction theorem gives the claim. In practice, one rarely writes deductions: by Corollary 2.6.2 below it suffices to check that the implication is valid.

2.6 The completeness theorem

Theorem 2.6.1 (Gödel, Henkin)**.** *Let $\mathcal{L}$ be a countable language and Γ a consistent set of $\mathcal{L}$-formulas. Then there are an $\mathcal{L}$-structure $\mathcal{A}$ with a countable universe and an assignment s such that $(\mathcal{A}, s)$ satisfies Γ.*

Proof. This is the method of [Hen49]; the theorem is due to [Gö30].

Step 1: new constants. Let $\mathcal{L}'$ be $\mathcal{L}$ together with new constant symbols $c_0, c_1, \ldots$. It is still countable, and Γ is consistent in $\mathcal{L}'$ by Lemma 2.5.12. From now on all formulas and deductions are in $\mathcal{L}'$.

Step 2: witnesses. Enumerate all pairs of a formula and a variable as $(\varphi_1, x_1), (\varphi_2, x_2), \ldots$. Define inductively

$$\theta_n = \big(\neg\forall x_n\, \varphi_n \to \neg\varphi_n[d_n/x_n]\big),$$

where d_n is the first of the new constants c_i that occurs neither in φ_n nor in $\theta_1, \ldots, \theta_{n-1}$. Let $\Theta = \{\theta_1, \theta_2, \ldots\}$. We claim that $\Gamma \cup \Theta$ is consistent. Otherwise, by Remark 2.5.4, there is a least m such that $\Gamma \cup \{\theta_1, \ldots, \theta_m\}$ is inconsistent, and $m \geq 1$. By Corollary 2.5.9, $\Gamma \cup \{\theta_1, \ldots, \theta_{m-1}\} \vdash \neg\theta_m$. The formula $\neg\theta_m$ tautologically implies both $\neg\forall x_m\, \varphi_m$ and $\varphi_m[d_m/x_m]$. The constant d_m occurs neither in Γ (whose formulas are in $\mathcal{L}$), nor in $\theta_1, \ldots, \theta_{m-1}$, nor in φ_m; so Corollary 2.5.11 gives $\Gamma \cup \{\theta_1, \ldots, \theta_{m-1}\} \vdash \forall x_m\, \varphi_m$. This set is therefore inconsistent, contradicting the minimality of m.

Step 3: a maximal consistent set. Enumerate all formulas as $\psi_0, \psi_1, \ldots$. Put $\Delta_0 = \Gamma \cup \Theta$, and $\Delta_{n+1} = \Delta_n \cup \{\psi_n\}$ if this set is consistent, $\Delta_{n+1} = \Delta_n \cup \{\neg\psi_n\}$ otherwise. If Δ_n is consistent, so is Δ_{n+1}: if both $\Delta_n \cup \{\psi_n\}$ and $\Delta_n \cup \{\neg\psi_n\}$ were inconsistent, then Δ_n would prove $\neg\psi_n$ and ψ_n by Corollary 2.5.9. Let $\Delta = \bigcup_n \Delta_n$. It is consistent (a deduction uses finitely many formulas), and for every formula ψ either $\psi \in \Delta$ or $\neg\psi \in \Delta$, but not both. Moreover, if $\Delta \vdash \psi$ then $\psi \in \Delta$: otherwise $\neg\psi \in \Delta$ and Δ would be inconsistent.

Step 4: the term structure. Let T be the set of terms of $\mathcal{L}'$; it is countable. For $t, t' \in T$ write $t \sim t'$ if $(t = t') \in \Delta$. Since Δ contains every formula it proves, Lemma 2.5.13 shows that $\sim$ is an equivalence relation and that

- if $t_i \sim s_i$ for all i, then $f(t_1, \ldots, t_n) \sim f(s_1, \ldots, s_n)$, and $R(t_1, \ldots, t_n) \in \Delta$ if and only if $R(s_1, \ldots, s_n) \in \Delta$.

Let $A = T/\!\sim$ and write $[t]$ for the class of t. Define an $\mathcal{L}'$-structure $\mathcal{A}$ with universe A by $c^{\mathcal{A}} = [c]$, $f^{\mathcal{A}}([t_1], \ldots, [t_n]) = [f(t_1, \ldots, t_n)]$ and $([t_1], \ldots, [t_n]) \in R^{\mathcal{A}}$ if and only if $R(t_1, \ldots, t_n) \in \Delta$; these are well defined by the displayed property. Let $s(x) = [x]$. By induction on terms, $t^{\mathcal{A}}[s] = [t]$ for every term t.

Step 5: the truth lemma. We show that for every formula φ,

$$\mathcal{A} \models \varphi[s] \iff \varphi \in \Delta,$$

by induction on the complexity $c(\varphi)$. For $t_1 = t_2$ both sides say $[t_1] = [t_2]$; for $R(t_1, \ldots, t_n)$ both sides say $R(t_1, \ldots, t_n) \in \Delta$. For $\neg\psi$: $\mathcal{A} \models \neg\psi[s]$ iff $\psi \notin \Delta$ iff $\neg\psi \in \Delta$. For $\psi \to \chi$: if $(\psi \to \chi) \in \Delta$ and $\psi \in \Delta$, then $\Delta \vdash \chi$, so $\chi \in \Delta$; conversely, if $\psi \notin \Delta$ then $\neg\psi \in \Delta$, and if $\chi \in \Delta$, in either case Δ proves $\psi \to \chi$ by Lemma 2.5.5. Hence $(\psi \to \chi) \in \Delta$ iff $\psi \notin \Delta$ or $\chi \in \Delta$, iff $\mathcal{A} \models (\psi \to \chi)[s]$.

Let $\varphi = \forall x\, \psi$. Suppose $\varphi \notin \Delta$. The pair (ψ, x) is (φ_n, x_n) for some n, and $\neg\forall x\, \psi \in \Delta$ and $\theta_n \in \Delta$, so $\neg\psi[d_n/x] \in \Delta$ and $\psi[d_n/x] \notin \Delta$. Since $c(\psi[d_n/x]) = c(\psi) < c(\varphi)$, the induction hypothesis gives $\mathcal{A} \not\models \psi[d_n/x][s]$, which by Lemma 2.4.6 means $\mathcal{A} \not\models \psi[s[x \mapsto [d_n]]]$. Hence $\mathcal{A} \not\models \varphi[s]$.

Suppose $\varphi \in \Delta$, and let $a \in A$, say $a = [t]$. Let ψ' be as in Lemma 2.5.14 for ψ, x and t. From $\vdash \psi \to \psi'$ we get $\vdash \forall x\, \psi \to \forall x\, \psi'$ (generalization and (A3)), so $\forall x\, \psi' \in \Delta$,

and by (A2) $\psi'[t/x] \in \Delta$. Since $c(\psi'[t/x]) = c(\psi) < c(\varphi)$, the induction hypothesis gives $\mathcal{A} \models \psi'[t/x][s]$, that is, $\mathcal{A} \models \psi'[s[x \mapsto a]]$ by Lemma 2.4.6. As $\vdash \psi' \to \psi$, Theorem 2.5.6 gives $\mathcal{A} \models \psi[s[x \mapsto a]]$. Since a was arbitrary, $\mathcal{A} \models \varphi[s]$.

Step 6: conclusion. Since $\Gamma \subseteq \Delta$, the pair $(\mathcal{A}, s)$ satisfies Γ. Forgetting the interpretations of the new constants gives an $\mathcal{L}$-structure which, with the same assignment, still satisfies Γ, because the satisfaction of an $\mathcal{L}$-formula only involves the symbols of $\mathcal{L}$. Its universe A is countable, as a quotient of T. □

Corollary 2.6.2 (Completeness theorem)**.** *Let $\mathcal{L}$ be countable. Then $\Gamma \models \varphi$ if and only if $\Gamma \vdash \varphi$.*

Proof. One direction is Theorem 2.5.6. Suppose $\Gamma \nvdash \varphi$. By Corollary 2.5.9, $\Gamma \cup \{\neg\varphi\}$ is consistent, so by Theorem 2.6.1 it is satisfied by some $(\mathcal{A}, s)$. Then $(\mathcal{A}, s)$ satisfies Γ but not φ, so $\Gamma \not\models \varphi$. □

Remark 2.6.3. The completeness theorem holds for languages of any cardinality, and the structure can then be taken of cardinality at most that of the language plus $\aleph_0$. The proof is the same, except that the enumerations of Steps 2 and 3 are replaced by transfinite ones, which requires the well-ordering theorem (Chapter 3); see [CK90, Section 2.2].

2.7 Compactness and the Löwenheim–Skolem theorem

Theorem 2.7.1 (Compactness theorem)**.** *Let $\mathcal{L}$ be countable and Γ a set of formulas. If every finite subset of Γ is satisfiable, then Γ is satisfiable.*

Proof. If Γ is not satisfiable, it is inconsistent by Theorem 2.6.1, so some finite subset Γ_0 is inconsistent (Remark 2.5.4), and Γ_0 is not satisfiable by Theorem 2.5.6. □

Theorem 2.7.2 (Löwenheim–Skolem)**.** *Let $\mathcal{L}$ be countable. Every satisfiable set of $\mathcal{L}$-formulas is satisfied by a structure with a countable (finite or countably infinite) universe. In particular, a theory with a model has a countable model.*

Proof. A satisfiable set is consistent by Theorem 2.5.6, and Theorem 2.6.1 produces a countable structure. □

Proposition 2.7.3. *Let $\mathcal{L}$ be countable and T a theory that has finite models of arbitrarily large size. Then T has an infinite model.*

Proof. For $n \geq 2$ let λ_n be the sentence $\exists x_1 \cdots \exists x_n \bigwedge_{1 \leq i < j \leq n} x_i \neq x_j$, which is true in a structure if and only if it has at least n elements. Every finite subset of $T \cup \{\lambda_n : n \geq 2\}$ is contained in $T \cup \{\lambda_2, \ldots, \lambda_N\}$ for some N, which is satisfied by a finite model of T with at least N elements. By Theorem 2.7.1, $T \cup \{\lambda_n : n \geq 2\}$ has a model, which is an infinite model of T. □

Counterexample 2.7.4. "Being finite" cannot be expressed in first-order logic: in any countable language there is no theory whose models are exactly the finite structures. Indeed, such a theory would have finite models of every size (in any language, every nonempty finite set carries a structure), hence an infinite model by Proposition 2.7.3.

Example 2.7.5. Let $\mathcal{N}$ be the structure $(\mathbb{N}, 0, S, +, \cdot)$ for the language of arithmetic, with $S(n) = n + 1$, and for $n \in \mathbb{N}$ let $\underline{n} = S(S(\cdots S(0)\cdots))$ with n symbols S. Add a new constant symbol c and consider $\mathrm{Th}(\mathcal{N}) \cup \{c \neq \underline{n} : n \in \mathbb{N}\}$. Every finite subset is satisfied by $\mathcal{N}$, interpreting c as a large enough number. By compactness there is a model $\mathcal{M}$ of $\mathrm{Th}(\mathcal{N})$ with an element $c^{\mathcal{M}}$ different from the values of all $\underline{n}$. Such a model is not isomorphic to $\mathcal{N}$ (in $\mathcal{N}$ every element is the value of some $\underline{n}$, and isomorphisms preserve the values of closed terms), although it satisfies exactly the same sentences. So first-order sentences cannot characterise $\mathcal{N}$ up to isomorphism.

Remark 2.7.6. Applied to set theory (Chapter 3), the Löwenheim–Skolem theorem says that if ZFC has a model, it has a countable model $\mathcal{M}$, even though ZFC proves that uncountable sets exist. There is no contradiction: an element of $\mathcal{M}$ that $\mathcal{M}$ believes to be uncountable has only countably many $\mathcal{M}$-elements, but no bijection with $\mathbb{N}$ exists *inside* $\mathcal{M}$. This is known as Skolem's paradox.

Remark 2.7.7. With the axiom of choice these results extend to languages and structures of any cardinality: Chapter 5 proves the compactness theorem for all languages, the downward and upward Löwenheim–Skolem theorems (a theory with an infinite model has models of every infinite cardinality at least that of the language), and Lindström's theorem, which characterises first-order logic by its compactness and Löwenheim–Skolem properties.

2.8 The incompleteness theorems

The completeness theorem says that the deductive calculus captures logical consequence: whatever is true in all models of a theory can be proved from it. The incompleteness theorems say something quite different, about *theories*: no reasonable theory of arithmetic decides every arithmetic sentence. The word "complete" is used in two senses here, and it is worth keeping them apart (Remark 2.8.5 below).

The incompleteness theorems are quoted without proof: their proofs require a precise theory of computability, which is not developed in this project. Complete proofs can be found in [Smi13] and [Sho67, Chapter 6]. We explain what they say, what they do not say, and why computability enters.

Definition 2.8.1. *Robinson arithmetic* Q is the theory in the language of arithmetic consisting of the sentences

$$\begin{aligned}
&\forall x\,(S(x) \neq 0), && \forall x \forall y\,(S(x) = S(y) \to x = y),\\
&\forall x\,(x \neq 0 \to \exists y\,(x = S(y))), && \forall x\,(x + 0 = x),\\
&\forall x \forall y\,(x + S(y) = S(x + y)), && \forall x\,(x \cdot 0 = 0),\\
&\forall x \forall y\,(x \cdot S(y) = x \cdot y + x).
\end{aligned}$$

Peano arithmetic PA consists of Q together with, for every formula φ whose free variables are among $x, y_1, \ldots, y_k$, the induction axiom $\forall y_1 \cdots \forall y_k \big((\varphi[0/x] \wedge \forall x\,(\varphi \to \varphi[S(x)/x])) \to \forall x\,\varphi\big)$.

Definition 2.8.2. A theory T is *complete* if for every sentence σ, either $T \vdash \sigma$ or $T \vdash \neg\sigma$. It is *computably axiomatized* if there is an algorithm deciding, for any sentence, whether it belongs to T.

Example 2.8.3. For every structure $\mathcal{A}$, the theory $\mathrm{Th}(\mathcal{A})$ is complete and consistent. For the structure $\mathcal{N}$ of Example 2.7.5, Theorem 2.8.7 implies that $\mathrm{Th}(\mathcal{N})$ is not computably axiomatized.

Example 2.8.4.

1. A finite theory, such as Q or the theory of groups, is computably axiomatized: to decide whether a sentence belongs to it, compare it with the finitely many axioms.

2. PA is infinite but computably axiomatized: a sentence belongs to PA if and only if it is one of the seven axioms of Q or has the shape of an induction axiom, and checking the shape is a mechanical inspection of the string of symbols.

3. The theory of algebraically closed fields of characteristic 0 (the field axioms, the sentences χ_p of Exercise 2.9.5, and for each $n \geq 1$ the sentence saying that every monic polynomial of degree n has a root) is computably axiomatized and infinite. It is also complete, by the Łoś–Vaught test (Chapter 5). This does not contradict the incompleteness theorem below, which concerns theories in the language of arithmetic that contain Q.

Remark 2.8.5. *Two meanings of "complete".* The *completeness theorem* (Corollary 2.6.2) is about the deductive calculus: $T \vdash \sigma$ if and only if σ is true in every model of T. A *complete theory* (Definition 2.8.2) is one that decides every sentence. The two notions are linked as follows: by the completeness theorem, a consistent theory T (in a countable language) fails to decide a sentence σ exactly when $T \nvdash \sigma$ and $T \nvdash \neg\sigma$, that is, when σ is true in some model of T and false in another. So T is complete if and only if all its models satisfy the same sentences.

Proposition 2.8.6. *Let $\mathcal{L}$ be countable and T a consistent theory. Then T is complete if and only if any two models of T satisfy the same sentences. If T is complete and computably axiomatized, then there is an algorithm deciding, for a sentence σ, whether $T \vdash \sigma$.*

Proof. If T is complete and $\mathcal{A}, \mathcal{B}$ are models, then for each sentence σ either $T \vdash \sigma$ or $T \vdash \neg\sigma$, and by soundness (Theorem 2.5.6) σ has the same truth value in $\mathcal{A}$ and $\mathcal{B}$. Conversely, if $T \nvdash \sigma$ and $T \nvdash \neg\sigma$, then by the completeness theorem $T \nvdash \sigma$ and $T \nvdash \neg\sigma$: there are a model in which σ is false and a model in which $\neg\sigma$ is false.

For the second statement, the deductions from T are finite strings of symbols, and whether a given string is a deduction from T can be checked mechanically, using an algorithm for membership in T to check the steps that cite axioms of T. So an algorithm can list all deductions from T, one after the other. Given σ, run through this list until a deduction of σ or of $\neg\sigma$ appears. As T is complete one of them appears after finitely many steps, and as T is consistent at most one of σ and $\neg\sigma$ is provable, so the answer is correct. □

The second half of this proposition shows where computability enters: a complete, computably axiomatized theory is decidable. So to prove that a theory is incomplete it suffices to prove that it is undecidable, and the first incompleteness theorem is, in essence, an undecidability theorem for arithmetic, proved by the diagonal argument of Remark 2.8.12 below.

Theorem 2.8.7 (First incompleteness theorem)**.** *Let T be a consistent, computably axiomatized theory in the language of arithmetic with $T \vdash \sigma$ for every $\sigma \in Q$. Then T is not complete.*

This is due to [Gö31], under a slightly stronger hypothesis than consistency, and in this form to [Ros36]; the theory Q is from [TMR53].

Remark 2.8.8. *What the first incompleteness theorem says, and what it does not say.*

1. It produces, for each such T, a sentence σ with $T \nvdash \sigma$ and $T \nvdash \neg\sigma$. By Proposition 2.8.6, T has a model in which σ is true and a model in which it is false. If $\mathcal{N}$ is a model of T, as for $T = \mathrm{PA}$, then one of $\sigma, \neg\sigma$ is true in $\mathcal{N}$ and not provable in T: there are true arithmetic statements that T does not prove, and the models in which they fail are nonstandard models, as in Example 2.7.5.

2. It does not say that some arithmetic truths are unprovable in an absolute sense: the unprovable sentence of T is provable in the larger theory $T \cup \{\sigma\}$. But this larger theory is again consistent and computably axiomatized, so it is incomplete in turn; no computably axiomatized extension of Q captures all of $\mathrm{Th}(\mathcal{N})$, which is Example 2.8.3.

3. The hypothesis "computably axiomatized" cannot be dropped: $\mathrm{Th}(\mathcal{N})$ itself is consistent, complete and contains Q. Nor can the hypothesis that T contains enough arithmetic: the theory of algebraically closed fields of characteristic 0 is complete (Example 2.8.4(3)).

4. The sentence σ produced by Gödel's proof says, via a coding of formulas and deductions by numbers, "I am not provable in T". Its truth in $\mathcal{N}$ is a consequence of the consistency of T, which leads to the second incompleteness theorem.

Theorem 2.8.9 (Second incompleteness theorem)**.** *Let T be a consistent, computably axiomatized theory in the language of arithmetic with $T \vdash \sigma$ for every $\sigma \in \mathrm{PA}$. Then $T \nvdash \mathrm{Con}_T$, where Con_T is the arithmetic sentence which expresses, via a Gödel numbering of formulas, that T is consistent.*

See [G̈31] and [Smi13].

Remark 2.8.10. Both theorems apply, with the same proofs, to any consistent computably axiomatized theory in which arithmetic can be interpreted, such as ZFC: if ZFC is consistent, it does not prove its own consistency [Kun11]. This affects the foundations of this project. In Chapter 3 we assume the existence of a Grothendieck universe, and a universe is a model of ZFC; so ZFC together with this assumption proves $\mathrm{Con}_{\mathrm{ZFC}}$. By the second incompleteness theorem, the existence of a universe can therefore not be proved in ZFC (if ZFC is consistent): it is a genuinely new axiom, of modest strength.

Example 2.8.11. *Natural independent statements.* The sentence of Gödel's proof is artificial, but later work found statements of ordinary mathematics that are independent.

1. *Goodstein sequences.* Write a number m in *hereditary base b*: as a sum of powers of b with coefficients less than b, with the exponents again written in hereditary base b, and so on. For instance $4 = 2^2$ and $26 = 2 \cdot 3^2 + 2 \cdot 3 + 2$ in hereditary bases 2 and 3. The Goodstein sequence of m starts with $m_2 = m$; to pass from m_b to m_{b+1}, write m_b in hereditary base b, replace every b by $b+1$, and subtract 1. Starting from $3 = 2 + 1$: replacing 2 by 3 gives 4, minus one is 3; then $3 = 3^1$ in base 3 becomes 4, minus one is 3; in base 4 the number 3 is a single digit, so the sequence continues $2, 1, 0$. Starting from $4 = 2^2$ the sequence grows quickly: 4, $3^3 - 1 = 26$, $2 \cdot 4^2 + 2 \cdot 4 + 2 - 1 = 41$, $2 \cdot 5^2 + 2 \cdot 5 + 1 - 1 = 60$, $2 \cdot 6^2 + 2 \cdot 6 - 1 = 83$, and so on. *Goodstein's theorem* [Goo44] says that every Goodstein sequence reaches 0. The proof replaces each base by the ordinal ω (Chapter 3): this turns the sequence into a strictly decreasing sequence of ordinals, which must be finite since ordinals are well-ordered. The theorem can be stated in the language of arithmetic and is true in $\mathcal{N}$, but it is not provable in PA [KP82]: any proof needs the well-ordering of ordinals beyond the reach of PA.

2. *The Paris–Harrington theorem*, a strengthening of the finite Ramsey theorem, is likewise true but not provable in PA [PH77].

3. In set theory, the continuum hypothesis is neither provable nor refutable in ZFC, if ZFC is consistent [G̈38, Coh63] (see Chapter 3). Unlike the examples above, its independence is proved by constructing models of ZFC in which it holds and models in which it fails, as in Proposition 2.8.6.

Remark 2.8.12. *Computability and Turing's theorem.* The informal notion of algorithm used in Definition 2.8.2 was made precise in the 1930s by Turing machines, recursive functions and the λ-calculus, which turned out to define the same class of computable functions (the *Church–Turing thesis* asserts that this class captures all effective procedures). *Turing's theorem* [Tur37] says that the *halting problem* is undecidable: there is no algorithm which, given a program P and an input x, decides whether P halts on x. The idea is a diagonal argument.

If H were such an algorithm, consider the program D which, on input a program P, runs H on (P, P) and loops forever if the answer is "halts", and halts otherwise; then D halts on input D if and only if it does not. The same diagonal mechanism, with provability in place of halting, underlies the first incompleteness theorem; and Turing's theorem implies that first-order validity is not decidable (Church and Turing's negative answer to Hilbert's *Entscheidungsproblem*), although by the completeness theorem the valid sentences can be listed by an algorithm.

2.9 Exercises

Exercise 2.9.1. Show that every function $\{0,1\}^n \to \{0,1\}$ with $n \geq 1$ is computed by a formula built from variables using only $\neg$ and $\wedge$ (as primitive connectives with their usual truth tables). Show that a formula built from variables using only $\wedge$, $\vee$ and $\to$ is satisfied by the truth assignment that is constantly 1, and deduce that these three connectives do not suffice to express negation.

Solution. By Lemma 2.2.8, $\varphi \vee \psi$ has the same truth table as $\neg(\neg\varphi \wedge \neg\psi)$ and $\varphi \to \psi$ the same as $\neg(\varphi \wedge \neg\psi)$; so the formula of Proposition 2.2.10 can be rewritten using $\neg$ and $\wedge$. For the second part, argue by induction: if v is constantly 1, then $\bar{v}$ is 1 on variables, and $1 \wedge 1$, $1 \vee 1$ and $1 \to 1$ all have value 1. Negation of a variable has value 0 under v, so it is not expressible. □

Exercise 2.9.2. Show that $\exists x \forall y\, \varphi \to \forall y \exists x\, \varphi$ is valid for every formula φ, and give a formula φ in the language of orders and a structure in which $\forall y \exists x\, \varphi \to \exists x \forall y\, \varphi$ is false.

Solution. Let $\mathcal{A}$ be a structure and s an assignment with $\mathcal{A} \models \exists x \forall y\, \varphi[s]$ (if x and y are the same variable the statement is trivial, so assume they are different). Then there is $a \in A$ with $\mathcal{A} \models \forall y\, \varphi[s[x \mapsto a]]$, that is, $\mathcal{A} \models \varphi[s[x \mapsto a][y \mapsto b]]$ for every $b \in A$. Given b, the element a witnesses $\mathcal{A} \models \exists x\, \varphi[s[y \mapsto b]]$, since $s[y \mapsto b][x \mapsto a] = s[x \mapsto a][y \mapsto b]$. So $\mathcal{A} \models \forall y \exists x\, \varphi[s]$. In words: one x that works for all y certainly gives, for each y, an x that works. The converse fails: in $(\mathbb{N}, <)$ with $\varphi = (y < x)$, the sentence $\forall y \exists x\, (y < x)$ is true (every number has a larger one) and $\exists x \forall y\, (y < x)$ is false (no number is larger than all numbers, itself included). □

Exercise 2.9.3. A *well-order* is a total order in which every nonempty subset has a least element. Show that there is no theory T in the language of orders whose models are exactly the well-ordered sets. (Hint: add constants $c_0, c_1, \ldots$ and the sentences $c_{n+1} < c_n$.)

Solution. Suppose T were such a theory, and let $T' = T \cup \{c_{n+1} < c_n : n \in \mathbb{N}\}$ in the language with the new constants. A finite subset of T' mentions only $c_0, \ldots, c_k$ for some k. The finite order $\{0, 1, \ldots, k\}$ is well-ordered, hence a model of T, and interpreting c_n as $k - n$ for $n \leq k$ and as 0 for $n > k$ satisfies the sentences $c_{n+1} < c_n$ with $n < k$. So every finite subset of T' is satisfiable, and by the compactness theorem (Theorem 2.7.1) T' has a model. Its underlying order is a model of T, hence well-ordered, but the set $\{c_n\}$ of interpretations of the constants has no least element, since $c_{n+1} < c_n$ for all n. This is a contradiction. The property "every nonempty subset has a least element" quantifies over subsets, which first-order logic cannot do. □

Exercise 2.9.4. A group is a *torsion group* if every element has finite order. Show that there is no theory in the language of groups whose models are exactly the torsion groups. Show, on the other hand, that for every $n \geq 1$ there is a single sentence whose models among groups are exactly the groups satisfying $g^n = e$ for all g.

Solution. Suppose T were such a theory. Add a constant c and let $T' = T \cup \{c^n \neq e : n \geq 1\}$, where c^n is the term $c \cdot c \cdots c$ with n factors. A finite subset of T' involves only $n < N$ for some N, and is satisfied by the cyclic group $\mathbb{Z}/N\mathbb{Z}$ (written multiplicatively) with c a generator: it is a torsion group, hence a model of T, and $c^n \neq e$ for $0 < n < N$. By compactness T' has a model, a torsion group in which c has infinite order, a contradiction. For the second statement take the sentence $\forall x\,(x^n = e)$ together with the group axioms. So "all elements have order dividing n" is first-order for each fixed n, but "all elements have finite order" would be the infinite disjunction over all n, which is not a formula. □

Exercise 2.9.5. Let T_0 be the theory of fields in the language of rings, and for a prime p let χ_p be the sentence $\underline{p} \neq 0$, where $\underline{p}$ is $1 + 1 + \cdots + 1$ with p summands. The models of $T_0 \cup \{\chi_p : p \text{ prime}\}$ are the fields of characteristic zero. Show that there is no single sentence σ such that the models of $T_0 \cup \{\sigma\}$ are exactly the fields of characteristic zero. Deduce that a sentence true in all fields of characteristic zero is true in all fields of sufficiently large prime characteristic.

Solution. We first prove the second statement. Let τ be true in all fields of characteristic zero, so that $T_0 \cup \{\chi_p : p \text{ prime}\} \cup \{\neg\tau\}$ has no model. By the compactness theorem some finite subset has no model; enlarging it, $T_0 \cup \{\chi_p : p \leq N\} \cup \{\neg\tau\}$ has no model for some N. A field of prime characteristic $q > N$ satisfies T_0 and χ_p for $p \leq N$, so it does not satisfy $\neg\tau$: τ is true in it.

Now suppose the models of $T_0 \cup \{\sigma\}$ are the fields of characteristic zero. Then σ is true in all fields of characteristic zero, hence, by what we just proved, in $\mathbb{F}_q = \mathbb{Z}/q\mathbb{Z}$ for a large prime q; so $\mathbb{F}_q$ is a model of $T_0 \cup \{\sigma\}$ of characteristic q, a contradiction. For example, the sentence "$1 + 1 \neq 0$" is true in characteristic zero and fails only in characteristic 2, in accordance with the statement. □

Exercise 2.9.6. Let T be a theory and σ a sentence with $T \models \sigma$. Show, without using deductions, that some finite subset $T_0 \subseteq T$ satisfies $T_0 \models \sigma$ (for $\mathcal{L}$ countable).

Solution. Otherwise, for every finite $T_0 \subseteq T$ the set $T_0 \cup \{\neg\sigma\}$ has a model. Then every finite subset of $T \cup \{\neg\sigma\}$ is satisfiable, so by Theorem 2.7.1 the whole set has a model, which is a model of T in which σ is false, contradicting $T \models \sigma$. □

Chapter 3

Set Theory

3.1 Introduction

This chapter sets up the foundations of the project: Zermelo–Fraenkel set theory with the axiom of choice (ZFC), formulated as a first-order theory in the sense of Chapter 2. We develop the basic constructions (relations, functions, quotients), the natural numbers, ordinals and cardinals, prove the equivalence of the axiom of choice with Zorn's lemma and the well-ordering theorem, and discuss Grothendieck universes, which provide the conventions on size used throughout the project (Notation 1.6.1).

Standard references are [Jec03] and [Kun11]; [Hal60] is an informal introduction.

Prerequisites. Chapter 2 (Mathematical Logic).

Remark 3.1.1. Set theory is the first-order theory, in the language with one binary relation symbol $\in$ (Example 2.3.5), whose axioms are listed in Definition 3.2.1. Its models have a universe whose elements are called *sets*. We write statements in ordinary mathematical language, but each of them can be translated into a formula of this language, and each "construction" is justified by the axioms. A *class* is given by a formula $\varphi(x)$, possibly with parameters, and we write $\{x : \varphi(x)\}$ for it; a class need not be a set. Statements about classes are abbreviations of statements about the corresponding formulas.

3.2 The axioms

We use the abbreviations $x \subseteq y$ for $\forall z\,(z \in x \to z \in y)$, $\forall z \in x\,\varphi$ for $\forall z\,(z \in x \to \varphi)$ and $\exists z \in x\,\varphi$ for $\exists z\,(z \in x \wedge \varphi)$.

Definition 3.2.1. The axioms of *Zermelo–Fraenkel set theory* ZF are the following sentences.

1. (Extensionality) $\forall x \forall y\,(\forall z\,(z \in x \leftrightarrow z \in y) \to x = y)$.
2. (Foundation) $\forall x\,(\exists y\,(y \in x) \to \exists y \in x\,\neg\exists z\,(z \in x \wedge z \in y))$.
3. (Separation) For every formula φ in which y is not free: $\forall w_1 \cdots \forall w_n \forall x \exists y \forall z\,(z \in y \leftrightarrow (z \in x \wedge \varphi))$, where $w_1, \ldots, w_n$ are the other free variables of φ.
4. (Pairing) $\forall x \forall y \exists z\,(x \in z \wedge y \in z)$.
5. (Union) $\forall F \exists A \forall Y \forall x\,((x \in Y \wedge Y \in F) \to x \in A)$.
6. (Replacement) For every formula φ in which B is not free: $\forall w_1 \cdots \forall w_n \forall A\,\big(\forall x \in A\,\exists! y\,\varphi \to \exists B\,\forall x \in A\,\exists y \in B\,\varphi\big)$, where $\exists! y\,\varphi$ abbreviates "there is exactly one y with φ" and $w_1, \ldots, w_n$ are the other free variables of φ.

7. (Infinity) $\exists x \left(\exists e \in x\, \forall z\, (z \notin e) \wedge \forall y \in x\, \exists s \in x\, \forall z\, (z \in s \leftrightarrow (z \in y \vee z = y))\right)$.

8. (Power set) $\forall x \exists y \forall z\, (z \subseteq x \to z \in y)$.

The theory ZFC consists of ZF together with the axiom of choice, Definition 3.7.1. The axioms of separation and replacement are *schemas*: they consist of one axiom for each formula φ.

Remark 3.2.2. Extensionality says that a set is determined by its elements. Separation says that for a set x and a property φ, the elements of x with the property form a set, written $\{z \in x : \varphi(z)\}$. Pairing, union and power set provide sets containing the pair $\{x, y\}$, the union of the elements of F, and the subsets of x; by separation and extensionality there are then unique sets with exactly these elements. Replacement says that the image of a set under a class function is contained in a set. Infinity provides a set containing $\emptyset$ and closed under $y \mapsto y \cup \{y\}$. Foundation says that every nonempty set has an $\in$-minimal element.

Lemma 3.2.3. *1. There is a unique set $\emptyset$ with no elements.*

2. For all sets x, y there are unique sets $\{x, y\}$, $\bigcup x$ and $\mathcal{P}(x)$ with, respectively, the elements x and y; the elements of elements of x; the subsets of x.

3. For all sets x, y there are unique sets $x \cup y$, $x \cap y$ and $x \setminus y$ with the usual elements. If F is a nonempty set, there is a unique set $\bigcap F$ whose elements are the sets belonging to every element of F.

Proof. Uniqueness always follows from extensionality. (1) Some set x exists, because $\exists x\, (x = x)$ is valid (structures are nonempty); apply separation with the formula $z \neq z$. (2) Apply separation to the sets given by pairing, union and power set, with the formulas $z = x \vee z = y$, $\exists Y \in x\, (z \in Y)$ and $z \subseteq x$. (3) Put $x \cup y = \bigcup \{x, y\}$, and use separation on x for $x \cap y$ and $x \setminus y$. If $X \in F$, put $\bigcap F = \{z \in X : \forall Y \in F\, (z \in Y)\}$. □

Proposition 3.2.4 (Russell)**.** *There is no set of which every set is an element.*

Proof. Suppose V is such a set and let $R = \{x \in V : x \notin x\}$. Then $R \in V$, so $R \in R$ if and only if $R \notin R$, a contradiction. □

Remark 3.2.5. So the class $\{x : x = x\}$ of all sets is not a set; such classes are called *proper classes*. Proposition 3.2.4 holds even without foundation. With foundation it is also immediate from Lemma 3.2.6.

Lemma 3.2.6. *There is no set x with $x \in x$, and there are no sets x, y with $x \in y$ and $y \in x$.*

Proof. If $x \in x$, the set $\{x\}$ has no $\in$-minimal element: its only element x satisfies $x \in x \cap \{x\}$. If $x \in y \in x$, the set $\{x, y\}$ has no $\in$-minimal element, since $y \in x \cap \{x, y\}$ and $x \in y \cap \{x, y\}$. □

3.3 Relations and functions

Definition 3.3.1. The *ordered pair* of a and b is $(a, b) = \{\{a\}, \{a, b\}\}$.

Lemma 3.3.2. *$(a, b) = (c, d)$ if and only if $a = c$ and $b = d$.*

Proof. Suppose $(a, b) = (c, d)$. If $a = b$, then $(a, b) = \{\{a\}\}$, so $\{c\} = \{c, d\} = \{a\}$ and $a = b = c = d$. Similarly if $c = d$. Assume $a \neq b$ and $c \neq d$. Then $\{a\}$ is the only one-element set in (a, b), and $\{c\}$ the only one in (c, d), so $\{a\} = \{c\}$ and $a = c$. The two-element sets $\{a, b\}$ and $\{c, d\} = \{a, d\}$ are then equal, and since $b \neq a$ we get $b = d$. The converse is clear. □

Definition 3.3.3. The *cartesian product* of sets A and B is $A \times B = \{p \in \mathcal{P}(\mathcal{P}(A \cup B)) : \exists a \in A\, \exists b \in B\, p = (a, b)\}$. It is a set by separation, and its elements are exactly the pairs (a, b) with $a \in A$, $b \in B$, because $\{a\}$ and $\{a, b\}$ are subsets of $A \cup B$.

Definition 3.3.4. A *relation* from A to B is a subset $R \subseteq A \times B$; we write aRb for $(a, b) \in R$. A *function* (or *map*) $f : A \to B$ is a relation $f \subseteq A \times B$ such that for every $a \in A$ there is exactly one $b \in B$ with $(a, b) \in f$, written $b = f(a)$. (Formally, a function records its target: it is the triple (A, B, f); we shall not insist on this.) For $X \subseteq A$ and $Y \subseteq B$, the *image* is $f[X] = \{f(a) : a \in X\}$ and the *preimage* is $f^{-1}[Y] = \{a \in A : f(a) \in Y\}$. The function is *injective*, *surjective* or *bijective* if it is so as a map in the usual sense. The composition $g \circ f$ of $f : A \to B$ and $g : B \to C$ is $\{(a, c) \in A \times C : \exists b\, ((a, b) \in f \wedge (b, c) \in g)\}$. The set of all functions from A to B is $B^A = \{f \in \mathcal{P}(A \times B) : f \text{ is a function } A \to B\}$.

Definition 3.3.5. A *family* $(x_i)_{i \in I}$ is a function $i \mapsto x_i$ with domain I. Its union is $\bigcup_{i \in I} x_i = \bigcup \{x_i : i \in I\}$, which is a set: the image of I under the family is a set by replacement and separation. Its *product* is the set $\prod_{i \in I} x_i = \{f \in (\bigcup_{i \in I} x_i)^I : f(i) \in x_i \text{ for all } i \in I\}$.

Lemma 3.3.6. *A function $f : A \to B$ is bijective if and only if there is a function $g : B \to A$ with $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$; such a g is unique.*

Proof. If f is bijective, $g = \{(b, a) : (a, b) \in f\}$ is a function $B \to A$ with the two properties. Conversely, $g \circ f = \mathrm{id}_A$ implies that f is injective and $f \circ g = \mathrm{id}_B$ that f is surjective. If g' also has the properties, then $g' = g' \circ f \circ g = g$. □

Definition 3.3.7. An *equivalence relation* on A is a relation $\sim$ from A to A that is reflexive, symmetric and transitive. The *class* of $a \in A$ is $[a] = \{x \in A : x \sim a\}$, and the *quotient* is the set $A/\!\sim = \{X \in \mathcal{P}(A) : \exists a \in A\, X = [a]\}$, with the *projection* $\pi : A \to A/\!\sim$, $a \mapsto [a]$. A *partition* of A is a set of nonempty, pairwise disjoint subsets of A whose union is A.

Proposition 3.3.8. *Let A be a set. The map $\sim \mapsto A/\!\sim$ is a bijection from the set of equivalence relations on A to the set of partitions of A. Its inverse sends a partition Π to the relation "x and y lie in the same element of Π".*

Proof. If $\sim$ is an equivalence relation, the classes are nonempty ($a \in [a]$) and cover A. If $z \in [a] \cap [b]$, then $z \sim a$ and $z \sim b$, so $a \sim b$ by symmetry and transitivity, and then $[a] = [b]$. So $A/\!\sim$ is a partition. Conversely, the relation defined by a partition Π is clearly an equivalence relation whose classes are the elements of Π, and $x \sim y$ if and only if x and y lie in the same class of $\sim$. So the two constructions are inverse to each other. □

Proposition 3.3.9. *Let $\sim$ be an equivalence relation on A and $f : A \to B$ a map such that $a \sim a'$ implies $f(a) = f(a')$. There is a unique map $\bar{f} : A/\!\sim \to B$ with $\bar{f} \circ \pi = f$.*

Proof. Uniqueness holds because π is surjective. Put $\bar{f} = \{(X, b) \in (A/\!\sim) \times B : \exists a \in X\, f(a) = b\}$. Every class X is nonempty, and all its elements have the same image under f, so $\bar{f}$ is a function, and $\bar{f}([a]) = f(a)$. □

Definition 3.3.10. A *partial order* on A is a relation $\leq$ that is reflexive, antisymmetric and transitive; it is *total* if $x \leq y$ or $y \leq x$ for all x, y. We write $x < y$ for $x \leq y \wedge x \neq y$. A *chain* is a totally ordered subset. For $X \subseteq A$, an *upper bound* of X is an $a \in A$ with $x \leq a$ for all $x \in X$; a *least upper bound* or *supremum* is an upper bound below all upper bounds. An element m is *maximal* if there is no x with $m < x$, and *greatest* if $x \leq m$ for all x. A *well-order* is a total order in which every nonempty subset has a least element. An *isomorphism* of ordered sets is a bijection f with $x \leq y \iff f(x) \leq f(y)$. For $a \in A$, the *initial segment* determined by a is $A_{<a} = \{x \in A : x < a\}$.

Counterexample 3.3.11. A maximal element need not be greatest, and a greatest element need not exist. In $\{\{0\},\{1\}\}$, ordered by inclusion, both elements are maximal and there is no greatest element.

Lemma 3.3.12. *Let W be a well-ordered set and $f : W \to W$ a strictly increasing map. Then $f(w) \geq w$ for all $w \in W$. Consequently, W is not isomorphic to any initial segment $W_{<a}$, and the only isomorphism $W \to W$ is the identity. If two well-ordered sets are isomorphic, the isomorphism is unique.*

Proof. If $\{w : f(w) < w\}$ is nonempty, let w be its least element. Then $f(w) < w$, so $f(f(w)) < f(w)$, and $f(w)$ is a smaller element of the same set; contradiction. An isomorphism $f : W \to W_{<a}$ would give a strictly increasing $W \to W$ with $f(a) < a$. If $f, g : W \to W'$ are isomorphisms, $g^{-1} \circ f$ and $f^{-1} \circ g$ are strictly increasing, so $f(w) \geq g(w)$ and $g(w) \geq f(w)$ for all w. □

3.4 The natural numbers

Definition 3.4.1. The *successor* of a set x is $S(x) = x \cup \{x\}$. A set I is *inductive* if $\emptyset \in I$ and $S(y) \in I$ for all $y \in I$. By the axiom of infinity there is an inductive set I_0, and we define

$$\omega = \{n \in I_0 : n \in I \text{ for every inductive set } I\}.$$

The elements of ω are the *natural numbers*; we write $\mathbb{N} = \omega$, $0 = \emptyset$, $1 = S(0)$, $2 = S(1)$, and so on.

Theorem 3.4.2 (Induction)**.** *The set ω is inductive and is contained in every inductive set; it does not depend on the choice of I_0. If $X \subseteq \omega$ contains 0 and $S(n) \in X$ for all $n \in X$, then $X = \omega$.*

Proof. Every inductive set contains $\emptyset$ and is closed under S, so the intersection ω has the same two properties; it is contained in every inductive set by definition. If I_1 is another inductive set, the set defined using I_1 is contained in I_0 and has the same elements. A set X as in the statement is inductive, so $\omega \subseteq X$. □

Remark 3.4.3. In this construction 0 is a natural number, and there is a good reason for it. The construction needs a first set to start from, and the only set that the axioms provide without any choices is $\emptyset$. With $0 = \emptyset$, each natural number is the set of its predecessors, $n = \{0, 1, \ldots, n-1\}$, so n is a set with exactly n elements: 0 counts the empty set, and "X has n elements" will mean that X is in bijection with n (Definition 3.6.4). Moreover 0 is the neutral element of addition (Example 3.4.7), so with $0 \in \mathbb{N}$ the natural numbers form a monoid under addition, and every natural number is a sum of ones, the empty sum being 0. For these reasons $\mathbb{N} = \{0, 1, 2, \ldots\}$ throughout the project. Many texts, especially in analysis and number theory, let $\mathbb{N}$ start at 1; when we need the positive integers we write $\mathbb{N}_{>0} = \{1, 2, 3, \ldots\}$, so a sequence indexed by $\mathbb{N}_{>0}$ starts with its term of index 1.

Lemma 3.4.4. *Every element of a natural number is a natural number, and every natural number n is* transitive*: if $y \in x \in n$ then $y \in n$.*

Proof. Let X be the set of $n \in \omega$ such that $n \subseteq \omega$ and n is transitive. Clearly $0 \in X$. If $n \in X$, then $S(n) = n \cup \{n\} \subseteq \omega$; and if $y \in x \in S(n)$, then either $x \in n$, so $y \in n$, or $x = n$, so $y \in n$; in both cases $y \in S(n)$. By Theorem 3.4.2, $X = \omega$. □

Proposition 3.4.5 (Peano axioms)**.** *For all $m, n \in \omega$: $S(n) \neq 0$, and $S(m) = S(n)$ implies $m = n$.*

Proof. $n \in S(n)$, so $S(n) \neq \emptyset$. Suppose $S(m) = S(n)$ and $m \neq n$. Since $m \in S(n)$, we get $m \in n$; similarly $n \in m$. This contradicts Lemma 3.2.6. □

Theorem 3.4.6 (Recursion)**.** *Let A be a set, $a \in A$ and $g\colon \omega \times A \to A$ a map. There is a unique map $f\colon \omega \to A$ with $f(0) = a$ and $f(S(n)) = g(n, f(n))$ for all $n \in \omega$.*

Proof. Call a map h *acceptable* if its domain is $S(k)$ for some $k \in \omega$, its values lie in A, $h(0) = a$, and $h(S(i)) = g(i, h(i))$ whenever $S(i) \in \mathrm{dom}(h)$. Acceptable maps are elements of $\mathcal{P}(\omega \times A)$, so they form a set H.

If $h, h' \in H$ and i lies in both domains, then $h(i) = h'(i)$. Indeed, let X be the set of $i \in \omega$ such that this holds whenever $i \in \mathrm{dom}(h) \cap \mathrm{dom}(h')$. Then $0 \in X$. If $i \in X$ and $S(i)$ lies in both domains, then so does i, because domains are transitive (Lemma 3.4.4), and $h(S(i)) = g(i, h(i)) = g(i, h'(i)) = h'(S(i))$. So $X = \omega$.

Hence $f = \bigcup H$ is a function with values in A. Every $n \in \omega$ lies in its domain: by induction there is an acceptable h_n with domain $S(n)$, since $\{(0, a)\}$ is acceptable with domain $S(0)$, and if h is acceptable with domain $S(n)$ then $h \cup \{(S(n), g(n, h(n)))\}$ is acceptable with domain $S(S(n))$. Then $f(0) = a$, and $f(S(n)) = h_{S(n)}(S(n)) = g(n, h_{S(n)}(n)) = g(n, f(n))$. If f' is another such map, induction shows $f(n) = f'(n)$ for all n. □

Example 3.4.7. By Theorem 3.4.6, for each $m \in \omega$ there are unique maps $n \mapsto m + n$ and $n \mapsto m \cdot n$ with $m + 0 = m$, $m + S(n) = S(m + n)$, $m \cdot 0 = 0$ and $m \cdot S(n) = m \cdot n + m$. So $(\omega, 0, S, +, \cdot)$ is a model of Robinson arithmetic Q (Definition 2.8.1). The laws of arithmetic are proved in Chapter 4.

3.5 Ordinals

Definition 3.5.1. A set x is *transitive* if every element of x is a subset of x. An *ordinal* is a transitive set α such that the relation $\{(\beta, \gamma) \in \alpha \times \alpha : \beta \in \gamma\}$ is a strict well-order on α: it is transitive, for all $\beta, \gamma \in \alpha$ exactly one of $\beta \in \gamma$, $\beta = \gamma$, $\gamma \in \beta$ holds, and every nonempty subset of α has an $\in$-least element. For ordinals we write $\beta < \gamma$ for $\beta \in \gamma$. The class of ordinals is denoted Ord.

Lemma 3.5.2. *Every element of an ordinal is an ordinal, and $\alpha \notin \alpha$ for every ordinal α.*

Proof. Let $\beta \in \alpha$. Then $\beta \subseteq \alpha$, so $\in$ is a strict well-order on β. If $\delta \in \gamma \in \beta$, then γ and δ lie in α, and transitivity of the order on α gives $\delta \in \beta$; so β is transitive. The last statement is Lemma 3.2.6. □

Lemma 3.5.3. *If α and β are ordinals with $\alpha \subsetneq \beta$, then $\alpha \in \beta$.*

Proof. Let γ be the least element of $\beta \setminus \alpha$. If $\delta \in \gamma$, then $\delta \in \beta$ and $\delta < \gamma$, so $\delta \in \alpha$ by minimality; hence $\gamma \subseteq \alpha$. If $\delta \in \alpha$, then $\delta \in \beta$; $\delta = \gamma$ is impossible since $\gamma \notin \alpha$, and $\gamma \in \delta$ is impossible since it would give $\gamma \in \alpha$ by transitivity of α; so $\delta \in \gamma$. Hence $\alpha = \gamma \in \beta$. □

Theorem 3.5.4. *For all ordinals α, β, exactly one of $\alpha < \beta$, $\alpha = \beta$, $\beta < \alpha$ holds. Every nonempty set X of ordinals has a least element, namely $\bigcap X$, and $\sup X = \bigcup X$ is an ordinal which is the least ordinal $\geq$ every element of X.*

Proof. The set $\gamma = \alpha \cap \beta$ is transitive and a subset of α, hence an ordinal. By Lemma 3.5.3, $\gamma = \alpha$ or $\gamma \in \alpha$, and $\gamma = \beta$ or $\gamma \in \beta$. If $\gamma \in \alpha$ and $\gamma \in \beta$, then $\gamma \in \gamma$, contradicting Lemma 3.5.2. So $\alpha \subseteq \beta$ or $\beta \subseteq \alpha$, and the first claim follows from Lemma 3.5.3; at most one of the three holds by Lemma 3.2.6.

Let $\alpha \in X$. If no element of X is smaller than α, then α is least. Otherwise $\alpha \cap X$ is a nonempty subset of α and has a least element, which is least in X. By the first part, the least element is contained in every element of X, so it equals $\bigcap X$. Finally $\bigcup X$ is a transitive set of ordinals; by the first part $\in$ is a strict total order on it, and every nonempty subset has a least element; so it is an ordinal. It contains each $\alpha \in X$ as a subset, and any ordinal containing all $\alpha \in X$ as subsets contains $\bigcup X$. □

Proposition 3.5.5 (Burali-Forti)**.** *There is no set containing all ordinals.*

Proof. If X were such a set, $\Omega = \{x \in X : x \text{ is an ordinal}\}$ would be a transitive set (Lemma 3.5.2) well-ordered by $\in$ (Theorem 3.5.4), hence an ordinal, so $\Omega \in \Omega$. □

Definition 3.5.6. For an ordinal α, the set $\alpha + 1 = S(\alpha)$ is an ordinal, the least ordinal greater than α. An ordinal is a *successor ordinal* if it is of the form $\alpha + 1$, and a *limit ordinal* if it is neither 0 nor a successor ordinal.

Proposition 3.5.7. *Every natural number is an ordinal, ω is an ordinal, and ω is the least limit ordinal. On ω the order $m < n$ (that is, $m \in n$) is a well-order.*

Proof. $\emptyset$ is an ordinal, and if α is an ordinal so is $S(\alpha)$ (it is transitive, and its elements are the ordinals in α together with α). By induction every natural number is an ordinal. By Lemma 3.4.4, ω is a transitive set of ordinals, hence an ordinal by Theorem 3.5.4. It is not 0, and not a successor, since $S(n) \in \omega$ for $n \in \omega$. If λ is a limit ordinal, then $0 \in \lambda$, and for $n \in \lambda$ we have $S(n) \leq \lambda$ and $S(n) \neq \lambda$, so $S(n) \in \lambda$; thus λ is inductive and $\omega \subseteq \lambda$. □

Theorem 3.5.8 (Transfinite induction)**.** *Let C be a class. Suppose that for every ordinal α, if all ordinals $\beta < \alpha$ belong to C, then α belongs to C. Then every ordinal belongs to C.*

Proof. Otherwise let α be an ordinal not in C. The set $\{\beta \in \alpha + 1 : \beta \notin C\}$ is nonempty, and its least element β satisfies the hypothesis, so $\beta \in C$; contradiction. □

Theorem 3.5.9 (Transfinite recursion)**.** *Let G be a class function, that is, a formula $\psi(x, y)$ such that for every x there is exactly one y with $\psi(x, y)$, written $y = G(x)$. There is a unique class function F defined on all ordinals such that $F(\alpha) = G(F|_\alpha)$ for every ordinal α, where $F|_\alpha = \{(\beta, F(\beta)) : \beta < \alpha\}$.*

Proof. As usual for class functions, the theorem is a schema: for each ψ we write down a formula defining F and prove its properties. Call a function f an *α-attempt* if its domain is the ordinal α and $f(\beta) = G(f|_\beta)$ for all $\beta < \alpha$. By transfinite induction on β, two attempts agree on the intersection of their domains.

We show by transfinite induction that an α-attempt exists for every α. Suppose that for each $\beta < \alpha$ there is a β-attempt f_β, which is unique. By replacement, $\{f_\beta : \beta < \alpha\}$ is a set. If $\alpha = \beta + 1$, then $f_\beta \cup \{(\beta, G(f_\beta))\}$ is an α-attempt. If α is 0 or a limit ordinal, then $\bigcup_{\beta<\alpha} f_\beta$ is a function (the f_β agree on common domains) with domain $\bigcup_{\beta<\alpha} \beta = \alpha$, and it is an α-attempt.

Define $F(\alpha) = G(f)$, where f is the α-attempt; then $F|_\alpha = f$ and $F(\alpha) = G(F|_\alpha)$. Uniqueness of F follows by transfinite induction. □

Theorem 3.5.10. *Every well-ordered set $(W, <)$ is isomorphic to a unique ordinal, by a unique isomorphism.*

Proof. Uniqueness follows from Lemma 3.3.12: two isomorphic ordinals are equal, since otherwise one is an initial segment of the other. Let A be the set of $w \in W$ such that $W_{<w}$ is isomorphic to some ordinal, necessarily unique, which we call α_w. By replacement, $\alpha = \{\alpha_w : w \in A\}$ is a set. If $v < w \in A$, the isomorphism $W_{<w} \to \alpha_w$ restricts to an isomorphism from $W_{<v}$ to an initial segment of α_w, which is an ordinal; so $v \in A$ and $\alpha_v < \alpha_w$.

Hence A is an initial segment of W or equal to W, α is a transitive set of ordinals, hence an ordinal, and $w \mapsto \alpha_w$ is an isomorphism $A \to \alpha$. If $A \neq W$, let w be the least element of $W \setminus A$; then $A = W_{<w}$, so $w \in A$, a contradiction. □

Definition 3.5.11. By transfinite recursion, define sets V_α for all ordinals by $V_0 = \emptyset$, $V_{\alpha+1} = \mathcal{P}(V_\alpha)$ and $V_\lambda = \bigcup_{\alpha<\lambda} V_\alpha$ for limit ordinals λ.

Lemma 3.5.12. *Every V_α is transitive, and $V_\beta \subseteq V_\alpha$ for $\beta \leq \alpha$. A set x lies in $V_{\alpha+1}$ if and only if $x \subseteq V_\alpha$.*

Proof. The last statement is the definition. We prove the first two statements together by transfinite induction on α. For $\alpha = 0$ they are clear, and for limit α they follow from the induction hypothesis. Let $\alpha = \gamma + 1$. If $x \in V_\alpha$ and $y \in x$, then $y \in V_\gamma$, so $y \subseteq V_\gamma$ by transitivity of V_γ, that is, $y \in V_\alpha$. Also $V_\gamma \subseteq V_\alpha$, because each element of V_γ is a subset of V_γ; combined with the induction hypothesis this gives $V_\beta \subseteq V_\alpha$ for all $\beta \leq \alpha$. □

Lemma 3.5.13. *Every set x is a subset of a transitive set.*

Proof. Apply Theorem 3.5.9 to the class function G with $G(f) = \bigcup f(n)$ if f is a function whose domain is $S(n)$ for some $n \in \omega$, and $G(f) = x$ otherwise. The resulting F satisfies $F(0) = x$ and $F(n+1) = \bigcup F(n)$ for $n \in \omega$. By replacement, $\{F(n) : n \in \omega\}$ is a set; let T be its union. Then $x \subseteq T$, and if $z \in y \in T$, say $y \in F(n)$, then $z \in F(n+1) \subseteq T$. □

Theorem 3.5.14. *Every set belongs to V_α for some ordinal α.*

Proof. Suppose x belongs to no V_α. Let T be a transitive set containing $\{x\}$ as a subset (Lemma 3.5.13) and let $Y = \{y \in T : y \notin V_\alpha \text{ for all } \alpha\}$; then $x \in Y$. By foundation there is $y \in Y$ with $y \cap Y = \emptyset$. Each $z \in y$ lies in T but not in Y, so there is a least ordinal α_z with $z \in V_{\alpha_z}$. By replacement, $\{\alpha_z : z \in y\}$ is a set; let α be its supremum. Then $y \subseteq V_\alpha$ by Lemma 3.5.12, so $y \in V_{\alpha+1}$, a contradiction. □

Definition 3.5.15. The *rank* $\operatorname{rk}(x)$ of a set x is the least ordinal α with $x \subseteq V_\alpha$.

3.6 Cardinality

Definition 3.6.1. Sets A and B are *equinumerous*, written $A \approx B$, if there is a bijection $A \to B$. We write $A \preceq B$ if there is an injection $A \to B$, and $A \prec B$ if $A \preceq B$ and not $A \approx B$.

Theorem 3.6.2 (Cantor–Schröder–Bernstein)**.** *If $A \preceq B$ and $B \preceq A$, then $A \approx B$.*

Proof. Let $f : A \to B$ and $g : B \to A$ be injections. Define $C_0 = A \setminus g[B]$ and $C_{n+1} = g[f[C_n]]$ by recursion, and let $C = \bigcup_n C_n$. Define $h : A \to B$ by $h(a) = f(a)$ if $a \in C$, and $h(a) = g^{-1}(a)$ if $a \notin C$; this makes sense because $a \notin C$ implies $a \in g[B]$, and g is injective.

h is surjective: let $b \in B$. If $g(b) \notin C$, then $h(g(b)) = b$. If $g(b) \in C$, then $g(b) \notin C_0$, so $g(b) \in C_{n+1}$ for some n, that is, $g(b) = g(f(c))$ with $c \in C_n$; then $b = f(c) = h(c)$.

h is injective: f and g^{-1} are injective, so it suffices to exclude $f(a) = g^{-1}(a')$ with $a \in C$ and $a' \notin C$. But then $a' = g(f(a)) \in C$ since $a \in C_n$ implies $g(f(a)) \in C_{n+1}$. □

Theorem 3.6.3 (Cantor)**.** *For every set A there is no surjection $A \to \mathcal{P}(A)$. In particular $A \prec \mathcal{P}(A)$.*

Proof. Let $f : A \to \mathcal{P}(A)$ and $D = \{a \in A : a \notin f(a)\}$. If $D = f(d)$, then $d \in D$ if and only if $d \notin D$. So f is not surjective. The map $a \mapsto \{a\}$ is an injection $A \to \mathcal{P}(A)$. □

Definition 3.6.4. A set is *finite* if it is equinumerous with some $n \in \omega$, and *infinite* otherwise. It is *countable* if $A \preceq \omega$, and *countably infinite* if $A \approx \omega$.

Proposition 3.6.5 (Pigeonhole principle)**.** *For $n \in \omega$ there is no injection $n+1 \to n$. Consequently, if $m \neq n$ are natural numbers then $m \not\approx n$, and ω is infinite.*

Proof. For $n = 0$ there is no map from $1 = \{0\}$ to $\emptyset$. Suppose there is no injection $n+1 \to n$, and let $f : n+2 \to n+1$ be injective. Let τ be the bijection of $n+1$ exchanging n and $f(n+1)$. Then $\tau \circ f$ is injective and sends $n+1$ to n, so it restricts to an injection $n+1 \to (n+1) \setminus \{n\} = n$, a contradiction. If $m < n$, then $m+1 \subseteq n$, so an injection $n \to m$ would give one $m+1 \to m$. Finally $\omega \approx n$ would give an injection $n+1 \to n$. □

Proposition 3.6.6. *Every subset of a natural number is finite, and every subset of ω is finite or countably infinite. A set is countable if and only if it is finite or countably infinite.*

Proof. By induction on n: subsets of 0 are finite, and a subset X of $n+1$ is $X \cap n$ or $(X \cap n) \cup \{n\}$; if $X \cap n \approx k$ then $(X \cap n) \cup \{n\} \approx k+1$.

Let $X \subseteq \omega$ be infinite. Then X has no upper bound in ω (otherwise $X \subseteq m$ for some m, and X would be finite). Define $e(0) = \min X$ and $e(n+1) = \min\{x \in X : x > e(n)\}$ by recursion. Then e is strictly increasing, hence injective, and $e(n) \geq n$ by induction. Given $x \in X$, let n be least with $e(n) \geq x$. If $n = 0$ then $x \leq e(0) = \min X \leq x$. If $n > 0$ then $e(n-1) < x$, so $e(n) \leq x$ by definition of $e(n)$. In both cases $x = e(n)$, so $e : \omega \to X$ is a bijection. The last statement follows: a set injecting into ω is equinumerous with a subset of ω. □

Definition 3.6.7. An ordinal κ is a *cardinal* if no $\beta < \kappa$ is equinumerous with κ. If A is equinumerous with some ordinal (for instance if A can be well-ordered, by Theorem 3.5.10), its *cardinality* $|A|$ is the least ordinal equinumerous with A; it is a cardinal.

Example 3.6.8. By Proposition 3.6.5, every natural number and ω are cardinals, and a finite set A has $|A| \in \omega$, its number of elements. The ordinal $\omega + 1$ is not a cardinal: $n \mapsto n+1$ for $n \in \omega$ and $\omega \mapsto 0$ define a bijection $\omega + 1 \to \omega$.

Theorem 3.6.9 (Hartogs)**.** *For every set A there is an ordinal $h(A)$ that does not inject into A; the least such ordinal is a cardinal. This uses no form of the axiom of choice.*

Proof. Let $\mathcal{W}$ be the set of pairs $(X, R) \in \mathcal{P}(A) \times \mathcal{P}(A \times A)$ such that R is a well-order on X. By Theorem 3.5.10 and replacement, the order types of these well-orders form a set H. An ordinal α injects into A if and only if $\alpha \in H$ (transport the order of α to the image). If $\alpha \in H$ and $\beta < \alpha$, then $\beta \in H$; so H is a transitive set of ordinals, hence an ordinal, and $H \notin H$. Put $h(A) = H$; it is the least ordinal not injecting into A. If some $\beta < H$ were equinumerous with H, then H would inject into A through β. □

Definition 3.6.10. By transfinite recursion let $\omega_0 = \omega$, $\omega_{\alpha+1} = h(\omega_\alpha)$ and $\omega_\lambda = \sup_{\alpha<\lambda} \omega_\alpha$ for limit λ. We also write $\aleph_\alpha$ for ω_α when we think of it as a cardinality.

Proposition 3.6.11. *The ω_α are cardinals, $\alpha < \beta$ implies $\omega_\alpha < \omega_\beta$, and every infinite cardinal equals ω_α for a unique α.*

Proof. Strict monotonicity is clear from the definition, because $\omega_\alpha < h(\omega_\alpha)$. By induction $\alpha \leq \omega_\alpha$. A supremum λ of cardinals is a cardinal: if $\beta < \lambda$ then $\beta < \kappa \leq \lambda$ for one of the cardinals κ, and $\beta \approx \lambda$ would give $\kappa \preceq \beta$ and hence $\kappa \approx \beta$ by Theorem 3.6.2. With Theorem 3.6.9, all ω_α are cardinals.

Let κ be an infinite cardinal, so $\omega \leq \kappa$, and let α be least with $\kappa \leq \omega_\alpha$ (it exists since $\kappa \leq \omega_\kappa$). If $\alpha = 0$, then $\kappa = \omega$. If α is a limit and $\kappa < \omega_\alpha$, then $\kappa < \omega_\beta$ for some $\beta < \alpha$, contradicting minimality. If $\alpha = \beta + 1$ and $\kappa < \omega_\alpha = h(\omega_\beta)$, then κ injects into ω_β, so $\kappa \preceq \omega_\beta \leq \kappa$ and $\kappa \approx \omega_\beta$ by Theorem 3.6.2; as both are cardinals, $\kappa = \omega_\beta$, contradicting the choice of α. □

Theorem 3.6.12 (Hessenberg). *For every infinite cardinal κ, $\kappa \times \kappa \approx \kappa$.*

Proof. Order pairs of ordinals by $(\alpha, \beta) \lhd (\gamma, \delta)$ if $\max(\alpha, \beta) < \max(\gamma, \delta)$, or the maxima are equal and $\alpha < \gamma$, or the maxima are equal, $\alpha = \gamma$ and $\beta < \delta$. On every set of pairs this is a strict total order, and every nonempty set of pairs has a least element (minimize the maximum, then the first coordinate, then the second). For an ordinal κ, the set $\kappa \times \kappa$ contains, with any pair, all $\lhd$-smaller pairs.

Suppose the theorem fails and let κ be the least infinite cardinal with $\kappa \times \kappa \not\approx \kappa$. Let θ be the order type of $(\kappa \times \kappa, \lhd)$ (Theorem 3.5.10). If $\theta \leq \kappa$, then $\kappa \times \kappa \preceq \kappa$, and since $\kappa \preceq \kappa \times \kappa$ via $\xi \mapsto (\xi, 0)$, Theorem 3.6.2 gives a contradiction. So $\kappa < \theta$, and there is a pair $(\xi, \eta) \in \kappa \times \kappa$ whose set of $\lhd$-predecessors P has order type κ. Let $\delta = \max(\xi, \eta) + 1$. An infinite cardinal is a limit ordinal: for an infinite ordinal α, the map $\alpha + 1 \to \alpha$ sending α to 0, each $n \in \omega$ to $n + 1$, and fixing all other elements, is a bijection, so $\alpha + 1$ is not a cardinal. Hence $\delta < \kappa$, and $P \subseteq \delta \times \delta$. If δ is finite, so is $\delta \times \delta$, contradicting $P \approx \kappa$. Otherwise $\mu = |\delta|$ is an infinite cardinal with $\mu \leq \delta < \kappa$, and by minimality of κ, $\kappa \approx P \preceq \delta \times \delta \approx \mu \times \mu \approx \mu$. Then $\kappa \preceq \mu \leq \kappa$, so $\kappa \approx \mu$, contradicting that κ is a cardinal. □

Corollary 3.6.13. *$\omega \times \omega$ is countably infinite, and so is ω^n for every $n \geq 1$. The set of finite sequences of elements of a countable set is countable.*

Proof. The first statement is Theorem 3.6.12; fix a bijection $p : \omega \times \omega \to \omega$. By recursion on n define bijections $b_n : \omega^n \to \omega$ for $n \geq 1$ by $b_1(s) = s(0)$ and $b_{n+1}(s) = p(b_n(s|_n), s(n))$. It suffices to treat sequences of natural numbers, since an injection $X \to \omega$ induces an injection on sequences. The map sending a sequence s of length $n \geq 1$ to $p(n, b_n(s))$, and the empty sequence to $p(0, 0)$, is an injection into ω. □

3.7 The axiom of choice

Definition 3.7.1. A *choice function* for a set F of nonempty sets is a function $c : F \to \bigcup F$ with $c(X) \in X$ for all $X \in F$. The *axiom of choice* (AC) states that every set of nonempty sets has a choice function.

Theorem 3.7.2. *In ZF, the following statements are equivalent.*

1. *The axiom of choice.*
2. *(Well-ordering theorem) Every set admits a well-order.*
3. *(Zorn's lemma) Let $(P, \leq)$ be a partially ordered set in which every chain has an upper bound. Then P has a maximal element.*

Proof. (1) $\Rightarrow$ (2). Let A be a set and c a choice function for $\mathcal{P}(A) \setminus \{\emptyset\}$. Fix some $\star \notin A$. By transfinite recursion let $F(\alpha) = c(A \setminus F[\alpha])$ if $A \setminus F[\alpha] \neq \emptyset$, and $F(\alpha) = \star$ otherwise, where $F[\alpha] = \{F(\beta) : \beta < \alpha\}$. If $F(\beta) = \star$, then $A \subseteq F[\beta]$, so $F(\alpha) = \star$ for all $\alpha > \beta$. Hence if $F(\alpha) \neq \star$, all $F(\beta)$ with $\beta \leq \alpha$ lie in A, and $F(\alpha) \notin F[\alpha]$; so these values are distinct. If $F(\alpha) \neq \star$ for all $\alpha < h(A)$, then F would inject $h(A)$ into A, contradicting Theorem 3.6.9. So there is a least α with $F(\alpha) = \star$, and then $F|_\alpha : \alpha \to A$ is a bijection; transport the order of α.

(2) $\Rightarrow$ (1). Well-order $\bigcup F$ and let $c(X)$ be the least element of X.

(1) $\Rightarrow$ (3). Suppose P has no maximal element. Then every chain C has a *strict* upper bound: an upper bound u of C is not maximal, so there is $v > u$. Let c be a choice function for $\mathcal{P}(P) \setminus \{\emptyset\}$ and $\star \notin P$. Define by transfinite recursion $F(\alpha) = c(S_\alpha)$, where S_α is the set of strict upper bounds of $F[\alpha]$, if $F[\alpha] \subseteq P$ is a chain and S_α is nonempty, and $F(\alpha) = \star$ otherwise. By induction on α, $F[\alpha]$ is a chain in P on which F is strictly increasing: if this

holds for all $\beta < \alpha$ then $F[\alpha]$ is a chain, so $S_\alpha \neq \emptyset$ and $F(\alpha)$ is greater than all $F(\beta)$, $\beta < \alpha$. Hence F injects $h(P)$ into P, contradicting Theorem 3.6.9.

(3) $\Rightarrow$ (1). Let F be a set of nonempty sets. Let P be the set of functions c with $\operatorname{dom}(c) \subseteq F$ and $c(X) \in X$ for all $X \in \operatorname{dom}(c)$, ordered by inclusion. The union of a chain in P is in P and is an upper bound. Let c be maximal. If $X \in F$ were not in the domain of c, choose $x \in X$; then $c \cup \{(X, x)\}$ would be larger. So c is a choice function for F. □

Remark 3.7.3. The well-ordering theorem was proved from the axiom of choice by Zermelo; see [Zer08]. Zorn's lemma is from [Zor35]. It is the form of choice used most often in algebra, for instance to show that every vector space has a basis or that every nonzero ring has a maximal ideal. Gödel showed that if ZF is consistent then so is ZFC [Gö38], and Cohen that the negation of AC is also consistent with ZF [Coh63]. From now on we work in ZFC.

Proposition 3.7.4. *Every set A has a cardinality $|A|$. For all sets, $A \approx B$ if and only if $|A| = |B|$, and $A \preceq B$ if and only if $|A| \leq |B|$. In particular any two sets are comparable: $A \preceq B$ or $B \preceq A$.*

Proof. By Theorem 3.7.2, every set can be well-ordered, so $|A|$ is defined. The first equivalence is clear. If $|A| \leq |B|$, then $|A| \subseteq |B|$ and $A \approx |A| \preceq |B| \approx B$. Conversely, if $A \preceq B$ and $|B| < |A|$, then $|A| \preceq |B| \subseteq |A|$ and Theorem 3.6.2 would give $|A| \approx |B|$, contradicting that $|A|$ is a cardinal. The last statement follows from Theorem 3.5.4. □

Proposition 3.7.5. *Let κ be an infinite cardinal and $(X_i)_{i \in I}$ a family with $|I| \leq \kappa$ and $|X_i| \leq \kappa$ for all i. Then $|\bigcup_{i \in I} X_i| \leq \kappa$. In particular a countable union of countable sets is countable.*

Proof. We may assume that all X_i are nonempty. By the axiom of choice there is a family $(g_i)_{i \in I}$ of surjections $g_i \colon \kappa \to X_i$ (choose one from each nonempty set of surjections). The map $(i, \xi) \mapsto g_i(\xi)$ is a surjection $q \colon I \times \kappa \to \bigcup_i X_i$. Since $I \times \kappa \preceq \kappa \times \kappa \approx \kappa$ by Theorem 3.6.12, there is an injection $j \colon I \times \kappa \to \kappa$. Sending $u \in \bigcup_i X_i$ to the least element of $j[q^{-1}[\{u\}]]$ is an injection $\bigcup_i X_i \to \kappa$, because the sets $q^{-1}[\{u\}]$ are nonempty and pairwise disjoint. □

Corollary 3.7.6. *If A and B are nonempty and at least one is infinite, then $|A \times B| = |A \sqcup B| = \max(|A|, |B|)$, where $A \sqcup B = (\{0\} \times A) \cup (\{1\} \times B)$.*

Proof. Let $\kappa = \max(|A|, |B|)$, an infinite cardinal, and say $|A| = \kappa$. By Proposition 3.7.4 and Theorem 3.6.12, $A \times B \preceq \kappa \times \kappa \approx \kappa$ and $A \sqcup B \preceq 2 \times \kappa \preceq \kappa \times \kappa \approx \kappa$. Conversely A injects into $A \sqcup B$, and into $A \times B$ via $a \mapsto (a, b_0)$ for a fixed $b_0 \in B$. Conclude with Theorem 3.6.2. □

Remark 3.7.7. We write 2^κ for $|\mathcal{P}(\kappa)|$; by Theorem 3.6.3, $2^\kappa > \kappa$. The *continuum hypothesis* (CH) asserts $2^{\aleph_0} = \aleph_1$. It is independent of ZFC: it can be neither proved [Coh63] nor refuted [Gö38]. In Chapter 4 we show $|\mathbb{R}| = 2^{\aleph_0}$.

Cantor's original formulation of the continuum hypothesis did not mention $\aleph_1$: he conjectured that every infinite set of real numbers is either countable or equinumerous with the whole line, that is, that there is no size strictly between that of $\mathbb{N}$ and that of $\mathbb{R}$. With the axiom of choice the two formulations are equivalent. Since $|\mathbb{R}| = 2^{\aleph_0} = |\mathcal{P}(\omega)|$ (Chapter 4), we state it for subsets of $\mathcal{P}(\omega)$.

Proposition 3.7.8. *The following are equivalent:*

1. $2^{\aleph_0} = \aleph_1$;
2. *there is no cardinal κ with $\aleph_0 < \kappa < 2^{\aleph_0}$;*
3. *every infinite subset of $\mathcal{P}(\omega)$ is either countable or equinumerous with $\mathcal{P}(\omega)$.*

Proof. First, $\aleph_1$ is the least cardinal larger than $\aleph_0$: by Proposition 3.6.11 an infinite cardinal κ is ω_α for a unique α, and $\kappa > \omega_0$ forces $\alpha \geq 1$, hence $\kappa \geq \omega_1$. Also $2^{\aleph_0} > \aleph_0$ by Theorem 3.6.3, so $2^{\aleph_0} \geq \aleph_1$.

(1)⇒(2): a cardinal $\kappa > \aleph_0$ satisfies $\kappa \geq \aleph_1 = 2^{\aleph_0}$.

(2)⇒(1): $\aleph_1$ is a cardinal with $\aleph_0 < \aleph_1 \leq 2^{\aleph_0}$; by (2) it is not strictly smaller than $2^{\aleph_0}$, so $\aleph_1 = 2^{\aleph_0}$.

(2)⇒(3): let $A \subseteq \mathcal{P}(\omega)$ be infinite. By Proposition 3.7.4, A has a cardinality $|A|$ with $|A| \leq 2^{\aleph_0}$, and $|A| \geq \aleph_0$, since an infinite cardinal is at least ω. If $|A| = \aleph_0$, A is countable; otherwise $\aleph_0 < |A| \leq 2^{\aleph_0}$, and by (2) $|A| = 2^{\aleph_0}$, that is, $A \approx \mathcal{P}(\omega)$.

(3)⇒(2): let $\aleph_0 < \kappa \leq 2^{\aleph_0}$. Then κ injects into $\mathcal{P}(\omega)$; its image A is an infinite subset with $|A| = \kappa > \aleph_0$, so it is not countable, and by (3) $\kappa = |A| = 2^{\aleph_0}$. □

Remark 3.7.9.

1. In terms of real numbers, (3) reads: *every uncountable set of real numbers can be put in bijection with* $\mathbb{R}$. Cantor proved this for closed sets, by showing that an uncountable closed set contains a nonempty perfect set (a closed set without isolated points), which has the cardinality of $\mathbb{R}$; the *Cantor–Bendixson theorem*. The same holds for Borel sets, and more generally for analytic sets (Alexandrov, Hausdorff, Suslin) [Jec03]. So no "reasonably definable" set of reals can be a counterexample to CH, and yet CH cannot be proved (Remark 3.7.7): a counterexample, if it exists, is a set that can only be produced by the axiom of choice.

2. The equivalence of (1) and (3) uses the axiom of choice, through Proposition 3.7.4: without it, a set of reals might be uncountable and still not comparable with $\aleph_1$, and the two formulations can come apart.

3. The *generalised continuum hypothesis* (GCH) asserts $2^{\aleph_\alpha} = \aleph_{\alpha+1}$ for all α; by the same argument it says that for every infinite cardinal κ there is no cardinal strictly between κ and 2^κ. It is also consistent with ZFC [Gö38], and so is its negation [Coh63].

Definition 3.7.10. Let α be a limit ordinal. A map $f : \beta \to \alpha$ is *cofinal* if its image is unbounded in α. The *cofinality* $\operatorname{cf}(\alpha)$ is the least ordinal β admitting a cofinal map $\beta \to \alpha$. An infinite cardinal κ is *regular* if $\operatorname{cf}(\kappa) = \kappa$, and *singular* otherwise.

Lemma 3.7.11. *Let κ be a regular cardinal and $X \subseteq \kappa$ a set with $|X| < \kappa$. Then* $\sup X < \kappa$.

Proof. Let $\mu = |X| < \kappa$ and let $f : \mu \to X$ be a bijection. If $\sup X = \kappa$, then f is a cofinal map $\mu \to \kappa$, so $\operatorname{cf}(\kappa) \leq \mu < \kappa$. □

Proposition 3.7.12. *The cardinal ω is regular, and every infinite successor cardinal $\omega_{\alpha+1}$ is regular.*

Proof. A map from a finite ordinal n to ω has a largest value, so it is not cofinal. Let $\kappa = \omega_{\alpha+1}$ and suppose $f : \beta \to \kappa$ is cofinal with $\beta < \kappa$. Then $|\beta| \leq \omega_\alpha$, every $f(\xi) < \kappa$ has $|f(\xi)| \leq \omega_\alpha$, and $\kappa = \bigcup_{\xi<\beta} f(\xi)$ because κ is a limit ordinal. By Proposition 3.7.5, $|\kappa| \leq \omega_\alpha$, a contradiction. □

Counterexample 3.7.13. Not every infinite cardinal is regular. The map $n \mapsto \omega_n$ is cofinal from ω to $\omega_\omega = \sup_n \omega_n$, so $\operatorname{cf}(\omega_\omega) = \omega < \omega_\omega$.

Definition 3.7.14. A cardinal κ is *strongly inaccessible* if it is uncountable, regular, and $2^\lambda < \kappa$ for every cardinal $\lambda < \kappa$.

3.8 Universes

Definition 3.8.1. A *Grothendieck universe* is a set $\mathcal{U}$ such that

1. if $x \in \mathcal{U}$ and $y \in x$, then $y \in \mathcal{U}$;
2. if $x, y \in \mathcal{U}$, then $\{x, y\} \in \mathcal{U}$;
3. if $x \in \mathcal{U}$, then $\mathcal{P}(x) \in \mathcal{U}$;
4. if $I \in \mathcal{U}$ and $(x_i)_{i \in I}$ is a family with $x_i \in \mathcal{U}$ for all i, then $\bigcup_{i \in I} x_i \in \mathcal{U}$.

This is the definition of [AGV72, Exposé I, Appendice].

Lemma 3.8.2. *Let $\mathcal{U}$ be a nonempty universe, $A, B, x, y \in \mathcal{U}$ and $I \in \mathcal{U}$.*

1. *$\emptyset \in \mathcal{U}$, and every subset of an element of $\mathcal{U}$ is in $\mathcal{U}$.*
2. *$\{x\}$, (x, y), $x \cup y$, $A \times B$ and B^A are in $\mathcal{U}$.*
3. *If $x_i \in \mathcal{U}$ for all $i \in I$, then $\prod_{i \in I} x_i \in \mathcal{U}$; and if $f : I \to \mathcal{U}$ is a map, the image $f[I]$ is in $\mathcal{U}$.*
4. *If $\sim$ is an equivalence relation on A, then $A/\sim \in \mathcal{U}$.*

Proof. (1) If $y \subseteq x \in \mathcal{U}$, then $y \in \mathcal{P}(x) \in \mathcal{U}$, so $y \in \mathcal{U}$. Taking any $x \in \mathcal{U}$ and $y = \emptyset$ gives $\emptyset \in \mathcal{U}$. (2) $\{x\} = \{x, x\}$ and $(x, y) = \{\{x\}, \{x, y\}\}$. Since $\{A, B\} \in \mathcal{U}$, the union of the family A, B indexed by it is $A \cup B \in \mathcal{U}$. Then $A \times B \subseteq \mathcal{P}(\mathcal{P}(A \cup B))$ and $B^A \subseteq \mathcal{P}(A \times B)$. (3) $\bigcup_i x_i \in \mathcal{U}$ and $\prod_i x_i \subseteq (\bigcup_i x_i)^I$. The image is $f[I] = \bigcup_{i \in I} \{f(i)\}$. (4) $A/\sim \subseteq \mathcal{P}(A)$. □

Theorem 3.8.3. *If κ is a strongly inaccessible cardinal, then V_κ is a Grothendieck universe and $\omega \in V_\kappa$.*

Proof. First, $|V_\alpha| < \kappa$ for all $\alpha < \kappa$, by transfinite induction: $V_0 = \emptyset$; $|V_{\alpha+1}| = 2^{|V_\alpha|} < \kappa$ by strong inaccessibility; and for a limit $\lambda < \kappa$, the cardinals $|V_\alpha|$, $\alpha < \lambda$, form a set of fewer than κ ordinals below κ, so their supremum μ is $< \kappa$ by Lemma 3.7.11, and $|V_\lambda| \leq \max(|\lambda|, \mu, \omega) < \kappa$ by Proposition 3.7.5.

Since κ is a limit ordinal (being an infinite cardinal), every $x \in V_\kappa$ lies in some $V_{\alpha+1}$ with $\alpha < \kappa$, so $x \subseteq V_\alpha$. We check the axioms. (1) holds because V_κ is transitive (Lemma 3.5.12). (2) If $x \subseteq V_\alpha$ and $y \subseteq V_\beta$ with $\alpha \leq \beta < \kappa$, then $x, y \in V_{\beta+1}$ and $\{x, y\} \in V_{\beta+2}$. (3) If $x \subseteq V_\alpha$ then $\mathcal{P}(x) \subseteq \mathcal{P}(V_\alpha) = V_{\alpha+1}$, so $\mathcal{P}(x) \in V_{\alpha+2}$. (4) Let $I \in V_\kappa$, so $|I| < \kappa$, and for $i \in I$ let $\alpha_i < \kappa$ be least with $x_i \subseteq V_{\alpha_i}$. The set $\{\alpha_i : i \in I\}$ has fewer than κ elements, so its supremum α is $< \kappa$ by Lemma 3.7.11. Then $\bigcup_i x_i \subseteq V_\alpha$, hence lies in $V_{\alpha+1}$. Finally, $\omega \subseteq V_\omega$, so $\omega \in V_{\omega+1}$, and $\omega + 1 < \kappa$ as κ is uncountable. □

Theorem 3.8.4. *A set $\mathcal{U}$ is a Grothendieck universe with $\omega \in \mathcal{U}$ if and only if $\mathcal{U} = V_\kappa$ for a strongly inaccessible cardinal κ.*

One direction is Theorem 3.8.3; for the other, see [AGV72, Exposé I, Appendice].

Remark 3.8.5. If κ is strongly inaccessible, the structure $(V_\kappa, \in)$ is a model of ZFC [Jec03, Chapter 12]. Hence ZFC cannot prove that a universe containing ω exists, if ZFC is consistent (Remark 2.8.10). Following [AGV72], we therefore add to ZFC the axiom that there is a Grothendieck universe containing ω, and we fix one, $\mathcal{U}$, as in Notation 1.6.1. Only this single universe is needed in this project.

Remark 3.8.6. A set is *small* if it is an element of $\mathcal{U}$. By Lemma 3.8.2, the small sets are closed under all the constructions of this chapter applied to small sets: subsets, pairs, products, sets of maps, quotients, power sets, and unions and products of families indexed by small sets. Since $\omega \in \mathcal{U}$, the number systems $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ constructed in Chapter 4 are small, and so is every object built from them by these constructions. A set that is merely equinumerous with a small set need not be small (it may even fail to be a subset of $\mathcal{U}$), but it is in bijection with one, which is all that matters in practice.

3.9 Exercises

Exercise 3.9.1. Let A, B, C be sets. Show that $f \mapsto (a \mapsto (b \mapsto f(a,b)))$ is a bijection $C^{A\times B} \to (C^B)^A$, and that $(A \times B) \times C \approx A \times (B \times C)$.

Exercise 3.9.2. Show that the axiom of choice is equivalent (in ZF) to the statement that every surjection $f : A \to B$ has a *section*, that is, a map $s : B \to A$ with $f \circ s = \mathrm{id}_B$.

Solution. Given AC and a surjection f, choose $s(b)$ from the nonempty set $f^{-1}[\{b\}]$. Conversely, given a set F of nonempty sets, let $A = \{(X, x) : X \in F, x \in X\}$ and let $f : A \to F$ be the first projection, which is surjective. If s is a section, then $X \mapsto$ (second coordinate of $s(X)$) is a choice function. □

Exercise 3.9.3. Using the axiom of choice, show that every infinite set has a countably infinite subset.

Exercise 3.9.4. Show that an ordinal α is a limit ordinal if and only if $\alpha \neq 0$ and $\alpha = \bigcup \alpha$.

Exercise 3.9.5. Show that V_ω is a Grothendieck universe which does not contain ω, and that every element of V_ω is a finite set. Show that $\emptyset$ is also a universe. (So the condition $\omega \in \mathcal{U}$ in Remark 3.8.5 is not automatic.)

Exercise 3.9.6. Show that ω^ω (the set of maps $\omega \to \omega$) and $\mathcal{P}(\omega)$ are equinumerous.

Exercise 3.9.7. Show that for every set A of sets there is a set not equinumerous with a subset of any element of A. Deduce that the class of all cardinals is not a set.

Chapter 4

Number Systems

4.1 Introduction

Starting from the natural numbers of Chapter 3, we construct the integers, the rational numbers, the real numbers and the complex numbers, and establish the properties used everywhere else: the laws of arithmetic, unique factorisation of integers, the least upper bound property of $\mathbb{R}$ and its uniqueness, and the basic properties of $\mathbb{C}$. Along the way we introduce the vocabulary of commutative rings, fields and ordered fields; general rings are studied in Chapter 8.

Classical references are [Lan30], which carries out these constructions in full detail, [Ded72] for the real numbers, and [Rud76, Chapter 1].

Prerequisites. Chapter 3 (Set Theory).

4.2 Arithmetic of the natural numbers

Recall from Example 3.4.7 that addition and multiplication on $\mathbb{N} = \omega$ are defined by $m+0 = m$, $m + S(n) = S(m + n)$, $m \cdot 0 = 0$ and $m \cdot S(n) = m \cdot n + m$. We have $m + 1 = S(m)$, since $m + S(0) = S(m + 0)$. All inductions below are instances of Theorem 3.4.2.

Proposition 4.2.1. *For all $m, n, k \in \mathbb{N}$:*

1. $(m + n) + k = m + (n + k)$;
2. $m + n = n + m$;
3. *if $m + k = n + k$, then $m = n$;*
4. *if $m + n = 0$, then $m = n = 0$.*

Proof. (1) By induction on k: for $k = 0$ both sides are $m+n$, and $(m+n)+S(k) = S((m+n)+k) = S(m + (n + k)) = m + S(n + k) = m + (n + S(k))$.

(2) First, $0 + n = n$ by induction on n. Next, $S(m) + n = S(m + n)$ by induction on n: $S(m) + 0 = S(m) = S(m + 0)$ and $S(m) + S(n) = S(S(m) + n) = S(S(m + n)) = S(m + S(n))$. Now induct on n: $m + 0 = m = 0 + m$, and $m + S(n) = S(m + n) = S(n + m) = S(n) + m$.

(3) By induction on k, using that S is injective (Proposition 3.4.5): $m + S(k) = n + S(k)$ means $S(m + k) = S(n + k)$.

(4) If $n = S(n')$ then $m + n = S(m + n') \neq 0$. So $n = 0$, and then $m = 0$. □

Proposition 4.2.2. *For all $m, n, k \in \mathbb{N}$:*

1. $m \cdot (n + k) = m \cdot n + m \cdot k$;

2. $(m \cdot n) \cdot k = m \cdot (n \cdot k)$;

3. $m \cdot n = n \cdot m$ *and* $1 \cdot n = n$;

4. *if* $m \cdot n = 0$, *then* $m = 0$ *or* $n = 0$.

Proof. (1) By induction on k: $m(n+0) = mn = mn + 0$, and $m(n + S(k)) = m \cdot S(n+k) = m(n+k) + m = (mn + mk) + m = mn + m \cdot S(k)$.

(2) By induction on k: $(mn) \cdot 0 = 0 = m(n \cdot 0)$, and, using (1), $(mn) \cdot S(k) = (mn)k + mn = m(nk) + mn = m(nk + n) = m(n \cdot S(k))$.

(3) By induction, $0 \cdot n = 0$ and $S(m) \cdot n = mn + n$; for the second, $S(m) \cdot S(n) = S(m)n + S(m) = (mn + n) + S(m)$ and $m \cdot S(n) + S(n) = (mn + m) + S(n)$, and both equal $S(mn + m + n)$ by Proposition 4.2.1. Then $mn = nm$ follows by induction on n, and $1 \cdot n = 0 \cdot n + n = n$.

(4) If $m = S(m')$ and $n = S(n')$, then $mn = mn' + m = S(mn' + m')$, which is not 0. □

Proposition 4.2.3. *For* $m, n \in \mathbb{N}$, *the following are equivalent:* $m \leq n$ *in the order of ordinals (Proposition 3.5.7); there is* $k \in \mathbb{N}$ *with* $m + k = n$. *This* k *is unique. Moreover, for all* $m, n, k \in \mathbb{N}$: *if* $m \leq n$ *then* $m + k \leq n + k$ *and* $mk \leq nk$; *and if* $mk = nk$ *with* $k \neq 0$ *then* $m = n$.

Proof. By induction on k, $m \leq m + k$: indeed $m + S(k) = S(m + k)$ contains $m + k$. Conversely we show by induction on n that $m \leq n$ implies $m + k = n$ for some k. For $n = 0$, $m = 0$ and $k = 0$. If $m \leq S(n)$, then $m = S(n)$ (take $k = 0$) or $m \leq n$, in which case $m + k = n$ for some k and $m + S(k) = S(n)$. Uniqueness of k is Proposition 4.2.1(3).

If $m + j = n$, then $(m + k) + j = n + k$ and $mk + jk = nk$, which gives the monotonicity statements. If $mk = nk$ with $k \neq 0$ and, say, $m < n$, write $n = m + j$ with $j \neq 0$; then $mk + jk = mk + 0$, so $jk = 0$, contradicting Proposition 4.2.2(4). □

Proposition 4.2.4 (Division with remainder)**.** *Let* $m, n \in \mathbb{N}$ *with* $n \geq 1$. *There are unique* $q, r \in \mathbb{N}$ *with* $m = qn + r$ *and* $r < n$.

Proof. Existence, by induction on m: for $m = 0$ take $q = r = 0$. If $m = qn + r$ with $r < n$, then $m + 1 = qn + (r + 1)$, and either $r + 1 < n$, or $r + 1 = n$ and $m + 1 = (q + 1)n + 0$. Uniqueness: if $qn + r = q'n + r'$ with $r, r' < n$ and $q < q'$, then $q'n + r' \geq (q + 1)n = qn + n > qn + r$, a contradiction; so $q = q'$, and $r = r'$ by cancellation. □

4.3 Commutative rings and fields

Definition 4.3.1. A *commutative ring* is a set R with elements $0, 1 \in R$ and maps $+, \cdot \colon R \times R \to R$, the *addition* $(a, b) \mapsto a + b$ and the *multiplication* $(a, b) \mapsto a \cdot b$, usually written ab, such that for all $a, b, c \in R$: $(a + b) + c = a + (b + c)$; $a + b = b + a$; $a + 0 = a$; there is $a' \in R$ with $a + a' = 0$; $(ab)c = a(bc)$; $ab = ba$; $1 \cdot a = a$; and $a(b + c) = ab + ac$. A *homomorphism* of commutative rings is a map f with $f(a + b) = f(a) + f(b)$, $f(ab) = f(a)f(b)$ and $f(1) = 1$. A commutative ring is an *integral domain* if $1 \neq 0$ and $ab = 0$ implies $a = 0$ or $b = 0$; it is a *field* if $1 \neq 0$ and every $a \neq 0$ has an element a^{-1} with $aa^{-1} = 1$.

Lemma 4.3.2. *In a commutative ring* R: *the element* a' *with* $a + a' = 0$ *is unique, and we write it* $-a$; $0 \cdot a = 0$; $(-a)b = -(ab)$; $(-a)(-b) = ab$. *A field is an integral domain, and in a field* a^{-1} *is unique.*

Proof. If $a + a' = 0 = a + a''$, then $a' = a' + (a + a'') = (a' + a) + a'' = a''$. Next $0 \cdot a = (0 + 0)a = 0 \cdot a + 0 \cdot a$, and adding $-(0 \cdot a)$ gives $0 \cdot a = 0$. Then $ab + (-a)b = (a + (-a))b = 0$, so $(-a)b = -(ab)$, and $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$. In a field, if $ab = 0$ and $a \neq 0$ then $b = a^{-1}ab = 0$. Uniqueness of inverses is proved like uniqueness of $-a$. □

Definition 4.3.3. An *ordered ring* is a commutative ring R with a total order $\leq$ such that $a \leq b$ implies $a + c \leq b + c$, and $0 \leq a$, $0 \leq b$ imply $0 \leq ab$. An *ordered field* is an ordered ring that is a field. An element a is *positive* if $a > 0$. The *absolute value* is $|a| = a$ if $a \geq 0$ and $|a| = -a$ otherwise.

Lemma 4.3.4. *Let R be an ordered ring with $1 \neq 0$ and $a, b, c \in R$.*

1. *$a < b$ if and only if $-b < -a$; and $a^2 \geq 0$, so $1 > 0$.*
2. *If $a \leq b$ and $c \geq 0$, then $ac \leq bc$.*
3. *$|ab| = |a||b|$ and $|a + b| \leq |a| + |b|$.*
4. *The elements $n \cdot 1 = 1 + \cdots + 1$ (n summands) are positive for $n \geq 1$; in particular they are pairwise distinct.*
5. *If R is a field and $0 < a < b$, then $0 < b^{-1} < a^{-1}$.*

Proof. (1) Add $-a - b$ to $a < b$. If $a \geq 0$ then $a^2 \geq 0$; if $a < 0$ then $-a > 0$ and $a^2 = (-a)(-a) \geq 0$. Then $1 = 1^2 \geq 0$ and $1 \neq 0$. (2) $b - a \geq 0$, so $(b - a)c \geq 0$. (3) Both sides of the first equation are $\pm ab$ with the sign making them non-negative. For the second, $-|a| \leq a \leq |a|$ and similarly for b, so $-(|a| + |b|) \leq a + b \leq |a| + |b|$. (4) By induction, $(n + 1) \cdot 1 = n \cdot 1 + 1 > n \cdot 1$. (5) $a^{-1} = a(a^{-1})^2 \geq 0$ and $a^{-1} \neq 0$; similarly $b^{-1} > 0$. Multiplying $a < b$ by $a^{-1}b^{-1} > 0$ gives $b^{-1} < a^{-1}$. □

4.4 The integers

Definition 4.4.1. On $\mathbb{N} \times \mathbb{N}$ let $(a, b) \sim (c, d)$ if $a + d = b + c$. This is an equivalence relation (transitivity: from $a + d = b + c$ and $c + f = d + e$ we get $a + f + (c + d) = b + e + (c + d)$, and we cancel). The set of *integers* is $\mathbb{Z} = (\mathbb{N} \times \mathbb{N})/\sim$; we write $[a, b]$ for the class of (a, b). Define

$$[a, b] + [c, d] = [a + c, b + d], \qquad [a, b] \cdot [c, d] = [ac + bd, ad + bc],$$

$0 = [0, 0]$, $1 = [1, 0]$, and $[a, b] \leq [c, d]$ if $a + d \leq b + c$.

Theorem 4.4.2. *The operations and the order on $\mathbb{Z}$ are well defined, and make $\mathbb{Z}$ an ordered integral domain. The map $\iota\colon \mathbb{N} \to \mathbb{Z}$, $n \mapsto [n, 0]$, is injective and preserves 0, 1, addition, multiplication and order. Every integer is of the form $\iota(n)$ or $-\iota(n)$ with $n \in \mathbb{N}$. Every nonempty set of integers that is bounded below has a least element.*

Proof. If $(a, b) \sim (a', b')$, that is, $a + b' = b + a'$, then $(a + c) + (b' + d) = (b + d) + (a' + c)$, so addition is well defined. For multiplication, multiplying $a + b' = b + a'$ by c and $b + a' = a + b'$ by d and adding gives $ac + bd + a'd + b'c = ad + bc + a'c + b'd$, which says $(ac + bd, ad + bc) \sim (a'c + b'd, a'd + b'c)$; by symmetry of the formula the same holds in the second argument. For the order, $a + d \leq b + c$ and $a + b' = b + a'$ give $a' + d + (a + b') \leq b' + c + (a + b')$, and we cancel.

The ring axioms reduce to identities in $\mathbb{N}$. For instance, $[b, a]$ is an additive inverse of $[a, b]$, and distributivity follows from $[a, b]([c, d] + [e, f]) = [a(c + e) + b(d + f), a(d + f) + b(c + e)]$, whose entries are those of $[ac + bd, ad + bc] + [ae + bf, af + be]$ by Propositions 4.2.1 and 4.2.2. The other axioms are verified in the same way.

If $a \geq b$, write $a = b + n$; then $[a, b] = [n, 0]$. If $a < b$, then $[a, b] = [0, n] = -[n, 0]$ with $b = a + n$. So every integer is $\iota(n)$ or $-\iota(n)$. The map ι preserves the operations, and $[m, 0] \leq [n, 0]$ if and only if $m \leq n$, so it is injective. The relation $\leq$ is a total order because

$\leq$ is one on $\mathbb{N}$. It is compatible with addition, and if $x, y \geq 0$ then $x = \iota(m)$, $y = \iota(n)$ and $xy = \iota(mn) \geq 0$. If $xy = 0$ with $x = \pm\iota(m)$, $y = \pm\iota(n)$, then $\iota(mn) = 0$, so $m = 0$ or $n = 0$.

Finally, if $X \subseteq \mathbb{Z}$ is nonempty and bounded below by c, then $\{x - c : x \in X\}$ is a nonempty subset of $\iota(\mathbb{N})$, which has a least element by Proposition 3.5.7. □

Notation 4.4.3. We identify $n \in \mathbb{N}$ with $\iota(n) \in \mathbb{Z}$, so that $\mathbb{N} \subseteq \mathbb{Z}$. For a commutative ring R, $n \in \mathbb{N}$ and $a \in R$, we write na for $a + \cdots + a$ (n summands) and $(-n)a = -(na)$.

4.5 Divisibility and prime numbers

Definition 4.5.1. Let $a, b \in \mathbb{Z}$. We say *a divides b*, written $a \mid b$, if $b = ac$ for some $c \in \mathbb{Z}$. An integer $p \geq 2$ is *prime* if its only positive divisors are 1 and p. A *greatest common divisor* of a and b, not both zero, is a positive common divisor d divisible by every common divisor.

Theorem 4.5.2 (Division with remainder)**.** *Let $a, b \in \mathbb{Z}$ with $b \neq 0$. There are unique $q, r \in \mathbb{Z}$ with $a = qb + r$ and $0 \leq r < |b|$.*

Proof. Uniqueness: if $qb + r = q'b + r'$ then $(q - q')b = r' - r$ and $|r' - r| < |b|$, so $q = q'$ and $r = r'$. Existence: the set $\{a - kb : k \in \mathbb{Z}\} \cap \mathbb{N}$ is nonempty (take $k = -|a|$ if $b > 0$ and $k = |a|$ if $b < 0$, for which $a - kb = a + |a||b| \geq 0$), so it has a least element $r = a - qb$. If $r \geq |b|$, then $r - |b| = a - (q \pm 1)b$ is a smaller element. □

Theorem 4.5.3 (Bézout)**.** *Let $a, b \in \mathbb{Z}$ be not both zero, and let d be the least positive element of $\{ax + by : x, y \in \mathbb{Z}\}$. Then d is a greatest common divisor of a and b, and it is the only one. We write $d = \gcd(a, b)$.*

Proof. The set contains $a^2 + b^2 > 0$, so d exists; write $d = ax_0 + by_0$. Divide: $a = qd + r$ with $0 \leq r < d$. Then $r = a(1 - qx_0) + b(-qy_0)$ lies in the set, so $r = 0$ by minimality, and $d \mid a$; similarly $d \mid b$. Every common divisor of a and b divides $ax_0 + by_0 = d$. If d' is another greatest common divisor, then $d \mid d'$ and $d' \mid d$, and both are positive, so $d = d'$. □

Lemma 4.5.4 (Euclid)**.** *Let p be prime and $a, b \in \mathbb{Z}$. If $p \mid ab$, then $p \mid a$ or $p \mid b$.*

Proof. If $p \nmid a$, then $\gcd(p, a) = 1$, since it is a positive divisor of p different from p. By Theorem 4.5.3, $1 = px + ay$, so $b = pbx + aby$ is divisible by p. □

Theorem 4.5.5 (Fundamental theorem of arithmetic)**.** *Every integer $n \geq 2$ is a product $n = p_1 p_2 \cdots p_k$ of primes, and if $n = q_1 \cdots q_l$ is another such product, then $k = l$ and the q_j are a permutation of the p_i.*

Proof. Existence: if some $n \geq 2$ is not a product of primes, take the least such n. It is not prime, so $n = ab$ with $2 \leq a, b < n$, and a, b are products of primes, hence so is n. Uniqueness, by induction on k: p_k divides $q_1 \cdots q_l$, so by Lemma 4.5.4 (and induction on l) it divides some q_j, and then $p_k = q_j$ since q_j is prime. Cancelling gives two factorisations of n/p_k and the induction hypothesis applies (if $k = 1$, then $n = p_1$ is prime and $l = 1$). □

Theorem 4.5.6 (Euclid)**.** *There are infinitely many prime numbers.*

Proof. Let $p_1, \ldots, p_k$ be primes and $N = p_1 \cdots p_k + 1 \geq 2$. By Theorem 4.5.5, some prime p divides N. If $p = p_i$, then p divides $N - p_1 \cdots p_k = 1$, which is impossible. So no finite set of primes contains all primes. □

Example 4.5.7. Repeated division with remainder computes greatest common divisors, since $\gcd(a,b) = \gcd(b,r)$ when $a = qb + r$. For $a = 252$, $b = 198$: $252 = 1 \cdot 198 + 54$, $198 = 3 \cdot 54 + 36$, $54 = 1 \cdot 36 + 18$, $36 = 2 \cdot 18$. So $\gcd(252, 198) = 18$, and going backwards, $18 = 54 - 36 = 4 \cdot 54 - 198 = 4 \cdot 252 - 5 \cdot 198$.

Remark 4.5.8. Unique factorisation is a special property of $\mathbb{Z}$. In the ring $\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\}$ one has $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$, and none of these four factors can be written as a product of two non-units. Such phenomena are studied in Chapters 8 and 14.

4.6 The rational numbers

Definition 4.6.1. Let $P = \{(a,b) \in \mathbb{Z} \times \mathbb{Z} : b > 0\}$ and $(a,b) \sim (c,d)$ if $ad = bc$. This is an equivalence relation (for transitivity, $ad = bc$ and $cf = de$ give $adf = bcf = bde$, and we cancel $d \neq 0$ in the integral domain $\mathbb{Z}$). The set of *rational numbers* is $\mathbb{Q} = P/\!\sim$, and a/b denotes the class of (a,b). Define

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \qquad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}, \qquad \frac{a}{b} \leq \frac{c}{d} \iff ad \leq bc.$$

Theorem 4.6.2. *The operations and the order on $\mathbb{Q}$ are well defined and make $\mathbb{Q}$ an ordered field. The map $\mathbb{Z} \to \mathbb{Q}$, $a \mapsto a/1$, is an injective ring homomorphism preserving the order, and $\mathbb{Q}$ is* Archimedean*: for every $x \in \mathbb{Q}$ there is $n \in \mathbb{N}$ with $x < n$.*

Proof. Well-definedness: if $ab' = a'b$, then $(ad + bc)(b'd) = (a'd + b'c)(bd)$ and $(ac)(b'd) = (a'c)(bd)$, and $ad \leq bc$ is equivalent to $ab'd \leq bb'c$ (multiply by $b' > 0$), that is to $a'bd \leq bb'c$, that is to $a'd \leq b'c$ (cancel $b > 0$). The field axioms are identities in $\mathbb{Z}$ as in Theorem 4.4.2; the zero is $0/1$, the unit $1/1$, $-(a/b) = (-a)/b$, and for $a \neq 0$ the inverse of a/b is b/a if $a > 0$ and $(-b)/(-a)$ if $a < 0$. The order is total, and compatible with the operations: $a/b \geq 0$ means $a \geq 0$, and a product or sum of such fractions again has non-negative numerator. The map $a \mapsto a/1$ is clearly an injective order-preserving homomorphism. Finally, for $x = a/b$ we have $x \leq |a| < |a| + 1$ because $b \geq 1$. □

Proposition 4.6.3. *Let K be an ordered field. There is a unique ring homomorphism $\mathbb{Q} \to K$; it is injective and order-preserving. We use it to regard $\mathbb{Q}$ as a subfield of K.*

Proof. By Lemma 4.3.4(4) the map $\mathbb{N} \to K$, $n \mapsto n \cdot 1$, is injective, and it preserves sums and products (by induction). Extend it to $\mathbb{Z}$ by $-n \mapsto -(n \cdot 1)$; this is an injective order-preserving ring homomorphism j. Define $f(a/b) = j(a)j(b)^{-1}$; this is well defined since $ad = bc$ gives $j(a)j(d) = j(b)j(c)$, and it is a ring homomorphism by the formulas of Definition 4.6.1. It preserves order: for $b, d > 0$, $a/b \leq c/d$ iff $ad \leq bc$ iff $j(a)j(d) \leq j(b)j(c)$ iff $j(a)j(b)^{-1} \leq j(c)j(d)^{-1}$, by Lemma 4.3.4(5). Hence it is injective. Uniqueness: a ring homomorphism g satisfies $g(1) = 1$, hence $g(n) = n \cdot 1$ and $g(-n) = -g(n)$, and $g(b)g(a/b) = g(a)$, so $g(a/b) = j(a)j(b)^{-1}$. □

Proposition 4.6.4. *There is no $x \in \mathbb{Q}$ with $x^2 = 2$.*

Proof. Suppose $(a/b)^2 = 2$ with $a, b > 0$, so $a^2 = 2b^2$. In the factorisations of Theorem 4.5.5, the prime 2 occurs an even number of times in a^2 and an odd number of times in $2b^2$, contradicting uniqueness. □

Counterexample 4.6.5. The set $A = \{x \in \mathbb{Q} : x \leq 0 \text{ or } x^2 < 2\}$ is nonempty and bounded above (by 2), but has no least upper bound in $\mathbb{Q}$. Indeed, for $p \in \mathbb{Q}$ with $p > 0$ put $q = p - \frac{p^2-2}{p+2} = \frac{2p+2}{p+2}$;

then $q^2 - 2 = \frac{2(p^2-2)}{(p+2)^2}$. If $p \in A$ and $p > 0$, then $p^2 < 2$, so $q > p$ and $q^2 < 2$: A has no largest element. If $p > 0$ is an upper bound of A, then $p^2 > 2$ by Proposition 4.6.4 (and $p^2 < 2$ is impossible since then $p \in A$ is largest), so $q < p$ and $q^2 > 2$, and q is a smaller upper bound. Hence no upper bound is least.

4.7 Complete ordered fields

Definition 4.7.1. An ordered field K is *complete* if every nonempty subset of K that is bounded above has a least upper bound (supremum) in K. It is *Archimedean* if for every $x \in K$ there is $n \in \mathbb{N}$ with $x < n$.

Lemma 4.7.2. *In a complete ordered field every nonempty subset bounded below has a greatest lower bound (infimum), namely* $\inf A = -\sup(-A)$. *If A, B are nonempty and bounded above, then* $\sup(A + B) = \sup A + \sup B$, *where* $A + B = \{a + b : a \in A, b \in B\}$; *if moreover A and B consist of positive elements, then* $\sup(AB) = \sup A \cdot \sup B$.

Proof. The first claim is clear since $x \mapsto -x$ reverses the order. Let $s = \sup A$, $t = \sup B$. Clearly $s + t$ is an upper bound of $A + B$. If $u < s + t$, then $u - t < s$, so there is $a \in A$ with $a > u - t$; then $u - a < t$ and there is $b \in B$ with $b > u - a$, so $a + b > u$. For products, st is an upper bound of AB by Lemma 4.3.4(2). If $0 < u < st$, then $u/t < s$, so some $a \in A$ has $a > u/t$, and then $u/a < t$ and some $b \in B$ has $b > u/a$, so $ab > u$. Upper bounds $u \le 0$ are not smaller than $st > 0$. □

Proposition 4.7.3. *A complete ordered field is Archimedean. In an Archimedean ordered field,* $\mathbb{Q}$ *is* dense*: for all $x < y$ there is $q \in \mathbb{Q}$ with $x < q < y$.*

Proof. If $\mathbb{N} \subseteq K$ were bounded above, let $s = \sup \mathbb{N}$. Then $s - 1$ is not an upper bound, so $n > s - 1$ for some $n \in \mathbb{N}$, and $n + 1 > s$; contradiction.

Let K be Archimedean and $x < y$. Choose $n \in \mathbb{N}$ with $n > (y - x)^{-1}$, so $ny > nx + 1$. The set of integers m with $m > nx$ is nonempty and bounded below (both by the Archimedean property applied to $\pm nx$), so it has a least element m by Theorem 4.4.2. Then $m - 1 \le nx$, so $nx < m \le nx + 1 < ny$, and $q = m/n$ works. □

Proposition 4.7.4. *Let K be a complete ordered field, $x \in K$ with $x \ge 0$, and $n \ge 1$. There is a unique $y \in K$ with $y \ge 0$ and $y^n = x$. We write $y = \sqrt[n]{x}$, or $\sqrt{x}$ if $n = 2$. In particular the non-negative elements of K are exactly the squares.*

Proof. Uniqueness: if $0 \le y_1 < y_2$ then $y_1^n < y_2^n$. For existence we may assume $x > 0$ and $n \ge 2$ (for $n = 1$ take $y = x$). For $0 < a < b$ we have $b^n - a^n = (b-a)(b^{n-1} + b^{n-2}a + \cdots + a^{n-1}) < (b - a)nb^{n-1}$. Let $E = \{t \in K : t \ge 0, t^n < x\}$. It contains 0, and is bounded above by $1 + x$ (if $t > 1 + x$ then $t^n \ge t > x$). Let $y = \sup E$; then $y > 0$, since $t = x/(1 + x)$ lies in E.

Suppose $y^n < x$. Choose h with $0 < h < 1$ and $h < (x - y^n)/(n(y + 1)^{n-1})$. Then $(y + h)^n - y^n < hn(y + h)^{n-1} < hn(y + 1)^{n-1} < x - y^n$, so $y + h \in E$, contradicting that y is an upper bound.

Suppose $y^n > x$. Let $k = (y^n - x)/(ny^{n-1})$; then $0 < k < y$. For $t \ge y - k$ we have $y^n - t^n \le y^n - (y - k)^n < kny^{n-1} = y^n - x$, so $t^n > x$ and $t \notin E$. Hence $y - k$ is an upper bound of E, contradicting that y is least. So $y^n = x$. □

Definition 4.7.5. Let K be an ordered field. A sequence $(x_n)_{n \in \mathbb{N}}$ in K *converges* to $x \in K$ if for every $\varepsilon > 0$ in K there is N with $|x_n - x| < \varepsilon$ for all $n \ge N$. It is a *Cauchy sequence* if for every $\varepsilon > 0$ there is N with $|x_n - x_m| < \varepsilon$ for all $m, n \ge N$.

Proposition 4.7.6. *In a complete ordered field every Cauchy sequence converges.*

Proof. Let (x_n) be Cauchy. Choosing N for $\varepsilon = 1$, all terms lie between $\min(x_0, \dots, x_N) - 1$ and $\max(x_0, \dots, x_N) + 1$, so the sequence is bounded. Let $s_n = \sup\{x_k : k \geq n\}$; this sequence is decreasing and bounded below, and we let $x = \inf_n s_n$. Given $\varepsilon > 0$, choose N with $|x_n - x_m| < \varepsilon$ for $m, n \geq N$. For $n \geq N$, every x_k with $k \geq n$ lies in $(x_n - \varepsilon, x_n + \varepsilon)$, so $x_n - \varepsilon \leq s_m \leq x_n + \varepsilon$ for all $m \geq n$; since the s_m decrease, $x = \inf_{m \geq n} s_m$ lies in $[x_n - \varepsilon, x_n + \varepsilon]$. Hence $|x_n - x| \leq \varepsilon$ for $n \geq N$. □

Theorem 4.7.7. *Let K and L be complete ordered fields. There is exactly one ring homomorphism $K \to L$, and it is an order-preserving isomorphism of fields.*

Proof. Regard $\mathbb{Q} \subseteq K$ and $\mathbb{Q} \subseteq L$ by Proposition 4.6.3. For $x \in K$ let $D(x) = \{q \in \mathbb{Q} : q < x\}$, which is nonempty and bounded above in $\mathbb{Q}$ by Proposition 4.7.3. Define $\varphi(x) = \sup_L D(x)$.

By density, $x = \sup_K D(x)$, and $x < y$ implies that $D(x) \subsetneq D(y)$ contains two rationals between x and y, so $\varphi(x) < \varphi(y)$. For $q \in \mathbb{Q}$, $\varphi(q) = q$. Every $y \in L$ is in the image: with $D_L(y) = \{q : q < y\}$ and $x = \sup_K D_L(y)$ we get $D(x) = D_L(y)$ by density in both fields, so $\varphi(x) = y$. So φ is an order-preserving bijection.

By density, $D(x + y) = D(x) + D(y)$, so Lemma 4.7.2 gives $\varphi(x + y) = \varphi(x) + \varphi(y)$. For $x, y > 0$, the positive elements of $D(xy)$ are the products of positive elements of $D(x)$ and $D(y)$, so $\varphi(xy) = \varphi(x)\varphi(y)$. Since $\varphi(-x) = -\varphi(x)$ (from additivity) and $\varphi(0) = 0$, multiplicativity follows for all signs. Hence φ is an isomorphism.

Let $\psi \colon K \to L$ be any ring homomorphism. By Proposition 4.7.4, $x \geq 0$ implies $x = y^2$ and so $\psi(x) = \psi(y)^2 \geq 0$; hence ψ is order-preserving. Also ψ is the identity on $\mathbb{Q}$ by Proposition 4.6.3. For $q \in D(x)$ we get $q = \psi(q) \leq \psi(x)$, so $\varphi(x) \leq \psi(x)$; applying this to $-x$ gives $\varphi(x) \geq \psi(x)$. So $\psi = \varphi$. □

4.8 The real numbers

Definition 4.8.1. Let C be the set of Cauchy sequences in $\mathbb{Q}$ and $\mathcal{N} \subseteq C$ the set of sequences converging to 0. For $x, y \in C$ write $x \sim y$ if $x - y \in \mathcal{N}$. The set of *real numbers* is $\mathbb{R} = C/{\sim}$. Operations are defined termwise: $[x] + [y] = [(x_n + y_n)_n]$ and $[x] \cdot [y] = [(x_n y_n)_n]$. A real number $[x]$ is *positive* if there are $\varepsilon \in \mathbb{Q}$, $\varepsilon > 0$, and N with $x_n > \varepsilon$ for all $n \geq N$, and $[x] < [y]$ means that $[y] - [x]$ is positive. A rational number q is identified with the class of the constant sequence $(q)_n$.

Lemma 4.8.2. 1. *Every $x \in C$ is bounded, and C is closed under termwise sums and products.*

2. *$\mathcal{N}$ is closed under sums, and $xz \in \mathcal{N}$ for $x \in C$ and $z \in \mathcal{N}$. In particular $\sim$ is an equivalence relation compatible with sums and products.*

3. *If $x \in C \setminus \mathcal{N}$, there are a rational $\varepsilon > 0$ and N such that either $x_n > \varepsilon$ for all $n \geq N$, or $x_n < -\varepsilon$ for all $n \geq N$.*

Proof. (1) Boundedness is proved as in Proposition 4.7.6. If $|x_n|, |y_n| \leq M$, then $|x_n y_n - x_m y_m| \leq M|y_n - y_m| + M|x_n - x_m|$, and similarly for sums. (2) If $|x_n| \leq M$ and $z \to 0$, then $|x_n z_n| \leq M|z_n| \to 0$. Sums are similar. If $x \sim x'$ and $y \sim y'$, then $xy - x'y' = x(y - y') + y'(x - x') \in \mathcal{N}$. (3) Since $x \notin \mathcal{N}$, there is a rational $\delta > 0$ such that $|x_n| \geq \delta$ for infinitely many n. Choose N with $|x_n - x_m| < \delta/2$ for $m, n \geq N$, and $m \geq N$ with $|x_m| \geq \delta$. If $x_m \geq \delta$ then $x_n > \delta/2$ for all $n \geq N$; if $x_m \leq -\delta$ then $x_n < -\delta/2$ for all $n \geq N$. Take $\varepsilon = \delta/2$. □

Theorem 4.8.3. *$\mathbb{R}$ is a complete ordered field, and $\mathbb{Q} \to \mathbb{R}$ is an injective order-preserving ring homomorphism. By Theorem 4.7.7, $\mathbb{R}$ is the unique complete ordered field up to unique isomorphism.*

Proof. By Lemma 4.8.2, the operations are well defined, and the ring axioms hold termwise. If $[x] \neq 0$, then by part (3) of the lemma there is N with $|x_n| > \varepsilon$ for $n \geq N$. Let $y_n = x_n^{-1}$ for $n \geq N$ and $y_n = 0$ otherwise; since $|y_n - y_m| = |x_m - x_n|/|x_n x_m| \leq \varepsilon^{-2}|x_m - x_n|$ for $m, n \geq N$, y is Cauchy, and $xy - 1$ is eventually 0. So $\mathbb{R}$ is a field.

Positivity is well defined: if $x_n > \varepsilon$ for $n \geq N$ and $z \in \mathcal{N}$, then $x_n + z_n > \varepsilon/2$ for large n. By part (3), every nonzero real number is either positive or its negative is positive, and not both; sums and products of positive numbers are positive. It follows that $<$ is a strict total order compatible with the ring operations, so $\mathbb{R}$ is an ordered field. The embedding of $\mathbb{Q}$ is clearly an injective order-preserving homomorphism. $\mathbb{R}$ is Archimedean: if $|x_n| \leq M \in \mathbb{N}$ for all n, then $[x] < M + 1$.

Convergence of rational approximations. Let $x \in C$. For rational $\varepsilon > 0$ choose N with $|x_n - x_m| < \varepsilon/2$ for $m, n \geq N$; then for $n \geq N$ the sequence $(\varepsilon - |x_n - x_m|)_m$ is eventually $> \varepsilon/2$, so $|[x] - x_n| < \varepsilon$ in $\mathbb{R}$. Thus the rational sequence (x_n) converges to $[x]$ in $\mathbb{R}$; by density of $\mathbb{Q}$ this holds for every real $\varepsilon > 0$.

Completeness. Let $A \subseteq \mathbb{R}$ be nonempty and bounded above. Let u_0 be the least integer that is an upper bound of A, and a_0 an integer smaller than some element of A; both exist by the Archimedean property and Theorem 4.4.2, and $a_0 < u_0$. Define rationals $a_n < u_n$ by recursion: let $m = (a_n + u_n)/2$; if m is an upper bound of A, put $(a_{n+1}, u_{n+1}) = (a_n, m)$, and otherwise $(a_{n+1}, u_{n+1}) = (m, u_n)$. Then every u_n is an upper bound, no a_n is, the sequences are monotone, and $u_n - a_n = (u_0 - a_0)/2^n \leq (u_0 - a_0)/n$ for $n \geq 1$. Hence (u_n) and (a_n) are Cauchy sequences in $\mathbb{Q}$ with the same class $s \in \mathbb{R}$, and both converge to s in $\mathbb{R}$. If $a \in A$, then $a \leq u_n$ for all n, hence $a \leq s$. If $t < s$, then $a_n > t$ for some n, and since a_n is not an upper bound, neither is t. So $s = \sup A$. □

Remark 4.8.4. Dedekind's construction [Ded72] defines a real number as a *cut*: a nonempty proper subset $\alpha \subsetneq \mathbb{Q}$ which is downward closed and has no largest element. The order is inclusion and the supremum of a bounded set of cuts is their union. It also produces a complete ordered field (Exercise 4.10.1), which by Theorem 4.7.7 is canonically isomorphic to the one constructed above. See [Rud76, Appendix to Chapter 1].

Notation 4.8.5. For $a \leq b$ in $\mathbb{R}$ we write $[a, b] = \{x : a \leq x \leq b\}$, $(a, b) = \{x : a < x < b\}$, and similarly $[a, b)$, $(a, b]$, $[a, \infty)$ and so on. We use the *extended real line* $[-\infty, \infty]$, obtained by adding a least and a greatest element, where convenient.

Proposition 4.8.6. $|\mathbb{R}| = 2^{\aleph_0}$. *In particular $\mathbb{R}$ is uncountable.*

Proof. The map $x \mapsto \{q \in \mathbb{Q} : q < x\}$ is injective by density of $\mathbb{Q}$, so $\mathbb{R} \preceq \mathcal{P}(\mathbb{Q})$. Since $\mathbb{Q}$ is countable (it injects into $\mathbb{Z}\times\mathbb{Z}$, which injects into $\mathbb{N}\times\mathbb{N}$, which is countable by Corollary 3.6.13), $\mathcal{P}(\mathbb{Q}) \preceq \mathcal{P}(\mathbb{N})$.

Conversely, for $S \subseteq \mathbb{N}$ let $f(S) = \sup\{\sum_{n\in S, n<N} 2 \cdot 3^{-(n+1)} : N \in \mathbb{N}\}$; the partial sums are at most $\sum_{n<N} 2 \cdot 3^{-(n+1)} = 1 - 3^{-N} < 1$. Let $S \neq T$ and let n be the least element of the symmetric difference, say $n \in S \setminus T$. The sums over S and T agree below n, the term $2 \cdot 3^{-(n+1)}$ occurs only for S, and the terms of T above n add up to at most $\sum_{k=n+1}^{M} 2 \cdot 3^{-(k+1)} < 3^{-(n+1)}$. Hence $f(S) - f(T) \geq 2 \cdot 3^{-(n+1)} - 3^{-(n+1)} > 0$, so f is injective. By Theorem 3.6.2, $\mathbb{R} \approx \mathcal{P}(\mathbb{N})$, and $\mathbb{R}$ is uncountable by Theorem 3.6.3. □

4.9 The complex numbers

Definition 4.9.1. The set of *complex numbers* is $\mathbb{C} = \mathbb{R} \times \mathbb{R}$ with

$$(a, b) + (c, d) = (a + c, b + d), \qquad (a, b)(c, d) = (ac - bd, ad + bc).$$

We identify $a \in \mathbb{R}$ with $(a,0)$ and write $i = (0,1)$, so that $(a,b) = a + bi$. For $z = a + bi$, the *real part* is $\operatorname{Re} z = a$, the *imaginary part* is $\operatorname{Im} z = b$, the *conjugate* is $\bar{z} = a - bi$, and the *absolute value* is $|z| = \sqrt{a^2 + b^2}$.

Theorem 4.9.2. *$\mathbb{C}$ is a field containing $\mathbb{R}$ as a subfield, and $i^2 = -1$. Every $z \in \mathbb{C}$ can be written uniquely as $a + bi$ with $a, b \in \mathbb{R}$. Complex conjugation is a field automorphism of $\mathbb{C}$ that is the identity exactly on $\mathbb{R}$, and for all $z, w \in \mathbb{C}$: $z\bar{z} = |z|^2$, $|zw| = |z||w|$, $|\operatorname{Re} z| \le |z|$ and $|z + w| \le |z| + |w|$.*

Proof. The ring axioms are direct computations; for instance the product is associative because both $((a,b)(c,d))(e,f)$ and $(a,b)((c,d)(e,f))$ equal $(ace - bde - adf - bcf,\ acf - bdf + ade + bce)$. The unit is $(1,0)$, and $(0,1)(0,1) = (-1,0)$. If $z = a + bi \neq 0$, then $a^2 + b^2 > 0$ and $z \cdot (a - bi)/(a^2 + b^2) = 1$. The map $a \mapsto (a,0)$ is an injective ring homomorphism. Conjugation preserves sums, and $\overline{(a + bi)(c + di)} = (ac - bd) - (ad + bc)i = (a - bi)(c - di)$; it is its own inverse and fixes $a + bi$ iff $b = 0$.

$z\bar{z} = a^2 + b^2 = |z|^2$. Then $|zw|^2 = zw\bar{z}\bar{w} = |z|^2|w|^2$, and both $|zw|$ and $|z||w|$ are non-negative, so they are equal by the uniqueness in Proposition 4.7.4. Clearly $|\operatorname{Re} z| = |a| \le \sqrt{a^2 + b^2}$. Finally $|z + w|^2 = |z|^2 + 2\operatorname{Re}(z\bar{w}) + |w|^2 \le |z|^2 + 2|z||w| + |w|^2 = (|z| + |w|)^2$. □

Proposition 4.9.3. *Every complex number has a square root: for $z = a + bi$, the number*

$$w = \sqrt{\tfrac{|z|+a}{2}} + \sigma\sqrt{\tfrac{|z|-a}{2}}\, i, \qquad \sigma = 1 \text{ if } b \ge 0,\ \sigma = -1 \text{ if } b < 0,$$

satisfies $w^2 = z$. Consequently every polynomial equation $z^2 + pz + q = 0$ with $p, q \in \mathbb{C}$ has a solution in $\mathbb{C}$.

Proof. Both radicands are non-negative since $|a| \le |z|$. We have $w^2 = \frac{|z|+a}{2} - \frac{|z|-a}{2} + 2\sigma\sqrt{\frac{|z|^2-a^2}{4}}\, i = a + \sigma|b|i = a + bi$. For the equation, take $z = (-p + w)/2$ where $w^2 = p^2 - 4q$. □

Counterexample 4.9.4. There is no order on $\mathbb{C}$ making it an ordered field. Indeed, in an ordered field squares are non-negative and $-1 < 0$ (Lemma 4.3.4), but $-1 = i^2$.

Remark 4.9.5. The fundamental theorem of algebra, which says that every nonconstant complex polynomial has a root, is proved algebraically in Chapter 9 and again analytically in Chapter 29. The polar form $z = |z|e^{i\theta}$ requires the exponential function, which is defined in Chapter 27.

4.10 Exercises

Exercise 4.10.1. Carry out Dedekind's construction of Remark 4.8.4: define the sum of cuts α, β as $\{a + b : a \in \alpha, b \in \beta\}$, define products first for cuts containing positive rationals, and show that the cuts form a complete ordered field.

Exercise 4.10.2. Show that the only ring homomorphism $\mathbb{R} \to \mathbb{R}$ is the identity.

Solution. This is the uniqueness statement of Theorem 4.7.7 with $K = L = \mathbb{R}$. □

Exercise 4.10.3. Show that the only ring homomorphisms $\mathbb{C} \to \mathbb{C}$ that are the identity on $\mathbb{R}$ are the identity and complex conjugation. (It is a much deeper fact, relevant to Chapter 75, that $\mathbb{C}$ has many other field automorphisms, which are not the identity on $\mathbb{R}$.)

Solution. Such a homomorphism σ is determined by $\sigma(i)$, because $\sigma(a + bi) = a + b\sigma(i)$. Since $\sigma(i)^2 = \sigma(-1) = -1$, and the only solutions of $w^2 = -1$ are $\pm i$ (if $w^2 = i^2$ then $(w - i)(w + i) = 0$), we get $\sigma(i) = \pm i$. □

Exercise 4.10.4. Let $n \geq 1$ and $m \geq 0$ be integers. Show that $\sqrt[n]{m}$ is rational if and only if m is the n-th power of an integer.

Exercise 4.10.5. Let R be a commutative ring. Show that there is exactly one ring homomorphism $\mathbb{Z} \to R$.

Exercise 4.10.6. A complex number is *algebraic* if it is a root of a nonzero polynomial with integer coefficients. Show that the set of algebraic numbers is countable, and deduce that there are complex (even real) numbers that are not algebraic.

Exercise 4.10.7. Show that for every $x \in [0, 1)$ there is a unique sequence $(d_n)_{n\geq 1}$ with $d_n \in \{0, 1, \ldots, 9\}$, not eventually equal to 9, such that $x = \sup_N \sum_{n=1}^{N} d_n 10^{-n}$.

Chapter 5

Model Theory

5.1 Introduction

Chapter 2 proved the compactness and Löwenheim–Skolem theorems for countable languages, as consequences of the completeness theorem, and without the axiom of choice. With the set theory of Chapter 3 (cardinals and choice) we can now say how large the models of a first-order theory can be. The downward Löwenheim–Skolem theorem (Theorem 5.2.5) produces small elementary substructures; ultraproducts and Łoś's theorem (Theorem 5.3.4) give the compactness theorem for languages of any size (Theorem 5.3.5); and together they give the upward Löwenheim–Skolem theorem (Theorem 5.4.1): a first-order theory with an infinite model has models of every infinite cardinality at least that of the language. So first-order logic cannot pin down an infinite structure up to isomorphism.

In the other direction, the Łoś–Vaught test (Theorem 5.4.3) shows that a theory with a unique model of some infinite cardinality is complete, and Cantor's back-and-forth argument (Theorem 5.4.5) supplies an example: the dense linear orders. Back-and-forth arguments are formalised by Ehrenfeucht–Fraïssé theory (Section 5.5), which leads to Lindström's theorem (Theorem 5.6.2): first-order logic is the strongest logic that has both the compactness and the Löwenheim–Skolem property. References: [CK90], [Mar02], [EFT94, Chapters XII and XIII].
Prerequisites. Chapter 2 (Mathematical Logic), Chapter 3 (Set Theory) and Chapter 4 (Number Systems).
On a first reading. No later chapter depends on this one, and it can be skipped entirely on the way to the Hodge conjecture.

Notation 5.1.1. We work in ZFC. Languages, formulas and structures are as in Chapter 2, but a language $\mathcal{L}$ may now have any cardinality; $|\mathcal{L}|$ is the cardinality of its set of constant, function and relation symbols, and $\|\mathcal{L}\| = \max(|\mathcal{L}|, \aleph_0)$. If the free variables of a formula φ are among the distinct variables $x_1, \dots, x_n$, we write $\varphi(x_1, \dots, x_n)$, and for elements $a_1, \dots, a_n$ of a structure $\mathcal{A}$ we write $\mathcal{A} \models \varphi[a_1, \dots, a_n]$ if $\mathcal{A} \models \varphi[s]$ for an assignment s with $s(x_i) = a_i$; by the coincidence lemma (Lemma 2.4.4) this does not depend on s. Tuples are written $\bar{a} = (a_1, \dots, a_n)$, and $\mathcal{A}$ also denotes the universe of $\mathcal{A}$. If $\mathcal{L} \subseteq \mathcal{L}'$ and $\mathcal{A}'$ is an $\mathcal{L}'$-structure, its *reduct* $\mathcal{A}'|_{\mathcal{L}}$ forgets the symbols not in $\mathcal{L}$, and $\mathcal{A}'$ is an *expansion* of $\mathcal{A}'|_{\mathcal{L}}$; by induction on formulas, an $\mathcal{L}$-formula has the same truth value in $\mathcal{A}'$ and in $\mathcal{A}'|_{\mathcal{L}}$.

Lemma 5.1.2. *A language $\mathcal{L}$ has at most $\|\mathcal{L}\|$ terms and at most $\|\mathcal{L}\|$ formulas.*

Proof. Let $\kappa = \|\mathcal{L}\|$. Let T_0 be the set of variables and constant symbols, and T_{k+1} the union of T_k with the set of terms $f(t_1, \dots, t_n)$ with $t_i \in T_k$. By induction $|T_k| \leq \kappa$: the new terms are indexed by the pairs of a function symbol and a finite tuple of elements of T_k, and there are at most $\kappa \cdot \kappa = \kappa$ of these, by Proposition 3.7.5 and Theorem 3.6.12. The union of the T_k contains

the variables and constants and is closed under the function symbols, so it contains every term, and it has cardinality at most κ by Proposition 3.7.5. The same argument, starting from the atomic formulas and closing under $\neg$, $\to$ and $\forall x$, bounds the formulas. □

5.2 Elementary substructures

Definition 5.2.1. Let $\mathcal{A}$ and $\mathcal{B}$ be $\mathcal{L}$-structures.

1. An *embedding* $h\colon \mathcal{A} \to \mathcal{B}$ is an injective map preserving constants, functions and relations as in Definition 2.4.10 (relations in both directions), but not necessarily surjective. If $\mathcal{A} \subseteq \mathcal{B}$ as sets and the inclusion is an embedding, $\mathcal{A}$ is a *substructure* of $\mathcal{B}$.

2. An embedding h is *elementary* if $\mathcal{A} \models \varphi[\bar{a}]$ if and only if $\mathcal{B} \models \varphi[h(\bar{a})]$ for every formula $\varphi(x_1, \dots, x_n)$ and every $\bar{a} \in \mathcal{A}^n$. A substructure whose inclusion is elementary is an *elementary substructure*, written $\mathcal{A} \preceq \mathcal{B}$; then $\mathcal{B}$ is an *elementary extension* of $\mathcal{A}$.

3. $\mathcal{A}$ and $\mathcal{B}$ are *elementarily equivalent*, $\mathcal{A} \equiv \mathcal{B}$, if they satisfy the same sentences, that is, $\mathrm{Th}(\mathcal{A}) = \mathrm{Th}(\mathcal{B})$.

A subset of $\mathcal{B}$ is the universe of a substructure if and only if it is nonempty, contains the constants $c^{\mathcal{B}}$ and is closed under the functions $f^{\mathcal{B}}$; the substructure is then unique. Isomorphisms are elementary (Proposition 2.4.11), and $\mathcal{A} \preceq \mathcal{B}$ implies $\mathcal{A} \equiv \mathcal{B}$, by taking $n = 0$. By induction on terms, a substructure computes the values of terms at elements of $\mathcal{A}$ as $\mathcal{B}$ does, so quantifier-free formulas have the same truth values in $\mathcal{A}$ and $\mathcal{B}$; the difference is in the quantifiers.

Example 5.2.2.

1. In the language of orders, $(\mathbb{N}, <)$ is a substructure of $(\mathbb{Z}, <)$ but not an elementary one: for $\varphi(x) = \exists y\, (y < x)$, $(\mathbb{Z}, <) \models \varphi[0]$ and $(\mathbb{N}, <) \not\models \varphi[0]$.

2. $\mathcal{A} \subseteq \mathcal{B}$ and $\mathcal{A} \equiv \mathcal{B}$ do not imply $\mathcal{A} \preceq \mathcal{B}$. The structure $(\mathbb{N}_{>0}, <)$ is a substructure of $(\mathbb{N}, <)$ and is isomorphic to it by $n \mapsto n - 1$, so the two are elementarily equivalent; but 1 has a predecessor in $\mathbb{N}$ and none in $\mathbb{N}_{>0}$, so the same formula φ shows that the inclusion is not elementary.

3. In the language of rings, $\mathbb{Z}$ is a substructure of $\mathbb{Q}$, and not even elementarily equivalent to it: the sentence $\exists y\, (y + y = 1)$ holds in $\mathbb{Q}$ and not in $\mathbb{Z}$.

The following test only involves truth in the larger structure.

Lemma 5.2.3 (Tarski–Vaught test)**.** *Let $\mathcal{A}$ be a substructure of $\mathcal{B}$. Then $\mathcal{A} \preceq \mathcal{B}$ if and only if, for every formula φ, every variable x and every assignment s with values in $\mathcal{A}$ such that $\mathcal{B} \models \exists x\, \varphi[s]$, there is $a \in \mathcal{A}$ with $\mathcal{B} \models \varphi[s[x \mapsto a]]$.*

Proof. If $\mathcal{A} \preceq \mathcal{B}$ and $\mathcal{B} \models \exists x\, \varphi[s]$, then $\mathcal{A} \models \exists x\, \varphi[s]$, so there is $a \in \mathcal{A}$ with $\mathcal{A} \models \varphi[s[x \mapsto a]]$, and then $\mathcal{B} \models \varphi[s[x \mapsto a]]$. Conversely, assume the condition. We show by induction on φ that $\mathcal{A} \models \varphi[s]$ if and only if $\mathcal{B} \models \varphi[s]$ for every assignment s with values in $\mathcal{A}$. Atomic formulas have the same truth values, as remarked above, and the cases $\neg$ and $\to$ follow from the induction hypothesis. Let $\varphi = \forall x\, \psi$. By induction, $\mathcal{A} \models \varphi[s]$ if and only if $\mathcal{B} \models \psi[s[x \mapsto a]]$ for all $a \in \mathcal{A}$. This holds if $\mathcal{B} \models \forall x\, \psi[s]$. If $\mathcal{B} \not\models \forall x\, \psi[s]$, then $\mathcal{B} \models \exists x\, \neg\psi[s]$, and the condition gives $a \in \mathcal{A}$ with $\mathcal{B} \models \neg\psi[s[x \mapsto a]]$; so both sides fail. □

Example 5.2.4. $(\mathbb{Q}, <) \preceq (\mathbb{R}, <)$. We check the Tarski–Vaught test. Let s take values in $\mathbb{Q}$, and suppose $(\mathbb{R}, <) \models \varphi[s[x \mapsto b]]$ with $b \in \mathbb{R}$. The free variables of $\exists x\, \varphi$ take finitely many values $q_1 < \cdots < q_m$ in $\mathbb{Q}$. If b is one of them we are done. Otherwise b lies in one of the open

intervals into which the q_j cut $\mathbb{R}$, and this interval contains a rational r (Proposition 4.7.3). There is an order-preserving bijection $\sigma\colon \mathbb{R} \to \mathbb{R}$ fixing all q_j with $\sigma(b) = r$: on an interval $[q_j, q_{j+1}]$ take the map that is affine on $[q_j, b]$ and on $[b, q_{j+1}]$, sending q_j, b, q_{j+1} to q_j, r, q_{j+1}, and the identity elsewhere; on an unbounded interval use an affine map of the half line fixing its endpoint, and if $m = 0$ a translation. By Proposition 2.4.11 applied to the automorphism σ, $(\mathbb{R}, <) \models \varphi[s[x \mapsto r]]$, since $\sigma \circ s[x \mapsto b]$ agrees with $s[x \mapsto r]$ on the free variables of φ. So $(\mathbb{Q}, <) \equiv (\mathbb{R}, <)$, although the two orders are not isomorphic, one being countable and the other not (Proposition 4.8.6).

Theorem 5.2.5 (Downward Löwenheim–Skolem)**.** *Let $\mathcal{B}$ be an $\mathcal{L}$-structure, $X \subseteq \mathcal{B}$, and κ a cardinal with $\|\mathcal{L}\| \leq \kappa$ and $|X| \leq \kappa \leq |\mathcal{B}|$. Then $\mathcal{B}$ has an elementary substructure $\mathcal{A}$ with $X \subseteq \mathcal{A}$ and $|\mathcal{A}| = \kappa$.*

Proof. Since $\kappa \leq |\mathcal{B}|$ there is a subset $Y \subseteq \mathcal{B}$ with $|Y| = \kappa$, and $X_0 = X \cup Y$ has cardinality κ (Corollary 3.7.6). Consider the triples $(\varphi, x, \bar{b})$ of a formula φ, a variable x and a tuple $\bar{b}$ of elements of $\mathcal{B}$ whose length is the number of free variables $y_1, \ldots, y_n$ of $\exists x\, \varphi$, listed in the order of their indices, such that $\mathcal{B} \models \exists x\, \varphi[\bar{b}]$. By the axiom of choice there is a function w choosing for each such triple a *witness* $w(\varphi, x, \bar{b}) \in \mathcal{B}$ with $\mathcal{B} \models \varphi[s]$ for the assignment sending x to the witness and y_i to b_i. Define X_{k+1} as the union of X_k with the witnesses of all triples whose tuple $\bar{b}$ has entries in X_k. There are at most κ such triples, by Lemma 5.1.2 and since X_k has at most κ finite tuples; so $|X_{k+1}| \leq \kappa$, and $A = \bigcup_k X_k$ has cardinality κ.

The set A is closed under the functions of $\mathcal{B}$: for $a_1, \ldots, a_n \in A$, all lying in some X_k, the formula $x = f(y_1, \ldots, y_n)$ has the unique witness $f^{\mathcal{B}}(a_1, \ldots, a_n)$, which lies in X_{k+1}. Likewise the witness of $x = c$ is $c^{\mathcal{B}} \in X_1$. So A is the universe of a substructure $\mathcal{A}$. It satisfies the Tarski–Vaught test: if s takes values in A and $\mathcal{B} \models \exists x\, \varphi[s]$, the finitely many values of s on the free variables of $\exists x\, \varphi$ lie in some X_k, and the corresponding witness lies in $X_{k+1} \subseteq A$. By Lemma 5.2.3, $\mathcal{A} \preceq \mathcal{B}$. □

For countable $\mathcal{L}$ and $\kappa = \aleph_0$ this refines Theorem 2.7.2: every infinite structure has a countable elementary substructure, which even contains any given countable subset. Some choice is necessary here: the proof chooses witnesses in an arbitrary structure $\mathcal{B}$, while the proof in Chapter 2 built a model from the syntax.

5.3 Ultraproducts and compactness

Definition 5.3.1. A *filter* on a set I is a set F of subsets of I with $I \in F$, $\emptyset \notin F$, and closed under finite intersections and under taking supersets. It is an *ultrafilter* if moreover, for every $X \subseteq I$, either $X \in F$ or $I \setminus X \in F$. For $i \in I$, the *principal* ultrafilter at i is $\{X \subseteq I : i \in X\}$. A family of subsets of I has the *finite intersection property* if all its finite intersections are nonempty.

Lemma 5.3.2. *1. If U is an ultrafilter, then $X \cup Y \in U$ if and only if $X \in U$ or $Y \in U$.*

2. Every family of subsets of I with the finite intersection property is contained in an ultrafilter.

Proof. (1) If $X \in U$ then $X \cup Y \in U$. If $X, Y \notin U$, then $I \setminus X$ and $I \setminus Y$ lie in U, and so does their intersection, which is disjoint from $X \cup Y$; so $X \cup Y \notin U$, as $\emptyset \notin U$.

(2) Let $\mathcal{S}$ be the family. The sets containing some finite intersection of members of $\mathcal{S}$ form a filter containing $\mathcal{S}$. Filters containing $\mathcal{S}$, ordered by inclusion, satisfy the hypothesis of Zorn's lemma (Theorem 3.7.2), since the union of a chain of filters is a filter. Let U be a maximal one and $X \subseteq I$. If X meets every member of U, the sets containing some $X \cap Y$ with $Y \in U$ form a filter containing U and X, so $X \in U$ by maximality. Otherwise $X \cap Y = \emptyset$ for some $Y \in U$, so $Y \subseteq I \setminus X$ and $I \setminus X \in U$. So U is an ultrafilter. □

Definition 5.3.3. Let $(\mathcal{A}_i)_{i\in I}$ be $\mathcal{L}$-structures and U an ultrafilter on I. On the product $\prod_i \mathcal{A}_i$ of the universes (nonempty by the axiom of choice), let $g \sim g'$ if $\{i : g(i) = g'(i)\} \in U$; this is an equivalence relation, transitivity coming from closure under intersections. The *ultraproduct* $\prod_i \mathcal{A}_i/U$ is the structure with universe the set of classes $[g]$, with $c = [(c^{\mathcal{A}_i})_i]$, $f([g_1], \dots, [g_n]) = [(f^{\mathcal{A}_i}(g_1(i), \dots, g_n(i)))_i]$, and $([g_1], \dots, [g_n]) \in R$ if and only if $\{i : (g_1(i), \dots, g_n(i)) \in R^{\mathcal{A}_i}\} \in U$. These are well defined: if $g_j \sim g'_j$ for all j, the set where $g_j(i) = g'_j(i)$ for all j is in U, and on it the two definitions agree. If all $\mathcal{A}_i$ equal $\mathcal{A}$, the ultraproduct is the *ultrapower* $\mathcal{A}^I/U$.

Theorem 5.3.4 (Łoś)**.** *Let s_i be an assignment in $\mathcal{A}_i$ for each i, and let s be the assignment in $\prod_i \mathcal{A}_i/U$ with $s(x) = [(s_i(x))_i]$. Then for every formula φ,*

$$\prod_i \mathcal{A}_i/U \models \varphi[s] \iff \{i \in I : \mathcal{A}_i \models \varphi[s_i]\} \in U.$$

Every assignment in the ultraproduct is of this form.

Proof. Every assignment is of this form: choose a representative of each value (axiom of choice). Write $\mathcal{P} = \prod_i \mathcal{A}_i/U$ and $S_\varphi = \{i : \mathcal{A}_i \models \varphi[s_i]\}$. By induction on terms, $t^{\mathcal{P}}[s] = [(t^{\mathcal{A}_i}[s_i])_i]$. For atomic formulas the claim is then the definition of $\sim$ and of $R^{\mathcal{P}}$. Since $S_{\neg\varphi} = I \setminus S_\varphi$, the case of $\neg$ follows from the ultrafilter property. Since $S_{\varphi\to\psi} = (I \setminus S_\varphi) \cup S_\psi$, the case of $\to$ follows from Lemma 5.3.2(1).

Let $\varphi = \forall x\, \psi$. For $g \in \prod_i \mathcal{A}_i$, the assignment $s[x \mapsto [g]]$ is obtained from the assignments $s_i[x \mapsto g(i)]$, so by induction $\mathcal{P} \models \psi[s[x \mapsto [g]]]$ if and only if $T_g = \{i : \mathcal{A}_i \models \psi[s_i[x \mapsto g(i)]]\} \in U$. If $S_\varphi \in U$, then $T_g \supseteq S_\varphi$ is in U for every g, so $\mathcal{P} \models \varphi[s]$. If $S_\varphi \notin U$, choose $g(i)$ with $\mathcal{A}_i \not\models \psi[s_i[x \mapsto g(i)]]$ for $i \notin S_\varphi$, and arbitrary for $i \in S_\varphi$. Then $T_g \subseteq S_\varphi$, so $T_g \notin U$ and $\mathcal{P} \not\models \varphi[s]$. □

In particular a structure $\mathcal{A}$ embeds elementarily into each of its ultrapowers, by $a \mapsto [(a)_i]$: for constant families the set in Łoś's theorem is I or $\emptyset$.

Theorem 5.3.5 (Compactness theorem)**.** *Let $\mathcal{L}$ be any language and Γ a set of $\mathcal{L}$-formulas. If every finite subset of Γ is satisfiable, then Γ is satisfiable.*

Proof. Let I be the set of finite subsets of Γ, and for $\Delta \in I$ choose a structure $\mathcal{A}_\Delta$ and an assignment s_Δ satisfying Δ. The sets $\widehat{\Delta} = \{\Delta' \in I : \Delta \subseteq \Delta'\}$ have the finite intersection property, since $\widehat{\Delta_1} \cap \dots \cap \widehat{\Delta_r}$ contains $\Delta_1 \cup \dots \cup \Delta_r$. Let U be an ultrafilter containing them (Lemma 5.3.2), and s the assignment in $\prod_\Delta \mathcal{A}_\Delta/U$ obtained from the s_Δ. For $\gamma \in \Gamma$, the set of Δ with $\mathcal{A}_\Delta \models \gamma[s_\Delta]$ contains $\widehat{\{\gamma\}} \in U$, so by Theorem 5.3.4 the ultraproduct satisfies γ under s. □

For countable languages this is Theorem 2.7.1, which was deduced from the completeness theorem; the proof above avoids formal deductions altogether, at the price of the axiom of choice.

5.4 Upward Löwenheim–Skolem and categoricity

Theorem 5.4.1 (Upward Löwenheim–Skolem)**.** *Let T be an $\mathcal{L}$-theory with an infinite model, and $\kappa \geq \|\mathcal{L}\|$ a cardinal. Then T has a model of cardinality exactly κ.*

Proof. Add new constant symbols c_α, $\alpha < \kappa$, to get a language $\mathcal{L}'$ with $\|\mathcal{L}'\| = \kappa$ (Corollary 3.7.6), and let $\Gamma = T \cup \{c_\alpha \neq c_\beta : \alpha < \beta < \kappa\}$. A finite subset of Γ mentions finitely many new constants; interpreting them as distinct elements of an infinite model of T (and the other

new constants arbitrarily) satisfies it. By Theorem 5.3.5, Γ has a model $\mathcal{B}$, and $\alpha \mapsto c_\alpha^{\mathcal{B}}$ is injective, so $|\mathcal{B}| \geq \kappa$. By Theorem 5.2.5 applied to $\mathcal{L}'$, $\mathcal{B}$ has an elementary substructure $\mathcal{A}$ of cardinality κ, which satisfies Γ like $\mathcal{B}$; its reduct to $\mathcal{L}$ is a model of T of cardinality κ. □

So a first-order theory with an infinite model never determines it up to isomorphism: it has models of arbitrarily large cardinality. Example 2.7.5 exhibited a different phenomenon, a non-standard model of the same (countable) cardinality.

Definition 5.4.2. Let κ be an infinite cardinal. A theory T is *κ-categorical* if it has a model of cardinality κ and any two models of cardinality κ are isomorphic. It is *complete* if for every sentence σ, either σ is true in all models of T or $\neg\sigma$ is. For countable languages this agrees with Definition 2.8.2, by the completeness theorem (Corollary 2.6.2).

Theorem 5.4.3 (Łoś–Vaught test)**.** *Let T be a satisfiable theory without finite models that is κ-categorical for some $\kappa \geq \|\mathcal{L}\|$. Then T is complete.*

Proof. Suppose σ is a sentence such that neither σ nor $\neg\sigma$ holds in all models of T. Then $T \cup \{\neg\sigma\}$ and $T \cup \{\sigma\}$ have models, which are infinite, so by Theorem 5.4.1 they have models $\mathcal{A}$ and $\mathcal{B}$ of cardinality κ. These are models of T, hence isomorphic, but $\mathcal{B} \models \sigma$ and $\mathcal{A} \not\models \sigma$, contradicting Proposition 2.4.11. □

Definition 5.4.4. The theory T_{DLO} of *dense linear orders without endpoints*, in the language of orders, consists of the sentences $\forall x\, \neg(x < x)$, $\forall x \forall y \forall z\, ((x < y \wedge y < z) \to x < z)$, $\forall x \forall y\, (x < y \vee x = y \vee y < x)$, $\forall x \forall y\, (x < y \to \exists z\, (x < z \wedge z < y))$, $\forall x \exists y\, (y < x)$ and $\forall x \exists y\, (x < y)$.

Its models are infinite: a model is nonempty and has no largest element. Examples are $(\mathbb{Q}, <)$, $(\mathbb{R}, <)$ and every open interval of $\mathbb{R}$.

Theorem 5.4.5 (Cantor)**.** *Any two countable dense linear orders without endpoints are isomorphic.*

Proof. Let A and B be such orders, enumerated as $a_0, a_1, \ldots$ and $b_0, b_1, \ldots$ (they are countably infinite). *Extension step*: if p is an order-preserving bijection from a finite subset of A onto a finite subset of B and $a \in A$, there is $b \in B$ such that $p \cup \{(a, b)\}$ is still order-preserving and injective. Indeed, if $a \in \operatorname{dom} p$ take $b = p(a)$; otherwise the elements of $\operatorname{dom} p$ smaller than a have images smaller than those of the elements larger than a, and we need b strictly between these two finite sets of images, which exists because B is dense and has no endpoints (and is arbitrary if $p = \emptyset$). The same holds with the roles of A and B exchanged. Starting from $p_0 = \emptyset$, let p_{2k+1} extend p_{2k} with a_k in its domain, and p_{2k+2} extend p_{2k+1} with b_k in its image (*back and forth*). The union of the p_k is an order-preserving bijection $A \to B$. □

Corollary 5.4.6. *T_{DLO} is $\aleph_0$-categorical and complete. In particular $(\mathbb{Q}, <) \equiv (\mathbb{R}, <)$, and a sentence in the language of orders holds in $(\mathbb{Q}, <)$ if and only if it holds in every dense linear order without endpoints.*

Proof. By Theorem 5.4.5, T_{DLO} is $\aleph_0$-categorical, its language is finite, and it has no finite models; apply Theorem 5.4.3 with $\kappa = \aleph_0$. □

Counterexample 5.4.7. T_{DLO} is not $2^{\aleph_0}$-categorical. The orders $(\mathbb{R}, <)$ and $(\mathbb{R} \setminus \{0\}, <)$ are dense without endpoints and have cardinality $2^{\aleph_0}$, but they are not isomorphic: in $\mathbb{R} \setminus \{0\}$ the set of negative numbers is nonempty and bounded above (by 1) but has no least upper bound, since every upper bound u is positive and $u/2$ is a smaller one, while in $\mathbb{R}$ every nonempty subset bounded above has a least upper bound (Theorem 4.8.3); an order isomorphism would preserve this property. By contrast, the theory of algebraically closed fields of a fixed characteristic is κ-categorical for every uncountable κ (such a field is determined by its transcendence degree, which equals its cardinality), hence complete by the Łoś–Vaught test; see [Mar02, Section 2.2] and Chapter 9.

5.5 Ehrenfeucht–Fraïssé theory

The proof of Cantor's theorem used only one feature of the orders: finite partial isomorphisms can always be extended, forth and back. We now make this precise and relate it to the quantifier depth of formulas. In this section and the next, languages are *relational*: they have no constant or function symbols. This is no real restriction, since an n-ary function can be replaced by its graph, an $(n+1)$-ary relation, and a constant by a unary relation holding at one point. In a relational language the terms are the variables.

Definition 5.5.1. The *quantifier rank* of a formula is defined by $\mathrm{qr}(\varphi) = 0$ for atomic φ, $\mathrm{qr}(\neg\varphi) = \mathrm{qr}(\varphi)$, $\mathrm{qr}(\varphi \to \psi) = \max(\mathrm{qr}(\varphi), \mathrm{qr}(\psi))$ and $\mathrm{qr}(\forall x\, \varphi) = \mathrm{qr}(\varphi) + 1$; so $\mathrm{qr}(\exists x\, \varphi) = \mathrm{qr}(\varphi) + 1$, and $\wedge$, $\vee$ take the maximum. For structures $\mathcal{A}$, $\mathcal{B}$, tuples $\bar{a} \in \mathcal{A}^m$, $\bar{b} \in \mathcal{B}^m$ and $n \geq 0$, write $(\mathcal{A}, \bar{a}) \equiv_n (\mathcal{B}, \bar{b})$ if $\mathcal{A} \models \varphi[\bar{a}] \iff \mathcal{B} \models \varphi[\bar{b}]$ for every formula $\varphi(x_1, \ldots, x_m)$ of quantifier rank at most n; for $m = 0$ write $\mathcal{A} \equiv_n \mathcal{B}$.

Definition 5.5.2. Let $\mathcal{A}$ and $\mathcal{B}$ be structures for a relational language. A *partial isomorphism* from $\mathcal{A}$ to $\mathcal{B}$ is an injective map p from a subset of $\mathcal{A}$ to $\mathcal{B}$ with $(a_1, \ldots, a_r) \in R^{\mathcal{A}} \iff (p(a_1), \ldots, p(a_r)) \in R^{\mathcal{B}}$ for every r-ary relation symbol R and all $a_j \in \mathrm{dom}\, p$. A set I of partial isomorphisms has the *forth property* with respect to a set J if for every $p \in I$ and $a \in \mathcal{A}$ there is $q \in J$ with $p \subseteq q$ and $a \in \mathrm{dom}\, q$, and the *back property* if for every $p \in I$ and $b \in \mathcal{B}$ there is $q \in J$ with $p \subseteq q$ and b in the image of q.

1. An *n-back-and-forth system* is a sequence $(I_k)_{0 \leq k \leq n}$ of nonempty sets of partial isomorphisms such that I_{k+1} has the forth and back properties with respect to I_k for $k < n$. We write $\mathcal{A} \cong_n \mathcal{B}$ if one exists, and $(\mathcal{A}, \bar{a}) \cong_n (\mathcal{B}, \bar{b})$ if one exists with $\bar{a} \mapsto \bar{b}$ (the map $a_j \mapsto b_j$) in I_n.

2. $\mathcal{A}$ and $\mathcal{B}$ are *partially isomorphic*, $\mathcal{A} \cong_p \mathcal{B}$, if there is a nonempty set I of partial isomorphisms with the forth and back properties with respect to itself.

For example, the finite order-preserving injections between two dense linear orders without endpoints show that any two such orders are partially isomorphic: this is the extension step in the proof of Theorem 5.4.5.

Lemma 5.5.3. *Countable partially isomorphic structures are isomorphic.*

Proof. Let I be as in the definition, and enumerate $\mathcal{A} = \{a_0, a_1, \ldots\}$ and $\mathcal{B} = \{b_0, b_1, \ldots\}$, with repetitions if the structures are finite. Choose $p_0 \in I$, then $p_{2k+1} \in I$ extending p_{2k} with a_k in its domain, and $p_{2k+2} \in I$ extending p_{2k+1} with b_k in its image. The union of the chain (p_k) is a bijection $\mathcal{A} \to \mathcal{B}$ which preserves the relations, since any finitely many elements lie in the domain of some p_k. □

Theorem 5.5.4 (Fraïssé)**.** *Let $(I_k)_{k \leq n}$ be an n-back-and-forth system from $\mathcal{A}$ to $\mathcal{B}$, let $k \leq n$, $p \in I_k$, φ a formula with $\mathrm{qr}(\varphi) \leq k$, and s an assignment in $\mathcal{A}$ whose values on the free variables of φ lie in $\mathrm{dom}\, p$. Then $\mathcal{A} \models \varphi[s]$ if and only if $\mathcal{B} \models \varphi[t]$ for any assignment t with $t(x) = p(s(x))$ for the free variables x of φ. Consequently $\mathcal{A} \cong_n \mathcal{B}$ implies $\mathcal{A} \equiv_n \mathcal{B}$, the same holds with tuples, and $\mathcal{A} \cong_p \mathcal{B}$ implies $\mathcal{A} \equiv \mathcal{B}$.*

Proof. Induction on φ. An atomic formula is $y = z$ or $R(y_1, \ldots, y_r)$ with variables y_j, and the claim holds because p is injective and preserves the relations. The cases $\neg$ and $\to$ follow from the induction hypothesis. Let $\varphi = \forall y\, \psi$, so $k \geq 1$ and $\mathrm{qr}(\psi) \leq k - 1$. Suppose $\mathcal{A} \models \varphi[s]$, and let $b \in \mathcal{B}$. By the back property there is $q \in I_{k-1}$ with $p \subseteq q$ and $b = q(a)$ for some a. Then $\mathcal{A} \models \psi[s[y \mapsto a]]$, the values of $s[y \mapsto a]$ on the free variables of ψ lie in $\mathrm{dom}\, q$, and $q(s[y \mapsto a](x)) = t[y \mapsto b](x)$ for these variables; by induction $\mathcal{B} \models \psi[t[y \mapsto b]]$. As b was

arbitrary, $\mathcal{B} \models \varphi[t]$. The converse uses the forth property in the same way. For sentences the condition on s is empty, which gives the consequences, taking $I_k = I$ for all k in the last one. □

Example 5.5.5. Let $\mathcal{L}$ be the empty language, so structures are nonempty sets. If $|\mathcal{A}| = |\mathcal{B}|$, or both have at least n elements, then $\mathcal{A} \cong_n \mathcal{B}$: let I_k be the set of injections from subsets of $\mathcal{A}$ of size at most $n - k$ into $\mathcal{B}$ (all injections with finite domain, if $|\mathcal{A}| = |\mathcal{B}|$ is finite); an injection with at most $n - k - 1$ elements in its domain can be extended by any point, since the other set has more than $n - k - 1$ elements. So a sentence of quantifier rank n cannot distinguish a set with n elements from one with $n + 1$, and no sentence of the empty language is true exactly in the finite sets of even cardinality.

For a *finite* relational language the converse of Fraïssé's theorem holds, and each class $\equiv_n$ is defined by one sentence. Fix an enumeration of all formulas of $\mathcal{L}$ (a countable set) and the variables $x_1, x_2, \ldots$; the conjunction $\bigwedge \Phi$ or disjunction $\bigvee \Phi$ of a finite nonempty set Φ of formulas is taken in the order of this enumeration.

Definition 5.5.6. Let $\mathcal{L}$ be a finite relational language. For a structure $\mathcal{A}$, a tuple $\bar{a} \in \mathcal{A}^m$ and n with $m + n \geq 1$, the *Hintikka formula* $\varphi^n_{\mathcal{A},\bar{a}}(x_1, \ldots, x_m)$ is defined by recursion on n. For $m \geq 1$, $\varphi^0_{\mathcal{A},\bar{a}}$ is the conjunction of the atomic formulas in the variables $x_1, \ldots, x_m$ that hold at $\bar{a}$ and the negations of those that do not; there are finitely many, as $\mathcal{L}$ is finite. Then

$$\varphi^{n+1}_{\mathcal{A},\bar{a}} = \varphi^0_{\mathcal{A},\bar{a}} \wedge \bigwedge_{a \in \mathcal{A}} \exists x_{m+1}\, \varphi^n_{\mathcal{A},\bar{a}a} \wedge \forall x_{m+1} \bigvee_{a \in \mathcal{A}} \varphi^n_{\mathcal{A},\bar{a}a},$$

where the conjunction and the disjunction are over the *set* of formulas $\{\varphi^n_{\mathcal{A},\bar{a}a} : a \in \mathcal{A}\}$, which is finite by Lemma 5.5.7, and the first conjunct is omitted if $m = 0$.

Lemma 5.5.7. *Let $\mathcal{L}$ be a finite relational language and $m + n \geq 1$. The set H^n_m of all formulas $\varphi^n_{\mathcal{A},\bar{a}}$, for all structures $\mathcal{A}$ and $\bar{a} \in \mathcal{A}^m$, is finite. Each has quantifier rank n, and $\mathcal{A} \models \varphi^n_{\mathcal{A},\bar{a}}[\bar{a}]$.*

Proof. Induction on n, for all m at once; this also shows that Definition 5.5.6 makes sense. The set H^0_m ($m \geq 1$) is finite, since $\varphi^0_{\mathcal{A},\bar{a}}$ is determined by the subset of the finitely many atomic formulas in $x_1, \ldots, x_m$ that hold at $\bar{a}$. The formula $\varphi^{n+1}_{\mathcal{A},\bar{a}}$ is determined by $\varphi^0_{\mathcal{A},\bar{a}}$ and by a subset of the finite set H^n_{m+1}, so H^{n+1}_m is finite. The statements about quantifier rank and truth follow by the same induction. □

Theorem 5.5.8 (Ehrenfeucht–Fraïssé)**.** *Let $\mathcal{L}$ be a finite relational language, $\bar{a} \in \mathcal{A}^m$, $\bar{b} \in \mathcal{B}^m$ and $m + n \geq 1$. The following are equivalent:*

(a) $\mathcal{B} \models \varphi^n_{\mathcal{A},\bar{a}}[\bar{b}]$*;*

(b) $(\mathcal{A}, \bar{a}) \cong_n (\mathcal{B}, \bar{b})$*;*

(c) $(\mathcal{A}, \bar{a}) \equiv_n (\mathcal{B}, \bar{b})$*.*

In particular, for $n \geq 1$ the class of structures $\mathcal{B}$ with $\mathcal{B} \equiv_n \mathcal{A}$ is the class of models of the single sentence $\varphi^n_{\mathcal{A}}$, and there are finitely many such classes.

Proof. (c) implies (a) by Lemma 5.5.7, and (b) implies (c) by Theorem 5.5.4. Assume (a). For $k \leq n$ let I_k be the set of maps $\bar{c} \mapsto \bar{d}$, where $\bar{c} \in \mathcal{A}^{m+n-k}$ and $\bar{d} \in \mathcal{B}^{m+n-k}$ extend $\bar{a}$ and $\bar{b}$ and $\mathcal{B} \models \varphi^k_{\mathcal{A},\bar{c}}[\bar{d}]$. Such a map is a well defined partial isomorphism: if the tuples are nonempty, $\varphi^k_{\mathcal{A},\bar{c}}$ contains the conjunct $\varphi^0_{\mathcal{A},\bar{c}}$, which records which equalities $x_i = x_j$ and which relations hold at $\bar{c}$, and these then hold at $\bar{d}$ as well; if they are empty, it is the empty

map. By (a), $\bar{a} \mapsto \bar{b}$ lies in I_n. Let $\bar{c} \mapsto \bar{d}$ be in I_{k+1}. For $a \in \mathcal{A}$, the conjunct $\exists x\, \varphi^k_{\mathcal{A},\bar{c}a}$ of $\varphi^{k+1}_{\mathcal{A},\bar{c}}$ gives b with $\mathcal{B} \models \varphi^k_{\mathcal{A},\bar{c}a}[\bar{d}b]$, so $\bar{c}a \mapsto \bar{d}b$ is in I_k: this is the forth property. For $b \in \mathcal{B}$, the conjunct $\forall x \bigvee_a \varphi^k_{\mathcal{A},\bar{c}a}$ gives a with $\bar{c}a \mapsto \bar{d}b$ in I_k: the back property. The forth property also shows inductively that all I_k are nonempty. So (b) holds. The last statement follows, with Lemma 5.5.7 for finiteness. □

5.6 Lindström's theorem

First-order logic is one of many possible logics: one can allow infinite conjunctions, quantifiers such as "there are uncountably many", or quantification over subsets. Lindström's theorem says that none of these extensions keeps both the compactness theorem and the Löwenheim–Skolem theorem.

Definition 5.6.1. A *logic* $\mathcal{S}$ assigns to each finite relational language $\mathcal{L}$ a set $\mathcal{S}[\mathcal{L}]$, whose elements are called $\mathcal{S}$*-sentences*, and a relation $\mathcal{A} \models_{\mathcal{S}} \psi$ between $\mathcal{L}$-structures and $\mathcal{S}$-sentences, such that:

1. (Isomorphism) isomorphic structures satisfy the same $\mathcal{S}$-sentences;
2. (Expansion) if $\mathcal{L} \subseteq \mathcal{L}'$, then $\mathcal{S}[\mathcal{L}] \subseteq \mathcal{S}[\mathcal{L}']$, and an $\mathcal{L}'$-structure satisfies $\psi \in \mathcal{S}[\mathcal{L}]$ if and only if its reduct to $\mathcal{L}$ does;
3. (Boolean operations) for $\psi, \chi \in \mathcal{S}[\mathcal{L}]$ there are $\neg\psi$ and $\psi \wedge \chi$ in $\mathcal{S}[\mathcal{L}]$ with the usual meaning;
4. (Relativisation) for $\psi \in \mathcal{S}[\mathcal{L}]$ and a unary relation symbol $P \notin \mathcal{L}$ there is $\psi^P \in \mathcal{S}[\mathcal{L} \cup \{P\}]$ such that an $(\mathcal{L} \cup \{P\})$-structure $\mathcal{A}$ with $P^{\mathcal{A}} \neq \emptyset$ satisfies ψ^P if and only if $\mathcal{A}^{(P)} \models_{\mathcal{S}} \psi$, where $\mathcal{A}^{(P)}$ is the $\mathcal{L}$-structure with universe $P^{\mathcal{A}}$ and the restrictions of the relations of $\mathcal{A}$;
5. (First-order logic) for every first-order $\mathcal{L}$-sentence σ there is $\sigma' \in \mathcal{S}[\mathcal{L}]$ with the same models.

The logic $\mathcal{S}$ has the *compactness property* if every set of $\mathcal{S}[\mathcal{L}]$-sentences whose finite subsets have models has a model, and the *Löwenheim–Skolem property* if every $\mathcal{S}$-sentence with a model has a countable model.

First-order logic is a logic in this sense, with relativisation defined by replacing each subformula $\forall x\, \varphi$ by $\forall x\, (P(x) \to \varphi)$; it has both properties, by Theorems 2.7.1 and 2.7.2. Infinitary logic with countable conjunctions has the Löwenheim–Skolem property but not the compactness property, and the logic with the quantifier "there are uncountably many" is compact for countable languages but fails the Löwenheim–Skolem property [EFT94, Chapter XIII].

Theorem 5.6.2 (Lindström)**.** *Let $\mathcal{S}$ be a logic with the compactness and the Löwenheim–Skolem properties. Then for every finite relational language $\mathcal{L}$, every $\psi \in \mathcal{S}[\mathcal{L}]$ has the same models as some first-order $\mathcal{L}$-sentence.*

Proof. Step 1. Suppose that $\psi \in \mathcal{S}[\mathcal{L}]$ has no first-order equivalent. We claim that for every $n \geq 1$ there are $\mathcal{L}$-structures $\mathcal{A}_n \models \psi$ and $\mathcal{B}_n \not\models \psi$ with $\mathcal{A}_n \cong_n \mathcal{B}_n$. Otherwise, for some n, every structure $\equiv_n$ to a model of ψ is a model of ψ (Theorem 5.5.8). Let σ be the disjunction of the finitely many sentences $\varphi^n_{\mathcal{A}} \in H^n_0$ with $\mathcal{A} \models \psi$ (or $\exists x\, (x \neq x)$ if ψ has no model). A model of ψ satisfies its own Hintikka sentence, hence σ; a model of σ is $\equiv_n$ to a model of ψ,

hence a model of ψ. So ψ has the same models as σ, a contradiction. Fix n-back-and-forth systems $(I^n_k)_{k\le n}$ from $\mathcal{A}_n$ to $\mathcal{B}_n$.

Step 2: coding. Let $\mathcal{L}^+ = \mathcal{L} \cup \{P, Q, U, W, C, <, I, G\}$ with P, Q, U, W, C unary, $<$ and I binary and G ternary. Let $\mathcal{M}_n$ be the $\mathcal{L}^+$-structure whose universe is the disjoint union of $\mathcal{A}_n$, $\mathcal{B}_n$, $\{0, 1, \dots, n\}$ and $\bigcup_k I^n_k$, with P, Q, U, W these four parts, $R^{\mathcal{M}_n} = R^{\mathcal{A}_n} \cup R^{\mathcal{B}_n}$ for $R \in \mathcal{L}$, $<$ the order of $\{0, \dots, n\}$, $C = \{n\}$, $I = \{(k, w) : w \in I^n_k\}$ and $G = \{(w, a, b) : w(a) = b\}$. Consider the following first-order $\mathcal{L}^+$-sentences:

(i) P, Q, U are nonempty, W is disjoint from P, and C holds at exactly one element, which lies in U;

(ii) $<$ is a linear order on U with a least element, and every other element of U has an immediate predecessor in U;

(iii) for every k in U there is w with $I(k, w)$, and $I(k, w)$ implies $U(k)$ and $W(w)$;

(iv) for every w in W, the relation $G(w, x, y)$ is the graph of an injective partial map from P to Q which preserves each relation of $\mathcal{L}$ in both directions (a finite conjunction, as $\mathcal{L}$ is finite);

(v) if k' is the immediate predecessor of k in U and $I(k, w)$, then for every a in P there is w' with $I(k', w')$, $\forall x \forall y\, (G(w, x, y) \to G(w', x, y))$ and $\exists y\, G(w', a, y)$; and symmetrically for every b in Q there is such a w' with $\exists x\, G(w', x, b)$.

Let $\chi \in \mathcal{S}[\mathcal{L}^+]$ be the conjunction of ψ^P, $(\neg\psi)^Q$ and the $\mathcal{S}$-versions of (i)–(v) (Definition 5.6.1(2)–(5)). Then $\mathcal{M}_n \models \chi$: the structures $\mathcal{M}^{(P)}_n = \mathcal{A}_n$ and $\mathcal{M}^{(Q)}_n = \mathcal{B}_n$ satisfy ψ and $\neg\psi$, and (v) holds because I^n_k has the forth and back properties with respect to I^n_{k-1}.

Step 3: compactness. For $j \in \mathbb{N}$ let λ_j be the first-order sentence saying that the element of C has at least j predecessors in U. A finite subset of $\{\chi\} \cup \{\lambda'_j : j \in \mathbb{N}\}$ is satisfied by $\mathcal{M}_n$ for n large, since n has n predecessors. By the compactness property there is a model $\mathcal{M}$. Call $k \in U^{\mathcal{M}}$ *infinite* if it has infinitely many predecessors; the element of $C^{\mathcal{M}}$ is infinite. An infinite k is not the least element, so it has an immediate predecessor k' by (ii), and k' is infinite, as every other predecessor of k is a predecessor of k'.

Step 4: Löwenheim–Skolem. Let J be a new unary relation symbol, and expand $\mathcal{M}$ by $J = \{w : I(k, w) \text{ for some infinite } k\}$. Let θ be the first-order sentence saying that J is nonempty and contained in W, and that for every w in J and every a in P (respectively b in Q) there is w' in J with $G(w, x, y) \to G(w', x, y)$ for all x, y and a in the domain (respectively b in the image) of w'. The expansion satisfies $\chi \wedge \theta'$: J is nonempty by (iii) applied to the element of C, and the extensions required by θ exist by (v), since the immediate predecessor of an infinite element is infinite. By the Löwenheim–Skolem property, $\chi \wedge \theta'$ has a countable model $\mathcal{N}$. Then $\mathcal{N}^{(P)}$ and $\mathcal{N}^{(Q)}$ are countable $\mathcal{L}$-structures with $\mathcal{N}^{(P)} \models \psi$ and $\mathcal{N}^{(Q)} \not\models \psi$, by relativisation, and by (iv) and θ the partial maps coded by the elements of $J^{\mathcal{N}}$ form a nonempty set of partial isomorphisms with the forth and back properties. By Lemma 5.5.3, $\mathcal{N}^{(P)} \cong \mathcal{N}^{(Q)}$, which contradicts the isomorphism property of $\mathcal{S}$. So ψ has a first-order equivalent. □

Remark 5.6.3. The theorem is from [Lin69]; we followed [EFT94, Chapter XIII]. It explains why first-order logic is the logic of this project: its two basic metatheorems, compactness and Löwenheim–Skolem, characterise it. The price is expressiveness: by Example 5.5.5 and Counterexample 2.7.4, finiteness and even cardinality are not first-order properties, and by Theorem 5.4.1 no infinite structure is characterised up to isomorphism.

5.7 Exercises

Exercise 5.7.1. Show that on an infinite set I there is an ultrafilter containing all complements of finite sets, and that such an ultrafilter is not principal. For such an ultrafilter U on $\mathbb{N}$, show that the ultrapower of the ordered field $(\mathbb{R}, 0, 1, +, \cdot, <)$ contains a positive element smaller than $1/n$ for every $n \in \mathbb{N}_{>0}$, and that it satisfies the same first-order sentences as $\mathbb{R}$.

Solution. The complements of finite sets have the finite intersection property, since a finite union of finite sets is finite and I is infinite; so they lie in an ultrafilter U (Lemma 5.3.2). It is not principal, as $I \setminus \{i\} \in U$ for each i. The element $\epsilon = [(1/(k+1))_k]$ is positive, and for $n \geq 1$ the set $\{k : 1/(k+1) < 1/n\}$ is cofinite, hence in U; by Theorem 5.3.4, $0 < \epsilon < 1/n$, where $1/n$ denotes the class of the constant family. The ultrapower is an elementary extension of $\mathbb{R}$ by the remark after Theorem 5.3.4. □

Exercise 5.7.2. Show that the theory of dense linear orders with a least and a greatest element (and at least two elements) is $\aleph_0$-categorical and complete.

Solution. Such an order is infinite, by density. Given two countable ones, the orders obtained by removing the endpoints are countable dense linear orders without endpoints, hence isomorphic (Theorem 5.4.5), and the isomorphism extends by sending endpoints to endpoints. The language is finite and there are no finite models, so Theorem 5.4.3 applies. □

Exercise 5.7.3. Show that connectedness of graphs is not first-order: there is no set of sentences in the language with one binary relation E whose models are exactly the connected graphs (symmetric irreflexive E in which any two vertices are joined by a finite path).

Solution. Suppose T were such a set. Add constants c, d and, for each n, the sentence δ_n saying that there is no path of length at most n from c to d. Every finite subset of $T \cup \{\delta_n : n\}$ is satisfied by a long path graph with c and d its endpoints, which is connected. By Theorem 5.3.5 there is a model, a connected graph in which c and d are joined by no path, a contradiction. □

Chapter 6

Categories

6.1 Introduction

Category theory is the language in which most of this project is written. This chapter introduces categories, functors and natural transformations, proves the Yoneda lemma, and develops equivalences, limits and colimits, adjoint functors, filtered colimits and additive categories. Abelian categories are treated in Chapter 15.

Categories, functors and natural transformations were introduced in [EML45], and adjoint functors in [Kan58]. Standard references are [ML98] and [Rie16]; [KS06] and the chapter on categories of [The] treat set-theoretic questions with universes, as we do.

Prerequisites. Chapter 3 (Set Theory), in particular the axiom of choice and Grothendieck universes.

Throughout, $\mathcal{U}$ is the universe fixed in Notation 1.6.1, and a set is *small* if it is an element of $\mathcal{U}$.

6.2 Categories

Definition 6.2.1. A *category* C consists of

1. a set $\mathrm{Ob}(C)$, whose elements are called the *objects* of C;
2. for all objects X, Y, a set $\mathrm{Hom}_C(X, Y)$, whose elements are called *morphisms* from X to Y;
3. for every object X, an element $\mathrm{id}_X \in \mathrm{Hom}_C(X, X)$, the *identity* of X;
4. for all objects X, Y, Z, a map

$$\mathrm{Hom}_C(Y, Z) \times \mathrm{Hom}_C(X, Y) \longrightarrow \mathrm{Hom}_C(X, Z), \qquad (g, f) \longmapsto g \circ f,$$

called *composition*,

such that the following hold:

1. (associativity) $h \circ (g \circ f) = (h \circ g) \circ f$ for all $f \in \mathrm{Hom}_C(X, Y)$, $g \in \mathrm{Hom}_C(Y, Z)$ and $h \in \mathrm{Hom}_C(Z, W)$;
2. (identities) $f \circ \mathrm{id}_X = f = \mathrm{id}_Y \circ f$ for all $f \in \mathrm{Hom}_C(X, Y)$;
3. the sets $\mathrm{Hom}_C(X, Y)$ are pairwise disjoint.

Notation 6.2.2. We write $X \in C$ instead of $X \in \mathrm{Ob}(C)$, and $f\colon X \to Y$ instead of $f \in \mathrm{Hom}_C(X, Y)$. By the last axiom of Definition 6.2.1, a morphism f determines its *source* X and its *target* Y. We often write gf for $g \circ f$, and $\mathrm{Hom}(X, Y)$ when the category is clear.

Definition 6.2.3. A category C is *locally small* if $\mathrm{Hom}_C(X, Y)$ is a small set for all $X, Y \in C$. It is *small* if it is locally small and $\mathrm{Ob}(C)$ is a small set.

Remark 6.2.4. There is no set of all sets, so "the category of all sets" cannot be a category in the sense of Definition 6.2.1. The universe $\mathcal{U}$ solves this problem: it is a set, it is closed under all the usual constructions of set theory (pairs, products, power sets, sets of maps, unions indexed by small sets), and it contains every set that appears in ordinary mathematics. So we can form the category of all *small* sets without leaving ZFC.

Example 6.2.5. The category Set has as objects the small sets, that is, $\mathrm{Ob}(\mathsf{Set}) = \mathcal{U}$. For small sets X and Y, the set $\mathrm{Hom}_{\mathsf{Set}}(X, Y)$ is the set of maps from X to Y, composition is composition of maps, and id_X is the identity map. Since $\mathcal{U}$ is closed under forming sets of maps, Set is locally small. It is not small, because $\mathcal{U} \notin \mathcal{U}$.

(Strictly speaking, a map $f\colon X \to Y$ should be recorded together with its target Y, so that the last axiom of Definition 6.2.1 holds. We do this silently here and in similar examples.)

Example 6.2.6. In the same way there are locally small categories of small groups Grp, of small abelian groups Ab, of small rings Ring, of small vector spaces over a field k, written Vect_k, and of small topological spaces Top. In each case the morphisms are the maps that preserve the relevant structure (group homomorphisms, ring homomorphisms, linear maps, continuous maps), and composition is composition of maps. These notions are defined in later chapters.

Definition 6.2.7. A *monoid* is a set M with a map $M \times M \to M$, written $(a, b) \mapsto a \cdot b$ or simply ab, which is associative, together with an element $e \in M$ such that $ea = a = ae$ for all $a \in M$. (For a monoid given by addition, such as $(\mathbb{N}, +)$, we write $a + b$ and 0 instead.) A *homomorphism of monoids* $\varphi\colon M \to N$ is a map with $\varphi(ab) = \varphi(a)\varphi(b)$ for all $a, b \in M$ and $\varphi(e) = e$. The small monoids and their homomorphisms form a locally small category Mon.

Example 6.2.8. Let M be a monoid. There is a category BM with a single object $*$, with $\mathrm{Hom}_{BM}(*, *) = M$, with composition given by the multiplication of M and with $\mathrm{id}_* = e$. The axioms of a category are exactly the axioms of a monoid. Conversely, if C is a category and $X \in C$, then $\mathrm{End}(X) = \mathrm{Hom}_C(X, X)$ is a monoid under composition.

Example 6.2.9. Let P be a set with a relation $\leq$ that is reflexive and transitive (a *preorder*). There is a category, also written P, whose objects are the elements of P, with $\mathrm{Hom}_P(x, y) = \{(x, y)\}$ if $x \leq y$ and $\mathrm{Hom}_P(x, y) = \emptyset$ otherwise. Composition is forced, because every set of morphisms has at most one element; it exists by transitivity, and the identities exist by reflexivity.

Example 6.2.10. Let S be a set. The *discrete category* on S has objects the elements of S, with $\mathrm{Hom}(s, s) = \{\mathrm{id}_s\}$ and $\mathrm{Hom}(s, t) = \emptyset$ for $s \neq t$. It is the category associated with the preorder "$s \leq t$ if and only if $s = t$".

Definition 6.2.11. Let C be a category. The *opposite category* C^{op} has the same objects as C and $\mathrm{Hom}_{C^{\mathrm{op}}}(X, Y) = \mathrm{Hom}_C(Y, X)$, with the same identities and with composition $g \circ_{\mathrm{op}} f = f \circ g$.

Remark 6.2.12. The axioms of a category are symmetric under reversing all morphisms, and $(C^{\mathrm{op}})^{\mathrm{op}} = C$. So every statement proved for all categories implies a *dual* statement, obtained by applying it to opposite categories.

Definition 6.2.13. A *subcategory* $\mathcal{D}$ of a category C consists of a subset $\mathrm{Ob}(\mathcal{D}) \subseteq \mathrm{Ob}(C)$ and subsets $\mathrm{Hom}_{\mathcal{D}}(X,Y) \subseteq \mathrm{Hom}_C(X,Y)$ for $X, Y \in \mathcal{D}$, which contain the identities and are closed under composition. It is a *full* subcategory if $\mathrm{Hom}_{\mathcal{D}}(X,Y) = \mathrm{Hom}_C(X,Y)$ for all $X, Y \in \mathcal{D}$.

Example 6.2.14. The category Ab is a full subcategory of Grp. The category whose objects are small sets and whose morphisms are the bijections is a subcategory of Set that is not full.

6.3 Isomorphisms, monomorphisms and epimorphisms

Definition 6.3.1. A morphism $f: X \to Y$ in a category C is an *isomorphism* if there is a morphism $g: Y \to X$ with $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$. Two objects X and Y are *isomorphic*, written $X \cong Y$, if there is an isomorphism $X \to Y$.

Lemma 6.3.2. *Let $f: X \to Y$ be an isomorphism. There is exactly one morphism $g: Y \to X$ with $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$.*

Proof. Suppose g and g' both have the property. Then $g = g \circ \mathrm{id}_Y = g \circ (f \circ g') = (g \circ f) \circ g' = \mathrm{id}_X \circ g' = g'$. □

Notation 6.3.3. The morphism of Lemma 6.3.2 is the *inverse* of f, written f^{-1}.

Definition 6.3.4. A morphism $f: X \to Y$ is a *monomorphism* if for every object W and all $g_1, g_2: W \to X$, the equality $f \circ g_1 = f \circ g_2$ implies $g_1 = g_2$. It is an *epimorphism* if for every object Z and all $h_1, h_2: Y \to Z$, the equality $h_1 \circ f = h_2 \circ f$ implies $h_1 = h_2$.

Lemma 6.3.5. *Let C be a category.*

1. *Every isomorphism is both a monomorphism and an epimorphism.*
2. *A composition of monomorphisms is a monomorphism, and a composition of epimorphisms is an epimorphism.*
3. *If $g \circ f$ is a monomorphism, then f is a monomorphism. If $g \circ f$ is an epimorphism, then g is an epimorphism.*

Proof. By Remark 6.2.12, a monomorphism in C is the same thing as an epimorphism in C^{op}, so in each part it suffices to prove the statement about monomorphisms.

(1) Let f be an isomorphism and suppose $f \circ g_1 = f \circ g_2$. Composing with f^{-1} on the left gives $g_1 = g_2$.

(2) Let $f: X \to Y$ and $f': Y \to Z$ be monomorphisms and suppose $f' \circ f \circ g_1 = f' \circ f \circ g_2$. Since f' is a monomorphism, $f \circ g_1 = f \circ g_2$, and since f is a monomorphism, $g_1 = g_2$.

(3) Suppose $f \circ g_1 = f \circ g_2$. Then $(g \circ f) \circ g_1 = (g \circ f) \circ g_2$, so $g_1 = g_2$. □

Lemma 6.3.6. *In the category* Set*, a map is a monomorphism if and only if it is injective, and an epimorphism if and only if it is surjective.*

Proof. Let $f: X \to Y$ be a map of small sets.

Suppose f is injective and $f \circ g_1 = f \circ g_2$ for maps $g_1, g_2: W \to X$. For every $w \in W$ we have $f(g_1(w)) = f(g_2(w))$, hence $g_1(w) = g_2(w)$; so $g_1 = g_2$. Conversely, suppose f is a monomorphism and $f(x_1) = f(x_2)$. Let $W = \{0\}$, which is a small set, and let $g_i: W \to X$ be the map with $g_i(0) = x_i$. Then $f \circ g_1 = f \circ g_2$, so $g_1 = g_2$, that is, $x_1 = x_2$.

Suppose f is surjective and $h_1 \circ f = h_2 \circ f$ for maps $h_1, h_2: Y \to Z$. Every $y \in Y$ is of the form $f(x)$, so $h_1(y) = h_1(f(x)) = h_2(f(x)) = h_2(y)$; hence $h_1 = h_2$. Conversely, suppose f is

an epimorphism. Let $Z = \{0, 1\}$, let $h_1 \colon Y \to Z$ be the map with $h_1(y) = 1$ if $y \in f(X)$ and $h_1(y) = 0$ otherwise, and let h_2 be the constant map with value 1. Then $h_1 \circ f = h_2 \circ f$, so $h_1 = h_2$, which says exactly that $f(X) = Y$. □

Counterexample 6.3.7. A morphism that is both a monomorphism and an epimorphism need not be an isomorphism. Let $P = \{0, 1\}$ with the usual order $0 \leq 1$, viewed as a category as in Example 6.2.9, and let $u \colon 0 \to 1$ be the unique morphism. Every set of morphisms of P has at most one element, so any two parallel morphisms are equal; hence every morphism of P is a monomorphism and an epimorphism. But u is not an isomorphism, because $\mathrm{Hom}_P(1, 0) = \emptyset$.

Counterexample 6.3.8. The same phenomenon occurs in categories of algebraic objects. Let $i \colon \mathbb{N} \to \mathbb{Z}$ be the inclusion, a homomorphism of monoids under addition. Then i is a monomorphism and an epimorphism in Mon, but not an isomorphism.

Indeed, if $u, v \colon M \to \mathbb{N}$ are homomorphisms with $i \circ u = i \circ v$, then $u = v$ because i is injective. Next, let $f, g \colon \mathbb{Z} \to M$ be homomorphisms with $f \circ i = g \circ i$, that is, $f(n) = g(n)$ for all $n \in \mathbb{N}$. Writing the operation of M multiplicatively, for every $n \in \mathbb{N}$ we have

$$f(-n) = f(-n)\, g(n+(-n)) = f(-n)\, g(n)\, g(-n) = f(-n)\, f(n)\, g(-n) = f(-n+n)\, g(-n) = g(-n),$$

where we used $g(0) = e$ and $f(0) = e$. Hence $f = g$. Finally, an isomorphism of monoids is in particular a bijection, and i is not surjective.

Remark 6.3.9. Counterexample 6.3.8 shows that being an epimorphism is not the same as being surjective, even in categories whose morphisms are maps. Whether a morphism is an epimorphism depends on the whole category, not only on the morphism itself.

6.4 Functors

Definition 6.4.1. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A *functor* $F \colon \mathcal{C} \to \mathcal{D}$ consists of a map $F \colon \mathrm{Ob}(\mathcal{C}) \to \mathrm{Ob}(\mathcal{D})$ and, for all $X, Y \in \mathcal{C}$, a map $F \colon \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(X), F(Y))$, such that $F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$ for all X and $F(g \circ f) = F(g) \circ F(f)$ for all composable morphisms f and g. A *contravariant functor* from $\mathcal{C}$ to $\mathcal{D}$ is a functor $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$.

Remark 6.4.2. Unwinding the definitions, a contravariant functor F from $\mathcal{C}$ to $\mathcal{D}$ assigns to each $f \colon X \to Y$ in $\mathcal{C}$ a morphism $F(f) \colon F(Y) \to F(X)$, and $F(g \circ f) = F(f) \circ F(g)$.

Example 6.4.3. Some functors.

1. Every category $\mathcal{C}$ has an identity functor $\mathrm{id}_{\mathcal{C}}$. If $F \colon \mathcal{C} \to \mathcal{D}$ and $G \colon \mathcal{D} \to \mathcal{E}$ are functors, then so is their composition $G \circ F$, defined on objects and on morphisms by composing maps.

2. Forgetful functors, such as $\mathsf{Grp} \to \mathsf{Set}$, which send a group to its underlying set and a homomorphism to its underlying map.

3. The covariant power set functor $\mathcal{P} \colon \mathsf{Set} \to \mathsf{Set}$ sends X to its power set $\mathcal{P}(X)$ and $f \colon X \to Y$ to the map $A \mapsto f(A)$. It is a functor because $(g \circ f)(A) = g(f(A))$.

4. The contravariant power set functor $\mathcal{P}^* \colon \mathsf{Set}^{\mathrm{op}} \to \mathsf{Set}$ sends X to $\mathcal{P}(X)$ and $f \colon X \to Y$ to the map $\mathcal{P}(Y) \to \mathcal{P}(X)$, $B \mapsto f^{-1}(B)$. It is a functor because $(g \circ f)^{-1}(C) = f^{-1}(g^{-1}(C))$.

5. If P and Q are preorders, viewed as categories, a functor $P \to Q$ is the same thing as a map $\varphi \colon P \to Q$ with $\varphi(x) \leq \varphi(y)$ whenever $x \leq y$.

6. If M and N are monoids, a functor $BM \to BN$ is the same thing as a homomorphism of monoids $M \to N$.

Definition 6.4.4. Let C be a locally small category and $X \in C$. The functor $h_X : C^{\mathrm{op}} \to \mathsf{Set}$ sends Y to $\mathrm{Hom}_C(Y, X)$ and a morphism $f : Y' \to Y$ of C to the map

$$h_X(f) \colon \mathrm{Hom}_C(Y, X) \to \mathrm{Hom}_C(Y', X), \qquad g \mapsto g \circ f.$$

Dually, the functor $h^X : C \to \mathsf{Set}$ sends Y to $\mathrm{Hom}_C(X, Y)$ and $f : Y \to Y'$ to $g \mapsto f \circ g$.

Lemma 6.4.5. *Let $F : C \to \mathcal{D}$ be a functor. If f is an isomorphism in C, then $F(f)$ is an isomorphism in $\mathcal{D}$, with inverse $F(f^{-1})$.*

Proof. If $f : X \to Y$, then $F(f^{-1}) \circ F(f) = F(f^{-1} \circ f) = F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$, and similarly $F(f) \circ F(f^{-1}) = \mathrm{id}_{F(Y)}$. □

Definition 6.4.6. A functor $F : C \to \mathcal{D}$ is *faithful* (respectively *full*) if for all $X, Y \in C$ the map $\mathrm{Hom}_C(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(X), F(Y))$ is injective (respectively surjective). It is *fully faithful* if it is both full and faithful. It is *essentially surjective* if every object of $\mathcal{D}$ is isomorphic to an object of the form $F(X)$.

Lemma 6.4.7. *Let $F : C \to \mathcal{D}$ be a fully faithful functor and $f : X \to Y$ a morphism of C. If $F(f)$ is an isomorphism, then f is an isomorphism. In particular, if $F(X) \cong F(Y)$ then $X \cong Y$.*

Proof. Let $h : F(Y) \to F(X)$ be the inverse of $F(f)$. Since F is full, there is $g : Y \to X$ with $F(g) = h$. Then $F(g \circ f) = h \circ F(f) = \mathrm{id}_{F(X)} = F(\mathrm{id}_X)$, and since F is faithful, $g \circ f = \mathrm{id}_X$. In the same way $f \circ g = \mathrm{id}_Y$.

For the second statement, let $\varphi : F(X) \to F(Y)$ be an isomorphism. Since F is full, $\varphi = F(f)$ for some $f : X \to Y$, and f is an isomorphism by the first statement. □

Counterexample 6.4.8. Faithfulness alone does not suffice in Lemma 6.4.7. Let P be the category $0 \leq 1$ of Counterexample 6.3.7, let $\mathbf{1}$ be the category with one object and one morphism, and let $F : P \to \mathbf{1}$ be the unique functor. Every set of morphisms of P has at most one element, so F is faithful. The morphism $u : 0 \to 1$ is not an isomorphism, but $F(u)$ is an identity, hence an isomorphism. (The functor F is not full: $\mathrm{Hom}_P(1, 0)$ is empty.)

Remark 6.4.9. A more natural instance of the same phenomenon is the forgetful functor $\mathsf{Top} \to \mathsf{Set}$, which is faithful. The identity map of $\mathbb{R}$, viewed as a map from $\mathbb{R}$ with the discrete topology to $\mathbb{R}$ with the Euclidean topology, is a continuous bijection, so its image in Set is an isomorphism; but it is not a homeomorphism.

6.5 Natural transformations

Definition 6.5.1. Let $F, G : C \to \mathcal{D}$ be functors. A *natural transformation* $\eta : F \Rightarrow G$ is a family of morphisms $\eta_X : F(X) \to G(X)$, one for each $X \in C$, such that $G(f) \circ \eta_X = \eta_Y \circ F(f)$ for every morphism $f : X \to Y$ of C; that is, such that the square

$$\begin{array}{ccc} F(X) & \xrightarrow{\eta_X} & G(X) \\ {\scriptstyle F(f)}\downarrow & & \downarrow{\scriptstyle G(f)} \\ F(Y) & \xrightarrow[\eta_Y]{} & G(Y) \end{array}$$

commutes. It is a *natural isomorphism* if every η_X is an isomorphism. We write $\mathrm{Nat}(F, G)$ for the set of natural transformations from F to G.

Lemma 6.5.2. *Let $\eta\colon F \Rightarrow G$ be a natural isomorphism. Then the morphisms $\eta_X^{-1}\colon G(X) \to F(X)$ form a natural transformation $\eta^{-1}\colon G \Rightarrow F$.*

Proof. Let $f\colon X \to Y$. Composing the equality $G(f) \circ \eta_X = \eta_Y \circ F(f)$ with η_Y^{-1} on the left and with η_X^{-1} on the right gives $\eta_Y^{-1} \circ G(f) = F(f) \circ \eta_X^{-1}$, which is the naturality of η^{-1}. □

Definition 6.5.3. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. If $\eta\colon F \Rightarrow G$ and $\theta\colon G \Rightarrow H$ are natural transformations, their composition $\theta \circ \eta\colon F \Rightarrow H$ is given by $(\theta \circ \eta)_X = \theta_X \circ \eta_X$. The *functor category* $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$ has as objects the functors $\mathcal{C} \to \mathcal{D}$, as morphisms the natural transformations, and this composition; the identity of F is the family $(\mathrm{id}_{F(X)})_X$. (Formally, a natural transformation $\eta\colon F \Rightarrow G$ is the triple $(F, G, (\eta_X)_X)$, so that the sets $\mathrm{Nat}(F, G)$ are pairwise disjoint.)

Lemma 6.5.4. *Definition 6.5.3 defines a category. A morphism η of $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$ is an isomorphism if and only if it is a natural isomorphism. If $\mathcal{C}$ is small and $\mathcal{D}$ is locally small, then $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$ is locally small.*

Proof. If η and θ are natural, then for $f\colon X \to Y$ we have $H(f) \circ \theta_X \circ \eta_X = \theta_Y \circ G(f) \circ \eta_X = \theta_Y \circ \eta_Y \circ F(f)$, so $\theta \circ \eta$ is natural. The family of identities is clearly natural. Associativity and the identity axioms hold componentwise, because they hold in $\mathcal{D}$.

If η is an isomorphism with inverse θ, then $\theta_X \circ \eta_X = \mathrm{id}_{F(X)}$ and $\eta_X \circ \theta_X = \mathrm{id}_{G(X)}$ for every X, so each η_X is an isomorphism. Conversely, if η is a natural isomorphism, then η^{-1} of Lemma 6.5.2 is a morphism of $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$ that is inverse to η.

Finally, $\mathrm{Nat}(F, G)$ is a subset of the product $\prod_{X \in \mathrm{Ob}(\mathcal{C})} \mathrm{Hom}_{\mathcal{D}}(F(X), G(X))$. If $\mathcal{C}$ is small and $\mathcal{D}$ is locally small, this is a product of small sets indexed by a small set, hence small, and so is every subset of it. □

Example 6.5.5. Let k be a field. For a vector space V over k, let V^* be its dual and $\mathrm{ev}_V\colon V \to V^{**}$ the map sending v to the linear form $\varphi \mapsto \varphi(v)$. The double dual is a functor $\mathsf{Vect}_k \to \mathsf{Vect}_k$, sending $f\colon V \to W$ to $f^{**}\colon \alpha \mapsto (\psi \mapsto \alpha(\psi \circ f))$, and the maps ev_V form a natural transformation $\mathrm{ev}\colon \mathrm{id} \Rightarrow (-)^{**}$: for $v \in V$ and $\psi \in W^*$,

$$f^{**}(\mathrm{ev}_V(v))(\psi) = \mathrm{ev}_V(v)(\psi \circ f) = \psi(f(v)) = \mathrm{ev}_W(f(v))(\psi).$$

On the full subcategory of finite-dimensional vector spaces it is a natural isomorphism, since there each ev_V is an isomorphism.

Remark 6.5.6. A finite-dimensional vector space V is also isomorphic to its dual V^*, but an isomorphism $V \to V^*$ depends on a choice, such as a basis. The notion of a natural transformation was introduced to make precise the difference between "canonical" isomorphisms, such as $V \cong V^{**}$, and isomorphisms that depend on choices.

Example 6.5.7. Let $n \geq 1$. For a commutative ring R, the determinant is a group homomorphism $\det_R\colon \mathrm{GL}_n(R) \to R^\times$. Both GL_n and $(-)^\times$ are functors from commutative rings to groups, and det is a natural transformation between them: if $\varphi\colon R \to S$ is a ring homomorphism and $A \in \mathrm{GL}_n(R)$, then $\det_S(\varphi(A)) = \varphi(\det_R(A))$, because the determinant is a polynomial with integer coefficients in the entries of the matrix.

6.6 The Yoneda lemma

In this section $\mathcal{C}$ is a locally small category, so that the functors $h_X\colon \mathcal{C}^{\mathrm{op}} \to \mathsf{Set}$ of Definition 6.4.4 are defined.

Theorem 6.6.1 (Yoneda lemma). *Let $X \in C$ and let $F\colon C^{\mathrm{op}} \to \mathsf{Set}$ be a functor. The map*

$$\Phi\colon\ \mathrm{Nat}(h_X, F) \longrightarrow F(X), \qquad \eta \longmapsto \eta_X(\mathrm{id}_X)$$

is a bijection. Its inverse sends $\xi \in F(X)$ to the natural transformation η^ξ given by $\eta^\xi_Y(g) = F(g)(\xi)$ for $Y \in C$ and $g \in \mathrm{Hom}_C(Y, X)$.

Proof. First, η^ξ is natural. Let $f\colon Y' \to Y$ be a morphism of C and $g \in \mathrm{Hom}_C(Y, X)$. Using that F is contravariant (Remark 6.4.2),

$$\eta^\xi_{Y'}(h_X(f)(g)) = \eta^\xi_{Y'}(g \circ f) = F(g \circ f)(\xi) = F(f)(F(g)(\xi)) = F(f)(\eta^\xi_Y(g)),$$

which is the naturality condition for f.

Second, $\Phi(\eta^\xi) = \eta^\xi_X(\mathrm{id}_X) = F(\mathrm{id}_X)(\xi) = \xi$.

Third, let $\eta\colon h_X \Rightarrow F$ be natural and $g\colon Y \to X$. The naturality of η with respect to g gives $\eta_Y(h_X(g)(\mathrm{id}_X)) = F(g)(\eta_X(\mathrm{id}_X))$. Since $h_X(g)(\mathrm{id}_X) = \mathrm{id}_X \circ g = g$, this says $\eta_Y(g) = F(g)(\Phi(\eta)) = \eta^{\Phi(\eta)}_Y(g)$. Hence $\eta = \eta^{\Phi(\eta)}$.

The second and third steps show that $\xi \mapsto \eta^\xi$ is inverse to Φ. □

Remark 6.6.2. The Yoneda lemma says that a natural transformation out of h_X is determined by what it does to the single element id_X, and that this element can be chosen freely. Informally: an object is determined by the morphisms into it.

Proposition 6.6.3. *The bijection Φ of Theorem 6.6.1 is natural in F and in X:*

1. *if $\theta\colon F \Rightarrow F'$ is a natural transformation, then $\Phi(\theta \circ \eta) = \theta_X(\Phi(\eta))$ for all $\eta \in \mathrm{Nat}(h_X, F)$;*
2. *if $a\colon X' \to X$ is a morphism and $h_a\colon h_{X'} \Rightarrow h_X$ is the natural transformation given by $(h_a)_Y(g) = a \circ g$, then $\Phi(\eta \circ h_a) = F(a)(\Phi(\eta))$ for all $\eta \in \mathrm{Nat}(h_X, F)$.*

Proof. We first check that h_a is natural: for $f\colon Y' \to Y$ and $g\colon Y \to X'$ we have $a \circ (g \circ f) = (a \circ g) \circ f$, that is, $(h_a)_{Y'}(h_{X'}(f)(g)) = h_X(f)((h_a)_Y(g))$.

(1) $\Phi(\theta \circ \eta) = \theta_X(\eta_X(\mathrm{id}_X)) = \theta_X(\Phi(\eta))$.

(2) $\Phi(\eta \circ h_a) = \eta_{X'}((h_a)_{X'}(\mathrm{id}_{X'})) = \eta_{X'}(a)$. By the third step of the proof of Theorem 6.6.1, $\eta_{X'}(a) = F(a)(\Phi(\eta))$. □

Corollary 6.6.4. *For all $X, X' \in C$, the map*

$$\mathrm{Hom}_C(X', X) \longrightarrow \mathrm{Nat}(h_{X'}, h_X), \qquad a \longmapsto h_a$$

is a bijection. Consequently, $X \mapsto h_X$, $a \mapsto h_a$ is a fully faithful functor $C \to \mathrm{Fun}(C^{\mathrm{op}}, \mathsf{Set})$, called the Yoneda embedding.

Proof. Apply Theorem 6.6.1 with X' in place of X and $F = h_X$. Then $\Phi(h_a) = (h_a)_{X'}(\mathrm{id}_{X'}) = a$, so the map $a \mapsto h_a$ is a right inverse of the bijection Φ, hence equal to its inverse and bijective. The identities $h_{\mathrm{id}_X} = \mathrm{id}_{h_X}$ and $h_{a \circ b} = h_a \circ h_b$ follow from the identity axiom and from associativity in C, so $X \mapsto h_X$ is a functor, and it is fully faithful by the first statement. □

Corollary 6.6.5. *Let $X, X' \in C$. If there is a natural isomorphism $h_{X'} \Rightarrow h_X$, then $X' \cong X$.*

Proof. By Lemma 6.5.4, a natural isomorphism $h_{X'} \Rightarrow h_X$ is an isomorphism in $\mathrm{Fun}(C^{\mathrm{op}}, \mathsf{Set})$. The Yoneda embedding is fully faithful by Corollary 6.6.4, so the claim follows from Lemma 6.4.7. □

Definition 6.6.6. A functor $F\colon \mathcal{C}^{\mathrm{op}} \to \mathsf{Set}$ is *representable* if there exist $X \in \mathcal{C}$ and a natural isomorphism $h_X \Rightarrow F$. By Theorem 6.6.1, such a natural isomorphism is η^ξ for a unique $\xi \in F(X)$; the pair (X, ξ) is said to *represent* F.

Example 6.6.7. The contravariant power set functor $\mathcal{P}^*$ of Example 6.4.3 is represented by the pair $(\{0,1\}, \{1\})$. Indeed, for a small set S the map $\eta_S^{\{1\}}$ sends $\chi\colon S \to \{0,1\}$ to $\mathcal{P}^*(\chi)(\{1\}) = \chi^{-1}(1)$, and this is a bijection $\mathrm{Hom}_{\mathsf{Set}}(S, \{0,1\}) \to \mathcal{P}(S)$: its inverse sends a subset $A \subseteq S$ to its indicator function.

6.7 Equivalences of categories

Definition 6.7.1. A functor $F\colon \mathcal{C} \to \mathcal{D}$ is an *equivalence of categories* if there are a functor $G\colon \mathcal{D} \to \mathcal{C}$ and natural isomorphisms $G \circ F \Rightarrow \mathrm{id}_{\mathcal{C}}$ and $F \circ G \Rightarrow \mathrm{id}_{\mathcal{D}}$; such a G is a *quasi-inverse* of F. It is an *isomorphism of categories* if moreover $G \circ F = \mathrm{id}_{\mathcal{C}}$ and $F \circ G = \mathrm{id}_{\mathcal{D}}$. Two categories are *equivalent* if there is an equivalence between them.

Lemma 6.7.2. *An equivalence of categories is fully faithful and essentially surjective.*

Proof. Let G, $\eta\colon GF \Rightarrow \mathrm{id}$ and $\varepsilon\colon FG \Rightarrow \mathrm{id}$ be as in Definition 6.7.1. Every $Y \in \mathcal{D}$ is isomorphic to $F(G(Y))$ via ε_Y, so F is essentially surjective. For $f\colon X \to X'$, naturality of η gives $f = \eta_{X'} \circ GF(f) \circ \eta_X^{-1}$; so $F(f)$ determines f and F is faithful. By symmetry G is faithful. Let $g\colon F(X) \to F(X')$ and put $f = \eta_{X'} \circ G(g) \circ \eta_X^{-1}$. By the same formula applied to f, $GF(f) = \eta_{X'}^{-1} \circ f \circ \eta_X = G(g)$, and since G is faithful, $F(f) = g$. So F is full. □

Theorem 6.7.3. *A functor $F\colon \mathcal{C} \to \mathcal{D}$ is an equivalence of categories if and only if it is fully faithful and essentially surjective.*

Proof. One direction is Lemma 6.7.2. Conversely, for each $Y \in \mathcal{D}$ the set of pairs (X, e) with $X \in \mathcal{C}$ and $e\colon F(X) \to Y$ an isomorphism is nonempty. By the axiom of choice (Definition 3.7.1) we may choose such a pair $(G(Y), \varepsilon_Y)$ for every Y. For $g\colon Y \to Y'$, let $G(g)$ be the unique morphism $G(Y) \to G(Y')$ with $F(G(g)) = \varepsilon_{Y'}^{-1} \circ g \circ \varepsilon_Y$; it exists and is unique because F is fully faithful. By uniqueness, G preserves identities and composition, and by construction $\varepsilon\colon FG \Rightarrow \mathrm{id}_{\mathcal{D}}$ is a natural isomorphism.

For $X \in \mathcal{C}$ let $\eta_X\colon GF(X) \to X$ be the unique morphism with $F(\eta_X) = \varepsilon_{F(X)}$. It is an isomorphism by Lemma 6.4.7. For $f\colon X \to X'$ we have $F(f \circ \eta_X) = F(f) \circ \varepsilon_{F(X)}$ and $F(\eta_{X'} \circ GF(f)) = \varepsilon_{F(X')} \circ \varepsilon_{F(X')}^{-1} \circ F(f) \circ \varepsilon_{F(X)}$, which are equal; as F is faithful, η is natural. □

Example 6.7.4. Let k be a field, and let Mat_k be the category whose objects are the natural numbers, whose morphisms $n \to m$ are the $m \times n$ matrices with entries in k, and whose composition is matrix multiplication. The functor from Mat_k to the category of finite-dimensional vector spaces sending n to k^n and a matrix to the corresponding linear map is fully faithful, and essentially surjective because every finite-dimensional vector space has a basis. So it is an equivalence: linear algebra of finite-dimensional spaces is "the same" as matrix algebra.

Counterexample 6.7.5. Equivalent categories need not be isomorphic. Let FinSet be the category of finite small sets and maps, and $\mathcal{N}$ its full subcategory whose objects are the natural numbers. Every finite set is isomorphic to a natural number (Definition 3.6.4), so the inclusion $\mathcal{N} \to \mathsf{FinSet}$ is fully faithful and essentially surjective, hence an equivalence. But an isomorphism of categories is a bijection on objects, and $\mathrm{Ob}(\mathcal{N}) = \omega$ is countable while $\mathrm{Ob}(\mathsf{FinSet})$ contains the singletons $\{x\}$ for all $x \in \mathcal{U}$, and $\mathcal{U}$ is uncountable since it contains the uncountable set $\mathbb{R}$ as a subset.

6.8 Limits

Definition 6.8.1. Let $\mathcal{I}$ be a small category and $F\colon \mathcal{I} \to C$ a functor, called a *diagram* of shape $\mathcal{I}$. A *cone* over F is an object L together with morphisms $\lambda_i\colon L \to F(i)$ for all $i \in \mathcal{I}$ such that $F(u) \circ \lambda_i = \lambda_j$ for every $u\colon i \to j$. A *limit* of F is a cone (L, λ) such that for every cone (M, μ) there is a unique morphism $m\colon M \to L$ with $\lambda_i \circ m = \mu_i$ for all i. We write $L = \lim F$ or $\lim_{i\in\mathcal{I}} F(i)$.

Lemma 6.8.2. *If (L, λ) and (L', λ') are limits of F, there is a unique morphism $m\colon L' \to L$ with $\lambda_i \circ m = \lambda'_i$ for all i, and it is an isomorphism.*

Proof. Existence and uniqueness of m is the definition. Let $m'\colon L \to L'$ be the morphism given by the universal property of L'. Then $m \circ m'$ and id_L are both morphisms of cones from (L, λ) to itself, so they are equal by uniqueness; similarly $m' \circ m = \mathrm{id}_{L'}$. □

Example 6.8.3.

1. If $\mathcal{I}$ is empty, a limit is a *terminal object*: an object T such that every object has exactly one morphism to T.
2. If $\mathcal{I}$ is a discrete category, a limit of $(X_i)_{i\in\mathcal{I}}$ is a *product* $\prod_i X_i$ with projections p_i.
3. If $\mathcal{I}$ has two objects $0, 1$ and two non-identity morphisms $u, v\colon 0 \to 1$, a diagram is a pair of parallel morphisms $f, g\colon X \to Y$, and a limit is an *equaliser*: a morphism $e\colon E \to X$ with $fe = ge$ through which every h with $fh = gh$ factors uniquely.
4. If $\mathcal{I}$ is $1 \to 0 \leftarrow 2$, a diagram is a pair $f\colon X \to Z$, $g\colon Y \to Z$, and a limit is a *fibre product* $X \times_Z Y$.
5. If $\mathcal{I}$ is the poset $\mathbb{N}$ with the opposite order, a limit of a sequence $\cdots \to X_2 \to X_1 \to X_0$ is an *inverse limit*.

Definition 6.8.4. A category *has limits* (respectively *finite limits*) if every diagram $F\colon \mathcal{I} \to C$ with $\mathcal{I}$ small (respectively with finitely many morphisms) has a limit. A functor $G\colon C \to \mathcal{D}$ *preserves* the limit (L, λ) of F if $(G(L), G(\lambda))$ is a limit of $G \circ F$.

Proposition 6.8.5. Set *has limits. For $F\colon \mathcal{I} \to \mathsf{Set}$ with $\mathcal{I}$ small, a limit is*

$$\lim F = \left\{(x_i) \in \prod_{i\in \mathrm{Ob}(\mathcal{I})} F(i) : F(u)(x_i) = x_j \text{ for all } u\colon i \to j\right\}$$

with the projections $\lambda_i((x_k)_k) = x_i$.

Proof. The set $\lim F$ is small by Lemma 3.8.2, and it is a cone by construction. If (M, μ) is a cone, the map $m(y) = (\mu_i(y))_i$ lands in $\lim F$, satisfies $\lambda_i \circ m = \mu_i$, and is the only such map. □

Theorem 6.8.6. *If C has products indexed by small sets and equalisers, then it has limits. If C has a terminal object, binary products and equalisers, then it has finite limits.*

Proof. Let $F\colon \mathcal{I} \to C$ be a diagram. Let $P = \prod_i F(i)$ over the objects of $\mathcal{I}$, and $Q = \prod_u F(t(u))$ over all morphisms $u\colon s(u) \to t(u)$ of $\mathcal{I}$, with projections p_i and q_u. Define $a, b\colon P \to Q$ by $q_u \circ a = F(u) \circ p_{s(u)}$ and $q_u \circ b = p_{t(u)}$. Let $e\colon L \to P$ be an equaliser of a and b, and $\lambda_i = p_i \circ e$. The equation $ae = be$ says exactly that $F(u) \circ \lambda_{s(u)} = \lambda_{t(u)}$ for all u, so (L, λ) is a cone. If (M, μ) is a cone, the morphism $\mu\colon M \to P$ with $p_i \circ \mu = \mu_i$ satisfies $a\mu = b\mu$, so it factors uniquely as $\mu = e \circ m$; and a morphism m satisfies $\lambda_i \circ m = \mu_i$ for all i if and only if $e \circ m = \mu$. So (L, λ) is a limit. If $\mathcal{I}$ is finite, only finite products are needed, and these can be built from binary products and a terminal object (the empty product) by induction. □

Proposition 6.8.7. *Let C be locally small, $F\colon \mathcal{I} \to C$ a diagram and (L, λ) a cone over F. Then (L, λ) is a limit if and only if for every $X \in C$ the map*

$$\mathrm{Hom}_C(X, L) \longrightarrow \lim_{i \in \mathcal{I}} \mathrm{Hom}_C(X, F(i)), \qquad m \longmapsto (\lambda_i \circ m)_i$$

is a bijection, where the right-hand side is the limit in Set *of Proposition 6.8.5. In particular the functors $h^X = \mathrm{Hom}_C(X, -)$ preserve all limits.*

Proof. An element of the right-hand side is exactly a cone with vertex X, so bijectivity for all X is the definition of a limit. □

Proposition 6.8.8. *Let $\mathcal{A}$ be a category and suppose that C has limits of shape $\mathcal{I}$. Then $\mathrm{Fun}(\mathcal{A}, C)$ has limits of shape $\mathcal{I}$, computed pointwise: for $F\colon \mathcal{I} \to \mathrm{Fun}(\mathcal{A}, C)$ there is a limit L with $L(a) = \lim_i F(i)(a)$ for all $a \in \mathcal{A}$.*

Proof. For each a choose a limit $(L(a), \lambda^a)$ of $i \mapsto F(i)(a)$. For $f\colon a \to a'$, the morphisms $F(i)(f) \circ \lambda^a_i$ form a cone over $i \mapsto F(i)(a')$ (by naturality of the $F(u)$), so there is a unique $L(f)\colon L(a) \to L(a')$ with $\lambda^{a'}_i \circ L(f) = F(i)(f) \circ \lambda^a_i$. Uniqueness shows that L is a functor and that the $\lambda_i = (\lambda^a_i)_a$ are natural transformations $L \Rightarrow F(i)$ forming a cone. If (M, μ) is a cone in the functor category, then for each a there is a unique m_a with $\lambda^a_i \circ m_a = \mu_{i,a}$, and uniqueness again shows that $m = (m_a)_a$ is natural; it is the unique morphism of cones. □

Counterexample 6.8.9. The category Field of small fields and ring homomorphisms does not have binary products. First, a ring homomorphism $f\colon K \to K'$ between fields is injective: if $f(a) = 0$ with $a \neq 0$ then $1 = f(a)f(a^{-1}) = 0$. Let $\mathbb{F}_2 = \{0, 1\}$ be the field with $1 + 1 = 0$. If P were a product of $\mathbb{Q}$ and $\mathbb{F}_2$, there would be homomorphisms $P \to \mathbb{Q}$ and $P \to \mathbb{F}_2$. The second one sends $1 + 1$ to 0, so $1 + 1 = 0$ in P by injectivity; but then the first one sends 0 to $2 \neq 0$.

6.9 Colimits

Definition 6.9.1. A *cocone* under a diagram $F\colon \mathcal{I} \to C$ is an object C with morphisms $\gamma_i\colon F(i) \to C$ such that $\gamma_j \circ F(u) = \gamma_i$ for all $u\colon i \to j$. A *colimit* is a cocone (C, γ) such that for every cocone (D, δ) there is a unique $m\colon C \to D$ with $m \circ \gamma_i = \delta_i$ for all i. We write $C = \mathrm{colim}\, F$.

Remark 6.9.2. A colimit of F is the same thing as a limit of the corresponding diagram $\mathcal{I}^{\mathrm{op}} \to C^{\mathrm{op}}$. By duality (Remark 6.2.12), all results of Section 6.8 have colimit versions: colimits are unique up to unique isomorphism, colimits can be built from coproducts and coequalisers, and (C, γ) is a colimit if and only if $\mathrm{Hom}(C, X) \to \lim_i \mathrm{Hom}(F(i), X)$ is bijective for all X. The colimit analogues of the examples in Example 6.8.3 are *initial objects*, *coproducts* $\coprod_i X_i$, *coequalisers*, *pushouts* $X \sqcup_Z Y$ and *direct limits*.

Proposition 6.9.3. Set *has colimits. For $F\colon \mathcal{I} \to$* Set *with $\mathcal{I}$ small, let $\sim$ be the smallest equivalence relation on $\coprod_i F(i) = \{(i, x) : i \in \mathrm{Ob}(\mathcal{I}), x \in F(i)\}$ with $(i, x) \sim (j, F(u)(x))$ for all $u\colon i \to j$ and $x \in F(i)$. Then $\mathrm{colim}\, F = (\coprod_i F(i))/\!\sim$, with $\gamma_i(x) = [(i, x)]$.*

Proof. The smallest equivalence relation containing a relation R exists: it is the intersection of all equivalence relations containing R, a subset of the square of the set. The maps γ_i form a cocone by construction. A cocone (D, δ) gives a map $\coprod_i F(i) \to D$, $(i, x) \mapsto \delta_i(x)$, whose fibres define an equivalence relation containing the generating relation; so it factors uniquely through the quotient (Proposition 3.3.9). □

Example 6.9.4. In Set, the coproduct is the disjoint union, and the initial object is $\emptyset$. In the category of abelian groups the coproduct of a family is its direct sum, and in the category of groups it is the free product; these are constructed in Chapter 7.

6.10 Adjoint functors

In this section all categories are locally small.

Definition 6.10.1. An *adjunction* between functors $F\colon \mathcal{C} \to \mathcal{D}$ and $G\colon \mathcal{D} \to \mathcal{C}$ is a family of bijections

$$\varphi_{X,Y}\colon \operatorname{Hom}_{\mathcal{D}}(F(X), Y) \longrightarrow \operatorname{Hom}_{\mathcal{C}}(X, G(Y))$$

for $X \in \mathcal{C}$, $Y \in \mathcal{D}$, natural in X and Y: for $f\colon X' \to X$, $g\colon F(X) \to Y$ and $h\colon Y \to Y'$, $\varphi(g \circ F(f)) = \varphi(g) \circ f$ and $\varphi(h \circ g) = G(h) \circ \varphi(g)$. We say F is *left adjoint* to G, and G *right adjoint* to F, written $F \dashv G$. The *unit* and *counit* are $\eta_X = \varphi(\mathrm{id}_{F(X)})\colon X \to GF(X)$ and $\varepsilon_Y = \varphi^{-1}(\mathrm{id}_{G(Y)})\colon FG(Y) \to Y$.

Proposition 6.10.2. *Let $F \dashv G$. Then $\eta\colon \mathrm{id}_{\mathcal{C}} \Rightarrow GF$ and $\varepsilon\colon FG \Rightarrow \mathrm{id}_{\mathcal{D}}$ are natural, $\varphi(g) = G(g) \circ \eta_X$, $\varphi^{-1}(f) = \varepsilon_Y \circ F(f)$, and the* triangle identities

$$\varepsilon_{F(X)} \circ F(\eta_X) = \mathrm{id}_{F(X)}, \qquad G(\varepsilon_Y) \circ \eta_{G(Y)} = \mathrm{id}_{G(Y)}$$

hold. Conversely, if $\eta\colon \mathrm{id} \Rightarrow GF$ and $\varepsilon\colon FG \Rightarrow \mathrm{id}$ are natural transformations satisfying the triangle identities, then $\varphi(g) = G(g) \circ \eta_X$ defines an adjunction with unit η and counit ε.

Proof. By naturality in Y, $\varphi(g) = \varphi(g \circ \mathrm{id}_{F(X)}) = G(g) \circ \varphi(\mathrm{id}_{F(X)}) = G(g) \circ \eta_X$. By naturality in X, for $f\colon X \to G(Y)$, $\varphi^{-1}(f) = \varphi^{-1}(\mathrm{id}_{G(Y)} \circ f) = \varphi^{-1}(\mathrm{id}_{G(Y)}) \circ F(f) = \varepsilon_Y \circ F(f)$. Naturality of η: for $f\colon X' \to X$, $GF(f) \circ \eta_{X'} = \varphi(F(f)) = \varphi(\mathrm{id}_{F(X)} \circ F(f)) = \eta_X \circ f$; naturality of ε is dual. The triangle identities are $\varphi^{-1}(\eta_X) = \mathrm{id}_{F(X)}$ and $\varphi(\varepsilon_Y) = \mathrm{id}_{G(Y)}$.

Conversely, define $\varphi(g) = G(g) \circ \eta_X$ and $\psi(f) = \varepsilon_Y \circ F(f)$. Then $\psi(\varphi(g)) = \varepsilon_Y \circ FG(g) \circ F(\eta_X) = g \circ \varepsilon_{F(X)} \circ F(\eta_X) = g$ by naturality of ε and the first triangle identity, and similarly $\varphi(\psi(f)) = f$. Naturality of φ in X follows from naturality of η, and in Y from functoriality of G. □

Proposition 6.10.3. *If G and G' are both right adjoint to F, there is a unique natural isomorphism $\theta\colon G \Rightarrow G'$ compatible with the adjunction bijections, that is, with $\varphi'_{X,Y} = \operatorname{Hom}(X, \theta_Y) \circ \varphi_{X,Y}$.*

Proof. For fixed Y, the bijections $\varphi'_{X,Y} \circ \varphi^{-1}_{X,Y}\colon \operatorname{Hom}(X, G(Y)) \to \operatorname{Hom}(X, G'(Y))$ are natural in X, so they form a natural isomorphism $h_{G(Y)} \Rightarrow h_{G'(Y)}$. By Corollary 6.6.4 it is h_{θ_Y} for a unique $\theta_Y\colon G(Y) \to G'(Y)$, which is an isomorphism because the Yoneda embedding is fully faithful (Lemma 6.4.7). Naturality in Y follows from naturality of φ and φ' in Y and faithfulness of the Yoneda embedding. □

Theorem 6.10.4. *A right adjoint preserves all limits that exist, and a left adjoint preserves all colimits that exist.*

Proof. Let $F \dashv G$ and let (L, λ) be a limit of $D\colon \mathcal{I} \to \mathcal{D}$. For every $X \in \mathcal{C}$ we have bijections, natural in X,

$$\operatorname{Hom}(X, G(L)) \cong \operatorname{Hom}(F(X), L) \cong \lim_i \operatorname{Hom}(F(X), D(i)) \cong \lim_i \operatorname{Hom}(X, G(D(i))),$$

by Proposition 6.8.7 in the middle. By naturality of φ in Y, the composite sends m to $(G(\lambda_i) \circ m)_i$. So $(G(L), G(\lambda))$ is a limit by Proposition 6.8.7. The statement about left adjoints follows by duality. □

Example 6.10.5. 1. *Free monoids.* For a small set S, let $W(S)$ be the set of finite sequences of elements of S, a monoid under concatenation with the empty sequence as unit. A map $f\colon S \to M$ to a monoid extends uniquely to a homomorphism $W(S) \to M$, namely $(s_1, \dots, s_n) \mapsto f(s_1) \cdots f(s_n)$. This gives an adjunction $W \dashv U$, where $U\colon \mathsf{Mon} \to \mathsf{Set}$ is the forgetful functor. Similarly, free groups, free abelian groups and free modules are left adjoints of forgetful functors (Chapters 7 and 8).

2. *Products and maps.* For a small set B, the functor $- \times B$ on Set is left adjoint to $(-)^B$: the bijection $\mathrm{Hom}(A \times B, C) \cong \mathrm{Hom}(A, C^B)$ of Exercise 3.9.1 is natural in A and C.

3. *Topology.* The forgetful functor $\mathsf{Top} \to \mathsf{Set}$ has a left adjoint (discrete topology) and a right adjoint (indiscrete topology).

Counterexample 6.10.6. The forgetful functor $U\colon \mathsf{Field} \to \mathsf{Set}$ has no left adjoint. A left adjoint L would preserve colimits by Theorem 6.10.4, in particular initial objects, so $L(\emptyset)$ would be an initial field. But an initial field would admit homomorphisms to $\mathbb{Q}$ and to $\mathbb{F}_2$, which is impossible by the argument of Counterexample 6.8.9.

6.11 Filtered colimits

Definition 6.11.1. A category $\mathcal{I}$ is *filtered* if it is nonempty, for all objects i, j there are an object k and morphisms $i \to k$ and $j \to k$, and for all morphisms $u, v\colon i \to j$ there is a morphism $w\colon j \to k$ with $w \circ u = w \circ v$. A *filtered colimit* is a colimit of a diagram whose shape is a small filtered category.

Example 6.11.2. A preordered set I is *directed* if it is nonempty and every two elements have a common upper bound. Viewed as a category (Example 6.2.9), a directed set is filtered, the third condition being automatic. For example $\mathbb{N}$ is directed, and the set of open neighbourhoods of a point in a topological space, ordered by reverse inclusion, is directed; colimits over it define stalks of sheaves (Chapter 25).

Lemma 6.11.3. *Let $\mathcal{I}$ be filtered.*

1. *For finitely many objects $i_1, \dots, i_n$ there is an object k with morphisms $i_m \to k$.*

2. *For finitely many morphisms $u_1, \dots, u_n\colon i \to j$ there is $w\colon j \to k$ with $wu_1 = wu_2 = \cdots = wu_n$.*

3. *For finitely many morphisms $u_m\colon i \to j_m$, $1 \le m \le n$, there are an object k and morphisms $w_m\colon j_m \to k$ with $w_1 u_1 = \cdots = w_n u_n$.*

Proof. (1) and (2) follow by induction on n from the definition; for (2), if $wu_1 = \cdots = wu_{n-1}$ with $w\colon j \to k$, choose $w'\colon k \to k'$ with $w'wu_{n-1} = w'wu_n$. (3) By (1) choose k' and $v_m\colon j_m \to k'$. The morphisms $v_m u_m\colon i \to k'$ are parallel, so by (2) there is $w\colon k' \to k$ with all $wv_m u_m$ equal; put $w_m = wv_m$. □

Proposition 6.11.4. *Let $\mathcal{I}$ be a small filtered category and $F\colon \mathcal{I} \to \mathsf{Set}$. On $\coprod_i F(i)$, the relation*

$$(i, x) \approx (j, y) \iff \textit{there are } u\colon i \to k \textit{ and } v\colon j \to k \textit{ with } F(u)(x) = F(v)(y)$$

is an equivalence relation, and $\operatorname{colim} F = (\coprod_i F(i))/{\approx}$. *In particular every element of* $\operatorname{colim} F$ *is of the form $\gamma_i(x)$, and $\gamma_i(x) = \gamma_i(x')$ if and only if $F(u)(x) = F(u)(x')$ for some $u\colon i \to k$.*

Proof. Reflexivity and symmetry are clear. For transitivity, let $F(u)(x) = F(v)(y)$ with $u\colon i \to k$, $v\colon j \to k$, and $F(v')(y) = F(w')(z)$ with $v'\colon j \to k'$, $w'\colon l \to k'$. By Lemma 6.11.3(3) applied to v and v', there are $a\colon k \to m$ and $b\colon k' \to m$ with $av = bv'$. Then $F(au)(x) = F(av)(y) = F(bv')(y) = F(bw')(z)$. The relation $\approx$ contains the generating relation of Proposition 6.9.3 (take $v = \mathrm{id}$), and it is contained in every equivalence relation containing it, since $(i,x) \sim (k, F(u)(x)) = (k, F(v)(y)) \sim (j,y)$. So it is the smallest one. For the last statement, if $F(u)(x) = F(v)(x')$ with $u, v\colon i \to k$, choose w with $wu = wv$ by filteredness; then $F(wu)(x) = F(wu)(x')$. □

Theorem 6.11.5. *Let $\mathcal{I}$ be a small filtered category, $\mathcal{J}$ a category with finitely many morphisms, and $F\colon \mathcal{I} \times \mathcal{J} \to \mathsf{Set}$ a functor. Then the canonical map*

$$\kappa\colon \operatorname*{colim}_{i\in\mathcal{I}} \lim_{j\in\mathcal{J}} F(i,j) \longrightarrow \lim_{j\in\mathcal{J}} \operatorname*{colim}_{i\in\mathcal{I}} F(i,j)$$

is a bijection. That is, in Set *filtered colimits commute with finite limits.*

Proof. We use the descriptions of Propositions 6.8.5 and 6.11.4. An element of the left side is represented by some i and a family $(x_j)_j$ with $x_j \in F(i,j)$ compatible with all morphisms of $\mathcal{J}$; the map κ sends it to the family of classes $([(i,x_j)])_j$.

Injectivity. Let two elements of the left side be represented by $(x_j) \in \lim_j F(i,j)$ and $(y_j) \in \lim_j F(i',j)$. By Lemma 6.11.3(1) and the functoriality of the limit in i, we may assume $i = i'$. Suppose $[(i,x_j)] = [(i,y_j)]$ for all j. By Proposition 6.11.4, for each of the finitely many objects j there is $u_j\colon i \to k_j$ with $F(u_j,j)(x_j) = F(u_j,j)(y_j)$. By Lemma 6.11.3(3) there are $w_j\colon k_j \to k$ with all $w_j u_j$ equal to one $u\colon i \to k$. Then $F(u,j)(x_j) = F(u,j)(y_j)$ for all j, so the two families have the same image in $\lim_j F(k,j)$ and define the same element of the colimit.

Surjectivity. Let $(\xi_j)_j$ be an element of the right side. Each ξ_j is represented by some $x_j \in F(i_j,j)$. Using Lemma 6.11.3(1) and replacing each x_j by its image, we may assume all i_j equal to one i. For each morphism $a\colon j \to j'$ of $\mathcal{J}$, compatibility of (ξ_j) says that $F(i,a)(x_j)$ and $x_{j'}$ have the same class in $\operatorname{colim}_i F(i,j')$, so there is $u_a\colon i \to k_a$ with $F(u_a,j')(F(i,a)(x_j)) = F(u_a,j')(x_{j'})$. There are finitely many a; by Lemma 6.11.3(3) we find $w_a\colon k_a \to k$ with all $w_a u_a$ equal to one $u\colon i \to k$. Put $y_j = F(u,j)(x_j) \in F(k,j)$. For every $a\colon j \to j'$, $F(k,a)(y_j) = F(u,j')(F(i,a)(x_j)) = F(w_a,j')F(u_a,j')(F(i,a)(x_j)) = F(u,j')(x_{j'}) = y_{j'}$, using that F is a functor on $\mathcal{I} \times \mathcal{J}$. So (y_j) lies in $\lim_j F(k,j)$, and its class maps to $([(k,y_j)])_j = (\xi_j)_j$. □

Counterexample 6.11.6. Filtered colimits do not commute with infinite products in Set. Let $\mathcal{I} = \mathbb{N}$, let $\mathcal{J}$ be the discrete category on $\mathbb{N}$, and let $F(n,j) = \{0,1,\dots,n\}$ with the inclusions as transition maps. Then $\operatorname{colim}_n F(n,j) = \mathbb{N}$ for every j, so the right side is the set of all sequences of natural numbers, whereas the left side $\operatorname{colim}_n \prod_j \{0,\dots,n\}$ consists of the bounded sequences. The sequence $j \mapsto j$ is not in the image of κ.

Counterexample 6.11.7. Colimits that are not filtered do not commute with finite limits either. Take $\mathcal{I}$ discrete with two objects and $\mathcal{J}$ discrete with two objects, and $F(i,j) = \{0\}$ for all i,j. The left side is a coproduct of two products of one-element sets, with 2 elements, whereas the right side is a product of two coproducts of two one-element sets, with 4 elements.

6.12 Additive categories

Definition 6.12.1. A category $\mathcal{A}$ is *preadditive* if every set $\operatorname{Hom}(X,Y)$ has the structure of an abelian group, written additively, such that composition is bilinear: $(f+f')\circ g = f\circ g + f'\circ g$ and $h\circ(f+f') = h\circ f + h\circ f'$. A *zero object* is an object that is both initial and terminal.

A preadditive category is *additive* if it has a zero object and binary products. A functor between preadditive categories is *additive* if the maps $\mathrm{Hom}(X,Y) \to \mathrm{Hom}(F(X),F(Y))$ are group homomorphisms.

Lemma 6.12.2. *Let $\mathcal{A}$ be preadditive and X_1, X_2, P objects with morphisms $i_m \colon X_m \to P$ and $p_m \colon P \to X_m$ such that*

$$p_1 i_1 = \mathrm{id}_{X_1}, \quad p_2 i_2 = \mathrm{id}_{X_2}, \quad p_2 i_1 = 0, \quad p_1 i_2 = 0, \quad i_1 p_1 + i_2 p_2 = \mathrm{id}_P.$$

Then (P, p_1, p_2) is a product and (P, i_1, i_2) is a coproduct of X_1 and X_2. Conversely, if $\mathcal{A}$ has a zero object and (P, p_1, p_2) is a product of X_1 and X_2, there are unique i_1, i_2 satisfying these equations.

Proof. Given $f_m \colon Y \to X_m$, the morphism $f = i_1 f_1 + i_2 f_2$ satisfies $p_m f = f_m$. If g also does, then $g = (i_1 p_1 + i_2 p_2) g = i_1 f_1 + i_2 f_2$. So P is a product; the coproduct statement is dual. Conversely, let $i_1 \colon X_1 \to P$ be the unique morphism with $p_1 i_1 = \mathrm{id}$ and $p_2 i_1 = 0$, and similarly i_2 (the zero morphism $0 \in \mathrm{Hom}(X,Y)$ exists since Hom sets are groups). Then $p_m(i_1 p_1 + i_2 p_2) = p_m$ for $m = 1, 2$, so $i_1 p_1 + i_2 p_2 = \mathrm{id}_P$ by the uniqueness in the universal property of the product. □

Definition 6.12.3. An object P with morphisms as in Lemma 6.12.2 is a *biproduct* or *direct sum* of X_1 and X_2, written $X_1 \oplus X_2$. In a preadditive category with a zero object, the *kernel* of $f \colon X \to Y$ is an equaliser of f and 0, and its *cokernel* is a coequaliser of f and 0.

Proposition 6.12.4. *Additive functors preserve biproducts and zero objects.*

Proof. The equations of Lemma 6.12.2 involve only composition, sums, identities and zero morphisms, all preserved by an additive functor F, so F sends a biproduct to a biproduct. An object Z is a zero object if and only if $\mathrm{id}_Z = 0$ (then every $\mathrm{Hom}(Z,Y)$ and $\mathrm{Hom}(Y,Z)$ is the zero group), and this is preserved by F. □

Example 6.12.5. The categories of abelian groups, of modules over a ring and of vector spaces over a field are additive, with the direct sum as biproduct (Chapter 8).

Counterexample 6.12.6. The category Set admits no preadditive structure: in a preadditive category every $\mathrm{Hom}(X,Y)$ is nonempty, since it contains 0, but $\mathrm{Hom}_{\mathsf{Set}}(\{0\}, \emptyset) = \emptyset$.

6.13 Exercises

Exercise 6.13.1. Show that a composition of faithful functors is faithful, and that a composition of full functors is full.

Solution. Let $F \colon \mathcal{C} \to \mathcal{D}$ and $G \colon \mathcal{D} \to \mathcal{E}$. The map $\mathrm{Hom}_{\mathcal{C}}(X,Y) \to \mathrm{Hom}_{\mathcal{E}}(GF(X), GF(Y))$ induced by $G \circ F$ is the composition of the maps induced by F and by G. A composition of injective maps is injective, and a composition of surjective maps is surjective. □

Exercise 6.13.2. Let $f \colon X \to Y$ be a morphism in a category. Show that if there is $r \colon Y \to X$ with $r \circ f = \mathrm{id}_X$, then f is a monomorphism, and if there is $s \colon Y \to X$ with $f \circ s = \mathrm{id}_Y$, then f is an epimorphism. Show that in Set every monomorphism with non-empty source has such an r. Show that the statement "every epimorphism in Set has such an s" is equivalent to the axiom of choice (for small sets).

Exercise 6.13.3. Let P and Q be preorders, viewed as categories, and let $F, G \colon P \to Q$ be functors. Show that there is a natural transformation $F \Rightarrow G$ if and only if $F(x) \leq G(x)$ for all $x \in P$, and that it is then unique.

Solution. Every set of morphisms of Q has at most one element, so a family $(\eta_x : F(x) \to G(x))_x$ exists if and only if $F(x) \leq G(x)$ for all x, it is unique if it exists, and every square of morphisms of Q commutes. □

Exercise 6.13.4. State and prove the analogue of Theorem 6.6.1 for functors $F : \mathcal{C} \to \mathsf{Set}$ and the functors h^X of Definition 6.4.4. (Hint: apply the theorem to $\mathcal{C}^{\mathrm{op}}$.)

Exercise 6.13.5. Suppose (X, ξ) and (X', ξ') both represent a functor $F : \mathcal{C}^{\mathrm{op}} \to \mathsf{Set}$. Show that there is a unique morphism $a : X \to X'$ with $F(a)(\xi') = \xi$, and that a is an isomorphism.

Exercise 6.13.6. Show that the inclusion $\mathbb{Z} \to \mathbb{Q}$ is an epimorphism in the category of rings, although it is not surjective.

Exercise 6.13.7. Show that in Set a morphism that is both a monomorphism and an epimorphism is an isomorphism, and that the same holds in the category of groups for monomorphisms that are surjective.

Exercise 6.13.8. Let P be a partially ordered set viewed as a category. Show that a product of a family of elements is a greatest lower bound, a coproduct is a least upper bound, and that every diagram in P has a limit if and only if every subset of P has a greatest lower bound.

Solution. A cone over a family (x_i) with vertex y exists if and only if $y \leq x_i$ for all i, and it is unique; a limit is a lower bound y such that every lower bound z satisfies $z \leq y$. Coproducts are dual. For a general diagram $F : \mathcal{I} \to P$ the cone condition only involves the objects, so a limit of F is a greatest lower bound of $\{F(i)\}$. □

Exercise 6.13.9. Let $F \dashv G$ with unit η and counit ε. Show that F is fully faithful if and only if η is a natural isomorphism, and that F is an equivalence if and only if both η and ε are natural isomorphisms. Conversely, show that every equivalence is part of an adjunction whose unit and counit are isomorphisms.

Exercise 6.13.10. Let $\mathcal{I}$, $\mathcal{J}$ be small and $F : \mathcal{I} \times \mathcal{J} \to \mathcal{C}$, where $\mathcal{C}$ has limits. Show that $\lim_i \lim_j F(i,j) \cong \lim_{(i,j)} F(i,j) \cong \lim_j \lim_i F(i,j)$.

Exercise 6.13.11. Show that a functor $G : \mathcal{D} \to \mathsf{Set}$ is representable (in the covariant sense of Exercise 6.13.4) if and only if it has a left adjoint F with $F(\{0\}) \cong X$ for some X, provided $\mathcal{D}$ has coproducts indexed by small sets. (Hint: $F(S) = \coprod_{s \in S} X$.)

Exercise 6.13.12. In an additive category, show that $f + g$ equals the composite $X \to X \oplus X \to Y \oplus Y \to Y$ of the diagonal, $f \oplus g$ and the codiagonal. Deduce that the abelian group structure on Hom sets of an additive category is determined by the category itself.

Part II

Algebra

Chapter 7

Groups

7.1 Introduction

Groups describe symmetry. This chapter develops the basic theory: subgroups, quotients and the isomorphism theorems, group actions and the Sylow theorems, symmetric and alternating groups, products, solvable and nilpotent groups, free groups and presentations, and the structure of finitely generated abelian groups. Groups appear throughout the project: as fundamental groups in topology (Chapter 19), as Galois groups (Chapter 9), as structure groups of vector bundles, and as Mumford–Tate groups of Hodge structures (Chapter 67).

References are [Lan02], [Rot95] and [DF04]; for free groups and amalgamated products see [Ser80] and [LS77].

Prerequisites. Chapter 3 (Set Theory), Chapter 4 (Number Systems) and Chapter 6 (Categories).

7.2 Groups and homomorphisms

Definition 7.2.1. A *group* is a set G together with a map $G \times G \to G$, called the *group law* and written $(a, b) \mapsto a \cdot b$, and an element $e \in G$, such that for all $a, b, c \in G$:

1. (associativity) $(a \cdot b) \cdot c = a \cdot (b \cdot c)$;
2. (neutral element) $e \cdot a = a \cdot e = a$;
3. (inverses) there is an element $a' \in G$ with $a \cdot a' = a' \cdot a = e$.

The group is *abelian* if $a \cdot b = b \cdot a$ for all $a, b \in G$. The *order* of G is its cardinality $|G|$.

Notation 7.2.2. Two notations are used for the group law.

- *Multiplicative notation*, used for all groups: the group law is written $a \cdot b$ or simply ab, the neutral element e (or 1), the inverse of a is written a^{-1} (it is unique by Lemma 7.2.3), and a^n denotes the n-th power (Lemma 7.2.9).
- *Additive notation*, used only for abelian groups: the group law is written $a + b$, the neutral element 0, the inverse $-a$, and na the n-th multiple.

Every statement in multiplicative notation applies to abelian groups written additively through the dictionary $ab \leftrightarrow a + b$, $e \leftrightarrow 0$, $a^{-1} \leftrightarrow -a$, $a^n \leftrightarrow na$. General results in this chapter are stated multiplicatively. For groups arising from a familiar structure we keep the familiar symbol: $+$ for $(\mathbb{Z}, +)$, multiplication for $\mathbb{Q}^\times$ or for invertible matrices, and composition

$\circ$ (written as juxtaposition) for groups of bijections. The symbol $*$ never denotes a group law; $G_1 * G_2$ is the free product of Definition 7.10.6. When the group law has to be named, we write $(G, \cdot)$ or $(G, +)$.

Lemma 7.2.3. *In a group the neutral element and inverses are unique,* $(ab)^{-1} = b^{-1}a^{-1}$, $(a^{-1})^{-1} = a$, *and* $ab = ac$ *or* $ba = ca$ *implies* $b = c$.

Proof. If e' is also neutral, $e = ee' = e'$. If b and c are inverses of a, then $b = b(ac) = (ba)c = c$; we write a^{-1} for the inverse. The element $b^{-1}a^{-1}$ is inverse to ab, and a is inverse to a^{-1}. Cancellation follows by multiplying with a^{-1}. □

Example 7.2.4.

1. $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$ and $(\mathbb{C}, +)$ are abelian groups, as are $\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}$, $\mathbb{R}^\times$ and $\mathbb{C}^\times$ under multiplication. More generally, for a commutative ring R the invertible elements form a group $R^\times$.
2. For a set X, the bijections $X \to X$ form the *symmetric group* $\mathrm{Sym}(X)$ under composition. We write $S_n = \mathrm{Sym}(\{1, \dots, n\})$.
3. For a field k, the invertible $n \times n$ matrices form the group $\mathrm{GL}_n(k)$ (Chapter 11).
4. The *trivial group* has one element.
5. A group with one object as a category (Example 6.2.8) is a category with one object in which every morphism is an isomorphism.

Definition 7.2.5. A *homomorphism* of groups is a map $f : G \to H$ with $f(ab) = f(a)f(b)$ for all $a, b \in G$. Its *kernel* is $\ker f = \{a \in G : f(a) = e\}$ and its *image* is $\mathrm{im}\, f = f[G]$. An *isomorphism* is a bijective homomorphism; an *automorphism* of G is an isomorphism $G \to G$, and they form a group $\mathrm{Aut}(G)$. Small groups and homomorphisms form a locally small category Grp, with the full subcategory Ab of abelian groups.

Lemma 7.2.6. *A homomorphism* f *satisfies* $f(e) = e$ *and* $f(a^{-1}) = f(a)^{-1}$*; it is injective if and only if* $\ker f = \{e\}$*; and the inverse of a bijective homomorphism is a homomorphism.*

Proof. $f(e) = f(e)f(e)$ gives $f(e) = e$ by cancellation, and then $f(a)f(a^{-1}) = f(e) = e$. If $\ker f$ is trivial and $f(a) = f(b)$, then $f(ab^{-1}) = e$, so $a = b$. If f is bijective with inverse g, then $f(g(x)g(y)) = xy$, so $g(xy) = g(x)g(y)$. □

Definition 7.2.7. A *subgroup* of G is a subset H containing e with $ab \in H$ and $a^{-1} \in H$ for all $a, b \in H$; it is a group with the restricted operation. We write $H \leq G$. For a subset $S \subseteq G$, the *subgroup generated by* S, written $\langle S \rangle$, is the intersection of all subgroups containing S.

Lemma 7.2.8. *The subgroup* $\langle S \rangle$ *consists of all finite products* $s_1^{\epsilon_1} \cdots s_k^{\epsilon_k}$ *with* $k \geq 0$, $s_i \in S$ *and* $\epsilon_i \in \{1, -1\}$ *(the empty product being* e*). Kernels and images of homomorphisms are subgroups.*

Proof. The set of such products contains S and is closed under products and inverses (Lemma 7.2.3), so it contains $\langle S \rangle$; and every subgroup containing S contains all such products. The second statement follows from Lemma 7.2.6. □

Lemma 7.2.9. *Let* G *be a group and* $g \in G$*. There is a unique homomorphism* $\mathbb{Z} \to G$ *sending* 1 *to* g*; we write* $n \mapsto g^n$*. Thus* $g^{m+n} = g^m g^n$ *and* $(g^m)^n = g^{mn}$ *for all* $m, n \in \mathbb{Z}$.

Proof. Uniqueness is clear since 1 generates $\mathbb{Z}$. For existence define g^n for $n \in \mathbb{N}$ by recursion ($g^0 = e$, $g^{n+1} = g^n g$) and put $g^{-n} = (g^{-1})^n$. One checks $g^{m+1} = g^m g$ for all $m \in \mathbb{Z}$ (for $m = -n - 1 < 0$ this reads $(g^{-1})^n = (g^{-1})^{n+1} g$, which holds since $g^{-1}g = e$), and then $g^{m+n} = g^m g^n$ follows by induction on $n \geq 0$ and on $-n \geq 0$. The last formula holds because $n \mapsto g^{mn}$ and $n \mapsto (g^m)^n$ are both homomorphisms sending 1 to g^m. □

Proposition 7.2.10. *Every subgroup of $\mathbb{Z}$ is of the form $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$ for a unique $n \in \mathbb{N}$.*

Proof. Let $H \leq \mathbb{Z}$. If $H = \{0\}$ take $n = 0$. Otherwise H contains a positive element, and we let n be the least one. Then $n\mathbb{Z} \subseteq H$, and for $h \in H$, division with remainder (Theorem 4.5.2) gives $h = qn + r$ with $0 \leq r < n$ and $r = h - qn \in H$, so $r = 0$. The number n is the least positive element of $n\mathbb{Z}$, hence unique. □

Definition 7.2.11. The *order* of $g \in G$ is the least $n \geq 1$ with $g^n = e$ if it exists, and ∞ otherwise. By Proposition 7.2.10, the kernel of $n \mapsto g^n$ is $\mathrm{ord}(g)\mathbb{Z}$ (with $\infty\mathbb{Z} := \{0\}$). A group is *cyclic* if it is generated by one element.

7.3 Cosets and Lagrange's theorem

Definition 7.3.1. Let $H \leq G$. A *left coset* of H is a set $gH = \{gh : h \in H\}$, and a right coset is a set Hg. The set of left cosets is G/H, and the *index* of H is $[G : H] = |G/H|$.

Lemma 7.3.2. *The left cosets of H are the classes of the equivalence relation $a \sim b \iff a^{-1}b \in H$; hence they partition G. Each left coset is in bijection with H, and $gH \mapsto Hg^{-1}$ is a bijection from left to right cosets.*

Proof. The relation is an equivalence relation because H is a subgroup, and the class of a is aH. The map $h \mapsto gh$ is a bijection $H \to gH$. Finally $aH = bH$ iff $a^{-1}b \in H$ iff $(a^{-1}b)^{-1} = b^{-1}a \in H$ iff $Ha^{-1} = Hb^{-1}$. □

Theorem 7.3.3 (Lagrange)**.** *Let G be finite and $H \leq G$. Then $|G| = [G : H] \cdot |H|$. In particular the order of H divides the order of G, and the order of every element divides $|G|$, so $g^{|G|} = e$ for all $g \in G$.*

Proof. By Lemma 7.3.2, G is the disjoint union of $[G : H]$ sets of size $|H|$. The subgroup $\langle g \rangle$ has order $\mathrm{ord}(g)$, since $g^0, \ldots, g^{\mathrm{ord}(g)-1}$ are distinct and every power g^n equals g^r with r the remainder of n. □

Corollary 7.3.4. *A group of prime order p is cyclic, generated by any element $\neq e$.*

Proof. If $g \neq e$, then $\mathrm{ord}(g) > 1$ divides p, so it equals p, and $\langle g \rangle = G$. □

Corollary 7.3.5 (Euler, Fermat, Wilson)**.** *Let $n \geq 2$, and write $a \equiv b \pmod{n}$ if $n \mid a - b$. Let U_n be the set of residue classes modulo n of the integers a with $\gcd(a, n) = 1$, and $\varphi(n) = |U_n|$ the number of $0 \leq a < n$ with $\gcd(a, n) = 1$.*

1. *Multiplication of representatives makes U_n an abelian group, with identity the class of 1.*
2. *(Euler) $a^{\varphi(n)} \equiv 1 \pmod{n}$ whenever $\gcd(a, n) = 1$.*
3. *(Fermat) If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$; and $a^p \equiv a \pmod{p}$ for every integer a.*
4. *(Wilson) If p is prime, then $(p-1)! \equiv -1 \pmod{p}$.*

Proof. (1) If $a \equiv a'$ and $b \equiv b'$, then $ab - a'b' = a(b - b') + b'(a - a')$ is divisible by n, so the product of classes is well defined; it is associative and commutative because multiplication of integers is. If $\gcd(a, n) = \gcd(b, n) = 1$ then $\gcd(ab, n) = 1$: a prime dividing n and ab divides a or b (Lemma 4.5.4). By Bézout (Theorem 4.5.3), $\gcd(a, n) = 1$ gives $ax + ny = 1$, so the class of x, which is again coprime to n, is an inverse of the class of a.

(2) By Theorem 7.3.3 applied to the group U_n of order $\varphi(n)$.

(3) For $n = p$ prime, every a with $0 < a < p$ is coprime to p, so $\varphi(p) = p - 1$ and (2) applies. Multiplying by a gives $a^p \equiv a$, which also holds trivially when $p \mid a$.

(4) An element x of U_p is its own inverse exactly when $p \mid x^2 - 1 = (x-1)(x+1)$, that is, by Lemma 4.5.4, when $x \equiv \pm 1$. The remaining elements of U_p fall into pairs $\{x, x^{-1}\}$ with product 1. So the product of all elements of U_p, which is the class of $(p-1)!$, is the class of $1 \cdot (-1) = -1$ (for $p = 2$, $1 \equiv -1$ and the statement is clear). □

Example 7.3.6. $\varphi(12) = 4$, with $U_{12} = \{1, 5, 7, 11\}$, and every element squares to 1; so U_{12} is not cyclic, although U_{12} has order 4 and $U_5 = \{1, 2, 3, 4\}$, also of order 4, is cyclic, generated by 2 ($2^2 = 4$, $2^3 = 8 \equiv 3$, $2^4 = 16 \equiv 1$). Fermat's little theorem gives, for instance, $2^{100} = (2^{10})^{10} \equiv 1 \pmod{11}$. Its converse fails: $561 = 3 \cdot 11 \cdot 17$ is not prime, yet $a^{560} \equiv 1 \pmod{561}$ for all a coprime to 561, because 2, 10 and 16 divide 560, so Fermat's theorem gives the congruence modulo 3, 11 and 17, hence modulo their product, the three primes being distinct (Theorem 4.5.5). Such numbers are called *Carmichael numbers*.

Proposition 7.3.7. *If $K \leq H \leq G$ and the indices are finite, then $[G : K] = [G : H][H : K]$.*

Proof. Let (g_i) represent the left cosets of H in G and (h_j) those of K in H. Then the $g_i h_j K$ are the left cosets of K in G, and they are distinct: $G = \bigsqcup_i g_i H = \bigsqcup_i \bigsqcup_j g_i h_j K$. □

7.4 Normal subgroups and quotients

Definition 7.4.1. A subgroup $N \leq G$ is *normal*, written $N \trianglelefteq G$, if $gNg^{-1} = N$ for all $g \in G$, equivalently $gN = Ng$ for all g. A group $G \neq \{e\}$ is *simple* if its only normal subgroups are $\{e\}$ and G.

Lemma 7.4.2. *Kernels of homomorphisms are normal. Every subgroup of an abelian group is normal. A subgroup of index 2 is normal.*

Proof. If $f(n) = e$ then $f(gng^{-1}) = f(g)f(g)^{-1} = e$. The second claim is clear. If $[G : H] = 2$, then for $g \notin H$ both gH and Hg are the complement of H. □

Theorem 7.4.3. *Let $N \trianglelefteq G$. The rule $(aN)(bN) = abN$ is a well-defined operation making G/N a group, the* quotient group, *and the projection $\pi \colon G \to G/N$ is a surjective homomorphism with kernel N. If $f \colon G \to H$ is a homomorphism with $N \subseteq \ker f$, there is a unique homomorphism $\bar{f} \colon G/N \to H$ with $\bar{f} \circ \pi = f$.*

Proof. If $aN = a'N$ and $bN = b'N$, write $a' = an, b' = bm$. Then $a'b' = anbm = ab(b^{-1}nb)m \in abN$ since N is normal. The group axioms follow from those of G, with neutral element N and $(aN)^{-1} = a^{-1}N$. Clearly π is a surjective homomorphism, and $\pi(a) = N$ iff $a \in N$. If $N \subseteq \ker f$ then f is constant on cosets, and $\bar{f}$ exists and is unique by Proposition 3.3.9; it is a homomorphism because π is surjective. □

Theorem 7.4.4 (Isomorphism theorems)**.**

1. *If $f \colon G \to H$ is a homomorphism, $\bar{f} \colon G/\ker f \to \operatorname{im} f$ is an isomorphism.*
2. *If $H \leq G$ and $N \trianglelefteq G$, then HN is a subgroup, $N \trianglelefteq HN$, $H \cap N \trianglelefteq H$, and $H/(H \cap N) \cong HN/N$.*
3. *If $N \leq M$ are normal subgroups of G, then $M/N \trianglelefteq G/N$ and $(G/N)/(M/N) \cong G/M$.*
4. *(Correspondence theorem) If $N \trianglelefteq G$, then $K \mapsto K/N$ is a bijection from the set of subgroups of G containing N to the set of subgroups of G/N, preserving inclusions, normality and indices.*

Proof. (1) $\bar{f}$ is surjective onto im f and has trivial kernel. (2) $HN = NH$ because $hN = Nh$, so HN is closed under products and inverses. The composite $H \to HN \to HN/N$ is surjective, since $hnN = hN$, with kernel $H \cap N$; apply (1). (3) The map $G/N \to G/M$, $gN \mapsto gM$, is a well-defined surjective homomorphism (Theorem 7.4.3) with kernel M/N; apply (1). (4) The inverse map sends $\bar{K} \leq G/N$ to $\pi^{-1}[\bar{K}]$; since π is surjective, $\pi[\pi^{-1}[\bar{K}]] = \bar{K}$, and $\pi^{-1}[\pi[K]] = KN = K$ when $N \subseteq K$. Normality is preserved because π is surjective, and $gK \mapsto (gN)(K/N)$ is a bijection $G/K \to (G/N)/(K/N)$. □

Example 7.4.5. For $n \geq 1$, the quotient $\mathbb{Z}/n\mathbb{Z}$ is a cyclic group of order n, with elements the classes of $0, 1, \ldots, n-1$. If $G = \langle g \rangle$ is cyclic, the homomorphism $\mathbb{Z} \to G$, $k \mapsto g^k$ is surjective with kernel $\mathrm{ord}(g)\mathbb{Z}$, so $G \cong \mathbb{Z}$ if g has infinite order and $G \cong \mathbb{Z}/n\mathbb{Z}$ if $\mathrm{ord}(g) = n$. Thus a cyclic group is determined up to isomorphism by its order. By the correspondence theorem the subgroups of $\mathbb{Z}/n\mathbb{Z}$ are the $d\mathbb{Z}/n\mathbb{Z}$ for $d \mid n$, one for each divisor.

Definition 7.4.6. The *center* of G is $Z(G) = \{z \in G : zg = gz \text{ for all } g\}$. For $g \in G$, *conjugation* $c_g(x) = gxg^{-1}$ is an automorphism; automorphisms of this form are *inner*. The *centraliser* of x is $C_G(x) = \{g : gx = xg\}$ and the *normaliser* of $H \leq G$ is $N_G(H) = \{g : gHg^{-1} = H\}$.

Lemma 7.4.7. *$Z(G)$ is a normal subgroup, and $g \mapsto c_g$ is a homomorphism $G \to \mathrm{Aut}(G)$ with kernel $Z(G)$. If $G/Z(G)$ is cyclic, then G is abelian.*

Proof. $c_{gh} = c_g \circ c_h$, and $c_g = \mathrm{id}$ iff $g \in Z(G)$. If $G/Z(G)$ is generated by $gZ(G)$, every element of G is $g^k z$ with $z \in Z(G)$, and such elements commute with each other. □

7.5 Products

Proposition 7.5.1. *For a family of groups $(G_i)_{i \in I}$, the set $\prod_i G_i$ with the componentwise operation is a group, and with the projections it is a product in* Grp *(Example 6.8.3). Let $N, M \trianglelefteq G$ with $N \cap M = \{e\}$ and $NM = G$. Then $nm = mn$ for $n \in N$, $m \in M$, and $N \times M \to G$, $(n, m) \mapsto nm$, is an isomorphism.*

Proof. The first statement is clear. For $n \in N$ and $m \in M$, the commutator $nmn^{-1}m^{-1}$ lies in N (as $mn^{-1}m^{-1} \in N$) and in M (as $nmn^{-1} \in M$), hence is e. So the map is a homomorphism; it is surjective since $NM = G$, and injective since $nm = e$ implies $n = m^{-1} \in N \cap M$. □

Definition 7.5.2. Let N and H be groups and $\varphi : H \to \mathrm{Aut}(N)$, $h \mapsto \varphi_h$, a homomorphism. The *semidirect product* $N \rtimes_\varphi H$ is the set $N \times H$ with the operation $(n, h)(n', h') = (n\varphi_h(n'), hh')$.

Proposition 7.5.3. *$N \rtimes_\varphi H$ is a group, in which $N \times \{e\}$ is a normal subgroup isomorphic to N, $\{e\} \times H$ is a subgroup isomorphic to H, and $(e, h)(n, e)(e, h)^{-1} = (\varphi_h(n), e)$. Conversely, if G is a group with $N \trianglelefteq G$ and $H \leq G$ such that $N \cap H = \{e\}$ and $NH = G$, then $(n, h) \mapsto nh$ is an isomorphism $N \rtimes_\varphi H \to G$, where $\varphi_h(n) = hnh^{-1}$.*

Proof. Associativity: $((n, h)(n', h'))(n'', h'') = (n\varphi_h(n')\varphi_{hh'}(n''), hh'h'')$ and $(n, h)((n', h')(n'', h'')) = (n\varphi_h(n'\varphi_{h'}(n'')), hh'h'')$ agree since φ is a homomorphism into $\mathrm{Aut}(N)$. The neutral element is (e, e) and $(n, h)^{-1} = (\varphi_{h^{-1}}(n^{-1}), h^{-1})$. The projection $(n, h) \mapsto h$ is a homomorphism with kernel $N \times \{e\}$, and the conjugation formula is a direct computation. For the converse, every $g \in G$ is uniquely nh (if $nh = n'h'$ then $n'^{-1}n = h'h^{-1} \in N \cap H$), and $nh \cdot n'h' = n(hn'h^{-1})hh'$, which is the product rule. □

Example 7.5.4. For $n \geq 3$ let $\varphi : \mathbb{Z}/2\mathbb{Z} \to \mathrm{Aut}(\mathbb{Z}/n\mathbb{Z})$ send the generator to $x \mapsto -x$. The group $D_n = \mathbb{Z}/n\mathbb{Z} \rtimes_\varphi \mathbb{Z}/2\mathbb{Z}$ of order $2n$ is the *dihedral group*; it is the group of symmetries of a regular n-gon (rotations and reflections). It is not abelian, since $(0, 1)(1, 0) = (-1, 1) \neq (1, 1) = (1, 0)(0, 1)$.

7.6 Group actions

Definition 7.6.1. An *action* of a group G on a set X is a map $G \times X \to X$, $(g, x) \mapsto gx$, with $ex = x$ and $g(hx) = (gh)x$; equivalently a homomorphism $G \to \mathrm{Sym}(X)$. The *orbit* of x is $Gx = \{gx : g \in G\}$, its *stabiliser* is $G_x = \{g : gx = x\}$, and $X^G = \{x : gx = x \text{ for all } g\}$ is the set of *fixed points*. The action is *transitive* if there is only one orbit, and *free* if all stabilisers are trivial.

Theorem 7.6.2 (Orbit–stabiliser)**.** *For an action of G on X and $x \in X$, the stabiliser G_x is a subgroup and $gG_x \mapsto gx$ is a bijection $G/G_x \to Gx$. In particular $|Gx| = [G : G_x]$, and the orbits partition X.*

Proof. $gG_x = hG_x$ iff $h^{-1}g \in G_x$ iff $gx = hx$, so the map is well defined and injective; it is surjective by definition. The relation "$y \in Gx$" is an equivalence relation since G is a group. □

Example 7.6.3.

1. G acts on itself by left multiplication, freely and transitively. This gives an injective homomorphism $G \to \mathrm{Sym}(G)$ (*Cayley's theorem*): every group is isomorphic to a group of permutations.
2. G acts on itself by conjugation; the orbits are the *conjugacy classes* and $G_x = C_G(x)$.
3. G acts on the set of its subgroups by conjugation, with stabiliser $N_G(H)$; so the number of conjugates of H is $[G : N_G(H)]$.
4. If $H \leq G$, then G acts transitively on G/H by $g(aH) = gaH$.

Corollary 7.6.4 (Class equation)**.** *Let G be finite and let $x_1, \dots, x_r$ represent the conjugacy classes with more than one element. Then $|G| = |Z(G)| + \sum_{i=1}^{r} [G : C_G(x_i)]$.*

Proof. The conjugacy classes partition G; the one-element classes are the elements of $Z(G)$, and the class of x_i has $[G : C_G(x_i)]$ elements by Theorem 7.6.2. □

Definition 7.6.5. Let p be a prime. A *p-group* is a finite group whose order is a power of p.

Lemma 7.6.6. *Let a p-group P act on a finite set X. Then $|X| \equiv |X^P| \pmod{p}$.*

Proof. Each orbit that is not a fixed point has size $[P : P_x] > 1$, a divisor of $|P|$, hence divisible by p. □

Proposition 7.6.7. *A nontrivial p-group has nontrivial center. Every group of order p^2 is abelian.*

Proof. Apply Lemma 7.6.6 to the conjugation action: $|Z(G)| \equiv |G| \equiv 0 \pmod{p}$ and $e \in Z(G)$, so $|Z(G)| \geq p$. If $|G| = p^2$, then $|G/Z(G)| \in \{1, p\}$, so $G/Z(G)$ is cyclic (Corollary 7.3.4), and G is abelian by Lemma 7.4.7. □

Theorem 7.6.8 (Cauchy)**.** *If a prime p divides the order of a finite group G, then G has an element of order p.*

Proof. Let $X = \{(g_1, \dots, g_p) \in G^p : g_1 g_2 \cdots g_p = e\}$. Since g_p is determined by the others, $|X| = |G|^{p-1} \equiv 0 \pmod{p}$. The group $\mathbb{Z}/p\mathbb{Z}$ acts on X by cyclic rotation: if $g_1 \cdots g_p = e$ then $g_2 \cdots g_p g_1 = g_1^{-1}(g_1 \cdots g_p)g_1 = e$. The fixed points are the tuples $(g, \dots, g)$ with $g^p = e$. By Lemma 7.6.6 their number is divisible by p, and $(e, \dots, e)$ is one of them, so there is a fixed point with $g \neq e$. □

7.7 The Sylow theorems

Definition 7.7.1. Let G be finite and $|G| = p^a m$ with p prime and $p \nmid m$. A *Sylow p-subgroup* of G is a subgroup of order p^a. We write $\mathrm{Syl}_p(G)$ for the set of them and $n_p = |\mathrm{Syl}_p(G)|$.

Theorem 7.7.2 (Sylow)**.** *Let G be a finite group, p a prime and $|G| = p^a m$ with $p \nmid m$.*

1. *Sylow p-subgroups exist.*
2. *Every p-subgroup of G is contained in a Sylow p-subgroup, and any two Sylow p-subgroups are conjugate.*
3. *n_p divides m and $n_p \equiv 1 \pmod{p}$.*

Proof. (1) By induction on $|G|$; if $a = 0$ there is nothing to prove. Suppose $p \mid |Z(G)|$. By Theorem 7.6.8 there is $z \in Z(G)$ of order p; $N = \langle z \rangle$ is normal, and G/N has order $p^{a-1}m$, so by induction it has a subgroup of order p^{a-1}, whose preimage in G has order p^a. Suppose $p \nmid |Z(G)|$. In the class equation (Corollary 7.6.4) the left side is divisible by p, so some $[G : C_G(x_i)]$ is not divisible by p. Then $C_G(x_i)$ is a proper subgroup of order divisible by p^a, and by induction it contains a subgroup of order p^a.

(2) Let P be a Sylow p-subgroup and Q a p-subgroup. Let Q act on G/P by left multiplication. Since $|G/P| = m \not\equiv 0 \pmod{p}$, Lemma 7.6.6 gives a fixed point gP. Then $QgP = gP$, so $g^{-1}Qg \subseteq P$ and $Q \subseteq gPg^{-1}$, which is a Sylow p-subgroup. If Q is itself Sylow, then $Q = gPg^{-1}$ by comparing orders.

(3) By (2), G acts transitively by conjugation on $\mathrm{Syl}_p(G)$, so $n_p = [G : N_G(P)]$, which divides $[G : P] = m$ since $P \leq N_G(P)$ (Proposition 7.3.7). Let P act on $\mathrm{Syl}_p(G)$ by conjugation. If Q is a fixed point, then $P \leq N_G(Q)$, so P and Q are Sylow p-subgroups of $N_G(Q)$, hence conjugate in $N_G(Q)$ by (2); as Q is normal in $N_G(Q)$, $P = Q$. So P is the only fixed point, and $n_p \equiv 1 \pmod{p}$ by Lemma 7.6.6. □

Corollary 7.7.3. *A Sylow p-subgroup is normal if and only if $n_p = 1$.*

Proof. By Theorem 7.7.2(2) the Sylow p-subgroups are the conjugates of one of them. □

Example 7.7.4. Every group G of order 15 is cyclic. Indeed n_3 divides 5 and is 1 modulo 3, so $n_3 = 1$; similarly $n_5 = 1$. The Sylow subgroups P_3, P_5 are normal, cyclic of prime order, and $P_3 \cap P_5 = \{e\}$ by Lagrange. Then $|P_3P_5| = 15$ (the map $P_3 \times P_5 \to G$ is injective), so $G \cong \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$ by Proposition 7.5.1, and the element $(1, 1)$ has order 15.

7.8 Symmetric and alternating groups

Definition 7.8.1. Let $n \geq 1$. For distinct $a_1, \ldots, a_k \in \{1, \ldots, n\}$, the *cycle* $(a_1\, a_2 \cdots a_k) \in S_n$ sends a_i to a_{i+1} for $i < k$, a_k to a_1, and fixes all other points. A cycle of length 2 is a *transposition*. Two cycles are *disjoint* if they move disjoint sets of points. Products in S_n are compositions, applied from right to left.

Proposition 7.8.2. *Every $\sigma \in S_n$ is a product of disjoint cycles of length ≥ 2, uniquely up to order; disjoint cycles commute. Every element of S_n is a product of transpositions. For $\tau \in S_n$, $\tau(a_1 \cdots a_k)\tau^{-1} = (\tau(a_1) \cdots \tau(a_k))$, so two permutations are conjugate if and only if they have the same cycle lengths.*

Proof. The orbits of $\langle\sigma\rangle$ on $\{1,\dots,n\}$ partition it; on an orbit $\{a,\sigma(a),\dots,\sigma^{k-1}(a)\}$ of size k, σ acts as the cycle $(a\,\sigma(a)\cdots\sigma^{k-1}(a))$, and σ is the product of these cycles for the orbits of size ≥ 2. Conversely the cycles in any such decomposition must be the orbits, which gives uniqueness. Disjoint cycles commute since they act on disjoint sets. We have $(a_1\cdots a_k)=(a_1\,a_k)(a_1\,a_{k-1})\cdots(a_1\,a_2)$, as one checks by evaluating both sides. The conjugation formula follows by evaluating both sides at $\tau(a_i)$ and at points not of this form. If σ and σ' have the same cycle lengths, a bijection matching their cycles gives τ with $\tau\sigma\tau^{-1}=\sigma'$. □

Proposition 7.8.3. *There is a unique homomorphism* $\operatorname{sgn}\colon S_n\to\{1,-1\}$ *sending every transposition to* -1 *(for* $n\geq 2$*). It is given by* $\operatorname{sgn}(\sigma)=\prod_{1\leq i<j\leq n}\frac{\sigma(j)-\sigma(i)}{j-i}$.

Proof. For $\sigma\in S_n$ let $s(\sigma)$ be the product in the statement, a rational number. The factors $\sigma(j)-\sigma(i)$ run, up to sign, over all differences $j-i$ with $i<j$, so $s(\sigma)=\pm 1$. For $\sigma,\tau\in S_n$,

$$s(\sigma\tau)=\prod_{i<j}\frac{\sigma(\tau(j))-\sigma(\tau(i))}{\tau(j)-\tau(i)}\cdot\prod_{i<j}\frac{\tau(j)-\tau(i)}{j-i}.$$

The expression $\frac{\sigma(b)-\sigma(a)}{b-a}$ is symmetric in a and b, and $\{\tau(i),\tau(j)\}$ runs over all two-element subsets as $\{i,j\}$ does; so the first product is $s(\sigma)$, and s is a homomorphism. For the transposition $(1\,2)$, the only factor that is negative is the one with $(i,j)=(1,2)$, so $s((1\,2))=-1$. All transpositions are conjugate by Proposition 7.8.2, hence have sign -1. Uniqueness holds because transpositions generate S_n. □

Definition 7.8.4. The *alternating group* A_n is the kernel of sgn. Permutations in A_n are *even*. For $n\geq 2$, $[S_n:A_n]=2$.

Lemma 7.8.5. *For* $n\geq 3$*, the group* A_n *is generated by the* 3*-cycles. For* $n\geq 5$*, any two* 3*-cycles are conjugate in* A_n.

Proof. A 3-cycle is even, being a product of two transpositions. Every even permutation is a product of an even number of transpositions, and for distinct a,b,c,d we have $(a\,b)(a\,c)=(a\,c\,b)$ and $(a\,b)(c\,d)=(a\,b\,c)(b\,c\,d)$, as one checks by evaluation. For the second claim, two 3-cycles $(a\,b\,c)$ and $(a'\,b'\,c')$ are conjugate by some $\tau\in S_n$. If τ is odd, choose $d,e\notin\{a,b,c\}$ distinct (possible since $n\geq 5$); then $\tau(d\,e)$ is even and also conjugates $(a\,b\,c)$ to $(a'\,b'\,c')$, because $(d\,e)$ commutes with $(a\,b\,c)$. □

Theorem 7.8.6. *For* $n\geq 5$ *the group* A_n *is simple.*

Proof. Let $N\trianglelefteq A_n$ with $N\neq\{e\}$. By Lemma 7.8.5 it suffices to show that N contains a 3-cycle. Choose $\sigma\in N\setminus\{e\}$ with the largest possible number of fixed points. For $\rho\in A_n$, the element $\tau=(\rho\sigma\rho^{-1})\sigma^{-1}$ lies in N.

Suppose σ moves exactly 4 points. Being even, it is then of the form $(a\,b)(c\,d)$. Choose $e\notin\{a,b,c,d\}$ and $\rho=(c\,d\,e)$. Then $\rho\sigma\rho^{-1}=(a\,b)(d\,e)$ and $\tau=(a\,b)(d\,e)(a\,b)(c\,d)=(d\,e)(c\,d)=(c\,e\,d)$, a 3-cycle in N.

Suppose σ moves at least 5 points. If σ has a cycle of length ≥ 3, write it as $(a\,b\,c\cdots)$ and choose further points d,e moved by σ, distinct from a,b,c. Otherwise σ is a product of at least three disjoint transpositions, and we write $\sigma=(a\,b)(c\,d)(e\,f)\cdots$. In both cases let $\rho=(c\,d\,e)$. Every point fixed by σ is fixed by ρ, hence by τ; and $\tau(\sigma(x))=\rho\sigma\rho^{-1}(x)$. In the first case $\tau(b)=\tau(\sigma(a))=\rho(\sigma(a))=\rho(b)=b$, and $\tau(c)=\tau(\sigma(b))=\rho(\sigma(b))=\rho(c)=d\neq c$. In the second case $\tau(b)=\rho\sigma(a)=b$, $\tau(a)=\rho\sigma(b)=a$, and $\tau(d)=\tau(\sigma(c))=\rho\sigma\rho^{-1}(c)=\rho(\sigma(e))=\rho(f)=f\neq d$. In both cases $\tau\neq e$, τ fixes all points fixed by σ and at least one point moved by σ, contradicting the choice of σ. So this case does not occur. The remaining possibility is that σ moves exactly 3 points (an even permutation cannot move exactly 2), and then σ is a 3-cycle. In all cases N contains a 3-cycle. □

Counterexample 7.8.7. The converse of Lagrange's theorem is false: A_4 has order 12 but no subgroup of order 6. Such a subgroup H would have index 2, so $g^2 \in H$ for all $g \in A_4$ (as G/H has order 2). But every 3-cycle is a square: $(a\,b\,c) = (a\,c\,b)^2$. So H would contain the eight 3-cycles of A_4.

Counterexample 7.8.8. Normality is not transitive. In A_4, the subgroup $V = \{e, (1\,2)(3\,4), (1\,3)(2\,4), (1\,4)(2\,3)\}$ is normal, being a union of conjugacy classes by Proposition 7.8.2. Its subgroup $K = \{e, (1\,2)(3\,4)\}$ is normal in the abelian group V. But $(1\,2\,3)(1\,2)(3\,4)(1\,2\,3)^{-1} = (2\,3)(1\,4) \notin K$, so K is not normal in A_4.

7.9 Solvable and nilpotent groups

Definition 7.9.1. The *commutator* of $a, b \in G$ is $[a, b] = aba^{-1}b^{-1}$. For subgroups H, K, $[H, K]$ is the subgroup generated by the $[h, k]$. The *derived subgroup* is $G' = [G, G]$, and the *derived series* is $G^{(0)} = G$, $G^{(i+1)} = [G^{(i)}, G^{(i)}]$. The group G is *solvable* if $G^{(n)} = \{e\}$ for some n.

Proposition 7.9.2. *$[G, G]$ is a normal subgroup, $G^{\mathrm{ab}} = G/[G, G]$ is abelian, and a normal subgroup N has abelian quotient G/N if and only if $[G, G] \subseteq N$. Every homomorphism from G to an abelian group factors uniquely through G^{ab}; thus $G \mapsto G^{\mathrm{ab}}$ is left adjoint to the inclusion* $\mathsf{Ab} \to \mathsf{Grp}$.

Proof. $g[a, b]g^{-1} = [gag^{-1}, gbg^{-1}]$, so $[G, G]$ is normal. The quotient G/N is abelian iff $abN = baN$ for all a, b, iff all commutators lie in N. A homomorphism f to an abelian group kills all commutators, so it factors uniquely by Theorem 7.4.3. The last statement is the resulting natural bijection $\mathrm{Hom}_{\mathsf{Grp}}(G, A) \cong \mathrm{Hom}_{\mathsf{Ab}}(G^{\mathrm{ab}}, A)$. □

Proposition 7.9.3.
1. *G is solvable if and only if there is a chain $\{e\} = G_n \trianglelefteq G_{n-1} \trianglelefteq \cdots \trianglelefteq G_0 = G$ with all G_i/G_{i+1} abelian.*
2. *Subgroups and quotients of solvable groups are solvable. If $N \trianglelefteq G$ and both N and G/N are solvable, then G is solvable.*
3. *p-groups are solvable, and S_n is not solvable for $n \geq 5$.*

Proof. (1) The derived series is such a chain. Conversely, given a chain, induction shows $G^{(i)} \subseteq G_i$, because G_i/G_{i+1} abelian gives $[G_i, G_i] \subseteq G_{i+1}$ (Proposition 7.9.2). (2) For $H \leq G$, $H^{(i)} \subseteq G^{(i)}$. For a surjection $f : G \to Q$, $f[G^{(i)}] = Q^{(i)}$. If $(G/N)^{(n)}$ is trivial then $G^{(n)} \subseteq N$, and then $G^{(n+m)} \subseteq N^{(m)}$ is trivial for large m. (3) By induction on $|G|$, using (2) and Proposition 7.6.7: $Z(G)$ is abelian and nontrivial, and $G/Z(G)$ is a smaller p-group. For $n \geq 5$, A_n is simple and not abelian, so $[A_n, A_n] = A_n$ and A_n, hence S_n, is not solvable. □

Example 7.9.4. S_4 is solvable: $\{e\} \trianglelefteq V \trianglelefteq A_4 \trianglelefteq S_4$, with V as in Counterexample 7.8.8, has quotients of orders 4, 3 and 2, which are abelian by Proposition 7.6.7. The failure of solvability of S_n for $n \geq 5$ is the group-theoretic reason why polynomial equations of degree ≥ 5 cannot in general be solved by radicals (Chapter 9).

Definition 7.9.5. The *upper central series* of G is defined by $Z_0 = \{e\}$ and $Z_{i+1}/Z_i = Z(G/Z_i)$ (using the correspondence theorem). The group is *nilpotent* if $Z_n = G$ for some n.

Proposition 7.9.6. *p-groups are nilpotent, and nilpotent groups are solvable. The group S_3 is solvable but not nilpotent.*

Proof. If G is a p-group and $Z_i \neq G$, then G/Z_i is a nontrivial p-group, so $Z_{i+1} \supsetneq Z_i$ by Proposition 7.6.7; hence $Z_n = G$ eventually. The chain $Z_0 \trianglelefteq Z_1 \trianglelefteq \cdots$ has abelian quotients, so nilpotent groups are solvable by Proposition 7.9.3. S_3 is solvable via $\{e\} \trianglelefteq A_3 \trianglelefteq S_3$, but $Z(S_3) = \{e\}$ (a non-identity permutation of $\{1, 2, 3\}$ does not commute with every transposition), so $Z_i = \{e\}$ for all i. □

Definition 7.9.7. A *composition series* of a group G is a chain $\{e\} = G_n \trianglelefteq G_{n-1} \trianglelefteq \cdots \trianglelefteq G_0 = G$ in which every factor G_i/G_{i+1} is simple; the factors are the *composition factors* and n is the *length*.

Theorem 7.9.8 (Jordan–Hölder)**.** *Every finite group has a composition series, and any two composition series of a finite group have the same length and the same composition factors up to isomorphism and permutation.*

Proof. Existence: by induction on $|G|$; if $G \neq \{e\}$, a normal subgroup $N \neq G$ of largest order has G/N simple by the correspondence theorem, and N has a composition series.

Uniqueness, by induction on $|G|$. Let (G_i) and (H_j) be composition series of lengths n and m. If $G_1 = H_1$, apply induction to G_1. Otherwise $H_1 \not\subseteq G_1$: if $H_1 \subsetneq G_1$, then G_1/H_1 would be a nontrivial proper normal subgroup of the simple group G/H_1. So G_1H_1 is a normal subgroup strictly containing G_1, and $G_1H_1 = G$ because G/G_1 is simple. Let $K = G_1 \cap H_1$. By Theorem 7.4.4(2), $G_1/K \cong G/H_1$ and $H_1/K \cong G/G_1$, both simple. Choose a composition series of K, of length l. Then $G_1 \trianglerighteq K \trianglerighteq \cdots$ is a composition series of G_1 of length $l+1$; by induction applied to G_1 it has the same length and factors as $G_1 \trianglerighteq G_2 \trianglerighteq \cdots$, so $n - 1 = l + 1$. Similarly $m - 1 = l + 1$. The factors of (G_i) are G/G_1, $G_1/K \cong G/H_1$ and those of K; the factors of (H_j) are G/H_1, $H_1/K \cong G/G_1$ and those of K. These agree up to permutation. □

7.10 Free groups and presentations

Definition 7.10.1. Let S be a set and $A = S \times \{1, -1\}$, whose elements we write s and s^{-1}; for $x = s^{\pm 1}$ write $x^{-1} = s^{\mp 1}$. A *word* is a finite sequence $x_1 \cdots x_k$ of elements of A, and it is *reduced* if no two adjacent letters are of the form xx^{-1}. Let $F(S)$ be the set of reduced words.

Theorem 7.10.2. *There is a unique group structure on $F(S)$ in which the product of reduced words u and v is obtained from the concatenation uv by repeatedly deleting adjacent pairs xx^{-1}. With the map $\iota\colon S \to F(S)$ sending s to the one-letter word, $F(S)$ is the* free group *on S: for every group G and map $f\colon S \to G$ there is a unique homomorphism $\bar{f}\colon F(S) \to G$ with $\bar{f} \circ \iota = f$. So F is left adjoint to the forgetful functor* Grp $\to$ Set.

Proof. For $x \in A$ define $L_x\colon F(S) \to F(S)$ by $L_x(w) = xw$ if w does not begin with x^{-1}, and $L_x(x^{-1}w') = w'$. Then $L_xL_{x^{-1}} = \mathrm{id}$ for all x, so each L_x is a bijection, and $L_{x^{-1}} = L_x^{-1}$. Let $\Phi \leq \mathrm{Sym}(F(S))$ be the subgroup generated by the L_x, and $\epsilon\colon \Phi \to F(S)$, $\pi \mapsto \pi(\emptyset)$, where $\emptyset$ is the empty word. By Lemma 7.2.8, every $\pi \in \Phi$ is a product $L_{y_1} \cdots L_{y_m}$; deleting adjacent factors $L_xL_{x^{-1}} = \mathrm{id}$ we may assume $y_1 \cdots y_m$ is reduced, and then $\pi(\emptyset) = y_1 \cdots y_m$. Hence π is determined by $\pi(\emptyset)$, and every reduced word $w = x_1 \cdots x_k$ equals $L_{x_1} \cdots L_{x_k}(\emptyset)$. So ϵ is a bijection, and we transport the group structure of Φ to $F(S)$. The product of u and v is then $L_{u_1} \cdots L_{u_k}(v)$, which is computed by the deletion process of the statement.

For $f\colon S \to G$ extend f to A by $f(s^{-1}) = f(s)^{-1}$ and define $\bar{f}(x_1 \cdots x_k) = f(x_1) \cdots f(x_k)$. The value of $f(x_1) \cdots f(x_k)$ on an arbitrary word does not change when an adjacent pair xx^{-1} is deleted, and the product in $F(S)$ is obtained from concatenation by such deletions; so $\bar{f}$ is a homomorphism. It is unique because $\iota(S)$ generates $F(S)$. □

Definition 7.10.3. Let S be a set and $R \subseteq F(S)$. The group with *presentation* $\langle S \mid R \rangle$ is $F(S)/N$, where N is the smallest normal subgroup containing R (the intersection of all normal subgroups containing R). Every group G has a presentation: take $S = G$ and R the kernel of $F(G) \to G$.

Proposition 7.10.4. *Let $G = \langle S \mid R \rangle$ and let $f\colon S \to H$ be a map to a group such that $\bar{f}(r) = e$ for all $r \in R$. Then there is a unique homomorphism $G \to H$ sending the class of each s to $f(s)$.*

Proof. $\ker \bar{f}$ is a normal subgroup containing R, hence containing N; apply Theorem 7.4.3. Uniqueness holds because the classes of S generate G. □

Example 7.10.5.
1. $\langle a \mid a^n \rangle \cong \mathbb{Z}/n\mathbb{Z}$: the homomorphism $F(\{a\}) \to \mathbb{Z}/n\mathbb{Z}$, $a \mapsto 1$, kills a^n and so induces a surjection $\langle a \mid a^n \rangle \to \mathbb{Z}/n\mathbb{Z}$; the source is generated by the class of a, which has order dividing n, so it has at most n elements.
2. $\langle a, b \mid aba^{-1}b^{-1} \rangle \cong \mathbb{Z}^2$: the map $a \mapsto (1,0)$, $b \mapsto (0,1)$ induces a surjection, and a homomorphism $\mathbb{Z}^2 \to \langle a, b \mid aba^{-1}b^{-1} \rangle$ inverse to it exists because the classes of a and b commute.
3. $D_n \cong \langle r, s \mid r^n, s^2, srsr \rangle$, by a similar argument.

Definition 7.10.6. Let $\alpha_1 \colon H \to G_1$ and $\alpha_2 \colon H \to G_2$ be homomorphisms. The *amalgamated free product* $G_1 *_H G_2$ is the group with generators the disjoint union of the sets G_1 and G_2 and relations xyz^{-1} whenever $xy = z$ holds in G_1 or in G_2, and $\alpha_1(h)\alpha_2(h)^{-1}$ for $h \in H$. If H is trivial, it is the *free product* $G_1 * G_2$.

Proposition 7.10.7. *With the evident homomorphisms $G_i \to G_1 *_H G_2$, the amalgamated free product is a pushout of $G_1 \leftarrow H \to G_2$ in* Grp. *In particular $G_1 * G_2$ is a coproduct, and* Grp *has all colimits.*

Proof. The relations of the first kind make the maps $G_i \to G_1 *_H G_2$ homomorphisms, and those of the second kind make the square commute. Given homomorphisms $f_i \colon G_i \to K$ with $f_1\alpha_1 = f_2\alpha_2$, the map on generators given by f_1 and f_2 kills all relations, so it induces a homomorphism by Proposition 7.10.4, which is unique since the images of G_1 and G_2 generate. Coproducts of arbitrary families are constructed in the same way, and coequalisers are quotients by normal subgroups; so Grp has all colimits by the dual of Theorem 6.8.6. □

Remark 7.10.8. If α_1 and α_2 are injective, the maps $G_i \to G_1 *_H G_2$ are injective, and every element has a normal form as an alternating product of coset representatives [Ser80, Chapter I, Theorem 1], [LS77, Chapter IV]. For example $\mathrm{SL}_2(\mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z} *_{\mathbb{Z}/2\mathbb{Z}} \mathbb{Z}/6\mathbb{Z}$ [Ser80, Chapter I, §4]. Amalgamated free products compute fundamental groups of unions of spaces via the Seifert–van Kampen theorem (Chapter 19).

7.11 Abelian groups

In this section groups are abelian and written additively. For $n \in \mathbb{Z}$ and $a \in A$ we write na for the n-th power of Lemma 7.2.9.

Definition 7.11.1. For a set S, let $\mathbb{Z}^{(S)}$ be the set of maps $S \to \mathbb{Z}$ that vanish outside a finite set, with pointwise addition, and e_s the map that is 1 at s and 0 elsewhere. An abelian group is *free* if it is isomorphic to some $\mathbb{Z}^{(S)}$, and a family (a_s) in A is a *basis* if the homomorphism $\mathbb{Z}^{(S)} \to A$, $e_s \mapsto a_s$, is an isomorphism. The *torsion subgroup* of A is $A_{\mathrm{tors}} = \{a : na = 0 \text{ for some } n \geq 1\}$, and A is *torsion-free* if $A_{\mathrm{tors}} = 0$.

Proposition 7.11.2. *For every abelian group A and map $f : S \to A$ there is a unique homomorphism $\mathbb{Z}^{(S)} \to A$ with $e_s \mapsto f(s)$, namely $\sum_s n_s e_s \mapsto \sum_s n_s f(s)$. If S is finite with n elements, $\mathbb{Z}^{(S)} \cong \mathbb{Z}^n$, and $\mathbb{Z}^n \cong \mathbb{Z}^m$ implies $n = m$.*

Proof. The formula defines a homomorphism, and it is forced. If $\mathbb{Z}^n \cong \mathbb{Z}^m$, then $\mathbb{Z}^n/2\mathbb{Z}^n \cong \mathbb{Z}^m/2\mathbb{Z}^m$, and these groups have 2^n and 2^m elements. □

Definition 7.11.3. The *rank* of a free abelian group $\mathbb{Z}^n$ is n; it is well defined by Proposition 7.11.2.

Proposition 7.11.4. *Every subgroup of $\mathbb{Z}^n$ is free of rank at most n.*

Proof. By induction on n; for $n = 0$ it is clear. Let $H \leq \mathbb{Z}^n$ and $p\colon \mathbb{Z}^n \to \mathbb{Z}$ the last coordinate. Then $H \cap \ker p$ is a subgroup of $\ker p \cong \mathbb{Z}^{n-1}$, hence free of rank $\leq n-1$ with some basis $(h_1, \dots, h_k)$. By Proposition 7.2.10, $p[H] = d\mathbb{Z}$. If $d = 0$, $H = H \cap \ker p$. Otherwise choose $h \in H$ with $p(h) = d$; every $x \in H$ is uniquely $x = (x - qh) + qh$ with $q = p(x)/d$ and $x - qh \in H \cap \ker p$, so $(h_1, \dots, h_k, h)$ is a basis of H. □

Theorem 7.11.5 (Smith normal form over $\mathbb{Z}$)**.** *Let M be an $m \times n$ matrix with integer entries. There are invertible integer matrices P and Q (with integer inverses) such that PMQ is diagonal, with diagonal entries $d_1, \dots, d_r, 0, \dots, 0$ where $d_i \geq 1$ and $d_1 \mid d_2 \mid \cdots \mid d_r$.*

Proof. Multiplying by invertible matrices on the left and right performs the elementary operations: swapping two rows or columns, multiplying a row or column by -1, and adding an integer multiple of one row (column) to another. We argue by induction on $m + n$; if $M = 0$ there is nothing to do. Among all matrices obtainable from M by elementary operations, choose one, M', whose smallest nonzero entry in absolute value, δ, is as small as possible; move it to position $(1, 1)$ and make it positive. Every entry a of the first row or column is divisible by δ: otherwise subtracting a multiple of the first column (row) would produce the remainder of a modulo δ, a smaller nonzero entry. So we can clear the first row and column. Every remaining entry a is also divisible by δ: otherwise adding its row to the first row and then subtracting a multiple of the first column would create a smaller nonzero entry. Apply induction to the remaining $(m-1) \times (n-1)$ block; the operations on it do not affect divisibility of its entries by δ, so $d_1 = \delta$ divides all later d_i. □

Theorem 7.11.6 (Structure of finitely generated abelian groups)**.** *Let A be a finitely generated abelian group. There are unique integers $r \geq 0$ and $d_1, \dots, d_k \geq 2$ with $d_1 \mid d_2 \mid \cdots \mid d_k$ such that*

$$A \cong \mathbb{Z}^r \oplus \mathbb{Z}/d_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/d_k\mathbb{Z}.$$

Here $A_{\mathrm{tors}} \cong \bigoplus_i \mathbb{Z}/d_i\mathbb{Z}$, and $A/A_{\mathrm{tors}} \cong \mathbb{Z}^r$.

Proof. Existence. Choose generators $a_1, \dots, a_n$, giving a surjection $\pi\colon \mathbb{Z}^n \to A$. By Proposition 7.11.4, $\ker \pi$ is free with a basis $v_1, \dots, v_m$, $m \leq n$. Let M be the $n \times m$ matrix with columns v_j. Changing the basis of $\mathbb{Z}^n$ corresponds to multiplying M by an invertible matrix on the left, and changing the basis of $\ker \pi$ to multiplying on the right. By Theorem 7.11.5 there are bases $f_1, \dots, f_n$ of $\mathbb{Z}^n$ and $w_1, \dots, w_m$ of $\ker \pi$ with $w_i = d_i f_i$ for $i \leq s$ and $w_i = 0$ for $i > s$; since the w_i are a basis, $s = m$ and all $d_i \geq 1$. Hence $A \cong \mathbb{Z}^n/\ker \pi \cong \bigoplus_{i \leq m} \mathbb{Z}/d_i\mathbb{Z} \oplus \mathbb{Z}^{n-m}$, and we discard the factors with $d_i = 1$.

Uniqueness. In such a decomposition the torsion subgroup is $T = \bigoplus \mathbb{Z}/d_i\mathbb{Z}$ and $A/T \cong \mathbb{Z}^r$, so r is determined by Proposition 7.11.2. For a prime p and $j \geq 0$, $p^j(\mathbb{Z}/d\mathbb{Z})/p^{j+1}(\mathbb{Z}/d\mathbb{Z})$ has order p if $p^{j+1} \mid d$ and order 1 otherwise. Hence $|p^jT/p^{j+1}T| = p^{t(p,j)}$, where $t(p, j)$ is the number of i with $p^{j+1} \mid d_i$. By the divisibility chain, these are the last $t(p, j)$ indices. So the numbers $t(p, j)$, which depend only on T, determine the exponent of p in every d_i, provided k is known; and $k = \max_p t(p, 0)$, since a prime dividing d_1 divides all d_i. □

Corollary 7.11.7. *If $\gcd(m, n) = 1$, then $\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \oplus \mathbb{Z}/n\mathbb{Z}$. Every finite abelian group is a direct sum of cyclic groups of prime power order.*

Proof. The element $(1, 1)$ of $\mathbb{Z}/m\mathbb{Z} \oplus \mathbb{Z}/n\mathbb{Z}$ has order $\operatorname{lcm}(m, n) = mn$, so it generates. Apply this to the prime factorisations of the d_i. □

Counterexample 7.11.8. The group $(\mathbb{Q}, +)$ is torsion-free but not free, and not finitely generated. Any two elements $a/b, c/d$ satisfy the nontrivial relation $bc \cdot (a/b) - ad \cdot (c/d) = 0$, so a basis would have one element, and $\mathbb{Q} \cong \mathbb{Z}$ is impossible since $\mathbb{Q}$ is divisible ($x = 2 \cdot (x/2)$) while $1 \in \mathbb{Z}$ is not twice anything. A finitely generated torsion-free group would be free by Theorem 7.11.6.

7.12 Exercises

Exercise 7.12.1 (Orbit-counting lemma)**.** Let a finite group G act on a finite set X. Show that the number of orbits is $\frac{1}{|G|}\sum_{g \in G} |X^g|$, where $X^g = \{x : gx = x\}$. Use it to count the colourings of the vertices of a square with two colours up to rotation. (This is often called *Burnside's lemma*, because it appears in Burnside's book on finite groups; it is due to Cauchy and Frobenius, and has been called "the lemma that is not Burnside's" [Neu79].)

Solution. Count the set $\{(g, x) : gx = x\}$ in two ways: it has $\sum_g |X^g|$ elements, and also $\sum_x |G_x| = \sum_x |G|/|Gx|$. Each orbit O contributes $\sum_{x \in O} |G|/|O| = |G|$ to the last sum. For the square, $G = \mathbb{Z}/4\mathbb{Z}$ acts on the $2^4 = 16$ colourings; the identity fixes 16, the rotations by $\pm\frac{\pi}{2}$ fix the 2 constant colourings, and the rotation by π fixes the 4 colourings with opposite vertices equal. So there are $\frac{1}{4}(16 + 2 + 2 + 4) = 6$ colourings up to rotation. □

Exercise 7.12.2. Show that every group of order at most 5 is abelian, and that S_3 is the smallest non-abelian group.

Exercise 7.12.3. Show that $[S_n, S_n] = A_n$ for $n \geq 2$ and $S_n^{\mathrm{ab}} \cong \mathbb{Z}/2\mathbb{Z}$.

Exercise 7.12.4. (Frattini argument.) Let $N \trianglelefteq G$ be finite and P a Sylow p-subgroup of N. Show that $G = N \cdot N_G(P)$.

Exercise 7.12.5. Let $p < q$ be primes with $p \nmid q - 1$. Show that every group of order pq is cyclic.

Exercise 7.12.6. Show that $F(S) \cong F(T)$ implies $|S| = |T|$ for finite sets S, T. (Hint: count homomorphisms to $\mathbb{Z}/2\mathbb{Z}$.)

Exercise 7.12.7. Show that the free group on two generators is not solvable. (Hint: construct a surjection onto S_5.)

Exercise 7.12.8. Let $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ with $i^2 = j^2 = k^2 = ijk = -1$. Show that every subgroup of Q_8 is normal but Q_8 is not abelian, and that Q_8 is not a semidirect product of two proper subgroups.

Chapter 8

Rings and Modules

8.1 Introduction

This chapter develops the basic theory of rings and modules: ideals and quotients, prime and maximal ideals, factorisation in integral domains, polynomial rings, modules and exact sequences, tensor products, algebras, and the structure of finitely generated modules over a principal ideal domain. Commutative rings were introduced in Section 4.3; the deeper theory of commutative rings is in Chapter 14.

References are [Lan02], [DF04] and, for modules over commutative rings, [AM69].

Prerequisites. Chapter 3 (Set Theory), Chapter 4 (Number Systems), Chapter 6 (Categories) and Chapter 7 (Groups).

8.2 Rings

Definition 8.2.1. A *ring* is a set R with two operations, addition $(a, b) \mapsto a + b$ and multiplication $(a, b) \mapsto a \cdot b$ (usually written ab), and elements 0, 1 such that $(R, +, 0)$ is an abelian group (in additive notation, Notation 7.2.2), multiplication is associative with neutral element 1, and both distributive laws $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$ hold. A *homomorphism* of rings is a map preserving $+$, $\cdot$ and 1. A *subring* is a subset containing 0 and 1 and closed under $+$, $-$ and $\cdot$. An element u is a *unit* if it has a two-sided inverse; the units form a group $R^\times$. A *division ring* is a ring with $1 \neq 0$ in which every nonzero element is a unit; a commutative division ring is a field. The rings form a category Ring, and the commutative rings a full subcategory CRing.

Remark 8.2.2. The computations of Lemma 4.3.2 hold in every ring. If $1 = 0$ then $a = 1 \cdot a = 0 \cdot a = 0$ for all a, so $R = \{0\}$ is the *zero ring*; it is the terminal object of Ring, and $\mathbb{Z}$ is the initial object (Exercise 4.10.5).

Example 8.2.3.

1. $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{Z}/n\mathbb{Z}$ with the multiplication $[a][b] = [ab]$, which is well defined since $a - a' \in n\mathbb{Z}$ implies $ab - a'b \in n\mathbb{Z}$.

2. For a ring R, the $n \times n$ matrices $M_n(R)$ form a ring under matrix addition and multiplication; it is not commutative for $n \geq 2$ if $1 \neq 0$.

3. For an abelian group A, the endomorphisms $\mathrm{End}(A)$ form a ring under pointwise addition and composition.

4. The product $\prod_i R_i$ of rings with componentwise operations is a product in Ring.

5. The *quaternions* $\mathbb{H}$ are the matrices $\begin{pmatrix} z & -\bar{w} \\ w & \bar{z} \end{pmatrix}$ with $z, w \in \mathbb{C}$. They form a subring of $M_2(\mathbb{C})$, and a division ring: the determinant of such a matrix is $|z|^2 + |w|^2$, and if it is nonzero the inverse $\frac{1}{|z|^2+|w|^2}\begin{pmatrix} \bar{z} & \bar{w} \\ -w & z \end{pmatrix}$ is again of this form. With $i = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$, $j = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $k = ij$ we get $i^2 = j^2 = k^2 = ijk = -1$, so $\mathbb{H}$ is not commutative.

8.3 Ideals and quotient rings

Definition 8.3.1. A *left ideal* of a ring R is an additive subgroup I with $ra \in I$ for $r \in R$, $a \in I$; right ideals are defined similarly, and a (two-sided) *ideal* is both. For a commutative ring these notions agree. The ideal generated by a subset S of a commutative ring is $(S) = \{r_1 s_1 + \cdots + r_n s_n : r_i \in R, s_i \in S\}$; an ideal of the form (a) is *principal*. For ideals I, J of a commutative ring, $I + J$, $I \cap J$ and the product IJ, generated by the products ab with $a \in I$, $b \in J$, are ideals, and $IJ \subseteq I \cap J$.

Theorem 8.3.2. *Let I be an ideal of R. The quotient group R/I is a ring with $(a + I)(b + I) = ab + I$, and $\pi\colon R \to R/I$ is a surjective ring homomorphism with kernel I. Every ring homomorphism $f\colon R \to S$ with $I \subseteq \ker f$ factors uniquely as $\bar{f} \circ \pi$. Kernels of ring homomorphisms are ideals, and $R/\ker f \cong \operatorname{im} f$. The ideals of R/I correspond to the ideals of R containing I via $J \mapsto J/I$, and $(R/I)/(J/I) \cong R/J$.*

Proof. If $a - a', b - b' \in I$, then $ab - a'b' = a(b - b') + (a - a')b' \in I$, so the product is well defined; the ring axioms follow from those of R. The remaining statements follow from Theorems 7.4.3 and 7.4.4 applied to the additive groups, together with the observation that the maps constructed there preserve products and that preimages and images of ideals under surjective ring homomorphisms are ideals. □

Theorem 8.3.3 (Chinese remainder theorem)**.** *Let R be a commutative ring and $I_1, \ldots, I_n$ ideals with $I_i + I_j = R$ for $i \neq j$. Then $I_1 \cap \cdots \cap I_n = I_1 \cdots I_n$, and the map $R \to \prod_i R/I_i$ induces an isomorphism $R/(I_1 \cap \cdots \cap I_n) \cong \prod_i R/I_i$.*

Proof. For $n = 2$: write $1 = a + b$ with $a \in I_1$, $b \in I_2$. If $x \in I_1 \cap I_2$ then $x = xa + xb \in I_1 I_2$. Given $r_1, r_2 \in R$, the element $r_1 b + r_2 a$ is congruent to r_1 modulo I_1 and to r_2 modulo I_2, so the map is surjective; its kernel is $I_1 \cap I_2$. For $n > 2$, note that $I_1 + I_2 \cdots I_n = R$: from $1 = a_j + b_j$ with $a_j \in I_1$, $b_j \in I_j$ we get $1 = \prod_{j \geq 2}(a_j + b_j) \in I_1 + I_2 \cdots I_n$. Now use induction. □

Example 8.3.4. If $n = p_1^{e_1} \cdots p_k^{e_k}$ with distinct primes p_i, then $\mathbb{Z}/n\mathbb{Z} \cong \prod_i \mathbb{Z}/p_i^{e_i}\mathbb{Z}$ as rings, since $(p_i^{e_i}) + (p_j^{e_j}) = \mathbb{Z}$ by Theorem 4.5.3.

8.4 Prime and maximal ideals

In this section all rings are commutative.

Definition 8.4.1. An ideal $\mathfrak{p} \neq R$ is *prime* if $ab \in \mathfrak{p}$ implies $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. An ideal $\mathfrak{m} \neq R$ is *maximal* if there is no ideal J with $\mathfrak{m} \subsetneq J \subsetneq R$.

Proposition 8.4.2. *An ideal I is prime if and only if R/I is an integral domain, and maximal if and only if R/I is a field. In particular maximal ideals are prime.*

Proof. The first statement is a restatement of the definitions. A commutative ring $K \neq 0$ is a field if and only if its only ideals are 0 and K: if $a \neq 0$ then $(a) = K$ means a is a unit. By the correspondence of ideals in Theorem 8.3.2, R/I has only the ideals 0 and R/I exactly when I is maximal. □

Theorem 8.4.3 (Krull)**.** *Every ideal $I \neq R$ is contained in a maximal ideal. In particular every nonzero ring has a maximal ideal.*

Proof. Let P be the set of ideals J with $I \subseteq J \neq R$, ordered by inclusion. It contains I. The union of a nonempty chain in P is an ideal, and it does not contain 1, so it lies in P. By Zorn's lemma (Theorem 3.7.2) P has a maximal element, which is a maximal ideal. □

Example 8.4.4. The ideals of $\mathbb{Z}$ are the $n\mathbb{Z}$ (Proposition 7.2.10). The prime ideals are 0 and $p\mathbb{Z}$ for p prime, by Lemma 4.5.4, and the maximal ideals are the $p\mathbb{Z}$. Hence $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is prime; we write $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$. The prime ideal 0 is not maximal.

Definition 8.4.5. The *characteristic* of a ring R is the $n \geq 0$ with $\ker(\mathbb{Z} \to R) = n\mathbb{Z}$.

Lemma 8.4.6. *The characteristic of an integral domain is* 0 *or a prime number.*

Proof. The image of $\mathbb{Z} \to R$ is a subring of a domain, hence a domain, so the kernel is a prime ideal of $\mathbb{Z}$. □

Proposition 8.4.7. *Let R be an integral domain. There is a field* $\mathrm{Frac}(R)$ *and an injective homomorphism $R \to \mathrm{Frac}(R)$ such that every injective homomorphism from R to a field K extends uniquely to a homomorphism* $\mathrm{Frac}(R) \to K$.

Proof. Construct $\mathrm{Frac}(R)$ as the set of classes a/b of pairs with $b \neq 0$ under $(a, b) \sim (c, d) \iff ad = bc$, with the operations of Definition 4.6.1; the verification of Theorem 4.6.2 only used that $\mathbb{Z}$ is an integral domain. Given $f : R \to K$ injective, $f(b) \neq 0$ for $b \neq 0$, and $a/b \mapsto f(a)f(b)^{-1}$ is the unique extension, as in Proposition 4.6.3. □

8.5 Factorisation in integral domains

In this section R is an integral domain.

Definition 8.5.1. Elements a, b are *associates* if $a = ub$ with $u \in R^\times$. A nonzero nonunit a is *irreducible* if $a = bc$ implies that b or c is a unit, and *prime* if $a \mid bc$ implies $a \mid b$ or $a \mid c$. The domain R is a *unique factorisation domain* (UFD) if every nonzero nonunit is a product of irreducible elements, and this product is unique up to order and associates. It is a *principal ideal domain* (PID) if every ideal is principal, and a *Euclidean domain* if there is a map $N : R \setminus \{0\} \to \mathbb{N}$ such that for all a and $b \neq 0$ there are q, r with $a = qb + r$ and $r = 0$ or $N(r) < N(b)$.

Lemma 8.5.2. *A nonzero element p is prime if and only if (p) is a prime ideal. Prime elements are irreducible.*

Proof. The first statement is a restatement. If p is prime and $p = bc$, then $p \mid b$ or $p \mid c$, say $b = pd$; then $p = pdc$, so $dc = 1$ and c is a unit. □

Lemma 8.5.3. *Suppose every nonzero nonunit of R is a product of irreducible elements. Then R is a UFD if and only if every irreducible element is prime.*

Proof. If irreducibles are prime, uniqueness follows as in the proof of Theorem 4.5.5: a prime dividing a product of irreducibles divides one of them, hence is associate to it, and we cancel. Conversely, let R be a UFD, p irreducible and $p \mid bc$, say $bc = pd$. Factoring b, c and d into irreducibles and comparing the two factorisations of bc shows that p is associate to an irreducible factor of b or of c (if b or c is a unit or zero the claim is immediate). □

Theorem 8.5.4. *A Euclidean domain is a PID, and a PID is a UFD.*

Proof. Let R be Euclidean and $I \neq 0$ an ideal; choose $b \in I \setminus \{0\}$ with $N(b)$ minimal. For $a \in I$ write $a = qb + r$; then $r \in I$, so $r = 0$ and $I = (b)$.

Let R be a PID. The union of a chain of ideals $(a_1) \subseteq (a_2) \subseteq \cdots$ is an ideal (a), and $a \in (a_n)$ for some n, so the chain is eventually constant. If some nonzero nonunit a were not a product of irreducibles, then a is not irreducible, so $a = bc$ with b, c nonunits, one of which, say a_1, is not a product of irreducibles; and $(a) \subsetneq (a_1)$. Repeating gives a strictly increasing chain of principal ideals, a contradiction. (The repeated choices are made using the axiom of choice in the form of Theorem 3.4.6 applied to a choice function.) Finally, if p is irreducible then (p) is maximal: if $(p) \subseteq (c) \neq R$ then $p = cd$ with c a nonunit, so d is a unit and $(c) = (p)$. Hence (p) is prime by Proposition 8.4.2, and R is a UFD by Lemma 8.5.3. □

Example 8.5.5. The ring $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\} \subseteq \mathbb{C}$ of *Gaussian integers* is Euclidean for $N(z) = |z|^2$. Let $a, b \in \mathbb{Z}[i]$, $b \neq 0$, and write $a/b = u + vi$ with $u, v \in \mathbb{Q}$. Choose integers m, n with $|u - m| \leq 1/2$ and $|v - n| \leq 1/2$, and put $q = m + ni$, $r = a - qb \in \mathbb{Z}[i]$. Then $N(r) = |b|^2|a/b - q|^2 \leq N(b)(1/4 + 1/4) < N(b)$.

Counterexample 8.5.6. The ring $R = \mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\} \subseteq \mathbb{C}$ is not a UFD. The norm $N(z) = |z|^2 = a^2 + 5b^2$ is multiplicative and takes values in $\mathbb{N}$, and the units are the elements of norm 1, namely ± 1. No element has norm 2 or 3. Hence 2 (norm 4) is irreducible: a factorisation into nonunits would have factors of norm 2. But 2 divides $6 = (1 + \sqrt{-5})(1 - \sqrt{-5})$ without dividing either factor, since $N(1 \pm \sqrt{-5}) = 6$ is not divisible by $N(2) = 4$. So 2 is irreducible but not prime.

8.6 Polynomial rings

Definition 8.6.1. Let R be a ring. The *polynomial ring* $R[x]$ is the set of sequences $(a_0, a_1, \ldots)$ in R with only finitely many nonzero terms, with termwise addition and the product $(a_i)(b_j) = (c_k)$, $c_k = \sum_{i+j=k} a_i b_j$. We identify $r \in R$ with $(r, 0, 0, \ldots)$ and write $x = (0, 1, 0, \ldots)$, so that every polynomial is uniquely $f = \sum_{i=0}^{n} a_i x^i$. If $f \neq 0$, its *degree* $\deg f$ is the largest i with $a_i \neq 0$, and $a_{\deg f}$ is the *leading coefficient*; f is *monic* if the leading coefficient is 1. The ring $R[x_1, \ldots, x_n]$ is defined inductively as $R[x_1, \ldots, x_{n-1}][x_n]$, and the *power series ring* $R[[x]]$ is defined like $R[x]$ but with all sequences allowed.

Proposition 8.6.2. *Let $\varphi\colon R \to S$ be a ring homomorphism and $s \in S$ an element commuting with all elements of $\varphi(R)$. There is a unique ring homomorphism $R[x] \to S$ extending φ and sending x to s, namely $\sum a_i x^i \mapsto \sum \varphi(a_i) s^i$. In particular, for R commutative, $R[x_1, \ldots, x_n]$ is the free commutative R-algebra on $x_1, \ldots, x_n$: homomorphisms to a commutative ring S extending φ correspond to n-tuples of elements of S.*

Proof. The formula is forced, and it preserves products because s commutes with the $\varphi(a_i)$: $\big(\sum \varphi(a_i) s^i\big)\big(\sum \varphi(b_j) s^j\big) = \sum_k \big(\sum_{i+j=k} \varphi(a_i)\varphi(b_j)\big) s^k$. The last statement follows by induction on n. □

Notation 8.6.3. For R commutative, $f \in R[x]$ and $a \in R$, we write $f(a)$ for the image of f under the homomorphism $R[x] \to R$ with $x \mapsto a$. An element a with $f(a) = 0$ is a *root* of f.

Proposition 8.6.4. *If R is an integral domain, then* $\deg(fg) = \deg f + \deg g$ *for nonzero f, g, so $R[x]$ is an integral domain and $R[x]^\times = R^\times$.*

Proof. The coefficient of $x^{\deg f + \deg g}$ in fg is the product of the leading coefficients, which is nonzero; all higher coefficients vanish. If $fg = 1$ then $\deg f = \deg g = 0$. □

Theorem 8.6.5 (Division with remainder)**.** *Let R be commutative and $g \in R[x]$ a polynomial whose leading coefficient is a unit. For every $f \in R[x]$ there are unique $q, r \in R[x]$ with $f = qg + r$ and $r = 0$ or $\deg r < \deg g$. Consequently, if k is a field then $k[x]$ is Euclidean with $N = \deg$, hence a PID and a UFD.*

Proof. Existence by induction on $\deg f$: if $f = 0$ or $\deg f < \deg g$ take $q = 0$. Otherwise, with a and b the leading coefficients of f and g, the polynomial $f - ab^{-1}x^{\deg f - \deg g}g$ has smaller degree or is zero. Uniqueness: if $qg + r = q'g + r'$ with $q \neq q'$, then $(q - q')g = r' - r$ has degree $\geq \deg g$, because the leading coefficient of g is a unit (so it is not a zero divisor), which is impossible. The last statement follows from Theorem 8.5.4. □

Corollary 8.6.6. *Let R be commutative, $f \in R[x]$ and $a \in R$. Then $f(a) = 0$ if and only if $(x - a) \mid f$. If R is an integral domain and $f \neq 0$, then f has at most* $\deg f$ *roots in R.*

Proof. Divide: $f = q(x - a) + r$ with $r \in R$, and $r = f(a)$. If R is a domain and $a_1, \ldots, a_m$ are distinct roots, induction shows $(x - a_1) \cdots (x - a_m) \mid f$: if $f = (x - a_1)g$ then $0 = f(a_j) = (a_j - a_1)g(a_j)$ gives $g(a_j) = 0$ for $j \geq 2$. So $m \leq \deg f$ by Proposition 8.6.4. □

Counterexample 8.6.7. The hypothesis that R is a domain is needed: in $\mathbb{Z}/8\mathbb{Z}$ the polynomial $x^2 - 1$ of degree 2 has the four roots $1, 3, 5, 7$.

Proposition 8.6.8. *Let R be commutative. A power series $f = \sum a_i x^i \in R[[x]]$ is a unit if and only if a_0 is a unit of R.*

Proof. If $fg = 1$ then $a_0 b_0 = 1$. Conversely, if a_0 is a unit define $b_0 = a_0^{-1}$ and recursively $b_k = -a_0^{-1} \sum_{i=1}^{k} a_i b_{k-i}$; then $\sum_{i+j=k} a_i b_j = 0$ for $k \geq 1$. □

Definition 8.6.9. Let R be a UFD. A *greatest common divisor* of elements of R is a common divisor divisible by every common divisor; it exists (take the product of the common prime factors with their minimal multiplicities) and is unique up to associates. The *content* $c(f)$ of a nonzero $f \in R[x]$ is a greatest common divisor of its coefficients, and f is *primitive* if $c(f)$ is a unit.

Lemma 8.6.10 (Gauss)**.** *Let R be a UFD. A product of primitive polynomials is primitive, and $c(fg) = c(f)c(g)$ up to associates.*

Proof. If f, g are primitive and a prime p divides all coefficients of fg, then the images of f and g in $(R/(p))[x]$ are nonzero but their product is zero; this is impossible because $R/(p)$ is a domain, hence so is $(R/(p))[x]$ (Proposition 8.6.4). In general write $f = c(f)f_0$ and $g = c(g)g_0$ with f_0, g_0 primitive; then $fg = c(f)c(g)f_0 g_0$ with $f_0 g_0$ primitive. □

Theorem 8.6.11. *Let R be a UFD with fraction field K.*

1. *A primitive polynomial $f \in R[x]$ of positive degree is irreducible in $R[x]$ if and only if it is irreducible in $K[x]$.*
2. *$R[x]$ is a UFD, whose prime elements are the prime elements of R and the primitive polynomials of positive degree that are irreducible in $K[x]$.*

Consequently $\mathbb{Z}[x_1, \ldots, x_n]$ and $k[x_1, \ldots, x_n]$ for a field k are UFDs.

Proof. Every $h \in K[x]$, $h \neq 0$, can be written $h = ah_0$ with $a \in K^\times$ and $h_0 \in R[x]$ primitive (clear denominators, then divide by the content), and if also $h = a'h_0'$ then $h_0' = uh_0$ with $u \in R^\times$: writing $a/a' = b/b'$ with $b, b' \in R$, $bh_0 = b'h_0'$ and comparing contents gives $b = b'u$.

(1) If $f = gh$ in $K[x]$ with $\deg g, \deg h > 0$, write $g = ag_0$, $h = bh_0$; then $f = ab\, g_0h_0$ with g_0h_0 primitive (Lemma 8.6.10), so $ab \in R^\times$ by the uniqueness just shown, and $f = (ab\, g_0)h_0$ is a factorisation in $R[x]$ into polynomials of positive degree. Conversely a factorisation in $R[x]$ of a primitive polynomial into nonunits has both factors of positive degree (a constant factor would divide the content), hence is a factorisation in $K[x]$.

(2) A prime p of R is prime in $R[x]$ since $R[x]/(p) \cong (R/(p))[x]$ is a domain. Let f be primitive of positive degree and irreducible in $K[x]$, and suppose $f \mid gh$ in $R[x]$ with $g, h \neq 0$. Since $K[x]$ is a PID, f is prime in $K[x]$, so say $g = fq$ with $q \in K[x]$. Write $q = aq_0$ with $a = c/c'$, $c, c' \in R$ and $q_0 \in R[x]$ primitive. Then $c'g = c\, f q_0$, and comparing contents using Lemma 8.6.10 gives $c = c'c(g)u$ with $u \in R^\times$, so $a \in R$ and $q \in R[x]$. Hence f is prime in $R[x]$.

Every irreducible element of $R[x]$ is of one of these two kinds: an irreducible of degree 0 is irreducible in R, hence prime in R; an irreducible f of positive degree is primitive (otherwise $c(f)$ would be a nonunit factor), hence irreducible in $K[x]$ by (1). Every nonzero nonunit of $R[x]$ is a product of irreducibles: write $f = c(f)f_0$, factor $c(f)$ in R, and factor f_0 in $K[x]$ into irreducibles, which by the first paragraph and (1) can be taken primitive in $R[x]$, the remaining constant being a unit of R by Lemma 8.6.10. By Lemma 8.5.3, $R[x]$ is a UFD. The last statement follows by induction from $\mathbb{Z}$ and k being UFDs. □

Proposition 8.6.12 (Eisenstein's criterion)**.** *Let R be a UFD with fraction field K, $f = a_nx^n + \cdots + a_0 \in R[x]$ with $n \geq 1$, and p a prime of R with $p \nmid a_n$, $p \mid a_i$ for $i < n$ and $p^2 \nmid a_0$. Then f is irreducible in $K[x]$.*

Proof. By Theorem 8.6.11(1) applied to $f/c(f)$ (note $p \nmid c(f)$) it suffices to exclude $f = gh$ in $R[x]$ with $\deg g, \deg h \geq 1$. Reducing modulo p gives $\bar{g}\bar{h} = \bar{a}_nx^n$ in $(R/(p))[x]$, which embeds in $L[x]$ for $L = \mathrm{Frac}(R/(p))$. By unique factorisation in $L[x]$, $\bar{g} = bx^k$ and $\bar{h} = cx^{n-k}$ with $k = \deg g$ and $n - k = \deg h$ both positive. So p divides the constant terms of g and h, and $p^2 \mid a_0$, a contradiction. □

Counterexample 8.6.13. A UFD need not be a PID: in $\mathbb{Z}[x]$ the ideal $(2, x)$ is not principal. If $(2, x) = (f)$, then $f \mid 2$, so $f = \pm 1$ or $f = \pm 2$. But $2 \nmid x$, and $1 \notin (2, x)$ since every element of $(2, x)$ has even constant term.

8.7 Modules

Definition 8.7.1. Let R be a ring. A (left) *R-module* is an abelian group M with a map $R \times M \to M, (r, m) \mapsto rm$, such that $r(m+m') = rm+rm'$, $(r+s)m = rm+sm$, $(rs)m = r(sm)$ and $1m = m$; equivalently, a ring homomorphism $R \to \mathrm{End}(M)$. A *homomorphism* (or R-linear map) is an additive map with $f(rm) = rf(m)$. *Submodules* are subgroups closed under the action. The set $\mathrm{Hom}_R(M, N)$ of homomorphisms is an abelian group, and an R-module if R is commutative. The small R-modules form a category R-Mod.

Example 8.7.2.

1. R is a module over itself, and its submodules are its left ideals.
2. A $\mathbb{Z}$-module is the same thing as an abelian group, by Lemma 7.2.9.
3. A module over a field k is a *vector space*.
4. A $k[x]$-module is a k-vector space V together with a k-linear map $T\colon V \to V$ (the action of x), by Proposition 8.6.2.

Proposition 8.7.3. *Quotients M/N by submodules are modules, and the isomorphism and correspondence theorems of Theorem 7.4.4 hold for modules. For a family $(M_i)_{i\in I}$, the product $\prod_i M_i$ and the* direct sum $\bigoplus_i M_i = \{(m_i) \in \prod_i M_i : m_i = 0 \text{ for almost all } i\}$ *are a product and a coproduct in R-*Mod*. The category R-*Mod* is additive (Definition 6.12.1), has kernels and cokernels, and has all limits and colimits.*

Proof. The action on M/N is $r(m+N) = rm + N$, well defined since N is a submodule; the maps in the isomorphism theorems are R-linear. The universal properties of $\prod$ and $\bigoplus$ are verified componentwise; for $\bigoplus$, a family $f_i\colon M_i \to N$ gives $(m_i) \mapsto \sum_i f_i(m_i)$, a finite sum. Hom sets are abelian groups with bilinear composition, 0 is a zero object, and finite products exist, so the category is additive. The kernel of f is $\ker f \to M$ and the cokernel is $N \to N/\operatorname{im} f$. Equalisers of f, g are kernels of $f - g$, so all limits exist by Theorem 6.8.6, and dually all colimits exist. □

Definition 8.7.4. For a set S, the *free module* $R^{(S)} = \bigoplus_{s\in S} R$ has basis vectors e_s. A family $(m_s)_{s\in S}$ in M *generates* M if every element is a finite R-linear combination of the m_s, is *linearly independent* if $\sum r_s m_s = 0$ implies all $r_s = 0$, and is a *basis* if both hold. M is *finitely generated* if it has a finite generating family.

Proposition 8.7.5. *For every module M and family $(m_s)_{s\in S}$ in M there is a unique homomorphism $R^{(S)} \to M$ with $e_s \mapsto m_s$. It is surjective iff the family generates, injective iff it is linearly independent. Every module is a quotient of a free module, and a finitely generated module is a quotient of some R^n.*

Proof. The map is $(r_s) \mapsto \sum r_s m_s$. The other statements are immediate; take the family of all elements of M (or a finite generating family). □

Definition 8.7.6. A sequence of homomorphisms $\cdots \to M_{i-1} \xrightarrow{f_{i-1}} M_i \xrightarrow{f_i} M_{i+1} \to \cdots$ is *exact* at M_i if $\operatorname{im} f_{i-1} = \ker f_i$, and exact if it is exact everywhere. A *short exact sequence* is an exact sequence $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$.

Proposition 8.7.7 (Splitting lemma)**.** *For a short exact sequence $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$ the following are equivalent:*

1. *there is $r\colon B \to A$ with $rf = \mathrm{id}_A$;*
2. *there is $s\colon C \to B$ with $gs = \mathrm{id}_C$;*
3. *there is an isomorphism $h\colon B \to A \oplus C$ with $hf(a) = (a, 0)$ and $g = \mathrm{pr}_C \circ h$.*

The sequence is then called split*. If C is free, the sequence splits.*

Proof. (1) ⇒ (3): $h = (r, g)$ is injective (if $r(b) = 0$ and $g(b) = 0$ then $b = f(a)$ and $a = rf(a) = 0$) and surjective (given (a, c), pick b with $g(b) = c$; then $b - fr(b) + f(a)$ maps to (a, c)). (2) ⇒ (3): the map $A \oplus C \to B$, $(a, c) \mapsto f(a) + s(c)$, is bijective by a similar check; take its inverse. (3) ⇒ (1), (2): take $r = \mathrm{pr}_A \circ h$ and $s = h^{-1} \circ \iota_C$. If C has a basis (c_s), choose preimages b_s and define $s(c_s) = b_s$ using Proposition 8.7.5. □

Counterexample 8.7.8. The sequence $0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$ of abelian groups is exact but not split, since $\mathbb{Z}$ has no element of order 2 and so cannot contain a copy of $\mathbb{Z}/2\mathbb{Z}$.

8.8 Vector spaces and rank

Lemma 8.8.1 (Exchange lemma)**.** *Let k be a field and V a vector space generated by $v_1, \dots, v_n$. Then any $n+1$ vectors $w_1, \dots, w_{n+1}$ in V are linearly dependent.*

Proof. Induction on n. For $n = 0$, $V = 0$ and $w_1 = 0$. For $n \geq 1$ write $w_j = \sum_{i=1}^{n} a_{ij} v_i$. If $a_{nj} = 0$ for all j, then $w_1, \dots, w_n$ lie in the span of $v_1, \dots, v_{n-1}$ and are dependent by induction. Otherwise, renumbering, let $a_{n,n+1} \neq 0$ and put $w'_j = w_j - a_{nj} a_{n,n+1}^{-1} w_{n+1}$ for $j \leq n$. These lie in the span of $v_1, \dots, v_{n-1}$, so by induction $\sum_{j \leq n} c_j w'_j = 0$ with some $c_j \neq 0$. Then $\sum_{j \leq n} c_j w_j - \left(\sum_{j \leq n} c_j a_{nj} a_{n,n+1}^{-1}\right) w_{n+1} = 0$ is a nontrivial relation. □

Theorem 8.8.2. *Let R be a nonzero commutative ring. If $R^m \cong R^n$ as R-modules, then $m = n$. In particular any two bases of a finitely generated vector space have the same number of elements.*

Proof. For a field, a basis of n elements generates, so by Lemma 8.8.1 every linearly independent family has at most n elements; hence $k^m \cong k^n$ implies $m \leq n$ and $n \leq m$. For general R, choose a maximal ideal $\mathfrak{m}$ (Theorem 8.4.3). An isomorphism $R^m \cong R^n$ maps $\mathfrak{m}R^m$ onto $\mathfrak{m}R^n$, so it induces an isomorphism $(R/\mathfrak{m})^m \cong R^m/\mathfrak{m}R^m \cong R^n/\mathfrak{m}R^n \cong (R/\mathfrak{m})^n$ of vector spaces over the field $R/\mathfrak{m}$. □

Definition 8.8.3. For R nonzero commutative, the *rank* of a free module R^n is n.

Theorem 8.8.4. *Let k be a field, V a vector space, $I \subseteq V$ a linearly independent subset and $S \supseteq I$ a generating subset. There is a basis B of V with $I \subseteq B \subseteq S$. In particular every vector space has a basis, every linearly independent subset extends to a basis, and every generating subset contains a basis.*

Proof. Let P be the set of linearly independent subsets J with $I \subseteq J \subseteq S$, ordered by inclusion. The union of a chain in P is linearly independent, since a linear relation involves finitely many vectors, which lie in one member of the chain. By Zorn's lemma (Theorem 3.7.2) there is a maximal $B \in P$. If some $s \in S$ were not a linear combination of B, then $B \cup \{s\}$ would be linearly independent, contradicting maximality. So every element of S, and hence of V, is a linear combination of B. □

Theorem 8.8.5. *Any two bases of a vector space V have the same cardinality, called the* dimension $\dim_k V$.

Proof. Let B and C be bases. If B is finite with n elements, then by Lemma 8.8.1 every linearly independent set has at most n elements, so C is finite, and $|B| = |C|$ by Theorem 8.8.2. Suppose both are infinite. Each $c \in C$ is a linear combination of a finite subset $F_c \subseteq B$. Every $b \in B$ lies in some F_c: otherwise b would be a linear combination of $\bigcup_c F_c \subseteq B \setminus \{b\}$, contradicting independence. So $B = \bigcup_{c \in C} F_c$ and $|B| \leq |C|$ by Proposition 3.7.5. By symmetry $|C| \leq |B|$, and Theorem 3.6.2 gives $|B| = |C|$. □

Counterexample 8.8.6. Theorem 8.8.2 fails for noncommutative rings. Let k be a field, $V = k^{(\mathbb{N})}$ and $R = \mathrm{End}_k(V)$. Splitting $\mathbb{N}$ into even and odd numbers gives $V \cong V \oplus V$. As a left R-module (by post-composition), $R = \mathrm{Hom}_k(V, V) \cong \mathrm{Hom}_k(V \oplus V, V) \cong \mathrm{Hom}_k(V, V) \oplus \mathrm{Hom}_k(V, V) = R^2$.

8.9 Tensor products

In this section R is a commutative ring.

Definition 8.9.1. Let M, N, P be R-modules. A map $\beta\colon M \times N \to P$ is *bilinear* if it is R-linear in each variable. A *tensor product* of M and N is a module $M \otimes_R N$ with a bilinear map $\otimes\colon M \times N \to M \otimes_R N$, $(m,n) \mapsto m \otimes n$, such that every bilinear $\beta\colon M \times N \to P$ factors as $\beta = \tilde{\beta} \circ \otimes$ for a unique linear $\tilde{\beta}$.

Proposition 8.9.2. *Tensor products exist and are unique up to unique isomorphism. The elements $m \otimes n$ generate $M \otimes_R N$.*

Proof. Uniqueness is the usual argument for universal properties. Let $F = R^{(M\times N)}$ and $Q \subseteq F$ the submodule generated by the elements $e_{(m+m',n)} - e_{(m,n)} - e_{(m',n)}$, $e_{(m,n+n')} - e_{(m,n)} - e_{(m,n')}$, $e_{(rm,n)} - re_{(m,n)}$ and $e_{(m,rn)} - re_{(m,n)}$. Put $M \otimes_R N = F/Q$ and $m \otimes n = e_{(m,n)} + Q$. A bilinear β induces a linear map $F \to P$, $e_{(m,n)} \mapsto \beta(m,n)$, vanishing on Q, hence a unique $\tilde{\beta}$. The elements $m \otimes n$ generate since the $e_{(m,n)}$ generate F. □

Proposition 8.9.3. *There are natural isomorphisms $R \otimes_R M \cong M$, $M \otimes N \cong N \otimes M$, $(M \otimes N) \otimes P \cong M \otimes (N \otimes P)$ and $(\bigoplus_i M_i) \otimes N \cong \bigoplus_i (M_i \otimes N)$. In particular $R^m \otimes R^n \cong R^{mn}$. For an ideal I, $(R/I) \otimes_R M \cong M/IM$.*

Proof. Each isomorphism is obtained by constructing maps in both directions from bilinear maps and checking that the composites fix generators $m \otimes n$. For instance $(r,m) \mapsto rm$ induces $R \otimes M \to M$, inverse to $m \mapsto 1 \otimes m$. For associativity, for fixed p the map $(m,n) \mapsto m \otimes (n \otimes p)$ is bilinear, giving $(m \otimes n) \mapsto m \otimes (n \otimes p)$; this is bilinear in $(m \otimes n, p)$, and an inverse is constructed symmetrically. For direct sums, $((m_i), n) \mapsto (m_i \otimes n)$ is bilinear, and the inverse is induced by the maps $M_i \otimes N \to (\bigoplus M_i) \otimes N$. Finally, $(\bar{r}, m) \mapsto \overline{rm}$ induces $R/I \otimes M \to M/IM$, with inverse $\bar{m} \mapsto \bar{1} \otimes m$, well defined since $1 \otimes am = \bar{a} \otimes m = 0$ for $a \in I$. □

Theorem 8.9.4 (Tensor–hom adjunction)**.** *For R-modules M, N, P there is a bijection, natural in all three variables,*

$$\mathrm{Hom}_R(M \otimes_R N, P) \cong \mathrm{Hom}_R(M, \mathrm{Hom}_R(N,P)), \qquad f \mapsto (m \mapsto (n \mapsto f(m \otimes n))).$$

So $- \otimes_R N$ is left adjoint to $\mathrm{Hom}_R(N,-)$. Consequently $- \otimes_R N$ preserves colimits, and for every exact sequence $A \to B \to C \to 0$ the sequence $A \otimes N \to B \otimes N \to C \otimes N \to 0$ is exact.

Proof. Both sides are in bijection with bilinear maps $M \times N \to P$, naturally. The functor $- \otimes_R N$ preserves colimits by Theorem 6.10.4. A sequence $A \xrightarrow{u} B \xrightarrow{v} C \to 0$ is exact if and only if v is a cokernel of u, that is, a coequaliser of u and 0; this is preserved. □

Example 8.9.5. $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/d\mathbb{Z}$ with $d = \gcd(m,n)$, since by Proposition 8.9.3 it is $(\mathbb{Z}/n\mathbb{Z})/m(\mathbb{Z}/n\mathbb{Z}) = \mathbb{Z}/(m,n) = \mathbb{Z}/d\mathbb{Z}$. In particular $\mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z}/3\mathbb{Z} = 0$, and $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} = 0$ since $x \otimes \bar{1} = nx/n \otimes \bar{1} = x/n \otimes \bar{n} = 0$.

Counterexample 8.9.6. Tensoring does not preserve injectivity. The map $2\colon \mathbb{Z} \to \mathbb{Z}$ is injective, but tensoring with $\mathbb{Z}/2\mathbb{Z}$ gives the zero map $\mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$. Similarly $\mathrm{Hom}_R(N,-)$ does not preserve surjectivity: $\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ is surjective but $\mathrm{Hom}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}) = 0 \to \mathrm{Hom}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ is not.

Definition 8.9.7. An R-module N is *flat* if $- \otimes_R N$ preserves injective homomorphisms. Free modules are flat, by Proposition 8.9.3.

Proposition 8.9.8 (Extension of scalars)**.** *Let $\varphi\colon R \to S$ be a homomorphism of commutative rings. For an R-module M, $S \otimes_R M$ is an S-module via $s(s' \otimes m) = ss' \otimes m$, and for every S-module N (an R-module via φ) there is a natural bijection $\mathrm{Hom}_S(S \otimes_R M, N) \cong \mathrm{Hom}_R(M,N)$, $f \mapsto (m \mapsto f(1 \otimes m))$.*

Proof. For $s \in S$, multiplication by s on S is R-linear, so it induces an endomorphism of $S \otimes_R M$; this gives the module structure. Given an R-linear $g\colon M \to N$, the map $(s, m) \mapsto sg(m)$ is R-bilinear and induces an S-linear map $S \otimes M \to N$; the two constructions are inverse because $1 \otimes m$ generate $S \otimes M$ as an S-module. □

8.10 Algebras and graded rings

Definition 8.10.1. Let R be commutative. An *R-algebra* is a ring A with a ring homomorphism $R \to A$ whose image lies in the center of A. Homomorphisms of R-algebras are ring homomorphisms compatible with the maps from R. A *graded ring* is a ring $A = \bigoplus_{n\in\mathbb{Z}} A_n$ (as abelian group) with $A_m A_n \subseteq A_{m+n}$; elements of A_n are *homogeneous of degree n*. A *graded module* over A is a module $M = \bigoplus_n M_n$ with $A_m M_n \subseteq M_{m+n}$. An ideal of a graded ring is *homogeneous* if it is generated by homogeneous elements.

Proposition 8.10.2. *Let A and B be commutative R-algebras. There is a unique ring structure on $A \otimes_R B$ with $(a \otimes b)(a' \otimes b') = aa' \otimes bb'$, and with the maps $a \mapsto a \otimes 1$ and $b \mapsto 1 \otimes b$ it is a coproduct of A and B in the category of commutative R-algebras.*

Proof. For fixed (a', b') the map $(a, b) \mapsto aa' \otimes bb'$ is bilinear, inducing $\mu_{a',b'}\colon A \otimes B \to A \otimes B$, and $(a', b') \mapsto \mu_{a',b'}$ is bilinear, inducing a bilinear multiplication on $A \otimes B$ with the required formula. Associativity, commutativity and the unit $1\otimes 1$ hold on generators, hence everywhere by bilinearity. Given algebra homomorphisms $f\colon A \to C$, $g\colon B \to C$ to a commutative algebra, the bilinear map $(a, b) \mapsto f(a)g(b)$ induces a linear map $A \otimes B \to C$, which is multiplicative on generators and hence a ring homomorphism; it is unique since the elements $a \otimes b = (a \otimes 1)(1 \otimes b)$ generate. □

Example 8.10.3. The polynomial ring $R[x_1, \dots, x_n]$ is graded by total degree, with A_d the span of the monomials of degree d. By Proposition 8.10.2 and Proposition 8.6.2, $R[x] \otimes_R R[y] \cong R[x, y]$. Graded rings of this kind are the homogeneous coordinate rings of projective varieties (Chapter 53).

Lemma 8.10.4. *An ideal I of a graded commutative ring A is homogeneous if and only if, for every $f \in I$, all homogeneous components of f lie in I.*

Proof. If I is generated by homogeneous g_j and $f = \sum_j a_j g_j \in I$, then the degree-n component of f is $\sum_j (a_j)_{n-\deg g_j} g_j \in I$. Conversely, if I contains the components of its elements, it is generated by these components. □

8.11 Modules over a principal ideal domain

In this section R is a principal ideal domain.

Proposition 8.11.1. *Every submodule of R^n is free of rank at most n.*

Proof. The proof of Proposition 7.11.4 applies verbatim, using that the image of a submodule under the last coordinate is an ideal (d) of R. □

Theorem 8.11.2 (Smith normal form)**.** *Let M be an $m \times n$ matrix over R. There are invertible matrices P, Q over R such that PMQ is diagonal with entries $d_1, \dots, d_r, 0, \dots, 0$, where $d_i \neq 0$ and $d_1 \mid d_2 \mid \cdots \mid d_r$.*

Proof. Since R is a UFD (Theorem 8.5.4), each nonzero $a \in R$ has a *length* $\ell(a)$, the number of prime factors counted with multiplicity; if $d \mid a$ and d is not associate to a, then $\ell(d) < \ell(a)$. Besides row and column swaps and adding multiples of one row (column) to another, we use the following operation: for entries a, b in the same row, in columns i and j, write $d = \gcd(a, b) = xa + yb$ (possible since $(a, b) = (d)$) and replace columns i, j by $x\,\mathrm{col}_i + y\,\mathrm{col}_j$ and $-(b/d)\mathrm{col}_i + (a/d)\mathrm{col}_j$. This is multiplication by an invertible matrix (its 2×2 block has determinant 1), and it replaces a, b by $d, 0$. The same operation exists for rows.

Now argue as in Theorem 7.11.5, with ℓ in place of the absolute value. Among all matrices obtained from M by these operations choose one whose nonzero entry of least length is as short as possible, and move it to position $(1, 1)$. If it did not divide an entry in the first row or column, the operation above would produce a proper divisor, of smaller length. So it divides these entries, and we clear the first row and column. If it did not divide some other entry, adding that row to the first row and applying the operation would again produce a shorter entry. Apply induction to the remaining block. □

Theorem 8.11.3 (Structure theorem)**.** *Let M be a finitely generated R-module. There are $r \geq 0$ and nonzero nonunits $d_1 \mid d_2 \mid \cdots \mid d_k$ in R with*

$$M \cong R^r \oplus R/(d_1) \oplus \cdots \oplus R/(d_k),$$

and r and the ideals $(d_1), \ldots, (d_k)$ are uniquely determined by M.

Proof. Existence is proved exactly as in Theorem 7.11.6, using Proposition 8.11.1 and Theorem 8.11.2. For uniqueness, the torsion submodule $T = \{m : am = 0 \text{ for some } a \neq 0\}$ is $\bigoplus R/(d_i)$, and $M/T \cong R^r$, so r is determined by Theorem 8.8.2. Let p be a prime of R and $j \geq 0$. For $d \neq 0$ with $v = v_p(d)$, the $R/(p)$-vector space $p^j(R/(d))/p^{j+1}(R/(d))$ is one-dimensional if $v \geq j + 1$ and zero otherwise: indeed $p^j(R/(d)) = (p^{\min(j,v)})/(d)$. Hence $\dim_{R/(p)} p^jT/p^{j+1}T$ is the number of i with $p^{j+1} \mid d_i$, which by Theorem 8.8.2 depends only on T. As in the proof of Theorem 7.11.6, these numbers determine k and the p-adic valuations of all d_i, hence the d_i up to units. □

Corollary 8.11.4. *Every finitely generated torsion R-module is a direct sum of modules $R/(p^e)$ with p prime and $e \geq 1$, and the multiset of the ideals (p^e) is uniquely determined.*

Proof. Apply Theorem 8.3.3 to $R/(d_i)$ with the prime factorisation of d_i; uniqueness follows from the uniqueness in Theorem 8.11.3, since the d_i are recovered from the prime powers by taking, for each p, the largest powers for d_k, the next largest for d_{k-1}, and so on. □

8.12 Exercises

Exercise 8.12.1. Let k be a field. Show that $M_n(k)$ has no two-sided ideals other than 0 and $M_n(k)$, although it has many left ideals for $n \geq 2$.

Exercise 8.12.2. Show that $\mathbb{Z}[x]/(x^2 + 1) \cong \mathbb{Z}[i]$, and that $\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$.

Exercise 8.12.3. Show that $(\mathbb{Z}/n\mathbb{Z})^\times$ consists of the classes of integers coprime to n, and deduce Fermat's little theorem: $a^{p-1} \equiv 1 \pmod{p}$ for p prime and $p \nmid a$.

Solution. If $\gcd(a, n) = 1$ then $ax + ny = 1$ by Theorem 4.5.3, so $[a][x] = 1$. If $d = \gcd(a, n) > 1$ then $[a][n/d] = 0$ with $[n/d] \neq 0$, so $[a]$ is a zero divisor and not a unit. For Fermat, apply Theorem 7.3.3 to the group $\mathbb{F}_p^\times$ of order $p - 1$. □

Exercise 8.12.4. Let p be prime. Show that $x^{p-1} + x^{p-2} + \cdots + 1$ is irreducible in $\mathbb{Q}[x]$ by applying Proposition 8.6.12 to its image under $x \mapsto x + 1$.

Exercise 8.12.5. Compute $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}$ and show that $\mathbb{Q}$ is a flat $\mathbb{Z}$-module which is not free.

Exercise 8.12.6. Let k be a field. Show that $k[x, y]$ is a UFD but not a PID.

Exercise 8.12.7. Let k be a field, V a finite-dimensional vector space and $T\colon V \to V$ linear. Using Example 8.7.2 and Theorem 8.11.3, show that $V \cong \bigoplus_i k[x]/(f_i)$ as $k[x]$-modules for monic $f_1 \mid \cdots \mid f_k$, and that f_k is the monic polynomial of least degree with $f_k(T) = 0$.

Chapter 9

Fields and Galois Theory

9.1 Introduction

This chapter studies field extensions: algebraic elements, splitting fields and algebraic closures, separability, the fundamental theorem of Galois theory, finite fields, cyclotomic fields and solvability by radicals. We give Artin's algebraic proof of the fundamental theorem of algebra, and we study transcendence bases, which show that $\mathbb{C}$ has an enormous group of automorphisms. These automorphisms reappear in the theory of absolute Hodge classes (Chapter 75).

References are [Lan02], [Art42] and [Ste15].

Prerequisites. Chapter 4 (Number Systems), Chapter 7 (Groups) and Chapter 8 (Rings and Modules).

9.2 Field extensions

Definition 9.2.1. A *field extension* L/K is a field L together with a subfield K (more generally, a homomorphism of fields $K \to L$, which is injective by Counterexample 6.8.9). Then L is a K-vector space, and its dimension $[L : K] = \dim_K L$ is the *degree* of the extension; the extension is *finite* if the degree is finite. For a subset $S \subseteq L$, $K(S)$ denotes the smallest subfield of L containing K and S, and $K[S]$ the smallest subring. A *K-homomorphism* between extensions of K is a field homomorphism that is the identity on K; $\mathrm{Aut}(L/K)$ is the group of K-automorphisms of L.

Theorem 9.2.2 (Tower law)**.** *Let $K \subseteq L \subseteq M$ be fields. If $(l_i)_{i\in I}$ is a basis of L over K and $(m_j)_{j\in J}$ a basis of M over L, then $(l_i m_j)_{(i,j)\in I\times J}$ is a basis of M over K. In particular $[M : K] = [M : L][L : K]$.*

Proof. Every $x \in M$ is $\sum_j a_j m_j$ with $a_j \in L$, and each $a_j = \sum_i c_{ij} l_i$ with $c_{ij} \in K$, so the $l_i m_j$ generate. If $\sum_{i,j} c_{ij} l_i m_j = 0$, then $\sum_j (\sum_i c_{ij} l_i) m_j = 0$ gives $\sum_i c_{ij} l_i = 0$ for all j, hence all $c_{ij} = 0$. □

Definition 9.2.3. Let L/K be an extension. An element $\alpha \in L$ is *algebraic* over K if $f(\alpha) = 0$ for some nonzero $f \in K[x]$, and *transcendental* otherwise. The extension is *algebraic* if all elements of L are algebraic over K.

Proposition 9.2.4. *Let $\alpha \in L$ be algebraic over K. The kernel of the evaluation homomorphism $K[x] \to L$, $f \mapsto f(\alpha)$, is generated by a unique monic irreducible polynomial m_α, the* minimal polynomial *of α. The evaluation induces an isomorphism $K[x]/(m_\alpha) \cong K[\alpha]$, so $K[\alpha] = K(\alpha)$ is a field with basis $1, \alpha, \ldots, \alpha^{d-1}$ over K, where $d = \deg m_\alpha = [K(\alpha) : K]$.*

Proof. The kernel is a nonzero ideal of the PID $K[x]$ (Theorem 8.6.5), hence generated by a unique monic polynomial. The image $K[\alpha]$ is a subring of a field, hence a domain, so the kernel is prime, and the generator is irreducible. Nonzero prime ideals of a PID are maximal (proof of Theorem 8.5.4), so $K[\alpha] \cong K[x]/(m_\alpha)$ is a field (Proposition 8.4.2). By division with remainder, the classes of $1, x, \dots, x^{d-1}$ form a basis of $K[x]/(m_\alpha)$. □

Proposition 9.2.5. *1. An element α is algebraic over K if and only if $[K(\alpha) : K] < \infty$. Every finite extension is algebraic.*

2. If $\alpha_1, \dots, \alpha_n$ are algebraic over K, then $K(\alpha_1, \dots, \alpha_n)/K$ is finite. The elements of L algebraic over K form a subfield of L.

3. If L/K and M/L are algebraic, then M/K is algebraic.

Proof. (1) If $[L : K] = n$ is finite and $\alpha \in L$, the $n+1$ elements $1, \alpha, \dots, \alpha^n$ are linearly dependent, which gives a nonzero polynomial with root α. The rest follows from Proposition 9.2.4. (2) Each α_i is algebraic over $K(\alpha_1, \dots, \alpha_{i-1})$, so the claim follows from the tower law. If α, β are algebraic, then $\alpha \pm \beta$, $\alpha\beta$ and α^{-1} lie in the finite extension $K(\alpha, \beta)$ and are algebraic by (1). (3) Let $\alpha \in M$ be a root of $c_n x^n + \cdots + c_0$ with $c_i \in L$ not all zero. Then α is algebraic over $K(c_0, \dots, c_n)$, which is finite over K by (2); so $K(c_0, \dots, c_n, \alpha)$ is finite over K and α is algebraic by (1). □

Example 9.2.6. $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$, since $x^2 - 2$ is irreducible by Proposition 4.6.4. The real number $\sqrt[3]{2}$ has minimal polynomial $x^3 - 2$, irreducible by Eisenstein's criterion (Proposition 8.6.12), so $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$. Hence $\sqrt[3]{2} \notin \mathbb{Q}(\sqrt{2})$, by the tower law. Transcendental numbers exist by Exercise 4.10.6.

Proposition 9.2.7 (Kronecker)**.** *Let $f \in K[x]$ be irreducible. Then $L = K[x]/(f)$ is an extension of K of degree* $\deg f$ *in which the class of x is a root of f.*

Proof. (f) is maximal, so L is a field; the composite $K \to K[x] \to L$ is injective, and $f(\bar{x}) = \overline{f(x)} = 0$. The degree is computed as in Proposition 9.2.4. □

9.3 Splitting fields and algebraic closures

Definition 9.3.1. Let $f \in K[x]$ be nonconstant. An extension L/K is a *splitting field* of f if $f = c(x - \alpha_1) \cdots (x - \alpha_n)$ in $L[x]$ and $L = K(\alpha_1, \dots, \alpha_n)$. A field is *algebraically closed* if every nonconstant polynomial over it has a root, and an *algebraic closure* of K is an algebraic extension that is algebraically closed.

Proposition 9.3.2. *Every nonconstant $f \in K[x]$ has a splitting field L, with $[L : K] \le (\deg f)!$.*

Proof. By induction on $n = \deg f$. Let g be an irreducible factor of f and $K_1 = K[x]/(g)$, in which f has a root α_1 and $[K_1 : K] \le n$. Over K_1, $f = (x - \alpha_1) f_1$ with $\deg f_1 = n - 1$; a splitting field of f_1 over K_1 is a splitting field of f over K of degree at most $n \cdot (n-1)!$. □

Theorem 9.3.3 (Extension of homomorphisms)**.** *Let $L = K(\alpha_1, \dots, \alpha_r)$ be a finite extension and $\tau : K \to M$ a field homomorphism. The number of homomorphisms $L \to M$ extending τ is at most $[L : K]$. For $i = 1, \dots, r$ let g_i be the minimal polynomial of α_i over $K(\alpha_1, \dots, \alpha_{i-1})$. If for every extension τ' of τ to $K(\alpha_1, \dots, \alpha_{i-1})$ the polynomial $g_i^{\tau'}$ splits into distinct linear factors in $M[x]$, then there are exactly $[L : K]$ extensions.*

Proof. By induction on r, using the tower law, it suffices to treat $r = 1$: $L = K(\alpha)$ with minimal polynomial g. An extension σ of τ is determined by $\sigma(\alpha)$, which is a root of g^τ (the polynomial obtained by applying τ to the coefficients). Conversely, for each root $\beta \in M$ of g^τ, the homomorphism $K[x] \to M$ extending τ with $x \mapsto \beta$ kills g and so induces an extension $K(\alpha) \cong K[x]/(g) \to M$. Hence the extensions correspond to the roots of g^τ in M, of which there are at most $\deg g = [K(\alpha) : K]$ by Corollary 8.6.6, with equality if g^τ splits into distinct linear factors. □

Corollary 9.3.4. *Let L and L' be splitting fields of $f \in K[x]$. Then there is a K-isomorphism $L \to L'$. For every finite extension L/K, $|\operatorname{Aut}(L/K)| \le [L : K]$.*

Proof. Write $L = K(\alpha_1, \dots, \alpha_n)$ with the α_i the roots of f. At each stage the minimal polynomial g_i divides f, and f splits in L', so $g_i^{\tau'}$ has a root in L' and the proof of Theorem 9.3.3 gives at least one extension of the inclusion $K \to L'$ to $\sigma : L \to L'$. It is injective, and its image contains all roots of f in L' (it contains n roots counted with multiplicity), so it is L'. The second statement is the first part of Theorem 9.3.3 with $M = L$. □

Theorem 9.3.5. *Every field K has an algebraic closure $\bar{K}$. If L/K is algebraic and $\tau : K \to M$ is a homomorphism into an algebraically closed field, then τ extends to $L \to M$. Any two algebraic closures of K are K-isomorphic.*

Proof. Existence (Artin). Let $\mathcal{F}$ be the set of monic nonconstant polynomials in $K[x]$, let $A = K[x_f : f \in \mathcal{F}]$ be the polynomial ring in variables indexed by $\mathcal{F}$ (the union of the polynomial rings in finitely many of them), and I the ideal generated by the $f(x_f)$. Then $I \neq A$: otherwise $1 = \sum_{i=1}^n g_i f_i(x_{f_i})$ for finitely many f_i; in a splitting field of $f_1 \cdots f_n$ choose roots β_i of f_i, and apply the homomorphism $x_{f_i} \mapsto \beta_i$, other variables $\mapsto 0$, to get $1 = 0$. Let $\mathfrak{m} \supseteq I$ be maximal (Theorem 8.4.3) and $E_1 = A/\mathfrak{m}$, a field extension of K in which every $f \in \mathcal{F}$ has a root. Repeating, we get $K = E_0 \subseteq E_1 \subseteq E_2 \subseteq \cdots$ where every nonconstant polynomial over E_n has a root in E_{n+1}. The union E is algebraically closed, since a polynomial over E has coefficients in some E_n. The elements of E algebraic over K form a field $\bar{K}$ (Proposition 9.2.5). If $g \in \bar{K}[x]$ is nonconstant, a root $\alpha \in E$ is algebraic over $\bar{K}$, hence over K, so $\alpha \in \bar{K}$. Thus $\bar{K}$ is an algebraic closure.

Extension. Consider pairs (F, σ) with $K \subseteq F \subseteq L$ a subfield and $\sigma : F \to M$ extending τ, ordered by extension. Chains have upper bounds (take unions), so by Zorn's lemma there is a maximal (F, σ). If $\alpha \in L \setminus F$, its minimal polynomial over F maps to a polynomial with a root in M, and Theorem 9.3.3 extends σ to $F(\alpha)$; contradiction. So $F = L$.

Uniqueness. If $\bar{K}$ and $\bar{K}'$ are algebraic closures, extend the inclusion $K \to \bar{K}'$ to $\sigma : \bar{K} \to \bar{K}'$. Its image is an algebraically closed field over which $\bar{K}'$ is algebraic, so every element of $\bar{K}'$ is a root of a polynomial that splits over $\sigma(\bar{K})$, and σ is surjective. □

9.4 Separability

Definition 9.4.1. A nonzero polynomial $f \in K[x]$ is *separable* if it has no repeated root in a splitting field. An algebraic element is *separable* if its minimal polynomial is, and an algebraic extension is separable if all its elements are. The *derivative* of $f = \sum a_i x^i$ is $f' = \sum i a_i x^{i-1}$; it satisfies $(fg)' = f'g + fg'$.

Proposition 9.4.2. *A nonzero polynomial f is separable if and only if $\gcd(f, f') = 1$ in $K[x]$. An irreducible polynomial is separable if and only if $f' \neq 0$. If $\operatorname{char} K = 0$, every irreducible polynomial is separable; if $\operatorname{char} K = p > 0$, an irreducible f is inseparable if and only if $f = g(x^p)$ for some $g \in K[x]$.*

Proof. In a splitting field, if $f = (x-\alpha)^2 h$ then $f'(\alpha) = 0$; if $f = (x-\alpha)h$ with $h(\alpha) \neq 0$ then $f'(\alpha) = h(\alpha) \neq 0$. So f has a repeated root iff f and f' have a common root in a splitting field, iff they have a common factor of positive degree in $K[x]$ (a common root α makes m_α a common factor; conversely a common factor has a root in the splitting field). If f is irreducible and $f' \neq 0$, the gcd divides f and has degree $<$ deg f, so it is 1. If $f' = 0$ then $\gcd(f, f') = f$. Finally $f' = 0$ iff $ia_i = 0$ for all i, which in characteristic 0 means f is constant, and in characteristic p means $a_i = 0$ unless $p \mid i$. □

Definition 9.4.3. A field K is *perfect* if it has characteristic 0, or characteristic p and the *Frobenius* $x \mapsto x^p$ is surjective. (The Frobenius is always an injective ring homomorphism in characteristic p, since p divides the binomial coefficients $\binom{p}{i}$ for $0 < i < p$.)

Proposition 9.4.4. *Over a perfect field every irreducible polynomial is separable, so every algebraic extension is separable. Finite fields are perfect.*

Proof. If $\operatorname{char} K = p$ and $f = \sum b_i x^{pi}$ with $b_i = c_i^p$, then $f = (\sum c_i x^i)^p$ is not irreducible. A finite field is perfect because an injective map from a finite set to itself is surjective. □

Counterexample 9.4.5. Let $K = \mathbb{F}_p(t)$ be the field of fractions of $\mathbb{F}_p[t]$. The polynomial $f = x^p - t$ is irreducible, by Eisenstein's criterion for the prime t of the UFD $\mathbb{F}_p[t]$, but $f' = 0$. In a splitting field, with $\beta^p = t$, we have $f = (x-\beta)^p$: a single root of multiplicity p. So K is not perfect and $K(\beta)/K$ is not separable.

9.5 Galois extensions

Definition 9.5.1. An algebraic extension L/K is *normal* if every irreducible polynomial in $K[x]$ with a root in L splits into linear factors in $L[x]$. For a group G of automorphisms of a field L, the *fixed field* is $L^G = \{x \in L : \sigma(x) = x \text{ for all } \sigma \in G\}$.

Lemma 9.5.2 (Artin)**.** *Let G be a finite group of automorphisms of a field L. Then $[L : L^G] \leq |G|$.*

Proof. Let $K = L^G$, $G = \{\sigma_1, \ldots, \sigma_n\}$ with $\sigma_1 = \mathrm{id}$, and let $\alpha_1, \ldots, \alpha_m \in L$ with $m > n$. The m vectors $(\sigma_i(\alpha_j))_i \in L^n$ are linearly dependent over L by Lemma 8.8.1, so the system $\sum_j \sigma_i(\alpha_j) x_j = 0$, $i = 1, \ldots, n$, has a nonzero solution $(x_j) \in L^m$. Choose one with the fewest nonzero entries and, scaling, with $x_{j_0} = 1$ for some j_0. For $\tau \in G$, applying τ gives $\sum_j \tau\sigma_i(\alpha_j)\tau(x_j) = 0$ for all i, and since $\tau\sigma_i$ runs over G as i does, $(\tau(x_j))_j$ is also a solution. The difference $(x_j - \tau(x_j))_j$ is a solution with fewer nonzero entries (the j_0 entry vanishes), so it is zero. Hence all $x_j \in K$, and the equation for $\sigma_1 = \mathrm{id}$ is a nontrivial K-linear relation $\sum_j \alpha_j x_j = 0$. So any $m > n$ elements are linearly dependent over K. □

Theorem 9.5.3. *For a finite extension L/K with $G = \mathrm{Aut}(L/K)$, the following are equivalent:*

1. $|G| = [L : K]$;
2. $L^G = K$;
3. *L is a splitting field over K of a separable polynomial;*
4. *L/K is normal and separable.*

Such an extension is called Galois*, and* $\mathrm{Gal}(L/K) = G$ *its* Galois group*. Moreover, if G is any finite group of automorphisms of a field L, then L/L^G is Galois with group G.*

Proof. For the last statement: $G \subseteq \mathrm{Aut}(L/L^G)$, so by Lemma 9.5.2 and Corollary 9.3.4, $[L : L^G] \leq |G| \leq |\mathrm{Aut}(L/L^G)| \leq [L : L^G]$.

(1) $\Rightarrow$ (2): $K \subseteq L^G$ and by the last statement $[L : L^G] = |G| = [L : K]$, so $L^G = K$ by the tower law.

(2) $\Rightarrow$ (4): Let $\alpha \in L$ and let $\alpha_1, \ldots, \alpha_r$ be the distinct elements of the orbit $G\alpha$. The polynomial $g = \prod_i (x - \alpha_i)$ has coefficients fixed by G, hence in K, and $g(\alpha) = 0$, so $m_\alpha \mid g$. Every $\alpha_i = \sigma(\alpha)$ is a root of m_α, so $g \mid m_\alpha$. Thus $m_\alpha = g$ splits in L with distinct roots.

(4) $\Rightarrow$ (3): Write $L = K(\beta_1, \ldots, \beta_s)$ and let f be the product of the distinct minimal polynomials of the β_j. It is separable (distinct monic irreducible polynomials have no common root), it splits in L by normality, and its roots generate L.

(3) $\Rightarrow$ (1): Apply Theorem 9.3.3 with $M = L$ and τ the inclusion $K \to L$: at every stage the minimal polynomial divides f, which splits into distinct linear factors in L. So there are exactly $[L : K]$ K-homomorphisms $L \to L$. Each is an injective K-linear map of the finite-dimensional K-vector space L to itself, hence bijective by Theorem 8.8.5 (its image is a subspace of the same finite dimension). So $|G| = [L : K]$. □

Theorem 9.5.4 (Fundamental theorem of Galois theory)**.** *Let L/K be a finite Galois extension with group G. The maps $H \mapsto L^H$ and $F \mapsto \mathrm{Gal}(L/F)$ are mutually inverse, inclusion-reversing bijections between subgroups $H \leq G$ and intermediate fields $K \subseteq F \subseteq L$. Moreover $[L : L^H] = |H|$ and $[L^H : K] = [G : H]$. An intermediate field F is Galois over K if and only if $\mathrm{Gal}(L/F)$ is normal in G, and then restriction induces an isomorphism $G/\mathrm{Gal}(L/F) \cong \mathrm{Gal}(F/K)$.*

Proof. If L is the splitting field of a separable $f \in K[x]$, it is also the splitting field of f over F, so L/F is Galois and $L^{\mathrm{Gal}(L/F)} = F$ by Theorem 9.5.3. For $H \leq G$, the last statement of that theorem gives $\mathrm{Gal}(L/L^H) = H$ and $[L : L^H] = |H|$; then $[L^H : K] = [L : K]/|H| = [G : H]$ by the tower law. The maps clearly reverse inclusions.

For $\sigma \in G$, $\mathrm{Gal}(L/\sigma(F)) = \sigma\mathrm{Gal}(L/F)\sigma^{-1}$. If F/K is Galois, it is normal, so σ maps each $\alpha \in F$ to a root of m_α, which lies in F; thus $\sigma(F) = F$ and $\mathrm{Gal}(L/F)$ is normal. Conversely, if $N = \mathrm{Gal}(L/F)$ is normal, then $\sigma(F) = L^{\sigma N \sigma^{-1}} = F$ for all σ, and restriction is a homomorphism $G \to \mathrm{Aut}(F/K)$ with kernel N. Its image has $[G : N] = [F : K]$ elements and lies in a group with at most $[F : K]$ elements (Corollary 9.3.4), so $|\mathrm{Aut}(F/K)| = [F : K]$, F/K is Galois and the restriction map is surjective. □

Example 9.5.5. The field $L = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ is the splitting field of $(x^2 - 2)(x^2 - 3)$, of degree 4 over $\mathbb{Q}$: if $\sqrt{3} = a + b\sqrt{2}$ with $a, b \in \mathbb{Q}$, squaring gives $2ab = 0$, and neither $a = 0$ nor $b = 0$ is possible by the argument of Proposition 4.6.4. Its Galois group consists of the four automorphisms $\sqrt{2} \mapsto \pm\sqrt{2}$, $\sqrt{3} \mapsto \pm\sqrt{3}$, so it is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. It has three subgroups of order 2, corresponding to the three quadratic subfields $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$ and $\mathbb{Q}(\sqrt{6})$.

Counterexample 9.5.6. The extension $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ of degree 3 is not Galois. A $\mathbb{Q}$-automorphism must send $\sqrt[3]{2}$ to a cube root of 2 in $\mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{R}$, and the only real cube root of 2 is $\sqrt[3]{2}$ (Proposition 4.7.4); so the automorphism group is trivial. The polynomial $x^3 - 2$ has a root in $\mathbb{Q}(\sqrt[3]{2})$ but does not split there, so the extension is not normal.

9.6 Finite fields

Lemma 9.6.1. *Every finite subgroup of the multiplicative group of a field is cyclic.*

Proof. Let G be such a subgroup. By Theorem 7.11.6, $G \cong \mathbb{Z}/d_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/d_k\mathbb{Z}$ with $d_1 \mid \cdots \mid d_k$, so every element satisfies $x^{d_k} = 1$. The polynomial $x^{d_k} - 1$ has at most d_k roots, so $|G| \leq d_k$ and $k \leq 1$. □

Theorem 9.6.2. *1. A finite field has $q = p^n$ elements, where p is its characteristic and n its degree over $\mathbb{F}_p$.*

2. For every prime power $q = p^n$ there is a field $\mathbb{F}_q$ with q elements, unique up to isomorphism: a splitting field of $x^q - x$ over $\mathbb{F}_p$, consisting of the roots of this polynomial.

3. $\mathbb{F}_{q^n}/\mathbb{F}_q$ is Galois with cyclic group of order n, generated by the Frobenius $x \mapsto x^q$. The subfields of $\mathbb{F}_{p^n}$ are the $\mathbb{F}_{p^m}$ with $m \mid n$.

Proof. (1) The characteristic is prime by Lemma 8.4.6, and F is a vector space of some finite dimension n over $\mathbb{F}_p$. (2) Let L be a splitting field of $f = x^q - x$. Since $f' = -1$, f has q distinct roots. The set of roots is closed under sums, products and inverses because $x \mapsto x^q$ is a ring homomorphism (a power of the Frobenius), so it is a subfield, and it equals L. If F has q elements, then $F^\times$ has order $q-1$, so $x^q = x$ for all $x \in F$ by Theorem 7.3.3; hence F is a splitting field of f and $F \cong L$ by Corollary 9.3.4. (3) $\mathbb{F}_{q^n}$ is the splitting field of the separable polynomial $x^{q^n} - x$ over $\mathbb{F}_q$, hence Galois of degree n. The Frobenius $\phi(x) = x^q$ fixes $\mathbb{F}_q$, and $\phi^k = \mathrm{id}$ means all q^n elements are roots of $x^{q^k} - x$, which requires $k \geq n$. So ϕ has order n and generates the Galois group. By the fundamental theorem, subfields correspond to subgroups of $\mathbb{Z}/n\mathbb{Z}$, one for each divisor. □

Theorem 9.6.3 (Primitive element theorem)**.** *Every finite separable extension L/K is* simple*: $L = K(\gamma)$ for some $\gamma \in L$.*

Proof. If K is finite, so is L, and a generator γ of the cyclic group $L^\times$ (Lemma 9.6.1) works. Let K be infinite. By induction on the number of generators it suffices to treat $L = K(\alpha, \beta)$. Let M be a splitting field of $m_\alpha m_\beta$ over L. The minimal polynomials over intermediate fields divide m_α or m_β, which are separable, so by Theorem 9.3.3 there are exactly $n = [L : K]$ distinct K-homomorphisms $\sigma_1, \ldots, \sigma_n \colon L \to M$. The polynomial $P(t) = \prod_{i<j}\big((\sigma_i(\alpha) - \sigma_j(\alpha)) + t(\sigma_i(\beta) - \sigma_j(\beta))\big) \in M[t]$ is nonzero, because $\sigma_i \neq \sigma_j$ differ on α or on β. It has finitely many roots, so there is $c \in K$ with $P(c) \neq 0$. Then $\gamma = \alpha + c\beta$ has n distinct images $\sigma_i(\gamma)$. The restrictions of the σ_i to $K(\gamma)$ are therefore distinct, so $[K(\gamma) : K] \geq n$ by Theorem 9.3.3, and $K(\gamma) = L$. □

9.7 The fundamental theorem of algebra

Lemma 9.7.1. *Every polynomial of odd degree in $\mathbb{R}[x]$ has a root in $\mathbb{R}$.*

Proof. We may assume $p = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ is monic, with n odd. Let $M = 1 + \sum_i |a_i|$. For $|x| \geq M$ we have $|a_{n-1}x^{n-1} + \cdots + a_0| \leq (M-1)|x|^{n-1} < |x|^n$, so $p(x)$ has the sign of x^n, that is, of x. On $[-M, M]$, the identity $x^i - y^i = (x-y)\sum_k x^k y^{i-1-k}$ gives $|p(x) - p(y)| \leq C|x-y|$ with $C = \sum_{i=1}^n i|a_i|M^{i-1}$ (with $a_n = 1$). Let $S = \{x \in [-M, M] : p(x) < 0\}$, which contains $-M$, and $c = \sup S$. If $p(c) < 0$, then $c < M$ and $p < 0$ on $[c, c + |p(c)|/C) \cap [-M, M]$, contradicting $c = \sup S$. If $p(c) > 0$, then $c > -M$ and $p > 0$ on $(c - p(c)/C, c]$, so $\sup S \leq c - p(c)/C$, again a contradiction. Hence $p(c) = 0$. □

Lemma 9.7.2. *Every group of order p^n with $n \geq 1$ has a subgroup of index p.*

Proof. By induction on n. By Proposition 7.6.7 and Theorem 7.6.8 there is $z \in Z(G)$ of order p. If $n = 1$ take $\{e\}$. Otherwise $G/\langle z\rangle$ has order p^{n-1} and, by induction, a subgroup of index p, whose preimage in G has index p. □

Theorem 9.7.3 (Fundamental theorem of algebra)**.** *The field $\mathbb{C}$ is algebraically closed.*

Proof. Let $f \in \mathbb{C}[x]$ be nonconstant, and let $\bar{f}$ be obtained by conjugating the coefficients. Then $g = (x^2+1)f\bar{f}$ lies in $\mathbb{R}[x]$. Let L be a splitting field of g over $\mathbb{C}$; it is also a splitting field of g over $\mathbb{R}$ (since $\mathbb{C} = \mathbb{R}(i)$ and i is a root of g), hence Galois over $\mathbb{R}$ as $\operatorname{char}\mathbb{R} = 0$. Let $G = \mathrm{Gal}(L/\mathbb{R})$ and let P be a Sylow 2-subgroup. The fixed field L^P has odd degree $[G : P]$ over $\mathbb{R}$; by Theorem 9.6.3 it is $\mathbb{R}(\gamma)$, and the minimal polynomial of γ is irreducible of odd degree, hence of degree 1 by Lemma 9.7.1. So $P = G$ is a 2-group, and so is $H = \mathrm{Gal}(L/\mathbb{C})$. If $H \neq \{e\}$, it has a subgroup of index 2 (Lemma 9.7.2), whose fixed field is an extension of $\mathbb{C}$ of degree 2, generated by a root of an irreducible quadratic polynomial over $\mathbb{C}$. This contradicts Proposition 4.9.3. So $L = \mathbb{C}$, and f has its roots in $\mathbb{C}$. □

Corollary 9.7.4. *Every nonconstant polynomial over $\mathbb{C}$ is a product of linear factors, and $\mathbb{C}$ is an algebraic closure of $\mathbb{R}$. The irreducible polynomials in $\mathbb{R}[x]$ are the linear ones and the quadratic ones without real roots.*

Proof. The first statement follows by induction on the degree. Since $[\mathbb{C} : \mathbb{R}] = 2$, the extension is algebraic. An irreducible real polynomial of degree ≥ 2 has a non-real root α; then $\bar{\alpha}$ is also a root, and $(x-\alpha)(x-\bar{\alpha}) \in \mathbb{R}[x]$ divides it. □

9.8 Cyclotomic fields

Definition 9.8.1. For $n \geq 1$, the n-th roots of unity in $\mathbb{C}$ form a group μ_n of order n (the polynomial $x^n - 1$ has derivative nx^{n-1}, so its n roots are distinct), cyclic by Lemma 9.6.1. Its generators are the *primitive* n-th roots of unity. The n-th *cyclotomic polynomial* is $\Phi_n = \prod_\zeta (x-\zeta)$ over the primitive n-th roots of unity ζ; it has degree $\varphi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times|$.

Theorem 9.8.2. *Φ_n lies in $\mathbb{Z}[x]$ and is irreducible over $\mathbb{Q}$. The extension $\mathbb{Q}(\zeta)/\mathbb{Q}$ for a primitive n-th root of unity ζ is Galois, and $\sigma \mapsto a_\sigma$, where $\sigma(\zeta) = \zeta^{a_\sigma}$, is an isomorphism $\mathrm{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$.*

Proof. Every n-th root of unity is a primitive d-th root of unity for exactly one $d \mid n$, so $x^n - 1 = \prod_{d|n} \Phi_d$. By induction on n, Φ_n is the quotient of $x^n - 1$ by a monic polynomial in $\mathbb{Z}[x]$, hence lies in $\mathbb{Z}[x]$ by division with remainder.

Let f be the minimal polynomial of ζ over $\mathbb{Q}$. Since $f \mid \Phi_n$ in $\mathbb{Q}[x]$, write $\Phi_n = fg$; by Lemma 8.6.10 (and since f, g are monic) $f, g \in \mathbb{Z}[x]$. Let p be a prime with $p \nmid n$. We claim $f(\zeta^p) = 0$. Otherwise $g(\zeta^p) = 0$, so $f(x) \mid g(x^p)$ in $\mathbb{Z}[x]$. Reducing modulo p and using $\bar{g}(x^p) = \bar{g}(x)^p$ in $\mathbb{F}_p[x]$, we find that $\bar{f}$ and $\bar{g}$ have a common irreducible factor, so $\bar{\Phi}_n = \bar{f}\bar{g}$, and hence $x^n - 1$, has a repeated root in a splitting field over $\mathbb{F}_p$. But the derivative nx^{n-1} is coprime to $x^n - 1$ in $\mathbb{F}_p[x]$, a contradiction. Every primitive root of unity is ζ^a with $\gcd(a, n) = 1$, and writing a as a product of primes not dividing n, repeated application of the claim shows that ζ^a is a root of f. So $f = \Phi_n$.

$\mathbb{Q}(\zeta)$ contains all n-th roots of unity (powers of ζ), so it is the splitting field of $x^n - 1$, hence Galois. Each σ maps ζ to a primitive root ζ^{a_σ}, and $\sigma \mapsto a_\sigma$ is an injective homomorphism into $(\mathbb{Z}/n\mathbb{Z})^\times$, surjective since $|\mathrm{Gal}| = \deg \Phi_n = \varphi(n)$. □

9.9 Solvability by radicals

In this section all fields have characteristic 0.

Definition 9.9.1. A *radical extension* of K is an extension E/K with a chain $K = E_0 \subseteq E_1 \subseteq \cdots \subseteq E_r = E$ where $E_{i+1} = E_i(a_i)$ and $a_i^{n_i} \in E_i$ for some $n_i \geq 1$. A polynomial $f \in K[x]$ is *solvable by radicals* if a splitting field of f is contained in a radical extension of K.

Lemma 9.9.2. *Let K contain a primitive n-th root of unity ω and let b in an extension satisfy $b^n = a \in K$. Then $K(b)/K$ is Galois with cyclic Galois group. For any K and $m \geq 1$, the splitting field of $x^m - 1$ over K is Galois with abelian Galois group.*

Proof. If $b = 0$ there is nothing to prove. Otherwise the roots of $x^n - a$ are the distinct $\omega^i b$, all in $K(b)$, so $K(b)$ is a splitting field of a separable polynomial. Each σ satisfies $\sigma(b) = \omega^{k_\sigma} b$, and since σ fixes ω, $\sigma \mapsto \omega^{k_\sigma}$ is an injective homomorphism into the cyclic group generated by ω. For the second statement, the roots of $x^m - 1$ form a cyclic group generated by some ζ (Lemma 9.6.1; they are distinct in characteristic 0), and $\sigma(\zeta) = \zeta^{a_\sigma}$ defines an injective homomorphism into the abelian group $(\mathbb{Z}/m\mathbb{Z})^\times$. □

Theorem 9.9.3 (Galois)**.** *If $f \in K[x]$ is solvable by radicals, then the Galois group of a splitting field of f over K is solvable.*

Proof. Let $L \subseteq E$ be a splitting field of f inside a radical extension with data a_i, n_i as in Definition 9.9.1. Let $m = n_0 \cdots n_{r-1}$, $h = (x^m - 1) \prod_i m_{a_i}$, and let N be a splitting field of h over E (Proposition 9.3.2). Since E is generated over K by roots of h, N is also a splitting field of h over K. Then N/K is Galois with group G, and N is generated over K by a primitive m-th root of unity ζ and the elements $\sigma(a_i)$, $\sigma \in G$, since the roots of m_{a_i} are the $\sigma(a_i)$ (proof of Theorem 9.5.3). Adjoin these generators in the order ζ, then all $\sigma(a_0)$, then all $\sigma(a_1)$, and so on, obtaining a chain $K = F_0 \subseteq F_1 = K(\zeta) \subseteq F_2 \subseteq \cdots \subseteq F_s = N$. Each $F_{j+1} = F_j(b)$ with F_1/F_0 abelian Galois, and for $j \geq 1$, $b = \sigma(a_i)$ with $b^{n_i} = \sigma(a_i^{n_i})$ in F_j (because $a_i^{n_i} \in E_i = K(a_0, \ldots, a_{i-1})$) and F_j containing a primitive n_i-th root of unity. By Lemma 9.9.2 each F_{j+1}/F_j is Galois with abelian group. By the fundamental theorem, $G_j = \mathrm{Gal}(N/F_j)$ satisfies $G_{j+1} \trianglelefteq G_j$ and $G_j/G_{j+1} \cong \mathrm{Gal}(F_{j+1}/F_j)$ is abelian; so G is solvable (Proposition 7.9.3), and so is its quotient $\mathrm{Gal}(L/K) \cong G/\mathrm{Gal}(N/L)$. □

Remark 9.9.4. The converse also holds: if the Galois group is solvable, then f is solvable by radicals. The proof uses Hilbert's Theorem 90 to show that cyclic extensions of fields containing enough roots of unity are generated by radicals; see [Lan02, Chapter VI, §7].

Lemma 9.9.5. *Let p be prime and $G \leq S_p$ a subgroup acting transitively on $\{1, \ldots, p\}$ and containing a transposition. Then $G = S_p$.*

Proof. The orbit has size p, so p divides $|G|$ (Theorem 7.6.2) and G contains an element of order p (Theorem 7.6.8), necessarily a p-cycle σ. Let $(a\ b) \in G$. Some power of σ sends a to b and is again a p-cycle; renumbering, we may assume it is $(1\,2\,\cdots\,p)$ and the transposition is $(1\,2)$. Conjugating $(1\,2)$ by powers of the p-cycle gives all $(i\ \ i{+}1)$, and these generate S_p since $(i\ j) = (j{-}1\ j)(i\ j{-}1)(j{-}1\ j)$ recursively. □

Example 9.9.6. The polynomial $f = x^5 - 4x + 2$ is not solvable by radicals over $\mathbb{Q}$. It is irreducible by Eisenstein's criterion at $p = 2$. Its roots in $\mathbb{C}$ (Theorem 9.7.3) satisfy $\sum \alpha_i = 0$ and $\sum_{i<j} \alpha_i \alpha_j = 0$ (the coefficients of x^4 and x^3 vanish), so $\sum \alpha_i^2 = 0$; hence not all roots are real. Since $f(-2) < 0 < f(0)$, $f(1) < 0 < f(2)$, the proof of Lemma 9.7.1 gives real roots in $(-2, 0)$, $(0, 1)$ and $(1, 2)$. Non-real roots come in conjugate pairs, so there are exactly 3 real and 2 non-real roots. Complex conjugation restricts to an element of the Galois group G of the splitting field in $\mathbb{C}$ that exchanges the two non-real roots and fixes the others: a transposition. G acts transitively on the roots (they are roots of one irreducible polynomial, and Theorem 9.3.3 moves any root to any other). By Lemma 9.9.5, $G \cong S_5$, which is not solvable (Proposition 7.9.3).

9.10 Transcendence bases

Definition 9.10.1. Let L/K be an extension. A subset $S \subseteq L$ is *algebraically independent* over K if for all distinct $s_1, \dots, s_n \in S$ the homomorphism $K[x_1, \dots, x_n] \to L$, $x_i \mapsto s_i$, is injective. A *transcendence basis* is an algebraically independent set S such that L is algebraic over $K(S)$. An extension is *purely transcendental* if $L = K(S)$ for an algebraically independent S.

Lemma 9.10.2. *Let $S \subseteq L$ be algebraically independent over K and $\alpha \in L$. Then $S \cup \{\alpha\}$ is algebraically independent if and only if $\alpha \notin S$ and α is transcendental over $K(S)$.*

Proof. A nonzero polynomial relation $P(s_1, \dots, s_n, \alpha) = 0$ with $P \notin K[x_1, \dots, x_n]$ is, after collecting powers of α, a nonzero polynomial over $K[s_1, \dots, s_n] \cong K[x_1, \dots, x_n]$ with root α; and conversely a nonzero polynomial over $K(S)$ with root α can be cleared of denominators to give such a relation. □

Theorem 9.10.3. *Every extension L/K has a transcendence basis, and any two have the same cardinality, the* transcendence degree $\operatorname{trdeg}_K L$. *More precisely, if $T \subseteq L$ is such that L is algebraic over $K(T)$ and $S_0 \subseteq T$ is algebraically independent, then there is a transcendence basis S with $S_0 \subseteq S \subseteq T$.*

Proof. By Zorn's lemma choose a maximal algebraically independent subset S with $S_0 \subseteq S \subseteq T$. By Lemma 9.10.2 every $t \in T$ is algebraic over $K(S)$, so $K(T)$ is algebraic over $K(S)$ and L is algebraic over $K(S)$ by Proposition 9.2.5.

We use the following fact: if F is a field, s is algebraic over $F(t)$ but not over F, then t is algebraic over $F(s)$. Indeed, clearing denominators gives a nonzero $P = \sum_k P_k(X)Y^k \in F[X, Y]$ with $P(s, t) = 0$; not all $P_k(s)$ vanish, since otherwise all $P_k = 0$ as s is transcendental over F; so t is a root of the nonzero polynomial $\sum_k P_k(s)Y^k$.

Let S and T be transcendence bases. Suppose first $T = \{t_1, \dots, t_n\}$ is finite and let $s_1, \dots, s_m \in S$ be distinct. Suppose that after $i < m$ steps L is algebraic over $K(s_1, \dots, s_i, T_i)$ with $T_i \subseteq T$ of size $n - i$. Then s_{i+1} is algebraic over this field; let $T' \subseteq T_i$ be minimal such that s_{i+1} is algebraic over $K(s_1, \dots, s_i, T')$. Since s_{i+1} is not algebraic over $K(s_1, \dots, s_i)$, T' is nonempty; pick $t \in T'$ and let $F = K(s_1, \dots, s_i, T' \setminus \{t\})$. By the fact above, t is algebraic over $F(s_{i+1})$, so with $T_{i+1} = T_i \setminus \{t\}$ the field $K(s_1, \dots, s_i, T_i)$ is algebraic over $K(s_1, \dots, s_{i+1}, T_{i+1})$, and so is L. In particular $T_i \neq \emptyset$ at every step $i < m$, so $m \leq n$. Hence S is finite with $|S| \leq |T|$, and by symmetry $|S| = |T|$.

If both are infinite, each $t \in T$ is algebraic over $K(S_t)$ for a finite $S_t \subseteq S$, and $S' = \bigcup_t S_t$ satisfies $|S'| \leq |T|$ by Proposition 3.7.5. Then L is algebraic over $K(S')$, so every $s \in S$ is algebraic over $K(S')$, which forces $S' = S$ by Lemma 9.10.2. So $|S| \leq |T|$, and by symmetry the cardinalities agree. □

Theorem 9.10.4. *Let Ω and Ω' be algebraically closed fields of the same characteristic, with prime fields $P \subseteq \Omega$ and $P' \subseteq \Omega'$, and $\operatorname{trdeg}_P \Omega = \operatorname{trdeg}_{P'} \Omega'$. Then every bijection between transcendence bases $B \to B'$ extends to an isomorphism $\Omega \to \Omega'$.*

Proof. The prime fields are isomorphic ($\mathbb{Q}$ or $\mathbb{F}_p$). A bijection $B \to B'$ induces an isomorphism of polynomial rings $P[B] \cong P'[B']$ and hence of their fields of fractions $P(B) \cong P'(B')$. The field Ω is an algebraic closure of $P(B)$ and Ω' one of $P'(B')$, so by Theorem 9.3.5 the isomorphism extends to an isomorphism $\Omega \to \Omega'$. □

Corollary 9.10.5. $\operatorname{trdeg}_{\mathbb{Q}} \mathbb{C} = 2^{\aleph_0}$. *If $\alpha, \beta \in \mathbb{C}$ are both transcendental over $\mathbb{Q}$, there is a field automorphism σ of $\mathbb{C}$ with $\sigma(\alpha) = \beta$. Every automorphism of the algebraic closure $\bar{\mathbb{Q}}$ of $\mathbb{Q}$ in $\mathbb{C}$ extends to an automorphism of $\mathbb{C}$.*

Proof. If B is a transcendence basis, $\mathbb{C}$ is algebraic over $\mathbb{Q}(B)$, and a field algebraic over a field of infinite cardinality κ has cardinality κ (each element is a root of one of κ polynomials, each with finitely many roots); $\mathbb{Q}(B)$ has cardinality $\max(|B|, \aleph_0)$. Since $|\mathbb{C}| = 2^{\aleph_0}$ (Proposition 4.8.6) is uncountable, $|B| = 2^{\aleph_0}$. Extend $\{\alpha\}$ and $\{\beta\}$ to transcendence bases B, B' (Theorem 9.10.3 with $T = \mathbb{C}$ and $S_0 = \{\alpha\}$, respectively $S_0 = \{\beta\}$); both have cardinality $2^{\aleph_0}$, and removing α and β leaves sets of the same cardinality, so there is a bijection $B \to B'$ with $\alpha \mapsto \beta$. Apply Theorem 9.10.4. For the last statement, choose a transcendence basis B of $\mathbb{C}$ over $\bar{\mathbb{Q}}$ (it is also one over $\mathbb{Q}$), extend the automorphism to $\bar{\mathbb{Q}}(B)$ by fixing B, and then to $\mathbb{C}$ by Theorem 9.3.5. □

Remark 9.10.6. The automorphisms of Corollary 9.10.5 are constructed with the axiom of choice, and apart from the identity and complex conjugation they are not continuous (Exercise 4.10.3 shows that the only ones fixing $\mathbb{R}$ are these two). They nevertheless act on algebraic varieties defined over $\mathbb{C}$ by acting on coefficients, and the question of how they interact with Hodge theory is the subject of Chapter 75.

Remark 9.10.7. For an algebraic extension L/K that is normal and separable but infinite, $\mathrm{Gal}(L/K)$ is the inverse limit of the finite groups $\mathrm{Gal}(F/K)$ over finite Galois subextensions F, and the fundamental theorem holds for closed subgroups with respect to the resulting profinite topology; see [Lan02, Chapter VI, §14]. Topological groups are introduced in Chapter 18.

9.11 Exercises

Exercise 9.11.1. Show that the splitting field of x^3-2 over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt[3]{2}, \omega)$ with $\omega = (-1+\sqrt{-3})/2$, that it has degree 6, and that its Galois group is isomorphic to S_3. Describe all intermediate fields.

Exercise 9.11.2. Show that if $\alpha \in \mathbb{R}$ lies in a tower $\mathbb{Q} = K_0 \subseteq \cdots \subseteq K_r$ with $[K_{i+1} : K_i] = 2$, then $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ is a power of 2. Deduce that $\sqrt[3]{2}$ is not in any such tower (impossibility of doubling the cube).

Exercise 9.11.3. Show that for every $n \geq 1$ there is an irreducible polynomial of degree n over $\mathbb{F}_p$, and that $x^{p^n} - x$ is the product of all monic irreducible polynomials over $\mathbb{F}_p$ of degree dividing n.

Exercise 9.11.4. Let G be a group and $\chi_1, \ldots, \chi_n \colon G \to L^\times$ distinct homomorphisms into the multiplicative group of a field. Show that the χ_i are linearly independent over L as maps $G \to L$.

Exercise 9.11.5. Let k be a field and $G \leq \mathrm{Aut}(k(t)/k)$ the group of order 2 generated by $t \mapsto 1/t$. Show that $k(t)^G = k(t + 1/t)$.

Exercise 9.11.6. Show that $\bar{\mathbb{Q}}$ is countable and that $\mathrm{trdeg}_{\bar{\mathbb{Q}}}\, \mathbb{C} = 2^{\aleph_0}$.

Chapter 10

Elementary Number Theory

10.1 Introduction

Number theory studies the integers, and above all the way they are built multiplicatively out of primes. The foundations are already in place: divisibility, greatest common divisors, Bézout's theorem and unique factorisation into primes were proved in Chapter 4, the theorems of Euler, Fermat and Wilson in Chapter 7, the Chinese remainder theorem in Chapters 7 and 8, and the structure of finite fields in Chapter 9. This chapter puts these tools to work on the classical questions of elementary number theory:

1. How does one compute with congruences, and which linear congruences and systems of congruences can be solved (Section 10.2)? As an application we explain the RSA cryptosystem.

2. For which n is the group of units modulo n cyclic, that is, when is there a *primitive root* modulo n (Section 10.3)?

3. Which integers are squares modulo a prime p? The answer is governed by the *law of quadratic reciprocity*, which relates the question "is p a square modulo q?" to the question "is q a square modulo p?" (Section 10.4).

4. Which integers are sums of two squares, and why is every positive integer a sum of four squares (Section 10.5)?

5. How do the multiplicative functions φ, the number of divisors and the sum of divisors behave, and how does Möbius inversion work (Section 10.6)? As an application we determine the even perfect numbers.

6. What are the primes of the Gaussian integers $\mathbb{Z}[i]$ and of the Eisenstein integers $\mathbb{Z}[\omega]$, in how many ways is n a sum of two squares, and how can unique factorisation fail (Section 10.7)?

7. Which primes are of the form $x^2 + 2y^2$, $x^2 + 5y^2$, $x^2 + 7y^2$? Gauss's reduction theory of binary quadratic forms and the class number answer such questions, and explain why $n^2 + n + 41$ is prime for $0 \le n \le 39$; this leads to the Heegner numbers $1, 2, 3, 7, 11, 19, 43, 67, 163$ (Section 10.8).

The theme running through the chapter is that questions about the integers become questions about finite groups and rings, $\mathbb{Z}/n\mathbb{Z}$ and its group of units, or about larger rings such as the Gaussian integers $\mathbb{Z}[i]$, where the answers are visible. This point of view leads to algebraic

number theory, and, through the zeta functions of varieties over finite fields, to the arithmetic side of Hodge theory (Chapters 65 and 76).

References: [IR90], [Ser73] and [HW08].

Prerequisites. Chapter 4 (Number Systems), Chapter 7 (Groups), Chapter 8 (Rings and Modules) and Chapter 9 (Fields and Galois Theory).

10.2 Congruences

Gauss introduced congruences in 1801 as a notation that lets one compute with remainders as if they were numbers. The idea is that many questions about integers only depend on remainders: an integer is even or odd according to its remainder modulo 2, the last digit of a number is its remainder modulo 10, and the day of the week of a date is a remainder modulo 7.

Definition 10.2.1. Let $n \geq 1$. Integers a and b are *congruent modulo* n, written $a \equiv b \pmod{n}$, if $n \mid a - b$. The *residue class* of a is $\bar{a} = a + n\mathbb{Z}$, and $\mathbb{Z}/n\mathbb{Z} = \{\bar{0}, \bar{1}, \ldots, \overline{n-1}\}$ is the quotient ring of $\mathbb{Z}$ by the ideal $n\mathbb{Z}$ (Theorem 8.3.2). We write $U_n = (\mathbb{Z}/n\mathbb{Z})^\times$ for its group of units and $\varphi(n) = |U_n|$; the function φ is *Euler's function*.

Congruence modulo n is an equivalence relation, its classes are the residue classes, and by division with remainder (Theorem 4.5.2) every integer is congruent to exactly one of $0, 1, \ldots, n-1$, its *least non-negative residue*. That $\mathbb{Z}/n\mathbb{Z}$ is a ring says that congruences can be added and multiplied: if $a \equiv a'$ and $b \equiv b'$, then $a + b \equiv a' + b'$ and $ab \equiv a'b' \pmod{n}$. Directly, $ab - a'b' = a(b - b') + (a - a')b'$. So in a computation modulo n we may at any moment replace a number by any other number congruent to it.

Example 10.2.2.

1. *Casting out nines.* Since $10 \equiv 1 \pmod{9}$, also $10^k \equiv 1$, so a number $\sum_k d_k 10^k$ with digits d_k is congruent modulo 9 to the sum $\sum_k d_k$ of its digits. For instance $2026 \equiv 2 + 0 + 2 + 6 = 10 \equiv 1 \pmod{9}$. Similarly $10 \equiv -1 \pmod{11}$, so a number is congruent modulo 11 to the alternating sum of its digits, starting from the last one: $2026 \equiv 6 - 2 + 0 - 2 = 2 \pmod{11}$.

2. *Squares modulo* 4 *and* 8. Every integer is congruent to $0, 1, 2$ or 3 modulo 4, with squares $0, 1, 0, 1$. So a square is congruent to 0 or 1 modulo 4, and a sum of two squares is never congruent to 3 modulo 4: none of $3, 7, 11, 2027$ is a sum of two squares. Similarly the squares of odd numbers are $\equiv 1 \pmod{8}$, since $(2m+1)^2 = 4m(m+1) + 1$ and $m(m+1)$ is even.

3. *Large powers.* To compute 3^{100} modulo 7: $3^2 = 9 \equiv 2$, $3^3 \equiv 6 \equiv -1$, so $3^6 \equiv 1$ and $3^{100} = (3^6)^{16} 3^4 \equiv 3^4 = 81 \equiv 4 \pmod{7}$.

The units of $\mathbb{Z}/n\mathbb{Z}$ are described by Bézout's theorem.

Proposition 10.2.3. *Let $n \geq 1$ and $a \in \mathbb{Z}$.*

1. *$\bar{a}$ is a unit of $\mathbb{Z}/n\mathbb{Z}$ if and only if $\gcd(a, n) = 1$. So $\varphi(n)$ is the number of $0 \leq a < n$ with $\gcd(a, n) = 1$, as in Corollary 7.3.5.*

2. *$\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is prime. For a prime p we write $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$.*

Proof. (1) If $\gcd(a, n) = 1$, Bézout's theorem (Theorem 4.5.3) gives $ax + ny = 1$, so $\bar{a}\bar{x} = \bar{1}$. Conversely, if $ax \equiv 1 \pmod{n}$, then $ax + ny = 1$ for some y, and every common divisor of a and n divides 1.

(2) $\mathbb{Z}/1\mathbb{Z}$ is the zero ring, which is not a field. For $n \geq 2$, $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if every $\bar{a} \neq 0$, that is every $1 \leq a \leq n-1$, is a unit, which by (1) means that $\gcd(a,n) = 1$ for $1 \leq a < n$. This holds if n is prime, since a positive divisor of n less than n is 1; and it fails if $n = bc$ with $1 < b < n$, since $\gcd(b,n) = b$. □

(The notation $\mathbb{F}_p$ is that of Theorem 9.6.2, of which $\mathbb{Z}/p\mathbb{Z}$ is the case $q = p$.) The proof of (1) is constructive: the inverse of a modulo n is found with the Euclidean algorithm (Example 4.5.7), run backwards to express 1 as $ax + ny$.

Proposition 10.2.4 (Linear congruences)**.** *Let $n \geq 1$, $a, b \in \mathbb{Z}$, and $d = \gcd(a,n)$ (with $d = n$ if $a = 0$). The congruence $ax \equiv b \pmod{n}$ has a solution if and only if $d \mid b$. In that case its solutions form exactly d residue classes modulo n, namely $x_0 + k\frac{n}{d}$ for $k = 0, 1, \ldots, d-1$, where x_0 is any one solution.*

Proof. If $ax - b = ny$, then d divides $ax - ny = b$. Conversely, if $d \mid b$, write $d = ax_1 + ny_1$ by Bézout's theorem; then $x_0 = x_1 b/d$ satisfies $ax_0 = b - ny_1 b/d \equiv b$. Let x_0 be a solution. Then x is a solution if and only if $n \mid a(x - x_0)$, that is, $\frac{n}{d} \mid \frac{a}{d}(x - x_0)$. Since $\gcd(\frac{a}{d}, \frac{n}{d}) = 1$ (a common divisor $e > 1$ would make de a common divisor of a and n), this means $\frac{n}{d} \mid x - x_0$, by Euclid's lemma in the form: if $m \mid uv$ and $\gcd(m,u) = 1$, then $m \mid v$ (from $1 = mr + us$, $v = mvr + uvs$). So the solutions are the $x \equiv x_0 \pmod{n/d}$, which modulo n are the d distinct classes $x_0 + kn/d$, $0 \leq k < d$. □

Example 10.2.5.

1. $6x \equiv 5 \pmod{10}$ has no solution, since $\gcd(6,10) = 2 \nmid 5$: indeed $6x$ is even and $10y + 5$ is odd.
2. $6x \equiv 4 \pmod{10}$: here $d = 2$, and dividing by 2 gives $3x \equiv 2 \pmod{5}$. The inverse of 3 modulo 5 is 2, so $x \equiv 4 \pmod{5}$, and modulo 10 the solutions are $x \equiv 4$ and $x \equiv 9$. Check: $24 = 2 \cdot 10 + 4$ and $54 = 5 \cdot 10 + 4$.
3. $17x \equiv 1 \pmod{60}$: the Euclidean algorithm gives $60 = 3 \cdot 17 + 9$, $17 = 9 + 8$, $9 = 8 + 1$, so $1 = 9 - 8 = 9 - (17 - 9) = 2 \cdot 9 - 17 = 2(60 - 3 \cdot 17) - 17 = 2 \cdot 60 - 7 \cdot 17$. Hence $x \equiv -7 \equiv 53 \pmod{60}$; indeed $17 \cdot 53 = 901 = 15 \cdot 60 + 1$.

Next, several congruences at once. A problem in the Chinese mathematical text of Sunzi (around the fourth century) asks for a number which leaves remainder 2 when divided by 3, remainder 3 when divided by 5 and remainder 2 when divided by 7.

Theorem 10.2.6 (Chinese remainder theorem for integers)**.** *Let $n_1, \ldots, n_k \geq 1$ be pairwise coprime and $N = n_1 \cdots n_k$. For all $a_1, \ldots, a_k \in \mathbb{Z}$ the system*

$$x \equiv a_1 \pmod{n_1}, \quad \ldots, \quad x \equiv a_k \pmod{n_k}$$

has a solution, unique modulo N. Explicitly, put $N_i = N/n_i$ and choose y_i with $N_i y_i \equiv 1 \pmod{n_i}$; then $x = \sum_i a_i N_i y_i$ is a solution. Consequently the map $\mathbb{Z}/N\mathbb{Z} \to \mathbb{Z}/n_1\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}$, $\bar{x} \mapsto (x \bmod n_1, \ldots, x \bmod n_k)$, is an isomorphism of rings, and restricts to an isomorphism of groups $U_N \cong U_{n_1} \times \cdots \times U_{n_k}$.

Proof. N_i is a product of integers coprime to n_i, so it is coprime to n_i (a prime dividing n_i and N_i would divide some n_j, $j \neq i$, by Lemma 4.5.4), and y_i exists by Proposition 10.2.3. For $j \neq i$, $n_j \mid N_i$; so $x \equiv a_j N_j y_j \equiv a_j \pmod{n_j}$. If x and x' are two solutions, each n_j divides $x - x'$, hence so does N: by induction, if $n_1 \cdots n_{j-1}$ and n_j divide m and are coprime, write $m = n_1 \cdots n_{j-1} m'$; then $n_j \mid n_1 \cdots n_{j-1} m'$ gives $n_j \mid m'$, by Euclid's lemma in the form used in the proof of Proposition 10.2.4. So the ring homomorphism of the statement is bijective. A ring isomorphism maps units onto units, and the units of a product of rings are the elements all of whose components are units. □

This is the special case $R = \mathbb{Z}$ of Theorem 8.3.3, made explicit.

Example 10.2.7. In Sunzi's problem $N = 105$, $N_1 = 35 \equiv 2 \pmod 3$, $N_2 = 21 \equiv 1 \pmod 5$ and $N_3 = 15 \equiv 1 \pmod 7$, so we may take $y_1 = 2$, $y_2 = 1$, $y_3 = 1$, and $x = 2 \cdot 35 \cdot 2 + 3 \cdot 21 + 2 \cdot 15 = 233 \equiv 23 \pmod{105}$. The answer is 23, and the solutions are the numbers $23 + 105m$.

Euler's function can now be computed.

Proposition 10.2.8. *1. If* $\gcd(m, n) = 1$, *then* $\varphi(mn) = \varphi(m)\varphi(n)$.

2. For a prime p *and* $k \geq 1$, $\varphi(p^k) = p^k - p^{k-1} = p^{k-1}(p-1)$.

3. If $n = p_1^{k_1} \cdots p_r^{k_r}$ *with distinct primes* p_i, *then* $\varphi(n) = n \prod_i \left(1 - \frac{1}{p_i}\right)$.

4. (Gauss) $\sum_{d|n} \varphi(d) = n$, *the sum running over the positive divisors of* n.

Proof. (1) is the isomorphism $U_{mn} \cong U_m \times U_n$ of Theorem 10.2.6. (2) An integer is coprime to p^k if and only if p does not divide it (its gcd with p^k is a power of p, by Theorem 4.5.5); among $0, \ldots, p^k - 1$ the multiples of p are the p^{k-1} numbers $0, p, \ldots, p^k - p$. (3) follows from (1) and (2), since the prime powers $p_i^{k_i}$ are pairwise coprime: $\varphi(n) = \prod_i p_i^{k_i}(1 - 1/p_i)$.

(4) Sort the integers $1 \leq a \leq n$ according to $\gcd(a, n)$, which is a divisor e of n. Those with $\gcd(a, n) = e$ are the $a = eb$ with $1 \leq b \leq n/e$ and $\gcd(b, n/e) = 1$, and there are $\varphi(n/e)$ of them ($\varphi(1) = 1$ counts $b = 1$). Hence $n = \sum_{e|n} \varphi(n/e) = \sum_{d|n} \varphi(d)$, as $d = n/e$ runs over the divisors of n when e does. □

Example 10.2.9. $\varphi(12) = 12(1 - \frac{1}{2})(1 - \frac{1}{3}) = 4$, the units modulo 12 being $1, 5, 7, 11$. The divisors of 12 are $1, 2, 3, 4, 6, 12$, and φ of them are $1, 1, 2, 2, 2, 4$, with sum 12. Also $\varphi(100) = 100 \cdot \frac{1}{2} \cdot \frac{4}{5} = 40$, so by Euler's theorem $a^{40} \equiv 1 \pmod{100}$ for a coprime to 10: for instance the last two digits of 3^{2026} are those of $3^{2026-50\cdot 40} = 3^{26}$; and $3^5 = 243 \equiv 43$, $3^{10} \equiv 43^2 = 1849 \equiv 49$, $3^{20} \equiv 49^2 = 2401 \equiv 1 \pmod{100}$, so $3^{26} \equiv 3^6 = 729 \equiv 29$. The last two digits of 3^{2026} are 29.

The last computation used *repeated squaring*: a^e modulo n is computed with about $2 \log_2 e$ multiplications of numbers less than n, by squaring repeatedly and multiplying together the squares a^{2^j} corresponding to the binary digits of e. By contrast no efficient method is known to factor a large integer into primes. The RSA cryptosystem (Rivest, Shamir and Adleman, 1977) is built on this asymmetry.

Proposition 10.2.10 (RSA)**.** *Let* $p \neq q$ *be primes,* $n = pq$, *and let* $e, d \geq 1$ *satisfy* $ed \equiv 1 \pmod{(p-1)(q-1)}$. *Then* $m^{ed} \equiv m \pmod n$ *for every integer* m.

Proof. By Theorem 10.2.6 it suffices to prove the congruence modulo p and modulo q; by symmetry, modulo p. Write $ed = 1 + k(p-1)(q-1)$ with $k \geq 0$. If $p \mid m$, both sides are $\equiv 0$. Otherwise $m^{p-1} \equiv 1 \pmod p$ by Fermat's theorem (Corollary 7.3.5), and $m^{ed} = m \cdot (m^{p-1})^{k(q-1)} \equiv m \pmod p$. □

In practice someone chooses two large primes p and q (several hundred digits each), publishes $n = pq$ and an exponent e coprime to $(p-1)(q-1)$, and keeps secret the inverse d of e modulo $(p-1)(q-1)$, computed with the Euclidean algorithm. Anyone can encrypt a message $0 \leq m < n$ as $c \equiv m^e \pmod n$, and by the proposition the owner decrypts it as $m \equiv c^d \pmod n$. Computing d from n and e is as hard as factoring n (Exercise 10.9.11 shows that knowing $(p-1)(q-1)$ already gives p and q). For a toy example take $p = 5$, $q = 11$, $n = 55$, $(p-1)(q-1) = 40$, $e = 3$ and $d = 27$, as $3 \cdot 27 = 81 \equiv 1 \pmod{40}$: the message $m = 2$ is encrypted as $c = 8$, and $8^{27} \equiv 2 \pmod{55}$.

10.3 Orders and primitive roots

The powers of 2 modulo 11 are

$$2,\ 4,\ 8,\ 5,\ 10,\ 9,\ 7,\ 3,\ 6,\ 1,$$

so every nonzero residue modulo 11 is a power of 2, and U_{11} is a cyclic group generated by 2. The powers of any a modulo 8, on the other hand, are only 1 and a, since $1^2 = 3^2 - 8 = 5^2 - 24 = 7^2 - 48 = 1$; so U_8, of order 4, is not cyclic. This section determines all n for which U_n is cyclic.

Definition 10.3.1. Let $n \geq 1$ and $\gcd(a, n) = 1$. The *order* of a modulo n, written $\mathrm{ord}_n(a)$, is the order of $\bar{a}$ in the group U_n: the least $k \geq 1$ with $a^k \equiv 1 \pmod{n}$. An integer g is a *primitive root* modulo n if $\mathrm{ord}_n(g) = \varphi(n)$, that is, if $\bar{g}$ generates U_n.

Lemma 10.3.2. *Let G be a group and $x \in G$ an element of finite order m.*

1. *$x^k = 1$ if and only if $m \mid k$.*
2. *For $k \geq 1$, the order of x^k is $m/\gcd(k, m)$.*
3. *If G is cyclic of order m generated by x, its generators are the x^k with $\gcd(k, m) = 1$, and there are $\varphi(m)$ of them. For each $d \mid m$, G has exactly one subgroup of order d, generated by $x^{m/d}$, and the elements of G whose order divides d are exactly the elements of this subgroup.*

In particular $\mathrm{ord}_n(a)$ divides $\varphi(n)$, and $a^k \equiv 1 \pmod{n}$ if and only if $\mathrm{ord}_n(a) \mid k$.

Proof. (1) Write $k = qm + r$ with $0 \leq r < m$. Then $x^k = (x^m)^q x^r = x^r$ (Lemma 7.2.9), which is 1 exactly when $r = 0$, by the minimality of m.

(2) Let $g = \gcd(k, m)$. By (1), $(x^k)^j = 1$ if and only if $m \mid kj$, that is, $\frac{m}{g} \mid \frac{k}{g} j$, that is, $\frac{m}{g} \mid j$, since $\gcd(\frac{k}{g}, \frac{m}{g}) = 1$ (as in the proof of Proposition 10.2.4). So the order of x^k is m/g.

(3) Every element is some x^k with $0 \leq k < m$, and by (2) it generates G, that is, has order m, if and only if $\gcd(k, m) = 1$; the number of such k is $\varphi(m)$. Let $d \mid m$. An element x^k has order dividing d if and only if $x^{kd} = 1$, that is, $m \mid kd$, that is, $\frac{m}{d} \mid k$; so these elements are the d powers of $x^{m/d}$, which form the subgroup generated by $x^{m/d}$, of order d by (2). Any subgroup H of order d consists of elements of order dividing d (Theorem 7.3.3), so H is contained in this subgroup and, having the same number of elements, equals it. The last assertion is (1) for $G = U_n$, with Lagrange's theorem. □

Theorem 10.3.3. *For every prime p the group $U_p = \mathbb{F}_p^\times$ is cyclic of order $p - 1$: there is a primitive root modulo p, and there are exactly $\varphi(p - 1)$ primitive roots among $1, \ldots, p - 1$.*

Proof. $\mathbb{F}_p^\times$ is a finite subgroup of the multiplicative group of the field $\mathbb{F}_p$, so it is cyclic by Lemma 9.6.1. The number of generators is $\varphi(p - 1)$ by Lemma 10.3.2(3). □

It is worth recalling why this is true, since the proof of Lemma 9.6.1 is short but indirect. The essential input is that a polynomial of degree d over a field has at most d roots (Corollary 8.6.6): so for each d there are at most d residues with $x^d \equiv 1 \pmod{p}$. In $\mathbb{Z}/8\mathbb{Z}$, which is not a field, the polynomial $x^2 - 1$ has the four roots $1, 3, 5, 7$, and this is exactly why U_8 is not cyclic. Exercise 10.9.5 gives a second proof of the theorem by counting with Proposition 10.2.8(4).

Example 10.3.4. 1. By the list above, 2 is a primitive root modulo 11. There are $\varphi(10) = 4$ primitive roots, the powers 2^k with $\gcd(k, 10) = 1$: $2^1 = 2$, $2^3 = 8$, $2^7 \equiv 7$ and $2^9 \equiv 6$.

2. Modulo 7, the powers of 3 are $3, 2, 6, 4, 5, 1$, so 3 is a primitive root; the powers of 2 are $2, 4, 1$, so 2 is not.

3. To find a primitive root modulo p in practice, one tests $g = 2, 3, \ldots$: by Lemma 10.3.2, g is a primitive root if and only if $g^{(p-1)/\ell} \not\equiv 1 \pmod{p}$ for every prime $\ell \mid p-1$, since a proper divisor of $p-1$ divides some $(p-1)/\ell$. For $p = 23$, $p-1 = 2 \cdot 11$; $2^{11} = 2048 = 89 \cdot 23 + 1$, so 2 is not a primitive root; $5^2 = 25 \equiv 2$ and $5^{11} = 5 \cdot (5^2)^5 \equiv 5 \cdot 2^5 = 160 \equiv -1 \pmod{23}$, so 5 is a primitive root modulo 23.

Remark 10.3.5. Whether 2 is a primitive root modulo infinitely many primes is not known. Artin conjectured in 1927 that every integer a that is neither -1 nor a perfect square is a primitive root modulo infinitely many primes; this is still open for every single value of a.

To pass from primes to prime powers we need a lemma on how powers of numbers close to 1 behave modulo powers of p.

Lemma 10.3.6. *Let p be a prime and $j \geq 1$, with $j \geq 2$ if $p = 2$. If $x \equiv 1 + p^j c \pmod{p^{j+1}}$ with $p \nmid c$, then $x^p \equiv 1 + p^{j+1}c \pmod{p^{j+2}}$. Consequently, under the same hypothesis, $x^{p^i} \equiv 1 + p^{j+i}c \pmod{p^{j+i+1}}$ for all $i \geq 0$.*

Proof. Write $x = 1 + p^j y$ with $y \equiv c \pmod{p}$. By the binomial theorem

$$x^p = 1 + p \cdot p^j y + \sum_{k=2}^{p} \binom{p}{k} p^{kj} y^k.$$

The second term is $p^{j+1}y \equiv p^{j+1}c \pmod{p^{j+2}}$. For $2 \leq k \leq p-1$, p divides $\binom{p}{k} = \frac{p!}{k!(p-k)!}$ (it divides the numerator and not the denominator, by Lemma 4.5.4), so the term is divisible by p^{1+kj}, and $1 + kj \geq 1 + 2j \geq j + 2$. The last term $p^{pj}y^p$ is divisible by p^{j+2} if $pj \geq j + 2$, that is, $(p-1)j \geq 2$, which holds if $p \geq 3$ or if $p = 2$ and $j \geq 2$. (For $p = 2$ there are no middle terms.) This proves the first statement; the second follows by induction on i, since x^{p^i} again satisfies the hypothesis, with $j + i$ in place of j. □

Theorem 10.3.7. *Let p be an odd prime and $k \geq 1$. Then U_{p^k} is cyclic. More precisely, if g is a primitive root modulo p with $g^{p-1} \not\equiv 1 \pmod{p^2}$, then g is a primitive root modulo p^k for every $k \geq 1$; and if g is any primitive root modulo p, then g or $g + p$ has this property.*

Proof. Let g be a primitive root modulo p. If $g^{p-1} \equiv 1 \pmod{p^2}$, then by the binomial theorem

$$(g+p)^{p-1} \equiv g^{p-1} + (p-1)g^{p-2}p \equiv 1 - pg^{p-2} \pmod{p^2},$$

which is $\not\equiv 1$ since $p \nmid g$; and $g + p \equiv g \pmod{p}$ is again a primitive root modulo p. So we may assume $g^{p-1} = 1 + pc$ with $p \nmid c$.

Let $d = \operatorname{ord}_{p^k}(g)$. Then $g^d \equiv 1 \pmod{p}$, so $p - 1 = \operatorname{ord}_p(g)$ divides d, and d divides $\varphi(p^k) = (p-1)p^{k-1}$ (Lemma 10.3.2, Proposition 10.2.8). Hence $d = (p-1)p^i$ for some $0 \leq i \leq k-1$. If $k \geq 2$ and $i \leq k-2$, then d divides $(p-1)p^{k-2}$, so $g^{(p-1)p^{k-2}} \equiv 1 \pmod{p^k}$; but Lemma 10.3.6 with $x = g^{p-1}$ and $j = 1$ gives $g^{(p-1)p^{k-2}} \equiv 1 + p^{k-1}c \pmod{p^k}$, which is $\not\equiv 1$. So $i = k-1$ and $d = \varphi(p^k)$. (For $k = 1$ there is nothing to prove.) □

Example 10.3.8. 3 is a primitive root modulo 7, and $3^6 = 729 = 14 \cdot 49 + 43 \not\equiv 1 \pmod{49}$. So 3 is a primitive root modulo 7^k for every k: for instance modulo 343 the powers of 3 run through all $\varphi(343) = 294$ residues prime to 7. The exceptional case of the theorem does occur, but rarely: $p = 29$, $g = 14$ is a primitive root modulo 29 with $14^{28} \equiv 1 \pmod{29^2}$ (Exercise 10.9.6).

Powers of 2 behave differently, as U_8 already showed.

Theorem 10.3.9. *U_2 and U_4 are cyclic, of orders 1 and 2. For $k \geq 3$, $\operatorname{ord}_{2^k}(5) = 2^{k-2}$, and the map*

$$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^{k-2}\mathbb{Z} \to U_{2^k}, \qquad (a, b) \mapsto (-1)^a 5^b,$$

is an isomorphism of groups. In particular U_{2^k} is not cyclic for $k \geq 3$.

Proof. $U_2 = \{\bar{1}\}$ and $U_4 = \{\bar{1}, \bar{3}\}$. Let $k \geq 3$. Since $5 = 1 + 2^2$, Lemma 10.3.6 with $p = 2$, $j = 2$, $c = 1$ gives $5^{2^i} \equiv 1 + 2^{i+2} \pmod{2^{i+3}}$. For $i = k-3$ this says $5^{2^{k-3}} \equiv 1 + 2^{k-1} \not\equiv 1 \pmod{2^k}$; for $i = k-2$ it gives $5^{2^{k-2}} \equiv 1 \pmod{2^k}$. So the order of 5 divides 2^{k-2} but not 2^{k-3}, and equals 2^{k-2}. The map of the statement is therefore a well defined homomorphism (both -1 and 5 have orders dividing 2 and 2^{k-2}, and U_{2^k} is abelian). It is injective: if $(-1)^a 5^b \equiv 1 \pmod{2^k}$, then reducing modulo 4, where $5 \equiv 1$, gives $(-1)^a \equiv 1 \pmod 4$, so a is even, and then $5^b \equiv 1$ gives $2^{k-2} \mid b$. Both groups have $2 \cdot 2^{k-2} = 2^{k-1} = \varphi(2^k)$ elements, so it is an isomorphism. Finally $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^{k-2}\mathbb{Z}$ has the three elements $(1,0)$, $(0, 2^{k-3})$, $(1, 2^{k-3})$ of order 2, while a cyclic group has at most one, by Lemma 10.3.2(3) with $d = 2$. □

Theorem 10.3.10 (Gauss)**.** *Let $n \geq 1$. There is a primitive root modulo n if and only if n is 1, 2, 4, p^k or $2p^k$, with p an odd prime and $k \geq 1$.*

Proof. For $n = 1, 2, 4$ and p^k this is Theorems 10.3.9 and 10.3.7, and $U_{2p^k} \cong U_2 \times U_{p^k} \cong U_{p^k}$ by Theorem 10.2.6, as U_2 is trivial. Conversely, a product $A \times B$ of two finite groups of even order is not cyclic: by Cauchy's theorem (Theorem 7.6.8) there are $x \in A$ and $y \in B$ of order 2, and then $(x, 1)$, $(1, y)$ and (x, y) are three elements of order 2, while a cyclic group has at most one. Now $\varphi(p^k) = p^{k-1}(p-1)$ is even for odd p, and $\varphi(2^k) = 2^{k-1}$ is even for $k \geq 2$.

Now let n be not of the listed forms. If n is a power of 2, then $n = 2^k$ with $k \geq 3$, and U_n is not cyclic by Theorem 10.3.9. Otherwise n has an odd prime factor p; write $n = p^k m$ with $k \geq 1$ and $p \nmid m$. Since n is neither p^k nor $2p^k$, we have $m \geq 3$. If m has an odd prime factor q, write $m = q^l m'$ with $q \nmid m'$; then $\varphi(m) = \varphi(q^l)\varphi(m')$ is even. If m has no odd prime factor, then $m = 2^a$ with $a \geq 2$, and $\varphi(m) = 2^{a-1}$ is even. In both cases $U_n \cong U_{p^k} \times U_m$ (Theorem 10.2.6) is a product of two groups of even order, hence not cyclic. □

A primitive root g modulo p turns multiplication into addition, as a logarithm does: every a prime to p is $a \equiv g^t$ for a unique t modulo $p-1$, the *index* or *discrete logarithm* of a. This immediately answers the question of which residues are k-th powers.

Proposition 10.3.11. *Let p be a prime, $k \geq 1$, $d = \gcd(k, p-1)$, and $p \nmid a$. The congruence $x^k \equiv a \pmod p$ has a solution if and only if $a^{(p-1)/d} \equiv 1 \pmod p$, and then it has exactly d solutions modulo p. The k-th powers form a subgroup of U_p of index d.*

Proof. Fix a primitive root g (Theorem 10.3.3); then $t \mapsto g^t$ is an isomorphism $\mathbb{Z}/(p-1)\mathbb{Z} \to U_p$, under which $x \mapsto x^k$ becomes $s \mapsto ks$. Write $a = g^t$ and $x = g^s$. The congruence becomes $ks \equiv t \pmod{p-1}$, which by Proposition 10.2.4 is solvable if and only if $d \mid t$, and then has d solutions s modulo $p-1$, that is, d solutions x modulo p. Finally $a^{(p-1)/d} = g^{t(p-1)/d} \equiv 1$ if and only if $p-1 \mid t(p-1)/d$, that is, $d \mid t$. The k-th powers are the image of the homomorphism $x \mapsto x^k$, whose kernel has d elements (the solutions for $a = 1$), so the image has $(p-1)/d$ elements and index d. □

For instance, for $p \equiv 2 \pmod 3$ every residue is a cube modulo p (as $d = \gcd(3, p-1) = 1$), while for $p \equiv 1 \pmod 3$ exactly a third of the nonzero residues are cubes. The case $k = 2$, where $d = 2$ for every odd prime, is the subject of the next section.

10.4 Quadratic residues and quadratic reciprocity

Which numbers are squares modulo a prime p? Modulo $p = 2$ every number is a square, so let p be odd. The squares of $1, 2, \dots, 10$ modulo 11 are $1, 4, 9, 5, 3, 3, 5, 9, 4, 1$: exactly half of the nonzero residues, $1, 3, 4, 5, 9$, are squares, each twice, as $x^2 = (-x)^2$. The question has a surprisingly deep answer, the law of quadratic reciprocity, which Euler conjectured and Gauss first proved in 1796.

Definition 10.4.1. Let p be an odd prime. An integer a with $p \nmid a$ is a *quadratic residue* modulo p if $x^2 \equiv a \pmod{p}$ has a solution, and a *quadratic nonresidue* otherwise. The *Legendre symbol* is

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } p \nmid a \text{ and } a \text{ is a quadratic residue modulo } p, \\ -1 & \text{if } a \text{ is a quadratic nonresidue modulo } p, \\ 0 & \text{if } p \mid a. \end{cases}$$

By definition $\left(\frac{a}{p}\right)$ depends only on a modulo p. The following criterion of Euler reduces its computation to a power, and shows that it is multiplicative in a.

Proposition 10.4.2 (Euler's criterion)**.** *Let p be an odd prime.*

1. *For every integer a, $\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}$.*
2. *There are exactly $\frac{p-1}{2}$ quadratic residues and $\frac{p-1}{2}$ quadratic nonresidues among $1, \dots, p-1$.*
3. *$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$ for all integers a, b. So the product of two residues or of two nonresidues is a residue, and the product of a residue and a nonresidue is a nonresidue.*

Proof. (1) If $p \mid a$ both sides are $\equiv 0$. Otherwise let $y = a^{(p-1)/2}$. Then $y^2 = a^{p-1} \equiv 1$ by Fermat's theorem, so $y \equiv \pm 1$, because $y^2 - 1 = (y-1)(y+1)$ and $\mathbb{F}_p$ is a field. By Proposition 10.3.11 with $k = 2$ and $d = \gcd(2, p-1) = 2$, a is a quadratic residue if and only if $y \equiv 1$. So $y \equiv 1$ when $\left(\frac{a}{p}\right) = 1$ and $y \equiv -1$ when $\left(\frac{a}{p}\right) = -1$.

(2) The squares form a subgroup of index $d = 2$ of U_p, by Proposition 10.3.11.

(3) By (1), both sides are congruent to $(ab)^{(p-1)/2} = a^{(p-1)/2} b^{(p-1)/2}$ modulo p. Both sides lie in $\{-1, 0, 1\}$, and distinct elements of this set are not congruent modulo $p \geq 3$; so they are equal. □

By (3), to compute $\left(\frac{a}{p}\right)$ it suffices to factor a and know $\left(\frac{-1}{p}\right)$, $\left(\frac{2}{p}\right)$ and $\left(\frac{q}{p}\right)$ for odd primes q. The first is immediate.

Corollary 10.4.3. *For an odd prime p, $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$: the congruence $x^2 \equiv -1 \pmod{p}$ is solvable if and only if $p \equiv 1 \pmod{4}$.*

Proof. By Euler's criterion $\left(\frac{-1}{p}\right) \equiv (-1)^{(p-1)/2} \pmod{p}$, and both sides are ± 1. The exponent $(p-1)/2$ is even exactly when $p \equiv 1 \pmod{4}$. □

For $p \equiv 1 \pmod{4}$ an explicit square root of -1 is $\left(\frac{p-1}{2}\right)!$ (Exercise 10.9.3); for instance modulo 13, $6! = 720 \equiv 5$ and $5^2 = 25 \equiv -1$. The corollary already has a pleasant consequence, an analogue of Euclid's theorem for a single residue class.

Proposition 10.4.4. *There are infinitely many primes $p \equiv 1 \pmod{4}$.*

Proof. Let $p_1, \dots, p_k$ be primes and $N = (2p_1 \cdots p_k)^2 + 1$. Let q be a prime dividing N (Theorem 4.5.5). Then q is odd, as N is odd, and $x = 2p_1 \cdots p_k$ satisfies $x^2 \equiv -1 \pmod{q}$; by Corollary 10.4.3, $q \equiv 1 \pmod{4}$. And $q \neq p_i$, since $p_i \mid N - 1$. So no finite list contains all primes $\equiv 1 \pmod{4}$. □

More generally, Dirichlet's theorem says that every residue class $a \bmod n$ with $\gcd(a,n) = 1$ contains infinitely many primes; its proof requires analysis [Ser73]. For the class of 3 modulo 4 see Exercise 10.9.1.

To compute $(\frac{2}{p})$ and $(\frac{q}{p})$ we use a lemma of Gauss, which counts signs in the proof of Euler's criterion.

Lemma 10.4.5 (Gauss's lemma)**.** *Let p be an odd prime, $h = \frac{p-1}{2}$, and $p \nmid a$. Let μ be the number of $k \in \{1, \dots, h\}$ for which the least non-negative residue of ka modulo p is greater than $\frac{p}{2}$. Then $\left(\frac{a}{p}\right) = (-1)^{\mu}$.*

Proof. For $1 \le k \le h$, let t_k be the least non-negative residue of ka; it is not 0, as $p \nmid ka$. Put $r_k = t_k$ and $\varepsilon_k = 1$ if $t_k < p/2$, and $r_k = p - t_k$ and $\varepsilon_k = -1$ if $t_k > p/2$. Then $ka \equiv \varepsilon_k r_k \pmod p$ with $r_k \in \{1, \dots, h\}$, and exactly μ of the signs ε_k are -1. The r_k are distinct: if $r_k = r_l$, then $ka \equiv \pm la$, so $k \equiv \pm l \pmod p$ (cancel the unit a); since $1 \le k, l \le h$ we have $0 < k + l < p$ and $|k - l| < p$, so $k \equiv -l$ is impossible and $k = l$. Hence $\{r_1, \dots, r_h\} = \{1, \dots, h\}$, and multiplying the congruences $ka \equiv \varepsilon_k r_k$ for $k = 1, \dots, h$ gives

$$a^h\, h! \equiv (-1)^{\mu}\, h! \pmod p.$$

Since $h!$ is prime to p, $a^h \equiv (-1)^{\mu}$, and Euler's criterion gives $(\frac{a}{p}) \equiv (-1)^{\mu}$; both sides are ± 1, so they are equal. □

Example 10.4.6. For $p = 11$, $h = 5$ and $a = 3$: the residues of $3, 6, 9, 12, 15$ are $3, 6, 9, 1, 4$, of which 6 and 9 exceed 5.5. So $\mu = 2$ and $(\frac{3}{11}) = 1$; indeed $5^2 = 25 \equiv 3$. For $a = 2$: the residues $2, 4, 6, 8, 10$ include $6, 8, 10$ above 5.5, so $\mu = 3$ and $(\frac{2}{11}) = -1$, in agreement with the list of squares modulo 11 above.

Proposition 10.4.7 (The second supplementary law)**.** *For an odd prime p, $\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}$: the number 2 is a quadratic residue modulo p if and only if $p \equiv \pm 1 \pmod 8$.*

Proof. For $a = 2$ the numbers $2k$, $1 \le k \le h$, already lie in $\{1, \dots, p-1\}$, and $2k > p/2$ exactly when $k > p/4$. So $\mu = h - \lfloor p/4 \rfloor$. Writing $p = 8m + r$ with $r \in \{1, 3, 5, 7\}$:

$$r = 1: \quad h = 4m,\ \lfloor p/4 \rfloor = 2m,\ \mu = 2m; \qquad r = 3: \quad h = 4m+1,\ \lfloor p/4 \rfloor = 2m,\ \mu = 2m+1;$$
$$r = 5: \quad h = 4m+2,\ \lfloor p/4 \rfloor = 2m+1,\ \mu = 2m+1; \qquad r = 7: \quad h = 4m+3,\ \lfloor p/4 \rfloor = 2m+1,\ \mu = 2m+2.$$

So μ is even exactly for $p \equiv \pm 1 \pmod 8$, and Gauss's lemma gives the first assertion in the form "$(\frac{2}{p}) = 1$ if and only if $p \equiv \pm 1 \pmod 8$". Finally $(p^2-1)/8$ is even for $p = 8m \pm 1$, where it equals $8m^2 \pm 2m$, and odd for $p = 8m \pm 3$, where it equals $8m^2 \pm 6m + 1$. □

For odd a the count μ in Gauss's lemma can be expressed through integer parts, which is the key to the reciprocity law.

Lemma 10.4.8 (Eisenstein)**.** *Let p be an odd prime, $h = \frac{p-1}{2}$, and a an odd integer with $p \nmid a$. Then, with μ as in Gauss's lemma,*

$$\mu \equiv \sum_{k=1}^{h} \left\lfloor \frac{ka}{p} \right\rfloor \pmod 2, \qquad \text{so} \qquad \left(\frac{a}{p}\right) = (-1)^{\sum_{k=1}^{h} \lfloor ka/p \rfloor}.$$

Proof. Divide: $ka = p\lfloor ka/p \rfloor + t_k$ with t_k the least non-negative residue. Let $b_1, \dots, b_\mu$ be the t_k greater than $p/2$ and $c_1, \dots, c_\lambda$ the others ($\lambda + \mu = h$). In the proof of Gauss's lemma we

saw that the numbers $c_1, \dots, c_\lambda, p - b_1, \dots, p - b_\mu$ are $1, \dots, h$ in some order. Summing the divisions and this rearrangement,

$$a \sum_{k=1}^{h} k = p \sum_{k=1}^{h} \left\lfloor \frac{ka}{p} \right\rfloor + \sum_i b_i + \sum_i c_i, \qquad \sum_{k=1}^{h} k = \sum_i c_i + \mu p - \sum_i b_i.$$

Subtracting, $(a-1)\sum_k k = p \sum_k \lfloor ka/p \rfloor - \mu p + 2\sum_i b_i$. Modulo 2, the left side is 0 because a is odd, and $p \equiv 1$; so $0 \equiv \sum_k \lfloor ka/p \rfloor - \mu$. The formula for the symbol is Gauss's lemma. □

Theorem 10.4.9 (Law of quadratic reciprocity)**.** *Let $p \neq q$ be odd primes. Then*

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}.$$

That is, $(\frac{p}{q}) = (\frac{q}{p})$ unless $p \equiv q \equiv 3 \pmod 4$, in which case $(\frac{p}{q}) = -(\frac{q}{p})$.

Proof. Let $h = \frac{p-1}{2}$ and $h' = \frac{q-1}{2}$, and count the points with integer coordinates in the rectangle $R = \{(x, y) : 1 \le x \le h,\ 1 \le y \le h'\}$; there are hh' of them. None lies on the line $py = qx$: if $py = qx$ with $1 \le x \le h$, then $p \mid qx$, so $p \mid x$ by Euclid's lemma, which is impossible for $0 < x < p$.

The points of R with $py < qx$, that is, below the line: for fixed $x \in \{1, \dots, h\}$ these are the y with $1 \le y < qx/p$, and all such y satisfy $y \le h'$, since $qx/p < q/2$ and so $y < q/2$, that is $y \le (q-1)/2$. Their number is $\lfloor qx/p \rfloor$ (as qx/p is not an integer). So there are $\sum_{x=1}^{h} \lfloor qx/p \rfloor$ points below the line, and symmetrically $\sum_{y=1}^{h'} \lfloor py/q \rfloor$ points above it. Hence

$$\sum_{x=1}^{h} \left\lfloor \frac{qx}{p} \right\rfloor + \sum_{y=1}^{h'} \left\lfloor \frac{py}{q} \right\rfloor = hh'.$$

By Eisenstein's lemma (Lemma 10.4.8), applied to the odd number q modulo p and to p modulo q, the left side is the sum of the exponents in $(\frac{q}{p}) = (-1)^{\sum \lfloor qx/p \rfloor}$ and $(\frac{p}{q}) = (-1)^{\sum \lfloor py/q \rfloor}$. So $(\frac{p}{q})(\frac{q}{p}) = (-1)^{hh'}$. Finally hh' is odd if and only if both h and h' are odd, that is, $p \equiv q \equiv 3 \pmod 4$. □

To see the proof at work, take $p = 11$ and $q = 7$, so $h = 5$ and $h' = 3$. The rectangle contains the 15 points (x, y) with $1 \le x \le 5$ and $1 \le y \le 3$. Below the line $11y = 7x$ lie, for $x = 1, \dots, 5$, respectively $\lfloor 7/11 \rfloor, \lfloor 14/11 \rfloor, \dots = 0, 1, 1, 2, 3$ points, 7 in all; to the left of it lie, for $y = 1, 2, 3$, respectively $\lfloor 11/7 \rfloor, \lfloor 22/7 \rfloor, \lfloor 33/7 \rfloor = 1, 3, 4$ points, 8 in all. So $(\frac{7}{11}) = (-1)^7 = -1$ and $(\frac{11}{7}) = (-1)^8 = 1$, and their product is $-1 = (-1)^{15}$. Directly: $(\frac{11}{7}) = (\frac{4}{7}) = 1$, and the squares modulo 11 are $1, 3, 4, 5, 9$, which do not include 7.

Together with Euler's criterion and the two supplementary laws, Corollary 10.4.3 and Proposition 10.4.7, reciprocity computes every Legendre symbol quickly, by repeatedly reducing the top modulo the bottom, factoring, and flipping.

Example 10.4.10.

1. *Is 30 a square modulo the prime 103?* $(\frac{30}{103}) = (\frac{2}{103})(\frac{3}{103})(\frac{5}{103})$. As $103 \equiv 7 \pmod 8$, $(\frac{2}{103}) = 1$. As $103 \equiv 3 \pmod 4$ and $3 \equiv 3 \pmod 4$, $(\frac{3}{103}) = -(\frac{103}{3}) = -(\frac{1}{3}) = -1$. As $5 \equiv 1 \pmod 4$, $(\frac{5}{103}) = (\frac{103}{5}) = (\frac{3}{5}) = -1$, the squares modulo 5 being 1 and 4. So $(\frac{30}{103}) = 1 \cdot (-1) \cdot (-1) = 1$: 30 is a square modulo 103. Indeed $37^2 = 1369 = 13 \cdot 103 + 30$. Note that reciprocity decides the question without finding the square root.

2. *For which primes is 3 a square?* For $p > 3$, $(\frac{3}{p}) = (-1)^{(p-1)/2}(\frac{p}{3})$, and $(\frac{p}{3}) = 1$ if and only if $p \equiv 1 \pmod 3$. So $(\frac{3}{p}) = 1$ if and only if $p \equiv 1 \pmod 4$ and $p \equiv 1 \pmod 3$, or $p \equiv 3 \pmod 4$ and $p \equiv 2 \pmod 3$; that is, by the Chinese remainder theorem, if and only if $p \equiv \pm 1 \pmod{12}$.

3. *For which primes is* 5 *a square?* For $p \neq 2, 5$, $(\frac{5}{p}) = (\frac{p}{5})$, which is 1 if and only if $p \equiv \pm 1 \pmod 5$. This is the remarkable feature of the law: whether q is a square modulo p depends only on p modulo $4q$, although the squares modulo p seem to have nothing to do with q.

Remark 10.4.11. Gauss called the law the *theorema aureum* and gave several proofs of it; the one above, by counting lattice points, is due to Eisenstein. Another proof uses Gauss sums: the sum $g = \sum_{a=1}^{q-1} (\frac{a}{q}) \zeta^a$, with ζ a primitive q-th root of unity, satisfies $g^2 = (-1)^{(q-1)/2} q$, so the cyclotomic field $\mathbb{Q}(\zeta)$ of Theorem 9.8.2 contains a square root of $\pm q$, and comparing g^p with g modulo p gives the law [IR90]. This is the beginning of a Galois theoretic explanation, in which $(\frac{p}{q})$ describes how the automorphism $\zeta \mapsto \zeta^p$ acts on that square root; its generalisation to all abelian extensions of number fields is class field theory.

10.5 Sums of squares

Which integers are sums of two squares? The first few are

$$0, 1, 2, 4, 5, 8, 9, 10, 13, 16, 17, 18, 20, 25, 26, 29, \ldots,$$

and among the odd primes, $5, 13, 17, 29, 37, 41$ appear, while $3, 7, 11, 19, 23, 31$ do not. By Example 10.2.2(2), no number $\equiv 3 \pmod 4$ is a sum of two squares. Fermat stated in 1640 that every prime $p \equiv 1 \pmod 4$ is one; Euler gave the first proof. We give a proof through the Gaussian integers, in which a sum of two squares factors: $a^2 + b^2 = (a + bi)(a - bi)$.

Recall from Example 8.5.5 that $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$ is a Euclidean domain for the norm $N(a + bi) = a^2 + b^2 = |a + bi|^2$, hence a PID and a UFD (Theorem 8.5.4). The norm is multiplicative, $N(zw) = |zw|^2 = N(z)N(w)$. Its units are the elements of norm 1, namely ± 1 and $\pm i$: if $zw = 1$ then $N(z)N(w) = 1$ in $\mathbb{N}$, so $N(z) = 1$; conversely $N(z) = z\bar{z} = 1$ gives $z^{-1} = \bar{z} \in \mathbb{Z}[i]$.

Theorem 10.5.1 (Fermat)**.** *An odd prime p is a sum of two squares if and only if $p \equiv 1 \pmod 4$. In that case the representation $p = a^2 + b^2$ is unique up to order and signs of a and b.*

Proof. If $p = a^2 + b^2$ then $p \not\equiv 3 \pmod 4$ by Example 10.2.2(2), so $p \equiv 1 \pmod 4$. Conversely let $p \equiv 1 \pmod 4$. By Corollary 10.4.3 there is $x \in \mathbb{Z}$ with $p \mid x^2 + 1 = (x + i)(x - i)$. But p divides neither factor in $\mathbb{Z}[i]$, since $(x \pm i)/p = x/p \pm i/p$ does not have integer coefficients. So p is not a prime element of $\mathbb{Z}[i]$, and since $\mathbb{Z}[i]$ is a UFD, p is not irreducible (Lemma 8.5.3): $p = zw$ with z, w not units. Then $p^2 = N(p) = N(z)N(w)$ with $N(z), N(w) > 1$, so $N(z) = p$, and writing $z = a + bi$, $p = a^2 + b^2$.

For uniqueness, let $p = a^2 + b^2 = c^2 + d^2$. Then $z = a + bi$ is irreducible: if $z = uv$, then $p = N(u)N(v)$, so one of $N(u)$, $N(v)$ is 1. In a UFD irreducible elements are prime, and z divides $p = (c + di)(c - di)$, so z divides $c + di$ or $c - di$; as both have norm $p = N(z)$, the quotient has norm 1 and is a unit. So $c + di$ or $c - di$ is $\pm(a + bi)$ or $\pm i(a + bi) = \mp(b - ai)$, and (c, d) is (a, b) or (b, a) up to order and signs. □

Example 10.5.2. $5 = 1^2 + 2^2$, $13 = 2^2 + 3^2$, $17 = 1^2 + 4^2$, $29 = 2^2 + 5^2$, $37 = 1^2 + 6^2$, $41 = 4^2 + 5^2$, and $1013 = 22^2 + 23^2$. The proof can be made into an algorithm: find x with $x^2 \equiv -1 \pmod p$, and compute a greatest common divisor of p and $x + i$ in $\mathbb{Z}[i]$ with the Euclidean algorithm. For $p = 13$, $x = 5$: dividing 13 by $5 + i$ gives $13/(5 + i) = 13(5 - i)/26 = (5 - i)/2$, rounded to $q = 2$ (or $3 - i$), with remainder $13 - 2(5 + i) = 3 - 2i$; and $(5 + i)/(3 - 2i) = (5 + i)(3 + 2i)/13 = (13 + 13i)/13 = 1 + i$ exactly. So the gcd is $3 - 2i$, and $13 = 3^2 + 2^2$.

Products of sums of two squares are sums of two squares, by the multiplicativity of the norm: writing out $N(zw) = N(z)N(w)$ for $z = a + bi$, $w = c + di$ gives the identity of Brahmagupta and Fibonacci,

$$(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2. \tag{10.1}$$

Theorem 10.5.3. *An integer $n \geq 1$ is a sum of two squares if and only if every prime $q \equiv 3 \pmod 4$ occurs in the prime factorisation of n with an even exponent.*

Proof. Suppose the condition holds. Then n is a product of factors $2 = 1^2 + 1^2$, primes $p \equiv 1 \pmod 4$, which are sums of two squares by Theorem 10.5.1, and squares $q^2 = q^2 + 0^2$; by (10.1) and induction on the number of factors, so is n.

Conversely let $n = a^2 + b^2$ and let $q \equiv 3 \pmod 4$ be a prime dividing n. We claim that $q \mid a$ and $q \mid b$. If $q \nmid b$, let b' be an inverse of b modulo q; then $a^2 \equiv -b^2$ gives $(ab')^2 \equiv -1 \pmod q$, contradicting Corollary 10.4.3. So $q \mid b$, hence $q \mid a^2$ and $q \mid a$. Then $q^2 \mid n$ and $n/q^2 = (a/q)^2 + (b/q)^2$ is again a sum of two squares. By induction on n, the exponent of q in n/q^2 is even, hence so is its exponent in n. □

Example 10.5.4. $2026 = 2 \cdot 1013$ with $1013 \equiv 1 \pmod 4$ prime, so 2026 is a sum of two squares; by (10.1) with $2 = 1^2 + 1^2$ and $1013 = 22^2 + 23^2$, $2026 = (22 - 23)^2 + (23 + 22)^2 = 1^2 + 45^2$. The number $65 = 5 \cdot 13$ has the two essentially different representations $1^2 + 8^2$ and $4^2 + 7^2$, coming from $(1 + 2i)(2 + 3i) = -4 + 7i$ and $(1 + 2i)(2 - 3i) = 8 + i$. On the other hand $21 = 3 \cdot 7$ and $2027 \equiv 3 \pmod 4$ are not sums of two squares, while $9 \cdot 7^2 = 441 = 21^2 + 0^2$ is.

Not every positive integer is a sum of three squares: no number $\equiv 7 \pmod 8$ is, since squares are $\equiv 0, 1, 4 \pmod 8$ and no three of these add up to 7. (Legendre and Gauss proved that the numbers that are not sums of three squares are exactly those of the form $4^a(8b + 7)$; we do not prove this, see [Ser73].) Four squares always suffice. The proof follows the same pattern as for two squares, with an identity of Euler in place of (10.1), and Fermat's method of descent in place of the Gaussian integers.

Lemma 10.5.5 (Euler's four-square identity)**.** *For all $a_1, \dots, a_4, b_1, \dots, b_4$ in a commutative ring,*

$$\Big(\sum_{i=1}^{4} a_i^2\Big)\Big(\sum_{i=1}^{4} b_i^2\Big) = z_1^2 + z_2^2 + z_3^2 + z_4^2,$$

where

$$z_1 = a_1b_1 + a_2b_2 + a_3b_3 + a_4b_4, \quad z_2 = a_1b_2 - a_2b_1 + a_3b_4 - a_4b_3,$$
$$z_3 = a_1b_3 - a_3b_1 + a_4b_2 - a_2b_4, \quad z_4 = a_1b_4 - a_4b_1 + a_2b_3 - a_3b_2.$$

Proof. Expand. Each z_k^2 contains four squares $a_i^2b_j^2$, and the sixteen squares that occur are all the $a_i^2b_j^2$, each once: z_1 contains the products a_ib_i, and z_2, z_3, z_4 contain the products a_ib_j for the pairs $\{i, j\}$ equal to $\{1, 2\}$ and $\{3, 4\}$, to $\{1, 3\}$ and $\{2, 4\}$, and to $\{1, 4\}$ and $\{2, 3\}$ respectively. The cross terms cancel in pairs; for instance the terms containing a_1a_2 are $2a_1a_2b_1b_2$ from z_1^2, $-2a_1a_2b_2b_1$ from z_2^2, $-2a_1a_2b_3b_4$ from z_3^2 and $2a_1a_2b_4b_3$ from z_4^2, with sum 0; the other five pairs a_ia_j are handled in the same way. (The identity expresses the multiplicativity of the norm of quaternions, $|xy|^2 = |x|^2|y|^2$.) □

Theorem 10.5.6 (Lagrange)**.** *Every positive integer is a sum of four squares of integers.*

Proof. By Lemma 10.5.5 and the factorisation into primes, and since $1 = 1^2 + 0^2 + 0^2 + 0^2$ and $2 = 1^2 + 1^2 + 0^2 + 0^2$, it suffices to treat an odd prime p.

Step 1: some multiple mp with $1 \le m < p$ is a sum of four squares. The $\frac{p+1}{2}$ numbers x^2, $0 \le x \le \frac{p-1}{2}$, are pairwise incongruent modulo p: if $x^2 \equiv x'^2$, then $p \mid (x-x')(x+x')$, and $0 \le x + x' < p$, $|x - x'| < p$, so $x = x'$. Likewise the $\frac{p+1}{2}$ numbers $-1 - y^2$, $0 \le y \le \frac{p-1}{2}$, are pairwise incongruent. Since $\frac{p+1}{2} + \frac{p+1}{2} > p$, some x^2 is congruent to some $-1 - y^2$, so $x^2 + y^2 + 1^2 + 0^2 = mp$ for some $m \ge 1$, and $mp \le 2\left(\frac{p-1}{2}\right)^2 + 1 < p^2$, so $m < p$.

Step 2: descent. Let m be the least positive integer such that mp is a sum of four squares; by Step 1, $m < p$. Suppose $m > 1$, and write $mp = x_1^2 + x_2^2 + x_3^2 + x_4^2$.

If m is even, then $\sum x_i^2$ is even, so an even number of the x_i are odd, and we may number them so that $x_1 \equiv x_2$ and $x_3 \equiv x_4 \pmod 2$. Then

$$\frac{m}{2}p = \left(\frac{x_1+x_2}{2}\right)^2 + \left(\frac{x_1-x_2}{2}\right)^2 + \left(\frac{x_3+x_4}{2}\right)^2 + \left(\frac{x_3-x_4}{2}\right)^2,$$

a sum of four squares of integers, contradicting the minimality of m.

If m is odd, $m \ge 3$, choose $y_i \equiv x_i \pmod m$ with $|y_i| < m/2$ (possible since m is odd). Then $\sum y_i^2 \equiv \sum x_i^2 = mp \equiv 0 \pmod m$, so $\sum y_i^2 = mr$ with $0 \le r$, and $mr < 4(m/2)^2 = m^2$, so $r < m$. If $r = 0$, all $y_i = 0$, so m divides every x_i, $m^2 \mid mp$ and $m \mid p$, impossible for $1 < m < p$. So $1 \le r < m$. By Lemma 10.5.5,

$$m^2 rp = \left(\sum x_i^2\right)\left(\sum y_i^2\right) = z_1^2 + z_2^2 + z_3^2 + z_4^2,$$

with the z_k formed from $a_i = x_i$, $b_i = y_i$. Since $y_i \equiv x_i \pmod m$, $z_1 \equiv \sum x_i^2 \equiv 0$, and $z_2 \equiv x_1x_2 - x_2x_1 + x_3x_4 - x_4x_3 = 0 \pmod m$, and likewise $z_3 \equiv z_4 \equiv 0$. So $rp = \sum (z_k/m)^2$ is a sum of four squares of integers with $1 \le r < m$, again contradicting the minimality of m. Hence $m = 1$. □

Example 10.5.7. $7 = 2^2 + 1^2 + 1^2 + 1^2$ needs four squares, as $7 \equiv 7 \pmod 8$. Also $2026 = 45^2 + 1^2 + 0^2 + 0^2$, $15 = 3^2 + 2^2 + 1^2 + 1^2$ and $2027 = 45^2 + 1^2 + 1^2 + 0^2$.

10.6 Arithmetic functions and Möbius inversion

Euler's function is one of many functions on the positive integers that are determined by their values on prime powers. This section develops the algebra of such functions.

Definition 10.6.1. An *arithmetic function* is a function $f : \{1, 2, 3, \ldots\} \to \mathbb{C}$. It is *multiplicative* if $f(1) = 1$ and $f(mn) = f(m)f(n)$ whenever $\gcd(m, n) = 1$, and *completely multiplicative* if moreover $f(mn) = f(m)f(n)$ for all m, n. The *Dirichlet convolution* of arithmetic functions f and g is

$$(f * g)(n) = \sum_{d \mid n} f(d)\, g\left(\frac{n}{d}\right) = \sum_{de = n} f(d)\, g(e),$$

the sums running over the positive divisors d of n, that is, over the pairs of positive integers (d, e) with $de = n$.

A multiplicative function is determined by its values at prime powers: $f(p_1^{k_1} \cdots p_r^{k_r}) = f(p_1^{k_1}) \cdots f(p_r^{k_r})$. We use the following functions:

- $\varepsilon(1) = 1$ and $\varepsilon(n) = 0$ for $n > 1$; $\mathbf{1}(n) = 1$ for all n; $\mathrm{N}(n) = n$. These are completely multiplicative.
- Euler's function φ, multiplicative by Proposition 10.2.8.

- $\tau(n)$, the number of positive divisors of n, and $\sigma(n)$, their sum. By definition $\tau = \mathbf{1} * \mathbf{1}$ and $\sigma = \mathrm{N} * \mathbf{1}$.
- The *Möbius function*: $\mu(1) = 1$, $\mu(n) = (-1)^r$ if n is a product of r distinct primes, and $\mu(n) = 0$ if n is divisible by the square of a prime. It is multiplicative but not completely multiplicative: $\mu(4) = 0 \neq \mu(2)^2$.

In this notation Proposition 10.2.8(4) reads $\varphi * \mathbf{1} = \mathrm{N}$.

Proposition 10.6.2. *1. Dirichlet convolution is commutative and associative, and ε is its identity element. With pointwise addition, the arithmetic functions form a commutative ring.*

2. An arithmetic function f has an inverse for $*$ if and only if $f(1) \neq 0$.

3. If f and g are multiplicative, so is $f * g$; and the inverse of a multiplicative function is multiplicative.

Proof. (1) Commutativity holds because $(f*g)(n) = \sum_{de=n} f(d)g(e)$ is symmetric in the roles of d and e. Associativity: both $((f*g)*h)(n)$ and $(f*(g*h))(n)$ equal $\sum_{cde=n} f(c)g(d)h(e)$, the sum over the triples of positive integers with product n. Further $(\varepsilon * f)(n) = \sum_{de=n} \varepsilon(d)f(e) = f(n)$. Distributivity over addition is clear.

(2) If $f * g = \varepsilon$, then $f(1)g(1) = 1$. Conversely, if $f(1) \neq 0$, define g recursively by $g(1) = 1/f(1)$ and, for $n > 1$,

$$g(n) = -\frac{1}{f(1)} \sum_{d \mid n,\ d>1} f(d)\, g\Big(\frac{n}{d}\Big),$$

which only involves values of g at numbers less than n. This says exactly that $(f * g)(1) = 1$ and $(f * g)(n) = 0$ for $n > 1$.

(3) Let $\gcd(m, n) = 1$. By unique factorisation, every divisor d of mn is uniquely a product $d = d_1 d_2$ with $d_1 \mid m$ and $d_2 \mid n$ (namely $d_1 = \gcd(d, m)$, $d_2 = \gcd(d, n)$), and then $\gcd(d_1, d_2) = \gcd(m/d_1, n/d_2) = 1$. Hence

$$(f * g)(mn) = \sum_{d_1 \mid m,\ d_2 \mid n} f(d_1) f(d_2)\, g\Big(\frac{m}{d_1}\Big) g\Big(\frac{n}{d_2}\Big) = (f * g)(m)\,(f * g)(n),$$

and $(f * g)(1) = 1$. For the inverse, let f be multiplicative, g its inverse, and let h be the multiplicative function with $h(p^k) = g(p^k)$ for all prime powers. Then $f * h$ is multiplicative by what we just proved, and $(f * h)(p^k) = \sum_{j=0}^{k} f(p^j)g(p^{k-j}) = (f * g)(p^k) = \varepsilon(p^k)$, because the divisors of p^k are the p^j. So $f * h$ and ε are multiplicative functions that agree on prime powers, hence $f * h = \varepsilon$ and $h = g$ by uniqueness of inverses. □

Proposition 10.6.3. *$\mu * \mathbf{1} = \varepsilon$; that is, $\sum_{d \mid n} \mu(d) = 0$ for every $n > 1$.*

Proof. Both sides are multiplicative (Proposition 10.6.2(3)), so it suffices to compare them at $n = p^k$, $k \geq 1$: the divisors are $1, p, \ldots, p^k$, and $\sum_{j=0}^{k} \mu(p^j) = 1 - 1 + 0 + \cdots + 0 = 0 = \varepsilon(p^k)$. □

Theorem 10.6.4 (Möbius inversion)**.** *For arithmetic functions f and g,*

$$g(n) = \sum_{d \mid n} f(d) \text{ for all } n \qquad \text{if and only if} \qquad f(n) = \sum_{d \mid n} \mu(d)\, g\Big(\frac{n}{d}\Big) \text{ for all } n.$$

Proof. The left side says $g = f * \mathbf{1}$, the right side $f = g * \mu$. Since μ is the inverse of $\mathbf{1}$ (Proposition 10.6.3), multiplying $g = f * \mathbf{1}$ by μ gives $g * \mu = f$, and multiplying $f = g * \mu$ by $\mathbf{1}$ gives $f * \mathbf{1} = g$. □

Example 10.6.5. 1. From $\varphi * \mathbf{1} = \mathrm{N}$, inversion gives $\varphi(n) = \sum_{d|n} \mu(d)\frac{n}{d} = n\sum_{d|n}\frac{\mu(d)}{d}$. Only squarefree d contribute, and for $n = p_1^{k_1}\cdots p_r^{k_r}$ the sum is $\prod_i(1-1/p_i)$, recovering Proposition 10.2.8(3).

2. τ and σ are multiplicative, being convolutions of multiplicative functions, and $\tau(p^k) = k+1$, $\sigma(p^k) = 1+p+\cdots+p^k = \frac{p^{k+1}-1}{p-1}$. So $\tau(n) = \prod_i(k_i+1)$ and $\sigma(n) = \prod_i \frac{p_i^{k_i+1}-1}{p_i-1}$. For instance $\tau(12) = 3\cdot 2 = 6$ and $\sigma(12) = 7\cdot 4 = 28 = 1+2+3+4+6+12$.

3. *Counting irreducible polynomials.* Let I_d be the number of monic irreducible polynomials of degree d over $\mathbb{F}_p$. We claim that $p^n = \sum_{d|n} dI_d$. By Theorem 9.6.2(2), $\mathbb{F}_{p^n}$ consists of the p^n roots of $x^{p^n} - x$, which are distinct. Each $\alpha \in \mathbb{F}_{p^n}$ has a minimal polynomial over $\mathbb{F}_p$ (Proposition 9.2.4), monic irreducible of some degree $d = [\mathbb{F}_p(\alpha) : \mathbb{F}_p]$, which divides n by the tower law (Theorem 9.2.2). Conversely, a monic irreducible f of degree $d \mid n$ has a root β in a field $\mathbb{F}_p(\beta) = \mathbb{F}_p[x]/(f)$ with p^d elements, so $\beta^{p^d} = \beta$ by Theorem 9.6.2(2), hence $\beta^{p^n} = \beta$ (apply $x \mapsto x^{p^d}$ n/d times), and f, being the minimal polynomial of β, divides $x^{p^n} - x$. As $x^{p^n} - x$ is the product of the $x - \alpha$ over the p^n distinct $\alpha \in \mathbb{F}_{p^n}$, f has exactly d distinct roots in $\mathbb{F}_{p^n}$, and these are exactly the elements with minimal polynomial f. Counting the elements of $\mathbb{F}_{p^n}$ according to their minimal polynomials gives the claim. Möbius inversion, applied to $g(n) = p^n$ and $f(d) = dI_d$, gives $nI_n = \sum_{d|n}\mu(d)p^{n/d}$. For $n = 2$: $I_2 = (p^2-p)/2$; for $p = 2$, $n = 4$: $I_4 = (16-4)/4 = 3$, namely x^4+x+1, x^4+x^3+1 and $x^4+x^3+x^2+x+1$. In particular $I_n > 0$ for all n: the term $d = 1$ of the sum is p^n, and the others have absolute value at most $p^{n/d} \le p^{n/2}$, so $nI_n \ge p^n - \sum_{e=0}^{\lfloor n/2\rfloor} p^e > p^n - p^{\lfloor n/2\rfloor+1} \ge 0$ for $n \ge 2$.

A number n is *perfect* if it is the sum of its divisors smaller than itself, that is, if $\sigma(n) = 2n$: for instance $6 = 1+2+3$, $28 = 1+2+4+7+14$, 496 and 8128. Euclid observed that $2^{k-1}(2^k-1)$ is perfect when $2^k - 1$ is prime; Euler proved the converse for even numbers.

Theorem 10.6.6 (Euclid, Euler)**.** *An even number n is perfect if and only if $n = 2^{k-1}(2^k-1)$ with $k \ge 2$ and $2^k - 1$ prime.*

Proof. If $q = 2^k - 1$ is prime, then by multiplicativity $\sigma(2^{k-1}q) = \sigma(2^{k-1})\sigma(q) = (2^k-1)(q+1) = (2^k-1)2^k = 2n$.

Conversely let n be even and perfect, and write $n = 2^{k-1}m$ with $k \ge 2$ and m odd. Then

$$2^k m = 2n = \sigma(n) = \sigma(2^{k-1})\sigma(m) = (2^k-1)\sigma(m).$$

Since $2^k - 1$ is odd, it is coprime to 2^k, so it divides m: $m = (2^k-1)M$, and then $\sigma(m) = 2^k M = m + M$. Now m and M are distinct divisors of m, as $2^k - 1 \ge 3$, and $\sigma(m) = m + M$ says that they are the only divisors of m. As 1 divides m, $M = 1$, and $m = 2^k - 1$ has only the divisors 1 and m: it is prime. □

Remark 10.6.7. Primes of the form $2^k - 1$ are called *Mersenne primes*; k must then be prime (Exercise 10.9.10), but this is not sufficient: $2^{11} - 1 = 2047 = 23\cdot 89$. It is unknown whether there are infinitely many Mersenne primes, and whether any odd perfect number exists.

10.7 Gaussian and Eisenstein integers

In Section 10.5 the Gaussian integers were a tool for proving Fermat's theorem on sums of two squares. Here we study the ring $\mathbb{Z}[i]$ for its own sake: we determine all its primes, which describes how each rational prime factors there, and we use the factorisation to count the

representations of an integer as a sum of two squares. The same method applied to the ring $\mathbb{Z}[\omega]$ of Eisenstein integers answers the analogous question for $x^2 + xy + y^2$. Finally, the ring $\mathbb{Z}[\sqrt{-5}]$ shows that unique factorisation, on which everything here rests, can fail; measuring this failure is one of the origins of algebraic number theory.

Recall that $\mathbb{Z}[i]$ is a Euclidean domain, hence a UFD, with multiplicative norm $N(a+bi) = a^2 + b^2$ and units ± 1, $\pm i$ (Section 10.5). Two elements are *associates* if they differ by a unit factor; in a UFD every nonzero nonunit is a product of primes, unique up to order and associates (Theorem 8.5.4), and irreducible elements are prime. An element whose norm is a rational prime is irreducible: if $z = uv$, then $N(u)N(v)$ is prime, so $N(u) = 1$ or $N(v) = 1$.

Theorem 10.7.1 (Gaussian primes)**.** *Up to associates, the primes of $\mathbb{Z}[i]$ are:*

1. $1 + i$, *with* $2 = -i(1+i)^2$;
2. *the rational primes* $q \equiv 3 \pmod 4$;
3. *for each rational prime* $p \equiv 1 \pmod 4$, *written* $p = a^2 + b^2$, *the two elements* $a + bi$ *and* $a - bi$, *which are not associates, with* $p = (a+bi)(a-bi)$.

Proof. The elements listed are prime. $N(1+i) = 2$ and $N(a \pm bi) = p$ are rational primes, so these elements are irreducible, hence prime. A rational prime $q \equiv 3 \pmod 4$ is irreducible in $\mathbb{Z}[i]$: if $q = zw$ with nonunits, then $q^2 = N(z)N(w)$ forces $N(z) = q$, so q would be a sum of two squares, contradicting Theorem 10.5.1. The associates of $a + bi$ are $a + bi$, $-a - bi$, $-b + ai$ and $b - ai$; if $a - bi$ were one of them, then $b = 0$, or $a = 0$, or $a = \pm b$, and $p = a^2 + b^2$ would be a square or even, which it is not.

Every prime is associate to one of them. Let π be a prime of $\mathbb{Z}[i]$. Then π divides $N(\pi) = \pi\bar{\pi}$, a positive integer greater than 1; factoring it into rational primes, π divides one of them, say p. Then $N(\pi)$ divides $N(p) = p^2$, so $N(\pi) = p$ or p^2. If $N(\pi) = p^2$, then p/π has norm 1, so π is associate to p, and p is prime in $\mathbb{Z}[i]$; this excludes $p = 2 = -i(1+i)^2$ and $p \equiv 1 \pmod 4$, which are products of two nonunits, so $p \equiv 3 \pmod 4$. If $N(\pi) = p$, then p is a sum of two squares, so $p = 2$ or $p \equiv 1 \pmod 4$. For $p = 2$, $\pi = x + yi$ with $x^2 + y^2 = 2$, so $x, y = \pm 1$ and π is associate to $1 + i$. For $p \equiv 1 \pmod 4$, π divides $p = (a+bi)(a-bi)$, hence divides one of the two factors, which has the same norm; so π is associate to $a + bi$ or to $a - bi$. □

So a rational prime p behaves in one of three ways in $\mathbb{Z}[i]$: it *ramifies* ($p = 2$ becomes a unit times the square of a prime), it stays *inert* ($p \equiv 3 \pmod 4$ remains prime), or it *splits* into two non-associate primes ($p \equiv 1 \pmod 4$). Which case occurs is decided by p modulo 4, that is, by the Legendre symbol $(\frac{-1}{p})$ (Corollary 10.4.3); this is the simplest instance of a *reciprocity law*, describing the factorisation of primes in an extension by congruence conditions.

Example 10.7.2. To factor $z = 7 + i$, factor its norm: $N(z) = 50 = 2 \cdot 5^2$. So z is divisible by $1 + i$ and by two primes of norm 5, each associate to $2 + i$ or $2 - i$. Indeed $(7+i)/(1+i) = (7+i)(1-i)/2 = 4 - 3i$, and $4 - 3i = -i(3+4i) = -i(2+i)^2$, so

$$7 + i = -i(1+i)(2+i)^2.$$

Note that $2 - i$ does not divide $7 + i$, although it divides $N(7+i)$: the two primes above 5 must be kept apart. As an application, $7 + i$ is not a square times a unit, although its norm 50 is twice a square.

The factorisation counts the representations of an integer as a sum of two squares, refining Theorem 10.5.3. Let $r_2(n)$ be the number of pairs $(x, y) \in \mathbb{Z}^2$ with $x^2 + y^2 = n$, signs and order counted; and let χ be the function on positive integers with $\chi(d) = 0$ for even d, $\chi(d) = 1$ for $d \equiv 1 \pmod 4$ and $\chi(d) = -1$ for $d \equiv 3 \pmod 4$. It is completely multiplicative, since the product of two odd numbers is $\equiv 1 \pmod 4$ exactly when both or neither are $\equiv 3 \pmod 4$.

Theorem 10.7.3 (Jacobi's two squares theorem)**.** *For every $n \geq 1$,*

$$r_2(n) = 4 \sum_{d \mid n} \chi(d) = 4\big(d_1(n) - d_3(n)\big),$$

where $d_1(n)$ and $d_3(n)$ are the numbers of positive divisors of n congruent to 1 and to 3 modulo 4.

Proof. $r_2(n)$ is the number of $z \in \mathbb{Z}[i]$ with $N(z) = n$. Write $n = 2^e \prod_j p_j^{a_j} \prod_k q_k^{b_k}$ with distinct primes $p_j \equiv 1$ and $q_k \equiv 3 \pmod 4$, and choose primes π_j with $p_j = \pi_j \bar{\pi}_j$ (Theorem 10.7.1). By unique factorisation, every nonzero z is uniquely

$$z = u(1+i)^{e'} \prod_j \pi_j^{s_j} \bar{\pi}_j^{t_j} \prod_k q_k^{c_k} \cdot w,$$

with u a unit and w a product of primes not dividing n, and then $N(z) = 2^{e'} \prod_j p_j^{s_j+t_j} \prod_k q_k^{2c_k} N(w)$. So $N(z) = n$ if and only if $w = 1$, $e' = e$, $s_j + t_j = a_j$ and $2c_k = b_k$. If some b_k is odd there is no solution; otherwise there are 4 choices of u and $a_j + 1$ choices of (s_j, t_j) for each j, so $r_2(n) = 4 \prod_j (a_j + 1)$.

On the other side, $\chi * \mathbf{1}$, $n \mapsto \sum_{d \mid n} \chi(d)$, is multiplicative (Proposition 10.6.2(3)), and on prime powers it takes the values $\sum_{i=0}^{e} \chi(2^i) = 1$, $\sum_{i=0}^{a} \chi(p^i) = a + 1$ for $p \equiv 1 \pmod 4$, and $\sum_{i=0}^{b} (-1)^i$, which is 1 for even b and 0 for odd b, for $q \equiv 3 \pmod 4$. The product of these values is $\prod_j (a_j + 1)$ if all b_k are even and 0 otherwise. □

Example 10.7.4. $r_2(5) = 4(2 - 0) = 8$: $(\pm 1, \pm 2)$, $(\pm 2, \pm 1)$. $r_2(25) = 4 \cdot 3 = 12$: $(0, \pm 5)$, $(\pm 5, 0)$, $(\pm 3, \pm 4)$, $(\pm 4, \pm 3)$. $r_2(9) = 4(2 - 1) = 4$: only $(\pm 3, 0)$ and $(0, \pm 3)$, as $9 = 3^2$ with 3 inert. $r_2(3) = 4(1 - 1) = 0$. A pretty consequence, which uses areas and is only sketched here: $\sum_{n \leq N} r_2(n)$ counts the points of $\mathbb{Z}^2$ in the disc of radius $\sqrt{N}$, about πN of them, while by the theorem it equals $4 \sum_{d \leq N} \chi(d) \lfloor N/d \rfloor$, which is about $4N(1 - \frac{1}{3} + \frac{1}{5} - \cdots)$; this is how Gauss's circle problem yields Leibniz's series $1 - \frac{1}{3} + \frac{1}{5} - \cdots = \frac{\pi}{4}$.

10.7.1 Eisenstein integers

Let $\omega = e^{2\pi i/3} = \frac{-1+\sqrt{-3}}{2}$, a root of $x^2 + x + 1$, so that $\omega^2 = -1 - \omega = \bar{\omega}$. The ring of *Eisenstein integers* is $\mathbb{Z}[\omega] = \{a + b\omega : a, b \in \mathbb{Z}\}$; it is closed under multiplication because $\omega^2 = -1 - \omega$. Its elements form the triangular lattice in $\mathbb{C}$, and the norm is

$$N(a + b\omega) = |a + b\omega|^2 = (a + b\omega)(a + b\bar{\omega}) = a^2 - ab + b^2,$$

using $\omega + \bar{\omega} = -1$ and $\omega\bar{\omega} = 1$. It is multiplicative, with values in $\mathbb{N}$. Note that $\sqrt{-3} = 1 + 2\omega \in \mathbb{Z}[\omega]$.

Proposition 10.7.5. *1. $\mathbb{Z}[\omega]$ is a Euclidean domain for N, hence a UFD.*

2. *Its units are the six elements ± 1, $\pm\omega$, $\pm\omega^2$ of norm 1, the sixth roots of unity.*
3. *Up to associates its primes are: $1 - \omega$, with $3 = -\omega^2(1 - \omega)^2$; the rational primes $q \equiv 2 \pmod 3$; and for each rational prime $p \equiv 1 \pmod 3$ two non-associate primes π, $\bar{\pi}$ with $p = \pi\bar{\pi}$.*

Proof. (1) Let $a, b \in \mathbb{Z}[\omega]$ with $b \neq 0$, and write $a/b = u + v\omega$ with $u, v \in \mathbb{R}$ (possible since 1 and ω are a basis of $\mathbb{C}$ over $\mathbb{R}$). Choose integers m, n with $|u - m|, |v - n| \leq \frac{1}{2}$, and put $q = m + n\omega$, $r = a - qb$. With $s = u - m$ and $t = v - n$, $|a/b - q|^2 = s^2 - st + t^2 \leq \frac{1}{4} + \frac{1}{4} + \frac{1}{4} < 1$, so $N(r) < N(b)$. A Euclidean domain is a UFD (Theorem 8.5.4).

(2) As for $\mathbb{Z}[i]$, the units are the elements of norm 1. From $4(a^2-ab+b^2) = (2a-b)^2+3b^2 = 4$ we get $b \in \{0, \pm 1\}$, and the solutions are ± 1, $\pm\omega$ and $\pm(1+\omega) = \mp\omega^2$.

(3) Since $1-\omega$ has norm $1+1+1=3$, it is prime, and $(1-\omega)^2 = 1-2\omega+\omega^2 = -3\omega$, so $3 = -\omega^2(1-\omega)^2$. The values of the norm modulo 3: $4N(a+b\omega) = (2a-b)^2+3b^2 \equiv (2a-b)^2 \pmod 3$, which is 0 or 1; as $4 \equiv 1$, no norm is $\equiv 2 \pmod 3$. Hence a rational prime $q \equiv 2 \pmod 3$ is prime in $\mathbb{Z}[\omega]$, by the argument used for $\mathbb{Z}[i]$. Let $p \equiv 1 \pmod 3$. By quadratic reciprocity (Theorem 10.4.9) and Corollary 10.4.3,

$$\left(\frac{-3}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{3}{p}\right) = (-1)^{\frac{p-1}{2}} \cdot (-1)^{\frac{p-1}{2}} \left(\frac{p}{3}\right) = \left(\frac{p}{3}\right) = \left(\frac{1}{3}\right) = 1,$$

using that the Legendre symbol is multiplicative (Proposition 10.4.2). So there is x with $p \mid x^2+3 = (x+\sqrt{-3})(x-\sqrt{-3})$. Now p divides neither factor: $(x \pm \sqrt{-3})/p = (x \pm 1)/p \pm (2/p)\omega$ does not lie in $\mathbb{Z}[\omega]$, as p is odd. So p is not prime, $p = \pi\pi'$ with nonunits, and $N(\pi) = p$, so π is prime and $p = \pi\bar{\pi}$.

$\pi = a+b\omega$ and $\bar{\pi}$ are not associates. Otherwise π divides $\bar{\pi}$, hence the integer $\pi+\bar{\pi} = 2a-b$ and the element $\pi - \bar{\pi} = b(\omega - \bar{\omega}) = b\sqrt{-3}$. If π divides an element z, then $p = N(\pi)$ divides $N(z)$; so $p \mid (2a-b)^2$ and $p \mid 3b^2$, hence $p \mid b$ (as $p \neq 3$), then $p \mid 2a$ and $p \mid a$. But then $p^2 \mid a^2 - ab + b^2 = p$, which is absurd.

Finally let π be any prime. As in Theorem 10.7.1, π divides a rational prime p, and $N(\pi) = p$ or p^2. If $N(\pi) = p^2$, then π is associate to p, so p is prime in $\mathbb{Z}[\omega]$; this excludes $p = 3$ and $p \equiv 1 \pmod 3$, which are products of two nonunits, so $p \equiv 2 \pmod 3$. If $N(\pi) = p$, then $p \not\equiv 2 \pmod 3$; for $p = 3$, π divides $(1-\omega)^2$, hence is associate to $1-\omega$, and for $p \equiv 1 \pmod 3$, π divides one of the two primes of norm p found above and is associate to it. □

Corollary 10.7.6. *A prime p can be written as $p = x^2+xy+y^2$ with $x, y \in \mathbb{Z}$ if and only if $p = 3$ or $p \equiv 1 \pmod 3$; and as $p = x^2+3y^2$ if and only if $p = 3$ or $p \equiv 1 \pmod 3$.*

Proof. $a^2 - ab + b^2 = x^2 + xy + y^2$ with $x = a$, $y = -b$, so the primes of the first form are the norms of elements of $\mathbb{Z}[\omega]$, which by the proposition are 3 and the $p \equiv 1 \pmod 3$. For the second form: $3 = 0^2+3\cdot 1^2$, and $x^2+3y^2 = N(x+y\sqrt{-3})$ is a norm, so the condition is necessary. Conversely let $p \equiv 1 \pmod 3$ and $p = N(a+b\omega)$. Among the six associates $\varepsilon(a+b\omega)$ choose one, $c+d\omega$, with d even: multiplying by ω sends $a+b\omega$ to $a\omega+b\omega^2 = -b+(a-b)\omega$, so the second coordinates of $a+b\omega$, $\omega(a+b\omega)$, $\omega^2(a+b\omega)$ are b, $a-b$, $-a$, and one of b, $a-b$, a is even. Then $c+d\omega = (c-d/2)+(d/2)\sqrt{-3}$, and $p = (c-d/2)^2+3(d/2)^2$. □

For example $7 = 2^2+2+1 = 2^2+3\cdot 1^2$, $13 = 3^2+3+1 = 1^2+3\cdot 2^2$, $31 = 5^2+5+1 = 2^2+3\cdot 3^2$, while 5, 11, 17 are of neither form. The ring $\mathbb{Z}[\omega]$ is the setting of Euler's proof of Fermat's last theorem for exponent 3, and its sixfold symmetry is that of the hexagonal lattice and of the elliptic curve $\mathbb{C}/\mathbb{Z}[\omega]$ (Chapter 61).

10.7.2 Failure of unique factorisation

Example 10.7.7. In $\mathbb{Z}[\sqrt{-5}] = \{a+b\sqrt{-5} : a, b \in \mathbb{Z}\}$, with multiplicative norm $N(a+b\sqrt{-5}) = a^2+5b^2$,

$$6 = 2\cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$$

are two factorisations into irreducible elements that do not correspond to each other. The norms of 2, 3, $1 \pm \sqrt{-5}$ are 4, 9, 6, 6. The units are ± 1, the only elements of norm 1. A proper factor of one of these four elements would have norm 2 or 3, but a^2+5b^2 never equals 2 or 3 (if $b \neq 0$ it is at least 5). So all four are irreducible, and 2 is not associate to $1 \pm \sqrt{-5}$, having a different norm. Hence $\mathbb{Z}[\sqrt{-5}]$ is not a UFD, and 2 is irreducible but not prime: it divides $(1+\sqrt{-5})(1-\sqrt{-5})$ but neither factor.

Kummer and Dedekind restored unique factorisation by passing from elements to *ideals*: in $\mathbb{Z}[\sqrt{-5}]$ the ideals $\mathfrak{p} = (2, 1+\sqrt{-5})$, $\mathfrak{q} = (3, 1+\sqrt{-5})$ and $\bar{\mathfrak{q}} = (3, 1-\sqrt{-5})$ are prime, and $(6) = \mathfrak{p}^2\mathfrak{q}\bar{\mathfrak{q}}$, with $(2) = \mathfrak{p}^2$, $(3) = \mathfrak{q}\bar{\mathfrak{q}}$ and $(1 \pm \sqrt{-5}) = \mathfrak{p}\mathfrak{q}$ or $\mathfrak{p}\bar{\mathfrak{q}}$; the two factorisations of 6 are two groupings of the same four prime ideals. The failure of unique factorisation of elements is measured by the *class group*, the group of ideals modulo principal ideals, which is finite; its order, the *class number*, is 2 for $\mathbb{Z}[\sqrt{-5}]$. We do not develop ideal theory here and only state these facts [IR90]; ideals of rings are studied in Chapter 14. The next section computes class numbers in the equivalent language of quadratic forms, which is how Gauss found them.

10.8 Binary quadratic forms

Fermat's theorem says which primes are $x^2 + y^2$, and Corollary 10.7.6 which are $x^2 + xy + y^2$ and $x^2 + 3y^2$. What about $x^2 + 5y^2$? The primes $29 = 3^2 + 5 \cdot 2^2$ and $41 = 6^2 + 5 \cdot 1^2$ are of this form, but 3 and 7 are not, although -20 is a square modulo 3 and 7. They are represented instead by the form $2x^2 + 2xy + 3y^2$, at $(x, y) = (0, 1)$ and $(1, 1)$. Lagrange and Gauss understood that one should study all forms of a given discriminant at once, up to changes of variables. This theory, reduction of forms and the class number, explains the examples, and leads to Euler's famous prime-producing polynomial $n^2 + n + 41$ and to the Heegner numbers.

Definition 10.8.1. A *binary quadratic form* is $f(x, y) = ax^2 + bxy + cy^2$ with $a, b, c \in \mathbb{Z}$, written $f = (a, b, c)$. Its *discriminant* is $D = b^2 - 4ac$. The form is *positive definite* if $f(x, y) > 0$ for all $(x, y) \in \mathbb{R}^2 \setminus \{0\}$. An integer m is *represented* by f if $m = f(x, y)$ with $x, y \in \mathbb{Z}$, and *properly* represented if moreover $\gcd(x, y) = 1$. Two forms f and g are *equivalent* if $g(x, y) = f(px + qy, rx + sy)$ for some integers p, q, r, s with $ps - qr = 1$, that is, for a matrix in $\mathrm{SL}_2(\mathbb{Z})$.

The discriminant satisfies $D \equiv b^2 \equiv 0$ or $1 \pmod 4$. Completing the square, $4a f(x, y) = (2ax + by)^2 - Dy^2$; so f is positive definite if and only if $D < 0$ and $a > 0$, and then $c > 0$ as well. Equivalence is an equivalence relation, because $\mathrm{SL}_2(\mathbb{Z})$ is a group; it preserves the discriminant, and equivalent forms represent the same integers, properly or not, since a matrix in $\mathrm{SL}_2(\mathbb{Z})$ permutes $\mathbb{Z}^2$ and the primitive vectors. In matrix terms, $f(x, y) = (x\ y)\, M_f\, (x\ y)^T$ with $M_f = \left(\begin{smallmatrix} a & b/2 \\ b/2 & c \end{smallmatrix}\right)$, $D = -4 \det M_f$, and substituting $(x, y)^T \mapsto P(x, y)^T$ replaces M_f by $P^T M_f P$, which has the same determinant. Two substitutions are used constantly:

$$(a, b, c) \sim (a, b+2at, at^2+bt+c) \quad \big(x \mapsto x+ty\big), \qquad (a, b, c) \sim (c, -b, a) \quad \big((x, y) \mapsto (-y, x)\big).$$

Definition 10.8.2. A positive definite form (a, b, c) is *reduced* if $|b| \le a \le c$, and $b \ge 0$ whenever $|b| = a$ or $a = c$.

Theorem 10.8.3 (Lagrange, Gauss)**.** *Every positive definite form is equivalent to exactly one reduced form. If (a, b, c) is reduced of discriminant D, then $a \le \sqrt{|D|/3}$; hence there are only finitely many reduced forms of a given discriminant.*

Proof. Existence. Let $f = (a, b, c)$ be positive definite. Using the first substitution with a suitable t we can achieve $|b| \le a$, since $b + 2at$ runs through a residue class modulo $2a$. If then $a > c$, the second substitution replaces a by the smaller positive integer c. Alternate these two steps; since a decreases strictly at each use of the second step, after finitely many steps $|b| \le a \le c$. Finally, if $b = -a$, the first substitution with $t = 1$ gives (a, a, c) (as $a - a + c = c$); and if $a = c$ and $b < 0$, the second gives $(a, -b, a)$. The result is reduced.

The bound. If $|b| \le a \le c$, then $|D| = 4ac - b^2 \ge 4a^2 - a^2 = 3a^2$.

Uniqueness. Let $f = (a, b, c)$ satisfy $|b| \le a \le c$. For integers x, y: if $|x| \ge |y|$, then $f(x, y) \ge ax^2 - |b||x||y| + cy^2 \ge (a - |b|)x^2 + cy^2$, and if $|y| \ge |x|$, similarly $f(x, y) \ge ax^2 + (c - |b|)y^2$. Hence $f(x, y) \ge a$ if $(x, y) \ne 0$, $f(x, y) \ge c$ if $y \ne 0$ (for $x = 0$ this is $cy^2 \ge c$, and for $xy \ne 0$ both bounds give at least $a - |b| + c \ge c$), and, when $|b| < a < c$, $f(x, y) > a$ for all $(x, y) \ne (\pm 1, 0)$, the bound for $xy \ne 0$ then being $a - |b| + c > c > a$.

Consequently a is the smallest nonzero value of f on $\mathbb{Z}^2$, and c is the smallest value $f(w)$ for w linearly independent of a vector v with $f(v) = a$. (For $v = (1, 0)$ the vectors w independent of v have $y \ne 0$, so $f(w) \ge c = f(0, 1)$. If $f(v) = a$ with $v = (x, y)$, $y \ne 0$, then $a = f(v) \ge c$, so $a = c$, every w has $f(w) \ge a = c$, and $w = (1, 0)$ is independent of v with $f(w) = c$.) These two numbers are unchanged by an equivalence, which maps $\mathbb{Z}^2$ bijectively to itself and preserves linear independence. So equivalent reduced forms have the same a and c, hence the same $|b| = \sqrt{D + 4ac}$.

It remains to exclude that (a, b, c) and $(a, -b, c)$, with $b > 0$, are both reduced and equivalent. If $(a, -b, c)$ is reduced, then $|b| < a < c$ by the sign condition. Suppose $(a, b, c)(x, y) = g(px + qy, rx + sy)$ with $g = (a, -b, c)$ and $ps - qr = 1$. Taking $(x, y) = (1, 0)$ gives $g(p, r) = a$, so $(p, r) = \pm(1, 0)$ by the strict inequality above, applied to g. Then $ps = 1$ gives $s = p = \pm 1$, and the substitution is $\pm(x + q'y, y)$ with $q' = \pm q$; as $g(-v) = g(v)$, (a, b, c) is obtained from g by the first substitution with $t = q'$, so $b = -b + 2aq'$. Then $2|b| = 2a|q'|$ with $0 < |b| < a$, which is impossible. □

The number of reduced forms of discriminant D is called the *class number* $h(D)$ (often one counts only *primitive* forms, with $\gcd(a, b, c) = 1$; for the discriminants below the two counts agree unless noted). The reduced forms are found by a finite search: $b \equiv D \pmod 2$, $|b| \le a \le \sqrt{|D|/3}$, and $c = (b^2 - D)/4a$ must be an integer with $c \ge a$.

Example 10.8.4.

1. $D = -4$: $a \le 1$, so $(a, b, c) = (1, 0, 1)$, and $h(-4) = 1$. $D = -3$: $(1, 1, 1)$. $D = -8$: $a \le 1$, $b = 0$, $(1, 0, 2)$. $D = -7$: $(1, 1, 2)$. So each of these discriminants has the single reduced form $x^2 + y^2$, $x^2 + xy + y^2$, $x^2 + 2y^2$, $x^2 + xy + 2y^2$ respectively.

2. $D = -12$: $a \le 2$; $a = 1$ gives $(1, 0, 3)$; $a = 2$ requires $b \in \{0, 2\}$ and $c = (b^2 + 12)/8$, which gives $(2, 2, 2)$. This form is not primitive and represents only even numbers; the primitive class number is $h(-12) = 1$.

3. $D = -20$: $a \le 2$; $(1, 0, 5)$, and for $a = 2$, $b \in \{0, 2\}$ with $c = (b^2 + 20)/8$, giving $(2, 2, 3)$. So $h(-20) = 2$, with the forms $x^2 + 5y^2$ and $2x^2 + 2xy + 3y^2$, matching the class number 2 of $\mathbb{Z}[\sqrt{-5}]$ in Example 10.7.7.

4. $D = -23$: $a \le 2$, b odd; $b = \pm 1$, $ac = 6$: $(1, 1, 6)$, $(2, 1, 3)$, $(2, -1, 3)$, all reduced (for $a = 2$ we have $|b| < a < c$, so both signs are allowed). So $h(-23) = 3$: the forms $2x^2 \pm xy + 3y^2$ are *improperly* equivalent (by $(x, y) \mapsto (x, -y)$) but not equivalent under $\mathrm{SL}_2(\mathbb{Z})$.

To connect forms with primes we need to know which forms represent a given number.

Lemma 10.8.5. *Let $D < 0$ and $m \ge 1$. Then m is properly represented by some form of discriminant D if and only if $D \equiv b^2 \pmod{4m}$ for some integer b. More precisely, if $D = b^2 - 4mc$, then m is properly represented by (m, b, c); and if $f(x_0, y_0) = m$ with $\gcd(x_0, y_0) = 1$, then f is equivalent to a form (m, b, c).*

Proof. If $D = b^2 - 4mc$, then (m, b, c) has discriminant D and $m = f(1, 0)$. Conversely let $f(x_0, y_0) = m$ with $\gcd(x_0, y_0) = 1$. By Bézout's identity choose q, s with $x_0 s - q y_0 = 1$; the substitution $(x, y) \mapsto (x_0 x + qy, y_0 x + sy)$ lies in $\mathrm{SL}_2(\mathbb{Z})$ and transforms f into a form g with $g(1, 0) = f(x_0, y_0) = m$, that is, $g = (m, b, c)$, and $D = b^2 - 4mc \equiv b^2 \pmod{4m}$. □

Corollary 10.8.6. *Let $D < 0$ be a discriminant such that the only reduced form of discriminant D is f_0. Then a prime p is represented by f_0 if and only if D is a square modulo $4p$. In particular:*

1. *(Fermat) an odd prime p is $x^2 + y^2$ if and only if $p \equiv 1 \pmod 4$;*
2. *an odd prime p is $x^2 + 2y^2$ if and only if $p \equiv 1, 3 \pmod 8$;*
3. *a prime $p \neq 7$ is $x^2 + xy + 2y^2$, equivalently $x^2 + 7y^2$ when p is odd, if and only if $(\frac{p}{7}) = 1$, that is, $p \equiv 1, 2, 4 \pmod 7$.*

Proof. A prime is properly represented whenever it is represented, since $f(x, y) = p$ forces $\gcd(x, y)^2 \mid p$. So by Lemma 10.8.5, p is represented by some form of discriminant D if and only if D is a square modulo $4p$; and every such form is equivalent to its reduced form f_0 (Theorem 10.8.3), which represents the same numbers.

For odd p, $D \equiv b^2 \pmod{4p}$ has a solution if and only if D is a square modulo p: given $D \equiv b^2 \pmod p$ we may change b by p to make $b \equiv D \pmod 2$, and then $b^2 \equiv D \pmod 4$ as well (both sides are $\equiv 0$ or $\equiv 1 \pmod 4$ according to their parity), so $b^2 \equiv D \pmod{4p}$ by the Chinese remainder theorem. Now (1) is $D = -4$, where -4 is a square modulo p if and only if -1 is (Corollary 10.4.3). For (2), $D = -8$ and $(\frac{-2}{p}) = (\frac{-1}{p})(\frac{2}{p})$ equals 1 exactly for $p \equiv 1, 3 \pmod 8$, by Corollary 10.4.3 and Proposition 10.4.7. For (3), $D = -7$; for odd $p \neq 7$, reciprocity gives $(\frac{-7}{p}) = (-1)^{\frac{p-1}{2}}(\frac{7}{p}) = (-1)^{\frac{p-1}{2}}(-1)^{\frac{p-1}{2}\cdot 3}(\frac{p}{7}) = (\frac{p}{7})$, and the squares modulo 7 are $1, 2, 4$. For $p = 2$, $-7 \equiv 1 = 1^2 \pmod 8$ and $2 = x^2 + xy + 2y^2$ with $(x, y) = (0, 1)$, consistent with $2 \equiv 2 \pmod 7$. Finally, for odd values, $4(x^2 + xy + 2y^2) = (2x + y)^2 + 7y^2$, and $p = x^2 + xy + 2y^2$ odd forces y even (if y were odd, $x^2 + xy = x(x + y)$ would be even and the value even), so $p = (x + y/2)^2 + 7(y/2)^2$; conversely $x^2 + 7y^2 = (x - y)^2 + (x - y)(2y) + 2(2y)^2$. □

When $h(D) > 1$ the corollary only says that p is represented by *one of* the reduced forms. For $D = -20$, an odd prime $p \neq 5$ is represented by $x^2 + 5y^2$ or by $2x^2 + 2xy + 3y^2$ exactly when $(\frac{-5}{p}) = 1$, that is (by reciprocity) when $p \equiv 1, 3, 7, 9 \pmod{20}$. Congruences separate the two forms: $x^2 + 5y^2$ takes odd prime values only $\equiv 1, 9 \pmod{20}$, and $2x^2 + 2xy + 3y^2$ only $\equiv 3, 7 \pmod{20}$ (Exercise 10.9.15). So $p = x^2 + 5y^2$ if and only if $p \equiv 1, 9 \pmod{20}$, as Euler conjectured. For $D = -23$ no such congruence description exists for $x^2 + xy + 6y^2$; the answer requires class field theory [Cox89].

10.8.1 Euler's polynomial and the Heegner numbers

Euler observed in 1772 that $n^2 + n + 41$ is prime for $n = 0, 1, \ldots, 39$: the values are $41, 43, 47, 53, 61, 71, \ldots, 1601$. (For $n = 40$ the value is 41^2.) The explanation is that -163, the discriminant of $x^2 + xy + 41y^2$, has class number one.

Theorem 10.8.7 (Rabinowitsch)**.** *Let p be a prime and $D = 1 - 4p$. The following are equivalent:*

1. *the only reduced form of discriminant D is $x^2 + xy + py^2$;*
2. *$n^2 + n + p$ is prime for $n = 0, 1, \ldots, p - 2$.*

Proof. Note $4(n^2 + n + p) = (2n + 1)^2 - D$.

(1) $\Rightarrow$ (2). Let $0 \leq n \leq p - 2$ and $N = n^2 + n + p$, and suppose N is not prime. Then it has a prime factor $q \leq \sqrt{N}$, and $N \leq (p - 2)^2 + (p - 2) + p = p^2 - 2p + 2 < p^2$, so $q < p$. Since $q \mid N$, $(2n + 1)^2 - D = 4N$ is divisible by $4q$, and $(q, 2n + 1, N/q)$ is a form of discriminant D representing q. By (1) and Theorem 10.8.3 it is equivalent to $x^2 + xy + py^2$, so $q = x^2 + xy + py^2$ for some integers x, y. If $y = 0$, $q = x^2$ is not prime. If $y \neq 0$,

$$q = \left(x + \frac{y}{2}\right)^2 + \frac{4p - 1}{4}y^2 \geq \frac{4p - 1}{4} > p - 1,$$

so $q \geq p$, a contradiction.

(2) $\Rightarrow$ (1). Let (a, b, c) be reduced of discriminant D. Then b is odd, $b = \pm(2n+1)$ with $n \geq 0$, and $ac = (b^2 - D)/4 = n^2 + n + p$. By Theorem 10.8.3, $2n + 1 = |b| \leq a \leq \sqrt{(4p-1)/3}$, so $n < \sqrt{p/3} \leq p - 2$ for $p \geq 3$, while for $p = 2$ the bound gives $n = 0$. So ac is prime by (2), and $a \leq c$ forces $a = 1$, hence $|b| \leq 1$, $b = 1$ (the sign condition for $|b| = a$), and $c = p$. □

Corollary 10.8.8. *$n^2 + n + 41$ is prime for $n = 0, 1, \ldots, 39$.*

Proof. We check condition (1) of Theorem 10.8.7 for $D = -163$. A reduced form (a, b, c) has b odd, $|b| \leq a \leq \sqrt{163/3} < 8$, and $ac = (b^2 + 163)/4$. For $b = \pm 1, \pm 3, \pm 5, \pm 7$ this is 41, 43, 47, 53, all prime; as $a \leq c$, $a = 1$, and then $|b| \leq 1$ and $b = 1$. So $(1, 1, 41)$ is the only reduced form. □

Forty values were checked by an argument involving eight. The same computation applies to $p = 2, 3, 5, 11, 17, 41$, with $D = -7, -11, -19, -43, -67, -163$ (Exercise 10.9.17).

Remark 10.8.9.

1. *The Heegner numbers.* The negative discriminants D for which every primitive positive definite form is equivalent to the principal one, $x^2 - \frac{D}{4}y^2$ or $x^2 + xy + \frac{1-D}{4}y^2$, are exactly

$$D = -3, -4, -7, -8, -11, -12, -16, -19, -27, -28, -43, -67, -163.$$

Equivalently, the imaginary quadratic fields $\mathbb{Q}(\sqrt{-d})$, $d > 0$ squarefree, whose ring of integers is a UFD are those with $d \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$, the *Heegner numbers*. Gauss conjectured that the list is complete; it was proved by Heegner in 1952, whose argument was long considered incomplete, and by Baker and Stark in 1966–1967 [Cox89]. Combined with Theorem 10.8.7, it determines the primes p for which $n^2 + n + p$ is prime for $0 \leq n \leq p - 2$: $1 - 4p$ must be one of the odd discriminants in the list, which gives $p \in \{2, 3, 5, 7, 11, 17, 41\}$, and $p = 7$ fails because of the non-primitive reduced form $(3, 3, 3)$ of discriminant -27 (indeed $1 + 1 + 7 = 9$). So the primes are $2, 3, 5, 11, 17, 41$, and Euler's 41 is the largest.

2. *An almost integer.* The number $e^{\pi\sqrt{163}} = 262537412640768743.99999999999925\ldots$ is extraordinarily close to the integer $640320^3 + 744$. The reason is again $h(-163) = 1$: the j-invariant of the elliptic curve $\mathbb{C}/\mathbb{Z}[\frac{1+\sqrt{-163}}{2}]$ is then an integer, equal to -640320^3, and the expansion $j(\tau) = e^{-2\pi i\tau} + 744 + 196884e^{2\pi i\tau} + \cdots$ at $\tau = \frac{1+\sqrt{-163}}{2}$, where $e^{2\pi i\tau} = -e^{-\pi\sqrt{163}}$ is tiny, gives $-e^{\pi\sqrt{163}} + 744 \approx -640320^3$ [Cox89]. This is the theory of complex multiplication, which connects class numbers with elliptic curves (Chapter 61).

3. *Class numbers grow.* Although $h(D) = 1$ only finitely often, $h(D) \to \infty$ as $D \to -\infty$ (Heilbronn, 1934); roughly, $h(D)$ is of size $\sqrt{|D|}$. The class number is also given by Dirichlet's *class number formula*, which for $D < -4$ fundamental reads $h(D) = -\frac{1}{|D|}\sum_{m=1}^{|D|}\left(\frac{D}{m}\right)m$ with the Kronecker symbol [Cox89]; for $D = -23$ it gives 3.

10.9 Exercises

Exercise 10.9.1. Show that there are infinitely many primes $p \equiv 3 \pmod 4$. (Consider $N = 4p_1 \cdots p_k - 1$.)

Solution. Let $p_1, \ldots, p_k$ be primes $\equiv 3 \pmod 4$ and $N = 4p_1 \cdots p_k - 1 \equiv 3 \pmod 4$. N is odd, so its prime factors are $\equiv 1$ or $3 \pmod 4$; if all were $\equiv 1$, their product N would be $\equiv 1 \pmod 4$. So some prime $q \equiv 3 \pmod 4$ divides N, and $q \neq p_i$, since $p_i \mid N + 1$. □

Exercise 10.9.2. Show that an integer $n \geq 2$ is prime if and only if $(n-1)! \equiv -1 \pmod{n}$.

Solution. If n is prime this is Wilson's theorem (Corollary 7.3.5). If $n = ab$ with $1 < a < n$, then a divides $(n-1)!$, so if $n \mid (n-1)! + 1$, then a divides $(n-1)! + 1$ and hence 1, which is absurd. □

Exercise 10.9.3. Let $p \equiv 1 \pmod{4}$ be prime. Show that $x = \left(\frac{p-1}{2}\right)!$ satisfies $x^2 \equiv -1 \pmod{p}$.

Solution. Let $h = \frac{p-1}{2}$. Pairing k with $p - k \equiv -k$, Wilson's theorem gives $-1 \equiv (p-1)! = \prod_{k=1}^{h} k(p-k) \equiv \prod_{k=1}^{h}(-k^2) = (-1)^h (h!)^2 \pmod{p}$. As h is even, $(h!)^2 \equiv -1$. □

Exercise 10.9.4. Show that $\varphi(n)$ is even for $n \geq 3$, and that $\varphi(n) \geq \sqrt{n/2}$ for all $n \geq 1$.

Solution. For $n \geq 3$ the map $a \mapsto n - a$ is an involution of the set of $1 \leq a < n$ prime to n without fixed points: $a = n - a$ would mean $n = 2a$, and then $\gcd(a, n) = a = 1$ forces $n = 2$. So the set has an even number of elements. For the inequality, by multiplicativity it suffices to note that $\varphi(p^k) = p^{k-1}(p-1) \geq \sqrt{p^k}$ for odd p (as $p - 1 \geq \sqrt{p}$ for $p \geq 3$ and $p^{k-1} \geq p^{(k-1)/2}$), and $\varphi(2^k) = 2^{k-1} \geq \sqrt{2^k/2}$; multiplying over the prime powers in n, with the factor $1/\sqrt{2}$ appearing at most once, gives $\varphi(n) \geq \sqrt{n/2}$. □

Exercise 10.9.5. Give a second proof of Theorem 10.3.3: for $d \mid p-1$ let $\psi(d)$ be the number of elements of order d in U_p. Show that $\psi(d) \leq \varphi(d)$, using that $x^d - 1$ has at most d roots in $\mathbb{F}_p$, and deduce from Proposition 10.2.8(4) that $\psi(d) = \varphi(d)$, in particular $\psi(p-1) \geq 1$.

Solution. If there is an element x of order d, its powers $1, x, \ldots, x^{d-1}$ are d distinct roots of $x^d - 1$, hence all of its roots (Corollary 8.6.6); so every element of order d is a power of x, and those of order d are the x^k with $\gcd(k, d) = 1$ (Lemma 10.3.2(2)), $\varphi(d)$ in number. So $\psi(d) \in \{0, \varphi(d)\}$. Every element has an order dividing $p - 1$, so $\sum_{d \mid p-1} \psi(d) = p - 1 = \sum_{d \mid p-1} \varphi(d)$. As $\psi(d) \leq \varphi(d)$ termwise, equality holds termwise. □

Exercise 10.9.6. Check that 14 is a primitive root modulo 29 and that $14^{28} \equiv 1 \pmod{29^2}$. Which of 14 and 43 is a primitive root modulo 29^2?

Solution. $29 - 1 = 28 = 2^2 \cdot 7$, so by Example 10.3.4(3) it suffices to check $14^{14} \not\equiv 1$ and $14^4 \not\equiv 1 \pmod{29}$. Modulo 29: $14^2 = 196 \equiv 22 \equiv -7$, $14^4 \equiv 49 \equiv 20$, $14^7 = 14^4 \cdot 14^2 \cdot 14 \equiv 20 \cdot (-7) \cdot 14 = -1960 \equiv -1960 + 68 \cdot 29 = 12$, and $14^{14} \equiv 144 \equiv -1$. So 14 is a primitive root. Modulo $841 = 29^2$: $14^2 = 196$, $14^4 = 38416 = 45 \cdot 841 + 571$, so $14^4 \equiv 571$; $14^7 = 14^4 \cdot 14^2 \cdot 14 \equiv 571 \cdot 196 \cdot 14$, and $571 \cdot 196 = 111916 = 133 \cdot 841 + 63$, so $14^7 \equiv 63 \cdot 14 = 882 \equiv 41$; $14^{14} \equiv 41^2 = 1681 = 2 \cdot 841 - 1 \equiv -1$, and $14^{28} \equiv 1 \pmod{841}$. So 14 is not a primitive root modulo 841, and by Theorem 10.3.7 $43 = 14 + 29$ is. □

Exercise 10.9.7. Show that for $k \geq 3$ the congruence $x^2 \equiv 1 \pmod{2^k}$ has exactly four solutions modulo 2^k, namely ± 1 and $\pm 1 + 2^{k-1}$, while for an odd prime p the congruence $x^2 \equiv 1 \pmod{p^k}$ has only the solutions ± 1. How many solutions does $x^2 \equiv 1 \pmod{n}$ have in general?

Solution. The solutions are the elements of order dividing 2 in U_{2^k}. By Theorem 10.3.9 this group is $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^{k-2}\mathbb{Z}$, which has four such elements; and ± 1, $\pm 1 + 2^{k-1}$ are four distinct solutions, as $(\pm 1 + 2^{k-1})^2 = 1 \pm 2^k + 2^{2k-2} \equiv 1$ for $k \geq 2$, so they are all. For odd p, U_{p^k} is cyclic (Theorem 10.3.7) of even order, and has exactly two elements of order dividing 2 (Lemma 10.3.2(3)), namely ± 1. By the Chinese remainder theorem, if $n = 2^a p_1^{k_1} \cdots p_r^{k_r}$ with distinct odd primes p_i, the number of solutions is 2^r times 1, 2 or 4 according as $a \leq 1$, $a = 2$ or $a \geq 3$. □

Exercise 10.9.8. Show that for a prime $p > 3$, $\left(\frac{-3}{p}\right) = 1$ if and only if $p \equiv 1 \pmod 3$. Deduce that there are infinitely many primes $p \equiv 1 \pmod 3$ (consider $N = (2p_1 \cdots p_k)^2 + 3$).

Solution. By Euler's criterion, Corollary 10.4.3 and reciprocity, $(\frac{-3}{p}) = (\frac{-1}{p})(\frac{3}{p}) = (-1)^{\frac{p-1}{2}} \cdot (-1)^{\frac{p-1}{2}}(\frac{p}{3}) = (\frac{p}{3})$, which is 1 exactly when $p \equiv 1 \pmod 3$. Given primes $p_1, \ldots, p_k \equiv 1 \pmod 3$, let $N = (2p_1 \cdots p_k)^2 + 3$, which is odd and not divisible by 3 (it is $\equiv 1 \pmod 3$, as $p_i \neq 3$ and $4 \equiv 1$). A prime $q \mid N$ satisfies $q > 3$ and $x^2 \equiv -3 \pmod q$ with $x = 2p_1 \cdots p_k$, so $q \equiv 1 \pmod 3$; and $q \neq p_i$, as otherwise $q \mid 3$. □

Exercise 10.9.9. For an odd $n = p_1 \cdots p_r$ (primes not necessarily distinct) and any integer a, the *Jacobi symbol* is $(\frac{a}{n}) = \prod_i (\frac{a}{p_i})$. Show that it is multiplicative in a and in n, that $(\frac{-1}{n}) = (-1)^{(n-1)/2}$, $(\frac{2}{n}) = (-1)^{(n^2-1)/8}$, and $(\frac{m}{n})(\frac{n}{m}) = (-1)^{\frac{m-1}{2}\cdot\frac{n-1}{2}}$ for coprime odd positive m, n. Use it to compute $(\frac{1001}{9907})$ without factoring 1001, given that 9907 is prime. Show by an example that $(\frac{a}{n}) = 1$ does not imply that a is a square modulo n.

Solution. Multiplicativity in n is the definition, and in a follows from Proposition 10.4.2(3). The functions $u \mapsto (-1)^{(u-1)/2}$ and $u \mapsto (-1)^{(u^2-1)/8}$ are multiplicative on odd integers: for odd u, v, $\frac{uv-1}{2} - \frac{u-1}{2} - \frac{v-1}{2} = \frac{(u-1)(v-1)}{2} \equiv 0 \pmod 2$, and $\frac{u^2v^2-1}{8} - \frac{u^2-1}{8} - \frac{v^2-1}{8} = \frac{(u^2-1)(v^2-1)}{8} \equiv 0 \pmod 2$, as $8 \mid u^2 - 1$ (Example 10.2.2(2)). So the two supplementary laws pass from primes to products. Likewise $(m, n) \mapsto (-1)^{\frac{m-1}{2}\frac{n-1}{2}}$ is multiplicative in each variable, so reciprocity for the Jacobi symbol follows from the prime case by writing both m and n as products of primes. Now $1001 \equiv 1 \pmod 4$, so $(\frac{1001}{9907}) = (\frac{9907}{1001}) = (\frac{898}{1001}) = (\frac{2}{1001})(\frac{449}{1001})$. As $1001 \equiv 1 \pmod 8$, $(\frac{2}{1001}) = 1$, and $449 \equiv 1 \pmod 4$ gives $(\frac{449}{1001}) = (\frac{1001}{449}) = (\frac{103}{449}) = (\frac{449}{103}) = (\frac{37}{103}) = (\frac{103}{37}) = (\frac{29}{37}) = (\frac{37}{29}) = (\frac{8}{29}) = (\frac{2}{29})^3 = -1$, as $29 \equiv 5 \pmod 8$. So $(\frac{1001}{9907}) = -1$, and 1001 is not a square modulo 9907. Finally $(\frac{2}{15}) = (\frac{2}{3})(\frac{2}{5}) = (-1)(-1) = 1$, but 2 is not a square modulo 15, since it is not a square modulo 3. □

Exercise 10.9.10. Show that if $2^k - 1$ is prime, then k is prime. Show that every prime divisor q of $2^p - 1$, for p an odd prime, satisfies $q \equiv 1 \pmod{2p}$, and use this to show that $2^{11} - 1$ is composite and $2^{13} - 1 = 8191$ is prime.

Solution. If $k = ab$ with $1 < a < k$, then $2^a - 1$ divides $2^{ab} - 1 = (2^a - 1)(2^{a(b-1)} + \cdots + 2^a + 1)$, and $1 < 2^a - 1 < 2^k - 1$. If $q \mid 2^p - 1$, then $2^p \equiv 1 \pmod q$, so $\mathrm{ord}_q(2)$ divides the prime p and is not 1; so it is p, and $p \mid q - 1$ by Lemma 10.3.2. As q is odd, $q \equiv 1 \pmod{2p}$. For $p = 11$ the candidates are $23, 67, 89, \ldots$, and $2047 = 23 \cdot 89$. For $p = 13$ the candidates are the primes $q \equiv 1 \pmod{26}$ with $q \leq \sqrt{8191} < 91$, namely 53 and 79; neither divides 8191 ($8191 = 154 \cdot 53 + 29 = 103 \cdot 79 + 54$), so 8191 is prime. □

Exercise 10.9.11. Let $n = pq$ with $p \neq q$ prime. Show that p and q can be computed from n and $\varphi(n)$, as the roots of a quadratic equation. Find p and q for $n = 2021$ and $\varphi(n) = 1932$.

Solution. $\varphi(n) = (p-1)(q-1) = n - (p+q) + 1$ gives $p + q = n + 1 - \varphi(n)$, and $pq = n$; so p and q are the roots of $X^2 - (n + 1 - \varphi(n))X + n$. For $n = 2021$: $p + q = 90$ and $pq = 2021$, so $X^2 - 90X + 2021 = (X - 43)(X - 47)$, and $2021 = 43 \cdot 47$. □

Exercise 10.9.12. Show that $\sum_{d|n} \mu(d)\tau(n/d) = 1$ and $\sum_{d|n} \mu(d)\sigma(n/d) = n$ for all n, and that $\sum_{d|n} |\mu(d)| = 2^r$ if n has r distinct prime factors.

Solution. $\mu * \tau = \mu * \mathbf{1} * \mathbf{1} = \varepsilon * \mathbf{1} = \mathbf{1}$ and $\mu * \sigma = \mu * \mathbf{1} * \mathrm{N} = \mathrm{N}$, by Propositions 10.6.2 and 10.6.3. The function $|\mu| * \mathbf{1}$ is multiplicative and takes the value $1 + 1 = 2$ at every prime power p^k, $k \geq 1$; so its value at n is 2^r. □

Exercise 10.9.13. Factor $11+7i$ into Gaussian primes, and list all representations of 65 as a sum of two squares, checking the count against Theorem 10.7.3.

Solution. $N(11+7i) = 170 = 2\cdot 5\cdot 17$. Dividing by $1+i$: $(11+7i)(1-i)/2 = 9-2i$, of norm 85. Of the primes $2\pm i$ of norm 5, $2-i$ divides: $(9-2i)(2+i)/5 = 4+i$, of norm 17. So $11+7i = (1+i)(2-i)(4+i)$. For $65 = 5\cdot 13$, $r_2(65) = 4(d_1-d_3) = 4\cdot 4 = 16$, the divisors $1,5,13,65$ all being $\equiv 1 \pmod 4$; the representations are $(\pm1,\pm8)$, $(\pm8,\pm1)$, $(\pm4,\pm7)$, $(\pm7,\pm4)$, coming from $65 = 1+64 = 16+49$. □

Exercise 10.9.14. Show that $\mathbb{Z}[\sqrt{-2}]$ is a Euclidean domain for $N(a+b\sqrt{-2}) = a^2+2b^2$, and deduce again that an odd prime p is of the form x^2+2y^2 if and only if $p \equiv 1,3 \pmod 8$. Show that $\mathbb{Z}[\sqrt{-3}]$ is not a UFD.

Solution. For $a/b = u+v\sqrt{-2}$ choose integers m,n with $|u-m|, |v-n| \le \frac12$; then $N(a/b - (m+n\sqrt{-2})) \le \frac14+\frac24 < 1$. If $(\frac{-2}{p}) = 1$, which by Corollary 10.4.3 and Proposition 10.4.7 means $p \equiv 1,3 \pmod 8$, then $p \mid x^2+2 = (x+\sqrt{-2})(x-\sqrt{-2})$ without dividing either factor, so p is not prime in the UFD $\mathbb{Z}[\sqrt{-2}]$ and $p = N(\pi) = a^2+2b^2$ as in Theorem 10.5.1. Conversely, if $p = x^2+2y^2$ then $p \nmid y$ and $(xy^{-1})^2 \equiv -2 \pmod p$. In $\mathbb{Z}[\sqrt{-3}]$, $4 = 2\cdot 2 = (1+\sqrt{-3})(1-\sqrt{-3})$; all these factors have norm 4, no element has norm 2 ($a^2+3b^2 \neq 2$), so they are irreducible, and 2 is not associate to $1\pm\sqrt{-3}$ since the only units are ±1. (In the larger ring $\mathbb{Z}[\omega] \ni \frac{1+\sqrt{-3}}{2}$ the factorisations agree: $1+\sqrt{-3} = 2(1+\omega)$, and $1+\omega$ is a unit.) □

Exercise 10.9.15. Show that an odd prime $p \neq 5$ of the form x^2+5y^2 is $\equiv 1,9 \pmod{20}$, and one of the form $2x^2+2xy+3y^2$ is $\equiv 3,7 \pmod{20}$. Using Lemma 10.8.5, Example 10.8.4(3) and quadratic reciprocity, conclude that $p = x^2+5y^2$ if and only if $p \equiv 1,9 \pmod{20}$.

Solution. If $p = x^2+5y^2$, then $p \equiv x^2 \equiv \pm1 \pmod 5$, and since p is odd exactly one of x, y is odd, so $p \equiv x^2+y^2 \equiv 1 \pmod 4$; hence $p \equiv 1,9 \pmod{20}$. If $p = 2x^2+2xy+3y^2$, then $2p = (2x+y)^2+5y^2 \equiv (2x+y)^2 \pmod 5$, so $2p \equiv \pm1$ and $p \equiv \pm2 \pmod 5$; and y is odd (else p is even), so one of x, $x+y$ is even and $p = 2x(x+y)+3y^2 \equiv 3 \pmod 4$; hence $p \equiv 3,7 \pmod{20}$. Now if $p \equiv 1,9 \pmod{20}$, then $(\frac{-5}{p}) = (\frac{-1}{p})(\frac{p}{5}) = 1\cdot 1$, so -20 is a square modulo $4p$ (as in the proof of Corollary 10.8.6), and p is represented by one of the two reduced forms of discriminant -20; by the congruences it is not the second. □

Exercise 10.9.16. Find all reduced forms of discriminants -15, -24 and -56, and check that $h(-15) = 2$, $h(-24) = 2$ and $h(-56) = 4$.

Solution. $D=-15$: $a \le 2$, b odd; $b = \pm1$, $ac = 4$: $(1,1,4)$ and $(2,1,2)$ (for $a = c = 2$ only $b \ge 0$ is allowed). $D=-24$: $a \le 2$, b even; $b = 0$, $ac = 6$: $(1,0,6)$ and $(2,0,3)$; $b = \pm2$ would need $a = 2$ and $c = 28/8$. $D=-56$: $a \le 4$, b even; $b = 0$, $ac = 14$: $(1,0,14)$, $(2,0,7)$; $b = \pm2$, $ac = 15$, $a \ge 2$: $(3,\pm2,5)$; $b = \pm4$, $ac = 18$, $a = 4$: no. So the reduced forms are $(1,0,14)$, $(2,0,7)$, $(3,2,5)$, $(3,-2,5)$. □

Exercise 10.9.17. Verify condition (1) of Theorem 10.8.7 for $p = 11$ and $p = 17$, so that n^2+n+11 and n^2+n+17 are prime for $0 \le n \le 9$ and $0 \le n \le 15$. Show that n^2+n+13 is not always prime for $0 \le n \le 11$ and find the corresponding non-principal reduced form of discriminant -51.

Solution. $p = 11$, $D = -43$: $a \le \sqrt{43/3} < 4$, $b \in \{\pm1,\pm3\}$, $ac = 11$ or 13, both prime, so $a = 1$ and the form is $(1,1,11)$. $p = 17$, $D = -67$: $a < 5$, $b \in \{\pm1,\pm3\}$, $ac = 17$ or 19, prime, so only $(1,1,17)$. For $p = 13$: $1+1+13 = 15 = 3\cdot 5$; correspondingly $D = -51$ has the reduced form $(3,3,5)$ (with $b = 3$, $ac = (9+51)/4 = 15$). □

Exercise 10.9.18. Let f be a positive definite form. Show that $f(x, y) = f(-x, -y)$, so that every number represented by f is represented an even number of times; and show that $x^2 + y^2$ represents n properly if and only if n is not divisible by 4 nor by any prime $q \equiv 3 \pmod 4$.

Solution. The first statement is clear. By Lemma 10.8.5 and $h(-4) = 1$, n is properly represented by $x^2 + y^2$ if and only if $-4 \equiv b^2 \pmod{4n}$ for some b, that is, $b = 2b'$ with $b'^2 \equiv -1 \pmod n$. This congruence is solvable modulo $q \equiv 3 \pmod 4$ never, modulo 4 never, modulo 2 always, and modulo powers p^k of primes $p \equiv 1 \pmod 4$ always: the unit group U_{p^k} is cyclic of order $p^{k-1}(p-1)$ (Theorem 10.3.7), which is divisible by 4, so it has an element x of order 4, and x^2 is the unique element of order 2, namely -1. By the Chinese remainder theorem (Theorem 10.2.6) it is solvable modulo n exactly under the stated condition. □

Chapter 11

Linear Algebra

11.1 Introduction

This chapter develops linear algebra over a field: linear maps and matrices, duality, determinants, linear systems (Gaussian elimination, the Rouché–Capelli theorem, Cramer's rule, and the computation of the rank by minors with Kronecker's bordered minors theorem), eigenvalues and canonical forms, bilinear forms, inner product spaces and the spectral theorem, and complex structures on real vector spaces, with the "two out of three" property of compatible inner products, symplectic forms and complex structures. The last topic is the linear algebra underlying Hodge theory: the decomposition $V_{\mathbb{C}} = V^{1,0} \oplus V^{0,1}$ of a real vector space with a complex structure is the pointwise version of the decomposition of forms on a complex manifold (Chapter 44) and of the Hodge decomposition (Chapter 48).

Bases and dimension were introduced in Section 8.8. References are [Hal74], [Axl15] and [Lan02, Chapters XIII–XV].

Prerequisites. Chapter 7 (Groups), Chapter 8 (Rings and Modules) and Chapter 9 (Fields and Galois Theory).

Throughout, k is a field and vector spaces are k-vector spaces, unless stated otherwise.

11.2 Linear maps and dimension

Definition 11.2.1. A *vector space* over a field k is a set V with an addition $V \times V \to V$, $(v, w) \mapsto v + w$, making V an abelian group with neutral element 0, and a scalar multiplication $k \times V \to V$, $(a, v) \mapsto av$, such that $a(v + w) = av + aw$, $(a + b)v = av + bv$, $(ab)v = a(bv)$ and $1v = v$; that is, a module over the ring k (Chapter 8). A *linear map* is a homomorphism of modules: $f(v + w) = f(v) + f(w)$ and $f(av) = a f(v)$. A *subspace* is a subset containing 0 and closed under addition and scalar multiplication.

From the axioms, $0v = 0$ (as $0v = (0 + 0)v = 0v + 0v$), $a0 = 0$, and $(-1)v = -v$. Bases and dimension are those of free modules (Section 8.8): every vector space has a basis, and all bases have the same cardinality, the dimension.

Example 11.2.2.

1. k^n, with the standard basis $e_1, \dots, e_n$; the matrices $M_{m\times n}(k)$, of dimension mn.
2. *Polynomials.* $k[x]$ is a vector space with basis $1, x, x^2, \dots$: every polynomial is uniquely a finite combination of monomials. It has infinite (countable) dimension. The polynomials of degree at most d form a subspace $k[x]_{\le d}$ of dimension $d + 1$; for distinct $a_0, \dots, a_d \in k$ the Lagrange polynomials $\prod_{j\neq i}(x - a_j)/(a_i - a_j)$ form another basis, in which the coordinates of p are its values $p(a_i)$.

3. *Function spaces.* For a set S, the functions $S \to k$ form a vector space k^S under pointwise operations; for S finite it is $k^{|S|}$, and for S infinite it has infinite dimension, since the indicator functions of points are independent. The continuous functions $C([0,1])$ form an infinite-dimensional real vector space: the monomials x^n are linearly independent, since a nonzero polynomial has finitely many roots. So do the sequences $k^{\mathbb{N}}$, which contain the subspace $k^{(\mathbb{N})}$ of finitely supported sequences, with basis the e_n; a basis of $k^{\mathbb{N}}$ itself exists by Zorn's lemma but cannot be written down.

4. $\mathbb{R}$ is a vector space over $\mathbb{Q}$ of uncountable dimension, since a countable-dimensional $\mathbb{Q}$-vector space is countable; $\mathbb{C}$ is a real vector space of dimension 2, with basis $1, i$.

5. The solutions of a homogeneous linear system $Ax = 0$, or of a homogeneous linear differential equation $y'' + y = 0$ (basis cos, sin), form vector spaces; the solutions of $Ax = b$ with $b \neq 0$ do not, as they do not contain 0.

Proposition 11.2.3. *Let V be a vector space and $W \subseteq V$ a subspace.*

1. *W has a* complement*: a subspace U with $V = W \oplus U$ (that is, $W + U = V$ and $W \cap U = 0$). Then $U \cong V/W$ and $\dim V = \dim W + \dim V/W$.*

2. *$\dim W \leq \dim V$, and if $\dim V$ is finite and $\dim W = \dim V$ then $W = V$.*

3. *If $U, W \subseteq V$ are finite-dimensional, $\dim(U + W) + \dim(U \cap W) = \dim U + \dim W$.*

Proof. (1) Extend a basis B_W of W to a basis B of V (Theorem 8.8.4) and let U be spanned by $B \setminus B_W$. The projection $V \to V/W$ restricts to an isomorphism $U \to V/W$, and $B = B_W \sqcup (B \setminus B_W)$ gives the formula (as cardinals). (2) The first claim follows from (1). If $\dim W = \dim V < \infty$, a basis of W has $\dim V$ elements and extends to a basis of V, which by Theorem 8.8.5 has no further elements. (3) Extend a basis of $U \cap W$ to bases of U and of W; the union of these bases is a basis of $U + W$. □

Theorem 11.2.4 (Rank–nullity)**.** *For a linear map $f : V \to W$, $V/\ker f \cong \operatorname{im} f$, so $\dim V = \dim \ker f + \dim \operatorname{im} f$. If $\dim V = \dim W < \infty$, then f is injective if and only if it is surjective.*

Proof. The isomorphism is the first isomorphism theorem (Proposition 8.7.3); combine with Proposition 11.2.3(1). If $\dim V = \dim W = n$, then f is injective iff $\dim \operatorname{im} f = n$ iff $\operatorname{im} f = W$ by Proposition 11.2.3(2). □

Counterexample 11.2.5. The last statement fails in infinite dimension. On $V = k^{(\mathbb{N})}$ the *shift* $e_n \mapsto e_{n+1}$ is injective but not surjective, and $e_0 \mapsto 0$, $e_{n+1} \mapsto e_n$ is surjective but not injective.

11.3 Matrices

Definition 11.3.1. Let V and W have ordered bases $\mathcal{B} = (v_1, \ldots, v_n)$ and $\mathcal{C} = (w_1, \ldots, w_m)$. The *matrix* of a linear map $f : V \to W$ is the $m \times n$ matrix $[f]_{\mathcal{C},\mathcal{B}} = (a_{ij})$ with $f(v_j) = \sum_i a_{ij} w_i$: its j-th column contains the coordinates of $f(v_j)$.

Example 11.3.2. 1. *Differentiation.* On the space $k[x]_{\leq 3}$ of polynomials of degree at most 3, with basis $(1, x, x^2, x^3)$, the derivative D sends $1 \mapsto 0$, $x \mapsto 1$, $x^2 \mapsto 2x$, $x^3 \mapsto 3x^2$; writing the coordinates of these images as columns,

$$[D] = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Since the fourth derivative of a cubic is zero, $[D]^4 = 0$, as one checks by multiplying.

2. *Projection and reflection in* $\mathbb{R}^2$. The orthogonal projection onto the line spanned by $(1,1)$ sends $e_1 \mapsto \frac{1}{2}(1,1)$ and $e_2 \mapsto \frac{1}{2}(1,1)$, so its matrix is $\frac{1}{2}\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$; the reflection in the same line exchanges e_1 and e_2, with matrix $\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$. In the basis $(1,1)$, $(1,-1)$ adapted to the line the two matrices become $\operatorname{diag}(1,0)$ and $\operatorname{diag}(1,-1)$: a good basis makes the matrix simple, which is the theme of the rest of the chapter.

3. *Rotations.* With the trigonometric functions (Chapter 27), the rotation of $\mathbb{R}^2$ by the angle θ has matrix $R_\theta = \left(\begin{smallmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{smallmatrix}\right)$, its columns being the images of e_1 and e_2. Composition corresponds to multiplication, and $R_\alpha R_\beta = R_{\alpha+\beta}$, read entry by entry, is the pair of addition formulas for $\cos(\alpha+\beta)$ and $\sin(\alpha+\beta)$.

Proposition 11.3.3. *The map* $f \mapsto [f]_{\mathcal{C},\mathcal{B}}$ *is an isomorphism of vector spaces* $\operatorname{Hom}_k(V,W) \to M_{m\times n}(k)$*, so* $\dim \operatorname{Hom}_k(V,W) = mn$*. Composition corresponds to matrix multiplication:* $[g\circ f]_{\mathcal{D},\mathcal{B}} = [g]_{\mathcal{D},\mathcal{C}}[f]_{\mathcal{C},\mathcal{B}}$*. If* $\mathcal{B}', \mathcal{C}'$ *are other bases and* $P = [\mathrm{id}]_{\mathcal{B},\mathcal{B}'}$, $Q = [\mathrm{id}]_{\mathcal{C},\mathcal{C}'}$*, then* $[f]_{\mathcal{C}',\mathcal{B}'} = Q^{-1}[f]_{\mathcal{C},\mathcal{B}}P$*. In particular the matrices of an endomorphism in different bases are* similar*:* $A' = P^{-1}AP$*.*

Proof. A linear map is determined by the images of a basis, which can be arbitrary (Proposition 8.7.5); this gives bijectivity, and linearity is clear. If $f(v_j) = \sum_i a_{ij}w_i$ and $g(w_i) = \sum_l b_{li}u_l$, then $g(f(v_j)) = \sum_l(\sum_i b_{li}a_{ij})u_l$. The change of basis formula follows by writing $f = \mathrm{id}_W \circ f \circ \mathrm{id}_V$ and using that $[\mathrm{id}]_{\mathcal{C}',\mathcal{C}} = Q^{-1}$. □

Remark 11.3.4. *Change of basis, change of coordinates, change of variables.* Let $\mathcal{B} = (v_1,\dots,v_n)$ and $\mathcal{B}' = (v_1',\dots,v_n')$ be bases of V, and $P = [\mathrm{id}]_{\mathcal{B},\mathcal{B}'}$, whose j-th column contains the coordinates of v_j' in $\mathcal{B}$: $v_j' = \sum_i p_{ij}v_i$. If a vector v has coordinate columns x in $\mathcal{B}$ and x' in $\mathcal{B}'$, then

$$x = Px', \qquad x' = P^{-1}x,$$

since $v = \sum_j x_j'v_j' = \sum_i(\sum_j p_{ij}x_j')v_i$. So the basis vectors change with P and the coordinates with P^{-1}, the "contravariant" behaviour of coordinates. This is a *passive* transformation: the vector stays where it is and only its description changes. An *active* transformation moves the vectors, for instance by the linear map f with $f(v_j) = v_j'$, whose matrix in $\mathcal{B}$ is P; it sends the vector with coordinates x to the vector with coordinates Px. The passive change by P and the active transformation by P act on coordinate columns by inverse matrices.

In linear algebra, a *change of variables* is the substitution $x = Px'$ in a formula, which is the same operation read from the side of the formula; the result depends on what kind of object the formula describes:

1. coordinates of vectors: $x' = P^{-1}x$;
2. linear forms $\varphi(v) = ax$, with a a row: $a' = aP$;
3. endomorphisms, $y = Ax$: $A' = P^{-1}AP$ (*similarity*, Proposition 11.3.3);
4. bilinear forms $B(v,w) = x^TGy$, with Gram matrix G (Section 11.9): $G' = P^TGP$ (*congruence*); for a quadratic form $q(x) = x^TGx$ this is the familiar substitution.

The rules (3) and (4) agree exactly when $P^{-1} = P^T$, that is, when P is orthogonal; this is why the spectral theorem (Theorem 11.10.6), which diagonalises a real symmetric matrix by an orthogonal P, diagonalises it as an endomorphism and as a quadratic form at the same time. In general the invariants differ: similar matrices have the same eigenvalues, congruent symmetric real matrices the same signature (Theorem 11.9.5) but not the same eigenvalues, as $\operatorname{diag}(1,4)$ and $I = P^T\operatorname{diag}(1,4)P$ with $P = \operatorname{diag}(1,\frac{1}{2})$ show.

Outside linear algebra, a change of variables is a substitution $x = \varphi(y)$ with φ a diffeomorphism, as in integrals, where the factor $|\det D\varphi|$ appears (Chapter 28), and a change of coordinates on a manifold is a transition map between charts (Chapter 33). Their derivatives at a point are linear changes of coordinates as above: components of tangent vectors transform like coordinates of vectors, with the Jacobian matrix in the role of P^{-1}, and differentials of functions like linear forms.

Definition 11.3.5. The *rank* of a linear map is $\dim \operatorname{im} f$. The *column rank* of a matrix $A \in M_{m\times n}(k)$ is the dimension of the span of its columns in k^m, that is, the rank of the map $k^n \to k^m$ it defines, and its *row rank* is the column rank of the transpose A^T.

11.4 Duality

Definition 11.4.1. The *dual* of V is $V^* = \operatorname{Hom}_k(V, k)$. For $f\colon V \to W$, the *transpose* is $f^*\colon W^* \to V^*$, $\varphi \mapsto \varphi \circ f$. For a subspace $W \subseteq V$, the *annihilator* is $W^\perp = \{\varphi \in V^* : \varphi|_W = 0\}$. If $(v_1, \dots, v_n)$ is a basis of V, the *dual basis* $(v_1^*, \dots, v_n^*)$ is given by $v_i^*(v_j) = \delta_{ij}$.

Proposition 11.4.2. *Let V, W be finite-dimensional.*

1. *The dual basis is a basis of V^*; so* $\dim V^* = \dim V$.
2. *The map* $\mathrm{ev}\colon V \to V^{**}$ *of Example 6.5.5 is an isomorphism.*
3. *In dual bases, the matrix of f^* is the transpose of the matrix of f.*
4. $\dim W^\perp = \dim V - \dim W$ *for $W \subseteq V$, and* $(\operatorname{im} f)^\perp = \ker f^*$.
5. $\operatorname{rank} f^* = \operatorname{rank} f$. *Hence row rank equals column rank for every matrix.*

Proof. (1) $\varphi = \sum_i \varphi(v_i) v_i^*$ for $\varphi \in V^*$, and the coefficients are determined by evaluating on the v_j. (2) ev is injective: if $v \neq 0$, extend v to a basis and take the corresponding dual basis element. Both sides have dimension n; apply Theorem 11.2.4. (3) If $f(v_j) = \sum_i a_{ij} w_i$ then $f^*(w_i^*)(v_j) = a_{ij}$, so $f^*(w_i^*) = \sum_j a_{ij} v_j^*$. (4) Extend a basis $v_1, \dots, v_r$ of W to a basis of V; then $W^\perp$ has basis $v_{r+1}^*, \dots, v_n^*$. And $\varphi \in \ker f^*$ iff $\varphi \circ f = 0$ iff φ vanishes on $\operatorname{im} f$. (5) By (4) and rank–nullity for f^*, $\operatorname{rank} f^* = \dim W - \dim \ker f^* = \dim W - \dim(\operatorname{im} f)^\perp = \dim \operatorname{im} f$. Apply (3). □

Counterexample 11.4.3. In infinite dimension $\mathrm{ev}\colon V \to V^{**}$ is injective (by the same argument) but not surjective. Let $k = \mathbb{Q}$ and $V = \mathbb{Q}^{(\mathbb{N})}$, which is countable. Then $V^* \cong \mathbb{Q}^{\mathbb{N}}$, which is uncountable (it contains the sequences with values in $\{0, 1\}$). Choosing a basis of V^* gives an injection $V^* \to V^{**}$ (send the basis to its dual family), so V^{**} is uncountable, and cannot be the image of the countable set V.

11.5 Determinants

In this section R is a commutative ring. For $A \in M_n(R)$ we write $A = (a_1 | \cdots | a_n)$ with columns $a_j \in R^n$.

Definition 11.5.1. A map $D\colon (R^n)^n \to R$ is *multilinear* if it is R-linear in each argument, and *alternating* if $D(a_1, \dots, a_n) = 0$ whenever $a_i = a_j$ for some $i \neq j$. The *determinant* of $A \in M_n(R)$ is

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma)\, a_{\sigma(1)1} a_{\sigma(2)2} \cdots a_{\sigma(n)n},$$

where sgn is the sign of Proposition 7.8.3.

Theorem 11.5.2. *The determinant, as a function of the columns, is multilinear and alternating with* $\det I = 1$. *Every multilinear alternating* $D\colon (R^n)^n \to R$ *satisfies* $D(a_1, \dots, a_n) = \det(a_1|\cdots|a_n) \cdot D(e_1, \dots, e_n)$.

Proof. Multilinearity is clear, and $\det I = 1$ since only $\sigma = \mathrm{id}$ contributes. If $a_i = a_j$ with $i \neq j$, let $\tau = (i\,j)$. The map $\sigma \mapsto \sigma\tau$ is a bijection from even to odd permutations, and the terms for σ and $\sigma\tau$ have the same product of entries (as columns i and j agree) and opposite signs; so the sum vanishes.

Let D be multilinear and alternating. First, D changes sign when two arguments are exchanged: expand $0 = D(\dots, a+b, \dots, a+b, \dots)$. Hence $D(e_{\sigma(1)}, \dots, e_{\sigma(n)}) = \mathrm{sgn}(\sigma) D(e_1, \dots, e_n)$ for $\sigma \in S_n$, writing σ as a product of transpositions. Writing $a_j = \sum_i a_{ij} e_i$ and expanding by multilinearity, $D(a_1, \dots, a_n) = \sum_\phi a_{\phi(1)1} \cdots a_{\phi(n)n} D(e_{\phi(1)}, \dots, e_{\phi(n)})$ over all maps $\phi\colon \{1, \dots, n\} \to \{1, \dots, n\}$. The terms with ϕ not injective vanish, and the others give $\det(a_1|\cdots|a_n) D(e_1, \dots, e_n)$. □

Proposition 11.5.3. *For* $A, B \in M_n(R)$*:*

1. $\det(AB) = \det A \det B$*;*
2. $\det A^T = \det A$*, so* det *is also multilinear and alternating in the rows;*
3. *(Laplace expansion) for each* j, $\det A = \sum_i (-1)^{i+j} a_{ij} \det A_{ij}$*, where* A_{ij} *is obtained by deleting row* i *and column* j*;*
4. *the* adjugate $\mathrm{adj}\, A$*, with* (i, j) *entry* $(-1)^{i+j} \det A_{ji}$*, satisfies* $A \,\mathrm{adj}\, A = \mathrm{adj}\, A\, A = \det(A) I$*;*
5. A *is invertible in* $M_n(R)$ *if and only if* $\det A \in R^\times$.

Proof. (1) The map $(b_1, \dots, b_n) \mapsto \det(Ab_1|\cdots|Ab_n)$ is multilinear and alternating, with value $\det A$ on $(e_1, \dots, e_n)$; apply Theorem 11.5.2. (2) Reindex the sum by $\sigma \mapsto \sigma^{-1}$, using $\mathrm{sgn}(\sigma^{-1}) = \mathrm{sgn}(\sigma)$. (3) Both sides are multilinear and alternating in the columns other than j, and linear in column j. Writing $a_j = \sum_i a_{ij} e_i$, it suffices to treat $a_j = e_i$. Moving row i to the top and column j to the front by $i - 1$ and $j - 1$ adjacent exchanges changes the determinant by $(-1)^{i+j}$ and gives a matrix with first column e_1, whose determinant is $\det A_{ij}$ by the Leibniz formula (only permutations fixing 1 contribute). (4) The (j, j') entry of $\mathrm{adj}(A)A$ is $\sum_i (-1)^{i+j} \det A_{ij} a_{ij'}$, which by (3) is the determinant of A with column j replaced by column j': this is $\det A$ if $j = j'$ and 0 otherwise. The other product is handled with (2). (5) If $AB = I$ then $\det A \det B = 1$. Conversely $(\det A)^{-1} \mathrm{adj}\, A$ is an inverse by (4). □

Example 11.5.4. 1. *A* 3×3 *determinant.* For $A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 1 & 2 & 0 \end{pmatrix}$, the Laplace expansion along the first column has two nonzero terms: $\det A = 2 \det \left(\begin{smallmatrix} -1 & 4 \\ 2 & 0 \end{smallmatrix}\right) + (-1)^{3+1} \cdot 1 \cdot \det \left(\begin{smallmatrix} 1 & 3 \\ -1 & 4 \end{smallmatrix}\right) = 2(-8) + 7 = -9$. Alternatively, subtracting $\frac{1}{2}$ of the first row from the third (which does not change the determinant, by multilinearity and the alternating property in the rows) gives the row $(0, \frac{3}{2}, -\frac{3}{2})$, and then adding $\frac{3}{2}$ times the second row gives $(0, 0, \frac{9}{2})$; the matrix is now upper triangular, and $\det A = 2 \cdot (-1) \cdot \frac{9}{2} = -9$.

2. *Triangular matrices.* The determinant of an upper triangular matrix is the product of its diagonal entries. In the Leibniz formula, a permutation with $\sigma(j) \leq j$ for all j is the identity ($\sigma(1) = 1$, then $\sigma(2) = 2$, and so on); so every $\sigma \neq \mathrm{id}$ has some j with $\sigma(j) > j$, and its term contains the entry $a_{\sigma(j)j}$, whose row index exceeds its column index: an entry below the diagonal, which is zero.

3. *The Vandermonde determinant.* For $x_1, \dots, x_n \in R$,
$$\det \begin{pmatrix} 1 & x_1 & \cdots & x_1^{n-1} \\ \vdots & & & \vdots \\ 1 & x_n & \cdots & x_n^{n-1} \end{pmatrix} = \prod_{i<j} (x_j - x_i).$$
By induction on n: subtract x_1 times column $j-1$ from column j, for $j = n, n-1, \dots, 2$ in this order. Row i becomes $(1, x_i - x_1, x_i(x_i - x_1), \dots, x_i^{n-2}(x_i - x_1))$, and the first row becomes $(1, 0, \dots, 0)$. Expanding along the first row and taking the factor $x_i - x_1$ out of row i for $i \geq 2$ leaves $\prod_{i \geq 2}(x_i - x_1)$ times the Vandermonde determinant of $x_2, \dots, x_n$. Over a field it is nonzero exactly when the x_i are distinct; this is why a polynomial of degree less than n is determined by its values at n distinct points.

4. *Geometric meaning.* Over $\mathbb{R}$, $|\det(a_1 | \cdots | a_n)|$ is the volume of the parallelepiped spanned by the columns (made precise with Lebesgue measure in Chapter 28), and the sign records whether the columns are positively oriented. For $n = 2$, $\det \left(\begin{smallmatrix} a & c \\ b & d \end{smallmatrix}\right) = ad - bc$ is the signed area of the parallelogram spanned by (a, b) and (c, d).

Definition 11.5.5. The determinant of an endomorphism f of a finite-dimensional vector space (or of a finitely generated free module over R) is the determinant of its matrix in any basis; it is well defined since $\det(P^{-1}AP) = \det A$. The group $\mathrm{GL}(V)$ of invertible endomorphisms contains $\mathrm{SL}(V) = \{f : \det f = 1\}$, and $\mathrm{GL}_n(R) = M_n(R)^\times$.

11.6 Linear systems, Gaussian elimination and the rank

A system of m linear equations in n unknowns $x_1, \dots, x_n$,
$$a_{i1}x_1 + \cdots + a_{in}x_n = b_i \qquad (1 \leq i \leq m),$$
is the single equation $Ax = b$ for the matrix $A = (a_{ij}) \in M_{m \times n}(k)$, the column $b \in k^m$ and the unknown column $x \in k^n$. Its theory is the theory of the linear map $x \mapsto Ax$: the system is solvable if and only if b lies in the image, and the solutions, when they exist, form a translate of the kernel. In practice one needs a procedure that decides solvability, finds the solutions and computes ranks; this is *Gaussian elimination*. For matrices depending on parameters it is often more convenient to compute the rank with determinants of square submatrices, the *minors*, and Kronecker's *bordered minors theorem* reduces the number of minors to be examined. In this section k is a field.

Definition 11.6.1. The *elementary row operations* on a matrix are: (i) exchanging two rows; (ii) multiplying a row by a nonzero scalar c; (iii) adding c times one row to another row, for $c \in k$. The *elementary matrices* are the matrices obtained from the identity matrix I_m by one elementary row operation.

Performing an elementary row operation on $A \in M_{m \times n}(k)$ is the same as multiplying A on the left by the corresponding elementary matrix E: row i of EA is (row i of E) times A, that is, the combination of the rows of A with the coefficients in row i of E. Every elementary matrix is invertible, its inverse being the elementary matrix of the inverse operation (exchange the same rows again; multiply by c^{-1}; add $-c$ times the row). Column operations are defined in the same way and correspond to multiplication on the right.

Lemma 11.6.2. *Let $B = PA$ with $P \in \mathrm{GL}_m(k)$, for instance a product of elementary matrices. Then A and B have the same kernel, the same row space and the same rank, and the columns of A and of B satisfy the same linear relations: $\sum_j x_j a_j = 0$ if and only if $\sum_j x_j b_j = 0$, where a_j and b_j are the j-th columns. In particular, a set of columns of A is linearly independent, or is a basis of the column space, if and only if the columns of B with the same indices are.*

Proof. $Ax = 0$ if and only if $PAx = 0$, since P is invertible; and $\sum_j x_j a_j = Ax$. The rows of B are combinations of the rows of A and, since $A = P^{-1}B$, conversely; so the row spaces agree. The rank is the dimension of the column space, which by rank–nullity (Theorem 11.2.4) is $n - \dim \ker A$. A set S of columns is independent if and only if no nontrivial relation with coefficients supported in S holds, and it is a basis of the column space if moreover every column is a combination of the columns in S, which again is a statement about relations. □

Definition 11.6.3. A matrix is in *row echelon form* if its nonzero rows come before its zero rows and the first nonzero entry of each nonzero row, its *pivot*, lies strictly to the right of the pivot of the row above. It is in *reduced* row echelon form if moreover every pivot is 1 and is the only nonzero entry of its column. The columns containing pivots are the *pivot columns*.

For instance

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

is in reduced row echelon form, with pivots in columns 1 and 3.

Theorem 11.6.4 (Gaussian elimination)**.** *Every $A \in M_{m\times n}(k)$ can be transformed by elementary row operations into a matrix R in reduced row echelon form; so $R = PA$ with P a product of elementary matrices. If R has r nonzero rows, then*

1. $\operatorname{rank} A = r$*;*
2. *the nonzero rows of R form a basis of the row space of A;*
3. *the columns of A with the indices of the pivot columns of R form a basis of the column space of A.*

Proof. Existence, by induction on the number m of rows. If $A = 0$ there is nothing to do. Otherwise let j be the first column with a nonzero entry. Exchange rows to bring a nonzero entry of column j to row 1, multiply row 1 by the inverse of this entry, and subtract suitable multiples of row 1 from the other rows; now column j is e_1, and the columns before it are zero. Apply the induction hypothesis to the matrix formed by rows $2, \dots, m$; operations on these rows do not change row 1, and keep the first j columns zero in these rows, so the result is in row echelon form with pivots equal to 1. Finally, for each pivot, from the last one to the first, subtract multiples of its row from the rows above to make the other entries of its column zero; this does not change the entries to the left of the pivot in those rows, since the row of the pivot is zero there.

Properties. Let the pivots of R be in columns $j_1 < \cdots < j_r$. Column j_s of R is the standard vector e_s, and every column of R has zeros below row r; so the pivot columns of R are independent and span the column space of R, which therefore has dimension r. By Lemma 11.6.2, (3) holds and $\operatorname{rank} A = \operatorname{rank} R = r$. The nonzero rows of R are independent, since a combination $\sum_s c_s(\text{row } s)$ has entry c_s in column j_s; they span the row space of R, which is that of A. □

Since the row rank of A is the dimension of its row space, (1) and (2) give another proof that row rank equals column rank (Proposition 11.4.2(5)). The proof is an algorithm, and the reduced row echelon form is unique (Exercise 11.12.8). To compute only the rank, a row echelon form, not necessarily reduced, suffices: its number of nonzero rows is the rank, by the same argument.

Theorem 11.6.5 (Rouché–Capelli)**.** *Let $A \in M_{m\times n}(k)$ and $b \in k^m$. The system $Ax = b$ has a solution if and only if* $\operatorname{rank} A = \operatorname{rank}(A \mid b)$*, where $(A \mid b)$ is the* augmented *matrix obtained by adding the column b. In that case, if x_0 is one solution, the set of all solutions is*

$$x_0 + \ker A = \{x_0 + y : Ay = 0\},$$

an affine subspace of dimension $n - \operatorname{rank} A$.

Proof. $Ax = \sum_j x_j a_j$, so the system is solvable if and only if b lies in the span of the columns of A, that is, if and only if adding b does not enlarge the column space; as the column space of $(A \mid b)$ contains that of A, this happens exactly when the dimensions agree. If $Ax_0 = b$, then $Ax = b$ if and only if $A(x - x_0) = 0$. Finally $\dim \ker A = n - \operatorname{rank} A$ by Theorem 11.2.4. □

To solve a system in practice, bring $(A \mid b)$ to reduced row echelon form $(R \mid c)$; the systems $Ax = b$ and $Rx = c$ have the same solutions, as $(R \mid c) = P(A \mid b)$ with P invertible. If the last column is a pivot column, some row reads $0 = 1$ and there is no solution; this is the case $\operatorname{rank}(A \mid b) > \operatorname{rank} A$. Otherwise the unknowns of the non-pivot columns, the *free variables*, can be chosen arbitrarily, and each equation of $Rx = c$ expresses one pivot unknown in terms of them.

Example 11.6.6. Consider the system

$$\begin{aligned} x_1 + 2x_2 + x_3 &= 1,\\ 2x_1 + 4x_2 + 2x_4 &= 2,\\ 3x_1 + 6x_2 + x_3 + 2x_4 &= 3. \end{aligned}$$

Subtracting 2 times the first row from the second and 3 times the first row from the third, then the new second row from the third, gives

$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 1\\ 2 & 4 & 0 & 2 & 2\\ 3 & 6 & 1 & 2 & 3 \end{array}\right) \to \left(\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 1\\ 0 & 0 & -2 & 2 & 0\\ 0 & 0 & -2 & 2 & 0 \end{array}\right) \to \left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 1\\ 0 & 0 & 1 & -1 & 0\\ 0 & 0 & 0 & 0 & 0 \end{array}\right),$$

where the last step multiplies the second row by $-\frac{1}{2}$ and subtracts it from the first. The rank of A and of $(A \mid b)$ is 2, the pivot columns are the first and the third, and the free variables are $x_2 = s$ and $x_4 = t$. The equations $x_1 + 2x_2 + x_4 = 1$ and $x_3 - x_4 = 0$ give the solutions

$$x = (1, 0, 0, 0) + s(-2, 1, 0, 0) + t(-1, 0, 1, 1), \qquad s, t \in k,$$

a plane in k^4, of dimension $4 - 2$. By Theorem 11.6.4(3), the first and third columns of A form a basis of its column space: indeed the second column is twice the first, and the fourth is the first minus the third, as the coefficients in the reduced form show. With right-hand side $(1, 2, 4)$ instead, the same operations produce the row $(0\ 0\ 0\ 0 \mid 1)$, and there is no solution: the third equation contradicts the sum of the first two.

For square systems with invertible matrix the solution has a closed formula in terms of determinants.

Proposition 11.6.7 (Cramer's rule)**.** *Let $A \in M_n(k)$ with $\det A \neq 0$ and $b \in k^n$. The unique solution of $Ax = b$ is given by*

$$x_i = \frac{\det A_i}{\det A} \qquad (1 \le i \le n),$$

where A_i is obtained from A by replacing its i-th column with b.

Proof. A is invertible with $A^{-1} = (\det A)^{-1} \operatorname{adj} A$ (Proposition 11.5.3(4)), so $x = A^{-1}b$ is the unique solution and $x_i = (\det A)^{-1} \sum_j (-1)^{i+j} \det(A_{ji})\, b_j$. Since A_i and A agree outside column i, deleting row j and column i from A_i gives A_{ji}, so the sum is the Laplace expansion of $\det A_i$ along its i-th column (Proposition 11.5.3(3)). □

For example, $2x + y = 3$, $x - y = 0$ has $\det A = -3$, $x = \det\left(\begin{smallmatrix} 3 & 1 \\ 0 & -1 \end{smallmatrix}\right)/(-3) = 1$ and $y = \det\left(\begin{smallmatrix} 2 & 3 \\ 1 & 0 \end{smallmatrix}\right)/(-3) = 1$. Cramer's rule is useful in theory, for instance to see that the solution depends rationally on the entries of A and b; for computations with large n, Gaussian elimination, with about n^3 operations, is far cheaper than evaluating $n + 1$ determinants of size n.

11.6.1 Rank and minors

Definition 11.6.8. Let $A \in M_{m\times n}(k)$. For sets $I \subseteq \{1, \dots, m\}$ and $J \subseteq \{1, \dots, n\}$ with $|I| = |J| = p$, the $p \times p$ matrix $A_{I,J} = (a_{ij})_{i\in I, j\in J}$, rows and columns in increasing order, is a *submatrix* of A, and $\det A_{I,J}$ is a *minor of order* p. If $M = \det A_{I,J}$, the minors $\det A_{I\cup\{i\},J\cup\{j\}}$ with $i \notin I$ and $j \notin J$ are the minors of order $p + 1$ *bordering* M.

Proposition 11.6.9. *The rank of $A \in M_{m\times n}(k)$ is the largest p such that A has a nonzero minor of order p (and 0 if $A = 0$).*

Proof. If $\det A_{I,J} \neq 0$, the columns of $A_{I,J}$ are independent, and then so are the columns of A with indices in J: a relation among them would restrict, in the rows I, to a relation among the columns of $A_{I,J}$. So $\operatorname{rank} A \geq p$. Conversely, let $r = \operatorname{rank} A$ and choose r independent columns, with index set J. The $m \times r$ matrix they form has column rank r, hence row rank r (Proposition 11.4.2(5)): some r of its rows, with index set I, are independent. Then $A_{I,J}$ is an $r \times r$ matrix of rank r, hence invertible, and $\det A_{I,J} \neq 0$ (Proposition 11.5.3(5)). □

Read literally, the proposition requires checking all minors of order $p + 1$ to show that the rank is at most p; a 4×5 matrix has $\binom{4}{3}\binom{5}{3} = 40$ minors of order 3. The next theorem shows that it suffices to check the minors bordering one nonzero minor of order p: there are $(m - p)(n - p)$ of them, 6 in this example if $p = 2$.

Theorem 11.6.10 (Kronecker's bordered minors theorem)**.** *Let $A \in M_{m\times n}(k)$ and let $M = \det A_{I,J}$ be a nonzero minor of order p. If all the minors of order $p + 1$ bordering M vanish, then $\operatorname{rank} A = p$, and the columns of A with indices in J form a basis of its column space (and the rows with indices in I a basis of its row space).*

Proof. As in the proof of Proposition 11.6.9, the columns a_l, $l \in J$, are independent. We show that every other column a_j, $j \notin J$, is a combination of them; then they form a basis of the column space, and $\operatorname{rank} A = p$. Fix $j \notin J$, and for each row index i consider the $(p+1)\times(p+1)$ matrix

$$N_i = \begin{pmatrix} A_{I,J} & (a_{i'j})_{i'\in I} \\ (a_{il})_{l\in J} & a_{ij} \end{pmatrix},$$

formed by the rows I followed by row i, and the columns J followed by column j; this ordering changes $\det N_i$ at most by a sign compared with the minor $\det A_{I\cup\{i\},J\cup\{j\}}$. We claim $\det N_i = 0$ for every i. If $i \notin I$, it is a bordering minor up to sign, and vanishes by hypothesis. If $i \in I$, the last row of N_i repeats one of the first p rows, and the determinant of a matrix with two equal rows is 0 (Proposition 11.5.3(2)).

Expand $\det N_i$ along its last row (the Laplace expansion along a row, which follows from Proposition 11.5.3(2) and (3)):

$$0 = \det N_i = \sum_{l\in J} c_l a_{il} + M a_{ij},$$

where $M = \det A_{I,J}$ is the cofactor of the corner entry a_{ij} and c_l is, up to sign, the determinant of the $p \times p$ matrix obtained by deleting the last row and the column of l. These cofactors only involve the rows I of A, so they *do not depend on* i. Since $M \neq 0$, $a_{ij} = -\sum_{l \in J} (c_l/M) a_{il}$ for every i, that is,

$$a_j = -\sum_{l \in J} \frac{c_l}{M}\, a_l.$$

The statement about rows follows by applying this to the transpose, whose minors are those of A (Proposition 11.5.3(2)): the transposes of the bordering minors are the bordering minors of M in A^T. □

This gives a procedure for computing the rank: find a nonzero entry (a nonzero minor of order 1); compute the minors bordering it; if all vanish the rank is 1, otherwise a nonzero one is a minor of order 2, whose bordering minors are computed next, and so on. The procedure stops at the rank, and at each step only the bordering minors of the current minor are needed.

Example 11.6.11. 1. The matrix of Example 11.6.6,

$$A = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 2 \\ 3 & 6 & 1 & 2 \end{pmatrix},$$

has the nonzero minor $M = \det\left(\begin{smallmatrix} 1 & 1 \\ 2 & 0 \end{smallmatrix}\right) = -2$ in rows $1, 2$ and columns $1, 3$. The bordering minors use row 3 and column 2 or 4:

$$\det \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 0 \\ 3 & 6 & 1 \end{pmatrix} = 0, \qquad \det \begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 2 \\ 3 & 1 & 2 \end{pmatrix} = 1 \cdot (0-2) - 1 \cdot (4-6) + 0 = 0,$$

the first because its second column is twice the first. So $\operatorname{rank} A = 2$ after computing two determinants of order 3, instead of all four.

2. *A rank depending on a parameter.* Let $t \in \mathbb{R}$ and

$$A_t = \begin{pmatrix} 1 & t & 1 \\ t & 1 & 1 \\ 1 & 1 & t \end{pmatrix}.$$

Expanding along the first row, $\det A_t = (t-1) - t(t^2-1) + (t-1) = (t-1)(2 - t - t^2) = -(t-1)^2(t+2)$. So $\operatorname{rank} A_t = 3$ for $t \neq 1, -2$. For $t = 1$ all entries equal 1, all minors of order 2 vanish, and the rank is 1. For $t = -2$ the minor $\det\left(\begin{smallmatrix} 1 & -2 \\ -2 & 1 \end{smallmatrix}\right) = -3$ is nonzero and the only bordering minor is $\det A_{-2} = 0$, so the rank is 2. Gaussian elimination would require dividing by expressions in t and keeping track of when they vanish; the minors give the answer directly.

Remark 11.6.12. Proposition 11.6.9 has consequences that are not obvious from the definition of the rank.

1. *The rank does not depend on the field.* If $k \subseteq L$ is a field extension and A has entries in k, its minors are the same whether computed in k or in L; so the rank of A over L equals its rank over k. For instance a real matrix has the same rank over $\mathbb{C}$, and a system with rational coefficients that has a real solution has a rational one (Theorem 11.6.5).

2. *The rank is lower semicontinuous.* For $k = \mathbb{R}$ or $\mathbb{C}$ the minors are polynomial, hence continuous, functions of the entries. So the set of matrices of rank at least r, where some minor of order r is nonzero, is open in $M_{m\times n}(k)$, and a small perturbation can only increase the rank; the set of matrices of rank at most r is closed, the common zero set of the minors of order $r+1$. These zero sets are the *determinantal varieties*, a basic family of examples in algebraic geometry.

11.7 Eigenvalues

In this section V is a vector space of finite dimension n and $f : V \to V$ is linear.

Definition 11.7.1. A scalar $\lambda \in k$ is an *eigenvalue* of f if $\ker(f - \lambda) \neq 0$; the nonzero elements of the *eigenspace* $V_\lambda = \ker(f - \lambda)$ are *eigenvectors*. The *characteristic polynomial* is $\chi_f(x) = \det(xI - A) \in k[x]$ for the matrix A of f in any basis; it is monic of degree n and independent of the basis. The *trace* of f is $\operatorname{tr} f = \sum_i a_{ii}$. The map f is *diagonalisable* if V has a basis of eigenvectors.

Proposition 11.7.2. *1. The eigenvalues of f are the roots of χ_f in k.*

2. $\chi_f(x) = x^n - (\operatorname{tr} f)x^{n-1} + \cdots + (-1)^n \det f$; in particular the trace is independent of the basis.

3. Eigenvectors with pairwise distinct eigenvalues are linearly independent.

Proof. (1) λ is an eigenvalue iff $\lambda - f$ is not injective iff $\det(\lambda I - A) = 0$, by Theorem 11.2.4 and Proposition 11.5.3(5). (2) In the Leibniz expansion of $\det(xI - A)$ only $\sigma = \mathrm{id}$ contributes to the coefficients of x^n and x^{n-1}, since any other permutation moves at least two indices; evaluate at $x = 0$ for the constant term. (3) Suppose $\sum_{i=1}^{m} c_i v_i = 0$ with m minimal and all $c_i \neq 0$. Applying $f - \lambda_m$ gives $\sum_{i<m} c_i(\lambda_i - \lambda_m)v_i = 0$, a shorter relation with nonzero coefficients. □

Theorem 11.7.3 (Cayley–Hamilton)**.** $\chi_f(f) = 0$.

Proof. Let A be the matrix of f. By Proposition 11.5.3(4) over $R = k[x]$, $(xI - A)\operatorname{adj}(xI - A) = \chi_f(x)I$. The entries of $\operatorname{adj}(xI - A)$ have degree $< n$, so we can write $\operatorname{adj}(xI - A) = \sum_{i=0}^{n-1} B_i x^i$ with $B_i \in M_n(k)$; write $\chi_f = \sum_{i=0}^{n} c_i x^i$. Comparing coefficients of x^i: $B_{i-1} - AB_i = c_i I$ for $0 \le i \le n$, with $B_{-1} = B_n = 0$. Multiplying the i-th equation by A^i on the left and summing, the left sides telescope to 0, so $\sum_i c_i A^i = 0$. □

Definition 11.7.4. The *minimal polynomial* m_f is the monic generator of the ideal $\{p \in k[x] : p(f) = 0\}$, which is nonzero by Theorem 11.7.3. So $m_f \mid \chi_f$.

Lemma 11.7.5 (Kernel lemma)**.** *If $p, q \in k[x]$ are coprime, then* $\ker(pq)(f) = \ker p(f) \oplus \ker q(f)$.

Proof. Write $1 = ap + bq$. For $v \in \ker(pq)(f)$, $v = a(f)p(f)v + b(f)q(f)v$, where the first term lies in $\ker q(f)$ (since $q(f)a(f)p(f)v = a(f)(pq)(f)v = 0$) and the second in $\ker p(f)$. If $v \in \ker p(f) \cap \ker q(f)$ then $v = a(f)p(f)v + b(f)q(f)v = 0$. □

Theorem 11.7.6. *f is diagonalisable if and only if m_f is a product of distinct monic linear factors. In that case $V = \bigoplus_\lambda V_\lambda$ over the eigenvalues. Commuting diagonalisable endomorphisms are simultaneously diagonalisable.*

Proof. If V has a basis of eigenvectors with distinct eigenvalues $\lambda_1, \ldots, \lambda_r$, then $\prod_i (f - \lambda_i) = 0$, so m_f divides $\prod_i (x - \lambda_i)$. Conversely if $m_f = \prod_i (x - \lambda_i)$ with distinct λ_i, the kernel lemma gives $V = \ker m_f(f) = \bigoplus_i \ker(f - \lambda_i)$. If g commutes with f, it preserves each V_λ, and its restriction to V_λ has minimal polynomial dividing m_g, hence is diagonalisable; choose eigenbases of these restrictions. □

Example 11.7.7. *Diagonalising a matrix, and what it is good for.* Let $A = \left(\begin{smallmatrix} 4 & 1 \\ 2 & 3 \end{smallmatrix}\right)$.

1. The characteristic polynomial is $\det(xI - A) = (x-4)(x-3) - 2 = x^2 - 7x + 10 = (x-2)(x-5)$, so the eigenvalues are 2 and 5; note $2 + 5 = 7 = \operatorname{tr} A$ and $2 \cdot 5 = 10 = \det A$, as Proposition 11.7.2(2) predicts.

2. Eigenvectors: $A - 2I = \left(\begin{smallmatrix} 2 & 1 \\ 2 & 1 \end{smallmatrix}\right)$ has kernel spanned by $(1, -2)$, and $A - 5I = \left(\begin{smallmatrix} -1 & 1 \\ 2 & -2 \end{smallmatrix}\right)$ by $(1, 1)$. They are independent (distinct eigenvalues), so with $P = \left(\begin{smallmatrix} 1 & 1 \\ -2 & 1 \end{smallmatrix}\right)$, whose columns are the eigenvectors, $P^{-1}AP = \operatorname{diag}(2, 5)$ by Proposition 11.3.3.

3. Powers become easy: $A^n = P \operatorname{diag}(2^n, 5^n) P^{-1}$, and with $P^{-1} = \frac{1}{3}\left(\begin{smallmatrix} 1 & -1 \\ 2 & 1 \end{smallmatrix}\right)$,

$$A^n = \frac{1}{3}\begin{pmatrix} 2^n + 2 \cdot 5^n & 5^n - 2^n \\ 2 \cdot 5^n - 2^{n+1} & 2^{n+1} + 5^n \end{pmatrix},$$

   which for $n = 1$ gives back A. For large n, $A^n \approx \frac{5^n}{3}\left(\begin{smallmatrix} 2 & 1 \\ 2 & 1 \end{smallmatrix}\right)$: the largest eigenvalue dominates.

4. *Fibonacci numbers.* $F_0 = 0$, $F_1 = 1$, $F_{n+1} = F_n + F_{n-1}$ means $(F_{n+1}, F_n)^T = M(F_n, F_{n-1})^T$ with $M = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Its characteristic polynomial $x^2 - x - 1$ has the real roots $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$, so M is diagonalisable over $\mathbb{R}$ and $F_n = a\varphi^n + b\psi^n$ for constants determined by F_0 and F_1: $a + b = 0$, $a\varphi + b\psi = 1$, so $a = -b = 1/\sqrt{5}$ and $F_n = (\varphi^n - \psi^n)/\sqrt{5}$ (Binet's formula). As $|\psi| < 1$, F_n is the integer closest to $\varphi^n/\sqrt{5}$.

5. *Cayley–Hamilton for* 2×2 *matrices* says $A^2 - (\operatorname{tr} A)A + (\det A)I = 0$; here $A^2 = \left(\begin{smallmatrix} 18 & 7 \\ 14 & 11 \end{smallmatrix}\right) = 7A - 10I$. If $\det A \neq 0$ it gives the inverse: $A^{-1} = \frac{1}{\det A}((\operatorname{tr} A)I - A) = \frac{1}{10}\left(\begin{smallmatrix} 3 & -1 \\ -2 & 4 \end{smallmatrix}\right)$, which is the adjugate formula.

Example 11.7.8. A *projection* is an endomorphism with $p^2 = p$. Its minimal polynomial divides $x^2 - x = x(x-1)$, a product of distinct linear factors, so by Theorem 11.7.6 p is diagonalisable with eigenvalues in $\{0, 1\}$, and $V = \ker p \oplus \operatorname{im} p$, with p the identity on $\operatorname{im} p = \ker(p - 1)$. So a projection is the same as a decomposition $V = U \oplus W$, p being the projection onto U along W. Similarly an *involution*, $s^2 = 1$, in characteristic $\neq 2$ is diagonalisable with eigenvalues ± 1, $V = \ker(s - 1) \oplus \ker(s + 1)$: a reflection in the first summand along the second.

Counterexample 11.7.9. The rotation $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has characteristic polynomial $x^2 + 1$ and no eigenvalue over $\mathbb{R}$, although it is diagonalisable over $\mathbb{C}$. The matrix $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has $m_N = x^2$ and is not diagonalisable over any field: its only eigenvalue is 0 and $\ker N$ is one-dimensional.

11.8 Canonical forms

Let $f : V \to V$ be as in the previous section. By Example 8.7.2, V is a $k[x]$-module V_f with x acting as f; it is finitely generated and torsion, being annihilated by χ_f.

Definition 11.8.1. The *companion matrix* of a monic $g = x^d + c_{d-1}x^{d-1} + \cdots + c_0$ is the $d \times d$ matrix $C(g)$ with 1's below the diagonal, last column $(-c_0, \ldots, -c_{d-1})^T$ and zeros elsewhere; it is the matrix of multiplication by x on $k[x]/(g)$ in the basis $1, x, \ldots, x^{d-1}$. The *Jordan block* $J_e(\lambda)$ is the $e \times e$ matrix with λ on the diagonal, 1 directly above the diagonal and zeros elsewhere.

Lemma 11.8.2. *The minimal and characteristic polynomials of $C(g)$ both equal g. The minimal and characteristic polynomials of $J_e(\lambda)$ both equal $(x-\lambda)^e$.*

Proof. On $k[x]/(g)$ a polynomial p acts as zero iff $p \cdot 1 = p$ lies in (g), so the minimal polynomial is g. The characteristic polynomial is monic of degree d and divisible by g (Definition 11.7.4), hence equal to g. The matrix $J_e(\lambda)$ is the matrix of x on $k[x]/((x-\lambda)^e)$ in the basis $(x-\lambda)^{e-1}, \dots, (x-\lambda), 1$, and the same argument applies. □

Theorem 11.8.3 (Rational canonical form)**.** *There are unique monic polynomials $f_1 \mid f_2 \mid \cdots \mid f_s$ of positive degree such that f has, in a suitable basis, the block diagonal matrix $C(f_1) \oplus \cdots \oplus C(f_s)$. Moreover $m_f = f_s$ and $\chi_f = f_1 \cdots f_s$. Two endomorphisms (or matrices) are similar if and only if they have the same f_i, the* invariant factors*.*

Proof. By Theorem 8.11.3, $V_f \cong \bigoplus_i k[x]/(f_i)$ with $f_1 \mid \cdots \mid f_s$ monic of positive degree (there is no free part since V_f is torsion), uniquely determined. Bases of the summands as in Definition 11.8.1 give the block form. An isomorphism $V_f \cong V_g$ of $k[x]$-modules is the same as a linear isomorphism h with $hf = gh$, so similarity corresponds to isomorphism of modules. The polynomial f_s kills every summand and is the minimal polynomial of the last one, so $m_f = f_s$; the characteristic polynomial of a block matrix is the product of those of the blocks, which gives χ_f by Lemma 11.8.2. □

Theorem 11.8.4 (Jordan normal form)**.** *Suppose χ_f splits into linear factors over k (for instance, k algebraically closed). Then f has, in a suitable basis, a block diagonal matrix $J_{e_1}(\lambda_1) \oplus \cdots \oplus J_{e_r}(\lambda_r)$, and the multiset of pairs (λ_i, e_i) is uniquely determined by f.*

Proof. By Corollary 8.11.4, $V_f \cong \bigoplus_i k[x]/(p_i^{e_i})$ with p_i monic irreducible, uniquely. Each p_i divides χ_f, hence is $x - \lambda_i$. Use the bases of Lemma 11.8.2. □

Example 11.8.5. A nilpotent endomorphism ($f^m = 0$ for some m) over any field has $\chi_f = x^n$ (its only possible eigenvalue is 0, and $m_f \mid x^m$), so it is similar to a direct sum of blocks $J_e(0)$; the sizes of the blocks form a partition of n, and they classify nilpotent endomorphisms up to similarity. Such nilpotent operators, and their Jordan decompositions, organise the limiting behaviour of variations of Hodge structure (Chapter 70).

Example 11.8.6. *Computing a Jordan basis.* The proof is not constructive as written, but the form is computed as follows. For each eigenvalue λ, the generalised eigenspace $V^{(\lambda)} = \ker(f-\lambda)^n$ is invariant, and $V = \bigoplus_\lambda V^{(\lambda)}$ (Lemma 11.7.5); on it $N = f - \lambda$ is nilpotent, and the number of Jordan blocks of size at least j is $\dim \ker N^j - \dim \ker N^{j-1}$ (Exercise 11.12.2). A block of size e corresponds to a *chain* $v, Nv, \dots, N^{e-1}v$ with $N^e v = 0$, where $v \in \ker N^e \setminus \ker N^{e-1}$; in the basis $N^{e-1}v, \dots, Nv, v$ the matrix of f is $J_e(\lambda)$. For example let

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}, \qquad N = A - 2I = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad N^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

and $N^3 = 0$. Here $\dim\ker N = 1$, $\dim\ker N^2 = 2$, $\dim\ker N^3 = 3$, so there is a single block $J_3(2)$. Take $v_3 = e_3 \notin \ker N^2$, $v_2 = Nv_3 = (1,1,0)$, $v_1 = Nv_2 = (1,0,0)$. Then $Av_1 = 2v_1$, $Av_2 = 2v_2 + v_1$, $Av_3 = 2v_3 + v_2$, so with $P = (v_1\ v_2\ v_3)$, $P^{-1}AP = J_3(2)$. By contrast $B = \operatorname{diag}(J_2(2), 2)$ has $\dim\ker(B-2) = 2$, two blocks, and the same characteristic polynomial $(x-2)^3$ as A; the dimensions of the kernels of the powers distinguish them.

Counterexample 11.8.7. The hypothesis that χ_f splits cannot be dropped. Over $\mathbb{R}$, the rotation matrix $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ is not similar to any block diagonal matrix of Jordan blocks.

Such a matrix is upper triangular, and the determinant of an upper triangular matrix is the product of its diagonal entries (in the Leibniz expansion every other permutation meets an entry below the diagonal), so its characteristic polynomial is $\prod_i (x - a_{ii})$, which splits over $\mathbb{R}$. But similar matrices have the same characteristic polynomial, and $\chi_A = x^2 + 1$ has no real root. The rational canonical form (Theorem 11.8.3) of A over $\mathbb{R}$ is the companion matrix $C(x^2+1)$, which is A itself; over $\mathbb{C}$, where $x^2 + 1 = (x - i)(x + i)$, A is similar to $J_1(i) \oplus J_1(-i)$.

Proposition 11.8.8. *An endomorphism f has a block diagonal matrix of Jordan blocks in some basis if and only if χ_f splits into linear factors over k. Every square matrix over k is similar to such a matrix if and only if k is algebraically closed.*

Proof. If χ_f splits, use Theorem 11.8.4. Conversely, a block diagonal matrix of Jordan blocks $J_{e_i}(\lambda_i)$ is upper triangular, so its characteristic polynomial is $\prod_i (x - \lambda_i)^{e_i}$, as in Counterexample 11.8.7; and χ_f is the characteristic polynomial of any matrix of f. For the second statement, if k is algebraically closed every χ_f splits. If k is not, some monic polynomial g of degree $d \geq 1$ does not split, and the companion matrix $C(g)$ has characteristic polynomial g (Lemma 11.8.2), so it is not similar to a matrix of Jordan blocks. □

When χ_f does not split, the primary decomposition still gives a canonical form. By Corollary 8.11.4, $V_f \cong \bigoplus_i k[x]/(p_i^{e_i})$ with p_i monic irreducible, uniquely, so f is similar to the block diagonal matrix $C(p_1^{e_1}) \oplus \cdots \oplus C(p_r^{e_r})$, the *primary rational canonical form*; for $p_i = x - \lambda_i$ one may use $J_{e_i}(\lambda_i)$ instead of $C(p_i^{e_i})$. Over $\mathbb{R}$ there is a convenient replacement for $C(p^e)$ as well.

Proposition 11.8.9 (Real Jordan form)**.** *For $a \in \mathbb{R}$ and $b > 0$ let $R = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ and let the* real Jordan block *$J_e(a, b)$ be the $2e \times 2e$ block matrix with R on the diagonal, the 2×2 identity matrix directly above the diagonal, and zeros elsewhere. Then $J_e(a, b)$ is similar to $C(p^e)$, where $p = (x-a)^2 + b^2$. Consequently every real square matrix is similar to a block diagonal matrix of blocks $J_e(\lambda)$ with $\lambda \in \mathbb{R}$ and $J_e(a, b)$ with $b > 0$, and the multiset of these blocks is uniquely determined.*

Proof. Write $M = J_e(a, b) = D + N$, where D is block diagonal with blocks R and N has the identity blocks above the diagonal. The matrices D and N commute, since both are built from 2×2 blocks that commute with each other; $N^e = 0$ and $N^{e-1} \neq 0$. The characteristic polynomial of R is p, so $p(R) = 0$ by Theorem 11.7.3 and $p(D) = 0$. Hence

$$p(M) = (D - aI + N)^2 + b^2 I = p(D) + 2(D - aI)N + N^2 = N\big(2(D - aI) + N\big).$$

The matrix $D - aI$ is block diagonal with invertible blocks $\left(\begin{smallmatrix} 0 & -b \\ b & 0 \end{smallmatrix}\right)$, and it commutes with the nilpotent matrix N; so $U = 2(D-aI)+N$ is invertible, with inverse $\sum_{j\geq 0}(-1)^j(2(D-aI))^{-j-1}N^j$, a finite sum. As N and U commute, $p(M)^j = N^j U^j$, which vanishes if and only if $j \geq e$. The minimal polynomial of M divides a power of the irreducible p, so it is p^e; the characteristic polynomial is $\chi_R^e = p^e$, since M is block upper triangular. By Theorem 11.8.3, the invariant factors of M have product p^e and the last one equals p^e, so there is only one, and M is similar to $C(p^e)$. The monic irreducible real polynomials are the $x - \lambda$ and the $(x - a)^2 + b^2$ with $b > 0$ (Theorem 9.7.3: a nonreal root $a + ib$ comes with its conjugate), so the last statement follows from the primary rational canonical form. □

For example, the rotation of Counterexample 11.8.7 is the real Jordan block $J_1(0, 1)$.

We turn to a decomposition that makes sense without any splitting. An endomorphism s is called *semisimple* if its minimal polynomial is separable (Definition 9.4.1), that is, has no repeated roots in a splitting field. We need that the minimal polynomial does not change under extension of the field.

Lemma 11.8.10. *Let L/k be a field extension and $A \in M_n(k)$. The minimal polynomial of A over L equals its minimal polynomial over k. In particular A is semisimple over k if and only if it is semisimple over L, and A is semisimple if and only if it is diagonalisable over a splitting field of m_A.*

Proof. Let d be the least integer such that $I, A, \dots, A^d$ are linearly dependent over k. Linear dependence of finitely many vectors of k^{n^2} over k or over L means that the matrix having them as columns has rank less than the number of vectors, that is, that all its maximal minors vanish; this does not depend on the field. So d is the same over k and over L, and the relation $A^d = -\sum_{j<d} c_j A^j$ is unique over L, because $I, \dots, A^{d-1}$ are independent over L; as it exists with $c_j \in k$, the minimal polynomials agree. The last statement follows from Theorem 11.7.6, applied over the splitting field. □

Theorem 11.8.11 (Jordan–Chevalley decomposition)**.** *Let k be a perfect field (Definition 9.4.3) and f an endomorphism of V. There are unique endomorphisms s and n of V with $f = s + n$, s semisimple, n nilpotent, and $sn = ns$. Moreover s and n are polynomials in f with coefficients in k.*

Proof. Step 1: existence when χ_f splits, over any field. Let $\chi_f = \prod_i (x-\lambda_i)^{e_i}$ with distinct λ_i. By the kernel lemma (Lemma 11.7.5), $V = \bigoplus_i V_i$ with $V_i = \ker(f-\lambda_i)^{e_i}$, since $\chi_f(f) = 0$. The ideals $((x-\lambda_i)^{e_i})$ are pairwise coprime, so by the Chinese remainder theorem (Theorem 8.3.3) there is $q \in k[x]$ with $q \equiv \lambda_i \bmod (x-\lambda_i)^{e_i}$ for all i. Then $s = q(f)$ acts on V_i as λ_i, so it is diagonalisable, hence semisimple, and $n = f - s$ acts on V_i as $f - \lambda_i$, which is nilpotent there; so n is nilpotent. Both are polynomials in f, so they commute.

Step 2: uniqueness, over any field, for decompositions as in Step 1. Let $f = s' + n'$ be another decomposition. Then s' and n' commute with $f = s' + n'$, hence with the polynomials s and n in f. Now $s - s' = n' - n$. The right side is nilpotent: n and n' commute, so $(n'-n)^{2m}$ is a sum of terms $n'^j n^{2m-j}$ which vanish for m large. The left side is semisimple: over a splitting field L of $m_s m_{s'}$, the commuting endomorphisms s and s' are diagonalisable (Lemma 11.8.10), hence simultaneously diagonalisable (Theorem 11.7.6), so $s - s'$ is diagonalisable over L. A diagonalisable nilpotent endomorphism is 0, as its eigenvalues are 0. So $s = s'$ and $n = n'$.

Step 3: k perfect. Let g be the product of the distinct monic irreducible factors of χ_f. Each is separable because k is perfect (Proposition 9.4.4), and distinct monic irreducible polynomials have no common root, so g is separable. Let L be a splitting field of g; then χ_f splits over L, and L/k is Galois with group G and $L^G = k$ (Theorem 9.5.3). Let $A \in M_n(k)$ be the matrix of f in a basis. By Steps 1 and 2 over L, $A = S + N$ uniquely, with $S = q(A)$ for some $q \in L[x]$. For $\sigma \in G$, applying σ to all entries is a ring automorphism of $M_n(L)$ fixing A, so $A = \sigma(S) + \sigma(N)$, with $\sigma(N)$ nilpotent, $\sigma(S)\sigma(N) = \sigma(N)\sigma(S)$, and $\sigma(S)$ semisimple: its minimal polynomial is $\sigma(m_S)$, which is separable since $a m_S + b m_S' = 1$ implies $\sigma(a)\sigma(m_S) + \sigma(b)\sigma(m_S)' = 1$ (Proposition 9.4.2). By uniqueness $\sigma(S) = S$, so the entries of S lie in $L^G = k$. Then S and $N = A - S$ have entries in k, and S is semisimple over k by Lemma 11.8.10. Any decomposition over k is one over L, so uniqueness over k follows from Step 2. Finally, with $d = \deg m_A$, the matrices $I, A, \dots, A^{d-1}$ form a basis of $L[A]$ over L (by the lemma, over L too the minimal polynomial has degree d); writing $S = \sum_j c_j A^j$ and applying σ shows $\sigma(c_j) = c_j$, so the c_j lie in k. □

Counterexample 11.8.12. Perfectness cannot be dropped. Let $k = \mathbb{F}_p(t)$ and f the endomorphism of k^p with matrix $C(x^p - t)$, whose minimal polynomial is $x^p - t$ (Lemma 11.8.2), irreducible but inseparable (Counterexample 9.4.5). As a $k[x]$-module, $V_f \cong k[x]/(x^p - t)$ is a field, so f has no invariant subspaces other than 0 and V: it is "irreducible", the opposite of nilpotent. Yet f is not semisimple, and it has no decomposition $f = s + n$ as in Theorem 11.8.11 with s and n defined over k. Indeed, over $L = k(\beta)$ with $\beta^p = t$ we have $x^p - t = (x-\beta)^p$, so χ_f splits and $f - \beta$ is nilpotent. A decomposition over k would be one over L, where by uniqueness (Steps 1 and 2 of the proof) $s = \beta I$; but $\beta \notin k$. So over imperfect

fields "irreducible" and "semisimple" diverge, and the semisimple part of f exists only after an inseparable extension.

11.9 Bilinear forms

Definition 11.9.1. A *bilinear form* on V is a bilinear map $B\colon V \times V \to k$. It is *symmetric* if $B(v,w) = B(w,v)$, *alternating* if $B(v,v) = 0$ for all v, and *nondegenerate* if $v \mapsto B(v,-)$ is an isomorphism $V \to V^*$ (for V finite-dimensional: if $B(v,w) = 0$ for all w implies $v = 0$). Its *Gram matrix* in a basis (v_i) is $(B(v_i,v_j))$. For a subspace W, $W^{\perp_B} = \{v : B(v,w) = 0 \text{ for all } w \in W\}$. A *quadratic form* is $q(v) = B(v,v)$ for a symmetric B.

In coordinates every bilinear form is given by a matrix: if $v = \sum_i x_i v_i$ and $w = \sum_j y_j v_j$, then $B(v,w) = \sum_{i,j} x_i y_j B(v_i,v_j) = x^T G y$, with G the Gram matrix. Conversely every $G \in M_n(k)$ defines the bilinear form $(x,y) \mapsto x^T G y$ on k^n; it is symmetric if and only if $G^T = G$, alternating (in characteristic $\neq 2$) if and only if $G^T = -G$, and nondegenerate if and only if $\det G \neq 0$, since $v \mapsto B(v,-)$ has matrix G^T in the basis (v_i) and the dual basis. Under a change of basis with matrix P the Gram matrix becomes $P^T G P$.

Lemma 11.9.2. *Let B be symmetric or alternating on a finite-dimensional V, and $W \subseteq V$ a subspace on which B is nondegenerate. Then $V = W \oplus W^{\perp_B}$. An alternating form B satisfies $B(v,w) = -B(w,v)$.*

Proof. $W \cap W^{\perp_B} = 0$ by nondegeneracy on W. Given $v \in V$, the functional $B(v,-)|_W$ equals $B(w,-)|_W$ for a unique $w \in W$, and then $v - w \in W^{\perp_B}$ (for alternating forms use $B(-,w) = -B(w,-)$). The last statement follows from expanding $B(v+w, v+w) = 0$. □

Theorem 11.9.3. *Let* $\operatorname{char} k \neq 2$ *and B symmetric on a finite-dimensional V. Then V has a basis (v_i) with $B(v_i,v_j) = 0$ for $i \neq j$.*

Proof. By induction on $\dim V$. If $B = 0$ any basis works. Otherwise some $q(v) \neq 0$: if $B(v,w) \neq 0$ then by the polarisation identity $2B(v,w) = q(v+w) - q(v) - q(w)$ one of $q(v+w)$, $q(v)$, $q(w)$ is nonzero. Then B is nondegenerate on kv, so $V = kv \oplus (kv)^{\perp_B}$ by Lemma 11.9.2; apply induction to $(kv)^{\perp_B}$. □

Counterexample 11.9.4. The hypothesis $\operatorname{char} k \neq 2$ is needed. On $\mathbb{F}_2^2$ the form $B((a,b),(c,d)) = ad + bc$ is symmetric and nondegenerate, but alternating ($B(v,v) = 2ab = 0$). A basis with $B(v_i,v_j) = 0$ for $i \neq j$ would make B zero.

Theorem 11.9.5 (Sylvester's law of inertia)**.** *Let B be a symmetric bilinear form on a real vector space V of dimension n. There is a basis in which the Gram matrix is diagonal with p entries 1, q entries -1 and $n-p-q$ entries 0, and the numbers p and q do not depend on the basis. The pair (p,q) is the* signature *of B.*

Proof. Take an orthogonal basis (Theorem 11.9.3) and rescale each v_i with $q(v_i) \neq 0$ by $|q(v_i)|^{-1/2}$ (Proposition 4.7.4). For uniqueness, p is the largest dimension of a subspace on which q is positive definite: q is positive definite on the span P of the v_i with $q(v_i) = 1$, and if U is a subspace on which q is positive definite, then U meets the span N of the other basis vectors, on which $q \leq 0$, only in 0; so $\dim U \leq n - \dim N = p$. Similarly for q. □

Theorem 11.9.6. *Let ω be a nondegenerate alternating form on a finite-dimensional V. Then* $\dim V = 2m$ *is even, and V has a* symplectic basis *$e_1, \dots, e_m, f_1, \dots, f_m$ with $\omega(e_i, f_j) = \delta_{ij}$ and $\omega(e_i, e_j) = \omega(f_i, f_j) = 0$.*

Proof. If $V \neq 0$, choose $e_1 \neq 0$, and f_1 with $\omega(e_1, f_1) = 1$ by nondegeneracy. Then e_1, f_1 are independent (as $\omega(e_1, e_1) = 0$), and ω is nondegenerate on their span W, whose Gram matrix is $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. By Lemma 11.9.2, $V = W \oplus W^{\perp_\omega}$, and ω is nondegenerate on $W^{\perp_\omega}$. Apply induction. □

11.10 Inner product spaces and the spectral theorem

Definition 11.10.1. Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. An *inner product* on a $\mathbb{K}$-vector space V is a map $\langle -, - \rangle \colon V \times V \to \mathbb{K}$, linear in the first variable, with $\langle w, v \rangle = \overline{\langle v, w \rangle}$ and $\langle v, v \rangle > 0$ for $v \neq 0$. (Over $\mathbb{C}$ it is then conjugate-linear in the second variable; such a form is also called a *Hermitian inner product*.) We write $\|v\| = \sqrt{\langle v, v \rangle}$. A family is *orthonormal* if $\langle v_i, v_j \rangle = \delta_{ij}$.

Proposition 11.10.2. $|\langle v, w \rangle| \leq \|v\|\|w\|$, *and* $\|v + w\| \leq \|v\| + \|w\|$.

Proof. If $w = 0$ this is clear. Otherwise let $u = v - \frac{\langle v, w \rangle}{\|w\|^2} w$; then $\langle u, w \rangle = 0$ and $0 \leq \|u\|^2 = \|v\|^2 - |\langle v, w \rangle|^2 / \|w\|^2$. For the triangle inequality, $\|v+w\|^2 = \|v\|^2 + 2\,\mathrm{Re}\langle v, w \rangle + \|w\|^2 \leq (\|v\| + \|w\|)^2$. □

Theorem 11.10.3 (Gram–Schmidt)**.** *If $v_1, v_2, \ldots$ is a finite or countable linearly independent family, there is an orthonormal family $u_1, u_2, \ldots$ with* $\mathrm{span}(u_1, \ldots, u_j) = \mathrm{span}(v_1, \ldots, v_j)$ *for all j. Every finite-dimensional inner product space has an orthonormal basis, and for every finite-dimensional subspace W, $V = W \oplus W^\perp$, where $W^\perp = \{v : \langle v, w \rangle = 0$ for all $w \in W\}$.*

Proof. Define recursively $u'_j = v_j - \sum_{i<j} \langle v_j, u_i \rangle u_i$, which is nonzero since $v_j \notin \mathrm{span}(u_1, \ldots, u_{j-1})$, and $u_j = u'_j / \|u'_j\|$. For the last claim, with an orthonormal basis (u_i) of W, write $v = \sum_i \langle v, u_i \rangle u_i + (v - \sum_i \langle v, u_i \rangle u_i)$; the second term lies in $W^\perp$, and $W \cap W^\perp = 0$ by positivity. □

Definition 11.10.4. Let V be a finite-dimensional inner product space and $f : V \to V$ linear. The *adjoint* $f^\dagger$ is the unique linear map with $\langle f(v), w \rangle = \langle v, f^\dagger(w) \rangle$ for all v, w. The map f is *self-adjoint* if $f^\dagger = f$, *normal* if $f f^\dagger = f^\dagger f$, and *unitary* (or *orthogonal*, if $\mathbb{K} = \mathbb{R}$) if $f^\dagger f = \mathrm{id}$. The unitary operators of $\mathbb{C}^n$ form the *unitary group* $\mathrm{U}(n)$, and the orthogonal operators of $\mathbb{R}^n$ the *orthogonal group* $\mathrm{O}(n)$.

Lemma 11.10.5. *The adjoint exists and is unique. In an orthonormal basis its matrix is the conjugate transpose of the matrix of f. If W is f-invariant then $W^\perp$ is $f^\dagger$-invariant.*

Proof. With an orthonormal basis (u_i), the map $f^\dagger(w) = \sum_i \langle w, f(u_i) \rangle u_i$ works, and a linear map is determined by the values $\langle v, f^\dagger(w) \rangle$. Its matrix entries are $\langle f^\dagger(u_j), u_i \rangle = \overline{\langle f(u_i), u_j \rangle}$. If $f(W) \subseteq W$ and $v \in W^\perp$, then $\langle f^\dagger(v), w \rangle = \langle v, f(w) \rangle = 0$ for $w \in W$. □

Theorem 11.10.6 (Spectral theorem)**.** *Let V be a finite-dimensional inner product space.*

1. *If $\mathbb{K} = \mathbb{C}$ and f is normal, then V has an orthonormal basis of eigenvectors of f.*
2. *If f is self-adjoint (over $\mathbb{R}$ or $\mathbb{C}$), then all its eigenvalues are real and V has an orthonormal basis of eigenvectors of f.*

Proof. For self-adjoint f and an eigenvector v with eigenvalue λ, $\lambda \|v\|^2 = \langle fv, v \rangle = \langle v, fv \rangle = \bar{\lambda} \|v\|^2$, so $\lambda \in \mathbb{R}$.

(1) By induction on $\dim V$. By Theorem 9.7.3, χ_f has a root, so f has an eigenvector v. Since f and $f^\dagger$ commute, $f^\dagger$ preserves the eigenspace V_λ, and its restriction there has an eigenvector u; then u is an eigenvector of both f and $f^\dagger$. Then $W = \mathbb{C}u$ is invariant under

both, so $W^\perp$ is invariant under $f^\dagger$ and $(f^\dagger)^\dagger = f$ (Lemma 11.10.5), and the restriction of f to $W^\perp$ is normal. Apply induction and normalise u.

(2) Over $\mathbb{C}$ this follows from (1). Over $\mathbb{R}$, choose an orthonormal basis and let A be the real symmetric matrix of f. As a complex matrix acting on $\mathbb{C}^n$ with the standard inner product, A is self-adjoint, so it has an eigenvalue, which is real, and hence a real root λ of χ_f; so f has an eigenvector $u \in V$. The line $\mathbb{R}u$ and its orthogonal complement are invariant under $f = f^\dagger$, and we conclude by induction. □

In matrix form: a real symmetric matrix A can be written $A = QDQ^T$ with Q orthogonal ($Q^TQ = I$) and D diagonal, the columns of Q being an orthonormal basis of eigenvectors and the diagonal of D the eigenvalues; so A is diagonalised by an orthonormal change of basis, for which $Q^{-1} = Q^T$ (Remark 11.3.4). Likewise a Hermitian or normal complex matrix is UDU^* with U unitary. For example $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ has $\chi_A = (x-1)(x-3)$, with unit eigenvectors $(1,-1)/\sqrt{2}$ and $(1,1)/\sqrt{2}$, and $A = QDQ^T$ with $Q = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ and $D = \operatorname{diag}(1,3)$. Eigenvectors for distinct eigenvalues of a self-adjoint map are automatically orthogonal: $\lambda\langle v,w\rangle = \langle fv,w\rangle = \langle v,fw\rangle = \mu\langle v,w\rangle$.

Counterexample 11.10.7. Normality is necessary in (1): the matrix N of Counterexample 11.7.9 is not diagonalisable at all. Over $\mathbb{R}$, normal is not enough: the rotation of that counterexample is orthogonal, hence normal, but has no real eigenvector.

Corollary 11.10.8. *A real symmetric (or complex Hermitian) matrix A defines an inner product $\langle v,w\rangle = w^*Av$ if and only if all its eigenvalues are positive. Every inner product on $\mathbb{R}^n$ (or $\mathbb{C}^n$) is of this form.*

Proof. Diagonalise $A = UDU^*$ with U unitary (Theorem 11.10.6); then $w^*Av = (U^*w)^*D(U^*v)$, which is positive definite iff all diagonal entries of D are positive. The matrix of an inner product in the standard basis is Hermitian. □

Proposition 11.10.9 (Singular value decomposition)**.** *Let $f : V \to W$ be a linear map between finite-dimensional inner product spaces over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, of rank r. There are orthonormal bases $(v_1,\dots,v_n)$ of V and $(w_1,\dots,w_m)$ of W and real numbers $\sigma_1 \geq \cdots \geq \sigma_r > 0$ with $f(v_i) = \sigma_i w_i$ for $i \leq r$ and $f(v_i) = 0$ for $i > r$. The σ_i^2 are the nonzero eigenvalues of $f^\dagger f$, so the* singular values *σ_i are determined by f; the largest one is the operator norm $\|f\| = \sup_{\|v\|=1}\|f(v)\|$.*

Proof. The endomorphism $f^\dagger f$ of V is self-adjoint and $\langle f^\dagger fv, v\rangle = \|f(v)\|^2 \geq 0$, so by Theorem 11.10.6 there is an orthonormal basis (v_i) of V with $f^\dagger f v_i = \lambda_i v_i$ and $\lambda_i \geq 0$; order it so that $\lambda_1 \geq \lambda_2 \geq \cdots$. Then $\langle f(v_i), f(v_j)\rangle = \langle f^\dagger f v_i, v_j\rangle = \lambda_i\delta_{ij}$, so the $f(v_i)$ are pairwise orthogonal, $f(v_i) = 0$ exactly when $\lambda_i = 0$, and the number of i with $\lambda_i > 0$ is the rank r. Put $\sigma_i = \sqrt{\lambda_i}$ and $w_i = f(v_i)/\sigma_i$ for $i \leq r$; these are orthonormal, and we extend them to an orthonormal basis of W (Theorem 11.10.3). For $v = \sum_i c_i v_i$ of norm one, $\|f(v)\|^2 = \sum_{i\leq r}\sigma_i^2|c_i|^2 \leq \sigma_1^2$, with equality at $v = v_1$. □

Example 11.10.10. For $f(x,y) = (x+y,0)$ on $\mathbb{R}^2$, $f^\dagger f$ has matrix $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ with eigenvalues 2 and 0 and eigenvectors $(1,1)/\sqrt{2}$ and $(1,-1)/\sqrt{2}$. So $\sigma_1 = \sqrt{2}$, $w_1 = (1,0)$, and $\|f\| = \sqrt{2}$, although the eigenvalues of f itself are 1 and 0: for non-normal maps the operator norm is not the largest absolute value of an eigenvalue.

11.11 Complex structures

Definition 11.11.1. Let V be a real vector space. Its *complexification* is $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$, a complex vector space with the *complex conjugation* $\overline{v \otimes z} = v \otimes \bar{z}$, which is $\mathbb{R}$-linear and conjugate-linear over $\mathbb{C}$. A *complex structure* on V is an $\mathbb{R}$-linear map $J\colon V \to V$ with $J^2 = -\mathrm{id}$.

Proposition 11.11.2. *Let J be a complex structure on a finite-dimensional real vector space V.*

1. *V becomes a complex vector space with $(a+bi)v = av + bJv$; hence $\dim_{\mathbb{R}} V = 2\dim_{\mathbb{C}} V$ is even.*
2. *Extend J $\mathbb{C}$-linearly to $V_{\mathbb{C}}$. Then $V_{\mathbb{C}} = V^{1,0} \oplus V^{0,1}$, where $V^{1,0} = \ker(J - i)$ and $V^{0,1} = \ker(J + i)$, and $\overline{V^{1,0}} = V^{0,1}$.*
3. *The map $V \to V^{1,0}$, $v \mapsto \frac{1}{2}(v - iJv)$, is an isomorphism of complex vector spaces from (V, J) to $V^{1,0}$, and $v \mapsto \frac{1}{2}(v + iJv)$ is a conjugate-linear isomorphism $V \to V^{0,1}$.*

Proof. (1) The axioms of a module over $\mathbb{C} = \mathbb{R}[x]/(x^2+1)$ follow from $J^2 = -1$ (Example 8.7.2 and Exercise 8.12.2); a complex basis (v_j) gives the real basis (v_j, Jv_j). (2) The polynomial $x^2 + 1 = (x - i)(x + i)$ kills J on $V_{\mathbb{C}}$, and the kernel lemma (Lemma 11.7.5) gives the decomposition. Conjugation commutes with J (which is real), so it exchanges the $\pm i$-eigenspaces. (3) $J(v - iJv) = Jv + iv = i(v - iJv)$, so the image lies in $V^{1,0}$, and J on V corresponds to multiplication by i. The map is injective because v is recovered as twice the real part, and both sides have complex dimension $\frac{1}{2}\dim_{\mathbb{R}} V$. The second map is the conjugate of the first. □

Proposition 11.11.3. *Let J be a complex structure on a real inner product space (V, g) with $g(Jv, Jw) = g(v, w)$. Put $\omega(v, w) = g(Jv, w)$ and $h = g - i\omega$. Then ω is a nondegenerate alternating form with $\omega(Jv, Jw) = \omega(v, w)$, and h is a Hermitian inner product on the complex vector space (V, J), whose real part is g. Conversely, g and J determine ω, and ω and J determine $g(v, w) = \omega(v, Jw)$.*

Proof. From J-invariance, $g(Jv, w) = g(J^2 v, Jw) = -g(v, Jw)$. Hence $\omega(w, v) = g(Jw, v) = -g(w, Jv) = -\omega(v, w)$, so ω is alternating; $\omega(Jv, Jw) = g(J(Jv), Jw) = g(Jv, w) = \omega(v, w)$ by J-invariance of g; and ω is nondegenerate because $\omega(v, Jv) = g(Jv, Jv) > 0$ for $v \neq 0$. For h: $h(v, w) = g(v, w) + ig(v, Jw)$. Then $h(Jv, w) = g(Jv, w) + ig(Jv, Jw) = -g(v, Jw) + ig(v, w) = ih(v, w)$, so h is complex linear in the first variable; $h(w, v) = g(v, w) - ig(v, Jw) = \overline{h(v, w)}$; and $h(v, v) = g(v, v) > 0$ for $v \neq 0$ since $g(v, Jv) = -g(Jv, v) = -g(v, Jv)$ vanishes. Finally $\omega(v, Jw) = g(Jv, Jw) = g(v, w)$. □

Example 11.11.4. On $\mathbb{R}^{2n}$ with coordinates $(x_1, y_1, \ldots, x_n, y_n)$, the standard complex structure is $J(\partial_{x_j}) = \partial_{y_j}$, $J(\partial_{y_j}) = -\partial_{x_j}$ (writing ∂_{x_j}, ∂_{y_j} for the standard basis vectors), which identifies $(\mathbb{R}^{2n}, J)$ with $\mathbb{C}^n$ via $x + iy$. The standard inner product is J-invariant, the associated form satisfies $\omega(\partial_{x_j}, \partial_{y_j}) = g(\partial_{y_j}, \partial_{y_j}) = 1$, so the ∂_{x_j}, ∂_{y_j} form a symplectic basis, and h is the standard Hermitian inner product of $\mathbb{C}^n$. The vectors $\partial_{z_j} = \frac{1}{2}(\partial_{x_j} - i\partial_{y_j})$ span $V^{1,0}$. These formulas reappear as the local model of Kähler manifolds (Chapter 47).

Proposition 11.11.3 relates three structures on V: an inner product g, a nondegenerate alternating form ω and a complex structure J. We call a triple (g, ω, J) of such structures *compatible* if $g(Jv, Jw) = g(v, w)$ and $\omega(v, w) = g(Jv, w)$ for all v, w. The next proposition says that any two members of a compatible triple determine the third, and says exactly which pairs occur. This "two out of three" property is the linear algebra behind Hermitian and Kähler manifolds (Chapter 47) and behind the compatible complex structures of symplectic geometry (Chapter 41).

Proposition 11.11.5 (Two out of three)**.** *Let V be a finite-dimensional real vector space.*

1. *In a compatible triple (g, ω, J) each member is determined by the other two:*

$$\omega(v,w) = g(Jv,w), \qquad g(v,w) = \omega(v,Jw), \qquad J \text{ is the unique endomorphism with } \omega(v,w) = g(Jv,w).$$

2. *A pair extends to a compatible triple exactly in the following cases.*
   (a) *An inner product g and a complex structure J: if and only if $g(Jv,Jw) = g(v,w)$.*
   (b) *A nondegenerate alternating form ω and a complex structure J: if and only if $\omega(Jv,Jw) = \omega(v,w)$ and $\omega(v,Jv) > 0$ for all $v \neq 0$.*
   (c) *An inner product g and a nondegenerate alternating form ω: let A be the endomorphism with $\omega(v,w) = g(Av,w)$; the pair extends if and only if $A^2 = -1$, equivalently, if and only if A is orthogonal for g; and then $J = A$.*

3. *Let (g, ω, J) be compatible, and $h = g - i\omega$ the Hermitian inner product of Proposition 11.11.3. An automorphism of V preserving two of g, ω, J (preserving J meaning $TJ = JT$) preserves the third. So the three groups $\mathrm{O}(V,g)$, $\mathrm{Sp}(V,\omega)$ and $\mathrm{GL}(V,J)$ of automorphisms preserving g, ω and J satisfy*

$$\mathrm{O}(V,g) \cap \mathrm{Sp}(V,\omega) = \mathrm{O}(V,g) \cap \mathrm{GL}(V,J) = \mathrm{Sp}(V,\omega) \cap \mathrm{GL}(V,J) = \mathrm{U}(V,h),$$

the unitary group of h on the complex vector space (V,J). For $\mathbb{R}^{2n} = \mathbb{C}^n$ with its standard structures this reads $\mathrm{O}(2n) \cap \mathrm{Sp}_{2n}(\mathbb{R}) = \mathrm{O}(2n) \cap \mathrm{GL}_n(\mathbb{C}) = \mathrm{Sp}_{2n}(\mathbb{R}) \cap \mathrm{GL}_n(\mathbb{C}) = \mathrm{U}(n)$.

Proof. (1) The first two formulas are in Proposition 11.11.3. If $g(Jv,w) = g(J'v,w)$ for all w, then $Jv = J'v$, as g is nondegenerate.

(2)(a) is Proposition 11.11.3. For (b), if the triple is compatible, ω is J-invariant by that proposition and $\omega(v,Jv) = g(Jv,Jv) > 0$ for $v \neq 0$. Conversely, define $g(v,w) = \omega(v,Jw)$. Using J-invariance of ω and $J^2 = -1$,

$$g(w,v) = \omega(w,Jv) = \omega(Jw,J^2v) = -\omega(Jw,v) = \omega(v,Jw) = g(v,w),$$

so g is symmetric; it is positive definite by hypothesis; $g(Jv,Jw) = \omega(Jv,J^2w) = -\omega(Jv,w) = \omega(w,Jv) = g(w,v) = g(v,w)$; and $g(Jv,w) = \omega(Jv,Jw) = \omega(v,w)$. So (g,ω,J) is compatible. For (c), $v \mapsto \omega(v,\cdot)$ is a linear form, represented through g by a unique vector Av (Proposition 11.4.2(1) and nondegeneracy of g), and A is linear. It is skew-adjoint: $g(Av,w) = \omega(v,w) = -\omega(w,v) = -g(Aw,v) = -g(v,Aw)$. If the pair extends, $J = A$ by (1), so $A^2 = -1$. Conversely, if $A^2 = -1$, then $J = A$ is a complex structure, $g(Jv,Jw) = -g(v,J^2w) = g(v,w)$ by skew-adjointness, and $\omega(v,w) = g(Jv,w)$ by definition. Finally, the adjoint of A is $-A$, so A is orthogonal, $A^*A = 1$, if and only if $-A^2 = 1$.

(3) Let T be an automorphism. If T preserves g and commutes with J, then $\omega(Tv,Tw) = g(JTv,Tw) = g(TJv,Tw) = g(Jv,w) = \omega(v,w)$. If T preserves ω and commutes with J, then $g(Tv,Tw) = \omega(Tv,JTw) = \omega(Tv,TJw) = \omega(v,Jw) = g(v,w)$. If T preserves g and ω, then for all v,w

$$g(JTv,Tw) = \omega(Tv,Tw) = \omega(v,w) = g(Jv,w) = g(TJv,Tw),$$

and since Tw runs through all of V, $JTv = TJv$. This proves the equality of the three intersections. An element of $\mathrm{GL}(V,J)$ is a complex linear automorphism of (V,J); it preserves g and ω if and only if it preserves $h = g - i\omega$, that is, if and only if it is unitary. For the standard structures on $\mathbb{R}^{2n}$, $\mathrm{GL}(V,J) = \mathrm{GL}_n(\mathbb{C})$ and $\mathrm{U}(V,h) = \mathrm{U}(n)$ (Example 11.11.4). □

Example 11.11.6. 1. On $\mathbb{R}^2$ with the standard inner product g and $\omega = c\,dx \wedge dy$, $c > 0$, the endomorphism A of (2)(c) is c times the rotation by $\pi/2$: $\omega(e_1, e_2) = c = g(Ae_1, e_2)$. So $A^2 = -c^2$, and the pair (g, ω) is compatible only for $c = 1$. For $c \neq 1$, rescaling g by c makes it compatible with ω and $J = A/c$. Similarly (ω, J) with $\omega = dx \wedge dy$ and J the rotation by $-\pi/2$ satisfies $\omega(Jv, Jw) = \omega(v, w)$ but $\omega(v, Jv) < 0$: the condition of positivity in (2)(b) singles out one of the two orientations.

2. *A pair that is not compatible.* On $\mathbb{R}^4$ with the standard inner product, let $\omega = dx^1 \wedge dy^1 + 2\,dx^2 \wedge dy^2$. Then A is the rotation by $\pi/2$ in the first plane and twice the rotation by $\pi/2$ in the second, $A^2 = \operatorname{diag}(-1, -1, -4, -4)$, and no complex structure is compatible with both g and ω.

Remark 11.11.7. An incompatible pair (g, ω) can be repaired canonically. With A as in (2)(c), $-A^2 = A^*A$ is self-adjoint and positive definite, so by the spectral theorem (Theorem 11.10.6) it has a unique positive definite square root P, which acts on each eigenspace of $-A^2$ as the positive square root of the eigenvalue; being a polynomial in A^2, P commutes with A. Put $J = AP^{-1}$ (the polar decomposition of A). Then $J^2 = A^2P^{-2} = -1$; using that P is self-adjoint, that A is skew-adjoint and that $AP = PA$,

$$\omega(Jv, Jw) = g(A^2P^{-1}v, AP^{-1}w) = -g(Pv, AP^{-1}w) = -g(v, Aw) = \omega(v, w), \qquad \omega(v, Jv) = g(Av, AP^{-1}v) = g($$

So J is compatible with ω in the sense of (2)(b), with the new inner product $\omega(v, Jw) = g(Pv, w)$, and $J = A$ when (g, ω) was already compatible. Since J depends only on g and ω, and continuously so, this produces compatible complex structures on every symplectic vector space, and indeed compatible almost complex structures on every symplectic manifold, from any Riemannian metric [MS17]. In Example 11.11.6(2), $P = \operatorname{diag}(1, 1, 2, 2)$ and J is the standard complex structure.

11.12 Exercises

Exercise 11.12.1. Show that $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for $A \in M_{m\times n}(k)$, $B \in M_{n\times m}(k)$, and deduce that similar matrices have the same trace. Show that if $\chi_f = \prod_i (x - \lambda_i)$ then $\operatorname{tr} f = \sum \lambda_i$ and $\det f = \prod \lambda_i$.

Exercise 11.12.2. Suppose χ_f splits. Show that the number of Jordan blocks $J_e(\lambda)$ with $e \geq j$ in the Jordan form of f is $\dim \ker(f - \lambda)^j - \dim \ker(f - \lambda)^{j-1}$.

Exercise 11.12.3. Let V be a finite-dimensional complex vector space and $\mathcal{F}$ a set of pairwise commuting endomorphisms. Show that there is a basis in which all elements of $\mathcal{F}$ are upper triangular.

Exercise 11.12.4. Show that every invertible complex matrix is uniquely $A = UP$ with U unitary and P Hermitian with positive eigenvalues.

Exercise 11.12.5. Let ω be a nondegenerate alternating form on V of dimension $2m$ and $\operatorname{Sp}(V, \omega)$ the group of $f \in \operatorname{GL}(V)$ with $\omega(fv, fw) = \omega(v, w)$. Show that $\det f = \pm 1$ for $f \in \operatorname{Sp}(V, \omega)$. (It is in fact always 1; this can be shown using the Pfaffian, Chapter 12.)

Exercise 11.12.6. Show that the set of complex structures J on $\mathbb{R}^2$ compatible with the standard inner product and with $\omega(e_1, e_2) > 0$ has exactly one element.

Exercise 11.12.7. Let $q(x_1, x_2) = x_1^2 + 4x_1x_2 + x_2^2$ on $\mathbb{R}^2$, with Gram matrix $G = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$.

1. Complete the square to find a change of variables $x = Py$ with $q = y_1^2 - 3y_2^2$, and check that $P^TGP = \operatorname{diag}(1,-3)$ but that $P^{-1}GP$ is not diagonal.

2. Find an orthogonal Q with $Q^TGQ = Q^{-1}GQ$ diagonal.

3. Compare the two diagonal forms with Sylvester's law of inertia, and explain why they are congruent but not similar.

Solution. (1) $q = (x_1 + 2x_2)^2 - 3x_2^2$, so $y_1 = x_1 + 2x_2$, $y_2 = x_2$, that is, $x = Py$ with $P = \left(\begin{smallmatrix} 1 & -2 \\ 0 & 1 \end{smallmatrix}\right)$. Then $GP = \left(\begin{smallmatrix} 1 & 0 \\ 2 & -3 \end{smallmatrix}\right)$, $P^TGP = \operatorname{diag}(1,-3)$, while $P^{-1} = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$ gives $P^{-1}GP = \left(\begin{smallmatrix} 5 & -6 \\ 2 & -3 \end{smallmatrix}\right)$. (2) The eigenvalues of G are 3 and -1, with unit eigenvectors $(1,1)/\sqrt{2}$ and $(1,-1)/\sqrt{2}$; with these as columns, Q is orthogonal and $Q^TGQ = Q^{-1}GQ = \operatorname{diag}(3,-1)$. (3) Both diagonal matrices have one positive and one negative entry, signature $(1,1)$, so they are congruent, as Sylvester's law requires; they are not similar, since their eigenvalues $\{1,-3\}$ and $\{3,-1\}$ differ. The matrix $P^{-1}GP$ of (1), similar to G, has eigenvalues 3 and -1, as it must. □

Exercise 11.12.8. Show that the reduced row echelon form R of a matrix A in Theorem 11.6.4 is unique: it depends only on A, not on the row operations used.

Solution. By Lemma 11.6.2 the columns of R satisfy the same linear relations as the columns of A. Column j of R is a pivot column exactly when it is not a combination of the columns before it (a pivot column e_s is not a combination of earlier columns, which vanish in row s; a non-pivot column has zeros below the row of the last pivot to its left, so it is a combination of the earlier pivot columns $e_1, \dots, e_s$). So the pivot positions are determined by A. A non-pivot column j of R equals $\sum_s c_s e_s$, the sum over the pivot columns e_s before it, and the coefficients c_s are the unique coefficients expressing column j of A as a combination of the pivot columns of A before it. So every column of R is determined by A. □

Exercise 11.12.9. For which $a \in \mathbb{R}$ does the system $x + y + z = 1$, $x + ay + z = 1$, $x + y + az = 2$ have solutions? Find them.

Solution. The matrix of the system has determinant $(a-1)^2$: subtracting the first row from the others gives rows $(0, a-1, 0)$ and $(0, 0, a-1)$. For $a \neq 1$ the solution is unique: the differences of the equations give $(a-1)y = 0$ and $(a-1)z = 1$, so $y = 0$, $z = 1/(a-1)$ and $x = 1 - 1/(a-1) = (a-2)/(a-1)$, as Cramer's rule also gives. For $a = 1$ the matrix has rank 1 and the augmented matrix has rank 2 (the minor $\det\left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ from the first and third rows, first and last columns, is 1), so by Theorem 11.6.5 there is no solution: the equations $x + y + z = 1$ and $x + y + z = 2$ contradict each other. □

Exercise 11.12.10. Compute the rank of

$$A = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & 0 \\ 2 & 1 & 1 & 2 \end{pmatrix}$$

with the bordered minors theorem, and find a basis of its column space.

Solution. The minor $M = \det\left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right) = -1$ in rows 1, 2 and columns 1, 2 is nonzero. Its four bordering minors use row 3 or 4 and column 3 or 4; they all vanish, because row 3 is row 1 minus row 2 and row 4 is row 1 plus row 2, so each bordering 3×3 matrix has dependent rows. So $\operatorname{rank} A = 2$, and the first two columns form a basis of the column space (indeed column 3 is column 1 minus column 2, and column 4 equals column 1). □

Exercise 11.12.11. Show that $\operatorname{rank}(AB) \leq \min(\operatorname{rank} A, \operatorname{rank} B)$ and $\operatorname{rank}(A + B) \leq \operatorname{rank} A + \operatorname{rank} B$ for matrices of compatible sizes, and that $\operatorname{rank}(PAQ) = \operatorname{rank} A$ for invertible P and Q. Deduce that every matrix of rank r can be written as $P\left(\begin{smallmatrix} I_r & 0 \\ 0 & 0 \end{smallmatrix}\right)Q$ with P, Q invertible.

Solution. The image of AB is contained in the image of A, and it is the image under A of the image of B, whose dimension is at most $\operatorname{rank} B$. The image of $A+B$ is contained in $\operatorname{im} A + \operatorname{im} B$. If P and Q are invertible, $\operatorname{rank}(PAQ) \leq \operatorname{rank} A = \operatorname{rank}(P^{-1}(PAQ)Q^{-1}) \leq \operatorname{rank}(PAQ)$. For the normal form, bring A to reduced row echelon form by row operations, then use column operations (multiplication on the right) to move the pivot columns to the front and to clear the remaining entries of the first r rows, which are multiples of the pivot columns $e_1, \ldots, e_r$. □

Chapter 12

Multilinear Algebra

12.1 Introduction

This chapter constructs the tensor, symmetric and exterior algebras of a vector space and studies the structures on them that underlie differential forms: alternating forms and interior products, the Pfaffian of a skew-symmetric matrix, orientations, the Hodge star operator of an inner product space, and the decomposition of the exterior algebra of a real vector space with a complex structure into forms of type (p, q). These are the pointwise versions of the constructions of Chapters 36, 40 and 44.

References are [Lan02, Chapters XVI and XIX] and [Gre78].

Prerequisites. Chapter 8 (Rings and Modules) and Chapter 11 (Linear Algebra).

Throughout, k is a field, vector spaces are over k and $\otimes = \otimes_k$. Tensor products were defined in Section 8.9.

12.2 Tensor products of vector spaces

Proposition 12.2.1. *Let $(v_i)_{i\in I}$ be a basis of V and $(w_j)_{j\in J}$ a basis of W. Then $(v_i \otimes w_j)$ is a basis of $V \otimes W$; in particular $\dim(V \otimes W) = \dim V \cdot \dim W$. For vector spaces $V_1, \dots, V_n, U$, multilinear maps $V_1 \times \cdots \times V_n \to U$ correspond bijectively to linear maps $V_1 \otimes \cdots \otimes V_n \to U$.*

Proof. $V \cong k^{(I)}$ and $W \cong k^{(J)}$, and by Proposition 8.9.3 $k^{(I)} \otimes k^{(J)} \cong \bigoplus_i (k \otimes k^{(J)}) \cong k^{(I\times J)}$, with $e_i \otimes e_j \mapsto e_{(i,j)}$. The second statement follows by induction from the definition of the tensor product and the associativity isomorphism. □

Proposition 12.2.2. *The linear map $V^* \otimes W \to \mathrm{Hom}_k(V, W)$, $\varphi \otimes w \mapsto (v \mapsto \varphi(v)w)$, is injective, with image the maps of finite rank; it is an isomorphism if V or W is finite-dimensional. For finite-dimensional V, W the map $V^* \otimes W^* \to (V \otimes W)^*$, $\varphi \otimes \psi \mapsto (v \otimes w \mapsto \varphi(v)\psi(w))$, is an isomorphism. Under $V^* \otimes V \cong \mathrm{End}(V)$, the* contraction *$\varphi \otimes v \mapsto \varphi(v)$ is the trace.*

Proof. Let (w_j) be a basis of W. An element of $V^* \otimes W$ is uniquely $\sum_j \varphi_j \otimes w_j$ with finitely many nonzero φ_j (Proposition 12.2.1 applied to a basis of V^*); it maps to $v \mapsto \sum_j \varphi_j(v)w_j$, which is zero only if all $\varphi_j = 0$. The image consists of maps with image in a finite-dimensional subspace, and every such map is of this form (choose a basis of W extending a basis of a finite-dimensional subspace containing the image). If V is finite-dimensional, every map has finite rank; if W is, likewise. For the second isomorphism, compare dual bases. Finally, with a basis (e_i) of V, $e_i^* \otimes e_j$ corresponds to the matrix unit E_{ji}, whose trace is $\delta_{ij} = e_i^*(e_j)$. □

Counterexample 12.2.3. Not every element of $V \otimes W$ is of the form $v \otimes w$. Under $k^2 \otimes k^2 \cong M_2(k)$ (sending $e_i \otimes e_j$ to the matrix unit E_{ij}), the element $v \otimes w$ corresponds to a matrix of rank at most 1, while $e_1 \otimes e_1 + e_2 \otimes e_2$ corresponds to the identity matrix, of rank 2.

12.3 The tensor, symmetric and exterior algebras

Definition 12.3.1. The *tensor algebra* of V is the graded k-algebra $T(V) = \bigoplus_{n\geq 0} T^n(V)$, with $T^0(V) = k$, $T^n(V) = V^{\otimes n}$, and multiplication induced by $(v_1 \otimes \cdots \otimes v_p)(w_1 \otimes \cdots \otimes w_q) = v_1 \otimes \cdots \otimes v_p \otimes w_1 \otimes \cdots \otimes w_q$. The *symmetric algebra* is $S(V) = T(V)/I_S$, where I_S is the two-sided ideal generated by the elements $v \otimes w - w \otimes v$, and the *exterior algebra* is $\Lambda(V) = T(V)/I_\Lambda$, where I_Λ is generated by the elements $v \otimes v$. The ideals are homogeneous, so both quotients are graded: $S(V) = \bigoplus_n S^n(V)$ and $\Lambda(V) = \bigoplus_n \Lambda^n(V)$. We write $v_1 \cdots v_n$ for the image of $v_1 \otimes \cdots \otimes v_n$ in $S(V)$ and $v_1 \wedge \cdots \wedge v_n$ for its image in $\Lambda(V)$.

Proposition 12.3.2. *Let A be an associative k-algebra and $f : V \to A$ linear.*

1. *f extends uniquely to an algebra homomorphism $T(V) \to A$.*
2. *If A is commutative, it extends uniquely to $S(V) \to A$.*
3. *If $f(v)^2 = 0$ for all $v \in V$, it extends uniquely to $\Lambda(V) \to A$.*

Moreover, n-linear maps $V^n \to U$ that are symmetric (respectively alternating) correspond to linear maps $S^n(V) \to U$ (respectively $\Lambda^n(V) \to U$).

Proof. (1) The n-linear map $(v_1, \ldots, v_n) \mapsto f(v_1) \cdots f(v_n)$ induces $T^n(V) \to A$, and the sum of these is an algebra homomorphism; it is unique since V generates $T(V)$. (2), (3) The homomorphism of (1) kills the generators of I_S, respectively I_Λ.

For the last statement, the degree-n part of I_Λ is spanned by the tensors $x \otimes v \otimes v \otimes y$ with x, y pure tensors, and similarly for I_S. A multilinear $\beta : V^n \to U$ vanishes on the tensors with two equal adjacent factors if and only if it is alternating: expanding $\beta(\ldots, v+w, v+w, \ldots) = 0$ shows it changes sign under adjacent transpositions, hence under all transpositions up to sign, so it vanishes whenever any two arguments are equal. The symmetric case is analogous. □

Proposition 12.3.3.

1. *In $\Lambda(V)$, $v \wedge w = -w \wedge v$ for $v, w \in V$, and $\alpha \wedge \beta = (-1)^{pq}\beta \wedge \alpha$ for $\alpha \in \Lambda^p(V)$, $\beta \in \Lambda^q(V)$.*
2. *$S(V)$ is commutative. If $(e_1, \ldots, e_n)$ is a basis of V, then $S(V) \cong k[x_1, \ldots, x_n]$ with $e_i \mapsto x_i$, and $S^d(V)$ corresponds to the homogeneous polynomials of degree d.*
3. *If $(e_1, \ldots, e_n)$ is a basis of V, the elements $e_I = e_{i_1} \wedge \cdots \wedge e_{i_p}$, for $I = \{i_1 < \cdots < i_p\} \subseteq \{1, \ldots, n\}$, form a basis of $\Lambda^p(V)$. So $\dim \Lambda^p(V) = \binom{n}{p}$, $\Lambda^p(V) = 0$ for $p > n$, and $\dim \Lambda(V) = 2^n$.*
4. *Vectors $v_1, \ldots, v_p$ are linearly independent if and only if $v_1 \wedge \cdots \wedge v_p \neq 0$.*
5. *For $f : V \to W$ linear there are induced maps $\Lambda^p f : \Lambda^p V \to \Lambda^p W$, $v_1 \wedge \cdots \wedge v_p \mapsto f(v_1) \wedge \cdots \wedge f(v_p)$, and if $\dim V = n$, $\Lambda^n f$ is multiplication by $\det f$ for $f : V \to V$.*

Proof. (1) $0 = (v+w)\wedge(v+w) = v \wedge w + w \wedge v$. Moving each of the q factors of a decomposable β past the p factors of α gives pq sign changes.

(2) Commutativity holds on generators. The two universal properties (Proposition 12.3.2(2) and Proposition 8.6.2) give inverse homomorphisms.

(3) By (1) and $e_i \wedge e_i = 0$, the e_I span $\Lambda^p(V)$. For independence, fix I and consider the map $V^p \to k$, $(v_1, \ldots, v_p) \mapsto \det(e_{i_a}^*(v_b))_{a,b}$. It is multilinear and alternating by Theorem 11.5.2, so it induces $\varphi_I : \Lambda^p(V) \to k$, and $\varphi_I(e_J) = \delta_{IJ}$. A relation $\sum_J c_J e_J = 0$ gives $c_I = 0$ after applying φ_I.

(4) If $v_1 = \sum_{i\geq 2} c_i v_i$, the product vanishes by (1) and $v_i \wedge v_i = 0$. If they are independent, extend them to a basis and use (3).

(5) The map exists by the universal property. For $V = W$ of dimension n with basis (e_i), $(v_1, \dots, v_n) \mapsto$ the coefficient of $f(v_1) \wedge \cdots \wedge f(v_n)$ on $e_1 \wedge \cdots \wedge e_n$ is multilinear alternating; by Theorem 11.5.2 applied to the columns of the matrix of f, $f(e_1) \wedge \cdots \wedge f(e_n) = \det(f)\, e_1 \wedge \cdots \wedge e_n$. □

Proposition 12.3.4. *Let $f : V \to W$ be linear with matrix $A = (a_{ij})$ in bases $(e_1, \dots, e_m)$ of V and $(e'_1, \dots, e'_n)$ of W. For subsets $I \subseteq \{1, \dots, n\}$, $J \subseteq \{1, \dots, m\}$ of size p, let $A_{I,J}$ be the $p \times p$ submatrix with rows I and columns J.*

1. *The matrix of $\Lambda^p f$ in the bases (e_J) and (e'_I) is $(\det A_{I,J})_{I,J}$: $\Lambda^p f(e_J) = \sum_I \det(A_{I,J})\, e'_I$.*
2. *(Cauchy–Binet) For an $n \times m$ matrix A and an $m \times n$ matrix B, $\det(AB) = \sum_S \det A_{\{1..n\},S} \det B_{S,\{1..n\}}$, the sum over the subsets $S \subseteq \{1, \dots, m\}$ of size n; in particular $\det(AB) = 0$ if $m < n$, and $\det(AB) = \det A \det B$ if $m = n$.*
3. *The rank of f is the largest p with $\Lambda^p f \neq 0$, that is, the largest size of a nonzero minor of A.*

Proof. (1) $\Lambda^p f(e_J) = f(e_{j_1}) \wedge \cdots \wedge f(e_{j_p})$, and its coefficient on e'_I is obtained by applying the functional φ_I of the proof of Proposition 12.3.3(3), which gives $\det(e'^*_{i_a}(f(e_{j_b})))_{a,b} = \det(a_{i_a j_b}) = \det A_{I,J}$.

(2) By functoriality $\Lambda^n(AB) = \Lambda^n A \circ \Lambda^n B$ on $\Lambda^n(k^n)$, which is one-dimensional with basis $e_{\{1..n\}}$, and $\Lambda^n(AB)$ is multiplication by $\det(AB)$ (Proposition 12.3.3(5)). By (1), $\Lambda^n B(e_{\{1..n\}}) = \sum_S \det B_{S,\{1..n\}} e_S$ and $\Lambda^n A(e_S) = \det A_{\{1..n\},S}\, e_{\{1..n\}}$. If $m < n$ there are no such S.

(3) The image of $\Lambda^p f$ is spanned by the products $f(v_1) \wedge \cdots \wedge f(v_p)$, so it is the image of $\Lambda^p(\operatorname{im} f) \to \Lambda^p W$, which is injective because a basis of $\operatorname{im} f$ extends to a basis of W and Proposition 12.3.3(3) applies to both. So $\Lambda^p f \neq 0$ if and only if $\Lambda^p(\operatorname{im} f) \neq 0$, that is, $p \leq \operatorname{rank} f$. Combine with (1). □

Counterexample 12.3.5. In characteristic 2 symmetric and alternating notions separate. Let $k = \mathbb{F}_2$ and $V = k$. The bilinear form $B(x, y) = xy$ satisfies $B(y, x) = B(x, y) = -B(x, y)$, so it is both symmetric and antisymmetric, but it is not alternating, since $B(1, 1) = 1$; so it does not factor through $\Lambda^2 V = 0$. Similarly the symmetrisation $S^2 V \to V \otimes V$, $vw \mapsto v \otimes w + w \otimes v$, sends the generator e^2 of $S^2 V$ to $2\, e \otimes e = 0$, so it is not injective. Factorials cause the same phenomena elsewhere: in Proposition 12.4.7 the characteristic enters through the factor $m!$ in $\omega^m = m!\, e^*_1 \wedge f^*_1 \wedge \cdots \wedge e^*_m \wedge f^*_m$, which vanishes in characteristic $p \leq m$.

12.4 Alternating forms

In this section V has finite dimension n.

Proposition 12.4.1. *The pairing $\Lambda^p(V^*) \times \Lambda^p(V) \to k$ determined by $\langle \varphi_1 \wedge \cdots \wedge \varphi_p, v_1 \wedge \cdots \wedge v_p \rangle = \det(\varphi_i(v_j))$ is well defined and perfect. Hence $\Lambda^p(V^*) \cong \Lambda^p(V)^*$, which by Proposition 12.3.2 is the space $\operatorname{Alt}^p(V)$ of alternating p-linear forms on V.*

Proof. The determinant is multilinear and alternating both in the φ_i and in the v_j (using Proposition 11.5.3(2)), so it induces a bilinear pairing. For a basis (e_i) with dual basis (e^*_i), $\langle e^*_I, e_J \rangle = \delta_{IJ}$, so the pairing is perfect. □

Notation 12.4.2. We identify $\Lambda^p(V^*)$ with $\operatorname{Alt}^p(V)$ via Proposition 12.4.1: an element $\alpha \in \Lambda^p(V^*)$ is evaluated on vectors as $\alpha(v_1, \dots, v_p) = \langle \alpha, v_1 \wedge \cdots \wedge v_p \rangle$. With this convention, $(\varphi_1 \wedge \cdots \wedge \varphi_p)(v_1, \dots, v_p) = \det(\varphi_i(v_j))$, without a factor $1/p!$.

Example 12.4.3. *The three descriptions of an alternating form, explicitly.* Let $e_1, \dots, e_n$ be a basis of V with dual basis $e_1^*, \dots, e_n^*$, and for $I = \{i_1 < \cdots < i_p\}$ write $e_I = e_{i_1} \wedge \cdots \wedge e_{i_p}$ and $e_I^* = e_{i_1}^* \wedge \cdots \wedge e_{i_p}^*$.

1. *From $\Lambda^p(V^*)$ to $\mathrm{Alt}^p(V)$.* An element $\alpha = \sum_I a_I e_I^*$ acts on vectors $v_j = \sum_i v_{ij} e_i$ by $\alpha(v_1, \dots, v_p) = \sum_I a_I \det(v_{i_k j})_{k,j}$, the sum of the coefficients times the $p \times p$ minors, with rows I, of the matrix of coordinates of the v_j.

2. *From $\mathrm{Alt}^p(V)$ to $\Lambda^p(V^*)$.* An alternating p-linear map A is $\sum_I A(e_{i_1}, \dots, e_{i_p})\, e_I^*$: the coefficients are the values on increasing tuples of basis vectors. Summing over all tuples instead, $A = \frac{1}{p!} \sum_{i_1,\dots,i_p} A(e_{i_1}, \dots, e_{i_p})\, e_{i_1}^* \wedge \cdots \wedge e_{i_p}^*$, since each increasing tuple appears $p!$ times, with the same sign in A and in the wedge product.

3. *As a linear form on $\Lambda^p V$.* α sends e_J to a_J, by Proposition 12.4.1; so (e_I^*) is the dual basis of (e_J).

For instance, on $\mathbb{R}^3$ let $\alpha = e_1^* \wedge e_2^* + 2e_2^* \wedge e_3^*$. As a bilinear map, $\alpha(v, w) = (v_1 w_2 - v_2 w_1) + 2(v_2 w_3 - v_3 w_2) = v^T A w$ with the skew-symmetric matrix A having $A_{12} = 1$, $A_{23} = 2$, $A_{13} = 0$; as a linear form on $\Lambda^2\mathbb{R}^3$ it sends $e_1 \wedge e_2 \mapsto 1$, $e_2 \wedge e_3 \mapsto 2$, $e_1 \wedge e_3 \mapsto 0$; and under the identification $\Lambda^2\mathbb{R}^3 \cong \mathbb{R}^3$ of the cross product ($e_2 \wedge e_3 \mapsto e_1$, $e_3 \wedge e_1 \mapsto e_2$, $e_1 \wedge e_2 \mapsto e_3$) it is the vector $(2, 0, 1)$, with $\alpha(v, w) = \langle (2, 0, 1), v \times w \rangle$. For differential forms the same dictionary holds at each point with $e_i^* = dx^i$ and $e_i = \partial_i$: $\omega = \sum_I \omega_I\, dx^I$ with $\omega_I = \omega(\partial_{i_1}, \dots, \partial_{i_p})$.

Conventions. Some books identify $\Lambda^p(V^*)$ with $\mathrm{Alt}^p(V)$ through the *alternation* map $\mathrm{Alt}(T)(v_1, \dots, v_p) = \frac{1}{p!} \sum_\sigma \mathrm{sgn}(\sigma) T(v_{\sigma(1)}, \dots, v_{\sigma(p)})$, which gives $(\varphi_1 \wedge \cdots \wedge \varphi_p)(v_1, \dots, v_p) = \frac{1}{p!} \det(\varphi_i(v_j))$. In our convention $\varphi_1 \wedge \cdots \wedge \varphi_p = p!\, \mathrm{Alt}(\varphi_1 \otimes \cdots \otimes \varphi_p)$, with no factor in the evaluation; for $p = 2$, $(\varphi \wedge \psi)(v, w) = \varphi(v)\psi(w) - \varphi(w)\psi(v)$ for us, and half of it in the other convention. Formulas such as the invariant formula for d (Chapter 36) change by such factors between books.

Proposition 12.4.4. *For $\alpha \in \Lambda^p(V^*)$ and $\beta \in \Lambda^q(V^*)$,*

$$(\alpha \wedge \beta)(v_1, \dots, v_{p+q}) = \sum_\sigma \mathrm{sgn}(\sigma)\, \alpha(v_{\sigma(1)}, \dots, v_{\sigma(p)})\, \beta(v_{\sigma(p+1)}, \dots, v_{\sigma(p+q)}),$$

the sum running over the shuffles*: permutations $\sigma \in S_{p+q}$ with $\sigma(1) < \cdots < \sigma(p)$ and $\sigma(p+1) < \cdots < \sigma(p+q)$.*

Proof. Both sides are bilinear in (α, β), so we may take $\alpha = \varphi_1 \wedge \cdots \wedge \varphi_p$ and $\beta = \varphi_{p+1} \wedge \cdots \wedge \varphi_{p+q}$. The left side is $\det M$ with $M = (\varphi_i(v_j))$. In the Leibniz formula $\det M = \sum_\pi \mathrm{sgn}(\pi) \prod_j M_{\pi(j)j}$, group the permutations π according to the set $A = \{j : \pi(j) \le p\}$ of columns sent to the first p rows. For fixed $A = \{a_1 < \cdots < a_p\}$ with complement $\{b_1 < \cdots < b_q\}$, let σ be the shuffle with $\sigma(i) = a_i$ and $\sigma(p + i) = b_i$. Every such π is uniquely $\pi = (\pi_1 \times \pi_2) \circ \sigma^{-1}$ with $\pi_1 \in S_p$ acting on $\{1, \dots, p\}$ and π_2 on $\{p + 1, \dots, p + q\}$, and $\mathrm{sgn}\, \pi = \mathrm{sgn}\, \pi_1\, \mathrm{sgn}\, \pi_2\, \mathrm{sgn}\, \sigma$. Summing over π_1 and π_2 gives $\mathrm{sgn}(\sigma) \det(\varphi_i(v_{a_j})) \det(\varphi_{p+i}(v_{b_j}))$, which is the term of σ on the right. □

Definition 12.4.5. For $v \in V$ and $\alpha \in \Lambda^p(V^*)$ with $p \ge 1$, the *interior product* $\iota_v \alpha \in \Lambda^{p-1}(V^*)$ is $(\iota_v \alpha)(v_2, \dots, v_p) = \alpha(v, v_2, \dots, v_p)$; for $p = 0$, $\iota_v \alpha = 0$.

Proposition 12.4.6. $\iota_v(\alpha \wedge \beta) = \iota_v \alpha \wedge \beta + (-1)^p \alpha \wedge \iota_v \beta$ *for $\alpha \in \Lambda^p(V^*)$, and $\iota_v \iota_w = -\iota_w \iota_v$.*

Proof. Expanding the determinant along the first column (Laplace expansion) gives $\iota_v(\varphi_1 \wedge \cdots \wedge \varphi_p) = \sum_{i=1}^p (-1)^{i-1} \varphi_i(v)\, \varphi_1 \wedge \cdots \widehat{\varphi_i} \cdots \wedge \varphi_p$. Applying this formula to a product of decomposable forms and splitting the sum according to whether i is among the first p factors gives the first identity. The second holds because $\alpha(w, v, \dots) = -\alpha(v, w, \dots)$. □

Proposition 12.4.7. *Suppose* $\operatorname{char} k = 0$*, and let* $\omega \in \Lambda^2(V^*)$ *be nondegenerate,* $\dim V = 2m$*. Then* $\omega^m = \omega \wedge \cdots \wedge \omega \neq 0$*, and every* $f \in \mathrm{GL}(V)$ *with* $\omega(fv, fw) = \omega(v, w)$ *has* $\det f = 1$*.*

Proof. With a symplectic basis (Theorem 11.9.6) and its dual basis, $\omega = \sum_i e_i^* \wedge f_i^*$, as both sides take the same values on pairs of basis vectors. The 2-forms $e_i^* \wedge f_i^*$ commute and square to zero, so $\omega^m = m!\, e_1^* \wedge f_1^* \wedge \cdots \wedge e_m^* \wedge f_m^* \neq 0$. The map $f^*\colon V^* \to V^*$, $\varphi \mapsto \varphi \circ f$, satisfies $(\Lambda^2 f^*)(\omega) = \omega$, hence $(\Lambda^{2m} f^*)(\omega^m) = \omega^m$; but $\Lambda^{2m} f^*$ is multiplication by $\det f^* = \det f$ (Proposition 12.3.3(5) and Proposition 11.4.2(3)). □

The same computation, applied to an arbitrary alternating form instead of a nondegenerate one, produces a square root of the determinant of a skew-symmetric matrix. In the rest of this section k is a field of characteristic 0, $V = k^{2m}$ with standard basis (e_i) and dual basis (e_i^*). To a skew-symmetric matrix $A \in M_{2m}(k)$, that is $A^T = -A$, attach the 2-form

$$\alpha_A = \sum_{i<j} A_{ij}\, e_i^* \wedge e_j^* = \frac{1}{2} \sum_{i,j} A_{ij}\, e_i^* \wedge e_j^* \in \Lambda^2(V^*).$$

The two expressions agree because $A_{ii} = 0$ and $A_{ji}\, e_j^* \wedge e_i^* = A_{ij}\, e_i^* \wedge e_j^*$. As an alternating form (Notation 12.4.2), $\alpha_A(v, w) = \sum_{i<j} A_{ij}(v_i w_j - v_j w_i) = \sum_{i,j} A_{ij} v_i w_j = v^T A w$. Since $\Lambda^{2m}(V^*)$ is one-dimensional, spanned by $e_1^* \wedge \cdots \wedge e_{2m}^*$ (Proposition 12.3.3(3)), the following definition makes sense.

Definition 12.4.8. The *Pfaffian* of a skew-symmetric matrix $A \in M_{2m}(k)$ is the scalar $\operatorname{Pf}(A)$ with

$$\alpha_A^m = \alpha_A \wedge \cdots \wedge \alpha_A = m!\operatorname{Pf}(A)\, e_1^* \wedge \cdots \wedge e_{2m}^*.$$

Proposition 12.4.9. *Let* $A \in M_{2m}(k)$ *be skew-symmetric.*

1. $\operatorname{Pf}(A) = \dfrac{1}{2^m m!} \displaystyle\sum_{\sigma \in S_{2m}} \operatorname{sgn}(\sigma)\, A_{\sigma(1)\sigma(2)} A_{\sigma(3)\sigma(4)} \cdots A_{\sigma(2m-1)\sigma(2m)}$*. So* Pf *is a homogeneous polynomial of degree* m *in the entries, with* $\operatorname{Pf}(\lambda A) = \lambda^m \operatorname{Pf}(A)$*; for instance*

$$\operatorname{Pf}\begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix} = a, \qquad \operatorname{Pf}(A) = A_{12}A_{34} - A_{13}A_{24} + A_{14}A_{23} \quad (m = 2).$$

2. $\operatorname{Pf}(g^T A g) = \det(g) \operatorname{Pf}(A)$ *for every* $g \in M_{2m}(k)$*. In particular* Pf *is invariant under conjugation by* $\mathrm{SO}(2m)$*, and changes sign under conjugation by an orthogonal matrix of determinant* -1*.*

3. *If* A *is block diagonal with blocks* $\begin{pmatrix} 0 & a_i \\ -a_i & 0 \end{pmatrix}$, $i = 1, \ldots, m$*, then* $\operatorname{Pf}(A) = a_1 a_2 \cdots a_m$*.*

4. $\operatorname{Pf}(A)^2 = \det A$*.*

Proof. (1) Expanding the m-th power of $\alpha_A = \frac{1}{2}\sum_{i,j} A_{ij} e_i^* \wedge e_j^*$ gives

$$\alpha_A^m = \frac{1}{2^m} \sum_{i_1, j_1, \ldots, i_m, j_m} A_{i_1 j_1} \cdots A_{i_m j_m}\, e_{i_1}^* \wedge e_{j_1}^* \wedge \cdots \wedge e_{i_m}^* \wedge e_{j_m}^*.$$

A wedge product of 1-forms with a repeated factor vanishes, so only the terms in which $(i_1, j_1, \ldots, i_m, j_m) = (\sigma(1), \ldots, \sigma(2m))$ for a permutation σ survive, and for them $e_{\sigma(1)}^* \wedge \cdots \wedge e_{\sigma(2m)}^* = \operatorname{sgn}(\sigma)\, e_1^* \wedge \cdots \wedge e_{2m}^*$ by Proposition 12.3.3(1). Comparing with the definition gives the formula. For $m = 1$ the two permutations give $\frac{1}{2}(A_{12} - A_{21}) = A_{12}$. In general the $2^m m!$

permutations that differ by reordering the m pairs (an even permutation of positions) and by swapping the two entries of some pairs (which changes both $\operatorname{sgn}\sigma$ and the sign of one factor) give the same term, so $\operatorname{Pf}(A)$ is the sum over the ways of splitting $\{1,\dots,2m\}$ into m pairs $\{a_l < b_l\}$ of $\pm\prod_l A_{a_l b_l}$, the sign being that of the permutation $(a_1, b_1, \dots, a_m, b_m)$. For $m = 2$ the three splittings $\{12,34\}$, $\{13,24\}$, $\{14,23\}$ give the permutations 1234, 1324, 1423, of signs $+1, -1, +1$.

(2) Let $g^*\colon V^* \to V^*$, $\varphi \mapsto \varphi \circ g$, where g acts on $V = k^{2m}$ by matrix multiplication. Then $g^* e_k^* = \sum_i g_{ki} e_i^*$, because $e_k^*(g e_i) = g_{ki}$. The induced map Λg^* on $\Lambda(V^*)$ is multiplicative, and

$$(\Lambda^2 g^*)(\alpha_A) = \frac{1}{2}\sum_{k,l} A_{kl}\Big(\sum_i g_{ki} e_i^*\Big) \wedge \Big(\sum_j g_{lj} e_j^*\Big) = \frac{1}{2}\sum_{i,j} (g^T A g)_{ij}\, e_i^* \wedge e_j^* = \alpha_{g^T A g},$$

$g^T A g$ being again skew-symmetric. Taking m-th powers, $\alpha^m_{g^T A g} = (\Lambda^{2m} g^*)(\alpha_A^m) = m!\operatorname{Pf}(A)\det(g)\, e_1^* \wedge \cdots \wedge e_{2m}^*$, since $\Lambda^{2m} g^*$ is multiplication by $\det g^* = \det g$ (Proposition 12.3.3(5) and Proposition 11.4.2(3)). For g orthogonal, $g^T A g = g^{-1} A g$.

(3) Here $\alpha_A = \sum_i a_i\, e_{2i-1}^* \wedge e_{2i}^*$. The 2-forms $e_{2i-1}^* \wedge e_{2i}^*$ commute and square to zero, so $\alpha_A^m = m!\, a_1 \cdots a_m\, e_1^* \wedge \cdots \wedge e_{2m}^*$, as in the proof of Proposition 12.4.7.

(4) Suppose first that A is invertible. Then $\omega(v,w) = v^T A w$ is a nondegenerate alternating form, and by Theorem 11.9.6 it has a symplectic basis $e_1', \dots, e_m', f_1', \dots, f_m'$. Let g be the matrix with columns $e_1', f_1', e_2', f_2', \dots, e_m', f_m'$ (in this order). The entries of $g^T A g$ are the values of ω on pairs of these vectors, so $g^T A g = J$ is block diagonal with blocks $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. By (3), $\operatorname{Pf}(J) = 1$, and $\det J = 1$. By (2), $\det(g)\operatorname{Pf}(A) = 1$, while $\det(g)^2 \det(A) = \det J = 1$; so $\operatorname{Pf}(A)^2 = \det(g)^{-2} = \det A$. If A is not invertible, choose $v \neq 0$ with $Av = 0$ and an invertible g with first column v. The first column of $g^T A g$ is $g^T A v = 0$, and by skew-symmetry so is its first row; hence $\alpha_{g^T A g}$ is a 2-form in $e_2^*, \dots, e_{2m}^*$ only, and its m-th power, of degree $2m$ in $2m-1$ linear forms, vanishes. So $\det(g)\operatorname{Pf}(A) = \operatorname{Pf}(g^T A g) = 0$, and $\operatorname{Pf}(A) = 0 = \det A$. □

Since Pf is a polynomial with rational coefficients, the identity in (2) is a polynomial identity in the entries of A and g; so it remains true when the entries of A are taken in any commutative $\mathbb{Q}$-algebra, for instance the algebra of differential forms of even degree on a manifold. This is how the Pfaffian is applied to curvature matrices in Chapter 39, where it produces the Euler form.

Example 12.4.10. For the 4×4 skew-symmetric matrix with $A_{12} = 1$, $A_{13} = 2$, $A_{14} = 3$, $A_{23} = 4$, $A_{24} = 5$, $A_{34} = 6$, the formula gives $\operatorname{Pf}(A) = 1\cdot6 - 2\cdot5 + 3\cdot4 = 8$, and indeed $\det A = 64$. A real skew-symmetric matrix has $\det A = \operatorname{Pf}(A)^2 \geq 0$; this explains why the determinant of a skew-symmetric matrix of even size is a perfect square, while one of odd size is 0 (as $\det A = \det A^T = \det(-A) = (-1)^n \det A$).

12.5 Orientations and the Hodge star

Definition 12.5.1. Let V be a real vector space of dimension $n \geq 1$. Two ordered bases are *equally oriented* if the change of basis matrix has positive determinant. This is an equivalence relation with two classes, the *orientations* of V; equivalently, an orientation is one of the two components $\{c\, e_1 \wedge \cdots \wedge e_n : c > 0\}$, $\{c\, e_1 \wedge \cdots \wedge e_n : c < 0\}$ of $\Lambda^n V \setminus \{0\}$.

Proposition 12.5.2. *Let (V, g) be a real inner product space of dimension n. There is a unique inner product on $\Lambda^p(V)$ with $\langle v_1 \wedge \cdots \wedge v_p, w_1 \wedge \cdots \wedge w_p\rangle = \det(g(v_i, w_j))$. If (e_i) is orthonormal, the e_I are orthonormal. If V is oriented, the element* $\operatorname{vol} = e_1 \wedge \cdots \wedge e_n$ *does not depend on the choice of an oriented orthonormal basis; it is the* volume element.

Proof. The formula is multilinear and alternating in the v_i and in the w_j, so it induces a bilinear form, which is symmetric by Proposition 11.5.3(2). On an orthonormal basis $\langle e_I, e_J\rangle = \delta_{IJ}$, so the form is positive definite. If (f_i) is another oriented orthonormal basis, $f_j = \sum_i a_{ij}e_i$ with $A \in \mathrm{O}(n)$ and $\det A > 0$, hence $\det A = 1$, and $f_1 \wedge \cdots \wedge f_n = \det(A)\, e_1 \wedge \cdots \wedge e_n$. □

Theorem 12.5.3. *Let (V, g) be an oriented real inner product space of dimension n. There is a unique linear map $*\colon \Lambda^p(V) \to \Lambda^{n-p}(V)$ with*

$$\alpha \wedge *\beta = \langle \alpha, \beta\rangle \,\mathrm{vol} \qquad \text{for all } \alpha, \beta \in \Lambda^p(V).$$

For an oriented orthonormal basis, $*e_I = \varepsilon_I e_{I^c}$ where $e_I \wedge e_{I^c} = \varepsilon_I \,\mathrm{vol}$, $\varepsilon_I = \pm 1$. The map $*$ is an isometry, $** = (-1)^{p(n-p)}$ on $\Lambda^p(V)$, and $\alpha \wedge *\beta = \beta \wedge *\alpha$.

Proof. The pairing $\Lambda^p(V) \times \Lambda^{n-p}(V) \to \Lambda^n(V) = \mathbb{R}\,\mathrm{vol}$ is perfect, since $e_I \wedge e_J$ is $\pm$vol if $J = I^c$ and 0 otherwise. Hence for each β there is a unique $*\beta$ representing the functional $\alpha \mapsto \langle \alpha, \beta\rangle$, and it depends linearly on β. The formula: $e_J \wedge \varepsilon_I e_{I^c}$ is $\varepsilon_I^2\,\mathrm{vol} = \mathrm{vol}$ if $J = I$ and 0 otherwise, which is $\langle e_J, e_I\rangle$vol. So $*$ maps the orthonormal basis (e_I) to $(\pm e_{I^c})$, hence is an isometry. Next, $** e_I = \varepsilon_I \varepsilon_{I^c} e_I$, and $e_{I^c} \wedge e_I = (-1)^{p(n-p)} e_I \wedge e_{I^c}$ by Proposition 12.3.3(1), so $\varepsilon_{I^c} = (-1)^{p(n-p)}\varepsilon_I$. The last identity follows from the symmetry of the inner product. □

Example 12.5.4. On $\mathbb{R}^3$ with the standard orientation, $*e_1 = e_2 \wedge e_3$, $*e_2 = e_3 \wedge e_1$, $*e_3 = e_1 \wedge e_2$, $*1 = \mathrm{vol}$ and $*\mathrm{vol} = 1$. The map $(v, w) \mapsto *(v \wedge w)$ is the cross product. On $\mathbb{R}^4$, $*$ is an involution of Λ^2 (since $(-1)^{2\cdot 2} = 1$), which splits into ± 1-eigenspaces of dimension 3: the self-dual and anti-self-dual 2-vectors.

12.6 Forms of type (p, q)

Let V be a real vector space of dimension $2n$ with a complex structure J and $W = V^* = \mathrm{Hom}_{\mathbb{R}}(V, \mathbb{R})$. Then $W_{\mathbb{C}} = W \otimes_{\mathbb{R}} \mathbb{C}$ is identified with $\mathrm{Hom}_{\mathbb{R}}(V, \mathbb{C})$, and $\Lambda^k(W_{\mathbb{C}})$ (exterior power over $\mathbb{C}$) with $\Lambda^k(W) \otimes_{\mathbb{R}} \mathbb{C}$, the complex-valued alternating k-forms on V.

Definition 12.6.1. Let $W^{1,0} \subseteq W_{\mathbb{C}}$ be the space of $\mathbb{R}$-linear maps $\varphi\colon V \to \mathbb{C}$ with $\varphi(Jv) = i\varphi(v)$, that is, the $\mathbb{C}$-linear maps $(V, J) \to \mathbb{C}$, and $W^{0,1} = \overline{W^{1,0}}$, the maps with $\varphi(Jv) = -i\varphi(v)$. For $p, q \geq 0$ let $\Lambda^{p,q} \subseteq \Lambda^{p+q}(W_{\mathbb{C}})$ be the span of the products $\alpha \wedge \beta$ with $\alpha \in \Lambda^p W^{1,0}$ and $\beta \in \Lambda^q W^{0,1}$. Its elements are the forms of *type* (p, q).

Proposition 12.6.2. *1. $W_{\mathbb{C}} = W^{1,0} \oplus W^{0,1}$, and both summands have complex dimension n.*

2. $\Lambda^k(W_{\mathbb{C}}) = \bigoplus_{p+q=k} \Lambda^{p,q}$, with $\dim_{\mathbb{C}} \Lambda^{p,q} = \binom{n}{p}\binom{n}{q}$, and $\overline{\Lambda^{p,q}} = \Lambda^{q,p}$.

3. A real 2-form $\alpha \in \Lambda^2(W)$ lies in $\Lambda^{1,1}$ if and only if $\alpha(Jv, Jw) = \alpha(v, w)$ for all v, w.

Proof. (1) The map $J^*\colon W \to W$, $\varphi \mapsto \varphi \circ J$, is a complex structure on W, and $W^{1,0}$, $W^{0,1}$ are the $\pm i$-eigenspaces of its complexification; apply Proposition 11.11.2. (2) Choose a basis $\zeta_1, \ldots, \zeta_n$ of $W^{1,0}$; then $\bar\zeta_1, \ldots, \bar\zeta_n$ is a basis of $W^{0,1}$ and together they form a basis of $W_{\mathbb{C}}$. By Proposition 12.3.3(3) the products $\zeta_I \wedge \bar\zeta_{I'}$ with $|I| + |I'| = k$ form a basis of $\Lambda^k(W_{\mathbb{C}})$ (after reordering factors, which only changes signs), and those with $|I| = p$, $|I'| = q$ span $\Lambda^{p,q}$. Complex conjugation, extended to forms, maps $\zeta_I \wedge \bar\zeta_{I'}$ to $\bar\zeta_I \wedge \zeta_{I'}$. (3) The map $\alpha \mapsto J^*\alpha$, $(J^*\alpha)(v, w) = \alpha(Jv, Jw)$, equals Λ^2 of J^* on $W_{\mathbb{C}}$. On $\Lambda^{2,0}$ it acts by $i \cdot i = -1$, on $\Lambda^{0,2}$ by $(-i)(-i) = -1$, and on $\Lambda^{1,1}$ by $i(-i) = 1$. So $\Lambda^{1,1}$ is exactly the fixed space of J^* on $\Lambda^2(W_{\mathbb{C}})$. □

Example 12.6.3. On $\mathbb{R}^{2n} = \mathbb{C}^n$ with the standard structure (Example 11.11.4) let dx_j, dy_j be the dual basis and $dz_j = dx_j + i\,dy_j$, $d\bar{z}_j = dx_j - i\,dy_j$. Then $dz_j \in W^{1,0}$, since $dz_j(J\partial_{x_j}) = dz_j(\partial_{y_j}) = i = i\,dz_j(\partial_{x_j})$ and similarly on ∂_{y_j}. The fundamental form of the standard inner product is

$$\omega = \sum_j dx_j \wedge dy_j = \frac{i}{2}\sum_j dz_j \wedge d\bar{z}_j,$$

because $dz_j \wedge d\bar{z}_j = -2i\,dx_j \wedge dy_j$. It is a real form of type $(1,1)$, in accordance with Proposition 12.6.2(3) and Proposition 11.11.3.

12.7 Exercises

Exercise 12.7.1. Show that $\dim S^d(k^n) = \binom{n+d-1}{d}$.

Exercise 12.7.2. Let $v \in V$ be nonzero and $\alpha \in \Lambda^p(V)$. Show that $v \wedge \alpha = 0$ if and only if $\alpha = v \wedge \beta$ for some $\beta \in \Lambda^{p-1}(V)$.

Exercise 12.7.3. Let $\operatorname{char} k \neq 2$. Show that $\omega \in \Lambda^2(V)$ is decomposable, $\omega = v \wedge w$, if and only if $\omega \wedge \omega = 0$.

Exercise 12.7.4. In the situation of Theorem 12.5.3, show that $\alpha \wedge *\alpha = \|\alpha\|^2$ vol, and compute $*$ on $\Lambda^1(\mathbb{R}^2)$: it is the rotation by 90 degrees.

Solution. The first identity is the definition with $\beta = \alpha$. On $\mathbb{R}^2$, $*e_1 = e_2$ and $*e_2 = -e_1$, since $e_1 \wedge e_2 = \text{vol}$ and $e_2 \wedge (-e_1) = \text{vol}$. This is the standard complex structure J. □

Exercise 12.7.5. Let $V = \mathbb{C}^n$. Show that the real (k,k)-forms, that is, $\alpha \in \Lambda^{k,k}$ with $\bar{\alpha} = \alpha$, form a real vector space of dimension $\binom{n}{k}^2$.

Exercise 12.7.6. With notation as in Section 12.6, let C act on $\Lambda^{p,q}$ as multiplication by i^{p-q}. Show that C commutes with complex conjugation, so preserves real forms, and that C is the map induced on $\Lambda(W_{\mathbb{C}})$ by J^*.

Exercise 12.7.7. Let $X \in M_m(k)$, $\operatorname{char} k = 0$, and $A = \begin{pmatrix} 0 & X \\ -X^T & 0 \end{pmatrix} \in M_{2m}(k)$. Show that $\operatorname{Pf}(A) = (-1)^{m(m-1)/2} \det X$, and check the case $m = 2$ against the formula of Proposition 12.4.9(1). Show also that Pf of a block diagonal matrix with skew-symmetric blocks A_1, A_2 of even sizes is $\operatorname{Pf}(A_1)\operatorname{Pf}(A_2)$.

Solution. Here $\alpha_A = \sum_{i,j} X_{ij}\, e_i^* \wedge e_{m+j}^* = \sum_{i=1}^m e_i^* \wedge \varphi_i$ with $\varphi_i = \sum_j X_{ij} e_{m+j}^*$. The 2-forms $e_i^* \wedge \varphi_i$ commute and square to zero, so $\alpha_A^m = m!\,(e_1^* \wedge \varphi_1) \wedge \cdots \wedge (e_m^* \wedge \varphi_m)$. Moving all the e_i^* to the front, φ_j must pass $e_{j+1}^*, \ldots, e_m^*$, which gives the sign $(-1)^{(m-1)+(m-2)+\cdots+0} = (-1)^{m(m-1)/2}$; and $\varphi_1 \wedge \cdots \wedge \varphi_m = \det(X)\, e_{m+1}^* \wedge \cdots \wedge e_{2m}^*$ by Proposition 12.3.3(5). So $\alpha_A^m = m!\,(-1)^{m(m-1)/2} \det(X)\, e_1^* \wedge \cdots \wedge e_{2m}^*$. For $m = 2$, $A_{12} = A_{34} = 0$, $A_{13} = X_{11}$, $A_{24} = X_{22}$, $A_{14} = X_{12}$, $A_{23} = X_{21}$, and the formula gives $-X_{11}X_{22} + X_{12}X_{21} = -\det X$, in accordance with $(-1)^1 = -1$. For the block diagonal matrix, of sizes $2m_1$ and $2m_2$, $\alpha_A = \alpha_{A_1} + \alpha_{A_2}$ with α_{A_1}, α_{A_2} commuting 2-forms in disjoint sets of variables. In $(\alpha_{A_1} + \alpha_{A_2})^{m_1+m_2}$ only the term $\binom{m_1+m_2}{m_1}\alpha_{A_1}^{m_1}\alpha_{A_2}^{m_2}$ survives, since a power of α_{A_i} above m_i vanishes, and it equals $\frac{(m_1+m_2)!}{m_1!m_2!}\, m_1!\operatorname{Pf}(A_1)\, m_2!\operatorname{Pf}(A_2)\, e_1^* \wedge \cdots \wedge e_{2m}^*$. □

Chapter 13

Representations of Groups and Lie Algebras

13.1 Introduction

A representation of a group or a Lie algebra is an action by linear maps. This chapter develops the basic theory: Schur's lemma, Maschke's theorem and character theory for finite groups, nilpotent and solvable Lie algebras with the theorems of Engel and Lie, and the complete classification of the finite-dimensional representations of the Lie algebra $\mathfrak{sl}_2(\mathbb{C})$. It then develops the structure theory of semisimple Lie algebras: Cartan's criteria and the Killing form, Weyl's theorem on complete reducibility, the root space decomposition, root systems and their classification by Dynkin diagrams, and highest weights (Sections 13.7–13.11). The classification for $\mathfrak{sl}_2$ is used repeatedly in Hodge theory: on the cohomology of a compact Kähler manifold the Lefschetz operator, its adjoint and the degree operator span a copy of $\mathfrak{sl}_2$, and the hard Lefschetz theorem and the primitive decomposition are consequences of the structure of $\mathfrak{sl}_2$-representations (Chapters 47 and 49).

References are [Ser77], [FH91] and [Hum72].

Prerequisites. Chapter 7 (Groups), Chapter 9 (Fields and Galois Theory), Chapter 11 (Linear Algebra) and Chapter 12 (Multilinear Algebra).

Throughout, k is a field and representations are on k-vector spaces.

13.2 Representations of groups

Definition 13.2.1. A *representation* of a group G is a vector space V with a homomorphism $\rho\colon G \to \mathrm{GL}(V)$; we write gv for $\rho(g)v$. A *homomorphism* of representations (or G-equivariant map) is a linear map f with $f(gv) = gf(v)$; the space of such maps is $\mathrm{Hom}_G(V, W)$. A *subrepresentation* is a G-stable subspace. V is *irreducible* if $V \neq 0$ and its only subrepresentations are 0 and V, and *completely reducible* (or *semisimple*) if it is a direct sum of irreducible subrepresentations. The *invariants* are $V^G = \{v : gv = v \text{ for all } g\}$.

Example 13.2.2. Given representations V, W: quotients V/U by subrepresentations, $V \oplus W$, $V \otimes W$ with $g(v \otimes w) = gv \otimes gw$, the dual V^* with $(g\varphi)(v) = \varphi(g^{-1}v)$, and $\mathrm{Hom}_k(V, W)$ with $(gf)(v) = gf(g^{-1}v)$ are representations, and $\mathrm{Hom}_k(V, W)^G = \mathrm{Hom}_G(V, W)$. The *trivial representation* is k with $gv = v$. If G acts on a finite set X, the *permutation representation* is k^X with $ge_x = e_{gx}$; for $X = G$ with the left action this is the *regular representation*.

Lemma 13.2.3 (Schur)**.** *Let V, W be irreducible representations. Every $f \in \mathrm{Hom}_G(V, W)$ is zero or an isomorphism. If k is algebraically closed and V is finite-dimensional, then $\mathrm{Hom}_G(V, V) = k \cdot \mathrm{id}$.*

Proof. $\ker f$ and $\operatorname{im} f$ are subrepresentations, so $\ker f \in \{0, V\}$ and $\operatorname{im} f \in \{0, W\}$. If k is algebraically closed and $f \in \operatorname{Hom}_G(V, V)$, f has an eigenvalue λ, and $f - \lambda \in \operatorname{Hom}_G(V, V)$ is not an isomorphism, hence zero. □

Theorem 13.2.4 (Maschke)**.** *Let G be finite with $\operatorname{char} k \nmid |G|$. Every subrepresentation U of a representation V has a G-stable complement. Hence every finite-dimensional representation is completely reducible.*

Proof. Choose a linear projection $\pi\colon V \to U$ (a linear map with $\pi|_U = \mathrm{id}$), using a complement of U (Proposition 11.2.3), and put $\pi_0 = \frac{1}{|G|}\sum_{g\in G} g\pi g^{-1}$. It is G-equivariant, has image in U, and restricts to the identity on U; so $\ker \pi_0$ is a G-stable complement of U. The second statement follows by induction on the dimension. □

Counterexample 13.2.5. Both hypotheses of Maschke's theorem are needed. The group $\mathbb{Z}$ acts on $\mathbb{C}^2$ by $n \mapsto \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$; the line $\mathbb{C}e_1$ is stable, but it has no stable complement, since a stable line would be spanned by an eigenvector and e_1 spans the only eigenline. Similarly $\mathbb{Z}/p\mathbb{Z}$ acts on $\mathbb{F}_p^2$ by the same matrices (with n read modulo p), and the representation is not completely reducible although $\mathbb{Z}/p\mathbb{Z}$ is finite; here p divides the order of the group.

13.3 Characters of finite groups

In this section G is finite, $k = \mathbb{C}$, and representations are finite-dimensional.

Definition 13.3.1. The *character* of a representation V is $\chi_V\colon G \to \mathbb{C}$, $\chi_V(g) = \operatorname{tr}(g|_V)$. A *class function* is a function $G \to \mathbb{C}$ constant on conjugacy classes. On class functions put the Hermitian inner product $\langle \phi, \psi \rangle = \frac{1}{|G|}\sum_{g\in G} \phi(g)\overline{\psi(g)}$.

Proposition 13.3.2. *Characters are class functions, $\chi_V(e) = \dim V$, $\chi_{V\oplus W} = \chi_V + \chi_W$, $\chi_{V\otimes W} = \chi_V\chi_W$, and $\chi_{V^*}(g) = \overline{\chi_V(g)} = \chi_V(g^{-1})$. Moreover $\dim V^G = \frac{1}{|G|}\sum_g \chi_V(g)$.*

Proof. The trace is invariant under conjugation, additive on direct sums and multiplicative on tensor products (compute in bases). Each g has finite order, so $g|_V$ satisfies $x^n - 1$ and is diagonalisable with roots of unity λ_i as eigenvalues (Theorem 11.7.6); then $\chi_{V^*}(g) = \sum \lambda_i^{-1} = \sum \bar{\lambda}_i$. Finally $p = \frac{1}{|G|}\sum_g g$ is a projection onto V^G (it is the identity on V^G and $hp = p$ for all h), and the trace of a projection is the dimension of its image. □

Theorem 13.3.3 (Orthogonality)**.** *For representations V, W, $\langle \chi_W, \chi_V \rangle = \dim \operatorname{Hom}_G(V, W)$. Consequently the characters of irreducible representations are orthonormal, a representation is determined up to isomorphism by its character, and V is irreducible if and only if $\langle \chi_V, \chi_V \rangle = 1$.*

Proof. $\operatorname{Hom}_G(V, W) = \operatorname{Hom}_{\mathbb{C}}(V, W)^G$, and $\operatorname{Hom}_{\mathbb{C}}(V, W) \cong V^* \otimes W$ as representations, so by Proposition 13.3.2 its dimension is $\frac{1}{|G|}\sum_g \overline{\chi_V(g)}\chi_W(g)$. For irreducible V, W, Schur's lemma gives $\dim \operatorname{Hom}_G(V, W) = 1$ if $V \cong W$ and 0 otherwise. By Maschke's theorem $V \cong \bigoplus_i V_i^{m_i}$ with distinct irreducible V_i, and $m_i = \langle \chi_V, \chi_{V_i} \rangle$ is determined by χ_V; also $\langle \chi_V, \chi_V \rangle = \sum m_i^2$. □

Theorem 13.3.4. *The characters of the irreducible representations of G form an orthonormal basis of the space of class functions. In particular the number of isomorphism classes of irreducible representations equals the number of conjugacy classes, and if $V_1, \dots, V_r$ are the irreducible representations, then $\sum_i (\dim V_i)^2 = |G|$.*

Proof. The regular representation has character $|G|$ at e and 0 elsewhere, so by Theorem 13.3.3 it contains each V_i with multiplicity $\langle \chi_{\text{reg}}, \chi_{V_i} \rangle = \dim V_i$; comparing dimensions gives $\sum (\dim V_i)^2 = |G|$. In particular there are finitely many irreducible representations up to isomorphism.

It remains to show that a class function ϕ orthogonal to all χ_{V_i} is zero. For a representation V let $T_V = \sum_g \overline{\phi(g)}\, g|_V$. Since ϕ is a class function, T_V commutes with all $h|_V$. If V is irreducible, $T_V = \lambda \mathrm{id}$ by Schur's lemma, and taking traces, $\lambda \dim V = \sum_g \overline{\phi(g)} \chi_V(g) = |G| \langle \chi_V, \phi \rangle = 0$. So $T_V = 0$ for every irreducible V, hence for every V by Maschke's theorem, in particular for the regular representation; but there $T(e_e) = \sum_g \overline{\phi(g)} e_g$, so $\phi = 0$. □

Example 13.3.5. S_3 has three conjugacy classes: $\{e\}$, the three transpositions and the two 3-cycles. Its irreducible representations are the trivial representation, the sign representation and the two-dimensional representation $U = \{x \in \mathbb{C}^3 : x_1 + x_2 + x_3 = 0\}$ (a subrepresentation of the permutation representation). Their characters take the values $(1, 1, 1)$, $(1, -1, 1)$ and $(2, 0, -1)$ on the three classes. One checks orthonormality directly: for instance $\langle \chi_U, \chi_U \rangle = \frac{1}{6}(4 + 0 + 2 \cdot 1) = 1$.

Proposition 13.3.6. *Let G be a finite abelian group. Every irreducible complex representation of G is one-dimensional, given by a homomorphism $\chi : G \to \mathbb{C}^\times$, and there are exactly $|G|$ of them. For $G = \mathbb{Z}/n\mathbb{Z}$ they are $\chi_j(k) = e^{2\pi i jk/n}$, $0 \le j < n$.*

Proof. Let V be irreducible. Each g commutes with the action of G, so by Schur's lemma (Lemma 13.2.3) it acts by a scalar; then every line in V is a subrepresentation, and $\dim V = 1$. The conjugacy classes of an abelian group are its elements, so by Theorem 13.3.4 there are $|G|$ irreducible representations. For $\mathbb{Z}/n\mathbb{Z}$, a homomorphism to $\mathbb{C}^\times$ is determined by the image ζ of 1, with $\zeta^n = 1$, and the n choices give the χ_j. □

Corollary 13.3.7. *The one-dimensional complex representations of a group G are the homomorphisms $G \to \mathbb{C}^\times$, and they all factor through the abelianisation $G^{\text{ab}} = G/[G, G]$ (Proposition 7.9.2). For finite G there are exactly $[G : [G, G]]$ of them.*

Proof. A one-dimensional representation is a homomorphism $G \to \mathrm{GL}_1(\mathbb{C}) = \mathbb{C}^\times$, and $\mathbb{C}^\times$ is abelian. So they correspond to the homomorphisms $G^{\text{ab}} \to \mathbb{C}^\times$, which by Proposition 13.3.6, applied to the finite abelian group G^{ab}, are $|G^{\text{ab}}|$ in number. □

For example $[S_n, S_n] = A_n$ for $n \ge 2$ (Exercise 7.12.3), so S_n has exactly two one-dimensional representations, the trivial one and the sign; a perfect group, such as A_5, has only the trivial one. The analogue for Lie algebras: one-dimensional representations of $\mathfrak{g}$ are the linear forms vanishing on $[\mathfrak{g}, \mathfrak{g}]$, so a Lie algebra with $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$, for instance a semisimple one, has only the trivial one.

Counterexample 13.3.8. The second statement of Schur's lemma needs an algebraically closed field. The group $\mathbb{Z}/4\mathbb{Z}$ acting on $\mathbb{R}^2$ by rotation by $\pi/2$, with generator acting by $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, is irreducible over $\mathbb{R}$, since J has no real eigenvector; but its endomorphisms, the matrices commuting with J, are the $aI + bJ$, a field isomorphic to $\mathbb{C}$, not $\mathbb{R} \cdot \mathrm{id}$. Over $\mathbb{C}$ the representation splits as $\chi_1 \oplus \chi_3$ in the notation of Proposition 13.3.6.

Definition 13.3.9. Let $H \subseteq G$ be a subgroup of a finite group and W a representation of H. The *induced representation* is $\mathrm{Ind}_H^G W = \{f : G \to W : f(hx) = h f(x) \text{ for } h \in H, x \in G\}$, with $(g f)(x) = f(xg)$. For a representation V of G, $\mathrm{Res}_H^G V$ is V viewed as a representation of H.

Proposition 13.3.10 (Frobenius reciprocity)**.** *In the situation of Definition 13.3.9:*

1. $\mathrm{Hom}_G(V, \mathrm{Ind}_H^G W) \cong \mathrm{Hom}_H(\mathrm{Res}_H^G V, W)$, *via* $\varphi \mapsto (v \mapsto \varphi(v)(e))$;

2. $\dim \mathrm{Ind}_H^G W = [G : H] \dim W$, *and*

$$\chi_{\mathrm{Ind}\, W}(g) = \frac{1}{|H|} \sum_{x \in G,\ xgx^{-1} \in H} \chi_W(xgx^{-1});$$

3. $\langle \chi_{\mathrm{Ind}\, W}, \chi_V \rangle_G = \langle \chi_W, \chi_{\mathrm{Res}\, V} \rangle_H$.

Proof. (1) If φ is G-equivariant, then $\psi(v) = \varphi(v)(e)$ satisfies $\psi(hv) = \varphi(hv)(e) = (h\varphi(v))(e) = \varphi(v)(h) = h\varphi(v)(e) = h\psi(v)$. Conversely, for $\psi \in \mathrm{Hom}_H(V, W)$ put $\varphi(v)(x) = \psi(xv)$; then $\varphi(v)(hx) = \psi(hxv) = h\varphi(v)(x)$, so $\varphi(v) \in \mathrm{Ind}\, W$, and $\varphi(gv)(x) = \psi(xgv) = \varphi(v)(xg) = (g\varphi(v))(x)$. The two constructions are inverse, since $\varphi(v)(x) = (x\varphi(v))(e) = \varphi(xv)(e)$.

(2) Choose representatives $x_1, \ldots, x_m$ of the cosets Hx, $m = [G : H]$. An element f is determined by the values $f(x_i)$, which are arbitrary, so $f \mapsto (f(x_i))_i$ is an isomorphism with W^m. For $g \in G$ write $x_i g = h_i x_{\sigma(i)}$ with $h_i \in H$ and σ a permutation; then $(gf)(x_i) = f(x_i g) = h_i f(x_{\sigma(i)})$. So in the decomposition W^m, g maps the $\sigma(i)$-th summand to the i-th by h_i, and only the indices with $\sigma(i) = i$ contribute to the trace, with $h_i = x_i g x_i^{-1} \in H$. Hence $\chi_{\mathrm{Ind}\, W}(g) = \sum_{i : x_i g x_i^{-1} \in H} \chi_W(x_i g x_i^{-1})$, and since χ_W is a class function on H, each coset Hx_i contributes $|H|$ equal terms to the sum in the statement.

(3) By Theorem 13.3.3, both sides are dimensions of the spaces in (1): the left side is $\dim \mathrm{Hom}_G(V, \mathrm{Ind}_H^G W)$ and the right side is $\dim \mathrm{Hom}_H(\mathrm{Res}_H^G V, W)$. □

Counterexample 13.3.11. The character table does not determine the group. Let G be a group of order 8 whose centre Z has order 2 and with $G/Z \cong (\mathbb{Z}/2\mathbb{Z})^2$; the dihedral group D_4 of symmetries of a square and the quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ are examples. For $x \notin Z$ the centraliser contains $\langle x \rangle Z$, of order 4, and is not G, so the class of x is $\{x, zx\}$, $Z = \{e, z\}$: there are five classes. For $x \notin Z$, $x^2 \in Z$ and every conjugate of x lies in xZ, because G/Z is abelian of exponent 2. The commutator subgroup is Z (G/Z is abelian and G is not), so the one-dimensional characters are the four characters of G/Z, equal to 1 at z; at each non-identity element of G/Z two of them take the value 1 and two the value -1; by Theorem 13.3.4 the fifth irreducible character χ has degree 2 ($4 + 2^2 = 8$), and comparing with the regular character, $2\chi = \chi_{\mathrm{reg}} - \sum \chi_{1\text{-dim}}$ gives $\chi(z) = -2$ and $\chi(x) = 0$ for $x \notin Z$. So the character tables of D_4 and Q_8 coincide, but the groups are not isomorphic: D_4 has five elements of order 2 and Q_8 only one.

13.4 Lie algebras

Definition 13.4.1. A *Lie algebra* over k is a vector space $\mathfrak{g}$ with a bilinear map $[-,-] \colon \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$, the *bracket*, with $[x, x] = 0$ and the *Jacobi identity* $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$. A *homomorphism* of Lie algebras is a linear map preserving brackets, an *ideal* is a subspace $\mathfrak{a}$ with $[\mathfrak{g}, \mathfrak{a}] \subseteq \mathfrak{a}$, and $\mathfrak{g}$ is *abelian* if its bracket is zero.

Example 13.4.2.

1. For an associative algebra A, the commutator $[x, y] = xy - yx$ makes A a Lie algebra. For $A = \mathrm{End}(V)$ this is $\mathfrak{gl}(V)$, and for $A = M_n(k)$ it is $\mathfrak{gl}_n(k)$.

2. $\mathfrak{sl}_n(k) = \{x \in \mathfrak{gl}_n(k) : \mathrm{tr}\, x = 0\}$ is a Lie subalgebra, since $\mathrm{tr}[x, y] = 0$ (Exercise 11.12.1).

3. The skew-symmetric matrices form $\mathfrak{so}_n(k)$, and more generally the x with $B(xv, w) + B(v, xw) = 0$ for a bilinear form B form a Lie subalgebra of $\mathfrak{gl}(V)$.

Definition 13.4.3. A *representation* of a Lie algebra $\mathfrak{g}$ is a vector space V with a Lie algebra homomorphism $\mathfrak{g} \to \mathfrak{gl}(V)$; we write xv for the action, so $[x,y]v = x(yv) - y(xv)$. Subrepresentations, irreducibility, homomorphisms and direct sums are defined as for groups. The tensor product $V \otimes W$ is a representation with $x(v \otimes w) = xv \otimes w + v \otimes xw$, and the dual with $(x\varphi)(v) = -\varphi(xv)$. The *adjoint representation* is $\mathfrak{g}$ itself with $x \cdot y = [x,y]$, a representation by the Jacobi identity.

Remark 13.4.4. Lie algebras arise as tangent spaces of Lie groups at the identity (Chapter 34), and representations of a Lie group differentiate to representations of its Lie algebra; the formula for the tensor product is the derivative of $g(v \otimes w) = gv \otimes gw$.

13.5 Nilpotent and solvable Lie algebras

The Lie algebras of upper triangular and of strictly upper triangular matrices are the models of two classes of Lie algebras, the solvable and the nilpotent ones. The theorems of Engel and Lie say that, conversely, a Lie algebra of nilpotent matrices can be put in strictly upper triangular form, and a solvable Lie algebra of matrices over $\mathbb{C}$ in upper triangular form, by one change of basis for all its elements at once. In this section all Lie algebras and representations are finite-dimensional. For subspaces $\mathfrak{a}, \mathfrak{b} \subseteq \mathfrak{g}$, $[\mathfrak{a},\mathfrak{b}]$ is the subspace spanned by the brackets $[x,y]$ with $x \in \mathfrak{a}$, $y \in \mathfrak{b}$, and the *centre* of $\mathfrak{g}$ is $\mathfrak{z}(\mathfrak{g}) = \{x : [x,y] = 0 \text{ for all } y\}$. We write $\operatorname{ad} x \colon \mathfrak{g} \to \mathfrak{g}$ for $y \mapsto [x,y]$, the adjoint representation of Definition 13.4.3.

Definition 13.5.1. The *derived series* of a Lie algebra $\mathfrak{g}$ is $D^0\mathfrak{g} = \mathfrak{g}$, $D^{i+1}\mathfrak{g} = [D^i\mathfrak{g}, D^i\mathfrak{g}]$, and its *lower central series* is $C^1\mathfrak{g} = \mathfrak{g}$, $C^{i+1}\mathfrak{g} = [\mathfrak{g}, C^i\mathfrak{g}]$. The Lie algebra $\mathfrak{g}$ is *solvable* if $D^m\mathfrak{g} = 0$ for some m, and *nilpotent* if $C^m\mathfrak{g} = 0$ for some m.

So $\mathfrak{g}$ is nilpotent when all iterated brackets $[x_1,[x_2,[\cdots,[x_{m-1},x_m]\cdots]]]$ of a fixed length m vanish; abelian Lie algebras are nilpotent with $m = 2$.

Lemma 13.5.2. *1. If $\mathfrak{a}$ and $\mathfrak{b}$ are ideals, so is $[\mathfrak{a},\mathfrak{b}]$. Hence all $D^i\mathfrak{g}$ and $C^i\mathfrak{g}$ are ideals, and $D^i\mathfrak{g} \subseteq C^{i+1}\mathfrak{g}$; in particular nilpotent Lie algebras are solvable.*

2. Subalgebras and homomorphic images of solvable (nilpotent) Lie algebras are solvable (nilpotent).

3. If $\mathfrak{a}$ is a solvable ideal of $\mathfrak{g}$ and $\mathfrak{g}/\mathfrak{a}$ is solvable, then $\mathfrak{g}$ is solvable.

4. A nonzero nilpotent Lie algebra has nonzero centre; and if $\mathfrak{g}/\mathfrak{z}(\mathfrak{g})$ is nilpotent, so is $\mathfrak{g}$.

Proof. (1) For $x \in \mathfrak{g}$, $a \in \mathfrak{a}$, $b \in \mathfrak{b}$ the Jacobi identity gives $[x,[a,b]] = [[x,a],b] + [a,[x,b]] \in [\mathfrak{a},\mathfrak{b}]$. By induction the terms of both series are ideals. Also $D^0\mathfrak{g} = C^1\mathfrak{g}$, and if $D^i\mathfrak{g} \subseteq C^{i+1}\mathfrak{g}$ then $D^{i+1}\mathfrak{g} = [D^i\mathfrak{g}, D^i\mathfrak{g}] \subseteq [\mathfrak{g}, C^{i+1}\mathfrak{g}] = C^{i+2}\mathfrak{g}$.

(2) For a subalgebra $\mathfrak{h}$, induction gives $D^i\mathfrak{h} \subseteq D^i\mathfrak{g}$ and $C^i\mathfrak{h} \subseteq C^i\mathfrak{g}$. For a surjective homomorphism $\varphi \colon \mathfrak{g} \to \mathfrak{g}'$, $\varphi([\mathfrak{a},\mathfrak{b}]) = [\varphi(\mathfrak{a}),\varphi(\mathfrak{b})]$, so $\varphi(D^i\mathfrak{g}) = D^i\mathfrak{g}'$ and $\varphi(C^i\mathfrak{g}) = C^i\mathfrak{g}'$.

(3) If $D^m(\mathfrak{g}/\mathfrak{a}) = 0$, then $D^m\mathfrak{g} \subseteq \mathfrak{a}$ by (2) applied to the projection, and by induction $D^{m+l}\mathfrak{g} \subseteq D^l\mathfrak{a}$, which vanishes for large l.

(4) Let m be least with $C^m\mathfrak{g} = 0$; as $\mathfrak{g} \neq 0$, $m \geq 2$ and $C^{m-1}\mathfrak{g} \neq 0$. Since $[\mathfrak{g}, C^{m-1}\mathfrak{g}] = C^m\mathfrak{g} = 0$, $C^{m-1}\mathfrak{g} \subseteq \mathfrak{z}(\mathfrak{g})$. If $C^m(\mathfrak{g}/\mathfrak{z}(\mathfrak{g})) = 0$, then $C^m\mathfrak{g} \subseteq \mathfrak{z}(\mathfrak{g})$ and $C^{m+1}\mathfrak{g} = [\mathfrak{g}, C^m\mathfrak{g}] = 0$. □

Example 13.5.3. Let $\mathfrak{b}_n \subseteq \mathfrak{gl}_n(k)$ be the upper triangular matrices, $\mathfrak{n}_n$ the strictly upper triangular ones and $\mathfrak{d}_n$ the diagonal ones; all three are Lie subalgebras.

1. $\mathfrak{n}_n$ is nilpotent. Let $\mathfrak{n}^{(j)}$ be the matrices (a_{il}) with $a_{il} = 0$ unless $l - i \geq j$, so that $\mathfrak{n}_n = \mathfrak{n}^{(1)}$ and $\mathfrak{n}^{(n)} = 0$. The entry $(AB)_{il} = \sum_m a_{im}b_{ml}$ of a product of $A \in \mathfrak{n}^{(j)}$ and $B \in \mathfrak{n}^{(j')}$ can be nonzero only if $m - i \geq j$ and $l - m \geq j'$ for some m, so $AB \in \mathfrak{n}^{(j+j')}$. Hence $[\mathfrak{n}^{(1)}, \mathfrak{n}^{(j)}] \subseteq \mathfrak{n}^{(j+1)}$, and $C^i\mathfrak{n}_n \subseteq \mathfrak{n}^{(i)}$ vanishes for $i \geq n$.

2. $\mathfrak{b}_n$ is solvable: the diagonal of a product of upper triangular matrices is the product of the diagonals, so the diagonal of a commutator vanishes and $[\mathfrak{b}_n, \mathfrak{b}_n] \subseteq \mathfrak{n}_n$, which is solvable by (1) and Lemma 13.5.2(1); so $D^{m+1}\mathfrak{b}_n \subseteq D^m\mathfrak{n}_n$ vanishes for large m. For $n \geq 2$ it is not nilpotent: with the matrix units E_{il}, $[E_{11}, E_{12}] = E_{12}$, so $E_{12} \in C^i\mathfrak{b}_n$ for all i.

3. The two-dimensional Lie algebra with basis x, y and $[x, y] = y$ (for instance $x = E_{11}$, $y = E_{12}$ in $\mathfrak{b}_2$) is solvable, with $D^1 = ky$ and $D^2 = 0$, but not nilpotent, since $C^i = ky$ for all $i \geq 2$. The *Heisenberg algebra* $\mathfrak{n}_3$, with basis E_{12}, E_{23}, E_{13} and the only nonzero bracket $[E_{12}, E_{23}] = E_{13}$, is nilpotent but not abelian.

4. If $2 \neq 0$ in k, $\mathfrak{sl}_2(k)$ is not solvable: its basis $e = E_{12}$, $f = E_{21}$, $h = E_{11} - E_{22}$ satisfies $[h, e] = 2e$, $[h, f] = -2f$ and $[e, f] = h$, so $[\mathfrak{sl}_2, \mathfrak{sl}_2] = \mathfrak{sl}_2$ and $D^i\mathfrak{sl}_2 = \mathfrak{sl}_2$ for all i.

Lemma 13.5.4. *If $x \in \mathfrak{gl}(V)$ is nilpotent, then* $\operatorname{ad} x \colon \mathfrak{gl}(V) \to \mathfrak{gl}(V)$ *is nilpotent.*

Proof. Write $\operatorname{ad} x = \lambda_x - \rho_x$ with $\lambda_x(y) = xy$ and $\rho_x(y) = yx$. These two linear maps commute, by associativity, and if $x^m = 0$ then $\lambda_x^m = \lambda_{x^m} = 0$ and $\rho_x^m = 0$. By the binomial formula for commuting maps, $(\lambda_x - \rho_x)^{2m} = \sum_i \binom{2m}{i} \lambda_x^i (-\rho_x)^{2m-i}$, and every term has $i \geq m$ or $2m - i \geq m$, hence vanishes. □

Theorem 13.5.5 (Engel's theorem, linear form)**.** *Let $V \neq 0$ be a finite-dimensional vector space over any field k, and $\mathfrak{g} \subseteq \mathfrak{gl}(V)$ a Lie subalgebra consisting of nilpotent endomorphisms. Then there is $v \neq 0$ in V with $xv = 0$ for all $x \in \mathfrak{g}$. Consequently V has a basis in which every element of $\mathfrak{g}$ is strictly upper triangular.*

Proof. Induction on $\dim \mathfrak{g}$. If $\mathfrak{g} = 0$ any $v \neq 0$ works, and if $\dim \mathfrak{g} = 1$, $\mathfrak{g} = kx$ with x nilpotent, and any nonzero $v \in \ker x$ works ($\ker x \neq 0$, as x is not invertible). Let $\dim \mathfrak{g} \geq 2$ and let $\mathfrak{h}$ be a proper subalgebra of maximal dimension (for instance 0 is proper).

Step 1: $\mathfrak{h}$ is an ideal of codimension 1. Since $[\mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{h}$, each $\operatorname{ad} h$, $h \in \mathfrak{h}$, preserves $\mathfrak{h}$ and induces an endomorphism $\overline{\operatorname{ad} h}$ of the nonzero space $\mathfrak{g}/\mathfrak{h}$; by the Jacobi identity $h \mapsto \overline{\operatorname{ad} h}$ is a homomorphism of Lie algebras $\mathfrak{h} \to \mathfrak{gl}(\mathfrak{g}/\mathfrak{h})$. Each $\operatorname{ad} h$ is nilpotent on $\mathfrak{gl}(V)$ (Lemma 13.5.4), hence on the invariant subspace $\mathfrak{g}$ and on the quotient $\mathfrak{g}/\mathfrak{h}$. So the image of $\mathfrak{h}$ is a Lie algebra of nilpotent endomorphisms of $\mathfrak{g}/\mathfrak{h}$ of dimension at most $\dim \mathfrak{h} < \dim \mathfrak{g}$, and by induction there is $y \in \mathfrak{g} \setminus \mathfrak{h}$ with $[h, y] \in \mathfrak{h}$ for all $h \in \mathfrak{h}$. Then $\mathfrak{h} + ky$ is a subalgebra strictly containing $\mathfrak{h}$, so it is $\mathfrak{g}$ by maximality; hence $\mathfrak{g} = \mathfrak{h} \oplus ky$ and $\mathfrak{h}$ is an ideal.

Step 2: a common null vector. By induction applied to $\mathfrak{h} \subseteq \mathfrak{gl}(V)$, the space $W = \{v \in V : hv = 0 \text{ for all } h \in \mathfrak{h}\}$ is nonzero. It is stable under y: for $w \in W$ and $h \in \mathfrak{h}$, $h(yw) = y(hw) + [h, y]w = 0$, as $[h, y] \in \mathfrak{h}$. The restriction of the nilpotent y to W is nilpotent, so it kills some $v \neq 0$ in W, and then $xv = 0$ for all $x \in \mathfrak{h} + ky = \mathfrak{g}$.

Step 3: the basis. Let v_1 be as above. Each $x \in \mathfrak{g}$ induces a nilpotent endomorphism of V/kv_1, and these form a Lie algebra of nilpotent endomorphisms; by induction on $\dim V$, V/kv_1 has a basis $\bar{v}_2, \dots, \bar{v}_d$ in which they are strictly upper triangular. Lifting it to $v_2, \dots, v_d$, we get $xv_1 = 0$ and $xv_j \in \operatorname{span}(v_1, \dots, v_{j-1})$, which is the claim. □

Corollary 13.5.6 (Engel's theorem)**.** *A Lie algebra $\mathfrak{g}$ is nilpotent if and only if* $\operatorname{ad} x$ *is nilpotent for every $x \in \mathfrak{g}$.*

Proof. If $C^m\mathfrak{g} = 0$, then $\operatorname{ad} x$ maps $C^i\mathfrak{g}$ into $C^{i+1}\mathfrak{g}$, so $(\operatorname{ad} x)^{m-1}\mathfrak{g} \subseteq C^m\mathfrak{g} = 0$. Conversely, the image $\operatorname{ad} \mathfrak{g} \subseteq \mathfrak{gl}(\mathfrak{g})$ is a Lie algebra of nilpotent endomorphisms. Theorem 13.5.5 gives a basis $v_1, \dots, v_d$ of $\mathfrak{g}$ such that, with $V_j = \operatorname{span}(v_1, \dots, v_j)$, $[\mathfrak{g}, V_j] \subseteq V_{j-1}$. By induction $C^i\mathfrak{g} \subseteq V_{d-i+1}$, since $C^1\mathfrak{g} = V_d$ and $C^{i+1}\mathfrak{g} = [\mathfrak{g}, C^i\mathfrak{g}] \subseteq [\mathfrak{g}, V_{d-i+1}] \subseteq V_{d-i}$. So $C^{d+1}\mathfrak{g} = 0$. □

Counterexample 13.5.7. 1. A nilpotent Lie algebra of matrices need not consist of nilpotent matrices: the diagonal matrices $\mathfrak{d}_n$ form an abelian, hence nilpotent, Lie algebra containing the identity. Engel's theorem is about the operators $\operatorname{ad} x$ on $\mathfrak{g}$, not about the matrices x on V.

2. In Theorem 13.5.5 it matters that $\mathfrak{g}$ is closed under brackets. The matrices $e = E_{12}$ and $f = E_{21}$ of $\mathfrak{sl}_2$ are nilpotent, but $\ker e = ke_1$ and $\ker f = ke_2$ meet only in 0; the Lie algebra they generate contains $[e, f] = h$, which is not nilpotent.

For Lie's theorem we need eigenvectors, hence an algebraically closed field, and characteristic 0. The key step is the following lemma.

Lemma 13.5.8 (Invariance lemma)**.** *Let k have characteristic 0, let $\mathfrak{g} \subseteq \mathfrak{gl}(V)$ be a Lie subalgebra, $\mathfrak{a}$ an ideal of $\mathfrak{g}$ and $\lambda\colon \mathfrak{a} \to k$ a linear form. Then $W = \{v \in V : av = \lambda(a)v \text{ for all } a \in \mathfrak{a}\}$ is stable under $\mathfrak{g}$.*

Proof. Let $w \in W$ and $x \in \mathfrak{g}$. For $a \in \mathfrak{a}$, $a(xw) = x(aw) + [a, x]w = \lambda(a)xw + \lambda([a, x])w$, as $[a, x] \in \mathfrak{a}$. So it suffices to show $\lambda([a, x]) = 0$ for all a, when $W \neq 0$. Fix $w \neq 0$ in W, let $W_i = \operatorname{span}(w, xw, \dots, x^{i-1}w)$ (with $W_0 = 0$), and let $m \geq 1$ be least with $W_{m+1} = W_m$; then $w, xw, \dots, x^{m-1}w$ is a basis of W_m, and $xW_m \subseteq W_m$.

We claim that $ax^iw \in \lambda(a)x^iw + W_i$ for all $a \in \mathfrak{a}$ and $i \geq 0$. For $i = 0$ this is $aw = \lambda(a)w$. If it holds for $i - 1$ and all a, then $ax^iw = x(ax^{i-1}w) + [a, x]x^{i-1}w$. By induction $ax^{i-1}w = \lambda(a)x^{i-1}w + u$ with $u \in W_{i-1}$, so $x(ax^{i-1}w) \in \lambda(a)x^iw + W_i$; and $[a, x]x^{i-1}w \in \lambda([a, x])x^{i-1}w + W_{i-1} \subseteq W_i$. This proves the claim.

So each $a \in \mathfrak{a}$ maps W_m into itself, by an upper triangular matrix in the basis $w, xw, \dots, x^{m-1}w$ with all diagonal entries $\lambda(a)$, and $\operatorname{tr}(a|_{W_m}) = m\lambda(a)$. Apply this to $[a, x] \in \mathfrak{a}$: since a and x both preserve W_m, $[a, x]|_{W_m}$ is the commutator of $a|_{W_m}$ and $x|_{W_m}$, which has trace 0. So $m\lambda([a, x]) = 0$, and $\lambda([a, x]) = 0$ as $m \neq 0$ in k. □

Theorem 13.5.9 (Lie's theorem)**.** *Let k be algebraically closed of characteristic 0, $V \neq 0$ a finite-dimensional vector space and $\mathfrak{g} \subseteq \mathfrak{gl}(V)$ a solvable Lie subalgebra. Then the elements of $\mathfrak{g}$ have a common eigenvector: there are $v \neq 0$ and a linear form λ on $\mathfrak{g}$ with $xv = \lambda(x)v$ for all $x \in \mathfrak{g}$. Consequently V has a basis in which every element of $\mathfrak{g}$ is upper triangular.*

Proof. Induction on $\dim \mathfrak{g}$; for $\mathfrak{g} = 0$ take any $v \neq 0$ and $\lambda = 0$. Let $\mathfrak{g} \neq 0$. Then $[\mathfrak{g}, \mathfrak{g}] \neq \mathfrak{g}$, since otherwise $D^i\mathfrak{g} = \mathfrak{g}$ for all i. Choose a subspace $\mathfrak{a}$ of codimension 1 containing $[\mathfrak{g}, \mathfrak{g}]$. It is an ideal, as $[\mathfrak{g}, \mathfrak{a}] \subseteq [\mathfrak{g}, \mathfrak{g}] \subseteq \mathfrak{a}$, and it is solvable (Lemma 13.5.2(2)). By induction there are $v_0 \neq 0$ and $\lambda\colon \mathfrak{a} \to k$ with $av_0 = \lambda(a)v_0$ for $a \in \mathfrak{a}$, so the space W of Lemma 13.5.8 is nonzero, and it is stable under $\mathfrak{g}$. Write $\mathfrak{g} = \mathfrak{a} \oplus kz$. As k is algebraically closed, $z|_W$ has an eigenvector $v \in W$, $zv = \mu v$. Then v is an eigenvector of every element of $\mathfrak{g}$, with λ extended by $\lambda(z) = \mu$. The basis is obtained as in Step 3 of the proof of Theorem 13.5.5, applying the first part to V/kv. □

Corollary 13.5.10. *Let k be algebraically closed of characteristic 0 and $\mathfrak{g}$ solvable.*

1. *Every irreducible representation of $\mathfrak{g}$ is one-dimensional.*

2. *There is a chain of ideals $0 = \mathfrak{g}_0 \subset \mathfrak{g}_1 \subset \cdots \subset \mathfrak{g}_d = \mathfrak{g}$ with $\dim \mathfrak{g}_i = i$.*

3. *$[\mathfrak{g}, \mathfrak{g}]$ is nilpotent. Conversely, over any field, a Lie algebra $\mathfrak{g}$ with $[\mathfrak{g}, \mathfrak{g}]$ nilpotent is solvable.*

Proof. (1) The image of $\mathfrak{g}$ in $\mathfrak{gl}(V)$ is solvable (Lemma 13.5.2(2)), so it has a common eigenvector v, and kv is a subrepresentation, hence equal to V. (2) Apply Theorem 13.5.9 to $\operatorname{ad} \mathfrak{g} \subseteq \mathfrak{gl}(\mathfrak{g})$: in the resulting basis $v_1, \dots, v_d$, the spans $\mathfrak{g}_i = \operatorname{span}(v_1, \dots, v_i)$ satisfy $[\mathfrak{g}, \mathfrak{g}_i] \subseteq \mathfrak{g}_i$. (3) In the

same basis every $\operatorname{ad} x$ is upper triangular, so every $\operatorname{ad}[x,y] = [\operatorname{ad} x, \operatorname{ad} y]$ is strictly upper triangular, as is every element of $\operatorname{ad}[\mathfrak{g},\mathfrak{g}]$; hence for $z \in [\mathfrak{g},\mathfrak{g}]$, $\operatorname{ad} z$ is nilpotent on $\mathfrak{g}$, and so on the invariant subspace $[\mathfrak{g},\mathfrak{g}]$. By Corollary 13.5.6, $[\mathfrak{g},\mathfrak{g}]$ is nilpotent. Conversely, if $[\mathfrak{g},\mathfrak{g}]$ is nilpotent it is solvable, and $\mathfrak{g}/[\mathfrak{g},\mathfrak{g}]$ is abelian, so $\mathfrak{g}$ is solvable by Lemma 13.5.2(3). □

Counterexample 13.5.11. Both hypotheses on k are needed in Theorem 13.5.9.

1. Over $\mathbb{R}$, the abelian Lie algebra spanned by the rotation generator $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ has no eigenvector in $\mathbb{R}^2$.
2. Let k be algebraically closed of characteristic $p > 0$, and $V = k^p$ with basis $e_0, \dots, e_{p-1}$, indices modulo p. Let $xe_i = ie_i$ and $ye_i = e_{i+1}$. Then $[x,y]e_i = (i+1)e_{i+1} - ie_{i+1} = ye_i$, so $[x,y] = y$ and $\mathfrak{g} = kx + ky$ is solvable (Example 13.5.3(3)). But x and y have no common eigenvector. An eigenvector $v = \sum_i c_i e_i$ of y with eigenvalue μ satisfies $c_{i-1} = \mu c_i$ for all i, hence $c_i = \mu^p c_i$ and $\mu^p = 1$, that is $(\mu - 1)^p = 0$ and $\mu = 1$; so v is a multiple of $\sum_i e_i$, and $xv = \sum_i ie_i$ is not a multiple of v since $p \geq 2$. In the proof of Lemma 13.5.8 the argument breaks down at $m\lambda([a,x]) = 0$, with $m = p$.

Remark 13.5.12. Every Lie algebra $\mathfrak{g}$ has a largest solvable ideal, the *radical* (the sum of two solvable ideals is solvable by Lemma 13.5.2(3), as $(\mathfrak{a}+\mathfrak{b})/\mathfrak{a} \cong \mathfrak{b}/(\mathfrak{a}\cap\mathfrak{b})$), and $\mathfrak{g}$ is *semisimple* if its radical is 0. Over a field of characteristic 0, *Cartan's criterion* characterises solvability by the trace form: $\mathfrak{g} \subseteq \mathfrak{gl}(V)$ is solvable if and only if $\operatorname{tr}(xy) = 0$ for all $x \in \mathfrak{g}$ and $y \in [\mathfrak{g},\mathfrak{g}]$; and $\mathfrak{g}$ is semisimple if and only if its Killing form $\operatorname{tr}(\operatorname{ad} x \operatorname{ad} y)$ is nondegenerate. Both are proved in Section 13.7 (Theorems 13.7.6 and 13.7.8), using Lie's theorem and the Jordan–Chevalley decomposition (Theorem 11.8.11). For example $\mathfrak{sl}_2(\mathbb{C})$ is semisimple, and Theorem 13.6.3 below is the first case of Weyl's theorem (Theorem 13.8.2).

13.6 Representations of $\mathfrak{sl}_2$

In this section $k = \mathbb{C}$ and $\mathfrak{sl}_2 = \mathfrak{sl}_2(\mathbb{C})$, with basis

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

satisfying $[h,e] = 2e$, $[h,f] = -2f$ and $[e,f] = h$. All representations are finite-dimensional.

Lemma 13.6.1. *Let V be a representation and for $\lambda \in \mathbb{C}$ let $V_\lambda = \ker(h-\lambda)$ and $V^{(\lambda)} = \bigcup_N \ker(h-\lambda)^N$ (the generalised eigenspace). Then e maps V_λ to $V_{\lambda+2}$ and $V^{(\lambda)}$ to $V^{(\lambda+2)}$, and f maps them to $V_{\lambda-2}$ and $V^{(\lambda-2)}$. Moreover, for $k \geq 1$,*

$$ef^k = f^k e + kf^{k-1}(h-k+1) \quad \text{as operators on } V.$$

Proof. From $he = eh+2e$ we get $(h-\lambda-2)e = e(h-\lambda)$, hence $(h-\lambda-2)^N e = e(h-\lambda)^N$; similarly for f. The identity holds for $k = 1$ since $ef = fe+h$. If it holds for k, then using $hf = fh-2f$, $ef^{k+1} = f^k ef + kf^{k-1}(h-k+1)f = f^k(fe+h) + kf^k(h-k-1) = f^{k+1}e + (k+1)f^k(h-k)$. □

Theorem 13.6.2. *For every $m \geq 0$ there is an irreducible representation $V(m)$ of dimension $m+1$ with a basis $v_0, \dots, v_m$ such that*

$$hv_j = (m-2j)v_j, \qquad fv_j = (j+1)v_{j+1}, \qquad ev_j = (m-j+1)v_{j-1},$$

with $v_{-1} = v_{m+1} = 0$. Every irreducible representation is isomorphic to exactly one $V(m)$. More precisely, if V is any representation and $v_0 \neq 0$ satisfies $ev_0 = 0$ and $hv_0 = \lambda v_0$, then $\lambda = m$ is a non-negative integer and the vectors $v_j = f^j v_0/j!$ span a subrepresentation isomorphic to $V(m)$.

Proof. The formulas define a representation: it suffices to check the three bracket relations on each v_j. For instance $(ef - fe)v_j = (j+1)(m-j)v_j - j(m-j+1)v_j = (m-2j)v_j = hv_j$, and $[h,e] = 2e$, $[h,f] = -2f$ hold because e raises and f lowers the h-eigenvalue by 2. It is irreducible: a nonzero subrepresentation is h-stable, so it contains some v_j (the eigenvalues $m-2j$ are distinct), then v_0 by applying e (the coefficients $m-j+1$ are nonzero for $1 \le j \le m$), then all v_j by applying f.

Now let v_0 be as in the last statement and $v_j = f^j v_0/j!$. By Lemma 13.6.1, $hv_j = (\lambda - 2j)v_j$, and $fv_j = (j+1)v_{j+1}$ by definition. By the identity of the lemma, $ev_j = \frac{1}{j!} j f^{j-1}(h-j+1)v_0 = (\lambda - j + 1)v_{j-1}$. The nonzero v_j have distinct h-eigenvalues, hence are linearly independent, so there is a largest m with $v_m \neq 0$, and then $v_{m+1} = 0$. Hence $0 = ev_{m+1} = (\lambda - m)v_m$, so $\lambda = m$, and the span of $v_0, \dots, v_m$ is a subrepresentation isomorphic to $V(m)$.

Every representation $V \neq 0$ contains such a v_0: h has an eigenvector w, with eigenvalue μ, and among $w, ew, e^2w, \dots$ (which lie in the distinct eigenspaces $V_{\mu+2i}$ or vanish) there is a last nonzero one. If V is irreducible, the subrepresentation generated by v_0 is all of V, so $V \cong V(m)$ with $m = \dim V - 1$. □

Theorem 13.6.3. *Every finite-dimensional representation of* $\mathfrak{sl}_2$ *is completely reducible, and* h *acts diagonalisably with integer eigenvalues.*

Proof. Let $C = ef + fe + \frac{1}{2}h^2$ (the *Casimir operator*). A direct computation with the relations shows that C commutes with e, f and h; for example $[C,e] = e[f,e] + [f,e]e + \frac{1}{2}(h[h,e] + [h,e]h) = -eh - he + eh + he = 0$. On $V(m)$, C is scalar by Lemma 13.2.3, and evaluating on v_0 gives $C = m + \frac{1}{2}m^2 = \frac{1}{2}m(m+2)$ (using $ef v_0 = mv_0$ and $ev_0 = 0$). These scalars are distinct for different $m \ge 0$.

Since C commutes with the action, V is the direct sum of the generalised eigenspaces of C, each of which is a subrepresentation (Lemma 11.7.5). By choosing an irreducible subrepresentation and passing to quotients repeatedly, every representation has a chain of subrepresentations with irreducible successive quotients, each isomorphic to some $V(m)$ by Theorem 13.6.2; C acts on the quotient $V(m)$ by $\frac{1}{2}m(m+2)$. Hence on the generalised c-eigenspace, all these quotients are isomorphic to the same $V(m)$ (with $c = \frac{1}{2}m(m+2)$). It therefore suffices to prove: if V has such a chain $0 = F_0 \subsetneq F_1 \subsetneq \cdots \subsetneq F_r = V$ with $F_i/F_{i-1} \cong V(m)$ for all i, then $V \cong V(m)^r$.

The eigenvalues of h on V are among $m, m-2, \dots, -m$, so $V^{(m+2)} = 0$ and e kills $V^{(m)}$ (Lemma 13.6.1). Also $\dim V^{(m)} = r$, since each quotient contributes one dimension. The map $f^m \colon V^{(m)} \to V^{(-m)}$ is injective: if $u \in V^{(m)}$ is nonzero, let i be least with $u \in F_i$; the image of u in $F_i/F_{i-1} \cong V(m)$ is a nonzero vector of weight m, on which f^m is nonzero, so $f^m u \notin F_{i-1}$. Now $f^{m+1}u \in V^{(-m-2)} = 0$, and by Lemma 13.6.1, $0 = ef^{m+1}u = f^{m+1}eu + (m+1)f^m(h-m)u = (m+1)f^m(h-m)u$. Since $(h-m)u \in V^{(m)}$ and f^m is injective there, $(h-m)u = 0$. So $h = m$ on $V^{(m)}$.

Let $u_1, \dots, u_r$ be a basis of $V^{(m)} = V_m$. Each u_i satisfies $eu_i = 0$ and $hu_i = mu_i$, so it generates a subrepresentation $W_i \cong V(m)$ by Theorem 13.6.2. The kernel K of $\bigoplus_i W_i \to V$ is a subrepresentation whose weight-m part is zero, since the u_i are independent. If $K \neq 0$, the last paragraph of the proof of Theorem 13.6.2, applied to K, gives $v_0 \in K$ with $v_0 \neq 0$ and $ev_0 = 0$. But in $V(m)$ the kernel of e is $\mathbb{C}v_0$, since $e(\sum_j c_j v_j) = \sum_{j \ge 1} c_j(m-j+1)v_{j-1}$ vanishes only if $c_j = 0$ for $j \ge 1$; so in $\bigoplus_i W_i \cong V(m)^r$ the kernel of e is the weight-m space, and v_0 would be a nonzero element of K of weight m. Hence $K = 0$, the sum is direct, and by dimension ($r(m+1) = \dim V$) it is all of V. In each W_i, h is diagonal with integer eigenvalues. □

Corollary 13.6.4. *Let* V *be a finite-dimensional representation of* $\mathfrak{sl}_2$*, with weight spaces* $V_\lambda = \ker(h - \lambda)$*,* $\lambda \in \mathbb{Z}$*. For every* $j \ge 0$ *the map* $f^j \colon V_j \to V_{-j}$ *is an isomorphism. Let* $P_j = V_j \cap \ker e$

(the primitive *vectors of weight j). Then*

$$V = \bigoplus_{j \geq 0} \bigoplus_{i=0}^{j} f^i P_j,$$

and f^i is injective on P_j for $0 \leq i \leq j$.

Proof. By Theorem 13.6.3 it suffices to treat $V = V(m)$, where $V_{m-2i} = \mathbb{C}v_i$, f^j maps $v_{(m-j)/2}$ to a nonzero multiple of $v_{(m+j)/2}$ when $m - j$ is even and non-negative, $P_m = \mathbb{C}v_0$, $P_j = 0$ for $j \neq m$, and $f^i P_m = \mathbb{C}v_i$. □

Remark 13.6.5. Theorem 13.6.3 is a special case of Weyl's theorem (Theorem 13.8.2): every finite-dimensional representation of a semisimple Lie algebra over a field of characteristic 0 is completely reducible. The irreducible representations are classified by their highest weights (Section 13.11), and the semisimple Lie algebras themselves by Dynkin diagrams (Section 13.10). The theory of $\mathfrak{sl}_2$ is also the main tool of that classification: every semisimple Lie algebra is built from copies of $\mathfrak{sl}_2$, one for each pair of roots $\pm\alpha$ (Proposition 13.9.5). See [Hum72] and [FH91].

13.7 Semisimple Lie algebras

By Lie's theorem a solvable Lie algebra over $\mathbb{C}$ is, in a suitable basis of any representation, an algebra of upper triangular matrices: it is as close to being abelian as a nonabelian Lie algebra can be. Semisimple Lie algebras are the opposite extreme: they have no nonzero solvable ideals at all. Every Lie algebra $\mathfrak{g}$ is an extension of the semisimple Lie algebra $\mathfrak{g}/\operatorname{rad}\mathfrak{g}$ by its largest solvable ideal $\operatorname{rad}\mathfrak{g}$ (Lemma 13.7.2), so the two classes together describe all Lie algebras. This section and the next four develop the structure of semisimple Lie algebras:

1. the Killing form and Cartan's criteria for solvability and semisimplicity; the decomposition into simple ideals; derivations; the abstract Jordan decomposition (this section);
2. Weyl's theorem on complete reducibility (Section 13.8);
3. the root space decomposition with respect to a Cartan subalgebra, built from copies of $\mathfrak{sl}_2$ (Section 13.9);
4. root systems, Weyl groups, Dynkin diagrams and the classification (Section 13.10);
5. weights and highest weights of representations (Section 13.11).

We follow [Hum72, Chapters II–III and VI]. Throughout these sections, k is a field of characteristic 0, so that $\mathbb{Q} \subseteq k$, and Lie algebras and representations are finite-dimensional.

Definition 13.7.1. Let $\mathfrak{g}$ be a Lie algebra.

1. The *radical* $\operatorname{rad}\mathfrak{g}$ is the largest solvable ideal of $\mathfrak{g}$ (it exists by Lemma 13.7.2 below). $\mathfrak{g}$ is *semisimple* if $\operatorname{rad}\mathfrak{g} = 0$, and *simple* if it is not abelian and its only ideals are 0 and $\mathfrak{g}$.
2. For a representation $\varphi\colon \mathfrak{g} \to \mathfrak{gl}(V)$, the *trace form* of φ is the symmetric bilinear form $\beta_\varphi(x, y) = \operatorname{tr}(\varphi(x)\varphi(y))$ on $\mathfrak{g}$. The trace form of the adjoint representation is the *Killing form*
$$\kappa(x, y) = \kappa_{\mathfrak{g}}(x, y) = \operatorname{tr}(\operatorname{ad} x \circ \operatorname{ad} y).$$

Lemma 13.7.2. *Let $\mathfrak{g}$ be a Lie algebra.*

1. *The sum of two solvable ideals is a solvable ideal. Hence* $\mathfrak{g}$ *has a largest solvable ideal* $\operatorname{rad}\mathfrak{g}$*, and* $\mathfrak{g}/\operatorname{rad}\mathfrak{g}$ *is semisimple.*

2. $\mathfrak{g}$ *is semisimple if and only if it has no nonzero abelian ideal. In particular a semisimple Lie algebra has centre* 0*. A simple Lie algebra* $\mathfrak{g}$ *satisfies* $[\mathfrak{g},\mathfrak{g}] = \mathfrak{g}$ *and is semisimple.*

3. *Every trace form is* invariant*:* $\beta_\varphi([x,y],z) = \beta_\varphi(x,[y,z])$*. Consequently the orthogonal* $\mathfrak{a}^\perp = \{x : \beta_\varphi(x,\mathfrak{a}) = 0\}$ *of an ideal* $\mathfrak{a}$ *is an ideal.*

4. *If* $\mathfrak{a}$ *is an ideal of* $\mathfrak{g}$*, the Killing form of the Lie algebra* $\mathfrak{a}$ *is the restriction of* $\kappa_\mathfrak{g}$ *to* $\mathfrak{a}\times\mathfrak{a}$*.*

Proof. (1) Let $\mathfrak{a}$, $\mathfrak{b}$ be solvable ideals. Then $\mathfrak{a}+\mathfrak{b}$ is an ideal, and $(\mathfrak{a}+\mathfrak{b})/\mathfrak{a} \cong \mathfrak{b}/(\mathfrak{a}\cap\mathfrak{b})$ is a homomorphic image of $\mathfrak{b}$, hence solvable (Lemma 13.5.2(2)); so $\mathfrak{a}+\mathfrak{b}$ is solvable by Lemma 13.5.2(3). A solvable ideal of maximal dimension therefore contains every solvable ideal. If $\bar{\mathfrak{a}}$ is a solvable ideal of $\mathfrak{g}/\operatorname{rad}\mathfrak{g}$, its preimage $\mathfrak{a}$ is an ideal of $\mathfrak{g}$ with $\mathfrak{a}/\operatorname{rad}\mathfrak{g}$ solvable, hence solvable, so $\mathfrak{a}\subseteq\operatorname{rad}\mathfrak{g}$ and $\bar{\mathfrak{a}}=0$.

(2) Abelian ideals are solvable. Conversely, if $\mathfrak{r} = \operatorname{rad}\mathfrak{g} \neq 0$, let m be largest with $D^m\mathfrak{r}\neq 0$. By Lemma 13.5.2(1), applied repeatedly, each $D^i\mathfrak{r}$ is an ideal of $\mathfrak{g}$, and $[D^m\mathfrak{r}, D^m\mathfrak{r}] = D^{m+1}\mathfrak{r} = 0$; so $D^m\mathfrak{r}$ is a nonzero abelian ideal. The centre is an abelian ideal. If $\mathfrak{g}$ is simple, $[\mathfrak{g},\mathfrak{g}]$ is an ideal, nonzero as $\mathfrak{g}$ is not abelian, so $[\mathfrak{g},\mathfrak{g}]=\mathfrak{g}$; an abelian ideal is then 0, since $\mathfrak{g}$ itself is not abelian.

(3) For endomorphisms X, Y, Z, $\operatorname{tr}([X,Y]Z) = \operatorname{tr}(XYZ) - \operatorname{tr}(YXZ) = \operatorname{tr}(XYZ) - \operatorname{tr}(XZY) = \operatorname{tr}(X[Y,Z])$, using $\operatorname{tr}(AB) = \operatorname{tr}(BA)$. Apply this to $X=\varphi(x)$, $Y=\varphi(y)$, $Z=\varphi(z)$, and use $\varphi([x,y]) = [\varphi(x),\varphi(y)]$. If $z\in\mathfrak{a}^\perp$, $x\in\mathfrak{g}$ and $a\in\mathfrak{a}$, then $\beta_\varphi([x,z],a) = -\beta_\varphi([z,x],a) = -\beta_\varphi(z,[x,a]) = 0$, since $[x,a]\in\mathfrak{a}$.

(4) If an endomorphism ϕ of a finite-dimensional space U maps U into a subspace W, then $\operatorname{tr}\phi = \operatorname{tr}(\phi|_W)$: in a basis of U extending one of W, the rows of the matrix of ϕ belonging to the added vectors are zero. For $x,y\in\mathfrak{a}$, $\operatorname{ad}x\circ\operatorname{ad}y$ maps $\mathfrak{g}$ into $\mathfrak{a}$, and its restriction to $\mathfrak{a}$ is $\operatorname{ad}_\mathfrak{a} x\circ\operatorname{ad}_\mathfrak{a} y$. □

Remark 13.7.3. *Levi's theorem* says that the extension $0\to\operatorname{rad}\mathfrak{g}\to\mathfrak{g}\to\mathfrak{g}/\operatorname{rad}\mathfrak{g}\to 0$ splits: $\mathfrak{g}$ contains a semisimple subalgebra $\mathfrak{s}$ mapping isomorphically onto $\mathfrak{g}/\operatorname{rad}\mathfrak{g}$, so that $\mathfrak{g} = \operatorname{rad}\mathfrak{g}\oplus\mathfrak{s}$ as vector spaces, with $\mathfrak{s}$ acting on the ideal $\operatorname{rad}\mathfrak{g}$; see [FH91, Appendix E]. For example $\mathfrak{gl}_n(k) = k\cdot I \oplus \mathfrak{sl}_n(k)$, where the radical $k\cdot I$ is the centre (Example 13.7.9). We do not use Levi's theorem.

The key tool is the following refinement of the Jordan–Chevalley decomposition, and its consequence for traces.

Lemma 13.7.4. *Let* k *be algebraically closed,* V *finite-dimensional,* $x\in\mathfrak{gl}(V)$*, and* $x = s+n$ *its Jordan–Chevalley decomposition (Theorem 11.8.11).*

1. *There are polynomials* $p,q\in k[T]$ *without constant term such that* $s=p(x)$ *and* $n=q(x)$*. Hence if* $A\subseteq B\subseteq V$ *are subspaces with* $x(B)\subseteq A$*, then* $s(B)\subseteq A$ *and* $n(B)\subseteq A$*.*

2. *If* $v_1,\dots,v_m$ *is a basis of* V *with* $sv_i = a_iv_i$ *and* E_{ij} *are the corresponding matrix units* $(E_{ij}v_l = \delta_{jl}v_i)$*, then* $\operatorname{ad}s(E_{ij}) = (a_i-a_j)E_{ij}$*.*

3. $\operatorname{ad}x = \operatorname{ad}s + \operatorname{ad}n$ *is the Jordan–Chevalley decomposition of the endomorphism* $\operatorname{ad}x$ *of* $\mathfrak{gl}(V)$*.*

Proof. (1) In Step 1 of the proof of Theorem 11.8.11, $s=p(x)$ for any polynomial p with $p\equiv\lambda_i \bmod (T-\lambda_i)^{e_i}$ for all the distinct eigenvalues λ_i; by uniqueness (Step 2) this s is the semisimple part. If some λ_i is 0, this congruence already gives $p(0)=0$. Otherwise T is coprime to all $(T-\lambda_i)^{e_i}$, and the Chinese remainder theorem (Theorem 8.3.3) gives a p satisfying in addition $p\equiv 0 \bmod T$. Put $q = T-p$. If $x(B)\subseteq A\subseteq B$, then $x(A)\subseteq x(B)\subseteq A$,

so $x^j(B) \subseteq A$ for all $j \geq 1$, and the same holds for the polynomials $p(x)$ and $q(x)$ without constant term.

(2) $(sE_{ij} - E_{ij}s)v_l = \delta_{jl}(a_i - a_j)v_i$.

(3) Over an algebraically closed field semisimple endomorphisms are diagonalisable (Lemma 11.8.10), so a basis as in (2) exists, and by (2) $\operatorname{ad} s$ is diagonalisable in the basis (E_{ij}). The endomorphism $\operatorname{ad} n$ is nilpotent by Lemma 13.5.4, and $[\operatorname{ad} s, \operatorname{ad} n] = \operatorname{ad}[s, n] = 0$ by the Jacobi identity. By the uniqueness in Theorem 11.8.11, $\operatorname{ad} s$ and $\operatorname{ad} n$ are the semisimple and nilpotent parts of $\operatorname{ad} x = \operatorname{ad} s + \operatorname{ad} n$. □

Lemma 13.7.5. *Let k be algebraically closed, V finite-dimensional, $A \subseteq B$ subspaces of $\mathfrak{gl}(V)$, and $M = \{x \in \mathfrak{gl}(V) : [x, B] \subseteq A\}$. If $x \in M$ satisfies $\operatorname{tr}(xy) = 0$ for all $y \in M$, then x is nilpotent.*

Proof. Let $x = s + n$ be the Jordan–Chevalley decomposition and $v_1, \dots, v_m$ a basis with $sv_i = a_i v_i$. Let $E \subseteq k$ be the $\mathbb{Q}$-vector space spanned by $a_1, \dots, a_m$. We show $E = 0$; then all a_i vanish, $s = 0$, and $x = n$ is nilpotent. As E is finite-dimensional over $\mathbb{Q}$, it suffices to show that every $\mathbb{Q}$-linear map $f : E \to \mathbb{Q}$ is zero.

Given f, let $y \in \mathfrak{gl}(V)$ be defined by $yv_i = f(a_i)v_i$. By Lemma 13.7.4(2), $\operatorname{ad} y(E_{ij}) = (f(a_i) - f(a_j))E_{ij}$ and $\operatorname{ad} s(E_{ij}) = (a_i - a_j)E_{ij}$. The finitely many numbers $a_i - a_j$ and 0 can be assigned the values $f(a_i) - f(a_j)$ and 0 consistently: if $a_i - a_j = a_{i'} - a_{j'}$, then $f(a_i) - f(a_j) = f(a_i - a_j) = f(a_{i'}) - f(a_{j'})$ by $\mathbb{Q}$-linearity, and $a_i - a_j = 0$ gives the value 0. By Lagrange interpolation there is $r \in k[T]$ with $r(0) = 0$ and $r(a_i - a_j) = f(a_i) - f(a_j)$ for all i, j; then $\operatorname{ad} y = r(\operatorname{ad} s)$, since both are diagonal in the basis (E_{ij}) with the same entries. By Lemma 13.7.4(3) and (1), applied to $\operatorname{ad} x$, $\operatorname{ad} s = p(\operatorname{ad} x)$ for a polynomial p without constant term; so $\operatorname{ad} y = r(p(\operatorname{ad} x))$ is a polynomial in $\operatorname{ad} x$ without constant term. As $\operatorname{ad} x$ maps B into A, so does $\operatorname{ad} y$ (Lemma 13.7.4(1)), that is, $y \in M$, and the hypothesis gives $\operatorname{tr}(xy) = 0$.

Now $\operatorname{tr}(xy) = \operatorname{tr}(sy) + \operatorname{tr}(ny)$. In the basis (v_i), sy is diagonal with entries $a_i f(a_i)$. Interpolating again, $y = g(s)$ for a polynomial g with $g(a_i) = f(a_i)$, so y commutes with n, which commutes with s; hence ny is nilpotent ($(ny)^N = n^N y^N$) and has trace 0. Therefore $\sum_i a_i f(a_i) = 0$. The left side is an element of E, since $f(a_i) \in \mathbb{Q}$; applying f gives $\sum_i f(a_i)^2 = 0$, a sum of squares of rational numbers. So $f(a_i) = 0$ for all i, and $f = 0$. □

Theorem 13.7.6 (Cartan's criterion)**.** *Let V be a finite-dimensional vector space and $\mathfrak{g} \subseteq \mathfrak{gl}(V)$ a Lie subalgebra such that $\operatorname{tr}(xy) = 0$ for all $x \in [\mathfrak{g}, \mathfrak{g}]$ and $y \in \mathfrak{g}$. Then $\mathfrak{g}$ is solvable.*

Proof. Reduction to k algebraically closed. Choose a basis, so that $\mathfrak{g} \subseteq \mathfrak{gl}_d(k) \subseteq \mathfrak{gl}_d(\bar{k})$ for an algebraic closure $\bar{k}$ (Theorem 9.3.5), and let $\bar{\mathfrak{g}}$ be the $\bar{k}$-span of $\mathfrak{g}$. It is a Lie subalgebra, as the bracket is bilinear; $[\bar{\mathfrak{g}}, \bar{\mathfrak{g}}]$ is the $\bar{k}$-span of $[\mathfrak{g}, \mathfrak{g}]$, so the hypothesis holds for $\bar{\mathfrak{g}}$ by bilinearity of $\operatorname{tr}(xy)$. If $\bar{\mathfrak{g}}$ is solvable, so is its subalgebra $\mathfrak{g}$. So we may assume k algebraically closed.

The case k algebraically closed. It suffices to show that every $x \in [\mathfrak{g}, \mathfrak{g}]$ is a nilpotent endomorphism of V. Indeed, then $\operatorname{ad} x$ is nilpotent on $\mathfrak{gl}(V)$ (Lemma 13.5.4), hence on its invariant subspace $[\mathfrak{g}, \mathfrak{g}]$, so $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent by Engel's theorem (Corollary 13.5.6), and $\mathfrak{g}$ is solvable by Corollary 13.5.10(3). Apply Lemma 13.7.5 with $A = [\mathfrak{g}, \mathfrak{g}]$ and $B = \mathfrak{g}$: then $M = \{y : [y, \mathfrak{g}] \subseteq [\mathfrak{g}, \mathfrak{g}]\}$ contains $\mathfrak{g}$. Let $x \in [\mathfrak{g}, \mathfrak{g}]$, a sum of brackets $[u, v]$ with $u, v \in \mathfrak{g}$, and $y \in M$. By the identity $\operatorname{tr}([u, v]y) = \operatorname{tr}(u[v, y])$ of Lemma 13.7.2(3), and since $[v, y] = -[y, v] \in [\mathfrak{g}, \mathfrak{g}]$, the hypothesis gives $\operatorname{tr}([u, v]y) = 0$. So $\operatorname{tr}(xy) = 0$ for all $y \in M$, and x is nilpotent by the lemma. □

Corollary 13.7.7. *A Lie algebra $\mathfrak{g}$ is solvable if and only if $\kappa(x, y) = 0$ for all $x \in [\mathfrak{g}, \mathfrak{g}]$ and $y \in \mathfrak{g}$.*

Proof. Suppose the Killing form vanishes on $[\mathfrak{g}, \mathfrak{g}] \times \mathfrak{g}$. Since $[\operatorname{ad} \mathfrak{g}, \operatorname{ad} \mathfrak{g}] = \operatorname{ad}[\mathfrak{g}, \mathfrak{g}]$, Cartan's criterion applied to $\operatorname{ad} \mathfrak{g} \subseteq \mathfrak{gl}(\mathfrak{g})$ shows that $\operatorname{ad} \mathfrak{g}$ is solvable. The kernel of ad is the centre, which is abelian, and $\mathfrak{g}/\mathfrak{z}(\mathfrak{g}) \cong \operatorname{ad} \mathfrak{g}$; so $\mathfrak{g}$ is solvable by Lemma 13.5.2(3).

Conversely, let $\mathfrak{g}$ be solvable, and let L be the $\bar{k}$-span of $\operatorname{ad}\mathfrak{g}$ in $\mathfrak{gl}_d(\bar{k})$, using a basis of $\mathfrak{g}$ and an algebraic closure $\bar{k}$. Then D^iL is the $\bar{k}$-span of $D^i(\operatorname{ad}\mathfrak{g}) = \operatorname{ad}(D^i\mathfrak{g})$, so L is solvable. By Lie's theorem (Theorem 13.5.9) there is a basis of $\bar{k}^d$ in which all elements of L are upper triangular; then the elements of $[L, L]$ are strictly upper triangular, and so is the product of an element of $[L, L]$ with an element of L. For $x \in [\mathfrak{g}, \mathfrak{g}]$ and $y \in \mathfrak{g}$, $\operatorname{ad} x \in [L, L]$, so $\kappa(x, y) = \operatorname{tr}(\operatorname{ad} x \operatorname{ad} y) = 0$. □

Theorem 13.7.8 (Cartan's criterion for semisimplicity)**.** *A Lie algebra is semisimple if and only if its Killing form is nondegenerate.*

Proof. Let $S = \mathfrak{g}^\perp = \{x : \kappa(x, \mathfrak{g}) = 0\}$ be the radical of the Killing form; it is an ideal by Lemma 13.7.2(3). The Killing form of S is the restriction of κ (Lemma 13.7.2(4)), which vanishes on $S \times S$; by Corollary 13.7.7, S is solvable, so $S \subseteq \operatorname{rad}\mathfrak{g}$. Hence if $\mathfrak{g}$ is semisimple, $S = 0$.

Conversely, suppose $\mathfrak{g}$ is not semisimple, and let $\mathfrak{a} \neq 0$ be an abelian ideal (Lemma 13.7.2(2)). For $a \in \mathfrak{a}$ and $y \in \mathfrak{g}$, $\operatorname{ad} a \operatorname{ad} y$ maps $\mathfrak{g}$ into $\mathfrak{a}$, and maps $\mathfrak{a}$ into $[a, \mathfrak{a}] = 0$; so $(\operatorname{ad} a \operatorname{ad} y)^2 = 0$ and $\kappa(a, y) = 0$. Thus $0 \neq \mathfrak{a} \subseteq S$, and κ is degenerate. □

Example 13.7.9. 1. *$\mathfrak{gl}_n$ and $\mathfrak{sl}_n$.* For $x, y \in \mathfrak{gl}_n(k)$, $\operatorname{ad} x \operatorname{ad} y(z) = xyz - xzy - yzx + zyx$. The map $z \mapsto xyz$ is left multiplication by xy, which acts on each of the n columns of z by xy, so its trace is $n \operatorname{tr}(xy)$; similarly $z \mapsto zyx$ has trace $n \operatorname{tr}(yx)$, and $z \mapsto xzy$ sends E_{ij} to $\sum_{l,m} x_{li}y_{jm}E_{lm}$, whose diagonal coefficient at E_{ij} is $x_{ii}y_{jj}$, so its trace is $\operatorname{tr} x \operatorname{tr} y$, as is that of $z \mapsto yzx$. Hence

$$\kappa_{\mathfrak{gl}_n}(x, y) = 2n \operatorname{tr}(xy) - 2 \operatorname{tr} x \operatorname{tr} y.$$

It is degenerate: $\kappa(I, y) = 2n \operatorname{tr} y - 2n \operatorname{tr} y = 0$. Indeed $k \cdot I$ is an abelian ideal, the centre. For $x, y \in \mathfrak{sl}_n(k)$, $\operatorname{ad} x \operatorname{ad} y$ maps $\mathfrak{gl}_n$ into $[\mathfrak{gl}_n, \mathfrak{gl}_n] \subseteq \mathfrak{sl}_n$, so by the argument of Lemma 13.7.2(4) its trace on $\mathfrak{gl}_n$ equals its trace on $\mathfrak{sl}_n$:

$$\kappa_{\mathfrak{sl}_n}(x, y) = 2n \operatorname{tr}(xy).$$

This form is nondegenerate for $n \geq 2$: if $x \in \mathfrak{sl}_n$ and $\operatorname{tr}(xy) = 0$ for all $y \in \mathfrak{sl}_n$, then $y = E_{ji}$ $(i \neq j)$ gives $x_{ij} = 0$, and $y = E_{ii} - E_{jj}$ gives $x_{ii} = x_{jj}$; so $x = cI$ with $nc = \operatorname{tr} x = 0$, and $x = 0$. Hence $\mathfrak{sl}_n(k)$ is semisimple. For $n = 2$, in the basis e, h, f of Section 13.6, $\kappa(e, f) = 4$, $\kappa(h, h) = 8$, and all other pairings of basis vectors vanish.

2. *A solvable example.* For the Lie algebra with basis x, y and $[x, y] = y$ (Example 13.5.3(3)), the matrices of $\operatorname{ad} x$ and $\operatorname{ad} y$ in the basis (x, y) are $\left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 0 & 0 \\ -1 & 0 \end{smallmatrix}\right)$, so $\kappa(x, x) = 1$ and $\kappa(x, y) = \kappa(y, y) = 0$. The form vanishes on $[\mathfrak{g}, \mathfrak{g}] \times \mathfrak{g} = ky \times \mathfrak{g}$, as Corollary 13.7.7 predicts, but it is not zero: the Killing form of a solvable Lie algebra need not vanish, in contrast with the nilpotent case (Exercise 13.12.8).

Theorem 13.7.10. *Let $\mathfrak{g}$ be semisimple. There are simple ideals $\mathfrak{g}_1, \ldots, \mathfrak{g}_t$ with $\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_t$, and $[\mathfrak{g}_i, \mathfrak{g}_j] = 0$ for $i \neq j$. Every simple ideal of $\mathfrak{g}$ is one of the $\mathfrak{g}_i$, and every ideal is a sum of some of them. In particular $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$, and ideals and quotients of $\mathfrak{g}$ are semisimple.*

Proof. Let $\mathfrak{a}$ be an ideal. Then $\mathfrak{a}^\perp$, the orthogonal for κ, is an ideal (Lemma 13.7.2(3)), and so is $\mathfrak{a} \cap \mathfrak{a}^\perp$, on which the Killing form of $\mathfrak{g}$ vanishes. By Lemma 13.7.2(4) and Corollary 13.7.7, $\mathfrak{a} \cap \mathfrak{a}^\perp$ is solvable, hence 0. Since κ is nondegenerate (Theorem 13.7.8), $\mathfrak{a}^\perp$ is the kernel of the map $\mathfrak{g} \to \mathfrak{a}^*$, $x \mapsto \kappa(x, -)|_{\mathfrak{a}}$, so $\dim \mathfrak{a}^\perp \geq \dim \mathfrak{g} - \dim \mathfrak{a}$. Hence $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{a}^\perp$, and $[\mathfrak{a}, \mathfrak{a}^\perp] \subseteq \mathfrak{a} \cap \mathfrak{a}^\perp = 0$. Consequently an ideal $\mathfrak{b}$ of the Lie algebra $\mathfrak{a}$ is an ideal of $\mathfrak{g}$, since $[\mathfrak{g}, \mathfrak{b}] = [\mathfrak{a}, \mathfrak{b}] + [\mathfrak{a}^\perp, \mathfrak{b}] = [\mathfrak{a}, \mathfrak{b}] \subseteq \mathfrak{b}$. In particular a solvable ideal of $\mathfrak{a}$ is a solvable ideal of $\mathfrak{g}$, hence 0: $\mathfrak{a}$ is semisimple, and so is $\mathfrak{g}/\mathfrak{a} \cong \mathfrak{a}^\perp$.

Existence of the decomposition, by induction on $\dim \mathfrak{g}$: if $\mathfrak{g} \neq 0$, let $\mathfrak{a}$ be a nonzero ideal of minimal dimension. Its ideals are ideals of $\mathfrak{g}$, so they are 0 and $\mathfrak{a}$; and $\mathfrak{a}$ is not abelian, being semisimple and nonzero. So $\mathfrak{a}$ is simple, and $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{a}^\perp$ with $\mathfrak{a}^\perp$ semisimple, to which the induction hypothesis applies; the simple ideals of $\mathfrak{a}^\perp$ are ideals of $\mathfrak{g}$.

Let $\mathfrak{b}$ be a simple ideal of $\mathfrak{g}$. Then $[\mathfrak{g}, \mathfrak{b}] = \mathfrak{b}$, since it is a nonzero ideal of $\mathfrak{b}$ (nonzero because the centre of $\mathfrak{g}$ is 0). So $\mathfrak{b} = \sum_i [\mathfrak{g}_i, \mathfrak{b}]$, and each $[\mathfrak{g}_i, \mathfrak{b}] \subseteq \mathfrak{g}_i \cap \mathfrak{b}$ is an ideal of $\mathfrak{g}$ contained in $\mathfrak{b}$, hence 0 or $\mathfrak{b}$. As the $\mathfrak{g}_i$ are independent, exactly one is $\mathfrak{b}$, and then $\mathfrak{b} \subseteq \mathfrak{g}_i$ is a nonzero ideal of the simple $\mathfrak{g}_i$, so $\mathfrak{b} = \mathfrak{g}_i$. An arbitrary ideal $\mathfrak{a}$ is semisimple, so by the existence part it is a sum of simple ideals of $\mathfrak{a}$, which are simple ideals of $\mathfrak{g}$, hence some of the $\mathfrak{g}_i$. Finally $[\mathfrak{g}, \mathfrak{g}] = \sum_i [\mathfrak{g}_i, \mathfrak{g}_i] = \sum_i \mathfrak{g}_i$ by Lemma 13.7.2(2). □

A *derivation* of a Lie algebra $\mathfrak{g}$ is a linear map $\delta \colon \mathfrak{g} \to \mathfrak{g}$ with $\delta[x, y] = [\delta x, y] + [x, \delta y]$. The derivations form a Lie subalgebra $\operatorname{Der} \mathfrak{g} \subseteq \mathfrak{gl}(\mathfrak{g})$, and by the Jacobi identity every $\operatorname{ad} x$ is a derivation, called *inner*. For $\delta \in \operatorname{Der} \mathfrak{g}$ and $x \in \mathfrak{g}$,

$$[\delta, \operatorname{ad} x] = \operatorname{ad}(\delta x),$$

since $[\delta, \operatorname{ad} x](y) = \delta[x, y] - [x, \delta y] = [\delta x, y]$; so $\operatorname{ad} \mathfrak{g}$ is an ideal of $\operatorname{Der} \mathfrak{g}$.

Theorem 13.7.11. *If $\mathfrak{g}$ is semisimple, every derivation of $\mathfrak{g}$ is inner, and $\operatorname{ad} \colon \mathfrak{g} \to \operatorname{Der} \mathfrak{g}$ is an isomorphism of Lie algebras.*

Proof. ad is injective, as the centre is 0. Let $D = \operatorname{Der} \mathfrak{g}$ and $I = \operatorname{ad} \mathfrak{g} \cong \mathfrak{g}$, an ideal of D. The Killing form of I is the restriction of κ_D (Lemma 13.7.2(4)), and it is nondegenerate since $I \cong \mathfrak{g}$ is semisimple; so $I \cap I^\perp = 0$, where $I^\perp$ is the orthogonal of I for κ_D, an ideal of D. Then $[I, I^\perp] \subseteq I \cap I^\perp = 0$, and for $\delta \in I^\perp$ and $x \in \mathfrak{g}$, $\operatorname{ad}(\delta x) = [\delta, \operatorname{ad} x] = 0$, so $\delta x = 0$: $I^\perp = 0$. As in the proof of Theorem 13.7.10, $\dim I^\perp \geq \dim D - \dim I$, so $D = I$. □

Proposition 13.7.12 (Abstract Jordan decomposition)**.** *Let k be algebraically closed and $\mathfrak{g}$ semisimple. Every $x \in \mathfrak{g}$ can be written uniquely as $x = x_s + x_n$ with $\operatorname{ad} x_s$ diagonalisable, $\operatorname{ad} x_n$ nilpotent and $[x_s, x_n] = 0$. Moreover $\operatorname{ad} x_s$ and $\operatorname{ad} x_n$ are the semisimple and nilpotent parts of $\operatorname{ad} x$, and every $y \in \mathfrak{g}$ commuting with x commutes with x_s and x_n.*

Proof. Let $\operatorname{ad} x = \sigma + \nu$ be the Jordan–Chevalley decomposition in $\operatorname{End}(\mathfrak{g})$. For $a \in k$ let $\mathfrak{g}_a = \ker(\operatorname{ad} x - a)^d$, $d = \dim \mathfrak{g}$; then $\mathfrak{g} = \bigoplus_a \mathfrak{g}_a$ and $\sigma = a$ on $\mathfrak{g}_a$ (Step 1 of the proof of Theorem 11.8.11). Since $\operatorname{ad} x$ is a derivation, $(\operatorname{ad} x - a - b)[y, z] = [(\operatorname{ad} x - a)y, z] + [y, (\operatorname{ad} x - b)z]$, and by induction

$$(\operatorname{ad} x - a - b)^m [y, z] = \sum_{i=0}^{m} \binom{m}{i} \big[(\operatorname{ad} x - a)^i y, (\operatorname{ad} x - b)^{m-i} z\big].$$

For $y \in \mathfrak{g}_a$, $z \in \mathfrak{g}_b$ and $m = 2d$ every term vanishes, so $[y, z] \in \mathfrak{g}_{a+b}$, and $\sigma[y, z] = (a+b)[y, z] = [\sigma y, z] + [y, \sigma z]$. By bilinearity σ is a derivation, so $\sigma = \operatorname{ad} x_s$ for a unique x_s (Theorem 13.7.11). Put $x_n = x - x_s$; then $\operatorname{ad} x_n = \nu$ is nilpotent, and $\operatorname{ad}[x_s, x_n] = [\sigma, \nu] = 0$ gives $[x_s, x_n] = 0$, as ad is injective. If $x = s' + n'$ is another such decomposition, then $\operatorname{ad} x = \operatorname{ad} s' + \operatorname{ad} n'$ is a Jordan–Chevalley decomposition, so $\operatorname{ad} s' = \sigma$ and $s' = x_s$. Finally, if $[x, y] = 0$, then $\operatorname{ad} y$ commutes with $\operatorname{ad} x$, hence with the polynomial σ in $\operatorname{ad} x$; so $\operatorname{ad}[y, x_s] = [\operatorname{ad} y, \sigma] = 0$ and $[y, x_s] = 0$. □

We call x *semisimple* if $x = x_s$, that is, if $\operatorname{ad} x$ is diagonalisable, and *nilpotent* if $x = x_n$. For $\mathfrak{g} \subseteq \mathfrak{gl}(V)$ this agrees with the usual Jordan decomposition of the matrix x (Corollary 13.8.4 below), but that requires Weyl's theorem.

To recognise semisimple Lie algebras of matrices, the following criterion is often quicker than computing the Killing form. A Lie algebra is *reductive* if it is the direct sum $\mathfrak{z}(\mathfrak{g}) \oplus \mathfrak{s}$ of its centre and a semisimple ideal $\mathfrak{s}$.

Proposition 13.7.13. *Let $k = \mathbb{R}$ or $\mathbb{C}$, and let $\mathfrak{g}$ be a Lie algebra over k with an inner product (Hermitian if $k = \mathbb{C}$) such that the adjoint of every $\operatorname{ad} x$ is of the form $\operatorname{ad} x'$ for some $x' \in \mathfrak{g}$. Then the orthogonal complement of an ideal is an ideal, $\mathfrak{g}$ is the orthogonal direct sum of its centre and of simple ideals, and $\mathfrak{g} = \mathfrak{z}(\mathfrak{g}) \oplus [\mathfrak{g}, \mathfrak{g}]$ with $[\mathfrak{g}, \mathfrak{g}]$ semisimple. In particular $\mathfrak{g}$ is reductive, and it is semisimple if its centre is 0.*

Proof. Let $\mathfrak{a}$ be an ideal, $z \in \mathfrak{a}^\perp$, $x \in \mathfrak{g}$ and $a \in \mathfrak{a}$. With $(\operatorname{ad} x)^* = \operatorname{ad} x'$, $\langle [x, z], a\rangle = \langle z, [x', a]\rangle = 0$, since $[x', a] \in \mathfrak{a}$. So $\mathfrak{a}^\perp$ is an ideal, and $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{a}^\perp$.

Every ideal $\mathfrak{b}$ is an orthogonal direct sum of minimal nonzero ideals of $\mathfrak{g}$, by induction on $\dim \mathfrak{b}$: if $\mathfrak{b} \neq 0$, it contains a minimal nonzero ideal $\mathfrak{a}$ of $\mathfrak{g}$, and $\mathfrak{b} = \mathfrak{a} \oplus (\mathfrak{b} \cap \mathfrak{a}^\perp)$, because for $y \in \mathfrak{b}$ the component of y in $\mathfrak{a}^\perp$ is y minus an element of $\mathfrak{a} \subseteq \mathfrak{b}$. So $\mathfrak{g} = \mathfrak{a}_1 \oplus \cdots \oplus \mathfrak{a}_m$ with minimal nonzero ideals $\mathfrak{a}_i$, and $[\mathfrak{a}_i, \mathfrak{a}_j] \subseteq \mathfrak{a}_i \cap \mathfrak{a}_j = 0$ for $i \neq j$. Hence an ideal $\mathfrak{c}$ of the Lie algebra $\mathfrak{a}_i$ is an ideal of $\mathfrak{g}$, as $[\mathfrak{g}, \mathfrak{c}] = [\mathfrak{a}_i, \mathfrak{c}]$, and by minimality $\mathfrak{a}_i$ has no ideals but 0 and itself. So each $\mathfrak{a}_i$ is either simple or abelian, and in the abelian case every subspace is an ideal, so $\dim \mathfrak{a}_i = 1$.

An element $\sum_i x_i$, $x_i \in \mathfrak{a}_i$, is central if and only if each x_i is central in $\mathfrak{a}_i$; the centre of a simple Lie algebra is 0, so $\mathfrak{z}(\mathfrak{g})$ is the sum of the abelian $\mathfrak{a}_i$. And $[\mathfrak{g}, \mathfrak{g}] = \sum_i [\mathfrak{a}_i, \mathfrak{a}_i]$ is the sum $\mathfrak{s}$ of the simple $\mathfrak{a}_i$ (Lemma 13.7.2(2)). It remains to see that $\mathfrak{s}$ is semisimple. If $\mathfrak{r}$ is an abelian ideal of $\mathfrak{s}$ and $\mathfrak{a}_i$ is one of its simple summands, then $\mathfrak{a}_i \cap \mathfrak{r}$ is an abelian ideal of $\mathfrak{a}_i$, hence 0, and so $[\mathfrak{a}_i, \mathfrak{r}] \subseteq \mathfrak{a}_i \cap \mathfrak{r} = 0$. Thus $\mathfrak{r}$ is central in $\mathfrak{s}$, whose centre is 0; so $\mathfrak{r} = 0$, and $\mathfrak{s}$ is semisimple by Lemma 13.7.2(2). □

The hypothesis holds for every complex Lie subalgebra $\mathfrak{g} \subseteq \mathfrak{gl}_n(\mathbb{C})$ that is closed under $x \mapsto x^* = \bar{x}^T$, with the Hermitian inner product $\langle x, y\rangle = \operatorname{tr}(xy^*) = \sum_{i,j} x_{ij}\bar{y}_{ij}$: then $(\operatorname{ad} x)^* = \operatorname{ad}(x^*)$, because

$$\langle [x, y], z\rangle = \operatorname{tr}(xyz^* - yxz^*) = \operatorname{tr}\big(y(z^*x - xz^*)\big) = \operatorname{tr}\big(y[x^*, z]^*\big) = \langle y, [x^*, z]\rangle.$$

It also holds for the Lie algebra of a compact Lie group, with an invariant inner product (Chapter 34).

Example 13.7.14. *The classical Lie algebras.* Let $J = \left(\begin{smallmatrix} 0 & I_n \\ -I_n & 0 \end{smallmatrix}\right)$ and let $\mathfrak{sp}_{2n}(k) = \{x \in \mathfrak{gl}_{2n}(k) : x^TJ + Jx = 0\}$, the Lie algebra of the bilinear form v^TJw in the sense of Example 13.4.2(3). Writing x in blocks of size n, $x^TJ + Jx = 0$ says exactly that

$$x = \begin{pmatrix} A & B \\ C & -A^T \end{pmatrix}, \qquad B = B^T, \quad C = C^T,$$

so $\dim \mathfrak{sp}_{2n} = n^2 + n(n+1) = n(2n+1)$. The Lie algebras $\mathfrak{sl}_n(\mathbb{C})$ ($n \geq 2$), $\mathfrak{so}_n(\mathbb{C})$ ($n \geq 3$) and $\mathfrak{sp}_{2n}(\mathbb{C})$ ($n \geq 1$) are semisimple, and $\mathfrak{gl}_n(\mathbb{C}) = \mathbb{C}I \oplus \mathfrak{sl}_n(\mathbb{C})$ is reductive. All four are closed under $x \mapsto x^*$: for $\mathfrak{so}_n$ the conjugate of a skew-symmetric matrix is skew-symmetric, and for $\mathfrak{sp}_{2n}$, conjugating $x^TJ + Jx = 0$ and multiplying by $J^{-1} = -J$ on both sides gives $\bar{x}J + J\bar{x}^T = 0$, which is the condition for x^*. By Proposition 13.7.13 it suffices to compute the centres.

1. $\mathfrak{sl}_n$: if x commutes with E_{ij}, $i \neq j$, comparing the entries in row i of $E_{ij}x = xE_{ij}$ gives $x_{jl} = 0$ for $l \neq j$ and $x_{jj} = x_{ii}$. So a central x is a scalar matrix cI with $nc = \operatorname{tr} x = 0$, and $x = 0$. The centre of $\mathfrak{gl}_n$ is $\mathbb{C}I$ by the same computation.

2. $\mathfrak{so}_n$, $n \geq 3$: a central x commutes with $F_{ij} = E_{ij} - E_{ji}$, so it preserves its image $\operatorname{span}(e_i, e_j)$. For distinct i, j, l it preserves $\operatorname{span}(e_i, e_j) \cap \operatorname{span}(e_i, e_l) = \mathbb{C}e_i$; so x is diagonal, and a diagonal skew-symmetric matrix is 0. (For $n = 2$, $\mathfrak{so}_2$ is one-dimensional, hence abelian.)

3. $\mathfrak{sp}_{2n}$: write a central x in blocks P, Q, R, S. Commuting with all $\operatorname{diag}(A, -A^T) \in \mathfrak{sp}_{2n}$ means $PA = AP$, $QA^T = -AQ$, $RA = -A^T R$ and $SA^T = A^T S$ for all $A \in \mathfrak{gl}_n$; so P and S are scalar, and $A = I$ gives $Q = R = 0$. Then $x = p\operatorname{diag}(I, -I)$, and commuting with $\left(\begin{smallmatrix} 0 & I \\ 0 & 0 \end{smallmatrix}\right) \in \mathfrak{sp}_{2n}$ gives $2p = 0$.

The real Lie algebras $\mathfrak{su}(n)$ ($n \geq 2$) and $\mathfrak{so}(n)$ ($n \geq 3$) are semisimple for the same reason, with the real inner product $\operatorname{Re}\operatorname{tr}(xy^*)$ and $x^* = -x$. These are the Lie algebras $\mathfrak{sl}_n$ ($n \geq 2$), $\mathfrak{so}_{2n+1}$ ($n \geq 1$), $\mathfrak{sp}_{2n}$ ($n \geq 1$) and $\mathfrak{so}_{2n}$ ($n \geq 2$) of the four infinite families in the classification (Section 13.10).

13.8 Weyl's theorem

For $\mathfrak{sl}_2$, complete reducibility was proved with the Casimir operator $C = ef + fe + \frac{1}{2}h^2$, which commutes with the action and separates the irreducible representations (Theorem 13.6.3). The same element exists for every semisimple Lie algebra; it is built from a nondegenerate invariant form, and it gives complete reducibility in general. In this section $\mathfrak{g}$ is semisimple and k has characteristic 0.

Lemma 13.8.1 (Casimir element)**.** *Let $\varphi\colon \mathfrak{g} \to \mathfrak{gl}(V)$ be a faithful (injective) representation. Then the trace form $\beta = \beta_\varphi$ is nondegenerate. Let $x_1, \dots, x_n$ be a basis of $\mathfrak{g}$ and $y_1, \dots, y_n$ the dual basis, $\beta(x_i, y_j) = \delta_{ij}$, and put*

$$c_\varphi = \sum_{i=1}^{n} \varphi(x_i)\varphi(y_i) \in \operatorname{End}(V).$$

Then c_φ commutes with $\varphi(x)$ for all $x \in \mathfrak{g}$, and $\operatorname{tr} c_\varphi = \dim \mathfrak{g}$.

Proof. The radical S of β is an ideal (Lemma 13.7.2(3)), and $\operatorname{tr}(\varphi(x)\varphi(y)) = 0$ for $x, y \in S$; by Cartan's criterion (Theorem 13.7.6) $\varphi(S) \cong S$ is solvable, so $S = 0$. For $x \in \mathfrak{g}$ write $[x, x_i] = \sum_j a_{ij} x_j$ and $[x, y_i] = \sum_j b_{ij} y_j$. By invariance, $a_{il} = \beta([x, x_i], y_l) = -\beta([x_i, x], y_l) = -\beta(x_i, [x, y_l]) = -b_{li}$. Using $[X, YZ] = [X, Y]Z + Y[X, Z]$ in $\operatorname{End}(V)$,

$$[\varphi(x), c_\varphi] = \sum_i \big([\varphi(x), \varphi(x_i)]\varphi(y_i) + \varphi(x_i)[\varphi(x), \varphi(y_i)]\big) = \sum_{i,j} a_{ij}\varphi(x_j)\varphi(y_i) + \sum_{i,j} b_{ij}\varphi(x_i)\varphi(y_j),$$

and renaming the indices in the second sum, it becomes $\sum_{i,j} b_{ji}\varphi(x_j)\varphi(y_i) = -\sum_{i,j} a_{ij}\varphi(x_j)\varphi(y_i)$. Finally $\operatorname{tr} c_\varphi = \sum_i \beta(x_i, y_i) = n$. □

If φ is not faithful, its kernel is an ideal, so $\mathfrak{g} = \ker\varphi \oplus \mathfrak{g}'$ where $\mathfrak{g}'$ is the sum of the other simple ideals (Theorem 13.7.10); $\varphi|_{\mathfrak{g}'}$ is faithful with the same image, and we define $c_\varphi = c_{\varphi|\mathfrak{g}'}$. It commutes with $\varphi(\mathfrak{g})$, and its trace is $\dim \varphi(\mathfrak{g})$. For $\mathfrak{sl}_2$ acting on k^2 the trace form is $\operatorname{tr}(xy)$, the dual basis of e, f, h is $f, e, \frac{1}{2}h$, and $c_\varphi = ef + fe + \frac{1}{2}h^2$ is the Casimir operator of Theorem 13.6.3.

Theorem 13.8.2 (Weyl)**.** *Every finite-dimensional representation of a semisimple Lie algebra over a field of characteristic 0 is completely reducible.*

Proof. It suffices to show that every subrepresentation W of a representation V has a complementary subrepresentation; complete reducibility then follows by induction on $\dim V$. First note that $\mathfrak{g}$ acts by 0 on every one-dimensional representation, since $\mathfrak{gl}_1$ is abelian and $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$.

Step 1: W of codimension 1. We argue by induction on $\dim V$. By the remark just made, $\mathfrak{g}V \subseteq W$. If $\varphi = 0$, any complement will do. Suppose first that W is irreducible, and let $c = c_\varphi$.

It maps V into W, being a sum of products of elements of $\varphi(\mathfrak{g})$, so $\operatorname{tr}(c|_W) = \operatorname{tr} c = \dim \varphi(\mathfrak{g}) \neq 0$ (Lemma 13.7.2(4)). Since c commutes with $\mathfrak{g}$, $\ker(c|_W)$ is a subrepresentation of W, not equal to W, hence 0. So $\ker c \cap W = 0$, while $\dim \ker c \geq \dim V - \dim W = 1$ because $c(V) \subseteq W$; hence $V = W \oplus \ker c$, and $\ker c$ is a subrepresentation. If W is not irreducible, let $0 \neq W' \subsetneq W$ be a subrepresentation. By induction, applied to $W/W' \subseteq V/W'$, there is a subrepresentation $\widetilde{W} \supseteq W'$ with $V/W' = W/W' \oplus \widetilde{W}/W'$ and $\dim \widetilde{W}/W' = 1$. By induction again, applied to $W' \subseteq \widetilde{W}$ (of codimension 1, and $\dim \widetilde{W} < \dim V$), $\widetilde{W} = W' \oplus X$ with X a subrepresentation of dimension 1. Then $X \cap W \subseteq \widetilde{W} \cap W = W'$, so $X \cap W = X \cap W' = 0$, and $V = W \oplus X$.

Step 2: general $W \neq 0$. The space $\operatorname{Hom}_k(V, W)$ is a representation with $(x \cdot f)(v) = xf(v) - f(xv)$ (it is $V^* \otimes W$, Definition 13.4.3). Let $\mathcal{V}$ be the subspace of maps f whose restriction to W is a scalar multiple $\lambda_f \mathrm{id}_W$, and $\mathcal{W} \subseteq \mathcal{V}$ those with $f|_W = 0$. For $f \in \mathcal{V}$ and $w \in W$, $(x \cdot f)(w) = \lambda_f xw - \lambda_f xw = 0$, so $\mathfrak{g}\mathcal{V} \subseteq \mathcal{W}$; in particular both are subrepresentations, and $\mathcal{W}$ has codimension 1 in $\mathcal{V}$, because $f \mapsto \lambda_f$ is onto (extend id_W by a linear projection). By Step 1, $\mathcal{V} = \mathcal{W} \oplus kf$ with kf a subrepresentation, on which $\mathfrak{g}$ acts by 0. We may take $\lambda_f = 1$. Then $x \cdot f = 0$ says $xf(v) = f(xv)$: $f : V \to W$ is a homomorphism of representations which is the identity on W, and $\ker f$ is a subrepresentation complementary to W. □

Counterexample 13.8.3. Semisimplicity is needed. The one-dimensional Lie algebra k acts on k^2 by $t \mapsto \left(\begin{smallmatrix} 0 & t \\ 0 & 0 \end{smallmatrix}\right)$; the line ke_1 is a subrepresentation, and a complementary one would be a second line killed by the nilpotent matrix, whose kernel is ke_1. The same happens for the adjoint representation of the solvable Lie algebra with $[x, y] = y$: the ideal ky has no complementary ideal, as a complement would be a line $k(x + cy)$ with $[y, x + cy] = -y$ in it.

Corollary 13.8.4 (Preservation of the Jordan decomposition)**.** *Let k be algebraically closed.*

1. *If $\mathfrak{g} \subseteq \mathfrak{gl}(V)$ is a semisimple Lie subalgebra and $x \in \mathfrak{g}$, the semisimple and nilpotent parts s, n of the endomorphism x lie in $\mathfrak{g}$, and they are the parts x_s, x_n of the abstract Jordan decomposition.*

2. *If $\varphi : \mathfrak{g} \to \mathfrak{gl}(V)$ is a representation of a semisimple Lie algebra, then $\varphi(x_s)$ and $\varphi(x_n)$ are the semisimple and nilpotent parts of $\varphi(x)$. In particular a semisimple element acts diagonalisably in every finite-dimensional representation.*

Proof. (1) By Lemma 13.7.4, $\operatorname{ad} s$ and $\operatorname{ad} n$ are polynomials without constant term in $\operatorname{ad} x$, which maps $\mathfrak{g}$ into itself; so s and n lie in the normaliser $N = \{y \in \mathfrak{gl}(V) : [y, \mathfrak{g}] \subseteq \mathfrak{g}\}$, a Lie subalgebra. For each subrepresentation $W \subseteq V$ let $L_W = \{y \in \mathfrak{gl}(V) : y(W) \subseteq W,\ \operatorname{tr}(y|_W) = 0\}$, also a Lie subalgebra. It contains $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$, as the restrictions of commutators are commutators, and it contains s and n: these are polynomials in x, so they preserve W, $n|_W$ is nilpotent, so of trace 0, and $s|_W = x|_W - n|_W$. Let $\mathfrak{g}' = N \cap \bigcap_W L_W$; it contains $\mathfrak{g}$, s and n, and $\mathfrak{g}$ is an ideal of $\mathfrak{g}'$. By Weyl's theorem, applied to the representation ad of $\mathfrak{g}$ on $\mathfrak{g}'$, $\mathfrak{g}' = \mathfrak{g} \oplus M$ with $[\mathfrak{g}, M] \subseteq M$; also $[\mathfrak{g}, M] \subseteq \mathfrak{g}$, so $[\mathfrak{g}, M] = 0$. Let $y \in M$ and let $W \subseteq V$ be an irreducible subrepresentation. Then $y|_W$ commutes with $\mathfrak{g}$, so it is a scalar by Schur's lemma (Lemma 13.2.3, whose proof applies to Lie algebras), and it has trace 0, so $y|_W = 0$. By Weyl's theorem V is a sum of irreducible subrepresentations, so $y = 0$. Hence $\mathfrak{g}' = \mathfrak{g}$ contains s and n. Now $\operatorname{ad}_{\mathfrak{g}} s$ is diagonalisable, as the restriction of the diagonalisable $\operatorname{ad} s$ to an invariant subspace, $\operatorname{ad}_{\mathfrak{g}} n$ is nilpotent, and $[s, n] = 0$; by uniqueness in Proposition 13.7.12, $s = x_s$ and $n = x_n$.

(2) The image $\varphi(\mathfrak{g}) \cong \mathfrak{g}/\ker\varphi$ is semisimple (Theorem 13.7.10). $\mathfrak{g}$ has a basis of eigenvectors of $\operatorname{ad} x_s$, and φ maps it onto a spanning set of $\varphi(\mathfrak{g})$ consisting of eigenvectors of $\operatorname{ad}\varphi(x_s)$; so $\operatorname{ad}\varphi(x_s)$ is diagonalisable on $\varphi(\mathfrak{g})$. Similarly $(\operatorname{ad}\varphi(x_n))^N \varphi(y) = \varphi((\operatorname{ad} x_n)^N y) = 0$ for large N, and $[\varphi(x_s), \varphi(x_n)] = 0$. By uniqueness, $\varphi(x) = \varphi(x_s) + \varphi(x_n)$ is the abstract Jordan decomposition in $\varphi(\mathfrak{g})$, and by (1) it is the Jordan decomposition of the endomorphism $\varphi(x)$. □

13.9 The root space decomposition

In $\mathfrak{sl}_n$ the diagonal matrices $\mathfrak{h}$ act on $\mathfrak{sl}_n$ by ad diagonalisably: $[h, E_{ij}] = (h_i - h_j)E_{ij}$ for $h = \operatorname{diag}(h_1, \dots, h_n)$. So $\mathfrak{sl}_n$ is the sum of $\mathfrak{h}$ and of the lines $\mathbb{C}E_{ij}$, on which $\mathfrak{h}$ acts through the linear forms $h \mapsto h_i - h_j$, the *roots*. This section shows that every semisimple Lie algebra has such a decomposition, and that the roots satisfy strong integrality conditions, all coming from the representation theory of $\mathfrak{sl}_2$. In this section $k = \mathbb{C}$ (the proofs work over any algebraically closed field of characteristic 0), and $\mathfrak{g}$ is semisimple, with Killing form κ.

Definition 13.9.1. A subalgebra $\mathfrak{t} \subseteq \mathfrak{g}$ is *toral* if all its elements are semisimple, that is, every $\operatorname{ad} x$, $x \in \mathfrak{t}$, is diagonalisable. A *Cartan subalgebra* of $\mathfrak{g}$ is a maximal toral subalgebra (one not contained in a larger toral subalgebra).

For semisimple Lie algebras over $\mathbb{C}$ this agrees with the general definition of a Cartan subalgebra as a nilpotent subalgebra equal to its own normaliser [Hum72, Section 15].

Lemma 13.9.2. *Toral subalgebras are abelian, and if $\mathfrak{g} \neq 0$ it has nonzero toral subalgebras. Hence Cartan subalgebras exist, and they are abelian and nonzero.*

Proof. Let $\mathfrak{t}$ be toral and $x \in \mathfrak{t}$. Since $\operatorname{ad}_{\mathfrak{t}} x$ is the restriction of the diagonalisable $\operatorname{ad} x$ to an invariant subspace, it is diagonalisable, and it suffices to show that it has no nonzero eigenvalue. Suppose $[x, y] = ay$ with $y \in \mathfrak{t}$ and $a \neq 0$. Then $\operatorname{ad}_{\mathfrak{t}} y(x) = -ay$ is a nonzero eigenvector of $\operatorname{ad}_{\mathfrak{t}} y$ with eigenvalue 0. But $\operatorname{ad}_{\mathfrak{t}} y$ is diagonalisable: writing $x = \sum_i v_i$ with $[y, v_i] = \lambda_i v_i$, the vector $[y, x] = \sum_{\lambda_i \neq 0} \lambda_i v_i$ lies in the sum of the eigenspaces with nonzero eigenvalues, which meets the kernel only in 0; contradiction.

If every $x \in \mathfrak{g}$ satisfied $x_s = 0$, every $\operatorname{ad} x$ would be nilpotent, and $\mathfrak{g}$ would be nilpotent by Engel's theorem (Corollary 13.5.6), hence solvable, hence 0. So some x_s is nonzero, and $\mathbb{C}x_s$ is toral. A toral subalgebra of maximal dimension is maximal. □

Fix a Cartan subalgebra $\mathfrak{h}$. Its elements commute, so the operators $\operatorname{ad} h$, $h \in \mathfrak{h}$, are commuting diagonalisable endomorphisms of $\mathfrak{g}$, and they are simultaneously diagonalisable: apply Theorem 11.7.6 to $\operatorname{ad} h_1$, then to the restrictions of $\operatorname{ad} h_2$ to its eigenspaces, and so on, for a basis $h_1, \dots, h_l$ of $\mathfrak{h}$. A common eigenvector x satisfies $[h, x] = \alpha(h)x$ for a linear form $\alpha \in \mathfrak{h}^*$. Hence

$$\mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_\alpha, \qquad \mathfrak{g}_\alpha = \{x \in \mathfrak{g} : [h, x] = \alpha(h)x \text{ for all } h \in \mathfrak{h}\}.$$

Here $\mathfrak{g}_0 = C_{\mathfrak{g}}(\mathfrak{h})$ is the centraliser of $\mathfrak{h}$, which contains $\mathfrak{h}$. The *roots* of $\mathfrak{g}$ with respect to $\mathfrak{h}$ are the nonzero α with $\mathfrak{g}_\alpha \neq 0$; they form a finite set $\Phi \subseteq \mathfrak{h}^*$, and $\mathfrak{g}_\alpha$ is the *root space* of α.

Proposition 13.9.3. *For $\alpha, \beta \in \mathfrak{h}^*$:*

1. $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}$;
2. *if $\alpha \neq 0$ and $x \in \mathfrak{g}_\alpha$, then $\operatorname{ad} x$ is nilpotent;*
3. *if $\alpha + \beta \neq 0$, then $\kappa(\mathfrak{g}_\alpha, \mathfrak{g}_\beta) = 0$;*
4. *the restriction of κ to $\mathfrak{g}_0$ is nondegenerate.*

Proof. (1) For $x \in \mathfrak{g}_\alpha$, $y \in \mathfrak{g}_\beta$, the Jacobi identity gives $[h, [x, y]] = [[h, x], y] + [x, [h, y]] = (\alpha(h) + \beta(h))[x, y]$. (2) By (1), $\operatorname{ad} x$ maps $\mathfrak{g}_\gamma$ into $\mathfrak{g}_{\gamma+\alpha}$, and only finitely many $\mathfrak{g}_\gamma$ are nonzero. (3) Choose h with $\alpha(h) + \beta(h) \neq 0$; by invariance, for $x \in \mathfrak{g}_\alpha$ and $y \in \mathfrak{g}_\beta$, $\alpha(h)\kappa(x, y) = \kappa([h, x], y) = -\kappa([x, h], y) = -\kappa(x, [h, y]) = -\beta(h)\kappa(x, y)$, so $\kappa(x, y) = 0$. (4) By (3), $\mathfrak{g}_0$ is orthogonal to all $\mathfrak{g}_\alpha$ with $\alpha \neq 0$; so an element of $\mathfrak{g}_0$ orthogonal to $\mathfrak{g}_0$ is orthogonal to $\mathfrak{g}$, hence zero (Theorem 13.7.8). □

Theorem 13.9.4. $C_\mathfrak{g}(\mathfrak{h}) = \mathfrak{h}$. *Hence* $\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha\in\Phi} \mathfrak{g}_\alpha$, *and the restriction of* κ *to* $\mathfrak{h}$ *is nondegenerate.*

Proof. Write $C = C_\mathfrak{g}(\mathfrak{h})$, a subalgebra by the Jacobi identity.

Step 1: C contains the semisimple and nilpotent parts of its elements. For $x \in C$, $\operatorname{ad} x$ maps $\mathfrak{h}$ to 0, and $\operatorname{ad} x_s$, $\operatorname{ad} x_n$ are polynomials without constant term in $\operatorname{ad} x$ (Proposition 13.7.12 and Lemma 13.7.4(1)); so they also kill $\mathfrak{h}$.

Step 2: the semisimple elements of C lie in $\mathfrak{h}$. If $s \in C$ is semisimple, $\mathfrak{h} + \mathbb{C}s$ is an abelian subalgebra, and its elements $h + cs$ have $\operatorname{ad}(h + cs) = \operatorname{ad} h + c \operatorname{ad} s$, a sum of commuting diagonalisable endomorphisms, which is diagonalisable (Theorem 11.7.6). So $\mathfrak{h} + \mathbb{C}s$ is toral, and $s \in \mathfrak{h}$ by maximality.

Step 3: $\kappa|_\mathfrak{h}$ is nondegenerate. Let $h \in \mathfrak{h}$ with $\kappa(h, \mathfrak{h}) = 0$, and $x \in C$. Then $x_s \in \mathfrak{h}$ by Steps 1 and 2, so $\kappa(h, x_s) = 0$. And $x_n \in C$, so $\operatorname{ad} x_n$ is nilpotent and commutes with $\operatorname{ad} h$; then $\operatorname{ad} h \operatorname{ad} x_n$ is nilpotent and $\kappa(h, x_n) = 0$. So $\kappa(h, C) = 0$, and $h = 0$ by Proposition 13.9.3(4).

Step 4: C is nilpotent. For $x \in C$, $\operatorname{ad}_C x_s = 0$ since $x_s \in \mathfrak{h}$, and $\operatorname{ad}_C x_n$ is nilpotent; so every $\operatorname{ad}_C x = \operatorname{ad}_C x_n$ is nilpotent, and C is nilpotent by Engel's theorem.

Step 5: $\mathfrak{h} \cap [C, C] = 0$. By invariance $\kappa(\mathfrak{h}, [C, C]) = \kappa([\mathfrak{h}, C], C) = 0$; apply Step 3.

Step 6: C is abelian. Otherwise $[C, C]$ is a nonzero ideal of the nilpotent Lie algebra C. The operators $\operatorname{ad}_C x$ restricted to $[C, C]$ are nilpotent, so by Theorem 13.5.5 they have a common null vector: there is $z \neq 0$ in $[C, C]$ with $[C, z] = 0$. By Steps 2 and 5, z is not semisimple, so $z_n \neq 0$; and $z_n \in C$ with $\operatorname{ad} z_n$ a polynomial in $\operatorname{ad} z$, which kills C, so $[z_n, C] = 0$. For $y \in C$, $\operatorname{ad} z_n$ is nilpotent and commutes with $\operatorname{ad} y$, so $\kappa(z_n, y) = 0$. This contradicts Proposition 13.9.3(4).

Step 7: $C = \mathfrak{h}$. Otherwise there is $x \in C \setminus \mathfrak{h}$, and $x_n = x - x_s \neq 0$ lies in C by Steps 1 and 2. Since C is abelian, $\operatorname{ad} x_n$ commutes with every $\operatorname{ad} y$, $y \in C$, so $\kappa(x_n, C) = 0$ as in Step 6; again a contradiction. □

Since $\kappa|_\mathfrak{h}$ is nondegenerate, every $\lambda \in \mathfrak{h}^*$ is $\kappa(t_\lambda, -)$ for a unique $t_\lambda \in \mathfrak{h}$, and $\lambda \mapsto t_\lambda$ is a linear isomorphism $\mathfrak{h}^* \to \mathfrak{h}$. We transport κ to the symmetric bilinear form $(\lambda, \mu) = \kappa(t_\lambda, t_\mu)$ on $\mathfrak{h}^*$, which is nondegenerate.

Proposition 13.9.5. 1. Φ *spans* $\mathfrak{h}^*$.

2. *If* $\alpha \in \Phi$, *then* $-\alpha \in \Phi$.

3. *If* $\alpha \in \Phi$, $x \in \mathfrak{g}_\alpha$ *and* $y \in \mathfrak{g}_{-\alpha}$, *then* $[x, y] = \kappa(x, y)t_\alpha$.

4. *If* $\alpha \in \Phi$, *then* $(\alpha, \alpha) = \alpha(t_\alpha) \neq 0$.

5. *If* $\alpha \in \Phi$ *and* $0 \neq x_\alpha \in \mathfrak{g}_\alpha$, *there is* $y_\alpha \in \mathfrak{g}_{-\alpha}$ *such that* x_α, y_α *and* $h_\alpha = [x_\alpha, y_\alpha]$ *span a subalgebra* S_α *isomorphic to* $\mathfrak{sl}_2$, *by* $x_\alpha \mapsto e$, $y_\alpha \mapsto f$, $h_\alpha \mapsto h$. *The element* $h_\alpha = 2t_\alpha/(\alpha, \alpha)$, *the* coroot *of* α, *does not depend on the choices,* $\alpha(h_\alpha) = 2$, *and* $h_{-\alpha} = -h_\alpha$.

Proof. (1) Otherwise some $h \neq 0$ in $\mathfrak{h}$ satisfies $\alpha(h) = 0$ for all $\alpha \in \Phi$. Then $[h, \mathfrak{g}_\alpha] = 0$ for all roots and $[h, \mathfrak{h}] = 0$, so h is central (Theorem 13.9.4), contradicting semisimplicity.

(2) If $-\alpha \notin \Phi$, then $\kappa(\mathfrak{g}_\alpha, \mathfrak{g}_\beta) = 0$ for all β with $\mathfrak{g}_\beta \neq 0$ (Proposition 13.9.3(3)), so $\mathfrak{g}_\alpha \perp \mathfrak{g}$ and $\mathfrak{g}_\alpha = 0$, which is absurd.

(3) $[x, y] \in \mathfrak{g}_0 = \mathfrak{h}$, and for $h \in \mathfrak{h}$, $\kappa(h, [x, y]) = \kappa([h, x], y) = \alpha(h)\kappa(x, y) = \kappa(h, \kappa(x, y)t_\alpha)$. As $\kappa|_\mathfrak{h}$ is nondegenerate, $[x, y] = \kappa(x, y)t_\alpha$.

(4) Suppose $\alpha(t_\alpha) = 0$. For $x \neq 0$ in $\mathfrak{g}_\alpha$, $\kappa(x, \mathfrak{g}_{-\alpha}) \neq 0$, since x is orthogonal to all other $\mathfrak{g}_\beta$ and κ is nondegenerate; choose $y \in \mathfrak{g}_{-\alpha}$ with $\kappa(x, y) = 1$, so that $[x, y] = t_\alpha$ by (3). Then $[t_\alpha, x] = \alpha(t_\alpha)x = 0$ and $[t_\alpha, y] = 0$, so $S = \operatorname{span}(x, y, t_\alpha)$ is a solvable subalgebra, with $[S, S] = \mathbb{C}t_\alpha$. By Lie's theorem applied to $\operatorname{ad} S \subseteq \mathfrak{gl}(\mathfrak{g})$, the element $\operatorname{ad} t_\alpha \in [\operatorname{ad} S, \operatorname{ad} S]$ is strictly upper triangular in a suitable basis, hence nilpotent. It is also diagonalisable, as $t_\alpha \in \mathfrak{h}$; so $\operatorname{ad} t_\alpha = 0$ and t_α is central, hence 0, contradicting $\alpha \neq 0$.

(5) Choose $y_\alpha \in \mathfrak{g}_{-\alpha}$ with $\kappa(x_\alpha, y_\alpha) = 2/(\alpha,\alpha)$, which is possible as in (4). By (3), $h_\alpha = [x_\alpha, y_\alpha] = 2t_\alpha/(\alpha,\alpha)$, so $\alpha(h_\alpha) = 2$, $[h_\alpha, x_\alpha] = 2x_\alpha$ and $[h_\alpha, y_\alpha] = -2y_\alpha$. The three elements are linearly independent (they lie in $\mathfrak{g}_\alpha$, $\mathfrak{g}_{-\alpha}$, $\mathfrak{h}$ and are nonzero), and their brackets are those of e, f, h. Since $t_{-\alpha} = -t_\alpha$, $h_{-\alpha} = -h_\alpha$. □

So $\mathfrak{g}$ is a representation of each $S_\alpha \cong \mathfrak{sl}_2$, by ad, and Section 13.6 applies. In such a representation M the eigenvalues of h_α, the *weights*, are integers, M is a direct sum of irreducible representations $V(m)$, and since $V(m)$ has the weights $m, m-2, \dots, -m$, each with multiplicity one, the number of summands is $\dim M_0 + \dim M_1$, where M_j is the j-eigenspace of h_α (Theorems 13.6.2 and 13.6.3).

Theorem 13.9.6. *Let $\alpha, \beta \in \Phi$.*

1. $\dim \mathfrak{g}_\alpha = 1$, *and the only multiples $c\alpha$, $c \in \mathbb{C}$, of α that are roots are α and $-\alpha$.*
2. $\beta(h_\alpha) \in \mathbb{Z}$, *and $\beta - \beta(h_\alpha)\alpha \in \Phi$.*
3. *Let $\beta \neq \pm\alpha$, and let r, q be the largest integers such that $\beta - r\alpha$ and $\beta + q\alpha$ are roots. Then $\beta + i\alpha \in \Phi$ for all $-r \le i \le q$ (the α-*string through β *is unbroken), and $\beta(h_\alpha) = r - q$.*
4. *If $\alpha + \beta \in \Phi$, then $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}$.*
5. $\mathfrak{h}$ *is spanned by the coroots h_α, and $\mathfrak{g}$ is generated as a Lie algebra by the root spaces.*

Proof. (1) Let $M = \mathfrak{h} \oplus \bigoplus_{c\neq 0} \mathfrak{g}_{c\alpha}$. By Proposition 13.9.3(1) it is stable under ad S_α, and h_α acts on $\mathfrak{g}_{c\alpha}$ by $c\alpha(h_\alpha) = 2c$. So $2c \in \mathbb{Z}$ whenever $c\alpha \in \Phi$, and the 0-weight space of M is $\mathfrak{h}$. Now $\mathfrak{h} = \ker\alpha \oplus \mathbb{C}h_\alpha$, and S_α acts by 0 on $\ker\alpha$: $[x_\alpha, h] = -\alpha(h)x_\alpha = 0$ and $[y_\alpha, h] = 0$ for $h \in \ker\alpha$. So $\ker\alpha \oplus S_\alpha$ is a subrepresentation of M, and by Theorem 13.6.3 it has a complementary subrepresentation M'. The 0-weight space of M is contained in $\ker\alpha \oplus S_\alpha$, so M' has no weight 0, hence no summand $V(m)$ with m even. Thus the even weights of M are those of $\ker\alpha \oplus S_\alpha$, namely 0 and ± 2; as $\mathfrak{g}_{2\alpha}$ would have weight 4, $2\alpha \notin \Phi$. This holds for every root, so $\frac{1}{2}\alpha$ is not a root either (α would be twice a root). Hence M has no weight 1, M' has no summand $V(m)$ with m odd, and $M' = 0$. So $M = \ker\alpha \oplus S_\alpha$: $\mathfrak{g}_\alpha = \mathbb{C}x_\alpha$, and $\mathfrak{g}_{c\alpha} = 0$ for $c \neq 0, \pm 1$.

(2) and (3) Let $\beta \neq \pm\alpha$ and $K = \bigoplus_{i\in\mathbb{Z}} \mathfrak{g}_{\beta+i\alpha}$, a subrepresentation of S_α. No $\beta + i\alpha$ is 0, by (1), so each summand is a root space, of dimension 1, or zero. The weight of $\mathfrak{g}_{\beta+i\alpha}$ is $\beta(h_\alpha) + 2i$; these are integers, so $\beta(h_\alpha) \in \mathbb{Z}$, they are distinct, and they all have the same parity. So $\dim K_0 + \dim K_1 = 1$, and K is irreducible, $K \cong V(m)$. The weights of $V(m)$ form the unbroken string $m, m-2, \dots, -m$, so the i with $\beta + i\alpha \in \Phi$ form an interval $[-r, q]$, and $m = \beta(h_\alpha) + 2q$, $-m = \beta(h_\alpha) - 2r$. Hence $\beta(h_\alpha) = r - q$, and $\beta - \beta(h_\alpha)\alpha = \beta + (q-r)\alpha$ is a root, as $-r \le q - r \le q$. For $\beta = \pm\alpha$, (2) holds since $\pm\alpha(h_\alpha) = \pm 2$ and $-\beta \in \Phi$.

(4) Since 2α and 0 are not roots, $\beta \neq \pm\alpha$, and $K \cong V(m)$ as above, with $\mathfrak{g}_\beta$ and $\mathfrak{g}_{\alpha+\beta}$ the weight spaces of weights $j = \beta(h_\alpha)$ and $j + 2 \le m$. In $V(m)$, e maps the weight space $m - 2i$ onto the weight space $m - 2i + 2$ for $1 \le i \le m$, since $ev_i = (m - i + 1)v_{i-1}$ (Theorem 13.6.2). So ad $x_\alpha : \mathfrak{g}_\beta \to \mathfrak{g}_{\alpha+\beta}$ is nonzero, and onto, both spaces being one-dimensional.

(5) The coroots are nonzero multiples of the t_α, and the t_α span $\mathfrak{h}$ because Φ spans $\mathfrak{h}^*$. As $h_\alpha = [x_\alpha, y_\alpha]$, the subalgebra generated by the root spaces contains $\mathfrak{h}$, hence everything. □

Corollary 13.9.7 (Dimension from the roots)**.** $\dim \mathfrak{g} = \dim \mathfrak{h} + |\Phi|$*: the dimension of a semisimple Lie algebra is its rank plus the number of its roots. The number of roots is even, as $\alpha \mapsto -\alpha$ is a fixed point free involution of Φ.*

Proof. $\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha\in\Phi} \mathfrak{g}_\alpha$ by Theorem 13.9.4, each $\mathfrak{g}_\alpha$ is a line by Theorem 13.9.6(1), and $-\Phi = \Phi$ by Proposition 13.9.5(2). □

For example $\mathfrak{sl}_n$ has rank $n-1$ and $n(n-1)$ roots, and $(n-1)+n(n-1) = n^2-1$ (Example 13.9.9).

Theorem 13.9.8. *1. For $h, h' \in \mathfrak{h}$, $\kappa(h,h') = \sum_{\gamma\in\Phi} \gamma(h)\gamma(h')$. For $\lambda \in \mathfrak{h}^*$ and $\alpha \in \Phi$, $\lambda(h_\alpha) = 2(\lambda,\alpha)/(\alpha,\alpha)$.*

2. *$(\alpha,\beta) \in \mathbb{Q}$ for all $\alpha, \beta \in \Phi$.*

3. *Let $E_\mathbb{Q}$ be the $\mathbb{Q}$-span of Φ in $\mathfrak{h}^*$. Every basis of $\mathfrak{h}^*$ consisting of roots is a basis of $E_\mathbb{Q}$ over $\mathbb{Q}$; so $\dim_\mathbb{Q} E_\mathbb{Q} = \dim \mathfrak{h}$, the* rank *of $\mathfrak{g}$.*

4. *The form $(\ ,\ )$ is positive definite on $E_\mathbb{Q}$, and extends to an inner product on the real vector space $E = \mathbb{R}\otimes_\mathbb{Q} E_\mathbb{Q}$.*

In E, with $\langle\beta,\alpha\rangle = 2(\beta,\alpha)/(\alpha,\alpha)$:

(R1) Φ is finite, spans E, and does not contain 0;

(R2) the only multiples of $\alpha \in \Phi$ in Φ are $\pm\alpha$;

(R3) for $\alpha,\beta \in \Phi$, $\sigma_\alpha(\beta) = \beta - \langle\beta,\alpha\rangle\alpha \in \Phi$;

(R4) for $\alpha,\beta \in \Phi$, $\langle\beta,\alpha\rangle \in \mathbb{Z}$.

Proof. (1) $\operatorname{ad} h \operatorname{ad} h'$ is 0 on $\mathfrak{h}$ and $\gamma(h)\gamma(h')$ on the line $\mathfrak{g}_\gamma$. And $\lambda(h_\alpha) = \kappa(t_\lambda, h_\alpha) = 2\kappa(t_\lambda,t_\alpha)/(\alpha,\alpha)$.

(2) By (1), $(\lambda,\mu) = \sum_\gamma \gamma(t_\lambda)\gamma(t_\mu) = \sum_\gamma(\gamma,\lambda)(\gamma,\mu)$. For $\beta\in\Phi$, dividing $(\beta,\beta) = \sum_\gamma(\gamma,\beta)^2$ by $(\beta,\beta)^2 \neq 0$ gives $1/(\beta,\beta) = \frac{1}{4}\sum_\gamma \gamma(h_\beta)^2$, which is rational by Theorem 13.9.6(2). So $(\beta,\beta)\in\mathbb{Q}$, and $(\alpha,\beta) = \frac{1}{2}\alpha(h_\beta)(\beta,\beta) \in \mathbb{Q}$.

(3) Let $\alpha_1,\dots,\alpha_l \in \Phi$ be a basis of $\mathfrak{h}^*$, and $\beta = \sum_i c_i\alpha_i \in \Phi$ with $c_i \in \mathbb{C}$. Then $(\beta,\alpha_j) = \sum_i c_i(\alpha_i,\alpha_j)$ for all j. The matrix $((\alpha_i,\alpha_j))$ has rational entries by (2) and is invertible, as $(\ ,\ )$ is nondegenerate; so by Cramer's rule the c_i are rational. Hence the α_i span $E_\mathbb{Q}$ over $\mathbb{Q}$, and they are linearly independent over $\mathbb{Q}$, even over $\mathbb{C}$.

(4) For $\lambda \in E_\mathbb{Q}$, the numbers (γ,λ) are rational, and $(\lambda,\lambda) = \sum_\gamma(\gamma,\lambda)^2 \geq 0$, with equality only if $(\gamma,\lambda) = 0$ for all roots γ, that is, $\lambda = 0$, since Φ spans $\mathfrak{h}^*$. In the basis (α_i) the Gram matrix G is a rational symmetric matrix with $x^TGx > 0$ for $x \in \mathbb{Q}^l \setminus 0$; by density of $\mathbb{Q}^l$ in $\mathbb{R}^l$ and continuity, $x^TGx \geq 0$ for all $x \in \mathbb{R}^l$. A real eigenvector x of G with eigenvalue μ (Theorem 11.10.6) gives $\mu|x|^2 = x^TGx \geq 0$, and $\mu \neq 0$ since $\det G \neq 0$; so all eigenvalues are positive, and G defines an inner product on $E = \mathbb{R}\otimes_\mathbb{Q} E_\mathbb{Q} = \mathbb{R}^l$.

(R1) is clear, (R2) is Theorem 13.9.6(1) (a real multiple of α lying in $E_\mathbb{Q}$ is a rational multiple), and (R3), (R4) are Theorem 13.9.6(2), since $\langle\beta,\alpha\rangle = \beta(h_\alpha)$ by (1). □

Example 13.9.9. *The roots of $\mathfrak{sl}_n$.* Let $\mathfrak{h} \subseteq \mathfrak{sl}_n(\mathbb{C})$ be the diagonal matrices of trace 0, and $\varepsilon_i \in \mathfrak{h}^*$ the i-th diagonal entry. Since $[h, E_{ij}] = (\varepsilon_i(h) - \varepsilon_j(h))E_{ij}$, $\mathfrak{h}$ is toral. Its centraliser is $\mathfrak{h}$: choosing h with distinct diagonal entries, $[h,x] = \sum_{i\neq j}(h_i - h_j)x_{ij}E_{ij}$ vanishes only if x is diagonal. A toral subalgebra containing $\mathfrak{h}$ is abelian (Lemma 13.9.2), hence contained in $C(\mathfrak{h}) = \mathfrak{h}$; so $\mathfrak{h}$ is a Cartan subalgebra. The roots are $\varepsilon_i - \varepsilon_j$, $i \neq j$, with root spaces $\mathbb{C}E_{ij}$; there are $n(n-1)$ of them, and the rank is $n-1$. With $\kappa(x,y) = 2n\operatorname{tr}(xy)$ (Example 13.7.9), for $\alpha = \varepsilon_i - \varepsilon_j$ one finds $t_\alpha = (E_{ii} - E_{jj})/2n$, $(\alpha,\alpha) = 1/n$ and $h_\alpha = E_{ii} - E_{jj}$, and S_α is the copy of $\mathfrak{sl}_2$ spanned by E_{ij}, E_{ji} and $E_{ii} - E_{jj}$. For distinct i, j, l, $(\varepsilon_i - \varepsilon_j, \varepsilon_j - \varepsilon_l) = -1/2n$. So all roots have the same length, and the angle between $\varepsilon_i - \varepsilon_j$ and $\varepsilon_j - \varepsilon_l$ is $2\pi/3$. Concretely, E is the hyperplane $x_1 + \cdots + x_n = 0$ of $\mathbb{R}^n$ with $\frac{1}{2n}$ times the standard inner product, and $\Phi = \{e_i - e_j : i \neq j\}$; this root system is called A_{n-1}. For $n = 2$ there are two roots $\pm\alpha$, with $\alpha(h) = 2$.

Counterexample 13.9.10. A maximal abelian subalgebra need not be a Cartan subalgebra. In $\mathfrak{sl}_2$, $[e, ae + bh + cf] = -2be + ch$ vanishes only for $b = c = 0$, so $\mathbb{C}e$ is a maximal abelian subalgebra; but it consists of nilpotent elements, and it is not toral. By contrast $\mathbb{C}h$ is a Cartan subalgebra. And over $\mathbb{R}$ the theory has to be modified: in $\mathfrak{so}(3)$, for the infinitesimal rotation $L_3 = E_{21} - E_{12}$ the operator $\operatorname{ad} L_3$ has eigenvalues $0, \pm i$, and in fact no nonzero element has $\operatorname{ad} x$ diagonalisable over $\mathbb{R}$; over $\mathbb{C}$, $\mathbb{C}L_3$ is a Cartan subalgebra of $\mathfrak{so}_3(\mathbb{C})$.

Remark 13.9.11. All Cartan subalgebras of a semisimple Lie algebra over $\mathbb{C}$ are conjugate under the group of automorphisms generated by the $\exp(\operatorname{ad} x)$, x nilpotent [Hum72, Section 16]. Hence the root system does not depend, up to isomorphism, on the choice of $\mathfrak{h}$, and the rank is an invariant of $\mathfrak{g}$.

13.10 Root systems and the classification

The properties (R1)–(R4) of Theorem 13.9.8 are purely Euclidean, and they turn out to determine the Lie algebra.

Definition 13.10.1. A *root system* in a finite-dimensional real inner product space E is a subset Φ satisfying (R1)–(R4) of Theorem 13.9.8, where for $\alpha \neq 0$

$$\langle \beta, \alpha \rangle = \frac{2(\beta, \alpha)}{(\alpha, \alpha)}, \qquad \sigma_\alpha(\beta) = \beta - \langle \beta, \alpha \rangle \alpha.$$

Its *rank* is $\dim E$. The *Weyl group* W is the subgroup of the orthogonal group of E generated by the reflections σ_α, $\alpha \in \Phi$. An isomorphism of root systems $(E, \Phi) \to (E', \Phi')$ is a linear isomorphism ϕ with $\phi(\Phi) = \Phi'$ and $\langle \phi\beta, \phi\alpha \rangle = \langle \beta, \alpha \rangle$. Φ is *irreducible* if it is not the union of two nonempty subsets orthogonal to each other.

σ_α is the orthogonal reflection in the hyperplane $\alpha^\perp$: it fixes $\alpha^\perp$ and sends α to $-\alpha$. By (R3) every $w \in W$ permutes Φ, and since Φ spans E, w is determined by this permutation; so W is finite. Note that $\langle \beta, \alpha \rangle$ is linear in β but not in α. The vectors $\alpha^\vee = 2\alpha/(\alpha, \alpha)$ form a root system as well, the *dual* root system; for the root system of a Lie algebra, $\alpha^\vee$ corresponds to the coroot h_α under the identification of $\mathfrak{h}$ with $\mathfrak{h}^*$.

Lemma 13.10.2. *Let $\alpha, \beta \in \Phi$ with $\beta \neq \pm\alpha$, and θ the angle between them. Then $\langle \alpha, \beta \rangle \langle \beta, \alpha \rangle = 4\cos^2\theta \in \{0, 1, 2, 3\}$. If $(\beta, \beta) \geq (\alpha, \alpha)$ and $(\alpha, \beta) \neq 0$, then $\langle \alpha, \beta \rangle = \pm 1$ and $(\beta, \beta)/(\alpha, \alpha) = |\langle \beta, \alpha \rangle| \in \{1, 2, 3\}$; so the possible pairs are*

$$\theta = \tfrac{\pi}{2},\ \tfrac{\pi}{3},\ \tfrac{2\pi}{3},\ \tfrac{\pi}{4},\ \tfrac{3\pi}{4},\ \tfrac{\pi}{6},\ \tfrac{5\pi}{6}, \qquad \frac{(\beta, \beta)}{(\alpha, \alpha)} = \text{any},\ 1,\ 1,\ 2,\ 2,\ 3,\ 3.$$

Moreover, if $(\alpha, \beta) > 0$ then $\alpha - \beta \in \Phi$, and if $(\alpha, \beta) < 0$ then $\alpha + \beta \in \Phi$.

Proof. $\langle \alpha, \beta \rangle \langle \beta, \alpha \rangle = 4(\alpha, \beta)^2/((\alpha, \alpha)(\beta, \beta)) = 4\cos^2\theta$ is an integer by (R4), and it is less than 4 because α and β are not proportional (by (R2)), so the Cauchy–Schwarz inequality is strict. The two integers $\langle \alpha, \beta \rangle$ and $\langle \beta, \alpha \rangle$ have the sign of (α, β), and $|\langle \alpha, \beta \rangle| \leq |\langle \beta, \alpha \rangle|$ when $(\beta, \beta) \geq (\alpha, \alpha)$; if their product is 1, 2 or 3, the smaller one is ± 1, and $\langle \beta, \alpha \rangle / \langle \alpha, \beta \rangle = (\beta, \beta)/(\alpha, \alpha)$. If $(\alpha, \beta) > 0$, one of $\langle \alpha, \beta \rangle, \langle \beta, \alpha \rangle$ is 1: in the first case $\sigma_\beta(\alpha) = \alpha - \beta \in \Phi$, and in the second $\sigma_\alpha(\beta) = \beta - \alpha \in \Phi$, so $\alpha - \beta = \sigma_{\alpha-\beta}(\beta - \alpha) \in \Phi$. If $(\alpha, \beta) < 0$, apply this to α and $-\beta = \sigma_\beta(\beta)$. □

Example 13.10.3. In rank 1 the only root system is $A_1 = \{\pm\alpha\}$, by (R2). In rank 2 there are four, up to isomorphism [Hum72, Section 9.3]; one checks (R1)–(R4) directly.

1. $A_1 \times A_1 = \{\pm e_1, \pm e_2\}$ in $\mathbb{R}^2$, reducible; it is the root system of $\mathfrak{sl}_2 \times \mathfrak{sl}_2$.

2. $A_2 = \{\pm(\varepsilon_1 - \varepsilon_2), \pm(\varepsilon_2 - \varepsilon_3), \pm(\varepsilon_1 - \varepsilon_3)\}$ in the plane $x_1 + x_2 + x_3 = 0$ of $\mathbb{R}^3$: six roots of equal length at angles $\pi/3$, the root system of $\mathfrak{sl}_3$ (Example 13.9.9).

3. $B_2 = \{\pm e_1, \pm e_2, \pm e_1 \pm e_2\}$: four short and four long roots, at angles $\pi/4$; it is also called C_2, the root system of $\mathfrak{sp}_4$ (Exercise 13.12.11).

4. G_2: in the plane $x_1 + x_2 + x_3 = 0$, the six short roots of A_2 together with the six long roots $\pm(2\varepsilon_1 - \varepsilon_2 - \varepsilon_3)$, $\pm(2\varepsilon_2 - \varepsilon_1 - \varepsilon_3)$, $\pm(2\varepsilon_3 - \varepsilon_1 - \varepsilon_2)$; consecutive roots make angles $\pi/6$, and long roots are $\sqrt{3}$ times longer than short ones.

Definition 13.10.4. A *base* of a root system Φ is a subset Δ that is a basis of E such that every root is $\beta = \sum_{\alpha\in\Delta} k_\alpha \alpha$ with integer coefficients k_α that are either all ≥ 0 or all ≤ 0. The elements of Δ are the *simple roots*, the roots with all $k_\alpha \geq 0$ are *positive* ($\beta > 0$), and they form a set Φ^+ with $\Phi = \Phi^+ \sqcup -\Phi^+$. The *height* of β is $\sum_\alpha k_\alpha$.

Theorem 13.10.5. *Let $\gamma \in E$ be* regular, *that is, $(\gamma, \alpha) \neq 0$ for all $\alpha \in \Phi$ (such γ exist, see the end of the proof). Let $\Phi^+(\gamma) = \{\alpha \in \Phi : (\gamma, \alpha) > 0\}$, and let $\Delta(\gamma)$ be the set of $\alpha \in \Phi^+(\gamma)$ that are not the sum of two elements of $\Phi^+(\gamma)$. Then $\Delta(\gamma)$ is a base, with positive roots $\Phi^+(\gamma)$, and $(\alpha, \beta) \leq 0$ for distinct $\alpha, \beta \in \Delta(\gamma)$.*

Proof. Every $\beta \in \Phi^+(\gamma)$ is a non-negative integer combination of $\Delta(\gamma)$. Otherwise choose such a β with (γ, β) minimal. It is not in $\Delta(\gamma)$, so $\beta = \beta_1 + \beta_2$ with $\beta_i \in \Phi^+(\gamma)$; as $(\gamma, \beta) = (\gamma, \beta_1) + (\gamma, \beta_2)$ with both terms positive, each $(\gamma, \beta_i) < (\gamma, \beta)$, so each β_i is such a combination, and so is β.

$(\alpha, \beta) \leq 0$ for distinct $\alpha, \beta \in \Delta(\gamma)$. Otherwise $\alpha - \beta \in \Phi$ by Lemma 13.10.2, and $\alpha - \beta$ or $\beta - \alpha$ lies in $\Phi^+(\gamma)$; then $\alpha = \beta + (\alpha - \beta)$ or $\beta = \alpha + (\beta - \alpha)$ is decomposable.

$\Delta(\gamma)$ is linearly independent. Suppose $\sum_\alpha r_\alpha \alpha = 0$, and separate the terms: $\varepsilon = \sum_{\alpha\in A} s_\alpha \alpha = \sum_{\beta\in B} t_\beta \beta$ with disjoint $A, B \subseteq \Delta(\gamma)$ and $s_\alpha, t_\beta > 0$. Then $(\varepsilon, \varepsilon) = \sum s_\alpha t_\beta (\alpha, \beta) \leq 0$, so $\varepsilon = 0$; and $0 = (\gamma, \varepsilon) = \sum s_\alpha (\gamma, \alpha)$ with all $(\gamma, \alpha) > 0$ forces $A = \emptyset$, and similarly $B = \emptyset$.

Since $\Phi = \Phi^+(\gamma) \sqcup -\Phi^+(\gamma)$ spans E, so does $\Delta(\gamma)$, and it is a base. Regular elements exist: in coordinates, $\gamma \mapsto \prod_{\alpha\in\Phi}(\gamma, \alpha)$ is a product of nonzero linear polynomials, hence a nonzero polynomial, and a nonzero real polynomial does not vanish everywhere on $\mathbb{R}^l$. □

Lemma 13.10.6. *Let Δ be a base and $\alpha \in \Delta$. Then σ_α permutes $\Phi^+ \setminus \{\alpha\}$. Consequently $\rho = \frac{1}{2}\sum_{\beta\in\Phi^+} \beta$ satisfies $\sigma_\alpha(\rho) = \rho - \alpha$ and $\langle \rho, \alpha \rangle = 1$ for every simple root α.*

Proof. Let $\beta = \sum_{\delta\in\Delta} k_\delta \delta \in \Phi^+$, $\beta \neq \alpha$. By (R2), β is not a multiple of α, so $k_\delta > 0$ for some $\delta \neq \alpha$. The coefficient of δ in $\sigma_\alpha(\beta) = \beta - \langle \beta, \alpha \rangle \alpha$ is still $k_\delta > 0$, so $\sigma_\alpha(\beta)$ is a positive root, different from α. As σ_α is injective and $\sigma_\alpha(\alpha) = -\alpha$, $\sigma_\alpha(2\rho) = 2\rho - 2\alpha$. Comparing with $\sigma_\alpha(\rho) = \rho - \langle \rho, \alpha \rangle \alpha$ gives $\langle \rho, \alpha \rangle = 1$. □

Further facts about bases, which we quote [Hum72, Sections 10.1 and 10.3]: every base is $\Delta(\gamma)$ for some regular γ; the Weyl group acts simply transitively on the set of bases and is generated by the simple reflections σ_α, $\alpha \in \Delta$; and every root is W-conjugate to a simple root.

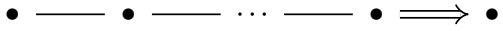
Definition 13.10.7. Let $\Delta = \{\alpha_1, \ldots, \alpha_l\}$ be a base. The *Cartan matrix* is the integer matrix $(\langle \alpha_i, \alpha_j \rangle)_{i,j}$. Its diagonal entries are 2, and by Theorem 13.10.5 and Lemma 13.10.2 its other entries lie in $\{0, -1, -2, -3\}$. The *Dynkin diagram* has a vertex for each simple root, and α_i and α_j ($i \neq j$) are joined by $\langle \alpha_i, \alpha_j \rangle \langle \alpha_j, \alpha_i \rangle \in \{0, 1, 2, 3\}$ edges; when two joined roots have different lengths, the edges carry an arrow pointing to the shorter one.

By Lemma 13.10.2 the Dynkin diagram determines the Cartan matrix, and the Cartan matrix determines the root system up to isomorphism, because Φ is the set of images of the simple roots under the group generated by the simple reflections, and these act by $\sigma_{\alpha_j}(\alpha_i) = \alpha_i - \langle \alpha_i, \alpha_j \rangle \alpha_j$ [Hum72, Section 11.1]. The root system is irreducible if and only if its Dynkin diagram is connected, and in general it is the orthogonal union of the irreducible root systems belonging to the connected components.

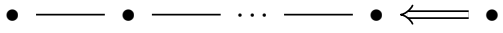
Theorem 13.10.8 (Classification of root systems)**.** *The connected Dynkin diagrams are those of the following list, and each belongs to exactly one irreducible root system up to isomorphism [Hum72, Sections 11.4 and 12.1].*

1. *A_l ($l \geq 1$), a chain of l vertices joined by single edges:*

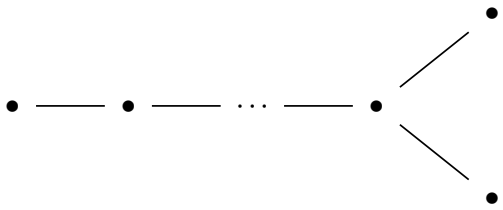
2. *B_l ($l \geq 2$), a chain whose last edge is double, with the arrow pointing to the last (short) root:*



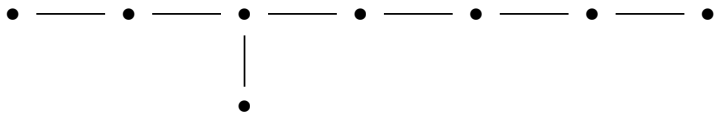
3. *C_l ($l \geq 3$), the same chain with the arrow reversed (the last root is long):*



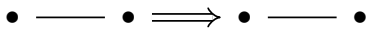
4. *D_l ($l \geq 4$), a chain of $l-2$ vertices whose last vertex is joined to two further vertices:*



5. *E_6, E_7, E_8, a chain of $5, 6, 7$ vertices with one more vertex joined to the third one; for E_8:*



6. *F_4, a chain of four vertices whose middle edge is double:*



7. *G_2, two vertices joined by a triple edge, with the arrow pointing to the short root.*

The restrictions on l avoid repetitions: $B_1 = C_1 = A_1$, $C_2 = B_2$, $D_2 = A_1 \times A_1$ and $D_3 = A_3$. The proof is an exercise in Euclidean geometry: the simple roots, normalised to unit vectors u_i, are linearly independent with $(u_i, u_j) \in \{0, -\frac{1}{2}, -\frac{1}{\sqrt{2}}, -\frac{\sqrt{3}}{2}\}$ for $i \neq j$, and positivity of $(\sum c_i u_i, \sum c_i u_i)$ for well-chosen coefficients excludes all other graphs.

Theorem 13.10.9 (Classification of semisimple Lie algebras)**.** *Let $\mathfrak{g}$ be a semisimple Lie algebra over $\mathbb{C}$ with Cartan subalgebra $\mathfrak{h}$ and root system Φ.*

1. *$\mathfrak{g}$ is simple if and only if Φ is irreducible; the simple ideals of $\mathfrak{g}$ correspond to the irreducible components of Φ.*

2. *Two semisimple Lie algebras are isomorphic if and only if their root systems are isomorphic.*

3. *Every root system is the root system of a semisimple Lie algebra.*

Consequently the simple Lie algebras over $\mathbb{C}$ are, up to isomorphism, exactly one for each diagram of Theorem 13.10.8: the classical algebras $\mathfrak{sl}_{l+1}$ (A_l), $\mathfrak{so}_{2l+1}$ (B_l), $\mathfrak{sp}_{2l}$ (C_l), $\mathfrak{so}_{2l}$ (D_l), of dimensions $l(l+2)$, $l(2l+1)$, $l(2l+1)$, $l(2l-1)$, and five exceptional algebras $\mathfrak{e}_6$, $\mathfrak{e}_7$, $\mathfrak{e}_8$, $\mathfrak{f}_4$, $\mathfrak{g}_2$ of dimensions 78, 133, 248, 52 and 14.

For (1) see [Hum72, Section 14.1], for (2) [Hum72, Sections 14 and 16], and for (3) [Hum72, Section 18], where $\mathfrak{g}$ is constructed by generators x_i, y_i, h_i ($1 \le i \le l$) and relations read off from the Cartan matrix (Serre's theorem). The dimension of $\mathfrak{g}$ is $l + |\Phi|$ (Corollary 13.9.7); the number of roots of each type can be computed from the Cartan matrix, as in Example 13.10.11 below.

Lemma 13.10.10. *Let Δ be a base. Every positive root β that is not simple has a simple root α with $\beta - \alpha \in \Phi^+$. Hence every positive root of height h is a sum $\alpha_{i_1} + \cdots + \alpha_{i_h}$ of simple roots such that every partial sum $\alpha_{i_1} + \cdots + \alpha_{i_j}$ is a root.*

Proof. Write $\beta = \sum_{\alpha\in\Delta} k_\alpha \alpha$ with $k_\alpha \ge 0$. If $(\beta, \alpha) \le 0$ for all $\alpha \in \Delta$, then $(\beta, \beta) = \sum k_\alpha(\beta, \alpha) \le 0$, which is impossible; so $(\beta, \alpha) > 0$ for some simple α, and $\beta - \alpha \in \Phi$ by Lemma 13.10.2 ($\beta \ne \alpha$ as β is not simple, and $\beta \ne -\alpha$ as $\beta > 0$). Since β is not a multiple of α, some k_δ with $\delta \ne \alpha$ is positive, and it is a coefficient of $\beta - \alpha$; so $\beta - \alpha$ is positive, of height one less. The second statement follows by induction on the height. □

Example 13.10.11. *Counting roots from the Cartan matrix.* By the lemma, the positive roots can be listed by height, starting from the simple roots: a root of height $h+1$ is $\beta + \alpha_i$ with β a root of height h and α_i simple. Whether $\beta + \alpha_i$ is a root is decided by the α_i-string through β (Theorem 13.9.6(3); the same holds for every root system [Hum72, Section 9.4]): if r is the largest integer with $\beta - r\alpha_i \in \Phi$, already known from the lower heights, then $q = r - \langle\beta, \alpha_i\rangle$, and $\beta + \alpha_i$ is a root if and only if $q \ge 1$. The numbers $\langle\beta, \alpha_i\rangle$ are computed from the Cartan matrix by linearity in β.

For G_2, with α short and β long, $\langle\alpha, \beta\rangle = -1$ and $\langle\beta, \alpha\rangle = -3$. Height 2: $\alpha+\beta$ (as $(\alpha, \beta) < 0$); 2α and 2β are not roots. Height 3: for $\alpha + \beta$, the α-string goes down to β, so $r = 1$, and $\langle\alpha + \beta, \alpha\rangle = 2 - 3 = -1$, so $q = 2$: $2\alpha + \beta$ and $3\alpha + \beta$ are roots; the β-string has $r = 1$ and $\langle\alpha + \beta, \beta\rangle = -1 + 2 = 1$, so $q = 0$ and $\alpha + 2\beta$ is not a root. Height 4: $3\alpha + \beta$; for $2\alpha + \beta$ the β-string has $r = 0$ and $\langle 2\alpha + \beta, \beta\rangle = -2 + 2 = 0$, so $2\alpha + 2\beta$ is not a root. Height 5: for $3\alpha + \beta$, $r = 0$ for β and $\langle 3\alpha + \beta, \beta\rangle = -3 + 2 = -1$, so $3\alpha + 2\beta$ is a root; $\langle 3\alpha + \beta, \alpha\rangle = 6 - 3 = 3 = r$ with $r = 3$, so $4\alpha + \beta$ is not. Height 6: $\langle 3\alpha + 2\beta, \alpha\rangle = 0$ with $r = 0$, and $\langle 3\alpha + 2\beta, \beta\rangle = -3 + 4 = 1$ with $r = 1$, so nothing more. So G_2 has the six positive roots α, β, $\alpha + \beta$, $2\alpha + \beta$, $3\alpha + \beta$, $3\alpha + 2\beta$, twelve roots in all, and $\dim \mathfrak{g}_2 = 2 + 12 = 14$ (Corollary 13.9.7).

The same computation for the other types gives the number of roots [Hum72, Section 12.1], and with Corollary 13.9.7 the dimension:

1. A_l: $l(l+1)$ roots, $\dim = l + l(l+1) = l(l+2) = (l+1)^2 - 1$, which is $\dim \mathfrak{sl}_{l+1}$;
2. B_l and C_l: $2l^2$ roots (for B_l: $\pm\varepsilon_i$ and $\pm\varepsilon_i \pm \varepsilon_j$, $i < j$), $\dim = l(2l+1)$, which is $\dim \mathfrak{so}_{2l+1} = \binom{2l+1}{2}$ and $\dim \mathfrak{sp}_{2l}$ (Example 13.7.14);
3. D_l: $2l(l-1)$ roots ($\pm\varepsilon_i \pm \varepsilon_j$, $i < j$), $\dim = l(2l-1) = \binom{2l}{2} = \dim \mathfrak{so}_{2l}$;
4. E_6, E_7, E_8, F_4, G_2: 72, 126, 240, 48, 12 roots, and dimensions 78, 133, 248, 52, 14.

Proposition 13.10.12 (Serre's relations)**.** *Let $\mathfrak{g}$ be semisimple with base $\Delta = \{\alpha_1, \ldots, \alpha_l\}$, and choose $e_i = x_{\alpha_i}$, $f_i = y_{\alpha_i}$, $h_i = h_{\alpha_i}$ as in Proposition 13.9.5. With $c_{ij} = \langle\alpha_j, \alpha_i\rangle$, the generators e_i, f_i, h_i of $\mathfrak{g}$ satisfy*

$$[h_i, h_j] = 0, \quad [e_i, f_j] = \delta_{ij} h_i, \quad [h_i, e_j] = c_{ij} e_j, \quad [h_i, f_j] = -c_{ij} f_j, \quad (\operatorname{ad} e_i)^{1-c_{ij}} e_j = 0 = (\operatorname{ad} f_i)^{1-c_{ij}} f_j \ (i \ne j).$$

Proof. $\mathfrak{h}$ is abelian. For $i \neq j$, $[e_i, f_j] \in \mathfrak{g}_{\alpha_i - \alpha_j} = 0$, since a difference of two simple roots has coefficients of both signs and is not a root; and $[e_i, f_i] = h_i$. Next $[h_i, e_j] = \alpha_j(h_i)e_j = \langle \alpha_j, \alpha_i \rangle e_j$ (Theorem 13.9.8(1)), and similarly for f_j. Finally, for $i \neq j$ the α_i-string through α_j has $r = 0$, as $\alpha_j - \alpha_i$ is not a root, so by Theorem 13.9.6(3) it is $\alpha_j, \dots, \alpha_j + q\alpha_i$ with $q = -c_{ij}$; hence $\alpha_j + (1 - c_{ij})\alpha_i$ is not a root, and $(\operatorname{ad} e_i)^{1-c_{ij}} e_j \in \mathfrak{g}_{\alpha_j + (1-c_{ij})\alpha_i} = 0$. The same argument with negative roots gives the relation for the f_i. □

Serre's theorem [Hum72, Section 18.3] says that these relations are a presentation: for every root system with Cartan matrix (c_{ij}), the Lie algebra generated by e_i, f_i, h_i with only these relations is finite-dimensional and semisimple, with the given root system. This gives part (3) of Theorem 13.10.9. For $\mathfrak{sl}_3$, with $e_1 = E_{12}$, $e_2 = E_{23}$, the relations say $[e_1, [e_1, e_2]] = [E_{12}, E_{13}] = 0$ and $[e_2, [e_2, e_1]] = -[E_{23}, E_{13}] = 0$, while $[e_1, e_2] = E_{13} \neq 0$.

13.11 Weights and highest weights

The classification of the representations of $\mathfrak{sl}_2$ by their highest weight m (Theorem 13.6.2) extends to every semisimple Lie algebra. In this section $\mathfrak{g}$ is semisimple over $\mathbb{C}$, $\mathfrak{h}$ is a Cartan subalgebra with roots Φ, Δ is a base, and representations are finite-dimensional. For $\lambda, \mu \in \mathfrak{h}^*$ write $\mu \leq \lambda$ if $\lambda - \mu$ is a sum of positive roots (possibly empty), and put $\langle \lambda, \alpha \rangle = \lambda(h_\alpha) = 2(\lambda, \alpha)/(\alpha, \alpha)$ (Theorem 13.9.8(1)).

Proposition 13.11.1. *Let V be a representation of $\mathfrak{g}$, and for $\lambda \in \mathfrak{h}^*$ let $V_\lambda = \{v : hv = \lambda(h)v \text{ for all } h \in \mathfrak{h}\}$.*

1. *$V = \bigoplus_\lambda V_\lambda$; the λ with $V_\lambda \neq 0$ are the* weights *of V, and* $\dim V_\lambda$ *is the* multiplicity *of λ.*
2. *$\mathfrak{g}_\alpha V_\lambda \subseteq V_{\lambda+\alpha}$.*
3. *Every weight λ is* integral*: $\langle \lambda, \alpha \rangle \in \mathbb{Z}$ for all $\alpha \in \Phi$.*
4. *The multiplicities are invariant under the Weyl group:* $\dim V_\lambda = \dim V_{\sigma_\alpha \lambda}$ *for $\alpha \in \Phi$.*

Proof. (1) The elements of $\mathfrak{h}$ are semisimple, so they act diagonalisably on V (Corollary 13.8.4(2)), and they commute; so they are simultaneously diagonalisable, as for the root space decomposition. (2) For $x \in \mathfrak{g}_\alpha$ and $v \in V_\lambda$, $hxv = xhv + [h, x]v = (\lambda(h) + \alpha(h))xv$. (3) V_λ is contained in the $\lambda(h_\alpha)$-eigenspace of h_α, and h_α has integer eigenvalues on the S_α-representation V (Theorem 13.6.3). (4) Let $m = \langle \lambda, \alpha \rangle$, and replacing λ by $\sigma_\alpha \lambda = \lambda - m\alpha$ if necessary, assume $m \geq 0$. The subspace $U = \bigoplus_{i \in \mathbb{Z}} V_{\lambda + i\alpha}$ is S_α-stable by (2), and $V_{\lambda+i\alpha}$ is its h_α-eigenspace for the eigenvalue $m + 2i$. By Corollary 13.6.4, y_α^m maps the eigenspace $U_m = V_\lambda$ isomorphically onto $U_{-m} = V_{\lambda - m\alpha} = V_{\sigma_\alpha \lambda}$. □

For the adjoint representation, the weights are the roots, of multiplicity 1, and 0, of multiplicity the rank. A *maximal vector* of weight λ is a nonzero $v \in V_\lambda$ with $\mathfrak{g}_\alpha v = 0$ for all $\alpha \in \Phi^+$. An integral weight λ is *dominant* if $\langle \lambda, \alpha \rangle \geq 0$ for all $\alpha \in \Delta$, hence for all $\alpha \in \Phi^+$.

Theorem 13.11.2 (Highest weight)**.** *Let V be an irreducible representation of $\mathfrak{g}$.*

1. *V has a maximal vector v^+, unique up to a scalar. Its weight λ, the* highest weight *of V, satisfies $\mu \leq \lambda$ for every weight μ of V, and* $\dim V_\lambda = 1$*. V is spanned by the vectors $y_1 y_2 \cdots y_k v^+$ with $k \geq 0$ and $y_i \in \mathfrak{g}_{-\beta_i}$, $\beta_i \in \Phi^+$.*
2. *The highest weight is dominant integral.*
3. *Two irreducible representations with the same highest weight are isomorphic.*

Proof. (1) The subspace $\mathfrak{n} = \bigoplus_{\alpha>0} \mathfrak{g}_\alpha$ is a subalgebra, since the sum of two positive roots is positive if it is a root, and by Proposition 13.9.3(1) the bracket $[\mathfrak{n}, \mathfrak{n}]$ involves only roots of height ≥ 2, and inductively the j-th term of the lower central series only roots of height $\geq j$; so $\mathfrak{n}$ is nilpotent. Hence $\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n}$ is a solvable subalgebra ($[\mathfrak{b}, \mathfrak{b}] \subseteq \mathfrak{n}$; Lemma 13.5.2). By Lie's theorem its image in $\mathfrak{gl}(V)$ has a common eigenvector v^+. Then $v^+ \in V_\lambda$ for some λ, and for $x \in \mathfrak{g}_\alpha$, $\alpha > 0$, the vector xv^+ lies in $V_{\lambda+\alpha} \cap \mathbb{C}v^+ = 0$. So v^+ is a maximal vector.

Let U_k be the span of the vectors $y_1 \cdots y_j v^+$ as in the statement with $j \leq k$, and $U = \bigcup_k U_k$. Clearly U is stable under the $\mathfrak{g}_{-\beta}$, $\beta > 0$, and under $\mathfrak{h}$, which acts on $y_1 \cdots y_j v^+$ by the weight $\lambda - \beta_1 - \cdots - \beta_j$. We show by induction on k that $xU_k \subseteq U_k$ for $x \in \mathfrak{g}_\alpha$, $\alpha > 0$; for $k = 0$, $xv^+ = 0$. For $w = y_1 u$ with $u \in U_{k-1}$, $xw = y_1(xu) + [x, y_1]u$; here $xu \in U_{k-1}$ by induction, and $[x, y_1] \in \mathfrak{g}_{\alpha-\beta_1}$ lies in $\mathfrak{h}$, in a negative root space, or in a positive root space, and in each case $[x, y_1]u \in U_k$ (by induction in the last case). So U is a nonzero subrepresentation, and $U = V$. All weights of V are therefore of the form $\lambda - \beta_1 - \cdots - \beta_j \leq \lambda$, and the weight λ occurs only for $j = 0$, since a nonempty sum of positive roots has positive height; so $V_\lambda = \mathbb{C}v^+$. If w is another maximal vector, of weight μ, the same argument gives $\lambda \leq \mu$, so $\mu = \lambda$ (the relation $\leq$ is antisymmetric, as heights add) and $w \in V_\lambda = \mathbb{C}v^+$.

(2) Integrality is Proposition 13.11.1(3). For $\alpha \in \Delta$, v^+ is killed by x_α and is an eigenvector of h_α with eigenvalue $\langle \lambda, \alpha \rangle$, which is therefore a non-negative integer by Theorem 13.6.2.

(3) Let V, V' be irreducible with maximal vectors v, v' of the same weight λ, and let $W \subseteq V \oplus V'$ be the subrepresentation generated by the maximal vector $w = (v, v')$. As in (1), W is spanned by the vectors $y_1 \cdots y_j w$, all its weights are $\leq \lambda$, and $W_\lambda = \mathbb{C}w$. A subrepresentation $W' \subseteq W$ is the sum of its intersections with the weight spaces, since $\mathfrak{h}$ acts diagonalisably; if $W'_\lambda \neq 0$ it contains w and equals W. So the sum of all proper subrepresentations has no component of weight λ, and it is the unique maximal proper subrepresentation N of W. The two projections $W \to V$ and $W \to V'$ are nonzero ($w \mapsto v, v'$), hence onto, as V and V' are irreducible; their kernels are maximal proper subrepresentations, hence equal to N. So $V \cong W/N \cong V'$. □

Remark 13.11.3. Conversely, for every dominant integral weight λ there is an irreducible representation $V(\lambda)$ with highest weight λ, constructed as the unique irreducible quotient of the Verma module $U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} \mathbb{C}_\lambda$ [Hum72, Section 21]. So the irreducible representations of $\mathfrak{g}$ correspond bijectively to the dominant integral weights, which are the $\sum_i n_i \omega_i$ with $n_i \in \mathbb{Z}_{\geq 0}$, where the *fundamental weights* ω_i are defined by $\langle \omega_i, \alpha_j \rangle = \delta_{ij}$ [Hum72, Section 13]. Its dimension is given by the *Weyl dimension formula* [Hum72, Section 24.3]

$$\dim V(\lambda) = \prod_{\alpha \in \Phi^+} \frac{(\lambda + \rho, \alpha)}{(\rho, \alpha)},$$

with ρ as in Lemma 13.10.6. For $\mathfrak{sl}_2$, with $\Phi^+ = \{\alpha\}$ and $\lambda = m\omega_1 = \frac{m}{2}\alpha$, the formula gives $m + 1$, and $V(\lambda)$ is the representation $V(m)$ of Section 13.6.

Example 13.11.4. For $\mathfrak{sl}_3$ take $\Delta = \{\alpha_1, \alpha_2\}$ with $\alpha_1 = \varepsilon_1 - \varepsilon_2$, $\alpha_2 = \varepsilon_2 - \varepsilon_3$; it is the base $\Delta(\gamma)$ of Theorem 13.10.5 for $\gamma = \varepsilon_1 - \varepsilon_3$, which pairs positively with α_1, α_2 and $\alpha_1 + \alpha_2$. The positive roots are α_1, α_2, $\alpha_1 + \alpha_2 = \varepsilon_1 - \varepsilon_3$, with root vectors E_{12}, E_{23}, E_{13}, the strictly upper triangular matrices, and $\mathfrak{b}$ consists of the upper triangular matrices of trace 0. The coroots are $h_{\alpha_1} = E_{11} - E_{22}$ and $h_{\alpha_2} = E_{22} - E_{33}$.

1. On $\mathbb{C}^3$, e_i has weight ε_i, and e_1 is a maximal vector: its highest weight is $\varepsilon_1 = \omega_1$, since $\varepsilon_1(h_{\alpha_1}) = 1$ and $\varepsilon_1(h_{\alpha_2}) = 0$.

2. On the dual $(\mathbb{C}^3)^*$ the weights are $-\varepsilon_i$, and the highest is $-\varepsilon_3 = \omega_2$.

3. On the adjoint representation the weights are the six roots and 0, with multiplicity 2; the maximal vector is E_{13}, of weight $\alpha_1 + \alpha_2 = \omega_1 + \omega_2$.

In each case the Weyl group S_3, which permutes the ε_i (Exercise 13.12.12), permutes the weights preserving multiplicities.

13.12 Exercises

Exercise 13.12.1. Let $\mathfrak{sl}_2$ act on $\mathbb{C}^2$ by matrix multiplication. Show that $S^m(\mathbb{C}^2) \cong V(m)$.

Exercise 13.12.2. Show that $V(m) \otimes V(n) \cong \bigoplus_{i=0}^{\min(m,n)} V(m+n-2i)$. (Hint: count weights.)

Solution. The vectors $v_a \otimes w_b$ form a basis of weight vectors, of weight $(m-2a)+(n-2b)$. So the weight $l = m+n-2s$ has multiplicity $c(s) = |\{(a,b) : 0 \le a \le m, 0 \le b \le n, a+b = s\}|$, and for $0 \le s \le (m+n)/2$ one finds $c(s) = \min(s, m, n) + 1$. By Theorem 13.6.3 the representation is a direct sum of $V(l)$'s, and the multiplicity of $V(l)$, $l \ge 0$, is $\dim V_l - \dim V_{l+2} = c(s) - c(s-1)$ (with $c(-1) = 0$), which is 1 for $s \le \min(m,n)$ and 0 otherwise. □

Exercise 13.12.3. Determine the character table of S_4.

Exercise 13.12.4. Show that a finite group is abelian if and only if all its irreducible complex representations are one-dimensional. (One direction is Proposition 13.3.6; for the other use Theorem 13.3.4.)

Exercise 13.12.5. Let $V = \bigoplus_{k=0}^{2n} V^k$ be a graded finite-dimensional vector space and $L\colon V^k \to V^{k+2}$ a linear map. Suppose there is $\Lambda\colon V^k \to V^{k-2}$ such that, with $H = (k-n)\mathrm{id}$ on V^k, the relations $[H, L] = 2L$, $[H, \Lambda] = -2\Lambda$, $[L, \Lambda] = H$ hold. Show that $L^j\colon V^{n-j} \to V^{n+j}$ is an isomorphism for all $j \ge 0$. (This is the algebraic form of the hard Lefschetz theorem.)

Exercise 13.12.6. Show that every Lie algebra of dimension 2 is either abelian or isomorphic to the Lie algebra with basis x, y and $[x, y] = y$ of Example 13.5.3(3).

Solution. Let u, v be a basis. The bracket is determined by $[u, v]$, as $[u, u] = [v, v] = 0$. If $[u, v] = 0$ the algebra is abelian. Otherwise $y = [u, v]$ spans $[\mathfrak{g}, \mathfrak{g}]$, which is one-dimensional. Complete y to a basis y, x'; then $[x', y] = cy$ for some $c \in k$, as $[x', y] \in [\mathfrak{g}, \mathfrak{g}]$, and $c \neq 0$ because $[x', y]$ spans $[\mathfrak{g}, \mathfrak{g}]$. So $x = x'/c$ satisfies $[x, y] = y$. □

Exercise 13.12.7. Let $\mathfrak{g}$ be the real Lie algebra $\mathbb{R}^3$ with the cross product as bracket (it is isomorphic to the Lie algebra $\mathfrak{so}(3)$ of the rotation group, Chapter 34). Show that $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$, so that $\mathfrak{g}$ is not solvable, and that it has no ideals other than 0 and itself.

Solution. $e_1 \times e_2 = e_3$, $e_2 \times e_3 = e_1$ and $e_3 \times e_1 = e_2$, so the brackets span everything and $D^i\mathfrak{g} = \mathfrak{g}$ for all i. The cross product satisfies the Jacobi identity, so $\mathfrak{g}$ is a Lie algebra. If $\mathfrak{a}$ is an ideal containing some $a \neq 0$, then $\mathfrak{a} \supseteq a \times \mathbb{R}^3$, and $a \times \mathbb{R}^3 = a^\perp$, since $v \mapsto a \times v$ has kernel $\mathbb{R}a$ and image in $a^\perp$. For an orthonormal basis u, w of $a^\perp$, $u \times w$ is a unit vector orthogonal to u and w, so $u \times w = \pm a/|a|$ lies in $\mathfrak{a}$. Hence $\mathfrak{a} \supseteq a^\perp + \mathbb{R}a = \mathbb{R}^3$. □

Exercise 13.12.8. Show that the Killing form $\mathrm{tr}(\mathrm{ad}\, x\ \mathrm{ad}\, y)$ of a nilpotent Lie algebra is zero.

Solution. By Corollary 13.5.6 and Theorem 13.5.5 applied to $\mathrm{ad}\,\mathfrak{g}$, there is a basis of $\mathfrak{g}$ in which every $\mathrm{ad}\, x$ is strictly upper triangular. A product of strictly upper triangular matrices is strictly upper triangular, so it has trace 0. □

Exercise 13.12.9. Compute the Gram matrix of the Killing form of $\mathfrak{sl}_2$ in the basis e, h, f, and the Casimir element $c_{\mathrm{ad}} \in \mathrm{End}(\mathfrak{sl}_2)$ of the adjoint representation (Lemma 13.8.1). Show that c_{ad} is $\frac{1}{4}$ times the Casimir operator C of Theorem 13.6.3 acting on $\mathfrak{sl}_2 \cong V(2)$, and deduce that $c_{\mathrm{ad}} = \mathrm{id}$, in agreement with $\operatorname{tr} c_{\mathrm{ad}} = \dim \mathfrak{sl}_2 = 3$.

Solution. In the basis (e, h, f), $\operatorname{ad} h = \operatorname{diag}(2, 0, -2)$, $\operatorname{ad} e$ sends $h \mapsto -2e$, $f \mapsto h$, and $\operatorname{ad} f$ sends $e \mapsto -h$, $h \mapsto 2f$. So $\kappa(h,h) = 8$, $\kappa(e,f) = \operatorname{tr}(\operatorname{ad} e \operatorname{ad} f) = 4$ (the composite sends $e \mapsto 2e$, $h \mapsto 2h$, $f \mapsto 0$), and the other pairings vanish by Proposition 13.9.3(3); the Gram matrix has 8 in the middle and 4 in the two corners (e,f), (f,e), in agreement with $\kappa = 4\operatorname{tr}(xy)$ (Example 13.7.9). The dual basis of (e, h, f) is $(\frac{1}{4}f, \frac{1}{8}h, \frac{1}{4}e)$, so

$$c_{\mathrm{ad}} = \tfrac{1}{4}\operatorname{ad} e \operatorname{ad} f + \tfrac{1}{8}(\operatorname{ad} h)^2 + \tfrac{1}{4}\operatorname{ad} f \operatorname{ad} e = \tfrac{1}{4}\big(\operatorname{ad} e \operatorname{ad} f + \operatorname{ad} f \operatorname{ad} e + \tfrac{1}{2}(\operatorname{ad} h)^2\big),$$

which is $\frac{1}{4}C$ in the adjoint representation. By the proof of Theorem 13.6.3, C acts on $V(m)$ by $\frac{1}{2}m(m+2)$, which is 4 for $m = 2$; so $c_{\mathrm{ad}} = \mathrm{id}$. □

Exercise 13.12.10. Show that $\mathfrak{sl}_n(\mathbb{C})$ is simple for $n \geq 2$. (Hint: an ideal is stable under $\operatorname{ad} \mathfrak{h}$, $\mathfrak{h}$ the diagonal matrices of trace 0.)

Solution. Let $\mathfrak{a} \neq 0$ be an ideal. It is stable under the commuting diagonalisable operators $\operatorname{ad} h$, $h \in \mathfrak{h}$, so it is the sum of its intersections with $\mathfrak{h}$ and with the root spaces $\mathbb{C}E_{ij}$ (Example 13.9.9). If $\mathfrak{a} \subseteq \mathfrak{h}$, take $h \neq 0$ in $\mathfrak{a}$; its diagonal entries are not all equal, since $\operatorname{tr} h = 0$, say $h_i \neq h_j$, and then $[h, E_{ij}] = (h_i - h_j)E_{ij} \in \mathfrak{a}$ is not in $\mathfrak{h}$, a contradiction. So $\mathfrak{a}$ contains some E_{ij}, $i \neq j$. Then it contains $[E_{ij}, E_{ji}] = E_{ii} - E_{jj}$ and $[E_{ji}, E_{ii} - E_{jj}] = 2E_{ji}$, and for $l \notin \{i, j\}$ also $[E_{li}, E_{ij}] = E_{lj}$ and $[E_{ij}, E_{jl}] = E_{il}$. Repeating with the new matrix units, $\mathfrak{a}$ contains all E_{lm} with $l \neq m$, hence all $E_{ll} - E_{mm} = [E_{lm}, E_{ml}]$, and $\mathfrak{a} = \mathfrak{sl}_n$. Finally $\mathfrak{sl}_n$ is not abelian. □

Exercise 13.12.11. Let $\mathfrak{h} \subseteq \mathfrak{sp}_4(\mathbb{C})$ be the diagonal matrices $h = \operatorname{diag}(a_1, a_2, -a_1, -a_2)$ (Example 13.7.14), and $\varepsilon_i(h) = a_i$. Show that $\mathfrak{h}$ is a Cartan subalgebra, that the roots are $\pm 2\varepsilon_1$, $\pm 2\varepsilon_2$, $\pm(\varepsilon_1 + \varepsilon_2)$, $\pm(\varepsilon_1 - \varepsilon_2)$, that $\kappa(h, h') = 12(a_1a_1' + a_2a_2')$ on $\mathfrak{h}$, and that $\Delta = \{\varepsilon_1 - \varepsilon_2, 2\varepsilon_2\}$ is a base with Cartan matrix $\left(\begin{smallmatrix} 2 & -1 \\ -2 & 2 \end{smallmatrix}\right)$. So the root system is $C_2 = B_2$.

Solution. Write $x = \left(\begin{smallmatrix} A & B \\ C & -A^T \end{smallmatrix}\right)$ and $h = \operatorname{diag}(D, -D)$ with $D = \operatorname{diag}(a_1, a_2)$. Then

$$[h, x] = \begin{pmatrix} [D, A] & DB + BD \\ -(DC + CD) & [D, A]' \end{pmatrix},$$

where $[D, A]' = -[D, A]^T$ is determined by the upper left block. The entries transform as $(DA - AD)_{ij} = (a_i - a_j)A_{ij}$, $(DB + BD)_{ij} = (a_i + a_j)B_{ij}$ and $(DC + CD)_{ij} = (a_i + a_j)C_{ij}$. So $\mathfrak{h}$ is toral, the root vectors are $A = E_{12}, E_{21}$ (roots $\pm(\varepsilon_1 - \varepsilon_2)$), the symmetric $B = E_{11}, E_{22}, E_{12} + E_{21}$ (roots $2\varepsilon_1, 2\varepsilon_2, \varepsilon_1 + \varepsilon_2$) and the corresponding C (their negatives). If $[h, x] = 0$ for h with $a_1, a_2, a_1 \pm a_2$ nonzero, then A is diagonal and $B = C = 0$, so $C(\mathfrak{h}) = \mathfrak{h}$, and $\mathfrak{h}$ is a Cartan subalgebra, as in Example 13.9.9. By Theorem 13.9.8(1),

$$\kappa(h, h') = 2\big[(a_1 - a_2)(a_1' - a_2') + 4a_1a_1' + 4a_2a_2' + (a_1 + a_2)(a_1' + a_2')\big] = 12(a_1a_1' + a_2a_2'),$$

so $(\ ,\ )$ is a multiple of the standard inner product in the coordinates $\varepsilon_1, \varepsilon_2$. With $\alpha_1 = \varepsilon_1 - \varepsilon_2$ and $\alpha_2 = 2\varepsilon_2$, the positive roots α_1, α_2, $\alpha_1 + \alpha_2 = \varepsilon_1 + \varepsilon_2$, $2\alpha_1 + \alpha_2 = 2\varepsilon_1$ are non-negative combinations, so Δ is a base, and $\langle \alpha_1, \alpha_2 \rangle = 2(-2)/4 = -1$, $\langle \alpha_2, \alpha_1 \rangle = 2(-2)/2 = -2$. The four long roots $\pm 2\varepsilon_i$ and the four short roots $\pm\varepsilon_1 \pm \varepsilon_2$ form, after rotating by $\pi/4$ and rescaling, the root system B_2 of Example 13.10.3. □

Exercise 13.12.12. Show that the Weyl group of the root system A_{n-1} of $\mathfrak{sl}_n$ (Example 13.9.9) is the symmetric group S_n, acting on $E = \{x \in \mathbb{R}^n : \sum_i x_i = 0\}$ by permuting the coordinates.

Solution. For $\alpha = e_i - e_j$ and $x \in E$, $\langle x, \alpha\rangle = 2(x\cdot\alpha)/(\alpha\cdot\alpha) = x_i - x_j$ (the factor $\frac{1}{2n}$ cancels), so $\sigma_\alpha(x) = x - (x_i - x_j)(e_i - e_j)$ exchanges the coordinates x_i and x_j. So W is the image of the group generated by the transpositions, which is S_n, and S_n acts faithfully on E for $n \geq 2$: a permutation fixing every $e_i - e_j$ is the identity. □

Exercise 13.12.13. Show that α-strings have length at most 4: in Theorem 13.9.6(3), $r + q \leq 3$. Find a string of length 4 in G_2.

Solution. Apply Theorem 13.9.6(3) to the bottom $\beta' = \beta - r\alpha$ of the string, which is the same string; for β' one has $r' = 0$ and $q' = r + q$, so $\langle\beta', \alpha\rangle = -(r+q)$. As $\beta' \neq \pm\alpha$ (β is not a multiple of α), Lemma 13.10.2 gives $|\langle\beta', \alpha\rangle| \leq 3$. In G_2, for the short root $\alpha = \varepsilon_1 - \varepsilon_2$ and the long root $\beta = -(2\varepsilon_1 - \varepsilon_2 - \varepsilon_3)$, the vectors β, $\beta+\alpha = \varepsilon_3 - \varepsilon_1$, $\beta+2\alpha = \varepsilon_3 - \varepsilon_2$ and $\beta+3\alpha = -(2\varepsilon_2 - \varepsilon_1 - \varepsilon_3)$ are roots, while $\beta - \alpha$ and $\beta + 4\alpha$ are not; accordingly $\langle\beta, \alpha\rangle = 2(-3)/2 = -3 = r - q$. □

Exercise 13.12.14. Using the Weyl dimension formula (Remark 13.11.3), check that $\dim V(\omega_1) = 3$ and $\dim V(\omega_1+\omega_2) = 8$ for $\mathfrak{sl}_3$, in agreement with Example 13.11.4, and compute $\dim V(2\omega_1)$. Identify $V(2\omega_1)$ with $S^2(\mathbb{C}^3)$.

Solution. All roots of A_2 have the same length; normalise $(\alpha_i, \alpha_i) = 2$, so $(\alpha_1, \alpha_2) = -1$, and $(\omega_i, \alpha_j) = \delta_{ij}$. By Lemma 13.10.6, $(\rho, \alpha_i) = 1$, so $\rho = \omega_1 + \omega_2$ and $(\rho, \alpha_1 + \alpha_2) = 2$. For $\lambda = a\omega_1 + b\omega_2$, the formula gives

$$\dim V(\lambda) = \frac{(a+1)}{1}\cdot\frac{(b+1)}{1}\cdot\frac{(a+b+2)}{2},$$

which is 3 for $(a,b) = (1,0)$, 8 for $(1,1)$, and 6 for $(2,0)$. The weights of $S^2(\mathbb{C}^3)$ are the $\varepsilon_i + \varepsilon_j$, $i \leq j$, with e_1^2 of weight $2\varepsilon_1 = 2\omega_1$ killed by the upper triangular matrices; its dimension is 6. The subrepresentation W generated by e_1^2 has $W_{2\omega_1} = \mathbb{C}e_1^2$, as in the proof of Theorem 13.11.2(3); in a decomposition $W = W_1 \oplus W_2$ into subrepresentations this line lies in one summand, which then contains e_1^2 and is all of W. So W is indecomposable, hence irreducible by Weyl's theorem, of highest weight $2\omega_1$, and of dimension 6; so $W = S^2(\mathbb{C}^3)$. □

Exercise 13.12.15. Show that Theorem 13.7.11 fails for the abelian Lie algebra k^2 and for the Heisenberg algebra $\mathfrak{n}_3$ (Example 13.5.3(3)): exhibit derivations that are not inner.

Solution. For an abelian Lie algebra every linear map is a derivation, while all inner derivations are 0. For $\mathfrak{n}_3$ with $[E_{12}, E_{23}] = E_{13}$, the linear map δ with $\delta E_{12} = E_{12}$, $\delta E_{23} = 0$, $\delta E_{13} = E_{13}$ is a derivation: on the only nonzero bracket, $\delta[E_{12}, E_{23}] = E_{13} = [\delta E_{12}, E_{23}] + [E_{12}, \delta E_{23}]$, and the brackets involving E_{13} vanish on both sides. An inner derivation $\operatorname{ad} x$ maps $\mathfrak{n}_3$ into $[\mathfrak{n}_3, \mathfrak{n}_3] = \mathbb{C}E_{13}$, while $\delta E_{12} = E_{12}$; so δ is not inner. □

Chapter 14

Commutative Algebra

14.1 Introduction

Commutative algebra is the local theory of algebraic geometry. This chapter develops the tools used in Part VIII and in the local theory of analytic functions (Chapter 31): localisation, Nakayama's lemma, Noetherian and Artinian rings, integral extensions and Noether normalisation, the Nullstellensatz, primary decomposition, dimension theory, regular local rings and Dedekind domains, completions, and Kähler differentials.

References are [AM69], [Mat86] and [Eis95].

Prerequisites. Chapter 8 (Rings and Modules), Chapter 9 (Fields and Galois Theory) and Chapter 11 (Linear Algebra).

In accordance with Notation 1.6.4, from now on all rings are commutative with 1.

14.2 Prime ideals and radicals

Definition 14.2.1. The *spectrum* $\operatorname{Spec} R$ is the set of prime ideals of R. For an ideal I let $V(I) = \{\mathfrak{p} \in \operatorname{Spec} R : I \subseteq \mathfrak{p}\}$, and for $f \in R$ let $D(f) = \{\mathfrak{p} : f \notin \mathfrak{p}\}$. The *nilradical* of R is the set of nilpotent elements, the *Jacobson radical* $J(R)$ is the intersection of all maximal ideals, and the *radical* of an ideal is $\sqrt{I} = \{f : f^n \in I \text{ for some } n\}$.

Proposition 14.2.2. *1. The nilradical is the intersection of all prime ideals, and $\sqrt{I}$ is the intersection of all primes containing I.*

2. *More precisely, if $S \subseteq R$ is closed under multiplication, $1 \in S$, and I is an ideal with $I \cap S = \emptyset$, then there is a prime $\mathfrak{p} \supseteq I$ with $\mathfrak{p} \cap S = \emptyset$.*

3. *$x \in J(R)$ if and only if $1 - xy$ is a unit for all $y \in R$.*

Proof. (2) The set of ideals containing I and disjoint from S is nonempty and closed under unions of chains, so it has a maximal element $\mathfrak{p}$ by Zorn's lemma. If $a, b \notin \mathfrak{p}$, then $\mathfrak{p} + (a)$ and $\mathfrak{p} + (b)$ meet S, say $s = p_1 + ra$ and $t = p_2 + r'b$; then $st \in \mathfrak{p} + (ab)$ lies in S, so $ab \notin \mathfrak{p}$. Also $1 \in S$, so $\mathfrak{p} \neq R$. (1) A nilpotent element lies in every prime. If f is not nilpotent, apply (2) with $S = \{f^n\}$ and $I = 0$. For $\sqrt{I}$, apply this to R/I. (3) If $x \in J(R)$ and $1 - xy$ is not a unit, it lies in a maximal ideal $\mathfrak{m}$ (Theorem 8.4.3), which also contains xy, hence 1. Conversely if $x \notin \mathfrak{m}$ for a maximal $\mathfrak{m}$, then $\mathfrak{m} + (x) = R$, so $1 = u + xy$ with $u \in \mathfrak{m}$, and $1 - xy = u$ is not a unit. □

Lemma 14.2.3 (Prime avoidance)**.** *If an ideal I is contained in the union of prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_n$, then $I \subseteq \mathfrak{p}_i$ for some i.*

Proof. By induction on n; suppose $I \not\subseteq \bigcup_{j\neq i} \mathfrak{p}_j$ for each i (otherwise use induction), and choose $x_i \in I$ with $x_i \notin \mathfrak{p}_j$ for $j \neq i$. Then $x_i \in \mathfrak{p}_i$ for all i. The element $y = x_1 + x_2 \cdots x_n$ lies in I but in no $\mathfrak{p}_i$: for $i = 1$, $x_1 \in \mathfrak{p}_1$ and $x_2 \cdots x_n \notin \mathfrak{p}_1$; for $i \geq 2$, $x_1 \notin \mathfrak{p}_i$ while $x_2 \cdots x_n \in \mathfrak{p}_i$. This contradicts $I \subseteq \mathfrak{p}_1 \cup \cdots \cup \mathfrak{p}_n$. □

14.3 Localisation

Definition 14.3.1. A subset $S \subseteq R$ is *multiplicative* if $1 \in S$ and $st \in S$ for $s, t \in S$. For an R-module M, the *localisation* $S^{-1}M$ is the set of equivalence classes m/s of pairs $(m, s) \in M \times S$ under $(m, s) \sim (m', s') \iff t(s'm - sm') = 0$ for some $t \in S$, with $m/s + m'/s' = (s'm + sm')/ss'$. For $M = R$ it is a ring with $(r/s)(r'/s') = rr'/ss'$, and $S^{-1}M$ is an $S^{-1}R$-module. For $f \in R$ we write R_f for $S^{-1}R$ with $S = \{f^n\}$, and for a prime $\mathfrak{p}$ we write $R_\mathfrak{p}$, $M_\mathfrak{p}$ for $S = R \setminus \mathfrak{p}$. A ring is *local* if it has exactly one maximal ideal.

Proposition 14.3.2. *1. $\sim$ is an equivalence relation and the operations are well defined. The map $R \to S^{-1}R$, $r \mapsto r/1$, is a ring homomorphism sending S to units, and every homomorphism $\varphi\colon R \to B$ sending S to units factors uniquely through it.*

2. If $M' \to M \to M''$ is exact, so is $S^{-1}M' \to S^{-1}M \to S^{-1}M''$. There is a natural isomorphism $S^{-1}R \otimes_R M \cong S^{-1}M$; hence $S^{-1}R$ is a flat R-module.

3. The primes of $S^{-1}R$ are the $S^{-1}\mathfrak{p}$ for primes $\mathfrak{p}$ of R with $\mathfrak{p} \cap S = \emptyset$, and this is an inclusion-preserving bijection with inverse $\mathfrak{q} \mapsto$ its preimage in R. In particular $R_\mathfrak{p}$ is local with maximal ideal $\mathfrak{p}R_\mathfrak{p}$.

4. A ring is local if and only if its nonunits form an ideal.

Proof. (1) Transitivity: from $t(s'm - sm') = 0$ and $t'(s''m' - s'm'') = 0$, multiplying by $t's''$ and ts and adding gives $tt's'(s''m - sm'') = 0$. The other verifications are similar. Given φ, the map $r/s \mapsto \varphi(r)\varphi(s)^{-1}$ is well defined since $t(s'r - sr') = 0$ implies $\varphi(s'r) = \varphi(sr')$, and it is forced. (2) Let $f\colon M' \to M$, $g\colon M \to M''$ with $\operatorname{im} f = \ker g$. If $g(m)/s = 0$, then $tg(m) = 0$ for some t, so $tm = f(m')$ and $m/s = f(m')/ts$. The map $S^{-1}R \times M \to S^{-1}M$, $(r/s, m) \mapsto rm/s$, is bilinear, and $m/s \mapsto (1/s) \otimes m$ is a well-defined inverse of the induced map: if $t(s'm - sm') = 0$ then $(1/s) \otimes m = (1/tss') \otimes ts'm = (1/tss') \otimes tsm' = (1/s') \otimes m'$. So $S^{-1}R \otimes -$ is exact, that is, $S^{-1}R$ is flat. (3) If $\mathfrak{q}$ is a prime of $S^{-1}R$, its preimage $\mathfrak{p}$ is prime and disjoint from S, and $\mathfrak{q} = S^{-1}\mathfrak{p}$ since $r/s \in \mathfrak{q}$ iff $r/1 \in \mathfrak{q}$. Conversely, for $\mathfrak{p}$ disjoint from S, $S^{-1}\mathfrak{p}$ is a proper ideal, it is prime because $(r/s)(r'/s') \in S^{-1}\mathfrak{p}$ gives $trr' \in \mathfrak{p}$ for some $t \in S$, and its preimage is $\mathfrak{p}$ by the same argument. For $R_\mathfrak{p}$, the primes correspond to primes contained in $\mathfrak{p}$. (4) If the nonunits form an ideal, it is the unique maximal ideal. If R is local with maximal ideal $\mathfrak{m}$, every nonunit lies in a maximal ideal, hence in $\mathfrak{m}$. □

Proposition 14.3.3. *For an R-module M the following are equivalent: $M = 0$; $M_\mathfrak{p} = 0$ for all primes $\mathfrak{p}$; $M_\mathfrak{m} = 0$ for all maximal ideals $\mathfrak{m}$. A homomorphism $M \to N$ is injective (surjective) if and only if all $M_\mathfrak{m} \to N_\mathfrak{m}$ are.*

Proof. If $x \in M$ is nonzero, its annihilator is a proper ideal, contained in a maximal ideal $\mathfrak{m}$, and then $x/1 \neq 0$ in $M_\mathfrak{m}$. The second statement follows by applying the first to kernels and cokernels, which commute with localisation by Proposition 14.3.2(2). □

Example 14.3.4. $\mathbb{Z}_{(p)}$ is the ring of rationals whose denominators are prime to p, a local ring with maximal ideal $p\mathbb{Z}_{(p)}$; $\mathbb{Z}_{(0)} = \mathbb{Q}$; and the localisation of $\mathbb{Z}$ at $f = p$ is $\mathbb{Z}[1/p]$, the ring of rationals whose denominators are powers of p (we avoid the notation $\mathbb{Z}_p$ for it, which is reserved for the p-adic integers). Localisation can kill elements: in $(\mathbb{Z}/6\mathbb{Z})_{(2)}$ the element 3 is a unit and $2 \cdot 3 = 0$, so $2/1 = 0$ and $(\mathbb{Z}/6\mathbb{Z})_{(2)} \cong \mathbb{Z}/2\mathbb{Z}$. If f is nilpotent, $R_f = 0$.

14.4 Nakayama's lemma

Lemma 14.4.1 (Determinant trick)**.** *Let M be an R-module generated by n elements, I an ideal and $\varphi\colon M \to M$ an R-linear map with $\varphi(M) \subseteq IM$. Then $\varphi^n + a_1\varphi^{n-1} + \cdots + a_n = 0$ for some $a_i \in I$.*

Proof. Let $x_1, \dots, x_n$ generate M and write $\varphi(x_i) = \sum_j a_{ij}x_j$ with $a_{ij} \in I$. View M as a module over the commutative ring $R[\varphi]$ of endomorphisms. The matrix $T = \varphi I - (a_{ij})$ over $R[\varphi]$ kills the column vector (x_j), so by Proposition 11.5.3(4) $\det(T)x_j = \operatorname{adj}(T)T(x) = 0$ for all j, that is, $\det T = 0$ on M. Expanding the determinant gives the claim. □

Theorem 14.4.2 (Nakayama's lemma)**.** *Let M be a finitely generated R-module and I an ideal with $IM = M$. Then there is $a \in 1 + I$ with $aM = 0$. If $I \subseteq J(R)$, then $M = 0$.*

Proof. Apply Lemma 14.4.1 to $\varphi = \mathrm{id}$: $a = 1 + a_1 + \cdots + a_n$ kills M. If $I \subseteq J(R)$, a is a unit by Proposition 14.2.2(3). □

Corollary 14.4.3. *Let $(R, \mathfrak{m}, k)$ be a local ring and M a finitely generated module. Elements $x_1, \dots, x_n \in M$ generate M if and only if their images generate the k-vector space $M/\mathfrak{m}M$. Every minimal generating set has $\dim_k M/\mathfrak{m}M$ elements.*

Proof. Let N be the submodule generated by the x_i. If their images generate $M/\mathfrak{m}M$ then $M = N + \mathfrak{m}M$, so $\mathfrak{m}(M/N) = M/N$ and $M/N = 0$ by Theorem 14.4.2. The last statement follows because a generating set of M maps to a spanning set of $M/\mathfrak{m}M$, and a minimal one to a basis. □

14.5 Noetherian and Artinian rings

Definition 14.5.1. A module is *Noetherian* if every increasing chain of submodules is eventually constant, and *Artinian* if every decreasing chain is. A ring is Noetherian (Artinian) if it is so as a module over itself.

Proposition 14.5.2.

1. *M is Noetherian if and only if every submodule is finitely generated, if and only if every nonempty set of submodules has a maximal element.*
2. *If $0 \to M' \to M \to M'' \to 0$ is exact, then M is Noetherian (Artinian) if and only if M' and M'' are.*
3. *Over a Noetherian ring, finitely generated modules are Noetherian; quotients and localisations of Noetherian rings are Noetherian.*

Proof. (1) If N is not finitely generated, choose $x_1, x_2, \dots \in N$ with $x_{n+1} \notin (x_1, \dots, x_n)$, a strictly increasing chain (using dependent choice). If every submodule is finitely generated, the union of a chain is generated by finitely many elements, which lie in one member. A nonempty set without maximal element contains a strictly increasing chain, and conversely a strictly increasing chain has no maximal element. (2) A chain in M gives chains in M' and M''; if both stabilise, so does the original, because $N_1 \subseteq N_2$ with $N_1 \cap M' = N_2 \cap M'$ and equal images in M'' implies $N_1 = N_2$. The converse is clear. (3) R^n is Noetherian by (2) and induction, and finitely generated modules are its quotients. Ideals of $S^{-1}R$ are of the form $S^{-1}I$ (by the argument of Proposition 14.3.2(3)), so they are finitely generated. □

Theorem 14.5.3 (Hilbert's basis theorem)**.** *If R is Noetherian, so is $R[x]$. Consequently every finitely generated R-algebra is Noetherian; in particular $k[x_1, \dots, x_n]$ and $\mathbb{Z}[x_1, \dots, x_n]$ are Noetherian.*

Proof. Let $I \subseteq R[x]$ be an ideal and $L_d \subseteq R$ the set of leading coefficients of elements of I of degree d, together with 0. These are ideals with $L_d \subseteq L_{d+1}$ (multiply by x), so $L_d = L_{d_0}$ for $d \geq d_0$. For each $d \leq d_0$ choose finitely many $f_{d,j} \in I$ of degree d whose leading coefficients generate L_d. Let I' be the ideal they generate. If $f \in I$ has degree d, some combination $g = \sum_j c_j x^{d-\min(d,d_0)} f_{\min(d,d_0),j}$ has the same degree and leading coefficient, so $f - g \in I$ has smaller degree; by induction on the degree $f \in I'$. So $I = I'$. The last statement follows from quotients (Proposition 14.5.2). □

Counterexample 14.5.4. The polynomial ring $k[x_1, x_2, \ldots]$ in infinitely many variables is not Noetherian: $(x_1) \subsetneq (x_1, x_2) \subsetneq \cdots$. It is a subring of its fraction field, which is Noetherian; so subrings of Noetherian rings need not be Noetherian.

Lemma 14.5.5. *Let R be a ring and $\mathfrak{m}_1, \ldots, \mathfrak{m}_n$ maximal ideals (not necessarily distinct) with $\mathfrak{m}_1 \cdots \mathfrak{m}_n = 0$. Then R is Noetherian if and only if it is Artinian.*

Proof. Consider the chain $R \supseteq \mathfrak{m}_1 \supseteq \mathfrak{m}_1\mathfrak{m}_2 \supseteq \cdots \supseteq \mathfrak{m}_1 \cdots \mathfrak{m}_n = 0$. Each quotient is a vector space over the field $R/\mathfrak{m}_i$, and for a vector space the ascending and the descending chain conditions are both equivalent to finite dimension. By Proposition 14.5.2(2) applied along the chain, R satisfies ACC iff all quotients are finite-dimensional iff R satisfies DCC. □

Theorem 14.5.6. *An Artinian ring R is Noetherian, every prime ideal of R is maximal, R has finitely many maximal ideals, its nilradical is nilpotent, and R is a finite product of local Artinian rings. (For the converse see Corollary 14.8.5.)*

Proof. Let R be Artinian. If $\mathfrak{p}$ is prime, $R/\mathfrak{p}$ is an Artinian domain; for $x \neq 0$ the chain $(x) \supseteq (x^2) \supseteq \cdots$ stabilises, $x^n = x^{n+1}y$, so $xy = 1$; hence $\mathfrak{p}$ is maximal. Among finite intersections of maximal ideals choose a minimal one, $\mathfrak{m}_1 \cap \cdots \cap \mathfrak{m}_n$; every maximal $\mathfrak{m}$ contains it, hence contains $\mathfrak{m}_1 \cdots \mathfrak{m}_n$ and so some $\mathfrak{m}_i$, and $\mathfrak{m} = \mathfrak{m}_i$. The nilradical $N = \mathfrak{m}_1 \cap \cdots \cap \mathfrak{m}_n$ satisfies $N^k = N^{k+1} = \cdots =: I$ for some k. If $I \neq 0$, choose an ideal J minimal with $IJ \neq 0$; there is $x \in J$ with $xI \neq 0$, so $J = (x)$ by minimality, and $(xI)I = xI \neq 0$ with $xI \subseteq (x)$ gives $xI = (x)$, so $x = xy$ with $y \in I$, hence $x = xy^m = 0$ as y is nilpotent; contradiction. So N is nilpotent, $(\mathfrak{m}_1 \cdots \mathfrak{m}_n)^k \subseteq N^k = 0$, and R is Noetherian by Lemma 14.5.5.

Finally, the ideals $\mathfrak{m}_i^k$ are pairwise comaximal and their product is 0, so $R \cong \prod_i R/\mathfrak{m}_i^k$ by Theorem 8.3.3, and $R/\mathfrak{m}_i^k$ is local with maximal ideal $\mathfrak{m}_i/\mathfrak{m}_i^k$. □

14.6 Integral extensions

Definition 14.6.1. Let $A \subseteq B$ be rings. An element $b \in B$ is *integral* over A if it is a root of a monic polynomial with coefficients in A. B is *integral* over A if all its elements are, and *finite* over A if it is a finitely generated A-module. The *integral closure* of A in B is the set of elements integral over A. A domain is *normal* (or integrally closed) if it equals its integral closure in its fraction field.

Proposition 14.6.2.

1. *b is integral over A if and only if $A[b]$ is a finite A-module, if and only if b lies in a subring $C \subseteq B$ that is a finite A-module.*
2. *The integral closure of A in B is a subring. If $A \subseteq B \subseteq C$ with C integral over B and B integral over A, then C is integral over A.*
3. *If $A \subseteq B$ are domains and B is integral over A, then B is a field if and only if A is.*
4. *A unique factorisation domain is normal.*

Proof. (1) A monic equation of degree n shows that $1, b, \dots, b^{n-1}$ generate $A[b]$. If $b \in C$ with C finite, apply Lemma 14.4.1 to multiplication by b on C with $I = A$; the resulting polynomial kills $1 \in C$, so it vanishes at b. (2) If b, b' are integral, $A[b, b']$ is finite over A, so $b \pm b'$ and bb' are integral by (1). If $c \in C$ satisfies a monic equation with coefficients $b_1, \dots, b_n \in B$, then $A[b_1, \dots, b_n, c]$ is finite over A by the tower of finite extensions, so c is integral over A. (3) If A is a field and $0 \neq b \in B$ satisfies $b^n + a_1 b^{n-1} + \cdots + a_n = 0$ with n minimal, then $a_n \neq 0$ (as B is a domain), and $b^{-1} = -a_n^{-1}(b^{n-1} + \cdots + a_{n-1}) \in B$. If B is a field and $0 \neq a \in A$, then $a^{-1} \in B$ is integral over A: $a^{-m} + c_1 a^{-m+1} + \cdots + c_m = 0$, so $a^{-1} = -(c_1 + c_2 a + \cdots + c_m a^{m-1}) \in A$. (4) If $x = r/s$ in lowest terms satisfies a monic equation of degree n, then $r^n = -s(a_1 r^{n-1} + \cdots + a_n s^{n-1})$, so every prime factor of s divides r; hence s is a unit. □

Theorem 14.6.3 (Lying over and going up)**.** *Let $A \subseteq B$ be an integral extension and $\mathfrak{p}$ a prime of A.*

1. *There is a prime $\mathfrak{q}$ of B with $\mathfrak{q} \cap A = \mathfrak{p}$.*
2. *If $\mathfrak{q} \subseteq \mathfrak{q}'$ are primes of B with $\mathfrak{q} \cap A = \mathfrak{q}' \cap A$, then $\mathfrak{q} = \mathfrak{q}'$ (incomparability). A prime $\mathfrak{q}$ is maximal if and only if $\mathfrak{q} \cap A$ is maximal.*
3. *If $\mathfrak{p} \subseteq \mathfrak{p}'$ are primes of A and $\mathfrak{q} \cap A = \mathfrak{p}$, there is $\mathfrak{q}' \supseteq \mathfrak{q}$ with $\mathfrak{q}' \cap A = \mathfrak{p}'$ (going up).*

Proof. For a prime $\mathfrak{q}$ of B over $\mathfrak{p}$, the extension $A/\mathfrak{p} \subseteq B/\mathfrak{q}$ is integral, so by Proposition 14.6.2(3) $\mathfrak{q}$ is maximal iff $\mathfrak{p}$ is. This gives the second part of (2).

(1) Let $S = A \setminus \mathfrak{p}$. Then $A_\mathfrak{p} \subseteq S^{-1}B$ is integral and $S^{-1}B \neq 0$. Let $\mathfrak{n}$ be a maximal ideal of $S^{-1}B$. Then $\mathfrak{n} \cap A_\mathfrak{p}$ is maximal by the first remark, hence equals $\mathfrak{p}A_\mathfrak{p}$, and the preimage $\mathfrak{q}$ of $\mathfrak{n}$ in B satisfies $\mathfrak{q} \cap A = \mathfrak{p}$. (2) Localising at $S = A \setminus \mathfrak{p}$ as in (1), $S^{-1}\mathfrak{q} \subseteq S^{-1}\mathfrak{q}'$ both lie over the maximal ideal $\mathfrak{p}A_\mathfrak{p}$, hence are maximal, hence equal, and so $\mathfrak{q} = \mathfrak{q}'$. (3) Apply (1) to the integral extension $A/\mathfrak{p} \subseteq B/\mathfrak{q}$ and the prime $\mathfrak{p}'/\mathfrak{p}$. □

Remark 14.6.4. If moreover A is a normal domain and B is a domain, the *going down* theorem holds: given $\mathfrak{p}' \subseteq \mathfrak{p}$ and $\mathfrak{q}$ over $\mathfrak{p}$, there is $\mathfrak{q}' \subseteq \mathfrak{q}$ over $\mathfrak{p}'$ [AM69, Theorem 5.16].

Theorem 14.6.5 (Noether normalisation)**.** *Let A be a finitely generated algebra over a field k. There are elements $y_1, \dots, y_d \in A$, algebraically independent over k, such that A is finite over $k[y_1, \dots, y_d]$.*

Proof. By induction on the number n of generators $x_1, \dots, x_n$. If they are algebraically independent, take $y_i = x_i$. Otherwise let $F \neq 0$ be a polynomial with $F(x_1, \dots, x_n) = 0$, and let r be larger than all exponents occurring in F. Put $z_i = x_i - x_n^{r^i}$ for $i < n$. Substituting $x_i = z_i + x_n^{r^i}$, a monomial $c\, x_1^{a_1} \cdots x_n^{a_n}$ of F becomes $c\, x_n^{a_n + a_1 r + \cdots + a_{n-1} r^{n-1}}$ plus terms of lower degree in x_n with coefficients in $k[z_1, \dots, z_{n-1}]$. The exponents $a_n + a_1 r + \cdots$ of distinct monomials are distinct (base r expansions), so exactly one monomial produces the highest power of x_n. Dividing by its coefficient, x_n satisfies a monic equation over $k[z_1, \dots, z_{n-1}]$. So $A = k[z_1, \dots, z_{n-1}][x_n]$ is finite over $k[z_1, \dots, z_{n-1}]$, which by induction is finite over a polynomial ring $k[y_1, \dots, y_d]$; conclude with Proposition 14.6.2(2). □

14.7 The Nullstellensatz

Theorem 14.7.1 (Zariski's lemma)**.** *Let k be a field and L a field that is finitely generated as a k-algebra. Then L is a finite extension of k. Consequently, for every finitely generated k-algebra A and maximal ideal $\mathfrak{m}$, $A/\mathfrak{m}$ is finite over k.*

Proof. By Theorem 14.6.5, L is finite over some $k[y_1, \dots, y_d]$, which is then a field by Proposition 14.6.2(3). A polynomial ring in $d \geq 1$ variables is not a field, so $d = 0$. □

Theorem 14.7.2 (Hilbert's Nullstellensatz). *Let k be algebraically closed. For $I \subseteq k[x_1, \dots, x_n]$ let $Z(I) = \{a \in k^n : f(a) = 0 \text{ for all } f \in I\}$.*

1. *The maximal ideals of $k[x_1, \dots, x_n]$ are the ideals $(x_1 - a_1, \dots, x_n - a_n)$, $a \in k^n$.*
2. *If $I \neq (1)$ then $Z(I) \neq \emptyset$.*
3. *If f vanishes on $Z(I)$, then $f \in \sqrt{I}$.*

Proof. (1) The ideal $(x_1 - a_1, \dots, x_n - a_n)$ is the kernel of evaluation at a, which is surjective onto k, so it is maximal. If $\mathfrak{m}$ is maximal, $k[x]/\mathfrak{m}$ is a finite extension of k by Theorem 14.7.1, hence equals k; so $x_i \equiv a_i$ modulo $\mathfrak{m}$ for some $a_i \in k$, and $\mathfrak{m}$ contains, hence equals, $(x - a)$. (2) A proper ideal lies in a maximal ideal $(x - a)$, and then $a \in Z(I)$. (3) (Rabinowitsch) We may assume $f \neq 0$. In $k[x_1, \dots, x_n, y]$ the ideal $J = Ik[x, y] + (1 - yf)$ has no zeros, since at a zero (a, b) we would have $a \in Z(I)$, so $f(a) = 0$ and $1 - bf(a) = 1$. By (2), $1 = \sum_i g_i h_i + g(1 - yf)$ with $h_i \in I$. Substituting $y = 1/f$ in $k(x_1, \dots, x_n)$ and multiplying by a high power f^N gives $f^N = \sum_i g_i' h_i \in I$. □

Counterexample 14.7.3. Algebraic closedness is needed: over $\mathbb{R}$, the ideal $(x^2 + 1) \subsetneq \mathbb{R}[x]$ is maximal but has no zeros in $\mathbb{R}$, and $\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$ is a finite extension of $\mathbb{R}$ different from $\mathbb{R}$.

Corollary 14.7.4. *In a finitely generated algebra A over a field, the nilradical equals the Jacobson radical.*

Proof. Let $f \in A$ be non-nilpotent. Then $A_f \neq 0$ has a maximal ideal $\mathfrak{n}$, and $A_f \cong A[t]/(tf - 1)$ is finitely generated, so $A_f/\mathfrak{n}$ is a finite field extension of k. The preimage $\mathfrak{m}$ of $\mathfrak{n}$ in A does not contain f, and $k \subseteq A/\mathfrak{m} \subseteq A_f/\mathfrak{n}$, so $A/\mathfrak{m}$ is a domain finite-dimensional over k, hence a field (Proposition 14.6.2(3)). So $f \notin J(A)$. □

14.8 Primary decomposition

In this section R is Noetherian.

Definition 14.8.1. Let M be an R-module. A prime $\mathfrak{p}$ is *associated* to M if $\mathfrak{p} = \operatorname{Ann}(x)$ for some $x \in M$; we write $\operatorname{Ass}(M)$. An ideal $\mathfrak{q} \neq R$ is *primary* if $xy \in \mathfrak{q}$ and $x \notin \mathfrak{q}$ imply $y^n \in \mathfrak{q}$ for some n. Then $\mathfrak{p} = \sqrt{\mathfrak{q}}$ is prime, and $\mathfrak{q}$ is called $\mathfrak{p}$-primary.

Proposition 14.8.2. *Let M be a finitely generated R-module.*

1. *Every maximal element of $\{\operatorname{Ann}(x) : 0 \neq x \in M\}$ is a prime ideal; so $M \neq 0$ implies $\operatorname{Ass}(M) \neq \emptyset$.*
2. *The set of zero divisors on M is $\bigcup_{\mathfrak{p} \in \operatorname{Ass}(M)} \mathfrak{p}$.*
3. *There is a chain $0 = M_0 \subseteq \cdots \subseteq M_n = M$ with $M_i/M_{i-1} \cong R/\mathfrak{p}_i$ for primes $\mathfrak{p}_i$, and $\operatorname{Ass}(M) \subseteq \{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}$; in particular $\operatorname{Ass}(M)$ is finite.*

Proof. (1) Let $I = \operatorname{Ann}(x)$ be maximal and $ab \in I$ with $b \notin I$. Then $bx \neq 0$ and $\operatorname{Ann}(bx) \supseteq I + (a)$, so $a \in I$ by maximality. Maximal elements exist by Proposition 14.5.2(1). (2) Elements of associated primes are clearly zero divisors. If $ax = 0$ with $x \neq 0$, choose $z \in Rx$, $z \neq 0$, with $\operatorname{Ann}(z)$ maximal among the annihilators of nonzero elements of Rx. The argument of (1), applied inside Rx, shows that $\operatorname{Ann}(z)$ is prime, and it contains $\operatorname{Ann}(x) \ni a$. (3) By (1), choose $M_1 = Rx_1 \cong R/\mathfrak{p}_1$, then apply (1) to M/M_1, and so on; the chain stops since M is Noetherian. If $0 \to M' \to M \to M'' \to 0$ is exact then $\operatorname{Ass}(M) \subseteq \operatorname{Ass}(M') \cup \operatorname{Ass}(M'')$: if $\operatorname{Ann}(x) = \mathfrak{p}$ and $Rx \cap M' \neq 0$, a nonzero $yx \in M'$ has annihilator $\mathfrak{p}$ (as $R/\mathfrak{p}$ is a domain); otherwise Rx maps isomorphically into M''. Finally $\operatorname{Ass}(R/\mathfrak{p}) = \{\mathfrak{p}\}$ since $R/\mathfrak{p}$ is a domain. □

Theorem 14.8.3 (Lasker–Noether). *Every ideal $I \neq R$ of a Noetherian ring is a finite intersection $I = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_r$ of primary ideals. One can choose the decomposition with the primes $\sqrt{\mathfrak{q}_i}$ distinct and no $\mathfrak{q}_i$ superfluous; the set of these primes is then $\operatorname{Ass}(R/I)$, independent of the decomposition. The minimal elements of $\operatorname{Ass}(R/I)$ are the minimal primes containing I.*

Proof. Call I *irreducible* if $I = J \cap J'$ implies $I = J$ or $I = J'$. If some ideal is not a finite intersection of irreducible ideals, a maximal such ideal is reducible, a contradiction; so every ideal is a finite intersection of irreducibles. An irreducible ideal I is primary: passing to R/I, suppose $xy = 0$ with $x \neq 0$. The chain $\operatorname{Ann}(y) \subseteq \operatorname{Ann}(y^2) \subseteq \cdots$ stabilises at n; then $(x) \cap (y^n) = 0$ (if $ax = by^n$ then $axy = 0 = by^{n+1}$, so $by^n = 0$), and irreducibility of 0 gives $y^n = 0$.

If $\mathfrak{q}$ is $\mathfrak{p}$-primary, then $\operatorname{Ass}(R/\mathfrak{q}) = \{\mathfrak{p}\}$: if $\operatorname{Ann}(\bar{x}) = \mathfrak{p}'$ with $x \notin \mathfrak{q}$, then $\mathfrak{p}'x \subseteq \mathfrak{q}$ gives $\mathfrak{p}' \subseteq \mathfrak{p}$, and every $a \in \mathfrak{p}$ has $a^n\bar{x} = 0$, so $a \in \mathfrak{p}'$. A finite intersection of $\mathfrak{p}$-primary ideals is $\mathfrak{p}$-primary, so we may merge components with the same prime, and delete superfluous ones. Then $R/I \hookrightarrow \bigoplus_i R/\mathfrak{q}_i$ gives $\operatorname{Ass}(R/I) \subseteq \{\mathfrak{p}_i\}$, by the argument in Proposition 14.8.2(3). Conversely, if $\mathfrak{q}_i$ is not superfluous, choose $x \in \bigcap_{j\neq i} \mathfrak{q}_j \setminus \mathfrak{q}_i$; then $\bar{x} \in R/I$ generates a submodule isomorphic to a submodule of $R/\mathfrak{q}_i$, whose associated primes are $\{\mathfrak{p}_i\}$, so $\mathfrak{p}_i \in \operatorname{Ass}(R/I)$. For the last statement, a prime contains $I = \bigcap \mathfrak{q}_i$ iff it contains some $\mathfrak{q}_i$ iff it contains some $\mathfrak{p}_i$. □

Example 14.8.4. In $k[x, y]$, $(x^2, xy) = (x) \cap (x^2, y)$, where (x) is prime and (x^2, y) is (x, y)-primary. The associated primes are (x) and the *embedded* prime $(x, y) \supsetneq (x)$. Geometrically, the ideal defines the line $x = 0$ with an extra infinitesimal structure at the origin.

Corollary 14.8.5. *A Noetherian ring in which every prime ideal is maximal is Artinian.*

Proof. By Theorem 14.8.3 applied to $I = 0$, the ring has finitely many minimal primes; since all primes are maximal, these are all the primes, say $\mathfrak{m}_1, \ldots, \mathfrak{m}_n$. Their intersection is the nilradical (Proposition 14.2.2), which is finitely generated, hence nilpotent: $(\mathfrak{m}_1 \cap \cdots \cap \mathfrak{m}_n)^k = 0$. Then $(\mathfrak{m}_1 \cdots \mathfrak{m}_n)^k = 0$, and the ring is Artinian by Lemma 14.5.5. □

14.9 Dimension theory

Definition 14.9.1. The *Krull dimension* $\dim R$ is the supremum of the lengths n of chains of primes $\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_n$. The *height* of a prime is $\operatorname{ht}\mathfrak{p} = \dim R_{\mathfrak{p}}$, the supremum of lengths of chains ending at $\mathfrak{p}$.

Proposition 14.9.2. *If B is integral over A, then $\dim B = \dim A$.*

Proof. A chain of primes of B contracts to a chain in A with strict inclusions by incomparability, and a chain in A lifts by lying over and going up (Theorem 14.6.3). □

Theorem 14.9.3. *Let A be a finitely generated k-algebra that is a domain. Then $\dim A = \operatorname{trdeg}_k \operatorname{Frac}(A)$. In particular $\dim k[x_1, \ldots, x_n] = n$.*

Proof. Let $t = \operatorname{trdeg}_k \operatorname{Frac}(A)$. By Theorem 14.6.5, A is finite over a polynomial ring $k[y_1, \ldots, y_d]$; then $\operatorname{Frac}(A)$ is algebraic over $k(y_1, \ldots, y_d)$, so $d = t$, and $\dim A = \dim k[y_1, \ldots, y_t]$ by Proposition 14.9.2. The chain $0 \subsetneq (y_1) \subsetneq (y_1, y_2) \subsetneq \cdots \subsetneq (y_1, \ldots, y_t)$ consists of primes (the quotients are polynomial rings), so $\dim \geq t$. We show $\dim A \leq t$ for all such A by induction on t; by the above we may assume $A = k[y_1, \ldots, y_t]$. Let $0 = \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_m$ be a chain. The domain $A/\mathfrak{p}_1$ is generated by the images of the y_i, which satisfy a nontrivial polynomial relation (any nonzero element of $\mathfrak{p}_1$), so its fraction field has transcendence degree $< t$ (Theorem 9.10.3). By induction the chain $\mathfrak{p}_1/\mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_m/\mathfrak{p}_1$ has length $m - 1 \leq t - 1$. □

Lemma 14.9.4. *A Noetherian local ring $(R, \mathfrak{m})$ in which $\mathfrak{m}$ is the only prime is Artinian, and $\mathfrak{m}^n = 0$ for some n.*

Proof. The nilradical is $\mathfrak{m}$ (Proposition 14.2.2), which is finitely generated, so $\mathfrak{m}^n = 0$; apply Lemma 14.5.5. □

Theorem 14.9.5 (Krull's principal ideal theorem)**.** *Let R be Noetherian, $x \in R$, and $\mathfrak{p}$ a prime minimal among primes containing x. Then $\operatorname{ht}\mathfrak{p} \le 1$. More generally, if $\mathfrak{p}$ is minimal over $(x_1, \dots, x_r)$, then $\operatorname{ht}\mathfrak{p} \le r$.*

Proof. Localising at $\mathfrak{p}$ we may assume $(R, \mathfrak{p})$ local with $\mathfrak{p}$ minimal over (x). Let $\mathfrak{q} \subsetneq \mathfrak{p}$ be prime; we show $\operatorname{ht}\mathfrak{q} = 0$. For $n \ge 1$ let $\mathfrak{q}^{(n)} = \mathfrak{q}^n R_{\mathfrak{q}} \cap R$. The ring $R/(x)$ has $\mathfrak{p}/(x)$ as its only prime, so it is Artinian (Lemma 14.9.4), and the chain $\mathfrak{q}^{(n)} + (x)$ stabilises: for large n, $\mathfrak{q}^{(n)} \subseteq \mathfrak{q}^{(n+1)} + (x)$. For $a \in \mathfrak{q}^{(n)}$ write $a = b + rx$ with $b \in \mathfrak{q}^{(n+1)}$; then $rx \in \mathfrak{q}^{(n)}$, and since $x \notin \mathfrak{q}$ (by minimality of $\mathfrak{p}$) and x is a unit in $R_{\mathfrak{q}}$, $r \in \mathfrak{q}^{(n)}$. Hence $\mathfrak{q}^{(n)} = \mathfrak{q}^{(n+1)} + x\mathfrak{q}^{(n)}$, and Nakayama's lemma gives $\mathfrak{q}^{(n)} = \mathfrak{q}^{(n+1)}$. Localising, $\mathfrak{q}^n R_{\mathfrak{q}} = \mathfrak{q}^{n+1} R_{\mathfrak{q}}$, and Nakayama in $R_{\mathfrak{q}}$ gives $\mathfrak{q}^n R_{\mathfrak{q}} = 0$. So every prime of $R_{\mathfrak{q}}$, which contains $0 = (\mathfrak{q}R_{\mathfrak{q}})^n$, equals $\mathfrak{q}R_{\mathfrak{q}}$, and $\operatorname{ht}\mathfrak{q} = 0$.

For the general statement, argue by induction on r, again with $(R, \mathfrak{p})$ local. Let $\mathfrak{q} \subsetneq \mathfrak{p}$ be a prime such that there is no prime strictly between them; it suffices to show $\operatorname{ht}\mathfrak{q} \le r - 1$. As $\mathfrak{p}$ is minimal over $(x_1, \dots, x_r)$, some x_i, say x_r, is not in $\mathfrak{q}$. Then $\mathfrak{p}$ is minimal over $\mathfrak{q} + (x_r)$, so $R/(\mathfrak{q} + (x_r))$ is Artinian and $x_i^N \in \mathfrak{q} + (x_r)$ for $i < r$: write $x_i^N = y_i + a_i x_r$ with $y_i \in \mathfrak{q}$. Every prime containing $(y_1, \dots, y_{r-1}, x_r)$ contains all x_i, so $\mathfrak{p}$ is minimal over $(y_1, \dots, y_{r-1}, x_r)$, hence $\mathfrak{p}/I$ is minimal over $(\bar{x}_r)$ in R/I, $I = (y_1, \dots, y_{r-1})$. By the case $r = 1$, $\operatorname{ht}(\mathfrak{p}/I) \le 1$, so $\mathfrak{q}/I$ is a minimal prime of R/I, that is, $\mathfrak{q}$ is minimal over I, and by induction $\operatorname{ht}\mathfrak{q} \le r - 1$. □

Corollary 14.9.6. *A Noetherian local ring $(R, \mathfrak{m}, k)$ has $\dim R \le \dim_k \mathfrak{m}/\mathfrak{m}^2 < \infty$.*

Proof. By Corollary 14.4.3, $\mathfrak{m}$ is generated by $\dim_k \mathfrak{m}/\mathfrak{m}^2$ elements, and $\operatorname{ht}\mathfrak{m} = \dim R$. □

14.10 Completions

Lemma 14.10.1 (Artin–Rees)**.** *Let R be Noetherian, I an ideal, M a finitely generated module and $N \subseteq M$ a submodule. There is c such that $I^n M \cap N = I^{n-c}(I^c M \cap N)$ for all $n \ge c$.*

Proof. The *Rees algebra* $R^* = \bigoplus_n I^n$ is a finitely generated R-algebra (generated by generators of I in degree 1), hence Noetherian, and $M^* = \bigoplus_n I^n M$ is a finitely generated graded R^*-module. Its graded submodule $N^* = \bigoplus_n (I^n M \cap N)$ is therefore finitely generated, by homogeneous elements of degrees at most some c. For $n \ge c$, this means $I^n M \cap N = \sum_{j \le c} I^{n-j}(I^j M \cap N) \subseteq I^{n-c}(I^c M \cap N)$, and the reverse inclusion is clear. □

Theorem 14.10.2 (Krull's intersection theorem)**.** *Let R be Noetherian, I an ideal and M finitely generated, and $N = \bigcap_n I^n M$. Then $IN = N$, so $aN = 0$ for some $a \in 1 + I$. If $I \subseteq J(R)$, for instance if R is local and $I \ne R$, then $\bigcap_n I^n M = 0$.*

Proof. By Lemma 14.10.1, $N = I^{c+1} M \cap N = I(I^c M \cap N) = IN$. Apply Theorem 14.4.2. □

Definition 14.10.3. The I-adic *completion* of a module M is $\hat{M} = \lim_n M/I^n M$, the inverse limit (Example 6.8.3) of the quotients. For $M = R$ it is a ring $\hat{R}$, and there is a natural map $M \to \hat{M}$, injective for R Noetherian local, $I = \mathfrak{m}$ and M finitely generated, by Theorem 14.10.2.

Example 14.10.4. The completion of $k[x_1, \dots, x_n]$ at $(x_1, \dots, x_n)$ is the power series ring $k[[x_1, \dots, x_n]]$, and the completion of $\mathbb{Z}$ at (p) is the ring $\mathbb{Z}_p$ of p-adic integers. For R Noetherian, $\hat{R}$ is Noetherian and flat over R, and completion is exact on finitely generated modules [AM69, Chapter 10].

14.11 Regular local rings and Dedekind domains

Definition 14.11.1. A Noetherian local ring $(R, \mathfrak{m}, k)$ is *regular* if $\dim R = \dim_k \mathfrak{m}/\mathfrak{m}^2$. A *discrete valuation ring* (DVR) is a domain R with a surjective map $v\colon \mathrm{Frac}(R)^\times \to \mathbb{Z}$ such that $v(xy) = v(x) + v(y)$, $v(x+y) \geq \min(v(x), v(y))$, and $R = \{0\} \cup \{x : v(x) \geq 0\}$. A *Dedekind domain* is a Noetherian normal domain of dimension at most 1.

Theorem 14.11.2. *For a Noetherian local domain $(R, \mathfrak{m}, k)$ of dimension 1, the following are equivalent: R is a DVR; R is normal; $\mathfrak{m}$ is principal; R is regular.*

Proof. DVR $\Rightarrow$ normal: if $x \notin R$ satisfies a monic equation $x^n + a_1 x^{n-1} + \cdots + a_n = 0$, then $v(x) < 0$ and $v(x^n)$ is smaller than the valuation of every other term, a contradiction.

Normal $\Rightarrow$ $\mathfrak{m}$ principal: the only primes are 0 and $\mathfrak{m}$, so for $0 \neq a \in \mathfrak{m}$, $R/(a)$ is Artinian and $\mathfrak{m}^n \subseteq (a)$, $\mathfrak{m}^{n-1} \not\subseteq (a)$ for some $n \geq 1$. Choose $b \in \mathfrak{m}^{n-1} \setminus (a)$ and let $x = a/b \in \mathrm{Frac}(R)$. Then $x^{-1} \notin R$ and $x^{-1}\mathfrak{m} \subseteq R$ (since $b\mathfrak{m} \subseteq \mathfrak{m}^n \subseteq (a)$). If $x^{-1}\mathfrak{m} \subseteq \mathfrak{m}$, apply Lemma 14.4.1 to multiplication by x^{-1} on the finitely generated module $\mathfrak{m}$ (with $I = R$): the resulting monic polynomial in x^{-1} kills the nonzero elements of $\mathfrak{m} \subseteq \mathrm{Frac}(R)$, hence vanishes at x^{-1}, so x^{-1} is integral over R, contradicting normality. So $x^{-1}\mathfrak{m}$ is an ideal not contained in $\mathfrak{m}$, that is, $x^{-1}\mathfrak{m} = R$, and $\mathfrak{m} = xR$.

$\mathfrak{m}$ principal $\Rightarrow$ DVR: let $\mathfrak{m} = (t)$. By Krull's intersection theorem (Theorem 14.10.2 below), $\bigcap_n \mathfrak{m}^n = 0$, so every $0 \neq r \in R$ is $r = ut^n$ with u a unit and n unique; put $v(ut^n/u't^{n'}) = n - n'$.

$\mathfrak{m}$ principal $\Leftrightarrow$ regular: $\dim_k \mathfrak{m}/\mathfrak{m}^2 \geq \dim R = 1$ by Corollary 14.9.6, and it equals 1 iff $\mathfrak{m}$ is principal by Corollary 14.4.3. □

Remark 14.11.3. Every regular local ring is a domain, and in fact a UFD [Mat86, Theorems 14.3 and 20.3]. The local rings $k[x_1, \ldots, x_n]_{(x_1,\ldots,x_n)}$ are regular of dimension n: the images of the x_i form a basis of $\mathfrak{m}/\mathfrak{m}^2$, and the dimension is at least n because of the chain $0 \subsetneq (x_1) \subsetneq \cdots$, and at most n by Corollary 14.9.6. Regularity is the algebraic counterpart of smoothness (Chapter 53).

Theorem 14.11.4. *In a Dedekind domain R every nonzero ideal I is uniquely a product $\mathfrak{p}_1^{n_1} \cdots \mathfrak{p}_r^{n_r}$ of distinct nonzero primes.*

Proof. If $I = R$ take the empty product. Otherwise let $I = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_r$ be a primary decomposition with distinct primes $\mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$, which are nonzero, hence maximal. The $\mathfrak{q}_i$ are pairwise comaximal (their radicals are distinct maximal ideals), so $I = \mathfrak{q}_1 \cdots \mathfrak{q}_r$ by Theorem 8.3.3. Each $R_{\mathfrak{p}_i}$ is a Noetherian local domain of dimension 1, and it is normal: if $x \in \mathrm{Frac}(R)$ satisfies a monic equation with coefficients $a_j/s_j \in R_{\mathfrak{p}_i}$, then with $s = \prod_j s_j$ the element sx satisfies a monic equation over R, so $sx \in R$ and $x \in R_{\mathfrak{p}_i}$. By Theorem 14.11.2 it is a DVR, so $\mathfrak{q}_i R_{\mathfrak{p}_i} = \mathfrak{p}_i^{n_i} R_{\mathfrak{p}_i}$ for some n_i. Both $\mathfrak{q}_i$ and $\mathfrak{p}_i^{n_i}$ are $\mathfrak{p}_i$-primary (an ideal whose radical is a maximal ideal $\mathfrak{m}$ is $\mathfrak{m}$-primary, because in the quotient every element outside $\mathfrak{m}$ is a unit), and a $\mathfrak{p}$-primary ideal $\mathfrak{q}$ is the preimage of $\mathfrak{q}R_{\mathfrak{p}}$ (if $sx \in \mathfrak{q}$ with $s \notin \mathfrak{p}$ then $x \in \mathfrak{q}$); so $\mathfrak{q}_i = \mathfrak{p}_i^{n_i}$. Uniqueness: localising a factorisation at $\mathfrak{p}$ gives $IR_{\mathfrak{p}} = \mathfrak{p}^n R_{\mathfrak{p}}$ with n the exponent of $\mathfrak{p}$, which determines n as $v_{\mathfrak{p}}$ of a generator. □

Example 14.11.5. The ring $\mathbb{Z}[\sqrt{-5}]$ is a Dedekind domain (it is the integral closure of $\mathbb{Z}$ in $\mathbb{Q}(\sqrt{-5})$, see Exercise 14.13.2). Although $6 = 2 \cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$ has two factorisations into irreducibles (Counterexample 8.5.6), the ideal (6) factors uniquely into primes: $(2) = \mathfrak{p}^2$ with $\mathfrak{p} = (2, 1+\sqrt{-5})$, and $(3) = \mathfrak{r}\bar{\mathfrak{r}}$ with $\mathfrak{r} = (3, 1+\sqrt{-5})$, $\bar{\mathfrak{r}} = (3, 1-\sqrt{-5})$, while $(1 \pm \sqrt{-5}) = \mathfrak{p}\mathfrak{r}$, $\mathfrak{p}\bar{\mathfrak{r}}$.

14.12 Kähler differentials

Definition 14.12.1. Let $A \to B$ be a ring homomorphism and M a B-module. An *A-derivation* $D\colon B \to M$ is an A-linear map with $D(bb') = bD(b') + b'D(b)$. The *module of Kähler differentials* $\Omega_{B/A}$ is the free B-module on symbols db, $b \in B$, modulo the relations $d(b+b') - db - db'$, $d(bb') - b\,db' - b'\,db$ and da for a in the image of A; the map $d\colon B \to \Omega_{B/A}$ is an A-derivation.

Proposition 14.12.2. *1.* $\mathrm{Hom}_B(\Omega_{B/A}, M) \cong \mathrm{Der}_A(B, M)$, $\varphi \mapsto \varphi \circ d$, *naturally in* M.

2. $\Omega_{A[x_1,\dots,x_n]/A}$ *is free with basis* $dx_1, \dots, dx_n$, *and* $df = \sum_i (\partial f/\partial x_i) dx_i$.

3. *For* $A \to B \to C$ *there is an exact sequence* $C \otimes_B \Omega_{B/A} \to \Omega_{C/A} \to \Omega_{C/B} \to 0$.

4. *If* $C = B/I$, *there is an exact sequence* $I/I^2 \xrightarrow{\delta} C \otimes_B \Omega_{B/A} \to \Omega_{C/A} \to 0$ *with* $\delta(\bar{f}) = 1 \otimes df$.

5. $S^{-1}\Omega_{B/A} \cong \Omega_{S^{-1}B/A}$ *for a multiplicative* $S \subseteq B$.

Proof. (1) is the universal property of the construction. (2) The formal partial derivatives define a derivation $A[x] \to \bigoplus A[x]\, e_i$, $f \mapsto \sum_i (\partial f/\partial x_i) e_i$; a derivation is determined by its values on the x_i, which can be arbitrary. By (1), Ω is free on the dx_i. (3) and (4): a sequence $X \to Y \to Z \to 0$ of C-modules is exact if and only if $0 \to \mathrm{Hom}_C(Z, M) \to \mathrm{Hom}_C(Y, M) \to \mathrm{Hom}_C(X, M)$ is exact for every C-module M (for the nontrivial direction take $M = Z/\mathrm{im}(Y)$ and $M = \mathrm{coker}(X \to Y)$). By (1), $\mathrm{Hom}_C(C \otimes_B \Omega_{B/A}, M) = \mathrm{Der}_A(B, M)$ and $\mathrm{Hom}_C(I/I^2, M) = \mathrm{Hom}_B(I, M)$, so it suffices to show that for every C-module M the sequences $0 \to \mathrm{Der}_B(C, M) \to \mathrm{Der}_A(C, M) \to \mathrm{Der}_A(B, M)$ and $0 \to \mathrm{Der}_A(C, M) \to \mathrm{Der}_A(B, M) \to \mathrm{Hom}_B(I, M)$ are exact, which is immediate: a derivation of C over A vanishing on B is a B-derivation; and an A-derivation of B into a C-module that vanishes on I factors through $C = B/I$ (it automatically vanishes on I^2). (5) Derivations $S^{-1}B \to M$ correspond to derivations $B \to M$ into $S^{-1}B$-modules by the quotient rule $D(b/s) = (sDb - bDs)/s^2$. □

Example 14.12.3. For $B = k[x, y]/(y^2 - x^3)$, Proposition 14.12.2(4) gives $\Omega_{B/k} = (B\,dx \oplus B\,dy)/(2y\,dy - 3x^2 dx)$. At the origin, $\Omega_{B/k} \otimes k$ has dimension 2 although B has dimension 1; this detects the singularity of the cusp. For a finite separable field extension L/K one has $\Omega_{L/K} = 0$: if $L = K(\gamma)$ with minimal polynomial f, then $0 = d(f(\gamma)) = f'(\gamma)\,d\gamma$ and $f'(\gamma) \neq 0$ (Theorem 9.6.3 and Proposition 9.4.2).

14.13 Exercises

Exercise 14.13.1. Describe $\mathrm{Spec}\,\mathbb{Z}$, $\mathrm{Spec}\,\mathbb{Z}[x]$ (in terms of prime numbers and irreducible polynomials) and $\mathrm{Spec}\,k[x, y]$ for k algebraically closed.

Exercise 14.13.2. Let $d \neq 0, 1$ be a squarefree integer with $d \equiv 2, 3 \pmod 4$. Show that the integral closure of $\mathbb{Z}$ in $\mathbb{Q}(\sqrt{d})$ is $\mathbb{Z}[\sqrt{d}]$, and deduce that $\mathbb{Z}[\sqrt{-5}]$ is a Dedekind domain.

Exercise 14.13.3. Let $(R, \mathfrak{m})$ be a local ring and M a finitely generated R-module that is a direct summand of a free module. Show that M is free.

Exercise 14.13.4. Let $K = \mathbb{F}_p(t)$ and $L = K(t^{1/p})$. Show that $\Omega_{L/K}$ is one-dimensional over L.

Exercise 14.13.5. (Hensel's lemma.) Let $f \in \mathbb{Z}_p[x]$ and $a \in \mathbb{Z}_p$ with $f(a) \equiv 0 \pmod p$ and $f'(a) \not\equiv 0 \pmod p$. Show that there is a unique $b \in \mathbb{Z}_p$ with $f(b) = 0$ and $b \equiv a \pmod p$.

Exercise 14.13.6. Let k be a field and $f \in k[x_1, \dots, x_n]$ irreducible. Show that $\dim k[x_1, \dots, x_n]/(f) = n - 1$.

Chapter 15

Homological Algebra

15.1 Introduction

Homological algebra measures the failure of exactness. Its basic objects are complexes and their cohomology; its basic tools are the long exact sequence and derived functors. Everything in this project that is called a cohomology theory (singular cohomology in Chapter 22, sheaf cohomology in Chapter 26, de Rham and Dolbeault cohomology in Chapters 37 and 46) is built with the machinery of this chapter.

We work in abelian categories, but prove the diagram lemmas for modules; by the Freyd–Mitchell embedding theorem they then hold in every abelian category (Remark 15.2.5). References are [Wei94], [CE56] and [GM03].

Prerequisites. Chapter 6 (Categories) and Chapter 8 (Rings and Modules).

15.2 Abelian categories

Definition 15.2.1. An additive category $\mathcal{A}$ (Definition 6.12.1) is *abelian* if every morphism has a kernel and a cokernel (Definition 6.12.3), every monomorphism is a kernel of its cokernel, and every epimorphism is a cokernel of its kernel. For $f\colon X \to Y$ the *image* is $\operatorname{im} f = \ker(\operatorname{coker} f)$. A sequence $X \xrightarrow{f} Y \xrightarrow{g} Z$ with $gf = 0$ is *exact* at Y if the induced morphism $\operatorname{im} f \to \ker g$ is an isomorphism. An additive functor between abelian categories is *left exact* if it sends exact sequences $0 \to X \to Y \to Z$ to exact sequences, *right exact* if it sends exact $X \to Y \to Z \to 0$ to exact sequences, and *exact* if both.

Example 15.2.2. For a ring R, the category R-Mod is abelian (Proposition 8.7.3): kernels and cokernels are the usual ones, an injective map is the kernel of the projection onto its cokernel, and a surjective map is the cokernel of the inclusion of its kernel. In particular Ab is abelian. The category of sheaves of abelian groups on a topological space is abelian (Chapter 25). By Theorem 8.9.4, $- \otimes_R N$ is right exact, and $\operatorname{Hom}_R(N, -)$ is left exact.

Lemma 15.2.3. *In an abelian category a morphism that is both a monomorphism and an epimorphism is an isomorphism.*

Proof. Let $f\colon X \to Y$ be both. Its cokernel is $Y \to 0$, since f is an epimorphism; as f is a monomorphism, it is a kernel of $Y \to 0$. But id_Y is also a kernel of $Y \to 0$, so f is an isomorphism by uniqueness of kernels (Lemma 6.8.2). □

Counterexample 15.2.4. The category $\mathcal{F}$ of finitely generated free abelian groups is additive and has kernels (Proposition 7.11.4), but it is not abelian. The morphism $2\colon \mathbb{Z} \to \mathbb{Z}$ is a

monomorphism, and it is an epimorphism in $\mathcal{F}$: if $g \circ 2 = 0$ for $g\colon \mathbb{Z} \to \mathbb{Z}^n$, then $2g(1) = 0$, so $g = 0$. It is not an isomorphism, contradicting Lemma 15.2.3.

Remark 15.2.5. By the Freyd–Mitchell embedding theorem, every small abelian category admits a fully faithful exact functor into R-Mod for some ring R [Wei94, Theorem 1.6.1]. Every statement about a finite diagram in an abelian category $\mathcal{A}$ that asserts exactness, or the existence of morphisms constructed from kernels, cokernels and compositions, involves only a small abelian subcategory of $\mathcal{A}$; hence it suffices to prove it for modules, where one may argue with elements ("diagram chasing"). We use this principle for Lemmas 15.4.1 and 15.4.2, and wherever we prove a statement about abelian categories by a chase.

15.3 Complexes

Definition 15.3.1. A *complex* (or cochain complex) $A^\bullet$ in an abelian category $\mathcal{A}$ is a family of objects A^n, $n \in \mathbb{Z}$, with morphisms $d^n\colon A^n \to A^{n+1}$ such that $d^{n+1}d^n = 0$. Its *cocycles*, *coboundaries* and *cohomology* are $Z^n = \ker d^n$, $B^n = \operatorname{im} d^{n-1}$ and $H^n(A^\bullet) = Z^n/B^n$. A *morphism of complexes* $f\colon A^\bullet \to B^\bullet$ is a family $f^n\colon A^n \to B^n$ with $df = fd$; it induces $H^n(f)$. It is a *quasi-isomorphism* if all $H^n(f)$ are isomorphisms. A complex is *acyclic* (or exact) if all H^n vanish, and *bounded below* if $A^n = 0$ for $n \ll 0$. The complexes form an abelian category $\mathrm{Ch}(\mathcal{A})$, with kernels and cokernels computed degreewise.

Notation 15.3.2. A *chain complex* $C_\bullet$ has differentials $d_n\colon C_n \to C_{n-1}$ and *homology* $H_n = \ker d_n/\operatorname{im} d_{n+1}$. Setting $C^n = C_{-n}$ turns it into a complex, with $H^n = H_{-n}$; all statements about complexes apply to chain complexes in this way. For a complex $A^\bullet$ and $k \in \mathbb{Z}$, the *shift* $A[k]$ has $A[k]^n = A^{n+k}$ and differential $(-1)^k d$.

Definition 15.3.3. Two morphisms $f, g\colon A^\bullet \to B^\bullet$ are *homotopic*, $f \simeq g$, if there are morphisms $h^n\colon A^n \to B^{n-1}$ with $f^n - g^n = d^{n-1}h^n + h^{n+1}d^n$. A morphism f is a *homotopy equivalence* if there is g with $gf \simeq \mathrm{id}$ and $fg \simeq \mathrm{id}$.

Lemma 15.3.4. *Homotopic morphisms induce the same maps on cohomology; homotopy equivalences are quasi-isomorphisms. Additive functors preserve homotopies.*

Proof. For a cocycle z, $f(z) - g(z) = dh(z) + hd(z) = dh(z)$ is a coboundary. The second statement follows, and the third because the defining identity is preserved by additive functors. □

Definition 15.3.5. The *mapping cone* of $f\colon A^\bullet \to B^\bullet$ is the complex $\mathrm{Cone}(f)$ with $\mathrm{Cone}(f)^n = A^{n+1} \oplus B^n$ and $d(a, b) = (-da, f(a) + db)$.

15.4 The long exact sequence

Lemma 15.4.1 (Snake lemma)**.** *Consider a commutative diagram of modules with exact rows*

$$\begin{array}{ccccccccc} & & A' & \xrightarrow{i'} & B' & \xrightarrow{p'} & C' & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle h} & & \\ 0 & \longrightarrow & A & \xrightarrow{i} & B & \xrightarrow{p} & C & & \end{array}$$

There is a homomorphism $\delta\colon \ker h \to \operatorname{coker} f$ *such that the sequence*

$$\ker f \to \ker g \to \ker h \xrightarrow{\delta} \operatorname{coker} f \to \operatorname{coker} g \to \operatorname{coker} h$$

is exact. If i' *is injective so is* $\ker f \to \ker g$*, and if* p *is surjective so is* $\operatorname{coker} g \to \operatorname{coker} h$*.*

Proof. Definition of δ. Let $c' \in \ker h$. Choose $b' \in B'$ with $p'(b') = c'$. Then $p(g(b')) = h(c') = 0$, so $g(b') = i(a)$ for a unique $a \in A$; put $\delta(c') = [a]$. If b' is replaced by $b' + i'(a')$, then a is replaced by $a + f(a')$, with the same class; so δ is well defined, and it is clearly a homomorphism.

Exactness at $\ker g$. The composite is zero. If $b' \in \ker g$ and $p'(b') = 0$, then $b' = i'(a')$ and $i(f(a')) = g(b') = 0$, so $f(a') = 0$.

Exactness at $\ker h$. If $c' = p'(b')$ with $g(b') = 0$, then $a = 0$ and $\delta(c') = 0$. Conversely if $\delta(c') = 0$, then $a = f(a'')$, and $b' - i'(a'')$ is a preimage of c' with $g(b' - i'(a'')) = i(a) - i(f(a'')) = 0$.

Exactness at $\operatorname{coker} f$. The image of $\delta(c')$ in $\operatorname{coker} g$ is the class of $i(a) = g(b')$, which is zero. Conversely, if $i(a) = g(b')$ for some b', then $c' = p'(b')$ satisfies $h(c') = p(g(b')) = pi(a) = 0$ and $\delta(c') = [a]$.

Exactness at $\operatorname{coker} g$. The composite is zero. If $b \in B$ maps to zero in $\operatorname{coker} h$, then $p(b) = h(p'(b'))$ for some b', so $p(b - g(b')) = 0$ and $b - g(b') = i(a)$; hence $[b]$ is the image of $[a]$.

The last two statements are clear. □

Lemma 15.4.2 (Five lemma)**.** *Consider a commutative diagram of modules with exact rows $A_1 \to A_2 \to A_3 \to A_4 \to A_5$ and $B_1 \to B_2 \to B_3 \to B_4 \to B_5$ and vertical maps $f_1, \dots, f_5$. If f_2 and f_4 are isomorphisms, f_1 is surjective and f_5 is injective, then f_3 is an isomorphism.*

Proof. Write $\alpha_i \colon A_i \to A_{i+1}$ and $\beta_i \colon B_i \to B_{i+1}$. *Injectivity:* let $f_3(x) = 0$. Then $f_4\alpha_3(x) = 0$, so $\alpha_3(x) = 0$ and $x = \alpha_2(y)$. Now $\beta_2 f_2(y) = 0$, so $f_2(y) = \beta_1(z)$ and $z = f_1(w)$. Then $f_2(y - \alpha_1(w)) = 0$, so $y = \alpha_1(w)$ and $x = \alpha_2\alpha_1(w) = 0$. *Surjectivity:* let $u \in B_3$. Choose $x \in A_4$ with $f_4(x) = \beta_3(u)$. Then $f_5\alpha_4(x) = \beta_4\beta_3(u) = 0$, so $\alpha_4(x) = 0$ and $x = \alpha_3(y)$. Then $\beta_3(u - f_3(y)) = 0$, so $u - f_3(y) = \beta_2(v)$ with $v = f_2(t)$, and $u = f_3(y + \alpha_2(t))$. □

Theorem 15.4.3 (Long exact sequence, or zig-zag lemma)**.** *A short exact sequence $0 \to A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{g} C^\bullet \to 0$ of complexes in an abelian category induces a long exact sequence*

$$\cdots \to H^n(A) \to H^n(B) \to H^n(C) \xrightarrow{\delta} H^{n+1}(A) \to H^{n+1}(B) \to \cdots,$$

natural in the short exact sequence. For $c \in Z^n(C)$ (in the case of modules), $\delta[c]$ is the class of a where $f(a) = d(b)$ and $g(b) = c$.

Proof. Applying the snake lemma to the vertical maps d^{n-1} between the short exact sequences in degrees $n-1$ and n, the cokernel sequence $A^n/B^n(A) \to B^n/B^n(B) \to C^n/B^n(C) \to 0$ is exact; applying it to d^{n+1}, the kernel sequence $0 \to Z^{n+1}(A) \to Z^{n+1}(B) \to Z^{n+1}(C)$ is exact. The differential d^n induces vertical maps from the first of these rows to the second, whose kernels are the H^n and whose cokernels are the H^{n+1}. Apply Lemma 15.4.1. Naturality follows from the explicit formula for δ. □

Remark 15.4.4. The name *zig-zag lemma* comes from the path that defines the connecting map δ. Start from a cocycle $c \in C^n$; go up to a preimage $b \in B^n$ with $g(b) = c$; go across by the differential to $d(b) \in B^{n+1}$, which satisfies $g(d(b)) = d(g(b)) = d(c) = 0$; and go down to the unique $a \in A^{n+1}$ with $f(a) = d(b)$. Then $\delta[c] = [a]$:

$$\begin{array}{ccccc} & & b & \xmapsto{g} & c \\ & & \Big\downarrow{\scriptstyle d} & & \Big\downarrow{\scriptstyle d} \\ a & \xmapsto{f} & d(b) & \xmapsto{g} & 0 \end{array}$$

The element a depends on the choice of b, but its class does not; Exercise 15.10.2 checks this and the exactness of the long sequence directly, by chasing elements through the diagram, without the snake lemma. In singular homology, where the short exact sequence is $0 \to C_\bullet(A) \to C_\bullet(X) \to C_\bullet(X, A) \to 0$, the recipe says: a relative cycle is a chain in X whose boundary lies in A, and δ sends it to that boundary.

Proposition 15.4.5. *For $f\colon A^\bullet \to B^\bullet$ there is a short exact sequence $0 \to B^\bullet \to \mathrm{Cone}(f) \to A[1]^\bullet \to 0$ whose connecting map $H^n(A[1]) = H^{n+1}(A) \to H^{n+1}(B)$ is $H^{n+1}(f)$. In particular f is a quasi-isomorphism if and only if $\mathrm{Cone}(f)$ is acyclic.*

Proof. The maps $b \mapsto (0, b)$ and $(a, b) \mapsto a$ are morphisms of complexes (with the sign convention of Notation 15.3.2 for $A[1]$). For a cocycle $a \in A^{n+1}$, the element $(a, 0)$ lifts it, and $d(a, 0) = (-da, f(a)) = (0, f(a))$; so $\delta[a] = [f(a)]$. The last statement follows from the long exact sequence. □

15.5 Projective and injective objects

Definition 15.5.1. An object P of an abelian category is *projective* if $\mathrm{Hom}(P, -)$ is exact, that is, every morphism $P \to Y$ lifts along every epimorphism $X \to Y$. An object I is *injective* if $\mathrm{Hom}(-, I)$ is exact, that is, every morphism $X \to I$ extends along every monomorphism $X \to Y$. The category has *enough projectives* (*injectives*) if every object is a quotient of a projective (a subobject of an injective).

Proposition 15.5.2. *An R-module is projective if and only if it is a direct summand of a free module. R-Mod has enough projectives.*

Proof. Free modules are projective by Proposition 8.7.5 (choose lifts of the images of a basis), and direct summands of projectives are projective. If P is projective, choose a surjection $F \to P$ from a free module; the identity of P lifts to a section, so P is a direct summand of F by Proposition 8.7.7. Every module is a quotient of a free module. □

Proposition 15.5.3 (Baer's criterion)**.** *An R-module I is injective if and only if every homomorphism $J \to I$ from a left ideal $J \subseteq R$ extends to R.*

Proof. Necessity is clear. Let $N \subseteq M$ and $\varphi\colon N \to I$. By Zorn's lemma choose a maximal extension $\psi\colon N' \to I$ with $N \subseteq N' \subseteq M$. If $m \in M \setminus N'$, let $J = \{r : rm \in N'\}$ and extend $r \mapsto \psi(rm)$ to $\theta\colon R \to I$. Then $n' + rm \mapsto \psi(n') + \theta(r)$ is a well-defined extension of ψ to $N' + Rm$ (if $n' + rm = 0$ then $r \in J$ and $\theta(r) = \psi(rm) = -\psi(n')$), contradicting maximality. □

Theorem 15.5.4. *R-Mod has enough injectives. An abelian group D is injective if and only if it is* divisible*: for every $x \in D$ and $n \geq 1$ there is y with $ny = x$.*

Proof. By Baer's criterion and Proposition 7.2.10, D is injective iff every homomorphism $n\mathbb{Z} \to D$ extends to $\mathbb{Z}$, iff divisible. Every abelian group M embeds in a divisible group: write $M = F/K$ with $F = \mathbb{Z}^{(S)}$ free; then $M \subseteq \mathbb{Q}^{(S)}/K$, which is divisible.

Let M be an R-module and $M \subseteq D$ an embedding of abelian groups with D divisible. Make $\mathrm{Hom}_\mathbb{Z}(R, D)$ an R-module by $(r\varphi)(s) = \varphi(sr)$. For every R-module N there is a natural bijection $\mathrm{Hom}_R(N, \mathrm{Hom}_\mathbb{Z}(R, D)) \cong \mathrm{Hom}_\mathbb{Z}(N, D)$, $\psi \mapsto (n \mapsto \psi(n)(1))$, with inverse $\chi \mapsto (n \mapsto (s \mapsto \chi(sn)))$. The right side is exact in N since D is injective, so $\mathrm{Hom}_\mathbb{Z}(R, D)$ is an injective R-module. The map $M \to \mathrm{Hom}_\mathbb{Z}(R, D)$, $m \mapsto (s \mapsto sm)$, is R-linear and injective. □

15.6 Derived functors

In this section $\mathcal{A}$ and $\mathcal{B}$ are abelian categories.

Definition 15.6.1. An *injective resolution* of an object X is an exact sequence $0 \to X \to I^0 \to I^1 \to \cdots$ with all I^n injective; we regard $I^\bullet$ as a complex with a quasi-isomorphism $X \to I^\bullet$ (X in degree 0). Projective resolutions $\cdots \to P_1 \to P_0 \to X \to 0$ are defined dually. If $\mathcal{A}$ has enough injectives, every object has an injective resolution: embed X in I^0, embed I^0/X in I^1, and so on.

Theorem 15.6.2 (Comparison theorem)**.** *Let $0 \to X \to E^\bullet$ be exact with $E^n = 0$ for $n < 0$, and let $0 \to Y \to I^\bullet$ be a complex with all I^n injective, $I^n = 0$ for $n < 0$. Every morphism $\varphi\colon X \to Y$ extends to a morphism of complexes $E^\bullet \to I^\bullet$, and any two extensions are homotopic.*

Proof. Write $\varepsilon\colon X \to E^0$ and $\eta\colon Y \to I^0$. The morphism $\eta\varphi$ extends along the monomorphism ε to $f^0\colon E^0 \to I^0$. Suppose $f^0, \dots, f^n$ are constructed commuting with the differentials (and with ε, η). Then df^n vanishes on the image of $E^{n-1} \to E^n$ (or of X when $n = 0$), so it factors through $E^n/\mathrm{im}(E^{n-1}) \hookrightarrow E^{n+1}$, and extends along this monomorphism to $f^{n+1}\colon E^{n+1} \to I^{n+1}$ by injectivity.

If f, g both extend φ, let $u = f - g$, which extends 0, so $u^0\varepsilon = 0$. Put $h^0 = 0\colon E^0 \to I^{-1} = 0$. Suppose $h^0, \dots, h^n$ are constructed with $u^k = dh^k + h^{k+1}d$ for all $k < n$. For $n \geq 1$,

$$(u^n - dh^n)d = u^n d - dh^n d = du^{n-1} - d(u^{n-1} - dh^{n-1}) = ddh^{n-1} = 0,$$

using that u commutes with the differentials and the identity for $k = n - 1$; for $n = 0$, $(u^0 - dh^0)\varepsilon = u^0\varepsilon = 0$. So $u^n - dh^n$ vanishes on the image of $E^{n-1} \to E^n$ (of $X \to E^0$ if $n = 0$), factors through the monomorphism $E^n/\mathrm{im} \hookrightarrow E^{n+1}$, and extends to $h^{n+1}\colon E^{n+1} \to I^n$ with $h^{n+1}d = u^n - dh^n$. □

Lemma 15.6.3 (Horseshoe lemma)**.** *Let $0 \to X' \to X \to X'' \to 0$ be exact, and let $X' \to I'^\bullet$, $X'' \to I''^\bullet$ be injective resolutions. There is an injective resolution $X \to I^\bullet$ with $I^n = I'^n \oplus I''^n$ such that $0 \to I'^\bullet \to I^\bullet \to I''^\bullet \to 0$ (with the inclusion and projection in each degree) is a short exact sequence of complexes compatible with the given sequence.*

Proof. Let $\varepsilon'\colon X' \to I'^0$, $\varepsilon''\colon X'' \to I''^0$, and $\iota\colon X' \to X$, $\pi\colon X \to X''$. Extend ε' along ι to $\lambda\colon X \to I'^0$ and define $\varepsilon = (\lambda, \varepsilon''\pi)\colon X \to I'^0 \oplus I''^0$. The diagram $0 \to X' \to X \to X'' \to 0$ over $0 \to I'^0 \to I^0 \to I''^0 \to 0$ commutes, and the snake lemma shows that ε is a monomorphism and that $0 \to \mathrm{coker}\,\varepsilon' \to \mathrm{coker}\,\varepsilon \to \mathrm{coker}\,\varepsilon'' \to 0$ is exact. Apply the same construction to this sequence and the resolutions $I'^{\geq 1}$, $I''^{\geq 1}$ of the cokernels, and continue by induction. □

Definition 15.6.4. Let $\mathcal{A}$ have enough injectives and $F\colon \mathcal{A} \to \mathcal{B}$ be a left exact functor. For each X choose an injective resolution $X \to I_X^\bullet$ and define the *right derived functors* $R^nF(X) = H^n(F(I_X^\bullet))$. A morphism $X \to Y$ induces a morphism of resolutions, unique up to homotopy (Theorem 15.6.2), hence a well-defined morphism $R^nF(X) \to R^nF(Y)$ by Lemma 15.3.4. Left derived functors L_nF of a right exact functor on a category with enough projectives are defined dually using projective resolutions.

Theorem 15.6.5. 1. *The R^nF are additive functors, independent of the choice of resolutions up to canonical isomorphism, with $R^nF = 0$ for $n < 0$ and $R^0F \cong F$.*

2. *$R^nF(I) = 0$ for $n > 0$ and I injective.*

3. *Every short exact sequence* $0 \to X' \to X \to X'' \to 0$ *induces a long exact sequence* $0 \to F(X') \to F(X) \to F(X'') \to R^1F(X') \to R^1F(X) \to \cdots$, *natural in the short exact sequence.*

Proof. (1) Two resolutions of X are related by morphisms extending id_X in both directions, whose composites are homotopic to the identities by the uniqueness part of Theorem 15.6.2; applying F gives homotopy equivalences and canonical isomorphisms on cohomology. Functoriality and additivity follow from uniqueness up to homotopy. $R^0F(X) = \ker(F(I^0) \to F(I^1)) = F(\ker(I^0 \to I^1)) = F(X)$ by left exactness. (2) Use the resolution $0 \to I \to I \to 0$. (3) Choose resolutions of X' and X'' and form the resolution of X given by Lemma 15.6.3. The sequence $0 \to I'^\bullet \to I^\bullet \to I''^\bullet \to 0$ is split in every degree, so it stays exact after applying the additive functor F. Apply Theorem 15.4.3. Naturality follows from the comparison theorem. □

Proposition 15.6.6. *Call J F-*acyclic *if $R^nF(J) = 0$ for all $n > 0$. If $0 \to X \to J^0 \to J^1 \to \cdots$ is exact with all J^n F-acyclic, then $R^nF(X) \cong H^n(F(J^\bullet))$ for all n.*

Proof. Let $K = \mathrm{coker}(X \to J^0) = \ker(J^1 \to J^2)$, so that $0 \to X \to J^0 \to K \to 0$ is exact and $K \to J^{\geq 1}$ is an F-acyclic resolution. The long exact sequence and acyclicity of J^0 give $R^nF(X) \cong R^{n-1}F(K)$ for $n \geq 2$, and $R^1F(X) \cong \mathrm{coker}(F(J^0) \to F(K))$. By left exactness $F(K) = \ker(F(J^1) \to F(J^2))$, so $R^1F(X) \cong H^1(F(J^\bullet))$. For $n \geq 2$ argue by induction on n, for all objects and resolutions simultaneously: $K \to J^1 \to J^2 \to \cdots$ is an F-acyclic resolution of K, with J^1 in degree 0, so $R^nF(X) \cong R^{n-1}F(K) \cong H^{n-1}(F(J^1) \to F(J^2) \to \cdots) = H^n(F(J^\bullet))$. The case $n = 0$ is left exactness. □

15.7 Double complexes

Definition 15.7.1. A *double complex* $C^{\bullet,\bullet}$ consists of objects $C^{p,q}$ with morphisms $d_h \colon C^{p,q} \to C^{p+1,q}$ and $d_v \colon C^{p,q} \to C^{p,q+1}$ such that $d_h^2 = 0$, $d_v^2 = 0$ and $d_h d_v = d_v d_h$. It is *first quadrant* if $C^{p,q} = 0$ unless $p, q \geq 0$. Its *total complex* is $\mathrm{Tot}^n = \bigoplus_{p+q=n} C^{p,q}$ with $d = d_h + (-1)^p d_v$ on $C^{p,q}$; then $d^2 = 0$.

Lemma 15.7.2. *Let C be a first-quadrant double complex of modules whose rows are exact in positive degrees: for each q, the complex $C^{0,q} \to C^{1,q} \to \cdots$ is exact at $C^{p,q}$ for all $p \geq 1$. Let $K^q = \ker(d_h \colon C^{0,q} \to C^{1,q})$, a complex with differential d_v. Then the inclusion $K^\bullet \to \mathrm{Tot}^\bullet(C)$ is a quasi-isomorphism.*

Proof. The inclusion is a morphism of complexes, since $d = d_v$ on K. *Surjectivity on cohomology.* Let $x = (x^{0,n}, x^{1,n-1}, \ldots, x^{n,0})$ be a cocycle of degree n. Let p be largest with $x^{p,n-p} \neq 0$. If $p \geq 1$, the component of dx in $C^{p+1,n-p}$ is $d_h x^{p,n-p} = 0$ (the other contribution comes from $x^{p+1,n-p-1} = 0$), so $x^{p,n-p} = d_h y$ with $y \in C^{p-1,n-p}$, and $x - dy$ is a cohomologous cocycle whose components vanish in positions $\geq p$. Repeating, we reach a cocycle concentrated in $C^{0,n}$; the vanishing of its $C^{1,n}$-component of d means it lies in K^n, and it is a d_v-cocycle.

Injectivity on cohomology. Let $k \in K^n$ with $k = dy$ for $y \in \mathrm{Tot}^{n-1}$. As above, if $p \geq 1$ is largest with $y^{p,n-1-p} \neq 0$, the component of dy in $C^{p+1,n-1-p}$ is $d_h y^{p,n-1-p}$, which is zero because $dy = k$ lives in $C^{0,n}$ (here $p + 1 \geq 2 > 0$); so $y^{p,n-1-p} = d_h z$ and we replace y by $y - dz$, which does not change dy. We reach $y \in C^{0,n-1}$ with $d_h y$ equal to the $C^{1,n-1}$-component of k, which is 0; so $y \in K^{n-1}$ and $k = d_v y$. □

15.8 Ext and Tor

In this section R is a ring and modules are left R-modules; for Tor we assume R commutative.

Definition 15.8.1. $\operatorname{Ext}^n_R(M,N) = R^n \operatorname{Hom}_R(M,-)(N)$, computed with an injective resolution of N, and $\operatorname{Tor}^R_n(M,N) = L_n(M \otimes_R -)(N)$, computed with a projective resolution of N.

Theorem 15.8.2 (Balance). *If $P_\bullet \to M$ is a projective resolution, then $\operatorname{Ext}^n_R(M,N) \cong H^n(\operatorname{Hom}_R(P_\bullet, N))$. Similarly $\operatorname{Tor}^R_n(M,N) \cong H_n(P_\bullet \otimes_R N)$, and $\operatorname{Tor}^R_n(M,N) \cong \operatorname{Tor}^R_n(N,M)$. Consequently $\operatorname{Ext}^n(-,N)$ are the right derived functors of the contravariant left exact functor $\operatorname{Hom}_R(-,N)$, and there are long exact sequences in both variables.*

Proof. Let $N \to I^\bullet$ be an injective resolution and $C^{p,q} = \operatorname{Hom}_R(P_p, I^q)$, a first-quadrant double complex. For fixed q the row $\operatorname{Hom}_R(P_\bullet, I^q)$ is exact in positive degrees with kernel $\operatorname{Hom}_R(M, I^q)$ in degree 0, because $\operatorname{Hom}_R(-, I^q)$ is exact and $P_\bullet \to M \to 0$ is exact. For fixed p the column $\operatorname{Hom}_R(P_p, I^\bullet)$ is exact in positive degrees with kernel $\operatorname{Hom}_R(P_p, N)$, because P_p is projective. By Lemma 15.7.2, applied to C and to its transpose, both $\operatorname{Hom}_R(M, I^\bullet)$ and $\operatorname{Hom}_R(P_\bullet, N)$ are quasi-isomorphic to $\operatorname{Tot}(C)$. The same argument with $P_p \otimes Q_q$ for projective resolutions P of M and Q of N (projective modules are flat, being summands of free modules) gives the statements for Tor, and symmetry follows from $P_p \otimes Q_q \cong Q_q \otimes P_p$. The long exact sequence in the first variable follows from the horseshoe lemma for projective resolutions and Theorem 15.4.3. □

Example 15.8.3. Using the projective resolution $0 \to \mathbb{Z} \xrightarrow{n} \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \to 0$, for every abelian group A: $\operatorname{Hom}(\mathbb{Z}/n\mathbb{Z}, A) = A[n] = \{a : na = 0\}$, $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) = A/nA$, $\mathbb{Z}/n\mathbb{Z} \otimes A = A/nA$ and $\operatorname{Tor}^{\mathbb{Z}}_1(\mathbb{Z}/n\mathbb{Z}, A) = A[n]$; all higher groups vanish. For instance $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$, although $\operatorname{Hom}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}) = 0$.

Proposition 15.8.4. *A module N is flat if and only if $\operatorname{Tor}^R_1(M,N) = 0$ for all M; then $\operatorname{Tor}^R_n(M,N) = 0$ for all $n \geq 1$. If R is Noetherian local with residue field k and N is finitely generated, then N is flat if and only if it is free, if and only if $\operatorname{Tor}^R_1(k,N) = 0$.*

Proof. If $0 \to M' \to M \to M'' \to 0$ is exact, the long exact sequence $\operatorname{Tor}_1(M'', N) \to M' \otimes N \to M \otimes N$ shows that vanishing of Tor_1 gives injectivity; conversely if N is flat, computing Tor with a projective resolution of M tensored with N (Theorem 15.8.2) gives an exact complex. For the second statement, free implies flat. Suppose $\operatorname{Tor}_1(k,N) = 0$. Choose a minimal set of generators, giving $0 \to K \to R^n \to N \to 0$ with $n = \dim_k N/\mathfrak{m}N$ (Corollary 14.4.3). Tensoring with k gives the exact sequence $0 \to K \otimes k \to k^n \to N \otimes k \to 0$, in which the last map is an isomorphism of n-dimensional spaces; so $K/\mathfrak{m}K = 0$, and $K = 0$ by Nakayama's lemma, as K is finitely generated (R is Noetherian). □

Proposition 15.8.5. *If R is a principal ideal domain, every submodule of a free R-module is free. Hence every R-module has a projective resolution of length at most 1, and $\operatorname{Ext}^n_R = \operatorname{Tor}^R_n = 0$ for $n \geq 2$.*

Proof. Let F be free with basis $(e_i)_{i \in I}$, where I is well-ordered (Theorem 3.7.2), and $M \subseteq F$. For $i \in I$ let $F_{\leq i}$ and $F_{<i}$ be spanned by the e_j with $j \leq i$, respectively $j < i$, let $M_i = M \cap F_{\leq i}$, and let $\pi_i \colon M_i \to R$ be the e_i-coordinate. Its image is an ideal (a_i); if $a_i \neq 0$ choose $m_i \in M_i$ with $\pi_i(m_i) = a_i$. The m_i are linearly independent: in a relation $\sum c_i m_i = 0$, the largest i with $c_i \neq 0$ gives $c_i a_i = 0$ on the e_i-coordinate. They generate M: otherwise let i be least such that M_i contains an element m not in their span. Then $\pi_i(m) = ra_i$, and $m - rm_i$ (or m itself if $a_i = 0$) lies in $M \cap F_{<i}$; having finite support, it lies in some M_j with $j < i$, hence in the span, and so does m. For the last statement, write M as a quotient of a free module; the kernel is free. □

Proposition 15.8.6. *Let M, N be R-modules. There is a bijection between $\operatorname{Ext}^1_R(M,N)$ and the set of equivalence classes of short exact sequences $0 \to N \to E \to M \to 0$, where two sequences are equivalent if there is an isomorphism of the middle terms compatible with the given maps; the zero element corresponds to the split sequence.*

Proof. Choose $0 \to K \xrightarrow{j} P \to M \to 0$ with P projective. By Theorem 15.8.2 and the long exact sequence, $\mathrm{Ext}^1(M,N) = \mathrm{coker}(\mathrm{Hom}(P,N) \to \mathrm{Hom}(K,N))$. Given an extension $0 \to N \to E \xrightarrow{q} M \to 0$, lift $P \to M$ to $\beta\colon P \to E$; then βj lands in N, and its class $\theta(E)$ does not depend on β (two lifts differ by a map $P \to N$). Conversely, given $\alpha\colon K \to N$, let E_α be the pushout of $N \xleftarrow{\alpha} K \xrightarrow{j} P$, that is, $(N \oplus P)/\{(\alpha(k), -j(k)) : k \in K\}$; it sits in an exact sequence $0 \to N \to E_\alpha \to M \to 0$, and α and $\alpha + \gamma j$ ($\gamma\colon P \to N$) give equivalent extensions via $E_\alpha \to E_{\alpha+\gamma j}$, $(n,p) \mapsto (n-\gamma(p), p)$, which maps the relation $(\alpha(k), -j(k))$ to $((\alpha+\gamma j)(k), -j(k))$. The lift $p \mapsto (0,p)$ of $P \to M$ to E_α restricts on K to $k \mapsto (0, j(k)) = (\alpha(k), 0)$, so $\theta(E_\alpha)$ is the class of α. For an extension E with lift β, the map $E_{\beta j} \to E$, $(n,p) \mapsto n + \beta(p)$, is well defined and an equivalence by the five lemma. Hence the constructions are inverse. The split extension admits a lift with $\beta j = 0$. □

Example 15.8.7. Let p be a prime. By Proposition 15.8.6, computed with the resolution $0 \to p\mathbb{Z} \xrightarrow{j} \mathbb{Z} \to \mathbb{Z}/p \to 0$: every $f\colon \mathbb{Z} \to \mathbb{Z}/p$ vanishes on $p\mathbb{Z}$, so $\mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/p, \mathbb{Z}/p) = \mathrm{Hom}(p\mathbb{Z}, \mathbb{Z}/p) \cong \mathbb{Z}/p$, via $\alpha \mapsto a = \alpha(p)$. For $a = 0$ the extension is split, $E = \mathbb{Z}/p \oplus \mathbb{Z}/p$. For $a \neq 0$ the pushout is $E_\alpha = (\mathbb{Z}/p \oplus \mathbb{Z})/\langle (a, -p) \rangle$, in which $(0,1)$ has order p^2: if $m(0,1) = k(a,-p)$, then $ka \equiv 0 \pmod p$ forces $p \mid k$ and $p^2 \mid m$. So $E_\alpha \cong \mathbb{Z}/p^2$, with $\mathbb{Z}/p$ embedded by $n \mapsto na^{-1}p$. The $p-1$ nonzero classes thus give pairwise inequivalent extensions whose middle terms are all isomorphic to $\mathbb{Z}/p^2$: equivalence of extensions remembers the maps, not only the middle group.

Counterexample 15.8.8. Projective, flat and injective are different conditions, already for abelian groups.

1. $\mathbb{Q}$ is flat, since $\mathbb{Q} = S^{-1}\mathbb{Z}$ with $S = \mathbb{Z} \setminus \{0\}$ (Proposition 14.3.2(2)), and injective, since it is divisible (Theorem 15.5.4). It is not projective: a projective abelian group is a direct summand, hence a subgroup, of a free group, hence free (Propositions 15.5.2 and 15.8.5); but $\mathrm{Hom}(\mathbb{Q}, \mathbb{Z}) = 0$, because the image of 1 under a homomorphism $\mathbb{Q} \to \mathbb{Z}$ is divisible by every $n \geq 1$ in $\mathbb{Z}$, while a nonzero free group has a nonzero coordinate map to $\mathbb{Z}$.

2. $\mathbb{Z}$ is projective, hence flat, but not injective, as it is not divisible.

3. $\mathbb{Z}/n$ for $n \geq 2$ is none of the three: it is not flat, since $\mathrm{Tor}_1(\mathbb{Z}/n, \mathbb{Z}/n) = \mathbb{Z}/n \neq 0$ (Example 15.8.3 and Proposition 15.8.4), hence not projective, and it is not divisible, hence not injective.

15.9 Künneth and universal coefficient theorems

In this section R is a principal ideal domain and we use chain complexes of R-modules. The tensor product of chain complexes is $(C \otimes D)_n = \bigoplus_{p+q=n} C_p \otimes D_q$ with $d(x \otimes y) = dx \otimes y + (-1)^p x \otimes dy$ for $x \in C_p$, and $\mathrm{Hom}(C, M)$ is the cochain complex $\mathrm{Hom}(C_n, M)$ with $(d\varphi)(x) = (-1)^{n+1}\varphi(dx)$ for $\varphi \in \mathrm{Hom}(C_n, M)$.

Theorem 15.9.1 (Künneth formula)**.** *Let C be a chain complex of free R-modules and D any chain complex. For every n there is a natural short exact sequence*

$$0 \to \bigoplus_{p+q=n} H_p(C) \otimes H_q(D) \to H_n(C \otimes D) \to \bigoplus_{p+q=n-1} \mathrm{Tor}_1^R(H_p(C), H_q(D)) \to 0,$$

where the first map sends $[x] \otimes [y]$ to $[x \otimes y]$. If D is also a complex of free modules, the sequence splits (non-naturally).

Proof. Let $Z_p = \ker d_p$ and $B_p = \operatorname{im} d_{p+1} \subseteq C_p$; they are free by Proposition 15.8.5, hence flat. Regard Z and B as complexes with zero differential, and let B' be the complex with $B'_p = B_{p-1}$ and zero differential. The sequences $0 \to Z_p \to C_p \xrightarrow{d} B_{p-1} \to 0$ form a short exact sequence of complexes $0 \to Z \to C \to B' \to 0$, split in each degree since B_{p-1} is free; so it stays exact after tensoring with D. As Z and B are flat with zero differential, $H_n(Z \otimes D) = \bigoplus Z_p \otimes H_q(D)$ and similarly for B. The long exact sequence reads

$$\cdots \to \bigoplus_{p+q=n} B_p \otimes H_q(D) \xrightarrow{\partial} \bigoplus_{p+q=n} Z_p \otimes H_q(D) \to H_n(C \otimes D) \to \bigoplus_{p+q=n-1} B_p \otimes H_q(D) \xrightarrow{\partial} \cdots,$$

and computing the connecting map with Theorem 15.4.3 shows that ∂ is induced by the inclusions $B_p \to Z_p$ (a class $b \otimes [y]$ lifts to $c \otimes y$ with $dc = b$, and $d(c \otimes y) = b \otimes y$). Hence $H_n(C \otimes D)$ is an extension of $\ker(\partial)$ in degree $n-1$ by $\operatorname{coker}(\partial)$ in degree n. Since $0 \to B_p \to Z_p \to H_p(C) \to 0$ is a free resolution, $\operatorname{coker}(B_p \otimes H_q(D) \to Z_p \otimes H_q(D)) = H_p(C) \otimes H_q(D)$ and $\ker = \operatorname{Tor}_1(H_p(C), H_q(D))$, by right exactness of $\otimes$ and Theorem 15.8.2.

If D is free as well, choose retractions $r_C \colon C_p \to Z_p$ (which exist since B_{p-1} is free) and r_D. The composites $C_p \to Z_p \to H_p(C)$ form a morphism of complexes $C \to H(C)$ (to the complex with zero differential), since $r_C(dx) = dx \in B_p$ maps to 0; similarly $D \to H(D)$. Their tensor product induces $H_n(C \otimes D) \to \bigoplus H_p(C) \otimes H_q(D)$, which is a left inverse of the first map. □

Theorem 15.9.2 (Universal coefficient theorem)**.** *Let C be a chain complex of free R-modules and M an R-module. There is a natural short exact sequence*

$$0 \to \operatorname{Ext}^1_R(H_{n-1}(C), M) \to H^n(\operatorname{Hom}(C, M)) \to \operatorname{Hom}_R(H_n(C), M) \to 0,$$

where the second map sends the class of a cocycle φ to its restriction to cycles. It splits non-naturally.

Proof. Apply $\operatorname{Hom}(-, M)$ to the degreewise split sequence $0 \to Z \to C \to B' \to 0$ of the previous proof. As Z and B have zero differential, the long exact sequence becomes $\cdots \to \operatorname{Hom}(B_{n-1}, M) \to H^n(\operatorname{Hom}(C, M)) \to \operatorname{Hom}(Z_n, M) \xrightarrow{\partial} \operatorname{Hom}(B_n, M) \to \cdots$, where ∂ is restriction along $B_n \subseteq Z_n$ (up to sign). From the free resolution $0 \to B_n \to Z_n \to H_n(C) \to 0$ we get $\ker(\partial) = \operatorname{Hom}(H_n(C), M)$ and $\operatorname{coker}(\operatorname{Hom}(Z_{n-1}, M) \to \operatorname{Hom}(B_{n-1}, M)) = \operatorname{Ext}^1(H_{n-1}(C), M)$, which gives the sequence. A splitting is obtained from a retraction $r \colon C_n \to Z_n$: a homomorphism $H_n(C) \to M$ composed with $C_n \to Z_n \to H_n(C)$ is a cocycle. □

Example 15.9.3. Let C be the complex $\cdots \to 0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \to 0$ in degrees 1 and 0, so $H_0 = \mathbb{Z}/2\mathbb{Z}$ and $H_1 = 0$. Then $\operatorname{Hom}(C, \mathbb{Z})$ is $\mathbb{Z} \xrightarrow{\pm 2} \mathbb{Z}$ in degrees 0 and 1, with $H^0 = 0$ and $H^1 = \mathbb{Z}/2\mathbb{Z}$. Indeed $\operatorname{Hom}(H_1, \mathbb{Z}) = 0$ and $\operatorname{Ext}^1(H_0, \mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}$: torsion in homology shifts up one degree in cohomology. This is the algebraic reason why $H^2(\mathbb{RP}^2, \mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ while $H_2(\mathbb{RP}^2, \mathbb{Z}) = 0$ (Chapter 22).

15.10 Exercises

Exercise 15.10.1. Prove the 3×3 lemma: in a commutative 3×3 diagram of modules with exact columns, if two of the rows are short exact sequences, so is the third.

Exercise 15.10.2. Let $0 \to A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{g} C^\bullet \to 0$ be a short exact sequence of complexes of modules. Prove Theorem 15.4.3 directly: show that the recipe of Remark 15.4.4 gives a well defined homomorphism $\delta \colon H^n(C) \to H^{n+1}(A)$, and that the long sequence is exact at $H^n(B)$, at $H^n(C)$ and at $H^{n+1}(A)$.

Solution. *a is a cocycle:* $f(d(a)) = d(f(a)) = d(d(b)) = 0$ and f is injective. *Independence of b:* another preimage of c is $b + f(a')$ with $a' \in A^n$, since $\ker g = \operatorname{im} f$; it gives $d(b) + f(d(a'))$, hence $a + d(a')$, which has the same class. *Independence of c in its class:* for $c + d(c')$, write $c' = g(b')$ and use the preimage $b + d(b')$, which gives the same $d(b)$ and the same a. So δ is well defined, and it is additive because sums of choices are choices for sums.

Exactness at $H^n(B)$. $g_* f_* = (gf)_* = 0$. If b is a cocycle with $g_*[b] = 0$, then $g(b) = d(c')$ with $c' = g(b')$, so $g(b - d(b')) = 0$ and $b - d(b') = f(a)$ for some a; then $f(d(a)) = d(b) = 0$, so a is a cocycle and $[b] = [b - d(b')] = f_*[a]$.

Exactness at $H^n(C)$. If $c = g(b)$ with b a cocycle, use this b: $d(b) = 0$, so $\delta[c] = 0$. Conversely, if $\delta[c] = 0$, then $a = d(a')$ and $b - f(a')$ is a cocycle, since $d(b - f(a')) = f(a) - f(d(a')) = 0$, with $g(b - f(a')) = c$; so $[c] \in \operatorname{im} g_*$.

Exactness at $H^{n+1}(A)$. $f_*\delta[c] = [f(a)] = [d(b)] = 0$. Conversely, if a is a cocycle with $f(a) = d(b)$ for some b, then $c = g(b)$ is a cocycle, since $d(c) = g(d(b)) = g(f(a)) = 0$, and by construction $\delta[c] = [a]$. □

Exercise 15.10.3. Show that $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Q}, \mathbb{Z}) \neq 0$. (Hint: use $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$.)

Exercise 15.10.4. Let R be a PID. Show that an R-module is flat if and only if it is torsion-free.

Exercise 15.10.5. Let $f : A^\bullet \to B^\bullet$ be a morphism of complexes. Show that f is homotopic to zero if and only if the sequence $0 \to B \to \operatorname{Cone}(f) \to A[1] \to 0$ splits as a sequence of complexes.

Exercise 15.10.6. Show that an abelian group is injective if and only if $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) = 0$ for all $n \geq 1$.

Solution. By Example 15.8.3, $\operatorname{Ext}^1(\mathbb{Z}/n\mathbb{Z}, A) = A/nA$, which vanishes for all n iff A is divisible, iff A is injective by Theorem 15.5.4. □

Chapter 16

Spectral Sequences

16.1 Introduction

A spectral sequence is a bookkeeping device for computing the cohomology of a filtered complex from the cohomology of its graded pieces, in a sequence of successive approximations. This chapter constructs the spectral sequence of a filtered complex with complete proofs, derives the two spectral sequences of a double complex, and proves the Grothendieck spectral sequence of a composition of functors. Later chapters use the Leray spectral sequence (Chapter 26), the Frölicher spectral sequence (Chapter 46) and the weight and Hodge spectral sequences of mixed Hodge theory (Chapter 74).

References are [Wei94, Chapter 5] and [McC01]. As in Chapter 15, we argue with elements; by Remark 15.2.5 the results hold in every abelian category.

Prerequisites. Chapter 15 (Homological Algebra).

16.2 Spectral sequences

Definition 16.2.1. A (cohomological) *spectral sequence* starting at page r_0 consists of objects $E_r^{p,q}$ for $p, q \in \mathbb{Z}$ and $r \geq r_0$, morphisms $d_r \colon E_r^{p,q} \to E_r^{p+r,q-r+1}$ with $d_r \circ d_r = 0$, and isomorphisms $E_{r+1}^{p,q} \cong \ker(d_r)/\operatorname{im}(d_r)$ at $E_r^{p,q}$. The object $E_r = \bigoplus_{p,q} E_r^{p,q}$ is the r-th *page*, and $p+q$ is the *total degree*. It is a *first-quadrant* spectral sequence if $E_{r_0}^{p,q} = 0$ unless $p, q \geq 0$.

Lemma 16.2.2. *In a first-quadrant spectral sequence $E_r^{p,q} = 0$ unless $p, q \geq 0$, for all r, and $E_r^{p,q} = E_{r+1}^{p,q}$ whenever $r > \max(p, q+1)$. The common value is denoted $E_\infty^{p,q}$.*

Proof. The first claim holds because $E_{r+1}^{p,q}$ is a subquotient of $E_r^{p,q}$. For $r > p$ the source $E_r^{p-r,q+r-1}$ of the incoming differential has negative first index, and for $r > q+1$ the target $E_r^{p+r,q-r+1}$ of the outgoing differential has negative second index; so both differentials vanish. □

Definition 16.2.3. A first-quadrant spectral sequence *converges* to objects H^n, written $E_{r_0}^{p,q} \Rightarrow H^{p+q}$, if each H^n has a finite decreasing filtration $H^n = F^0H^n \supseteq F^1H^n \supseteq \cdots \supseteq F^{n+1}H^n = 0$ with isomorphisms $E_\infty^{p,q} \cong F^pH^{p+q}/F^{p+1}H^{p+q}$. The spectral sequence *degenerates* at page r if $d_s = 0$ for all $s \geq r$, so that $E_r = E_\infty$.

Proposition 16.2.4. *Let $E_2^{p,q} \Rightarrow H^{p+q}$ be a convergent first-quadrant spectral sequence.*

1. *There are* edge maps $E_2^{n,0} \to H^n$ *and* $H^n \to E_2^{0,n}$*: the first is the composite $E_2^{n,0} \twoheadrightarrow E_\infty^{n,0} = F^nH^n \hookrightarrow H^n$, the second $H^n \twoheadrightarrow H^n/F^1H^n = E_\infty^{0,n} \hookrightarrow E_2^{0,n}$.*

2. *(Five-term exact sequence) The sequence* $0 \to E_2^{1,0} \to H^1 \to E_2^{0,1} \xrightarrow{d_2} E_2^{2,0} \to H^2$ *is exact.*

3. *If* $E_2^{p,q} = 0$ *for* $q \neq 0$*, then* $H^n \cong E_2^{n,0}$*. If* $E_2^{p,q} = 0$ *for* $p \neq 0$*, then* $H^n \cong E_2^{0,n}$*.*

Proof. (1) Since nothing is in negative degrees, $E_{r+1}^{n,0}$ is a quotient of $E_r^{n,0}$ (no outgoing differential lands in $q < 0$) and $E_{r+1}^{0,n}$ a subobject of $E_r^{0,n}$ (no incoming differential starts at $p < 0$). Also $F^{n+1}H^n = 0$, so $E_\infty^{n,0} = F^nH^n$, and $F^0H^n = H^n$, so $E_\infty^{0,n} = H^n/F^1H^n$. (2) In total degree 1, H^1 has the filtration $0 \subseteq F^1H^1 \subseteq H^1$ with $F^1H^1 = E_\infty^{1,0} = E_2^{1,0}$ (no differential reaches or leaves $E_r^{1,0}$ nontrivially: incoming would start at $p = 1 - r < 0$, outgoing lands at $q = 1 - r < 0$) and $H^1/F^1H^1 = E_\infty^{0,1} = \ker(d_2 \colon E_2^{0,1} \to E_2^{2,0})$ (for $r \geq 3$ all differentials at $(0,1)$ vanish). This gives exactness at $E_2^{1,0}$, H^1 and $E_2^{0,1}$. Next $E_\infty^{2,0} = E_3^{2,0} = E_2^{2,0}/\mathrm{im}(d_2)$, and it injects into H^2 as F^2H^2; so the kernel of $E_2^{2,0} \to H^2$ is $\mathrm{im}(d_2)$. (3) All but one of the graded pieces $E_\infty^{p,n-p}$ of H^n vanish, and all differentials vanish because either the source or the target is zero. □

Proposition 16.2.5. *Let* $E_2^{p,q} \Rightarrow H^{p+q}$ *be a convergent first-quadrant spectral sequence with* $E_2^{p,q} = 0$ *unless* $q \in \{0, s\}$*, for some* $s \geq 1$*. Then there is a long exact sequence*

$$\cdots \to E_2^{n,0} \to H^n \to E_2^{n-s,s} \xrightarrow{d_{s+1}} E_2^{n+1,0} \to H^{n+1} \to \cdots.$$

Proof. The only possibly nonzero differentials are $d_{s+1} \colon E_{s+1}^{p,s} \to E_{s+1}^{p+s+1,0}$, and $E_{s+1} = E_2$. So $E_\infty^{n,0} = \mathrm{coker}(d_{s+1} \colon E_2^{n-s-1,s} \to E_2^{n,0})$ and $E_\infty^{n-s,s} = \ker(d_{s+1} \colon E_2^{n-s,s} \to E_2^{n+1,0})$, and H^n is an extension $0 \to E_\infty^{n,0} \to H^n \to E_\infty^{n-s,s} \to 0$. Splicing these sequences gives the claim. □

Proposition 16.2.6. *Let* E *be a spectral sequence of finite-dimensional vector spaces over a field, with only finitely many nonzero* $E_{r_0}^{p,q}$*. Then* $\chi(E_r) = \sum_{p,q}(-1)^{p+q} \dim E_r^{p,q}$ *does not depend on* r*, and if* $E_{r_0}^{p,q} \Rightarrow H^{p+q}$ *converges, then* $\chi(E_{r_0}) = \sum_n (-1)^n \dim H^n$*. In particular* $\dim H^n \leq \sum_{p+q=n} \dim E_r^{p,q}$ *for every* r*.*

Proof. Grading E_r by total degree $n = p + q$, the differential d_r raises n by one, so (E_r, d_r) is a finite complex of finite-dimensional spaces with cohomology E_{r+1}. For such a complex $C^\bullet$, rank–nullity gives $\dim C^n = \dim Z^n + \dim B^{n+1}$ and $\dim H^n = \dim Z^n - \dim B^n$, so $\sum_n (-1)^n \dim C^n = \sum_n (-1)^n \dim H^n$. Hence $\chi(E_{r+1}) = \chi(E_r)$. Only finitely many positions are nonzero, so the pages stabilise, and convergence gives $\dim H^n = \sum_p \dim E_\infty^{p,n-p}$. The inequality holds because each $E_\infty^{p,q}$ is a subquotient of $E_r^{p,q}$. □

Counterexample 16.2.7. The limit of a spectral sequence determines the abutment only up to extensions. Let K and K' be the complexes concentrated in degree 0, with zero differential, given by $K^0 = \mathbb{Z}/4$ with the filtration $F^0 = \mathbb{Z}/4 \supseteq F^1 = 2\mathbb{Z}/4 \supseteq F^2 = 0$, and $K'^0 = \mathbb{Z}/2 \oplus \mathbb{Z}/2$ with $F^0 = K'^0 \supseteq F^1 = 0 \oplus \mathbb{Z}/2 \supseteq F^2 = 0$. In both cases all differentials vanish and the spectral sequence of the filtered complex (Section 16.3) has $E_0^{0,0} = \mathbb{Z}/2$ and $E_0^{1,-1} = \mathbb{Z}/2$ on every page. The spectral sequences are isomorphic, but the abutments $\mathbb{Z}/4$ and $\mathbb{Z}/2 \oplus \mathbb{Z}/2$ are not. Over a field this cannot happen, since every extension of vector spaces splits and $H^n \cong \bigoplus_p E_\infty^{p,n-p}$.

16.3 The spectral sequence of a filtered complex

Definition 16.3.1. A *filtered complex* is a complex $K^\bullet$ with subcomplexes $\cdots \supseteq F^pK \supseteq F^{p+1}K \supseteq \cdots$. The filtration is *finite in each degree* if for every n there are $a \leq b$ with $F^aK^n = K^n$ and $F^bK^n = 0$. The *associated graded* complexes are $\mathrm{gr}^pK = F^pK/F^{p+1}K$, and the induced filtration on cohomology is $F^pH^n(K) = \mathrm{im}(H^n(F^pK) \to H^n(K))$.

Definition 16.3.2. For a filtered complex K we abbreviate $F^p = F^pK$ and, for $r \geq 0$,

$$Z_r^p = F^p \cap d^{-1}(F^{p+r}), \qquad B_r^p = F^p \cap d(F^{p-r+1}), \qquad E_r^p = Z_r^p/(B_r^p + Z_{r-1}^{p+1}),$$

with the convention $Z_{-1}^{p+1} = F^{p+1}$. These are graded by the degree of K, and we write $E_r^{p,q}$ for the part of E_r^p in degree $p+q$. Let $d_r \colon E_r^p \to E_r^{p+r}$ be induced by d.

Lemma 16.3.3. *The inclusions $B_r^p \subseteq Z_r^p$ and $Z_{r-1}^{p+1} \subseteq Z_r^p$ hold, so E_r^p is defined, and d_r is a well-defined morphism with $d_r^2 = 0$, of bidegree $(r, 1-r)$. Moreover $E_0^p = \mathrm{gr}^p K$ with d_0 induced by d, and $E_1^{p,q} = H^{p+q}(\mathrm{gr}^p K)$.*

Proof. Elements of B_r^p are boundaries, hence cycles; and $x \in F^{p+1}$ with $dx \in F^{p+r}$ lies in Z_r^p. If $x \in Z_r^p$ then $dx \in F^{p+r}$ and $d(dx) = 0$, so $dx \in Z_r^{p+r}$. We must check that d maps $B_r^p + Z_{r-1}^{p+1}$ into $B_r^{p+r} + Z_{r-1}^{p+r+1}$: $d(B_r^p) = 0$, and for $x \in Z_{r-1}^{p+1}$ we have $dx \in F^{p+r} \cap d(F^{p+1}) = B_r^{p+r}$. Clearly $d_r^2 = 0$. For $r = 0$: $Z_0^p = F^p$ and $B_0^p = d(F^{p+1}) \subseteq F^{p+1}$, so $E_0^p = F^p/F^{p+1}$. For $r = 1$: Z_1^p consists of the elements of F^p whose image in gr^p is a cocycle, $Z_0^{p+1} = F^{p+1}$ and $B_1^p = d(F^p)$, so $E_1^p = Z_1^p/(d(F^p) + F^{p+1})$ is the cohomology of $\mathrm{gr}^p K$. □

Theorem 16.3.4. *For every filtered complex, the objects $E_r^{p,q}$ and differentials d_r, $r \geq 0$, form a spectral sequence: $H(E_r, d_r) \cong E_{r+1}$. If the filtration is finite in each degree, then for fixed (p,q) and all sufficiently large r, $E_r^{p,q} = E_\infty^{p,q}$ with*

$$E_\infty^{p,q} \cong F^pH^{p+q}(K)/F^{p+1}H^{p+q}(K).$$

In particular, if $F^0K = K$ and $F^{n+1}K^n = 0$ for all n, the spectral sequence is first-quadrant (from E_1 on, since $E_1^{p,q}$ vanishes for $p < 0$ or $q < 0$) and $E_1^{p,q} = H^{p+q}(\mathrm{gr}^p K) \Rightarrow H^{p+q}(K)$.

Proof. Kernel of d_r at E_r^p. Let $x \in Z_r^p$ with $dx \in B_r^{p+r} + Z_{r-1}^{p+r+1}$, say $dx = dy + z$ with $y \in F^{p+1}$, $dy \in F^{p+r}$ and $z \in F^{p+r+1}$. Then $y \in Z_{r-1}^{p+1}$, and $x - y \in F^p$ with $d(x-y) = z \in F^{p+r+1}$, so $x - y \in Z_{r+1}^p$. Conversely, Z_{r+1}^p maps into the kernel, since $dx \in F^{p+r+1}$ and $d(dx) = 0$ give $dx \in Z_{r-1}^{p+r+1}$. Hence the kernel is $(Z_{r+1}^p + Z_{r-1}^{p+1})/(B_r^p + Z_{r-1}^{p+1})$.

Image of d_r in E_r^p. It consists of the classes of dx with $x \in Z_r^{p-r}$, that is of the elements of $F^p \cap d(F^{p-r}) = B_{r+1}^p$. Since $B_r^p \subseteq B_{r+1}^p$, the image is $(B_{r+1}^p + Z_{r-1}^{p+1})/(B_r^p + Z_{r-1}^{p+1})$.

Cohomology. Therefore $H(E_r)^p = (Z_{r+1}^p + Z_{r-1}^{p+1})/(B_{r+1}^p + Z_{r-1}^{p+1})$. The map $Z_{r+1}^p \to H(E_r)^p$ is surjective, with kernel $Z_{r+1}^p \cap (B_{r+1}^p + Z_{r-1}^{p+1}) = B_{r+1}^p + (Z_{r+1}^p \cap Z_{r-1}^{p+1})$ (as $B_{r+1}^p \subseteq Z_{r+1}^p$). Now $Z_{r+1}^p \cap Z_{r-1}^{p+1} = F^{p+1} \cap d^{-1}(F^{p+r+1}) = Z_r^{p+1}$. So $H(E_r)^p \cong Z_{r+1}^p/(B_{r+1}^p + Z_r^{p+1}) = E_{r+1}^p$.

Convergence. Fix n and p and let Z, B denote the cocycles and coboundaries of K. If the filtration is finite in degrees $n-1$, n and $n+1$, then for large r we have $F^{p+r}K^{n+1} = 0$ and $F^{p-r+1}K^{n-1} = K^{n-1}$, so in degree n: $Z_r^p = F^p \cap Z$, $Z_{r-1}^{p+1} = F^{p+1} \cap Z$ and $B_r^p = F^p \cap B$. Hence $E_\infty^p = (F^p \cap Z)/((F^p \cap B) + (F^{p+1} \cap Z))$. On the other hand $F^pH^n = (F^p \cap Z + B)/B$, and

$$\frac{F^pH^n}{F^{p+1}H^n} \cong \frac{F^p \cap Z}{(F^p \cap Z) \cap (F^{p+1} \cap Z + B)} = \frac{F^p \cap Z}{F^{p+1} \cap Z + F^p \cap B},$$

using the modular law. For the last statement, $\mathrm{gr}^p K = 0$ for $p < 0$ and $\mathrm{gr}^p K^n = 0$ for $p > n$, so $E_1^{p,q} = 0$ unless $p \geq 0$ and $q \geq 0$, and F^pH^n runs from H^n to 0. □

16.4 Double complexes

Theorem 16.4.1. *Let C be a first-quadrant double complex (Definition 15.7.1) and $T = \mathrm{Tot}(C)$. There are two convergent first-quadrant spectral sequences*

$$ {}^{I}E_2^{p,q} = H_h^p(H_v^q(C)) \Rightarrow H^{p+q}(T), \qquad {}^{II}E_2^{p,q} = H_v^p(H_h^q(C)) \Rightarrow H^{p+q}(T), $$

where $H_v^q(C)$ is the cohomology of the columns (a complex in p via d_h), and $H_h^q(C)$ that of the rows (a complex in the other index via d_v).

Proof. Filter T by columns: ${}^{I}F^pT^n = \bigoplus_{i \geq p} C^{i,n-i}$. These are subcomplexes, $F^0T = T$ and $F^{n+1}T^n = 0$. The graded piece gr^pT is the column $C^{p,\bullet}$ (shifted by p) with differential $\pm d_v$, so ${}^{I}E_1^{p,q} = H_v^q(C^{p,\bullet})$ by Theorem 16.3.4. The differential d_1 is the connecting map of $0 \to \mathrm{gr}^{p+1} \to F^p/F^{p+2} \to \mathrm{gr}^p \to 0$; a cocycle $x \in C^{p,q}$ of d_v lifts to itself, and its total differential is $d_h x \in C^{p+1,q}$, so $d_1 = H_v^q(d_h)$ and ${}^{I}E_2^{p,q} = H_h^p(H_v^q(C))$. Convergence follows from Theorem 16.3.4. Filtering by rows, ${}^{II}F^pT^n = \bigoplus_{j \geq p} C^{n-j,j}$, gives the other spectral sequence in the same way. □

Example 16.4.2. If the rows of C are exact in positive degrees, with kernel complex $K^\bullet$ in column 0, then ${}^{II}E_1^{p,q} = H_h^q(C^{\bullet,p})$ vanishes for $q > 0$ and equals K^p for $q = 0$. So ${}^{II}E_2^{p,0} = H^p(K)$, all other terms vanish, and $H^n(T) \cong H^n(K)$ by Proposition 16.2.4(3). This recovers Lemma 15.7.2.

16.5 Exact couples

Definition 16.5.1. An *exact couple* consists of objects D, E and morphisms $i\colon D \to D$, $j\colon D \to E$, $k\colon E \to D$ such that the triangle $D \xrightarrow{i} D \xrightarrow{j} E \xrightarrow{k} D$ is exact at each vertex. Put $d = jk\colon E \to E$; then $d^2 = j(kj)k = 0$.

Proposition 16.5.2. *Given an exact couple, let $D' = i(D)$, $E' = \ker d / \mathrm{im}\, d$, $i' = i|_{D'}$, $j'(i(x)) = [j(x)]$ and $k'([e]) = k(e)$. Then (D', E', i', j', k') is an exact couple, the* derived couple. *Iterating gives a spectral sequence $E^{(r)}$ with differentials $d^{(r)} = j^{(r)}k^{(r)}$.*

Proof. j' is well defined: if $i(x) = i(x')$ then $x - x' = k(e)$ and $j(x - x') = jk(e) = d(e)$; and $j(x)$ is a d-cocycle since $dj(x) = jkj(x) = 0$. k' is well defined: if $e \in \ker d$ then $jk(e) = 0$ so $k(e) \in \ker j = \mathrm{im}\, i = D'$, and $k(d(e')) = kjk(e') = 0$. Exactness at D' (between i' and j'): $j'(i(x)) = 0$ iff $j(x) = jk(e)$ iff $x - k(e) \in \ker j = \mathrm{im}\, i$, iff $i(x) \in i(\mathrm{im}\, i) + ik(E) = i(D')$ since $ik = 0$. Exactness at E': $k'[e] = 0$ iff $k(e) = 0$ iff $e = j(x)$, iff $[e] = j'(i(x))$. Exactness at the other D' (between k' and i'): $i'(i(x)) = 0$ iff $i(x) \in \ker i = \mathrm{im}\, k$, iff $i(x) = k(e)$ for some e, which is then a d-cocycle since $jk(e) = ji(x) = 0$; that is, iff $i(x) \in \mathrm{im}\, k'$. □

Remark 16.5.3. A filtered complex gives an exact couple with $D = \bigoplus_{p,n} H^n(F^pK)$, $E = \bigoplus_{p,n} H^n(\mathrm{gr}^pK)$, i induced by the inclusions $F^{p+1} \subseteq F^p$, and j, k from the long exact sequences of $0 \to F^{p+1} \to F^p \to \mathrm{gr}^p \to 0$. The resulting spectral sequence is the one of Theorem 16.3.4, from E_1 on [Wei94, Section 5.9].

16.6 The Grothendieck spectral sequence

Definition 16.6.1. Let $A^\bullet$ be a complex with $A^n = 0$ for $n < 0$ in an abelian category with enough injectives. A *Cartan–Eilenberg resolution* of A is a first-quadrant double complex

$I^{\bullet,\bullet}$ of injectives with a morphism $A^p \to I^{p,0}$ for each p, such that for every p the columns give injective resolutions $A^p \to I^{p,\bullet}$, $Z^p(A) \to Z_h^p(I)^\bullet$, $B^p(A) \to B_h^p(I)^\bullet$ and $H^p(A) \to H_h^p(I)^\bullet$, where Z_h, B_h, H_h denote horizontal cocycles, coboundaries and cohomology.

Lemma 16.6.2. *Every such complex A has a Cartan–Eilenberg resolution.*

Proof. For each p choose injective resolutions $B^p(A) \to J_B^{p,\bullet}$ and $H^p(A) \to J_H^{p,\bullet}$. The horseshoe lemma (Lemma 15.6.3) applied to $0 \to B^p \to Z^p \to H^p \to 0$ gives a resolution $J_Z^{p,\bullet}$ of Z^p with $J_Z^{p,q} = J_B^{p,q} \oplus J_H^{p,q}$, and applied to $0 \to Z^p \to A^p \to B^{p+1} \to 0$ gives a resolution $I^{p,\bullet}$ of A^p with $I^{p,q} = J_Z^{p,q} \oplus J_B^{p+1,q}$. Define $d_h \colon I^{p,q} \to I^{p+1,q}$ as the composite $I^{p,q} \to J_B^{p+1,q} \to J_Z^{p+1,q} \to I^{p+1,q}$ of the projection and inclusions provided by the horseshoe lemma. Then $d_h^2 = 0$ (the projection $I^{p+1,q} \to J_B^{p+2,q}$ kills $J_Z^{p+1,q}$), d_h commutes with the vertical differentials since all maps involved are morphisms of complexes, and it covers $d \colon A^p \to A^{p+1}$. By construction the horizontal cocycles, coboundaries and cohomology in column p are $J_Z^{p,\bullet}$, $J_B^{p,\bullet}$ and $J_H^{p,\bullet}$. □

Theorem 16.6.3 (Grothendieck spectral sequence)**.** *Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories, $\mathcal{A}$ and $\mathcal{B}$ with enough injectives, and $F \colon \mathcal{A} \to \mathcal{B}$, $G \colon \mathcal{B} \to \mathcal{C}$ left exact functors such that F maps injective objects to G-acyclic objects. For every $X \in \mathcal{A}$ there is a convergent first-quadrant spectral sequence*

$$E_2^{p,q} = R^pG(R^qF(X)) \Rightarrow R^{p+q}(G \circ F)(X).$$

Proof. Let $X \to A^\bullet$ be an injective resolution in $\mathcal{A}$, and let $I^{\bullet,\bullet}$ be a Cartan–Eilenberg resolution of the complex $F(A^\bullet)$ in $\mathcal{B}$. Consider the double complex $G(I)$ and its total complex T.

First filtration (by columns). For each p, the column $I^{p,\bullet}$ is an injective resolution of the G-acyclic object $F(A^p)$, so $H_v^q(G(I^{p,\bullet})) = R^qG(F(A^p))$ is $G(F(A^p))$ for $q = 0$ and 0 for $q > 0$. So ${}^{I}E_2^{p,q} = 0$ for $q > 0$ and ${}^{I}E_2^{p,0} = H^p(GF(A^\bullet)) = R^p(GF)(X)$, and $H^n(T) \cong R^n(GF)(X)$ by Proposition 16.2.4(3).

Second filtration (by rows). Fix a row index j. In the row $I^{\bullet,j}$ the sequences $0 \to B_h^{q,j} \to Z_h^{q,j} \to H_h^{q,j} \to 0$ and $0 \to Z_h^{q,j} \to I^{q,j} \to B_h^{q+1,j} \to 0$ consist of injective objects (they are terms of injective resolutions), hence split; so G commutes with taking horizontal cocycles, coboundaries and cohomology of this row, and $H_h^q(G(I^{\bullet,j})) = G(H_h^{q,j}(I))$. For fixed q, the column $H_h^{q,\bullet}(I)$ is an injective resolution of $H^q(F(A^\bullet)) = R^qF(X)$. By Theorem 16.4.1, ${}^{II}E_2^{p,q} = H_v^p(H_h^q(G(I))) = H^p(G(H_h^{q,\bullet}(I))) = R^pG(R^qF(X))$, and it converges to $H^{p+q}(T) \cong R^{p+q}(GF)(X)$. □

16.7 Exercises

Exercise 16.7.1. State and prove the analogue of Proposition 16.2.5 for a spectral sequence concentrated in two columns $p \in \{0, s\}$.

Exercise 16.7.2. Let $0 \to A^\bullet \to B^\bullet \to C^\bullet \to 0$ be a short exact sequence of bounded below complexes, viewed as a double complex with three columns. Using both spectral sequences of Theorem 16.4.1 (after shifting so that the double complex is first-quadrant), show that if two of A, B, C are acyclic, then so is the third.

Exercise 16.7.3. Use Theorem 16.4.1 to give a second proof of Theorem 15.8.2.

Exercise 16.7.4. Let $R \to S$ be a homomorphism of commutative rings, M an S-module and N an R-module. Construct a spectral sequence $\operatorname{Tor}_p^S(M, \operatorname{Tor}_q^R(S, N)) \Rightarrow \operatorname{Tor}_{p+q}^R(M, N)$ (homological indexing).

Chapter 17

Derived Categories

17.1 Introduction

The derived category of an abelian category is obtained from complexes by declaring quasi-isomorphisms to be isomorphisms. It is the natural home of derived functors: instead of a sequence of functors R^nF one has a single functor RF on complexes, compositions of derived functors are derived functors of compositions, and hypercohomology of complexes (for instance of the de Rham complex of a manifold) is defined without choices. Filtered derived categories and strict morphisms, discussed at the end, are the framework of Deligne's mixed Hodge theory (Chapter 74).

Some foundational verifications (the octahedral axiom, and the construction of localisations by roofs) are quoted from [Wei94, Chapter 10], [GM03, Chapter III] and [KS06]; everything used about derived functors is proved.

Prerequisites. Chapter 6 (Categories), Chapter 15 (Homological Algebra) and Chapter 16 (Spectral Sequences).

Throughout, $\mathcal{A}$ is an abelian category, complexes are as in Definition 15.3.1, and we argue with elements as justified by Remark 15.2.5.

17.2 The homotopy category

Definition 17.2.1. The *homotopy category* $K(\mathcal{A})$ has the complexes as objects and homotopy classes of morphisms of complexes (Definition 15.3.3) as morphisms. $K^+(\mathcal{A})$ is its full subcategory of bounded below complexes. The *shift* $X \mapsto X[1]$ is an automorphism of $K(\mathcal{A})$. A *distinguished triangle* in $K(\mathcal{A})$ is a diagram $X \to Y \to Z \to X[1]$ isomorphic (in $K(\mathcal{A})$) to one of the form $X \xrightarrow{f} Y \xrightarrow{i} \operatorname{Cone}(f) \xrightarrow{p} X[1]$, with i and p the maps of Proposition 15.4.5.

Remark 17.2.2. Homotopy is compatible with sums and compositions, so $K(\mathcal{A})$ is an additive category, and cohomology $H^n\colon K(\mathcal{A}) \to \mathcal{A}$ is a well-defined functor by Lemma 15.3.4.

Definition 17.2.3. A *triangulated category* is an additive category $\mathcal{T}$ with an additive automorphism $X \mapsto X[1]$ and a class of *distinguished triangles* $X \to Y \to Z \to X[1]$ such that:

(TR1) $X \xrightarrow{\mathrm{id}} X \to 0 \to X[1]$ is distinguished; triangles isomorphic to distinguished ones are distinguished; every morphism $X \to Y$ is the first morphism of a distinguished triangle.

(TR2) $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$ is distinguished if and only if $Y \xrightarrow{g} Z \xrightarrow{h} X[1] \xrightarrow{-f[1]} Y[1]$ is.

(TR3) Given distinguished triangles $X \to Y \to Z \to X[1]$ and $X' \to Y' \to Z' \to X'[1]$ and morphisms $u\colon X \to X'$, $v\colon Y \to Y'$ making the first square commute, there is $w\colon Z \to Z'$ making the whole diagram commute.

(TR4) (Octahedral axiom) For composable $X \xrightarrow{f} Y \xrightarrow{g} Z$ and distinguished triangles on f, g and gf with third vertices Z', X', Y', there is a distinguished triangle $Z' \to Y' \to X' \to Z'[1]$ compatible with the given morphisms.

An additive functor between triangulated categories commuting with the shifts and preserving distinguished triangles is *exact* (or triangulated).

Theorem 17.2.4. *$K(\mathcal{A})$ and $K^+(\mathcal{A})$ are triangulated categories.*

Sketch of proof. (TR1): $\mathrm{Cone}(\mathrm{id}_X)$ is contractible, with contracting homotopy $(a,b) \mapsto (b,0)$, so it is isomorphic to 0 in $K(\mathcal{A})$; the other two parts hold by definition. (TR3): if h is a homotopy with $vf - f'u = dh + hd$, then $(a,b) \mapsto (u(a), v(b) + h(a))$ is a morphism of complexes $\mathrm{Cone}(f) \to \mathrm{Cone}(f')$ (a direct check using the formula for the differential of the cone in Definition 15.3.5) compatible with the maps of the triangles. The omitted verifications are (TR2), which amounts to an explicit homotopy equivalence $\mathrm{Cone}(i) \simeq X[1]$ for $i\colon Y \to \mathrm{Cone}(f)$, and the octahedral axiom (TR4); see [Wei94, Sections 10.1–10.2]. □

Proposition 17.2.5. *Let $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$ be a distinguished triangle in a triangulated category $\mathcal{T}$ and W an object. Then $gf = 0$, and the sequence $\mathrm{Hom}(W,X) \to \mathrm{Hom}(W,Y) \to \mathrm{Hom}(W,Z)$ is exact; so by (TR2) there is a long exact sequence $\cdots \to \mathrm{Hom}(W,X) \to \mathrm{Hom}(W,Y) \to \mathrm{Hom}(W,Z) \to \mathrm{Hom}(W,X[1]) \to \cdots$. Similarly for $\mathrm{Hom}(-,W)$. For $\mathcal{T} = K(\mathcal{A})$, the functor H^0 sends distinguished triangles to long exact sequences.*

Proof. Apply (TR3) to the triangles $X \xrightarrow{\mathrm{id}} X \to 0 \to X[1]$ and $X \to Y \to Z \to X[1]$ with $u = \mathrm{id}$, $v = f$: the resulting w satisfies $gf = w \circ 0 = 0$. Let $\varphi\colon W \to Y$ with $g\varphi = 0$. By (TR1) and (TR2), the rotated triangles $W \to 0 \to W[1] \xrightarrow{-\mathrm{id}} W[1]$ and $Y \xrightarrow{g} Z \xrightarrow{h} X[1] \xrightarrow{-f[1]} Y[1]$ are distinguished. The square formed by $\varphi\colon W \to Y$ and $0\colon 0 \to Z$ commutes because $g\varphi = 0$, so (TR3) gives $w\colon W[1] \to X[1]$ with $(-f[1]) \circ w = \varphi[1] \circ (-\mathrm{id})$. Then $\psi = w[-1]$ satisfies $f\psi = \varphi$. The dual statement is proved in the same way. For $K(\mathcal{A})$ the last claim follows from the long exact sequence of the cone (Proposition 15.4.5) and invariance under isomorphism. □

17.3 The derived category

Definition 17.3.1. A class S of morphisms in a category C is a *multiplicative system* if it contains the identities and is closed under composition, and:

1. for $s\colon X' \to X$ in S and $f\colon Y \to X$ there are $t\colon Y' \to Y$ in S and $g\colon Y' \to X'$ with $ft = sg$, and dually;
2. if $f, g\colon X \to Y$ and $s\colon Y \to Y'$ in S satisfy $sf = sg$, then $ft = gt$ for some $t\colon X' \to X$ in S, and dually.

Theorem 17.3.2 (Localisation)**.** *Let S be a multiplicative system in C. There is a category $C[S^{-1}]$ with the same objects and a functor $Q\colon C \to C[S^{-1}]$ that is the identity on objects, sends S to isomorphisms, and is universal: every functor sending S to isomorphisms factors uniquely through Q. Morphisms $X \to Y$ in $C[S^{-1}]$ are equivalence classes of* roofs $X \xleftarrow{s} X' \xrightarrow{f} Y$ *with $s \in S$,*

representing $Q(f)Q(s)^{-1}$, and for $f, g\colon X \to Y$ in $\mathcal{C}$ one has $Q(f) = Q(g)$ if and only if $ft = gt$ for some $t \in S$. If $\mathcal{C}$ is additive (triangulated) and S is compatible with the structure, then $\mathcal{C}[S^{-1}]$ is additive (triangulated), with distinguished triangles the images of those of $\mathcal{C}$.

This is proved in [GM03, Chapter III] and [KS06, Chapter 7]; for the triangulated case see [Wei94, Section 10.3].

Proposition 17.3.3. *The quasi-isomorphisms form a multiplicative system in $K(\mathcal{A})$ and in $K^+(\mathcal{A})$.*

Proof. Identities and composites of quasi-isomorphisms are quasi-isomorphisms. Note that a morphism is a quasi-isomorphism iff its cone is acyclic (Proposition 15.4.5), and by Proposition 17.2.5 the cohomology of the three vertices of a distinguished triangle forms a long exact sequence.

(1) Let $s\colon X' \to X$ be a quasi-isomorphism and $f\colon Y \to X$. Complete s to a triangle $X' \to X \xrightarrow{p} C \to X'[1]$, so C is acyclic, and complete $pf\colon Y \to C$ to a triangle $Y' \xrightarrow{t} Y \xrightarrow{pf} C \to Y'[1]$ (using (TR1) and (TR2)). Then t is a quasi-isomorphism since C is acyclic, and (TR3) applied to the rotated triangles gives $g\colon Y' \to X'$ with $sg = ft$. The dual statement is proved in the same way.

(2) Suppose $s(f - g) = 0$ with $s\colon Y \to Y'$ a quasi-isomorphism. Complete s to a triangle $C[-1] \xrightarrow{k} Y \xrightarrow{s} Y' \to C$ with C acyclic. By Proposition 17.2.5, $f - g = k\varphi$ for some $\varphi\colon X \to C[-1]$. Complete φ to a triangle $X' \xrightarrow{t} X \xrightarrow{\varphi} C[-1] \to X'[1]$; then t is a quasi-isomorphism and $(f - g)t = k\varphi t = 0$. Dually. □

Definition 17.3.4. The *derived category* is $D(\mathcal{A}) = K(\mathcal{A})[\mathrm{Qis}^{-1}]$, and $D^+(\mathcal{A}) = K^+(\mathcal{A})[\mathrm{Qis}^{-1}]$. By Theorem 17.3.2 and Proposition 17.3.3 they are triangulated, H^n factors through them, and a morphism in $D(\mathcal{A})$ is an isomorphism if and only if it induces isomorphisms on all H^n.

Remark 17.3.5. The morphism classes of $D(\mathcal{A})$ need not be small sets for an arbitrary locally small $\mathcal{A}$, since roofs involve all complexes. For $\mathcal{A}$ with enough injectives this problem disappears for $D^+(\mathcal{A})$ by Corollary 17.4.2, which computes morphisms using a single complex of injectives.

17.4 Injective resolutions of complexes

Lemma 17.4.1. *Let X be an acyclic complex and I a bounded below complex of injective objects. Every morphism of complexes $f : X \to I$ is homotopic to zero.*

Proof. Let $I^n = 0$ for $n < n_0$; put $h^n = 0\colon X^n \to I^{n-1}$ for $n \le n_0$. Suppose h^k is defined for $k \le n$ with $f^{k-1} = dh^{k-1} + h^k d$ for $k \le n$. Let $g = f^n - dh^n\colon X^n \to I^n$. Then $gd = df^{n-1} - dh^n d = df^{n-1} - d(f^{n-1} - dh^{n-1}) = 0$, so g vanishes on $\mathrm{im}(X^{n-1} \to X^n) = \ker(X^n \to X^{n+1})$. Hence g factors through $X^n/\ker \cong \mathrm{im}(X^n \to X^{n+1}) \subseteq X^{n+1}$ and extends, by injectivity of I^n, to $h^{n+1}\colon X^{n+1} \to I^n$ with $h^{n+1}d = g$, that is, $f^n = dh^n + h^{n+1}d$. □

Corollary 17.4.2. *Let I be a bounded below complex of injectives. For every quasi-isomorphism $s\colon X' \to X$, composition with s is a bijection $\mathrm{Hom}_{K(\mathcal{A})}(X, I) \to \mathrm{Hom}_{K(\mathcal{A})}(X', I)$. Consequently $\mathrm{Hom}_{D(\mathcal{A})}(X, I) = \mathrm{Hom}_{K(\mathcal{A})}(X, I)$, and every quasi-isomorphism $I \to I'$ between bounded below complexes of injectives is a homotopy equivalence.*

Proof. Complete s to a distinguished triangle $X' \to X \to C \to X'[1]$ with C acyclic. By Lemma 17.4.1, $\mathrm{Hom}_K(C, I) = \mathrm{Hom}_K(C[-1], I) = 0$ ($C[-1]$ is also acyclic), and the long exact sequence of Proposition 17.2.5 gives the bijection. For the second statement, we use the description of morphisms in Theorem 17.3.2. The map $Q\colon \mathrm{Hom}_K(X, I) \to \mathrm{Hom}_D(X, I)$ is

surjective: a roof $X \xleftarrow{s} X' \xrightarrow{f} I$ has $f = f's$ for some $f' \colon X \to I$, and then $Q(f)Q(s)^{-1} = Q(f')$. It is injective: if $Q(f_1) = Q(f_2)$, then $f_1 t = f_2 t$ for a quasi-isomorphism t, so $f_1 = f_2$ by the bijection just proved. For the last, a quasi-isomorphism $I \to I'$ is an isomorphism in $D(\mathcal{A})$, hence by the second statement an isomorphism in $K(\mathcal{A})$. □

Proposition 17.4.3. *If $\mathcal{A}$ has enough injectives, every bounded below complex X admits a quasi-isomorphism $X \to I$ to a bounded below complex of injectives. Hence $K^+(\operatorname{Inj}\mathcal{A}) \to D^+(\mathcal{A})$ is an equivalence of categories.*

Proof. Shifting, assume $X^n = 0$ for $n < 0$. Let $J^{\bullet,\bullet}$ be a Cartan–Eilenberg resolution of X (Lemma 16.6.2) and $I = \operatorname{Tot}(J)$, a bounded below complex of injectives (each I^n is a finite sum). The augmentations $X^p \to J^{p,0}$ define a morphism $X \to I$. For each p the column $0 \to X^p \to J^{p,0} \to J^{p,1} \to \cdots$ is exact, so by Lemma 15.7.2 (applied with the roles of rows and columns exchanged) the map $X \to I$ is a quasi-isomorphism. (The total complex of the transposed double complex is isomorphic to I by the signs $(-1)^{pq}$, and the compatibility of $X^p \to J^{p,0}$ with the differentials holds because the column differential kills the image of X^p.) The functor $K^+(\operatorname{Inj}\mathcal{A}) \to D^+(\mathcal{A})$ is fully faithful by Corollary 17.4.2 and essentially surjective by the first statement. □

Definition 17.4.4. Let $\mathcal{A}$ have enough injectives and $F \colon \mathcal{A} \to \mathcal{B}$ be an additive functor to an abelian category. For each bounded below complex X choose a quasi-isomorphism $X \to I_X$ as in Proposition 17.4.3. The *right derived functor* is $RF \colon D^+(\mathcal{A}) \to D^+(\mathcal{B})$, $RF(X) = F(I_X)$ (with F applied termwise), on morphisms via $\operatorname{Hom}_D(X, Y) \cong \operatorname{Hom}_K(I_X, I_Y)$. For $X \in D^+(\mathcal{A})$ and $n \in \mathbb{Z}$, $R^nF(X) = H^n(RF(X))$ is the n-th *hypercohomology* of F at X.

Proposition 17.4.5. *RF is an exact functor of triangulated categories, independent of the choices up to canonical isomorphism. If F is left exact and $X \in \mathcal{A}$ is viewed as a complex in degree 0, then $H^n(RF(X)) = R^nF(X)$ as in Definition 15.6.4. For $X, Y \in \mathcal{A}$ and $n \in \mathbb{Z}$, $\operatorname{Hom}_{D(\mathcal{A})}(X, Y[n]) \cong R^n \operatorname{Hom}(X, -)(Y)$; in particular $\operatorname{Hom}_{D(R\text{-}\mathsf{Mod})}(M, N[n]) \cong \operatorname{Ext}^n_R(M, N)$ for R-modules M, N.*

Proof. Independence of the choices follows from uniqueness of I_X up to unique isomorphism in $K(\mathcal{A})$ (Corollary 17.4.2). For exactness, a distinguished triangle of $D^+(\mathcal{A})$ is isomorphic to one of the form $X \xrightarrow{g} Y \to \operatorname{Cone}(g) \to X[1]$ with g a morphism of bounded below complexes. By Corollary 17.4.2 there is $g_I \colon I_X \to I_Y$ with $g_I \iota_X = \iota_Y g$ in $K(\mathcal{A})$, where ι denotes the resolutions, and (TR3) in $K(\mathcal{A})$ gives a morphism of triangles (ι_X, ι_Y, w) from the triangle of g to that of g_I. By Proposition 17.2.5, the five lemma (Lemma 15.4.2) shows that w is a quasi-isomorphism. So the triangle is isomorphic in $D(\mathcal{A})$ to $I_X \to I_Y \to \operatorname{Cone}(g_I) \to I_X[1]$, whose vertices are bounded below complexes of injectives, and RF sends it to $F(I_X) \to F(I_Y) \to F(\operatorname{Cone}(g_I)) \to F(I_X)[1]$. Since F is additive and applied termwise, $F(\operatorname{Cone}(g_I)) = \operatorname{Cone}(F(g_I))$ with the corresponding maps, so this triangle is distinguished.

For $X \in \mathcal{A}$, an injective resolution is a quasi-isomorphism $X \to I$, so $H^n(RF(X)) = H^n(F(I)) = R^nF(X)$. Finally $\operatorname{Hom}_D(X, Y[n]) = \operatorname{Hom}_K(X, I_Y[n])$ by Corollary 17.4.2. A morphism of complexes $X \to I_Y[n]$ is a morphism $\varphi \colon X \to I_Y^n$ with $d\varphi = 0$, that is, an element of $\ker(\operatorname{Hom}(X, I_Y^n) \to \operatorname{Hom}(X, I_Y^{n+1}))$, and it is homotopic to zero if and only if $\varphi = dh$ for some $h \colon X \to I_Y^{n-1}$. So $\operatorname{Hom}_K(X, I_Y[n]) = H^n(\operatorname{Hom}(X, I_Y)) = R^n \operatorname{Hom}(X, -)(Y)$. For modules this is $\operatorname{Ext}^n_R(M, N)$ by Definition 15.8.1, as R-Mod has enough injectives (Theorem 15.5.4). □

Lemma 17.4.6. *Let $\mathcal{B}$ have enough injectives, $G \colon \mathcal{B} \to \mathcal{C}$ be left exact, and Y a bounded below acyclic complex of G-acyclic objects (Proposition 15.6.6). Then $G(Y)$ is acyclic. Consequently, if Y is any bounded below complex of G-acyclic objects, then $RG(Y) \cong G(Y)$ in $D^+(\mathcal{C})$.*

Proof. Let $Z^n = \ker(Y^n \to Y^{n+1})$. Acyclicity gives short exact sequences $0 \to Z^n \to Y^n \to Z^{n+1} \to 0$, and $Z^n = 0$ for $n \ll 0$. By induction and the long exact sequence of Theorem 15.6.5, all Z^n are G-acyclic, so G preserves the exactness of these sequences ($R^1G(Z^n) = 0$). Since $G(Z^{n+1}) \to G(Y^{n+1})$ is a monomorphism, the kernel of $G(Y^n) \to G(Y^{n+1})$ is $G(Z^n)$, which is the image of $G(Y^{n-1}) \to G(Z^n) \to G(Y^n)$; so $G(Y)$ is acyclic. For the second statement, let $Y \to I$ be a quasi-isomorphism to a complex of injectives; its cone is a bounded below acyclic complex (Proposition 15.4.5) of G-acyclic objects (termwise $Y^{n+1} \oplus I^n$, and R^kG is additive). By the first statement $G(\mathrm{Cone}) = \mathrm{Cone}(G(Y) \to G(I))$ is acyclic, so $G(Y) \to G(I) = RG(Y)$ is a quasi-isomorphism. □

Theorem 17.4.7. *Let $F\colon \mathcal{A} \to \mathcal{B}$ and $G\colon \mathcal{B} \to \mathcal{C}$ be left exact, with $\mathcal{A}$ and $\mathcal{B}$ having enough injectives, and suppose F sends injectives to G-acyclic objects. Then $R(G \circ F) \cong RG \circ RF$.*

Proof. For $X \in D^+(\mathcal{A})$ with resolution $X \to I_X$, $RF(X) = F(I_X)$ is a bounded below complex of G-acyclic objects, so $RG(RF(X)) \cong G(F(I_X)) = R(GF)(X)$ by Lemma 17.4.6. □

Proposition 17.4.8 (Hypercohomology spectral sequences)**.** *Let F be left exact, $\mathcal{A}$ with enough injectives, and X a complex with $X^n = 0$ for $n < 0$. There are convergent first-quadrant spectral sequences*

$$'E_1^{p,q} = R^qF(X^p) \Rightarrow R^{p+q}F(X), \qquad ''E_2^{p,q} = R^pF(H^q(X)) \Rightarrow R^{p+q}F(X).$$

Proof. Let J be a Cartan–Eilenberg resolution of X; by the proof of Proposition 17.4.3, $RF(X) = F(\mathrm{Tot}(J)) = \mathrm{Tot}(F(J))$. Filtering by columns as in the proof of Theorem 16.4.1, $'E_1^{p,q} = H_v^q(F(J^{p,\bullet})) = R^qF(X^p)$, since $J^{p,\bullet}$ is an injective resolution of X^p. Filtering by rows, the rows of J are split (second filtration in the proof of Theorem 16.6.3), so $H_h^q(F(J)) = F(H_h^q(J))$, and $H_h^q(J)^\bullet$ is an injective resolution of $H^q(X)$; hence $''E_2^{p,q} = H^p(F(H_h^q(J))) = R^pF(H^q(X))$. Convergence is Theorem 16.4.1. □

17.5 t-structures

Definition 17.5.1. For a complex X and $n \in \mathbb{Z}$ the *truncations* are

$$\tau_{\le n}X = (\cdots \to X^{n-1} \to Z^n(X) \to 0 \to \cdots), \qquad \tau_{\ge n}X = (\cdots \to 0 \to X^n/B^n(X) \to X^{n+1} \to \cdots).$$

Proposition 17.5.2. *There are natural morphisms $\tau_{\le n}X \to X \to \tau_{\ge n}X$, and $H^k(\tau_{\le n}X) = H^k(X)$ for $k \le n$ and 0 otherwise, while $H^k(\tau_{\ge n}X) = H^k(X)$ for $k \ge n$ and 0 otherwise. There is a distinguished triangle $\tau_{\le n}X \to X \to \tau_{\ge n+1}X \to (\tau_{\le n}X)[1]$ in $D(\mathcal{A})$.*

Proof. The cohomology computations are immediate from the definitions. The morphism $X \to \tau_{\ge n+1}X$ is surjective with kernel the complex $K = (\cdots \to X^{n-1} \to X^n \to B^{n+1}(X) \to 0 \to \cdots)$, and $\tau_{\le n}X \to K$ is a quasi-isomorphism (both have cohomology $H^k(X)$ for $k \le n$ and zero above, $H^{n+1}(K) = 0$ as $X^n \to B^{n+1}$ is onto). Finally, a short exact sequence of complexes $0 \to K \xrightarrow{\iota} X \xrightarrow{\pi} Q \to 0$ gives a distinguished triangle $K \to X \to Q \to K[1]$ in $D(\mathcal{A})$: the map $\mathrm{Cone}(\iota) \to Q$, $(a, b) \mapsto \pi(b)$, is a surjective morphism of complexes whose kernel $K^{n+1} \oplus K^n$ is $\mathrm{Cone}(\mathrm{id}_K)$, which is contractible (proof of Theorem 17.2.4); by the long exact sequence (Theorem 15.4.3) it is a quasi-isomorphism, so the triangle of ι is isomorphic in $D(\mathcal{A})$ to $K \to X \to Q \to K[1]$. Apply this to $K \to X \to \tau_{\ge n+1}X$ and replace K by $\tau_{\le n}X$. □

Remark 17.5.3. The full subcategories $D^{\le 0}$ and $D^{\ge 0}$ of complexes with cohomology in degrees ≤ 0, respectively ≥ 0, form the *standard t-structure* on $D(\mathcal{A})$: $\mathrm{Hom}(D^{\le 0}, D^{\ge 1}) = 0$, $D^{\le 0}[1] \subseteq D^{\le 0}$, and every object sits in a triangle as in Proposition 17.5.2. Its *heart* $D^{\le 0} \cap D^{\ge 0}$ is

equivalent to $\mathcal{A}$. Abstract t-structures, and in particular the perverse t-structure on derived categories of sheaves, were introduced in [BBD82]; perverse sheaves and M. Saito's mixed Hodge modules are the modern setting for the Hodge theory of singular varieties.

17.6 Filtered complexes and strictness

Definition 17.6.1. A *filtered object* (A, F) is an object with a decreasing filtration F^pA, *finite* if $F^aA = A$ and $F^bA = 0$ for some a, b. A morphism $f : (A, F) \to (B, F)$ is *filtered* if $f(F^pA) \subseteq F^pB$, and *strict* if $f(F^pA) = f(A) \cap F^pB$ for all p. A morphism of filtered complexes is a *filtered quasi-isomorphism* if it induces quasi-isomorphisms on all gr^p.

Counterexample 17.6.2. Filtered morphisms need not be strict, and the category of filtered vector spaces is not abelian. Let $A = B = k$ with $F^0A = A$, $F^1A = 0$ and $F^0B = F^1B = B$, $F^2B = 0$. The identity $A \to B$ is filtered and bijective, but not strict ($f(F^1A) = 0 \neq F^1B \cap B = k$), and it is not an isomorphism of filtered objects, because its inverse is not filtered. Its kernel and cokernel (with the induced filtrations) are zero, so it is a monomorphism and an epimorphism; by Lemma 15.2.3 the category is not abelian.

Proposition 17.6.3 (Deligne)**.** *Let K be a complex with a filtration that is finite in each degree. The spectral sequence of Theorem 16.3.4 degenerates at E_1 (that is, $d_r = 0$ for all $r \geq 1$) if and only if every differential $d\colon K^n \to K^{n+1}$ is strict. In this case $H^n(\mathrm{gr}^pK) \cong \mathrm{gr}^pH^n(K)$.*

Proof. We use the notation of Definition 16.3.2; strictness of d means $F^p \cap d(K) = d(F^p)$.

Suppose d is strict. Let $x \in Z_r^p$ with $r \geq 1$. Then $dx \in F^{p+1} \cap d(K) = d(F^{p+1})$, so $dx = dy$ with $y \in F^{p+1}$. Since $dy = dx \in F^{p+r}$, $y \in Z_{r-1}^{p+1}$, so x and the cocycle $x - y$ have the same class in E_r^p. Hence every class in E_r^p is represented by a cocycle of K, and $d_r = 0$.

Suppose all d_r, $r \geq 1$, vanish. Fix a degree n. We show $F^p \cap d(K^{n-1}) \subseteq d(F^pK^{n-1})$ for all p by descending induction on p (for p large, $F^pK^n = 0$). Let $y = dx \in F^p$ with $x \in F^aK^{n-1}$; since the filtration is finite in degree $n-1$, it suffices to treat all a, by descending induction on a. If $a \geq p$ there is nothing to show. If $a < p$, then $x \in Z_{p-a}^a$, and $d_{p-a}[x] = [dx] = 0$ in E_{p-a}^p, so $dx \in B_{p-a}^p + Z_{p-a-1}^{p+1}$: $dx = dx' + z$ with $x' \in F^{a+1}$, $dx' \in F^p$, and $z \in F^{p+1}$ with $z = d(x - x') \in d(K)$. By induction on a (as $x' \in F^{a+1}$), $dx' \in d(F^p)$; by induction on p, $z \in d(F^{p+1}) \subseteq d(F^p)$. So $y \in d(F^p)$. The last statement follows from $E_1 = E_\infty$ and Theorem 16.3.4. □

Remark 17.6.4. Localising the homotopy category of filtered complexes at filtered quasi-isomorphisms gives the *filtered derived category* $DF(\mathcal{A})$. Deligne used it to construct mixed Hodge structures on the cohomology of algebraic varieties [Del71]; the degeneration of the Hodge-to-de Rham spectral sequence of a compact Kähler manifold (Chapter 48) is, by Proposition 17.6.3, the strictness of the de Rham differential with respect to the Hodge filtration.

17.7 Exercises

Exercise 17.7.1. Show that in a triangulated category, if (u, v, w) is a morphism of distinguished triangles and two of u, v, w are isomorphisms, so is the third.

Solution. Let (u, v, w) be a morphism from $X \to Y \to Z \to X[1]$ to $X' \to Y' \to Z' \to X'[1]$ with u, v isomorphisms, and let W be any object. Applying $\mathrm{Hom}(W, -)$ and Proposition 17.2.5 gives two long exact sequences of abelian groups connected by the maps induced by u, v,

w, $u[1]$, $v[1]$. The five lemma (Lemma 15.4.2) shows that $w_*\colon \operatorname{Hom}(W,Z) \to \operatorname{Hom}(W,Z')$ is bijective for all W. Taking $W = Z'$ gives $s\colon Z' \to Z$ with $ws = \mathrm{id}$; then $w(sw) = w$, and injectivity of w_* for $W = Z$ gives $sw = \mathrm{id}$. The other cases follow by rotating the triangles with (TR2). □

Exercise 17.7.2. Let k be a field. Show that every complex X of k-vector spaces is isomorphic in $D(\mathsf{Vect}_k)$ to the complex $\bigoplus_n H^n(X)[-n]$ with zero differential.

Solution. Every subspace has a complement. Choose complements $X^n = Z^n \oplus C^n$ and $Z^n = B^n \oplus H^n$, where $H^n \cong H^n(X)$. The subcomplex $H = \bigoplus_n H^n$ has zero differential. Choose linear maps $r^n\colon X^n \to H^n$ that are the identity on H^n and vanish on $B^n \oplus C^n$. Then $r\colon X \to H$ is a morphism of complexes: $r(dx) = 0$ because $dx \in B^{n+1}$. It induces the identity on cohomology, so it is a quasi-isomorphism, hence an isomorphism in $D(\mathsf{Vect}_k)$. □

Exercise 17.7.3. Using Proposition 17.4.5, define the Yoneda product $\operatorname{Ext}^m(Y,Z)\times\operatorname{Ext}^n(X,Y) \to \operatorname{Ext}^{m+n}(X,Z)$ as composition in $D(\mathcal{A})$ and show that it is associative.

Exercise 17.7.4. Show that $D(\mathcal{A})$ is not abelian unless every short exact sequence in $\mathcal{A}$ splits. (Hint: a monomorphism in a triangulated category is split.)

Solution. Let $f\colon X \to Y$ be a monomorphism in a triangulated category, and complete it to a triangle $Z[-1] \xrightarrow{h} X \xrightarrow{f} Y \to Z$. Then $fh = 0$, so $h = 0$. Applying Proposition 17.2.5 with $\operatorname{Hom}(-,X)$ shows that $\operatorname{Hom}(Y,X) \to \operatorname{Hom}(X,X)$ is surjective (the next term maps through $h = 0$), so f has a retraction. Dually, epimorphisms have sections.

Suppose $D(\mathcal{A})$ is abelian and let $0 \to A \to B \xrightarrow{g} C \to 0$ be exact in $\mathcal{A}$. As in the proof of Proposition 17.5.2, it gives a distinguished triangle $A \to B \xrightarrow{g} C \xrightarrow{\delta} A[1]$. Factor $\delta = me$ with $e\colon C \to I$ an epimorphism and $m\colon I \to A[1]$ a monomorphism. Both split, so I is a direct summand of C and of $A[1]$ in $D(\mathcal{A})$. Cohomology is additive, so $H^k(I)$ vanishes for $k \neq 0$ (as a summand of C) and for $k \neq -1$ (as a summand of $A[1]$). Hence I is acyclic, $I \cong 0$ and $\delta = 0$. By Proposition 17.2.5, the sequence $\operatorname{Hom}(C,B) \to \operatorname{Hom}(C,C) \to \operatorname{Hom}(C,A[1])$ is exact and $\mathrm{id}_C \mapsto \delta = 0$, so $g\sigma = \mathrm{id}_C$ for some $\sigma\colon C \to B$ in $D(\mathcal{A})$. Applying H^0 gives $g \circ H^0(\sigma) = \mathrm{id}_C$ in $\mathcal{A}$, so the sequence splits. □

Part III

Topology

Chapter 18

General Topology

18.1 Introduction

Topology axiomatises the notions of nearness and continuity. This chapter develops the point-set topology used in the rest of the project: metric spaces for analysis, quotients and compactness for geometry, and paracompactness and partitions of unity, which are the basic tool for passing from local to global constructions on manifolds. The main theorems are Tychonoff's theorem (Theorem 18.6.8), the characterisation of compact metric spaces (Theorem 18.7.4), the Baire category theorem (Theorem 18.7.7), Urysohn's lemma and the Tietze extension theorem (Theorems 18.8.4 and 18.8.5), and the existence of partitions of unity (Theorem 18.10.6). Standard references are [Mun00] and [Eng89].
Prerequisites. Chapter 3 (Set Theory), Chapter 4 (Number Systems), Chapter 6 (Categories) and Chapter 11 (Linear Algebra).

18.2 Topological spaces and continuous maps

Why open sets. In analysis, "x_n tends to x" and "f is continuous at x" are expressed with distances: $|x_n - x| < \varepsilon$ eventually, $|f(y) - f(x)| < \varepsilon$ when $|y - x| < \delta$. Looking closely, the actual numbers ε and δ do not matter, only the sets $\{y : |y - x| < \delta\}$ and, more precisely, the sets that contain such a ball around each of their points, the *open* sets. For instance $f : \mathbb{R} \to \mathbb{R}$ is continuous if and only if the preimage of every open set is open (Proposition 18.3.5 below). Open sets have three basic properties, all following from the definition: $\emptyset$ and the whole space are open; a union of open sets is open (a ball around a point of the union lies in one of the sets); and the intersection of *two* open sets is open (around a point of the intersection take the smaller of two balls). An infinite intersection need not be open: the intervals $(-\frac{1}{n}, \frac{1}{n})$ are open and their intersection is $\{0\}$, which contains no ball. A topology is a choice of "open sets" with exactly these three properties, and nothing else; this is enough to speak of limits and continuity, and it applies to spaces where no natural distance exists, such as quotient spaces, spaces of functions with pointwise convergence, or the Zariski topology of algebraic geometry.

Definition 18.2.1. A *topology* on a set X is a set $\mathcal{T}$ of subsets of X, called *open* sets, such that $\emptyset$ and X are open, every union of open sets is open, and every finite intersection of open sets is open. A *topological space* is a set with a topology. A subset $C \subseteq X$ is *closed* if $X \setminus C$ is open. A *neighbourhood* of a point x (or of a subset A) is a set containing an open set containing x (or A). If $\mathcal{T} \subseteq \mathcal{T}'$ are two topologies on X, then $\mathcal{T}'$ is *finer* and $\mathcal{T}$ is *coarser*.

Example 18.2.2. 1. The *discrete* topology $\mathcal{P}(X)$ and the *indiscrete* topology $\{\emptyset, X\}$ are the finest and the coarsest topologies on X.

2. The *cofinite* topology on X consists of $\emptyset$ and the complements of finite sets. The *cocountable* topology consists of $\emptyset$ and the complements of countable sets.

3. The *Sierpiński space* is $\{0,1\}$ with open sets $\emptyset$, $\{1\}$ and $\{0,1\}$.

4. On a two-point set $\{a,b\}$ there are exactly four topologies: the indiscrete one, the discrete one, and the two Sierpiński topologies $\{\emptyset,\{a\},\{a,b\}\}$ and $\{\emptyset,\{b\},\{a,b\}\}$. Indeed a topology must contain $\emptyset$ and $\{a,b\}$, and any choice of which singletons to add satisfies the axioms. The family $\{\emptyset,\{a\},\{b\}\}$ is not a topology, since the union $\{a\}\cup\{b\}$ and the whole space are missing.

5. The *metric topology* of $\mathbb{R}$ (Section 18.3): U is open if every $x\in U$ has an $\varepsilon>0$ with $(x-\varepsilon,x+\varepsilon)\subseteq U$. Open intervals and their unions are open; $[0,1)$ is not open (no interval around 0 fits inside) and not closed (its complement $(-\infty,0)\cup[1,\infty)$ is not open at 1). So "not open" does not mean "closed": a set can be both ($\emptyset$, $\mathbb{R}$), one of the two, or neither.

A finer topology has more open sets. The finer the topology on X, the easier it is for maps *from* X to be continuous and the harder for maps *to* X; every map from a discrete space and every map to an indiscrete space is continuous (Definition 18.2.12 below).

Definition 18.2.3. Let A be a subset of a space X. Its *interior* $\operatorname{int}A$ is the union of all open sets contained in A, its *closure* $\overline{A}$ is the intersection of all closed sets containing A, and its *boundary* is $\partial A=\overline{A}\setminus\operatorname{int}A$. The set A is *dense* if $\overline{A}=X$.

Lemma 18.2.4. *Let $A,B\subseteq X$. Then $\operatorname{int}A$ is the largest open subset of A and $\overline{A}$ is the smallest closed set containing A; $X\setminus\overline{A}=\operatorname{int}(X\setminus A)$; a point x lies in $\overline{A}$ if and only if every open set containing x meets A; and $\overline{A\cup B}=\overline{A}\cup\overline{B}$.*

Proof. The first two statements hold because unions of open sets are open and intersections of closed sets are closed. Taking complements turns closed sets containing A into open sets contained in $X\setminus A$, which gives the third statement. So $x\notin\overline{A}$ if and only if x lies in an open set contained in $X\setminus A$, that is, an open set not meeting A. Finally $\overline{A}\cup\overline{B}$ is closed and contains $A\cup B$, so it contains $\overline{A\cup B}$; the reverse inclusion holds as $\overline{A},\overline{B}\subseteq\overline{A\cup B}$. □

Definition 18.2.5. Let $A\subseteq X$ and $x\in X$. Then x is an *adherent point* of A if every neighbourhood of x meets A (that is, $x\in\overline{A}$), an *accumulation point* of A if every neighbourhood of x meets $A\setminus\{x\}$, and an *isolated point* of A if $x\in A$ and some neighbourhood of x meets A only in x. The set A' of accumulation points is the *derived set*. A set without isolated points is *dense in itself*, and a closed set without isolated points is *perfect*.

The term *limit point* is used by some authors for accumulation points and by others for adherent points (limits of nets or, in metric spaces, of sequences in A); we avoid it.

Proposition 18.2.6. *Let $A\subseteq X$.*

1. *$\overline{A}=A\cup A'$, and every point of $\overline{A}$ is an accumulation point of A or an isolated point of A, but not both.*

2. *A is closed if and only if $A'\subseteq A$; A is open if and only if it is a neighbourhood of each of its points.*

3. *$\partial A=\overline{A}\cap\overline{X\setminus A}$ is closed; A is closed if and only if $\partial A\subseteq A$, and open if and only if $\partial A\cap A=\emptyset$; $\overline{A}=A\cup\partial A$ and $\operatorname{int}A=A\setminus\partial A$.*

4. *A is dense if and only if it meets every nonempty open set, if and only if* $\operatorname{int}(X \setminus A) = \emptyset$.

5. *If X is Hausdorff, $x \in A'$ if and only if every neighbourhood of x contains infinitely many points of A; in particular finite sets have no accumulation points.*

6. *A point $x \in X$ is isolated in X if and only if $\{x\}$ is open; a dense set contains all isolated points of X.*

Proof. (1) A point of $\overline{A}$ not in A has neighbourhoods meeting $A = A \setminus \{x\}$; a point of A either has a neighbourhood meeting A only in x (isolated) or not (accumulation point), and the two cases exclude each other. Conversely $A' \subseteq \overline{A}$ by Lemma 18.2.4. (2) The first statement follows from (1). If A is open it is a neighbourhood of its points; if A contains an open $U_x \ni x$ for every $x \in A$, then $A = \bigcup_x U_x$ is open. (3) $\overline{X \setminus A} = X \setminus \operatorname{int} A$ by Lemma 18.2.4, so $\overline{A} \cap \overline{X \setminus A} = \overline{A} \setminus \operatorname{int} A$; it is an intersection of closed sets. The other statements follow from $\operatorname{int} A \subseteq A \subseteq \overline{A}$. (4) By Lemma 18.2.4, $\overline{A} = X$ if and only if every point lies in no open set missing A, if and only if no nonempty open set misses A; and $X \setminus \overline{A} = \operatorname{int}(X \setminus A)$. (5) If a neighbourhood U of x meets $A \setminus \{x\}$ only in finitely many points $a_1, \dots, a_k$, choose by the Hausdorff property open sets $U_i \ni x$ not containing a_i; then $U \cap U_1 \cap \cdots \cap U_k$ is a neighbourhood of x missing $A \setminus \{x\}$. (6) The first statement is the definition, and an open $\{x\}$ must meet a dense set by (4). □

Example 18.2.7. In $\mathbb{R}$: for $A = \{1/n : n \geq 1\}$, every point of A is isolated, $A' = \{0\}$, and $\overline{A} = A \cup \{0\}$; so A is neither open nor closed. $\mathbb{Z}$ has no accumulation points and is closed. $\mathbb{Q}$ is dense, with $\mathbb{Q}' = \mathbb{R}$, and has empty interior, so $\partial\mathbb{Q} = \mathbb{R}$. A closed interval $[a, b]$ $(a < b)$ is perfect. In the indiscrete topology on a set with at least two points, every point is an accumulation point of every nonempty set other than a singleton, and (5) fails: the space is not Hausdorff.

Example 18.2.8. In $\mathbb{R}$ with its usual topology:

1. For $A = [0, 1)$: $\operatorname{int} A = (0, 1)$, $\overline{A} = [0, 1]$ and $\partial A = \{0, 1\}$. For $A = (0, 1) \cup \{2\}$: $\operatorname{int} A = (0, 1)$, $\overline{A} = [0, 1] \cup \{2\}$, and $\partial A = \{0, 1, 2\}$; the isolated point 2 belongs to the boundary although it belongs to A.

2. For $A = \mathbb{Q}$: $\operatorname{int} \mathbb{Q} = \emptyset$ (every interval contains irrationals), $\overline{\mathbb{Q}} = \mathbb{R}$ (every interval contains rationals), and $\partial\mathbb{Q} = \mathbb{R}$.

3. The closure does not commute with intersections: for $A = (0, 1)$ and $B = (1, 2)$, $\overline{A \cap B} = \emptyset$ while $\overline{A} \cap \overline{B} = \{1\}$; in general only $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$ holds. It commutes with finite unions (Lemma 18.2.4) but not with infinite ones: the sets $\{\frac{1}{n}\}$ are closed, while their union has closure $\{\frac{1}{n}\} \cup \{0\}$.

In the discrete topology every set is open and closed, so $\operatorname{int} A = A = \overline{A}$ and $\partial A = \emptyset$. In the indiscrete topology on X, a nonempty $A \neq X$ has $\operatorname{int} A = \emptyset$ and $\overline{A} = X$: the topology is too coarse to distinguish the points of A from the others.

Definition 18.2.9. A *basis* of a topology $\mathcal{T}$ is a subset $\mathcal{B} \subseteq \mathcal{T}$ such that every open set is a union of members of $\mathcal{B}$. A *subbasis* is a subset $\mathcal{S} \subseteq \mathcal{T}$ such that the finite intersections of members of $\mathcal{S}$ (the empty intersection being X) form a basis.

Lemma 18.2.10. *1. A set $\mathcal{B}$ of subsets of X is a basis of some topology if and only if $\mathcal{B}$ covers X and for all $B_1, B_2 \in \mathcal{B}$ and $x \in B_1 \cap B_2$ there is $B_3 \in \mathcal{B}$ with $x \in B_3 \subseteq B_1 \cap B_2$. This topology consists of the unions of members of $\mathcal{B}$.*

2. *For every set $\mathcal{S}$ of subsets of X there is a coarsest topology containing $\mathcal{S}$, the topology* generated *by $\mathcal{S}$, and $\mathcal{S}$ is a subbasis of it.*

Proof. (1) If $\mathcal{B}$ is a basis, then X is a union of members and $B_1 \cap B_2$ is open, hence a union of members, which gives the two conditions. Conversely, let $\mathcal{T}$ be the set of unions of members of $\mathcal{B}$. It contains $\emptyset$ (the empty union) and X (first condition), and is closed under unions. If $U = \bigcup_i B_i$ and $V = \bigcup_j B'_j$, then $U \cap V = \bigcup_{i,j} B_i \cap B'_j$, and each $B_i \cap B'_j$ is a union of members of $\mathcal{B}$ by the second condition. By induction $\mathcal{T}$ is closed under finite intersections.

(2) The set $\mathcal{B}$ of finite intersections of members of $\mathcal{S}$ contains X and is closed under finite intersections, so it satisfies the conditions of (1). The topology it generates contains $\mathcal{S}$, and every topology containing $\mathcal{S}$ contains $\mathcal{B}$ and all unions of its members. □

Example 18.2.11.

1. In a metric space the open balls form a basis (Section 18.3). In $\mathbb{R}$ already the intervals (p, q) with rational endpoints form a basis: if $x \in U$ open, there is $(x - \varepsilon, x + \varepsilon) \subseteq U$, and rationals $p \in (x - \varepsilon, x)$, $q \in (x, x + \varepsilon)$ give $x \in (p, q) \subseteq U$. This basis is countable, so $\mathbb{R}$ is second countable.

2. The rays $(-\infty, a)$ and (b, ∞) form a subbasis of the topology of $\mathbb{R}$ which is not a basis: the interval $(0, 1) = (-\infty, 1) \cap (0, \infty)$ is open but not a union of rays.

3. The half-open intervals $[a, b)$ satisfy the conditions of Lemma 18.2.10(1), so they are a basis of a topology, the *Sorgenfrey line*; it is finer than the usual topology, since $(a, b) = \bigcup_n [a + \frac{1}{n}, b)$, and strictly finer, since $[0, 1)$ is open in it.

Definition 18.2.12. A map $f : X \to Y$ between spaces is *continuous* if $f^{-1}(V)$ is open for every open $V \subseteq Y$. It is a *homeomorphism* if it is bijective and both f and f^{-1} are continuous. It is *open* (*closed*) if it maps open (closed) sets to open (closed) sets.

Proposition 18.2.13. *For a map $f : X \to Y$ the following are equivalent:*

1. *f is continuous;*
2. *$f^{-1}(C)$ is closed for every closed $C \subseteq Y$;*
3. *$f(\overline{A}) \subseteq \overline{f(A)}$ for every $A \subseteq X$;*
4. *for every $x \in X$ and every open $V \ni f(x)$ there is an open $U \ni x$ with $f(U) \subseteq V$;*
5. *$f^{-1}(S)$ is open for every member S of a subbasis of Y.*

Compositions of continuous maps are continuous, so spaces and continuous maps form a category Top.

Proof. (1)⇔(2) since $f^{-1}(Y \setminus C) = X \setminus f^{-1}(C)$. (2)⇒(3): $f^{-1}(\overline{f(A)})$ is closed and contains A, hence contains $\overline{A}$. (3)⇒(2): if C is closed and $A = f^{-1}(C)$, then $f(\overline{A}) \subseteq \overline{f(A)} \subseteq C$, so $\overline{A} \subseteq A$. (1)⇒(4): take $U = f^{-1}(V)$. (4)⇒(1): $f^{-1}(V)$ is the union of the open sets U provided by (4) for its points. (1)⇔(5): preimages commute with unions and intersections, and every open set is a union of finite intersections of members of the subbasis. The last statement follows from $(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W))$. □

Example 18.2.14.

1. Constant maps are continuous: the preimage of any set is $\emptyset$ or the whole space. The identity of X is continuous from a finer to a coarser topology, and only in that direction.

2. The floor function $\lfloor \cdot \rfloor : \mathbb{R} \to \mathbb{R}$ is not continuous: the preimage of the open interval $(-\frac{1}{2}, \frac{1}{2})$ is $[0, 1)$, which is not open. In terms of (4), at $x = 0$ the open set $V = (-\frac{1}{2}, \frac{1}{2}) \ni 0 = \lfloor 0 \rfloor$ contains $\lfloor U \rfloor$ for no open $U \ni 0$, since U contains negative numbers. It is continuous at every non-integer, where it is locally constant.

3. The indicator function $\mathbf{1}_{\mathbb{Q}}$ is continuous at no point, since every interval contains rational and irrational points; restricted to the subspace $\mathbb{Q}$ it is constant, hence continuous. Continuity depends on the topologies on both sides, and restricting a map to a subspace can make it continuous.

Counterexample 18.2.15. A continuous bijection need not be a homeomorphism: the identity map from X with the discrete topology to X with the indiscrete topology is continuous, and its inverse is not if X has at least two points.

Proposition 18.2.16. *For a bijection $f : X \to Y$ the following are equivalent:*

1. *f is a homeomorphism;*
2. *f is continuous and open;*
3. *f is continuous and closed;*
4. *$f(\overline{A}) = \overline{f(A)}$ for every $A \subseteq X$;*
5. *$U \subseteq X$ is open if and only if $f(U)$ is open.*

Proof. For a bijection, $(f^{-1})^{-1}(U) = f(U)$, so f^{-1} is continuous if and only if f is open, if and only if (taking complements) f is closed; this gives (1)⇔(2)⇔(3), and (5) is (2) written out. (1)⇒(4): by Proposition 18.2.13(3) applied to f and to f^{-1}, $f(\overline{A}) \subseteq \overline{f(A)}$ and $f^{-1}(\overline{f(A)}) \subseteq \overline{A}$. (4)⇒(3): f is continuous by Proposition 18.2.13(3), and if C is closed, $f(C) = f(\overline{C}) = \overline{f(C)}$ is closed. □

So a homeomorphism is a bijection of the points that is also a bijection of the open sets: every topological property (being open, closed, dense, compact, connected, Hausdorff, ...) is preserved. Two further criteria are proved later: a continuous bijection from a compact space to a Hausdorff space is a homeomorphism (Proposition 18.6.3(4)), and by Brouwer's *invariance of domain* a continuous injection from an open subset of $\mathbb{R}^n$ to $\mathbb{R}^n$ is open, hence a homeomorphism onto an open image [Hat02, Theorem 2B.3]. Examples: $x \mapsto x/(1-|x|)$ is a homeomorphism from the open ball $B(0,1) \subseteq \mathbb{R}^n$ onto $\mathbb{R}^n$, with inverse $y \mapsto y/(1+|y|)$, so all open balls are homeomorphic to $\mathbb{R}^n$; $t \mapsto e^{2\pi i t}$ is a continuous bijection $[0,1) \to S^1$ that is not a homeomorphism, since S^1 is compact and $[0,1)$ is not; and $\mathbb{R}$ is not homeomorphic to $\mathbb{R}^2$, since removing a point disconnects $\mathbb{R}$ but not $\mathbb{R}^2$ (Section 18.5).

Definition 18.2.17. A space X is *Hausdorff* if any two distinct points have disjoint open neighbourhoods. A sequence $(x_n)_{n\in\mathbb{N}}$ in X *converges* to x if every open set containing x contains x_n for all sufficiently large n.

Lemma 18.2.18. *In a Hausdorff space every sequence has at most one limit, and every finite subset is closed.*

Proof. If $x \neq y$ are limits, take disjoint open $U \ni x$ and $V \ni y$; for large n, $x_n \in U \cap V$, a contradiction. For a point x, every $y \neq x$ has an open neighbourhood not containing x, so $X \setminus \{x\}$ is open; finite unions of closed sets are closed. □

Counterexample 18.2.19. The cofinite topology on an infinite set X is not Hausdorff: two nonempty open sets have finite complements, and the union of these complements is not all of X, so they meet. In this space every injective sequence converges to every point, since an open set misses only finitely many of its terms.

18.3 Metric spaces

Definition 18.3.1. A *metric* on a set X is a map $d\colon X \times X \to \mathbb{R}$ such that for all $x, y, z \in X$: $d(x,y) \geq 0$ with equality if and only if $x = y$; $d(x,y) = d(y,x)$; and $d(x,z) \leq d(x,y) + d(y,z)$ (the *triangle inequality*). The *open ball* of radius $r > 0$ around x is $B(x,r) = \{y : d(x,y) < r\}$. A set A is *bounded* if $A \subseteq B(x,r)$ for some x and r.

Lemma 18.3.2. *The open balls of a metric space form a basis of a topology, the* metric topology*: U is open if and only if for every $x \in U$ there is $\varepsilon > 0$ with $B(x,\varepsilon) \subseteq U$. It is Hausdorff. A space whose topology is the metric topology of some metric is* metrisable*.*

Proof. If $x \in B(a,r) \cap B(b,s)$, let $\varepsilon = \min(r - d(a,x), s - d(b,x)) > 0$. By the triangle inequality $B(x,\varepsilon)$ is contained in both balls, so Lemma 18.2.10 applies; the description of open sets follows. If $x \neq y$ and $r = d(x,y)$, the balls $B(x, r/2)$ and $B(y, r/2)$ are disjoint by the triangle inequality. □

Definition 18.3.3. Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, with absolute value $|\cdot|$. A *norm* on a $\mathbb{K}$-vector space V is a map $\|\cdot\|\colon V \to \mathbb{R}$ such that $\|v\| = 0$ only for $v = 0$, $\|\lambda v\| = |\lambda|\,\|v\|$ and $\|v + w\| \leq \|v\| + \|w\|$ (so $\|v\| \geq 0$). Then $d(v,w) = \|v - w\|$ is a metric. On $\mathbb{K}^n$ we use the *Euclidean norm* $\|x\|_2 = (\sum_i |x_i|^2)^{1/2}$, the norm of the standard inner product (a norm by Proposition 11.10.2), and the *maximum norm* $\|x\|_\infty = \max_i |x_i|$.

Lemma 18.3.4. *If d, d' are metrics on X with $c\,d \leq d' \leq C\,d$ for constants $c, C > 0$, they define the same topology. In particular, since $\|x\|_\infty \leq \|x\|_2 \leq \sqrt{n}\,\|x\|_\infty$, the two norms on $\mathbb{K}^n$ define the same topology. On $\mathbb{R}$ it is the topology with basis the open intervals, the* standard topology*.*

Proof. $B_{d'}(x, c\varepsilon) \subseteq B_d(x,\varepsilon)$ and $B_d(x, \varepsilon/C) \subseteq B_{d'}(x,\varepsilon)$, so the criterion of Lemma 18.3.2 gives the same open sets. The norm inequalities hold because $\max_i |x_i|^2 \leq \sum_i |x_i|^2 \leq n \max_i |x_i|^2$. On $\mathbb{R}$ the balls are the bounded open intervals $(a - r, a + r)$, and every bounded open interval is such a ball. □

Proposition 18.3.5. *Let $f\colon X \to Y$ be a map of metric spaces.*

1. *f is continuous if and only if for every x and $\varepsilon > 0$ there is $\delta > 0$ with $d(f(x), f(x')) < \varepsilon$ whenever $d(x,x') < \delta$.*
2. *For $A \subseteq X$, a point x lies in $\overline{A}$ if and only if it is the limit of a sequence in A.*
3. *f is continuous if and only if $f(x_n) \to f(x)$ whenever $x_n \to x$.*

Proof. (1) is Proposition 18.2.13(4) with balls. (2) If $x \in \overline{A}$, choose $a_n \in A \cap B(x, 1/(n+1))$ by Lemma 18.2.4; then $a_n \to x$. Conversely, if $a_n \to x$, every open set containing x contains some a_n. (3) If f is continuous and $x_n \to x$, every open $V \ni f(x)$ has open preimage containing x, hence containing x_n for large n. If f is not continuous at x, there are $\varepsilon > 0$ and x_n with $d(x,x_n) < 1/(n+1)$ and $d(f(x), f(x_n)) \geq \varepsilon$; then $x_n \to x$ but $f(x_n) \not\to f(x)$. □

Remark 18.3.6. *Limits of maps.* Let $f\colon X \to Y$ be a map of topological spaces and $x_0 \in X$ a point that is not isolated, that is, every neighbourhood of x_0 contains points other than x_0. One writes $\lim_{x \to x_0} f(x) = y$ if for every neighbourhood V of y there is a neighbourhood U of x_0 with $f(U \setminus \{x_0\}) \subseteq V$; the value $f(x_0)$ plays no role. If Y is Hausdorff the limit is unique: limits $y \neq y'$ with disjoint neighbourhoods V, V' would give neighbourhoods U, U' with $f(U \cap U' \setminus \{x_0\}) \subseteq V \cap V' = \emptyset$, although $U \cap U'$ contains points other than x_0. By Proposition 18.2.13(4), applied at the single point x_0, f is *continuous at* x_0 exactly when

$$\lim_{x \to x_0} f(x) = f(x_0),$$

the condition at x_0 itself being automatic; and f is continuous if and only if it is continuous at every point. At an isolated point every map is continuous, and the limit is not defined (every y would satisfy the condition). For metric spaces, $\lim_{x\to x_0} f(x) = y$ means that for every $\varepsilon > 0$ there is $\delta > 0$ with $d(f(x), y) < \varepsilon$ whenever $0 < d(x, x_0) < \delta$, and, as in the proof of (3) above, it is equivalent to $f(x_n) \to y$ for every sequence $x_n \to x_0$ with $x_n \neq x_0$. For $X \subseteq \mathbb{R}$ this is the limit of real analysis (Chapter 27).

Example 18.3.7. *The closed ball and the closure of the ball.* In a metric space the closed ball $\bar{B}(x, r) = \{y : d(x, y) \leq r\}$ is closed (its complement is open by the triangle inequality), so it contains the closure $\overline{B(x, r)}$ of the open ball. They can differ: for the discrete metric ($d(x, y) = 1$ for $x \neq y$) on a set with at least two points, $B(x, 1) = \{x\}$ is closed, while $\bar{B}(x, 1)$ is the whole space. In a normed space, for $r > 0$ they agree: if $\|y - x\| = r$, the points $x + t(y - x)$, $0 < t < 1$, lie in $B(x, r)$ and tend to y. Similarly the boundary of $B(x, r)$ is contained in the sphere $\{d(x, y) = r\}$, with equality in normed spaces but not in general.

Counterexample 18.3.8. In general spaces sequences do not detect closures. Give $\mathbb{R}$ the cocountable topology. If $x_n \to x$, the set $C = \{x_n : x_n \neq x\}$ is countable, so $\mathbb{R} \setminus C$ is an open set containing x, and hence $x_n = x$ for large n. Now $A = \mathbb{R} \setminus \{0\}$ is uncountable (Proposition 4.8.6), so the only closed set containing it is $\mathbb{R}$ and $0 \in \overline{A}$; but no sequence in A converges to 0. (Nets or filters replace sequences in general [Eng89, Chapter 1].)

Proposition 18.3.9. *Addition and multiplication $\mathbb{K} \times \mathbb{K} \to \mathbb{K}$ are continuous for the metric $\|(x, y) - (a, b)\|_\infty$, and so is $x \mapsto 1/x$ on $\mathbb{K} \setminus \{0\}$.*

Proof. We have $|(x + y) - (a + b)| \leq |x - a| + |y - b|$, and if $|x - a|, |y - b| < \delta \leq 1$ then $|xy - ab| \leq |x|\,|y - b| + |b|\,|x - a| \leq (|a| + 1 + |b|)\delta$. If $|x - a| < |a|/2$ then $|x| > |a|/2$ and $|1/x - 1/a| = |x - a|/(|x|\,|a|) \leq 2|x - a|/|a|^2$. In each case the criterion of Proposition 18.3.5(1) holds. □

Proposition 18.3.10. *Let X be a space, (Y, d) a metric space and $f_n \colon X \to Y$ continuous maps converging* uniformly *to f: for every $\varepsilon > 0$ there is N with $d(f_n(x), f(x)) < \varepsilon$ for all $n \geq N$ and all x. Then f is continuous.*

Proof. Let $x \in X$ and $\varepsilon > 0$. Choose n with $d(f_n, f) < \varepsilon/3$ everywhere, and an open $U \ni x$ with $d(f_n(x), f_n(x')) < \varepsilon/3$ for $x' \in U$. Then $d(f(x), f(x')) < \varepsilon$ for $x' \in U$, and Proposition 18.2.13(4) applies. □

Definition 18.3.11. A space is *first countable* if every point x has countably many open neighbourhoods such that every open set containing x contains one of them; *second countable* if its topology has a countable basis; *separable* if it has a countable dense subset; and *Lindelöf* if every open cover has a countable subcover.

Proposition 18.3.12. *Metric spaces are first countable. Second countable spaces are separable and Lindelöf. A separable metric space is second countable. In particular $\mathbb{R}^n$ and $\mathbb{C}^n$ are second countable.*

Proof. The balls $B(x, 1/(n + 1))$ show first countability. Let $\mathcal{B}$ be a countable basis. Choosing a point in each nonempty member gives a countable set meeting every nonempty open set, hence dense by Lemma 18.2.4. Given an open cover $\mathcal{U}$, for each $B \in \mathcal{B}$ contained in some member of $\mathcal{U}$ choose one such member U_B; these countably many sets cover, because a point $x \in U \in \mathcal{U}$ lies in some $B \in \mathcal{B}$ with $B \subseteq U$, and then $x \in U_B$. If D is a countable dense subset of a metric space, the balls $B(q, 1/n)$ with $q \in D$, $n \geq 1$ form a basis: if $x \in U$ with U open, choose n with $B(x, 2/n) \subseteq U$ and $q \in D \cap B(x, 1/n)$; then $x \in B(q, 1/n) \subseteq B(x, 2/n) \subseteq U$. Finally $\mathbb{Q}^n$ and $(\mathbb{Q} + i\mathbb{Q})^n$ are countable (Corollary 3.6.13) and dense, by density of $\mathbb{Q}$ in $\mathbb{R}$ (Proposition 4.7.3). □

Counterexample 18.3.13 (The Sorgenfrey line)**.** The half-open intervals $[a, b)$ form a basis of a topology on $\mathbb{R}$, the *Sorgenfrey line*, finer than the Euclidean topology. It is first countable (the sets $[x, x + 1/n)$ form a neighbourhood basis at x) and separable ($\mathbb{Q}$ is dense, as every $[a, b)$ contains a rational number). It is not second countable: if $\mathcal{B}$ is a basis, then for each x some $B_x \in \mathcal{B}$ satisfies $x \in B_x \subseteq [x, x + 1)$, so $x = \min B_x$, and $x \mapsto B_x$ is injective from the uncountable set $\mathbb{R}$ to $\mathcal{B}$. By Proposition 18.3.12, it is therefore not metrisable: separable metric spaces are second countable.

18.4 Subspaces, products and quotients

Definition 18.4.1. Let X be a set. For a family of maps $f_i \colon X \to X_i$ into spaces, the *initial topology* is the topology generated by the sets $f_i^{-1}(U)$, $U \subseteq X_i$ open. For a family of maps $g_i \colon X_i \to X$ from spaces, the *final topology* consists of the sets $U \subseteq X$ such that $g_i^{-1}(U)$ is open for all i.

Proposition 18.4.2. *With the initial topology, a map $h \colon Z \to X$ from a space is continuous if and only if every $f_i \circ h$ is continuous. With the final topology, a map $h \colon X \to Z$ to a space is continuous if and only if every $h \circ g_i$ is continuous.*

Proof. The final topology is a topology because preimages commute with unions and intersections. For the initial topology, $h^{-1}(f_i^{-1}(U)) = (f_i h)^{-1}(U)$ and Proposition 18.2.13(5) give the claim. For the final topology, $h^{-1}(W)$ is open if and only if $g_i^{-1}(h^{-1}(W)) = (hg_i)^{-1}(W)$ is open for all i. □

Definition 18.4.3. The *subspace topology* on $A \subseteq X$ is the initial topology for the inclusion: its open sets are the sets $U \cap A$ with U open in X, and its closed sets are the sets $C \cap A$ with C closed in X. An *embedding* is a map $f \colon X \to Y$ that is a homeomorphism onto $f(X)$ with the subspace topology.

Example 18.4.4.

1. In $A = [0, 1] \subseteq \mathbb{R}$ the set $[0, \frac{1}{2}) = (-1, \frac{1}{2}) \cap A$ is open, although it is not open in $\mathbb{R}$: "open" is relative to the ambient space. Similarly $[0, 1]$ is open and closed in itself. If A is open in X, the open subsets of A are exactly the open subsets of X contained in A; if A is closed, the same holds for closed subsets.

2. $\mathbb{Z} \subseteq \mathbb{R}$ has the discrete topology: $\{n\} = (n - \frac{1}{2}, n + \frac{1}{2}) \cap \mathbb{Z}$ is open. The subspace $\{0\} \cup \{\frac{1}{n} : n \geq 1\}$ is not discrete, as every open set containing 0 contains some $\frac{1}{n}$; and $\mathbb{Q}$ is not discrete either. So two subspaces can be in bijection without being homeomorphic: $\mathbb{Z}$ and $\{0\} \cup \{\frac{1}{n}\}$ are both countable.

3. The inclusion $A \to X$ of a subspace is continuous, and a map $Z \to A$ is continuous if and only if its composite with the inclusion is (Proposition 18.4.2).

Lemma 18.4.5 (Pasting lemma)**.** *Let $f \colon X \to Y$ be a map and suppose $X = \bigcup_i A_i$, where either all A_i are open, or there are finitely many A_i and all are closed. If every restriction $f|_{A_i}$ is continuous, then f is continuous.*

Proof. For $V \subseteq Y$ we have $f^{-1}(V) = \bigcup_i (f|_{A_i})^{-1}(V)$. In the first case, for V open each $(f|_{A_i})^{-1}(V)$ is open in the open set A_i, hence open in X. In the second case, for V closed each term is closed in the closed set A_i, hence closed in X, and a finite union of closed sets is closed. □

Definition 18.4.6. The *product* of a family of spaces $(X_i)_{i \in I}$ is the set $\prod_i X_i$ with the initial topology for the projections $p_i \colon \prod_j X_j \to X_i$, the *product topology*. The *disjoint union* $\coprod_i X_i$ carries the final topology for the inclusions.

Proposition 18.4.7. *1. The sets $\prod_i U_i$, with $U_i \subseteq X_i$ open and $U_i = X_i$ for all but finitely many i, form a basis of the product topology.*

2. $\prod_i X_i$ with the projections is a product in Top*, and $\coprod_i X_i$ with the inclusions is a coproduct, in the sense of Example 6.8.3.*

3. The projections are open maps. A product of Hausdorff spaces is Hausdorff, and so is a subspace of a Hausdorff space.

4. The product topology on $\mathbb{K}^n = \mathbb{K} \times \cdots \times \mathbb{K}$ is the topology of the Euclidean norm.

5. For every space X, sums and products of continuous functions $X \to \mathbb{K}$ are continuous, and so is $1/g$ for continuous $g \colon X \to \mathbb{K} \setminus \{0\}$.

Proof. (1) These sets are the finite intersections of the subbasic sets $p_i^{-1}(U_i)$. (2) By Proposition 18.4.2, a map $h \colon Z \to \prod_i X_i$ is continuous if and only if all its components $p_i h$ are; so continuous families $Z \to X_i$ correspond bijectively to continuous maps $Z \to \prod_i X_i$. Dually for disjoint unions. (3) The image of a basic open set under p_i is U_i, and images commute with unions. If $x \neq y$ in the product, then $x_i \neq y_i$ for some i, and preimages under p_i of disjoint neighbourhoods of x_i, y_i separate x, y. For subspaces intersect with A. (4) By Lemma 18.3.4 we may use $\|\cdot\|_\infty$, whose balls are products of balls; and every basic open set $\prod_i U_i$ contains, around each of its points a, the $\|\cdot\|_\infty$-ball of radius $\min_i \varepsilon_i$, where $B(a_i, \varepsilon_i) \subseteq U_i$. (5) If f, g are continuous, then $(f, g) \colon X \to \mathbb{K}^2$ is continuous by (2), and $\mathbb{K}^2$ carries the metric topology of $\|\cdot\|_\infty$ by (4); compose with the maps of Proposition 18.3.9. □

Counterexample 18.4.8. The *box topology* on $\prod_i X_i$, with basis all products of open sets, is finer than the product topology and does not have the universal property of the product. The map $\mathbb{R} \to \mathbb{R}^{\mathbb{N}}$, $t \mapsto (t, t, \ldots)$, has continuous components, but the preimage of the box $\prod_n (-1/(n+1), 1/(n+1))$ is $\{0\}$, which is not open.

Definition 18.4.9. A surjective map $q \colon X \to Y$ from a space is a *quotient map* if Y carries the final topology: $V \subseteq Y$ is open if and only if $q^{-1}(V)$ is open. For an equivalence relation $\sim$ on X, the *quotient space* $X/\!\sim$ is the set of classes with the final topology for the projection. For $A \subseteq X$, X/A denotes the quotient identifying all points of A with each other.

Proposition 18.4.10. *Let $q \colon X \to Y$ be a quotient map. A map $h \colon Y \to Z$ is continuous if and only if $h \circ q$ is continuous; hence every continuous map $g \colon X \to Z$ that is constant on the fibres of q factors uniquely as $g = \bar{g} \circ q$ with $\bar{g}$ continuous. A continuous surjection that is open or closed is a quotient map.*

Proof. The first statement is Proposition 18.4.2; $\bar{g}$ is well defined as a map of sets because g is constant on fibres. If $f \colon X \to Y$ is a continuous open surjection and $f^{-1}(V)$ is open, then $V = f(f^{-1}(V))$ is open. The closed case is the same with closed sets, using $f^{-1}(Y \setminus V) = X \setminus f^{-1}(V)$. □

Example 18.4.11. 1. The *unit sphere* $S^n = \{x \in \mathbb{R}^{n+1} : \|x\|_2 = 1\}$ is a subspace of $\mathbb{R}^{n+1}$, and the *real projective space* is $\mathbb{RP}^n = S^n/(x \sim -x)$.

2. The *cylinder* and the *torus* are the quotients of $[0,1]^2$ identifying $(0, t) \sim (1, t)$, respectively identifying in addition $(s, 0) \sim (s, 1)$. The *cone* on X is $(X \times [0,1])/(X \times \{1\})$.

3. Two spaces X, Y with a subspace $A \subseteq X$ and a continuous map $f \colon A \to Y$ give the *adjunction space* $X \cup_f Y = (X \sqcup Y)/(a \sim f(a))$, the space obtained by *attaching* X to Y along f. These are the building blocks of CW complexes (Chapter 19).

Counterexample 18.4.12. Quotients of Hausdorff spaces need not be Hausdorff.

1. Let $x \sim y$ on $\mathbb{R}$ if $x - y \in \mathbb{Q}$. If $V \subseteq \mathbb{R}/\sim$ is open and nonempty, then $W = q^{-1}(V)$ is a nonempty open set with $W + \mathbb{Q} = W$. Its complement is closed and invariant under translation by $\mathbb{Q}$; if it contained a point y it would contain the dense set $y + \mathbb{Q}$, hence be all of $\mathbb{R}$. So $W = \mathbb{R}$, and the quotient is indiscrete.

2. (The line with two origins.) Let $X = (\mathbb{R} \times \{1, 2\})/\sim$ with $(t, 1) \sim (t, 2)$ for $t \neq 0$. The projection q is open, since for U open $q^{-1}(q(U))$ is the union of U and the swap of $U \setminus (\{0\} \times \{1, 2\})$; so q restricts to homeomorphisms of each copy of $\mathbb{R}$ onto open subsets, and every point of X has an open neighbourhood homeomorphic to $\mathbb{R}$. But the two origins $q(0, 1)$ and $q(0, 2)$ have no disjoint neighbourhoods, since preimages of neighbourhoods contain $(-\varepsilon, \varepsilon) \times \{1\}$ and $(-\varepsilon, \varepsilon) \times \{2\}$, whose images share $q(\varepsilon/2, 1)$.

18.5 Connectedness

Definition 18.5.1. A space X is *connected* if it is nonempty and is not the union of two disjoint nonempty open subsets. A subset is connected if it is connected as a subspace.

Lemma 18.5.2. *For a nonempty space X the following are equivalent: X is connected; the only subsets of X that are both open and closed are $\emptyset$ and X; every continuous map from X to the discrete space $\{0, 1\}$ is constant.*

Proof. A decomposition $X = U \sqcup V$ into disjoint open sets is the same as a set $U = X \setminus V$ that is open and closed, and the same as the continuous map equal to 0 on U and 1 on V. □

Theorem 18.5.3. *A nonempty subset $I \subseteq \mathbb{R}$ is connected if and only if it is an interval: $a, b \in I$ and $a < c < b$ imply $c \in I$.*

Proof. If I is not an interval, with $a, b \in I$, $a < c < b$ and $c \notin I$, then $I \cap (-\infty, c)$ and $I \cap (c, \infty)$ disconnect I.

Let I be an interval and suppose $I = U \sqcup V$ with U, V nonempty and open in I. Choose $a \in U$ and $b \in V$; exchanging the roles of U and V if necessary, $a < b$. Then $[a, b] \subseteq I$. Let $s = \sup(U \cap [a, b])$, which exists by completeness of $\mathbb{R}$, and $s \in [a, b] \subseteq I$. Suppose $s \in U$. Then $s < b$, and since U is open in I there is $\varepsilon > 0$ with $(s - \varepsilon, s + \varepsilon) \cap I \subseteq U$; the points of $(s, \min(s + \varepsilon, b))$ lie in $U \cap [a, b]$, contradicting the definition of s. Suppose $s \in V$. Then $s > a$, and there is $\varepsilon > 0$ with $(s - \varepsilon, s + \varepsilon) \cap I \subseteq V$. So $U \cap [a, b] \subseteq [a, \max(a, s - \varepsilon)]$, whose supremum is smaller than s, again a contradiction. □

Proposition 18.5.4. 1. *Continuous images of connected spaces are connected.*

2. *If (A_i) is a family of connected subsets of X with a common point, then $\bigcup_i A_i$ is connected.*

3. *If $A \subseteq X$ is connected and $A \subseteq B \subseteq \overline{A}$, then B is connected.*

4. *A product of connected spaces is connected.*

Proof. We use Lemma 18.5.2. (1) If $f : X \to Y$ is surjective and $h : Y \to \{0, 1\}$ is continuous, then hf is constant, so h is. (2) A continuous map $\bigcup_i A_i \to \{0, 1\}$ is constant on each A_i, with the value it takes at the common point. (3) A continuous $h : B \to \{0, 1\}$ is constant, say equal to c, on A; then $h^{-1}(c)$ is closed in B and contains A, so it contains the closure of A in B, which is B. (4) For two factors, fix (x_0, y_0). For each y, the set $(X \times \{y\}) \cup (\{x_0\} \times Y)$ is connected by (2), both pieces being homeomorphic to X or Y and sharing (x_0, y). All these sets contain (x_0, y_0) and cover $X \times Y$, so (2) applies; by induction finite products are

connected. For an arbitrary product, fix a point a and let D be the set of points that differ from a in finitely many coordinates. For each finite set F of indices, the points agreeing with a outside F form a subspace homeomorphic to $\prod_{i\in F} X_i$, which is connected and contains a; so D is connected by (2). Every basic open set of Proposition 18.4.7(1) meets D, so D is dense and the product is connected by (3). □

Corollary 18.5.5 (Intermediate value theorem)**.** *If X is connected and $f\colon X \to \mathbb{R}$ is continuous, then $f(X)$ is an interval. In particular, a continuous $f\colon [a,b] \to \mathbb{R}$ takes every value between $f(a)$ and $f(b)$.*

Proof. Combine Theorem 18.5.3 and Proposition 18.5.4(1). □

Definition 18.5.6. The *connected component* of $x \in X$ is the union of all connected subsets containing x. A *path* from x to y is a continuous map $\gamma\colon [0,1] \to X$ with $\gamma(0) = x$ and $\gamma(1) = y$. The space X is *path-connected* if it is nonempty and any two points are joined by a path. It is *locally connected* (*locally path-connected*) if every neighbourhood of every point contains a connected (path-connected) open neighbourhood.

Proposition 18.5.7. 1. *The connected components are connected, closed, and partition X. In a locally connected space they are open.*

2. *Path-connected spaces are connected. The relation "joined by a path" is an equivalence relation, whose classes are the* path components*.*

3. *In a locally path-connected space the path components are open, and they coincide with the connected components. In particular, a connected open subset of $\mathbb{R}^n$ is path-connected.*

Proof. (1) The component C of x is connected by Proposition 18.5.4(2), and $\overline{C}$ is connected by (3), so $C = \overline{C}$. If two components meet, their union is connected, so they are equal. If X is locally connected and $y \in C$, a connected open neighbourhood of y lies in the component of y, which is C.

(2) If γ is a path from x to y and δ one from y to z, then $\gamma * \delta$, equal to $\gamma(2t)$ on $[0,1/2]$ and to $\delta(2t-1)$ on $[1/2,1]$, is continuous by the pasting lemma (Lemma 18.4.5); reversing $t \mapsto 1-t$ gives symmetry. If X is path-connected and $x \in X$, then X is the union of the images of paths starting at x, which are connected by Theorem 18.5.3 and Proposition 18.5.4(1), so X is connected by Proposition 18.5.4(2).

(3) If y lies in the path component P and U is a path-connected open neighbourhood of y, then $U \subseteq P$; so path components are open. Each connected component C is the disjoint union of the path components it contains, which are open; by connectedness there is only one. Open balls in $\mathbb{R}^n$ are path-connected (use straight segments), so open subsets of $\mathbb{R}^n$ are locally path-connected. □

Counterexample 18.5.8 (Topologist's sine curve)**.** Connected does not imply path-connected. Let $f\colon (0,1] \to [-1,1]$ be the continuous function with $f(1/n) = (-1)^n$ for $n \geq 1$ that is affine on each interval $[1/(n+1), 1/n]$ (continuous by the pasting lemma applied on the open sets $(1/(n+1), 1/(n-1))$ and $(1/2,1]$); classically one uses $\sin(1/x)$ instead. Let G be its graph and $T = G \cup (\{0\} \times [-1,1]) \subseteq \mathbb{R}^2$. The map $x \mapsto (x, f(x))$ is continuous, so G is connected. By Corollary 18.5.5, for each $y \in [-1,1]$ and n there is $x_n \in [1/(n+1), 1/n]$ with $f(x_n) = y$, and $(x_n, y) \to (0, y)$; so $G \subseteq T \subseteq \overline{G}$ and T is connected.

Suppose $\gamma = (x, y)\colon [0,1] \to T$ is a path from $(0,0)$ to $(1, f(1))$. The set of t with $x(t) = 0$ is closed and contains 0; let t_0 be its largest element, so $t_0 < 1$ and $\gamma(t) \in G$ for $t > t_0$. By continuity there is $\delta > 0$ with $t_0 + \delta \leq 1$ and $|y(t) - y(t_0)| < 1/2$ for $t \in [t_0, t_0 + \delta]$. Let $u = x(t_0 + \delta) > 0$ and choose n with $1/n < u$. By Corollary 18.5.5 there are $t, t' \in (t_0, t_0 + \delta]$

with $x(t) = 1/n$ and $x(t') = 1/(n+1)$, so $y(t)$ and $y(t')$ are 1 and -1 in some order; both lie within $1/2$ of $y(t_0)$, which is impossible.

Example 18.5.9. $\mathbb{R}$ and $\mathbb{R}^2$ are not homeomorphic. A homeomorphism would restrict to a homeomorphism between $\mathbb{R} \setminus \{a\}$, which is not connected, and $\mathbb{R}^2 \setminus \{b\}$, which is path-connected: two points $p, q \neq b$ are joined by the segment $[p, q]$ if it avoids b, and otherwise by the segments $[p, r]$ and $[r, q]$ for any r not on the line through p and q. Similarly S^n is path-connected for $n \geq 1$, being a continuous image of $\mathbb{R}^{n+1} \setminus \{0\}$ under $x \mapsto x/\|x\|_2$.

18.6 Compactness

Definition 18.6.1. An *open cover* of a space X is a family of open sets whose union is X. The space X is *compact* if every open cover has a finite subcover. A subset is compact if it is compact as a subspace, that is, if every cover of it by open subsets of X has a finite subcover.

Compactness is the topological substitute for finiteness: many statements about finite sets, such as "a real function on it has a maximum", extend to compact spaces, because one can reduce to finitely many open sets. To prove that a space is *not* compact one exhibits a single open cover without finite subcover; to prove that it is compact one has to handle every cover.

Example 18.6.2.

1. Every finite space is compact, and so is every space with finitely many open sets. An infinite discrete space is not: the singletons cover it.

2. $(0, 1]$ is not compact: the open sets $(\frac{1}{n}, 1]$ cover it, and finitely many of them have union $(\frac{1}{N}, 1]$ for the largest N used, which misses $\frac{1}{2N}$. Likewise $\mathbb{R}$ is not compact (cover by $(-n, n)$), and $\mathbb{Q} \cap [0, 1]$ is not (cover by the sets $\{q : |q - \alpha| > \frac{1}{n}\}$ for an irrational $\alpha \in (0, 1)$).

3. $K = \{0\} \cup \{\frac{1}{n} : n \geq 1\}$ is compact: given an open cover, some member U contains 0, hence all $\frac{1}{n}$ with $n > N$ for some N, and finitely many further members cover $1, \frac{1}{2}, \ldots, \frac{1}{N}$. A convergent sequence together with its limit is always compact, by the same argument.

4. Every set with the cofinite topology is compact: one member of a cover misses only finitely many points, each of which lies in some member.

5. In a non-Hausdorff space compact sets need not be closed: in the Sierpiński space $\{1\}$ is compact but not closed, and in an indiscrete space every subset is compact.

Proposition 18.6.3. *1. A closed subset of a compact space is compact.*

2. In a Hausdorff space, disjoint compact subsets K, L have disjoint open neighbourhoods; in particular compact subsets are closed.

3. Continuous images of compact spaces are compact.

4. A continuous map from a compact space to a Hausdorff space is closed. If it is bijective, it is a homeomorphism; if it is injective, it is an embedding; if it is surjective, it is a quotient map.

Proof. (1) An open cover of $C \subseteq X$, together with $X \setminus C$, covers X. (2) First let $L = \{y\}$. For each $k \in K$ choose disjoint open $U_k \ni k$ and $V_k \ni y$; finitely many $U_{k_1}, \ldots, U_{k_m}$ cover K, and $U = \bigcup_j U_{k_j}$ and $V = \bigcap_j V_{k_j}$ are disjoint. For general L, do this for each $y \in L$ and use a finite subcover of L by the resulting V's in the same way. (3) Preimages of an open cover of $f(X)$

cover X. (4) Closed subsets are compact by (1), their images are compact by (3), hence closed by (2). A closed continuous bijection has continuous inverse; for an injection apply this to $X \to f(X)$, and for a surjection use Proposition 18.4.10. □

Proposition 18.6.4. *If X and Y are compact, so is $X \times Y$.*

Proof. Let $\mathcal{W}$ be an open cover of $X \times Y$ and fix $y \in Y$. For each $x \in X$ choose $W_x \in \mathcal{W}$ containing (x, y) and a basic open set $U_x \times V_x$ with $(x, y) \in U_x \times V_x \subseteq W_x$. Finitely many $U_{x_1}, \dots, U_{x_m}$ cover X; let $N_y = \bigcap_j V_{x_j}$. Then $X \times N_y$ is covered by $W_{x_1}, \dots, W_{x_m}$ (the *tube lemma*). Finitely many N_y cover Y, and the corresponding finitely many members of $\mathcal{W}$ cover $X \times Y$. □

Theorem 18.6.5 (Heine–Borel)**.** *Every closed bounded interval $[a, b] \subseteq \mathbb{R}$ is compact. A subset of $\mathbb{R}^n$ (or $\mathbb{C}^n = \mathbb{R}^{2n}$) is compact if and only if it is closed and bounded.*

Proof. Let $\mathcal{U}$ be an open cover of $[a, b]$ and S the set of $t \in [a, b]$ such that $[a, t]$ is covered by finitely many members. Then $a \in S$; let $s = \sup S$. Choose $U \in \mathcal{U}$ with $s \in U$ and $\varepsilon > 0$ with $(s - \varepsilon, s + \varepsilon) \cap [a, b] \subseteq U$, and $t \in S$ with $t > s - \varepsilon$. A finite cover of $[a, t]$ together with U covers $[a, t']$ for $t' = \min(s + \varepsilon/2, b)$, so $t' \in S$ and $t' \leq s$. Hence $s = b \in S$.

For the second statement, a compact $K \subseteq \mathbb{R}^n$ is closed by Proposition 18.6.3(2) and bounded, since the balls $B(0, m)$ cover it. A closed bounded set lies in some cube $[-R, R]^n$, which is compact as a finite product of compact spaces by Proposition 18.6.4 (with the product topology, which is the Euclidean one by Proposition 18.4.7(4)); so it is compact by Proposition 18.6.3(1). □

Corollary 18.6.6 (Extreme value theorem)**.** *A continuous function $f : X \to \mathbb{R}$ on a nonempty compact space attains a maximum and a minimum.*

Proof. $f(X)$ is compact, hence closed and bounded, so it contains its supremum and infimum, which lie in its closure. □

Lemma 18.6.7 (Alexander subbasis lemma)**.** *Let $\mathcal{S}$ be a subbasis of the topology of X. If every cover of X by members of $\mathcal{S}$ has a finite subcover, then X is compact.*

Proof. Suppose not. The set of open covers of X without finite subcover is nonempty, and ordered by inclusion every chain in it has an upper bound, its union: a finite subcover of the union would lie in one member of the chain. By Zorn's lemma (Theorem 3.7.2) there is a maximal open cover $\mathcal{C}$ without finite subcover. The members of $\mathcal{C} \cap \mathcal{S}$ do not cover X, since otherwise they would have a finite subcover. Choose x not covered by them and $U \in \mathcal{C}$ with $x \in U$. There are $S_1, \dots, S_k \in \mathcal{S}$ with $x \in S_1 \cap \cdots \cap S_k \subseteq U$, and no S_j belongs to $\mathcal{C}$. By maximality, $\mathcal{C} \cup \{S_j\}$ has a finite subcover, so $X = S_j \cup Y_j$ with Y_j a finite union of members of $\mathcal{C}$. Then $X = (S_1 \cap \cdots \cap S_k) \cup Y_1 \cup \cdots \cup Y_k \subseteq U \cup Y_1 \cup \cdots \cup Y_k$ is covered by finitely many members of $\mathcal{C}$, a contradiction. (If $k = 0$, then $U = X$ is itself a finite subcover.) □

Theorem 18.6.8 (Tychonoff)**.** *Every product of compact spaces is compact.*

Proof. Let $X = \prod_i X_i$. If some X_i is empty, so is X. Otherwise, by Lemma 18.6.7 it suffices to consider a cover $\mathcal{S}$ of X by subbasic sets $p_i^{-1}(U)$. For each i let $\mathcal{U}_i$ be the set of open $U \subseteq X_i$ with $p_i^{-1}(U) \in \mathcal{S}$. If for some i the family $\mathcal{U}_i$ covers X_i, then a finite subcover of $\mathcal{U}_i$ gives a finite subcover of $\mathcal{S}$. Otherwise choose, by the axiom of choice, $x_i \in X_i$ not covered by $\mathcal{U}_i$ for each i; then the point $(x_i)_i$ is not covered by $\mathcal{S}$, a contradiction. □

Remark 18.6.9. Tychonoff's theorem implies the axiom of choice over ZF [Kel50], so the use of choice in its proof cannot be avoided.

Example 18.6.10. The sphere S^n is compact by Theorem 18.6.5, and so is $\mathbb{RP}^n$, a continuous image of S^n. The quotient $[0,4]/\{0,4\}$ is homeomorphic to S^1. Indeed, let $Q = \{x \in \mathbb{R}^2 : \|x\|_\infty = 1\}$ be the boundary of the square. The maps $x \mapsto x/\|x\|_2$ and $x \mapsto x/\|x\|_\infty$ are mutually inverse continuous maps $Q \to S^1$ and $S^1 \to Q$. The map $[0,4] \to Q$ running affinely along the four sides, from $(1,-1)$ to $(1,1)$, $(-1,1)$, $(-1,-1)$ and back, is a continuous surjection identifying only 0 and 4; the induced continuous bijection $[0,4]/\{0,4\} \to Q$ from a compact space to a Hausdorff space is a homeomorphism by Proposition 18.6.3(4).

Counterexample 18.6.11.

1. An infinite set X with the metric $d(x,y) = 1$ for $x \neq y$ is closed and bounded in itself but not compact: its topology is discrete, and the cover by singletons has no finite subcover.

2. In the cofinite topology on $\mathbb{N}$ every subset is compact (any member of a cover contains all but finitely many points), and the set of even numbers is compact but not closed. Hausdorffness in Proposition 18.6.3(2) is needed.

18.7 Complete and compact metric spaces

Definition 18.7.1. A sequence (x_n) in a metric space is *Cauchy* if for every $\varepsilon > 0$ there is N with $d(x_m, x_n) < \varepsilon$ for $m, n \geq N$. The space is *complete* if every Cauchy sequence converges. It is *totally bounded* if for every $\varepsilon > 0$ it is covered by finitely many balls of radius ε. A *Banach space* is a normed space that is complete for the metric of its norm.

Proposition 18.7.2. *1. $\mathbb{R}^n$ and $\mathbb{C}^n$ are complete, and so are all their closed subsets.*

2. A complete subspace of a metric space is closed; a closed subspace of a complete space is complete.

3. For a nonempty set X and a complete metric space (Y,d), the set $B(X,Y)$ of maps with bounded image, with $d_\infty(f,g) = \sup_x d(f(x),g(x))$, is a complete metric space. If X is a space, the subset $C_b(X,Y)$ of continuous maps is closed, hence complete. In particular $C_b(X,\mathbb{K})$ with the norm $\|f\|_\infty = \sup|f|$ is a Banach space.

Proof. (2) If $A \subseteq X$ is complete and $x \in \overline{A}$, a sequence in A converging to x (Proposition 18.3.5) is Cauchy, so it converges in A, and by uniqueness of limits $x \in A$. If X is complete and A closed, a Cauchy sequence in A converges in X, and its limit lies in $\overline{A} = A$.

(1) With the maximum norm, a sequence is Cauchy if and only if each coordinate sequence is, and converges if and only if each coordinate does. Real Cauchy sequences converge by Proposition 4.7.6, and complex ones because real and imaginary parts are Cauchy. The second part follows from (2).

(3) For bounded f, g, $d_\infty(f,g)$ is finite by the triangle inequality, and d_∞ is a metric. Let (f_n) be Cauchy. For each x, $(f_n(x))$ is Cauchy in Y; let $f(x)$ be its limit. Given $\varepsilon > 0$ choose N with $d(f_n(x), f_m(x)) < \varepsilon$ for all x and $m, n \geq N$. Since $|d(a,b) - d(a,c)| \leq d(b,c)$, letting $m \to \infty$ gives $d(f_n(x), f(x)) \leq \varepsilon$ for all x and $n \geq N$. So f is bounded and $f_n \to f$ in d_∞. Uniform limits of continuous maps are continuous (Proposition 18.3.10), so $C_b(X,Y)$ is closed. □

Lemma 18.7.3 (Lebesgue number lemma)**.** *Let X be a metric space in which every sequence has a convergent subsequence, and $\mathcal{U}$ an open cover. There is $\delta > 0$ such that every ball of radius δ is contained in some member of $\mathcal{U}$.*

Proof. Otherwise choose x_n such that $B(x_n, 1/n)$ lies in no member of $\mathcal{U}$, and a subsequence $x_{n_k} \to x$. Let $x \in U \in \mathcal{U}$ and $B(x, 2\varepsilon) \subseteq U$. For k large, $d(x_{n_k}, x) < \varepsilon$ and $1/n_k < \varepsilon$, so $B(x_{n_k}, 1/n_k) \subseteq B(x, 2\varepsilon) \subseteq U$, a contradiction. □

Theorem 18.7.4. *For a metric space X the following are equivalent:*

1. *X is compact;*
2. *every sequence in X has a convergent subsequence (X is* sequentially compact*);*
3. *X is complete and totally bounded.*

Proof. (1)⇒(2): Let (x_n) have no convergent subsequence. Then every $x \in X$ has an open ball containing x_n for only finitely many n: otherwise we could choose $n_1 < n_2 < \cdots$ with $d(x_{n_k}, x) < 1/k$. Finitely many such balls cover X, but together they contain x_n for only finitely many n, a contradiction.

(2)⇒(3): A Cauchy sequence with a convergent subsequence converges to the same limit. If X were not totally bounded, for some $\varepsilon > 0$ we could choose inductively $x_n \notin \bigcup_{m<n} B(x_m, \varepsilon)$; this sequence has no Cauchy subsequence.

(3)⇒(2): Let (x_n) be a sequence. Cover X by finitely many balls of radius 1; one of them contains x_n for infinitely many n. Among these indices, cover by balls of radius $1/2$ and choose again, and so on; this gives nested infinite index sets $N_1 \supseteq N_2 \supseteq \cdots$ such that $d(x_m, x_n) < 2/k$ for $m, n \in N_k$. Choosing $n_k \in N_k$ increasing gives a Cauchy subsequence, which converges.

(2)⇒(1): Let $\mathcal{U}$ be an open cover and δ a Lebesgue number (Lemma 18.7.3). By (2)⇒(3), X is covered by finitely many balls of radius δ, each contained in a member of $\mathcal{U}$. □

Corollary 18.7.5. *A continuous map $f\colon X \to Y$ from a compact metric space to a metric space is* uniformly continuous*: for every $\varepsilon > 0$ there is $\delta > 0$ such that $d(x, x') < \delta$ implies $d(f(x), f(x')) < \varepsilon$.*

Proof. Let δ be a Lebesgue number of the cover by the sets $f^{-1}(B(y, \varepsilon/2))$. If $d(x, x') < \delta$, then $x, x' \in B(x, \delta)$ lie in one such set, so $d(f(x), f(x')) < \varepsilon$. □

Theorem 18.7.6 (Banach fixed point theorem)**.** *Let X be a nonempty complete metric space and $T\colon X \to X$ a* contraction*: $d(Tx, Ty) \le c\, d(x, y)$ for some $0 \le c < 1$. Then T has a unique fixed point, and for every x_0 the sequence $x_{n+1} = Tx_n$ converges to it.*

Proof. By induction $d(x_{n+1}, x_n) \le c^n d(x_1, x_0)$, so for $m > n$, $d(x_m, x_n) \le \sum_{k=n}^{m-1} c^k d(x_1, x_0) = c^n \frac{1-c^{m-n}}{1-c} d(x_1, x_0) \le \frac{c^n}{1-c} d(x_1, x_0)$. Since $c^n \to 0$ (if $c > 0$, write $1/c = 1 + h$ with $h > 0$; then $c^{-n} \ge 1 + nh$), the sequence is Cauchy; let x be its limit. A contraction is continuous, so $Tx = \lim Tx_n = \lim x_{n+1} = x$. If also $Ty = y$, then $d(x, y) \le c\, d(x, y)$, so $x = y$. □

Theorem 18.7.7 (Baire category theorem)**.** *In a complete metric space X, a countable intersection of dense open sets is dense. Equivalently, a countable union of closed sets with empty interior has empty interior.*

Proof. Let $U_1, U_2, \ldots$ be dense open and W nonempty open. Since $W \cap U_1$ is nonempty and open, it contains a ball $B(x_1, 2r_1)$ with $r_1 < 1$, and then the closed ball $\bar{B}(x_1, r_1) = \{y : d(x_1, y) \le r_1\}$. Inductively, $B(x_n, r_n) \cap U_{n+1}$ is nonempty and open, so it contains $\bar{B}(x_{n+1}, r_{n+1})$ with $r_{n+1} < 1/(n+1)$. For $m \ge n$, $x_m \in B(x_n, r_n)$, so (x_n) is Cauchy; its limit x lies in every closed set $\bar{B}(x_n, r_n)$, hence in $W \cap \bigcap_n U_n$. The second statement is the first for the complements. □

Corollary 18.7.8. *A nonempty complete metric space in which no point is isolated (no singleton is open) is uncountable.*

Proof. If $X = \{x_1, x_2, \ldots\}$, then X is the countable union of the closed sets $\{x_n\}$, which have empty interior, contradicting Theorem 18.7.7. □

Counterexample 18.7.9. Completeness is not a topological property: $t \mapsto t/(1-|t|)$ is a homeomorphism $(-1,1) \to \mathbb{R}$ with inverse $s \mapsto s/(1+|s|)$, but $(-1,1)$ is not complete. And $\mathbb{Q}$, which is not complete, is a countable union of closed sets with empty interior, so the Baire category theorem fails for it.

18.8 Separation axioms, Urysohn and Tietze

Definition 18.8.1. A space X is T_1 if its points are closed. A T_1 space is *regular* if a point x and a closed set $C \not\ni x$ have disjoint open neighbourhoods, and *normal* if any two disjoint closed sets have disjoint open neighbourhoods. So normal $\Rightarrow$ regular $\Rightarrow$ Hausdorff $\Rightarrow T_1$.

Lemma 18.8.2. *A T_1 space X is normal if and only if for every closed C and open $U \supseteq C$ there is an open V with $C \subseteq V \subseteq \overline{V} \subseteq U$. It is regular if and only if this holds for C a point.*

Proof. If X is normal, separate C and $X \setminus U$ by disjoint open V and W; then $\overline{V} \subseteq X \setminus W \subseteq U$. Conversely, for disjoint closed C, D choose V for $U = X \setminus D$; then V and $X \setminus \overline{V}$ separate them. The regular case is identical. □

Proposition 18.8.3. *Metric spaces and compact Hausdorff spaces are normal.*

Proof. For a nonempty subset A of a metric space let $d(x,A) = \inf_{a\in A} d(x,a)$. Then $|d(x,A) - d(y,A)| \le d(x,y)$, so $d(-,A)$ is continuous, and $d(x,A) = 0$ if and only if $x \in \overline{A}$. For disjoint nonempty closed A, B, the continuous function (Proposition 18.4.7(5)) $g(x) = d(x,A) - d(x,B)$ is negative on A and positive on B, so $\{g < 0\}$ and $\{g > 0\}$ separate them. For compact Hausdorff spaces use Propositions 18.6.3(1) and (2). □

Theorem 18.8.4 (Urysohn's lemma)**.** *Let A, B be disjoint closed subsets of a normal space X. There is a continuous $f\colon X \to [0,1]$ with $f = 0$ on A and $f = 1$ on B.*

Proof. Let D be the set of dyadic rationals $k/2^n$ in $[0,1]$. We construct open sets U_r, $r \in D$, with $A \subseteq U_0$, $U_1 = X \setminus B$ and $\overline{U_r} \subseteq U_s$ for $r < s$. Choose U_0 with $A \subseteq U_0 \subseteq \overline{U_0} \subseteq U_1$ by Lemma 18.8.2. If U_r is defined for all r with denominator 2^n, then for $r = (2k+1)/2^{n+1}$ choose U_r with $\overline{U_{k/2^n}} \subseteq U_r \subseteq \overline{U_r} \subseteq U_{(k+1)/2^n}$. The required inclusions for $r < s$ with denominators dividing 2^{n+1} follow: if one of them is new, pass through the neighbouring old index, and if both are new, there is an old index strictly between them.

Define $f(x) = \inf(\{r \in D : x \in U_r\} \cup \{1\})$. Then $f = 0$ on A, and $f = 1$ on B since B meets no U_r. If $x \in \overline{U_r}$, then $x \in U_s$ for all $s > r$ in D, so $f(x) \le r$ by density of D. If $x \notin U_r$, then $x \notin U_s$ for $s \le r$, so $f(x) \ge r$. Hence for $a \in \mathbb{R}$,

$$f^{-1}([0,a)) = \bigcup_{r\in D,\, r<a} U_r \quad (a \le 1), \qquad f^{-1}((a,1]) = \bigcup_{r\in D,\, r>a} (X \setminus \overline{U_r}) \quad (a \ge 0).$$

For the second equality, if $f(x) > a$ choose $r \in D$ with $a < r < f(x)$; then $x \notin \overline{U_r}$. These sets are open, and the intervals $[0,a)$, $(a,1]$ form a subbasis of $[0,1]$, so f is continuous. □

Theorem 18.8.5 (Tietze extension theorem)**.** *Let A be a closed subset of a normal space X. Every continuous $f\colon A \to [-1,1]$ extends to a continuous $F\colon X \to [-1,1]$, and every continuous $f\colon A \to \mathbb{R}$ extends to a continuous $F\colon X \to \mathbb{R}$.*

Proof. Step 1. If $g\colon A \to [-c,c]$ is continuous, there is a continuous $h\colon X \to [-c/3, c/3]$ with $|g - h| \le 2c/3$ on A: the sets $g^{-1}([-c,-c/3])$ and $g^{-1}([c/3,c])$ are disjoint and closed in X, and Urysohn's lemma (rescaled) gives h equal to $-c/3$ on the first and $c/3$ on the second; check $|g-h| \le 2c/3$ separately on the two sets and on the rest of A, where $|g| < c/3$.

Step 2. Put $g_0 = f$ and, given $g_n\colon A \to [-(2/3)^n, (2/3)^n]$, let h_n be as in Step 1 and $g_{n+1} = g_n - h_n|_A$. Then $|h_n| \le \frac{1}{3}(\frac{2}{3})^n$, so $F = \sum_n h_n$ converges uniformly and is continuous by Proposition 18.3.10, with $|F| \le \sum_n \frac{1}{3}(\frac{2}{3})^n = 1$. On A, $f - \sum_{n<N} h_n = g_N \to 0$, so $F|_A = f$.

Step 3. For $f\colon A \to \mathbb{R}$, compose with the homeomorphism $\mathbb{R} \to (-1,1)$ of Counterexample 18.7.9 and extend to $F\colon X \to [-1,1]$ by Step 2. The set $E = F^{-1}(\{-1,1\})$ is closed and disjoint from A; by Urysohn's lemma choose $\varphi\colon X \to [0,1]$ with $\varphi = 0$ on E and $\varphi = 1$ on A. Then φF extends f and takes values in $(-1,1)$; compose back. □

Counterexample 18.8.6. Let $K = \{1/n : n \ge 1\}$ and give $\mathbb{R}$ the topology with basis the open intervals (a,b) and the sets $(a,b) \setminus K$ (the conditions of Lemma 18.2.10 are clear). It is finer than the standard topology, hence Hausdorff, and K is closed. Suppose $U \supseteq K$ and $V \ni 0$ are disjoint open sets. Since every interval around 0 meets K, V contains a set $(a,b) \setminus K$ with $a < 0 < b$. Choose n with $1/n < b$. The only basic sets containing $1/n$ are intervals, so $U \supseteq (c,d) \ni 1/n$, and $(c,d) \cap (a,b)$ contains points not in K, which lie in $U \cap V$. So this Hausdorff space is not regular.

Lemma 18.8.7 (Tychonoff)**.** *A regular space in which every open cover has a countable subcover (a* Lindelöf *space), for instance a regular second countable space, is normal.*

Proof. Let A, B be disjoint closed sets. For $a \in A$, regularity and Lemma 18.8.2 give an open $U \ni a$ with $\overline{U} \cap B = \emptyset$. As A is closed in a Lindelöf space, it is Lindelöf, so countably many such sets $U_1, U_2, \ldots$ cover A; similarly open $V_1, V_2, \ldots$ cover B with $\overline{V_k} \cap A = \emptyset$. Put $U_n' = U_n \setminus \bigcup_{k\le n} \overline{V_k}$ and $V_n' = V_n \setminus \bigcup_{k\le n} \overline{U_k}$, which are open, and $U' = \bigcup_n U_n'$, $V' = \bigcup_n V_n'$. Then $A \subseteq U'$, since a point of A lies in some U_n and in no $\overline{V_k}$; similarly $B \subseteq V'$. If $x \in U_n' \cap V_m'$ with $n \le m$, then $x \in U_n$ but $x \notin \overline{U_n}$, a contradiction, and the case $n > m$ is symmetric. So $U' \cap V' = \emptyset$. Second countable spaces are Lindelöf by Proposition 18.3.12. □

Theorem 18.8.8 (Urysohn metrisation theorem)**.** *Every regular second countable space is metrisable: its topology is induced by a metric.*

Proof. Let X be regular with a countable basis (B_n); by Lemma 18.8.7 it is normal. For each pair (n,m) with $\overline{B_n} \subseteq B_m$ choose, by Urysohn's lemma (Theorem 18.8.4), a continuous $f_{n,m}\colon X \to [0,1]$ equal to 1 on $\overline{B_n}$ and to 0 outside B_m. Enumerate these functions as $f_1, f_2, \ldots$. They separate points from closed sets: if $x \in U$ with U open, choose m with $x \in B_m \subseteq U$, and by regularity and Lemma 18.8.2 an open V with $x \in V \subseteq \overline{V} \subseteq B_m$, and then n with $x \in B_n \subseteq V$; the corresponding f_k has $f_k(x) = 1$ and vanishes outside U.

Define $d(x,y) = \sum_k 2^{-k}|f_k(x) - f_k(y)|$. It is symmetric and satisfies the triangle inequality, and $d(x,y) > 0$ for $x \ne y$, because X is T_1, so $X \setminus \{y\}$ is an open neighbourhood of x, and some f_k equals 1 at x and vanishes outside it, so $f_k(x) = 1 \ne 0 = f_k(y)$. Each $d(x,-)$ is continuous, being a uniformly convergent series of continuous functions (Proposition 18.3.10), so d-balls are open. Conversely, if $x \in U$ with U open and f_k is as above, then $d(x,y) < 2^{-k}$ implies $|f_k(y) - 1| < 1$, so $f_k(y) > 0$ and $y \in U$: the d-ball of radius 2^{-k} about x lies in U. So d induces the topology of X. □

Remark 18.8.9. Topological manifolds are regular (indeed locally compact Hausdorff) and second countable, hence metrisable (Chapter 20). Regularity cannot be weakened to the Hausdorff property: the space of Counterexample 18.8.6 is second countable (a basis is given by the intervals with rational endpoints and their differences with K) and Hausdorff, but not regular, hence not metrisable, since metric spaces are normal (Proposition 18.8.3).

18.9 Local compactness and proper maps

Definition 18.9.1. A space is *locally compact* if every point has a compact neighbourhood. A continuous map $f\colon X \to Y$ is *proper* if $f^{-1}(K)$ is compact for every compact $K \subseteq Y$. The *support* of a function $f\colon X \to \mathbb{K}$ is $\operatorname{supp} f = \overline{\{f \neq 0\}}$, and $C_c(X)$ denotes the continuous functions with compact support.

Definition 18.9.2. Let X be a Hausdorff space. The *one-point compactification* is $X^+ = X \sqcup \{\infty\}$, whose open sets are the open subsets of X and the sets $(X \setminus K) \cup \{\infty\}$ with $K \subseteq X$ compact.

Theorem 18.9.3. *Let X be Hausdorff. Then X^+ is a compact space containing X as an open subspace, and X^+ is Hausdorff if and only if X is locally compact.*

Proof. Compact subsets of X are closed (Proposition 18.6.3), so the sets $X \setminus K$ are open. The open sets form a topology: $\bigcup_i ((X \setminus K_i) \cup \{\infty\}) = (X \setminus \bigcap_i K_i) \cup \{\infty\}$ and $U \cup (X \setminus K) \cup \{\infty\} = (X \setminus (K \setminus U)) \cup \{\infty\}$, where $\bigcap_i K_i$ and $K \setminus U$ are closed subsets of a compact set, hence compact; and $U \cap ((X \setminus K) \cup \{\infty\}) = U \cap (X \setminus K)$ and $((X \setminus K) \cup \{\infty\}) \cap ((X \setminus L) \cup \{\infty\}) = (X \setminus (K \cup L)) \cup \{\infty\}$. The traces on X of the open sets are open in X, so X is an open subspace. Given an open cover of X^+, a member $(X \setminus K) \cup \{\infty\}$ contains ∞, and finitely many members cover K.

If X is locally compact, points of X are separated in X, and $x \in X$ with a compact neighbourhood K is separated from ∞ by $\operatorname{int} K$ and $(X \setminus K) \cup \{\infty\}$. Conversely, if x and ∞ have disjoint neighbourhoods U and $(X \setminus K) \cup \{\infty\}$, then $U \subseteq K$, so K is a compact neighbourhood of x. □

Example 18.9.4. $(\mathbb{R}^n)^+ \cong S^n$. Let $N = (0, \dots, 0, 1)$ and define $\tau\colon \mathbb{R}^n \to S^n$ by $\tau(y) = (2y, \|y\|^2 - 1)/(\|y\|^2 + 1)$. A direct computation shows $\|\tau(y)\| = 1$, that τ is a bijection onto $S^n \setminus \{N\}$ with inverse the *stereographic projection* $x \mapsto (x_1, \dots, x_n)/(1 - x_{n+1})$, and that $\|\tau(y) - N\| = 2/\sqrt{\|y\|^2 + 1}$. Extend τ by $\tau(\infty) = N$. It is continuous at ∞: for $\varepsilon > 0$, τ maps the complement of a closed ball of large radius R, which is a compact set, into $B(N, \varepsilon)$. So $\tau\colon (\mathbb{R}^n)^+ \to S^n$ is a continuous bijection from a compact space to a Hausdorff space, hence a homeomorphism.

Lemma 18.9.5. *Let X be locally compact Hausdorff, $K \subseteq X$ compact and $U \supseteq K$ open. There is an open V with compact closure such that $K \subseteq V \subseteq \overline{V} \subseteq U$, and there is $f \in C_c(X)$ with values in $[0, 1]$, $f = 1$ on K and $\operatorname{supp} f \subseteq U$.*

Proof. By Theorem 18.9.3 and Proposition 18.8.3, X^+ is normal. The set K is closed in X^+, $X^+ \setminus U$ is closed, and they are disjoint. By Lemma 18.8.2 there is an open V with $K \subseteq V$ and $\overline{V} \subseteq U$ (closure in X^+); then $\overline{V}$ is a closed subset of X^+ not containing ∞, so it is compact and equals the closure of V in X. Urysohn's lemma (Theorem 18.8.4) in X^+ gives a continuous $f\colon X^+ \to [0, 1]$ equal to 1 on K and 0 on $X^+ \setminus V$. Its restriction to X has support contained in the compact set $\overline{V} \subseteq U$, so the support is compact. □

Proposition 18.9.6. *Let $f\colon X \to Y$ be a continuous map to a locally compact Hausdorff space. If f is proper, then f is closed. A continuous map from a compact space to a Hausdorff space is proper.*

Proof. Let $C \subseteq X$ be closed and $y \in \overline{f(C)}$. Choose a compact neighbourhood K of y. Then $C \cap f^{-1}(K)$ is compact, so $f(C \cap f^{-1}(K)) = f(C) \cap K$ is compact, hence closed. Every open $W \ni y$ gives an open set $W \cap \operatorname{int} K \ni y$, which meets $f(C)$; so $y \in \overline{f(C) \cap K} = f(C) \cap K$. For the last statement, compact subsets of a Hausdorff space are closed, and closed subsets of compact spaces are compact. □

Counterexample 18.9.7. $\mathbb{Q}$ is not locally compact. A compact neighbourhood K of 0 in $\mathbb{Q}$ would contain $[-\varepsilon, \varepsilon] \cap \mathbb{Q}$ for some $\varepsilon > 0$, which is closed in K and hence compact. But choosing an irrational $\alpha \in (0, \varepsilon)$, the open sets $(\mathbb{Q} \cap (-\infty, \alpha - 1/n)) \cup (\mathbb{Q} \cap (\alpha + 1/n, \infty))$, $n \geq 1$, form an increasing family that covers $[-\varepsilon, \varepsilon] \cap \mathbb{Q}$ without finite subcover.

18.10 Paracompactness and partitions of unity

Definition 18.10.1. A family $(A_i)_{i \in I}$ of subsets of X is *locally finite* if every point has a neighbourhood meeting A_i for only finitely many i. A *refinement* of a cover (U_i) is a cover (V_j) such that each V_j is contained in some U_i. A space is *paracompact* if every open cover has a locally finite open refinement.

Lemma 18.10.2. *If (A_i) is locally finite, then so is $(\overline{A_i})$, and $\overline{\bigcup_i A_i} = \bigcup_i \overline{A_i}$. In particular a union of a locally finite family of closed sets is closed. If (f_i) are continuous functions $X \to \mathbb{K}$ whose supports form a locally finite family, then $\sum_i f_i$ is a well-defined continuous function.*

Proof. An open set meets $\overline{A_i}$ if and only if it meets A_i. Let $x \in \overline{\bigcup_i A_i}$ and let W be an open neighbourhood of x meeting only $A_{i_1}, \ldots, A_{i_k}$. Every open $W' \ni x$ satisfies $W' \cap W \cap \bigcup_i A_i \neq \emptyset$, so W' meets $A_{i_1} \cup \cdots \cup A_{i_k}$; hence $x \in \overline{A_{i_1} \cup \cdots \cup A_{i_k}} = \overline{A_{i_1}} \cup \cdots \cup \overline{A_{i_k}}$ by Lemma 18.2.4. For the sum, each point has an open neighbourhood on which all but finitely many f_i vanish, and on it the sum is a finite sum of continuous functions (Proposition 18.4.7(5)); conclude by the pasting lemma. □

Proposition 18.10.3. *A paracompact Hausdorff space is normal.*

Proof. Regularity. Let C be closed and $x \notin C$. For $c \in C$ choose disjoint open $U_c \ni c$ and $W_c \ni x$. Let $\mathcal{V}$ be a locally finite open refinement of the cover $\{U_c\} \cup \{X \setminus C\}$, and $\mathcal{V}'$ the members meeting C; each is contained in some U_c, so its closure is contained in $X \setminus W_c$ and does not contain x. Then $V = \bigcup \mathcal{V}'$ is an open set containing C with $\overline{V} = \bigcup_{V' \in \mathcal{V}'} \overline{V'} \not\ni x$ by Lemma 18.10.2. So V and $X \setminus \overline{V}$ separate C and x.

Normality. The same argument, with x replaced by a closed set D disjoint from C and the sets U_c, W_c provided by regularity, separates C and D. □

Lemma 18.10.4 (Shrinking lemma)**.** *Let X be normal and $(U_i)_{i \in I}$ an open cover that is* point-finite *(each point lies in finitely many U_i), for instance locally finite. There is an open cover $(V_i)_{i \in I}$ with $\overline{V_i} \subseteq U_i$ for all i.*

Proof. Consider pairs $(J, (V_j)_{j \in J})$ with $J \subseteq I$, V_j open and $\overline{V_j} \subseteq U_j$, such that $(V_j)_{j \in J} \cup (U_i)_{i \notin J}$ covers X, ordered by extension. The pair $(\emptyset, ())$ is one. Given a chain, let J^* be the union of the index sets, with the V_j defined compatibly. Let $x \in X$ and F the finite nonempty set of i with $x \in U_i$. If $F \not\subseteq J^*$, x is covered by some U_i with $i \notin J^*$. Otherwise $F \subseteq J$ for some member (J, V) of the chain; there x is covered by some V_j, $j \in J$, since $x \in U_i$ implies $i \in F \subseteq J$. So the union is an upper bound, and Zorn's lemma gives a maximal pair (J, V). If $i \notin J$, the set $C = X \setminus (\bigcup_{j \in J} V_j \cup \bigcup_{k \notin J, k \neq i} U_k)$ is closed and contained in U_i; by Lemma 18.8.2 choose open V_i with $C \subseteq V_i \subseteq \overline{V_i} \subseteq U_i$, contradicting maximality. So $J = I$. □

Definition 18.10.5. A *partition of unity* on X is a family of continuous functions $\psi_i \colon X \to [0, 1]$ whose supports form a locally finite family and with $\sum_i \psi_i = 1$. It is *subordinate* to an open cover $(U_i)_{i \in I}$ (with the same index set) if $\operatorname{supp} \psi_i \subseteq U_i$ for all i.

Theorem 18.10.6. *Let X be paracompact Hausdorff. Every open cover $(U_i)_{i \in I}$ has a subordinate partition of unity.*

Proof. Let $(W_k)_{k\in K}$ be a locally finite open refinement, with $W_k \subseteq U_{\varphi(k)}$ for a chosen map $\varphi\colon K \to I$. The space is normal (Proposition 18.10.3), so by Lemma 18.10.4 there is an open cover (V_k) with $\overline{V_k} \subseteq W_k$, and by Lemma 18.8.2 open V_k' with $\overline{V_k} \subseteq V_k' \subseteq \overline{V_k'} \subseteq W_k$. By Urysohn's lemma choose $f_k\colon X \to [0,1]$ equal to 1 on $\overline{V_k}$ and 0 outside V_k', so $\operatorname{supp} f_k \subseteq \overline{V_k'} \subseteq W_k$. By Lemma 18.10.2, $f = \sum_k f_k$ is continuous, and $f \geq 1$ because the V_k cover. The functions $g_k = f_k / f$ are continuous with supports in W_k and sum 1. Finally let $\psi_i = \sum_{\varphi(k)=i} g_k$. Its support is contained in $\bigcup_{\varphi(k)=i} \overline{V_k'}$, which is closed by Lemma 18.10.2 and contained in U_i; the supports of the ψ_i are locally finite because each point has a neighbourhood meeting only finitely many W_k; and $\sum_i \psi_i = \sum_k g_k = 1$. □

Proposition 18.10.7. *Let X be locally compact Hausdorff and second countable. There are compact sets $K_1 \subseteq K_2 \subseteq \cdots$ with $K_n \subseteq \operatorname{int} K_{n+1}$ and $\bigcup_n K_n = X$ (an* exhaustion by compact sets*), and X is paracompact.*

Proof. Let $\mathcal{B}$ be a countable basis, and $\mathcal{B}'$ the members with compact closure. $\mathcal{B}'$ covers X: if x has a compact neighbourhood K and $x \in B \subseteq \operatorname{int} K$ with $B \in \mathcal{B}$, then $\overline{B} \subseteq K$ is compact. Enumerate $\mathcal{B}' = \{B_1, B_2, \ldots\}$ (with repetitions if finite; if $X = \emptyset$ there is nothing to prove). Let $K_1 = \overline{B_1}$, and given K_n choose $m_n \geq n+1$ with $K_n \subseteq B_1 \cup \cdots \cup B_{m_n}$ and let $K_{n+1} = \overline{B_1} \cup \cdots \cup \overline{B_{m_n}}$. Then $K_n \subseteq \operatorname{int} K_{n+1}$ and $B_n \subseteq K_{n+1}$.

Put $K_0 = K_{-1} = \emptyset$ and let $\mathcal{U}$ be an open cover. For $n \geq 1$ the compact set $L_n = K_n \setminus \operatorname{int} K_{n-1}$ is contained in the open set $O_n = \operatorname{int} K_{n+1} \setminus K_{n-2}$. Cover L_n by finitely many sets $U \cap O_n$ with $U \in \mathcal{U}$, and let $\mathcal{V}$ be the collection of all these sets for all n. It is an open refinement of $\mathcal{U}$, and it covers X because a point lies in L_n for the least n with $x \in K_n$. A point x lies in $\operatorname{int} K_m$ for some m, and the sets of $\mathcal{V}$ coming from L_n lie in $X \setminus K_{n-2}$, so they meet $\operatorname{int} K_m$ only if $n \leq m+1$. Hence $\mathcal{V}$ is locally finite. □

Theorem 18.10.8 (Stone)**.** *Every metric space is paracompact.*

Proof. This is [Sto48]; we give the proof of [Rud69]. Let $(U_s)_{s\in S}$ be an open cover and well-order S (Theorem 3.7.2). For $n \geq 1$ define inductively $V_{s,n}$ as the union of the balls $B(x, 2^{-n})$ over all x such that: (i) s is the least index with $x \in U_s$; (ii) $x \notin V_{t,j}$ for all t and all $j < n$; (iii) $B(x, 3\cdot 2^{-n}) \subseteq U_s$. Each $V_{s,n}$ is open and contained in U_s by (iii).

Cover. For $x \in X$ let s be least with $x \in U_s$ and n with $B(x, 3\cdot 2^{-n}) \subseteq U_s$. Either $x \in V_{t,j}$ for some $j < n$, or x satisfies (i)–(iii) and $x \in V_{s,n}$.

Local finiteness. Let $x \in X$, let s be least such that $x \in V_{s,n}$ for some n, fix such n, and choose $j \geq 1$ with $B(x, 2^{-j}) \subseteq V_{s,n}$. We claim that $B(x, 2^{-n-j})$ meets only finitely many $V_{t,i}$. If $i \geq n+j$, a ball $B(y, 2^{-i})$ in $V_{t,i}$ has $y \notin V_{s,n}$ by (ii) (as $n < i$), so $d(x,y) \geq 2^{-j}$; since $2^{-i} + 2^{-n-j} \leq 2^{-j}$, it does not meet $B(x, 2^{-n-j})$. If $i < n+j$, then $B(x, 2^{-n-j})$ meets $V_{t,i}$ for at most one t: suppose it contains $p \in B(y, 2^{-i}) \subseteq V_{t,i}$ and $q \in B(z, 2^{-i}) \subseteq V_{t',i}$ with $t < t'$ and centres y, z as in the definition. By (iii) $B(y, 3\cdot 2^{-i}) \subseteq U_t$, and by (i) $z \notin U_t$, so $d(y,z) \geq 3\cdot 2^{-i}$. But $d(y,z) < 2^{-i} + 2\cdot 2^{-n-j} + 2^{-i} \leq 3\cdot 2^{-i}$, since $2^{-n-j} \leq 2^{-i-1}$. So only finitely many pairs (t,i) occur. □

Remark 18.10.9. Topological manifolds (Hausdorff, second countable and locally homeomorphic to $\mathbb{R}^n$) are locally compact, hence paracompact by Proposition 18.10.7, and admit partitions of unity; smooth partitions of unity are constructed in Chapter 33. Without second countability this fails: the long line is a Hausdorff space locally homeomorphic to $\mathbb{R}$ that is not paracompact [Eng89]. Paracompactness is also the natural hypothesis under which Čech cohomology computes sheaf cohomology (Chapter 26).

18.11 Exercises

Exercise 18.11.1. Show that X is Hausdorff if and only if the diagonal $\Delta = \{(x, x)\}$ is closed in $X \times X$. Deduce that if $f, g\colon Z \to X$ are continuous and X is Hausdorff, then $\{z : f(z) = g(z)\}$ is closed, and that the graph of a continuous map to a Hausdorff space is closed.

Solution. $(x, y) \notin \Delta$ has a basic neighbourhood $U \times V$ disjoint from Δ if and only if $U \cap V = \emptyset$. So $X \times X \setminus \Delta$ is open if and only if distinct points have disjoint neighbourhoods. The set $\{f = g\}$ is the preimage of Δ under the continuous map $(f, g)\colon Z \to X \times X$, and the graph of $f\colon Y \to X$ is $\{(y, x) : f(y) = x\}$, the equaliser of $f \circ p_1$ and p_2. □

Exercise 18.11.2. Show that $\overline{\prod_i A_i} = \prod_i \overline{A_i}$ in a product of spaces.

Solution. A point x lies in the closure of $\prod_i A_i$ if and only if every basic open set $\prod_i U_i \ni x$ meets $\prod_i A_i$, that is, $U_i \cap A_i \neq \emptyset$ for the finitely many i with $U_i \neq X_i$. This holds for all such sets if and only if $x_i \in \overline{A_i}$ for every i. □

Exercise 18.11.3. Show that $\mathbb{RP}^n$ is a compact Hausdorff space, and that the projection $S^n \to \mathbb{RP}^n$ is open.

Solution. If $U \subseteq S^n$ is open, $q^{-1}(q(U)) = U \cup (-U)$ is open, so q is open. Given $[x] \neq [y]$, we have $x \neq \pm y$; choose $\varepsilon < \min(\|x - y\|, \|x + y\|)/2$. The open sets $U = (B(x, \varepsilon) \cup B(-x, \varepsilon)) \cap S^n$ and $V = (B(y, \varepsilon) \cup B(-y, \varepsilon)) \cap S^n$ are invariant under $x \mapsto -x$ and disjoint, so $q(U)$ and $q(V)$ are disjoint open neighbourhoods. Compactness was shown in Example 18.6.10. □

Exercise 18.11.4. Prove the Baire category theorem for locally compact Hausdorff spaces.

Solution. Let $U_1, U_2, \ldots$ be dense open and W nonempty open. By Lemma 18.9.5, choose nonempty open sets V_n with compact closure and $\overline{V_1} \subseteq W \cap U_1$, $\overline{V_{n+1}} \subseteq V_n \cap U_{n+1}$. The compact sets $\overline{V_n}$ are nested and nonempty, so their intersection is nonempty: otherwise the open sets $\overline{V_1} \setminus \overline{V_n}$ would cover $\overline{V_1}$ without finite subcover. A point of the intersection lies in $W \cap \bigcap_n U_n$. □

Exercise 18.11.5. Let X be a metric space. Show that $x \mapsto d(x, -) - d(x_0, -)$ is an isometric embedding of X into the Banach space $C_b(X, \mathbb{R})$, for a fixed $x_0 \in X$, and deduce that X has a *completion*: a complete metric space containing X as a dense subspace.

Solution. The function $\phi_x = d(x, -) - d(x_0, -)$ is bounded by $d(x, x_0)$ and continuous. By the triangle inequality $|\phi_x(z) - \phi_y(z)| \leq d(x, y)$ with equality at $z = y$, so $\|\phi_x - \phi_y\|_\infty = d(x, y)$. The closure of the image in the complete space $C_b(X, \mathbb{R})$ (Proposition 18.7.2) is complete and contains X as a dense subspace. □

Exercise 18.11.6. Show that the one-point compactification of $\mathbb{N}$ (discrete) is homeomorphic to $\{0\} \cup \{1/n : n \geq 1\} \subseteq \mathbb{R}$, and that $(0, 1)^+$ is homeomorphic to S^1.

Exercise 18.11.7. Let $f\colon X \to Y$ be a continuous map between locally compact Hausdorff spaces. Show that f is proper if and only if its extension $f^+\colon X^+ \to Y^+$ with $f^+(\infty) = \infty$ is continuous.

Exercise 18.11.8. Let $q\colon X \to Y$ be a quotient map and K a compact Hausdorff space. Show that $q \times \mathrm{id}_K\colon X \times K \to Y \times K$ is a quotient map. (Hint: use the tube lemma from the proof of Proposition 18.6.4.)

Exercise 18.11.9. Show that $\operatorname{int}(A \cap B) = \operatorname{int} A \cap \operatorname{int} B$ but only $\operatorname{int}(A \cup B) \supseteq \operatorname{int} A \cup \operatorname{int} B$, with an example where the inclusion is strict.

Solution. $\operatorname{int} A \cap \operatorname{int} B$ is open and contained in $A \cap B$, hence in $\operatorname{int}(A \cap B)$; conversely $\operatorname{int}(A \cap B)$ is an open subset of A and of B. For unions, $\operatorname{int} A \cup \operatorname{int} B$ is an open subset of $A \cup B$. For $A = \mathbb{Q}$ and $B = \mathbb{R} \setminus \mathbb{Q}$ in $\mathbb{R}$, both interiors are empty while $\operatorname{int}(A \cup B) = \mathbb{R}$. □

Exercise 18.11.10. Show that $\partial A = \emptyset$ if and only if A is both open and closed, and deduce that a space is connected (Section 18.5) if and only if every subset other than $\emptyset$ and X has nonempty boundary.

Solution. $\partial A = \overline{A} \setminus \operatorname{int} A$ is empty if and only if $\overline{A} = \operatorname{int} A$, and since $\operatorname{int} A \subseteq A \subseteq \overline{A}$ this means $\operatorname{int} A = A = \overline{A}$, that is, A is open and closed. A space is connected exactly when its only clopen subsets are $\emptyset$ and X. □

Exercise 18.11.11. Let $f : X \to Y$ be continuous and surjective and $A \subseteq X$ dense. Show that $f(A)$ is dense in Y. Show also that two continuous maps $f, g : X \to Y$ to a Hausdorff space that agree on a dense subset are equal; give an example showing that the Hausdorff condition is needed.

Solution. $Y = f(X) = f(\overline{A}) \subseteq \overline{f(A)}$ by Proposition 18.2.13(3). The set $\{x : f(x) = g(x)\}$ is the preimage of the diagonal under (f, g), which is closed as Y is Hausdorff (Exercise 18.11.1); it contains the dense set, hence is everything. For Y the Sierpiński space $\{0, 1\}$ and $X = [0, 1]$, the constant map 1 and the map that is 0 at 0 and 1 elsewhere are continuous (the preimage of the open set $\{1\}$ is X, respectively $(0, 1]$) and agree on the dense set $(0, 1]$. □

Exercise 18.11.12. Let A be a nonempty subset of a metric space and $d(x, A) = \inf_{a \in A} d(x, a)$. Show that $|d(x, A) - d(y, A)| \leq d(x, y)$, so that $x \mapsto d(x, A)$ is continuous, and that $\overline{A} = \{x : d(x, A) = 0\}$. Deduce that every closed set in a metric space is the zero set of a continuous function, and that every open set is a countable union of closed sets.

Solution. For $a \in A$, $d(x, A) \leq d(x, a) \leq d(x, y) + d(y, a)$; taking the infimum over a gives $d(x, A) \leq d(x, y) + d(y, A)$, and symmetrically. If $d(x, A) = 0$ there are $a_n \in A$ with $d(x, a_n) \to 0$, so $x \in \overline{A}$; conversely if $d(x, A) = r > 0$, the ball $B(x, r)$ misses A. For C closed, $C = \{d(\cdot, C) = 0\}$, and an open U is the union of the closed sets $\{x : d(x, X \setminus U) \geq \frac{1}{n}\}$ (if $U \neq X$). □

Exercise 18.11.13. Let $S = \{0, 1\}$ be the Sierpiński space, with $\{1\}$ open. Show that continuous maps $X \to S$ correspond bijectively to open subsets of X, via $f \mapsto f^{-1}(1)$; and that continuous maps $X \to \{0, 1\}$ with the discrete topology correspond to subsets of X that are open and closed.

Solution. A map $f : X \to S$ is continuous if and only if the preimages of the three open sets $\emptyset$, $\{1\}$, S are open, that is, if and only if $f^{-1}(1)$ is open; and every subset U is $f^{-1}(1)$ for exactly one map, $f = \mathbf{1}_U$. For the discrete topology also $f^{-1}(0) = X \setminus f^{-1}(1)$ must be open. □

Chapter 19

Homotopy and Covering Spaces

19.1 Introduction

Homotopy theory studies spaces up to continuous deformation. Its first invariant is the fundamental group, which we compute for the circle using covering spaces (Theorem 19.4.6) and for spaces glued from simpler pieces using the Seifert–van Kampen theorem (Theorem 19.5.1). Covering spaces of a reasonable space are classified by subgroups of its fundamental group (Theorem 19.6.6), in close analogy with Galois theory. We then introduce CW complexes and the higher homotopy groups. Our main references are [Hat02] and [Spa66].

Prerequisites. Chapter 18 (General Topology) and Chapter 7 (Groups).

Throughout, $I = [0,1]$, $D^n = \{x \in \mathbb{R}^n : \|x\|_2 \le 1\}$ is the closed unit ball and $S^{n-1} = \partial D^n$ the unit sphere. A *map* of spaces means a continuous map.

19.2 Homotopy

Definition 19.2.1. A *homotopy* between maps $f, g\colon X \to Y$ is a map $H\colon X \times I \to Y$ with $H(x,0) = f(x)$ and $H(x,1) = g(x)$; we write $f \simeq g$ and $H_t = H(-,t)$. It is a homotopy *relative to* $A \subseteq X$ if $H_t(a) = f(a)$ for all $a \in A$ and $t \in I$. A map $f\colon X \to Y$ is a *homotopy equivalence* if there is $g\colon Y \to X$ with $gf \simeq \mathrm{id}_X$ and $fg \simeq \mathrm{id}_Y$; then X and Y are *homotopy equivalent*, $X \simeq Y$. A space is *contractible* if it is homotopy equivalent to a point. A subspace $A \subseteq X$ is a *retract* if there is a map $r\colon X \to A$ with $r|_A = \mathrm{id}_A$, and a *deformation retract* if moreover $ir \simeq \mathrm{id}_X$ relative to A, where $i\colon A \to X$ is the inclusion.

Lemma 19.2.2. *Homotopy (relative to A) is an equivalence relation on maps $X \to Y$, compatible with composition: if $f \simeq f'$ and $g \simeq g'$, then $gf \simeq g'f'$. Hence spaces and homotopy classes of maps form a category* hTop*, and homotopy equivalences are its isomorphisms.*

Proof. Reflexivity: $H(x,t) = f(x)$. Symmetry: $H(x,1-t)$. Transitivity: if H goes from f to g and K from g to h, the map equal to $H(x,2t)$ for $t \le 1/2$ and $K(x,2t-1)$ for $t \ge 1/2$ is continuous by the pasting lemma (Lemma 18.4.5). If H goes from f to f' and K from g to g', then $(x,t) \mapsto K(H(x,t),t)$ goes from gf to $g'f'$; it is continuous as the composite of $(x,t) \mapsto (H(x,t),t)$ and K. Relative homotopies are preserved by these constructions. □

Example 19.2.3. 1. A convex subset $C \subseteq \mathbb{R}^n$ is contractible: for $c \in C$, the *straight-line homotopy* $H(x,t) = (1-t)x + tc$ goes from id_C to the constant map, and it is continuous by Proposition 18.4.7(5). More generally, any two maps $f, g\colon X \to C$ are homotopic via $(1-t)f + tg$, relative to the set where they agree.

2. S^{n-1} is a deformation retract of $\mathbb{R}^n \setminus \{0\}$ and of $D^n \setminus \{0\}$, via $H(x,t) = (1-t)x + tx/\|x\|$, which never vanishes.

Lemma 19.2.4. *If $q\colon X \to Y$ is a quotient map and K is a compact Hausdorff space, then $q \times \mathrm{id}_K\colon X \times K \to Y \times K$ is a quotient map. If $U \subseteq Y$ is open, then $q^{-1}(U) \to U$ is a quotient map.*

Proof. Let $W \subseteq Y \times K$ with $W' = (q \times \mathrm{id})^{-1}(W)$ open, and let $(y_0, k_0) \in W$ with $y_0 = q(x_0)$. The set $\{k : (x_0, k) \in W'\}$ is open and contains k_0; since K is regular (Proposition 18.8.3), there is an open $N \ni k_0$ whose closure $\overline{N}$ lies in it. Let $U' = \{x \in X : \{x\} \times \overline{N} \subseteq W'\}$, so $x_0 \in U'$. The set U' is open: for $x \in U'$, each (x,k) with $k \in \overline{N}$ has a basic neighbourhood $M_k \times O_k \subseteq W'$, finitely many O_k cover the compact set $\overline{N}$, and the intersection M of the corresponding M_k satisfies $M \times \overline{N} \subseteq W'$. It is saturated: if $q(x) = q(x')$ and $x \in U'$, then (x',k) and (x,k) have the same image, and W' is a union of fibres of $q \times \mathrm{id}$. So $q(U')$ is open, and $(y_0, k_0) \in q(U') \times N \subseteq W$. Hence W is open. For the second statement, if $V \subseteq U$ and $q^{-1}(V)$ is open in $q^{-1}(U)$, then it is open in X, so V is open in Y. □

Remark 19.2.5. A *deformation* of a space X is a homotopy $H\colon X \times I \to X$ starting at $H_0 = \mathrm{id}_X$: we watch X move continuously inside itself. By Definition 19.2.1, $A \subseteq X$ is a deformation retract if and only if there is a deformation H of X that ends in A ($H_1(X) \subseteq A$) and keeps every point of A fixed at all times ($H_t(a) = a$ for $a \in A$ and $t \in I$); then $r = H_1$, viewed as a map $X \to A$, is a retraction, and H is called a *deformation retraction*. Terminology varies: what we call a deformation retract is also called a *strong* deformation retract, and some authors call A a deformation retract as soon as there is a retraction r with $ir \simeq \mathrm{id}_X$, the homotopy not being required to fix A; we call this a *weak* deformation retract. Counterexample 19.2.9 shows that the two notions differ.

Lemma 19.2.6. 1. *If A is a deformation retract of X with retraction r, then the inclusion $i\colon A \to X$ and r are mutually inverse homotopy equivalences.*

2. *(Transitivity) If $A \subseteq B \subseteq X$, B is a deformation retract of X and A is a deformation retract of B, then A is a deformation retract of X.*

3. *(Products) If A is a deformation retract of X and B one of Y, then $A \times B$ is a deformation retract of $X \times Y$.*

Proof. (1) $ri = \mathrm{id}_A$ and $ir \simeq \mathrm{id}_X$. (2) Let H be a deformation retraction of X onto B with retraction r_B, and K one of B onto A with retraction r_A. Let $F(x,t) = H(x,2t)$ for $t \le 1/2$ and $F(x,t) = K(r_B(x), 2t-1)$ for $t \ge 1/2$; the two formulas agree at $t = 1/2$, where both give $r_B(x)$, so F is continuous by the pasting lemma (Lemma 18.4.5). It starts at id_X, ends at $r_A r_B(x) \in A$, and fixes every $a \in A$ at all times, since H fixes $B \supseteq A$, $r_B(a) = a$ and K fixes A. (3) If H and K are deformation retractions, so is $((x,y),t) \mapsto (H(x,t), K(y,t))$. □

Definition 19.2.7. The *mapping cylinder* of a map $f\colon X \to Y$ is the quotient $M_f = ((X \times I) \sqcup Y)/\!\sim$, where $(x,1) \sim f(x)$ for $x \in X$. We write $[x,s]$ for the class of (x,s) and $[y]$ for that of y, and $j\colon X \to M_f$, $j(x) = [x,0]$.

Example 19.2.8. 1. The cylinder $X \times I$ deformation retracts onto $X \times \{0\}$, by $H((x,s),t) = (x,(1-t)s)$.

2. *The Möbius band onto its core circle.* Let $M = ([0,1] \times [-1,1])/\!\sim$ with $(0,y) \sim (1,-y)$, and let C be the image of $[0,1] \times \{0\}$, a circle. The homotopy $([x,y],t) \mapsto [x,(1-t)y]$ is well defined, because $(0,(1-t)y) \sim (1,-(1-t)y)$, and continuous, because the product of the quotient map with id_I is a quotient map (Lemma 19.2.4). It fixes C and ends in C, so C is a deformation retract of M and $M \simeq S^1$.

3. *Every map is an inclusion up to homotopy.* Let $f\colon X \to Y$ be a map. The maps $i\colon Y \to M_f$, $y \mapsto [y]$, and $r\colon M_f \to Y$, $[x,s] \mapsto f(x)$, $[y] \mapsto y$, are well defined and continuous, and $ri = \mathrm{id}_Y$, so i is a homeomorphism onto its image, which we identify with Y. The homotopy $H([x,s],t) = [x, s + t(1-s)]$, $H([y],t) = [y]$ is well defined (at $s = 1$ both formulas give $[f(x)]$), continuous by Lemma 19.2.4, starts at the identity, fixes Y and ends at ir. So Y is a deformation retract of M_f, r is a homotopy equivalence by Lemma 19.2.6, and $f = r \circ j$: every map factors as a map j into a larger space followed by a homotopy equivalence.

4. *The punctured torus onto two circles.* Let $T = [0,1]^2/\sim$ be the torus, with $(x,0) \sim (x,1)$ and $(0,y) \sim (1,y)$, let $c = (1/2,1/2)$, and let $p \in T$ be its image. For $z \in Q = [0,1]^2 \setminus \{c\}$ let $\rho(z) = c + (z-c)/(2\|z-c\|_\infty)$, the point where the ray from c through z leaves the square, and $H(z,t) = (1-t)z + t\rho(z)$, which stays on that ray, hence in Q. Since H fixes the boundary of the square pointwise, it respects the identifications, and it induces a deformation retraction of $T \setminus \{p\}$ onto the image of the boundary; it is continuous because $Q \to T \setminus \{p\}$ is a quotient map (the second statement of Lemma 19.2.4, as Q is open and saturated), and so is its product with id_I. The image of the boundary is the union of the two circles given by the bottom and the left side, meeting in one point: a wedge of two circles (Example 19.5.4). The same argument shows that a closed orientable surface of genus g minus a point deformation retracts onto a wedge of $2g$ circles.

Counterexample 19.2.9. *A weak deformation retract that is not a deformation retract.* Let $C \subseteq \mathbb{R}^2$ be the *comb space*

$$C = ([0,1] \times \{0\}) \cup (\{0\} \times [0,1]) \cup \textstyle\bigcup_{n \geq 1} (\{1/n\} \times [0,1]),$$

and $p = (0,1)$. The space C is contractible: $(x,y,t) \mapsto (x,(1-t)y)$ deforms it into the segment $[0,1] \times \{0\}$, which contracts to $(0,0)$, and the constant map at $(0,0)$ is homotopic to the constant map at p along the path $\{0\} \times [0,1]$. So $\{p\}$ is a weak deformation retract of C. But it is not a deformation retract. Suppose H is a deformation of C fixing p at all times and ending at the constant map p. Let $U = \{z \in C : \|z - p\| < 1/2\}$. Since $H(p,t) = p$ for all t and I is compact, there is a neighbourhood V of p in C with $H(V \times I) \subseteq U$ (as in the proof of Lemma 19.2.4). Choose n with $q = (1/n, 1) \in V$. Then $t \mapsto H(q,t)$ is a path in U from q to p. But $U \subseteq (\{0\} \cup \{1/n : n \geq 1\}) \times (1/2, 1]$, and the first coordinate of a path in U is a continuous map from I into the set $\{0\} \cup \{1/n : n \geq 1\}$, which contains no interval, so it is constant by the intermediate value theorem. This contradicts that the path joins q to p.

19.3 The fundamental group

Definition 19.3.1. For paths $\gamma, \delta\colon I \to X$ with $\gamma(1) = \delta(0)$, the *concatenation* $\gamma * \delta$ is $\gamma(2s)$ for $s \leq 1/2$ and $\delta(2s-1)$ for $s \geq 1/2$ (first γ, then δ), and the *reverse* of γ is $\bar\gamma(s) = \gamma(1-s)$. We write $\gamma \simeq \delta$ for homotopy relative to $\{0,1\}$ and $[\gamma]$ for the class. A *loop* at x_0 is a path with $\gamma(0) = \gamma(1) = x_0$, and c_{x_0} denotes the constant loop.

Lemma 19.3.2. *Concatenation is compatible with $\simeq$, and for composable paths:*

1. *$\gamma \circ \varphi \simeq \gamma$ for every map $\varphi\colon I \to I$ with $\varphi(0) = 0$ and $\varphi(1) = 1$;*
2. *$(\gamma * \delta) * \epsilon \simeq \gamma * (\delta * \epsilon)$, and $c_{\gamma(0)} * \gamma \simeq \gamma \simeq \gamma * c_{\gamma(1)}$;*
3. *$\gamma * \bar\gamma \simeq c_{\gamma(0)}$ and $\bar\gamma * \gamma \simeq c_{\gamma(1)}$.*

Proof. If H and K are homotopies $\gamma \simeq \gamma'$ and $\delta \simeq \delta'$, then $H(-,t) * K(-,t)$ is a homotopy $\gamma * \delta \simeq \gamma' * \delta'$. (1) $H(s,t) = \gamma((1-t)\varphi(s) + ts)$. (2) Each side is the other composed with a piecewise affine φ as in (1): for instance $(\gamma * \delta) * \epsilon = (\gamma * (\delta * \epsilon)) \circ \varphi$ with φ affine on $[0, \frac{1}{4}]$, $[\frac{1}{4}, \frac{1}{2}]$, $[\frac{1}{2}, 1]$ mapping them to $[0, \frac{1}{2}]$, $[\frac{1}{2}, \frac{3}{4}]$, $[\frac{3}{4}, 1]$. (3) $H(s,t) = \gamma(2s(1-t))$ for $s \le 1/2$ and $H(s,t) = \gamma((2-2s)(1-t))$ for $s \ge 1/2$ is a homotopy from $\gamma * \bar{\gamma}$ to $c_{\gamma(0)}$; apply this to $\bar{\gamma}$ for the second statement. □

Definition 19.3.3. The *fundamental group* $\pi_1(X, x_0)$ is the set of classes of loops at x_0 with the multiplication $[\gamma][\delta] = [\gamma * \delta]$, which is a group by Lemma 19.3.2, with identity $[c_{x_0}]$ and inverse $[\gamma]^{-1} = [\bar{\gamma}]$. (The group is written multiplicatively; note that $[\gamma][\delta]$ means "first γ".) A map $f\colon (X, x_0) \to (Y, y_0)$ of pointed spaces induces a homomorphism $f_*[\gamma] = [f \circ \gamma]$, and $(gf)_* = g_* f_*$, $\mathrm{id}_* = \mathrm{id}$. A path-connected space X is *simply connected* if $\pi_1(X, x_0)$ is trivial.

Lemma 19.3.4. *A path h from x_0 to x_1 induces an isomorphism $\beta_h\colon \pi_1(X, x_0) \to \pi_1(X, x_1)$, $[\gamma] \mapsto [\bar{h} * \gamma * h]$. If H is a homotopy from f to $g\colon X \to Y$ and $h(t) = H(x_0, t)$, then $g_* = \beta_h \circ f_*\colon \pi_1(X, x_0) \to \pi_1(Y, g(x_0))$. Consequently a homotopy equivalence induces isomorphisms on fundamental groups, and contractible spaces are simply connected.*

Proof. β_h is a homomorphism by Lemma 19.3.2, with inverse $\beta_{\bar{h}}$. For the second statement let γ be a loop at x_0 and $F(s,t) = H(\gamma(s), t)$ on the square I^2. In the convex set I^2, the path along the bottom and then the right side and the path along the left side and then the top have the same endpoints, so they are homotopic relative to the endpoints (Example 19.2.3(1)). Composing with F gives $(f \circ \gamma) * h \simeq h * (g \circ \gamma)$, that is, $g_*[\gamma] = [\bar{h} * (f \circ \gamma) * h]$.

Let f be a homotopy equivalence with inverse g. Since $gf \simeq \mathrm{id}_X$, the composite $\pi_1(X, x_0) \xrightarrow{f_*} \pi_1(Y, f(x_0)) \xrightarrow{g_*} \pi_1(X, gf(x_0))$ is β_h, a bijection. Since $fg \simeq \mathrm{id}_Y$, the composite $\pi_1(Y, f(x_0)) \xrightarrow{g_*} \pi_1(X, gf(x_0)) \xrightarrow{f_*} \pi_1(Y, fgf(x_0))$ is also a bijection. So this g_* is injective and surjective, hence bijective, and then $f_* = g_*^{-1}(g_* f_*)$ is bijective. □

Proposition 19.3.5. *$\pi_1(X \times Y, (x_0, y_0)) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$ via the projections.*

Proof. Loops and homotopies in $X \times Y$ are pairs of loops and homotopies in X and Y (Proposition 18.4.7(2)). □

19.4 Covering spaces and the circle

Definition 19.4.1. A *covering map* is a surjective map $p\colon E \to B$ such that every $b \in B$ has an open neighbourhood U that is *evenly covered*: $p^{-1}(U)$ is a disjoint union of open sets (the *sheets* over U), each mapped homeomorphically onto U by p. A *lift* of a map $f\colon Y \to B$ is a map $\tilde{f}\colon Y \to E$ with $p\tilde{f} = f$.

Lemma 19.4.2 (Unique lifting)**.** *Let $p\colon E \to B$ be a covering map and Y connected. Two lifts of $f\colon Y \to B$ that agree at one point are equal.*

Proof. Let $\tilde{f}_1, \tilde{f}_2$ be lifts, $y \in Y$, U an evenly covered neighbourhood of $f(y)$, and $\tilde{U}_1, \tilde{U}_2$ the sheets containing $\tilde{f}_1(y), \tilde{f}_2(y)$. On the open neighbourhood $N = \tilde{f}_1^{-1}(\tilde{U}_1) \cap \tilde{f}_2^{-1}(\tilde{U}_2)$ of y: if $\tilde{f}_1(y) = \tilde{f}_2(y)$, then $\tilde{U}_1 = \tilde{U}_2$ and, p being injective on it, $\tilde{f}_1 = \tilde{f}_2$ on N; if $\tilde{f}_1(y) \ne \tilde{f}_2(y)$, then $\tilde{U}_1 \ne \tilde{U}_2$ (as p is injective on sheets), the sheets are disjoint and $\tilde{f}_1 \ne \tilde{f}_2$ on N. So the set where the lifts agree is open and closed, and nonempty, hence all of Y. □

Proposition 19.4.3 (Homotopy lifting)**.** *Let $p\colon E \to B$ be a covering map, $F\colon Y \times I \to B$ a homotopy and $\tilde{F}_0\colon Y \to E$ a lift of $F(-,0)$. There is a unique lift $\tilde{F}\colon Y \times I \to E$ of F with $\tilde{F}(-,0) = \tilde{F}_0$. In particular (for Y a point) every path in B has a unique lift with a given initial point.*

Proof. Fix $y_0 \in Y$. Each (y_0, t) has a neighbourhood $N_t \times J_t$ mapped by F into an evenly covered set. By compactness of I there are an open $N \ni y_0$ and a partition $0 = t_0 < t_1 < \cdots < t_m = 1$ with $F(N \times [t_i, t_{i+1}]) \subseteq U_i$, U_i evenly covered (intersect finitely many N_t, and take a partition subordinate to a Lebesgue number of the cover (J_t), Lemma 18.7.3). Suppose a lift $\tilde{F}$ has been constructed on $N \times [0, t_i]$ (for $i = 0$ this is $\tilde{F}_0$ on $N \times \{0\}$). Let $\tilde{U}_i$ be the sheet over U_i containing $\tilde{F}(y_0, t_i)$; replacing N by the open set $\{y \in N : \tilde{F}(y, t_i) \in \tilde{U}_i\}$, define $\tilde{F} = (p|_{\tilde{U}_i})^{-1} \circ F$ on $N \times [t_i, t_{i+1}]$; this agrees on $N \times \{t_i\}$ with the previous definition, and the pasting lemma gives a lift on $N \times [0, t_{i+1}]$. After m steps we have a lift on $N_{y_0} \times I$ for an open $N_{y_0} \ni y_0$.

For each y, the restrictions to $\{y\} \times I$ of two lifts agreeing at $(y, 0)$ coincide, by Lemma 19.4.2. Hence the lifts on the sets $N_{y_0} \times I$ agree on overlaps, glue to a continuous lift on $Y \times I$ (pasting lemma for the open cover), and this lift is unique. □

Corollary 19.4.4. *Let $p \colon E \to B$ be a covering map, $e_0 \in E$ and $b_0 = p(e_0)$.*

1. *If $\gamma \simeq \gamma'$ are paths starting at b_0, their lifts starting at e_0 have the same endpoint and are homotopic.*

2. *$p_* \colon \pi_1(E, e_0) \to \pi_1(B, b_0)$ is injective, and its image consists of the classes of loops whose lift starting at e_0 is a loop.*

3. *The* monodromy *$e \cdot [\gamma] = \tilde{\gamma}_e(1)$, where $\tilde{\gamma}_e$ is the lift of γ starting at e, is a right action of $\pi_1(B, b_0)$ on the fibre $p^{-1}(b_0)$. If E is path-connected, the action is transitive and the stabiliser of e_0 is $p_*\pi_1(E, e_0)$; so the fibre is in bijection with the set of right cosets of $p_*\pi_1(E, e_0)$.*

Proof. (1) Lift a homotopy F from γ to γ' with $\tilde{F}(-, 0) = \tilde{\gamma}$ (Proposition 19.4.3). The paths $t \mapsto \tilde{F}(0, t)$ and $t \mapsto \tilde{F}(1, t)$ lift constant paths, so they are constant by uniqueness (Lemma 19.4.2); hence $\tilde{F}(-, 1)$ is the lift of γ' starting at e_0 and has the same endpoint as $\tilde{\gamma}$. (2) If $p \circ \tilde{\gamma} \simeq c_{b_0}$ for a loop $\tilde{\gamma}$, lifting the homotopy gives $\tilde{\gamma} \simeq c_{e_0}$ by (1). If the lift of γ is a loop $\tilde{\gamma}$, then $[\gamma] = p_*[\tilde{\gamma}]$; conversely the lift of $p \circ \tilde{\gamma}$ is $\tilde{\gamma}$, and (1) handles homotopic loops. (3) The action is well defined by (1), and the lift of $\gamma * \delta$ starting at e is $\tilde{\gamma}_e * \tilde{\delta}_{e \cdot [\gamma]}$, so $e \cdot ([\gamma][\delta]) = (e \cdot [\gamma]) \cdot [\delta]$. If η is a path from e_0 to e, then $e_0 \cdot [p \circ \eta] = e$. The stabiliser statement is (2). □

Proposition 19.4.5. *Let $Q = \{x \in \mathbb{R}^2 : \|x\|_\infty = 1\}$ and let $P \colon \mathbb{R} \to Q$ be the map of period 4 that on $[0, 4]$ runs affinely through the vertices $(1, -1)$, $(1, 1)$, $(-1, 1)$, $(-1, -1)$, $(1, -1)$ at the times $0, 1, 2, 3, 4$. Let $r \colon Q \to S^1$, $x \mapsto x/\|x\|_2$, and $p = r \circ P$. Then p is a covering map, $p(t) = p(t')$ if and only if $t - t' \in 4\mathbb{Z}$, and $p(t + 4) = p(t)$.*

Proof. P is continuous by the pasting lemma (applied to the closed intervals $[k, k + 1]$ inside each bounded open interval), and r is a homeomorphism with inverse $x \mapsto x/\|x\|_\infty$ (Example 18.6.10). By construction P is injective on $[a, a + 4)$ for every a and has period 4, which gives the second statement. For $a \in \mathbb{R}$, the restriction $P|_{[a,a+4]}$ is a closed map (Proposition 18.6.3(4)). If $V \subseteq (a, a + 4)$ is open, then $P(V) = Q \setminus P([a, a + 4] \setminus V)$, because P is surjective, injective on $(a, a + 4)$ and identifies only a and $a + 4$; so $P(V)$ is open, and P maps $(a, a + 4)$ homeomorphically onto $Q \setminus \{P(a)\}$. Given $q \in Q$, choose a with $P(a) \neq q$; then $U = Q \setminus \{P(a)\}$ is an open neighbourhood of q with $P^{-1}(U) = \bigsqcup_{k \in \mathbb{Z}} (a + 4k, a + 4k + 4)$, each interval mapped homeomorphically onto U. Composing with r proves that p is a covering map. □

Theorem 19.4.6. *Let $b_0 = p(0)$ and $\omega(s) = p(4s)$. Then $\pi_1(S^1, b_0) \cong \mathbb{Z}$, with $[\omega]$ corresponding to 1. Hence $\pi_1(S^1, x) \cong \mathbb{Z}$ for every $x \in S^1$.*

Proof. The space $\mathbb{R}$ is contractible, hence simply connected (Lemma 19.3.4), and path-connected. By Corollary 19.4.4(3), $[\gamma] \mapsto 0 \cdot [\gamma]$ is a bijection from $\pi_1(S^1, b_0)$ onto $p^{-1}(b_0) = 4\mathbb{Z}$. If the lifts of γ and δ starting at 0 end at $4m$ and $4n$, then $t \mapsto \tilde{\delta}_0(t) + 4m$ lifts δ (as p has period 4) and starts at $4m$, so the lift of $\gamma * \delta$ ends at $4(m+n)$. So $[\gamma] \mapsto \tilde{\gamma}_0(1)/4$ is an isomorphism onto $\mathbb{Z}$, and $\tilde{\omega}_0(s) = 4s$ ends at 4. The last statement follows from Lemma 19.3.4. □

Remark 19.4.7. With the exponential function of Chapter 27, the usual covering is $\mathbb{R} \to S^1$, $t \mapsto (\cos 2\pi t, \sin 2\pi t)$. We used a piecewise affine model so as not to depend on analysis. A map $f\colon S^1 \to S^1$ induces multiplication by an integer $\deg f$ on $\pi_1(S^1) \cong \mathbb{Z}$ (after identifying basepoints by a path, which does not matter since $\mathbb{Z}$ is abelian), its *degree*.

Corollary 19.4.8 (Brouwer fixed point theorem in dimension 2)**.** *There is no retraction $D^2 \to S^1$, and every map $f\colon D^2 \to D^2$ has a fixed point.*

Proof. If $r\colon D^2 \to S^1$ were a retraction with inclusion i, then $r_* i_* = \mathrm{id}$ on $\pi_1(S^1) \cong \mathbb{Z}$ would factor through $\pi_1(D^2) = 0$. If f had no fixed point, let $u(x) = x - f(x) \neq 0$ and $r(x) = x + \lambda(x)u(x)$, where

$$\lambda(x) = \frac{-\langle x, u\rangle + \sqrt{\langle x, u\rangle^2 + \|u\|^2(1 - \|x\|^2)}}{\|u\|^2}$$

is the non-negative solution of $\|x + \lambda u\|^2 = 1$. The square root is continuous on $[0, \infty)$ since $|\sqrt{a} - \sqrt{b}| \leq \sqrt{|a - b|}$, so r is continuous, and $r(x) \in S^1$. If $\|x\| = 1$, then $\langle x, u\rangle = 1 - \langle x, f(x)\rangle \geq 0$ by Cauchy–Schwarz, so $\lambda(x) = 0$ and $r(x) = x$. This contradicts the first statement. □

Counterexample 19.4.9. A retract need not be a deformation retract, not even a weak one. A point $\{b\}$ of S^1 is a retract (take the constant map), but not a weak deformation retract: otherwise the inclusion would be a homotopy equivalence (as in Lemma 19.2.6(1)), so $\pi_1(S^1) \cong \pi_1(\{b\}) = 0$ by Lemma 19.3.4, contradicting Theorem 19.4.6. Likewise the circle $S^1 \times \{b\}$ is a retract of the torus $S^1 \times S^1$ (via the first projection), but not a weak deformation retract, since $\pi_1(S^1 \times S^1) \cong \mathbb{Z}^2$ (Proposition 19.3.5) is not isomorphic to $\mathbb{Z}$ ($\mathbb{Z}^2/2\mathbb{Z}^2$ has four elements and $\mathbb{Z}/2\mathbb{Z}$ two). And by Corollary 19.4.8, S^1 is not even a retract of D^2, although it is a deformation retract of $D^2 \setminus \{0\}$ (Example 19.2.3(2)).

Example 19.4.10.

1. For $n \geq 1$ the map $S^1 \to S^1$, $p(t) \mapsto p(nt)$, is well defined and is a covering map with n sheets; it corresponds to the subgroup $n\mathbb{Z} \subseteq \mathbb{Z}$ in Theorem 19.6.6.
2. The projection $q\colon S^n \to \mathbb{RP}^n$ is a two-sheeted covering: for $x \in S^n$ the open hemisphere $W = \{y \in S^n : \langle x, y\rangle > 0\}$ is mapped injectively by q, $q^{-1}(q(W)) = W \sqcup (-W)$ is open, so $q(W)$ is open and $q|_W$, $q|_{-W}$ are homeomorphisms onto it (q is open, since $q^{-1}(q(V)) = V \cup (-V)$). For $n \geq 2$, S^n is simply connected (Corollary 19.5.2), so $\pi_1(\mathbb{RP}^n) \cong \mathbb{Z}/2\mathbb{Z}$ by Corollary 19.4.4(3).

19.5 The Seifert–van Kampen theorem

Theorem 19.5.1 (Seifert–van Kampen)**.** *Let $X = U \cup V$ with U, V open and U, V, $U \cap V$ path-connected, and $x_0 \in U \cap V$. The homomorphism*

$$\Phi\colon \pi_1(U, x_0) *_{\pi_1(U \cap V, x_0)} \pi_1(V, x_0) \to \pi_1(X, x_0)$$

induced by the inclusions (Proposition 7.10.7) is an isomorphism.

Proof. Write G for the amalgamated product and ι_U, ι_V for the maps from $\pi_1(U)$, $\pi_1(V)$ to G; by Definition 7.10.6, G is generated by their images. For loops $\gamma_1, \dots, \gamma_k$ at x_0 let $\gamma_1 \cdots \gamma_k$ denote the loop running through γ_i on $[(i-1)/k, i/k]$; it is homotopic to the iterated concatenation by Lemma 19.3.2(1).

Surjectivity. Let γ be a loop at x_0. By the Lebesgue number lemma there is a partition $0 = s_0 < \cdots < s_m = 1$ such that each $\gamma([s_{i-1}, s_i])$ lies in a set $A_i \in \{U, V\}$; merging adjacent intervals with the same label, we may assume $A_i \neq A_{i+1}$, so $\gamma(s_i) \in U \cap V$ for $0 < i < m$. Choose paths g_i in $U \cap V$ from x_0 to $\gamma(s_i)$, with g_0, g_m constant. Then $\gamma \simeq \prod_i (g_{i-1} * \gamma|_{[s_{i-1}, s_i]} * \bar{g}_i)$ (with the restriction reparametrised to I), a product of loops in U or V.

Injectivity. Let $\gamma_1, \dots, \gamma_k$ and $\gamma'_1, \dots, \gamma'_l$ be loops at x_0, each in U or V (label them A_i and A'_j), with $F \colon I \times I \to X$ a homotopy from $\gamma_1 \cdots \gamma_k$ to $\gamma'_1 \cdots \gamma'_l$ relative to $\{0, 1\}$. We show $\prod_i \iota_{A_i}[\gamma_i] = \prod_j \iota_{A'_j}[\gamma'_j]$ in G. This proves that $\ker \Phi$ is trivial: an element of the kernel can be written as $\prod_i \iota_{A_i}[\gamma_i]$, and $\gamma_1 \cdots \gamma_k$ is then homotopic to the constant loop, which we take as γ'_1 with $l = 1$.

Choose N divisible by k and l such that each small square R of the grid of mesh $1/N$ is mapped by F into U or V (Lebesgue number lemma), and fix a label $A_R \in \{U, V\}$ with $F(R) \subseteq A_R$. For each grid vertex v let S_v be the intersection of the labels of the squares containing v and, for v on the bottom (top) edge in the domain of γ_i (γ'_j), also of A_i (A'_j). Then S_v is U, V or $U \cap V$, so it is path-connected, and it contains $F(v)$. Choose a path g_v in S_v from x_0 to $F(v)$, constant if $F(v) = x_0$ and v lies on the boundary of the square. For an edge e from v to w let $\ell_e = g_v * F|_e * \bar{g}_w$, a loop in the intersection of the labels of the (one or two) squares containing e. Put $\phi(e) = \iota_{A_R}[\ell_e]$ for a square R containing e; this does not depend on R, because if two squares with different labels contain e, then ℓ_e lies in $U \cap V$ and the two images agree in G by the amalgamation relations. For an edge path P let $\phi(P)$ be the product of the $\phi(e)$ along P.

For a square R with lower left vertex a, lower right b, upper left c and upper right d, all four loops ℓ_e of its edges lie in A_R, and $F|_R$ gives, in the convex set R, a homotopy relative to endpoints in A_R between $F|_{[a,b]} * F|_{[b,d]}$ and $F|_{[a,c]} * F|_{[c,d]}$. Hence $[\ell_{ab}][\ell_{bd}] = [g_a * F|_{[a,b]} * F|_{[b,d]} * \bar{g}_d] = [\ell_{ac}][\ell_{cd}]$ in $\pi_1(A_R)$. So replacing the two lower-right edges of a square by its two upper-left edges does not change $\phi(P)$, and moving square by square the path P_0 along the bottom and right sides of I^2 to the path P_1 along the left and top sides gives $\phi(P_0) = \phi(P_1)$. On the vertical sides F is constant at x_0 and all g_v are constant, so those edges contribute 1. The bottom edges in the domain of γ_i have loops ℓ_e in A_i, and their product in $\pi_1(A_i)$ is $[\gamma_i]$ (the paths g_v cancel, and g_v is constant at the ends of the domain); since each ℓ_e also lies in A_R, $\iota_{A_R}[\ell_e] = \iota_{A_i}[\ell_e]$. Hence $\phi(P_0) = \prod_i \iota_{A_i}[\gamma_i]$, and likewise $\phi(P_1) = \prod_j \iota_{A'_j}[\gamma'_j]$. □

Corollary 19.5.2. *For $n \geq 2$ the sphere S^n is simply connected. If $X = U \cup V$ as in Theorem 19.5.1 with $U \cap V$ simply connected, then $\pi_1(X) \cong \pi_1(U) * \pi_1(V)$; if V is simply connected, then $\pi_1(U) \to \pi_1(X)$ is surjective with kernel the normal subgroup generated by the image of $\pi_1(U \cap V)$.*

Proof. Let $U = S^n \setminus \{N\}$ and $V = S^n \setminus \{-N\}$. By stereographic projection (Example 18.9.4), U and V are homeomorphic to $\mathbb{R}^n$, hence contractible, and $U \cap V$ is homeomorphic to $\mathbb{R}^n \setminus \{0\}$, which is path-connected for $n \geq 2$. So $\pi_1(S^n) \cong 1 *_{\pi_1(U \cap V)} 1 = 1$. The other statements follow from the presentation of the amalgamated product (Definition 7.10.6): the relations from $\pi_1(U \cap V)$ are trivial in the first case, and in the second they identify the image of each $h \in \pi_1(U \cap V)$ in $\pi_1(U)$ with 1. □

Example 19.5.3. Combining what we have proved so far:

1. A convex set, or any contractible space, has trivial fundamental group (Lemma 19.3.4).

2. $\pi_1(\mathbb{R}^n \setminus \{0\}) \cong \pi_1(S^{n-1})$, since S^{n-1} is a deformation retract (Example 19.2.3(2)); this is $\mathbb{Z}$ for $n = 2$ and trivial for $n \geq 3$ (Theorem 19.4.6 and Corollary 19.5.2). So $\mathbb{R}^2$ and $\mathbb{R}^n$, $n \geq 3$, are not homeomorphic: a homeomorphism would restrict to a homeomorphism of the complements of a point and its image.

3. The torus $T^2 = S^1 \times S^1$ has $\pi_1 \cong \mathbb{Z}^2$ by Proposition 19.3.5, generated by the two circles $S^1 \times \{*\}$ and $\{*\} \times S^1$; these commute, and the square relation $aba^{-1}b^{-1} = 1$ is the boundary of the square from which the torus is glued. Similarly $\pi_1(T^n) \cong \mathbb{Z}^n$, $\pi_1(S^1 \times S^2) \cong \mathbb{Z}$, and the solid torus $S^1 \times D^2$ and the Möbius band (Example 19.2.8(2)) have $\pi_1 \cong \mathbb{Z}$.

4. The circle is not a retract of the disc, and the identity of S^1 is not null-homotopic: a retraction $D^2 \to S^1$ would make $\pi_1(S^1) \to \pi_1(D^2) = 0$ injective (Exercise 19.9.1). This is the argument behind Corollary 19.4.8.

Example 19.5.4. Let $X = S^1 \vee S^1$ be the wedge of two circles, the quotient of $S^1 \sqcup S^1$ identifying the two basepoints b_0 to a point x_0. Let U (V) be the union of the first (second) circle with the second (first) circle minus the point $-b_0$. The open arc $S^1 \setminus \{-b_0\}$ deformation retracts onto b_0 (it is homeomorphic to an open interval by Proposition 19.4.5), so U and V deformation retract onto the circles (Lemma 19.2.4 ensures continuity of the homotopies on the quotient), and $U \cap V$ is contractible. By Corollary 19.5.2, $\pi_1(S^1 \vee S^1) \cong \mathbb{Z} * \mathbb{Z}$, the free group on two generators. It is not abelian, although $\pi_1(S^1 \times S^1) \cong \mathbb{Z}^2$ by Proposition 19.3.5.

Proposition 19.5.5 (Attaching cells)**.** *Let X be path-connected, $n \geq 2$, $\varphi\colon S^{n-1} \to X$ a map, and $Y = X \cup_\varphi D^n$ the space obtained by attaching an n-cell (Example 18.4.11). Let $z_0 \in S^{n-1}$ and $x_0 = \varphi(z_0)$. The inclusion induces a surjection $\pi_1(X, x_0) \to \pi_1(Y, x_0)$. For $n = 2$ its kernel is the normal subgroup generated by $[\varphi \circ \omega]$, where ω is the loop of Theorem 19.4.6 based at z_0; for $n \geq 3$ it is an isomorphism.*

Proof. Let $q\colon X \sqcup D^n \to Y$ be the quotient map. For $C \subseteq X$ closed, $q^{-1}(q(C)) = C \sqcup \varphi^{-1}(C)$ is closed, so $q|_X$ is an embedding onto a closed subspace, which we identify with X; similarly q maps the open ball $\operatorname{int} D^n$ homeomorphically onto an open subset. Let $V = q(\operatorname{int} D^n)$, which is contractible, and $U = Y \setminus \{q(0)\}$. On $q^{-1}(U) \times I = (X \times I) \sqcup ((D^n \setminus \{0\}) \times I)$ define H as the projection on $X \times I$ and as $q((1-t)z + tz/\|z\|)$ on the second part; it respects the identifications, so by Lemma 19.2.4 it induces a homotopy on $U \times I$, which is a deformation retraction of U onto X. The set $U \cap V$ is homeomorphic to $\operatorname{int} D^n \setminus \{0\}$, which is path-connected and deformation retracts onto the sphere of radius $1/2$; so $\pi_1(U \cap V)$ is $\mathbb{Z}$ for $n = 2$ (generated by $c(s) = q(\omega(s)/2)$) and trivial for $n \geq 3$ (Corollary 19.5.2).

Work at the basepoint $y_1 = q(z_0/2) \in U \cap V$. By Corollary 19.5.2, $\pi_1(U, y_1) \to \pi_1(Y, y_1)$ is surjective with kernel the normal subgroup generated by the image of $\pi_1(U \cap V, y_1)$. Transport to x_0 along the path $\delta(t) = H(y_1, t)$ from y_1 to x_0: the inclusion $X \to U$ is a homotopy equivalence, and by Lemma 19.3.4 applied to the homotopy H, the class $[c]$ corresponds to $[\bar{\delta} * c * \delta] = [H_1 \circ c] = [\varphi \circ \omega]$ in $\pi_1(X, x_0)$. Since the change-of-basepoint isomorphisms β_δ in U and Y are compatible with the inclusion, the statement follows. □

19.6 Classification of covering spaces

In this section B is path-connected and locally path-connected. Then a covering space E is locally path-connected, being locally homeomorphic to B, and if E is connected it is path-connected (Proposition 18.5.7).

Proposition 19.6.1 (Lifting criterion)**.** *Let $p\colon (E,e_0)\to(B,b_0)$ be a covering map, Y path-connected and locally path-connected, and $f\colon (Y,y_0)\to(B,b_0)$ a map. A lift $\tilde f\colon (Y,y_0)\to(E,e_0)$ exists if and only if $f_*\pi_1(Y,y_0)\subseteq p_*\pi_1(E,e_0)$, and it is then unique.*

Proof. Necessity follows from $f_*=p_*\tilde f_*$, and uniqueness from Lemma 19.4.2. Conversely, for $y\in Y$ choose a path η from y_0 to y and let $\tilde f(y)$ be the endpoint of the lift of $f\circ\eta$ starting at e_0. If η' is another such path, the loop $h=(f\circ\eta')*(f\circ\bar\eta)$ represents an element of $f_*\pi_1(Y)\subseteq p_*\pi_1(E)$, so its lift starting at e_0 is a loop (Corollary 19.4.4(2)). This lift is the lift of $f\circ\eta'$, ending at some e', followed by the lift of $f\circ\bar\eta$ starting at e', which therefore ends at e_0; reversing it gives the lift of $f\circ\eta$ starting at e_0, which ends at e'. So $\tilde f$ is well defined, and $p\tilde f=f$.

Continuity: let $y\in Y$, U an evenly covered neighbourhood of $f(y)$ and $\tilde U$ the sheet containing $\tilde f(y)$. Choose a path-connected open $W\ni y$ with $f(W)\subseteq U$. For $y'\in W$, using $\eta*\sigma$ with σ a path in W from y to y', the lift of $f\circ\sigma$ starting at $\tilde f(y)$ is $(p|_{\tilde U})^{-1}\circ f\circ\sigma$. So $\tilde f=(p|_{\tilde U})^{-1}\circ f$ on W. □

Definition 19.6.2. Two coverings $p\colon E\to B$ and $p'\colon E'\to B$ are *isomorphic* if there is a homeomorphism $\tau\colon E\to E'$ with $p'\tau=p$. The automorphisms of a covering form the group $\mathrm{Deck}(p)$ of *deck transformations*. A covering with E path-connected is *normal* (or Galois) if $\mathrm{Deck}(p)$ acts transitively on the fibres, and *universal* if E is simply connected.

Proposition 19.6.3. *Let $p\colon E\to B$ be a covering with E path-connected, $e_0\in p^{-1}(b_0)$ and $H=p_*\pi_1(E,e_0)$.*

1. *For $e\in p^{-1}(b_0)$ there is a deck transformation τ with $\tau(e_0)=e$ if and only if $p_*\pi_1(E,e)=H$. For $e=e_0\cdot[\gamma]$ one has $p_*\pi_1(E,e)=[\gamma]^{-1}H[\gamma]$.*

2. *The covering is normal if and only if H is a normal subgroup, and then $\mathrm{Deck}(p)\cong\pi_1(B,b_0)/H$. In particular, for a universal covering, $\mathrm{Deck}(p)\cong\pi_1(B,b_0)$.*

Proof. (1) If $p_*\pi_1(E,e)=H$, the lifting criterion gives maps $\tau,\tau'\colon E\to E$ over B with $\tau(e_0)=e$ and $\tau'(e)=e_0$; then $\tau'\tau$ and $\tau\tau'$ are lifts of p fixing a point, hence identities. The converse is clear. If $\tilde\gamma$ is the lift of γ from e_0 to e, then $[\tilde\alpha]\mapsto[\bar{\tilde\gamma}*\tilde\alpha*\tilde\gamma]$ is an isomorphism $\pi_1(E,e_0)\to\pi_1(E,e)$, whence the formula.

(2) The first statement follows from (1) and transitivity of monodromy (Corollary 19.4.4). A deck transformation commutes with monodromy, $\tau(e\cdot[\gamma])=\tau(e)\cdot[\gamma]$, since $\tau\circ\tilde\gamma_e$ lifts γ from $\tau(e)$. For H normal let $\Psi[\gamma]$ be the deck transformation with $\Psi[\gamma](e_0)=e_0\cdot[\gamma]$. Then $\Psi([\gamma][\delta])(e_0)=(e_0\cdot[\gamma])\cdot[\delta]=\Psi[\gamma](e_0\cdot[\delta])=\Psi[\gamma]\Psi[\delta](e_0)$, so Ψ is a homomorphism (deck transformations agreeing at a point are equal). It is surjective by (1), and its kernel is the stabiliser H of e_0. □

Definition 19.6.4. B is *semilocally simply connected* if every point b has a neighbourhood U such that $\pi_1(U,b)\to\pi_1(B,b)$ is trivial.

Theorem 19.6.5. *If B is path-connected, locally path-connected and semilocally simply connected, then B has a universal covering.*

Proof. Let $\tilde B$ be the set of classes $[\gamma]$ of paths starting at b_0 (up to homotopy relative to endpoints), and $p[\gamma]=\gamma(1)$. Let $\mathcal{U}$ be the set of path-connected open $U\subseteq B$ with $\pi_1(U)\to\pi_1(B)$ trivial (at any, hence every, basepoint in U). Every path-connected open subset of a member of $\mathcal{U}$ is in $\mathcal{U}$, so by the hypotheses $\mathcal{U}$ is a basis of B. For $U\in\mathcal{U}$ and $\gamma(1)\in U$ let $U_{[\gamma]}=\{[\gamma*\eta]:\eta\text{ a path in }U\text{ starting at }\gamma(1)\}$. Then $p\colon U_{[\gamma]}\to U$ is bijective: surjective as U is path-connected, and injective since for two paths η,η' in U with the same endpoints, $\eta*\bar\eta'$ is null-homotopic in B. If $[\gamma']\in U_{[\gamma]}$ then $U_{[\gamma']}=U_{[\gamma]}$. Hence if

$[\gamma''] \in U_{[\gamma]} \cap V_{[\gamma']}$, choosing $W \in \mathcal{U}$ with $\gamma''(1) \in W \subseteq U \cap V$ gives $[\gamma''] \in W_{[\gamma'']} \subseteq U_{[\gamma]} \cap V_{[\gamma']}$; so the sets $U_{[\gamma]}$ form a basis of a topology on $\tilde{B}$ (Lemma 18.2.10).

For $V \subseteq U$ in $\mathcal{U}$, p maps the basic open sets $V_{[\gamma']} \subseteq U_{[\gamma]}$ onto V, and $p^{-1}(V) \cap U_{[\gamma]} = V_{[\gamma']}$ for $[\gamma'] \in U_{[\gamma]}$ with $\gamma'(1) \in V$. So $p|_{U_{[\gamma]}}$ is a homeomorphism onto U. The set $p^{-1}(U)$ is the union of the sets $U_{[\gamma]}$ with $\gamma(1) \in U$, and two of them are equal or disjoint. So p is a covering map.

For $[\gamma] \in \tilde{B}$, let $\gamma_t(s) = \gamma(ts)$. The path $t \mapsto [\gamma_t]$ is continuous: near t_0, with $\gamma([t_0, t]) \subseteq U \in \mathcal{U}$, it equals $(p|_{U_{[\gamma_{t_0}]}})^{-1} \circ \gamma$. It is the lift of γ starting at $\tilde{b}_0 = [c_{b_0}]$ and ends at $[\gamma]$. So $\tilde{B}$ is path-connected, and a loop γ whose lift is a loop satisfies $[\gamma] = [c_{b_0}]$. By Corollary 19.4.4(2), $p_*\pi_1(\tilde{B}) = 1$, and as p_* is injective, $\tilde{B}$ is simply connected. □

Theorem 19.6.6 (Classification of coverings)**.** *Let B be path-connected, locally path-connected and semilocally simply connected, and $b_0 \in B$. Sending $p\colon (E, e_0) \to (B, b_0)$ to $p_*\pi_1(E, e_0)$ is a bijection between isomorphism classes of pointed coverings with E path-connected and subgroups of $\pi_1(B, b_0)$. Forgetting basepoints, isomorphism classes of coverings with E path-connected correspond to conjugacy classes of subgroups.*

Proof. Injectivity: if $p_*\pi_1(E, e_0) = p'_*\pi_1(E', e'_0)$, the lifting criterion gives pointed maps $E \to E'$ and $E' \to E$ over B, whose composites are identities by uniqueness of lifts.

Surjectivity: let $H \subseteq \pi_1(B, b_0)$ and $\tilde{B}$ be as in the proof of Theorem 19.6.5. Put $[\gamma] \sim [\gamma']$ if $\gamma(1) = \gamma'(1)$ and $[\gamma * \bar{\gamma}'] \in H$, an equivalence relation, and let $E_H = \tilde{B}/\!\sim$. If $[\gamma] \sim [\gamma']$ then $[\gamma * \eta] \sim [\gamma' * \eta]$, so the saturation of a basic set $U_{[\gamma]}$ is the union of the sets $U_{[\gamma']}$ with $[\gamma'] \sim [\gamma]$; hence the projection $\tilde{B} \to E_H$ is open, and the images of the sets $U_{[\gamma]}$ are open, pairwise equal or disjoint, and mapped bijectively (as distinct points of $U_{[\gamma]}$ have distinct endpoints), continuously and openly onto U. So $E_H \to B$ is a covering, and E_H is path-connected. The lift of a loop γ at b_0 starting at the class of $[c_{b_0}]$ is $t \mapsto$ class of $[\gamma_t]$, which is a loop if and only if $[\gamma] \in H$; by Corollary 19.4.4(2) the subgroup of E_H is H.

Changing the basepoint within the fibre replaces the subgroup by a conjugate, and every conjugate arises (Proposition 19.6.3); an isomorphism of unpointed coverings carries e_0 to some point of the fibre, which gives the last statement. □

Example 19.6.7. *Monodromy.* For a covering $p\colon E \to B$ and $b \in B$, lifting loops defines an action of $\pi_1(B, b)$ on the fibre $F = p^{-1}(b)$: $[\gamma]$ sends $e \in F$ to the endpoint of the lift of γ starting at e (Corollary 19.4.4), the *monodromy* of the covering. It is transitive when E is path-connected, and the stabiliser of e is $p_*\pi_1(E, e)$; so the fibre is $\pi_1(B, b)/p_*\pi_1(E, e)$, and the classification theorem says that connected coverings correspond to transitive π_1-sets. For the n-sheeted covering $S^1 \to S^1$, $p(t) \mapsto p(nt)$ (Example 19.4.10(1)), the fibre over $b_0 = p(0)$ consists of the points $p(4k/n)$, $0 \le k < n$, and the lift of the generator $\omega(s) = p(4s)$ starting at $p(4k/n)$ is $s \mapsto p(4(k + s)/n)$, which ends at $p(4(k + 1)/n)$: the generator acts by the cyclic permutation of the n points. For $S^n \to \mathbb{RP}^n$ the nontrivial element of $\pi_1(\mathbb{RP}^n) = \mathbb{Z}/2\mathbb{Z}$ exchanges the two antipodal points of a fibre. In complex analysis the same phenomenon appears for multivalued functions: continuing a branch of $\sqrt[n]{z}$ around the origin multiplies it by $e^{2\pi i/n}$, which is the monodromy of the covering $z \mapsto z^n$ of $\mathbb{C} \setminus 0$ (Chapter 29).

Counterexample 19.6.8. Semilocal simple connectivity is necessary: a space with a simply connected covering $p\colon E \to B$ is semilocally simply connected, because a loop in an evenly covered U lifts to a loop in a sheet, which is null-homotopic in E. The *Hawaiian earring*, the union of the circles of radius $1/n$ centred at $(1/n, 0)$ in $\mathbb{R}^2$, is path-connected and locally path-connected but not semilocally simply connected (every neighbourhood of the origin contains a whole circle, which is a retract of the earring and hence not null-homotopic in it), so it has no universal covering [Hat02].

19.7 CW complexes

Definition 19.7.1. A *CW complex* is a space X with an increasing sequence of subspaces $X^0 \subseteq X^1 \subseteq \cdots$, its *skeleta*, such that X^0 is discrete, X^n is obtained from X^{n-1} by attaching a family of n-cells, $X^n = (X^{n-1} \sqcup \coprod_\alpha D^n_\alpha)/(z \sim \varphi_\alpha(z))$ for maps $\varphi_\alpha \colon S^{n-1} \to X^{n-1}$, and $X = \bigcup_n X^n$ carries the final topology for the inclusions of the X^n (a set is open if and only if its intersection with every X^n is open). The images e^n_α of the open balls are the *open cells*, the maps $\Phi_\alpha \colon D^n_\alpha \to X$ are the *characteristic maps*, and X is *finite* if there are finitely many cells and *finite-dimensional* if $X = X^n$ for some n. A *subcomplex* is a closed union of cells containing, with each cell, the image of its characteristic map; (X, A) is then a *CW pair*.

Example 19.7.2.

1. S^n has a CW structure with one 0-cell and one n-cell attached by the constant map, since $D^n/S^{n-1} \cong S^n$. It also has a structure with two cells in each dimension $0, \dots, n$ (the hemispheres), compatible with the antipodal map.

2. Passing to the quotient by the antipodal map in the second structure gives $\mathbb{RP}^n$ one cell in each dimension $0, \dots, n$, with $\mathbb{RP}^n = \mathbb{RP}^{n-1} \cup e^n$ attached along the double covering $S^{n-1} \to \mathbb{RP}^{n-1}$. Similarly $\mathbb{CP}^n = \mathbb{CP}^{n-1} \cup e^{2n}$ has one cell in each even dimension (Example 19.7.3).

3. A graph is a 1-dimensional CW complex. The torus is obtained from $S^1 \vee S^1$ by attaching a 2-cell along the loop $aba^{-1}b^{-1}$, and the closed orientable surface of genus g from a wedge of $2g$ circles along $a_1b_1a_1^{-1}b_1^{-1} \cdots a_gb_ga_g^{-1}b_g^{-1}$. By Proposition 19.5.5 and Example 19.5.4 (extended to $2g$ circles), its fundamental group is $\langle a_1, b_1, \dots, a_g, b_g \mid \prod_i a_ib_ia_i^{-1}b_i^{-1} \rangle$.

Example 19.7.3. *Complex projective space, its cells and its fundamental group.* Let $\mathbb{CP}^n = (\mathbb{C}^{n+1} \setminus \{0\})/\mathbb{C}^\times$ be the space of complex lines in $\mathbb{C}^{n+1}$, with the quotient topology, and write $[z] = [z_0 : \cdots : z_n]$.

1. $\mathbb{CP}^n$ is compact and Hausdorff. It is compact, being the image of the compact sphere S^{2n+1}. The map $[z] \mapsto zz^*/\|z\|^2$ to the space of complex $(n+1) \times (n+1)$ matrices is well defined and continuous (Proposition 18.4.10), and injective, since $zz^*/\|z\|^2$ is the orthogonal projection onto the line $\mathbb{C}z$; a space with a continuous injection into a Hausdorff space is Hausdorff.

2. $\mathbb{CP}^n = \mathbb{CP}^{n-1} \cup e^{2n}$. The map $\mathbb{CP}^{n-1} \to \mathbb{CP}^n$, $[z'] \mapsto [z' : 0]$, is a continuous injection from a compact space to a Hausdorff space, hence a homeomorphism onto the closed set $\{z_n = 0\}$ (Proposition 18.6.3(4)). Let $\Phi \colon D^{2n} \to \mathbb{CP}^n$, $\Phi(w) = [w_1 : \cdots : w_n : (1 - \|w\|^2)^{1/2}]$ for $w \in \mathbb{C}^n = \mathbb{R}^{2n}$ with $\|w\| \le 1$. A point $[z]$ with $z_n \neq 0$ has exactly one representative with $\|z\| = 1$ and z_n real and positive, so Φ maps the open disc bijectively onto $\{z_n \neq 0\}$; on the boundary sphere S^{2n-1} it is the quotient map onto $\mathbb{CP}^{n-1}$. So Φ induces a continuous bijection $\mathbb{CP}^{n-1} \cup_{\Phi|_{S^{2n-1}}} D^{2n} \to \mathbb{CP}^n$ from a compact space onto a Hausdorff space, which is a homeomorphism. By induction, $\mathbb{CP}^n$ is a CW complex with one cell in each dimension $0, 2, \dots, 2n$. For $n = 1$, a 2-cell is attached to a point, so $\mathbb{CP}^1 \cong D^2/S^1 \cong S^2$ (Example 19.7.2(1)); an explicit homeomorphism is given in Chapter 20.

3. $\mathbb{CP}^n$ is simply connected. $\mathbb{CP}^0$ is a point, and attaching a 2-cell to a point gives a simply connected space, as π_1 of the point surjects onto it; attaching cells of dimension $2n \ge 4$ does not change π_1 (Proposition 19.5.5). So $\pi_1(\mathbb{CP}^n) = 1$ for all n by induction.

4. The same induction for $\mathbb{RP}^n = \mathbb{RP}^{n-1} \cup e^n$ does *not* give simple connectivity, because it starts from a circle. The map $S^1 \to S^1$, $z \mapsto z^2$, induces a continuous bijection

$\mathbb{RP}^1 = S^1/\{\pm 1\} \to S^1$, hence a homeomorphism, and in this identification the 2-cell of $\mathbb{RP}^2$ is attached along the double covering $z \mapsto z^2$, a loop representing $2 \in \mathbb{Z} \cong \pi_1(S^1)$ (Theorem 19.4.6). So $\pi_1(\mathbb{RP}^1) \cong \mathbb{Z}$, $\pi_1(\mathbb{RP}^2) \cong \mathbb{Z}/2\mathbb{Z}$ by Proposition 19.5.5, and the cells of dimension $n \geq 3$ do not change this: $\pi_1(\mathbb{RP}^n) \cong \mathbb{Z}/2\mathbb{Z}$ for $n \geq 2$, as found with covering spaces in Example 19.4.10(2).

Theorem 19.7.4. *Let X be a CW complex.*

1. *X is Hausdorff, and in fact paracompact, and it is locally contractible.*
2. *Every compact subset of X lies in a finite subcomplex.*
3. *(Homotopy extension property) If (X, A) is a CW pair, then $X \times \{0\} \cup A \times I$ is a deformation retract of $X \times I$; so every map $X \times \{0\} \cup A \times I \to Y$ extends to $X \times I$.*
4. *The inclusion $X^2 \to X$ induces an isomorphism on fundamental groups (at a 0-cell).*

These are [Hat02, Appendix, and Propositions 0.16 and 1.26]. Part (4) follows from Proposition 19.5.5 applied to the finite subcomplexes containing a given loop or homotopy, using (2).

Theorem 19.7.5 (Cellular approximation)**.** *Every map $f : X \to Y$ of CW complexes is homotopic to a* cellular *map ($f(X^n) \subseteq Y^n$ for all n), by a homotopy relative to any subcomplex on which f is already cellular.*

This is [Hat02, Theorem 4.8].

19.8 Higher homotopy groups and fibrations

Definition 19.8.1. For $n \geq 1$, $\pi_n(X, x_0)$ is the set of homotopy classes, relative to ∂I^n, of maps $f : (I^n, \partial I^n) \to (X, x_0)$, with $[f] + [g] = [f *_1 g]$, where $(f *_1 g)(s) = f(2s_1, s_2, \ldots, s_n)$ for $s_1 \leq 1/2$ and $g(2s_1 - 1, s_2, \ldots, s_n)$ for $s_1 \geq 1/2$. We write the operation additively for $n \geq 2$ and multiplicatively for $n = 1$, where it is the fundamental group. Also $\pi_0(X, x_0)$ is the pointed set of path components.

Proposition 19.8.2. *$\pi_n(X, x_0)$ is a group, abelian for $n \geq 2$. A pointed map induces homomorphisms f_*, pointed homotopic maps induce the same homomorphisms, and $\pi_n(X \times Y) \cong \pi_n(X) \times \pi_n(Y)$.*

Proof. The proofs of Lemma 19.3.2 apply verbatim in the first coordinate, the other coordinates being carried along. For $n \geq 2$ define $*_2$ in the same way using the second coordinate; it also induces a well-defined operation on π_n with the same identity e, the class of the constant map. For maps, $(a *_1 b) *_2 (c *_1 d) = (a *_2 c) *_1 (b *_2 d)$ holds exactly (both sides are a, b, c, d on the four quarters of $I^2 \times I^{n-2}$, rescaled). Hence on classes

$$a *_2 b = (a *_1 e) *_2 (e *_1 b) = (a *_2 e) *_1 (e *_2 b) = a *_1 b, \quad a *_1 b = (e *_1 a) *_2 (b *_1 e) = (e *_2 b) *_1 (a *_2 e) = b *_1 a$$

(the Eckmann–Hilton argument). The remaining statements are clear. □

Proposition 19.8.3. *A covering map $p : E \to B$ induces isomorphisms $\pi_n(E, e_0) \to \pi_n(B, b_0)$ for $n \geq 2$. In particular $\pi_n(S^1) = 0$ and $\pi_n(\mathbb{RP}^m) \cong \pi_n(S^m)$ for $n \geq 2$.*

Proof. For $n \geq 2$, ∂I^n is homeomorphic to S^{n-1} (by $x \mapsto x/\|x\|_2$ after centring, as in Example 18.6.10), hence path-connected. Let $f : (I^n, \partial I^n) \to (B, b_0)$. The cube is simply connected and locally path-connected, so f has a lift $\tilde{f}$ with $\tilde{f}(0) = e_0$ (Proposition 19.6.1); $\tilde{f}(\partial I^n)$ is a connected subset of the fibre $p^{-1}(b_0)$, which is discrete, so $\tilde{f}(\partial I^n) = e_0$. So p_* is

surjective. If $p \circ \tilde{f}$ is null-homotopic relative to ∂I^n, lift the homotopy (Proposition 19.4.3); the image of the connected set $\partial I^n \times I$ lies in the fibre, hence is e_0, and at time 1 the lift of the constant map is constant. So p_* is injective. Apply this to $\mathbb{R} \to S^1$ ($\mathbb{R}$ is contractible) and $S^m \to \mathbb{RP}^m$. □

Definition 19.8.4. A map $p\colon E \to B$ has the *homotopy lifting property* with respect to Y if for every homotopy $F\colon Y \times I \to B$ and lift $\tilde{F}_0$ of F_0 there is a lift $\tilde{F}$ of F with $\tilde{F}(-,0) = \tilde{F}_0$ (not necessarily unique). It is a *Hurewicz fibration* if this holds for all Y, and a *Serre fibration* if it holds for all cubes $Y = I^k$. The fibre over b_0 is $F = p^{-1}(b_0)$. For a pair (X, A) with $x_0 \in A$ and $n \geq 1$, the *relative homotopy group* $\pi_n(X, A, x_0)$ is the set of homotopy classes of maps $(I^n, \partial I^n, J^{n-1}) \to (X, A, x_0)$, where $J^{n-1} = \overline{\partial I^n \setminus I^{n-1} \times \{0\}}$; it is a group for $n \geq 2$.

Theorem 19.8.5 (Long exact sequences)**.** 1. *For a pair (X, A) there is a long exact sequence*
$$\cdots \to \pi_n(A) \to \pi_n(X) \to \pi_n(X, A) \xrightarrow{\partial} \pi_{n-1}(A) \to \cdots \to \pi_0(X), \text{ with } \partial[f] = [f|_{I^{n-1}\times\{0\}}].$$

2. *For a Serre fibration $p\colon E \to B$ with B path-connected, fibre F and $x_0 \in F$, the map $p_*\colon \pi_n(E, F, x_0) \to \pi_n(B, b_0)$ is an isomorphism for $n \geq 1$, which gives a long exact sequence*
$$\cdots \to \pi_n(F) \to \pi_n(E) \to \pi_n(B) \to \pi_{n-1}(F) \to \cdots \to \pi_0(E) \to 0.$$

Sketch of proof. (1) Exactness at each term is a direct manipulation of maps of cubes, based on the *compression criterion*: a map $f\colon (I^n, \partial I^n, J^{n-1}) \to (X, A, x_0)$ represents 0 if and only if it is homotopic relative to ∂I^n to a map into A. (2) Surjectivity of p_* lifts a map $(I^n, \partial I^n) \to (B, b_0)$, viewed as a homotopy of maps $I^{n-1} \to B$, starting from the constant lift; injectivity lifts a homotopy in the same way, using the homotopy lifting property for $I^{n-1} \times I$ relative to J^{n-1} (which holds for Serre fibrations because $(I^n, J^{n-1}) \cong (I^n, I^{n-1} \times \{0\})$). Substituting into (1) for the pair (E, F) gives the sequence. The omitted verifications are in [Hat02, Theorems 4.3 and 4.41]. □

Theorem 19.8.6 (Whitehead)**.** *A map between connected CW complexes that induces isomorphisms on all homotopy groups π_n, $n \geq 1$, is a homotopy equivalence.*

This is [Hat02, Theorem 4.5].

Counterexample 19.8.7. The theorem needs a *map* inducing the isomorphisms: abstractly isomorphic homotopy groups are not enough. Let $X = \mathbb{RP}^2 \times S^3$ and $Y = S^2 \times \mathbb{RP}^3$. Both have fundamental group $\mathbb{Z}/2\mathbb{Z}$ (Proposition 19.3.5 and Example 19.4.10(2), with $\pi_1(S^k) = 0$ for $k \geq 2$), and both have the universal covering $S^2 \times S^3$, so by Proposition 19.8.3 $\pi_n(X) \cong \pi_n(S^2 \times S^3) \cong \pi_n(Y)$ for all $n \geq 2$. But they are not homotopy equivalent: their second homology groups, which are homotopy invariants, are $H_2(X;\mathbb{Z}) = 0$ and $H_2(Y;\mathbb{Z}) \cong \mathbb{Z}$, by the Künneth formula (Chapter 21); see [Hat02, Example 4.35]. Also X is non-orientable and Y is orientable.

Remark 19.8.8. Higher homotopy groups are hard to compute. By cellular approximation, $\pi_k(S^n) = 0$ for $k < n$ (a map $S^k \to S^n$ is homotopic to a cellular map, which misses the open n-cell and so factors through a point). The *Hopf fibration* $S^3 \to S^2 = \mathbb{CP}^1$, $(z_1, z_2) \mapsto [z_1 : z_2]$, has fibre S^1, and its long exact sequence gives $\pi_3(S^2) \cong \pi_3(S^3) \cong \mathbb{Z}$, although $H_3(S^2) = 0$. Homology (Chapter 21) is much more computable, and the Hurewicz theorem relates the two.

19.9 Exercises

Exercise 19.9.1. Show that if $A \subseteq X$ is a retract, then $i_*\colon \pi_1(A,a) \to \pi_1(X,a)$ is injective, and if A is a deformation retract it is an isomorphism.

Solution. $r_*i_* = (ri)_* = \mathrm{id}$, so i_* is injective. For a deformation retract, $ir \simeq \mathrm{id}_X$ relative to A, so i is a homotopy equivalence and Lemma 19.3.4 applies (the path h is constant since $a \in A$ is fixed). □

Exercise 19.9.2. Let G be a topological group (a group with continuous multiplication and inversion) with identity 1. Show that $\pi_1(G,1)$ is abelian, and that the group law of π_1 coincides with $[\gamma][\delta] = [s \mapsto \gamma(s)\delta(s)]$.

Solution. Let $\gamma \cdot \delta$ denote the pointwise product. For loops, $(\gamma * \delta) \cdot (\gamma' * \delta') = (\gamma \cdot \gamma') * (\delta \cdot \delta')$ exactly, and the constant loop is a two-sided unit for both operations on classes. The Eckmann–Hilton argument of Proposition 19.8.2 shows that the two operations coincide and are commutative. □

Exercise 19.9.3. Show that every map $S^n \to S^1$ with $n \geq 2$ is null-homotopic, and that every map $S^1 \to S^1$ of degree 0 is null-homotopic.

Solution. In both cases $f_*\pi_1 = 0$ (S^n is simply connected for $n \geq 2$; for degree 0 by definition), so f lifts to $\tilde{f}\colon S^n \to \mathbb{R}$ by the lifting criterion. Then $f = p \circ \tilde{f} \simeq p \circ c$ via $p \circ ((1-t)\tilde{f} + tc)$, where c is a constant. □

Exercise 19.9.4. Show that the deck transformation group of $p\colon \mathbb{R} \to S^1$ consists of the translations $t \mapsto t + 4k$, $k \in \mathbb{Z}$, and deduce again $\pi_1(S^1) \cong \mathbb{Z}$ from Proposition 19.6.3.

Solution. Translations by multiples of 4 commute with p, since p has period 4. Conversely, if $\tau\colon \mathbb{R} \to \mathbb{R}$ is continuous with $p \circ \tau = p$, then $\tau(t) - t \in 4\mathbb{Z}$ for every t (Proposition 19.4.5), so $(\tau(t) - t)/4$ is a continuous integer-valued function on the connected space $\mathbb{R}$, hence constant. So the deck group is $\mathbb{Z}$. Since $\mathbb{R}$ is simply connected, p is a universal covering, and Proposition 19.6.3 identifies $\pi_1(S^1)$ with the deck group. □

Exercise 19.9.5. (Borsuk–Ulam in dimension 2.) Show that for every map $f\colon S^2 \to \mathbb{R}^2$ there is x with $f(x) = f(-x)$. (Hint: otherwise $g(x) = (f(x) - f(-x))/\|f(x) - f(-x)\|$ is an odd map $S^2 \to S^1$; restrict to the equator and compare lifts to $\mathbb{R}$ along a half circle.)

Solution. Suppose $f(x) \neq f(-x)$ for all x, and let g be as in the hint, so $g(-x) = -g(x)$. Let $h\colon S^1 \to S^1$ be the restriction of g to the equator. Let $\gamma\colon I \to S^1$ be a loop going once around the equator with $\gamma(s + \frac{1}{2}) = -\gamma(s)$ for $0 \leq s \leq \frac{1}{2}$, and $\tilde{h}$ a lift of $h \circ \gamma$ through p.

On the one hand, γ is null-homotopic in the upper hemisphere, which is homeomorphic to a disc by vertical projection, so $h \circ \gamma = g \circ \gamma$ is null-homotopic in S^1, and by Corollary 19.4.4(2) its lift is a loop: $\tilde{h}(1) = \tilde{h}(0)$. On the other hand: The covering p of Proposition 19.4.5 satisfies $p(t+2) = -p(t)$ (the point $P(t+2)$ is opposite to $P(t)$ on the square). Since h is odd, $p(\tilde{h}(s + \frac{1}{2})) = -p(\tilde{h}(s)) = p(\tilde{h}(s) + 2)$, so $\tilde{h}(s + \frac{1}{2}) - \tilde{h}(s) - 2 \in 4\mathbb{Z}$ for $s \in [0, \frac{1}{2}]$, and by continuity it is a constant $4k$. Then $\tilde{h}(1) - \tilde{h}(0) = 2(2 + 4k) = 4(1 + 2k)$, which is not 0. This contradiction proves the claim. (In the language of Theorem 19.4.6: an odd map $S^1 \to S^1$ has odd degree.) (For instance, at any moment there are two antipodal points on the Earth with the same temperature and pressure.) □

Exercise 19.9.6. Find all connected two-sheeted coverings of $S^1 \vee S^1$, and describe the covering corresponding to the kernel of $\mathbb{Z} * \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Solution. By Theorem 19.6.6, connected two-sheeted coverings correspond to subgroups of index 2 of $\pi_1(S^1 \vee S^1) = F(a, b)$ (Example 19.5.4), up to conjugacy. A subgroup H of index 2 is normal (its left cosets and its right cosets are both H and its complement), so it is the kernel of a surjection $F(a, b) \to F(a, b)/H \cong \mathbb{Z}/2\mathbb{Z}$; such a homomorphism is determined by the images of a and b, which are not both zero, and different homomorphisms to $\mathbb{Z}/2\mathbb{Z}$ have different kernels. So there are three such subgroups. The coverings are graphs with two vertices, each with one incoming and one outgoing a-edge and b-edge, mapping to the wedge. For $a \mapsto 1, b \mapsto 0$: the two a-edges join the two vertices in a circle, and each vertex carries a b-loop. For $a \mapsto 0, b \mapsto 1$: the same with a and b exchanged. For $a, b \mapsto 1$: all four edges join the two vertices. The kernel of $F(a, b) \to \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ is normal of index 4, and the covering is the *Cayley graph* of $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$: four vertices, the elements of the group, with an a-edge from g to $g + (1, 0)$ and a b-edge from g to $g + (0, 1)$. It has 4 vertices and 8 edges; the fundamental group of a connected graph is free on the edges outside a maximal tree [Hat02, Proposition 1A.2], so this subgroup is free of rank $8 - 3 = 5$. □

Chapter 20

Topological Manifolds and Group Actions

20.1 Introduction

A topological manifold is a space that looks locally like $\mathbb{R}^n$. The definition has three axioms: locally Euclidean, Hausdorff and second countable. This chapter explains why each is there, by counterexamples showing that none follows from the other two (Section 20.5), and develops the two main ways of producing manifolds from old ones: quotients by group actions (Sections 20.2 to 20.4) and gluing (Section 20.8). The quotients of spheres by the unit elements of $\mathbb{R}$, $\mathbb{C}$ and the quaternions give the projective spaces and the Hopf fibrations $S^{2n+1} \to \mathbb{CP}^n$ and $S^{4n+3} \to \mathbb{HP}^n$, with their consequences for the homotopy groups of spheres (Section 20.7). Smooth structures come in Chapter 33, and homological invariants such as the dimension and the orientation in Chapter 23. References: [Mun00], [Hat02], [Lee13].

Prerequisites. Chapter 18 (General Topology), Chapter 19 (Homotopy and Covering Spaces), Chapter 7 (Groups) and, for the quaternions, Chapter 8 (Rings and Modules).

Conventions. Groups are written multiplicatively, with identity element e, unless stated otherwise; $\mathbb{R}^n$ and $\mathbb{Z}^n$ are groups under addition. A *neighbourhood* of a point is an open set containing it.

20.2 Topological groups

Definition 20.2.1. A *topological group* is a group G with a topology such that the multiplication $G \times G \to G, (g, h) \mapsto gh$, and the inversion $G \to G, g \mapsto g^{-1}$, are continuous. A *homomorphism* of topological groups is a continuous group homomorphism.

Example 20.2.2.

1. $(\mathbb{R}^n, +)$, $(\mathbb{C}, +)$, the multiplicative groups $\mathbb{R}^\times$, $\mathbb{R}_{>0}$ and $\mathbb{C}^\times$, and the circle $S^1 = \{z \in \mathbb{C} : |z| = 1\}$, with the Euclidean topologies: the operations are polynomial or rational.

2. $\mathrm{GL}_n(\mathbb{R})$ and $\mathrm{GL}_n(\mathbb{C})$, open subsets of the spaces of matrices $\mathbb{R}^{n^2}$ and $\mathbb{C}^{n^2}$ (as the complement of the zero set of det); multiplication is polynomial in the entries and inversion is rational by Cramer's rule. Their subgroups SL_n, $\mathrm{O}(n) = \{A : A^T A = I\}$, $\mathrm{SO}(n)$, $\mathrm{U}(n) = \{A : \bar{A}^T A = I\}$ and $\mathrm{SU}(n)$, with the subspace topologies. $\mathrm{O}(n)$ and $\mathrm{U}(n)$ are compact, being closed and bounded (their columns are unit vectors).

3. Any group with the discrete topology; products of topological groups.

4. $(\mathbb{Q}, +)$ with the subspace topology, a topological group that is neither discrete nor locally compact.

Proposition 20.2.3. *Let G be a topological group.*

1. *For $g \in G$, left translation $h \mapsto gh$, right translation and inversion are homeomorphisms of G.*
2. *An open subgroup H is also closed.*
3. *Let U be a neighbourhood of e with $U = U^{-1}$. The subgroup generated by U is open. If G is connected, it is G.*
4. *The connected component G^0 of e is a closed normal subgroup.*
5. *G is Hausdorff if and only if $\{e\}$ is closed.*

Proof. (1) Left translation by g is continuous, as the composite of $h \mapsto (g, h)$ and multiplication, with continuous inverse left translation by g^{-1}; similarly on the right, and inversion is its own inverse. (2) The complement of H is the union of the cosets gH, $g \notin H$, which are open by (1). (3) The subgroup H generated by U is the union of the sets $u_1 \cdots u_k U$ with $u_i \in U$, which are open by (1). By (2) it is also closed, and nonempty, so $H = G$ if G is connected. (4) G^0G^0 is the image of the connected set $G^0 \times G^0$ under multiplication, hence connected, and it contains e; so $G^0G^0 \subseteq G^0$. Similarly $(G^0)^{-1} \subseteq G^0$ and $gG^0g^{-1} \subseteq G^0$ for all g. Components are closed. (5) If G is Hausdorff, points are closed. Conversely, if $\{e\}$ is closed, so is the diagonal $\{(g, h) : gh^{-1} = e\}$, the preimage of $\{e\}$ under the continuous map $(g, h) \mapsto gh^{-1}$. If $g \neq h$, then (g, h) lies in the open complement of the diagonal, so some product neighbourhood $A \times B$ of (g, h) misses the diagonal, that is, $A \cap B = \emptyset$. □

Counterexample 20.2.4. Any group with the indiscrete topology, whose only open sets are $\emptyset$ and G, is a topological group, and it is not Hausdorff if $|G| > 1$. So (5) is a genuine condition. More generally, for any normal subgroup N the quotient G/N with the quotient topology is a topological group, and it is Hausdorff if and only if N is closed (Exercise 20.9.1).

20.3 Group actions and orbit spaces

Definition 20.3.1. A (left) *action* of a topological group G on a space X is a continuous map $G \times X \to X$, $(g, x) \mapsto g \cdot x = gx$, with $ex = x$ and $g(hx) = (gh)x$. The *orbit* of x is $Gx = \{gx : g \in G\}$, the *stabiliser* is $G_x = \{g : gx = x\}$, and the *orbit space* X/G is the set of orbits with the quotient topology for the projection $\pi\colon X \to X/G$. The action is *free* if all stabilisers are trivial, and *transitive* if there is only one orbit.

Each $g \in G$ acts by a homeomorphism of X, with inverse the action of g^{-1}. The orbits partition X, and $gG_xg^{-1} = G_{gx}$.

Proposition 20.3.2. *Let G act on X, with $\pi\colon X \to X/G$.*

1. *π is open: for U open, $\pi^{-1}(\pi(U)) = \bigcup_{g \in G} gU$ is open.*
2. *X/G is Hausdorff if and only if the* orbit relation *$R = \{(x, gx) : x \in X, g \in G\}$ is closed in $X \times X$.*
3. *If X is second countable, so is X/G; if X is compact, connected or path-connected, so is X/G.*
4. *If G is compact and X Hausdorff, then X/G is Hausdorff, π is closed, and for each x the map $G/G_x \to Gx$, $gG_x \mapsto gx$, is a homeomorphism.*

Proof. (1) Each gU is open. (2) Suppose R is closed and $\pi(x) \neq \pi(y)$. Then $(x, y) \notin R$, so there are open $U \ni x$ and $V \ni y$ with $(U \times V) \cap R = \emptyset$. The sets $\pi(U)$ and $\pi(V)$ are open by (1) and disjoint: if $\pi(u) = \pi(v)$ then $(u, v) \in R$. Conversely, if X/G is Hausdorff, its diagonal is closed (a point (a, b) with $a \neq b$ has the neighbourhood $A \times B$, for disjoint neighbourhoods $A \ni a$, $B \ni b$, which misses the diagonal), and R is its preimage under the continuous map $\pi \times \pi$. (3) The images of a countable basis form a countable basis, since π is open, continuous and surjective; continuous images of compact, connected or path-connected spaces have the same property.

(4) We first show that projections along a compact factor are closed: if K is compact, Y any space and $C \subseteq K \times Y$ closed, then $\mathrm{pr}_Y(C)$ is closed. Indeed, if $y \notin \mathrm{pr}_Y(C)$, each $k \in K$ has a product neighbourhood $A_k \times B_k$ of (k, y) disjoint from C; finitely many $A_{k_1}, \ldots, A_{k_r}$ cover K, and $B = \bigcap_j B_{k_j}$ is a neighbourhood of y with $(K \times B) \cap C = \emptyset$, so B misses $\mathrm{pr}_Y(C)$. Now let $C \subseteq X$ be closed. Then $\pi^{-1}(\pi(C)) = GC$ is the projection to X of $\{(g, x) : g^{-1}x \in C\}$, which is closed in $G \times X$; so GC is closed, and π is a closed map. Next, as X is Hausdorff its diagonal Δ_X is closed (as in (2)), and R is the projection to $X \times X$ of the closed set $\{(g, x, y) : (gx, y) \in \Delta_X\} \subseteq G \times X \times X$; so R is closed and X/G is Hausdorff by (2). Finally $g \mapsto gx$ induces a continuous bijection from G/G_x, which is compact as the image of G, to the Hausdorff space Gx; it is a homeomorphism by Proposition 18.6.3(4). □

For the trivial group $G = \{e\}$, $X/G = X$ and the orbit relation is the diagonal $\Delta_X = \{(x, x)\}$, so (2) says: *X is Hausdorff if and only if the diagonal is closed in $X \times X$*. This is the most useful form of the Hausdorff property: for instance, if $f, g \colon Z \to X$ are continuous and X is Hausdorff, the set $\{z : f(z) = g(z)\}$, the preimage of Δ_X under (f, g), is closed.

Example 20.3.3.

1. $\mathbb{Z}$ acts on $\mathbb{R}$ by translations. The map $t \mapsto e^{2\pi i t}$ induces a continuous bijection $\mathbb{R}/\mathbb{Z} \to S^1$, which is a homeomorphism because $\mathbb{R}/\mathbb{Z}$ is compact, the image of $[0, 1]$. Likewise $\mathbb{R}^n/\mathbb{Z}^n \cong (S^1)^n = T^n$, the n-torus.

2. $\{\pm 1\}$ acts on S^n by $x \mapsto \pm x$, and $S^n/\{\pm 1\} = \mathbb{RP}^n$, the real projective space.

3. $\mathbb{R}^\times$ acts on $\mathbb{R}^{n+1} \setminus \{0\}$ by scalar multiplication, and the orbit space is again $\mathbb{RP}^n$; $\mathbb{C}^\times$ acting on $\mathbb{C}^{n+1} \setminus \{0\}$ gives the complex projective space $\mathbb{CP}^n$, and S^1 acting on $S^{2n+1} \subseteq \mathbb{C}^{n+1}$ gives $\mathbb{CP}^n$ as well (Proposition 20.6.3).

4. $\mathrm{O}(n)$ acts transitively on S^{n-1}, with the stabiliser of e_n equal to $\mathrm{O}(n-1)$ (embedded in the upper left corner). By Proposition 20.3.2(4), $S^{n-1} \cong \mathrm{O}(n)/\mathrm{O}(n-1)$; similarly $S^{2n-1} \cong \mathrm{U}(n)/\mathrm{U}(n-1)$.

Lemma 20.3.4 (Kronecker)**.** *If $\alpha \in \mathbb{R}$ is irrational, the set $\{n\alpha - \lfloor n\alpha \rfloor : n \in \mathbb{Z}\}$ of fractional parts is dense in $[0, 1]$.*

Proof. Let $N \geq 1$. The $N + 1$ fractional parts of $0, \alpha, \ldots, N\alpha$ lie in $[0, 1)$, which is the union of the N intervals $[j/N, (j+1)/N)$, so two of them, of $k\alpha$ and $l\alpha$ with $k < l$, lie in the same interval. Then $m = l - k$ satisfies $0 < |m\alpha - p| < 1/N$ for some integer p; the first inequality holds because α is irrational. Replacing m, p by $-m, -p$ if necessary, we may assume that $\beta = m\alpha - p$ satisfies $0 < \beta < 1/N$. The numbers $j\beta$, $j = 0, 1, 2, \ldots$, increase in steps of $\beta < 1/N$, so every interval $[c, c + 1/N) \subseteq [0, 1)$ contains some $j\beta$; and for $0 \leq j\beta < 1$, $j\beta = (jm)\alpha - jp$ is the fractional part of $(jm)\alpha$. As N is arbitrary, the set is dense. □

Counterexample 20.3.5. Orbit spaces can be very bad. Let α be irrational and let $\mathbb{Z}$ act on S^1 by $n \cdot z = e^{2\pi i n \alpha} z$. By Lemma 20.3.4 every orbit is dense. If $U \subseteq S^1$ is open, saturated and nonempty, then for every $y \in S^1$ the orbit of y meets U, so $y \in U$; hence $U = S^1$. The orbit space $S^1/\mathbb{Z}$ is therefore an uncountable set with the indiscrete topology: it is not Hausdorff, and every map from it to a Hausdorff space is constant. The action is free.

20.4 Covering space actions and quotients

From now on in this section G is a group acting on X by homeomorphisms, that is, a topological group with the discrete topology.

Definition 20.4.1. An action of a discrete group G on X is

1. a *covering space action* if every $x \in X$ has a neighbourhood U with $gU \cap U = \emptyset$ for all $g \neq e$;
2. *proper* if for every compact $K \subseteq X$ the set $\{g \in G : gK \cap K \neq \emptyset\}$ is finite.

A covering space action is free. An action of a finite group is proper.

Theorem 20.4.2. *Let G act on X by a covering space action. Then $\pi\colon X \to X/G$ is a covering map. If X is path-connected, then π is a normal covering whose deck transformation group is G, and $\pi_1(X/G)/\pi_*\pi_1(X) \cong G$; in particular $\pi_1(X/G) \cong G$ if X is simply connected.*

Proof. Let $x \in X$ and U as in Definition 20.4.1. Then $\pi^{-1}(\pi(U)) = \bigsqcup_{g\in G} gU$, a disjoint union of open sets: if $gU \cap hU \neq \emptyset$ then $h^{-1}gU \cap U \neq \emptyset$, so $g = h$. The restriction of π to gU is continuous, open (Proposition 20.3.2(1)) and injective, since $\pi(gu) = \pi(gu')$ means $gu' = hgu$ for some h, whence $g^{-1}hg = e$ and $u = u'$. So $\pi|_{gU}$ is a homeomorphism onto the open set $\pi(U)$, and $\pi(U)$ is evenly covered.

Each g is a deck transformation, since $\pi(gx) = \pi(x)$, and G acts transitively on each fibre $\pi^{-1}(\pi(x)) = Gx$. If X is path-connected, a deck transformation is determined by the image of one point (Chapter 19), so every deck transformation τ equals the g with $gx = \tau(x)$, and $g \mapsto (x \mapsto gx)$ is an isomorphism $G \to \mathrm{Deck}(\pi)$, injective because the action is free. The deck group acts transitively on the fibres, so the covering is normal, and Proposition 19.6.3(2) gives $\pi_1(X/G)/\pi_*\pi_1(X) \cong \mathrm{Deck}(\pi) \cong G$. □

Theorem 20.4.3. *Let a discrete group G act freely and properly on a locally compact Hausdorff space X. Then the action is a covering space action and X/G is Hausdorff.*

Proof. Let $x \in X$ and let K be a compact neighbourhood of x (containing an open neighbourhood of x). The set $F = \{g \neq e : gK \cap K \neq \emptyset\}$ is finite. For $g \in F$, $gx \neq x$ as the action is free, so there are disjoint open sets $V_g \ni x$ and $W_g \ni gx$. Let $U = \operatorname{int} K \cap \bigcap_{g\in F}(V_g \cap g^{-1}W_g)$, an open neighbourhood of x. For $g \neq e$ not in F, $gU \cap U \subseteq gK \cap K = \emptyset$; for $g \in F$, $gU \subseteq W_g$ and $U \subseteq V_g$, which are disjoint. So the action is a covering space action.

For the Hausdorff property we show that the orbit relation R is closed (Proposition 20.3.2(2)). Let $(x, y) \notin R$ and choose compact neighbourhoods K_x of x and K_y of y. With $K = K_x \cup K_y$, the set $F = \{g : gK_x \cap K_y \neq \emptyset\}$ is contained in $\{g : gK \cap K \neq \emptyset\}$, hence finite. For $g \in F$, $gx \neq y$, so there are disjoint open $A_g \ni gx$ and $B_g \ni y$. Put $U = \operatorname{int} K_x \cap \bigcap_{g\in F} g^{-1}A_g$ and $V = \operatorname{int} K_y \cap \bigcap_{g\in F} B_g$. For every g, $gU \cap V = \emptyset$: if $g \notin F$ because $gK_x \cap K_y = \emptyset$, and if $g \in F$ because $gU \subseteq A_g$ and $V \subseteq B_g$. Hence $(U \times V) \cap R = \emptyset$. □

Counterexample 20.4.4. A covering space action need not have a Hausdorff quotient. Let $\mathbb{Z}$ act on $X = \mathbb{R}^2 \setminus \{0\}$ by $n \cdot (x, y) = (2^n x, 2^{-n} y)$. It is a covering space action: a point with $x \neq 0$ has the neighbourhood $\{(x', y') : |x|/\sqrt{2} < |x'| < \sqrt{2}|x|\}$, which is moved off itself by every $n \neq 0$, and similarly if $y \neq 0$. But the points $p = (1, 0)$ and $q = (0, 1)$ lie in different orbits, while $(1, 2^{-k}) \to p$ and $(-k) \cdot (1, 2^{-k}) = (2^{-k}, 1) \to q$; so (p, q) lies in the closure of the orbit relation, which is not closed, and $X/\mathbb{Z}$ is not Hausdorff. The action is free but not proper: the compact set $K = \{(x, y) : \max(|x|, |y|) = 1\}$ meets its translates by all n.

Example 20.4.5. 1. $\mathbb{Z}^n$ acts freely and properly on $\mathbb{R}^n$ by translations; $T^n = \mathbb{R}^n/\mathbb{Z}^n$ and $\pi_1(T^n) \cong \mathbb{Z}^n$, as $\mathbb{R}^n$ is simply connected.

2. $\{\pm 1\}$ acts freely on S^n, so $\pi_1(\mathbb{RP}^n) \cong \mathbb{Z}/2$ for $n \geq 2$, since S^n is simply connected for $n \geq 2$ (Chapter 19).

3. (Lens spaces) Let $p \geq 2$ and q be coprime integers, $\zeta = e^{2\pi i/p}$. The group $\mathbb{Z}/p$ acts freely on $S^3 \subseteq \mathbb{C}^2$ by $k \cdot (z_1, z_2) = (\zeta^k z_1, \zeta^{qk} z_2)$: if $\zeta^k z_1 = z_1$ and $\zeta^{qk} z_2 = z_2$ with $(z_1, z_2) \neq 0$, then $\zeta^k = 1$ or $\zeta^{qk} = 1$, and either forces $p \mid k$ as q is prime to p. The quotient $L(p, q) = S^3/(\mathbb{Z}/p)$ is the *lens space*, with $\pi_1 \cong \mathbb{Z}/p$.

4. (Klein bottle) The transformations $a(x, y) = (x + \frac{1}{2}, -y)$ and $b(x, y) = (x, y + 1)$ of $\mathbb{R}^2$ generate a group G acting freely and properly, with $aba^{-1} = b^{-1}$; the quotient $\mathbb{R}^2/G$ is the Klein bottle, with $\pi_1 \cong G$, a non-abelian group.

20.5 Topological manifolds

Definition 20.5.1. Let $n \geq 0$. A space M is *locally Euclidean of dimension n* if every point has a neighbourhood homeomorphic to an open subset of $\mathbb{R}^n$ (equivalently, to $\mathbb{R}^n$ itself, since every point of an open subset of $\mathbb{R}^n$ has a ball neighbourhood, and balls are homeomorphic to $\mathbb{R}^n$). A *topological manifold of dimension n*, or *n-manifold*, is a space that is

1. locally Euclidean of dimension n,

2. Hausdorff, and

3. second countable.

A *chart* is a homeomorphism $\varphi \colon U \to \varphi(U)$ from an open $U \subseteq M$ onto an open subset of $\mathbb{R}^n$.

That the dimension of a nonempty manifold is determined by the space is a theorem, proved with homology (Chapter 23). The next three counterexamples show that none of the three axioms follows from the other two.

Counterexample 20.5.2 (Not Hausdorff)**.** The *line with two origins* L is the quotient of $\mathbb{R} \times \{1, 2\}$ by $(t, 1) \sim (t, 2)$ for $t \neq 0$ (Counterexample 18.4.12). The quotient map is open, the two copies of $\mathbb{R}$ map homeomorphically onto open subsets covering L, so L is locally Euclidean of dimension 1 and second countable. But the two origins $0_1, 0_2$ have no disjoint neighbourhoods: every neighbourhood of 0_i contains the images of $(-\varepsilon, \varepsilon) \setminus \{0\}$ for some $\varepsilon > 0$. In L the sequence $1/n$ converges to both origins, so limits are not unique.

Counterexample 20.5.3 (Not second countable)**.** Let I be an uncountable set with the discrete topology, and $M = I \times \mathbb{R}$, a disjoint union of uncountably many lines. It is locally Euclidean of dimension 1 and Hausdorff. It is not second countable: a basis must contain, for each $i \in I$, a nonempty open subset of $\{i\} \times \mathbb{R}$, and these are pairwise distinct. Connected examples also exist: the *long line* and the *Prüfer surface* are connected, Hausdorff and locally Euclidean of dimensions 1 and 2 but not second countable [Mun00]. Without second countability, partitions of unity (Theorem 18.10.6) and embeddings in $\mathbb{R}^N$ can fail.

Lemma 20.5.4. *Let $n \geq 2$, $W \subseteq \mathbb{R}^n$ open and connected, and $p \in W$. Then $W \setminus \{p\}$ is connected. For $n = 1$, removing a point from an open interval leaves exactly two components.*

Proof. $\mathbb{R}^n \setminus \{0\}$ is path-connected for $n \geq 2$: two points a, b are joined by the segment $[a, b]$ unless 0 lies on it, and then by the segments $[a, c]$ and $[c, b]$ for any c not on the line through a and b. Hence every punctured ball $B \setminus \{p\}$, homeomorphic to $\mathbb{R}^n \setminus \{0\}$, is path-connected. Choose a ball B with $p \in B \subseteq W$ and let A be the set of points of $W \setminus \{p\}$ that can be joined to a point of $B \setminus \{p\}$ by a path in $W \setminus \{p\}$. Every point of $W \setminus \{p\}$ has a ball neighbourhood

in $W \setminus \{p\}$, which is path-connected; so A and its complement in $W \setminus \{p\}$ are open. If the complement were nonempty, then $W = (A \cup B) \cup (W \setminus (A \cup \{p\}))$ would be a decomposition into disjoint nonempty open sets ($A \cup B$ is open, and $B \setminus \{p\} \subseteq A$), contradicting connectedness. So $W \setminus \{p\} = A$ is path-connected. The case $n = 1$ is clear, as connected subsets of $\mathbb{R}$ are intervals (Theorem 18.5.3). □

Counterexample 20.5.5 (Not locally Euclidean)**.** The union $X = \{(x, y) \in \mathbb{R}^2 : xy = 0\}$ of the two coordinate axes is Hausdorff and second countable, being a subspace of $\mathbb{R}^2$, but not locally Euclidean at the origin. Suppose $\varphi \colon U \to \varphi(U) \subseteq \mathbb{R}^n$ were a chart at 0. Choose $\varepsilon > 0$ with $V = X \cap (-\varepsilon, \varepsilon)^2 \subseteq U$. Then V is a cross with four open arms, open in X and connected, and $V \setminus \{0\}$ has exactly four components, the arms. So $W = \varphi(V)$ is a connected open subset of $\mathbb{R}^n$, and $W \setminus \{\varphi(0)\}$, homeomorphic to $V \setminus \{0\}$, has four components. This contradicts Lemma 20.5.4 for $n \geq 1$, and $n = 0$ is impossible because $\{0\}$ is not open in X.

Similarly the closed half-line $[0, +\infty)$ is not locally Euclidean at 0. If φ were a chart at 0 with values in $\mathbb{R}^n$: $n = 0$ is impossible as before; for $n = 1$, a connected neighbourhood $[0, \varepsilon) \subseteq U$ would map to an open interval, which is disconnected by removing the image of 0, while $(0, \varepsilon)$ is connected; for $n \geq 2$, a small interval $J \subseteq (0, \varepsilon)$ would map to a connected open subset of $\mathbb{R}^n$ that stays connected after removing a point, while J does not. It is a manifold with boundary (Definition 20.5.9).

Proposition 20.5.6. *Let M be an n-manifold.*

1. *M is locally compact and locally path-connected. Its connected components are open, coincide with its path components, and are countable in number.*
2. *Every open subset of M is an n-manifold.*
3. *M is paracompact and admits partitions of unity subordinate to any open cover.*
4. *M has a countable cover by open sets B_i with homeomorphisms $B_i \cong \mathbb{R}^n$ (*coordinate balls*).*

Proof. (1) Each point has a neighbourhood homeomorphic to $\mathbb{R}^n$, and closed balls and open balls in $\mathbb{R}^n$ give compact and path-connected neighbourhoods. In a locally path-connected space, path components are open, hence also closed, and so they are the components. They are disjoint open sets, each containing an element of a countable basis, so there are countably many. (2) The three axioms pass to open subspaces. (3) By (1) and Hausdorffness M is locally compact Hausdorff and second countable, so Proposition 18.10.7 and Theorem 18.10.6 apply. (4) Let $\mathcal{B}$ be a countable basis and $\mathcal{B}'$ the set of $W \in \mathcal{B}$ contained in the domain of some chart; for each $W \in \mathcal{B}'$ fix a chart φ_W whose domain contains W. The sets $\varphi_W^{-1}(B)$, for $W \in \mathcal{B}'$ and B an open ball with rational centre and radius contained in $\varphi_W(W)$, are countably many coordinate balls. They cover M: a point x lies in a chart domain, hence in some $W \in \mathcal{B}$ inside it, so $W \in \mathcal{B}'$, and $\varphi_W(x)$ lies in a rational ball contained in the open set $\varphi_W(W)$. □

Theorem 20.5.7. *The fundamental group of a connected manifold is countable.*

Proof. Let $\{B_i\}$ be a countable cover of M by coordinate balls (Proposition 20.5.6(4)); each B_i is simply connected, being homeomorphic to $\mathbb{R}^n$. For each pair i, j, the open set $B_i \cap B_j$ has countably many path components (by Proposition 20.5.6(1)); choose a point in each, and let S be the countable set of all chosen points, together with a base point p. For any two points $s, s' \in S$ lying in a common B_i, choose a path $\sigma^i_{s,s'}$ in B_i from s to s'; these are countably many paths. We claim that every loop γ at p is homotopic to a finite concatenation of such paths; since there are countably many finite concatenations, this proves the theorem.

By the Lebesgue number lemma there is a partition $0 = t_0 < t_1 < \cdots < t_k = 1$ with $\gamma([t_{m-1}, t_m]) \subseteq B_{i_m}$. We may take $p \in B_{i_1} \cap B_{i_k}$. For $0 < m < k$, the point $\gamma(t_m)$ lies in

$B_{i_m} \cap B_{i_{m+1}}$, in a path component containing some $s_m \in S$; choose a path τ_m in that component from $\gamma(t_m)$ to s_m, and let τ_0, τ_k be constant at $p = s_0 = s_k$. Then γ is homotopic to the concatenation over m of the paths $\bar{\tau}_{m-1} \cdot \gamma|_{[t_{m-1},t_m]} \cdot \tau_m$, each of which runs in B_{i_m} from s_{m-1} to s_m. As B_{i_m} is simply connected, each is homotopic relative to its endpoints to $\sigma^{i_m}_{s_{m-1},s_m}$. □

Corollary 20.5.8. *If M is a connected n-manifold and $p\colon E \to M$ a covering map with E connected, then E is an n-manifold.*

Proof. E is locally homeomorphic to M, hence locally Euclidean. It is Hausdorff: two points in different fibres are separated by preimages of disjoint neighbourhoods in M, and two points in the same fibre lie in different sheets over an evenly covered neighbourhood. The fibres are in bijection with the cosets of $p_*\pi_1(E)$ in the countable group $\pi_1(M)$ (Chapter 19), hence countable. If $\{W_k\}$ is a countable basis of M consisting of evenly covered sets, then the sheets over the W_k form a countable family of open sets, and the sheets over basic open subsets of the W_k give a countable basis of E. □

If M is a smooth manifold, E has a unique smooth structure for which p is a local diffeomorphism, obtained by composing the charts of M with the local inverses of p (Chapter 33); this is how $S^n \to \mathbb{RP}^n$, $\mathbb{R}^n \to T^n$ and universal coverings of smooth manifolds are treated as smooth maps.

Definition 20.5.9. Let $\mathbb{H}^n = \{x \in \mathbb{R}^n : x_n \geq 0\}$. An *$n$-manifold with boundary* is a Hausdorff, second countable space in which every point has a neighbourhood homeomorphic to an open subset of $\mathbb{H}^n$. A point is an *interior point* if it has a neighbourhood homeomorphic to $\mathbb{R}^n$, and a *boundary point* otherwise; the boundary is written ∂M.

A point sent by a chart to a point with $x_n > 0$ is interior. That a point sent by some chart to $\{x_n = 0\}$ is not interior, so that ∂M is the set of such points and ∂M is an $(n-1)$-manifold, uses invariance of domain (Chapter 23); for $n = 1$ it follows from Lemma 20.5.4.

Example 20.5.10. The closed disc D^n is an n-manifold with boundary S^{n-1}; $[0,1]$ has boundary $\{0,1\}$; the *Möbius band*, the quotient of $[0,1] \times [-1,1]$ by $(0,t) \sim (1,-t)$, has boundary a single circle.

20.6 Constructions of manifolds

Proposition 20.6.1. 1. *Open subsets of n-manifolds and countable disjoint unions of n-manifolds are n-manifolds.*

2. *If M is an m-manifold and N an n-manifold, $M \times N$ is an $(m+n)$-manifold.*

3. *If a discrete group G acts freely and properly on an n-manifold M, then M/G is an n-manifold, and $M \to M/G$ is a covering map.*

4. *Connected covering spaces of connected n-manifolds are n-manifolds.*

Proof. (1) Clear. (2) Products of charts are charts, $\mathbb{R}^m \times \mathbb{R}^n = \mathbb{R}^{m+n}$, and products of Hausdorff or second countable spaces are Hausdorff or second countable. (3) By Theorem 20.4.3 and Proposition 20.5.6(1) the quotient is Hausdorff and π is a covering map, so M/G is locally homeomorphic to M; it is second countable by Proposition 20.3.2(3). (4) is Corollary 20.5.8. □

Example 20.6.2. 1. $\mathbb{R}^n$; open subsets such as $\mathrm{GL}_n(\mathbb{R}) \subseteq \mathbb{R}^{n^2}$.

2. The sphere $S^n \subseteq \mathbb{R}^{n+1}$, with the two stereographic charts (Example 18.9.4); it is Hausdorff and second countable as a subspace of $\mathbb{R}^{n+1}$.

3. The graph $\{(x, f(x))\}$ of a continuous map $f\colon U \to \mathbb{R}^k$, $U \subseteq \mathbb{R}^n$ open, is an n-manifold, homeomorphic to U by the projection.

4. The torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$, $\mathbb{RP}^n = S^n/\{\pm 1\}$, the lens spaces and the Klein bottle of Example 20.4.5, by Proposition 20.6.1(3).

Proposition 20.6.3. *$\mathbb{CP}^n = (\mathbb{C}^{n+1} \setminus \{0\})/\mathbb{C}^\times$ is a compact $2n$-manifold, homeomorphic to S^{2n+1}/S^1. The sets $U_i = \{[z] : z_i \neq 0\}$ are open and $\varphi_i([z]) = (z_0/z_i, \dots, \widehat{z_i/z_i}, \dots, z_n/z_i)$ is a homeomorphism $U_i \to \mathbb{C}^n$. The same holds for $\mathbb{RP}^n$, with $\mathbb{R}$ in place of $\mathbb{C}$.*

Proof. The inclusion $S^{2n+1} \to \mathbb{C}^{n+1} \setminus \{0\}$ induces a continuous map $S^{2n+1}/S^1 \to \mathbb{CP}^n$, which is bijective since every line meets the sphere in a circle, and has continuous inverse induced by $z \mapsto z/|z|$. By Proposition 20.3.2(3),(4) applied to the compact group S^1, $\mathbb{CP}^n$ is compact, Hausdorff and second countable. The set U_i is open because its preimage $\{z_i \neq 0\}$ is. The map φ_i is well defined and continuous (it is induced by a continuous map constant on orbits), with continuous inverse $w \mapsto [w_0 : \cdots : 1 : \cdots : w_n]$, the 1 in position i. □

Example 20.6.4. $\mathbb{CP}^1$ is homeomorphic to S^2, by an explicit map. Identify $\mathbb{R}^3$ with $\mathbb{C} \times \mathbb{R}$ and let

$$h\colon \mathbb{CP}^1 \to S^2, \qquad h([z_0 : z_1]) = \frac{(2\bar{z}_0 z_1,\ |z_1|^2 - |z_0|^2)}{|z_0|^2 + |z_1|^2}.$$

Multiplying (z_0, z_1) by $\lambda \in \mathbb{C}^\times$ multiplies numerator and denominator by $|\lambda|^2$, so h is well defined, and it is continuous since it is induced by a continuous map on $\mathbb{C}^2 \setminus \{0\}$. It takes values in S^2, because $|2\bar{z}_0 z_1|^2 + (|z_1|^2 - |z_0|^2)^2 = (|z_0|^2 + |z_1|^2)^2$. In the chart $U_0 = \{z_0 \neq 0\}$, with $w = z_1/z_0$,

$$h([1 : w]) = \frac{(2w,\ |w|^2 - 1)}{|w|^2 + 1},$$

which is the inverse of the stereographic projection from $N = (0, 1)$ (Example 18.9.4, with $\mathbb{R}^2 = \mathbb{C}$). So h maps U_0 bijectively onto $S^2 \setminus \{N\}$, and the remaining point $[0 : 1]$ goes to N. Thus h is a continuous bijection from the compact space $\mathbb{CP}^1$ to the Hausdorff space S^2, hence a homeomorphism: $\mathbb{CP}^1$ is the one-point compactification of the chart $U_0 \cong \mathbb{C}$, the *Riemann sphere*. The composite $S^3 \to \mathbb{CP}^1 \to S^2$, $(z_0, z_1) \mapsto (2\bar{z}_0 z_1, |z_1|^2 - |z_0|^2)$, is the Hopf map of Remark 19.8.8. For $\mathbb{RP}^1$ the same construction gives $\mathbb{RP}^1 \cong S^1$, the one-point compactification of $\mathbb{R}$.

20.7 Hopf fibrations

The quotient map $S^{2n+1} \to \mathbb{CP}^n$ of Proposition 20.6.3 has all its fibres equal to circles, and near each point it looks like a product. Replacing $\mathbb{C}$ by $\mathbb{R}$ or by the quaternions gives a whole family of such maps, the *Hopf fibrations*: $S^n \to \mathbb{RP}^n$ with fibre S^0, $S^{2n+1} \to \mathbb{CP}^n$ with fibre S^1, and $S^{4n+3} \to \mathbb{HP}^n$ with fibre S^3. For $n = 1$ the base is a sphere, and one gets the maps $S^1 \to S^1$, $S^3 \to S^2$ and $S^7 \to S^4$, to which the octonions add $S^{15} \to S^8$. They are the first nontrivial examples of fibre bundles, and through the long exact sequence of homotopy groups they show that a sphere can have nonzero homotopy groups above its dimension.

Definition 20.7.1. A *fibre bundle* with fibre F is a continuous map $p\colon E \to B$ such that every point of B has an open neighbourhood U with a homeomorphism $\phi\colon U \times F \to p^{-1}(U)$ satisfying $p \circ \phi = \mathrm{pr}_U$, a *local trivialisation*.

A covering space is a fibre bundle with discrete fibre, and a vector bundle (Chapter 35) is a fibre bundle with fibre a vector space and linear transition maps.

Theorem 20.7.2. *A fibre bundle $p\colon E \to B$ is a Serre fibration (Definition 19.8.4). Hence, for B path-connected and $F = p^{-1}(b_0)$, there is a long exact sequence $\cdots \to \pi_k(F) \to \pi_k(E) \to \pi_k(B) \to \pi_{k-1}(F) \to \cdots$ (Theorem 19.8.5(2)).*

This is [Hat02, Proposition 4.48]: the homotopy lifting property for cubes is local on the base, as for coverings, and over a set U where E is trivial a homotopy into $U \times F$ is lifted by keeping the F-component of the initial lift constant; the work lies in subdividing the cube so that each piece maps into a trivialising set.

Let $\mathbb{K}$ be $\mathbb{R}$, $\mathbb{C}$ or the quaternions $\mathbb{H}$ (Example 8.2.3), of real dimension $d = 1, 2$ or 4. Each has a conjugation $\lambda \mapsto \bar\lambda$ and the multiplicative norm $|\lambda| = (\lambda\bar\lambda)^{1/2}$; for $\mathbb{H}$, viewed as matrices $\left(\begin{smallmatrix} z & -\bar w \\ w & \bar z \end{smallmatrix}\right)$, $\bar\lambda$ is the conjugate transpose and $|\lambda|^2$ is the determinant $|z|^2 + |w|^2$, so $|\lambda\mu| = |\lambda||\mu|$, and $\mathbb{H} \cong \mathbb{R}^4$ isometrically. The unit elements form the group S^{d-1}: $S^0 = \{\pm 1\}$, S^1, and $S^3 = \mathrm{SU}(2)$. On $\mathbb{K}^{n+1}$, with the norm $|q|^2 = \sum_i |q_i|^2$, the group $\mathbb{K}^\times$ acts by multiplication on the right, $q \mapsto q\lambda = (q_0\lambda, \dots, q_n\lambda)$, preserving lines, and $|q\lambda| = |q||\lambda|$. The *projective space*

$$\mathbb{KP}^n = (\mathbb{K}^{n+1} \setminus \{0\})/\mathbb{K}^\times$$

is the set of right $\mathbb{K}$-lines in $\mathbb{K}^{n+1}$; we write $[q] = [q_0 : \cdots : q_n]$. For $\mathbb{K} = \mathbb{H}$, which is not commutative, the side matters: right lines are preserved by left multiplication by matrices, and coordinates $q_j q_i^{-1}$ are invariant under $q \mapsto q\lambda$.

Proposition 20.7.3 (Hopf fibrations)**.** *Let $\mathbb{K} = \mathbb{R}, \mathbb{C}$ or $\mathbb{H}$, of real dimension d, and $N = d(n+1)-1$, so that S^N is the unit sphere of $\mathbb{K}^{n+1}$.*

1. *$\mathbb{KP}^n$ is a compact manifold of dimension dn, homeomorphic to S^N/S^{d-1}, with the charts $\varphi_i\colon U_i = \{[q] : q_i \neq 0\} \to \mathbb{K}^n$, $\varphi_i([q]) = (q_0 q_i^{-1}, \dots, \widehat{q_i q_i^{-1}}, \dots, q_n q_i^{-1})$.*
2. *The projection $p\colon S^N \to \mathbb{KP}^n$, $p(q) = [q]$, is a fibre bundle with fibre S^{d-1}: over U_i the map $\phi_i\colon U_i \times S^{d-1} \to p^{-1}(U_i)$, $\phi_i([q], \lambda) = \sigma_i([q])\lambda$ with $\sigma_i([q]) = q\, q_i^{-1}|q_i|/|q|$, is a local trivialisation.*

Proof. (1) The proof of Proposition 20.6.3 applies word for word. Every right line meets S^N in an orbit qS^{d-1}, so $S^N/S^{d-1} \to \mathbb{KP}^n$ is a continuous bijection with continuous inverse induced by $q \mapsto q/|q|$, and $\mathbb{KP}^n$ is compact, Hausdorff and second countable as the orbit space of a compact group (Proposition 20.3.2). The map φ_i is well defined, as $(q_j\lambda)(q_i\lambda)^{-1} = q_j\lambda\lambda^{-1}q_i^{-1} = q_j q_i^{-1}$, continuous, and has the continuous inverse $w \mapsto [w_0 : \cdots : 1 : \cdots : w_n]$ with 1 in position i.

(2) The map σ_i is well defined on U_i: replacing q by $q\mu$ replaces $q q_i^{-1}$ by $q\mu\mu^{-1}q_i^{-1} = q q_i^{-1}$ and does not change $|q_i|/|q|$. Its value is the unique unit vector of the line $[q]$ whose i-th coordinate is a positive real number, namely $|q_i|/|q|$; in particular $|\sigma_i([q])| = 1$ and $p \circ \sigma_i = \mathrm{id}$, so $p(\phi_i([q], \lambda)) = [q]$. The map ϕ_i is continuous, and it has the continuous inverse $x \mapsto ([x], |x_i|^{-1}x_i)$ on $p^{-1}(U_i)$: for $x \in S^N$ with $x_i \neq 0$, $\sigma_i([x]) = x\, x_i^{-1}|x_i|$, so $\sigma_i([x])\,|x_i|^{-1}x_i = x$, and conversely the i-th coordinate of $\sigma_i([q])\lambda$ is $\frac{|q_i|}{|q|}\lambda$, which recovers λ by normalising. □

Proposition 20.7.4. *$\mathbb{HP}^1$ is homeomorphic to S^4, by $h([q_0 : q_1]) = \big(2q_0\bar q_1, |q_0|^2 - |q_1|^2\big)/(|q_0|^2 + |q_1|^2) \in \mathbb{H} \times \mathbb{R} = \mathbb{R}^5$. So the Hopf fibration for $\mathbb{H}$ and $n = 1$ is a fibre bundle $S^7 \to S^4$ with fibre S^3, and for $\mathbb{C}$ it is $S^3 \to S^2$ with fibre S^1 (Example 20.6.4).*

Proof. Replacing (q_0, q_1) by $(q_0\lambda, q_1\lambda)$ multiplies $q_0\bar q_1$ by $\lambda\bar\lambda = |\lambda|^2$ (as $\overline{q_1\lambda} = \bar\lambda\bar q_1$) and the denominator by $|\lambda|^2$, so h is well defined, and it is continuous. It takes values in S^4, since

$|2q_0\bar{q}_1|^2 + (|q_0|^2 - |q_1|^2)^2 = (|q_0|^2 + |q_1|^2)^2$ by multiplicativity of the norm. On $U_1 = \{q_1 \neq 0\}$, with $w = q_0 q_1^{-1}$, $[q_0 : q_1] = [w : 1]$ and $h([w : 1]) = (2w, |w|^2 - 1)/(|w|^2 + 1)$, the inverse of the stereographic projection $S^4 \setminus \{N\} \to \mathbb{R}^4 = \mathbb{H}$ from $N = (0, 1)$ (Example 18.9.4). So h maps U_1 bijectively onto $S^4 \setminus \{N\}$, and the remaining point $[1 : 0]$ to N. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism. □

For the real numbers the Hopf fibration $S^n \to \mathbb{RP}^n$ is the two-sheeted covering of Example 20.4.5, and for $n = 1$, with $\mathbb{RP}^1 \cong S^1$, it is the map $z \mapsto z^2$ of the circle. The octonions, an eight-dimensional normed algebra that is not associative, give in the same way a fibre bundle $S^{15} \to S^8$ with fibre S^7 [Hat02, Section 4.2].

Corollary 20.7.5. *1. $\pi_1(\mathbb{CP}^n) = 0$, $\pi_2(\mathbb{CP}^n) \cong \mathbb{Z}$, and $\pi_k(\mathbb{CP}^n) \cong \pi_k(S^{2n+1})$ for $k \geq 3$. In particular $\pi_k(S^2) \cong \pi_k(S^3)$ for $k \geq 3$, so $\pi_3(S^2) \cong \pi_3(S^3)$.*

2. $\pi_k(\mathbb{HP}^n) \cong \pi_{k-1}(S^3)$ for $k \leq 4n + 2$; in particular $\mathbb{HP}^n$ is 3-connected, $\pi_4(\mathbb{HP}^n) \cong \pi_3(S^3)$, and $\pi_k(S^4) \cong \pi_{k-1}(S^3)$ for $k \leq 6$.

3. The map $\pi_7(S^7) \to \pi_7(S^4)$ induced by the Hopf fibration is injective.

Proof. We use the long exact sequences of Theorem 20.7.2, together with $\pi_k(S^m) = 0$ for $k < m$ (cellular approximation, Theorem 19.7.5, as in Remark 19.8.8), $\pi_1(S^1) \cong \mathbb{Z}$ and $\pi_k(S^1) = 0$ for $k \geq 2$ (Proposition 19.8.3). (1) For $S^1 \to S^{2n+1} \to \mathbb{CP}^n$ with $n \geq 1$, the segment $\pi_k(S^1) \to \pi_k(S^{2n+1}) \to \pi_k(\mathbb{CP}^n) \to \pi_{k-1}(S^1)$ has outer terms 0 for $k \geq 3$. For $k = 2$, $\pi_2(S^{2n+1}) = 0 \to \pi_2(\mathbb{CP}^n) \to \pi_1(S^1) \to \pi_1(S^{2n+1}) = 0$, and for $k = 1$, $0 = \pi_1(S^{2n+1}) \to \pi_1(\mathbb{CP}^n) \to \pi_0(S^1) = 0$. Take $n = 1$ and $\mathbb{CP}^1 \cong S^2$ for the last statement. (2) For $S^3 \to S^{4n+3} \to \mathbb{HP}^n$, if $k \leq 4n + 2$ the groups $\pi_k(S^{4n+3})$ and $\pi_{k-1}(S^{4n+3})$ vanish, so $\pi_k(\mathbb{HP}^n) \to \pi_{k-1}(S^3)$ is an isomorphism; and $\pi_{k-1}(S^3) = 0$ for $k \leq 3$. Take $n = 1$ and $\mathbb{HP}^1 \cong S^4$ for the last statement. (3) In the segment $\pi_7(S^3) \to \pi_7(S^7) \to \pi_7(S^4)$, the first map is induced by the inclusion of a fibre $S^3 \to S^7$, which is not surjective; as S^7 minus a point is homeomorphic to $\mathbb{R}^7$, which is contractible, the inclusion is null-homotopic and induces 0. By exactness the second map is injective. □

By the Hurewicz theorem (Chapter 21), $\pi_n(S^n) \cong \mathbb{Z}$, so the corollary gives $\pi_3(S^2) \cong \mathbb{Z}$, generated by the Hopf map itself, $\pi_4(S^4) \cong \mathbb{Z}$, and a copy of $\mathbb{Z}$ in $\pi_7(S^4)$: unlike homology, the homotopy groups of a sphere do not vanish above its dimension. The Hopf maps $S^3 \to S^2$, $S^7 \to S^4$ and $S^{15} \to S^8$ are also the maps of Hopf invariant one (Chapter 24), whose mapping cones are the projective planes $\mathbb{CP}^2$, $\mathbb{HP}^2$ and the octonionic plane. Geometrically, the fibres of $S^3 \to S^2$ are great circles, the intersections of S^3 with the complex lines of $\mathbb{C}^2$; any two of them are linked, and the preimages of the two closed hemispheres are the solid tori $\{|z_0| \leq |z_1|\}$ and $\{|z_0| \geq |z_1|\}$, which decompose S^3 (Exercise 20.9.5). The bundle $S^{2n+1} \to \mathbb{CP}^n$ is the unit circle bundle of the tautological line bundle $O(-1)$, whose nonzero first Chern class shows again that it is not trivial (Chapters 35 and 39).

20.8 Gluing

Lemma 20.8.1. *Let X and Y be spaces, $U \subseteq X$ and $V \subseteq Y$ open, and $f : U \to V$ a homeomorphism. Let $Z = X \cup_f Y$ be the quotient of $X \sqcup Y$ by $u \sim f(u)$ for $u \in U$.*

1. X and Y map homeomorphically onto open subsets of Z covering Z.

2. If X and Y are Hausdorff, then Z is Hausdorff if and only if the graph $\Gamma_f = \{(u, f(u))\}$ is closed in $X \times Y$.

3. *If X and Y are n-manifolds and Γ_f is closed, then Z is an n-manifold.*

Proof. (1) The saturation of an open $A \subseteq X$ is $A \sqcup f(A \cap U)$, which is open, so the quotient map q is open, and it is injective on X and on Y. (2) Two points of Z coming from X are separated by images of disjoint open subsets of X, and similarly for Y. The remaining case is $q(x) \neq q(y)$ with $x \in X \setminus U$, $y \in Y \setminus V$; then $(x, y) \notin \Gamma_f$. If Γ_f is closed, there are open $A \ni x$, $B \ni y$ with $(A \times B) \cap \Gamma_f = \emptyset$, that is $f(A \cap U) \cap B = \emptyset$, and $q(A)$, $q(B)$ are disjoint neighbourhoods. Conversely, if $(x, y) \in \bar{\Gamma}_f \setminus \Gamma_f$, then every pair of neighbourhoods $A \ni x$, $B \ni y$ has $f(A \cap U) \cap B \neq \emptyset$, so $q(A) \cap q(B) \neq \emptyset$; and $q(x) \neq q(y)$ (if $x \in U$ then $y \neq f(x)$ since $(x, y) \notin \Gamma_f$; if $x \notin U$ and $y \notin V$ they are not identified; and $x \notin U$, $y \in V$ is also not identified). (3) Locally Euclidean by (1), Hausdorff by (2), and second countable as the union of two open second countable subspaces. □

Counterexample 20.8.2. The line with two origins is $\mathbb{R} \cup_f \mathbb{R}$ with $U = V = \mathbb{R} \setminus \{0\}$ and $f = \mathrm{id}$. The graph of f is the diagonal minus the origin, whose closure in $\mathbb{R} \times \mathbb{R}$ contains $(0, 0)$; in accordance with Lemma 20.8.1(2), the result is not Hausdorff (Counterexample 20.5.2).

Definition 20.8.3. Let M and N be connected n-manifolds, $n \geq 2$, and let $\varphi \colon U \to \mathbb{R}^n$, $\psi \colon V \to \mathbb{R}^n$ be charts onto $\mathbb{R}^n$. Let $X = M \setminus \varphi^{-1}(\bar{B}_1)$ and $Y = N \setminus \psi^{-1}(\bar{B}_1)$, where $\bar{B}_r = \{|x| \leq r\}$, and let f be the homeomorphism from $\varphi^{-1}(\{1 < |x| < 2\}) \subseteq X$ onto $\psi^{-1}(\{1 < |y| < 2\}) \subseteq Y$ given in coordinates by $x \mapsto (3 - |x|)x/|x|$, which exchanges the two boundary spheres of the annulus. The *connected sum* is $M\#N = X \cup_f Y$.

Proposition 20.8.4. *$M\#N$ is a connected n-manifold, compact if M and N are.*

Proof. X and Y are n-manifolds by Proposition 20.6.1(1). We check that the graph of f is closed in $X \times Y$. Let $(x_k, f(x_k)) \to (x, y) \in X \times Y$ with $1 < |\varphi(x_k)| < 2$; sequences suffice, as $X \times Y$ is first countable. The points $\varphi(x_k)$ lie in the compact set $\{1 \leq |w| \leq 2\}$, so a subsequence converges to some w there, and then the corresponding subsequence of (x_k) converges to $\varphi^{-1}(w)$; by uniqueness of limits in the Hausdorff space M, $x = \varphi^{-1}(w)$, and $|w| > 1$ because $x \in X$. In the same way $y = \psi^{-1}(w')$ with $|w'| > 1$ a limit point of $\psi(f(x_k))$, whose norms are $3 - |\varphi(x_k)|$; so $|w'| = 3 - |w|$, which forces $|w| < 2$. Hence x lies in the domain of f, and $y = f(x)$ by continuity of f. By Lemma 20.8.1, $M\#N$ is an n-manifold.

X is connected: the set $E = \varphi^{-1}(\{|w| > 1\})$ is connected, being homeomorphic to $S^{n-1} \times (1, \infty)$ with S^{n-1} path-connected for $n \geq 2$ (as the image of $\mathbb{R}^n \setminus \{0\}$ under $w \mapsto w/|w|$). If $X = A \sqcup A'$ with A, A' open, we may assume $E \subseteq A$; then $M = (A \cup \varphi^{-1}(B_2)) \sqcup A'$ is a decomposition into disjoint open sets, since the points of $\varphi^{-1}(B_2)$ in X lie in E; as M is connected, $A' = \emptyset$. Similarly Y is connected, and the images of X and Y in $M\#N$ meet, so $M\#N$ is connected. If M and N are compact, the closed subsets $M \setminus \varphi^{-1}(B_{3/2}) \subseteq X$ and $N \setminus \psi^{-1}(B_{3/2}) \subseteq Y$ are compact, and their images cover $M\#N$, because a point of X with $1 < |\varphi| < 3/2$ is identified with a point of Y with $3/2 < |\psi| < 2$. □

Remark 20.8.5. The homeomorphism type of $M\#N$ can depend on the choice of charts, through orientations: for orientable manifolds the two choices of relative orientation can give different results. For surfaces, and more generally up to these orientation choices, the result is independent of the charts, but the proof in the topological category uses deep results; in the smooth category it follows from the disc theorem of Palais. We shall only use connected sums of surfaces, where $T^2\#T^2$ is the closed orientable surface of genus 2 and $\mathbb{RP}^2\#\mathbb{RP}^2$ is the Klein bottle.

Theorem 20.8.6 (Classification of surfaces)**.** *Every compact connected 2-manifold is homeomorphic to exactly one of: the sphere S^2, a connected sum $T^2\#\cdots\#T^2$ of $g \geq 1$ tori, or a connected sum $\mathbb{RP}^2\#\cdots\#\mathbb{RP}^2$ of $k \geq 1$ projective planes.*

This is [Mun00, Chapter 12]. The surfaces in the list are distinguished by their fundamental groups (their abelianisations are $\mathbb{Z}^{2g}$ and $\mathbb{Z}^{k-1} \oplus \mathbb{Z}/2$), and every compact surface is homeomorphic to one obtained by identifying the edges of a polygon in pairs, which is the route to the proof. In dimension 1 the classification is simpler: every connected 1-manifold is homeomorphic to $\mathbb{R}$ or to S^1 [Mil97, Appendix].

20.9 Exercises

Exercise 20.9.1. Let N be a normal subgroup of a topological group G. Show that G/N with the quotient topology is a topological group, and that it is Hausdorff if and only if N is closed.

Solution. The quotient map $\pi\colon G \to G/N$ is open, by the argument of Proposition 20.3.2(1) for the action of N by right translation. So $\pi \times \pi$ is open and surjective, hence a quotient map, and multiplication $G/N \times G/N \to G/N$ is continuous because its composite with $\pi \times \pi$ is. Inversion is similar. By Proposition 20.2.3(5), G/N is Hausdorff if and only if $\{eN\}$ is closed, that is, if and only if $\pi^{-1}(eN) = N$ is closed (the complement of $\{eN\}$ is open if and only if its preimage $G \setminus N$ is). □

Exercise 20.9.2. Show that a compact Hausdorff space that is locally Euclidean is second countable, hence a manifold.

Solution. Finitely many chart domains $U_1, \dots, U_k$ cover the space, and each is second countable, being homeomorphic to an open subset of $\mathbb{R}^n$. The union of countable bases of the U_i is a countable basis. □

Exercise 20.9.3. Show that the Möbius band B of Example 20.5.10 is the quotient of $\mathbb{R} \times [-1,1]$ by the free proper action of $\mathbb{Z}$ given by $n \cdot (x,t) = (x+n, (-1)^n t)$, and deduce that $B \setminus \partial B$ is a 2-manifold whose fundamental group is $\mathbb{Z}$.

Solution. Every orbit meets $[0,1] \times [-1,1]$, in one point or, for $x \in \mathbb{Z}$, in the two points $(0,t)$ and $(1,-t)$; the induced continuous bijection from the band to the orbit space is a homeomorphism since the band is compact and the orbit space Hausdorff (Theorem 20.4.3, which applies because $K \cap (K+n) = \emptyset$ for compact K and large $|n|$). Restricted to $\mathbb{R} \times (-1,1)$, which is simply connected and a 2-manifold, Theorem 20.4.2 gives $\pi_1(B \setminus \partial B) \cong \mathbb{Z}$. □

Exercise 20.9.4. Let $\mathbb{Z}$ act on $\mathbb{R}^n \setminus \{0\}$ by $k \cdot x = 2^k x$. Show that the action is free and proper, and that the quotient is homeomorphic to $S^{n-1} \times S^1$.

Solution. The map $\mathbb{R}^n \setminus \{0\} \to S^{n-1} \times \mathbb{R}$, $x \mapsto (x/|x|, \log_2 |x|)$, is a homeomorphism carrying the action to $k \cdot (u,s) = (u, s+k)$, which is free and proper on $S^{n-1} \times \mathbb{R}$ with quotient $S^{n-1} \times (\mathbb{R}/\mathbb{Z}) \cong S^{n-1} \times S^1$ (Example 20.3.3(1)). The action on $S^{n-1} \times \mathbb{R}$ is proper because a compact set lies in some $S^{n-1} \times [-r, r]$, which meets its translate by k only if $|k| \le 2r$. For $n = 2$ the quotient is a torus. For $n \ge 3$, $\mathbb{R}^n \setminus \{0\} \cong S^{n-1} \times \mathbb{R}$ is simply connected, S^{n-1} being simply connected (Chapter 19), so the quotient has $\pi_1 \cong \mathbb{Z}$ by Theorem 20.4.2. □

Exercise 20.9.5. Let $h\colon S^3 \to S^2$, $h(z_0, z_1) = (2\bar{z}_0 z_1, |z_1|^2 - |z_0|^2)$, be the Hopf map (Example 20.6.4). Show that its fibres are the great circles $\{\lambda z : |\lambda| = 1\}$, that the fibres over the poles $(0, \pm 1)$ are the circles $\{z_0 = 0\}$ and $\{z_1 = 0\}$, and that the preimage of the closed upper hemisphere is $\{|z_0| \le |z_1|\}$, which is homeomorphic to the solid torus $D^2 \times S^1$. Deduce that S^3 is the union of two solid tori glued along their boundary torus.

Solution. h is the composite of $p\colon S^3 \to \mathbb{CP}^1$ with the homeomorphism of Example 20.6.4, so its fibres are those of p, the orbits $\{\lambda z\}$ of S^1; each is the intersection of S^3 with the complex line $\mathbb{C}z$, a real 2-plane through 0, hence a great circle. The last coordinate is 1 exactly when $|z_1|^2 - |z_0|^2 = 1 = |z_0|^2 + |z_1|^2$, that is $z_0 = 0$, and -1 exactly when $z_1 = 0$. It is ≥ 0 exactly when $|z_1| \geq |z_0|$, that is $|z_0|^2 \leq \frac{1}{2}$. On this set $z_1 \neq 0$, and $(z_0, z_1) \mapsto (\sqrt{2}\, z_0, z_1/|z_1|)$ is a homeomorphism onto $D^2 \times S^1$, with inverse $(a, \mu) \mapsto (a/\sqrt{2}, \mu(1 - |a|^2/2)^{1/2})$. The lower hemisphere gives the solid torus $\{|z_0| \geq |z_1|\}$ in the same way, and the two meet in the torus $|z_0| = |z_1| = 1/\sqrt{2}$, the preimage of the equator. □

Chapter 21

Singular Homology

21.1 Introduction

Singular homology assigns to every space X abelian groups $H_n(X)$, functorial in X and invariant under homotopy. Unlike homotopy groups, they are computable: the long exact sequence of a pair, excision (Theorem 21.5.4) and the Mayer–Vietoris sequence reduce computations to simple pieces, and for CW complexes cellular homology (Theorem 21.7.3) computes them from the cells. We follow [Hat02, Chapter 2]; see also [Spa66].

Prerequisites. Chapter 19 (Homotopy and Covering Spaces) and Chapter 15 (Homological Algebra).

Abelian groups are written additively. Chain complexes are as in Notation 15.3.2, with differential ∂ of degree -1.

21.2 Singular chains

Definition 21.2.1. The *standard n-simplex* is $\Delta^n = \{(t_0, \dots, t_n) \in \mathbb{R}^{n+1} : t_i \geq 0, \sum_i t_i = 1\}$, with vertices the standard basis vectors $e_0, \dots, e_n$. For points $w_0, \dots, w_k$ of some $\mathbb{R}^N$ we write $[w_0, \dots, w_k] \colon \Delta^k \to \mathbb{R}^N$ for the affine map $(t_i) \mapsto \sum_i t_i w_i$. The *face maps* are $\delta^i = [e_0, \dots, \widehat{e_i}, \dots, e_n] \colon \Delta^{n-1} \to \Delta^n$, where the hat means omission.

Definition 21.2.2. A *singular n-simplex* in a space X is a map $\sigma \colon \Delta^n \to X$. The group $C_n(X)$ of *singular n-chains* is the free abelian group (Definition 7.11.1) with basis the singular n-simplices; its elements are finite formal sums $\sum_i n_i \sigma_i$ with $n_i \in \mathbb{Z}$. For $n < 0$, $C_n(X) = 0$. The *boundary* $\partial \colon C_n(X) \to C_{n-1}(X)$ is the homomorphism with

$$\partial \sigma = \sum_{i=0}^{n} (-1)^i \sigma \circ \delta^i.$$

The *singular homology* of X is $H_n(X) = \ker(\partial \colon C_n \to C_{n-1}) / \operatorname{im}(\partial \colon C_{n+1} \to C_n)$. A map $f \colon X \to Y$ induces $f_\# \colon C_n(X) \to C_n(Y)$, $\sigma \mapsto f \circ \sigma$, and $f_* \colon H_n(X) \to H_n(Y)$.

What the definition says in low dimensions. A singular 0-simplex is a point of X. A singular 1-simplex is a path, via $\Delta^1 \cong I$, $(t_0, t_1) \mapsto t_1$, and $\partial \sigma = \sigma(e_1) - \sigma(e_0)$ is "endpoint minus starting point". A singular 2-simplex σ is a map of a triangle, with three edges $\tau_i = \sigma \circ \delta^i \colon \tau_2$ runs from $\sigma(e_0)$ to $\sigma(e_1)$, τ_0 from $\sigma(e_1)$ to $\sigma(e_2)$, and τ_1 from $\sigma(e_0)$ to $\sigma(e_2)$. So $\partial \sigma = \tau_2 + \tau_0 - \tau_1$ is the boundary of the triangle traversed once: along τ_2, then τ_0, then back along τ_1. The signs in the definition of ∂ are chosen so that this works in every dimension, and so that $\partial \partial = 0$: the boundary of a boundary is empty.

Example 21.2.3. 1. A loop γ, regarded as a 1-simplex, is a cycle: $\partial\gamma = \gamma(1) - \gamma(0) = 0$. More generally, if $f_1, \dots, f_k$ are paths such that each f_i ends where f_{i+1} starts and f_k ends where f_1 starts, then $f_1 + \cdots + f_k$ is a cycle, a "closed polygon" of paths, since the endpoints cancel in pairs.

2. The constant 2-simplex c_2 at a point x has all three edges equal to the constant 1-simplex c_1, so $\partial c_2 = c_1 - c_1 + c_1 = c_1$: a constant path is a boundary, and is zero in $H_1(X)$.

3. The 1-cycles that are boundaries are those that "bound a surface" made of triangles. So $H_1(X)$ counts closed curves in X that are not filled in by surfaces, with multiplicities and signs. In the circle, the cycle formed by the upper and the lower half circle is not a boundary (Example 21.6.2 below), while in a disc every 1-cycle is a boundary (Theorem 21.3.1 below). The precise relation with closed curves up to homotopy is Theorem 21.9.1.

Lemma 21.2.4. *$\partial\partial = 0$, and $f_\#$ is a morphism of chain complexes. Hence H_n is a functor* $\mathsf{Top} \to \mathsf{Ab}$.

Proof. For $i < j$ one has $\delta^j\delta^i = \delta^i\delta^{j-1}$ as maps $\Delta^{n-2} \to \Delta^n$: both are the affine map omitting e_i and e_j. So

$$\partial\partial\sigma = \sum_j \sum_i (-1)^{i+j} \sigma\delta^j\delta^i = \sum_{i<j} (-1)^{i+j} \sigma\delta^i\delta^{j-1} + \sum_{i\geq j} (-1)^{i+j} \sigma\delta^j\delta^i,$$

and substituting $(i, j-1) \mapsto (j', i')$ in the first sum shows that it is the negative of the second. Clearly $\partial f_\# = f_\#\partial$. □

Proposition 21.2.5. *1. If (X_α) are the path components of X, then $H_n(X) \cong \bigoplus_\alpha H_n(X_\alpha)$.*

2. If X is path-connected and nonempty, then $H_0(X) \cong \mathbb{Z}$, via $\sum n_i x_i \mapsto \sum n_i$.

3. For a point, $H_0 = \mathbb{Z}$ and $H_n = 0$ for $n \neq 0$.

Proof. (1) The image of a singular simplex is path-connected, so it lies in one X_α, and $C_\bullet(X) = \bigoplus_\alpha C_\bullet(X_\alpha)$ as chain complexes. (2) Singular 0-simplices are points and $\partial\sigma = \sigma(e_1) - \sigma(e_0)$ for a 1-simplex, a path. So the *augmentation* $\varepsilon(\sum n_i x_i) = \sum n_i$ vanishes on boundaries and is surjective. If $\varepsilon(\sum n_i x_i) = 0$, fix x_0 and paths γ_i from x_0 to x_i; then $\sum n_i x_i = \sum n_i(x_i - x_0) = \partial(\sum n_i\gamma_i)$. (3) There is exactly one singular n-simplex σ_n for each n, and $\partial\sigma_n = \sum_{i=0}^n (-1)^i \sigma_{n-1}$ is 0 for n odd and σ_{n-1} for $n \geq 2$ even. So the complex is $\cdots \to \mathbb{Z} \xrightarrow{1} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{1} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0$, with homology $\mathbb{Z}$ in degree 0 only. □

Definition 21.2.6. The *reduced homology* $\tilde{H}_n(X)$ of a nonempty space is the homology of the augmented complex $\cdots \to C_1(X) \to C_0(X) \xrightarrow{\varepsilon} \mathbb{Z} \to 0$, with $\mathbb{Z}$ in degree -1. So $\tilde{H}_n(X) = H_n(X)$ for $n > 0$, $H_0(X) \cong \tilde{H}_0(X) \oplus \mathbb{Z}$, and $\tilde{H}_n(\mathrm{pt}) = 0$ for all n.

21.3 Homotopy invariance

Theorem 21.3.1. *If $f \simeq g\colon X \to Y$, then $f_\#$ and $g_\#$ are chain homotopic, so $f_* = g_*\colon H_n(X) \to H_n(Y)$. Consequently homotopy equivalences induce isomorphisms, and contractible spaces have $\tilde{H}_\bullet = 0$.*

Proof. Let $F\colon X \times I \to Y$ be a homotopy from f to g. In $\Delta^n \times I \subseteq \mathbb{R}^{n+2}$ let $v_i = (e_i, 0)$ and $w_i = (e_i, 1)$. Define the *prism operator* $P\colon C_n(X) \to C_{n+1}(Y)$ by

$$P(\sigma) = \sum_{i=0}^n (-1)^i F \circ (\sigma \times \mathrm{id}_I) \circ [v_0, \dots, v_i, w_i, \dots, w_n].$$

Write $[\dots]$ for $F \circ (\sigma \times \mathrm{id}) \circ [\dots]$. Then

$$\partial P(\sigma) = \sum_{j \le i} (-1)^{i+j} [v_0, \dots, \widehat{v_j}, \dots, v_i, w_i, \dots, w_n] + \sum_{j \ge i} (-1)^{i+j+1} [v_0, \dots, v_i, w_i, \dots, \widehat{w_j}, \dots, w_n].$$

In the terms with $i = j$, the term $[v_0, \dots, v_{i-1}, w_i, \dots, w_n]$ of the first sum cancels the term with index $i-1$ of the second sum, for $1 \le i \le n$. What remains of them is $[w_0, \dots, w_n] = g \circ \sigma$ (first sum, $i = 0$) and $-[v_0, \dots, v_n] = -f \circ \sigma$ (second sum, $i = n$). Since $\sigma \circ \delta^j$ corresponds to the vertices with index j omitted,

$$P(\partial\sigma) = \sum_{i < j} (-1)^{i+j} [v_0, \dots, v_i, w_i, \dots, \widehat{w_j}, \dots, w_n] + \sum_{i > j} (-1)^{i-1+j} [v_0, \dots, \widehat{v_j}, \dots, v_i, w_i, \dots, w_n],$$

which is the negative of the terms with $i \neq j$ above. Hence $\partial P + P\partial = g_\# - f_\#$, and chain homotopic maps induce the same map on homology (Lemma 15.3.4). The rest follows from functoriality and Proposition 21.2.5(3). □

21.4 Relative homology and exact sequences

Definition 21.4.1. For a subspace $A \subseteq X$, $C_n(X, A) = C_n(X)/C_n(A)$ with the induced boundary, and $H_n(X, A)$ is its homology, the *relative homology*. A map of pairs $f : (X, A) \to (Y, B)$ (a map with $f(A) \subseteq B$) induces $f_* : H_n(X, A) \to H_n(Y, B)$, and homotopic maps of pairs (through maps of pairs) induce the same maps, since the prism operator preserves chains in A.

Theorem 21.4.2. *For $B \subseteq A \subseteq X$ there is a natural long exact sequence*

$$\cdots \to H_n(A, B) \to H_n(X, B) \to H_n(X, A) \xrightarrow{\partial} H_{n-1}(A, B) \to \cdots \to H_0(X, A) \to 0.$$

For $B = \emptyset$ this is the long exact sequence of the pair (X, A), where $\partial[c] = [\partial c]$ for a relative cycle c. The same holds with reduced homology when $B = \emptyset$ and $A \neq \emptyset$.

Proof. The sequence $0 \to C(A, B) \to C(X, B) \to C(X, A) \to 0$ of chain complexes is exact. Apply Theorem 15.4.3. For reduced homology use the augmented complexes, whose quotient is still $C(X, A)$. □

Example 21.4.3. For $x_0 \in X$, the long exact sequence of (X, x_0) in reduced homology gives $H_n(X, x_0) \cong \tilde{H}_n(X)$. If A is a deformation retract of X, then $H_n(X, A) = 0$ for all n, by exactness of the long exact sequence, since $H_n(A) \to H_n(X)$ is an isomorphism for all n (Theorem 21.3.1).

Example 21.4.4. *What relative cycles are.* A chain $c \in C_n(X)$ represents a class in $H_n(X, A)$ when $\partial c \in C_{n-1}(A)$: it is closed "up to A". The simplest case is $X = I = [0, 1]$, $A = \partial I = \{0, 1\}$. Since I is contractible, $\tilde{H}_\bullet(I) = 0$, and the reduced long exact sequence $\tilde{H}_n(I) \to H_n(I, \partial I) \xrightarrow{\partial} \tilde{H}_{n-1}(\partial I) \to \tilde{H}_{n-1}(I)$ shows that ∂ is an isomorphism for every n. By Proposition 21.2.5, $\tilde{H}_\bullet(\partial I)$ is zero except in degree 0, where it is the kernel of the augmentation $\mathbb{Z}\{0\} \oplus \mathbb{Z}\{1\} \to \mathbb{Z}$, generated by $\{1\} - \{0\}$. So $H_n(I, \partial I)$ is $\mathbb{Z}$ for $n = 1$ and 0 otherwise. A generator is the identity path $\iota(t) = t$: it is not a cycle in I, but its boundary $\{1\} - \{0\}$ lies in ∂I, and $\partial[\iota] = [\{1\} - \{0\}]$ is the generator. This is the one-dimensional case of $H_n(D^n, S^{n-1}) \cong \mathbb{Z}$ (Proposition 21.6.4 below).

21.5 Excision and Mayer–Vietoris

Definition 21.5.1. Let $\mathcal{U}$ be a family of subspaces of X whose interiors cover X. Then $C_n^{\mathcal{U}}(X) \subseteq C_n(X)$ is the subgroup generated by singular simplices with image in some $U \in \mathcal{U}$. It is a subcomplex.

Lemma 21.5.2. *There are homomorphisms $S\colon C_n(X) \to C_n(X)$ and $T\colon C_n(X) \to C_{n+1}(X)$, natural in X ($f_\# S = S f_\#$, $f_\# T = T f_\#$), with $\partial S = S\partial$ and $\partial T + T\partial = 1 - S$, such that $S(\sigma)$ and $T(\sigma)$ are combinations of simplices with image in $\sigma(\Delta^n)$. Moreover $S^m(\sigma) = \sigma_\#(S^m \iota_n)$ with $\iota_n = [e_0, \dots, e_n]$, and $S^m \iota_n$ is a combination of affine simplices of diameter at most $(n/(n+1))^m \sqrt{2}$.*

Proof. Linear chains. For a convex $Y \subseteq \mathbb{R}^N$ let $LC_n(Y) \subseteq C_n(Y)$ be generated by the affine simplices $[w_0, \dots, w_n]$; it is a subcomplex. For $b \in Y$ the *cone* $b\colon LC_n(Y) \to LC_{n+1}(Y)$, $[w_0, \dots, w_n] \mapsto [b, w_0, \dots, w_n]$, satisfies $\partial b(\lambda) = \lambda - b(\partial\lambda)$ for $n \geq 1$, and $\partial b(\lambda) = \lambda - \varepsilon(\lambda)[b]$ for $n = 0$, by the formula for ∂. For an affine simplex λ let b_λ be its barycentre $\frac{1}{n+1}\sum_i w_i$. Define $S = \mathrm{id}$ on LC_0 and $S(\lambda) = b_\lambda(S\partial\lambda)$ for $n \geq 1$; define $T = 0$ on LC_0 and $T(\lambda) = b_\lambda(\lambda - T\partial\lambda)$ for $n \geq 1$.

By induction on n: $\partial S\lambda = S\partial\lambda - b_\lambda(\partial S\partial\lambda) = S\partial\lambda - b_\lambda(S\partial\partial\lambda) = S\partial\lambda$ for $n \geq 2$, and for $n = 1$, $\partial S\lambda = S\partial\lambda - \varepsilon(\partial\lambda)[b_\lambda] = S\partial\lambda$. Similarly, using $\partial T\partial\lambda = \partial\lambda - S\partial\lambda$ by induction, $\partial T\lambda = \lambda - T\partial\lambda - b_\lambda(\partial\lambda - \partial T\partial\lambda) = \lambda - T\partial\lambda - b_\lambda(S\partial\lambda) = \lambda - T\partial\lambda - S\lambda$. Affine maps preserve barycentres and cones, so S and T commute with $g_\#$ for affine maps g.

General chains. Put $S\sigma = \sigma_\# S\iota_n$ and $T\sigma = \sigma_\# T\iota_n$. Naturality is clear, and since $\sigma \circ \delta^i = \sigma_\#\delta^i$ with δ^i affine, $\partial S\sigma = \sigma_\# S\partial\iota_n = \sum_i (-1)^i \sigma_\#\delta^i_\# S\iota_{n-1} = S\partial\sigma$, and likewise for T. By naturality $S^m\sigma = \sigma_\# S^m\iota_n$.

Diameters. The diameter of an affine simplex is the largest distance between its vertices, since $|\sum_i t_i w_i - y| \leq \max_i |w_i - y|$. The simplices of $S\lambda$ are of the form $[b_\lambda, u_0, \dots, u_{n-1}]$ with $[u_0, \dots, u_{n-1}]$ a simplex of S applied to a face of λ. For a vertex w_i, $|b_\lambda - w_i| = |\sum_j (w_j - w_i)|/(n+1) \leq \frac{n}{n+1}\operatorname{diam}\lambda$, so all points of λ are within this distance of b_λ; and by induction the simplices of S of a face have diameter at most $\frac{n-1}{n}\operatorname{diam}\lambda$. So the simplices of $S\lambda$ have diameter at most $\frac{n}{n+1}\operatorname{diam}\lambda$. Iterate, using $\operatorname{diam}\Delta^n = \sqrt{2}$. □

Proposition 21.5.3. *The inclusion $\iota\colon C^{\mathcal{U}}(X) \to C(X)$ is a chain homotopy equivalence. Hence it induces isomorphisms on homology.*

Proof. For a singular simplex $\sigma\colon \Delta^n \to X$ let $m(\sigma)$ be the least $m \geq 0$ such that every subset of Δ^n of diameter at most $(n/(n+1))^m\sqrt{2}$ lies in $\sigma^{-1}(\operatorname{int} U)$ for some $U \in \mathcal{U}$. It exists by the Lebesgue number lemma (Lemma 18.7.3), and by Lemma 21.5.2, $S^i\sigma \in C^{\mathcal{U}}(X)$ for $i \geq m(\sigma)$. The face maps are isometric embeddings and $\frac{n-1}{n} < \frac{n}{n+1}$, so $m(\sigma \circ \delta^j) \leq m(\sigma)$.

Let $D_m = \sum_{0 \leq i < m} TS^i$, so that $\partial D_m + D_m\partial = \sum_i (1 - S)S^i = 1 - S^m$. Define $D\sigma = D_{m(\sigma)}\sigma$ and $\rho = 1 - \partial D - D\partial$. Then ρ is a chain map, and

$$\rho(\sigma) = S^{m(\sigma)}\sigma + (D_{m(\sigma)} - D)\partial\sigma = S^{m(\sigma)}\sigma + \sum_j (-1)^j \sum_{m(\sigma\delta^j) \leq i < m(\sigma)} TS^i(\sigma\delta^j)$$

lies in $C^{\mathcal{U}}(X)$, since T preserves $C^{\mathcal{U}}(X)$. So $\rho\colon C(X) \to C^{\mathcal{U}}(X)$ satisfies $1 - \iota\rho = \partial D + D\partial$. As D maps $C^{\mathcal{U}}(X)$ into itself, the same identity shows $1 - \rho\iota = \partial D + D\partial$ on $C^{\mathcal{U}}(X)$. □

Theorem 21.5.4 (Excision)**.** *Let $Z \subseteq A \subseteq X$ with $\overline{Z} \subseteq \operatorname{int} A$. Then the inclusion $(X \setminus Z, A \setminus Z) \to (X, A)$ induces isomorphisms on all H_n. Equivalently, if $A, B \subseteq X$ with $\operatorname{int} A \cup \operatorname{int} B = X$, then $H_n(B, A \cap B) \to H_n(X, A)$ is an isomorphism.*

Proof. The two statements correspond via $B = X \setminus Z$, since $\operatorname{int} B = X \setminus \overline{Z}$. Let $\mathcal{U} = \{A, B\}$. The map $C(B)/C(A \cap B) \to C^{\mathcal{U}}(X)/C(A)$ is an isomorphism of complexes: it is surjective as every generator of $C^{\mathcal{U}}$ lies in $C(A)$ or $C(B)$, and its kernel is $C(B) \cap C(A) = C(A \cap B)$. The map $C^{\mathcal{U}}(X)/C(A) \to C(X)/C(A)$ induces isomorphisms on homology, by Proposition 21.5.3 and the five lemma applied to the long exact sequences of $0 \to C(A) \to C^{\mathcal{U}}(X) \to C^{\mathcal{U}}(X)/C(A) \to 0$ and $0 \to C(A) \to C(X) \to C(X, A) \to 0$. □

Theorem 21.5.5 (Mayer–Vietoris)**.** *If $A, B \subseteq X$ with $\operatorname{int} A \cup \operatorname{int} B = X$, there is a natural long exact sequence*

$$\cdots \to H_n(A \cap B) \xrightarrow{(i_*, -j_*)} H_n(A) \oplus H_n(B) \xrightarrow{k_* + l_*} H_n(X) \xrightarrow{\partial} H_{n-1}(A \cap B) \to \cdots,$$

where i, j, k, l are the inclusions, and similarly in reduced homology if $A \cap B \neq \emptyset$.

Proof. The sequence $0 \to C(A \cap B) \xrightarrow{(x \mapsto (x, -x))} C(A) \oplus C(B) \xrightarrow{((x,y) \mapsto x+y)} C^{\{A,B\}}(X) \to 0$ is exact: the kernel of the second map consists of pairs $(x, -x)$ with $x \in C(A) \cap C(B) = C(A \cap B)$. Take the long exact sequence and replace $H_n(C^{\{A,B\}}(X))$ by $H_n(X)$ using Proposition 21.5.3. For reduced homology augment each complex by $\mathbb{Z}$, $\mathbb{Z} \oplus \mathbb{Z}$, $\mathbb{Z}$. □

Remark 21.5.6. *How to compute the connecting map.* By Proposition 21.5.3, every class in $H_n(X)$ is represented by a cycle z that is a sum of simplices lying in A or in B; write $z = a + b$ with $a \in C_n(A)$ and $b \in C_n(B)$ (simplices in $A \cap B$ may go to either side). Then $\partial a = -\partial b$ lies in $C_{n-1}(A) \cap C_{n-1}(B) = C_{n-1}(A \cap B)$, it is a cycle, and

$$\partial[z] = [\partial a] \in H_{n-1}(A \cap B).$$

Indeed, in the description of the connecting map in Theorem 15.4.3, (a, b) is a preimage of z, and its boundary $(\partial a, \partial b) = (\partial a, -\partial a)$ is the image of ∂a. In words: cut the cycle into a piece in A and a piece in B; the common boundary of the pieces is the image of the cycle.

21.6 Spheres, good pairs and degree

Theorem 21.6.1. *For $n \geq 0$, $\tilde{H}_k(S^n) \cong \mathbb{Z}$ for $k = n$ and 0 otherwise. For $n \geq 1$ the Mayer–Vietoris connecting map is an isomorphism $\tilde{H}_k(S^n) \to \tilde{H}_{k-1}(S^{n-1})$, where $S^{n-1} \subseteq S^n$ is the equator $\{x_{n+1} = 0\}$.*

Proof. For $n = 0$, S^0 is two points and $\tilde{H}_0(S^0) \cong \mathbb{Z}$, generated by the difference of the two points. For $n \geq 1$ let $A = \{x \in S^n : x_{n+1} > -1/2\}$ and $B = \{x_{n+1} < 1/2\}$. By stereographic projection from $\mp e_{n+1}$ (Example 18.9.4), A and B are homeomorphic to open balls, hence contractible, and $A \cap B$ deformation retracts onto the equator via $H(x, t) = ((1-t)x + t(x', 0))/\|(1-t)x + t(x', 0)\|$, where $x = (x', x_{n+1})$ with $x' \neq 0$. The reduced Mayer–Vietoris sequence gives the isomorphism, and induction gives the result. □

Example 21.6.2. *A generator of $H_1(S^1)$.* For $n = 1$ the proof uses $A = \{x_2 > -\frac{1}{2}\}$ and $B = \{x_2 < \frac{1}{2}\}$ in $S^1 \subseteq \mathbb{R}^2$; their intersection consists of two arcs, one containing $p = (1, 0)$ and one containing $q = (-1, 0)$. Let

$$a(t) = \left(1 - 2t, 2\sqrt{t(1-t)}\right), \qquad b(t) = \left(2t - 1, -2\sqrt{t(1-t)}\right), \qquad t \in I,$$

the upper half circle from p to q and the lower half circle from q to p (indeed $(1-2t)^2 + 4t(1-t) = 1$). Then a lies in A, b lies in B, and $z = a + b$ is a cycle, as in Example 21.2.3(1). By Remark 21.5.6, $\partial[z] = [\partial a] = [\{q\} - \{p\}]$, which generates $\tilde{H}_0(A \cap B) \cong \mathbb{Z}$, as p and q lie in different path

components. Since ∂ is an isomorphism, $[z]$ generates $H_1(S^1) \cong \mathbb{Z}$. In particular z is not a boundary: the circle has a hole that the cycle z goes around once. Neither half a nor b is a cycle by itself; the connecting map detects precisely how the two halves are glued.

Definition 21.6.3. A pair (X, A) is *good* if A is a nonempty closed subspace that is a deformation retract of some open neighbourhood V of A in X.

Proposition 21.6.4. *For a good pair, the quotient map $q\colon (X, A) \to (X/A, A/A)$ induces isomorphisms $H_n(X, A) \cong H_n(X/A, A/A) \cong \tilde{H}_n(X/A)$. For example, for $n \geq 1$, (D^n, S^{n-1}) is good and $H_k(D^n, S^{n-1}) \cong \tilde{H}_k(S^n) \cong \tilde{H}_{k-1}(S^{n-1})$.*

Proof. Consider the commutative diagram

$$\begin{array}{ccccc} H_n(X,A) & \longrightarrow & H_n(X,V) & \longleftarrow & H_n(X\setminus A, V\setminus A) \\ \downarrow{\scriptstyle q_*} & & \downarrow{\scriptstyle q_*} & & \downarrow{\scriptstyle q_*} \\ H_n(X/A, A/A) & \longrightarrow & H_n(X/A, V/A) & \longleftarrow & H_n(X/A\setminus A/A, V/A\setminus A/A) \end{array}$$

The upper left map is an isomorphism by the long exact sequence of the triple (X, V, A), since $H_n(V, A) = 0$ (Example 21.4.3). The upper right map is an isomorphism by excision, as $\overline{A} = A \subseteq V = \operatorname{int} V$. Since V is open and saturated, the deformation retraction of V induces one of V/A onto the point A/A (Lemma 19.2.4), so the lower maps are isomorphisms for the same reasons. The right vertical map is induced by a homeomorphism of pairs, because q restricts to a bijective quotient map $X \setminus A \to X/A \setminus A/A$ (again by Lemma 19.2.4, as $X \setminus A$ is open and saturated). So the left vertical map is an isomorphism. The last isomorphism is Example 21.4.3. For (D^n, S^{n-1}) take $V = D^n \setminus \{0\}$ (Example 19.2.3); D^n/S^{n-1} is compact, and the map $D^n \to S^n$, $x \mapsto (2\sqrt{1 - \|x\|^2}\, x, 2\|x\|^2 - 1)$, induces a continuous bijection $D^n/S^{n-1} \to S^n$, hence a homeomorphism. The last isomorphism follows from the long exact sequence of (D^n, S^{n-1}). □

Definition 21.6.5. The *degree* of a map $f\colon S^n \to S^n$ is the integer $\deg f$ such that f_* is multiplication by $\deg f$ on $\tilde{H}_n(S^n) \cong \mathbb{Z}$.

Proposition 21.6.6.

1. $\deg \mathrm{id} = 1$, $\deg(fg) = \deg f \deg g$, *and homotopic maps have the same degree; a map that is not surjective has degree* 0.
2. *A reflection $x \mapsto (x_1, \ldots, -x_i, \ldots, x_{n+1})$ has degree -1, and the antipodal map has degree $(-1)^{n+1}$.*
3. *A map $f\colon S^n \to S^n$ without fixed points is homotopic to the antipodal map, so* $\deg f = (-1)^{n+1}$.

Proof. (1) Functoriality and homotopy invariance; a non-surjective map factors through $S^n \setminus \{\mathrm{pt}\} \cong \mathbb{R}^n$, which is contractible.

(2) Let r be the reflection in the first coordinate. For $n = 0$ it swaps the two points and acts by -1 on $\tilde{H}_0(S^0)$. For $n \geq 1$, r preserves the sets A, B of the proof of Theorem 21.6.1 and commutes with the deformation retraction of $A \cap B$ onto the equator, on which it is again the reflection in the first coordinate. By naturality of the Mayer–Vietoris sequence, $\deg r$ on S^n equals $\deg r$ on S^{n-1}, hence equals -1. A reflection in the i-th coordinate is $\pi r \pi^{-1}$ for a coordinate permutation π, so has degree -1 by (1). The antipodal map is the composite of $n + 1$ reflections.

(3) $(1 - t)f(x) - tx$ never vanishes: otherwise $f(x) = \frac{t}{1-t}x$ with $\|f(x)\| = \|x\| = 1$ forces $f(x) = x$. Normalising gives a homotopy from f to $-\mathrm{id}$. □

Corollary 21.6.7.

1. *(Brouwer) There is no retraction $D^{n+1} \to S^n$, and every map $D^{n+1} \to D^{n+1}$ has a fixed point.*

2. *(Hairy ball theorem) S^n has a continuous nowhere vanishing tangent vector field ($v\colon S^n \to \mathbb{R}^{n+1}$ with $\langle v(x), x\rangle = 0$) if and only if n is odd.*

3. *(Invariance of dimension) If nonempty open subsets $U \subseteq \mathbb{R}^m$ and $V \subseteq \mathbb{R}^n$ are homeomorphic, then $m = n$.*

Proof. (1) $\tilde{H}_n(S^n) \neq 0$ while D^{n+1} is contractible, so the identity of $\tilde{H}_n(S^n)$ cannot factor through $\tilde{H}_n(D^{n+1}) = 0$. The formula in the proof of Corollary 19.4.8 turns a fixed point free map into a retraction.

(2) If $n = 2k-1$, $v(x) = (-x_2, x_1, \ldots, -x_{2k}, x_{2k-1})$ works. Conversely, if v never vanishes, let $u = v/\|v\|$. Since $x \perp u(x)$ are unit vectors, $(1-t)x + tu(x)$ and $(1-t)u(x) - tx$ never vanish, and normalising gives homotopies $\mathrm{id} \simeq u \simeq -\mathrm{id}$. So $1 = (-1)^{n+1}$ and n is odd.

(3) Let $x \in U$. By excision (removing the closed set $\mathbb{R}^m \setminus U$ from $\mathbb{R}^m \setminus \{x\}$) and the long exact sequence, $H_k(U, U\setminus\{x\}) \cong H_k(\mathbb{R}^m, \mathbb{R}^m\setminus\{x\}) \cong \tilde{H}_{k-1}(\mathbb{R}^m\setminus\{x\}) \cong \tilde{H}_{k-1}(S^{m-1})$ for $m \geq 1$, which is nonzero exactly for $k = m$. A homeomorphism $U \to V$ gives isomorphisms of these local homology groups. For $m = 0$ or $n = 0$ the spaces are points, and a nonempty open subset of $\mathbb{R}^n$ is a point only if $n = 0$. □

21.7 Cellular homology

Lemma 21.7.1. *Let X be a CW complex (Definition 19.7.1).*

1. *$H_k(X^n, X^{n-1})$ is zero for $k \neq n$, and free abelian with basis the classes e_α of the characteristic maps $\Phi_\alpha\colon (D^n, S^{n-1}) \to (X^n, X^{n-1})$ applied to a fixed generator of $H_n(D^n, S^{n-1}) \cong \mathbb{Z}$, for $k = n$.*

2. *$H_k(X^n) = 0$ for $k > n$.*

3. *The inclusion $X^n \to X$ induces an isomorphism $H_k(X^n) \to H_k(X)$ for $k < n$.*

Proof. (1) X^{n-1} is closed in X^n (as in the proof of Proposition 19.5.5), and it is a deformation retract of the open set obtained by removing the centres of the n-cells, via the radial deformation of $D^n \setminus \{0\}$ onto S^{n-1}, which descends to the quotient by Lemma 19.2.4. So (X^n, X^{n-1}) is a good pair, and so is $(\coprod_\alpha D^n_\alpha, \coprod_\alpha S^{n-1}_\alpha)$. The characteristic maps induce a continuous bijection $\coprod_\alpha D^n_\alpha / \coprod_\alpha S^{n-1}_\alpha \to X^n/X^{n-1}$, which is a homeomorphism: a set whose preimage in $\coprod_\alpha D^n_\alpha$ is open has open preimage in $X^{n-1} \sqcup \coprod_\alpha D^n_\alpha$, since that preimage meets X^{n-1} in $\emptyset$ or X^{n-1}. By Proposition 21.6.4 and its naturality, $H_k(X^n, X^{n-1}) \cong H_k(\coprod_\alpha D^n_\alpha, \coprod_\alpha S^{n-1}_\alpha) = \bigoplus_\alpha H_k(D^n, S^{n-1})$, which is $\bigoplus_\alpha \mathbb{Z}$ for $k = n$ and 0 otherwise. (For $n = 0$, $X^{-1} = \emptyset$ and $H_k(X^0)$ is computed by Proposition 21.2.5.)

(2) By the exact sequence $H_k(X^{n-1}) \to H_k(X^n) \to H_k(X^n, X^{n-1})$ and induction on n, starting from $X^{-1} = \emptyset$.

(3) For $k < n$ the exact sequence $H_{k+1}(X^{n+1}, X^n) \to H_k(X^n) \to H_k(X^{n+1}) \to H_k(X^{n+1}, X^n)$ has outer terms zero, so $H_k(X^n) \cong H_k(X^m)$ for all $m \geq n$. A singular chain has compact image, which lies in a finite subcomplex (Theorem 19.7.4(2)), hence in some X^m. So every cycle of X is a cycle of some X^m, and a cycle of X^n that bounds in X bounds in some X^m; this gives surjectivity and injectivity. □

Definition 21.7.2. The *cellular chain complex* of a CW complex X has $C^{CW}_n(X) = H_n(X^n, X^{n-1})$ and differential d_n the composite $H_n(X^n, X^{n-1}) \xrightarrow{\partial} H_{n-1}(X^{n-1}) \xrightarrow{j_{n-1}} H_{n-1}(X^{n-1}, X^{n-2})$. Then $d_{n-1}d_n = 0$ because it contains two consecutive maps $\partial \circ j_{n-1}$ of the long exact sequence of (X^{n-1}, X^{n-2}). Its homology is the *cellular homology* $H^{CW}_n(X)$.

Theorem 21.7.3. *$H^{CW}_n(X) \cong H_n(X)$ for every CW complex X.*

Proof. Write $j_n\colon H_n(X^n) \to H_n(X^n, X^{n-1})$, which is injective since $H_n(X^{n-1}) = 0$ by Lemma 21.7.1(2), and $\partial_{n+1}\colon H_{n+1}(X^{n+1}, X^n) \to H_n(X^n)$, so $d_{n+1} = j_n\partial_{n+1}$. Since j_n is injective, $\ker d_n = \ker(j_{n-1}\partial_n) = \ker\partial_n = \operatorname{im} j_n$, and j_n maps $\operatorname{im}\partial_{n+1}$ isomorphically onto $\operatorname{im} d_{n+1}$. Hence $H_n^{CW}(X) \cong \operatorname{im} j_n/\operatorname{im} d_{n+1} \cong H_n(X^n)/\operatorname{im}\partial_{n+1}$. By the exact sequence $H_{n+1}(X^{n+1}, X^n) \xrightarrow{\partial_{n+1}} H_n(X^n) \to H_n(X^{n+1}) \to H_n(X^{n+1}, X^n) = 0$, this is $H_n(X^{n+1})$, which is $H_n(X)$ by Lemma 21.7.1(3). □

Theorem 21.7.4 (Cellular boundary formula)**.** *For $n \geq 2$, $d_n(e_\alpha^n) = \sum_\beta d_{\alpha\beta}e_\beta^{n-1}$, where $d_{\alpha\beta}$ is the degree of the composite $S^{n-1} \xrightarrow{\varphi_\alpha} X^{n-1} \to X^{n-1}/(X^{n-1} \setminus e_\beta^{n-1}) \cong S^{n-1}$ of the attaching map with the map collapsing the complement of the cell e_β^{n-1} (identified with S^{n-1} via Φ_β). For $n = 1$, $d_1(e_\alpha^1)$ is the difference of the endpoint and the starting point of the 1-cell.*

This is [Hat02, Section 2.2, Cellular Boundary Formula]; the proof is a diagram chase using naturality of the long exact sequences for the characteristic maps.

Example 21.7.5.

1. If all cells of X are in even dimensions, all d_n vanish and $H_n(X)$ is free with basis the n-cells. For $\mathbb{CP}^n$ (one cell in each even dimension $\leq 2n$), $H_k(\mathbb{CP}^n) \cong \mathbb{Z}$ for k even with $0 \leq k \leq 2n$, and 0 otherwise.
2. For the closed orientable surface Σ_g of genus g (Example 19.7.2), $d_1 = 0$ (one 0-cell) and $d_2 = 0$ (each 1-cell appears twice in the attaching word with opposite signs), so $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}^{2g}$, $H_2 = \mathbb{Z}$.
3. For $\mathbb{RP}^n$ with one cell in each dimension, d_k is multiplication by $1 + (-1)^k$ [Hat02, Example 2.42]. So $H_k(\mathbb{RP}^n)$ is $\mathbb{Z}$ for $k = 0$, $\mathbb{Z}/2\mathbb{Z}$ for k odd with $0 < k < n$, $\mathbb{Z}$ for $k = n$ odd, and 0 otherwise.

Definition 21.7.6. The *Euler characteristic* of a finite CW complex with c_n cells of dimension n is $\chi(X) = \sum_n (-1)^n c_n$.

Proposition 21.7.7. *$\chi(X) = \sum_n (-1)^n \dim_{\mathbb{Q}}(H_n(X) \otimes \mathbb{Q})$. In particular $\chi(X)$ depends only on the homotopy type of X.*

Proof. $\mathbb{Q} = S^{-1}\mathbb{Z}$ is a flat $\mathbb{Z}$-module (Proposition 14.3.2), so $H_n(C^{CW} \otimes \mathbb{Q}) \cong H_n(C^{CW}) \otimes \mathbb{Q} \cong H_n(X) \otimes \mathbb{Q}$. For a bounded complex V of finite-dimensional vector spaces, rank–nullity (Theorem 11.2.4) gives $\dim V_n = \dim Z_n + \dim B_{n-1}$ and $\dim H_n = \dim Z_n - \dim B_n$, so $\sum(-1)^n \dim V_n = \sum(-1)^n \dim H_n$. Apply this to $C^{CW} \otimes \mathbb{Q}$, with $\dim = c_n$. □

Example 21.7.8. $\chi(S^n) = 1 + (-1)^n$, $\chi(\Sigma_g) = 2 - 2g$, $\chi(\mathbb{CP}^n) = n + 1$, $\chi(\mathbb{RP}^n) = 1$ for n even and 0 for n odd. As $\chi(\Sigma_g)$ determines g, surfaces of different genus are not homotopy equivalent.

21.8 Homology with coefficients

Definition 21.8.1. For an abelian group G, $C_n(X, A; G) = C_n(X, A) \otimes_{\mathbb{Z}} G$, and $H_n(X, A; G)$ is its homology. For a ring R these are R-modules.

Theorem 21.8.2.

1. *Homotopy invariance, the long exact sequences, excision and Mayer–Vietoris hold for $H_\bullet(-; G)$, and cellular homology with coefficients, computed from $C_\bullet^{CW}(X) \otimes G$, agrees with $H_\bullet(X; G)$.*
2. *(Universal coefficients) There is a natural short exact sequence $0 \to H_n(X, A) \otimes G \to H_n(X, A; G) \to \operatorname{Tor}_1^{\mathbb{Z}}(H_{n-1}(X, A), G) \to 0$. It splits, non-naturally [Hat02, Theorem 3A.3].*

Proof. (1) The complexes $C(X, A)$ are free, so the short exact sequences of chain complexes used above are degreewise split and remain exact after $\otimes G$; chain homotopies remain chain homotopies. The arguments therefore apply verbatim, and the proof of Theorem 21.7.3 goes through with $H_n(X^n, X^{n-1}; G) \cong C_n^{CW}(X) \otimes G$. (2) Apply the Künneth formula (Theorem 15.9.1) to the free complex $C(X, A)$ and the complex G concentrated in degree 0. The splitting is quoted. □

Example 21.8.3. With coefficients in $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$ all cellular differentials of $\mathbb{RP}^n$ vanish, since $1 + (-1)^k$ is even; so $H_k(\mathbb{RP}^n; \mathbb{F}_2) \cong \mathbb{F}_2$ for $0 \le k \le n$. With coefficients in a field F of characteristic 0, $H_\bullet(X; F) \cong H_\bullet(X) \otimes F$, because F is a $\mathbb{Q}$-vector space, hence a direct sum of copies of the flat module $\mathbb{Q}$, so $\operatorname{Tor}_1^{\mathbb{Z}}(-, F) = 0$ (Proposition 15.8.4).

21.9 The first homology group

Theorem 21.9.1 (Hurewicz in degree one)**.** *Let X be path-connected and $x_0 \in X$. Regarding a loop as a singular 1-simplex via $\Delta^1 \cong I$, $(t_0, t_1) \mapsto t_1$, gives a well-defined homomorphism $h\colon \pi_1(X, x_0) \to H_1(X)$, which induces an isomorphism $\pi_1(X, x_0)^{\mathrm{ab}} \cong H_1(X)$ from the abelianisation (Proposition 7.9.2).*

Proof. For paths we write $f \sim g$ if $f - g \in C_1(X)$ is a boundary. (i) A constant path c satisfies $c \sim 0$: it is the boundary of the constant 2-simplex. (ii) If $f \simeq g$ relative to endpoints via $F\colon I^2 \to X$, divide I^2 into two triangles along the diagonal and compose with F; the boundary of the difference of the two resulting 2-simplices is $f - g$ plus constant paths, so $f \sim g$. (iii) $f * g \sim f + g$: the 2-simplex $\sigma = (f * g) \circ \pi$, where $\pi\colon \Delta^2 \to I$ is the affine map with $e_0 \mapsto 0$, $e_1 \mapsto 1/2$, $e_2 \mapsto 1$, has $\partial\sigma = g - f * g + f$. (iv) Hence $f + \bar{f} \sim f * \bar{f} \sim c \sim 0$. So h is well defined and a homomorphism, and it factors through π_1^{ab} as $H_1(X)$ is abelian.

Choose for each $x \in X$ a path γ_x from x_0 to x, with γ_{x_0} constant. For a singular 1-simplex τ let $L(\tau) = [\gamma_{\tau(0)} * \tau * \bar{\gamma}_{\tau(1)}]$ in $\pi_1(X, x_0)^{\mathrm{ab}}$, written additively, and extend L linearly to $C_1(X)$. For a singular 2-simplex σ with edges τ_0, τ_1, τ_2 (so $\partial\sigma = \tau_0 - \tau_1 + \tau_2$), $L(\partial\sigma)$ is the class of the loop $\gamma_{\sigma(e_0)} * \tau_2 * \tau_0 * \bar{\tau}_1 * \bar{\gamma}_{\sigma(e_0)}$, which is null-homotopic because the loop around the boundary of Δ^2 is null-homotopic in the contractible space Δ^2. So L induces $\bar{L}\colon H_1(X) \to \pi_1(X, x_0)^{\mathrm{ab}}$. Clearly $\bar{L}h = \mathrm{id}$. For a cycle $z = \sum n_i \tau_i$, by (iii) and (iv), $hL(z) = \sum n_i(\gamma_{\tau_i(0)} + \tau_i - \gamma_{\tau_i(1)}) = z - \Gamma(\partial z) = z$ in $H_1(X)$, where $\Gamma\colon C_0(X) \to C_1(X)$ is the homomorphism $x \mapsto \gamma_x$. So h and $\bar{L}$ are inverse isomorphisms. □

Remark 21.9.2. Homotopy invariance, the long exact sequence, excision, additivity over disjoint unions and the value on a point are the *Eilenberg–Steenrod axioms*. On CW complexes they determine homology, as the proof of Theorem 21.7.3 shows. Other constructions satisfying them, such as simplicial, Čech or sheaf-theoretic homology on suitable spaces, agree with singular homology there; the comparison with de Rham cohomology is Chapter 37.

21.10 Exercises

Exercise 21.10.1. Show that S^m and S^n are homotopy equivalent only if $m = n$, and that $\mathbb{R}^m \setminus \{0\}$ and $\mathbb{R}^n \setminus \{0\}$ are homeomorphic only if $m = n$.

Solution. By Theorem 21.6.1, $\tilde{H}_\bullet(S^m)$ is nonzero only in degree m. The spaces $\mathbb{R}^m \setminus \{0\}$ and S^{m-1} are homotopy equivalent (Example 19.2.3). □

Exercise 21.10.2. The *suspension* SX is the quotient of $X \times [-1, 1]$ collapsing $X \times \{1\}$ and $X \times \{-1\}$ to two points. Show that $\tilde{H}_{n+1}(SX) \cong \tilde{H}_n(X)$ for all n.

Solution. Let A and B be the images of $X \times (-1/2, 1]$ and $X \times [-1, 1/2)$. They are contractible (slide to the cone points; the homotopies descend by Lemma 19.2.4), and $A \cap B \cong X \times (-1/2, 1/2) \simeq X$. The reduced Mayer–Vietoris sequence gives the isomorphism. □

Exercise 21.10.3. Show that $\tilde{H}_n(\bigvee_\alpha X_\alpha) \cong \bigoplus_\alpha \tilde{H}_n(X_\alpha)$ if the basepoints x_α are such that each (X_α, x_α) is a good pair. Compute $H_\bullet$ of a wedge of k circles.

Solution. The pair $(\coprod X_\alpha, \coprod \{x_\alpha\})$ is good with quotient $\bigvee X_\alpha$, so $\tilde{H}_n(\bigvee X_\alpha) \cong H_n(\coprod X_\alpha, \coprod \{x_\alpha\}) = \bigoplus H_n(X_\alpha, x_\alpha) = \bigoplus \tilde{H}_n(X_\alpha)$. For k circles, $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}^k$, and $H_n = 0$ for $n \geq 2$; this agrees with Theorem 21.9.1, since the abelianisation of a free group of rank k is $\mathbb{Z}^k$. □

Exercise 21.10.4. Using Theorem 21.9.1 and Remark 19.4.7, show that the degree of a map $S^1 \to S^1$ defined via π_1 agrees with the homological degree.

Exercise 21.10.5. Let X be a finite CW complex and $f : X \to X$ a cellular map. Define the Lefschetz number $\Lambda(f) = \sum_n (-1)^n \operatorname{tr}(f_* \mid H_n(X; \mathbb{Q}))$. Show that $\Lambda(f) = \sum_n (-1)^n \operatorname{tr}(f_\# \mid C_n^{CW}(X) \otimes \mathbb{Q})$, and deduce that $\Lambda(\mathrm{id}) = \chi(X)$.

Solution. A cellular map sends X^n into X^n, so it is a map of pairs $(X^n, X^{n-1}) \to (X^n, X^{n-1})$ and induces maps $f_\#$ on $C_n^{CW}(X)$; they commute with the differentials, which are composites of maps of long exact sequences. The isomorphism $H_n^{CW}(X) \cong H_n(X)$ of Theorem 21.7.3 is built from maps induced by inclusions and connecting maps, so it is compatible with f; the same holds with coefficients in $\mathbb{Q}$ (Theorem 21.8.2).

It remains to prove the *Hopf trace formula*: if V is a bounded complex of finite-dimensional vector spaces and φ a chain map from V to itself, then $\sum_n (-1)^n \operatorname{tr}(\varphi \mid V_n) = \sum_n (-1)^n \operatorname{tr}(\varphi \mid H_n(V))$. The subspaces $B_n \subseteq Z_n \subseteq V_n$ are preserved by φ. Choosing a basis of B_n, extending it to Z_n and then to V_n, the matrix of φ on V_n is block upper triangular, with diagonal blocks the matrices of φ on B_n, on $Z_n/B_n = H_n$ and on V_n/Z_n. The differential induces an isomorphism $V_n/Z_n \cong B_{n-1}$ compatible with φ, so $\operatorname{tr}(\varphi \mid V_n) = \operatorname{tr}(\varphi \mid B_n) + \operatorname{tr}(\varphi \mid H_n) + \operatorname{tr}(\varphi \mid B_{n-1})$. In the alternating sum the terms with B cancel, as in the proof of Proposition 21.7.7. For $f = \mathrm{id}$, the trace on $C_n^{CW}(X) \otimes \mathbb{Q} \cong \mathbb{Q}^{c_n}$ is c_n, so $\Lambda(\mathrm{id}) = \chi(X)$. □

Exercise 21.10.6. The Klein bottle K is obtained from $S^1 \vee S^1$ by attaching a 2-cell along $aba^{-1}b$. Compute $\pi_1(K)$, $H_\bullet(K)$ and $H_\bullet(K; \mathbb{F}_2)$.

Solution. By Proposition 19.5.5 and Example 19.5.4, $\pi_1(K) \cong \langle a, b \mid aba^{-1}b \rangle$. The cellular chain complex has one 0-cell, two 1-cells a, b and one 2-cell e. Since there is one 0-cell, $d_1 = 0$. The degrees in the cellular boundary formula are the sums of the exponents of each letter in the attaching word, as in Example 21.7.5(2) [Hat02, Section 2.2]: a occurs with exponents $+1$ and -1, and b twice with exponent $+1$. So $d_2(e) = 2b$, and

$$H_0(K) \cong \mathbb{Z}, \qquad H_1(K) \cong \mathbb{Z}^2/\langle (0,2) \rangle \cong \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}, \qquad H_2(K) = \ker d_2 = 0.$$

This agrees with Theorem 21.9.1: abelianising the relation $aba^{-1}b = 1$ gives $2b = 0$. With coefficients in $\mathbb{F}_2$ the differential $d_2 = 2b$ becomes 0, so $H_k(K; \mathbb{F}_2)$ is $\mathbb{F}_2$, $\mathbb{F}_2^2$, $\mathbb{F}_2$ for $k = 0, 1, 2$; for instance $H_2(K; \mathbb{F}_2) \cong \operatorname{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{F}_2) \cong \mathbb{F}_2$ by universal coefficients. Both computations give $\chi(K) = 1 - 2 + 1 = 0$. Unlike the torus, $H_2(K; \mathbb{Z}) = 0$: the 2-cell is not a cycle, which reflects the fact that K is not orientable. □

Exercise 21.10.7. Let n be even.

1. Show that for every map $f : S^n \to S^n$ there is a point x with $f(x) = x$ or $f(x) = -x$.

2. Show that if a group G acts on S^n by homeomorphisms *freely*, that is, no $g \neq 1$ has a fixed point, then G has at most two elements. (The antipodal map gives a free action of $\mathbb{Z}/2\mathbb{Z}$, with quotient $\mathbb{RP}^n$.)

For odd n both statements fail: on $S^1 \subseteq \mathbb{C}$, the rotation by a right angle moves every point to a point that is neither equal nor opposite to it, and rotations by multiples of $2\pi/k$ give a free action of $\mathbb{Z}/k\mathbb{Z}$.

Solution. (1) Let α be the antipodal map, of degree $(-1)^{n+1} = -1$ (Proposition 21.6.6(2)). If f has no fixed point, then $\deg f = -1$ by Proposition 21.6.6(3). If $f(x) \neq -x$ for all x, then $\alpha \circ f$ has no fixed point, so $-\deg f = \deg(\alpha f) = -1$ and $\deg f = 1$. The two cannot both hold.

(2) Each $g \in G$ acts by a homeomorphism, and $\deg g \cdot \deg g^{-1} = \deg \mathrm{id} = 1$, so $\deg g = \pm 1$, and $g \mapsto \deg g$ is a homomorphism $G \to \{\pm 1\}$ by Proposition 21.6.6(1). If $g \neq 1$, then g has no fixed point, so $\deg g = -1$. Hence the homomorphism is injective, and $|G| \leq 2$. □

Chapter 22

Singular Cohomology

22.1 Introduction

Cohomology is obtained by dualising singular chains. As groups it carries the same information as homology, by the universal coefficient theorem, but it has a natural ring structure given by the cup product, which distinguishes spaces with the same homology (Example 22.5.2). Cohomology classes are the objects of the Hodge conjecture: a Hodge class is a rational cohomology class of a special type, and the conjecture asks whether it is a combination of classes of subvarieties. The cap product relates cohomology and homology and leads to Poincaré duality (Chapter 23). We follow [Hat02, Chapter 3].

Prerequisites. Chapter 21 (Singular Homology).

Throughout, R is a commutative ring and G an abelian group, written additively.

22.2 Cochains and cohomology

Definition 22.2.1. For a pair (X, A) let $C^n(X, A; G) = \mathrm{Hom}_{\mathbb{Z}}(C_n(X, A), G)$, the group of *singular n-cochains*, with coboundary $\delta\colon C^n \to C^{n+1}$, $\delta\varphi = \varphi \circ \partial$. So $C^n(X, A; G)$ is the group of functions from singular n-simplices of X to G vanishing on simplices in A, and $(\delta\varphi)(\sigma) = \sum_i (-1)^i \varphi(\sigma \circ \delta^i)$. The *singular cohomology* $H^n(X, A; G)$ is the cohomology of this complex, and $H^n(X; G) = H^n(X, \emptyset; G)$. A map of pairs f induces $f^{\#} = \mathrm{Hom}(f_{\#}, G)$ and $f^*\colon H^n(Y, B; G) \to H^n(X, A; G)$, contravariantly. Reduced cohomology $\tilde{H}^n(X; G)$ is obtained by dualising the augmented chain complex. We write $H^n(X)$ for $H^n(X; \mathbb{Z})$. The *Kronecker pairing* $\langle -, -\rangle\colon H^n(X, A; G) \times H_n(X, A) \to G$ is $\langle [\varphi], [c]\rangle = \varphi(c)$.

Theorem 22.2.2. *1. Homotopic maps of pairs induce the same map on cohomology.*

2. *For $B \subseteq A \subseteq X$ there is a natural long exact sequence $\cdots \to H^n(X, A; G) \to H^n(X, B; G) \to H^n(A, B; G) \xrightarrow{\delta} H^{n+1}(X, A; G) \to \cdots$, and $\langle \delta\alpha, c\rangle = \langle \alpha, \partial c\rangle$ for $\alpha \in H^n(A; G)$ and $c \in H_{n+1}(X, A)$.*

3. *(Excision) If $Z \subseteq A \subseteq X$ with $\overline{Z} \subseteq \operatorname{int} A$, then $H^n(X, A; G) \to H^n(X \setminus Z, A \setminus Z; G)$ is an isomorphism.*

4. *(Mayer–Vietoris) If $\operatorname{int} A \cup \operatorname{int} B = X$, there is a long exact sequence $\cdots \to H^n(X; G) \to H^n(A; G) \oplus H^n(B; G) \to H^n(A \cap B; G) \to H^{n+1}(X; G) \to \cdots$.*

5. *$H^0(X; G)$ is the group of functions from the set of path components of X to G.*

Proof. Applying $\mathrm{Hom}(-, G)$ turns chain homotopies into cochain homotopies, which gives (1) from Theorem 21.3.1. The sequences $0 \to C_n(A, B) \to C_n(X, B) \to C_n(X, A) \to 0$ and

$0 \to C_n(A \cap B) \to C_n(A) \oplus C_n(B) \to C_n^{\{A,B\}}(X) \to 0$ consist of free abelian groups and split degreewise, since the quotients are free on the simplices not in the subspace. So they remain exact after $\mathrm{Hom}(-, G)$, and Theorem 15.4.3 gives the long exact sequences. The connecting map sends $[\varphi]$ to the class of $\delta\tilde{\varphi}$ for an extension $\tilde{\varphi}$ of φ to X, which gives the formula for the pairing. The chain homotopy equivalence $C^{\mathcal{U}}(X) \to C(X)$ of Proposition 21.5.3 dualises to a cochain homotopy equivalence, which gives (3) as in Theorem 21.5.4 and allows the replacement of $H^n(\mathrm{Hom}(C^{\{A,B\}}(X), G))$ by $H^n(X; G)$ in (4). For (5), a 0-cocycle is a function φ on points with $\varphi(\gamma(1)) - \varphi(\gamma(0)) = 0$ for every path γ. □

Theorem 22.2.3 (Universal coefficients)**.** *There is a natural short exact sequence*

$$0 \to \mathrm{Ext}^1_{\mathbb{Z}}(H_{n-1}(X, A), G) \to H^n(X, A; G) \to \mathrm{Hom}(H_n(X, A), G) \to 0,$$

where the second map is $\alpha \mapsto \langle \alpha, - \rangle$. It splits, non-naturally. If $G = F$ is a field and $A = \emptyset$, then $H^n(X; F) \cong \mathrm{Hom}_F(H_n(X; F), F)$.

Proof. The first statement is Theorem 15.9.2 for the free complex $C(X, A)$. For a field, $\mathrm{Hom}_{\mathbb{Z}}(C_n(X), F) = \mathrm{Hom}_F(C_n(X) \otimes F, F)$, and $\mathrm{Hom}_F(-, F)$ is exact, so it commutes with taking cohomology. □

Lemma 22.2.4. $\mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}, G) = 0$ *and* $\mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/m\mathbb{Z}, G) \cong G/mG$*;* $\mathrm{Ext}^1_{\mathbb{Z}}$ *commutes with finite direct sums. Hence, if $H_{n-1}(X)$ and $H_n(X)$ are finitely generated, $H^n(X; \mathbb{Z}) \cong \mathbb{Z}^{b_n} \oplus T_{n-1}$, where b_n is the rank of $H_n(X)$ and T_{n-1} the torsion subgroup of $H_{n-1}(X)$.*

Proof. Use the projective resolution $0 \to \mathbb{Z} \xrightarrow{m} \mathbb{Z} \to \mathbb{Z}/m\mathbb{Z} \to 0$ and Theorem 15.8.2: $\mathrm{Ext}^1(\mathbb{Z}/m\mathbb{Z}, G)$ is the cokernel of $G \xrightarrow{m} G$. Additivity follows from additivity of Hom and of derived functors. For the last statement, write $H_n(X) \cong \mathbb{Z}^{b_n} \oplus T_n$ and $H_{n-1}(X) \cong \mathbb{Z}^{b_{n-1}} \oplus T_{n-1}$ with T_n, T_{n-1} finite direct sums of groups $\mathbb{Z}/m\mathbb{Z}$ (Theorem 7.11.6). By the split universal coefficient sequence, $H^n(X) \cong \mathrm{Hom}(H_n(X), \mathbb{Z}) \oplus \mathrm{Ext}^1(H_{n-1}(X), \mathbb{Z})$; here $\mathrm{Hom}(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}) = 0$ gives $\mathrm{Hom}(H_n(X), \mathbb{Z}) \cong \mathbb{Z}^{b_n}$, and $\mathrm{Ext}^1(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}) \cong \mathbb{Z}/m\mathbb{Z}$ gives $\mathrm{Ext}^1(H_{n-1}(X), \mathbb{Z}) \cong T_{n-1}$. □

Example 22.2.5.

1. $\tilde{H}^k(S^n; G) \cong G$ for $k = n$ and 0 otherwise.
2. By Example 21.7.5 and Lemma 22.2.4, $H^k(\mathbb{RP}^n; \mathbb{Z})$ is $\mathbb{Z}$ for $k = 0$, $\mathbb{Z}/2\mathbb{Z}$ for k even with $0 < k \leq n$, $\mathbb{Z}$ for $k = n$ odd, and 0 otherwise: torsion moves up one degree compared with homology. With coefficients in $\mathbb{F}_2$, $H^k(\mathbb{RP}^n; \mathbb{F}_2) \cong \mathbb{F}_2$ for $0 \leq k \leq n$.

Theorem 22.2.6 (Cellular cohomology)**.** *For a CW complex X, $H^n(X; G)$ is isomorphic to the cohomology of the complex* $\mathrm{Hom}(C^{CW}_\bullet(X), G)$ *with the transposed differentials.*

Proof. By the universal coefficient theorem and Lemma 21.7.1: $H^k(X^n, X^{n-1}; G)$ is $\mathrm{Hom}(C_n^{CW}(X), G)$ for $k = n$ and 0 otherwise; $H^k(X^n; G) = 0$ for $k > n$, because $H_k(X^n) = 0$ for $k > n$ and $H_n(X^n)$ is a subgroup of the free group $C_n^{CW}(X)$, hence free (Proposition 15.8.5). From the long exact sequence of (X, X^n) and Lemma 21.7.1(3), $H_k(X, X^n) = 0$ for $k \leq n$, hence $H^k(X, X^n; G) = 0$ for $k \leq n$ and $H^k(X; G) \to H^k(X^n; G)$ is an isomorphism for $k < n$. The proof of Theorem 21.7.3 now applies with all arrows reversed. The coboundary $H^n(X^n, X^{n-1}) \to H^n(X^n) \to H^{n+1}(X^{n+1}, X^n)$ is the transpose of d_{n+1} by the formula $\langle \delta\alpha, c \rangle = \langle \alpha, \partial c \rangle$. □

Corollary 22.2.7. *Let X be a CW complex with finitely many cells in each dimension and no cells of odd dimension. Then $H^k(X; G) = 0$ for k odd, and $H^k(X; G) \cong G^{c_k}$ for k even, where c_k is the number of k-cells. The same holds for homology, and $\chi(X) = \sum_k c_k$ when X is finite.*

Proof. The cellular chain complex has $C_k^{CW}(X) = 0$ for k odd, so all its differentials vanish, being maps into or out of zero groups. By Theorem 22.2.6 the cohomology is $\mathrm{Hom}(C_k^{CW}(X), G) = \mathrm{Hom}(\mathbb{Z}^{c_k}, G) = G^{c_k}$ in each even degree k, and 0 in odd degrees; the homology statement is Theorem 21.7.3. The Euler characteristic is the alternating sum of the ranks. □

Example 22.2.8. $\mathbb{CP}^n$ has one cell in each even dimension $0, 2, \dots, 2n$ (Example 21.7.5), so $H^{2k}(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z}$ for $0 \le k \le n$ and all other groups vanish, and $\chi(\mathbb{CP}^n) = n + 1$. The product $S^2 \times S^2$, with cells e^0, $e^2 \times e^0$, $e^0 \times e^2$, $e^2 \times e^2$, has $H^0 = H^4 = \mathbb{Z}$ and $H^2 = \mathbb{Z}^2$. For $\mathbb{RP}^n$, which has one cell in each dimension $\le n$, the corollary does not apply, and indeed its cohomology has torsion (Example 22.2.5).

22.3 The cup product

Definition 22.3.1. For $\varphi \in C^k(X; R)$ and $\psi \in C^l(X; R)$, the *cup product* $\varphi \smile \psi \in C^{k+l}(X; R)$ is

$$(\varphi \smile \psi)(\sigma) = \varphi(\sigma \circ [e_0, \dots, e_k]) \cdot \psi(\sigma \circ [e_k, \dots, e_{k+l}]),$$

the product being taken in R: the "front k-face" times the "back l-face". We write $\sigma|[e_0, \dots, e_k]$ for $\sigma \circ [e_0, \dots, e_k]$.

Lemma 22.3.2. $\delta(\varphi \smile \psi) = \delta\varphi \smile \psi + (-1)^k \varphi \smile \delta\psi$. *The cup product of cochains is associative, R-bilinear, has unit the* 0*-cochain* 1 *with value* 1 *on every point, and is natural:* $f^\#(\varphi \smile \psi) = f^\#\varphi \smile f^\#\psi$.

Proof. Let $\sigma\colon \Delta^{k+l+1} \to X$ and $n = k + l + 1$. Then

$$(\delta\varphi \smile \psi)(\sigma) = \sum_{i=0}^{k+1} (-1)^i \varphi(\sigma|[e_0, \dots, \widehat{e_i}, \dots, e_{k+1}]) \psi(\sigma|[e_{k+1}, \dots, e_n]),$$

$$(-1)^k (\varphi \smile \delta\psi)(\sigma) = \sum_{i=k}^{n} (-1)^i \varphi(\sigma|[e_0, \dots, e_k]) \psi(\sigma|[e_k, \dots, \widehat{e_i}, \dots, e_n]).$$

The term $i = k + 1$ of the first sum cancels the term $i = k$ of the second. The remaining terms are $\sum_i (-1)^i (\varphi \smile \psi)(\sigma|[e_0, \dots, \widehat{e_i}, \dots, e_n])$, which is $(\varphi \smile \psi)(\partial\sigma) = \delta(\varphi \smile \psi)(\sigma)$. The other statements are immediate from the definition. □

Definition 22.3.3. By Lemma 22.3.2, the cup product of two cocycles is a cocycle, and the product of a cocycle and a coboundary is a coboundary. It therefore induces $\smile\colon H^k(X; R) \times H^l(X; R) \to H^{k+l}(X; R)$, making $H^*(X; R) = \bigoplus_k H^k(X; R)$ an associative graded R-algebra with unit, the *cohomology ring*, and f^* a homomorphism of graded rings. If φ vanishes on chains in A so does $\varphi \smile \psi$, which gives a product $H^k(X, A; R) \times H^l(X; R) \to H^{k+l}(X, A; R)$.

Theorem 22.3.4 (Graded commutativity)**.** *For $\alpha \in H^k(X; R)$ and $\beta \in H^l(X; R)$, $\alpha \smile \beta = (-1)^{kl} \beta \smile \alpha$.*

Sketch of proof. Let $\bar\sigma = \sigma \circ [e_n, \dots, e_0]$ be σ with its vertices in reverse order, and $\rho(\sigma) = \varepsilon_n \bar\sigma$ with $\varepsilon_n = (-1)^{n(n+1)/2}$. A direct computation shows that ρ is a chain map and that $\rho^\#\varphi \smile \rho^\#\psi = (-1)^{kl} \rho^\#(\psi \smile \varphi)$, using $\varepsilon_{k+l} = (-1)^{kl} \varepsilon_k \varepsilon_l$. The omitted verification is that ρ is chain homotopic to the identity, by the homotopy $P(\sigma) = \sum_i (-1)^i \varepsilon_{n-i}\, \sigma \circ \pi \circ [v_0, \dots, v_i, w_n, \dots, w_i]$, where $v_j = (e_j, 0)$ and $w_j = (e_j, 1)$ are the vertices of $\Delta^n \times I$ and $\pi\colon \Delta^n \times I \to \Delta^n$ is the projection; see [Hat02, Theorem 3.14]. Then $\rho^* = \mathrm{id}$ on cohomology, and the formula follows. □

Example 22.3.5. For a wedge $X \vee Y$ of CW complexes (at 0-cells), the inclusions induce an isomorphism of rings $\tilde{H}^*(X \vee Y; R) \cong \tilde{H}^*(X; R) \times \tilde{H}^*(Y; R)$, where products between the two factors vanish: the isomorphism of groups follows from Exercise 21.10.3 and the universal coefficient theorem, and it is induced by the restrictions to X and Y, which are ring homomorphisms. In particular all cup products $\tilde{H}^k(S^m) \times \tilde{H}^l(S^m) \to \tilde{H}^{k+l}(S^m)$ vanish, for degree reasons.

22.4 The Künneth formula

Definition 22.4.1. The *cross product* of $a \in H^k(X; R)$ and $b \in H^l(Y; R)$ is $a \times b = p_X^*(a) \smile p_Y^*(b) \in H^{k+l}(X \times Y; R)$, where p_X, p_Y are the projections. It is R-bilinear and induces $H^*(X; R) \otimes_R H^*(Y; R) \to H^*(X \times Y; R)$, a ring homomorphism for the product $(a \otimes b)(c \otimes d) = (-1)^{|b||c|} ac \otimes bd$ on the tensor product.

Theorem 22.4.2 (Künneth formula)**.** *Let X and Y be CW complexes such that $H^k(Y; R)$ is a finitely generated free R-module for every k (for instance R a field and Y a finite complex). Then the cross product $H^*(X; R) \otimes_R H^*(Y; R) \to H^*(X \times Y; R)$ is an isomorphism of graded rings.*

This is [Hat02, Theorem 3.15]. The ring homomorphism property follows from graded commutativity: $(a \times b)(c \times d) = p_X^* a \, p_Y^* b \, p_X^* c \, p_Y^* d = (-1)^{|b||c|} p_X^*(ac) \, p_Y^*(bd)$. The homology Künneth formula for fields, and the Eilenberg–Zilber theorem $C_\bullet(X \times Y) \simeq C_\bullet(X) \otimes C_\bullet(Y)$ underlying it, are in [Hat02, Section 3.B].

Example 22.4.3. $H^*(S^1; \mathbb{Z}) = \Lambda_{\mathbb{Z}}[x]$ with $|x| = 1$ (an exterior algebra, as $x^2 \in H^2(S^1) = 0$). By induction and the Künneth formula, the cohomology ring of the torus $T^n = (S^1)^n$ is the exterior algebra $\Lambda_{\mathbb{Z}}[x_1, \dots, x_n]$ with $|x_i| = 1$, so $H^k(T^n) \cong \mathbb{Z}^{\binom{n}{k}}$. Similarly $H^*(S^2 \times S^2; \mathbb{Z}) = \mathbb{Z}[a, b]/(a^2, b^2)$ with $|a| = |b| = 2$.

22.5 Cohomology rings of projective spaces

Theorem 22.5.1. *$H^*(\mathbb{RP}^n; \mathbb{F}_2) \cong \mathbb{F}_2[\alpha]/(\alpha^{n+1})$ with $|\alpha| = 1$, and $H^*(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z}[\alpha]/(\alpha^{n+1})$ with $|\alpha| = 2$. Also $H^*(\mathbb{RP}^\infty; \mathbb{F}_2) = \mathbb{F}_2[\alpha]$ and $H^*(\mathbb{CP}^\infty; \mathbb{Z}) = \mathbb{Z}[\alpha]$.*

This is [Hat02, Theorem 3.12]; the additive structure was computed above, and the content is that the generator in each degree is a power of α. A proof via Poincaré duality is given in Chapter 23.

Example 22.5.2. $\mathbb{CP}^2$ and $S^2 \vee S^4$ have the same homology and cohomology groups ($\mathbb{Z}$ in degrees $0, 2, 4$), but they are not homotopy equivalent: in $H^*(\mathbb{CP}^2)$ the square of a generator of H^2 generates H^4, whereas in $H^*(S^2 \vee S^4)$ it is zero by Example 22.3.5. Likewise $S^2 \times S^2$ and $S^2 \vee S^2 \vee S^4$ have the same groups but different rings.

Corollary 22.5.3 (Borsuk–Ulam)**.** *There is no map $g \colon S^n \to S^{n-1}$ with $g(-x) = -g(x)$ for all x. Consequently every map $f \colon S^n \to \mathbb{R}^n$ identifies some pair of antipodal points.*

Proof. For $n = 1$ an odd map $S^1 \to S^0$ cannot exist, since its image would be a single point. Let $n \geq 2$ and suppose g exists; it induces $\bar{g} \colon \mathbb{RP}^n \to \mathbb{RP}^{n-1}$. For a path-connected space X, the universal coefficient theorem and Theorem 21.9.1 give $H^1(X; \mathbb{F}_2) \cong \mathrm{Hom}(H_1(X), \mathbb{F}_2) \cong \mathrm{Hom}(\pi_1(X), \mathbb{F}_2)$, naturally. For $m \geq 1$ let $w_m \colon \pi_1(\mathbb{RP}^m) \to \mathbb{F}_2$ be the monodromy of the two-sheeted covering $S^m \to \mathbb{RP}^m$ (Example 19.4.10): $w_m[\gamma] = 0$ if and only if the lifts of γ are loops. It is a homomorphism (Corollary 19.4.4(3)) and nonzero. Since $\pi_1(\mathbb{RP}^n) \cong \mathbb{Z}/2\mathbb{Z}$, w_n corresponds to the generator α of $H^1(\mathbb{RP}^n; \mathbb{F}_2)$, and $\pi_1(\mathbb{RP}^n)$ is generated by the image of a

path γ in S^n from x to $-x$. Its image under $\bar{g}$ is the image of the path $g \circ \gamma$ from $g(x)$ to $-g(x)$, whose lift is not a loop; so $w_{n-1}(\bar{g}_*[\gamma]) = 1 = w_n[\gamma]$ and $\bar{g}^*(w_{n-1}) = \alpha$. Then $\bar{g}^*(w_{n-1}^n) = \alpha^n \neq 0$ by Theorem 22.5.1, but $w_{n-1}^n \in H^n(\mathbb{RP}^{n-1}; \mathbb{F}_2) = 0$, a contradiction. If $f(x) \neq f(-x)$ for all x, then $g(x) = (f(x) - f(-x))/\|f(x) - f(-x)\|$ is an odd map $S^n \to S^{n-1}$. □

22.6 Cup length and coverings by contractible sets

Lemma 22.6.1. *Let A, B be open subsets of X. The cup product of cochains induces a product $H^k(X, A; R) \times H^l(X, B; R) \to H^{k+l}(X, A \cup B; R)$, compatible with the absolute cup product under the maps $H^\bullet(X, C; R) \to H^\bullet(X; R)$ for $C = A, B, A \cup B$.*

Proof. If φ vanishes on simplices in A and ψ on simplices in B, then $\varphi \smile \psi$ vanishes on simplices in A (because of its front face) and on simplices in B (because of its back face), hence on the subcomplex $C(A) + C(B) \subseteq C(A \cup B)$. By Lemma 22.3.2, this gives a product with values in the cohomology of $\mathrm{Hom}(C(X)/(C(A) + C(B)), R)$. It remains to identify this cohomology with $H^\bullet(X, A \cup B; R)$. The inclusion $C(A) + C(B) \to C(A \cup B)$ is a chain homotopy equivalence by Proposition 21.5.3, applied to the open cover $\{A, B\}$ of $A \cup B$, so it induces isomorphisms on cohomology with coefficients in R. The sequences $0 \to C(A) + C(B) \to C(X) \to C(X)/(C(A) + C(B)) \to 0$ and $0 \to C(A \cup B) \to C(X) \to C(X, A \cup B) \to 0$ split degreewise, the quotients being free on the simplices not in the subcomplex, so they give long exact cohomology sequences after applying $\mathrm{Hom}(-, R)$, and the inclusions give a map between them. The five lemma (Lemma 15.4.2) shows that $H^\bullet(X, A \cup B; R) \to H^\bullet(\mathrm{Hom}(C(X)/(C(A) + C(B)), R))$ is an isomorphism. Compatibility with the absolute product holds on cochains. □

Proposition 22.6.2. *Let X be covered by open sets $U_1, \dots, U_n$ such that each inclusion $U_i \to X$ is homotopic to a constant map. Then $\alpha_1 \smile \cdots \smile \alpha_n = 0$ for all $\alpha_i \in H^{k_i}(X; R)$ with $k_i \geq 1$.*

Proof. The restriction $H^k(X; R) \to H^k(U_i; R)$ is induced by the inclusion, which is homotopic to a map factoring through a point, so it is zero for $k \geq 1$, as $H^k(\mathrm{pt}; R) = 0$ (Theorem 22.2.2(1)). By the long exact sequence of the pair (X, U_i), each α_i lifts to $\tilde{\alpha}_i \in H^{k_i}(X, U_i; R)$. By Lemma 22.6.1, applied $n - 1$ times, $\tilde{\alpha}_1 \smile \cdots \smile \tilde{\alpha}_n$ lies in $H^\bullet(X, U_1 \cup \cdots \cup U_n; R) = H^\bullet(X, X; R) = 0$ and maps to $\alpha_1 \smile \cdots \smile \alpha_n$. □

Corollary 22.6.3.

1. *The torus T^n cannot be covered by n open sets that are contractible in T^n, but it can be covered by $n + 1$ of them.*
2. *$\mathbb{CP}^n$ cannot be covered by n open sets contractible in $\mathbb{CP}^n$, and it is covered by the $n+1$ standard affine charts, each homeomorphic to $\mathbb{C}^n$.*
3. *$\mathbb{RP}^n$ cannot be covered by n open sets contractible in $\mathbb{RP}^n$.*

Proof. By Proposition 22.6.2 it suffices to exhibit n classes of positive degree with nonzero product: $x_1 \smile \cdots \smile x_n$ generates $H^n(T^n)$ (Example 22.4.3), and $\alpha^n \neq 0$ in $H^{2n}(\mathbb{CP}^n; \mathbb{Z})$ and in $H^n(\mathbb{RP}^n; \mathbb{F}_2)$ (Theorem 22.5.1). For the covering of $T^n = \mathbb{R}^n/\mathbb{Z}^n$ by $n + 1$ sets, let U_j, $0 \leq j \leq n$, be the image of the open set of points $x \in \mathbb{R}^n$ whose coordinates avoid the interval $[\frac{j}{n+1} + \frac{1}{4(n+1)}, \frac{j}{n+1} + \frac{3}{4(n+1)}]$ modulo 1: each coordinate then ranges in an open interval of $\mathbb{R}/\mathbb{Z}$, so U_j is homeomorphic to an open cube, which is contractible. Every x lies in some U_j, because its n coordinates can meet at most n of the $n + 1$ disjoint intervals. □

Remark 22.6.4. The least m such that X is covered by $m + 1$ open sets contractible in X is the *Lusternik–Schnirelmann category* of X. Proposition 22.6.2 bounds it from below by the *cup length*,

the largest number of classes of positive degree with nonzero product. By Corollary 22.6.3 the category of T^n is n. Lusternik and Schnirelmann introduced this invariant to bound from below the number of critical points of smooth functions on closed manifolds.

22.7 The cap product

Definition 22.7.1. For $\sigma\colon \Delta^n \to X$ and $\varphi \in C^k(X;R)$ with $k \le n$, the *cap product* is

$$\sigma \frown \varphi = \varphi(\sigma|[e_0,\dots,e_k]) \cdot \sigma|[e_k,\dots,e_n] \in C_{n-k}(X;R),$$

extended R-linearly to $C_n(X;R) = C_n(X) \otimes R$.

Proposition 22.7.2. *1. $\partial(\sigma \frown \varphi) = (-1)^k(\partial\sigma \frown \varphi - \sigma \frown \delta\varphi)$, so the cap product induces $\frown\colon H_n(X;R) \times H^k(X;R) \to H_{n-k}(X;R)$, and relative versions $H_n(X,A;R) \times H^k(X;R) \to H_{n-k}(X,A;R)$ and $H_n(X,A;R) \times H^k(X,A;R) \to H_{n-k}(X;R)$.*

2. (Naturality) $f_*(c) \frown \varphi = f_*(c \frown f^*\varphi)$ for $f\colon X \to Y$, $c \in H_n(X;R)$ and $\varphi \in H^k(Y;R)$.

3. $\psi(c \frown \varphi) = (\varphi \smile \psi)(c)$ for $\psi \in C^{n-k}(X;R)$, and $(c \frown \varphi) \frown \psi = c \frown (\varphi \smile \psi)$.

Proof. (1) We have $\partial(\sigma \frown \varphi) = \sum_{i\ge k}(-1)^{i-k}\varphi(\sigma|[e_0,\dots,e_k])\,\sigma|[e_k,\dots,\widehat{e_i},\dots,e_n]$, while

$$\partial\sigma \frown \varphi = \sum_{i\le k}(-1)^i\varphi(\sigma|[e_0,\dots,\widehat{e_i},\dots,e_{k+1}])\,\sigma|[e_{k+1},\dots,e_n] + \sum_{i\ge k+1}(-1)^i\varphi(\sigma|[e_0,\dots,e_k])\,\sigma|[e_k,\dots,\widehat{e_i},\dots,e_n]$$

and $\sigma \frown \delta\varphi = \sum_{i\le k+1}(-1)^i\varphi(\sigma|[e_0,\dots,\widehat{e_i},\dots,e_{k+1}])\,\sigma|[e_{k+1},\dots,e_n]$. Subtracting, what remains is the second sum together with the term $-(-1)^{k+1}\varphi(\sigma|[e_0,\dots,e_k])\,\sigma|[e_{k+1},\dots,e_n]$, which is the term $i = k$; this is $(-1)^k\partial(\sigma \frown \varphi)$. Hence cycle $\frown$ cocycle is a cycle, boundary $\frown$ cocycle and cycle $\frown$ coboundary are boundaries. For the relative versions, $\sigma \frown \varphi$ lies in $C(A)$ if σ does, and vanishes if φ vanishes on A and σ lies in A. (2) and (3) are immediate from the definitions: $\psi(\sigma \frown \varphi) = \varphi(\sigma|[e_0,\dots,e_k])\psi(\sigma|[e_k,\dots,e_n]) = (\varphi \smile \psi)(\sigma)$. □

Remark 22.7.3. For a closed oriented n-manifold M with fundamental class $[M] \in H_n(M;\mathbb{Z})$, the map $H^k(M;R) \to H_{n-k}(M;R)$, $\alpha \mapsto [M] \frown \alpha$, is an isomorphism: this is Poincaré duality (Chapter 23). Combined with Proposition 22.7.2(3) it shows that the cup product pairing $H^k(M;F) \times H^{n-k}(M;F) \to F$, $(\alpha,\beta) \mapsto \langle \alpha \smile \beta, [M]\rangle$, is nondegenerate over a field F.

22.8 Exercises

Exercise 22.8.1. Compute $H^k(\mathbb{RP}^n;\mathbb{Z}/4\mathbb{Z})$ from the cellular cochain complex, and check the result with the universal coefficient theorem.

Solution. The cellular cochain complex is $\mathbb{Z}/4 \to \mathbb{Z}/4 \to \cdots$ with differential $\mathrm{Hom}(d_{k+1},\mathbb{Z}/4)$, multiplication by $1 + (-1)^{k+1}$: zero out of even degrees and multiplication by 2 out of odd degrees. So $H^0 = \mathbb{Z}/4$; for $0 < k < n$, $H^k \cong \mathbb{Z}/2$ (for k odd the kernel of 2 is $2\mathbb{Z}/4$ and the image of the previous zero map is 0; for k even the kernel is all of $\mathbb{Z}/4$ and the image is $2\mathbb{Z}/4$); and H^n is $\mathbb{Z}/4$ for n odd and $\mathbb{Z}/2$ for n even. With universal coefficients, for $0 < k < n$ odd: $\mathrm{Hom}(\mathbb{Z}/2,\mathbb{Z}/4) \oplus \mathrm{Ext}(0,\mathbb{Z}/4) = \mathbb{Z}/2$; for k even: $\mathrm{Hom}(0,\mathbb{Z}/4) \oplus \mathrm{Ext}(\mathbb{Z}/2,\mathbb{Z}/4) = \mathbb{Z}/4/2\mathbb{Z}/4 = \mathbb{Z}/2$. □

Exercise 22.8.2. Let Σ_g be the closed orientable surface of genus g. Using the map $\Sigma_g \to \bigvee_{i=1}^g T^2$ collapsing $g-1$ circles, show that $H^1(\Sigma_g) \cong \mathbb{Z}^{2g}$ has a basis a_i, b_i with $a_i \smile b_i = -b_i \smile a_i$ a generator of $H^2(\Sigma_g) \cong \mathbb{Z}$, and all other products of basis elements zero.

Exercise 22.8.3. For a map $f\colon S^{2n-1} \to S^n$ with $n \geq 2$ let $X_f = S^n \cup_f D^{2n}$. Show that $H^n(X_f) \cong \mathbb{Z} \cdot a$ and $H^{2n}(X_f) \cong \mathbb{Z} \cdot b$ and define the Hopf invariant $h(f)$ by $a^2 = h(f)b$. Show that $h(f) = 0$ if n is odd, and that h of the Hopf map $S^3 \to S^2$ is ± 1, using $X_f = \mathbb{CP}^2$.

Exercise 22.8.4. Show that $S^2 \vee S^4$ is not a retract of $\mathbb{CP}^2$, even up to homotopy: there are no maps $i\colon S^2 \vee S^4 \to \mathbb{CP}^2$ and $r\colon \mathbb{CP}^2 \to S^2 \vee S^4$ with $ri \simeq \mathrm{id}$. Show also that there is no map $\mathbb{CP}^2 \to S^2$ inducing an isomorphism on H^2.

Solution. A map $f\colon \mathbb{CP}^2 \to S^2$ with f^* an isomorphism on H^2 would give $f^*(x)^2 = f^*(x^2) = 0$ for a generator x of $H^2(S^2)$, while $f^*(x)$ generates $H^2(\mathbb{CP}^2)$ and its square generates $H^4(\mathbb{CP}^2)$. If $S^2 \vee S^4 \xrightarrow{i} \mathbb{CP}^2 \xrightarrow{r} S^2 \vee S^4$ composed to the identity (up to homotopy), then i^* would be surjective, and r^* injective; a generator $u \in H^2(S^2 \vee S^4)$ would satisfy $r^*(u)^2 = r^*(u^2) = 0$ with $r^*(u) = \pm$ a generator of $H^2(\mathbb{CP}^2)$ (as r^* is injective on $H^2 \cong \mathbb{Z}$ and $i^*r^* = \mathrm{id}$ forces r^* to be split injective), contradicting $\alpha^2 \neq 0$. □

Chapter 23

Topological Manifolds and Poincaré Duality

23.1 Introduction

A closed oriented n-manifold M has a fundamental class $[M] \in H_n(M;\mathbb{Z})$, and capping with it is an isomorphism $H^k(M;R) \cong H_{n-k}(M;R)$ (Theorem 23.6.2). Duality is what turns a closed oriented submanifold, or a complex subvariety, of dimension d into a cohomology class of degree $n-d$, and it makes the cup product pairing between complementary degrees nondegenerate. We follow [Hat02, Section 3.3].

Prerequisites. Chapter 22 (Singular Cohomology).

For a space X and $A \subseteq X$ we write $H_i(X \mid A;R) = H_i(X, X \setminus A;R)$ and $H^i(X \mid A;R) = H^i(X, X \setminus A;R)$, and $H_i(X \mid x)$ for $A = \{x\}$.

23.2 Topological manifolds

Definition 23.2.1. A *topological n-manifold* is a Hausdorff, second countable space in which every point has an open neighbourhood homeomorphic to $\mathbb{R}^n$ (Definition 20.5.1, where the role of each axiom is discussed). A *chart* is such a neighbourhood with a homeomorphism to $\mathbb{R}^n$. A manifold is *closed* if it is compact. A *manifold with boundary* is defined in the same way with neighbourhoods homeomorphic to open subsets of the half-space $\{x \in \mathbb{R}^n : x_n \geq 0\}$; its *boundary* ∂M consists of the points that correspond to points with $x_n = 0$.

Lemma 23.2.2. *For a point x of an n-manifold M, $H_i(M \mid x;R) \cong R$ for $i = n$ and 0 otherwise. In particular the dimension of a nonempty manifold is well defined. For an open ball B of finite radius in a chart and $x \in B$, the restriction $H_i(M \mid B;R) \to H_i(M \mid x;R)$ is an isomorphism.*

Proof. Let $V \ni x$ be a chart. The closure of B in V is compact, hence closed in M, so excision of $M \setminus V$ gives $H_i(M \mid B) \cong H_i(V \mid B) \cong H_i(\mathbb{R}^n \mid B)$, and likewise for x. Both $\mathbb{R}^n \setminus B$ and $\mathbb{R}^n \setminus \{x\}$ deformation retract onto a sphere centred at x of radius larger than that of B, by radial projection from x; so the restriction is an isomorphism, and $H_i(\mathbb{R}^n \mid x) \cong \tilde{H}_{i-1}(\mathbb{R}^n \setminus \{x\}) \cong \tilde{H}_{i-1}(S^{n-1};R)$ by the long exact sequence. The dimension is well defined by Corollary 21.6.7(3). □

Example 23.2.3. $\mathbb{R}^n$, open subsets of manifolds, S^n (charts by stereographic projection), and products of manifolds (of dimension $m+n$) are manifolds; hence so is the torus T^n. The projective spaces $\mathbb{RP}^n$ and $\mathbb{CP}^n$ are closed manifolds of dimensions n and $2n$ (Chapter 44), and so are the surfaces Σ_g. The disc D^n is a manifold with boundary S^{n-1}. The line with two origins (Counterexample 18.4.12) is locally Euclidean but not Hausdorff.

23.3 Orientations

Definition 23.3.1. Let M be an n-manifold. A *coordinate ball* is an open ball of finite radius in a chart. Let M_R be the set of pairs (x, α) with $x \in M$ and $\alpha \in H_n(M \mid x; R)$, with $p(x, \alpha) = x$. For a coordinate ball B and $\alpha_B \in H_n(M \mid B; R)$ let $U(\alpha_B)$ be the set of pairs (x, α_x) with $x \in B$ and α_x the image of α_B. An *R-orientation* of M is a continuous section $x \mapsto \mu_x$ of p such that each μ_x generates $H_n(M \mid x; R) \cong R$ as an R-module. A $\mathbb{Z}$-orientation is an *orientation*, and M is *(R-)orientable* if it has one. The *orientation double covering* $\tilde{M} \subseteq M_{\mathbb{Z}}$ consists of the pairs (x, α) with α a generator of $H_n(M \mid x; \mathbb{Z}) \cong \mathbb{Z}$.

Proposition 23.3.2. *The sets $U(\alpha_B)$ form a basis of a topology on M_R for which p is a covering map, and $\tilde{M} \to M$ is a two-sheeted covering. Let M be connected.*

1. *M is orientable if and only if $\tilde{M}$ has two components; then M has exactly two orientations.*
2. *If $\pi_1(M)$ has no subgroup of index 2, for instance if M is simply connected, then M is orientable.*
3. *M is $\mathbb{F}_2$-orientable, and an orientable manifold is R-orientable for every R.*

Proof. If $(x, \alpha) \in U(\alpha_B) \cap U(\beta_{B'})$, choose a coordinate ball $B'' \ni x$ inside $B \cap B'$. The images of α_B and $\beta_{B'}$ in $H_n(M \mid B'')$ both restrict to α, so they are equal by Lemma 23.2.2, and their U is contained in both sets. By the same lemma, p maps $U(\alpha_B)$ bijectively onto B, and $p^{-1}(B)$ is the disjoint union of the sets $U(\alpha_B)$, $\alpha_B \in H_n(M \mid B)$. For $B' \subseteq B$, $p^{-1}(B') \cap U(\alpha_B) = U(\alpha_{B'})$ with $\alpha_{B'}$ the restriction, and $p(U(\alpha_{B'})) = B'$; so $p|_{U(\alpha_B)}$ is a homeomorphism onto B, and p is a covering map. The generators form a union of sheets, two over each coordinate ball.

(1) A section s of a covering over a connected space has open image (near $s(x)$ it lies in one sheet, by continuity) and closed image (it is the set of points e with $s(p(e)) = e$, and the covering space is Hausdorff because M is and distinct sheets are disjoint open sets). So an orientation μ gives two components $\{\mu_x\}$ and $\{-\mu_x\}$ of $\tilde{M}$. Conversely, each sheet over a coordinate ball is connected and so lies in a single component; hence the number of points of a component C of $\tilde{M}$ over x is locally constant, hence constant; if $\tilde{M}$ is not connected it is 1, and $C \to M$ is a bijective covering map, hence a homeomorphism whose inverse is an orientation. The orientations are the sections of $\tilde{M}$, which are the inverses of the two components.

(2) If M is not orientable, $\tilde{M}$ is a connected two-sheeted covering, and $p_*\pi_1(\tilde{M})$ has index 2 by Corollary 19.4.4(3).

(3) $H_n(M \mid x; \mathbb{F}_2)$ has a unique generator, and $x \mapsto$ this generator is continuous since it lies in $U(\alpha_B)$ for the generator α_B. The coefficient homomorphism $\mathbb{Z} \to R$ induces a continuous map $M_{\mathbb{Z}} \to M_R$ over M sending generators to generators. □

23.4 The fundamental class

Lemma 23.4.1 (Relative Mayer–Vietoris)**.** *Let Y_1, Y_2 be open subsets of a space X. There are natural long exact sequences*

$$\cdots \to H_i(X, Y_1 \cap Y_2) \to H_i(X, Y_1) \oplus H_i(X, Y_2) \to H_i(X, Y_1 \cup Y_2) \to H_{i-1}(X, Y_1 \cap Y_2) \to \cdots$$

and $\cdots \to H^i(X, Y_1 \cup Y_2) \to H^i(X, Y_1) \oplus H^i(X, Y_2) \to H^i(X, Y_1 \cap Y_2) \to H^{i+1}(X, Y_1 \cup Y_2) \to \cdots$, with any coefficients.

Proof. The sequence $0 \to C(X)/C(Y_1 \cap Y_2) \to C(X)/C(Y_1) \oplus C(X)/C(Y_2) \to C(X)/(C(Y_1) + C(Y_2)) \to 0$, with maps $x \mapsto (x, -x)$ and $(x, y) \mapsto x + y$, is exact, because $C(Y_1) \cap C(Y_2) = C(Y_1 \cap Y_2)$. All terms are free abelian groups on sets of simplices, so it is degreewise

split, and it remains exact after tensoring with R or applying $\mathrm{Hom}(-, R)$. The inclusion $C(Y_1) + C(Y_2) \to C(Y_1 \cup Y_2)$ is a quasi-isomorphism by Proposition 21.5.3 (the Y_i are open in $Y_1 \cup Y_2$), so by the five lemma the last complex computes $H_\bullet(X, Y_1 \cup Y_2)$; dually in cohomology. □

Lemma 23.4.2. *Let M be an n-manifold and $A \subseteq M$ compact.*

1. *If $x \mapsto \alpha_x$ is a continuous section of M_R over A, there is a unique $\alpha_A \in H_n(M \mid A; R)$ restricting to α_x for every $x \in A$.*
2. *$H_i(M \mid A; R) = 0$ for $i > n$.*

Proof. We omit R from the notation.

Step 1: if the lemma holds for compact A, B and $A \cap B$, it holds for $A \cup B$. Apply Lemma 23.4.1 with $Y_1 = M \setminus A$ and $Y_2 = M \setminus B$:

$$H_{i+1}(M \mid A \cap B) \to H_i(M \mid A \cup B) \xrightarrow{\Phi} H_i(M \mid A) \oplus H_i(M \mid B) \xrightarrow{\Psi} H_i(M \mid A \cap B),$$

where Φ consists of restrictions and Ψ is the difference of restrictions (up to the signs of the maps). For $i > n$ the outer terms vanish, giving (2). For $i = n$, Φ is injective, which gives uniqueness since restrictions to points factor through A or B. For existence, α_A and α_B restrict to elements of $H_n(M \mid A \cap B)$ that both restrict to α_x at every point, hence agree by uniqueness for $A \cap B$; so (α_A, α_B) lies in the kernel of Ψ and comes from some $\alpha_{A \cup B}$.

Step 2: $M = \mathbb{R}^n$ and A compact convex. Choose $x \in A$. Both $\mathbb{R}^n \setminus A$ and $\mathbb{R}^n \setminus \{x\}$ deformation retract onto a large sphere centred at x (moving points along rays from x, which leave the convex set A once), so $H_i(\mathbb{R}^n \mid A) \to H_i(\mathbb{R}^n \mid x)$ is an isomorphism, giving (2). For (1), let α_A be the element restricting to α_x. Choosing a coordinate ball $B \supseteq A$, the image of α_A at points of A is a continuous section (it comes from $H_n(\mathbb{R}^n \mid B) \cong H_n(\mathbb{R}^n \mid A)$). It agrees with the given section at x, so everywhere, since A is connected (Lemma 19.4.2).

Step 3: $M = \mathbb{R}^n$ and $A = A_1 \cup \cdots \cup A_m$ with each A_j compact convex. Induction on m using Step 1, since $(A_1 \cup \cdots \cup A_{m-1}) \cap A_m = \bigcup_{j<m} (A_j \cap A_m)$ is a union of $m - 1$ compact convex sets.

Step 4: $M = \mathbb{R}^n$ and A arbitrary compact. Let $\alpha \in H_i(\mathbb{R}^n \mid A)$ be represented by a chain z with ∂z a chain in $\mathbb{R}^n \setminus A$. The images of the simplices of ∂z form a compact set C disjoint from A. Cover $\mathbb{R}^n$ by the closed cubes of a grid of side smaller than the distance from A to C divided by $\sqrt{n}$, and let K be the union of those meeting A; then $A \subseteq K$, $K \cap C = \emptyset$, and z defines $\alpha_K \in H_i(\mathbb{R}^n \mid K)$ mapping to α. If $i > n$, $\alpha_K = 0$ by Step 3, so $\alpha = 0$. If $i = n$ and α restricts to 0 at every point of A, then α_K restricts to 0 at every point of K: each cube Q meets A, and the restriction of α_K to Q is determined by its value at a point of $Q \cap A$ (Step 2). By uniqueness in Step 3, $\alpha_K = 0$, so $\alpha = 0$; this is uniqueness. For existence, $M_R \to \mathbb{R}^n$ is a trivial covering $\mathbb{R}^n \times R$ (the restrictions from a large ball are isomorphisms), so a section over A is a locally constant function $A \to R$. By compactness it takes finitely many values, on disjoint compact sets A_r, which have positive distances from each other. Choose the grid so that twice the diameter of a cube is smaller than these distances. Then two cubes meeting different sets A_r are disjoint, so the unions K_r of the cubes meeting A_r are disjoint closed sets covering K, and the section extends to the locally constant function equal to r on K_r. Step 3 gives α_K, and its image in $H_n(\mathbb{R}^n \mid A)$ is the required class.

Step 5: general M. A compact $A \subseteq M$ is a finite union $A_1 \cup \cdots \cup A_m$ of compact sets each contained in a chart $V_j \cong \mathbb{R}^n$. By excision, $H_i(M \mid A_j) \cong H_i(V_j \mid A_j)$, compatibly with restrictions to points, so the lemma holds for sets contained in a chart by Step 4. Conclude by induction on m using Step 1, as $(A_1 \cup \cdots \cup A_{m-1}) \cap A_m$ is contained in the chart V_m. □

Theorem 23.4.3. *Let M be a closed connected n-manifold. Then $H_i(M; R) = 0$ for $i > n$, and the restriction $H_n(M; R) \to H_n(M \mid x; R) \cong R$ is injective for every x. Its image is R if M is R-orientable,*

and $\{r \in R : 2r = 0\}$ *otherwise. If* M *is* R*-oriented by* μ*, the unique class* $[M] \in H_n(M;R)$ *restricting to* μ_x *for all* x *is the* fundamental class*. In particular* $H_n(M;\mathbb{Z}) \cong \mathbb{Z}$ *if* M *is orientable and* 0 *otherwise.*

Proof. Apply Lemma 23.4.2 with $A = M$, noting $H_i(M \mid M) = H_i(M)$: classes in $H_n(M)$ correspond to sections of M_R over M. A section over the connected space M is determined by its value at x, by Lemma 19.4.2. The set M_r of pairs (y, α) with $\alpha = \pm r\mu_y$ for a local generator μ_y is a union of components of M_R (it is open and closed, as can be checked on the sets $U(\alpha_B)$). If $r = -r$, then $M_r \to M$ is a homeomorphism, and there is a section with value r. If $r \neq -r$, then $M_r \cong \tilde{M}$ (identify $(y, \pm r\mu_y)$ with $(y, \pm\mu_y)$), which has a section if and only if M is orientable. Writing a general element of R as r times a generator gives the description of the image; for R-orientability, an R-orientation takes values $u\mu_y$ with u a unit, and $u \neq -u$ unless $2 = 0$ in R. □

23.5 Cohomology with compact supports

Lemma 23.5.1. *Let* I *be a directed set (Example 6.11.2). Colimits of diagrams* $I \to \mathsf{Ab}$ *exist, their underlying sets are the colimits of the underlying sets (Proposition 6.11.4), and they are exact: a colimit of exact sequences is exact. Hence they commute with homology of chain complexes. If* $\phi\colon J \to I$ *is an order-preserving map of directed sets with* cofinal *image (every* $i \in I$ *has some* j *with* $i \leq \phi(j)$*), then the induced map* $\operatorname{colim}_J A_{\phi(j)} \to \operatorname{colim}_I A_i$ *is an isomorphism.*

Proof. On the set $\operatorname{colim}_i A_i = (\coprod_i A_i)/\!\approx$, define $[a_i] + [b_j] = [u(a_i) + v(b_j)]$ after mapping both to a common k; this is well defined by Proposition 6.11.4 and makes the colimit an abelian group with the universal property in Ab. If $A_i \xrightarrow{f_i} B_i \xrightarrow{g_i} C_i$ are exact and $[b_i]$ maps to 0, then $g_i(b_i)$ becomes 0 in some C_k; replacing i by k we may assume $g_i(b_i) = 0$, so $b_i = f_i(a_i)$. So the colimit sequence is exact. Homology is kernel modulo image, hence preserved. For ϕ with cofinal image, every element of colim_I comes from some A_i and hence from $A_{\phi(j)}$ with $\phi(j) \geq i$. If an element of $A_{\phi(j)}$ vanishes in colim_I, it vanishes in some A_i with $i \geq \phi(j)$; choose j'' with $\phi(j'') \geq i$ and $j' \geq j, j''$; then it vanishes in $A_{\phi(j')}$, hence in colim_J. □

Definition 23.5.2. The *cohomology with compact supports* of a Hausdorff space X is $H^k_c(X;R) = \operatorname{colim}_K H^k(X \mid K;R)$, the colimit over the compact subsets $K \subseteq X$ directed by inclusion, along the restriction maps $H^k(X \mid K) \to H^k(X \mid L)$ for $K \subseteq L$. If X is compact, $H^k_c(X;R) = H^k(X;R)$. For an open $U \subseteq X$ and compact $K \subseteq U$, excision gives $H^k(U \mid K) \cong H^k(X \mid K)$; these isomorphisms induce a map $H^k_c(U;R) \to H^k_c(X;R)$ (extension by zero).

Example 23.5.3. The open balls centred at 0 are cofinal among compact subsets of $\mathbb{R}^n$, and for balls $B \subseteq B'$ the restriction $H^k(\mathbb{R}^n \mid B) \to H^k(\mathbb{R}^n \mid B')$ is an isomorphism, as in Lemma 23.2.2. So $H^k_c(\mathbb{R}^n;R) \cong R$ for $k = n$ and 0 otherwise, whereas $H^k(\mathbb{R}^n;R)$ is R in degree 0. Compactly supported cohomology is not a homotopy invariant.

Definition 23.5.4. Let M be an R-oriented n-manifold. For compact K let $\mu_K \in H_n(M \mid K;R)$ be the class of Lemma 23.4.2 restricting to the orientation. For $K \subseteq L$, μ_L restricts to μ_K, and naturality of the cap product $H_n(M \mid K) \times H^k(M \mid K) \to H_{n-k}(M)$ (Proposition 22.7.2) shows that the maps $\varphi \mapsto \mu_K \frown \varphi$ are compatible. They induce the *duality map* $D_M\colon H^k_c(M;R) \to H_{n-k}(M;R)$. If M is closed, $D_M(\varphi) = [M] \frown \varphi$.

23.6 Poincaré duality

Lemma 23.6.1. *Let* $M = U \cup V$ *with* U, V *open. There are long exact Mayer–Vietoris sequences*

$$\cdots \to H^k_c(U \cap V) \to H^k_c(U) \oplus H^k_c(V) \to H^k_c(M) \to H^{k+1}_c(U \cap V) \to \cdots$$

and $\cdots \to H_{n-k}(U \cap V) \to H_{n-k}(U) \oplus H_{n-k}(V) \to H_{n-k}(M) \to H_{n-k-1}(U \cap V) \to \cdots$, *and the duality maps* $D_{U\cap V}$, $D_U \oplus D_V$, D_M *form a diagram between them that commutes up to sign.*

Proof. For compact $K \subseteq U$ and $L \subseteq V$, Lemma 23.4.1 with $Y_1 = M \setminus K$, $Y_2 = M \setminus L$ and excision give an exact sequence $\cdots \to H^k(U \cap V \mid K \cap L) \to H^k(U \mid K) \oplus H^k(V \mid L) \to H^k(M \mid K \cup L) \to H^{k+1}(U \cap V \mid K \cap L) \to \cdots$, natural in the pair (K, L). Take the colimit over such pairs (Lemma 23.5.1). The sets $K \cap L$ are cofinal among compact subsets of $U \cap V$ (take $K = L = C$), and the sets $K \cup L$ are cofinal among compact subsets of M: given compact $C \subseteq M$, the disjoint closed subsets $C \setminus V$ and $C \setminus U$ of the normal space C can be separated, giving an open $W \subseteq C$ with $C \setminus V \subseteq W \subseteq \overline{W} \subseteq U$, and then $K = \overline{W} \subseteq U$ and $L = C \setminus W \subseteq V$ are compact with $K \cup L = C$. This gives the first sequence; the second is Theorem 21.5.5. Commutativity of the squares not involving connecting maps is naturality of the cap product. Commutativity up to sign of the remaining square is [Hat02, Lemma 3.36]. □

Theorem 23.6.2 (Poincaré duality)**.** *For every R-oriented n-manifold M, the duality map $D_M \colon H^k_c(M; R) \to H_{n-k}(M; R)$ is an isomorphism for all k. In particular, if M is closed, $H^k(M; R) \to H_{n-k}(M; R)$, $\alpha \mapsto [M] \frown \alpha$, is an isomorphism.*

Proof. Open subsets of M inherit R-orientations, and the duality maps are compatible with the maps $H^k_c(U) \to H^k_c(M)$ and $H_{n-k}(U) \to H_{n-k}(M)$ by naturality of the cap product. We omit R.

(A) If D_U, D_V and $D_{U\cap V}$ are isomorphisms, so is $D_{U\cup V}$. By Lemma 23.6.1 and the five lemma (signs do not affect it).

(B) If $M = \bigcup_i U_i$ for a family of open sets directed by inclusion and all D_{U_i} are isomorphisms, so is D_M. Every compact subset of M lies in some U_i, so by cofinality $H^k_c(M) = \operatorname{colim}_i H^k_c(U_i)$. Every singular chain has compact image, so $H_{n-k}(M) = \operatorname{colim}_i H_{n-k}(U_i)$. The duality maps are compatible, so $D_M = \operatorname{colim}_i D_{U_i}$ is an isomorphism.

(C) $M = U \subseteq \mathbb{R}^n$ convex open. If $K \subseteq U$ is compact, cover it by finitely many closed balls $D_1, \dots, D_m \subseteq U$; their convex hull is the image of the compact set $\Delta^{m-1} \times D_1 \times \cdots \times D_m$ under $(t, x) \mapsto \sum_j t_j x_j$, hence a compact convex subset of U containing K. So compact convex sets C are cofinal, and for $x \in C$ the restriction $H^k(U \mid C) \to H^k(U \mid x)$ is an isomorphism (excise to $\mathbb{R}^n$ and argue as in Step 2 of the proof of Lemma 23.4.2). Hence $H^k_c(U)$ is 0 for $k \neq n$, and so is $H_{n-k}(U)$, as U is contractible. For $k = n$, $H^n(U \mid C) \cong \operatorname{Hom}(H_n(U \mid C; \mathbb{Z}), R)$ by universal coefficients, and the composite of D_U with the augmentation $H_0(U) \cong R$ sends φ to $\varphi(\mu_C)$ by Proposition 22.7.2(3) (with $\psi = 1$). Since μ_C is a unit times the image of a generator of $H_n(U \mid C; \mathbb{Z}) \cong \mathbb{Z}$, this is an isomorphism.

(D) $M = U \subseteq \mathbb{R}^n$ open. Finite unions of convex open sets satisfy duality by (C) and (A), by induction on the number of sets, since $(U_1 \cup \cdots \cup U_{m-1}) \cap U_m$ is a union of $m - 1$ convex open sets. The open set U is the union of countably many open balls $B_1, B_2, \dots$ (Proposition 18.3.12), hence the increasing union of the sets $B_1 \cup \cdots \cup B_m$; apply (B).

(E) General M. By Zorn's lemma and (B), there is an open $U \subseteq M$ maximal among open sets for which duality holds. If $U \neq M$, choose a chart $V \cong \mathbb{R}^n$ not contained in U. Duality holds for V and for $U \cap V$, which is an open subset of $\mathbb{R}^n$, by (D); so it holds for $U \cup V$ by (A), contradicting maximality. □

Corollary 23.6.3. *Let M be a closed F-oriented n-manifold over a field F. The pairing $H^k(M; F) \times H^{n-k}(M; F) \to F$, $(\alpha, \beta) \mapsto \langle \alpha \smile \beta, [M] \rangle$, is nondegenerate: $\alpha \mapsto \langle \alpha \smile -, [M] \rangle$ is injective, and it is an isomorphism $H^k(M; F) \cong \operatorname{Hom}_F(H^{n-k}(M; F), F)$ when these spaces are finite-dimensional (which is always the case [Hat02, Appendix]).*

Proof. By Proposition 22.7.2(3), $\langle \alpha \smile \beta, [M] \rangle = \langle \beta, [M] \frown \alpha \rangle$. By the universal coefficient theorem over a field, $H^{n-k}(M; F) = \operatorname{Hom}_F(H_{n-k}(M; F), F)$, so $\langle \alpha \smile -, [M] \rangle$ is the functional

"evaluate on $[M] \frown \alpha$" on this dual space. Since $\alpha \mapsto [M] \frown \alpha$ is injective, and a nonzero vector is detected by the dual space, the map is injective; for finite dimensions it is bijective by counting dimensions. □

Corollary 23.6.4. *Let M be a closed n-manifold whose integral homology is finitely generated, and define $\chi(M) = \sum_i (-1)^i \operatorname{rank} H_i(M;\mathbb{Z})$. Then $\chi(M) = \sum_i (-1)^i \dim_F H_i(M;F)$ for every field F, and $\chi(M) = 0$ if n is odd.*

Proof. By the universal coefficient theorem (Theorem 21.8.2), $\dim_F H_i(M;F) = \operatorname{rank} H_i(M) + t_i + t_{i-1}$, where t_i is the number of cyclic summands of $H_i(M)$ of order divisible by the characteristic of F (these contribute to $H_i \otimes F$ and to $\operatorname{Tor}_1(H_i, F)$). The t terms cancel in the alternating sum. Take $F = \mathbb{F}_2$, for which M is oriented (Proposition 23.3.2). By duality and universal coefficients, $\dim H_i(M;\mathbb{F}_2) = \dim H^{n-i}(M;\mathbb{F}_2) = \dim H_{n-i}(M;\mathbb{F}_2)$, so $\chi(M) = (-1)^n \chi(M)$. □

Definition 23.6.5. Let M be a closed oriented manifold of dimension $4m$. The *intersection form* is the symmetric bilinear form $Q(\alpha, \beta) = \langle \alpha \smile \beta, [M] \rangle$ on $H^{2m}(M;\mathbb{R})$; it is nondegenerate by Corollary 23.6.3, and symmetric by Theorem 22.3.4. Its *signature* $\sigma(M)$ is the number of positive minus the number of negative eigenvalues. On the free group $H^{2m}(M;\mathbb{Z})/\text{torsion}$ the integral form is unimodular (determinant ± 1) [Hat02, Proposition 3.38]. In dimension $4m+2$ the form on H^{2m+1} is skew-symmetric and nondegenerate.

Example 23.6.6.

1. For $\mathbb{CP}^2$ with the orientation for which $\langle \alpha^2, [\mathbb{CP}^2] \rangle = 1$, $Q = (1)$ and $\sigma = 1$. For $S^2 \times S^2$, in the basis a, b of Example 22.4.3, $Q = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $\sigma = 0$. The connected sum $\mathbb{CP}^2 \# \overline{\mathbb{CP}^2}$ (with reversed orientation on the second summand) has the same Betti numbers as $S^2 \times S^2$, but its integral form is $\operatorname{diag}(1, -1)$, which takes odd values, whereas the form of $S^2 \times S^2$ takes only even values on the diagonal; so the forms are not isomorphic over $\mathbb{Z}$ and the two manifolds are not homotopy equivalent (Counterexample 23.6.7).

2. (Cohomology of $\mathbb{CP}^n$ via duality.) The inclusion $\mathbb{CP}^{n-1} \to \mathbb{CP}^n$ is an isomorphism on H^i for $i \le 2n-2$ by cellular cohomology. By induction on n, α^{n-1} generates $H^{2n-2}(\mathbb{CP}^n)$. Since $\mathbb{CP}^n$ is a closed simply connected manifold (Example 19.7.3), it is orientable, and by unimodularity of the pairing $H^2 \times H^{2n-2} \to H^{2n} \cong \mathbb{Z}$ between free groups of rank 1, $\alpha \smile \alpha^{n-1}$ generates H^{2n}. This proves the part of Theorem 22.5.1 about $\mathbb{CP}^n$; the argument for $\mathbb{RP}^n$ with $\mathbb{F}_2$-coefficients is the same.

Counterexample 23.6.7. Homology groups do not determine a closed simply connected 4-manifold up to homotopy equivalence; the cup product is needed. The manifolds S^4, $\mathbb{CP}^2$ and $S^2 \times S^2$ are closed and simply connected (Corollary 19.5.2, Example 19.7.3 and Proposition 19.3.5), but they are told apart already by H_2, which is $0, \mathbb{Z}$ and $\mathbb{Z}^2$. The interesting pair is $M = S^2 \times S^2$ and $N = \mathbb{CP}^2 \# \overline{\mathbb{CP}^2}$ of Example 23.6.6(1).

1. *N is simply connected.* If X is a closed simply connected n-manifold with $n \ge 3$ and X' is X minus an open coordinate ball, then X is obtained from X' by attaching an n-cell, so $\pi_1(X') \cong \pi_1(X) = 1$ (Proposition 19.5.5). The connected sum is the union of two such pieces along a sphere S^3; thickening them slightly, they become open sets that deformation retract onto the pieces and meet in $S^3 \times (-1, 1)$, which is simply connected, so $\pi_1(N) = 1$ by Corollary 19.5.2.

2. *Same homology.* Both have homology $\mathbb{Z}, 0, \mathbb{Z}^2, 0, \mathbb{Z}$ in degrees $0, \dots, 4$. For N: collapsing the gluing sphere S^3 gives a map $N \to N/S^3 = \mathbb{CP}^2 \vee \overline{\mathbb{CP}^2}$ which, by the long exact sequence of the pair (N, S^3) and $\tilde{H}^k(S^3) = 0$ for $k \le 2$, is an isomorphism on H^k for

$k \leq 2$; on H^4 it sends the generator of each summand (the class pairing to 1 with its fundamental class) to the generator of $H^4(N;\mathbb{Z})$, since the collapse map has degree one on each summand. In the wedge the product of classes from different summands vanishes (Example 22.3.5). This gives the form $\mathrm{diag}(1,-1)$ on $H^2(N;\mathbb{Z}) \cong \mathbb{Z}^2$.

3. *Not homotopy equivalent.* A homotopy equivalence $f : N \to M$ induces an isomorphism of cohomology rings, which is ± 1 on $H^4 \cong \mathbb{Z}$; so f^* would identify the integral intersection form of M with $\pm$ that of N. But $Q_M(x,x) = 2pq$ is even for every $x = pa + qb \in H^2(M;\mathbb{Z})$, while $Q_N(e,e) = \pm 1$ for the generator e coming from $\mathbb{CP}^2$.

The same argument separates $S^2 \times S^2$ from $\mathbb{CP}^2 \# \mathbb{CP}^2$, whose form $\mathrm{diag}(1,1)$ is moreover definite.

Remark 23.6.8. Compare the case where the homology is that of a sphere. Let M be a closed simply connected 4-manifold with $H_*(M) \cong H_*(S^4)$. Then $\pi_2(M) = \pi_3(M) = 0$ and $\pi_4(M) \cong H_4(M) \cong \mathbb{Z}$ by the Hurewicz theorem, and a map $S^4 \to M$ representing a generator induces isomorphisms on all homology groups; as M has the homotopy type of a CW complex, it is a homotopy equivalence by Whitehead's theorem in its homological form [Hat02, Section 4.2 and Appendix]. So a simply connected homology 4-sphere is a homotopy 4-sphere, and Freedman proved that it is homeomorphic to S^4, the topological Poincaré conjecture in dimension 4 [Fre82]; whether it must be *diffeomorphic* to S^4 is the smooth Poincaré conjecture in dimension 4, which is open. Counterexample 23.6.7 shows that as soon as $H_2 \neq 0$ homology is not enough. What is enough is the cup product: by theorems of Whitehead and Milnor, the homotopy type of a closed simply connected 4-manifold is determined by its integral intersection form [Whi49], [Mil58], and by Freedman its homeomorphism type is determined by the form together with one invariant in $\mathbb{Z}/2\mathbb{Z}$.

23.7 Other dualities

Theorem 23.7.1 (Lefschetz duality)**.** *Let M be a compact R-orientable n-manifold with boundary. There is a fundamental class $[M] \in H_n(M, \partial M; R)$ restricting to local orientations, and cap product with it gives isomorphisms $H^k(M, \partial M; R) \cong H_{n-k}(M;R)$ and $H^k(M;R) \cong H_{n-k}(M, \partial M; R)$.*

This is [Hat02, Theorem 3.43]; the proof uses a collar neighbourhood of ∂M to reduce to Theorem 23.6.2 for the interior of M.

Theorem 23.7.2 (Alexander duality)**.** *Let $K \subsetneq S^n$ be a compact, locally contractible, nonempty subspace. Then $\tilde{H}_i(S^n \setminus K;\mathbb{Z}) \cong \tilde{H}^{n-i-1}(K;\mathbb{Z})$ for all i.*

This is [Hat02, Theorem 3.44]. The idea: $\tilde{H}_i(S^n \setminus K) \cong H_c^{n-i}(S^n \setminus K)$ by duality, and $H_c^{n-i}(S^n \setminus K)$ is the colimit of $H^{n-i}(S^n, U)$ over open neighbourhoods U of K, which for locally contractible K is $H^{n-i}(S^n, K) \cong \tilde{H}^{n-i-1}(K)$.

Corollary 23.7.3 (Jordan–Brouwer separation theorem)**.** *If $K \subseteq S^n$ is homeomorphic to S^{n-1}, then $S^n \setminus K$ has exactly two path components. If K is homeomorphic to D^k, then $S^n \setminus K$ is path-connected.*

Proof. By Theorem 23.7.2, $\tilde{H}_0(S^n \setminus K) \cong \tilde{H}^{n-1}(S^{n-1}) \cong \mathbb{Z}$, respectively $\tilde{H}^{n-1}(D^k) = 0$. Since $S^n \setminus K$ is an open subset of a manifold, its path components are its components, and $\tilde{H}_0$ is free of rank one less than their number (Proposition 21.2.5). □

23.8 Borel–Moore homology and fundamental classes

Remark 23.8.1. For a locally compact space X, *Borel–Moore homology* $H_i^{BM}(X;R)$ is the homology of locally finite, possibly infinite, singular chains; for reasonable X (for instance a closed subspace of a manifold that is locally contractible, or a CW pair) it can be computed as $H_i(X^+, \{\infty\}; R)$, where X^+ is the one-point compactification. It is covariant for proper maps, contravariant for open inclusions, and a manifold M oriented over R has a fundamental class $[M] \in H_n^{BM}(M;R)$ even when it is not compact, with $H^k(M;R) \cong H_{n-k}^{BM}(M;R)$. If Z is a closed subset of an R-oriented n-manifold M, then

$$H_i^{BM}(Z;R) \cong H^{n-i}(M, M \setminus Z; R),$$

which for $Z = M$ is the duality above. See [Bre97, Chapter V], [Ive86] and [Ful98, Chapter 19].

Remark 23.8.2. These isomorphisms produce cohomology classes from subspaces. If $N \subseteq M$ is a closed oriented submanifold of dimension d (or, more generally, a closed subspace of "dimension" d with a fundamental class $[N] \in H_d^{BM}(N;R)$), its image under $H_d^{BM}(N) \cong H^{n-d}(M, M \setminus N) \to H^{n-d}(M)$ is the *cohomology class of* N. When N and M are compact it is the Poincaré dual of $i_*[N] \in H_d(M)$. A complex subvariety Z of complex dimension d of a complex manifold of dimension m has a fundamental class in $H_{2d}^{BM}(Z;\mathbb{Z})$ because its singular locus has real codimension at least 2; this gives the *cycle class* $[Z] \in H^{2m-2d}(X;\mathbb{Z})$ of Chapter 63, and the Hodge conjecture asks which rational classes are combinations of such $[Z]$. Under these identifications, cup product corresponds to intersection of cycles meeting transversally.

23.9 Exercises

Exercise 23.9.1. Let M be a closed connected n-manifold with finitely generated homology. Show that the torsion subgroup of $H_{n-1}(M;\mathbb{Z})$ is 0 if M is orientable and $\mathbb{Z}/2\mathbb{Z}$ otherwise.

Solution. By universal coefficients, $H_n(M;\mathbb{Z}/m) \cong H_n(M) \otimes \mathbb{Z}/m \oplus \operatorname{Tor}_1(H_{n-1}(M), \mathbb{Z}/m)$, and $\operatorname{Tor}_1(\mathbb{Z}/a, \mathbb{Z}/m) \cong \mathbb{Z}/\gcd(a,m)$. If M is orientable, $H_n(M) = \mathbb{Z}$ and $H_n(M;\mathbb{Z}/m) \cong \mathbb{Z}/m$ (Theorem 23.4.3), so all $\operatorname{Tor}_1(H_{n-1}, \mathbb{Z}/m)$ vanish and H_{n-1} is torsion-free. Otherwise $H_n(M) = 0$ and $H_n(M;\mathbb{Z}/m) \cong \{r : 2r = 0\}$: for m an odd prime this is 0, so there is no odd torsion; for $m = 2$ it is $\mathbb{Z}/2$, so there is exactly one cyclic summand of even order; for $m = 4$ it is $\mathbb{Z}/2$, which forces that summand to be $\mathbb{Z}/2$. □

Exercise 23.9.2. Show that $\mathbb{RP}^n$ is orientable if and only if n is odd.

Solution. By Example 21.7.5, $H_n(\mathbb{RP}^n;\mathbb{Z})$ is $\mathbb{Z}$ for n odd and 0 for n even; apply Theorem 23.4.3. □

Exercise 23.9.3. Show that a closed simply connected 3-manifold has the homology of S^3.

Solution. It is orientable (Proposition 23.3.2), so $H_3 \cong \mathbb{Z}$ and $H_i = 0$ for $i > 3$. $H_1 = \pi_1^{\mathrm{ab}} = 0$. By duality $H_2 \cong H^1 \cong \operatorname{Hom}(H_1, \mathbb{Z}) \oplus \operatorname{Ext}^1(H_0, \mathbb{Z}) = 0$. □

Exercise 23.9.4. Show that $\sigma(M) = 0$ if M admits an orientation-reversing self-homeomorphism, and compute the intersection form of T^4.

Exercise 23.9.5. Let $K \subseteq S^3$ be homeomorphic to S^1. Show that $H_1(S^3 \setminus K;\mathbb{Z}) \cong \mathbb{Z}$ and $H_2(S^3 \setminus K;\mathbb{Z}) = 0$.

Chapter 24

Cohomology Operations

24.1 Introduction

A cohomology operation is a natural transformation between cohomology functors, such as $x \mapsto x^2$. Because cohomology is represented by Eilenberg–MacLane spaces, operations are the same as cohomology classes of these spaces (Proposition 24.3.2). The Steenrod squares are the most important operations with $\mathbb{F}_2$-coefficients; they refine the cup product and give obstructions that cup products cannot see, such as the theorem that a map of Hopf invariant one $S^{2n-1} \to S^n$ exists only if n is a power of 2 (Corollary 24.4.4). For the Hodge conjecture, operations show that some integral cohomology classes of smooth projective varieties cannot be classes of algebraic cycles (Remark 24.6.2). This chapter is partly a survey: the existence theorems are quoted from [Hat02], [SE62] and [MT68], and what follows from them is proved.

Prerequisites. Chapter 23 (Topological Manifolds and Poincaré Duality) and Chapter 19 (Homotopy and Covering Spaces).

On a first reading. Later chapters use this one only for the Stiefel–Whitney classes and Wu's formula in Chapter 39; it can be skipped until then.

24.2 Eilenberg–MacLane spaces

Definition 24.2.1. Let G be a group and $n \geq 1$, with G abelian if $n \geq 2$. An *Eilenberg–MacLane space* $K(G, n)$ is a connected CW complex with $\pi_n(K(G,n)) \cong G$ and $\pi_i(K(G,n)) = 0$ for $i \neq n$.

Theorem 24.2.2. *For every $n \geq 1$ and every group G (abelian if $n \geq 2$) there is an Eilenberg–MacLane space $K(G,n)$, which can be chosen with a single 0-cell and no other cells of dimension less than n; any two are homotopy equivalent.*

For the proof, which attaches cells to a wedge of n-spheres to realise a presentation of G and then kills the higher homotopy groups, see [Hat02, Section 4.2].

Example 24.2.3.

1. S^1 is a $K(\mathbb{Z}, 1)$, since its universal covering $\mathbb{R}$ is contractible and $\pi_i(S^1) \cong \pi_i(\mathbb{R}) = 0$ for $i \geq 2$ (Proposition 19.8.3). More generally, a connected CW complex with contractible universal covering is a $K(\pi_1, 1)$; this applies to the tori T^n and to the surfaces Σ_g with $g \geq 1$.

2. $\mathbb{RP}^\infty = \bigcup_n \mathbb{RP}^n$ is a $K(\mathbb{Z}/2\mathbb{Z}, 1)$ and $\mathbb{CP}^\infty$ is a $K(\mathbb{Z}, 2)$: they are the quotients of the contractible space $S^\infty = \bigcup_n S^n$ by $\{\pm 1\}$ and by the circle group, and the long exact sequence of the fibration $S^1 \to S^\infty \to \mathbb{CP}^\infty$ gives $\pi_2(\mathbb{CP}^\infty) \cong \pi_1(S^1)$ [Hat02, Section 4.2].

Theorem 24.2.4 (Representability)**.** *Let G be abelian and $n \geq 1$. There is a class $\iota_n \in H^n(K(G,n);G)$ such that for every CW complex X the map $\langle X, K(G,n)\rangle \to H^n(X;G)$, $[f] \mapsto f^*(\iota_n)$, is a bijection, natural in X.*

This is [Hat02, Theorem 4.57]. The class ι_n corresponds under $H^n(K(G,n);G) \cong \mathrm{Hom}(H_n(K(G,n)),G) \cong \mathrm{Hom}(\pi_n(K(G,n)),G) = \mathrm{Hom}(G,G)$ (universal coefficients and the Hurewicz theorem) to the identity.

24.3 Cohomology operations

Definition 24.3.1. A *cohomology operation* of type $(G,m;H,n)$ is a family of maps of sets $\theta_X\colon H^m(X;G) \to H^n(X;H)$, one for each space X, such that $f^*\theta_Y = \theta_X f^*$ for every map $f\colon X \to Y$. It is *additive* if every θ_X is a homomorphism.

Proposition 24.3.2. *Restricted to pointed CW complexes, operations of type $(G,m;H,n)$ correspond bijectively to elements of $H^n(K(G,m);H)$, via $\theta \mapsto \theta(\iota_m)$.*

Proof. On the category whose objects are pointed CW complexes and whose morphisms are pointed homotopy classes of maps, $H^m(-;G)$ is represented by $(K(G,m),\iota_m)$, by Theorem 24.2.4 and Definition 6.6.6, and $H^n(-;H)$ is a functor to Set by homotopy invariance. Operations are natural transformations between them, and the Yoneda lemma (Theorem 6.6.1) identifies these with $H^n(K(G,m);H)$. Explicitly, $u \in H^n(K(G,m);H)$ gives $\theta(x) = f_x^*(u)$, where $f_x\colon X \to K(G,m)$ is the map with $f_x^*(\iota_m) = x$. □

Example 24.3.3.

1. For a commutative ring R, $x \mapsto x^k$ is an operation of type $(R,m;R,km)$, corresponding to ι_m^k; it is not additive in general.
2. A homomorphism $G \to H$ induces an additive operation of type $(G,m;H,m)$, for example the reduction $\rho\colon H^m(X;\mathbb{Z}) \to H^m(X;\mathbb{Z}/p\mathbb{Z})$.
3. Since $H^n(K(G,m);H) = 0$ for $0 < n < m$ ($K(G,m)$ may be chosen with no cells in dimensions $0 < i < m$), there are no nonzero operations lowering degree, except constant ones for $n = 0$.

Proposition 24.3.4 (Bockstein homomorphisms)**.** *Let p be a prime. The short exact sequences $0 \to \mathbb{Z} \xrightarrow{p} \mathbb{Z} \to \mathbb{Z}/p \to 0$ and $0 \to \mathbb{Z}/p \to \mathbb{Z}/p^2 \to \mathbb{Z}/p \to 0$ induce connecting homomorphisms $\tilde\beta\colon H^n(X;\mathbb{Z}/p) \to H^{n+1}(X;\mathbb{Z})$ and $\beta\colon H^n(X;\mathbb{Z}/p) \to H^{n+1}(X;\mathbb{Z}/p)$, the* Bockstein *operations. They satisfy $\beta = \rho\tilde\beta$, $\beta\beta = 0$, the image of $\tilde\beta$ is the p-torsion subgroup $\{y : py = 0\}$ of $H^{n+1}(X;\mathbb{Z})$, and the kernel of $\tilde\beta$ is the image of the reduction $\rho\colon H^n(X;\mathbb{Z}) \to H^n(X;\mathbb{Z}/p)$.*

Proof. Since each $C_n(X)$ is free, $\mathrm{Hom}(C_n(X),-)$ is exact, so a short exact sequence of coefficient groups gives a short exact sequence of cochain complexes and a long exact sequence (Theorem 15.4.3): $H^n(X;\mathbb{Z}) \xrightarrow{p} H^n(X;\mathbb{Z}) \xrightarrow{\rho} H^n(X;\mathbb{Z}/p) \xrightarrow{\tilde\beta} H^{n+1}(X;\mathbb{Z}) \xrightarrow{p} H^{n+1}(X;\mathbb{Z})$. Exactness gives the descriptions of the image and the kernel of $\tilde\beta$. The reduction $\mathbb{Z} \to \mathbb{Z}/p^2$ gives a morphism from the first short exact sequence to the second, and naturality of connecting homomorphisms gives $\beta = \rho\tilde\beta$. Finally $\beta\beta = \rho\tilde\beta\rho\tilde\beta = 0$ because $\tilde\beta\rho = 0$. □

24.4 Steenrod squares

Theorem 24.4.1 (Steenrod)**.** *There are unique additive operations $\mathrm{Sq}^i\colon H^n(X;\mathbb{F}_2) \to H^{n+i}(X;\mathbb{F}_2)$, $i \geq 0$, defined for all spaces and pairs and natural, such that:*

1. $\mathrm{Sq}^0 = \mathrm{id}$;
2. $\mathrm{Sq}^n(x) = x^2$ *if* $|x| = n$, *and* $\mathrm{Sq}^i(x) = 0$ *if* $i > |x|$;
3. *(Cartan formula)* $\mathrm{Sq}^i(xy) = \sum_{j+k=i} \mathrm{Sq}^j(x)\mathrm{Sq}^k(y)$;
4. *they commute with the suspension isomorphisms* $\tilde{H}^n(X) \cong \tilde{H}^{n+1}(SX)$ *and with the connecting homomorphisms of pairs.*

Moreover Sq^1 *is the Bockstein* β *for* $p = 2$, *and for* $a < 2b$ *the* Adem relations

$$\mathrm{Sq}^a\mathrm{Sq}^b = \sum_{c=0}^{\lfloor a/2 \rfloor} \binom{b-c-1}{a-2c} \mathrm{Sq}^{a+b-c}\mathrm{Sq}^c$$

hold, with binomial coefficients taken modulo 2.

See [SE62] and [Hat02, Section 4.L]. Writing $\mathrm{Sq} = \sum_i \mathrm{Sq}^i$ (a finite sum on each class), the Cartan formula says that $\mathrm{Sq}\colon H^*(X;\mathbb{F}_2) \to H^*(X;\mathbb{F}_2)$ is a ring homomorphism. There are analogous *reduced power operations* $P^i\colon H^n(X;\mathbb{F}_p) \to H^{n+2i(p-1)}(X;\mathbb{F}_p)$ for odd primes p, with $P^i(x) = x^p$ for $|x| = 2i$.

Proposition 24.4.2. *In* $H^*(\mathbb{RP}^\infty;\mathbb{F}_2) = \mathbb{F}_2[\alpha]$, $\mathrm{Sq}^i(\alpha^n) = \binom{n}{i}\alpha^{n+i}$. *In* $H^*(\mathbb{CP}^\infty;\mathbb{F}_2) = \mathbb{F}_2[x]$ *with* $|x| = 2$, $\mathrm{Sq}^{2i}(x^n) = \binom{n}{i}x^{n+i}$ *and* $\mathrm{Sq}^{2i+1} = 0$.

Proof. By axioms (1) and (2), $\mathrm{Sq}(\alpha) = \alpha + \alpha^2$. Since Sq is a ring homomorphism, $\mathrm{Sq}(\alpha^n) = (\alpha+\alpha^2)^n = \alpha^n(1+\alpha)^n = \sum_i \binom{n}{i}\alpha^{n+i}$; compare degrees. For $\mathbb{CP}^\infty$, $\mathrm{Sq}^1(x) \in H^3 = 0$, so $\mathrm{Sq}(x) = x + x^2$ and the same computation applies, all odd squares vanishing for degree reasons. The rings are those of Theorem 22.5.1. □

Lemma 24.4.3. *If* $i \geq 1$ *is not a power of* 2, *then* Sq^i *is* decomposable*: it is a sum of composites* $\mathrm{Sq}^a\mathrm{Sq}^b$ *with* $0 < a, b < i$.

Proof. Choose k with $2^k < i < 2^{k+1}$ and let $a = i - 2^k$, so $0 < a < 2^k$ and $a < 2 \cdot 2^k$. In $\mathbb{F}_2[t]$ we have $(1+t)^{2^k} = 1 + t^{2^k}$ (the Frobenius map is a ring homomorphism), hence $(1+t)^{2^k-1} = (1+t^{2^k})/(1+t) = 1 + t + \cdots + t^{2^k-1}$, so $\binom{2^k-1}{a}$ is odd. The Adem relation for $\mathrm{Sq}^a\mathrm{Sq}^{2^k}$ has $c = 0$ term $\binom{2^k-1}{a}\mathrm{Sq}^i = \mathrm{Sq}^i$, so $\mathrm{Sq}^i = \mathrm{Sq}^a\mathrm{Sq}^{2^k} + \sum_{c\geq 1}\binom{2^k-c-1}{a-2c}\mathrm{Sq}^{i-c}\mathrm{Sq}^c$, and all indices lie strictly between 0 and i. □

Corollary 24.4.4 (Hopf invariant one)**.** *Let* $n \geq 2$ *and* $f\colon S^{2n-1} \to S^n$, *and let* $X_f = S^n \cup_f D^{2n}$, *with* $H^n(X_f;\mathbb{Z}) = \mathbb{Z}a$ *and* $H^{2n}(X_f;\mathbb{Z}) = \mathbb{Z}b$. *Define the* Hopf invariant $h(f)$ *by* $a^2 = h(f)b$. *If* $h(f)$ *is odd, then* n *is a power of* 2.

Proof. X_f has cells in dimensions $0, n, 2n$, so by cellular cohomology its integral cohomology is free of rank one in these degrees and zero elsewhere, and the reductions $\bar{a}, \bar{b}$ generate the $\mathbb{F}_2$-cohomology. Reduction is a ring homomorphism, so $\mathrm{Sq}^n(\bar{a}) = \bar{a}^2 = h(f)\bar{b} \neq 0$. If n were not a power of 2, Lemma 24.4.3 would write $\mathrm{Sq}^n(\bar{a})$ as a sum of terms $\mathrm{Sq}^a\mathrm{Sq}^b(\bar{a})$ with $0 < b < n$, and $\mathrm{Sq}^b(\bar{a}) \in H^{n+b}(X_f;\mathbb{F}_2) = 0$. □

Remark 24.4.5. The Hopf maps $S^3 \to S^2$, $S^7 \to S^4$ and $S^{15} \to S^8$ have Hopf invariant one. Adams proved, using secondary cohomology operations, that maps of Hopf invariant one exist only for $n = 1, 2, 4, 8$ [Ada60]. Consequences include the facts that $\mathbb{R}^n$ admits a bilinear multiplication without zero divisors only for $n = 1, 2, 4, 8$, and that S^{n-1} is an H-space only for $n = 1, 2, 4, 8$.

24.5 Cohomology of Eilenberg–MacLane spaces

Definition 24.5.1. A sequence $I = (i_1, \dots, i_r)$ of positive integers is *admissible* if $i_j \geq 2i_{j+1}$ for all j. Its *degree* is $\sum_j i_j$, its *excess* is $e(I) = \sum_j (i_j - 2i_{j+1}) = i_1 - i_2 - \cdots - i_r$ (with $i_{r+1} = 0$), and $\mathrm{Sq}^I = \mathrm{Sq}^{i_1} \cdots \mathrm{Sq}^{i_r}$. The empty sequence is admissible with excess 0.

Theorem 24.5.2 (Serre)**.** *1. The* Steenrod algebra $\mathcal{A}_2$*, the algebra of operations $H^*(-;\mathbb{F}_2) \to H^*(-;\mathbb{F}_2)$ generated by the Sq^i modulo the Adem relations, has the admissible monomials Sq^I as an $\mathbb{F}_2$-basis.*

2. $H^*(K(\mathbb{Z}/2, n);\mathbb{F}_2)$ is the polynomial algebra on the classes $\mathrm{Sq}^I \iota_n$ with I admissible and $e(I) < n$.

3. $H^*(K(\mathbb{Z}, n);\mathbb{F}_2)$ is the polynomial algebra on the classes $\mathrm{Sq}^I \iota_n$ with I admissible, $e(I) < n$ and $i_r \neq 1$.

This is [Ser53]; see also [MT68].

Example 24.5.3. For $n = 3$, the admissible sequences of excess less than 3 not ending in 1 are $\emptyset$, (2), (4, 2), (8, 4, 2), ... So $H^*(K(\mathbb{Z}, 3);\mathbb{F}_2) = \mathbb{F}_2[\iota_3, \mathrm{Sq}^2\iota_3, \mathrm{Sq}^4\mathrm{Sq}^2\iota_3, \dots]$ with generators in degrees $3, 5, 9, 17, \dots$. In low degrees: $H^3 = \langle \iota_3 \rangle$, $H^4 = 0$, $H^5 = \langle \mathrm{Sq}^2\iota_3 \rangle$, $H^6 = \langle \iota_3^2 \rangle$, where $\iota_3^2 = \mathrm{Sq}^3\iota_3 = \mathrm{Sq}^1\mathrm{Sq}^2\iota_3$ by the Adem relation $\mathrm{Sq}^1\mathrm{Sq}^2 = \mathrm{Sq}^3$. Since $\mathrm{Sq}^1 = \rho\tilde{\beta}$ is nonzero on $\mathrm{Sq}^2\iota_3$, the integral class $\tilde{\beta}\mathrm{Sq}^2\iota_3 \in H^6(K(\mathbb{Z}, 3);\mathbb{Z})$ is a nonzero 2-torsion class. In general $x \mapsto \tilde{\beta}\mathrm{Sq}^2\rho(x)$ is an integral operation raising degree by 3, defined in every degree, whose reduction modulo 2 is $\rho\tilde{\beta}\mathrm{Sq}^2\rho(x) = \mathrm{Sq}^1\mathrm{Sq}^2\rho(x) = \mathrm{Sq}^3\rho(x)$; it appears in Remark 24.6.2.

24.6 Complex cobordism and algebraic cycles

Remark 24.6.1. *Complex cobordism MU^** is a generalised cohomology theory: it satisfies all Eilenberg–Steenrod axioms except the dimension axiom. Its coefficient ring is $MU^* = MU^*(\mathrm{pt}) \cong \mathbb{Z}[x_1, x_2, \dots]$ with x_i in degree $-2i$, generated geometrically by classes of complex projective spaces and hypersurfaces [Sto68], and Quillen showed that it carries the universal one-dimensional commutative formal group law [Qui69]. For a compact complex manifold X there is a natural map $MU^*(X) \to H^*(X;\mathbb{Z})$, and a closed complex submanifold $Z \subseteq X$ of codimension c defines a class in $MU^{2c}(X)$ whose image is the cohomology class of Z (Chapter 23). See [Rav86] for the homotopy-theoretic side and [Tho54] for the origins of cobordism.

Remark 24.6.2. Atiyah and Hirzebruch showed that on a smooth projective complex variety X the integral cohomology class of every algebraic cycle lies in the image of $MU^*(X) \to H^*(X;\mathbb{Z})$, using resolution of singularities, and that cohomology operations of odd degree vanish on this image; in particular $\mathrm{Sq}^3\rho(x) = 0$ for the class x of a codimension-2 cycle. They constructed smooth projective varieties with torsion classes $x \in H^4(X;\mathbb{Z})$ with $\mathrm{Sq}^3\rho(x) \neq 0$. Such a class is a Hodge class (torsion classes are Hodge classes trivially), but it is not the class of an algebraic cycle, which disproves the Hodge conjecture with integral coefficients [AH62]. Totaro refined this by showing that the cycle class map factors as $CH^*(X) \to MU^*(X) \otimes_{MU^*} \mathbb{Z} \to H^*(X;\mathbb{Z})$ [Tot97]. These results are developed in Chapter 73.

24.7 Exercises

Exercise 24.7.1. Deduce from the Adem relations that $\mathrm{Sq}^1\mathrm{Sq}^1 = 0$, $\mathrm{Sq}^1\mathrm{Sq}^2 = \mathrm{Sq}^3$ and $\mathrm{Sq}^2\mathrm{Sq}^2 = \mathrm{Sq}^3\mathrm{Sq}^1$.

Solution. For $(a,b) = (1,1)$ the only term is $c = 0$ with coefficient $\binom{0}{1} = 0$. For $(1,2)$: $\binom{1}{1}\mathrm{Sq}^3\mathrm{Sq}^0 = \mathrm{Sq}^3$. For $(2,2)$: $c = 0$ gives $\binom{1}{2} = 0$ and $c = 1$ gives $\binom{0}{0}\mathrm{Sq}^3\mathrm{Sq}^1$. □

Exercise 24.7.2. Show that there is no space X with $H^*(X;\mathbb{F}_2) \cong \mathbb{F}_2[x]$ and $|x| = 3$.

Solution. We would have $x^2 = \mathrm{Sq}^3 x = \mathrm{Sq}^1\mathrm{Sq}^2 x$, and $\mathrm{Sq}^2 x \in H^5(X;\mathbb{F}_2) = 0$, so $x^2 = 0$, contradicting that H^* is a polynomial ring. (More generally, the argument of Corollary 24.4.4 shows that $|x|$ must be a power of 2, and by Adams' theorem (Remark 24.4.5) only $|x| = 1, 2, 4, 8$ can occur.) □

Exercise 24.7.3. Show that $\beta(\alpha^n) = n\alpha^{n+1}$ in $H^*(\mathbb{RP}^\infty;\mathbb{F}_2)$, directly from the cellular cochain complex, and compare with Proposition 24.4.2.

Exercise 24.7.4. Show that the attaching map $S^3 \to S^2$ of the 4-cell of $\mathbb{CP}^2$ is not null-homotopic, using Sq^2 or the cup product, and deduce that $\pi_3(S^2) \neq 0$.

Solution. If it were null-homotopic, $\mathbb{CP}^2$ would be homotopy equivalent to $S^2 \vee S^4$ (attaching along homotopic maps gives homotopy equivalent spaces [Hat02, Proposition 0.18]), whose cup squares in degree 2 vanish; but $\mathrm{Sq}^2(x) = x^2 \neq 0$ in $H^4(\mathbb{CP}^2;\mathbb{F}_2)$ by Theorem 22.5.1. □

Part IV

Sheaves

Chapter 25

Sheaves on Topological Spaces

25.1 Introduction

A sheaf on a space records data that are defined locally and glue uniquely: continuous, smooth or holomorphic functions, sections of a bundle, solutions of a differential equation. Sheaves are the language in which complex and algebraic geometry and Hodge theory are written: the cohomology of sheaves (Chapter 26) unifies singular, de Rham and Dolbeault cohomology, and the comparison between them is the starting point of Hodge theory. This chapter develops the basic theory: sheaves and stalks, sheafification, the abelian category of sheaves, direct and inverse images, local systems, and sheaves of modules. References are [God58], [Bre97], [KS90], [Har77, Chapter II] and [The].

Prerequisites. Chapter 18 (General Topology), Chapter 6 (Categories), Chapter 15 (Homological Algebra) and Chapter 19 (Homotopy and Covering Spaces).

Throughout, X is a topological space and $\mathrm{Open}(X)$ is the set of open subsets of X, ordered by inclusion and viewed as a category.

25.2 Presheaves and sheaves

Definition 25.2.1. A *presheaf* $\mathcal{F}$ on X with values in a category $\mathcal{C}$ (sets, abelian groups, rings, modules) is a functor $\mathcal{F}\colon \mathrm{Open}(X)^{\mathrm{op}} \to \mathcal{C}$: an object $\mathcal{F}(U)$, the *sections* over U, for each open U, and *restriction maps* $\mathcal{F}(U) \to \mathcal{F}(V)$, $s \mapsto s|_V$, for $V \subseteq U$, with $s|_U = s$ and $(s|_V)|_W = s|_W$. We also write $\Gamma(U, \mathcal{F}) = \mathcal{F}(U)$. A *morphism* of presheaves is a natural transformation. A presheaf of sets (or abelian groups, ...) is a *sheaf* if for every open cover $U = \bigcup_i U_i$:

1. (Locality) if $s, t \in \mathcal{F}(U)$ satisfy $s|_{U_i} = t|_{U_i}$ for all i, then $s = t$;
2. (Gluing) if $s_i \in \mathcal{F}(U_i)$ satisfy $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all i, j, there is $s \in \mathcal{F}(U)$ with $s|_{U_i} = s_i$ for all i.

Sheaves form full subcategories $\mathrm{Sh}(X) \subseteq \mathrm{PSh}(X)$ and $\mathrm{Ab}(X) \subseteq \mathrm{PAb}(X)$. Applied to the empty cover of $\emptyset$, the axioms say that $\mathcal{F}(\emptyset)$ is a single point (the zero group).

Example 25.2.2.

1. For a space Y, $U \mapsto \{\text{continuous maps } U \to Y\}$ is a sheaf of sets, because continuity is local (pasting lemma, Lemma 18.4.5). For $Y = \mathbb{R}$ or $\mathbb{C}$ it is a sheaf of rings C_X. Likewise smooth functions on a manifold and holomorphic functions on an open subset of $\mathbb{C}$ form sheaves of rings.
2. For a continuous map $p\colon E \to X$, the continuous sections $s\colon U \to E$ with $ps = \mathrm{id}_U$ form a sheaf of sets, the *sheaf of sections* of p.

3. Let A be a set with at least two elements. The *constant presheaf* $U \mapsto A$ (with $\emptyset \mapsto$ a point) is not a sheaf if X has two disjoint nonempty open subsets U_1, U_2: sections $a \neq b$ on them do not glue to a section on $U_1 \sqcup U_2$. The *constant sheaf* A_X is $U \mapsto \{\text{locally constant maps } U \to A\}$.

4. For $x \in X$ and a set (group) A, the *skyscraper sheaf* x_*A has $x_*A(U) = A$ if $x \in U$ and a point otherwise.

5. The presheaf of bounded continuous functions on $\mathbb{R}$ satisfies locality but not gluing.

Definition 25.2.3. The *stalk* of a presheaf $\mathcal{F}$ at x is $\mathcal{F}_x = \operatorname{colim}_{U \ni x} \mathcal{F}(U)$, the colimit over the open neighbourhoods of x directed by reverse inclusion (Example 6.11.2). By Proposition 6.11.4, its elements are *germs* s_x of sections $s \in \mathcal{F}(U)$, $x \in U$, where $s_x = t_x$ for $t \in \mathcal{F}(V)$ if and only if $s|_W = t|_W$ for some open $W \ni x$ in $U \cap V$. For presheaves of abelian groups the stalk is an abelian group (Lemma 23.5.1). A morphism φ induces maps $\varphi_x \colon \mathcal{F}_x \to \mathcal{G}_x$.

Proposition 25.2.4. *Let $\varphi \colon \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves.*

1. *Two sections $s, t \in \mathcal{F}(U)$ with $s_x = t_x$ for all $x \in U$ are equal.*

2. *φ_U is injective for all U if and only if φ_x is injective for all x.*

3. *φ is an isomorphism if and only if φ_x is bijective for all x.*

Proof. (1) Each $x \in U$ has an open neighbourhood $W_x \subseteq U$ with $s|_{W_x} = t|_{W_x}$; apply locality to the cover (W_x). (2) If all φ_x are injective and $\varphi(s) = \varphi(t)$, then $\varphi(s)_x = \varphi(t)_x$ gives $s_x = t_x$ for all x, so $s = t$ by (1). Conversely, if $\varphi_x(s_x) = \varphi_x(t_x)$ with $s, t \in \mathcal{F}(U)$, then $\varphi(s)|_W = \varphi(t)|_W$ for some $W \ni x$, so $s|_W = t|_W$ and $s_x = t_x$. (3) An isomorphism induces bijections on stalks. Conversely, by (2) each φ_U is injective. Let $t \in \mathcal{G}(U)$. For each $x \in U$ choose $s^{(x)} \in \mathcal{F}(W_x)$ with $\varphi(s^{(x)})_x = t_x$; shrinking W_x, $\varphi(s^{(x)}) = t|_{W_x}$. On $W_x \cap W_y$, $\varphi(s^{(x)}) = \varphi(s^{(y)})$, so $s^{(x)} = s^{(y)}$ there by injectivity. By gluing there is s with $s|_{W_x} = s^{(x)}$, and $\varphi(s) = t$ by locality. □

Proposition 25.2.5 (Gluing sheaves)**.** *Let $(U_i)_{i \in I}$ be an open cover of X, let $\mathcal{F}_i$ be a sheaf on U_i, and write $U_{ij} = U_i \cap U_j$, $U_{ijk} = U_i \cap U_j \cap U_k$. Let $\varphi_{ij} \colon \mathcal{F}_j|_{U_{ij}} \to \mathcal{F}_i|_{U_{ij}}$ be isomorphisms with $\varphi_{ii} = \mathrm{id}$ and $\varphi_{ij} \circ \varphi_{jk} = \varphi_{ik}$ on U_{ijk} (the* cocycle condition*). Then there is a sheaf $\mathcal{F}$ on X with isomorphisms $\psi_i \colon \mathcal{F}|_{U_i} \to \mathcal{F}_i$ such that $\psi_i = \varphi_{ij} \circ \psi_j$ on U_{ij}, and $(\mathcal{F}, (\psi_i))$ is unique up to unique isomorphism.*

Proof. For U open let $\mathcal{F}(U)$ be the set of families $(s_i)_i$ with $s_i \in \mathcal{F}_i(U \cap U_i)$ and $s_i = \varphi_{ij}(s_j)$ on $U \cap U_{ij}$ for all i, j, with componentwise restriction. The sheaf axioms hold componentwise, since each $\mathcal{F}_i$ is a sheaf and the compatibility conditions can be checked locally (by locality of the $\mathcal{F}_i$). For $V \subseteq U_i$ define $\psi_i((s_j)_j) = s_i$. Its inverse sends $t \in \mathcal{F}_i(V)$ to the family $s_j = \varphi_{ji}(t|_{V \cap U_j})$; this family is compatible because $\varphi_{jk}(s_k) = \varphi_{jk}\varphi_{ki}(t) = \varphi_{ji}(t) = s_j$, and $s_i = t$ because $\varphi_{ii} = \mathrm{id}$. Conversely a compatible family is determined by its i-th member, as $s_j = \varphi_{ji}(s_i)$; so ψ_i is an isomorphism, and $\psi_i = \varphi_{ij}\psi_j$ is the compatibility condition. If $(\mathcal{G}, (\psi_i'))$ is another solution, the isomorphisms $\psi_i^{-1}\psi_i' \colon \mathcal{G}|_{U_i} \to \mathcal{F}|_{U_i}$ agree on the overlaps, since $\psi_i^{-1}\psi_i' = \psi_j^{-1}\varphi_{ij}^{-1}\varphi_{ij}\psi_j' = \psi_j^{-1}\psi_j'$; morphisms of sheaves that agree on overlaps glue uniquely, by the sheaf axioms of the target applied to the images of sections, so they define the unique isomorphism $\mathcal{G} \to \mathcal{F}$ compatible with the ψ's. □

Example 25.2.6. Vector bundles are the basic instance: on a cover trivialising a bundle, the sheaves of sections $\mathcal{F}_i \cong C^r_{U_i}$ are glued by the transition functions, and the cocycle condition of Proposition 25.2.5 is the cocycle condition for transition functions. Local systems (Section 25.5) are glued in the same way from constant sheaves, with locally constant transition functions.

25.3 Sheafification

Theorem 25.3.1. *Let $\mathcal{F}$ be a presheaf of sets (abelian groups). For U open let $\mathcal{F}^\#(U)$ be the set of functions s assigning to each $x \in U$ a germ $s(x) \in \mathcal{F}_x$ such that every $x \in U$ has an open neighbourhood $W \subseteq U$ and $t \in \mathcal{F}(W)$ with $s(y) = t_y$ for all $y \in W$. With restriction of functions, $\mathcal{F}^\#$ is a sheaf, the* sheafification *of $\mathcal{F}$, and $\theta\colon \mathcal{F} \to \mathcal{F}^\#$, $t \mapsto (y \mapsto t_y)$, has the following properties.*

1. *$\theta_x\colon \mathcal{F}_x \to \mathcal{F}_x^\#$ is bijective for all x.*
2. *Every morphism $\varphi\colon \mathcal{F} \to \mathcal{G}$ to a sheaf factors uniquely as $\varphi = \psi \circ \theta$. So sheafification is left adjoint to the inclusion of sheaves into presheaves.*
3. *If $\mathcal{F}$ is a sheaf, θ is an isomorphism.*

Proof. The conditions defining $\mathcal{F}^\#(U)$ are local, so $\mathcal{F}^\#$ is a sheaf; for abelian groups, the pointwise operations preserve the condition.

(1) Surjectivity: an element of $\mathcal{F}_x^\#$ is the germ of some $s \in \mathcal{F}^\#(U)$, which near x equals $\theta(t)$, so the germ is $\theta_x(t_x)$. Injectivity: if $\theta(t)_x = \theta(t')_x$, then in particular $t_x = \theta(t)(x) = \theta(t')(x) = t'_x$.

(2) Let $s \in \mathcal{F}^\#(U)$, and choose an open cover (W_i) of U and $t_i \in \mathcal{F}(W_i)$ with $s|_{W_i} = \theta(t_i)$. On $W_i \cap W_j$ the germs of t_i and t_j agree, hence so do the germs of $\varphi(t_i)$ and $\varphi(t_j)$, and by Proposition 25.2.4(1) $\varphi(t_i) = \varphi(t_j)$ there. Let $\psi(s)$ be the section of $\mathcal{G}$ obtained by gluing. It does not depend on the choices (two choices agree on a common refinement), and ψ is a morphism with $\psi\theta = \varphi$. Uniqueness: any ψ' with $\psi'\theta = \varphi$ agrees with ψ on each $\theta(t_i)$, hence on s by locality.

(3) By (1) and Proposition 25.2.4(3), applied to the morphism θ of sheaves. □

Proposition 25.3.2. $\mathrm{Ab}(X)$ *is an abelian category. The kernel of $\varphi\colon \mathcal{F} \to \mathcal{G}$ is $U \mapsto \ker \varphi_U$; its cokernel is the sheafification of the presheaf $U \mapsto \operatorname{coker} \varphi_U$. Taking stalks is exact:* $(\ker \varphi)_x = \ker \varphi_x$ *and* $(\operatorname{coker} \varphi)_x = \operatorname{coker} \varphi_x$. *A sequence of sheaves is exact if and only if it is exact on all stalks. The functor $\Gamma(U, -)$ is left exact but not exact in general.*

Proof. Sums of morphisms and finite direct sums $(\mathcal{F} \oplus \mathcal{G})(U) = \mathcal{F}(U) \oplus \mathcal{G}(U)$ make $\mathrm{Ab}(X)$ additive. The presheaf kernel is a sheaf and is a kernel in $\mathrm{Ab}(X)$. By Theorem 25.3.1(2), the sheafified presheaf cokernel is a cokernel. Stalks are directed colimits, which are exact (Lemma 23.5.1), and sheafification does not change stalks; this gives the formulas for stalks. If φ is a monomorphism, the canonical morphism from $\mathcal{F}$ to the kernel of its cokernel is bijective on stalks, because Ab is abelian; so it is an isomorphism by Proposition 25.2.4; similarly for epimorphisms. Images and exactness are defined by kernels and cokernels, so they are computed on stalks. Left exactness of Γ holds because kernels are computed sectionwise.

For the failure of right exactness, let $X = \mathbb{R}$, $p \neq q$ in X, and consider $\mathbb{Z}_X \to p_*\mathbb{Z} \oplus q_*\mathbb{Z}$, sending a locally constant function to its values at p and q (when defined). On stalks it is surjective: at p it is $\mathbb{Z} \to \mathbb{Z}$, the identity, and at points $x \neq p, q$ the target stalk is 0, since x has a neighbourhood containing neither p nor q. On global sections it is $\mathbb{Z} \to \mathbb{Z} \oplus \mathbb{Z}$, $n \mapsto (n, n)$, because $\mathbb{R}$ is connected, and this is not surjective. □

Counterexample 25.3.3. Surjectivity on stalks is a local lifting property, and global sections can fail to lift for topological reasons. Let $T = \mathbb{R}/\mathbb{Z}$ with quotient map $q\colon \mathbb{R} \to T$. The covering $p\colon \mathbb{R} \to S^1$ of Proposition 19.4.5 induces a continuous bijection $\mathbb{R}/4\mathbb{Z} \to S^1$ from a compact space to a Hausdorff space, hence a homeomorphism (Proposition 18.6.3), so

$T \cong \mathbb{R}/4\mathbb{Z} \cong S^1$ and q is a covering map. On any space X, let $C_{\mathbb{R}}$ and C_T be the sheaves of continuous maps to $\mathbb{R}$ and to the abelian group T, and consider

$$0 \to \mathbb{Z}_X \to C_{\mathbb{R}} \xrightarrow{f \mapsto q \circ f} C_T \to 0.$$

The kernel of $f \mapsto q \circ f$ consists of the continuous maps to $\mathbb{Z}$, that is, $\mathbb{Z}_X$. On stalks the map is surjective: a continuous $g\colon W \to T$ near x takes a small neighbourhood of x into an evenly covered open set, over which q has a continuous inverse on each sheet. So the sequence is exact (Proposition 25.3.2). But for $X = T$, the identity $\mathrm{id}_T \in C_T(T)$ is not of the form $q \circ f$: a continuous $f : T \to \mathbb{R}$ with $q \circ f = \mathrm{id}$ would be a lift of id_T through q, which by the lifting criterion (Proposition 19.6.1) requires $\pi_1(T) \subseteq q_*\pi_1(\mathbb{R}) = 0$, whereas $\pi_1(T) \cong \pi_1(S^1) \cong \mathbb{Z}$ (Theorem 19.4.6). So $\Gamma(T, -)$ is not right exact on this sequence; the obstruction will reappear as the nonzero group $H^1(S^1, \mathbb{Z}) \cong \mathbb{Z}$ (Chapter 26).

25.4 Direct and inverse images

Definition 25.4.1. Let $f : X \to Y$ be continuous. The *direct image* of a sheaf $\mathcal{F}$ on X is the sheaf $f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}V)$ on Y. The *inverse image* of a sheaf $\mathcal{G}$ on Y is the sheafification $f^{-1}\mathcal{G}$ of the presheaf $f_p^{-1}\mathcal{G}(U) = \operatorname{colim}_{V \supseteq f(U)} \mathcal{G}(V)$, the colimit over open $V \supseteq f(U)$ directed by reverse inclusion. For $i : Z \to X$ the inclusion of a subspace, $\mathcal{F}|_Z = i^{-1}\mathcal{F}$ is the *restriction*; for Z open, $\mathcal{F}|_Z(W) = \mathcal{F}(W)$.

Proposition 25.4.2. *1. $(f^{-1}\mathcal{G})_x \cong \mathcal{G}_{f(x)}$; hence f^{-1} is exact on sheaves of abelian groups. The functor f_* is left exact.*

2. There is a bijection $\mathrm{Hom}_{\mathrm{Sh}(X)}(f^{-1}\mathcal{G}, \mathcal{F}) \cong \mathrm{Hom}_{\mathrm{Sh}(Y)}(\mathcal{G}, f_*\mathcal{F})$, natural in $\mathcal{F}$ and $\mathcal{G}$: f^{-1} is left adjoint to f_*.

3. $(g \circ f)_* = g_* \circ f_*$ and $(g \circ f)^{-1} \cong f^{-1} \circ g^{-1}$.

Proof. (1) The pairs (U, V) with $x \in U$ and $V \supseteq f(U)$ map cofinally to the open neighbourhoods V of $f(x)$ (take $U = f^{-1}V$), so $(f_p^{-1}\mathcal{G})_x = \operatorname{colim}_{V \ni f(x)} \mathcal{G}(V)$ by Lemma 23.5.1, and sheafification preserves stalks. Exactness follows from Proposition 25.3.2. Kernels of sheaves are computed sectionwise, so f_* preserves them.

(2) By Theorem 25.3.1(2), morphisms $f^{-1}\mathcal{G} \to \mathcal{F}$ are morphisms of presheaves $f_p^{-1}\mathcal{G} \to \mathcal{F}$, that is, families of maps $\mathcal{G}(V) \to \mathcal{F}(U)$ for $V \supseteq f(U)$, compatible with restrictions in U and V. Such a family determines maps $\mathcal{G}(V) \to \mathcal{F}(f^{-1}V)$ (take $U = f^{-1}V$), compatible with restrictions, that is, a morphism $\mathcal{G} \to f_*\mathcal{F}$. Conversely a morphism $\psi : \mathcal{G} \to f_*\mathcal{F}$ gives $\mathcal{G}(V) \xrightarrow{\psi_V} \mathcal{F}(f^{-1}V) \to \mathcal{F}(U)$ by restriction, as $U \subseteq f^{-1}V$. These constructions are mutually inverse and natural.

(3) The first equality is clear; the second follows from (2) and uniqueness of adjoints (Proposition 6.10.3). □

Definition 25.4.3. Let $j : U \to X$ be the inclusion of an open subset and $\mathcal{F}$ a sheaf of abelian groups on U. The *extension by zero* $j_!\mathcal{F}$ is the sheafification of the presheaf $W \mapsto \mathcal{F}(W)$ if $W \subseteq U$ and $W \mapsto 0$ otherwise.

Proposition 25.4.4. *$(j_!\mathcal{F})_x = \mathcal{F}_x$ for $x \in U$ and 0 for $x \notin U$. If $Z = X \setminus U$ with inclusion i, then for every sheaf of abelian groups $\mathcal{F}$ on X there is a short exact sequence $0 \to j_!(\mathcal{F}|_U) \to \mathcal{F} \to i_*(\mathcal{F}|_Z) \to 0$.*

Proof. For $x \in U$ the open neighbourhoods contained in U are cofinal; for $x \notin U$ no neighbourhood is contained in U. The first map is induced, via Theorem 25.3.1(2), by the inclusions $\mathcal{F}(W) \to \mathcal{F}(W)$ for $W \subseteq U$; the second is the unit $\mathcal{F} \to i_* i^{-1}\mathcal{F}$ of the adjunction. On stalks at $x \in U$, the sequence is $0 \to \mathcal{F}_x \to \mathcal{F}_x \to 0 \to 0$, since x has a neighbourhood missing the closed set Z. At $z \in Z$, the open subsets $W \cap Z$ are cofinal among neighbourhoods of z in Z, so $(i_* i^{-1}\mathcal{F})_z = (i^{-1}\mathcal{F})_z = \mathcal{F}_z$, and the sequence is $0 \to 0 \to \mathcal{F}_z \to \mathcal{F}_z \to 0$. □

25.5 Locally constant sheaves and local systems

Definition 25.5.1. A sheaf $\mathcal{F}$ of sets is *locally constant* if every point has an open neighbourhood U with $\mathcal{F}|_U$ isomorphic to a constant sheaf A_U. A *local system* of K-vector spaces (for a field K) is a locally constant sheaf of K-vector spaces whose stalks are finite-dimensional; local systems of abelian groups or modules are defined similarly.

Proposition 25.5.2. *Let $\mathcal{F}$ be a sheaf of sets on X. The* étalé space $E_{\mathcal{F}} = \bigsqcup_x \mathcal{F}_x$, *with the topology whose basis is the sets $\{s_x : x \in U\}$ for $s \in \mathcal{F}(U)$, has a local homeomorphism $p\colon E_{\mathcal{F}} \to X$, and $\mathcal{F}$ is isomorphic to the sheaf of sections of p. If $\mathcal{F}$ is locally constant and X is locally connected, p is a covering map (except that fibres may be empty), and conversely the sheaf of sections of such a covering is locally constant.*

Proof. If $s_x = t_x$ for $s \in \mathcal{F}(U)$, $t \in \mathcal{F}(V)$, the two basic sets contain the basic set of $s|_W$ for some $W \ni x$; so these sets form a basis, and p maps the basic set of s homeomorphically onto U, with inverse $\hat{s}\colon x \mapsto s_x$. The map $s \mapsto \hat{s}$ from $\mathcal{F}(U)$ to continuous sections over U is injective by Proposition 25.2.4(1); it is surjective, because a continuous section σ is locally of the form $\hat{s}$ (near x, σ lies in the basic set of some s with $s_x = \sigma(x)$), and these local s glue. If $\mathcal{F}|_U \cong A_U$ with U connected, then $p^{-1}(U) \cong U \times A$ with A discrete, because the germs of the constant sections $a \in A$ give disjoint open sets covering $p^{-1}(U)$. Conversely, over a connected evenly covered open set, sections of a covering correspond to locally constant choices of sheets. □

Theorem 25.5.3. *Let X be path-connected, locally path-connected and semilocally simply connected, and $x \in X$. Taking the stalk at x, with the monodromy action, gives equivalences of categories between*

1. *locally constant sheaves of sets on X and sets with a right action of $\pi_1(X, x)$;*
2. *local systems of K-vector spaces on X and finite-dimensional right $K[\pi_1(X, x)]$-modules, that is, representations $\rho\colon \pi_1(X, x)^{\mathrm{op}} \to \mathrm{GL}(\mathcal{F}_x)$.*

Sketch of proof. By Proposition 25.5.2, locally constant sheaves correspond to covering spaces (with possibly empty fibres), and morphisms of sheaves to maps of coverings. A covering is the disjoint union of its components, which are connected coverings of X, and the fibre over x with its monodromy action (Corollary 19.4.4) is a right $\pi_1(X, x)$-set whose orbits correspond to the components. For connected coverings, the classification theorem (Theorem 19.6.6) identifies isomorphism classes with transitive π_1-sets $H\backslash\pi_1(X, x)$, and the lifting criterion (Proposition 19.6.1) identifies maps of coverings with equivariant maps of fibres; this gives (1). The omitted verification is that these identifications are functorial; see [Sza09, Chapter 2]. For (2), a local system is a locally constant sheaf of sets with a K-vector space structure on the stalks compatible with monodromy, and the monodromy action is then K-linear. □

Example 25.5.4. 1. On a simply connected space every local system is constant.

2. On S^1, the rank one local systems of $\mathbb{C}$-vector spaces correspond to $\lambda \in \mathbb{C}^\times$, the monodromy around the generator of $\pi_1(S^1) \cong \mathbb{Z}$. For $\lambda \neq 1$ the local system $\mathcal{L}_\lambda$ has no nonzero global sections, since a global section is a vector fixed by monodromy. The real local system with monodromy -1 is the sheaf of locally constant sections of the Möbius band, viewed as a line bundle over S^1.

3. For a manifold M, the *orientation sheaf* or_M, with stalks $H_n(M \mid x; \mathbb{Z})$, is the local system of rank one associated with the orientation double covering (Proposition 23.3.2); it is constant if and only if M is orientable.

4. For a family of spaces $f : E \to B$ that is locally trivial, the cohomology groups of the fibres form local systems $R^k f_* \mathbb{Z}$ on B (Chapter 26); their monodromy is a central object in the theory of variations of Hodge structure (Chapter 70).

25.6 Ringed spaces and sheaves of modules

Definition 25.6.1. A *ringed space* is a pair $(X, \mathcal{O}_X)$ of a space and a sheaf of commutative rings. It is *locally ringed* if all stalks $\mathcal{O}_{X,x}$ are local rings. A *morphism* $(f, f^\#) \colon (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a continuous map f with a morphism of sheaves of rings $f^\# \colon \mathcal{O}_Y \to f_* \mathcal{O}_X$; for locally ringed spaces one requires the induced maps $\mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$ to be local homomorphisms. An *$\mathcal{O}_X$-module* is a sheaf of abelian groups $\mathcal{F}$ with $\mathcal{O}_X(U)$-module structures on $\mathcal{F}(U)$ compatible with restrictions.

Example 25.6.2. A space with its sheaf of continuous real functions, a smooth manifold with smooth functions (Chapter 33), a complex manifold with holomorphic functions (Chapter 44) and a scheme (Chapter 54) are locally ringed spaces: in each case the germs vanishing at x form the unique maximal ideal of the stalk, since a germ not vanishing at x is invertible near x.

Definition 25.6.3. Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{O}_X$-modules. The *tensor product* $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$ is the sheafification of $U \mapsto \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U)$. The sheaf $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$ has sections over U the morphisms $\mathcal{F}|_U \to \mathcal{G}|_U$ of $\mathcal{O}_U$-modules, and $\mathcal{F}^\vee = \mathcal{H}om(\mathcal{F}, \mathcal{O}_X)$ is the *dual*. An $\mathcal{O}_X$-module is *locally free of rank r* if every point has an open neighbourhood U with $\mathcal{F}|_U \cong \mathcal{O}_U^{\oplus r}$, and *invertible* if it is locally free of rank 1. For a morphism f of ringed spaces and an $\mathcal{O}_Y$-module $\mathcal{G}$, the *pullback* is $f^*\mathcal{G} = f^{-1}\mathcal{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$.

Proposition 25.6.4. *1. $\mathcal{H}om(\mathcal{F}, \mathcal{G})$ is a sheaf, and $(\mathcal{F} \otimes \mathcal{G})_x \cong \mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} \mathcal{G}_x$.*

2. *If $\mathcal{E}$ and $\mathcal{E}'$ are locally free of ranks r and r', then $\mathcal{E} \otimes \mathcal{E}'$ is locally free of rank rr', $\mathcal{E}^\vee$ is locally free of rank r, and the evaluation map $\mathcal{E} \otimes \mathcal{E}^\vee \to \mathcal{O}_X$ is an isomorphism if $r = 1$.*

3. *The isomorphism classes of invertible sheaves form an abelian group* $\mathrm{Pic}(X)$ *under $\otimes$, with identity $\mathcal{O}_X$ and inverse $\mathcal{L}^\vee$.*

Proof. (1) Morphisms defined on the members of an open cover and agreeing on overlaps glue uniquely, because sections of $\mathcal{G}$ do. For the stalk, sheafification does not change stalks, so it suffices to compare $\operatorname{colim}_{U \ni x} \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U)$ with $\mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} \mathcal{G}_x$. The map $[s \otimes t] \mapsto s_x \otimes t_x$ is well defined; an inverse sends $s_x \otimes t_x$ to the class of $s|_W \otimes t|_W$ on a common neighbourhood W, which is well defined because every relation defining the tensor product over $\mathcal{O}_{X,x}$ involves finitely many germs and already holds over some neighbourhood of x [Har77, Chapter II, Section 5].

(2) All statements are local, so we may assume $\mathcal{E} = \mathcal{O}^r$ and $\mathcal{E}' = \mathcal{O}^{r'}$. Then $\mathcal{O}^r \otimes \mathcal{O}^{r'} \cong \mathcal{O}^{rr'}$ (the presheaf tensor product is already a sheaf in this case), $\mathcal{H}om(\mathcal{O}^r, \mathcal{O}) \cong \mathcal{O}^r$, and for

$r = 1$ evaluation $\mathcal{O} \otimes \mathcal{O} \to \mathcal{O}$ is multiplication, an isomorphism. Since the evaluation map is a globally defined morphism, and being an isomorphism can be checked on stalks (Proposition 25.2.4), it is an isomorphism.

(3) Tensor product is associative and commutative up to natural isomorphism, $\mathcal{O}_X \otimes \mathcal{L} \cong \mathcal{L}$, and (2) gives inverses. □

Remark 25.6.5. If $E \to X$ is a (real or complex, topological, smooth or holomorphic) vector bundle of rank r, its sheaf of sections is a locally free module of rank r over the corresponding sheaf of functions, and every locally free sheaf arises in this way, uniquely up to isomorphism (Chapter 35). Under this correspondence, $\mathrm{Pic}(X)$ is the group of line bundles; for a complex manifold it is computed by sheaf cohomology via the exponential sequence (Chapter 45).

25.7 Exercises

Exercise 25.7.1. Show that the sheafification of the constant presheaf with value A is the constant sheaf A_X of locally constant functions, and that $A_X = \pi^{-1}A$ for $\pi\colon X \to \mathrm{pt}$.

Solution. The stalks of the constant presheaf are all A, and a section of the sheafification is a function $U \to A$ locally equal to a constant, that is, a locally constant function. For the second statement, $\pi_p^{-1}A(U) = A$ for $U \neq \emptyset$. □

Exercise 25.7.2. Let $i\colon \{x\} \to X$. Show that $x_*A = i_*A$ and $\mathrm{Hom}(\mathcal{F}, x_*A) \cong \mathrm{Hom}(\mathcal{F}_x, A)$ for every sheaf $\mathcal{F}$ of abelian groups.

Solution. The first statement is the definition. By Proposition 25.4.2(2) and (1), $\mathrm{Hom}(\mathcal{F}, i_*A) \cong \mathrm{Hom}(i^{-1}\mathcal{F}, A) = \mathrm{Hom}(\mathcal{F}_x, A)$, since a sheaf on a point is a single group. □

Exercise 25.7.3. Give an example of a continuous map f and a sheaf $\mathcal{F}$ such that $(f_*\mathcal{F})_y$ is not isomorphic to the global sections of $\mathcal{F}$ on the fibre $f^{-1}(y)$.

Exercise 25.7.4. Let $X = \mathbb{C} \setminus \{0\}$ with the sheaf $\mathcal{O}$ of holomorphic functions and $\mathcal{O}^\times$ of nowhere vanishing ones. Accepting that holomorphic functions have local logarithms of nonvanishing functions but z has no global logarithm, show that $\exp\colon \mathcal{O} \to \mathcal{O}^\times$, $f \mapsto e^{2\pi i f}$, is an epimorphism of sheaves but not surjective on global sections, and identify its kernel.

Exercise 25.7.5. For $j\colon U \to X$ open, show that $j_!$ is left adjoint to j^{-1} on sheaves of abelian groups, and that j^{-1} is left adjoint to j_*.

Chapter 26

Sheaf Cohomology

26.1 Introduction

The global sections functor $\Gamma(X, -)$ is left exact but not exact (Proposition 25.3.2), and sheaf cohomology $H^i(X, \mathcal{F})$ is its right derived functor. It can be computed by several kinds of acyclic resolutions: flasque, soft or fine sheaves (Theorem 26.4.5), Čech complexes of acyclic covers (Theorem 26.5.3), and hypercohomology of complexes. On reasonable spaces the cohomology of a constant sheaf is singular cohomology (Theorem 26.6.1); on a manifold the de Rham complex is a fine resolution of the constant sheaf, which gives the de Rham theorem (Chapter 37). References: [Har77, Chapter III], [God58], [Bre97], [KS90] and [Voi02b, Chapter 4].

Prerequisites. Chapter 25 (Sheaves on Topological Spaces), Chapter 16 (Spectral Sequences), Chapter 17 (Derived Categories) and Chapter 22 (Singular Cohomology).

26.2 Cohomology as a derived functor

Lemma 26.2.1. *Let $L\colon \mathcal{B} \to \mathcal{A}$ be an exact functor between abelian categories with right adjoint $R\colon \mathcal{A} \to \mathcal{B}$. Then R preserves injective objects. In particular f_* preserves injective sheaves for every continuous f, and so does restriction j^{-1} to an open subset $j\colon U \to X$.*

Proof. For I injective, $\mathrm{Hom}_{\mathcal{B}}(-, R(I)) \cong \mathrm{Hom}_{\mathcal{A}}(L(-), I)$ is a composite of exact functors, hence exact. For f_* use Proposition 25.4.2. For j^{-1} we show that $j_!$ is an exact left adjoint: a morphism $j_!\mathcal{F} \to \mathcal{G}$ is, by Theorem 25.3.1, a compatible family of maps $\mathcal{F}(W) \to \mathcal{G}(W)$ for $W \subseteq U$, that is, a morphism $\mathcal{F} \to \mathcal{G}|_U$; and $j_!$ is exact on stalks (Proposition 25.4.4). □

Theorem 26.2.2. *The category $\mathrm{Ab}(X)$ has enough injectives, and so does the category of $\mathcal{O}_X$-modules on a ringed space.*

Proof. For each x embed $\mathcal{F}_x$ into an injective abelian group I_x (Theorem 15.5.4). The skyscraper sheaf x_*I_x is injective, since $\mathrm{Hom}(\mathcal{G}, x_*I_x) \cong \mathrm{Hom}(\mathcal{G}_x, I_x)$ (Exercise 25.7.2) is exact in $\mathcal{G}$. A product of injective objects is injective, and $\mathcal{F} \to \prod_x x_*I_x$ is a monomorphism, being injective on stalks. For $\mathcal{O}_X$-modules use injective $\mathcal{O}_{X,x}$-modules instead. □

Definition 26.2.3. The *sheaf cohomology* groups are $H^i(X, \mathcal{F}) = R^i\Gamma(X, -)(\mathcal{F})$ (Definition 15.6.4). For a complex $\mathcal{K}^\bullet$ of sheaves, bounded below, the *hypercohomology* is $\mathbb{H}^i(X, \mathcal{K}^\bullet) = H^i(R\Gamma(X, \mathcal{K}^\bullet))$ (Definition 17.4.4). For a continuous $f\colon X \to Y$, the *higher direct images* are $R^i f_*\mathcal{F}$.

Remark 26.2.4. For an $\mathcal{O}_X$-module $\mathcal{F}$, derived functors computed in $\mathcal{O}_X$-modules and in $\mathrm{Ab}(X)$ agree, because injective $\mathcal{O}_X$-modules are flasque (Lemma 26.3.2 applies verbatim with $j_!\mathcal{O}_U \to \mathcal{O}_X$) and flasque sheaves are acyclic in both settings.

26.3 Flasque sheaves

Definition 26.3.1. A sheaf $\mathcal{F}$ is *flasque* if all restriction maps $\mathcal{F}(U) \to \mathcal{F}(V)$ are surjective.

Lemma 26.3.2. *Injective sheaves of abelian groups are flasque. Direct images and restrictions to open subsets of flasque sheaves are flasque, and so are products.*

Proof. Let I be injective and $V \subseteq U$ open, with inclusions j_V, j_U into X. The morphism $j_{V!}\mathbb{Z}_V \to j_{U!}\mathbb{Z}_U$ is injective on stalks, so $\mathrm{Hom}(j_{U!}\mathbb{Z}_U, I) \to \mathrm{Hom}(j_{V!}\mathbb{Z}_V, I)$ is surjective. By the adjunction in the proof of Lemma 26.2.1, $\mathrm{Hom}(j_{U!}\mathbb{Z}_U, I) = \mathrm{Hom}(\mathbb{Z}_U, I|_U) = I(U)$, a morphism from the constant sheaf being determined by the image of 1; so $I(U) \to I(V)$ is surjective. The other statements are immediate from the definitions. □

Proposition 26.3.3. *Let $0 \to \mathcal{F}' \to \mathcal{F} \xrightarrow{\pi} \mathcal{F}'' \to 0$ be exact with $\mathcal{F}'$ flasque. Then $\mathcal{F}(U) \to \mathcal{F}''(U)$ is surjective for all U. If also $\mathcal{F}$ is flasque, so is $\mathcal{F}''$. Flasque sheaves are Γ-acyclic, so $H^i(X, \mathcal{F})$ can be computed with flasque resolutions.*

Proof. We may take $U = X$. Let $s \in \mathcal{F}''(X)$, and consider pairs (V, t) with V open, $t \in \mathcal{F}(V)$ and $\pi(t) = s|_V$, ordered by extension. A chain has an upper bound by gluing, and it is nonempty, since s lifts locally. Let (V, t) be maximal and suppose $x \notin V$. Choose $W \ni x$ and $t' \in \mathcal{F}(W)$ lifting $s|_W$. On $V \cap W$, $t - t'$ lies in $\mathcal{F}'(V \cap W)$, which extends to $u \in \mathcal{F}'(X)$ since $\mathcal{F}'$ is flasque. Then t and $t' + u|_W$ agree on $V \cap W$ and glue to a lift over $V \cup W$, contradicting maximality. If $\mathcal{F}$ is flasque and $s'' \in \mathcal{F}''(V)$, lift it to $\mathcal{F}(V)$, extend to $\mathcal{F}(X)$ and map to $\mathcal{F}''(X)$.

Let $\mathcal{F}$ be flasque and embed it into an injective I with cokernel $\mathcal{G}$, which is flasque by the above and Lemma 26.3.2. The long exact sequence gives $H^1(X, \mathcal{F}) = \mathrm{coker}(I(X) \to \mathcal{G}(X)) = 0$ and $H^i(X, \mathcal{F}) \cong H^{i-1}(X, \mathcal{G})$ for $i \geq 2$, which vanishes by induction on i for all flasque sheaves simultaneously. The last statement is Proposition 15.6.6. □

Example 26.3.4. For every sheaf $\mathcal{F}$, the sheaf $C^0(\mathcal{F}) = \prod_x x_*\mathcal{F}_x$ of possibly discontinuous sections is flasque, and $\mathcal{F} \to C^0(\mathcal{F})$ is injective. Repeating with the cokernel gives the *Godement resolution*, a canonical flasque resolution, functorial and exact in $\mathcal{F}$.

26.4 Soft and fine sheaves

In this section X is paracompact Hausdorff. For a subset $Z \subseteq X$ we write $\mathcal{F}(Z) = \Gamma(Z, \mathcal{F}|_Z)$; by Proposition 25.5.2 this is the set of continuous sections over Z of the étalé space of $\mathcal{F}$. Sections over the members of a locally finite closed cover of Z that agree on overlaps glue, by the pasting lemma applied locally.

Lemma 26.4.1. *Let $Z \subseteq X$ be closed. Every $s \in \mathcal{F}(Z)$ extends to a section over some open neighbourhood of Z.*

Proof. Each $z \in Z$ has an open neighbourhood W_z in X and $t_z \in \mathcal{F}(W_z)$ with germs $s(y)$ at all $y \in W_z \cap Z$. Take a locally finite open refinement $(U_i)_{i \in I}$ of the cover $\{W_z\} \cup \{X \setminus Z\}$, choosing for each i either a section $t_i = t_z|_{U_i}$ (first kind) or $U_i \subseteq X \setminus Z$ (second kind), and closed sets $\overline{V_i} \subseteq U_i$ with (V_i) still covering (Lemma 18.10.4, Proposition 18.10.3). For $x \in X$

let $J(x) = \{i : x \in \overline{V_i}\}$, a finite set. Let W be the set of x such that $J(x)$ contains only indices of the first kind and the germs $(t_i)_x$, $i \in J(x)$, all agree.

$Z \subseteq W$, since for $z \in Z$ all $i \in J(z)$ are of the first kind and $(t_i)_z = s(z)$. W is open: for $x \in W$ choose an open $N \ni x$ that meets only finitely many $\overline{V_i}$, avoids those with $i \notin J(x)$, lies in $\bigcap_{i\in J(x)} U_i$, and on which the sections t_i, $i \in J(x)$, agree; then $J(y) \subseteq J(x)$ for $y \in N$, so $N \subseteq W$. On such an $N = N_x$, put $t^{(x)} = t_i|_{N_x}$ for any $i \in J(x)$ (nonempty, since the V_i cover). At $y \in N_x \cap N_{x'}$ both $t^{(x)}$ and $t^{(x')}$ have the germ $(t_k)_y$ for $k \in J(y)$, so they agree, and they glue to $t \in \mathcal{F}(W)$ with $t_z = s(z)$ on Z. □

Definition 26.4.2. A sheaf $\mathcal{F}$ on X is *soft* if $\mathcal{F}(X) \to \mathcal{F}(Z)$ is surjective for every closed Z. The *support* of an endomorphism η of $\mathcal{F}$ is the closure of $\{x : \eta_x \neq 0\}$. A sheaf of abelian groups is *fine* if for every locally finite open cover (U_i) there are endomorphisms η_i with $\operatorname{supp} \eta_i \subseteq U_i$ and $\sum_i \eta_i = \mathrm{id}$ (a *partition of unity* of $\mathcal{F}$).

Example 26.4.3. The sheaf C_X of continuous real functions is fine, by partitions of unity (Theorem 18.10.6): take η_i to be multiplication by ψ_i. More generally every sheaf of modules over C_X is fine, and on a smooth manifold every module over the sheaf of smooth functions is fine, using smooth partitions of unity (Chapter 33). The constant sheaf $\mathbb{Z}_X$ on a connected space with more than one point is not soft.

Lemma 26.4.4. 1. *Flasque sheaves and fine sheaves are soft.*

2. *If $0 \to \mathcal{F}' \to \mathcal{F} \xrightarrow{\pi} \mathcal{F}'' \to 0$ is exact and $\mathcal{F}'$ is soft, then $\mathcal{F}(X) \to \mathcal{F}''(X)$ is surjective.*

3. *If moreover $\mathcal{F}$ is soft, then $\mathcal{F}''$ is soft.*

Proof. (1) Let $s \in \mathcal{F}(Z)$ and extend it to an open $W \supseteq Z$ by Lemma 26.4.1. If $\mathcal{F}$ is flasque, extend further to X. If $\mathcal{F}$ is fine, take $\eta_1 + \eta_2 = \mathrm{id}$ for the cover $\{W, X \setminus Z\}$. The section $\eta_1(s)$ of $\mathcal{F}(W)$ vanishes on $W \setminus \operatorname{supp} \eta_1$, so it glues with 0 on $X \setminus \operatorname{supp} \eta_1$ to a global section t; at $z \in Z$, $t_z = s_z - (\eta_2)_z(s_z) = s_z$ since $z \notin \operatorname{supp} \eta_2$.

(2) Choose a locally finite open cover (U_i) with lifts $s_i \in \mathcal{F}(U_i)$ of $s|_{U_i}$ for a given $s \in \mathcal{F}''(X)$, and closed $\overline{V_i} \subseteq U_i$ covering X. Consider pairs (J, t) with $J \subseteq I$ and $t \in \mathcal{F}(Z_J)$ lifting s, where $Z_J = \bigcup_{i\in J} \overline{V_i}$, which is closed (Lemma 18.10.2). Chains have upper bounds by gluing along the locally finite closed cover, so Zorn's lemma gives a maximal (J, t). If $i \notin J$, the difference $t - s_i$ on $Z_J \cap \overline{V_i}$ is a section of $\mathcal{F}'$ (its image under π is 0), which extends to $u \in \mathcal{F}'(X)$. Then t and $s_i + u$ agree on $Z_J \cap \overline{V_i}$ and glue to a lift over $Z_{J\cup\{i\}}$, a contradiction. So $J = I$.

(3) A closed subspace Z is paracompact Hausdorff, the restricted sequence on Z is exact, and $\mathcal{F}'|_Z$ is soft. By (2) applied on Z, $\mathcal{F}(Z) \to \mathcal{F}''(Z)$ is surjective, and $\mathcal{F}(X) \to \mathcal{F}(Z)$ is surjective. □

Theorem 26.4.5. *On a paracompact Hausdorff space, soft sheaves, in particular fine sheaves, are Γ-acyclic. Hence cohomology can be computed with soft or fine resolutions.*

Proof. Identical to the proof of Proposition 26.3.3, using Lemma 26.4.4 and the fact that injective sheaves are flasque, hence soft. □

Corollary 26.4.6. *Let X be paracompact Hausdorff. Then $H^i(X, \mathcal{F}) = 0$ for $i > 0$ for every sheaf of C_X-modules, for instance the sheaf of sections of a real vector bundle.*

Proof. Such a sheaf is fine (Example 26.4.3) and is its own resolution. □

26.5 Čech cohomology

Definition 26.5.1. Let $\mathcal{U} = (U_i)_{i\in I}$ be an open cover of X and $\mathcal{F}$ a presheaf of abelian groups. Write $U_{i_0\cdots i_p} = U_{i_0} \cap \cdots \cap U_{i_p}$. An *alternating Čech p-cochain* is a family $\alpha = (\alpha_{i_0\cdots i_p})$, indexed by all $(i_0, \ldots, i_p) \in I^{p+1}$, with $\alpha_{i_0\cdots i_p} \in \mathcal{F}(U_{i_0\cdots i_p})$, which changes sign when two indices are exchanged and vanishes when two indices are equal. They form a group $\check{C}^p(\mathcal{U}, \mathcal{F})$, with differential

$$(d\alpha)_{i_0\cdots i_{p+1}} = \sum_{k=0}^{p+1} (-1)^k \alpha_{i_0\cdots \widehat{i_k}\cdots i_{p+1}}|_{U_{i_0\cdots i_{p+1}}}.$$

Its cohomology is the *Čech cohomology* $\check{H}^p(\mathcal{U}, \mathcal{F})$. Choosing a well-ordering of I, $\check{C}^p(\mathcal{U}, \mathcal{F}) \cong \prod_{i_0<\cdots<i_p} \mathcal{F}(U_{i_0\cdots i_p})$. For a sheaf, $\check{H}^0(\mathcal{U}, \mathcal{F}) = \mathcal{F}(X)$ by the sheaf axioms. The *sheafified Čech complex* $\check{\mathcal{C}}^p(\mathcal{U}, \mathcal{F})$ is the sheaf $V \mapsto \check{C}^p(\mathcal{U} \cap V, \mathcal{F}|_V)$, with augmentation $\mathcal{F} \to \check{\mathcal{C}}^0$ given by restriction.

Lemma 26.5.2. *For every sheaf $\mathcal{F}$ and open cover $\mathcal{U}$, the sequence $0 \to \mathcal{F} \to \check{\mathcal{C}}^0(\mathcal{U}, \mathcal{F}) \to \check{\mathcal{C}}^1(\mathcal{U}, \mathcal{F}) \to \cdots$ is exact. If $\mathcal{F}$ is flasque, then $\check{H}^p(\mathcal{U}, \mathcal{F}) = 0$ for $p > 0$.*

Proof. Let $x \in X$ and choose j with $x \in U_j$. For open $V \subseteq U_j$ define $k\colon \check{C}^p(\mathcal{U} \cap V, \mathcal{F}) \to \check{C}^{p-1}(\mathcal{U} \cap V, \mathcal{F})$ by $(k\alpha)_{i_0\cdots i_{p-1}} = \alpha_{ji_0\cdots i_{p-1}}$, which makes sense since $V \cap U_{ji_0\cdots i_{p-1}} = V \cap U_{i_0\cdots i_{p-1}}$, and k = restriction to $V \cap U_j = V$ in degree 0. Then $(dk\alpha)_{i_0\cdots i_p} = \sum_k (-1)^k \alpha_{ji_0\cdots \widehat{i_k}\cdots i_p}$ and $(kd\alpha)_{i_0\cdots i_p} = \alpha_{i_0\cdots i_p} - \sum_k (-1)^k \alpha_{ji_0\cdots \widehat{i_k}\cdots i_p}$, so $dk + kd = \mathrm{id}$, and similarly in degree 0 with the augmentation. Passing to the colimit over $V \ni x$, the complex of stalks is contractible, hence exact.

If $\mathcal{F}$ is flasque, each $\check{\mathcal{C}}^p(\mathcal{U}, \mathcal{F})$ is a product of sheaves $(j_{i_0\cdots i_p})_*(\mathcal{F}|_{U_{i_0\cdots i_p}})$, hence flasque (Lemma 26.3.2); so the resolution is flasque, its global sections $\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F})$ compute $H^\bullet(X, \mathcal{F})$ (Proposition 26.3.3), and this vanishes in positive degrees. □

Theorem 26.5.3 (Leray)**.** *Let $\mathcal{U}$ be an open cover of X such that $H^q(U_{i_0\cdots i_p}, \mathcal{F}) = 0$ for all $q > 0$ and all finite intersections. Then $\check{H}^p(\mathcal{U}, \mathcal{F}) \cong H^p(X, \mathcal{F})$ for all p.*

Proof. Let $\mathcal{F} \to I^\bullet$ be an injective resolution and consider the first-quadrant double complex $C^{p,q} = \check{C}^p(\mathcal{U}, I^q)$. For fixed q, the row $\check{C}^\bullet(\mathcal{U}, I^q)$ is exact in positive degrees with kernel $I^q(X)$ in degree 0, by Lemmas 26.5.2 and 26.3.2. For fixed p, the column is $\prod_{i_0<\cdots<i_p} I^\bullet(U_{i_0\cdots i_p})$; since $I^\bullet|_U$ is an injective resolution of $\mathcal{F}|_U$ (Lemma 26.2.1) and products of abelian groups are exact, its cohomology is $\prod H^q(U_{i_0\cdots i_p}, \mathcal{F})$, which is 0 for $q > 0$ and $\check{C}^p(\mathcal{U}, \mathcal{F})$ for $q = 0$. By Lemma 15.7.2, applied once to rows and once to columns, both $\Gamma(X, I^\bullet)$ and $\check{C}^\bullet(\mathcal{U}, \mathcal{F})$ are quasi-isomorphic to the total complex. Taking cohomology gives the theorem. □

Remark 26.5.4. Refinements of covers induce maps on Čech cohomology, well defined up to homotopy, and the colimit over all covers is the Čech cohomology $\check{H}^p(X, \mathcal{F})$. There is always a natural map $\check{H}^p(X, \mathcal{F}) \to H^p(X, \mathcal{F})$, which is an isomorphism for $p \le 1$, and for all p if X is paracompact Hausdorff [God58, Chapter II, Section 5]. Cohomology classes in $\check{H}^1(\mathcal{U}, \mathcal{F})$ are represented by cocycles (g_{ij}) with $g_{ij} + g_{jk} = g_{ik}$; for nonabelian sheaves of groups such as $\mathrm{GL}_n(\mathcal{O}_X)$ the same formula (written multiplicatively) classifies locally free sheaves of rank n (Chapter 35).

26.6 Comparison with singular cohomology

Theorem 26.6.1. *Let X be a paracompact Hausdorff, locally contractible space (for instance a manifold or a CW complex) and A an abelian group. Then $H^p(X, A_X) \cong H^p(X; A)$, the singular cohomology, naturally in X (and compatibly with cup products [Bre97, Chapter III]).*

Sketch of proof. Let $\mathcal{S}^p$ be the sheafification of the presheaf $U \mapsto C^p(U; A)$ of singular cochains. Since every point has arbitrarily small contractible neighbourhoods, the stalks of the cohomology presheaves $U \mapsto H^p(U; A)$ vanish for $p > 0$ and equal A for $p = 0$, so $0 \to A_X \to \mathcal{S}^\bullet$ is a resolution (sheafification preserves stalks and stalks are exact). The omitted verifications are that the sheaves $\mathcal{S}^p$ are soft on a paracompact space, and that $\Gamma(X, \mathcal{S}^\bullet) = C^\bullet(X; A)/C_0^\bullet(X; A)$, where $C_0^\bullet$ consists of cochains vanishing on sufficiently small simplices of some open cover, which is acyclic by the subdivision argument of Proposition 21.5.3; see [Bre97, Chapter III]. Then Theorem 26.4.5 gives $H^p(X, A_X) \cong H^p(C^\bullet(X; A)/C_0^\bullet) \cong H^p(X; A)$. □

Example 26.6.2. $H^p(\mathbb{R}^n, \mathbb{Z}) = 0$ for $p > 0$, $H^p(S^n, \mathbb{Z})$ is $\mathbb{Z}$ for $p = 0, n$, and on a closed oriented surface of genus g, $H^1(\Sigma_g, \mathbb{Z}) \cong \mathbb{Z}^{2g}$. On spaces that are not locally contractible sheaf and singular cohomology can differ: for the topologist's sine curve T of Counterexample 18.5.8, singular cohomology sees two path components, $H^0(T; \mathbb{Z}) = \mathbb{Z}^2$, while $H^0(T, \mathbb{Z}_T) = \mathbb{Z}$ since T is connected.

Example 26.6.3. Cover S^1 by two open arcs U, V with $U \cap V = W_1 \sqcup W_2$, two disjoint arcs. Let $\mathcal{L}$ be the local system of rank one with monodromy -1, obtained by gluing $\mathbb{Z}_U$ and $\mathbb{Z}_V$ by the identity over W_1 and by -1 over W_2 (Proposition 25.2.5 and Definition 25.5.1). The arcs and their intersections are homeomorphic to intervals, so the constant sheaf $\mathbb{Z}$ has no higher cohomology on them (Theorem 26.6.1), and Leray's theorem (Theorem 26.5.3) lets us compute with the cover $\{U, V\}$. In the trivialisation over V, the Čech complex is $\mathbb{Z}^2 \to \mathbb{Z}^2$, $(a, b) \mapsto (b - a, b + a)$, one coordinate for each W_i. Its kernel is 0 and its image is $\{(x, y) : x \equiv y \pmod 2\}$, so

$$H^0(S^1, \mathcal{L}) = 0, \qquad H^1(S^1, \mathcal{L}) \cong \mathbb{Z}/2.$$

For the constant sheaf the differential is $(a, b) \mapsto (b - a, b - a)$, which gives $H^0 = H^1 = \mathbb{Z}$, in accordance with Theorem 26.6.1. With coefficients in $\mathbb{Q}$ the twisted local system has $H^0 = H^1 = 0$, since the determinant 2 of the differential becomes invertible: a nontrivial local system can have vanishing rational cohomology on a space whose ordinary cohomology does not vanish.

26.7 Hypercohomology and the Leray spectral sequence

Proposition 26.7.1. *Let $\mathcal{K}^\bullet$ be a complex of sheaves with $\mathcal{K}^p = 0$ for $p < 0$.*

1. *A quasi-isomorphism $\mathcal{K}^\bullet \to \mathcal{L}^\bullet$ induces isomorphisms on hypercohomology.*
2. *There are convergent spectral sequences $E_1^{p,q} = H^q(X, \mathcal{K}^p) \Rightarrow \mathbb{H}^{p+q}(X, \mathcal{K}^\bullet)$ and $E_2^{p,q} = H^p(X, \mathcal{H}^q(\mathcal{K}^\bullet)) \Rightarrow \mathbb{H}^{p+q}(X, \mathcal{K}^\bullet)$, where $\mathcal{H}^q$ are the cohomology sheaves.*
3. *If every $\mathcal{K}^p$ is Γ-acyclic, $\mathbb{H}^p(X, \mathcal{K}^\bullet) = H^p(\Gamma(X, \mathcal{K}^\bullet))$. If $\mathcal{F} \to \mathcal{K}^\bullet$ is a resolution, then $\mathbb{H}^p(X, \mathcal{K}^\bullet) = H^p(X, \mathcal{F})$.*

Proof. (1) Quasi-isomorphisms are isomorphisms in the derived category, and $R\Gamma$ is a functor on it. (2) is Proposition 17.4.8. (3) The first statement is Lemma 17.4.6, and the second follows from (1) applied to $\mathcal{F} \to \mathcal{K}^\bullet$, with $\mathcal{F}$ in degree 0. □

Remark 26.7.2. On a complex manifold X the holomorphic de Rham complex $\Omega_X^\bullet$ is a resolution of $\mathbb{C}_X$ (the holomorphic Poincaré lemma), so $H^k(X;\mathbb{C}) \cong \mathbb{H}^k(X, \Omega_X^\bullet)$, and the first spectral sequence of Proposition 26.7.1 becomes the *Hodge-to-de Rham* (Frölicher) spectral sequence $E_1^{p,q} = H^q(X, \Omega_X^p) \Rightarrow H^{p+q}(X;\mathbb{C})$. Its degeneration for compact Kähler manifolds is a form of the Hodge decomposition (Chapter 48).

Theorem 26.7.3 (Leray spectral sequence)**.** *Let $f : X \to Y$ be continuous and $\mathcal{F}$ a sheaf of abelian groups on X. Then $R^q f_*\mathcal{F}$ is the sheafification of the presheaf $V \mapsto H^q(f^{-1}V, \mathcal{F})$, and there is a convergent spectral sequence $E_2^{p,q} = H^p(Y, R^q f_*\mathcal{F}) \Rightarrow H^{p+q}(X, \mathcal{F})$.*

Proof. Let $\mathcal{F} \to I^\bullet$ be an injective resolution. Then $R^q f_*\mathcal{F} = \mathcal{H}^q(f_* I^\bullet)$, the sheafification of the presheaf of cohomology groups $V \mapsto H^q(I^\bullet(f^{-1}V))$, and $H^q(I^\bullet(f^{-1}V)) = H^q(f^{-1}V, \mathcal{F})$ because $I^\bullet|_{f^{-1}V}$ is an injective resolution (Lemma 26.2.1). Since $\Gamma(X, -) = \Gamma(Y, -) \circ f_*$ and f_* preserves injectives, which are $\Gamma(Y, -)$-acyclic, the Grothendieck spectral sequence (Theorem 16.6.3) applies. □

Example 26.7.4. If f is a closed embedding, f_* is exact (compute stalks as in Proposition 25.4.4), so $H^p(Y, f_*\mathcal{F}) \cong H^p(X, \mathcal{F})$. For a locally trivial fibre bundle $f : E \to B$ with fibre F over a locally contractible paracompact base, $R^q f_*\mathbb{Z}_E$ is the local system with stalks $H^q(F;\mathbb{Z})$, and the Leray spectral sequence becomes the *Serre spectral sequence* $E_2^{p,q} = H^p(B, \mathcal{H}^q(F)) \Rightarrow H^{p+q}(E;\mathbb{Z})$ with local coefficients [Bre97].

26.8 Cohomology with compact supports

Definition 26.8.1. Let X be locally compact Hausdorff. The *support* of a section $s \in \mathcal{F}(X)$ is the closed set $\{x : s_x \neq 0\}$. Let $\Gamma_c(X, \mathcal{F})$ be the group of sections with compact support, a left exact functor, and $H_c^i(X, \mathcal{F}) = R^i\Gamma_c(X, \mathcal{F})$.

Remark 26.8.2. Flasque sheaves are Γ_c-acyclic, and on locally compact Hausdorff spaces that are countable at infinity, so are soft sheaves; so H_c^* can be computed by fine resolutions. For a manifold M and an abelian group A, $H_c^i(M, A_M)$ agrees with the compactly supported singular cohomology of Definition 23.5.2, and Poincaré duality generalises to *Verdier duality* $R\Gamma_c(X, \mathcal{F})^\vee \cong R\operatorname{Hom}(\mathcal{F}, \omega_X)$ in the derived category, with $\omega_X = or_X[n]$ for an n-manifold [KS90, Chapter III], [Ive86].

26.9 Exercises

Exercise 26.9.1. Compute $\check{H}^\bullet(\mathcal{U}, \mathbb{Z}_{S^1})$ for the cover of S^1 by two open arcs whose intersection has two components, and check with Theorems 26.5.3 and 26.6.1 that $H^1(S^1, \mathbb{Z}) \cong \mathbb{Z}$.

Solution. With $I = \{0, 1\}$, $\check{C}^0 = \mathbb{Z}_{S^1}(U_0) \oplus \mathbb{Z}_{S^1}(U_1) = \mathbb{Z}^2$, $\check{C}^1 = \mathbb{Z}_{S^1}(U_0 \cap U_1) = \mathbb{Z}^2$ and $d(a, b) = (b - a, b - a)$. So $\check{H}^0 = \mathbb{Z}$ and $\check{H}^1 = \mathbb{Z}^2/\{(c, c)\} \cong \mathbb{Z}$. The arcs and the components of their intersection are contractible, hence have no higher cohomology with constant coefficients, so Leray's theorem applies. □

Exercise 26.9.2. Show that skyscraper sheaves are flasque, and that $H^i(X, x_*A) = 0$ for $i > 0$.

Solution. Restriction maps are identities or maps to 0. Flasque sheaves are acyclic (Proposition 26.3.3). □

Exercise 26.9.3. Let X be an irreducible space (every nonempty open subset is dense). Show that every constant sheaf on X is flasque, and deduce $H^i(X, A_X) = 0$ for $i > 0$. (This applies to algebraic varieties with the Zariski topology, and shows that sheaf cohomology of constant sheaves in that topology does not see the classical topology.)

Solution. Nonempty open subsets of an irreducible space are connected (two disjoint nonempty open subsets would contradict density), so locally constant functions on them are constant, and $A_X(U) = A$ for $U \neq \emptyset$, with identity restriction maps. □

Exercise 26.9.4. For open $U, V \subseteq X$ and a sheaf $\mathcal{F}$, construct a long exact Mayer–Vietoris sequence $\cdots \to H^i(U \cup V, \mathcal{F}) \to H^i(U, \mathcal{F}) \oplus H^i(V, \mathcal{F}) \to H^i(U \cap V, \mathcal{F}) \to H^{i+1}(U \cup V, \mathcal{F}) \to \cdots$. (Hint: for flasque I, $0 \to I(U \cup V) \to I(U) \oplus I(V) \to I(U \cap V) \to 0$ is exact.)

Part V

Analysis

Chapter 27

Real Analysis

27.1 Introduction

This chapter develops calculus in one and several real variables with complete proofs, starting slowly from limits of sequences and of functions: the sequential characterisation of limits, continuous functions and the theorems of Bolzano, Weierstrass and Heine–Cantor, the mean value theorems, de l'Hôpital's rule, Taylor's formula and convexity; then series and power series, the exponential and trigonometric functions (including a definition of π), the Riemann integral, uniform convergence of sequences of functions (with a continuous nowhere differentiable function), derivatives of maps between Euclidean spaces, the inverse and implicit function theorems, extrema in several variables with Lagrange multipliers and the Karush–Kuhn–Tucker conditions of optimisation, ordinary differential equations, and the approximation and compactness theorems of Stone–Weierstrass and Arzelà–Ascoli. The topological background (metric spaces, compactness, completeness, uniform convergence, the Banach fixed point theorem) is in Chapter 18. References: [Rud76], [Die60], [Spi65], [Lan93] and [Har02].

Prerequisites. Chapter 18 (General Topology), Chapter 4 (Number Systems) and Chapter 11 (Linear Algebra); Theorem 27.7.3 also uses Chapter 19 (Homotopy and Covering Spaces).

Throughout, $\mathbb{K}$ is $\mathbb{R}$ or $\mathbb{C}$, and $\mathbb{R}^n$ and $\mathbb{C}^n$ carry the Euclidean norm unless otherwise stated.

27.2 Limits of sequences

We recall that a real sequence $(a_n)_{n\in\mathbb{N}}$ *converges* to $a \in \mathbb{R}$ if for every $\varepsilon > 0$ there is N with $|a_n - a| < \varepsilon$ for all $n \geq N$ (Definition 4.7.5). This is convergence in the metric space $\mathbb{R}$ (Chapter 18). In this section we develop the theory of limits of sequences directly from this definition.

Definition 27.2.1. Let (a_n) be a real sequence.

1. If (a_n) converges to a we write $a_n \to a$ or $\lim_n a_n = a$, and call a the *limit*.
2. (a_n) *diverges to* $+\infty$, written $a_n \to +\infty$, if for every $M \in \mathbb{R}$ there is N with $a_n > M$ for all $n \geq N$; similarly $a_n \to -\infty$ if for every M eventually $a_n < M$.
3. (a_n) is *regular* if it converges or diverges to $\pm\infty$, and *oscillating* otherwise.
4. (a_n) is *bounded* if there is C with $|a_n| \leq C$ for all n; *bounded above* or *below* are defined similarly.

5. (a_n) is *increasing* if $a_n \le a_{n+1}$ for all n, *strictly increasing* if $a_n < a_{n+1}$, and similarly *decreasing*; it is *monotone* if it is increasing or decreasing.

6. A *subsequence* of (a_n) is a sequence $(a_{n_k})_{k\in\mathbb{N}}$ with $n_0 < n_1 < n_2 < \cdots$.

A property holds *eventually*, or *for large* n, if there is N such that it holds for all $n \ge N$.

Proposition 27.2.2. *Let (a_n) be a real sequence.*

1. *A sequence has at most one limit.*

2. *A convergent sequence is bounded.*

3. *(Permanence of sign) If $a_n \to a$ and $a > 0$, then $a_n > a/2 > 0$ for large n.*

4. *Changing, adding or removing finitely many terms does not affect convergence or the limit.*

5. *If $a_n \to a$, every subsequence converges to a; the same holds for divergence to $\pm\infty$.*

Proof. (1) Suppose $a_n \to a$ and $a_n \to b$ with $a \ne b$, and let $\varepsilon = |a-b|/2$. For n larger than both thresholds, $|a-b| \le |a-a_n| + |a_n - b| < 2\varepsilon = |a-b|$, a contradiction. (2) Take $\varepsilon = 1$: for $n \ge N$, $|a_n| \le |a_n - a| + |a| < |a| + 1$, so $|a_n| \le \max(|a_0|, \ldots, |a_{N-1}|, |a| + 1)$ for all n. (3) Take $\varepsilon = a/2$: then $a_n > a - a/2 = a/2$ for large n. (4) The definitions only involve the terms with $n \ge N$, and N may be enlarged. (5) By induction $n_k \ge k$, so $k \ge N$ implies $n_k \ge N$. □

Lemma 27.2.3 (Bernoulli's inequality)**.** *For $h \ge -1$ and $n \in \mathbb{N}$, $(1+h)^n \ge 1 + nh$.*

Proof. For $n = 0$ both sides are 1. If $(1+h)^n \ge 1 + nh$, then, as $1 + h \ge 0$, $(1+h)^{n+1} \ge (1+nh)(1+h) = 1 + (n+1)h + nh^2 \ge 1 + (n+1)h$. □

Proposition 27.2.4 (Algebra of limits)**.** *Let $a_n \to a$ and $b_n \to b$ be real sequences and $c \in \mathbb{R}$. Then $a_n + b_n \to a + b$, $ca_n \to ca$, $a_n b_n \to ab$ and $|a_n| \to |a|$. If $b \ne 0$, then $b_n \ne 0$ for large n and $a_n/b_n \to a/b$.*

Proof. Given $\varepsilon > 0$, for large n we have $|a_n - a| < \varepsilon/2$ and $|b_n - b| < \varepsilon/2$, so $|(a_n + b_n) - (a+b)| < \varepsilon$. The reverse triangle inequality $||a_n| - |a|| \le |a_n - a|$ gives $|a_n| \to |a|$, and $|ca_n - ca| = |c|\,|a_n - a|$ gives $ca_n \to ca$. For the product, let C bound $|a_n|$ (Proposition 27.2.2(2)); then

$$|a_n b_n - ab| \le |a_n|\,|b_n - b| + |b|\,|a_n - a| \le C|b_n - b| + |b|\,|a_n - a|,$$

which is less than ε for large n. For the quotient it suffices to show $1/b_n \to 1/b$. As $|b_n| \to |b| > 0$, permanence of sign gives $|b_n| > |b|/2$ for large n, and then $|1/b_n - 1/b| = |b - b_n|/(|b_n|\,|b|) \le 2|b_n - b|/|b|^2$, which tends to 0. □

Proposition 27.2.5 (Comparison and squeeze)**.** *Let (a_n), (b_n), (c_n) be real sequences.*

1. *If $a_n \le b_n$ for large n, $a_n \to a$ and $b_n \to b$, then $a \le b$.*

2. *(Squeeze theorem) If $a_n \le c_n \le b_n$ for large n and $a_n \to L$, $b_n \to L$, then $c_n \to L$.*

3. *If $a_n \le b_n$ for large n and $a_n \to +\infty$, then $b_n \to +\infty$.*

4. *If $|a_n| \le b_n$ for large n and $b_n \to 0$, then $a_n \to 0$. In particular, if (a_n) is bounded and $b_n \to 0$, then $a_n b_n \to 0$.*

Proof. (1) If $a > b$, then $a_n - b_n \to a - b > 0$ by Proposition 27.2.4, so $a_n - b_n > 0$ for large n by permanence of sign, a contradiction. (2) Given $\varepsilon > 0$, for large n we have $L - \varepsilon < a_n \le c_n \le b_n < L + \varepsilon$. (3) Eventually $b_n \ge a_n > M$. (4) Apply (2) to $-b_n \le a_n \le b_n$; for the last statement, $|a_n b_n| \le C|b_n|$. □

Counterexample 27.2.6. Strict inequalities do not pass to the limit: $1/(n+1) > 0$ for all n, but the limit is 0. The convergence $|a_n| \to |a|$ does not imply $a_n \to a$ unless $a = 0$: the sequence $a_n = (-1)^n$ oscillates although $|a_n| \to 1$.

Proposition 27.2.7 (Infinite limits)**.** *Let $a_n \to +\infty$.*

1. *If (b_n) is bounded below, then $a_n + b_n \to +\infty$.*
2. *If $b_n \geq c$ for large n, for some $c > 0$, then $a_n b_n \to +\infty$.*
3. *$a_n > 0$ for large n, and $1/a_n \to 0$. Conversely, if $b_n > 0$ and $b_n \to 0$, then $1/b_n \to +\infty$.*

Proof. (1) If $b_n \geq -C$, then for any M, eventually $a_n > M + C$, so $a_n + b_n > M$. (2) Eventually $a_n > M/c$ with $M > 0$, so $a_n b_n > M$. (3) Eventually $a_n > 1/\varepsilon > 0$, so $0 < 1/a_n < \varepsilon$. Conversely, eventually $0 < b_n < 1/M$ for $M > 0$, so $1/b_n > M$. □

Counterexample 27.2.8 (Indeterminate forms)**.** If $a_n \to +\infty$ and $b_n \to -\infty$, the limits alone do not determine the behaviour of $a_n + b_n$: with $a_n = n$, the choices $b_n = -n + c$, $b_n = -n/2$, $b_n = -2n$ and $b_n = -n + (-1)^n$ give the limit c, the limit $+\infty$, the limit $-\infty$, and an oscillating sequence. Similarly, if $a_n \to 0$ and $b_n \to +\infty$, then with $a_n = 1/(n+1)$ the choices $b_n = c(n+1)$, $(n+1)^2$ and $(-1)^n(n+1)$ show that $a_n b_n$ can converge to any c, diverge, or oscillate. The same happens for the forms ∞/∞ and $0/0$.

Proposition 27.2.9 (Fundamental limits)**.**

1. *For $q \in \mathbb{R}$: $q^n \to 0$ if $|q| < 1$, $q^n \to 1$ if $q = 1$, $q^n \to +\infty$ if $q > 1$, and (q^n) oscillates if $q \leq -1$.*
2. *For $a > 0$, $a^{1/n} \to 1$, and $n^{1/n} \to 1$.*
3. *For $k \in \mathbb{N}$ and $a > 1$, $n^k/a^n \to 0$: exponentials beat powers.*
4. *For every $a \in \mathbb{R}$, $a^n/n! \to 0$, and $n!/n^n \to 0$.*

Proof. (1) If $q > 1$, write $q = 1 + h$ with $h > 0$; by Lemma 27.2.3, $q^n \geq 1 + nh \to +\infty$. If $0 < |q| < 1$, then $|q|^{-n} \to +\infty$, so $|q^n| \to 0$ by Proposition 27.2.7(3), and $q^n \to 0$ by Proposition 27.2.5(4). If $q = -1$ the terms alternate between ± 1; if $q < -1$, $|q^n| \to +\infty$ and the signs alternate, so the sequence is unbounded above and below and has no limit. (2) Let $a \geq 1$ and $a^{1/n} = 1 + h_n$ with $h_n \geq 0$ (the n-th root exists by Proposition 4.7.4). By Bernoulli, $a \geq 1 + nh_n$, so $0 \leq h_n \leq (a-1)/n \to 0$. For $a < 1$ apply this to $1/a$ and use Proposition 27.2.4. For $n^{1/n} = 1 + h_n$ with $n \geq 2$, the binomial formula gives $n \geq \binom{n}{2} h_n^2$, so $0 \leq h_n \leq \sqrt{2/(n-1)} \to 0$. (3) Write $a = 1 + h$. For $n \geq 2k+2$ the binomial formula gives $a^n \geq \binom{n}{k+1} h^{k+1} \geq \frac{(n/2)^{k+1}}{(k+1)!} h^{k+1}$, since each of the $k+1$ factors $n, n-1, \ldots, n-k$ is at least $n/2$. Hence $0 \leq n^k/a^n \leq 2^{k+1}(k+1)!/(h^{k+1}n) \to 0$. (4) Choose an integer $N > 2|a|$. For $n \geq N$, $|a|^n/n! \leq \frac{|a|^N}{N!} 2^{N-n}$, which tends to 0 by (1). Finally $n!/n^n = \frac{1}{n} \cdot \frac{2}{n} \cdots \frac{n}{n} \leq \frac{1}{n}$. □

Proposition 27.2.10. *Every monotone sequence is regular. An increasing sequence converges to its supremum if it is bounded above, and diverges to $+\infty$ otherwise; a decreasing sequence converges to its infimum if it is bounded below, and diverges to $-\infty$ otherwise.*

Proof. Let (a_n) be increasing. If it is bounded above, let $s = \sup_n a_n$ (Definition 4.7.1). For $\varepsilon > 0$, $s - \varepsilon$ is not an upper bound, so $a_N > s - \varepsilon$ for some N, and then $s - \varepsilon < a_N \leq a_n \leq s$ for $n \geq N$. If it is not bounded above, for every M some $a_N > M$, and then $a_n \geq a_N > M$ for $n \geq N$. The decreasing case follows by passing to $(-a_n)$. □

Lemma 27.2.11. *Every real sequence has a monotone subsequence.*

Proof. Call n a *peak* if $a_n \geq a_m$ for all $m > n$. If there are infinitely many peaks $n_0 < n_1 < \cdots$, then $a_{n_0} \geq a_{n_1} \geq \cdots$ is a decreasing subsequence. Otherwise let n_0 be larger than all peaks. As n_0 is not a peak, there is $n_1 > n_0$ with $a_{n_1} > a_{n_0}$; as n_1 is not a peak, there is $n_2 > n_1$ with $a_{n_2} > a_{n_1}$; continuing (by recursion), we obtain a strictly increasing subsequence. □

Theorem 27.2.12 (Bolzano–Weierstrass)**.** *Every bounded real sequence has a convergent subsequence. Every real sequence has a regular subsequence: a sequence that is unbounded above has a subsequence diverging to $+\infty$.*

Proof. By Lemma 27.2.11 there is a monotone subsequence, which is regular by Proposition 27.2.10, and convergent if the sequence is bounded. If (a_n) is unbounded above, choose recursively $n_{k+1} > n_k$ with $a_{n_{k+1}} > k + 1$, which is possible because only finitely many terms have index $\leq n_k$. □

Theorem 27.2.13 (Cauchy criterion)**.** *A real sequence converges if and only if it is a Cauchy sequence.*

Proof. If $a_n \to a$, then $|a_n - a_m| \leq |a_n - a| + |a - a_m| < \varepsilon$ when both are within $\varepsilon/2$ of a. Conversely, let (a_n) be Cauchy. Taking $\varepsilon = 1$ shows that (a_n) is bounded, so by Theorem 27.2.12 a subsequence converges to some a. Given $\varepsilon > 0$, choose N with $|a_n - a_m| < \varepsilon/2$ for $m, n \geq N$, and k with $n_k \geq N$ and $|a_{n_k} - a| < \varepsilon/2$. Then $|a_n - a| \leq |a_n - a_{n_k}| + |a_{n_k} - a| < \varepsilon$ for $n \geq N$. (This is a second proof of Proposition 4.7.6 for $\mathbb{R}$.) □

Counterexample 27.2.14. It is not enough that consecutive terms get close. The partial sums $H_n = \sum_{k=1}^{n} 1/k$ of the harmonic series satisfy $H_{n+1} - H_n = 1/(n+1) \to 0$, but $H_{2n} - H_n = \sum_{k=n+1}^{2n} 1/k \geq n \cdot \frac{1}{2n} = \frac{1}{2}$, so (H_n) is not Cauchy; being increasing, it diverges to $+\infty$. Likewise $\sqrt{n+1} - \sqrt{n} = 1/(\sqrt{n+1} + \sqrt{n}) \to 0$ while $\sqrt{n} \to +\infty$.

Proposition 27.2.15 (Subsequence criterion)**.**

1. *$a_n \to a$ if and only if every subsequence of (a_n) has a further subsequence converging to a.*
2. *A bounded sequence all of whose convergent subsequences have the same limit a converges to a.*

Proof. (1) One direction is Proposition 27.2.2(5). Conversely, if $a_n \not\to a$, there is $\varepsilon > 0$ such that $|a_n - a| \geq \varepsilon$ for infinitely many n; these terms form a subsequence none of whose subsequences converges to a. (2) By Theorem 27.2.12, every subsequence is bounded and has a convergent subsequence, whose limit is a by hypothesis; apply (1). □

Proposition 27.2.16 (Stolz–Cesàro)**.** *Let (b_n) be strictly increasing with $b_n \to +\infty$, and let (a_n) be a real sequence with $\frac{a_{n+1}-a_n}{b_{n+1}-b_n} \to L \in \mathbb{R}$. Then $a_n/b_n \to L$. In particular (Cesàro), if $a_n \to L$, then the averages $\frac{1}{n}(a_1 + \cdots + a_n) \to L$.*

Proof. Let $\varepsilon > 0$ and choose N such that $b_N > 0$ and $|(a_{k+1} - a_k) - L(b_{k+1} - b_k)| \leq \varepsilon(b_{k+1} - b_k)$ for $k \geq N$. Summing over $N \leq k < n$ gives $|a_n - a_N - L(b_n - b_N)| \leq \varepsilon(b_n - b_N) \leq \varepsilon b_n$. Dividing by $b_n > 0$,

$$\left|\frac{a_n}{b_n} - L\right| \leq \varepsilon + \frac{|a_N - Lb_N|}{b_n},$$

and the last term is less than ε for large n, as $b_n \to +\infty$. For Cesàro, apply this to $a'_n = a_1 + \cdots + a_n$ and $b_n = n$. □

Counterexample 27.2.17. The converse of Cesàro's theorem fails: $a_n = (-1)^n$ does not converge, but its averages tend to 0, since the partial sums are 0 or -1.

27.3 Limits of functions

To treat finite and infinite limits at once we use the *extended real line* $\bar{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$, ordered so that $-\infty < x < +\infty$ for $x \in \mathbb{R}$.

Definition 27.3.1. For $a \in \bar{\mathbb{R}}$ and $\delta > 0$, the *basic neighbourhood* $U_\delta(a)$ is $(a - \delta, a + \delta)$ if $a \in \mathbb{R}$, $(1/\delta, +\infty)$ if $a = +\infty$, and $(-\infty, -1/\delta)$ if $a = -\infty$. Let $D \subseteq \mathbb{R}$. A point $a \in \bar{\mathbb{R}}$ is an *accumulation point* of D if every $U_\delta(a)$ contains a point of $D \setminus \{a\}$. A point of D that is not an accumulation point is *isolated*.

A sequence (x_n) tends to $a \in \bar{\mathbb{R}}$ if and only if for every $\delta > 0$, eventually $x_n \in U_\delta(a)$: for $a \in \mathbb{R}$ this is convergence, and for $a = \pm\infty$ divergence to $\pm\infty$. Every $a \in \bar{\mathbb{R}}$ that is an accumulation point of D is the limit of a sequence in $D \setminus \{a\}$: choose $x_n \in U_{1/(n+1)}(a) \cap D \setminus \{a\}$.

Definition 27.3.2. Let $f : D \to \mathbb{R}$, let $a \in \bar{\mathbb{R}}$ be an accumulation point of D, and $L \in \bar{\mathbb{R}}$. We write $\lim_{x \to a} f(x) = L$, or $f(x) \to L$ as $x \to a$, if for every $\varepsilon > 0$ there is $\delta > 0$ such that $f(x) \in U_\varepsilon(L)$ for all $x \in D \cap U_\delta(a)$ with $x \neq a$. For $a, L \in \mathbb{R}$ this reads: for every $\varepsilon > 0$ there is $\delta > 0$ with $|f(x) - L| < \varepsilon$ whenever $x \in D$ and $0 < |x - a| < \delta$. The *right limit* $\lim_{x \to a^+} f(x)$ is the limit of the restriction of f to $D \cap (a, +\infty)$, when a is an accumulation point of that set, and the *left limit* $\lim_{x \to a^-} f(x)$ is defined similarly. The same definitions apply to $\mathbb{C}$- or $\mathbb{R}^m$-valued functions and finite limits, with $|\cdot|$ the modulus or the Euclidean norm.

The value of f at a, if $a \in D$, plays no role in the definition.

Proposition 27.3.3. *Let $f : D \to \mathbb{R}$ and let $a \in \bar{\mathbb{R}}$ be an accumulation point of D.*

1. *The limit of f at a, if it exists, is unique.*
2. *If $\lim_{x \to a} f(x) = L \in \mathbb{R}$, then f is bounded on $D \cap U_\delta(a) \setminus \{a\}$ for some $\delta > 0$; if $L > 0$, then $f(x) > L/2$ for $x \in D \cap U_\delta(a) \setminus \{a\}$ for some $\delta > 0$.*
3. *If $a \in \mathbb{R}$ is an accumulation point of both $D \cap (a, +\infty)$ and $D \cap (-\infty, a)$, then $\lim_{x \to a} f(x) = L$ if and only if both one-sided limits exist and equal L.*

Proof. (1) If $L \neq L'$ are both limits, choose ε with $U_\varepsilon(L) \cap U_\varepsilon(L') = \emptyset$, and δ working for both; since a is an accumulation point, some $x \in D \cap U_\delta(a) \setminus \{a\}$ exists, and $f(x)$ lies in both neighbourhoods. (2) Take $\varepsilon = 1$, respectively $\varepsilon = L/2$. (3) If both one-sided limits equal L, take the smaller of the two δ; the converse is clear. □

Theorem 27.3.4 (Sequential characterisation of limits)**.** *Let $f : D \to \mathbb{R}$, let $a \in \bar{\mathbb{R}}$ be an accumulation point of D, and $L \in \bar{\mathbb{R}}$. The following are equivalent:*

1. *$\lim_{x \to a} f(x) = L$;*
2. *for every sequence (x_n) in $D \setminus \{a\}$ with $x_n \to a$, we have $f(x_n) \to L$.*

Proof. (1) $\Rightarrow$ (2). Let $\varepsilon > 0$ and choose δ as in Definition 27.3.2. Eventually $x_n \in U_\delta(a)$, and $x_n \in D \setminus \{a\}$, so eventually $f(x_n) \in U_\varepsilon(L)$.

(2) $\Rightarrow$ (1). Suppose (1) fails. Then there is $\varepsilon > 0$ such that for every $\delta > 0$ some $x \in D \cap U_\delta(a) \setminus \{a\}$ has $f(x) \notin U_\varepsilon(L)$. Choosing such a point x_n for $\delta = 1/(n+1)$, for each n (this uses the axiom of countable choice), we obtain a sequence in $D \setminus \{a\}$ with $x_n \to a$ and $f(x_n) \notin U_\varepsilon(L)$ for all n, so $f(x_n) \not\to L$, contradicting (2). □

Theorem 27.3.4, sometimes called the *bridge theorem*, transfers every result about limits of sequences to limits of functions, and it is the standard way to show that a limit does *not* exist: exhibit two sequences tending to a along which f has different limits.

Corollary 27.3.5 (Rules for limits of functions)**.** *Let $f, g, h\colon D \to \mathbb{R}$ and let $a \in \bar{\mathbb{R}}$ be an accumulation point of D.*

1. *If $f(x) \to L$ and $g(x) \to M$ with $L, M \in \mathbb{R}$, then $f+g \to L+M$, $fg \to LM$, and, if $M \neq 0$, $f/g \to L/M$ (where f/g is defined near a).*
2. *If $f \leq g$ near a and $f \to L$, $g \to M$, then $L \leq M$.*
3. *(Squeeze) If $f \leq h \leq g$ near a and $f(x) \to L$, $g(x) \to L$, then $h(x) \to L$.*
4. *The rules for infinite limits of Proposition 27.2.7 hold for functions.*

Proof. Apply Theorem 27.3.4 and Propositions 27.2.4, 27.2.5 and 27.2.7 to $f(x_n)$, $g(x_n)$, $h(x_n)$ for sequences $x_n \to a$ in $D \setminus \{a\}$. For the quotient, $g(x) \neq 0$ near a by Proposition 27.3.3(2) applied to $|g|$. □

Proposition 27.3.6 (Limits of composite functions)**.** *Let $f\colon D \to \mathbb{R}$ and $g\colon E \to \mathbb{R}$ with $f(D) \subseteq E$. Suppose $\lim_{x\to a} f(x) = b$ and $\lim_{y\to b} g(y) = L$, and that one of the following holds:*

1. *$f(x) \neq b$ for all $x \in D \setminus \{a\}$ near a;*
2. *$b \in E$ and $g(b) = L$.*

Then $\lim_{x\to a} g(f(x)) = L$.

Proof. Let $x_n \to a$ in $D \setminus \{a\}$. Then $f(x_n) \to b$ by Theorem 27.3.4. In case (1), eventually $f(x_n) \in E \setminus \{b\}$, so $g(f(x_n)) \to L$ by the same theorem. In case (2), given $\varepsilon > 0$ choose η with $g(y) \in U_\varepsilon(L)$ for $y \in E \cap U_\eta(b)$, including $y = b$; eventually $f(x_n) \in U_\eta(b)$, so $g(f(x_n)) \in U_\varepsilon(L)$. Apply the theorem again. □

Counterexample 27.3.7. Neither hypothesis can be dropped. Let $f(x) = 0$ for all x, and $g(y) = 0$ for $y \neq 0$, $g(0) = 1$. Then $\lim_{x\to 0} f(x) = 0$ and $\lim_{y\to 0} g(y) = 0$, but $g(f(x)) = 1$ for all x.

Proposition 27.3.8 (Cauchy criterion for functions)**.** *Let $f\colon D \to \mathbb{R}$ and let $a \in \bar{\mathbb{R}}$ be an accumulation point of D. Then $\lim_{x\to a} f(x)$ exists in $\mathbb{R}$ if and only if for every $\varepsilon > 0$ there is $\delta > 0$ with $|f(x) - f(y)| < \varepsilon$ for all $x, y \in D \cap U_\delta(a) \setminus \{a\}$.*

Proof. If $f(x) \to L$, take δ for $\varepsilon/2$ and use the triangle inequality. Conversely, for every sequence $x_n \to a$ in $D \setminus \{a\}$, the sequence $(f(x_n))$ is Cauchy, hence converges (Theorem 27.2.13). If $x_n \to a$ and $y_n \to a$ are two such sequences, the interleaved sequence $x_0, y_0, x_1, y_1, \ldots$ also tends to a, so $(f(x_n))$ and $(f(y_n))$ have the same limit, being subsequences of a convergent sequence. So all these limits equal a common L, and $f(x) \to L$ by Theorem 27.3.4. □

Proposition 27.3.9 (Monotone functions)**.** *Let $f\colon (a,b) \to \mathbb{R}$ be increasing, with $-\infty \leq a < b \leq +\infty$. For every $c \in (a,b)$ the one-sided limits exist and*

$$\lim_{x\to c^-} f(x) = \sup_{a<x<c} f(x) \leq f(c) \leq \inf_{c<x<b} f(x) = \lim_{x\to c^+} f(x),$$

and $\lim_{x\to b^-} f(x) = \sup f \in \mathbb{R} \cup \{+\infty\}$, $\lim_{x\to a^+} f(x) = \inf f \in \mathbb{R} \cup \{-\infty\}$. The analogous statements hold for decreasing functions.

Proof. The set $\{f(x) : a < x < c\}$ is bounded above by $f(c)$; let s be its supremum, so $s \leq f(c)$. Given $\varepsilon > 0$ there is $x_0 < c$ with $f(x_0) > s - \varepsilon$, and for $x_0 < x < c$, $s - \varepsilon < f(x_0) \leq f(x) \leq s$. So $\lim_{x\to c^-} f(x) = s$. The other statements are proved in the same way (for $b = +\infty$ or an unbounded supremum, use neighbourhoods of $+\infty$), and the decreasing case follows by passing to $-f$. □

Counterexample 27.3.10.

1. The sign function $\operatorname{sgn}(x) = x/|x|$ on $\mathbb{R} \setminus \{0\}$ has one-sided limits ± 1 at 0, and no limit there.
2. $1/x$ has no limit at 0: its one-sided limits are $+\infty$ and $-\infty$.
3. The *Dirichlet function* $\mathbf{1}_{\mathbb{Q}}$, equal to 1 on $\mathbb{Q}$ and 0 on $\mathbb{R} \setminus \mathbb{Q}$, has no limit at any $a \in \mathbb{R}$. Both $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are dense in $\mathbb{R}$: $\mathbb{Q}$ by Proposition 4.7.3, and $\mathbb{R} \setminus \mathbb{Q}$ because $q\sqrt{2}$ is irrational for rational $q \neq 0$ (Proposition 4.6.4) and $x/\sqrt{2} < q < y/\sqrt{2}$ is possible for $x < y$. So there are rational $x_n \to a$ and irrational $y_n \to a$, all $\neq a$, and $\mathbf{1}_{\mathbb{Q}}(x_n) \to 1$, $\mathbf{1}_{\mathbb{Q}}(y_n) \to 0$.

27.4 Continuous functions of a real variable

Definition 27.4.1. Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$. Then f is *continuous at* $a \in D$ if for every $\varepsilon > 0$ there is $\delta > 0$ with $|f(x) - f(a)| < \varepsilon$ for all $x \in D$ with $|x - a| < \delta$, and *continuous* if it is continuous at every point of D.

This is continuity for the metric of $\mathbb{R}$ restricted to D (Proposition 18.3.5). If $a \in D$ is an accumulation point of D, then f is continuous at a if and only if $\lim_{x \to a} f(x) = f(a)$; if a is isolated, every f is continuous at a.

Proposition 27.4.2. *Let $f, g : D \to \mathbb{R}$ and $a \in D$.*

1. *f is continuous at a if and only if $f(x_n) \to f(a)$ for every sequence (x_n) in D with $x_n \to a$.*
2. *If f and g are continuous at a, so are $f + g$, fg, $|f|$, $\max(f, g)$, $\min(f, g)$, and f/g if $g(a) \neq 0$.*
3. *If f is continuous at a and $h : E \to \mathbb{R}$, with $f(D) \subseteq E$, is continuous at $f(a)$, then $h \circ f$ is continuous at a.*
4. *Polynomials are continuous on $\mathbb{R}$, and rational functions are continuous where the denominator does not vanish.*

Proof. (1) The proof of Theorem 27.3.4 applies verbatim, now allowing $x_n = a$. (2) follows from (1) and Proposition 27.2.4, since $\max(f, g) = \frac{1}{2}(f + g + |f - g|)$ and $\min(f, g) = \frac{1}{2}(f + g - |f - g|)$. (3) If $x_n \to a$, then $f(x_n) \to f(a)$ and so $h(f(x_n)) \to h(f(a))$. (4) Constants and the identity are continuous; apply (2). □

Definition 27.4.3. Let f be defined on an open interval containing a, and discontinuous at a. The discontinuity is *removable* if $\lim_{x \to a} f(x)$ exists in $\mathbb{R}$ (and differs from $f(a)$); it is a *jump* if both one-sided limits exist in $\mathbb{R}$ and differ; otherwise it is *essential*.

Example 27.4.4. The function equal to 1 for $x \neq 0$ and 0 at 0 has a removable discontinuity at 0. The function equal to $\operatorname{sgn}(x)$ for $x \neq 0$ and 0 at 0 has a jump. The function equal to $1/x$ for $x \neq 0$ and 0 at 0 has an essential discontinuity, and the Dirichlet function $\mathbf{1}_{\mathbb{Q}}$ has an essential discontinuity at every point (Counterexample 27.3.10).

Theorem 27.4.5. *Let $f : I \to \mathbb{R}$ be monotone on an interval I. Then every discontinuity of f at an interior point of I is a jump, and the set of points of discontinuity of f is countable.*

Proof. We may assume f increasing. At an interior point c the one-sided limits $f(c^-) \leq f(c) \leq f(c^+)$ exist by Proposition 27.3.9, and f is continuous at c if and only if $f(c^-) = f(c^+)$. So every interior discontinuity is a jump, with nonempty open interval $J_c = (f(c^-), f(c^+))$. If $c < c'$ are interior discontinuity points and $c < x < c'$, then $f(c^+) \leq f(x) \leq f(c'^-)$, so

$J_c \cap J_{c'} = \emptyset$. The set $\mathbb{Q}$ is countable (see the proof of Proposition 4.8.6); fix an injection $\iota\colon \mathbb{Q} \to \mathbb{N}$ and send c to the rational number $q \in J_c$ with the smallest $\iota(q)$, which exists as J_c contains rational numbers. This map is injective, since the J_c are disjoint, so the interior discontinuities form a countable set. Adding the at most two endpoints of I keeps it countable. □

Example 27.4.6 (Thomae's function)**.** Let $T(x) = 1/q$ if $x = p/q$ with $p \in \mathbb{Z}$, $q \geq 1$ coprime, and $T(x) = 0$ for irrational x. Then T is continuous at every irrational number and discontinuous at every rational number.

Indeed, at a rational a, $T(a) > 0$, but there are irrational $y_n \to a$ (Counterexample 27.3.10) with $T(y_n) = 0$. At an irrational a, let $\varepsilon > 0$. The rationals p/q in $(a-1, a+1)$ with $q \leq 1/\varepsilon$ are finitely many, and none equals a; let δ be smaller than 1 and than their distances to a. For $|x - a| < \delta$, either x is irrational and $T(x) = 0$, or $x = p/q$ with $q > 1/\varepsilon$, so $|T(x) - T(a)| = 1/q < \varepsilon$.

Remark 27.4.7. There is no function $\mathbb{R} \to \mathbb{R}$ that is continuous exactly at the rational numbers: the set of points of continuity of any function is a countable intersection of open sets, and $\mathbb{Q}$ is not, by the Baire category theorem (Theorem 18.7.7). By contrast, $x\mathbf{1}_{\mathbb{Q}}(x)$ is continuous exactly at 0 (Exercise 27.15.10).

Theorem 27.4.8 (Bolzano's theorem on zeros)**.** *Let $f\colon [a,b] \to \mathbb{R}$ be continuous with $f(a) < 0 < f(b)$. Then $f(c) = 0$ for some $c \in (a,b)$.*

Proof. We construct intervals by bisection. Let $a_0 = a$, $b_0 = b$. Given $[a_n, b_n]$ with $f(a_n) < 0 \leq f(b_n)$, let $m = (a_n + b_n)/2$; if $f(m) < 0$ put $a_{n+1} = m$, $b_{n+1} = b_n$, and otherwise $a_{n+1} = a_n$, $b_{n+1} = m$. Then $f(a_{n+1}) < 0 \leq f(b_{n+1})$, (a_n) is increasing, (b_n) is decreasing, and $b_n - a_n = (b-a)/2^n \to 0$. By Proposition 27.2.10 both converge, to the same limit $c \in [a,b]$. By continuity, $f(c) = \lim f(a_n) \leq 0$ and $f(c) = \lim f(b_n) \geq 0$ (Proposition 27.2.5(1)), so $f(c) = 0$, and $c \neq a, b$. □

Corollary 27.4.9. *Let I be an interval and $f\colon I \to \mathbb{R}$ continuous.*

1. *(Intermediate value theorem) f takes every value between any two of its values, so $f(I)$ is an interval.*
2. *If f is injective, it is strictly monotone.*
3. *Every continuous map $f\colon [a,b] \to [a,b]$ has a fixed point.*
4. *Every real polynomial of odd degree has a real root.*

Proof. (1) If $f(x) < y < f(z)$, apply Theorem 27.4.8 to $f - y$ on the interval between x and z (to $y - f$ if $x > z$). This is also the case $X = I$ of Corollary 18.5.5. (2) Let $a < b$ in I, and assume $f(a) < f(b)$ (otherwise replace f by $-f$). Let $x < y$ in I and put $h(t) = f((1-t)b + ty) - f((1-t)a + tx)$ for $t \in [0,1]$. Since $(1-t)a + tx < (1-t)b + ty$ and f is injective, $h(t) \neq 0$ for all t. As h is continuous and $h(0) > 0$, (1) gives $h(1) = f(y) - f(x) > 0$. So f is strictly increasing. (3) $g(x) = f(x) - x$ satisfies $g(a) \geq 0 \geq g(b)$; if neither is 0, apply (1). (4) We may take $p(x) = x^{2k+1} + c_{2k}x^{2k} + \cdots + c_0$. For $|x| \geq 1 + \sum |c_i|$ we have $|c_{2k}x^{2k} + \cdots + c_0| \leq \sum |c_i|\,|x|^{2k} < |x|^{2k+1}$, so $p(x)$ has the sign of x there. Apply (1). □

Theorem 27.4.10 (Weierstrass)**.** *A continuous function $f\colon [a,b] \to \mathbb{R}$ is bounded and attains its maximum and its minimum.*

Proof. If f were unbounded, there would be $x_n \in [a,b]$ with $|f(x_n)| > n$. By Theorem 27.2.12 a subsequence converges to some c, and $c \in [a,b]$ by Proposition 27.2.5(1). Then $f(x_{n_k}) \to f(c)$, so $(f(x_{n_k}))$ is bounded, a contradiction. Let $s = \sup f$ and choose x_n with $f(x_n) > s - 1/(n+1)$; a subsequence converges to some $c \in [a,b]$, and $f(c) = \lim f(x_{n_k}) = s$ by the squeeze theorem. The minimum is the maximum of $-f$. (This is the case $X = [a,b]$ of Corollary 18.6.6.) □

Counterexample 27.4.11. Each hypothesis of Theorem 27.4.10 is needed: x on $(0,1)$ attains neither its supremum nor its infimum; $1/x$ on $(0,1]$ is unbounded; x on $[0,+\infty)$ is unbounded; and the function on $[0,1]$ equal to x for $x<1$ and to 0 at 1 has supremum 1, which is not attained.

Definition 27.4.12. A function $f: D \to \mathbb{R}$ is *uniformly continuous* if for every $\varepsilon > 0$ there is $\delta > 0$ with $|f(x) - f(y)| < \varepsilon$ for all $x, y \in D$ with $|x - y| < \delta$. It is *Lipschitz* with constant $K \geq 0$ if $|f(x) - f(y)| \leq K|x - y|$ for all $x, y \in D$.

Proposition 27.4.13. *Let $f: D \to \mathbb{R}$.*

1. *Lipschitz functions are uniformly continuous, and uniformly continuous functions are continuous.*
2. *f is uniformly continuous if and only if $f(x_n) - f(y_n) \to 0$ for all sequences (x_n), (y_n) in D with $x_n - y_n \to 0$.*
3. *(Heine–Cantor) If $D = [a,b]$ and f is continuous, then f is uniformly continuous.*
4. *A uniformly continuous function maps Cauchy sequences to Cauchy sequences. A uniformly continuous $f: (a,b) \to \mathbb{R}$ on a bounded interval extends to a continuous function on $[a,b]$.*

Proof. (1) Take $\delta = \varepsilon/K$ (any δ if $K = 0$); the second statement is clear. (2) If f is uniformly continuous and $x_n - y_n \to 0$, eventually $|x_n - y_n| < \delta$ and so $|f(x_n) - f(y_n)| < \varepsilon$. If f is not uniformly continuous, there is $\varepsilon > 0$ and, for each n, points x_n, y_n with $|x_n - y_n| < 1/(n+1)$ and $|f(x_n) - f(y_n)| \geq \varepsilon$. (3) If not, take x_n, y_n as in the last sentence. By Theorem 27.2.12, $x_{n_k} \to c \in [a,b]$, and then $y_{n_k} \to c$. By continuity $f(x_{n_k}) - f(y_{n_k}) \to f(c) - f(c) = 0$, a contradiction. (4) If (x_n) is Cauchy and δ corresponds to ε, eventually $|x_n - x_m| < \delta$, so $|f(x_n) - f(x_m)| < \varepsilon$. For the extension to a: for any sequence $x_n \to a$ in (a,b), $(f(x_n))$ is Cauchy, hence converges; by (2), two such sequences give the same limit (as $x_n - y_n \to 0$), say L. By Theorem 27.3.4, $\lim_{x \to a^+} f(x) = L$, and setting $f(a) = L$ gives a function continuous at a; similarly at b. □

Counterexample 27.4.14.

1. $f(x) = x^2$ on $\mathbb{R}$ is continuous but not uniformly continuous: with $x_n = n+1$ and $y_n = n + 1 + 1/(n+1)$, $x_n - y_n \to 0$ but $f(y_n) - f(x_n) \geq 2$.
2. $f(x) = 1/x$ on $(0,1)$ is continuous but not uniformly continuous: with $x_n = 1/(n+2)$ and $y_n = 1/(n+3)$, $f(y_n) - f(x_n) = 1$; it has no continuous extension to $[0,1)$, in accordance with Proposition 27.4.13(4).
3. $f(x) = \sqrt{x}$ on $[0,+\infty)$ is uniformly continuous, since $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x-y|}$, but not Lipschitz, since $\sqrt{x}/x \to +\infty$ as $x \to 0^+$. So the implications in (1) are strict.

27.5 Upper and lower limits, and series

Definition 27.5.1. For a real sequence (a_n) let $\limsup_n a_n = \inf_n \sup_{k \geq n} a_k$ and $\liminf_n a_n = \sup_n \inf_{k \geq n} a_k$, with values in the extended real line.

Proposition 27.5.2.

1. *A bounded monotone real sequence converges, to its supremum or infimum.*
2. *(a_n) converges to $a \in \mathbb{R}$ if and only if $\limsup a_n = \liminf a_n = a$.*
3. *(Bolzano–Weierstrass) Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.*

Proof. (1) is Proposition 27.2.10. (2) The sequences $\sup_{k\geq n} a_k$ and $\inf_{k\geq n} a_k$ are monotone and bracket a_n. If both converge to a, so does a_n; conversely, if $|a_k - a| < \varepsilon$ for $k \geq N$, both lie within ε of a for $n \geq N$. (3) A bounded set lies in a closed cube, which is compact (Theorem 18.6.5) and hence sequentially compact (Theorem 18.7.4). □

Definition 27.5.3. Let V be a normed space. A series $\sum_n v_n$ *converges* to s if the partial sums $s_N = \sum_{n\leq N} v_n$ converge to s, and it *converges absolutely* if $\sum_n \|v_n\| < \infty$.

Proposition 27.5.4. *Let V be a Banach space.*

1. *An absolutely convergent series converges, and every rearrangement $\sum_n v_{\sigma(n)}$ (σ a bijection of $\mathbb{N}$) converges to the same sum.*
2. *(Comparison) If $\|v_n\| \leq c_n$ and $\sum c_n < \infty$, then $\sum v_n$ converges absolutely.*
3. *(Root and ratio tests) If $\limsup \|v_n\|^{1/n} < 1$, or $v_n \neq 0$ and $\limsup \|v_{n+1}\|/\|v_n\| < 1$, then $\sum v_n$ converges absolutely. If $\limsup \|v_n\|^{1/n} > 1$, it diverges.*
4. *(Geometric series) For $z \in \mathbb{C}$ with $|z| < 1$, $\sum_n z^n = 1/(1-z)$.*
5. *(Weierstrass M-test) If $f_n \colon X \to V$ satisfy $\sup_x \|f_n(x)\| \leq M_n$ with $\sum M_n < \infty$, then $\sum f_n$ converges uniformly.*

Proof. (1) $\|s_M - s_N\| \leq \sum_{N<n\leq M} \|v_n\|$, so the partial sums are Cauchy. For a rearrangement, given ε choose N with $\sum_{n\geq N} \|v_n\| < \varepsilon$ and then M with $\{0,\dots,N-1\} \subseteq \sigma(\{0,\dots,M\})$; for $m \geq M$, $\|\sum_{k\leq m} v_{\sigma(k)} - s_{N-1}\| \leq \sum_{n\geq N} \|v_n\| < \varepsilon$, and $\|s_{N-1} - s\| \leq \varepsilon$. (2) follows from (1). (3) If $\limsup \|v_n\|^{1/n} < q < 1$, then $\|v_n\| \leq q^n$ for large n; compare with $\sum q^n$ using (4). If the ratio limsup is $< q < 1$, then $\|v_n\| \leq Cq^n$ for large n. If $\limsup \|v_n\|^{1/n} > 1$, then $\|v_n\| > 1$ infinitely often, so $v_n \not\to 0$ and the partial sums are not Cauchy. (4) $(1-z)\sum_{n\leq N} z^n = 1 - z^{N+1}$, and $|z|^N \to 0$: if $0 < |z| < 1$ write $1/|z| = 1 + h$, so $|z|^{-N} \geq 1 + Nh$. (5) The partial sums are bounded and uniformly Cauchy, and the space $B(X, V)$ of bounded maps is complete (Proposition 18.7.2). □

Proposition 27.5.5 (More convergence tests)**.** *Let (a_n), (b_n) be sequences of real numbers.*

1. *If $\sum a_n$ converges, then $a_n \to 0$. The converse fails: the harmonic series $\sum_{n\geq 1} 1/n$ diverges.*
2. *(Limit comparison) If $a_n, b_n > 0$ and $a_n/b_n \to L \in (0,\infty)$, then $\sum a_n$ and $\sum b_n$ both converge or both diverge. In particular this holds if $a_n \sim b_n$, that is, $a_n/b_n \to 1$.*
3. *(Cauchy condensation) If $a_1 \geq a_2 \geq \cdots \geq 0$, then $\sum_{n\geq 1} a_n$ converges if and only if $\sum_{k\geq 0} 2^k a_{2^k}$ converges. Consequently $\sum_{n\geq 1} n^{-p}$ converges if and only if $p > 1$, and $\sum_{n\geq 2} 1/(n(\log n)^p)$ converges if and only if $p > 1$.*
4. *(Ratio and root) If $a_n > 0$, then $\liminf \frac{a_{n+1}}{a_n} \leq \liminf a_n^{1/n} \leq \limsup a_n^{1/n} \leq \limsup \frac{a_{n+1}}{a_n}$; so whenever the ratio test applies, the root test does too. If $a_{n+1} \geq a_n > 0$ for large n, the series diverges. If the limit of the ratios or of the roots is 1, no conclusion is possible: $\sum 1/n$ diverges and $\sum 1/n^2$ converges, and both have limit 1.*
5. *(Dirichlet) If $b_n \in \mathbb{C}$ have bounded partial sums $B_n = \sum_{k\leq n} b_k$ and $a_n \geq a_{n+1} \to 0$, then $\sum a_n b_n$ converges. For $b_n = (-1)^n$ this is the alternating series test (Lemma 27.5.7 below).*

Proof. (1) $a_n = s_n - s_{n-1} \to s - s = 0$. For the harmonic series, $s_{2n} - s_n = \sum_{n<k\leq 2n} \frac{1}{k} \geq n \cdot \frac{1}{2n} = \frac{1}{2}$, so the partial sums are not Cauchy.

(2) A series of positive terms converges if and only if its partial sums are bounded (Proposition 27.5.2). For large n, $\frac{L}{2}b_n \le a_n \le 2Lb_n$, so the partial sums are bounded for both series or for neither.

(3) For $2^k \le n < 2^{k+1}$ there are 2^k terms, each between $a_{2^{k+1}}$ and a_{2^k}, so $\frac{1}{2}2^{k+1}a_{2^{k+1}} \le \sum_{n=2^k}^{2^{k+1}-1} a_n \le 2^k a_{2^k}$. Summing over $k \le K$ shows that the partial sums of the two series bound each other, up to a factor 2 and the term a_1. For $a_n = n^{-p}$ with $p > 0$, $2^k a_{2^k} = (2^{1-p})^k$, a geometric series converging if and only if $p > 1$; for $p \le 0$ the terms do not tend to 0. For $a_n = 1/(n(\log n)^p)$, $2^k a_{2^k} = (k\log 2)^{-p}$, which reduces to the previous case.

(4) If $\limsup a_{n+1}/a_n < q$, then $a_{n+1} \le qa_n$ for $n \ge N$, so $a_n \le Cq^n$ and $a_n^{1/n} \le C^{1/n}q \to q$; hence $\limsup a_n^{1/n} \le q$. The inequality for the $\liminf$ is proved in the same way. If a_n is eventually nondecreasing and positive, it does not tend to 0.

(5) Summation by parts: for $m \le N$, $\sum_{n=m}^{N} a_n b_n = a_N B_N - a_m B_{m-1} + \sum_{n=m}^{N-1}(a_n - a_{n+1})B_n$. With $|B_n| \le M$ and $a_n - a_{n+1} \ge 0$, the absolute value is at most $M(a_N + a_m + (a_m - a_N)) = 2Ma_m$, which tends to 0; so the partial sums are Cauchy. □

Example 27.5.6. *Combining tests.* The tests are sufficient conditions, and in practice one combines them: first replace a_n by a simpler sequence with the same behaviour (limit comparison with an asymptotic equivalent), then apply the ratio, root or condensation test to it.

1. $\sum n!/n^n$: the ratio is $(n/(n+1))^n = (1+1/n)^{-n} \to 1/e < 1$, so the series converges.

2. $\sum \sin(1/n)$ diverges and $\sum(1-\cos(1/n))$ converges, by limit comparison with $\sum 1/n$ and $\sum 1/(2n^2)$, since $\sin t/t \to 1$ and $(1-\cos t)/t^2 \to \frac{1}{2}$ as $t \to 0$.

3. $\sqrt{n+1} - \sqrt{n} = 1/(\sqrt{n+1} + \sqrt{n}) \sim 1/(2\sqrt{n})$; so $\sum(\sqrt{n+1} - \sqrt{n})$ diverges (its partial sums are $\sqrt{N+1} - 1$) and $\sum(\sqrt{n+1} - \sqrt{n})/n$ converges, by comparison with $\sum n^{-3/2}$.

4. $\sum_{n\ge1} \sin(n)/n$ converges by Dirichlet's test: from $2\sin\frac{1}{2}\sin k = \cos(k-\frac{1}{2}) - \cos(k+\frac{1}{2})$ the partial sums of $\sum \sin k$ telescope and are bounded by $1/\sin\frac{1}{2}$. It does not converge absolutely, since $|\sin n| \ge \sin^2 n = \frac{1}{2}(1-\cos 2n)$ and $\sum \cos(2n)/n$ converges by the same argument while $\sum 1/n$ diverges.

5. $\sum n^2/2^n$ converges by the root test ($n^{2/n} \to 1$), and $\sum 2^n/n^2$ diverges since its terms tend to ∞.

Lemma 27.5.7 (Alternating series)**.** *If $t_0 \ge t_1 \ge \cdots \ge 0$ and $t_k \to 0$, then $S = \sum_k (-1)^k t_k$ converges and $S_{2m+1} \le S \le S_{2m}$ for all m, where S_N are the partial sums.*

Proof. $S_{2m+2} - S_{2m} = -t_{2m+1} + t_{2m+2} \le 0$ and $S_{2m+3} - S_{2m+1} = t_{2m+2} - t_{2m+3} \ge 0$, and $S_{2m+1} = S_{2m} - t_{2m+1} \le S_{2m}$. So the even partial sums decrease, the odd ones increase, every odd one is below every even one, and their difference tends to 0. □

Rearranging series. A finite sum does not depend on the order of its terms, and by Proposition 27.5.4(1) neither does the sum of an absolutely convergent series. For series that converge but not absolutely the order matters, and Riemann discovered that it matters as much as it possibly could: by reordering the terms one can make the sum anything one likes. A series $\sum_n a_n$ of real numbers is *conditionally convergent* if it converges but $\sum_n |a_n| = \infty$; the alternating harmonic series $\sum_{n\ge1}(-1)^{n+1}/n$ is the basic example (Lemma 27.5.7 and Proposition 27.5.5(1)). Write $a^+ = \max(a, 0)$ and $a^- = \max(-a, 0)$ for the positive and negative parts of a real number, so that $a = a^+ - a^-$ and $|a| = a^+ + a^-$.

Lemma 27.5.8. *If $\sum_n a_n$ is a conditionally convergent series of real numbers, then $a_n \to 0$ and $\sum_n a_n^+ = \sum_n a_n^- = \infty$. In particular infinitely many a_n are positive and infinitely many are negative.*

Proof. $a_n \to 0$ by Proposition 27.5.5(1). If $\sum_n a_n^+$ were finite, then $\sum_n a_n^- = \sum_n (a_n^+ - a_n)$ would converge, being the difference of two convergent series, and so would $\sum_n |a_n| = \sum_n (a_n^+ + a_n^-)$, a contradiction. The same argument applies to $\sum_n a_n^-$. A series of non-negative terms with only finitely many nonzero terms is finite, so infinitely many a_n^+ and infinitely many a_n^- are nonzero. □

So the positive terms alone have infinite sum, and so do the negative terms alone; the series converges only because they are interleaved so as to cancel. This is what a rearrangement can exploit.

Theorem 27.5.9 (Riemann's rearrangement theorem)**.** *Let $\sum_n a_n$ be a conditionally convergent series of real numbers, and let $-\infty \le \alpha \le \beta \le \infty$. There is a bijection $\sigma : \mathbb{N} \to \mathbb{N}$ such that the partial sums $T_N = \sum_{n \le N} a_{\sigma(n)}$ satisfy*

$$\liminf_{N\to\infty} T_N = \alpha \qquad \text{and} \qquad \limsup_{N\to\infty} T_N = \beta.$$

In particular, for every $c \in \mathbb{R}$ some rearrangement converges to c, and some rearrangements diverge to ∞ and to $-\infty$.

Proof. Targets. Choose real numbers $\alpha_k \to \alpha$ and $\beta_k \to \beta$ with $\alpha_k < \beta_k$ and $\alpha_k < \beta_{k+1}$ for all $k \ge k_0$: put $\beta_k = \beta + \frac{1}{k}$, k or $-k$ according as β is finite, $+\infty$ or $-\infty$, and $\alpha_k = \alpha - \frac{1}{k}$, $k-1$ or $-k-2$ according as α is finite, $+\infty$ or $-\infty$. One checks the inequalities case by case for k large (recall $\alpha \le \beta$; for instance if $\alpha = \beta = -\infty$, then $\alpha_k = -k-2 < -k-1 = \beta_{k+1}$).

Construction. Let $p_1 < p_2 < \cdots$ be the indices n with $a_n \ge 0$ and $q_1 < q_2 < \cdots$ those with $a_n < 0$; both lists are infinite by Lemma 27.5.8, and together they contain every index once. We build the rearrangement in stages $k = k_0, k_0+1, \ldots$, each consisting of an *up step* and a *down step*. In the up step of stage k we append the next unused terms a_{p_j}, in order, as long as the running sum is $\le \beta_k$, stopping as soon as it exceeds β_k. In the down step we append the next unused terms a_{q_j}, in order, until the running sum is $< \alpha_k$. Each step stops after finitely many terms: the unused positive terms are all but finitely many of the a_n^+, so their sum is still infinite by Lemma 27.5.8, and likewise for the negative terms.

Every index is used. A down step adds at least one term, because it starts with a sum $> \beta_k > \alpha_k$; an up step of stage $k+1$ adds at least one term, because it starts with a sum $< \alpha_k < \beta_{k+1}$. So infinitely many p_j and infinitely many q_j are used, and since each list is used in order, all of them are. Hence the order in which the terms are appended defines a bijection $\sigma : \mathbb{N} \to \mathbb{N}$.

The partial sums. Let U_k be the running sum at the end of the up step of stage k and L_k the one at the end of the down step. If the up step added terms, then just before its last term a_{u_k} the sum was $\le \beta_k$, so

$$\beta_k < U_k \le \beta_k + a_{u_k}, \qquad \text{and similarly} \qquad \alpha_k + a_{l_k} \le L_k < \alpha_k,$$

with $a_{l_k} < 0$ the last term of the down step (for $k > k_0$ the up step always adds terms, as shown above). The indices u_k and l_k are distinct for distinct k, so they tend to ∞, and $a_{u_k}, a_{l_k} \to 0$ because $a_n \to 0$. Hence $U_k - \beta_k \to 0$ and $L_k - \alpha_k \to 0$, so $U_k \to \beta$ and $L_k \to \alpha$ (in the extended sense when α or β is infinite). Within an up step the running sum increases, and within a down step it decreases; so for N in the up step of stage k we have $L_{k-1} \le T_N \le U_k$, and for N in the down step $L_k \le T_N \le U_k$. Therefore $\limsup_N T_N \le \limsup_k U_k = \beta$ and $\liminf_N T_N \ge \liminf_k L_k = \alpha$; and the values $T_N = U_k$ and $T_N = L_k$ occur for every k, so equality holds in both. The last statement is the case $\alpha = \beta$. □

Corollary 27.5.10. *A series $\sum_n a_n$ of real numbers is absolutely convergent if and only if every rearrangement of it converges, and if and only if all its rearrangements converge to the same sum.*

Proof. If $\sum_n |a_n| < \infty$, every rearrangement converges to the same sum by Proposition 27.5.4(1). If $\sum_n |a_n| = \infty$, either the series diverges, and then the identity rearrangement does not converge, or it is conditionally convergent, and then Theorem 27.5.9 with $\alpha = \beta = \infty$ gives a rearrangement that diverges. □

Example 27.5.11. 1. Let $S = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$. By Lemma 27.5.7, $\frac{1}{2} = S_1 \le S \le S_0 = 1$, so $S \neq 0$. Take one positive term followed by two negative ones:

$$1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \frac{1}{5} - \frac{1}{10} - \frac{1}{12} + \cdots$$

Every term of S occurs exactly once: the odd denominators in order, and the even denominators in order, two at a time. The k-th group of three terms is $\frac{1}{2k-1} - \frac{1}{4k-2} - \frac{1}{4k} = \frac{1}{4k-2} - \frac{1}{4k} = \frac{1}{2}\left(\frac{1}{2k-1} - \frac{1}{2k}\right)$. So the partial sums after $3m$ terms are half the partial sums of S after $2m$ terms, and tend to $\frac{1}{2}S$; the partial sums in between differ from these by at most two terms, which tend to 0. So this rearrangement converges to $\frac{1}{2}S \neq S$.

2. The proof of the theorem is an algorithm. To rearrange S so that it converges to $c = 2$: take $1 + \frac{1}{3} + \frac{1}{5} + \cdots$ until the sum exceeds 2, which happens at $1 + \frac{1}{3} + \cdots + \frac{1}{15} \approx 2.02$ (eight terms), then subtract $\frac{1}{2}$ to get $\approx 1.52 < 2$, then add $\frac{1}{17} + \frac{1}{19} + \cdots$ until the sum exceeds 2 again, and so on. The positive terms are used much faster than the negative ones. This is how it must be: a rearrangement with a larger sum takes more positive terms per negative term, just as in (1) taking fewer positive terms per negative term gave a smaller sum.

3. For complex series the theorem fails as stated: the rearrangements of $\sum_n i(-1)^{n+1}/n$ have as sums exactly the points of the imaginary axis $i\mathbb{R}$, by the theorem applied to the real series $\sum_n (-1)^{n+1}/n$. For series in $\mathbb{R}^m$ the possible sums of rearrangements form a more complicated set, which can be a line in the plane even when the series does not lie on a line, as the example $\sum_n \left((-1)^{n+1}/n, (-1)^{n+1}/n + 1/n^2\right)$ shows: the second coordinate differs from the first by an absolutely convergent series, whose sum does not change, so every sum lies on the line $y = x + \sum_n 1/n^2$, and every point of it occurs.

Proposition 27.5.12 (Cauchy product)**.** *If $\sum a_n$ and $\sum b_n$ are absolutely convergent series in $\mathbb{C}$, then $c_n = \sum_{k=0}^{n} a_k b_{n-k}$ defines an absolutely convergent series with $\sum c_n = (\sum a_n)(\sum b_n)$.*

Proof. Let $A = \sum |a_n|$ and $B = \sum |b_n|$. Then $\sum_{n \le N} |c_n| \le AB$, so $\sum c_n$ converges absolutely. The difference $\sum_{n \le 2N} c_n - (\sum_{k \le N} a_k)(\sum_{l \le N} b_l)$ is the sum of $a_k b_l$ over $k + l \le 2N$ with $k > N$ or $l > N$, bounded by $B \sum_{k>N} |a_k| + A \sum_{l>N} |b_l| \to 0$. □

Theorem 27.5.13. *Let $a_n \in \mathbb{C}$ and $R = 1/\limsup_n |a_n|^{1/n} \in [0, \infty]$, the* radius of convergence. *The power series $\sum_n a_n z^n$ converges absolutely for $|z| < R$, uniformly on $|z| \le r$ for every $r < R$, and diverges for $|z| > R$. The series $\sum_n n a_n z^{n-1}$ and $\sum_n n^2 a_n z^n$ have the same radius of convergence.*

Proof. For $|z| \le r < R$ choose $r < q < R$; then $|a_n| \le q^{-n}$ for large n, so $|a_n z^n| \le (r/q)^n$, and the M-test applies. For $|z| > R$, $\limsup |a_n z^n|^{1/n} > 1$. Finally $n^{1/n} \to 1$: writing $n^{1/n} = 1 + h_n$, the binomial formula gives $n \ge \binom{n}{2} h_n^2$, so $h_n^2 \le 2/(n-1)$. Hence $\limsup |n^k a_n|^{1/n} = \limsup |a_n|^{1/n}$. □

27.6 Differentiation in one variable

Definition 27.6.1. Let $I \subseteq \mathbb{R}$ be an interval with more than one point and $f : I \to \mathbb{K}^m$. Then f is *differentiable* at $a \in I$ with derivative $f'(a)$ if $(f(x) - f(a))/(x - a) \to f'(a)$ as $x \to a$ in $I \setminus \{a\}$. Equivalently, there is a function $\phi : I \to \mathbb{K}^m$, continuous at a, with $f(x) - f(a) = \phi(x)(x - a)$; then $f'(a) = \phi(a)$. The same definition with I replaced by an open subset of $\mathbb{C}$ and x complex defines the *complex derivative*.

Proposition 27.6.2. *Differentiable functions are continuous. Sums, products and quotients (where the denominator does not vanish) of differentiable scalar functions are differentiable with the usual formulas, and if f is differentiable at a and g at $f(a)$, then $(g \circ f)'(a) = g'(f(a))f'(a)$.*

Proof. Write $f(x) - f(a) = \phi(x)(x - a)$ and $g(y) - g(b) = \psi(y)(y - b)$ with ϕ, ψ continuous at $a, b = f(a)$. Continuity is clear. For products, $fg(x) - fg(a) = (\phi(x)g(x) + f(a)\chi(x))(x - a)$ if $g(x) - g(a) = \chi(x)(x - a)$. For $1/f$, $1/f(x) - 1/f(a) = -\phi(x)(x - a)/(f(x)f(a))$. For composites, $g(f(x)) - g(f(a)) = \psi(f(x))\phi(x)(x - a)$ and $\psi \circ f$ is continuous at a. Evaluate the continuous factors at a. □

Definition 27.6.3. Let $f : I \to \mathbb{R}$ on an interval I. Then f has a *local maximum* at $c \in I$ if there is $\delta > 0$ with $f(x) \leq f(c)$ for all $x \in I$ with $|x - c| < \delta$, a *strict* local maximum if $f(x) < f(c)$ for these $x \neq c$, and similarly for (strict) local minima. A *local extremum* is a local maximum or minimum.

Lemma 27.6.4 (Fermat)**.** *If $f : I \to \mathbb{R}$ has a local extremum at an interior point c of I and is differentiable at c, then $f'(c) = 0$.*

Proof. Say c is a local maximum. For $x > c$ close to c, $(f(x) - f(c))/(x - c) \leq 0$, so $f'(c) \leq 0$ by Corollary 27.3.5(2); for $x < c$ the quotient is ≥ 0, so $f'(c) \geq 0$. □

Counterexample 27.6.5. The converse fails: $f(x) = x^3$ has $f'(0) = 0$ and no extremum at 0. The hypothesis that c is interior is needed: $f(x) = x$ on $[0, 1]$ has a maximum at 1 with $f'(1) = 1$. And $|x|$ has a minimum at 0, where it is not differentiable.

Theorem 27.6.6 (Mean value theorem)**.** *Let $f, g : [a, b] \to \mathbb{R}$ be continuous and differentiable on (a, b).*

1. *(Rolle) If $f(a) = f(b)$, then $f'(\xi) = 0$ for some $\xi \in (a, b)$.*
2. *There is $\xi \in (a, b)$ with $f(b) - f(a) = f'(\xi)(b - a)$, and (Cauchy) some η with $(f(b) - f(a))g'(\eta) = (g(b) - g(a))f'(\eta)$.*
3. *If $f' = 0$ on (a, b), f is constant; if $f' \geq 0$ (> 0), f is increasing (strictly increasing).*

Proof. (1) By Corollary 18.6.6, f attains a maximum and a minimum; if both are at endpoints, f is constant. Otherwise one is at some $\xi \in (a, b)$, and $f'(\xi) = 0$ by Lemma 27.6.4. (2) Apply (1) to $f(x) - \frac{f(b)-f(a)}{b-a}(x - a)$, and to $(f(b) - f(a))g(x) - (g(b) - g(a))f(x)$. (3) Apply (2) on subintervals. □

Remark 27.6.7. Both hypotheses of (3) matter. On a set that is not an interval a function with zero derivative need not be constant: $f = 0$ on $(0, 1)$ and $f = 1$ on $(2, 3)$. And "$f' = 0$" must hold at every point: the Cantor function (Chapter 28) is continuous, nondecreasing and nonconstant on $[0, 1]$, with $f' = 0$ outside a set of measure zero. On the other hand finitely many exceptional points do not matter: if f is continuous on $[a, b]$ and $f' = 0$ except at $c_1 < \cdots < c_k$, applying (3) on each interval between consecutive points and using continuity shows that f is constant.

Proposition 27.6.8. *Let $f : I \to \mathbb{R}$ be continuous and strictly monotone. Then $J = f(I)$ is an interval, $f^{-1}: J \to I$ is continuous, and if f is differentiable at a with $f'(a) \neq 0$, then f^{-1} is differentiable at $f(a)$ with derivative $1/f'(a)$.*

Proof. J is an interval by Corollary 18.5.5. A strictly monotone bijection of intervals maps subintervals onto subintervals, so f^{-1} maps open subintervals of J (relative to J) to open subintervals of I, and is continuous. Write $f(x) - f(a) = \phi(x)(x - a)$; strict monotonicity gives $\phi(x) \neq 0$ for $x \neq a$, and $\phi(a) = f'(a) \neq 0$. So $f^{-1}(y) - a = (y - f(a))/\phi(f^{-1}(y))$ with $1/(\phi \circ f^{-1})$ continuous at $f(a)$. □

Proposition 27.6.9 (Limits of derivatives)**.** *Let $f : [a, b) \to \mathbb{R}$ be continuous, differentiable on (a, b), and suppose $\lim_{x\to a^+} f'(x) = L \in \mathbb{R}$. Then f is differentiable at a (from the right) with $f'(a) = L$. Consequently, if f is differentiable on an open interval containing c and f' has one-sided limits at c, both equal $f'(c)$: a derivative has neither removable nor jump discontinuities.*

Proof. For $a < x < b$, the mean value theorem gives $\xi_x \in (a, x)$ with $(f(x) - f(a))/(x - a) = f'(\xi_x)$. If $x_n \to a^+$, then $\xi_{x_n} \to a^+$ by the squeeze theorem, so $f'(\xi_{x_n}) \to L$; by Theorem 27.3.4, the difference quotient tends to L. For the second statement, apply this on each side of c. □

Theorem 27.6.10 (de l'Hôpital)**.** *Let $f, g : (a, b) \to \mathbb{R}$ be differentiable, with $-\infty \leq a < b \leq +\infty$, and $g'(x) \neq 0$ for all x. Suppose that $\lim_{x\to a^+} f'(x)/g'(x) = L \in \bar{\mathbb{R}}$ and that either*

1. *$\lim_{x\to a^+} f(x) = \lim_{x\to a^+} g(x) = 0$, or*
2. *$\lim_{x\to a^+} |g(x)| = +\infty$ and $L \in \mathbb{R}$.*

Then $\lim_{x\to a^+} f(x)/g(x) = L$. The same holds for left limits at b, and hence for two-sided limits.

Proof. Since $g' \neq 0$, g is injective on (a, b) by Rolle's theorem, so $g(x) \neq g(y)$ for $x \neq y$, and by the Cauchy mean value theorem (Theorem 27.6.6(2)), for $x < y$ in (a, b) there is $\xi \in (x, y)$ with

$$\frac{f(x) - f(y)}{g(x) - g(y)} = \frac{f'(\xi)}{g'(\xi)}. \tag{27.1}$$

Case (1) with $a \in \mathbb{R}$: extend f and g by $f(a) = g(a) = 0$; they become continuous on $[a, b)$, and (27.1) holds for $x = a$ as well, with $g(y) \neq 0$ for $y > a$ by Rolle's theorem. Let $y_n \to a^+$ and choose $\xi_n \in (a, y_n)$ with $f(y_n)/g(y_n) = f'(\xi_n)/g'(\xi_n)$. Then $\xi_n \to a^+$, so $f(y_n)/g(y_n) \to L$, and Theorem 27.3.4 gives the claim. If $a = -\infty$, apply the case just proved to $F(t) = f(-1/t)$ and $G(t) = g(-1/t)$ on an interval $(0, \eta)$, noting $F'(t)/G'(t) = f'(-1/t)/g'(-1/t)$ and using Proposition 27.3.6.

Case (2). Let $\varepsilon > 0$ and choose $c \in (a, b)$ with $|f'(\xi)/g'(\xi) - L| < \varepsilon$ for $a < \xi < c$. Fix $y \in (a, c)$. For $a < x < y$, (27.1) gives $Q(x) = (f(x) - f(y))/(g(x) - g(y))$ with $|Q(x) - L| < \varepsilon$. Since $|g(x)| \to +\infty$, $g(x) \neq 0$ for x close to a, and

$$\frac{f(x)}{g(x)} - L = \big(Q(x) - L\big)\Big(1 - \frac{g(y)}{g(x)}\Big) - L\frac{g(y)}{g(x)} + \frac{f(y)}{g(x)}.$$

As $x \to a^+$ with y fixed, the last two terms tend to 0 and $1 - g(y)/g(x) \to 1$, so $|f(x)/g(x) - L| < 2\varepsilon$ for x close to a. Left limits at b follow by applying the result to $x \mapsto f(-x)$, $g(-x)$. □

Theorem 27.6.11 (Darboux)**.** *Let $f : [a, b] \to \mathbb{R}$ be differentiable, with one-sided derivatives at the endpoints. Then f' takes every value between $f'(a)$ and $f'(b)$.*

Proof. Let $f'(a) < y < f'(b)$ (the other case follows by passing to $-f$) and $g(x) = f(x) - yx$. Then $g'(a) < 0 < g'(b)$. By Theorem 27.4.10, g attains a minimum at some $c \in [a, b]$. As $g'(a) < 0$, $(g(x) - g(a))/(x - a) < 0$ for $x > a$ close to a (Proposition 27.3.3(2)), so $g(x) < g(a)$ and $c \neq a$; similarly $c \neq b$. By Lemma 27.6.4, $g'(c) = 0$, that is, $f'(c) = y$. □

Theorem 27.6.12 (Taylor). *Let $f : [a, x] \to \mathbb{R}$ (or $[x, a]$) have n continuous derivatives, with $f^{(n+1)}$ existing on the open interval. Then there is ξ strictly between a and x with*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1}.$$

Proof. Let $g(t) = f(x) - \sum_{k\le n} f^{(k)}(t)(x-t)^k/k! - M(x-t)^{n+1}/(n+1)!$, with M chosen so that $g(a) = 0$. Also $g(x) = 0$. In the derivative the sum telescopes: $g'(t) = -f^{(n+1)}(t)(x-t)^n/n! + M(x-t)^n/n!$. By Rolle's theorem $g'(\xi) = 0$ for some ξ between a and x, so $M = f^{(n+1)}(\xi)$. □

Proposition 27.6.13 (Taylor's formula with Peano remainder). *Let $n \ge 1$, let $f : I \to \mathbb{R}$ be $n-1$ times differentiable on an interval I, and suppose $f^{(n)}(a)$ exists at a point $a \in I$. Let $T_n(x) = \sum_{k=0}^{n} f^{(k)}(a)(x-a)^k/k!$. Then*

$$\lim_{x\to a} \frac{f(x) - T_n(x)}{(x-a)^n} = 0,$$

that is, $f(x) = T_n(x) + o((x-a)^n)$ as $x \to a$.

Proof. Let $R = f - T_n$. Then R is $n-1$ times differentiable, $R^{(k)}(a) = 0$ for $0 \le k \le n$, and each $R^{(k)}$ with $k \le n-1$ is continuous at a. For $n = 1$ the claim is $(R(x)-R(a))/(x-a) \to R'(a) = 0$. For $n \ge 2$, apply Theorem 27.6.10(1) $n-1$ times, on each side of a, to the quotients $R^{(j)}(x)/(x-a)^{n-j}$, $0 \le j \le n-2$, whose numerators and denominators tend to 0 and whose denominators have nonvanishing derivatives for $x \ne a$. It remains to see that the last quotient tends to 0:

$$\frac{R^{(n-1)}(x)}{n!\,(x-a)} = \frac{1}{n!} \cdot \frac{R^{(n-1)}(x) - R^{(n-1)}(a)}{x-a} \longrightarrow \frac{R^{(n)}(a)}{n!} = 0.$$

□

Corollary 27.6.14 (Second derivative test). *Let f be differentiable on an open interval containing c, with $f'(c) = 0$, and suppose $f''(c)$ exists. If $f''(c) > 0$, then f has a strict local minimum at c; if $f''(c) < 0$, a strict local maximum.*

Proof. By Proposition 27.6.13 with $n = 2$, $(f(x)-f(c))/(x-c)^2 \to f''(c)/2 > 0$, so $f(x)-f(c) > 0$ for $x \ne c$ close to c (Proposition 27.3.3(2)). □

Counterexample 27.6.15. If $f''(c) = 0$ the test says nothing: x^4, $-x^4$ and x^3 all have vanishing first and second derivatives at 0, and have a minimum, a maximum and no extremum there.

Definition 27.6.16. A function $f : I \to \mathbb{R}$ on an interval is *convex* if $f((1-t)x + ty) \le (1-t)f(x) + tf(y)$ for all $x, y \in I$ and $t \in [0, 1]$, and *strictly convex* if the inequality is strict whenever $x \ne y$ and $0 < t < 1$. It is *(strictly) concave* if $-f$ is (strictly) convex.

Lemma 27.6.17 (Three slopes). *A function $f : I \to \mathbb{R}$ is convex if and only if for all $x < y < z$ in I,*

$$\frac{f(y) - f(x)}{y - x} \le \frac{f(z) - f(x)}{z - x} \le \frac{f(z) - f(y)}{z - y}.$$

Proof. Write $y = (1-t)x + tz$ with $t = (y-x)/(z-x) \in (0, 1)$. Convexity at this triple says $(z-x)f(y) \le (z-y)f(x) + (y-x)f(z)$. Multiplying out, each of the two displayed inequalities is equivalent to this one. Since every $t \in (0, 1)$ and $x < z$ arise, and the cases $t \in \{0, 1\}$ or $x = y$ are trivial, the lemma follows (for $x > y$ exchange the roles). □

Proposition 27.6.18. *Let $f : I \to \mathbb{R}$ on an open interval I.*

1. *If f is convex, it is continuous, it has one-sided derivatives $f'_-(x) \le f'_+(x)$ at every point, and $f'_+(x) \le \frac{f(y)-f(x)}{y-x} \le f'_-(y)$ for $x < y$.*
2. *If f is differentiable, the following are equivalent: f is convex; f' is increasing; $f(y) \ge f(x) + f'(x)(y-x)$ for all $x, y \in I$.*
3. *If f is twice differentiable, f is convex if and only if $f'' \ge 0$; if $f'' > 0$, f is strictly convex.*
4. *(Jensen) If f is convex, $x_1, \dots, x_k \in I$, $t_i \ge 0$ and $\sum t_i = 1$, then $f(\sum t_i x_i) \le \sum t_i f(x_i)$.*

Proof. (1) By Lemma 27.6.17, the slope $s(x,y) = (f(y) - f(x))/(y-x)$ is increasing in each variable. For fixed x, $s(x,y)$ decreases as $y \downarrow x$ and is bounded below by $s(w,x)$ for any $w < x$ in I (which exists, I being open); so $f'_+(x) = \lim_{y \to x^+} s(x,y)$ exists by Proposition 27.3.9, and $f'_+(x) \ge s(w,x)$. Letting $w \uparrow x$ gives $f'_-(x)$, with $f'_-(x) \le f'_+(x)$. The inequality for $x < y$ follows from the monotonicity of slopes. The existence of one-sided derivatives implies continuity, as $f(y) - f(x) = s(x,y)(y-x) \to 0$. (2) If f is convex and differentiable, (1) gives $f'(x) = f'_+(x) \le f'_-(y) = f'(y)$ for $x < y$. If f' is increasing and $y > x$, the mean value theorem gives $f(y) - f(x) = f'(\xi)(y-x) \ge f'(x)(y-x)$; for $y < x$, $f'(\xi) \le f'(x)$ and $y - x < 0$ give the same inequality. Finally, if the tangent inequality holds, let $z = (1-t)x + ty$; adding $(1-t)$ times $f(x) \ge f(z) + f'(z)(x-z)$ and t times $f(y) \ge f(z) + f'(z)(y-z)$ gives $(1-t)f(x) + tf(y) \ge f(z)$. (3) $f'' \ge 0$ if and only if f' is increasing (Theorem 27.6.6(3) and, conversely, f'' = lim of nonnegative quotients); apply (2). If $f'' > 0$, f' is strictly increasing and the inequalities in the proof of (2) become strict for $y \ne x$, hence so does the final one for $x \ne y$, $0 < t < 1$. (4) Induction on k: for $t_k < 1$, $\sum t_i x_i = (1-t_k) \sum_{i<k} \frac{t_i}{1-t_k} x_i + t_k x_k$, and the inner point lies in I because I is an interval. □

Counterexample 27.6.19. A convex function need not be differentiable: $|x|$ has $f'_-(0) = -1 < 1 = f'_+(0)$. On a closed interval a convex function need not be continuous at the endpoints: the function equal to 0 on $[0,1)$ and to 1 at 1 is convex. The converse of the last statement of (3) fails: x^4 is strictly convex, but its second derivative vanishes at 0.

Theorem 27.6.20. *Let $f(z) = \sum_n a_n z^n$ have radius of convergence $R > 0$. Then f is complex differentiable on $\{|z| < R\}$ with $f'(z) = \sum_n n a_n z^{n-1}$. Hence f has complex derivatives of all orders there, and $a_n = f^{(n)}(0)/n!$. The same holds for real z in $(-R, R)$.*

Proof. Let $g(z) = \sum n a_n z^{n-1}$, which has radius R by Theorem 27.5.13. Fix $|z| < \rho < R$ and $h \ne 0$ with $|z| + |h| \le \rho$. By the binomial formula, $(z+h)^n - z^n - nz^{n-1}h = \sum_{k \ge 2} \binom{n}{k} z^{n-k} h^k$, and $\binom{n}{k} \le \binom{n}{2}\binom{n-2}{k-2}$ for $k \ge 2$, so its absolute value is at most $\binom{n}{2}|h|^2(|z| + |h|)^{n-2} \le n^2 |h|^2 \rho^{n-2}$. Hence $|(f(z+h) - f(z))/h - g(z)| \le |h| \sum_n n^2 |a_n| \rho^{n-2}$, and the last series converges. Iterating gives all derivatives, and evaluating at 0 gives a_n. □

27.7 The exponential and trigonometric functions

Definition 27.7.1. For $z \in \mathbb{C}$ let $\exp(z) = e^z = \sum_{n \ge 0} z^n/n!$ (radius of convergence ∞, by the ratio test), and for $x \in \mathbb{R}$ let $\cos x = \operatorname{Re} e^{ix}$ and $\sin x = \operatorname{Im} e^{ix}$, so that $\cos x = \sum_k (-1)^k x^{2k}/(2k)!$ and $\sin x = \sum_k (-1)^k x^{2k+1}/(2k+1)!$.

Proposition 27.7.2. *1. $e^{z+w} = e^z e^w$, $e^z \ne 0$, $\overline{e^z} = e^{\bar z}$, and $\exp' = \exp$.*

2. *$\exp$ restricts to a strictly increasing bijection $\mathbb{R} \to (0, \infty)$, whose inverse $\log$ is differentiable with $\log'(x) = 1/x$ and $\log(xy) = \log x + \log y$.*
3. *$\cos^2 x + \sin^2 x = 1$, $\cos' = -\sin$, $\sin' = \cos$, and $\cos(x+y) = \cos x \cos y - \sin x \sin y$, $\sin(x+y) = \sin x \cos y + \cos x \sin y$.*

Proof. (1) By Proposition 27.5.12 and the binomial formula, $e^z e^w = \sum_n \sum_k \frac{z^k w^{n-k}}{k!(n-k)!} = \sum_n (z+w)^n/n!$. So $e^z e^{-z} = 1$. Conjugation is continuous and commutes with the partial sums. The derivative is Theorem 27.6.20. (2) For real x, $e^x = (e^{x/2})^2 > 0$, so exp is strictly increasing by Theorem 27.6.6. For $x \geq 0$, $e^x \geq 1 + x$, and $e^{-x} = 1/e^x$; so exp is unbounded above and takes values arbitrarily close to 0, and by the intermediate value theorem its image is $(0, \infty)$. Apply Proposition 27.6.8. (3) $|e^{ix}|^2 = e^{ix}\overline{e^{ix}} = e^{ix}e^{-ix} = 1$; $(e^{ix})' = ie^{ix}$ by Theorem 27.6.20 applied to the real power series $\sum_n i^n x^n/n!$; the addition formulas are the real and imaginary parts of $e^{i(x+y)} = e^{ix}e^{iy}$. □

Theorem 27.7.3. *The function* cos *has a unique zero in* $(0, 2)$*; call it* $\pi/2$*. Then* $e^{i\pi/2} = i$*, the map* $t \mapsto e^{it}$ *is a surjective continuous homomorphism* $\mathbb{R} \to S^1 = \{|z| = 1\}$ *with kernel* $2\pi\mathbb{Z}$*, and* $p(t) = e^{2\pi i t}$ *is a covering map* $\mathbb{R} \to S^1$*. Every* $z \neq 0$ *is* $re^{i\theta}$ *with* $r = |z|$ *and* θ *unique modulo* 2π*.*

Proof. For $0 < x \leq 2$ the terms $x^{2k+1}/(2k+1)!$ decrease, since consecutive ratios are $x^2/((2k+2)(2k+3)) < 1$; by Lemma 27.5.7, $\sin x \geq x - x^3/6 > 0$. So cos is strictly decreasing on $[0, 2]$. The terms $4^k/(2k)!$ of $\cos 2$ decrease from $k = 1$ on, so $\cos 2 \leq 1 - 2 + 16/24 < 0$, while $\cos 0 = 1$; by the intermediate value theorem there is a unique zero $\pi/2 \in (0, 2)$. Then $\sin(\pi/2) = 1$, so $e^{i\pi/2} = i$ and $e^{2\pi i} = i^4 = 1$.

Kernel: if $0 < t < 2\pi$, write $e^{it/4} = u + iv$ with $u, v > 0$ (as $0 < t/4 < \pi/2$). If $e^{it} = (u+iv)^4 = 1$, then $\operatorname{Im}(u+iv)^4 = 4uv(u^2 - v^2) = 0$ forces $u^2 = v^2 = 1/2$, and then $\operatorname{Re}(u+iv)^4 = u^4 - 6u^2v^2 + v^4 = -1$, a contradiction. With periodicity, the kernel is $2\pi\mathbb{Z}$.

Surjectivity: for $a + ib \in S^1$ with $a, b \geq 0$, choose $t \in [0, \pi/2]$ with $\cos t = a$ (intermediate value theorem); then $\sin t = \sqrt{1 - a^2} = b$. The other quadrants are obtained by multiplying by powers of $i = e^{i\pi/2}$. For the covering property, p is injective on $[a, a+1)$ and identifies only a and $a + 1$ in $[a, a+1]$, so the argument of Proposition 19.4.5 applies verbatim. The polar form follows by applying this to $z/|z|$. □

Remark 27.7.4. The proof of Theorem 19.4.6 uses only that $\mathbb{R}$ is simply connected and that the covering map has period 4 with fibres the translates of $4\mathbb{Z}$; it applies verbatim to $t \mapsto e^{2\pi i t}$, with period 1, so $\pi_1(S^1) \cong \mathbb{Z}$ is generated by the loop $t \mapsto e^{2\pi i t}$. For $x > 0$ and $s \in \mathbb{C}$ we define $x^s = e^{s \log x}$; then $\frac{d}{dx} x^s = s x^{s-1}$.

Proposition 27.7.5 (Notable limits)**.**

1. $\lim_{x\to 0} \frac{e^x - 1}{x} = 1$, $\lim_{x\to 0} \frac{\log(1+x)}{x} = 1$, $\lim_{x\to 0} \frac{\sin x}{x} = 1$ *and* $\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}$.
2. *For every* $x \in \mathbb{R}$, $(1 + x/n)^n \to e^x$*; in particular* $(1 + 1/n)^n \to e = \exp(1)$.
3. *For* $k \in \mathbb{N}$ *and* $a > 0$*:* $x^k e^{-ax} \to 0$ *and* $(\log x)/x^a \to 0$ *as* $x \to +\infty$*, and* $x^a \log x \to 0$ *as* $x \to 0^+$.

Proof. (1) The first three are the derivatives of exp at 0, of log at 1 and of sin at 0 (Proposition 27.7.2), which are $e^0 = 1$, $1/1$ and $\cos 0 = 1$. For the last, apply Theorem 27.6.10(1): $\frac{1-\cos x}{x^2}$ has the same limit as $\frac{\sin x}{2x} \to \frac{1}{2}$. (2) For $x = 0$ this is clear. Otherwise, for $n > |x|$, $1 + x/n > 0$ and $(1 + x/n)^n = \exp(n \log(1 + x/n))$, with $n \log(1 + x/n) = x \cdot \frac{\log(1+x/n)}{x/n} \to x$ by (1) and Theorem 27.3.4 (as $x/n \to 0$ and $x/n \neq 0$). By continuity of exp, $(1+x/n)^n \to e^x$. (3) For $x > 0$, $e^{ax} \geq (ax)^{k+1}/(k+1)!$, one term of the exponential series, so $0 < x^k e^{-ax} \leq (k+1)!/(a^{k+1}x) \to 0$. For $x = e^t$, $(\log x)/x^a = te^{-at}$, which tends to 0 as $t \to +\infty$ by the case $k = 1$; since $\log x \to +\infty$ as $x \to +\infty$, Proposition 27.3.6 applies. Similarly $x^a \log x = -te^{-at}$ for $x = e^{-t}$, and $t \to +\infty$ as $x \to 0^+$. □

Counterexample 27.7.6. 1. $\sin(1/x)$ has no limit at 0: along $x_n = 1/(2\pi n + \pi/2)$ it is 1, along $y_n = 1/(\pi n)$ it is 0 ($n \geq 1$), and Theorem 27.3.4 applies. Since $x_n - y_n \to 0$,

$\sin(1/x)$ is continuous but not uniformly continuous on $(0,1)$. On the other hand $x\sin(1/x) \to 0$ by the squeeze theorem, as $|x\sin(1/x)| \le |x|$.

2. A derivative need not be continuous. Let $f(x) = x^2\sin(1/x)$ for $x \neq 0$ and $f(0) = 0$. Then $f'(0) = \lim_{x\to 0} x\sin(1/x) = 0$, and $f'(x) = 2x\sin(1/x) - \cos(1/x)$ for $x \neq 0$, which has no limit at 0. In accordance with Proposition 27.6.9 and Theorem 27.6.11, the discontinuity of f' at 0 is essential.

3. The converse of de l'Hôpital's rule fails. With f as in (2) and $g(x) = x$, $f(x)/g(x) = x\sin(1/x) \to 0$, while $f'(x)/g'(x) = 2x\sin(1/x) - \cos(1/x)$ has no limit as $x \to 0$.

Corollary 27.7.7 (Inequality of arithmetic and geometric means)**.** *For $x_1, \dots, x_k > 0$ and weights $t_i \ge 0$ with $\sum t_i = 1$, $\prod_i x_i^{t_i} \le \sum_i t_i x_i$. In particular $\sqrt[k]{x_1\cdots x_k} \le \frac{1}{k}(x_1 + \cdots + x_k)$, with equality if and only if all x_i are equal.*

Proof. $\log'' = -1/x^2 < 0$, so log is concave (Proposition 27.6.18(3)), and by Jensen's inequality $\sum t_i \log x_i \le \log \sum t_i x_i$. Apply exp, which is increasing. For the equality statement, let $m = \frac{1}{k}\sum x_i$. As $\log'$ is strictly decreasing, the mean value theorem gives $\log x < \log m + (x-m)/m$ for every $x > 0$ with $x \neq m$ (the tangent inequality of Proposition 27.6.18(2) for $-\log$, made strict). Averaging over $x = x_1, \dots, x_k$ gives $\frac{1}{k}\sum \log x_i < \log m$ unless all $x_i = m$. □

Lemma 27.7.8. *The function $\beta(t) = e^{-1/t}$ for $t > 0$ and $\beta(t) = 0$ for $t \le 0$ has derivatives of all orders on $\mathbb{R}$, all vanishing at 0. The function $\gamma(t) = \beta(t)/(\beta(t) + \beta(1-t))$ has derivatives of all orders, takes values in $[0,1]$, and equals 0 for $t \le 0$ and 1 for $t \ge 1$.*

Proof. By induction, on $t > 0$, $\beta^{(k)}(t) = p_k(1/t)e^{-1/t}$ for a polynomial p_k. Since $e^s \ge s^m/m!$ for $s > 0$, $s^j e^{-s} \to 0$ as $s \to \infty$; so $\beta^{(k)}(t) \to 0$ as $t \to 0^+$, and the difference quotients of $\beta^{(k)}$ at 0 tend to 0, which by induction shows $\beta^{(k+1)}(0) = 0$. The denominator of γ never vanishes (one of t, $1-t$ is positive), so γ has derivatives of all orders by Proposition 27.6.2, and the other properties are clear. □

Definition 27.7.9. A function $f : U \to \mathbb{R}$ on an open set $U \subseteq \mathbb{R}$ is *(real) analytic* if every $a \in U$ has a neighbourhood on which $f(x) = \sum_n c_n(x-a)^n$ for a convergent power series.

Proposition 27.7.10. *1. If $f(x) = \sum_n a_n x^n$ has radius of convergence $R > 0$ and $|b| < R$, then $f(b+h) = \sum_k \frac{f^{(k)}(b)}{k!}h^k$ for $|h| < R - |b|$. So f is analytic on $(-R, R)$.*

2. *An analytic function is smooth, and near each point it equals its Taylor series there.*

3. *(Identity theorem) Let f be analytic on an open interval I. If all derivatives of f vanish at one point, or if the zeros of f have an accumulation point in I, then $f = 0$ on I.*

Proof. (1) $\sum_n \sum_{k\le n} |a_n|\binom{n}{k}|b|^{n-k}|h|^k = \sum_n |a_n|(|b|+|h|)^n < \infty$, so the double series $\sum_{n,k} a_n\binom{n}{k}b^{n-k}h^k$ converges absolutely and may be summed in any order (Proposition 27.5.4(1)): summing over k first gives $f(b+h)$, and summing over n first gives $\sum_k c_k h^k$ with $c_k = \sum_{n\ge k} a_n\binom{n}{k}b^{n-k}$, which is $f^{(k)}(b)/k!$ by Theorem 27.6.20. (2) follows from Theorem 27.6.20 applied to the local expansions.

(3) Let S be the set of points of I where all derivatives of f vanish. It is closed in I, as the derivatives are continuous, and open: near $x \in S$, f equals its Taylor series at x, which is 0, so $f = 0$ near x. As I is connected (Theorem 18.5.3), $S = \emptyset$ or $S = I$. If the zeros accumulate at $a \in I$ and $f(x) = \sum c_n(x-a)^n$ near a with some $c_n \neq 0$, let m be least with $c_m \neq 0$; then $f(x) = (x-a)^m g(x)$ with g continuous and $g(a) = c_m \neq 0$, so f has no zeros near a other than a, a contradiction. So all derivatives vanish at a, and $S = I$. □

Polynomials, exp, sin and cos are analytic on $\mathbb{R}$, by (1), with radius ∞. The function $1/(1+x^2)$ is smooth on $\mathbb{R}$, but its Taylor series at 0, $\sum(-1)^n x^{2n}$, has radius only 1: the obstruction is invisible on $\mathbb{R}$, and comes from the complex points $\pm i$ where $1/(1+z^2)$ is not defined; it is nevertheless analytic on all of $\mathbb{R}$, as holomorphic functions are analytic (Chapter 29). The identity theorem fails for smooth functions:

Counterexample 27.7.11. The function β of Lemma 27.7.8 has all derivatives zero at 0, so its Taylor series at 0 is zero, but $\beta(t) \neq 0$ for $t > 0$. Smooth functions need not be given by their Taylor series; complex differentiable functions are (Chapter 29).

27.8 The Riemann integral

Definition 27.8.1. Let $f\colon [a,b] \to \mathbb{R}$ be bounded. For a partition $P\colon a = t_0 < \cdots < t_k = b$ let $U(f,P) = \sum_i (t_i - t_{i-1}) \sup_{[t_{i-1},t_i]} f$ and $L(f,P)$ the same with inf. The function is *Riemann integrable* if $\sup_P L(f,P) = \inf_P U(f,P)$, and this common value is $\int_a^b f = \int_a^b f(t)\,dt$. A $\mathbb{C}^m$-valued function is integrable if its real and imaginary components are, with the integral taken componentwise. We set $\int_b^a f = -\int_a^b f$.

Proposition 27.8.2. *1. If P' refines P, then $L(f,P) \le L(f,P') \le U(f,P') \le U(f,P)$; and $L(f,P) \le U(f,Q)$ for all P, Q. So f is integrable if and only if for every $\varepsilon > 0$ there is P with $U(f,P) - L(f,P) < \varepsilon$.*

2. Continuous functions are integrable, and $\sum_i (t_i - t_{i-1}) f(s_i) \to \int_a^b f$ for any choice of $s_i \in [t_{i-1}, t_i]$ as $\max_i (t_i - t_{i-1}) \to 0$.

3. For continuous f, g: the integral is linear, $f \le g$ implies $\int f \le \int g$, $\int_a^c f = \int_a^b f + \int_b^c f$, $|\int_a^b f| \le \int_a^b |f|$ for $\mathbb{C}^m$-valued f with the Euclidean norm, and if $f_n \to f$ uniformly, $\int f_n \to \int f$.

Proof. (1) Adding a point to a partition can only increase infima and decrease suprema on the pieces; for P, Q compare both with $P \cup Q$. (2) By uniform continuity (Corollary 18.7.5), for $\varepsilon > 0$ there is δ such that f varies by less than ε on intervals of length $< \delta$; for partitions of mesh $< \delta$, $U - L \le \varepsilon(b-a)$, and Riemann sums lie between L and U. (3) Linearity and additivity follow from (2), monotonicity from $L(f,P) \le L(g,P)$, and uniform limits from $|\int (f_n - f)| \le (b-a) \sup |f_n - f|$. For the norm inequality let $v = \int f \in \mathbb{C}^m$; then $\|v\|^2 = \operatorname{Re}\langle \int f, v\rangle = \int \operatorname{Re}\langle f(t), v\rangle\,dt \le \int \|f\|\,\|v\|$ by Cauchy–Schwarz (Proposition 11.10.2). □

Theorem 27.8.3 (Fundamental theorem of calculus)**.** *Let $f\colon [a,b] \to \mathbb{C}^m$ be continuous. Then $F(x) = \int_a^x f$ is differentiable with $F' = f$. If $G\colon [a,b] \to \mathbb{C}^m$ is continuously differentiable, then $\int_a^b G' = G(b) - G(a)$.*

Proof. $|(F(x+h) - F(x))/h - f(x)| = |\frac{1}{h}\int_x^{x+h} (f - f(x))| \le \sup_{|t-x|\le|h|} |f(t) - f(x)| \to 0$. For the second statement, $\int_a^x G' - G(x)$ has derivative 0, hence is constant by Theorem 27.6.6 applied to real and imaginary components. □

Corollary 27.8.4. *For continuously differentiable f, g and $\varphi\colon [c,d] \to [a,b]$: $\int_a^b f'g = [fg]_a^b - \int_a^b fg'$ and $\int_c^d f(\varphi(s))\varphi'(s)\,ds = \int_{\varphi(c)}^{\varphi(d)} f$ for continuous f. Moreover $\|G(b) - G(a)\| \le (b-a) \sup \|G'\|$ for G continuously differentiable.*

Proof. Apply Theorem 27.8.3 to fg, to $F \circ \varphi$ with $F(x) = \int_a^x f$, and use the norm inequality of Proposition 27.8.2. □

The first formula of the corollary is *integration by parts*: it moves the derivative from one factor to the other, at the price of a boundary term. It is useful whenever differentiating one factor simplifies it while integrating the other does not make it worse.

Example 27.8.5.

1. With $f(x) = e^x$ and $g(x) = x$, so that $f'g = xe^x$: $\int_0^1 xe^x\,dx = [xe^x]_0^1 - \int_0^1 e^x\,dx = e - (e-1) = 1$.
2. With $f(x) = x$ and $g = \log$, so that $f'g = \log x$ and $fg' = 1$: $\int_1^e \log x\,dx = [x\log x]_1^e - \int_1^e 1\,dx = e - (e-1) = 1$.
3. *Wallis' integrals.* Let $I_n = \int_0^{\pi/2} \sin^n x\,dx$, so $I_0 = \frac{\pi}{2}$ and $I_1 = 1$. For $n \geq 2$ take $f(x) = -\cos x$ and $g(x) = \sin^{n-1} x$: the boundary term $[-\cos x \sin^{n-1} x]_0^{\pi/2}$ vanishes, and
$$I_n = (n-1)\int_0^{\pi/2} \cos^2 x \sin^{n-2} x\,dx = (n-1)(I_{n-2} - I_n), \qquad \text{so} \qquad I_n = \frac{n-1}{n} I_{n-2}.$$
Hence $I_{2m} = \frac{1\cdot 3\cdots(2m-1)}{2\cdot 4\cdots(2m)} \cdot \frac{\pi}{2}$ and $I_{2m+1} = \frac{2\cdot 4\cdots(2m)}{3\cdot 5\cdots(2m+1)}$.
4. *Periodic functions.* If f and g are continuously differentiable and 2π-periodic, the boundary term $[fg]_0^{2\pi}$ vanishes, and $\int_0^{2\pi} f'g = -\int_0^{2\pi} fg'$: without boundary, the derivative can be moved from one factor to the other at the cost of a sign. In several variables this becomes $\int (\partial_i f)g = -\int f\,\partial_i g$ for compactly supported functions (Chapter 28), and on a region with boundary the boundary term becomes an integral over the boundary (Green's identities, Chapter 40).

Proposition 27.8.6 (Mean value theorem for integrals)**.** *Let $f, g\colon [a,b] \to \mathbb{R}$ be continuous with $g \geq 0$. Then there is $c \in [a,b]$ with $\int_a^b fg = f(c)\int_a^b g$. In particular $\int_a^b f = f(c)(b-a)$ for some c: the average value of a continuous function is attained.*

Proof. Let m, M be the minimum and maximum of f (Corollary 18.6.6). Then $mg \leq fg \leq Mg$, so $m\int g \leq \int fg \leq M\int g$. If $\int g = 0$ any c works; otherwise $\int fg / \int g \in [m,M]$ is a value of f by the intermediate value theorem (Corollary 18.5.5). □

The hypotheses matter: for $g(x) = x$ on $[-1,1]$ and $f = g$, $\int fg = \frac{2}{3}$ while $\int g = 0$; and for vector-valued f the statement fails, since $f(t) = e^{it}$ on $[0, 2\pi]$ has integral 0 but never vanishes.

Example 27.8.7. *Computing integrals.* The fundamental theorem of calculus reduces integration to finding antiderivatives, and the rules of Corollary 27.8.4 are the main tools.

1. *Substitution.* With $u = 1 + x^2$, $du = 2x\,dx$: $\int_0^1 \frac{x}{1+x^2}\,dx = \frac{1}{2}\int_1^2 \frac{du}{u} = \frac{1}{2}\log 2$.
2. *Partial fractions.* $\frac{1}{x^2-1} = \frac{1}{2}\left(\frac{1}{x-1} - \frac{1}{x+1}\right)$, so $\int_2^3 \frac{dx}{x^2-1} = \frac{1}{2}\left[\log\frac{x-1}{x+1}\right]_2^3 = \frac{1}{2}\log\frac{3}{2}$. In general every rational function is a polynomial plus a combination of terms $(x-a)^{-k}$ and $(Ax+B)((x-p)^2+q^2)^{-k}$, whose antiderivatives involve only rational functions, log and arctan.
3. *Trigonometric substitution.* With $x = \sin\theta$: $\int_{-1}^1 \sqrt{1-x^2}\,dx = \int_{-\pi/2}^{\pi/2} \cos^2\theta\,d\theta = \frac{\pi}{2}$, the area of a half disc; here $\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$.

4. *Symmetry.* $\int_{-a}^{a} f = 0$ for odd f, as the substitution $x \mapsto -x$ shows; for instance $\int_{-1}^{1} x^3 e^{x^2}\,dx = 0$ without computing an antiderivative.

5. *Recursion by parts*, as for Wallis' integrals in Example 27.8.5: $\int_0^1 x^n e^x\,dx = e - n\int_0^1 x^{n-1}e^x\,dx$.

Many integrals have no elementary antiderivative, such as $\int e^{-x^2}\,dx$; they are computed by other means, such as polar coordinates for $\int_{\mathbb{R}} e^{-x^2} = \sqrt{\pi}$ and differentiation with respect to a parameter (Chapter 28).

Theorem 27.8.8 (Stirling's formula)**.** *$n! \sim \sqrt{2\pi n}\,(n/e)^n$, that is, $n!\,e^n/(n^{n+1/2}\sqrt{2\pi}) \to 1$ as $n \to \infty$.*

Proof. Step 1: $a_n = n!\,e^n/n^{n+1/2}$ converges to some $C > 0$. For $t \geq 0$, $t - \frac{t^2}{2} \leq \log(1+t) \leq t - \frac{t^2}{2} + \frac{t^3}{3}$: the differences vanish at 0 and have derivatives $\frac{t^2}{1+t} \geq 0$ and $\frac{t^3}{1+t} \geq 0$. Now $\log(a_n/a_{n+1}) = (n+\frac{1}{2})\log(1+\frac{1}{n}) - 1$, and with $t = 1/n$ the two bounds give $-\frac{1}{4n^2} \leq \log(a_n/a_{n+1}) \leq \frac{1}{12n^2} + \frac{1}{6n^3} \leq \frac{1}{4n^2}$. So $\sum_n \log(a_n/a_{n+1})$ converges absolutely, $\log a_n$ converges, and $a_n \to C > 0$.

Step 2: $C = \sqrt{2\pi}$. With Wallis' integrals I_k (Example 27.8.5), $\sin^{2m+1} \leq \sin^{2m} \leq \sin^{2m-1}$ on $[0, \frac{\pi}{2}]$ gives $I_{2m+1} \leq I_{2m} \leq I_{2m-1} = \frac{2m+1}{2m} I_{2m+1}$, so $I_{2m}/I_{2m+1} \to 1$. Using $2\cdot 4\cdots 2m = 2^m m!$ and $1\cdot 3\cdots(2m-1) = (2m)!/(2^m m!)$, this says

$$\frac{\pi}{2} = \lim_{m\to\infty} \frac{2^{4m}(m!)^4}{((2m)!)^2(2m+1)}$$

(Wallis' product). Substituting $m! = a_m m^{m+1/2}e^{-m}$ and $(2m)! = a_{2m}(2m)^{2m+1/2}e^{-2m}$, the right side is $\frac{a_m^4}{a_{2m}^2}\cdot\frac{m}{2(2m+1)} \to \frac{C^2}{4}$. Hence $C^2 = 2\pi$. □

For example $10! = 3628800$, while $\sqrt{20\pi}(10/e)^{10} \approx 3598696$, an error of less than 1%; the relative error is about $1/(12n)$, in accordance with the bound in Step 1.

Proposition 27.8.9. *Let $f\colon [c,d]\times[a,b] \to \mathbb{C}$ be continuous.*

1. *$x \mapsto \int_a^b f(x,t)\,dt$ is continuous, and if $\partial_x f$ exists and is continuous on $[c,d]\times[a,b]$, it is differentiable with derivative $\int_a^b \partial_x f(x,t)\,dt$.*

2. *(Fubini for continuous functions) $\int_c^d \int_a^b f(x,t)\,dt\,dx = \int_a^b \int_c^d f(x,t)\,dx\,dt$.*

Proof. (1) Continuity follows from uniform continuity of f. For the derivative, by the mean value theorem applied to real and imaginary parts, $(f(x+h,t) - f(x,t))/h - \partial_x f(x,t)$ is bounded by the oscillation of $\partial_x f$ over $|s-x| \leq |h|$, which tends to 0 uniformly in t by uniform continuity. (2) Let $\Phi(y) = \int_c^y \int_a^b f\,dt\,dx$ and $\Psi(y) = \int_a^b \int_c^y f\,dx\,dt$. By Theorem 27.8.3 and (1) (applied to $g(y,t) = \int_c^y f(x,t)\,dx$, with $\partial_y g = f$ continuous), $\Phi' = \Psi' = \int_a^b f(y,t)\,dt$, and $\Phi(c) = \Psi(c) = 0$. □

27.9 Sequences of functions

Definition 27.9.1. Let D be a set and $f_n, f\colon D \to \mathbb{K}^m$. Then $f_n \to f$ *pointwise* if $f_n(x) \to f(x)$ for every $x \in D$, and *uniformly* if $\|f_n - f\|_\infty = \sup_{x\in D}\|f_n(x) - f(x)\| \to 0$.

The difference is the order of quantifiers: for pointwise convergence the threshold N may depend on x, for uniform convergence it may not. Uniform convergence implies pointwise convergence.

Proposition 27.9.2. *1. (Cauchy criterion) (f_n) converges uniformly on D if and only if for every $\varepsilon > 0$ there is N with $\|f_n - f_m\|_\infty < \varepsilon$ for $m, n \geq N$.*

2. If D is a metric space, the f_n are continuous and $f_n \to f$ uniformly, then f is continuous.

3. If $f_n \to f$ uniformly on $[a, b]$ and the f_n are continuous, then $\int_a^b f_n \to \int_a^b f$.

Proof. (1) If $f_n \to f$ uniformly, use the triangle inequality. Conversely, for each x the sequence $(f_n(x))$ is Cauchy, hence converges to some $f(x)$ (Theorem 27.2.13, componentwise). Letting $m \to \infty$ in $\|f_n(x) - f_m(x)\| < \varepsilon$ gives $\|f_n(x) - f(x)\| \leq \varepsilon$ for $n \geq N$ and all x. (2) is Proposition 18.3.10. (3) is Proposition 27.8.2(3). □

Counterexample 27.9.3. 1. $f_n(x) = x^n$ on $[0, 1]$ converges pointwise to the function equal to 0 on $[0, 1)$ and 1 at 1, which is discontinuous; so the convergence is not uniform. On $[0, 1)$ the limit 0 is continuous, but $\sup_{[0,1)} x^n = 1$, so the convergence is still not uniform. On $[0, r]$ with $r < 1$ it is uniform, as $\sup = r^n \to 0$.

2. Pointwise convergence does not commute with integration. Let $f_n(x) = nx(1 - x^2)^n$ on $[0, 1]$. For $0 < x \leq 1$, $f_n(x) \to 0$ by Proposition 27.2.9(3) with $a = (1-x^2)^{-1}$ (or trivially if $x = 1$), and $f_n(0) = 0$. But, substituting $u = 1 - x^2$, $\int_0^1 f_n = \frac{n}{2} \int_0^1 u^n \, du = \frac{n}{2(n+1)} \to \frac{1}{2} \neq 0$.

3. Uniform convergence does not commute with differentiation. $f_n(x) = x^n/n$ converges uniformly to 0 on $[0, 1]$, since $|f_n| \leq 1/n$, but $f_n'(1) = 1$ for all n.

Theorem 27.9.4. *Let $f_n : [a, b] \to \mathbb{R}$ be continuously differentiable. Suppose that (f_n') converges uniformly to g and that $(f_n(x_0))$ converges for some $x_0 \in [a, b]$. Then (f_n) converges uniformly to a continuously differentiable f with $f' = g$.*

Proof. The function g is continuous by Proposition 27.9.2(2). Let $L = \lim f_n(x_0)$ and $f(x) = L + \int_{x_0}^x g$, which is continuously differentiable with $f' = g$ by Theorem 27.8.3. By the same theorem $f_n(x) = f_n(x_0) + \int_{x_0}^x f_n'$, so $|f_n(x) - f(x)| \leq |f_n(x_0) - L| + (b - a)\|f_n' - g\|_\infty$, which tends to 0 uniformly in x. □

Theorem 27.9.5 (Dini)**.** *Let K be a compact metric space and $f_n : K \to \mathbb{R}$ continuous functions with $f_1 \leq f_2 \leq \cdots$, converging pointwise to a continuous function f. Then the convergence is uniform.*

Proof. Let $g_n = f - f_n$, continuous, ≥ 0, decreasing in n and tending to 0 pointwise. For $\varepsilon > 0$ the sets $K_n = \{x : g_n(x) \geq \varepsilon\}$ are closed, decreasing, and have empty intersection. By compactness some K_N is empty (the complements form an open cover with a finite subcover, and the K_n decrease), and then $0 \leq g_n < \varepsilon$ on K for $n \geq N$. □

Counterexample 27.9.6. Each hypothesis of Dini's theorem is needed. On the noncompact $[0, 1)$, $f_n(x) = -x^n$ increases to the continuous function 0, not uniformly. On $[0, 1]$, $f_n(x) = -x^n$ increases to a discontinuous limit, not uniformly. And without monotonicity, the continuous functions $f_n(x) = \max(0, 1 - |nx - 2|)$, bumps of height 1 moving towards 0, converge pointwise to 0 on $[0, 1]$ but not uniformly.

Corollary 27.9.7 (Series of functions)**.** *Let $u_n : [a, b] \to \mathbb{R}$ be continuous and suppose $\sum_n \|u_n\|_\infty < \infty$. Then $\sum_n u_n$ converges uniformly to a continuous function, and $\int_a^b \sum_n u_n = \sum_n \int_a^b u_n$. If moreover the u_n are continuously differentiable and $\sum_n \|u_n'\|_\infty < \infty$, then $\sum_n u_n$ is continuously differentiable with derivative $\sum_n u_n'$.*

Proof. Apply the Weierstrass M-test (Proposition 27.5.4(5)) and Propositions 27.9.2(2),(3) to the partial sums, and Theorem 27.9.4 to the partial sums and their derivatives. □

Theorem 27.9.8 (A continuous nowhere differentiable function)**.** *Let $d(x)$ be the distance from $x \in \mathbb{R}$ to the nearest integer. The* Takagi function $T(x) = \sum_{m \geq 0} 2^{-m} d(2^m x)$ *is continuous on $\mathbb{R}$ and differentiable at no point.*

Proof. Since $0 \leq d \leq \frac{1}{2}$ and d is continuous (it is 1-Lipschitz), the series converges uniformly by the M-test, and T is continuous.

We first note: if f is differentiable at x and $u_n \leq x \leq v_n$ with $u_n < v_n$ and $v_n - u_n \to 0$, then $\frac{f(v_n)-f(u_n)}{v_n-u_n} \to f'(x)$. Indeed, with $\lambda_n = (v_n - x)/(v_n - u_n) \in [0,1]$,

$$\frac{f(v_n) - f(u_n)}{v_n - u_n} = \lambda_n \frac{f(v_n) - f(x)}{v_n - x} + (1 - \lambda_n) \frac{f(x) - f(u_n)}{x - u_n},$$

where a term with weight 0 (when $v_n = x$ or $u_n = x$) is omitted; both quotients tend to $f'(x)$, and so does their convex combination.

Fix x, and for $n \geq 1$ let $u_n = k_n/2^n \leq x < v_n = (k_n + 1)/2^n$ with $k_n \in \mathbb{Z}$. For $m \geq n$, $2^m u_n$ and $2^m v_n$ are integers, so $d(2^m u_n) = d(2^m v_n) = 0$. For $m < n$, the numbers $2^m u_n$ and $2^m v_n$ are consecutive multiples of $2^{m-n} \leq \frac{1}{2}$, so they lie in a common interval $[\frac{j}{2}, \frac{j+1}{2}]$, on which d is affine with slope ± 1. Hence

$$q_n = \frac{T(v_n) - T(u_n)}{v_n - u_n} = \sum_{m=0}^{n-1} 2^{-m} \frac{d(2^m v_n) - d(2^m u_n)}{v_n - u_n} = \sum_{m=0}^{n-1} \varepsilon_{m,n}, \qquad \varepsilon_{m,n} \in \{1, -1\}.$$

So q_n is an integer of the same parity as n, and (q_n) cannot converge: consecutive terms differ by an odd integer, so $|q_{n+1} - q_n| \geq 1$. By the preliminary remark, T is not differentiable at x. □

27.10 Derivatives in several variables

Lemma 27.10.1. *All norms on a finite-dimensional real or complex vector space are equivalent (they define the same topology), and every linear map $A \colon V \to W$ between finite-dimensional normed spaces is continuous, with finite* operator norm $\|A\| = \sup_{\|v\| \leq 1} \|Av\|$*. Then $\|Av\| \leq \|A\| \, \|v\|$ and $\|AB\| \leq \|A\| \, \|B\|$.*

Proof. Identify V with $\mathbb{R}^n$ by a basis and let $\|\cdot\|$ be a norm. By the triangle inequality and Cauchy–Schwarz, $\|x\| \leq \sum |x_i| \, \|e_i\| \leq C\|x\|_2$, so $\|\cdot\|$ is continuous for the Euclidean topology. On the compact unit sphere of $\|\cdot\|_2$ it attains a minimum $c > 0$, so $c\|x\|_2 \leq \|x\| \leq C\|x\|_2$, and Lemma 18.3.4 applies. A linear map is Lipschitz for Euclidean norms, since $\|Ax\| \leq \sum |x_i| \, \|Ae_i\|$. The inequalities follow from the definition. □

Definition 27.10.2. Let $U \subseteq \mathbb{R}^n$ be open and $f \colon U \to \mathbb{R}^m$. Then f is *differentiable* at $a \in U$ if there is a linear map $Df(a) \colon \mathbb{R}^n \to \mathbb{R}^m$ with

$$f(a + h) = f(a) + Df(a)h + r(h), \qquad \|r(h)\|/\|h\| \to 0 \text{ as } h \to 0.$$

The *partial derivatives* are $\partial_j f(a) = \frac{d}{dt} f(a + te_j)|_{t=0}$. The map f is C^1 if it is differentiable everywhere and $Df \colon U \to \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m)$ is continuous; C^k if Df is C^{k-1}; and *smooth* (C^∞) if it is C^k for all k. A *C^k-diffeomorphism* is a bijection between open sets which is C^k with C^k inverse.

Proposition 27.10.3. *1. $Df(a)$ is unique, f is continuous at a, and the matrix of $Df(a)$ is the* Jacobian matrix $(\partial_j f_i(a))$.

2. *(Chain rule) If f is differentiable at a and g at $f(a)$, then $g \circ f$ is differentiable at a and $D(g \circ f)(a) = Dg(f(a)) \circ Df(a)$.*

3. *If all partial derivatives of f exist near a and are continuous at a, then f is differentiable at a. So f is C^1 if and only if its partial derivatives exist and are continuous, and C^k if and only if all partial derivatives of order at most k exist and are continuous.*

4. *(Mean value inequality) If f is C^1 and the segment $[x, y]$ lies in U, then $\|f(y) - f(x)\| \leq \sup_{z\in[x,y]} \|Df(z)\| \, \|y - x\|$.*

Proof. (1) If A and B both work, $\|(A - B)h\|/\|h\| \to 0$, so $A = B$ by linearity. Taking $h = te_j$ gives the partial derivatives. (2) Let $b = f(a)$, $k = f(a + h) - b = Df(a)h + r_f(h)$. Then $g(f(a + h)) - g(b) - Dg(b)Df(a)h = Dg(b)r_f(h) + r_g(k)$. For small h, $\|k\| \leq (\|Df(a)\| + 1)\|h\|$, so both terms are $o(\|h\|)$. (3) We may assume $m = 1$. Write $f(a + h) - f(a) = \sum_j [f(a + h_1e_1 + \cdots + h_je_j) - f(a + h_1e_1 + \cdots + h_{j-1}e_{j-1})]$. By the mean value theorem the j-th term is $h_j\partial_j f(\xi_j)$ with ξ_j within $\|h\|$ of a, and $\partial_j f(\xi_j) \to \partial_j f(a)$. The statement about C^k follows by induction. (4) Let $g(t) = f(x + t(y - x))$; by the chain rule $g'(t) = Df(x + t(y - x))(y - x)$, and Corollary 27.8.4 applies. □

Counterexample 27.10.4. Let $f(x, y) = xy/(x^2 + y^2)$ for $(x, y) \neq 0$ and $f(0) = 0$. Both partial derivatives exist everywhere (f vanishes on the axes), but $f(t, t) = 1/2$ for $t \neq 0$, so f is not continuous at 0, let alone differentiable.

Theorem 27.10.5 (Schwarz)**.** *If $f : U \to \mathbb{R}$ is C^2, then $\partial_i\partial_j f = \partial_j\partial_i f$.*

Proof. Restricting to the plane of the i-th and j-th coordinates, we may assume $n = 2$. For a rectangle $R = [x_0, x_1] \times [y_0, y_1] \subseteq U$, Theorem 27.8.3 applied twice gives $\int_{y_0}^{y_1}\int_{x_0}^{x_1} \partial_x\partial_y f \, dx \, dy = f(x_1, y_1) - f(x_0, y_1) - f(x_1, y_0) + f(x_0, y_0)$, and the same for $\int_{x_0}^{x_1}\int_{y_0}^{y_1} \partial_y\partial_x f \, dy \, dx$. By Proposition 27.8.9(2), the continuous function $g = \partial_x\partial_y f - \partial_y\partial_x f$ has integral 0 over every small rectangle. If $g(p) > 0$, then $g > g(p)/2$ on a small rectangle around p, and its integral is positive, a contradiction. □

Counterexample 27.10.6. For $f(x, y) = xy(x^2 - y^2)/(x^2 + y^2)$, $f(0) = 0$, one computes $\partial_x f(0, y) = -y$ and $\partial_y f(x, 0) = x$, so $\partial_y\partial_x f(0) = -1 \neq 1 = \partial_x\partial_y f(0)$. The second partial derivatives exist but are not continuous at 0.

Theorem 27.10.7 (Taylor's theorem in several variables)**.** *Let $f : U \to \mathbb{R}$ be C^{k+1} and $[a, a + h] \subseteq U$. With multi-index notation $\partial^\alpha = \partial_1^{\alpha_1} \cdots \partial_n^{\alpha_n}$, $h^\alpha = h_1^{\alpha_1} \cdots h_n^{\alpha_n}$ and $\alpha! = \alpha_1! \cdots \alpha_n!$, there is $\xi \in (a, a + h)$ with*

$$f(a + h) = \sum_{|\alpha|\leq k} \frac{\partial^\alpha f(a)}{\alpha!} h^\alpha + \sum_{|\alpha|=k+1} \frac{\partial^\alpha f(\xi)}{\alpha!} h^\alpha.$$

Proof. Apply Theorem 27.6.12 to $g(t) = f(a + th)$ on $[0, 1]$. By the chain rule and Theorem 27.10.5, $g^{(j)}(t) = \sum_{|\alpha|=j} \frac{j!}{\alpha!} \partial^\alpha f(a + th)h^\alpha$ (the multinomial formula). □

Corollary 27.10.8. *For $0 < r < R$ there is a smooth function $\mathbb{R}^n \to [0, 1]$ equal to 1 on the closed ball of radius r and to 0 outside the open ball of radius R.*

Proof. Take $x \mapsto 1 - \gamma((\|x\|^2 - r^2)/(R^2 - r^2))$ with γ as in Lemma 27.7.8; it is smooth by the chain rule, since $\|x\|^2$ is a polynomial. □

27.11 The inverse and implicit function theorems

Lemma 27.11.1. *If $A \in \mathrm{End}(\mathbb{R}^n)$ has $\|A\| < 1$, then $1 - A$ is invertible with inverse $\sum_k A^k$ and $\|(1-A)^{-1}\| \le 1/(1-\|A\|)$. The group $\mathrm{GL}_n(\mathbb{R})$ is open in $M_n(\mathbb{R})$, and inversion $\mathrm{GL}_n(\mathbb{R}) \to \mathrm{GL}_n(\mathbb{R})$ is smooth.*

Proof. The series converges absolutely in the complete space $M_n(\mathbb{R})$ (Proposition 27.5.4), and $(1-A)\sum_{k\le N} A^k = 1 - A^{N+1} \to 1$. The determinant is a polynomial in the entries, hence continuous, and $\mathrm{GL}_n(\mathbb{R}) = \det^{-1}(\mathbb{R} \setminus \{0\})$. By the adjugate formula (Proposition 11.5.3), the entries of A^{-1} are polynomials in the entries of A divided by $\det A$, so inversion is smooth. □

Theorem 27.11.2 (Inverse function theorem)**.** *Let $U \subseteq \mathbb{R}^n$ be open, $f : U \to \mathbb{R}^n$ of class C^k with $k \ge 1$, and $a \in U$ with $Df(a)$ invertible. There are open sets $V \ni a$ and $W \ni f(a)$ such that $f : V \to W$ is a C^k-diffeomorphism, and $D(f^{-1})(f(x)) = Df(x)^{-1}$.*

Proof. Replacing f by $x \mapsto Df(a)^{-1}(f(a+x) - f(a))$, we may assume $a = 0$, $f(0) = 0$ and $Df(0) = 1$. Let $g(x) = x - f(x)$, so $Dg(0) = 0$; choose $r > 0$ with $\overline{B}(0,r) \subseteq U$ and $\|Dg(x)\| \le 1/2$ there. By the mean value inequality, $\|g(x) - g(x')\| \le \frac{1}{2}\|x - x'\|$ on $\overline{B}(0,r)$, and $\|g(x)\| \le r/2$.

For $y \in W = B(0, r/2)$, the map $T_y(x) = y + g(x)$ sends $\overline{B}(0,r)$ into $B(0,r)$ and is a contraction with constant $1/2$. By the Banach fixed point theorem (Theorem 18.7.6) there is a unique $x \in \overline{B}(0,r)$ with $x = y + g(x)$, that is, $f(x) = y$, and $x \in B(0,r)$. Let $V = B(0,r) \cap f^{-1}(W)$, open; then $f : V \to W$ is bijective. From $x - x' = f(x) - f(x') + g(x) - g(x')$ we get $\|x - x'\| \le 2\|f(x) - f(x')\|$, so f^{-1} is continuous. For $x \in V$, $Df(x) = 1 - Dg(x)$ is invertible with $\|Df(x)^{-1}\| \le 2$ (Lemma 27.11.1).

Let $y = f(x)$ and $y + \kappa = f(x+h)$ with $x, x+h \in V$, so $\|h\| \le 2\|\kappa\|$. Then $f^{-1}(y+\kappa) - f^{-1}(y) - Df(x)^{-1}\kappa = -Df(x)^{-1}(f(x+h) - f(x) - Df(x)h)$, whose norm is at most $2\|r(h)\| = o(\|\kappa\|)$. So f^{-1} is differentiable with $D(f^{-1}) = \iota \circ Df \circ f^{-1}$, where ι is inversion. If f^{-1} is C^j with $j < k$, this composite is C^j, so f^{-1} is C^{j+1}; by induction f^{-1} is C^k (composites of C^j maps are C^j by the chain rule and induction). □

Theorem 27.11.3 (Implicit function theorem)**.** *Let $U \subseteq \mathbb{R}^n \times \mathbb{R}^m$ be open, $F : U \to \mathbb{R}^m$ of class C^k, $(a,b) \in U$ with $F(a,b) = 0$, and suppose the partial derivative $D_y F(a,b) : \mathbb{R}^m \to \mathbb{R}^m$ is invertible. There are open sets $A \ni a$ and $B \ni b$ and a C^k map $g : A \to B$ such that for $(x,y) \in A \times B$: $F(x,y) = 0$ if and only if $y = g(x)$.*

Proof. The map $\Phi(x,y) = (x, F(x,y))$ has invertible derivative at (a,b), so by Theorem 27.11.2 it restricts to a C^k-diffeomorphism of an open set, which we may shrink to $A' \times B$, onto an open set containing $(a,0)$. Its inverse has the form $(x,z) \mapsto (x, \psi(x,z))$. Let $A = \{x : (x,0) \in \Phi(A' \times B)\}$, open, and $g(x) = \psi(x,0)$. □

Theorem 27.11.4 (Local submersions and immersions)**.** *Let $f : U \to \mathbb{R}^m$ be C^k, $U \subseteq \mathbb{R}^n$ open, $a \in U$.*

1. *If $Df(a)$ is surjective, there is a C^k-diffeomorphism ψ from a neighbourhood of a onto an open subset of $\mathbb{R}^n$ with $f \circ \psi^{-1}(x_1, \dots, x_n) = (x_1, \dots, x_m)$.*
2. *If $Df(a)$ is injective, there is a C^k-diffeomorphism ϕ from a neighbourhood of $f(a)$ onto an open subset of $\mathbb{R}^m$ with $\phi \circ f(x) = (x_1, \dots, x_n, 0, \dots, 0)$ near a.*

Proof. We may assume $a = 0$ and $f(0) = 0$. (1) After permuting coordinates, the first m columns of the Jacobian are linearly independent. Then $\psi(x) = (f(x), x_{m+1}, \dots, x_n)$ has invertible derivative at 0, and $f \circ \psi^{-1}$ is the projection. (2) After permuting target coordinates, the first n rows of the Jacobian are independent. Let $\Theta(x,y) = f(x) + (0,y)$ for $x \in U$, $y \in \mathbb{R}^{m-n}$; $D\Theta(0)$ is invertible, and $\phi = \Theta^{-1}$ satisfies $\phi(f(x)) = (x,0)$. □

Remark 27.11.5. More generally, if Df has constant rank r near a, there are local diffeomorphisms in source and target in which f becomes $(x_1, \dots, x_n) \mapsto (x_1, \dots, x_r, 0, \dots, 0)$ (the *constant rank theorem* [Lee13, Theorem 4.12]). All results of this section hold with the same proofs over $\mathbb{C}$ for complex differentiable (holomorphic) maps, once holomorphic functions are known to be C^∞ (Chapter 31).

27.12 Extrema in several variables, Lagrange multipliers and the Karush–Kuhn–Tucker conditions

In one variable, Fermat's lemma and the second derivative test (Lemma 27.6.4, Corollary 27.6.14) locate the extrema of a function. This section does the same in several variables, first for functions on open sets, then for functions restricted to a set defined by equations (Lagrange multipliers), and finally to a set defined by equations and inequalities (the Karush–Kuhn–Tucker conditions), which is the basic situation of optimisation theory. The tools are the chain rule, Taylor's formula and the implicit function theorem of the previous sections, together with one piece of convex geometry, Farkas' lemma.

27.12.1 Gradient and Hessian

Let $U \subseteq \mathbb{R}^n$ be open and $f : U \to \mathbb{R}$ differentiable at a. The *gradient* of f at a is the vector $\nabla f(a) = (\partial_1 f(a), \dots, \partial_n f(a)) \in \mathbb{R}^n$; by Proposition 27.10.3(1) it represents the derivative through the standard inner product,

$$Df(a)h = \langle \nabla f(a), h \rangle \qquad (h \in \mathbb{R}^n).$$

If $\nabla f(a) \neq 0$, the Cauchy–Schwarz inequality shows that among unit vectors h the *directional derivative* $\langle \nabla f(a), h \rangle$ is largest for $h = \nabla f(a)/\|\nabla f(a)\|$: the gradient points in the direction of steepest increase, and its length is the rate of increase in that direction. It is orthogonal to the level set through a, in the sense that $\langle \nabla f(a), \gamma'(0) \rangle = (f \circ \gamma)'(0) = 0$ for every differentiable curve γ with $\gamma(0) = a$ on which f is constant.

If f is C^2, its *Hessian* at a is the matrix $D^2 f(a) = (\partial_i \partial_j f(a))_{i,j}$, which is symmetric by Schwarz's theorem (Theorem 27.10.5). We write $D^2 f(a)(h,h) = h^T D^2 f(a) h = \sum_{i,j} \partial_i \partial_j f(a) h_i h_j$ for the associated quadratic form. With this notation Taylor's theorem (Theorem 27.10.7) for $k = 1$ reads

$$f(a+h) = f(a) + \langle \nabla f(a), h \rangle + \tfrac{1}{2} D^2 f(\xi)(h,h) \qquad \text{for some } \xi \in (a, a+h), \tag{27.2}$$

since $\sum_{|\alpha|=2} \frac{\partial^\alpha f}{\alpha!} h^\alpha = \frac{1}{2} \sum_{i,j} \partial_i \partial_j f \, h_i h_j$ (a diagonal term $\alpha = 2e_i$ has $\alpha! = 2$ and occurs once, an off-diagonal pair $\{i,j\}$ has $\alpha! = 1$ and occurs twice in the double sum). Along a line, if $g(t) = f(a + th)$, the chain rule gives

$$g'(t) = \langle \nabla f(a+th), h \rangle, \qquad g''(t) = D^2 f(a+th)(h,h). \tag{27.3}$$

A real symmetric matrix H, or its quadratic form, is *positive definite* if $h^T H h > 0$ for all $h \neq 0$, *positive semidefinite* if $h^T H h \geq 0$ for all h, *negative (semi)definite* if $-H$ is positive (semi)definite, and *indefinite* if $h^T H h$ takes both signs. By the spectral theorem (Theorem 11.10.6), $H = Q D Q^T$ with Q orthogonal, and $h^T H h = \sum_i \lambda_i y_i^2$ with $y = Q^T h$: so H is positive definite if and only if all its eigenvalues are positive, positive semidefinite if and only if they are all ≥ 0, and indefinite if and only if it has eigenvalues of both signs.

Definition 27.12.1. Let $S \subseteq \mathbb{R}^n$ and $f : S \to \mathbb{R}$. Then f has a *local minimum* at $a \in S$ if there is $\delta > 0$ with $f(x) \geq f(a)$ for all $x \in S$ with $\|x - a\| < \delta$, and a *strict local minimum* if moreover $f(x) > f(a)$ for these $x \neq a$; similarly for (strict) local maxima. A *global minimum* on S is a point a with $f(x) \geq f(a)$ for all $x \in S$. When S is given by equations or inequalities and f is the restriction of a function defined on a larger open set, one speaks of a *constrained* extremum. A point a of an open set U where f is differentiable is a *critical point* of f if $\nabla f(a) = 0$.

Proposition 27.12.2. *Let $U \subseteq \mathbb{R}^n$ be open, $f : U \to \mathbb{R}$ of class C^2, and $a \in U$.*

1. *If f has a local minimum at a, then $\nabla f(a) = 0$ and $D^2 f(a)$ is positive semidefinite.*
2. *If $\nabla f(a) = 0$ and $D^2 f(a)$ is positive definite, then f has a strict local minimum at a.*
3. *If $\nabla f(a) = 0$ and $D^2 f(a)$ is indefinite, then f has no local extremum at a (a* saddle point*).*

The same holds for maxima with "negative" in place of "positive".

Proof. (1) Let $h \in \mathbb{R}^n$ and $g(t) = f(a + th)$, defined and C^2 on an interval around 0, with a local minimum at 0. By Lemma 27.6.4 and (27.3), $0 = g'(0) = \langle \nabla f(a), h \rangle$; as h is arbitrary, $\nabla f(a) = 0$. If $g''(0) = D^2 f(a)(h, h)$ were negative, g would have a strict local maximum at 0 by Corollary 27.6.14, so $g(t) < g(0) \leq g(t)$ for small $t \neq 0$, which is absurd. Hence $D^2 f(a)(h, h) \geq 0$.

(2) The unit sphere $\{\|u\| = 1\}$ is compact (Theorem 18.6.5) and $u \mapsto D^2 f(a)(u, u)$ is continuous and positive on it, so it has a minimum $m > 0$ (Corollary 18.6.6). For $\|u\| = 1$, $|D^2 f(\xi)(u, u) - D^2 f(a)(u, u)| \leq \sum_{i,j} |\partial_i \partial_j f(\xi) - \partial_i \partial_j f(a)|$, which is $< m/2$ for $\|\xi - a\| < \delta$ with δ small, by continuity of the second derivatives. Then for $0 < \|h\| < \delta$ (and $\overline{B}(a, \delta) \subseteq U$), (27.2) with $\nabla f(a) = 0$ gives

$$f(a + h) - f(a) = \tfrac{1}{2}\|h\|^2 D^2 f(\xi)\left(\tfrac{h}{\|h\|}, \tfrac{h}{\|h\|}\right) \geq \tfrac{m}{4}\|h\|^2 > 0.$$

(3) By (1), applied to f and to $-f$, a local minimum forces $D^2 f(a)$ positive semidefinite and a local maximum forces it negative semidefinite; an indefinite form is neither. □

Example 27.12.3. For $n = 2$ write $D^2 f(a) = \left(\begin{smallmatrix} A & B \\ B & C \end{smallmatrix}\right)$. Completing the square, $Ah_1^2 + 2Bh_1h_2 + Ch_2^2 = A(h_1 + \frac{B}{A}h_2)^2 + \frac{AC - B^2}{A}h_2^2$ when $A \neq 0$. So the Hessian is positive definite if and only if $A > 0$ and $AC - B^2 > 0$, negative definite if and only if $A < 0$ and $AC - B^2 > 0$, and indefinite if $AC - B^2 < 0$ (the determinant is the product of the two eigenvalues). For instance:

1. $f(x, y) = x^2 + xy + y^2 - 3x$ has the single critical point $(2, -1)$, with $A = C = 2$, $B = 1$, $AC - B^2 = 3$: a strict local minimum, in fact global, as f minus its value there is a positive definite quadratic form in $(x - 2, y + 1)$.
2. $f(x, y) = x^3 - 3x + y^2$ has critical points $(\pm 1, 0)$, with $D^2 f = \operatorname{diag}(6x, 2)$: a strict local minimum at $(1, 0)$ and a saddle point at $(-1, 0)$. The local minimum is not global, since $f(x, 0) \to -\infty$ as $x \to -\infty$.
3. $x^2 + y^4$, $x^2 - y^4$ and $x^2 + y^3$ all have the critical point 0 with the semidefinite Hessian $\operatorname{diag}(2, 0)$, and they have a minimum, a saddle and a saddle there: when the Hessian is degenerate the test says nothing, as in Counterexample 27.6.15.

Counterexample 27.12.4. Checking all lines through a is not enough. Let $f(x, y) = (y - x^2)(y - 3x^2)$. On every line through 0 the restriction of f has a strict local minimum at 0: on the x-axis $f = 3x^4$, and on $y = mx$ we get $f = m^2x^2 - 4mx^3 + 3x^4$, whose second derivative at 0 is $2m^2 > 0$ for $m \neq 0$ (and on the y-axis $f = y^2$). But $f < 0$ between the two parabolas, at $(x, 2x^2)$, arbitrarily close to 0; so 0 is not a local minimum. Here $D^2 f(0) = \operatorname{diag}(0, 2)$ is only semidefinite, as Proposition 27.12.2(2) requires.

27.12.2 Lagrange multipliers

Suppose we want the extrema of $f(x,y)$ on the curve $M = \{h(x,y) = 0\}$. Walking along M, the value of f stops increasing at a point a where M does not cross the level curve of f through a but touches it. There the two curves have the same tangent line, so their normals $\nabla f(a)$ and $\nabla h(a)$ are parallel: $\nabla f(a) = \lambda \nabla h(a)$ for some number λ, the *Lagrange multiplier*. In this way a constrained problem in n unknowns becomes an unconstrained system of $n+1$ equations ($\nabla f = \lambda \nabla h$ and $h = 0$) in $n+1$ unknowns. To prove this we need curves inside the constraint set with a prescribed velocity; they are provided by the submersion theorem.

Lemma 27.12.5. *Let $U \subseteq \mathbb{R}^n$ be open, $h = (h_1, \dots, h_q) \colon U \to \mathbb{R}^q$ of class C^k with $k \geq 1$, and $a \in U$ with $h(a) = 0$ and $Dh(a)$ surjective, that is, with $\nabla h_1(a), \dots, \nabla h_q(a)$ linearly independent. For every $d \in \ker Dh(a)$ there are $\varepsilon > 0$ and a C^k curve $x \colon (-\varepsilon, \varepsilon) \to U$ with $x(0) = a$, $x'(0) = d$ and $h(x(t)) = 0$ for all t.*

Proof. By Theorem 27.11.4(1) there is a C^k-diffeomorphism ψ from an open $V \ni a$ onto an open $W \subseteq \mathbb{R}^n$ with $h \circ \psi^{-1} = \pi$, where $\pi(z) = (z_1, \dots, z_q)$. (If $q = 0$ there is nothing to prove: take $x(t) = a + td$.) Thus $h = \pi \circ \psi$ on V, so $\pi(\psi(a)) = h(a) = 0$ and, by the chain rule, $\pi(D\psi(a)d) = Dh(a)d = 0$. Put $v = D\psi(a)d$ and $x(t) = \psi^{-1}(\psi(a) + tv)$, defined for $|t| < \varepsilon$ since W is open. Then $h(x(t)) = \pi(\psi(a)) + t\pi(v) = 0$, and by the chain rule and Theorem 27.11.2, $x'(0) = D\psi(a)^{-1}v = d$. □

Theorem 27.12.6 (Lagrange multipliers)**.** *Let $U \subseteq \mathbb{R}^n$ be open, $f \colon U \to \mathbb{R}$ and $h = (h_1, \dots, h_q) \colon U \to \mathbb{R}^q$ of class C^1, and $M = \{x \in U : h(x) = 0\}$. Let $a \in M$ be a point where $\nabla h_1(a), \dots, \nabla h_q(a)$ are linearly independent. If $f|_M$ has a local extremum at a, there are unique numbers $\lambda_1, \dots, \lambda_q$ with*

$$\nabla f(a) = \lambda_1 \nabla h_1(a) + \cdots + \lambda_q \nabla h_q(a). \tag{27.4}$$

Equivalently, (a, λ) is a critical point of the Lagrangian $L(x, \lambda) = f(x) - \sum_j \lambda_j h_j(x)$ *on $U \times \mathbb{R}^q$.*

Proof. Let $d \in \ker Dh(a)$, and let $x(t)$ be the curve of Lemma 27.12.5. Since $x(t) \in M$ and $x(t) \to a$, the function $f \circ x$ has a local extremum at $t = 0$, so by Lemma 27.6.4 and the chain rule $0 = (f \circ x)'(0) = \langle \nabla f(a), d \rangle$. Thus $\nabla f(a)$ is orthogonal to $\ker Dh(a)$. Now $\ker Dh(a)$ is exactly the orthogonal complement of the span R of the rows $\nabla h_j(a)$ of the Jacobian matrix, so $\nabla f(a) \in (R^\perp)^\perp = R$ (in a finite-dimensional inner product space $(R^\perp)^\perp = R$, since $R \subseteq (R^\perp)^\perp$ and both have dimension $\dim R$). Uniqueness of the λ_j is the linear independence of the $\nabla h_j(a)$. Finally, $\partial L/\partial x_i = 0$ for all i is (27.4), and $\partial L/\partial \lambda_j = -h_j = 0$ is the constraint. □

The space $\ker Dh(a)$ is the *tangent space* of M at a: by the lemma it is the set of velocities of curves in M through a, and by the submersion theorem M is, near a, a submanifold of dimension $n - q$ in the sense of Chapter 33. The theorem says that at a constrained extremum the gradient of f is normal to M. The linear independence hypothesis cannot be dropped:

Counterexample 27.12.7. Minimise $f(x,y) = x$ on the cusp $M = \{h = 0\}$, $h(x,y) = y^2 - x^3$. On M, $x^3 = y^2 \geq 0$, so $x \geq 0$ and the minimum is at $(0,0)$. But $\nabla f = (1,0)$, while $\nabla h = (-3x^2, 2y)$ vanishes at the origin, so no λ satisfies $\nabla f(0) = \lambda \nabla h(0)$. Solving the Lagrange system $1 = -3\lambda x^2$, $0 = 2\lambda y$, $y^2 = x^3$ gives no solution at all: the method finds nothing, and the minimum sits at the singular point of M. In practice one must examine separately the points of M where the gradients of the constraints are dependent.

Example 27.12.8. 1. *Extrema of $x + y$ on the circle $x^2 + y^2 = 1$.* The circle is compact, so both extrema exist (Corollary 18.6.6), and $\nabla h = (2x, 2y) \neq 0$ on it. The system $(1,1) = \lambda(2x, 2y)$ gives $x = y = \frac{1}{2\lambda}$, and $x^2 + y^2 = 1$ gives $(x,y) = \pm(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$: maximum $\sqrt{2}$, minimum $-\sqrt{2}$.

2. *The inequality of the means.* Maximise $f(x) = x_1 x_2 \cdots x_k$ on the compact set $\Delta = \{x : x_i \geq 0, \sum_i x_i = k\}$. The maximum is positive (take all $x_i = 1$), so it is attained at a point with all $x_i > 0$, which is a local maximum of f on $M = \{x \in U : \sum_i x_i = k\}$ with $U = \{x_i > 0\}$. With $h = \sum_i x_i - k$, $\nabla h = (1, \ldots, 1)$ and $\partial_i f = f(x)/x_i$, so $f(x)/x_i = \lambda$ for all i: all x_i are equal, hence equal to 1, and $f \leq 1$ on Δ. By homogeneity, $\sqrt[k]{x_1 \cdots x_k} \leq \frac{1}{k} \sum_i x_i$ for $x_i \geq 0$, which recovers Corollary 27.7.7 by a quite different route.

3. *Eigenvectors as critical points.* Let A be a real symmetric $n \times n$ matrix and $q(x) = x^T A x$, so that $\nabla q(x) = 2Ax$. On the compact unit sphere, $h(x) = \|x\|^2 - 1$ with $\nabla h(x) = 2x \neq 0$, the function q has a maximum at some u, and (27.4) reads $Au = \lambda u$: *u is an eigenvector*, and $\lambda = u^T A u$ is the maximum of q on the sphere, the largest eigenvalue. This proves, without the fundamental theorem of algebra, that a real symmetric matrix has a real eigenvector, which is the key step of Theorem 11.10.6(2). More generally the critical points of the *Rayleigh quotient* $x^T Ax / x^T x$ on $\mathbb{R}^n \setminus \{0\}$ are exactly the eigenvectors.

4. *Nearest points.* Let $h \colon \mathbb{R}^n \to \mathbb{R}$ be C^1, $M = \{h = 0\}$ nonempty with $\nabla h \neq 0$ on M, and $p \in \mathbb{R}^n$. Since M is closed, $f(x) = \|x - p\|^2$ attains a minimum on M (on the compact set $M \cap \overline{B}(p, R)$ for R large). At the minimum a, $2(a - p) = \lambda \nabla h(a)$: *the segment from p to the nearest point is normal to M*. For the hyperplane $\langle n, x \rangle = c$ this gives $a = p + tn$ with $t = (c - \langle n, p \rangle)/\|n\|^2$, and the familiar distance $|c - \langle n, p \rangle|/\|n\|$.

The multiplier has a meaning of its own: it is the rate at which the optimal value changes when the constraint is relaxed.

Proposition 27.12.9. *Let $f, h \colon U \to \mathbb{R}$ be C^1 on an open $U \subseteq \mathbb{R}^n$, J an open interval, and $c \mapsto x(c)$ a differentiable map $J \to U$ such that for every $c \in J$, $h(x(c)) = c$ and $\nabla f(x(c)) = \lambda(c) \nabla h(x(c))$ for some number $\lambda(c)$. Then $V(c) = f(x(c))$ is differentiable on J and $V'(c) = \lambda(c)$.*

Proof. By the chain rule, $V'(c) = \langle \nabla f(x(c)), x'(c) \rangle = \lambda(c) \langle \nabla h(x(c)), x'(c) \rangle = \lambda(c) \frac{d}{dc} h(x(c)) = \lambda(c)$. □

For example, the largest product xy with $x + y = c$ is $V(c) = c^2/4$, at $x = y = c/2$, where $(y, x) = \frac{c}{2}(1, 1)$: indeed $\lambda = c/2 = V'(c)$. In economics, where f is a profit and $h \leq c$ a budget, λ is the *shadow price* of the resource: the extra profit that one more unit of it would bring. In mechanics, where Lagrange introduced the method in his *Mécanique analytique* (1788), $\lambda \nabla h$ is the force by which the constraint keeps the system on M.

27.12.3 Second-order conditions

A critical point of the Lagrangian need not be an extremum, and as in Proposition 27.12.2 one looks at second derivatives. The correct quadratic form is not the Hessian of f but that of the Lagrangian, and only in tangent directions: the curvature of M contributes through the multipliers.

Theorem 27.12.10. *Let f and $h = (h_1, \ldots, h_q)$ be C^2 on the open set $U \subseteq \mathbb{R}^n$, $M = \{h = 0\}$, and $a \in M$ with $\nabla f(a) = \sum_j \lambda_j \nabla h_j(a)$. Put $L = f - \sum_j \lambda_j h_j$ (with these fixed λ_j) and $T = \ker Dh(a)$.*

1. *If $\nabla h_1(a), \ldots, \nabla h_q(a)$ are linearly independent and $f|_M$ has a local minimum at a, then $D^2 L(a)(d, d) \geq 0$ for all $d \in T$.*

2. *If $D^2 L(a)(d, d) > 0$ for all nonzero $d \in T$, then $f|_M$ has a strict local minimum at a.*

Proof. Note that $\nabla L(a) = 0$ and that $L = f$ on M.

(1) Let $d \in T$ and let $x(t)$ be the C^2 curve of Lemma 27.12.5, and $\varphi(t) = f(x(t)) = L(x(t))$. Differentiating $\varphi'(t) = \sum_i \partial_i L(x(t))x_i'(t)$ once more,

$$\varphi''(0) = \sum_{i,j} \partial_j\partial_i L(a)x_i'(0)x_j'(0) + \sum_i \partial_i L(a)x_i''(0) = D^2L(a)(d,d),$$

because $\nabla L(a) = 0$; this is where the Lagrangian, rather than f, is needed, since $x''(0)$ is unknown. Now φ has a local minimum at 0, and as in the proof of Proposition 27.12.2(1), $\varphi''(0) \geq 0$.

(2) Suppose not. Then there are $x_k \in M$ with $x_k \neq a$, $x_k \to a$ and $f(x_k) \leq f(a)$. Write $x_k = a + t_k d_k$ with $t_k = \|x_k - a\| \to 0$ and $\|d_k\| = 1$; passing to a subsequence (the unit sphere is compact), $d_k \to d$ with $\|d\| = 1$. From $h(x_k) = h(a) = 0$ we get $0 = (h(a + t_k d_k) - h(a))/t_k \to Dh(a)d$, so $d \in T$. By (27.2) for L, with $\nabla L(a) = 0$,

$$0 \geq f(x_k) - f(a) = L(x_k) - L(a) = \tfrac{1}{2}t_k^2 D^2L(\xi_k)(d_k, d_k)$$

with $\xi_k \to a$. Dividing by $t_k^2/2$ and letting $k \to \infty$, $D^2L(a)(d,d) \leq 0$, contradicting the hypothesis. □

Example 27.12.11. Let M be the parabola $y = x^2$, $h(x,y) = y - x^2$, and $f(x,y) = y + cx^2$ with a constant c. At $a = 0$, $\nabla f = (0,1) = 1 \cdot \nabla h(0)$, so $\lambda = 1$ and $T = \mathbb{R}(1,0)$. Then $D^2L = D^2f - D^2h = \operatorname{diag}(2c+2, 0)$, so $D^2L(0)(e_1,e_1) = 2(1+c)$: by the theorem, 0 is a strict local minimum of $f|_M$ for $c > -1$ and a strict local maximum for $c < -1$. This is correct, since $f = (1+c)x^2$ on M. The Hessian of f alone gives $D^2f(0)(e_1,e_1) = 2c$, which has the wrong sign for $-1 < c < 0$: the term $-\lambda D^2h$ records the bending of M.

Example 27.12.12 (The bordered Hessian in two variables)**.** Let $n = 2$, $q = 1$, $\nabla h(a) = (h_x, h_y) \neq 0$ and $D^2L(a) = \left(\begin{smallmatrix} L_{xx} & L_{xy} \\ L_{xy} & L_{yy} \end{smallmatrix}\right)$. Then T is spanned by $d = (-h_y, h_x)$ and

$$D^2L(a)(d,d) = L_{xx}h_y^2 - 2L_{xy}h_xh_y + L_{yy}h_x^2 = -\det\begin{pmatrix} 0 & h_x & h_y \\ h_x & L_{xx} & L_{xy} \\ h_y & L_{xy} & L_{yy} \end{pmatrix},$$

as one sees by expanding the determinant along the first row. So if the *bordered Hessian* on the right has negative determinant, a is a strict local minimum of $f|_M$, and if it has positive determinant, a strict local maximum. In Example 27.12.8(1), at $\pm(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ we have $\lambda = \pm\frac{1}{\sqrt{2}}$, $D^2L = -2\lambda I$ and $(h_x, h_y) = \pm(\sqrt{2}, \sqrt{2})$; the determinant is $-(-2\lambda)(h_x^2 + h_y^2) = 8\lambda$, positive at the maximum and negative at the minimum.

27.12.4 Inequality constraints and Farkas' lemma

Many problems constrain the unknowns by inequalities: quantities are nonnegative, budgets are bounded. Consider

$$\text{minimise } f(x) \quad \text{subject to} \quad g_i(x) \leq 0 \ (1 \leq i \leq p), \qquad h_j(x) = 0 \ (1 \leq j \leq q), \tag{27.5}$$

where $f, g_i, h_j \colon U \to \mathbb{R}$ are C^1 on an open $U \subseteq \mathbb{R}^n$; the *feasible set* is the set F of $x \in U$ satisfying the constraints. At a feasible point a, the inequality $g_i \leq 0$ is *active* if $g_i(a) = 0$; let $I(a)$ be the set of active indices. An inactive constraint, $g_i(a) < 0$, still holds near a and plays no role locally; an active one behaves like a wall.

To see what to expect, take one inequality $g \leq 0$ and no equations. If the minimum a is interior ($g(a) < 0$), then $\nabla f(a) = 0$. If $g(a) = 0$, the minimum lies on the wall $\{g = 0\}$,

so $\nabla f(a) = \lambda \nabla g(a)$ by Theorem 27.12.6; and now the *sign* of λ is determined: moving into the feasible region, in the direction $-\nabla g(a)$, must not decrease f, so $\nabla f(a)$ points into the feasible region and $-\nabla f(a) = \mu \nabla g(a)$ with $\mu \geq 0$. The general statement needs a lemma of linear algebra over the ordered field $\mathbb{R}$.

Lemma 27.12.13 (Farkas)**.** *Let $v_1, \dots, v_r, b \in \mathbb{R}^n$. Exactly one of the following holds:*

1. *$b = c_1 v_1 + \cdots + c_r v_r$ with all $c_i \geq 0$;*
2. *there is $w \in \mathbb{R}^n$ with $\langle w, v_i \rangle \leq 0$ for all i and $\langle w, b \rangle > 0$.*

Geometrically: either b lies in the convex cone $C = \{\sum_i c_i v_i : c_i \geq 0\}$, or a hyperplane through 0, with normal w, separates b from C.

Proof. Both cannot hold, since then $\langle w, b \rangle = \sum_i c_i \langle w, v_i \rangle \leq 0$. Assume (1) fails, that is, $b \notin C$.

Step 1: C is closed. First, every $y \in C$ is a nonnegative combination of a linearly independent subset of the v_i (*Carathéodory's lemma*). Indeed, choose $y = \sum_{i \in S} c_i v_i$ with all $c_i > 0$ and S as small as possible. If the v_i with $i \in S$ were dependent, there would be a relation $\sum_{i \in S} \gamma_i v_i = 0$ with some $\gamma_i > 0$ (change the sign of γ if necessary); with $t = \min\{c_i/\gamma_i : \gamma_i > 0\}$, $y = \sum_{i \in S}(c_i - t\gamma_i)v_i$ has nonnegative coefficients, one of which vanishes, contradicting the minimality of S. So C is the union of the finitely many cones C_S spanned by linearly independent subsets S, and it suffices that each C_S is closed. The map $\mathbb{R}^S \to \mathbb{R}^n$, $c \mapsto \sum_{i \in S} c_i v_i$, is injective and linear, so on the compact unit sphere of $\mathbb{R}^S$ its norm has a positive minimum m, and $\|\sum_i c_i v_i\| \geq m\|c\|$. If $\sum_i c_i^{(k)} v_i \to y$ with $c^{(k)} \geq 0$, then $(c^{(k)})$ is a Cauchy sequence, with a limit $c \geq 0$, and $y = \sum_i c_i v_i \in C_S$.

Step 2: a nearest point. The continuous function $x \mapsto \|b - x\|$ attains a minimum on the compact set $C \cap \overline{B}(b, \|b\|)$, which contains 0; the minimum point p is then a nearest point to b in all of C.

Step 3: separation. Let $w = b - p$, nonzero since $b \notin C$. For $c \in C$ and $0 < t \leq 1$, $(1-t)p + tc \in C$, so $\|w - t(c-p)\|^2 \geq \|w\|^2$, that is, $-2t\langle w, c - p\rangle + t^2\|c - p\|^2 \geq 0$; dividing by t and letting $t \to 0$, $\langle w, c - p\rangle \leq 0$. With $c = 0$ and $c = 2p$ this gives $\langle w, p\rangle = 0$, hence $\langle w, c\rangle \leq 0$ for all $c \in C$, in particular $\langle w, v_i\rangle \leq 0$. Finally $\langle w, b\rangle = \langle w, w + p\rangle = \|w\|^2 > 0$. □

Theorem 27.12.14 (Karush–Kuhn–Tucker)**.** *In the situation (27.5), let $a \in F$ be a local minimum of f on F, and assume the* linear independence constraint qualification*: the gradients $\nabla g_i(a)$ for $i \in I(a)$ and $\nabla h_j(a)$ for $1 \leq j \leq q$ are linearly independent. Then there are unique numbers $\mu_1, \dots, \mu_p$ and $\lambda_1, \dots, \lambda_q$ with*

$$\nabla f(a) + \sum_{i=1}^{p} \mu_i \nabla g_i(a) + \sum_{j=1}^{q} \lambda_j \nabla h_j(a) = 0, \qquad \mu_i \geq 0, \qquad \mu_i g_i(a) = 0 \quad (1 \leq i \leq p). \tag{27.6}$$

The last condition, *complementary slackness*, says that $\mu_i = 0$ unless the i-th constraint is active. The signs of the λ_j are not restricted, and for $p = 0$ the theorem is Theorem 27.12.6 for minima (with $-\lambda_j$ in place of λ_j).

Proof. Write $I = I(a)$.

Step 1. Let $d \in \mathbb{R}^n$ with $\langle \nabla g_i(a), d\rangle < 0$ for $i \in I$ and $Dh(a)d = 0$. We show $\langle \nabla f(a), d\rangle \geq 0$. The $\nabla h_j(a)$ are linearly independent, so Lemma 27.12.5 gives a C^1 curve $x(t)$ with $x(0) = a$, $x'(0) = d$ and $h(x(t)) = 0$. For $i \in I$, $\varphi_i(t) = g_i(x(t))$ has $\varphi_i(0) = 0$ and $\varphi_i'(0) = \langle \nabla g_i(a), d\rangle < 0$, so $\varphi_i(t) < 0$ for small $t > 0$; for $i \notin I$, $g_i(x(t)) < 0$ for small t by continuity. So $x(t) \in F$ for small $t > 0$, and since $x(t) \to a$, $f(x(t)) \geq f(a)$ for small $t > 0$. Hence $\langle \nabla f(a), d\rangle = \lim_{t \to 0^+}(f(x(t)) - f(a))/t \geq 0$.

Step 2. The same holds if only $\langle \nabla g_i(a), d\rangle \le 0$ for $i \in I$ and $Dh(a)d = 0$. Indeed, the linear map $d \mapsto ((\langle \nabla g_i(a), d\rangle)_{i\in I}, Dh(a)d)$ has linearly independent rows, hence is surjective, so there is d_0 with $\langle \nabla g_i(a), d_0\rangle = -1$ for $i \in I$ and $Dh(a)d_0 = 0$. For $\varepsilon > 0$, $d + \varepsilon d_0$ satisfies the hypothesis of Step 1, so $\langle \nabla f(a), d + \varepsilon d_0\rangle \ge 0$; let $\varepsilon \to 0$.

Step 3. Apply Lemma 27.12.13 to $b = -\nabla f(a)$ and the vectors $\nabla g_i(a)$ ($i \in I$), $\nabla h_j(a)$ and $-\nabla h_j(a)$ ($1 \le j \le q$). Alternative (2) would give w with $\langle \nabla g_i(a), w\rangle \le 0$ for $i \in I$, $\langle \pm\nabla h_j(a), w\rangle \le 0$, that is, $Dh(a)w = 0$, and $\langle \nabla f(a), w\rangle < 0$, contradicting Step 2. So alternative (1) holds: $-\nabla f(a) = \sum_{i\in I} \mu_i \nabla g_i(a) + \sum_j (c_j^+ - c_j^-)\nabla h_j(a)$ with $\mu_i, c_j^\pm \ge 0$. Put $\lambda_j = c_j^+ - c_j^-$ and $\mu_i = 0$ for $i \notin I$; this gives (27.6). Uniqueness follows from the linear independence, since $\mu_i = 0$ is forced for $i \notin I$. □

The conditions were found by W. Karush in his 1939 master's thesis and, independently, by H. W. Kuhn and A. W. Tucker in 1951, at the start of nonlinear programming. In the language of the introduction to this subsection: at a constrained minimum, $-\nabla f(a)$, the direction of steepest descent, lies in the cone spanned by the outward normals $\nabla g_i(a)$ of the active walls (and the normal space of the equations), so that every feasible direction is uphill to first order.

Counterexample 27.12.15. Without a constraint qualification the conclusion can fail. Minimise $f(x, y) = x$ subject to $g_1 = y - x^3 \le 0$ and $g_2 = -y \le 0$. Feasibility means $0 \le y \le x^3$, so $x \ge 0$, and the minimum is at $(0, 0)$, where both constraints are active. There $\nabla f = (1, 0)$, $\nabla g_1 = (0, 1)$ and $\nabla g_2 = (0, -1)$, and $(1, 0) + \mu_1(0, 1) + \mu_2(0, -1) = 0$ is impossible. The two active gradients are dependent: the feasible set has a cusp at the origin, and the linearised constraints $\langle \nabla g_i, d\rangle \le 0$, which allow every $d = (d_1, 0)$, do not describe the feasible directions, which have $d_1 \ge 0$.

Remark 27.12.16. Reading the proof again shows that weaker hypotheses suffice for the existence of multipliers (not for their uniqueness, which used the full linear independence).

1. *Mangasarian–Fromovitz condition:* the $\nabla h_j(a)$ are linearly independent and there is d_0 with $Dh(a)d_0 = 0$ and $\langle \nabla g_i(a), d_0\rangle < 0$ for $i \in I(a)$. Step 1 used only the independence of the $\nabla h_j(a)$, Step 2 only the existence of such a d_0, and Step 3 nothing else.

2. *Affine constraints:* if all g_i and h_j are affine, no qualification is needed. Indeed, for d with $\langle \nabla g_i(a), d\rangle \le 0$ ($i \in I(a)$) and $Dh(a)d = 0$, the points $a + td$ are feasible for small $t > 0$ (for active i, $g_i(a + td) = t\langle \nabla g_i(a), d\rangle \le 0$), so $\langle \nabla f(a), d\rangle \ge 0$ directly, and Step 3 applies.

27.12.5 Convex problems

The conditions of the Karush–Kuhn–Tucker theorem are necessary. For convex problems they are also sufficient, and give global minima. A function f on a convex set $K \subseteq \mathbb{R}^n$ is *convex* if $f((1 - t)x + ty) \le (1 - t)f(x) + tf(y)$ for all $x, y \in K$ and $t \in [0, 1]$, that is, if its restriction to every segment is convex in the sense of Definition 27.6.16, and *strictly convex* if the inequality is strict whenever $x \ne y$ and $0 < t < 1$. A sublevel set $\{x \in K : f(x) \le c\}$ of a convex function is convex, and so is the zero set of an affine function; hence the feasible set of (27.5) is convex when U is convex, the g_i are convex and the h_j are affine.

Lemma 27.12.17. *Let $U \subseteq \mathbb{R}^n$ be open and convex, and $f : U \to \mathbb{R}$ convex and C^1. Then $f(y) \ge f(x) + \langle \nabla f(x), y - x\rangle$ for all $x, y \in U$. If f is C^2 with positive semidefinite Hessian everywhere, then f is convex.*

Proof. We may assume $x \neq y$. As U is open, the points $x - \varepsilon(y-x)$ and $y + \varepsilon(y-x)$ lie in U for small $\varepsilon > 0$, and as U is convex so does the segment joining them. Hence $\varphi(t) = f(x + t(y-x))$ is defined on the open interval $J = (-\varepsilon, 1+\varepsilon)$, it is convex there (being the restriction of f to a segment), and it is differentiable with $\varphi'(0) = \langle \nabla f(x), y - x\rangle$ by the chain rule. By Proposition 27.6.18(2), $\varphi(1) \geq \varphi(0) + \varphi'(0)$, which is the inequality. For the second statement, $\varphi''(t) = D^2 f(x + t(y-x))(y-x, y-x) \geq 0$ on J by (27.3), so φ is convex by Proposition 27.6.18(3); this is convexity of f on the segment $[x, y]$. □

Theorem 27.12.18. *In (27.5), let U be convex, f and $g_1, \dots, g_p$ convex and $h_1, \dots, h_q$ affine. If $a \in F$ and numbers μ_i, λ_j satisfy (27.6), then a is a global minimum of f on F. If f is strictly convex, it is the only one.*

Proof. Let $x \in F$. By Lemma 27.12.17 and (27.6),

$$f(x) - f(a) \geq \langle \nabla f(a), x - a\rangle = -\sum_i \mu_i \langle \nabla g_i(a), x - a\rangle - \sum_j \lambda_j \langle \nabla h_j(a), x - a\rangle.$$

Since h_j is affine, $\langle \nabla h_j(a), x - a\rangle = h_j(x) - h_j(a) = 0$. By the lemma again and complementary slackness, $\mu_i \langle \nabla g_i(a), x - a\rangle \leq \mu_i (g_i(x) - g_i(a)) = \mu_i g_i(x) \leq 0$. Hence $f(x) \geq f(a)$. If f is strictly convex and $f(x) = f(a)$ with $x \neq a$, then the midpoint $m = \frac{1}{2}(x + a)$ is feasible (F is convex) and $f(m) < \frac{1}{2} f(x) + \frac{1}{2} f(a) = f(a)$, a contradiction. □

Example 27.12.19 (Water-filling)**.** Given $\alpha_1, \dots, \alpha_n > 0$ and $P > 0$, maximise $\sum_i \log(\alpha_i + x_i)$ subject to $x_i \geq 0$ and $\sum_i x_i = P$. (In information theory, x_i is the power sent through the i-th of n parallel channels with noise level α_i, and $\frac{1}{2}\log(1 + x_i/\alpha_i)$ is its capacity, so this maximises the total capacity for a total power P.) Minimise the strictly convex function $f(x) = -\sum_i \log(\alpha_i + x_i)$ on $U = \{x_i > -\alpha_i\}$, with $g_i = -x_i$ and $h = \sum_i x_i - P$. The conditions (27.6) read

$$-\frac{1}{\alpha_i + x_i} - \mu_i + \lambda = 0, \qquad \mu_i \geq 0, \qquad \mu_i x_i = 0.$$

So $\lambda \geq 1/(\alpha_i + x_i) > 0$; put $\nu = 1/\lambda$. If $x_i > 0$, then $\mu_i = 0$ and $x_i = \nu - \alpha_i$; if $x_i = 0$, then $\lambda \geq 1/\alpha_i$, that is, $\alpha_i \geq \nu$. In both cases

$$x_i = \max(0, \nu - \alpha_i),$$

and ν is fixed by $\sum_i \max(0, \nu - \alpha_i) = P$; the left side is continuous and strictly increasing in $\nu \geq \min_i \alpha_i$, from 0 to ∞, so there is exactly one such ν (Theorem 27.4.8). Conversely this x, with $\lambda = 1/\nu$ and $\mu_i = \lambda - 1/(\alpha_i + x_i) \geq 0$, satisfies the conditions, so by Theorem 27.12.18 it is the unique optimum. The picture: a vessel whose floor has height α_i over the i-th cell is filled with a volume P of water; the water settles at level ν, and x_i is the depth over the i-th cell. The noisiest channels get nothing.

Example 27.12.20 (Maximum entropy and the Boltzmann distribution)**.** Let $E_1, \dots, E_N$ be real numbers (the energies of the states of a physical system), not all equal, and $\min_i E_i < E < \max_i E_i$. Among probability vectors p with $p_i > 0$, $\sum_i p_i = 1$ and mean energy $\sum_i p_i E_i = E$, which one has the largest *entropy* $H(p) = -\sum_i p_i \log p_i$? The function $-H(p) = \sum_i \phi(p_i)$, $\phi(t) = t \log t$, is strictly convex on $U = \{p_i > 0\}$: $\phi''(t) = 1/t > 0$, so ϕ is strictly convex (Proposition 27.6.18(3)), and for $p \neq p'$ the convexity inequality is strict in a coordinate where they differ. The two constraints are affine. The conditions (27.6) for minimising $-H$ read $\log p_i + 1 + \lambda_1 + \lambda_2 E_i = 0$, so

$$p_i = \frac{e^{-\beta E_i}}{Z(\beta)}, \qquad Z(\beta) = \sum_k e^{-\beta E_k},$$

with $\beta = \lambda_2$, after normalising. This is the *Boltzmann (Gibbs) distribution* of statistical mechanics, β being the inverse temperature and Z the *partition function*. It remains to choose β with mean energy E. The mean $m(\beta) = \sum_k E_k e^{-\beta E_k}/Z(\beta)$ is continuous, has derivative $m'(\beta) = -(\sum_k E_k^2 p_k - m(\beta)^2) < 0$, minus the variance of the energy, which is positive since the E_k are not all equal, and tends to $\min E_k$ as $\beta \to +\infty$ and to $\max E_k$ as $\beta \to -\infty$. So exactly one β gives $m(\beta) = E$, and by Theorem 27.12.18 the corresponding p is the unique entropy maximiser. Without the energy constraint ($\lambda_2 = 0$) the maximiser is the uniform distribution, with $H = \log N$.

Remark 27.12.21 (Duality). For (27.5) define the Lagrangian $L(x, \mu, \lambda) = f(x) + \sum_i \mu_i g_i(x) + \sum_j \lambda_j h_j(x)$ and the *dual function* $\theta(\mu, \lambda) = \inf_{x \in U} L(x, \mu, \lambda)$. For $\mu \geq 0$ and $x \in F$, $L(x, \mu, \lambda) \leq f(x)$, so

$$\theta(\mu, \lambda) \leq \inf_F f \qquad (\mu \geq 0) \qquad (\textit{weak duality}) :$$

every choice of multipliers gives a lower bound for the minimum. In the convex situation of Theorem 27.12.18, if (a, μ, λ) satisfies the conditions (27.6), then $x \mapsto L(x, \mu, \lambda)$ is convex with vanishing gradient at a, so by Lemma 27.12.17 it is minimal at a, and $\theta(\mu, \lambda) = L(a, \mu, \lambda) = f(a)$ by complementary slackness and $h(a) = 0$: the bound is attained, and the multipliers solve the *dual problem* of maximising θ over $\mu \geq 0$. For convex problems with a strictly feasible point (*Slater's condition*) the maximum of the dual equals the minimum of the primal even without assuming that the minimum is attained [BV04, Chapter 5]; for linear programmes this is the duality theorem of linear programming, a consequence of Farkas' lemma. The point $(a, (\mu, \lambda))$ is then a saddle point of L: a minimum in x and a maximum in (μ, λ) with $\mu \geq 0$.

27.13 Ordinary differential equations

Definition 27.13.1. Let $D \subseteq \mathbb{R} \times \mathbb{R}^n$ be open and $F \colon D \to \mathbb{R}^n$ continuous. F is *locally Lipschitz in x* if every point of D has a neighbourhood N and L with $\|F(t, x) - F(t, y)\| \leq L\|x - y\|$ for $(t, x), (t, y) \in N$; by the mean value inequality this holds if F is C^1. A *solution* of $x' = F(t, x)$ is a differentiable map $\varphi \colon J \to \mathbb{R}^n$ on an interval with $(t, \varphi(t)) \in D$ and $\varphi'(t) = F(t, \varphi(t))$. By Theorem 27.8.3, this is equivalent to continuity of φ and $\varphi(t) = \varphi(t_0) + \int_{t_0}^t F(s, \varphi(s))\, ds$.

Lemma 27.13.2 (Grönwall)**.** *If $u \colon [t_0, t_1] \to [0, \infty)$ is continuous and $u(t) \leq C + L \int_{t_0}^t u$ with $C, L \geq 0$, then $u(t) \leq C e^{L(t - t_0)}$.*

Proof. Let $v(t) = C + L \int_{t_0}^t u$. Then $v' = Lu \leq Lv$, so $(e^{-L(t-t_0)} v)' \leq 0$, and $u \leq v \leq v(t_0) e^{L(t-t_0)} = C e^{L(t-t_0)}$. □

Theorem 27.13.3 (Picard–Lindelöf)**.** *Let F be continuous and locally Lipschitz in x, and $(t_0, x_0) \in D$.*

1. *There is $\delta > 0$ and a solution on $[t_0 - \delta, t_0 + \delta]$ with $\varphi(t_0) = x_0$.*
2. *Two solutions on intervals $J_1, J_2 \ni t_0$ with the same value at t_0 agree on $J_1 \cap J_2$.*
3. *There is a unique* maximal *solution, defined on an open interval (α, β); if $\beta < \infty$, then $(t, \varphi(t))$ leaves every compact subset of D as $t \to \beta$, and similarly at α.*

Proof. (1) Choose $a, b > 0$ such that $R = [t_0 - a, t_0 + a] \times \overline{B}(x_0, b) \subseteq D$ and F is L-Lipschitz in x on R; let $M = \sup_R \|F\|$ (finite by compactness) and $\delta = \min(a, b/(M+1), 1/(2L+1))$. On the complete metric space X of continuous maps $[t_0 - \delta, t_0 + \delta] \to \overline{B}(x_0, b)$ with the supremum

metric (Proposition 18.7.2), $T\varphi(t) = x_0 + \int_{t_0}^{t} F(s,\varphi(s))\,ds$ satisfies $\|T\varphi(t) - x_0\| \le M\delta \le b$ and $\|T\varphi - T\psi\| \le L\delta\|\varphi - \psi\| \le \frac{1}{2}\|\varphi - \psi\|$. Its fixed point is a solution.

(2) The set of $t \in J_1 \cap J_2$ where the solutions agree is closed and contains t_0. If they agree at t_1, near $(t_1, \varphi(t_1))$ F is L-Lipschitz, and $u = \|\varphi_1 - \varphi_2\|$ satisfies $u(t) \le L\left|\int_{t_1}^{t} u\right|$ for t near t_1, so $u = 0$ near t_1 by Lemma 27.13.2 (with $C = 0$, on both sides). So the set is open, and equals the connected set $J_1 \cap J_2$.

(3) By (2), the solutions with $\varphi(t_0) = x_0$ glue to a solution on the union of their intervals, which is maximal; the interval is open because a solution on a closed end could be continued by (1). Suppose $\beta < \infty$ and $(t_k, \varphi(t_k)) \in K$ for a compact $K \subseteq D$ and $t_k \to \beta$. By compactness there are a, b, L, M as in (1) valid for all rectangles centred at points of K (cover K by finitely many rectangles of half the size), hence a uniform δ. For $t_k > \beta - \delta/2$, the solution through $(t_k, \varphi(t_k))$ exists up to $t_k + \delta > \beta$ and extends φ by (2), a contradiction. □

Counterexample 27.13.4.

1. Without the Lipschitz condition uniqueness fails: $x' = 3x^{2/3}$, $x(0) = 0$ has the solutions $x = 0$ and $x = t^3$.

2. Solutions need not exist for all time: $x' = x^2$, $x(0) = 1$ has maximal solution $x = 1/(1-t)$ on $(-\infty, 1)$.

Proposition 27.13.5. *1. If φ, ψ are solutions on $[t_0, t_1]$ and $\|F(t,\varphi(t)) - F(t,\psi(t))\| \le L\|\varphi(t) - \psi(t)\|$ for all t, then $\|\varphi(t) - \psi(t)\| \le \|\varphi(t_0) - \psi(t_0)\| e^{L(t-t_0)}$.*

2. (Linear equations) If $A\colon I \to M_n(\mathbb{R})$ and $b\colon I \to \mathbb{R}^n$ are continuous on an open interval I, the maximal solutions of $x' = A(t)x + b(t)$ are defined on all of I.

Proof. (1) $u = \|\varphi - \psi\|$ satisfies $u(t) \le u(t_0) + L\int_{t_0}^{t} u$; apply Lemma 27.13.2. (2) On a compact subinterval $[c, d] \ni t_0$ with $\|A\| \le L$ and $\|b\| \le B$, $\|\varphi(t)\| \le \|\varphi(t_0)\| + B(d-c) + L\int \|\varphi\|$, so φ is bounded there by Grönwall, its graph stays in a compact set, and by Theorem 27.13.3(3) the maximal interval contains $[c, d]$. □

Theorem 27.13.6 (Smooth dependence)**.** *If F is C^k with $k \ge 1$, the* flow *$\Phi(t, t_0, x_0)$, the value at t of the maximal solution with $\varphi(t_0) = x_0$, is defined on an open subset of $\mathbb{R} \times D$ and is C^k there. Its derivative in x_0 solves the linearised equation $Y' = D_x F(t,\varphi(t))Y$, $Y(t_0) = 1$.*

This is [Har02, Chapter V]; see also [Lan93]. Continuity of Φ follows from Proposition 27.13.5(1) and the uniform δ in the proof of Theorem 27.13.3.

27.14 Stone–Weierstrass and Arzelà–Ascoli

Lemma 27.14.1. *The polynomials $p_0 = 0$, $p_{k+1}(t) = p_k(t) + (t^2 - p_k(t)^2)/2$ converge uniformly to $|t|$ on $[-1, 1]$, and $p_k(0) = 0$.*

Proof. By induction $0 \le p_k(t) \le p_{k+1}(t) \le |t|$: indeed $|t| - p_{k+1} = (|t| - p_k)(1 - (|t| + p_k)/2)$ with $0 \le (|t| + p_k)/2 \le |t| \le 1$, and $p_{k+1} - p_k = (t^2 - p_k^2)/2 \ge 0$. Iterating, $0 \le |t| - p_k(t) \le |t|(1 - |t|/2)^k$, and the maximum of $s(1 - s/2)^k$ on $[0, 1]$ is at most $2/(k+1)$ (it is attained at $s = 2/(k+1)$ when $k \ge 1$). □

Theorem 27.14.2 (Stone–Weierstrass)**.** *Let K be a compact Hausdorff space and $\mathcal{A} \subseteq C(K, \mathbb{R})$ a subalgebra containing the constants and separating points. Then $\mathcal{A}$ is dense for the supremum norm. The same holds for $\mathcal{A} \subseteq C(K, \mathbb{C})$ if $\mathcal{A}$ is also closed under complex conjugation.*

Proof. The closure $\bar{\mathcal{A}}$ is a closed subalgebra. If $f \in \bar{\mathcal{A}}$ with $\|f\| \le c$, then $|f| = c \lim_k p_k(f/c) \in \bar{\mathcal{A}}$ by Lemma 27.14.1; hence $\max(f,g) = (f+g+|f-g|)/2$ and $\min(f,g)$ lie in $\bar{\mathcal{A}}$. For $x \neq y$ and reals α, β there is $h \in \mathcal{A}$ with $h(x) = \alpha$, $h(y) = \beta$ (take $h = \alpha + (\beta-\alpha)(u-u(x))/(u(y)-u(x))$ with u separating x and y).

Let $f \in C(K,\mathbb{R})$ and $\varepsilon > 0$. Fix x. For each y choose $h_y \in \mathcal{A}$ with $h_y(x) = f(x)$ and $h_y(y) = f(y)$; the open sets $\{h_y < f + \varepsilon\}$ cover K, and the minimum g_x of finitely many of these h_y lies in $\bar{\mathcal{A}}$, with $g_x(x) = f(x)$ and $g_x < f + \varepsilon$. The open sets $\{g_x > f - \varepsilon\}$ cover K, and the maximum g of finitely many g_x lies in $\bar{\mathcal{A}}$ and satisfies $\|g - f\| < \varepsilon$. For the complex case, the real parts $(f + \bar{f})/2$ of elements of $\mathcal{A}$ form a real subalgebra with the same properties, which is dense in $C(K,\mathbb{R})$; so $\mathcal{A}$ is dense in $C(K,\mathbb{C})$. □

Corollary 27.14.3. *Polynomials are dense in $C(K,\mathbb{R})$ for every compact $K \subseteq \mathbb{R}^n$, and trigonometric polynomials $\sum_{|k|\le N} c_k e^{2\pi i k t}$ are dense in the space of continuous functions $\mathbb{R} \to \mathbb{C}$ of period 1.*

Proof. The first statement is immediate. For the second, continuous functions of period 1 are continuous functions on $S^1 = \mathbb{R}/\mathbb{Z}$ (Theorem 27.7.3), and the polynomials in z and $\bar{z}$ restricted to S^1 form a conjugation-closed separating algebra; on S^1, $z^k \bar{z}^l = z^{k-l}$ corresponds to $e^{2\pi i(k-l)t}$. □

Theorem 27.14.4 (Arzelà–Ascoli)**.** *Let K be a compact Hausdorff space and $\mathcal{F} \subseteq C(K,\mathbb{R}^m)$ a family that is* equicontinuous *(for every x and $\varepsilon > 0$ there is a neighbourhood U of x with $\|f(x') - f(x)\| < \varepsilon$ for all $x' \in U$ and $f \in \mathcal{F}$) and pointwise bounded. Then $\mathcal{F}$ has compact closure in $C(K,\mathbb{R}^m)$ with the supremum metric.*

Proof. $C(K,\mathbb{R}^m)$ is complete, so by Theorem 18.7.4 it suffices to show that $\mathcal{F}$ is totally bounded. Let $\varepsilon > 0$. Choose finitely many points $x_1, \dots, x_k$ with neighbourhoods U_i as in the definition covering K. The set $\{(f(x_1), \dots, f(x_k)) : f \in \mathcal{F}\}$ is bounded in $\mathbb{R}^{mk}$, hence totally bounded, so there are $f_1, \dots, f_N \in \mathcal{F}$ such that every $f \in \mathcal{F}$ satisfies $\max_i \|f(x_i) - f_j(x_i)\| < \varepsilon$ for some j. For $x \in U_i$, $\|f(x) - f_j(x)\| \le \|f(x) - f(x_i)\| + \|f(x_i) - f_j(x_i)\| + \|f_j(x_i) - f_j(x)\| < 3\varepsilon$. □

27.15 Exercises

Exercise 27.15.1. Show that $e = \exp(1)$ is irrational.

Solution. If $e = p/q$, then $x = q!(e - \sum_{k\le q} 1/k!)$ is a positive integer, but $x = \sum_{k>q} q!/k! < \sum_{j\ge 1}(q+1)^{-j} = 1/q \le 1$. □

Exercise 27.15.2. Show that $\sum 1/n$ diverges while $\sum (-1)^n/n$ converges, and that the latter series can be rearranged to converge to any given real number.

Solution. $\sum_{2^j < n \le 2^{j+1}} 1/n \ge 1/2$, so the partial sums are unbounded; the alternating series converges by Lemma 27.5.7. For a target c, add positive terms $1/(2k)$ until the partial sum exceeds c, then negative terms until it is below c, and so on; since both the positive and the negative parts diverge and the terms tend to 0, the partial sums converge to c. This is the special case $\alpha = \beta = c$ of Theorem 27.5.9, where the details are written out. □

Exercise 27.15.3. Let $\sum_n a_n$ be a convergent series of real numbers and $\sigma : \mathbb{N} \to \mathbb{N}$ a bijection that moves every index by a bounded amount: $|\sigma(n) - n| \le K$ for all n. Show that $\sum_n a_{\sigma(n)}$ converges to the same sum. Deduce that the rearrangement of Example 27.5.11(1) moves some indices arbitrarily far.

Solution. Let $S_N = \sum_{n\le N} a_n$ and $T_N = \sum_{n\le N} a_{\sigma(n)} = \sum_{m\in\sigma(\{0,\dots,N\})} a_m$. If $m \le N-K$, then $\sigma^{-1}(m) \le m+K \le N$, so $m \in \sigma(\{0,\dots,N\})$; and every $m \in \sigma(\{0,\dots,N\})$ satisfies $m \le N+K$. So $\{0,\dots,N\}$ and $\sigma(\{0,\dots,N\})$ differ only in indices between $N-K$ and $N+K$, and $|T_N - S_N| \le \sum_{N-K<m\le N+K} |a_m|$, a sum of at most $2K$ terms, which tends to 0 as $a_m \to 0$. Hence $T_N - S_N \to 0$. The rearrangement of the example changes the sum, so it cannot move indices by a bounded amount; indeed its $3m$-th term is $-1/(4m)$, the $4m$-th term of S, moved by m places. □

Exercise 27.15.4. Show that $f_n(x) = \sqrt{x^2 + 1/n}$ are smooth and converge uniformly on $\mathbb{R}$ to $|x|$, which is not differentiable. Prove that if $f_n \to f$ pointwise and $f_n' \to g$ uniformly on an interval, with f_n continuously differentiable, then f is differentiable with $f' = g$.

Solution. $0 \le \sqrt{x^2+1/n} - |x| = \frac{1/n}{\sqrt{x^2+1/n}+|x|} \le 1/\sqrt{n}$. For the second part, $f_n(x) = f_n(a) + \int_a^x f_n'$ converges to $f(a) + \int_a^x g$, so $f(x) = f(a) + \int_a^x g$ and Theorem 27.8.3 applies. □

Exercise 27.15.5. Show that $(r,\theta) \mapsto (r\cos\theta, r\sin\theta)$ is a local diffeomorphism $(0,\infty)\times\mathbb{R} \to \mathbb{R}^2 \setminus \{0\}$, but not injective, and compute its Jacobian determinant.

Exercise 27.15.6. (Peano's theorem.) Show that if F is only continuous, $x' = F(t,x)$ still has local solutions, by applying the Arzelà–Ascoli theorem to the Euler polygons φ_k with step $1/k$.

Exercise 27.15.7. Show that a compact subset of $C(K,\mathbb{R}^m)$ is equicontinuous and pointwise bounded.

Exercise 27.15.8. Compute $\lim_n \frac{n^2+1}{2n^2-n}$, $\lim_n(\sqrt{n^2+n} - n)$ and $\lim_n (2^n + 3^n)^{1/n}$.

Solution. Dividing by n^2, the first is $\frac{1+1/n^2}{2-1/n} \to \frac{1}{2}$. For the second, $\sqrt{n^2+n} - n = \frac{n}{\sqrt{n^2+n}+n} = \frac{1}{\sqrt{1+1/n}+1} \to \frac{1}{2}$. For the third, $3 \le (2^n+3^n)^{1/n} \le (2\cdot 3^n)^{1/n} = 3\cdot 2^{1/n} \to 3$ (Proposition 27.2.9(2)), so the limit is 3 by the squeeze theorem. □

Exercise 27.15.9. Show that $\limsup_n a_n$ is the largest element of $\bar{\mathbb{R}}$ that is a limit of a subsequence of (a_n), and $\liminf_n a_n$ the smallest.

Solution. Let $S_n = \sup_{k\ge n} a_k$, a decreasing sequence with limit $L = \limsup a_n$ (Proposition 27.2.10). If $a_{n_k} \to \ell$, then $a_{n_k} \le S_n$ for $n_k \ge n$, so $\ell \le S_n$ for all n, and $\ell \le L$. Conversely, if $L \in \mathbb{R}$, choose recursively $n_k > n_{k-1}$ with $S_{n_{k-1}+1} - \frac{1}{k} < a_{n_k} \le S_{n_{k-1}+1}$; both bounds tend to L, so $a_{n_k} \to L$ by the squeeze theorem. If $L = +\infty$ the sequence is unbounded above and Theorem 27.2.12 applies; if $L = -\infty$ then $a_n \le S_n \to -\infty$. The statement for lim inf follows by passing to $(-a_n)$. □

Exercise 27.15.10. Show that $f(x) = x\mathbf{1}_{\mathbb{Q}}(x)$ is continuous at 0 and discontinuous at every $a \neq 0$.

Solution. $|f(x)| \le |x|$, so $f(x) \to 0 = f(0)$ as $x \to 0$. For $a \neq 0$, choose rational $x_n \to a$ and irrational $y_n \to a$ (Counterexample 27.3.10); then $f(x_n) = x_n \to a$ and $f(y_n) = 0$, so f has no limit at a (Theorem 27.3.4), in particular it is not continuous there. □

Exercise 27.15.11. Let $f:\mathbb{R}\to\mathbb{R}$ be continuous with $f(x+y) = f(x)+f(y)$ for all x, y. Show that $f(x) = cx$ with $c = f(1)$.

Solution. $f(0) = f(0) + f(0)$ gives $f(0) = 0$, then $f(-x) = -f(x)$, and by induction $f(nx) = nf(x)$ for $n \in \mathbb{N}$, hence for $n \in \mathbb{Z}$. For $x = p/q$ with $q \geq 1$, $qf(p/q) = f(p) = pc$, so $f(x) = cx$ on $\mathbb{Q}$. For real x, choose rational $x_n \to x$ (density of $\mathbb{Q}$); by continuity $f(x) = \lim f(x_n) = \lim cx_n = cx$. □

Exercise 27.15.12. Show that a continuous periodic function $f : \mathbb{R} \to \mathbb{R}$ is bounded and uniformly continuous.

Solution. Let $p > 0$ be a period. On $[0, 2p]$, f is bounded and uniformly continuous (Theorem 27.4.10 and Proposition 27.4.13(3)), and every value of f is taken on $[0, p]$. Given ε, let $\delta < p$ work on $[0, 2p]$. If $|x - y| < \delta$, translating both by the same multiple of p we may assume $0 \leq \min(x, y) < p$, and then $x, y \in [0, 2p]$, so $|f(x) - f(y)| < \varepsilon$. □

Exercise 27.15.13. Show that $n(a^{1/n} - 1) \to \log a$ for every $a > 0$.

Solution. Let $t = \log a$, so $a^{1/n} = e^{t/n}$. If $t = 0$ both sides vanish. Otherwise $n(e^{t/n} - 1) = t \cdot \frac{e^{t/n}-1}{t/n} \to t$ by Proposition 27.7.5(1) and Theorem 27.3.4, since $t/n \to 0$ and $t/n \neq 0$. □

Exercise 27.15.14. Compute $\int_0^\pi x^2 \sin x \, dx$, and show by integration by parts and induction on n that $\int_0^1 x^m(1 - x)^n \, dx = \frac{m!\, n!}{(m+n+1)!}$ for integers $m, n \geq 0$.

Solution. Twice by parts: $\int_0^\pi x^2 \sin x \, dx = [-x^2 \cos x]_0^\pi + 2\int_0^\pi x \cos x \, dx = \pi^2 + 2\big([x \sin x]_0^\pi - \int_0^\pi \sin x \, dx\big) = \pi^2 + 2(0 - 2) = \pi^2 - 4$. For the second, let $B(m, n)$ be the integral. For $n \geq 1$, take $f(x) = x^{m+1}/(m + 1)$ and $g(x) = (1 - x)^n$; the boundary term vanishes at both ends, and $B(m, n) = \frac{n}{m+1} B(m + 1, n - 1)$. Iterating, $B(m, n) = \frac{n(n-1)\cdots 1}{(m+1)(m+2)\cdots(m+n)} B(m + n, 0)$, and $B(m + n, 0) = \frac{1}{m+n+1}$, which gives $\frac{m!\, n!}{(m+n+1)!}$. □

Exercise 27.15.15. Let $f : I \to \mathbb{R}$ be continuous on an interval and *midpoint convex*: $f(\frac{x+y}{2}) \leq \frac{f(x)+f(y)}{2}$ for all $x, y \in I$. Show that f is convex. (Without continuity this fails: a $\mathbb{Q}$-linear map $\mathbb{R} \to \mathbb{R}$ that is not $\mathbb{R}$-linear, which exists by choosing a basis of $\mathbb{R}$ over $\mathbb{Q}$, is midpoint affine but not convex.)

Solution. By induction on k, $f(\sum_i x_i/2^k) \leq \sum_i f(x_i)/2^k$ for 2^k points $x_i \in I$ (split them into two halves and apply midpoint convexity to the two averages). Given x, y and a dyadic $t = j/2^k \in [0, 1]$, take j copies of x and $2^k - j$ copies of y: $f(tx + (1 - t)y) \leq t f(x) + (1 - t) f(y)$. Dyadic numbers are dense in $[0, 1]$, and both sides are continuous in t, so the inequality holds for all $t \in [0, 1]$. □

Exercise 27.15.16. Decide whether the following series converge: $\sum_{n\geq 1} \frac{n^n}{n!\, 3^n}$, $\sum_{n\geq 2} \frac{1}{(\log n)^{\log n}}$, $\sum_{n\geq 1} \big(1 - \frac{1}{n}\big)^{n^2}$, $\sum_{n\geq 2} \frac{(-1)^n}{\sqrt{n}+(-1)^n}$.

Solution. (1) The ratio is $\frac{1}{3}(1 + \frac{1}{n})^n \to \frac{e}{3} < 1$: converges. (2) $(\log n)^{\log n} = n^{\log \log n} \geq n^2$ once $\log \log n \geq 2$, so it converges by comparison with $\sum n^{-2}$. (3) The n-th root is $(1 - \frac{1}{n})^n \to e^{-1} < 1$: converges. (4) $\frac{(-1)^n}{\sqrt{n}+(-1)^n} = \frac{(-1)^n}{\sqrt{n}} - \frac{1}{\sqrt{n}(\sqrt{n}+(-1)^n)}$; the first series converges (alternating), the second diverges by limit comparison with $\sum \frac{1}{n}$; so the series diverges, although its terms are alternating and tend to 0 (they are not monotone in absolute value). □

Exercise 27.15.17. Find the largest volume of a box with edges parallel to the coordinate axes inscribed in the ellipsoid $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$ $(a, b, c > 0)$.

Solution. A vertex (x, y, z) with $x, y, z > 0$ on the ellipsoid determines the box, of volume $V = 8xyz$. The volume is a continuous function on the compact part of the ellipsoid with $x, y, z \geq 0$, positive somewhere and zero when a coordinate vanishes, so its maximum is attained with $x, y, z > 0$, where $\nabla h = (2x/a^2, 2y/b^2, 2z/c^2) \neq 0$. Theorem 27.12.6 gives $8yz = 2\lambda x/a^2$, $8xz = 2\lambda y/b^2$, $8xy = 2\lambda z/c^2$. Multiplying by x, y, z respectively, $x^2/a^2 = y^2/b^2 = z^2/c^2 = 4xyz/\lambda$, and the constraint makes each equal to $\frac{1}{3}$. So $(x, y, z) = (a, b, c)/\sqrt{3}$ and the maximal volume is $8abc/(3\sqrt{3})$, a fraction $2/(\sqrt{3}\pi) \approx 0.37$ of the volume $\frac{4}{3}\pi abc$ of the ellipsoid. □

Exercise 27.15.18. Find the points of the hyperbola $xy = 1$ nearest to the origin, and confirm with Example 27.12.12 that they are strict local minima of the distance.

Solution. Minimise $f = x^2 + y^2$ on $M = \{h = 0\}$, $h = xy - 1$; M is closed and $f \to \infty$ along M, so a minimum exists, and $\nabla h = (y, x) \neq 0$ on M. The equations $2x = \lambda y$, $2y = \lambda x$ give $2x^2 = \lambda xy = 2y^2$, so $y = \pm x$, and $xy = 1$ forces $y = x = \pm 1$, with $\lambda = 2$: the nearest points are $\pm(1, 1)$, at distance $\sqrt{2}$. With $L = x^2 + y^2 - 2(xy - 1)$, $L_{xx} = L_{yy} = 2$, $L_{xy} = -2$, and at $(1, 1)$, $(h_x, h_y) = (1, 1)$; the bordered determinant is $-(L_{xx} - 2L_{xy} + L_{yy}) = -8 < 0$, a strict local minimum. Note that $D^2 f = 2I$ alone would also suggest a minimum here, but $D^2 L$ is only semidefinite on $\mathbb{R}^2$; its restriction to the tangent line $\mathbb{R}(-1, 1)$ is what counts. □

Exercise 27.15.19. Prove *Hadamard's inequality*: for a real $n \times n$ matrix A with columns $a_1, \ldots, a_n$, $|\det A| \leq \|a_1\| \cdots \|a_n\|$, with equality for nonzero columns exactly when the columns are pairwise orthogonal. (Geometrically: among parallelepipeds with given edge lengths, the rectangular box has the largest volume.) Hint: maximise det on the set of matrices with unit columns.

Solution. If a column vanishes both sides are 0; otherwise divide each column by its length, which divides both sides by $\prod_j \|a_j\|$. So we must show $\det A \leq 1$ when all $\|a_j\| = 1$. The set S of such matrices is compact in $\mathbb{R}^{n^2}$, so det has a maximum on S at some A, and the maximum is $\geq \det I = 1 > 0$. The constraints $h_j(A) = \|a_j\|^2 - 1$ have gradients $2a_j$ placed in the j-th block of coordinates, which are nonzero and lie in different blocks, hence are independent. Expanding along the j-th column, $\partial \det/\partial a_{ij} = C_{ij}$, the (i, j) cofactor (Proposition 11.5.3). Theorem 27.12.6 gives $C_{ij} = 2\lambda_j a_{ij}$, that is, $C = 2A\Lambda$ with $\Lambda = \operatorname{diag}(\lambda_j)$ and $C = (C_{ij})$. Since $\det A \neq 0$, the adjugate formula $A\,C^T = \det(A)\, I$ gives $C = \det(A)(A^{-1})^T$, so $\det(A)\, I = A^T C = 2A^T A\Lambda$. Hence Λ is invertible and $A^T A = \det(A)(2\Lambda)^{-1}$ is diagonal: the columns of A are orthogonal unit vectors, A is orthogonal, and $\det A = 1$ at the maximum. For the equality case, if the columns are orthogonal, $|\det A|^2 = \det(A^T A) = \prod_j \|a_j\|^2$; conversely, equality means that the normalised matrix (with columns $a_j/\|a_j\|$, possibly after changing the sign of one column) is a maximum point, hence has orthogonal columns by the argument above. □

Exercise 27.15.20. Let $y \in \mathbb{R}^n$. Show that the point of the simplex $\Delta = \{x \in \mathbb{R}^n : x_i \geq 0, \sum_i x_i = 1\}$ nearest to y is $x_i = \max(0, y_i - \tau)$, where τ is the unique number with $\sum_i \max(0, y_i - \tau) = 1$. Compute it for $y = (0.8, 0.6, -0.5)$.

Solution. Minimise the strictly convex function $f(x) = \frac{1}{2}\|x - y\|^2$ subject to $g_i = -x_i \leq 0$ and $h = \sum_i x_i - 1 = 0$. The conditions (27.6) read $x_i - y_i - \mu_i + \lambda = 0$, $\mu_i \geq 0$, $\mu_i x_i = 0$. If $x_i > 0$ then $x_i = y_i - \lambda$; if $x_i = 0$ then $y_i - \lambda = -\mu_i \leq 0$. In both cases $x_i = \max(0, y_i - \lambda)$, and $\sum_i x_i = 1$ determines $\lambda = \tau$: the function $t \mapsto \sum_i \max(0, y_i - t)$ is continuous, decreasing, strictly so where positive, equal to 0 for $t \geq \max_i y_i$ and tending to ∞ as $t \to -\infty$. Conversely, this x with $\mu_i = \max(0, \tau - y_i)$ satisfies the conditions, so it is the unique minimum by Theorem 27.12.18 (like the water-filling of Example 27.12.19, with the vessel upside down).

For $y = (0.8, 0.6, -0.5)$, trying $x_3 = 0$: $(0.8 - \tau) + (0.6 - \tau) = 1$ gives $\tau = 0.2$, and indeed $-0.5 - 0.2 < 0$; so $x = (0.6, 0.4, 0)$. □

Chapter 28

Measure and Integration

28.1 Introduction

The Lebesgue integral extends the Riemann integral to a class of functions that is closed under the limits needed in analysis: the monotone and dominated convergence theorems (Theorems 28.4.3 and 28.4.8) and the completeness of the spaces L^p (Theorem 28.5.3) are the basis of the functional analysis used to prove the Hodge theorem. This chapter constructs Lebesgue measure, proves the convergence theorems, Littlewood's three principles (approximation of sets by boxes, Lusin's theorem, Egorov's theorem), the Lebesgue differentiation theorem with its consequences (the density theorem and the fundamental theorem of calculus for Lebesgue integrals, with the Cantor function as a counterexample), independence and the two Borel–Cantelli lemmas, with applications to the digits of random numbers and Borel's normal number theorem (Theorem 28.7.9), Fubini's theorem with the layer cake formula and the Hardy–Littlewood maximal theorem, the change of variables formula, the Radon–Nikodym theorem, and the characterisation of indefinite integrals as the absolutely continuous functions. It then treats Radon measures as positive linear functionals (the Riesz–Markov theorem, Theorem 28.14.6, and its version for signed and vector-valued functionals, Theorem 28.14.9), weak convergence of measures and its compactness theorem, and the differentiation of measures with respect to Lebesgue measure, with Lebesgue's theorem on the differentiability of monotone functions. It ends with an introduction to geometric measure theory: Hausdorff measures and Hausdorff dimension (with the Cantor set as example), box-counting dimension and self-similar sets such as the Sierpiński gasket and the Koch curve, whose dimension is computed by Moran's equation (Theorem 28.18.10), Rademacher's theorem on the differentiability of Lipschitz maps, the area formula, which computes the k-dimensional measure of parametrised sets, the coarea formula with integration in polar coordinates, and densities and rectifiable sets (Sections 28.17–28.22). References: [Rud87] and [Fol99].

Prerequisites. Chapter 27 (Real Analysis).

We use the extended half-line $[0, \infty]$ with $a + \infty = \infty$ and $0 \cdot \infty = 0$.

28.2 Measures

Why σ-algebras. We want to assign a size to subsets of $\mathbb{R}$ extending the length of intervals, additive on disjoint sets, and invariant under translation. Two features of the definition below need explanation. First, additivity is required for *countable* disjoint unions, not only finite ones: this is what makes limits work (the convergence theorems of Section 28.4), and it is natural, since for instance $[0, 1) = \bigsqcup_n [1 - 2^{-n}, 1 - 2^{-n-1})$ should have length $\sum_n 2^{-n-1} = 1$.

Second, the size is defined only on a family of "measurable" sets, not on all subsets: by the Vitali construction (Counterexample 28.3.7) there is no countably additive, translation invariant length defined on all subsets of $\mathbb{R}$. The family of measurable sets must be closed under the operations used in analysis, complements and countable unions; this is a σ-algebra.

Definition 28.2.1. A *σ-algebra* on a set X is a set $\mathcal{A}$ of subsets containing X and closed under complements and countable unions. The σ-algebra *generated* by a family of subsets is the intersection of all σ-algebras containing it; for a topological space, the *Borel* σ-algebra $\mathcal{B}(X)$ is generated by the open sets. A *measure* on $(X, \mathcal{A})$ is a map $\mu\colon \mathcal{A} \to [0, \infty]$ with $\mu(\emptyset) = 0$ and $\mu(\bigsqcup_k A_k) = \sum_k \mu(A_k)$ for countable disjoint families. It is *finite* if $\mu(X) < \infty$, *σ-finite* if X is a countable union of sets of finite measure, and *complete* if subsets of measure zero sets are measurable. A property holds *almost everywhere* (a.e.) if it holds outside a set of measure zero.

Example 28.2.2.

1. On any set X, $\{\emptyset, X\}$ and the power set $\mathcal{P}(X)$ are σ-algebras, the smallest and the largest. If $X = A_1 \sqcup \cdots \sqcup A_k$ is a finite partition, the unions of some of the A_i form a σ-algebra with 2^k elements; on a finite set every σ-algebra is of this form, generated by its minimal nonempty members.

2. The sets that are countable or have countable complement form a σ-algebra on any set: complements exchange the two kinds, and a countable union is countable unless one of the sets is cocountable, in which case the union is cocountable. On $\mathbb{R}$ it is strictly smaller than the Borel σ-algebra, since $[0, 1]$ is neither countable nor cocountable.

3. The Borel σ-algebra of $\mathbb{R}$ contains the open sets, the closed sets, the countable unions of closed sets (such as $\mathbb{Q}$ or $[0, 1)$) and the countable intersections of open sets (such as $\mathbb{R} \setminus \mathbb{Q}$), and much more; it is generated already by the intervals (a, b), or by the rays (a, ∞).

Example 28.2.3.

1. *Counting measure* on $\mathcal{P}(X)$: $\mu(A)$ is the number of elements of A, or ∞. It is σ-finite if and only if X is countable.

2. *Dirac measure* at x: $\delta_x(A) = 1$ if $x \in A$ and 0 otherwise; more generally $\sum_i c_i \delta_{x_i}$ with $c_i \geq 0$, a *discrete* measure. A fair die is the probability measure $\frac{1}{6}\sum_{k=1}^{6} \delta_k$ on $\{1, \ldots, 6\}$, and the probability of an even result is $\mu(\{2, 4, 6\}) = \frac{1}{2}$.

3. If μ is a measure and E a measurable set, $\mu_E(A) = \mu(A \cap E)$ is a measure, the restriction of μ to E; and if $f \geq 0$ is measurable, $A \mapsto \int_A f \, d\mu$ is a measure (Section 28.4), the measure with *density* f.

4. Lebesgue measure on $\mathbb{R}^n$ (Section 28.3), which assigns to boxes their volume.

Countable additivity forces continuity properties that finite additivity does not give (Proposition 28.2.4 below). For instance, for counting measure on $\mathbb{N}$ the sets $A_n = \{n, n+1, \ldots\}$ decrease to $\emptyset$ while $\mu(A_n) = \infty$ for all n: the hypothesis $\mu(A_1) < \infty$ in continuity from above cannot be dropped.

Proposition 28.2.4. *Let μ be a measure. If $A \subseteq B$, then $\mu(A) \leq \mu(B)$; $\mu(\bigcup_k A_k) \leq \sum_k \mu(A_k)$; if $A_1 \subseteq A_2 \subseteq \cdots$, then $\mu(\bigcup_k A_k) = \lim_k \mu(A_k)$; and if $A_1 \supseteq A_2 \supseteq \cdots$ with $\mu(A_1) < \infty$, then $\mu(\bigcap_k A_k) = \lim_k \mu(A_k)$.*

Proof. $\mu(B) = \mu(A) + \mu(B \setminus A)$. Replacing A_k by $A_k \setminus \bigcup_{j<k} A_j$ gives disjoint sets with the same union and smaller measures. For increasing sets, $\bigcup_k A_k = \bigsqcup_k (A_k \setminus A_{k-1})$, and the partial sums of the measures are $\mu(A_k)$. For decreasing sets apply this to $A_1 \setminus A_k$. □

Definition 28.2.5. A map $f : (X, \mathcal{A}) \to (Y, \mathcal{B})$ is *measurable* if $f^{-1}(B) \in \mathcal{A}$ for $B \in \mathcal{B}$. Functions to $[-\infty, \infty]$, $\mathbb{R}$ or $\mathbb{C}$ are measurable for the Borel σ-algebras. A *simple function* is a finite combination $\sum_i a_i \mathbf{1}_{A_i}$ of indicator functions of measurable sets.

Measurability of functions is the condition that makes integrals of f meaningful: one needs to measure the sets $\{a < f \le b\}$, on which f is approximately constant. Continuous functions on a topological space are Borel measurable, since preimages of open sets are open; monotone functions on $\mathbb{R}$ are Borel measurable, since $\{f > a\}$ is an interval; and an indicator function $\mathbf{1}_A$ is measurable if and only if A is measurable, since $\{\mathbf{1}_A > a\}$ is $\emptyset$, A or X.

Proposition 28.2.6.
1. *$f : X \to [-\infty, \infty]$ is measurable if and only if $\{f > a\}$ is measurable for all $a \in \mathbb{R}$.*
2. *If $f_1, \dots, f_k : X \to \mathbb{R}$ are measurable and $\Phi : \mathbb{R}^k \to \mathbb{R}$ is continuous, $\Phi(f_1, \dots, f_k)$ is measurable; so sums and products of measurable functions are measurable, and $f : X \to \mathbb{C}$ is measurable if and only if its real and imaginary parts are.*
3. *Pointwise* sup, inf, lim sup *and* lim inf *of sequences of measurable functions are measurable.*
4. *Every measurable $f : X \to [0, \infty]$ is the pointwise limit of an increasing sequence of simple functions $0 \le s_1 \le s_2 \le \cdots$.*

Proof. (1) The sets $(a, \infty]$ generate the Borel σ-algebra of $[-\infty, \infty]$, and the sets B with $f^{-1}(B)$ measurable form a σ-algebra. (2) Open subsets of $\mathbb{R}^k$ are countable unions of products of open intervals, and $(f_1, \dots, f_k)^{-1}$ of such a product is a finite intersection of measurable sets; so $(f_1, \dots, f_k)$ is measurable for Borel sets, and so is its composite with Φ. (3) $\{\sup_n f_n > a\} = \bigcup_n \{f_n > a\}$, $\inf_n f_n = -\sup_n(-f_n)$, and $\limsup f_n = \inf_n \sup_{k \ge n} f_k$. (4) Let $s_n = \min(n, 2^{-n}\lfloor 2^n f \rfloor)$. □

28.3 Outer measures and Lebesgue measure

Definition 28.3.1. An *outer measure* on X is a map $\mu^* : \mathcal{P}(X) \to [0, \infty]$ with $\mu^*(\emptyset) = 0$, monotone and countably subadditive. A set A is *μ^*-measurable* if $\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \setminus A)$ for all $E \subseteq X$.

Theorem 28.3.2 (Carathéodory)**.** *The μ^*-measurable sets form a σ-algebra $\mathcal{M}$, and $\mu^*|_{\mathcal{M}}$ is a complete measure.*

Proof. By subadditivity, A is measurable if and only if $\mu^*(E \cap A) + \mu^*(E \setminus A) \le \mu^*(E)$ for all E. The definition is symmetric in A and $X \setminus A$. For measurable A, B and any E,

$$\mu^*(E\cap(A\cup B))+\mu^*(E\setminus(A\cup B)) \le \mu^*(E\cap A)+\mu^*((E\setminus A)\cap B)+\mu^*((E\setminus A)\setminus B) = \mu^*(E\cap A)+\mu^*(E\setminus A) = \mu^*(E),$$

so $\mathcal{M}$ is closed under finite unions. Let $A_1, A_2, \ldots \in \mathcal{M}$ be disjoint, $B_n = A_1 \cup \cdots \cup A_n$ and $B = \bigcup_n A_n$. Measurability of A_n gives $\mu^*(E \cap B_n) = \mu^*(E \cap A_n) + \mu^*(E \cap B_{n-1})$, so $\mu^*(E \cap B_n) = \sum_{j \le n} \mu^*(E \cap A_j)$, and $\mu^*(E) = \mu^*(E \cap B_n) + \mu^*(E \setminus B_n) \ge \sum_{j \le n} \mu^*(E \cap A_j) + \mu^*(E \setminus B)$. Letting $n \to \infty$ and using subadditivity, $\mu^*(E) \ge \sum_j \mu^*(E \cap A_j) + \mu^*(E \setminus B) \ge \mu^*(E \cap B) + \mu^*(E \setminus B)$. So $B \in \mathcal{M}$, and $E = B$ gives countable additivity. General countable unions reduce to disjoint ones. If $\mu^*(A) = 0$, then $\mu^*(E \cap A) + \mu^*(E \setminus A) \le \mu^*(E)$, so A is measurable. □

Definition 28.3.3. A *box* in $\mathbb{R}^n$ is a product $Q = \prod_i [a_i, b_i]$ with volume $v(Q) = \prod_i (b_i - a_i)$. The *Lebesgue outer measure* of $A \subseteq \mathbb{R}^n$ is $\lambda^*(A) = \inf\{\sum_k v(Q_k) : A \subseteq \bigcup_k Q_k\}$, the infimum over countable covers by boxes. The λ^*-measurable sets are *Lebesgue measurable*, and $\lambda = \lambda^n$ is *Lebesgue measure*.

Theorem 28.3.4. *λ^* is an outer measure, and $\lambda^*(Q) = v(Q)$ for boxes. Every Borel set is Lebesgue measurable, so λ is a complete measure on a σ-algebra containing $\mathcal{B}(\mathbb{R}^n)$. It is translation invariant and* regular*: for measurable E, $\lambda(E) = \inf\{\lambda(U) : U \supseteq E \text{ open}\} = \sup\{\lambda(K) : K \subseteq E \text{ compact}\}$.*

Proof. Countable subadditivity: given covers of A_k with total volume $\leq \lambda^*(A_k) + \varepsilon 2^{-k}$, their union covers $\bigcup_k A_k$.

Volume of a box. Clearly $\lambda^*(Q) \leq v(Q)$. Let $Q \subseteq \bigcup_k Q_k$ and $\varepsilon > 0$. Enlarge each Q_k to a box Q_k' whose interior contains Q_k, with $v(Q_k') \leq v(Q_k) + \varepsilon 2^{-k}$. By compactness finitely many interiors cover Q, say those of $Q_1', \dots, Q_N'$. The endpoints of Q and of the Q_k' in each coordinate define a grid dividing Q into boxes C with disjoint interiors, and $v(Q) = \sum_C v(C)$ (volume is multiplicative and length is additive). The centre of each C lies in some Q_k', and then $C \subseteq Q_k'$ because no grid line crosses the interior of C. The cells contained in Q_k' have disjoint interiors, so their volumes add up to at most $v(Q_k')$. Hence $v(Q) \leq \sum_k v(Q_k') \leq \sum_k v(Q_k) + \varepsilon$.

Borel sets. The half-spaces $H = \{x_i \leq c\}$ generate $\mathcal{B}(\mathbb{R}^n)$, since open sets are countable unions of boxes with rational vertices. Given E and a cover by boxes Q_k with $\sum v(Q_k) \leq \lambda^*(E) + \varepsilon$, the boxes $Q_k \cap H$ and $Q_k \cap \{x_i \geq c\}$ cover $E \cap H$ and $E \setminus H$ and have total volume $\sum_k v(Q_k)$. So $\lambda^*(E \cap H) + \lambda^*(E \setminus H) \leq \lambda^*(E) + \varepsilon$, and H is measurable by Theorem 28.3.2.

Invariance and regularity. Translation maps boxes to boxes of the same volume. Enlarging the covering boxes slightly into open boxes gives outer regularity. For inner regularity, first let E be bounded, contained in a closed box Q. Given ε, choose open $U \supseteq Q \setminus E$ with $\lambda(U) \leq \lambda(Q \setminus E) + \varepsilon$; then $K = Q \setminus U \subseteq E$ is compact with $\lambda(E \setminus K) \leq \lambda(U \setminus (Q \setminus E)) \leq \varepsilon$. For general E, apply this to $E \cap [-m, m]^n$ and use continuity from below. □

Definition 28.3.5. Let μ be a measure on the Borel sets of a topological space X (or on a σ-algebra containing them). It is *locally finite* if every point has a neighbourhood of finite measure; *outer regular* if $\mu(E) = \inf\{\mu(U) : U \supseteq E \text{ open}\}$, *inner regular* if $\mu(E) = \sup\{\mu(K) : K \subseteq E \text{ compact}\}$, for all measurable E, and *regular* if it is both. A locally finite regular measure on a locally compact Hausdorff space is a *Radon measure*.

Example 28.3.6. Lebesgue measure is locally finite and regular (Theorem 28.3.4), so it is a Radon measure, and so is a Dirac measure δ_x. Counting measure on $\mathbb{Z}$ is Radon, but counting measure on $\mathbb{R}$ is neither locally finite nor outer regular: $\{0\}$ has measure 1, while every open set containing 0 is infinite and has measure ∞. The 1-dimensional Hausdorff measure on $\mathbb{R}^2$ (Section 28.17) is not locally finite either, since every open set has infinite length.

Counterexample 28.3.7. There are subsets of $\mathbb{R}$ that are not Lebesgue measurable. Choose (by the axiom of choice) a set $V \subseteq [0, 1]$ containing exactly one element of each class of $\mathbb{R}/\mathbb{Q}$. The translates $V + q$, $q \in \mathbb{Q} \cap [-1, 1]$, are disjoint and their union lies between $[0, 1]$ and $[-1, 2]$. If V were measurable, translation invariance and countable additivity would give $1 \leq \sum_q \lambda(V) \leq 3$, which is impossible whether $\lambda(V) = 0$ or $\lambda(V) > 0$.

Proposition 28.3.8. *Let F be the free group on two generators a, b (Chapter 7), and for $x \in \{a, a^{-1}, b, b^{-1}\}$ let $W(x)$ be the set of reduced words beginning with x. Then*

$$F = \{e\} \sqcup W(a) \sqcup W(a^{-1}) \sqcup W(b) \sqcup W(b^{-1}), \qquad F = W(a) \sqcup aW(a^{-1}) = W(b) \sqcup bW(b^{-1}).$$

So F is paradoxical*: it is the disjoint union of four pieces, two of which can be translated to give F again, and so can the other two.*

Proof. The first decomposition is the classification of reduced words by their first letter. If $w \notin W(a)$, then $a^{-1}w$ is reduced and begins with a^{-1}, so $w = a(a^{-1}w) \in aW(a^{-1})$. If $u \in W(a^{-1})$, $u = a^{-1}u'$ with u' reduced and not beginning with a, so $au = u' \notin W(a)$. Hence $aW(a^{-1})$ is exactly the complement of $W(a)$; the same for b. □

Remark 28.3.9. *The Banach–Tarski paradox* [BT24]: the closed unit ball of $\mathbb{R}^3$ is the disjoint union of finitely many pieces which can be moved by rigid motions to form two disjoint copies of the ball. The proof transfers Proposition 28.3.8 to geometry: the rotation group SO(3) contains a free group on two generators (for instance the rotations by $\arccos\frac{1}{3}$ about two perpendicular axes); it acts freely on the sphere minus a countable set, and choosing, by the axiom of choice, one point in each orbit transports the paradoxical decomposition of F to the sphere and then, along radii, to the ball, the countable exceptional sets being absorbed by a further trick. The pieces cannot be Lebesgue measurable, since rigid motions preserve measure and $\lambda(B) \neq 2\lambda(B)$; so the theorem says that, as for the Vitali set (Counterexample 28.3.7), no rotation-invariant notion of volume, even finitely additive, extends to all subsets of $\mathbb{R}^3$. On the line, Vitali's argument uses countable additivity in an essential way.

Lemma 28.3.10 (Dynkin)**.** *Let $\mathcal{P}$ be a family of subsets of X closed under finite intersections, and $\mathcal{L} \supseteq \mathcal{P}$ a family containing X, closed under complements and countable disjoint unions. Then $\mathcal{L}$ contains the σ-algebra generated by $\mathcal{P}$. Consequently two measures that agree on $\mathcal{P}$ and are finite on an increasing sequence of sets of $\mathcal{P}$ exhausting X agree on $\sigma(\mathcal{P})$.*

Proof. Let $\mathcal{L}_0$ be the smallest such family containing $\mathcal{P}$. For $A \in \mathcal{P}$, the sets B with $A \cap B \in \mathcal{L}_0$ form such a family (for complements use $A \setminus B = X \setminus ((X \setminus A) \sqcup (A \cap B))$) containing $\mathcal{P}$, hence $\mathcal{L}_0$. Repeating with $A \in \mathcal{L}_0$ shows that $\mathcal{L}_0$ is closed under finite intersections, hence under all countable unions (disjointify), so it is a σ-algebra. For the consequence, if $\mu = \nu$ on $\mathcal{P}$ and $\mu(X_m) = \nu(X_m) < \infty$, the sets B with $\mu(B \cap X_m) = \nu(B \cap X_m)$ form such a family; let $m \to \infty$. □

28.4 The integral

Definition 28.4.1. For a simple function $s = \sum_i a_i \mathbf{1}_{A_i} \geq 0$ with disjoint A_i, $\int s\, d\mu = \sum_i a_i \mu(A_i)$ (independent of the representation, by passing to a common refinement). For measurable $f : X \to [0, \infty]$, $\int f\, d\mu = \sup\{\int s\, d\mu : 0 \leq s \leq f \text{ simple}\}$, and $\int_E f = \int f \mathbf{1}_E$. A measurable $f : X \to \mathbb{C}$ is *integrable* if $\int |f| < \infty$; then $\int f = \int (\operatorname{Re} f)^+ - \int (\operatorname{Re} f)^- + i(\int (\operatorname{Im} f)^+ - \int (\operatorname{Im} f)^-)$, where $g^\pm = \max(\pm g, 0)$.

The idea of the definition. The Riemann integral cuts the *domain* into intervals; the Lebesgue integral cuts the *range*: for $f \geq 0$ and levels $0 = y_0 < y_1 < \cdots$, the simple function $\sum_i y_i \mathbf{1}_{\{y_i \leq f < y_{i+1}\}}$ lies below f, its integral is $\sum_i y_i\, \mu(\{y_i \leq f < y_{i+1}\})$, and refining the levels approximates $\int f$ from below. This only needs the measures of the level sets, which is why it works on any measure space and for very irregular functions.

Example 28.4.2.

1. *Counting measure on $\mathbb{N}$.* For $f : \mathbb{N} \to [0, \infty]$, $\int f\, d\mu = \sum_n f(n)$. If some $f(n) = \infty$ both sides are ∞. Otherwise the functions $s_N = f \mathbf{1}_{\{0,\ldots,N\}}$ are simple and satisfy $\int s_N = \sum_{n \leq N} f(n)$ and increase to f, so the monotone convergence theorem below gives the claim; a function $f : \mathbb{N} \to \mathbb{C}$ is integrable exactly when $\sum_n |f(n)| < \infty$. So series are integrals, and theorems about integrals specialise to theorems about series.
2. *Dirac measure.* $\int f\, d\delta_x = f(x)$: a simple function $s \leq f$ has $\int s\, d\delta_x = s(x) \leq f(x)$, and the simple function $f(x)\mathbf{1}_{\{x\}}$ attains it.
3. *The indicator of the rationals.* $\int_{[0,1]} \mathbf{1}_{\mathbb{Q}}\, d\lambda = \lambda(\mathbb{Q} \cap [0,1]) = 0$, as countable sets have Lebesgue measure 0. The Riemann integral of $\mathbf{1}_{\mathbb{Q}}$ does not exist: every upper sum is 1 and every lower sum is 0.

Theorem 28.4.3 (Monotone convergence)**.** *If $0 \leq f_1 \leq f_2 \leq \cdots$ are measurable and $f = \lim f_n$, then $\int f = \lim \int f_n$.*

Proof. The integrals increase and are at most $\int f$. Let $0 \le s \le f$ be simple and $0 < c < 1$, and $E_n = \{f_n \ge cs\}$, which increase to X. Then $\int f_n \ge c \int_{E_n} s$, and $A \mapsto \int_A s$ is a measure, so $\int_{E_n} s \to \int s$ (Proposition 28.2.4). Hence $\lim \int f_n \ge c \int s$ for all such c and s. □

Corollary 28.4.4. *1. For measurable $f, g \ge 0$ and $a \ge 0$, $\int (af + g) = a \int f + \int g$, and $\int \sum_n f_n = \sum_n \int f_n$.*

2. (Fatou) For measurable $f_n \ge 0$, $\int \liminf f_n \le \liminf \int f_n$.

3. The integral is linear on integrable functions and $|\int f| \le \int |f|$.

4. For measurable $f \ge 0$, $\int f = 0$ if and only if $f = 0$ a.e.; and $\mu(\{f \ge t\}) \le \frac{1}{t} \int f$ for $t > 0$ (Markov's inequality).

Proof. (1) For simple functions additivity is clear; approximate by increasing simple functions (Proposition 28.2.6(4)) and use monotone convergence. (2) Apply monotone convergence to $g_n = \inf_{k \ge n} f_k$, which satisfy $\int g_n \le \inf_{k \ge n} \int f_k$. (3) Linearity follows from (1) by decomposing into positive and negative parts. For complex f, write $\int f = |\int f| e^{i\theta}$; then $|\int f| = \int \mathrm{Re}(e^{-i\theta} f) \le \int |f|$. (4) $t\mathbf{1}_{\{f \ge t\}} \le f$; and $\{f > 0\} = \bigcup_n \{f \ge 1/n\}$. □

Remark 28.4.5. *Names.* The inequality $\mu(\{f \ge t\}) \le \frac{1}{t} \int f$ is usually called *Markov's inequality*, and some books call it Chebyshev's. *Chebyshev's inequality* proper is the special case applied to $(f - m)^2$: for a probability measure, $f \in L^2$ with mean $m = \int f$ and variance $\sigma^2 = \int (f - m)^2$,

$$\mu(\{|f - m| \ge t\}) = \mu(\{(f - m)^2 \ge t^2\}) \le \frac{\sigma^2}{t^2}.$$

More generally $\mu(\{|f| \ge t\}) \le t^{-p} \int |f|^p$ for $p > 0$. The proof is one line, $t\mathbf{1}_{\{f \ge t\}} \le f$, and it is often all one needs: for instance it shows that convergence in L^p implies convergence in measure (Proposition 28.6.3).

Proposition 28.4.6 (Jensen's inequality)**.** *Let μ be a probability measure, $I \subseteq \mathbb{R}$ an open interval, $\varphi : I \to \mathbb{R}$ convex, and f an integrable function with values in I. Then $\int f \in I$, and*

$$\varphi\Big(\int f \, d\mu\Big) \le \int \varphi \circ f \, d\mu,$$

where the right side is defined in $(-\infty, \infty]$.

Proof. Convex functions. For $s < t < u$ in I, writing $t = \lambda s + (1 - \lambda)u$, convexity $\varphi(t) \le \lambda\varphi(s) + (1 - \lambda)\varphi(u)$ is equivalent to each of the inequalities between slopes

$$\frac{\varphi(t) - \varphi(s)}{t - s} \le \frac{\varphi(u) - \varphi(s)}{u - s} \le \frac{\varphi(u) - \varphi(t)}{u - t}.$$

So for $c \in I$ the slope $(\varphi(t) - \varphi(c))/(t - c)$ is nondecreasing in $t \ne c$; with $m = \sup_{t<c}(\varphi(t) - \varphi(c))/(t - c) \le \inf_{t>c}(\varphi(t) - \varphi(c))/(t - c)$, a finite number, $\varphi(t) \ge \varphi(c) + m(t - c)$ for all $t \in I$: the graph lies above a line through $(c, \varphi(c))$. The same monotonicity shows that φ has finite one-sided derivatives, so it is continuous, and $\varphi \circ f$ is measurable.

Jensen. Let $c = \int f$. If I is bounded above by b, then $c \le b$, and $c = b$ would force $\int (b - f) = 0$ with $b - f > 0$, which is impossible; similarly at the lower end, so $c \in I$. With m as above, $\varphi(f(x)) \ge \varphi(c) + m(f(x) - c)$; the right side is integrable, so $\int \varphi \circ f$ is defined in $(-\infty, \infty]$, and integrating, using $\mu(X) = 1$, gives $\int \varphi \circ f \ge \varphi(c) + m(c - c) = \varphi(c)$. □

Example 28.4.7. 1. With the convex function $\varphi(t) = |t|^p$ on $\mathbb{R}$, $p \geq 1$, applied to $|f|$: $(\int |f|)^p \leq \int |f|^p$ for a probability measure.

2. With $\varphi = \exp$: $\exp(\int f) \leq \int e^f$.

3. *Arithmetic and geometric means.* Let μ give mass $\frac{1}{n}$ to each of n points, where f takes the values $a_1, \ldots, a_n > 0$, and $\varphi = -\log$. Then $-\log(\frac{1}{n}\sum a_i) \leq -\frac{1}{n}\sum \log a_i$, that is, $(a_1 \cdots a_n)^{1/n} \leq \frac{1}{n}(a_1 + \cdots + a_n)$.

The hypothesis $\mu(X) = 1$ matters: for Lebesgue measure on $[0, 2]$ and $f = 1$, $\varphi(t) = t^2$ gives $\varphi(\int f) = 4 > 2 = \int \varphi(f)$.

Theorem 28.4.8 (Dominated convergence)**.** *Let f_n be measurable with $f_n \to f$ a.e. and $|f_n| \leq g$ a.e. for an integrable g. Then f is integrable and $\int |f_n - f| \to 0$; in particular $\int f_n \to \int f$.*

Proof. Modifying on a null set, we may assume convergence and domination everywhere. Then $|f| \leq g$, and Fatou's lemma applied to $2g - |f_n - f| \geq 0$ gives $\int 2g \leq \int 2g - \limsup \int |f_n - f|$. □

Proposition 28.4.9. *For continuous $f : [a, b] \to \mathbb{C}$, the Lebesgue integral $\int_{[a,b]} f \, d\lambda$ equals the Riemann integral.*

Proof. For a partition P with mesh tending to 0, the step functions $\sum_i \min_{[t_{i-1},t_i]} \operatorname{Re} f \cdot \mathbf{1}_{[t_{i-1},t_i)}$ converge to $\operatorname{Re} f$ uniformly, and their Lebesgue integrals are the lower sums (Theorem 28.3.4). Apply dominated convergence, and treat the imaginary part in the same way. □

Proposition 28.4.10. *Every bounded Riemann integrable function $f : [a, b] \to \mathbb{R}$ (Definition 27.8.1) is Lebesgue measurable and Lebesgue integrable, with the same integral.*

Proof. Choose partitions $P_1, P_2, \ldots$, each refining the previous one, with $U(f, P_k) - L(f, P_k) < 1/k$ (Proposition 27.8.2(1)). Let ϕ_k and ψ_k be the step functions equal, on the interior of each interval of P_k, to the infimum and supremum of f there, and to f at the finitely many partition points. Then $\phi_k \leq f \leq \psi_k$, ϕ_k increases and ψ_k decreases (refinement), and their Lebesgue integrals are $L(f, P_k)$ and $U(f, P_k)$, since points have measure zero. The limits $\phi = \lim \phi_k$ and $\psi = \lim \psi_k$ are measurable, $\phi \leq f \leq \psi$, and by monotone convergence $\int(\psi - \phi) = \lim(U(f, P_k) - L(f, P_k)) = 0$. So $\psi = \phi$ a.e. (Corollary 28.4.4(4)), hence $f = \phi$ a.e., and f is measurable because Lebesgue measure is complete. Finally $\int f \, d\lambda = \lim \int \phi_k \, d\lambda = \lim L(f, P_k)$, which is the Riemann integral. □

Counterexample 28.4.11. The converse fails. The Dirichlet function $\mathbf{1}_{\mathbb{Q}}$ on $[0, 1]$ is Lebesgue integrable with integral 0, since $\mathbb{Q}$ is countable and hence of measure zero. It is not Riemann integrable: every nondegenerate interval contains rational and irrational points, so $L(\mathbf{1}_{\mathbb{Q}}, P) = 0$ and $U(\mathbf{1}_{\mathbb{Q}}, P) = 1$ for every partition P. It is the increasing pointwise limit of the Riemann integrable functions $\mathbf{1}_{\{q_1,\ldots,q_n\}}$, each of integral 0, for an enumeration of $\mathbb{Q} \cap [0, 1]$: the Riemann integral does not behave well under monotone limits, which is what the Lebesgue integral repairs (Theorem 28.4.3).

Proposition 28.4.12 (Improper integrals)**.** *Let $f : [a, \infty) \to \mathbb{R}$ be continuous, and call $\lim_{b\to\infty} \int_a^b f$, when it exists, the* improper (Riemann) integral $\int_a^\infty f$.

1. *If $f \geq 0$, then $\int_{[a,\infty)} f \, d\lambda = \lim_{b\to\infty} \int_a^b f$ in $[0, \infty]$.*

2. *$f \in L^1([a, \infty))$ if and only if $\int_a^\infty |f| < \infty$, and then the improper integral exists and equals the Lebesgue integral.*

3. *The improper integral can exist without f being Lebesgue integrable:* $\int_\pi^\infty \frac{\sin x}{x}\,dx$ *converges, but* $\int_\pi^\infty \frac{|\sin x|}{x}\,dx = \infty$.

The same holds for continuous functions on $(a,b]$ with a singularity at a; for instance $\int_0^1 x^{-p}\,dx$ is finite if and only if $p < 1$.

Proof. (1) $f\mathbf{1}_{[a,b]}$ increases to f as $b \to \infty$ (along any sequence $b_n \uparrow \infty$), and $\int_{[a,b]} f\,d\lambda = \int_a^b f$ (Proposition 28.4.9); apply monotone convergence. (2) The first statement is (1) for $|f|$. Then $|f\mathbf{1}_{[a,b]}| \le |f|$, and dominated convergence gives $\int_a^b f \to \int f\,d\lambda$. (3) On $[k\pi,(k+1)\pi]$ the integral of $|\sin x|/x$ is at least $\frac{2}{(k+1)\pi}$, and $\sum_k \frac{1}{k+1} = \infty$. The integrals $c_k = \int_{k\pi}^{(k+1)\pi} \frac{\sin x}{x}\,dx$ alternate in sign, decrease in absolute value (substitute $x \mapsto x+\pi$) and tend to 0, so $\sum_k c_k$ converges (Lemma 27.5.7); and for b between $k\pi$ and $(k+1)\pi$, $\int_\pi^b$ differs from $\sum_{j<k} c_j$ by at most $|c_k| \to 0$. For x^{-p}, $\int_\varepsilon^1 x^{-p}\,dx = \frac{1-\varepsilon^{1-p}}{1-p}$ for $p \ne 1$ and $-\log\varepsilon$ for $p = 1$. □

So for nonnegative functions the improper integral is the Lebesgue integral, and in general Lebesgue integrability is absolute convergence of the improper integral; the convergent improper integrals of non-integrable functions are the analogue of conditionally convergent series.

Proposition 28.4.13 (Differentiation under the integral sign)**.** *Let (X,μ) be a measure space, $I \subseteq \mathbb{R}$ an open interval, and $f\colon X \times I \to \mathbb{R}$ such that $f(\cdot,t)$ is integrable for every t, $f(x,\cdot)$ is differentiable for every x, and $|\partial_t f(x,t)| \le g(x)$ for an integrable g independent of t. Then $F(t) = \int_X f(x,t)\,d\mu(x)$ is differentiable, with $F'(t) = \int_X \partial_t f(x,t)\,d\mu(x)$.*

Proof. For $h_n \to 0$, $(f(x,t+h_n) - f(x,t))/h_n \to \partial_t f(x,t)$, and by the mean value theorem these quotients are bounded by $g(x)$. So $\partial_t f(\cdot,t)$ is measurable, as a limit of measurable functions, and dominated convergence gives $(F(t+h_n) - F(t))/h_n \to \int \partial_t f(x,t)\,d\mu$. □

Example 28.4.14. *Feynman's trick.* Introducing a parameter and differentiating under the integral sign computes integrals that have no elementary antiderivative.

1. $F(t) = \int_0^1 \frac{x^t-1}{\log x}\,dx$ for $t > -1$: here $\partial_t \frac{x^t-1}{\log x} = x^t$, dominated by x^s for $t \ge s > -1$, so $F'(t) = \int_0^1 x^t\,dx = \frac{1}{t+1}$, and $F(0) = 0$ gives $F(t) = \log(1+t)$.

2. $F(t) = \int_{\mathbb{R}} e^{-x^2}\cos(tx)\,dx$. Differentiating, with domination by $|x|e^{-x^2}$, and integrating by parts, $F'(t) = -\int_{\mathbb{R}} xe^{-x^2}\sin(tx)\,dx = -\frac{t}{2}F(t)$; so $F(t) = F(0)e^{-t^2/4} = \sqrt{\pi}\,e^{-t^2/4}$ by Example 28.11.3. This is the Fourier transform of the Gaussian.

3. *The Dirichlet integral* $\int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2}$ (an improper integral, Proposition 28.4.12(3)). For $t > 0$ let $F(t) = \int_0^\infty e^{-tx}\frac{\sin x}{x}\,dx$, a Lebesgue integral. On $t \ge \delta > 0$ the derivative $-e^{-tx}\sin x$ is dominated by $e^{-\delta x}$, so $F'(t) = -\int_0^\infty e^{-tx}\sin x\,dx = -\operatorname{Im}\int_0^\infty e^{-(t-i)x}\,dx = -\operatorname{Im}\frac{1}{t-i} = -\frac{1}{1+t^2}$. Since $|\sin x/x| \le 1$, $|F(t)| \le 1/t \to 0$ as $t \to \infty$; hence $F(t) = \frac{\pi}{2} - \arctan t$, with arctan the inverse of $\tan\colon (-\frac{\pi}{2},\frac{\pi}{2}) \to \mathbb{R}$. Finally $F(t) \to \int_0^\infty \frac{\sin x}{x}$ as $t \to 0^+$: for a nonincreasing C^1 function $u \ge 0$, integrating by parts, $|\int_R^S u(x)\sin x\,dx| \le 2u(R)$; with $u(x) = e^{-tx}/x$ this bounds the tails beyond R by $2/R$ for all $t \ge 0$, while $|\int_0^R (e^{-tx}-1)\frac{\sin x}{x}\,dx| \le \int_0^R tx\,dx = \frac{tR^2}{2}$. Choosing first R large and then t small gives the limit.

28.5 Lebesgue spaces

Definition 28.5.1. For $1 \le p < \infty$, $L^p(\mu)$ is the space of measurable $f : X \to \mathbb{C}$ with $\|f\|_p = (\int |f|^p)^{1/p} < \infty$, modulo equality a.e. $L^\infty(\mu)$ consists of the essentially bounded functions with $\|f\|_\infty = \inf\{c : |f| \le c \text{ a.e.}\}$, modulo equality a.e. On $L^2(\mu)$ we have the inner product $\langle f, g\rangle = \int f\bar{g}$.

Proposition 28.5.2. *Let $1 \le p, q \le \infty$ with $1/p + 1/q = 1$. Then $\int |fg| \le \|f\|_p \|g\|_q$ (Hölder) and $\|f + g\|_p \le \|f\|_p + \|g\|_p$ (Minkowski). So $L^p(\mu)$ is a normed space.*

Proof. The cases $p = 1$ or ∞ are clear. For $1 < p < \infty$ and $a, b > 0$, concavity of log ($\log'' < 0$) gives $\log(\frac{a^p}{p} + \frac{b^q}{q}) \ge \log a + \log b$, that is, $ab \le a^p/p + b^q/q$. If $\|f\|_p = \|g\|_q = 1$, integrating $|fg| \le |f|^p/p + |g|^q/q$ gives $\int |fg| \le 1$; the general case follows by scaling. For Minkowski, $|f + g|^p \le |f||f + g|^{p-1} + |g||f + g|^{p-1}$ and Hölder with $(p - 1)q = p$ give $\|f + g\|_p^p \le (\|f\|_p + \|g\|_p)\|f + g\|_p^{p-1}$; also $|f + g|^p \le 2^p(|f|^p + |g|^p)$ shows $\|f + g\|_p < \infty$. Functions with $\|f\|_p = 0$ vanish a.e. by Corollary 28.4.4(4). □

Theorem 28.5.3 (Riesz–Fischer)**.** *$L^p(\mu)$ is a Banach space for $1 \le p \le \infty$, and every convergent sequence in L^p, $p < \infty$, has a subsequence converging a.e.*

Proof. Let $p < \infty$ and (f_n) Cauchy. Choose $n_1 < n_2 < \cdots$ with $\|f_{n_{k+1}} - f_{n_k}\|_p < 2^{-k}$, and let $g = \sum_k |f_{n_{k+1}} - f_{n_k}|$. By monotone convergence and Minkowski, $\|g\|_p \le 1$, so $g < \infty$ a.e., and $f = f_{n_1} + \sum_k (f_{n_{k+1}} - f_{n_k})$ converges absolutely a.e. Since $|f - f_{n_k}| \le \sum_{j \ge k} |f_{n_{j+1}} - f_{n_j}|$, Minkowski gives $\|f - f_{n_k}\|_p \le 2^{1-k}$, and a Cauchy sequence with a convergent subsequence converges. For $p = \infty$, outside a null set $|f_n - f_m| \le \|f_n - f_m\|_\infty$ for all n, m, so (f_n) converges uniformly there. □

Proposition 28.5.4. *For $1 \le p < \infty$, the continuous functions with compact support are dense in $L^p(\mathbb{R}^n, \lambda)$.*

Proof. By Proposition 28.2.6(4) and dominated convergence, simple functions $\sum a_i \mathbf{1}_{E_i}$ with $\lambda(E_i) < \infty$ are dense. Given E with $\lambda(E) < \infty$ and $\varepsilon > 0$, choose compact $K \subseteq E \subseteq U$ open with $\lambda(U \setminus K) < \varepsilon$ (Theorem 28.3.4) and, by Lemma 18.9.5, a continuous $\phi : \mathbb{R}^n \to [0, 1]$ with compact support in U and $\phi = 1$ on K. Then $\|\mathbf{1}_E - \phi\|_p^p \le \lambda(U \setminus K) < \varepsilon$. □

28.6 Modes of convergence

Definition 28.6.1. Let f_n, f be measurable functions on a measure space (X, μ) with values in $\mathbb{C}$. Then $f_n \to f$ *almost everywhere* if $f_n(x) \to f(x)$ for x outside a null set; *in measure* if $\mu(\{|f_n - f| > \varepsilon\}) \to 0$ for every $\varepsilon > 0$; and *in L^p* if $\|f_n - f\|_p \to 0$.

Theorem 28.6.2 (Egorov)**.** *Let $\mu(X) < \infty$ and $f_n \to f$ almost everywhere. For every $\delta > 0$ there is a measurable E with $\mu(X \setminus E) < \delta$ such that $f_n \to f$ uniformly on E.*

Proof. For $k, n \ge 1$ let $E_{k,n} = \bigcup_{m \ge n} \{|f_m - f| \ge 1/k\}$. For fixed k these sets decrease in n, and their intersection consists of points where $|f_m - f| \ge 1/k$ for infinitely many m, a null set. As $\mu(X) < \infty$, $\mu(E_{k,n}) \to 0$ as $n \to \infty$ (Proposition 28.2.4). Choose n_k with $\mu(E_{k,n_k}) < \delta 2^{-k}$ and let $E = X \setminus \bigcup_k E_{k,n_k}$. Then $\mu(X \setminus E) < \delta$, and on E, $|f_m - f| < 1/k$ for all $m \ge n_k$, for every k. □

Proposition 28.6.3.

1. *If $f_n \to f$ in L^p for some $1 \le p < \infty$, then $f_n \to f$ in measure.*
2. *If $f_n \to f$ in measure, a subsequence converges to f almost everywhere.*

3. *If $\mu(X) < \infty$ and $f_n \to f$ almost everywhere, then $f_n \to f$ in measure.*

Proof. (1) By Markov's inequality (Corollary 28.4.4(4)) applied to $|f_n - f|^p$, $\mu(\{|f_n - f| > \varepsilon\}) \le \varepsilon^{-p}\|f_n - f\|_p^p$. (2) Choose $n_1 < n_2 < \cdots$ with $\mu(A_k) < 2^{-k}$, where $A_k = \{|f_{n_k} - f| > 2^{-k}\}$. The set A of points lying in infinitely many A_k is contained in $\bigcup_{k\ge K} A_k$ for every K, so $\mu(A) \le \sum_{k\ge K} 2^{-k} = 2^{1-K}$ for all K, and $\mu(A) = 0$. For $x \notin A$, $|f_{n_k}(x) - f(x)| \le 2^{-k}$ for large k. (3) Given $\varepsilon, \delta > 0$, Egorov's theorem gives E with $\mu(X \setminus E) < \delta$ and $|f_n - f| < \varepsilon$ on E for large n, so $\mu(\{|f_n - f| > \varepsilon\}) < \delta$ for large n. □

Counterexample 28.6.4. No other implications hold in general. Take $X = [0,1]$ or $\mathbb{R}$ with Lebesgue measure.

1. $f_n = n\mathbf{1}_{(0,1/n)}$ on $[0,1]$ converges to 0 everywhere, but $\int f_n = 1$: almost everywhere convergence and convergence in measure do not imply L^1 convergence, even on a finite measure space.

2. $f_n = \mathbf{1}_{[n,n+1]}$ on $\mathbb{R}$ converges to 0 everywhere but not in measure, as $\lambda(\{f_n > \frac{1}{2}\}) = 1$: part (3) of Proposition 28.6.3 and Egorov's theorem need $\mu(X) < \infty$.

3. (The typewriter sequence.) For $n = 2^k + j$ with $0 \le j < 2^k$ let $f_n = \mathbf{1}_{[j2^{-k},(j+1)2^{-k}]}$ on $[0,1]$. Then $\|f_n\|_1 = 2^{-k} \to 0$, so $f_n \to 0$ in L^1 and in measure; but every $x \in [0,1]$ lies in one of the intervals for each k and outside most of them, so $f_n(x) = 1$ and $f_n(x) = 0$ both happen infinitely often, and $(f_n(x))$ converges for no x. The subsequence in part (2) of Proposition 28.6.3 is necessary.

28.7 Independence and the Borel–Cantelli lemmas

Measure theory is also the language of probability. A *probability space* is a measure space $(\Omega, \mathcal{F}, P)$ with $P(\Omega) = 1$; its measurable sets are called *events*, measurable functions are *random variables*, and "almost everywhere" becomes *almost surely*. A typical question asks whether infinitely many of a sequence of events happen. For events $A_1, A_2, \ldots$ the event "A_k happens for infinitely many k" is

$$\limsup_k A_k = \bigcap_{n\ge1} \bigcup_{k\ge n} A_k = \{\omega : \omega \in A_k \text{ for infinitely many } k\},$$

and its complement is $\liminf_k A_k^c = \bigcup_n \bigcap_{k\ge n} A_k^c$, "from some index on, no A_k happens". The two Borel–Cantelli lemmas decide, in many cases, whether $\limsup_k A_k$ has probability 0 or 1. The first needs nothing; the second needs independence, which we now define.

Definition 28.7.1. Events $A_1, A_2, \ldots$ (finitely or countably many) in a probability space are *independent* if

$$P(A_{k_1} \cap A_{k_2} \cap \cdots \cap A_{k_m}) = P(A_{k_1})P(A_{k_2}) \cdots P(A_{k_m})$$

for all m and all distinct indices $k_1, \ldots, k_m$.

The condition must be imposed on all subfamilies, not only on pairs.

Example 28.7.2. Toss two fair coins: $\Omega = \{HH, HT, TH, TT\}$, each point with probability $\frac{1}{4}$. Let A = "the first coin shows heads", B = "the second coin shows heads", C = "the two coins agree". Each event has probability $\frac{1}{2}$, and each intersection of two of them has probability $\frac{1}{4}$; so they are pairwise independent. But $A \cap B \cap C = \{HH\}$ has probability $\frac{1}{4} \ne \frac{1}{8}$: if two of the events happen, so does the third.

Lemma 28.7.3. *If $A_1, A_2, \ldots$ are independent, so are the events obtained by replacing any of the A_k by their complements.*

Proof. It suffices to replace one event, say A_1, by A_1^c; replacing several is done one at a time. Let $k_2, \ldots, k_m$ be distinct indices different from 1 and $B = A_{k_2} \cap \cdots \cap A_{k_m}$ (with $B = \Omega$ if $m = 1$). Then $P(A_1^c \cap B) = P(B) - P(A_1 \cap B) = P(B) - P(A_1)P(B) = P(A_1^c)P(B)$, and $P(B)$ is the product of the $P(A_{k_j})$. Intersections not involving A_1^c are unchanged. □

The basic example of independence lives on $[0,1)$ with Lebesgue measure: the binary digits of a number chosen at random behave like independent tosses of a fair coin. For $x \in [0,1)$ and $k \geq 1$ let

$$d_k(x) = \lfloor 2^k x \rfloor - 2\lfloor 2^{k-1} x \rfloor \in \{0,1\},$$

the k-th digit of the binary expansion $x = \sum_k d_k(x) 2^{-k}$ that does not end with infinitely many 1s. For $K \geq 1$ the integer $\lfloor 2^K x \rfloor$ has binary digits $d_1(x), \ldots, d_K(x)$, so $d_1(x) = \delta_1, \ldots, d_K(x) = \delta_K$ holds exactly on the interval $[N2^{-K}, (N+1)2^{-K})$ with $N = \sum_{k \leq K} \delta_k 2^{K-k}$, of length 2^{-K}. Call a set $E \subseteq [0,1)$ *determined by* a finite set S of positions if whether $x \in E$ depends only on the digits $d_k(x)$, $k \in S$.

Lemma 28.7.4. 1. *For a finite set S of positions and prescribed digits $\delta_k \in \{0,1\}$, $k \in S$, the set $\{x : d_k(x) = \delta_k \text{ for } k \in S\}$ has measure $2^{-|S|}$.*

2. *If $E_1, \ldots, E_m$ are determined by pairwise disjoint finite sets of positions $S_1, \ldots, S_m$, then $\lambda(E_1 \cap \cdots \cap E_m) = \lambda(E_1) \cdots \lambda(E_m)$. Consequently, events $E_1, E_2, \ldots$ determined by pairwise disjoint finite sets of positions are independent; in particular the events $\{d_k = 1\}$, $k \geq 1$, are independent of probability $\frac{1}{2}$.*

3. *If $g_i = G_i\big((d_k)_{k \in S_i}\big)$ for functions G_i of finitely many digits and pairwise disjoint $S_1, \ldots, S_m$, then $\int_0^1 g_1 \cdots g_m \, d\lambda = \prod_i \int_0^1 g_i \, d\lambda$.*

Proof. (1) Let $K = \max S$ (the case $S = \emptyset$ is trivial). The set is the disjoint union, over the $2^{K-|S|}$ choices of the digits at the positions in $\{1, \ldots, K\} \setminus S$, of intervals of length 2^{-K}, so its measure is $2^{K-|S|}2^{-K}$.

(2) A set determined by S_i is a disjoint union of n_i sets as in (1), one for each admissible pattern of digits on S_i, so $\lambda(E_i) = n_i 2^{-|S_i|}$. With $S = S_1 \cup \cdots \cup S_m$, the intersection $E_1 \cap \cdots \cap E_m$ is the disjoint union of the sets of (1) for the patterns on S whose restriction to each S_i is admissible for E_i; there are $n_1 \cdots n_m$ of them, so its measure is $n_1 \cdots n_m 2^{-|S|} = \prod_i n_i 2^{-|S_i|}$. A subfamily of the E_k is again determined by disjoint sets of positions, which gives independence.

(3) Each g_i is a finite linear combination of indicator functions of sets determined by S_i; expand the product and apply (2) to each term. □

Theorem 28.7.5 (Borel–Cantelli lemmas)**.** *Let $A_1, A_2, \ldots$ be events in a probability space.*

1. *If $\sum_k P(A_k) < \infty$, then $P(\limsup_k A_k) = 0$: almost surely only finitely many A_k happen.*

2. *If the A_k are independent and $\sum_k P(A_k) = \infty$, then $P(\limsup_k A_k) = 1$: almost surely infinitely many A_k happen.*

Proof. (1) For every n, $\limsup_k A_k \subseteq \bigcup_{k \geq n} A_k$, so $P(\limsup_k A_k) \leq \sum_{k \geq n} P(A_k)$, the tail of a convergent series, which tends to 0 as $n \to \infty$.

(2) We show that the complement $\bigcup_n \bigcap_{k \geq n} A_k^c$ is null; it suffices to show $P(\bigcap_{k \geq n} A_k^c) = 0$ for each n. For $N \geq n$, the events A_k^c are independent (Lemma 28.7.3), and $1 - t \leq e^{-t}$ for real t (the exponential lies above its tangent line at 0), so

$$P\Big(\bigcap_{k=n}^{N} A_k^c\Big) = \prod_{k=n}^{N} \big(1 - P(A_k)\big) \leq \exp\Big(-\sum_{k=n}^{N} P(A_k)\Big) \longrightarrow 0 \qquad (N \to \infty).$$

These sets decrease in N to $\bigcap_{k\ge n} A_k^c$, so by continuity of measures (Proposition 28.2.4) its probability is 0. □

For independent events $\limsup_k A_k$ therefore has probability 0 or 1, never anything in between: a first instance of a *zero-one law*. The first lemma was already used, without the name, in the proof of Proposition 28.6.3(2).

Counterexample 28.7.6. Independence cannot be dropped in (2). On $[0,1)$ let $A_k = [0, 1/k)$. Then $\sum_k \lambda(A_k) = \sum_k 1/k = \infty$, but the sets decrease, so $\limsup_k A_k = \bigcap_k A_k = \{0\}$, a null set. The events are far from independent: $\lambda(A_j \cap A_k) = 1/\max(j,k)$, not $1/(jk)$. Even more simply, $A_k = A$ for all k with $\lambda(A) = \frac{1}{2}$ gives $\limsup_k A_k = A$, of probability $\frac{1}{2}$.

Example 28.7.7. *Every pattern appears infinitely often.* Fix a block $\beta = (\beta_1, \dots, \beta_m)$ of binary digits. For $j \ge 0$ let B_j be the event that the digits at positions $jm+1, \dots, jm+m$ form the block β. These events are determined by the disjoint sets of positions $\{jm+1, \dots, jm+m\}$, so they are independent, each of probability 2^{-m} (Lemma 28.7.4). Since $\sum_j 2^{-m} = \infty$, almost every $x \in [0,1)$ lies in infinitely many B_j. There are countably many blocks, so for almost every x every finite block of digits occurs infinitely often in the binary expansion of x. (Encoding letters by blocks of digits, this is the "infinite monkey theorem": a random infinite text contains every book infinitely often.)

The probability of a long block is small, and this limits how long the runs of 1s in a random expansion can be. Let $R_n(x)$ be the length of the run of 1s starting at position n: the largest $r \ge 0$ with $d_n(x) = \cdots = d_{n+r-1}(x) = 1$, which is finite because the expansion does not end with infinitely many 1s.

Proposition 28.7.8 (Long runs). *For almost every $x \in [0,1)$,* $\displaystyle\limsup_{n\to\infty} \frac{R_n(x)}{\log_2 n} = 1.$

Proof. Upper bound. For $r > 0$, $R_n \ge r$ means that the $\lceil r \rceil$ digits from position n on are 1, so $\lambda(R_n \ge r) = 2^{-\lceil r \rceil} \le 2^{-r}$ by Lemma 28.7.4(1). For $\varepsilon > 0$ and $r = (1+\varepsilon)\log_2 n$ this is at most $n^{-1-\varepsilon}$, which is summable. By Theorem 28.7.5(1), for almost every x we have $R_n(x) < (1+\varepsilon)\log_2 n$ for all large n, so the lim sup is at most $1+\varepsilon$. Let $\varepsilon = 1/m$, $m = 1, 2, \dots$, and discard countably many null sets.

Lower bound. We need independent events, so we use disjoint windows. For $j \ge 2$ let $L_j = \lfloor \log_2 j \rfloor \ge 1$, $s_2 = 1$ and $s_{j+1} = s_j + L_j$, so that the windows $W_j = \{s_j, \dots, s_j + L_j - 1\}$ are disjoint. Let E_j be the event that all digits in W_j are 1; then $E_j \subseteq \{R_{s_j} \ge L_j\}$, and $\lambda(E_j) = 2^{-L_j} \ge 1/j$. The E_j are independent (Lemma 28.7.4(2)) and $\sum_j 1/j = \infty$, so by Theorem 28.7.5(2) almost every x lies in infinitely many E_j. Now $s_j = 1 + \sum_{2\le i<j} L_i \le j \log_2 j$, so $\log_2 s_j \le \log_2 j + \log_2 \log_2 j$, while $L_j \ge \log_2 j - 1$. Hence for such x,

$$\frac{R_{s_j}(x)}{\log_2 s_j} \ge \frac{\log_2 j - 1}{\log_2 j + \log_2 \log_2 j}$$

for infinitely many j, and the right side tends to 1. So the lim sup is at least 1. □

So among the first n tosses of a fair coin one should expect a run of heads of length about $\log_2 n$, and no essentially longer one: about 10 in a thousand tosses, 20 in a million. This is a form of the *Erdős–Rényi law of long runs*; people asked to write down "random" sequences typically avoid such long runs, which is a simple way to detect fake randomness.

The most famous application of the first lemma is Borel's theorem that almost every number is *normal*: its digits appear with the frequencies one expects. We prove the simplest case.

Theorem 28.7.9 (Borel's normal number theorem)**.** *For almost every $x \in [0,1)$,*

$$\lim_{n\to\infty} \frac{d_1(x) + d_2(x) + \cdots + d_n(x)}{n} = \frac{1}{2}.$$

Proof. Let $\rho_k = 2d_k - 1 \in \{-1, 1\}$ and $S_n = \rho_1 + \cdots + \rho_n$; the claim is $S_n/n \to 0$ almost everywhere.

Step 1: the fourth moment. Expanding, $\int_0^1 S_n^4 = \sum_{i,j,k,l \le n} \int_0^1 \rho_i\rho_j\rho_k\rho_l$. Since $\rho^2 = 1$, a product $\rho_i\rho_j\rho_k\rho_l$ in which some index occurs an odd number of times reduces to a product ρ_a or $\rho_a\rho_b$ ($a \neq b$), whose integral is $\int \rho_a \cdot \int \rho_b = 0$ by Lemma 28.7.4(3), because $\int_0^1 \rho_a = \frac{1}{2} - \frac{1}{2} = 0$. The remaining terms equal 1: either all four indices are equal (n terms), or they take two different values, each twice ($3n(n-1)$ terms: choose which of the three pairings of the four places carry equal indices, then the two values in order). So $\int_0^1 S_n^4 = n + 3n(n-1) \le 3n^2$.

Step 2: Borel–Cantelli. Let $\varepsilon > 0$. By Markov's inequality (Corollary 28.4.4(4)) applied to S_n^4,

$$\lambda\big(|S_n| \ge \varepsilon n\big) \le \frac{1}{\varepsilon^4 n^4} \int_0^1 S_n^4 \le \frac{3}{\varepsilon^4 n^2},$$

which is summable in n. By Theorem 28.7.5(1), for almost every x we have $|S_n(x)| < \varepsilon n$ for all large n. Taking $\varepsilon = 1/m$ for all m and discarding countably many null sets, $S_n/n \to 0$ almost everywhere. Since $d_k = (1 + \rho_k)/2$, the average of the first n digits is $\frac{1}{2} + S_n/(2n) \to \frac{1}{2}$. □

Remark 28.7.10. The same argument, applied to the blocks of Example 28.7.7 and to expansions in any base $b \ge 2$, shows that almost every real number is *normal in every base*: in base b, each block of m digits occurs with limiting frequency b^{-m} [Bor09]. The theorem is the strong law of large numbers for fair coin tosses. It is a striking example of an existence proof by measure: almost all numbers are normal, yet it is not known whether $\sqrt{2}$, e or π is normal in any base. Explicit normal numbers can be constructed, such as Champernowne's number $0.123456789101112\ldots$, obtained by writing the positive integers one after another, which is normal in base 10 [Cha33].

Example 28.7.11. *Almost no number is very well approximable.* By Dirichlet's approximation theorem (Exercise 28.23.24), every irrational x has infinitely many approximations p/q with $|x - p/q| < 1/q^2$. The exponent 2 cannot be improved for almost all x. Fix $\varepsilon > 0$ and for $q \ge 1$ let A_q be the set of $x \in [0,1]$ for which some integer p satisfies $|x - p/q| < q^{-2-\varepsilon}$. Only $p = -1, 0, \ldots, q+1$ can occur, so A_q is covered by $q+3$ intervals of length $2q^{-2-\varepsilon}$, and $\lambda(A_q) \le 2(q+3)q^{-2-\varepsilon} \le 8q^{-1-\varepsilon}$, which is summable. By Theorem 28.7.5(1), almost every x lies in only finitely many A_q, and hence has only finitely many approximations with $|x - p/q| < q^{-2-\varepsilon}$. In particular the *Liouville numbers*, the irrational x that for every n have an approximation p/q with $q \ge 2$ and $|x - p/q| < q^{-n}$, form a null set. An example is $\sum_{k\ge1} 10^{-k!}$: cutting the series after the term $k = m$ gives p/q with $q = 10^{m!}$ and error less than $2 \cdot 10^{-(m+1)!} = 2q^{-(m+1)}$.

28.8 Littlewood's principles and Lusin's theorem

Littlewood summarised the theory of Lebesgue measure on $\mathbb{R}^n$ in three principles: every measurable set of finite measure is nearly a finite union of boxes; every measurable function is nearly continuous; every convergent sequence of measurable functions is nearly uniformly convergent. The third is Egorov's theorem (Theorem 28.6.2); this section makes the first two precise. Throughout, λ is Lebesgue measure on $\mathbb{R}^n$ and $E \mathbin{\triangle} A = (E \setminus A) \cup (A \setminus E)$ is the symmetric difference.

Lemma 28.8.1. *Every open set $W \subseteq \mathbb{R}^n$ is a countable union of closed* dyadic cubes $\prod_i [k_i 2^{-j}, (k_i + 1)2^{-j}]$ *($j \in \mathbb{N}$, $k_i \in \mathbb{Z}$) with pairwise disjoint interiors, and $\lambda(W)$ is the sum of their volumes.*

Proof. Two dyadic cubes of side at most 1 are either nested or have disjoint interiors, and a dyadic cube of side at most 1 is contained in only finitely many dyadic cubes of side at most 1. Every $x \in W$ lies in a dyadic cube of side at most 1 contained in W, since W contains a ball around x. Let (Q_k) be the dyadic cubes of side at most 1 contained in W that are maximal for inclusion among these; every dyadic cube of side at most 1 in W lies in one of them, so they cover W, and distinct maximal cubes have disjoint interiors. There are countably many dyadic cubes. The boundary of a cube is a finite union of boxes of volume 0, a null set (Theorem 28.3.4), so the interiors, which are disjoint, have the same measures as the cubes, and $\lambda(W) = \sum_k \lambda(Q_k)$ by countable additivity. □

Proposition 28.8.2 (First principle)**.** *Let $E \subseteq \mathbb{R}^n$ be Lebesgue measurable with $\lambda(E) < \infty$, and $\varepsilon > 0$. There is a finite union A of boxes with $\lambda(E \triangle A) < \varepsilon$.*

Proof. By Theorem 28.3.4 there is an open $W \supseteq E$ with $\lambda(W \setminus E) < \varepsilon/2$. Write $W = \bigcup_k Q_k$ as in Lemma 28.8.1. Since $\sum_k \lambda(Q_k) = \lambda(W) < \infty$, there is N with $\sum_{k>N} \lambda(Q_k) < \varepsilon/2$. Let $A = Q_1 \cup \cdots \cup Q_N$. Then $\lambda(A \setminus E) \le \lambda(W \setminus E) < \varepsilon/2$ and $\lambda(E \setminus A) \le \lambda(W \setminus A) \le \sum_{k>N} \lambda(Q_k) < \varepsilon/2$. □

Theorem 28.8.3 (Lusin)**.** *Let $E \subseteq \mathbb{R}^n$ be Lebesgue measurable with $\lambda(E) < \infty$, let $f : E \to \mathbb{C}$ be measurable, and $\varepsilon > 0$. There is a compact set $K \subseteq E$ with $\lambda(E \setminus K) < \varepsilon$ such that the restriction $f|_K$ is continuous. Moreover there is $g \in C_c(\mathbb{R}^n)$ with $g = f$ on K, and if $|f| \le c$ on E we can take $|g| \le c$.*

Proof. Step 1: indicator functions. Let $f = \mathbf{1}_A$ with $A \subseteq E$ measurable. By inner regularity (Theorem 28.3.4) there are compact sets $K_1 \subseteq A$ and $K_2 \subseteq E \setminus A$ with $\lambda(A \setminus K_1) < \varepsilon/2$ and $\lambda((E \setminus A) \setminus K_2) < \varepsilon/2$. Then $K = K_1 \cup K_2$ is compact with $\lambda(E \setminus K) < \varepsilon$. As K_1 and K_2 are disjoint and closed, each is open in K, and $f|_K$ is constant on each, hence continuous.

Step 2: simple functions. If $f = \sum_{i=1}^m a_i \mathbf{1}_{A_i}$, apply Step 1 to each A_i with ε/m and intersect the compact sets.

Step 3: real functions. By Proposition 28.2.6(4), applied to f^+ and f^-, there are simple functions s_k with $s_k \to f$ pointwise on E. By Step 2 there are compact $K_k \subseteq E$ with $\lambda(E \setminus K_k) < \varepsilon 2^{-k-2}$ and $s_k|_{K_k}$ continuous. By Egorov's theorem (Theorem 28.6.2, on the finite measure space E) there is a measurable $F \subseteq E$ with $\lambda(E \setminus F) < \varepsilon/4$ on which $s_k \to f$ uniformly, and by inner regularity a compact $K_0 \subseteq F$ with $\lambda(F \setminus K_0) < \varepsilon/4$. The compact set $K = K_0 \cap \bigcap_k K_k$ satisfies $\lambda(E \setminus K) < \varepsilon/4 + \varepsilon/4 + \sum_k \varepsilon 2^{-k-2} \le \varepsilon$, and on it the continuous functions $s_k|_K$ converge uniformly to $f|_K$, which is therefore continuous (Proposition 18.3.10). For complex f, treat the real and imaginary parts with $\varepsilon/2$ each and intersect.

Extension. The space $\mathbb{R}^n$ is metric, hence normal (Proposition 18.8.3), and K is closed, so by the Tietze extension theorem (Theorem 18.8.5) the real and imaginary parts of $f|_K$ extend to continuous functions on $\mathbb{R}^n$; let h be the resulting extension of $f|_K$. If $|f| \le c$, replace h by $r \circ h$, where $r(z) = z$ for $|z| \le c$ and $r(z) = cz/|z|$ otherwise is a continuous retraction of $\mathbb{C}$ onto the closed disc of radius c. Finally choose $\phi \in C_c(\mathbb{R}^n)$ with values in $[0,1]$ and $\phi = 1$ on K (Lemma 18.9.5), and let $g = \phi h$. □

Example 28.8.4. Lusin's theorem says that the *restriction* of f to K is continuous, not that f is continuous at the points of K. The Dirichlet function $\mathbf{1}_{\mathbb{Q}}$ on $[0,1]$ is discontinuous at every point. Enumerate $\mathbb{Q} \cap [0,1] = \{q_1, q_2, \ldots\}$ and let U be the union of the intervals $(q_k - \varepsilon 2^{-k-2}, q_k + \varepsilon 2^{-k-2})$. Then $\lambda(U) \le \varepsilon/2$, the set $K = [0,1] \setminus U$ is compact with $\lambda([0,1] \setminus K) < \varepsilon$, and $\mathbf{1}_{\mathbb{Q}}$ vanishes on K, so its restriction to K is continuous.

Corollary 28.8.5. *Every measurable function $f : \mathbb{R}^n \to \mathbb{C}$ is the almost everywhere limit of a sequence of continuous functions.*

Proof. For $k \geq 1$ apply Theorem 28.8.3 to the restriction of f to the ball $B(0,k)$ with $\varepsilon = 2^{-k}$, to get $g_k \in C_c(\mathbb{R}^n)$ equal to f on $B(0,k)$ outside a set N_k of measure less than 2^{-k}. By monotone convergence, $\int \sum_k \mathbf{1}_{N_k} = \sum_k \lambda(N_k) < \infty$, so $\sum_k \mathbf{1}_{N_k} < \infty$ almost everywhere: almost every x lies in only finitely many N_k. For such x and all large $k > \|x\|$, $g_k(x) = f(x)$. □

28.9 Differentiation of integrals

For a continuous function f on $\mathbb{R}$, the fundamental theorem of calculus says that $\frac{1}{2r}\int_{x-r}^{x+r} f \to f(x)$. We show that the averages of an integrable function over small balls converge to the function almost everywhere. Let $B(x,r)$ be the open Euclidean ball. The map $y \mapsto x + ry$ multiplies the volume of boxes by r^n, hence Lebesgue measure by r^n, so $\lambda(B(x,r)) = r^n \lambda(B(0,1))$. A measurable function is *locally integrable*, $f \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$, if $\int_K |f| < \infty$ for every compact K.

Lemma 28.9.1 (Vitali covering lemma)**.** *Every finite family of open balls $B_1, \dots, B_m$ in $\mathbb{R}^n$ has a pairwise disjoint subfamily $B_{i_1}, \dots, B_{i_k}$ with $\bigcup_i B_i \subseteq \bigcup_j 3B_{i_j}$, where $3B(x,r) = B(x,3r)$. Consequently $\lambda(\bigcup_i B_i) \leq 3^n \sum_j \lambda(B_{i_j})$.*

Proof. Choose a ball of largest radius, discard all the balls that meet it, choose a ball of largest radius among the remaining ones, and so on. The chosen balls are pairwise disjoint. A discarded ball $B = B(c,r)$ meets a chosen ball $B' = B(c',r')$ with $r' \geq r$, namely the one chosen at the step where B was discarded. If $z \in B \cap B'$ and $y \in B$, then $\|y - c'\| \leq \|y - c\| + \|c - z\| + \|z - c'\| < r + r + r' \leq 3r'$, so $B \subseteq 3B'$. The inequality follows from $\lambda(3B') = 3^n \lambda(B')$ and subadditivity. □

Definition 28.9.2. For $f \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$, $x \in \mathbb{R}^n$ and $r > 0$, the *average* of f over $B(x,r)$ is $A_r f(x) = \frac{1}{\lambda(B(x,r))}\int_{B(x,r)} f \, d\lambda$, and the *Hardy–Littlewood maximal function* is $Mf(x) = \sup_{r>0} A_r|f|(x) \in [0,\infty]$.

Theorem 28.9.3 (Hardy–Littlewood maximal inequality)**.** *Let $f \in L^1(\mathbb{R}^n)$ and $t > 0$. The set $\{Mf > t\}$ is open, and $\lambda(\{Mf > t\}) \leq \frac{3^n}{t}\|f\|_1$.*

Proof. If $A_r|f|(x) > t$, choose $s > r$ with $(r/s)^n A_r|f|(x) > t$. For $\|x' - x\| < s - r$ we have $B(x',s) \supseteq B(x,r)$, so $A_s|f|(x') \geq (r/s)^n A_r|f|(x) > t$; hence $\{Mf > t\}$ is open. Let $K \subseteq \{Mf > t\}$ be compact. Every $x \in K$ is the centre of a ball B_x with $\int_{B_x} |f| > t\lambda(B_x)$, and finitely many of these balls cover K. By Lemma 28.9.1 they contain disjoint balls $B_1, \dots, B_k$ with

$$\lambda(K) \leq 3^n \sum_j \lambda(B_j) < \frac{3^n}{t} \sum_j \int_{B_j} |f| \leq \frac{3^n}{t}\|f\|_1.$$

By inner regularity (Theorem 28.3.4), the same bound holds for $\{Mf > t\}$. □

Theorem 28.9.4 (Lebesgue differentiation theorem)**.** *Let $f \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$. For almost every $x \in \mathbb{R}^n$,*

$$\lim_{r \to 0} \frac{1}{\lambda(B(x,r))} \int_{B(x,r)} |f(y) - f(x)| \, d\lambda(y) = 0,$$

and in particular $A_r f(x) \to f(x)$ as $r \to 0$.

Proof. For $x \in B(0,R)$ and $r < 1$ the quantity only involves f on $B(0,R+1)$, so, replacing f by $f\mathbf{1}_{B(0,R+1)}$ and letting R run through $\mathbb{N}$, we may assume $f \in L^1(\mathbb{R}^n)$. Let $\Omega_f(x) = \limsup_{r\to 0} A_r|f - f(x)|(x)$. Fix $t > 0$ and $\varepsilon > 0$, and choose $g \in C_c(\mathbb{R}^n)$ with $\|f - g\|_1 < \varepsilon$ (Proposition 28.5.4). For the continuous g, $A_r|g - g(x)|(x) \le \sup_{B(x,r)} |g - g(x)| \to 0$, so $\Omega_g = 0$. With $h = f - g$ we have $|f(y) - f(x)| \le |h(y)| + |h(x)| + |g(y) - g(x)|$, hence $\Omega_f(x) \le Mh(x) + |h(x)|$, and

$$\{\Omega_f > 2t\} \subseteq \{Mh > t\} \cup \{|h| > t\}.$$

By Theorem 28.9.3 and Markov's inequality (Corollary 28.4.4(4)), the right side has measure at most $3^n\varepsilon/t + \varepsilon/t$. As ε is arbitrary, $\{\Omega_f > 2t\}$ has outer measure 0, so it is a null set (Lebesgue measure is complete), and so is the union of these sets for $t = 1/k$, $k \ge 1$. Finally $|A_r f(x) - f(x)| \le A_r|f - f(x)|(x)$. □

The points where the conclusion holds are the *Lebesgue points* of f; changing f on a null set does not change the averages, so almost every point is a Lebesgue point of every representative.

Corollary 28.9.5 (Lebesgue density theorem)**.** *Let $A \subseteq \mathbb{R}^n$ be Lebesgue measurable. Then $\lambda(A \cap B(x,r))/\lambda(B(x,r)) \to 1$ as $r \to 0$ for almost every $x \in A$, and $\to 0$ for almost every $x \notin A$.*

Proof. Apply Theorem 28.9.4 to the bounded, hence locally integrable, function $\mathbf{1}_A$, whose average over $B(x,r)$ is the ratio in question. □

Corollary 28.9.6 (Fundamental theorem of calculus for Lebesgue integrals)**.** *Let $f \in L^1([a,b])$ and $F(x) = \int_{[a,x]} f\,d\lambda$. Then F is continuous, and at almost every $x \in (a,b)$ it is differentiable with $F'(x) = f(x)$.*

Proof. Extend f by 0 to $\mathbb{R}$. For $y \to x$, $|F(y) - F(x)| \le \int \mathbf{1}_{[x,y]}|f| \to 0$ by dominated convergence, where $[x,y]$ means the interval between x and y. For $h \ne 0$ let I_h be the closed interval between x and $x+h$, of length $|h|$. Then $F(x+h) - F(x) = \pm\int_{I_h} f$, with the sign of h, so in both cases $\frac{F(x+h)-F(x)}{h} - f(x) = \frac{1}{|h|}\int_{I_h}(f - f(x))$. As $I_h \subseteq \overline{B}(x,|h|)$, an interval of length $2|h|$ whose endpoints are null,

$$\left|\frac{F(x+h) - F(x)}{h} - f(x)\right| \le \frac{1}{|h|}\int_{B(x,|h|)} |f - f(x)| = 2A_{|h|}|f - f(x)|(x),$$

which tends to 0 at every Lebesgue point of f. □

Counterexample 28.9.7. *The Cantor function.* A continuous nondecreasing function can have derivative 0 almost everywhere without being constant, so the formula $F(b) - F(a) = \int_a^b F'$ fails for such functions. Let $C \subseteq [0,1]$ be the Cantor set, the set of numbers $\sum_{k\ge1} a_k 3^{-k}$ with all $a_k \in \{0,2\}$; these expansions are unique, and C is compact of measure zero, being covered, for each k, by 2^k intervals of length 3^{-k} (Exercise 28.23.1). Define $c\colon C \to [0,1]$ by $c(\sum_k a_k 3^{-k}) = \sum_k (a_k/2)2^{-k}$.

1. c is nondecreasing and surjective. If $x < y$ in C, their digits first differ at some k with $a_k = 0$ and $b_k = 2$ (a difference $2\cdot 3^{-k}$ in position k cannot be compensated by the later digits, which contribute at most $\sum_{j>k} 2\cdot 3^{-j} = 3^{-k}$); then $c(y) - c(x) \ge 2^{-k} - \sum_{j>k} 2^{-j} = 0$. Every $t \in [0,1]$ has a binary expansion $\sum_k b_k 2^{-k}$, and $t = c(\sum_k 2b_k 3^{-k})$.

2. Let $F(x) = \sup\{c(y) : y \in C, y \le x\}$ for $x \in [0,1]$. Then F is nondecreasing, $F = c$ on C, and $F([0,1]) = [0,1]$. A nondecreasing function whose image is an interval has no jumps, so F is continuous.

3. The complement $[0,1] \setminus C$ is a countable union of open intervals (α, β) with $\alpha \in C$, removed in the construction; on each of them $F = c(\alpha)$ is constant, as C has no point in (α, β). So $F' = 0$ on $[0,1] \setminus C$, that is, almost everywhere, while $F(1) - F(0) = 1$.

So F is not the integral of its derivative. The functions that are integrals of their derivatives are exactly the *absolutely continuous* ones [Rud87, Chapter 7]. Note also that the continuous map F sends the null set C onto $[0,1]$.

Counterexample 28.9.8. *A Lebesgue measurable set that is not a Borel set.* First, every $A \subseteq \mathbb{R}$ with $\lambda^*(A) > 0$ contains a non-measurable subset. Let V be as in Counterexample 28.3.7. If S is a measurable subset of a translate $V + q$, $q \in \mathbb{Q}$, then the sets $S - q + q'$, $q' \in \mathbb{Q} \cap [-1,1]$, are pairwise disjoint (they lie in the disjoint translates $V + q'$), contained in $[-1,2]$, and all of measure $\lambda(S)$; so $\lambda(S) = 0$. The sets $A \cap (V + q)$, $q \in \mathbb{Q}$, cover A, since every real number differs from an element of V by a rational; if they were all measurable, A would be a countable union of null sets.

Now let F be the Cantor function and $h(x) = x + F(x)$ on $[0,1]$. It is continuous and strictly increasing from $[0,1]$ onto $[0,2]$, hence a homeomorphism. It maps each interval (α, β) of $[0,1] \setminus C$ onto an interval of the same length, since F is constant there; these images are disjoint, so $\lambda(h([0,1] \setminus C)) = 1$ and $\lambda(h(C)) = 1$. Choose a non-measurable $N \subseteq h(C)$. Then $h^{-1}(N) \subseteq C$ is a subset of a null set, hence Lebesgue measurable. It is not a Borel set: otherwise N, its preimage under the continuous map h^{-1}, would be Borel, hence measurable. In particular, a composite of a Lebesgue measurable function with a continuous one need not be Lebesgue measurable: $\mathbf{1}_{h^{-1}(N)} \circ h^{-1} = \mathbf{1}_N$.

28.10 Product measures and Fubini's theorem

Definition 28.10.1. For measurable spaces $(X, \mathcal{A})$ and $(Y, \mathcal{B})$, the *product σ-algebra* $\mathcal{A} \otimes \mathcal{B}$ on $X \times Y$ is generated by the *rectangles* $A \times B$. For $E \subseteq X \times Y$, the sections are $E_x = \{y : (x,y) \in E\}$ and $E^y = \{x : (x,y) \in E\}$.

Theorem 28.10.2. *Let μ and ν be σ-finite measures on X and Y. For $E \in \mathcal{A} \otimes \mathcal{B}$, the sections are measurable, the functions $x \mapsto \nu(E_x)$ and $y \mapsto \mu(E^y)$ are measurable, and*

$$(\mu \times \nu)(E) = \int \nu(E_x)\, d\mu(x) = \int \mu(E^y)\, d\nu(y)$$

defines the unique measure on $\mathcal{A} \otimes \mathcal{B}$ with $(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$.

Proof. Assume first that μ and ν are finite. The sets E for which all claims hold (including equality of the two integrals) contain the rectangles, which are closed under finite intersections, and they are closed under complements and countable disjoint unions (by additivity and monotone convergence). Measurability of sections holds for all $E \in \mathcal{A} \otimes \mathcal{B}$ since the sets with measurable sections form a σ-algebra. By Lemma 28.3.10 the claims hold on $\mathcal{A} \otimes \mathcal{B}$, and countable additivity follows from monotone convergence. For σ-finite measures write $X \times Y$ as an increasing union of $X_m \times Y_m$ with finite measures and pass to the limit. Uniqueness follows from Lemma 28.3.10. □

Theorem 28.10.3 (Tonelli and Fubini)**.** *Let μ, ν be σ-finite.*

1. *If $f : X \times Y \to [0, \infty]$ is measurable, then $x \mapsto \int f(x,y)\, d\nu(y)$ is measurable and $\int_{X \times Y} f\, d(\mu \times \nu) = \int \left(\int f(x,y)\, d\nu(y) \right) d\mu(x) = \int \left(\int f(x,y)\, d\mu(x) \right) d\nu(y)$.*
2. *If $f \in L^1(\mu \times \nu)$, then $f(x,-) \in L^1(\nu)$ for a.e. x, the function $x \mapsto \int f(x,y)\, d\nu(y)$ (defined a.e.) is integrable, and the same equalities hold.*

Proof. (1) For indicator functions this is Theorem 28.10.2; it extends to simple functions by linearity and to all $f \geq 0$ by monotone convergence. (2) Apply (1) to $|f|$ to see that $\int |f(x,y)|\,dv(y) < \infty$ for a.e. x, and then to the positive and negative parts of the real and imaginary parts. □

Counterexample 28.10.4. Integrability is needed in (2). On $\mathbb{N}\times\mathbb{N}$ with counting measures let $f(n,n) = 1$, $f(n,n+1) = -1$ and $f = 0$ otherwise. Then $\sum_m \sum_n f(n,m) = 1$, since only column $m = 0$ has nonzero sum, whereas $\sum_n \sum_m f(n,m) = 0$.

Remark 28.10.5. λ^{m+n} and $\lambda^m \times \lambda^n$ agree on boxes, hence on Borel sets by Lemma 28.3.10, and λ^{m+n} is the completion of $\lambda^m \times \lambda^n$. So Fubini's theorem applies to Lebesgue measure on $\mathbb{R}^{m+n} = \mathbb{R}^m \times \mathbb{R}^n$ for Borel functions, and for Lebesgue measurable functions with "for almost every x" qualifications.

Proposition 28.10.6 (Layer cake formula)**.** *Let μ be a σ-finite measure on X and $f : X \to [0,\infty]$ measurable. Then*

$$\int_X f\,d\mu = \int_0^\infty \mu(\{f > t\})\,dt.$$

More generally, if $\psi\colon [0,\infty) \to [0,\infty)$ is Borel measurable and $\phi(s) = \int_0^s \psi(t)\,dt$ (with $\phi(\infty) = \int_0^\infty \psi$), then $\int_X \phi\circ f\,d\mu = \int_0^\infty \psi(t)\,\mu(\{f > t\})\,dt$; in particular $\int_X f^p\,d\mu = p\int_0^\infty t^{p-1}\mu(\{f > t\})\,dt$ for $p > 0$.

Proof. The set $E = \{(x,t) \in X \times [0,\infty) : t < f(x)\}$ is measurable for the product σ-algebra, because $t < f(x)$ holds exactly when $t < q < f(x)$ for some rational q, so $E = \bigcup_{q\in\mathbb{Q}_{>0}} \{f > q\} \times [0,q)$. Apply Tonelli's theorem (Theorem 28.10.3(1)) to the non-negative measurable function $(x,t) \mapsto \psi(t)\mathbf{1}_E(x,t)$ on $X\times[0,\infty)$ with $\mu\times\lambda$. Integrating first in t gives $\int_0^{f(x)} \psi(t)\,dt = \phi(f(x))$, integrating first in x gives $\psi(t)\mu(\{f > t\})$. The first formula is the case $\psi = 1$, and the last one the case $\psi(t) = pt^{p-1}$. □

The name describes the picture: the region under the graph of f is cut into horizontal layers $\{f > t\}$ of thickness dt. As an application we show that the maximal function of an L^p function is in L^p for $p > 1$, although it is never in L^1 (Exercise 28.23.4).

Corollary 28.10.7 (Hardy–Littlewood maximal theorem)**.** *For $1 < p \leq \infty$ and $f \in L^p(\mathbb{R}^n)$, $\|Mf\|_p \leq C_{n,p}\|f\|_p$, with $C_{n,\infty} = 1$ and $C_{n,p}^p = 2^p 3^n p/(p-1)$.*

Proof. For $p = \infty$, every average of $|f|$ is at most $\|f\|_\infty$. Let $p < \infty$ and $t > 0$, and split $f = f_1 + f_2$ with $f_1 = f\mathbf{1}_{\{|f|>t/2\}}$. Then $f_1 \in L^1$, since $|f_1| \leq (2/t)^{p-1}|f|^p$, and $Mf \leq Mf_1 + Mf_2 \leq Mf_1 + t/2$, so $\{Mf > t\} \subseteq \{Mf_1 > t/2\}$, and by Theorem 28.9.3, $\lambda(\{Mf > t\}) \leq \frac{2\cdot 3^n}{t}\int_{\{|f|>t/2\}} |f|$. By Proposition 28.10.6 and Tonelli's theorem,

$$\int (Mf)^p = p\int_0^\infty t^{p-1}\lambda(\{Mf > t\})\,dt \leq 2\cdot 3^n p \int |f(x)| \int_0^{2|f(x)|} t^{p-2}\,dt\,d\lambda(x) = \frac{2\cdot 3^n p}{p-1}\int |f|(2|f|)^{p-1},$$

which is $\frac{2^p 3^n p}{p-1}\|f\|_p^p$. □

28.11 Change of variables

Lemma 28.11.1. *For $T \in \mathrm{GL}_n(\mathbb{R})$ and Borel $E \subseteq \mathbb{R}^n$, $\lambda(T(E)) = |\det T|\,\lambda(E)$.*

Proof. Both sides are multiplicative in T, and every invertible matrix is a product of *elementary* matrices (by Gaussian elimination): permutations of coordinates, scalings of one coordinate by $c \neq 0$, and shears $x_i \mapsto x_i + cx_j$ $(i \neq j)$. Permutations and scalings map boxes to boxes, multiplying volumes by $|\det T|$, so they multiply outer measure by $|\det T|$. A shear preserves each line parallel to the i-th axis and acts on it by a translation; by Theorem 28.10.3 and Remark 28.10.5, integrating the one-dimensional measures of these sections shows $\lambda(T(E)) = \lambda(E)$, and $\det T = 1$. □

Theorem 28.11.2 (Change of variables)**.** *Let $U, V \subseteq \mathbb{R}^n$ be open and $\phi\colon U \to V$ a C^1-diffeomorphism. For every measurable $f : V \to [0,\infty]$, and every $f \in L^1(V)$,*

$$\int_V f\, d\lambda = \int_U f(\phi(x))\,|\det D\phi(x)|\, d\lambda(x).$$

Proof. We use the norm $\|x\|_\infty$ and the corresponding operator norm; the mean value inequality (Proposition 27.10.3(4)) holds for it with the same proof. Cubes are balls $Q = \{x : \|x-a\|_\infty \leq h\}$.

Step 1. Let $Q \subseteq U$ be a cube with centre a and $T \in \mathrm{GL}_n(\mathbb{R})$, and let $M = \sup_{y\in Q} \|T^{-1}D\phi(y)\|$. By the mean value inequality applied to $T^{-1}\phi$, $\phi(Q) \subseteq \phi(a) + T(\{\|z\|_\infty \leq Mh\})$, so $\lambda(\phi(Q)) \leq |\det T| M^n \lambda(Q)$ by Lemma 28.11.1.

Step 2: $\lambda(\phi(Q)) \leq \int_Q |\det D\phi|$ for cubes $Q \subseteq U$. Let $\varepsilon > 0$. Since $D\phi$ is uniformly continuous on Q and $\|D\phi^{-1}\|$ is bounded there, we can divide Q into subcubes Q_j with centres x_j such that $\|D\phi(x_j)^{-1}D\phi(y)\| \leq 1+\varepsilon$ for $y \in Q_j$, and $|\det D\phi|$ varies by less than ε on each Q_j. Step 1 with $T = D\phi(x_j)$ gives $\lambda(\phi(Q)) \leq \sum_j \lambda(\phi(Q_j)) \leq (1+\varepsilon)^n \sum_j |\det D\phi(x_j)|\lambda(Q_j) \leq (1+\varepsilon)^n(\int_Q |\det D\phi| + \varepsilon\lambda(Q))$.

Step 3: $\lambda(\phi(E)) \leq \int_E |\det D\phi|$ for measurable $E \subseteq U$. An open $W \subseteq U$ is a countable union of cubes with disjoint interiors (dyadic cubes), whose boundaries are null, so the inequality holds for W. Let $E \subseteq K$ with $K \subseteq U$ compact, and $C = \sup$ of $|\det D\phi|$ on a compact neighbourhood $N \subseteq U$ of K. By regularity choose open W with $E \subseteq W \subseteq N$ and $\lambda(W \setminus E) < \delta$; then $\lambda(\phi(E)) \leq \lambda(\phi(W)) \leq \int_E |\det D\phi| + C\delta$. This shows that ϕ maps null sets to null sets, hence Lebesgue measurable sets to Lebesgue measurable sets (a measurable set is a Borel set up to a null set), and gives the inequality for E in a compact subset; exhaust U by compact sets for general E.

Step 4. By Step 3, $\int_V \mathbf{1}_{\phi(E)} \leq \int_U \mathbf{1}_{\phi(E)} \circ \phi\, |\det D\phi|$; by linearity and monotone convergence, $\int_V f \leq \int_U (f\circ\phi)|\det D\phi|$ for all measurable $f \geq 0$. Applying this to ϕ^{-1} and the function $(f\circ\phi)|\det D\phi|$ on U, and using $D\phi(\phi^{-1}(y))D(\phi^{-1})(y) = 1$, gives the reverse inequality. The L^1 case follows by linearity. □

Example 28.11.3. With polar coordinates $(r,\theta) \mapsto (r\cos\theta, r\sin\theta)$, a diffeomorphism $(0,\infty)\times(0,2\pi) \to \mathbb{R}^2$ minus a closed half-line (a null set), with Jacobian determinant r, Tonelli's theorem gives $(\int_{\mathbb{R}} e^{-x^2}dx)^2 = \int_{\mathbb{R}^2} e^{-\|x\|^2} = \int_0^{2\pi}\int_0^\infty e^{-r^2} r\, dr\, d\theta = \pi$. So $\int_{\mathbb{R}} e^{-x^2}\, dx = \sqrt{\pi}$.

Fubini's theorem reduces integration by parts in several variables to one variable. When one of the functions has compact support there is no boundary term.

Proposition 28.11.4 (Integration by parts in $\mathbb{R}^n$)**.** *Let $f, g\colon \mathbb{R}^n \to \mathbb{C}$ be continuously differentiable, and suppose that one of them has compact support. Then for each i*

$$\int_{\mathbb{R}^n} (\partial_i f)\, g\, d\lambda = -\int_{\mathbb{R}^n} f\, \partial_i g\, d\lambda.$$

Consequently, for a continuously differentiable vector field F and a continuously differentiable function g, one of them compactly supported, $\int \operatorname{div} F\, g\, d\lambda = -\int \langle F, \nabla g\rangle\, d\lambda$, where $\operatorname{div} F = \sum_i \partial_i F^i$ and $\nabla g = (\partial_1 g, \dots, \partial_n g)$; and for f of class C^2, $\int (\sum_i \partial_i^2 f)\, g\, d\lambda = -\int \langle \nabla f, \nabla g\rangle\, d\lambda$.

Proof. The product $h = fg$ is continuously differentiable with compact support, contained in a cube $[-R,R]^n$, and $\partial_i h = (\partial_i f)g + f\,\partial_i g$ by the product rule; all these functions are continuous with compact support, hence integrable. By Fubini's theorem (Theorem 28.10.3) we may integrate first in the variable x_i, the others being fixed. For fixed values of the other variables, $t \mapsto h(\dots,t,\dots)$ is continuously differentiable and vanishes for $|t| \ge R$, so by the fundamental theorem of calculus (Theorem 27.8.3) $\int_{-R}^{R} \partial_i h\, dx_i = h(\dots,R,\dots) - h(\dots,-R,\dots) = 0$. Hence $\int \partial_i h\, d\lambda = 0$, which is the formula. For the vector field, apply it to F^i and g and sum over i; for the last formula take $F = \nabla f$. □

The formula says that the operator ∂_i is *anti-self-adjoint* for the L^2 inner product on compactly supported functions: $\langle \partial_i f, \bar g\rangle = -\langle f, \partial_i \bar g\rangle$. It is the reason why the derivative of a distribution is defined by $(\partial_i T)(\varphi) = -T(\partial_i \varphi)$ (Chapter 32), and, applied twice, it shows that $-\sum_i \partial_i^2$ is a non-negative operator: $\int (-\sum_i \partial_i^2 f)\bar f\, d\lambda = \int |\nabla f|^2\, d\lambda \ge 0$ for compactly supported f of class C^2.

28.12 The Radon–Nikodym and Riesz representation theorems

Lemma 28.12.1. *Let H be a complex inner product space that is complete (a* Hilbert space*), for instance $L^2(\mu)$. Every continuous linear functional $\Lambda\colon H \to \mathbb{C}$ is of the form $\Lambda(f) = \langle f, g\rangle$ for a unique $g \in H$.*

Proof. If $\Lambda = 0$ take $g = 0$. Otherwise $N = \ker\Lambda$ is a closed proper subspace. Choose $u \notin N$ and let $d = \inf_{v\in N}\|u - v\| > 0$. For $v_k \in N$ with $\|u - v_k\| \to d$, the parallelogram law $\|v_k - v_l\|^2 = 2\|u - v_k\|^2 + 2\|u - v_l\|^2 - 4\|u - \frac{v_k+v_l}{2}\|^2$ shows that (v_k) is Cauchy; its limit v satisfies $\|u - v\| = d$. For $w \in N$ and $t \in \mathbb{C}$, $\|u - v - tw\|^2 \ge d^2$ forces $\langle u - v, w\rangle = 0$. So $z = u - v \ne 0$ is orthogonal to N. For $f \in H$, $\Lambda(f)z - \Lambda(z)f \in N$, so $\Lambda(f)\|z\|^2 = \Lambda(z)\langle f, z\rangle$, and $g = \overline{\Lambda(z)}z/\|z\|^2$ works. Uniqueness: if $\langle f, g - g'\rangle = 0$ for all f, take $f = g - g'$. □

Theorem 28.12.2 (Radon–Nikodym)**.** *Let μ and ν be σ-finite measures on $(X,\mathcal{A})$ with $\nu \ll \mu$, meaning $\mu(E) = 0 \Rightarrow \nu(E) = 0$. There is a measurable $h \ge 0$, unique a.e., with $\nu(E) = \int_E h\, d\mu$ for all E.*

Proof. Assume μ and ν finite; the σ-finite case follows by decomposing X. Let $\rho = \mu + \nu$. On $L^2(\rho)$, $f \mapsto \int f\, d\nu$ is continuous, since $|\int f\, d\nu| \le \int |f|\, d\rho \le \rho(X)^{1/2}\|f\|_2$. By Lemma 28.12.1 there is g with $\int f\, d\nu = \int f g\, d\rho$ for all $f \in L^2(\rho)$, and g may be taken real. Taking $f = \mathbf{1}_E$ gives $\nu(E) = \int_E g\, d\rho$, so $g \ge 0$ ρ-a.e.; and $\int_E (1-g)\, d\nu = \int_E g\, d\mu$ (for bounded f, $\int f(1-g)\, d\nu = \int f g\, d\mu$). For $E = \{g \ge 1\}$ the left side is ≤ 0 and the right side is $\ge \mu(E)$, so $\mu(E) = 0$ and $\nu(E) = 0$. Applying the identity to $f = \mathbf{1}_E(1 + g + \cdots + g^k)$ gives $\int_E (1 - g^{k+1})\, d\nu = \int_E (g + \cdots + g^{k+1})\, d\mu$. As $k \to \infty$, monotone convergence gives $\nu(E) = \int_E h\, d\mu$ with $h = g/(1-g)$. If h' also works, $\int_E (h - h')\, d\mu = 0$ for $E = \{h > h'\}$ and $\{h < h'\}$ (intersected with sets of finite measure), so $h = h'$ a.e. □

Two measures μ, ν on $(X,\mathcal{A})$ are *mutually singular*, $\nu \perp \mu$, if there is $S \in \mathcal{A}$ with $\mu(S) = 0$ and $\nu(X \setminus S) = 0$: they live on disjoint sets.

Theorem 28.12.3 (Lebesgue decomposition)**.** *Let μ, ν be σ-finite measures on $(X,\mathcal{A})$. There are unique measures $\nu_a \ll \mu$ and $\nu_s \perp \mu$ with $\nu = \nu_a + \nu_s$; and $\nu_a = h\,\mu$ for a measurable $h \ge 0$.*

Proof. As in Theorem 28.12.2 we may assume μ, ν finite. Let $\rho = \mu + \nu$. Since $\nu \ll \rho$, Theorem 28.12.2 gives $g \ge 0$ with $\nu(E) = \int_E g\, d\rho$; from $\nu(E) \le \rho(E)$ for all E, $g \le 1$ ρ-almost

everywhere, and we may take $0 \le g \le 1$. Writing $\int_E g\,d\rho = \int_E g\,d\mu + \int_E g\,d\nu$ gives

$$\int_E (1-g)\,d\nu = \int_E g\,d\mu \qquad (E \in \mathcal{A}).$$

Let $S = \{g = 1\}$. For $E = S$ the left side vanishes, so $\int_S g\,d\mu = \mu(S) = 0$. Put $\nu_s(E) = \nu(E \cap S)$ and $\nu_a(E) = \nu(E \setminus S)$; then $\nu_s \perp \mu$. If $\mu(E) = 0$, the identity gives $\int_E (1-g)\,d\nu = 0$, so $1 - g = 0$ ν-almost everywhere on E, that is, $\nu(E \setminus S) = 0$: $\nu_a \ll \mu$, and $\nu_a = h\,\mu$ by Theorem 28.12.2. For uniqueness, let $\nu = \nu_a' + \nu_s'$ be another decomposition, with ν_s' living on a μ-null set S'. On subsets of $X \setminus (S \cup S')$ both singular parts vanish, so $\nu_a = \nu = \nu_a'$; on subsets of the μ-null set $S \cup S'$ both absolutely continuous parts vanish. Hence $\nu_a = \nu_a'$, and $\nu_s = \nu - \nu_a = \nu_s'$ (ν being finite). □

Example 28.12.4. With respect to Lebesgue measure λ on $\mathbb{R}$: $\nu = \delta_0 + \mathbf{1}_{[0,1]}\lambda$ has $\nu_a = \mathbf{1}_{[0,1]}\lambda$ and $\nu_s = \delta_0$, living on $\{0\}$. The measure μ_F of the Cantor function (Lemma 28.13.3) is singular, living on the Cantor set, but has no atoms. In general the singular part splits further into a *discrete* part, a countable sum of point masses, and a *singular continuous* part without atoms, like μ_F: the atoms of a σ-finite measure form a countable set, and the discrete part is the restriction to it.

28.13 Absolutely continuous functions

By Corollary 28.9.6, an indefinite integral $F(x) = F(a) + \int_a^x f$ of an integrable function satisfies $F' = f$ almost everywhere. Conversely, when is a function the integral of its derivative? Continuity and differentiability almost everywhere are not enough, by the Cantor function (Counterexample 28.9.7). The right condition is absolute continuity.

Definition 28.13.1. A function $F\colon [a,b] \to \mathbb{C}$ is *absolutely continuous* if for every $\varepsilon > 0$ there is $\delta > 0$ such that $\sum_k |F(b_k) - F(a_k)| < \varepsilon$ for every finite family of pairwise disjoint intervals $(a_k, b_k) \subseteq [a,b]$ with $\sum_k (b_k - a_k) < \delta$. For real F, the *total variation* of F on $[a,x]$ is $V_F(x) = \sup \sum_i |F(t_i) - F(t_{i-1})|$, the supremum over the partitions $a = t_0 < \cdots < t_m = x$.

Lemma 28.13.2. *1. For $f \in L^1([a,b])$, $F(x) = \int_{[a,x]} f\,d\lambda$ is absolutely continuous; more precisely, for every $\varepsilon > 0$ there is $\delta > 0$ with $\int_E |f| < \varepsilon$ whenever $\lambda(E) < \delta$.*

2. If F is real and absolutely continuous, then V_F is finite, nondecreasing and absolutely continuous, and so is $V_F - F$. So $F = V_F - (V_F - F)$ is a difference of nondecreasing absolutely continuous functions.

Proof. (1) By monotone convergence $\int_{\{|f|>N\}} |f| \to 0$ as $N \to \infty$; choose N with $\int_{\{|f|>N\}} |f| < \varepsilon/2$ and $\delta = \varepsilon/(2N)$. Then $\int_E |f| \le \int_{\{|f|>N\}} |f| + N\lambda(E) < \varepsilon$. For disjoint intervals of total length $< \delta$, $\sum_k |F(b_k) - F(a_k)| \le \int_{\bigcup_k (a_k,b_k)} |f| < \varepsilon$.

(2) Take δ for $\varepsilon = 1$ and cut $[a,b]$ into N intervals of length $< \delta$; refining a partition by these cut points does not decrease the sum, and the sum over each of the N intervals is < 1, so $V_F(b) \le N$. For $x < y$, $V_F(y) - V_F(x)$ is the total variation of F on $[x,y]$ (a partition of $[a,y]$ refined by x splits into partitions of $[a,x]$ and $[x,y]$), so V_F is nondecreasing and $V_F(y) - V_F(x) \ge |F(y) - F(x)| \ge F(y) - F(x)$, which makes $V_F - F$ nondecreasing. If (a_k, b_k) are disjoint with $\sum_k (b_k - a_k) < \delta$ for the δ of ε, then any partitions of the intervals $[a_k, b_k]$ together give disjoint intervals of total length $< \delta$, so $\sum_k (V_F(b_k) - V_F(a_k)) \le \varepsilon$: V_F is absolutely continuous, and so is $V_F - F$. □

Lemma 28.13.3. *Let $F\colon [a,b] \to \mathbb{R}$ be continuous and nondecreasing. There is a finite Borel measure μ_F on $[a,b]$ with $\mu_F([a,x]) = F(x) - F(a)$ for all x; so $\mu_F((x,y]) = F(y) - F(x)$ and points have measure 0. If F is absolutely continuous, then μ_F is absolutely continuous with respect to λ.*

Proof. Let $c = F(a)$ and $d = F(b)$. For $t \in [c,d]$ the set $\{x : F(x) = t\}$ is nonempty by the intermediate value theorem and closed, so it has a least element $G(t)$. The map $G\colon [c,d] \to [a,b]$ is nondecreasing, hence Borel measurable (preimages of intervals are intervals), and $G(t) \le x$ if and only if $t \le F(x)$: if $t \le F(x)$ some $y \le x$ has $F(y) = t$, and if $G(t) \le x$ then $t = F(G(t)) \le F(x)$. Let $\mu_F(E) = \lambda(\{t \in [c,d] : G(t) \in E\})$ for Borel $E \subseteq [a,b]$, the image of Lebesgue measure under G. Then $\mu_F([a,x]) = \lambda([c,F(x)]) = F(x) - c$.

Let F be absolutely continuous and E a Borel set with $\lambda(E) = 0$. Given $\varepsilon > 0$, take δ from the definition and an open $U \supseteq E$ with $\lambda(U) < \delta$ (Theorem 28.3.4). The open set $U \cap (a,b)$ is the disjoint union of countably many open intervals (a_k, b_k), its connected components, with $\sum_k (b_k - a_k) < \delta$. For every N, $\sum_{k\le N} \mu_F((a_k,b_k)) = \sum_{k\le N}(F(b_k) - F(a_k)) < \varepsilon$, so $\mu_F(U \cap (a,b)) \le \varepsilon$ and $\mu_F(E) \le \varepsilon$, the endpoints a, b having measure 0. As ε is arbitrary, $\mu_F(E) = 0$. □

Theorem 28.13.4 (Fundamental theorem of calculus for absolutely continuous functions)**.** *Let $F\colon [a,b] \to \mathbb{C}$ be absolutely continuous. Then F is differentiable almost everywhere, $F' \in L^1([a,b])$, and*

$$F(x) = F(a) + \int_a^x F'(t)\,dt \qquad \text{for all } x \in [a,b].$$

Conversely, every function of the form $F(a) + \int_a^x f$ with $f \in L^1([a,b])$ is absolutely continuous. So the absolutely continuous functions are exactly the indefinite Lebesgue integrals.

Proof. The converse is Lemma 28.13.2(1). For the direct statement, treat real and imaginary parts separately, and by Lemma 28.13.2(2) assume that F is nondecreasing, the statement being linear in F. By Lemma 28.13.3, μ_F is a finite measure with $\mu_F \ll \lambda$ on the Borel sets of $[a,b]$, so by the Radon–Nikodym theorem (Theorem 28.12.2) there is an integrable $h \ge 0$ with $\mu_F(E) = \int_E h\,d\lambda$. Hence $F(x) - F(a) = \mu_F([a,x]) = \int_a^x h$, and by Corollary 28.9.6, F is differentiable almost everywhere with $F' = h$. □

Example 28.13.5. A Lipschitz function, $|F(x) - F(y)| \le L|x-y|$, is absolutely continuous (take $\delta = \varepsilon/L$), so it is differentiable almost everywhere and is the integral of its derivative; for instance $|x| = \int_0^x \operatorname{sgn}$. The Cantor function is continuous and nondecreasing but not absolutely continuous: it maps the Cantor set, of measure 0, onto $[0,1]$, whereas the proof of Lemma 28.13.3 shows that an absolutely continuous nondecreasing function gives measure 0 to null sets, and $\mu_F(C) = F(1) - F(0) = 1$ for the Cantor function because F is constant on the complement of C. The function $x\sin(1/x)$ (and 0 at 0) on $[0,1]$ is continuous but not of bounded variation, hence not absolutely continuous.

28.14 Radon measures and the Riesz representation theorem

A measure assigns numbers to sets; integration against it assigns numbers to functions, $f \mapsto \int f\,d\mu$, and this assignment is linear and positive. On $\mathbb{R}^n$ the Riemann integral of continuous functions with compact support is such an assignment, and Lebesgue measure can be recovered from it: $\lambda([0,1]^n)$ is the infimum of the integrals of the continuous functions that are at least 1 on the cube. The Riesz representation theorem says that this always works: every positive linear functional on continuous functions with compact support is integration against exactly one measure of a reasonable kind. This point of view, *measures as linear functionals*, is basic in geometric measure theory, in probability and in the theory of partial

differential equations. It allows one to take limits of measures (Section 28.15), and it is the model for distributions, which are linear functionals on smooth functions (Chapter 32).

Throughout this section X is a locally compact Hausdorff space with a countable basis; for instance $\mathbb{R}^n$, an open or a closed subset of $\mathbb{R}^n$, or a manifold. Such a space is metrisable: it is regular, because for a point x outside a closed set F, Lemma 18.9.5 applied to $\{x\} \subseteq X \setminus F$ gives an open $V \ni x$ with $\overline{V} \cap F = \emptyset$, and a regular space with a countable basis is metrisable (Theorem 18.8.8). It has an exhaustion by compact sets $K_1 \subseteq K_2 \subseteq \cdots$ with $K_m \subseteq \operatorname{int} K_{m+1}$ (Proposition 18.10.7); every compact $K \subseteq X$ lies in some K_m, since the increasing open sets $\operatorname{int} K_m$ cover K.

We write $C_c(X)$ for the real vector space of continuous functions $f : X \to \mathbb{R}$ with compact support, and, for an open $V \subseteq X$,

$$f \prec V \quad \text{if} \quad f \in C_c(X), \quad 0 \le f \le 1 \quad \text{and} \quad \operatorname{supp} f \subseteq V.$$

Continuous functions are Borel measurable. If μ is a Borel measure that is finite on compact sets, every $f \in C_c(X)$ is integrable, with $|\int f \, d\mu| \le \|f\|_\infty \, \mu(\operatorname{supp} f)$.

Lemma 28.14.1. *Let ν be a finite Borel measure on a metrisable space X. For every Borel set E and every $\varepsilon > 0$ there are a closed set F and an open set U with $F \subseteq E \subseteq U$ and $\nu(U \setminus F) < \varepsilon$.*

Proof. Fix a metric d inducing the topology, and let $\mathcal{G}$ be the family of Borel sets E with the stated property. A closed set E belongs to $\mathcal{G}$: take $F = E$ and $U_k = \{x : d(x, E) < 1/k\}$, which are open, decrease, and have intersection E because E is closed; as ν is finite, $\nu(U_k \setminus E) \to 0$ (Proposition 28.2.4). The family $\mathcal{G}$ is closed under complements, since $F \subseteq E \subseteq U$ gives $X \setminus U \subseteq X \setminus E \subseteq X \setminus F$ with the same difference $U \setminus F$. It is closed under countable unions: if $F_k \subseteq E_k \subseteq U_k$ with $\nu(U_k \setminus F_k) < \varepsilon 2^{-k-1}$, then $U = \bigcup_k U_k$ is open and $F' = \bigcup_k F_k$ satisfies $\nu(U \setminus F') < \varepsilon/2$; the closed sets $F_1 \cup \cdots \cup F_N$ increase to F', so one of them, F, has $\nu(F' \setminus F) < \varepsilon/2$, and then $F \subseteq \bigcup_k E_k \subseteq U$ and $\nu(U \setminus F) < \varepsilon$. So $\mathcal{G}$ is a σ-algebra containing the closed sets, hence all Borel sets. □

Proposition 28.14.2. *Let X be a locally compact Hausdorff space with a countable basis and μ a Borel measure on X that is finite on compact sets. Then μ is σ-finite and regular, hence a Radon measure (Definition 28.3.5): for every Borel set E,*

$$\mu(E) = \inf\{\mu(U) : U \supseteq E \text{ open}\} = \sup\{\mu(K) : K \subseteq E \text{ compact}\}.$$

Proof. X is the union of the compact sets K_m of an exhaustion, which have finite measure, so μ is σ-finite; and every point has a neighbourhood $\operatorname{int} K_{m+1} \subseteq K_{m+1}$ of finite measure. The measures $\nu_m(A) = \mu(A \cap \operatorname{int} K_m)$ are finite, and Lemma 28.14.1 applies to them, X being metrisable.

Outer regularity. Let E be a Borel set with $\mu(E) < \infty$ (otherwise there is nothing to prove) and $\varepsilon > 0$. Choose open sets $U_m \supseteq E$ with $\nu_m(U_m \setminus E) < \varepsilon 2^{-m}$, and let $U = \bigcup_m (U_m \cap \operatorname{int} K_m)$. It is open and contains $E = \bigcup_m (E \cap \operatorname{int} K_m)$, and $\mu(U \setminus E) \le \sum_m \mu\big((U_m \cap \operatorname{int} K_m) \setminus E\big) = \sum_m \nu_m(U_m \setminus E) < \varepsilon$. So $\mu(U) < \mu(E) + \varepsilon$.

Inner regularity. Let E be a Borel set and $t < \mu(E)$. By continuity from below, $\mu(E \cap \operatorname{int} K_m) > t$ for some m. Choose a closed $F \subseteq E$ with $\nu_m(E \setminus F) < \mu(E \cap \operatorname{int} K_m) - t$. Then $K = F \cap K_m$ is compact, contained in E, and $\mu(K) \ge \mu(F \cap \operatorname{int} K_m) = \nu_m(E) - \nu_m(E \setminus F) > t$. □

So on $\mathbb{R}^n$ and on manifolds, *Radon measure* simply means *Borel measure finite on compact sets*. For general locally compact Hausdorff spaces this is false, and the theory below has to be set up with more care [Rud87, Chapter 2]; we shall not need that generality.

Lemma 28.14.3. *Let X be a locally compact Hausdorff space, $K \subseteq X$ compact and $V_1, \ldots, V_n$ open sets covering K. There are $h_i \prec V_i$ with $\sum_i h_i \le 1$ on X and $\sum_i h_i = 1$ on K.*

Proof. Each $x \in K$ lies in some V_i, and Lemma 18.9.5 gives an open $W_x \ni x$ with compact closure $\overline{W_x} \subseteq V_i$. Finitely many W_x cover K; let H_i be the union of the closures of those among them for which the chosen i was i. Then $H_i \subseteq V_i$ is compact and $K \subseteq H_1 \cup \cdots \cup H_n$. Lemma 18.9.5 gives $g_i \prec V_i$ with $g_i = 1$ on H_i. Put $h_1 = g_1$ and $h_k = (1 - g_1)\cdots(1 - g_{k-1})g_k$. Then $0 \le h_k \le g_k$, so $h_k \prec V_k$, and by induction on m,

$$h_1 + \cdots + h_m = 1 - (1 - g_1)(1 - g_2)\cdots(1 - g_m),$$

since $1 - \prod_{k\le m}(1 - g_k) + h_{m+1} = 1 - \prod_{k\le m+1}(1 - g_k)$. For $m = n$ the right side lies in $[0, 1]$, and it equals 1 on each H_i, where the factor $1 - g_i$ vanishes. □

Definition 28.14.4. A linear map $\Lambda\colon C_c(X) \to \mathbb{R}$ is *positive* if $\Lambda(f) \ge 0$ whenever $f \ge 0$.

A positive functional is monotone: $f \le g$ implies $\Lambda f \le \Lambda g$. It is also automatically continuous in the following sense. If K is compact and $\phi_K \in C_c(X)$ satisfies $0 \le \phi_K \le 1$ and $\phi_K = 1$ on K (Lemma 18.9.5), then every f with $\operatorname{supp} f \subseteq K$ satisfies $-\|f\|_\infty \phi_K \le f \le \|f\|_\infty \phi_K$, hence

$$|\Lambda f| \le \Lambda(\phi_K)\,\|f\|_\infty.$$

So uniform convergence of functions supported in a fixed compact set implies convergence of the values of Λ.

Example 28.14.5.

1. *Point masses.* $\Lambda f = f(x_0)$ is positive, and $\Lambda f = \int f\, d\delta_{x_0}$. More generally, for points $x_1, x_2, \ldots$ of X with no accumulation point and numbers $c_k \ge 0$, the finite sum $\Lambda f = \sum_k c_k f(x_k)$ is integration against $\sum_k c_k \delta_{x_k}$.

2. *Densities.* For $g \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$ with $g \ge 0$, $\Lambda f = \int f g\, d\lambda$ is positive; it is integration against the measure $g\lambda$, $E \mapsto \int_E g\, d\lambda$. For $g = 1$ this is the Riemann integral of f (Proposition 28.4.9, together with Fubini's theorem for the iterated integrals).

3. *Arc length on a circle.* On $X = \mathbb{R}^2$, $\Lambda f = \int_0^{2\pi} f(\cos t, \sin t)\, dt$ is positive. The corresponding measure σ (Theorem 28.14.6 below) lives on the unit circle S^1: if K is compact and disjoint from S^1, a function $f \prec \mathbb{R}^2 \setminus S^1$ with $f = 1$ on K gives $\sigma(K) \le \int f\, d\sigma = \Lambda f = 0$, and by inner regularity $\sigma(\mathbb{R}^2 \setminus S^1) = 0$. Since $\lambda^2(S^1) = 0$, the measures σ and λ^2 are mutually singular. The measure of an arc is its length (Exercise 28.23.13), and σ is the restriction to S^1 of the one-dimensional Hausdorff measure of Section 28.17.

4. *Push-forward.* More generally, for a continuous map $\varphi\colon [a, b] \to X$, the functional $\Lambda f = \int_a^b f(\varphi(t))\, dt$ is integration against the push-forward $\varphi_*\lambda$, $E \mapsto \lambda(\{t \in [a, b] : \varphi(t) \in E\})$, which is a Borel measure, finite because $\lambda([a, b]) < \infty$.

Theorem 28.14.6 (Riesz–Markov)**.** *Let X be a locally compact Hausdorff space with a countable basis, and $\Lambda\colon C_c(X) \to \mathbb{R}$ a positive linear functional. There is a unique Borel measure μ on X, finite on compact sets, with*

$$\Lambda f = \int_X f\, d\mu \qquad \text{for all } f \in C_c(X).$$

It is a Radon measure, and for open V and compact K,

$$\mu(V) = \sup\{\Lambda f : f \prec V\}, \qquad \mu(K) = \inf\{\Lambda f : f \in C_c(X),\ f \ge \mathbf{1}_K\}.$$

Proof. The proof follows [Rud87, Theorem 2.14]. Motivated by the formulas to be proved, define $\mu(V) = \sup\{\Lambda f : f \prec V\}$ for open V, and for an arbitrary $E \subseteq X$

$$\mu^*(E) = \inf\{\mu(V) : V \supseteq E \text{ open}\}.$$

If $V \subseteq W$ are open, $f \prec V$ implies $f \prec W$, so $\mu(V) \le \mu(W)$; hence $\mu^*(V) = \mu(V)$ for open V, and μ^* is monotone.

Step 1: μ^* is an outer measure. First let $V = \bigcup_i V_i$ be a countable union of open sets, and $f \prec V$. The compact set $\operatorname{supp} f$ is covered by finitely many $V_1, \dots, V_n$, and Lemma 28.14.3 gives $h_i \prec V_i$ with $\sum_i h_i = 1$ on $\operatorname{supp} f$. Then $f = \sum_i h_i f$ with $h_i f \prec V_i$, so $\Lambda f = \sum_i \Lambda(h_i f) \le \sum_i \mu(V_i)$; taking the supremum over f, $\mu(V) \le \sum_i \mu(V_i)$. Now let $E = \bigcup_i E_i$ be arbitrary, with $\sum_i \mu^*(E_i) < \infty$, and $\varepsilon > 0$. Choose open $V_i \supseteq E_i$ with $\mu(V_i) \le \mu^*(E_i) + \varepsilon 2^{-i}$; then $\mu^*(E) \le \mu(\bigcup_i V_i) \le \sum_i \mu^*(E_i) + \varepsilon$. Finally $\mu^*(\emptyset) = 0$, as $f \prec \emptyset$ forces $f = 0$.

Step 2: compact sets. Let K be compact. If $f \in C_c(X)$ and $f \ge \mathbf{1}_K$, and $0 < \alpha < 1$, then $V_\alpha = \{f > \alpha\}$ is an open set containing K, and every $g \prec V_\alpha$ satisfies $\alpha g \le f$; so $\mu^*(K) \le \mu(V_\alpha) \le \Lambda f/\alpha$, and letting $\alpha \to 1$, $\mu^*(K) \le \Lambda f$. Conversely, for an open $V \supseteq K$, Lemma 18.9.5 gives $f \prec V$ with $f = 1$ on K, and $\Lambda f \le \mu(V)$. So $\mu^*(K) = \inf\{\Lambda f : f \ge \mathbf{1}_K\}$, which is finite.

Step 3: Borel sets are μ^*-measurable. By Carathéodory's theorem (Theorem 28.3.2) the μ^*-measurable sets form a σ-algebra on which μ^* is a measure, so it suffices to show that every open U is μ^*-measurable: $\mu^*(E \cap U) + \mu^*(E \setminus U) \le \mu^*(E)$ for every E with $\mu^*(E) < \infty$. Let $\varepsilon > 0$ and choose an open $V \supseteq E$ with $\mu(V) \le \mu^*(E) + \varepsilon$. Choose $f \prec V \cap U$ with $\Lambda f > \mu(V \cap U) - \varepsilon$; its support K is a compact subset of U, so $V \setminus K$ is open and we may choose $g \prec V \setminus K$ with $\Lambda g > \mu(V \setminus K) - \varepsilon$. The functions f and g have disjoint supports, so $f + g \prec V$, and

$$\mu^*(E) + \varepsilon \ge \mu(V) \ge \Lambda f + \Lambda g > \mu(V \cap U) + \mu(V \setminus K) - 2\varepsilon \ge \mu^*(E \cap U) + \mu^*(E \setminus U) - 2\varepsilon,$$

because $E \cap U \subseteq V \cap U$ and $E \setminus U \subseteq V \setminus K$. Let $\varepsilon \to 0$.

So the restriction of μ^* to the Borel sets is a Borel measure μ, finite on compact sets by Step 2, and by construction it satisfies the two formulas of the statement.

Step 4: $\Lambda f = \int f \, d\mu$. Both sides are linear in f, so it suffices to show $\Lambda f \le \int f \, d\mu$ for every $f \in C_c(X)$: applied to $-f$ it gives the reverse inequality. Let $K = \operatorname{supp} f$, choose $a < b$ with $f(X) \subseteq [a, b]$, let $\varepsilon > 0$, and choose $y_0 < a < y_1 < \cdots < y_n = b$ with $y_i - y_{i-1} < \varepsilon$. The Borel sets $E_i = \{y_{i-1} < f \le y_i\} \cap K$ are disjoint with union K. By the definition of μ^* there are open sets $V_i \supseteq E_i$ with $\mu(V_i) < \mu(E_i) + \varepsilon/n$; replacing V_i by $V_i \cap \{f < y_i + \varepsilon\}$ we may assume $f < y_i + \varepsilon$ on V_i. Lemma 28.14.3 gives $h_i \prec V_i$ with $\sum_i h_i = 1$ on K. Then $f = \sum_i h_i f$, and $h_i f \le (y_i + \varepsilon) h_i$. By Step 2, $\sum_i \Lambda h_i = \Lambda(\sum_i h_i) \ge \mu(K)$, and $\Lambda h_i \le \mu(V_i)$. As $|a| + y_i + \varepsilon > 0$,

$$\begin{aligned}
\Lambda f = \sum_i \Lambda(h_i f) &\le \sum_i (y_i + \varepsilon) \Lambda h_i = \sum_i (|a| + y_i + \varepsilon) \Lambda h_i - |a| \sum_i \Lambda h_i \\
&\le \sum_i (|a| + y_i + \varepsilon) \Big(\mu(E_i) + \frac{\varepsilon}{n} \Big) - |a| \mu(K) \\
&\le \sum_i (y_i - \varepsilon) \mu(E_i) + 2\varepsilon \mu(K) + \varepsilon(|a| + b + \varepsilon),
\end{aligned}$$

using $\sum_i \mu(E_i) = \mu(K)$ and $y_i \le b$. On E_i we have $f > y_{i-1} > y_i - \varepsilon$, so $\sum_i (y_i - \varepsilon) \mu(E_i) \le \int f \, d\mu$. Letting $\varepsilon \to 0$ gives $\Lambda f \le \int f \, d\mu$.

Step 5: regularity and uniqueness. μ is a Radon measure by Proposition 28.14.2. Let μ' be another Borel measure, finite on compact sets, representing Λ, and K compact. If $f \ge \mathbf{1}_K$, then $\mu'(K) \le \int f \, d\mu' = \Lambda f$, so $\mu'(K) \le \mu(K)$ by Step 2. If $V \supseteq K$ is open and $f \prec V$ with $f = 1$ on K, then $\mu(K) \le \Lambda f = \int f \, d\mu' \le \mu'(V)$, and by outer regularity of μ' (Proposition 28.14.2) $\mu(K) \le \mu'(K)$. So μ and μ' agree on compact sets, hence, by inner regularity, on all Borel sets. □

Remark 28.14.7. 1. For $X = \mathbb{R}^n$ and Λ the Riemann integral, the theorem produces Lebesgue measure on the Borel sets; the uniqueness statement is a convenient way to

identify measures, as in Exercise 28.23.13. On a smooth manifold, an orientation and a volume form, or more generally a density, define a positive functional by integration, and hence a Radon measure (Chapter 36).

2. The theorem holds for every locally compact Hausdorff space, but then the measure is only outer regular and inner regular on open sets and on sets of finite measure [Rud87, Theorem 2.14 and Theorem 2.18].

3. If X is compact, $1 \in C_c(X)$, and $|\Lambda f| \le \Lambda(1)\|f\|_\infty = \mu(X)\|f\|_\infty$: a positive functional on $C(X)$ is bounded with norm $\mu(X)$. Every bounded linear functional on $C(X)$ is integration against a *signed* Radon measure, a difference of two finite Radon measures [Rud87, Theorem 6.19] (for the spaces of this section, see Corollary 28.14.10 below); this identifies the dual space of $C(X)$, a theme of Chapter 30.

28.14.1 Signed and vector-valued functionals

Many natural linear functionals are not positive. The functional $\varphi \mapsto \varphi(0) - \varphi(1)$ on $C_c(\mathbb{R})$ is the difference of two Dirac measures; $\varphi \mapsto \int \varphi h \, d\lambda$ with h changing sign is the difference of the measures $h^+\lambda$ and $h^-\lambda$. In geometric measure theory one also meets functionals on *vector fields*: for the unit circle $S^1 \subseteq \mathbb{R}^2$ with outward normal $n(x) = x$, the *flux* $F \mapsto \int_0^{2\pi} \langle F(\cos t, \sin t), (\cos t, \sin t)\rangle \, dt$ through the circle is a linear functional on continuous vector fields $F\colon \mathbb{R}^2 \to \mathbb{R}^2$ with compact support. The Riesz representation theorem extends to all these cases: a functional that is merely bounded on functions of bounded size supported in a fixed compact set is integration against a measure, after pairing with a field of unit vectors.

Let H be a finite-dimensional real vector space with an inner product $\langle\,,\rangle$ and norm $|\,|$; for instance $H = \mathbb{R}^m$ with the standard inner product. Let $C_c(X, H)$ be the vector space of continuous maps $f\colon X \to H$ with compact support. For $f \in C_c(X, H)$ the function $|f|$ belongs to $C_c(X)$. The map $r\colon H \to H$, $r(v) = v$ for $|v| \le 1$ and $r(v) = v/|v|$ for $|v| > 1$, the nearest point projection onto the closed unit ball, is continuous, takes values in the ball, and moves no point farther from the ball: $|r(u) - w| \le |u - w|$ whenever $|w| \le 1$. To see this, we may assume $|u| > 1$; then $|u - w|^2 - |r(u) - w|^2 = |u|^2 - 1 - 2(|u| - 1)\langle u/|u|, w\rangle \ge |u|^2 - 1 - 2(|u| - 1) = (|u| - 1)^2 \ge 0$.

Lemma 28.14.8. *Let μ be a Radon measure on X.*

1. *For $1 \le p < \infty$, $C_c(X)$ is dense in $L^p(\mu)$.*

2. *Let $w\colon X \to H$ be Borel with $|w| \le 1$, E a Borel set with $\mu(E) < \infty$, and $V \supseteq E$ open. For every $\varepsilon > 0$ there is $g \in C_c(X, H)$ with $|g| \le 1$, $\operatorname{supp} g \subseteq V$ and $\int |g - \mathbf{1}_E w| \, d\mu < \varepsilon$.*

Proof. (1) As in Proposition 28.5.4, simple functions $\sum a_i \mathbf{1}_{E_i}$ with $\mu(E_i) < \infty$ are dense in $L^p(\mu)$, by Proposition 28.2.6(4) and dominated convergence. Given E with $\mu(E) < \infty$ and $\varepsilon > 0$, regularity of μ gives a compact K and an open U with $K \subseteq E \subseteq U$ and $\mu(U \setminus K) < \varepsilon$, and Lemma 18.9.5 gives $\phi \in C_c(X)$ with $0 \le \phi \le 1$, $\phi = 1$ on K and $\operatorname{supp}\phi \subseteq U$. Then $\int |\mathbf{1}_E - \phi|^p \, d\mu \le \mu(U \setminus K) < \varepsilon$.

(2) Fix an orthonormal basis $e_1, \dots, e_m$ of H and write $w = \sum_j w_j e_j$. By (1) choose $\phi_j \in C_c(X)$ with $\int |\phi_j - \mathbf{1}_E w_j| \, d\mu < \varepsilon/(2m)$, and put $g_0 = \sum_j \phi_j e_j$, so $\int |g_0 - \mathbf{1}_E w| \, d\mu < \varepsilon/2$. Then $g_1 = r \circ g_0$ is continuous with compact support, $|g_1| \le 1$, and $|g_1 - \mathbf{1}_E w| \le |g_0 - \mathbf{1}_E w|$ pointwise, as $|\mathbf{1}_E w| \le 1$. Choose a compact $K \subseteq E$ with $\mu(E \setminus K) < \varepsilon/4$ and $\psi \in C_c(X)$ with $0 \le \psi \le 1$, $\psi = 1$ on K and $\operatorname{supp}\psi \subseteq V$, and let $g = \psi g_1$. On K, $g = g_1$; outside E, $|g - \mathbf{1}_E w| = |g| \le |g_1| = |g_1 - \mathbf{1}_E w|$; on $E \setminus K$, $|g - w| \le 2$. Hence $\int |g - \mathbf{1}_E w| \, d\mu \le \int |g_1 - \mathbf{1}_E w| \, d\mu + 2\mu(E \setminus K) < \varepsilon$. ∎

Theorem 28.14.9 (Riesz representation theorem for vector-valued functions)**.** *Let $L\colon C_c(X,H)\to\mathbb{R}$ be linear and suppose that for every compact $K\subseteq X$*

$$M_K=\sup\bigl\{L(f):f\in C_c(X,H),\ |f|\le 1,\ \operatorname{supp} f\subseteq K\bigr\}<\infty.$$

Then there are a Radon measure μ on X and a Borel map $\nu\colon X\to H$ with $|\nu|=1$ μ-almost everywhere such that

$$L(f)=\int_X\langle f,\nu\rangle\,d\mu\qquad\text{for all } f\in C_c(X,H).$$

Moreover, for every open $V\subseteq X$,

$$\mu(V)=\sup\bigl\{L(f):f\in C_c(X,H),\ |f|\le 1,\ \operatorname{supp} f\subseteq V\bigr\},\tag{28.1}$$

and the pair (μ,ν) is unique: μ is unique, and ν is unique up to μ-null sets. The measure μ is the total variation measure *of L.*

Proof. The idea is to manufacture a positive functional out of L, namely the largest value of L on fields dominated by a given function, and to apply the Riesz–Markov theorem to it.

Step 1: a positive functional on nonnegative functions. For $\varphi\in C_c(X)$ with $\varphi\ge 0$ let

$$\lambda(\varphi)=\sup\bigl\{L(g):g\in C_c(X,H),\ |g|\le\varphi\bigr\}.$$

Taking $g=0$ shows $\lambda(\varphi)\ge 0$. If $|g|\le\varphi$, then $\operatorname{supp} g\subseteq K=\operatorname{supp}\varphi$, and if $\varphi\neq 0$, $g/\|\varphi\|_\infty$ is admissible in the definition of M_K; so $L(g)\le\|\varphi\|_\infty M_K$ and $\lambda(\varphi)<\infty$. Clearly $\lambda(c\varphi)=c\lambda(\varphi)$ for $c>0$, as $g\mapsto cg$ matches the admissible fields. We show $\lambda(\varphi_1+\varphi_2)=\lambda(\varphi_1)+\lambda(\varphi_2)$ for $\varphi_1,\varphi_2\ge 0$. If $|g_i|\le\varphi_i$, then $|g_1+g_2|\le\varphi_1+\varphi_2$, so $L(g_1)+L(g_2)\le\lambda(\varphi_1+\varphi_2)$; taking suprema gives "$\ge$". Conversely let $|g|\le\varphi_1+\varphi_2$, and define $g_i=\varphi_i g/(\varphi_1+\varphi_2)$ where $\varphi_1+\varphi_2>0$ and $g_i=0$ elsewhere. Then $|g_i|\le\varphi_i$, and g_i is continuous: this is clear where $\varphi_1+\varphi_2>0$, and at a point x_0 where $\varphi_1+\varphi_2=0$ we have $|g_i|\le\varphi_i$ everywhere and $\varphi_i(x_0)=0$, so $g_i(x)\to 0=g_i(x_0)$ as $x\to x_0$. Since $g=0$ where $\varphi_1+\varphi_2=0$, $g=g_1+g_2$, and $L(g)=L(g_1)+L(g_2)\le\lambda(\varphi_1)+\lambda(\varphi_2)$.

Step 2: extension and the measure μ. For $\varphi\in C_c(X)$ put $\lambda(\varphi)=\lambda(\varphi^+)-\lambda(\varphi^-)$, where $\varphi^\pm=\max(\pm\varphi,0)\in C_c(X)$. If $\varphi=a-b$ with $a,b\ge 0$ in $C_c(X)$, then $\varphi^++b=\varphi^-+a$, and Step 1 gives $\lambda(\varphi)=\lambda(a)-\lambda(b)$. Writing $\varphi+\psi=(\varphi^++\psi^+)-(\varphi^-+\psi^-)$ shows that λ is additive; it is homogeneous for $c\ge 0$ by Step 1, and $\lambda(-\varphi)=-\lambda(\varphi)$ since $(-\varphi)^\pm=\varphi^\mp$. So λ is a positive linear functional on $C_c(X)$, and the Riesz–Markov theorem (Theorem 28.14.6) gives a Radon measure μ with $\lambda(\varphi)=\int\varphi\,d\mu$.

Step 3: $|L(g)|\le\int|g|\,d\mu$. For $g\in C_c(X,H)$, g is admissible for $\lambda(|g|)$, so $L(g)\le\lambda(|g|)=\int|g|\,d\mu$; apply this also to $-g$.

Step 4: construction of ν. Fix an orthonormal basis $e_1,\dots,e_m$ of H and let $L_j(\varphi)=L(\varphi e_j)$ for $\varphi\in C_c(X)$. By Step 3, $|L_j(\varphi)|\le\int|\varphi|\,d\mu$, so $\Lambda_j(\varphi)=\int\varphi\,d\mu+L_j(\varphi)$ is a positive linear functional on $C_c(X)$. By the Riesz–Markov theorem there is a Radon measure σ_j with $\Lambda_j(\varphi)=\int\varphi\,d\sigma_j$. For open V and $\varphi\prec V$, $\Lambda_j(\varphi)\le 2\int\varphi\,d\mu\le 2\mu(V)$, so $\sigma_j(V)\le 2\mu(V)$ by the formula for $\sigma_j(V)$ in Theorem 28.14.6; by outer regularity of μ, $\sigma_j(E)\le 2\mu(E)$ for all Borel E. Both measures are σ-finite (Proposition 28.14.2) and $\sigma_j\ll\mu$, so by the Radon–Nikodym theorem (Theorem 28.12.2) $\sigma_j(E)=\int_E h_j\,d\mu$ for a Borel function $h_j\ge 0$. Since $\int_E h_j\,d\mu\le 2\mu(E)$ for all E, applied to $E=\{h_j>2\}\cap K_m$ for the compact exhaustion (K_m), $h_j\le 2$ almost everywhere, and we may change h_j on a null set so that $0\le h_j\le 2$ everywhere. By the usual extension from indicator functions to simple functions and limits, $\int\varphi\,d\sigma_j=\int\varphi h_j\,d\mu$ for $\varphi\in C_c(X)$. Put $\nu_j=h_j-1\in[-1,1]$ and $\nu=\sum_j\nu_j e_j$, a Borel map with $|\nu|\le\sqrt{m}$. Then $L_j(\varphi)=\int\varphi\,\nu_j\,d\mu$, and for $f\in C_c(X,H)$, writing $f=\sum_j\langle f,e_j\rangle e_j$ with $\langle f,e_j\rangle\in C_c(X)$,

$$L(f)=\sum_j L_j\bigl(\langle f,e_j\rangle\bigr)=\int\sum_j\langle f,e_j\rangle\nu_j\,d\mu=\int\langle f,\nu\rangle\,d\mu.$$

Step 5: $|\nu| \le 1$ almost everywhere. Let $E = \{|\nu| > 1\} \cap K_m$ and $w = \nu/|\nu|$ on E, $w = 0$ elsewhere. For $\varepsilon > 0$, Lemma 28.14.8(2) gives $g \in C_c(X, H)$ with $|g| \le 1$ and $\int |g - \mathbf{1}_E w|\, d\mu < \varepsilon$. Then, by Step 3,

$$\int_E |\nu|\, d\mu = \int \langle \mathbf{1}_E w, \nu \rangle\, d\mu \le L(g) + \sqrt{m}\,\varepsilon \le \int |g|\, d\mu + \sqrt{m}\,\varepsilon \le \mu(E) + (1 + \sqrt{m})\varepsilon.$$

So $\int_E (|\nu| - 1)\, d\mu \le 0$, while $|\nu| - 1 > 0$ on E; hence $\mu(E) = 0$, for every m.

Step 6: $|\nu| \ge 1$ almost everywhere. For $\varphi \ge 0$ in $C_c(X)$ and $|g| \le \varphi$, $L(g) = \int \langle g, \nu \rangle\, d\mu \le \int \varphi |\nu|\, d\mu$; taking the supremum, $\int \varphi\, d\mu = \lambda(\varphi) \le \int \varphi |\nu|\, d\mu$, that is, $\int \varphi(1 - |\nu|)\, d\mu \le 0$. Given a compact K, choose such φ with $\varphi = 1$ on K and $0 \le \varphi \le 1$ (Lemma 18.9.5); as $1 - |\nu| \ge 0$ almost everywhere by Step 5, $\int_K (1 - |\nu|)\, d\mu \le \int \varphi(1 - |\nu|)\, d\mu \le 0$, so $|\nu| = 1$ almost everywhere on K, and on $X = \bigcup_m K_m$.

Step 7: the formula (28.1) and uniqueness. Let (μ', ν') be any pair with the properties of the theorem; we show that $\mu'(V)$ equals the right side $s(V)$ of (28.1). If $|f| \le 1$, then $L(f) = \int \langle f, \nu' \rangle\, d\mu' \le \int_V |f|\, d\mu' \le \mu'(V)$ for $\operatorname{supp} f \subseteq V$, so $s(V) \le \mu'(V)$. Conversely, let $K \subseteq V$ be compact and $\varepsilon > 0$. Lemma 28.14.8(2), applied to μ', $w = \nu'$ and $E = K$, gives g with $|g| \le 1$, $\operatorname{supp} g \subseteq V$ and $\int |g - \mathbf{1}_K \nu'|\, d\mu' < \varepsilon$, so $L(g) \ge \int_K |\nu'|^2\, d\mu' - \varepsilon = \mu'(K) - \varepsilon$. By inner regularity $s(V) \ge \mu'(V)$. Hence $\mu'(V) = s(V)$ for all open V; this proves (28.1) for μ, and shows that μ' and μ agree on open sets, hence on all Borel sets by outer regularity. Finally $\int \langle f, \nu - \nu' \rangle\, d\mu = 0$ for all $f \in C_c(X, H)$. Let $E = \{\nu \ne \nu'\} \cap K_m$ and $w = (\nu - \nu')/|\nu - \nu'|$ on E, 0 elsewhere; approximating $\mathbf{1}_E w$ by g as in Lemma 28.14.8(2) and using $|\nu - \nu'| \le 2$ almost everywhere gives $\int_E |\nu - \nu'|\, d\mu \le 2\varepsilon$ for every ε, so $\nu = \nu'$ almost everywhere on E, and on X. □

For a positive functional on $C_c(X) = C_c(X, \mathbb{R})$ the theorem gives back the Riesz–Markov measure with $\nu = 1$. For general functionals on $C_c(X)$ it gives the decomposition into a positive and a negative part.

Corollary 28.14.10. *Let $L\colon C_c(X) \to \mathbb{R}$ be linear with $\sup\{|L(\varphi)| : |\varphi| \le 1, \operatorname{supp}\varphi \subseteq K\} < \infty$ for every compact K. Then there are Radon measures μ^+ and μ^-, concentrated on disjoint Borel sets X^+ and X^-, with*

$$L(\varphi) = \int \varphi\, d\mu^+ - \int \varphi\, d\mu^- \qquad (\varphi \in C_c(X)).$$

If X is compact, every bounded linear functional L on $C(X)$ (with the norm $\|\ \|_\infty$) has this form with $\mu^\pm$ finite, and $\|L\| = \mu^+(X) + \mu^-(X)$.

Proof. Apply Theorem 28.14.9 with $H = \mathbb{R}$. Then $\nu = \pm 1$ almost everywhere; let $X^\pm = \{\nu = \pm 1\}$ and $\mu^\pm(E) = \mu(E \cap X^\pm)$, Borel measures finite on compact sets, hence Radon by Proposition 28.14.2. Then $\int \varphi\nu\, d\mu = \int \varphi\, d\mu^+ - \int \varphi\, d\mu^-$. If X is compact and L is bounded, the hypothesis holds with the constant $\|L\|$, $\mu^\pm(X) \le \mu(X) < \infty$, and (28.1) with $V = X$ gives $\mu(X) = \sup\{L(\varphi) : \|\varphi\|_\infty \le 1\} = \|L\|$. □

So the dual space of $C(X)$, for a compact metrisable X, is the space of signed Radon measures with the total variation norm; this proves the statement quoted in Remark 28.14.7(3) for the spaces of this section. The decomposition $X = X^+ \cup X^-$ is the *Hahn decomposition* and $\mu^\pm$ the *Jordan decomposition* of the signed measure $\mu^+ - \mu^-$.

Example 28.14.11. 1. For $L(\varphi) = \varphi(0) - \varphi(1)$ on $C_c(\mathbb{R})$, $\mu = \delta_0 + \delta_1$, $\nu(0) = 1$ and $\nu(1) = -1$. Indeed, for open V, the supremum in (28.1) is the number of the points $0, 1$ lying in V, attained by functions equal to 1 near 0 and -1 near 1.

2. For $h \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$ and $L(\varphi) = \int \varphi h \, d\lambda$, $\mu = |h|\lambda$ and $\nu = h/|h|$ where $h \neq 0$ (and $\nu = 1$ elsewhere, a μ-null set). The total variation of L on V is $\int_V |h| \, d\lambda$, the L^1 norm.

3. For the flux $L(F) = \int_0^{2\pi} \langle F(\cos t, \sin t), (\cos t, \sin t)\rangle \, dt = \int \langle F, n\rangle \, d\sigma$, with σ the arc length measure of Example 28.14.5(3), $\mu = \sigma$ and $\nu = n$ on S^1: the theorem recovers the curve and its unit normal from the functional. For a C^1 field F, the divergence theorem in the plane gives $L(F) = \int_{|x|<1} \operatorname{div} F \, d\lambda^2$.

Remark 28.14.12. *Sets of finite perimeter.* Example 28.14.11(3) suggests a definition of boundary measure for arbitrary sets. For a Lebesgue measurable $E \subseteq \mathbb{R}^n$ consider $L_E(F) = \int_E \operatorname{div} F \, d\lambda^n$ for C^1 vector fields F with compact support. If $\sup\{L_E(F) : |F| \le 1, \operatorname{supp} F \subseteq K\} < \infty$ for every compact K, then L_E extends by uniform approximation to $C_c(\mathbb{R}^n, \mathbb{R}^n)$, and Theorem 28.14.9 gives a Radon measure μ_E, the *perimeter measure*, and a unit vector field ν_E with $\int_E \operatorname{div} F = \int \langle F, \nu_E\rangle \, d\mu_E$: a divergence theorem for E, valid with no regularity of ∂E assumed. For a bounded open set with smooth boundary, μ_E is the $(n-1)$-dimensional Hausdorff measure $\mathcal{H}^{n-1}$ (Section 28.17) on ∂E and ν_E is the outer normal. Such sets E are the sets of *finite perimeter* of Caccioppoli and De Giorgi, and the same construction with $\int u \operatorname{div} F$ defines functions of *bounded variation*; De Giorgi's structure theorem shows that μ_E is $\mathcal{H}^{n-1}$ restricted to a countably $(n-1)$-rectifiable "reduced boundary" [EG92, Chapter 5]. This is the setting in which the isoperimetric problem and minimal surfaces are solved in geometric measure theory.

28.15 Weak convergence of measures

Sequences of measures appear whenever something discrete approximates something continuous, or something smooth approximates something singular. Riemann sums are integrals against finitely many point masses that approximate Lebesgue measure; in probability the distributions of normalised sums of random variables approximate the Gaussian distribution; smooth bumps of total mass 1 concentrating at a point approximate the Dirac measure; and in geometric measure theory one studies limits of the area measures of a sequence of surfaces, for instance of a minimising sequence for the area. Convergence $\mu_k(A) \to \mu(A)$ for all Borel sets A is far too strong for these purposes (Example 28.15.2(1)). The useful notion tests the measures against continuous functions. As in Section 28.14, X is a locally compact Hausdorff space with a countable basis.

Definition 28.15.1. A sequence of Radon measures μ_k on X *converges weakly* to a Radon measure μ, written $\mu_k \rightharpoonup \mu$, if

$$\int_X f \, d\mu_k \to \int_X f \, d\mu \qquad \text{for every } f \in C_c(X).$$

The limit is unique, by the uniqueness in Theorem 28.14.6. In probability theory this is called *vague* convergence, and *weak* convergence of probability measures means testing against all bounded continuous functions; the two agree when no mass is lost (Exercise 28.23.14). In functional analysis it is the weak-$*$ convergence of functionals on $C_c(X)$.

Example 28.15.2. 1. *Moving point masses.* If $x_k \to x$ in X, then $\delta_{x_k} \rightharpoonup \delta_x$, since $f(x_k) \to f(x)$ for continuous f. On $\mathbb{R}$, $\delta_{1/k} \rightharpoonup \delta_0$, but $\delta_{1/k}(\{0\}) = 0$ does not converge to $\delta_0(\{0\}) = 1$, and $\delta_{1/k}((0,1)) = 1$ does not converge to $\delta_0((0,1)) = 0$. So weak convergence does not imply convergence on compact sets, nor on open sets; Proposition 28.15.3 gives the inequalities that do hold.

2. *Riemann sums.* On $\mathbb{R}$ let $\mu_k = \frac{1}{k}\sum_{j=1}^{k} \delta_{j/k}$. Then $\mu_k \rightharpoonup \mathbf{1}_{[0,1]}\lambda$. Indeed $f \in C_c(\mathbb{R})$ is uniformly continuous, being continuous with compact support; with $\omega(\delta) = \sup\{|f(s)-f(t)| : |s-t| \le \delta\}$, which tends to 0 with δ,

$$\left|\int f\,d\mu_k - \int_0^1 f\,d\lambda\right| = \left|\sum_{j=1}^{k} \int_{(j-1)/k}^{j/k} \big(f(j/k) - f(t)\big)\,dt\right| \le \omega(1/k) \to 0.$$

Each μ_k lives on a finite set, of Lebesgue measure 0, while the limit is absolutely continuous with respect to λ.

3. *Approximate identities.* Let $\phi \ge 0$ be continuous on $\mathbb{R}$ with support in $[-1,1]$ and $\int \phi\,d\lambda = 1$, and $\phi_k(x) = k\phi(kx)$. Then $\int \phi_k\,d\lambda = 1$ (Theorem 28.11.2), ϕ_k vanishes outside $[-1/k, 1/k]$, and for $f \in C_c(\mathbb{R})$

$$\left|\int f\phi_k\,d\lambda - f(0)\right| = \left|\int \big(f(x) - f(0)\big)\phi_k(x)\,dx\right| \le \omega(1/k),$$

so $\phi_k\lambda \rightharpoonup \delta_0$: absolutely continuous measures converge to a singular one.

4. *Loss of mass.* On $\mathbb{R}$, $\delta_k \rightharpoonup 0$, since every $f \in C_c(\mathbb{R})$ vanishes at k for k large; and $\frac{1}{k}\mathbf{1}_{[0,k]}\lambda \rightharpoonup 0$, since $|\int_0^k f\,d\lambda|/k \le \|f\|_1/k$. All these measures have total mass 1, the limit has mass 0: mass can escape to infinity, by moving away or by spreading out.

5. *Oscillation.* Let $g \ge 0$ be continuous on $\mathbb{R}$ and periodic of period 2π, with mean value $\bar{g} = \frac{1}{2\pi}\int_0^{2\pi} g\,d\lambda$. Then $g(kx)\lambda \rightharpoonup \bar{g}\lambda$; for instance $(1+\sin kx)\lambda \rightharpoonup \lambda$, although the densities $1+\sin kx$ do not converge pointwise (at $x = \pi/2$ they run through $2,1,0,1,2,\ldots$). To see this, let $f \in C_c(\mathbb{R})$ with support in $[a,b]$ and $\varepsilon > 0$, and choose $a = a_0 < \cdots < a_N = b$ with $|f(s) - f(a_j)| \le \varepsilon$ for $s \in [a_{j-1}, a_j]$. An interval $[k\alpha, k\beta]$ consists of $\lfloor k(\beta-\alpha)/2\pi \rfloor$ full periods and a remainder shorter than 2π, so $\int_\alpha^\beta g(kx)\,dx = \frac{1}{k}\int_{k\alpha}^{k\beta} g(t)\,dt = (\beta-\alpha)\bar{g} + r$ with $|r| \le 4\pi\|g\|_\infty/k$. Replacing f by $f(a_j)$ on $[a_{j-1}, a_j]$ costs at most $\varepsilon(b-a)\|g\|_\infty$ in $\int fg(kx)\,dx$ and in $\bar{g}\int f\,dx$, so

$$\left|\int f(x)g(kx)\,dx - \bar{g}\int f(x)\,dx\right| \le 2\varepsilon(b-a)\|g\|_\infty + \frac{4\pi N\|f\|_\infty\|g\|_\infty}{k},$$

whose lim sup as $k \to \infty$ is at most $2\varepsilon(b-a)\|g\|_\infty$. Fast oscillations average out; this is the measure-theoretic content of the Riemann–Lebesgue lemma.

Proposition 28.15.3. *Let $\mu_k \rightharpoonup \mu$ be Radon measures on X.*

1. *$\mu(U) \le \liminf_k \mu_k(U)$ for every open U.*
2. *$\mu(K) \ge \limsup_k \mu_k(K)$ for every compact K.*
3. *$\mu_k(A) \to \mu(A)$ for every Borel set A with compact closure and $\mu(\partial A) = 0$.*

Proof. (1) By inner regularity, $\mu(U)$ is the supremum of $\mu(K)$ over compact $K \subseteq U$; for such K, Lemma 18.9.5 gives $f \prec U$ with $f = 1$ on K, and $\mu(K) \le \int f\,d\mu \le \mu(U)$. So $\mu(U) = \sup\{\int f\,d\mu : f \prec U\}$. For $f \prec U$, $\int f\,d\mu = \lim_k \int f\,d\mu_k \le \liminf_k \mu_k(U)$.

(2) Let $V \supseteq K$ be open. By Lemma 18.9.5 there is $f \prec V$ with $f = 1$ on K, and $\limsup_k \mu_k(K) \le \lim_k \int f\,d\mu_k = \int f\,d\mu \le \mu(V)$. By outer regularity, the infimum of $\mu(V)$ over such V is $\mu(K)$.

(3) The interior $\operatorname{int} A$ and the closure $\overline{A}$ differ by ∂A, so $\mu(\operatorname{int} A) = \mu(A) = \mu(\overline{A})$. By (1) and (2), $\mu(A) = \mu(\operatorname{int} A) \le \liminf_k \mu_k(\operatorname{int} A) \le \liminf_k \mu_k(A) \le \limsup_k \mu_k(A) \le \limsup_k \mu_k(\overline{A}) \le \mu(\overline{A}) = \mu(A)$. □

In Example 28.15.2(1) the inequalities are strict: $\delta_0((0,1)) = 0 < 1$ and $\delta_0(\{0\}) = 1 > 0$; the sets $(0,1)$ and $\{0\}$ have the point 0, of δ_0-measure 1, in their boundary. In (2) of the same example, (3) says that the proportion of the points j/k lying in an interval $[\alpha,\beta] \subseteq [0,1]$ tends to $\beta - \alpha$.

The main theorem about weak convergence is a compactness theorem: bounded mass on compact sets suffices for a convergent subsequence. Its proof needs a countable supply of test functions.

Lemma 28.15.4. *Let $K_1 \subseteq K_2 \subseteq \cdots$ be an exhaustion of X by compact sets with $K_m \subseteq \operatorname{int} K_{m+1}$. There is a countable set $\mathcal{D} \subseteq C_c(X)$ such that for every m, every $f \in C_c(X)$ with $\operatorname{supp} f \subseteq K_m$ and every $\varepsilon > 0$ there is $g \in \mathcal{D}$ with $\operatorname{supp} g \subseteq K_{m+1}$ and $\|f - g\|_\infty < \varepsilon$.*

Proof. First, $C(L,\mathbb{R})$ has a countable dense subset for every compact metric space (L,d). Indeed L has a countable dense subset $\{x_1, x_2, \ldots\}$ (for each j, finitely many balls of radius $1/j$ cover L; take their centres). The functions $u_i(y) = d(x_i, y)$ separate points: if $y \neq z$ and $d(x_i,y) < d(y,z)/2$, then $d(x_i,z) > d(y,z)/2 > d(x_i,y)$. By the Stone–Weierstrass theorem (Theorem 27.14.2) the polynomials in finitely many of the u_i are dense in $C(L,\mathbb{R})$, and so are those with rational coefficients, which form a countable set, since the u_i are bounded and real coefficients can be approximated by rational ones.

Each K_{m+1} is a compact metric space, X being metrisable; let $\mathcal{D}_m \subseteq C(K_{m+1},\mathbb{R})$ be countable and dense. By Lemma 18.9.5 choose $\chi_m \in C_c(X)$ with values in $[0,1]$, $\chi_m = 1$ on K_m and $\operatorname{supp}\chi_m \subseteq \operatorname{int} K_{m+1}$. For $u \in \mathcal{D}_m$ the function $\chi_m u$, extended by 0 outside K_{m+1}, is continuous on X: it is continuous on the open set $\operatorname{int} K_{m+1}$ and vanishes on the open set $X \setminus \operatorname{supp}\chi_m$, and these two open sets cover X. Let $\mathcal{D}$ be the set of all these functions, for all m. Given f with support in K_m, choose $u \in \mathcal{D}_m$ with $|f - u| < \varepsilon$ on K_{m+1}; then $f = \chi_m f$, so $|f - \chi_m u| = \chi_m |f - u| < \varepsilon$ on K_{m+1}, and both functions vanish outside K_{m+1}. □

Theorem 28.15.5 (Weak compactness of Radon measures)**.** *Let (μ_k) be a sequence of Radon measures on X with $\sup_k \mu_k(K) < \infty$ for every compact $K \subseteq X$. Then a subsequence converges weakly to a Radon measure.*

Proof. Let (K_m) be an exhaustion as in Lemma 28.15.4, $C_m = \sup_k \mu_k(K_m)$, and $\mathcal{D} = \{g_1, g_2, \ldots\}$ the countable set of the lemma. For each i the sequence $\int g_i \, d\mu_k$ is bounded, by $\|g_i\|_\infty \sup_k \mu_k(\operatorname{supp} g_i)$, so it has a convergent subsequence (Theorem 27.2.12). Choose a subsequence along which $\int g_1 \, d\mu_k$ converges, a further subsequence along which $\int g_2 \, d\mu_k$ converges, and so on; the diagonal sequence, the k-th term of the k-th subsequence, is a subsequence of each of them from its i-th term on. Renaming it (μ_k), $\int g \, d\mu_k$ converges for every $g \in \mathcal{D}$.

Let $f \in C_c(X)$, with support in some K_m, and $\varepsilon > 0$. Choose $g \in \mathcal{D}$ with support in K_{m+1} and $\|f - g\|_\infty < \varepsilon$. Then

$$\left| \int f \, d\mu_k - \int f \, d\mu_l \right| \leq 2\varepsilon C_{m+1} + \left| \int g \, d\mu_k - \int g \, d\mu_l \right|,$$

so $\int f \, d\mu_k$ is a Cauchy sequence. Its limit Λf is linear and positive in f, so by Theorem 28.14.6 it is $\int f \, d\mu$ for a Radon measure μ, and $\mu_k \rightharpoonup \mu$. □

Corollary 28.15.6. *Let X be a compact metric space. Every sequence of Borel probability measures on X has a subsequence converging weakly to a Borel probability measure.*

Proof. A compact metric space is locally compact and Hausdorff, and it has a countable basis: the balls of radius $1/j$ centred at the points of finite sets of such balls covering X. Every Borel measure on X that is finite is a Radon measure (Proposition 28.14.2), and $\mu_k(X) = 1$ bounds

the masses. Theorem 28.15.5 gives a weakly convergent subsequence, and its limit μ satisfies $\mu(X) = \int 1\,d\mu = \lim \int 1\,d\mu_k = 1$, as $1 \in C_c(X)$ for compact X. □

Remark 28.15.7.

1. The hypothesis cannot be dropped: $k\delta_0$ has no weakly convergent subsequence on $\mathbb{R}$. On non-compact spaces the limit of probability measures may have smaller mass, by Example 28.15.2(4); by Proposition 28.15.3(1) applied to $U = X$, mass can only be lost, never gained.

2. Theorem 28.15.5 is a case of the Banach–Alaoglu theorem: bounded sets of linear functionals are compact for the weak-$*$ topology (Chapter 30).

3. In the calculus of variations and in geometric measure theory this is the basic source of limits. A minimising sequence of surfaces for the area, all contained in a fixed ball, has area measures with bounded mass, and a subsequence of these converges weakly to a Radon measure; the hard part, the *regularity theory*, is to show that the limit is again the area measure of a surface, perhaps with singularities. For curves, Exercise 28.23.15 shows that length can jump in the limit.

28.16 Differentiation of measures

The Lebesgue differentiation theorem (Theorem 28.9.4) says that for $h \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$ the averages $\frac{1}{\lambda(B(x,r))}\int_{B(x,r)} h$ tend to $h(x)$ for almost every x. In terms of the measure $\nu = h\lambda$ it says that $\nu(B(x,r))/\lambda(B(x,r)) \to h(x)$: the density of ν can be computed by shrinking balls. We now do this for an arbitrary Radon measure ν on $\mathbb{R}^n$. It has a Lebesgue decomposition $\nu = h\lambda + \nu_s$ with respect to λ (Theorem 28.12.3, which applies since ν is σ-finite by Proposition 28.14.2), and the shrinking balls recover h and do not see the singular part, except on a null set. In this section λ denotes Lebesgue measure on the Borel sets of $\mathbb{R}^n$.

Definition 28.16.1. Let ν be a Radon measure on $\mathbb{R}^n$. The *upper* and *lower derivatives* of ν with respect to λ at x are

$$\overline{D}\nu(x) = \limsup_{r\to 0} \frac{\nu(B(x,r))}{\lambda(B(x,r))}, \qquad \underline{D}\nu(x) = \liminf_{r\to 0} \frac{\nu(B(x,r))}{\lambda(B(x,r))},$$

in $[0,\infty]$. If they agree, their common value is the *derivative* $D\nu(x)$.

Example 28.16.2. For $\nu = \delta_0$ on $\mathbb{R}^n$, $\nu(B(x,r)) = 0$ for $r \le \|x\|$, so $D\nu(x) = 0$ for $x \neq 0$, while $D\nu(0) = \lim_r 1/\lambda(B(0,r)) = \infty$. For the measure σ on the circle (Example 28.14.5(3)), $D\sigma = 0$ outside S^1, because the balls around a point at positive distance from S^1 eventually miss S^1; at a point of S^1 the ball $B(x,r)$ meets S^1 in an arc of length about $2r$, so $\sigma(B(x,r))/\pi r^2$ behaves like $2/(\pi r)$ and $D\sigma = \infty$. In both cases the derivative is 0 almost everywhere, in accordance with the singularity of these measures.

Theorem 28.16.3 (Differentiation of measures)**.** *Let ν be a Radon measure on $\mathbb{R}^n$, and $\nu = h\lambda + \nu_s$ its Lebesgue decomposition, with $h \ge 0$ Borel and $\nu_s \perp \lambda$. Then $h \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$, and for λ-almost every x*

$$\lim_{r\to 0} \frac{1}{\lambda(B(x,r))}\Big(\int_{B(x,r)} |h(y) - h(x)|\,d\lambda(y) + \nu_s(B(x,r))\Big) = 0.$$

In particular $D\nu = h$ almost everywhere, and $\nu \perp \lambda$ if and only if $D\nu = 0$ almost everywhere.

Proof. For compact K, $\int_K h\,d\lambda \le \nu(K) < \infty$, so $h \in L^1_{\mathrm{loc}}$, and by the Lebesgue differentiation theorem (Theorem 28.9.4) the first term tends to 0 almost everywhere. It remains to show

that $\overline{D}\nu_s = 0$ almost everywhere. Let S be a Borel set with $\lambda(S) = 0$ and $\nu_s(\mathbb{R}^n \setminus S) = 0$. The measure ν_s is a Borel measure finite on compact sets ($\nu_s \le \nu$), so it is a Radon measure (Proposition 28.14.2). Fix $t > 0$ and $R > 0$, and let

$$A = \{x \in B(0,R) \setminus S : \overline{D}\nu_s(x) > t\}.$$

We show $\lambda^*(A) = 0$. Let $\varepsilon > 0$. The Borel set $B(0,R) \setminus S$ has ν_s-measure 0, so by outer regularity it is contained in an open set U with $\nu_s(U) < \varepsilon$. Let G be the union of all open balls B with $B \subseteq U$ and $\nu_s(B) > t\lambda(B)$. It is open and contains A: for $x \in A$ we have $x \in U$, so $B(x,r) \subseteq U$ for all small r, and $\nu_s(B(x,r)) > t\lambda(B(x,r))$ for some of these r. Let $K \subseteq G$ be compact. Finitely many of these balls cover K, and by the Vitali covering lemma (Lemma 28.9.1) they contain pairwise disjoint balls $B_1, \dots, B_k$ with

$$\lambda(K) \le 3^n \sum_j \lambda(B_j) < \frac{3^n}{t} \sum_j \nu_s(B_j) \le \frac{3^n}{t}\nu_s(U) < \frac{3^n \varepsilon}{t}.$$

By inner regularity of λ, $\lambda^*(A) \le \lambda(G) \le 3^n\varepsilon/t$, and as ε is arbitrary, A is a null set. Taking the union over $t = 1/j$ and $R = j$, $j \ge 1$, and adding the null set S, we find that $\overline{D}\nu_s = 0$ outside a null set.

Finally, $\nu(B(x,r))/\lambda(B(x,r))$ differs from the average of h over $B(x,r)$ by $\nu_s(B(x,r))/\lambda(B(x,r))$, and the average differs from $h(x)$ by at most the average of $|h - h(x)|$; so $D\nu = h$ almost everywhere. If $\nu \perp \lambda$, then $h = 0$ almost everywhere by uniqueness of the decomposition, so $D\nu = 0$ almost everywhere; conversely, if $D\nu = 0$ almost everywhere, then $h = 0$ almost everywhere and $\nu = \nu_s$. □

So the Radon–Nikodym derivative of the absolutely continuous part, an object obtained in Theorem 28.12.2 by an abstract Hilbert space argument, can be computed as a limit of ratios. Balls are not essential: any sets that shrink to x and occupy a fixed proportion of a ball around x will do.

Corollary 28.16.4. *Let $\nu = h\lambda + \nu_s$ be as in Theorem 28.16.3 and $c > 0$. For almost every x the following holds: if E_k are Borel sets with $E_k \subseteq B(x, r_k)$, $r_k \to 0$ and $\lambda(E_k) \ge c\lambda(B(x,r_k))$, then $\nu(E_k)/\lambda(E_k) \to h(x)$.*

Proof. Take x as in the theorem. Then

$$\left|\frac{\nu(E_k)}{\lambda(E_k)} - h(x)\right| \le \frac{1}{\lambda(E_k)}\Big(\int_{E_k} |h-h(x)|\,d\lambda + \nu_s(E_k)\Big) \le \frac{1}{c\lambda(B(x,r_k))}\Big(\int_{B(x,r_k)} |h-h(x)|\,d\lambda + \nu_s(B(x,r_k))\Big),$$

which tends to 0. □

Corollary 28.16.5 (Lebesgue's theorem on monotone functions)**.** *Let $F\colon [a,b] \to \mathbb{R}$ be continuous and nondecreasing. Then F is differentiable almost everywhere, $F' \ge 0$ is integrable, and*

$$\int_a^b F'\,d\lambda \le F(b) - F(a),$$

with equality if and only if F is absolutely continuous.

Proof. Let μ_F be the measure of Lemma 28.13.3, and $\nu(E) = \mu_F(E \cap [a,b])$ for Borel $E \subseteq \mathbb{R}$; it is a finite Borel measure on $\mathbb{R}$, hence a Radon measure. Let $\nu = h\lambda + \nu_s$ be its Lebesgue decomposition, and let $x \in (a,b)$ be a point where Corollary 28.16.4 holds with $c = 1/4$. For $0 < r < \min(b-x, x-a)$, the intervals $(x, x+r]$ and $(x-r, x]$ have length $r = \frac{1}{4}\lambda(B(x,2r))$ and lie in $B(x,2r)$, and $\nu((x,x+r]) = F(x+r) - F(x)$, $\nu((x-r,x]) = F(x) - F(x-r)$. So both

one-sided difference quotients of F at x tend to $h(x)$, and $F'(x) = h(x)$. Hence $F' = h$ almost everywhere on $[a, b]$, and

$$\int_a^b F' \, d\lambda = \int_{[a,b]} h \, d\lambda \le \nu([a, b]) = F(b) - F(a),$$

with equality if and only if $\nu_s([a, b]) = 0$, that is, $\nu_s = 0$. In that case $F(x) - F(a) = \nu([a, x]) = \int_a^x h \, d\lambda$ for all x, so F is absolutely continuous by Theorem 28.13.4. Conversely, if F is absolutely continuous, the same theorem gives $F(b) - F(a) = \int_a^b F'$. □

For the Cantor function (Counterexample 28.9.7) the inequality is strict: μ_F is singular, so $F' = 0$ almost everywhere, as we saw directly, while $F(1) - F(0) = 1$. The corollary holds for all nondecreasing functions, continuous or not: a nondecreasing function has at most countably many jumps, and the jumps contribute a further singular part to its measure [Rud87, Chapter 7]. Since a function of bounded variation is a difference of two nondecreasing functions, it is differentiable almost everywhere as well.

Remark 28.16.6. The proof used Lebesgue measure through the Vitali covering lemma, that is, through the fact that $\lambda(3B) = 3^n \lambda(B)$. To differentiate a Radon measure ν with respect to an arbitrary Radon measure μ on $\mathbb{R}^n$, which may give very different masses to B and $3B$, one uses instead the *Besicovitch covering theorem*: from every family of balls whose centres form a bounded set A one can extract a subfamily covering A that splits into a number of disjoint subfamilies depending only on n. With it one shows that $\nu(B(x, r))/\mu(B(x, r))$ converges μ-almost everywhere to the density of the part of ν absolutely continuous with respect to μ, and tends to ∞ almost everywhere for the singular part [EG92, Chapter 1]. This is the setting of geometric measure theory, where μ is often a Hausdorff measure restricted to a set.

28.17 Hausdorff measures

Lebesgue measure λ^n measures n-dimensional volume in $\mathbb{R}^n$, but it does not see lower-dimensional sets: a curve in the plane, or a surface in space, has measure 0. How long is a curve, and what is the area of a surface, without choosing a parametrisation? And how big is the Cantor set, which has measure 0 and is uncountable? Hausdorff measures answer these questions at once: for every $s \ge 0$ there is an s-dimensional measure $\mathcal{H}^s$ on any metric space, defined by covering a set with small sets C and adding up $(\operatorname{diam} C)^s$. This and the following sections are an introduction to *geometric measure theory*: Hausdorff measure and dimension, self-similar sets, differentiability of Lipschitz maps, the area formula, which computes $\mathcal{H}^k$ of parametrised k-dimensional sets, the coarea formula, which slices sets by level sets, and densities and rectifiable sets. We follow [EG92, Chapters 2–3]; the comprehensive reference is [Fed69].

Throughout, X is a metric space, $\operatorname{diam} C = \sup\{d(x, y) : x, y \in C\}$ for $C \ne \emptyset$, and $\operatorname{diam} \emptyset = 0$. For $s \ge 0$ let $\alpha(s) = \pi^{s/2}/(2^s \Gamma(\frac{s}{2} + 1))$, with Γ as in Exercise 28.23.7; for an integer n, $\alpha(n)$ is the volume of a ball of diameter 1 in $\mathbb{R}^n$. The constant only serves to make $\mathcal{H}^n$ equal to Lebesgue measure on $\mathbb{R}^n$ (Theorem 28.17.5); in the sums below the empty set contributes 0 and every nonempty set C contributes $\alpha(s)(\operatorname{diam} C)^s$, with the convention $(\operatorname{diam} C)^0 = 1$, also when C is a single point.

Definition 28.17.1. For $A \subseteq X$, $s \ge 0$ and $\delta > 0$ let

$$\mathcal{H}^s_\delta(A) = \inf\Big\{ \sum_k \alpha(s)(\operatorname{diam} C_k)^s : A \subseteq \bigcup_k C_k, \ \operatorname{diam} C_k \le \delta \Big\},$$

the infimum over countable covers by sets of diameter at most δ. As δ decreases, fewer covers are allowed and $\mathcal{H}^s_\delta(A)$ increases; the *s-dimensional Hausdorff (outer) measure* is $\mathcal{H}^s(A) = \lim_{\delta\to 0} \mathcal{H}^s_\delta(A) = \sup_{\delta>0} \mathcal{H}^s_\delta(A)$.

The restriction $\operatorname{diam} C_k \le \delta$ is essential: with arbitrary covers, a long segment could be covered by a single set and would get the same size as a short one, while small sets are forced to follow the shape of A. Replacing each C_k by its closure does not change its diameter, so closed covers give the same infimum. And $\mathcal{H}^s(A)$ only depends on the metric space A: a cover of A by subsets of X gives the cover by the sets $C_k \cap A$, of smaller diameters. In particular an isometry $A \to B$ between subsets of two metric spaces preserves $\mathcal{H}^s$.

Lemma 28.17.2 (Carathéodory's criterion)**.** *Let μ^* be an outer measure on X which is* metric*: $\mu^*(A \cup B) = \mu^*(A) + \mu^*(B)$ whenever $d(A,B) = \inf\{d(a,b) : a \in A, b \in B\} > 0$. Then every Borel set is μ^*-measurable.*

Proof. The measurable sets form a σ-algebra (Theorem 28.3.2), so it suffices to show that a closed set F is measurable, that is, $\mu^*(E\cap F)+\mu^*(E\setminus F) \le \mu^*(E)$ for every E with $\mu^*(E) < \infty$. Let $E_j = \{x \in E : d(x,F) \ge 1/j\}$. Since $d(E\cap F, E_j) \ge 1/j$, $\mu^*(E\cap F) + \mu^*(E_j) = \mu^*((E\cap F)\cup E_j) \le \mu^*(E)$, and it remains to show $\mu^*(E_j) \to \mu^*(E\setminus F)$. Let $R_j = \{x \in E : \frac{1}{j+1} \le d(x,F) < \frac{1}{j}\}$. As F is closed, $E\setminus F = E_j \cup \bigcup_{i\ge j} R_i$. If $i \ge l+2$, then $d(R_i,R_l) > 0$, so by induction $\sum_{i\le N} \mu^*(R_{2i}) = \mu^*(\bigcup_{i\le N} R_{2i}) \le \mu^*(E)$, and similarly for the odd indices. Hence $\sum_i \mu^*(R_i) < \infty$, and $\mu^*(E\setminus F) \le \mu^*(E_j) + \sum_{i\ge j} \mu^*(R_i)$, where the tail tends to 0. With $\mu^*(E_j) \le \mu^*(E\setminus F)$ this gives the limit. □

This is *Carathéodory's criterion*; it is a different statement from Carathéodory's theorem (Theorem 28.3.2), which it uses: the theorem builds a measure from any outer measure, and the criterion says that for a metric outer measure the resulting σ-algebra contains the Borel sets.

Proposition 28.17.3.

1. *$\mathcal{H}^s$ is a metric outer measure, so its restriction to the Borel sets is a measure.*
2. *$\mathcal{H}^0$ is the counting measure.*
3. *If $f : A \to Y$ is Lipschitz with constant L, then $\mathcal{H}^s(f(A)) \le L^s\mathcal{H}^s(A)$. For $X = \mathbb{R}^n$, $\mathcal{H}^s$ is translation invariant and $\mathcal{H}^s(rA) = r^s\mathcal{H}^s(A)$ for $r > 0$.*
4. *If $\mathcal{H}^s(A) < \infty$ and $t > s$, then $\mathcal{H}^t(A) = 0$. If $\mathcal{H}^t(A) > 0$ and $s < t$, then $\mathcal{H}^s(A) = \infty$.*

Proof. (1) Each $\mathcal{H}^s_\delta$ is an outer measure: it is monotone, and given covers of the A_k with sums at most $\mathcal{H}^s_\delta(A_k) + \varepsilon 2^{-k}$, their union covers $\bigcup_k A_k$. Then $\mathcal{H}^s(\bigcup_k A_k) = \sup_\delta \mathcal{H}^s_\delta(\bigcup_k A_k) \le \sup_\delta \sum_k \mathcal{H}^s_\delta(A_k) \le \sum_k \mathcal{H}^s(A_k)$. If $d(A,B) > 0$ and $\delta < d(A,B)$, no set of diameter at most δ meets both A and B, so a δ-cover of $A\cup B$ splits into covers of A and of B, and $\mathcal{H}^s_\delta(A\cup B) \ge \mathcal{H}^s_\delta(A) + \mathcal{H}^s_\delta(B)$; let $\delta \to 0$. Apply Lemma 28.17.2.

(2) $\alpha(0) = 1$. A set of N points needs at least N nonempty sets of diameter less than the smallest distance between the points, and the N singletons suffice; so $\mathcal{H}^0_\delta$ of it is N for small δ. An infinite set contains finite sets of every size.

(3) If (C_k) is a δ-cover of A, then $(f(C_k\cap A))$ is an $L\delta$-cover of $f(A)$ with $\operatorname{diam} f(C_k\cap A) \le L \operatorname{diam} C_k$; so $\mathcal{H}^s_{L\delta}(f(A)) \le L^s\mathcal{H}^s_\delta(A)$. Translations are isometries, and $x \mapsto rx$ multiplies diameters by r, with inverse $x \mapsto x/r$.

(4) For a δ-cover, $\sum_k \alpha(t)(\operatorname{diam} C_k)^t \le \frac{\alpha(t)}{\alpha(s)}\delta^{t-s} \sum_k \alpha(s)(\operatorname{diam} C_k)^s$, so $\mathcal{H}^t_\delta(A) \le \frac{\alpha(t)}{\alpha(s)}\delta^{t-s}\mathcal{H}^s_\delta(A) \le \frac{\alpha(t)}{\alpha(s)}\delta^{t-s}\mathcal{H}^s(A) \to 0$. The second statement is the contrapositive. □

Remark 28.17.4. 1. *Covers may not exist.* With the convention $\inf\emptyset = +\infty$, $\mathcal{H}^s_\delta(A) = \infty$ when A has no countable cover by sets of diameter at most δ. In a separable metric space this never happens: if $\{x_k\}$ is a countable dense set, the balls $B(x_k, \delta/2)$ have diameter at most δ and cover X. In an uncountable set with the discrete metric ($d(x, y) = 1$ for $x \neq y$), sets of diameter less than 1 are points, so an uncountable set has no countable δ-cover for $\delta < 1$, and $\mathcal{H}^s$ of it is ∞ for every s, consistently with (2) for $s = 0$. All subsets of $\mathbb{R}^n$ and of manifolds are separable.

2. *Borel regularity.* $\mathcal{H}^s$ is *Borel regular*: every A is contained in a Borel set B with $\mathcal{H}^s(B) = \mathcal{H}^s(A)$. We may assume $\mathcal{H}^s(A) < \infty$. For each j choose a cover of A by closed sets $C^{(j)}_k$ of diameter at most $1/j$ with $\sum_k \alpha(s)(\operatorname{diam} C^{(j)}_k)^s \leq \mathcal{H}^s_{1/j}(A) + 1/j$, and let $B = \bigcap_j \bigcup_k C^{(j)}_k$, a Borel set containing A. Then $\mathcal{H}^s_{1/j}(B) \leq \mathcal{H}^s_{1/j}(A) + 1/j$ for all j, so $\mathcal{H}^s(B) \leq \mathcal{H}^s(A)$. On the other hand $\mathcal{H}^s$ is in general not locally finite (Example 28.3.6), hence not a Radon measure on the whole space; restricted to a set of finite $\mathcal{H}^s$-measure it is [EG92, Section 1.1].

By (4), for each A there is a unique $s_0 \in [0, \infty]$ with $\mathcal{H}^s(A) = \infty$ for $s < s_0$ and $\mathcal{H}^s(A) = 0$ for $s > s_0$; it is the *Hausdorff dimension* $\dim_{\mathcal{H}} A = \inf\{s : \mathcal{H}^s(A) = 0\}$. At $s = s_0$ the measure can be anything in $[0, \infty]$. By (3), Lipschitz maps do not increase Hausdorff dimension, and bi-Lipschitz maps preserve it.

Theorem 28.17.5. *On the Borel sets of $\mathbb{R}^n$, $\mathcal{H}^n = \lambda^n$. Consequently the subsets of $\mathbb{R}^n$ have Hausdorff dimension at most n, and nonempty open subsets have dimension n.*

Proof. We show that $\mathcal{H}^n = c\lambda^n$ for a constant $0 < c < \infty$; that $c = 1$ is the content of the *isodiametric inequality* $\lambda^n(C) \leq \alpha(n)(\operatorname{diam} C)^n$, which says that among sets of given diameter the balls have the largest volume, together with Vitali's covering theorem, and we refer to [EG92, Section 2.2] for these two steps.

The constant. Let $Q = [0, 1)^n$. Covering Q by m^n cubes of side $1/m$ and diameter $\sqrt{n}/m$ gives $\mathcal{H}^n_{\sqrt{n}/m}(Q) \leq \alpha(n)n^{n/2}$, so $\mathcal{H}^n(Q) < \infty$. Conversely, a set C of diameter d lies in a closed box with sides d (each coordinate varies by at most d on C), so $\lambda^*(C) \leq d^n$; for any cover of Q, $\sum_k \alpha(n)(\operatorname{diam} C_k)^n \geq \alpha(n) \sum_k \lambda^*(C_k) \geq \alpha(n)$, so $c = \mathcal{H}^n(Q) \geq \alpha(n) > 0$.

Comparison. $\mathcal{H}^n$ is a translation invariant measure on the Borel sets (Proposition 28.17.3). For $j \geq 0$, Q is the disjoint union of 2^{jn} translates of $[0, 2^{-j})^n$, so every half-open dyadic cube $\prod_i [a_i 2^{-j}, (a_i + 1)2^{-j})$ ($a_i \in \mathbb{Z}$, $j \in \mathbb{Z}$) has $\mathcal{H}^n$-measure $c2^{-jn} = c\lambda^n$ of it; for $j < 0$ use that such a cube is a disjoint union of 2^{-jn} unit cubes. Fix $m \geq 0$ and $P = [-2^m, 2^m)^n$. The dyadic cubes contained in P, together with P, form a family closed under finite intersections, on which $\mathcal{H}^n$ and $c\lambda^n$ agree and are finite. By Lemma 28.3.10 the two measures agree on the σ-algebra it generates, which contains every open subset of the interior of P (a countable union of dyadic cubes), hence every Borel subset of the interior. The boundary of P is a finite union of pieces of hyperplanes, which are null for both measures: a piece $[0, 1]^{n-1} \times \{0\}$ is covered by j^{n-1} cubes of diameter $\sqrt{n}/j$, with $\sum \alpha(n)(\operatorname{diam})^n \to 0$ as $j \to \infty$. So the two measures agree on the Borel subsets of P, and letting $m \to \infty$, on all Borel sets.

The last statement follows from Proposition 28.17.3(4): $\mathcal{H}^n$ is finite on bounded sets and $\mathbb{R}^n$ is a countable union of them, so $\mathcal{H}^t = 0$ for $t > n$, while a nonempty open set has positive $\mathcal{H}^n$-measure. □

Example 28.17.6. *The Cantor set has Hausdorff dimension $s = \log 2/\log 3$, and $\alpha(s)/2 \leq \mathcal{H}^s(C) \leq \alpha(s)$;* note $3^s = 2$. Let $C \subseteq [0, 1]$ be as in Counterexample 28.9.7. For $k \geq 0$ and digits $a_1, \dots, a_k \in \{0, 2\}$ let $I_{a_1 \cdots a_k}$ be the closed interval with left end $\sum_{j \leq k} a_j 3^{-j}$ and length 3^{-k}; there are 2^k of them, and C is contained in their union C_k, since the digits beyond the k-th contribute at most 3^{-k}.

1. *Upper bound.* The 2^k intervals form a 3^{-k}-cover of C with $\sum \alpha(s)3^{-ks} = \alpha(s)2^k3^{-ks} = \alpha(s)$. So $\mathcal{H}^s(C) \le \alpha(s)$.

2. *Separation.* Two of these intervals of the same generation k are at distance at least 3^{-k}: if their digit strings first differ at position $j \le k$, their left ends differ by at least $2\cdot 3^{-j} - \sum_{j<i\le k} 2\cdot 3^{-i} = 3^{-j} + 3^{-k} \ge 2\cdot 3^{-k}$. So a set of diameter less than 3^{-k} meets at most one of them.

3. *The Cantor measure.* Let $\mu = \mu_F$ be the measure of Lemma 28.13.3 for the Cantor function F. Since F is constant on each interval of $[0,1]\setminus C$, $\mu([0,1]\setminus C) = 0$ and $\mu(C) = F(1) - F(0) = 1$. On the generation-k interval I with left end $a = \sum_{j\le k} a_j 3^{-j}$ and right end $a + \sum_{j>k} 2\cdot 3^{-j}$, $\mu(I) = F(a+3^{-k}) - F(a) = \sum_{j>k} 2^{-j} = 2^{-k}$, by the formula for c on C.

4. *Lower bound.* Let U be a closed set with $0 < \operatorname{diam} U = d < 1$, and k with $3^{-k-1} \le d < 3^{-k}$. By (2), U meets at most one generation-k interval I, and $\mu(U) \le \mu(I) + \mu([0,1]\setminus C) = 2^{-k} = 3^s 3^{-(k+1)s} \le 2d^s$; for a single point, $\mu(U) = 0$. For a cover of C by closed sets U_i of diameter less than 1, $1 = \mu(C) \le \sum_i \mu(U_i \cap [0,1]) \le 2\sum_i (\operatorname{diam} U_i)^s$. Hence $\mathcal{H}^s_\delta(C) \ge \alpha(s)/2$ for $\delta < 1$.

The argument in (4), bounding the mass that a measure on A gives to small sets, is the *mass distribution principle*, the standard way to bound Hausdorff measures from below. So the Cantor set is too small to have positive length but too large to have zero s-dimensional measure for any $s < \log 2/\log 3 \approx 0.63$.

28.18 Self-similar sets

The Cantor set C is the union of two copies of itself, $C/3$ and $C/3 + 2/3$, each scaled by the factor $\frac{1}{3}$. If we only knew that $0 < \mathcal{H}^s(C) < \infty$ for some s, the scaling property of Proposition 28.17.3(3) would already give the value of s: the two copies are disjoint Borel sets, so $\mathcal{H}^s(C) = 2\cdot 3^{-s}\mathcal{H}^s(C)$, and $3^s = 2$. Many sets that come up in analysis and dynamics, usually called *fractals*, are built in the same way from finitely many scaled copies of themselves, and their dimension can be read off from the scaling factors. This section makes the heuristic rigorous: we construct such sets in general (Theorem 28.18.5), and compute their dimension when the copies are disjoint (Theorem 28.18.10). Along the way we meet a second notion of dimension, which is cruder than the Hausdorff dimension but easier to compute. References: [Fal90], [Hut81].

28.18.1 Box-counting dimension

Hausdorff measure allows covers by sets of very different sizes, and this is what makes it a measure. If instead we cover a set by sets of one fixed size δ and only count them, we get a simpler quantity. A segment of length 1 needs about $1/\delta$ sets of diameter δ, a square about δ^{-2}, a cube about δ^{-3}: the exponent is the dimension.

Definition 28.18.1. For a nonempty bounded $A \subseteq \mathbb{R}^n$ and $\delta > 0$, let $N_\delta(A)$ be the smallest number of sets of diameter at most δ that cover A. The *lower* and *upper box-counting dimensions* of A are

$$\underline{\dim}_B A = \liminf_{\delta\to 0} \frac{\log N_\delta(A)}{-\log\delta}, \qquad \overline{\dim}_B A = \limsup_{\delta\to 0} \frac{\log N_\delta(A)}{-\log\delta}.$$

If they are equal, their common value is the *box-counting dimension* $\dim_B A$.

The number $N_\delta(A)$ is finite: A lies in a cube, which is a finite union of small cubes. The definition says that $N_\delta(A)$ is roughly $\delta^{-\dim_B A}$; more precisely, $\overline{\dim}_B A < t$ if and only if $N_\delta(A) \le \delta^{-t'}$ for some $t' < t$ and all small δ.

Proposition 28.18.2. *Let $A, A_1, A_2, \ldots$ be subsets of $\mathbb{R}^n$.*

1. *(*Countable stability*)* $\dim_{\mathcal{H}} \bigcup_k A_k = \sup_k \dim_{\mathcal{H}} A_k$. *In particular countable sets have Hausdorff dimension* 0.

2. *If A is nonempty and bounded, $N_\delta(A) = N_\delta(\overline{A})$, so A and its closure have the same box-counting dimensions.*

3. *If A is nonempty and bounded,* $\dim_{\mathcal{H}} A \le \underline{\dim}_B A \le \overline{\dim}_B A \le n$.

Proof. (1) Since $\mathcal{H}^s$ is monotone, $\dim_{\mathcal{H}} A_k \le \dim_{\mathcal{H}} \bigcup_k A_k$ for every k. Conversely, if $s > \sup_k \dim_{\mathcal{H}} A_k$, then $\mathcal{H}^s(A_k) = 0$ for every k, and $\mathcal{H}^s(\bigcup_k A_k) = 0$ by countable subadditivity; so $\dim_{\mathcal{H}} \bigcup_k A_k \le s$. A point has $\mathcal{H}^s$-measure 0 for every $s > 0$, since it is covered by one set of diameter 0; so a point has dimension 0, and so has a countable set.

(2) A cover of A by sets of diameter at most δ gives the cover of $\overline{A}$ by their closures, which have the same diameters, since a finite union of closed sets is closed. So $N_\delta(\overline{A}) \le N_\delta(A)$, and the other inequality holds as $A \subseteq \overline{A}$.

(3) The middle inequality is clear. For the last one, let A lie in a cube of side ℓ. Dividing it into m^n cubes of side ℓ/m, of diameter $\ell\sqrt{n}/m$, with m the smallest integer $\ge \ell\sqrt{n}/\delta$, gives $N_\delta(A) \le m^n \le (\ell\sqrt{n}/\delta + 1)^n$, and $\log N_\delta(A)/(-\log\delta) \le n \log(\ell\sqrt{n} + \delta)/(-\log\delta) + n$, whose lim sup as $\delta \to 0$ is n. For the first inequality, let $s > t > \underline{\dim}_B A$. There are $\delta_j \to 0$ with $\log N_{\delta_j}(A) < t(-\log\delta_j)$, that is, $N_{\delta_j}(A) < \delta_j^{-t}$. Covering A by $N_{\delta_j}(A)$ sets of diameter at most δ_j gives

$$\mathcal{H}^s_{\delta_j}(A) \le \alpha(s) N_{\delta_j}(A) \delta_j^s \le \alpha(s) \delta_j^{s-t} \to 0.$$

As $\mathcal{H}^s_\delta(A)$ increases when δ decreases, $\mathcal{H}^s(A) = \lim_j \mathcal{H}^s_{\delta_j}(A) = 0$, so $\dim_{\mathcal{H}} A \le s$. Since $s > \underline{\dim}_B A$ was arbitrary, the inequality follows. □

By (2), box-counting dimension cannot see the difference between a set and its closure: $\mathbb{Q} \cap [0,1]$ has box-counting dimension 1, like $[0,1]$, while its Hausdorff dimension is 0 by (1). So box-counting dimension is not countably stable, and in general it is not attached to a measure. The inequality in (3) can be strict even for compact sets.

Example 28.18.3. *A compact countable set of box-counting dimension $\frac{1}{2}$.* Let $E = \{0\} \cup \{1/m : m \ge 1\}$. It is countable, so $\dim_{\mathcal{H}} E = 0$. We show that $\dim_B E = \frac{1}{2}$. Let $0 < \delta < 1$.

1. *Lower bound.* Let $m = \lfloor \delta^{-1/2} \rfloor \ge 1$. For $2 \le j \le m$ the consecutive points $1/(j-1)$ and $1/j$ are at distance $\frac{1}{j(j-1)} > \frac{1}{m^2} \ge \delta$. So the points $1, \frac{1}{2}, \ldots, \frac{1}{m}$ are pairwise more than δ apart, a set of diameter at most δ contains at most one of them, and $N_\delta(E) \ge m > \delta^{-1/2} - 1$.

2. *Upper bound.* Let $m = \lceil \delta^{-1/2} \rceil$. The interval $[0, \frac{1}{m}]$, which contains all points $1/j$ with $j \ge m$, is covered by $\lceil \frac{1}{m\delta} \rceil$ intervals of length δ, and the remaining $m-1$ points are covered by singletons. Since $m - 1 < \delta^{-1/2}$ and $\frac{1}{m\delta} \le \delta^{-1/2}$, $N_\delta(E) \le 2\delta^{-1/2} + 1 \le 3\delta^{-1/2}$.

Taking logarithms, $\log(\delta^{-1/2} - 1)/(-\log\delta)$ and $(\log 3 + \frac{1}{2}(-\log\delta))/(-\log\delta)$ both tend to $\frac{1}{2}$ as $\delta \to 0$. The reason is that the points $1/j$ are spread out for j small and crowded for j large: at scale δ one sees about $\delta^{-1/2}$ separate points, and the rest look like an interval of length about $\delta^{1/2}$.

28.18.2 Iterated function systems

A map $f\colon \mathbb{R}^n \to \mathbb{R}^n$ is a *similarity with ratio* $r > 0$ if $|f(x) - f(y)| = r|x - y|$ for all x, y; for example $f(x) = rOx + b$ with O orthogonal (every similarity has this form, but we do not need this). A similarity with ratio r multiplies diameters by r, and by Proposition 28.17.3(3), applied to f and to its inverse, $\mathcal{H}^s(f(A)) = r^s\mathcal{H}^s(A)$ for every A and s.

Definition 28.18.4. An *iterated function system* (of similarities) is a finite family $f_1, \dots, f_N$ of similarities of $\mathbb{R}^n$ with ratios $r_1, \dots, r_N \in (0,1)$, where $N \geq 2$. For a set A write $F(A) = f_1(A) \cup \cdots \cup f_N(A)$. A nonempty compact set K with $F(K) = K$ is called the *attractor* of the system, or a *self-similar set*.

It is convenient to use words in the alphabet $\{1, \dots, N\}$. For a word $w = w_1 \cdots w_k$ of length $|w| = k$ let

$$f_w = f_{w_1} \circ f_{w_2} \circ \cdots \circ f_{w_k}, \qquad r_w = r_{w_1} r_{w_2} \cdots r_{w_k},$$

so that f_w is a similarity with ratio r_w; for the empty word $\varnothing$, $f_\varnothing$ is the identity and $r_\varnothing = 1$. Write wi for the word w followed by the letter i, and $r = \max_i r_i < 1$, so that $r_w \leq r^{|w|}$. For an infinite sequence $\omega = \omega_1\omega_2\cdots$ of letters, $\omega|k = \omega_1 \cdots \omega_k$ is its beginning of length k. Since $F(A) = \bigcup_i f_i(A)$, induction on k gives $F^k(A) = \bigcup_{|w|=k} f_w(A)$.

Theorem 28.18.5 (Hutchinson)**.** *Every iterated function system has a unique attractor K. Moreover:*

1. *If B is a nonempty compact set with $F(B) \subseteq B$, then $K = \bigcap_k F^k(B)$; in particular $K \subseteq B$.*
2. *Let $K_w = f_w(K)$. Then $K = \bigcup_{|w|=k} K_w$ for every k, and $K_w = \bigcup_i K_{wi}$, with $\operatorname{diam} K_w = r_w \operatorname{diam} K$.*
3. *Every $x \in K$ has an* address*: an infinite sequence ω with $x \in K_{\omega|k}$ for all k. For every infinite sequence ω, $\bigcap_k K_{\omega|k}$ is a single point.*

Proof. Existence, and (1). First, a compact set B with $F(B) \subseteq B$ exists: let $R = \max_i |f_i(0)|/(1-r)$ and let B be the closed ball of radius R about 0. For $|x| \leq R$, $|f_i(x)| \leq |f_i(x) - f_i(0)| + |f_i(0)| \leq rR + (1-r)R = R$. Now let B be any nonempty compact set with $F(B) \subseteq B$. Applying F, which preserves inclusions, gives $B \supseteq F(B) \supseteq F^2(B) \supseteq \cdots$, a decreasing sequence of nonempty compact sets (finite unions of continuous images of compact sets). Their intersection K is nonempty and compact. To see $F(K) = K$, note that $f_i(K) = \bigcap_k f_i(F^k(B))$ since f_i is injective, so

$$F(K) = \bigcup_i \bigcap_k f_i(F^k(B)) \qquad \text{and} \qquad K = \bigcap_k F^{k+1}(B) = \bigcap_k \bigcup_i f_i(F^k(B)).$$

The first set is contained in the second. Conversely, if x lies in $\bigcup_i f_i(F^k(B))$ for every k, then there is one i with $x \in f_i(F^k(B))$ for infinitely many k, and so for all k, as these sets decrease with k. So the two sets are equal.

Uniqueness. Let A and K be nonempty compact sets with $F(A) = A$ and $F(K) = K$, and $D = \max\{|a - y| : a \in A, y \in K\}$. Fix k. Since $A = F^k(A)$, every $a \in A$ is $f_w(a')$ for a word w of length k and some $a' \in A$. For any $y \in K$, the point $f_w(y)$ lies in $F^k(K) = K$, and $|a - f_w(y)| = r_w|a' - y| \leq r^k D$. So every point of A is within r^kD of K, for every k; as K is closed, $A \subseteq K$. By symmetry $K \subseteq A$.

(2) The first statement is $K = F^k(K)$, the second is $f_w(K) = f_w(F(K)) = \bigcup_i f_w(f_i(K))$, and the third holds because f_w multiplies distances by r_w.

(3) Since $K = \bigcup_i f_i(K)$, a point $x \in K$ can be written $x = f_{\omega_1}(x_1)$ with $x_1 \in K$; then $x_1 = f_{\omega_2}(x_2)$ with $x_2 \in K$, and so on. Then $x = f_{\omega|k}(x_k) \in K_{\omega|k}$ for all k. For any ω, the sets $K_{\omega|k}$ are nonempty, compact, decreasing by (2), and of diameter at most $r^k \operatorname{diam} K \to 0$; so their intersection is one point. □

Remark 28.18.6. Hutchinson's original proof [Hut81] is an application of the Banach fixed point theorem. The nonempty compact subsets of $\mathbb{R}^n$ form a complete metric space for the *Hausdorff distance* $d_H(A,B) = \inf\{\varepsilon > 0 : A \subseteq B_\varepsilon, B \subseteq A_\varepsilon\}$, where A_ε is the set of points at distance less than ε from A, and F is a contraction with constant r for this distance; the attractor is its fixed point, and $F^k(B) \to K$ for every nonempty compact B. This also explains the name: starting from any picture B, repeatedly replacing it by N smaller copies of itself converges to K. The uniqueness proof above is the contraction argument in disguise.

Example 28.18.7. 1. *The Cantor set.* $f_1(x) = x/3$, $f_2(x) = x/3 + 2/3$ on $\mathbb{R}$. The Cantor set C, the set of $\sum_k a_k 3^{-k}$ with all $a_k \in \{0,2\}$, satisfies $f_1(C) = \{a_1 = 0\}$ and $f_2(C) = \{a_1 = 2\}$, so $F(C) = C$ and C is the attractor. The sets C_w are the parts of C in the intervals I_w of Example 28.17.6, and the address of a point is its sequence of ternary digits, with 1 and 2 standing for the digits 0 and 2.

2. *An interval.* $f_1(x) = x/2$, $f_2(x) = x/2 + 1/2$. Here $F([0,1]) = [0,1]$, so the attractor is $[0,1]$; addresses are binary expansions, and the dyadic rationals have two addresses.

3. *The Sierpiński gasket.* Let p_1, p_2, p_3 be the vertices of a triangle T in $\mathbb{R}^2$ and $f_i(x) = (x+p_i)/2$. Then $f_i(T)$ is the triangle with vertex p_i and the midpoints of the two sides through p_i, so $F(T) \subseteq T$ and the attractor is $S = \bigcap_k F^k(T)$ by (1): at each step every triangle is replaced by its three corner triangles, and the open middle one is removed.

4. *The Koch curve.* In $\mathbb{R}^2 = \mathbb{C}$ let $\zeta = e^{i\pi/3}$ and

$$f_1(z) = \tfrac{z}{3}, \quad f_2(z) = \tfrac{\zeta z+1}{3}, \quad f_3(z) = \tfrac{\bar\zeta z}{3} + \tfrac12 + \tfrac{i\sqrt3}{6}, \quad f_4(z) = \tfrac{z+2}{3}.$$

   They map the segment $[0,1]$ to the four sides of the polygon $0, \frac13, \frac12 + \frac{i\sqrt3}{6}, \frac23, 1$, whose middle part is the tip of an equilateral triangle erected on the middle third of $[0,1]$: for instance $f_2(1) = \frac{1+\zeta}{3} = \frac12 + \frac{i\sqrt3}{6} = f_3(0)$ and $f_3(1) = \frac{\bar\zeta}{3} + \frac12 + \frac{i\sqrt3}{6} = \frac23 = f_4(0)$. The attractor is the limit of the polygons $F^k([0,1])$, in the sense of Remark 28.18.6.

5. *The four-corner Cantor set.* For $a \in \{0, \frac34\}^2$ let $f_a(x) = \frac{x}{4} + a$ on $\mathbb{R}^2$. The four squares $f_a([0,1]^2)$ are the corner squares of side $\frac14$ of the unit square, so the attractor $K_4 \subseteq [0,1]^2$ is obtained by repeatedly replacing every square by its four corner squares of a quarter of the size. It contains the fixed points $(0,0)$ and $(1,1)$ of $f_{(0,0)}$ and $f_{(3/4,3/4)}$, so $\operatorname{diam} K_4 = \sqrt2$.

28.18.3 The similarity dimension

The function $s \mapsto \sum_i r_i^s$ is continuous and strictly decreasing on $[0,\infty)$, equal to $N \geq 2$ at $s = 0$ and tending to 0 as $s \to \infty$. So there is a unique $s > 0$ with

$$r_1^s + r_2^s + \cdots + r_N^s = 1, \tag{28.2}$$

the *similarity dimension* of the system. If all ratios equal ρ, then $N\rho^s = 1$ and $s = \log N/\log(1/\rho)$: for the Cantor set $s = \log 2/\log 3$, for the interval of Example 28.18.7(2) $s = 1$, for the Sierpiński gasket $s = \log 3/\log 2 \approx 1.585$, for the Koch curve $s = \log 4/\log 3 \approx 1.262$, and for the four-corner set $s = \log 4/\log 4 = 1$. To cover K efficiently by the pieces K_w one should use pieces of comparable sizes, not of equal word length; the following lemma organises this.

Lemma 28.18.8. *For $0 < \rho < 1$ let Q_ρ be the set of nonempty words w with $r_w \le \rho < r_{w^-}$, where w^- is w with its last letter removed. Then every infinite sequence ω has exactly one beginning $\omega|k$ in Q_ρ, the set Q_ρ is finite, and*

$$\sum_{w\in Q_\rho} r_w^s = 1, \qquad \#Q_\rho \le (\rho\, r_{\min})^{-s}, \qquad K = \bigcup_{w\in Q_\rho} K_w,$$

where s is the similarity dimension and $r_{\min} = \min_i r_i$.

Proof. Along ω the numbers $r_{\omega|k}$ decrease strictly from $r_{\omega|0} = 1 > \rho$ to 0, so there is exactly one $k \ge 1$ with $r_{\omega|k} \le \rho < r_{\omega|(k-1)}$. Let k_0 be such that $r^{k_0} \le \rho$. Every word w of length at least $k_0 + 1$ has $r_{w^-} \le r^{k_0} \le \rho$, so it is not in Q_ρ; hence Q_ρ is finite. Every word v of length k_0 has $r_v \le \rho$, so, applying the first statement to any infinite sequence beginning with v, v has exactly one beginning in Q_ρ. Grouping the words of length k_0 according to this beginning, and using that the words of length k_0 extending a given w are the wu with $|u| = k_0 - |w|$,

$$1 = \Big(\sum_i r_i^s\Big)^{k_0} = \sum_{|v|=k_0} r_v^s = \sum_{w\in Q_\rho}\ \sum_{|u|=k_0-|w|} r_w^s r_u^s = \sum_{w\in Q_\rho} r_w^s \Big(\sum_i r_i^s\Big)^{k_0-|w|} = \sum_{w\in Q_\rho} r_w^s.$$

Each $w \in Q_\rho$ has $r_w = r_{w^-} r_{w_{|w|}} > \rho\, r_{\min}$, so $1 \ge \#Q_\rho\,(\rho\, r_{\min})^s$. Finally, if $x \in K$ has address ω and $\omega|k \in Q_\rho$, then $x \in K_{\omega|k}$. □

Proposition 28.18.9. *Let K be the attractor of an iterated function system with similarity dimension s. Then $\mathcal{H}^s(K) \le \alpha(s)(\operatorname{diam} K)^s$, and $\dim_{\mathcal{H}} K \le \overline{\dim}_B K \le s$.*

Proof. We may assume $d = \operatorname{diam} K > 0$. For $0 < \rho < 1$ the sets K_w, $w \in Q_\rho$, cover K by Lemma 28.18.8 and have diameter $r_w d \le \rho d$, so

$$\mathcal{H}^s_{\rho d}(K) \le \sum_{w\in Q_\rho} \alpha(s)(r_w d)^s = \alpha(s) d^s,$$

and $\rho \to 0$ gives the first claim. With $\delta = \rho d$ they also show $N_\delta(K) \le \#Q_\rho \le (r_{\min}\delta/d)^{-s}$ for $0 < \delta < d$, so $\log N_\delta(K)/(-\log\delta) \le s + s\log(d/r_{\min})/(-\log\delta)$, whose lim sup is s. The remaining inequality is Proposition 28.18.2(3). □

The upper bound can be far from the truth when the pieces overlap: for $f_1(x) = f_2(x) = x/2$ the attractor is the point $\{0\}$, while $s = 1$. The lower bound needs a condition which prevents such overlaps. The simplest one is that the pieces are disjoint.

Theorem 28.18.10. *Let K be the attractor of an iterated function system with similarity dimension s, and suppose that the sets $f_1(K), \dots, f_N(K)$ are pairwise disjoint (the* strong separation condition*). Then*

$$\alpha(s)(r_{\min}\eta)^s \le \mathcal{H}^s(K) \le \alpha(s)(\operatorname{diam} K)^s, \qquad \eta = \min_{i\ne j} d(f_i(K), f_j(K)) > 0,$$

and $\dim_{\mathcal{H}} K = \dim_B K = s$.

Proof. Step 1: separation at all scales. $\eta > 0$ because the $f_i(K)$ are disjoint compact sets. For a word w and $i \ne j$, f_w multiplies distances by r_w, so $d(K_{wi}, K_{wj}) = r_w d(f_i(K), f_j(K)) \ge r_w\eta$. Let $x, y \in K$ with addresses ω and ω', and suppose these first differ at position j; let $u = \omega|(j-1)$. Then $x \in K_{u\omega_j}$ and $y \in K_{u\omega'_j}$, so

$$|x - y| \ge r_{\omega|(j-1)}\,\eta. \tag{28.3}$$

In particular, for each k the N^k sets K_w with $|w| = k$ are pairwise disjoint, and each of them is open in K, its complement in K being the finite union of the others, which is compact.

Step 2: the natural measure. Put $p_w = r_w^s$, so that $\sum_i p_{wi} = p_w \sum_i r_i^s = p_w$, and by induction $\sum_{|u|=l} p_{wu} = p_w$ for every l. Fix $x_0 \in K$, put $x_w = f_w(x_0) \in K_w$, and for $\varphi \in C(K)$ let

$$S_k(\varphi) = \sum_{|w|=k} p_w \, \varphi(x_w).$$

Let $\omega_\varphi(t) = \sup\{|\varphi(x) - \varphi(y)| : |x - y| \le t\}$, which tends to 0 as $t \to 0$ since φ is uniformly continuous on the compact set K. For $l > k$, regrouping the words of length l by their beginnings of length k,

$$|S_l(\varphi) - S_k(\varphi)| = \Big| \sum_{|w|=k} \sum_{|u|=l-k} p_{wu} \big(\varphi(x_{wu}) - \varphi(x_w)\big) \Big| \le \omega_\varphi(r^k \operatorname{diam} K) \sum_{|w|=k} p_w = \omega_\varphi(r^k \operatorname{diam} K),$$

since x_{wu} and x_w both lie in K_w. So $(S_k(\varphi))_k$ is a Cauchy sequence, and $\Lambda(\varphi) = \lim_k S_k(\varphi)$ is a positive linear functional on $C(K) = C_c(K)$ with $\Lambda(1) = 1$. By the Riesz–Markov theorem (Theorem 28.14.6) there is a Borel probability measure μ on K with $\int \varphi \, d\mu = \Lambda(\varphi)$. By Step 1 the indicator function $\mathbf{1}_{K_w}$ is continuous on K, and for $l \ge k = |w|$ the points x_v with $|v| = l$ lying in K_w are those with v beginning with w. Hence

$$\mu(K_w) = \lim_l S_l(\mathbf{1}_{K_w}) = \sum_{|u|=l-k} p_{wu} = p_w = r_w^s.$$

Step 3: the mass of small sets. Let U be a closed set with $d = \operatorname{diam} U < \eta$; we show $\mu(U \cap K) \le (d/(r_{\min}\eta))^s$. This is clear if $U \cap K = \emptyset$. Let $x \in U \cap K$ have address ω. If $d = 0$, then $\mu(U \cap K) = \mu(\{x\}) \le \mu(K_{\omega|k}) = r_{\omega|k}^s \to 0$. If $d > 0$, the numbers $r_{\omega|k}\eta$ decrease from $\eta > d$ to 0, so there is a least $k \ge 1$ with $r_{\omega|k}\eta \le d$, and $r_{\omega|(k-1)}\eta > d$. Every $y \in U \cap K$ lies in $K_{\omega|(k-1)}$: otherwise an address of y first differs from ω at a position $j \le k-1$, and (28.3) gives $|x - y| \ge r_{\omega|(j-1)}\eta \ge r_{\omega|(k-1)}\eta > d$, which is impossible. Therefore

$$\mu(U \cap K) \le \mu(K_{\omega|(k-1)}) = r_{\omega|(k-1)}^s \le \left(\frac{d}{r_{\min}\eta}\right)^s,$$

the last step because $d \ge r_{\omega|k}\eta = r_{\omega|(k-1)} r_{\omega_k}\eta \ge r_{\omega|(k-1)} r_{\min}\eta$.

Step 4: conclusion. Let $\delta < \eta$ and let (U_i) be a cover of K by closed sets of diameter at most δ; closed covers suffice to compute $\mathcal{H}^s_\delta$. By Step 3,

$$1 = \mu(K) \le \sum_i \mu(U_i \cap K) \le (r_{\min}\eta)^{-s} \sum_i (\operatorname{diam} U_i)^s,$$

so $\mathcal{H}^s_\delta(K) \ge \alpha(s)(r_{\min}\eta)^s$, and the lower bound follows. With Proposition 28.18.9, $0 < \mathcal{H}^s(K) < \infty$, so $\dim_{\mathcal{H}} K = s$, and $s = \dim_{\mathcal{H}} K \le \underline{\dim}_B K \le \overline{\dim}_B K \le s$ by Proposition 28.18.2(3). □

Step 3 is the mass distribution principle of Example 28.17.6 again; the new ingredient is that the measure is built for us by the Riesz–Markov theorem, once we know what it should give to the pieces K_w. The measure μ is in fact the normalised restriction of $\mathcal{H}^s$ to K (Exercise 28.23.20).

Example 28.18.11. 1. *The Cantor set.* $f_1(C) \subseteq [0, \frac{1}{3}]$ and $f_2(C) \subseteq [\frac{2}{3}, 1]$ are disjoint, at distance $\eta = \frac{1}{3}$, the distance between the points $\frac{1}{3}$ and $\frac{2}{3}$ of C. The theorem gives $\dim_{\mathcal{H}} C = \dim_B C = \log 2/\log 3$ and $\alpha(s)/4 \le \mathcal{H}^s(C) \le \alpha(s)$, as $(r_{\min}\eta)^s = 9^{-s} = \frac{1}{4}$; the direct argument of Example 28.17.6 gave the better lower bound $\alpha(s)/2$. In fact $\mathcal{H}^s(C) = \alpha(s)$ [Fal90], but this requires a finer argument.

2. *A totally disconnected set of dimension* 1. For the four-corner set K_4 of Example 28.18.7(5) the four pieces lie in the four corner squares of side $\frac{1}{4}$, which are pairwise at distance at least $\frac{1}{2}$. So $\dim_{\mathcal{H}} K_4 = 1$ and $0 < \mathcal{H}^1(K_4) \le \alpha(1)\sqrt{2} = \sqrt{2}$ (note $\alpha(1) = 1$). Yet K_4 contains no segment, indeed no connected set with more than one point: the sets $(K_4)_w$ are open and closed in K_4 by Step 1, and every point has such neighbourhoods of arbitrarily small diameter. So K_4 has the dimension and the finite positive length of a curve, but none of its geometry; Example 28.18.12 and Section 28.22 return to this.

3. *Products.* $C \times C \subseteq \mathbb{R}^2$ is the attractor of the four maps $(x, y) \mapsto (x/3 + a, y/3 + b)$ with $a, b \in \{0, \frac{2}{3}\}$ (by uniqueness, since $C \times C$ is compact and invariant), whose images lie in the four corner squares of side $\frac{1}{3}$. So $\dim_{\mathcal{H}}(C \times C) = \log 4/\log 3 = 2 \dim_{\mathcal{H}} C$.

4. *Middle-λ Cantor sets.* Removing open middle intervals of relative length $\lambda \in (0, 1)$ instead of $\frac{1}{3}$ gives the attractor of $x \mapsto \rho x$ and $x \mapsto \rho x + 1 - \rho$ with $\rho = (1 - \lambda)/2$, of dimension $\log 2/\log(2/(1 - \lambda))$. As λ runs through $(0, 1)$ this takes every value in $(0, 1)$.

Example 28.18.12. *Shadows of the four-corner set.* Let K_4 be as in Example 28.18.7(5). Its projection $\pi_1(x, y) = x$ to the first axis satisfies $\pi_1 \circ f_a = g_{a_1} \circ \pi_1$ with $g_0(t) = t/4$ and $g_{3/4}(t) = t/4 + \frac{3}{4}$. So $\pi_1(K_4)$ is a nonempty compact set with $\pi_1(K_4) = g_0(\pi_1(K_4)) \cup g_{3/4}(\pi_1(K_4))$, the attractor of $\{g_0, g_{3/4}\}$, whose pieces lie in $[0, \frac{1}{4}]$ and $[\frac{3}{4}, 1]$. By Theorem 28.18.10 it has dimension $\log 2/\log 4 = \frac{1}{2}$, hence length 0. Now project instead by $\pi(x, y) = x + y/2$. Then $\pi \circ f_a = h_a \circ \pi$ with $h_a(t) = t/4 + \pi(a)$, and $\pi(a)$ runs through $0, \frac{3}{8}, \frac{3}{4}, \frac{9}{8}$ as a runs through $(0, 0), (0, \frac{3}{4}), (\frac{3}{4}, 0), (\frac{3}{4}, \frac{3}{4})$. The interval $[0, \frac{3}{2}]$ satisfies

$$\bigcup_a h_a\big([0, \tfrac{3}{2}]\big) = [0, \tfrac{3}{8}] \cup [\tfrac{3}{8}, \tfrac{3}{4}] \cup [\tfrac{3}{4}, \tfrac{9}{8}] \cup [\tfrac{9}{8}, \tfrac{3}{2}] = [0, \tfrac{3}{2}],$$

so by uniqueness of attractors $\pi(K_4) = [0, \frac{3}{2}]$. The orthogonal projection onto the line spanned by $(2, 1)$ is $\frac{2}{\sqrt{5}}\pi$ up to an isometry of the line, so the shadow of K_4 in this direction has length $\frac{3}{\sqrt{5}}$, while in the direction of the first axis it has length 0. As orthogonal projections are 1-Lipschitz, this also gives

$$\frac{3}{\sqrt{5}} \le \mathcal{H}^1(K_4) \le \sqrt{2},$$

that is, $1.341\ldots \le \mathcal{H}^1(K_4) \le 1.414\ldots$, a much better lower bound than Theorem 28.18.10 gives.

Remark 28.18.13. *Touching pieces.* For the Sierpiński gasket and the Koch curve the pieces $f_i(K)$ touch, so Theorem 28.18.10 does not apply, and Proposition 28.18.9 only gives the upper bounds $\log 3/\log 2$ and $\log 4/\log 3$ for their dimensions. Equality holds under the weaker *open set condition*: there is a nonempty bounded open set V with $f_i(V) \subseteq V$ and the $f_i(V)$ pairwise disjoint (for the gasket, the open triangle; for the Koch curve, the open isosceles triangle with base $[0, 1]$ and apex $\frac{1}{2} + \frac{i\sqrt{3}}{6}$). Under this condition $0 < \mathcal{H}^s(K) < \infty$ and $\dim_{\mathcal{H}} K = \dim_B K = s$ [Hut81], [Fal90, Chapter 9]. The proof follows the same plan: the natural measure exists without any condition, and the open set condition replaces (28.3) by a volume argument showing that a ball of radius ρ meets a bounded number of pieces K_w, $w \in Q_\rho$.

28.19 Lipschitz maps and Rademacher's theorem

A Lipschitz function of one variable is absolutely continuous, hence differentiable almost everywhere (Example 28.13.5). Rademacher's theorem extends this to several variables. It is

the reason why Lipschitz maps are the natural class of maps in geometric measure theory: they are just regular enough to have a derivative almost everywhere, and they are preserved by the operations of the theory (distance functions, compositions, restrictions, inverses of bi-Lipschitz maps), which smooth maps are not.

Lemma 28.19.1. *If $g \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$ and $\int g\varphi \, d\lambda = 0$ for all compactly supported functions φ of class C^1, then $g = 0$ almost everywhere.*

Proof. Let $\psi(s) = 3s^2 - 2s^3$ for $0 \le s \le 1$, $\psi = 0$ for $s < 0$ and $\psi = 1$ for $s > 1$, a C^1 function. For a ball $B(x,r)$ and $j > 1/r$ let $\varphi_j(y) = \psi(j(r - \|y - x\|))$; it is C^1, since it is constant near $y = x$, takes values in $[0,1]$, vanishes outside $B(x,r)$, and $\varphi_j \to \mathbf{1}_{B(x,r)}$ pointwise. By dominated convergence $\int_{B(x,r)} g = 0$ for all balls, and by the Lebesgue differentiation theorem (Theorem 28.9.4), $g(x) = \lim_{r\to 0} A_r g(x) = 0$ for almost every x. □

Theorem 28.19.2 (Rademacher)**.** *Every Lipschitz map $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at almost every point of $\mathbb{R}^n$.*

Proof. It suffices to treat $m = 1$, applying the result to the components. Let L be a Lipschitz constant. For a unit vector v and $x \in \mathbb{R}^n$ let $D_v f(x) = \lim_{t\to 0}(f(x + tv) - f(x))/t$ where the limit exists.

Step 1: for each v, $D_v f(x)$ exists for almost every x. As the difference quotients are continuous in $t \neq 0$, the limit exists if and only if the lim sup and lim inf over rational $t \to 0$ are equal and finite, so the set N_v where it fails is Borel, and $D_v f$ is Borel measurable where defined, with $|D_v f| \le L$. Choose an orthogonal map R with $Re_1 = v$; it preserves λ (Lemma 28.11.1), and $f \circ R$ is Lipschitz. For each $y \in \mathbb{R}^{n-1}$, $t \mapsto f(R(t,y))$ is a Lipschitz function of one variable, differentiable almost everywhere (Example 28.13.5); so every line parallel to e_1 meets $R^{-1}(N_v)$ in a null set, and by Tonelli's theorem (Theorem 28.10.3) $\lambda(N_v) = \lambda(R^{-1}(N_v)) = 0$. In particular the gradient $\nabla f = (D_{e_1} f, \dots, D_{e_n} f)$ exists almost everywhere, and $|\nabla f| \le \sqrt{n}L$.

Step 2: $D_v f = \langle v, \nabla f\rangle$ almost everywhere. Let φ be C^1 with compact support. By translation invariance,

$$\int \frac{f(x + tv) - f(x)}{t}\varphi(x)\, dx = -\int f(x)\frac{\varphi(x) - \varphi(x - tv)}{t}\, dx.$$

Let $t = 1/j \to 0$. On the left the integrands converge almost everywhere to $D_v f\, \varphi$ and are bounded by $L|\varphi|$; on the right they converge to $f\, D_v\varphi = f\langle v, \nabla\varphi\rangle$ and are bounded by a constant times the indicator of a compact set, by the mean value inequality and the continuity of f. By dominated convergence, $\int D_v f\, \varphi = -\int f\langle v, \nabla\varphi\rangle = -\sum_i v_i \int f\, \partial_i\varphi = \sum_i v_i \int D_{e_i} f\, \varphi$, the last equality being the first one for $v = e_i$. So $\int (D_v f - \langle v, \nabla f\rangle)\varphi = 0$ for all φ, and Lemma 28.19.1 applies.

Step 3: differentiability. Let (v_k) be a dense sequence in the unit sphere, and let A be the set of points where ∇f and all $D_{v_k} f$ exist and $D_{v_k} f = \langle v_k, \nabla f\rangle$; its complement is a countable union of null sets. Fix $x \in A$, and for a unit vector v and $t \neq 0$ put $Q(v,t) = (f(x + tv) - f(x))/t - \langle v, \nabla f(x)\rangle$. Then $|Q(v,t) - Q(v',t)| \le (L + \sqrt{n}L)|v - v'|$. Given $\varepsilon > 0$, by compactness of the sphere there is N such that every unit vector is within $\varepsilon/(2(1 + \sqrt{n})L)$ of some v_k with $k \le N$, and there is $\delta > 0$ with $|Q(v_k,t)| < \varepsilon/2$ for $0 < |t| < \delta$ and $k \le N$. Then $|Q(v,t)| < \varepsilon$ for all unit vectors v and $0 < |t| < \delta$, that is, $|f(x + h) - f(x) - \langle \nabla f(x), h\rangle| \le \varepsilon|h|$ for $|h| < \delta$. So f is differentiable at x, with derivative $h \mapsto \langle \nabla f(x), h\rangle$. □

Counterexample 28.19.3. 1. Continuity is not enough: the Takagi function (Theorem 27.9.8) is continuous and differentiable nowhere, so it is not Lipschitz on any interval.

2. The exceptional null set can be dense. Let $q_1, q_2, \dots$ be an enumeration of $\mathbb{Q}$ and $f(x) = \sum_j 2^{-j}|x - q_j|$, a 1-Lipschitz function on $\mathbb{R}$. For $h > 0$,

$$\frac{f(q_k + h) - f(q_k)}{h} - \frac{f(q_k) - f(q_k - h)}{h} = \sum_j 2^{-j} \frac{|q_k - q_j + h| + |q_k - q_j - h| - 2|q_k - q_j|}{h} \geq 2^{1-k},$$

since every term is ≥ 0 by the triangle inequality and the term $j = k$ is 2^{1-k}. If f were differentiable at q_k the left side would tend to 0; so f is differentiable at no rational point.

3. Existence of all directional derivatives at a point does not imply differentiability there, even for Lipschitz functions: $f(x, y) = x^2 y/(x^2 + y^2)$, $f(0, 0) = 0$, is Lipschitz, has $D_v f(0) = f(v)$ for every unit vector v, and $v \mapsto f(v)$ is not linear. Step 3 of the proof is where the Lipschitz condition turns directional derivatives into a derivative.

28.20 The area formula

For a linear map $T\colon \mathbb{R}^n \to \mathbb{R}^n$, $\lambda^n(T(A)) = |\det T|\lambda^n(A)$. For a linear map $L\colon \mathbb{R}^k \to \mathbb{R}^m$ with $k \leq m$ the image of a cube is a k-dimensional parallelepiped in $\mathbb{R}^m$, and its k-dimensional volume is measured by $\mathcal{H}^k$. The factor replacing $|\det T|$ is the Jacobian of L.

Definition 28.20.1. The *Jacobian* of a linear map $L\colon \mathbb{R}^k \to \mathbb{R}^m$, $k \leq m$, is $JL = \sqrt{\det(L^T L)}$. For a map f differentiable at x, $Jf(x) = J(Df(x))$.

Lemma 28.20.2. *Let $L\colon \mathbb{R}^k \to \mathbb{R}^m$ be linear, $k \leq m$.*

1. *$(JL)^2$ is the sum of the squares of the $k \times k$ minors of the matrix of L, and $JL = 0$ if and only if L is not injective. For $k = m$, $JL = |\det L|$.*

2. *If L is injective, $L = O \circ S$ with $S = (L^T L)^{1/2}$ positive definite symmetric on $\mathbb{R}^k$, $\det S = JL$, and $O\colon \mathbb{R}^k \to \mathbb{R}^m$ an isometry onto $L(\mathbb{R}^k)$.*

3. *$\mathcal{H}^k(L(A)) = JL\,\lambda^k(A)$ for every Lebesgue measurable $A \subseteq \mathbb{R}^k$.*

Proof. (1) By Cauchy–Binet (Proposition 12.3.4(2)) applied to L^T and L, $\det(L^T L) = \sum_S \det(L_S)^2$, where L_S is the $k \times k$ submatrix with rows S. By Proposition 12.3.4(3), some minor is nonzero if and only if L has rank k. For $k = m$, $\det(L^T L) = (\det L)^2$.

(2) $L^T L$ is symmetric with $\langle L^T L v, v\rangle = |Lv|^2 > 0$ for $v \neq 0$, so by the spectral theorem (Theorem 11.10.6) it has a positive definite square root S, with $\det S = \sqrt{\det L^T L}$. Then $O = LS^{-1}$ satisfies $O^T O = S^{-1} L^T L S^{-1} = I$, so $|Ov| = |v|$, and $O(\mathbb{R}^k) = L(\mathbb{R}^k)$.

(3) If L is not injective, $L(\mathbb{R}^k)$ lies in a linear subspace W of dimension $k - 1$; an isometry $\mathbb{R}^{k-1} \to W$ and Theorem 28.17.5 give $\mathcal{H}^{k-1}(W \cap B) < \infty$ for every ball B, so $\mathcal{H}^k(W) = 0$ by Proposition 28.17.3(4), and both sides vanish. If L is injective, O is an isometry from $\mathbb{R}^k$ onto $L(\mathbb{R}^k)$, which preserves $\mathcal{H}^k$, so $\mathcal{H}^k(L(A)) = \mathcal{H}^k(S(A)) = \lambda^k(S(A)) = |\det S|\lambda^k(A)$ by Theorem 28.17.5 and Lemma 28.11.1; for Lebesgue measurable A write $A = B \cup N$ with B Borel and N contained in a Borel null set, which S maps to a null set. □

So JL is the k-dimensional volume of the parallelepiped spanned by $Le_1, \dots, Le_k$, the image of the unit cube. For a curve, $J\gamma(t) = |\gamma'(t)|$; for a parametrised surface $x(u, v)$ in $\mathbb{R}^3$, $Jx = |x_u \times x_v|$, since the three 2×2 minors of Dx are the components of $x_u \times x_v$ up to sign.

Theorem 28.20.3 (Area formula)**.** *Let $U \subseteq \mathbb{R}^k$ be open and $f\colon U \to \mathbb{R}^m$ an injective C^1 map whose differential $Df(x)$ is injective at every point. Then for every Lebesgue measurable $A \subseteq U$ the set $f(A)$ is $\mathcal{H}^k$-measurable, and*

$$\mathcal{H}^k(f(A)) = \int_A Jf \, d\lambda^k.$$

Consequently $\int_{f(U)} g \, d\mathcal{H}^k = \int_U g(f(x))\, Jf(x)\, d\lambda^k(x)$ for every Borel function $g \geq 0$ on $\mathbb{R}^m$.

Proof. For an injective linear $L\colon \mathbb{R}^k \to \mathbb{R}^m$ let $m(L) = \min_{|v|=1} |Lv| > 0$. Fix $t > 1$.

Step 1: a decomposition of U. Call a pair (B, L) of an open ball $B \subseteq U$ with rational centre and radius and an injective linear map L with rational matrix *good* if, for all $x, y \in B$,

$$t^{-1}|L(x-y)| \leq |f(x)-f(y)| \leq t|L(x-y)| \qquad \text{and} \qquad t^{-k}JL \leq Jf(x) \leq t^k JL. \tag{28.4}$$

Every $x_0 \in U$ lies in the ball of some good pair. Indeed, let $\mu_0 = m(Df(x_0)) > 0$ and $\eta = (1-t^{-1})\mu_0/2$. Since $|m(L) - m(L')| \leq \|L - L'\|$, J is continuous, and $Jf(x_0) > 0$, there is L with rational matrix such that $\|Df(x_0) - L\| < \min(\eta, \mu_0)/2$ and $t^{-k}JL < Jf(x_0) < t^k JL$; then $m(L) \geq \mu_0/2$, so $\eta \leq (1-t^{-1})m(L)$. By continuity of Df and of J there is a rational ball $B \ni x_0$ on which $\|Df - L\| \leq \eta$ and still $t^{-k}JL \leq Jf \leq t^k JL$. For $x, y \in B$, by the mean value inequality (Proposition 27.10.3(4)) applied to $f - L$ on the convex set B, $|f(x)-f(y)-L(x-y)| \leq \eta|x-y| \leq (1-t^{-1})|L(x-y)|$, which gives $t^{-1}|L(x-y)| \leq |f(x)-f(y)| \leq (2-t^{-1})|L(x-y)| \leq t|L(x-y)|$. Enumerate the countably many good pairs as (B_j, L_j), and let $E_j = B_j \setminus \bigcup_{i<j} B_i$; these are disjoint Borel sets covering U.

Step 2: estimates on one piece. Let $A \subseteq E_j$ be Lebesgue measurable. By (28.4) the map $L_j(x) \mapsto f(x)$, from $L_j(B_j)$ to $f(B_j)$, is t-Lipschitz, and so is its inverse. By Proposition 28.17.3(3) and Lemma 28.20.2(3),

$$t^{-k}JL_j\, \lambda^k(A) \leq \mathcal{H}^k(f(A)) \leq t^k JL_j\, \lambda^k(A), \qquad t^{-k}JL_j\, \lambda^k(A) \leq \int_A Jf \leq t^k JL_j\, \lambda^k(A),$$

the second pair by (28.4). Hence $t^{-2k}\int_A Jf \leq \mathcal{H}^k(f(A)) \leq t^{2k}\int_A Jf$; here $\mathcal{H}^k$ is the outer measure.

Step 3: measurability. A Lebesgue measurable $A \subseteq U$ is $K \cup N$ with K a countable union of compact sets and N null (Theorem 28.3.4). Then $f(K)$ is a countable union of compact sets, hence Borel, and $\mathcal{H}^k(f(N)) \leq \sum_j \mathcal{H}^k(f(N \cap E_j)) = 0$ by Step 2. So $f(A)$ differs from a Borel set by an $\mathcal{H}^k$-null set, and it is $\mathcal{H}^k$-measurable (Theorem 28.3.2).

Step 4. For measurable $A \subseteq U$, the sets $f(A \cap E_j)$ are measurable and disjoint, since f is injective. Summing Step 2 over j, $t^{-2k}\int_A Jf \leq \mathcal{H}^k(f(A)) \leq t^{2k}\int_A Jf$, and $t > 1$ was arbitrary. For a Borel set $G \subseteq \mathbb{R}^m$, applying this to $A = f^{-1}(G)$ gives the integral formula for $g = \mathbf{1}_G$; it follows for simple functions by linearity, and for Borel $g \geq 0$ by monotone convergence (Theorem 28.4.3). □

For $k = m$ the area formula is the change of variables formula (Theorem 28.11.2), with $\mathcal{H}^m = \lambda^m$. The proof is the same idea: on small pieces f is almost linear, and linear maps multiply measure by the Jacobian.

Example 28.20.4.

1. *Length of curves.* For an injective C^1 curve $\gamma\colon (a,b) \to \mathbb{R}^m$ with $\gamma' \neq 0$, $\mathcal{H}^1(\gamma((a,b))) = \int_a^b |\gamma'(t)|\, dt$. So the length of a curve, defined by parametrisations, is the one-dimensional Hausdorff measure of its image, which depends only on the set.

2. *Area of graphs.* For $h\colon U \to \mathbb{R}$ of class C^1, $U \subseteq \mathbb{R}^k$ open, the graph map $f(x) = (x, h(x))$ is injective with injective differential, and $Jf = \sqrt{\det(I + \nabla h\, \nabla h^T)} = \sqrt{1 + |\nabla h|^2}$, since $I + ww^T$ is the identity on $w^\perp$ and multiplies w by $1 + |w|^2$. So the graph has area $\int_U \sqrt{1 + |\nabla h|^2}\, d\lambda^k$.

3. *The area of the sphere.* The upper hemisphere of S^2 is the graph of $h(x) = \sqrt{1-|x|^2}$ over the open unit disc, with $|\nabla h|^2 = |x|^2/(1-|x|^2)$ and $Jf = (1-|x|^2)^{-1/2}$. In polar coordinates (Example 28.11.3), $\mathcal{H}^2 = \int_0^{2\pi}\int_0^1 r(1-r^2)^{-1/2}\,dr\,d\theta = 2\pi$. The equator has finite $\mathcal{H}^1$-measure by (1), so it is $\mathcal{H}^2$-null (Proposition 28.17.3(4)), and $\mathcal{H}^2(S^2) = 4\pi$.

Counterexample 28.20.5. Injectivity is needed: for $f(t) = (\cos t, \sin t)$ on $(0, 4\pi)$, $\int Jf = 4\pi$, while the image, the circle minus one point, has $\mathcal{H}^1$-measure 2π. The integral counts the points of the image with multiplicity: in general $\int_A Jf = \int_{\mathbb{R}^m} \#(A \cap f^{-1}(y))\,d\mathcal{H}^k(y)$ (Remark 28.20.6).

Remark 28.20.6. *The general area and coarea formulas.* By Rademacher's theorem a Lipschitz map $f\colon \mathbb{R}^k \to \mathbb{R}^m$ has a Jacobian almost everywhere, and the area formula holds in the generality

$$\int_A Jf\,d\lambda^k = \int_{\mathbb{R}^m} \#\big(A \cap f^{-1}(y)\big)\,d\mathcal{H}^k(y) \qquad (k \le m),$$

for Lebesgue measurable A, where # is the number of points [EG92, Section 3.3], [Fed69, Section 3.2]. For $k \ge m$ there is a dual statement, the *coarea formula*: with $Jf = \sqrt{\det(Df\,Df^T)}$,

$$\int_A Jf\,d\lambda^k = \int_{\mathbb{R}^m} \mathcal{H}^{k-m}\big(A \cap f^{-1}(y)\big)\,d\lambda^m(y) \qquad (k \ge m),$$

which slices A by the level sets of f instead of parametrising it [EG92, Section 3.4]. For $f(x) = |x|$ on $\mathbb{R}^n$, $Jf = |\nabla f| = 1$ away from 0, and the coarea formula gives integration in spherical shells, $\int_{\mathbb{R}^n} g\,d\lambda = \int_0^\infty \big(\int_{|x|=r} g\,d\mathcal{H}^{n-1}\big)dr$; for a C^1 function u it gives $\int |\nabla u| = \int_{\mathbb{R}} \mathcal{H}^{n-1}(\{u = s\})\,ds$, the starting point of the isoperimetric and Sobolev inequalities. A set is *k-rectifiable* if it is covered, up to an $\mathcal{H}^k$-null set, by countably many Lipschitz images of subsets of $\mathbb{R}^k$; these sets, which include all k-dimensional submanifolds, are the objects on which geometric measure theory does calculus, and the area formula is the bridge between them and Lebesgue measure.

28.21 The coarea formula

The area formula measures a set by parametrising it. The coarea formula measures it by slicing. The model is Fubini's theorem: the volume of $A \subseteq \mathbb{R}^n$ is the integral of the $(n-1)$-dimensional volumes of the slices $A \cap \{x_n = t\}$. If the slices are the level sets $f^{-1}(t)$ of a function f which are not parallel hyperplanes, the slab between the levels t and $t + dt$ has thickness about $dt/|\nabla f|$: where the levels are crowded, the slab is thin. So the volume of the slab is about $\mathcal{H}^{n-1}(\text{slice})\,dt/|\nabla f|$, and integrating $|\nabla f|$ instead of 1 compensates exactly:

$$\int_A |\nabla f|\,d\lambda^n = \int_{\mathbb{R}} \mathcal{H}^{n-1}\big(A \cap f^{-1}(t)\big)\,dt.$$

We prove this for C^1 functions without critical points; the general statement is in Remark 28.20.6. Throughout, $n \ge 2$.

Lemma 28.21.1. *Let $v_1, \dots, v_{n-1} \in \mathbb{R}^n$ be linearly independent, V the $n \times (n-1)$ matrix with these columns, v a unit vector orthogonal to all v_i, and $w \in \mathbb{R}^n$. Then*

$$|\det(v_1, \dots, v_{n-1}, w)| = JV \cdot |\langle w, v\rangle|.$$

In words: the volume of a parallelepiped is the $(n-1)$-dimensional volume of its base times its height. Recall that $JV = \sqrt{\det(V^TV)}$ is the $(n-1)$-dimensional volume of the base (Lemma 28.20.2).

Proof. Write $w = u + \langle w, v\rangle v$ with u in the span of the v_i. The determinant is linear in the last column and vanishes when the columns are dependent, so $\det(V, w) = \langle w, v\rangle \det(V, v)$. For the matrix $M = (V, v)$, $M^T M$ is block diagonal with blocks $V^T V$ and $\langle v, v\rangle = 1$, because $V^T v = 0$. Hence $(\det M)^2 = \det(M^T M) = \det(V^T V) = (JV)^2$. □

Theorem 28.21.2 (Coarea formula)**.** *Let $U \subseteq \mathbb{R}^n$ be open and $f\colon U \to \mathbb{R}$ of class C^1 with $\nabla f(x) \neq 0$ for all $x \in U$. For every Borel set $A \subseteq U$ the function $t \mapsto \mathcal{H}^{n-1}(A \cap f^{-1}(t))$ is Lebesgue measurable, and*

$$\int_A |\nabla f|\, d\lambda^n = \int_{\mathbb{R}} \mathcal{H}^{n-1}\big(A \cap f^{-1}(t)\big)\, dt.$$

More generally, for every Borel function $g\colon U \to [0, \infty]$,

$$\int_U g\, |\nabla f|\, d\lambda^n = \int_{\mathbb{R}} \Big(\int_{f^{-1}(t)} g\, d\mathcal{H}^{n-1} \Big) dt. \tag{28.5}$$

Proof. Step 1: local coordinates adapted to f. Let $x_0 \in U$. Some partial derivative $\partial_i f(x_0)$ is nonzero. A permutation of the coordinates is an isometry of $\mathbb{R}^n$, so it preserves λ^n (Lemma 28.11.1), $\mathcal{H}^{n-1}$ and $|\nabla f|$; so we may assume $\partial_n f(x_0) \neq 0$. The map $\psi(x) = (x_1, \dots, x_{n-1}, f(x))$ has $\det D\psi(x_0) = \partial_n f(x_0) \neq 0$. By the inverse function theorem (Theorem 27.11.2) there are open sets $V \ni x_0$ and W such that $\psi\colon V \to W$ is a C^1-diffeomorphism. Let $\phi = \psi^{-1}\colon W \to V$, and write points of $W \subseteq \mathbb{R}^{n-1} \times \mathbb{R}$ as (y, t). Since $\psi(\phi(y, t)) = (y, t)$,

$$f(\phi(y, t)) = t \qquad \text{for } (y, t) \in W.$$

For fixed t let $W_t = \{y : (y, t) \in W\}$, open in $\mathbb{R}^{n-1}$, and $\phi_t = \phi(-, t)\colon W_t \to \mathbb{R}^n$. It is injective and C^1, and $D\phi_t$ is injective, as its columns are the first $n-1$ columns of the invertible matrix $D\phi$. Its image is $V \cap f^{-1}(t)$: a point $x \in V$ has $f(x) = t$ if and only if $\psi(x) = (y, t)$ for some y.

Step 2: the Jacobian identity. We claim $|\det D\phi(y, t)|\,|\nabla f(\phi(y, t))| = J\phi_t(y)$. Differentiating $f(\phi(y, t)) = t$ with the chain rule gives $\langle \nabla f, \partial_{y_i}\phi\rangle = 0$ for $i < n$ and $\langle \nabla f, \partial_t \phi\rangle = 1$, all evaluated at $\phi(y, t)$ resp. (y, t). So $\nu = \nabla f / |\nabla f|$ is a unit vector orthogonal to the columns $\partial_{y_i}\phi$ of $D\phi_t$, and Lemma 28.21.1 with $w = \partial_t \phi$ gives

$$|\det D\phi| = J\phi_t \cdot |\langle \partial_t \phi, \nu\rangle| = \frac{J\phi_t}{|\nabla f|}.$$

Step 3: the formula on V. Let $A \subseteq V$ be Borel, and $h = \mathbf{1}_A \circ \phi$, a Borel function on W. By the change of variables formula (Theorem 28.11.2) for $\phi\colon W \to V$ and Step 2,

$$\int_A |\nabla f|\, d\lambda^n = \int_W h(y, t)\, |\nabla f(\phi(y, t))|\, |\det D\phi(y, t)|\, d\lambda^n(y, t) = \int_W h(y, t)\, J\phi_t(y)\, d\lambda^n(y, t).$$

The integrand, extended by 0 outside W, is a nonnegative Borel function on $\mathbb{R}^{n-1} \times \mathbb{R}$. By Tonelli's theorem (Theorem 28.10.3 and Remark 28.10.5) the function $t \mapsto \int_{W_t} h(y, t) J\phi_t(y)\, dy$ is measurable and the last integral equals $\int_{\mathbb{R}} \big(\int_{W_t} h(y, t) J\phi_t(y)\, dy \big) dt$. By the area formula (Theorem 28.20.3) for ϕ_t and the Borel set $B_t = \{y \in W_t : \phi(y, t) \in A\}$, whose image is $A \cap f^{-1}(t)$ by Step 1,

$$\int_{W_t} h(y, t) J\phi_t(y)\, dy = \int_{B_t} J\phi_t\, d\lambda^{n-1} = \mathcal{H}^{n-1}\big(A \cap f^{-1}(t)\big).$$

This proves the theorem for Borel subsets of V.

Step 4: globalisation. U is a countable union of compact sets, each covered by finitely many of the open sets V of Step 1; so $U = \bigcup_j V_j$ for countably many such sets. Let $E_j = V_j \setminus \bigcup_{i<j} V_i$,

disjoint Borel sets covering U. For a Borel set $A \subseteq U$, apply Step 3 to $A \cap E_j \subseteq V_j$ and sum over j. On the left, $\sum_j \int_{A\cap E_j} |\nabla f| = \int_A |\nabla f|$ by monotone convergence. On the right, for every t the sets $A \cap E_j \cap f^{-1}(t)$ are disjoint Borel sets with union $A \cap f^{-1}(t)$, and $\mathcal{H}^{n-1}$ is a measure on Borel sets (Proposition 28.17.3), so $\mathcal{H}^{n-1}(A \cap f^{-1}(t)) = \sum_j \mathcal{H}^{n-1}(A \cap E_j \cap f^{-1}(t))$, a sum of measurable functions of t; integrate with monotone convergence (Theorem 28.4.3). Finally, (28.5) is the first formula for $g = \mathbf{1}_A$; it follows for simple Borel functions by linearity and for all Borel $g \geq 0$ by monotone convergence, applied on both sides. □

For $f(x) = x_n$ the coarea formula is Fubini's theorem. Its most important special case is integration in polar coordinates.

Corollary 28.21.3 (Polar coordinates)**.** *Let $S^{n-1} = \{x \in \mathbb{R}^n : |x| = 1\}$ and $\omega_n = \lambda^n(B(0,1))$. Then $\mathcal{H}^{n-1}(S^{n-1}) = n\omega_n$, and for every Borel function $g\colon \mathbb{R}^n \to [0,\infty]$,*

$$\int_{\mathbb{R}^n} g\, d\lambda^n = \int_0^\infty \Big(\int_{|x|=r} g\, d\mathcal{H}^{n-1} \Big) dr = \int_0^\infty r^{n-1} \Big(\int_{S^{n-1}} g(r\theta)\, d\mathcal{H}^{n-1}(\theta) \Big) dr.$$

In particular, if $g(x) = h(|x|)$ is radial, $\int_{\mathbb{R}^n} g\, d\lambda^n = n\omega_n \int_0^\infty h(r) r^{n-1}\, dr$.

Proof. Apply Theorem 28.21.2 to $f(x) = |x|$ on $U = \mathbb{R}^n \setminus \{0\}$. Here $\nabla f(x) = x/|x|$, so $|\nabla f| = 1$, and the point 0 is a null set. This gives the first equality. For the second, $\theta \mapsto r\theta$ is a similarity with ratio r from S^{n-1} onto the sphere of radius r, so $\mathcal{H}^{n-1}(\{|x| = r\} \cap G) = r^{n-1}\mathcal{H}^{n-1}(S^{n-1} \cap r^{-1}G)$ for Borel sets G (Proposition 28.17.3(3)), which is the second equality for $g = \mathbf{1}_G$; pass to simple functions and to limits by monotone convergence. For $g = \mathbf{1}_{B(0,1)}$ we get $\omega_n = \mathcal{H}^{n-1}(S^{n-1}) \int_0^1 r^{n-1}\, dr = \mathcal{H}^{n-1}(S^{n-1})/n$. □

Example 28.21.4. 1. *Spheres.* $\mathcal{H}^1(S^1) = 2\omega_2 = 2\pi$ and $\mathcal{H}^2(S^2) = 3\omega_3 = 3 \cdot \frac{4}{3}\pi = 4\pi$, in agreement with Example 28.20.4. The formula $\mathcal{H}^{n-1}(S^{n-1}) = n\omega_n$ says that the volume of the ball, $\omega_n R^n$, has derivative $n\omega_n R^{n-1}$ in R equal to the area of the sphere of radius R: adding a thin shell of thickness dR adds volume (area)$\cdot dR$.

2. *Integrability of powers.* For $a \in \mathbb{R}$, $\int_{|x|<1} |x|^{-a}\, dx = n\omega_n \int_0^1 r^{n-1-a}\, dr$ is finite if and only if $a < n$, and $\int_{|x|>1} |x|^{-a}\, dx = n\omega_n \int_1^\infty r^{n-1-a}\, dr$ is finite if and only if $a > n$. So $|x|^{-a}$ is never integrable on all of $\mathbb{R}^n$, and the borderline exponent is the dimension, as for $\sum_{k\in\mathbb{Z}^n\setminus 0} |k|^{-a}$.

3. *The volume of balls.* By Fubini's theorem and Example 28.11.3, $\int_{\mathbb{R}^n} e^{-|x|^2} dx = (\int_{\mathbb{R}} e^{-t^2} dt)^n = \pi^{n/2}$. By the corollary and the substitution $u = r^2$ it also equals $n\omega_n \int_0^\infty e^{-r^2} r^{n-1}\, dr = \frac{n\omega_n}{2} \int_0^\infty e^{-u} u^{n/2-1}\, du = \frac{n\omega_n}{2}\Gamma(\frac{n}{2})$. Hence $\omega_n = \pi^{n/2}/\Gamma(\frac{n}{2}+1)$, which solves Exercise 28.23.7: $\omega_1 = 2$, $\omega_2 = \pi$, $\omega_3 = \frac{4}{3}\pi$, $\omega_4 = \frac{\pi^2}{2}$, and $\omega_n \to 0$ very fast as $n \to \infty$.

Remark 28.21.5. 1. *Critical points.* For a Lipschitz f the coarea formula holds without excluding critical points [EG92, Section 3.4]. On the critical set $Z = \{\nabla f = 0\}$ the left side vanishes, so the formula contains the statement that $\mathcal{H}^{n-1}(Z \cap f^{-1}(t)) = 0$ for almost every t. This is remarkable because the set of critical values $f(Z)$ can have positive measure: Sard's theorem, which says it is null, needs f of class C^n, and Whitney constructed a C^1 function on $\mathbb{R}^2$ that is not constant on a connected arc of critical points [Whi35]; its critical values fill an interval, yet each level set meets Z in an $\mathcal{H}^1$-null set.

2. *Isoperimetry and Sobolev inequalities.* For $u \in C_c^1(\mathbb{R}^n)$ the coarea formula reads $\int |\nabla u| = \int_{\mathbb{R}} \mathcal{H}^{n-1}(\{u = t\})\,dt$ (in the general version), and $\{u = t\}$ is, for almost every $t > 0$, the boundary of $\{u > t\}$. The *isoperimetric inequality* $\mathcal{H}^{n-1}(\partial E) \geq n\omega_n^{1/n}\lambda^n(E)^{(n-1)/n}$, with equality for balls, then gives the *Sobolev inequality* $\|u\|_{n/(n-1)} \leq (n\omega_n^{1/n})^{-1}\|\nabla u\|_1$ with its best constant; conversely, the Sobolev inequality applied to smoothed indicator functions gives back the isoperimetric inequality. This equivalence between a geometric and an analytic inequality is due to Federer and Fleming and to Maz'ya [FF60].

28.22 Densities and rectifiability

By the Lebesgue density theorem (Corollary 28.9.5) a measurable set $A \subseteq \mathbb{R}^n$ fills almost all of every small ball centred at almost every one of its points. For a set of dimension $s < n$ the analogous question is whether $\mathcal{H}^s(A \cap \overline{B}(x,r))$ is comparable to $\alpha(s)(2r)^s$, the s-dimensional "volume" of a ball of radius r. For smooth curves and surfaces the answer is yes, with ratio tending to 1; for the Cantor set the ratio oscillates forever. This difference turns out to be the difference between the sets on which one can do calculus, the rectifiable ones, and the others. Throughout, X is a metric space and $\overline{B}(x,r) = \{y : d(x,y) \leq r\}$.

Definition 28.22.1. For $A \subseteq X$, $s > 0$ and $x \in X$, the *upper* and *lower s-dimensional densities* of A at x are

$$\Theta^{*s}(A,x) = \limsup_{r\to 0} \frac{\mathcal{H}^s(A \cap \overline{B}(x,r))}{\alpha(s)(2r)^s}, \qquad \Theta^s_*(A,x) = \liminf_{r\to 0} \frac{\mathcal{H}^s(A \cap \overline{B}(x,r))}{\alpha(s)(2r)^s}.$$

If they are equal, the common value is the *density* $\Theta^s(A,x)$.

The normalisation is chosen so that flat pieces have density 1. For $A \subseteq \mathbb{R}^n$ Lebesgue measurable and $s = n$, $\mathcal{H}^n = \lambda^n$ and $\alpha(n)(2r)^n = \lambda^n(\overline{B}(x,r))$, so by the density theorem $\Theta^n(A,x) = 1$ for almost every $x \in A$ and $\Theta^n(A,x) = 0$ for almost every $x \notin A$ (spheres are null, so closed and open balls give the same ratios).

Example 28.22.2. *Curves have density* 1. Let $\gamma\colon (a,b) \to \mathbb{R}^m$ be an injective C^1 map with $\gamma' \neq 0$ which is a homeomorphism onto its image Γ. Then $\Theta^1(\Gamma,x) = 1$ for every $x \in \Gamma$. Let $x = \gamma(t_0)$, $c = |\gamma'(t_0)| > 0$, and $0 < \varepsilon < 1$. Choose $\delta > 0$ with $[t_0 - \delta, t_0 + \delta] \subseteq (a,b)$ and $|\gamma'(t) - \gamma'(t_0)| \leq \varepsilon c$ there. By the mean value inequality, applied to $\gamma(t) - \gamma'(t_0)(t - t_0)$,

$$(1-\varepsilon)c|t - t_0| \leq |\gamma(t) - x| \leq (1+\varepsilon)c|t - t_0| \qquad (|t - t_0| \leq \delta),$$

and $(1-\varepsilon)c \leq |\gamma'(t)| \leq (1+\varepsilon)c$ there. Since γ^{-1} is continuous at x, there is $r_0 \in (0, (1-\varepsilon)c\delta]$ such that $|\gamma(t) - x| \leq r_0$ implies $|t - t_0| \leq \delta$. For $r < r_0$ the set $I_r = \{t : |\gamma(t) - x| \leq r\}$ therefore satisfies

$$\left[t_0 - \tfrac{r}{(1+\varepsilon)c}, t_0 + \tfrac{r}{(1+\varepsilon)c}\right] \subseteq I_r \subseteq \left[t_0 - \tfrac{r}{(1-\varepsilon)c}, t_0 + \tfrac{r}{(1-\varepsilon)c}\right].$$

By the area formula (Theorem 28.20.3), $\mathcal{H}^1(\Gamma \cap \overline{B}(x,r)) = \int_{I_r} |\gamma'|$ lies between $2r\frac{1-\varepsilon}{1+\varepsilon}$ and $2r\frac{1+\varepsilon}{1-\varepsilon}$. As $\alpha(1) = 1$ and ε was arbitrary, the ratio tends to 1. The homeomorphism condition matters: for a curve shaped like the digit 6, whose end $\gamma(t)$, $t \to b$, runs into an interior point x from one side, a small ball around x contains the arc of the curve through x, of length about $2r$, and the end arc, of length about r, and $\Theta^1(\Gamma,x) = \frac{3}{2}$.

Example 28.22.3. *The Cantor set has no density at* 0. Let C be the Cantor set, $s = \log 2/\log 3$, and $c = \mathcal{H}^s(C)$, so that $\alpha(s)/2 \leq c \leq \alpha(s)$ (Example 28.17.6). For $k \geq 0$, $C \cap [0, 3^{-k}] = 3^{-k}C$ (the piece C_w for $w = 1\cdots 1$ of length k, in the notation of Example 28.18.7), so $\mathcal{H}^s(C \cap [0,3^{-k}]) =$

$3^{-ks}c = 2^{-k}c$. Moreover C does not meet the gap $(3^{-k-1}, 2\cdot 3^{-k-1})$, so $C \cap [0, 2\cdot 3^{-k-1}]$ is $C \cap [0, 3^{-k-1}]$ plus the single point $2\cdot 3^{-k-1}$. Hence, with $r = 3^{-k}$ and $r = 2\cdot 3^{-k-1}$ respectively,

$$\frac{\mathcal{H}^s(C \cap \overline{B}(0, 3^{-k}))}{\alpha(s)(2\cdot 3^{-k})^s} = \frac{2^{-k}c}{\alpha(s)2^s 2^{-k}} = \frac{c}{\alpha(s)2^s}, \qquad \frac{\mathcal{H}^s(C \cap \overline{B}(0, 2\cdot 3^{-k-1}))}{\alpha(s)(4\cdot 3^{-k-1})^s} = \frac{2^{-k-1}c}{\alpha(s)4^s 2^{-k-1}} = \frac{c}{\alpha(s)4^s}.$$

So $\Theta^{*s}(C, 0) \geq c/(\alpha(s)2^s)$ and $\Theta^s_*(C, 0) \leq c/(\alpha(s)4^s)$, which is smaller by the factor $2^s \approx 1.55$; as $c > 0$, the density does not exist. The same computation works at the left endpoint $f_w(0)$ of every interval I_w: for $r < 3^{-|w|}$ the ball $\overline{B}(f_w(0), r)$ meets C in $f_w(C \cap [0, 3^{|w|}r])$, because C has a gap of length at least $3^{-|w|}$ to the left of I_w (Example 28.17.6(2)), and $\mathcal{H}^s$ scales by $3^{-|w|s}$ under f_w. These endpoints are dense in C.

For a general set we cannot expect the density to exist, but the upper density is bounded below almost everywhere, by a short covering argument that uses nothing but the definition of $\mathcal{H}^s$.

Theorem 28.22.4. *Let $A \subseteq X$ with $\mathcal{H}^s(A) < \infty$, $s > 0$. Then $\Theta^{*s}(A, x) \geq 2^{-s}$ for $\mathcal{H}^s$-almost every $x \in A$.*

Proof. For $0 < t < 2^{-s}$ and $k \geq 1$ let

$$A_{t,k} = \left\{x \in A : \mathcal{H}^s(A \cap \overline{B}(x, r)) \leq t\,\alpha(s)(2r)^s \text{ for all } 0 < r < 1/k\right\}.$$

A point $x \in A$ with $\Theta^{*s}(A, x) < 2^{-s}$ lies in some $A_{t,k}$ with t rational; so it suffices to show $\mathcal{H}^s(A_{t,k}) = 0$ for all t and k, by countable subadditivity.

Fix t, k, let $0 < \delta < 1/k$ and $\varepsilon > 0$, and choose a cover (C_i) of $A_{t,k}$ by sets of diameter at most δ with $\sum_i \alpha(s)(\operatorname{diam} C_i)^s \leq \mathcal{H}^s_\delta(A_{t,k}) + \varepsilon$. We may discard the C_i that do not meet $A_{t,k}$. For the others pick $x_i \in C_i \cap A_{t,k}$ and let $d_i = \operatorname{diam} C_i$. If $d_i > 0$, then $C_i \subseteq \overline{B}(x_i, d_i)$ with $d_i < 1/k$, so by the definition of $A_{t,k}$

$$\mathcal{H}^s(A_{t,k} \cap C_i) \leq \mathcal{H}^s(A \cap \overline{B}(x_i, d_i)) \leq t\,\alpha(s)(2d_i)^s = t2^s\,\alpha(s)d_i^s.$$

If $d_i = 0$, then C_i is a point and both sides are 0. Summing, and using countable subadditivity of $\mathcal{H}^s$,

$$\mathcal{H}^s(A_{t,k}) \leq \sum_i \mathcal{H}^s(A_{t,k} \cap C_i) \leq t2^s\left(\mathcal{H}^s_\delta(A_{t,k}) + \varepsilon\right) \leq t2^s\left(\mathcal{H}^s(A_{t,k}) + \varepsilon\right).$$

Let $\varepsilon \to 0$. Since $\mathcal{H}^s(A_{t,k}) \leq \mathcal{H}^s(A) < \infty$ and $t2^s < 1$, this forces $\mathcal{H}^s(A_{t,k}) = 0$. □

The bound 2^{-s} comes from the comparison $C_i \subseteq \overline{B}(x_i, \operatorname{diam} C_i)$, which loses a factor 2 in the radius; in $\mathbb{R}^n$, finer covering theorems give the complementary inequalities, which we quote.

Theorem 28.22.5. *Let $A \subseteq \mathbb{R}^n$ be $\mathcal{H}^s$-measurable with $\mathcal{H}^s(A) < \infty$. Then $2^{-s} \leq \Theta^{*s}(A, x) \leq 1$ for $\mathcal{H}^s$-almost every $x \in A$, and $\Theta^{*s}(A, x) = 0$ for $\mathcal{H}^s$-almost every $x \in \mathbb{R}^n \setminus A$ [EG92, Section 2.3], [Fal85].*

So every set of finite s-dimensional measure is, at almost all of its points, at least somewhat dense and at most as dense as a flat piece. Whether the density exists, that is, whether the lower density equals the upper one, is a deep geometric property, and it is related to the following class of sets.

Definition 28.22.6. Let $k \geq 1$ be an integer. A set $A \subseteq \mathbb{R}^n$ is *countably k-rectifiable* if there are Lipschitz maps $g_j \colon E_j \to \mathbb{R}^n$, $E_j \subseteq \mathbb{R}^k$, $j = 1, 2, \ldots$, such that

$$\mathcal{H}^k\Big(A \setminus \bigcup_j g_j(E_j)\Big) = 0.$$

It is *purely k-unrectifiable* if $\mathcal{H}^k(A \cap g(E)) = 0$ for every Lipschitz map $g \colon E \to \mathbb{R}^n$ with $E \subseteq \mathbb{R}^k$.

So a rectifiable set is, up to a null set, a countable union of Lipschitz images of pieces of $\mathbb{R}^k$, and on such pieces Rademacher's theorem and the area formula give tangent planes and a calculus. A purely unrectifiable set meets each Lipschitz image of a piece of $\mathbb{R}^k$ only in a null set.

Proposition 28.22.7.

1. *Subsets and countable unions of countably k-rectifiable sets are countably k-rectifiable.*
2. *A countably k-rectifiable set A has $\dim_{\mathcal{H}} A \leq k$, and $A \cap \bigcup_j g_j(E_j)$ has σ-finite $\mathcal{H}^k$-measure.*
3. *If $U \subseteq \mathbb{R}^k$ is open and $g \colon U \to \mathbb{R}^n$ is of class C^1, then $g(U)$ is countably k-rectifiable.*
4. *If A is countably k-rectifiable and B is purely k-unrectifiable, then $\mathcal{H}^k(A \cap B) = 0$.*

Proof. (1) is clear from the definition: a subset can use the same maps, and countably many countable families of maps form a countable family.

(2) Let L_j be a Lipschitz constant of g_j. For $m \geq 1$, Proposition 28.17.3(3) and Theorem 28.17.5 give $\mathcal{H}^k(g_j(E_j \cap [-m,m]^k)) \leq L_j^k \mathcal{H}^k([-m,m]^k) = L_j^k (2m)^k < \infty$. So $\bigcup_j g_j(E_j)$ is a countable union of sets of finite $\mathcal{H}^k$-measure, each of which has $\mathcal{H}^t$-measure 0 for $t > k$ (Proposition 28.17.3(4)); the remaining part of A is $\mathcal{H}^k$-null, hence also $\mathcal{H}^t$-null for $t > k$.

(3) U is the union of the countably many closed balls $\overline{B} \subseteq U$ with rational centres and radii. On such a ball, which is compact and convex, $\|Dg\|$ is bounded, say by L, and the mean value inequality (Proposition 27.10.3(4)) shows that g is L-Lipschitz on $\overline{B}$.

(4) $A \cap B$ is contained in the union of the $\mathcal{H}^k$-null set $A \setminus \bigcup_j g_j(E_j)$ and the sets $B \cap g_j(E_j)$, which are $\mathcal{H}^k$-null by assumption. □

By (3), images of C^1 maps, such as the curves of Example 28.22.2, graphs of C^1 or Lipschitz functions, and k-dimensional submanifolds of $\mathbb{R}^n$ (which are covered by countably many images of charts), are countably k-rectifiable. At the other extreme, the four-corner Cantor set K_4 of Example 28.18.11(2), which has $0 < \mathcal{H}^1(K_4) < \infty$, is purely 1-unrectifiable [Fal85], [Mat95]. The two classes are distinguished by densities and by projections, and these are among the deepest results of the subject.

Theorem 28.22.8. *Let $A \subseteq \mathbb{R}^n$ be $\mathcal{H}^k$-measurable with $0 < \mathcal{H}^k(A) < \infty$, k an integer with $1 \leq k < n$.*

1. *(*Densities*) A is countably k-rectifiable if and only if $\Theta^k(A, x) = 1$ for $\mathcal{H}^k$-almost every $x \in A$, and if and only if $\Theta^k(A, x)$ exists and is positive for $\mathcal{H}^k$-almost every $x \in A$.*
2. *(*Projections, Besicovitch–Federer*) If A is countably k-rectifiable, then its orthogonal projections to almost all k-dimensional linear subspaces have positive $\mathcal{H}^k$-measure. If A is purely k-unrectifiable, then its projections to almost all k-dimensional linear subspaces are $\mathcal{H}^k$-null.*

Here "almost all" refers to the rotation-invariant measure on the set of k-dimensional subspaces; for $k = 1$, $n = 2$ it means almost all angles. See [Mat95], [Fed69]; the last statement of (1) is Preiss's theorem [Pre87].

Remark 28.22.9.

1. *Non-integral dimensions.* If $0 < \mathcal{H}^s(A) < \infty$ and the density $\Theta^s(A, x)$ exists and is positive at almost every point of A, then s is an integer (Marstrand, see [Mat95]). So sets of non-integral dimension, such as the Cantor set (Example 28.22.3) and the self-similar sets of Section 28.18, are always irregular: their densities oscillate.

2. *The four-corner set again.* Example 28.18.12 computed two projections of K_4: in the direction of the axes it has length 0, as Theorem 28.22.8(2) predicts for almost all directions, while in the direction of $(2, 1)$ it is an interval. So the exceptional set of directions in the Besicovitch–Federer theorem is not empty. For a segment the situation is reversed: its projection has positive length in every direction except the perpendicular one.

3. *Besicovitch and Kakeya sets.* A *Besicovitch set* is a compact subset of $\mathbb{R}^n$ that contains a segment of length 1 in every direction. Besicovitch showed that such sets exist in $\mathbb{R}^2$ with Lebesgue measure 0; correspondingly, a needle of length 1 can be turned around completely inside a region of arbitrarily small area (*Kakeya's needle problem*). Every Besicovitch set in $\mathbb{R}^2$ has Hausdorff dimension 2 (Davies), see [Fal85]. Whether Besicovitch sets in $\mathbb{R}^n$ must have Hausdorff dimension n is the *Kakeya conjecture*, a central problem linking geometric measure theory with harmonic analysis and analytic number theory.

28.23 Exercises

Exercise 28.23.1. Show that the Cantor set, obtained from $[0, 1]$ by repeatedly removing open middle thirds, is compact, uncountable and of Lebesgue measure zero.

Solution. It is an intersection of closed subsets of $[0, 1]$, hence compact. After k steps it is covered by 2^k intervals of length 3^{-k}, so its measure is at most $(2/3)^k$. Its points are the numbers $\sum_k a_k 3^{-k}$ with $a_k \in \{0, 2\}$, and distinct sequences give distinct numbers, so it is in bijection with $\{0, 1\}^{\mathbb{N}}$, which is uncountable (Theorem 3.6.3). □

Exercise 28.23.2. (Borel–Cantelli.) If $\sum_k \mu(A_k) < \infty$, show that almost every point lies in only finitely many A_k.

Solution. By monotone convergence $\int \sum_k \mathbf{1}_{A_k} = \sum_k \mu(A_k) < \infty$, so $\sum_k \mathbf{1}_{A_k} < \infty$ a.e. □

Exercise 28.23.3. Show that there is no Lebesgue measurable $A \subseteq \mathbb{R}$ with $\lambda(A \cap I) = \frac{1}{2}\lambda(I)$ for every bounded interval I.

Solution. For such A, $\lambda(A \cap B(x, r))/\lambda(B(x, r)) = \frac{1}{2}$ for all x and r. By Corollary 28.9.5 this ratio tends to 1 at almost every point of A and to 0 at almost every point outside A; so A and $\mathbb{R} \setminus A$ are both null, which is impossible. □

Exercise 28.23.4. Show that $Mf \notin L^1(\mathbb{R}^n)$ for every $f \in L^1(\mathbb{R}^n)$ that is not zero almost everywhere; so the maximal inequality cannot be improved to $\|Mf\|_1 \le C\|f\|_1$.

Solution. Choose R with $a = \int_{B(0,R)} |f| > 0$. For $\|x\| \ge R$, the ball $B(x, 2\|x\|)$ contains $B(0, R)$, so $Mf(x) \ge a/\lambda(B(x, 2\|x\|)) = a/(2^n\|x\|^n\lambda(B(0, 1)))$. The function $\|x\|^{-n}$ is not integrable on $\{\|x\| \ge R\}$: the shells $2^k R \le \|x\| < 2^{k+1}R$ have measure $(2^n - 1)2^{kn}R^n\lambda(B(0, 1))$, and on each of them $\|x\|^{-n} \ge 2^{-(k+1)n}R^{-n}$, so each contributes at least the positive constant $(1 - 2^{-n})\lambda(B(0, 1))$ to the integral. □

Exercise 28.23.5. Show that $L^q(\mu) \subseteq L^p(\mu)$ for $p \le q$ if $\mu(X) < \infty$, and that there is no inclusion in either direction for Lebesgue measure on $\mathbb{R}$.

Solution. By Hölder with exponents q/p and its conjugate, $\int |f|^p \le (\int |f|^q)^{p/q} \mu(X)^{1-p/q}$. On $\mathbb{R}$, $|x|^{-1/2}\mathbf{1}_{[-1,1]}$ is in L^1 but not L^2, and $(1+|x|)^{-1}$ is in L^2 but not L^1. □

Exercise 28.23.6. For $f \in L^1(\mathbb{R}^n)$ and $g \in L^p(\mathbb{R}^n)$, show using Fubini's theorem that $(f*g)(x) = \int f(x-y)g(y)\,dy$ is defined a.e. and $\|f*g\|_p \le \|f\|_1 \|g\|_p$.

Exercise 28.23.7. Show that the volume of the unit ball in $\mathbb{R}^n$ is $\omega_n = \pi^{n/2}/\Gamma(\frac{n}{2}+1)$, where $\Gamma(s) = \int_0^\infty t^{s-1}e^{-t}\,dt$, by computing $\int_{\mathbb{R}^n} e^{-\|x\|^2}$ in two ways.

Exercise 28.23.8. Let $K \subseteq [0,1]$ be the set obtained by removing the open middle half $(\frac{1}{4}, \frac{3}{4})$ of $[0,1]$, then the open middle halves of the two remaining intervals, and so on; equivalently, the numbers $\sum_{k\ge1} a_k 4^{-k}$ with all $a_k \in \{0,3\}$. Show that $\dim_{\mathcal{H}} K = \frac{1}{2}$.

Solution. Follow Example 28.17.6 with 3 replaced by 4 and the digits $\{0,2\}$ by $\{0,3\}$; now $s = \log 2/\log 4 = \frac{1}{2}$, so $4^s = 2$. The 2^k intervals of generation k have length 4^{-k}, which gives $\mathcal{H}^{1/2}(K) \le \alpha(\frac{1}{2})2^k 4^{-k/2} = \alpha(\frac{1}{2})$. Two of them first differing at position $j \le k$ have left ends differing by at least $3\cdot 4^{-j} - \sum_{j<i\le k} 3\cdot 4^{-i} \ge 2\cdot 4^{-j}$, so they are at distance at least 4^{-k}. The function $F(\sum_k a_k 4^{-k}) = \sum_k (a_k/3)2^{-k}$ on K, extended to $[0,1]$ as in Counterexample 28.9.7, is continuous and nondecreasing, and its measure μ_F (Lemma 28.13.3) gives mass 2^{-k} to each interval of generation k and mass 0 to $[0,1]\setminus K$. A closed set of diameter d with $4^{-k-1} \le d < 4^{-k}$ meets at most one interval of generation k, so its μ_F-measure is at most $2^{-k} = 2\cdot 4^{-(k+1)/2} \le 2d^{1/2}$, and as in the example $\mathcal{H}^{1/2}(K) \ge \alpha(\frac{1}{2})/2 > 0$. □

Exercise 28.23.9. Let $h\colon \mathbb{R}^n \to \mathbb{R}$ be Lipschitz with constant L and $U \subseteq \mathbb{R}^n$ a bounded open set. Show that the graph $\Gamma = \{(x, h(x)) : x \in U\}$ satisfies $\lambda^n(U) \le \mathcal{H}^n(\Gamma) \le (1+L^2)^{n/2}\lambda^n(U)$, so that $\dim_{\mathcal{H}} \Gamma = n$, even though h need not be differentiable everywhere.

Solution. The map $g(x) = (x, h(x))$ satisfies $|g(x)-g(y)|^2 = |x-y|^2 + |h(x)-h(y)|^2 \le (1+L^2)|x-y|^2$, so it is $\sqrt{1+L^2}$-Lipschitz, and its inverse, the projection $\Gamma \to U$, is 1-Lipschitz. By Proposition 28.17.3(3) and Theorem 28.17.5, $\lambda^n(U) = \mathcal{H}^n(U) \le \mathcal{H}^n(\Gamma) \le (1+L^2)^{n/2}\mathcal{H}^n(U)$. Both bounds are positive and finite, so $\mathcal{H}^s(\Gamma)$ is ∞ for $s < n$ and 0 for $s > n$ by Proposition 28.17.3(4). (By the general area formula of Remark 28.20.6 and Rademacher's theorem, $\mathcal{H}^n(\Gamma) = \int_U \sqrt{1+|\nabla h|^2}$.) □

Exercise 28.23.10. For $R > r > 0$ let $x(u,v) = ((R + r\cos u)\cos v, (R + r\cos u)\sin v, r\sin u)$. Show that x is injective with injective differential on $(0,2\pi)^2$, that $Jx = r(R + r\cos u)$, and that the torus $T = x(\mathbb{R}^2)$ has $\mathcal{H}^2(T) = 4\pi^2 Rr$.

Solution. $x_u = (-r\sin u\cos v, -r\sin u\sin v, r\cos u)$ and $x_v = (-(R+r\cos u)\sin v, (R+r\cos u)\cos v, 0)$ are orthogonal, of lengths r and $R + r\cos u > 0$, so $Jx = |x_u \times x_v| = r(R + r\cos u)$ and Dx is injective. The point $x(u,v)$ determines v modulo 2π (its projection to the plane $z = 0$ has argument v) and then u modulo 2π, so x is injective on $(0,2\pi)^2$. Its image is T minus the two circles $u = 0$ and $v = 0$, which have finite $\mathcal{H}^1$-measure and are therefore $\mathcal{H}^2$-null. By the area formula, $\mathcal{H}^2(T) = \int_0^{2\pi}\int_0^{2\pi} r(R + r\cos u)\,du\,dv = 4\pi^2 Rr$. □

Exercise 28.23.11. Let $(X, \mathcal{A}, \mu)$ be a complete measure space, f measurable and $g = f$ almost everywhere. Show that g is measurable. Deduce that for Lebesgue measure, changing a measurable function on a null set gives a measurable function, so that "measurable function defined almost everywhere" makes sense.

Solution. Let $N \in \mathcal{A}$ be a null set with $f = g$ outside N. For $a \in \mathbb{R}$, $\{g > a\} = (\{f > a\} \setminus N) \cup (\{g > a\} \cap N)$. The first set is measurable, and the second is a subset of the null set N, hence measurable by completeness. So g is measurable (Proposition 28.2.6(1)). Lebesgue measure is complete (Theorem 28.3.4). □

Exercise 28.23.12. For $A \subseteq [0,1]$ define the *inner measure* $\lambda_*(A) = \sup\{\lambda(K) : K \subseteq A \text{ compact}\}$, approximating A from inside instead of from outside. Show that $\lambda_*(A) = 1 - \lambda^*([0,1] \setminus A)$, that $\lambda_*(A) \le \lambda^*(A)$, and that A is Lebesgue measurable if and only if $\lambda_*(A) = \lambda^*(A)$. (This was Lebesgue's original definition of measurability.)

Solution. Compact subsets $K \subseteq A$ correspond to relatively open sets $V = [0,1] \setminus K \supseteq [0,1] \setminus A$, with $\lambda(K) = 1 - \lambda(V)$; and $\lambda^*(B)$ is the infimum of $\lambda(U \cap [0,1])$ over open $U \supseteq B$ for $B \subseteq [0,1]$ (enlarge the covering boxes, as in the proof of Theorem 28.3.4). This gives the formula. If $K \subseteq A \subseteq U$ then $\lambda(K) \le \lambda^*(A) \le \lambda(U)$, so $\lambda_* \le \lambda^*$. If A is measurable, equality is inner regularity. Conversely, if $\lambda_*(A) = \lambda^*(A)$, choose compact $K_n \subseteq A$ and open $U_n \supseteq A$ with $\lambda(K_n) \to \lambda_*(A)$ and $\lambda(U_n) \to \lambda^*(A)$; then $F = \bigcup K_n \subseteq A \subseteq G = \bigcap U_n$ are Borel sets with $\lambda(G \setminus F) = \lambda(G) - \lambda(F) = 0$, so $A = F \cup (A \setminus F)$ with $A \setminus F \subseteq G \setminus F$ null, and A is measurable by completeness. □

Exercise 28.23.13. Let σ be the measure on $\mathbb{R}^2$ of Example 28.14.5(3), and $\varphi(t) = (\cos t, \sin t)$. Show that $\sigma(E) = \lambda(\{t \in [0, 2\pi] : \varphi(t) \in E\})$ for every Borel set E, and deduce that an arc $\{\varphi(t) : \alpha < t < \beta\}$ with $0 \le \alpha < \beta \le 2\pi$ has measure $\beta - \alpha$.

Solution. The right side defines a finite Borel measure ν, the push-forward of λ restricted to $[0, 2\pi]$, since preimages under the continuous φ of Borel sets are Borel. For $f = \mathbf{1}_E$, $\int f \, d\nu = \int_{[0,2\pi]} f \circ \varphi \, d\lambda$ by definition; by linearity this holds for simple functions, by monotone convergence for measurable $f \ge 0$, and then for $f \in C_c(\mathbb{R}^2)$. By Proposition 28.4.9, $\int_{[0,2\pi]} f \circ \varphi \, d\lambda = \Lambda f$. So ν represents Λ, and $\nu = \sigma$ by the uniqueness in Theorem 28.14.6. Since φ is injective on $[0, 2\pi)$ and $\varphi(2\pi) = \varphi(0)$, the set of $t \in [0, 2\pi]$ with $\varphi(t)$ in the arc is (α, β), of measure $\beta - \alpha$. □

Exercise 28.23.14. Let $\mu_k \rightharpoonup \mu$ be Radon measures on X (locally compact Hausdorff with a countable basis).

1. Show that $\mu(X) \le \liminf_k \mu_k(X)$, and that $\mu_k(X) \to \mu(X)$ if X is compact.
2. Suppose that the μ_k and μ are probability measures. Show that $\int f \, d\mu_k \to \int f \, d\mu$ for every bounded continuous $f : X \to \mathbb{R}$.

Solution. (1) The first statement is Proposition 28.15.3(1) with $U = X$; if X is compact, $1 \in C_c(X)$. (2) Let $\varepsilon > 0$. By inner regularity choose a compact K with $\mu(K) > 1 - \varepsilon$, and $\phi \in C_c(X)$ with $0 \le \phi \le 1$ and $\phi = 1$ on K. Then $\int \phi \, d\mu_k \to \int \phi \, d\mu > 1 - \varepsilon$, so $\int (1 - \phi) \, d\mu_k < 2\varepsilon$ for k large, while $\int (1 - \phi) \, d\mu < \varepsilon$. As $f\phi \in C_c(X)$,

$$\left| \int f \, d\mu_k - \int f \, d\mu \right| \le \left| \int f\phi \, d\mu_k - \int f\phi \, d\mu \right| + \|f\|_\infty \left(\int (1 - \phi) \, d\mu_k + \int (1 - \phi) \, d\mu \right),$$

whose lim sup is at most $3\varepsilon \|f\|_\infty$. So when no mass is lost, testing against $C_c(X)$ and against bounded continuous functions give the same notion of convergence. □

Exercise 28.23.15. *Length is not continuous.* For $k \ge 1$ let $\gamma_k(x) = (x, \frac{1}{k} \sin kx)$ for $x \in [0, 2\pi]$, and let μ_k be the length measure of γ_k, the Radon measure on $\mathbb{R}^2$ with

$$\int f \, d\mu_k = \int_0^{2\pi} f(\gamma_k(x)) \, |\gamma_k'(x)| \, dx = \int_0^{2\pi} f\left(x, \tfrac{1}{k} \sin kx\right) \sqrt{1 + \cos^2 kx} \, dx.$$

Show that $\gamma_k \to \gamma_0$, $\gamma_0(x) = (x, 0)$, uniformly, that $\mu_k \rightharpoonup c\mu_0$ with $c = \frac{1}{2\pi} \int_0^{2\pi} \sqrt{1 + \cos^2 t} \, dt$, where μ_0 is the length measure of the segment, and that $c > 1$. So the curves γ_k have length $2\pi c$, which does not converge to the length 2π of the limit curve.

Solution. $|\gamma_k(x) - \gamma_0(x)| \le 1/k$. Let $f \in C_c(\mathbb{R}^2)$, with modulus of uniform continuity ω. Then $|f(\gamma_k(x)) - f(x,0)| \le \omega(1/k)$, so $\int f\,d\mu_k$ differs from $\int_0^{2\pi} f(x,0)g(kx)\,dx$, where $g(t) = \sqrt{1+\cos^2 t}$, by at most $2\pi\sqrt{2}\,\omega(1/k)$. The argument of Example 28.15.2(5), applied on $[0,2\pi]$ to the continuous function $x \mapsto f(x,0)$ (only its continuity on $[0,2\pi]$ was used), shows that $\int_0^{2\pi} f(x,0)g(kx)\,dx \to \bar{g}\int_0^{2\pi} f(x,0)\,dx = c\int f\,d\mu_0$. Since $g \ge 1$, with equality only where $\cos t = 0$, and g is continuous, $c > 1$. The length of γ_k is $\int_0^{2\pi} g(kx)\,dx = \frac{1}{k}\int_0^{2\pi k} g = 2\pi c$, as the interval consists of k periods. In general length is only *lower semicontinuous* under uniform convergence of curves: the limit can be shorter, never longer. □

Exercise 28.23.16. Let μ be a Radon measure on $\mathbb{R}^n$ and $r > 0$. Show that $x \mapsto \mu(B(x,r))$ is lower semicontinuous and $x \mapsto \mu(\overline{B}(x,r))$ is upper semicontinuous, and that for fixed x the sphere $\{y : \|y-x\| = r\}$ has μ-measure 0 for all but countably many r.

Solution. Let $x_k \to x$. If $\|y-x\| < r$, then $\|y-x_k\| < r$ for k large, so $\mathbf{1}_{B(x,r)} \le \liminf_k \mathbf{1}_{B(x_k,r)}$, and Fatou's lemma (Corollary 28.4.4) gives $\mu(B(x,r)) \le \liminf_k \mu(B(x_k,r))$. If $\|y-x_k\| \le r$ for infinitely many k, then $\|y-x\| \le r$; so $\limsup_k \mathbf{1}_{\overline{B}(x_k,r)} \le \mathbf{1}_{\overline{B}(x,r)}$, and Fatou's lemma applied to the non-negative functions $\mathbf{1}_{\overline{B}(x,r+1)} - \mathbf{1}_{\overline{B}(x_k,r)}$ (k large), whose integrals are finite, gives $\limsup_k \mu(\overline{B}(x_k,r)) \le \mu(\overline{B}(x,r))$. The spheres of different radii are disjoint; those of radius less than R with measure at least $1/m$ number at most $m\mu(B(x,R))$, so only countably many spheres have positive measure. □

Exercise 28.23.17. Show directly from Example 28.17.6(2), without Theorem 28.18.10, that the Cantor set has box-counting dimension $\log 2/\log 3$.

Solution. Let $3^{-k-1} \le \delta < 3^{-k}$. The 2^{k+1} intervals of generation $k+1$ have length $3^{-k-1} \le \delta$ and cover C, so $N_\delta(C) \le 2^{k+1}$. A set of diameter at most $\delta < 3^{-k}$ meets at most one of the 2^k intervals of generation k, and each of them contains points of C, so $N_\delta(C) \ge 2^k$. Since $k\log 3 < -\log\delta \le (k+1)\log 3$,

$$\frac{k\log 2}{(k+1)\log 3} \le \frac{\log N_\delta(C)}{-\log\delta} \le \frac{(k+1)\log 2}{k\log 3} \qquad (k \ge 1),$$

and both bounds tend to $\log 2/\log 3$ as $\delta \to 0$, that is, as $k \to \infty$. □

Exercise 28.23.18. Let C be the Cantor set. Show that $C + C = \{x+y : x, y \in C\}$ is the interval $[0,2]$. (Hint: $C \times C$ is the attractor of Example 28.18.11(3); project it as in Example 28.18.12.) So the set $C \times C$, of dimension $\log 4/\log 3 < 2$, has a projection of positive length, while its projections to the axes are copies of C, of length 0.

Solution. Let $\pi(x,y) = x + y$ and $f_{a,b}(x,y) = (x/3 + a, y/3 + b)$ for $a, b \in \{0, \frac{2}{3}\}$. Then $\pi \circ f_{a,b} = g_{a+b} \circ \pi$ with $g_c(t) = t/3 + c$, and $a + b$ takes the values $0, \frac{2}{3}, \frac{4}{3}$. Since $C \times C = \bigcup_{a,b} f_{a,b}(C \times C)$, its image $\pi(C \times C) = C + C$ is a nonempty compact set with $C + C = g_0(C + C) \cup g_{2/3}(C + C) \cup g_{4/3}(C + C)$, that is, the attractor of $\{g_0, g_{2/3}, g_{4/3}\}$. The interval $[0,2]$ is also invariant: $g_0([0,2]) \cup g_{2/3}([0,2]) \cup g_{4/3}([0,2]) = [0, \frac{2}{3}] \cup [\frac{2}{3}, \frac{4}{3}] \cup [\frac{4}{3}, 2]$. By the uniqueness in Theorem 28.18.5, $C + C = [0,2]$. (More generally, by Marstrand's projection theorem a Borel set $A \subseteq \mathbb{R}^2$ with $\dim_{\mathcal{H}} A > 1$ has projections of positive length in almost every direction [Mat95].) □

Exercise 28.23.19. Show that for every $s \in [0,1]$ there is a compact set $K \subseteq [0,1]$ with $\dim_{\mathcal{H}} K = s$ and $0 < \mathcal{H}^s(K) < \infty$.

Solution. For $s = 0$ take a point, for $s = 1$ the interval $[0,1]$ (Theorem 28.17.5). For $0 < s < 1$ let $\rho = 2^{-1/s}$, so $0 < \rho < \frac{1}{2}$, and consider $f_1(x) = \rho x$, $f_2(x) = \rho x + 1 - \rho$. Since $f_1([0,1]) \cup f_2([0,1]) = [0,\rho] \cup [1-\rho, 1] \subseteq [0,1]$, the attractor K lies in $[0,1]$ (Theorem 28.18.5(1)), and the pieces $f_1(K) \subseteq [0,\rho]$ and $f_2(K) \subseteq [1-\rho, 1]$ are disjoint because $\rho < 1 - \rho$. The similarity dimension solves $2\rho^t = 1$, so it is s, and Theorem 28.18.10 applies. □

Exercise 28.23.20. Let K be the attractor of an iterated function system satisfying the strong separation condition, with similarity dimension s, and let μ be the measure constructed in the proof of Theorem 28.18.10.

1. Show that $\mathcal{H}^s(K_w) = r_w^s \mathcal{H}^s(K)$ for every word w, and explain why this, together with $0 < \mathcal{H}^s(K) < \infty$, forces (28.2).

2. Show that two Borel probability measures on K which agree on all the sets K_w are equal. Deduce that $\mu(B) = \mathcal{H}^s(B)/\mathcal{H}^s(K)$ for every Borel set $B \subseteq K$.

Solution. (1) f_w is a similarity with ratio r_w, so $\mathcal{H}^s(f_w(K)) = r_w^s \mathcal{H}^s(K)$. The sets $K_1, \dots, K_N$ are disjoint compact sets with union K, so $\mathcal{H}^s(K) = \sum_i \mathcal{H}^s(K_i) = \left(\sum_i r_i^s\right)\mathcal{H}^s(K)$, and dividing by $\mathcal{H}^s(K) \in (0,\infty)$ gives $\sum_i r_i^s = 1$. This is the heuristic of the introduction to Section 28.18.

(2) Let $\mathcal{P}$ consist of $\emptyset$ and the sets K_w, including $K = K_\varnothing$. Two of the K_w are either nested (if one word begins with the other, by Theorem 28.18.5(2)) or disjoint (if the words first differ at some position, by Step 1 of the proof of Theorem 28.18.10), so $\mathcal{P}$ is closed under finite intersections. Every relatively open $G \subseteq K$ is a countable union of sets $K_w \subseteq G$: if $x \in G$ has address ω, then $x \in K_{\omega|k}$ and $\operatorname{diam} K_{\omega|k} \le r^k \operatorname{diam} K$, so $K_{\omega|k} \subseteq G$ for k large; and there are only countably many words. So $\sigma(\mathcal{P})$ contains the open subsets of K, hence all Borel subsets, and Lemma 28.3.10 shows that two probability measures agreeing on $\mathcal{P}$ agree on all Borel sets. Finally, $B \mapsto \mathcal{H}^s(B)/\mathcal{H}^s(K)$ is a Borel probability measure on K (Proposition 28.17.3(1) and Theorem 28.18.10) which by (1) gives the value $r_w^s = \mu(K_w)$ to every K_w. □

Exercise 28.23.21. Let S be the Sierpiński gasket of a triangle T with perimeter P (Example 28.18.7(3)). Show:

1. $\lambda^2(S) = 0$;

2. two distinct triangles $f_v(T)$, $f_w(T)$ with $|v| = |w|$ meet in at most one point;

3. S contains the boundary ∂T, and more generally the boundaries of all triangles $f_w(T)$;

4. $\mathcal{H}^1(S) = \infty$, so that $1 \le \dim_{\mathcal{H}} S \le \log 3/\log 2$.

Solution. (1) $F^k(T)$ is the union of the 3^k triangles $f_w(T)$, $|w| = k$, which are images of T under similarities of ratio 2^{-k}, so of area $4^{-k}\lambda^2(T)$ (Proposition 28.17.3(3) and Theorem 28.17.5). Hence $\lambda^2(S) \le \lambda^2(F^k(T)) \le (\frac{3}{4})^k \lambda^2(T) \to 0$.

(2) For $i \ne j$ the corner triangles $f_i(T)$ and $f_j(T)$ meet only in the midpoint of the side $p_i p_j$. If $v \ne w$ first differ at position j and u is their common beginning of length $j-1$, then $f_v(T) \subseteq f_u(f_{v_j}(T))$ and $f_w(T) \subseteq f_u(f_{w_j}(T))$, and these meet in the image under the injective map f_u of a single point.

(3) The side $[p_i, p_j]$ is the union of $[p_i, m]$ and $[m, p_j]$, m its midpoint, which are sides of $f_i(T)$ and $f_j(T)$. So $\partial T \subseteq F(\partial T)$, and inductively $\partial T \subseteq F(\partial T) \subseteq F^2(\partial T) \subseteq \cdots$, since F preserves inclusions. As $F^k(\partial T) \subseteq F^k(T)$, $\partial T \subseteq \bigcap_k F^k(T) = S$ (Theorem 28.18.5(1)). Then $f_w(\partial T) \subseteq f_w(S) \subseteq S$.

(4) A segment of length ℓ is isometric to an interval, so it has $\mathcal{H}^1$-measure ℓ (Theorem 28.17.5). By (3), S contains the boundaries of the 3^k triangles of generation k, each the union of three segments of total length $2^{-k}P$. By (2) these 3^{k+1} segments meet each

other only in finitely many points, which are $\mathcal{H}^1$-null; so, $\mathcal{H}^1$ being a measure on Borel sets, $\mathcal{H}^1(S) \geq 3^k 2^{-k} P = (\frac{3}{2})^k P$ for every k. Hence $\mathcal{H}^1(S) = \infty$ and $\dim_{\mathcal{H}} S \geq 1$; the upper bound is Proposition 28.18.9, with $3 \cdot 2^{-s} = 1$. (By Remark 28.18.13, $\dim_{\mathcal{H}} S = \log 3/\log 2$.) □

Exercise 28.23.22. Let $h\colon [0,1] \to \mathbb{R}$ satisfy $|h(x) - h(y)| \leq c|x-y|^\alpha$ for some $c > 0$ and $0 < \alpha \leq 1$ (a *Hölder continuous* function), and let $G = \{(x, h(x)) : x \in [0,1]\}$ be its graph. Show that $1 \leq \dim_{\mathcal{H}} G \leq \overline{\dim}_B G \leq 2 - \alpha$.

Solution. Projection to the first coordinate is 1-Lipschitz and maps G onto $[0,1]$, so $\mathcal{H}^1(G) \geq 1$ and $\dim_{\mathcal{H}} G \geq 1$. For $m \geq 1$ divide $[0,1]$ into m intervals I_j of length $\frac{1}{m}$. On I_j the values of h lie in an interval of length at most $cm^{-\alpha}$, which is covered by at most $cm^{1-\alpha} + 1$ intervals of length $\frac{1}{m}$; so the part of G over I_j is covered by at most $cm^{1-\alpha} + 1$ squares of side $\frac{1}{m}$, and G by at most $m(cm^{1-\alpha} + 1) \leq (c+1)m^{2-\alpha}$ of them. The squares have diameter $\sqrt{2}/m$. Given $0 < \delta < \sqrt{2}$, choose m with $\sqrt{2}/(m+1) \leq \delta < \sqrt{2}/m$; then $N_\delta(G) \leq (c+1)(m+1)^{2-\alpha}$ and $-\log\delta > \log(m/\sqrt{2})$, so

$$\frac{\log N_\delta(G)}{-\log\delta} \leq \frac{\log(c+1) + (2-\alpha)\log(m+1)}{\log m - \log\sqrt{2}} \to 2 - \alpha \qquad (\delta \to 0).$$

The middle inequality is Proposition 28.18.2(3). For $\alpha = 1$, a Lipschitz function, the graph has dimension 1, in accordance with Exercise 28.23.9; rough functions can have graphs of dimension close to 2. □

Exercise 28.23.23. Let $g\colon \mathbb{R}^n \to \mathbb{R}$ be continuous, $n \geq 2$, and $\Phi(R) = \int_{B(0,R)} g\, d\lambda^n$. Show that Φ is differentiable on $(0,\infty)$ with $\Phi'(R) = \int_{|x|=R} g\, d\mathcal{H}^{n-1}$, and that the averages $\frac{1}{n\omega_n R^{n-1}} \int_{|x|=R} g\, d\mathcal{H}^{n-1}$ tend to $g(0)$ as $R \to 0$.

Solution. Apply Corollary 28.21.3 to the positive and negative parts of $g\mathbf{1}_{B(0,R)}$, which are bounded Borel functions vanishing outside a bounded set; subtracting, $\Phi(R) = \int_0^R \eta(r)\, dr$ with $\eta(r) = r^{n-1} \int_{S^{n-1}} g(r\theta)\, d\mathcal{H}^{n-1}(\theta)$. As $r \to r_0$, $g(r\theta) \to g(r_0\theta)$ uniformly in $\theta \in S^{n-1}$, by uniform continuity of g on a compact ball, and $\mathcal{H}^{n-1}(S^{n-1}) = n\omega_n < \infty$; so η is continuous on $[0,\infty)$, and $\Phi'(R) = \eta(R)$ by the fundamental theorem of calculus (Theorem 27.8.3). By the scaling in the proof of Corollary 28.21.3, $\eta(R) = \int_{|x|=R} g\, d\mathcal{H}^{n-1}$. Finally $\eta(R)/(n\omega_n R^{n-1}) - g(0) = \frac{1}{n\omega_n} \int_{S^{n-1}} (g(R\theta) - g(0))\, d\mathcal{H}^{n-1}(\theta)$, which is at most $\sup_{|x|=R} |g(x) - g(0)| \to 0$ in absolute value. □

Exercise 28.23.24. (*Dirichlet's approximation theorem.*) Let $x \in \mathbb{R}$ and $N \geq 1$. Using the pigeonhole principle for the $N+1$ numbers $kx - \lfloor kx \rfloor \in [0,1)$, $k = 0, \ldots, N$, show that there are integers p, q with $1 \leq q \leq N$ and $|qx - p| < 1/N$. Deduce that an irrational x has infinitely many approximations p/q with $|x - p/q| < 1/q^2$.

Solution. Divide $[0,1)$ into the N intervals $[\frac{i}{N}, \frac{i+1}{N})$. Two of the $N+1$ numbers, for $k < l$ say, lie in the same interval, so $|(l-k)x - (\lfloor lx \rfloor - \lfloor kx \rfloor)| < 1/N$; take $q = l - k$ and $p = \lfloor lx \rfloor - \lfloor kx \rfloor$. Then $|x - p/q| < 1/(qN) \leq 1/q^2$, and also $|x - p/q| < 1/N$. Writing p/q in lowest terms p'/q' with $q' \leq q$ keeps $|x - p'/q'| < 1/q^2 \leq 1/q'^2$. If x is irrational, no fraction equals x, so a finite set of fractions has positive minimal distance from x; since letting $N \to \infty$ produces fractions arbitrarily close to x, we obtain infinitely many distinct ones. □

Exercise 28.23.25. For $x \in [0,1)$ let $e_k(x) = \lfloor 10^k x \rfloor - 10\lfloor 10^{k-1} x \rfloor$ be its decimal digits. Show that for almost every x:

1. every finite block of decimal digits occurs infinitely often among the $e_k(x)$;

2. the frequency $\frac{1}{n}\#\{k \le n : e_k(x) = 7\}$ tends to $\frac{1}{10}$.

Solution. The analogue of Lemma 28.7.4 holds with the same proof: prescribing the digits at the positions of a finite set S defines a set of measure $10^{-|S|}$, a union of decimal intervals of length 10^{-K}, $K = \max S$, and events determined by disjoint sets of positions are independent, with the product formula for integrals. (1) is then proved as in Example 28.7.7, with probability 10^{-m} for a block of length m. For (2) let $\rho_k = \mathbf{1}_{\{e_k=7\}} - \frac{1}{10}$, so $\int \rho_k = 0$ and $|\rho_k| \le 1$, and $S_n = \rho_1 + \cdots + \rho_n$. In $\int S_n^4 = \sum \int \rho_i\rho_j\rho_k\rho_l$, a term in which some index occurs exactly once vanishes by the product formula. The remaining terms have all indices equal (n terms) or two values each occurring twice ($3n(n-1)$ terms), and each is at most 1 in absolute value; so $\int S_n^4 \le 3n^2$, and Step 2 of the proof of Theorem 28.7.9 applies verbatim. □

Exercise 28.23.26. Let $L(F) = F_1(0) + 2F_2(0)$ for $F = (F_1, F_2) \in C_c(\mathbb{R}, \mathbb{R}^2)$. Find the measure μ and the field ν of Theorem 28.14.9, and check (28.1) directly.

Solution. $L(F) = \langle F(0), (1,2)\rangle$, so $L(F) = \int \langle F, \nu\rangle \, d\mu$ with $\mu = \sqrt{5}\,\delta_0$ and $\nu(0) = (1,2)/\sqrt{5}$, a unit vector. For an open $V \ni 0$ and $|F| \le 1$ with support in V, the Cauchy–Schwarz inequality gives $L(F) \le |(1,2)| = \sqrt{5}$, with equality when $F(0) = (1,2)/\sqrt{5}$, which is possible for $F = \psi\,(1,2)/\sqrt{5}$ with $\psi \prec V$, $\psi(0) = 1$; for $0 \notin V$ the supremum is 0. So the total variation is $\sqrt{5}\,\delta_0$, not $\delta_0 + 2\delta_0 = 3\delta_0$: the components cannot be estimated separately, as $|F| \le 1$ couples them. □

Chapter 29

Complex Analysis

29.1 Introduction

A function of one complex variable that is differentiable once is analytic, and is determined by its values on any set with a limit point. This rigidity, which has no analogue in real analysis, is the source of the special features of complex geometry: on a compact complex manifold there are few holomorphic functions, and the interplay between holomorphic and topological invariants leads to Hodge theory. This chapter proves Cauchy's theorem in homotopy form (Theorem 29.3.3), the Cauchy integral formula and its consequences, the residue theorem, and the Riemann mapping theorem (Theorem 29.6.5). The last section studies the Riemann sphere $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$: stereographic projection, the rational functions as the meromorphic functions on $\hat{\mathbb{C}}$ (Theorem 29.7.5), and the Möbius transformations, with the cross-ratio and their classification into parabolic, elliptic, hyperbolic and loxodromic ones (Theorem 29.7.14). References: [Ahl78], [SS03], [Con78] and [Rud87, Chapters 10–16].

Prerequisites. Chapter 27 (Real Analysis) and Chapter 19 (Homotopy and Covering Spaces).

We write $D(a, r) = \{z : |z - a| < r\}$, $\overline{D}(a, r)$ for its closure, and $\mathbb{D} = D(0, 1)$. All domains (= nonempty connected open subsets of $\mathbb{C}$) are denoted U, V.

29.2 Holomorphic functions

Definition 29.2.1. Let $U \subseteq \mathbb{C}$ be open. A function $f : U \to \mathbb{C}$ is *holomorphic* if the limit $f'(a) = \lim_{h \to 0} (f(a + h) - f(a))/h$, $h \in \mathbb{C}^\times$, exists for every $a \in U$. An *entire* function is one holomorphic on $\mathbb{C}$.

The definition looks exactly like the one for real functions, but h now tends to 0 through all complex numbers, from every direction at once, and the difference quotient must have the same limit along all of them. This single requirement turns out to be extraordinarily strong: it implies that f has derivatives of all orders and is given by its Taylor series (Theorem 29.4.2 and Corollary 29.4.3). Before seeing why, let us compute a few derivatives directly from the definition.

Example 29.2.2. Let $a \in \mathbb{C}$ and $h \in \mathbb{C}^\times$ small.

1. $f(z) = z^n$, $n \geq 1$. By the binomial theorem $(a + h)^n - a^n = na^{n-1}h + \sum_{k=2}^{n} \binom{n}{k} a^{n-k} h^k$, so the difference quotient is $na^{n-1} + \sum_{k=2}^{n} \binom{n}{k} a^{n-k} h^{k-1} \to na^{n-1}$. So $(z^n)' = nz^{n-1}$, as over $\mathbb{R}$.

2. $f(z) = 1/z$ on $\mathbb{C}^\times$. For $a \neq 0$ and $|h| < |a|$, $\frac{1}{h}\left(\frac{1}{a+h} - \frac{1}{a}\right) = \frac{-1}{a(a+h)} \to -\frac{1}{a^2}$.

3. $f(z) = \bar{z}$. The difference quotient is $\bar{h}/h$, which equals 1 for real h and -1 for purely imaginary h. So the limit exists at no point: $\bar{z}$ is a perfectly smooth map of the plane, an isometry even, but it is nowhere complex differentiable. The same happens for $\operatorname{Re} z$, whose quotient $\operatorname{Re} h/h$ is 1 on the real axis and 0 on the imaginary axis.

4. $f(z) = |z|^2 = z\bar{z}$. Here $|a+h|^2 - |a|^2 = \bar{a}h + a\bar{h} + |h|^2$, so the quotient is $\bar{a} + a\,\bar{h}/h + \bar{h}$. For $a = 0$ it is $\bar{h} \to 0$; for $a \neq 0$ the middle term $a\,\bar{h}/h$ has no limit, by (3). So $|z|^2$ is complex differentiable at 0 and nowhere else. A function can be complex differentiable at a single point; "holomorphic" asks for it at every point of an open set, and all the theory below depends on this.

Proposition 29.2.3 (Cauchy–Riemann)**.** *Identify $\mathbb{C}$ with $\mathbb{R}^2$ and write $f = u + iv$. Then f is holomorphic at a if and only if f is differentiable at a as a map of real variables (Definition 27.10.2) and its derivative is complex linear, which holds if and only if*

$$\partial_x u = \partial_y v, \qquad \partial_y u = -\partial_x v \quad \text{at } a.$$

Then $f'(a) = \partial_x u + i\partial_x v$. Sums, products, quotients and composites of holomorphic functions are holomorphic, with the usual rules.

Proof. Complex differentiability at a says $f(a+h) = f(a) + ch + o(|h|)$ with $c = f'(a) \in \mathbb{C}$, which is real differentiability with $Df(a)$ the real linear map $h \mapsto ch$, that is, with matrix $\left(\begin{smallmatrix} \alpha & -\beta \\ \beta & \alpha \end{smallmatrix}\right)$ for $c = \alpha + i\beta$. Comparing with the Jacobian matrix $\left(\begin{smallmatrix} \partial_x u & \partial_y u \\ \partial_x v & \partial_y v \end{smallmatrix}\right)$ gives the equations. The rules follow as in Proposition 27.6.2, whose proofs are valid over $\mathbb{C}$. □

Example 29.2.4. Polynomials are entire; $1/z$ is holomorphic on $\mathbb{C}^\times$; power series are holomorphic in their disc of convergence, with the termwise derivative (Theorem 27.6.20); exp is entire with $\exp' = \exp$. The function $\bar{z}$ is nowhere holomorphic, since $\partial_x u = 1 \neq -1 = \partial_y v$.

The Cauchy–Riemann equations have a geometric meaning. A real linear map $\mathbb{R}^2 \to \mathbb{R}^2$ is complex linear exactly when its matrix is $\left(\begin{smallmatrix} \alpha & -\beta \\ \beta & \alpha \end{smallmatrix}\right)$, that is, when it is multiplication by $c = \alpha + i\beta = |c|e^{i\varphi}$: a rotation by φ followed by a dilation by $|c|$. So near a point where $f'(a) \neq 0$, a holomorphic map looks, to first order, like a rotation and a dilation, which preserve angles and orientation. Maps with this property are called *conformal*, and the converse also holds.

Proposition 29.2.5. *1. Let f be holomorphic near a with $f'(a) \neq 0$, and let γ_1, γ_2 be C^1 curves with $\gamma_1(0) = \gamma_2(0) = a$ and nonzero tangent vectors $\gamma_1'(0), \gamma_2'(0)$. Then $(f \circ \gamma_j)'(0) = f'(a)\gamma_j'(0)$. Hence the image curves have tangent vectors rotated by the same angle $\arg f'(a)$ and stretched by the same factor $|f'(a)|$, and the oriented angle between them equals that between γ_1 and γ_2.*

2. Conversely, let f be real differentiable at a with $A = Df(a)$ invertible, and suppose that A preserves oriented angles, that is, Aw/Av and w/v have the same argument for all nonzero $v, w \in \mathbb{C}$. Then f is complex differentiable at a.

Proof. (1) By the chain rule (Proposition 27.6.2) and Proposition 29.2.3, $(f \circ \gamma_j)'(0) = Df(a)\gamma_j'(0) = f'(a)\gamma_j'(0)$. The quotient of the two image tangent vectors is $\gamma_2'(0)/\gamma_1'(0)$, the same as for the original ones, and the oriented angle from one vector to another is the argument of this quotient.

(2) Let $p = A(1) \neq 0$ and $q = A(i)$. The oriented angle from 1 to i is $\frac{\pi}{2}$, so q/p has argument $\frac{\pi}{2}$: $q = itp$ for some $t > 0$. The oriented angle from 1 to $1+i$ is $\frac{\pi}{4}$, so $A(1+i)/A(1) =$

$(p+q)/p = 1+it$ has argument $\frac{\pi}{4}$, which forces $t = 1$. Hence $A(x+iy) = xp + yip = (x+iy)p$: A is multiplication by p, so it is complex linear, and f is complex differentiable at a with $f'(a) = p$ by Proposition 29.2.3. □

Example 29.2.6. 1. *Critical points.* For $f(z) = z^2$ and $a = 0$, where $f'(0) = 0$, angles are not preserved but doubled: the ray $\{re^{i\theta} : r > 0\}$ is mapped to the ray $\{r^2 e^{2i\theta}\}$. In general, near a zero of order m of f' the map multiplies angles by $m+1$; this follows from the local form of holomorphic maps (Theorem 29.5.12).

2. *The exponential.* $e^{x+iy} = e^x e^{iy}$ maps the vertical line $x = x_0$ onto the circle of radius e^{x_0}, and the horizontal line $y = y_0$ onto the ray of argument y_0. The lines of the grid meet at right angles, and so do their images, circles and rays from the origin, as Proposition 29.2.5 predicts, since $\exp' = \exp$ never vanishes.

3. *Reflection.* $z \mapsto \bar{z}$ preserves the size of angles but reverses their orientation, and it is not holomorphic (Example 29.2.2(3)). The argument in (2) adapts: an invertible real linear map that preserves unoriented angles is either complex linear or of the form $z \mapsto p\bar{z}$, so the maps that preserve unoriented angles, at points where the derivative is invertible, are the holomorphic and the *antiholomorphic* ones (those f with $\bar{f}$ holomorphic).

Definition 29.2.7. A *path* is a piecewise continuously differentiable map $\gamma\colon [a,b] \to \mathbb{C}$; it is *closed* if $\gamma(a) = \gamma(b)$. Its image is γ^* and its length is $L(\gamma) = \int_a^b |\gamma'(t)|\,dt$. For continuous f on γ^*,

$$\int_\gamma f(z)\,dz = \int_a^b f(\gamma(t))\gamma'(t)\,dt.$$

Lemma 29.2.8. *The integral is unchanged by increasing C^1 reparametrisations, changes sign under reversal, and $|\int_\gamma f| \le L(\gamma)\sup_{\gamma^*}|f|$. If $f = F'$ for a holomorphic F on an open set containing γ^*, then $\int_\gamma f = F(\gamma(b)) - F(\gamma(a))$; in particular $\int_\gamma f = 0$ for closed γ. For the circle $\gamma(t) = a + re^{2\pi it}$, $t \in [0,1]$, one has $\int_\gamma (z-a)^n\,dz = 0$ for $n \neq -1$ and $= 2\pi i$ for $n = -1$.*

Proof. Reparametrisation and reversal follow from the substitution rule (Corollary 27.8.4), the estimate from Proposition 27.8.2. If $f = F'$, then $(F\circ\gamma)' = f(\gamma)\gamma'$ on each C^1 piece, so the integral telescopes by the fundamental theorem of calculus. For the circle and $n \neq -1$, $(z-a)^n$ has the primitive $(z-a)^{n+1}/(n+1)$; for $n = -1$ the integral is $\int_0^1 2\pi i\,dt$. □

Example 29.2.9. 1. *A function without primitive.* Let $\gamma(t) = e^{2\pi it}$, $t \in [0,1]$, be the unit circle. On it $\bar{z} = e^{-2\pi it}$ and $\gamma'(t) = 2\pi i e^{2\pi it}$, so $\int_\gamma \bar{z}\,dz = \int_0^1 2\pi i\,dt = 2\pi i \neq 0$. By Lemma 29.2.8, $\bar{z}$ is not the derivative of a holomorphic function on any open set containing the circle.

2. *Dependence on the path.* Integrate $\bar{z}$ from 0 to $1+i$ in two ways. Along the segment $\gamma(t) = t(1+i)$: $\int_0^1 t(1-i)(1+i)\,dt = \int_0^1 2t\,dt = 1$. Along the segment from 0 to 1 followed by the segment from 1 to $1+i$, parametrised by t and $1+it$: $\int_0^1 t\,dt + \int_0^1 (1-it)\,i\,dt = \frac{1}{2} + \left(i + \frac{1}{2}\right) = 1+i$. The results differ. For a function with a primitive, such as $z^2 = (z^3/3)'$, every path from 0 to $1+i$ gives $(1+i)^3/3$.

3. *A holomorphic function without primitive.* $1/z$ is holomorphic on $\mathbb{C}^\times$, yet $\int_\gamma dz/z = 2\pi i$ over the unit circle (Lemma 29.2.8); so $1/z$ has no primitive on $\mathbb{C}^\times$, that is, there is no holomorphic logarithm on $\mathbb{C}^\times$. What goes wrong is the hole at 0: Cauchy's theorem below says that integrals of holomorphic functions over closed paths vanish when the path can be shrunk to a point inside the domain.

29.3 Cauchy's theorem

Example 29.2.9 shows that integrals over closed paths need not vanish, either because the function is not holomorphic ($\bar{z}$) or because the domain has a hole ($1/z$). Cauchy's theorem says that these are the only obstructions. For a function $f = u + iv$ of class C^1 on a neighbourhood of a triangle T there is a quick argument, using Green's formula $\int_{\partial T}(P\,dx + Q\,dy) = \iint_T (\partial_x Q - \partial_y P)\,dx\,dy$ (proved in Chapter 36 in the language of forms) with $P = f$ and $Q = i f$, since $dz = dx + i\,dy$:

$$\int_{\partial T} f\,dz = \iint_T \left(i\,\partial_x f - \partial_y f\right) dx\,dy = 0,$$

because the Cauchy–Riemann equations say exactly that $\partial_y f = i\,\partial_x f$ (Proposition 29.2.3: $\partial_y f = \partial_y u + i\partial_y v = -\partial_x v + i\partial_x u$). Goursat's theorem proves the same conclusion assuming only that f' exists, without its continuity, by subdividing the triangle and using the linear approximation of f at one point. This is what later allows us to prove, rather than assume, that holomorphic functions are smooth.

Theorem 29.3.1 (Goursat)**.** *Let U be open, $p \in U$, and $f : U \to \mathbb{C}$ continuous on U and holomorphic on $U \setminus \{p\}$. Then $\int_{\partial T} f = 0$ for every closed triangle $T \subseteq U$, where ∂T is its boundary traversed once.*

Proof. First assume f is holomorphic on all of U. Let $I = \int_{\partial T} f$. Joining the midpoints of the sides divides T into four triangles whose boundary integrals add up to I (the inner edges cancel), so one of them, T_1, has $|\int_{\partial T_1} f| \geq |I|/4$. Iterating gives nested triangles T_n with $\operatorname{diam} T_n = 2^{-n} \operatorname{diam} T$, $L(\partial T_n) = 2^{-n} L(\partial T)$ and $|\int_{\partial T_n} f| \geq 4^{-n}|I|$. Their intersection is a point z_0. Given $\varepsilon > 0$, holomorphy at z_0 gives δ with $|f(z) - f(z_0) - f'(z_0)(z - z_0)| \leq \varepsilon |z - z_0|$ for $|z - z_0| < \delta$. The affine function $f(z_0) + f'(z_0)(z - z_0)$ has a primitive, so its integral over ∂T_n vanishes, and for large n, $|\int_{\partial T_n} f| \leq \varepsilon \operatorname{diam}(T_n) L(\partial T_n) = \varepsilon 4^{-n} \operatorname{diam}(T) L(\partial T)$. Hence $|I| \leq \varepsilon \operatorname{diam}(T) L(\partial T)$ for all ε.

If f is only continuous at p: if $p \notin T$ we are done. If p is a vertex of T, write T as the union of a small triangle T' with vertex p and two triangles not containing p; then $|\int_{\partial T} f| = |\int_{\partial T'} f| \leq L(\partial T') \sup_T |f| \to 0$ as T' shrinks. For general $p \in T$, join p to the vertices. □

Corollary 29.3.2. *Let U be convex, $p \in U$, and f continuous on U and holomorphic on $U \setminus \{p\}$. Then f has a primitive on U, and $\int_\gamma f = 0$ for every closed path γ in U.*

Proof. Fix $z_0 \in U$ and set $F(z) = \int_{[z_0, z]} f$, the integral over the segment. For $z, z + h \in U$, the triangle with vertices $z_0, z, z + h$ lies in U, so $F(z + h) - F(z) = \int_{[z, z+h]} f$ by Theorem 29.3.1. Hence $|F(z + h) - F(z) - f(z)h| \leq |h| \sup_{[z,z+h]} |f - f(z)| = o(|h|)$ by continuity, so $F' = f$. Apply Lemma 29.2.8. □

Theorem 29.3.3 (Cauchy's theorem, homotopy form)**.** *Let f be holomorphic on an open $U \subseteq \mathbb{C}$ and let γ_0, γ_1 be paths in U that are homotopic in U, either as closed paths or with fixed endpoints. Then $\int_{\gamma_0} f = \int_{\gamma_1} f$. In particular $\int_\gamma f = 0$ for every closed path that is null-homotopic in U, and every holomorphic function on a simply connected open set has a primitive.*

Proof. Let $H : [0,1]^2 \to U$ be the homotopy, $H(s, 0) = \gamma_0(s)$, $H(s, 1) = \gamma_1(s)$. Its image is compact, so there is $\rho > 0$ with $D(H(s,t), \rho) \subseteq U$ for all (s, t), and by uniform continuity there is N with $|H(s,t) - H(s',t')| < \rho$ whenever $|s - s'|, |t - t'| \leq 1/N$. Put $z_{jk} = H(j/N, k/N)$ and let $D_{jk} = D(z_{jk}, \rho)$, so that H maps the square $[j/N, (j+1)/N] \times [k/N, (k+1)/N]$ into D_{jk}. Let P_k be the closed polygonal path through $z_{0k}, z_{1k}, \ldots, z_{Nk}$.

Each segment $[z_{jk}, z_{j+1,k}]$ and the corresponding piece of $H(-, k/N)$ lie in the convex set D_{jk}, where f has a primitive by Corollary 29.3.2; so the integrals of f over them agree, and $\int_{\gamma_0} f = \int_{P_0} f$, $\int_{\gamma_1} f = \int_{P_N} f$ (for γ_0, γ_1 piecewise C^1; the intermediate paths are only continuous, and the polygonal paths replace them). The closed polygon with vertices $z_{jk}, z_{j+1,k}, z_{j+1,k+1}, z_{j,k+1}$ lies in the convex set D_{jk}, so the integral of f over it vanishes. Summing over j, the contributions of the segments $[z_{jk}, z_{j,k+1}]$ cancel in pairs (for closed homotopies because $z_{0k} = z_{Nk}$; for fixed endpoints because those segments are constant), so $\int_{P_k} f = \int_{P_{k+1}} f$. Hence $\int_{P_0} f = \int_{P_N} f$. The last statement follows by fixing $z_0 \in U$ and setting $F(z) = \int_\gamma f$ for any path from z_0 to z, which is independent of the path since U is simply connected; then $F' = f$ as in Corollary 29.3.2. □

Definition 29.3.4. For a closed path γ and $a \notin \gamma^*$, the *winding number* is $n(\gamma, a) = \frac{1}{2\pi i} \int_\gamma \frac{dz}{z-a}$.

Proposition 29.3.5. *$n(\gamma, a)$ is an integer; it is constant on connected components of $\mathbb{C} \setminus \gamma^*$ and zero on the unbounded component. Homotopic closed paths in $\mathbb{C} \setminus \{a\}$ have the same winding number about a, and for the circle $\gamma(t) = a + re^{2\pi i t}$ one has $n(\gamma, a) = 1$.*

Proof. Let $\gamma\colon [0,1] \to \mathbb{C}$ and $\varphi(t) = \int_0^t \frac{\gamma'(s)}{\gamma(s)-a}\, ds$. Then $(e^{-\varphi}(\gamma - a))' = 0$ on each C^1 piece, so $e^{\varphi(1)} = (\gamma(1)-a)/(\gamma(0)-a) = 1$ and $\varphi(1) \in 2\pi i\mathbb{Z}$ by Theorem 27.7.3. The function $a \mapsto n(\gamma, a)$ is continuous on $\mathbb{C} \setminus \gamma^*$ and integer valued, hence locally constant, and it tends to 0 as $|a| \to \infty$. Homotopy invariance is Theorem 29.3.3 applied to $z \mapsto 1/(z-a)$ on $\mathbb{C} \setminus \{a\}$, and the circle was computed in Lemma 29.2.8. □

29.4 The Cauchy integral formula

Theorem 29.4.1. *Let U be convex, f holomorphic on U, γ a closed path in U and $a \in U \setminus \gamma^*$. Then*

$$n(\gamma, a) f(a) = \frac{1}{2\pi i} \int_\gamma \frac{f(z)}{z-a}\, dz.$$

In particular, if $\overline{D}(a, r) \subseteq U$ and γ is the circle $\partial D(a, r)$ traversed once, then $f(w) = \frac{1}{2\pi i} \int_\gamma \frac{f(z)}{z-w} dz$ for $w \in D(a, r)$.

Proof. Define $g(z) = (f(z) - f(a))/(z-a)$ for $z \neq a$ and $g(a) = f'(a)$. Then g is continuous on U and holomorphic on $U \setminus \{a\}$, so $\int_\gamma g = 0$ by Corollary 29.3.2. For the second statement, note that $n(\gamma, w) = n(\gamma, a) = 1$ for $w \in D(a, r)$ by Proposition 29.3.5, and apply the formula in the convex set $D(a, r')$ for $r < r'$ with $\overline{D}(a, r') \subseteq U$. □

Theorem 29.4.2. *Let f be holomorphic on an open U and $D(a, R) \subseteq U$. Then $f(w) = \sum_{n \geq 0} c_n (w - a)^n$ on $D(a, R)$, with $c_n = \frac{1}{2\pi i} \int_{\partial D(a,r)} \frac{f(z)}{(z-a)^{n+1}} dz = \frac{f^{(n)}(a)}{n!}$ for any $0 < r < R$. In particular holomorphic functions have holomorphic derivatives of all orders, and $|f^{(n)}(a)| \leq n!\, r^{-n} \sup_{|z-a|=r} |f|$ (*Cauchy estimates*).*

Proof. Fix $|w - a| < r < R$. For $|z - a| = r$, $\frac{1}{z-w} = \sum_{n \geq 0} \frac{(w-a)^n}{(z-a)^{n+1}}$, converging uniformly in z since $|w - a|/r < 1$. Multiplying by $f(z)/2\pi i$ and integrating term by term (uniform convergence, Proposition 27.8.2) gives the expansion with the stated c_n, by Theorem 29.4.1. A power series is infinitely complex differentiable with $c_n = f^{(n)}(a)/n!$ (Theorem 27.6.20), and the estimate follows from Lemma 29.2.8. □

The last statement of Theorem 29.4.2 deserves a closer look, because nothing like it is true for real functions. Definition 29.2.1 only asks that $f'(a)$ exists at every point; it does not ask that f' be continuous. For a function of one real variable this would be a much weaker requirement than being of class C^1: $x^2 \sin(1/x)$, extended by 0 at 0, is differentiable everywhere with a discontinuous derivative, and its antiderivatives are differentiable twice but not three times. For complex differentiability the hierarchy collapses: one derivative gives all of them, and they are all continuous.

Corollary 29.4.3. *Let $f = u + iv$ be holomorphic on an open $U \subseteq \mathbb{C}$, and identify $\mathbb{C}$ with $\mathbb{R}^2$, $z = x + iy$.*

1. *The derivatives $f', f'', \dots, f^{(k)}, \dots$ exist and are holomorphic, hence continuous, on U.*
2. *As a function of the two real variables x, y, f is of class C^∞, and for all $a, b \geq 0$*

$$\partial_x^a \partial_y^b f = i^b f^{(a+b)}.$$

   In particular u and v are of class C^∞.

3. *u and v are* harmonic*: $\partial_x^2 u + \partial_y^2 u = 0$ and $\partial_x^2 v + \partial_y^2 v = 0$ on U.*

Proof. (1) By Theorem 29.4.2, applied on discs $D(a, R) \subseteq U$, the derivative of a holomorphic function is holomorphic. By induction on k every $f^{(k)}$ exists and is holomorphic. A function that is complex differentiable at a point is continuous there, since $f(a+h) - f(a) = h \cdot (f(a+h) - f(a))/h \to 0 \cdot f'(a) = 0$.

(2) Let g be holomorphic on U. By Proposition 29.2.3, g is real differentiable and $Dg(a)$ is multiplication by $g'(a)$. The partial derivatives are the values of $Dg(a)$ on the basis vectors 1 and i of $\mathbb{R}^2 = \mathbb{C}$:

$$\partial_x g(a) = g'(a) \cdot 1 = g'(a), \qquad \partial_y g(a) = g'(a) \cdot i = i g'(a).$$

So ∂_x and ∂_y map holomorphic functions to holomorphic functions, acting as $g \mapsto g'$ and $g \mapsto ig'$. Applying this $a + b$ times, starting from $g = f$ and using (1), shows that every partial derivative $\partial_x^a \partial_y^b f$ (in any order of differentiation) exists and equals $i^b f^{(a+b)}$, which is continuous by (1). Hence f is C^∞ by definition, and so are its real and imaginary parts $u = \operatorname{Re} f$ and $v = \operatorname{Im} f$, whose partial derivatives are the real and imaginary parts of those of f.

(3) By (2), $\partial_x^2 f + \partial_y^2 f = f'' + i^2 f'' = 0$. The real and imaginary parts of this equation are $\Delta u = 0$ and $\Delta v = 0$, where $\Delta = \partial_x^2 + \partial_y^2$, since ∂_x and ∂_y commute with taking real and imaginary parts. □

Remark 29.4.4. *Where does the regularity come from?* The proof of Corollary 29.4.3 passes through four steps, and it is worth seeing what each one uses.

1. *Goursat's theorem* (Theorem 29.3.1) uses complex differentiability only at a single point z_0 for each triangle, through the linear approximation $f(z) = f(z_0) + f'(z_0)(z - z_0) + o(|z - z_0|)$. No continuity of f' is needed. This is the crucial step: the classical proof of Cauchy's theorem by Green's formula, $\int_{\partial T} f\, dz = \iint_T (\partial_x f + i\partial_y f)\, i\, dx\, dy$, needs f to be of class C^1, and would give the result only for C^1 functions.

2. From vanishing integrals over triangles one gets a primitive on discs (Corollary 29.3.2), and from that the Cauchy integral formula (Theorem 29.4.1), applied to the auxiliary function $g(z) = (f(z) - f(a))/(z - a)$, which is continuous at a and holomorphic elsewhere. This is why Goursat's theorem allows one exceptional point.

3. The Cauchy formula $f(w) = \frac{1}{2\pi i}\int_{|z-a|=r} \frac{f(z)}{z-w}\,dz$ expresses f inside the disc as an average of its values on the circle against the kernel $1/(z-w)$, which is infinitely differentiable in w. All the regularity of the kernel passes to f: one may differentiate under the integral sign, or, as in Theorem 29.4.2, expand the kernel in a geometric series and obtain a power series.

4. Power series are infinitely differentiable (Theorem 27.6.20).

So the equation $\partial f/\partial \bar z = 0$ (the Cauchy–Riemann equations, see below) forces smoothness. This is the first instance of *elliptic regularity*: solutions of elliptic differential equations are as smooth as the equation allows, whatever regularity was assumed a priori; Chapter 42 proves this in general, and it is what makes harmonic forms smooth in the Hodge theorem.

It is often convenient to package the two real partial derivatives into the *Wirtinger derivatives*

$$\frac{\partial}{\partial z} = \frac{1}{2}\Big(\partial_x - i\partial_y\Big), \qquad \frac{\partial}{\partial \bar z} = \frac{1}{2}\Big(\partial_x + i\partial_y\Big),$$

defined for real differentiable functions. For $f = u + iv$ the real and imaginary parts of $2\,\partial f/\partial \bar z = (\partial_x u - \partial_y v) + i(\partial_x v + \partial_y u)$ are the two Cauchy–Riemann expressions, so a real differentiable f is holomorphic exactly when $\partial f/\partial \bar z = 0$, and then $\partial f/\partial z = f'$. On C^2 functions

$$4\frac{\partial}{\partial z}\frac{\partial}{\partial \bar z} = (\partial_x - i\partial_y)(\partial_x + i\partial_y) = \partial_x^2 + \partial_y^2 = \Delta,$$

since ∂_x and ∂_y commute on C^2 functions (Theorem 27.10.5). This gives a second proof of part (3): $\Delta f = 4\frac{\partial}{\partial z}\big(\frac{\partial f}{\partial \bar z}\big) = 0$. The Wirtinger derivatives satisfy the product rule, and $\partial \bar g/\partial \bar z = \overline{\partial g/\partial z}$, as one checks from the definitions.

Example 29.4.5. 1. From $z^2 = (x^2-y^2) + i\,2xy$, the functions x^2-y^2 and $2xy$ are harmonic; from $e^z = e^x\cos y + ie^x\sin y$, so are $e^x\cos y$ and $e^x\sin y$. The level curves of u and v meet at right angles wherever $f' \neq 0$: by the Cauchy–Riemann equations, $\nabla u = (\partial_x u, \partial_y u)$ and $\nabla v = (-\partial_y u, \partial_x u)$ are orthogonal vectors of the same length $|f'|$.

2. $\log|f|$ *is harmonic where* $f \neq 0$. On a disc $D \subseteq U$ on which f has no zeros, f'/f is holomorphic, so it has a primitive g (Corollary 29.3.2); then $(fe^{-g})' = (f' - fg')e^{-g} = 0$, so $f = ce^g$ with a constant $c = e^{c_0} \neq 0$, and $\log|f| = \operatorname{Re}(g + c_0)$ is harmonic. For example $\log|z| = \frac{1}{2}\log(x^2+y^2)$ is harmonic on $\mathbb{C}^\times$, although it is not the real part of a holomorphic function g on all of $\mathbb{C}^\times$: by the Cauchy–Riemann equations $g' = \partial_x \operatorname{Re} g - i\partial_y \operatorname{Re} g$, which for $\operatorname{Re} g = \log|z|$ is $(x - iy)/(x^2+y^2) = 1/z$, and $1/z$ has no primitive on $\mathbb{C}^\times$, its integral over the unit circle being $2\pi i$ (Lemma 29.2.8).

3. $|f|^2$ *is not harmonic, but subharmonic.* With the product rule, $\frac{\partial}{\partial \bar z}(f\bar f) = f\,\overline{f'}$, because $\partial f/\partial \bar z = 0$ and $\partial \bar f/\partial \bar z = \overline{f'}$; differentiating again, $\frac{\partial}{\partial z}\big(f\,\overline{f'}\big) = f'\,\overline{f'}$, since $\overline{f'}$ has $\partial/\partial z$ equal to $\overline{\partial f'/\partial \bar z} = 0$. Hence $\Delta|f|^2 = 4|f'|^2 \geq 0$. For $f(z) = z$ this is $\Delta(x^2+y^2) = 4$.

Counterexample 29.4.6. The Cauchy–Riemann equations at a point do not imply complex differentiability there; Proposition 29.2.3 also requires real differentiability. Let $f(x+iy) = \sqrt{|xy|}$, so $u = \sqrt{|xy|}$ and $v = 0$. Since u vanishes on both axes, $\partial_x u(0) = \partial_y u(0) = 0$, and the Cauchy–Riemann equations hold at 0. But for $h = t(1+i)$ with $t \in \mathbb{R}^\times$, $(f(h) - f(0))/h = |t|/(t(1+i))$, which is $\frac{1}{1+i}$ for $t > 0$ and $-\frac{1}{1+i}$ for $t < 0$; so $f'(0)$ does not exist. Existence of the partial derivatives only concerns the two directions of the axes, while complex differentiability concerns all directions at once.

Corollary 29.4.7. *1. (Liouville) A bounded entire function is constant; more generally, if $|f(z)| \le C(1+|z|)^m$ for an entire f, then f is a polynomial of degree at most m.*

2. (Fundamental theorem of algebra) Every nonconstant polynomial over $\mathbb{C}$ has a root (cf. Chapter 9).

3. (Morera) A continuous function on an open set whose integral over the boundary of every closed triangle contained in the set vanishes is holomorphic.

4. (Weierstrass) If holomorphic functions f_n on U converge uniformly on compact subsets to f, then f is holomorphic and $f_n^{(k)} \to f^{(k)}$ uniformly on compact subsets.

Proof. (1) By the Cauchy estimates with $r \to \infty$, $f^{(n)}(a) = 0$ for $n > m$ and all a. (2) If p has no root, $1/p$ is entire and tends to 0 at infinity, hence is bounded and constant. (3) Locally on a disc the hypothesis gives a primitive F as in Corollary 29.3.2, and F is holomorphic, hence so is $F' = f$ by Theorem 29.4.2. (4) The uniform limit is continuous and satisfies Morera's condition by Theorem 29.3.1 and uniform convergence; convergence of derivatives follows from the Cauchy integral formula for $f^{(k)}$ on circles inside U. □

Theorem 29.4.8 (Identity theorem)**.** *Let f be holomorphic on a domain U. If the zero set of f has a limit point in U, then $f \equiv 0$. Consequently two holomorphic functions agreeing on a set with a limit point in U are equal, and the zeros of a nonzero f are isolated, each of finite order m with $f(z) = (z-a)^m g(z)$, g holomorphic and $g(a) \neq 0$.*

Proof. Let A be the set of $a \in U$ with $f^{(n)}(a) = 0$ for all n; it is closed in U, and open because f vanishes identically near such a point by Theorem 29.4.2. If $z_k \to a$ with $z_k \neq a$ and $f(z_k) = 0$, and if some derivative at a were nonzero, then with m minimal, $f(z) = (z-a)^m g(z)$ with $g(a) \neq 0$ by the power series expansion, so f would have no zeros near a other than a. Hence $a \in A$, and $A = U$ by connectedness. □

Theorem 29.4.9 (Maximum modulus principle)**.** *Let f be holomorphic on a domain U. If $|f|$ has a local maximum in U, then f is constant. If moreover U is bounded and f extends continuously to $\overline{U}$, then $|f|$ attains its maximum on ∂U.*

Proof. Theorem 29.4.1 on circles gives the mean value property $f(a) = \int_0^1 f(a + re^{2\pi it})\,dt$. If $|f| \le |f(a)|$ on $D(a,R)$, then for $r < R$, $|f(a)| \le \int_0^1 |f(a+re^{2\pi it})|\,dt \le |f(a)|$, so $|f| = |f(a)|$ on every such circle by continuity, and $|f|$ is constant on $D(a,R)$. Then $f\bar{f}$ is constant; if $f(a) \neq 0$ this makes $\bar{f} = c/f$ holomorphic, so both f and $\bar{f}$ satisfy the Cauchy–Riemann equations and $f' = 0$ on the disc. In either case f is constant on $D(a,R)$, hence on U by Theorem 29.4.8. The last statement follows from compactness of $\overline{U}$. □

29.5 Singularities, Laurent series and residues

Theorem 29.5.1 (Laurent expansion)**.** *Let f be holomorphic on the annulus $A = \{r < |z-a| < R\}$, $0 \le r < R \le \infty$. There are unique $c_n \in \mathbb{C}$, $n \in \mathbb{Z}$, with $f(z) = \sum_{n\in\mathbb{Z}} c_n (z-a)^n$, converging absolutely and uniformly on compact subsets of A, and $c_n = \frac{1}{2\pi i}\int_{\partial D(a,\rho)} \frac{f(z)}{(z-a)^{n+1}}\,dz$ for any $r < \rho < R$.*

Proof. Fix w with $r < r_1 < |w-a| < r_2 < R$. The function $g(z) = (f(z) - f(w))/(z-w)$, extended by $f'(w)$ at w, is continuous on A and holomorphic off w, hence holomorphic by Corollary 29.4.7(3) and Theorem 29.3.1. The circles $\partial D(a,r_1)$ and $\partial D(a,r_2)$ are homotopic

as closed paths in A (move radially), so $\int_{\partial D(a,r_2)} g = \int_{\partial D(a,r_1)} g$ by Theorem 29.3.3. Since $n(\partial D(a,r_2), w) = 1$ and $n(\partial D(a,r_1), w) = 0$, this gives

$$f(w) = \frac{1}{2\pi i}\int_{\partial D(a,r_2)} \frac{f(z)}{z-w}dz - \frac{1}{2\pi i}\int_{\partial D(a,r_1)} \frac{f(z)}{z-w}dz.$$

In the first integral expand $\frac{1}{z-w} = \sum_{n\geq 0} \frac{(w-a)^n}{(z-a)^{n+1}}$, in the second $\frac{-1}{z-w} = \sum_{n\geq 1} \frac{(z-a)^{n-1}}{(w-a)^n}$, both uniformly convergent on the respective circles, and integrate term by term. The resulting coefficients are independent of the radius, again by homotopy invariance. Uniqueness follows by integrating the series against $(z-a)^{-n-1}$ term by term. □

Definition 29.5.2. Let f be holomorphic on a punctured disc $D(a,R)\setminus\{a\}$, with Laurent coefficients c_n. The singularity at a is *removable* if $c_n = 0$ for all $n < 0$, a *pole of order* m if $c_{-m} \neq 0$ and $c_n = 0$ for $n < -m$, and *essential* otherwise. The *residue* is $\operatorname{Res}_a f = c_{-1}$. A function holomorphic on U except for poles at a discrete closed subset is *meromorphic*; the *order* $\operatorname{ord}_a f$ is m at a zero of order m and $-m$ at a pole of order m.

Proposition 29.5.3. *Let f be holomorphic on $D(a,R)\setminus\{a\}$.*

1. *(Riemann) If f is bounded near a, the singularity is removable.*
2. *The singularity is a pole if and only if $|f(z)| \to \infty$ as $z \to a$.*
3. *(Casorati–Weierstrass) If the singularity is essential, $f(D(a,\varepsilon)\setminus\{a\})$ is dense in $\mathbb{C}$ for every $\varepsilon > 0$.*

Proof. (1) If $|f| \leq M$ near a, then $|c_{-n}| \leq M\rho^n$ for small ρ by Theorem 29.5.1, so $c_{-n} = 0$ for $n \geq 1$. (2) If f has a pole of order m, then $(z-a)^m f(z)$ extends holomorphically with nonzero value at a, so $|f| \to \infty$. Conversely, if $|f| \to \infty$, then $1/f$ is holomorphic and nonvanishing near a with limit 0, so by (1) it extends with a zero of some order m at a, and $f = 1/(1/f)$ has a pole of order m. (3) If the image of some punctured disc misses $D(w,\delta)$, then $g = 1/(f-w)$ is bounded there, hence extends holomorphically by (1); then $f = w + 1/g$ has a removable singularity or a pole by (2). □

Example 29.5.4. The function $f(z) = e^{1/z}$ on $\mathbb{C}\setminus\{0\}$ has Laurent series $\sum_{n\geq 0} z^{-n}/n!$, so 0 is an essential singularity. More than Casorati–Weierstrass holds here: every $w \neq 0$ is taken infinitely often in every punctured disc around 0. Indeed, if $w = e^c$, the points $z_k = 1/(c + 2\pi i k)$, $k \in \mathbb{Z}$, satisfy $f(z_k) = w$ and tend to 0. The value 0 is never taken. On the real axis, $e^{1/x} \to \infty$ as $x \to 0^+$ and $\to 0$ as $x \to 0^-$, while along the imaginary axis $|e^{1/(iy)}| = 1$: the behaviour near an essential singularity depends on the direction of approach.

Counterexample 29.5.5. Real analysis does not see the radius of convergence. The function $g(x) = 1/(1+x^2)$ is real-analytic on all of $\mathbb{R}$, but its Taylor series $\sum_n (-1)^n x^{2n}$ at 0 has radius of convergence 1. The reason is complex: g extends to the meromorphic function $1/(1+z^2)$ with poles at $\pm i$, at distance 1 from 0. By Theorem 29.4.2 the series converges on $D(0,1)$, and it cannot converge on a larger disc: its sum would be holomorphic there, equal to $1/(1+z^2)$ on $D(0,1)$ by the identity theorem, hence bounded near i, whereas $|1/(1+z^2)| \to \infty$ as $z \to i$. Similarly $x \mapsto e^{-1/x^2}$ (extended by 0) is smooth on $\mathbb{R}$ with all derivatives zero at 0, so it is not real-analytic there; as a complex function, e^{-1/z^2} has an essential singularity at 0, and it is unbounded along the imaginary axis.

Lemma 29.5.6. *Let h be holomorphic on an open U and define $g\colon U\times U \to \mathbb{C}$ by $g(z,w) = (h(z)-h(w))/(z-w)$ for $z \neq w$ and $g(z,z) = h'(z)$. Then g is continuous, and holomorphic in each variable separately.*

Proof. Holomorphy in each variable off the diagonal is clear, and at $z = w$ the singularity is removable by Proposition 29.5.3(1), since g is bounded near it by the estimate below. For continuity at (a, a), take $\overline{D}(a, r) \subseteq U$; for $z, w \in D(a, r)$ the segment $[w, z]$ lies in the disc and $g(z, w) = \int_0^1 h'(w + t(z - w))\, dt$ (by the fundamental theorem of calculus applied to $t \mapsto h(w + t(z - w))$, also valid when $z = w$). Since h' is continuous, hence uniformly continuous on $\overline{D}(a, r/2)$, this expression is continuous in (z, w) there. □

Theorem 29.5.7 (Cauchy's theorem, homology form)**.** *Let h be holomorphic on an open $U \subseteq \mathbb{C}$ and let γ be a closed path in U with $n(\gamma, z) = 0$ for every $z \in \mathbb{C} \setminus U$. Then $\int_\gamma h = 0$, and $n(\gamma, w)h(w) = \frac{1}{2\pi i} \int_\gamma \frac{h(z)}{z-w}\, dz$ for every $w \in U \setminus \gamma^*$.*

Proof. (Dixon.) With g as in Lemma 29.5.6, let $\varphi(w) = \frac{1}{2\pi i} \int_\gamma g(z, w)\, dz$ for $w \in U$. It is continuous, since g is uniformly continuous on $\gamma^* \times K$ for compact $K \subseteq U$, and holomorphic: for a closed triangle $T \subseteq U$, Fubini's theorem for continuous functions (Proposition 27.8.9) and Theorem 29.3.1 give $\int_{\partial T} \varphi(w)\, dw = \frac{1}{2\pi i} \int_\gamma \left(\int_{\partial T} g(z, w)\, dw \right) dz = 0$, so Morera's theorem applies. Let $V = \{w \in \mathbb{C} \setminus \gamma^* : n(\gamma, w) = 0\}$, an open set, and $\psi(w) = \frac{1}{2\pi i} \int_\gamma \frac{h(z)}{z-w}\, dz$ on V, which is holomorphic (same argument). For $w \in U \cap V$, $\varphi(w) = \psi(w) - n(\gamma, w)h(w) = \psi(w)$. By hypothesis $U \cup V = \mathbb{C}$, so φ and ψ define an entire function F. The unbounded component of $\mathbb{C} \setminus \gamma^*$ lies in V, and $\psi(w) \to 0$ as $|w| \to \infty$; so F is bounded, hence constant equal to 0 by Liouville's theorem (Corollary 29.4.7). Thus $\varphi \equiv 0$, which is the integral formula. Applying it to $z \mapsto (z - w)h(z)$ at the point w gives $0 = \frac{1}{2\pi i} \int_\gamma h$. □

Theorem 29.5.8 (Residue theorem)**.** *Let U be simply connected and open, $S \subseteq U$ discrete and closed in U, and f holomorphic on $U \setminus S$. For every closed path γ in $U \setminus S$,*

$$\frac{1}{2\pi i} \int_\gamma f = \sum_{a \in S} n(\gamma, a) \operatorname{Res}_a f,$$

a finite sum.

Proof. Let $W = \{z \in \mathbb{C} \setminus \gamma^* : n(\gamma, z) \neq 0\}$, a union of bounded components of $\mathbb{C} \setminus \gamma^*$, so $K = \overline{W} \subseteq W \cup \gamma^*$ is compact. If $z \notin U$, then $1/(w - z)$ is holomorphic on the simply connected U, hence has a primitive, so $n(\gamma, z) = 0$; therefore $K \subseteq U$, and $S \cap K$ is finite, say $\{a_1, \dots, a_k\}$, because S is discrete and closed in U. Let $P_j(z) = \sum_{n \geq 1} c_{-n}^{(j)} (z - a_j)^{-n}$ be the principal part of f at a_j. It converges locally uniformly on $\mathbb{C} \setminus \{a_j\}$, because the Laurent series of f converges on the punctured disc and the principal part, as a power series in $1/(z - a_j)$, then has infinite radius of convergence; so P_j is holomorphic there. The function $h = f - \sum_j P_j$ has removable singularities at $a_1, \dots, a_k$, hence extends holomorphically to $U' = U \setminus (S \setminus \{a_1, \dots, a_k\})$. For $z \notin U'$ we have $n(\gamma, z) = 0$: either $z \notin U$, or $z \in S \setminus \{a_1, \dots, a_k\}$, which was excluded from K. So $\int_\gamma h = 0$ by Theorem 29.5.7. Finally $\int_\gamma P_j = 2\pi i\, n(\gamma, a_j) c_{-1}^{(j)}$, since the terms with $n \geq 2$ have primitives, and $n(\gamma, a) = 0$ for the remaining $a \in S$. □

Example 29.5.9. $\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi$. Integrate $f(z) = 1/(1 + z^2)$ over the boundary of the half-disc $\{|z| \leq R, \operatorname{Im} z \geq 0\}$: the only singularity inside is the simple pole at i with residue $\lim_{z \to i} (z - i) f(z) = 1/(2i)$, so the integral is π. The contribution of the semicircle is at most $\pi R/(R^2 - 1) \to 0$, and the integral over $[-R, R]$ tends to the improper integral.

Theorem 29.5.10 (Argument principle and Rouché)**.** *Let U be simply connected, f meromorphic on U and γ a closed path in U avoiding the zeros and poles of f. Then $\frac{1}{2\pi i} \int_\gamma \frac{f'}{f} = \sum_a n(\gamma, a) \operatorname{ord}_a f$. If g is holomorphic on U and $|f - g| < |f|$ on γ^*, with f holomorphic, then f and g have the same number of zeros counted with multiplicity and weighted by $n(\gamma, -)$.*

Proof. Near a zero or pole, $f(z) = (z-a)^m h(z)$ with $h(a) \neq 0$ and $m = \operatorname{ord}_a f$, so $f'/f = m/(z-a) + h'/h$ has residue m at a; apply Theorem 29.5.8. For Rouché's theorem, the functions $f_t = f + t(g-f)$, $t \in [0,1]$, have no zeros on γ^* because $|f_t| \geq |f| - |g-f| > 0$; the integer $N(t) = \frac{1}{2\pi i}\int_\gamma f_t'/f_t$ depends continuously on t (the integrand is continuous in (t,z) and γ^* is compact), hence is constant. □

Example 29.5.11. *More proofs of the fundamental theorem of algebra.* Let $p(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0$, $n \geq 1$. Besides the algebraic proof (Theorem 9.7.3, by Galois theory and the intermediate value theorem) and Liouville's theorem (Corollary 29.4.7(2)):

1. *By Rouché's theorem, or by winding numbers.* On the circle $|z| = R$ with $R > 1 + \sum_i |a_i|$, $|p(z) - z^n| \leq \sum_i |a_i| R^i < R^n = |z^n|$. By the theorem, p has as many zeros in $|z| < R$ as z^n, namely n, counted with multiplicity. Topologically: the loops $t \mapsto p(Re^{it})$ and $t \mapsto R^n e^{int}$ in $\mathbb{C} \setminus 0$ are homotopic, so they have the same winding number n about 0; if p had no zeros, $t \mapsto p(re^{it})$, $0 \leq r \leq R$, would be a homotopy in $\mathbb{C} \setminus 0$ to a constant loop, of winding number 0.

2. *By the minimum modulus (d'Alembert–Argand).* Since $|p(z)| \geq |z|^n(1 - \sum_i |a_i||z|^{i-n}) \to \infty$, $|p|$ attains a minimum on $\mathbb{C}$ (on a large closed disc by compactness, and outside it $|p| > |p(0)|$). Suppose the minimum is attained at z_0 with $p(z_0) \neq 0$. Expanding in h, $p(z_0 + h) = p(z_0) + ch^k + h^{k+1}q(h)$ with $c \neq 0$, $k \geq 1$ and q a polynomial. Choose ζ with $c\zeta^k = -p(z_0)$ (k-th roots exist by the polar form). For $0 < t < 1$, $p(z_0 + t\zeta) = (1-t^k)p(z_0) + t^{k+1}\zeta^{k+1}q(t\zeta)$, so $|p(z_0 + t\zeta)| \leq (1-t^k)|p(z_0)| + Ct^{k+1} < |p(z_0)|$ for small $t > 0$, contradicting minimality. This proof uses only that $\mathbb{C}$ has k-th roots and that continuous functions on compact sets have minima.

Theorem 29.5.12 (Open mapping theorem)**.** *A nonconstant holomorphic function on a domain is an open map. If it is injective, its derivative has no zeros and its inverse is holomorphic.*

Proof. Let $w_0 = f(z_0)$. By Theorem 29.4.8 the zero of $f - w_0$ at z_0 has finite order $m \geq 1$ and is isolated, so there is r with $|f - w_0| \geq \delta > 0$ on $|z - z_0| = r$ and no other zero in $\overline{D}(z_0, r)$. For $|w - w_0| < \delta$, Rouché's theorem applied to $f - w_0$ and $f - w$ on the circle shows that $f - w$ has m zeros in $D(z_0, r)$; so $D(w_0, \delta) \subseteq f(D(z_0, r))$. If f is injective, then $m = 1$ at every point, that is, $f' \neq 0$; and f^{-1} is continuous because f is open, and differentiable with derivative $1/f'(f^{-1})$ by the computation of Proposition 27.6.8, valid over $\mathbb{C}$. □

29.6 Conformal maps and the Riemann mapping theorem

Lemma 29.6.1 (Schwarz)**.** *Let $f : \mathbb{D} \to \mathbb{D}$ be holomorphic with $f(0) = 0$. Then $|f(z)| \leq |z|$ and $|f'(0)| \leq 1$, with equality for some $z \neq 0$ or in the second inequality only if $f(z) = \lambda z$ with $|\lambda| = 1$.*

Proof. The function $g(z) = f(z)/z$, extended by $f'(0)$ at 0, is holomorphic on $\mathbb{D}$ (Proposition 29.5.3(1)). On $|z| = r$, $|g| \leq 1/r$, so by the maximum principle $|g| \leq 1/r$ on $D(0, r)$; letting $r \to 1$ gives $|g| \leq 1$. Equality at an interior point forces g to be a constant of modulus 1. □

Proposition 29.6.2. *For $a \in \mathbb{D}$ the Möbius map $\phi_a(z) = \frac{z-a}{1-\bar{a}z}$ is a biholomorphic self-map of $\mathbb{D}$ with $\phi_a(a) = 0$ and inverse ϕ_{-a}. Every biholomorphic self-map of $\mathbb{D}$ is $z \mapsto \lambda\phi_a(z)$ with $|\lambda| = 1$.*

Proof. For $|z| < 1$ and $|a| < 1$, $|1 - \bar{a}z|^2 - |z - a|^2 = (1 - |a|^2)(1 - |z|^2) > 0$, so ϕ_a maps $\mathbb{D}$ into $\mathbb{D}$, and a computation gives $\phi_{-a}\phi_a = \mathrm{id}$. If F is a biholomorphic self-map with $F(a) = 0$, then $g = F \circ \phi_{-a}$ fixes 0, and Lemma 29.6.1 applied to g and g^{-1} gives $|g'(0)| = 1$, hence $g(z) = \lambda z$. □

Theorem 29.6.3 (Montel)**.** *Let $\mathcal{F}$ be a family of holomorphic functions on an open U that is uniformly bounded on every compact subset. Then every sequence in $\mathcal{F}$ has a subsequence converging uniformly on compact subsets to a holomorphic function.*

Proof. Let $K \subseteq U$ be compact and 2ρ the distance from K to $\mathbb{C} \setminus U$ (or 1 if $U = \mathbb{C}$), and let K' be the set of points at distance $\le \rho$ from K, a compact subset of U. By the Cauchy estimates, $|f'| \le \rho^{-1} \sup_{K'} |f|$ on K for all $f \in \mathcal{F}$, so $\mathcal{F}$ is equicontinuous on K (mean value inequality). By the Arzelà–Ascoli theorem (Theorem 27.14.4) a sequence has a subsequence converging uniformly on K. Exhausting U by compact sets $K_1 \subseteq K_2 \subseteq \cdots$ and taking a diagonal subsequence gives uniform convergence on all compact subsets; the limit is holomorphic by Corollary 29.4.7(4). □

Theorem 29.6.4 (Hurwitz)**.** *Let $f_n \to f$ uniformly on compact subsets of a domain U, with f_n holomorphic and f nonconstant. If every f_n is nowhere zero, so is f; if every f_n is injective, so is f.*

Proof. Suppose $f(z_0) = 0$. Since $f \not\equiv 0$, there is r with $\overline{D}(z_0, r) \subseteq U$ and $f \neq 0$ on the circle $|z - z_0| = r$, so $|f| \ge \delta > 0$ there. Then $f_n \to f$ and $f_n' \to f'$ uniformly on the circle, so $\frac{1}{2\pi i} \int f_n'/f_n \to \frac{1}{2\pi i} \int f'/f \ge 1$, and by Theorem 29.5.10 f_n has a zero inside for large n. For injectivity, if $f(z_1) = f(z_2)$ with $z_1 \neq z_2$, apply the first part on a punctured neighbourhood of z_2 to the functions $f_n - f_n(z_1)$, which are nowhere zero there. □

Theorem 29.6.5 (Riemann mapping theorem)**.** *Every simply connected domain $U \subsetneq \mathbb{C}$ is biholomorphic to $\mathbb{D}$. More precisely, for $z_0 \in U$ there is a unique biholomorphic $\psi \colon U \to \mathbb{D}$ with $\psi(z_0) = 0$ and $\psi'(z_0) > 0$.*

Proof. Let $\mathcal{F}$ be the set of injective holomorphic maps $U \to \mathbb{D}$ sending z_0 to 0.

$\mathcal{F} \neq \emptyset$. Choose $b \notin U$. Since $z - b$ is nowhere zero on the simply connected U, the function $(z-b)'/(z-b)$ has a primitive h (Theorem 29.3.3); adjusting by a constant, $e^h = z - b$. Let $\phi = e^{h/2}$, so $\phi^2 = z - b$. Then ϕ is injective, and $\phi(z_1) = -\phi(z_2)$ implies $z_1 = z_2$ and $\phi(z_1) = 0$, which is impossible; so $\phi(U)$ and $-\phi(U)$ are disjoint. By Theorem 29.5.12, $\phi(U)$ contains a disc $D(w_0, \varrho)$, hence $|\phi(z) + w_0| \ge \varrho$ on U, and $z \mapsto \varrho/(\phi(z) + w_0)$ maps U injectively into $\mathbb{D}$. Composing with a suitable ϕ_a (Proposition 29.6.2) gives an element of $\mathcal{F}$.

An extremal element. Injective holomorphic maps have nonvanishing derivative, so $\eta = \sup_{\psi \in \mathcal{F}} |\psi'(z_0)| > 0$. Take $\psi_n \in \mathcal{F}$ with $|\psi_n'(z_0)| \to \eta$; by Theorem 29.6.3 a subsequence converges uniformly on compact subsets to a holomorphic ψ with $|\psi'(z_0)| = \eta > 0$, so ψ is nonconstant, hence injective by Theorem 29.6.4, with $\psi(z_0) = 0$ and $|\psi| \le 1$; by the open mapping theorem $\psi(U) \subseteq \mathbb{D}$. So $\psi \in \mathcal{F}$ and $|\psi'(z_0)| = \eta$.

Surjectivity. Suppose $\alpha \in \mathbb{D} \setminus \psi(U)$. Then $\phi_\alpha \circ \psi$ is nowhere zero on U, so as above it has a holomorphic square root g, which is injective because g^2 is. Let $F = \phi_{g(z_0)} \circ g \in \mathcal{F}$. Writing $s(w) = w^2$, we have $\psi = \phi_{-\alpha} \circ s \circ \phi_{-g(z_0)} \circ F = \Phi \circ F$ with $\Phi \colon \mathbb{D} \to \mathbb{D}$ holomorphic and $\Phi(0) = 0$. As s is not injective, Φ is not injective, so Φ is not a rotation and $|\Phi'(0)| < 1$ by Lemma 29.6.1. Then $\eta = |\psi'(z_0)| = |\Phi'(0)|\,|F'(z_0)| < |F'(z_0)| \le \eta$, a contradiction. So ψ is a biholomorphism onto $\mathbb{D}$, and multiplying by a unimodular constant makes $\psi'(z_0) > 0$.

Uniqueness. If ψ_1, ψ_2 both work, $\psi_2 \circ \psi_1^{-1}$ is an automorphism of $\mathbb{D}$ fixing 0 with positive derivative there, hence the identity by Proposition 29.6.2. □

Remark 29.6.6. A real function u on an open $U \subseteq \mathbb{C}$ is *harmonic* if it is C^2 with $\partial_x^2 u + \partial_y^2 u = 0$. Real and imaginary parts of holomorphic functions are harmonic (Corollary 29.4.3). Conversely, on a simply connected U every harmonic u is the real part of a holomorphic function: the function $\partial_x u - i\partial_y u$ is holomorphic (its Cauchy–Riemann equations are the harmonicity of u and the symmetry of second derivatives), so it has a primitive f, and $\operatorname{Re} f$ differs from u by a constant. Harmonic functions therefore satisfy the mean value property and the maximum principle; this is the one-variable case of the Hodge theory of Chapter 43.

29.7 The Riemann sphere

Many statements of complex analysis become simpler once the point ∞ is added to the plane: a pole is a point where a function takes the value ∞, lines are circles through ∞, and the rational functions are exactly the functions that are meromorphic everywhere, ∞ included. The resulting space is a sphere, the first example of a compact Riemann surface; its symmetries, the Möbius transformations, form the group $\mathrm{PSL}_2(\mathbb{C})$, which reappears in hyperbolic geometry, in the theory of modular forms and in the classification of Riemann surfaces.

29.7.1 The sphere and stereographic projection

Definition 29.7.1. The *Riemann sphere* is the one-point compactification $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ of $\mathbb{C}$ (Definition 18.9.2). The neighbourhoods of ∞ are the sets containing $\{|z| > R\} \cup \{\infty\}$ for some R, so $z_n \to \infty$ in $\hat{\mathbb{C}}$ if and only if $|z_n| \to \infty$.

Identifying $\mathbb{C}$ with $\mathbb{R}^2$ by $z = x + iy \mapsto (x, y)$, Example 18.9.4 gives a homeomorphism $\tau\colon \hat{\mathbb{C}} \to S^2 \subseteq \mathbb{R}^3$,

$$\tau(z) = \frac{(2x, 2y, |z|^2 - 1)}{|z|^2 + 1} = \frac{(2\operatorname{Re} z, 2\operatorname{Im} z, |z|^2 - 1)}{|z|^2 + 1}, \qquad \tau(\infty) = N = (0, 0, 1),$$

whose inverse is the stereographic projection from the north pole N: the point $\tau(z)$ is the second intersection of S^2 with the line through N and $(x, y, 0)$. The unit circle is mapped to the equator, the unit disc $\mathbb{D}$ to the southern hemisphere, 0 to the south pole and ∞ to the north pole.

Proposition 29.7.2. *1. (*Chordal distance*) For $z, w \in \mathbb{C}$,*

$$\|\tau(z) - \tau(w)\| = \frac{2|z - w|}{\sqrt{(1 + |z|^2)(1 + |w|^2)}}, \qquad \|\tau(z) - N\| = \frac{2}{\sqrt{1 + |z|^2}}.$$

2. The map $z \mapsto 1/z$, with $0 \mapsto \infty \mapsto 0$, corresponds to the rotation by the angle π about the x_1-axis: $\tau(1/z) = (x_1, -x_2, -x_3)$ if $\tau(z) = (x_1, x_2, x_3)$. In particular it preserves the chordal distance.

Proof. (1) Since $\|\tau(z)\| = \|\tau(w)\| = 1$, $\|\tau(z) - \tau(w)\|^2 = 2 - 2\langle\tau(z), \tau(w)\rangle$, and

$$2 - 2\langle\tau(z), \tau(w)\rangle = 2\,\frac{(|z|^2 + 1)(|w|^2 + 1) - 4\operatorname{Re}(z\bar{w}) - (|z|^2 - 1)(|w|^2 - 1)}{(|z|^2 + 1)(|w|^2 + 1)} = \frac{4(|z|^2 + |w|^2 - 2\operatorname{Re}(z\bar{w}))}{(|z|^2 + 1)(|w|^2 + 1)},$$

which is $4|z - w|^2/((1 + |z|^2)(1 + |w|^2))$. The second formula is in Example 18.9.4.

(2) For $z \neq 0$, $1/z = \bar{z}/|z|^2$ has real part $x/|z|^2$ and imaginary part $-y/|z|^2$, so

$$\tau(1/z) = \frac{(2x/|z|^2, -2y/|z|^2, 1/|z|^2 - 1)}{1/|z|^2 + 1} = \frac{(2x, -2y, 1 - |z|^2)}{1 + |z|^2},$$

which is $(x_1, -x_2, -x_3)$; and τ maps 0 and ∞ to the south and north poles, which the rotation exchanges. □

A *generalised circle* in $\hat{\mathbb{C}}$ is either a circle in $\mathbb{C}$ or a set $L \cup \{\infty\}$ with L a line in $\mathbb{C}$. Every generalised circle is the set of solutions in $\hat{\mathbb{C}}$ of an equation

$$A|z|^2 + 2\beta x + 2\gamma y + C = 0, \qquad A, \beta, \gamma, C \in \mathbb{R}, \quad \beta^2 + \gamma^2 > AC, \tag{29.1}$$

where ∞ counts as a solution when $A = 0$: for $A \neq 0$ this is the circle with centre $-(\beta + i\gamma)/A$ and radius $\sqrt{\beta^2 + \gamma^2 - AC}/|A|$, and for $A = 0$ a line. Through any three distinct points of $\hat{\mathbb{C}}$ passes exactly one generalised circle: the circle through three non-collinear points of $\mathbb{C}$, the line through three collinear ones, and the line through the two finite points if one of them is ∞.

Proposition 29.7.3. *The map τ induces a bijection between the generalised circles in $\hat{\mathbb{C}}$ and the circles on S^2, that is, the intersections of S^2 with planes at distance less than 1 from the origin. The lines correspond to the circles through N.*

Proof. A plane is $\{ax_1 + bx_2 + cx_3 = d\}$ with $a^2 + b^2 + c^2 = 1$, and it meets S^2 in a circle exactly when $|d| < 1$. The point $\tau(z)$, $z \in \mathbb{C}$, lies on it if and only if $2ax + 2by + c(|z|^2 - 1) = d(|z|^2 + 1)$, that is,

$$(c - d)|z|^2 + 2ax + 2by - (c + d) = 0,$$

an equation of the form (29.1) with $A = c - d$, $\beta = a$, $\gamma = b$, $C = -(c + d)$, for which $\beta^2 + \gamma^2 - AC = a^2 + b^2 + c^2 - d^2 = 1 - d^2 > 0$. The point $N = \tau(\infty)$ lies on the plane if and only if $c = d$, that is, $A = 0$. Conversely, given (A, β, γ, C) with $\beta^2 + \gamma^2 > AC$, the numbers $a = s\beta$, $b = s\gamma$, $c = s(A - C)/2$, $d = -s(A + C)/2$ satisfy the equations above, and $a^2 + b^2 + c^2 - d^2 = s^2(\beta^2 + \gamma^2 - AC) > 0$; choosing $s > 0$ with $a^2 + b^2 + c^2 = 1$ gives a plane with $1 - d^2 > 0$ whose circle corresponds to the given generalised circle. □

Stereographic projection also preserves angles between curves; equivalently, the round metric of S^2 pulls back to the conformal metric $4|dz|^2/(1 + |z|^2)^2$ (Chapter 40).

29.7.2 Meromorphic functions on the sphere

The behaviour of a function at ∞ is defined through the coordinate $w = 1/z$.

Definition 29.7.4. Let f be holomorphic on $\{|z| > R\}$, and $g(w) = f(1/w)$ on $0 < |w| < 1/R$. Then f is *holomorphic at* ∞, has a *pole*, a *zero* or an *essential singularity* at ∞, if g has the corresponding property at 0, and $\operatorname{ord}_\infty f = \operatorname{ord}_0 g$. In terms of the Laurent expansion $f(z) = \sum_{n\in\mathbb{Z}} c_n z^n$ on $|z| > R$ (Theorem 29.5.1), where $g(w) = \sum_n c_n w^{-n}$: f is holomorphic at ∞ if $c_n = 0$ for $n > 0$, and has a pole of order m if $c_m \neq 0$ and $c_n = 0$ for $n > m$. The *residue at* ∞ is

$$\operatorname{Res}_\infty f = -\frac{1}{2\pi i}\int_{|z|=\rho} f(z)\,dz = -c_{-1} \qquad (\rho > R),$$

the circle traversed counterclockwise. A function f on $\hat{\mathbb{C}}$ is *meromorphic* if it is meromorphic on $\mathbb{C}$ and holomorphic or has a pole at ∞; it is regarded as a map $\hat{\mathbb{C}} \to \hat{\mathbb{C}}$ by sending the poles to ∞, which is continuous by Proposition 29.5.3(2).

The residue at ∞ is not the residue of g at 0: it belongs to the differential $f(z)\,dz = -f(1/w)w^{-2}\,dw$, and $\operatorname{Res}_\infty f = \operatorname{Res}_0(-g(w)/w^2)$. For instance $f(z) = 1/z$ is holomorphic at ∞, with a zero of order 1 there, but $\operatorname{Res}_\infty(1/z) = -1$. The minus sign is there because the circle $|z| = \rho$, traversed counterclockwise, goes around ∞ clockwise.

A rational function $R = p/q$, with p, q coprime polynomials and $q \neq 0$, is meromorphic on $\hat{\mathbb{C}}$: near ∞, $R(z) = z^{\deg p - \deg q} h(1/z)$ with h holomorphic near 0 and $h(0) \neq 0$ (the ratio of the leading coefficients), so

$$\operatorname{ord}_\infty \frac{p}{q} = \deg q - \deg p. \tag{29.2}$$

The *degree* of R is $\deg R = \max(\deg p, \deg q)$.

Theorem 29.7.5. *The meromorphic functions on $\hat{\mathbb{C}}$ are exactly the rational functions. More precisely, if f is meromorphic on $\hat{\mathbb{C}}$, with poles $a_1, \dots, a_k$ in $\mathbb{C}$, principal parts $P_j(z) = \sum_{n=1}^{m_j} c_{-n}^{(j)}(z-a_j)^{-n}$ there, and principal part $P_\infty(z) = \sum_{n=1}^{m} c_n z^n$ at ∞ (the positive part of its Laurent expansion at ∞), then*

$$f = c + P_\infty + P_1 + \cdots + P_k$$

for a constant c. This is the partial fraction decomposition *of f, and it is unique.*

Proof. Let $f = \sum_n c_n z^n$ on $|z| > R$. Since f is not essential at ∞, $c_n = 0$ for large n, so P_∞ is a polynomial. The poles of f in $\mathbb{C}$ lie in the compact disc $|z| \le R$, where f is holomorphic outside a discrete closed set of poles, so there are finitely many, $a_1, \dots, a_k$. Each P_j is a rational function, holomorphic on $\hat{\mathbb{C}} \setminus \{a_j\}$ and vanishing at ∞. The function $h = f - P_\infty - \sum_j P_j$ has removable singularities at the a_j, so it extends to an entire function. As $z \to \infty$, $f(z) - P_\infty(z) = \sum_{n \le 0} c_n z^n \to c_0$ and $P_j(z) \to 0$, so h is bounded outside a large disc, hence bounded on $\mathbb{C}$, and constant by Liouville's theorem (Corollary 29.4.7). So f is rational. Uniqueness: the principal parts, and hence c, are determined by f. □

So a meromorphic function on $\hat{\mathbb{C}}$ is determined, up to an additive constant, by its principal parts, and up to a multiplicative constant by its zeros and poles: if f and g have the same zeros and poles with the same orders, f/g is meromorphic on $\hat{\mathbb{C}}$ with only removable singularities, so it has no principal parts and is constant by the theorem. This is the simplest instance of the interplay of local data and global holomorphic functions studied for general compact Riemann surfaces in Chapter 59.

Proposition 29.7.6. *Let R be a nonconstant rational function of degree d. For every $w \in \hat{\mathbb{C}}$, the equation $R(z) = w$ has exactly d solutions in $\hat{\mathbb{C}}$, counted with multiplicity; here the multiplicity of a solution a is $\operatorname{ord}_a(R - w)$ if a and w are finite, and is defined through the coordinates $1/z$ near $z = \infty$ and $1/R$ near $w = \infty$ otherwise. Consequently $\sum_{a \in \hat{\mathbb{C}}} \operatorname{ord}_a R = 0$: a rational function has as many zeros as poles on $\hat{\mathbb{C}}$, counted with multiplicity.*

Proof. Write $R = p/q$ with p, q coprime. Let $w \in \mathbb{C}$. The finite solutions of $R(z) = w$ are the roots of $p - wq$: at a root a of $p - wq$ we have $q(a) \neq 0$, as otherwise $p(a) = 0$ too, and then $\operatorname{ord}_a(R - w) = \operatorname{ord}_a(p - wq)$. By the fundamental theorem of algebra, these multiplicities add up to $e = \deg(p - wq) \le d$. The point ∞ is a solution exactly when $R - w = (p - wq)/q$ vanishes at ∞, with multiplicity $\deg q - e$ by (29.2). If $\deg p > \deg q$, then $e = \deg p = d$ and $R(\infty) = \infty \neq w$. If $\deg p \le \deg q = d$, then either $e = d$, and $R - w$ does not vanish at ∞; or $e < d$, and ∞ is a solution of multiplicity $d - e$. In all cases the total is d. For $w = \infty$, the solutions are the poles: the roots of q, with multiplicities adding up to $\deg q$, and ∞ if $\deg p > \deg q$, with multiplicity $\deg p - \deg q$; the total is again d. Finally $\sum_a \operatorname{ord}_a R$ is the number of zeros minus the number of poles, $d - d = 0$. □

Proposition 29.7.7. *For every rational function R, $\sum_{a \in \hat{\mathbb{C}}} \operatorname{Res}_a R = 0$, the sum running over ∞ and the poles of R in $\mathbb{C}$.*

Proof. Choose ρ larger than the absolute values of all poles of R in $\mathbb{C}$. By the residue theorem in the simply connected set $\mathbb{C}$ (Theorem 29.5.8), $\frac{1}{2\pi i}\int_{|z|=\rho} R(z)\,dz = \sum_{a \in \mathbb{C}} \operatorname{Res}_a R$, and the left side is $-\operatorname{Res}_\infty R$ by definition. □

Example 29.7.8. 1. The residue at ∞ often computes a contour integral at once. For $n \ge 2$, $\int_{|z|=2} \frac{dz}{z^n - 1} = 0$, because $\frac{1}{z^n - 1} = z^{-n}(1 + z^{-n} + z^{-2n} + \cdots)$ for $|z| > 1$ has $c_{-1} = 0$; computing the n residues at the roots of unity would be longer. Similarly $\int_{|z|=2} \frac{z^{n-1}\,dz}{z^n - 1} = 2\pi i$, which is also $2\pi i$ times the number of roots of $z^n - 1$ in the disc, by the argument principle (Theorem 29.5.10).

2. Applying the proposition to $R(z) = \frac{1}{z(z-1)}$: the residues at 0 and 1 are -1 and 1, and indeed $R = z^{-2}(1 + z^{-1} + \cdots)$ has $\operatorname{Res}_\infty R = 0$.

3. A polynomial of degree $d \geq 1$ is a rational function of degree d with a single pole, of order d, at ∞; Proposition 29.7.6 says it takes every value d times in $\mathbb{C}$, which is the fundamental theorem of algebra.

29.7.3 Möbius transformations

Definition 29.7.9. For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2(\mathbb{C})$, the *Möbius transformation* $M_A \colon \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ is

$$M_A(z) = \frac{az + b}{cz + d},$$

with $M_A(-d/c) = \infty$ and $M_A(\infty) = a/c$ if $c \neq 0$, and $M_A(\infty) = \infty$ if $c = 0$.

It is convenient to write points of $\hat{\mathbb{C}}$ in *homogeneous coordinates*: the nonzero vector $(z_1, z_2) \in \mathbb{C}^2$ represents $z_1/z_2 \in \hat{\mathbb{C}}$ (with $z_1/0 = \infty$), and two vectors represent the same point if and only if they are proportional. So $\hat{\mathbb{C}}$ is the set of lines through 0 in $\mathbb{C}^2$, the *complex projective line* $\mathbb{P}^1(\mathbb{C})$, and M_A is the map induced by the linear map A: if (z_1, z_2) represents z, then $A(z_1, z_2) = (az_1 + bz_2, cz_1 + dz_2)$ represents $M_A(z)$. (The cases with ∞ in the definition are exactly what this description gives.)

Proposition 29.7.10. *1. $M_A M_B = M_{AB}$, and $M_A = \mathrm{id}$ if and only if A is a scalar matrix. So the Möbius transformations form a group, isomorphic to $\mathrm{PGL}_2(\mathbb{C}) = \mathrm{GL}_2(\mathbb{C})/\mathbb{C}^*$, and also to $\mathrm{PSL}_2(\mathbb{C}) = \mathrm{SL}_2(\mathbb{C})/\{\pm 1\}$.*

2. Each M_A is a homeomorphism of $\hat{\mathbb{C}}$ and a rational function of degree 1; its derivative $\det A/(cz + d)^2$ never vanishes on $\mathbb{C} \setminus \{-d/c\}$. Conversely every rational function of degree 1, and every bijective meromorphic function $\hat{\mathbb{C}} \to \hat{\mathbb{C}}$, is a Möbius transformation.

3. The Möbius transformations are generated by the translations $z \mapsto z + b$, the maps $z \mapsto az$ ($a \neq 0$) and the inversion $z \mapsto 1/z$.

4. Möbius transformations map generalised circles to generalised circles.

Proof. (1) The first statement is the description in homogeneous coordinates: AB induces the composite of the maps induced by B and A. If $M_A = \mathrm{id}$, every line in $\mathbb{C}^2$ is mapped to itself, so every nonzero vector is an eigenvector of A; then e_1, e_2 and $e_1 + e_2$ are eigenvectors, which forces the two eigenvalues for e_1, e_2 to agree, and A is scalar. So $A \mapsto M_A$ induces an isomorphism from $\mathrm{GL}_2(\mathbb{C})/\mathbb{C}^*$ onto the group of Möbius transformations. Every A is a scalar multiple of a matrix of determinant 1 (divide by a square root of $\det A$), and a scalar matrix of determinant 1 is ± 1, which gives the second description.

(2) M_A is bijective with inverse $M_{A^{-1}}$. It is continuous on $\hat{\mathbb{C}}$: away from $-d/c$ and ∞ this is clear, $|M_A(z)| \to \infty$ as $z \to -d/c$ since the numerator does not vanish there ($a(-d/c) + b = -\det A/c$), and $M_A(z) \to a/c$ (or ∞ if $c = 0$) as $z \to \infty$. A continuous bijection of the compact Hausdorff space $\hat{\mathbb{C}}$ is a homeomorphism. The polynomials $az + b$ and $cz + d$ are coprime, since $ad - bc \neq 0$, so M_A has degree 1, and the formula for the derivative is the quotient rule. Conversely, a rational function $(az + b)/(cz + d)$ of degree 1 in lowest terms has $ad - bc \neq 0$, since otherwise the numerator and denominator would be proportional. If f is a meromorphic bijection of $\hat{\mathbb{C}}$, it is rational (Theorem 29.7.5) of some degree $d \geq 1$, and by Proposition 29.7.6 each value has a single preimage, of multiplicity d. If $d \geq 2$, then f' would vanish at every point of $\mathbb{C}$ that is not a pole and not mapped to ∞, which is impossible for a nonconstant function. So $d = 1$.

(3) If $c = 0$, $M_A(z) = (a/d)z + b/d$. If $c \neq 0$, $M_A(z) = \frac{a}{c} - \frac{\det A}{c(cz+d)}$, the composite of $z \mapsto cz + d$, $z \mapsto 1/z$, multiplication by $-\det A/c$ and translation by a/c.

(4) By (3) it suffices to check the generators. Translations and the maps $z \mapsto az$ are similarities of the plane, which map circles to circles and lines to lines, and fix ∞. The inversion corresponds under τ to a rotation of S^2 (Proposition 29.7.2(2)), which maps plane sections of S^2 to plane sections; by Proposition 29.7.3 it maps generalised circles to generalised circles. □

The proof of (4) by the sphere avoids a computation; directly, substituting $z = 1/u$ in (29.1) and multiplying by $|u|^2$ gives $A + 2\beta \operatorname{Re} u - 2\gamma \operatorname{Im} u + C|u|^2 = 0$, again of the same form. Under inversion, circles through 0 become lines, and lines not through 0 become circles through 0.

Theorem 29.7.11. *Given distinct points $z_1, z_2, z_3 \in \hat{\mathbb{C}}$ and distinct points $w_1, w_2, w_3 \in \hat{\mathbb{C}}$, there is exactly one Möbius transformation M with $M(z_i) = w_i$ for $i = 1, 2, 3$.*

Proof. If $z_1, z_2, z_3 \in \mathbb{C}$, the map

$$S(z) = \frac{(z - z_1)(z_2 - z_3)}{(z - z_3)(z_2 - z_1)}$$

is Möbius, with $S(z_1) = 0$, $S(z_2) = 1$, $S(z_3) = \infty$. If one of the z_i is ∞, delete the two factors containing it: $S(z) = \frac{z_2 - z_3}{z - z_3}$, $\frac{z - z_1}{z - z_3}$ or $\frac{z - z_1}{z_2 - z_1}$ for z_1, z_2 or $z_3 = \infty$. Doing the same for the w_i gives T, and $M = T^{-1}S$ has the required property. For uniqueness it suffices to show that a Möbius map fixing 0, 1, ∞ is the identity: from $M(\infty) = \infty$, $c = 0$, so $M(z) = \alpha z + \beta$; then $M(0) = 0$ gives $\beta = 0$ and $M(1) = 1$ gives $\alpha = 1$. □

Definition 29.7.12. For distinct $z_1, z_2, z_3 \in \hat{\mathbb{C}}$ and $z \in \hat{\mathbb{C}}$, the *cross-ratio* (z, z_1, z_2, z_3) is $S(z)$, for the unique Möbius S with $S(z_1) = 0$, $S(z_2) = 1$, $S(z_3) = \infty$; for finite points

$$(z, z_1, z_2, z_3) = \frac{(z - z_1)(z_2 - z_3)}{(z - z_3)(z_2 - z_1)}.$$

Corollary 29.7.13. *1. For every Möbius transformation M, $(Mz, Mz_1, Mz_2, Mz_3) = (z, z_1, z_2, z_3)$.*

2. Four distinct points z, z_1, z_2, z_3 lie on a generalised circle if and only if (z, z_1, z_2, z_3) is real.

Proof. (1) SM^{-1} sends Mz_1, Mz_2, Mz_3 to 0, 1, ∞, so it is the map defining the left side, and its value at Mz is $S(z)$. (2) Let C be the generalised circle through z_1, z_2, z_3. By Proposition 29.7.10(4), applied to S and S^{-1}, S maps C onto the generalised circle through 0, 1, ∞, which is $\mathbb{R} \cup \{\infty\}$. So $z \in C$ if and only if $S(z) \in \mathbb{R} \cup \{\infty\}$, and $S(z) \neq \infty$ since $z \neq z_3$. □

29.7.4 Fixed points and conjugacy classes

A point $z = z_1/z_2$ is fixed by M_A if and only if (z_1, z_2) is an eigenvector of A. Since A has one or two eigenlines unless it is scalar, a Möbius transformation other than the identity has one or two fixed points in $\hat{\mathbb{C}}$. Normalising $\det A = 1$, the trace $\operatorname{tr} A$ is determined by M_A up to sign, so $(\operatorname{tr} A)^2$ is an invariant of M_A, unchanged under conjugation $M \mapsto NMN^{-1}$.

Theorem 29.7.14. *Let $M = M_A \neq \mathrm{id}$ with $\det A = 1$, and $t = \operatorname{tr} A$.*

1. *If $t^2 = 4$, M has exactly one fixed point and is conjugate to the translation $z \mapsto z + 1$; M is called* parabolic.

2. *If $t^2 \neq 4$, M has exactly two fixed points and is conjugate to $z \mapsto kz$ for some $k \neq 0, 1$, determined up to $k \leftrightarrow k^{-1}$ by $k + 2 + k^{-1} = t^2$. Then M is called* elliptic *if $|k| = 1$, equivalently $t \in (-2, 2)$;* hyperbolic *if $k > 0$, equivalently $t \in \mathbb{R}$ with $|t| > 2$; and* loxodromic *otherwise, equivalently $t \notin \mathbb{R}$.*

Two Möbius transformations other than the identity are conjugate if and only if they have the same invariant t^2.

Proof. Conjugating M_A by M_P replaces A by PAP^{-1}. By the Jordan normal form (Theorem 11.8.4), a non-scalar $A \in \mathrm{SL}_2(\mathbb{C})$ is conjugate either to $\left(\begin{smallmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{smallmatrix}\right)$ with $\lambda \neq \lambda^{-1}$, or to $\left(\begin{smallmatrix} \lambda & 1 \\ 0 & \lambda \end{smallmatrix}\right)$ with $\lambda^2 = 1$; a diagonal matrix with $\lambda = \lambda^{-1}$ would be scalar. In the second case $t = 2\lambda = \pm 2$, M is conjugate to $z \mapsto z \pm 1$, and $z \mapsto z - 1$ is conjugate to $z \mapsto z + 1$ by $z \mapsto -z$; the only fixed point of a translation is ∞. In the first case M is conjugate to $z \mapsto kz$ with $k = \lambda^2 \neq 1$, which fixes 0 and ∞, and $t^2 = (\lambda + \lambda^{-1})^2 = k + 2 + k^{-1}$; here $t^2 \neq 4$ since $k \neq 1$. Exchanging the eigenvectors replaces k by k^{-1}. The eigenvalues λ, λ^{-1} are the roots of $X^2 - tX + 1$, and $k = \lambda^2$. If $t \in (-2, 2)$, the roots are non-real and conjugate with product 1, so $|\lambda| = 1$ and $|k| = 1$; conversely $|\lambda| = 1$ gives $t = \lambda + \bar{\lambda} = 2 \operatorname{Re} \lambda \in [-2, 2]$, and $t \neq \pm 2$ as $t^2 \neq 4$. If t is real with $|t| > 2$, the roots are real, so $k > 0$; conversely $k > 0$ means $\lambda \in \mathbb{R}$, so $t = \lambda + \lambda^{-1}$ is real, and $|t| > 2$ since $\lambda \neq \pm 1$. As $t^2 \neq 4$, a real t falls in one of these two cases, so the remaining, loxodromic, case is $t \notin \mathbb{R}$. Finally, transformations with the same t^2 are conjugate, since both are conjugate to the same normal form ($z \mapsto z + 1$, or $z \mapsto kz$ with k determined up to inversion, and $z \mapsto k^{-1}z$ is conjugate to $z \mapsto kz$ by $z \mapsto 1/z$). □

The names describe the dynamics. A hyperbolic map $z \mapsto kz$, $k > 1$, pushes every point away from the repelling fixed point 0 towards the attracting fixed point ∞ along rays; an elliptic map $z \mapsto e^{i\theta}z$ rotates around its two fixed points, and corresponds to a rotation of the sphere about the axis through them; a loxodromic map spirals from one fixed point to the other; and a parabolic map moves points along circles tangent at the single fixed point. For example $z \mapsto 1/z$, with matrix $i\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ of determinant 1 and trace 0, is elliptic of order 2 with fixed points ± 1, in accordance with Proposition 29.7.2(2).

29.7.5 The disc, the half-plane and the sphere

Three subgroups of the Möbius group correspond to the three simply connected Riemann surfaces of the uniformisation theorem (Chapter 44): the sphere, the plane and the disc.

Proposition 29.7.15. *Let $\mathbb{H} = \{\operatorname{Im} z > 0\}$.*

1. *The* Cayley transform *$\kappa(z) = \frac{z-i}{z+i}$ maps $\mathbb{H}$ biholomorphically onto $\mathbb{D}$, and $\mathbb{R} \cup \{\infty\}$ onto the unit circle.*

2. *For $A \in \mathrm{SL}_2(\mathbb{R})$, $\operatorname{Im} M_A(z) = \operatorname{Im} z / |cz + d|^2$, so M_A maps $\mathbb{H}$ onto itself.*

3. *Every biholomorphic self-map of $\mathbb{H}$ is M_A for some $A \in \mathrm{SL}_2(\mathbb{R})$, unique up to sign; so $\mathrm{Aut}(\mathbb{H}) \cong \mathrm{PSL}_2(\mathbb{R})$.*

Proof. (1) $|\kappa(z)| < 1$ if and only if $|z - i| < |z + i|$, that is, z is closer to i than to $-i$, that is, $\operatorname{Im} z > 0$; similarly $|\kappa(z)| = 1$ exactly on $\mathbb{R}$, and $\kappa(\infty) = 1$. Since κ is a bijection of $\hat{\mathbb{C}}$, it maps $\mathbb{H}$ onto $\mathbb{D}$, and it is holomorphic with holomorphic inverse $M_{A^{-1}}$.

(2) $(az + b)(c\bar{z} + d) = ac|z|^2 + adz + bc\bar{z} + bd$ has imaginary part $(ad - bc) \operatorname{Im} z = \operatorname{Im} z$, and $\operatorname{Im} M_A(z) = \operatorname{Im}\big((az + b)(c\bar{z} + d)\big)/|cz + d|^2$. So M_A maps $\mathbb{H}$ into itself, and so does its inverse $M_{A^{-1}}$, $A^{-1} \in \mathrm{SL}_2(\mathbb{R})$.

(3) Let f be a biholomorphic self-map of $\mathbb{H}$. Then $\kappa f \kappa^{-1}$ is a biholomorphic self-map of $\mathbb{D}$, hence a Möbius transformation by Proposition 29.6.2, and so $f = M$ is Möbius. As

a homeomorphism of $\hat{\mathbb{C}}$ mapping $\mathbb{H}$ onto itself, M maps the boundary $\mathbb{R} \cup \{\infty\}$ of $\mathbb{H}$ onto itself. Choose three distinct real points z_i; the $w_i = M(z_i)$ lie in $\mathbb{R} \cup \{\infty\}$, and in the proof of Theorem 29.7.11 the maps S and T have real coefficients, so $M = T^{-1}S = M_A$ with $A \in \mathrm{GL}_2(\mathbb{R})$. By the computation in (2), $\operatorname{Im} M_A(z) = \det A \operatorname{Im} z/|cz + d|^2$, so $\det A > 0$, and dividing A by $\sqrt{\det A}$ gives $A \in \mathrm{SL}_2(\mathbb{R})$. Finally $M_A = M_B$ with $A, B \in \mathrm{SL}_2(\mathbb{R})$ forces $A = \pm B$ by Proposition 29.7.10(1). □

So the three groups are: all of $\mathrm{PSL}_2(\mathbb{C})$ for the sphere (Proposition 29.7.10(2)); the affine maps $z \mapsto az + b$, the Möbius maps fixing ∞, for the plane (Exercise 29.8.2); and $\mathrm{PSL}_2(\mathbb{R}) \cong \mathrm{PSU}(1,1)$ for the half-plane or the disc. The subgroup $\mathrm{PSL}_2(\mathbb{Z})$ acting on $\mathbb{H}$ gives the moduli space of elliptic curves (Chapter 51).

The rotations of the sphere form a fourth, compact, subgroup. In homogeneous coordinates the chordal distance of Proposition 29.7.2 reads

$$\|\tau(z) - \tau(w)\| = \frac{2|z_1 w_2 - z_2 w_1|}{\|(z_1, z_2)\| \, \|(w_1, w_2)\|} \qquad (z = z_1/z_2,\ w = w_1/w_2),$$

as one sees by taking $z_2 = w_2 = 1$ (both sides are unchanged by rescaling either vector, and the case of ∞ follows by continuity). The numerator is $|\det|$ of the matrix with columns (z_1, z_2) and (w_1, w_2), so a matrix $A \in \mathrm{SU}(2)$, which preserves determinants and norms, gives a Möbius transformation M_A preserving the chordal distance, an isometry of S^2. These isometries are rotations, and every rotation arises in this way; this is the double covering $\mathrm{SU}(2) \to \mathrm{SO}(3)$ of Chapter 34. The matrices in $\mathrm{SU}(2)$ are those of the form $\left(\begin{smallmatrix} a & -\bar{b} \\ b & \bar{a} \end{smallmatrix}\right)$ with $|a|^2 + |b|^2 = 1$; for instance $z \mapsto e^{i\theta} z$ is the rotation by θ about the vertical axis.

Remark 29.7.16.

1. *Riemann surfaces.* With the two charts z and $1/z$, $\hat{\mathbb{C}}$ is a compact Riemann surface, biholomorphic to the projective line $\mathbb{CP}^1$; meromorphic functions on $\hat{\mathbb{C}}$ are the holomorphic maps $\hat{\mathbb{C}} \to \hat{\mathbb{C}}$ other than the constant ∞, and the Möbius transformations are its automorphisms (Chapter 44). It is the only compact Riemann surface of genus 0.

2. *Riemann–Hurwitz.* A rational function of degree d has $2d - 2$ critical points on $\hat{\mathbb{C}}$, counted with multiplicity; for a polynomial of degree d these are the $d - 1$ zeros of p' and the point ∞, of multiplicity $d - 1$. This is the Riemann–Hurwitz formula for maps $\hat{\mathbb{C}} \to \hat{\mathbb{C}}$ [For81].

3. *Iteration.* Iterating a rational function of degree $d \geq 2$ on $\hat{\mathbb{C}}$ is the subject of complex dynamics, with its Julia and Fatou sets; the family $z \mapsto z^2 + c$ gives the Mandelbrot set. For Möbius transformations iteration is simple, by Theorem 29.7.14: the iterates of $z \mapsto kz$ are $z \mapsto k^n z$, and those of $z \mapsto z + 1$ are $z \mapsto z + n$.

29.8 Exercises

Exercise 29.8.1. Show that $\mathbb{D}$ and $\mathbb{C}$ are not biholomorphic, although they are homeomorphic.

Solution. A biholomorphism $\mathbb{C} \to \mathbb{D}$ would be a bounded entire function, hence constant (Corollary 29.4.7). They are homeomorphic by $z \mapsto z/(1 + |z|)$ (Counterexample 18.7.9). □

Exercise 29.8.2. Show that every biholomorphic map $\mathbb{C} \to \mathbb{C}$ is $z \mapsto az + b$ with $a \neq 0$.

Solution. Let f be such a map and consider $g(z) = f(1/z)$ on $\mathbb{C}^\times$. If 0 were an essential singularity of g, then by Casorati–Weierstrass the image $f(\{|z| > 1\})$ would be dense, while $f(\mathbb{D})$ is open and disjoint from it, a contradiction. So g has a pole or removable singularity at 0, which means $|f(z)| \leq C|z|^m$ for large $|z|$; by Liouville's theorem f is a polynomial, and it is injective only if $\deg f = 1$. □

Exercise 29.8.3. Use Rouché's theorem to show that $z^5 + 3z + 1$ has exactly one root in $\mathbb{D}$ and all five roots in $D(0, 2)$.

Exercise 29.8.4. Compute $\int_0^{2\pi} \frac{dt}{a+\cos t}$ for $a > 1$ by writing $z = e^{it}$ and using the residue theorem.

Exercise 29.8.5. Give an example of a domain U and a nowhere vanishing holomorphic f on U with no holomorphic logarithm, and relate this to $\pi_1(U)$.

Exercise 29.8.6. Find the Möbius transformation sending $1, i, -1$ to $0, 1, \infty$, and show that it maps $\mathbb{D}$ onto $\mathbb{H}$. Find a Möbius transformation mapping the half-plane $\{\operatorname{Re} z > 0\}$ onto $\mathbb{D}$ with $1 \mapsto 0$.

Solution. By Theorem 29.7.11, $S(z) = \frac{(z-1)(i+1)}{(z+1)(i-1)}$, and $\frac{1+i}{i-1} = -i$, so $S(z) = i\frac{1-z}{1+z}$. It maps the unit circle, through $1, i, -1$, onto $\mathbb{R} \cup \{\infty\}$, so it maps $\mathbb{D}$ onto one of the two half-planes; as $S(0) = i$, it is $\mathbb{H}$. For the second part, $z \mapsto iz$ maps $\{\operatorname{Re} z > 0\}$ onto $\mathbb{H}$, and then the Cayley transform gives $\frac{iz-i}{iz+i} = \frac{z-1}{z+1}$, which sends 1 to 0. □

Exercise 29.8.7. Let $\lambda = (z, z_1, z_2, z_3)$. Show that permuting the four points changes the cross-ratio into one of λ, $1-\lambda$, $1/\lambda$, $1/(1-\lambda)$, $(\lambda-1)/\lambda$, $\lambda/(\lambda-1)$, and that each value is obtained from four of the 24 permutations.

Solution. We may assume the four points are finite (move them by a Möbius map, Corollary 29.7.13(1)). The permutations $(z\, z_1)(z_2\, z_3)$, $(z\, z_2)(z_1\, z_3)$ and $(z\, z_3)(z_1\, z_2)$ leave the cross-ratio unchanged: for instance $(z_1, z, z_3, z_2) = \frac{(z_1-z)(z_3-z_2)}{(z_1-z_2)(z_3-z)}$, and numerator and denominator are those of (z, z_1, z_2, z_3), each with two signs changed; the other two are similar. With the identity they form a normal subgroup V of order 4, and every permutation is the product of an element of V and a permutation fixing the first entry, so it suffices to permute z_1, z_2, z_3. Moving them to $0, 1, \infty$, so that $z = \lambda$, and using the formulas of Theorem 29.7.11 for points at ∞:

$$(\lambda, 0, 1, \infty) = \lambda, \quad (\lambda, 1, 0, \infty) = 1-\lambda, \quad (\lambda, 0, \infty, 1) = \frac{\lambda}{\lambda-1}, \quad (\lambda, 1, \infty, 0) = \frac{\lambda-1}{\lambda}, \quad (\lambda, \infty, 0, 1) = \frac{1}{1-\lambda}, \quad (\lambda,$$

Each value is obtained from a coset of V, that is, from four permutations. □

Exercise 29.8.8. Classify the following Möbius transformations as parabolic, elliptic, hyperbolic or loxodromic, and find their fixed points: $z \mapsto \frac{z}{z+1}$, $z \mapsto \frac{2z+1}{z+1}$, $z \mapsto 2iz$, $z \mapsto -\frac{1}{z}$, $z \mapsto \frac{z-i}{z+i}$. Show that a Möbius transformation of finite order $n \geq 2$ is elliptic and conjugate to $z \mapsto e^{2\pi ij/n}z$ for some j prime to n.

Solution. Compute $t^2 = (\operatorname{tr} A)^2/\det A$, which is the invariant for the normalised matrix. $\frac{z}{z+1}$: $t^2 = 4$, parabolic, fixed point 0. $\frac{2z+1}{z+1}$: $t^2 = 9$, hyperbolic, fixed points $z^2 - z - 1 = 0$, $z = (1 \pm \sqrt{5})/2$. $2iz$: $t^2 = (1+2i)^2/(2i) = (-3+4i)/(2i) = 2 + \frac{3}{2}i$, loxodromic, fixed points $0, \infty$. $-1/z$: $t^2 = 0$, elliptic, fixed points $\pm i$. $\frac{z-i}{z+i}$: $t^2 = 4/(2i) = -2i$, loxodromic, fixed points the roots of $z^2 + (i-1)z + i = 0$. If $M^n = \mathrm{id}$ with $M \neq \mathrm{id}$, then M is not parabolic (the iterates of $z+1$ are $z+n$), so it is conjugate to $z \mapsto kz$ with $k^n = 1$, $k \neq 1$, which is elliptic; replacing n by the exact order, k is a primitive n-th root of unity. □

Exercise 29.8.9. Let $J(z) = \frac{1}{2}(z + z^{-1})$. Show that J is a rational function of degree 2 with critical points ± 1, that it maps the circle $|z| = r \neq 1$ onto an ellipse with foci ± 1 and the unit circle onto $[-1, 1]$, and that it maps $\mathbb{D} \setminus \{0\}$ biholomorphically onto $\mathbb{C} \setminus [-1, 1]$. (This *Joukowski map* was used to model the flow around aircraft wings.)

Solution. $J = (z^2+1)/(2z)$ has degree 2, and $J'(z) = \frac{1}{2}(1 - z^{-2})$ vanishes at ± 1. For $z = re^{i\theta}$, $J(z) = \frac{1}{2}(r+r^{-1})\cos\theta + \frac{i}{2}(r-r^{-1})\sin\theta$, an ellipse with semi-axes $a = \frac{1}{2}(r+r^{-1})$ and $b = \frac{1}{2}|r-r^{-1}|$, so $a^2 - b^2 = 1$ and the foci are ± 1; for $r = 1$ it is $\cos\theta \in [-1, 1]$. Next $J(z) = J(w)$ if and only if $(z-w)(1-\frac{1}{zw}) = 0$, that is, $w = z$ or $w = 1/z$; so J is injective on $\mathbb{D} \setminus \{0\}$. Every $w \in \mathbb{C}$ is taken twice counted with multiplicity (Proposition 29.7.6), by z and $1/z$; if $w \notin [-1, 1]$, neither lies on the unit circle, so exactly one lies in $\mathbb{D} \setminus \{0\}$. The inverse is holomorphic, since $J' \neq 0$ on $\mathbb{D} \setminus \{0\}$. □

Exercise 29.8.10. Let $a \neq b$ in $\mathbb{C}$ and $k > 0$. Show that $\{z : |z-a| = k|z-b|\}$ is a circle for $k \neq 1$ and the perpendicular bisector of a and b for $k = 1$ (a *circle of Apollonius*). Deduce that the images of the circles $|w| = k$ under the Möbius map $w \mapsto \frac{a-bw}{1-w}$, inverse of $z \mapsto \frac{z-a}{z-b}$, are these circles.

Solution. Squaring, $|z|^2 - 2\operatorname{Re}(z\bar{a}) + |a|^2 = k^2(|z|^2 - 2\operatorname{Re}(z\bar{b}) + |b|^2)$, that is, $(1-k^2)|z|^2 - 2\operatorname{Re}\big(z(\bar{a} - k^2\bar{b})\big) + |a|^2 - k^2|b|^2 = 0$, which is of the form (29.1); it is a line, the bisector, if $k = 1$. For $k \neq 1$ the set is nonempty and not a point (it meets the line through a and b in two points, dividing the segment internally and externally in the ratio k), so it is a circle. The second statement is the first, since $|w| = k$ means $|z-a| = k|z-b|$ for z the preimage of w. □

Exercise 29.8.11. Compute $\int_{|z|=3} \frac{z^5\,dz}{(z-1)(z^5+2)}$ using the residue at ∞.

Solution. All poles lie in $|z| < 3$. For large $|z|$, $\frac{z^5}{(z-1)(z^5+2)} = \frac{1}{z-1} \cdot \frac{1}{1+2z^{-5}} = z^{-1}(1 + z^{-1} + \cdots)(1 - 2z^{-5} + \cdots)$, whose coefficient of z^{-1} is 1. So $\operatorname{Res}_\infty = -1$ and, by Proposition 29.7.7, the integral is $2\pi i \sum_{a\in\mathbb{C}} \operatorname{Res}_a = -2\pi i \operatorname{Res}_\infty = 2\pi i$. □

Exercise 29.8.12. Let $f = u + iv$ be holomorphic on a connected open set U. Show that f is constant if any of the following holds: (1) u is constant; (2) f takes only real values; (3) $|f|$ is constant; (4) $\bar{f}$ is also holomorphic.

Solution. If $f' = 0$ on U, then f is constant on every disc in U (Lemma 29.2.8, integrating along segments), so the set where f takes a given value is open, as is its complement, and f is constant since U is connected. (1) If u is constant, $\partial_x u = \partial_y u = 0$, so $\partial_x v = -\partial_y u = 0$ by the Cauchy–Riemann equations, and $f' = \partial_x u + i\partial_x v = 0$. (2) Then $v = 0$, and $-if = v - iu$ has constant real part; apply (1). (3) If $|f|^2$ is constant, then $0 = \Delta|f|^2 = 4|f'|^2$ by Example 29.4.5(3). (4) Then $f + \bar{f} = 2u$ is holomorphic and real valued; by (2) u is constant, and (1) applies. □

Exercise 29.8.13. (*Cauchy–Riemann equations in polar coordinates.*) Let $f = u + iv$ be real differentiable on an open set $U \subseteq \mathbb{C}^\times$, and write $z = re^{i\theta}$. Show that f is holomorphic if and only if

$$\partial_r u = \frac{1}{r}\partial_\theta v, \qquad \partial_r v = -\frac{1}{r}\partial_\theta u,$$

and that then $f'(z) = e^{-i\theta}\partial_r f$. Deduce that the principal logarithm $\operatorname{Log} z = \log r + i\theta$, $-\pi < \theta < \pi$, is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$ with derivative $1/z$.

Solution. By the chain rule, $\partial_r = \cos\theta\,\partial_x + \sin\theta\,\partial_y$ and $\partial_\theta = -r\sin\theta\,\partial_x + r\cos\theta\,\partial_y$ on functions of $(x, y) = (r\cos\theta, r\sin\theta)$. Hence

$$\partial_\theta f - ir\,\partial_r f = r\big((-\sin\theta - i\cos\theta)\partial_x f + (\cos\theta - i\sin\theta)\partial_y f\big) = -ire^{-i\theta}\big(\partial_x f + i\,\partial_y f\big).$$

The Cauchy–Riemann equations say $\partial_x f + i\,\partial_y f = 0$ (its real and imaginary parts are $\partial_x u - \partial_y v$ and $\partial_x v + \partial_y u$). So f is holomorphic if and only if $\partial_\theta f = ir\,\partial_r f$, whose real and imaginary parts are the stated equations. Then $\partial_r f = \cos\theta\, f' + \sin\theta\, if' = e^{i\theta} f'$. For Log, which is real differentiable because (r, θ) are smooth functions of (x, y) on the slit plane, $u = \log r$ and $v = \theta$ give $\partial_r u = \frac{1}{r} = \frac{1}{r}\partial_\theta v$ and $\partial_r v = 0 = -\frac{1}{r}\partial_\theta u$, and $\operatorname{Log}'(z) = e^{-i\theta}/r = 1/z$. □

Exercise 29.8.14. For integers $m, n \geq 0$ compute $\int_\gamma z^n \bar{z}^m \, dz$ over the unit circle $\gamma(t) = e^{2\pi i t}$. Deduce that a polynomial $p(z, \bar{z}) = \sum c_{nm} z^n \bar{z}^m$ satisfies $\int_\gamma p \, dz = 2\pi i \sum_n c_{n,n+1}$; in particular this integral can be nonzero only if p really involves $\bar{z}$.

Solution. On the circle $\bar{z} = 1/z$, so the integrand is z^{n-m}, and by Lemma 29.2.8 the integral is $2\pi i$ if $n - m = -1$ and 0 otherwise. The formula for p follows by linearity, and if p does not involve $\bar{z}$ then $c_{n,n+1} = 0$ for all n. □

Chapter 30

Functional Analysis

30.1 Introduction

Hodge theory rests on an analytic theorem: on a compact Riemannian manifold every cohomology class has a unique harmonic representative. The proof solves the equation $\Delta u = f$ in a space of generalised functions and then shows that the solution is smooth. The functional analysis needed for the first step is in this chapter: Hilbert spaces and orthogonal projections, the consequences of Baire's theorem, compact operators, the spectral theorem for compact self-adjoint operators (Theorem 30.6.3) and the Fredholm alternative (Theorem 30.6.4). Around this core we develop the classical theory: the applications of Baire's theorem, among them a continuous function whose Fourier series diverges (Theorem 30.5.5); the spectrum of a bounded operator and the spectral radius formula (Theorem 30.7.3); the separation of convex sets, weak closures and the direct method of the calculus of variations (Section 30.10); and dual spaces and reflexivity (Section 30.11). The guiding theme is what survives from finite-dimensional linear algebra and what does not: in infinite dimensions linear maps need not be continuous, norms need not be equivalent, closed bounded sets need not be compact, injective operators need not be surjective, and bases have to be replaced by orthonormal bases, whose expansions are convergent series. We also introduce Fréchet spaces, such as the space of smooth functions, whose topology is given by countably many seminorms; they are the natural home of distributions (Chapter 32). References: [Bre11], [Rud91], [RS80].

Prerequisites. Chapter 28 (Measure and Integration), Chapter 11 (Linear Algebra) and Chapter 29 (Complex Analysis).

All vector spaces are over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

30.2 Banach spaces and bounded operators

Definition 30.2.1. A linear map $T\colon X \to Y$ between normed spaces is *bounded* if $\|T\| = \sup_{\|x\|\le 1} \|Tx\| < \infty$; equivalently (as for finite dimensions, Lemma 27.10.1) T is continuous. The bounded operators form a normed space $B(X,Y)$, and $X^* = B(X,\mathbb{K})$ is the *dual space*.

Proposition 30.2.2. *1. If Y is complete, so is $B(X,Y)$; in particular X^* is always a Banach space.*

2. $C(K,\mathbb{K})$ with the supremum norm (K compact Hausdorff), $L^p(\mu)$ for $1 \le p \le \infty$, and the sequence spaces ℓ^p are Banach spaces.

3. (Riesz) If the closed unit ball of a normed space X is compact, then X is finite-dimensional.

Proof. (1) A Cauchy sequence (T_n) is pointwise Cauchy, and the pointwise limit T is linear with $\|T - T_n\| \le \limsup_m \|T_m - T_n\| \to 0$. (2) Proposition 18.7.2 and Theorem 28.5.3. (3) Suppose X is infinite-dimensional. We construct unit vectors x_n with $\|x_n - x_m\| \ge 1/2$: given the finite-dimensional, hence closed (all norms on it are equivalent and it is complete), subspace Y_n spanned by $x_1, \dots, x_n$, choose $x \notin Y_n$ and $y \in Y_n$ with $\|x - y\| \le 2\operatorname{dist}(x, Y_n)$; then $x_{n+1} = (x - y)/\|x - y\|$ has distance at least $1/2$ from Y_n. Such a sequence has no convergent subsequence. □

Example 30.2.3.

1. On $\mathbb{K}^n$ the norms $\|x\|_p = (\sum_i |x_i|^p)^{1/p}$ $(1 \le p < \infty)$ and $\|x\|_\infty = \max_i |x_i|$ are all equivalent (Lemma 27.10.1); for instance $\|x\|_\infty \le \|x\|_2 \le \|x\|_1 \le n\|x\|_\infty$. Their unit balls are a cube, a ball and a cross-polytope.

2. On $C([0,1])$ the norms $\|f\|_\infty$ and $\|f\|_1 = \int_0^1 |f|$ are not equivalent: $\|f\|_1 \le \|f\|_\infty$, but $f_n(x) = x^n$ has $\|f_n\|_\infty = 1$ and $\|f_n\|_1 = 1/(n+1)$. So $f_n \to 0$ in one norm and not in the other, and $C([0,1])$ is complete for $\|\cdot\|_\infty$ but not for $\|\cdot\|_1$ (Exercise 30.12.5).

3. *Norms of operators.* A matrix $A\colon \mathbb{K}^n \to \mathbb{K}^m$ with Euclidean norms has $\|A\|$ = its largest singular value, the square root of the largest eigenvalue of A^*A (Proposition 11.10.9). For $g \in L^\infty([0,1])$ the multiplication operator $M_g f = gf$ on $L^2([0,1])$ has $\|M_g\| = \|g\|_\infty$ (Exercise 30.12.6). For a continuous kernel k on $[0,1]^2$, $(Tf)(x) = \int_0^1 k(x,y)f(y)\,dy$ is bounded on $C([0,1])$ with $\|T\| \le \sup_x \int_0^1 |k(x,y)|\,dy$.

Lemma 30.2.4. *A normed space X is complete if and only if every absolutely convergent series ($\sum \|x_n\| < \infty$) converges. If X is a Banach space and $M \subseteq X$ a closed subspace, the quotient X/M with $\|x + M\| = \operatorname{dist}(x, M)$ is a Banach space, and the projection $X \to X/M$ has norm at most* 1.

Proof. If X is complete, the partial sums of an absolutely convergent series are Cauchy (Proposition 27.5.4(1)). Conversely, given a Cauchy sequence choose $n_1 < n_2 < \cdots$ with $\|x_{n_{k+1}} - x_{n_k}\| < 2^{-k}$; the series $x_{n_1} + \sum_k (x_{n_{k+1}} - x_{n_k})$ converges absolutely, so the subsequence converges, and then so does the Cauchy sequence. For the quotient, $\|x + M\| = 0$ forces $x \in \overline{M} = M$, and the other axioms are clear, so it is a norm. If $\sum \|x_n + M\| < \infty$, choose $m_n \in M$ with $\|x_n - m_n\| \le \|x_n + M\| + 2^{-n}$; then $\sum (x_n - m_n)$ converges absolutely in X, to some x, and $\sum (x_n + M)$ converges to $x + M$. □

Remark 30.2.5. A normed space is *separable* if it has a countable dense subset. ℓ^p and $L^p([0,1])$ for $p < \infty$, and $C([0,1])$, are separable (finitely supported sequences with rational entries; step functions or polynomials with rational coefficients, Corollary 27.14.3). ℓ^∞ is not: the indicator sequences $\mathbf{1}_S$ of the uncountably many subsets $S \subseteq \mathbb{N}$ are at mutual distance 1. A *Hamel basis*, a basis in the sense of linear algebra, of an infinite-dimensional Banach space is uncountable: if $e_1, e_2, \dots$ were one, X would be the countable union of the closed subspaces $\operatorname{span}(e_1, \dots, e_n)$, which have empty interior, contradicting Baire's theorem (Theorem 18.7.7). This is why in analysis one works with expansions $x = \sum_n c_n e_n$ as convergent series instead: orthonormal bases in Hilbert spaces (Theorem 30.3.6) and, more generally, *Schauder bases*.

30.3 Hilbert spaces

Definition 30.3.1. A *Hilbert space* is a vector space with an inner product (Definition 11.10.1) that is complete for the norm $\|x\| = \langle x, x\rangle^{1/2}$. Vectors are *orthogonal* if $\langle x, y\rangle = 0$; for $S \subseteq H$, $S^\perp = \{x : \langle x, s\rangle = 0 \ \forall s \in S\}$, a closed subspace.

Example 30.3.2. $L^2(\mu)$ with $\langle f, g\rangle = \int f\bar{g}$ (Theorem 28.5.3); ℓ^2; finite-dimensional inner product spaces. The space $C([0,1])$ with the L^2 inner product is not complete, and L^2 is its completion (Exercise 18.11.5).

Theorem 30.3.3. *Let H be a Hilbert space and $C \subseteq H$ nonempty, closed and convex. Every $x \in H$ has a unique nearest point $P_C x \in C$. If M is a closed subspace, P_M is linear with $\|P_M\| \le 1$, $x - P_M x \in M^\perp$, and $H = M \oplus M^\perp$. Moreover $(M^\perp)^\perp = M$, and $M^\perp = 0$ if and only if M is dense.*

Proof. Let $d = \inf_{c\in C}\|x - c\|$ and $c_k \in C$ with $\|x - c_k\| \to d$. By the parallelogram law, $\|c_k - c_l\|^2 = 2\|x - c_k\|^2 + 2\|x - c_l\|^2 - 4\|x - \frac{c_k+c_l}{2}\|^2 \le 2\|x - c_k\|^2 + 2\|x - c_l\|^2 - 4d^2 \to 0$, using $\frac{c_k+c_l}{2} \in C$. So (c_k) is Cauchy and converges to some $c \in C$ with $\|x - c\| = d$; uniqueness follows from the same identity. For a closed subspace M and $m \in M$, $t \in \mathbb{K}$, the inequality $\|x - P_M x - tm\| \ge \|x - P_M x\|$ forces $\langle x - P_M x, m\rangle = 0$. Hence $x = P_M x + (x - P_M x) \in M + M^\perp$, and $M \cap M^\perp = 0$, which also gives linearity and $\|P_M\| \le 1$ (Pythagoras). Finally $M \subseteq (M^\perp)^\perp$, and if $x \in (M^\perp)^\perp$ then $x - P_M x \in M^\perp \cap (M^\perp)^\perp = 0$. If M is a subspace, $\overline{M}^\perp = M^\perp$, so $M^\perp = 0$ if and only if $\overline{M} = H$. □

Theorem 30.3.4 (Riesz representation)**.** *For every $\Lambda \in H^*$ there is a unique $y \in H$ with $\Lambda(x) = \langle x, y\rangle$ for all x, and $\|\Lambda\| = \|y\|$. Consequently every bounded operator $T\colon H \to H$ has a unique* adjoint *T^* with $\langle Tx, y\rangle = \langle x, T^*y\rangle$; it satisfies $\|T^*\| = \|T\|$ and $(\ker T)^\perp = \overline{\operatorname{im} T^*}$.*

Proof. Existence and uniqueness of y are Lemma 28.12.1, whose proof used only completeness; the norm equality follows from Cauchy–Schwarz and $\Lambda(y) = \|y\|^2$. For fixed y, $x \mapsto \langle Tx, y\rangle$ is a bounded functional, so it is $\langle x, T^*y\rangle$ for a unique T^*y, and T^* is linear and bounded with $\|T^*\| = \|T\|$ (from $\|T\| = \sup_{\|x\|,\|y\|\le 1}|\langle Tx, y\rangle|$, which is symmetric in T and T^*). Finally $x \in \ker T$ if and only if $\langle x, T^*y\rangle = 0$ for all y, that is, $x \in (\operatorname{im} T^*)^\perp$; take orthogonal complements. □

Proposition 30.3.5 (Bessel's inequality)**.** *Let $(e_i)_{i\in I}$ be an orthonormal family in an inner product space H. For every $x \in H$,*

$$\sum_{i\in I} |\langle x, e_i\rangle|^2 \le \|x\|^2;$$

in particular only countably many coefficients $\langle x, e_i\rangle$ are nonzero, and $\langle x, e_n\rangle \to 0$ for an orthonormal sequence.

Proof. For a finite $J \subseteq I$ let $y = \sum_{i\in J}\langle x, e_i\rangle e_i$. Then $\langle x - y, e_j\rangle = 0$ for $j \in J$, so $x - y \perp y$ and, by Pythagoras, $\|x\|^2 = \|y\|^2 + \|x - y\|^2 \ge \|y\|^2 = \sum_{i\in J}|\langle x, e_i\rangle|^2$. A sum of nonnegative terms is the supremum of its finite partial sums. For each k at most $k\|x\|^2$ indices have $|\langle x, e_i\rangle|^2 \ge 1/k$, so the set of nonzero coefficients is a countable union of finite sets. □

The vector y in the proof is the best approximation of x by combinations of e_i, $i \in J$: $\|x - y\|^2 = \|x\|^2 - \sum_{i\in J}|\langle x, e_i\rangle|^2$, and equality in Bessel's inequality for all x is Parseval's identity, which characterises orthonormal bases.

Theorem 30.3.6 (Orthonormal bases)**.** *Let $(e_i)_{i\in I}$ be an orthonormal family in a Hilbert space H. The following are equivalent: its span is dense; $x = \sum_i\langle x, e_i\rangle e_i$ for all x (with at most countably many nonzero terms, in any order); $\|x\|^2 = \sum_i|\langle x, e_i\rangle|^2$ for all x (Parseval). Such a family is an* orthonormal basis*; every Hilbert space has one, and H is separable if and only if it has a countable one, in which case $H \cong \ell^2(I)$.*

Proof. For finite $J \subseteq I$, $P_J x = \sum_{i\in J}\langle x, e_i\rangle e_i$ is the orthogonal projection onto the span of $(e_i)_{i\in J}$, so $\sum_{i\in J}|\langle x, e_i\rangle|^2 \le \|x\|^2$ (Bessel); hence at most countably many coefficients are nonzero and, by completeness, $\sum_i\langle x, e_i\rangle e_i$ converges (its partial sums are Cauchy by Pythagoras), to the

projection of x onto the closed span M. So the three conditions each say that $M = H$, using $\|x\|^2 = \|P_M x\|^2 + \|x - P_M x\|^2$. Zorn's lemma gives a maximal orthonormal family, whose closed span M satisfies $M^\perp = 0$ (a unit vector in $M^\perp$ would enlarge the family), so $M = H$. If H is separable, an orthonormal family is discrete ($\|e_i - e_j\| = \sqrt{2}$) hence countable. The map $x \mapsto (\langle x, e_i\rangle)$ is then an isometric isomorphism onto $\ell^2(I)$. □

Example 30.3.7. The functions $e_n(t) = e^{2\pi i n t}$, $n \in \mathbb{Z}$, form an orthonormal basis of $L^2(S^1)$, where $S^1 = \mathbb{R}/\mathbb{Z}$ with Lebesgue measure: they are orthonormal by direct computation, their span (the trigonometric polynomials) is dense in $C(S^1)$ for the supremum norm (Corollary 27.14.3), hence in L^2, and $C(S^1)$ is dense in $L^2(S^1)$: by Proposition 28.5.4 the continuous functions with compact support in $(0,1)$ are dense in $L^2([0,1])$, and they extend to continuous functions on S^1. So every $f \in L^2(S^1)$ is the L^2-limit of its Fourier series $\sum_n \hat{f}(n) e_n$ with $\hat{f}(n) = \int_0^1 f(t) e^{-2\pi i n t}\, dt$, and $\|f\|_2^2 = \sum_n |\hat{f}(n)|^2$.

Example 30.3.8. *Orthonormal bases of $L^2([0,1])$.* Up to the null set $\{1\}$, $[0,1)$ is the circle $\mathbb{R}/\mathbb{Z}$, so Example 30.3.7 gives the basis $e^{2\pi i n t}$, $n \in \mathbb{Z}$.

1. *Real Fourier basis.* Taking real and imaginary parts, $1, \sqrt{2}\cos(2\pi n t), \sqrt{2}\sin(2\pi n t)$ $(n \geq 1)$ is an orthonormal basis of the real space $L^2([0,1])$: these functions are orthonormal ($\int_0^1 \cos^2(2\pi n t)\, dt = \frac{1}{2}$), and they span the same space of trigonometric polynomials, over $\mathbb{C}$, as the $e^{\pm 2\pi i n t}$.

2. *Legendre polynomials.* Gram–Schmidt (Theorem 11.10.3) applied to $1, x, x^2, \ldots$ gives orthonormal polynomials $P_0, P_1, P_2, \ldots$ with $\deg P_n = n$. With $\langle f, g\rangle = \int_0^1 fg$: $P_0 = 1$; $x - \langle x, 1\rangle = x - \frac{1}{2}$ has norm^2 $\int_0^1 (x - \frac{1}{2})^2 = \frac{1}{12}$, so $P_1 = \sqrt{3}(2x-1)$; and $x^2 - \langle x^2, P_0\rangle P_0 - \langle x^2, P_1\rangle P_1 = x^2 - \frac{1}{3} - \frac{\sqrt{3}}{6}\sqrt{3}(2x-1) = x^2 - x + \frac{1}{6}$, whose norm^2 is $\frac{1}{180}$, so $P_2 = \sqrt{5}(6x^2 - 6x + 1)$. They form an orthonormal basis: their span is the space of all polynomials, which is dense in $C([0,1])$ for the supremum norm (Corollary 27.14.3), hence in L^2 (since $\|f\|_2 \leq \|f\|_\infty$ and $C([0,1])$ is dense in L^2, Proposition 28.5.4), and Theorem 30.3.6 applies.

3. *The Haar basis.* Let $h = \mathbf{1}_{[0,1/2)} - \mathbf{1}_{[1/2,1)}$ and $h_{n,k}(t) = 2^{n/2} h(2^n t - k)$ for $n \geq 0$, $0 \leq k < 2^n$. Together with 1 these are orthonormal: two of them either have disjoint supports, or the support of one lies in an interval where the other is constant, and then the integral of the first vanishes. The 2^N functions 1 and $h_{n,k}$ with $n < N$ lie in the 2^N-dimensional space of functions constant on the dyadic intervals $[j2^{-N}, (j+1)2^{-N})$ and are independent, so they span it. Such step functions approximate continuous functions uniformly, so the span is dense, and this is an orthonormal basis; unlike the Fourier basis it consists of functions with small support, the starting point of wavelets.

Example 30.3.9. *Parseval and $\sum 1/n^2$.* For $f(t) = t$ on $[0,1)$, $\hat{f}(0) = \frac{1}{2}$ and, integrating by parts, $\hat{f}(n) = \int_0^1 t e^{-2\pi i n t}\, dt = \left[\frac{t e^{-2\pi i n t}}{-2\pi i n}\right]_0^1 = \frac{i}{2\pi n}$ for $n \neq 0$. Parseval's identity gives $\frac{1}{3} = \int_0^1 t^2\, dt = \frac{1}{4} + \sum_{n \neq 0} \frac{1}{4\pi^2 n^2}$, that is, $\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$. Note that the Fourier series of f converges to f in L^2, not pointwise at $t = 0$, where the 1-periodic extension of f jumps.

30.4 The Hahn–Banach and Baire theorems

Theorem 30.4.1 (Hahn–Banach). *Let X be a vector space over $\mathbb{K}$ and $p\colon X \to \mathbb{R}$ with $p(x+y) \leq p(x) + p(y)$ and $p(\lambda x) = |\lambda| p(x)$ (a seminorm). Every linear functional ℓ on a subspace Y with*

$|\ell| \le p$ on Y extends to a linear functional on X with $|\ell| \le p$. Consequently, for a normed X and $x_0 \neq 0$ there is $\Lambda \in X^*$ with $\|\Lambda\| = 1$ and $\Lambda(x_0) = \|x_0\|$; in particular X^* separates points and $\|x\| = \sup_{\|\Lambda\|\le 1} |\Lambda(x)|$.

Proof. Real case: consider pairs (Z, m) of a subspace $Z \supseteq Y$ and an extension $m \le p$, ordered by extension; chains have upper bounds, so Zorn's lemma gives a maximal (Z, m). If $x \notin Z$, then for $z, z' \in Z$, $m(z) + m(z') = m(z + z') \le p(z + x) + p(z' - x)$, so $\sup_{z'}(m(z') - p(z' - x)) \le \inf_z(p(z + x) - m(z))$; choosing c between them extends m to $Z + \mathbb{K}x$ by $m(z + tx) = m(z) + tc$, contradicting maximality. Complex case: apply the real case to $\operatorname{Re}\ell$ and set $\ell(x) = (\operatorname{Re}\ell)(x) - i(\operatorname{Re}\ell)(ix)$. For the consequence take $Y = \mathbb{K}x_0$, $\ell(tx_0) = t\|x_0\|$ and $p = \|\cdot\|$. □

Theorem 30.4.2. *Let X, Y be Banach spaces.*

1. *(Uniform boundedness) If $\mathcal{T} \subseteq B(X, Y)$ satisfies $\sup_{T\in\mathcal{T}} \|Tx\| < \infty$ for every x, then $\sup_{T\in\mathcal{T}} \|T\| < \infty$.*

2. *(Open mapping) A surjective $T \in B(X, Y)$ is open; if it is bijective, T^{-1} is bounded.*

3. *(Closed graph) A linear $T\colon X \to Y$ whose graph is closed in $X \times Y$ is bounded.*

Proof. (1) The sets $F_n = \{x : \|Tx\| \le n \ \forall T \in \mathcal{T}\}$ are closed with union X, so by Baire's theorem (Theorem 18.7.7) some F_n contains a ball $\overline{B}(x_0, r)$; then $\|Tx\| \le 2n/r$ for $\|x\| \le 1$, using $x = ((x_0 + rx) - x_0)/r$.

(2) By Baire's theorem applied to $Y = \bigcup_n \overline{T(B(0, n))}$, the closure of $T(B(0, 1))$ contains a ball $B(y_0, \delta)$; by symmetry and convexity it contains $B(0, \delta)$. A standard iteration removes the closure: given $\|y\| < \delta$, choose x_1 with $\|x_1\| < 1$ and $\|y - Tx_1\| < \delta/2$, then x_2 with $\|x_2\| < 1/2$ and $\|y - Tx_1 - Tx_2\| < \delta/4$, and so on; the series $\sum x_k$ converges to some x with $\|x\| < 2$ and $Tx = y$. So $T(B(0, 2)) \supseteq B(0, \delta)$, which gives openness by translation and scaling.

(3) The graph G is a Banach space, and the projection $G \to X$ is bounded and bijective, so its inverse $x \mapsto (x, Tx)$ is bounded by (2). □

Counterexample 30.4.3.

1. *An unbounded operator.* Differentiation $D\colon C^1([0, 1]) \to C([0, 1])$, both with the supremum norm, is linear but not bounded: $\|x^n\|_\infty = 1$ and $\|D(x^n)\|_\infty = n$. It is closed (if $f_k \to f$ and $f_k' \to g$ uniformly, then $f' = g$, Theorem 27.9.4), which does not contradict the closed graph theorem because $C^1([0, 1])$ is not complete for the supremum norm.

2. *An incomplete normed space.* The polynomials, with the supremum norm on $[0, 1]$: the partial sums of $\sum x^k/k!$ form a Cauchy sequence whose uniform limit e^x is not a polynomial, since all its derivatives are nonzero.

3. *Completeness is needed for uniform boundedness.* On the space c_{00} of finitely supported sequences with the supremum norm, the functionals $T_n(x) = nx_n$ satisfy $\sup_n |T_n x| < \infty$ for each x, but $\|T_n\| = n$.

4. *Discontinuous linear functionals.* Every infinite-dimensional normed space X has one. Choose linearly independent unit vectors $e_1, e_2, \ldots$ and extend them, by Zorn's lemma (Theorem 3.7.2), to a maximal linearly independent set B, which is a basis of X as a vector space. The linear functional with $\ell(e_n) = n$ and $\ell = 0$ on $B \setminus \{e_1, e_2, \ldots\}$ is unbounded. In finite dimensions every linear functional is continuous (Lemma 27.10.1).

30.5 Applications of the Baire theorems

The three theorems of Theorem 30.4.2 turn qualitative information (a limit exists for every vector, a map is bijective, a graph is closed) into quantitative information (a uniform bound). We give some classical consequences, culminating in the existence of continuous functions whose Fourier series diverge.

Corollary 30.5.1. *Let X be a vector space that is complete for two norms $\|\cdot\|_1$ and $\|\cdot\|_2$, with $\|x\|_1 \le C\|x\|_2$ for all x. Then the two norms are equivalent: $\|x\|_2 \le C'\|x\|_1$ for some C'.*

Proof. The identity map $(X, \|\cdot\|_2) \to (X, \|\cdot\|_1)$ is bounded and bijective between Banach spaces, so its inverse is bounded by the open mapping theorem (Theorem 30.4.2(2)). □

For instance, $C^1([0,1])$ is complete both for $\|f\|_\infty + \|f'\|_\infty$ and for $|f(0)| + \|f'\|_\infty$ (Exercise 30.12.10), and the second norm is bounded by the first, so they are equivalent. On the other hand $C([0,1])$ is complete for $\|\cdot\|_\infty$ and not for $\|\cdot\|_1 \le \|\cdot\|_\infty$, and these norms are not equivalent (Example 30.2.3(2)): completeness of both norms cannot be dropped.

Corollary 30.5.2 (Hellinger–Toeplitz)**.** *Let H be a Hilbert space and $T\colon H \to H$ a linear map, not assumed continuous, with $\langle Tx, y\rangle = \langle x, Ty\rangle$ for all $x, y \in H$. Then T is bounded.*

Proof. We show that the graph of T is closed and apply the closed graph theorem (Theorem 30.4.2(3)). Let $x_n \to x$ and $Tx_n \to z$. For every y, $\langle z, y\rangle = \lim_n \langle Tx_n, y\rangle = \lim_n \langle x_n, Ty\rangle = \langle x, Ty\rangle = \langle Tx, y\rangle$, so $z = Tx$. □

The operators of quantum mechanics, such as the position operator $f(x) \mapsto xf(x)$ or the momentum operator $f \mapsto -if'$ on $L^2(\mathbb{R})$, are symmetric and unbounded. By the Hellinger–Toeplitz theorem they cannot be defined on all of $L^2(\mathbb{R})$: they are only defined on dense subspaces, which is the origin of the theory of unbounded operators.

Corollary 30.5.3 (Banach–Steinhaus)**.** *Let X, Y be Banach spaces and $T_n \in B(X,Y)$ such that $Tx = \lim_n T_n x$ exists for every $x \in X$. Then $\sup_n \|T_n\| < \infty$, and T is a bounded operator with $\|T\| \le \liminf_n \|T_n\|$.*

Proof. Convergent sequences are bounded, so $\sup_n \|T_n x\| < \infty$ for every x, and the uniform boundedness principle (Theorem 30.4.2(1)) gives $\sup_n \|T_n\| < \infty$. The limit T is clearly linear, and $\|Tx\| = \lim_n \|T_n x\| \le \liminf_n \|T_n\|\,\|x\|$. □

30.5.1 Divergence of Fourier series

The Fourier series of $f \in L^2(S^1)$ converges to f in L^2 (Example 30.3.7), but this says nothing about convergence at a given point. For a continuous f one might hope that the Fourier series converges at every point. The uniform boundedness principle shows that this fails. Throughout, $S^1 = \mathbb{R}/\mathbb{Z}$, functions on S^1 are 1-periodic functions on $\mathbb{R}$, and for $f \in L^1(S^1)$ we write

$$\hat f(n) = \int_0^1 f(t)e^{-2\pi int}\,dt, \qquad S_N f(x) = \sum_{n=-N}^{N} \hat f(n) e^{2\pi inx}.$$

Integrals of 1-periodic functions over an interval of length 1 do not depend on the interval, and are invariant under $t \mapsto t + c$ and $t \mapsto -t$.

Lemma 30.5.4 (The Dirichlet kernel)**.** *For $f \in L^1(S^1)$, $S_N f(x) = \int_0^1 f(x-t)D_N(t)\,dt$, where*

$$D_N(t) = \sum_{n=-N}^{N} e^{2\pi int} = \frac{\sin((2N+1)\pi t)}{\sin(\pi t)} \quad (t \notin \mathbb{Z}), \qquad D_N(t) = 2N+1 \quad (t \in \mathbb{Z}).$$

The Lebesgue constants $L_N = \int_0^1 |D_N(t)|\,dt$ *satisfy* $L_N \geq \frac{4}{\pi^2}\sum_{k=1}^{N}\frac{1}{k}$*; in particular* $L_N \to \infty$*.*

Proof. Inserting the definition of $\hat{f}(n)$, $S_N f(x) = \int_0^1 f(s)\sum_{|n|\leq N} e^{2\pi i n(x-s)}\,ds = \int_0^1 f(s)D_N(x-s)\,ds$, and the substitution $t = x - s$ gives the first formula. For $q = e^{2\pi i t} \neq 1$, the geometric sum is

$$\sum_{n=-N}^{N} q^n = \frac{q^{N+1} - q^{-N}}{q-1} = \frac{q^{N+1/2} - q^{-N-1/2}}{q^{1/2} - q^{-1/2}} = \frac{\sin((2N+1)\pi t)}{\sin(\pi t)},$$

with $q^{1/2} = e^{\pi i t}$ and $e^{i\theta} - e^{-i\theta} = 2i\sin\theta$. Since D_N is even and 1-periodic, $L_N = 2\int_0^{1/2} |D_N(t)|\,dt$. On $(0, \frac{1}{2}]$ we have $0 < \sin(\pi t) \leq \pi t$, so, substituting $u = (2N+1)t$,

$$L_N \geq 2\int_0^{1/2} \frac{|\sin((2N+1)\pi t)|}{\pi t}\,dt = \frac{2}{\pi}\int_0^{N+1/2} \frac{|\sin(\pi u)|}{u}\,du \geq \frac{2}{\pi}\sum_{k=0}^{N-1}\frac{1}{k+1}\int_k^{k+1} |\sin(\pi u)|\,du = \frac{4}{\pi^2}\sum_{k=1}^{N}\frac{1}{k},$$

as $\int_k^{k+1} |\sin(\pi u)|\,du = 2/\pi$. The harmonic series diverges, so $L_N \to \infty$. □

The numbers L_N grow like $\frac{4}{\pi^2}\log N$. The kernel D_N is large near 0 and oscillates with slowly decaying amplitude elsewhere; its integral is $\int_0^1 D_N = 1$, but the integral of its absolute value is unbounded. This is what makes pointwise convergence delicate.

Theorem 30.5.5 (du Bois-Reymond)**.** *There is a continuous function f on S^1 whose Fourier series diverges at 0:* $\sup_N |S_N f(0)| = \infty$*. More precisely, the set of continuous functions with* $\sup_N |S_N f(0)| = \infty$ *contains a countable intersection of dense open subsets of $C(S^1)$, and in particular it is dense.*

Proof. Consider the Banach space $C(S^1)$ with the supremum norm and the linear functionals $\Lambda_N(f) = S_N f(0) = \int_0^1 f(-t)D_N(t)\,dt = \int_0^1 f(t)D_N(t)\,dt$, using that D_N is even. Clearly $|\Lambda_N f| \leq L_N\|f\|_\infty$. We claim $\|\Lambda_N\| = L_N$. The function $g = \operatorname{sign} D_N$ takes the values ± 1 except at the finitely many zeros $j/(2N+1)$ of D_N in $[0,1]$, and is constant between consecutive zeros. Let f_k be the continuous 1-periodic function that equals g outside the intervals of length $1/k$ centred at the zeros, and is linear on these intervals; then $|f_k| \leq 1$ and $f_k \to g$ except at the zeros, so $\Lambda_N(f_k) \to \int_0^1 gD_N = L_N$ by dominated convergence (Theorem 28.4.8). Hence $\|\Lambda_N\| = L_N \to \infty$ by Lemma 30.5.4.

Now let $F_n = \{f \in C(S^1) : |\Lambda_N f| \leq n \text{ for all } N\}$, a closed set. If some F_n had nonempty interior, the argument in the proof of the uniform boundedness principle (Theorem 30.4.2(1)) would give $\sup_N \|\Lambda_N\| < \infty$, which is false. So each F_n is closed with empty interior, its complement U_n is dense and open, and $\bigcap_n U_n$, which is the set of f with $\sup_N |S_N f(0)| = \infty$, is dense by Baire's theorem (Theorem 18.7.7). □

Translating, for every point x there are continuous functions whose Fourier series diverges at x, and with a little more work one finds continuous functions whose Fourier series diverges on a dense set of points. The proof is not constructive, although explicit examples exist. On the positive side:

1. If f is differentiable at x, or merely satisfies $|f(t) - f(x)| \leq C|t - x|$ near x, then $S_N f(x) \to f(x)$ (Exercise 30.12.12).

2. The averages $\sigma_N f = \frac{1}{N+1}(S_0 f + \cdots + S_N f)$ converge uniformly to f for every continuous f (Fejér's theorem, Exercise 30.12.11): averaging replaces D_N by a positive kernel.

3. For $f \in L^2(S^1)$ the Fourier series converges almost everywhere; this deep theorem is due to Carleson [Car66]. For L^1 it fails: Kolmogorov constructed an integrable function whose Fourier series diverges almost everywhere [Kol23].

Proposition 30.5.6 (Riemann–Lebesgue lemma)**.** *For $f \in L^1(S^1)$, $|\hat{f}(n)| \le \|f\|_1$ and $\hat{f}(n) \to 0$ as $|n| \to \infty$. The map $\mathcal{F} : L^1(S^1) \to c_0(\mathbb{Z})$, $f \mapsto (\hat{f}(n))_{n\in\mathbb{Z}}$, into the Banach space of sequences tending to 0 with the supremum norm, is linear, bounded and injective.*

Proof. The bound is clear from $|e^{-2\pi int}| = 1$. The trigonometric polynomials are dense in $L^1(S^1)$: continuous functions are dense in $L^1([0,1])$, as in Example 30.3.7 (Proposition 28.5.4 with $p = 1$), and trigonometric polynomials approximate continuous functions on S^1 uniformly (Corollary 27.14.3), hence in L^1 since $\|\cdot\|_1 \le \|\cdot\|_\infty$. Given $\varepsilon > 0$, choose a trigonometric polynomial p with $\|f - p\|_1 < \varepsilon$; then $\hat{p}(n) = 0$ for $|n|$ large, and for such n, $|\hat{f}(n)| \le |\hat{p}(n)| + \|f - p\|_1 < \varepsilon$. So $\mathcal{F}$ takes values in $c_0(\mathbb{Z})$, which is a closed subspace of the Banach space $\ell^\infty(\mathbb{Z})$, and $\|\mathcal{F}\| \le 1$.

For injectivity let $\hat{f} = 0$. Then $\int_0^1 fp = 0$ for every trigonometric polynomial p, and by uniform approximation $\int_0^1 f\phi = 0$ for every continuous ϕ on S^1, since $|\int f(\phi - p)| \le \|f\|_1\|\phi - p\|_\infty$. For $0 < a < b < 1$ there are continuous functions ϕ_k with $0 \le \phi_k \le 1$, supported in $(0,1)$, converging pointwise to $\mathbf{1}_{[a,b]}$ (piecewise linear), so $\int_a^b f = 0$ by dominated convergence. Then the averages $\frac{1}{2r}\int_{x-r}^{x+r} f$ vanish for $0 < x - r < x + r < 1$, and the Lebesgue differentiation theorem (Theorem 28.9.4) gives $f = 0$ almost everywhere. □

Proposition 30.5.7. *The map $\mathcal{F} : L^1(S^1) \to c_0(\mathbb{Z})$ is not surjective: there are sequences tending to 0 that are not the Fourier coefficients of any integrable function.*

Proof. If $\mathcal{F}$ were surjective, it would be a bounded bijection between Banach spaces, so by the open mapping theorem there would be a constant C with $\|f\|_1 \le C \sup_n |\hat{f}(n)|$ for all $f \in L^1(S^1)$. Apply this to $f = D_N$: its Fourier coefficients are 1 for $|n| \le N$ and 0 otherwise, so $L_N = \|D_N\|_1 \le C$ for all N, contradicting Lemma 30.5.4. □

In contrast, on L^2 the Fourier transform is an isometry onto $\ell^2(\mathbb{Z})$ (Theorem 30.3.6): on L^2 the coefficients are characterised by $\sum |c_n|^2 < \infty$, while no such description of the Fourier coefficients of integrable functions is known.

30.6 Compact operators

Definition 30.6.1. $T \in B(X, Y)$ is *compact* if the closure of $T(B(0,1))$ is compact; equivalently, every bounded sequence (x_n) has a subsequence with (Tx_{n_k}) convergent.

Proposition 30.6.2.

1. *Compact operators form a closed subspace of $B(X, Y)$, and STR is compact if T is compact and S, R are bounded.*
2. *Operators of finite rank are compact; on Hilbert spaces, T is compact if and only if it is a norm limit of finite rank operators.*
3. *If H is a Hilbert space, $T \in B(H)$ is compact if and only if T^* is.*

Proof. (1) Sums and scalar multiples are clear, and compositions with bounded operators map bounded sets to bounded sets. If $T_n \to T$ with T_n compact, then $T(B)$ is totally bounded: given ε, choose n with $\|T - T_n\| < \varepsilon$ and cover $T_n(B)$ by finitely many ε-balls; so $\overline{T(B)}$ is compact (Theorem 18.7.4). (2) A bounded set in a finite-dimensional space is

relatively compact (Theorem 18.6.5). Conversely let T be compact and $\varepsilon > 0$; cover $\overline{T(B)}$ by finitely many balls $B(y_i, \varepsilon)$, let P be the orthogonal projection onto the span of the y_i (Theorem 30.3.3); then $\|Tx - PTx\| \le \|Tx - y_i\| < \varepsilon$ for a suitable i, since P is a projection and $\|Tx - PTx\| = \operatorname{dist}(Tx, \operatorname{im} P)$. So $\|T - PT\| \le \varepsilon$ with PT of finite rank. (3) follows from (2), since $(PT)^* = T^*P$ has finite rank and $\|T^* - (PT)^*\| = \|T - PT\|$. □

Theorem 30.6.3 (Spectral theorem for compact self-adjoint operators)**.** *Let $H \neq 0$ be a Hilbert space and $T \in B(H)$ compact and self-adjoint ($T = T^*$).*

1. *$\|T\| = \sup_{\|x\|=1} |\langle Tx, x\rangle|$, and $\|T\|$ or $-\|T\|$ is an eigenvalue of T.*
2. *H has an orthonormal basis consisting of eigenvectors of T. The nonzero eigenvalues are real, form a finite or countable set with 0 as the only possible accumulation point, and each nonzero eigenvalue has finite-dimensional eigenspace.*

Proof. (1) Let $s = \sup_{\|x\|=1} |\langle Tx, x\rangle| \le \|T\|$. For x, y with $\|x\|, \|y\| \le 1$, the identity $\langle T(x + y), x + y\rangle - \langle T(x - y), x - y\rangle = 4\operatorname{Re}\langle Tx, y\rangle$ (using self-adjointness) and the parallelogram law give $4\operatorname{Re}\langle Tx, y\rangle \le s(\|x + y\|^2 + \|x - y\|^2) = 2s(\|x\|^2 + \|y\|^2) \le 4s$, so $\|T\| = \sup |\langle Tx, y\rangle| \le s$. Choose unit vectors x_n with $\langle Tx_n, x_n\rangle \to \lambda$ where $|\lambda| = \|T\|$ (note $\langle Tx, x\rangle$ is real). Then $\|Tx_n - \lambda x_n\|^2 = \|Tx_n\|^2 - 2\lambda\langle Tx_n, x_n\rangle + \lambda^2 \le 2\lambda^2 - 2\lambda\langle Tx_n, x_n\rangle \to 0$. By compactness, after passing to a subsequence $Tx_n \to y$; if $\lambda \neq 0$ then $x_n \to y/\lambda$, so y/λ is a unit vector with $Ty/\lambda = \lambda y/\lambda$. If $\lambda = 0$ then $T = 0$ and every vector is an eigenvector.

(2) Eigenvalues are real and eigenvectors for distinct eigenvalues are orthogonal, as in finite dimensions. If $|\lambda| \ge \varepsilon > 0$, an orthonormal family in the λ-eigenspace is mapped by T to a family of vectors at mutual distance $|\lambda|\sqrt{2}$, so it must be finite by compactness; the same argument shows that only finitely many eigenvalues satisfy $|\lambda| \ge \varepsilon$. Let M be the closed span of all eigenvectors. It is T-invariant, hence so is $M^\perp$ (if $x \in M^\perp$ and $m \in M$, $\langle Tx, m\rangle = \langle x, Tm\rangle = 0$), and $T|_{M^\perp}$ is compact and self-adjoint; by (1) it has an eigenvector unless $M^\perp = 0$ or $T|_{M^\perp} = 0$. An eigenvector in $M^\perp$ is impossible, and if $T|_{M^\perp} = 0$ then $M^\perp$ consists of eigenvectors for the eigenvalue 0, so $M^\perp \subseteq M$ and $M^\perp = 0$. Hence $M = H$, and an orthonormal basis of each eigenspace, together, is an orthonormal basis of H by Theorem 30.3.6. □

Theorem 30.6.4 (Fredholm alternative)**.** *Let H be a Hilbert space, $K \in B(H)$ compact and $T = I - K$.*

1. *$\ker T$ is finite-dimensional and $\operatorname{im} T$ is closed, with $\operatorname{im} T = (\ker T^*)^\perp$.*
2. *T is injective if and only if it is surjective.*
3. *$\dim \ker T = \dim \ker T^*$; so the equation $Tx = y$ is solvable if and only if $y \perp \ker T^*$, and the solution is unique up to $\ker T$, which has the same finite dimension as $\ker T^*$.*

Proof. (1) On $\ker T$ we have $K = I$, so the unit ball of $\ker T$ is compact and $\ker T$ is finite-dimensional (Proposition 30.2.2(3)). For closedness, let $M = (\ker T)^\perp$; it suffices to show $\|Tx\| \ge c\|x\|$ on M for some $c > 0$, since then images of Cauchy sequences are Cauchy. If not, there are unit vectors $x_n \in M$ with $Tx_n \to 0$; passing to a subsequence, $Kx_n \to y$, so $x_n = Tx_n + Kx_n \to y$ with $\|y\| = 1$, $y \in M$ and $Ty = 0$, a contradiction. Then $\operatorname{im} T$ is closed and $\operatorname{im} T = \overline{\operatorname{im} T} = (\ker T^*)^\perp$ by Theorem 30.3.4 applied to T^*.

(2) Suppose T is injective but not surjective. Then $H \supsetneq TH \supsetneq T^2H \supsetneq \cdots$ is a strictly decreasing chain of closed subspaces (each T^kH is closed by (1) applied to $I - K_k$ with $K_k = I - T^k$ compact, since $T^k = (I - K)^k = I - K_k$ with K_k a sum of products involving K). Choose unit vectors $u_k \in T^kH \cap (T^{k+1}H)^\perp$. For $l > k$, $Ku_k - Ku_l = u_k - (Tu_k + u_l - Tu_l)$ and

$Tu_k + u_l - Tu_l \in T^{k+1}H$, so $\|Ku_k - Ku_l\| \geq 1$, contradicting compactness. Conversely, if T is surjective, then T^* is injective: indeed $\ker T^* = (\operatorname{im} T)^\perp = 0$; applying the previous argument to $T^* = I - K^*$ (with K^* compact) shows T^* is surjective, hence $\ker T = (\operatorname{im} T^*)^\perp = 0$.

(3) This is the statement that the index of $I - K$ is zero; see [Bre11, Theorem 6.6]. □

Remark 30.6.5. $T \in B(X, Y)$ is *Fredholm* if $\ker T$ and $\operatorname{coker} T = Y/\operatorname{im} T$ are finite-dimensional; then $\operatorname{im} T$ is closed, and the *index* $\operatorname{ind} T = \dim \ker T - \dim \operatorname{coker} T$ is unchanged by compact perturbations and is locally constant in T [Bre11, Chapter 6]. Elliptic operators on compact manifolds are Fredholm (Chapter 42); the index is then computed by the Atiyah–Singer theorem, and its vanishing for the Laplacian is what produces harmonic representatives.

Counterexample 30.6.6. Self-adjointness is needed in the spectral theorem.

1. The *Volterra operator* $Vf(x) = \int_0^x f(t)\,dt$ on $L^2([0,1])$ is compact but has no eigenvalues. It is compact: by the Cauchy–Schwarz inequality $|Vf(x) - Vf(y)| \leq |x-y|^{1/2}\|f\|_2$ and $|Vf(x)| \leq \|f\|_2$, so V maps the unit ball to a bounded equicontinuous set of continuous functions, which is relatively compact in $C([0,1])$ by the Arzelà–Ascoli theorem (Theorem 27.14.4), hence in L^2. If $Vf = \lambda f$ with $\lambda \neq 0$, then $f = \lambda^{-1}Vf$ is continuous, hence continuously differentiable, with $f' = \lambda^{-1}f$ and $f(0) = 0$, so $f = 0$; and $Vf = 0$ forces $\int_0^x f = 0$ for all x, hence $f = 0$ almost everywhere.
2. The *right shift* $S(x_1, x_2, \ldots) = (0, x_1, x_2, \ldots)$ on ℓ^2 is an isometry without eigenvalues ($Sx = \lambda x$ gives $\lambda x_1 = 0$, $\lambda x_{k+1} = x_k$, hence $x = 0$), and it is not surjective. Its adjoint, the left shift, has every λ with $|\lambda| < 1$ as an eigenvalue, with eigenvector $(1, \lambda, \lambda^2, \ldots)$. So an injective bounded operator on an infinite-dimensional space need not be surjective, unlike in finite dimensions.

30.7 The spectrum of a bounded operator

In finite dimensions an operator T fails to be invertible exactly at its eigenvalues. In infinite dimensions $T - \lambda$ can be injective without being surjective, and the right generalisation of the set of eigenvalues is the spectrum. Throughout this section X is a complex Banach space and $B(X) = B(X, X)$; note that $\|ST\| \leq \|S\|\,\|T\|$, so $\|T^n\| \leq \|T\|^n$.

Lemma 30.7.1 (Neumann series)**.** *Let $T \in B(X)$ with $\|T\| < 1$. Then $I - T$ is invertible in $B(X)$, with*

$$(I-T)^{-1} = \sum_{n \geq 0} T^n, \qquad \|(I-T)^{-1}\| \leq \frac{1}{1 - \|T\|}.$$

More generally, if $S \in B(X)$ is invertible and $\|R\| < \|S^{-1}\|^{-1}$, then $S - R$ is invertible, and $(S-R)^{-1} \to S^{-1}$ as $R \to 0$. So the invertible operators form an open subset of $B(X)$, on which inversion is continuous.

Proof. The series converges absolutely, $\sum \|T^n\| \leq \sum \|T\|^n = (1 - \|T\|)^{-1}$, so it converges in the Banach space $B(X)$ (Proposition 30.2.2(1) and Lemma 30.2.4). Its sum A satisfies $(I-T)\sum_{n \leq N} T^n = I - T^{N+1} = (\sum_{n \leq N} T^n)(I - T)$, and $T^{N+1} \to 0$, so $(I-T)A = I = A(I-T)$. For the second statement write $S - R = S(I - S^{-1}R)$ with $\|S^{-1}R\| \leq \|S^{-1}\|\,\|R\| < 1$; then $(S-R)^{-1} = \sum_{n \geq 0}(S^{-1}R)^n S^{-1}$, and $\|(S-R)^{-1} - S^{-1}\| \leq \sum_{n \geq 1}(\|S^{-1}\|\,\|R\|)^n \|S^{-1}\| \to 0$ as $R \to 0$. □

Definition 30.7.2. Let $T \in B(X)$. The *resolvent set* $\rho(T)$ is the set of $\lambda \in \mathbb{C}$ such that $T - \lambda = T - \lambda I$ is bijective; then its inverse $R(\lambda) = (T - \lambda)^{-1}$, the *resolvent*, is bounded by the open mapping theorem (Theorem 30.4.2(2)). The *spectrum* is $\sigma(T) = \mathbb{C} \setminus \rho(T)$, and the *spectral radius* is $r(T) = \sup\{|\lambda| : \lambda \in \sigma(T)\}$. The *point spectrum* $\sigma_p(T) \subseteq \sigma(T)$ is the set of eigenvalues, the λ for which $T - \lambda$ is not injective.

If $\dim X < \infty$, an injective operator is bijective, so $\sigma(T) = \sigma_p(T)$ is the set of eigenvalues, the roots of the characteristic polynomial.

Theorem 30.7.3. *Let $X \neq 0$ be a complex Banach space and $T \in B(X)$.*

1. *The spectrum $\sigma(T)$ is compact and contained in the disc $\{|\lambda| \leq \|T\|\}$.*
2. *(*Resolvent identity*) $R(\lambda) - R(\mu) = (\lambda - \mu)R(\lambda)R(\mu)$ for $\lambda, \mu \in \rho(T)$, and for every bounded linear functional Φ on $B(X)$, the function $\lambda \mapsto \Phi(R(\lambda))$ is holomorphic on the open set $\rho(T)$.*
3. *(*Gelfand*) $\sigma(T)$ is nonempty.*
4. *(*Spectral radius formula*) $r(T) = \lim_{n\to\infty} \|T^n\|^{1/n} = \inf_{n\geq 1} \|T^n\|^{1/n}$.*

Proof. (1) If $|\lambda| > \|T\|$, then $T - \lambda = -\lambda(I - \lambda^{-1}T)$ is invertible by Lemma 30.7.1. If $\lambda \in \rho(T)$ and $|\mu - \lambda| < \|R(\lambda)\|^{-1}$, then $T - \mu = (T - \lambda) - (\mu - \lambda)I$ is invertible by the same lemma. So $\rho(T)$ is open and $\sigma(T)$ is closed and bounded, hence compact (Theorem 18.6.5).

(2) $R(\lambda) - R(\mu) = R(\lambda)\big((T - \mu) - (T - \lambda)\big)R(\mu) = (\lambda - \mu)R(\lambda)R(\mu)$. As $\mu \to \lambda$, $R(\mu) \to R(\lambda)$ by the continuity of inversion (Lemma 30.7.1), so $\frac{R(\mu)-R(\lambda)}{\mu-\lambda} = R(\mu)R(\lambda) \to R(\lambda)^2$ in norm, and $\Phi(R(\mu))$ has the complex derivative $\Phi(R(\lambda)^2)$ at λ.

(3) Suppose $\sigma(T) = \emptyset$. For $|\lambda| \geq 2\|T\|$, the Neumann series gives $R(\lambda) = -\lambda^{-1} \sum_{n\geq 0} \lambda^{-n} T^n$, so $\|R(\lambda)\| \leq |\lambda|^{-1} \sum_n 2^{-n} = 2/|\lambda|$, which tends to 0 as $|\lambda| \to \infty$. For $\Phi \in B(X)^*$, the entire function $\Phi(R(\lambda))$ is therefore bounded (by continuity on the compact disc $|\lambda| \leq 2\|T\| + 1$, and by $2\|\Phi\|/|\lambda|$ outside), hence constant by Liouville's theorem (Corollary 29.4.7), and the constant is its limit 0 at infinity. As this holds for every Φ, the Hahn–Banach theorem (Theorem 30.4.1), applied in the normed space $B(X)$, gives $R(0) = 0$. But $R(0) = T^{-1}$ is invertible, and an invertible operator on $X \neq 0$ is not zero.

(4) First, $r(T) \leq \|T^n\|^{1/n}$ for every $n \geq 1$. Indeed, if $\lambda \in \sigma(T)$, then $\lambda^n \in \sigma(T^n)$: we have $T^n - \lambda^n = (T - \lambda)P = P(T - \lambda)$ with $P = \sum_{k=0}^{n-1} \lambda^{n-1-k} T^k$, and if $T^n - \lambda^n$ had an inverse Q, then PQ would be a right inverse and QP a left inverse of $T - \lambda$, which would be bijective. So $|\lambda|^n \leq \|T^n\|$ by (1). Hence $r(T) \leq \inf_n \|T^n\|^{1/n} \leq \liminf_n \|T^n\|^{1/n}$.

Second, $\limsup_n \|T^n\|^{1/n} \leq r(T)$. Let $\Phi \in B(X)^*$, and consider $g(w) = \Phi(R(1/w))$ for $0 < |w| < 1/r(T)$ (where $1/r(T) = \infty$ if $r(T) = 0$), holomorphic by (2). For $0 < |w| < 1/\|T\|$ the Neumann series gives

$$g(w) = -\sum_{n\geq 0} \Phi(T^n) w^{n+1},$$

a convergent power series, so g extends holomorphically to $w = 0$ by $g(0) = 0$, and is then holomorphic on the whole disc $D(0, 1/r(T))$. By Theorem 29.4.2, its Taylor series at 0 converges on this whole disc; its coefficients are $g^{(n+1)}(0)/(n+1)! = -\Phi(T^n)$, the coefficients of the series above. So for every λ with $|\lambda| > r(T)$ the series $\sum_n \Phi(T^n)\lambda^{-n-1}$ converges, and in particular $\sup_n |\Phi(T^n/\lambda^{n+1})| < \infty$ for every $\Phi \in B(X)^*$. Now apply the uniform boundedness principle to the family of bounded linear functionals $\Phi \mapsto \Phi(T^n/\lambda^{n+1})$ on the Banach space $B(X)^*$: their norms are $\|T^n\|/|\lambda|^{n+1}$ by the Hahn–Banach theorem, so $\|T^n\| \leq C|\lambda|^{n+1}$ for some C and all n. Hence $\limsup_n \|T^n\|^{1/n} \leq |\lambda|$, and letting $|\lambda|$ decrease to $r(T)$ gives the claim. Together with the first part, the limit exists and equals $r(T) = \inf_n \|T^n\|^{1/n}$. □

The proof of (4) is a nice interplay of the three main tools of this chapter: complex analysis enters through the radius of convergence of the Taylor series of the resolvent, and the uniform boundedness principle converts the bounds on each $\Phi(T^n)$ into a bound on $\|T^n\|$. Note that $r(T)$ can be much smaller than $\|T\|$.

Example 30.7.4. 1. *Matrices.* For $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $\sigma(A) = \{0\}$ and $r(A) = 0 < 1 = \|A\|$, in accordance with $A^2 = 0$. For a matrix, $r(A)$ is the largest absolute value of an eigenvalue, and the spectral radius formula says that $\|A^n\|$ grows like $r(A)^n$, up to subexponential factors: for $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, $A^n = \left(\begin{smallmatrix} 1 & n \\ 0 & 1 \end{smallmatrix}\right)$ has norm between n and $n+2$, and $\|A^n\|^{1/n} \to 1$.

2. *Multiplication operators.* Let g be a continuous function on $[0,1]$ and $M_g f = gf$ on $L^2([0,1])$. Then $\sigma(M_g) = g([0,1])$. If $\lambda \notin g([0,1])$, then $1/(g-\lambda)$ is continuous, hence bounded, and $M_{1/(g-\lambda)}$ is the inverse of $M_g - \lambda$. If $\lambda = g(x_0)$, then for every k there is an interval $I_k \ni x_0$ of positive length on which $|g - \lambda| < 1/k$, and $f_k = \mathbf{1}_{I_k}/\|\mathbf{1}_{I_k}\|_2$ satisfies $\|(M_g - \lambda)f_k\|_2 \le 1/k$ with $\|f_k\|_2 = 1$; so $M_g - \lambda$ has no bounded inverse, and $\lambda \in \sigma(M_g)$. The eigenvalues are the λ for which $\{g = \lambda\}$ has positive measure. For $g(x) = x$, the spectrum of the self-adjoint operator M_x is the whole interval $[0,1]$, and there are no eigenvalues at all.

3. *Shifts.* For the left shift $L(x_1, x_2, \ldots) = (x_2, x_3, \ldots)$ on ℓ^2, every λ with $|\lambda| < 1$ is an eigenvalue (Counterexample 30.6.6(2)), and $\|L\| = 1$; as $\sigma(L)$ is closed, $\sigma(L)$ is the closed unit disc. The right shift $S = L^*$ has the same spectrum (Exercise 30.12.14), although it has no eigenvalues.

4. *The Volterra operator* $V f(x) = \int_0^x f(t)\,dt$ on $L^2([0,1])$. By induction and Fubini's theorem,

$$V^n f(x) = \int_0^x \frac{(x-t)^{n-1}}{(n-1)!} f(t)\,dt,$$

so $|V^n f(x)| \le \|f\|_1/(n-1)! \le \|f\|_2/(n-1)!$ and $\|V^n\| \le 1/(n-1)!$. For every $K \ge 1$, the factors $K+1, \ldots, n-1$ of $(n-1)!$ are at least K, so $((n-1)!)^{1/n} \ge K^{(n-1-K)/n} \to K$; hence $\|V^n\|^{1/n} \to 0$ and $r(V) = 0$. So $\sigma(V) = \{0\}$, nonempty as Gelfand's theorem requires: V is not surjective, since Vf is always continuous. For $\lambda \neq 0$ the inverse of $V - \lambda$ is a Neumann series, which solves Volterra's integral equation (Exercise 30.12.13).

For compact operators on Hilbert spaces the spectrum is as close to the finite-dimensional case as possible.

Corollary 30.7.5. *Let H be an infinite-dimensional complex Hilbert space and $K \in B(H)$ compact. Then $0 \in \sigma(K)$, every nonzero $\lambda \in \sigma(K)$ is an eigenvalue with finite-dimensional eigenspace, and for every $\varepsilon > 0$ only finitely many points of $\sigma(K)$ satisfy $|\lambda| \ge \varepsilon$. So $\sigma(K)$ is countable, and 0 is its only possible accumulation point.*

Proof. If $0 \notin \sigma(K)$, then $I = KK^{-1}$ would be compact (Proposition 30.6.2(1)), contradicting Proposition 30.2.2(3). For $\lambda \neq 0$, $K - \lambda = -\lambda(I - \lambda^{-1}K)$ with $\lambda^{-1}K$ compact, so by the Fredholm alternative (Theorem 30.6.4) it is bijective as soon as it is injective, and its kernel is finite-dimensional. Suppose there were infinitely many distinct eigenvalues $\lambda_1, \lambda_2, \ldots$ with $|\lambda_n| \ge \varepsilon$, with eigenvectors x_n. Eigenvectors for distinct eigenvalues are linearly independent (as in finite dimensions: applying $K - \lambda_j$ to a shortest nontrivial linear relation $\sum c_i x_i = 0$, for an index j occurring in it, would give a shorter one). Let $M_n = \operatorname{span}(x_1, \ldots, x_n)$, and choose unit vectors $y_n \in M_n \cap M_{n-1}^{\perp}$. Since $(K - \lambda_n)x_j = (\lambda_j - \lambda_n)x_j$, we have $(K - \lambda_n)M_n \subseteq M_{n-1}$. For $m < n$,

$$Ky_n - Ky_m = \lambda_n y_n + \big((K - \lambda_n)y_n - Ky_m\big),$$

where the bracket lies in M_{n-1} and $y_n \perp M_{n-1}$, so $\|Ky_n - Ky_m\| \ge |\lambda_n| \ge \varepsilon$. So (Ky_n) has no convergent subsequence, contradicting compactness. □

Proposition 30.7.6. *Let $H \neq 0$ be a complex Hilbert space, $T \in B(H)$ self-adjoint, and $m = \inf_{\|x\|=1}\langle Tx, x\rangle$, $M = \sup_{\|x\|=1}\langle Tx, x\rangle$. Then $\sigma(T) \subseteq [m, M]$, both m and M lie in $\sigma(T)$, and $r(T) = \|T\| = \max(|m|, |M|)$.*

Proof. The numbers $\langle Tx,x\rangle = \langle x,Tx\rangle = \overline{\langle Tx,x\rangle}$ are real. Call an operator A *bounded below* if $\|Ax\| \geq c\|x\|$ for some $c > 0$; then A is injective with closed image (the preimages of a Cauchy sequence in the image form a Cauchy sequence). Let $\lambda = \alpha + i\beta$ with $\beta \neq 0$. Then $\|(T-\lambda)x\|\,\|x\| \geq |\operatorname{Im}\langle (T-\lambda)x,x\rangle| = |\beta|\,\|x\|^2$, so $T-\lambda$ is bounded below; its image is closed, and dense because its orthogonal complement is $\ker(T-\lambda)^* = \ker(T-\bar\lambda) = 0$ (Theorem 30.3.4), the same estimate applying to $\bar\lambda$. So $\lambda \in \rho(T)$. If $\lambda > M$ is real, then $\langle(\lambda - T)x,x\rangle \geq (\lambda - M)\|x\|^2$, and the same argument shows $\lambda \in \rho(T)$; similarly for $\lambda < m$. So $\sigma(T) \subseteq [m,M]$.

The sesquilinear form $[x,y] = \langle (M-T)x,y\rangle$ is Hermitian and positive semidefinite, so it satisfies the Cauchy–Schwarz inequality $|[x,y]|^2 \leq [x,x][y,y]$: choosing $|c| = 1$ with $c[y,x] = |[x,y]|$, the real quadratic polynomial $[x+tcy, x+tcy] = [x,x] + 2t|[x,y]| + t^2[y,y]$ is nonnegative for all real t, which forces $[x,y] = 0$ if $[y,y] = 0$, and a nonpositive discriminant otherwise. Choose unit vectors x_n with $\langle Tx_n,x_n\rangle \to M$, so $[x_n,x_n] \to 0$. With $y = (M-T)x_n$,

$$\|(M-T)x_n\|^4 = [x_n,y]^2 \leq [x_n,x_n]\,[y,y] \leq [x_n,x_n]\,\|M-T\|\,\|(M-T)x_n\|^2,$$

so $(M-T)x_n \to 0$ while $\|x_n\| = 1$. Thus $M-T$ is not bounded below and has no bounded inverse: $M \in \sigma(T)$. Applying this to $-T$ gives $m \in \sigma(T)$. Finally $\|T\| = \sup_{\|x\|=1} |\langle Tx,x\rangle| = \max(|m|,|M|)$ by the first step of the proof of Theorem 30.6.3(1), which did not use compactness, and $\max(|m|,|M|) \leq r(T) \leq \|T\|$ since $m, M \in \sigma(T)$. □

Remark 30.7.7. The spectral theorem for compact self-adjoint operators (Theorem 30.6.3) diagonalises T in an orthonormal basis. A general bounded self-adjoint operator, such as M_x in Example 30.7.4(2), may have no eigenvectors at all, and the correct generalisation is: *every bounded self-adjoint operator on a separable Hilbert space is unitarily equivalent to a multiplication operator*. That is, there are a finite measure space (Ω,μ), a bounded real measurable function g on Ω and a unitary $U\colon H \to L^2(\Omega,\mu)$ with $UTU^{-1} = M_g$ [RS80, Chapter VII]. It yields a *functional calculus*: $f(T) = U^{-1}M_{f\circ g}U$ for bounded Borel functions f on $\sigma(T)$, with $\|f(T)\| \leq \sup_{\sigma(T)}|f|$ (equality for continuous f) and $\sigma(f(T)) = f(\sigma(T))$ for continuous f; for instance $\sqrt{T}$ for $T \geq 0$, e^{itT}, or the spectral projections $\mathbf{1}_{(-\infty,c]}(T)$. The same holds for normal operators ($TT^* = T^*T$), with complex g, and, with more work, for the unbounded self-adjoint operators of quantum mechanics and for the Laplacian [RS80, Chapter VIII].

30.8 Fréchet spaces

The space $C^\infty(U)$ of smooth functions on an open set $U \subseteq \mathbb{R}^n$ has a natural notion of convergence, uniform convergence of all derivatives on compact sets, which does not come from a norm. It is described by countably many seminorms.

Definition 30.8.1. Let X be a vector space with a sequence of seminorms $p_1 \leq p_2 \leq \cdots$ (Theorem 30.4.1) that *separates points*: $p_j(x) = 0$ for all j only if $x = 0$. The sets $x + \{y : p_j(y) < \varepsilon\}$ form a basis of a topology, in which $x_k \to x$ if and only if $p_j(x_k - x) \to 0$ for every j. It is induced by the translation invariant metric

$$d(x,y) = \sum_{j\geq 1} 2^{-j}\min\bigl(1, p_j(x-y)\bigr),$$

and X is a *Fréchet space* if this metric is complete.

That the sets form a basis and that d induces the topology follow from the inequalities: $d(x,y) < 2^{-j}\varepsilon$ implies $p_j(x-y) < \varepsilon$ for $\varepsilon \leq 1$, and conversely $p_j(x-y) < \varepsilon$ implies $d(x,y) < \varepsilon + 2^{-j}$ (the terms with index $\leq j$ are at most $2^{-i}\varepsilon$ since the seminorms increase, and the tail is at most 2^{-j}). Banach spaces are Fréchet spaces with $p_j = \|\cdot\|$ for all j.

Example 30.8.2.

1. $C(\mathbb{R})$ with $p_j(f) = \sup_{[-j,j]} |f|$: convergence is uniform convergence on compact sets, and completeness follows from that of $C([-j, j])$ for each j.

2. $C^\infty(U)$, with an exhaustion $K_1 \subseteq K_2 \subseteq \cdots$ of U by compact sets ($K_j \subseteq \operatorname{int} K_{j+1}$, $\bigcup K_j = U$) and $p_j(f) = \sum_{|\alpha|\le j} \sup_{K_j} |\partial^\alpha f|$. A Cauchy sequence has all derivatives uniformly Cauchy on each K_j; the limits are continuous, and they are the derivatives of the limit of the functions, by the theorem on uniform convergence of derivatives (Theorem 27.9.4) applied in each variable. So $C^\infty(U)$ is a Fréchet space.

3. The *Schwartz space* $\mathcal{S}(\mathbb{R}^n)$ of smooth functions all of whose derivatives decay faster than every power, with $p_j(f) = \sup_x (1 + |x|)^j \sum_{|\alpha|\le j} |\partial^\alpha f(x)|$; it contains $e^{-|x|^2}$ and is the domain of the Fourier transform in Chapter 32.

Proposition 30.8.3. *The topology of $C(\mathbb{R})$ in Example 30.8.2(1) is not induced by any norm.*

Proof. Suppose a norm induces it, with unit ball W. Then W contains a basic neighbourhood $\{f : p_j(f) < \varepsilon\}$, and the neighbourhood $U = \{f : p_{j+1}(f) < 1\}$ of 0 contains a ball tW, $t > 0$, of the norm. Let ϕ be continuous, supported in $(j, j + 1)$, with maximum 1. For every $c > 0$, $p_j(c\phi) = 0$, so $c\phi \in W$ and $tc\phi \in tW \subseteq U$; but $p_{j+1}(tc\phi) = tc$, which exceeds 1 for c large. Contradiction. □

The Baire category theorem applies to Fréchet spaces, since they are complete metric spaces, and the uniform boundedness, open mapping and closed graph theorems hold for them with the same proofs [Rud91, Chapter 2]. A linear map $T\colon X \to Y$ of Fréchet spaces, with seminorms (p_k) and (q_j), is continuous if and only if for every j there are k and C with $q_j(Tx) \le Cp_k(x)$ [Rud91, Chapter 1]; for example $\partial/\partial x^i\colon C^\infty(U) \to C^\infty(U)$ is continuous, with $p_j(\partial_i f) \le p_{j+1}(f)$, although differentiation is unbounded for every norm on C^1 (Counterexample 30.4.3).

30.9 Weak topologies

Definition 30.9.1. The *weak topology* on a normed space X is the initial topology (Definition 18.4.1) for all $\Lambda \in X^*$; $x_n \rightharpoonup x$ weakly means $\Lambda(x_n) \to \Lambda(x)$ for all Λ. The *weak-* topology* on X^* is the initial topology for the evaluations at points of X.

Theorem 30.9.2 (Banach–Alaoglu)**.** *The closed unit ball of X^* is weak-* compact. If H is a separable Hilbert space, every bounded sequence in H has a weakly convergent subsequence.*

Proof. Map $\Lambda \mapsto (\Lambda(x))_{x\in X}$ into $\prod_{x\in X} \overline{D}(0, \|x\|)$, which is compact by Tychonoff's theorem (Theorem 18.6.8). The image consists of the families satisfying the (closed) linearity conditions $\lambda_{ax+by} = a\lambda_x + b\lambda_y$, so it is a closed subset, hence compact, and the map is a homeomorphism onto its image by definition of the weak-* topology. For the second statement let (x_n) be bounded and (y_k) a dense sequence in H. By a diagonal argument, after passing to a subsequence $\langle x_n, y_k\rangle$ converges for every k, hence $\langle x_n, y\rangle$ converges for every y (uniformly approximate y by the y_k, using boundedness). The limit is a bounded conjugate-linear functional of y, so it is $\langle x, y\rangle$ for some x by Theorem 30.3.4. □

Counterexample 30.9.3. Weak convergence does not imply norm convergence: an orthonormal sequence (e_n) in a Hilbert space satisfies $e_n \rightharpoonup 0$ by Bessel's inequality, while $\|e_n\| = 1$. By Proposition 30.2.2(3), the closed unit ball of an infinite-dimensional space is never norm compact, although it is weakly compact for Hilbert spaces.

30.10 Convexity and the geometric Hahn–Banach theorem

In the plane, two disjoint convex sets, one of them open, can be separated by a line. In a normed space X the role of lines is played by the real *hyperplanes* $\{x : \operatorname{Re}\Lambda(x) = c\}$ for $0 \neq \Lambda \in X^*$ and $c \in \mathbb{R}$, and the separation theorem is a geometric form of the Hahn–Banach theorem. It needs the following version of the extension theorem, in which the seminorm is replaced by a *sublinear* function.

Lemma 30.10.1. *Let X be a real vector space and $p\colon X \to \mathbb{R}$* sublinear*: $p(x+y) \le p(x) + p(y)$ and $p(tx) = tp(x)$ for $t \ge 0$. If ℓ is a linear functional on a subspace Y with $\ell \le p$ on Y, there is a linear functional L on X extending ℓ with $L \le p$ on X.*

Proof. This is the real case of the proof of Theorem 30.4.1, which uses only these two properties; we repeat it. By Zorn's lemma there is a maximal extension (Z, m) with $m \le p$ on Z, extensions being ordered by inclusion of graphs. Suppose $x \notin Z$. For $z, z' \in Z$, $m(z)+m(z') = m(z+z') \le p(z+x) + p(z'-x)$, so $a = \sup_{z'\in Z}(m(z') - p(z'-x)) \le b = \inf_{z\in Z}(p(z+x) - m(z))$. Choose $c \in [a, b]$ and put $m(z+tx) = m(z) + tc$. For $t > 0$, $m(z) + tc = t(m(z/t) + c) \le tp(z/t + x) = p(z + tx)$ since $c \le b$; for $t = -s < 0$, $m(z) - sc = s(m(z/s) - c) \le sp(z/s - x) = p(z - sx)$ since $c \ge a$. This contradicts maximality. □

Lemma 30.10.2 (Minkowski functional)**.** *Let C be an open convex subset of a real normed space X with $0 \in C$, and $p_C(x) = \inf\{t > 0 : x \in tC\}$. Then p_C is sublinear, $0 \le p_C(x) \le \|x\|/r$ if $B(0, r) \subseteq C$, and $C = \{x : p_C(x) < 1\}$.*

Proof. If $t > \|x\|/r$, then $\|x/t\| < r$, so $x \in tC$; hence $p_C(x) \le \|x\|/r$. If $x \in tC$ and $t' > t$, then $x/t' = \frac{t}{t'}(x/t) + (1 - \frac{t}{t'})0 \in C$ by convexity, so $x \in t'C$. Since $x \in tC$ if and only if $sx \in stC$, $p_C(sx) = sp_C(x)$ for $s > 0$ (and trivially for $s = 0$). Given $\varepsilon > 0$, put $s = p_C(x) + \varepsilon$ and $s' = p_C(y) + \varepsilon$; then $x/s, y/s' \in C$ and

$$\frac{x+y}{s+s'} = \frac{s}{s+s'} \cdot \frac{x}{s} + \frac{s'}{s+s'} \cdot \frac{y}{s'} \in C,$$

so $p_C(x+y) \le p_C(x) + p_C(y) + 2\varepsilon$. If $x \in C$, then $(1+\delta)x \in C$ for small $\delta > 0$ as C is open, so $p_C(x) \le 1/(1+\delta) < 1$; if $p_C(x) < 1$, then $x \in tC$ for some $t < 1$, hence $x \in C$. □

For the unit ball, p_C is the norm; in general p_C is the "norm" whose unit ball is C, possibly asymmetric.

Theorem 30.10.3 (Separation of convex sets)**.** *Let A and B be nonempty disjoint convex subsets of a normed space X over $\mathbb{K}$.*

1. *If A is open, there are $\Lambda \in X^*$ and $c \in \mathbb{R}$ with $\operatorname{Re}\Lambda(a) < c \le \operatorname{Re}\Lambda(b)$ for all $a \in A$, $b \in B$.*
2. *If A is compact and B is closed, there are $\Lambda \in X^*$ and $c_1 < c_2$ with $\operatorname{Re}\Lambda(a) \le c_1 < c_2 \le \operatorname{Re}\Lambda(b)$ for all $a \in A$, $b \in B$.*

Proof. Suppose first $\mathbb{K} = \mathbb{R}$. (1) Fix $a_0 \in A$, $b_0 \in B$, put $x_0 = b_0 - a_0$ and $C = A - B + x_0 = \{a - b + x_0\}$. Then C is convex, open (a union of translates of the open set A), contains 0, and does not contain x_0, since $A \cap B = \emptyset$. Choose $r > 0$ with $B(0, r) \subseteq C$, and let $p = p_C$ be the Minkowski functional of C (Lemma 30.10.2), so $p(x_0) \ge 1$. On $\mathbb{R}x_0$ define $\ell(tx_0) = t$; then $\ell(tx_0) \le tp(x_0) = p(tx_0)$ for $t \ge 0$, and $\ell(tx_0) \le 0 \le p(tx_0)$ for $t < 0$. By Lemma 30.10.1, ℓ extends to a linear $\Lambda \le p$ on X. Then $\Lambda(x) \le p(x) \le \|x\|/r$ and $-\Lambda(x) = \Lambda(-x) \le \|x\|/r$, so $\Lambda \in X^*$. For $a \in A$ and $b \in B$, $\Lambda(a) - \Lambda(b) + 1 = \Lambda(a - b + x_0) \le p(a - b + x_0) < 1$, so $\Lambda(a) < \Lambda(b)$. Let $c = \inf_{b\in B} \Lambda(b)$, so $\Lambda(a) \le c \le \Lambda(b)$. Since $\Lambda \neq 0$, choose v with $\Lambda(v) = 1$; for $a \in A$, $a + sv \in A$ for small $s > 0$, so $\Lambda(a) + s \le c$, and $\Lambda(a) < c$.

(2) The continuous function $a \mapsto \operatorname{dist}(a, B)$ is positive on the compact set A, since B is closed, so it has a positive minimum δ. The set $A' = A + B(0, \delta/2)$ is open, convex and disjoint from B. By (1) there are $\Lambda \neq 0$ and c with $\Lambda(a+u) < c \leq \Lambda(b)$ for $a \in A$, $\|u\| < \delta/2$, $b \in B$. Taking the supremum over u gives $\Lambda(a) + \frac{\delta}{2}\|\Lambda\| \leq c$, so (2) holds with $c_1 = \sup_A \Lambda \leq c - \frac{\delta}{2}\|\Lambda\|$ and $c_2 = c$.

If $\mathbb{K} = \mathbb{C}$, regard X as a real normed space, and let $\Lambda_{\mathbb{R}}$ be the real functional just constructed. Then $\Lambda(x) = \Lambda_{\mathbb{R}}(x) - i\Lambda_{\mathbb{R}}(ix)$ is complex linear, as $\Lambda(ix) = \Lambda_{\mathbb{R}}(ix) + i\Lambda_{\mathbb{R}}(x) = i\Lambda(x)$, with $\operatorname{Re}\Lambda = \Lambda_{\mathbb{R}}$; and it is bounded, since for $|\Lambda(x)| = e^{-i\theta}\Lambda(x) = \Lambda(e^{-i\theta}x)$ this number is real, hence equal to $\Lambda_{\mathbb{R}}(e^{-i\theta}x) \leq \|\Lambda_{\mathbb{R}}\|\,\|x\|$. □

The compactness in (2) cannot be weakened to closedness: in $\mathbb{R}^2$ the closed convex sets $\{y \leq 0\}$ and $\{y \geq e^x\}$ are disjoint, and every line separating them is horizontal, with $\{y = 0\}$ the only one; no line separates them strictly.

Corollary 30.10.4 (Mazur)**.** *Let X be a normed space.*

1. *A convex subset of X is weakly closed if and only if it is norm closed.*
2. *If $x_n \rightharpoonup x$ weakly, then for every k there is a convex combination y_k of $x_k, x_{k+1}, \ldots$ (a finite one) with $\|y_k - x\| < 1/k$; in particular $y_k \to x$ in norm.*
3. *If $x_n \rightharpoonup x$, then $\|x\| \leq \liminf_n \|x_n\|$.*
4. *A subspace $M \subseteq X$ is dense if and only if every $\Lambda \in X^*$ vanishing on M is zero.*

Proof. (1) Every $\Lambda \in X^*$ is norm continuous, so weakly open sets are norm open, and weakly closed sets norm closed. Conversely let C be convex and norm closed, and $x \notin C$. By Theorem 30.10.3(2) applied to $\{x\}$ and C there are Λ and c_2 with $\operatorname{Re}\Lambda(x) < c_2 \leq \operatorname{Re}\Lambda(y)$ for $y \in C$; the weakly open set $\{y : \operatorname{Re}\Lambda(y) < c_2\}$ contains x and misses C. So the complement of C is weakly open.

(2) The norm closure K of the convex hull of $\{x_n : n \geq k\}$ is convex (the closure of a convex set is convex) and norm closed, hence weakly closed by (1); it contains x_n for $n \geq k$, hence the weak limit x (if $x \notin K$, a weakly open neighbourhood of x missing K would contain x_n for large n). So x is a norm limit of convex combinations of the x_n, $n \geq k$.

(3) By Theorem 30.4.1 there is Λ with $\|\Lambda\| = 1$ and $\Lambda(x) = \|x\|$; then $\|x\| = \lim_n |\Lambda(x_n)| \leq \liminf_n \|x_n\|$.

(4) If M is dense and $\Lambda|_M = 0$, then $\Lambda = 0$ by continuity. If $\overline{M} \neq X$, choose $x \notin \overline{M}$ and separate $\{x\}$ from the closed subspace $\overline{M}$ as in (1): $\operatorname{Re}\Lambda(x) < c_2 \leq \operatorname{Re}\Lambda(m)$ for $m \in \overline{M}$. Applying this to tm for all real t shows $\operatorname{Re}\Lambda = 0$ on $\overline{M}$, hence $\Lambda = 0$ on $\overline{M}$ (for $\mathbb{K} = \mathbb{C}$ use $\Lambda(m) = \operatorname{Re}\Lambda(m) - i\operatorname{Re}\Lambda(im)$); and $\Lambda \neq 0$ because $\operatorname{Re}\Lambda(x) < c_2 \leq 0$. □

Example 30.10.5. *The weak closure of the sphere.* Let X be an infinite-dimensional normed space, $S = \{\|x\| = 1\}$ and $\overline{B} = \{\|x\| \leq 1\}$. Then the weak closure of S is $\overline{B}$. Indeed $\overline{B}$ is convex and norm closed, hence weakly closed by Mazur's theorem, and contains S. Conversely, let $\|x\| < 1$ and let $V = \{y : |\Lambda_i(y - x)| < \varepsilon,\ i = 1, \ldots, k\}$ be a basic weak neighbourhood of x. The subspace $N = \bigcap_i \ker \Lambda_i$ is the kernel of the linear map $X \to \mathbb{K}^k$, $y \mapsto (\Lambda_i(y))_i$, so it has codimension at most k and contains some $v \neq 0$. The continuous function $t \mapsto \|x + tv\|$ on $[0, \infty)$ is less than 1 at $t = 0$ and tends to infinity, so it takes the value 1 at some t, and $x + tv \in S \cap V$. So the norm-closed set S is not weakly closed: for instance an orthonormal sequence in a Hilbert space lies on S and converges weakly to 0 (Counterexample 30.9.3). This also shows that in (3) of the corollary the inequality can be strict.

Proposition 30.10.6 (The direct method)**.** *Let H be a separable Hilbert space and $F\colon H \to \mathbb{R}$ a continuous convex function with $F(x) \to \infty$ as $\|x\| \to \infty$. Then F attains its minimum.*

Proof. Let $m = \inf F$, possibly $-\infty$ a priori, and choose x_n with $F(x_n) \to m$. Since F is coercive, (x_n) is bounded, and by the Banach–Alaoglu theorem for Hilbert spaces (Theorem 30.9.2) a subsequence converges weakly to some x. For every $t > m$ the sublevel set $\{F \le t\}$ is convex (by convexity of F) and norm closed (by continuity), hence weakly closed by Corollary 30.10.4(1); it contains x_n for n large, hence the weak limit x. So $F(x) \le t$ for every $t > m$, and $F(x) = m$; in particular $m > -\infty$. □

This is the abstract form of *Dirichlet's principle*: to solve a boundary value problem, minimise an energy such as $\int |\nabla u|^2$ over a suitable Hilbert space. Weak compactness gives a weak limit of a minimising sequence, convexity makes the energy weakly lower semicontinuous, and it remains to show that the minimiser is a smooth solution, which is a regularity question. In Hodge theory the harmonic representative of a cohomology class is likewise the closed form of least energy in its class (Chapter 43). The theorem holds more generally in reflexive Banach spaces, for instance in L^p with $1 < p < \infty$, by the results quoted in Remark 30.11.6. Without convexity the conclusion fails: on ℓ^2 the function $F(x) = (\|x\|^2 - 1)^2 + \sum_n 2^{-n}|x_n|^2$ is continuous and coercive, and $F(e_N) = 2^{-N} \to 0$, but $F > 0$ everywhere, since $F(x) = 0$ would require both $\|x\| = 1$ and $x = 0$.

30.11 Dual spaces and reflexivity

To understand a Banach space one often determines its dual. For sequence spaces this is elementary. Let $c_0 \subseteq \ell^\infty$ be the closed subspace of sequences tending to 0, and for $1 \le p \le \infty$ let q be the *conjugate exponent*, $1/p + 1/q = 1$ (with $q = \infty$ for $p = 1$ and $q = 1$ for $p = \infty$). For a sequence y, let $\Lambda_y(x) = \sum_n x_n y_n$ whenever this converges absolutely.

Proposition 30.11.1. *1. For $1 \le p < \infty$, the map $y \mapsto \Lambda_y$ is an isometric isomorphism $\ell^q \to (\ell^p)^*$.*

2. The map $y \mapsto \Lambda_y$ is an isometric isomorphism $\ell^1 \to (c_0)^*$.

Proof. (1) By Hölder's inequality for the counting measure (Proposition 28.5.2), $|\Lambda_y(x)| \le \|x\|_p \|y\|_q$, so $\Lambda_y \in (\ell^p)^*$ with $\|\Lambda_y\| \le \|y\|_q$, and $y \mapsto \Lambda_y$ is linear and injective, as $\Lambda_y(e_n) = y_n$ for the unit vectors e_n. Let $\Lambda \in (\ell^p)^*$ and $y_n = \Lambda(e_n)$; we show $\|y\|_q \le \|\Lambda\|$ and $\Lambda = \Lambda_y$. If $p = 1$, $|y_n| = |\Lambda(e_n)| \le \|\Lambda\|$, so $\|y\|_\infty \le \|\Lambda\|$. If $1 < p < \infty$, so $q < \infty$ and $(q-1)p = q$, let $x_n^{(N)} = |y_n|^{q-2}\bar{y}_n$ for $n \le N$ (and 0 if $y_n = 0$ or $n > N$). Then

$$\sum_{n \le N} |y_n|^q = \Lambda(x^{(N)}) \le \|\Lambda\|\,\|x^{(N)}\|_p = \|\Lambda\| \Big(\sum_{n \le N} |y_n|^{(q-1)p} \Big)^{1/p} = \|\Lambda\| \Big(\sum_{n \le N} |y_n|^q \Big)^{1/p},$$

so $(\sum_{n \le N} |y_n|^q)^{1/q} \le \|\Lambda\|$ for all N, and $\|y\|_q \le \|\Lambda\|$. In both cases Λ and Λ_y are continuous and agree on the finitely supported sequences, which are dense in ℓ^p for $p < \infty$ (for $x \in \ell^p$ the tail $\sum_{n>N} |x_n|^p$ tends to 0); so $\Lambda = \Lambda_y$, and $\|\Lambda\| \le \|y\|_q \le \|\Lambda\|$.

(2) In the same way, $|\Lambda_y(x)| \le \|x\|_\infty \|y\|_1$ for $x \in c_0$ and $y \in \ell^1$. Given $\Lambda \in c_0^*$, put $y_n = \Lambda(e_n)$ and $x_n^{(N)} = \bar{y}_n/|y_n|$ for $n \le N$, $y_n \ne 0$ (and 0 otherwise); then $\|x^{(N)}\|_\infty \le 1$ and $\sum_{n \le N} |y_n| = \Lambda(x^{(N)}) \le \|\Lambda\|$. The finitely supported sequences are dense in c_0, so again $\Lambda = \Lambda_y$ and $\|\Lambda\| = \|y\|_1$. □

The dual of ℓ^∞ is strictly larger than ℓ^1. The subspace c_0 is closed and not dense in ℓ^∞, since $\mathbf{1} = (1, 1, \ldots)$ has distance 1 from it, so by Corollary 30.10.4(4) there is $0 \ne \Lambda \in (\ell^\infty)^*$ vanishing on c_0. It is not of the form Λ_y with $y \in \ell^1$, since then $y_n = \Lambda(e_n) = 0$ for all n. Such functionals are invisible on every finite part of a sequence, like the *Banach limits* of

Exercise 30.12.16; they are produced by the Hahn–Banach theorem, that is, by Zorn's lemma, and no explicit formula for one is known.

Lemma 30.11.2. *Let M be a closed subspace of a normed space X and $x_0 \notin M$, with $d = \operatorname{dist}(x_0, M)$. There is $\Lambda \in X^*$ with $\|\Lambda\| = 1$, $\Lambda|_M = 0$ and $\Lambda(x_0) = d$.*

Proof. The quotient X/M with $\|x + M\| = \operatorname{dist}(x, M)$ is a normed space (the proof of Lemma 30.2.4 that this is a norm did not use completeness), and the projection $\pi\colon X \to X/M$ has norm at most 1. As $\|x_0 + M\| = d > 0$, the Hahn–Banach theorem gives $\lambda \in (X/M)^*$ with $\|\lambda\| = 1$ and $\lambda(x_0 + M) = d$. Then $\Lambda = \lambda \circ \pi$ vanishes on M, has $\|\Lambda\| \le 1$ and $\Lambda(x_0) = d$; and for $m \in M$, $d = |\Lambda(x_0 - m)| \le \|\Lambda\|\,\|x_0 - m\|$, so taking the infimum over m, $d \le \|\Lambda\| d$ and $\|\Lambda\| = 1$. □

Lemma 30.11.3. *If the dual X^* of a normed space X is separable, then X is separable.*

Proof. We may assume $X \neq 0$. Let (Λ_n) be dense in X^*, and choose unit vectors x_n with $|\Lambda_n(x_n)| \ge \|\Lambda_n\|/2$. The closed span M of the x_n is separable (finite combinations with coefficients in $\mathbb{Q}$ or $\mathbb{Q} + i\mathbb{Q}$ are dense in it). If $M \neq X$, Lemma 30.11.2 gives Λ with $\|\Lambda\| = 1$ and $\Lambda|_M = 0$. Choose n with $\|\Lambda - \Lambda_n\| < \frac{1}{3}$. Then $\|\Lambda_n\| > \frac{2}{3}$, so $|\Lambda_n(x_n)| > \frac{1}{3}$; but $|\Lambda_n(x_n)| = |(\Lambda_n - \Lambda)(x_n)| < \frac{1}{3}$. So $M = X$. □

The converse fails: ℓ^1 is separable, but its dual ℓ^∞ (Proposition 30.11.1) is not (Remark 30.2.5).

Definition 30.11.4. The *bidual* of a normed space X is $X^{**} = (X^*)^*$, and the *canonical map* $J\colon X \to X^{**}$ is $(Jx)(\Lambda) = \Lambda(x)$. By Theorem 30.4.1, $\|Jx\| = \sup_{\|\Lambda\| \le 1} |\Lambda(x)| = \|x\|$, so J is a linear isometry. X is *reflexive* if J is surjective. A reflexive space is complete, being isometric to the Banach space X^{**}.

Proposition 30.11.5. *1. Finite-dimensional normed spaces and Hilbert spaces are reflexive.*

2. ℓ^p is reflexive for $1 < p < \infty$.

3. A Banach space X is reflexive if and only if X^* is reflexive.

4. c_0, ℓ^1 and ℓ^∞ are not reflexive.

Proof. (1) If $\dim X < \infty$, then $\dim X^{**} = \dim X$ and the injective map J is bijective. Let H be a Hilbert space, and $\Phi\colon H \to H^*$, $\Phi(y) = \langle \cdot, y\rangle$, the conjugate-linear bijection of Theorem 30.3.4. Given $\xi \in H^{**}$, the map $y \mapsto \overline{\xi(\Phi(y))}$ is linear (as $\Phi(ty) = \bar{t}\Phi(y)$) and bounded, so it is $\langle y, x\rangle$ for some $x \in H$. Then $\xi(\Phi(y)) = \overline{\langle y, x\rangle} = \langle x, y\rangle = \Phi(y)(x) = (Jx)(\Phi(y))$ for all y, and as Φ is onto, $\xi = Jx$.

(2) Let $\xi \in (\ell^p)^{**}$. By Proposition 30.11.1(1), $y \mapsto \xi(\Lambda_y)$ is a bounded linear functional on ℓ^q, and since $1 < q < \infty$ the same proposition gives $x \in \ell^p$ with $\xi(\Lambda_y) = \sum_n y_n x_n = \Lambda_y(x) = (Jx)(\Lambda_y)$. As every element of $(\ell^p)^*$ is some Λ_y, $\xi = Jx$.

(3) Write J_X and J_{X^*} for the canonical maps. Suppose X reflexive and $\varphi \in X^{***}$. Then $\Lambda = \varphi \circ J_X \in X^*$, and for $\xi = J_X x \in X^{**}$, $\varphi(\xi) = \Lambda(x) = \xi(\Lambda) = (J_{X^*}\Lambda)(\xi)$; so $\varphi = J_{X^*}\Lambda$, and X^* is reflexive. Conversely, suppose X^* reflexive and $J_X(X) \neq X^{**}$. Since X is complete and J_X is an isometry, $J_X(X)$ is closed, and Lemma 30.11.2 gives $0 \neq \varphi \in X^{***}$ vanishing on $J_X(X)$. Write $\varphi = J_{X^*}\Lambda$; then $0 = \varphi(J_X x) = (J_X x)(\Lambda) = \Lambda(x)$ for all x, so $\Lambda = 0$ and $\varphi = 0$, a contradiction.

(4) By Proposition 30.11.1, every $\xi \in c_0^{**}$ is determined by the sequence $z_n = \xi(\Lambda_{e_n})$, which is bounded, $|z_n| \le \|\xi\|$, and every bounded sequence z arises, from $\xi(\Lambda_y) = \sum_n y_n z_n$. For $x \in c_0$, $(Jx)(\Lambda_{e_n}) = x_n$. So the element ξ with $z = (1, 1, \ldots)$ is not in $J(c_0)$, and c_0 is not reflexive. If ℓ^1 were reflexive, then its bidual $(\ell^1)^{**} \cong (\ell^\infty)^*$ would be isometric to the separable space ℓ^1, so ℓ^∞ would be separable by Lemma 30.11.3, which it is not. Finally $\ell^\infty \cong (\ell^1)^*$, so by (3) it is not reflexive either. □

Remark 30.11.6.

1. *Duals of L^p.* For a σ-finite measure space and $1 \le p < \infty$, every bounded linear functional on $L^p(\mu)$ is $f \mapsto \int f g \, d\mu$ for a unique $g \in L^q(\mu)$, and $(L^p)^* \cong L^q$ isometrically; hence L^p is reflexive for $1 < p < \infty$. The spaces $L^1([0,1])$ and $L^\infty([0,1])$ are not reflexive. See [Bre11, Chapter 4]; for $p = 2$ this is the Riesz representation theorem.

2. *The dual of $C(K)$.* For a compact metric space K, $C(K)^*$ is the space of signed (or complex) finite Borel measures with the total variation norm; for positive functionals this is the Riesz–Markov theorem (Theorem 28.14.6, Remark 28.14.7). Weak-$*$ convergence in $C(K)^*$ is then weak convergence of measures, and Theorem 28.15.5 is a case of the Banach–Alaoglu theorem.

3. *Weak compactness.* A Banach space is reflexive if and only if its closed unit ball is weakly compact (Kakutani), and then every bounded sequence has a weakly convergent subsequence (Eberlein–Šmulian). Every uniformly convex space, such as L^p with $1 < p < \infty$, is reflexive (Milman–Pettis). See [Bre11, Chapter 3]. This is why the direct method (Proposition 30.10.6) works in L^p and in the Sobolev spaces built from it, and fails in L^1: there the integrable functions $n\mathbf{1}_{[0,1/n]}$, of norm 1, concentrate at a point and have no weakly convergent subsequence in L^1, their limit being the Dirac measure, which lies outside L^1.

30.12 Exercises

Exercise 30.12.1. Show that $(ST)^* = T^*S^*$, $T^{**} = T$ and $\|T^*T\| = \|T\|^2$ for bounded operators on a Hilbert space.

Solution. The first two follow from uniqueness of adjoints. For the last, $\|T^*T\| \le \|T\|^2$, and $\|Tx\|^2 = \langle T^*Tx, x\rangle \le \|T^*T\|\,\|x\|^2$. □

Exercise 30.12.2. Let $k \in L^2([0,1]^2)$ and $(Tf)(x) = \int_0^1 k(x,y) f(y)\, dy$. Show that T is a compact operator on $L^2([0,1])$ with $\|T\| \le \|k\|_2$, and that T^* has kernel $\overline{k(y,x)}$.

Solution. Boundedness follows from Cauchy–Schwarz and Fubini's theorem. Kernels of the form $\sum_i \phi_i(x)\psi_i(y)$ give finite rank operators and are dense in $L^2([0,1]^2)$ (products of an orthonormal basis of $L^2([0,1])$ form an orthonormal basis of $L^2([0,1]^2)$), so T is a limit of finite rank operators. The formula for T^* is Fubini's theorem. □

Exercise 30.12.3. Show that the functional $f \mapsto f(0)$ on $C([-1,1])$ is not continuous for the L^2 norm, and deduce that it does not extend to $L^2([-1,1])$.

Exercise 30.12.4. Let T be compact self-adjoint with eigenvalues λ_k and orthonormal eigenvectors e_k. Show that $Tx = \sum_k \lambda_k \langle x, e_k\rangle e_k$, that T is a norm limit of the finite rank operators obtained by truncating this sum, and that $\|T\| = \max_k |\lambda_k|$.

Exercise 30.12.5. Show that $C([0,1])$ with the norm $\|f\|_1 = \int_0^1 |f|$ is not complete, by exhibiting a Cauchy sequence converging in L^1 to $\mathbf{1}_{[1/2,1]}$.

Solution. Let f_n be 0 on $[0, \frac{1}{2} - \frac{1}{n}]$, 1 on $[\frac{1}{2}, 1]$ and linear in between. Then $\|f_n - f_m\|_1 \le \frac{1}{\min(n,m)}$, and $\|f_n - \mathbf{1}_{[1/2,1]}\|_1 \le \frac{1}{n}$. If $f_n \to f$ in $\|\cdot\|_1$ with f continuous, then $f = \mathbf{1}_{[1/2,1]}$ almost everywhere, which is impossible: f would be 0 on $[0, \frac{1}{2})$ and 1 on $(\frac{1}{2}, 1]$ by continuity, and could not be continuous at $\frac{1}{2}$. □

Exercise 30.12.6. Let $g \in L^\infty([0,1])$. Show that $M_g f = gf$ defines a bounded operator on $L^2([0,1])$ with $\|M_g\| = \|g\|_\infty$, that $M_g^* = M_{\bar g}$, and that M_g is compact only if $g = 0$.

Solution. $\|gf\|_2 \le \|g\|_\infty \|f\|_2$. For $\varepsilon > 0$ the set $E = \{|g| > \|g\|_\infty - \varepsilon\}$ has positive measure, and $f = \mathbf{1}_E$ gives $\|gf\|_2 \ge (\|g\|_\infty - \varepsilon)\|f\|_2$. $\langle gf, h\rangle = \int gf\bar h = \langle f, \bar g h\rangle$. If $g \neq 0$, some $E = \{|g| > \delta\}$ has positive measure, and $L^2(E)$ is infinite-dimensional (it contains the indicator functions of disjoint subsets of E of positive measure, which exist since $t \mapsto \lambda(E \cap [0,t])$ is continuous). On $L^2(E)$, M_g has the bounded inverse $M_{1/g}$, so if it were compact the identity of $L^2(E)$ would be compact, contradicting Proposition 30.2.2(3). □

Exercise 30.12.7. Let (e_n) be an orthonormal sequence in a Hilbert space and $(c_n) \in \ell^2$. Show that $\sum_n c_n e_n$ converges, that its norm is $(\sum |c_n|^2)^{1/2}$, and that $c_n = \langle x, e_n\rangle$ for its sum x (the Riesz–Fischer theorem for orthonormal sequences).

Solution. By Pythagoras $\|\sum_{n=M}^N c_n e_n\|^2 = \sum_{n=M}^N |c_n|^2$, so the partial sums are Cauchy and converge by completeness, with the stated norm in the limit. By continuity of the inner product, $\langle x, e_m\rangle = \lim_N \sum_{n\le N} c_n \langle e_n, e_m\rangle = c_m$. □

Exercise 30.12.8. Compute the next orthonormal polynomial P_3 of Example 30.3.8(2), and find the polynomial q of degree at most 2 minimising $\int_0^1 (e^t - q(t))^2\,dt$.

Solution. Gram–Schmidt on x^3 gives $P_3 = \sqrt{7}(20x^3 - 30x^2 + 12x - 1)$: one checks that $20x^3 - 30x^2 + 12x - 1$ is orthogonal to 1, x, x^2 (its integrals against them are $5 - 10 + 6 - 1 = 0$, $4 - 7.5 + 4 - 0.5 = 0$ and $\frac{20}{6} - 6 + 3 - \frac{1}{3} = 0$) and has norm^2 $\frac{1}{7}$. The best approximation is the orthogonal projection $q = \sum_{n\le 2} \langle e^t, P_n\rangle P_n$, with $\langle e^t, P_0\rangle = e - 1$, $\langle e^t, P_1\rangle = \sqrt{3}(3 - e)$ and $\langle e^t, P_2\rangle = \sqrt{5}(7e - 19)$, using $\int_0^1 te^t = 1$ and $\int_0^1 t^2 e^t = e - 2$. □

Exercise 30.12.9. Show that a sequence of polynomials converging in $C^\infty(\mathbb{R})$ (Example 30.8.2(2)) to a function f need not have bounded degrees, but that if the degrees are bounded by d, then f is a polynomial of degree at most d. Deduce that the polynomials of degree at most d form a closed subspace, and that all polynomials form a dense, non-closed subspace of $C^\infty(\mathbb{R})$.

Solution. The partial sums of $\sum x^k/k!$ converge to e^x in $C^\infty(\mathbb{R})$, as all derivatives converge uniformly on compact sets. If $\deg p_k \le d$, then $p_k^{(d+1)} = 0$ converges to $f^{(d+1)}$, so $f^{(d+1)} = 0$ and f is a polynomial of degree at most d (Theorem 27.6.6 applied repeatedly). Density: by the Weierstrass theorem applied to $f^{(j)}$ on $[-R,R]$ and integrating j times, every smooth function is approximated with its first j derivatives on $[-R,R]$ by polynomials; the space is not closed since e^x is not a polynomial. □

Exercise 30.12.10. Show that $C^1([0,1])$ is complete for the norm $\|f\|_\infty + \|f'\|_\infty$ and for the norm $|f(0)| + \|f'\|_\infty$, and deduce from Corollary 30.5.1 that these norms are equivalent. Find an explicit constant.

Solution. If (f_k) is Cauchy for the first norm, then $f_k \to f$ and $f_k' \to g$ uniformly, and $f' = g$ by Theorem 27.9.4, so $f \in C^1$ and $f_k \to f$ in the norm. For the second norm, if (f_k) is Cauchy then $f_k(0) \to a$ and $f_k' \to g$ uniformly, and $f_k(x) = f_k(0) + \int_0^x f_k'$ converges uniformly to $f(x) = a + \int_0^x g$, which is C^1 with $f' = g$. Clearly $|f(0)| + \|f'\|_\infty \le \|f\|_\infty + \|f'\|_\infty$, so the corollary applies. Explicitly, $|f(x)| \le |f(0)| + \|f'\|_\infty$ by the mean value theorem, so $\|f\|_\infty + \|f'\|_\infty \le 2(|f(0)| + \|f'\|_\infty)$. □

Exercise 30.12.11. For $f \in C(S^1)$ let $\sigma_N f = \frac{1}{N+1}\sum_{k=0}^{N} S_k f$. Show that $\sigma_N f(x) = \int_0^1 f(x-t)F_N(t)\,dt$ with the *Fejér kernel*

$$F_N(t) = \frac{1}{N+1}\sum_{k=0}^{N} D_k(t) = \frac{1}{N+1}\Big(\frac{\sin((N+1)\pi t)}{\sin(\pi t)}\Big)^2,$$

and deduce Fejér's theorem: $\sigma_N f \to f$ uniformly. Conclude again that trigonometric polynomials are dense in $C(S^1)$.

Solution. The integral formula follows from Lemma 30.5.4 by averaging. With $s = \pi t$, $2\sin s\sin((2k+1)s) = \cos(2ks) - \cos((2k+2)s)$, so the sum telescopes: $\sum_{k=0}^{N}\sin((2k+1)s) = \frac{1-\cos(2(N+1)s)}{2\sin s} = \frac{\sin^2((N+1)s)}{\sin s}$; dividing by $\sin s$ gives the formula for F_N. Hence $F_N \geq 0$, $\int_0^1 F_N = 1$ (as $\int_0^1 D_k = 1$), and for $0 < \delta \leq |t| \leq \frac{1}{2}$, $F_N(t) \leq \frac{1}{(N+1)\sin^2(\pi\delta)}$. Given $\varepsilon > 0$, uniform continuity gives δ with $|f(x-t) - f(x)| < \varepsilon$ for $|t| < \delta$, and then, integrating over $[-\frac{1}{2}, \frac{1}{2}]$,

$$|\sigma_N f(x) - f(x)| \leq \int F_N(t)|f(x-t) - f(x)|\,dt \leq \varepsilon + \frac{2\|f\|_\infty}{(N+1)\sin^2(\pi\delta)},$$

which is less than 2ε for N large, uniformly in x. Each $\sigma_N f$ is a trigonometric polynomial. □

Exercise 30.12.12. Let $f \in L^1(S^1)$ and suppose $t \mapsto (f(t) - f(0))/t$ is integrable on $[-\frac{1}{2}, \frac{1}{2}]$; for instance, f is continuous and differentiable at 0, or Lipschitz near 0. Show that $S_N f(0) \to f(0)$ (*Dini's criterion*). Deduce that the Fourier series of $f(t) = t$ on $[-\frac{1}{2}, \frac{1}{2})$, extended periodically, converges to t for $|t| < \frac{1}{2}$, and evaluate it at $t = \frac{1}{4}$ to obtain $1 - \frac{1}{3} + \frac{1}{5} - \cdots = \frac{\pi}{4}$.

Solution. Since $\int_0^1 D_N = 1$ and D_N is even, $S_N f(0) - f(0) = \int_{-1/2}^{1/2}(f(t) - f(0))D_N(t)\,dt = \int_{-1/2}^{1/2} g(t)\sin((2N+1)\pi t)\,dt$ with $g(t) = (f(t) - f(0))/\sin(\pi t)$, which is integrable because $|\sin(\pi t)| \geq 2|t|$ on $[-\frac{1}{2}, \frac{1}{2}]$ (concavity of sin on $[0, \frac{\pi}{2}]$). Writing $\sin((2N+1)\pi t) = \frac{1}{2i}(e^{\pi i t}e^{2\pi i N t} - e^{-\pi i t}e^{-2\pi i N t})$, the integral is a combination of Fourier coefficients $\hat{h}_\pm(\mp N)$ of the integrable functions $h_\pm(t) = g(t)e^{\pm\pi i t}$ on $[-\frac{1}{2}, \frac{1}{2})$, which tend to 0 by the Riemann–Lebesgue lemma (Proposition 30.5.6). If f is differentiable at 0, the quotient $(f(t) - f(0))/t$ is bounded for $|t| < \delta$, for some $\delta > 0$, and for $|t| \geq \delta$ it is at most $(|f(t)| + |f(0)|)/\delta$, which is integrable; so the hypothesis holds. For $f(t) = t$: $\hat{f}(0) = 0$ and, integrating by parts, $\hat{f}(n) = \frac{(-1)^n i}{2\pi n}$ for $n \neq 0$, so the Fourier series is $\sum_{n\geq 1}\frac{(-1)^{n+1}}{\pi n}\sin(2\pi n t)$. At an interior point x, apply the criterion to the translate $f(\cdot + x)$, whose Fourier coefficients are $\hat{f}(n)e^{2\pi i n x}$. At $t = \frac{1}{4}$, $\sin(2\pi n/4)$ is 0 for even n and $(-1)^{(n-1)/2}$ for odd n, giving $\frac{1}{4} = \frac{1}{\pi}(1 - \frac{1}{3} + \frac{1}{5} - \cdots)$. □

Exercise 30.12.13. Let V be the Volterra operator. Show that for every $\lambda \in \mathbb{C}$ and $g \in L^2([0,1])$ the integral equation $f(x) - \lambda\int_0^x f(t)\,dt = g(x)$ has the unique solution $f = \sum_{n\geq 0}\lambda^n V^n g$ in $L^2([0,1])$, and compute it for $g = 1$.

Solution. By Example 30.7.4(4), $\|\lambda^n V^n\| \leq |\lambda|^n/(n-1)!$, so the series $\sum\lambda^n V^n$ converges absolutely in $B(L^2)$, and as in Lemma 30.7.1 its sum is the inverse of $I - \lambda V$. For $g = 1$, $V^n 1 = x^n/n!$, so $f(x) = \sum_n \lambda^n x^n/n! = e^{\lambda x}$; indeed $e^{\lambda x} - \lambda\int_0^x e^{\lambda t}\,dt = 1$. □

Exercise 30.12.14. Let H be a complex Hilbert space and $T \in B(H)$. Show that $\sigma(T^*) = \{\bar{\lambda} : \lambda \in \sigma(T)\}$, that the spectrum of a unitary operator lies on the unit circle, and that the right shift on ℓ^2 has the closed unit disc as its spectrum.

Solution. If A is invertible, then $(A^{-1})^*A^* = (AA^{-1})^* = I$ and $A^*(A^{-1})^* = I$, so A^* is invertible; the converse follows from $A^{**} = A$ (Exercise 30.12.1). Apply this to $A = T - \lambda$, with $A^* = T^* - \bar{\lambda}$. If U is unitary, $\|U\| = \|U^{-1}\| = 1$, so $\sigma(U)$ and $\sigma(U^{-1})$ lie in the closed unit disc, and $0 \notin \sigma(U)$. For $\lambda \neq 0$, $U - \lambda = -\lambda U(U^{-1} - \lambda^{-1})$, so $\lambda \in \sigma(U)$ implies $\lambda^{-1} \in \sigma(U^{-1})$ and $|\lambda| \geq 1$. The right shift is the adjoint of the left shift, whose spectrum is the closed unit disc (Example 30.7.4(3)), which is invariant under conjugation. □

Exercise 30.12.15. Show that the unit vectors e_n converge weakly to 0 in c_0 and in ℓ^p for $1 < p < \infty$, but not in ℓ^1.

Solution. By Proposition 30.11.1, every functional on ℓ^p ($1 < p < \infty$) is Λ_y with $y \in \ell^q$, $q < \infty$, and every functional on c_0 is Λ_y with $y \in \ell^1$; in both cases $\Lambda_y(e_n) = y_n \to 0$. On ℓ^1, the functional $\Lambda_{\mathbf{1}}$ given by $\mathbf{1} = (1, 1, \ldots) \in \ell^\infty$ has $\Lambda_{\mathbf{1}}(e_n) = 1$ for all n. □

Exercise 30.12.16. Let ℓ^∞ be the real space of bounded real sequences and $(Sx)_n = x_{n+1}$ the left shift. Show that there is a linear functional L on ℓ^∞, a *Banach limit*, with $\liminf_n x_n \leq L(x) \leq \limsup_n x_n$, $L(Sx) = L(x)$ for all x, and hence $L(x) = \lim_n x_n$ for convergent x. Compute $L(1, 0, 1, 0, \ldots)$, and show that L is not of the form Λ_y with $y \in \ell^1$. (Hint: apply Lemma 30.10.1 with $p(x) = \limsup_n \frac{x_1 + \cdots + x_n}{n}$.)

Solution. p is sublinear, and on the subspace c of convergent sequences $p(x) = \lim x_n$, since Cesàro means of a convergent sequence converge to the same limit. Extend lim from c to a linear $L \leq p$. Then $L(x) \leq p(x) \leq \limsup x_n$ and $-L(x) = L(-x) \leq \limsup(-x_n) = -\liminf x_n$. Moreover $p(x - Sx) = \limsup_n \frac{x_1 - x_{n+1}}{n} = 0$ for bounded x, and likewise $p(Sx - x) = 0$, so $L(x - Sx) \leq 0$ and $L(Sx - x) \leq 0$, that is, $L(Sx) = L(x)$. For $x = (1, 0, 1, 0, \ldots)$, $x + Sx = \mathbf{1}$, so $2L(x) = L(\mathbf{1}) = 1$ and $L(x) = \frac{1}{2}$. If $L = \Lambda_y$ with $y \in \ell^1$, then $y_n = L(e_n) = 0$, as e_n converges to 0; so $L = 0$, contradicting $L(\mathbf{1}) = 1$. □

Exercise 30.12.17. Let X be a reflexive Banach space and $M \subseteq X$ a closed subspace. Show that M is reflexive. (Hint: given $\xi \in M^{**}$, define $\eta \in X^{**}$ by $\eta(\Lambda) = \xi(\Lambda|_M)$, write $\eta = Jx$, and use Lemma 30.11.2 to show $x \in M$.)

Solution. Restriction $X^* \to M^*$ has norm at most 1, so η is bounded, and $\eta = Jx$ for some $x \in X$. If $x \notin M$, the lemma gives $\Lambda \in X^*$ vanishing on M with $\Lambda(x) \neq 0$; but $\Lambda(x) = \eta(\Lambda) = \xi(\Lambda|_M) = \xi(0) = 0$. So $x \in M$. For $\lambda \in M^*$, extend it to $\Lambda \in X^*$ by the Hahn–Banach theorem; then $\xi(\lambda) = \xi(\Lambda|_M) = \eta(\Lambda) = \Lambda(x) = \lambda(x)$. So $\xi = J_M x$. □

Chapter 31

Several Complex Variables

31.1 Introduction

In several variables holomorphic functions are still given by convergent power series, but new phenomena appear: functions extend across small holes (Theorem 31.2.9), the local ring of convergent power series is a Noetherian unique factorisation domain (Theorem 31.3.4), and the equation $\bar\partial u = f$ is solvable on polydiscs (Theorem 31.4.2), which is the local statement behind Dolbeault cohomology and the Hodge decomposition. References: [Hör90b], [GR65], [GH78, Chapter 0].

Prerequisites. Chapter 29 (Complex Analysis) and Chapter 14 (Commutative Algebra).

For $a \in \mathbb{C}^n$ and $r = (r_1, \dots, r_n)$ with $r_j > 0$, the *polydisc* is $\Delta(a, r) = \{z : |z_j - a_j| < r_j\}$ and its *distinguished boundary* is $\{|z_j - a_j| = r_j \ \forall j\}$. We write $\partial/\partial z_j = \frac{1}{2}(\partial_{x_j} - i\partial_{y_j})$ and $\partial/\partial \bar z_j = \frac{1}{2}(\partial_{x_j} + i\partial_{y_j})$, so that a C^1 function is holomorphic in z_j if and only if $\partial f/\partial \bar z_j = 0$ (Proposition 29.2.3).

31.2 Holomorphic functions of several variables

Definition 31.2.1. Let $U \subseteq \mathbb{C}^n$ be open. A function $f : U \to \mathbb{C}$ is *holomorphic* if it is continuous and holomorphic in each variable separately. The set of such functions is $\mathcal{O}(U)$.

Theorem 31.2.2. *Let f be holomorphic on a neighbourhood of the closure of $\Delta = \Delta(a, r)$. Then for $z \in \Delta$*

$$f(z) = \frac{1}{(2\pi i)^n} \int_{|\zeta_n - a_n| = r_n} \cdots \int_{|\zeta_1 - a_1| = r_1} \frac{f(\zeta)}{(\zeta_1 - z_1)\cdots(\zeta_n - z_n)}\, d\zeta_1 \cdots d\zeta_n,$$

and $f(z) = \sum_{\alpha \in \mathbb{N}^n} c_\alpha (z-a)^\alpha$ with $c_\alpha = \partial^\alpha f(a)/\alpha!$, the series converging absolutely and uniformly on compact subsets of Δ. In particular holomorphic functions are C^∞, all their partial derivatives are holomorphic, and $|c_\alpha| \le \sup |f| \cdot r^{-\alpha}$ (Cauchy estimates).

Proof. Apply the one-variable Cauchy formula (Theorem 29.4.1) in each variable in turn, using continuity to integrate iterated integrals (Proposition 27.8.9). Expanding each factor $1/(\zeta_j - z_j)$ in a geometric series, as in Theorem 29.4.2, and multiplying out gives the power series, uniformly convergent on compact subsets; term-by-term differentiation identifies the coefficients. Smoothness and holomorphy of the derivatives follow from differentiating the integral formula under the integral sign. □

Proposition 31.2.3. *Let $U \subseteq \mathbb{C}^n$ be a domain.*

1. *(Identity theorem) If $f \in \mathcal{O}(U)$ vanishes on a nonempty open subset, then $f = 0$.*

2. *(Maximum principle) If $|f|$ has a local maximum in U, then f is constant.*

3. *If $f_k \in \mathcal{O}(U)$ converge uniformly on compact subsets, the limit is holomorphic, and all partial derivatives converge uniformly on compact subsets.*

Proof. (1) The set where all derivatives $\partial^\alpha f$ vanish is open by Theorem 31.2.2 and closed, hence is U. (2) Restrict to complex lines through the point and apply Theorem 29.4.9: f is constant on a polydisc, hence on U by (1). (3) Separate holomorphy passes to uniform limits (Corollary 29.4.7), and convergence of derivatives follows from the Cauchy estimates applied to $f_k - f_l$ on slightly smaller polydiscs. □

Lemma 31.2.4. *Let A be a complex $n \times n$ matrix and $A_\mathbb{R}$ the real-linear map of $\mathbb{R}^{2n} = \mathbb{C}^n$ it defines. Then $\det A_\mathbb{R} = |\det A|^2$. In particular $A_\mathbb{R}$ is invertible if and only if A is, and a holomorphic map with invertible complex Jacobian preserves orientation.*

Proof. The map $A \mapsto A_\mathbb{R}$ is multiplicative, so both sides are invariant under conjugation $A \mapsto PAP^{-1}$ by an invertible complex matrix. By induction on n, choosing an eigenvector (which exists because $\mathbb{C}$ is algebraically closed, Theorem 9.7.3) and completing it to a basis, A is conjugate to an upper triangular matrix T with diagonal entries $\lambda_1, \dots, \lambda_n$. In the real basis $e_1, ie_1, \dots, e_n, ie_n$, the matrix of $T_\mathbb{R}$ is block upper triangular with 2×2 diagonal blocks $\begin{pmatrix} a_j & -b_j \\ b_j & a_j \end{pmatrix}$, the matrices of multiplication by $\lambda_j = a_j + ib_j$, of determinant $a_j^2 + b_j^2 = |\lambda_j|^2$. So $\det T_\mathbb{R} = \prod_j |\lambda_j|^2 = |\det T|^2$. □

Proposition 31.2.5 (Holomorphic inverse and implicit function theorems)**.** 1. *Let $f : U \to \mathbb{C}^n$ be holomorphic on an open $U \subseteq \mathbb{C}^n$ (that is, each component is holomorphic), and let $a \in U$ with $\det(\partial f_i/\partial z_j(a)) \neq 0$. Then there are open neighbourhoods V of a and W of $f(a)$ such that $f : V \to W$ is bijective with holomorphic inverse.*

2. *Let $F : U \to \mathbb{C}^m$ be holomorphic on an open $U \subseteq \mathbb{C}^n \times \mathbb{C}^m$, with $F(a,b) = 0$ and $\det(\partial F_i/\partial w_j(a,b)) \neq 0$, where w are the coordinates on $\mathbb{C}^m$. Then there are neighbourhoods V of a and W of b and a holomorphic $g : V \to W$ such that $\{(z,w) \in V \times W : F(z,w) = 0\}$ is the graph of g.*

Proof. (1) By Theorem 31.2.2 f is C^∞, and by the Cauchy–Riemann equations (Proposition 29.2.3) in each variable its real derivative at z is the real-linear map underlying the complex Jacobian $J(z) = (\partial f_i/\partial z_j(z))$. By Lemma 31.2.4 it is invertible at a, so the inverse function theorem (Theorem 27.11.2) gives V, W and a C^∞ inverse $g : W \to V$. Differentiating $f \circ g = \mathrm{id}$, the real derivative of g at w is $J(g(w))^{-1}$, the inverse of a $\mathbb{C}$-linear map, hence $\mathbb{C}$-linear. A C^1 map whose real derivative is everywhere $\mathbb{C}$-linear satisfies the Cauchy–Riemann equations in each variable, so g is holomorphic (Definition 31.2.1).

(2) Apply (1) to $\Phi(z,w) = (z, F(z,w))$, whose complex Jacobian at (a,b) is block triangular with determinant $\det(\partial F_i/\partial w_j(a,b)) \neq 0$. Its local holomorphic inverse has the form $\Psi(z,u) = (z, h(z,u))$, and $g(z) = h(z,0)$ is holomorphic, with $F(z,w) = 0$ for (z,w) near (a,b) exactly when $w = g(z)$. □

Corollary 31.2.6 (Liouville)**.** *Let f be holomorphic on $\mathbb{C}^n$ with $|f(z)| \leq C(1 + \|z\|)^N$ for all z. Then f is a polynomial of degree at most N. In particular bounded holomorphic functions on $\mathbb{C}^n$ are constant.*

Proof. Theorem 31.2.2, applied on the polydiscs $\Delta(0, (r, \dots, r))$ for all r, shows that the Taylor series $\sum_\alpha c_\alpha z^\alpha$ at 0 converges to f on all of $\mathbb{C}^n$, with $|c_\alpha| \leq C(1 + \sqrt{n}\, r)^N r^{-|\alpha|}$. For $|\alpha| > N$ the right side tends to 0 as $r \to \infty$, so $c_\alpha = 0$. □

Example 31.2.7. Domains of convergence of power series in several variables need not be polydiscs. The series $\sum_{k\geq 0}(z_1z_2)^k$ converges, to $1/(1-z_1z_2)$, exactly on $\{|z_1z_2|<1\}$, an unbounded domain containing both coordinate axes. It is invariant under $(z_1,z_2)\mapsto(e^{i\theta_1}z_1,e^{i\theta_2}z_2)$ (a *Reinhardt domain*), as is the domain of convergence of every power series centred at 0, because convergence of $\sum c_\alpha z^\alpha$ depends only on the moduli $|z_j|$.

Counterexample 31.2.8. The continuity assumption in Definition 31.2.1 is needed for real variables but not, by a theorem of Hartogs, for complex ones [Hör90b]. On $\mathbb{R}^2$ let $f(x,y)=xy/(x^2+y^2)$ and $f(0,0)=0$. For each fixed y, $x\mapsto f(x,y)$ is real-analytic (it is 0 if $y=0$, and a rational function with nonvanishing denominator otherwise), and likewise in y. But $f(t,t)=\frac{1}{2}$ for $t\neq 0$, so f is not continuous at the origin.

Theorem 31.2.9 (Hartogs)**.** *Let $n\geq 2$, let $\Delta=\Delta(0,r)\subseteq\mathbb{C}^n$ be a polydisc and $K\subseteq\Delta$ a compact subset such that $\Delta\setminus K$ is connected and $K\subseteq\Delta(0,s)$ for some s with $s_j<r_j$. Then every $f\in\mathcal{O}(\Delta\setminus K)$ extends to a holomorphic function on Δ. In particular holomorphic functions of $n\geq 2$ variables have no isolated singularities and no isolated zeros.*

Proof. Write $z=(z',z_n)$ and choose ρ with $s_n<\rho<r_n$. For $z'\in\Delta'(0,r')$ and $|z_n|<\rho$ set

$$F(z',z_n)=\frac{1}{2\pi i}\int_{|\zeta|=\rho}\frac{f(z',\zeta)}{\zeta-z_n}\,d\zeta,$$

which makes sense because $(z',\zeta)\notin K$ when $|\zeta|=\rho>s_n$. Differentiating under the integral sign, F is holomorphic in z_n, and it is continuous and holomorphic in each variable z_j' because f is. So F is holomorphic on $\Delta'\times\Delta(0,\rho)$, where $\Delta'=\Delta'(0,r')$. If $|z_j'|>s_j$ for some j, the whole disc $\{z'\}\times\Delta(0,\rho)$ avoids K, so the one-variable Cauchy formula gives $F(z',z_n)=f(z',z_n)$ there.

Now fix z_n with $s_n<|z_n|<\rho$. The functions $z'\mapsto F(z',z_n)$ and $z'\mapsto f(z',z_n)$ are both holomorphic on the connected set Δ' (the slice avoids K, since $|z_n|>s_n$) and agree on a nonempty open subset, hence agree on Δ' by Proposition 31.2.3. So $F=f$ on $\Delta'\times\{s_n<|z_n|<\rho\}$, which is exactly the overlap of $\Delta'\times\Delta(0,\rho)$ and $\Delta\cap\{|z_n|>s_n\}$. These two open sets cover Δ because $K\subseteq\{|z_n|<s_n\}$, so F and f glue to a holomorphic function G on Δ. Finally $G=f$ on all of $\Delta\setminus K$, since both are holomorphic there, $\Delta\setminus K$ is connected, and they agree on a nonempty open subset. □

31.3 The Weierstrass theorems

Definition 31.3.1. Let $\mathcal{O}_n=\mathcal{O}_{\mathbb{C}^n,0}$ be the ring of germs at 0 of holomorphic functions, that is, of convergent power series $\sum_\alpha c_\alpha z^\alpha$. A germ f is *regular of order k in z_n* if $f(0,\dots,0,z_n)=cz_n^k+O(z_n^{k+1})$ with $c\neq 0$. A *Weierstrass polynomial* of degree k is $W=z_n^k+a_{k-1}(z')z_n^{k-1}+\cdots+a_0(z')$ with $a_j\in\mathcal{O}_{n-1}$ and $a_j(0)=0$.

Lemma 31.3.2. *$\mathcal{O}_n$ is an integral domain and a local ring with maximal ideal $\mathfrak{m}=\{f:f(0)=0\}$; a germ is a unit if and only if $f(0)\neq 0$. After a linear change of coordinates, any nonzero germ becomes regular of some finite order in z_n.*

Proof. Germs with $f(0)\neq 0$ are invertible, since $1/f$ is holomorphic near 0; so the nonunits form the ideal $\mathfrak{m}$ and the ring is local (Proposition 14.3.2(4)). If $fg=0$ with $f\neq 0$, then g vanishes on a nonempty open subset of the complement of the zero set of f, which is dense, so $g=0$ by the identity theorem. For the last statement, choose $v\in\mathbb{C}^n$ with $f(tv)\neq 0$ for small $t\neq 0$ (possible since the zero set of f is not a neighbourhood of 0), and a linear change of coordinates taking v to e_n; then $t\mapsto f(tv)$ has a zero of finite order at 0. □

Theorem 31.3.3 (Weierstrass preparation and division)**.** *Let $f \in \mathcal{O}_n$ be regular of order k in z_n.*

1. *(Preparation) $f = u\,W$ with $u \in \mathcal{O}_n$ a unit and W a Weierstrass polynomial of degree k, both unique.*
2. *(Division) Every $g \in \mathcal{O}_n$ can be written uniquely as $g = q f + \rho$ with $q \in \mathcal{O}_n$ and $\rho \in \mathcal{O}_{n-1}[z_n]$ of degree $< k$.*

Proof. Choose $\varepsilon > 0$ such that $z_n \mapsto f(0, z_n)$ has $z_n = 0$ as its only zero in $|z_n| \le \varepsilon$, and then, by continuity of f and the argument principle (Theorem 29.5.10), $\delta > 0$ such that for $|z'| < \delta$ the function $z_n \mapsto f(z', z_n)$ has no zero with $|z_n| = \varepsilon$ and exactly k zeros $w_1(z'), \dots, w_k(z')$ in $|z_n| < \varepsilon$, counted with multiplicity.

(1) Let $W(z', z_n) = \prod_{j=1}^{k} (z_n - w_j(z'))$, whose coefficients are, up to sign, the elementary symmetric functions $e_m(z')$ of the roots. The power sums $p_m(z') = \sum_j w_j(z')^m = \frac{1}{2\pi i}\int_{|\zeta|=\varepsilon} \zeta^m \frac{\partial_\zeta f(z',\zeta)}{f(z',\zeta)}\, d\zeta$ are holomorphic in z' (differentiate under the integral sign), by the residue computation in the proof of Theorem 29.5.10. From $E(t) = \prod_j (1 - w_j t) = \sum_m (-1)^m e_m t^m$ we get $E'(t)/E(t) = -\sum_{m \ge 1} p_m t^{m-1}$, so comparing coefficients expresses each e_m as a polynomial with rational coefficients in $p_1, \dots, p_m$; hence the e_m, and therefore the coefficients of W, are holomorphic. Since all roots tend to 0 as $z' \to 0$, $W(0, z_n) = z_n^k$ and W is a Weierstrass polynomial. The quotient $u = f/W$ extends holomorphically across the zeros in each slice (the zeros of f and W agree with multiplicities), and $u(z', z_n) = \frac{1}{2\pi i}\int_{|\zeta|=\varepsilon} \frac{f(z',\zeta)}{W(z',\zeta)} \frac{d\zeta}{\zeta - z_n}$ shows that u is holomorphic in both variables; it has no zeros, so it is a unit. If $f = uW = u'W'$ with W, W' Weierstrass polynomials, then W and W' have the same zeros with multiplicities in each slice, hence are equal.

(2) By (1) we may assume $f = W$ is a Weierstrass polynomial. Define

$$q(z) = \frac{1}{2\pi i}\int_{|\zeta|=\varepsilon} \frac{g(z', \zeta)}{W(z', \zeta)} \cdot \frac{d\zeta}{\zeta - z_n}, \qquad \rho = g - qW,$$

so q is holomorphic. Using the one-variable Cauchy formula for g,

$$\rho(z) = \frac{1}{2\pi i}\int_{|\zeta|=\varepsilon} g(z', \zeta)\, \frac{W(z', \zeta) - W(z', z_n)}{W(z', \zeta)\,(\zeta - z_n)}\, d\zeta,$$

and since W is monic of degree k in the last variable, $(W(z', \zeta) - W(z', z_n))/(\zeta - z_n)$ is a polynomial in z_n of degree $k-1$ whose coefficients are holomorphic in (z', ζ). So $\rho \in \mathcal{O}_{n-1}[z_n]$ has degree $< k$. For uniqueness, if $q_1 W + \rho_1 = q_2 W + \rho_2$, then $(q_1 - q_2)W = \rho_2 - \rho_1$; for fixed small z' the left-hand side has at least k zeros in $|z_n| < \varepsilon$ with multiplicity, while the right-hand side is a polynomial of degree $< k$, so both sides vanish. For general $f = uW$ take $q = u^{-1} q_W$. □

Theorem 31.3.4. *$\mathcal{O}_n$ is a Noetherian local ring and a unique factorisation domain, of Krull dimension n.*

Proof. Induction on n; $\mathcal{O}_0 = \mathbb{C}$ is a field. Suppose $\mathcal{O}_{n-1}$ is Noetherian and a UFD. By Hilbert's basis theorem (Theorem 14.5.3), $\mathcal{O}_{n-1}[z_n]$ is Noetherian, and by Theorem 8.6.11 it is a UFD.

Noetherian. Let $I \subseteq \mathcal{O}_n$ be a nonzero ideal and $f \in I$ nonzero; after a linear change of coordinates f is regular of some order k in z_n (Lemma 31.3.2). By Weierstrass division, I is generated by f together with $I \cap \mathcal{O}_{n-1}[z_n]$, which is finitely generated because $\mathcal{O}_{n-1}[z_n]$ is Noetherian.

Factorial. Let $f \in \mathcal{O}_n$ be nonzero and not a unit; after a change of coordinates and by preparation, $f = uW$ with W a Weierstrass polynomial. Factor $W = W_1 \cdots W_m$ into

irreducibles in $\mathcal{O}_{n-1}[z_n]$; by Gauss's lemma (Lemma 8.6.10) and induction each factor may be taken to be a Weierstrass polynomial. Uniqueness of the factorisation in $\mathcal{O}_n$ follows from uniqueness in $\mathcal{O}_{n-1}[z_n]$ together with the uniqueness statement in Theorem 31.3.3: a factorisation in $\mathcal{O}_n$ of a Weierstrass polynomial produces, after preparation of each factor, a factorisation into Weierstrass polynomials, and irreducible Weierstrass polynomials are irreducible in $\mathcal{O}_n$ (a factorisation in $\mathcal{O}_n$ can be normalised in the same way).

Dimension. The chain of primes $0 \subsetneq (z_1) \subsetneq (z_1, z_2) \subsetneq \cdots \subsetneq (z_1, \dots, z_n) = \mathfrak{m}$ has length n, so $\dim \mathcal{O}_n \geq n$; equality holds because $\mathfrak{m}$ is generated by the n elements $z_1, \dots, z_n$ (Corollary 14.9.6). □

31.4 The Dolbeault–Grothendieck lemma

Lemma 31.4.1 (Cauchy transform)**.** *Let $g\colon \mathbb{C} \to \mathbb{C}$ be C^∞ with compact support and $u(z) = \frac{1}{\pi}\int_{\mathbb{C}} \frac{g(z-w)}{w}\, dA(w)$, where dA is Lebesgue measure. Then u is C^∞ and $\partial u/\partial \bar z = g$. If g depends smoothly (or holomorphically) on additional parameters, so does u.*

Proof. The function $1/|w|$ is integrable near 0, so the integral converges and, by dominated convergence and the smoothness of g, we may differentiate under the integral sign: u is C^∞ and $\partial u/\partial \bar z = \frac{1}{\pi}\int (\partial g/\partial \bar z)(z-w) w^{-1}\, dA(w)$. Write $w = \rho e^{i\theta}$, so $dA = \rho\, d\rho\, d\theta$ and $w^{-1} dA = e^{-i\theta} d\rho\, d\theta$, and set $F(\rho,\theta) = g(z - \rho e^{i\theta})$. From $\partial_\rho F = -(e^{i\theta} g_z + e^{-i\theta} g_{\bar z})$ and $\partial_\theta F = i\rho(-e^{i\theta} g_z + e^{-i\theta} g_{\bar z})$ we get $e^{-i\theta} g_{\bar z} = -\frac{1}{2}\partial_\rho F + \frac{1}{2i\rho}\partial_\theta F$. Hence

$$\frac{\partial u}{\partial \bar z} = \frac{1}{\pi}\int_0^{2\pi}\int_0^{\infty} \left(-\tfrac{1}{2}\partial_\rho F + \tfrac{1}{2i\rho}\partial_\theta F\right) d\rho\, d\theta = \frac{1}{\pi}\int_0^{2\pi} \tfrac{1}{2} g(z)\, d\theta = g(z),$$

since $\int_0^\infty \partial_\rho F\, d\rho = -g(z)$ by the fundamental theorem of calculus and compact support, and $\int_0^{2\pi} \partial_\theta F\, d\theta = 0$ by periodicity. Dependence on parameters follows in the same way from differentiation under the integral sign. □

Theorem 31.4.2 ($\bar\partial$-Poincaré lemma)**.** *Let $\Delta' \subseteq \Delta \subseteq \mathbb{C}^n$ be polydiscs centred at 0 with $\overline{\Delta'} \subseteq \Delta$, and let $f_1, \dots, f_n$ be C^∞ functions on Δ with*

$$\frac{\partial f_j}{\partial \bar z_l} = \frac{\partial f_l}{\partial \bar z_j} \qquad \text{for all } j, l.$$

Then there is a C^∞ function u on Δ' with $\partial u/\partial \bar z_j = f_j$ for all j. In the language of differential forms (Chapter 46), every $\bar\partial$-closed $(0,1)$-form on Δ is $\bar\partial$-exact on Δ', and the same holds for (p,q)-forms.

Proof. We prove by descending induction on k the following statement: if $f_j = 0$ for $j > k$, then on any polydisc Δ'' with $\overline{\Delta''} \subseteq \Delta$ there is u with $\partial u/\partial \bar z_j = f_j$ for all j. For $k = 0$ take $u = 0$. Let $k \geq 1$ and choose a C^∞ function $\chi(z_k)$ with compact support in the k-th disc of Δ, equal to 1 on a neighbourhood of the closure of the k-th disc of Δ''. Put $g = \chi f_k$, extended by zero, and

$$v(z) = \frac{1}{\pi}\int_{\mathbb{C}} \frac{g(z_1, \dots, z_{k-1}, z_k - w, z_{k+1}, \dots, z_n)}{w}\, dA(w),$$

which is C^∞ on Δ by Lemma 31.4.1 and satisfies $\partial v/\partial \bar z_k = g$, so $\partial v/\partial \bar z_k = f_k$ on Δ''. For $j > k$ we have $\partial g/\partial \bar z_j = \chi\, \partial f_k/\partial \bar z_j = \chi\, \partial f_j/\partial \bar z_k = 0$ (because χ depends only on z_k and $f_j = 0$), so $\partial v/\partial \bar z_j = 0$ by differentiating under the integral sign.

Now set $f_j' = f_j - \partial v/\partial \bar z_j$. These are C^∞ on Δ, satisfy the same compatibility conditions (second derivatives commute, Theorem 27.10.5), and $f_j' = 0$ for $j \geq k$ on Δ''. By the induction

hypothesis applied on Δ'' (with $k-1$), there is u_0 on any polydisc compactly contained in Δ'', in particular on Δ' if we choose Δ'' between them, with $\partial u_0/\partial \bar z_j = f_j'$; then $u = u_0 + v$ works. □

31.5 Analytic sets

Definition 31.5.1. A subset A of an open $U \subseteq \mathbb{C}^n$ is *analytic* if every point of U has a neighbourhood V such that $A \cap V$ is the common zero set of finitely many holomorphic functions on V. A point $a \in A$ is *regular* of dimension d if near a the set A is, after a biholomorphic change of coordinates, an open subset of $\mathbb{C}^d \times \{0\}$; otherwise it is *singular*.

Example 31.5.2. The zero set of a single nonzero $f \in \mathcal{O}(U)$ is an analytic *hypersurface*; by Theorem 31.3.3 it is, in suitable local coordinates, the zero set of a Weierstrass polynomial, hence a finite branched covering of a polydisc in $\mathbb{C}^{n-1}$. The set $\{z_1 z_2 = 0\} \subseteq \mathbb{C}^2$ is analytic with singular point the origin; $\{z_1^2 = z_2^3\}$ is analytic, irreducible, with a cusp at the origin.

Theorem 31.5.3. *Let $A \subseteq U$ be analytic.*

1. *The regular points of A are dense in A, and the singular locus A_{sing} is a closed analytic subset of U contained in A, nowhere dense in A.*
2. *Locally, A has finitely many irreducible components, and an irreducible germ of dimension d admits a local projection to a polydisc in $\mathbb{C}^d$ that is a finite branched covering, branched over a proper analytic subset.*
3. *(Riemann extension) If $A \subseteq U$ is analytic of dimension $\le n-1$ and f is holomorphic on $U \setminus A$ and locally bounded near A, then f extends holomorphically to U.*

These are proved in [GR65, Chapters II–III] and [GR84, Chapters 1–6]; the key tools are the Weierstrass theorems and the local parametrisation theorem. Part (3) for a hypersurface follows from the one-variable Riemann extension theorem (Proposition 29.5.3) applied on lines transverse to A, together with Theorem 31.2.2.

Remark 31.5.4. The sheaf $\mathcal{O}_X$ of holomorphic functions on a complex manifold is *coherent* (Oka's theorem), and so is the ideal sheaf of an analytic subset (Cartan); these are the foundations of the theory of coherent analytic sheaves used in Chapter 58, where they are compared with the algebraic theory [GR84].

31.6 Exercises

Exercise 31.6.1. Show that $\{z \in \mathbb{C}^2 : |z_1| < 1, |z_2| < 1\} \setminus \{0\}$ and $\mathbb{C}^2 \setminus \{0\}$ have the property that every holomorphic function extends to the full polydisc, respectively to $\mathbb{C}^2$, and deduce that these domains are not biholomorphic to any domain in $\mathbb{C}$.

Solution. Extension is Theorem 31.2.9 with $K = \{0\}$. In $\mathbb{C}$, the function $1/z$ is holomorphic on the punctured disc and does not extend; so no biholomorphism with a punctured planar domain can exist, since biholomorphisms preserve the algebra of holomorphic functions. □

Exercise 31.6.2. Apply the Weierstrass preparation theorem to $f(z_1, z_2) = z_2^2 - z_1$ and to $f(z_1, z_2) = z_1 z_2$ at the origin, and explain why a linear change of coordinates is needed in the second case.

Exercise 31.6.3. Let f be C^∞ on a neighbourhood of the closed unit disc. Show that there is a C^∞ function u on the open disc with $\partial u/\partial \bar{z} = f$, and that u is unique up to adding a holomorphic function.

Solution. Multiply f by a cut-off function equal to 1 on a slightly smaller disc and apply Lemma 31.4.1; exhaust the disc. The difference of two solutions satisfies $\partial u/\partial \bar{z} = 0$, hence is holomorphic (Proposition 29.2.3). □

Exercise 31.6.4. Show that a holomorphic function on a bounded domain $U \subseteq \mathbb{C}^n$, continuous on $\overline{U}$, attains the maximum of its modulus on ∂U, and give an example showing that for a polydisc the maximum is already attained on the distinguished boundary.

Chapter 32

Distributions and Sobolev Spaces

32.1 Introduction

To solve $\Delta u = f$ one first finds a weak solution in a space of distributions and then proves that it is smooth. This chapter sets up the two tools that make this work: the Fourier transform, which turns differentiation into multiplication, and the Sobolev spaces H^s, in which the elliptic estimate (Theorem 32.5.3) and Rellich's compactness theorem (Theorem 32.4.5) hold. On a compact manifold these local statements are patched together with partitions of unity to prove the Hodge theorem (Chapter 43). References: [Hör90a], [Rud91, Chapters 6–7], [Bre11], [War83, Chapter 6].

Prerequisites. Chapter 30 (Functional Analysis).

Multi-index notation is as in Theorem 27.10.7; we write $D^\alpha = \partial^\alpha$ and $\langle \xi \rangle = (1 + |\xi|^2)^{1/2}$.

32.2 Distributions

Definition 32.2.1. Let $U \subseteq \mathbb{R}^n$ be open and $\mathcal{D}(U) = C_c^\infty(U)$. A sequence $\varphi_k \to \varphi$ in $\mathcal{D}(U)$ means: all φ_k have support in a fixed compact subset of U, and $D^\alpha \varphi_k \to D^\alpha \varphi$ uniformly for every α. A *distribution* is a linear map $T\colon \mathcal{D}(U) \to \mathbb{C}$ such that $T(\varphi_k) \to T(\varphi)$ whenever $\varphi_k \to \varphi$; equivalently, for every compact $K \subseteq U$ there are C and N with $|T(\varphi)| \le C \sum_{|\alpha| \le N} \sup |D^\alpha \varphi|$ for all φ supported in K. The space of distributions is $\mathcal{D}'(U)$, and $T_k \to T$ means $T_k(\varphi) \to T(\varphi)$ for every φ.

Example 32.2.2.

1. Every $f \in L^1_{\mathrm{loc}}(U)$ defines $T_f(\varphi) = \int f\varphi$, and $T_f = T_g$ if and only if $f = g$ a.e.; so $\mathcal{D}'(U)$ contains the locally integrable functions.
2. The *Dirac distribution* $\delta_a(\varphi) = \varphi(a)$ is not of this form: if $\delta_0 = T_f$, then $\int f\varphi = 0$ for all φ supported away from 0, so $f = 0$ a.e., while $\delta_0 \neq 0$.
3. $\operatorname{pv} \frac{1}{x}(\varphi) = \lim_{\varepsilon \to 0} \int_{|x| > \varepsilon} \frac{\varphi(x)}{x}\, dx$ is a distribution on $\mathbb{R}$ of order 1.

Definition 32.2.3. For $T \in \mathcal{D}'(U)$, α a multi-index and $g \in C^\infty(U)$, set $(D^\alpha T)(\varphi) = (-1)^{|\alpha|} T(D^\alpha \varphi)$ and $(gT)(\varphi) = T(g\varphi)$. The *support* of T is the complement of the largest open set on which T vanishes.

Proposition 32.2.4. *These are distributions; $D^\alpha D^\beta T = D^{\alpha+\beta} T$; for $f \in C^k(U)$ the distributional derivatives of order $\le k$ are the classical ones; and differentiation is continuous: $T_k \to T$ implies $D^\alpha T_k \to D^\alpha T$.*

Proof. Continuity is clear from the definition, and the formula for $D^\alpha T$ is forced by integration by parts, which for $f \in C^k$ and $\varphi \in C_c^\infty$ gives $\int (D^\alpha f)\varphi = (-1)^{|\alpha|} \int f D^\alpha \varphi$ (Fubini's theorem

and the fundamental theorem of calculus in one variable at a time, Proposition 27.8.9). The other statements are immediate. □

Example 32.2.5. The Heaviside function $H = \mathbf{1}_{[0,\infty)}$ on $\mathbb{R}$ has $H' = \delta_0$, since $-\int_0^\infty \varphi' = \varphi(0)$. Hence $\delta_0'(\varphi) = -\varphi'(0)$, and the function $\log|x|$ has distributional derivative $\operatorname{pv} \frac{1}{x}$.

Proposition 32.2.6. *Let $I \subseteq \mathbb{R}$ be an open interval. A distribution T on I with $T' = 0$ is a constant function: $T = T_c$ for some $c \in \mathbb{C}$.*

Proof. Fix $\psi \in \mathcal{D}(I)$ with $\int \psi = 1$ and put $c = T(\psi)$. For $\varphi \in \mathcal{D}(I)$, the function $\varphi - (\int \varphi)\psi$ has integral 0 and compact support in I, so it is the derivative of $\chi(x) = \int_{-\infty}^x (\varphi - (\int \varphi)\psi)$, which also has compact support in I (it vanishes to the left of both supports, and to the right because the total integral is 0). Hence $T(\varphi) - (\int \varphi)c = T(\chi') = -T'(\chi) = 0$, that is, $T(\varphi) = \int c\varphi$. □

Proposition 32.2.7. *Let $\rho \in L^1(\mathbb{R}^n)$ with $\int \rho = 1$ and $\rho_\varepsilon(x) = \varepsilon^{-n}\rho(x/\varepsilon)$. Then $T_{\rho_\varepsilon} \to \delta_0$ in $\mathcal{D}'(\mathbb{R}^n)$ as $\varepsilon \to 0$.*

Proof. For $\varphi \in \mathcal{D}(\mathbb{R}^n)$, the substitution $x = \varepsilon y$ gives $\int \rho_\varepsilon \varphi - \varphi(0) = \int \rho(y)(\varphi(\varepsilon y) - \varphi(0))\,dy$. The integrand tends to 0 pointwise and is bounded by $2\sup|\varphi|\,|\rho|$, so the integral tends to 0 by dominated convergence (Theorem 28.4.8). □

32.3 The Fourier transform

Definition 32.3.1. The *Schwartz space* $\mathcal{S}(\mathbb{R}^n)$ consists of the $f \in C^\infty(\mathbb{R}^n)$ with $p_{\alpha\beta}(f) = \sup_x |x^\alpha D^\beta f(x)| < \infty$ for all α, β; convergence in $\mathcal{S}$ means convergence in all these seminorms. Its continuous dual $\mathcal{S}'(\mathbb{R}^n)$ is the space of *tempered distributions*. For $f \in \mathcal{S}$,

$$\hat{f}(\xi) = \mathcal{F}f(\xi) = \int_{\mathbb{R}^n} f(x)e^{-2\pi i x\cdot\xi}\,dx.$$

Theorem 32.3.2. *$\mathcal{F}$ maps $\mathcal{S}$ continuously to $\mathcal{S}$, with $\mathcal{F}(D^\alpha f)(\xi) = (2\pi i\xi)^\alpha \hat{f}(\xi)$ and $\mathcal{F}(x^\alpha f)(\xi) = (-2\pi i)^{-|\alpha|}D^\alpha \hat{f}(\xi)$. Moreover:*

1. *$\mathcal{F}(e^{-\pi|x|^2}) = e^{-\pi|\xi|^2}$;*
2. *(Inversion) $f(x) = \int \hat{f}(\xi)e^{2\pi i x\cdot\xi}\,d\xi$, so $\mathcal{F}$ is a bijection of $\mathcal{S}$ with inverse $\mathcal{F}^{-1}g(x) = \hat{g}(-x)$;*
3. *(Plancherel) $\int f\bar{g} = \int \hat{f}\bar{\hat{g}}$; in particular $\|\hat{f}\|_2 = \|f\|_2$, and $\mathcal{F}$ extends to a unitary operator on $L^2(\mathbb{R}^n)$.*

Proof. Differentiation under the integral sign and integration by parts give the two formulas, and they imply that $\mathcal{F}f$ is smooth and rapidly decreasing, with continuity of $\mathcal{F}$ in the seminorms.

(1) For $n = 1$ let $g(\xi) = \int e^{-\pi x^2}e^{-2\pi i x\xi}dx$. Differentiating under the integral and integrating by parts, $g'(\xi) = -2\pi\xi g(\xi)$, so $g(\xi) = g(0)e^{-\pi\xi^2}$, and $g(0) = 1$ by Example 28.11.3. The general case follows by Fubini's theorem, since $e^{-\pi|x|^2} = \prod_j e^{-\pi x_j^2}$.

(2) For $f, g \in \mathcal{S}$, Fubini's theorem gives the multiplication formula $\int \hat{f}g = \int f\hat{g}$. Apply it with $g(\xi) = e^{2\pi i x\cdot\xi}e^{-\pi\varepsilon^2|\xi|^2}$, whose Fourier transform is $\varepsilon^{-n}e^{-\pi|x-y|^2/\varepsilon^2}$ by (1) and the scaling and translation rules. The right-hand side is a convolution of f with an approximate identity, which converges to $f(x)$ as $\varepsilon \to 0$ by uniform continuity, and the left-hand side converges to $\int \hat{f}(\xi)e^{2\pi i x\cdot\xi}d\xi$ by dominated convergence.

(3) Let $h = \overline{\hat{g}}$; then $\hat{h} = \bar{g}$ by (2) and $\overline{\mathcal{F}\bar{u}} = \mathcal{F}^{-1}u$. The multiplication formula with f and h gives $\int \hat{f}\overline{\hat{g}} = \int f\bar{g}$. The space $\mathcal{S}$ is dense in L^2: it contains C_c^∞, and every continuous function with compact support is a uniform limit of its convolutions with a smooth approximate identity (Exercise 28.23.6), which lie in C_c^∞, while such functions are dense in L^2 by Proposition 28.5.4. Hence $\mathcal{F}$ extends uniquely to an isometry of L^2, which is surjective because its image contains the dense subspace $\mathcal{S}$. □

Proposition 32.3.3. *For $f \in L^1(\mathbb{R}^n)$ the integral $\hat{f}(\xi) = \int f(x)e^{-2\pi ix\cdot\xi}\,dx$ defines a continuous function with $\|\hat{f}\|_\infty \le \|f\|_1$ and $\hat{f}(\xi) \to 0$ as $|\xi| \to \infty$ (the* Riemann–Lebesgue lemma*). For $f, g \in L^1$, $\widehat{f * g} = \hat{f}\,\hat{g}$.*

Proof. The bound is clear, and continuity follows from dominated convergence. For $f \in \mathcal{S}$, $\hat{f} \in \mathcal{S}$ tends to 0 at infinity. In general choose $f_k \in \mathcal{S}$ with $\|f - f_k\|_1 \to 0$ (C_c^∞ is dense in L^1, by the argument in the proof of Theorem 32.3.2(3)); then $\hat{f}_k \to \hat{f}$ uniformly, and a uniform limit of functions tending to 0 at infinity tends to 0 at infinity. For the convolution, $f * g \in L^1$ (Exercise 28.23.6), and by Fubini's theorem $\int\int f(x-y)g(y)e^{-2\pi ix\cdot\xi}\,dy\,dx = \int g(y)e^{-2\pi iy\cdot\xi}\int f(x-y)e^{-2\pi i(x-y)\cdot\xi}\,dx\,dy = \hat{f}(\xi)\hat{g}(\xi)$. □

Example 32.3.4.

1. The indicator of $[-\frac{1}{2}, \frac{1}{2}]$ has $\hat{f}(\xi) = \int_{-1/2}^{1/2} e^{-2\pi ix\xi}\,dx = \frac{\sin\pi\xi}{\pi\xi}$, which is continuous and tends to 0, but is not integrable: by the inversion theorem it cannot be, since f is not continuous.
2. $f(x) = e^{-2\pi|x|}$ on $\mathbb{R}$: $\hat{f}(\xi) = 2\int_0^\infty e^{-2\pi x}\cos(2\pi x\xi)\,dx = 2\operatorname{Re}\frac{1}{2\pi(1-i\xi)} = \frac{1}{\pi(1+\xi^2)}$. A kink of f at 0 gives a decay $|\xi|^{-2}$ only; smoothness of f corresponds to decay of $\hat{f}$, and decay of f to smoothness of $\hat{f}$.
3. Scaling and translation: $\widehat{f(\lambda x)}(\xi) = \lambda^{-n}\hat{f}(\xi/\lambda)$ and $\widehat{f(x-a)}(\xi) = e^{-2\pi ia\cdot\xi}\hat{f}(\xi)$. With Theorem 32.3.2(1), $\widehat{e^{-\pi t|x|^2}} = t^{-n/2}e^{-\pi|\xi|^2/t}$ for $t > 0$: the narrower the Gaussian, the wider its transform.

Proposition 32.3.5 (Poisson summation)**.** *For $f \in \mathcal{S}(\mathbb{R})$, $\sum_{n\in\mathbb{Z}} f(x+n) = \sum_{k\in\mathbb{Z}} \hat{f}(k)e^{2\pi ikx}$ for all x; in particular $\sum_n f(n) = \sum_k \hat{f}(k)$.*

Proof. The periodisation $P(x) = \sum_n f(x+n)$ converges uniformly with all derivatives on compact sets, as f and its derivatives decay faster than $|x|^{-2}$; so P is a smooth function on $\mathbb{R}/\mathbb{Z}$. Its Fourier coefficients are $\int_0^1 P(x)e^{-2\pi ikx}\,dx = \sum_n \int_n^{n+1} f(y)e^{-2\pi iky}\,dy = \hat{f}(k)$. These decay rapidly, so the Fourier series converges absolutely and uniformly to a continuous function, which equals P almost everywhere, being its L^2 limit (Example 30.3.7), hence everywhere. □

For $f(x) = e^{-\pi tx^2}$ this gives the transformation formula of the theta function, $\sum_n e^{-\pi tn^2} = t^{-1/2}\sum_k e^{-\pi k^2/t}$ for $t > 0$, which relates large and small t and is the starting point of the functional equation of the Riemann zeta function.

Proposition 32.3.6 (Heisenberg's uncertainty principle)**.** *For $f \in \mathcal{S}(\mathbb{R})$, $\|f\|_2^2 \le 4\pi\|xf\|_2\,\|\xi\hat{f}\|_2$. So f and $\hat{f}$ cannot both be concentrated near 0.*

Proof. Integrating by parts, $\|f\|_2^2 = \int |f|^2\,dx = -\int x\frac{d}{dx}|f|^2\,dx = -2\operatorname{Re}\int x\bar{f}f'\,dx \le 2\|xf\|_2\|f'\|_2$ by Cauchy–Schwarz, and $\|f'\|_2 = \|2\pi\xi\hat{f}\|_2$ by Plancherel and Theorem 32.3.2. □

Equality holds for the Gaussians $e^{-\pi tx^2}$; after normalising $\|f\|_2 = 1$, the inequality says that the variances of $|f|^2$ and $|\hat{f}|^2$ have product at least $1/(16\pi^2)$, the mathematical content of the uncertainty relation of quantum mechanics.

Definition 32.3.7. For $T \in \mathcal{S}'$, $\hat{T}(\varphi) = T(\hat{\varphi})$. By Theorem 32.3.2 this defines a linear bijection of $\mathcal{S}'$ extending the transform on $\mathcal{S}$ and on L^2, with $\mathcal{F}(D^\alpha T) = (2\pi i\xi)^\alpha \hat{T}$.

Example 32.3.8. $\hat{\delta}_0 = 1$ and $\hat{1} = \delta_0$; the Fourier transform of a polynomial is a combination of derivatives of δ_0. Compactly supported distributions are tempered, and so are the functions in $L^p(\mathbb{R}^n)$ for $1 \le p \le \infty$.

32.4 Sobolev spaces

Definition 32.4.1. For $s \in \mathbb{R}$, $H^s(\mathbb{R}^n) = \{u \in \mathcal{S}' : \hat{u} \in L^1_{\text{loc}} \text{ and } \|u\|_s^2 = \int \langle\xi\rangle^{2s}|\hat{u}(\xi)|^2\, d\xi < \infty\}$, a Hilbert space with the evident inner product. On the torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$, a distribution u has Fourier coefficients $\hat{u}(k) = u(e^{-2\pi ik\cdot x})$, $k \in \mathbb{Z}^n$, and $H^s(T^n) = \{u : \|u\|_s^2 = \sum_{k\in\mathbb{Z}^n}\langle k\rangle^{2s}|\hat{u}(k)|^2 < \infty\}$.

Proposition 32.4.2. *$H^0 = L^2$, and for a non-negative integer m, H^m is the space of $u \in L^2$ whose distributional derivatives of order $\le m$ lie in L^2, with an equivalent norm $(\sum_{|\alpha|\le m}\|D^\alpha u\|_2^2)^{1/2}$. The spaces H^s decrease in s, C^∞ functions (with compact support, respectively on the torus) are dense in each, and the pairing $(u, v) \mapsto \int u\bar{v}$ identifies H^{-s} with the dual of H^s.*

Proof. $H^0 = L^2$ is Plancherel's theorem (Theorem 32.3.2), or Parseval's identity on the torus (Example 30.3.7). For integer m, $\mathcal{F}(D^\alpha u) = (2\pi i\xi)^\alpha \hat{u}$, and $\sum_{|\alpha|\le m}|(2\pi\xi)^\alpha|^2$ is bounded above and below by multiples of $\langle\xi\rangle^{2m}$. Density follows by truncating and smoothing the Fourier transform, and duality from the Riesz representation theorem applied to the weighted L^2 spaces. □

Theorem 32.4.3 (Sobolev embedding)**.** *If $s > k + n/2$, then every $u \in H^s$ (on $\mathbb{R}^n$ or on T^n) has a representative of class C^k, and $\|u\|_{C^k} \le C\|u\|_s$. Hence $\bigcap_s H^s$ consists of the smooth functions.*

Proof. On the torus, for $|\alpha| \le k$, $\sum_k |(2\pi ik)^\alpha \hat{u}(k)| \le \left(\sum_k \langle k\rangle^{2s}|\hat{u}(k)|^2\right)^{1/2}\left(\sum_k \langle k\rangle^{2(|\alpha|-s)}\right)^{1/2}$ by Cauchy–Schwarz, and the second factor is finite because $2(s - |\alpha|) > n$ (compare with $\int\langle\xi\rangle^{2(|\alpha|-s)}d\xi$, which converges by polar coordinates). So the Fourier series of $D^\alpha u$ converges absolutely and uniformly, and u is C^k with the stated bound. On $\mathbb{R}^n$ the same computation with integrals shows that $\widehat{D^\alpha u} \in L^1$, so $D^\alpha u$ is continuous and bounded by Fourier inversion. □

The embedding theorem trades $n/2$ derivatives in L^2 for continuity. With L^p norms one can do better, and the exponents are dictated by scaling: for $u_\lambda(x) = u(\lambda x)$, $\|u_\lambda\|_q = \lambda^{-n/q}\|u\|_q$ and $\|\nabla u_\lambda\|_p = \lambda^{1-n/p}\|\nabla u\|_p$, so an inequality $\|u\|_q \le C\|\nabla u\|_p$ for all $u \in C_c^1(\mathbb{R}^n)$ can only hold if $-n/q = 1 - n/p$, that is, $q = p^* = np/(n-p)$, the *Sobolev exponent*, which requires $p < n$.

Theorem 32.4.4 (Gagliardo–Nirenberg–Sobolev)**.** *Let $n \ge 2$ and $1 \le p < n$. For $u \in C_c^1(\mathbb{R}^n)$, $\|u\|_{n/(n-1)} \le \|\nabla u\|_1$, and $\|u\|_{p^*} \le C_{n,p}\|\nabla u\|_p$ with $p^* = np/(n-p)$.*

Sketch of proof. Case $p = 1$, $n = 2$. By the fundamental theorem of calculus in each variable, $|u(x_1, x_2)| \le \int_{\mathbb{R}}|\partial_1 u(y_1, x_2)|\, dy_1 =: g_1(x_2)$ and $|u(x_1, x_2)| \le g_2(x_1)$ similarly. So $|u(x)|^2 \le g_1(x_2)g_2(x_1)$, and integrating, $\int|u|^2 \le \int g_1 \int g_2 = \|\partial_1 u\|_1\|\partial_2 u\|_1 \le \|\nabla u\|_1^2$. For general n the same bounds give $|u(x)|^{n/(n-1)} \le \prod_i g_i^{1/(n-1)}$, where g_i does not depend on x_i, and one integrates in $x_1, \dots, x_n$ in turn, using at each step the generalised Hölder inequality $\int\prod_{j=1}^{n-1}|h_j| \le \prod_j \|h_j\|_{n-1}$; this omitted computation gives $\int|u|^{n/(n-1)} \le \prod_i \|\partial_i u\|_1^{1/(n-1)}$ [Bre11, Chapter 9].

From $p = 1$ to $p > 1$. Apply the case $p = 1$ to $v = |u|^\gamma$ with $\gamma = p(n-1)/(n-p) > 1$, which is C^1 with $|\nabla v| \le \gamma |u|^{\gamma-1}|\nabla u|$. By Hölder's inequality (Proposition 28.5.2) with $p' = p/(p-1)$,

$$\left(\int |u|^{\gamma n/(n-1)}\right)^{(n-1)/n} \le \gamma \int |u|^{\gamma-1}|\nabla u| \le \gamma \left(\int |u|^{(\gamma-1)p'}\right)^{1/p'} \|\nabla u\|_p.$$

The choice of γ makes $\gamma n/(n-1) = (\gamma-1)p' = p^*$, and dividing gives $\|u\|_{p^*}^{\gamma - p^*/p'} \le \gamma \|\nabla u\|_p$ with $\gamma - p^*/p' = 1$. □

For $p = n$ the inequality fails, $p^* = \infty$: in $\mathbb{R}^2$ the function $\log\log(1/|x|)$, cut off near $|x| = e^{-1}$, has $\nabla u \in L^2$ but is unbounded near 0. For $p > n$, functions with $\nabla u \in L^p$ are Hölder continuous (Morrey's inequality [Bre11, Chapter 9]), the L^p analogue of Theorem 32.4.3.

Theorem 32.4.5 (Rellich)**.** *For $s > t$ the inclusion $H^s(T^n) \to H^t(T^n)$ is a compact operator.*

Proof. Let P_N be the truncation $\widehat{P_N u}(k) = \hat{u}(k)$ for $|k| \le N$ and 0 otherwise, a finite rank operator. For $u \in H^s$,

$$\|u - P_N u\|_t^2 = \sum_{|k|>N} \langle k\rangle^{2t}|\hat{u}(k)|^2 \le \langle N\rangle^{2(t-s)} \sum_{|k|>N} \langle k\rangle^{2s}|\hat{u}(k)|^2 \le \langle N\rangle^{2(t-s)}\|u\|_s^2,$$

so the inclusion is the norm limit of finite rank operators, hence compact (Proposition 30.6.2). □

Counterexample 32.4.6. Rellich's theorem fails on $\mathbb{R}^n$: translates $u_j(x) = u(x-j)$ of a fixed nonzero $u \in C_c^\infty$ have constant H^s norms and no subsequence converging in H^t, since they are eventually supported in disjoint sets. Compactness holds for the inclusions $H_K^s \to H^t$ of the subspaces of distributions supported in a fixed compact K, which is the form used on manifolds.

32.5 Elliptic estimates

Definition 32.5.1. Let $P(D) = \sum_{|\alpha|\le m} a_\alpha D^\alpha$ with constant coefficients $a_\alpha \in \mathbb{C}$. Its *principal symbol* is $\sigma(\xi) = \sum_{|\alpha|=m} a_\alpha (2\pi i\xi)^\alpha$, and P is *elliptic* if $\sigma(\xi) \ne 0$ for $\xi \ne 0$.

Example 32.5.2. The Laplacian $\Delta = -\sum_j \partial_j^2$ has symbol $4\pi^2|\xi|^2$ and is elliptic; the heat operator $\partial_t - \Delta$ and the wave operator are not.

Theorem 32.5.3 (Elliptic estimate and regularity)**.** *Let $P(D)$ be elliptic of order m with constant coefficients, on the torus T^n. For every s there is C with*

$$\|u\|_{s+m} \le C\left(\|P(D)u\|_s + \|u\|_s\right) \qquad \text{for all } u \in H^{s+m}(T^n).$$

Moreover, if $u \in H^t(T^n)$ for some t and $P(D)u \in H^s(T^n)$, then $u \in H^{s+m}(T^n)$; in particular $P(D)u \in C^\infty$ implies $u \in C^\infty$.

Proof. On the Fourier side $P(D)$ is multiplication by $p(k) = \sum_{|\alpha|\le m} a_\alpha (2\pi i k)^\alpha$. By homogeneity and compactness of the unit sphere, $|\sigma(\xi)| \ge c|\xi|^m$ for some $c > 0$, and the lower-order terms are $O(|\xi|^{m-1})$, so there are $c' > 0$ and R with $|p(k)| \ge c'\langle k\rangle^m$ for $|k| \ge R$. Hence

$$\sum_{|k|\ge R} \langle k\rangle^{2(s+m)}|\hat{u}(k)|^2 \le c'^{-2} \sum_{|k|\ge R} \langle k\rangle^{2s}|p(k)\hat{u}(k)|^2 \le c'^{-2}\|P(D)u\|_s^2,$$

while the finitely many terms with $|k| < R$ are bounded by $\langle R\rangle^{2m}\|u\|_s^2$. This gives the estimate. For regularity, the same computation shows that $\sum_k \langle k\rangle^{2(s+m)}|\hat{u}(k)|^2$ is finite whenever $\sum_k \langle k\rangle^{2s}|p(k)\hat{u}(k)|^2$ is; smoothness follows from Theorem 32.4.3, since then $u \in H^s$ for all s. □

Remark 32.5.4. For operators with variable coefficients the estimate is proved by freezing the coefficients at a point, using the constant-coefficient case and absorbing the error terms; on a compact manifold one patches the local estimates with a partition of unity. The result is that an elliptic operator P of order m between sections of vector bundles over a compact manifold satisfies $\|u\|_{s+m} \le C(\|Pu\|_s + \|u\|_s)$, is Fredholm as an operator $H^{s+m} \to H^s$ by Rellich's theorem, and has smooth kernel; this is proved in Chapter 42 and applied to the Laplacian in Chapter 43. The general theory is in [Hör90a, Chapters 7–10] and [War83, Chapter 6].

32.6 Exercises

Exercise 32.6.1. Show that $\varepsilon^{-n}\rho(x/\varepsilon) \to \delta_0$ in $\mathcal{D}'(\mathbb{R}^n)$ as $\varepsilon \to 0$, for any $\rho \in C_c^\infty$ with $\int \rho = 1$.

Solution. $\int \varepsilon^{-n}\rho(x/\varepsilon)\varphi(x)\,dx = \int \rho(y)\varphi(\varepsilon y)\,dy \to \varphi(0)\int\rho$ by dominated convergence. □

Exercise 32.6.2. Show that $E(x) = \tfrac{1}{2}|x|$ satisfies $-E'' = -\delta_0$ on $\mathbb{R}$, and that on $\mathbb{R}^n$ with $n \ge 3$ the function $c_n|x|^{2-n}$ is a fundamental solution of the Laplacian for a suitable constant c_n.

Exercise 32.6.3. Show that $H^{n/2}(T^n)$ is not contained in $C^0(T^n)$, so the condition $s > n/2$ in Theorem 32.4.3 is sharp.

Exercise 32.6.4. Using the Fourier transform, solve the heat equation $\partial_t u = -\Delta u$ on T^n with initial datum $u_0 \in H^s$, and show that $u(t) \in C^\infty$ for every $t > 0$.

Solution. $\hat u(k,t) = e^{-4\pi^2|k|^2 t}\hat u_0(k)$ solves the equation coefficientwise, and for $t > 0$ the factors decay faster than any power of $\langle k\rangle$, so $u(t) \in H^r$ for all r and Theorem 32.4.3 applies. □

Part VI

Differential Geometry

Chapter 33

Smooth Manifolds

33.1 Introduction

A topological manifold (Chapter 20) looks locally like $\mathbb{R}^n$, but only as a topological space: nothing in its definition says which functions on it are differentiable. A *smooth manifold* is a topological manifold together with exactly this missing information, and the way to supply it is simple. Around every point there are charts, homeomorphisms $\varphi\colon U \to \varphi(U) \subseteq \mathbb{R}^n$, and we would like to call a function f smooth near a point of U if $f \circ \varphi^{-1}$ is smooth, as a function on an open subset of $\mathbb{R}^n$. If ψ is a second chart, then $f \circ \psi^{-1} = (f \circ \varphi^{-1}) \circ (\varphi \circ \psi^{-1})$, so the two charts give the same answer as soon as the *transition map* $\varphi \circ \psi^{-1}$ and its inverse are smooth. A smooth manifold is therefore a topological manifold with a chosen family of charts whose transition maps are smooth, and the whole chapter is built on this definition.

The chapter goes slowly, in the following order.

1. Charts, atlases and smooth structures (Section 33.2), with many examples: spheres, projective spaces, products, and two different smooth structures on $\mathbb{R}$.
2. Smooth maps and diffeomorphisms (Section 33.3), and the proof that smoothness does not depend on the charts used to test it.
3. Tangent vectors (Section 33.4). There are three natural definitions: velocities of curves, directional derivatives, and vectors of components that change by the Jacobian matrix under a change of coordinates. We take directional derivatives as the definition and prove that the three agree.
4. The tangent bundle and vector fields (Section 33.5).
5. Submanifolds (Section 33.6): immersions, submersions, and the regular value theorem (Theorem 33.6.11), which produces most examples of manifolds, with counterexamples marking its limits; and Sard's theorem.
6. Quotients by group actions (Theorem 33.7.1) and Grassmannians (Section 33.7).
7. Smooth partitions of unity (Theorem 33.8.2), which glue local constructions into global ones (Section 33.8).
8. Flows of vector fields, the Lie bracket, the Frobenius theorem and the Whitney embedding theorem (Section 33.9).

Complex manifolds are defined in the same way with holomorphic transition maps (Chapter 44). References: [Lee13], [War83], [Hir76].

Prerequisites. Chapter 27 (Real Analysis), Chapter 18 (General Topology), Chapter 20 (Topological Manifolds and Group Actions) and Chapter 28 (Measure and Integration).

Notation 33.1.1. *Smooth* means C^∞ (Definition 27.10.2); everything in this chapter works for C^k, $k \geq 1$, with the obvious changes. In general statements, points of $\mathbb{R}^n$ are written $x = (x^1, \ldots, x^n)$, with upper indices; in concrete examples, where powers occur, we write $x = (x_1, \ldots, x_n)$ instead. The standard basis of $\mathbb{R}^n$ is $e_1, \ldots, e_n$. For a smooth map $F\colon W \to \mathbb{R}^m$ on an open set $W \subseteq \mathbb{R}^n$, $\partial_i F = \partial F/\partial x^i$ is its i-th partial derivative and $DF(a)\colon \mathbb{R}^n \to \mathbb{R}^m$ its derivative at a, a linear map whose matrix is the Jacobian matrix $(\partial_j F^i(a))_{i,j}$. We write $\langle x, y\rangle = \sum_i x^i y^i$, $\|x\| = \langle x, x\rangle^{1/2}$, $B(a, r)$ for the open ball and $\overline{B}(a, r)$ for the closed ball. A *diffeomorphism* between open subsets of Euclidean spaces is a smooth bijection whose inverse is smooth.

33.2 Charts, atlases and smooth structures

Throughout this section M is a topological manifold of dimension n: a Hausdorff, second countable space in which every point has an open neighbourhood homeomorphic to an open subset of $\mathbb{R}^n$ (Definition 20.5.1).

Definition 33.2.1. Let M be a topological n-manifold.

1. A *chart* is a pair (U, φ) of an open set $U \subseteq M$ and a homeomorphism $\varphi\colon U \to \varphi(U)$ onto an open subset of $\mathbb{R}^n$. Writing $\varphi(q) = (x^1(q), \ldots, x^n(q))$ defines functions $x^i\colon U \to \mathbb{R}$, the *coordinates* of the chart, and we also write $\varphi = (x^1, \ldots, x^n)$. The chart is *centred* at p if $\varphi(p) = 0$.

2. Two charts (U, φ) and (V, ψ) are *compatible* if the *transition map*
$$\psi \circ \varphi^{-1}\colon \varphi(U \cap V) \to \psi(U \cap V)$$
is a diffeomorphism (if $U \cap V = \emptyset$ there is nothing to check). Its inverse is the transition map $\varphi \circ \psi^{-1}$, so compatibility is a symmetric relation.

3. A *smooth atlas* is a set of pairwise compatible charts whose domains cover M.

4. A *smooth structure* is a *maximal* smooth atlas: one that is not contained in any strictly larger smooth atlas.

5. A *smooth manifold* of dimension n is a topological n-manifold together with a smooth structure $\mathcal{A}$. The charts in $\mathcal{A}$ are its *smooth charts*, or simply its charts. We usually omit $\mathcal{A}$ from the notation.

The sets $\varphi(U \cap V)$ and $\psi(U \cap V)$ are open in $\mathbb{R}^n$, because $U \cap V$ is open and φ, ψ are homeomorphisms onto open sets; so it makes sense to ask whether the transition map is smooth. It is always a homeomorphism, being a composite of homeomorphisms. Compatibility asks for smoothness in both directions, and one direction does not imply the other: see Example 33.2.5(3).

Remark 33.2.2. Here is why compatibility is the right condition. Let $f\colon M \to \mathbb{R}$ be a function and let p lie in the domains of two charts (U, φ) and (V, ψ). If $f \circ \varphi^{-1}$ is smooth near $\varphi(p)$, then near $\psi(p)$
$$f \circ \psi^{-1} = (f \circ \varphi^{-1}) \circ (\varphi \circ \psi^{-1}),$$
which is smooth by the chain rule (Proposition 27.10.3) if the transition map $\varphi \circ \psi^{-1}$ is smooth; and symmetrically with the roles of φ and ψ exchanged. So for compatible charts the two tests for smoothness of f near p give the same answer. For incompatible charts they can give different answers (Example 33.2.5(3)).

A smooth atlas is the data one writes down in practice; a smooth structure is what one wants to work with, because it contains *all* the charts that are compatible with the given ones, and so lets us shrink charts or change coordinates freely (Lemma 33.2.6). The next lemma shows that the two carry the same information.

Lemma 33.2.3. *Every smooth atlas $\mathcal{A}$ on a topological manifold is contained in a unique maximal smooth atlas, namely the set $\mathcal{A}^{\max}$ of all charts compatible with every chart of $\mathcal{A}$.*

Proof. First, $\mathcal{A}^{\max}$ is a smooth atlas. It contains $\mathcal{A}$, since the charts of $\mathcal{A}$ are pairwise compatible, so its domains cover M. Let (U, φ) and (V, ψ) be in $\mathcal{A}^{\max}$; we show that $\psi \circ \varphi^{-1}$ is smooth on $\varphi(U \cap V)$. Smoothness is a local property, so it suffices to check it near $\varphi(p)$ for each $p \in U \cap V$. Choose a chart (W, χ) of $\mathcal{A}$ with $p \in W$. On the open set $\varphi(U \cap V \cap W)$, which contains $\varphi(p)$,

$$\psi \circ \varphi^{-1} = (\psi \circ \chi^{-1}) \circ (\chi \circ \varphi^{-1}),$$

and both factors are smooth because φ and ψ are compatible with χ; so $\psi \circ \varphi^{-1}$ is smooth near $\varphi(p)$ by the chain rule. Exchanging φ and ψ shows that the inverse $\varphi \circ \psi^{-1}$ is smooth as well, so φ and ψ are compatible.

Second, $\mathcal{A}^{\max}$ is maximal: if $\mathcal{B} \supseteq \mathcal{A}^{\max}$ is a smooth atlas, every chart of $\mathcal{B}$ is compatible with the charts of $\mathcal{A} \subseteq \mathcal{B}$, hence lies in $\mathcal{A}^{\max}$, so $\mathcal{B} = \mathcal{A}^{\max}$. Third, it is the only maximal atlas containing $\mathcal{A}$: if $\mathcal{B}$ is one, its charts are compatible with those of $\mathcal{A}$, so $\mathcal{B} \subseteq \mathcal{A}^{\max}$, and maximality of $\mathcal{B}$ gives equality. □

We say that $\mathcal{A}$ *generates* the smooth structure $\mathcal{A}^{\max}$.

Corollary 33.2.4. 1. *Let $\mathcal{A}$ be a smooth structure and $\mathcal{A}_0 \subseteq \mathcal{A}$ a smooth atlas. A chart belongs to $\mathcal{A}$ if and only if it is compatible with every chart of $\mathcal{A}_0$.*

2. *Two smooth atlases $\mathcal{A}$ and $\mathcal{B}$ generate the same smooth structure if and only if every chart of $\mathcal{A}$ is compatible with every chart of $\mathcal{B}$.*

Proof. (1) The smooth structure $\mathcal{A}$ is a maximal atlas containing $\mathcal{A}_0$, so $\mathcal{A} = \mathcal{A}_0^{\max}$ by the uniqueness in Lemma 33.2.3, and $\mathcal{A}_0^{\max}$ consists of the charts compatible with every chart of $\mathcal{A}_0$.

(2) If $\mathcal{A}^{\max} = \mathcal{B}^{\max}$, the charts of $\mathcal{B}$ lie in $\mathcal{A}^{\max}$, so they are compatible with the charts of $\mathcal{A}$. Conversely, if every chart of $\mathcal{B}$ is compatible with every chart of $\mathcal{A}$, then $\mathcal{B} \subseteq \mathcal{A}^{\max}$, so $\mathcal{A}^{\max}$ is a maximal atlas containing $\mathcal{B}$, and $\mathcal{A}^{\max} = \mathcal{B}^{\max}$ by uniqueness. □

Example 33.2.5. Let $M = \mathbb{R}$.

1. The single chart $(\mathbb{R}, \mathrm{id})$ is a smooth atlas, since the only transition map is id. The smooth structure it generates is the *standard* smooth structure on $\mathbb{R}$. In the same way $(\mathbb{R}^n, \mathrm{id})$ generates the standard smooth structure on $\mathbb{R}^n$, and whenever we speak of $\mathbb{R}^n$ as a smooth manifold we mean this one.

2. The map $\varphi(t) = t^3$ is a homeomorphism $\mathbb{R} \to \mathbb{R}$: it is continuous, strictly increasing and surjective, and its inverse $s \mapsto s^{1/3}$ is continuous. So $(\mathbb{R}, \varphi)$ is a chart, and by itself it is a smooth atlas.

3. The charts id and φ are *not* compatible. The transition map $\varphi \circ \mathrm{id}^{-1}(t) = t^3$ is smooth, but its inverse $\mathrm{id} \circ \varphi^{-1}(s) = s^{1/3}$ is not differentiable at 0: its difference quotient $s^{1/3}/s = s^{-2/3}$ tends to $+\infty$ as $s \to 0$. By Corollary 33.2.4(2), the atlases $\{\mathrm{id}\}$ and $\{\varphi\}$ generate two different smooth structures on $\mathbb{R}$. They have different smooth functions in the sense of Remark 33.2.2: the function $f(t) = t$ satisfies $f \circ \mathrm{id}^{-1} = f$, which is smooth, while $f \circ \varphi^{-1}(s) = s^{1/3}$ is not.

4. The chart $\chi(t) = t + t^3$ is compatible with id. Indeed χ is smooth with $\chi'(t) = 1 + 3t^2 > 0$, so it is strictly increasing, and it is surjective since it tends to $\pm\infty$ at $\pm\infty$. Its inverse is smooth near every point by the inverse function theorem (Theorem 27.11.2), because χ' never vanishes. So χ belongs to the standard smooth structure of $\mathbb{R}$.

We shall see in Example 33.3.5 that the two smooth structures of (3), although different, are diffeomorphic.

Lemma 33.2.6. *Let M be a smooth manifold and (U, φ) a chart.*

1. *If $W \subseteq U$ is open, then $(W, \varphi|_W)$ is a chart.*
2. *If $\theta\colon \varphi(U) \to \Omega$ is a diffeomorphism onto an open set $\Omega \subseteq \mathbb{R}^n$, then $(U, \theta \circ \varphi)$ is a chart.*
3. *Every $p \in M$ lies in the domain of a chart (U, φ) centred at p with $\varphi(U) = B(0, 1)$, and of a chart centred at p with $\varphi(U) = \mathbb{R}^n$. These charts can be taken with U inside any given open neighbourhood of p.*

Proof. By Corollary 33.2.4(1), it suffices in (1) and (2) to show that the new chart is compatible with every chart (V, ψ) of the smooth structure.

(1) The map $\varphi|_W$ is a homeomorphism onto the open set $\varphi(W)$. Its transition maps with ψ are the restrictions of $\psi \circ \varphi^{-1}$ and $\varphi \circ \psi^{-1}$ to the open sets $\varphi(W \cap V)$ and $\psi(W \cap V)$, so they are smooth.

(2) The map $\theta \circ \varphi$ is a homeomorphism onto Ω. Its transition maps with ψ are $\psi \circ (\theta \circ \varphi)^{-1} = (\psi \circ \varphi^{-1}) \circ \theta^{-1}$ and $(\theta \circ \varphi) \circ \psi^{-1} = \theta \circ (\varphi \circ \psi^{-1})$, which are smooth by the chain rule.

(3) Let (U, φ) be a chart with $p \in U$, and W a given open neighbourhood of p. By (1) we may replace U by $U \cap W$. Choose $r > 0$ with $B(\varphi(p), r) \subseteq \varphi(U)$, restrict φ to $\varphi^{-1}(B(\varphi(p), r))$ by (1), and compose with the diffeomorphism $x \mapsto (x - \varphi(p))/r$ of $B(\varphi(p), r)$ onto $B(0, 1)$ by (2). This gives a chart centred at p with image $B(0, 1)$. Finally the map $x \mapsto x/(1 - \|x\|^2)^{1/2}$ is a diffeomorphism $B(0, 1) \to \mathbb{R}^n$: it is smooth because $1 - \|x\|^2 > 0$ on the ball, and the smooth map $y \mapsto y/(1 + \|y\|^2)^{1/2}$ is its inverse, since for $y = x/(1 - \|x\|^2)^{1/2}$ we have $1 + \|y\|^2 = 1/(1 - \|x\|^2)$. Composing with it gives a chart centred at p with image $\mathbb{R}^n$. □

Often we are given a set, not yet a topological space, together with candidate charts. The following lemma builds the topology and the smooth structure at once.

Lemma 33.2.7 (Smooth manifold chart lemma)**.** *Let M be a set and $(U_\alpha, \varphi_\alpha)_{\alpha \in A}$ a family of subsets $U_\alpha \subseteq M$ with injections $\varphi_\alpha\colon U_\alpha \to \mathbb{R}^n$ such that*

1. *each $\varphi_\alpha(U_\alpha)$ and each $\varphi_\alpha(U_\alpha \cap U_\beta)$ is open in $\mathbb{R}^n$;*
2. *each $\varphi_\beta \circ \varphi_\alpha^{-1}\colon \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$ is smooth;*
3. *countably many of the U_α cover M;*
4. *for $p \neq q$ in M, either some U_α contains both, or there are disjoint $U_\alpha \ni p$ and $U_\beta \ni q$.*

Then M has a unique topology and smooth structure for which the $(U_\alpha, \varphi_\alpha)$ are smooth charts.

Proof. The topology. Declare $W \subseteq M$ open if $\varphi_\alpha(W \cap U_\alpha)$ is open in $\mathbb{R}^n$ for every α. This is a topology: $\emptyset$ and M are open by (1); since each φ_α is injective, φ_α commutes with finite intersections, and images always commute with unions.

Each U_β is open and φ_β is a homeomorphism onto $\varphi_\beta(U_\beta)$. The set U_β is open since $\varphi_\alpha(U_\beta \cap U_\alpha)$ is open by (1). By (2), applied to the pair (α, β) and to the pair (β, α), the transition map

$\tau = \varphi_\alpha \circ \varphi_\beta^{-1} \colon \varphi_\beta(U_\alpha \cap U_\beta) \to \varphi_\alpha(U_\alpha \cap U_\beta)$ is a smooth bijection with smooth inverse, in particular a homeomorphism between open subsets of $\mathbb{R}^n$. Now let $V \subseteq U_\beta$. For every α,

$$\varphi_\alpha(V \cap U_\alpha) = \tau\big(\varphi_\beta(V) \cap \varphi_\beta(U_\alpha \cap U_\beta)\big).$$

If $\varphi_\beta(V)$ is open, the right-hand side is the image under the homeomorphism τ of an open subset of $\varphi_\beta(U_\alpha \cap U_\beta)$, so it is open in the open set $\varphi_\alpha(U_\alpha \cap U_\beta)$, hence in $\mathbb{R}^n$; so V is open in M. Conversely, if V is open in M, then taking $\alpha = \beta$ in the definition shows that $\varphi_\beta(V)$ is open. So φ_β maps the open subsets of U_β exactly onto the open subsets of $\varphi_\beta(U_\beta)$: it is a homeomorphism. In particular M is locally Euclidean of dimension n.

Hausdorff and second countable. Let $p \neq q$. If both lie in some U_α, they have disjoint open neighbourhoods in U_α, which is homeomorphic to a subset of $\mathbb{R}^n$ and hence Hausdorff; these are open in M because U_α is. Otherwise (4) gives disjoint open sets $U_\alpha \ni p$ and $U_\beta \ni q$. Each U_α is second countable, being homeomorphic to a subspace of $\mathbb{R}^n$ (Proposition 18.3.12); if countably many $U_{\alpha_1}, U_{\alpha_2}, \ldots$ cover M as in (3), the union of countable bases of the U_{α_i} is a countable basis of M.

The smooth structure. By (2) the $(U_\alpha, \varphi_\alpha)$ are pairwise compatible charts covering M, so they form a smooth atlas, which generates a unique smooth structure (Lemma 33.2.3).

Uniqueness of the topology. In any topology for which the U_α are open and the φ_α are homeomorphisms onto open subsets of $\mathbb{R}^n$, a set W is open if and only if every $W \cap U_\alpha$ is open (the U_α form an open cover), that is, if and only if every $\varphi_\alpha(W \cap U_\alpha)$ is open. This is the topology defined above. □

Counterexample 33.2.8. Condition (4) is needed: the two charts of the line with two origins (Counterexample 20.5.2), each the identity on a copy of $\mathbb{R}$, satisfy (1)–(3), with transition map the identity of $\mathbb{R} \setminus \{0\}$, but the two origins lie in no common chart and in no disjoint charts, and the resulting space is not Hausdorff.

Example 33.2.9.

1. *Euclidean spaces, vector spaces and open subsets.* $\mathbb{R}^n$ has its standard smooth structure (Example 33.2.5). A real vector space V of dimension n has a canonical smooth structure: every linear isomorphism $V \to \mathbb{R}^n$ is a chart, and two of them differ by an invertible linear map of $\mathbb{R}^n$, which is smooth with smooth inverse. If W is an open subset of a smooth manifold M, the charts $(U \cap W, \varphi|_{U\cap W})$, for (U, φ) a chart of M, form a smooth atlas of W: their transition maps are restrictions of transition maps of M to open sets. With this structure W is an *open submanifold* of M. For example, the general linear group $\mathrm{GL}_n(\mathbb{R}) = \{A \in M_n(\mathbb{R}) : \det A \neq 0\}$ is open in $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$, since det is a polynomial in the entries, hence continuous; so it is a smooth manifold of dimension n^2.

2. *Spheres.* Let $S^n = \{x \in \mathbb{R}^{n+1} : \|x\| = 1\}$, $N = (0, \ldots, 0, 1)$ and $S = -N$. The stereographic projections

$$\sigma_N(x) = \frac{(x_1, \ldots, x_n)}{1 - x_{n+1}} \quad (x \neq N), \qquad \sigma_S(x) = \frac{(x_1, \ldots, x_n)}{1 + x_{n+1}} \quad (x \neq S)$$

are homeomorphisms $S^n \setminus \{N\} \to \mathbb{R}^n$ and $S^n \setminus \{S\} \to \mathbb{R}^n$ (Example 18.9.4, and its mirror image under $x_{n+1} \mapsto -x_{n+1}$). The inverse of σ_N is $\tau(y) = (2y, \|y\|^2 - 1)/(\|y\|^2 + 1)$, and $\sigma_N(S^n \setminus \{N, S\}) = \mathbb{R}^n \setminus \{0\}$ because $\sigma_N(S) = 0$. For $y \neq 0$, the last coordinate of $\tau(y)$ gives $1 + x_{n+1} = 2\|y\|^2/(\|y\|^2 + 1)$, so

$$\sigma_S \circ \sigma_N^{-1}(y) = \frac{2y/(\|y\|^2 + 1)}{2\|y\|^2/(\|y\|^2 + 1)} = \frac{y}{\|y\|^2}.$$

This map is smooth on $\mathbb{R}^n \setminus \{0\}$ and is its own inverse, so the two charts are compatible and form a smooth atlas of S^n.

3. *Projective spaces.* Let $\mathbb{RP}^n = (\mathbb{R}^{n+1} \setminus \{0\})/\mathbb{R}^\times$, with points written $[x] = [x_0 : \cdots : x_n]$. By Proposition 20.6.3 it is a topological n-manifold, the sets $U_i = \{[x] : x_i \neq 0\}$ $(0 \le i \le n)$ are open, and the maps

$$\varphi_i([x]) = \left(\frac{x_0}{x_i}, \ldots, \widehat{\frac{x_i}{x_i}}, \ldots, \frac{x_n}{x_i}\right)$$

(the hat means that the entry is omitted) are homeomorphisms $U_i \to \mathbb{R}^n$. The inverse $\varphi_i^{-1}(u)$ is the class of the vector obtained from $u \in \mathbb{R}^n$ by inserting 1 in position i. For $i \neq j$, $\varphi_i(U_i \cap U_j)$ is the set of u whose entry in the position of x_j is nonzero, which is open, and $\varphi_j \circ \varphi_i^{-1}(u)$ is obtained by dividing this vector by that entry and omitting position j. Its components are of the form u_k/u_l or $1/u_l$ with $u_l \neq 0$, so it is smooth, and the (U_i, φ_i) form a smooth atlas. For $n = 1$ there are two charts, $[x_0 : x_1] \mapsto x_1/x_0$ and $[x_0 : x_1] \mapsto x_0/x_1$, with transition map $u \mapsto 1/u$ on $\mathbb{R} \setminus \{0\}$. In the same way the complex projective space $\mathbb{CP}^n$ is a smooth $2n$-manifold, whose transition maps are even holomorphic (Chapter 44).

4. *Products.* Let M and N be smooth manifolds of dimensions m and n. Then $M \times N$ is a topological $(m+n)$-manifold (Proposition 20.6.1(2)). For charts (U, φ) of M and (V, ψ) of N, the map $\varphi \times \psi \colon U \times V \to \varphi(U) \times \psi(V) \subseteq \mathbb{R}^{m+n}$ is a homeomorphism onto an open set, these *product charts* cover $M \times N$, and the transition map between two of them is $(\varphi' \times \psi') \circ (\varphi \times \psi)^{-1} = (\varphi' \circ \varphi^{-1}) \times (\psi' \circ \psi^{-1})$, which is smooth with smooth inverse. So the product charts form a smooth atlas. For example the torus $T^n = S^1 \times \cdots \times S^1$ is a smooth n-manifold.

5. *Graphs.* Let $W \subseteq \mathbb{R}^n$ be open and $f \colon W \to \mathbb{R}^k$ smooth. The graph $\Gamma_f = \{(x, f(x)) : x \in W\} \subseteq \mathbb{R}^{n+k}$ is Hausdorff and second countable as a subspace of $\mathbb{R}^{n+k}$, and the projection $(x, y) \mapsto x$ is a homeomorphism $\Gamma_f \to W$, with continuous inverse $x \mapsto (x, f(x))$. So this projection is a chart, and by itself a smooth atlas. For instance the parabola $\{y = x^2\}$ and the paraboloid $\{z = x^2 + y^2\}$ are smooth manifolds.

33.3 Smooth maps

We now define smooth maps between smooth manifolds by testing in charts, and show that the test does not depend on the charts chosen.

Definition 33.3.1. Let M and N be smooth manifolds of dimensions m and n. A map $f \colon M \to N$ is *smooth* if every $p \in M$ has charts (U, φ) of M with $p \in U$ and (V, ψ) of N with $f(U) \subseteq V$ such that the *local representative*

$$\psi \circ f \circ \varphi^{-1} \colon \varphi(U) \to \psi(V) \subseteq \mathbb{R}^n$$

is smooth. A *diffeomorphism* is a smooth bijection whose inverse is smooth; M and N are *diffeomorphic* if there is a diffeomorphism $M \to N$. We write $C^\infty(M)$ for the set of smooth functions $M \to \mathbb{R}$, where $\mathbb{R}$ has its standard smooth structure.

Lemma 33.3.2. *Let $f \colon M \to N$ be a smooth map.*

1. *f is continuous.*

2. *For* all *charts (U, φ) of M and (V, ψ) of N, the local representative $\psi \circ f \circ \varphi^{-1}$ is smooth on the open set $\varphi(U \cap f^{-1}(V))$.*

Conversely, a continuous map $f\colon M \to N$ is smooth as soon as its local representatives are smooth for all charts of some atlas of M and some atlas of N. In particular, a map between open subsets of Euclidean spaces is smooth in the sense of Definition 33.3.1 if and only if it is smooth in the usual sense.

Proof. (1) With charts as in Definition 33.3.1, $f|_U = \psi^{-1} \circ (\psi \circ f \circ \varphi^{-1}) \circ \varphi$ is continuous on the open set U. Continuity is a local property, so f is continuous.

(2) The set $U \cap f^{-1}(V)$ is open by (1), so $\varphi(U \cap f^{-1}(V))$ is open. Let $p \in U \cap f^{-1}(V)$, and choose charts (U', φ') at p and (V', ψ') at $f(p)$ with $f(U') \subseteq V'$ and $\psi' \circ f \circ \varphi'^{-1}$ smooth. On the open set $\varphi(U \cap U' \cap f^{-1}(V))$, which contains $\varphi(p)$,

$$\psi \circ f \circ \varphi^{-1} = (\psi \circ \psi'^{-1}) \circ (\psi' \circ f \circ \varphi'^{-1}) \circ (\varphi' \circ \varphi^{-1}),$$

a composite of two transition maps and a local representative, all smooth. So $\psi \circ f \circ \varphi^{-1}$ is smooth near every point of its domain.

For the converse, let $p \in M$; choose a chart (V, ψ) of the given atlas of N at $f(p)$ and a chart (U, φ) of the given atlas of M at p. By Lemma 33.2.6(1), φ restricted to the open set $U \cap f^{-1}(V)$ is a chart, and it maps into V, with smooth local representative. For open subsets of Euclidean spaces use the identity charts: the local representative of f is then f itself. □

Proposition 33.3.3. *Let L, M, N be smooth manifolds.*

1. *Identity maps, constant maps, and composites of smooth maps are smooth.*
2. *Smoothness is local: $f\colon M \to N$ is smooth if and only if every point of M has an open neighbourhood W such that $f|_W$ is smooth, W being an open submanifold. If $W' \subseteq N$ is open and $f(M) \subseteq W'$, then f is smooth as a map to N if and only if it is smooth as a map to W'.*
3. *If (U, φ) is a chart of M, then $\varphi\colon U \to \varphi(U)$ is a diffeomorphism. Conversely, if $U \subseteq M$ is open and $\varphi\colon U \to \Omega$ is a diffeomorphism onto an open $\Omega \subseteq \mathbb{R}^n$, then (U, φ) is a chart of M.*
4. *$C^\infty(M)$ is an $\mathbb{R}$-algebra under pointwise operations, and if $f \in C^\infty(M)$ has no zeros then $1/f \in C^\infty(M)$. A map $f = (f_1, \dots, f_k)\colon M \to \mathbb{R}^k$ is smooth if and only if every f_i lies in $C^\infty(M)$.*
5. *The projections $M \times N \to M$ and $M \times N \to N$ are smooth, and a map $f = (f_1, f_2)\colon L \to M \times N$ is smooth if and only if $f_1\colon L \to M$ and $f_2\colon L \to N$ are smooth.*

Proof. We use Lemma 33.3.2 throughout: local representatives may be computed in any charts.

(1) The local representatives of id_M are the transition maps $\psi \circ \varphi^{-1}$, which are smooth, and those of a constant map are constant. Let $f\colon L \to M$ and $g\colon M \to N$ be smooth, and $p \in L$. Choose a chart (W, χ) of N at $g(f(p))$, a chart (V, ψ) of M at $f(p)$ with $V \subseteq g^{-1}(W)$, and a chart (U, φ) of L at p with $U \subseteq f^{-1}(V)$; this is possible by continuity and Lemma 33.2.6(1). Then $\chi \circ (g \circ f) \circ \varphi^{-1} = (\chi \circ g \circ \psi^{-1}) \circ (\psi \circ f \circ \varphi^{-1})$ on $\varphi(U)$, which is smooth by the chain rule.

(2) The charts of the open submanifold W are the charts of M with domain inside W (Lemma 33.2.6(1) and Example 33.2.9(1)), and the definition of smoothness only involves charts around each point; this proves the first statement. For the second, the charts of N with domain inside W' are the charts of W', and near each point we may use such charts.

(3) The local representative of $\varphi\colon U \to \varphi(U)$ with respect to the charts φ of U and id of $\varphi(U)$ is $\mathrm{id} \circ \varphi \circ \varphi^{-1} = \mathrm{id}$, and similarly for φ^{-1}. Conversely, let $\varphi\colon U \to \Omega$ be a diffeomorphism. It is a homeomorphism by Lemma 33.3.2(1). For any chart (V, ψ) of M, the transition map $\psi \circ \varphi^{-1}$ is the local representative of the smooth map $\varphi^{-1}\colon \Omega \to U$ in the

charts id and $\psi|_{U\cap V}$, and $\varphi\circ\psi^{-1}$ is the local representative of φ in the charts $\psi|_{U\cap V}$ and id; both are smooth. So φ is compatible with every chart, and it is a chart by Corollary 33.2.4(1).

(4) In a chart, $(f+g)\circ\varphi^{-1} = f\circ\varphi^{-1} + g\circ\varphi^{-1}$ and $(fg)\circ\varphi^{-1} = (f\circ\varphi^{-1})(g\circ\varphi^{-1})$, and $(1/f)\circ\varphi^{-1} = 1/(f\circ\varphi^{-1})$; these are smooth. With the identity chart of $\mathbb{R}^k$, the local representative of f is $(f_1\circ\varphi^{-1},\dots,f_k\circ\varphi^{-1})$, which is smooth if and only if its components are.

(5) In product charts the projection $M\times N\to M$ is $(x,y)\mapsto x$. A map $f=(f_1,f_2)$ is continuous if and only if f_1 and f_2 are. If they are, let $p\in L$, let (U,φ) and (V,ψ) be charts at $f_1(p)$ and $f_2(p)$, and let χ be a chart at p with domain inside $f^{-1}(U\times V)$. Then the local representative $(\varphi\times\psi)\circ f\circ\chi^{-1} = (\varphi\circ f_1\circ\chi^{-1}, \psi\circ f_2\circ\chi^{-1})$ is smooth if and only if both components are. If f is smooth, so are f_1 and f_2, as composites with the projections. □

So smooth manifolds and smooth maps form a category, whose isomorphisms are the diffeomorphisms.

Example 33.3.4.

1. The inclusion $\iota\colon S^n\to\mathbb{R}^{n+1}$ is smooth: in the chart σ_N it is $\iota\circ\sigma_N^{-1}(y) = (2y, \|y\|^2-1)/(\|y\|^2+1)$, and in the chart σ_S it is $(2y, 1-\|y\|^2)/(\|y\|^2+1)$. Consequently the restriction to S^n of a smooth function on an open subset of $\mathbb{R}^{n+1}$ containing S^n is smooth, being a composite of smooth maps.

2. The quotient map $q\colon\mathbb{R}^{n+1}\setminus\{0\}\to\mathbb{RP}^n$, $x\mapsto[x]$, is smooth: on the open set $q^{-1}(U_i) = \{x_i\neq 0\}$ its local representative is $\varphi_i\circ q(x) = (x_j/x_i)_{j\neq i}$. So the map $S^n\to\mathbb{RP}^n$, $x\mapsto[x]$, is smooth, as the composite of q with $\iota\colon S^n\to\mathbb{R}^{n+1}\setminus\{0\}$ (Proposition 33.3.3(2)).

3. The map $e\colon\mathbb{R}\to S^1$, $e(t) = (\cos t, \sin t)$, is smooth: $\sigma_N\circ e(t) = \cos t/(1-\sin t)$ where $\sin t\neq 1$, and $\sigma_S\circ e(t) = \cos t/(1+\sin t)$ where $\sin t\neq -1$, and these two open sets cover $\mathbb{R}$.

4. Multiplication $\mathrm{GL}_n(\mathbb{R})\times\mathrm{GL}_n(\mathbb{R})\to\mathrm{GL}_n(\mathbb{R})$ and inversion $\mathrm{GL}_n(\mathbb{R})\to\mathrm{GL}_n(\mathbb{R})$ are smooth. The entries of AB are polynomials in the entries of A and B, and $A^{-1} = \mathrm{adj}(A)/\det(A)$ (Proposition 11.5.3(4)), whose entries are polynomials divided by the nowhere vanishing polynomial det.

Example 33.3.5. Let $\mathbb{R}_{\mathrm{std}}$ be $\mathbb{R}$ with the standard structure and $\mathbb{R}_3$ be $\mathbb{R}$ with the structure generated by the chart $\varphi(t) = t^3$ (Example 33.2.5). The map $F\colon\mathbb{R}_{\mathrm{std}}\to\mathbb{R}_3$, $F(t) = t^{1/3}$, is a diffeomorphism: its local representative in the charts id and φ is $\varphi\circ F(t) = t$, and that of F^{-1} is $F^{-1}\circ\varphi^{-1}(s) = (s^{1/3})^3 = s$. So different smooth structures can be diffeomorphic; the diffeomorphism F is not differentiable as a map $\mathbb{R}\to\mathbb{R}$ in the usual sense, which is no contradiction, because the smooth structure on its target is not the standard one.

Counterexample 33.3.6. A smooth homeomorphism need not be a diffeomorphism: $t\mapsto t^3$ is a smooth homeomorphism $\mathbb{R}_{\mathrm{std}}\to\mathbb{R}_{\mathrm{std}}$, but its inverse $s\mapsto s^{1/3}$ is not smooth. The inverse function theorem on manifolds (Theorem 33.4.14) explains what goes wrong: the derivative vanishes at 0.

Counterexample 33.3.7. The same topological manifold can carry non-diffeomorphic smooth structures: $\mathbb{R}^4$ has uncountably many, and S^7 has 28 up to orientation-preserving diffeomorphism (Milnor); in dimensions ≤ 3 the smooth structure is unique up to diffeomorphism. By contrast, the two structures on $\mathbb{R}$ of Example 33.3.5 are different but diffeomorphic. See [Hir76] and [Mil97].

Proposition 33.3.8. *A map $f\colon M\to N$ is smooth if and only if it is continuous and, for every open $V\subseteq N$ and every $g\in C^\infty(V)$, the function $g\circ f$ is smooth on the open set $f^{-1}(V)$.*

Proof. If f is smooth, then $f|_{f^{-1}(V)}\colon f^{-1}(V) \to V$ is smooth (Proposition 33.3.3(2)), and so is its composite with g. Conversely, let $p \in M$ and let $(V, \psi = (y^1, \dots, y^n))$ be a chart at $f(p)$. The coordinates y^j are smooth on V, being the components of the diffeomorphism ψ (Proposition 33.3.3(3) and (4)). By hypothesis the $y^j \circ f$ are smooth on $f^{-1}(V)$, so $\psi \circ f$ is smooth there, and so is its composite with φ^{-1} for every chart φ of $f^{-1}(V)$. □

Remark 33.3.9. Smoothness of functions is a local property, so $V \mapsto C^\infty(V)$ is a sheaf of $\mathbb{R}$-algebras C^∞_M on M, and (M, C^∞_M) is a locally ringed space (Example 25.6.2). Proposition 33.3.8 says that smooth maps are exactly the continuous maps that pull back smooth functions to smooth functions. This point of view, in which a space is described by its functions, is the one taken for complex manifolds and schemes later on.

The following bump functions are the basic tool for passing between local and global statements.

Lemma 33.3.10. *Let M be a smooth manifold, $p \in M$ and W an open neighbourhood of p. There is $\chi \in C^\infty(M)$ with values in $[0,1]$, equal to 1 on a neighbourhood of p, whose support* $\operatorname{supp}\chi = \overline{\{\chi \neq 0\}}$ *is compact and contained in W.*

Proof. Choose a chart (U, φ) centred at p with $U \subseteq W$ (Lemma 33.2.6(3)) and $r > 0$ with $\overline{B}(0, 3r) \subseteq \varphi(U)$. By Corollary 27.10.8 there is a smooth $b\colon \mathbb{R}^n \to [0,1]$ equal to 1 on $\overline{B}(0,r)$ and to 0 outside $B(0,2r)$. Define $\chi = b \circ \varphi$ on U and $\chi = 0$ on $M \setminus U$. The set $K = \varphi^{-1}(\overline{B}(0,2r))$ is compact, as a continuous image of a compact set, hence closed in the Hausdorff space M (Proposition 18.6.3). Now χ is smooth on the open set U, and it vanishes on the open set $M \setminus K$; since $K \subseteq U$ these two open sets cover M, so χ is smooth (Proposition 33.3.3(2)). Its support lies in the closed set $K \subseteq W$, so it is compact, and $\chi = 1$ on the neighbourhood $\varphi^{-1}(B(0,r))$ of p. □

33.4 Tangent vectors

For a surface $S \subseteq \mathbb{R}^3$, the tangent plane at p is the set of velocities $\gamma'(0)$ of curves in S through p, a linear subspace of $\mathbb{R}^3$. An abstract manifold has no ambient space in which such velocities live, and there are three ways around this:

(a) regard a tangent vector as a curve through p, two curves being identified when they have the same velocity in a chart;

(b) regard a tangent vector as the operation "derivative in this direction" on functions;

(c) regard a tangent vector as a vector of $\mathbb{R}^n$ in each chart, the vectors attached to two charts being related by the Jacobian matrix of the transition map.

We take (b) as the definition, because it is visibly independent of all choices and visibly a vector space, and then prove that (a) and (c) give the same thing (Propositions 33.4.7 and 33.4.10). The model is $\mathbb{R}^n$: a vector $v \in \mathbb{R}^n$ gives the directional derivative $f \mapsto \frac{d}{dt} f(a+tv)|_{t=0} = \sum_i v^i \partial_i f(a)$ at a, which is linear in f and satisfies the product rule.

Definition 33.4.1. Let M be a smooth manifold and $p \in M$. A *derivation at p* is a linear map $v\colon C^\infty(M) \to \mathbb{R}$ satisfying the Leibniz rule

$$v(fg) = v(f)g(p) + f(p)v(g).$$

The *tangent space* T_pM is the real vector space of derivations at p; its elements are *tangent vectors* at p. A smooth map $f\colon M \to N$ induces the *differential* $d_pf\colon T_pM \to T_{f(p)}N$, $(d_pf(v))(g) = v(g \circ f)$ for $g \in C^\infty(N)$.

The differential is well defined: $d_p f(v)$ is linear, and it satisfies the Leibniz rule at $f(p)$, because $v((g\circ f)(h\circ f)) = v(g\circ f)h(f(p)) + g(f(p))v(h\circ f)$. It is clearly linear in v.

Lemma 33.4.2. *Let $v \in T_pM$ and $f, g \in C^\infty(M)$.*

1. *$v(c) = 0$ for every constant function c.*
2. *If $f(p) = g(p) = 0$, then $v(fg) = 0$.*
3. *(Locality) If $f = g$ on a neighbourhood of p, then $v(f) = v(g)$.*

Proof. (1) The Leibniz rule gives $v(1) = v(1\cdot 1) = 2v(1)$, so $v(1) = 0$, and $v(c) = cv(1) = 0$ by linearity. (2) is the Leibniz rule. (3) The function $h = f - g$ vanishes on an open neighbourhood W of p. Choose χ as in Lemma 33.3.10 with support in W and $\chi(p) = 1$. Then $\chi h = 0$ everywhere (on W because $h = 0$, off W because $\chi = 0$), so $0 = v(\chi h) = v(\chi)h(p) + \chi(p)v(h) = v(h)$. □

Proposition 33.4.3. *Let $W \subseteq M$ be open, $p \in W$, and $\iota\colon W \to M$ the inclusion. Then $d_p\iota\colon T_pW \to T_pM$ is an isomorphism.*

We use it to identify T_pW with T_pM; in particular a tangent vector at p can be applied to functions defined only near p.

Proof. Fix χ as in Lemma 33.3.10 with compact support inside W and $\chi = 1$ near p. For $g \in C^\infty(W)$, let $\widetilde{\chi g}$ be the function equal to χg on W and to 0 outside W. It is smooth on M, because it is smooth on W and vanishes on the open set $M \setminus \operatorname{supp}\chi$, and these two open sets cover M. It agrees with g near p.

Injectivity. Suppose $d_p\iota(v) = 0$, that is, $v(f|_W) = 0$ for all $f \in C^\infty(M)$. For $g \in C^\infty(W)$, the functions g and $\widetilde{\chi g}|_W$ agree near p, so $v(g) = v(\widetilde{\chi g}|_W) = 0$ by Lemma 33.4.2(3). So $v = 0$.

Surjectivity. Let $w \in T_pM$, and define $v(g) = w(\widetilde{\chi g})$ for $g \in C^\infty(W)$. It is linear. It satisfies the Leibniz rule: the functions $\widetilde{\chi gh}$ and $\widetilde{\chi g}\,\widetilde{\chi h}$ agree near p (both equal gh there), so by locality $v(gh) = w(\widetilde{\chi g}\,\widetilde{\chi h}) = v(g)h(p) + g(p)v(h)$. Finally, for $f \in C^\infty(M)$ the functions $\widetilde{\chi f|_W}$ and f agree near p, so $d_p\iota(v)(f) = v(f|_W) = w(f)$. □

Lemma 33.4.4 (Hadamard)**.** *Let $B = B(a,r) \subseteq \mathbb{R}^n$ and $f \in C^\infty(B)$. There are $g_1, \dots, g_n \in C^\infty(B)$ with $g_i(a) = \partial_i f(a)$ and*

$$f(x) = f(a) + \sum_{i=1}^{n} (x^i - a^i) g_i(x) \qquad (x \in B).$$

Proof. For $x \in B$ the segment from a to x lies in B. By the chain rule, $h(t) = f(a + t(x-a))$ has derivative $h'(t) = \sum_i (x^i - a^i)\partial_i f(a + t(x-a))$, so by the fundamental theorem of calculus (Theorem 27.8.3)

$$f(x) - f(a) = \int_0^1 h'(t)\,dt = \sum_i (x^i - a^i) g_i(x), \qquad g_i(x) = \int_0^1 \partial_i f(a + t(x-a))\,dt,$$

and $g_i(a) = \partial_i f(a)$. It remains to show that the g_i are smooth. Consider the functions $K(x) = \int_0^1 t^m \ell(a + t(x-a))\,dt$ with $m \geq 0$ and $\ell \in C^\infty(B)$. Each is continuous: for $x_0 \in B$ choose a closed ball $\overline{B}' \subseteq B$ around x_0; the integrand is uniformly continuous on the compact set $\overline{B}' \times [0,1]$, so $K(x) \to K(x_0)$ as $x \to x_0$. Each has partial derivatives, of the same form: by Proposition 27.8.9(1), applied in the variable x^j with the other coordinates fixed, $\partial_j K(x) = \int_0^1 t^{m+1}(\partial_j \ell)(a + t(x-a))\,dt$. By induction, all partial derivatives of all orders of K exist and are continuous, so K is smooth (Proposition 27.10.3(3)). The g_i are of this form, with $m = 0$ and $\ell = \partial_i f$. □

Proposition 33.4.5. *Let $U \subseteq \mathbb{R}^n$ be open and $a \in U$. For $v \in \mathbb{R}^n$ let $D_v|_a \in T_aU$ be the directional derivative $D_v|_a(f) = \sum_i v^i \partial_i f(a)$. Then $v \mapsto D_v|_a$ is an isomorphism $\mathbb{R}^n \to T_aU$, which sends e_i to $\partial/\partial x^i|_a \colon f \mapsto \partial_i f(a)$. If $F \colon U \to V \subseteq \mathbb{R}^m$ is smooth, then $d_aF(D_v|_a) = D_{DF(a)v}|_{F(a)}$: under these isomorphisms the differential is the derivative $DF(a)$, with the Jacobian matrix.*

Proof. $D_v|_a$ is linear in f and satisfies the Leibniz rule by the product rule, and it is linear in v. It is injective in v, since $D_v|_a(x^j) = v^j$ for the coordinate function x^j. For surjectivity, let $w \in T_aU$ and put $v^i = w(x^i)$. Let $f \in C^\infty(U)$, choose $r > 0$ with $B = B(a,r) \subseteq U$, and write $f = f(a) + \sum_i (x^i - a^i) g_i$ on B (Lemma 33.4.4). By Proposition 33.4.3 we may apply w to functions on B, and by the Leibniz rule and Lemma 33.4.2(1),

$$w(f) = w(f(a)) + \sum_i \big(w(x^i - a^i) g_i(a) + (a^i - a^i) w(g_i) \big) = \sum_i v^i \partial_i f(a).$$

So $w = D_v|_a$. Finally, for $g \in C^\infty(V)$ the chain rule gives $d_aF(D_v|_a)(g) = D_v|_a(g \circ F) = \sum_j v^j \sum_i \partial_i g(F(a)) \partial_j F^i(a) = \sum_i (DF(a)v)^i \partial_i g(F(a))$. □

Proposition 33.4.6. *Let M be a smooth n-manifold and $p \in M$.*

1. *Derivations are local: if $f = g$ near p then $v(f) = v(g)$; and $T_pU = T_pM$ for every open $U \ni p$. So T_pM may be computed in any chart.*
2. *If $(U, \varphi = (x^1, \dots, x^n))$ is a chart at p, the derivations*

$$\left.\frac{\partial}{\partial x^i}\right|_p \colon f \mapsto \partial_i (f \circ \varphi^{-1})(\varphi(p))$$

   form a basis of T_pM, and every $v \in T_pM$ is $v = \sum_i v(x^i)\, \partial/\partial x^i|_p$. In particular $\dim T_pM = n$.
3. *$d_p(g \circ f) = d_{f(p)}g \circ d_p f$ and $d_p\mathrm{id} = \mathrm{id}$. In charts (x^j) at p and (y^i) at $f(p)$, with local representative $F = \psi \circ f \circ \varphi^{-1}$,*

$$d_p f\left(\left.\frac{\partial}{\partial x^j}\right|_p\right) = \sum_i \partial_j F^i(\varphi(p)) \left.\frac{\partial}{\partial y^i}\right|_{f(p)},$$

   so the matrix of $d_p f$ is the Jacobian matrix of F. A diffeomorphism has invertible differentials.

Proof. We first prove the chain rule of (3). For $h \in C^\infty$ of the target, $d_p(g \circ f)(v)(h) = v(h \circ g \circ f) = d_p f(v)(h \circ g) = d_{f(p)}g(d_p f(v))(h)$, and $d_p\mathrm{id}(v)(h) = v(h)$. If f is a diffeomorphism, applying this to $f^{-1} \circ f = \mathrm{id}$ and $f \circ f^{-1} = \mathrm{id}$ shows that $d_p f$ is invertible.

(1) is Lemma 33.4.2(3) and Proposition 33.4.3.

(2) By Proposition 33.3.3(3), $\varphi \colon U \to \varphi(U)$ is a diffeomorphism, so by the chain rule $d_{\varphi(p)}(\varphi^{-1}) \colon T_{\varphi(p)}\varphi(U) \to T_pU = T_pM$ is an isomorphism. By Proposition 33.4.5 the vectors $\partial/\partial x^i|_{\varphi(p)}$ form a basis of $T_{\varphi(p)}\varphi(U)$, and $d_{\varphi(p)}(\varphi^{-1})$ sends $\partial/\partial x^i|_{\varphi(p)}$ to $f \mapsto \partial_i(f \circ \varphi^{-1})(\varphi(p))$, which is $\partial/\partial x^i|_p$. So these form a basis of T_pM. Since $x^j \circ \varphi^{-1}$ is the j-th coordinate function of $\mathbb{R}^n$, $\partial/\partial x^i|_p(x^j) = \delta_{ij}$; so if $v = \sum_i c^i \partial/\partial x^i|_p$, applying v to x^j gives $c^j = v(x^j)$.

(3) By (2), the coefficient of $\partial/\partial y^i|_{f(p)}$ in $d_p f(\partial/\partial x^j|_p)$ is $d_p f(\partial/\partial x^j|_p)(y^i) = \partial/\partial x^j|_p(y^i \circ f) = \partial_j(y^i \circ f \circ \varphi^{-1})(\varphi(p)) = \partial_j F^i(\varphi(p))$. □

Proposition 33.4.7. *Let $(x^1, \dots, x^n)$ and $(y^1, \dots, y^n)$ be the coordinates of two charts φ, ψ at p, and write $\partial y^j/\partial x^i(p) = \partial/\partial x^i|_p(y^j) = \partial_i(y^j \circ \varphi^{-1})(\varphi(p))$, the entries of the Jacobian matrix of the transition map $\psi \circ \varphi^{-1}$ at $\varphi(p)$. Then*

$$\left.\frac{\partial}{\partial x^i}\right|_p = \sum_j \frac{\partial y^j}{\partial x^i}(p) \left.\frac{\partial}{\partial y^j}\right|_p,$$

and if $v = \sum_i a^i\, \partial/\partial x^i|_p = \sum_j b^j\, \partial/\partial y^j|_p$, then $b^j = \sum_i \frac{\partial y^j}{\partial x^i}(p)\, a^i$.

Proof. By Proposition 33.4.6(2) applied to the chart ψ, the coefficient of $\partial/\partial y^j|_p$ in $\partial/\partial x^i|_p$ is $\partial/\partial x^i|_p(y^j)$. The formula for the components follows by linearity. □

So the vector $(a^1, \dots, a^n)$ of components of v in the chart φ and the vector $(b^1, \dots, b^n)$ of components in ψ are related by the Jacobian matrix $D(\psi \circ \varphi^{-1})(\varphi(p))$. This is description (c) above: a tangent vector is the same thing as a rule that assigns to each chart at p a vector of $\mathbb{R}^n$, in such a way that the vectors attached to any two charts are related by the Jacobian matrix of the transition map.

Example 33.4.8. Let $U = \mathbb{R}^2 \setminus \{(x, 0) : x \le 0\}$. The map $P(r, \theta) = (r\cos\theta, r\sin\theta)$ is a bijection $(0, \infty) \times (-\pi, \pi) \to U$, since every point of U has a unique angle in $(-\pi, \pi)$, and its Jacobian determinant is $r > 0$ (compare Exercise 27.15.5). By the inverse function theorem its inverse is smooth, so $(r, \theta) = P^{-1}$ is a chart of $\mathbb{R}^2$ (Proposition 33.3.3(3)), the *polar coordinates*. With $x = r\cos\theta$ and $y = r\sin\theta$, Proposition 33.4.7 gives

$$\frac{\partial}{\partial r} = \cos\theta\,\frac{\partial}{\partial x} + \sin\theta\,\frac{\partial}{\partial y} = \frac{x}{r}\frac{\partial}{\partial x} + \frac{y}{r}\frac{\partial}{\partial y}, \qquad \frac{\partial}{\partial \theta} = -r\sin\theta\,\frac{\partial}{\partial x} + r\cos\theta\,\frac{\partial}{\partial y} = -y\frac{\partial}{\partial x} + x\frac{\partial}{\partial y}.$$

So $\partial/\partial r$ is the unit radial vector and $\partial/\partial\theta$ is the velocity of rotation about the origin. Note that $\partial/\partial r$ depends on the whole chart (r, θ), not only on the function r: in the chart (r, y), defined where $x > 0$, the vector $\partial/\partial r$ is $(r/x)\,\partial/\partial x$, a different vector. The formula $dr = \cos\theta\,dx + \sin\theta\,dy$ that one often meets is a different statement, about the differential of the function r, a covector rather than a vector; the two are compared in Chapter 36.

Now description (a). A *smooth curve* in M is a smooth map $\gamma\colon J \to M$ from an open interval $J \subseteq \mathbb{R}$.

Proposition 33.4.9. *For a smooth curve $\gamma\colon J \to M$ and $t_0 \in J$, the* velocity $\gamma'(t_0) = d_{t_0}\gamma(d/dt|_{t_0}) \in T_{\gamma(t_0)}M$ *is the derivation $\gamma'(t_0)f = (f \circ \gamma)'(t_0)$. In a chart,*

$$\gamma'(t_0) = \sum_i (x^i \circ \gamma)'(t_0)\frac{\partial}{\partial x^i}\Big|_{\gamma(t_0)}.$$

Every $v \in T_pM$ is the velocity $\gamma'(0)$ of a smooth curve with $\gamma(0) = p$, and for smooth $f\colon M \to N$ we have $d_p f(\gamma'(0)) = (f \circ \gamma)'(0)$.

Proof. By definition $\gamma'(t_0)f = d/dt|_{t_0}(f \circ \gamma) = (f \circ \gamma)'(t_0)$. The chart formula follows from Proposition 33.4.6(2), since $\gamma'(t_0)(x^i) = (x^i \circ \gamma)'(t_0)$. Given $v = \sum_i v^i \partial/\partial x^i|_p$ in a chart φ centred at p, let $\gamma(t) = \varphi^{-1}(tv^1, \dots, tv^n)$ for $|t|$ small enough that $tv \in \varphi(U)$. Then $x^i \circ \gamma(t) = tv^i$, so $\gamma'(0) = v$ by the chart formula. Finally $d_p f(\gamma'(0)) = d_p f \circ d_0\gamma(d/dt|_0) = d_0(f \circ \gamma)(d/dt|_0) = (f \circ \gamma)'(0)$ by the chain rule. □

Proposition 33.4.10. *Let γ and δ be smooth curves defined near 0 with $\gamma(0) = \delta(0) = p$. The following are equivalent: (a) $\gamma'(0) = \delta'(0)$; (b) $(\varphi \circ \gamma)'(0) = (\varphi \circ \delta)'(0)$ for some chart φ at p; (c) the same holds for every chart at p. Consequently $[\gamma] \mapsto \gamma'(0)$ is a bijection from the set of smooth curves through p at time 0, modulo the equivalence relation (b), onto T_pM.*

Proof. In a chart $\varphi = (x^i)$, the vector $(\varphi \circ \gamma)'(0) \in \mathbb{R}^n$ is the vector of components of $\gamma'(0)$ in the basis $\partial/\partial x^i|_p$ (Proposition 33.4.9), and similarly for δ. Since the $\partial/\partial x^i|_p$ form a basis, (a) is equivalent to (b) for this chart, whichever chart is chosen; this proves the equivalences. The map is well defined and injective by the equivalence of (a) and (b), and surjective by Proposition 33.4.9. □

Remark 33.4.11. We now have three equivalent descriptions of T_pM: derivations at p (the definition); classes of curves through p (Proposition 33.4.10); and vectors of components that change by the Jacobian matrix of the transition map (Proposition 33.4.7). Each is useful: derivations for algebra (vector fields as derivations, the Lie bracket), curves for geometry (velocities, flows), components for computation.

Corollary 33.4.12. *If nonempty smooth manifolds M and N are diffeomorphic, then* $\dim M = \dim N$.

Proof. A diffeomorphism f gives an isomorphism $d_p f : T_pM \to T_{f(p)}N$ (Proposition 33.4.6(3)), and these spaces have dimensions $\dim M$ and $\dim N$ by Proposition 33.4.6(2). □

The topological statement, that homeomorphic nonempty open subsets of $\mathbb{R}^m$ and $\mathbb{R}^n$ force $m = n$, is harder and needs homology (Corollary 21.6.7(3)); here linear algebra suffices because smooth structures come with tangent spaces.

Proposition 33.4.13. *Let M and N be smooth manifolds with projections pr_M, pr_N from $M \times N$. For $(p, q) \in M \times N$, the map*

$$(d\,\mathrm{pr}_M, d\,\mathrm{pr}_N) \colon T_{(p,q)}(M \times N) \to T_pM \oplus T_qN$$

is an isomorphism. Its inverse sends (v, w) to $d_p\iota_q(v) + d_q\iota_p(w)$, where $\iota_q(x) = (x, q)$ and $\iota_p(y) = (p, y)$.

Proof. In product charts $(x^1, \dots, x^m, y^1, \dots, y^n)$ the projection pr_M is $(x, y) \mapsto x$, so its differential sends $\partial/\partial x^i$ to $\partial/\partial x^i$ and $\partial/\partial y^j$ to 0 (Proposition 33.4.6(3)); similarly for pr_N. So the map sends a basis to a basis. The maps ι_q and ι_p are smooth (Proposition 33.3.3(5)), and in the same charts $d\iota_q(\partial/\partial x^i) = \partial/\partial x^i$ and $d\iota_p(\partial/\partial y^j) = \partial/\partial y^j$, which gives the inverse. □

The inverse function theorem of analysis now carries over to manifolds.

Theorem 33.4.14 (Inverse function theorem on manifolds)**.** *Let $f : M \to N$ be smooth and $p \in M$ with $d_p f$ an isomorphism. There are open sets $U \ni p$ and $V \ni f(p)$ such that $f|_U : U \to V$ is a diffeomorphism. Consequently, a smooth bijection whose differential is everywhere an isomorphism is a diffeomorphism.*

Proof. Choose charts (U_0, φ) at p and (V_0, ψ) at $f(p)$ with $f(U_0) \subseteq V_0$, and let $F = \psi \circ f \circ \varphi^{-1}$. By Proposition 33.4.6(3), $DF(\varphi(p))$ is the matrix of $d_p f$, so it is invertible (and $\dim M = \dim N$). By Theorem 27.11.2 there are open sets $A \ni \varphi(p)$ in $\varphi(U_0)$ and $B \ni F(\varphi(p))$ such that $F : A \to B$ is a diffeomorphism. Put $U = \varphi^{-1}(A)$ and $V = \psi^{-1}(B)$. Then $f|_U = \psi^{-1} \circ F \circ \varphi$ is a bijection $U \to V$, and its inverse $\varphi^{-1} \circ F^{-1} \circ \psi$ is smooth as a composite of smooth maps. For the last statement, the inverse of such a bijection is smooth near every point, hence smooth. □

A smooth map whose differential is everywhere an isomorphism is called a *local diffeomorphism*.

33.5 The tangent bundle and vector fields

A vector field should assign to every point p a tangent vector at p, varying smoothly with p. To say what "smoothly" means we assemble all tangent spaces into one manifold.

Definition 33.5.1. The *tangent bundle* of a smooth n-manifold M is the disjoint union $TM = \bigsqcup_{p \in M} T_pM$, with the projection $\pi : TM \to M$, $\pi(v) = p$ for $v \in T_pM$. A chart $(U, \varphi = (x^i))$ of M gives the *bundle chart*

$$\widetilde{\varphi} \colon \pi^{-1}(U) \to \varphi(U) \times \mathbb{R}^n, \qquad v \mapsto \big(\varphi(\pi(v)), v(x^1), \dots, v(x^n)\big),$$

which records the point and the components of v.

Proposition 33.5.2. *There is a unique topology and smooth structure on TM for which the bundle charts are charts. With it, TM is a smooth manifold of dimension $2n$ and π is smooth. The transition map between the bundle charts of φ and ψ is*

$$(x,\xi) \mapsto \big(\tau(x), D\tau(x)\xi\big), \qquad \tau = \psi \circ \varphi^{-1}.$$

Proof. We check the conditions of Lemma 33.2.7 for the bundle charts of all charts of M. Each $\widetilde{\varphi}$ is a bijection onto the open set $\varphi(U)\times\mathbb{R}^n$ by Proposition 33.4.6(2), and $\widetilde{\varphi}(\pi^{-1}(U)\cap\pi^{-1}(V)) = \varphi(U\cap V)\times\mathbb{R}^n$ is open; this is (1). The transition map is as stated, because the components of a vector change by the Jacobian matrix of τ (Proposition 33.4.7); its entries are smooth functions of (x,ξ), which gives (2). Since M is second countable, hence Lindelöf (Proposition 18.3.12), countably many chart domains U cover M, and the corresponding $\pi^{-1}(U)$ cover TM; this is (3). For (4), let $v \neq w$ in TM. If $\pi(v) = \pi(w)$, both lie in one $\pi^{-1}(U)$. Otherwise $\pi(v)$ and $\pi(w)$ have disjoint open neighbourhoods in M, which we may take to be domains of charts (Lemma 33.2.6(1)), and their preimages under π are disjoint domains of bundle charts. Finally, in bundle charts π is $(x,\xi)\mapsto x$, which is smooth. □

The differentials of a smooth map at all points fit together into one smooth map, and the chain rule of Proposition 33.4.6(3) becomes a statement about these maps.

Proposition 33.5.3. *Let $f\colon M\to N$ be smooth. The* tangent map *$df\colon TM\to TN$, $df(v) = d_{\pi(v)}f(v)$, is smooth. In charts φ of M and ψ of N, with local representative $F = \psi\circ f\circ\varphi^{-1}$, and the corresponding bundle charts, it is*

$$(x,\xi)\mapsto \big(F(x), DF(x)\xi\big).$$

The chain rule holds globally: $d(g\circ f) = dg\circ df$ for smooth $g\colon N\to P$, and $d(\mathrm{id}_M) = \mathrm{id}_{TM}$. So $M\mapsto TM$, $f\mapsto df$ is a functor from smooth manifolds to smooth manifolds, the tangent functor, *and if f is a diffeomorphism, so is df, with inverse $d(f^{-1})$.*

Proof. Let $v\in T_pM$ with p in the domain U of φ and $f(p)$ in the domain V of ψ, and let $x = \varphi(p)$ and $\xi = (v(x^1),\dots,v(x^m))$ be the coordinates of v in the bundle chart. Then $\psi(f(p)) = F(x)$, and by Proposition 33.4.6(3) the components of $d_pf(v)$ in the basis $\partial/\partial y^i|_{f(p)}$ are the entries of $DF(x)\xi$. So on the open set $\pi^{-1}(U\cap f^{-1}(V))$ the map df is $\widetilde{\psi}^{-1}\circ G\circ\widetilde{\varphi}$ with $G(x,\xi) = (F(x), DF(x)\xi)$, which is smooth because the entries of DF are smooth. These open sets cover TM, so df is continuous, and smooth by Lemma 33.3.2. The chain rule and $d(\mathrm{id}_M) = \mathrm{id}_{TM}$ hold at every point by Proposition 33.4.6(3), hence globally; for a diffeomorphism f they give $d(f^{-1})\circ df = \mathrm{id}_{TM}$ and $df\circ d(f^{-1}) = \mathrm{id}_{TN}$. □

In the charts of the proposition, the global chain rule for $d(g\circ f)$ is the chain rule of calculus: the local representative of $g\circ f$ is $G\circ F$, and $D(G\circ F)(x) = DG(F(x))\,DF(x)$ is the product of the Jacobian matrices (Proposition 27.10.3(2)).

Definition 33.5.4. A *vector field* on M is a smooth map $X\colon M\to TM$ with $\pi\circ X = \mathrm{id}_M$; we write $X_p = X(p)\in T_pM$. The set of vector fields is $\mathfrak{X}(M)$. A vector field acts on functions by $(Xf)(p) = X_p(f)$. In a chart, $X = \sum_i X^i\,\partial/\partial x^i$ on U, with component functions $X^i = X(x^i)\colon U\to\mathbb{R}$.

Lemma 33.5.5. *A map $X\colon M\to TM$ with $\pi\circ X = \mathrm{id}_M$ is smooth, hence a vector field, if and only if its components X^i are smooth in every chart of some atlas. For a vector field X and $f\in C^\infty(M)$, the function Xf is smooth; in a chart, $Xf = \sum_i X^i\,\partial f/\partial x^i$ with $\partial f/\partial x^i(q) = \partial/\partial x^i|_q(f)$. Sums of vector fields and products of vector fields with smooth functions are vector fields, so $\mathfrak{X}(M)$ is a module over $C^\infty(M)$.*

Proof. In the chart φ of M and the bundle chart $\widetilde{\varphi}$, the local representative of X is $x \mapsto (x, X^1(\varphi^{-1}(x)), \dots, X^n(\varphi^{-1}(x)))$; so X is smooth on U if and only if the X^i are smooth on U, and by Lemma 33.3.2 it suffices to check this for the charts of one atlas. The formula for Xf is Proposition 33.4.6(2), and $\partial f/\partial x^i = \partial_i(f \circ \varphi^{-1}) \circ \varphi$ is smooth. The last statement holds because components add and multiply pointwise. □

Example 33.5.6.

1. On a chart domain U, the *coordinate vector fields* $\partial/\partial x^i$ are vector fields, with components δ_{ij}.
2. On $\mathbb{R}^n$, the constant vector fields $\sum_i a^i \partial/\partial x^i$ and the *Euler vector field* $E = \sum_i x^i \partial/\partial x^i$, which points radially outwards, with length $\|x\|$.
3. On $\mathbb{R}^2$, the *rotation vector field* $R = -y\,\partial/\partial x + x\,\partial/\partial y$. On $U = \mathbb{R}^2 \setminus \{(x, 0) : x \le 0\}$ it equals $\partial/\partial\theta$ in polar coordinates (Example 33.4.8).
4. On S^1 there is a nowhere vanishing vector field. The map $e(t) = (\cos t, \sin t)$ is smooth (Example 33.3.4(3)), and its differential never vanishes, because the composite with the inclusion $S^1 \to \mathbb{R}^2$ has derivative $(-\sin t, \cos t) \neq 0$. As $\dim S^1 = 1$, e is a local diffeomorphism, and by Theorem 33.4.14 its restrictions to $(-\pi, \pi)$ and to $(0, 2\pi)$ are diffeomorphisms onto $S^1 \setminus \{(-1, 0)\}$ and $S^1 \setminus \{(1, 0)\}$. Their inverses θ and θ' are charts (the *angle charts*). On the overlap, $\theta' = \theta$ where $\theta > 0$ and $\theta' = \theta + 2\pi$ where $\theta < 0$; the transition map has derivative 1, so $\partial/\partial\theta = \partial/\partial\theta'$ on the overlap by Proposition 33.4.7. So the two coordinate vector fields glue to a vector field $\partial/\partial\theta$ on S^1, which vanishes nowhere.

Proposition 33.5.7. *The map sending X to the operator $f \mapsto Xf$ is a bijection from $\mathfrak{X}(M)$ onto the set of* derivations *of the $\mathbb{R}$-algebra $C^\infty(M)$, that is, of the linear maps $D\colon C^\infty(M) \to C^\infty(M)$ with $D(fg) = (Df)g + f(Dg)$.*

Proof. For a vector field X, $f \mapsto Xf$ is linear, maps into $C^\infty(M)$ by Lemma 33.5.5, and satisfies the Leibniz rule because each X_p does. If $Xf = 0$ for all f, then each X_p is the zero derivation, so $X = 0$; hence the map is injective. Conversely, let D be a derivation, and define $X_p(f) = (Df)(p)$; each X_p is a derivation at p. We show that X is smooth. Let q lie in a chart $(U, (x^i))$, and choose χ as in Lemma 33.3.10 with support in U and $\chi = 1$ on a neighbourhood V of q. The functions $\widetilde{x^i}$, equal to χx^i on U and to 0 outside U, lie in $C^\infty(M)$ and agree with x^i on V. For $q' \in V$, locality gives $X^i(q') = X_{q'}(x^i) = X_{q'}(\widetilde{x^i}) = (D\widetilde{x^i})(q')$, so the components of X are smooth on V. So X is a vector field, and $Xf = Df$ by construction. □

Example 33.5.8. *From vector fields to derivations and back.* A vector field $X = \sum_i X^i \partial_i$ on an open subset of $\mathbb{R}^n$ acts on functions by $Xf = \sum_i X^i\, \partial f/\partial x^i$; conversely a derivation D is the vector field whose components are its values on the coordinate functions,

$$D = \sum_i (Dx^i) \frac{\partial}{\partial x^i},$$

because both sides are derivations and agree on the x^i, and a derivation is determined by its values on the coordinates (Proposition 33.4.5 at each point). For example the rotation field $X = -y\,\partial_x + x\,\partial_y$ on $\mathbb{R}^2$ is the derivation $f \mapsto -y\,f_x + x\,f_y$, and from the derivation one recovers $Xx = -y$, $Xy = x$. In polar coordinates $X = \partial_\theta$, since $X(r) = (-yx + xy)/r = 0$ and $X(\theta) = 1$, where θ is a local angle function with $d\theta = (x\,dy - y\,dx)/r^2$. By contrast, $f \mapsto \partial^2 f/\partial x^2$ is linear but not a derivation, since $(fg)_{xx} = f_{xx}g + 2f_x g_x + f g_{xx}$; second order operators are not vector fields.

33.6 Submanifolds

Most manifolds met in practice sit inside a Euclidean space or another manifold: spheres, surfaces in $\mathbb{R}^3$, matrix groups such as $\mathrm{O}(n)$. A k-dimensional submanifold is a subset that looks, in suitable charts of the ambient manifold, like $\mathbb{R}^k \times \{0\} \subseteq \mathbb{R}^n$. The main tools to produce submanifolds are graphs (Proposition 33.6.4) and level sets of submersions (Theorem 33.6.11).

Definition 33.6.1. Let $f\colon M \to N$ be smooth. It is an *immersion* if every $d_p f$ is injective, a *submersion* if every $d_p f$ is surjective, and an *embedding* if it is an immersion that is a homeomorphism onto its image (with the subspace topology). A subset $S \subseteq M^n$ is a *submanifold* of dimension k if every $p \in S$ lies in a chart (U, φ) of M with

$$\varphi(U \cap S) = \varphi(U) \cap (\mathbb{R}^k \times \{0\});$$

such a chart is a *slice chart* for S. The number $n - k$ is the *codimension* of S.

Example 33.6.2.

1. The inclusion $\mathbb{R}^k \to \mathbb{R}^n$, $x \mapsto (x, 0)$, is an immersion and an embedding; the projection $\mathbb{R}^n \to \mathbb{R}^k$ onto the first coordinates is a submersion. The projection $M \times N \to M$ is a submersion (Proposition 33.4.13).
2. A curve $\gamma\colon J \to M$ is an immersion if and only if its velocity never vanishes. So $t \mapsto (\cos t, \sin t) \in \mathbb{R}^2$ is an immersion, and $t \mapsto (t^2, t^3)$ is not, since its velocity vanishes at $t = 0$.
3. The function $f(x) = \|x\|^2$ on $\mathbb{R}^n$ is a submersion on $\mathbb{R}^n \setminus \{0\}$, since $d_x f(v) = 2\langle x, v\rangle$, and not at 0.
4. A local diffeomorphism, such as $e\colon \mathbb{R} \to S^1$ of Example 33.5.6(4), is both an immersion and a submersion.

Lemma 33.6.3. *Let $S \subseteq M^n$ be a submanifold of dimension k, with the subspace topology, and let $\mathrm{pr}\colon \mathbb{R}^n \to \mathbb{R}^k$ be the projection onto the first k coordinates. Then S is a topological k-manifold, the maps $\mathrm{pr} \circ \varphi|_{U\cap S}$, for slice charts (U, φ), form a smooth atlas of S, and with the smooth structure it generates the inclusion $S \to M$ is an embedding. In these charts the inclusion is $x \mapsto (x, 0)$.*

Proof. S is Hausdorff and second countable, as a subspace of M. For a slice chart, $\varphi|_{U\cap S}$ is a homeomorphism of the open subset $U \cap S$ of S onto $\varphi(U) \cap (\mathbb{R}^k \times \{0\})$, which is open in $\mathbb{R}^k \times \{0\} \cong \mathbb{R}^k$; so $\mathrm{pr} \circ \varphi|_{U\cap S}$ is a chart of the topological manifold S. For two slice charts φ and ψ, the transition map of the restricted charts is $x \mapsto \mathrm{pr}(\psi \circ \varphi^{-1}(x, 0))$ on the open set $\{x \in \mathbb{R}^k : (x, 0) \in \varphi(U \cap V)\}$, which is smooth; so these charts form a smooth atlas. In the restricted chart on S and the chart φ on M, the inclusion is $x \mapsto (x, 0)$, which is smooth with injective derivative. So the inclusion is an immersion, and it is a homeomorphism onto its image because S has the subspace topology. □

From now on a submanifold carries this smooth structure. We identify T_pS with its image under the injective map $d_p\iota\colon T_pS \to T_pM$, where ι is the inclusion; in a slice chart, T_pS is spanned by $\partial/\partial x^1|_p, \ldots, \partial/\partial x^k|_p$.

Proposition 33.6.4. *Let $W \subseteq \mathbb{R}^k$ be open and $g\colon W \to \mathbb{R}^{n-k}$ smooth. Then the graph $\Gamma_g = \{(x, g(x)) : x \in W\}$ is a k-dimensional submanifold of $\mathbb{R}^n$, and the projection $\Gamma_g \to W$ is a diffeomorphism.*

Proof. The map $\Phi(x, y) = (x, y - g(x))$ is a diffeomorphism of the open set $W \times \mathbb{R}^{n-k}$ onto itself, with inverse $(x, z) \mapsto (x, z + g(x))$, so it is a chart of $\mathbb{R}^n$ (Proposition 33.3.3(3)). It maps Γ_g onto $W \times \{0\}$, so it is a slice chart around every point of Γ_g. The restricted chart $\mathrm{pr} \circ \Phi|_{\Gamma_g}$ is the projection $(x, g(x)) \mapsto x$, which is therefore a diffeomorphism onto W. □

Proposition 33.6.5. *Let $S \subseteq N$ be a submanifold and $f : M \to N$ a smooth map with $f(M) \subseteq S$. Then f is smooth as a map $M \to S$. The restriction of a smooth map $M \to N$ to a submanifold of M is smooth.*

Proof. As a map to S, f is continuous, since S has the subspace topology. Let $p \in M$ and let (V, ψ) be a slice chart at $f(p)$; choose a chart (U, φ) at p with $f(U) \subseteq V$. The local representative of $f : M \to S$ in the charts φ and $\mathrm{pr} \circ \psi|_{V \cap S}$ is $\mathrm{pr} \circ (\psi \circ f \circ \varphi^{-1})$, which is smooth. The second statement holds because the restriction is the composite with the smooth inclusion. □

Example 33.6.6. The sphere S^n is an n-dimensional submanifold of $\mathbb{R}^{n+1}$. Let $p \in S^n$ and choose i with $p_i \neq 0$, say $p_i > 0$. On the open set $U = \{x : x_i > 0, \sum_{j \neq i} x_j^2 < 1\}$, the sphere is the graph $x_i = g(x')$ of the smooth function $g(x') = (1 - \|x'\|^2)^{1/2}$ of the other coordinates $x' = (x_j)_{j \neq i}$, which ranges over the open unit ball B of $\mathbb{R}^n$. The chart $x \mapsto (x', x_i - g(x'))$ of Proposition 33.6.4, restricted to U, maps $U \cap S^n$ onto $B \times \{0\}$ and U onto a set whose intersection with $\mathbb{R}^n \times \{0\}$ is $B \times \{0\}$, since $g > 0$; so it is a slice chart at p. The case $p_i < 0$ is the same with the negative square root. The structure obtained in this way agrees with the one given by the stereographic charts (Example 33.2.9(2)): the map σ_N is the restriction to $S^n \setminus \{N\}$ of the smooth map $x \mapsto (x_1, \dots, x_n)/(1 - x_{n+1})$ on $\{x_{n+1} \neq 1\}$, and its inverse τ is smooth as a map to $\mathbb{R}^{n+1}$ with image in S^n; by Proposition 33.6.5 both are smooth for the submanifold structure. So σ_N, and likewise σ_S, is a diffeomorphism onto $\mathbb{R}^n$, hence a chart of the submanifold structure (Proposition 33.3.3(3)), and the two structures coincide by Corollary 33.2.4(1).

The next proposition says that, up to a change of coordinates, immersions and submersions look like the linear models of Example 33.6.2(1).

Proposition 33.6.7. *Let $f : M^m \to N^n$ be smooth and $p \in M$.*

1. *If $d_p f$ is injective, there are charts around p and $f(p)$ in which f is $x \mapsto (x, 0)$; if $d_p f$ is surjective, there are charts in which f is $(x, y) \mapsto x$. In particular an immersion is injective on a neighbourhood of each point, and a submersion is an open map.*

2. *The image of an embedding f is a submanifold, and f is a diffeomorphism onto it. Conversely, a submanifold is a smooth manifold for which the inclusion is an embedding (Lemma 33.6.3).*

3. *An injective immersion of a compact manifold is an embedding.*

Proof. (1) Choose charts (U, φ) at p and (V, ψ) at $f(p)$ with $f(U) \subseteq V$, and let $F = \psi \circ f \circ \varphi^{-1}$. By Proposition 33.4.6(3), $DF(\varphi(p))$ is the matrix of $d_p f$, so it is injective, respectively surjective, when $d_p f$ is. If it is surjective, Theorem 27.11.4(1) gives a diffeomorphism θ from a neighbourhood of $\varphi(p)$ onto an open subset of $\mathbb{R}^m$ with $F \circ \theta^{-1}(x^1, \dots, x^m) = (x^1, \dots, x^n)$. By Lemma 33.2.6, $\theta \circ \varphi$, restricted to the preimage of that neighbourhood, is a chart, and in the charts $\theta \circ \varphi$ and ψ the local representative of f is $F \circ \theta^{-1}$, the projection. If $DF(\varphi(p))$ is injective, Theorem 27.11.4(2) gives a diffeomorphism θ from a neighbourhood Ω of $F(\varphi(p))$ onto an open subset of $\mathbb{R}^n$ with $\theta \circ F(x) = (x, 0)$ near $\varphi(p)$. Then $\theta \circ \psi$ is a chart on $\psi^{-1}(\Omega)$; shrinking U so that $F(\varphi(U)) \subseteq \Omega$ and $\theta \circ F(x) = (x, 0)$ on $\varphi(U)$, the local representative in the charts φ and $\theta \circ \psi$ is $x \mapsto (x, 0)$. The map $x \mapsto (x, 0)$ is injective and the projection is open; since charts are homeomorphisms, the last statements follow.

(2) Let $f : M^k \to N^n$ be an embedding and $p \in M$. By (1) there are charts (U, φ) at p and (V, ψ) at $f(p)$ with $f(U) \subseteq V$ and $\psi \circ f \circ \varphi^{-1}(x) = (x, 0)$ on $\varphi(U)$; so $\psi(f(U)) = \varphi(U) \times \{0\}$. This alone does not make ψ a slice chart, because other parts of $f(M)$ may pass through V, as happens for a figure eight at its crossing point. But f is a homeomorphism onto its

image, so $f(U)$ is open in $f(M)$: there is an open $W \subseteq N$ with $f(U) = W \cap f(M)$. Let $V' = V \cap W \cap \psi^{-1}(\varphi(U) \times \mathbb{R}^{n-k})$, an open neighbourhood of $f(p)$. Every point of $V' \cap f(M)$ lies in $W \cap f(M) = f(U)$, so $\psi(V' \cap f(M)) = \psi(V') \cap (\varphi(U) \times \{0\})$. Conversely, if $(x, 0) \in \psi(V')$, then $x \in \varphi(U)$ by the choice of V', and $(x, 0) = \psi(f(\varphi^{-1}(x)))$ lies in $\psi(V' \cap f(M))$. So $\psi(V' \cap f(M)) = \psi(V') \cap (\mathbb{R}^k \times \{0\})$, and $\psi|_{V'}$ is a slice chart. In the charts φ and $\mathrm{pr} \circ \psi|_{V' \cap f(M)}$, the map $f : M \to f(M)$ is the identity near $\varphi(p)$; so f is a bijective local diffeomorphism onto $f(M)$, hence a diffeomorphism (Theorem 33.4.14).

(3) A continuous injection from a compact space to a Hausdorff space is a homeomorphism onto its image (Proposition 18.6.3(4)). □

Proposition 33.6.8. *An injective immersion $f : M \to N$ that is proper (preimages of compact sets are compact) is an embedding, and its image is a closed submanifold.*

Proof. N is locally compact and Hausdorff, so the proper continuous map f is closed (Proposition 18.9.6). A closed continuous injection is a homeomorphism onto its image, which is closed: the image of a closed subset $C \subseteq M$ is $f(C)$, closed in N and hence in $f(M)$. So f is an embedding, and $f(M)$ is a submanifold by Proposition 33.6.7(2). □

Counterexample 33.6.9. Each hypothesis is needed. The map $t \mapsto (\cos 2t, \sin 2t)$ from $S^1 = \mathbb{R}/2\pi\mathbb{Z}$ to $\mathbb{R}^2$ is a proper immersion, but not injective, as it winds twice around the circle, and it is not an embedding. The curve $t \mapsto (t^2, t^3)$ is injective, proper and smooth, but not an immersion, as its velocity vanishes at $t = 0$. And injective immersions that are not proper need not be embeddings: the figure eight and the dense line on the torus of Counterexample 33.6.17 below.

Example 33.6.10. *Homeomorphic to a manifold, but not a submanifold.* Let $\Gamma = \{(x, |x|) : x \in \mathbb{R}\} \subseteq \mathbb{R}^2$, the graph of $|x|$, and $C = \{(x, y, z) : z = \sqrt{x^2 + y^2}\} \subseteq \mathbb{R}^3$, the cone.

1. *They are smooth manifolds.* The projections $\Gamma \to \mathbb{R}$, $(x, |x|) \mapsto x$, and $C \to \mathbb{R}^2$, $(x, y, z) \mapsto (x, y)$, are homeomorphisms (continuous bijections with continuous inverses $x \mapsto (x, |x|)$ and $(x, y) \mapsto (x, y, \sqrt{x^2 + y^2})$). So Γ and C are topological manifolds, and each of them carries a smooth structure: a homeomorphism $h : X \to N$ onto a smooth manifold transports the atlas of N to the atlas $\{\varphi \circ h\}$ of X, whose transition maps are those of N; for this structure h is a diffeomorphism. Being homeomorphic to a smooth manifold is a topological property, and so is "admitting a smooth structure".

2. *They are not submanifolds.* A subset of $\mathbb{R}^n$ has a preferred smooth structure only when it is a submanifold (Definition 33.6.1): then the slice charts give the unique structure for which the inclusion is an embedding (Proposition 33.6.7(2)). Neither Γ nor C is a submanifold near the origin. Suppose $S \subseteq \mathbb{R}^n$ is a k-dimensional submanifold with a slice chart φ at $0 \in S$, and put $T = d(\varphi^{-1})_{\varphi(0)}(\mathbb{R}^k \times \{0\})$, a k-dimensional subspace. If $c : [0, \varepsilon) \to \mathbb{R}^n$ is a curve with values in S, $c(0) = 0$, differentiable from the right at 0, then $c'(0) \in T$: the curve $\varphi \circ c$ takes values in $\mathbb{R}^k \times \{0\}$, so its right derivative lies in this closed subspace, and $c'(0) = d(\varphi^{-1})((\varphi \circ c)'(0))$ by the chain rule. For Γ the rays $t \mapsto (t, t)$ and $t \mapsto (-t, t)$, $t \geq 0$, have velocities $(1, 1)$ and $(-1, 1)$, which span $\mathbb{R}^2$, so no line T contains them: Γ is not a 1-dimensional submanifold. For C the rays $t \mapsto (tu, t)$ with $u \in S^1$ have velocities $(u, 1)$, which span $\mathbb{R}^3$, so C is not a 2-dimensional submanifold. (Neither is a submanifold of any other dimension, since it is a topological manifold of dimension 1, respectively 2.)

3. *What goes wrong with the transported structure.* For the smooth structure on Γ transported from $\mathbb{R}$, the inclusion $\Gamma \to \mathbb{R}^2$ is, in the chart x, the map $x \mapsto (x, |x|)$, which is not differentiable at 0. Any other smooth structure for which the inclusion is smooth makes

it a non-immersion at the origin, since an injective immersion that is a homeomorphism onto its image is an embedding, whose image would be a submanifold; for instance $t \mapsto (t^3, |t|^3)$ is C^2 but not an immersion at 0. So Γ is a smooth manifold, but not with "the structure induced from $\mathbb{R}^2$", because there is no such structure: the corner is invisible to topology and visible to calculus.

By contrast the graph of a smooth function $f : U \to \mathbb{R}^m$ is always a submanifold (Proposition 33.6.4), and the projection to U is then a diffeomorphism for the induced structure.

Theorem 33.6.11 (Regular value theorem)**.** *Let $f : M^n \to N^m$ be smooth and $q \in N$ a* regular value*: $d_p f$ is surjective for every $p \in f^{-1}(q)$. Then $f^{-1}(q)$ is a submanifold of dimension $n - m$, with $T_p(f^{-1}(q)) = \ker d_p f$.*

Proof. Let $S = f^{-1}(q)$ and $p \in S$. By Proposition 33.6.7(1) there are charts (U, φ) at p and (V, ψ) at q with $f(U) \subseteq V$ in which f is $(x^1, \dots, x^n) \mapsto (x^1, \dots, x^m)$. With $c = \psi(q) \in \mathbb{R}^m$, $\varphi(U \cap S) = \{x \in \varphi(U) : (x^1, \dots, x^m) = c\}$. Compose φ with the translation $x \mapsto x - (c, 0)$ and then with the linear map that moves the last $n - m$ coordinates to the front (Lemma 33.2.6(2)). In the new chart φ', $\varphi'(U \cap S) = \varphi'(U) \cap (\mathbb{R}^{n-m} \times \{0\})$: it is a slice chart. So S is a submanifold of dimension $n - m$. Since f is constant on S, the chain rule gives $d_p f \circ d_p \iota = 0$, so $T_p S \subseteq \ker d_p f$. Both spaces have dimension $n - m$, the second by rank–nullity (Theorem 11.2.4), so they are equal. □

If q is not in the image of f, it is a regular value and $f^{-1}(q) = \emptyset$.

On the identification of tangent spaces. For an embedding, or more generally an immersion, $f : M \to N$, the differential $d_x f : T_x M \to T_{f(x)} N$ is injective, and one identifies $T_x M$ with the subspace $d_x f(T_x M)$ of $T_{f(x)} N$; in the language of derivations, $v \in T_x M$ corresponds to the derivation $g \mapsto v(g \circ f)$ of functions on N. For a submanifold $S \subseteq N$ with inclusion ι, this is how $T_p S$ is regarded as a subspace of $T_p N$: as $d_p \iota(T_p S)$, the derivations $g \mapsto v(g|_S)$. The equality $T_p(f^{-1}(q)) = \ker d_p f$ in the theorem means precisely $d_p \iota(T_p S) = \ker d_p f$, which is what the proof shows. For a submanifold S of $\mathbb{R}^N$, Proposition 33.4.9 identifies $T_p S \subseteq T_p \mathbb{R}^N = \mathbb{R}^N$ with the set of velocities $\gamma'(0)$ of smooth curves γ in S with $\gamma(0) = p$, which is the tangent space of elementary geometry.

Proposition 33.6.12 (Fibre products)**.** *Let $f : M \to P$ and $g : N \to P$ be smooth with g a submersion. Then*

$$M \times_P N = \{(m, n) \in M \times N : f(m) = g(n)\}$$

is a submanifold of $M \times N$ of dimension $\dim M + \dim N - \dim P$, with $T_{(m,n)}(M \times_P N) = \{(u, v) \in T_m M \oplus T_n N : d_m f(u) = d_n g(v)\}$. The projections to M and N are smooth, the projection to M is a submersion, and for every manifold L the smooth maps $L \to M \times_P N$ correspond to pairs of smooth maps $a : L \to M$, $b : L \to N$ with $f \circ a = g \circ b$.

Proof. Let $(m, n) \in M \times_P N$ and $p = g(n)$. By Proposition 33.6.7(1) there are charts in which g is the projection $(x, y) \mapsto x$ from an open subset of $\mathbb{R}^{\dim P} \times \mathbb{R}^{\dim N - \dim P}$, and a chart (U, φ) at m with $f(U)$ inside the chart of P; write $\tilde{f}$ for the local representative of f. In these coordinates the fibre product is $\{(z, (x, y)) : x = \tilde{f}(z)\}$, which after reordering the factors as (z, y, x) is the graph of the smooth map $(z, y) \mapsto \tilde{f}(z)$. By Proposition 33.6.4 a graph is a slice in suitable coordinates, so these coordinates give a slice chart of $M \times N$ at (m, n), and $M \times_P N$ is a submanifold of dimension $\dim M + (\dim N - \dim P)$. In the corresponding chart of $M \times_P N$ the projection to M is $(z, y) \mapsto z$, a submersion. Differentiating $f \circ \mathrm{pr}_M = g \circ \mathrm{pr}_N$ on $M \times_P N$ shows that its tangent space is contained in the space described, which has the same dimension, because $(u, v) \mapsto d_m f(u) - d_n g(v)$ is a surjective linear map $T_m M \oplus T_n N \to T_p P$ (already $v \mapsto -d_n g(v)$ is surjective). The universal property follows from Proposition 33.6.5:

a pair (a, b) with $f \circ a = g \circ b$ is a smooth map $L \to M \times N$ with image in the submanifold $M \times_P N$. □

Example 33.6.13. 1. *Spheres and quadrics.* $S^n = f^{-1}(1)$ for $f(x) = \|x\|^2$, with $d_x f(v) = 2\langle x, v\rangle \neq 0$ on S^n; so $T_x S^n = \ker d_x f = x^\perp$. Similarly, for $g(x, y, z) = x^2 + y^2 - z^2$ on $\mathbb{R}^3$ the only critical point is 0, so every $c \neq 0$ is a regular value, and the hyperboloids $g^{-1}(c)$ are submanifolds of dimension 2.

2. *The orthogonal group.* Let Sym_n be the space of symmetric $n \times n$ matrices, of dimension $n(n+1)/2$, and $F\colon M_n(\mathbb{R}) \to \mathrm{Sym}_n$, $F(A) = A^T A$. Then $\mathrm{O}(n) = F^{-1}(I)$. Since $F(A+tX) = A^T A + t(A^T X + X^T A) + t^2 X^T X$, the differential is $d_A F(X) = A^T X + X^T A$. For $A \in \mathrm{O}(n)$ it is surjective onto Sym_n: given a symmetric S, the matrix $X = AS/2$ gives $A^T X + X^T A = S/2 + S/2 = S$. So $\mathrm{O}(n)$ is a submanifold of dimension $n^2 - n(n+1)/2 = n(n-1)/2$, and its tangent space at I is $\ker d_I F = \{X : X + X^T = 0\}$, the skew-symmetric matrices.

3. *The special linear group.* For $A \in \mathrm{GL}_n(\mathbb{R})$, $\det(A+tX) = \det(A)\det(I+tA^{-1}X)$, and $\det(I+tY) = 1 + t\,\mathrm{tr}\,Y + O(t^2)$ by expanding the determinant; so $d_A \det(X) = \det(A)\,\mathrm{tr}(A^{-1}X)$. For $A \in \mathrm{SL}_n(\mathbb{R}) = \det^{-1}(1)$, taking $X = A$ gives $d_A \det(A) = n \neq 0$, so 1 is a regular value, $\mathrm{SL}_n(\mathbb{R})$ is a submanifold of dimension $n^2 - 1$, and its tangent space at I consists of the matrices of trace 0.

Example 33.6.14. If $S \subseteq \mathbb{R}^N$ is a submanifold and $X\colon S \to \mathbb{R}^N$ is a smooth map with $X(p) \in T_p S$ for all p, then X is a vector field on S: in a slice chart $\psi = (y^1, \ldots, y^N)$ of $\mathbb{R}^N$, the components of X in the restricted chart $(y^1, \ldots, y^k)$ of S are $X(p)(y^i) = \sum_j X_j(p)\,\partial_j y^i(p)$, which are smooth. On the odd-dimensional sphere $S^{2m-1} \subseteq \mathbb{R}^{2m}$, the map

$$X(x) = (-x_2, x_1, -x_4, x_3, \ldots, -x_{2m}, x_{2m-1})$$

satisfies $\langle X(x), x\rangle = 0$, so $X(x) \in T_x S^{2m-1}$ by Example 33.6.13(1), and $\|X(x)\| = 1$. So odd spheres carry nowhere vanishing vector fields. Even spheres do not, by the hairy ball theorem (Corollary 21.6.7(2)).

Counterexample 33.6.15. At a critical value the level set can fail to be a submanifold, or not. For $f(x, y) = xy$ on $\mathbb{R}^2$, 0 is a critical value ($df = 0$ at the origin) and $f^{-1}(0)$ is the union of the coordinate axes, which is not even a topological manifold (Counterexample 20.5.5). For $f(x, y) = x^2$, 0 is also a critical value, but $f^{-1}(0)$ is the y-axis, a submanifold. So the regular value theorem gives a sufficient condition only. Likewise the cone $C = g^{-1}(0)$ for $g = x^2 + y^2 - z^2$ is not a 2-dimensional submanifold near its vertex 0. Otherwise 0 would have a neighbourhood D in C homeomorphic to an open ball of $\mathbb{R}^2$, and $D \setminus \{0\}$ would be connected (Lemma 20.5.4). But $D \setminus \{0\}$ contains points with $z > 0$ and points with $z < 0$ (the points $(t, 0, \pm t) \in C$ tend to 0), and it is the disjoint union of its open subsets $\{z > 0\}$ and $\{z < 0\}$, because $z = 0$ on C only at the vertex.

Lemma 33.6.16. *Let $S \subseteq \mathbb{R}^2$ be a 1-dimensional submanifold and $p \in S$. Then near p, S is the graph of a smooth function over one of the coordinate axes: there are $i \in \{1, 2\}$, an open interval J and a smooth $g\colon J \to \mathbb{R}$ such that near p, $S = \{(s, g(s)) : s \in J\}$ if $i = 1$, or $S = \{(g(s), s) : s \in J\}$ if $i = 2$.*

Proof. The inclusion $\iota\colon S \to \mathbb{R}^2$ is an embedding (Proposition 33.6.7(2)), so $d_p\iota(T_p S)$ is a line, and for some coordinate projection pr_i the composite $\mathrm{pr}_i \circ \iota\colon S \to \mathbb{R}$ has nonzero differential at p. By the inverse function theorem (Theorem 33.4.14) it restricts to a diffeomorphism from a neighbourhood of p in S onto an open interval J; composing its inverse with the other coordinate gives g. □

Counterexample 33.6.17. 1. *An injective immersion that is not an embedding.* Let $\beta(t) = (\sin 2t, \sin t)$ for $t \in (-\pi, \pi)$. Then $\beta'(t) = (2\cos 2t, \cos t)$ never vanishes, since $\cos t = 0$ forces $\cos 2t = -1$. It is injective: if $\beta(s) = \beta(t)$ with $s \neq t$ in $(-\pi, \pi)$, then $\sin s = \sin t$ gives $s = \pm\pi - t$, so $\sin 2s = -\sin 2t$, and $\sin 2s = \sin 2t$ forces $\sin 2t = 0$, that is $t \in \{0, \pm\pi/2\}$; but then $s = \pm\pi - t$ is either $\pm\pi \notin (-\pi, \pi)$ or equal to t, a contradiction. The image, a figure eight, equals $\beta([-\pi, \pi])$ because $\beta(\pm\pi) = \beta(0)$, so it is compact; since $(-\pi, \pi)$ is not compact, β is not a homeomorphism onto its image.

2. *A dense immersed line.* Let α be irrational and $\beta\colon \mathbb{R} \to T^2 = S^1 \times S^1$, $\beta(t) = (e^{2\pi i t}, e^{2\pi i \alpha t})$. It is an immersion, and injective: $\beta(s) = \beta(t)$ forces $s - t \in \mathbb{Z}$ and $\alpha(s - t) \in \mathbb{Z}$, so $s = t$. The closure of $\beta(\mathbb{R})$ contains $\{1\} \times S^1$, since the points $\beta(n) = (1, e^{2\pi i \alpha n})$, $n \in \mathbb{Z}$, are dense there by Lemma 20.3.4; and it is invariant under multiplication by $\beta(t)$, since β is a homomorphism and multiplication is a homeomorphism; so it contains $\{e^{2\pi i t}\} \times S^1$ for all t, and $\beta(\mathbb{R})$ is dense. For $z = e^{2\pi i t_0}$, $\beta(\mathbb{R}) \cap (\{z\} \times S^1) = \beta(t_0 + \mathbb{Z})$ is countable. If $\beta(\mathbb{R})$ were a submanifold, it would be locally closed, being locally a slice in a chart, and a dense locally closed set is open (it is open in its closure). But a nonempty open subset of T^2 meets some circle $\{z\} \times S^1$ in a nonempty open subset of that circle, which is uncountable. So $\beta(\mathbb{R})$ is not a submanifold.

3. *A smooth homeomorphism onto a non-submanifold.* The map $\gamma(t) = (t^2, t^3)$ is smooth and a homeomorphism of $\mathbb{R}$ onto the cusp $C = \{y^2 = x^3\}$, with inverse $(x, y) \mapsto y^{1/3}$, but $\gamma'(0) = 0$. The cusp is not a submanifold near 0: by Lemma 33.6.16 it would be a graph over one axis near 0, but over the x-axis it has two points $(x, \pm x^{3/2})$ for small $x > 0$, and over the y-axis it is $x = y^{2/3}$, which is not differentiable at 0.

Proposition 33.6.18. *Let $S \subseteq \mathbb{R}^N$ and $p \in S$, and let $0 \leq k \leq N$. The following are equivalent.*

1. *(Slice) There is a chart (U, φ) of $\mathbb{R}^N$ at p with $\varphi(U \cap S) = \varphi(U) \cap (\mathbb{R}^k \times \{0\})$.*

2. *(Graph) After permuting the coordinates of $\mathbb{R}^N$ there are open sets $\Omega \subseteq \mathbb{R}^k$ and $W \ni p$ in $\mathbb{R}^N$ and a smooth $g\colon \Omega \to \mathbb{R}^{N-k}$ with $W \cap S = \{(u, g(u)) : u \in \Omega\}$.*

3. *(Parametrisation) There are an open $\Omega \subseteq \mathbb{R}^k$, an open $W \ni p$ in $\mathbb{R}^N$ and a smooth $x\colon \Omega \to \mathbb{R}^N$ with $x(\Omega) = W \cap S$, such that x is a homeomorphism onto $W \cap S$ with the subspace topology and $d_u x$ is injective for every $u \in \Omega$.*

4. *(Regular level set) There are an open $W \ni p$ in $\mathbb{R}^N$ and a smooth $F\colon W \to \mathbb{R}^{N-k}$ with $W \cap S = F^{-1}(0)$ and $d_q F$ surjective for every $q \in W \cap S$.*

So S is a k-dimensional submanifold of $\mathbb{R}^N$ if and only if one of these holds at every point of S.

Proof. (1) $\Rightarrow$ (4). Let F be the last $N - k$ coordinates of φ on $W = U$; then $F^{-1}(0) = U \cap S$ and $d_q F$ is surjective, being the last $N - k$ rows of the invertible matrix of $d_q \varphi$.

(4) $\Rightarrow$ (2). By the implicit function theorem (Theorem 27.11.3), applied after permuting coordinates so that the last $N - k$ columns of $DF(p)$ are independent, there are open sets $\Omega \times B \ni p$ inside W and a smooth $g\colon \Omega \to B$ such that, for $(u, v) \in \Omega \times B$, $F(u, v) = 0$ if and only if $v = g(u)$.

(2) $\Rightarrow$ (3). Take $x(u) = (u, g(u))$, whose differential is injective and whose inverse is the projection to the first k coordinates, a continuous map.

(3) $\Rightarrow$ (1). The map $x\colon \Omega \to \mathbb{R}^N$ is an injective immersion and a homeomorphism onto its image, so it is an embedding, and its image $W \cap S$ is a k-dimensional submanifold of $\mathbb{R}^N$ (Proposition 33.6.7(2)); a slice chart at p for $W \cap S$, intersected with W, is one for S. □

Remark 33.6.19. A *regular surface* in $\mathbb{R}^3$, in the sense of do Carmo [dC76, Section 2.2], is a subset $S \subseteq \mathbb{R}^3$ such that every $p \in S$ has a *parametrisation*: an open $\Omega \subseteq \mathbb{R}^2$, a neighbourhood W of p in $\mathbb{R}^3$ and a smooth $x\colon \Omega \to \mathbb{R}^3$ with image $W \cap S$, which is a homeomorphism onto $W \cap S$ and has injective differential everywhere. This is condition (3) of Proposition 33.6.18 with $k = 2$ and $N = 3$, so regular surfaces are exactly the 2-dimensional submanifolds of $\mathbb{R}^3$, and the charts of the submanifold structure are the inverses x^{-1} of the parametrisations (Lemma 33.6.3); the change of parameters between two parametrisations is the transition map of these charts, and is smooth. Both conditions on x are needed: dropping injectivity of $d_u x$ allows the cusp, and dropping the requirement that x be a homeomorphism onto its image allows the figure eight, neither of which is a submanifold (Counterexample 33.6.17).

Definition 33.6.20. A subset A of a smooth n-manifold has *measure zero* if $\varphi(A \cap U)$ has Lebesgue measure zero for every chart; by the change of variables formula (Theorem 28.11.2) it suffices to check this for the charts of one atlas.

Theorem 33.6.21 (Sard)**.** *Let $f : M \to N$ be smooth. The set of critical values of f (the images of points where $d_p f$ is not surjective) has measure zero in N; in particular regular values are dense. If $\dim M < \dim N$, the image of f has measure zero.*

Proof. The last statement follows from the first, since then every point is critical; alternatively, compose with the projection $\mathbb{R}^{\dim N} \to \mathbb{R}^{\dim M}$ and use that a smooth map is locally Lipschitz, so it maps sets of measure zero to sets of measure zero and $\mathbb{R}^{\dim M} \times \{0\}$ has measure zero. The general case is [Mil97, Chapter 2] or [Lee13, Chapter 6]; the proof is a careful induction on the order of vanishing of the derivatives, using Taylor's theorem and a covering argument in charts, together with the fact that a countable union of sets of measure zero has measure zero (Proposition 28.2.4) and that manifolds are covered by countably many charts. □

Remark 33.6.22. *The "in particular" in Sard's theorem.* A set of measure zero has dense complement: a nonempty open set $W \subseteq N$ contains, in a chart, a ball, which has positive Lebesgue measure, so W is not of measure zero and meets the complement. Hence the regular values, the complement of the critical values, are dense. The notion of measure zero is defined without any measure on N, and it does not depend on the charts. If N carries a measure given in every chart by a positive continuous density with respect to Lebesgue measure, as the Riemannian measure of a metric does (Chapter 40), then its null sets are exactly the sets of measure zero, because such a density is bounded above and below on compact subsets of the chart; so Sard's theorem says that the critical values form a null set for any such measure.

33.7 Constructions of smooth manifolds

Open subsets, products and regular level sets of smooth manifolds are smooth manifolds (Example 33.2.9 and Theorem 33.6.11). Two further constructions are quotients by group actions and spaces of linear subspaces. In both, the difficulty is to find charts on a set that is not given as a subset of a known manifold.

Theorem 33.7.1. *Let a discrete group G act freely and properly on a smooth manifold M by diffeomorphisms. Then M/G has a unique smooth structure for which $\pi\colon M \to M/G$ is a local diffeomorphism. A map $f : M/G \to N$ is smooth if and only if $f \circ \pi$ is smooth.*

Proof. By Proposition 20.6.1(3), M/G is a topological manifold and π is a covering map. Every $x \in M$ has a neighbourhood U, which we may take inside the domain of a chart φ, with $gU \cap U = \emptyset$ for $g \neq e$ (Theorem 20.4.3); $\pi|_U$ is a homeomorphism onto the open set

$\pi(U)$, and we use $\varphi \circ (\pi|_U)^{-1}$ as a chart. For two such charts, from (U, φ) and (V, ψ), and a point $\pi(u) = \pi(v)$ in $\pi(U) \cap \pi(V)$ with $u \in U$, $v \in V$, we have $v = gu$ for a unique g, and for u' near u, the unique point of V in the orbit of u' is gu' (as $gu' \in V$ for u' near u). So near $\varphi(u)$ the transition map is $\psi \circ g \circ \varphi^{-1}$, which is smooth, g being a diffeomorphism. These charts form a smooth atlas, and in them π is given by $\varphi \circ (\pi|_U)^{-1} \circ \pi \circ \varphi^{-1} = \mathrm{id}$, so π is a local diffeomorphism. Uniqueness: if π is a local diffeomorphism for some smooth structure, the maps $(\pi|_U)^{-1}$ are smooth, so the charts above belong to that structure. Finally, if $f \circ \pi$ is smooth then $f = (f \circ \pi) \circ (\pi|_U)^{-1}$ is smooth on each $\pi(U)$. □

Example 33.7.2. The torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$, the projective space $\mathbb{RP}^n = S^n/\{\pm 1\}$, the lens spaces $S^3/(\mathbb{Z}/p)$ and the Klein bottle (Example 20.4.5) are smooth manifolds, with smooth covering maps from $\mathbb{R}^n$, S^n, S^3 and $\mathbb{R}^2$. The smooth structure on $\mathbb{RP}^n$ agrees with the one given by the charts of Example 33.2.9(3). Indeed, $\varphi_i \circ \pi$ is $x \mapsto (x_j/x_i)_{j \neq i}$ on $S^n \cap \{x_i \neq 0\}$, which is smooth, so φ_i is smooth for the quotient structure; and its inverse is $u \mapsto \pi(\hat{u}/\|\hat{u}\|)$, where $\hat{u}$ is u with 1 inserted in position i, a composite of smooth maps. So each φ_i is a diffeomorphism onto $\mathbb{R}^n$, hence a chart of the quotient structure (Proposition 33.3.3(3)). By the same argument, a covering space of a smooth manifold has a unique smooth structure making the covering map a local diffeomorphism.

Lemma 33.7.3. *Two linear subspaces $W, W' \subseteq \mathbb{R}^n$ of the same dimension k have a common complement: a subspace Q with $W \oplus Q = W' \oplus Q = \mathbb{R}^n$.*

Proof. Induction on $n - k$. If $k = n$, take $Q = 0$. Otherwise W and W' are proper subspaces, and $\mathbb{R}^n \neq W \cup W'$: if $w \in W \setminus W'$ and $w' \in W' \setminus W$ (if one contains the other they are equal, and any vector outside W works), then $w + w'$ lies in neither. Choose $v \notin W \cup W'$. By induction $W \oplus \mathbb{R}v$ and $W' \oplus \mathbb{R}v$ have a common complement Q', and $Q = Q' \oplus \mathbb{R}v$ is a common complement of W and W', by counting dimensions. □

Proposition 33.7.4. *The set $\mathrm{Gr}_k(\mathbb{R}^n)$ of k-dimensional linear subspaces of $\mathbb{R}^n$ has a natural structure of smooth manifold of dimension $k(n-k)$, with charts defined as follows. For a decomposition $\mathbb{R}^n = P \oplus Q$ with $\dim P = k$, let $U_Q = \{W : W \cap Q = 0\}$; every $W \in U_Q$ is the graph $\{x + Ax : x \in P\}$ of a unique linear map $A \colon P \to Q$, and $\varphi_{P,Q}(W) = A \in \mathrm{Hom}(P, Q) \cong \mathbb{R}^{k(n-k)}$. Moreover $\mathrm{Gr}_1(\mathbb{R}^{n+1}) = \mathbb{RP}^n$.*

Proof. If $W \cap Q = 0$, then $W \oplus Q = \mathbb{R}^n$ by dimensions, so the projection $W \to P$ along Q is an isomorphism and W is the graph of $A = \mathrm{pr}_Q \circ (\mathrm{pr}_P|_W)^{-1}$; conversely every graph lies in U_Q. So $\varphi_{P,Q} \colon U_Q \to \mathrm{Hom}(P, Q)$ is a bijection. We verify the conditions of Lemma 33.2.7. For another decomposition $P' \oplus Q'$, the graph W of A lies in $U_{Q'}$ if and only if the linear map $\mathrm{pr}_{P'} \circ (\mathrm{id} + A) \colon P \to P'$ is invertible, an open condition on A (nonvanishing of a determinant in bases); and then W is the graph of $A' = \mathrm{pr}_{Q'} \circ (\mathrm{id} + A) \circ (\mathrm{pr}_{P'} \circ (\mathrm{id} + A))^{-1}$, whose entries are rational functions of those of A with nonvanishing denominators, by Cramer's rule. This gives (1) and (2). The $\binom{n}{k}$ charts with P and Q spanned by standard basis vectors cover, since every W has a complement spanned by standard basis vectors (extend a basis of W by standard basis vectors); this gives (3). For (4), two subspaces W, W' have a common complement Q (Lemma 33.7.3), so both lie in U_Q. For $k = 1$, a line is determined by a nonzero vector up to scaling, and the charts $\varphi_{\langle e_i \rangle, Q_i}$ with Q_i spanned by the other basis vectors are the charts of $\mathbb{RP}^n$. □

Remark 33.7.5. $\mathrm{Gr}_k(\mathbb{R}^n)$ is compact: the map from the compact group $\mathrm{O}(n)$ sending a matrix to the span of its first k columns is continuous (in the charts above it is given by rational functions) and surjective. It is the orbit space $\mathrm{O}(n)/(\mathrm{O}(k) \times \mathrm{O}(n-k))$ (Proposition 20.3.2(4)). The complex Grassmannian $\mathrm{Gr}_k(\mathbb{C}^n)$ is constructed in the same way, with holomorphic transition maps (Chapter 44).

33.8 Partitions of unity

Many objects on a manifold are easy to write down in a single chart: a function, a metric, a vector field. A *partition of unity* is a family of smooth functions $\psi_i \geq 0$ with $\sum_i \psi_i = 1$, each supported in one chart; given local objects s_i, one on each chart, the sum $\sum_i \psi_i s_i$ is then a global object, provided the objects can be added and multiplied by functions. The point of this section is that smooth partitions of unity always exist.

Example 33.8.1. On $\mathbb{R}$, take the open cover $U_1 = (-\infty, 1)$, $U_2 = (0, \infty)$, and let γ be the function of Lemma 27.7.8, smooth, equal to 0 for $t \leq 0$ and to 1 for $t \geq 1$. Put $\psi_2(t) = \gamma(3t - 1)$ and $\psi_1 = 1 - \psi_2$. Then $\psi_2 = 0$ for $t \leq 1/3$ and $\psi_2 = 1$ for $t \geq 2/3$, so $\operatorname{supp} \psi_2 \subseteq [1/3, \infty) \subseteq U_2$ and $\operatorname{supp} \psi_1 \subseteq (-\infty, 2/3] \subseteq U_1$. So (ψ_1, ψ_2) is a smooth partition of unity subordinate to the cover. With analytic functions this is impossible: an analytic function on $\mathbb{R}$ that vanishes on an interval vanishes everywhere, so ψ_2 cannot be analytic.

Theorem 33.8.2. *Let M be a smooth manifold and $(U_i)_{i \in I}$ an open cover. There is a smooth partition of unity subordinate to it: smooth functions $\psi_i \colon M \to [0, 1]$ with $\operatorname{supp} \psi_i \subseteq U_i$, locally finite supports, and $\sum_i \psi_i = 1$.*

Proof. Step 1: a locally finite cover by coordinate balls. The manifold M is locally compact, Hausdorff and second countable, so it has an exhaustion by compact sets $K_1 \subseteq K_2 \subseteq \cdots$ with $K_j \subseteq \operatorname{int} K_{j+1}$ (Proposition 18.10.7); put $K_0 = K_{-1} = \emptyset$. For $j \geq 0$ the compact set $A_j = K_{j+1} \setminus \operatorname{int} K_j$ lies in the open set $W_j = \operatorname{int} K_{j+2} \setminus K_{j-1}$. For each $q \in A_j$ choose a chart (U, φ) centred at q with $U \subseteq W_j \cap U_i$ for some i (Lemma 33.2.6(3)), and $r > 0$ with $\overline{B}(0, 3r) \subseteq \varphi(U)$. Finitely many of the sets $\varphi^{-1}(B(0, r))$ cover A_j. Collecting these charts for all j gives a countable family $(U_k, \varphi_k, r_k)_k$ and indices $i(k)$ with $U_k \subseteq U_{i(k)}$, such that the sets $B'_k = \varphi_k^{-1}(B(0, r_k))$ cover $M = \bigcup_j A_j$. The family (U_k) is locally finite: a point lies in $\operatorname{int} K_{j+1}$ for some j, and this open set meets W_l only if $K_{l-1} \not\supseteq K_{j+1}$, that is, only for $l \leq j + 1$, and each l contributes finitely many charts.

Step 2: bump functions. As in the proof of Lemma 33.3.10, there are smooth $\chi_k \colon M \to [0, 1]$ equal to 1 on B'_k with support in $\varphi_k^{-1}(\overline{B}(0, 2r_k)) \subseteq U_k$.

Step 3: normalise. The supports of the χ_k form a locally finite family, so $\chi = \sum_k \chi_k$ is smooth: near each point it is a finite sum. It satisfies $\chi \geq 1$, because the B'_k cover M. So the functions $g_k = \chi_k / \chi$ are smooth, take values in $[0, 1]$, have $\operatorname{supp} g_k = \operatorname{supp} \chi_k \subseteq U_{i(k)}$, and sum to 1.

Step 4: regroup. For $i \in I$ put $\psi_i = \sum_{k : i(k) = i} g_k$, a locally finite sum of smooth functions, hence smooth. By Lemma 18.10.2, $\operatorname{supp} \psi_i$ is the union of the $\operatorname{supp} g_k$ with $i(k) = i$, so it lies in U_i, and the supports of the ψ_i form a locally finite family. Finally $\sum_i \psi_i = \sum_k g_k = 1$. □

Corollary 33.8.3.

1. *For a closed $A \subseteq M$ and an open $U \supseteq A$ there is a smooth $\chi \colon M \to [0, 1]$ equal to 1 on A with support in U.*
2. *Every smooth function on a closed subset (that is, a function that is locally the restriction of a smooth function) extends to a smooth function on M.*
3. *M admits a proper smooth function $M \to \mathbb{R}$.*

Proof. (1) Take the partition of unity (ψ_U, ψ_0) subordinate to the cover $\{U, M \setminus A\}$ and put $\chi = \psi_U$. Its support lies in U, and on A we have $\psi_0 = 0$, because $\operatorname{supp} \psi_0 \subseteq M \setminus A$, so $\chi = 1$ there.

(2) Let A be closed and $f \colon A \to \mathbb{R}$ such that every $a \in A$ has an open neighbourhood W_a and a smooth $f_a \colon W_a \to \mathbb{R}$ with $f_a = f$ on $W_a \cap A$. Take a partition of unity $(\psi_a)_{a \in A} \cup (\psi_0)$ subordinate to the cover $(W_a)_{a \in A} \cup (M \setminus A)$, and put $F = \sum_a \psi_a f_a$, where each term is extended

by 0 outside W_a (it is smooth, since it vanishes near every point outside $\operatorname{supp}\psi_a$). The sum is locally finite, so F is smooth. For $x \in A$, $\psi_0(x) = 0$, so $F(x) = \sum_a \psi_a(x) f(x) = f(x)$.

(3) Every point has a neighbourhood with compact closure (a small coordinate ball), and countably many of them, $V_1, V_2, \ldots$, cover M (Proposition 18.3.12). Let (ψ_k) be a partition of unity subordinate to (V_k) and $f = \sum_k k\psi_k$, which is smooth. If $x \notin V_1 \cup \cdots \cup V_m$, then $\psi_k(x) = 0$ for $k \le m$, so $f(x) = \sum_{k>m} k\psi_k(x) \ge (m+1)\sum_k \psi_k(x) > m$. So $f^{-1}([-m, m])$ is a closed subset of the compact set $\overline{V_1} \cup \cdots \cup \overline{V_m}$, hence compact, and f is proper. □

Remark 33.8.4. The same argument produces global objects out of local ones whenever the objects in question can be averaged: Riemannian metrics (Chapter 40), connections (Chapter 38) and Hermitian metrics on bundles all exist on any smooth manifold for this reason, while there is no such argument for complex structures or for nowhere vanishing vector fields, which are subject to topological obstructions. Nor is there one for holomorphic objects: holomorphic functions, like analytic ones, cannot vanish on an open set without vanishing on the whole connected component (Example 33.8.1), which is why complex geometry needs sheaf cohomology where smooth geometry uses partitions of unity.

33.9 Vector fields and flows

A vector field X can be followed: an *integral curve* of X is a smooth curve γ with $\gamma'(t) = X_{\gamma(t)}$ for all t. In a chart this is a system of ordinary differential equations, so integral curves exist and are unique; the *flow* of X moves every point along its integral curve.

Example 33.9.1. We write $\Phi(t, p)$ for the point reached at time t by the integral curve starting at p.

1. On $\mathbb{R}^n$, the constant field $X = \sum_i a^i \partial/\partial x^i$ has flow $\Phi(t, x) = x + ta$, a translation.
2. The Euler field $E = \sum_i x^i \partial/\partial x^i$ has integral curves with $\gamma' = \gamma$, so $\Phi(t, x) = e^t x$, a dilation.
3. The rotation field $R = -y\,\partial/\partial x + x\,\partial/\partial y$ on $\mathbb{R}^2$ has flow $\Phi(t, (x, y)) = (x\cos t - y\sin t, x\sin t + y\cos t)$, the rotation by the angle t: differentiating in t gives $(-x\sin t - y\cos t, x\cos t - y\sin t)$, which is R at the point $\Phi(t, (x, y))$.
4. On $\mathbb{R}$, the field $X = x^2\, d/dx$ has integral curves with $\gamma' = \gamma^2$ and $\gamma(0) = p$, namely $\gamma(t) = p/(1 - tp)$. For $p > 0$ it is defined only for $t < 1/p$ and goes to infinity in finite time; for $p < 0$ only for $t > 1/p$. So the flow is defined on $\mathcal{D} = \{(t, p) : tp < 1\}$, not on all of $\mathbb{R} \times \mathbb{R}$.
5. On the open interval $(0, 1)$, the field d/dx has flow $x + t$, defined only while $0 < x + t < 1$: removing points from a manifold can also stop the flow. On a compact manifold neither phenomenon can occur (Theorem 33.9.2).

Theorem 33.9.2. *Let $X \in \mathfrak{X}(M)$. For every $p \in M$ there is a unique maximal* integral curve *$\gamma_p : (\alpha_p, \beta_p) \to M$ with $\gamma_p(0) = p$ and $\gamma_p'(t) = X_{\gamma_p(t)}$. The set $\mathcal{D} = \{(t, p) : \alpha_p < t < \beta_p\}$ is open in $\mathbb{R} \times M$, the* flow *$\Phi(t, p) = \gamma_p(t)$ is smooth, and $\Phi(t + s, p) = \Phi(t, \Phi(s, p))$ whenever both sides are defined. If M is compact, or X has compact support, then $\mathcal{D} = \mathbb{R} \times M$ and $t \mapsto \Phi(t, -)$ is a homomorphism from $\mathbb{R}$ to the diffeomorphism group.*

Proof. Local existence and smoothness. Let (U, φ) be a chart, write $X = \sum_i X^i \partial/\partial x^i$ on U, and let $F = (X^1, \ldots, X^n) \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}^n$, which is smooth. By the chart formula for velocities (Proposition 33.4.9), a curve γ in U is an integral curve of X if and only if $c = \varphi \circ \gamma$ solves

$c' = F(c)$. By Theorems 27.13.3 and 27.13.6, for every $q \in U$ there are $\varepsilon > 0$ and an open $V \ni q$ in U such that for $p \in V$ the solution with $c(0) = \varphi(p)$ exists on $(-\varepsilon, \varepsilon)$, and $(t, p) \mapsto \varphi^{-1}(c(t))$ is smooth on $(-\varepsilon, \varepsilon) \times V$.

Uniqueness. Let $\gamma_1, \gamma_2 \colon J \to M$ be integral curves on an open interval J with $\gamma_1(t_0) = \gamma_2(t_0)$. The set $\{t \in J : \gamma_1(t) = \gamma_2(t)\}$ is nonempty; it is closed in J, because M is Hausdorff (if $\gamma_1(t) \neq \gamma_2(t)$, disjoint neighbourhoods and continuity show that they differ near t); and it is open, because near a point where they agree both are solutions of the same equation in a chart with the same value, so they agree nearby by Theorem 27.13.3(2). As J is connected, $\gamma_1 = \gamma_2$.

Maximal curves and the group law. The integral curves with value p at 0 agree on their common domains, so together they define one integral curve γ_p on the union (α_p, β_p) of their domains, which is maximal by construction. For $s \in (\alpha_p, \beta_p)$ and $q = \gamma_p(s)$, the curve $t \mapsto \gamma_p(t + s)$ on $(\alpha_p - s, \beta_p - s)$ is an integral curve with value q at 0, so it is a restriction of γ_q; symmetrically, $t \mapsto \gamma_q(t - s)$ is an integral curve with value p at 0, so it is a restriction of γ_p. Hence $(\alpha_q, \beta_q) = (\alpha_p - s, \beta_p - s)$ and $\gamma_q(t) = \gamma_p(t + s)$, which is the group law.

$\mathcal{D}$ is open and Φ is smooth. Let $(t_0, p_0) \in \mathcal{D}$, say with $t_0 \geq 0$ (for $t_0 < 0$ replace X by $-X$, whose flow is $\Phi(-t, p)$). Let T be the set of $t \in [0, t_0]$ for which there are $\delta > 0$ and an open $U \ni p_0$ with $(-\delta, t + \delta) \times U \subseteq \mathcal{D}$ and Φ smooth there. By local existence $0 \in T$, and T is an interval; let $\tau = \sup T \leq t_0$ and $q = \gamma_{p_0}(\tau)$. Take ε and V for q as in the first step. By continuity of γ_{p_0} there is $t_1 \in T$ with $\tau - \varepsilon/2 < t_1 \leq \tau$ and $\gamma_{p_0}(t_1) \in V$; let δ and U witness $t_1 \in T$, and shrink U so that $\Phi(t_1, U) \subseteq V$. For $p \in U$, the group law shows that $(t, p) \in \mathcal{D}$ and $\Phi(t, p) = \Phi(t - t_1, \Phi(t_1, p))$ for $|t - t_1| < \varepsilon$, and the right-hand side is smooth in (t, p). So Φ is smooth on $(-\delta, t_1 + \varepsilon) \times U \subseteq \mathcal{D}$, the union of two open sets on each of which it is smooth. Since $t_1 + \varepsilon > \tau + \varepsilon/2$, every $t \leq \min(t_0, \tau + \varepsilon/4)$ lies in T; as $\tau = \sup T$, this forces $\tau = t_0$, and then $t_0 \in T$. So $\mathcal{D}$ contains a neighbourhood of (t_0, p_0) on which Φ is smooth.

Completeness. Suppose X has compact support K; this holds if M is compact. Outside K, $X = 0$ and the constant curves are the integral curves, defined for all time. By local existence and compactness, finitely many open sets V with numbers ε_V cover K; let ε be the smallest of these numbers. Then $(-\varepsilon, \varepsilon) \times M \subseteq \mathcal{D}$. If $\beta_p < \infty$ for some p, choose $s \in (\beta_p - \varepsilon, \beta_p)$; then γ_q with $q = \gamma_p(s)$ is defined on $(-\varepsilon, \varepsilon)$, and by the group law γ_p extends to $(\alpha_p, s + \varepsilon)$, which contains β_p, contradicting maximality. So $\beta_p = \infty$, and similarly $\alpha_p = -\infty$. Then the maps $\Phi_t = \Phi(t, -)$ satisfy $\Phi_t \circ \Phi_s = \Phi_{t+s}$ and $\Phi_0 = \mathrm{id}$, so each Φ_t is a diffeomorphism with inverse Φ_{-t}. □

Remark 33.9.3. The Hausdorff condition was used for uniqueness, and it is needed: on the line with two origins (Counterexample 33.2.8) the field d/dx, defined in both charts, has two different integral curves starting at -1, one through each origin.

Proposition 33.9.4 (Naturality of integral curves and flows)**.** *Let $F \colon M \to N$ be smooth and let $X \in \mathfrak{X}(M)$, $Y \in \mathfrak{X}(N)$ satisfy $d_pF(X_p) = Y_{F(p)}$ for all p (they are F-*related*). Then F maps integral curves of X to integral curves of Y: if γ is an integral curve of X, then $F \circ \gamma$ is one of Y. Consequently $F(\Phi_t(p)) = \Psi_t(F(p))$ whenever $\Phi_t(p)$ is defined, where Φ and Ψ are the flows of X and Y; in particular, if F is a diffeomorphism and $Y = F_*X$, then $\Psi_t = F \circ \Phi_t \circ F^{-1}$.*

Proof. By the chain rule (Proposition 33.4.9), $(F \circ \gamma)'(t) = d_{\gamma(t)}F(\gamma'(t)) = d_{\gamma(t)}F(X_{\gamma(t)}) = Y_{F(\gamma(t))}$, so $F \circ \gamma$ is an integral curve of Y. Applying this to γ_p gives an integral curve of Y through $F(p)$ at time 0, which is therefore the restriction of $\gamma_{F(p)}$ (Theorem 33.9.2); evaluating at t gives $F(\Phi_t(p)) = \Psi_t(F(p))$. If F is a diffeomorphism, X and F_*X are F-related by definition, and the formula reads $\Psi_t = F \circ \Phi_t \circ F^{-1}$. □

For two vector fields X and Y, the composite $f \mapsto X(Yf)$ is not a vector field: in coordinates it involves second derivatives of f. But the second derivatives cancel in the

commutator.

Definition 33.9.5. For $X, Y \in \mathfrak{X}(M)$, the *Lie bracket* is the vector field with $[X,Y]f = X(Yf) - Y(Xf)$.

Example 33.9.6. On $\mathbb{R}^2$, $[\partial/\partial x, x\,\partial/\partial y]f = \partial_x(x\partial_y f) - x\partial_y\partial_x f = \partial_y f$, so $[\partial/\partial x, x\,\partial/\partial y] = \partial/\partial y$. For the Euler field E and the rotation field R of Example 33.5.6, the coordinate formula below gives $[E,R] = 0$: the i-th component $\sum_j (E^j\partial_j R^i - R^j\partial_j E^i)$ is $(-y,x) - (-y,x) = 0$. Correspondingly, dilations commute with rotations (Example 33.9.1), while the flows of $\partial/\partial x$ and $x\,\partial/\partial y$ do not commute (Exercise 33.10.4).

Brackets of coordinate fields vanish: in a chart, $[\partial_i, \partial_j]f = \partial_i\partial_j f - \partial_j\partial_i f = 0$ by the symmetry of second derivatives (Theorem 27.10.5). So n vector fields can be the coordinate fields of a chart only if they commute, and conversely commuting fields that are linearly independent at a point are the coordinate fields of a chart near it, by the argument in the proof of Theorem 33.9.8 below. The fields ∂_x and $x\,\partial_y$ of the example are independent everywhere but are not coordinate fields of any chart, since their bracket ∂_y is not zero. Note also that the coordinate field $\partial/\partial x^1$ depends on the whole chart, not only on the function x^1: in the coordinates $(u,v) = (x, y-x)$ of $\mathbb{R}^2$, $\partial/\partial u = \partial_x + \partial_y \neq \partial_x$, although $u = x$.

Proposition 33.9.7. *$[X,Y]$ is a vector field; the bracket is bilinear, antisymmetric, satisfies the Jacobi identity and $[fX, gY] = fg[X,Y] + f(Xg)Y - g(Yf)X$, so $\mathfrak{X}(M)$ is a Lie algebra. In coordinates, $[X,Y] = \sum_{i,j}(X^j\partial_j Y^i - Y^j\partial_j X^i)\partial_i$. Moreover $[X,Y] = \frac{d}{dt}\Big|_{t=0}(\Phi_{-t})_*Y$, the* Lie derivative L_XY, *where Φ is the flow of X and $((\Phi_{-t})_*Y)_p = d\Phi_{-t}(Y_{\Phi_t(p)})$; and $[X,Y] = 0$ if and only if the flows of X and Y commute.*

Proof. $[X,Y]$ is a vector field. The operator $f \mapsto X(Yf) - Y(Xf)$ is linear on $C^\infty(M)$, and $X(Y(fg)) = X((Yf)g + f(Yg)) = (XYf)g + (Yf)(Xg) + (Xf)(Yg) + f(XYg)$. Subtracting the same expression with X and Y exchanged, the mixed terms cancel and $[X,Y](fg) = ([X,Y]f)g + f([X,Y]g)$. So $[X,Y]$ is a derivation of $C^\infty(M)$, hence a vector field by Proposition 33.5.7.

Identities and coordinates. Bilinearity and antisymmetry are clear. The Jacobi identity $[X,[Y,Z]]+[Y,[Z,X]]+[Z,[X,Y]] = 0$ holds because, applied to f, the left side expands into twelve terms of the form $ABCf$ which cancel in pairs. The formula for $[fX,gY]$ follows from $fX(gYh) = fg\,XYh + f(Xg)Yh$ and the same with the roles exchanged. The component $[X,Y]^i = [X,Y]x^i = X(Y^i) - Y(X^i)$ is $\sum_j(X^j\partial_j Y^i - Y^j\partial_j X^i)$ by Lemma 33.5.5.

Lie derivative. Work in a chart around p and write $\Phi(t,x)$ for the flow in coordinates, smooth near $(0,p)$, with $\Phi(0,x) = x$ and $\partial_t\Phi(t,x) = X(\Phi(t,x))$. Let $A(t) = D_x\Phi(t,p)$ be the derivative in x. Then $A(0) = I$, and $A'(0) = D_x\partial_t\Phi(0,p) = DX(p)$ because mixed partial derivatives commute (Theorem 27.10.5). Differentiating $\Phi(-t,\Phi(t,x)) = x$ in x shows that the matrix of $d\Phi_{-t}$ at $\Phi_t(p)$ is $A(t)^{-1}$. So in coordinates $((\Phi_{-t})_*Y)_p = A(t)^{-1}Y(\Phi(t,p))$, whose derivative at $t=0$ is $-A'(0)Y(p) + DY(p)X(p) = DY(p)X(p) - DX(p)Y(p)$; its i-th component is $\sum_j(X^j\partial_j Y^i - Y^j\partial_j X^i)(p)$, which is $[X,Y]^i(p)$.

Commuting flows. Let Ψ be the flow of Y. If $\Phi_t \circ \Psi_s = \Psi_s \circ \Phi_t$ wherever defined, differentiating in s at $s=0$ gives $d\Phi_t(Y_q) = Y_{\Phi_t(q)}$, so $(\Phi_{-t})_*Y = Y$ for all t and $[X,Y] = L_XY = 0$. Conversely, suppose $[X,Y] = 0$, fix p, and let $V(t) = ((\Phi_{-t})_*Y)_p \in T_pM$. By the group law, $V(t_0+h) = d\Phi_{-t_0}\big(((\Phi_{-h})_*Y)_{\Phi_{t_0}(p)}\big)$, so $V'(t_0) = d\Phi_{-t_0}\big((L_XY)_{\Phi_{t_0}(p)}\big) = 0$. So V is constant: $d\Phi_t(Y_p) = Y_{\Phi_t(p)}$ for all p and t. Then, for fixed t, the curve $s \mapsto \Phi_t(\Psi_s(p))$ has velocity $d\Phi_t(Y_{\Psi_s(p)}) = Y_{\Phi_t(\Psi_s(p))}$, so it is an integral curve of Y starting at $\Phi_t(p)$, and by uniqueness it equals $s \mapsto \Psi_s(\Phi_t(p))$. □

Theorem 33.9.8 (Frobenius)**.** *Let $D \subseteq TM$ be a smooth distribution of rank k (a subbundle, Chapter 35) that is* involutive*: $[X, Y]$ is a section of D whenever X and Y are. Then every point has a chart in which D is spanned by $\partial/\partial x^1, \dots, \partial/\partial x^k$; the manifold is locally a union of k-dimensional* integral manifolds*.*

Sketch of proof. Choose local sections $X_1, \dots, X_k$ spanning D near p; after a linear change we may assume $X_i = \partial/\partial x^i + \sum_{j>k} a_i^j \partial/\partial x^j$, and involutivity forces $[X_i, X_j] = 0$ (the bracket lies in D and has no $\partial/\partial x^i$ components for $i \le k$). Commuting vector fields have commuting flows (Proposition 33.9.7), and $(t_1, \dots, t_k, y) \mapsto \Phi^1_{t_1} \circ \cdots \circ \Phi^k_{t_k}(y)$ is a diffeomorphism onto a neighbourhood of p taking the coordinate fields to the X_i. The omitted verification is that this map has invertible differential and defines a chart; see [Lee13, Chapter 19] or [War83, Chapter 1]. □

Theorem 33.9.9 (Whitney embedding, compact case)**.** *Every compact smooth n-manifold embeds in $\mathbb{R}^N$ for some N. (In fact $N = 2n + 1$ suffices, and $2n$ for $n > 0$; see [Hir76, Chapter 1].)*

Proof. Cover M by finitely many charts $\varphi_i \colon U_i \to \mathbb{R}^n$, $i = 1, \dots, m$, with smaller open sets V_i still covering and bump functions χ_i equal to 1 on $\overline{V_i}$ with support in U_i (Corollary 33.8.3). The map $F = (\chi_1\varphi_1, \dots, \chi_m\varphi_m, \chi_1, \dots, \chi_m) \colon M \to \mathbb{R}^{mn+m}$ is smooth; it is an immersion, since on V_i the component $\chi_i\varphi_i = \varphi_i$ has injective differential, and it is injective: if $F(p) = F(q)$, choose i with $p \in V_i$; then $\chi_i(q) = \chi_i(p) = 1$, so $q \in U_i$ and $\varphi_i(q) = \varphi_i(p)$, whence $p = q$. By Proposition 33.6.7(3) it is an embedding. □

Theorem 33.9.10 (Whitney)**.** *Every smooth n-manifold admits a proper embedding into $\mathbb{R}^{2n+1}$ and an immersion into $\mathbb{R}^{2n}$; for $n > 0$ it even embeds into $\mathbb{R}^{2n}$.*

Sketch of proof. First one embeds M properly into some $\mathbb{R}^N$: take a countable locally finite atlas with charts φ_i and a subordinate partition of unity ψ_i (Theorem 33.8.2), and use the maps $\psi_i\varphi_i$ and ψ_i as in the proof of Theorem 33.9.9, together with a proper function $M \to \mathbb{R}$ (Corollary 33.8.3(3)) as one more coordinate; properness of that coordinate makes the embedding proper. Then one lowers the dimension: if $N > 2n + 1$, the set of directions $v \in S^{N-1}$ for which the orthogonal projection along v fails to be an injective immersion is the image of two smooth maps from manifolds of dimensions $2n$ and $2n - 1$ (pairs of distinct points, and nonzero tangent vectors), hence has measure zero by Sard's theorem (Theorem 33.6.21); so a good direction exists, and the projection of a proper embedding along it is again one. The omitted verifications are that the projected map is again proper and an embedding, and the harder statement that $\mathbb{R}^{2n}$ suffices; see [Hir76, Chapter 1] and [Lee13, Chapter 6]. □

33.10 Exercises

Exercise 33.10.1. Show that $T_x S^n = x^\perp \subseteq \mathbb{R}^{n+1}$ under the standard identification, and that TS^1 is diffeomorphic to $S^1 \times \mathbb{R}$.

Solution. The first statement is Theorem 33.6.11 applied to $\|x\|^2$. For S^1, the vector field $X_{(a,b)} = (-b, a)$ is nowhere zero (Example 33.6.14), and $(p, t) \mapsto tX_p$ is a smooth bijection $S^1 \times \mathbb{R} \to TS^1$. In an angle chart θ and its bundle chart it is $(\theta, t) \mapsto (\theta, t)$, since $X = \partial/\partial\theta$; so it is a diffeomorphism. □

Exercise 33.10.2. Let M be a compact manifold with a nowhere vanishing vector field X. Show that the flow of X is defined for all time and gives a smooth action of $\mathbb{R}$ on M, and that M has Euler characteristic 0 (using the Poincaré–Hopf theorem, Chapter 39).

Exercise 33.10.3. Give an example of an injective immersion $\mathbb{R} \to \mathbb{R}^2$ that is not an embedding, and of an immersion of S^1 in $\mathbb{R}^2$ that is not injective.

Exercise 33.10.4. Compute $[X, Y]$ for $X = \partial_x$, $Y = x\partial_y$ on $\mathbb{R}^2$, and verify that the corresponding flows do not commute.

Solution. $[X, Y] = \partial_y$ (Example 33.9.6). The flow of X is $\Phi_t(x, y) = (x + t, y)$ and that of Y is $\Psi_s(x, y) = (x, y + sx)$. Then $\Phi_t(\Psi_s(0, 0)) = (t, 0)$ while $\Psi_s(\Phi_t(0, 0)) = (t, st)$: the two orders differ by a translation in y. □

Exercise 33.10.5. Show that $W \mapsto W^\perp$ is a diffeomorphism $\mathrm{Gr}_k(\mathbb{R}^n) \to \mathrm{Gr}_{n-k}(\mathbb{R}^n)$.

Solution. It is a bijection, its own inverse up to exchanging k and $n - k$. Use charts with P and Q spanned by complementary sets of standard basis vectors, so $Q = P^\perp$. If W is the graph of $A\colon P \to Q$, then $y + By$ with $y \in Q$ and $B\colon Q \to P$ is orthogonal to all $x + Ax$ if and only if $\langle x, By\rangle + \langle Ax, y\rangle = 0$ for all x, that is, $B = -A^T$. So $W^\perp$ is the graph of $-A^T$, and in the charts $\varphi_{P,Q}$ and $\varphi_{Q,P}$ the map is the linear map $A \mapsto -A^T$. □

Exercise 33.10.6. Show that the smooth structure on $\mathbb{R}/\mathbb{Z}$ given by Theorem 33.7.1 makes $t \mapsto e^{2\pi i t}$ a diffeomorphism $\mathbb{R}/\mathbb{Z} \to S^1$, where $S^1 \subseteq \mathbb{R}^2$ has the structure of a regular level set.

Solution. The induced map $\mathbb{R}/\mathbb{Z} \to S^1$ is a bijection, and it is smooth because its composite with π is $t \mapsto (\cos 2\pi t, \sin 2\pi t)$, which is smooth into $\mathbb{R}^2$ and hence into S^1 (Proposition 33.6.5). Its differential is nonzero, since the velocity $2\pi(-\sin 2\pi t, \cos 2\pi t)$ is nonzero and π is a local diffeomorphism; as both manifolds have dimension 1, the differential is an isomorphism. So the map is a diffeomorphism by Theorem 33.4.14. □

Exercise 33.10.7. For odd $k \geq 1$, let $\mathcal{A}_k$ be the smooth structure on $\mathbb{R}$ generated by the chart $t \mapsto t^k$. Show that $\mathcal{A}_k \neq \mathcal{A}_l$ for $k \neq l$, but that all these structures are diffeomorphic.

Solution. For odd k, $t \mapsto t^k$ is a strictly increasing bijection with continuous inverse, so it is a chart. Suppose $\mathcal{A}_k = \mathcal{A}_l$ with $k < l$. Then the transition maps $s \mapsto s^{l/k}$ and $s \mapsto s^{k/l}$ are smooth and inverse to each other, so by the chain rule the product of their derivatives at 0 is 1. But the derivative of $s^{l/k}$ at 0 is $\lim_{s\to 0} s^{l/k}/s = 0$, since $l/k > 1$, a contradiction. A diffeomorphism $(\mathbb{R}, \mathcal{A}_k) \to (\mathbb{R}, \mathcal{A}_l)$ is $F(t) = t^{k/l}$, taking odd roots: its local representative is $s \mapsto ((s^{1/k})^{k/l})^l = s$, as in Example 33.3.5. □

Exercise 33.10.8. Show that $F([x_0 : x_1]) = (x_0^2 - x_1^2, 2x_0x_1)/(x_0^2 + x_1^2)$ is a well defined diffeomorphism $\mathbb{RP}^1 \to S^1$.

Solution. The formula is unchanged when (x_0, x_1) is multiplied by $\lambda \neq 0$, so F is well defined; in complex notation $z = x_0 + ix_1$ it is $z^2/|z|^2$. It is surjective, since every complex number of modulus 1 is a square of one, and injective, since $(z/|z|)^2 = (w/|w|)^2$ forces $z/|z| = \pm w/|w|$, that is $[z] = [w]$. In the chart $u = x_1/x_0$, $F([1 : u]) = (1 - u^2, 2u)/(1 + u^2)$, a smooth map $\mathbb{R} \to S^1 \setminus \{(-1, 0)\}$ (Proposition 33.6.5) with smooth inverse $(a, b) \mapsto b/(1 + a)$: indeed $1 + a = 2/(1+u^2)$ and $b/(1+a) = u$. Similarly, in the chart $w = x_0/x_1$, $F([w : 1]) = (w^2-1, 2w)/(w^2+1)$ with inverse $(a, b) \mapsto b/(1 - a)$ on $S^1 \setminus \{(1, 0)\}$. So F is smooth, and its inverse is smooth on each of two open sets covering S^1. □

Exercise 33.10.9. Show that the torus of revolution $T = \{(x, y, z) : ((x^2 + y^2)^{1/2} - 2)^2 + z^2 = 1\}$ is a 2-dimensional submanifold of $\mathbb{R}^3$.

Solution. On the open set $\Omega = \{(x, y) \neq (0, 0)\} \times \mathbb{R}$, the function $f = (r - 2)^2 + z^2$, with $r = (x^2 + y^2)^{1/2}$, is smooth, and $T = f^{-1}(1)$ lies in Ω, because $f = 4 + z^2 > 1$ where $r = 0$. Its differential is $(2(r - 2)x/r, 2(r - 2)y/r, 2z)$, which vanishes only where $z = 0$ and $r = 2$; there $f = 0 \neq 1$. So 1 is a regular value of $f|_\Omega$, and T is a submanifold of dimension 2 by Theorem 33.6.11. □

Exercise 33.10.10. Show that a map $f : M \to N$ is smooth if and only if it is continuous and $g \circ f \in C^\infty(M)$ for every $g \in C^\infty(N)$: global functions suffice in Proposition 33.3.8.

Solution. One direction is clear. Conversely, let $p \in M$, let $(V, (y^1, \dots, y^n))$ be a chart at $f(p)$, and choose χ as in Lemma 33.3.10 with support in V and $\chi = 1$ on an open $V' \ni f(p)$. The functions g_j equal to χy^j on V and to 0 elsewhere lie in $C^\infty(N)$, so $g_j \circ f$ is smooth; on the open set $f^{-1}(V')$ it equals $y^j \circ f$. So the local representative of f near p is smooth. □

Exercise 33.10.11. Let V be a finite-dimensional real vector space with its canonical smooth structure (Example 33.2.9(1)) and $v \in V$. Show that $w \mapsto (f \mapsto \frac{d}{dt} f(v + tw)|_{t=0})$ is an isomorphism $V \to T_v V$, and that under these isomorphisms the differential of a linear map $L : V \to V'$ is L itself.

Solution. Choose a linear isomorphism $V \cong \mathbb{R}^n$, which is a chart. In it the map becomes $w \mapsto D_w|_v$, an isomorphism by Proposition 33.4.5; the definition does not refer to the chart. For linear L, $d_v L(D_w|_v)(g) = \frac{d}{dt} g(Lv + tLw)|_{t=0} = D_{Lw}|_{Lv}(g)$. □

Chapter 34

Lie Groups

34.1 Introduction

A Lie group is a group that is also a smooth manifold, with smooth group operations: the rotations of space, the invertible matrices, the circle, the tori. Such a group is curved, but it is homogeneous (every point looks like every other, via translations), so everything about it near one point is known near all points. Near the identity it is described by a vector space, the tangent space $\mathfrak{g}$ at the identity, and the group law leaves a trace on $\mathfrak{g}$: a bilinear bracket making $\mathfrak{g}$ a Lie algebra. The exponential map $\exp\colon \mathfrak{g} \to G$ carries this linear object back into the group. For groups of matrices, $\mathfrak{g}$ is a space of matrices, the bracket is the commutator $XY - YX$ and exp is the exponential series $\sum_k X^k/k!$, and the reader may keep these examples in mind throughout.

Many spaces of Hodge theory are homogeneous spaces of Lie groups: projective spaces, Grassmannians, flag manifolds, and the period domains that classify Hodge structures (Chapter 70). The chapter goes as follows.

1. Lie groups, the classical matrix groups with their dimensions, and the Lie algebra of a Lie group (Section 34.2).

2. The matrix exponential and the exponential map of a general Lie group; the adjoint representation; connected abelian Lie groups (Section 34.3).

3. Cartan's closed subgroup theorem (Theorem 34.4.1) and homogeneous spaces (Theorem 34.4.4), with the groups SU(2) and SO(3) worked out (Section 34.4).

4. Compact Lie groups: Haar measure and averaging; their Lie algebras are reductive, with negative definite Killing form on the semisimple part (Section 34.5).

References: [Kna02], [BtD85], [War83].
Prerequisites. Chapter 33 (Smooth Manifolds) and Chapter 13 (Representations of Groups and Lie Algebras).

Notation 34.1.1. The identity element of a group is e. For $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, $M_n(\mathbb{K})$ is the space of $n \times n$ matrices, I the identity matrix, A^T the transpose and $A^* = \overline{A}^T$ the conjugate transpose; tr is the trace. As a real vector space $M_n(\mathbb{C}) \cong \mathbb{R}^{2n^2}$, and $M_n(\mathbb{K})$ has the operator norm $\|A\| = \sup_{\|v\|\le 1} \|Av\|$, which satisfies $\|AB\| \le \|A\|\|B\|$. The commutator of two matrices is $[X, Y] = XY - YX$. Smooth maps and tangent vectors are as in Chapter 33; in particular $T_A M_n(\mathbb{K}) = M_n(\mathbb{K})$ for the vector space $M_n(\mathbb{K})$ (Exercise 33.10.11).

34.2 Lie groups and their Lie algebras

Definition 34.2.1. A *Lie group* is a group G with a smooth structure for which multiplication $m\colon G \times G \to G$, $m(g,h) = gh$, and inversion $i\colon G \to G$, $i(g) = g^{-1}$, are smooth. A *homomorphism of Lie groups* is a smooth group homomorphism. For $g \in G$, *left translation*, *right translation* and *conjugation* are $L_g(x) = gx$, $R_g(x) = xg$ and $c_g(x) = gxg^{-1}$.

The maps L_g, R_g and c_g are diffeomorphisms: L_g is the composite of $x \mapsto (g,x)$ and m, so it is smooth (Proposition 33.3.3), and its inverse is $L_{g^{-1}}$; similarly for R_g, and $c_g = L_g \circ R_{g^{-1}}$. In particular $d_e(L_g)\colon T_eG \to T_gG$ is an isomorphism, so the whole tangent bundle is determined by the single space T_eG.

Lemma 34.2.2. *Let G be a Lie group. Under the identification $T_{(g,h)}(G \times G) = T_gG \oplus T_hG$ of Proposition 33.4.13,*

$$d_{(g,h)}m(v,w) = d_g(R_h)(v) + d_h(L_g)(w).$$

In particular $d_{(e,e)}m(X,Y) = X + Y$ and $d_e i(X) = -X$ for $X, Y \in T_eG$.

Proof. By Proposition 33.4.13, (v,w) corresponds to $d\iota_h(v) + d\iota_g(w)$ with $\iota_h(x) = (x,h)$ and $\iota_g(y) = (g,y)$. Since $m \circ \iota_h = R_h$ and $m \circ \iota_g = L_g$, the chain rule gives the formula. For the inverse, the map $g \mapsto m(g, i(g))$ is constant, so differentiating at e gives $0 = X + d_e i(X)$. □

Example 34.2.3.

1. $(\mathbb{R}^n, +)$, where multiplication $(x,y) \mapsto x + y$ and inversion $x \mapsto -x$ are linear, hence smooth. Any finite-dimensional real vector space is a Lie group in the same way.

2. The circle $S^1 = \{z \in \mathbb{C} : |z| = 1\}$ under multiplication. Complex multiplication $\mathbb{C} \times \mathbb{C} \to \mathbb{C}$ is a polynomial map of the real coordinates, hence smooth; its restriction to the submanifold $S^1 \times S^1$ is smooth and takes values in S^1, so it is smooth as a map to S^1 (Proposition 33.6.5). Inversion is $z \mapsto \overline{z}$, the restriction of a linear map.

3. The general linear groups $\mathrm{GL}_n(\mathbb{R})$ and $\mathrm{GL}_n(\mathbb{C})$, open subsets of $M_n(\mathbb{R})$ and $M_n(\mathbb{C})$, with multiplication polynomial in the entries and inversion given by the adjugate divided by the determinant (Example 33.3.4(4)). They have dimensions n^2 and $2n^2$.

4. Products $G \times H$ of Lie groups, with componentwise operations (Proposition 33.3.3(5)). So the torus $T^n = S^1 \times \cdots \times S^1$ is a Lie group. The torus $\mathbb{R}^n/\mathbb{Z}^n$ is also a Lie group (use Theorem 33.7.1: the operations lift to the smooth operations of $\mathbb{R}^n$), isomorphic to T^n by Exercise 33.10.6.

5. The identity component G^0 of a Lie group is an open normal subgroup (Propositions 20.5.6(1) and 20.2.3(4)), hence a Lie group as an open submanifold. The quotient G/G^0 is discrete.

Most Lie groups arise as subgroups of the groups in (3). The following proposition makes them Lie groups.

Proposition 34.2.4. *Let G be a Lie group and $H \subseteq G$ a subgroup that is a submanifold. Then H, with its submanifold structure, is a Lie group, the inclusion $H \to G$ is a homomorphism of Lie groups, and H is closed in G.*

Proof. The product $H \times H$ is a submanifold of $G \times G$ (products of slice charts are slice charts). The multiplication of G, restricted to $H \times H$, is smooth, and it takes values in H, so it is smooth as a map $H \times H \to H$ (Proposition 33.6.5); similarly for inversion. The inclusion is smooth by Lemma 33.6.3.

For closedness, let $\overline{H}$ be the closure of H, which is a subgroup, because multiplication and inversion are continuous. We show that H is open in $\overline{H}$. Let $h \in H$ and let (U, φ) be a slice chart at h; then $U \cap H$ is closed in U, being the preimage of the closed set $\mathbb{R}^k \times \{0\}$. If $x \in U \cap \overline{H}$, every neighbourhood of x inside U meets H, hence meets $U \cap H$, so x lies in the closure of $U \cap H$ in U, which is $U \cap H$. So $U \cap \overline{H} = U \cap H$, and H is open in $\overline{H}$. An open subgroup of a topological group is closed (Proposition 20.2.3(2)), so H is closed in $\overline{H}$, that is, $H = \overline{H}$. □

Example 34.2.5. The *classical groups* are the following subgroups of $\mathrm{GL}_n(\mathbb{R})$ and $\mathrm{GL}_n(\mathbb{C})$. Each is a submanifold, by the regular value theorem (Theorem 33.6.11), hence a Lie group by Proposition 34.2.4; we compute its dimension and its tangent space at I.

1. The *special linear group* $\mathrm{SL}_n(\mathbb{R}) = \{\det A = 1\}$ has dimension $n^2 - 1$ and $T_I\mathrm{SL}_n(\mathbb{R}) = \{X : \operatorname{tr} X = 0\}$ (Example 33.6.13(3)). The same computation over $\mathbb{C}$, with $\det\colon M_n(\mathbb{C}) \to \mathbb{C}$, shows that $\mathrm{SL}_n(\mathbb{C})$ has real dimension $2n^2 - 2$: at $A \in \mathrm{SL}_n(\mathbb{C})$, $d_A \det(A) = n$ and $d_A \det(iA) = in$, so $d_A \det$ is onto $\mathbb{C}$.

2. The *orthogonal group* $\mathrm{O}(n) = \{A \in M_n(\mathbb{R}) : A^TA = I\}$ has dimension $n(n-1)/2$, and $T_I\mathrm{O}(n) = \{X : X^T = -X\}$ (Example 33.6.13(2)). It is compact: it is closed, and bounded because its columns are unit vectors. For $A \in \mathrm{O}(n)$, $\det(A)^2 = \det(A^TA) = 1$, so $\det A = \pm 1$; the *special orthogonal group* $\mathrm{SO}(n) = \{A \in \mathrm{O}(n) : \det A = 1\}$ is open and closed in $\mathrm{O}(n)$, so it is a compact Lie group of the same dimension, with the same tangent space at I.

3. The *unitary group* $\mathrm{U}(n) = \{A \in M_n(\mathbb{C}) : A^*A = I\}$. Let Herm_n be the real vector space of Hermitian matrices ($S^* = S$), of real dimension n^2 (n real diagonal entries and $n(n-1)/2$ complex entries above the diagonal), and $F(A) = A^*A$, a smooth map $M_n(\mathbb{C}) \to \mathrm{Herm}_n$. Since $F(A + tX) = A^*A + t(A^*X + X^*A) + t^2X^*X$, $d_AF(X) = A^*X + X^*A$. For $A \in \mathrm{U}(n)$ and S Hermitian, $X = AS/2$ gives $d_AF(X) = S$, so I is a regular value. Hence $\mathrm{U}(n)$ is a submanifold of dimension $2n^2 - n^2 = n^2$, with $T_I\mathrm{U}(n) = \{X : X^* = -X\}$, the *skew-Hermitian* matrices. It is compact, like $\mathrm{O}(n)$. For $n = 1$, $\mathrm{U}(1) = S^1$.

4. The *special unitary group* $\mathrm{SU}(n) = \{A \in \mathrm{U}(n) : \det A = 1\}$. The determinant is a homomorphism of Lie groups $\mathrm{U}(n) \to \mathrm{U}(1) = S^1$ (for A unitary, $|\det A|^2 = \det(A^*A) = 1$). Its differential at I is $X \mapsto \operatorname{tr} X$, because $\det(I + tX) = 1 + t \operatorname{tr} X + O(t^2)$; on skew-Hermitian matrices it takes values in $i\mathbb{R} = T_1S^1$, and it is onto $i\mathbb{R}$ since $\operatorname{tr}(i E_{11}) = i$, where E_{11} is the matrix unit. At any $A \in \mathrm{SU}(n)$ we have $\det \circ L_A = \det$ on $\mathrm{U}(n)$, so $d_A \det \circ d_I(L_A) = d_I \det$ is surjective as well. So 1 is a regular value of $\det\colon \mathrm{U}(n) \to S^1$, and $\mathrm{SU}(n)$ is a compact submanifold of $\mathrm{U}(n)$ of dimension $n^2 - 1$, with $T_I\mathrm{SU}(n) = \{X : X^* = -X, \operatorname{tr} X = 0\}$.

In the smallest cases, $\mathrm{SO}(2)$ is the group of rotations $\left(\begin{smallmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{smallmatrix}\right)$, isomorphic to S^1, and $\mathrm{O}(2)$ is the union of $\mathrm{SO}(2)$ and the reflections; $\mathrm{SU}(2)$ is diffeomorphic to S^3 (Example 34.2.11).

We now attach a Lie algebra to a Lie group. The idea is to use the translations to spread a tangent vector at e over the whole group, obtaining a vector field, and to take the bracket of vector fields (Definition 33.9.5).

Definition 34.2.6. A vector field X on G is *left-invariant* if $d(L_g)(X_h) = X_{gh}$ for all $g, h \in G$. The *Lie algebra* of G is $\mathfrak{g} = T_eG$, with the bracket transported from the left-invariant vector fields: for $X, Y \in \mathfrak{g}$, $[X, Y] = [\widetilde{X}, \widetilde{Y}]_e$, where $\widetilde{X}$ is the left-invariant vector field with value X at e (Proposition 34.2.8).

For a diffeomorphism $F\colon M \to N$ and a vector field X on M, the *push-forward* F_*X is the vector field $(F_*X)_{F(p)} = d_pF(X_p)$ on N; so X is left-invariant exactly when $(L_g)_*X = X$ for all g. More generally, for a smooth map $\varphi\colon M \to N$, vector fields X on M and Y on N are *φ-related* if $d_p\varphi(X_p) = Y_{\varphi(p)}$ for all p.

Lemma 34.2.7. *Let $\varphi\colon M \to N$ be smooth. Then X and Y are φ-related if and only if $X(f\circ\varphi) = (Yf)\circ\varphi$ for all $f \in C^\infty(N)$. If X_1, X_2 are φ-related to Y_1, Y_2, then $[X_1, X_2]$ is φ-related to $[Y_1, Y_2]$. In particular, for a diffeomorphism F, $F_*[X_1, X_2] = [F_*X_1, F_*X_2]$.*

Proof. At p, $X_p(f\circ\varphi) = d_p\varphi(X_p)(f)$ and $(Yf)(\varphi(p)) = Y_{\varphi(p)}(f)$; two tangent vectors at $\varphi(p)$ are equal if they agree on all f, which proves the first statement. Then $[X_1, X_2](f\circ\varphi) = X_1((Y_2f)\circ\varphi) - X_2((Y_1f)\circ\varphi) = (Y_1Y_2f - Y_2Y_1f)\circ\varphi$. For a diffeomorphism F, X and F_*X are F-related, and F_*X is the only vector field F-related to X. □

Proposition 34.2.8. *The map $X \mapsto X_e$ is a bijection from left-invariant vector fields to T_eG, its inverse sending v to the field $g \mapsto d(L_g)_ev$; left-invariant fields are smooth and closed under the Lie bracket. Hence $\mathfrak{g}$ is a Lie algebra of dimension $\dim G$ (Definition 13.4.1), and a homomorphism $\varphi\colon G \to H$ induces a homomorphism of Lie algebras $d_e\varphi\colon \mathfrak{g} \to \mathfrak{h}$.*

Proof. The bijection. A left-invariant field satisfies $X_g = d(L_g)(X_e)$, so it is determined by X_e. Conversely, for $v \in T_eG$ put $\widetilde{v}_g = d_e(L_g)(v)$. It is left-invariant by the chain rule, since $L_h\circ L_g = L_{hg}$. It is smooth: let γ be a smooth curve with $\gamma(0) = e$ and $\gamma'(0) = v$ (Proposition 33.4.9). For $f \in C^\infty(G)$, $(\widetilde{v}f)(g) = v(f\circ L_g) = \frac{\partial}{\partial t}\big|_{t=0} f(g\gamma(t))$, the partial derivative in t at $t = 0$ of the smooth function $(g,t) \mapsto f(g\gamma(t))$; computed in charts, it is a smooth function of g. So $f \mapsto \widetilde{v}f$ is a derivation of $C^\infty(G)$, and $\widetilde{v}$ is the corresponding vector field (Proposition 33.5.7).

Brackets. If X and Y are left-invariant, then $(L_g)_*[X,Y] = [(L_g)_*X, (L_g)_*Y] = [X,Y]$ by Lemma 34.2.7, so $[X,Y]$ is left-invariant. Thus the left-invariant fields form a Lie subalgebra of $\mathfrak{X}(G)$ (Proposition 33.9.7), and transporting the bracket along the linear bijection $X \mapsto X_e$ makes $\mathfrak{g}$ a Lie algebra of dimension $\dim T_eG = \dim G$.

Homomorphisms. Let $v \in \mathfrak{g}$ and $w = d_e\varphi(v)$. Since $\varphi\circ L_g = L_{\varphi(g)}\circ\varphi$, $d\varphi(\widetilde{v}_g) = d\varphi\circ d(L_g)(v) = d(L_{\varphi(g)})(w) = \widetilde{w}_{\varphi(g)}$: the fields $\widetilde{v}$ and $\widetilde{w}$ are φ-related. By Lemma 34.2.7, $[\widetilde{v}_1, \widetilde{v}_2]$ is φ-related to $[\widetilde{w}_1, \widetilde{w}_2]$, and evaluating at e gives $d_e\varphi([v_1, v_2]) = [d_e\varphi(v_1), d_e\varphi(v_2)]$. □

Example 34.2.9. For $G = \mathrm{GL}_n(\mathbb{K})$ the Lie algebra is $\mathfrak{gl}_n(\mathbb{K}) = M_n(\mathbb{K})$ with the commutator $[X,Y] = XY - YX$. Indeed L_A is the restriction of the linear map $B \mapsto AB$, so its differential is again $B \mapsto AB$, and the left-invariant field with value X at I is $\widetilde{X}_A = AX$. In the linear coordinates of $M_n(\mathbb{K})$, the coordinate formula of Proposition 33.9.7 reads $[\widetilde{X}, \widetilde{Y}](A) = D\widetilde{Y}(A)[\widetilde{X}(A)] - D\widetilde{X}(A)[\widetilde{Y}(A)]$; since $\widetilde{Y}(A) = AY$ is linear in A, $D\widetilde{Y}(A)[B] = BY$, so $[\widetilde{X}, \widetilde{Y}](A) = AXY - AYX$, and at $A = I$ this is $XY - YX$.

If $G \subseteq \mathrm{GL}_n(\mathbb{K})$ is a subgroup and a submanifold, the inclusion is a homomorphism of Lie groups (Proposition 34.2.4) with injective differential, so by Proposition 34.2.8 its Lie algebra is the subspace $T_IG \subseteq M_n(\mathbb{K})$ with the commutator. By Example 34.2.5:

$$\mathfrak{sl}_n(\mathbb{K}) = \{X : \operatorname{tr} X = 0\}, \qquad \mathfrak{o}(n) = \mathfrak{so}(n) = \{X \in M_n(\mathbb{R}) : X^T = -X\},$$
$$\mathfrak{u}(n) = \{X \in M_n(\mathbb{C}) : X^* = -X\}, \qquad \mathfrak{su}(n) = \{X \in \mathfrak{u}(n) : \operatorname{tr} X = 0\}.$$

One can check directly that these subspaces are closed under the commutator: $\operatorname{tr}(XY - YX) = 0$ always, and if $X^T = -X$ and $Y^T = -Y$, then $(XY - YX)^T = Y^TX^T - X^TY^T = YX - XY$.

Example 34.2.10. The Lie algebra $\mathfrak{so}(3)$ of skew-symmetric 3×3 matrices is $\mathbb{R}^3$ with the cross product. For $a\in\mathbb{R}^3$ let L_a be the skew-symmetric matrix with $L_a v = a\times v$ for all $v\in\mathbb{R}^3$:

$$L_a = \begin{pmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{pmatrix}.$$

Every skew-symmetric matrix has this form, so $a\mapsto L_a$ is a linear isomorphism $\mathbb{R}^3\to\mathfrak{so}(3)$. By the Jacobi identity for the cross product, $a\times(b\times v)-b\times(a\times v)=(a\times b)\times v$, so $[L_a,L_b]=L_{a\times b}$. In the basis $L_1=L_{e_1}, L_2=L_{e_2}, L_3=L_{e_3}$: $[L_1,L_2]=L_3$, $[L_2,L_3]=L_1$, $[L_3,L_1]=L_2$.

Example 34.2.11. The group SU(2) is diffeomorphic to S^3, and its Lie algebra is isomorphic to $\mathfrak{so}(3)$. A matrix $A\in M_2(\mathbb{C})$ lies in SU(2) if and only if its columns form an orthonormal basis of $\mathbb{C}^2$ and $\det A=1$. If the first column is (a,b) with $|a|^2+|b|^2=1$, the second is a unit vector orthogonal to it, hence $\lambda(-\overline{b},\overline{a})$ with $|\lambda|=1$, and then $\det A=\lambda$. So

$$\mathrm{SU}(2)=\left\{\begin{pmatrix} a & -\overline{b} \\ b & \overline{a}\end{pmatrix} : |a|^2+|b|^2=1\right\},$$

and $(a,b)\mapsto A$ is a bijection $S^3\to\mathrm{SU}(2)$, where $S^3\subseteq\mathbb{C}^2=\mathbb{R}^4$. It is smooth (the restriction of a real-linear map, with values in SU(2)), and so is its inverse, $A\mapsto$ first column. The Lie algebra $\mathfrak{su}(2)$ has the basis

$$I_1=\begin{pmatrix} i & 0 \\ 0 & -i\end{pmatrix},\qquad I_2=\begin{pmatrix} 0 & 1 \\ -1 & 0\end{pmatrix},\qquad I_3=\begin{pmatrix} 0 & i \\ i & 0\end{pmatrix},$$

since every traceless skew-Hermitian matrix is $\left(\begin{smallmatrix} ix & y+iz \\ -y+iz & -ix\end{smallmatrix}\right) = xI_1+yI_2+zI_3$. A direct computation gives $I_1^2=I_2^2=I_3^2=-I$ and $I_1I_2=-I_2I_1=I_3$, $I_2I_3=-I_3I_2=I_1$, $I_3I_1=-I_1I_3=I_2$ (the quaternion relations). So $[I_1,I_2]=2I_3$ and cyclically, and the basis $e_k=I_k/2$ satisfies $[e_1,e_2]=e_3$, $[e_2,e_3]=e_1$, $[e_3,e_1]=e_2$. Comparing with Example 34.2.10, $e_k\mapsto L_k$ is an isomorphism of Lie algebras $\mathfrak{su}(2)\to\mathfrak{so}(3)$. It comes from a homomorphism of groups $\mathrm{SU}(2)\to\mathrm{SO}(3)$, which is a double covering (Exercise 34.6.2).

34.3 The exponential map

A *one-parameter subgroup* of G is a homomorphism of Lie groups $\mathbb{R}\to G$. For matrix groups these are given by the exponential series, which we treat first.

Lemma 34.3.1. *For $X\in M_n(\mathbb{K})$ the series $e^X=\sum_{k\ge 0}X^k/k!$ converges absolutely, and:*

1. *if $XY=YX$, then $e^{X+Y}=e^Xe^Y$; in particular e^X is invertible with inverse e^{-X};*
2. *$t\mapsto e^{tX}$ is a one-parameter subgroup $\mathbb{R}\to\mathrm{GL}_n(\mathbb{K})$, with derivative $\frac{d}{dt}e^{tX}=Xe^{tX}=e^{tX}X$;*
3. *$e^{AXA^{-1}}=Ae^XA^{-1}$ for A invertible, $(e^X)^T=e^{X^T}$ and $(e^X)^*=e^{X^*}$;*
4. *$\det e^X=e^{\operatorname{tr}X}$.*

Proof. Since $\|X^k/k!\|\le\|X\|^k/k!$ and $\sum_k\|X\|^k/k!=e^{\|X\|}$, the series converges absolutely in the complete space $M_n(\mathbb{K})$ (Proposition 27.5.4).

(1) As X and Y commute, $(X+Y)^k=\sum_{i+j=k}\frac{k!}{i!j!}X^iY^j$. So, with $a=\|X\|$ and $b=\|Y\|$,

$$\Big\|\sum_{k\le N}\frac{(X+Y)^k}{k!}-\sum_{i\le N}\frac{X^i}{i!}\sum_{j\le N}\frac{Y^j}{j!}\Big\| = \Big\|\sum_{\substack{i,j\le N\\ i+j>N}}\frac{X^iY^j}{i!j!}\Big\| \le \sum_{i+j>N}\frac{a^ib^j}{i!j!} = e^{a+b}-\sum_{k\le N}\frac{(a+b)^k}{k!},$$

which tends to 0 as $N \to \infty$. Hence $e^{X+Y} = e^X e^Y$. Taking $Y = -X$ gives $e^X e^{-X} = e^0 = I$.

(2) By (1), $e^{sX}e^{tX} = e^{(s+t)X}$. Each entry of e^{tX} is a power series $\sum_k (X^k)_{ij}t^k/k!$ in t with infinite radius of convergence, so it is smooth, and differentiating term by term (Theorem 27.6.20) gives $\sum_{k\geq 1} t^{k-1}X^k/(k-1)! = Xe^{tX}$, which also equals $e^{tX}X$.

(3) $(AXA^{-1})^k = AX^kA^{-1}$, and transposition and conjugate transposition are continuous and linear, with $(X^k)^T = (X^T)^k$; sum the series.

(4) Let $f(t) = \det e^{tX}$. By (2), $f(t+h) = f(t)\det(e^{hX})$, and $e^{hX} = I + hX + O(h^2)$, so expanding the determinant as a sum of products of entries, $\det e^{hX} = 1 + h\operatorname{tr} X + O(h^2)$. Hence $f'(t) = f(t)\operatorname{tr} X$ and $f(0) = 1$, so $(f(t)e^{-t\operatorname{tr} X})' = 0$ and $f(t) = e^{t\operatorname{tr} X}$. Take $t = 1$. □

Example 34.3.2.

1. For a diagonal matrix, $e^{\operatorname{diag}(\lambda_1,\dots,\lambda_n)} = \operatorname{diag}(e^{\lambda_1},\dots,e^{\lambda_n})$.
2. For $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $N^2 = 0$, so $e^{tN} = I + tN = \left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right)$.
3. For $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, $J^2 = -I$, so separating even and odd powers, $e^{tJ} = \cos t\, I + \sin t\, J = \left(\begin{smallmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{smallmatrix}\right)$, the rotation by t. Similarly e^{tL_3}, with L_3 as in Example 34.2.10, is the rotation by the angle t about the e_3-axis.
4. The exponential maps each Lie algebra of Example 34.2.9 into the corresponding group. If $X^T = -X$, then X and X^T commute, so $(e^X)^T e^X = e^{X^T+X} = I$, and $\det e^X = e^{\operatorname{tr} X} = e^0 = 1$: so $e^X \in \mathrm{SO}(n)$. In the same way $e^X \in \mathrm{U}(n)$ for skew-Hermitian X, $e^X \in \mathrm{SL}_n$ for $\operatorname{tr} X = 0$, and $e^X \in \mathrm{SU}(n)$ for $X \in \mathfrak{su}(n)$.

For a general Lie group we use the flows of left-invariant vector fields (Theorem 33.9.2) in place of the series.

Theorem 34.3.3. *For $X \in \mathfrak{g}$ the left-invariant field X is complete, and its flow through e is a homomorphism $\gamma_X\colon \mathbb{R} \to G$; conversely every homomorphism $\mathbb{R} \to G$ arises this way. Setting $\exp(X) = \gamma_X(1)$ defines a smooth map $\exp\colon \mathfrak{g} \to G$ with $d_0\exp = \mathrm{id}$, so $\exp$ is a diffeomorphism from a neighbourhood of 0 onto a neighbourhood of e. It satisfies $\exp((s+t)X) = \exp(sX)\exp(tX)$, $\exp(-X) = \exp(X)^{-1}$, and $\varphi(\exp X) = \exp(d_e\varphi(X))$ for every homomorphism φ. For matrix groups, $\exp(X) = \sum_k X^k/k!$.*

Proof. Write $\widetilde{X}$ for the left-invariant field with value X at e.

Step 1: left translates of integral curves are integral curves. If γ is an integral curve of $\widetilde{X}$, so is $L_g \circ \gamma$, since $(L_g \circ \gamma)'(t) = d(L_g)(\widetilde{X}_{\gamma(t)}) = \widetilde{X}_{g\gamma(t)}$ by left-invariance.

Step 2: completeness. By Theorem 33.9.2, the integral curve γ with $\gamma(0) = e$ is defined on some $(-\varepsilon, \varepsilon)$. By Step 1, $t \mapsto g\gamma(t)$ is the integral curve through g, so every maximal integral curve is defined at least on $(-\varepsilon, \varepsilon)$. If the maximal integral curve γ_g through g were defined only up to $\beta < \infty$, then for $s \in (\beta - \varepsilon, \beta)$ the integral curve through $\gamma_g(s)$, defined on $(-\varepsilon, \varepsilon)$, would extend γ_g beyond β (by the group law of the flow), a contradiction. So all integral curves are defined on $\mathbb{R}$; let γ_X be the one with $\gamma_X(0) = e$.

Step 3: γ_X is a homomorphism. For fixed s, the curves $t \mapsto \gamma_X(s)\gamma_X(t)$ (Step 1) and $t \mapsto \gamma_X(s+t)$ are integral curves with the same value $\gamma_X(s)$ at $t = 0$, so they coincide by uniqueness of integral curves.

Step 4: every one-parameter subgroup is some γ_X. Let $\gamma\colon \mathbb{R} \to G$ be a homomorphism of Lie groups and $X = \gamma'(0)$. Since $\gamma(t+h) = L_{\gamma(t)}(\gamma(h))$, differentiating in h at $h = 0$ gives $\gamma'(t) = d(L_{\gamma(t)})(X) = \widetilde{X}_{\gamma(t)}$. So γ is the integral curve of $\widetilde{X}$ through e, that is, $\gamma = \gamma_X$.

Step 5: smoothness of exp. Choose a basis $X_1, \dots, X_n$ of $\mathfrak{g}$, and consider on $G \times \mathfrak{g}$ the vector field $Z_{(g,\sum_i a^i X_i)} = (\sum_i a^i(\widetilde{X}_i)_g, 0)$. In product charts its components are the a^i times the components of the smooth fields $\widetilde{X}_i$, so Z is smooth. Its integral curve through (e, X) is $t \mapsto (\gamma_X(t), X)$, and more generally that through (g, X) is $t \mapsto (g\gamma_X(t), X)$, defined for all t.

By Theorem 33.9.2 its flow Φ is smooth on $\mathbb{R} \times G \times \mathfrak{g}$, so $(t, X) \mapsto \gamma_X(t)$, the first component of $\Phi(t, (e, X))$, is smooth, and in particular $\exp(X) = \gamma_X(1)$ is smooth.

Step 6: the group laws. For fixed s, $t \mapsto \gamma_X(st)$ is a one-parameter subgroup with velocity sX at 0, so it equals γ_{sX} by Step 4. Hence $\exp(sX) = \gamma_X(s)$, and Step 3 gives $\exp((s+t)X) = \exp(sX)\exp(tX)$ and, with $t = -s = -1$, $\exp(-X) = \exp(X)^{-1}$.

Step 7: the differential at 0. Identifying $T_0\mathfrak{g}$ with $\mathfrak{g}$, the vector X is the velocity of $t \mapsto tX$, so $d_0 \exp(X) = \frac{d}{dt}\big|_{t=0} \exp(tX) = \gamma_X'(0) = X$. As $d_0 \exp = \mathrm{id}$ is an isomorphism, exp is a diffeomorphism near 0 (Theorem 33.4.14).

Step 8: naturality. For a homomorphism $\varphi\colon G \to H$, $t \mapsto \varphi(\exp tX)$ is a one-parameter subgroup of H with velocity $d_e\varphi(X)$ at 0, so it equals $t \mapsto \exp(t\, d_e\varphi(X))$ by Step 4; take $t = 1$.

Step 9: matrix groups. In $\mathrm{GL}_n(\mathbb{K})$, $t \mapsto e^{tX}$ is a one-parameter subgroup with velocity X at 0 (Lemma 34.3.1(2)), so $\exp(X) = e^X$ by Step 4. For a subgroup $G \subseteq \mathrm{GL}_n(\mathbb{K})$ that is a submanifold, Step 8 applied to the inclusion shows that the exponential map of G is the restriction of that of $\mathrm{GL}_n(\mathbb{K})$. □

Example 34.3.4.

1. For $G = (\mathbb{R}^n, +)$, the one-parameter subgroups are $t \mapsto tX$, and $\exp = \mathrm{id}$. For the torus $\mathbb{R}^n/\mathbb{Z}^n$, exp is the projection $\mathbb{R}^n \to \mathbb{R}^n/\mathbb{Z}^n$: it is surjective, a homomorphism, and not injective.

2. For $S^1 = \mathrm{U}(1)$, $\mathfrak{u}(1) = i\mathbb{R}$ and $\exp(it) = e^{it}$. For $\mathrm{SO}(2)$, $\exp(tJ)$ is the rotation by t (Example 34.3.2(3)); in both cases exp is surjective with kernel $2\pi\mathbb{Z}$ (times i, respectively J).

3. For $\mathrm{SL}_2(\mathbb{R})$, exp is not surjective (Exercise 34.6.1), although it is a diffeomorphism near 0.

Definition 34.3.5. The *adjoint representation* is $\mathrm{Ad}\colon G \to \mathrm{GL}(\mathfrak{g})$, $\mathrm{Ad}(g) = d_e(c_g)$ where $c_g(x) = gxg^{-1}$; its differential is $\mathrm{ad} = d_e \mathrm{Ad}\colon \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$.

For $G \subseteq \mathrm{GL}_n(\mathbb{K})$, c_A is the restriction of the linear map $X \mapsto AXA^{-1}$, so $\mathrm{Ad}(A)X = AXA^{-1}$; and differentiating $\mathrm{Ad}(e^{tX})Y = e^{tX}Ye^{-tX}$ at $t = 0$ gives $\mathrm{ad}(X)Y = XY - YX$.

Proposition 34.3.6. Ad *is a homomorphism of Lie groups, each* $\mathrm{Ad}(g)$ *is an automorphism of the Lie algebra* $\mathfrak{g}$, $\mathrm{ad}(X)(Y) = [X, Y]$, *and* $\mathrm{Ad}(\exp X) = e^{\mathrm{ad}\, X}$.

Proof. Since $c_{gh} = c_g \circ c_h$ and $c_e = \mathrm{id}$, the chain rule shows that Ad is a homomorphism into $\mathrm{GL}(\mathfrak{g})$. Each c_g is an automorphism of the Lie group G, so $\mathrm{Ad}(g) = d_e c_g$ preserves brackets (Proposition 34.2.8). To see that Ad is smooth, note that $\mathrm{Ad}(g)X = \frac{\partial}{\partial t}\big|_{t=0} c_g(\exp tX)$, the partial derivative at $t = 0$ of the smooth map $(g, t) \mapsto g\exp(tX)g^{-1}$; in a chart around e its components are smooth functions of g, so the matrix entries of $\mathrm{Ad}(g)$ in a basis of $\mathfrak{g}$ are smooth.

For the bracket, let $\widetilde{X}, \widetilde{Y}$ be the left-invariant fields. By Step 1 of the proof of Theorem 34.3.3, the integral curve of $\widetilde{X}$ through g is $t \mapsto g\exp(tX)$, so the flow of $\widetilde{X}$ is $\Phi_t = R_{\exp(tX)}$. By Proposition 33.9.7,

$$[X, Y] = [\widetilde{X}, \widetilde{Y}]_e = \frac{d}{dt}\Big|_{t=0} d(R_{\exp(-tX)})\big(\widetilde{Y}_{\exp(tX)}\big) = \frac{d}{dt}\Big|_{t=0} d(R_{\exp(-tX)})\, d(L_{\exp(tX)})(Y) = \frac{d}{dt}\Big|_{t=0} \mathrm{Ad}(\exp tX)Y,$$

because $R_{\exp(-tX)} \circ L_{\exp(tX)} = c_{\exp(tX)}$. The curve $t \mapsto \mathrm{Ad}(\exp tX)$ in $\mathrm{GL}(\mathfrak{g})$ has velocity $d_e \mathrm{Ad}(X) = \mathrm{ad}(X)$ at $t = 0$, and evaluation at Y is linear, so the right-hand side is $\mathrm{ad}(X)(Y)$. Finally, Ad is a homomorphism $G \to \mathrm{GL}(\mathfrak{g})$, and the exponential map of $\mathrm{GL}(\mathfrak{g}) \cong \mathrm{GL}_n(\mathbb{R})$ is the matrix exponential (Theorem 34.3.3), so naturality of exp gives $\mathrm{Ad}(\exp X) = \exp(d_e \mathrm{Ad}(X)) = e^{\mathrm{ad}\, X}$. □

The next lemma says that to first order the group law, read through exp, is addition in $\mathfrak{g}$.

Lemma 34.3.7. *For $X, Y \in \mathfrak{g}$, $\lim_{k\to\infty}\big(\exp(X/k)\exp(Y/k)\big)^k = \exp(X+Y)$.*

Proof. Let exp map an open neighbourhood U_0 of 0 diffeomorphically onto an open neighbourhood V_0 of e, with inverse $\log\colon V_0 \to U_0$. For (A, B) in a neighbourhood of $(0,0)$ we have $\exp(A)\exp(B) \in V_0$, and $\mu(A,B) = \log(\exp(A)\exp(B))$ is smooth. By Lemma 34.2.2 and $d_0\exp = \mathrm{id}$, its differential at $(0,0)$ is $(A,B) \mapsto A + B$. So $\mu(sX, sY) = s(X+Y) + r(s)$ with $r(s)/s \to 0$ as $s \to 0$. Since $\exp(Z)^k = \exp(kZ)$ by the group laws,

$$\big(\exp(X/k)\exp(Y/k)\big)^k = \exp\big(\mu(X/k, Y/k)\big)^k = \exp\big(X + Y + k\,r(1/k)\big),$$

and $k\,r(1/k) \to 0$; by continuity of exp the limit is $\exp(X+Y)$. □

Remark 34.3.8. Going one order further, $\exp(sX)\exp(sY) = \exp\big(s(X+Y) + \frac{s^2}{2}[X,Y] + O(s^3)\big)$, and in fact $\log(\exp X \exp Y)$ is given near 0 by a universal series in iterated brackets of X and Y, the *Baker–Campbell–Hausdorff formula* [Kna02, Chapter 1]. So near the identity the group law is determined by the bracket; this is why the Lie algebra captures the local structure of G. We do not need these facts.

Proposition 34.3.9. *Let G be a connected Lie group.*

1. *G is generated by $\exp(\mathfrak{g})$, indeed by $\exp(U)$ for any neighbourhood U of 0.*
2. *A homomorphism of Lie groups $\varphi\colon G \to H$ is determined by $d_e\varphi$.*
3. *If G is abelian, then $\exp\colon \mathfrak{g} \to G$ is a surjective homomorphism of groups, and $\mathfrak{g}$ is abelian.*

Proof. (1) By Theorem 34.3.3, $\exp(U)$ contains an open neighbourhood V of e, and replacing V by $V \cap V^{-1}$ we may take $V = V^{-1}$. By Proposition 20.2.3(3), V generates the connected group G.

(2) $\varphi(\exp X) = \exp(d_e\varphi(X))$, so $d_e\varphi$ determines φ on $\exp(\mathfrak{g})$, which generates G by (1).

(3) If G is abelian, every c_g is the identity, so $\mathrm{Ad}(g) = \mathrm{id}$ for all g, hence $\mathrm{ad} = d_e\,\mathrm{Ad} = 0$, and $[X,Y] = \mathrm{ad}(X)(Y) = 0$ by Proposition 34.3.6. For $X, Y \in \mathfrak{g}$, the curve $t \mapsto \exp(tX)\exp(tY)$ is a homomorphism $\mathbb{R} \to G$, because G is abelian, and its velocity at 0 is $X + Y$ by Lemma 34.2.2; so it equals $\exp(t(X+Y))$ by Theorem 34.3.3. So exp is a homomorphism, and its image is a subgroup containing a neighbourhood of e, hence all of G by (1). □

Lemma 34.3.10. *A discrete subgroup Γ of $\mathbb{R}^n$ is of the form $\mathbb{Z}v_1 \oplus \cdots \oplus \mathbb{Z}v_b$ for linearly independent vectors $v_1, \ldots, v_b$.*

Proof. Induction on the dimension of the span V of Γ; the case $V = 0$ is trivial. As Γ is discrete, it meets every bounded set in a finite set (a bounded infinite set would have an accumulation point, and differences of nearby points of Γ are small nonzero elements of Γ, while discreteness gives a ball around 0 containing no other element of Γ). Choose a basis $w_1, \ldots, w_b$ of V contained in Γ, let W be the span of $w_1, \ldots, w_{b-1}$, and $\Gamma' = \Gamma \cap W$, which by induction is $\mathbb{Z}v_1 \oplus \cdots \oplus \mathbb{Z}v_{b-1}$ with v_i spanning W. The elements of Γ of the form $\sum_{i<b} s_i v_i + s w_b$ with $s_i \in [0,1]$ and $s \in (0,1]$ form a finite nonempty set (it contains w_b); choose one, v_b, with the smallest s, say s_0. For $\gamma \in \Gamma$, write $\gamma = \sum_{i<b} c_i v_i + c w_b$ and let $k = \lfloor c/s_0 \rfloor$. Then $\gamma - k v_b$ has w_b-coefficient in $[0, s_0)$, and subtracting a suitable element of Γ' moves its other coefficients into $[0,1)$; by minimality of s_0 the w_b-coefficient is 0, so $\gamma - k v_b \in \Gamma'$. Hence $\Gamma = \Gamma' \oplus \mathbb{Z}v_b$. □

Theorem 34.3.11. *Every connected abelian Lie group is isomorphic to $\mathbb{R}^a \times T^b$ for some a, b; in particular every compact connected abelian Lie group is a torus.*

Proof. Let $n = \dim G$ and identify $\mathfrak{g}$ with $\mathbb{R}^n$ by a basis. By Proposition 34.3.9(3), $\exp\colon \mathbb{R}^n \to G$ is a surjective homomorphism. It is a diffeomorphism near 0, hence a local diffeomorphism everywhere, since $\exp(X+Z) = \exp(X)\exp(Z)$, that is, $\exp \circ (Z \mapsto X+Z) = L_{\exp X} \circ \exp$. Its kernel Γ is a subgroup with $\Gamma \cap U = \{0\}$ for a neighbourhood U of 0 on which exp is injective; so Γ is discrete, because $(\gamma + U) \cap \Gamma = \{\gamma\}$ for $\gamma \in \Gamma$. By Lemma 34.3.10, $\Gamma = \mathbb{Z}v_1 \oplus \cdots \oplus \mathbb{Z}v_b$; completing to a basis $v_1, \dots, v_n$ of $\mathbb{R}^n$ gives a linear isomorphism taking Γ to $\mathbb{Z}^b \times \{0\}$, hence $\mathbb{R}^n/\Gamma \cong (\mathbb{R}^b/\mathbb{Z}^b) \times \mathbb{R}^{n-b} = T^b \times \mathbb{R}^{n-b}$. Here Γ acts freely and properly on $\mathbb{R}^n$ by translations (a compact set meets only finitely many of its translates by Γ, since Γ meets bounded sets in finite sets), so $\mathbb{R}^n/\Gamma$ is a Lie group by Theorem 33.7.1. The induced map $\mathbb{R}^n/\Gamma \to G$ is a bijective homomorphism and a local diffeomorphism, hence an isomorphism of Lie groups (Theorem 33.4.14). If G is compact, so is $T^b \times \mathbb{R}^{n-b}$, and $n - b = 0$. □

Counterexample 34.3.12. Connectedness is needed: $\mathrm{O}(2)$ has the abelian Lie algebra $\mathfrak{o}(2) \cong \mathbb{R}$, but it is not abelian, as a reflection does not commute with a rotation by an angle other than 0 or π.

34.4 Subgroups and homogeneous spaces

In the examples so far, we proved that a subgroup is a Lie group by exhibiting it as a regular level set. Cartan's theorem makes this unnecessary: every *closed* subgroup is a submanifold. For instance, stabilisers of points under smooth actions and kernels of homomorphisms are closed, hence Lie groups.

Theorem 34.4.1 (Cartan)**.** *A closed subgroup H of a Lie group G is an embedded submanifold, hence a Lie group, with Lie algebra $\mathfrak{h} = \{X \in \mathfrak{g} : \exp(tX) \in H \text{ for all } t \in \mathbb{R}\}$.*

Proof. Step 1: $\mathfrak{h}$ is a linear subspace. It is closed under multiplication by scalars by definition. If $X, Y \in \mathfrak{h}$ and $t \in \mathbb{R}$, then $\exp(tX/k)\exp(tY/k) \in H$ for all k, so $\exp(t(X+Y)) = \lim_k (\exp(tX/k)\exp(tY/k))^k$ (Lemma 34.3.7) lies in the closed set H. So $X + Y \in \mathfrak{h}$.

Step 2: exp *of a complement avoids H near 0.* Choose a linear complement $\mathfrak{m}$ with $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$ and a norm on $\mathfrak{m}$. We claim that there is a neighbourhood B of 0 in $\mathfrak{m}$ with $\exp(Y) \notin H$ for $Y \in B \setminus \{0\}$. Otherwise there are $Y_k \in \mathfrak{m} \setminus \{0\}$ with $Y_k \to 0$ and $\exp Y_k \in H$. The unit sphere of $\mathfrak{m}$ is compact, so after passing to a subsequence $Y_k/\|Y_k\| \to Y$ with $\|Y\| = 1$. Let $t \in \mathbb{R}$ and $m_k = \lfloor t/\|Y_k\| \rfloor \in \mathbb{Z}$. Then $|m_k\|Y_k\| - t| \le \|Y_k\| \to 0$, so $m_k Y_k = (m_k\|Y_k\|)(Y_k/\|Y_k\|) \to tY$, while $\exp(m_k Y_k) = (\exp Y_k)^{m_k} \in H$. As H is closed, $\exp(tY) \in H$. This holds for all t, so $Y \in \mathfrak{h} \cap \mathfrak{m} = 0$, contradicting $\|Y\| = 1$.

Step 3: a slice chart at e. The map $\Psi\colon \mathfrak{h} \times \mathfrak{m} \to G$, $\Psi(X,Y) = \exp(X)\exp(Y)$, has differential $(X,Y) \mapsto X + Y$ at $(0,0)$ (Lemma 34.2.2 and $d_0 \exp = \mathrm{id}$), an isomorphism $\mathfrak{h} \oplus \mathfrak{m} \to \mathfrak{g}$. So Ψ maps a neighbourhood $A \times B'$ of $(0,0)$ diffeomorphically onto an open neighbourhood V of e (Theorem 33.4.14); shrinking, we may assume $B' \subseteq B$. Then $V \cap H = \Psi(A \times \{0\})$. Indeed $\Psi(X,0) = \exp X \in H$ for $X \in \mathfrak{h}$; conversely, if $\exp(X)\exp(Y) \in H$ with $(X,Y) \in A \times B'$, then $\exp(Y) \in H$, because $\exp X \in H$ and H is a subgroup, so $Y = 0$ by Step 2. Choosing bases, $\Psi^{-1}\colon V \to A \times B' \subseteq \mathbb{R}^k \times \mathbb{R}^{n-k}$ is a chart of G (Proposition 33.3.3(3)) with $\Psi^{-1}(V \cap H) = \Psi^{-1}(V) \cap (\mathbb{R}^k \times \{0\})$: a slice chart at e.

Step 4: slice charts everywhere. For $h \in H$, the diffeomorphism L_h maps H onto H, so $\Psi^{-1} \circ L_{h^{-1}}$, defined on hV, is a slice chart at h. So H is a submanifold of dimension $\dim \mathfrak{h}$, and a Lie group by Proposition 34.2.4. Its tangent space at e is the image of the slice directions, $d_{(0,0)}\Psi(\mathfrak{h} \times \{0\}) = \mathfrak{h}$. By Proposition 34.2.8, applied to the inclusion, $\mathfrak{h}$ is a Lie subalgebra of $\mathfrak{g}$ and is the Lie algebra of H. □

Example 34.4.2. 1. The classical groups of Example 34.2.5 are closed subgroups of GL_n, defined by polynomial equations; Cartan's theorem gives again that they are Lie groups, and the formula for $\mathfrak{h}$ gives their Lie algebras. For instance $\mathfrak{u}(n) = \{X : e^{tX} \in \mathrm{U}(n) \text{ for all } t\}$: if $X^* = -X$ this holds by Example 34.3.2(4), and conversely differentiating $(e^{tX})^* e^{tX} = I$ at $t = 0$ gives $X^* + X = 0$.

2. The kernel of a homomorphism of Lie groups is closed, hence a Lie group. The centre $Z(G) = \{g : gx = xg \text{ for all } x\}$ is closed, hence a Lie group.

3. The diagonal matrices in $\mathrm{U}(n)$ form a closed subgroup isomorphic to the torus T^n, with Lie algebra the diagonal matrices with imaginary entries.

To build smooth quotients G/H we need one general fact about submersions.

Lemma 34.4.3. *Let $\pi\colon M \to N$ be a surjective submersion and $f\colon N \to P$ a map. Then f is smooth if and only if $f \circ \pi$ is smooth. Every point of N has a neighbourhood W with a smooth* local section *$s\colon W \to M$, $\pi \circ s = \mathrm{id}_W$.*

Proof. Let $q \in N$ and choose p with $\pi(p) = q$. By Proposition 33.6.7(1) there are charts (U, φ) at p and (V, ψ) at q in which π is $(x, y) \mapsto x$. Write $\varphi(p) = (x_0, y_0)$ and choose balls $B_1 \ni x_0$ and $B_2 \ni y_0$ with $B_1 \times B_2 \subseteq \varphi(U)$. Then $W = \psi^{-1}(B_1)$ is an open neighbourhood of q, and $s = \varphi^{-1}(\psi(\cdot), y_0)$ is a smooth map $W \to M$ with $\pi \circ s = \mathrm{id}_W$. If $f \circ \pi$ is smooth, then $f|_W = (f \circ \pi) \circ s$ is smooth, and f is smooth because q was arbitrary. The converse is clear. □

Theorem 34.4.4. *Let $H \subseteq G$ be a closed subgroup. The quotient G/H has a unique smooth structure such that the projection $\pi\colon G \to G/H$ is a submersion; it is a manifold of dimension* $\dim G - \dim H$ *on which G acts smoothly and transitively. If a Lie group G acts smoothly and transitively on a manifold M and H is the stabiliser of $p \in M$, then $G/H \to M$, $gH \mapsto gp$, is a G-equivariant diffeomorphism.*

Proof. Topology. G/H is the orbit space of the action of H on G by right multiplication, with the quotient topology, so π is open and G/H is second countable (Proposition 20.3.2(1) and (3)). It is Hausdorff by Proposition 20.3.2(2), since the orbit relation $\{(g, g') : g^{-1}g' \in H\}$ is the preimage of the closed set H under the continuous map $(g, g') \mapsto g^{-1}g'$.

Charts. Let $\mathfrak{h}$ and $\mathfrak{m}$ be as in the proof of Theorem 34.4.1, with $B \subseteq \mathfrak{m}$ as in its Step 2. As there, the map $\Psi'(Y, X) = \exp(Y)\exp(X)$ maps a neighbourhood $B' \times A$ of $(0, 0)$ in $\mathfrak{m} \times \mathfrak{h}$, with $B' \subseteq B$, diffeomorphically onto an open $V \ni e$, and $V \cap H = \Psi'(\{0\} \times A)$ (if $\exp(Y)\exp(X) \in H$ then $\exp(Y) \in H$, so $Y = 0$). By continuity choose a neighbourhood $B_0 \subseteq B'$ of 0 with $\exp(B_0)^{-1}\exp(B_0) \subseteq V$. For $g \in G$ define $\kappa_g\colon B_0 \to G/H$, $\kappa_g(Y) = \pi(g \exp Y)$.

- κ_g is injective: if $\pi(g \exp Y) = \pi(g \exp Y')$, then $h = \exp(-Y)\exp(Y') \in H \cap V$, so $h = \exp X$ with $X \in A$, and $\Psi'(Y', 0) = \exp Y' = \exp(Y)\exp(X) = \Psi'(Y, X)$; injectivity of Ψ' gives $Y = Y'$.

- κ_g is open: for $W \subseteq B_0$ open, $\kappa_g(W) = \pi(g\,\Psi'(W \times A))$, the image under the open map π of an open subset of G.

So κ_g is a homeomorphism onto an open subset of G/H, and $\varphi_g = \kappa_g^{-1}$ is a chart with values in $\mathfrak{m} \cong \mathbb{R}^{n-k}$, where $n = \dim G$ and $k = \dim H$. These charts cover G/H. The transition maps are smooth: let $Y_0 \in B_0$ with $\kappa_g(Y_0) = \kappa_{g'}(Y_0')$, so $g \exp Y_0 = g' \exp(Y_0')h_0$ for some $h_0 \in H$. For Y near Y_0, the element $g'^{-1}g \exp(Y)h_0^{-1}$ lies in the open set $\Psi'(B_0 \times A)$, since it equals $\exp Y_0'$ at $Y = Y_0$; write it as $\Psi'(Y'(Y), X(Y))$, where $(Y'(Y), X(Y))$ depends smoothly on Y. Then

$g \exp Y = g' \exp(Y'(Y)) \exp(X(Y))h_0$, so $\kappa_g(Y) = \kappa_{g'}(Y'(Y))$, and the transition map $\varphi_{g'} \circ \kappa_g$ is $Y \mapsto Y'(Y)$ near Y_0, which is smooth. So G/H is a smooth manifold of dimension $n - k$.

π is a submersion. In the chart $(Y, X) \mapsto g\Psi'(Y, X)$ of G (a diffeomorphism onto the open set $g\Psi'(B_0 \times A)$) and the chart φ_g of G/H, the map π is $(Y, X) \mapsto Y$.

Uniqueness and the action. If another smooth structure on G/H makes π a submersion, then by Lemma 34.4.3, applied in both directions to the identity map of G/H (whose composite with π is π), the identity is a diffeomorphism between the two structures, so they coincide. The action $G \times G/H \to G/H$, $(g', gH) \mapsto g'gH$, is smooth by the same lemma, because its composite with the surjective submersion $\mathrm{id} \times \pi \colon G \times G \to G \times G/H$ is $\pi \circ m$. It is transitive.

Homogeneous manifolds. Let G act smoothly and transitively on M, and let $f \colon G \to M$, $f(g) = gp$. The stabiliser $H = f^{-1}(p)$ is a closed subgroup, and f induces a bijection $\overline{f} \colon G/H \to M$, which is smooth by Lemma 34.4.3 and G-equivariant. For $g \in G$, $f \circ L_g = \theta_g \circ f$, where θ_g is the diffeomorphism $x \mapsto gx$ of M; so $d_g f$ and $d_e f$ have the same rank. If $d_e f$ were not surjective, no $d_g f$ would be, every point of $M = f(G)$ would be a critical value, and M would have measure zero in itself by Sard's theorem (Theorem 33.6.21), which is false, as the image of a chart is a nonempty open subset of Euclidean space. So f is a submersion, and so is $\overline{f}$, since $f = \overline{f} \circ \pi$. A submersion that is injective has bijective differentials: in the local form $(x, y) \mapsto x$ of Proposition 33.6.7(1), injectivity forces the variable y to be absent. So $\overline{f}$ is a bijective local diffeomorphism, hence a diffeomorphism (Theorem 33.4.14). □

Example 34.4.5.

1. *Spheres.* $\mathrm{SO}(n+1)$ acts smoothly on S^n by matrix multiplication. The action is transitive: a unit vector v extends to an orthonormal basis $v, v_2, \ldots, v_{n+1}$, and after changing the sign of v_{n+1} if necessary the matrix with these columns lies in $\mathrm{SO}(n+1)$ and maps e_1 to v (for $n \geq 1$). The stabiliser of e_1 consists of the matrices $\left(\begin{smallmatrix} 1 & 0 \\ 0 & B \end{smallmatrix}\right)$ with $B \in \mathrm{SO}(n)$. So $S^n \cong \mathrm{SO}(n+1)/\mathrm{SO}(n)$, and in the same way $S^n = \mathrm{O}(n+1)/\mathrm{O}(n)$ and, for $S^{2n+1} \subseteq \mathbb{C}^{n+1}$, $S^{2n+1} = \mathrm{U}(n+1)/\mathrm{U}(n) = \mathrm{SU}(n+1)/\mathrm{SU}(n)$.

2. *Projective spaces and Grassmannians.* $\mathrm{O}(n)$ acts transitively on $\mathrm{Gr}_k(\mathbb{R}^n)$, since orthonormal bases of W and $W^\perp$ together form an orthogonal matrix mapping the span of $e_1, \ldots, e_k$ onto W. The action is smooth: in the charts of Proposition 33.7.4 it is given by rational functions, by the same computation as for the transition maps. The stabiliser of $\langle e_1, \ldots, e_k \rangle$ is $\mathrm{O}(k) \times \mathrm{O}(n-k)$. So $\mathrm{Gr}_k(\mathbb{R}^n) = \mathrm{O}(n)/(\mathrm{O}(k) \times \mathrm{O}(n-k))$, compact of dimension $k(n-k)$, and in particular $\mathbb{RP}^n = \mathrm{O}(n+1)/(\mathrm{O}(1) \times \mathrm{O}(n))$. Similarly $\mathbb{CP}^n = \mathrm{U}(n+1)/(\mathrm{U}(1) \times \mathrm{U}(n))$ and $\mathrm{Gr}_k(\mathbb{C}^n) = \mathrm{U}(n)/(\mathrm{U}(k) \times \mathrm{U}(n-k))$.

3. *Flag manifolds.* $\mathrm{U}(n)$ acts transitively on the flags $0 \subset V_1 \subset \cdots \subset V_r = \mathbb{C}^n$ with $\dim V_i/V_{i-1} = n_i$, with stabiliser $\mathrm{U}(n_1) \times \cdots \times \mathrm{U}(n_r)$; these flag manifolds and their open subsets are the period domains parametrising Hodge structures (Chapter 70).

4. *Noncompact examples.* $\mathrm{GL}_n(\mathbb{R})$ acts on the open set of positive definite symmetric matrices by $g \cdot S = gSg^T$, transitively (by the spectral theorem $S = QDQ^T$ with Q orthogonal and D diagonal positive, so $S = g \cdot I$ with $g = QD^{1/2}$), and the stabiliser of I is $\mathrm{O}(n)$; so this space is $\mathrm{GL}_n(\mathbb{R})/\mathrm{O}(n)$. The group $\mathrm{SL}_2(\mathbb{R})$ acts on the upper half plane $\mathbb{H} = \{\mathrm{Im}\,\tau > 0\}$ by $\tau \mapsto (a\tau + b)/(c\tau + d)$; the matrix $\left(\begin{smallmatrix} \sqrt{y} & x/\sqrt{y} \\ 0 & 1/\sqrt{y} \end{smallmatrix}\right)$ maps i to $x + iy$, and $(ai + b)/(ci + d) = i$ means $a = d$ and $b = -c$, which together with $ad - bc = 1$ describes $\mathrm{SO}(2)$. So $\mathbb{H} = \mathrm{SL}_2(\mathbb{R})/\mathrm{SO}(2)$, the period domain for Hodge structures of weight one and rank two (Chapter 66).

Proposition 34.4.6. *Let H be a closed subgroup of a Lie group G. If H and G/H are connected, so is G. Consequently $\mathrm{SO}(n)$, $\mathrm{U}(n)$, $\mathrm{SU}(n)$ and $\mathrm{GL}_n(\mathbb{C})$ are connected, while $\mathrm{O}(n)$ and $\mathrm{GL}_n(\mathbb{R})$ have exactly two components.*

Proof. If $G = A \sqcup B$ with A, B open, each coset gH, homeomorphic to H, is connected and so lies in A or in B. Hence A and B are unions of cosets, and $\pi(A)$, $\pi(B)$ are disjoint open sets covering G/H (the projection is open, Proposition 20.3.2(1)); as G/H is connected, one of them is empty. For the examples, $\mathrm{SO}(1)$ and $\mathrm{U}(1) = S^1$ are connected, and $\mathrm{SO}(n)/\mathrm{SO}(n-1) \cong S^{n-1}$ and $\mathrm{U}(n)/\mathrm{U}(n-1) \cong S^{2n-1}$ (Example 34.4.5) are connected for $n \geq 2$, so induction applies; similarly for $\mathrm{SU}(n)$, using $\mathrm{SU}(1) = \{1\}$. For $\mathrm{GL}_n(\mathbb{C})$, given A invertible, $t \mapsto \det((1-t)I + tA)$ is a nonzero polynomial with finitely many roots in $\mathbb{C}$, so a path from 0 to 1 in $\mathbb{C}$ avoiding them gives a path in $\mathrm{GL}_n(\mathbb{C})$ from I to A. The sign of the determinant separates $\mathrm{O}(n)$ and $\mathrm{GL}_n(\mathbb{R})$ into two open sets, and each is connected: $\mathrm{O}(n)^- = r\,\mathrm{SO}(n)$ for a reflection r, and $\mathrm{GL}_n^+(\mathbb{R})$ retracts onto $\mathrm{SO}(n)$ by the Gram–Schmidt process, which is continuous. □

34.5 Compact Lie groups

On a finite group one averages by summing over the group and dividing by its order. On a compact Lie group the sum is replaced by an integral against an invariant probability measure.

Theorem 34.5.1 (Haar measure)**.** *On a compact Lie group G there is a unique regular Borel probability measure μ that is invariant under all left translations; it is also right invariant and invariant under inversion. Consequently $\int_G f(gx)\,d\mu(x) = \int_G f(x)\,d\mu(x)$ for all g and all continuous f.*

This is [BtD85, Chapter I]; the measure can be constructed by integrating a left-invariant volume form (Chapter 36), which exists because $\Lambda^{\dim G}\mathfrak{g}^*$ is one-dimensional, and normalising. Uniqueness follows by convolving two invariant measures.

Example 34.5.2. For a finite group, μ gives each element mass $1/|G|$. For $S^1 = \mathbb{R}/2\pi\mathbb{Z}$, μ is $d\theta/2\pi$: Lebesgue measure on $[0, 2\pi)$ divided by 2π, which is invariant under rotations $\theta \mapsto \theta + \theta_0$. For the torus $\mathbb{R}^n/\mathbb{Z}^n$ it is Lebesgue measure on $[0,1)^n$.

Corollary 34.5.3. *Every finite-dimensional representation of a compact Lie group G on a complex vector space admits a G-invariant Hermitian inner product, and is a direct sum of irreducible subrepresentations.*

Proof. Average any Hermitian inner product: $\langle v, w\rangle_G = \int_G \langle gv, gw\rangle\,d\mu(g)$, which is Hermitian, positive definite and invariant. The orthogonal complement of a subrepresentation is then a subrepresentation, and induction on the dimension gives the decomposition, exactly as in Maschke's theorem (Theorem 13.2.4). □

Proposition 34.5.4. *If G is compact, $\mathfrak{g}$ has an inner product for which every $\mathrm{Ad}(g)$ is orthogonal and every $\mathrm{ad}(X)$ is skew-symmetric. Consequently the Killing form $B(X,Y) = \mathrm{tr}(\mathrm{ad}\,X\ \mathrm{ad}\,Y)$ is negative semidefinite.*

Proof. Averaging an inner product over Haar measure (Theorem 34.5.1) as in Corollary 34.5.3 gives an Ad-invariant inner product; differentiating $\langle \mathrm{Ad}(\exp tX)Y, \mathrm{Ad}(\exp tX)Z\rangle = \langle Y, Z\rangle$ at $t = 0$ gives skew-symmetry of $\mathrm{ad}\,X$ (Proposition 34.3.6). For a skew-symmetric A, $\mathrm{tr}(A^2) = -\mathrm{tr}(A^TA) \leq 0$. □

For $\mathrm{U}(n)$ the invariant inner product can be written down directly: $\langle X, Y\rangle = -\mathrm{tr}(XY)$ on $\mathfrak{u}(n)$ is positive definite, since $-\mathrm{tr}(X^2) = \mathrm{tr}(X^*X) = \sum_{i,j}|x_{ij}|^2$, and it is invariant under $\mathrm{Ad}(A)X = AXA^{-1}$ because the trace is invariant under conjugation.

Proposition 34.5.5. *Let G be a compact Lie group. Then $\mathfrak{g} = \mathfrak{z}(\mathfrak{g}) \oplus [\mathfrak{g},\mathfrak{g}]$, the ideal $[\mathfrak{g},\mathfrak{g}]$ is semisimple, and its Killing form, which is the restriction of the Killing form B of $\mathfrak{g}$, is negative definite. Conversely, a real Lie algebra with negative definite Killing form is semisimple.*

Proof. Choose an inner product as in Proposition 34.5.4. The adjoint of $\operatorname{ad} X$ is $-\operatorname{ad} X = \operatorname{ad}(-X)$, so Proposition 13.7.13 applies: $\mathfrak{g} = \mathfrak{z}(\mathfrak{g}) \oplus [\mathfrak{g}, \mathfrak{g}]$ with $[\mathfrak{g}, \mathfrak{g}]$ semisimple. The Killing form of the ideal $[\mathfrak{g}, \mathfrak{g}]$ is the restriction of B (Lemma 13.7.2(4)). For $X \in \mathfrak{g}$, $B(X, X) = \operatorname{tr}((\operatorname{ad} X)^2) = -\operatorname{tr}((\operatorname{ad} X)^T \operatorname{ad} X) \leq 0$, with equality only if $\operatorname{ad} X = 0$, that is, $X \in \mathfrak{z}(\mathfrak{g})$; as $[\mathfrak{g}, \mathfrak{g}] \cap \mathfrak{z}(\mathfrak{g}) = 0$, B is negative definite on $[\mathfrak{g}, \mathfrak{g}]$. The converse is Cartan's criterion, Theorem 13.7.8, since a definite form is nondegenerate. □

Example 34.5.6. $\mathfrak{u}(n) = i\mathbb{R} \cdot I \oplus \mathfrak{su}(n)$, where $i\mathbb{R} \cdot I$ is the centre and $\mathfrak{su}(n)$ is semisimple for $n \geq 2$; $\mathfrak{so}(n)$ is semisimple for $n \geq 3$ (Example 13.7.14). The torus T^n has abelian Lie algebra, with $[\mathfrak{g}, \mathfrak{g}] = 0$. Not every real semisimple Lie algebra comes from a compact group in this way: $\mathfrak{sl}_2(\mathbb{R})$ is semisimple, but its Killing form $4\operatorname{tr}(XY)$ (Example 13.7.9 with $k = \mathbb{R}$) takes the value 8 on $X = \operatorname{diag}(1, -1)$, so by the proposition $\mathfrak{sl}_2(\mathbb{R})$ is not the Lie algebra of a compact group. Its Killing form has signature $(2, 1)$: it is negative on the rotation generator $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, the Lie algebra of the compact subgroup SO(2).

Remark 34.5.7. Deeper structure theory, which we quote when needed: a connected Lie group has a unique simply connected covering group, and Lie algebra homomorphisms integrate to group homomorphisms from the simply connected group (Lie's second theorem); Lie subalgebras correspond to connected Lie subgroups; every compact connected Lie group is a quotient of a product of a torus and simply connected compact simple groups, which are classified by their Dynkin diagrams; and every connected compact Lie group is the maximal compact subgroup of a unique complex reductive group, its *complexification* (for instance $U(n) \subseteq \mathrm{GL}_n(\mathbb{C})$). See [Kna02] and [BtD85]. The unitary trick, which deduces complete reducibility of representations of complex reductive groups from Corollary 34.5.3, is used in Chapter 67.

34.6 Exercises

Exercise 34.6.1. Show that $\exp\colon \mathfrak{sl}_2(\mathbb{R}) \to \mathrm{SL}_2(\mathbb{R})$ is not surjective, although exp is surjective for compact connected Lie groups and for $\mathrm{GL}_n(\mathbb{C})$.

Solution. If $X \in \mathfrak{sl}_2(\mathbb{R})$ has eigenvalues $\pm\lambda$, then e^X has eigenvalues $e^{\pm\lambda}$ (triangularise X over $\mathbb{C}$ and use Lemma 34.3.1(3)). If these are real and distinct, then λ is real (for $\lambda = i\theta$, $e^{\pm i\theta}$ is real only if it equals ± 1, and then the two eigenvalues coincide), so both are positive. The matrix $\operatorname{diag}(-2, -1/2) \in \mathrm{SL}_2(\mathbb{R})$ has distinct negative eigenvalues, so it is not in the image. For $\mathrm{GL}_n(\mathbb{C})$: by the Jordan form it suffices to treat a block $\lambda(I + N)$ with $\lambda \neq 0$ and N nilpotent; choose c with $e^c = \lambda$ and put $L = \sum_{k \geq 1} (-1)^{k+1} N^k / k$, a finite sum. The identity $\exp(\log(1 + x)) = 1 + x$ of formal power series gives $e^L = I + N$, and L commutes with cI, so $e^{cI+L} = \lambda(I + N)$. For compact connected groups, surjectivity follows from the theory of maximal tori [BtD85, Chapter IV]. □

Exercise 34.6.2. Show that SU(2) is diffeomorphic to S^3, that its Lie algebra is $\mathfrak{su}(2) \cong \mathbb{R}^3$ with the cross product, and that the adjoint representation gives a double covering $\mathrm{SU}(2) \to \mathrm{SO}(3)$.

Solution. The first two statements are Example 34.2.11, together with Example 34.2.10. For the third, the form $\langle X, Y \rangle = -2\operatorname{tr}(XY)$ on $\mathfrak{su}(2)$ makes the basis e_1, e_2, e_3 of Example 34.2.11 orthonormal (for instance $\operatorname{tr}(e_1^2) = -1/2$ and $\operatorname{tr}(e_1 e_2) = \operatorname{tr}(I_3)/4 = 0$), and it is invariant under $\operatorname{Ad}(A)X = AXA^{-1}$. So in this basis Ad is a homomorphism $\mathrm{SU}(2) \to \mathrm{O}(3)$; since $\mathrm{SU}(2) \cong S^3$ is connected and $\det \circ \operatorname{Ad}$ is continuous with values ± 1 and value 1 at I, it maps into SO(3). Its kernel consists of the A commuting with I_1, I_2, I_3; these together with I span $M_2(\mathbb{C})$ over $\mathbb{C}$, so $A = \lambda I$, and $\det A = \lambda^2 = 1$ gives $A = \pm I$. Its differential $\operatorname{ad}\colon \mathfrak{su}(2) \to \mathfrak{so}(3)$ is injective, because $\operatorname{ad} X = 0$ means that X commutes with e_1, e_2, e_3, which forces $X = 0$

by the bracket relations; so it is an isomorphism, both spaces having dimension 3. As Ad is a homomorphism, its differential is then an isomorphism everywhere, so Ad is a local diffeomorphism. Its image is open, and compact hence closed, so it is all of the connected group SO(3) (Proposition 34.4.6). The induced bijection $\mathrm{SU}(2)/\{\pm I\} \to \mathrm{SO}(3)$ is a local diffeomorphism, hence a diffeomorphism, where $\mathrm{SU}(2)/\{\pm I\}$ is the quotient of S^3 by the free action of $\{\pm 1\}$, with covering map of degree 2 (Theorem 33.7.1). In particular $\mathrm{SO}(3) \cong \mathbb{RP}^3$. □

Exercise 34.6.3. Show that the image of the homomorphism $\mathbb{R} \to T^2$, $t \mapsto (t, \alpha t) \bmod \mathbb{Z}^2$, with α irrational, is a dense subgroup that is not closed and not an embedded submanifold; so "closed" cannot be dropped in Theorem 34.4.1.

Solution. Identifying $\mathbb{R}/\mathbb{Z}$ with S^1, this is the map of Counterexample 33.6.17(2): its image is a dense subgroup, and not a submanifold. It is not closed, since it is dense and not all of T^2 (it meets the circle $\{0\} \times S^1$ in a countable set). □

Exercise 34.6.4. Verify that $\mathrm{Gr}(k, n) = O(n)/(O(k) \times O(n-k))$ has dimension $k(n-k)$ and is connected, and identify $\mathrm{Gr}(1, n)$ with $\mathbb{RP}^{n-1}$.

Solution. By Theorem 34.4.4 and Example 34.2.5(2), the dimension is $\frac{n(n-1)}{2} - \frac{k(k-1)}{2} - \frac{(n-k)(n-k-1)}{2} = \frac{n^2-k^2-(n-k)^2}{2} = k(n-k)$. It is connected, being the image of the connected group $\mathrm{SO}(n)$, which already acts transitively for $0 < k < n$ (change the sign of one basis vector of $W^\perp$). For $k = 1$ it is the space of lines, $\mathbb{RP}^{n-1}$ (Proposition 33.7.4). □

Exercise 34.6.5. Show that $\exp\colon \mathfrak{so}(3) \to \mathrm{SO}(3)$ is surjective: every rotation of $\mathbb{R}^3$ is $e^{\theta L_a}$ for a unit vector a and an angle θ, with L_a as in Example 34.2.10.

Solution. Let $A \in \mathrm{SO}(3)$. Its characteristic polynomial is real of degree 3, and its complex eigenvalues have modulus 1 and product 1; the nonreal ones come in conjugate pairs, so 1 is an eigenvalue. Let a be a unit vector with $Aa = a$. Then A preserves the plane $a^\perp$ and acts on it by an element of SO(2), a rotation by some angle θ. In an orthonormal basis $a, b, a \times b$, the matrix L_a acts on $a^\perp$ as J of Example 34.3.2(3) and kills a, so $e^{\theta L_a}$ fixes a and rotates $a^\perp$ by θ; hence $e^{\theta L_a} = A$. □

Exercise 34.6.6. Let N be the group of real upper triangular 3×3 matrices with 1 on the diagonal (the *Heisenberg group*). Show that N is a closed subgroup of $\mathrm{GL}_3(\mathbb{R})$ with Lie algebra $\mathfrak{n}$ the strictly upper triangular matrices, compute the bracket on $\mathfrak{n}$, and show that $\exp\colon \mathfrak{n} \to N$ is a diffeomorphism.

Solution. N is defined by polynomial equations on the entries, so it is closed, and it is a subgroup. For $X \in \mathfrak{n}$ we have $X^3 = 0$, so $e^X = I + X + X^2/2 \in N$; conversely, if $e^{tX} \in N$ for all t, differentiating at 0 shows $X \in \mathfrak{n}$. So $\mathfrak{n}$ is the Lie algebra (Theorem 34.4.1). With X having entries x_{12}, x_{13}, x_{23}, the bracket $[X, Y]$ has the single entry $x_{12}y_{23} - y_{12}x_{23}$ in position $(1, 3)$. The inverse of exp is $\log(I + M) = M - M^2/2$ for M strictly upper triangular, since $M^3 = 0$; both maps are polynomial, hence smooth, and they are inverse to each other by the formal identities $\log(\exp x) = x$ and $\exp(\log(1 + x)) = 1 + x$. □

Exercise 34.6.7. Let G be a connected Lie group. Show that its centre is the kernel of Ad, and that its Lie algebra is the centre $\{X : [X, Y] = 0 \text{ for all } Y\}$ of $\mathfrak{g}$.

Solution. By naturality of exp, $c_g(\exp X) = \exp(\mathrm{Ad}(g)X)$. If $\mathrm{Ad}(g) = \mathrm{id}$, then g commutes with $\exp(\mathfrak{g})$, which generates G (Proposition 34.3.9), so g is central; conversely, if g is central then $c_g = \mathrm{id}$ and $\mathrm{Ad}(g) = \mathrm{id}$. By Theorem 34.4.1, the Lie algebra of the centre is $\{X : \exp(tX) \in Z(G) \text{ for all } t\} = \{X : \mathrm{Ad}(\exp tX) = \mathrm{id} \text{ for all } t\}$. Since $\mathrm{Ad}(\exp tX) = e^{t\,\mathrm{ad}\,X}$ (Proposition 34.3.6), this holds for all t if and only if $\mathrm{ad}\,X = 0$, by differentiating at $t = 0$. □

Exercise 34.6.8. Show that the Killing form of $\mathfrak{su}(n)$ is $B(X,Y) = 2n\,\mathrm{tr}(XY)$, and check directly that it is negative definite. For $\mathfrak{so}(3)$, identified with $\mathbb{R}^3$ with the cross product (Example 34.2.10), show that $B(x,y) = -2\,x\cdot y$.

Solution. Every $Z \in \mathfrak{sl}_n(\mathbb{C})$ is uniquely $X + iY$ with $X, Y \in \mathfrak{su}(n)$, namely $X = \frac{1}{2}(Z - Z^*)$ and $Y = \frac{1}{2i}(Z + Z^*)$, so a real basis of $\mathfrak{su}(n)$ is a complex basis of $\mathfrak{sl}_n(\mathbb{C})$. In such a basis, for $X \in \mathfrak{su}(n)$ the matrix of $\operatorname{ad} X$ on $\mathfrak{su}(n)$ and on $\mathfrak{sl}_n(\mathbb{C})$ is the same, so the Killing form of $\mathfrak{su}(n)$ is the restriction of that of $\mathfrak{sl}_n(\mathbb{C})$, which is $2n\,\mathrm{tr}(XY)$ (Example 13.7.9). For $X \neq 0$ skew-Hermitian, $\mathrm{tr}(X^2) = -\mathrm{tr}(XX^*) = -\sum_{i,j}|x_{ij}|^2 < 0$. For $\mathfrak{so}(3)$ with $[x,y] = x \times y$, in the standard basis $\operatorname{ad} e_1$ maps $e_1 \mapsto 0$, $e_2 \mapsto e_3$, $e_3 \mapsto -e_2$, so $(\operatorname{ad} e_1)^2 = \mathrm{diag}(0,-1,-1)$ and $B(e_1,e_1) = -2$; similarly for e_2, e_3, and $B(e_i,e_j) = 0$ for $i \neq j$, because $\operatorname{ad} e_i \operatorname{ad} e_j$ maps e_j and the third basis vector e_l to 0 (as $e_j \times e_l = \pm e_i$) and e_i to $e_i \times (e_j \times e_i) = e_j$, so it has zero diagonal. So $B(x,y) = -2\,x\cdot y$. □

Exercise 34.6.9.

1. Let G be a group with a smooth structure for which the multiplication m is smooth. Show that the inversion is automatically smooth, so that in Definition 34.2.1 the condition on i can be omitted. (Hint: consider $\Phi\colon G\times G \to G\times G$, $\Phi(g,h) = (g,gh)$.)

2. Show that the analogous statement for topological groups fails: on the Sorgenfrey line $\mathbb{R}_S$ (Counterexample 18.3.13), addition $\mathbb{R}_S \times \mathbb{R}_S \to \mathbb{R}_S$ is continuous, but $x \mapsto -x$ is not.

Solution. (1) Each L_g is smooth, being the composite of $x \mapsto (g,x)$ with m, and it has the smooth inverse $L_{g^{-1}}$; so $d_h(L_g)$ is an isomorphism. The formula of Lemma 34.2.2 for dm only uses the smoothness of m, and it gives $d_{(g,h)}\Phi(v,w) = (v, d_g(R_h)(v) + d_h(L_g)(w))$. This is injective: if it vanishes, then $v = 0$ and $d_h(L_g)(w) = 0$, so $w = 0$. So $d\Phi$ is everywhere an isomorphism. Φ is bijective, with inverse $(g,k) \mapsto (g, g^{-1}k)$, so by the inverse function theorem (Theorem 33.4.14) it is a diffeomorphism. Then $i(g) = \mathrm{pr}_2(\Phi^{-1}(g,e))$ is a composite of smooth maps.

(2) Let $x + y \in [c,d)$. For $\varepsilon > 0$ with $x + y + 2\varepsilon \leq d$, the open set $[x, x+\varepsilon) \times [y, y+\varepsilon)$ of $\mathbb{R}_S \times \mathbb{R}_S$ contains (x,y) and is mapped into $[x+y, x+y+2\varepsilon) \subseteq [c,d)$; so the preimage of a basic open set is open, and addition is continuous. But the preimage of the open set $[0,1)$ under $x \mapsto -x$ is $(-1,0]$, which contains 0 and no interval $[0,\varepsilon)$, so it is not open. Thus $\mathbb{R}_S$ is a group with continuous multiplication that is not a topological group. The difference with (1) is that a smooth structure gives the inverse function theorem, which has no topological counterpart. □

Chapter 35

Vector Bundles

35.1 Introduction

A vector bundle is a family of vector spaces E_p, one for each point p of a manifold M, glued together smoothly. The model is the tangent bundle: at each point of a surface in $\mathbb{R}^3$ there is a tangent plane, and as the point moves the plane moves with it. Locally such a family looks like a product $U \times \mathbb{K}^r$, a *local trivialisation*; globally it may be twisted, as the Möbius band is a twisted version of the cylinder. The twisting is recorded by *transition functions*: where two trivialisations overlap, they differ by a smoothly varying invertible matrix.

Vector bundles appear everywhere in this book. Tangent and cotangent bundles and the bundles of differential forms are the setting of analysis on manifolds; sections of bundles are "twisted functions", which is how the line bundles of complex and algebraic geometry encode divisors and linear systems; and the amount of twisting is measured by characteristic classes, the first of which, the first Chern class of a complex line bundle, is constructed at the end of this chapter.

The chapter goes as follows.

1. Definitions, examples, and the description of bundles by cocycles of transition functions (Section 35.2), with the Möbius bundle, the tangent bundle and the tautological bundles worked out.

2. Sections and frames, and the dictionary between vector bundles and locally free sheaves.

3. The linear algebra of bundles: direct sums, tensor products, duals, Hom bundles, exterior powers, pullbacks; metrics, subbundles, kernels and normal bundles (Section 35.3).

4. Complex line bundles: their classification by $H^2(M;\mathbb{Z})$, and an elementary classification on the sphere by the winding number of a transition function (Section 35.4).

5. Parallelizable manifolds, whose tangent bundle is trivial: Lie groups, S^3, $S^1 \times S^n$, and the theorem that S^n is parallelizable only for $n = 0, 1, 3, 7$ (Section 35.5).

References: [Lee13], [MS74], [GH78, Chapter 0].

Prerequisites. Chapter 33 (Smooth Manifolds), Chapter 12 (Multilinear Algebra) and Chapter 25 (Sheaves on Topological Spaces).

Conventions. $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, and M is a smooth manifold. Vectors of $\mathbb{K}^r$ are columns, and $e_1, \dots, e_r$ is the standard basis. $\mathrm{GL}_r(\mathbb{K})$ is the group of invertible $r \times r$ matrices, an open subset of $M_r(\mathbb{K})$.

35.2 Vector bundles and cocycles

Definition 35.2.1. A *vector bundle* of rank r over M is a smooth map $\pi\colon E \to M$ together with $\mathbb{K}$-vector space structures on the fibres $E_p = \pi^{-1}(p)$, such that every point of M has an open neighbourhood U and a diffeomorphism $\Phi\colon \pi^{-1}(U) \to U \times \mathbb{K}^r$ (a *local trivialisation*) with $\mathrm{pr}_1 \circ \Phi = \pi$, linear on each fibre. A *morphism* of vector bundles over M is a smooth $f : E \to F$ with $\pi_F \circ f = \pi_E$, linear on fibres; *line bundle* means rank 1.

Unwinding the definition: E is a manifold, π maps it onto M, each fibre E_p is a vector space of dimension r, and over a neighbourhood of each point the manifold E is identified with $U \times \mathbb{K}^r$ in such a way that the fibre E_p goes to $\{p\} \times \mathbb{K}^r$ by a linear isomorphism $\Phi_p : E_p \to \mathbb{K}^r$. The rank is part of the data; one could allow it to vary, but it is locally constant (because of the local trivialisations), so on a connected manifold it is constant. An *isomorphism* is a morphism with an inverse morphism, and E is *trivial* if it is isomorphic to the *trivial bundle* $M \times \mathbb{K}^r$ with the projection to M. A complex vector bundle of rank r is also a real one of rank $2r$, by forgetting the complex structure.

Lemma 35.2.2. *A morphism of vector bundles over M that is bijective on every fibre is an isomorphism.*

Proof. The inverse map is linear on fibres; we must show it is smooth. This is local, so we may assume both bundles are trivial, $U \times \mathbb{K}^r$, and the morphism is $(p, v) \mapsto (p, A(p)v)$ for a smooth map $A\colon U \to M_r(\mathbb{K})$ (its entries are the components of the images of the constant sections e_j). By hypothesis $A(p)$ is invertible, and $A(p)^{-1}$ is smooth in p by Cramer's rule, since the entries of the inverse are polynomials in the entries of A divided by $\det A \neq 0$. So the inverse $(p, w) \mapsto (p, A(p)^{-1}w)$ is smooth. □

Example 35.2.3. 1. *The trivial bundle $M \times \mathbb{K}^r$.*

2. *The tangent bundle TM* (Definition 33.5.1). A chart $\varphi = (x^1, \dots, x^n)$ on U gives the local trivialisation $T_pM \to \mathbb{R}^n$, $v \mapsto (dx^1(v), \dots, dx^n(v))$, sending $\sum_i v^i \partial/\partial x^i$ to $(v^1, \dots, v^n)$. Its dual, the *cotangent bundle* T^*M, has fibres $T_p^*M = (T_pM)^*$.

3. *The Möbius bundle* over $S^1 = \mathbb{R}/\mathbb{Z}$ is the quotient of $\mathbb{R} \times \mathbb{R}$ by $(x, t) \sim (x + 1, -t)$, with the projection $[x, t] \mapsto [x]$. Equivalently, it is obtained from $[0, 1] \times \mathbb{R}$ by gluing $(0, t)$ to $(1, -t)$. Over each interval $\{[x] : a < x < a + 1\}$ it is trivialised by $[x, t] \mapsto ([x], t)$ with $a < x < a + 1$.

4. *The tautological line bundles.* On $\mathbb{RP}^n$ and on $\mathbb{CP}^n$, $\mathcal{O}(-1) = \{(\ell, v) : v \in \ell\} \subseteq \mathbb{KP}^n \times \mathbb{K}^{n+1}$, whose fibre over a line ℓ is the line itself (Exercise 35.6.3).

5. *The normal bundle $N_{S/M} = (TM|_S)/TS$* of a submanifold $S \subseteq M$ (Proposition 35.3.7); for $S \subseteq \mathbb{R}^N$ it can be realised as $\{(p, v) \in S \times \mathbb{R}^N : v \perp T_pS\}$.

Proposition 35.2.4. *The tangent bundle of a Lie group G is trivial: $G \times \mathfrak{g} \to TG$, $(g, v) \mapsto d(L_g)_e(v)$, is an isomorphism of vector bundles. In particular S^1, $S^3 \cong \mathrm{SU}(2)$ and the tori T^n have trivial tangent bundles.*

Proof. The map is smooth, as the differential of multiplication restricted to $G \times T_eG$, it covers the identity of G, and it is a linear isomorphism on each fibre since L_g is a diffeomorphism. By Lemma 35.2.2 it is an isomorphism. For S^3 note that $\mathrm{SU}(2) = \{\begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix} : |a|^2 + |b|^2 = 1\}$ is diffeomorphic to $S^3 \subseteq \mathbb{C}^2$. □

Counterexample 35.2.5. The Möbius bundle E over S^1 (Example 35.2.3) is not trivial. A nowhere vanishing section would correspond to a continuous $s\colon [0,1] \to \mathbb{R}\setminus\{0\}$ with $s(1) = -s(0)$, and by the intermediate value theorem s would vanish somewhere; and a real line bundle with a nowhere vanishing section s is trivial, by $(p,t) \mapsto ts(p)$. The tautological line bundle of $\mathbb{RP}^1$ is isomorphic to E: parametrising $\mathbb{RP}^1$ by $\theta \in [0,\pi]$, with θ corresponding to the line through $u(\theta) = (\cos\theta, \sin\theta)$, the section u trivialises the bundle over $[0,\pi]$ and $u(\pi) = -u(0)$, which is the gluing of E. Hence the tautological line bundle of $\mathbb{RP}^n$, which restricts to it on $\mathbb{RP}^1 \subseteq \mathbb{RP}^n$, is nontrivial as well.

Two local trivialisations Φ_i and Φ_j of the same bundle can be compared on the overlap of their domains: $\Phi_i \circ \Phi_j^{-1}$ maps (p,v) to $(p, g_{ij}(p)v)$, where $g_{ij}(p) = \Phi_{i,p} \circ \Phi_{j,p}^{-1}$ is the invertible matrix that converts the coordinates of a vector of E_p in the trivialisation j into its coordinates in the trivialisation i. These matrices are the transition functions, and they determine the bundle.

Proposition 35.2.6. *Let (U_i) be an open cover of M.*

1. *A vector bundle with trivialisations Φ_i over U_i determines* transition functions $g_{ij}\colon U_i \cap U_j \to \mathrm{GL}_r(\mathbb{K})$*, smooth, with $\Phi_i\Phi_j^{-1}(p,v) = (p, g_{ij}(p)v)$; they satisfy the* cocycle condition *$g_{ij}g_{jk} = g_{ik}$ on triple intersections, and $g_{ii} = 1$.*

2. *Conversely, every such family (g_{ij}) arises from a vector bundle, unique up to isomorphism, namely $E = (\bigsqcup_i U_i \times \mathbb{K}^r)/\!\sim$ with $(p,v)_j \sim (p, g_{ij}(p)v)_i$.*

3. *Two cocycles (g_{ij}), (g'_{ij}) define isomorphic bundles if and only if there are smooth $h_i\colon U_i \to \mathrm{GL}_r(\mathbb{K})$ with $g'_{ij} = h_i g_{ij} h_j^{-1}$. Hence isomorphism classes of rank r bundles correspond to $\check{H}^1(M, \underline{\mathrm{GL}_r})$, Čech cohomology with values in the sheaf of smooth $\mathrm{GL}_r(\mathbb{K})$-valued functions (Remark 26.5.4).*

Proof. (1) The map $\Phi_i\Phi_j^{-1}$ is a diffeomorphism of $(U_i \cap U_j) \times \mathbb{K}^r$ that commutes with the projection to $U_i \cap U_j$ and is linear on fibres, so it has the form $(p,v) \mapsto (p, g_{ij}(p)v)$ with $g_{ij}(p)$ invertible. The entries of g_{ij} are smooth: the k-th column of $g_{ij}(p)$ is the second component of $\Phi_i\Phi_j^{-1}(p, e_k)$, a smooth function of p. On $U_i \cap U_j \cap U_k$, $\Phi_i\Phi_k^{-1} = (\Phi_i\Phi_j^{-1})(\Phi_j\Phi_k^{-1})$, which is $g_{ik} = g_{ij}g_{jk}$, and $g_{ii} = 1$. In particular $g_{ji} = g_{ij}^{-1}$.

(2) The relation $\sim$ is reflexive because $g_{ii} = 1$, symmetric because $g_{ji} = g_{ij}^{-1}$, and transitive because of the cocycle condition. Let E be the quotient, $\pi[p,v]_i = p$, and give E_p the vector space structure transported from $\mathbb{K}^r$ by $v \mapsto [p,v]_i$ for any i with $p \in U_i$; this does not depend on i, as the identifications differ by the linear maps $g_{ij}(p)$. The maps $\Phi_i\colon \pi^{-1}(U_i) \to U_i \times \mathbb{K}^r$, $[p,v]_i \mapsto (p,v)$, are bijective, and their transition maps $(p,v) \mapsto (p, g_{ij}(p)v)$ are diffeomorphisms of the open sets $(U_i \cap U_j) \times \mathbb{K}^r$. Composing with charts of M, the Φ_i form an atlas of E (Hausdorff and second countable because M is and the cover can be taken countable), for which π is smooth and the Φ_i are local trivialisations with transition functions g_{ij}. If a bundle E' has trivialisations Φ'_i with the same transition functions, then $\Phi_i'^{-1}\Phi_i$ on $\pi^{-1}(U_i)$ agree on overlaps and glue to an isomorphism $E \to E'$.

(3) An isomorphism $f : E \to E'$ is given in the trivialisations by $\Phi'_i f \Phi_i^{-1}(p,v) = (p, h_i(p)v)$ with h_i smooth and invertible (as in (1)). On $U_i \cap U_j$, computing $\Phi'_i f \Phi_j^{-1}$ in two ways gives $h_i g_{ij} = g'_{ij} h_j$. Conversely, given such h_i, the maps $\Phi_i'^{-1} h_i \Phi_i$ agree on overlaps and define an isomorphism, which is smooth with smooth inverse. The relation $g'_{ij} = h_i g_{ij} h_j^{-1}$ is exactly the statement that the two Čech cocycles are cohomologous. □

Example 35.2.7. 1. *The tangent bundle.* For the trivialisations of TM given by charts φ and ψ (Example 35.2.3(2)), a vector with components ξ in the chart φ has components $D\tau(\varphi(p))\xi$ in the chart ψ, where $\tau = \psi \circ \varphi^{-1}$ (Proposition 33.5.2). So the transition functions of TM are the Jacobian matrices of the transition maps of the atlas, and the cocycle condition is the chain rule.

2. *The sphere.* For the two stereographic charts of S^n (Example 18.9.4), the transition map is the inversion $\tau(y) = y/|y|^2$ of $\mathbb{R}^n \setminus \{0\}$, whose Jacobian matrix is

$$D\tau(y) = \frac{1}{|y|^2}\Big(I - \frac{2yy^T}{|y|^2}\Big),$$

the reflection in the hyperplane $y^\perp$ divided by $|y|^2$. It has determinant $-|y|^{-2n}$: the two charts induce opposite orientations. This single matrix-valued function on $\mathbb{R}^n \setminus \{0\}$ describes TS^n completely.

3. *The Möbius bundle.* With the two trivialisations of Example 35.2.3(3) over the intervals $0 < x < 1$ and $\frac{1}{2} < x < \frac{3}{2}$, the overlap consists of two arcs; on one the transition function is 1 and on the other it is -1. A cocycle with values in $\{\pm 1\} \subseteq \mathrm{GL}_1(\mathbb{R})$ describes a bundle with locally constant transition functions, a *flat* bundle (Chapter 38).

4. *The tautological line bundle of $\mathbb{CP}^1$.* Over $U_0 = \{[1 : z]\}$ and $U_1 = \{[w : 1]\}$ the line ℓ has the basis vectors $s_0 = (1, z)$ and $s_1 = (w, 1)$, which trivialise $\mathcal{O}(-1)$. On $U_0 \cap U_1$, where $w = 1/z$, $s_1 = z^{-1}s_0$. So a vector $v = a\,s_0 = b\,s_1$ has coordinates related by $a = z^{-1}b$: the transition function is $g_{01} = z^{-1}$, a function $\mathbb{C}^\times \to \mathrm{GL}_1(\mathbb{C}) = \mathbb{C}^\times$.

Definition 35.2.8. A *section* of E over $U \subseteq M$ open is a smooth $s : U \to E$ with $\pi \circ s = \mathrm{id}$; they form a $C^\infty(U)$-module $\Gamma(U, E)$, and $U \mapsto \Gamma(U, E)$ is a sheaf $\mathcal{E}$ of C^∞_M-modules. A *frame* over U is a tuple of sections that is a basis of each fibre.

A section picks a vector in each fibre, smoothly. In a trivialisation Φ over U a section is the same as a smooth function $U \to \mathbb{K}^r$, namely $\Phi \circ s$; so sections are "functions twisted by the bundle". The sections of the trivial line bundle are the functions, the sections of TM are the vector fields, and those of T^*M are the 1-forms. Every bundle has the zero section, and many others: if $v \in E_p$, a bump function times a local section through v gives a global section with value v at p.

Example 35.2.9. 1. *Sections of the Möbius bundle* are the smooth functions $f : \mathbb{R} \to \mathbb{R}$ with $f(x + 1) = -f(x)$, such as $\sin(\pi x)$, which vanishes exactly at the integers. Every section vanishes somewhere (Counterexample 35.2.5).

2. *Sections of a bundle given by a cocycle* are families of functions $s_i : U_i \to \mathbb{K}^r$ with $s_i = g_{ij}s_j$ on $U_i \cap U_j$. For $\mathcal{O}(-1)$ on $\mathbb{CP}^1$, with Example 35.2.7(4), a section is a pair of functions a on U_0 and b on U_1 with $a = z^{-1}b$; holomorphic sections would be power series $a(z)$ and $b(w)$ with $a(z) = z^{-1}b(1/z)$, and comparing the powers of z shows that only $a = b = 0$ is possible. So $\mathcal{O}(-1)$ has no nonzero holomorphic section, although it has many smooth ones (Chapter 45).

Proposition 35.2.10. *$\mathcal{E}$ is locally free of rank r (Definition 25.6.3): a trivialisation over U is the same as a frame over U, and gives $\mathcal{E}|_U \cong (C^\infty_U)^r$. Conversely every locally free C^∞_M-module of rank r is the sheaf of sections of a vector bundle, unique up to isomorphism, and this correspondence is an equivalence between vector bundles and locally free sheaves.*

Proof. A trivialisation Φ gives the frame $s_i(p) = \Phi^{-1}(p, e_i)$, and conversely a frame defines $\Phi(v) = (p, (v^1, \dots, v^r))$ where $v = \sum v^i s_i(p)$; smoothness in both directions follows because the coefficients are smooth functions (they are the components in a trivialisation). Given a locally free sheaf $\mathcal{F}$, choose isomorphisms $\mathcal{F}|_{U_i} \cong (C^\infty_{U_i})^r$; on overlaps they differ by automorphisms of $(C^\infty)^r$, that is, by matrices g_{ij} of smooth functions with smooth inverse, which form a cocycle. The bundle of Proposition 35.2.6(2) has $\mathcal{F}$ as its sheaf of sections. Morphisms correspond in the same way, since a morphism of bundles is determined by its action on sections. □

35.3 Operations on bundles

Every construction of linear algebra that is applied to vector spaces can be applied fibre by fibre to vector bundles. The principle is always the same: apply the construction to the transition functions, and use Proposition 35.2.6(2).

Proposition 35.3.1. *Let E, F be vector bundles over M of ranks r, s, with cocycles (g_{ij}), (h_{ij}), and let $f : N \to M$ be smooth.*

1. *There are vector bundles $E \oplus F$, $E \otimes F$, $E^\vee$, $\operatorname{Hom}(E, F)$, $\Lambda^k E$ and $\operatorname{Sym}^k E$ whose fibres are the corresponding constructions on the fibres, with cocycles $g_{ij} \oplus h_{ij}$, $g_{ij} \otimes h_{ij}$, $(g_{ij}^T)^{-1}$, and so on. Their sheaves of sections are the corresponding operations on locally free sheaves.*
2. *The* pullback *f^*E, with fibre $(f^*E)_q = E_{f(q)}$ and cocycle $g_{ij} \circ f$ for the cover $(f^{-1}U_i)$, is a vector bundle on N; f^* is functorial and $(g \circ f)^* \cong f^* g^*$.*
3. *$\det E = \Lambda^r E$ is a line bundle with cocycle $\det g_{ij}$.*

Explicitly, the cocycles are: for $E \oplus F$ the block matrices $\left(\begin{smallmatrix} g_{ij} & 0 \\ 0 & h_{ij} \end{smallmatrix}\right)$; for $E \otimes F$ the Kronecker products $g_{ij} \otimes h_{ij}$, which act on $v \otimes w$ by $g_{ij}v \otimes h_{ij}w$; for the dual $E^\vee$, $(g_{ij}^T)^{-1}$, because a linear form with coordinates ξ (a row vector ξ^T) must satisfy $\xi_i^T v_i = \xi_j^T v_j$ when $v_i = g_{ij} v_j$; for $\operatorname{Hom}(E, F)$, the map $A \mapsto h_{ij} A g_{ij}^{-1}$ on matrices; for $\Lambda^k E$ the matrices of $k \times k$ minors of g_{ij}; and for $\det E$ the determinants.

Proof. In each case the fibrewise construction is a functor on finite-dimensional vector spaces and isomorphisms, and the transition matrices of the new bundle are obtained from the g_{ij} and h_{ij} by applying this functor. Their entries are polynomials in the entries of g_{ij}, h_{ij}, possibly divided by $\det g_{ij}$, so they are smooth; and functoriality turns the cocycle conditions for (g_{ij}) and (h_{ij}) into the cocycle condition for the new family. By Proposition 35.2.6(2) there is a bundle with these transition functions, and its fibre over p is canonically the construction applied to E_p and F_p, since the identifications with $\mathbb{K}^r$ and $\mathbb{K}^s$ over different U_i differ exactly by the transition matrices. The sheaf of sections is computed locally, where everything is trivial. For the pullback, the functions $g_{ij} \circ f$ are smooth and satisfy the cocycle condition on the cover $(f^{-1}U_i)$, and $(f^*E)_q = E_{f(q)}$ canonically; functoriality holds because the cocycle of $(g \circ f)^*E$ is that of $f^*(g^*E)$. For the determinant, Λ^r of an $r \times r$ matrix is multiplication by its determinant. □

Example 35.3.2. 1. The cotangent bundle T^*M has transition functions $(D\tau^T)^{-1}$, the inverse transposes of the Jacobian matrices; the bundle $\Lambda^k T^*M$ of k-forms has the $k \times k$ minors of these, and the bundle $\Lambda^n T^*M$ of top degree forms has transition functions $\det(D\tau)^{-1}$. It is trivial when the atlas can be chosen with $\det D\tau > 0$, which is the orientability of M (Chapter 36).

2. The line bundles $\mathcal{O}(k)$ on $\mathbb{CP}^1$: $\mathcal{O}(1) = \mathcal{O}(-1)^\vee$ has transition function z, and $\mathcal{O}(k) = \mathcal{O}(1)^{\otimes k}$ for $k \geq 0$, $\mathcal{O}(k) = \mathcal{O}(-1)^{\otimes(-k)}$ for $k < 0$, has transition function z^k with respect to the covering U_0, U_1 (for line bundles, tensor products multiply the transition functions, and duals invert them).

3. *Restrictions and constant maps.* For the inclusion $\iota\colon S \to M$ of a submanifold, $\iota^*E = E|_S$ is the restriction. For a constant map $f : N \to M$ with value p, $f^*E = N \times E_p$ is trivial.

Proposition 35.3.3. *Let $\pi\colon E \to M$ be a vector bundle and $f : N \to M$ smooth. The fibre product*

$$N \times_M E = \{(q, e) \in N \times E : f(q) = \pi(e)\}$$

is a submanifold of $N \times E$ (Proposition 33.6.12), and the projection to N, with the vector space structures of the fibres $E_{f(q)}$, is a vector bundle isomorphic to the pullback f^*E of Proposition 35.3.1(2). It has the universal property of the fibre product: for a manifold L, the smooth maps $L \to f^*E$ correspond to pairs of smooth maps $a\colon L \to N$ and $b\colon L \to E$ with $f \circ a = \pi \circ b$. In particular the sections of f^*E over N correspond to the smooth maps $s\colon N \to E$ with $\pi \circ s = f$, and a section t of E pulls back to the section $f^*t = t \circ f$.

Proof. A vector bundle projection is a submersion, since in a local trivialisation it is the projection $U \times \mathbb{K}^r \to U$, so Proposition 33.6.12 applies and gives the submanifold and the universal property. If $\Phi\colon \pi^{-1}(U) \to U \times \mathbb{K}^r$ is a local trivialisation of E, then $(q, e) \mapsto (q, \mathrm{pr}_2\Phi(e))$ is a diffeomorphism from the part of $N \times_M E$ over $f^{-1}(U)$ onto $f^{-1}(U) \times \mathbb{K}^r$, linear on fibres; so the projection to N is a vector bundle, and its transition functions for these trivialisations are the $g_{ij} \circ f$, which are those of f^*E. Two vector bundles with the same cocycle are isomorphic (Proposition 35.2.6). □

So a section of f^*E is "a section of E along f": for example, for a curve γ in M, a section of γ^*TM is a vector field along the curve, such as its velocity, and the differential df of a smooth map is a section of $T^*N \otimes f^*TM$.

Proposition 35.3.4. *Every vector bundle over a manifold admits a metric: a smooth family of inner products (Hermitian inner products in the complex case) on the fibres. Consequently:*

1. *every subbundle has a complement, and every short exact sequence $0 \to E' \to E \to E'' \to 0$ of vector bundles splits;*

2. *$E \cong E^\vee$ for real bundles, and $E^\vee \cong \overline{E}$ for complex bundles with Hermitian metric.*

Here a *subbundle* of E is a subset $E' \subseteq E$ such that every point has a neighbourhood with a frame $s_1, \dots, s_r$ of E for which $s_1, \dots, s_k$ is a frame of E'; then E' is a vector bundle of rank k, and the quotient E/E', with fibres E_p/E'_p, is a vector bundle with the local frames given by the images of $s_{k+1}, \dots, s_r$. The *conjugate* $\overline{E}$ of a complex bundle is the same real bundle with the conjugate complex structure.

Proof. Choose trivialisations over a cover (U_i) and a subordinate partition of unity (ψ_i) (Theorem 33.8.2); the sum $\sum_i \psi_i\langle -, -\rangle_i$ of the pulled-back standard inner products is a metric, since a convex combination of inner products is an inner product. Given a subbundle $E' \subseteq E$, the orthogonal complement $E'^\perp$ is a subbundle (in a local frame adapted to E', apply Gram–Schmidt, Theorem 11.10.3, which depends smoothly on the frame) and $E = E' \oplus E'^\perp$; a surjection $E \to E''$ then splits by inverting $E'^\perp \to E''$. Statement (2) is the fibrewise isomorphism $v \mapsto \langle -, v\rangle$, which is smooth. □

With a metric one can always choose *orthonormal* local frames, by Gram–Schmidt; then the transition functions are orthogonal (unitary) matrices. So every real bundle can be described by a cocycle with values in $\mathrm{O}(r)$, and every complex one by a cocycle with values in $\mathrm{U}(r)$: the structure group can be *reduced* to a compact group.

Proposition 35.3.5. *1. A vector bundle E of rank r is trivial if and only if it has r sections $s_1, \dots, s_r$ that are linearly independent at every point (a* global frame*).*

2. For a real line bundle L, $L \otimes L$ is trivial. For a complex line bundle L, $L \otimes \overline{L}$ is trivial, so $\overline{L} \cong L^\vee$.

In particular the isomorphism classes of real line bundles form a group under $\otimes$ in which every element has order at most 2*.*

Proof. (1) A frame gives the morphism $M \times \mathbb{K}^r \to E$, $(p, a) \mapsto \sum_i a_i s_i(p)$, which is bijective on each fibre, hence an isomorphism (Lemma 35.2.2); conversely the standard basis vectors are a frame of the trivial bundle. (2) Choose a metric (Proposition 35.3.4). In the real case the map $L \otimes L \to M \times \mathbb{R}$, $v \otimes w \mapsto \langle v, w\rangle$, is fibrewise a nonzero linear map between lines, hence an isomorphism; smoothness is checked in a local frame. In the complex case use $v \otimes \bar{w} \mapsto \langle v, w\rangle$ on $L \otimes \overline{L}$. The group structure on line bundles is Proposition 35.3.1 and the uniqueness of tensor products up to isomorphism, with the trivial bundle as unit and $L^\vee$ as inverse, since $L \otimes L^\vee \cong \mathrm{End}(L)$ has the nowhere vanishing section id. □

Example 35.3.6. For the Möbius bundle E over S^1, $E \otimes E$ is trivial by Proposition 35.3.5, although E is not (Counterexample 35.2.5). Directly: $E \otimes E$ is glued from $[0,1] \times \mathbb{R}$ by $(0, t) \sim (1, (-1)^2 t) = (1, t)$, which is the trivial bundle. Likewise $E \oplus E$ is trivial: it is glued by $-\mathrm{id}_{\mathbb{R}^2}$, the rotation by π, and the rotations by πs, $0 \le s \le 1$, connect it to the identity. Concretely, the sections $x \mapsto (\cos \pi x, \sin \pi x)$ and $x \mapsto (-\sin \pi x, \cos \pi x)$ of $E \oplus E$ (functions $\mathbb{R} \to \mathbb{R}^2$ with $f(x+1) = -f(x)$) form a global frame. The same argument shows that a real bundle glued from $[0,1] \times \mathbb{R}^r$ by a matrix A is trivial when A can be joined to the identity by a path in $\mathrm{GL}_r(\mathbb{R})$: if A_t is such a path, constant near its ends, with $A_0 = \mathrm{id}$ and $A_1 = A$, the sections $t \mapsto A_t e_i$ satisfy the gluing condition and form a global frame.

Proposition 35.3.7. *Let $f : E \to F$ be a morphism of vector bundles over M whose rank on the fibres is constant. Then* $\ker f \subseteq E$ *and* $\operatorname{im} f \subseteq F$ *are subbundles, and* $\operatorname{coker} f = F/\operatorname{im} f$ *is a vector bundle. For a submanifold $S \subseteq M$ the normal bundle $N_{S/M} = (TM|_S)/TS$ is a vector bundle, and if $S = \phi^{-1}(q)$ for a regular value q of a smooth $\phi : M \to N$, then $d\phi$ induces an isomorphism $N_{S/M} \cong S \times T_q N$, so $N_{S/M}$ is trivial.*

Proof. The question is local, so let $E = U \times \mathbb{K}^r$, $F = U \times \mathbb{K}^s$ and f a smooth family of matrices $A(p)$ of constant rank k. Near p_0 some $k \times k$ minor of A is nonzero; after permuting bases, $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$ with A_{11} invertible near p_0. The first k columns of A are linearly independent and, as the rank is k, span the image; so they form a frame of $\operatorname{im} f$, which is therefore a subbundle. The vectors $\begin{pmatrix} -A_{11}^{-1} A_{12} w \\ w \end{pmatrix}$, $w \in \mathbb{K}^{r-k}$, lie in the kernel (the last $s - k$ rows are combinations of the first k), and they span it by dimension count; they form a frame of $\ker f$ depending smoothly on p. Completing a frame of $\operatorname{im} f$ by constant vectors to a frame of F gives a frame of $\operatorname{coker} f$. For the normal bundle apply this to the inclusion $TS \to TM|_S$, of constant rank $\dim S$. If $S = \phi^{-1}(q)$ with q regular, $d\phi : TM|_S \to S \times T_q N$ is surjective with kernel TS (Theorem 33.6.11), so it induces the stated isomorphism. □

Counterexample 35.3.8. The rank must be constant. The morphism $f : \mathbb{R} \times \mathbb{R} \to \mathbb{R} \times \mathbb{R}$, $(x, t) \mapsto (x, xt)$, of trivial line bundles over $\mathbb{R}$ has rank 1 for $x \neq 0$ and 0 at $x = 0$. Its kernel is the zero vector over each $x \neq 0$ and the whole line over $x = 0$, and its image is the whole

line over each $x \neq 0$ and the zero vector over $x = 0$. In both cases the dimension of the fibres jumps, so neither is a subbundle.

Example 35.3.9. TS^n is *stably trivial*: $TS^n \oplus (S^n \times \mathbb{R}) \cong S^n \times \mathbb{R}^{n+1}$, because the normal bundle of $S^n = f^{-1}(1)$, $f(x) = |x|^2$, is trivial by Proposition 35.3.7, and $TS^n \oplus N_{S^n/\mathbb{R}^{n+1}} \cong T\mathbb{R}^{n+1}|_{S^n}$ by Proposition 35.3.4(1). Concretely, TS^n is the kernel of the constant rank morphism $S^n \times \mathbb{R}^{n+1} \to S^n \times \mathbb{R}$, $(x, v) \mapsto (x, \langle x, v\rangle)$, and the normal line is spanned by the section $x \mapsto x$. Nevertheless TS^2 is not trivial, since S^2 has no nowhere vanishing vector field (the hairy ball theorem, Corollary 21.6.7); so a bundle is not determined by its stable class.

Example 35.3.10. On $\mathbb{CP}^n$ there is an exact sequence of holomorphic vector bundles $0 \to \mathcal{O} \to \mathcal{O}(1)^{\oplus(n+1)} \to T\mathbb{CP}^n \to 0$ (the *Euler sequence*), where $\mathcal{O}(1) = \mathcal{O}(-1)^\vee$ (Chapter 45). Over the smooth category it splits by Proposition 35.3.4, but it does not split holomorphically: holomorphic splittings are much rarer, which is why holomorphic bundles carry more information than smooth ones.

35.4 Line bundles and the first Chern class

Definition 35.4.1. The *Picard group* $\mathrm{Pic}^\infty(M)$ is the set of isomorphism classes of smooth complex line bundles with the tensor product; by Proposition 25.6.4 it is a group, with inverse the dual.

For line bundles the transition functions are nowhere vanishing functions $g_{ij} : U_i \cap U_j \to \mathbb{C}^\times$, which commute, and the tensor product multiplies them. So $\mathrm{Pic}^\infty(M)$ is the first Čech cohomology group of the sheaf of nowhere vanishing smooth functions, and it can be computed with the exponential function.

Theorem 35.4.2. *Let $\underline{\mathbb{C}}^\times_M$ be the sheaf of smooth nowhere vanishing complex functions. There are natural isomorphisms*

$$\mathrm{Pic}^\infty(M) \cong H^1(M, \underline{\mathbb{C}}^\times_M) \xrightarrow{\sim} H^2(M; \mathbb{Z}),$$

the second induced by the connecting homomorphism of the exponential sequence $0 \to \mathbb{Z} \to \underline{\mathbb{C}}_M \xrightarrow{\exp(2\pi i\,\cdot)} \underline{\mathbb{C}}^\times_M \to 0$. *The resulting class $c_1(L) \in H^2(M; \mathbb{Z})$ is the* first Chern class*; it satisfies $c_1(L \otimes L') = c_1(L) + c_1(L')$, $c_1(f^*L) = f^*c_1(L)$, and L is trivial if and only if $c_1(L) = 0$.*

Proof. Line bundles are classified by $\check{H}^1$ of the sheaf of smooth nonvanishing functions by Proposition 35.2.6(3), and Čech and sheaf cohomology agree in degree one (Remark 26.5.4); the group structures agree because the cocycle of a tensor product is the product of the cocycles. The exponential sequence is exact as a sequence of sheaves: surjectivity holds because a nonvanishing function has a smooth logarithm on any set where it is defined and contractible, for instance a small ball, and the kernel of $\exp(2\pi i\,\cdot)$ is the constant sheaf $\mathbb{Z}$ (Theorem 27.7.3). Its long exact sequence reads

$$H^1(M, \underline{\mathbb{C}}_M) \to H^1(M, \underline{\mathbb{C}}^\times_M) \to H^2(M, \mathbb{Z}_M) \to H^2(M, \underline{\mathbb{C}}_M),$$

and the outer terms vanish because $\underline{\mathbb{C}}_M$ is a sheaf of modules over C_M, hence fine, hence acyclic (Corollary 26.4.6); manifolds are paracompact (Proposition 18.10.7). Finally $H^2(M, \mathbb{Z}_M) \cong H^2(M; \mathbb{Z})$ by Theorem 26.6.1, since manifolds are locally contractible. Additivity and naturality follow from the corresponding properties of the cocycles and of the connecting homomorphism, and triviality of L means exactly that its class in $H^1(M, \underline{\mathbb{C}}^\times_M)$ vanishes. □

In particular every complex line bundle on a manifold with $H^2(M;\mathbb{Z}) = 0$ is trivial: on $\mathbb{R}^n$, on a disc, on S^1, on S^3. On the sphere S^2 the theorem gives $\mathrm{Pic}^\infty(S^2) \cong \mathbb{Z}$, and in this case the classification can be made completely explicit, by the winding number of a transition function.

For a continuous loop $c\colon [0,1] \to \mathbb{C}^\times$, $c(0) = c(1)$, choose a continuous logarithm L along it, $e^{L(t)} = c(t)$; one exists because $\exp\colon \mathbb{C} \to \mathbb{C}^\times$ is a covering map (local inverses are the branches of the logarithm on slit planes) and paths lift (Proposition 19.4.3). The *winding number* $\mathrm{wind}(c) = (L(1) - L(0))/2\pi i$ is an integer, as $e^{L(1)} = e^{L(0)}$, and does not depend on L, since two lifts differ by a constant in $2\pi i\mathbb{Z}$. For a function $g\colon \mathbb{C}^\times \to \mathbb{C}^\times$ we write $\mathrm{wind}(g)$ for the winding number of $t \mapsto g(e^{2\pi i t})$ along the unit circle.

Lemma 35.4.3. *1.* $\mathrm{wind}(c_1c_2) = \mathrm{wind}(c_1) + \mathrm{wind}(c_2)$, *and* $\mathrm{wind}(z^k) = k$.

2. *Loops that are homotopic through loops in* $\mathbb{C}^\times$ *have the same winding number. In particular* $\mathrm{wind}(g) = 0$ *if* g *extends to a continuous nowhere vanishing function on the closed unit disc, or on* $\{|z| \geq 1\} \cup \{\infty\} \subseteq \mathbb{CP}^1$.

3. *A continuous* $g\colon \mathbb{C}^\times \to \mathbb{C}^\times$ *with* $\mathrm{wind}(g) = 0$ *has a continuous logarithm on* $\mathbb{C}^\times$, *smooth if* g *is smooth.*

Proof. (1) If L_1, L_2 are logarithms along c_1, c_2, then $L_1 + L_2$ is one along c_1c_2. For z^k along $e^{2\pi i t}$ take $L(t) = 2\pi i k t$. (2) Let $H(s,t)$ be a homotopy of loops. Lifting H through exp (Proposition 19.4.3) gives $\tilde{H}$ with $e^{\tilde{H}} = H$, and $s \mapsto (\tilde{H}(s,1) - \tilde{H}(s,0))/2\pi i$ is a continuous integer-valued function, hence constant. If g extends to the closed unit disc, the loops $t \mapsto g(\rho e^{2\pi i t})$, $0 \leq \rho \leq 1$, form a homotopy through loops in $\mathbb{C}^\times$ from the constant loop $g(0)$ to the given one; if it extends to $\{|z| \geq 1\} \cup \{\infty\}$, use the loops $t \mapsto g(\sigma^{-1}e^{2\pi i t})$, $0 < \sigma \leq 1$, with the constant loop $g(\infty)$ at $\sigma = 0$. (3) The function $G(r,t) = g(re^{2\pi i t})$ on $(0,\infty)\times\mathbb{R}$, which is simply connected, lifts through exp to a continuous Λ with $e^\Lambda = G$ (Proposition 19.6.1). The difference $\Lambda(r,t+1) - \Lambda(r,t)$ lies in $2\pi i\mathbb{Z}$ and depends continuously on (r,t), so it is a constant, equal to $2\pi i$ times the winding number of the circle of radius 1, which is 0. So $\Lambda(r,t)$ depends only on $re^{2\pi i t}$, and defines a continuous logarithm of g on $\mathbb{C}^\times$; it is smooth when g is, because exp is a local diffeomorphism. □

Proposition 35.4.4. *Let* L *be a complex line bundle on* $\mathbb{CP}^1 = U_0 \cup U_1$, *with* $U_0 = \{[1:z]\} \cong \mathbb{C}$ *and* $U_1 = \{[w:1]\}$, $w = 1/z$ *on the overlap. Then* L *has frames* s_0 *over* U_0 *and* s_1 *over* U_1, *and on* $U_0 \cap U_1 \cong \mathbb{C}^\times$ *(coordinate* z*) we can write* $s_1 = g\,s_0$ *with* $g\colon \mathbb{C}^\times \to \mathbb{C}^\times$. *The integer*

$$\deg L = \mathrm{wind}(g)$$

does not depend on the frames, $\deg(L \otimes L') = \deg L + \deg L'$, *every integer is the degree of some line bundle, and* $L \cong L'$ *if and only if* $\deg L = \deg L'$. *So* $\deg\colon \mathrm{Pic}^\infty(\mathbb{CP}^1) \to \mathbb{Z}$ *is an isomorphism.*

Proof. The frames exist because U_0 and U_1 are diffeomorphic to $\mathbb{R}^2$, over which every complex line bundle is trivial, as $H^2(\mathbb{R}^2;\mathbb{Z}) = 0$ (Theorem 35.4.2). Other frames are $s_0' = h_0 s_0$ and $s_1' = h_1 s_1$ with $h_i\colon U_i \to \mathbb{C}^\times$, and then $g' = h_1 g h_0^{-1}$; by Lemma 35.4.3(1),(2), $\mathrm{wind}(g') = \mathrm{wind}(g) + \mathrm{wind}(h_1) - \mathrm{wind}(h_0) = \mathrm{wind}(g)$, as h_0 is defined on all of $U_0 = \mathbb{C}$ and h_1 on $U_1 = \mathbb{CP}^1 \setminus \{0\}$, which contains $\{|z| \geq 1\} \cup \{\infty\}$. For the tensor product, $s_1 \otimes s_1' = g g'\, s_0 \otimes s_0'$, so degrees add. The bundle with transition function $g = z^k$, obtained by Proposition 35.2.6(2), has degree k. If $\deg L = \deg L'$, with transition functions g and g', then g'/g has winding number 0, hence a smooth logarithm f on $\mathbb{C}^\times$ (Lemma 35.4.3(3)). Choose a partition of unity $\psi_0 + \psi_1 = 1$ subordinate to $\{U_0, U_1\}$ and put $f_0 = -\psi_1 f$ on U_0 and $f_1 = \psi_0 f$ on U_1, each extended by 0 where the cut-off vanishes; then $f_1 - f_0 = f$ on the overlap, and $h_i = e^{f_i}$ satisfy $h_1 g h_0^{-1} = g e^f = g'$. So after changing the frames of L by h_0 and h_1, the two bundles have the same transition function, and they are isomorphic by Proposition 35.2.6(3). □

Example 35.4.5. 1. The tautological bundle has $s_1 = z^{-1}s_0$ (Example 35.2.7(4)), so $\deg \mathcal{O}(-1) = -1$, and $\deg \mathcal{O}(k) = k$.

2. The tangent bundle $T\mathbb{CP}^1$, a complex line bundle with the frames $s_0 = \partial/\partial z$ and $s_1 = \partial/\partial w$: from $z = 1/w$, $\partial/\partial w = (dz/dw)\,\partial/\partial z = -w^{-2}\,\partial/\partial z = -z^2\,\partial/\partial z$, so $g = -z^2$ and $\deg T\mathbb{CP}^1 = 2$. Hence $T\mathbb{CP}^1 \cong \mathcal{O}(2)$, and TS^2 is not trivial, which gives another proof of the hairy ball theorem for S^2. The cotangent bundle has degree -2.

3. The degree is the first Chern class: under $H^2(\mathbb{CP}^1;\mathbb{Z}) \cong \mathbb{Z}$ given by the complex orientation, $c_1(L) = \deg L$. We do not verify this here by computing the connecting homomorphism; it follows from Chern–Weil theory (Chapter 39), and in Chapter 38 the curvature of a connection on $\mathcal{O}(-1)$ is computed to integrate to -1. The degree 2 of $T\mathbb{CP}^1$ is the Euler characteristic of S^2, an instance of the Gauss–Bonnet theorem.

Example 35.4.6. $H^2(S^2;\mathbb{Z}) \cong \mathbb{Z}$, so the complex line bundles on $S^2 = \mathbb{CP}^1$ are classified by an integer, the degree; the tautological bundle $\mathcal{O}(-1)$ has $c_1 = -1$ with the standard orientation. On $\mathbb{CP}^n$, $H^2 \cong \mathbb{Z}$ is generated by $c_1(\mathcal{O}(1))$. Every complex line bundle on a surface of genus g is classified by its degree in $H^2 \cong \mathbb{Z}$, while on a simply connected manifold with $H^2 = 0$, such as S^3 or S^4, every complex line bundle is trivial. Real line bundles are classified in the same way by $H^1(M;\mathbb{Z}/2\mathbb{Z})$, using the sheaf of nonvanishing real functions and the sequence $0 \to \mathbb{Z}/2 \to \underline{\mathbb{R}}^\times \to \underline{\mathbb{R}}^\times_{>0} \to 0$; the resulting class is the first Stiefel–Whitney class $w_1(L)$, which vanishes if and only if L is orientable.

Remark 35.4.7. For bundles of higher rank the classification is not by a single class: isomorphism classes of rank r complex bundles on M correspond to homotopy classes of maps $M \to \mathrm{BU}(r)$, and the Chern classes $c_k(E) \in H^{2k}(M;\mathbb{Z})$ are pulled back from universal classes on the classifying space. They are constructed and computed in Chapter 39; for line bundles c_1 agrees with the class above.

35.5 Parallelizable manifolds

The most natural question about the tangent bundle is whether it is trivial. A trivialisation $TM \cong M \times \mathbb{R}^n$ lets one compare tangent vectors at different points, and decide when two of them are "parallel" independently of any path, which is the origin of the name.

Definition 35.5.1. A smooth n-manifold M is *parallelizable* if its tangent bundle is trivial; equivalently (Proposition 35.3.5(1)), if there are vector fields $X_1, \dots, X_n$ on M that are linearly independent at every point, a *global frame*.

Then every vector field is uniquely $\sum_i f^i X_i$ with smooth functions f^i, the components of a field are defined on all of M, and a vector at one point is transported to every other point by keeping its components constant.

Example 35.5.2. 1. $\mathbb{R}^n$ and its open subsets, with the frame $\partial_1, \dots, \partial_n$; the torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$, since the fields ∂_i are invariant under translations and descend to T^n.

2. *Lie groups.* If $v_1, \dots, v_n$ is a basis of T_eG, the left-invariant fields $X_i(g) = d_e(L_g)(v_i)$ are smooth (Proposition 34.2.8) and linearly independent at every g, because d_eL_g is an isomorphism, L_g being a diffeomorphism. So every Lie group is parallelizable: S^1, $\mathrm{GL}_n(\mathbb{R})$, $\mathrm{SO}(n)$, $\mathrm{U}(n)$, and $\mathrm{SU}(2) \cong S^3$ (Example 34.2.11).

3. *The sphere S^3*, explicitly. Identify $\mathbb{R}^4$ with the quaternions, $x = x_0 + x_1 i + x_2 j + x_3 k$. Right multiplication by i, j, k gives

$$xi = (-x_1, x_0, x_3, -x_2), \qquad xj = (-x_2, -x_3, x_0, x_1), \qquad xk = (-x_3, x_2, -x_1, x_0),$$

and for $|x| = 1$ the four vectors x, xi, xj, xk are orthonormal (a direct check of the inner products, for instance $\langle x, xi\rangle = -x_0x_1 + x_1x_0 + x_2x_3 - x_3x_2 = 0$). So xi, xj, xk are three vector fields tangent to S^3 (orthogonal to x) and orthonormal at every point: a global frame.

4. *Products.* By Proposition 33.4.13, $T_{(p,q)}(M \times N) = T_pM \oplus T_qN$; so if M and N are parallelizable, the frames of M and of N, regarded as vector fields on $M \times N$, form a global frame of $M \times N$.

5. *The manifolds $S^1 \times S^n$* are parallelizable for every n, although S^2 is not. With θ the coordinate of S^1, and $x \in S^n \subseteq \mathbb{R}^{n+1}$, the $n+1$ fields

$$Y_i(\theta, x) = \langle e_i, x\rangle\, \partial_\theta + \big(e_i - \langle e_i, x\rangle x\big), \qquad i = 1, \dots, n+1,$$

are tangent to $S^1 \times S^n$ (the second part is the tangential part of e_i), and they are linearly independent at every point: if $\sum_i c_iY_i = 0$ with $c = \sum_i c_ie_i$, then $\langle c, x\rangle = 0$ and $c - \langle c, x\rangle x = 0$, so $c = 0$. This is the stable triviality of TS^n (Example 35.3.9) made explicit: the trivial line $\mathbb{R}\,\partial_\theta$ of the circle plays the role of the normal line $\mathbb{R}x$.

The sphere S^2 is not parallelizable, as it has no nowhere vanishing vector field at all, and more generally the even-dimensional spheres S^{2m} are not, by the hairy ball theorem (Corollary 21.6.7(2)); for S^2 this also follows from $TS^2 \cong T\mathbb{CP}^1$ having degree 2 (Example 35.4.5). The odd-dimensional spheres have a nowhere vanishing field, $x \mapsto (-x_2, x_1, -x_4, x_3, \dots)$, but one field is far from a frame, and the answer for spheres is a deep theorem.

Theorem 35.5.3. *The sphere S^n is parallelizable if and only if $n = 0, 1, 3$ or 7.*

For $n = 1$ and $n = 3$ this is Example 35.5.2, and for $n = 7$ the same construction with the octonions, an eight-dimensional normed algebra that is not associative, gives seven orthonormal fields. That no other sphere is parallelizable was proved independently by Bott and Milnor [BM58] and by Kervaire [Ker58]; it is closely related to the theorem that $\mathbb{R}^n$ carries a bilinear multiplication without zero divisors only for $n = 1, 2, 4, 8$, and to Adams' theorem on the Hopf invariant [Ada60] (Corollary 24.4.4 is its first step).

Remark 35.5.4. Several necessary conditions for parallelizability appear in later chapters. A parallelizable manifold is orientable: the dual coframe of a global frame gives a nowhere vanishing n-form (Chapter 36). A compact parallelizable manifold has a nowhere vanishing vector field, so its Euler characteristic is 0 by the Poincaré–Hopf theorem, and all the characteristic classes of its tangent bundle vanish (Chapter 39); this rules out, for example, all compact orientable surfaces except the torus and all complex projective spaces. These conditions are not sufficient: S^5 is orientable, has Euler characteristic 0 and, being stably parallelizable, vanishing characteristic classes, but it is not parallelizable by Theorem 35.5.3. Finally, a global frame (X_i) defines a connection by declaring the X_i parallel, $\nabla(\sum_i f^iX_i) = \sum_i df^i \otimes X_i$; it is flat, its parallel transport does not depend on the path, and conversely a flat connection with trivial holonomy on a connected manifold produces a global parallel frame (Chapter 38).

35.6 Exercises

Exercise 35.6.1. Show that the Möbius bundle $L \to S^1$ is nontrivial, that $L \otimes L$ is trivial, and that L has no nowhere vanishing section.

Solution. A nowhere vanishing section would give a trivialisation; on the other hand a section is a continuous function $f\colon [0,1] \to \mathbb{R}$ with $f(1) = -f(0)$, which must vanish somewhere by the intermediate value theorem. The transition function of $L \otimes L$ is the square of that of L, namely 1. □

Exercise 35.6.2. Show that TS^2 is not trivial, using that a trivialisation would give a nowhere vanishing vector field (Corollary 21.6.7).

Solution. A trivialisation $TS^2 \cong S^2 \times \mathbb{R}^2$ gives the global frame $\Phi^{-1}(p, e_1)$, $\Phi^{-1}(p, e_2)$, and in particular the nowhere vanishing vector field $\Phi^{-1}(p, e_1)$, which does not exist. For another proof, see Example 35.4.5(2). □

Exercise 35.6.3. Show that the tautological line bundle on $\mathbb{CP}^n$ is $\mathcal{O}(-1) = \{(\ell, v) : v \in \ell\} \subseteq \mathbb{CP}^n \times \mathbb{C}^{n+1}$, that it is a submanifold and a line bundle, and compute its transition functions for the standard cover.

Solution. On $U_i = \{x_i \neq 0\}$ the map $\ell \mapsto (\ell, (x_0/x_i, \dots, 1, \dots, x_n/x_i))$ is a nowhere vanishing section s_i, giving a trivialisation; then $s_j = (x_i/x_j)s_i$, so $g_{ij} = x_i/x_j$ as functions on $U_i \cap U_j$. □

Exercise 35.6.4. Let $S \subseteq M$ be a submanifold with normal bundle $N = (TM|_S)/TS$. Show that if $S = f^{-1}(q)$ for a submersion f, then $N \cong S \times \mathbb{R}^m$ is trivial, and that the normal bundle of $\mathbb{RP}^{n-1} \subseteq \mathbb{RP}^n$ is the tautological bundle.

Solution. The first statement is Proposition 35.3.7. For the second, $\mathbb{RP}^{n-1} = \{x_n = 0\}$. A tangent vector to $\mathbb{RP}^n$ at a line ℓ is the same as a linear map $\ell \to \ell^\perp$ (the derivative of a curve of lines ℓ_t, measured by how a vector of ℓ moves orthogonally), so $T_\ell \mathbb{RP}^n = \mathrm{Hom}(\ell, \ell^\perp)$. For $\ell \subseteq \mathbb{R}^n = \{x_n = 0\}$, the tangent space of $\mathbb{RP}^{n-1}$ is $\mathrm{Hom}(\ell, \ell^\perp \cap \mathbb{R}^n)$, and the quotient is $\mathrm{Hom}(\ell, \mathbb{R}e_n) \cong \ell^\vee$. So the normal bundle is the dual of the tautological bundle, which is isomorphic to it by Proposition 35.3.4(2). □

Exercise 35.6.5. Compute the Jacobian matrix of the inversion $\tau(y) = y/|y|^2$ of $\mathbb{R}^n \setminus \{0\}$ and verify the formula of Example 35.2.7(2), including $\det D\tau(y) = -|y|^{-2n}$. For $n = 2$, write $y = x_1 + ix_2$ and check that $\tau(y) = 1/\bar{y}$.

Solution. $\partial_j(y_i/|y|^2) = \delta_{ij}/|y|^2 - 2y_iy_j/|y|^4$, which is the formula. The matrix $I - 2uu^T$ with $u = y/|y|$ is the reflection in $u^\perp$, of determinant -1, so $\det D\tau = (|y|^{-2})^n \cdot (-1)$. For $n = 2$, $1/\bar{y} = y/(y\bar{y}) = y/|y|^2$. □

Exercise 35.6.6. Let $f\colon \mathbb{CP}^1 \to \mathbb{CP}^1$ be $[x_0 : x_1] \mapsto [x_0^k : x_1^k]$ for $k \geq 1$. Show that $f^*\mathcal{O}(-1) \cong \mathcal{O}(-k)$, by computing the transition function of the pulled back bundle.

Solution. In the coordinate z of U_0, $f(z) = z^k$, and f maps U_0 to U_0 and U_1 to U_1. The pulled back frames $s_0 \circ f$ and $s_1 \circ f$ satisfy $s_1 \circ f = (z^k)^{-1}\, s_0 \circ f$, by Example 35.2.7(4) evaluated at $f(z)$. So the transition function is z^{-k}, of winding number $-k$, and $f^*\mathcal{O}(-1) \cong \mathcal{O}(-k)$ by Proposition 35.4.4. The degree is multiplied by the degree k of the map. □

Exercise 35.6.7. Show that for a real line bundle L the following are equivalent: L is trivial; L has a nowhere vanishing section; the transition functions of some trivialising cover can be chosen positive. Deduce that the tautological bundle of $\mathbb{RP}^n$ is not trivial for $n \geq 1$, and that every real line bundle over a simply connected manifold is trivial (using the classification by $H^1(M; \mathbb{Z}/2)$ of Example 35.4.6).

Solution. A nowhere vanishing section s gives the trivialisation $(p,t) \mapsto ts(p)$; conversely $p \mapsto \Phi^{-1}(p,1)$. With positive transition functions, choose a partition of unity and glue the local sections $\Phi_i^{-1}(p,1)$: $s = \sum_i \psi_i \Phi_i^{-1}(\cdot,1)$ is nowhere zero, because in any trivialisation the summands are positive multiples of each other. Conversely, with a nowhere vanishing section, rescale the trivialisations so that it becomes the constant 1; then all transition functions are 1. For $\mathbb{RP}^n$ the tautological bundle restricts on $\mathbb{RP}^1$ to the Möbius bundle (Counterexample 35.2.5). On a simply connected M, $H^1(M;\mathbb{Z}/2) = \mathrm{Hom}(\pi_1(M),\mathbb{Z}/2) = 0$. □

Exercise 35.6.8. Show directly that $\mathrm{Hom}(E,F)$, defined as the bundle with fibres $\mathrm{Hom}(E_p,F_p)$ and transition functions $A \mapsto h_{ij} A g_{ij}^{-1}$, is isomorphic to $E^\vee \otimes F$, and that its sections are exactly the morphisms of vector bundles $E \to F$.

Solution. Fibrewise $E_p^\vee \otimes F_p \to \mathrm{Hom}(E_p,F_p)$, $\phi \otimes w \mapsto (v \mapsto \phi(v)w)$, is an isomorphism (Proposition 12.2.2); in trivialisations it is a fixed linear isomorphism $\mathbb{K}^{r\vee} \otimes \mathbb{K}^s \to M_{s\times r}(\mathbb{K})$ intertwining the transition functions, since $(g^T)^{-1}\xi \otimes hw \mapsto hw\,\xi^T g^{-1}$. So it is an isomorphism of bundles (Lemma 35.2.2). A section is a smoothly varying family of linear maps $E_p \to F_p$, which is a morphism of bundles: in trivialisations it is a smooth matrix-valued function. □

Exercise 35.6.9. Let E be the Möbius bundle. Show that $E \oplus (S^1 \times \mathbb{R})$ is not trivial but that $E \oplus E$ is, by describing sections as functions on $\mathbb{R}$ with the appropriate periodicity.

Solution. Sections of $E \oplus (S^1 \times \mathbb{R})$ are pairs (f,g) with $f(x+1) = -f(x)$ and $g(x+1) = g(x)$. If it were trivial, its determinant line bundle $\det(E \oplus \mathbb{R}) \cong E$ (Proposition 35.3.1(3)) would be trivial. For $E \oplus E$, the sections $(\cos \pi x, \sin \pi x)$ and $(-\sin \pi x, \cos \pi x)$ are anti-periodic, hence sections, and linearly independent everywhere (Example 35.3.6). □

Exercise 35.6.10. Show that $\mathbb{RP}^3 = S^3/\{\pm 1\}$ is parallelizable, by checking that the vector fields xi, xj, xk of Example 35.5.2(3) descend to $\mathbb{RP}^3$. Show also that $T^2 \times S^2$ is parallelizable.

Solution. Let $a(x) = -x$ be the antipodal map and $\pi\colon S^3 \to \mathbb{RP}^3$ the projection, a local diffeomorphism with $\pi \circ a = \pi$. For $X(x) = xi$, $d_x a(X(x)) = -xi = (-x)i = X(a(x))$, so the vectors $d_x\pi(X(x))$ and $d_{-x}\pi(X(-x)) = d_{-x}\pi(d_x a(X(x))) = d_x(\pi \circ a)(X(x))$ agree: $Y(\pi(x)) = d_x\pi(X(x))$ is a well-defined vector field on $\mathbb{RP}^3$, smooth because π has smooth local inverses. The same holds for xj and xk, and the three fields are linearly independent at every point because $d_x\pi$ is an isomorphism. (Alternatively, $\mathbb{RP}^3$ is diffeomorphic to the Lie group SO(3).) Finally $T^2 \times S^2 = S^1 \times (S^1 \times S^2)$ is a product of parallelizable manifolds, by Example 35.5.2(4) and (5). □

Chapter 36

Differential Forms and Integration

36.1 Introduction

Differential forms are the integrands of geometry. A k-form is a quantity that can be integrated over oriented k-dimensional submanifolds: a 1-form over curves (the work of a force field along a path), a 2-form over surfaces (the flux of a field through a surface), an n-form over an n-manifold (a volume). Why forms, and not functions, are the right integrands is explained by the change of variables formula. Under a change of coordinates $x = \tau(y)$, the integral $\int f \, dx^1 \cdots dx^n$ becomes $\int (f \circ \tau) \, |\det D\tau| \, dy^1 \cdots dy^n$, with the *absolute value* of the Jacobian determinant, while the n-form $f \, dx^1 \wedge \cdots \wedge dx^n$ becomes $(f \circ \tau) \det D\tau \, dy^1 \wedge \cdots \wedge dy^n$, with the determinant itself. The two rules agree for orientation-preserving changes of coordinates, and this is why an n-form can be integrated over an oriented manifold without any further choice, such as coordinates or a metric. (A metric allows one to integrate functions instead: Chapter 40.)

The *exterior derivative* d turns k-forms into $(k+1)$-forms. On $\mathbb{R}^3$ it unifies the gradient, the curl and the divergence of vector calculus, it satisfies $d \circ d = 0$ (which contains $\operatorname{curl} \operatorname{grad} = 0$ and $\operatorname{div} \operatorname{curl} = 0$), and it makes *Stokes' theorem* (Theorem 36.4.8) true: the integral of $d\omega$ over a manifold with boundary is the integral of ω over the boundary. This one statement contains the fundamental theorem of calculus and the theorems of Green, Gauss and Stokes. Everything in Hodge theory is written in this language: de Rham cohomology (Chapter 37), harmonic forms (Chapter 43), and the decomposition of forms on a complex manifold into types (Chapter 46).

The chapter goes as follows.

1. Differential forms, their three descriptions (sections of a bundle, alternating maps on tangent vectors, coefficients in charts) and their behaviour under changes of coordinates; computations of wedge products and pullbacks (Section 36.2).

2. The exterior derivative, constructed and characterised step by step; the interior product, the Lie derivative and Cartan's formula; closed and exact forms and line integrals.

3. Orientations, in terms of forms, of atlases and of unit normal fields (Section 36.3).

4. Integration of forms, manifolds with boundary and Stokes' theorem, with worked computations (Section 36.4).

References: [Lee13], [War83], [Spi65].
Prerequisites. Chapter 35 (Vector Bundles) and Chapter 28 (Measure and Integration).

36.2 Differential forms

Definition 36.2.1. A *differential k-form* on a smooth manifold M is a smooth section of $\Lambda^k T^*M$; the space of k-forms is $\Omega^k(M)$, and $\Omega^\bullet(M) = \bigoplus_k \Omega^k(M)$ with the wedge product (Proposition 12.3.3) is a graded-commutative algebra over $C^\infty(M)$. By Proposition 36.2.5 below, a k-form is the same as a smoothly varying family of alternating k-linear maps $(T_pM)^k \to \mathbb{R}$, or as a family of coefficient functions in charts transforming in a prescribed way. For $f : N \to M$ smooth, the *pullback* $f^*\omega$ is defined by $(f^*\omega)_p(v_1, \dots, v_k) = \omega_{f(p)}(d_p f\, v_1, \dots, d_p f\, v_k)$; it is an algebra homomorphism and $(g \circ f)^* = f^* g^*$.

In a chart $(U, (x^1, \dots, x^n))$, dx^i is the differential of the coordinate function x^i, so $dx^i(\partial/\partial x^j) = \partial x^i/\partial x^j = \delta_{ij}$: the $dx^i|_p$ form the basis of T_p^*M dual to the basis $\partial/\partial x^j|_p$ of T_pM. For a multi-index $I = (i_1 < \cdots < i_k)$ we write $dx^I = dx^{i_1} \wedge \cdots \wedge dx^{i_k}$; for a sequence $(i_1, \dots, i_k)$ that is not increasing, $dx^{i_1} \wedge \cdots \wedge dx^{i_k}$ is 0 if two indices coincide and $\pm dx^I$ otherwise, with the sign of the permutation that sorts the indices.

Remark 36.2.2. The most important 1-forms are the differentials of functions. For $f \in C^\infty(M)$, the differential $d_p f : T_pM \to T_{f(p)}\mathbb{R}$ (Definition 33.4.1) takes values in the tangent space of $\mathbb{R}$, which is identified with $\mathbb{R}$ by $w \mapsto w(t)$, the value of the derivation w on the coordinate function t of $\mathbb{R}$ (that is, $w = w(t)\, d/dt$). With this identification

$$(df)_p(v) = d_p f(v)(t) = v(t \circ f) = v(f), \qquad \text{so} \qquad df(X) = X(f) = Xf$$

for a vector field X: the differential of f evaluated on X is the derivative of f in the direction X. In a chart, $df(\partial/\partial x^j) = \partial f/\partial x^j$, so $df = \sum_j \frac{\partial f}{\partial x^j} dx^j$; for $f = x^i$ this gives the statement $dx^i(\partial/\partial x^j) = \delta_{ij}$ above. The 1-form df is smooth, since its coefficients are, and $d(fg) = f\, dg + g\, df$ by the Leibniz rule for tangent vectors.

Remark 36.2.3. *Pullback, and what a smooth map induces.* The definition of $f^*\omega$ says: to evaluate $f^*\omega$ on tangent vectors at $p \in N$, push the vectors forward to $f(p)$ with the differential $d_p f$ and evaluate ω there. Since $d_p f$ is linear, $(f^*\omega)_p$ is again alternating and multilinear, and f^* is linear, compatible with wedge products ($f^*(\omega \wedge \eta) = f^*\omega \wedge f^*\eta$, as both sides are computed pointwise from the same multilinear algebra) and functorial: $(g \circ f)^* = f^* g^*$ by the chain rule $d(g \circ f) = dg \circ df$, and $\mathrm{id}^* = \mathrm{id}$. A smooth map $f : N \to M$ induces maps in two directions:

1. *forward* (covariantly): tangent vectors, $d_p f : T_pN \to T_{f(p)}M$, and curves, $\gamma \mapsto f \circ \gamma$;
2. *backward* (contravariantly): functions, $f^* g = g \circ f$; covectors and forms, $f^*\omega$; covariant tensors such as metrics, $(f^*g)(v, w) = g(df\, v, df\, w)$, which is again a metric if and only if f is an immersion (Remark 36.2.4); vector bundles and their sections, f^*E (Chapter 35); and de Rham cohomology (Chapter 37).

Vector fields are neither: a vector field X on N cannot be pushed forward in general (if f is not injective, two points with the same image may give different vectors, and if f is not surjective there is nothing to define off the image), nor pulled back. A diffeomorphism transports everything in both directions, $f_* X = df \circ X \circ f^{-1}$; for a general f one only has the notion of *f-related* fields, $df(X_p) = Y_{f(p)}$, and then $f^*(\omega(Y_1, \dots, Y_k)) = (f^*\omega)(X_1, \dots, X_k)$. This asymmetry is why differential forms, not vector fields, are the objects that can be integrated over images of maps and that form a contravariant functor, cohomology.

Remark 36.2.4. For $f : N \to (M, g)$ with g a Riemannian metric, $(f^*g)_p(v, v) = |d_p f\, v|^2$ vanishes for $v \neq 0$ exactly when $v \in \ker d_p f$; so f^*g is positive definite, a metric, if and only if every $d_p f$ is injective. For the inclusion of a submanifold this is the induced metric (Chapter 40).

Proposition 36.2.5. *Let M be a smooth n-manifold. For each p, the fibre $\Lambda^k T^*_p M$ is the space of alternating k-linear maps $(T_pM)^k \to \mathbb{R}$ (Proposition 12.4.1 and Notation 12.4.2). For a family $\omega = (\omega_p)_{p\in M}$ of such maps the following are equivalent:*

1. *ω is a smooth section of $\Lambda^k T^*M$, that is, a k-form;*
2. *for all smooth vector fields $X_1, \dots, X_k$ on an open set U, the function $p \mapsto \omega_p(X_1(p), \dots, X_k(p))$ is smooth on U;*
3. *in every chart of some atlas, the* coefficients *$\omega_I(p) = \omega_p(\partial/\partial x^{i_1}|_p, \dots, \partial/\partial x^{i_k}|_p)$ are smooth.*

In a chart $\omega = \sum_I \omega_I\, dx^I$, uniquely. If $(y^1, \dots, y^n)$ is a second chart, then on the overlap

$$dx^I = \sum_J \det\Big(\frac{\partial x^I}{\partial y^J}\Big) dy^J, \qquad \omega^{(y)}_J = \sum_I \omega^{(x)}_I \det\Big(\frac{\partial x^I}{\partial y^J}\Big),$$

where $\partial x^I/\partial y^J$ is the $k \times k$ matrix $(\partial x^{i_a}/\partial y^{j_b})_{a,b}$ and the superscripts indicate the chart. For $k = n$ this reads $\omega^{(y)} = \omega^{(x)} \det(\partial x/\partial y)$, with the Jacobian determinant of the transition map. Conversely, functions $\omega^{(x)}_I$ given on the charts of an atlas and satisfying this rule on all overlaps come from a unique k-form.

Proof. The expression in a chart. Since the $dx^i|_p$ are the basis dual to the $\partial/\partial x^i|_p$, the $dx^I|_p$ form a basis of $\Lambda^k T^*_p M$ (Proposition 12.3.3(3)), and for increasing I and J, $dx^I(\partial/\partial x^{j_1}, \dots, \partial/\partial x^{j_k}) = \det(dx^{i_a}(\partial/\partial x^{j_b}))_{a,b}$, which is 1 if $I = J$ and 0 otherwise (Notation 12.4.2). Evaluating $\omega_p = \sum_I c_I\, dx^I|_p$ on $(\partial/\partial x^{j_1}, \dots, \partial/\partial x^{j_k})$ therefore gives $c_J = \omega_J(p)$: the coefficients are the ω_J, and they are unique.

(1) $\Leftrightarrow$ (3). The smooth structure of $\Lambda^k T^*M$ (Proposition 35.3.1) is the one for which the frames (dx^I) of charts are local trivialisations; a section is smooth if and only if its coefficients in these frames are smooth. (3) $\Rightarrow$ (2). On a chart domain write $X_b = \sum_j X^j_b\, \partial/\partial x^j$. By multilinearity $\omega(X_1, \dots, X_k) = \sum_I \omega_I\, dx^I(X_1, \dots, X_k)$, and $dx^I(X_1, \dots, X_k) = \det(X^{i_a}_b)_{a,b}$ is a polynomial in the smooth functions X^j_b. (2) $\Rightarrow$ (3). Take the X_b to be coordinate vector fields.

Change of chart. By the first part applied to the chart y, the coefficient of dy^J in dx^I is $dx^I(\partial/\partial y^{j_1}, \dots, \partial/\partial y^{j_k}) = \det(dx^{i_a}(\partial/\partial y^{j_b})) = \det(\partial x^{i_a}/\partial y^{j_b})$, since $dx^i(\partial/\partial y^j) = \partial x^i/\partial y^j$ is the derivative of the function x^i in the direction $\partial/\partial y^j$. Substituting into $\omega = \sum_I \omega^{(x)}_I\, dx^I$ and comparing coefficients gives the rule for $\omega^{(y)}_J$. For $k = n$ there is one multi-index, $(1, \dots, n)$. Conversely, if coefficients satisfying the rule are given on the charts of an atlas, define $\omega_p = \sum_I \omega^{(x)}_I(p)\, dx^I|_p$ using any chart at p; the rule says exactly that this does not depend on the chart, and ω is smooth by (3). □

Two operations are used constantly: the wedge product of forms, computed pointwise with the rules of the exterior algebra ($dx^i \wedge dx^j = -dx^j \wedge dx^i$, so $dx^i \wedge dx^i = 0$), and the pullback, computed by substituting: for $f\colon N \to M$, $f^*(\sum_I \omega_I\, dx^{i_1} \wedge \cdots \wedge dx^{i_k}) = \sum_I (\omega_I \circ f)\, d(x^{i_1} \circ f) \wedge \cdots \wedge d(x^{i_k} \circ f)$, since f^* is an algebra homomorphism with $f^*(dg) = d(g \circ f)$ for functions (the chain rule).

Example 36.2.6. 1. *A wedge product.* On $\mathbb{R}^3$, for $\alpha = x\, dx + y\, dy$ and $\beta = dx + z\, dz$,

$$\alpha \wedge \beta = x\, dx \wedge dx + xz\, dx \wedge dz + y\, dy \wedge dx + yz\, dy \wedge dz = -y\, dx \wedge dy + xz\, dx \wedge dz + yz\, dy \wedge dz.$$

On the vectors e_1, e_2 it takes the value $\alpha(e_1)\beta(e_2) - \alpha(e_2)\beta(e_1) = x \cdot 0 - y \cdot 1 = -y$, in accordance with the coefficient of $dx \wedge dy$.

2. *Pullback to a curve.* For the helix $\gamma(t) = (\cos t, \sin t, t)$ and $\omega = z\,dx + x\,dy$, $\gamma^*\omega = t\,d(\cos t) + \cos t\,d(\sin t) = (\cos^2 t - t\sin t)\,dt$.

3. *Pullback to the sphere.* Let $x(\vartheta,\varphi) = (\sin\vartheta\cos\varphi, \sin\vartheta\sin\varphi, \cos\vartheta)$ and $\omega = x\,dy\wedge dz + y\,dz\wedge dx + z\,dx\wedge dy$. By Proposition 36.2.5, the pullback of $dy\wedge dz$ is the Jacobian minor $\partial(y,z)/\partial(\vartheta,\varphi)$ times $d\vartheta\wedge d\varphi$, and similarly for the others:

   $$x^*(dy\wedge dz) = \sin^2\vartheta\cos\varphi\,d\vartheta\wedge d\varphi, \quad x^*(dz\wedge dx) = \sin^2\vartheta\sin\varphi\,d\vartheta\wedge d\varphi, \quad x^*(dx\wedge dy) = \sin\vartheta\cos\vartheta\,d\vartheta\wedge d\varphi.$$

   So $x^*\omega = (\sin^3\vartheta\cos^2\varphi + \sin^3\vartheta\sin^2\varphi + \sin\vartheta\cos^2\vartheta)\,d\vartheta\wedge d\varphi = \sin\vartheta\,d\vartheta\wedge d\varphi$, the area form of the unit sphere in spherical coordinates.

4. *Spherical coordinates in space.* For $(r,\vartheta,\varphi)\mapsto r\,x(\vartheta,\varphi)$, the pullback of $dx\wedge dy\wedge dz$ is the Jacobian determinant times $dr\wedge d\vartheta\wedge d\varphi$, namely $r^2\sin\vartheta\,dr\wedge d\vartheta\wedge d\varphi$ (Exercise 36.5.6).

Example 36.2.7. In the polar coordinates (r,θ) of Example 33.4.8, with $x = r\cos\theta$ and $y = r\sin\theta$, the rule of Proposition 36.2.5 for $k = 1$ gives $dx = \cos\theta\,dr - r\sin\theta\,d\theta$ and $dy = \sin\theta\,dr + r\cos\theta\,d\theta$. Solving,

$$dr = \cos\theta\,dx + \sin\theta\,dy = \frac{x\,dx + y\,dy}{r}, \qquad d\theta = \frac{-\sin\theta\,dx + \cos\theta\,dy}{r} = \frac{x\,dy - y\,dx}{x^2+y^2},$$

the second being the angle form of Counterexample 36.2.18. Compare with the tangent vectors $\partial/\partial r = \cos\theta\,\partial/\partial x + \sin\theta\,\partial/\partial y$ and $\partial/\partial\theta = -r\sin\theta\,\partial/\partial x + r\cos\theta\,\partial/\partial y$. The formula $dr = \cos\theta\,dx + \sin\theta\,dy$ found in many texts is about the covector dr, not the vector $\partial/\partial r$: vectors change by the Jacobian matrix $P = \partial(x,y)/\partial(r,\theta)$ (its columns give $\partial/\partial r$ and $\partial/\partial\theta$), and covectors by its inverse (the rows of P^{-1} give dr and $d\theta$). The coefficients agree for r and not for θ because P is a rotation followed by $\mathrm{diag}(1,r)$: $\partial/\partial r$ is a unit vector orthogonal to the circles $r = \text{const}$, so the Euclidean metric identifies it with dr, while $\partial/\partial\theta$ has length r and corresponds to $r^2\,d\theta$. For 2-forms the rule gives

$$dx\wedge dy = \det\frac{\partial(x,y)}{\partial(r,\theta)}\,dr\wedge d\theta = r\,dr\wedge d\theta,$$

which is the change of variables $\int f\,dx\,dy = \int f\,r\,dr\,d\theta$ of Example 28.11.3, in the language of forms (Proposition 36.4.2).

Theorem 36.2.8. *There is a unique family of linear maps $d\colon \Omega^k(M)\to\Omega^{k+1}(M)$, for all M and k, such that: df is the differential of f for $f\in\Omega^0(M) = C^\infty(M)$; $d(\alpha\wedge\beta) = d\alpha\wedge\beta + (-1)^{|\alpha|}\alpha\wedge d\beta$; and $d\circ d = 0$. It is local, given in a chart by $d(\sum_I \omega_I\,dx^I) = \sum_I d\omega_I\wedge dx^I$, and natural: $f^*(d\omega) = d(f^*\omega)$. Moreover, for vector fields,*

$$d\omega(X_0,\dots,X_k) = \sum_i (-1)^i X_i\left(\omega(X_0,\dots,\widehat{X_i},\dots,X_k)\right) + \sum_{i<j}(-1)^{i+j}\omega([X_i,X_j],X_0,\dots,\widehat{X_i},\dots,\widehat{X_j},\dots,X_k).$$

Proof. We write $\partial_l = \partial/\partial x^l$ in a chart, and $J\setminus j_m$ for the multi-index $J = (j_0<\cdots<j_k)$ with j_m removed.

Step 1: the formula in a chart and its coefficients. On a chart domain U define

$$d_U\Big(\sum_I \omega_I\,dx^I\Big) = \sum_I d\omega_I\wedge dx^I = \sum_I\sum_{l=1}^n \partial_l\omega_I\,dx^l\wedge dx^I.$$

Then $d_U(f\,dx^{i_1}\wedge\cdots\wedge dx^{i_k}) = df\wedge dx^{i_1}\wedge\cdots\wedge dx^{i_k}$ also for sequences that are not increasing, since both sides vanish for repeated indices and change by the same sign when the indices are sorted. We compute the coefficients of $d_U\omega$: for $J=(j_0<\cdots<j_k)$,

$$(d_U\omega)(\partial_{j_0},\dots,\partial_{j_k}) = \sum_{m=0}^{k}(-1)^m\,\partial_{j_m}\omega_{J\setminus j_m}.$$

Indeed the form $dx^l\wedge dx^I$ vanishes on $(\partial_{j_0},\dots,\partial_{j_k})$ unless $l\notin I$ and $\{l\}\cup I = J$ as sets, that is, unless $l=j_m$ and $I = J\setminus j_m$ for some m; and then $dx^{j_m}\wedge dx^{J\setminus j_m} = (-1)^m\,dx^J$, moving dx^{j_m} past the m forms $dx^{j_0},\dots,dx^{j_{m-1}}$, which takes the value $(-1)^m$. Conversely, collecting the terms, $\sum_J\big(\sum_m(-1)^m\partial_{j_m}\omega_{J\setminus j_m}\big)dx^J = \sum_I\sum_l\partial_l\omega_I\,dx^l\wedge dx^I$.

Step 2: the three rules hold in a chart. For a function, $d_Uf = \sum_l\partial_l f\,dx^l$ is the differential df, by the dual basis property. For the Leibniz rule it suffices by linearity to take $\alpha = f\,dx^I$ with $|I|=k$ and $\beta = g\,dx^J$; then $\alpha\wedge\beta = fg\,dx^I\wedge dx^J$ and

$$d_U(\alpha\wedge\beta) = (g\,df+f\,dg)\wedge dx^I\wedge dx^J = (df\wedge dx^I)\wedge(g\,dx^J) + (-1)^k(f\,dx^I)\wedge(dg\wedge dx^J),$$

moving the 1-form dg past the k-form dx^I. Finally $d_Ud_U(f\,dx^I) = \sum_{m,l}\partial_m\partial_l f\,dx^m\wedge dx^l\wedge dx^I = 0$: the terms (m,l) and (l,m) cancel because $\partial_m\partial_l f = \partial_l\partial_m f$ (Theorem 27.10.5) while $dx^m\wedge dx^l = -dx^l\wedge dx^m$, and the terms with $m=l$ vanish.

Step 3: uniqueness and locality. Let D be any family of linear maps with the three properties. If ω vanishes on an open set $U\ni p$, choose a bump function χ with support in U and $\chi(p)=1$ (Lemma 33.3.10); then $\chi\omega=0$, so $0 = D(\chi\omega) = d\chi\wedge\omega+\chi\,D\omega$, and at p this gives $(D\omega)_p = 0$. So $D\omega$ near p depends only on ω near p. Now let p lie in a chart domain U, choose bump functions χ and $\tilde\chi$ with support in U, with $\chi=1$ near p and $\tilde\chi=1$ near the support of χ (Corollary 33.8.3(1), applied to a compact neighbourhood of $\operatorname{supp}\chi$ in U). Then $\chi\omega = \sum_I(\chi\omega_I)\,d(\tilde\chi x^{i_1})\wedge\cdots\wedge d(\tilde\chi x^{i_k})$ is a form on M built from global functions, and the Leibniz rule and $D\circ D=0$ give $D(\chi\omega) = \sum_I d(\chi\omega_I)\wedge d(\tilde\chi x^{i_1})\wedge\cdots\wedge d(\tilde\chi x^{i_k})$, since $D(d(\tilde\chi x^i)) = DD(\tilde\chi x^i) = 0$. Near p this is $\sum_I d\omega_I\wedge dx^I$, and $D\omega = D(\chi\omega)$ near p by locality. So D is given by the coordinate formula in every chart, and is unique.

Step 4: existence, and what happens on the intersection of two charts. Define $d\omega$ on each chart domain U by d_U. On the intersection of two chart domains U and V, the operator d_V satisfies the three rules (Step 2, on the manifold $U\cap V$), so, writing $\omega = \sum_I\omega_I\,dx^I$ in the coordinates of U and using $d_V(dx^i) = d_V(d_Vx^i) = 0$ and the Leibniz rule,

$$d_V\Big(\sum_I\omega_I\,dx^I\Big) = \sum_I d_V\omega_I\wedge dx^I = \sum_I d\omega_I\wedge dx^I = d_U\Big(\sum_I\omega_I\,dx^I\Big).$$

So the local definitions agree on overlaps and glue to an operator d on M, which satisfies the three rules because they are local.

Step 5: naturality. For a function g on M, $f^*(dg) = d(g\circ f)$ by the chain rule. Locally $\omega = \sum_I\omega_I\,dx^{i_1}\wedge\cdots\wedge dx^{i_k}$, so $f^*\omega = \sum_I(\omega_I\circ f)\,d(x^{i_1}\circ f)\wedge\cdots\wedge d(x^{i_k}\circ f)$, and by the Leibniz rule and $d\circ d=0$ on N, $d(f^*\omega) = \sum_I d(\omega_I\circ f)\wedge d(x^{i_1}\circ f)\wedge\cdots\wedge d(x^{i_k}\circ f) = f^*\big(\sum_I d\omega_I\wedge dx^I\big) = f^*(d\omega)$.

Step 6: the invariant formula. Call the right-hand side $T(X_0,\dots,X_k)$. It is local in the X_i. It is $C^\infty(M)$-linear in X_0: replacing X_0 by fX_0, the term $i=0$ becomes f times itself; the terms $i\ge1$ acquire the extra summands $(-1)^i(X_if)\,\omega(X_0,\dots,\widehat{X_i},\dots,X_k)$; and the bracket terms with $i=0$ acquire, since $[fX_0,X_j] = f[X_0,X_j] - (X_jf)X_0$, the extra summands $-(-1)^j(X_jf)\,\omega(X_0,X_1,\dots,\widehat{X_j},\dots,X_k)$, which cancel the previous ones. A similar check shows that T changes sign when two arguments are exchanged, so it is $C^\infty(M)$-linear in every argument, like $d\omega$. Two such expressions agree if they agree on coordinate fields, and by

alternation it suffices to take $\partial_{j_0},\dots,\partial_{j_k}$ with $j_0<\cdots<j_k$. For coordinate fields all brackets vanish, and $T(\partial_{j_0},\dots,\partial_{j_k})=\sum_m(-1)^m\partial_{j_m}\omega_{J\setminus j_m}$, which is $d\omega(\partial_{j_0},\dots,\partial_{j_k})$ by Step 1. □

Remark 36.2.9. For a function f, $df=\sum_j(\partial f/\partial x^j)\,dx^j$ (Remark 36.2.2) is a 1-form, a section of the cotangent bundle T^*M; it should not be confused with the tangent map $Tf\colon TM\to T\mathbb{R}$, although at each point both are the linear map $v\mapsto v(f)$. The identity $d(df)=0$ is, in coordinates, the symmetry of second derivatives:

$$d(df)=\sum_j d\Big(\frac{\partial f}{\partial x^j}\Big)\wedge dx^j=\sum_{i,j}\frac{\partial^2 f}{\partial x^i\partial x^j}\,dx^i\wedge dx^j=\sum_{i<j}\Big(\frac{\partial^2 f}{\partial x^i\partial x^j}-\frac{\partial^2 f}{\partial x^j\partial x^i}\Big)dx^i\wedge dx^j=0$$

(Theorem 27.10.5). So a 1-form $\omega=\sum_j\omega_j\,dx^j$ can be exact only if $\partial_i\omega_j=\partial_j\omega_i$; this necessary condition is $d\omega=0$, and whether it is sufficient is the question answered by de Rham cohomology (Chapter 37).

Example 36.2.10. For $k=1$ the invariant formula reads $d\omega(X,Y)=X\omega(Y)-Y\omega(X)-\omega([X,Y])$, and in coordinates on $\mathbb{R}^2$, for $\omega=P\,dx+Q\,dy$, $d\omega=(\partial_xQ-\partial_yP)\,dx\wedge dy$: Step 1 with $J=(1,2)$ gives the coefficient $\partial_1\omega_2-\partial_2\omega_1$. This is the integrand of Green's theorem, which is Stokes' theorem (Theorem 36.4.8) for a domain in the plane.

Definition 36.2.11. For $X\in\mathfrak{X}(M)$ the *interior product* $\iota_X\colon\Omega^k\to\Omega^{k-1}$ is $(\iota_X\omega)(X_1,\dots,X_{k-1})=\omega(X,X_1,\dots,X_{k-1})$ (Definition 12.4.5), and the *Lie derivative* is $L_X\omega=\frac{d}{dt}\big|_{t=0}\Phi_t^*\omega$, where Φ is the flow of X.

The interior product inserts X as the first argument; on a function it is 0, and on a 1-form $\iota_X\alpha=\alpha(X)$. By Proposition 12.4.6 it is an antiderivation: $\iota_X(\alpha\wedge\beta)=\iota_X\alpha\wedge\beta+(-1)^k\alpha\wedge\iota_X\beta$ for a k-form α. For instance $\iota_{\partial_x}(dx\wedge dy)=dy$ and $\iota_{\partial_y}(dx\wedge dy)=-dx$.

The Lie derivative measures how ω changes along the flow of X. The flow is in general only defined for small $|t|$ near each point (Theorem 33.9.2), but this suffices: $(\Phi_t^*\omega)_p$ involves only the values of ω near $\Phi_t(p)$, it is defined for $|t|$ small, and it depends smoothly on (t,p), as its coefficients in a chart are built from the coefficients of ω and the derivatives of the flow. So $L_X\omega$ is again a k-form. On a function, $L_Xf=\frac{d}{dt}f(\Phi_t(p))|_{t=0}=Xf$.

Example 36.2.12. On $\mathbb{R}^2$ let $X=x\,\partial_x+y\,\partial_y$, whose flow is $\Phi_t(p)=e^tp$. Then $\Phi_t^*(dx\wedge dy)=d(e^tx)\wedge d(e^ty)=e^{2t}dx\wedge dy$, so $L_X(dx\wedge dy)=2\,dx\wedge dy$: the flow expands areas at the rate 2. For the rotation field $Y=-y\,\partial_x+x\,\partial_y$, the flow consists of rotations, which preserve $dx\wedge dy$, so $L_Y(dx\wedge dy)=0$. Cartan's formula below gives the same results by pure algebra: $\iota_X(dx\wedge dy)=x\,dy-y\,dx$, whose differential is $2\,dx\wedge dy$, and $\iota_Y(dx\wedge dy)=-y\,dy-x\,dx=-d(\frac12(x^2+y^2))$, whose differential is 0.

Proposition 36.2.13 (Cartan's formulas)**.** *$L_X=d\iota_X+\iota_Xd$ on forms; L_X commutes with d; $\frac{d}{dt}\Phi_t^*\omega=\Phi_t^*L_X\omega$; and $L_{[X,Y]}=L_XL_Y-L_YL_X$.*

Proof. Step 1: L_X is a local derivation of degree 0 commuting with d. Since $\Phi_t^*(\alpha\wedge\beta)=\Phi_t^*\alpha\wedge\Phi_t^*\beta$, differentiating at $t=0$ (the product rule, applied pointwise to the coefficients) gives $L_X(\alpha\wedge\beta)=L_X\alpha\wedge\beta+\alpha\wedge L_X\beta$. Since $\Phi_t^*d=d\Phi_t^*$ (naturality of d, Theorem 36.2.8) and derivatives in t and in the coordinates commute (the coefficients are smooth in (t,p)), $L_Xd=dL_X$. It is local, as noted above.

Step 2: $D=d\iota_X+\iota_Xd$ is a local derivation of degree 0 commuting with d. For a k-form α, using the antiderivation rules for d and for ι_X,

$$d\iota_X(\alpha\wedge\beta)=d\iota_X\alpha\wedge\beta+(-1)^{k-1}\iota_X\alpha\wedge d\beta+(-1)^kd\alpha\wedge\iota_X\beta+\alpha\wedge d\iota_X\beta,$$
$$\iota_Xd(\alpha\wedge\beta)=\iota_Xd\alpha\wedge\beta+(-1)^{k+1}d\alpha\wedge\iota_X\beta+(-1)^k\iota_X\alpha\wedge d\beta+\alpha\wedge\iota_Xd\beta,$$

and adding, the middle terms cancel: $D(\alpha\wedge\beta) = D\alpha\wedge\beta+\alpha\wedge D\beta$. Moreover $dD = d\iota_X d = Dd$, since $d\circ d = 0$.

Step 3: $L_X = D$. On a function, $L_X f = Xf = df(X) = \iota_X df = Df$, as $\iota_X f = 0$. On an exact 1-form df, both give $d(Xf)$, since both commute with d. Locally every form is a sum of forms $f\,dg_1\wedge\cdots\wedge dg_k$ (in a chart, $\omega_I\,dx^{i_1}\wedge\cdots\wedge dx^{i_k}$), and two local derivations of degree 0 that agree on functions and on exact 1-forms agree on such products, hence everywhere.

Step 4: the derivative at time t. By the group law, near a point and for small s, $\Phi_{t+s} = \Phi_s\circ\Phi_t$, so $\Phi_{t+s}^*\omega = \Phi_t^*(\Phi_s^*\omega)$, and differentiating at $s = 0$ gives $\frac{d}{dt}\Phi_t^*\omega = \Phi_t^* L_X\omega$ (Φ_t^* is linear and acts pointwise, so it commutes with $\frac{d}{ds}$).

Step 5: brackets. $L_XL_Y - L_YL_X$ is again a local derivation of degree 0 commuting with d, and on functions it is $XYf - YXf = [X,Y]f = L_{[X,Y]}f$. As in Step 3 the two derivations agree everywhere. □

By Step 4, the flow of X preserves a form ω, that is $\Phi_t^*\omega = \omega$ for all t, if and only if $L_X\omega = 0$. For example, the flow of a vector field on $\mathbb{R}^n$ preserves the volume form $dx^1\wedge\cdots\wedge dx^n$ exactly when $\operatorname{div} X = \sum_i \partial_i X^i = 0$ (Exercise 36.5.3); in dimension 2 the Hamiltonian vector fields are such (Exercise 36.5.9).

Example 36.2.14. On $\mathbb{R}^3$ the Euclidean metric and orientation identify 1-forms and 2-forms with vector fields, by $F = (F_1,F_2,F_3)\mapsto F_1\,dx + F_2\,dy + F_3\,dz$ and $F\mapsto F_1\,dy\wedge dz + F_2\,dz\wedge dx + F_3\,dx\wedge dy$, and 3-forms with functions. Under these identifications d on 0-, 1- and 2-forms is the gradient, the curl and the divergence, and $d\circ d = 0$ becomes $\operatorname{curl}\operatorname{grad} = 0$ and $\operatorname{div}\operatorname{curl} = 0$. Indeed $df = f_x\,dx + f_y\,dy + f_z\,dz$; for a 1-form,

$$d(F_1\,dx+F_2\,dy+F_3\,dz) = (\partial_yF_3-\partial_zF_2)\,dy\wedge dz+(\partial_zF_1-\partial_xF_3)\,dz\wedge dx+(\partial_xF_2-\partial_yF_1)\,dx\wedge dy,$$

since, for instance, $dF_1\wedge dx = \partial_yF_1\,dy\wedge dx + \partial_zF_1\,dz\wedge dx$; and for a 2-form, $d(F_1\,dy\wedge dz + F_2\,dz\wedge dx + F_3\,dx\wedge dy) = (\partial_xF_1+\partial_yF_2+\partial_zF_3)\,dx\wedge dy\wedge dz$, because only the derivative in the missing variable survives in each term.

Definition 36.2.15. A form ω is *closed* if $d\omega = 0$ and *exact* if $\omega = d\eta$ for some form η. Exact forms are closed, since $d\circ d = 0$.

Definition 36.2.16. For a 1-form α on M and a piecewise smooth curve $\gamma\colon[a,b]\to M$, the *line integral* is $\int_\gamma\alpha = \int_a^b \alpha_{\gamma(t)}(\dot\gamma(t))\,dt$, the sum of the integrals of $\gamma^*\alpha$ over the smooth pieces.

Proposition 36.2.17.
1. *The line integral is unchanged under orientation-preserving reparametrisation $\gamma\circ\phi$, $\phi\colon[c,d]\to[a,b]$ increasing and piecewise smooth with $\phi(c) = a$, $\phi(d) = b$, and changes sign under reversal.*
2. *(Fundamental theorem for line integrals) $\int_\gamma df = f(\gamma(b)) - f(\gamma(a))$.*
3. *On a connected manifold, a 1-form α is exact if and only if $\int_\gamma\alpha = 0$ for every closed piecewise smooth curve γ.*

Proof. (1) is the substitution rule (Corollary 27.8.4) on each piece, as $(\gamma\circ\phi)^*\alpha = \phi^*\gamma^*\alpha$. (2) $\gamma^*df = d(f\circ\gamma) = (f\circ\gamma)'\,dt$, and Theorem 27.8.3 applies on each piece; the intermediate values cancel. (3) If $\alpha = df$, closed curves give 0 by (2). Conversely, fix p_0 and define $f(p) = \int_\gamma\alpha$ for any piecewise smooth curve γ from p_0 to p; such curves exist since M is connected and locally path-connected by piecewise smooth curves (use straight segments in charts), and the value does not depend on γ, because for two curves γ, γ' the concatenation of γ with the reverse of γ' is closed. In a chart around p, with p at the origin, computing f along γ followed by the segment from p to $p+se_i$ gives $f(p+se_i) - f(p) = \int_0^s \alpha_i(p+ue_i)\,du$, so $\partial_i f = \alpha_i$ by Theorem 27.8.3; hence f is smooth and $df = \alpha$. □

Counterexample 36.2.18. Closed forms need not be exact. On $M = \mathbb{R}^2 \setminus \{0\}$, the *angle form* $\alpha = (x\,dy - y\,dx)/(x^2+y^2)$ is closed, as a direct computation shows, but along the unit circle $\gamma(t) = (\cos t, \sin t)$, $\gamma^*\alpha = (\cos^2 t + \sin^2 t)\,dt = dt$, so $\int_\gamma \alpha = 2\pi \neq 0$, and α is not exact by Proposition 36.2.17(3). On the half-plane $\{x > 0\}$ it is exact, equal to $d\arctan(y/x)$: exactness is a global question, measured by de Rham cohomology (Chapter 37).

36.3 Orientations

Definition 36.3.1. An *orientation* of a smooth n-manifold M is an equivalence class of nowhere vanishing n-forms, two being equivalent if they differ by a positive function; M is *orientable* if such a form exists, and *oriented* once one is chosen. A chart is *positively oriented* if $dx^1 \wedge \cdots \wedge dx^n$ represents the orientation.

Proposition 36.3.2. *M is orientable if and only if it has an atlas whose transition maps have positive Jacobian determinant, if and only if the line bundle* $\det T^*M$ *is trivial. For connected M there are then exactly two orientations. This agrees with the topological notion of Definition 23.3.1.*

Proof. A nowhere vanishing n-form μ gives such an atlas: take charts with $\mu = f\,dx^1\wedge\cdots\wedge dx^n$, $f > 0$ (compose with a reflection if necessary), and on overlaps the transition Jacobian is the ratio of two positive functions. Conversely, given such an atlas, patch the forms $dx^1 \wedge \cdots \wedge dx^n$ with a partition of unity: the sum is nowhere vanishing because in each chart it is a positive combination of positive multiples of $dx^1 \wedge \cdots \wedge dx^n$. A nowhere vanishing section of the line bundle $\det T^*M = \Lambda^n T^*M$ is the same as a trivialisation. For the last statement, in a positively oriented chart a generator of $H_n(M \mid p; \mathbb{Z})$ is determined by the orientation of $\mathbb{R}^n$, and this choice is locally constant, giving a section of the orientation covering; conversely such a section determines a compatible atlas. □

Counterexample 36.3.3. The open Möbius band is not orientable. Let $\mathbb{Z}$ act on $\mathbb{R} \times (-1,1)$ by $g^k(x,t) = (x+k, (-1)^k t)$, freely and properly, and let B be the smooth quotient (Theorem 33.7.1). A nowhere vanishing 2-form μ on B would pull back to a nowhere vanishing form $f\,dx \wedge dt$ on $\mathbb{R}\times(-1,1)$ invariant under g. As $g^*(f\,dx\wedge dt) = (f\circ g)\,dx \wedge d(-t) = -(f\circ g)\,dx\wedge dt$, invariance means $f \circ g = -f$. But f is continuous and nowhere zero on the connected set $\mathbb{R}\times(-1,1)$, hence of constant sign (Corollary 18.5.5), which contradicts $f(g(x,t)) = -f(x,t)$.

For surfaces in $\mathbb{R}^3$ one usually thinks of an orientation as a choice of side, that is, of a unit normal vector varying continuously along the surface. This is the same notion.

Proposition 36.3.4. *Let $S \subseteq \mathbb{R}^{n+1}$ be an n-dimensional submanifold. A* unit normal field *is a continuous map $\nu\colon S \to \mathbb{R}^{n+1}$ with $|\nu(p)| = 1$ and $\nu(p) \perp T_pS$ for all p. Unit normal fields correspond bijectively to orientations of S: to ν corresponds the orientation of the nowhere vanishing n-form $\mu_p(v_1,\dots,v_n) = \det(\nu(p), v_1, \dots, v_n)$, for which a basis $(v_1,\dots,v_n)$ of T_pS is positive exactly when $(\nu(p), v_1, \dots, v_n)$ is a positive basis of $\mathbb{R}^{n+1}$. In particular S is orientable if and only if it has a unit normal field, and every unit normal field is smooth. For a surface in $\mathbb{R}^3$, a basis (v,w) of T_pS is positive exactly when $v \times w$ is a positive multiple of $\nu(p)$.*

Proof. Near a point of S write S as a regular level set $F^{-1}(0)$ with F real valued (Proposition 33.6.18(4)). Then $T_pS = \ker d_pF = \nabla F(p)^\perp$ (Theorem 33.6.11), so the normal line at p is spanned by $\nabla F(p) \neq 0$, and a unit normal field is there $\varepsilon\,\nabla F/|\nabla F|$ with $\varepsilon = \pm 1$ continuous, hence locally constant; so ν is smooth. Given ν, μ is a smooth n-form on S (in a parametrisation its coefficient is $\det(\nu, x_1, \dots, x_n)$), and it vanishes nowhere, because $\nu(p) \notin T_pS$, so $(\nu(p), v_1, \dots, v_n)$ is a basis of $\mathbb{R}^{n+1}$ whenever (v_i) is one of T_pS. Conversely, given an orientation, exactly one of the two unit normals $\pm\nu(p)$ makes $\det(\nu(p), v_1, \dots, v_n) > 0$ for

positive bases (v_i), the choice not depending on the positive basis, as two positive bases differ by a matrix of positive determinant. Near a point, in the notation above, $\nu = \varepsilon \nabla F/|\nabla F|$ with ε the sign of $\det(\nabla F, \partial_1, \dots, \partial_n)$ for the coordinate fields of a positive chart, a continuous nowhere zero function; so ν is continuous. The two constructions are inverse to each other. For $n = 2$, $\det(\nu, v, w) = \langle \nu, v \times w \rangle$ and $v \times w$ is normal to $T_p S$. □

Remark 36.3.5. So a hypersurface is orientable when a unit normal can be carried continuously around every closed curve and comes back to itself. The regular level sets $F^{-1}(c) \subseteq \mathbb{R}^{n+1}$ are orientable, with $\nu = \nabla F/|\nabla F|$; for instance spheres, with the outward normal. For the Möbius band embedded in $\mathbb{R}^3$ as the image of $x(s,t) = ((1 + t\cos\frac{s}{2})\cos s, (1 + t\cos\frac{s}{2})\sin s, t\sin\frac{s}{2})$, $|t| < \frac{1}{2}$, the normal along the central circle $t = 0$ is $x_s \times x_t = (\cos s \sin\frac{s}{2}, \sin s \sin\frac{s}{2}, -\cos\frac{s}{2})$, which is $(0,0,-1)$ at $s = 0$ and $(0,0,1)$ at $s = 2\pi$, at the same point $(1,0,0)$: going once around reverses the normal, so there is no unit normal field, in accordance with Counterexample 36.3.3. The induced orientation of a boundary (Definition 36.4.5 below) is the same idea with the outward normal: for a domain in $\mathbb{R}^3$ bounded by a surface, the boundary is oriented by the outward unit normal, which is the convention of the divergence theorem.

36.4 Integration and Stokes' theorem

Let $\tau\colon V \to U$ be a diffeomorphism between open subsets of $\mathbb{R}^n$, with coordinates y on V and x on U. For a function f on U, the change of variables formula (Theorem 28.11.2) says $\int_U f\, d\lambda = \int_V (f \circ \tau)\,|\det D\tau|\, d\lambda$, while the pullback of the n-form $f\, dx^1 \wedge \cdots \wedge dx^n$ is $(f\circ\tau)\det D\tau\, dy^1 \wedge \cdots \wedge dy^n$ (Proposition 36.2.5). So if we integrate an n-form by integrating its coefficient, the answer does not change under orientation-preserving changes of coordinates ($\det D\tau > 0$) and changes sign under orientation-reversing ones. This is the whole reason for the following definition.

Definition 36.4.1. Let M be an oriented n-manifold and ω an n-form with compact support contained in a positively oriented chart $\varphi\colon U \to \mathbb{R}^n$. Write $(\varphi^{-1})^*\omega = f\, dx^1 \wedge \cdots \wedge dx^n$ and set $\int_M \omega = \int_{\varphi(U)} f\, d\lambda$. For general compactly supported ω, choose a partition of unity (ψ_i) subordinate to a cover by positively oriented charts and set $\int_M \omega = \sum_i \int_M \psi_i \omega$.

Proposition 36.4.2. *The integral is well defined, linear, and satisfies $\int_M \omega = \int_N f^*\omega$ for an orientation-preserving diffeomorphism $f\colon N \to M$; it changes sign if the orientation is reversed.*

Proof. One chart. Let ω have compact support in the domains of two positively oriented charts φ and ψ, and $\tau = \varphi \circ \psi^{-1}$, a diffeomorphism between open subsets of $\mathbb{R}^n$ with $\det D\tau > 0$. If $(\varphi^{-1})^*\omega = f\, dx$, with $dx = dx^1 \wedge \cdots \wedge dx^n$, then $(\psi^{-1})^*\omega = \tau^*(\varphi^{-1})^*\omega = (f\circ\tau)\det D\tau\, dy$, and by the change of variables formula $\int (f\circ\tau)\det D\tau\, d\lambda = \int (f\circ\tau)|\det D\tau|\, d\lambda = \int f\, d\lambda$. (The supports lie in the common domain, and the integrands vanish outside it.)

Partitions of unity. The sum in the definition is finite, because the compact support of ω meets only finitely many supports of a locally finite partition of unity, and each term is defined by the first case. For two partitions (ψ_i) and (χ_j), each $\psi_i\chi_j\omega$ is supported in the domains of the charts of both, so $\sum_i \int \psi_i\omega = \sum_{i,j} \int \chi_j\psi_i\omega = \sum_j \int \chi_j\omega$, using the linearity of the one-chart integral.

Properties. Linearity follows from that of the one-chart integral. If $f\colon N \to M$ is an orientation-preserving diffeomorphism, compositions $\varphi \circ f$ of positive charts of M with f are positive charts of N, and $\psi_i \circ f$ is a partition of unity; the coefficient of $f^*\omega$ in the chart $\varphi \circ f$ is that of ω in φ. So the defining sums agree. Reversing the orientation replaces each positive chart φ by $r \circ \varphi$, with the reflection $r(x^1, \dots, x^n) = (-x^1, x^2, \dots, x^n)$, which changes the sign of every one-chart integral. □

In practice one does not use partitions of unity: one parametrises M up to a set of measure zero.

Proposition 36.4.3. *Let M be an oriented n-manifold, $U \subseteq \mathbb{R}^n$ open and $x\colon U \to M$ an orientation-preserving diffeomorphism onto an open subset whose complement $M \setminus x(U)$ has measure zero, that is, its image in every chart has Lebesgue measure 0. For a compactly supported n-form ω write $x^*\omega = f\, du^1 \wedge \cdots \wedge du^n$. Then f is integrable and*

$$\int_M \omega = \int_U f\, d\lambda.$$

Proof. Choose a partition of unity (ψ_i) subordinate to positively oriented charts (U_i, φ_i); finitely many ψ_i meet the support of ω. In the one-chart integral of $\psi_i\omega$ we may remove the null set $\varphi_i(U_i \setminus x(U))$. On the rest, $\tau_i = \varphi_i \circ x$ is an orientation-preserving diffeomorphism from $x^{-1}(U_i)$ onto $\varphi_i(U_i \cap x(U))$, so by the change of variables formula, as in the proof of Proposition 36.4.2, $\int_M \psi_i\omega = \int_{x^{-1}(U_i)} f_i\, d\lambda$, where $x^*(\psi_i\omega) = f_i\, du$. Summing over the finitely many i, and using $\sum_i f_i = f$ on U, gives the claim; each f_i is integrable, so f is. □

Example 36.4.4.

1. *Curves.* If $\gamma\colon (a,b) \to M$ is an embedding onto a curve C, oriented by γ, then $\int_C \alpha = \int_a^b \alpha(\dot\gamma(t))\, dt$ is the line integral of Definition 36.2.16.

2. *The sphere.* Orient S^2 by its outward normal (Proposition 36.3.4); the spherical coordinates $x(\vartheta, \varphi)$ on $(0,\pi) \times (0, 2\pi)$ are positively oriented, since $x_\vartheta \times x_\varphi = \sin\vartheta\, x$ points outward, and they miss only a closed half great circle, which has measure zero (it is a segment in the chart given by $\varphi \in (-\pi, \pi)$, together with two points). By Example 36.2.6(3) and Proposition 36.4.3,

$$\int_{S^2} x\, dy \wedge dz + y\, dz \wedge dx + z\, dx \wedge dy = \int_0^{2\pi}\int_0^{\pi} \sin\vartheta\, d\vartheta\, d\varphi = 4\pi.$$

   With the inner normal the same integral is -4π.

3. *The angle form.* $\int_{S^1} \alpha = 2\pi$ for $\alpha = (x\, dy - y\, dx)/(x^2 + y^2)$ and the counterclockwise orientation of $S^1 \subseteq \mathbb{R}^2$ (Counterexample 36.2.18).

Definition 36.4.5. A *smooth n-manifold with boundary* is defined as in Definition 33.2.1 with charts taking values in open subsets of the half-space $\mathbb{H}^n = \{x \in \mathbb{R}^n : x^n \geq 0\}$; the *boundary* ∂M is the set of points corresponding to $x^n = 0$, a smooth $(n-1)$-manifold. If M is oriented, ∂M carries the *induced orientation*, defined by declaring a chart of ∂M positively oriented when the n-form $dx^1 \wedge \cdots \wedge dx^n$ of a positively oriented chart of M is $(-1)^n\, \nu \wedge (dx^1 \wedge \cdots \wedge dx^{n-1})$ for an outward-pointing covector ν; concretely, $(x^1, \ldots, x^{n-1})$ is positively oriented on ∂M when $(-1)^n(x^1, \ldots, x^n)$ is on M.

Here a map defined on an open subset of $\mathbb{H}^n$ (open for the subspace topology) is *smooth* if near each point it is the restriction of a smooth map defined on an open subset of $\mathbb{R}^n$; the transition maps between charts are required to be smooth in this sense. Tangent spaces, vector fields and forms are defined as for manifolds; at a boundary point the tangent space is still n-dimensional, and a tangent vector is *outward-pointing* if in a chart its n-th component is negative. The rule for the induced orientation says: a basis $(v_1, \ldots, v_{n-1})$ of $T_p\partial M$ is positive if and only if $(w, v_1, \ldots, v_{n-1})$ is a positive basis of T_pM for an outward vector w (*outward vector first*). Indeed, in a positive chart $w = -\partial_n$ is outward, and $\det(-e_n, e_1, \ldots, e_{n-1}) = (-1)^n$.

Lemma 36.4.6. *Whether a point of a manifold with boundary lies on the boundary does not depend on the chart. The restrictions of the charts to ∂M form an atlas of an $(n-1)$-manifold without boundary, and $M \setminus \partial M$ is an n-manifold without boundary.*

Proof. Suppose a chart φ sends p to a point with $x^n = 0$ and a chart ψ sends it to a point with $x^n > 0$. Shrinking the domain, ψ maps onto an open subset W of $\mathbb{R}^n$ contained in $\{x^n > 0\}$, and $\tau = \varphi \circ \psi^{-1} \colon W \to \mathbb{H}^n$ is smooth with a smooth inverse, which near $\varphi(p)$ extends to a smooth map σ on an open subset of $\mathbb{R}^n$. From $\sigma \circ \tau = \mathrm{id}$ near $\psi(p)$, the chain rule gives $D\sigma(\varphi(p))D\tau(\psi(p)) = I$, so $D\tau(\psi(p))$ is invertible. By the inverse function theorem (Theorem 27.11.2), τ maps a neighbourhood of $\psi(p)$ onto an open subset of $\mathbb{R}^n$ containing $\varphi(p)$; but every neighbourhood of $\varphi(p)$ in $\mathbb{R}^n$ contains points with $x^n < 0$, which are not in $\mathbb{H}^n$. This contradiction proves the first statement. The restrictions of the transition maps to $\{x^n = 0\} = \mathbb{R}^{n-1}$ are smooth with smooth inverses, which gives the atlas of ∂M, and the charts restricted to the interior points take values in the open set $\{x^n > 0\}$ of $\mathbb{R}^n$. □

Example 36.4.7.

1. $M = [a, b]$, with the orientation of $\mathbb{R}$. At b the outward direction is $+\partial_t$ and at a it is $-\partial_t$; a point is oriented by a sign, and "outward vector first" gives $\partial[a, b] = \{b\} - \{a\}$: the sign $+1$ at b and -1 at a.

2. The closed unit disc $\bar{D} \subseteq \mathbb{R}^2$: at a boundary point x the vector x points outward, and (x, v) is a positive basis of $\mathbb{R}^2$ exactly when v is a positive multiple of $(-x_2, x_1)$. So the boundary circle is oriented counterclockwise: the disc is on the left.

3. The closed ball in $\mathbb{R}^3$: its boundary S^2 is oriented by the outward normal, in the sense of Proposition 36.3.4.

4. The half-space $\mathbb{H}^n$ itself: $\partial\mathbb{H}^n = \mathbb{R}^{n-1}$ with $(-1)^n$ times its standard orientation; for $n = 2$ the boundary of the upper half-plane is the real line oriented from left to right.

Theorem 36.4.8 (Stokes)**.** *Let M be an oriented n-manifold with boundary and ω an $(n-1)$-form with compact support. Then*

$$\int_M d\omega = \int_{\partial M} \omega,$$

where ∂M has the induced orientation. If $\partial M = \emptyset$, the integral of an exact compactly supported form vanishes.

Proof. Reduction to one chart. Let (ψ_i) be a partition of unity subordinate to positively oriented charts. Only finitely many $\psi_i\omega$ are nonzero, $\omega = \sum_i \psi_i\omega$ and $d\omega = \sum_i d(\psi_i\omega)$. Both sides of the formula are linear in ω, so it suffices to prove it for each $\psi_i\omega$, that is, for ω supported in a single positively oriented chart. Via the chart, M becomes an open subset of $\mathbb{R}^n$ or of $\mathbb{H}^n$, and extending ω by 0, we may assume $M = \mathbb{R}^n$ or $M = \mathbb{H}^n$ and ω has compact support.

The computation. Write $\omega = \sum_i (-1)^{i-1} f_i \, dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n$; then $d\omega = (\sum_i \partial_i f_i)\, dx^1 \wedge \cdots \wedge dx^n$, because $df_i \wedge dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n = (-1)^{i-1}\partial_i f_i \, dx^1 \wedge \cdots \wedge dx^n$. By Fubini's theorem (Theorem 28.10.3) we may integrate $\partial_i f_i$ first in the variable x^i. For $i < n$, and also for $i = n$ when $M = \mathbb{R}^n$, the integral over the whole line vanishes by the fundamental theorem of calculus, as f_i has compact support. So $\int_{\mathbb{R}^n} d\omega = 0 = \int_{\partial\mathbb{R}^n} \omega$. For $M = \mathbb{H}^n$ the only surviving term is

$$\int_{\mathbb{H}^n} \partial_n f_n \, d\lambda = \int_{\mathbb{R}^{n-1}} \left(\int_0^\infty \partial_n f_n \, dx^n \right) dx^1 \cdots dx^{n-1} = -\int_{\mathbb{R}^{n-1}} f_n(x', 0)\, dx'.$$

On $\partial\mathbb{H}^n = \{x^n = 0\}$ the forms containing dx^n restrict to 0, so $\omega|_{\partial\mathbb{H}^n} = (-1)^{n-1} f_n \, dx^1 \wedge \cdots \wedge dx^{n-1}$, and the induced orientation is $(-1)^n$ times the standard one (Example 36.4.7(4)). Hence $\int_{\partial\mathbb{H}^n} \omega = (-1)^n(-1)^{n-1} \int f_n(x', 0)\, dx' = -\int f_n(x', 0)\, dx'$, which is the left side. □

Counterexample 36.4.9. Stokes' theorem needs compact support. On $M = \mathbb{R}$, without boundary, the 0-form arctan has $\int_{\mathbb{R}} d\arctan = \int_{\mathbb{R}} \frac{dx}{1+x^2} = \pi \neq 0$. On $M = [0, +\infty)$, with boundary the point 0 carrying the induced orientation -1, the constant function 1 has $\int_M d1 = 0$ while $\int_{\partial M} 1 = -1$.

Example 36.4.10. For $M = [a, b]$ with its standard orientation, $\partial M = \{b\} - \{a\}$ with the induced orientation, and Stokes' theorem for a function f is the fundamental theorem of calculus, $\int_a^b f' = f(b) - f(a)$. For a compact region $M \subseteq \mathbb{R}^2$ with smooth boundary and $\omega = P\,dx + Q\,dy$, it is Green's formula $\int_M (\partial_x Q - \partial_y P)\,dx\,dy = \int_{\partial M} P\,dx + Q\,dy$, the boundary being traversed with M on the left.

Example 36.4.11. For $M \subseteq \mathbb{R}^3$ a compact region with smooth boundary, Stokes' theorem specialises to the divergence theorem $\int_M \operatorname{div} F\,dV = \int_{\partial M} \langle F, \nu \rangle\,dA$ and to the classical Stokes and Green formulas, after identifying vector fields with 1- or 2-forms using the Euclidean metric and orientation (Theorem 12.5.3). On a compact oriented manifold without boundary, $\int_M d\omega = 0$, so integration descends to de Rham cohomology (Chapter 37).

Example 36.4.12.

1. *Area by a line integral.* For the unit disc $\bar{D}$ and $\omega = x\,dy$, $d\omega = dx \wedge dy$, so $\int_{\partial \bar{D}} x\,dy = \int_{\bar{D}} dx \wedge dy = \pi$. Directly, with $x = \cos t$, $y = \sin t$: $\int_0^{2\pi} \cos^2 t\,dt = \pi$. More generally the area of a plane region is $\frac{1}{2}\int_{\partial R} (x\,dy - y\,dx)$.

2. *Volume by a flux.* For the unit ball B and $\omega = x\,dy \wedge dz + y\,dz \wedge dx + z\,dx \wedge dy$, $d\omega = 3\,dx \wedge dy \wedge dz$, and Stokes' theorem gives $\int_{S^2} \omega = 3\operatorname{vol}(B) = 4\pi$, which is the value computed directly in Example 36.4.4(2). So $\operatorname{vol}(B) = 4\pi/3$.

3. *A closed form on an annulus.* For the annulus $A = \{1 \leq |x| \leq 2\}$ and the angle form α, $d\alpha = 0$ and ∂A is the outer circle counterclockwise and the inner circle clockwise, so $\int_{\partial A} \alpha = 2\pi - 2\pi = 0$, as it must be. The punctured disc is not compact, and there Stokes' theorem does not apply: the closed form α has integral 2π over the boundary circle.

Corollary 36.4.13. *On a compact oriented n-manifold M without boundary, $\int_M : \Omega^n(M) \to \mathbb{R}$ is surjective, and $\int_M \mu > 0$ for any μ representing the orientation. Hence M carries a* volume form, *and any two differ by a positive function.*

Proof. In a positively oriented chart the integral of a nonnegative bump n-form is positive; a form representing the orientation is positive in every positively oriented chart, so its integral is positive by the definition via a partition of unity. □

36.5 Exercises

Exercise 36.5.1. Compute the pullback of $dx \wedge dy$ under polar coordinates $(r, \theta) \mapsto (r\cos\theta, r\sin\theta)$, and the exterior derivative of $\alpha = (x\,dy - y\,dx)/(x^2 + y^2)$ on $\mathbb{R}^2 \setminus \{0\}$. Show that α is closed but not exact, by integrating over the unit circle.

Solution. The pullback is $r\,dr \wedge d\theta$. A computation gives $d\alpha = 0$, while $\int_{S^1} \alpha = 2\pi \neq 0$; an exact form would integrate to 0 over the closed curve S^1 by Stokes' theorem. □

Exercise 36.5.2. Show that $\mathbb{RP}^n$, $n \geq 1$, is orientable if and only if n is odd, by computing the effect of the antipodal map on a volume form of S^n (compare Exercise 23.9.2).

Solution. Let $\Omega = dx^1 \wedge \cdots \wedge dx^{n+1}$ and $\omega = (\iota_x \Omega)|_{S^n} = \sum_i (-1)^{i-1} x^i\,dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^{n+1}$, restricted to S^n. For a basis $v_1, \ldots, v_n$ of $T_x S^n = x^\perp$, $\omega_x(v_1, \ldots, v_n) = \det(x, v_1, \ldots, v_n) \neq 0$,

so ω is a volume form. The antipodal map $a(x) = -x$ changes the sign of every x^i and of every dx^j, so each term of ω is multiplied by $(-1)^{n+1}$: $a^*\omega = (-1)^{n+1}\omega$. Let $\pi\colon S^n \to \mathbb{RP}^n$ be the projection, a local diffeomorphism with $\pi \circ a = \pi$.

If n is odd, $a^*\omega = \omega$. Define an n-form η on $\mathbb{RP}^n$ by $\eta = (\sigma^{-1})^*\omega$ on $\pi(U)$, where $U \subseteq S^n$ is open with $U \cap a(U) = \emptyset$ and $\sigma = \pi|_U$; the other local inverse is $\pi|_{a(U)} = \sigma \circ a^{-1}$ on $a(U)$, and it gives the same form since $a^*\omega = \omega$. So η is well defined, nowhere zero, and $\mathbb{RP}^n$ is orientable (Proposition 36.3.2). If n is even and η is a nowhere vanishing n-form on $\mathbb{RP}^n$, then $\pi^*\eta = f\,\omega$ with f nowhere zero, hence of constant sign since S^n is connected; but $a^*\pi^*\eta = \pi^*\eta$ gives $(f \circ a)(-1)^{n+1}\omega = f\,\omega$, that is, $f \circ a = -f$, a contradiction. (For $n = 0$, $\mathbb{RP}^0$ is a point, which is orientable.) □

Exercise 36.5.3. Let μ be a volume form and X a vector field. Show that $L_X\mu = (\operatorname{div} X)\mu$ defines the divergence, that $\operatorname{div} X = \sum_i \partial_i X^i$ in coordinates with $\mu = dx^1 \wedge \cdots \wedge dx^n$, and that the flow of X preserves μ if and only if $\operatorname{div} X = 0$.

Solution. $L_X\mu$ is an n-form, hence a function times μ. By Cartan's formula $L_X\mu = d\,\iota_X\mu$, and computing $\iota_X\mu$ in coordinates gives $\sum_i (-1)^{i-1} X^i\, dx^1 \wedge \cdots \widehat{dx^i} \cdots \wedge dx^n$, whose differential is $(\sum_i \partial_i X^i)\mu$. The flow preserves μ if and only if $\frac{d}{dt}\Phi_t^*\mu = \Phi_t^* L_X\mu$ vanishes. □

Exercise 36.5.4. Let $f\colon M \to N$ be a smooth map between compact oriented n-manifolds without boundary. Show that $\int_M f^*\omega = \deg(f) \int_N \omega$ for a suitable integer $\deg(f)$ independent of ω, using Sard's theorem and the local form of f near regular values.

Exercise 36.5.5. On $\mathbb{R}^3$ let $\omega = yz\,dx + zx\,dy + xy\,dz$. Show that $d\omega = 0$ and find f with $df = \omega$. Compute also $d(e^{xy}\,dx \wedge dz)$.

Solution. $d\omega = (z\,dy + y\,dz) \wedge dx + (z\,dx + x\,dz) \wedge dy + (y\,dx + x\,dy) \wedge dz$, and the six terms cancel in pairs, for example $z\,dy \wedge dx + z\,dx \wedge dy = 0$. A primitive is $f = xyz$. Next, $d(e^{xy}dx \wedge dz) = (ye^{xy}dx + xe^{xy}dy) \wedge dx \wedge dz = xe^{xy}\,dy \wedge dx \wedge dz = -xe^{xy}\,dx \wedge dy \wedge dz$. □

Exercise 36.5.6. Show that the pullback of $dx \wedge dy \wedge dz$ under spherical coordinates $(r, \vartheta, \varphi) \mapsto (r\sin\vartheta\cos\varphi, r\sin\vartheta\sin\varphi, r\cos\vartheta)$ is $r^2\sin\vartheta\, dr \wedge d\vartheta \wedge d\varphi$, and compute the volume of the ball of radius R as the integral of $dx \wedge dy \wedge dz$.

Solution. The pullback is the Jacobian determinant times $dr \wedge d\vartheta \wedge d\varphi$. Expanding the 3×3 determinant along the row of $z = r\cos\vartheta$, whose entries are $\cos\vartheta$, $-r\sin\vartheta$, 0, gives $\cos\vartheta \cdot r^2\sin\vartheta\cos\vartheta + r\sin\vartheta \cdot r\sin^2\vartheta = r^2\sin\vartheta$. By Proposition 36.4.3, the volume is $\int_0^{2\pi}\int_0^{\pi}\int_0^{R} r^2\sin\vartheta\, dr\, d\vartheta\, d\varphi = \frac{4}{3}\pi R^3$. □

Exercise 36.5.7. Verify Stokes' theorem for $\omega = x\,dy - y\,dx$ on the closed upper hemisphere $H = S^2 \cap \{z \geq 0\}$, oriented by the outward normal, by computing both sides.

Solution. $d\omega = 2\,dx \wedge dy$. The parametrisation $(u, v) \mapsto (u, v, \sqrt{1 - u^2 - v^2})$ of the open upper hemisphere by the open unit disc is positively oriented, as its normal $x_u \times x_v$ has positive third component, and the pullback of $dx \wedge dy$ is $du \wedge dv$; so $\int_H d\omega = 2\pi$ (Proposition 36.4.3; the equator has measure zero in H). The boundary is the equator. At $(1, 0, 0)$ an outward vector of H is $-e_3$, and $(-e_3, e_2)$ is positive for the outward normal e_1, since $\det(e_1, -e_3, e_2) = 1$; so the equator is traversed counterclockwise seen from above, and $\int_{\partial H} x\,dy - y\,dx = \int_0^{2\pi}(\cos^2 t + \sin^2 t)\,dt = 2\pi$. □

Exercise 36.5.8. On $\mathbb{R}^3$ let $X = x\,\partial_y - y\,\partial_x$ be the infinitesimal rotation about the z-axis. Compute L_X of dx, dy, dz, $x\,dy - y\,dx$ and $dx \wedge dy$, and interpret the results.

Solution. $L_X dx = d(Xx) = d(-y) = -dy$, $L_X dy = d(Xy) = dx$, $L_X dz = 0$. Then $L_X(x\,dy) = (Xx)\,dy + x\,L_X dy = -y\,dy + x\,dx$ and $L_X(y\,dx) = x\,dx - y\,dy$, so $L_X(x\,dy - y\,dx) = 0$, and $L_X(dx \wedge dy) = -dy \wedge dy + dx \wedge dx = 0$. The rotations preserve the angular form and the area form of the planes orthogonal to the axis, and the height z. □

Exercise 36.5.9. For a smooth function H on $\mathbb{R}^2$ let $X_H = \partial_y H\,\partial_x - \partial_x H\,\partial_y$. Show that $\iota_{X_H}(dx \wedge dy) = dH$, that $X_H H = 0$, and deduce that the flow of X_H preserves areas and the level sets of H.

Solution. $\iota_{X_H}(dx \wedge dy) = \partial_y H\,dy + \partial_x H\,dx = dH$. By Cartan's formula $L_{X_H}(dx \wedge dy) = d\iota_{X_H}(dx \wedge dy) = d\,dH = 0$, so the flow preserves $dx \wedge dy$ (Proposition 36.2.13). And $X_H H = \partial_y H\,\partial_x H - \partial_x H\,\partial_y H = 0$, so H is constant along the flow lines. This is the two-dimensional case of Liouville's theorem in Hamiltonian mechanics. □

Exercise 36.5.10. Let $C = \{x^2 + y^2 = 1\} \subseteq \mathbb{R}^3$ be the cylinder, with the parametrisation $(\theta, z) \mapsto (\cos\theta, \sin\theta, z)$ and inclusion ι. Compute ι^* of $dx \wedge dy$, $dx \wedge dz$, $dy \wedge dz$ and $x\,dy \wedge dz - y\,dx \wedge dz$, and of the 1-forms $x\,dy - y\,dx$ and dz. Which of these forms restrict to zero on C, and why is that visible without computation for $dx \wedge dy$?

Solution. $\iota^* dx = -\sin\theta\,d\theta$, $\iota^* dy = \cos\theta\,d\theta$, $\iota^* dz = dz$. So $\iota^*(dx \wedge dy) = 0$, $\iota^*(dx \wedge dz) = -\sin\theta\,d\theta \wedge dz$, $\iota^*(dy \wedge dz) = \cos\theta\,d\theta \wedge dz$, $\iota^*(x\,dy \wedge dz - y\,dx \wedge dz) = (\cos^2\theta + \sin^2\theta)\,d\theta \wedge dz = d\theta \wedge dz$, the area form of the cylinder, $\iota^*(x\,dy - y\,dx) = d\theta$ and $\iota^* dz = dz$. A form restricts to zero exactly when it vanishes on all tuples of tangent vectors of C. At each point the tangent plane of C is spanned by ∂_z and a horizontal vector w, and $dx \wedge dy(w, \partial_z) = 0$ since $dx(\partial_z) = dy(\partial_z) = 0$; by bilinearity $dx \wedge dy$ vanishes on the whole tangent plane. Geometrically, $dx \wedge dy$ measures the area of the projection to the (x, y)-plane, which maps C onto a circle, of area zero. Note that $\iota^*(x\,dy - y\,dx) = d\theta$ is closed but not exact on C, although $x\,dy - y\,dx$ is not closed on $\mathbb{R}^3$. □

Chapter 37

de Rham Cohomology

37.1 Introduction

Every exact form is closed, because $d \circ d = 0$. The converse is a question about the shape of the manifold. On $\mathbb{R}^n$, and on any convex open set, every closed form of positive degree is exact (the Poincaré lemma). On the punctured plane $\mathbb{R}^2 \setminus \{0\}$ the angle form $(x\,dy - y\,dx)/(x^2 + y^2)$ is closed but not exact: its integral around the origin is 2π, while the integral of an exact form around a closed curve is 0. The obstruction comes from the hole at the origin. *De Rham cohomology*, closed forms modulo exact forms, measures such obstructions in every degree:

$$H^k_{\mathrm{dR}}(M) = \frac{\{\text{closed } k\text{-forms}\}}{\{\text{exact } k\text{-forms}\}}.$$

The de Rham theorem (Theorem 37.3.1) says that this analytic invariant is the singular cohomology of M with real coefficients, a topological invariant. This is the bridge between analysis and topology on which Hodge theory is built: the Hodge theorem selects a canonical harmonic representative in each de Rham class (Chapter 43), and on a complex manifold the type decomposition of harmonic forms gives the Hodge decomposition (Chapter 48).

The chapter goes as follows.

1. The de Rham complex, first examples, homotopy invariance with an explicit homotopy operator, and the Poincaré lemma with an explicit formula for primitives (Section 37.2).
2. The Mayer–Vietoris sequence, with its connecting homomorphism made explicit, and the computation of the cohomology of spheres, punctured planes and projective spaces.
3. The de Rham theorem, periods of 1-forms, the Künneth formula, Poincaré duality, and the cohomology of compact Lie groups (Section 37.3).
4. Currents, the forms with distribution coefficients, which contain the submanifolds and are the natural home of cycle classes (Section 37.4).

References: [BT82], [War83], [GH78, Chapter 0].

Prerequisites. Chapter 36 (Differential Forms and Integration), Chapter 26 (Sheaf Cohomology), Chapter 23 (Topological Manifolds and Poincaré Duality) and Chapter 32 (Distributions and Sobolev Spaces).

37.2 The de Rham complex

Definition 37.2.1. The *de Rham complex* of a smooth manifold M is $0 \to \Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \cdots$, a complex by Theorem 36.2.8. Its cohomology $H^k_{\mathrm{dR}}(M) = \ker(d\colon \Omega^k \to$

$\Omega^{k+1})/\mathrm{im}(d\colon \Omega^{k-1} \to \Omega^k)$ is the *de Rham cohomology*; its elements are classes $[\omega]$ of *closed* forms modulo *exact* ones. The wedge product makes $H^\bullet_{\mathrm{dR}}(M)$ a graded-commutative $\mathbb{R}$-algebra, and a smooth map $f\colon N \to M$ induces an algebra homomorphism $f^*\colon H^\bullet_{\mathrm{dR}}(M) \to H^\bullet_{\mathrm{dR}}(N)$. Replacing $\Omega^\bullet(M)$ by the compactly supported forms $\Omega^\bullet_c(M)$ gives $H^\bullet_{\mathrm{dR},c}(M)$, functorial for open inclusions (extension by zero) and for proper maps.

Proposition 37.2.2. *$H^0_{\mathrm{dR}}(M) \cong \mathbb{R}^{\pi_0(M)}$, the locally constant functions. The wedge product is well defined on classes, and f^* commutes with d, hence descends to cohomology.*

Proof. A function with $df = 0$ has vanishing differential, hence is constant on connected components (Theorem 27.6.6 in charts, and connectedness). If α, β are closed then $d(\alpha \wedge \beta) = 0$, and if $\alpha = d\gamma$ then $\alpha \wedge \beta = d(\gamma \wedge \beta)$, because $d(\gamma \wedge \beta) = d\gamma \wedge \beta \pm \gamma \wedge d\beta$ and $d\beta = 0$; naturality of d is Theorem 36.2.8. □

Example 37.2.3.

1. *A point.* On the 0-dimensional manifold $\{p\}$, $\Omega^0 = \mathbb{R}$ (the functions are the constants) and $\Omega^k = 0$ for $k \geq 1$. So $H^0_{\mathrm{dR}}(\{p\}) = \mathbb{R}$ and $H^k_{\mathrm{dR}}(\{p\}) = 0$ for $k \geq 1$; more generally a 0-dimensional manifold, a countable discrete set S, has $H^0 = \mathbb{R}^S$ and nothing else. By homotopy invariance (Theorem 37.2.4 below) every contractible manifold, such as $\mathbb{R}^n$ or a ball, has the de Rham cohomology of a point: this is the Poincaré lemma.

2. *The real line.* Every 1-form on $\mathbb{R}$ is $f\,dx$, and it is exact: $f\,dx = dF$ with $F(x) = \int_0^x f$. So $H^0_{\mathrm{dR}}(\mathbb{R}) = \mathbb{R}$ and $H^k_{\mathrm{dR}}(\mathbb{R}) = 0$ for $k \geq 1$ (there are no forms of degree ≥ 2).

3. *The circle.* On $S^1 = \mathbb{R}/2\pi\mathbb{Z}$ every 1-form is closed, and is $f(\theta)\,d\theta$ with f periodic. It is exact if and only if $\int_0^{2\pi} f = 0$: then $F(\theta) = \int_0^\theta f$ is periodic and $dF = f\,d\theta$; conversely $\int_{S^1} dF = 0$. So $\int_{S^1}\colon H^1_{\mathrm{dR}}(S^1) \to \mathbb{R}$ is an isomorphism, and $[d\theta]$ is a basis. Here $d\theta$ is a closed 1-form which is not the differential of a function, since θ is not a function on S^1.

4. *The punctured plane.* The angle form $\alpha = (x\,dy - y\,dx)/(x^2 + y^2)$ is closed and not exact on $\mathbb{R}^2 \setminus \{0\}$ (Counterexample 36.2.18); so $H^1_{\mathrm{dR}}(\mathbb{R}^2 \setminus \{0\}) \neq 0$. It is the pullback of $d\theta$ by $(x, y) \mapsto (x, y)/r \in S^1$, and we will see that it spans H^1 (Example 37.3.4).

Theorem 37.2.4 (Homotopy invariance and the Poincaré lemma)**.** *Let $i_0, i_1\colon M \to M \times \mathbb{R}$ be $p \mapsto (p, 0)$ and $p \mapsto (p, 1)$. There is a linear map $K\colon \Omega^k(M \times \mathbb{R}) \to \Omega^{k-1}(M)$ with*

$$i_1^*\omega - i_0^*\omega = d(K\omega) + K(d\omega).$$

Consequently smoothly homotopic maps induce the same map on de Rham cohomology, homotopy equivalent manifolds have isomorphic de Rham cohomology, and $H^k_{\mathrm{dR}}(U) = 0$ for $k > 0$ and $H^0_{\mathrm{dR}}(U) = \mathbb{R}$ for every convex open $U \subseteq \mathbb{R}^n$ (the Poincaré lemma*).*

Proof. The decomposition. Let t be the coordinate of $\mathbb{R}$. In local coordinates x on M and t, a k-form on $M \times \mathbb{R}$ is a sum of terms $f(x, t)\,dx^I$ and $g(x, t)\,dt \wedge dx^J$; so it can be written uniquely as

$$\omega = \alpha_t + dt \wedge \beta_t,$$

where α_t is a k-form and β_t a $(k-1)$-form on M, depending smoothly on the parameter t (their coefficients are $f(\cdot, t)$ and $g(\cdot, t)$); this is independent of the coordinates on M. For instance $i_s^*\omega = \alpha_s$, since $i_s^*dt = 0$. Write d_M for the exterior derivative in the M-directions at fixed t, and ∂_t for the derivative of the coefficients in t. Then $d = d_M + dt \wedge \partial_t$ on forms without dt, and

$$d\omega = d_M\alpha_t + dt \wedge \partial_t\alpha_t + d(dt \wedge \beta_t) = d_M\alpha_t + dt \wedge (\partial_t\alpha_t - d_M\beta_t),$$

because $d(dt \wedge \beta_t) = -dt \wedge d\beta_t = -dt \wedge (d_M\beta_t + dt \wedge \partial_t\beta_t) = -dt \wedge d_M\beta_t$.

The homotopy operator. Define $K\omega = \int_0^1 \beta_t\, dt$, the $(k-1)$-form on M whose coefficients are the integrals over $t \in [0,1]$ of those of β_t. By the formula for $d\omega$, its dt-part is $\partial_t\alpha_t - d_M\beta_t$, so $K(d\omega) = \int_0^1 (\partial_t\alpha_t - d_M\beta_t)\, dt$. On the other hand $d(K\omega) = \int_0^1 d_M\beta_t\, dt$, by differentiation under the integral sign (Proposition 27.8.9). Adding, and using the fundamental theorem of calculus for the coefficients,

$$d(K\omega) + K(d\omega) = \int_0^1 \partial_t\alpha_t\, dt = \alpha_1 - \alpha_0 = i_1^*\omega - i_0^*\omega.$$

If $F\colon M \times \mathbb{R} \to N$ is smooth with $F \circ i_0 = f$ and $F \circ i_1 = g$, then for closed ω on N, $g^*\omega - f^*\omega = d(KF^*\omega)$, so $f^* = g^*$ on cohomology. Continuous homotopies may be replaced by smooth ones: continuous maps between manifolds are homotopic to smooth maps, and continuously homotopic smooth maps are smoothly homotopic [Hir76, Chapter 2]. A convex U is smoothly contractible, by the straight-line homotopy, so its de Rham cohomology is that of a point. □

Remark 37.2.5. *Smooth and continuous homotopies.* Here a homotopy between smooth maps $f, g\colon M \to N$ is a smooth map $F\colon M \times \mathbb{R} \to N$ with $F(\cdot,0) = f$ and $F(\cdot,1) = g$, a *smooth homotopy*, and a smooth homotopy equivalence is defined with smooth maps and smooth homotopies. Nothing is lost: by the Whitney approximation theorem every continuous map between smooth manifolds is homotopic to a smooth one, and two smooth maps that are homotopic through continuous maps are smoothly homotopic [Lee13, Chapter 6]. So homotopic in the sense of Chapter 19 and smoothly homotopic agree for smooth maps, and homotopy equivalent manifolds have isomorphic de Rham cohomology, in accordance with the de Rham theorem below, which identifies de Rham cohomology with singular cohomology, a homotopy invariant.

The homotopy operator gives explicit primitives. For a star-shaped open set $U \subseteq \mathbb{R}^n$ (with respect to 0), let $X = \sum_i x^i\partial_i$ be the position vector field.

Corollary 37.2.6 (Explicit Poincaré lemma)**.** *Let $U \subseteq \mathbb{R}^n$ be star-shaped with respect to 0 and ω a closed k-form on U, $k \geq 1$. Then $\omega = d\eta$ with*

$$\eta_x = \int_0^1 t^{k-1}\,(\iota_X\omega)_{tx}\, dt, \qquad \textit{that is,} \qquad \eta = \sum_I \Big(\int_0^1 t^{k-1}\omega_I(tx)\, dt\Big)\iota_X(dx^I),$$

where $\iota_X(dx^{i_1} \wedge \cdots \wedge dx^{i_k}) = \sum_m (-1)^{m-1}x^{i_m}\, dx^{i_1} \wedge \cdots \wedge \widehat{dx^{i_m}} \wedge \cdots \wedge dx^{i_k}$.

Proof. Let $\rho\colon \mathbb{R} \to [0,1]$ be smooth and nondecreasing, with $\rho = 0$ on $(-\infty, 0]$ and $\rho = 1$ on $[1,\infty)$, and apply Theorem 37.2.4 to $F(x,t) = \rho(t)x$, a smooth map $U \times \mathbb{R} \to U$ since U is star-shaped. Since $F^*dx^i = \rho(t)\, dx^i + \rho'(t)x^i\, dt$, the terms of $F^*(\omega_I\, dx^{i_1} \wedge \cdots \wedge dx^{i_k})$ containing dt are $\omega_I(\rho x)\,\rho^{k-1}\rho' \sum_m x^{i_m}\, dx^{i_1} \wedge \cdots \wedge dt \wedge \cdots \wedge dx^{i_k}$, with dt in the m-th place; moving dt to the front gives the sign $(-1)^{m-1}$. So $\beta_t = \rho(t)^{k-1}\rho'(t)(\iota_X\omega)_{\rho(t)x}$ in the notation of the proof of the theorem, and the substitution $s = \rho(t)$ gives $K(F^*\omega) = \int_0^1 s^{k-1}(\iota_X\omega)_{sx}\, ds = \eta$. As $F(\cdot,1) = \mathrm{id}$, $F(\cdot,0)$ is constant, whose pullback kills forms of positive degree, and $d\omega = 0$, the homotopy formula gives $\omega = d\eta$. □

Example 37.2.7.

1. For the closed 1-form $\omega = 2xy\, dx + x^2\, dy$ on $\mathbb{R}^2$, $\iota_X\omega = 2x^2y + x^2y = 3x^2y$, so $\eta = \int_0^1 3t^2\, dt \cdot x^2y = x^2y$, and indeed $d(x^2y) = \omega$.

2. For $\omega = dx \wedge dy$ on $\mathbb{R}^2$, $\iota_X\omega = x\, dy - y\, dx$ and $\eta = \int_0^1 t\, dt\,(x\, dy - y\, dx) = \frac{1}{2}(x\, dy - y\, dx)$, the primitive used in Example 36.4.12(1) to compute areas.

3. For the volume form $dx\wedge dy\wedge dz$ of $\mathbb{R}^3$ the formula gives $\frac{1}{3}(x\,dy\wedge dz + y\,dz\wedge dx + z\,dx\wedge dy)$.

4. The slit plane $\mathbb{R}^2 \setminus \{(x,0) : x \le 0\}$ does not contain 0, but it is star-shaped with respect to $(1,0)$: a segment from $(1,0)$ to a point off the x-axis meets the axis only at $(1,0)$. So the angle form is exact there; a primitive is the polar angle with values in $(-\pi,\pi)$.

Theorem 37.2.8 (Mayer–Vietoris)**.** *Let $M = U \cup V$ with U, V open. There is a long exact sequence*

$$\cdots \to H^k_{\mathrm{dR}}(M) \to H^k_{\mathrm{dR}}(U) \oplus H^k_{\mathrm{dR}}(V) \to H^k_{\mathrm{dR}}(U \cap V) \xrightarrow{\delta} H^{k+1}_{\mathrm{dR}}(M) \to \cdots$$

Proof. The sequence of complexes $0 \to \Omega^\bullet(M) \to \Omega^\bullet(U) \oplus \Omega^\bullet(V) \to \Omega^\bullet(U \cap V) \to 0$, with maps $\omega \mapsto (\omega|_U, \omega|_V)$ and $(\alpha,\beta) \mapsto \alpha|_{U\cap V} - \beta|_{U\cap V}$, is exact. Injectivity: a form vanishing on U and on V vanishes. Exactness in the middle: if $\alpha = \beta$ on $U \cap V$, they glue to a form on M. Surjectivity: for a partition of unity $\{\psi_U, \psi_V\}$ subordinate to $\{U, V\}$ (Theorem 33.8.2) and $\gamma \in \Omega^\bullet(U \cap V)$, the forms $\psi_V\gamma$ on U and $-\psi_U\gamma$ on V, both extended by zero (the support of ψ_V does not meet $U \setminus V$, and similarly for ψ_U), have difference γ on $U \cap V$. The maps commute with d, so Theorem 15.4.3 gives the long exact sequence. Following the construction of the connecting homomorphism, for a closed γ on $U \cap V$ we lift it to $(\psi_V\gamma, -\psi_U\gamma)$, apply d, and observe that $d(\psi_V\gamma)$ and $-d(\psi_U\gamma)$ agree on $U \cap V$, since $d(\psi_U\gamma + \psi_V\gamma) = d\gamma = 0$. So

$$\delta[\gamma] = [\omega], \qquad \omega = \begin{cases} d(\psi_V\gamma) = d\psi_V \wedge \gamma & \text{on } U, \\ -d(\psi_U\gamma) = -d\psi_U \wedge \gamma & \text{on } V, \end{cases}$$

a closed form on M supported in $U \cap V$. □

Example 37.2.9. $H^k_{\mathrm{dR}}(S^n) \cong \mathbb{R}$ for $k = 0, n$ and 0 otherwise ($n \ge 1$). Let $U = S^n \setminus \{N\}$ and $V = S^n \setminus \{S\}$ for the north and south poles. Both are diffeomorphic to $\mathbb{R}^n$ by stereographic projection, so their cohomology is $\mathbb{R}$ in degree 0 and 0 otherwise; and $U \cap V$ is diffeomorphic to $\mathbb{R}^n \setminus \{0\}$, which is homotopy equivalent to S^{n-1} (by $x \mapsto x/|x|$, with the inclusion of the unit sphere as homotopy inverse, and the homotopy $(x,t) \mapsto x/((1-t)+t|x|)$), so $H^\bullet_{\mathrm{dR}}(U \cap V) \cong H^\bullet_{\mathrm{dR}}(S^{n-1})$ by Theorem 37.2.4. The Mayer–Vietoris sequence splits into pieces:

1. for $k \ge 1$ the terms $H^k(U) \oplus H^k(V)$ vanish, so $\delta\colon H^k_{\mathrm{dR}}(U \cap V) \to H^{k+1}_{\mathrm{dR}}(S^n)$ is an isomorphism, that is $H^{k+1}(S^n) \cong H^k(S^{n-1})$;

2. in low degrees, $0 \to H^0(S^n) \to H^0(U) \oplus H^0(V) = \mathbb{R}^2 \to H^0(U \cap V) \to H^1(S^n) \to 0$.

For $n = 1$, $U \cap V$ has two components, so $H^0(U \cap V) = \mathbb{R}^2$, the map $\mathbb{R}^2 \to \mathbb{R}^2$, $(a,b) \mapsto (a-b, a-b)$, has rank 1, and $H^1(S^1) \cong \mathbb{R}$, $H^0(S^1) = \mathbb{R}$. For $n \ge 2$, $U \cap V$ is connected, the map $\mathbb{R}^2 \to \mathbb{R}$ is surjective, and $H^1(S^n) = 0$, $H^0(S^n) = \mathbb{R}$. Induction on n with (1) gives the result: $H^n(S^n) \cong H^{n-1}(S^{n-1}) \cong \cdots \cong H^1(S^1) \cong \mathbb{R}$, and all other groups vanish. For the circle, the generator of H^1 is the form $d\theta$, which is closed but not exact; its integral over S^1 is 2π (Exercise 36.5.1). In general $\int_M\colon H^n_{\mathrm{dR}}(M) \to \mathbb{R}$ is well defined for compact oriented M by Stokes' theorem (Theorem 36.4.8).

The connecting homomorphism can be seen at work on the circle. Take U and V two open arcs covering S^1, whose intersection consists of two arcs A and B, and let γ be the locally constant function equal to 1 on A and 0 on B. Then $\delta[\gamma]$ is represented by $d\psi_V$ on A (and 0 elsewhere), a 1-form supported in A. Near the end of A where V stops, $\psi_V = 0$, as the support of ψ_V lies in V; near the end where U stops, $\psi_U = 0$, so $\psi_V = 1$. Hence $\int_A d\psi_V = \pm 1$, and $\delta[\gamma]$ is a nonzero multiple of the generator of $H^1(S^1)$: a "bump" 1-form concentrated on one arc.

37.3 The de Rham theorem

Theorem 37.3.1 (de Rham)**.** *For every smooth manifold M there are natural isomorphisms*

$$H^k_{\mathrm{dR}}(M) \cong H^k(M, \mathbb{R}_M) \cong H^k(M; \mathbb{R}),$$

the first to sheaf cohomology of the constant sheaf, the second to singular cohomology with real coefficients. Under these isomorphisms the wedge product corresponds to the cup product, and the isomorphism is induced by integration: the class of a closed k-form ω corresponds to the singular cochain $\sigma \mapsto \int_{\Delta^k} \sigma^*\omega$ on smooth simplices.

Proof. Let $\underline{\Omega}^k$ be the sheaf $U \mapsto \Omega^k(U)$. The sequence of sheaves

$$0 \to \mathbb{R}_M \to \underline{\Omega}^0 \xrightarrow{d} \underline{\Omega}^1 \xrightarrow{d} \cdots$$

is exact: exactness at $\underline{\Omega}^0$ is Proposition 37.2.2, and exactness at $\underline{\Omega}^k$ for $k \geq 1$ holds on stalks, because every point has a basis of convex coordinate neighbourhoods, on which closed forms are exact by the Poincaré lemma (Theorem 37.2.4). Each $\underline{\Omega}^k$ is a module over the sheaf of smooth functions, hence fine, hence acyclic on the paracompact space M (Example 26.4.3 and Theorem 26.4.5). So this is an acyclic resolution of $\mathbb{R}_M$ and Proposition 26.7.1(3) gives $H^k(M, \mathbb{R}_M) \cong H^k(\Gamma(M, \underline{\Omega}^\bullet)) = H^k_{\mathrm{dR}}(M)$. The second isomorphism is Theorem 26.6.1, since manifolds are locally contractible and paracompact, and it is compatible with products. The description by integration over smooth simplices is the standard comparison of the two resolutions; see [War83, Chapter 5] or [Bre97, Chapter III]. □

Corollary 37.3.2. *For a compact manifold M the spaces $H^k_{\mathrm{dR}}(M)$ are finite-dimensional, they vanish for $k > \dim M$, and $\dim H^k_{\mathrm{dR}}(M) = b_k(M)$, the k-th Betti number. For a compact oriented n-manifold, $H^n_{\mathrm{dR}}(M) \cong \mathbb{R}$ for each connected component, with the isomorphism given by integration.*

Proof. Compact manifolds have finitely generated singular homology [Hat02, Appendix], so Theorem 37.3.1 and the universal coefficient theorem give finite dimension and the identification with Betti numbers; the vanishing above the dimension holds because there are no nonzero forms of degree $> n$. For the top degree, $H^n(M; \mathbb{R}) \cong H_0(M; \mathbb{R}) \cong \mathbb{R}$ per component by Poincaré duality (Theorem 23.6.2), and integration is nonzero on the class of a volume form (Corollary 36.4.13). □

Explicitly, if $M_1, \ldots, M_c$ are the connected components of the compact oriented manifold M (finitely many, as they are open and cover the compact M), then $\Omega^\bullet(M) = \prod_i \Omega^\bullet(M_i)$, so $H^n_{\mathrm{dR}}(M) = \bigoplus_i H^n_{\mathrm{dR}}(M_i)$, and the isomorphism of the corollary is

$$H^n_{\mathrm{dR}}(M) \to \mathbb{R}^c, \qquad [\omega] \mapsto \Big(\int_{M_1} \omega, \ldots, \int_{M_c} \omega \Big).$$

So a top degree form on M is exact if and only if its integral over *each* component vanishes; the vanishing of the total integral $\int_M \omega$ is not enough when M is disconnected. For example, on $M = S^2 \sqcup S^2$ the form equal to the area form on the first sphere and to minus the area form on the second has total integral 0 but is not exact.

Proposition 37.3.3. *Let M be a connected manifold and $x_0 \in M$. There is an isomorphism*

$$H^1_{\mathrm{dR}}(M) \xrightarrow{\sim} \mathrm{Hom}(\pi_1(M, x_0), \mathbb{R}),$$

sending $[\omega]$ to the homomorphism whose value on the class of a piecewise smooth loop γ is the period $\int_\gamma \omega$*. In particular $H^1_{\mathrm{dR}}(M) = 0$ if M is simply connected, and a closed 1-form is exact if and only if all its periods vanish.*

Proof. Compose the isomorphism $H^1_{\mathrm{dR}}(M) \cong H^1(M;\mathbb{R})$ of Theorem 37.3.1, given by integration over smooth simplices, with $H^1(M;\mathbb{R}) \cong \mathrm{Hom}_{\mathbb{R}}(H_1(M;\mathbb{R}),\mathbb{R}) = \mathrm{Hom}(H_1(M),\mathbb{R})$ (Theorem 22.2.3 over the field $\mathbb{R}$) and with $\mathrm{Hom}(H_1(M),\mathbb{R}) \cong \mathrm{Hom}(\pi_1(M,x_0),\mathbb{R})$, which holds by the Hurewicz theorem (Theorem 21.9.1) because homomorphisms to the abelian group $\mathbb{R}$ factor through the abelianisation. The composite is the period map.

The last statement can also be seen directly. If $\omega = df$, then $\int_\gamma \omega = f(\gamma(1)) - f(\gamma(0)) = 0$ for loops. Conversely, if all periods vanish, $f(x) = \int_{x_0}^{x} \omega$ along any piecewise smooth path is well defined, since two paths differ by a loop, and in a chart around x that is a ball, integrating along segments shows $df = \omega$, by the Poincaré lemma on the ball (Theorem 37.2.4): there $\omega = dg$, and $f - g$ is constant. □

Example 37.3.4. On $M = \mathbb{R}^2 \setminus \{0\}$ the *angle form* $\omega = \frac{x\,dy - y\,dx}{x^2+y^2}$ is closed, and its period on the unit circle is 2π, so it is not exact. Since $\pi_1(M) \cong \mathbb{Z}$, Proposition 37.3.3 gives $H^1_{\mathrm{dR}}(M) = \mathbb{R}[\omega]$: a closed 1-form η on M is exact if and only if $\int_{S^1} \eta = 0$, and in general $\eta - \frac{1}{2\pi}\big(\int_{S^1}\eta\big)\omega$ is exact. On the simply connected $\mathbb{R}^3 \setminus \{0\}$, by contrast, every closed 1-form is exact.

Counterexample 37.3.5. De Rham cohomology of a noncompact manifold can be infinite-dimensional. On $M = \mathbb{C} \setminus \mathbb{Z}$ let α_n be the angle form centred at n, the pullback of the angle form of Counterexample 36.2.18 by $z \mapsto z - n$. It is closed, and its integral over the circle γ_m of radius $\frac{1}{2}$ around m is 2π if $m = n$ and 0 otherwise (for $m \neq n$ the circle lies in a half-plane, where α_n is exact). If $\sum_n c_n\alpha_n = df$ for finitely many nonzero c_n, integrating over γ_m gives $2\pi c_m = 0$ (Proposition 36.2.17). So the classes $[\alpha_n]$, $n \in \mathbb{Z}$, are linearly independent in $H^1_{\mathrm{dR}}(M)$.

Proposition 37.3.6. *Let a finite group G act freely on M by diffeomorphisms, and $\pi\colon M \to M/G$ the quotient (Theorem 33.7.1). Then π^* induces an isomorphism $H^k_{\mathrm{dR}}(M/G) \cong H^k_{\mathrm{dR}}(M)^G$ onto the classes fixed by G.*

Proof. As π is a local diffeomorphism and $\pi \circ g = \pi$, π^* identifies $\Omega^\bullet(M/G)$ with the complex $\Omega^\bullet(M)^G$ of G-invariant forms: an invariant form descends, because locally π has smooth inverses that differ by elements of G. The averaging operator $A = \frac{1}{|G|}\sum_{g\in G} g^*$ commutes with d and is a projection of $\Omega^\bullet(M)$ onto $\Omega^\bullet(M)^G$. The map $H^k(\Omega^\bullet(M)^G) \to H^k_{\mathrm{dR}}(M)^G$ is injective: if an invariant ω equals $d\eta$, then $\omega = A\omega = d(A\eta)$ with $A\eta$ invariant. It is surjective: if $[\omega]$ is G-fixed, then $A\omega$ is invariant, closed, and cohomologous to ω, since each $g^*\omega$ is. □

Example 37.3.7. $H^k_{\mathrm{dR}}(\mathbb{RP}^n) \cong \mathbb{R}$ for $k = 0$, and for $k = n$ if n is odd; it vanishes otherwise. By Proposition 37.3.6 and Example 37.2.9, only the action of the antipodal map a on $H^n_{\mathrm{dR}}(S^n) \cong \mathbb{R}$ matters. The volume form of S^n is $\iota_X\Omega$ restricted to S^n, where $\Omega = dx^1 \wedge \cdots \wedge dx^{n+1}$ and $X = \sum x^i\partial_i$; as a is the restriction of $-\mathrm{id}$, $a^*\Omega = (-1)^{n+1}\Omega$ and a preserves X, so a^* acts on H^n by $(-1)^{n+1}$, and the invariant part is H^n for n odd and 0 for n even. In particular $\mathbb{RP}^n$ is not orientable for n even, by Corollary 37.3.2.

Theorem 37.3.8 (Künneth formula)**.** *If M has a finite good cover, the map $\bigoplus_{p+q=k} H^p_{\mathrm{dR}}(M) \otimes H^q_{\mathrm{dR}}(N) \to H^k_{\mathrm{dR}}(M\times N)$, $[\alpha]\otimes[\beta] \mapsto [\mathrm{pr}_M^*\alpha \wedge \mathrm{pr}_N^*\beta]$, is an isomorphism for every manifold N.*

Proof. Write ψ_M for the map. It is natural in M for restrictions to open subsets. If $M \cong \mathbb{R}^m$, then $\mathrm{pr}_N\colon M\times N \to N$ is a homotopy equivalence and $H^\bullet_{\mathrm{dR}}(M) = \mathbb{R}$ in degree 0, so ψ_M is pr_N^*, an isomorphism by Theorem 37.2.4. For open $U, V \subseteq M$, tensoring the Mayer–Vietoris sequence of U, V with the vector space $H^q_{\mathrm{dR}}(N)$ and summing over q gives an exact sequence, which ψ maps to the Mayer–Vietoris sequence of $U\times N$, $V\times N$. The squares involving restrictions commute, and so does the one with the connecting maps: with a partition of

unity $\{\rho_U, \rho_V\}$ on $U \cup V$, the connecting maps are $[\omega] \mapsto [d(\rho_V \omega)]$ on both sides, and for closed β, $d(\mathrm{pr}_M^*(\rho_V\omega) \wedge \mathrm{pr}_N^*\beta) = \mathrm{pr}_M^* d(\rho_V\omega) \wedge \mathrm{pr}_N^*\beta$. So if ψ is an isomorphism for U, V and $U \cap V$, it is one for $U \cup V$ by the five lemma (Lemma 15.4.2). Now let $U_1, \dots, U_k$ be a finite good cover of M; we show by induction on k that ψ is an isomorphism for every open set covered by k open sets forming a good cover. For $k = 1$ this is the case $\mathbb{R}^m$. For the induction step apply the five lemma to $W = U_1 \cup \cdots \cup U_{k-1}$ and U_k, noting that $W \cap U_k$ is covered by the $k-1$ sets $U_i \cap U_k$, which form a good cover. □

Corollary 37.3.9. *$H^\bullet_{\mathrm{dR}}(T^n)$ is the exterior algebra on the classes of $dx^1, \dots, dx^n$, so $b_k(T^n) = \binom{n}{k}$; and $H^\bullet_{\mathrm{dR}}(S^1 \times S^2)$ has dimensions $1, 1, 1, 1$.*

Proof. By Example 37.2.9, $H^\bullet(S^1)$ is spanned by 1 and $[d\theta]$, and $T^n = (S^1)^n$ has a finite good cover; Theorem 37.3.8 and induction give the tensor product of n copies of $\mathbb{R} \oplus \mathbb{R}[d\theta]$, which is the exterior algebra because $d\theta \wedge d\theta = 0$ and the wedge product is graded commutative. The second statement is the same computation with $H^\bullet(S^2)$. □

Theorem 37.3.10 (Poincaré duality in de Rham cohomology)**.** *Let M be an oriented n-manifold. The pairing*

$$H^k_{\mathrm{dR}}(M) \times H^{n-k}_{\mathrm{dR},c}(M) \to \mathbb{R}, \qquad ([\alpha], [\beta]) \mapsto \int_M \alpha \wedge \beta,$$

is well defined and nondegenerate if M has a finite good cover (for instance if M is compact); in that case $H^k_{\mathrm{dR}}(M) \cong H^{n-k}_{\mathrm{dR},c}(M)^*$. For compact oriented M this is the statement that $\int_M \alpha \wedge \beta$ is a perfect pairing between $H^k_{\mathrm{dR}}(M)$ and $H^{n-k}_{\mathrm{dR}}(M)$.

Proof. The pairing is well defined by Stokes' theorem: if α is closed and $\beta = d\gamma$ has compact support, then $\alpha \wedge \beta = \pm d(\alpha \wedge \gamma)$ has compact support and integrates to zero. Nondegeneracy is proved by induction on the number of sets in a finite good cover, one in which all nonempty intersections are diffeomorphic to $\mathbb{R}^n$ (such covers exist on any manifold, and finite ones on compact manifolds), comparing the Mayer–Vietoris sequences of $H^\bullet_{\mathrm{dR}}$ and $H^\bullet_{\mathrm{dR},c}$ with the five lemma; see [BT82, Chapter I, Section 5]. Alternatively it follows from Theorem 37.3.1 and topological Poincaré duality (Corollary 23.6.3), since the wedge product corresponds to the cup product. □

Theorem 37.3.11 (Cartan–Chevalley–Eilenberg)**.** *Let G be a compact connected Lie group. Every de Rham class of G has a unique representative that is invariant under left and right translations, and the bi-invariant forms are closed. Consequently $H^\bullet_{\mathrm{dR}}(G) \cong (\Lambda^\bullet \mathfrak{g}^*)^G$, the space of $\mathrm{Ad}(G)$-invariant alternating forms on the Lie algebra.*

Proof. $G \times G$ acts on G by $(a, b) \cdot x = axb^{-1}$; let μ be the Haar probability measure on the compact connected group $K = G \times G$ (Theorem 34.5.1) and, for a form ω, let $A\omega = \int_K k^*\omega \, d\mu(k)$, defined pointwise; $A\omega$ is smooth and bi-invariant, A commutes with d (differentiation under the integral sign), and $A\omega = \omega$ for bi-invariant ω.

Averaging does not change classes. Let ω be closed. Each $k \in K$ is connected to the identity by a smooth path, so $k^*\omega$ is cohomologous to ω (Theorem 37.2.4). For a closed form η of complementary degree, Fubini's theorem gives $\int_G A\omega \wedge \eta = \int_K (\int_G k^*\omega \wedge \eta)\, d\mu(k) = \int_G \omega \wedge \eta$, since the inner integral depends only on the class of $k^*\omega$ (Stokes). By Poincaré duality (Theorem 37.3.10), $[A\omega] = [\omega]$.

Bi-invariant forms are closed. The inversion $\iota(x) = x^{-1}$ satisfies $\iota \circ L_a = R_{a^{-1}} \circ \iota$, so it maps bi-invariant forms to bi-invariant forms, and $d_e\iota = -\mathrm{id}$ (differentiate $\exp(tX)^{-1} = \exp(-tX)$). A bi-invariant form is determined by its value at e, so $\iota^*\omega = (-1)^k\omega$ for a bi-invariant k-form. As $d\omega$ is bi-invariant of degree $k+1$, $(-1)^{k+1}d\omega = \iota^* d\omega = d\iota^*\omega = (-1)^k d\omega$, so $d\omega = 0$.

Uniqueness. If a bi-invariant ω is exact, $\omega = d\eta$, then $\omega = A\omega = d(A\eta) = 0$, because $A\eta$ is bi-invariant, hence closed. So bi-invariant forms map isomorphically onto $H^\bullet_{\mathrm{dR}}(G)$. A left-invariant form is determined by its value $\omega_e \in \Lambda^\bullet \mathfrak{g}^*$, and it is also right-invariant if and only if ω_e is invariant under $\mathrm{Ad}(g) = d_e(L_g \circ R_{g^{-1}})$. □

Example 37.3.12. For $G = \mathrm{SU}(2)$, $\mathfrak{g} = \mathfrak{su}(2) \cong \mathbb{R}^3$ and Ad maps SU(2) onto SO(3) (Exercise 34.6.2). The SO(3)-invariants in $\Lambda^\bullet \mathbb{R}^3$ are the constants and the multiples of the determinant $e_1^* \wedge e_2^* \wedge e_3^*$, since Λ^1 and $\Lambda^2 \cong \mathbb{R}^3$ are irreducible nontrivial representations. So $H^\bullet(\mathrm{SU}(2))$ is $\mathbb{R}$ in degrees 0 and 3, in accordance with $\mathrm{SU}(2) \cong S^3$.

37.4 Currents

Definition 37.4.1. Let M be an oriented n-manifold. A *k-current* is a linear functional T on $\Omega_c^{n-k}(M)$ that is continuous in the sense of distributions: $T(\varphi_j) \to 0$ whenever the coefficients of φ_j, in any chart, have supports in a fixed compact set and converge to 0 together with all derivatives. Currents form a space $\mathcal{D}'_k(M)$, and dT is defined by $(dT)(\varphi) = (-1)^{k+1} T(d\varphi)$, so that $d \circ d = 0$.

Example 37.4.2.

1. A locally integrable k-form η defines a current $T_\eta(\varphi) = \int_M \eta \wedge \varphi$, and $dT_\eta = T_{d\eta}$ for smooth η by Stokes' theorem; so currents generalise forms in the same way as distributions generalise functions (Chapter 32).

2. A compact oriented k-dimensional submanifold $Z \subseteq M$ defines the *current of integration* $[Z](\varphi) = \int_Z \varphi$ for $\varphi \in \Omega_c^k(M)$. Stokes' theorem gives $d[Z] = 0$ when $\partial Z = \emptyset$, so $[Z]$ has a cohomology class.

Theorem 37.4.3. *On an oriented manifold the complex of currents computes de Rham cohomology: the inclusion of smooth forms into currents is a quasi-isomorphism. Under this identification and Poincaré duality, the class of $[Z]$ for a compact oriented k-dimensional submanifold Z of a compact oriented n-manifold M is the Poincaré dual of Z: it is the unique class $\eta_Z \in H^{n-k}_{\mathrm{dR}}(M)$ with*

$$\int_Z \varphi = \int_M \eta_Z \wedge \varphi \qquad \text{for all closed } \varphi \in \Omega^k(M).$$

This is proved in [dR84, Chapter IV] and [GH78, Chapter 0, Section 4]; the first statement is a sheaf-theoretic argument like the proof of Theorem 37.3.1, applied to the fine resolution of $\mathbb{R}_M$ by sheaves of currents, and the second follows by comparing both sides with the topological Poincaré dual (Remark 23.8.2).

Remark 37.4.4. Currents are the natural home for the cycle classes of Hodge theory: an analytic subvariety Z of a compact complex manifold, possibly singular, defines a current of integration because its singular locus has measure zero and $\int_{Z_{\mathrm{reg}}} \varphi$ converges (Chapter 63); its class in $H^{2n-2k}(X;\mathbb{Z})$ is the cycle class whose Hodge-theoretic characterisation the Hodge conjecture concerns.

37.5 Exercises

Exercise 37.5.1. Compute $H^\bullet_{\mathrm{dR}}(T^n)$ and show that it is the exterior algebra on H^1, generated by the forms $d\theta^1, \ldots, d\theta^n$.

Solution. By Theorem 37.3.1 and Example 22.4.3, $H^\bullet_{\mathrm{dR}}(T^n) \cong \Lambda^\bullet \mathbb{R}^n$. The classes of the closed forms $d\theta^i$, which are not exact since $\int_{S^1_i} d\theta^i = 2\pi$, generate: their products are the $\binom{n}{k}$ linearly independent classes of $d\theta^{i_1} \wedge \cdots \wedge d\theta^{i_k}$, which can be detected by integration over the corresponding subtori. □

Exercise 37.5.2. Show that $H^0_{\mathrm{dR},c}(\mathbb{R}) = 0$ and $H^1_{\mathrm{dR},c}(\mathbb{R}) \cong \mathbb{R}$, with the isomorphism given by integration; so compactly supported de Rham cohomology is not homotopy invariant.

Solution. A compactly supported function with $df = 0$ is constant, hence zero. The map $\int_{\mathbb{R}} \colon \Omega^1_c(\mathbb{R}) \to \mathbb{R}$ is surjective, and $f\,dx$ integrates to zero if and only if $F(x) = \int_{-\infty}^x f$ has compact support, in which case $f\,dx = dF$ is exact. □

Exercise 37.5.3. Let $f \colon M \to N$ be a smooth map of compact connected oriented n-manifolds. Show that $f^* \colon H^n_{\mathrm{dR}}(N) \to H^n_{\mathrm{dR}}(M)$ is multiplication by the degree of f, and that a map of nonzero degree is surjective.

Exercise 37.5.4. Prove the Künneth formula $H^\bullet_{\mathrm{dR}}(M \times N) \cong H^\bullet_{\mathrm{dR}}(M) \otimes H^\bullet_{\mathrm{dR}}(N)$ for manifolds with finite good covers, by induction with Mayer–Vietoris.

Exercise 37.5.5. Let $M = \mathbb{R}^2 \setminus \{p_1, \ldots, p_k\}$ for distinct points p_i. Show that $H^1_{\mathrm{dR}}(M) \cong \mathbb{R}^k$, with a basis given by the angle forms α_i centred at the p_i, and that a closed 1-form η on M is exact if and only if its integrals over small circles around all the p_i vanish.

Solution. After a rotation we may assume the p_i have distinct first coordinates, p_1 having the largest. Let U be M intersected with $\{x < a\}$ and V be M intersected with $\{x > b\}$, where $b < a$ are chosen so that p_1 is the only point with first coordinate $> b$ and none has first coordinate in $[b, a]$. Then $U \cap V$ is a strip, which is contractible, so Mayer–Vietoris gives $H^1(M) \cong H^1(U) \oplus H^1(V)$. The half-planes are diffeomorphic to $\mathbb{R}^2$, so V is a plane minus one point, with $H^1 \cong \mathbb{R}$ spanned by α_1 (Example 37.3.4), and U is a plane minus $k-1$ points; induction gives $\mathbb{R}^k$. The integral of α_i over a small circle around p_j is $2\pi\delta_{ij}$ (away from p_i, α_i is exact on a disc), so the map $[\eta] \mapsto (\int_{C_j} \eta)_j$ is surjective onto $\mathbb{R}^k$, hence an isomorphism, and the $[\alpha_i]$ form a basis. □

Exercise 37.5.6. On $\mathbb{R}^3 \setminus \{0\}$ let $\sigma = (x\,dy \wedge dz + y\,dz \wedge dx + z\,dx \wedge dy)/r^3$, $r = (x^2+y^2+z^2)^{1/2}$ (the *solid angle form*). Show that σ is closed, that $\int_{S^2} \sigma = 4\pi$, and deduce that $H^2_{\mathrm{dR}}(\mathbb{R}^3 \setminus \{0\}) = \mathbb{R}[\sigma]$ and that σ is not exact.

Solution. By Example 36.2.14, $d\sigma = \operatorname{div}(x/r^3, y/r^3, z/r^3)\,dx \wedge dy \wedge dz$, and $\partial_x(x/r^3) = 1/r^3 - 3x^2/r^5$, so the divergence is $3/r^3 - 3r^2/r^5 = 0$. On the unit sphere $r = 1$ and σ restricts to the area form (Example 36.4.4(2)), so $\int_{S^2} \sigma = 4\pi$. An exact form would have integral 0 over the closed surface S^2 (Theorem 36.4.8). Finally $\mathbb{R}^3 \setminus \{0\}$ is homotopy equivalent to S^2, so its H^2 is one-dimensional (Example 37.2.9), spanned by the nonzero class $[\sigma]$. □

Exercise 37.5.7. Decide whether the 2-form $\omega = z\,dx \wedge dy + x\,dy \wedge dz + y\,dz \wedge dx$ on $\mathbb{R}^3$ is closed, and find a primitive of the closed form $\omega' = dx \wedge dy + dy \wedge dz$ using Corollary 37.2.6.

Solution. $d\omega = \partial_z z\,dz \wedge dx \wedge dy + \partial_x x\,dx \wedge dy \wedge dz + \partial_y y\,dy \wedge dz \wedge dx = 3\,dx \wedge dy \wedge dz \neq 0$, so ω is not closed and has no primitive. For ω', which has constant coefficients and is closed, $\iota_X \omega' = x\,dy - y\,dx + y\,dz - z\,dy$ and $\eta = \frac{1}{2}(x\,dy - y\,dx + y\,dz - z\,dy)$; then $d\eta = \frac{1}{2}(2\,dx \wedge dy + 2\,dy \wedge dz) = \omega'$. □

Chapter 38

Connections and Curvature

38.1 Introduction

To differentiate a function f on a manifold along a curve γ one compares the values $f(\gamma(t))$ and $f(\gamma(0))$, which are numbers. For a section s of a vector bundle E the values $s(\gamma(t)) \in E_{\gamma(t)}$ and $s(\gamma(0)) \in E_{\gamma(0)}$ lie in *different* vector spaces, and there is no canonical way to subtract them. A local trivialisation identifies nearby fibres with $\mathbb{K}^r$, and then one can differentiate the components of s; but the result depends on the trivialisation, because changing it multiplies the components by a matrix of functions, whose derivative enters the answer. The simplest example is a vector field Y on the sphere $S^2 \subseteq \mathbb{R}^3$: its derivative in $\mathbb{R}^3$ along a curve on the sphere is in general not tangent to the sphere, so it is not a candidate for a derivative of Y within the tangent bundle of S^2.

A *connection* is the extra structure that removes this ambiguity: a rule ∇ that assigns to a section s and a tangent vector v a derivative $\nabla_v s$ in the same fibre, subject to the Leibniz rule. Everything else follows from it. Solving $\nabla_{\dot\gamma} s = 0$ along curves gives *parallel transport*, a way of moving vectors from fibre to fibre along a path, which in general depends on the path; going around loops gives the *holonomy group*. The *curvature* measures the failure of covariant derivatives to commute, $F(X, Y) = \nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X,Y]}$; unlike the connection itself it is a tensor, an $\mathrm{End}(E)$-valued 2-form, and it is the infinitesimal form of the holonomy around small loops (Proposition 38.5.7). Connections with zero curvature are the same as local systems (Theorem 38.5.10), which is the differential-geometric form of the monodromy correspondence of Chapter 25 and the starting point of the theory of variations of Hodge structure (Chapter 70), where the Gauss–Manin connection appears. The curvature is also the source of the characteristic classes of Chapter 39, and for the tangent bundle of a Riemannian manifold it is the Riemann curvature tensor of Chapter 40.

The chapter goes as follows.

1. Forms with values in a vector bundle, the objects in which connections and curvature take their values; the tensoriality lemma, which says when an operator on sections is given pointwise (Section 38.2).

2. Connections, in the two equivalent languages $\nabla\colon \Gamma(E) \to \Omega^1(M, E)$ and $(X, s) \mapsto \nabla_X s$; connection forms in local frames and how they change; existence; the basic examples, among them the connection of a subbundle of a trivial bundle; induced connections on duals, tensor products and Hom bundles (Section 38.3).

3. The exterior covariant derivative and the curvature, with complete computations for the sphere and for the tautological line bundle of $\mathbb{CP}^1$ (Section 38.4).

4. Parallel transport, holonomy, the geometric meaning of curvature, and flat connections (Section 38.5).

5. Metric connections (Section 38.6).

6. Vertical and horizontal subspaces: a connection as a choice of horizontal subspaces in the tangent spaces of the total space, parallel transport as horizontal lifting of curves, and curvature as the obstruction to integrating the horizontal distribution (Section 38.7).

7. Principal bundles, principal connections and connection forms, with the Hopf fibration as example (Section 38.8).

References: [KN63], [MS74], [GH78, Chapter 0].
Prerequisites. Chapter 35 (Vector Bundles), Chapter 36 (Differential Forms and Integration) and Chapter 34 (Lie Groups).
Conventions. $\mathbb{K}$ is $\mathbb{R}$ or $\mathbb{C}$, M is a smooth manifold of dimension n, and $E \to M$ is a $\mathbb{K}$-vector bundle of rank r. For an open $U \subseteq M$, $\Gamma(U, E)$ is the space of smooth sections over U, and $\Gamma(E) = \Gamma(M, E)$; $\mathfrak{X}(M)$ is the space of vector fields. A *frame* of E over U is a tuple $e = (e_1, \dots, e_r)$ of sections that is a basis of every fibre (Definition 35.2.8). When $\mathbb{K} = \mathbb{C}$, differential forms have complex values: $\Omega^k(M)$ then means $\Omega^k(M) \otimes_{\mathbb{R}} \mathbb{C}$, and d is extended $\mathbb{C}$-linearly. We write sums explicitly, and in computations with frames we use matrix notation: a section $s = \sum_j s^j e_j$ is identified with the column vector $(s^1, \dots, s^r)^T$ of its components.

38.2 Bundle-valued forms

A connection takes a section s and produces, at each point p, a linear map $T_pM \to E_p$, $v \mapsto \nabla_v s$, the derivative of s in the direction v; the curvature takes two tangent vectors and produces an endomorphism of E_p. Such objects are forms with values in a vector bundle, and this section explains what they are, in several equivalent ways.

Definition 38.2.1. The space of *E-valued k-forms* on M is

$$\Omega^k(M, E) = \Gamma(M, \Lambda^k T^*M \otimes E),$$

the smooth sections of the tensor product of the bundle of k-forms with E (Proposition 35.3.1). So $\Omega^0(M, E) = \Gamma(M, E)$, and for the trivial line bundle $\Omega^k(M, M \times \mathbb{K}) = \Omega^k(M)$. We write $\Omega^k(U, E)$ for the forms over an open set U.

The following lemma is used constantly. It explains when an operator that takes sections to sections is really a pointwise, linear algebra operation.

Lemma 38.2.2 (Tensoriality)**.** *Let $E_1, \dots, E_k$ and F be vector bundles over M, and let $A\colon \Gamma(E_1) \times \cdots \times \Gamma(E_k) \to \Gamma(F)$ be $C^\infty(M)$-linear in each argument: $A(\dots, fs + gt, \dots) = fA(\dots, s, \dots) + gA(\dots, t, \dots)$ for functions f, g. Then the value $A(s_1, \dots, s_k)(p)$ depends only on the vectors $s_1(p), \dots, s_k(p)$. So there are multilinear maps $A_p\colon (E_1)_p \times \cdots \times (E_k)_p \to F_p$ with $A(s_1, \dots, s_k)(p) = A_p(s_1(p), \dots, s_k(p))$, and they form a smooth section of* $\operatorname{Hom}(E_1 \otimes \cdots \otimes E_k, F)$.

Proof. Locality. Suppose s_1 vanishes on an open set $U \ni p$. By Lemma 33.3.10 there is a function χ with $\chi(p) = 1$ and support in U. Then $\chi s_1 = 0$, so $0 = A(\chi s_1, s_2, \dots) = \chi A(s_1, s_2, \dots)$, and evaluating at p gives $A(s_1, \dots)(p) = 0$.

Pointwise dependence. Suppose $s_1(p) = 0$. Let $(e_1, \dots, e_m)$ be a frame of E_1 on a neighbourhood U of p, and write $s_1 = \sum_j a^j e_j$ on U with smooth a^j vanishing at p. Choose χ as above, equal to 1 on a neighbourhood V of p, with support in U. The sections $\tilde{e}_j = \chi e_j$ and the

functions $\tilde{a}^j = \chi a^j$, extended by 0 outside U, are smooth on M, and $\sum_j \tilde{a}^j \tilde{e}_j = \chi^2 s_1$ agrees with s_1 on V. By locality and linearity, $A(s_1, \ldots)(p) = A(\sum_j \tilde{a}^j \tilde{e}_j, \ldots)(p) = \sum_j \tilde{a}^j(p) A(\tilde{e}_j, \ldots)(p) = 0$. By linearity in each argument, $A(s_1, \ldots, s_k)(p)$ depends only on the values $s_i(p)$.

The maps A_p. Every vector $v \in (E_i)_p$ is the value at p of a global section, for instance of $\chi \sum_j v^j e_j$ with constant coefficients v^j. So $A_p(v_1, \ldots, v_k) = A(s_1, \ldots, s_k)(p)$, for any sections with $s_i(p) = v_i$, is well defined and multilinear. In local frames of the E_i and of F, the matrix coefficients of A_p are the components of A applied to (cut-off) frame sections, which are smooth; so $p \mapsto A_p$ is a smooth section of $\operatorname{Hom}(E_1 \otimes \cdots \otimes E_k, F)$. □

Proposition 38.2.3. *For an E-valued k-form ω and a point p, the value $\omega_p \in \Lambda^k T_p^* M \otimes E_p$ is the same as an alternating k-linear map $(T_pM)^k \to E_p$, by $(\alpha \otimes e)(v_1, \ldots, v_k) = \alpha(v_1, \ldots, v_k)\, e$. Moreover the following descriptions of $\Omega^k(M, E)$ are equivalent.*

1. *Smooth sections ω of $\Lambda^k T^*M \otimes E$.*
2. *Families of alternating k-linear maps $\omega_p \colon (T_pM)^k \to E_p$ such that, for all vector fields $X_1, \ldots, X_k$ on an open set U, the section $p \mapsto \omega_p(X_1(p), \ldots, X_k(p))$ of E over U is smooth.*
3. *Maps $\omega \colon \mathfrak{X}(M)^k \to \Gamma(E)$ that are alternating and $C^\infty(M)$-linear in each argument.*
4. *Locally, for a frame $(e_1, \ldots, e_r)$ of E over U: expressions $\omega|_U = \sum_i \omega^i \otimes e_i$ with ordinary k-forms $\omega^i \in \Omega^k(U)$, which are uniquely determined by ω and are smooth exactly when ω is. Equivalently $\omega(X_1, \ldots, X_k) = \sum_i \omega^i(X_1, \ldots, X_k)\, e_i$.*

Proof. Pointwise. For finite-dimensional V and W, $\Lambda^k V^* \otimes W \cong \operatorname{Hom}(\Lambda^k V, W)$ by Propositions 12.4.1 and 12.2.2, and linear maps $\Lambda^k V \to W$ are the same as alternating k-linear maps $V^k \to W$ by the universal property of the exterior power (Proposition 12.3.2). With the convention of Notation 12.4.2, $\alpha \otimes e$ corresponds to $(v_1, \ldots, v_k) \mapsto \alpha(v_1, \ldots, v_k)e$.

(1) ⇔ (4). If (e_i) is a frame of E over U and (dx^I) the frame of $\Lambda^k T^*M$ given by a chart on U, the products $dx^I \otimes e_i$ form a frame of $\Lambda^k T^*M \otimes E$ over U (Proposition 35.3.1), and a section is smooth if and only if its coefficients ω_I^i in this frame are smooth. Grouping, $\omega = \sum_i \omega^i \otimes e_i$ with $\omega^i = \sum_I \omega_I^i\, dx^I$; the ω^i are unique because at each point $\Lambda^k T_p^* M \otimes E_p = \bigoplus_i \Lambda^k T_p^* M \otimes \mathbb{K} e_i(p)$, and ω is smooth exactly when all the ω^i are.

(2) ⇔ (4). In a frame, $\omega(X_1, \ldots, X_k) = \sum_i \omega^i(X_1, \ldots, X_k) e_i$, which is smooth for all vector fields if and only if every $\omega^i(X_1, \ldots, X_k)$ is; by Proposition 36.2.5 this holds if and only if the ω^i are smooth forms.

(1) ⇔ (3). A form ω gives $(X_1, \ldots, X_k) \mapsto \omega(X_1, \ldots, X_k)$, which is alternating and $C^\infty(M)$-linear because it is computed pointwise. Conversely, a map as in (3) is given pointwise by alternating multilinear maps ω_p (Lemma 38.2.2, applied with $E_i = TM$), which satisfy (2). □

Example 38.2.4.

1. On the trivial bundle $M \times \mathbb{K}^r$ with its standard frame, an E-valued k-form is a column vector $(\omega^1, \ldots, \omega^r)^T$ of ordinary k-forms.
2. A TM-valued 1-form is at each point a linear map $T_pM \to T_pM$: $\Omega^1(M, TM)$ is the space of endomorphisms of the tangent bundle. The identity is the TM-valued 1-form $\sum_i dx^i \otimes \partial/\partial x^i$ in every chart; the shape operator of a hypersurface (Chapter 40) is another example.
3. An $\operatorname{End}(E)$-valued k-form takes k tangent vectors to an endomorphism of the fibre. In a frame (e_i) it is an $r \times r$ matrix (ω_j^i) of k-forms, with $\omega(X_1, \ldots, X_k)e_j = \sum_i \omega_j^i(X_1, \ldots, X_k)e_i$. The curvature of a connection will be such a form with $k = 2$.
4. The differential of a smooth map $f \colon N \to M$ is at each point a linear map $T_qN \to T_{f(q)}M = (f^*TM)_q$, so df is an f^*TM-valued 1-form on N.

Products. Forms with values in bundles can be multiplied whenever their values can be multiplied. The recipe is always the same: at a point, a bundle-valued form is a sum of products $\alpha \otimes s$ of an ordinary form and a vector, and one multiplies the form parts with the wedge product and the vector parts with whatever product the values have. Three cases are needed in this chapter: an ordinary form times an E-valued form, an $\mathrm{End}(E)$-valued form acting on an E-valued form (the connection form acting on components, or the curvature acting on a section), and the product of two $\mathrm{End}(E)$-valued forms (as in $\theta \wedge \theta$).

Definition 38.2.5. Let $\alpha \in \Omega^k(M)$, $\omega \in \Omega^l(M, E)$, $\theta \in \Omega^k(M, \mathrm{End}\, E)$ and $\eta \in \Omega^l(M, \mathrm{End}\, E)$. Their *wedge products* are the forms $\alpha \wedge \omega \in \Omega^{k+l}(M, E)$, $\theta \wedge \omega \in \Omega^{k+l}(M, E)$ and $\theta \wedge \eta \in \Omega^{k+l}(M, \mathrm{End}\, E)$ given at each point on decomposable elements by

$$\alpha \wedge (\beta \otimes s) = (\alpha \wedge \beta) \otimes s, \qquad (\gamma \otimes A) \wedge (\beta \otimes s) = (\gamma \wedge \beta) \otimes As, \qquad (\gamma \otimes A) \wedge (\beta \otimes B) = (\gamma \wedge \beta) \otimes AB,$$

for ordinary forms γ, β, vectors $s \in E_p$ and endomorphisms $A, B \in \mathrm{End}(E_p)$. For a section $s \in \Omega^0(M, E)$ we write $\alpha \wedge s = \alpha \otimes s$, and $\theta \wedge s = \theta s$ is the E-valued k-form $(X_1, \dots, X_k) \mapsto \theta(X_1, \dots, X_k)s$.

The formulas are given on decomposable elements only, and one has to check that they determine the products. At a point p, with $V = T_pM$, the map $(\gamma, A, \beta, s) \mapsto (\gamma \wedge \beta) \otimes As$ is linear in each of its four arguments; by the universal property of the tensor product (Definition 8.9.1), applied twice, it induces a well-defined bilinear map $(\Lambda^k V^* \otimes \mathrm{End}\, E_p) \times (\Lambda^l V^* \otimes E_p) \to \Lambda^{k+l} V^* \otimes E_p$, and similarly in the other two cases. So the products are defined point by point, and they are smooth because in a frame they are given by the following formulas.

Proposition 38.2.6. *Let $e = (e_1, \dots, e_r)$ be a frame of E over U. Write $\omega = \sum_j \omega^j \otimes e_j$, and write the $\mathrm{End}(E)$-valued forms as matrices of ordinary forms, $\theta e_j = \sum_i \theta^i_j e_i$ and $\eta e_j = \sum_i \eta^i_j e_i$ (Example 38.2.4(3)).*

1. *The products are matrix products in which the entries are multiplied with the wedge product:*
$$(\alpha \wedge \omega)^i = \alpha \wedge \omega^i, \qquad (\theta \wedge \omega)^i = \sum_j \theta^i_j \wedge \omega^j, \qquad (\theta \wedge \eta)^i_j = \sum_k \theta^i_k \wedge \eta^k_j.$$
2. *In row and column notation, $\omega = e\,\omega$, where e is the row of the frame and ω the column of components. If $e' = eg$ is another frame, with $g \colon U \to \mathrm{GL}_r(\mathbb{K})$ smooth, that is $e'_j = \sum_i e_i g^i_j$, then the components change by $\omega' = g^{-1}\omega$ and $\theta' = g^{-1}\theta g$, and the formulas of (1) give the same products in both frames: $\theta' \wedge \omega' = g^{-1}(\theta \wedge \omega)$ and $\theta' \wedge \eta' = g^{-1}(\theta \wedge \eta)g$.*
3. *For 1-forms $\theta \in \Omega^1(M, \mathrm{End}\, E)$ and $\omega \in \Omega^1(M, E)$ and vector fields X, Y,*
$$(\theta \wedge \omega)(X, Y) = \theta(X)\omega(Y) - \theta(Y)\omega(X), \qquad (\theta \wedge \theta)(X, Y) = \theta(X)\theta(Y) - \theta(Y)\theta(X) = [\theta(X), \theta(Y)].$$
4. *The products are associative, $(\theta \wedge \eta) \wedge \omega = \theta \wedge (\eta \wedge \omega)$, and ordinary forms commute with bundle-valued forms up to the usual sign: $\alpha \wedge (\theta \wedge \omega) = (-1)^{kk'} \theta \wedge (\alpha \wedge \omega)$ for α of degree k and θ of degree k'. Two $\mathrm{End}(E)$-valued forms do not commute in general.*

Proof. (1) Let E^i_j be the endomorphism of $E|_U$ with $E^i_j e_j = e_i$ and $E^i_j e_k = 0$ for $k \neq j$, the matrix unit, so that $\theta = \sum_{i,j} \theta^i_j \otimes E^i_j$. By bilinearity, $\theta \wedge \omega = \sum_{i,j,k} (\theta^i_j \wedge \omega^k) \otimes E^i_j e_k = \sum_{i,j} (\theta^i_j \wedge \omega^j) \otimes e_i$, and since $E^i_k E^l_j = \delta_{kl} E^i_j$, $\theta \wedge \eta = \sum_{i,j,k} (\theta^i_k \wedge \eta^k_j) \otimes E^i_j$. The first formula is immediate.

(2) $e'(g^{-1}\omega) = e\, g g^{-1} \omega = e\, \omega$, so by uniqueness of components $\omega' = g^{-1}\omega$: the frame is multiplied by g on the right and the components by g^{-1} on the left, and the two changes

cancel. For an endomorphism-valued form, in row notation $\theta(e) = e\,\theta$ (applying θ to each e_j), hence $\theta(e') = \theta(e)g = e\,\theta g = e'g^{-1}\theta g$, so $\theta' = g^{-1}\theta g$. The entries of g are functions, which can be moved through wedge products, so $\theta' \wedge \omega' = g^{-1}\theta g \wedge g^{-1}\omega = g^{-1}(\theta \wedge \omega)$, which is the column of components of $\theta \wedge \omega$ in the frame e'; similarly for $\theta \wedge \eta$.

(3) For ordinary 1-forms, $(\gamma \wedge \beta)(X, Y) = \gamma(X)\beta(Y) - \gamma(Y)\beta(X)$ (Proposition 12.4.4). Applying this to each entry of (1), $(\theta \wedge \omega)^i(X, Y) = \sum_j \big(\theta^i_j(X)\omega^j(Y) - \theta^i_j(Y)\omega^j(X)\big)$, which is the i-th component of $\theta(X)\omega(Y) - \theta(Y)\omega(X)$. The same computation gives the second formula.

(4) In a frame, associativity is that of the wedge product and of matrix multiplication, and the sign rule is that of ordinary forms, applied entry by entry. □

Part (2) is the pattern for checking that a formula written in a frame defines a global object: compute how both sides transform under $e' = eg$ and check that they transform in the same way. The products pass this test because they are computed point by point. The exterior derivative of components does not pass it (Remark 38.2.8).

Example 38.2.7. 1. On the trivial bundle $\mathbb{R}^2 \times \mathbb{R}^2$ over $\mathbb{R}^2$, with its standard frame, let $\theta = \left(\begin{smallmatrix} 0 & dx \\ dy & 0 \end{smallmatrix}\right)$ and $\omega = \left(\begin{smallmatrix} dx \\ dy \end{smallmatrix}\right)$. Then

$$\theta \wedge \omega = \begin{pmatrix} dx \wedge dy \\ dy \wedge dx \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} dx \wedge dy, \qquad \theta \wedge \theta = \begin{pmatrix} dx \wedge dy & 0 \\ 0 & dy \wedge dx \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} dx \wedge dy.$$

Check with (3): $\theta(\partial_x) = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $\theta(\partial_y) = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$, whose commutator is $\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, the value of $\theta \wedge \theta$ on (∂_x, ∂_y). So for a matrix of 1-forms $\theta \wedge \theta$ need not vanish, unlike $\gamma \wedge \gamma = 0$ for a single 1-form; it vanishes for $r = 1$, and more generally whenever all the matrices $\theta(X)$ commute.

2. The order matters: $\theta \wedge \eta$ and $\eta \wedge \theta$ differ in general even for 1-forms. For $\eta = \left(\begin{smallmatrix} dx & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and θ as in (1), $\theta \wedge \eta = \left(\begin{smallmatrix} 0 & 0 \\ dy \wedge dx & 0 \end{smallmatrix}\right)$ and $\eta \wedge \theta = \left(\begin{smallmatrix} 0 & dx \wedge dx \\ 0 & 0 \end{smallmatrix}\right) = 0$.

3. A 2-form $F \in \Omega^2(M, \operatorname{End} E)$ and a section s give the E-valued 2-form $F \wedge s = Fs$, $(X, Y) \mapsto F(X, Y)s$. This is how the curvature of a connection acts on sections (Proposition 38.4.3).

Remark 38.2.8. There is no canonical exterior derivative on $\Omega^k(M, E)$: unlike the products, "differentiate the components" is not a pointwise operation, and it depends on the frame. Use the row and column notation of Proposition 38.2.6(2): $\omega = e\,\omega$, and in the frame e define $d_e\omega = e\,d\omega$, differentiating the column of components. In another frame $e' = eg$ the same form is $\omega = e'\omega'$ with $\omega' = g^{-1}\omega$; one may picture this as inserting $gg^{-1} = I$ between frame and components, $e\,\omega = (eg)(g^{-1}\omega)$. The two changes cancel in ω, but d only sees the one in the components:

$$d_{e'}\omega = e'd\omega' = e\,g\,d(g^{-1}\omega) = e\,g\big(g^{-1}d\omega + d(g^{-1}) \wedge \omega\big) = d_e\omega - e\,(dg\,g^{-1}) \wedge \omega,$$

using $d(g^{-1}) = -g^{-1}dg\,g^{-1}$, which follows by differentiating $gg^{-1} = I$. So d_e and $d_{e'}$ agree on all forms only if $dg = 0$, that is, if the change of frame is locally constant. The simplest instance: on the trivial line bundle $M \times \mathbb{R}$ take the frame $e = 1$ and the frame $e' = f$ for a nowhere vanishing function f. The section $s = f$ has component f in the frame e and component 1 in the frame e', so $d_e s = df \otimes 1$ while $d_{e'}s = 0$. A connection supplies the missing correction term, a matrix of 1-forms θ in each frame, with the transformation rule $\theta' = g^{-1}\theta g + g^{-1}dg$ that exactly compensates the term $e\,(dg\,g^{-1}) \wedge \omega$ (Proposition 38.3.4 and Proposition 38.4.1).

38.3 Connections

How should one differentiate a section s of a vector bundle? On a trivial bundle s is a vector-valued function and has a derivative ds, but Remark 38.2.8 shows that "differentiate the components" depends on the frame. Instead of choosing a formula, we list the properties a derivative of sections should have: it should take a section to something that eats a tangent vector and returns a vector of the fibre (an E-valued 1-form), be additive, and obey the product rule with functions. A connection is anything with these properties.

Definition 38.3.1. A *connection* (or *covariant derivative*) on a vector bundle $E \to M$ is a $\mathbb{K}$-linear map $\nabla\colon \Gamma(M,E) \to \Omega^1(M,E)$ that satisfies the Leibniz rule

$$\nabla(fs) = df \otimes s + f\nabla s, \qquad f \in C^\infty(M),\ s \in \Gamma(M,E).$$

In the Leibniz rule both sides are E-valued 1-forms: $df \otimes s$ is the product of the 1-form df with the section s (Definition 38.2.5), and $f\nabla s$ is the E-valued 1-form ∇s multiplied by the function f. The model is the trivial connection d on $M \times \mathbb{K}^r$, for which the rule is the product rule $d(fs) = df\, s + f\, ds$ (Example 38.3.3(1) below).

At a point p, the value $(\nabla s)_p$ of the E-valued 1-form ∇s is a linear map $T_pM \to E_p$ (Proposition 38.2.3). For a tangent vector $v \in T_pM$ and a vector field X we write

$$\nabla_v s = (\nabla s)_p(v) \in E_p, \qquad \nabla_X s = (\nabla s)(X) \in \Gamma(M,E), \quad (\nabla_X s)(p) = \nabla_{X(p)} s,$$

the *covariant derivative* of s in the direction v, respectively along X. Its geometric meaning, the rate of change of s along a curve with velocity v after transporting its values back to the fibre E_p by parallel transport, is Proposition 38.5.4 below. In Riemannian geometry and in many books a connection is introduced directly as an operation $(X,s) \mapsto \nabla_X s$, and for $E = TM$ one writes $\nabla_X Y$. The next proposition shows that the two points of view are equivalent, and translates the Leibniz rule into the language of ∇_X.

Proposition 38.3.2. *Let ∇ be a connection. The map $\mathfrak{X}(M) \times \Gamma(E) \to \Gamma(E)$, $(X,s) \mapsto \nabla_X s$, has the properties*

1. *$\nabla_{fX+gY} s = f\nabla_X s + g\nabla_Y s$ for functions f, g: it is $C^\infty(M)$-linear in X;*
2. *$\nabla_X(s+t) = \nabla_X s + \nabla_X t$ and $\nabla_X(cs) = c\nabla_X s$ for $c \in \mathbb{K}$;*
3. *$\nabla_X(fs) = (Xf)s + f\nabla_X s$.*

Conversely, every map $(X,s) \mapsto D_X s$ with these three properties is $D_X s = \nabla_X s$ for a unique connection ∇. In particular $(\nabla_X s)(p)$ depends only on $X(p)$, so $\nabla_v s$ is defined for a single tangent vector v.

Proof. (1) For fixed s, ∇s is an E-valued 1-form, and evaluating it on vector fields is $C^\infty(M)$-linear (Proposition 38.2.3(3)): $(\nabla s)(fX + gY) = f(\nabla s)(X) + g(\nabla s)(Y)$.

(2) This is the $\mathbb{K}$-linearity of $s \mapsto \nabla s$, evaluated on X.

(3) Evaluate the Leibniz rule on X. For the first term, $(df \otimes s)(X) = df(X)\, s$, by the identification of $(\alpha \otimes e)(v)$ with $\alpha(v)e$ in Proposition 38.2.3, and $df(X) = Xf$, the derivative of f along X (Remark 36.2.2). For the second, $(f\nabla s)(X) = f\,(\nabla s)(X) = f\nabla_X s$. So $\nabla_X(fs) = (Xf)s + f\nabla_X s$.

Conversely, let D have properties (1)–(3). For fixed s the map $X \mapsto D_X s$ is $C^\infty(M)$-linear by (1), so by Proposition 38.2.3(3) there is a unique E-valued 1-form, call it ∇s, with $(\nabla s)(X) = D_X s$ for all X. The map $s \mapsto \nabla s$ is $\mathbb{K}$-linear by (2) and the uniqueness. For the Leibniz rule, the two E-valued 1-forms $\nabla(fs)$ and $df \otimes s + f\nabla s$ take the same value on every

vector field X, namely $D_X(fs) = (Xf)s + fD_Xs$ by (3) and the computation in the first part; an E-valued 1-form is determined by its values on vector fields (Proposition 38.2.3), so they are equal. The connection is unique because ∇s is determined by the values D_Xs. Finally, $X \mapsto \nabla_X s$ is $C^\infty(M)$-linear, so its value at p depends only on $X(p)$ by Lemma 38.2.2. □

Note the asymmetry: $\nabla_X s$ is C^∞-linear in X but not in s. It depends only on the value of X at a point, but on the values of s near the point (in fact, on its values along a curve with velocity $X(p)$; Remark 38.5.1).

Example 38.3.3. 1. *The trivial connection.* A section of the trivial bundle $\pi\colon M \times \mathbb{K}^r \to M$ is a smooth map $s\colon M \to M \times \mathbb{K}^r$ with $\pi \circ s = \mathrm{id}$, that is, $s(p) = (p, \sigma(p))$ for a smooth function $\sigma\colon M \to \mathbb{K}^r$; the first component carries no information, and one identifies s with σ. So a section is a vector-valued function, whose components s^i are the coefficients of s in the standard frame $\varepsilon_i(p) = (p, e_i)$: $s = \sum_i s^i \varepsilon_i$. We write it as a column, $s = (s^1, \dots, s^r)^T$; the transpose only says that the row written inline is to be read as a column, as in the row and column notation of Proposition 38.2.6(2), where the frame is a row and the components form a column, $s = \varepsilon\, s$. Columns are used so that a matrix A acts on the left, As, as in (2) below. Then $ds = (ds^1, \dots, ds^r)^T$, a column of 1-forms, is a connection: it is linear, and $d(fs) = (d(fs^i))_i = (df\, s^i + f\, ds^i)_i = df\, s + f\, ds$. Here $\nabla_X s = X(s) = (Xs^1, \dots, Xs^r)^T$ is the componentwise derivative.

2. *Adding a matrix of* 1*-forms.* For an $r \times r$ matrix A of 1-forms on M, $\nabla s = ds + As$ is a connection on $M \times \mathbb{K}^r$, since the extra term is $C^\infty(M)$-linear in s: $A(fs) = fAs$. We will see that every connection on a trivial bundle has this form.

3. *The Euclidean connection.* On $T\mathbb{R}^n = \mathbb{R}^n \times \mathbb{R}^n$ the trivial connection is $\nabla_X Y = DY(X) = \sum_i X(Y^i)\partial_i$, the directional derivative of the components of Y.

4. *Subbundles of trivial bundles.* Let $E \subseteq M \times \mathbb{R}^N$ be a subbundle, so each E_p is a subspace of $\mathbb{R}^N$, and let $P_p\colon \mathbb{R}^N \to E_p$ be the orthogonal projection. Locally E has a frame, and Gram–Schmidt turns it into an orthonormal frame $(u_1, \dots, u_r)$; then $P = \sum_i u_i u_i^T$, so P is a smooth matrix-valued function. A section of E is a function $s\colon M \to \mathbb{R}^N$ with $s(p) \in E_p$, and

$$\nabla s = P(ds), \qquad \text{that is,} \qquad \nabla_X s = P(X(s)),$$

is a connection: $\nabla(fs) = P(df\, s + f\, ds) = df \otimes s + fP(ds)$, since $Ps = s$. For the tangent bundle $TS^n \subseteq S^n \times \mathbb{R}^{n+1}$ this says: differentiate the vector field in $\mathbb{R}^{n+1}$ and project back onto the tangent space. This is the Levi-Civita connection of the round sphere, and more generally, for the tangent bundle of any submanifold of $\mathbb{R}^N$, the projection connection is the Levi-Civita connection of the induced metric (Chapter 40, where geodesics turn out to be the curves whose acceleration is normal to the submanifold). The same construction applies to the tautological line bundles of $\mathbb{RP}^n$ and $\mathbb{CP}^n$ (with the Hermitian projection $P(w) = \sum_i \langle w, u_i\rangle u_i$ in the complex case), to the tangent and normal bundles of any submanifold of $\mathbb{R}^N$, and to the tautological bundles of Grassmannians.

Proposition 38.3.4. *Let ∇ be a connection on E.*

1. Locality. *If s vanishes on an open set U, so does ∇s. Consequently ∇ induces a connection on $E|_U$ for every open U, with $(\nabla s)|_U = \nabla(s|_U)$.*

2. Connection forms. *If $e = (e_1, \dots, e_r)$ is a frame over U, there is a unique matrix $\theta = (\theta^i_j)$ of 1-forms on U, the* connection form *of ∇ in the frame e, with $\nabla e_j = \sum_i \theta^i_j \otimes e_i$. For $s = \sum_j s^j e_j$,*

$$\nabla s = \sum_i \Big(ds^i + \sum_j \theta^i_j s^j \Big) \otimes e_i, \qquad \textit{in vector notation} \quad \nabla s = ds + \theta s.$$

3. Change of frame. *If $e' = eg$ is another frame over U, with $g\colon U \to \mathrm{GL}_r(\mathbb{K})$ smooth, then*

$$\theta' = g^{-1}\theta g + g^{-1} dg.$$

4. Existence. *Every vector bundle over a manifold admits a connection.*

5. The space of connections. *If ∇ is a connection and $A \in \Omega^1(M, \operatorname{End} E)$, then $\nabla + A$ is a connection, and every connection is $\nabla + A$ for a unique A: the connections form an affine space over $\Omega^1(M, \operatorname{End} E)$.*

Proof. (1) Let $s = 0$ on U, $p \in U$, and χ a bump function with $\chi = 1$ near p and support in U (Lemma 33.3.10). Then $\chi s = 0$, and $0 = \nabla(\chi s) = d\chi \otimes s + \chi \nabla s$; at p, $s(p) = 0$ and $\chi(p) = 1$, so $(\nabla s)_p = 0$. For a section s over U and $p \in U$ define $(\nabla^U s)_p = (\nabla(\chi s))_p$, where χs is extended by 0 outside U and χ is as above; by locality this does not depend on χ, it is smooth (near p one χ works for all nearby points), and it satisfies the Leibniz rule. It is the only connection on $E|_U$ compatible with restriction, again by locality.

(2) By (1) we may work on U. Each ∇e_j is an E-valued 1-form, so by Proposition 38.2.3(4) it is uniquely $\sum_i \theta^i_j \otimes e_i$ with 1-forms θ^i_j. By linearity and the Leibniz rule, $\nabla(\sum_j s^j e_j) = \sum_j (ds^j \otimes e_j + s^j \sum_i \theta^i_j \otimes e_i)$, which is the formula.

(3) Compute $\nabla e'_j$ in two ways. On the one hand $e'_j = \sum_k e_k g^k_j$ is a combination of the e_k with function coefficients, so by linearity and the Leibniz rule

$$\nabla e'_j = \sum_k \big(dg^k_j \otimes e_k + g^k_j \nabla e_k \big) = \sum_k \Big(dg^k_j \otimes e_k + g^k_j \sum_i \theta^i_k \otimes e_i \Big) = \sum_i (dg + \theta g)^i_j \otimes e_i.$$

On the other hand, by definition of θ' and $e'_l = \sum_i e_i g^i_l$, $\nabla e'_j = \sum_l \theta'^l_j \otimes e'_l = \sum_i (g\theta')^i_j \otimes e_i$. The coefficients in the frame e are unique (Proposition 38.2.3(4)), so $g\theta' = dg + \theta g$, that is $\theta' = g^{-1}\theta g + g^{-1} dg$. In the row notation of Proposition 38.2.6(2) the computation reads $\nabla(eg) = (\nabla e)g + e\, dg = e(\theta g + dg)$ and $\nabla e' = e'\theta' = e\, g\theta'$.

(4) Choose an open cover (U_α) of M by sets over which E has a frame e_α, and a partition of unity (ψ_α) subordinate to it (Theorem 33.8.2). On U_α let ∇^α be the connection with connection form 0 in the frame e_α, that is, $\nabla^\alpha(\sum_j s^j e_{\alpha,j}) = \sum_j ds^j \otimes e_{\alpha,j}$; it is the trivial connection of Example 38.3.3(1), transported by the frame, so it satisfies the Leibniz rule on $E|_{U_\alpha}$. Put $\nabla s = \sum_\alpha \psi_\alpha \nabla^\alpha(s|_{U_\alpha})$. Each term $\psi_\alpha \nabla^\alpha(s|_{U_\alpha})$ is an E-valued 1-form on U_α vanishing outside the support of ψ_α, which is closed in M and contained in U_α, so extended by 0 it is smooth on M; and the sum is locally finite, so it is a smooth E-valued 1-form. It is linear, and $\nabla(fs) = \sum_\alpha \psi_\alpha (df \otimes s + f \nabla^\alpha s) = df \otimes s + f \nabla s$, because $\sum_\alpha \psi_\alpha = 1$. (A general affine combination of connections fails the Leibniz rule; the coefficients must add up to 1.)

(5) $(\nabla + A)(fs) = df \otimes s + f \nabla s + fAs$, so $\nabla + A$ is a connection. If ∇' is another connection, $D = \nabla' - \nabla$ satisfies $D(fs) = fD(s)$: the terms $df \otimes s$ cancel. So $(X, s) \mapsto D(s)(X)$ is $C^\infty(M)$-linear in both arguments, and by Lemma 38.2.2 it is given by a section of $\operatorname{Hom}(TM \otimes E, E) = T^*M \otimes \operatorname{End} E$, that is, by some $A \in \Omega^1(M, \operatorname{End} E)$. □

By (2) and (5), a connection on a trivial bundle $M \times \mathbb{K}^r$ is $d + \theta$ for a matrix θ of 1-forms on M, its connection form in the standard frame. The connection form is not intrinsic: the

transformation rule (3) shows that the same connection can have connection form 0 in one frame and $g^{-1}dg \neq 0$ in another. For instance, on the trivial line bundle over $\mathbb{R}$ with the connection d, the frame $e' = e^x e$ has connection form $e^{-x}d(e^x) = dx$. Changing frames is a *gauge transformation* in the language of physics.

Example 38.3.5. 1. *The sphere.* On S^2 minus the poles use the spherical coordinates $x(\vartheta,\varphi) = (\sin\vartheta\cos\varphi, \sin\vartheta\sin\varphi, \cos\vartheta)$ and the orthonormal frame $e_1 = x_\vartheta = (\cos\vartheta\cos\varphi, \cos\vartheta\sin\varphi, -$ $e_2 = x_\varphi/\sin\vartheta = (-\sin\varphi, \cos\varphi, 0)$ of TS^2. For the projection connection of Example 38.3.3(4) we differentiate in $\mathbb{R}^3$ and project onto the tangent plane, which amounts to dropping the component along the normal x. Since $\partial_\vartheta e_1 = -x$, $\partial_\varphi e_1 = \cos\vartheta\, e_2$ and $\partial_\varphi e_2 = -(\cos\varphi, \sin\varphi, 0) = -(\sin\vartheta\, x + \cos\vartheta\, e_1)$,

$$de_1 = -x\, d\vartheta + \cos\vartheta\, e_2\, d\varphi, \qquad de_2 = -(\sin\vartheta\, x + \cos\vartheta\, e_1)\, d\varphi,$$

and dropping the terms along x,

$$\nabla e_1 = \cos\vartheta\, d\varphi \otimes e_2, \qquad \nabla e_2 = -\cos\vartheta\, d\varphi \otimes e_1.$$

So the connection form is $\theta = \left(\begin{smallmatrix} 0 & -\cos\vartheta\, d\varphi \\ \cos\vartheta\, d\varphi & 0 \end{smallmatrix}\right)$, a skew-symmetric matrix, in accordance with Proposition 38.6.2 below.

2. *The tautological line bundle of* $\mathbb{CP}^1$. Let $L \subseteq \mathbb{CP}^1 \times \mathbb{C}^2$ have fibre $L_\ell = \ell$ over the line ℓ, with the Hermitian inner product of $\mathbb{C}^2$ and the projection connection. Over $U_0 = \{[1:z]\}$ the section $s(z) = (1, z)$ is a frame, $ds = (0,1)\, dz$, and projecting onto the line spanned by s,

$$\nabla s = \frac{\langle (0,1), s\rangle}{|s|^2}\, dz \otimes s = \frac{\bar z\, dz}{1+|z|^2} \otimes s, \qquad \theta = \frac{\bar z\, dz}{1+|z|^2}.$$

A connection on E induces connections on all the bundles built from E by linear algebra, by requiring the Leibniz rule for the natural pairings and products.

Proposition 38.3.6. *Let ∇ and ∇' be connections on E and E'. The following formulas define connections, uniquely characterised by the Leibniz rules they express.*

1. *On $E^\vee$: $(\nabla_X\phi)(s) = X(\phi(s)) - \phi(\nabla_X s)$. In the dual frame $(e^1, \dots, e^r)$ of a frame e the connection form is $-\theta^T$.*
2. *On $E \oplus E'$: $\nabla_X(s, s') = (\nabla_X s, \nabla'_X s')$, with connection form $\left(\begin{smallmatrix} \theta & 0 \\ 0 & \theta' \end{smallmatrix}\right)$.*
3. *On $E \otimes E'$: $\nabla_X(s \otimes s') = \nabla_X s \otimes s' + s \otimes \nabla'_X s'$, with connection form $\theta \otimes 1 + 1 \otimes \theta'$ in the frame $(e_i \otimes e'_k)$.*
4. *On $\operatorname{Hom}(E, E')$: $(\nabla_X A)(s) = \nabla'_X(As) - A(\nabla_X s)$; in frames, the matrix of ∇A is $dA + \theta' A - A\theta$. On $\operatorname{End} E$ this is $\nabla A = dA + [\theta, A]$ in matrix form, that is, $\nabla_X A = [\nabla_X, A]$.*
5. *On $\Lambda^k E$: $\nabla_X(s_1 \wedge \cdots \wedge s_k) = \sum_i s_1 \wedge \cdots \wedge \nabla_X s_i \wedge \cdots \wedge s_k$; on the determinant line bundle $\det E = \Lambda^r E$ the connection form in the frame $e_1 \wedge \cdots \wedge e_r$ is $\operatorname{tr}\theta$.*

Proof. (1) The right side is $C^\infty(M)$-linear in X, and in s: replacing s by fs gives $X(f\phi(s)) - \phi((Xf)s + f\nabla_X s) = f(X(\phi(s)) - \phi(\nabla_X s))$. By Lemma 38.2.2 it defines a section $\nabla_X\phi$ of $E^\vee$, and the three properties of Proposition 38.3.2 are checked directly; for instance $(\nabla_X(f\phi))(s) = X(f\phi(s)) - f\phi(\nabla_X s) = (Xf)\phi(s) + f(\nabla_X\phi)(s)$. In the dual frame, $(\nabla e^j)(e_k) = d(\delta^j_k) - e^j(\nabla e_k) = -\theta^j_k$, so $\nabla e^j = -\sum_k \theta^j_k \otimes e^k$, whose matrix is $-\theta^T$. (2) is clear.

(3) Define ∇ on $E \otimes E'$ over a set with frames e, e' by the connection form $\theta\otimes 1 + 1\otimes\theta'$, that is, by $\nabla(e_i \otimes e'_k) = \nabla e_i \otimes e'_k + e_i \otimes \nabla' e'_k$ and the Leibniz rule. For sections $s = \sum_i s^i e_i$ and $s' = \sum_k s'^k e'_k$,

expanding $\nabla(\sum_{i,k} s^i s'^k e_i \otimes e'_k)$ with $d(s^i s'^k) = ds^i\, s'^k + s^i\, ds'^k$ gives $\nabla s \otimes s' + s \otimes \nabla' s'$. So the product rule holds, and since it determines ∇ on the products of frame sections, which span the fibres, the local definitions do not depend on the frames and glue.

(4) $\mathrm{Hom}(E, E') = E^\vee \otimes E'$, and the connection of (1) and (3) gives the stated formula: for $A = \phi \otimes s'$, $\nabla_X(A)(s) = (\nabla_X \phi)(s)s' + \phi(s)\nabla'_X s' = \nabla'_X(\phi(s)s') - \phi(\nabla_X s)s'$. In frames, A is a matrix of functions, $As = A\sigma$ for the component vector σ of s, and $\nabla'(A\sigma) - A(d\sigma + \theta\sigma) = (dA)\sigma + A\, d\sigma + \theta' A\sigma - A\, d\sigma - A\theta\sigma$.

(5) As in (3), define ∇ on $\Lambda^k E$ in the frame $(e_{i_1} \wedge \cdots \wedge e_{i_k})$ by the stated formula on the frame elements and the Leibniz rule; expanding $s_1 \wedge \cdots \wedge s_k$ in the frame shows that the formula holds for all sections, so the definition does not depend on the frame. For $k = r$, $\nabla(e_1 \wedge \cdots \wedge e_r) = \sum_i e_1 \wedge \cdots \wedge (\sum_l \theta^l_i e_l) \wedge \cdots \wedge e_r = (\sum_i \theta^i_i)\, e_1 \wedge \cdots \wedge e_r$, since the terms with $l \neq i$ contain a repeated factor. □

38.4 Curvature

For the trivial connection on a trivial bundle, second derivatives commute: $X(Y(s)) - Y(X(s)) = [X, Y](s)$, the Lie bracket measuring exactly the failure of X and Y to commute as derivations. For a general connection the difference $\nabla_X \nabla_Y s - \nabla_Y \nabla_X s - \nabla_{[X,Y]} s$ need not vanish, and it turns out to be a tensor: the curvature. The cleanest way to see this is to extend ∇ to bundle-valued forms, as d extends from functions to forms, and to observe that the extension does not square to zero.

Proposition 38.4.1 (Exterior covariant derivative)**.** *Let ∇ be a connection on E. There is a unique family of $\mathbb{K}$-linear maps $d_\nabla\colon \Omega^k(M, E) \to \Omega^{k+1}(M, E)$, $k \geq 0$, such that $d_\nabla = \nabla$ on $\Omega^0(M, E) = \Gamma(E)$ and*

$$d_\nabla(\alpha \wedge \omega) = d\alpha \wedge \omega + (-1)^k \alpha \wedge d_\nabla \omega \qquad (\alpha \in \Omega^k(M),\ \omega \in \Omega^l(M, E)).$$

In a frame with connection form θ, writing $\omega = \sum_i \omega^i \otimes e_i$,

$$d_\nabla \omega = \sum_i \Big(d\omega^i + \sum_j \theta^i_j \wedge \omega^j \Big) \otimes e_i, \qquad \textit{in vector notation} \quad d_\nabla \omega = d\omega + \theta \wedge \omega.$$

On vector fields, for $\omega \in \Omega^k(M, E)$,

$$(d_\nabla \omega)(X_0, \ldots, X_k) = \sum_i (-1)^i \nabla_{X_i}\big(\omega(X_0, \ldots, \widehat{X_i}, \ldots, X_k)\big) + \sum_{i<j} (-1)^{i+j} \omega([X_i, X_j], X_0, \ldots, \widehat{X_i}, \ldots, \widehat{X_j}, \ldots, X_k);$$

for $k = 1$, $(d_\nabla \omega)(X, Y) = \nabla_X(\omega(Y)) - \nabla_Y(\omega(X)) - \omega([X, Y])$.

Proof. Uniqueness. As for the exterior derivative (Theorem 36.2.8), the Leibniz rule makes d_∇ local: if $\omega = 0$ near p, choose a bump χ equal to 1 near p with $\chi\omega = 0$; then $0 = d_\nabla(\chi\omega) = d\chi \wedge \omega + \chi d_\nabla \omega$, and at p this is $(d_\nabla \omega)_p$. So it suffices to compute in a frame, where $\omega = \sum_i \omega^i \wedge e_i$ with the e_i viewed as E-valued 0-forms. The Leibniz rule forces

$$d_\nabla \omega = \sum_i \big(d\omega^i \wedge e_i + (-1)^k \omega^i \wedge \nabla e_i \big) = \sum_i d\omega^i\, e_i + \sum_{i,j} (-1)^k \omega^i \wedge \theta^j_i\, e_j = \sum_j \Big(d\omega^j + \sum_i \theta^j_i \wedge \omega^i \Big) e_j,$$

using $\omega^i \wedge \theta^j_i = (-1)^k \theta^j_i \wedge \omega^i$ for the k-form ω^i and the 1-form θ^j_i.

Existence. Define d_∇ on a frame domain by the formula $d\omega + \theta \wedge \omega$. It does not depend on the frame: if $e' = eg$, then $\omega = g\omega'$ and $\theta' = g^{-1}\theta g + g^{-1} dg$ (Proposition 38.3.4(3)), and

$$g(d\omega' + \theta' \wedge \omega') = g\, d\omega' + \theta \wedge g\omega' + dg \wedge \omega' = d(g\omega') + \theta \wedge (g\omega') = d\omega + \theta \wedge \omega,$$

which says that the two definitions give the same E-valued form. So the local definitions glue. For $k = 0$ the formula is that of ∇ (Proposition 38.3.4(2)), and the Leibniz rule holds in a frame: $d(\alpha\omega^i) + \sum_j \theta^i_j \wedge \alpha\omega^j = d\alpha \wedge \omega^i + (-1)^k\alpha \wedge (d\omega^i + \sum_j \theta^i_j \wedge \omega^j)$, since $\theta^i_j \wedge \alpha = (-1)^k \alpha \wedge \theta^i_j$.

The formula with vector fields. Both sides are local and $\mathbb{K}$-linear in ω, and locally ω is a sum of forms $\beta \otimes s$ with $\beta \in \Omega^k$ and s a section. For such a form, the left side is $(d\beta \otimes s + (-1)^k \beta \wedge \nabla s)(X_0, \dots, X_k)$. On the right side, $\nabla_{X_i}(\beta(\dots)s) = X_i(\beta(\dots))s + \beta(\dots)\nabla_{X_i} s$; the terms of the first kind, together with the bracket terms, give $d\beta(X_0, \dots, X_k)\, s$ by the formula of Theorem 36.2.8. The remaining terms $\sum_i (-1)^i \beta(X_0, \dots, \widehat{X_i}, \dots, X_k)\nabla_{X_i} s$ equal $(-1)^k(\beta \wedge \nabla s)(X_0, \dots, X_k)$: by Proposition 12.4.4, the shuffle that moves X_i to the last place has sign $(-1)^{k-i}$, and $(-1)^k(-1)^{k-i} = (-1)^i$. □

Definition 38.4.2. The *exterior covariant derivative* of ∇ is the operator d_∇ of Proposition 38.4.1; it is determined by $d_\nabla(\alpha \otimes s) = d\alpha \otimes s + (-1)^k \alpha \wedge \nabla s$ for $\alpha \in \Omega^k(M)$ and $s \in \Gamma(E)$. The *curvature* of ∇ is $F_\nabla = d_\nabla \circ \nabla \colon \Gamma(M, E) \to \Omega^2(M, E)$, also written F or R.

Proposition 38.4.3. *F_∇ is $C^\infty(M)$-linear, hence an element of $\Omega^2(M, \operatorname{End} E)$, and*

$$F_\nabla(X, Y)s = \nabla_X \nabla_Y s - \nabla_Y \nabla_X s - \nabla_{[X,Y]} s.$$

In a frame with connection form θ one has $F = d\theta + \theta \wedge \theta$, and under a change of frame $F' = g^{-1}Fg$. Moreover $d_\nabla F = 0$ in the sense that $dF + \theta \wedge F - F \wedge \theta = 0$ (the Bianchi identity*), and $d_\nabla \circ d_\nabla = F_\nabla \wedge -$ on $\Omega^\bullet(M, E)$.*

Proof. Tensoriality. Using the Leibniz rules of ∇ and d_∇ and $d(df) = 0$,

$$F(fs) = d_\nabla(df \otimes s + f\nabla s) = \big(d(df) \otimes s - df \wedge \nabla s\big) + \big(df \wedge \nabla s + f\, d_\nabla \nabla s\big) = fF(s).$$

So $(X, Y, s) \mapsto F(s)(X, Y)$ is $C^\infty(M)$-linear in all three arguments, and by Lemma 38.2.2 it is given by an element of $\Omega^2(M, \operatorname{End} E)$: at each point, $F(v, w)$ is an endomorphism of E_p, alternating in v and w.

The formula with vector fields is the case $k = 1$ of Proposition 38.4.1 applied to $\omega = \nabla s$: $F(s)(X, Y) = \nabla_X((\nabla s)(Y)) - \nabla_Y((\nabla s)(X)) - (\nabla s)([X, Y])$.

In a frame, $Fe_j = d_\nabla(\sum_i \theta^i_j \otimes e_i) = \sum_i (d\theta^i_j + \sum_k \theta^i_k \wedge \theta^k_j) \otimes e_i$ by the frame formula for d_∇, so the matrix of F (with $Fe_j = \sum_i F^i_j e_i$) is $d\theta + \theta \wedge \theta$. Since F is a tensor, its matrices in two frames are related as the matrices of any endomorphism: $Fe'_j = \sum_k g^k_j Fe_k$, so $F' = g^{-1}Fg$. (This can also be checked from the transformation rule of θ.)

Bianchi. For matrices of 1-forms, $d(\theta \wedge \theta) = d\theta \wedge \theta - \theta \wedge d\theta$, so $dF = d\theta \wedge \theta - \theta \wedge d\theta = (F - \theta \wedge \theta) \wedge \theta - \theta \wedge (F - \theta \wedge \theta) = F \wedge \theta - \theta \wedge F$. The induced connection on $\operatorname{End} E$ (Proposition 38.3.6(4)) has exterior derivative $d_\nabla A = dA + \theta \wedge A - (-1)^k A \wedge \theta$ on $\operatorname{End} E$-valued k-forms, by the same argument as in Proposition 38.4.1; for $k = 2$ the Bianchi identity says $d_\nabla F = 0$.

d^2_∇. In a frame, $d_\nabla d_\nabla \omega = d(d\omega + \theta \wedge \omega) + \theta \wedge (d\omega + \theta \wedge \omega) = d\theta \wedge \omega - \theta \wedge d\omega + \theta \wedge d\omega + \theta \wedge \theta \wedge \omega = F \wedge \omega$. □

So d_∇ is a "derivative" on bundle-valued forms that squares to zero exactly when the curvature vanishes; the de Rham complex of a flat bundle (Chapter 37) is built from it. The term $\nabla_{[X,Y]}$ in the curvature is what makes it a tensor: for coordinate vector fields, $[\partial_i, \partial_j] = 0$ and $F(\partial_i, \partial_j) = \nabla_i \nabla_j - \nabla_j \nabla_i$ is the commutator of covariant partial derivatives; in coordinates $F(\partial_i, \partial_j) = \partial_i \theta_j - \partial_j \theta_i + [\theta_i, \theta_j]$, where $\theta = \sum_i \theta_i\, dx^i$ with matrix-valued functions θ_i.

Example 38.4.4. 1. *Trivial bundles and line bundles.* The trivial connection d has $F = 0$. The connection $d + A$ on $M \times \mathbb{K}^r$ has $F = dA + A \wedge A$; for a line bundle the connection form is a single 1-form A, and $F = dA$. On a manifold of dimension 1 every connection is flat, since there are no nonzero 2-forms.

2. *The sphere.* With the connection form of Example 38.3.5(1), all entries of θ are multiples of $d\varphi$, so $\theta \wedge \theta = 0$, and $d(\cos\vartheta\, d\varphi) = -\sin\vartheta\, d\vartheta \wedge d\varphi$. With the area form $dA = \sin\vartheta\, d\vartheta \wedge d\varphi$ (for which $dA(e_1, e_2) = 1$),

$$F = d\theta = \begin{pmatrix} 0 & dA \\ -dA & 0 \end{pmatrix}, \qquad F(e_1,e_2)e_1 = -e_2, \quad F(e_1,e_2)e_2 = e_1.$$

So $F = dA \otimes J$, where J is the rotation of each tangent plane by $-\pi/2$ ($Je_1 = -e_2$, $Je_2 = e_1$). Although θ was only defined away from the poles, F is a global tensor, and $dA \otimes J$ is indeed defined on all of S^2. The number $\langle F(e_1,e_2)e_2, e_1\rangle = 1$ is the Gaussian curvature of the unit sphere (Chapter 40); since $F \neq 0$, no neighbourhood of any point of S^2 has a frame of parallel vector fields.

3. *The tautological line bundle of* $\mathbb{CP}^1$. With $\theta = \bar z\, dz/(1+|z|^2)$ from Example 38.3.5(2), and writing $z = x + iy$,

$$F = d\theta = \frac{\partial}{\partial \bar z}\Big(\frac{\bar z}{1+|z|^2}\Big)\, d\bar z \wedge dz = \frac{d\bar z \wedge dz}{(1+|z|^2)^2} = \frac{2i\, dx \wedge dy}{(1+x^2+y^2)^2},$$

using $\partial_{\bar z}(\bar z(1+z\bar z)^{-1}) = (1+|z|^2)^{-2}$ and $d\bar z \wedge dz = 2i\, dx \wedge dy$. The real 2-form $\frac{i}{2\pi}F = -\frac{1}{\pi}\, dx \wedge dy/(1+x^2+y^2)^2$ integrates over $U_0 \cong \mathbb{C}$, which is $\mathbb{CP}^1$ minus a point, to $-\frac{1}{\pi}\int_0^\infty \frac{2\pi r\, dr}{(1+r^2)^2} = -1$. This integer is the first Chern class of the tautological bundle (Chapter 39); that the integral of a curvature form is an integer independent of the connection is the beginning of Chern–Weil theory.

Proposition 38.4.5. *Let ∇ be a connection on $E \to M$ with curvature F, and ∇' a connection on $E' \to M$ with curvature F'.*

1. *For smooth $f : N \to M$ there is a unique connection $f^*\nabla$ on f^*E with $(f^*\nabla)(f^*s) = f^*(\nabla s)$ for sections s of E; its curvature is f^*F.*

2. *The connection $\nabla \otimes 1 + 1 \otimes \nabla'$ on $E \otimes E'$ has curvature $F \otimes 1 + 1 \otimes F'$, and the dual connection on $E^\vee$, $(\nabla^\vee\phi)(s) = d(\phi(s)) - \phi(\nabla s)$, has curvature $-F^T$.*

3. *For a line bundle, ∇ and $\nabla + \alpha$, with α a 1-form, have curvatures differing by $d\alpha$; so the curvature of a line bundle is a closed 2-form whose class does not depend on the connection.*

Here $f^*s = s \circ f$ is the pulled back section (Proposition 35.3.3), and $f^*(\nabla s)$ is the f^*E-valued 1-form $v \mapsto (\nabla s)(df(v))$ on N. Concretely, $(f^*\nabla)_v(s \circ f) = \nabla_{df(v)}s$ for $v \in T_qN$: the pullback connection differentiates in the direction of the image vector. In a frame e of E with connection form θ, the pulled back frame $e \circ f$ has connection form $f^*\theta$.

Proof. (1) Over a frame domain U of E with connection form θ, define a connection on $f^*E|_{f^{-1}U}$ by the connection form $f^*\theta$ in the frame $(e_j \circ f)$. If $e' = eg$ is another frame, the pulled back frames are related by $g \circ f$, and pulling back the transformation rule gives $f^*\theta' = (g\circ f)^{-1}(f^*\theta)(g\circ f) + (g\circ f)^{-1}d(g\circ f)$, since f^* commutes with d and with products. So the local connections agree on overlaps and glue. For $s = \sum_j s^j e_j$, $f^*s = \sum_j (s^j \circ f)(e_j \circ f)$, and $(f^*\nabla)(f^*s) = \sum_i (d(s^i\circ f) + \sum_j (f^*\theta^i_j)(s^j \circ f)) \otimes (e_i \circ f) = f^*(\nabla s)$. Uniqueness holds because the condition fixes $f^*\nabla$ on the pulled back frames, which span f^*E locally over $C^\infty(N)$. The curvature, computed in the pulled back frame, is $d(f^*\theta) + f^*\theta \wedge f^*\theta = f^*(d\theta + \theta\wedge\theta) = f^*F$.

(2) In frames (e_j), (e'_k) with connection forms θ, θ', the frame $(e_j \otimes e'_k)$ has connection form $\theta \otimes 1 + 1 \otimes \theta'$ (Proposition 38.3.6), and as the two summands act on different factors, they commute: $(\theta\otimes 1)\wedge(1\otimes\theta') + (1\otimes\theta')\wedge(\theta\otimes 1) = 0$ by graded commutativity of the 1-form

entries. So d of the connection form plus its square is $(d\theta + \theta \wedge \theta) \otimes 1 + 1 \otimes (d\theta' + \theta' \wedge \theta')$. The dual frame has connection form $-\theta^T$, and $d(-\theta^T) + \theta^T \wedge \theta^T = -(d\theta + \theta \wedge \theta)^T$, as $(\theta \wedge \theta)^T = -\theta^T \wedge \theta^T$ for matrices of 1-forms.

(3) For rank one, θ is a 1-form and $\theta \wedge \theta = 0$, so $F = d\theta$ in every frame, and replacing θ by $\theta + \alpha$ adds $d\alpha$. Closedness is $dF = F \wedge \theta - \theta \wedge F = 0$, the Bianchi identity for commuting 1×1 matrices. □

38.5 Parallel transport, holonomy and flatness

A connection lets us say when a section is "constant" along a curve: when its covariant derivative in the direction of the curve vanishes. Solving this differential equation moves vectors from one fibre to another along the curve. First we make precise what the derivative along a curve means.

Remark 38.5.1. A connection differentiates sections of E over M in the direction of vector fields on M, while along a curve $\gamma\colon I \to M$ the direction $\dot\gamma$ and the section to be differentiated are only defined along the curve. The notation $\nabla_{\dot\gamma} s$ is justified as follows.

1. *The direction may be a single vector.* For a section s of E, ∇s is an E-valued 1-form, so $\nabla_v s = (\nabla s)_p(v) \in E_p$ is defined for every tangent vector $v \in T_pM$; equivalently, $\nabla_X s$ at p depends only on X_p, because $\nabla_{fX} s = f \nabla_X s$. So for a section s defined near $\gamma(I)$, $\nabla_{\dot\gamma(t)} s$ makes sense without extending $\dot\gamma$ to a vector field.

2. *Only the values of s along a curve matter.* In a local frame, $\nabla_v s = \sum_i \big(v(s^i) + \sum_j \theta^i_j(v) s^j\big) e_i$ (Proposition 38.3.4), and $v(s^i) = (s^i \circ c)'(0)$ for any curve c with $\dot c(0) = v$. So $\nabla_v s$ depends only on the values of s along such a curve.

3. *Sections along a curve need not extend.* A *section along* γ is a section of γ^*E, that is, a smooth map $t \mapsto s(t) \in E_{\gamma(t)}$; for instance the velocity $\dot\gamma$ is a section of γ^*TM. It need not be of the form $S \circ \gamma$ for a section S of E, even near one point. For $\gamma(t) = (t^2, 0)$ in $\mathbb{R}^2$ we have $\gamma(t) = \gamma(-t)$ but $\dot\gamma(t) = (2t, 0) \neq \dot\gamma(-t)$ for $t \neq 0$, so no vector field X near the origin satisfies $X(\gamma(t)) = \dot\gamma(t)$; the same happens at every self-intersection of a curve, where two velocities sit at one point.

The precise definition therefore uses the pullback connection of Proposition 38.4.5(1): for a section s along γ,

$$\nabla_{\dot\gamma} s = (\gamma^*\nabla)_{d/dt}\, s,$$

a derivative on the interval I, where d/dt is a genuine vector field. By the defining property $(\gamma^*\nabla)(\gamma^* S) = \gamma^*(\nabla S)$, $(\gamma^*\nabla)_{d/dt}(S \circ \gamma) = (\nabla S)(\dot\gamma) = \nabla_{\dot\gamma} S$ in the sense of (1), and in a local frame $(\gamma^*\nabla)_{d/dt} s = \sum_i \big(\dot s^i + \sum_j \theta^i_j(\dot\gamma) s^j\big) e_i$. When s does extend to a section S near $\gamma(t)$, the two meanings of $\nabla_{\dot\gamma} s$ agree, whatever the extension, by (2).

Definition 38.5.2. A section s along a smooth curve $\gamma\colon [0,1] \to M$, that is, a section of γ^*E, is *parallel* if $\nabla_{\dot\gamma} s = 0$, in the sense of Remark 38.5.1. *Parallel transport* along γ is the map $P_\gamma\colon E_{\gamma(0)} \to E_{\gamma(1)}$ sending v to $s(1)$, where s is the parallel section with $s(0) = v$. For piecewise smooth curves, transport along the smooth pieces is composed.

Proposition 38.5.3. *Parallel sections along γ exist and are unique for each initial value, so P_γ is a well-defined linear isomorphism, depending smoothly on the data and satisfying $P_{\gamma * \delta} = P_\delta \circ P_\gamma$ and $P_{\bar\gamma} = P_\gamma^{-1}$.*

Proof. In a local frame the equation $\nabla_{\dot\gamma} s = 0$ reads

$$\dot s^i(t) + \sum_j \theta^i_j(\dot\gamma(t))\, s^j(t) = 0, \qquad i = 1, \dots, r,$$

a linear system of ordinary differential equations with smooth coefficients. By Proposition 27.13.5(2) its solutions exist on the whole interval on which γ stays in the frame domain, are unique, and depend linearly on the initial value, and smoothly on parameters (Theorem 27.13.6). The interval $[0,1]$ is covered by finitely many subintervals on each of which γ stays in one frame domain; solving successively on them, using the end value of one piece as initial value of the next, gives a unique parallel section on $[0,1]$. Uniqueness does not depend on the frames, because the equation $\nabla_{\dot\gamma} s = 0$ does not. So P_γ is well defined and linear. Running the curve backwards, $\bar\gamma(t) = \gamma(1-t)$, the section $t \mapsto s(1-t)$ is parallel along $\bar\gamma$ (the equation is linear in $\dot\gamma$), so $P_{\bar\gamma} P_\gamma = \mathrm{id}$ and P_γ is invertible; the composition rule follows from uniqueness. The same argument shows that P_γ does not change under an increasing reparametrisation of γ. □

So far the connection came first and parallel transport was derived from it. The two notions are equivalent: the connection can be recovered from its parallel transport, and this is the geometric meaning of the covariant derivative. Vectors in different fibres $E_{\gamma(t)}$ and $E_{\gamma(0)}$ cannot be subtracted, so a section cannot be differentiated by a difference quotient; parallel transport brings $s(\gamma(t))$ back to the fibre over $\gamma(0)$, where the difference quotient makes sense, and the covariant derivative is its limit.

Proposition 38.5.4 (The connection from parallel transport)**.** *Let $\gamma\colon (-\varepsilon,\varepsilon) \to M$ be a smooth curve with $\gamma(0) = p$ and $\dot\gamma(0) = v$, and for each t let $P_t\colon E_p \to E_{\gamma(t)}$ be parallel transport along γ, $P_t(w) = u(t)$ where u is the parallel section along γ with $u(0) = w$. Then for every section s of E defined near p, and more generally for every section s along γ,*

$$\nabla_v s = \frac{d}{dt}\Big|_{t=0} P_t^{-1}\big(s(\gamma(t))\big) = \lim_{t\to 0} \frac{P_t^{-1}\big(s(\gamma(t))\big) - s(p)}{t},$$

a derivative of a curve in the single vector space E_p. (For a section along γ, read $s(t)$ for $s(\gamma(t))$ and $\nabla_{\dot\gamma} s$ at $t = 0$ for $\nabla_v s$.)

Proof. Parallel sections along γ exist on the whole interval and depend linearly on the initial value (Proposition 38.5.3, whose proof applies to any interval). Choose a basis $w_1, \dots, w_r$ of E_p and let $u_i(t) = P_t(w_i)$, the parallel sections with $u_i(0) = w_i$. Since each P_t is an isomorphism, the u_i form a frame of γ^*E, so $s(\gamma(t)) = \sum_i a^i(t) u_i(t)$ with smooth functions a^i. Then $P_t^{-1}(s(\gamma(t))) = \sum_i a^i(t) w_i$, a curve in E_p with derivative $\sum_i \dot a^i(0) w_i$ at $t = 0$. On the other hand, by the Leibniz rule for the connection $\gamma^*\nabla$ along γ and $\nabla_{\dot\gamma} u_i = 0$,

$$\nabla_{\dot\gamma}(s \circ \gamma) = \sum_i \big(\dot a^i u_i + a^i \nabla_{\dot\gamma} u_i\big) = \sum_i \dot a^i(t) u_i(t),$$

which at $t = 0$ is $\sum_i \dot a^i(0) w_i$. Finally $\nabla_{\dot\gamma}(s\circ\gamma)$ at $t = 0$ is $\nabla_v s$, by Remark 38.5.1. The same computation applies to a section along γ. □

In particular a section is parallel along a curve exactly when $P_t^{-1}(s(\gamma(t)))$ is constant, that is, when $s(\gamma(t)) = P_t(s(p))$: the covariant derivative measures how fast s moves away from the parallel transport of its initial value. In a local frame e with connection form θ, the parallel transport equation $\dot u + \theta(\dot\gamma) u = 0$ gives, for the matrix of P_t in the frames $e(p)$ and $e(\gamma(t))$,

$$P_t = I - t\,\theta(v) + O(t^2), \qquad P_t^{-1} = I + t\,\theta(v) + O(t^2),$$

so $P_t^{-1}(s(\gamma(t))) = s(p) + t\big(ds(v) + \theta(v)s(p)\big) + O(t^2)$, and the proposition returns the frame formula $\nabla_v s = ds(v) + \theta(v)s$ of Proposition 38.3.4(2). The connection form is the infinitesimal parallel transport, and the curvature is the infinitesimal holonomy around small loops (Proposition 38.5.7 below). For the trivial connection on a trivial bundle, $P_t = I$ and the formula is the ordinary derivative.

The formula has the same shape as the Lie derivative of a vector field, $(L_X Y)_p = \frac{d}{dt}\big|_{t=0} d\Phi_{-t}(Y_{\Phi_t(p)}) = [X,Y]_p$ for the flow Φ of X, where vectors at $\Phi_t(p)$ are brought back to p by the differential of the flow instead of by parallel transport. The two differ in an essential way: parallel transport along γ only depends on the curve through p with velocity v to first order, so $\nabla_v s$ only depends on v, while the flow of X near p depends on X near p, and $L_X Y$ at p depends on the derivatives of X at p (indeed $L_{fX}Y = fL_XY - (Yf)X$).

Proposition 38.5.5 (Homotopy invariance of pullbacks)**.** *Let $E \to N$ be a vector bundle and $F\colon M \times \mathbb{R} \to N$ a smooth map, and put $f = F(-,0)$ and $g = F(-,1)$. Then $f^*E \cong g^*E$ as vector bundles on M. In particular every vector bundle on a star-shaped open subset of $\mathbb{R}^n$, for instance on $\mathbb{R}^n$ or on a ball, is trivial.*

Proof. Choose a connection on $E' = F^*E$ (Proposition 38.3.4). For $x \in M$ let $\gamma_x(t) = (x,t)$, $0 \le t \le 1$, and let $P_x\colon E'_{(x,0)} \to E'_{(x,1)}$ be parallel transport along γ_x. Since $E'_{(x,0)} = E_{f(x)} = (f^*E)_x$ and $E'_{(x,1)} = (g^*E)_x$, the P_x are linear isomorphisms $(f^*E)_x \to (g^*E)_x$, and they depend smoothly on x (Proposition 38.5.3), so they form an isomorphism of vector bundles $f^*E \to g^*E$. For a star-shaped U, with centre p, apply this to $F(x,t) = p + t(x-p)$ and $N = U$: then $g = \mathrm{id}$ and f is constant, so $E = \mathrm{id}^*E \cong f^*E = U \times E_p$ is trivial. □

Example 38.5.6.

1. For the trivial connection d on $M \times \mathbb{K}^r$ the parallel sections along any curve are the constant ones, and $P_\gamma = \mathrm{id}$.

2. *Around a latitude of the sphere.* Let $c(\varphi) = x(\vartheta_0, \varphi)$, $\varphi \in [0, 2\pi]$, be the circle of latitude at polar angle ϑ_0, with the frame and connection form of Example 38.3.5(1). For $V = ae_1 + be_2$, $\nabla_{\dot c} e_1 = \cos\vartheta_0\, e_2$ and $\nabla_{\dot c} e_2 = -\cos\vartheta_0\, e_1$, so $\nabla_{\dot c} V = (a' - b\cos\vartheta_0)e_1 + (b' + a\cos\vartheta_0)e_2$. With $\kappa = \cos\vartheta_0$ the parallel fields are

$$a(\varphi) = a_0\cos(\kappa\varphi) + b_0\sin(\kappa\varphi), \qquad b(\varphi) = -a_0\sin(\kappa\varphi) + b_0\cos(\kappa\varphi)\,:$$

relative to the frame (e_1, e_2), which itself comes back to its initial value, the vector rotates by the angle $-\kappa\varphi$. After one turn it is rotated by $-2\pi\cos\vartheta_0$, which modulo 2π equals $2\pi(1-\cos\vartheta_0)$, the area of the spherical cap $\{\vartheta < \vartheta_0\}$ enclosed by the circle. On the equator, $\vartheta_0 = \pi/2$, the transport is trivial, and near the north pole the rotation is small, as the enclosed area is. This is the Gauss–Bonnet theorem in its simplest form (Chapter 40), and an instance of the next proposition.

The next result is the geometric meaning of curvature: transport around a small loop differs from the identity by the curvature times the enclosed area.

Proposition 38.5.7 (Holonomy of small loops)**.** *Let $x^1, \dots, x^n$ be coordinates centred at p, and for small $\varepsilon > 0$ let γ_ε be the loop at p that runs around the square with vertices 0, εe_1, $\varepsilon(e_1+e_2)$, εe_2 in the (x^1, x^2)-plane, in this order. Then*

$$P_{\gamma_\varepsilon} = \mathrm{id} - \varepsilon^2 F(\partial_1, \partial_2)_p + O(\varepsilon^3).$$

Proof. Work in a frame near p, with $\theta = \sum_i \theta_i\, dx^i$, and write θ_i, $\partial_j\theta_i$ for the values at p. Transport along a segment $t \mapsto q + tu$, u a unit coordinate vector $\pm e_k$, solves $\dot s = -\theta(u)s$ with $\theta(u) = \pm\theta_k(q+tu)$. Differentiating the equation, $\ddot s = -(\partial_u\theta(u))s + \theta(u)^2 s$, so by Taylor's

formula (the solution depends smoothly on q and ε) transport along the segment of length ε is $T = \mathrm{id} - \varepsilon\theta(u)(q) + \frac{\varepsilon^2}{2}\big(\theta(u)^2 - \partial_u\theta(u)\big)(q) + O(\varepsilon^3)$. For the four sides, expanding the values at the starting points 0, εe_1, $\varepsilon(e_1+e_2)$, εe_2 around p to first order,

$$\begin{aligned} T_1 &= \mathrm{id} - \varepsilon\theta_1 + \tfrac{\varepsilon^2}{2}(\theta_1^2 - \partial_1\theta_1),\\ T_2 &= \mathrm{id} - \varepsilon\theta_2 - \varepsilon^2\partial_1\theta_2 + \tfrac{\varepsilon^2}{2}(\theta_2^2 - \partial_2\theta_2),\\ T_3 &= \mathrm{id} + \varepsilon\theta_1 + \varepsilon^2(\partial_1\theta_1 + \partial_2\theta_1) + \tfrac{\varepsilon^2}{2}(\theta_1^2 - \partial_1\theta_1),\\ T_4 &= \mathrm{id} + \varepsilon\theta_2 + \varepsilon^2\partial_2\theta_2 + \tfrac{\varepsilon^2}{2}(\theta_2^2 - \partial_2\theta_2), \end{aligned}$$

up to $O(\varepsilon^3)$; for T_3 and T_4 the direction is $-e_1$, $-e_2$, so $\theta(u) = -\theta_1$, $-\theta_2$ and $\partial_u\theta(u) = \partial_1\theta_1$, $\partial_2\theta_2$. In $P_{\gamma_\varepsilon} = T_4T_3T_2T_1$ the terms of order ε cancel. The terms of order ε^2 coming from a single factor add up to $\theta_1^2 + \theta_2^2 - \partial_1\theta_2 + \partial_2\theta_1$, and the products of two first-order terms, taken in the order of the factors, add up to $\theta_2\theta_1 - \theta_2^2 - \theta_2\theta_1 - \theta_1\theta_2 - \theta_1^2 + \theta_2\theta_1 = -\theta_1^2 - \theta_2^2 - \theta_1\theta_2 + \theta_2\theta_1$. The total coefficient of ε^2 is $-(\partial_1\theta_2 - \partial_2\theta_1 + \theta_1\theta_2 - \theta_2\theta_1) = -F(\partial_1,\partial_2)_p$, by the coordinate formula for F after Proposition 38.4.3. □

For a line bundle the formula is exact and global in the following sense.

Proposition 38.5.8. *Let L be a line bundle with connection ∇ and curvature F (a closed 2-form), let $\bar D$ be the closed unit disc, $h\colon B \to M$ a smooth map defined on an open disc $B \supseteq \bar D$, and let γ be the loop $h|_{\partial\bar D}$, traversed counterclockwise. Then parallel transport around γ is multiplication by*

$$P_\gamma = \exp\Big(-\int_{\bar D} h^*F\Big).$$

Proof. The bundle h^*L over the star-shaped B is trivial (Proposition 38.5.5), and parallel transport along γ for ∇ is parallel transport along the boundary circle for $h^*\nabla$, because $(h\circ c)^*\nabla = c^*(h^*\nabla)$ for curves c in the disc (both satisfy the characterisation of Proposition 38.4.5(1)). In a frame over B, $h^*\nabla = d + \alpha$ for a 1-form α, and a parallel section along the boundary circle c solves the scalar equation $\dot s + \alpha(\dot c)s = 0$, so $s(1) = \exp(-\int_c \alpha)\, s(0)$. By Stokes' theorem (Theorem 36.4.8), $\int_c \alpha = \int_{\bar D} d\alpha = \int_{\bar D} h^*F$, as $h^*F = d\alpha$ by Proposition 38.4.5(1),(3). □

Definition 38.5.9. The *holonomy group* of ∇ at p is $\mathrm{Hol}_p(\nabla) = \{P_\gamma : \gamma \text{ a loop at } p\} \subseteq \mathrm{GL}(E_p)$, a subgroup. A connection is *flat* if $F_\nabla = 0$.

It is a subgroup by Proposition 38.5.3: the product of P_γ and P_δ is transport along the concatenated loop, and the inverse is transport along the reversed loop. If σ is a path from p to q, then $\mathrm{Hol}_q = P_\sigma \mathrm{Hol}_p P_\sigma^{-1}$, so on a connected manifold the holonomy groups at different points are conjugate. For the sphere, Example 38.5.6(2) shows that the holonomy contains rotations by all angles $2\pi(1-\cos\vartheta_0)$, hence it is the whole rotation group $\mathrm{SO}(2)$ of the tangent plane.

Theorem 38.5.10. *For a connection ∇ on $E \to M$ the following are equivalent:*

1. *$F_\nabla = 0$;*
2. *every point has a neighbourhood with a frame of parallel sections (a* flat frame*), in which $\nabla = d$;*
3. *parallel transport along a path depends only on its homotopy class with fixed endpoints.*

In that case the sheaf $\mathcal{E}^\nabla$ of parallel sections is a local system of rank r with $\mathcal{E}^\nabla \otimes_{\mathbb{R}_M} C^\infty_M \cong \mathcal{E}$, and on a connected M this gives an equivalence between flat bundles and representations of $\pi_1(M,p)$, the representation being the holonomy

$$\pi_1(M,p) \to \mathrm{GL}(E_p), \qquad [\gamma] \mapsto P_\gamma^{-1}.$$

Proof. (2)⇒(1) is clear, since d has zero curvature, and curvature is local.

(1)⇒(2). Work over a coordinate ball U with a frame, so that $E|_U = U \times \mathbb{K}^r$ and $\nabla = d + \theta$ with $\theta = \sum_i \theta_i\, dx^i$, a matrix of 1-forms. We look for sections $s\colon U \to \mathbb{K}^r$ with $\partial_i s + \theta_i s = 0$ for all i. Their graphs are submanifolds of $U \times \mathbb{K}^r$ whose tangent spaces are spanned by the vector fields

$$X_i = \frac{\partial}{\partial x^i} - \sum_{j,k} (\theta_i)^k_j v^j \frac{\partial}{\partial v^k}, \qquad i = 1, \dots, n,$$

where $v^1, \dots, v^r$ are the coordinates of $\mathbb{K}^r$ (real and imaginary parts separately if $\mathbb{K} = \mathbb{C}$). These span a distribution of rank n. The coefficients of X_j in the $\partial/\partial v$ directions are linear in v, and a direct computation of the brackets gives

$$[X_i, X_j] = -\sum_{k,l} \big(\partial_i \theta_j - \partial_j \theta_i + \theta_i \theta_j - \theta_j \theta_i\big)^k_l v^l \frac{\partial}{\partial v^k} = -\sum_{k,l} (F_{ij})^k_l v^l \frac{\partial}{\partial v^k},$$

with $F_{ij} = F(\partial_i, \partial_j)$, where $F = \sum_{i<j} F_{ij}\, dx^i \wedge dx^j$: the terms ∂_i applied to the coefficients of X_j give $-\partial_i \theta_j$, and the bracket of the two vertical parts gives the commutator of the matrices. So the distribution is involutive if and only if $F = 0$. If $F = 0$, the Frobenius theorem (Theorem 33.9.8) gives integral manifolds; each is transverse to the fibres $\{x\} \times \mathbb{K}^r$, hence locally the graph of a map $s\colon U' \to \mathbb{K}^r$, and the condition that the graph is an integral manifold is exactly $\partial_i s + \theta_i s = 0$, that is, $\nabla s = 0$. Taking the integral manifolds through $(p, v_1), \dots, (p, v_r)$ for a basis (v_i) gives r parallel sections near p, which form a frame because they are linearly independent at p, hence nearby.

(2)⇒(3). Let $H\colon [0,1]^2 \to M$ be a homotopy with fixed endpoints from γ_0 to γ_1. Cover the square by finitely many small squares each mapped by H into a neighbourhood carrying a flat frame, as in the proof of Theorem 29.3.3. In a flat frame parallel transport along any path is the identity matrix, so the transport along the two edge paths of each small square agree; moving across the squares one at a time shows that the transports along γ_0 and γ_1 coincide.

(3)⇒(2). On a simply connected neighbourhood U of a point p, define $s_i(q) = P_\gamma(v_i)$ for a path γ from p to q in U and a basis (v_i) of E_p; this is well defined by (3), smooth by smooth dependence of solutions of differential equations, and gives a frame of parallel sections, in which the connection form vanishes.

For the last statement, parallel sections form a sheaf that is locally constant by (2), hence a local system, and a local frame of parallel sections is a frame of E, which gives the isomorphism with $\mathcal{E}$. Conversely a local system $\mathcal{L}$ determines the bundle $E = \mathcal{L} \otimes_{\mathbb{R}_M} C^\infty_M$ with the connection d in any flat frame; these constructions are inverse to each other. The correspondence with representations of π_1 is Theorem 25.5.3, and the monodromy of the local system of parallel sections is given by parallel transport; the inverse appears because monodromy was defined as a right action. □

In a flat frame the transition functions between two flat frames are locally constant (both are parallel), so a flat bundle is one whose cocycle can be chosen locally constant; this is the situation of Remark 38.2.8, in which componentwise differentiation is well defined.

Example 38.5.11. Flat connections on the trivial line bundle over $S^1 = \mathbb{R}/2\pi\mathbb{Z}$: every connection is $d + a(\theta)\, d\theta$, it is flat, and parallel transport around the circle is multiplication by $\exp(-\int_0^{2\pi} a)$. Two such connections are isomorphic (by a gauge transformation $g\colon S^1 \to \mathbb{C}^\times$, which changes $a\, d\theta$ by $g^{-1} dg$) if and only if their holonomies agree, because the 1-forms $g^{-1} dg$ are exactly those with $\int_{S^1} g^{-1} dg \in 2\pi i \mathbb{Z}$ (a nonvanishing function has a continuous logarithm along the circle whose increase is $2\pi i$ times its winding number). So flat line

bundles with connection on S^1 are classified by the holonomy in $\mathbb{C}^\times$, in accordance with Theorem 38.5.10, since $\pi_1(S^1) = \mathbb{Z}$.

Counterexample 38.5.12. A flat connection can have nontrivial holonomy, and a bundle with trivial holonomy around contractible loops need not be trivial.

1. In Example 38.5.11, $d + a\,d\theta$ with $\int_0^{2\pi} a \notin 2\pi i\mathbb{Z}$ is flat but has no global parallel section: the holonomy is not 1. Flatness only gives parallel frames *locally*.

2. The Möbius bundle over S^1 (Example 35.2.3) has a flat connection, as its transition function -1 is locally constant, with holonomy -1 around the circle; it is not trivial.

3. Curvature is a property of the connection, not of the bundle: the trivial line bundle on $\mathbb{R}^2$ carries the flat connection d and the connection $d + x\,dy$ of curvature $dx \wedge dy$. Only the topology of the bundle constrains the curvature, through characteristic classes (Chapter 39); for instance the tautological bundle of $\mathbb{CP}^1$ admits no flat connection (Exercise 38.9.2).

38.6 Metric connections

Definition 38.6.1. Let E carry a Euclidean metric $\langle -, - \rangle$ (or a Hermitian metric in the complex case). A connection is *metric* if $d\langle s, t\rangle = \langle \nabla s, t\rangle + \langle s, \nabla t\rangle$, that is, $X\langle s, t\rangle = \langle \nabla_X s, t\rangle + \langle s, \nabla_X t\rangle$ for all vector fields X and sections s, t.

Proposition 38.6.2. *Metric connections exist, and ∇ is metric if and only if parallel transport is an isometry, if and only if in every orthonormal local frame the connection form is skew-symmetric (skew-Hermitian in the complex case); then the curvature takes values in the skew-symmetric (skew-Hermitian) endomorphisms, that is, in the Lie algebra of the structure group.*

Proof. Existence: patch the connections with connection form 0 in orthonormal local frames with a partition of unity, as in Proposition 38.3.4(4); each of them is metric, since in an orthonormal frame $\langle s, t\rangle = \sum_i s^i \bar{t}^i$ and $d\langle s, t\rangle = \sum_i (ds^i\,\bar{t}^i + s^i\,d\bar{t}^i)$, and the metric condition is preserved by combinations with coefficients adding up to 1. If ∇ is metric and s, t are parallel along γ, then $\frac{d}{dt}\langle s, t\rangle = \langle \nabla_{\dot\gamma} s, t\rangle + \langle s, \nabla_{\dot\gamma} t\rangle = 0$ (the metric condition along the curve follows from the one on M by pulling back), so transport is an isometry. Conversely, suppose transport is an isometry, and let $v \in T_pM$, γ a curve with $\dot\gamma(0) = v$, and (u_i) a parallel frame along γ; the $\langle u_i, u_j\rangle$ are constant. Writing $s \circ \gamma = \sum_i a^i u_i$ and $t \circ \gamma = \sum_j b^j u_j$, we get $\nabla_{\dot\gamma} s = \sum_i \dot a^i u_i$, and $\frac{d}{dt}\langle s, t\rangle = \sum_{i,j} (\dot a^i \bar b^j + a^i \dot{\bar b}^j)\langle u_i, u_j\rangle = \langle \nabla_{\dot\gamma} s, t\rangle + \langle s, \nabla_{\dot\gamma} t\rangle$, which at $t = 0$ is the metric condition in the direction v. In an orthonormal frame, $\langle e_i, e_j\rangle = \delta_{ij}$, and the metric condition for $s = e_j$, $t = e_i$ reads $0 = \theta^i_j + \overline{\theta^j_i}$. Finally, if θ is skew-symmetric (skew-Hermitian), so are $d\theta$ and $\theta \wedge \theta$, as $(\theta \wedge \theta)^T = -\theta^T \wedge \theta^T$; hence so is $F = d\theta + \theta \wedge \theta$. □

Example 38.6.3. The projection connection $\nabla s = P(ds)$ of a subbundle $E \subseteq M \times \mathbb{R}^N$ (Example 38.3.3(4)) is metric for the metric induced from $\mathbb{R}^N$: for sections s, t of E, $d\langle s, t\rangle = \langle ds, t\rangle + \langle s, dt\rangle = \langle P\,ds, t\rangle + \langle s, P\,dt\rangle$, because $t = Pt$, $s = Ps$ and P is self-adjoint. So parallel transport on the sphere is by rotations, as in Example 38.5.6(2), and the connection form of Example 38.3.5(1) is skew-symmetric. On the tangent bundle of a Riemannian manifold there are many metric connections; among them exactly one is *torsion-free*, $\nabla_X Y - \nabla_Y X = [X, Y]$, the Levi-Civita connection (Chapter 40).

Remark 38.6.4. *What torsion measures.* For a connection ∇ on TM the torsion $T(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y]$ is a tensor (Exercise 38.9.7), and in coordinates it vanishes if and only if $\Gamma^k_{ij} = \Gamma^k_{ji}$. It is invisible to geodesics: two connections ∇ and $\nabla' = \nabla + A$, with A a TM-valued tensor, have the same geodesics if and only if $A(v,v) = 0$ for all v, that is, A is antisymmetric, since $\nabla'_{\dot\gamma}\dot\gamma = \nabla_{\dot\gamma}\dot\gamma + A(\dot\gamma,\dot\gamma)$. Taking $A = -\frac{1}{2}T$ gives a torsion-free connection with the same geodesics. For example on $\mathbb{R}^3$, $\nabla_X Y = DY(X) + \frac{1}{2}X \times Y$ is a metric connection (the cross product is antisymmetric for the inner product) whose geodesics are the straight lines, with torsion $T(X,Y) = X \times Y$: parallel transport along a line rotates vectors about it. Geometrically, torsion measures the failure of small parallelograms built from geodesics to close up, to second order; the Levi-Civita connection is the unique metric connection for which they close.

Example 38.6.5. On a complex vector bundle with a Hermitian metric over a complex manifold, there is a unique connection that is metric and compatible with the holomorphic structure ($\nabla^{0,1} = \bar\partial$): the *Chern connection* (Chapter 45). Its curvature is a $(1,1)$-form, and for line bundles $\frac{i}{2\pi}F$ represents the first Chern class in de Rham cohomology, which is the differential-geometric form of Theorem 35.4.2. The projection connection on the tautological line bundle of $\mathbb{CP}^1$ (Example 38.4.4(3)) is of this kind.

38.7 Vertical and horizontal subspaces

The total space E of a vector bundle is itself a manifold, of dimension $n + r$ (or $n + 2r$ if $\mathbb{K} = \mathbb{C}$), and at each point $e \in E$ its tangent space contains a distinguished subspace: the directions tangent to the fibre, along which the base point does not move. These are the *vertical* directions. There is no distinguished complementary *horizontal* subspace: "moving the base point while keeping the vector constant" means nothing until one says how to compare vectors in different fibres, which is the problem a connection solves. In fact a connection is the same thing as a (linear) choice of horizontal subspaces, and in this language parallel transport becomes horizontal lifting of curves and the curvature becomes the failure of the horizontal subspaces to fit together into submanifolds. When $\mathbb{K} = \mathbb{C}$ we regard E as a real manifold; in both cases the tangent space of $\mathbb{K}^r$ at any point is $\mathbb{K}^r$.

Local coordinates on E come from frames. A frame $e = (e_1, \dots, e_r)$ over U gives the diffeomorphism

$$\Phi_e \colon U \times \mathbb{K}^r \to E|_U, \qquad (x,v) \mapsto \sum_j v^j e_j(x)$$

(Proposition 35.3.5), and a tangent vector of E at $\Phi_e(x,v)$ is written (ξ,η) with $\xi \in T_xU$ and $\eta \in \mathbb{K}^r$. For another frame $e' = eg$, $\Phi_{e'}(x,v') = \Phi_e(x, g(x)v')$, so the coordinates change by $(x,v') \mapsto (x, gv')$, and tangent vectors by

$$(\xi,\eta') \longmapsto \big(\xi, dg(\xi)v' + g\eta'\big). \tag{38.1}$$

Definition 38.7.1. Let $\pi \colon E \to M$ be a vector bundle. The *vertical subspace* at $e \in E$ is $V_e = \ker(d_e\pi) \subseteq T_eE$, and the *vertical lift* is the linear map

$$\mathrm{vl}_e \colon E_{\pi(e)} \to V_e, \qquad \mathrm{vl}_e(w) = \frac{d}{dt}\Big|_{t=0}(e + tw).$$

In the coordinates of a frame, $\pi \circ \Phi_e(x,v) = x$, so $d\pi(\xi,\eta) = \xi$ and $V_e = \{(0,\eta)\}$; and $e + tw$ has coordinates $(x, v + tw)$, where we write w also for the column of components of w, so $\mathrm{vl}_e(w) = (0,w)$. Hence vl_e is an isomorphism, $V = \bigcup_e V_e$ is a smooth subbundle of TE (spanned by the coordinate fields of $\mathbb{K}^r$), and vl is an isomorphism of vector bundles

$\pi^*E \to V$ over E. The tangent vectors $(\xi, 0)$, by contrast, are not intrinsic: by (38.1) the vector $(\xi, 0)$ in the frame e' becomes $(\xi, dg(\xi)v')$ in the frame e. They are the velocities of the curves along which the components stay constant, and this depends on the frame, exactly as in Remark 38.2.8.

Definition 38.7.2. A *horizontal distribution* on E is a smooth subbundle $H \subseteq TE$ with $T_eE = H_e \oplus V_e$ for all $e \in E$. Then $d_e\pi\colon H_e \to T_{\pi(e)}M$ is an isomorphism. The *horizontal lift* of $v \in T_pM$ to $e \in E_p$ is the unique $v^h_e \in H_e$ with $d\pi(v^h_e) = v$, and the horizontal lift of a vector field X on M is the vector field X^h on E with $X^h_e = (X_{\pi(e)})^h_e$. A curve in E is *horizontal* if its velocity lies in H. Writing a tangent vector as $Y = Y^H + Y^V$ with $Y^H \in H$ and $Y^V \in V$, the *connection map* is

$$K\colon TE \to E, \qquad K(Y) = \mathrm{vl}_e^{-1}(Y^V) \quad \text{for } Y \in T_eE.$$

X^h is smooth, because $d\pi$ restricted to H is an isomorphism of vector bundles $H \to \pi^*TM$ (Lemma 35.2.2). Equivalently, a horizontal distribution is a splitting of the exact sequence of vector bundles over E

$$0 \to \pi^*E \xrightarrow{\mathrm{vl}} TE \xrightarrow{d\pi} \pi^*TM \to 0.$$

Such splittings always exist (Proposition 35.3.4), but a general one has nothing to do with the linear structure of the fibres; the connections are the splittings compatible with it (Proposition 38.7.4).

Proposition 38.7.3 (The horizontal distribution of a connection)**.** *Let ∇ be a connection on E. For a frame e over U with connection form θ, put*

$$H_{\Phi_e(x,v)} = \{(\xi, -\theta_x(\xi)v) : \xi \in T_xU\}.$$

1. *This does not depend on the frame, and defines a horizontal distribution $H = H^\nabla$ on E. In coordinates it is spanned by the vector fields $X_i(x, v) = (\partial_i, -\theta_i(x)v)$, where $\theta = \sum_i \theta_i\, dx^i$.*

2. *For a section s and $v \in T_pM$,*

$$d_ps(v) = v^h_{s(p)} + \mathrm{vl}_{s(p)}(\nabla_v s), \qquad \textit{that is,} \qquad \nabla_v s = K(d_ps(v)).$$

   So $\nabla_v s$ is the vertical part of the derivative of s, and s is parallel if and only if ds takes values in H, that is, if the submanifold $s(M) \subseteq E$ is tangent to H.

3. *A curve u in E is horizontal if and only if it is a parallel section along the curve $\gamma = \pi \circ u$. Hence every curve γ in M has a unique horizontal lift starting at a given $e \in E_{\gamma(0)}$, namely $t \mapsto P_{\gamma|[0,t]}(e)$.*

4. *∇ is determined by H^∇.*

Proof. (1) Let $e' = eg$ be another frame, with connection form $\theta' = g^{-1}\theta g + g^{-1}dg$ (Proposition 38.3.4(3)). By (38.1) the vector $(\xi, -\theta'(\xi)v')$ in the coordinates of e' is, in those of e, (ξ, η) with

$$\eta = dg(\xi)v' - g\big(g^{-1}\theta(\xi)g + g^{-1}dg(\xi)\big)v' = -\theta(\xi)gv' = -\theta(\xi)v,$$

since $v = gv'$. So the two definitions agree. The fields X_i are smooth and span H, and $H_e \cap V_e = 0$, because $(\xi, -\theta(\xi)v)$ is vertical only for $\xi = 0$; as $\dim H_e = n$, $T_eE = H_e \oplus V_e$.

(2) In a frame, $s = \Phi_e(x, \sigma(x))$ with σ the column of components, and

$$d_ps(v) = (v, d\sigma(v)) = (v, -\theta(v)\sigma(p)) + (0, d\sigma(v) + \theta(v)\sigma(p)).$$

The first term lies in $H_{s(p)}$ and projects to v, so it is $v^h_{s(p)}$; the second is $\mathrm{vl}_{s(p)}(\nabla_v s)$, because $\nabla_v s$ has components $d\sigma(v) + \theta(v)\sigma$ (Proposition 38.3.4(2)).

(3) Locally $u(t) = \Phi_e(\gamma(t), \sigma(t))$, and $\dot u = (\dot\gamma, \dot\sigma)$ lies in H if and only if $\dot\sigma + \theta(\dot\gamma)\sigma = 0$, which is the equation $\nabla_{\dot\gamma} u = 0$ in the frame (Remark 38.5.1). The rest follows from Proposition 38.5.3.

(4) By (2), $\nabla_v s = K(ds(v))$, and K is defined by H alone. □

So the derivative $ds(v) \in T_{s(p)}E$ of a section is always defined, without any choice; what the connection provides is the splitting of $ds(v)$ into a horizontal part, which only remembers v, and a vertical part, which is a vector of the fibre and is the covariant derivative. The vector fields X_i are the ones used in the proof of Theorem 38.5.10: they are the horizontal lifts of the coordinate fields.

Proposition 38.7.4. *A horizontal distribution H on E is H^∇ for a connection ∇ if and only if it is invariant under the fibrewise multiplications $m_c\colon E \to E$, $m_c(e) = ce$, for all $c \in \mathbb{K}$: $dm_c(H_e) \subseteq H_{ce}$.*

Proof. In the coordinates of a frame $m_c(x, v) = (x, cv)$ and $dm_c(\xi, \eta) = (\xi, c\eta)$; so $dm_c(\xi, -\theta(\xi)v) = (\xi, -\theta(\xi)(cv))$, and H^∇ is invariant. Conversely, let H be invariant. Since $H_{(x,v)}$ is a complement of $V = \{(0, \eta)\}$, it is the graph of a linear map: $H_{(x,v)} = \{(\xi, -\Gamma_{(x,v)}(\xi))\}$ with $\Gamma_{(x,v)}\colon T_xU \to \mathbb{K}^r$ linear, and Γ is smooth in (x, v), since $-\Gamma_{(x,v)}(\partial_i)$ is the $\mathbb{K}^r$-component of the smooth vector field $(\partial_i)^h$. Invariance says $\Gamma_{(x,cv)} = c\Gamma_{(x,v)}$ for all $c \in \mathbb{K}$, including $c = 0$. Fix x and ξ, and let $f(v) = \Gamma_{(x,v)}(\xi)$, a smooth map $\mathbb{K}^r \to \mathbb{K}^r$ with $f(cv) = cf(v)$. Differentiating $f(tv) = tf(v)$ at $t = 0$, for real t, gives $f(v) = df_0(v)$, so f is $\mathbb{R}$-linear, and if $\mathbb{K} = \mathbb{C}$ also $f(iv) = if(v)$, so f is $\mathbb{C}$-linear. Hence $\Gamma_{(x,v)}(\xi) = \theta_x(\xi)v$ for a matrix $\theta_x(\xi)$, which is linear in ξ and smooth in x, being the derivative at $v = 0$ of the smooth map $v \mapsto \Gamma_{(x,v)}(\xi)$. So θ is a matrix of 1-forms, and over U, $H = H^{d+\theta}$. Now define $\nabla_v s = K(d_p s(v))$ with the connection map K of H. Over each frame domain this is the connection $d + \theta$, by Proposition 38.7.3(2); so ∇ is a well-defined connection with $H^\nabla = H$. □

Smoothness along the zero section matters: the function $f(x, y) = x^3/(x^2+y^2)$, $f(0, 0) = 0$, satisfies $f(cv) = cf(v)$ for real c but is not linear; it is not differentiable at 0. Accordingly, horizontal distributions defined only away from the zero section, or on general fibre bundles, give *nonlinear* parallel transport; they are the *Ehresmann connections*, and the linear ones are the connections of this chapter.

Proposition 38.7.5 (Curvature and integrability)**.** *Let ∇ be a connection with curvature F and horizontal distribution H. For vector fields X, Y on M and $e \in E$,*

$$[X^h, Y^h]_e = [X, Y]^h_e - \mathrm{vl}_e\big(F(X, Y)e\big).$$

Hence H is involutive if and only if $F = 0$, and then its integral manifolds are, locally, the images of parallel sections.

Proof. In a frame, write $X = \sum_i X^i\partial_i$ and $Y = \sum_j Y^j\partial_j$; then $X^h = \sum_i X^i X_i$ and $Y^h = \sum_j Y^j X_j$ with the fields X_i of Proposition 38.7.3(1), where the coefficients are functions of x only. Since the ∂_x-component of X^h is X, $X^h(Y^j) = X(Y^j)$, and

$$[X^h, Y^h] = \sum_{i,j} X^iY^j[X_i, X_j] + \sum_j \big(X(Y^j) - Y(X^j)\big)X_j = \sum_{i,j} X^iY^j[X_i, X_j] + [X, Y]^h.$$

In the proof of Theorem 38.5.10 we computed $[X_i, X_j] = (0, -F_{ij}v)$ with $F_{ij} = F(\partial_i, \partial_j)$, and $\sum_{i,j} X^iY^jF_{ij} = F(X, Y)$, whose matrix applied to the components v of e is $F(X, Y)e$; so the vertical term is $-\mathrm{vl}_e(F(X, Y)e)$. If $F = 0$, the brackets of the horizontal lifts of the coordinate fields are horizontal, and so are the brackets of their combinations with smooth coefficients,

as $[fA, gB] = fg[A,B] + f(Ag)B - g(Bf)A$; so H is involutive. Conversely, if H is involutive the vertical part of $[X^h, Y^h]$ vanishes, so $F(X,Y)e = 0$ for all e. An integral manifold N has $TN = H$, which is complementary to V; so $\pi|_N$ is a local diffeomorphism, and locally $N = s(U)$ for a section s with ds taking values in H, a parallel section by Proposition 38.7.3(2). □

Example 38.7.6. 1. *The trivial connection.* For d on $M \times \mathbb{K}^r$, $H_{(x,v)} = T_xM \times 0$: the horizontal curves are those with constant second component, and the integral manifolds of H are the slices $M \times \{v\}$, the graphs of the constant sections.

2. *Subbundles of trivial bundles.* Let $E \subseteq M \times \mathbb{R}^N$ be a subbundle with the projection connection $\nabla s = P(ds)$ (Example 38.3.3(4)). The total space E is a submanifold of $M \times \mathbb{R}^N$, $T_{(p,w)}E \subseteq T_pM \times \mathbb{R}^N$, and $\mathrm{vl}_{(p,w)}(u) = (0,u)$, so $V_{(p,w)} = 0 \times E_p$. We claim

$$H_{(p,w)} = T_{(p,w)}E \cap (T_pM \times E_p^\perp) :$$

the horizontal vectors are the tangent vectors whose $\mathbb{R}^N$-component is orthogonal to the fibre. Indeed, for $v \in T_pM$ choose a section $s = (x, \sigma(x))$ with $s(p) = w$; by Proposition 38.7.3(2), $v^h = ds(v) - \mathrm{vl}(\nabla_v s) = (v, d\sigma(v) - P\,d\sigma(v))$, whose second component lies in $E_p^\perp$. So H is contained in the right side, which meets V only in 0 and therefore has dimension at most n. For $TS^n = \{(x,w) : x \cdot w = 0\} \subseteq S^n \times \mathbb{R}^{n+1}$, the tangent vectors at (x,w) are the (ξ,η) with $\xi \cdot x = 0$ and $\eta \cdot x + w \cdot \xi = 0$, and the horizontal ones have $\eta \in \mathbb{R}x$:

$$H_{(x,w)} = \{(\xi, -(w \cdot \xi)\, x) : \xi \perp x\}.$$

For a great circle $\gamma(t) = \cos t\, x + \sin t\, y$ ($x \perp y$ unit vectors), the curve $u = (\gamma, \dot\gamma)$ in TS^n has velocity $(\dot\gamma, \ddot\gamma) = (\dot\gamma, -\gamma)$, which is horizontal since $-(\dot\gamma \cdot \dot\gamma)\gamma = -\gamma$. So the velocity field of a great circle is parallel along it: great circles are geodesics (Chapter 40).

3. *The tautological line bundle and the Hopf fibration.* For the tautological bundle $L \subseteq \mathbb{CP}^1 \times \mathbb{C}^2$ with the projection connection, the map $(\ell, w) \mapsto w$ identifies L minus the zero section with $\mathbb{C}^2 \setminus 0$ (the inverse is $w \mapsto (\mathbb{C}w, w)$), and tangent vectors (v,u) with their second components u. By (2), with the Hermitian inner product,

$$V_w = \mathbb{C}w, \qquad H_w = (\mathbb{C}w)^\perp = \{u \in \mathbb{C}^2 : \langle u, w\rangle = 0\}.$$

On the unit sphere $S^3 \subseteq \mathbb{C}^2 \setminus 0$, the circle bundle of L, whose projection to $\mathbb{CP}^1$ is the Hopf map (Section 20.7), $T_zS^3 = \{u : \mathrm{Re}\langle u, z\rangle = 0\} = i\mathbb{R}z \oplus (\mathbb{C}z)^\perp$: the tangent line $i\mathbb{R}z$ of the Hopf circle through z, and the horizontal plane H_z orthogonal to it. So the horizontal lifts of vector fields on $\mathbb{CP}^1$ are tangent to S^3, and parallel transport preserves S^3, as it must for a metric connection. The curvature is not zero (Example 38.4.4(3)), so by Proposition 38.7.5 the planes $(\mathbb{C}z)^\perp$ are not tangent to any family of surfaces in S^3: the brackets of horizontal fields have a component along the Hopf circles. This plane field is the *standard contact structure* of S^3.

38.8 Principal bundles

Vector bundles are often better understood through their bundles of frames, on which the structure group acts. This point of view is used in Chern–Weil theory and in gauge theory; we give the definitions, the correspondence between horizontal distributions and connection forms, and the Hopf fibration as example, and sketch the rest.

Definition 38.8.1. Let G be a Lie group. A *principal G-bundle* is a smooth surjection $\pi\colon P \to M$ with a free right action of G preserving the fibres and acting simply transitively on them, which is locally trivial: every point of M has a neighbourhood U with a G-equivariant diffeomorphism $\pi^{-1}(U) \cong U \times G$ over U. As for vector bundles, principal bundles are classified by cocycles $g_{ij}\colon U_i \cap U_j \to G$, that is, by $\check{H}^1(M, \underline{G})$.

Example 38.8.2. The frame bundle $\mathrm{Fr}(E)$ of a rank r real vector bundle, whose fibre over p is the set of bases of E_p, is a principal $\mathrm{GL}_r(\mathbb{R})$-bundle, and E is recovered as the associated bundle $\mathrm{Fr}(E) \times_{\mathrm{GL}_r} \mathbb{R}^r$; a metric reduces the structure group to $O(r)$, an orientation to GL_r^+, and a complex structure to $\mathrm{GL}_{r/2}(\mathbb{C})$. Covering spaces are principal bundles with discrete structure group. The oriented orthonormal frames of TS^2 form a principal SO(2)-bundle which is diffeomorphic to SO(3): an oriented orthonormal frame (v, w) at $x \in S^2$ is the same as the rotation matrix with columns x, v, w. The Hopf map $S^3 \to S^2 = \mathbb{CP}^1$, $(z_0, z_1) \mapsto [z_0 : z_1]$, is a principal $U(1)$-bundle, the unit circle bundle of the tautological line bundle.

The language of horizontal subspaces carries over to principal bundles, where it is the standard definition of a connection. Let $\pi\colon P \to M$ be a principal G-bundle, write $R_g(p) = pg$ for the action, and let $\mathfrak{g}$ be the Lie algebra of G. As for vector bundles, the *vertical subspace* at p is $V_p = \ker d_p\pi$, the tangent space of the fibre through p. For $\xi \in \mathfrak{g}$ the *fundamental vector field* $\xi^\#$ on P is

$$\xi^\#_p = \frac{d}{dt}\Big|_{t=0} p \exp(t\xi),$$

the velocity of the orbit of the one-parameter subgroup $\exp(t\xi)$; its flow is $R_{\exp(t\xi)}$.

Definition 38.8.3. A *principal connection* on P is a smooth subbundle $H \subseteq TP$ with $T_pP = H_p \oplus V_p$ for all p which is invariant under the action: $dR_g(H_p) = H_{pg}$. Its *connection form* is the $\mathfrak{g}$-valued 1-form ω with $\omega_p(Y) = \xi$ when $Y - \xi^\#_p \in H_p$.

Proposition 38.8.4. *1. For every $p \in P$, $\xi \mapsto \xi^\#_p$ is a linear isomorphism $\mathfrak{g} \to V_p$, and $dR_g(\xi^\#_p) = (\mathrm{Ad}(g^{-1})\xi)^\#_{pg}$.*

2. The connection form of a principal connection is a well-defined smooth $\mathfrak{g}$-valued 1-form, with

$$\omega(\xi^\#) = \xi \quad (\xi \in \mathfrak{g}), \qquad R_g^*\omega = \mathrm{Ad}(g^{-1}) \circ \omega \quad (g \in G).$$

3. Conversely, every $\mathfrak{g}$-valued 1-form ω with these two properties is the connection form of exactly one principal connection, namely $H = \ker\omega$.

Proof. (1) $\xi^\#_p$ is the differential at e of the orbit map $g \mapsto pg$ applied to ξ, so it is linear in ξ, and it is vertical since $\pi(p\exp(t\xi)) = \pi(p)$. If $\xi^\#_p = 0$, the curve $c(t) = p\exp(t\xi)$, an integral curve of $\xi^\#$ (its velocity at t is $\xi^\#_{c(t)}$, by $\exp((t+s)\xi) = \exp(t\xi)\exp(s\xi)$), and the constant curve at p are integral curves of $\xi^\#$ through p, so they coincide: $p\exp(t\xi) = p$, hence $\exp(t\xi) = e$ for all t, as the action is free, and $\xi = 0$ because exp is injective near 0 (Theorem 34.3.3). In a local trivialisation the fibre is G, so $\dim V_p = \dim G$, and the map is an isomorphism. Finally $p\exp(t\xi)g = pg \cdot g^{-1}\exp(t\xi)g = pg\exp(t\,\mathrm{Ad}(g^{-1})\xi)$, by Theorem 34.3.3 applied to the automorphism $x \mapsto g^{-1}xg$, whose differential is $\mathrm{Ad}(g^{-1})$; differentiate at $t = 0$.

(2) By (1) and $T_pP = H_p \oplus V_p$, every $Y \in T_pP$ is uniquely $Y^H + \xi^\#_p$, so ω is well defined and linear on each T_pP. For a basis $\xi_1, \dots, \xi_m$ of $\mathfrak{g}$, the fields $\xi^\#_a$ form a frame of V, which together with a local frame of H is a local frame of TP, and the components of ω are coefficients of vector fields in this frame, hence smooth. Clearly $\omega(\xi^\#) = \xi$. For $Y = Y^H + \xi^\#_p$, $dR_g(Y) = dR_g(Y^H) + (\mathrm{Ad}(g^{-1})\xi)^\#_{pg}$ with $dR_g(Y^H) \in H_{pg}$, so $\omega_{pg}(dR_gY) = \mathrm{Ad}(g^{-1})\omega_p(Y)$.

(3) By $\omega(\xi^\#) = \xi$, ω_p restricted to V_p is the inverse of the isomorphism of (1); so ω_p is onto, and $\ker \omega_p$ is a complement of V_p. The kernel of the morphism $\omega\colon TP \to P \times \mathfrak{g}$, of constant rank $\dim \mathfrak{g}$, is a smooth subbundle (Proposition 35.3.7), and it is invariant, since $\omega(dR_g Y) = \mathrm{Ad}(g^{-1})\omega(Y)$. Its connection form is ω, because $Y - \omega(Y)^\# \in \ker \omega$. A principal connection is the kernel of its connection form, which gives uniqueness. □

Example 38.8.5. *The Hopf bundle.* The group $U(1)$ acts freely on $S^3 \subseteq \mathbb{C}^2$ by $z \cdot \lambda = \lambda z$, and $S^3 \to \mathbb{CP}^1$ is a principal $U(1)$-bundle (Example 38.8.2). The Lie algebra of $U(1)$ is $i\mathbb{R}$, exp is the exponential function, and Ad is trivial, as $U(1)$ is abelian. For $\xi \in i\mathbb{R}$, $\xi_z^\# = \frac{d}{dt} e^{t\xi} z\big|_{t=0} = \xi z$. Define

$$\omega_z(u) = \langle u, z\rangle = u_0\bar z_0 + u_1 \bar z_1, \qquad u \in T_z S^3 = \{u : \mathrm{Re}\langle u, z\rangle = 0\},$$

the restriction of the 1-form $\bar z_0\, dz_0 + \bar z_1\, dz_1$; it takes values in $i\mathbb{R}$. Then $\omega(\xi_z^\#) = \xi|z|^2 = \xi$, and $(R_\lambda^*\omega)_z(u) = \omega_{\lambda z}(\lambda u) = |\lambda|^2\langle u, z\rangle = \omega_z(u)$. So ω is a connection form, and its horizontal subspaces are $\ker \omega_z = (\mathbb{C}z)^\perp$: the principal connection is the horizontal distribution of the projection connection on the tautological line bundle found in Example 38.7.6(3). A local section σ of $S^3 \to \mathbb{CP}^1$ is a local frame of the tautological bundle with $|\sigma| = 1$, and $\sigma^*\omega = \langle d\sigma, \sigma\rangle$ is its connection form for the projection connection, because the orthogonal projection onto the line $\mathbb{C}\sigma$ is $P(w) = \langle w, \sigma\rangle\sigma$, so $\nabla\sigma = P(d\sigma) = \langle d\sigma, \sigma\rangle \otimes \sigma$. For $\sigma = (1, z)/(1 + |z|^2)^{1/2}$ over U_0 this gives

$$\sigma^*\omega = \frac{\bar z\, dz - z\, d\bar z}{2(1 + |z|^2)},$$

which is $\theta + d\log(1 + |z|^2)^{-1/2}$, with $\theta = \bar z\, dz/(1 + |z|^2)$ the connection form in the frame $(1, z)$ (Example 38.3.5(2)), as the change of frame formula of Proposition 38.3.4(3) predicts. Its differential $d\sigma^*\omega = d\theta$ is the curvature computed in Example 38.4.4(3).

Remark 38.8.6. By Proposition 38.8.4, a principal connection is the same as a G-equivariant splitting of $0 \to V \to TP \to \pi^*TM \to 0$, or as a $\mathfrak{g}$-valued 1-form ω on P with $\omega(\xi^\#) = \xi$, which says that ω restricts to the Maurer–Cartan form on the fibres, and $R_g^*\omega = \mathrm{Ad}(g^{-1})\omega$. Its curvature is $\Omega = d\omega + \frac{1}{2}[\omega, \omega]$, and connections on P correspond to connections on all associated vector bundles, compatible with the structure group. This is the framework of gauge theory and of Chern–Weil theory, which we use in Chapter 39; see [KN63, Chapters I–II]. For a vector bundle, a local frame is a local section of the frame bundle, and pulling back ω by it gives the connection form θ of Proposition 38.3.4; the transformation rule $\theta' = g^{-1}\theta g + g^{-1}dg$ is the pullback of the equivariance of ω.

38.9 Exercises

Exercise 38.9.1. Compute the connection and curvature forms of the Levi-Civita connection of the sphere of radius ρ in $\mathbb{R}^3$, that is, of the projection connection of its tangent bundle, in the frame of Example 38.3.5(1), and verify that the curvature is $F = \rho^{-2}\, dA \otimes J$, where dA is the area form, so that the integral of the Gaussian curvature ρ^{-2} is 4π (compare the Gauss–Bonnet theorem, Chapter 39).

Solution. Scaling by ρ does not change the unit vectors e_1, e_2 or the normal, so the computation of Example 38.3.5(1) is unchanged: $\theta_1^2 = \cos\vartheta\, d\varphi$ and $F_2^1 = \sin\vartheta\, d\vartheta \wedge d\varphi$. The area form is now $dA = \rho^2 \sin\vartheta\, d\vartheta \wedge d\varphi$, so $F = \rho^{-2} dA \otimes J$, and $\int \rho^{-2}\, dA = \rho^{-2} \cdot 4\pi\rho^2 = 4\pi$. □

Exercise 38.9.2. Give an example of a flat bundle that is not trivial, and of a bundle admitting no flat connection.

Solution. The Möbius bundle over S^1 is flat (its transition function is locally constant) but not trivial (Exercise 35.6.1). The tautological bundle $\mathcal{O}(-1)$ on $\mathbb{CP}^1$ admits no flat connection: by Theorem 38.5.10 a flat bundle on a simply connected manifold is determined by a representation of the trivial group, hence is trivial, while $\mathcal{O}(-1)$ is not trivial because $c_1(\mathcal{O}(-1)) \neq 0$ (Example 35.4.6). □

Exercise 38.9.3. Show that for a line bundle with connection, the holonomy is given by $\exp(-\int_\Sigma F)$ for a loop bounding a surface Σ, and deduce that flat line bundles on a simply connected manifold are trivial.

Solution. For a disc this is Proposition 38.5.8; for a surface Σ with one boundary circle over which the pulled back bundle is trivial the same proof applies. If $F = 0$ and M is simply connected, the holonomy is trivial, since every loop bounds a disc (up to homotopy, which does not change the transport of a flat connection by Theorem 38.5.10); so transport from a fixed point along any path gives a well-defined nowhere vanishing parallel section, which trivialises the bundle. □

Exercise 38.9.4. Verify that a connection on E induces connections on $E^\vee$ and $E \otimes F$, and that the curvature of $E \otimes F$ is $F_E \otimes 1 + 1 \otimes F_F$.

Solution. This is Proposition 38.3.6(1),(3) and Proposition 38.4.5(2). With vector fields, for $s \otimes s'$: $\nabla_X \nabla_Y (s \otimes s') = \nabla_X \nabla_Y s \otimes s' + \nabla_Y s \otimes \nabla_X s' + \nabla_X s \otimes \nabla_Y s' + s \otimes \nabla_X \nabla_Y s'$; the middle terms are symmetric in X and Y and cancel in $F(X, Y)$, which leaves $F_E(X, Y)s \otimes s' + s \otimes F_F(X, Y)s'$. □

Exercise 38.9.5. Let $E \subseteq M \times \mathbb{R}^N$ be a subbundle with orthogonal projection P and the projection connection $\nabla s = P\, ds$. Show that $P\, dP\, P = 0$ and that the curvature is $F(s) = P\, dP \wedge dP\, s$ for sections s of E, where dP is the matrix of 1-forms of the derivatives of P. Recover Example 38.4.4(2).

Solution. Differentiating $P^2 = P$ gives $dP = dP\, P + P\, dP$; multiplying by P on both sides gives $P\, dP\, P = 2P\, dP\, P$, so $P\, dP\, P = 0$. For a section $s = Ps$, extend ∇ to E-valued 1-forms by $d_\nabla \omega = P\, d\omega$; then $F(s) = P\, d(P\, ds) = P\, dP \wedge ds$, and $ds = d(Ps) = dP\, s + P\, ds$, so $F(s) = P\, dP \wedge dP\, s + P\, dP\, P \wedge ds = P\, dP \wedge dP\, s$. For the sphere, $P = I - xx^T$ and $dP = -dx\, x^T - x\, dx^T$, and evaluating on e_1, e_2 gives the formula of the example. □

Exercise 38.9.6. On the trivial real line bundle over $\mathbb{R}^2$, show that the connections $d + x\, dy$ and $d - y\, dx$ have the same curvature, and find a function $g\colon \mathbb{R}^2 \to \mathbb{R}_{>0}$ such that the frame $e' = ge$ transforms one connection form into the other.

Solution. Both curvatures are $d(x\, dy) = dx \wedge dy = d(-y\, dx)$. In the frame $e' = ge$ the connection form $x\, dy$ becomes $x\, dy + g^{-1} dg = x\, dy + d \log g$ (Proposition 38.3.4(3), with commuting 1×1 matrices). We need $d \log g = -y\, dx - x\, dy = -d(xy)$, so $g = e^{-xy}$. □

Exercise 38.9.7. For a connection ∇ on TM define the *torsion* $T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y]$. Show that T is a tensor, a TM-valued 2-form, and that in coordinates, with $\nabla_{\partial_i} \partial_j = \sum_k \Gamma_{ij}^k \partial_k$, it vanishes if and only if $\Gamma_{ij}^k = \Gamma_{ji}^k$. Show that the projection connection on TS^n is torsion-free.

Solution. $T(fX, Y) = f\nabla_X Y - (Yf)X - f\nabla_Y X - f[X, Y] + (Yf)X = fT(X, Y)$, and T is antisymmetric, so it is C^∞-linear in both arguments and is a tensor by Lemma 38.2.2. On coordinate fields $[\partial_i, \partial_j] = 0$ and $T(\partial_i, \partial_j) = \sum_k (\Gamma_{ij}^k - \Gamma_{ji}^k)\partial_k$. For TS^n, $\nabla_X Y - \nabla_Y X = P(X(Y) - Y(X)) = P([X, Y]) = [X, Y]$, where $X(Y)$ is the derivative in $\mathbb{R}^{n+1}$ of the extended fields and the bracket of fields tangent to S^n is tangent to it. □

Exercise 38.9.8. Show that every vector bundle with connection over an interval $I \subseteq \mathbb{R}$ has a frame of parallel sections, and deduce again that every vector bundle over an interval is trivial.

Solution. Fix $t_0 \in I$ and a basis (v_i) of the fibre over t_0. The parallel sections along the curve $t \mapsto t$ with initial values v_i exist on all of I (Proposition 38.5.3, applied on compact subintervals) and are linearly independent everywhere, since parallel transport is an isomorphism; so they form a global frame. The curvature plays no role: on a 1-dimensional base every connection is flat. □

Exercise 38.9.9.

1. On the trivial line bundle over $\mathbb{R}$ with the connection $\nabla s = ds + a(x)s\,dx$, compute parallel transport along $\gamma(t) = x_0 + t$ and verify Proposition 38.5.4 directly.
2. Using Proposition 38.5.4, give a short proof that a connection whose parallel transports are isometries for a metric on E is metric.

Solution. (1) The parallel sections along γ solve $\dot{u} + a(x_0 + t)u = 0$, so $P_t(w) = e^{-A(t)}w$ with $A(t) = \int_{x_0}^{x_0+t} a(x)\,dx$. Then $P_t^{-1}(s(x_0 + t)) = e^{A(t)}s(x_0 + t)$, whose derivative at $t = 0$ is $a(x_0)s(x_0) + s'(x_0)$, which is $\nabla_{\partial_x} s$ at x_0. (2) Let s, s' be sections, $v \in T_pM$ and γ a curve with $\dot{\gamma}(0) = v$. If each P_t is an isometry, $\langle s, s'\rangle(\gamma(t)) = \langle P_t^{-1}s(\gamma(t)), P_t^{-1}s'(\gamma(t))\rangle$ is the inner product of two curves in the fixed space E_p. Differentiating at $t = 0$ with the product rule and the proposition gives $v\langle s, s'\rangle = \langle \nabla_v s, s'\rangle + \langle s, \nabla_v s'\rangle$, the metric condition. □

Chapter 39

Characteristic Classes

39.1 Introduction

Characteristic classes measure how far a vector bundle is from being trivial. They are cohomology classes $c(E) \in H^\bullet(M)$ attached to vector bundles, natural under pullback ($c(f^*E) = f^*c(E)$) and trivial on trivial bundles; so a bundle with a nonzero characteristic class is not trivial, and in many cases the classes detect more precise obstructions, such as the impossibility of finding a nowhere vanishing section. The first example is the first Chern class of a complex line bundle (Theorem 35.4.2), which classifies line bundles completely.

There are two ways to construct them. The topological way uses the Thom isomorphism or classifying spaces. The differential-geometric way, *Chern–Weil theory*, starts from a connection: its curvature F is a matrix of 2-forms, and applying to it a polynomial that is invariant under conjugation gives a closed form whose cohomology class does not depend on the connection. For the tautological line bundle of $\mathbb{CP}^1$ this was already visible in Chapter 38, where the curvature of a natural connection integrates to $\int \frac{i}{2\pi} F = -1$, a number that cannot change when the connection changes; and for the tangent bundle of a surface the same mechanism is the Gauss–Bonnet theorem, $\int K \, dA = 2\pi\chi$.

Characteristic classes are central to the Hodge conjecture: Chern classes of algebraic vector bundles are Hodge classes, the Chern character appears in the Riemann–Roch theorem, and the class of a divisor is the first Chern class of a line bundle (Chapter 57).

The chapter goes as follows.

1. Chern–Weil theory: invariant polynomials, the closedness and independence of the Chern–Weil forms with complete proofs, Chern forms and Chern classes, computed for line bundles on $\mathbb{CP}^1$ and for the tangent bundle of the sphere (Section 39.2).
2. The axioms for Chern classes, the splitting principle, the Chern character and the Todd class (Section 39.3).
3. The Thom isomorphism and the Euler class in de Rham cohomology (Section 39.4).
4. Euler, Pontryagin and Stiefel–Whitney classes (Section 39.5).
5. The Euler form of a metric connection and Chern's intrinsic proof of the Gauss–Bonnet–Chern theorem: the integral of the Euler form counts the zeros of a section, with signs (Section 39.6).
6. Wu's formula (Section 39.7).

References: [MS74], [BT82], [Hir66], [GH78, Chapter 1].

Prerequisites. Chapter 38 (Connections and Curvature), Chapter 37 (de Rham Cohomology) and Chapter 24 (Cohomology Operations).

39.2 Chern–Weil theory

Definition 39.2.1. An *invariant polynomial* of degree k is a symmetric k-linear form $\tilde{P}$ on $\mathfrak{gl}_r(\mathbb{C}) = M_r(\mathbb{C})$ with $\tilde{P}(gA_1g^{-1}, \dots, gA_kg^{-1}) = \tilde{P}(A_1, \dots, A_k)$ for $g \in \mathrm{GL}_r(\mathbb{C})$; we write $P(A) = \tilde{P}(A, \dots, A)$. Differentiating the invariance along $g = \exp(tB)$ gives

$$\sum_{i=1}^{k} \tilde{P}(A_1, \dots, [B, A_i], \dots, A_k) = 0.$$

Multilinear forms are extended to matrices of differential forms by multiplying the entries with the wedge product; then $\tilde{P}(\alpha_1, \dots, \alpha_k)$ is a form of degree $\sum \deg \alpha_i$, and it is symmetric in the graded sense.

Example 39.2.2.

1. The trace, $P(A) = \operatorname{tr} A$, of degree 1, and more generally $P_k(A) = \operatorname{tr}(A^k)$, with polarisation $\tilde{P}_k(A_1, \dots, A_k) = \frac{1}{k!}\sum_{\sigma \in S_k} \operatorname{tr}(A_{\sigma(1)} \cdots A_{\sigma(k)})$; invariance follows from $\operatorname{tr}(gBg^{-1}) = \operatorname{tr} B$.

2. The determinant, of degree r, and more generally the coefficients $\sigma_k(A)$ of the characteristic polynomial, $\det(I + tA) = \sum_{k=0}^{r} \sigma_k(A) t^k$; they are invariant because $\det(I + tgAg^{-1}) = \det(g(I + tA)g^{-1}) = \det(I + tA)$. In terms of the eigenvalues λ_j of A, $\sigma_k(A)$ is the k-th elementary symmetric function and $\operatorname{tr}(A^k) = \sum_j \lambda_j^k$, the k-th power sum. So $\sigma_1 = \operatorname{tr}$, $\sigma_2(A) = \frac{1}{2}((\operatorname{tr} A)^2 - \operatorname{tr}(A^2))$, and every invariant polynomial is a polynomial in the σ_k (a conjugation invariant function of a diagonalisable matrix is a symmetric function of its eigenvalues, and diagonalisable matrices are dense).

3. On the Lie algebra $\mathfrak{so}(2m)$ of skew-symmetric matrices there is one more invariant for $\mathrm{SO}(2m)$ (not for GL, nor for $\mathrm{O}(2m)$), the *Pfaffian* (Definition 12.4.8), with $\operatorname{Pf}(A)^2 = \det A$ and $\operatorname{Pf}(g^T A g) = \det(g)\operatorname{Pf}(A)$; for $m = 1$, $\operatorname{Pf}\left(\begin{smallmatrix} 0 & a \\ -a & 0 \end{smallmatrix}\right) = a$. It gives the Euler form of Section 39.6.

The key computation in Chern–Weil theory is that invariant polynomials, applied to matrices of forms, turn covariant derivatives into ordinary ones.

Lemma 39.2.3. *Let $\tilde{P}$ be an invariant symmetric k-linear form, θ an $r \times r$ matrix of 1-forms on an open set, and $\alpha_1, \dots, \alpha_k$ matrices of forms of degrees $d_1, \dots, d_k$. Put $D\alpha = d\alpha + \theta \wedge \alpha - (-1)^{\deg \alpha} \alpha \wedge \theta$. Then*

$$d\tilde{P}(\alpha_1, \dots, \alpha_k) = \sum_{i=1}^{k} (-1)^{d_1 + \cdots + d_{i-1}} \tilde{P}(\alpha_1, \dots, D\alpha_i, \dots, \alpha_k).$$

Proof. Since $\tilde{P}$ is multilinear in the matrix entries with constant coefficients, and the entries are multiplied with the wedge product, the Leibniz rule for d gives $d\tilde{P}(\alpha_1, \dots, \alpha_k) = \sum_i (-1)^{d_1 + \cdots + d_{i-1}} \tilde{P}(\alpha_1, \dots, d\alpha_i, \dots, \alpha_k)$. It remains to show that the terms with θ add up to zero. Write $\theta = \sum_j dx^j \theta_j$ with matrices of functions θ_j. Then $\theta \wedge \alpha_i = \sum_j dx^j \wedge \theta_j \alpha_i$ and $(-1)^{d_i} \alpha_i \wedge \theta = \sum_j (-1)^{d_i}(-1)^{d_i} dx^j \wedge \alpha_i \theta_j = \sum_j dx^j \wedge \alpha_i \theta_j$, so the extra term in $D\alpha_i$ is $\sum_j dx^j \wedge [\theta_j, \alpha_i]$. Moving dx^j in front of $\alpha_1, \dots, \alpha_{i-1}$ produces exactly the sign $(-1)^{d_1 + \cdots + d_{i-1}}$, so the sum of the extra terms is $\sum_j dx^j \wedge \sum_i \tilde{P}(\alpha_1, \dots, [\theta_j, \alpha_i], \dots, \alpha_k)$. Each inner sum vanishes: writing each α_i as a sum of products of matrices of functions with scalar forms and using multilinearity, it reduces to the infinitesimal invariance $\sum_i \tilde{P}(A_1, \dots, [B, A_i], \dots, A_k) = 0$ of Definition 39.2.1, since the scalar forms can be moved out of $\tilde{P}$ (with the same signs in every term). □

Theorem 39.2.4 (Chern–Weil). *Let $E \to M$ be a complex vector bundle of rank r, ∇ a connection with curvature F, and P an invariant polynomial of degree k. Then:*

1. *$P(F)$ is a well-defined global $2k$-form (computed in any local frame) and $dP(F) = 0$;*
2. *the class $[P(F)] \in H^{2k}_{\mathrm{dR}}(M)$ does not depend on ∇;*
3. *the resulting map from invariant polynomials to $H^\bullet_{\mathrm{dR}}(M)$ is natural: $[P(F_{f^*\nabla})] = f^*[P(F_\nabla)]$ for smooth $f : N \to M$.*

Proof. (1) Under a change of frame the curvature matrix transforms by conjugation, $F' = g^{-1}Fg$ (Proposition 38.4.3), and P is invariant under conjugation (for matrices of 2-forms, whose entries commute, the invariance of P on matrices of numbers implies it pointwise); so $P(F)$ is the same in all frames and defines a global form. By Lemma 39.2.3 with all $\alpha_i = F$, $dP(F) = \sum_i \tilde{P}(F, \dots, DF, \dots, F)$, and $DF = dF + \theta \wedge F - F \wedge \theta = 0$ by the Bianchi identity (Proposition 38.4.3). So $P(F)$ is closed.

(2) Let ∇_0, ∇_1 be connections and $\alpha = \nabla_1 - \nabla_0 \in \Omega^1(M, \operatorname{End} E)$. The family $\nabla_t = \nabla_0 + t\alpha$ has curvature $F_t = F_0 + t\, d_{\nabla_0}\alpha + t^2 \alpha \wedge \alpha$, so $\frac{d}{dt}F_t = d_{\nabla_t}\alpha$. Hence

$$\frac{d}{dt}P(F_t) = k\,\tilde{P}(d_{\nabla_t}\alpha, F_t, \dots, F_t) = k\,d\,\tilde{P}(\alpha, F_t, \dots, F_t),$$

using the formula of (1) and $d_{\nabla_t}F_t = 0$. Integrating in t from 0 to 1 gives $P(F_1) - P(F_0) = d\left(k\int_0^1 \tilde{P}(\alpha, F_t, \dots, F_t)\,dt\right)$, an exact form.

(3) The pullback connection has curvature f^*F, and P commutes with pullback. □

Definition 39.2.5. The *Chern forms* of (E, ∇) are defined by

$$\det\left(I + \frac{i}{2\pi}F\right) = 1 + c_1(E,\nabla) + c_2(E,\nabla) + \cdots, \qquad c_k(E,\nabla) \in \Omega^{2k}(M),$$

and the *Chern classes* in de Rham cohomology are $c_k(E) = [c_k(E,\nabla)] \in H^{2k}_{\mathrm{dR}}(M)$, independent of ∇ by Theorem 39.2.4. The *total Chern class* is $c(E) = \sum_k c_k(E)$.

Expanding the determinant with Example 39.2.2(2), the first Chern forms are

$$c_1(E,\nabla) = \frac{i}{2\pi}\operatorname{tr}F, \qquad c_2(E,\nabla) = \frac{1}{8\pi^2}\big(\operatorname{tr}(F\wedge F) - \operatorname{tr}F \wedge \operatorname{tr}F\big), \qquad c_r(E,\nabla) = \det\Big(\frac{i}{2\pi}F\Big),$$

using $\sigma_2(A) = \frac{1}{2}((\operatorname{tr}A)^2 - \operatorname{tr}A^2)$ with $A = \frac{i}{2\pi}F$ and $(\frac{i}{2\pi})^2 = -\frac{1}{4\pi^2}$. For a line bundle F is a single 2-form, and $c_1 = \frac{i}{2\pi}F$ is the only Chern form. The factor $\frac{i}{2\pi}$ makes the classes real (for a Hermitian metric and a metric connection F is skew-Hermitian, so $\frac{i}{2\pi}F$ is Hermitian) and integral (Proposition 39.2.7(3)).

Example 39.2.6.

1. *The tautological bundle of $\mathbb{CP}^1$.* With the projection connection, $F = 2i\,dx \wedge dy/(1+|z|^2)^2$ in the chart $z = x + iy$ (Example 38.4.4(3)), so $c_1(\mathcal{O}(-1), \nabla) = -\frac{1}{\pi}\,dx \wedge dy/(1+|z|^2)^2$, and $\int_{\mathbb{CP}^1} c_1 = -1$.
2. *The tangent bundle of S^2.* The rotation by $+\pi/2$ in each tangent plane makes TS^2 a complex line bundle, with i acting as this rotation. The projection connection commutes with the rotation, and its curvature is $dA \otimes J$ with J the rotation by $-\pi/2$ (Example 38.4.4(2)), that is $F = -i\,dA$ as a complex 1×1 matrix. So $c_1(TS^2, \nabla) = \frac{i}{2\pi}(-i)\,dA = \frac{1}{2\pi}dA$ and $\int_{S^2} c_1 = \frac{4\pi}{2\pi} = 2$. This is the degree of $T\mathbb{CP}^1$ found in Example 35.4.5(2), and the Euler characteristic of S^2: the Gauss–Bonnet theorem.

3. *Flat and trivial bundles.* A flat connection has $F = 0$, so all Chern forms vanish. On a trivial bundle the connection $d + A$ has $F = dA + A \wedge A$ and $c_1 = \frac{i}{2\pi}\operatorname{tr}(dA + A \wedge A) = \frac{i}{2\pi} d \operatorname{tr} A$, since $\operatorname{tr}(A \wedge A) = 0$ (the entries of $A \wedge A$ on the diagonal are $\sum_k A^i_k \wedge A^k_i$, and the terms for (i,k) and (k,i) cancel); so c_1 is exact, as it must be.

Proposition 39.2.7. 1. *$c_0 = 1$ and $c_k(E) = 0$ for $k > r$; $c_k(f^*E) = f^*c_k(E)$.*

2. *$c(E \oplus E') = c(E)c(E')$ (the* Whitney formula*).*

3. *For a line bundle L with Hermitian metric and Chern connection, $c_1(L) = [\frac{i}{2\pi}F]$, which is the image in $H^2(M;\mathbb{R})$ of the integral class of Theorem 35.4.2.*

4. *$c_k(\overline{E}) = (-1)^k c_k(E)$, and for the dual $c_k(E^\vee) = (-1)^k c_k(E)$.*

Proof. (1) The coefficients of $\det(I + tA)$ are the elementary symmetric functions of the eigenvalues, so they vanish above degree r; naturality is Theorem 39.2.4(3). (2) Choose connections ∇, ∇'; their direct sum has block diagonal curvature, and the determinant of a block diagonal matrix is the product of the determinants. (4) A conjugate-linear isomorphism $E \cong \overline{E}$ carries a connection to the conjugate one with curvature $\overline{F}$, and $\det(I + \frac{i}{2\pi}\overline{F}) = \overline{\det(I - \frac{i}{2\pi}F)}$; comparing degrees gives the sign, and $E^\vee \cong \overline{E}$ for a Hermitian metric (Proposition 35.3.4).

(3) For a line bundle the curvature is a 2-form F and $c_1(L,\nabla) = \frac{i}{2\pi}F$. To identify the class, represent L by a cocycle (g_{ij}) and choose a partition of unity; the Čech–de Rham comparison ([BT82, Chapter II]) identifies the image of the integral class under $H^2(M;\mathbb{Z}) \to H^2_{\mathrm{dR}}(M)$ with the class of the 2-form obtained from $\frac{1}{2\pi i} d \log g_{ij}$, which is exactly $\frac{i}{2\pi}F$ for a connection built from the same trivialisations; see [GH78, Chapter 1]. □

Example 39.2.8. For the tautological bundle $\mathcal{O}(-1)$ on $\mathbb{CP}^1 = S^2$ with the metric induced from $\mathbb{C}^2$, the Chern connection has curvature $F = -\frac{dz \wedge d\bar z}{(1+|z|^2)^2}$ in the chart z, and $\int_{\mathbb{CP}^1} \frac{i}{2\pi}F = -1$; so $c_1(\mathcal{O}(-1))$ is -1 times the positive generator of $H^2(\mathbb{CP}^1;\mathbb{Z}) \cong \mathbb{Z}$. Consequently $c_1(\mathcal{O}(1)) = h$ is the positive generator, and $c(T\mathbb{CP}^n) = (1+h)^{n+1}$ by the Euler sequence (Example 35.3.10) and the Whitney formula; for instance $c_1(T\mathbb{CP}^n) = (n+1)h$.

39.3 Chern classes: axioms and the splitting principle

Theorem 39.3.1. *There is a unique way to assign to every complex vector bundle E over a paracompact space classes $c_k(E) \in H^{2k}(M;\mathbb{Z})$ such that: $c_0 = 1$; $c_k(f^*E) = f^*c_k(E)$; $c(E \oplus E') = c(E)c(E')$; and $c_1(\mathcal{O}(-1))$ is the negative generator of $H^2(\mathbb{CP}^1;\mathbb{Z})$. Their images in de Rham cohomology are the Chern–Weil classes of Definition 39.2.5.*

This is [MS74, Chapter 14], where the classes are constructed from the Thom isomorphism or from the cohomology of projective bundles; the comparison with Chern–Weil theory is [MS74, Appendix C] or [GH78, Chapter 1].

Theorem 39.3.2 (Splitting principle)**.** *For every complex vector bundle $E \to M$ of rank r there is a space $F(E)$ (the flag bundle) and a map $\pi\colon F(E) \to M$ such that π^* is injective on cohomology and π^*E is a direct sum of line bundles $L_1 \oplus \cdots \oplus L_r$. Writing $x_i = c_1(L_i)$, the* Chern roots*, one has $c(E) = \prod_i (1 + x_i)$, and every symmetric polynomial identity in the x_i gives an identity between characteristic classes.*

This is [BT82, Chapter 21] or [MS74, Chapter 14]: one takes the bundle of full flags in the fibres, whose cohomology is a free module over $H^\bullet(M)$ by the Leray–Hirsch theorem, and the pullback of E acquires a filtration with line bundle quotients, which splits after a further pullback or, over $\mathbb{R}$, by choosing a metric.

Definition 39.3.3. With Chern roots as above, the *Chern character* and the *Todd class* are

$$\operatorname{ch}(E) = \sum_i e^{x_i} = r + c_1 + \tfrac{1}{2}(c_1^2 - 2c_2) + \cdots, \qquad \operatorname{td}(E) = \prod_i \frac{x_i}{1 - e^{-x_i}} = 1 + \tfrac{1}{2}c_1 + \tfrac{1}{12}(c_1^2 + c_2) + \cdots,$$

elements of $H^\bullet(M;\mathbb{Q})$, well defined because they are symmetric power series in the roots, hence polynomials in the Chern classes.

Proposition 39.3.4. $\operatorname{ch}(E \oplus E') = \operatorname{ch}(E) + \operatorname{ch}(E')$ *and* $\operatorname{ch}(E \otimes E') = \operatorname{ch}(E)\operatorname{ch}(E')$*; the Todd class is multiplicative in short exact sequences. Hence* ch *induces a ring homomorphism from the K-theory of* M *to* $H^{\text{even}}(M;\mathbb{Q})$.

Proof. By the splitting principle we may assume $E = \bigoplus L_i$ and $E' = \bigoplus L'_j$; then the Chern roots of $E \oplus E'$ are the x_i and y_j, and those of $E \otimes E'$ are the $x_i + y_j$, because $c_1(L \otimes L') = c_1(L) + c_1(L')$ (Theorem 35.4.2). The statements follow from $e^{x+y} = e^x e^y$ and from multiplicativity of products over the roots. Short exact sequences of bundles split topologically (Proposition 35.3.4), which gives the last statements. □

Example 39.3.5. On $\mathbb{CP}^1$ let $h \in H^2(\mathbb{CP}^1;\mathbb{Z})$ be the positive generator, so $\int h = 1$ and $h^2 = 0$. Then $c_1(\mathcal{O}(k)) = kh$, $\operatorname{ch}(\mathcal{O}(k)) = e^{kh} = 1 + kh$, and $T\mathbb{CP}^1 \cong \mathcal{O}(2)$ has $\operatorname{td}(T\mathbb{CP}^1) = 1 + \frac{1}{2}c_1 = 1 + h$. The Riemann–Roch formula (Remark 39.5.3) then predicts $\chi(\mathbb{CP}^1, \mathcal{O}(k)) = \int (1 + kh)(1 + h) = k + 1$, which is the number of homogeneous polynomials of degree k in two variables when $k \geq 0$, the dimension of the space of holomorphic sections of $\mathcal{O}(k)$ (Chapter 45).

39.4 The Thom isomorphism in de Rham cohomology

In this section $\pi\colon E \to M$ is a smooth real vector bundle of rank n over a manifold M of dimension m. It is *oriented* if it has an atlas of local trivialisations $E|_U \cong U \times \mathbb{R}^n$ whose transition functions take values in $\mathrm{GL}_n^+(\mathbb{R})$, the matrices of positive determinant; such trivialisations are called oriented.

Definition 39.4.1. A form ω on E has *compact vertical support* if $\operatorname{supp}\omega \cap \pi^{-1}(K)$ is compact for every compact $K \subseteq M$. These forms form a subcomplex $\Omega^\bullet_{\mathrm{cv}}(E)$ of $\Omega^\bullet(E)$, whose cohomology is written $H^\bullet_{\mathrm{cv}}(E)$. Restriction to $E|_U$ for $U \subseteq M$ open preserves compact vertical support.

Let $t = (t_1, \ldots, t_n)$ be the fibre coordinates of an oriented trivialisation $E|_U \cong U \times \mathbb{R}^n$. Every $\omega \in \Omega^k_{\mathrm{cv}}(E|_U)$ is uniquely a sum $\omega = \omega' + \omega''$, where ω'' is the part containing $dt_1 \wedge \cdots \wedge dt_n$ and ω' contains fewer than n of the dt_j. Locally on U, in coordinates x of M, $\omega'' = \sum_I f_I(x,t)\,dx^I \wedge dt_1 \wedge \cdots \wedge dt_n$.

Definition 39.4.2. For an oriented bundle, *integration along the fibres* $\pi_*\colon \Omega^k_{\mathrm{cv}}(E) \to \Omega^{k-n}(M)$ is defined in an oriented trivialisation by $\pi_*\omega' = 0$ and

$$\pi_*\Big(\sum_I f_I(x,t)\,dx^I \wedge dt_1 \wedge \cdots \wedge dt_n\Big) = \sum_I \Big(\int_{\mathbb{R}^n} f_I(x,t)\,dt\Big)dx^I.$$

Proposition 39.4.3. *Integration along the fibres is well defined, commutes with restriction to open subsets of* M *and with* d, $d\pi_* = \pi_* d$*, and satisfies the* projection formula $\pi_*(\pi^*\alpha \wedge \omega) = \alpha \wedge \pi_*\omega$ *for* $\alpha \in \Omega^\bullet(M)$ *and* $\omega \in \Omega^\bullet_{\mathrm{cv}}(E)$*. Reversing the orientation of* E *changes the sign of* π_*.

Proof. The integrals converge, since $f_I(x,-)$ has compact support, locally uniformly in x. A change of oriented trivialisation $t' = g(x)t$ with $\det g > 0$ gives $dt'_1 \wedge \cdots \wedge dt'_n = \det g(x)\,dt_1 \wedge \cdots \wedge dt_n$ plus terms containing some dx^i and fewer dt_j; so the parts ω' and

ω'' are the same in both trivialisations, and the integrals agree by the change of variables formula (Theorem 28.11.2), with Jacobian $\det g(x) = |\det g(x)|$. A change of coordinates in M does not affect the fibre integrals. So π_* is well defined, and it clearly commutes with restrictions. For the projection formula, $\pi^*\alpha \wedge \omega'$ has fewer than n factors dt_j, and $\pi^*(dx^K) \wedge f\,dx^I \wedge dt_1 \wedge \cdots \wedge dt_n = f\,dx^K \wedge dx^I \wedge dt_1 \wedge \cdots \wedge dt_n$; by linearity in α over $C^\infty(M)$ this gives the formula. For d: if $\omega = f\,dx^I \wedge dt_J$ with $|J| < n$, the part of $d\omega$ with n factors dt is a sum of terms $\pm\partial_{t_j} f\,dx^I \wedge dt_1 \wedge \cdots \wedge dt_n$, whose fibre integrals vanish because $\int_{\mathbb{R}} \partial_{t_j} f\,dt_j = 0$ for compactly supported f (Theorem 27.8.3); so $\pi_* d\omega = 0 = d\pi_*\omega$. If $\omega = f\,dx^I \wedge dt_1 \wedge \cdots \wedge dt_n$, then $d\omega = \sum_i \partial_{x_i} f\,dx^i \wedge dx^I \wedge dt_1 \wedge \cdots \wedge dt_n$, and $\pi_* d\omega = \sum_i (\int \partial_{x_i} f\,dt)\,dx^i \wedge dx^I = d\pi_*\omega$ by differentiation under the integral sign (Proposition 27.8.9). Reversing the orientation replaces $dt_1 \wedge \cdots \wedge dt_n$ by $-dt_1 \wedge \cdots \wedge dt_n$ in an oriented trivialisation. □

Lemma 39.4.4. *For every manifold N and $n \geq 0$, integration along the fibres of the trivial bundle $N \times \mathbb{R}^n \to N$ induces isomorphisms $H^k_{\mathrm{cv}}(N \times \mathbb{R}^n) \to H^{k-n}(N)$.*

Proof. We treat $n = 1$ in a form that allows induction. Let $N' = N \times \mathbb{R}^{n-1}$ and consider the projection $p\colon N' \times \mathbb{R} \to N'$ with fibre coordinate t, and the subcomplexes $C^\bullet \subseteq \Omega^\bullet(N' \times \mathbb{R})$ and $C'^\bullet \subseteq \Omega^\bullet(N')$ of forms whose supports meet the preimage of every compact subset of N in a compact set. Every $\omega \in C^k$ is uniquely $\omega = \alpha_t + \beta_t \wedge dt$, with α_t, β_t families of forms on N' depending smoothly on t and without dt, and $p_*\omega = \int_{\mathbb{R}} \beta_t\,dt$, which lies in C'^{k-1}. Choose $e \in C^\infty_c(\mathbb{R})$ with $\int e = 1$, let $E(t) = \int_{-\infty}^t e$, and define

$$e_*(\beta) = p^*\beta \wedge e(t)\,dt, \qquad K\omega = \int_{-\infty}^t \beta_u\,du - E(t)\int_{-\infty}^{\infty} \beta_u\,du.$$

Then $e_*\colon C'^{k-1} \to C^k$ commutes with d (as $d(e\,dt) = 0$), $p_*e_* = \mathrm{id}$, and $K\omega$ lies in C^{k-1}: it vanishes for t below the support of ω, and also above it, where $\int_{-\infty}^t \beta_u\,du = \int_{-\infty}^{\infty} \beta_u\,du$ and $E(t) = 1$ (using that the t-supports are uniformly bounded over compact subsets of N). We claim that on C^k,

$$dK\omega - Kd\omega = (-1)^{k-1}(\omega - e_*p_*\omega).$$

If $\omega = \alpha_t$ has no dt, then $K\omega = 0$, $e_*p_*\omega = 0$, and the dt-part of $d\omega$ is $(-1)^k \partial_t\alpha_t \wedge dt$, so $Kd\omega = (-1)^k \int_{-\infty}^t \partial_u\alpha_u\,du = (-1)^k\alpha_t$ (the second term vanishes as $\int_{\mathbb{R}} \partial_u\alpha_u\,du = 0$), which gives the claim. If $\omega = \beta_t \wedge dt$, write $d_{N'}$ for the exterior derivative along N' and $g_t = K\omega$. Then $d\omega = d_{N'}\beta_t \wedge dt$, so $Kd\omega = d_{N'}g_t$ by differentiation under the integral sign, while $dK\omega = d_{N'}g_t + (-1)^{k-1}\partial_t g_t \wedge dt$ with $\partial_t g_t = \beta_t - e(t)\int_{\mathbb{R}} \beta_u\,du$. Hence $dK\omega - Kd\omega = (-1)^{k-1}(\beta_t \wedge dt - e(t)(\int_{\mathbb{R}} \beta_u\,du) \wedge dt) = (-1)^{k-1}(\omega - e_*p_*\omega)$, proving the claim. So p_* and e_* are inverse isomorphisms on cohomology.

For $n = 1$ this is the lemma. For general n, integration over $\mathbb{R}^n$ is the composite of integration over the last coordinate, which maps the complex of forms on $N \times \mathbb{R}^n$ with compact vertical support over N to the corresponding complex on $N \times \mathbb{R}^{n-1}$, and integration over $\mathbb{R}^{n-1}$; by Fubini's theorem (in the form of Proposition 27.8.9) and the orientation convention $dt_1 \wedge \cdots \wedge dt_n = (dt_1 \wedge \cdots \wedge dt_{n-1}) \wedge dt_n$, and induction on n, the lemma follows. □

Theorem 39.4.5 (Thom isomorphism)**.** *Let $\pi\colon E \to M$ be an oriented vector bundle of rank n over a manifold with a finite good cover (for instance a compact manifold). Then $\pi_*\colon H^k_{\mathrm{cv}}(E) \to H^{k-n}(M)$ is an isomorphism for every k. There is a unique class $\Phi \in H^n_{\mathrm{cv}}(E)$ with $\pi_*\Phi = 1$, the* Thom class*, and the inverse isomorphism is $\alpha \mapsto \pi^*\alpha \wedge \Phi$.*

Proof. For open $U, V \subseteq M$ and a partition of unity $\{\rho_U, \rho_V\}$ subordinate to $\{U, V\}$ on $U \cup V$, the sequence $0 \to \Omega_{\mathrm{cv}}(E|_{U\cup V}) \to \Omega_{\mathrm{cv}}(E|_U) \oplus \Omega_{\mathrm{cv}}(E|_V) \to \Omega_{\mathrm{cv}}(E|_{U\cap V}) \to 0$ is exact, exactly as in Theorem 37.2.8, using the functions $\pi^*\rho_U$, $\pi^*\rho_V$: multiplying a form on $E|_{U\cap V}$ with

compact vertical support by $\pi^*\rho_V$ and extending by zero gives a form on $E|_U$ with compact vertical support, because a compact $K \subseteq U$ meets $\operatorname{supp}\rho_V$ in a compact subset of $U \cap V$. By Proposition 39.4.3, π_* maps this sequence to the Mayer–Vietoris sequence of U, V in de Rham cohomology, commuting with restrictions, and it commutes with the connecting maps, which are given on both sides by $\omega \mapsto d(\pi^*\rho_V\,\omega)$ and $\beta \mapsto d(\rho_V\beta)$, since $\pi_* d(\pi^*\rho_V\,\omega) = d(\rho_V \pi_*\omega)$ by the projection formula. So the long exact sequences form a commutative ladder.

Choose a finite good cover $\{U_1, \dots, U_N\}$ of M subordinate to a cover by trivialising open sets (possible by [BT82, Chapter I, Section 5]): all nonempty finite intersections of the U_i are diffeomorphic to $\mathbb{R}^m$ and E is trivial over each. We show by induction on k that π_* is an isomorphism over any union of k open sets forming such a good cover. For $k = 1$ this is Lemma 39.4.4. For the induction step write the union as $W \cup U$ with W a union of $k-1$ of the sets; then $W \cap U$ is the union of the $k-1$ sets $U_i \cap U$, which again form such a cover. By induction π_* is an isomorphism over W, U and $W \cap U$, and by the five lemma (Lemma 15.4.2) over $W \cup U$.

The class $\Phi = \pi_*^{-1}(1)$ is unique. By the projection formula $\pi_*(\pi^*\alpha \wedge \Phi) = \alpha \wedge \pi_*\Phi = \alpha$, so $\alpha \mapsto \pi^*\alpha \wedge \Phi$ is the inverse of π_*. □

Definition 39.4.6. For E and M as in Theorem 39.4.5, the *Euler class* of E in de Rham cohomology is $e(E) = s^*\Phi \in H^n(M)$, where $s\colon M \to E$ is the zero section and Φ is viewed as a class in $H^n(E)$.

Proposition 39.4.7. *Let M be compact and E, F oriented bundles over M of ranks n, n'.*

1. *$e(f^*E) = f^*e(E)$ for a smooth map $f\colon N \to M$ of compact manifolds, and reversing the orientation of E changes the sign of $e(E)$.*
2. *If n is odd, $e(E) = 0$.*
3. *If E has a nowhere vanishing section, $e(E) = 0$.*
4. *$e(E \oplus F) = e(E) \wedge e(F)$, with $E \oplus F$ oriented by an oriented frame of E followed by one of F.*

Proof. We use that a closed form $\phi \in \Omega^n_{\mathrm{cv}}(E)$ represents Φ if and only if $\pi_*\phi = 1$: $\pi_*\phi$ is a closed 0-form, that is, locally constant, and it represents $\pi_*[\phi]$. (1) The bundle map $\tilde f\colon f^*E \to E$ is an orientation-preserving isomorphism on fibres, so $\tilde f^*\phi$ has compact vertical support and, computing in pulled back trivialisations, fibre integral $f^*\pi_*\phi = 1$; hence it represents the Thom class of f^*E, and $s_N^*\tilde f^*\phi = f^*s^*\phi$. The orientation statement follows from the last statement of Proposition 39.4.3. (2) Let $a(v) = -v$. In oriented trivialisations the change of variables $t \mapsto -t$ has Jacobian $(-1)^n = -1$, so $\pi_* a^*\phi = -\pi_*\phi = -1$, and $a^*\phi$ represents $-\Phi$. As $a \circ s = s$, $e(E) = [s^*\phi] = [s^*a^*\phi] = -e(E)$. (3) Choose a metric on E and let σ be a section with $|\sigma| \geq c > 0$ (by compactness). For $\lambda > 0$ let $m_\lambda(v) = \lambda v$; then $m_\lambda^*\phi$ has fibre integral 1 (Jacobian $\lambda^n > 0$), so it represents Φ, and its support is $\lambda^{-1}\operatorname{supp}\phi$. As $\operatorname{supp}\phi$ lies in some $\{|v| \leq R\}$, for $\lambda > R/c$ the form $m_\lambda^*\phi$ vanishes along $\sigma(M)$, so $\sigma^* m_\lambda^*\phi = 0$. The sections s and σ are homotopic through $t\sigma$, so $e(E) = s^*[m_\lambda^*\phi] = \sigma^*[m_\lambda^*\phi] = 0$ by homotopy invariance (Theorem 37.2.4). (4) Let $p_1\colon E \oplus F \to E$ and $p_2\colon E \oplus F \to F$ be the projections and ϕ, ψ Thom forms. Then $p_1^*\phi \wedge p_2^*\psi$ has compact vertical support, and in product trivialisations with fibre coordinates (t, u) its component containing $dt_1 \wedge \cdots \wedge dt_n \wedge du_1 \wedge \cdots \wedge du_{n'}$ is the product of the components of ϕ and ψ of top vertical degree, so by Fubini its fibre integral is $\pi_*\phi \cdot \pi_*\psi = 1$. Hence it represents the Thom class of $E \oplus F$, and pulling back by the zero section, which is (s_E, s_F), gives $e(E \oplus F) = e(E) \wedge e(F)$. □

Remark 39.4.8. The Thom class links Poincaré duality and characteristic classes. If $S \subseteq M$ is a compact oriented submanifold of a compact oriented manifold, with normal bundle

ν identified with a tubular neighbourhood of S, the Thom form of ν, extended by zero, represents the Poincaré dual of S in M; restricted to S it gives $e(\nu)$, the *self-intersection* of S. Applied to the diagonal of $M \times M$, whose normal bundle is TM, this yields $\int_M e(TM) = \chi(M)$, part (3) of Theorem 39.5.1 [BT82].

39.5 Euler, Pontryagin and Stiefel–Whitney classes

Theorem 39.5.1. *Let $E \to M$ be an oriented real vector bundle of rank n over a compact manifold.*

1. *(Thom isomorphism) $H^k(M;\mathbb{Z}) \cong H^{k+n}(E, E \setminus 0;\mathbb{Z})$, given by cup product with a* Thom class *$u \in H^n(E, E \setminus 0;\mathbb{Z})$ restricting to a generator on each fibre.*

2. *The* Euler class *$e(E) \in H^n(M;\mathbb{Z})$ is the restriction of u to the zero section. It is natural, satisfies $e(E \oplus E') = e(E)e(E')$, vanishes if E has a nowhere vanishing section, and for a complex bundle of rank r one has $e(E) = c_r(E)$.*

3. *(Gauss–Bonnet–Chern and Poincaré–Hopf) For a compact oriented n-manifold, $\int_M e(TM) = \chi(M)$, and the sum of the indices of the zeros of a vector field with isolated zeros equals $\chi(M)$.*

These are [MS74, Chapters 9–11] and [BT82, Chapter 6]. With real coefficients, (1) is Theorem 39.4.5 (whose proof works over any manifold with a finite good cover), and the de Rham image of the Euler class is the class of Definition 39.4.6, whose properties were proved in Proposition 39.4.7. In the Chern–Weil picture, the Pfaffian of the curvature of a metric connection, divided by $(2\pi)^{n/2}$, is a closed form whose integral over M is computed in Section 39.6 by Chern's intrinsic argument: it is the sum of the indices of any vector field with isolated zeros, hence $\chi(M)$ by the Poincaré–Hopf theorem in (3).

Definition 39.5.2. For a real vector bundle E, the *Pontryagin classes* are $p_k(E) = (-1)^k c_{2k}(E \otimes \mathbb{C}) \in H^{4k}(M;\mathbb{Z})$, and the *Stiefel–Whitney classes* $w_k(E) \in H^k(M;\mathbb{Z}/2\mathbb{Z})$ are the mod 2 characteristic classes determined by the same axioms with $\mathbb{Z}/2$ coefficients; $w_1(E) = 0$ if and only if E is orientable.

Remark 39.5.3. Hirzebruch's signature theorem expresses the signature of a compact oriented $4m$-manifold (Definition 23.6.5) as $\int_M L(p_1, \dots, p_m)$ for a universal polynomial in the Pontryagin classes, and the Riemann–Roch–Hirzebruch theorem computes the holomorphic Euler characteristic of a holomorphic bundle on a compact complex manifold as $\int_X \operatorname{ch}(E)\operatorname{td}(TX)$ [Hir66]. Both are special cases of the Atiyah–Singer index theorem (Chapter 64), and both are used in Chapter 50 and Chapter 62.

Remark 39.5.4. Isomorphism classes of rank r complex bundles over a paracompact space M correspond to homotopy classes of maps $M \to \mathrm{BU}(r)$, where $\mathrm{BU}(r)$ is the infinite Grassmannian $\mathrm{Gr}_r(\mathbb{C}^\infty)$, and $H^\bullet(\mathrm{BU}(r);\mathbb{Z}) = \mathbb{Z}[c_1, \dots, c_r]$ is a polynomial ring on the universal Chern classes; every characteristic class is a pullback of a universal one [MS74, Chapter 14]. For line bundles, $\mathrm{BU}(1) = \mathbb{CP}^\infty = K(\mathbb{Z}, 2)$, which recovers Theorem 35.4.2 and Theorem 24.2.4.

Example 39.5.5. On a smooth projective complex variety, the Chern classes of an algebraic vector bundle are algebraic cycle classes, hence Hodge classes; for a line bundle, $c_1(L)$ is the class of a divisor (Chapter 57), and the Lefschetz theorem on $(1,1)$-classes (Chapter 69) says conversely that every integral $(1,1)$-class is of this form. This is the case $p = 1$ of the Hodge conjecture, and it is the only case known in general.

39.6 The Euler form and the Gauss–Bonnet–Chern theorem

Chern–Weil theory for complex bundles uses polynomials that are invariant under all of $\mathrm{GL}_r(\mathbb{C})$. A real oriented vector bundle with a metric has more structure: in oriented orthonormal frames the curvature of a metric connection is a skew-symmetric matrix of 2-forms, and the changes of frame take values in $\mathrm{SO}(n)$. Skew-symmetric matrices have an invariant that is not a polynomial in the coefficients of the characteristic polynomial, the Pfaffian (Definition 12.4.8), a square root of the determinant. Applied to the curvature it gives a closed form, the *Euler form*. This section proves, following Chern [Che44], that when the rank of the bundle equals the dimension of the base, the integral of the Euler form counts the zeros of any section, with signs:

$$\int_M e(\nabla) = \sum_{s(z)=0} \operatorname{ind}_z(s).$$

For the tangent bundle the right side is the Euler characteristic by the Poincaré–Hopf theorem, and the formula becomes the *Gauss–Bonnet–Chern theorem* $\int_M e(TM,\nabla) = \chi(M)$, the higher-dimensional Gauss–Bonnet theorem. Chern's proof is *intrinsic*: earlier proofs, by Allendoerfer, Fenchel and Weil, used isometric embeddings of M, or of pieces of it, into Euclidean spaces, while Chern only computes on the unit sphere bundle of E, where the pullback of the Euler form becomes exact, with an explicit primitive. The surface case, in which the whole argument fits in a few lines, is worked out in Example 39.6.11, and it may help to read it first.

On a first reading one can read the definitions and the statements, the example of surfaces, and Corollary 39.6.13; the transgression computation (Lemma 39.6.9 and Proposition 39.6.10) is not used elsewhere.

Throughout, $E \to M$ is an oriented real vector bundle of rank n with a metric $\langle\ ,\ \rangle$ on its fibres, and ∇ is a metric connection (Proposition 38.6.2). Near every point there are *oriented orthonormal frames* $e = (e_1, \dots, e_n)$, obtained from any oriented frame by the Gram–Schmidt process. Since such a frame is orthonormal we write its connection form and curvature with lower indices, $\theta_{ij} = \theta^i_j$ and $F_{ij} = F^i_j$, so that

$$\nabla e_j = \sum_i \theta_{ij} \otimes e_i, \qquad F_{ij} = d\theta_{ij} + \sum_k \theta_{ik} \wedge \theta_{kj}, \qquad dF = F \wedge \theta - \theta \wedge F$$

(Propositions 38.3.4 and 38.4.3). Both matrices are skew-symmetric, $\theta_{ji} = -\theta_{ij}$ and $F_{ji} = -F_{ij}$, by Proposition 38.6.2. Another oriented orthonormal frame is $e' = eg$ with $g\colon U \to \mathrm{SO}(n)$, and then $\theta' = g^{-1}\theta g + g^{-1}dg$ and $F' = g^{-1}Fg = g^TFg$.

Lemma 39.6.1. *Let $x \in M$ and let $(b_1, \dots, b_n)$ be an oriented orthonormal basis of E_x. There is an oriented orthonormal frame e on a neighbourhood of x with $e_i(x) = b_i$ and whose connection form vanishes at x, $\theta(x) = 0$.*

Proof. Let f be an oriented orthonormal frame on a coordinate neighbourhood of x, with coordinates $x^1, \dots, x^d$ vanishing at x, and connection form $\theta_f = \sum_k \theta_k\, dx^k$ with skew-symmetric matrices of functions θ_k. Let $A_k = \theta_k(x)$ and let $h \in \mathrm{SO}(n)$ be the matrix with $f(x)h = (b_1, \dots, b_n)$. Put $g = \exp\bigl(-\sum_k x^k A_k\bigr)h$. The exponential of a skew-symmetric matrix B is orthogonal, as $\exp(B)^T = \exp(-B) = \exp(B)^{-1}$, with determinant $e^{\operatorname{tr} B} = 1$; so g takes values in $\mathrm{SO}(n)$ and $e = fg$ is an oriented orthonormal frame with $e(x) = f(x)h = b$. At x we have $g(x) = h$ and $dg(x) = -\sum_k A_k\, dx^k\, h = -\theta_f(x)h$, so $\theta_e(x) = h^{-1}\theta_f(x)h + h^{-1}dg(x) = 0$. □

Such a frame is the analogue of normal coordinates: at the single point x the connection looks like the trivial one, while the curvature, which is a tensor, cannot be removed. It simplifies computations of quantities that do not depend on the frame: the second Bianchi identity says that $dF(x) = 0$ in such a frame.

Let $n = 2m$ be even. Since 2-forms commute, the Pfaffian can be applied to the skew-symmetric matrix of 2-forms F, using the formula of Proposition 12.4.9(1):

$$\mathrm{Pf}(F) = \frac{1}{2^m m!} \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma)\, F_{\sigma(1)\sigma(2)} \wedge F_{\sigma(3)\sigma(4)} \wedge \cdots \wedge F_{\sigma(n-1)\sigma(n)}.$$

Definition 39.6.2. For $n = 2m$, the *Euler form* of (E, ∇) is the n-form

$$e(\nabla) = \mathrm{Pf}\Big(\frac{F}{2\pi}\Big) = \frac{1}{(2\pi)^m}\mathrm{Pf}(F) \in \Omega^n(M),$$

computed in any oriented orthonormal frame. For $n = 2$, $e(\nabla) = \frac{1}{2\pi}F_{12}$; for $n = 4$, $e(\nabla) = \frac{1}{4\pi^2}(F_{12} \wedge F_{34} - F_{13} \wedge F_{24} + F_{14} \wedge F_{23})$.

Proposition 39.6.3. *Let $n = 2m$.*

1. *$e(\nabla)$ does not depend on the oriented orthonormal frame, so it is a global n-form, and reversing the orientation of E changes its sign.*
2. *$e(\nabla)$ is closed.*
3. *Its class in $H^n_{\mathrm{dR}}(M)$ does not depend on the metric connection ∇.*
4. *It is natural: $e(f^*\nabla) = f^*e(\nabla)$ for a smooth map $f : N \to M$.*
5. *If E is the underlying oriented real bundle of a complex line bundle L with a Hermitian metric h, with the metric $\mathrm{Re}\, h$, and ∇ is a metric connection on L, then $e(\nabla) = c_1(L, \nabla)$.*

Proof. (1) For $e' = eg$ with g in $\mathrm{SO}(n)$, $\mathrm{Pf}(F') = \mathrm{Pf}(g^T F g) = \det(g)\mathrm{Pf}(F) = \mathrm{Pf}(F)$ by Proposition 12.4.9(2), which applies to matrices with entries in the commutative algebra of even forms at a point. Reversing the orientation replaces oriented frames by frames eg with $\det g = -1$.

(2) $\mathrm{Pf}(F)$ is a polynomial with constant coefficients in the 2-forms F_{ij}, which commute, so by the Leibniz rule $d\,\mathrm{Pf}(F)$ is a sum of terms each containing one factor dF_{ij}. At a point x compute in a frame as in Lemma 39.6.1: there $dF(x) = (F \wedge \theta - \theta \wedge F)(x) = 0$, so $d\,\mathrm{Pf}(F)$ vanishes at x. As x is arbitrary and $\mathrm{Pf}(F)$ does not depend on the frame, $d\,\mathrm{Pf}(F) = 0$.

(3) Let $\widetilde{\mathrm{Pf}}$ be the polarisation of Pf, the symmetric m-linear form on skew-symmetric matrices with $\widetilde{\mathrm{Pf}}(A, \dots, A) = \mathrm{Pf}(A)$; it is given by a universal formula in terms of Pf (for instance $m!\,\widetilde{\mathrm{Pf}}(A_1, \dots, A_m)$ is the coefficient of $t_1 t_2 \cdots t_m$ in $\mathrm{Pf}(t_1A_1 + \cdots + t_mA_m)$), so it is invariant under $A_i \mapsto g^T A_i g$ for $g \in \mathrm{SO}(n)$. Differentiating this invariance along $g = \exp(tB)$ with B skew-symmetric, for which $\frac{d}{dt}\big|_{t=0} \exp(tB)^T A \exp(tB) = AB - BA$, gives $\sum_i \widetilde{\mathrm{Pf}}(A_1, \dots, [B, A_i], \dots, A_m) = 0$ for skew-symmetric B and A_i. The proof of Lemma 39.2.3 uses only this infinitesimal invariance, for B the components of the connection form, and these are skew-symmetric for a metric connection in an orthonormal frame. So that lemma holds for $\widetilde{\mathrm{Pf}}$ and skew-symmetric matrices of forms, and the proof of Theorem 39.2.4(2) applies word for word: for metric connections ∇_0, ∇_1 the connections $\nabla_t = \nabla_0 + t(\nabla_1 - \nabla_0)$ are metric, the matrix of $\alpha = \nabla_1 - \nabla_0$ in an orthonormal frame is skew-symmetric, and $\mathrm{Pf}(F_1) - \mathrm{Pf}(F_0) = d\big(m \int_0^1 \widetilde{\mathrm{Pf}}(\alpha, F_t, \dots, F_t)\, dt\big)$, where the form in brackets does not depend on the oriented orthonormal frame by the invariance of $\widetilde{\mathrm{Pf}}$.

(4) The pullback of an oriented orthonormal frame is one for f^*E, with connection form $f^*\theta$ and curvature f^*F.

(5) Let s be a local frame of L with $h(s,s) = 1$. Then $(e_1, e_2) = (s, is)$ is an oriented orthonormal frame of E for $\operatorname{Re} h$. The connection form of ∇ on L in the frame s is $i\omega$ for a real 1-form ω (it is skew-Hermitian), so $\nabla e_1 = i\omega \otimes s = \omega \otimes e_2$ and $\nabla e_2 = i\nabla s = -\omega \otimes e_1$. Hence $\theta_{21} = \omega$, $\theta_{12} = -\omega$, and $F_{12} = d\theta_{12} = -d\omega$ (for $n = 2$ the term $\theta \wedge \theta$ has zero off-diagonal entries). The curvature of L is $d(i\omega) = i\, d\omega$, so $c_1(L, \nabla) = \frac{i}{2\pi}\, i\, d\omega = -\frac{1}{2\pi} d\omega = \frac{1}{2\pi} F_{12} = e(\nabla)$. □

By (5) and Example 39.2.6(2), for the round sphere S^2 the Euler form of the projection connection on TS^2 is $\frac{1}{2\pi} dA$, with integral $2 = \chi(S^2)$. The rest of the section shows that this is no accident.

Next we define the index of an isolated zero of a section, which generalises the winding number of a vector field in the plane around a zero.

Write σ for the $(n-1)$-form

$$\sigma = \sum_{i=1}^{n} (-1)^{i-1} y^i \, dy^1 \wedge \cdots \wedge \widehat{dy^i} \wedge \cdots \wedge dy^n$$

on $\mathbb{R}^n$, and also for its restriction to the unit sphere S^{n-1}, oriented as the boundary of the unit ball. At y it is the interior product of $dy^1 \wedge \cdots \wedge dy^n$ with the position vector y: $\sigma_y(v_2, \ldots, v_n) = \det(y, v_2, \ldots, v_n)$. Hence σ restricts to the area form of S^{n-1}, and for a linear map $h \in \mathrm{O}(n)$, which maps S^{n-1} to itself, $h^*\sigma = \det(h)\, \sigma$, since $\det(hy, hv_2, \ldots, hv_n) = \det(h) \det(y, v_2, \ldots, v_n)$. Write $|S^{n-1}| = \int_{S^{n-1}} \sigma$ for the area of the unit sphere.

Lemma 39.6.4. *$|S^{n-1}| = n\,|B^n|$, where $|B^n|$ is the volume of the unit ball, and $|B^{2m}| = \pi^m/m!$. Hence $|S^{2m-1}| = 2\pi^m/(m-1)!$; for instance $|S^1| = 2\pi$ and $|S^3| = 2\pi^2$.*

Proof. On $\mathbb{R}^n$, $d\sigma = \sum_i (-1)^{i-1} dy^i \wedge dy^1 \wedge \cdots \wedge \widehat{dy^i} \wedge \cdots \wedge dy^n = n\, dy^1 \wedge \cdots \wedge dy^n$, so Stokes' theorem (Theorem 36.4.8) on the unit ball gives $\int_{S^{n-1}} \sigma = n|B^n|$. For the volumes, slice $B^{n+2} \subseteq \mathbb{R}^2 \times \mathbb{R}^n$ by the first two coordinates: for $w \in \mathbb{R}^2$ with $|w| < 1$ the slice is a ball of radius $(1 - |w|^2)^{1/2}$ in $\mathbb{R}^n$, of volume $|B^n|(1-|w|^2)^{n/2}$ (a linear change of variables, Lemma 28.11.1). By Fubini's theorem (Theorem 28.10.3) and polar coordinates in the plane (Example 28.11.3),

$$|B^{n+2}| = |B^n| \int_0^{2\pi} \int_0^1 (1 - r^2)^{n/2} r \, dr \, d\varphi = |B^n| \, \frac{2\pi}{n+2}.$$

Starting from $|B^2| = \pi$, induction gives $|B^{2m}| = \pi^m/m!$, and then $|S^{2m-1}| = 2m\,\pi^m/m! = 2\pi^m/(m-1)!$. □

Now let M be oriented of dimension n, equal to the rank of E, and let s be a section of E with an isolated zero z. Choose a positively oriented chart $\varphi\colon V \to \mathbb{R}^n$ with $\varphi(z) = 0$ and a positively oriented frame e of E on V, not necessarily orthonormal, and write $s = \sum_i s^i e_i$. For small $\varepsilon > 0$ let $\bar{B}_\varepsilon = \varphi^{-1}(\{|u| \le \varepsilon\})$, let B_ε be its interior, assume z is the only zero of s in $\bar{B}_\varepsilon$, and let $S_\varepsilon = \partial \bar{B}_\varepsilon$, oriented as the boundary of $\bar{B}_\varepsilon$. On $V \setminus \{z\}$ define $f = (s^1, \ldots, s^n)/|(s^1, \ldots, s^n)|$, with values in S^{n-1}.

Definition 39.6.5. The *index* of s at the isolated zero z is

$$\operatorname{ind}_z(s) = \frac{1}{|S^{n-1}|} \int_{S_\varepsilon} f^*\sigma.$$

For $E = TM$ this is the index of a vector field at an isolated zero.

Lemma 39.6.6. *1. The index does not depend on ε nor on the positively oriented frame e.*

2. If $V \subseteq \mathbb{R}^n$, $E = V \times \mathbb{R}^n$ with the standard frame and $s(u) = hu$ near $z = 0$ for some $h \in \mathrm{O}(n)$, then $\mathrm{ind}_0(s) = \det h$. In particular a source, $s(u) = u$, has index 1, and a sink, $s(u) = -u$, has index $(-1)^n$.

Proof. Two facts are used. First, $f^*\sigma$ is closed on $V \setminus \{z\}$, being the pullback of a form of top degree on S^{n-1}. Second, if $f_t \colon S_\varepsilon \to S^{n-1}$, $t \in [0,1]$, is a smooth homotopy, then $\int_{S_\varepsilon} f_0^*\sigma = \int_{S_\varepsilon} f_1^*\sigma$: by Theorem 37.2.4 the closed forms $f_0^*\sigma$ and $f_1^*\sigma$ differ by an exact form, whose integral over the compact manifold S_ε without boundary vanishes by Stokes' theorem.

(1) For $\varepsilon' < \varepsilon$, the region $\varepsilon' \le |\varphi| \le \varepsilon$ is a compact manifold with boundary S_ε and $S_{\varepsilon'}$ with the opposite orientation, so Stokes' theorem and $d(f^*\sigma) = 0$ give the same integral over both spheres. If $e' = eg$ is another positively oriented frame, with g taking values in invertible matrices of positive determinant, then the components are related by $(s^i) = g\,(s'^i)$. Suppose first that both frames are orthonormal, so that g takes values in $\mathrm{SO}(n)$. Then $f' = g^T f$, and $f_t(q) = g(\varphi^{-1}(t\varphi(q)))^T f(q)$, $t \in [0,1]$, is a smooth homotopy on S_ε from $f_0 = g(z)^T \circ f$ to $f_1 = f'$. Since $g(z)^T \in \mathrm{SO}(n)$, $f_0^*\sigma = f^*(g(z)^{T*}\sigma) = f^*\sigma$, and the claim follows. In general, apply the Gram–Schmidt process to e: it gives an oriented orthonormal frame $e' = eg$ with g upper triangular with positive diagonal entries, so $(s^i) = g\,(s'^i)$. The matrices $g_t = (1-t)I + tg$ are again upper triangular with positive diagonal, hence invertible, and $f_t = g_t(s'^i)/|g_t(s'^i)|$ is a homotopy from the map f' computed in e' to the map f computed in e.

(2) Here $f(u) = hu/|u|$, and on S_ε this is h composed with the dilation $u \mapsto u/\varepsilon$ onto S^{n-1}, which preserves orientations. So $\int_{S_\varepsilon} f^*\sigma = \int_{S^{n-1}} h^*\sigma = \det(h)|S^{n-1}|$. The sink is $h = -I$, of determinant $(-1)^n$. □

The index is the degree of $f \colon S_\varepsilon \to S^{n-1}$ (Exercise 37.5.3), so it is an integer, and it does not depend on the chart either [Mil97, Chapter 6]; neither fact is needed below, where any chart may be used at each zero. For a vector field in the plane, f is the direction of the field along a small circle, and the index is the number of turns it makes: for $X(z) = z^k$ (in complex notation) it is k, and for $X(z) = \bar{z}^k$ it is $-k$ (Exercise 39.8.9).

We now come to Chern's construction, which takes place on the unit sphere bundle of E.

The *unit sphere bundle* of E is $SE = \{v \in E : |v| = 1\}$, with projection $\pi \colon SE \to M$. Over an open set U with an oriented orthonormal frame e, the map $U \times S^{n-1} \to SE|_U$, $(x, y) \mapsto \sum_i y^i e_i(x)$, is a bijection; for another such frame $e' = eg$ the transition map is $(x, y) \mapsto (x, g(x)^T y)$, a diffeomorphism. So SE is a manifold of dimension $\dim M + n - 1$ (a closed submanifold of E), π is a submersion, and the fibre S_xE is the unit sphere of E_x. On $SE|_U$ we have the functions $y^1, \dots, y^n$, with $\sum_i (y^i)^2 = 1$, and we define the 1-forms

$$\eta^i = dy^i + \sum_j \theta_{ij}\, y^j, \qquad i = 1, \dots, n,$$

where, here and below, forms on M are pulled back to SE by π without changing the notation. The meaning of η is the following. On SE the bundle π^*E has the *tautological section* $v \mapsto v$, which is $\sum_i y^i\, e_i$ in the frame π^*e, and η^i are the components of its covariant derivative for the pullback connection (Proposition 38.3.4(2)): η measures how the point v of the sphere bundle moves compared with parallel transport.

Lemma 39.6.7. *1. $\sum_i y^i \eta^i = 0$; in particular $\eta^1 \wedge \cdots \wedge \eta^n = 0$.*

2. For another oriented orthonormal frame $e' = eg$, $y' = g^T y$ and $\eta' = g^T \eta$ (as columns, g being composed with π).

3. $d\eta^i = \sum_j F_{ij}\, y^j - \sum_j \theta_{ij} \wedge \eta^j$.

4. *On a fibre S_xE, identified with S^{n-1} by the frame at x, η^i restricts to dy^i, and F and θ restrict to 0.*

Proof. (1) $\sum_i y^i\eta^i = \frac{1}{2}d\big(\sum_i (y^i)^2\big) + \sum_{i,j}\theta_{ij}y^iy^j = 0$, the first term because $\sum_i (y^i)^2 = 1$ and the second because θ is skew-symmetric. At each point $y \neq 0$, so the η^i are linearly dependent and their wedge product vanishes (Proposition 12.3.3(4)). (2) From $\sum_i y^i e_i = \sum_j y'^j e'_j = \sum_{i,j} y'^j g_{ij} e_i$ we get $y = gy'$, so $y' = g^{-1}y = g^T y$. Then, in vector notation, $\eta' = dy' + \theta' y' = d(g^{-1})y + g^{-1}dy + (g^{-1}\theta g + g^{-1}dg)g^{-1}y = g^{-1}(dy + \theta y)$, because $d(g^{-1}) = -g^{-1}dg\, g^{-1}$. (3) $d\eta^i = \sum_j d\theta_{ij}\, y^j - \sum_j \theta_{ij}\wedge dy^j$. Substitute $d\theta_{ij} = F_{ij} - \sum_k \theta_{ik}\wedge\theta_{kj}$ and $dy^j = \eta^j - \sum_k \theta_{jk}y^k$: the two terms $\mp\sum_{j,k}\theta_{ik}\wedge\theta_{kj}\, y^j$ cancel, leaving the formula. (4) The composite of the inclusion of S_xE with π is constant, so pulled back forms from M of positive degree restrict to zero. □

Now we build forms on SE out of y, η and F, contracting all indices with the sign symbol $\varepsilon_{i_1\dots i_n}$, which is the sign of the permutation $(i_1,\dots,i_n)$ of $(1,\dots,n)$, and 0 if two indices are equal. Let $n = 2m$, and for $0 \le k \le m-1$ put

$$\Phi_k = \sum \varepsilon_{i_1\dots i_n}\, y^{i_1}\,\eta^{i_2}\wedge\cdots\wedge\eta^{i_{n-2k}}\wedge F_{i_{n-2k+1}i_{n-2k+2}}\wedge\cdots\wedge F_{i_{n-1}i_n},$$

with $n-2k-1$ factors η and k factors F, a form of degree $n-1$; and for $-1 \le k \le m-1$ put

$$\Psi_k = \sum \varepsilon_{i_1\dots i_n}\,\eta^{i_1}\wedge\cdots\wedge\eta^{i_{n-2k-2}}\wedge F_{i_{n-2k-1}i_{n-2k}}\wedge\cdots\wedge F_{i_{n-1}i_n},$$

with $n-2k-2$ factors η and $k+1$ factors F, a form of degree n. All sums run over all $(i_1,\dots,i_n)$.

Lemma 39.6.8. *1. The forms Φ_k and Ψ_k do not depend on the oriented orthonormal frame, so they are global forms on SE.*

2. *$\Psi_{-1} = 0$ and $\Psi_{m-1} = 2^m m!\,\pi^*\mathrm{Pf}(F) = (4\pi)^m m!\,\pi^* e(\nabla)$.*

3. *On a fibre $S_xE \cong S^{n-1}$, Φ_k restricts to 0 for $k \ge 1$, and Φ_0 restricts to $(n-1)!\,\sigma$.*

Proof. (1) By Lemma 39.6.7(2), in the frame eg we have $y'^i = \sum_a g_{ai}y^a$, $\eta'^i = \sum_a g_{ai}\eta^a$ and $F'_{ij} = \sum_{a,b} g_{ai}g_{bj}F_{ab}$. Substituting, every index i_l in Φ'_k becomes a sum over an index a_l with a factor $g_{a_l i_l}$, and the order of the factors is unchanged; so Φ'_k is the same expression as Φ_k with $\varepsilon_{a_1\dots a_n}$ replaced by $\sum_{i_1,\dots,i_n}\varepsilon_{i_1\dots i_n}g_{a_1i_1}\cdots g_{a_ni_n}$. By the Leibniz formula this is the determinant of the matrix whose rows are the rows $a_1,\dots,a_n$ of g, that is $\varepsilon_{a_1\dots a_n}\det g = \varepsilon_{a_1\dots a_n}$. The same argument applies to Ψ_k. (2) $\Psi_{-1} = \sum\varepsilon_{i_1\dots i_n}\eta^{i_1}\wedge\cdots\wedge\eta^{i_n} = n!\,\eta^1\wedge\cdots\wedge\eta^n = 0$ by Lemma 39.6.7(1). And $\Psi_{m-1} = \sum_\sigma \mathrm{sgn}(\sigma)F_{\sigma(1)\sigma(2)}\wedge\cdots\wedge F_{\sigma(n-1)\sigma(n)} = 2^m m!\,\mathrm{Pf}(F)$. (3) By Lemma 39.6.7(4) the factors F restrict to 0 and η^i to dy^i. So Φ_0 restricts to $\sum\varepsilon_{i_1\dots i_n}y^{i_1}dy^{i_2}\wedge\cdots\wedge dy^{i_n}$. For fixed $i_1 = i$, the $(n-1)!$ orderings of the remaining indices all give $\varepsilon_{i,1,\dots,\hat{\imath},\dots,n}\,dy^1\wedge\cdots\wedge\widehat{dy^i}\wedge\cdots\wedge dy^n$, and $\varepsilon_{i,1,\dots,\hat{\imath},\dots,n} = (-1)^{i-1}$; the result is $(n-1)!\,\sigma$. □

Lemma 39.6.9 (Chern)**.** *For $0 \le k \le m-1$,*

$$d\Phi_k = \Psi_{k-1} - \frac{2m-2k-1}{2k+2}\,\Psi_k.$$

Proof. Both sides are global forms on SE, so it suffices to compare them at an arbitrary point $v \in SE$, in a frame adapted to v. Let $x = \pi(v)$ and choose, by Lemma 39.6.1, an oriented orthonormal frame near x with $\theta(x) = 0$ and $e_n(x) = v$. Then at the point v:

- $y = (0,\dots,0,1)$, and the pulled back forms θ_{ij} vanish (their value at v is $\theta_{ij}(x)\circ d\pi = 0$);

- hence $\eta^i = dy^i$, and $\eta^n = dy^n = 0$, because $\sum_i y^i dy^i = \frac{1}{2} d \sum_i (y^i)^2 = 0$ and $y = (0, \dots, 0, 1)$;
- $d\eta^i = \sum_j F_{ij} y^j = F_{in}$ by Lemma 39.6.7(3), and $dF = F \wedge \theta - \theta \wedge F = 0$.

Write $r = n - 2k$ and $G = F_{i_{r+1} i_{r+2}} \wedge \cdots \wedge F_{i_{n-1} i_n}$ for the product of the k factors F in Φ_k. By the Leibniz rule,

$$d\big(y^{i_1}\eta^{i_2}\wedge\cdots\wedge\eta^{i_r}\wedge G\big) = dy^{i_1}\wedge\eta^{i_2}\wedge\cdots\wedge\eta^{i_r}\wedge G + y^{i_1}\sum_{l=2}^{r}(-1)^l\,\eta^{i_2}\wedge\cdots\wedge d\eta^{i_l}\wedge\cdots\wedge\eta^{i_r}\wedge G \pm y^{i_1}\eta^{i_2}\wedge\cdots\wedge\eta^{i_r}\wedge dG,$$

since $d\eta^{i_l}$ is preceded by $l-2$ factors of degree 1. We evaluate the three resulting sums at v.

The terms with dG vanish at v, since $dF = 0$ there.

The terms with dy^{i_1}. At v, $dy^{i_1} = \eta^{i_1}$, so their sum is $\sum \varepsilon_{i_1 \dots i_n} \eta^{i_1} \wedge \cdots \wedge \eta^{i_r} \wedge G$, which has $n - 2k$ factors η and k factors F: it is Ψ_{k-1} at v.

The terms with $d\eta^{i_l}$. At v, $y^{i_1} = \delta_{i_1 n}$ and $d\eta^{i_l} = F_{i_l n}$. Moving the 2-form $F_{i_l n}$ to the right of the remaining factors η costs no sign, so these terms add up to

$$\sum_{l=2}^{r}(-1)^l \sum \varepsilon_{n\, i_2 \dots i_n}\, \eta^{i_2} \wedge \cdots \widehat{\eta^{i_l}} \cdots \wedge \eta^{i_r} \wedge F_{i_l n} \wedge G.$$

In the l-th term rename the indices so that i_l comes last among $i_2, \dots, i_r$; this moves it past $r - l$ indices and multiplies ε by $(-1)^{r-l}$. Since r is even, every l contributes $(-1)^l (-1)^{r-l} = 1$ times the same sum, so the terms with $d\eta$ give $(r-1)\,T$ with

$$T = \sum \varepsilon_{n\, j_1 \dots j_{r-2}\, a\, j_{r+1} \dots j_n}\, \eta^{j_1} \wedge \cdots \wedge \eta^{j_{r-2}} \wedge F_{an} \wedge F_{j_{r+1} j_{r+2}} \wedge \cdots \wedge F_{j_{n-1} j_n}.$$

Now look at Ψ_k at v. Its factors η carry the indices in the positions $1, \dots, r-2$, and since $\eta^n = 0$ at v, the index n must occur in one of the $k+1$ factors F. Exchanging two pairs of positions among the factors F is an even permutation of the indices and does not change the product of the commuting 2-forms; exchanging the two positions of one pair changes the sign of ε and of that factor F. So the terms in which n occupies any of the $2(k+1)$ positions of the factors F add up to the same as those in which it occupies position r, and

$$\Psi_k = 2(k+1) \sum \varepsilon_{j_1 \dots j_{r-2}\, a\, n\, j_{r+1} \dots j_n}\, \eta^{j_1} \wedge \cdots \wedge \eta^{j_{r-2}} \wedge F_{an} \wedge F_{j_{r+1} j_{r+2}} \wedge \cdots \wedge F_{j_{n-1} j_n} \quad \text{at } v.$$

Moving n from the first position to position r passes $r-1$ indices, an odd number, so $\varepsilon_{n\, j_1 \dots j_{r-2}\, a \dots} = -\varepsilon_{j_1 \dots j_{r-2}\, a\, n \dots}$ and $T = -\Psi_k/(2(k+1))$ at v. Altogether, at v we get $d\Phi_k = \Psi_{k-1} - \frac{r-1}{2(k+1)}\Psi_k$, and $r - 1 = 2m - 2k - 1$. □

Write $(2m-1)!! = 1 \cdot 3 \cdot 5 \cdots (2m-1)$.

Proposition 39.6.10 (Chern). *Let $n = 2m$, $c_k = \frac{2m-2k-1}{2k+2}$, $a_0 = 1$ and $a_k = c_0 c_1 \cdots c_{k-1}$, and put*

$$\Pi = -\frac{1}{(2\pi)^m (2m-1)!!} \sum_{k=0}^{m-1} a_k \Phi_k \in \Omega^{n-1}(SE).$$

Then $d\Pi = \pi^* e(\nabla)$, and the restriction of Π to every fibre $S_x E \cong S^{n-1}$ is $-\sigma/|S^{n-1}|$; in particular $\int_{S_x E} \Pi = -1$.

Proof. By Lemma 39.6.9,

$$d\sum_{k=0}^{m-1} a_k\Phi_k = \sum_{k=0}^{m-1} a_k\Psi_{k-1} - \sum_{k=0}^{m-1} a_kc_k\Psi_k = \sum_{k=0}^{m-2}(a_{k+1} - a_kc_k)\Psi_k - a_{m-1}c_{m-1}\Psi_{m-1} = -a_m\Psi_{m-1},$$

using $\Psi_{-1} = 0$ and $a_{k+1} = a_kc_k$. Here $a_m = \prod_{k=0}^{m-1}\frac{2m-2k-1}{2k+2} = \frac{(2m-1)(2m-3)\cdots 1}{2^m m!} = \frac{(2m-1)!!}{2^m m!}$, and $\Psi_{m-1} = 2^m m!(2\pi)^m\pi^*e(\nabla)$ by Lemma 39.6.8(2). So $d\sum_k a_k\Phi_k = -(2m-1)!!(2\pi)^m\pi^*e(\nabla)$, which is $d\Pi = \pi^*e(\nabla)$. On a fibre only Φ_0 survives, with $a_0 = 1$, by Lemma 39.6.8(3), so Π restricts to $-\frac{(2m-1)!}{(2\pi)^m(2m-1)!!}\sigma$. Now $(2m-1)!/(2m-1)!! = 2\cdot 4\cdots(2m-2) = 2^{m-1}(m-1)!$, and by Lemma 39.6.4, $2^{m-1}(m-1)!/(2\pi)^m = (m-1)!/(2\pi^m) = 1/|S^{2m-1}|$. □

So the pullback of the Euler form to the sphere bundle is exact, although the Euler form itself is in general not exact on M; a form Π with these two properties is called a *global angular form* [BT82, Section 11], because on each fibre it is (minus) the normalised angular form of the sphere. As a consequence, $\pi^*d\,e(\nabla) = d\,d\Pi = 0$, and π^* is injective on forms because π is a submersion; this gives a second proof that $e(\nabla)$ is closed.

Example 39.6.11. Let $n = 2$, so $m = 1$, $\Pi = -\frac{1}{2\pi}\Phi_0$ and $\Phi_0 = y^1\eta^2 - y^2\eta^1$. Over a set U with an oriented orthonormal frame, a point of $SE|_U$ is $v = \cos\phi\, e_1 + \sin\phi\, e_2$, where the angle ϕ is a local function on $SE|_U$ defined up to adding constant multiples of 2π, so $d\phi$ is well defined on $SE|_U$. Then $y = (\cos\phi, \sin\phi)$, and with $\theta_{21} = -\theta_{12}$,

$$\eta^1 = -\sin\phi\, d\phi + \theta_{12}\sin\phi, \qquad \eta^2 = \cos\phi\, d\phi + \theta_{21}\cos\phi, \qquad \Phi_0 = d\phi + \theta_{21}.$$

So $\Pi = -\frac{1}{2\pi}(d\phi + \theta_{21})$ and $d\Pi = -\frac{1}{2\pi}d\theta_{21} = \frac{1}{2\pi}F_{12} = \pi^*e(\nabla)$, as $F_{12} = d\theta_{12}$ for $n = 2$. The form $d\phi + \theta_{21}$ measures the turning of the unit vector v relative to parallel transport: along a curve, v is parallel exactly when $\dot\phi = -\theta_{21}(\dot\gamma)$, since $\nabla e_1 = \theta_{21}\otimes e_2$.

The proof of Theorem 39.6.12 below becomes, in this case, the following classical argument. Let s be a section with isolated zeros, and on $M\setminus\{s=0\}$ use the frame $e_1 = s/|s|$, e_2 its rotation by $+\pi/2$. Then $s/|s|$ has angle $\phi = 0$ in this frame, so $\int_{M\setminus\cup B_\varepsilon} e(\nabla) = \frac{1}{2\pi}\sum_z\int_{S_\varepsilon(z)}\theta_{21}$, by Stokes' theorem and $\frac{1}{2\pi}F_{12} = -\frac{1}{2\pi}d\theta_{21}$. Near a zero z compare with a frame (e_1', e_2') defined on all of $B_\varepsilon(z)$: if $e_1 = \cos\psi\, e_1' + \sin\psi\, e_2'$, the connection forms are related by $\theta_{21} = \theta_{21}' + d\psi$ (Exercise 39.8.11). The integral of $d\psi$ over $S_\varepsilon(z)$ is $2\pi\,\mathrm{ind}_z(s)$, since ψ is the angle of s in the frame e', and the integral of the smooth form θ_{21}' tends to 0 with ε. So $\int_M e(\nabla) = \sum_z \mathrm{ind}_z(s)$.

Theorem 39.6.12 (Chern)**.** *Let M be a compact oriented manifold of even dimension $n = 2m$, $E\to M$ an oriented real vector bundle of rank n with a metric and a metric connection ∇, and s a section of E whose zeros are isolated. Then s has finitely many zeros, and*

$$\int_M e(\nabla) = \sum_{s(z)=0}\mathrm{ind}_z(s).$$

Proof. The zero set Z of s is closed in the compact space M and discrete, hence finite. For each $z\in Z$ choose a positively oriented chart φ_z centred at z and an oriented orthonormal frame e of E on its domain, and for small $\varepsilon > 0$ let $\bar B_\varepsilon(z)$ and $S_\varepsilon(z)$ be as in Definition 39.6.5, with the balls pairwise disjoint. The section $\hat s = s/|s|$ of SE over $M\setminus Z$ is a smooth map $\hat s\colon M\setminus Z\to SE$ with $\pi\circ\hat s = \mathrm{id}$, so $\hat s^*\pi^*e(\nabla) = e(\nabla)$. Let $M_\varepsilon = M\setminus\bigcup_z B_\varepsilon(z)$, a compact manifold with boundary the union of the spheres $S_\varepsilon(z)$, each with the orientation opposite to its orientation as the boundary of $\bar B_\varepsilon(z)$. By Proposition 39.6.10 and Stokes' theorem,

$$\int_{M_\varepsilon} e(\nabla) = \int_{M_\varepsilon}\hat s^*d\Pi = \int_{M_\varepsilon} d(\hat s^*\Pi) = -\sum_{z\in Z}\int_{S_\varepsilon(z)}\hat s^*\Pi.$$

As $\varepsilon \to 0$ the left side tends to $\int_M e(\nabla)$, since $e(\nabla)$ is a continuous form and the volumes of the balls in the charts tend to 0. It remains to show that $\int_{S_\varepsilon(z)} \hat{s}^*\Pi \to -\operatorname{ind}_z(s)$ for each zero z.

Fix z, write $S_\varepsilon = S_\varepsilon(z)$, and use the frame e to identify SE over the chart domain with the product of the domain and S^{n-1}. In this identification $\hat{s}(q) = (q, f(q))$, with f as in Definition 39.6.5. Define

$$H\colon S_\varepsilon \times [0,1] \to SE, \qquad H(q,t) = \big(\varphi_z^{-1}(t\varphi_z(q)),\ f(q)\big),$$

so that $H(q,1) = \hat{s}(q)$ and $H(q,0) = \iota_z(f(q))$, where $\iota_z\colon S^{n-1} \to S_zE$ is the identification of the fibre given by the frame at z. By Stokes' theorem on the compact manifold $S_\varepsilon \times [0,1]$, whose boundary is $S_\varepsilon \times \{1\}$ and $S_\varepsilon \times \{0\}$ with opposite orientations,

$$\int_{S_\varepsilon} \hat{s}^*\Pi - \int_{S_\varepsilon} f^*\iota_z^*\Pi = \pm \int_{S_\varepsilon \times [0,1]} H^* d\Pi = \pm \int_{S_\varepsilon \times [0,1]} (\pi \circ H)^* e(\nabla).$$

The map $\pi \circ H(q,t) = \varphi_z^{-1}(t\varphi_z(q))$ restricts to a diffeomorphism from $S_\varepsilon \times (0,1]$ onto $\bar{B}_\varepsilon(z) \setminus \{z\}$ (in the chart it is $(u,t) \mapsto tu$, with inverse $w \mapsto (\varepsilon w/|w|, |w|/\varepsilon)$), and $S_\varepsilon \times \{0\}$ has measure zero; so the right side is $\pm \int_{B_\varepsilon(z)} e(\nabla)$, which tends to 0 with ε. On the other hand, by Proposition 39.6.10, $\iota_z^*\Pi = -\sigma/|S^{n-1}|$, so $\int_{S_\varepsilon} f^*\iota_z^*\Pi = -\frac{1}{|S^{n-1}|}\int_{S_\varepsilon} f^*\sigma = -\operatorname{ind}_z(s)$ by Lemma 39.6.6(1). Hence $\int_{S_\varepsilon} \hat{s}^*\Pi \to -\operatorname{ind}_z(s)$, and $\int_M e(\nabla) = \sum_z \operatorname{ind}_z(s)$. □

The theorem has a striking structure: the left side does not depend on the section, and the right side does not depend on the metric or the connection. So both are invariants of the oriented bundle E alone; the left side shows it is the integral of a characteristic form, the right side that it is an integer. A section without zeros forces it to vanish, in accordance with Proposition 39.4.7(3). The de Rham class of $e(\nabla)$ is the Euler class of Definition 39.4.6 [MS74, Appendix C]; we do not need this.

Corollary 39.6.13 (Gauss–Bonnet–Chern)**.** *Let M be a compact oriented manifold of dimension $2m$ with a Riemannian metric, and ∇ a metric connection on TM, for instance the Levi-Civita connection.*

1. *For every vector field X on M with isolated zeros, $\int_M e(TM,\nabla) = \sum_z \operatorname{ind}_z(X)$.*
2. *(Hopf) The sum of the indices of a vector field with isolated zeros is the same for all such vector fields.*
3. *$\int_M e(TM,\nabla) = \chi(M)$. In particular the integral does not depend on the metric, and if M admits a nowhere vanishing vector field, then $\chi(M) = 0$.*

Proof. (1) is Theorem 39.6.12 for $E = TM$, and (2) follows because the left side of (1) does not involve X. For (3), the sum of the indices equals $\chi(M)$ by the Poincaré–Hopf theorem, Theorem 39.5.1(3), applied to any vector field with isolated zeros (for instance the gradient of a Morse function [Mil63]). A nowhere vanishing vector field has no zeros, so the sum of its indices is 0. □

Example 39.6.14. Let M be a parallelizable manifold (Definition 35.5.1), with a global frame $(X_1, \dots, X_n)$. The connection on TM for which the X_i are parallel, with connection form 0 in this frame, is flat, so all the Chern–Weil forms of $TM \otimes \mathbb{C}$ vanish, and so do the Pontryagin classes of TM in de Rham cohomology. Choosing the metric for which the frame is orthonormal, this connection is also metric (Proposition 38.6.2), so its Euler form is 0; if M is compact, oriented and of even dimension, Corollary 39.6.13 gives $\chi(M) = 0$. This also follows directly from Corollary 39.6.13(3), since X_1 vanishes nowhere. So S^{2m}, $\mathbb{CP}^n$ (with $\chi = n+1$) and the compact orientable surfaces of genus $g \neq 1$ (with $\chi = 2-2g$) are not

parallelizable. The converse fails: the spheres are stably parallelizable (Example 35.3.9), so by the Whitney formula $c(E \oplus \mathbb{C}) = c(E)$ of Theorem 39.3.1 their Pontryagin classes vanish, and the odd-dimensional ones have $\chi = 0$; yet S^5 is not parallelizable (Theorem 35.5.3). Characteristic classes detect the nontriviality of TS^{2m}, through the Euler class, but not that of TS^5.

For surfaces, $e(TM, \nabla) = \frac{1}{2\pi} F_{12}$, and for the Levi-Civita connection $F_{12} = K\, dA$ with K the Gaussian curvature: this is the Gauss–Bonnet theorem $\int_M K\, dA = 2\pi\chi(M)$, proved again in Chapter 40 together with its higher-dimensional Riemannian form. There the curvature of the round sphere S^{2m} is computed, and Corollary 39.6.13 gives a formula for its volume.

39.7 Stiefel–Whitney classes and Wu's formula

With coefficients in $\mathbb{F}_2$ no orientations are needed, and the Steenrod squares (Theorem 24.4.1) produce the Stiefel–Whitney classes directly from the Thom class. For the tangent bundle of a closed manifold this has a striking consequence, Wu's formula: the Stiefel–Whitney classes of TM are determined by the cohomology ring $H^*(M;\mathbb{F}_2)$ with its Steenrod squares, so they are homotopy invariants, although TM is not. In this section H^* means $H^*(-;\mathbb{F}_2)$, and $\mathrm{Sq} = \sum_i \mathrm{Sq}^i$.

Theorem 39.7.1 (Thom). *Let $\pi\colon E \to M$ be a real vector bundle of rank n over a manifold, and E_0 the complement of the zero section.*

1. *(Thom isomorphism mod 2) There is a unique class $u \in H^n(E, E_0)$ restricting to the nonzero class on every fibre, and $\Phi(x) = \pi^*(x) \smile u$ is an isomorphism $H^k(M) \to H^{k+n}(E, E_0)$ for all k.*

2. *The classes $w_i(E) = \Phi^{-1}(\mathrm{Sq}^i u) \in H^i(M)$ are the Stiefel–Whitney classes of Definition 39.5.2: $w_0 = 1$, $w_i = 0$ for $i > n$, they are natural, the total class $w = \sum_i w_i$ satisfies $w(E \oplus E') = w(E)w(E')$, and w_1 of the tautological line bundle on $\mathbb{RP}^1$ is nonzero. Equivalently, $\mathrm{Sq}(u) = \pi^* w(E) \smile u$.*

This is [MS74, Chapters 8 and 10]; it is Thom's definition of the Stiefel–Whitney classes. The properties in (2) follow from the axioms of Theorem 24.4.1: $w_0 = 1$ from $\mathrm{Sq}^0 = \mathrm{id}$, $w_i = 0$ for $i > n$ from $\mathrm{Sq}^i u = 0$ for $i > \deg u$, and the Whitney formula from the Cartan formula, as the Thom class of $E \oplus E'$ is the cross product of the Thom classes.

Definition 39.7.2. Let M be a closed n-manifold and $[M] \in H_n(M)$ its fundamental class (every manifold is $\mathbb{F}_2$-oriented, Proposition 23.3.2(3)). For each i, the map $H^{n-i}(M) \to \mathbb{F}_2$, $x \mapsto \langle \mathrm{Sq}^i(x), [M]\rangle$, is linear, so by Poincaré duality (Corollary 23.6.3) there is a unique class $v_i \in H^i(M)$ with

$$\langle v_i \smile x, [M]\rangle = \langle \mathrm{Sq}^i(x), [M]\rangle \quad \text{for all } x \in H^{n-i}(M).$$

The v_i are the *Wu classes* of M, and $v = 1 + v_1 + v_2 + \cdots$ is the *total Wu class*.

Since $\mathrm{Sq}^0 = \mathrm{id}$, $v_0 = 1$. If $2i > n$, then $\mathrm{Sq}^i(x) = 0$ for x of degree $n - i < i$, so $v_i = 0$: only $v_0, \ldots, v_{\lfloor n/2 \rfloor}$ can be nonzero.

Theorem 39.7.3 (Wu's formula). *Let M be a closed smooth n-manifold. Then $w(TM) = \mathrm{Sq}(v)$, that is, $w_k(TM) = \sum_i \mathrm{Sq}^{k-i}(v_i)$ for every k.*

Sketch of proof. Choose a basis (b_j) of $H^*(M)$ and the dual basis $(b_j^\#)$ for the cup pairing, $\langle b_j \smile b_l^\#, [M]\rangle = \delta_{jl}$ (Corollary 23.6.3); then every $y \in H^*(M)$ is $y = \sum_j \langle y \smile b_j, [M]\rangle\, b_j^\#$. We quote two geometric facts [MS74, Chapter 11], which are the omitted verifications. First, the class $U = \sum_j b_j \times b_j^\# \in H^n(M \times M)$ (Künneth, Theorem 22.4.2) is the Poincaré dual of the diagonal. Second, a tubular neighbourhood of the diagonal is isomorphic to TM, and under this isomorphism and excision U corresponds to the Thom class of TM; with Theorem 39.7.1(2) this gives $\mathrm{Sq}(U) = (1 \times w(TM)) \smile U$.

The rest is algebra. Let $\Theta\colon H^*(M \times M) \to H^*(M)$ be the linear map with $\Theta(a \times b) = \langle a, [M]\rangle\, b$ (the slant product with $[M]$ in the first factor), so that $\Theta((1\times w) \smile Z) = w \smile \Theta(Z)$. Since $\langle b_j, [M]\rangle = \langle 1 \smile b_j, [M]\rangle$, $\Theta(U) = \sum_j \langle 1 \smile b_j, [M]\rangle b_j^\# = 1$, so $\Theta(\mathrm{Sq}(U)) = w(TM)$. On the other hand, by the Cartan formula $\mathrm{Sq}(U) = \sum_j \mathrm{Sq}(b_j) \times \mathrm{Sq}(b_j^\#)$, and $\langle \mathrm{Sq}(b_j), [M]\rangle = \langle \mathrm{Sq}^{n-\deg b_j}(b_j), [M]\rangle = \langle v \smile b_j, [M]\rangle$ by the definition of the Wu classes; so $\Theta(\mathrm{Sq}(U)) = \sum_j \langle v \smile b_j, [M]\rangle \mathrm{Sq}(b_j^\#) = \mathrm{Sq}\big(\sum_j \langle v \smile b_j, [M]\rangle b_j^\#\big) = \mathrm{Sq}(v)$. □

Corollary 39.7.4. *Let M and N be closed connected smooth n-manifolds and $h\colon M \to N$ a homotopy equivalence. Then $h^*w(TN) = w(TM)$, and M and N have the same Stiefel–Whitney numbers $\langle w_{i_1}(T-)\cdots w_{i_r}(T-), [-]\rangle$. Moreover $w_1(TM) = v_1$, so M is orientable if and only if $\mathrm{Sq}^1\colon H^{n-1}(M) \to H^n(M)$ is zero.*

Proof. The map h^* is an isomorphism of rings commuting with the Steenrod squares (naturality), and $h_*[M] = [N]$, because h_* is an isomorphism $H_n(M) \cong \mathbb{F}_2 \to H_n(N) \cong \mathbb{F}_2$ (Theorem 23.4.3). For $x \in H^{n-i}(N)$, $\langle h^*(v_i(N)) \smile h^*x, [M]\rangle = \langle v_i(N) \smile x, [N]\rangle = \langle \mathrm{Sq}^i(x), [N]\rangle = \langle \mathrm{Sq}^i(h^*x), [M]\rangle$, and h^* is surjective; so $h^*v(N) = v(M)$, and $h^*w(TN) = h^*\mathrm{Sq}(v(N)) = \mathrm{Sq}(v(M)) = w(TM)$ by Theorem 39.7.3. The Stiefel–Whitney numbers agree because $\langle h^*y, [M]\rangle = \langle y, h_*[M]\rangle$. Next, $w_1 = \mathrm{Sq}^1(v_0) + \mathrm{Sq}^0(v_1) = v_1$, as $\mathrm{Sq}^1(1) = 0$. By definition $v_1 = 0$ if and only if $\langle \mathrm{Sq}^1(x), [M]\rangle = 0$ for all $x \in H^{n-1}(M)$, that is, since $\langle -, [M]\rangle$ is an isomorphism $H^n(M) \to \mathbb{F}_2$, if and only if $\mathrm{Sq}^1 = 0$ on $H^{n-1}(M)$; and $w_1(TM) = 0$ if and only if TM, equivalently M, is orientable (Definition 39.5.2). □

Lemma 39.7.5. *Let V be a finite-dimensional vector space over $\mathbb{F}_2$ with a nondegenerate symmetric bilinear form B. There is a unique $c \in V$ with $B(x,x) = B(c,x)$ for all x, and $B(c,c) \equiv \dim V \pmod 2$.*

Proof. The map $x \mapsto B(x,x)$ is linear, because $B(x+y, x+y) = B(x,x) + B(y,y) + 2B(x,y)$, so c exists and is unique by nondegeneracy. We argue by induction on $\dim V$, the case $V = 0$ being trivial. If $B(x,x) = 0$ for all x, then $c = 0$ and B is alternating. Choose $x \neq 0$ and y with $B(x,y) = 1$; the plane P spanned by x and y has Gram matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, so B is nondegenerate on P, $V = P \oplus P^{\perp_B}$ (Lemma 11.9.2), B is nondegenerate and alternating on $P^{\perp_B}$, and by induction $\dim V$ is even, as required. Otherwise choose x with $B(x,x) = 1$; then $V = \mathbb{F}_2 x \oplus W$ with $W = x^{\perp_B}$, B is nondegenerate on W, and if c' is the element of W given by induction, $c = x + c'$ works: for $y = \lambda x + w$, $B(y,y) = \lambda + B(w,w) = \lambda + B(c',w) = B(x+c', y)$. Then $B(c,c) = 1 + B(c',c') \equiv 1 + \dim W = \dim V$. □

Corollary 39.7.6. *For a closed smooth n-manifold M, $\langle w_n(TM), [M]\rangle \equiv \chi(M) \pmod 2$.*

Proof. We may assume M connected. By Theorem 39.7.3, $w_n = \sum_i \mathrm{Sq}^{n-i}(v_i)$. For $i < n/2$ the term vanishes, because $n - i > i = \deg v_i$, and for $i > n/2$ we have $v_i = 0$. If n is odd nothing remains, so $w_n = 0$, and $\chi(M) = 0$ by Corollary 23.6.4 (closed manifolds have finitely generated homology [Hat02, Appendix]). If $n = 2k$, then $w_n = \mathrm{Sq}^k(v_k) = v_k^2$. The cup pairing $B(x,y) = \langle x \smile y, [M]\rangle$ on $V = H^k(M)$ is symmetric (mod 2 there are no signs) and nondegenerate, and $B(x,x) = \langle \mathrm{Sq}^k(x), [M]\rangle = B(v_k, x)$; so v_k is the element c of

Lemma 39.7.5, and $\langle w_n, [M]\rangle = B(v_k, v_k) \equiv \dim H^k(M)$. Finally $\chi(M) = \sum_i (-1)^i \dim H^i(M)$ (Corollary 23.6.4 and universal coefficients over a field), and $\dim H^i(M) = \dim H^{n-i}(M)$ by Poincaré duality; so the terms with $i \neq k$ cancel in pairs modulo 2, and $\chi(M) \equiv \dim H^k(M)$. □

Example 39.7.7. Let $M = \mathbb{RP}^n$, with $H^*(M) = \mathbb{F}_2[\alpha]/(\alpha^{n+1})$, $[M]$ pairing to 1 with α^n, and $\mathrm{Sq}^i(\alpha^m) = \binom{m}{i}\alpha^{m+i}$ (Proposition 24.4.2, restricted from $\mathbb{RP}^\infty$). Then $v_i = \binom{n-i}{i}\alpha^i$, since $\mathrm{Sq}^i(\alpha^{n-i}) = \binom{n-i}{i}\alpha^n$, and $\mathrm{Sq}(\alpha^i) = \alpha^i(1+\alpha)^i$.

1. $n = 2$: $v = 1 + \alpha$ and $w = 1 + \alpha(1+\alpha) = 1 + \alpha + \alpha^2$. So $\mathbb{RP}^2$ is not orientable ($w_1 \neq 0$), and $w_2 \neq 0$ matches $\chi(\mathbb{RP}^2) = 1$.

2. $n = 3$: $v_1 = \binom{2}{1}\alpha = 0$, so $v = 1$ and $w = 1$: all Stiefel–Whitney classes of $\mathbb{RP}^3$ vanish. This is consistent with $\mathbb{RP}^3 \cong \mathrm{SO}(3)$ being a Lie group, hence parallelisable.

3. $n = 4$: $v = 1 + \alpha + \alpha^2$ and $w = 1 + \alpha(1+\alpha) + \alpha^2(1+\alpha)^2 = 1 + \alpha + \alpha^4$. So $w_4[\mathbb{RP}^4] = 1$, and $\mathbb{RP}^4$ has a nonzero Stiefel–Whitney number; by a theorem of Pontryagin, the Stiefel–Whitney numbers of a boundary vanish, so $\mathbb{RP}^4$ is not the boundary of any compact 5-manifold [MS74, Chapter 4]. (Thom proved the converse: a closed manifold whose Stiefel–Whitney numbers all vanish is a boundary [Tho54].)

In all three cases $w = (1+\alpha)^{n+1}$, which is the general formula for $\mathbb{RP}^n$ [MS74, Chapter 4].

39.8 Exercises

Exercise 39.8.1. Using the Euler sequence, compute $c(T\mathbb{CP}^2)$ and $\int_{\mathbb{CP}^2} c_2(T\mathbb{CP}^2)$, and check that the latter is $\chi(\mathbb{CP}^2) = 3$.

Solution. $c(T\mathbb{CP}^2) = (1+h)^3 = 1 + 3h + 3h^2$, so $c_2 = 3h^2$ and $\int_{\mathbb{CP}^2} c_2 = 3$ since $\int h^2 = 1$. By Theorem 39.5.1, $c_2(T\mathbb{CP}^2) = e(T\mathbb{CP}^2)$ integrates to $\chi(\mathbb{CP}^2) = 3$ (Example 21.7.8). □

Exercise 39.8.2. Show that a bundle with a flat connection has vanishing Chern classes in de Rham cohomology, and give an example where the integral Chern classes are nonzero.

Solution. Chern–Weil forms are computed from the curvature, which is zero. A flat line bundle on a space with torsion in $H^2(M;\mathbb{Z})$, for instance a nontrivial real line bundle on $\mathbb{RP}^2$ complexified, has torsion $c_1 \neq 0$ mapping to zero in de Rham cohomology. □

Exercise 39.8.3. Show that $\mathrm{ch}(L) = e^{c_1(L)}$ for a line bundle, and compute ch and td of $T\mathbb{CP}^1$.

Exercise 39.8.4. Show that S^4 admits no almost complex structure, using that c_1 would lie in $H^2(S^4) = 0$ and $c_2 = e(TS^4)$ would have to satisfy $\int_{S^4} c_2 = \chi(S^4) = 2$ while also $c_1^2 = p_1 + 2c_2$ with $p_1(TS^4) = 0$.

Exercise 39.8.5. On $\mathbb{CP}^1$ compute the total Chern class of $\mathcal{O}(a) \oplus \mathcal{O}(b)$, and show that $\mathcal{O}(1) \oplus \mathcal{O}(-1)$ and $\mathcal{O} \oplus \mathcal{O}$ have the same Chern classes.

Solution. By the Whitney formula $c = (1+ah)(1+bh) = 1 + (a+b)h$, as $h^2 = 0$. For $a = 1$, $b = -1$ this is 1, the total Chern class of the trivial bundle. (The two bundles are in fact isomorphic as smooth bundles, but not as holomorphic ones: the first has a holomorphic section vanishing at one point and the second does not.) □

Exercise 39.8.6. Show that $c_1(E) = c_1(\det E)$ for every complex vector bundle E, using the connection induced on $\det E$ (Proposition 38.3.6(5)).

Solution. If ∇ has connection form θ in a frame e, the induced connection on $\det E$ has connection form $\operatorname{tr}\theta$ in the frame $e_1\wedge\cdots\wedge e_r$, so its curvature is $d\operatorname{tr}\theta = \operatorname{tr}(d\theta) = \operatorname{tr}(d\theta+\theta\wedge\theta) = \operatorname{tr}F$, using $\operatorname{tr}(\theta\wedge\theta)=0$. Hence $c_1(\det E,\nabla^{\det}) = \frac{i}{2\pi}\operatorname{tr}F = c_1(E,\nabla)$. □

Exercise 39.8.7. Let $f:\mathbb{CP}^1\to\mathbb{CP}^1$ be $z\mapsto z^k$ ($k\geq 1$) and ∇ the projection connection on $\mathcal{O}(-1)$. Show that $\int_{\mathbb{CP}^1} c_1(f^*\mathcal{O}(-1), f^*\nabla) = -k$, by computing the integral of $f^*c_1(\mathcal{O}(-1),\nabla)$, and compare with Exercise 35.6.6.

Solution. The curvature of $f^*\nabla$ is f^*F (Proposition 38.4.5(1)), so the Chern form is $f^*\omega$ with $\omega = c_1(\mathcal{O}(-1),\nabla)$. On $\mathbb{C}^\times$, f is a k-sheeted covering and a local diffeomorphism preserving the orientation (it is holomorphic). Cut $\mathbb{C}^\times$ into the k open sectors $\{2\pi j/k < \arg z < 2\pi(j+1)/k\}$; each is mapped diffeomorphically, preserving orientation, onto $\mathbb{C}$ minus a ray, and the rays and the points $0,\infty$ have measure zero. By Proposition 36.4.3 each sector contributes $\int_{\mathbb{CP}^1}\omega = -1$, so the total is $-k$, in accordance with $f^*\mathcal{O}(-1)\cong\mathcal{O}(-k)$. □

Exercise 39.8.8. Verify the formula $c_2 = \frac{1}{8\pi^2}(\operatorname{tr}(F\wedge F) - \operatorname{tr}F\wedge\operatorname{tr}F)$ for a direct sum of two line bundles with curvatures F_1, F_2, and check that it equals $c_1(L_1)\wedge c_1(L_2)$.

Solution. Here $F = \operatorname{diag}(F_1,F_2)$, $\operatorname{tr}(F\wedge F) = F_1\wedge F_1 + F_2\wedge F_2$ and $\operatorname{tr}F\wedge\operatorname{tr}F = (F_1+F_2)\wedge(F_1+F_2)$, so the difference is $-2F_1\wedge F_2$ (the 2-forms commute) and $c_2 = -\frac{1}{4\pi^2}F_1\wedge F_2 = \frac{i}{2\pi}F_1\wedge\frac{i}{2\pi}F_2$, as predicted by the Whitney formula. □

Exercise 39.8.9. Using Definition 39.6.5 with the standard frame of $\mathbb{R}^2=\mathbb{C}$, show that the vector field $X(z)=z^k$ (that is, $X = \operatorname{Re}(z^k)\partial_x + \operatorname{Im}(z^k)\partial_y$) has index k at 0 for $k\geq 1$, and that $X(z)=\bar z^k$ has index $-k$. Which of these are gradients of functions?

Solution. On S^1, $\sigma = y^1dy^2 - y^2dy^1$, and on the circle $z=\varepsilon e^{i\varphi}$, $0\leq\varphi\leq 2\pi$, positively oriented as the boundary of the disc, $z^k/|z^k| = e^{ik\varphi}$, so $f(\varphi) = (\cos k\varphi, \sin k\varphi)$ and $f^*\sigma = k(\cos^2 k\varphi + \sin^2 k\varphi)\,d\varphi = k\,d\varphi$, with integral $2\pi k$; dividing by $|S^1| = 2\pi$ gives k. For $\bar z^k$, $f(\varphi) = (\cos k\varphi, -\sin k\varphi)$ and $f^*\sigma = -k\,d\varphi$, giving $-k$. A gradient field $\nabla F = F_x\partial_x + F_y\partial_y$ corresponds to the complex function $F_x + iF_y$. For $k=1$, $\bar z = x-iy$ is the gradient of $\frac12(x^2-y^2)$ (a saddle), and z that of $\frac12(x^2+y^2)$. In general $\bar z^k$ is the gradient of $\frac{1}{k+1}\operatorname{Re}(z^{k+1})$, since for a holomorphic $h = u+iv$ the gradient of u is $u_x + iu_y = u_x - iv_x = \overline{h'}$; for $k=2$ this is the monkey saddle $\frac13\operatorname{Re}(z^3) = \frac13(x^3-3xy^2)$. On the other hand z^k for $k\geq 2$ is not a gradient: $F_x = \operatorname{Re}z^k$ and $F_y = \operatorname{Im}z^k$ would require $\partial_y\operatorname{Re}z^k = \partial_x\operatorname{Im}z^k$, but by the Cauchy–Riemann equations $\partial_y\operatorname{Re}z^k = -\partial_x\operatorname{Im}z^k$, so both vanish, which fails for $k\geq 2$ (for $k=2$, $\operatorname{Im}z^2 = 2xy$ has $\partial_x = 2y\neq 0$). □

Exercise 39.8.10. Let $A\in\mathrm{GL}_n(\mathbb{R})$ and let $s(u) = Au$ on $\mathbb{R}^n$, with the standard frame. Show that $\operatorname{ind}_0(s) = \operatorname{sgn}\det A$. (Write $A = hP$ with h orthogonal and P symmetric positive definite.)

Solution. A^TA is symmetric positive definite, so by the spectral theorem it has a symmetric positive definite square root P, and $h = AP^{-1}$ satisfies $h^Th = P^{-1}A^TAP^{-1} = I$. The matrices $P_t = (1-t)I + tP$ are positive definite, so $hP_tu\neq 0$ for $u\neq 0$, and $f_t(u) = hP_tu/|hP_tu|$ is a homotopy on S_ε from $hu/|u|$ to $Au/|Au|$. As in the proof of Lemma 39.6.6, homotopic maps give the same integral, so by Lemma 39.6.6(2) the index is $\det h = \det A/\det P$, which has the sign of $\det A$. □

Exercise 39.8.11. Let (e_1', e_2') be an oriented orthonormal frame of an oriented rank 2 bundle with a metric connection, ψ a smooth function, and $e_1 = \cos\psi\,e_1' + \sin\psi\,e_2'$, $e_2 = -\sin\psi\,e_1' + \cos\psi\,e_2'$. Show that $\theta_{21} = \theta'_{21} + d\psi$, and deduce again that F_{12} does not depend on the frame. Compare with the transformation rule $\theta = g^{-1}\theta' g + g^{-1}dg$.

Solution. $\nabla e_1 = d\psi \otimes (-\sin\psi\, e'_1 + \cos\psi\, e'_2) + \cos\psi\, \nabla e'_1 + \sin\psi\, \nabla e'_2$, and $\nabla e'_1 = \theta'_{21} \otimes e'_2$, $\nabla e'_2 = -\theta'_{21} \otimes e'_1$. So $\nabla e_1 = (d\psi + \theta'_{21}) \otimes e_2$, that is $\theta_{21} = \theta'_{21} + d\psi$. Then $F_{12} = d\theta_{12} = -d\theta_{21} = -d\theta'_{21} = F'_{12}$. In matrix form g is the rotation by ψ, $g^{-1}\theta' g = \theta'$ because rotations of the plane commute, and $g^{-1}dg$ is the rotation generator $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ times $d\psi$, whose $(2,1)$ entry is $d\psi$. □

Exercise 39.8.12. On $S^2 = \mathbb{CP}^1$ with the charts z and $w = 1/z$, let X be the vector field equal to ∂_x in the chart $z = x + iy$. Show that X extends smoothly to the point $w = 0$, where it vanishes, and has index 2 there. Deduce from Corollary 39.6.13(1) that $\int_{S^2} e(TS^2, \nabla) = 2$ for every metric and every metric connection, without using the Poincaré–Hopf theorem.

Solution. In complex notation X is the constant field 1 in the chart z: its flow is $z \mapsto z + t$. In the chart $w = 1/z$ this flow is $w \mapsto w/(1 + tw)$, whose derivative at $t = 0$ is $-w^2$. So in the chart w the field is $X(w) = -w^2$, which is smooth, also at $w = 0$, where it vanishes. The chart w is positively oriented, because $z \mapsto 1/z$ is holomorphic, hence orientation-preserving (its Jacobian determinant is $|-1/z^2|^2 > 0$). On the circle $w = \varepsilon e^{i\varphi}$ the direction of $-w^2$ is $-e^{2i\varphi} = e^{i(2\varphi+\pi)}$, which turns twice: the index is 2, by the computation of Exercise 39.8.9, using Lemma 39.6.6(1) to work with the coordinate frame. As X has no other zeros, Corollary 39.6.13(1) gives $\int_{S^2} e(TS^2, \nabla) = 2$. □

Exercise 39.8.13. Let E_1, E_2 be oriented bundles of even ranks with metrics and metric connections ∇_1, ∇_2, and give $E_1 \oplus E_2$ the orthogonal sum metric, the direct sum connection and the orientation of an oriented frame of E_1 followed by one of E_2. Show that $e(\nabla_1 \oplus \nabla_2) = e(\nabla_1) \wedge e(\nabla_2)$. For a compact oriented surface S with a metric, deduce that $\int_{S\times S} e(T(S\times S), \nabla) = \chi(S)^2$ for the product metric and the direct sum of the Levi-Civita connections.

Solution. In the frame of $E_1 \oplus E_2$ made of oriented orthonormal frames of E_1 and E_2, the connection form and the curvature are block diagonal, $F = F_1 \oplus F_2$. By Exercise 12.7.7, applied to matrices of commuting even forms, $\mathrm{Pf}(F) = \mathrm{Pf}(F_1) \wedge \mathrm{Pf}(F_2)$, and the factors $(2\pi)^{-m}$ split accordingly. For $S \times S$, $T(S \times S) = p_1^* TS \oplus p_2^* TS$, and the pullback connections have Euler forms $p_i^* e(TS, \nabla)$ (Proposition 39.6.3(4)), so $\int_{S\times S} p_1^* e \wedge p_2^* e = (\int_S e)^2$ by Fubini, and $\int_S e(TS, \nabla) = \chi(S)$ by Corollary 39.6.13. □

Chapter 40

Riemannian Geometry

40.1 Introduction

A Riemannian metric turns a manifold into a metric space, produces a canonical connection (Theorem 40.2.8) and, together with an orientation, defines the Hodge star, the codifferential and the Laplacian (Definition 40.15.2). Harmonic forms are the kernel of that Laplacian, and the Hodge theorem (Chapter 43) says that on a compact manifold they represent de Rham cohomology. Along the way we develop the Riemannian measure and the volume form, which make surface integrals and the divergence theorem instances of integration on manifolds; the gradient, Hessian, divergence and Laplacian of a Riemannian manifold; the local and global theory of geodesics: covariant derivatives along curves, the Gauss lemma, the fact that geodesics are locally the shortest curves, and the Hopf–Rinow theorem, all with complete proofs; parallel transport and holonomy, with the holonomy principle, and the theorems of Ambrose–Singer, de Rham and Berger; Jacobi fields, conjugate points and the first comparison theorems; Cartan's method of moving frames, with which we compute the curvature of hyperbolic space, of conformally flat metrics and of hypersurfaces, and the Gauss–Bonnet–Chern theorem in Riemannian form; the first and second variation of energy, with Jacobi's theorem that geodesics stop minimising after a conjugate point, a complete proof of the Bonnet–Myers theorem, the theorems of Weinstein and Synge on manifolds of positive curvature, and the splitting theorem; the geometry of submanifolds, with the first and second fundamental forms, the Weingarten equations and the equations of Gauss and Codazzi; the classical global theorems on surfaces: Gauss–Bonnet for regions with boundary, the curvature of conformal metrics, and the theorems of Minding, Hadamard, Liebmann and Hilbert; minimal surfaces, with the first and second variation of area, the area-minimising property of minimal graphs and Bernstein's theorem; and isometries and Killing fields, with Bochner's theorem, the theorem of Killing and Hopf classifying the spaces of constant curvature, and the statements of Cheng's maximal diameter theorem and of the sphere theorem. References: [dC92], [Lee18], [KN63], [Joy00], [War83], [GH78, Chapter 0].

Prerequisites. Chapter 38 (Connections and Curvature).

On a first reading. The chapters on Hodge theory use only the metric, the volume form, normal coordinates, the curvature tensor and the Hodge star and Laplacian (Sections 40.2, 40.3, 40.6, 40.8 and 40.15); holonomy is the background of Chapter 52. The sections on moving frames, variations of energy, submanifolds, surfaces, minimal surfaces, Killing fields and space forms develop Riemannian geometry for its own sake and can be skipped on a first reading.

Conventions. (M, g) is a Riemannian manifold of dimension n, and we write $\langle v, w\rangle = g(v, w)$ and $|v| = \langle v, v\rangle^{1/2}$. Curves are piecewise smooth unless stated otherwise. In local coordinates we use the summation convention: an index repeated once up and once down is summed

over.

40.2 Metrics and the Levi-Civita connection

Definition 40.2.1. A *Riemannian metric* on a smooth manifold M is a smooth section g of $\mathrm{Sym}^2 T^*M$ that is an inner product on each tangent space; (M, g) is a *Riemannian manifold*. The *length* of a piecewise smooth curve is $L(\gamma) = \int_a^b |\dot\gamma(t)|\,dt$ with $|v| = g(v,v)^{1/2}$, and $d(p,q)$ is the infimum of the lengths of curves from p to q. An *isometry* is a diffeomorphism preserving g.

Proposition 40.2.2. *Every smooth manifold admits a Riemannian metric, and d is a metric inducing the topology of M.*

Proof. Patch the Euclidean metrics of charts with a partition of unity (Theorem 33.8.2); a convex combination of inner products is an inner product. Symmetry and the triangle inequality for d are clear, and $d(p,q) > 0$ for $p \neq q$ because in a chart around p the metric is comparable to the Euclidean one on a compact neighbourhood, so curves leaving a small ball have length bounded below. The same comparison shows that the d-balls form a basis of the topology. □

Remark 40.2.3. *Which distances are Riemannian?* By the proposition every smooth manifold, being metrisable, carries Riemannian metrics whose distance induces its topology. A given metric d_0 inducing the topology is in general not of this form. A Riemannian distance d is a *length metric*: if γ is a curve from p to q with $L(\gamma) < d(p,q) + \varepsilon$ and m is the point dividing it into two arcs of equal length, then $d(p,m), d(m,q) \le L(\gamma)/2$, so m is an approximate midpoint. On $\mathbb{R}$ the metric $d_0(x,y) = |x-y|^{1/2}$ induces the usual topology but has no approximate midpoints: $d_0(0,m) + d_0(m,1) = \sqrt{|m|} + \sqrt{|1-m|} \ge 1$, with values close to 1 only for m near 0 or 1, where one of the two terms is close to $1 = d_0(0,1)$. So d_0 is not the distance of any Riemannian metric. Similarly the bounded metric $\min(|x-y|, 1)$ on $\mathbb{R}$ is not Riemannian, since in a length metric points at distance 2 have a point at distance about 1 from both.

Remark 40.2.4. *The musical isomorphisms.* A metric identifies each tangent space with its dual: $v \mapsto v^\flat = g(v,-)$ is an isomorphism $T_pM \to T_p^*M$, with inverse $\alpha \mapsto \alpha^\sharp$, and these give isomorphisms of vector bundles $\flat\colon TM \to T^*M$ and $\sharp\colon T^*M \to TM$ ("lowering and raising indices"). In coordinates, with (g^{ij}) the inverse matrix of (g_{ij}),

$$\Big(\sum_i v^i \partial_i\Big)^\flat = \sum_{i,j} g_{ij} v^i\, dx^j, \qquad \Big(\sum_i \alpha_i\, dx^i\Big)^\sharp = \sum_{i,j} g^{ij} \alpha_i\, \partial_j.$$

The gradient is $\operatorname{grad} f = (df)^\sharp = \sum_{i,j} g^{ij} \partial_i f\, \partial_j$ (Section 40.5), and the metric on T^*M making $\flat$ an isometry is $\langle \alpha, \beta\rangle = \sum g^{ij}\alpha_i\beta_j$. For the Euclidean metric $g_{ij} = \delta_{ij}$ the musical isomorphisms are invisible, which is why in vector calculus vectors and covectors are not distinguished. In polar coordinates on $\mathbb{R}^2$, $g = dr^2 + r^2 d\theta^2$, so $(\partial_\theta)^\flat = r^2\, d\theta$, $(d\theta)^\sharp = r^{-2}\partial_\theta$, and $\operatorname{grad} f = \partial_r f\, \partial_r + r^{-2}\partial_\theta f\, \partial_\theta$; the coefficient r^{-2}, not r^{-1}, appears because ∂_θ has length r.

Example 40.2.5. The Euclidean metric on $\mathbb{R}^n$; the induced metric on a submanifold, for instance the round metric on $S^n \subseteq \mathbb{R}^{n+1}$; the hyperbolic metric $(dx^2 + dy^2)/y^2$ on the upper half plane; a left-invariant metric on a Lie group, obtained from any inner product on $\mathfrak{g}$, which is bi-invariant when G is compact (average over Haar measure, Theorem 34.5.1); and the Fubini–Study metric on $\mathbb{CP}^n$ (Chapter 47).

Proposition 40.2.6. *Let a discrete group G act freely and properly on M by isometries of g. Then M/G carries a unique Riemannian metric for which $\pi\colon M \to M/G$ is a local isometry.*

Proof. By Theorem 33.7.1, π is a local diffeomorphism. For $x \in M$ define the metric on $T_{\pi(x)}(M/G)$ by transporting g_x through $d_x\pi$. This does not depend on the choice of x in its orbit: $\pi \circ h = \pi$ for $h \in G$, so $d_{hx}\pi \circ d_x h = d_x\pi$, and $d_x h$ is an isometry $T_xM \to T_{hx}M$. On an open set $\pi(U)$ over which π has the smooth inverse $(\pi|_U)^{-1}$ the metric is the pullback of g by this inverse, hence smooth. Uniqueness is clear. □

Example 40.2.7. The *flat torus* $\mathbb{R}^n/\mathbb{Z}^n$ and the *flat Klein bottle* $\mathbb{R}^2/G$ (Example 20.4.5), since translations and the maps $(x, y) \mapsto (x + \frac{1}{2}, -y)$ are Euclidean isometries; and the round metric on $\mathbb{RP}^n = S^n/\{\pm 1\}$, as $-\mathrm{id}$ is an isometry of S^n.

Theorem 40.2.8 (Fundamental theorem of Riemannian geometry)**.** *On a Riemannian manifold there is a unique connection ∇ on TM that is metric ($Xg(Y,Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$) and torsion-free ($\nabla_X Y - \nabla_Y X = [X, Y]$): the* Levi-Civita connection*. It is determined by the Koszul formula*

$$2g(\nabla_X Y, Z) = Xg(Y,Z) + Yg(X,Z) - Zg(X,Y) + g([X,Y],Z) - g([X,Z],Y) - g([Y,Z],X).$$

In coordinates, $\nabla_{\partial_i}\partial_j = \sum_k \Gamma_{ij}^k \partial_k$ with $\Gamma_{ij}^k = \frac{1}{2}\sum_l g^{kl}(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$.

Proof. If ∇ is metric and torsion-free, expanding the three terms $Xg(Y,Z)$, $Yg(X,Z)$, $-Zg(X,Y)$ with the metric property and using torsion-freeness to replace $\nabla_X Y - \nabla_Y X$ by $[X, Y]$ gives the Koszul formula; since g is nondegenerate, the formula determines $\nabla_X Y$, which proves uniqueness. Conversely, define $\nabla_X Y$ by the Koszul formula. The right-hand side is $\mathbb{R}$-bilinear, C^∞-linear in X and Z and satisfies the Leibniz rule in Y (the terms with derivatives of f in $Y \mapsto fY$ combine to $2(Xf)g(Y,Z)$), so ∇ is a connection; exchanging X and Y shows it is torsion-free, and adding the formulas for $g(\nabla_X Y, Z)$ and $g(Y, \nabla_X Z)$ gives $Xg(Y,Z)$. The coordinate formula follows by taking X, Y, Z coordinate fields, whose brackets vanish. □

In coordinates, torsion-freeness says $\Gamma_{ij}^k = \Gamma_{ji}^k$, and the metric property for coordinate fields says

$$\partial_i g_{jk} = \Gamma_{ij}^l g_{lk} + \Gamma_{ik}^l g_{jl}. \tag{40.1}$$

Definition 40.2.9. A *geodesic* is a curve with $\nabla_{\dot\gamma}\dot\gamma = 0$, where the velocity $\dot\gamma$ is differentiated along γ as in Remark 38.5.1 (it need not extend to a vector field on M; see Remark 40.4.2); in coordinates, $\ddot\gamma^k + \sum_{i,j}\Gamma_{ij}^k \dot\gamma^i \dot\gamma^j = 0$. By Theorem 27.13.3 there is a unique maximal geodesic with given initial position and velocity, depending smoothly on them; the *exponential map* $\exp_p(v) = \gamma_v(1)$ is defined on a neighbourhood of $0 \in T_pM$.

40.3 Volume and integration

A metric measures the lengths of tangent vectors, hence the volumes of small parallelepipeds, and so it determines how to integrate functions over the manifold. For a surface in $\mathbb{R}^3$ this is the classical surface integral $\iint f\, dA$; the point of this section is that it is an instance of the integration of forms of Chapter 36, and that the same formulas work for submanifolds of any dimension. In a chart (U, φ) with coordinates $x^1, \dots, x^n$ we write $g_{ij} = g(\partial_i, \partial_j)$ and $\det g$ for the determinant of the matrix (g_{ij}), a smooth positive function on U; λ is Lebesgue measure.

Lemma 40.3.1. *Let $y^1, \dots, y^n$ be the coordinates of a second chart, with metric coefficients $\tilde g_{ab} = g(\partial/\partial y^a, \partial/\partial y^b)$. Then on the intersection of the two chart domains*

$$\tilde g_{ab} = \sum_{i,j} \frac{\partial x^i}{\partial y^a}\frac{\partial x^j}{\partial y^b}\, g_{ij}, \qquad \sqrt{\det \tilde g} = \left|\det\left(\frac{\partial x^i}{\partial y^a}\right)\right| \sqrt{\det g}.$$

Proof. By Proposition 33.4.7, $\partial/\partial y^a = \sum_i (\partial x^i/\partial y^a)\,\partial/\partial x^i$; expanding g bilinearly gives the first formula. In matrices it reads $\tilde g = P^T g P$ with $P = (\partial x^i/\partial y^a)$, and taking determinants, $\det \tilde g = (\det P)^2 \det g$. □

Proposition 40.3.2 (Riemannian measure)**.** *Let (M, g) be a Riemannian manifold. There is a unique measure μ_g on the Borel sets of M such that for every chart (U, φ) and every Borel set $A \subseteq U$,*

$$\mu_g(A) = \int_{\varphi(A)} \sqrt{\det g} \circ \varphi^{-1}\, d\lambda.$$

Consequently $\int_A f\, d\mu_g = \int_{\varphi(A)} (f\sqrt{\det g}) \circ \varphi^{-1}\, d\lambda$ for every measurable $f \geq 0$ and every integrable f on A. We write dV for μ_g, and dA and ds in dimensions 2 and 1 (the volume*, the* area *and the* arc length*), and* $\operatorname{vol}(M) = \mu_g(M)$.

Proof. For a chart (U, φ) the formula defines a measure μ_U on the Borel subsets of U, the image under the homeomorphism φ^{-1} of the measure with density $\sqrt{\det g} \circ \varphi^{-1}$ on $\varphi(U)$. Two such measures agree where both are defined: let A be a Borel subset of $U \cap V$ for a second chart (V, ψ) with coordinates y, and let $\tau = \varphi \circ \psi^{-1}$, whose Jacobian matrix is $(\partial x^i/\partial y^a)$. By Lemma 40.3.1, $\sqrt{\det \tilde g} \circ \psi^{-1} = (\sqrt{\det g} \circ \varphi^{-1} \circ \tau)\,|\det D\tau|$, so by the change of variables formula (Theorem 28.11.2) applied to $\tau\colon \psi(A) \to \varphi(A)$,

$$\int_{\psi(A)} \sqrt{\det \tilde g} \circ \psi^{-1}\, d\lambda = \int_{\psi(A)} (\sqrt{\det g} \circ \varphi^{-1} \circ \tau)\,|\det D\tau|\, d\lambda = \int_{\varphi(A)} \sqrt{\det g} \circ \varphi^{-1}\, d\lambda.$$

Cover M by countably many chart domains $U_1, U_2, \ldots$ (M is second countable), let $B_1 = U_1$ and $B_k = U_k \setminus (U_1 \cup \cdots \cup U_{k-1})$, and put $\mu_g(A) = \sum_k \mu_{U_k}(A \cap B_k)$. This is a measure, being a countable sum of measures. If A is a Borel subset of a chart domain U, the compatibility just proved gives $\mu_{U_k}(A \cap B_k) = \mu_U(A \cap B_k)$, so $\mu_g(A) = \sum_k \mu_U(A \cap B_k) = \mu_U(A)$. Conversely, any measure with the stated property assigns to A the value $\sum_k \mu_{U_k}(A \cap B_k)$, which proves uniqueness. The formula for integrals follows from the one for sets, first for simple functions, then by monotone convergence (Theorem 28.4.3) for $f \geq 0$, and by linearity for integrable f. □

Proposition 40.3.3 (Riemannian volume form)**.** *Let (M, g) be an oriented Riemannian n-manifold. There is a unique n-form* vol_g *with* $\operatorname{vol}_g(e_1, \ldots, e_n) = 1$ *for every positively oriented orthonormal basis $(e_1, \ldots, e_n)$ of a tangent space. In a positively oriented chart*

$$\operatorname{vol}_g = \sqrt{\det g}\; dx^1 \wedge \cdots \wedge dx^n,$$

and $\int_M f \operatorname{vol}_g = \int_M f\, d\mu_g$ for every continuous function f with compact support.

Proof. Let p lie in a positively oriented chart, and let $(e_1, \ldots, e_n)$ be a positively oriented orthonormal basis of T_pM. Write $e_a = \sum_i P^i_a \partial_i$ with an invertible matrix $P = (P^i_a)$. Then $\delta_{ab} = \langle e_a, e_b\rangle = (P^T g P)_{ab}$, so $(\det P)^2 \det g = 1$, and $\det P > 0$ because both bases are positively oriented; hence $\det P = 1/\sqrt{\det g}$ and

$$\big(\sqrt{\det g}\; dx^1 \wedge \cdots \wedge dx^n\big)(e_1, \ldots, e_n) = \sqrt{\det g}\, \det\big(dx^i(e_a)\big) = \sqrt{\det g}\, \det P = 1.$$

Since $\Lambda^n T^*_pM$ is one-dimensional, an n-form at p is determined by its value on one basis. So on each positively oriented chart domain the smooth form $\sqrt{\det g}\, dx^1 \wedge \cdots \wedge dx^n$ is the only form with the required property; the forms of two charts therefore agree on the overlap, and together they define the smooth n-form vol_g, which is unique. If f is supported in a positively oriented chart, both integrals equal $\int (f\sqrt{\det g}) \circ \varphi^{-1}\, d\lambda$, by Definition 36.4.1 and Proposition 40.3.2; in general write $f = \sum_i \psi_i f$ with a partition of unity subordinate to positively oriented charts and use linearity of both sides. □

For a surface vol_g is the *area form* dA. By Proposition 36.3.2, a non-orientable manifold, such as the Möbius band or $\mathbb{RP}^2$, has no nowhere vanishing n-form, hence no volume form; but it still has a Riemannian measure, so areas and integrals of functions make sense on it. The integral of a function needs a metric, the integral of an n-form needs an orientation, and the volume form converts one into the other.

Proposition 40.3.4 (Volume of parametrised submanifolds)**.** *Let $M \subseteq \mathbb{R}^N$ be a k-dimensional submanifold with the induced metric, and $x\colon \Omega \to M$ a parametrisation (Proposition 33.6.18(3)), with $x_i = \partial x/\partial u^i$. Then for every measurable $f \geq 0$ on $x(\Omega)$,*

$$\int_{x(\Omega)} f\,d\mu = \int_\Omega f(x(u))\,\mathcal{J}(u)\,du, \qquad \mathcal{J} = \sqrt{\det\bigl(\langle x_i, x_j\rangle\bigr)} = \Bigl(\sum_I \Bigl(\det \frac{\partial x^I}{\partial u}\Bigr)^2\Bigr)^{1/2},$$

where I runs over the subsets $\{i_1 < \cdots < i_k\}$ of $\{1, \dots, N\}$, and $\partial x^I/\partial u$ is the $k \times k$ matrix formed by the rows $i_1, \dots, i_k$ of the Jacobian matrix Dx. In particular:

1. *for a curve ($k = 1$), $\mathcal{J} = |x'(t)|$, and $\int f\,ds = \int f(x(t))\,|x'(t)|\,dt$;*
2. *for a surface in $\mathbb{R}^3$, with $E = |x_u|^2$, $F = \langle x_u, x_v\rangle$ and $G = |x_v|^2$, $\mathcal{J} = \sqrt{EG - F^2} = |x_u \times x_v|$;*
3. *for a hypersurface ($k = N - 1$) with unit normal ν, $\mathcal{J} = |\det(\nu, x_1, \dots, x_k)|$;*
4. *for the graph of a smooth function $\phi\colon \Omega \to \mathbb{R}$, parametrised by $x(u) = (u, \phi(u))$, $\mathcal{J} = \sqrt{1 + |\nabla\phi|^2}$.*

Proof. The inverse x^{-1} is a chart of M (Lemma 33.6.3), whose coordinate vector field $\partial/\partial u^i$ is x_i; so $g_{ij} = \langle x_i, x_j\rangle$ and the formula is Proposition 40.3.2. For the second expression of $\mathcal{J}$, Proposition 12.5.2 gives $\det(\langle x_i, x_j\rangle) = |x_1 \wedge \cdots \wedge x_k|^2$ in $\Lambda^k\mathbb{R}^N$. Writing $x_j = \sum_i (\partial x^i/\partial u^j) e_i$ and expanding by multilinearity, the terms with a repeated e_i vanish, and collecting the terms with the same set of indices I (the Leibniz formula for the determinant) gives $x_1 \wedge \cdots \wedge x_k = \sum_I \det(\partial x^I/\partial u)\, e_I$; as the e_I are orthonormal, $|x_1 \wedge \cdots \wedge x_k|^2 = \sum_I \det(\partial x^I/\partial u)^2$.

(1) is the case of a single vector. (2) $EG - F^2 = |x_u|^2|x_v|^2 - \langle x_u, x_v\rangle^2 = |x_u \times x_v|^2$ (Lagrange's identity); equivalently, the three 2×2 minors of Dx are, up to sign, the components of $x_u \times x_v$. (3) The matrix P with columns $\nu, x_1, \dots, x_k$ satisfies $P^T P = \left(\begin{smallmatrix} 1 & 0 \\ 0 & (\langle x_i, x_j\rangle)\end{smallmatrix}\right)$, since $|\nu| = 1$ and $\langle \nu, x_i\rangle = 0$; so $(\det P)^2 = \det(\langle x_i, x_j\rangle)$. (4) Here $x_i = (e_i, \partial_i\phi)$, so $\langle x_i, x_j\rangle = \delta_{ij} + \partial_i\phi\,\partial_j\phi$: this is the matrix $I + ww^T$ with $w = \nabla\phi$, which is the identity on $w^\perp$ and multiplies w by $1 + |w|^2$, so its determinant is $1 + |w|^2$. □

Remark 40.3.5. In classical surface theory [dC76, Chapter 2] the area of a region R contained in the image of a parametrisation x, and the integral of a function over it, are defined by

$$\iint_R f\,dA = \iint_{x^{-1}(R)} f(x(u,v))\,|x_u \times x_v|\,du\,dv,$$

and one checks by hand that this does not depend on the parametrisation. By Proposition 40.3.4(2) this is the integral with respect to the Riemannian measure of the induced metric: surface integration is integration on a manifold, and nothing new. The independence of the parametrisation is Lemma 40.3.1 together with the change of variables formula, and a region that is not contained in one parametrisation is cut into pieces, or treated with a partition of unity, as in Definition 36.4.1. When the surface is oriented, $\iint_R f\,dA = \int_R f\,\mathrm{vol}_S$ is the integral of the 2-form $f\,\mathrm{vol}_S$ (Proposition 40.3.3). The same holds in every dimension and codimension: the length of a curve (Definition 40.2.1) is its one-dimensional volume, by (1), and the k-dimensional volume of a piece of a k-dimensional submanifold of $\mathbb{R}^N$ is the integral of the square root of the sum of the squares of the $k \times k$ minors.

Proposition 40.3.6 (Riemannian measure and Hausdorff measure)**.** *Let $M \subseteq \mathbb{R}^N$ be a k-dimensional submanifold with the induced metric. Then $\mu_g(A) = \mathcal{H}^k(A)$ for every Borel set $A \subseteq M$, where $\mathcal{H}^k$ is the k-dimensional Hausdorff measure of $\mathbb{R}^N$ (Definition 28.17.1). In particular the volume of M depends only on M as a metric subspace of $\mathbb{R}^N$, and a nonempty M has Hausdorff dimension k.*

Proof. A submanifold is locally closed (each point has a neighbourhood W in $\mathbb{R}^N$ with $M \cap W$ closed in W, by the slice charts), so M and its Borel subsets are Borel in $\mathbb{R}^N$. Let $x\colon \Omega \to M$ be a parametrisation; it is an injective smooth map with injective differential and a homeomorphism onto the open set $x(\Omega) \subseteq M$, and its function $\mathcal{J} = \sqrt{\det(Dx^T Dx)}$ of Proposition 40.3.4 is the Jacobian Jx of Definition 28.20.1. For a Borel set $A \subseteq x(\Omega)$, Proposition 40.3.4 with $f = \mathbf{1}_A$ and the area formula (Theorem 28.20.3) give

$$\mu_g(A) = \int_{x^{-1}(A)} Jx \, d\lambda = \mathcal{H}^k(x(x^{-1}(A))) = \mathcal{H}^k(A).$$

Cover M by countably many such sets $V_i = x_i(\Omega_i)$ and let $B_i = V_i \setminus \bigcup_{j<i} V_j$; then $\mu_g(A) = \sum_i \mu_g(A \cap B_i) = \sum_i \mathcal{H}^k(A \cap B_i) = \mathcal{H}^k(A)$. Finally, if $M \neq \emptyset$, $\mathcal{H}^k(M) > 0$, and M is a countable union of compact sets of finite $\mathcal{H}^k$-measure, so Proposition 28.17.3(4) gives $\dim_{\mathcal{H}} M = k$. □

So the Riemannian volume of a submanifold of $\mathbb{R}^N$, defined through charts and the metric, is the most naive notion of k-dimensional size: cover the set by small pieces and add up $\alpha(k)(\text{diam})^k$. This is the point of view of geometric measure theory, where the area formula (Theorem 28.20.3) replaces the computation in charts, and which extends the notion of k-dimensional volume to sets that are not manifolds, such as rectifiable sets and the singular varieties of algebraic geometry.

For hypersurfaces of $\mathbb{R}^{n+1}$ the volume form has a simple expression through the unit normal, and it turns integrals of n-forms into flux integrals. We write $\Omega = dx^1 \wedge \cdots \wedge dx^{n+1}$ for the standard volume form of $\mathbb{R}^{n+1}$, and, for a vector field F defined along M (for instance the unit normal ν), $(\iota_F \Omega)|_M$ for the n-form on M whose value at p is the restriction of $\iota_{F(p)}\Omega_p$ to T_pM.

Proposition 40.3.7 (Volume form of a hypersurface; flux)**.** *Let $M \subseteq \mathbb{R}^{n+1}$ be a hypersurface oriented by a unit normal field ν (Proposition 36.3.4). Then*

$$\operatorname{vol}_M = (\iota_\nu \Omega)|_M = \Big(\sum_{i=1}^{n+1} (-1)^{i-1} \nu^i \, dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^{n+1} \Big)\Big|_M,$$

and for every vector field F along M, $(\iota_F \Omega)|_M = \langle F, \nu \rangle \operatorname{vol}_M$. In $\mathbb{R}^3$, for a surface S: $dA = (\nu^1 \, dy \wedge dz + \nu^2 \, dz \wedge dx + \nu^3 \, dx \wedge dy)|_S$, and the flux *of F through S is*

$$\int_S F^1 \, dy \wedge dz + F^2 \, dz \wedge dx + F^3 \, dx \wedge dy = \iint_S \langle F, \nu \rangle \, dA.$$

Proof. By Proposition 36.3.4 the orientation of M is that of the form $(v_1, \ldots, v_n) \mapsto \det(\nu, v_1, \ldots, v_n) = \Omega(\nu, v_1, \ldots, v_n) = (\iota_\nu \Omega)(v_1, \ldots, v_n)$. If $(e_1, \ldots, e_n)$ is a positively oriented orthonormal basis of T_pM, then $(\nu(p), e_1, \ldots, e_n)$ is an orthonormal basis of $\mathbb{R}^{n+1}$, positive by the same proposition, so its determinant is 1. By the uniqueness in Proposition 40.3.3, $(\iota_\nu \Omega)|_M = \operatorname{vol}_M$. The coordinate expression is $\iota_\nu(dx^1 \wedge \cdots \wedge dx^{n+1}) = \sum_i (-1)^{i-1} dx^i(\nu) \, dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^{n+1}$, which follows from Proposition 12.4.6 by induction on the number of factors. For F, write $F = \langle F, \nu \rangle \nu + F^\top$ with $F^\top$ tangent to M. For $v_1, \ldots, v_n \in T_pM$ the $n+1$ vectors $F^\top, v_1, \ldots, v_n$ lie in the n-dimensional space T_pM, hence are linearly dependent, and $\Omega(F^\top, v_1, \ldots, v_n) = 0$;

so $(\iota_F\Omega)|_M = \langle F, \nu\rangle(\iota_\nu\Omega)|_M$. In $\mathbb{R}^3$, $\iota_\nu(dx\wedge dy\wedge dz) = \nu^1\,dy\wedge dz - \nu^2\,dx\wedge dz + \nu^3\,dx\wedge dy$ and $-dx\wedge dz = dz\wedge dx$; the flux formula is the integral of $(\iota_F\Omega)|_S = \langle F,\nu\rangle\,dA$, by Proposition 40.3.3. □

Remark 40.3.8. It is worth spelling out what the flux formula means, because the classical notation hides it. The integrand $F^1\,dy\wedge dz + F^2\,dz\wedge dx + F^3\,dx\wedge dy$ is a 2-form on $\mathbb{R}^3$ (or on a neighbourhood of S), and "$|_S$" means its pullback ι^* by the inclusion $\iota\colon S\to\mathbb{R}^3$ (Definition 36.2.1): the 2-form on S whose value on $v, w\in T_pS$ is the value of the original form on the same vectors, now regarded as vectors of $\mathbb{R}^3$. It is this pullback that one integrates over S.

In a positively oriented parametrisation $x(u,v) = (x(u,v), y(u,v), z(u,v))$ of S, pulling back is substitution: $x^*dy = d(y\circ x) = y_u\,du + y_v\,dv$, and so on, so that

$$x^*(dy\wedge dz) = (y_uz_v - y_vz_u)\,du\wedge dv = \frac{\partial(y,z)}{\partial(u,v)}\,du\wedge dv,$$

and similarly $x^*(dz\wedge dx) = \frac{\partial(z,x)}{\partial(u,v)}\,du\wedge dv$ and $x^*(dx\wedge dy) = \frac{\partial(x,y)}{\partial(u,v)}\,du\wedge dv$. These three 2×2 minors of the Jacobian matrix are exactly the components of the cross product $x_u\times x_v$. Hence

$$x^*\big(F^1\,dy\wedge dz + F^2\,dz\wedge dx + F^3\,dx\wedge dy\big) = \langle F\circ x, x_u\times x_v\rangle\,du\wedge dv = \langle F\circ x,\nu\rangle\,|x_u\times x_v|\,du\wedge dv,$$

because $x_u\times x_v = |x_u\times x_v|\,\nu$ for a parametrisation that is positively oriented with respect to ν (by Proposition 36.3.4, (x_u, x_v) is positive exactly when $\det(\nu, x_u, x_v) = \langle \nu, x_u\times x_v\rangle > 0$). Since $|x_u\times x_v|\,du\,dv$ is the area element (Proposition 40.3.4), integrating gives the flux formula again: this is a second proof of the proposition in $\mathbb{R}^3$, by direct computation in a chart.

This also explains the classical notation. One writes $d\vec{\Sigma} = \nu\,dA$ for the "vector surface element" and $\iint_S F\cdot d\vec{\Sigma} = \iint_S F_1\,dy\,dz + F_2\,dz\,dx + F_3\,dx\,dy$, as in the usual statement $\iint_S \operatorname{curl} F\cdot d\vec{\Sigma} = \oint_{\partial S} F\cdot d\vec{r}$ of Stokes' theorem. The symbols $dy\,dz$ there are not mere notation: they are the pullbacks to S of the 2-forms $dy\wedge dz$, and so on, and the identity $d\vec{\Sigma} = \nu\,dA$ is the statement of the proposition, $(dy\wedge dz, dz\wedge dx, dx\wedge dy)|_S = \nu\operatorname{vol}_S$. Geometrically, $(dy\wedge dz)|_S(v,w)$ is the signed area of the projection to the (y,z)-plane of the parallelogram spanned by v and w, while $\operatorname{vol}_S(v,w)$ is the area of the parallelogram itself; the components of ν are the ratios between the two, the cosines of the angles between the tangent plane and the coordinate planes.

Example 40.3.9. Let S be the graph of a smooth function h over a region $R\subseteq\mathbb{R}^2$, with the parametrisation $x(u,v) = (u,v,h(u,v))$ and the upward normal. The minors are $\frac{\partial(y,z)}{\partial(u,v)} = \det\left(\begin{smallmatrix}0 & 1\\ h_u & h_v\end{smallmatrix}\right) = -h_u$, $\frac{\partial(z,x)}{\partial(u,v)} = -h_v$ and $\frac{\partial(x,y)}{\partial(u,v)} = 1$, so $x_u\times x_v = (-h_u, -h_v, 1)$, $\nu = (-h_u,-h_v,1)/(1+h_u^2+h_v^2)^{1/2}$, and the flux of F through S is

$$\iint_S\langle F,\nu\rangle\,dA = \iint_R\big(-F^1h_u - F^2h_v + F^3\big)(u,v,h(u,v))\,du\,dv.$$

For instance the flux of $F = (0,0,1)$ through any graph over R is the area of R: the vertical field only sees the projection of the surface to the horizontal plane, $(dx\wedge dy)|_S = \nu^3\,dA$. For the upper unit hemisphere, $h = (1-u^2-v^2)^{1/2}$ on the open unit disc, and the flux of $F(x) = x$ is $\iint(u^2/h + v^2/h + h)\,du\,dv = \iint du\,dv/h = \int_0^{2\pi}\int_0^1 \frac{r\,dr\,d\varphi}{(1-r^2)^{1/2}} = 2\pi$, which is also $\iint_S\langle x,\nu\rangle\,dA = \iint_S 1\,dA$, the area of the hemisphere, as $\nu(x) = x$ on the sphere.

Proposition 40.3.7 has a version in every codimension, which is the starting point of the theory of currents in geometric measure theory. For a hypersurface the orientation

is described by the unit normal ν; for a k-dimensional submanifold of $\mathbb{R}^N$ there is no preferred normal vector, but there is a preferred *tangent k-vector*. Let $M \subseteq \mathbb{R}^N$ be an oriented k-dimensional submanifold. For $x \in M$ let

$$\xi(x) = \tau_1 \wedge \cdots \wedge \tau_k \in \Lambda^k \mathbb{R}^N,$$

where $(\tau_1, \ldots, \tau_k)$ is a positively oriented orthonormal basis of $T_x M$. By Proposition 12.5.2, applied to the oriented inner product space $T_x M$, $\xi(x)$ does not depend on the choice of the basis (it is the volume element of $T_x M$, regarded in $\Lambda^k \mathbb{R}^N$), and $|\xi(x)| = 1$. It is the *orienting k-vector* of M: it determines both the tangent space, as the set of v with $v \wedge \xi(x) = 0$, and its orientation. A k-form ω on an open set of $\mathbb{R}^N$ can be paired with it pointwise, via the pairing $\langle\,,\rangle\colon \Lambda^k(\mathbb{R}^N)^* \times \Lambda^k \mathbb{R}^N \to \mathbb{R}$ of Proposition 12.4.1, for which $\langle \omega_x, v_1 \wedge \cdots \wedge v_k \rangle = \omega_x(v_1, \ldots, v_k)$.

Proposition 40.3.10. *Let $M \subseteq \mathbb{R}^N$ be an oriented k-dimensional submanifold with orienting k-vector ξ, contained in an open set U, and let ω be a k-form on U. Then*

$$\omega|_M = \langle \omega, \xi \rangle \operatorname{vol}_M,$$

where $\omega|_M$ is the pullback of ω by the inclusion and $\langle \omega, \xi \rangle$ is the function $x \mapsto \langle \omega_x, \xi(x) \rangle$ on M. Consequently, if $\omega|_M$ has compact support,

$$\int_M \omega = \int_M \langle \omega, \xi \rangle \, d\mu_g = \int_M \langle \omega(x), \xi(x) \rangle \, d\mathcal{H}^k(x).$$

Proof. Fix $x \in M$ and a positively oriented orthonormal basis $(\tau_1, \ldots, \tau_k)$ of $T_x M$. The space $\Lambda^k T_x^* M$ of alternating k-forms on the k-dimensional space $T_x M$ is one-dimensional, spanned by $(\operatorname{vol}_M)_x$, so $(\omega|_M)_x = c\,(\operatorname{vol}_M)_x$ for a number c. Evaluating both sides on $(\tau_1, \ldots, \tau_k)$, and using $\operatorname{vol}_M(\tau_1, \ldots, \tau_k) = 1$ (Proposition 40.3.3) and the definition of the pullback,

$$c = (\omega|_M)_x(\tau_1, \ldots, \tau_k) = \omega_x(\tau_1, \ldots, \tau_k) = \langle \omega_x, \tau_1 \wedge \cdots \wedge \tau_k \rangle = \langle \omega_x, \xi(x) \rangle.$$

This proves the identity of forms. Integrating, $\int_M \omega = \int_M \langle \omega, \xi \rangle \operatorname{vol}_M = \int_M \langle \omega, \xi \rangle \, d\mu_g$ by Proposition 40.3.3 (the function $\langle \omega, \xi \rangle$ is continuous with compact support on M, since ξ is continuous, as it can be computed from a local positively oriented frame by the Gram–Schmidt process), and $\mu_g = \mathcal{H}^k$ on M by Proposition 40.3.6. □

The same computation in a chart makes each factor visible. Let $x\colon \Omega \to M$ be a positively oriented parametrisation with $x_i = \partial x / \partial u^i$. Then $x_1 \wedge \cdots \wedge x_k$ is a positive multiple of ξ (Gram–Schmidt does not change the orientation), and its length is $\mathcal{J} = \sqrt{\det(\langle x_i, x_j \rangle)}$ by Proposition 12.5.2; so $x_1 \wedge \cdots \wedge x_k = \mathcal{J}\, \xi \circ x$. On the other hand, by the definition of the integral of a form,

$$\int_{x(\Omega)} \omega = \int_\Omega (x^* \omega)(\partial_1, \ldots, \partial_k)\, du = \int_\Omega \omega_{x(u)}(x_1, \ldots, x_k)\, du = \int_\Omega \langle \omega_{x(u)}, \xi(x(u)) \rangle \, \mathcal{J}(u)\, du,$$

and $\mathcal{J}\, du$ is the volume element (Proposition 40.3.4). So the form ω splits into two factors: the number $\langle \omega, \xi \rangle$, which measures how much of ω is "seen" by the oriented tangent plane, and the volume $\mathcal{J}\, du$, which only depends on the metric.

Example 40.3.11.

1. For a hypersurface $M \subseteq \mathbb{R}^{n+1}$ oriented by ν and $\omega = \iota_F \Omega$, $\langle \omega, \xi \rangle = \Omega(F, \tau_1, \ldots, \tau_n) = \det(F, \tau_1, \ldots, \tau_n) = \langle F, \nu \rangle$, because $(\nu, \tau_1, \ldots, \tau_n)$ is a positive orthonormal basis of $\mathbb{R}^{n+1}$; this is Proposition 40.3.7 again.
2. For an oriented curve with unit tangent T, $\xi = T$ and $\int_C \omega = \int_C \omega(T)\, ds$: the line integral of a 1-form is the integral of its tangential component with respect to arc length.

3. For a 2-plane $M \subseteq \mathbb{R}^4$ spanned by orthonormal τ_1, τ_2 and $\omega = dx^1 \wedge dx^2$, $\langle \omega, \xi \rangle = \det\left(\begin{smallmatrix} \tau_1^1 & \tau_2^1 \\ \tau_1^2 & \tau_2^2 \end{smallmatrix}\right)$ is the factor by which the orthogonal projection to the (x^1, x^2)-plane multiplies oriented areas on M. In general $\xi = \sum_I \xi^I e_I$ and $|\langle dx^I, \xi \rangle| = |\xi^I| \le 1$, with equality only for the coordinate planes; this is the inequality behind *calibrations* (Section 40.13).

Remark 40.3.12. *Currents.* The proposition says that the functional $[\![M]\!]\colon \omega \mapsto \int_M \omega$ on compactly supported k-forms is

$$[\![M]\!](\omega) = \int \langle \omega(x), \xi(x) \rangle \, d\mu(x), \qquad \mu = \mathcal{H}^k \text{ restricted to } M,$$

an integral of the pairing of ω with a unit k-vector field against a measure. De Rham and Federer–Fleming call a continuous linear functional on compactly supported smooth k-forms a *k-current* [FF60]. If a current T has locally finite mass, that is, $\sup\{T(\omega) : |\omega| \le 1, \operatorname{supp}\omega \subseteq K\} < \infty$ for every compact K, then it extends to continuous forms, and the Riesz representation theorem for vector-valued functions (Theorem 28.14.9, with $H = \Lambda^k(\mathbb{R}^N)^*$, identified with $\Lambda^k\mathbb{R}^N$ by matching the orthonormal bases dx^I and e_I) writes it in exactly this form: $T(\omega) = \int \langle \omega, \xi \rangle \, d\mu$ with a Radon measure $\mu = \|T\|$, the *mass measure*, and a Borel field ξ of unit k-vectors. For $T = [\![M]\!]$ one recovers $\mu = \mathcal{H}^k$ on M and the orienting k-vector; for general currents ξ need not be a simple wedge product and μ need not live on a manifold, and this is the generality in which geometric measure theory solves Plateau's problem.

Corollary 40.3.13 (Divergence theorem)**.** *Let $D \subseteq \mathbb{R}^{n+1}$ be a compact subset that is a smooth $(n+1)$-manifold with boundary (Definition 36.4.5), with the charts given by charts of $\mathbb{R}^{n+1}$, for instance a closed ball. Let ν be the outward unit normal along ∂D, and F a smooth vector field on a neighbourhood of D, with divergence* $\operatorname{div} F = \sum_i \partial_i F^i$*. Then*

$$\int_D \operatorname{div} F \, d\lambda = \int_{\partial D} \langle F, \nu \rangle \, dA.$$

Proof. By Proposition 12.4.6, $\iota_F\Omega = \sum_i (-1)^{i-1} F^i \, dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^{n+1}$, and $d(\iota_F\Omega) = \sum_i \partial_i F^i \, \Omega = (\operatorname{div} F)\,\Omega$, as in the proof of Theorem 36.4.8. Stokes' theorem gives $\int_D (\operatorname{div} F)\,\Omega = \int_{\partial D} \iota_F\Omega$, where D carries the standard orientation of $\mathbb{R}^{n+1}$, so that the left side is $\int_D \operatorname{div} F \, d\lambda$, and ∂D the induced orientation. We check that the induced orientation is the one defined by ν; then the right side is $\int_{\partial D} \langle F, \nu \rangle \, dA$ by Proposition 40.3.7. In a positively oriented chart in which D is $\{x^{n+1} \ge 0\}$, the vector $w = -\partial_{n+1}$ points outward, and $\det(w, \partial_1, \dots, \partial_n) = (-1)^{n+1}$ with respect to the chart; by Definition 36.4.5, $(x^1, \dots, x^n)$ is positively oriented on ∂D exactly when $(-1)^{n+1} = 1$. So a basis $(v_1, \dots, v_n)$ of $T_p\partial D$ is positive for the induced orientation if and only if $(w, v_1, \dots, v_n)$ is a positive basis of $\mathbb{R}^{n+1}$. The same holds with ν in place of w, because $w = c\nu + t$ with $c > 0$ and t tangent to ∂D, so that $\det(w, v_1, \dots, v_n) = c \det(\nu, v_1, \dots, v_n)$. By Proposition 36.3.4 this is the orientation defined by ν. □

The divergence theorem is the many-variable version of the fundamental theorem of calculus, and it gives integration by parts on a region with a boundary term. Write $\Delta_0 = \sum_i \partial_i^2$ for the Euclidean Laplacian with the sign of analysis (so $\Delta_0 = -\Delta$ for the Hodge Laplacian of Definition 40.15.2), ∇f for the gradient, and $\partial_\nu f = \langle \nabla f, \nu \rangle$ for the outward normal derivative.

Corollary 40.3.14 (Integration by parts and Green's identities)**.** *Let D and ν be as in Corollary 40.3.13, and f, g smooth functions on a neighbourhood of D. Then*

1. $\displaystyle\int_D (\partial_i f)\, g \, d\lambda = \int_{\partial D} f\, g\, \nu^i \, dA - \int_D f\, \partial_i g \, d\lambda;$

2. *(first Green identity)* $\displaystyle\int_D \big(g\,\Delta_0 f + \langle \nabla g, \nabla f\rangle\big)\,d\lambda = \int_{\partial D} g\,\partial_\nu f\,dA$;

3. *(second Green identity)* $\displaystyle\int_D \big(g\,\Delta_0 f - f\,\Delta_0 g\big)\,d\lambda = \int_{\partial D} \big(g\,\partial_\nu f - f\,\partial_\nu g\big)\,dA$.

Proof. (1) Apply Corollary 40.3.13 to $F = f g\, e_i$: $\operatorname{div} F = \partial_i(fg) = (\partial_i f)g + f\,\partial_i g$ and $\langle F, \nu\rangle = f g\, \nu^i$. (2) Apply it to $F = g\nabla f$: $\operatorname{div} F = \sum_i \partial_i(g\,\partial_i f) = \langle \nabla g, \nabla f\rangle + g\,\Delta_0 f$ and $\langle F, \nu\rangle = g\,\partial_\nu f$. (3) Subtract from (2) the same identity with f and g exchanged; the symmetric terms $\langle \nabla g, \nabla f\rangle$ cancel. □

For $n+1 = 1$ and $D = [a, b]$, $\nu = -1$ at a and $+1$ at b, and (1) is the integration by parts of one variable (Corollary 27.8.4); when f or g vanishes near ∂D, it is Proposition 28.11.4. With $g = 1$, (2) says that the total flux of ∇f through the boundary is $\int_D \Delta_0 f$; with $g = f$ it gives $\int_D |\nabla f|^2 = \int_{\partial D} f\,\partial_\nu f - \int_D f\,\Delta_0 f$, the starting point of the energy methods for the Laplace equation (Exercise 40.17.21).

Stokes' theorem for 1-forms on surfaces gives the two other classical theorems of vector calculus, which Example 36.4.11 mentioned without formulas.

Corollary 40.3.15 (Green's theorem)**.** *Let $D \subseteq \mathbb{R}^2$ be a compact region with smooth boundary, as in Corollary 40.3.13, with ∂D oriented as the boundary of D (so that D lies to the left of the direction of motion), and P, Q smooth functions near D. Then*

$$\oint_{\partial D} P\,dx + Q\,dy = \iint_D \Big(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\Big)dx\,dy.$$

In particular the area of D is $\frac{1}{2}\oint_{\partial D} x\,dy - y\,dx$.

Proof. By Theorem 36.2.8, $d(P\,dx + Q\,dy) = P_y\,dy\wedge dx + Q_x\,dx\wedge dy = (Q_x - P_y)\,dx\wedge dy$, and Stokes' theorem (Theorem 36.4.8) gives the formula, the integral of the 2-form $(Q_x - P_y)\,dx\wedge dy$ over D with its standard orientation being the Lebesgue integral. In the boundary orientation, an outward vector w followed by a positive tangent vector T is a positive basis of $\mathbb{R}^2$ (Definition 36.4.5), so T is w rotated by $+\pi/2$ and D is on the left. For the area take $P = -y/2$, $Q = x/2$, with $Q_x - P_y = 1$. □

For the ellipse $x = a\cos t$, $y = b\sin t$, $0 \le t \le 2\pi$, the area formula gives $\frac{1}{2}\int_0^{2\pi}(ab\cos^2 t + ab\sin^2 t)\,dt = \pi ab$. Green's theorem is also the divergence theorem in the plane: if $T = (T^1, T^2)$ is the positive unit tangent of ∂D, the outward normal is $\nu = (T^2, -T^1)$, so for $F = (Q, -P)$ one has $\langle F, \nu\rangle = PT^1 + QT^2$ and $\operatorname{div} F = Q_x - P_y$: the flux of F out of D is the circulation of (P, Q) along ∂D.

Corollary 40.3.16 (Stokes' theorem of vector calculus)**.** *Let $\Sigma \subseteq \mathbb{R}^3$ be a surface oriented by a unit normal field ν, and $S \subseteq \Sigma$ a compact region with smooth boundary (a 2-manifold with boundary in the charts of Σ). Orient ∂S as the boundary of S: a tangent vector T of ∂S is positive when $\nu \times T$ points into S, that is, walking along ∂S with one's head in the direction of ν, the region S is on the left. Then for a smooth vector field F on a neighbourhood of S,*

$$\iint_S \langle \operatorname{curl} F, \nu\rangle\,dA = \oint_{\partial S} F^1\,dx + F^2\,dy + F^3\,dz = \int_{\partial S} \langle F, T\rangle\,ds,$$

where T is the positive unit tangent and ds the arc length.

Proof. Let $\alpha = F^1\,dx + F^2\,dy + F^3\,dz$ and $\iota\colon S \to \mathbb{R}^3$ the inclusion. By Example 36.2.14, $d\alpha = G^1\,dy\wedge dz + G^2\,dz\wedge dx + G^3\,dx\wedge dy$ with $G = \operatorname{curl} F$, and by Proposition 40.3.7, applied to the surface Σ, $\iota^*d\alpha = \langle G, \nu\rangle \operatorname{vol}_S$. Since pullback commutes with d (Theorem 36.2.8), Stokes' theorem on the compact manifold with boundary S gives $\iint_S \langle G,\nu\rangle\,dA = \int_S d(\iota^*\alpha) = \int_{\partial S}\iota^*\alpha$, and the last integral is the line integral of α along ∂S in its boundary orientation (Definition 36.2.16), which is $\int\langle F, T\rangle\,ds$ in a parametrisation by arc length. It remains to describe the boundary orientation. By Definition 36.4.5, T is positive when (w, T) is a positive basis of T_pS for an outward vector w, and by Proposition 36.3.4 this means $\det(\nu, w, T) > 0$. For w and T unit and orthogonal (which we may assume), (ν, w, T) is then a positive orthonormal basis of $\mathbb{R}^3$, so $\nu\times T = -w$, which points into S. □

Example 40.3.17. Let S be the closed upper unit hemisphere, with the outward normal $\nu(x) = x$, and $F = (-y, x, 0)$, the rotation field, with $\operatorname{curl} F = (0,0,2)$. The boundary is the equator, and its boundary orientation is counterclockwise seen from above: at $(1,0,0)$ the inward direction in S is $(0,0,1)$, and $\nu\times T = (1,0,0)\times(0,1,0) = (0,0,1)$ for $T = (0,1,0)$. With $\gamma(t) = (\cos t, \sin t, 0)$, the right side is $\int_0^{2\pi}(\sin^2 t + \cos^2 t)\,dt = 2\pi$. The left side is $\iint_S 2z\,dA$, and by Example 40.3.9 (with $F = (0,0,2)$) it is twice the area of the unit disc, 2π. The same value is obtained for every surface with boundary the equator and the corresponding orientation, for instance the flat unit disc with $\nu = (0,0,1)$, where Stokes' theorem is Green's theorem: the flux of a curl through a surface only depends on its boundary, because $\operatorname{curl} F\cdot d\vec{\Sigma}$ is an exact 2-form.

Example 40.3.18.

1. *The sphere.* The map $x(\theta,\varphi) = r(\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$ on $(0,\pi)\times(0,2\pi)$ parametrises the sphere of radius r minus a closed half great circle. That half circle has area 0: without its two endpoints it is the segment $\varphi = 0$ for the same formula on $(0,\pi)\times(-\pi,\pi)$, a Lebesgue null set, and points have measure 0. Here $E = r^2$, $F = 0$, $G = r^2\sin^2\theta$, so $dA = r^2\sin\theta\,d\theta\,d\varphi$ and the area is $\int_0^{2\pi}\int_0^{\pi} r^2\sin\theta\,d\theta\,d\varphi = 4\pi r^2$.

2. *The torus of revolution.* For $0 < b < a$, $x(u,v) = ((a+b\cos u)\cos v, (a+b\cos u)\sin v, b\sin u)$ has $E = b^2$, $F = 0$ and $G = (a+b\cos u)^2$, so the area is $\int_0^{2\pi}\int_0^{2\pi} b(a+b\cos u)\,du\,dv = 4\pi^2 ab$: the length $2\pi b$ of the rotated circle times the length $2\pi a$ of the path of its centre.

3. *Graphs.* The upper hemisphere is the graph of $\phi = \sqrt{1-u^2-v^2}$ over the open unit disc, with $1+|\nabla\phi|^2 = 1/(1-u^2-v^2)$. In polar coordinates its area is $\int_0^{2\pi}\int_0^1 \rho(1-\rho^2)^{-1/2}\,d\rho\,d\theta = 2\pi$, half of 4π.

4. *Spheres and balls.* For $F(x) = x$ on $\mathbb{R}^{n+1}$, $\operatorname{div} F = n+1$, and on the unit sphere $\nu(x) = x$ and $\langle F,\nu\rangle = 1$. By Corollary 40.3.13, $\operatorname{vol}(S^n) = (n+1)\,\lambda(B^{n+1})$: for instance $2\pi = 2\cdot\pi$ and $4\pi = 3\cdot\frac{4\pi}{3}$. With the volume $\lambda(B^m) = \pi^{m/2}/\Gamma(\frac{m}{2}+1)$ of the unit ball (Exercise 28.23.7) this gives $\operatorname{vol}(S^n) = 2\pi^{(n+1)/2}/\Gamma(\frac{n+1}{2})$, using $\Gamma(s+1) = s\Gamma(s)$: for instance $\operatorname{vol}(S^3) = 2\pi^2$.

40.4 Covariant derivatives along curves

A *vector field along* a smooth curve $\gamma\colon I\to M$ is a smooth map $V\colon I\to TM$ with $V(t)\in T_{\gamma(t)}M$; for instance the velocity $\dot\gamma$, or $Y\circ\gamma$ for a vector field Y on M.

Proposition 40.4.1. *There is a unique operator D_t sending vector fields along γ to vector fields along γ such that D_t is $\mathbb{R}$-linear, $D_t(fV) = \dot f V + f D_tV$ for smooth $f\colon I\to\mathbb{R}$, and $D_t(Y\circ\gamma) = \nabla_{\dot\gamma}Y$ for every vector field Y defined near $\gamma(I)$. In coordinates, for $V = V^k\partial_k$,*

$$D_tV = \big(\dot V^k + \Gamma^k_{ij}(\gamma)\dot\gamma^i V^j\big)\partial_k.$$

Moreover $\frac{d}{dt}\langle V, W\rangle = \langle D_tV, W\rangle + \langle V, D_tW\rangle$. *A vector field along* γ *is parallel in the sense of Definition 38.5.2 if and only if* $D_tV = 0$.

Proof. In a chart, $V = V^k(\partial_k \circ \gamma)$, and the axioms force $D_tV = \dot{V}^k\partial_k + V^k\nabla_{\dot\gamma}\partial_k = (\dot{V}^k + \dot\gamma^i V^j\Gamma^k_{ij})\partial_k$, since $\nabla_{\dot\gamma}\partial_j = \dot\gamma^i\Gamma^k_{ij}\partial_k$. This proves uniqueness. Conversely, the formula satisfies the axioms in each chart (for $Y = Y^k\partial_k$, $\frac{d}{dt}Y^k(\gamma) = \dot\gamma^i\partial_iY^k$ matches the coordinate expression of $\nabla_{\dot\gamma}Y$), and by uniqueness the local definitions agree on overlaps. For the metric property, differentiate $\langle V, W\rangle = g_{jk}(\gamma)V^jW^k$: the derivative of $g_{jk}(\gamma)$ is $\dot\gamma^i\partial_ig_{jk}$, and by (40.1) the result equals $\langle D_tV, W\rangle + \langle V, D_tW\rangle$. The last statement is the local expression of $\nabla_{\dot\gamma}s = 0$ in the coordinate frame. □

Remark 40.4.2. In the language of Chapter 38, D_t is the pullback connection $\gamma^*\nabla$ on the bundle γ^*TM over I, differentiated in the direction of the vector field d/dt on I. Indeed $(\gamma^*\nabla)_{d/dt}$ is $\mathbb{R}$-linear, satisfies the Leibniz rule for functions of t, and sends $Y \circ \gamma$ to $(\nabla Y)(\dot\gamma) = \nabla_{\dot\gamma}Y$ (Proposition 38.4.5(1)), so it equals D_t by the uniqueness in Proposition 40.4.1. This is the meaning of $\nabla_{\dot\gamma}\dot\gamma$ in Definition 40.2.9: it is $D_t\dot\gamma$. It could not be defined as ∇_XX for a vector field X, because the velocity need not be the restriction of a vector field on M: the curve $\gamma(t) = (t^2, 0)$ passes through each point $(t^2, 0)$ twice, with the opposite velocities $(\pm 2t, 0)$ (Remark 38.5.1). When $\dot\gamma$ does extend near $\gamma(t_0)$ to a vector field X, that is $X(\gamma(t)) = \dot\gamma(t)$ for t near t_0, then $D_t\dot\gamma = D_t(X \circ \gamma) = \nabla_{\dot\gamma}X = (\nabla_XX)(\gamma(t))$ there, whichever extension is chosen. For example, a unit speed great circle c on S^2 is the integral curve of the rotation field $X(x) = a \times x$ about its axis a, and $D_t\dot c = (\nabla_XX)(c)$.

For a submanifold M of $\mathbb{R}^N$ with the induced metric there is a concrete description: regarding V as a function $I \to \mathbb{R}^N$, D_tV is the orthogonal projection of the ordinary derivative $V'(t)$ onto $T_{\gamma(t)}M$, and a curve is a geodesic exactly when its acceleration γ'' is normal to M (Corollary 40.11.3 below, with examples).

Lemma 40.4.3 (Symmetry lemma)**.** *Let* $f : (-\varepsilon, \varepsilon) \times (a, b) \to M$ *be smooth, and write* D_s, D_t *for the covariant derivatives along the curves* $s \mapsto f(s, t)$ *and* $t \mapsto f(s, t)$. *Then* $D_s\partial_tf = D_t\partial_sf$.

Proof. In coordinates both sides equal $(\partial_s\partial_tf^k + \Gamma^k_{ij}\partial_sf^i\partial_tf^j)\partial_k$, by the symmetry $\Gamma^k_{ij} = \Gamma^k_{ji}$. □

Proposition 40.4.4. *A curve* γ *is a geodesic (Definition 40.2.9) if and only if* $D_t\dot\gamma = 0$. *Geodesics have constant speed, and parallel transport along any curve is a linear isometry.*

Proof. The coordinate expression of $D_t\dot\gamma$ is the geodesic equation. If $D_t\dot\gamma = 0$ then $\frac{d}{dt}|\dot\gamma|^2 = 2\langle D_t\dot\gamma, \dot\gamma\rangle = 0$. If V and W are parallel, $\langle V, W\rangle$ is constant by Proposition 40.4.1. □

40.5 Gradient, Hessian, divergence and Laplacian

The differential df of a function is defined on every manifold, but the gradient, the vector "pointing in the direction of steepest ascent", and the second derivative of f as a symmetric bilinear form are not: the first needs a metric to turn covectors into vectors, the second a connection to differentiate df. On a Riemannian manifold both are available, together with the divergence of a vector field and the Laplacian of a function, and in Euclidean space they reduce to the familiar operators of vector calculus. In this section (M, g) is a Riemannian manifold with Levi-Civita connection ∇, and $\langle\ ,\ \rangle = g$.

Definition 40.5.1. Let f be a smooth function on M.

1. The *gradient* of f is the vector field $\operatorname{grad} f$ with $\langle \operatorname{grad} f, X\rangle = df(X) = Xf$ for all vector fields X; it is the image of df under the isomorphism $T^*M \cong TM$ given by the metric (Proposition 35.3.4).

2. The *Hessian* of f is the covariant derivative of the 1-form df, the tensor $\operatorname{Hess} f = \nabla df$:
$$\operatorname{Hess} f(X,Y) = (\nabla_X df)(Y) = X(Yf) - (\nabla_X Y)f.$$
3. The *divergence* of a vector field X is the trace of the endomorphism $v \mapsto \nabla_v X$ of each tangent space: $\operatorname{div} X = \sum_i \langle \nabla_{e_i} X, e_i\rangle$ for an orthonormal basis (e_i).
4. The *Laplacian* of f (with the sign of analysis) is $\Delta_g f = \operatorname{div}\operatorname{grad} f$.

The formula for the Hessian is the Leibniz rule for the induced connection on 1-forms (Proposition 38.3.6(1)), $(\nabla_X\alpha)(Y) = X(\alpha(Y)) - \alpha(\nabla_X Y)$, with $\alpha = df$ and $df(Y) = Yf$ (Remark 36.2.2). Without the correction term, $X(Yf)$ alone is not a tensor in X and Y: it is not $C^\infty(M)$-linear in Y, as $X(hYf) = (Xh)(Yf) + hX(Yf)$.

Proposition 40.5.2. *Let f be smooth and X a vector field on M.*

1. *In coordinates,* $\operatorname{grad} f = \sum_{i,j} g^{ij}\partial_j f\,\partial_i$, $\operatorname{Hess} f(\partial_i,\partial_j) = \partial_i\partial_j f - \sum_k \Gamma^k_{ij}\partial_k f$, $\operatorname{div} X = \frac{1}{\sqrt{\det g}}\sum_i \partial_i\big(\sqrt{\det g}\,X^i\big)$ *and* $\Delta_g f = \frac{1}{\sqrt{\det g}}\sum_{i,j}\partial_i\big(\sqrt{\det g}\,g^{ij}\partial_j f\big)$.
2. *At a point where $df \neq 0$, $\operatorname{grad} f$ is orthogonal to the level set of f through it, and among unit vectors v the derivative $v(f)$ is largest for $v = \operatorname{grad} f/|\operatorname{grad} f|$, with maximum value $|\operatorname{grad} f|$.*
3. $\operatorname{Hess} f$ *is symmetric, and* $\operatorname{Hess} f(X,Y) = \langle \nabla_X \operatorname{grad} f, Y\rangle$; *hence* $\Delta_g f = \operatorname{tr}_g \operatorname{Hess} f = \sum_i \operatorname{Hess} f(e_i,e_i)$.
4. *For a geodesic γ, $(f\circ\gamma)''(t) = \operatorname{Hess} f(\dot\gamma(t),\dot\gamma(t))$. At a critical point p of f, $\operatorname{Hess} f_p(v,w) = \sum_{i,j} v^i w^j \partial_i\partial_j f(p)$ in any chart: there the Hessian does not depend on the metric.*
5. *If M is oriented, $d(\iota_X \operatorname{vol}_g) = (\operatorname{div} X)\operatorname{vol}_g$.*

Proof. (1) Writing $\operatorname{grad} f = \sum_i G^i\partial_i$, the definition gives $\sum_i g_{ij}G^i = \partial_j f$, so $G^i = \sum_j g^{ij}\partial_j f$. The Hessian formula is the definition with $\nabla_{\partial_i}\partial_j = \sum_k \Gamma^k_{ij}\partial_k$. For the divergence, $\nabla_{\partial_i} X = \sum_k(\partial_i X^k + \sum_j X^j\Gamma^k_{ij})\partial_k$, so its trace is $\sum_i \partial_i X^i + \sum_{i,j}\Gamma^i_{ij}X^j$; the trace does not depend on the basis, so it may be computed in the coordinate basis. By the formula of Theorem 40.2.8, $\sum_i \Gamma^i_{ij} = \frac12\sum_{i,l} g^{il}\partial_j g_{il} = \frac12\operatorname{tr}(g^{-1}\partial_j g)$, the other two terms cancelling by the symmetry of g. For an invertible matrix $A(t)$, $\det A(t+h) = \det A(t)\det\big(I + A(t)^{-1}(A(t+h)-A(t))\big)$ and $\det(I+hB) = 1 + h\operatorname{tr}B + O(h^2)$ by the Leibniz formula for the determinant, so $\partial_j \det g = \det g\operatorname{tr}(g^{-1}\partial_j g)$ and $\sum_i\Gamma^i_{ij} = \partial_j \log\sqrt{\det g}$. Hence $\operatorname{div} X = \sum_i\partial_i X^i + \sum_j X^j\partial_j\log\sqrt{\det g} = \frac{1}{\sqrt{\det g}}\sum_i\partial_i(\sqrt{\det g}\,X^i)$. The formula for Δ_g combines this with the one for the gradient.

(2) For v tangent to the level set, $\langle\operatorname{grad} f, v\rangle = v(f) = 0$, since f is constant along a curve in the level set. For $|v| = 1$, $v(f) = \langle\operatorname{grad} f, v\rangle \le |\operatorname{grad} f|$ by the Cauchy–Schwarz inequality, with equality exactly for $v = \operatorname{grad} f/|\operatorname{grad} f|$.

(3) $\operatorname{Hess} f(X,Y) - \operatorname{Hess} f(Y,X) = X(Yf) - Y(Xf) - (\nabla_X Y - \nabla_Y X)f = [X,Y]f - [X,Y]f = 0$, by torsion-freeness. Since the connection is metric, $X\langle\operatorname{grad} f, Y\rangle = \langle\nabla_X\operatorname{grad} f, Y\rangle + \langle\operatorname{grad} f, \nabla_X Y\rangle$, that is $X(Yf) = \langle\nabla_X\operatorname{grad} f, Y\rangle + (\nabla_X Y)f$, which is the second formula. Taking the trace in an orthonormal basis, $\sum_i \operatorname{Hess} f(e_i,e_i) = \sum_i\langle\nabla_{e_i}\operatorname{grad} f, e_i\rangle = \operatorname{div}\operatorname{grad} f$.

(4) $(f\circ\gamma)' = df(\dot\gamma) = \langle\operatorname{grad} f\circ\gamma, \dot\gamma\rangle$, and differentiating with the metric property of D_t (Proposition 40.4.1), $(f\circ\gamma)'' = \langle\nabla_{\dot\gamma}\operatorname{grad} f, \dot\gamma\rangle + \langle\operatorname{grad} f, D_t\dot\gamma\rangle = \operatorname{Hess} f(\dot\gamma,\dot\gamma)$, as $D_t\dot\gamma = 0$. At a critical point the terms $\Gamma^k_{ij}\partial_k f$ of (1) vanish.

(5) By Cartan's formula (Proposition 36.2.13) and $d\,\mathrm{vol}_g = 0$, $d(\iota_X \mathrm{vol}_g) = L_X \mathrm{vol}_g$. In a positively oriented chart $\iota_X \mathrm{vol}_g = \sum_i (-1)^{i-1} \sqrt{\det g}\, X^i\, dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n$ (Proposition 40.3.3 and Proposition 12.4.6), whose exterior derivative is $\sum_i \partial_i(\sqrt{\det g}\, X^i)\, dx^1 \wedge \cdots \wedge dx^n = (\operatorname{div} X)\,\mathrm{vol}_g$ by (1). □

By (4), at a critical point p the second derivative test of calculus is coordinate free: if $\operatorname{Hess} f_p$ is positive definite, f has a strict local minimum at p (the test in any chart, where $\operatorname{Hess} f_p$ is the matrix of second derivatives), and conversely at a local minimum it is positive semidefinite (apply the one-variable test to $f \circ \gamma$ for geodesics γ through p). Away from critical points the Hessian depends on the metric through the Christoffel symbols. A function is *convex* along all geodesics exactly when $\operatorname{Hess} f \geq 0$, again by (4). The Hodge Laplacian of Definition 40.15.2 on functions is $-\Delta_g$ (compare (1) with Proposition 40.15.3(5)): geometers prefer the sign that makes the Laplacian a non-negative operator, analysts the sign that gives $\sum_i \partial_i^2$ on $\mathbb{R}^n$.

Example 40.5.3.

1. *Euclidean space.* For $g = \sum_i dx_i^2$ all Christoffel symbols vanish, and $\operatorname{grad} f = (\partial_1 f, \ldots, \partial_n f)$, $\operatorname{Hess} f = (\partial_i \partial_j f)$, $\operatorname{div} X = \sum_i \partial_i X^i$ and $\Delta_g = \sum_i \partial_i^2$.

2. *Polar coordinates.* For $g = dr^2 + r^2 d\theta^2$ on the plane minus the origin, $\det g = r^2$, $g^{rr} = 1$, $g^{\theta\theta} = r^{-2}$, so

$$\operatorname{grad} f = f_r\, \partial_r + \frac{f_\theta}{r^2}\partial_\theta, \qquad \Delta_g f = \frac{1}{r}\partial_r(r f_r) + \frac{1}{r^2} f_{\theta\theta} = f_{rr} + \frac{1}{r} f_r + \frac{1}{r^2} f_{\theta\theta}.$$

   In terms of the unit vectors $e_r = \partial_r$, $e_\theta = r^{-1}\partial_\theta$ the gradient is $f_r e_r + \frac{1}{r} f_\theta e_\theta$, the formula of vector calculus. On the round sphere with $g = d\vartheta^2 + \sin^2\vartheta\, d\varphi^2$, the same computation gives $\Delta_g f = \frac{1}{\sin\vartheta}\partial_\vartheta(\sin\vartheta\, f_\vartheta) + \frac{1}{\sin^2\vartheta} f_{\varphi\varphi}$.

3. *Conformal metrics.* For $g' = e^{2u} g$, $\operatorname{grad}' f = e^{-2u} \operatorname{grad} f$: the gradient depends on the metric, while df does not. In dimension 2 also $\Delta_{g'} f = e^{-2u} \Delta_g f$, since $\sqrt{\det g'}\, g'^{ij} = \sqrt{\det g}\, g^{ij}$; so harmonic functions on a surface depend only on the conformal class of the metric. For the hyperbolic half-plane, $\Delta_g = y^2(\partial_x^2 + \partial_y^2)$, and y^s satisfies $\Delta_g y^s = s(s-1) y^s$.

4. *Distance from the origin* in $\mathbb{R}^n$: for $r = |x|$ on $\mathbb{R}^n \setminus \{0\}$, $\operatorname{grad} r = x/r$, a unit vector field, and $\operatorname{Hess} r = \frac{1}{r}\big(I - \frac{x x^T}{r^2}\big)$: it vanishes in the radial direction, along which r is linear, and equals $\frac{1}{r}$ times the identity on the tangent spaces of the spheres $\{r = \text{const}\}$, whose curvature it measures. Hence $\Delta_g r = \frac{n-1}{r}$, and $\Delta_g(r^{2-n}) = 0$ for $n \geq 3$ (Exercise 40.17.24); for $n = 3$, $-\frac{1}{4\pi r}$ is the fundamental solution of the Laplacian (Chapter 42).

The divergence theorem of Corollary 40.3.13 holds on any Riemannian manifold with boundary, with the divergence of the metric. Let M be an oriented Riemannian manifold with boundary and ν the outward unit normal along ∂M: the unit vector orthogonal to $T(\partial M)$ pointing out of M.

Proposition 40.5.4 (Divergence theorem and Green's formula)**.** *Let M be a compact oriented Riemannian manifold with boundary, with the induced metric and the boundary orientation on ∂M. For a vector field X and smooth functions f, h,*

$$\int_M \operatorname{div} X\, \mathrm{vol}_g = \int_{\partial M} \langle X, \nu\rangle\, \mathrm{vol}_{\partial M}, \qquad \int_M \big(h\, \Delta_g f + \langle \operatorname{grad} h, \operatorname{grad} f\rangle\big)\mathrm{vol}_g = \int_{\partial M} h\, \langle \operatorname{grad} f, \nu\rangle\, \mathrm{vol}_{\partial M}.$$

In particular, if $\partial M = \emptyset$, then $\int_M \operatorname{div} X\, \mathrm{vol}_g = 0$ and $\int_M h\, \Delta_g f\, \mathrm{vol}_g = -\int_M \langle \operatorname{grad} h, \operatorname{grad} f\rangle\, \mathrm{vol}_g$.

Proof. By Proposition 40.5.2(5) and Stokes' theorem (Theorem 36.4.8), $\int_M \operatorname{div} X \operatorname{vol}_g = \int_{\partial M} \iota_X \operatorname{vol}_g$. As in the proof of Proposition 40.3.7, write $X = \langle X, \nu\rangle \nu + X^\top$ with $X^\top$ tangent to ∂M; then $\iota_{X^\top}\operatorname{vol}_g$ vanishes on $T(\partial M)$, because n vectors in the $(n-1)$-dimensional space $T_p(\partial M)$ are linearly dependent. And $\iota_\nu \operatorname{vol}_g$ restricted to ∂M is $\operatorname{vol}_{\partial M}$: if $(e_2, \dots, e_n)$ is an orthonormal basis of $T_p(\partial M)$ that is positive for the boundary orientation, then $(\nu, e_2, \dots, e_n)$ is a positive orthonormal basis of T_pM (the outward vector comes first, Definition 36.4.5, and ν differs from any outward vector by a positive multiple plus a tangent vector, as in the proof of Corollary 40.3.13), so $(\iota_\nu \operatorname{vol}_g)(e_2, \dots, e_n) = 1$. Green's formula is the divergence theorem for $X = h \operatorname{grad} f$, with $\operatorname{div}(h \operatorname{grad} f) = \langle \operatorname{grad} h, \operatorname{grad} f\rangle + h\,\Delta_g f$, by the Leibniz rule $\nabla_v(hY) = v(h)Y + h\nabla_v Y$ inside the trace. □

40.6 The exponential map and minimising geodesics

Geodesics are the projections to M of the integral curves of a vector field on TM, the *geodesic spray*, which in the coordinates (x, v) of TM is $v^k\partial_{x^k} - \Gamma^k_{ij}(x)v^iv^j\partial_{v^k}$. By Theorem 33.9.2, the set $\mathcal{E} \subseteq TM$ of vectors v for which γ_v is defined on $[0, 1]$ is open, and $\exp\colon \mathcal{E} \to M$ is smooth. Since $\gamma_{cv}(t) = \gamma_v(ct)$, each $\mathcal{E}_p = \mathcal{E} \cap T_pM$ is star-shaped about 0, and $\exp_p(tv) = \gamma_v(t)$.

Proposition 40.6.1 (The geodesic spray and the Sasaki metric)**.** *Let (M, g) be a Riemannian manifold, $\pi\colon TM \to M$, and $T(TM) = H \oplus V$ the splitting into the horizontal distribution of the Levi-Civita connection and the vertical subbundle (Proposition 38.7.3), with connection map K.*

1. *The vector field G on TM with $G(v) = v^h_v$, the horizontal lift of $v \in T_pM$ to the point v itself, is smooth, and its integral curves are the velocity curves $t \mapsto \dot\gamma(t)$ of the geodesics. So the flow of G, the* geodesic flow*, is $\Phi_t(v) = \dot\gamma_v(t)$.*

2. *$g_S(A, B) = g(d\pi\, A, d\pi\, B) + g(KA, KB)$ is a Riemannian metric on TM, the* Sasaki metric*, for which $H \perp V$, $d\pi\colon H_v \to T_pM$ and $K\colon V_v \to T_pM$ are isometries; in particular π is a* Riemannian submersion*.*

Proof. (1) In coordinates (x, v) on TM, by Proposition 38.7.3(1), $G(x, v) = (v, -\theta_x(v)v) = \sum_k v^k\partial_{x^k} - \sum_{i,j,k}\Gamma^k_{ij}v^iv^j\partial_{v^k}$, which is smooth. A curve $u(t) = (x(t), v(t))$ is an integral curve if and only if $\dot x = v$ and $\dot v^k = -\sum \Gamma^k_{ij}v^iv^j$, that is, $\ddot x^k + \sum \Gamma^k_{ij}\dot x^i\dot x^j = 0$ with $v = \dot x$: the geodesic equation of Definition 40.2.9. Equivalently, by Proposition 38.7.3(3), the curve $\dot\gamma$ over γ is horizontal if and only if it is parallel, $D_t\dot\gamma = 0$, and its projection has velocity $\dot\gamma$ exactly when it is an integral curve of G.

(2) g_S is symmetric and smooth. If $g_S(A, A) = 0$ then $d\pi\, A = 0$, so A is vertical, and $KA = 0$, so $A = 0$ (K is injective on V, being the inverse of the vertical lift); hence g_S is positive definite. For $A \in H$ and $B \in V$, $KA = 0$ and $d\pi\, B = 0$, so $g_S(A, B) = 0$. The other statements follow from the definition. □

So geodesics are the projections of the integral curves of a single vector field on TM, so their properties (existence, uniqueness, smooth dependence) are those of flows (Theorem 33.9.2); on T^*M the same flow is Hamiltonian (Chapter 41).

Proposition 40.6.2. *$d_0\exp_p = \mathrm{id}$, so $\exp_p$ is a diffeomorphism from a neighbourhood of 0 onto a neighbourhood of p, giving* normal coordinates *in which $g_{ij}(p) = \delta_{ij}$ and $\Gamma^k_{ij}(p) = 0$. Geodesics through p are the images of lines through 0.*

Proof. Since $\exp_p(tv) = \gamma_v(t)$, the differential of $\exp_p$ at 0 sends v to $\dot\gamma_v(0) = v$, and the inverse function theorem applies. In normal coordinates for an orthonormal basis of T_pM,

$g_{ij}(p) = \delta_{ij}$, and the lines $t \mapsto tv$ are geodesics, so the geodesic equation at $t = 0$ gives $\Gamma^k_{ij}(p)v^iv^j = 0$ for all v; as Γ^k_{ij} is symmetric in i, j, it vanishes at p. □

Theorem 40.6.3 (Gauss lemma)**.** *Let $v \in \mathcal{E}_p$ and $w \in T_pM$, and identify $T_v(T_pM)$ with T_pM. Then $\langle d_v \exp_p(v), d_v \exp_p(w)\rangle = \langle v, w\rangle$.*

Proof. Since $\mathcal{E}_p$ is open and star-shaped, $f(s,t) = \exp_p(t(v + sw))$ is defined and smooth for $|s|$ small and $t \in [0,1]$. Let $T = \partial_t f$ and $S = \partial_s f$. For fixed s, $t \mapsto f(s,t)$ is the geodesic with initial velocity $v + sw$, so $D_tT = 0$ and $|T|^2 = |v + sw|^2$ for all t. Using Proposition 40.4.1 and Lemma 40.4.3,

$$\partial_t\langle S, T\rangle = \langle D_tS, T\rangle + \langle S, D_tT\rangle = \langle D_sT, T\rangle = \tfrac{1}{2}\partial_s|T|^2 = \tfrac{1}{2}\partial_s|v + sw|^2 = \langle v + sw, w\rangle.$$

At $s = 0$ this is $\langle v, w\rangle$. Also $f(s,0) = p$, so $S(s,0) = 0$. Integrating, $\langle S, T\rangle(0,1) = \langle v, w\rangle$. Finally $T(0,1) = \frac{d}{dt}\exp_p(tv)|_{t=1} = d_v\exp_p(v)$ and $S(0,1) = \frac{d}{ds}\exp_p(v + sw)|_{s=0} = d_v\exp_p(w)$. □

Theorem 40.6.4 (Geodesics are locally minimising)**.** *Let $\varepsilon > 0$ be such that $\exp_p$ is a diffeomorphism from $B(0,\varepsilon) \subseteq T_pM$ onto an open set U (a* normal ball*).*

1. *For $|v| < \varepsilon$, every curve from p to $\exp_p(v)$ has length $\geq |v|$, with equality if and only if it is a monotone reparametrisation of the radial geodesic $t \mapsto \exp_p(tv)$, $t \in [0,1]$.*
2. *$d(p, \exp_p(v)) = |v|$ for $|v| < \varepsilon$, and $\exp_p(B(0,r))$ is the metric ball $\{x : d(p,x) < r\}$ for $0 < r \leq \varepsilon$.*

Proof. For $x \in U$ let $\rho(x) = |\exp_p^{-1}(x)|$, a continuous function, smooth on $U \setminus \{p\}$. Let $c\colon [a,b] \to U \setminus \{p\}$ be a piecewise smooth curve and write $c(t) = \exp_p(\rho(t)u(t))$ with $\rho(t) = \rho(c(t)) > 0$ and $|u(t)| = 1$. Then $\dot c = d\exp_p(\dot\rho u + \rho\dot u)$ at the point ρu, with $\langle u, \dot u\rangle = 0$. By the Gauss lemma, $d\exp_p(u)$ has length 1 and is orthogonal to $d\exp_p(\rho\dot u)$ (both at ρu), so

$$|\dot c|^2 = \dot\rho^2 + |d\exp_p(\rho\dot u)|^2 \geq \dot\rho^2, \qquad \text{hence} \qquad L(c) \geq \int_a^b |\dot\rho|\,dt \geq \rho(b) - \rho(a). \tag{40.2}$$

Now let $c\colon [a,b] \to M$ go from p to $q = \exp_p(v)$, $0 < |v| = r < \varepsilon$. Suppose first that c stays in U. Let a' be the last time with $c(a') = p$; applying (40.2) on $[a'', b]$ for $a'' \downarrow a'$ gives $L(c) \geq r$. If $L(c) = r$, then c is constant on $[a, a']$, and on $(a', b]$ equality holds in (40.2): $\dot\rho \geq 0$ and $d\exp_p(\rho\dot u) = 0$, so $\dot u = 0$ ($d\exp_p$ being injective on $B(0,\varepsilon)$), u is constant and c is a monotone reparametrisation of the radial geodesic. Conversely such reparametrisations have length r. If c leaves U, choose $r < r' < \varepsilon$. The set $K = \exp_p(\bar B(0,r'))$ is compact, hence closed, and contains p; so c leaves K, and by continuity $\rho(c(t))$ takes the value r' at some first time t_2 with $c([a,t_2]) \subseteq K$. The first case applied to $c|_{[a,t_2]}$ gives $L(c) \geq r' > r$. This proves (1).

For (2), the radial geodesic has length $|v|$, so $d(p, \exp_p(v)) = |v|$ by (1). The ball $\exp_p(B(0,r))$ lies in the metric ball of radius r. Conversely, if $d(p,x) < r$, choose a curve c from p to x with $L(c) < r' < r$; if $x \notin \exp_p(B(0,r'))$, the argument above shows $L(c) \geq r'$, a contradiction. □

Remark 40.6.5. Every point also has a *geodesically convex* neighbourhood, in which any two points are joined by a unique minimising geodesic, contained in the neighbourhood [dC92, Chapter 3].

Lemma 40.6.6 (Uniformly normal neighbourhoods)**.** *Every $q \in M$ has a neighbourhood W and a $\delta > 0$ such that for every $x \in W$, $\exp_x$ is defined on $B(0,\delta) \subseteq T_xM$, is a diffeomorphism from it onto an open set, and $\exp_x(B(0,\delta)) \supseteq W$.*

Proof. Consider $E\colon \mathcal{E} \to M \times M$, $E(v) = (\pi(v), \exp(v))$. In coordinates (x, v) near $0_q \in TM$, its differential at 0_q is $\begin{pmatrix} I & 0 \\ I & I \end{pmatrix}$, since $\exp(0_x) = x$ and $d_0 \exp_x = \mathrm{id}$; so by the inverse function theorem E is a diffeomorphism from a neighbourhood O of 0_q onto a neighbourhood of (q, q). Shrinking, we may take $O = \{v \in T_xM : x \in W', |v| < \delta\}$ for a neighbourhood W' of q and some $\delta > 0$ (the norm is continuous on TM), and choose a neighbourhood $W \subseteq W'$ of q with $W \times W \subseteq E(O)$. For $x \in W$, $\exp_x$ is injective on $B(0, \delta)$ with invertible differential, hence a diffeomorphism onto an open set; and for $y \in W$, $(x, y) = E(v)$ with $v \in O$, that is $y = \exp_x(v)$ with $|v| < \delta$. □

Lemma 40.6.7. *Let $c\colon [a, b] \to M$ be a piecewise smooth curve of constant speed with $L(c) = d(c(a), c(b))$. Then c is a geodesic; in particular it is smooth.*

Proof. Every subarc of c is also minimising, since otherwise replacing it by a shorter curve would shorten c. Let $t_0 \in [a, b]$, and take W and δ for $q = c(t_0)$ as in Lemma 40.6.6. For $t_1 < t_0 < t_2$ close to t_0 (or $t_1 = t_0$ at the endpoints), $c([t_1, t_2]) \subseteq W$ and $L(c|_{[t_1,t_2]}) < \delta$. Then $c|_{[t_1,t_2]}$ is a minimising curve from $x = c(t_1)$ to a point of the normal ball $\exp_x(B(0, \delta))$, so by Theorem 40.6.4(1) it is a monotone reparametrisation of a radial geodesic from x, and having constant speed, it is that geodesic, affinely reparametrised. So c is a geodesic near every t_0. □

Theorem 40.6.8 (Hopf–Rinow)**.** *For a connected Riemannian manifold (M, g) the following are equivalent: (M, d) is a complete metric space; the geodesics are defined for all time; closed bounded subsets are compact. In that case any two points are joined by a minimising geodesic. In particular compact Riemannian manifolds are geodesically complete.*

Proof. We show more: the three conditions are also equivalent to (d) $\exp_p$ *is defined on all of T_pM for some p*; and (d) implies that every q is joined to p by a minimising geodesic.

Step 1: (d) gives minimising geodesics from p. Let $q \neq p$, $r = d(p, q)$, and choose $0 < \delta < r$ with $\exp_p$ a diffeomorphism on $B(0, \delta')$ for some $\delta' > \delta$. The *geodesic sphere* $S = \exp_p(\{|w| = \delta\})$ is compact; let $x_0 = \exp_p(\delta u) \in S$, $|u| = 1$, minimise $d(\cdot, q)$ on S. Every curve from p to q meets S (by the argument in the proof of Theorem 40.6.4), and points of S have distance δ from p, so $r = \delta + d(x_0, q)$. Let $\gamma(t) = \exp_p(tu)$, defined for all t by (d), and let $T = \{t \in [\delta, r] : d(\gamma(t), q) = r - t\}$, a closed set containing δ. Let $t_0 = \max T$ and suppose $t_0 < r$. Apply the same construction at $\gamma(t_0)$: for small $\eta > 0$ there is x_1 on the geodesic sphere of radius η about $\gamma(t_0)$ with $d(\gamma(t_0), q) = \eta + d(x_1, q)$, so $d(x_1, q) = r - t_0 - \eta$ and $d(p, x_1) \geq d(p, q) - d(x_1, q) = t_0 + \eta$. The curve going along γ from p to $\gamma(t_0)$ and then along the radial geodesic to x_1 has length $t_0 + \eta$, so it is minimising, and by Lemma 40.6.7 (after parametrising by arc length) it is a geodesic, hence equal to γ on $[0, t_0 + \eta]$. So $x_1 = \gamma(t_0 + \eta)$ and $t_0 + \eta \in T$, a contradiction. Hence $t_0 = r$, $\gamma(r) = q$ and $L(\gamma|_{[0,r]}) = r$.

Step 2: the equivalences. If closed bounded sets are compact, a Cauchy sequence, being bounded, has a convergent subsequence, hence converges; so M is complete. If M is complete, let $\gamma\colon [0, b) \to M$ be a maximal unit-speed geodesic with $b < \infty$. For $t_k \uparrow b$, $d(\gamma(t_k), \gamma(t_l)) \leq |t_k - t_l|$, so $(\gamma(t_k))$ is Cauchy and converges to some q. Take W, δ for q as in Lemma 40.6.6; then geodesics with initial point in W and speed 1 are defined on $(-\delta, \delta)$. Choose $t_k > b - \delta$ with $\gamma(t_k) \in W$; the geodesic with initial velocity $\dot\gamma(t_k)$ is defined up to time $t_k + \delta > b$ and extends γ by uniqueness, a contradiction. So geodesics are defined for all time, which implies (d). Finally, assume (d) and let K be closed and bounded, say $K \subseteq \{x : d(p, x) \leq R\}$. By Step 1, this ball is contained in $\exp_p(\bar B(0, R))$, which is compact; so K is compact. The last statements follow from Step 1 applied at every point, and from the compactness of closed bounded subsets of a compact space. □

Remark 40.6.9. The proof of Step 2 shows that in *any* metric space the Heine–Borel property (closed bounded sets are compact) implies completeness. The converse is false for general metric spaces: $\mathbb{R}$ with the complete metric $\min(|x-y|,1)$ is bounded and closed in itself but not compact, and the closed unit ball of an infinite-dimensional Hilbert space is closed, bounded and complete but not compact (Chapter 30). What the Hopf–Rinow theorem adds is that for the distance of a Riemannian metric, which is defined by lengths of curves on a locally compact space, completeness does imply the Heine–Borel property; and geodesic completeness, the second condition, has no meaning in a general metric space.

Counterexample 40.6.10. Completeness is needed. In $M = \mathbb{R}^2 \setminus \{0\}$ with the Euclidean metric, the geodesic $t \mapsto (1-t, 0)$ is not defined at $t = 1$, the Cauchy sequence $(1/k, 0)$ has no limit, and the set $\{x \in M : |x| \le 1\}$ is closed in M and bounded but not compact. The points $(1,0)$ and $(-1,0)$ have distance 2, the infimum of the lengths of curves avoiding the origin, but no curve of length 2 joins them in M, since in $\mathbb{R}^2$ the only such curve is the segment through the origin.

Example 40.6.11. *The exponential of a Lie group and the Riemannian exponential.* Let G be a Lie group with a bi-invariant metric, for instance a compact group (Example 40.2.5). Then the geodesics through e are the one-parameter subgroups, and the Riemannian exponential $\exp_e \colon T_eG \to G$ is the exponential map of the Lie group (Theorem 34.3.3). Indeed, for left-invariant fields X, Y, Z the functions $g(Y,Z)$ etc. are constant, and the metric on $\mathfrak{g}$ is Ad-invariant, since $\mathrm{Ad}(h) = d(L_h \circ R_{h^{-1}})$ is the differential of an isometry; differentiating, as in the proof of Proposition 34.5.4, gives $g([A,B],C) = -g(B,[A,C])$; the Koszul formula (Theorem 40.2.8) then reduces to $2g(\nabla_X Y, Z) = g([X,Y],Z) - g([X,Z],Y) - g([Y,Z],X) = g([X,Y],Z)$, that is,

$$\nabla_X Y = \tfrac{1}{2}[X,Y].$$

In particular $\nabla_X X = 0$, so the integral curves of X, which are $t \mapsto g\exp(tX)$, are geodesics. For a left-invariant metric that is not bi-invariant this fails. On the group of affine maps $x \mapsto ax + b$ $(a > 0)$, identified with the half plane $\{(a,b) : a > 0\}$ with the product $(a,b)(a',b') = (aa', ab' + b)$, left translations are the maps $(a',b') \mapsto (aa', ab' + b)$, which preserve the hyperbolic metric $(da^2 + db^2)/a^2$; so this metric is left-invariant. The one-parameter subgroup $c(t) = (e^t, e^t - 1)$ is not a geodesic: the Christoffel symbols of this metric are $\Gamma^a_{aa} = -\frac{1}{a}$, $\Gamma^a_{bb} = \frac{1}{a}$, $\Gamma^b_{ab} = \Gamma^b_{ba} = -\frac{1}{a}$ and 0 otherwise, and the first geodesic equation $\ddot{a} - \dot{a}^2/a + \dot{b}^2/a = 0$ gives $e^t \neq 0$ along c.

Example 40.6.12. *Geodesics and parallel transport in the model spaces.*

1. *The sphere.* Along a great circle $\gamma(t) = \cos t\, e_1 + \sin t\, e_2$ in S^2, with the projection connection, $D_t\dot\gamma = P(\ddot\gamma) = P(-\gamma) = 0$, so great circles are geodesics; and the constant field e_3, orthogonal to the plane of the circle, is tangent to S^2 along γ with $D_t e_3 = P(0) = 0$. So parallel transport along the great circle by an angle θ is the rotation of $\mathbb{R}^3$ by θ about e_3, restricted to the tangent planes: it keeps the angle with the circle constant. Since through every point and in every direction there is a great circle, these are all the geodesics (by uniqueness, Definition 40.2.9).

2. *The hyperbolic plane.* For $(dx^2 + dy^2)/y^2$ on $\{y > 0\}$, the formula of Theorem 40.2.8 gives $\Gamma^x_{xy} = \Gamma^x_{yx} = \Gamma^y_{yy} = -\frac{1}{y}$, $\Gamma^y_{xx} = \frac{1}{y}$ and the others 0, so the geodesic equations are

$$\ddot{x} - \tfrac{2}{y}\dot{x}\dot{y} = 0, \qquad \ddot{y} + \tfrac{1}{y}(\dot{x}^2 - \dot{y}^2) = 0.$$

   The vertical lines $t \mapsto (c, e^t)$ and the semicircles $t \mapsto (c + r\tanh t, r\,\mathrm{sech}\, t)$ centred on the real axis solve them, with unit speed: for the semicircle, $\dot{x} = r\,\mathrm{sech}^2 t$, $\dot{y} =$

$-r\operatorname{sech} t\tanh t$, $\dot{x}^2+\dot{y}^2 = r^2\operatorname{sech}^2 t = y^2$, and substituting, both equations hold (for instance $\ddot{x} = -2r\operatorname{sech}^2 t\tanh t = \frac{2}{y}\dot{x}\dot{y}$). Through every point and in every direction passes exactly one vertical line or semicircle centred on the axis (the centre is where the normal to the direction meets the axis), so by uniqueness these are all the geodesics. In particular $d(i, iy) = |\log y|$, the length of the vertical segment, which is minimising since the projection $(x, y) \mapsto (0, y)$ does not increase lengths.

Remark 40.6.13. Minimising geodesics need not be unique: on the round sphere every half great circle joins two antipodal points, and has length π. And geodesics stop minimising: a great circle arc longer than π is a geodesic but not the shortest curve between its endpoints. Both phenomena are explained by conjugate points (Section 40.8).

40.7 Parallel transport and holonomy

The *holonomy group* $\mathrm{Hol}_p(g) \subseteq \mathrm{GL}(T_pM)$ of (M, g) at p is the holonomy group of the Levi-Civita connection (Definition 38.5.9), for piecewise smooth loops at p. The *restricted holonomy group* $\mathrm{Hol}_p^0(g)$ is the subgroup of transports along null-homotopic loops.

Proposition 40.7.1. *Let (M, g) be connected and $p, q \in M$.*

1. $\mathrm{Hol}_p(g) \subseteq \mathrm{O}(T_pM, g_p)$.
2. *For a curve σ from p to q, $\mathrm{Hol}_q(g) = P_\sigma \mathrm{Hol}_p(g) P_\sigma^{-1}$; so the holonomy groups at different points are conjugate.*
3. $\mathrm{Hol}_p^0(g)$ *is a normal subgroup of* $\mathrm{Hol}_p(g)$, *and* $[\gamma] \mapsto P_\gamma^{-1}\mathrm{Hol}_p^0(g)$ *is a surjective homomorphism* $\pi_1(M, p) \to \mathrm{Hol}_p(g)/\mathrm{Hol}_p^0(g)$. *In particular* $\mathrm{Hol}_p = \mathrm{Hol}_p^0$ *if M is simply connected.*
4. *M is orientable if and only if* $\mathrm{Hol}_p(g) \subseteq \mathrm{SO}(T_pM)$.
5. $\mathrm{Hol}_p^0(g)$ *is trivial if and only if g is flat, $R = 0$.*

Proof. (1) is Proposition 40.4.4. (2) Loops γ at p correspond bijectively to loops $\bar{\sigma} * \gamma * \sigma$ at q, with transport $P_\sigma P_\gamma P_\sigma^{-1}$ (Proposition 38.5.3). (3) If β is null-homotopic and α is a loop at p, then $\bar{\alpha} * \beta * \alpha$ is null-homotopic, with transport $P_\alpha P_\beta P_\alpha^{-1}$; so Hol_p^0 is normal. If $\gamma \simeq \gamma'$, then $\bar{\gamma} * \gamma'$ is null-homotopic and $P_{\gamma'} P_\gamma^{-1} \in \mathrm{Hol}_p^0$, so the map is well defined on homotopy classes, and it is multiplicative since $P_{\gamma*\delta}^{-1} = P_\gamma^{-1} P_\delta^{-1}$. Surjectivity is clear. (4) If M is oriented, a parallel frame along a curve is continuous, so the set of times at which it is positively oriented is open and closed; hence parallel transport preserves orientations. Conversely, if $\mathrm{Hol}_p \subseteq \mathrm{SO}(T_pM)$, fix an orientation of T_pM and transport it along curves from p; by (2) the result does not depend on the curve, and near each point the transported frames along radial geodesics form a smooth local frame, so the orientation is continuous. (5) By Theorem 38.5.10, $R = 0$ if and only if parallel transport depends only on the homotopy class of the path with fixed endpoints, which is equivalent to the triviality of transport around null-homotopic loops. □

Theorem 40.7.2 (Holonomy principle)**.** *Let (M, g) be connected and let E be a tensor bundle $T^{\otimes a} \otimes (T^*)^{\otimes b}$ of M with the connection induced by ∇. A tensor $\tau_p \in E_p$ is the value at p of a parallel section of E if and only if it is invariant under* $\mathrm{Hol}_p(g)$, *acting on E_p through its action on T_pM; the parallel section is then unique. Similarly, a subspace $V \subseteq T_pM$ is the fibre at p of a parallel subbundle of TM if and only if it is* $\mathrm{Hol}_p(g)$*-invariant.*

Proof. Parallel transport on E is induced by that on TM. If τ is parallel, then along every loop γ at p the section $\tau \circ \gamma$ is parallel, so $P_\gamma \tau_p = \tau_p$. Conversely, if τ_p is invariant, define $\tau_q = P_\sigma \tau_p$ for any curve σ from p to q. If σ' is another such curve, $P_{\sigma'}^{-1} P_\sigma$ is the transport along the loop $\sigma * \bar{\sigma}'$ at p, which lies in Hol_p and fixes τ_p; so τ_q is well defined. Near a point q_0, τ_q is the transport of τ_{q_0} along the radial geodesic from q_0 to q in a normal ball, which depends smoothly on q by smooth dependence of solutions of differential equations; so τ is smooth. For every curve c, $\tau \circ c$ is the transport of $\tau_{c(0)}$ along c, hence parallel; so $\nabla \tau = 0$. Uniqueness holds because a parallel section is determined by its value at one point of a connected manifold. The statement for subbundles is proved in the same way. □

Example 40.7.3.

1. On $\mathbb{R}^n$ the Christoffel symbols vanish in the standard coordinates, so parallel transport is the identity and $\mathrm{Hol} = \{1\}$; the same holds for the flat torus, since parallel transport in the quotient is induced from $\mathbb{R}^n$ and translations have trivial differential.

2. For the flat Klein bottle $\mathbb{R}^2/G$ (Example 40.2.7), a loop at $\pi(\tilde{x})$ lifts to a curve from $\tilde{x}$ to $h\tilde{x}$ for some $h \in G$; transport in $\mathbb{R}^2$ is the identity, and $d\pi_{h\tilde{x}} = d\pi_{\tilde{x}} \circ (dh)^{-1}$, so the holonomy is $(dh)^{-1}$, the inverse of the linear part of h. Hence $\mathrm{Hol} = \{1, \mathrm{diag}(1,-1)\}$: the Klein bottle is flat, $\mathrm{Hol}^0 = \{1\}$, but $\mathrm{Hol} \neq \{1\}$, and $\mathrm{Hol} \not\subseteq \mathrm{SO}(2)$, in accordance with non-orientability.

3. For the round sphere S^2, $\mathrm{Hol}_p = \mathrm{Hol}_p^0$ is a nontrivial subgroup of $\mathrm{SO}(2)$ by Proposition 40.7.1(3),(4),(5), as S^2 is simply connected, orientable and not flat. Since Hol_p^0 is a connected Lie subgroup (Theorem 40.7.4 below), it is $\mathrm{SO}(2)$. Concretely, transport around the boundary of the octant $\{x, y, z \geq 0\}$ rotates tangent vectors by $\pi/2$, the area enclosed.

Theorem 40.7.4 (Ambrose–Singer)**.** *$\mathrm{Hol}_p(g)$ is a Lie subgroup of $\mathrm{O}(T_pM)$ with identity component $\mathrm{Hol}_p^0(g)$, and the Lie algebra of $\mathrm{Hol}_p(g)$ is spanned by the endomorphisms $P_\sigma^{-1} \circ R(P_\sigma u, P_\sigma v) \circ P_\sigma$ of T_pM, for all curves σ starting at p and all $u, v \in T_pM$.*

This is [KN63, Chapter II]; see also [Joy00]. The theorem makes Proposition 40.7.1(5) quantitative: curvature measures infinitesimal holonomy.

Theorem 40.7.5 (de Rham decomposition)**.** *If the holonomy representation of a connected Riemannian manifold on T_pM is reducible, $T_pM = V_1 \oplus V_2$ with both summands Hol_p-invariant and nonzero, then M is locally isometric to a Riemannian product. If moreover M is complete and simply connected, it is globally isometric to a product $M_1 \times M_2$ of Riemannian manifolds with $T_pM_i = V_i$.*

Sketch of proof. By Theorem 40.7.2, V_1 and V_2 extend to parallel, mutually orthogonal distributions D_1 and D_2. A parallel distribution is involutive, as $[X, Y] = \nabla_X Y - \nabla_Y X$ lies in D_i when X, Y do, so by the Frobenius theorem (Theorem 33.9.8) there are local coordinates (x, y) with D_1 spanned by the ∂_x and D_2 by the ∂_y; the omitted verification is that in these coordinates the metric has the product form $g_1(x) + g_2(y)$, which follows from the parallelism of D_1 and D_2. The global statement is [KN63, Chapter IV]. □

Theorem 40.7.6 (Berger)**.** *Let (M, g) be a simply connected Riemannian manifold of dimension n that is not locally a product (its holonomy representation is irreducible) and not locally symmetric ($\nabla R \neq 0$). Then $\mathrm{Hol}(g)$ is one of: $\mathrm{SO}(n)$; $\mathrm{U}(m)$ with $n = 2m \geq 4$ (Kähler metrics); $\mathrm{SU}(m)$ with $n = 2m \geq 4$ (Ricci-flat Kähler, Calabi–Yau); $\mathrm{Sp}(m)$ with $n = 4m \geq 8$ (hyper-Kähler); $\mathrm{Sp}(m)\mathrm{Sp}(1)$ with $n = 4m \geq 8$ (quaternionic Kähler); G_2 with $n = 7$; $\mathrm{Spin}(7)$ with $n = 8$.*

This is [Ber55]; Berger's list also contained Spin(9) in dimension 16, which was later shown to occur only for symmetric metrics. See [Joy00] for the modern treatment and for the existence of metrics with holonomy G_2 and Spin(7).

Remark 40.7.7. By the holonomy principle, the reduction of the holonomy group to a subgroup of $\mathrm{O}(n)$ is the same as the existence of a parallel tensor with that stabiliser. For instance $\mathrm{Hol} \subseteq \mathrm{U}(m)$ if and only if there is a parallel orthogonal almost complex structure J, that is, (M, g, J) is Kähler (Chapter 47); and $\mathrm{Hol} \subseteq \mathrm{SU}(m)$ if and only if there is moreover a parallel complex volume form, which for a simply connected Kähler manifold is equivalent to Ricci-flatness (Chapter 52). This is the dictionary between Riemannian holonomy and complex geometry that underlies Part VII.

40.8 Curvature

Definition 40.8.1. The *Riemann curvature tensor* is the curvature of the Levi-Civita connection (Proposition 38.4.3), $R(X,Y)Z = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z$, viewed as the 4-tensor $R(X,Y,Z,W) = g(R(X,Y)Z,W)$. For a plane $\sigma \subseteq T_pM$ with orthonormal basis u, v, the *sectional curvature* is $K(\sigma) = R(u,v,v,u)$; the *Ricci tensor* is $\mathrm{Ric}(X,Y) = \sum_i R(e_i,X,Y,e_i)$ for an orthonormal frame, and the *scalar curvature* is $s = \sum_i \mathrm{Ric}(e_i,e_i)$.

Proposition 40.8.2. *R is tensorial and satisfies $R(X,Y,Z,W) = -R(Y,X,Z,W) = -R(X,Y,W,Z) = R(Z,W,X,Y)$ and the first Bianchi identity $R(X,Y)Z + R(Y,Z)X + R(Z,X)Y = 0$. The sectional curvatures determine R.*

Proof. Tensoriality and the first antisymmetry are general facts about curvature. The second follows by polarising $R(X,Y,Z,Z) = 0$, which holds because the connection is metric: $0 = XY\,g(Z,Z)/2 - \cdots$ expands to $g(R(X,Y)Z,Z)$. The first Bianchi identity follows from torsion-freeness and the Jacobi identity, and the symmetry in the pairs is a formal consequence of the other three, obtained by adding the Bianchi identities for the four cyclic permutations. The last statement is the standard polarisation argument [dC92, Chapter 4]. □

Example 40.8.3. Euclidean space has $R = 0$; the round sphere of radius 1 has constant sectional curvature 1 and $\mathrm{Ric} = (n-1)g$; hyperbolic space has constant curvature -1. A Lie group with a bi-invariant metric has $R(X,Y)Z = \tfrac{1}{4}[[X,Y],Z]$ for left-invariant fields, hence non-negative sectional curvature. A metric has constant sectional curvature κ if and only if $R(X,Y)Z = \kappa(\langle Y,Z\rangle X - \langle X,Z\rangle Y)$, since the right-hand side has the symmetries of Proposition 40.8.2 and sectional curvature κ.

Proposition 40.8.4. *Let (M,g) have dimension n and constant sectional curvature κ. Then*

$$\mathrm{Ric} = (n-1)\kappa\, g, \qquad s = n(n-1)\kappa.$$

For instance the unit sphere has $\mathrm{Ric} = (n-1)g$ and $s = n(n-1)$, and hyperbolic space has $\mathrm{Ric} = -(n-1)g$. Every surface ($n = 2$), constant curvature or not, has $\mathrm{Ric} = Kg$ and $s = 2K$, where K is the Gaussian curvature.

Proof. Let (e_i) be an orthonormal basis of T_pM. By the formula of Example 40.8.3,

$$\mathrm{Ric}(X,Y) = \sum_i \langle R(e_i,X)Y, e_i\rangle = \kappa \sum_i \big(\langle X,Y\rangle\langle e_i,e_i\rangle - \langle e_i,Y\rangle\langle X,e_i\rangle\big) = \kappa\big(n\langle X,Y\rangle - \langle X,Y\rangle\big),$$

using $\sum_i \langle e_i,Y\rangle\langle X,e_i\rangle = \langle X,Y\rangle$ (expand X in the basis). Taking the trace once more gives $s = \sum_i (n-1)\kappa = n(n-1)\kappa$. For a surface, the only plane in T_pM is T_pM itself, so the sectional curvature is constant on each tangent space, equal to $K(p)$, and the computation applies pointwise with $n = 2$. □

Remark 40.8.5. The sphere of radius r has constant curvature $1/r^2$ (Section 40.9 below), so its Ricci tensor is $\frac{n-1}{r^2}g$. In a space of constant curvature κ, the Ricci curvature in the direction of a unit vector v is $\mathrm{Ric}(v,v) = (n-1)\kappa$: it is the sum of the $n-1$ sectional curvatures of the planes spanned by v and the vectors of an orthonormal basis of $v^\perp$, each equal to κ. A metric with $\mathrm{Ric} = \lambda g$ for a constant λ is called *Einstein*; so constant curvature metrics are Einstein. The converse fails in general: for the product $S^2 \times S^2$ of two unit spheres, the Koszul formula shows that the Levi-Civita connection is the product of the connections of the factors, so the curvature tensor is the sum of the curvature tensors of the factors. Hence a plane spanned by a vector tangent to the first factor and one tangent to the second has sectional curvature 0, while $\mathrm{Ric} = g$, each factor contributing $\mathrm{Ric} = g$ on its own tangent vectors. In dimension 2 the Ricci tensor carries exactly the same information as the Gaussian curvature.

Lemma 40.8.6. *For f as in Lemma 40.4.3 and a vector field V along f, $D_sD_tV - D_tD_sV = R(\partial_s f, \partial_t f)V$.*

Proof. Write $V = V^k(\partial_k \circ f)$. In D_sD_tV the terms containing derivatives of the V^k are symmetric in s and t, so the left side is $V^k(D_sD_t - D_tD_s)(\partial_k \circ f)$. Now $D_t(\partial_k \circ f) = \partial_t f^i (\nabla_{\partial_i}\partial_k) \circ f$, and $D_sD_t(\partial_k \circ f) = \partial_s\partial_t f^i (\nabla_{\partial_i}\partial_k)\circ f + \partial_t f^i \partial_s f^j (\nabla_{\partial_j}\nabla_{\partial_i}\partial_k)\circ f$. Antisymmetrising and using $[\partial_i, \partial_j] = 0$ gives $\partial_s f^j \partial_t f^i R(\partial_j, \partial_i)\partial_k = R(\partial_s f, \partial_t f)\partial_k$. □

Definition 40.8.7. A *Jacobi field* along a geodesic γ is a vector field J along γ with $D_t^2 J + R(J, \dot\gamma)\dot\gamma = 0$. A point $\gamma(b)$ is *conjugate* to $\gamma(a)$ along γ if there is a nonzero Jacobi field vanishing at a and b.

The Jacobi equation is a linear second order equation, so a Jacobi field is determined by $J(a)$ and $D_tJ(a)$, and the Jacobi fields along γ form a vector space of dimension $2n$.

Proposition 40.8.8.

1. *If $f(s,t)$ is a family of geodesics $t \mapsto f(s,t)$, then $J = \partial_s f(0,-)$ is a Jacobi field along $\gamma = f(0,-)$.*
2. *For $v \in \mathcal{E}_p$ and $w \in T_pM$, $J(t) = d_{tv}\exp_p(tw)$ is the Jacobi field along $\gamma(t) = \exp_p(tv)$ with $J(0) = 0$ and $D_tJ(0) = w$.*
3. *$d_v\exp_p$ is singular if and only if $\exp_p(v)$ is conjugate to p along $t \mapsto \exp_p(tv)$, $t \in [0,1]$.*

Proof. (1) With $T = \partial_t f$ and $S = \partial_s f$, Lemmas 40.4.3 and 40.8.6 give $D_tD_tS = D_tD_sT = D_sD_tT + R(T,S)T = R(T,S)T = -R(S,T)T$, since $D_tT = 0$. (2) Apply (1) to $f(s,t) = \exp_p(t(v+sw))$: $\partial_s f(0,t) = d_{tv}\exp_p(tw)$. It vanishes at $t = 0$, and $D_tS(0,0) = D_sT(0,0) = \frac{d}{ds}(v+sw) = w$, the covariant derivative along the constant curve $s \mapsto p$ being the ordinary derivative in T_pM. (3) By (2) and uniqueness, the Jacobi fields vanishing at 0 are exactly the $d_{tv}\exp_p(tw)$, and at $t = 1$ this is $d_v\exp_p(w)$. □

Example 40.8.9. If g has constant sectional curvature κ and γ has unit speed, then for a parallel unit field $E \perp \dot\gamma$, $R(E,\dot\gamma)\dot\gamma = \kappa E$ by Example 40.8.3, and the Jacobi fields normal to γ with $J(0) = 0$ are $J(t) = s_\kappa(t)E(t)$ with $s_\kappa(t) = t$, $\sin(\sqrt{\kappa}t)/\sqrt{\kappa}$ or $\sinh(\sqrt{-\kappa}t)/\sqrt{-\kappa}$ for $\kappa = 0$, > 0, < 0. On the unit sphere they vanish again at $t = \pi$: antipodal points are conjugate, which explains Remark 40.6.13. For $\kappa \le 0$ there are no conjugate points.

Theorem 40.8.10 (Cartan–Hadamard)**.** *Let (M,g) be complete and connected with sectional curvature $K \le 0$. Then for every p, $\exp_p : T_pM \to M$ is a covering map; if M is simply connected, it is a diffeomorphism, so M is diffeomorphic to $\mathbb{R}^n$.*

Sketch of proof. Let J be a Jacobi field along a geodesic γ with $J(0) = 0$ and $D_tJ(0) \neq 0$, and $\phi = |J|^2$. Then $\phi'' = 2|D_tJ|^2 - 2\langle R(J,\dot\gamma)\dot\gamma, J\rangle = 2|D_tJ|^2 - 2K(J,\dot\gamma)(|J|^2|\dot\gamma|^2 - \langle J,\dot\gamma\rangle^2) \geq 0$, where $K(J,\dot\gamma)$ is the sectional curvature of the plane they span (if they are independent). So ϕ is convex with $\phi(0) = \phi'(0) = 0$ and $\phi''(0) > 0$, hence positive for $t > 0$: there are no conjugate points, and $\exp_p$ is a local diffeomorphism on T_pM (defined everywhere by Theorem 40.6.8) by Proposition 40.8.8(3). The omitted verification is that a local diffeomorphism from a complete Riemannian manifold (here T_pM with the pulled back metric, whose radial lines are geodesics) that is a local isometry is a covering map [dC92, Chapter 7]. □

Theorem 40.8.11 (Gauss–Bonnet)**.** *Let M be a compact oriented Riemannian surface, with Gaussian curvature K (the sectional curvature of its tangent planes) and area form dA (Proposition 40.3.3). Then*

$$\int_M K\,dA = 2\pi\chi(M).$$

Proof. Let J be the rotation by $+\pi/2$ in each tangent plane, for the metric and the orientation: $Je_1 = e_2$ for every oriented orthonormal frame (e_1, e_2). It is a smooth endomorphism with $J^2 = -1$, so TM becomes a complex line bundle, with i acting as J, whose underlying oriented real bundle is TM. Near a point choose an oriented orthonormal frame (e_1, e_2) and define the 1-form ω by $\nabla_X e_1 = \omega(X)e_2$; since ∇ is metric, $\langle \nabla_X e_1, e_1\rangle = 0$ and $\nabla_X e_2 = -\omega(X)e_1$. Hence $\nabla J = J\nabla$, so ∇ is a connection on the complex line bundle, and in the complex frame e_1 it reads $\nabla e_1 = \omega \otimes Je_1 = i\omega \otimes e_1$: its connection form is $\theta = i\omega$ and its curvature is $F = d\theta = i\,d\omega$ (Proposition 38.4.3).

We compare $d\omega$ with K. From the definition of the curvature tensor,

$$R(X,Y)e_1 = \nabla_X(\omega(Y)e_2) - \nabla_Y(\omega(X)e_2) - \omega([X,Y])e_2 = \big(X\omega(Y) - Y\omega(X) - \omega([X,Y])\big)e_2 = d\omega(X,Y)\,e_2,$$

the terms $\mp\omega(X)\omega(Y)e_1$ cancelling. By Definition 40.8.1 and Proposition 40.8.2, $K = R(e_1,e_2,e_2,e_1) = -R(e_1,e_2,e_1,e_2) = -\langle R(e_1,e_2)e_1, e_2\rangle = -d\omega(e_1,e_2)$. So $d\omega = -K\,e^1 \wedge e^2 = -K\,dA$ and $F = -iK\,dA$.

By Chern–Weil theory (Theorem 39.2.4 and Definition 39.2.5), $c_1(TM)$ is represented by $\frac{i}{2\pi}F = \frac{1}{2\pi}K\,dA$ for this connection. For a complex line bundle the Euler class of the underlying oriented real bundle is c_1 (Theorem 39.5.1(2)), and $\int_M e(TM) = \chi(M)$ by Theorem 39.5.1(3). Hence $\frac{1}{2\pi}\int_M K\,dA = \chi(M)$. □

Corollary 40.8.12. *Let Σ_g be the closed orientable surface of genus g, so that $\chi(\Sigma_g) = 2 - 2g$ (Example 21.7.5).*

1. *The integral of the Gaussian curvature does not depend on the metric: for the round unit sphere, $K = 1$ and the area is $4\pi = 2\pi \cdot 2$.*

2. *If Σ_g carries a metric with $K > 0$ everywhere, then $g = 0$; if it carries a metric with $K \geq 0$ everywhere, then $g \leq 1$; if it carries one with $K < 0$ everywhere, then $g \geq 2$.*

3. *The torus admits flat metrics, as $\mathbb{R}^2/\mathbb{Z}^2$, but no metric with $K > 0$ everywhere or $K < 0$ everywhere.*

Proof. (1) is Theorem 40.8.11. (2) The sign of $\int_M K\,dA = 2\pi(2-2g)$ is positive, non-negative or negative accordingly, and $2 - 2g > 0$ forces $g = 0$, $2 - 2g \geq 0$ forces $g \leq 1$, and $2 - 2g < 0$ forces $g \geq 2$. (3) The torus has $\chi = 0$, so K cannot have constant nonzero sign, while the flat metric induced from $\mathbb{R}^2$ has $K = 0$ (Example 40.8.3). □

40.9 Moving frames

The curvature of a metric given by an explicit formula can be computed from the Christoffel symbols, but this is laborious: there are $n^2(n+1)/2$ symbols, each a combination of derivatives of the g_{ij}, and the curvature involves their derivatives and their products. Élie Cartan's method of *moving frames* replaces coordinates by orthonormal frames, and the Christoffel symbols by a skew-symmetric matrix of 1-forms, which is found by solving a system of linear equations between differential forms, usually by inspection. The curvature then follows from the formula $F = d\theta + \theta \wedge \theta$ of Proposition 38.4.3. In this section we develop the method and use it to compute the curvature of surfaces in orthogonal coordinates, of conformally flat metrics in every dimension (in particular hyperbolic space has constant curvature -1), and of hypersurfaces of Euclidean space, where Cartan's lemma gives the symmetry of the second fundamental form and the Gauss equation in a few lines. Finally the Gauss–Bonnet–Chern theorem of Chapter 39 is written in terms of the curvature forms, and applied to spaces of constant curvature.

On a first reading this section can be skipped; later sections use only the curvature of hyperbolic space (Corollary 40.9.6).

Let (M, g) be a Riemannian manifold of dimension n and $U \subseteq M$ open. An *orthonormal frame* on U is a frame $(e_1, \dots, e_n)$ of $TM|_U$ with $\langle e_i, e_j \rangle = \delta_{ij}$; one exists near every point, obtained by applying the Gram–Schmidt process to the coordinate fields of a chart. Its *dual coframe* is the n-tuple of 1-forms $(\omega^1, \dots, \omega^n)$ with $\omega^i(e_j) = \delta_{ij}$. Then $\omega^i(X) = \langle X, e_i \rangle$ and $g = \sum_i \omega^i \otimes \omega^i$, usually written $g = \sum_i (\omega^i)^2$; conversely, if 1-forms $\omega^1, \dots, \omega^n$ on U, linearly independent at every point, satisfy $g = \sum_i (\omega^i)^2$, they are the dual coframe of the dual frame, which is orthonormal. So to apply the method to a metric written as a sum of squares, $g = \sum_i (\omega^i)^2$, there is nothing to compute: the coframe can be read off. Following Cartan we write the connection form of the Levi-Civita connection in the frame e (Proposition 38.3.4) as (ω^i_j) and its curvature as (Ω^i_j), so that

$$\nabla_X e_j = \sum_i \omega^i_j(X)\, e_i, \qquad R(X,Y)e_j = \sum_i \Omega^i_j(X,Y)\, e_i;$$

equivalently $\omega^i_j(X) = \langle \nabla_X e_j, e_i \rangle$ and $\Omega^i_j(X,Y) = R(X,Y,e_j,e_i)$ (Definition 40.8.1). For a surface with an oriented orthonormal frame, the 1-form ω of the proof of Theorem 40.8.11, defined by $\nabla_X e_1 = \omega(X) e_2$, is ω^2_1.

We need one algebraic lemma about 1-forms.

Lemma 40.9.1 (Cartan's lemma)**.** *Let $\omega^1, \dots, \omega^n$ be 1-forms on an n-manifold, linearly independent at every point, and let $\alpha_1, \dots, \alpha_p$, $p \le n$, be 1-forms with $\sum_{j=1}^p \alpha_j \wedge \omega^j = 0$. Then $\alpha_j = \sum_{k=1}^p A_{jk}\, \omega^k$ with smooth functions $A_{jk} = A_{kj}$.*

Proof. Let (e_k) be the dual frame and $A_{jk} = \alpha_j(e_k)$, so that $\alpha_j = \sum_{k=1}^n A_{jk}\omega^k$, with smooth A_{jk}. Then $0 = \sum_{j \le p} \sum_{k \le n} A_{jk}\, \omega^k \wedge \omega^j$. The 2-forms $\omega^a \wedge \omega^b$ with $a < b$ form a basis of the 2-forms at each point (Proposition 12.3.3(3)). For $a < b \le p$ the coefficient of $\omega^a \wedge \omega^b$ is $A_{ba} - A_{ab}$, which therefore vanishes; for $a \le p < b$ it is $-A_{ab}$, since the term with $j = b$ does not occur, and therefore $A_{ab} = 0$ for $b > p$. □

Theorem 40.9.2 (Cartan's structure equations)**.** *Let (ω^i) be the dual coframe of an orthonormal frame (e_i) on U, with connection forms ω^i_j and curvature forms Ω^i_j.*

1. $\omega^j_i = -\omega^i_j$.
2. *(First structure equation)* $d\omega^i = -\sum_j \omega^i_j \wedge \omega^j$.

3. *(Second structure equation)* $\Omega^i_j = d\omega^i_j + \sum_k \omega^i_k \wedge \omega^k_j$, *and* $\Omega^j_i = -\Omega^i_j$.

4. $\Omega^i_j = \sum_{k<l} R(e_k, e_l, e_j, e_i)\, \omega^k \wedge \omega^l$. *In particular the sectional curvature of the plane spanned by* e_k *and* e_l *is* $K(e_k, e_l) = \Omega^k_l(e_k, e_l)$, *and* g *has constant sectional curvature* κ *on* U *if and only if* $\Omega^i_j = \kappa\, \omega^i \wedge \omega^j$ *for all* i, j.

5. *(Bianchi identities)* $\sum_j \Omega^i_j \wedge \omega^j = 0$ *and* $d\Omega^i_j = \sum_k (\Omega^i_k \wedge \omega^k_j - \omega^i_k \wedge \Omega^k_j)$.

Moreover (1) and (2) characterise the connection forms: if (η^i_j) *is a matrix of* 1*-forms on* U *with* $\eta^j_i = -\eta^i_j$ *and* $d\omega^i = -\sum_j \eta^i_j \wedge \omega^j$ *for all* i, *then* $\eta^i_j = \omega^i_j$.

Proof. (1) The connection is metric: $0 = X\langle e_i, e_j\rangle = \langle \nabla_X e_i, e_j\rangle + \langle e_i, \nabla_X e_j\rangle = \omega^j_i(X) + \omega^i_j(X)$.

(2) This is the vanishing of the torsion. For vector fields X and $Y = \sum_j \omega^j(Y) e_j$, $\nabla_X Y = \sum_j X(\omega^j(Y)) e_j + \sum_{i,j} \omega^j(Y)\omega^i_j(X) e_i$, so $\omega^i(\nabla_X Y) = X(\omega^i(Y)) + \sum_j \omega^i_j(X)\omega^j(Y)$. Hence

$$\omega^i([X,Y]) = \omega^i(\nabla_X Y - \nabla_Y X) = X(\omega^i(Y)) - Y(\omega^i(X)) + \sum_j \big(\omega^i_j(X)\omega^j(Y) - \omega^i_j(Y)\omega^j(X)\big).$$

The last sum is $\sum_j (\omega^i_j \wedge \omega^j)(X,Y)$, and by the invariant formula $d\omega^i(X,Y) = X(\omega^i(Y)) - Y(\omega^i(X)) - \omega^i([X,Y])$ (Theorem 36.2.8), the identity says $0 = d\omega^i(X,Y) + \sum_j (\omega^i_j \wedge \omega^j)(X,Y)$.

(3) The curvature of the Levi-Civita connection is R (Definition 40.8.1), and the formula is that of Proposition 38.4.3 in the frame e. For a skew-symmetric matrix of 1-forms θ, $d\theta$ is skew-symmetric and so is $\theta \wedge \theta$: $\sum_k \theta^j_k \wedge \theta^k_i = \sum_k \theta^k_j \wedge \theta^i_k = -\sum_k \theta^i_k \wedge \theta^k_j$.

(4) A 2-form α equals $\sum_{k<l} \alpha(e_k, e_l)\, \omega^k \wedge \omega^l$, as both sides agree on pairs of frame vectors, and $\Omega^i_j(e_k, e_l) = R(e_k, e_l, e_j, e_i)$. With $i = k$, $j = l$: $\Omega^k_l(e_k, e_l) = R(e_k, e_l, e_l, e_k) = K(e_k, e_l)$. By Example 40.8.3, constant curvature κ means $R(X,Y)Z = \kappa(\langle Y, Z\rangle X - \langle X, Z\rangle Y)$; taking $Z = e_j$ and pairing with e_i this says $\Omega^i_j(X,Y) = \kappa\big(\omega^i(X)\omega^j(Y) - \omega^j(X)\omega^i(Y)\big) = \kappa(\omega^i \wedge \omega^j)(X,Y)$, and conversely these identities for all i, j give the formula for $R(X,Y)Z$ by linearity in Z.

(5) Differentiating (2), $0 = dd\omega^i = -\sum_j d\omega^i_j \wedge \omega^j + \sum_j \omega^i_j \wedge d\omega^j = -\sum_k d\omega^i_k \wedge \omega^k - \sum_{j,k} \omega^i_j \wedge \omega^j_k \wedge \omega^k$, which is $-\sum_k \Omega^i_k \wedge \omega^k$ by (3). The second identity is the Bianchi identity $dF = F \wedge \theta - \theta \wedge F$ of Proposition 38.4.3.

Finally, let $\alpha^i_j = \eta^i_j - \omega^i_j$. Then $\alpha^j_i = -\alpha^i_j$ and, subtracting (2), $\sum_j \alpha^i_j \wedge \omega^j = 0$ for each i. By Lemma 40.9.1, $\alpha^i_j = \sum_k A^i_{jk}\omega^k$ with $A^i_{jk} = A^i_{kj}$, while $A^j_{ik} = -A^i_{jk}$. A triple of indices with a symmetry in the last two and an antisymmetry in the first two vanishes:

$$A^i_{jk} = -A^j_{ik} = -A^j_{ki} = A^k_{ji} = A^k_{ij} = -A^i_{kj} = -A^i_{jk}.$$

So $A = 0$ and $\eta^i_j = \omega^i_j$. □

The last statement is what makes the method work. Given a metric written as $g = \sum_i (\omega^i)^2$, one computes the $d\omega^i$, finds by inspection a skew-symmetric matrix of 1-forms ω^i_j satisfying the first structure equation, and knows that it is the Levi-Civita connection; the second structure equation then gives the curvature forms, and (4) reads off the curvature. Equation (2) says in another language that the identity endomorphism $\sum_i e_i \otimes \omega^i \in \Omega^1(U, TM)$ has $d_\nabla(\mathrm{id}) = 0$, by the frame formula $d_\nabla \omega = d\omega + \theta \wedge \omega$ of Proposition 38.4.1: the torsion is the exterior covariant derivative of the identity.

On a surface there is a single connection form $\omega^1_2 = -\omega^2_1$, and everything becomes explicit.

Proposition 40.9.3. *Let (e_1, e_2) be an orthonormal frame on an open subset of a Riemannian surface, with dual coframe (ω^1, ω^2).*

1. $\omega^1_2 = -d\omega^1(e_1, e_2)\,\omega^1 - d\omega^2(e_1, e_2)\,\omega^2$.
2. $d\omega^1_2 = K\,\omega^1 \wedge \omega^2$, *where K is the Gaussian curvature.*
3. *(Orthogonal coordinates) If $g = E\,du^2 + G\,dv^2$ with positive functions E, G, then*
$$K = -\frac{1}{\sqrt{EG}}\left[\frac{\partial}{\partial v}\left(\frac{(\sqrt{E})_v}{\sqrt{G}}\right) + \frac{\partial}{\partial u}\left(\frac{(\sqrt{G})_u}{\sqrt{E}}\right)\right] = -\frac{1}{2\sqrt{EG}}\left[\frac{\partial}{\partial v}\left(\frac{E_v}{\sqrt{EG}}\right) + \frac{\partial}{\partial u}\left(\frac{G_u}{\sqrt{EG}}\right)\right].$$
4. *(Geodesic polar coordinates) If $g = dr^2 + f(r)^2 d\varphi^2$ with $f > 0$, then $K = -f''/f$.*

Proof. (1) Write $\omega^1_2 = a\,\omega^1 + b\,\omega^2$. The first structure equation gives $d\omega^1 = -\omega^1_2 \wedge \omega^2 = -a\,\omega^1 \wedge \omega^2$ and $d\omega^2 = -\omega^2_1 \wedge \omega^1 = \omega^1_2 \wedge \omega^1 = -b\,\omega^1 \wedge \omega^2$; evaluate on (e_1, e_2). (2) In the second structure equation for Ω^1_2 the quadratic terms $\omega^1_1 \wedge \omega^1_2$ and $\omega^1_2 \wedge \omega^2_2$ vanish, as $\omega^1_1 = \omega^2_2 = 0$; so $\Omega^1_2 = d\omega^1_2$, and $K = \Omega^1_2(e_1, e_2)$ by Theorem 40.9.2(4). (3) Here $\omega^1 = \sqrt{E}\,du$ and $\omega^2 = \sqrt{G}\,dv$, so $d\omega^1 = (\sqrt{E})_v\,dv \wedge du = -\frac{(\sqrt{E})_v}{\sqrt{EG}}\,\omega^1 \wedge \omega^2$ and $d\omega^2 = (\sqrt{G})_u\,du \wedge dv = \frac{(\sqrt{G})_u}{\sqrt{EG}}\,\omega^1 \wedge \omega^2$. By (1),
$$\omega^1_2 = \frac{(\sqrt{E})_v}{\sqrt{EG}}\,\omega^1 - \frac{(\sqrt{G})_u}{\sqrt{EG}}\,\omega^2 = \frac{(\sqrt{E})_v}{\sqrt{G}}\,du - \frac{(\sqrt{G})_u}{\sqrt{E}}\,dv,$$
and $d\omega^1_2 = -\left[\left(\frac{(\sqrt{E})_v}{\sqrt{G}}\right)_v + \left(\frac{(\sqrt{G})_u}{\sqrt{E}}\right)_u\right] du \wedge dv$, while $K\,\omega^1 \wedge \omega^2 = K\sqrt{EG}\,du \wedge dv$; compare with (2). The second expression for K, the classical one [dC76, Chapter 4], follows from $(\sqrt{E})_v = E_v/(2\sqrt{E})$, which gives $(\sqrt{E})_v/\sqrt{G} = E_v/(2\sqrt{EG})$, and likewise $(\sqrt{G})_u/\sqrt{E} = G_u/(2\sqrt{EG})$: the factor $\frac{1}{2}$ appears when the derivatives are taken of E and G rather than of their square roots. (4) is (3) with $(u, v) = (r, \varphi)$, $E = 1$ and $G = f^2$: $K = -\frac{1}{f}(f')'$. □

Example 40.9.4. 1. *The round sphere.* In spherical coordinates $g = d\vartheta^2 + \sin^2\vartheta\,d\varphi^2$, so $K = -(\sin\vartheta)''/\sin\vartheta = 1$. Here $\omega^1 = d\vartheta$, $\omega^2 = \sin\vartheta\,d\varphi$, $d\omega^1 = 0$ and $d\omega^2 = \cos\vartheta\,d\vartheta \wedge d\varphi = \cot\vartheta\,\omega^1 \wedge \omega^2$, so (1) gives $\omega^1_2 = -\cos\vartheta\,d\varphi$, in accordance with Example 38.3.5, and $d\omega^1_2 = \sin\vartheta\,d\vartheta \wedge d\varphi = \omega^1 \wedge \omega^2$.

2. *Polar coordinates.* $dr^2 + r^2 d\varphi^2$ (the plane) has $K = 0$, and $dr^2 + \sinh^2 r\,d\varphi^2$ has $K = -\sinh r/\sinh r = -1$.

3. *Surfaces of revolution.* Rotating a curve $u \mapsto (f(u), h(u))$ in the half-plane $\{(\rho, z) : \rho > 0\}$, parametrised by arc length ($f'^2 + h'^2 = 1$), about the z-axis gives the surface $x(u, v) = (f(u)\cos v, f(u)\sin v, h(u))$ with metric $du^2 + f(u)^2 dv^2$, so $K = -f''/f$ and $dA = f\,du \wedge dv$. For the torus with radii $R > r > 0$, $f(u) = R + r\cos(u/r)$ and $h(u) = r\sin(u/r)$, so
$$K = \frac{\cos(u/r)}{r(R + r\cos(u/r))}, \qquad \iint_T K\,dA = \int_0^{2\pi}\int_0^{2\pi r} \frac{\cos(u/r)}{r}\,du\,dv = 0 = 2\pi\chi(T),$$
with $K > 0$ on the outer half of the torus and $K < 0$ on the inner half, in accordance with the Gauss–Bonnet theorem.

In higher dimensions the method is just as efficient for metrics that are a function times the Euclidean metric.

Proposition 40.9.5 (Conformally flat metrics)**.** *Let $U \subseteq \mathbb{R}^n$ be open, $u\colon U \to \mathbb{R}$ smooth, and $g = e^{2u}(dx_1^2 + \cdots + dx_n^2)$, with the orthonormal coframe $\omega^i = e^u\,dx_i$. Write $u_i = \partial u/\partial x_i$, $u_{ik} = \partial^2 u/\partial x_i \partial x_k$, $|\nabla u|^2 = \sum_k u_k^2$, and $\alpha_i = du_i - u_i\,du = \sum_k (u_{ik} - u_i u_k)\,dx_k$. Then*

$$\omega_j^i = u_j\,dx_i - u_i\,dx_j, \qquad \Omega_j^i = dx_j \wedge \alpha_i - dx_i \wedge \alpha_j - |\nabla u|^2\,dx_i \wedge dx_j.$$

In particular, if $u_{ik} - u_i u_k = \lambda\,\delta_{ik}$ for a function λ, then $\Omega_j^i = -e^{-2u}(2\lambda + |\nabla u|^2)\,\omega^i \wedge \omega^j$, so at each point all planes have the same sectional curvature $-e^{-2u}(2\lambda + |\nabla u|^2)$. For $n = 2$, $\Omega_2^1 = -e^{-2u}\Delta u\;\omega^1 \wedge \omega^2$, where $\Delta u = u_{11} + u_{22}$.

Proof. $d\omega^i = e^u\,du \wedge dx_i = \sum_j u_j e^u\,dx_j \wedge dx_i$. The proposed forms are skew-symmetric, and

$$-\sum_j \omega_j^i \wedge \omega^j = -e^u \sum_j (u_j\,dx_i - u_i\,dx_j) \wedge dx_j = -e^u \sum_j u_j\,dx_i \wedge dx_j = d\omega^i,$$

so they are the connection forms by Theorem 40.9.2. For the curvature, $d\omega_j^i = du_j \wedge dx_i - du_i \wedge dx_j$, and

$$\sum_k \omega_k^i \wedge \omega_j^k = \sum_k (u_k\,dx_i - u_i\,dx_k) \wedge (u_j\,dx_k - u_k\,dx_j) = u_j\,dx_i \wedge du - |\nabla u|^2\,dx_i \wedge dx_j + u_i\,du \wedge dx_j,$$

the term $-u_i u_j \sum_k dx_k \wedge dx_k$ being zero. Adding, $\Omega_j^i = dx_i \wedge (u_j\,du - du_j) + (u_i\,du - du_i) \wedge dx_j - |\nabla u|^2\,dx_i \wedge dx_j$, which is the formula. If $\alpha_i = \lambda\,dx_i$ for all i, then $\Omega_j^i = \lambda\,dx_j \wedge dx_i - \lambda\,dx_i \wedge dx_j - |\nabla u|^2 dx_i \wedge dx_j = -(2\lambda + |\nabla u|^2)e^{-2u}\omega^i \wedge \omega^j$. For $n = 2$, the terms $u_{12} - u_1 u_2$ of α_1 and α_2 contribute $dx_2 \wedge dx_2 = 0$ and $dx_1 \wedge dx_1 = 0$, so $\Omega_2^1 = -\big((u_{11} - u_1^2) + (u_{22} - u_2^2) + |\nabla u|^2\big)dx_1 \wedge dx_2 = -\Delta u\,dx_1 \wedge dx_2$. □

For $n = 2$ this is the formula $K = -e^{-2u}\Delta u$ proved with Christoffel symbols in Proposition 40.12.7 below.

Corollary 40.9.6. *1. The* upper half-space $H^n = \{x \in \mathbb{R}^n : x_n > 0\}$ *with the metric $(dx_1^2 + \cdots + dx_n^2)/x_n^2$ has constant sectional curvature -1.*

2. The ball $B^n = \{|x| < 1\}$ *with the metric $4(dx_1^2 + \cdots + dx_n^2)/(1 - |x|^2)^2$ has constant sectional curvature -1.*

3. $\mathbb{R}^n$ with the metric $4(dx_1^2 + \cdots + dx_n^2)/(1 + |x|^2)^2$ has constant sectional curvature 1. It is the pullback of the round metric of S^n by the inverse of the stereographic projection (Exercise 40.17.15).

Multiplying these metrics by a constant $c > 0$ multiplies the curvatures by $1/c$.

Proof. We apply Proposition 40.9.5. (1) Here $u = -\log x_n$, $u_i = -\delta_{in}/x_n$ and $u_{ik} = \delta_{in}\delta_{kn}/x_n^2 = u_i u_k$, so $\lambda = 0$, $|\nabla u|^2 = 1/x_n^2$ and the curvature is $-x_n^2 \cdot x_n^{-2} = -1$. (2) Let $s = 1 - |x|^2$, so $u = \log 2 - \log s$, $u_i = 2x_i/s$ and $u_{ik} = 2\delta_{ik}/s + 4x_i x_k/s^2$. So $u_{ik} - u_i u_k = (2/s)\delta_{ik}$, $|\nabla u|^2 = 4|x|^2/s^2$, and $2\lambda + |\nabla u|^2 = (4s + 4|x|^2)/s^2 = 4/s^2$; with $e^{-2u} = s^2/4$ the curvature is -1. (3) Now $s = 1 + |x|^2$, $u = \log 2 - \log s$, $u_i = -2x_i/s$, $u_{ik} = -2\delta_{ik}/s + 4x_i x_k/s^2$, so $\lambda = -2/s$ and $2\lambda + |\nabla u|^2 = (-4s + 4|x|^2)/s^2 = -4/s^2$; the curvature is 1. Finally, if (ω^i) is an orthonormal coframe for g, then $(\sqrt{c}\,\omega^i)$ is one for cg; the same forms ω_j^i satisfy the first structure equation for it, as that equation is linear in the coframe, so the connection and curvature forms are unchanged, and $\Omega_j^i = \kappa\,\omega^i \wedge \omega^j = \frac{\kappa}{c}(\sqrt{c}\,\omega^i) \wedge (\sqrt{c}\,\omega^j)$. □

Next, hypersurfaces. For $M \subseteq \mathbb{R}^{n+1}$ the frame is completed by a unit normal, and the structure equations of $\mathbb{R}^{n+1}$ restricted to M contain the second fundamental form.

Proposition 40.9.7. *Let $M \subseteq \mathbb{R}^{n+1}$ be a hypersurface with the induced metric, $U \subseteq M$ open with an orthonormal frame $(e_1, \dots, e_n)$ and a unit normal field $e_{n+1} = \nu$. Regard the e_A, $A = 1, \dots, n+1$, and the inclusion $x\colon U \to \mathbb{R}^{n+1}$ as vector-valued functions, and define 1-forms on U by*

$$dx = \sum_{i=1}^{n} \omega^i\, e_i, \qquad de_B = \sum_{A=1}^{n+1} \omega_B^A\, e_A.$$

1. *(ω^i) is the dual coframe of (e_i), $\omega_A^B = -\omega_B^A$, and for $i, j \le n$ the ω_j^i are the Levi-Civita connection forms of M.*
2. *$\omega_i^{n+1} = \sum_j h_{ij}\,\omega^j$ with $h_{ij} = h(e_i, e_j)$, the scalar second fundamental form (Definition 40.11.9), and $h_{ij} = h_{ji}$.*
3. *(Gauss equation) $\Omega_j^i = \omega_i^{n+1} \wedge \omega_j^{n+1} = \sum_{k,l} h_{ik}h_{jl}\,\omega^k \wedge \omega^l$, so $K(e_i, e_j) = h_{ii}h_{jj} - h_{ij}^2$. For a surface in $\mathbb{R}^3$, $K = \det(h_{ij})$.*

Proof. (1) For X tangent to M, $dx(X) = X$, so $\omega^i(X) = \langle X, e_i\rangle$; the normal component of dx vanishes. Since $\langle e_A, e_B\rangle$ is constant, $0 = \langle de_A, e_B\rangle + \langle e_A, de_B\rangle = \omega_A^B + \omega_B^A$. For a 1-form β and a vector-valued function e, $d(\beta\, e) = d\beta\, e - \beta \wedge de$ (apply the Leibniz rule to each component). So

$$0 = d(dx) = \sum_i \big(d\omega^i\, e_i - \omega^i \wedge de_i\big) = \sum_A \Big(d\omega^A - \sum_i \omega^i \wedge \omega_i^A\Big) e_A,$$

where $\omega^{n+1} = 0$. For $A = j \le n$ this is $d\omega^j = -\sum_i \omega_i^j \wedge \omega^i$, the first structure equation for the skew-symmetric matrix $(\omega_i^j)_{i,j\le n}$, so by Theorem 40.9.2 it is the Levi-Civita connection form of M. (2) For $A = n+1$ the same identity gives $\sum_i \omega_i^{n+1} \wedge \omega^i = 0$, so by Lemma 40.9.1 $\omega_i^{n+1} = \sum_j h_{ij}\omega^j$ with $h_{ij} = h_{ji}$. And $h_{ij} = \omega_i^{n+1}(e_j) = \langle de_i(e_j), \nu\rangle = \langle \overline{\nabla}_{e_j} e_i, \nu\rangle = h(e_j, e_i)$, where $\overline{\nabla}$ is the derivative in $\mathbb{R}^{n+1}$: this is a second proof of the symmetry of the second fundamental form. (3) Similarly $0 = d(de_B) = \sum_A \big(d\omega_B^A\, e_A - \omega_B^A \wedge de_A\big) = \sum_C \big(d\omega_B^C - \sum_A \omega_B^A \wedge \omega_A^C\big) e_C$, that is $d\omega_B^C + \sum_A \omega_A^C \wedge \omega_B^A = 0$: the curvature of $\mathbb{R}^{n+1}$ vanishes. For $C = i$, $B = j \le n$, separating the term $A = n+1$ and using (1) and the second structure equation on M, $\Omega_j^i = -\omega_{n+1}^i \wedge \omega_j^{n+1} = \omega_i^{n+1} \wedge \omega_j^{n+1}$. Evaluating on (e_i, e_j) gives $K(e_i, e_j) = h_{ii}h_{jj} - h_{ij}h_{ji}$. □

This recovers the Theorema Egregium (Corollary 40.11.13) with almost no computation.

Example 40.9.8. For the sphere $S^n(r) = \{|x| = r\} \subseteq \mathbb{R}^{n+1}$ take $\nu = x/r$. Then $de_{n+1} = dx/r = \sum_i (\omega^i/r) e_i$, so $\omega_{n+1}^i = \omega^i/r$ and $\omega_i^{n+1} = -\omega^i/r$, that is $h_{ij} = -\delta_{ij}/r$. By the Gauss equation $\Omega_j^i = \frac{1}{r^2}\omega^i \wedge \omega^j$: the sphere of radius r has constant sectional curvature $1/r^2$, in every dimension, and without computing a single Christoffel symbol.

Finally we return to the Gauss–Bonnet–Chern theorem. Let M be oriented of dimension $n = 2m$. In an oriented orthonormal frame the Levi-Civita curvature forms Ω_j^i form a skew-symmetric matrix of 2-forms, and its Pfaffian (Definition 12.4.8) divided by $(2\pi)^m$ is the Euler form $e(TM, \nabla)$ of Definition 39.6.2. In dimension 2 it is $\frac{1}{2\pi}\Omega_2^1 = \frac{1}{2\pi} K\, dA$, by Proposition 40.9.3(2) and Proposition 40.3.3; in dimension 4 it is $\frac{1}{4\pi^2}(\Omega_2^1 \wedge \Omega_4^3 - \Omega_3^1 \wedge \Omega_4^2 + \Omega_4^1 \wedge \Omega_3^2)$.

Corollary 40.9.9 (Gauss–Bonnet–Chern)**.** *Let M be a compact oriented Riemannian manifold of dimension $2m$.*

1. *$\chi(M) = \dfrac{1}{(2\pi)^m} \displaystyle\int_M \mathrm{Pf}(\Omega)$, and this equals the sum of the indices of any vector field on M with isolated zeros.*

2. *If M has constant sectional curvature κ, then* $\chi(M) = \dfrac{(2m-1)!!\,\kappa^m}{(2\pi)^m}\operatorname{vol}(M)$, *where* $(2m-1)!! = 1\cdot 3\cdots(2m-1)$.

3. $\operatorname{vol}(S^{2m}) = 2(2\pi)^m/(2m-1)!!$ *for the round unit sphere; for instance* $\operatorname{vol}(S^2) = 4\pi$ *and* $\operatorname{vol}(S^4) = 8\pi^2/3$.

4. *A compact oriented $2m$-manifold with a metric of constant curvature -1 has $(-1)^m\chi(M) > 0$ and volume $(2\pi)^m|\chi(M)|/(2m-1)!!$; for surfaces of genus $g \geq 2$ the area is $4\pi(g-1)$.*

Proof. (1) is Corollary 39.6.13 for the Levi-Civita connection. (2) By Theorem 40.9.2(4), $\Omega^i_j = \kappa\,\omega^i \wedge \omega^j$, so by Proposition 12.4.9(1)

$$\operatorname{Pf}(\Omega) = \frac{\kappa^m}{2^m m!}\sum_{\sigma\in S_{2m}} \operatorname{sgn}(\sigma)\,\omega^{\sigma(1)}\wedge\cdots\wedge\omega^{\sigma(2m)} = \frac{(2m)!}{2^m m!}\kappa^m\,\omega^1\wedge\cdots\wedge\omega^{2m},$$

and $\omega^1\wedge\cdots\wedge\omega^{2m}$ is the volume form (Proposition 40.3.3), while $(2m)!/(2^m m!) = (2m-1)!!$. (3) follows from (2) with $\kappa = 1$ (Example 40.9.8) and $\chi(S^{2m}) = 2$ (Theorem 21.6.1). (4) follows from (2) with $\kappa = -1$, since the volume is positive; for a surface of genus g, $\chi = 2-2g$. □

So the volume of a compact manifold of even dimension with constant curvature -1 is determined by its Euler characteristic; for surfaces this will reappear in Chapter 44, through the uniformisation theorem. In the opposite direction, (2) shows that a compact oriented manifold of even dimension with $\chi(M) = 0$, such as a torus or $S^1\times S^3$, carries no metric of constant curvature $\kappa \neq 0$, and that the sign of χ restricts the possible signs of κ.

40.10 The first and second variation of energy

Geodesics are the critical points of the energy of curves, and the second derivative of the energy along a geodesic is a quadratic form, the index form, in which the curvature appears. This section proves both variation formulas and uses the second one to show that geodesics stop minimising after a conjugate point (Jacobi's theorem) and that positive Ricci curvature bounds the diameter (the Bonnet–Myers theorem).

Let $\gamma\colon[a,b]\to M$ be a piecewise smooth curve. A *variation* of γ is a continuous map $f\colon(-\varepsilon,\varepsilon)\times[a,b]\to M$ with $f(0,t) = \gamma(t)$, together with a partition $a = t_0 < t_1 < \cdots < t_k = b$ such that f is smooth on each $(-\varepsilon,\varepsilon)\times[t_{i-1},t_i]$; it is *proper* if $f(s,a) = \gamma(a)$ and $f(s,b) = \gamma(b)$ for all s. We write $f_s = f(s,-)$, and $V(t) = \partial_s f(0,t)$ for the *variation field*, a continuous vector field along γ, smooth on each $[t_{i-1},t_i]$ (*piecewise smooth*); it vanishes at a and b if f is proper. The *energy* of a piecewise smooth curve $c\colon[a,b]\to M$ is $E(c) = \frac12\int_a^b |\dot c|^2\,dt$. On each piece we use the covariant derivatives D_s and D_t of Lemma 40.4.3.

Every piecewise smooth field V along γ is the variation field of a variation, for instance of

$$f(s,t) = \exp_{\gamma(t)}(sV(t)).$$

Indeed the set of (s,t) with $sV(t)$ in the domain $\mathcal{E}$ of exp is open and contains $\{0\}\times[a,b]$, so by compactness it contains $(-\varepsilon,\varepsilon)\times[a,b]$ for some $\varepsilon > 0$; and $\partial_s f(0,t) = d_0\exp_{\gamma(t)}(V(t)) = V(t)$. This variation is proper if $V(a) = V(b) = 0$, and its curves $s\mapsto f(s,t)$ are geodesics, so $D_s\partial_s f = 0$.

Lemma 40.10.1. *For a piecewise smooth curve $c\colon[a,b]\to M$, $L(c)^2 \leq 2(b-a)E(c)$, with equality if and only if c has constant speed. Consequently, if γ has constant speed and $E(c) < E(\gamma)$ for a curve c with the same interval, then $L(c) < L(\gamma)$.*

Proof. By the Cauchy–Schwarz inequality, $L(c)^2 = \left(\int_a^b 1 \cdot |\dot c|\,dt\right)^2 \le (b-a)\int_a^b |\dot c|^2\,dt = 2(b-a)E(c)$, with equality if and only if $|\dot c|$ is constant. If γ has constant speed and $E(c) < E(\gamma)$, then $L(c)^2 \le 2(b-a)E(c) < 2(b-a)E(\gamma) = L(\gamma)^2$. □

Proposition 40.10.2 (First variation formula)**.** *Let f be a variation of a piecewise smooth curve γ, with variation field V, and suppose that γ is smooth on each $[t_{i-1}, t_i]$. Write $\Delta_i\dot\gamma = \dot\gamma(t_i^+) - \dot\gamma(t_i^-)$ for the jump of the velocity at t_i, $0 < i < k$. Then $s \mapsto E(f_s)$ is smooth, and*

$$\frac{d}{ds}\Big|_{s=0} E(f_s) = \langle V, \dot\gamma\rangle\Big|_a^b - \sum_{i=1}^{k-1}\langle V(t_i), \Delta_i\dot\gamma\rangle - \int_a^b \langle V, D_t\dot\gamma\rangle\,dt.$$

Consequently γ is a critical point of the energy for all proper variations, that is, $\frac{d}{ds}|_{s=0}E(f_s) = 0$ for every proper f, if and only if γ is a geodesic.

Proof. On each piece the integrand of $E(f_s) = \sum_i \frac12\int_{t_{i-1}}^{t_i}|\partial_t f|^2\,dt$ is smooth on a compact set, so we may differentiate under the integral sign. By Proposition 40.4.1 and Lemma 40.4.3,

$$\frac{\partial}{\partial s}\frac12|\partial_t f|^2 = \langle D_s\partial_t f, \partial_t f\rangle = \langle D_t\partial_s f, \partial_t f\rangle = \frac{\partial}{\partial t}\langle \partial_s f, \partial_t f\rangle - \langle \partial_s f, D_t\partial_t f\rangle.$$

At $s = 0$ this is $\partial_t\langle V, \dot\gamma\rangle - \langle V, D_t\dot\gamma\rangle$. Integrating over $[t_{i-1}, t_i]$ gives $\langle V(t_i), \dot\gamma(t_i^-)\rangle - \langle V(t_{i-1}), \dot\gamma(t_{i-1}^+)\rangle - \int_{t_{i-1}}^{t_i}\langle V, D_t\dot\gamma\rangle\,dt$, and summing over i gives the formula, since V is continuous.

If γ is a geodesic, it is smooth, so there are no jumps, $D_t\dot\gamma = 0$, and $V(a) = V(b) = 0$ for proper variations; so the derivative vanishes. Conversely, suppose it vanishes for all proper variations. The function $\phi(t) = \prod_i (t - t_i)^2$ is smooth, positive except at the t_i, and $V = \phi\, D_t\dot\gamma$ is a piecewise smooth field vanishing at all the t_i; for the variation $\exp_{\gamma(t)}(sV(t))$ the formula gives $0 = -\int_a^b \phi|D_t\dot\gamma|^2\,dt$, so $D_t\dot\gamma = 0$ on each piece. Next choose a piecewise smooth V with $V(a) = V(b) = 0$ and $V(t_i) = \Delta_i\dot\gamma$, for instance a sum of products of bump functions with parallel fields along γ; then $0 = -\sum_i |\Delta_i\dot\gamma|^2$. So $\dot\gamma$ is continuous, and γ is a geodesic on each piece whose velocities match at the t_i; by uniqueness of geodesics with given initial velocity, γ is a single geodesic. □

In particular a curve of constant speed that minimises length between its endpoints minimises energy (Lemma 40.10.1), hence is a geodesic, which is Lemma 40.6.7 again.

Reading the first variation formula. Each of its three terms describes one way of making a curve shorter.

1. The integral $-\int\langle V, D_t\dot\gamma\rangle\,dt$: if the curve bends, pushing it in the direction of its acceleration $D_t\dot\gamma$ (towards the inside of the bend, $V = D_t\dot\gamma$) decreases the energy at the rate $-\int|D_t\dot\gamma|^2\,dt$. Think of a bent wire being pulled straight. A geodesic is a curve with nothing to straighten: the geodesic equation $D_t\dot\gamma = 0$ is the Euler–Lagrange equation of the energy.

2. The jump terms $-\langle V(t_i), \Delta_i\dot\gamma\rangle$: at a corner, pushing the corner point in the direction of $\Delta_i\dot\gamma = \dot\gamma(t_i^+) - \dot\gamma(t_i^-)$, which points into the angle, "cuts the corner" and decreases the energy.

3. The boundary term $\langle V, \dot\gamma\rangle|_a^b$: moving the endpoints along the curve changes the energy even if the curve is a geodesic, by lengthening or shortening it at the ends. It disappears for proper variations, and also when the endpoints move orthogonally to $\dot\gamma$; this is why a shortest curve from a point to a submanifold meets the submanifold orthogonally.

In $\mathbb{R}^n$, with $D_t\dot\gamma = \ddot\gamma$, the formula is the integration by parts $\frac{d}{ds}\frac12\int|\dot\gamma + s\dot V|^2\,dt\big|_{s=0} = \int\langle\dot V,\dot\gamma\rangle\,dt = \langle V,\dot\gamma\rangle|_a^b - \int\langle V,\ddot\gamma\rangle\,dt$; the covariant derivative is what makes the same computation meaningful on a manifold.

Definition 40.10.3. Let $\gamma\colon[a,b]\to M$ be a geodesic. The *index form* of γ is the symmetric bilinear form on piecewise smooth vector fields along γ

$$I(V,W) = \int_a^b \big(\langle D_tV, D_tW\rangle - R(V,\dot\gamma,\dot\gamma,W)\big)\,dt.$$

It is symmetric because $R(V,\dot\gamma,\dot\gamma,W) = R(\dot\gamma,W,V,\dot\gamma) = R(W,\dot\gamma,\dot\gamma,V)$ by Proposition 40.8.2. If V is smooth on the pieces $[t_{i-1},t_i]$, integrating $\frac{d}{dt}\langle D_tV,W\rangle = \langle D_t^2V,W\rangle + \langle D_tV,D_tW\rangle$ over each piece gives

$$I(V,W) = \langle D_tV,W\rangle\Big|_a^b - \sum_{i=1}^{k-1}\langle\Delta_iD_tV,W(t_i)\rangle - \int_a^b\langle D_t^2V + R(V,\dot\gamma)\dot\gamma,W\rangle\,dt, \tag{40.3}$$

with $\Delta_iD_tV = D_tV(t_i^+) - D_tV(t_i^-)$. In particular $I(J,W)=0$ for a Jacobi field J and every W vanishing at a and b.

Proposition 40.10.4 (Second variation formula)**.** *Let $\gamma\colon[a,b]\to M$ be a geodesic and f a variation of γ with variation field V, and let $A(t) = D_s\partial_sf(0,t)$, the acceleration at $s=0$ of the curve $s\mapsto f(s,t)$. Then*

$$\frac{d^2}{ds^2}\Big|_{s=0}E(f_s) = I(V,V) + \langle A,\dot\gamma\rangle\Big|_a^b = \int_a^b\big(|D_tV|^2 - R(V,\dot\gamma,\dot\gamma,V)\big)\,dt + \langle A,\dot\gamma\rangle\Big|_a^b.$$

The boundary term vanishes if f is proper, and also if the curves $s\mapsto f(s,a)$ and $s\mapsto f(s,b)$ are geodesics, for instance for $f(s,t) = \exp_{\gamma(t)}(sV(t))$.

Proof. As in the proof of Proposition 40.10.2, on each piece $\frac{d}{ds}E = \int\langle D_t\partial_sf,\partial_tf\rangle\,dt$, and differentiating again,

$$\frac{d^2}{ds^2}E(f_s) = \sum_i\int_{t_{i-1}}^{t_i}\big(\langle D_sD_t\partial_sf,\partial_tf\rangle + \langle D_t\partial_sf,D_s\partial_tf\rangle\big)\,dt.$$

At $s=0$, $D_s\partial_tf = D_t\partial_sf = D_tV$, so the second term is $|D_tV|^2$. By Lemma 40.8.6, $D_sD_t\partial_sf = D_tD_s\partial_sf + R(\partial_sf,\partial_tf)\partial_sf$, and at $s=0$, using $D_t\dot\gamma=0$,

$$\langle D_sD_t\partial_sf,\dot\gamma\rangle = \frac{\partial}{\partial t}\langle D_s\partial_sf,\dot\gamma\rangle + R(V,\dot\gamma,V,\dot\gamma) = \frac{\partial}{\partial t}\langle D_s\partial_sf,\dot\gamma\rangle - R(V,\dot\gamma,\dot\gamma,V).$$

The field A at t_i is the acceleration of the curve $s\mapsto f(s,t_i)$, which is the same curve for both pieces meeting at t_i; so A is continuous, and so is $\dot\gamma$. Hence the integrals of $\partial_t\langle A,\dot\gamma\rangle$ add up to $\langle A,\dot\gamma\rangle|_a^b$. If f is proper, the curves $s\mapsto f(s,a)$ and $s\mapsto f(s,b)$ are constant, so $A(a)=A(b)=0$; if they are geodesics, $A(a)=A(b)=0$ by Proposition 40.4.4. □

Remark 40.10.5. *Reading the second variation formula.* The first variation vanishes at a geodesic, so whether nearby curves are longer or shorter is decided by the second derivative, a competition between two terms.

1. $\int|D_tV|^2\,dt\ge0$: a variation field that changes along the curve forces the nearby curves to wiggle, and wiggling costs energy. In $\mathbb{R}^n$ this is the only term, so $\frac{d^2}{ds^2}E\ge0$ for every proper variation: straight segments are local minima of the energy.

2. $-\int R(V,\dot\gamma,\dot\gamma,V)\,dt$: if γ has unit speed and V is orthogonal to $\dot\gamma$, then $R(V,\dot\gamma,\dot\gamma,V) = K(V,\dot\gamma)|V|^2$, the sectional curvature of the plane spanned by V and $\dot\gamma$. Positive curvature makes nearby geodesics converge, like the meridians of a sphere meeting at the poles, so pushing the curve sideways can make it *shorter*; negative curvature makes geodesics diverge and only helps the minimising property.

A computation on the unit sphere. Let $\gamma\colon[0,L]\to S^n$ be a unit speed great circle arc, E a parallel unit field along γ orthogonal to $\dot\gamma$, and $V=\phi E$ with $\phi(t)=\sin(\pi t/L)$, which vanishes at the endpoints. Then $D_tV=\phi' E$ and $R(V,\dot\gamma,\dot\gamma,V)=\phi^2$, as $K=1$, so for the proper variation $\exp_{\gamma(t)}(sV(t))$,

$$\frac{d^2}{ds^2}\Big|_{s=0}E(f_s)=\int_0^L\Big(\frac{\pi^2}{L^2}\cos^2\frac{\pi t}{L}-\sin^2\frac{\pi t}{L}\Big)\,dt=\frac{L}{2}\Big(\frac{\pi^2}{L^2}-1\Big),$$

using $\int_0^L\cos^2(\pi t/L)\,dt=\int_0^L\sin^2(\pi t/L)\,dt=L/2$. This is negative exactly when $L>\pi$: a great circle arc longer than half a great circle can be shortened by pushing it sideways, which is the content of Theorem 40.10.6 below in this example. The same computation with constant curvature $\kappa>0$ gives $\frac{L}{2}(\pi^2/L^2-\kappa)$, negative for $L>\pi/\sqrt{\kappa}$; this is the computation behind the Bonnet–Myers theorem below, where the Ricci curvature appears because one sums such computations over $n-1$ orthonormal directions E.

Theorem 40.10.6 (Jacobi)**.** *Let $\gamma\colon[0,L]\to M$ be a geodesic, and suppose that $\gamma(t_0)$ is conjugate to $\gamma(0)$ along γ for some $0<t_0<L$. Then there are piecewise smooth curves from $\gamma(0)$ to $\gamma(L)$, arbitrarily close to γ, that are shorter than γ. In particular γ is not a minimising geodesic.*

Proof. Let J be a nonzero Jacobi field on $[0,t_0]$ with $J(0)=J(t_0)=0$. Then $D_tJ(t_0)\neq0$, as a Jacobi field is determined by its value and derivative at one point. Let $V=J$ on $[0,t_0]$ and $V=0$ on $[t_0,L]$, a piecewise smooth field vanishing at 0 and L. In (40.3) for V and a piecewise smooth U vanishing at 0 and L, the integral vanishes, because J satisfies the Jacobi equation on $[0,t_0]$ and $V=0$ on $[t_0,L]$, and the only remaining term comes from the jump $\Delta D_tV=-D_tJ(t_0)$ at t_0; so $I(V,U)=\langle D_tJ(t_0),U(t_0)\rangle$. In particular $I(V,V)=\langle D_tJ(t_0),J(t_0)\rangle=0$. Let P be the parallel field along γ with $P(t_0)=-D_tJ(t_0)$, and $W=\sin(\pi t/L)P/\sin(\pi t_0/L)$, which vanishes at 0 and L and equals $-D_tJ(t_0)$ at t_0. Then $I(V,W)=-|D_tJ(t_0)|^2<0$. Hence for small $\delta>0$,

$$I(V+\delta W,V+\delta W)=2\delta I(V,W)+\delta^2I(W,W)<0.$$

The proper variation $f(s,t)=\exp_{\gamma(t)}(s(V+\delta W)(t))$ has $\frac{d}{ds}E(f_s)=0$ at $s=0$ (Proposition 40.10.2) and $\frac{d^2}{ds^2}E(f_s)<0$ at $s=0$ (Proposition 40.10.4), so $E(f_s)<E(\gamma)$ for all small $s\neq0$, and $L(f_s)<L(\gamma)$ by Lemma 40.10.1, as the geodesic γ has constant speed. □

For example, a great circle on the unit sphere S^n starting at p meets the conjugate point $-p$ at time π (Example 40.8.9), and so it stops minimising after time π, in accordance with Remark 40.6.13.

Theorem 40.10.7 (Bonnet–Myers)**.** *Let (M,g) be complete and connected with* $\mathrm{Ric}\geq(n-1)\kappa g$ *for some $\kappa>0$. Then M has diameter at most $\pi/\sqrt{\kappa}$; hence M is compact, and $\pi_1(M)$ is finite.*

Proof. Let $p,q\in M$ and $L=d(p,q)$, and suppose $L>\pi/\sqrt{\kappa}$. By Theorem 40.6.8 there is a minimising geodesic $\gamma\colon[0,L]\to M$ of unit speed from p to q. Let $E_1,\dots,E_{n-1}$ be parallel fields along γ that together with $\dot\gamma$ form an orthonormal basis at one point, hence at every point (Proposition 40.4.4), and put $V_i(t)=\sin(\pi t/L)E_i(t)$. Then $D_tV_i=(\pi/L)\cos(\pi t/L)E_i$, and since $\sum_i R(E_i,\dot\gamma,\dot\gamma,E_i)=\mathrm{Ric}(\dot\gamma,\dot\gamma)\geq(n-1)\kappa$ (Definition 40.8.1),

$$\sum_{i=1}^{n-1}I(V_i,V_i)=\int_0^L\Big((n-1)\frac{\pi^2}{L^2}\cos^2\frac{\pi t}{L}-\sin^2\frac{\pi t}{L}\,\mathrm{Ric}(\dot\gamma,\dot\gamma)\Big)dt\leq(n-1)\frac{L}{2}\Big(\frac{\pi^2}{L^2}-\kappa\Big)<0,$$

using $\int_0^L \cos^2(\pi t/L)\,dt = \int_0^L \sin^2(\pi t/L)\,dt = L/2$. So $I(V_i, V_i) < 0$ for some i, and as in the proof of Theorem 40.10.6 the proper variation $\exp_{\gamma(t)}(sV_i(t))$ gives curves from p to q shorter than $L = d(p, q)$, which is absurd. Hence every two points have distance at most $\pi/\sqrt{\kappa}$, and M, being closed and bounded in itself, is compact by Theorem 40.6.8.

For the fundamental group, let $\pi\colon \widetilde{M} \to M$ be a universal covering (Theorem 19.6.5). It is a manifold (Proposition 20.6.1(4)) with a unique smooth structure making π a local diffeomorphism (Example 33.7.2), and $\tilde{g} = \pi^* g$ makes π a local isometry, so $\widetilde{M}$ satisfies the same bound on the Ricci curvature. In charts y of M and $y \circ \pi$ of $\widetilde{M}$ the metrics have the same coefficients, hence the same geodesic equations; so π and its local inverses map geodesics to geodesics. Consequently $\widetilde{M}$ is complete: if $\tilde{\gamma}$ is a geodesic of $\widetilde{M}$, the geodesic $\pi \circ \tilde{\gamma}$ extends to $\mathbb{R}$ by completeness of M, its lift through $\tilde{\gamma}(0)$ (Proposition 19.4.3) is locally a geodesic followed by a local inverse of π, hence a geodesic, and it extends $\tilde{\gamma}$ to $\mathbb{R}$. By the first part $\widetilde{M}$ is compact, so the fibre $\pi^{-1}(p)$, which is closed and discrete, is finite. By Corollary 19.4.4(3), $\pi_1(M, p)$ acts on this fibre transitively with trivial stabiliser, since $\widetilde{M}$ is simply connected; so $\pi_1(M, p)$ is finite. □

For example the round sphere of radius $1/\sqrt{\kappa}$ has $\mathrm{Ric} = (n-1)\kappa g$ and diameter exactly $\pi/\sqrt{\kappa}$, so the bound is sharp; Cheng's theorem (Theorem 40.16.18) says that it is the only case of equality. The hypothesis cannot be weakened to $\mathrm{Ric} > 0$: the paraboloid $z = x^2 + y^2$ is complete, being closed in $\mathbb{R}^3$, and has Gaussian curvature $4/(1+4x^2+4y^2)^2 > 0$ (Section 40.11), but it is not compact; its curvature tends to 0 at infinity.

Remark 40.10.8. Compactness in Theorem 40.10.7 comes from the *uniform* bound $\mathrm{Ric} \geq (n-1)\kappa g$ with $\kappa > 0$, not from positivity alone: a non-compact manifold can carry complete metrics with $K > 0$, such as the paraboloid, but no complete metric with $\mathrm{Ric} \geq \delta g$ for a constant $\delta > 0$. On a compact manifold, positive curvature is automatically bounded below by a positive constant. Complete non-compact manifolds with $K > 0$ are nevertheless very restricted: by the theorem of Gromoll and Meyer [GM69] they are diffeomorphic to $\mathbb{R}^n$, in particular contractible and simply connected. More generally, by the soul theorem of Cheeger and Gromoll [CG72a], a complete non-compact manifold with $K \geq 0$ is diffeomorphic to the normal bundle of a compact totally geodesic submanifold, its *soul*, and Perelman [Per94] proved that the soul is a point as soon as all sectional curvatures are positive at one point. Neither completeness nor strict positivity can be dropped: the sphere minus two antipodal points has $K = 1$ and $\pi_1 = \mathbb{Z}$ (Counterexample 40.10.11 below), and the flat cylinder $S^1 \times \mathbb{R}$ is complete with $K = 0$ and $\pi_1 = \mathbb{Z}$; its soul is a circle.

In the converse direction, the theorem is an obstruction: a connected manifold that is not compact, or that is compact with infinite fundamental group, carries no complete metric with $\mathrm{Ric} \geq \delta g$, $\delta > 0$. For a compact manifold every metric is complete and a positive continuous function is bounded below, so: *a compact manifold with infinite fundamental group, such as the torus T^n or $S^1 \times S^{n-1}$, has no metric of positive Ricci curvature at all.*

The second variation formula also controls the topology of compact manifolds of positive curvature through their isometries. The key step is a fixed point theorem.

Theorem 40.10.9 (Weinstein)**.** *Let M be a compact connected oriented Riemannian manifold of dimension n with positive sectional curvature, and F an isometry of M that preserves the orientation if n is even and reverses it if n is odd. Then F has a fixed point.*

Proof. Suppose not. The function $x \mapsto d(x, F(x))$ is continuous and positive on the compact M; let x_0 be a minimum point, $L = d(x_0, F(x_0)) > 0$, and $\gamma\colon [0, L] \to M$ a minimising geodesic of unit speed from x_0 to $F(x_0)$ (Theorem 40.6.8). An isometry maps geodesics to

geodesics, since in charts y and $y \circ F$ the metric has the same coefficients (as in the proof of Theorem 40.10.7), and it preserves lengths of curves, hence distances.

Step 1: $d_{x_0}F(\dot\gamma(0)) = \dot\gamma(L)$. Let $m = \gamma(L/2)$. Then $d(m, F(m)) \le d(m, F(x_0)) + d(F(x_0), F(m)) = L/2 + d(x_0, m) = L$, and $d(m, F(m)) \ge L$ by the choice of x_0. So the curve that follows γ from m to $F(x_0)$ and then $F \circ \gamma$ from $F(x_0)$ to $F(m)$ has unit speed and length $d(m, F(m))$; by Lemma 40.6.7 it is a geodesic, hence smooth at $F(x_0)$, where its two velocities are $\dot\gamma(L)$ and $d_{x_0}F(\dot\gamma(0))$.

Step 2: a closed-up parallel field. Let $P\colon T_{x_0}M \to T_{F(x_0)}M$ be parallel transport along γ. It is a linear isometry (Proposition 40.4.4) and preserves orientations, as in the proof of Proposition 40.7.1(4). So $B = P^{-1} \circ d_{x_0}F$ is an orthogonal map of $T_{x_0}M$ with $\det B = 1$ if F preserves the orientation and $\det B = -1$ if it reverses it; in both cases of the theorem $\det B = (-1)^n$. As $\dot\gamma$ is parallel, $P(\dot\gamma(0)) = \dot\gamma(L)$, so B fixes $\dot\gamma(0)$ by Step 1, and preserves the orthogonal complement E of $\dot\gamma(0)$, of dimension $m = n - 1$, with $\det(B|_E) = (-1)^{m+1}$. An orthogonal map C of an m-dimensional space with $\det C = (-1)^{m+1}$ has the eigenvalue 1: $\det(C - I) = \det C \det(I - C^T) = \det C \det(I - C) = (-1)^{m+1}(-1)^m \det(C - I) = -\det(C - I)$. So there is a unit vector $v \in E$ with $Bv = v$. Let V be the parallel field along γ with $V(0) = v$; then $|V| = 1$, $V \perp \dot\gamma$, and $V(L) = Pv = d_{x_0}F(v)$.

Step 3: a shorter curve. Let $f(s,t) = \exp_{\gamma(t)}(sV(t))$ and $x_s = \exp_{x_0}(sv)$. Then $f(s,0) = x_s$ and, since F maps geodesics to geodesics, $f(s,L) = \exp_{F(x_0)}(s\, d_{x_0}F(v)) = F(x_s)$. By Proposition 40.10.2 the first derivative of $E(f_s)$ at $s = 0$ is $\langle V, \dot\gamma\rangle\big|_0^L = 0$. By Proposition 40.10.4, whose boundary term vanishes because the curves $s \mapsto f(s,t)$ are geodesics, and since $D_t V = 0$,

$$\frac{d^2}{ds^2}\Big|_{s=0} E(f_s) = -\int_0^L R(V, \dot\gamma, \dot\gamma, V)\, dt < 0,$$

the integrand being the sectional curvature of the plane spanned by the orthonormal vectors V and $\dot\gamma$. So $E(f_s) < E(\gamma)$ for small $s \ne 0$, and $L(f_s) < L$ by Lemma 40.10.1. But f_s joins x_s to $F(x_s)$, so $L(f_s) \ge d(x_s, F(x_s)) \ge L$, a contradiction. □

Corollary 40.10.10 (Synge)**.** *Let M be a compact connected Riemannian manifold with positive sectional curvature.*

1. *If n is even and M is orientable, then M is simply connected.*
2. *If n is odd, then M is orientable.*

Proof. (1) The sectional curvature $K(u,v) = R(u,v,v,u)$ is continuous and positive on the compact set of orthonormal pairs of tangent vectors, so $K \ge k$ for some $k > 0$, and $\mathrm{Ric} \ge (n-1)kg$. Let $\pi\colon \widetilde M \to M$ be a universal covering with the metric π^*g. As in the proof of Theorem 40.10.7, $\widetilde M$ is complete, hence compact by that theorem, and it has positive sectional curvature. Orient $\widetilde M$ by pulling back a volume form μ of M by the local diffeomorphism π. A deck transformation τ satisfies $\pi \circ \tau = \pi$, so it is an isometry and $\tau^*\pi^*\mu = \pi^*\mu$: it preserves the orientation. By Theorem 40.10.9, τ has a fixed point, so it is the identity, by the uniqueness of lifts (Proposition 19.6.1). Hence $\mathrm{Deck}(\pi)$ is trivial, and so is $\pi_1(M) \cong \mathrm{Deck}(\pi)$ (Proposition 19.6.3).

(2) Suppose M is not orientable, and let $p\colon \widehat M \to M$ be the orientation double covering (Definition 23.3.1), which is connected by Proposition 23.3.2(1). It is compact, a smooth manifold for which p is a local diffeomorphism (Example 33.7.2), and it has positive sectional curvature for p^*g. It is orientable: at a point (x, α), p identifies the local homology of $\widehat M$ with $H_n(M \mid x; \mathbb{Z})$ (a sheet over a coordinate ball is mapped homeomorphically), and $(x, \alpha) \mapsto \alpha$ is a continuous choice of generators, that is an orientation (Proposition 36.3.2). The deck transformation $\tau(x, \alpha) = (x, -\alpha)$ is an isometry of p^*g without fixed points, and it

reverses this orientation, because $p \circ \tau = p$ carries the generator α at (x, α) to α at $(x, -\alpha)$, where the chosen generator is $-\alpha$. By Theorem 40.10.9 with n odd, τ has a fixed point, a contradiction. □

Counterexample 40.10.11. Every hypothesis of Theorem 40.10.9 and Corollary 40.10.10 is needed.

1. *The parity conditions in Weinstein's theorem.* The antipodal map $x \mapsto -x$ of the unit sphere S^n is an isometry without fixed points. It is the restriction of $-I$, of determinant $(-1)^{n+1}$, so it preserves the orientation of S^n when n is odd and reverses it when n is even: exactly the cases excluded by the theorem.

2. *Orientability in (1).* $\mathbb{RP}^n = S^n/\{\pm 1\}$ with the round metric has $K = 1$ and $\pi_1 = \mathbb{Z}/2$. For n even it is not orientable (Exercise 36.5.2).

3. *Even dimension in (1).* For n odd, $\mathbb{RP}^n$ is orientable with $K = 1$ and $\pi_1 = \mathbb{Z}/2$; so are the lens spaces $S^3/(\mathbb{Z}/p)$ (Example 20.4.5), with $\pi_1 = \mathbb{Z}/p$, as the group acts by unitary, hence orthogonal, maps.

4. *Odd dimension in (2).* $\mathbb{RP}^2$ has $K = 1$ and is not orientable.

5. *Positive curvature, not just $K \geq 0$.* The flat torus $\mathbb{R}^2/\mathbb{Z}^2$ is compact and orientable of even dimension, with $\pi_1 = \mathbb{Z}^2$. The product $\mathbb{RP}^2 \times S^1$ of the round $\mathbb{RP}^2$ and a circle is compact of dimension 3 with $K \geq 0$ (the curvature tensor of a Riemannian product is the sum of those of the factors, and the circle is flat), and it is not orientable: by Exercise 40.17.8 its holonomy contains $h \times 1$ for $h \in \mathrm{Hol}(\mathbb{RP}^2)$ with $\det h = -1$, which exists by Proposition 40.7.1(4).

6. *Compactness in (1).* The sphere minus two antipodal points, $S^2 \setminus \{\pm e_3\}$, has $K = 1$ and is orientable of even dimension, but by stereographic projection it is homeomorphic to $\mathbb{R}^2 \setminus \{0\}$, with $\pi_1 = \mathbb{Z}$.

7. *Compactness in (2).* Let $A = \mathrm{diag}(-1, -1, -1, 1)$ acting on $U = S^3 \setminus \{\pm e_4\}$. Its fixed points on S^3 are $\pm e_4$, so the group $\{1, A\}$ acts freely, and properly as it is finite, by isometries; $\det A = -1$, so A reverses the orientation of S^3 (defined by the outward normal, which A preserves). The quotient $U/\{1, A\}$ has $K = 1$ (Proposition 40.2.6) and dimension 3, and it is not orientable: an orientation would pull back to an orientation of the connected U preserved by A.

Example 40.10.12. $\mathbb{RP}^2 \times \mathbb{RP}^2$ carries no metric of positive sectional curvature. Its fundamental group $\mathbb{Z}/2 \times \mathbb{Z}/2$ (Proposition 19.3.5) is nontrivial, so by Corollary 40.10.10(1) it could only be non-orientable. But then its orientation double covering would be a compact orientable 4-manifold of positive curvature whose fundamental group has order 2, a subgroup of index 2 of $\mathbb{Z}/2 \times \mathbb{Z}/2$ (Corollary 19.4.4), contradicting (1). Whether $S^2 \times S^2$ carries a metric of positive sectional curvature is a famous open question of Hopf; Synge's theorem does not decide it, as $S^2 \times S^2$ is simply connected.

Remark 40.10.13. Theorems 40.8.10 and 40.10.7 are the simplest *comparison theorems*, which compare a manifold with a space of constant curvature through its Jacobi fields. The Rauch comparison theorem [dC92, Chapter 10] and the Bishop–Gromov volume comparison [Lee18] are their refinements, and they lead to the sphere theorems and to Gromov's compactness theorem.

A different kind of rigidity concerns straight lines. A *ray* in a Riemannian manifold is a unit speed geodesic $\gamma\colon [0,\infty) \to M$ with $d(\gamma(s),\gamma(t)) = |s-t|$ for all s,t, and a *line* is a unit speed geodesic $\gamma\colon \mathbb{R} \to M$ with the same property for all $s,t \in \mathbb{R}$: it minimises length between any two of its points.

Proposition 40.10.14. *Let M be complete and connected.*

1. *If M is not compact, every point is the starting point of a ray.*
2. *If there are a compact $C \subseteq M$ and points p_k, q_k with $d(p_k, C) \to \infty$ and $d(q_k, C) \to \infty$ such that every curve from p_k to q_k meets C, then M contains a line.*

Proof. (1) By Theorem 40.6.8 closed bounded sets are compact, so M is unbounded: choose q_k with $L_k = d(p,q_k) \to \infty$ and minimising unit speed geodesics $\gamma_k(t) = \exp_p(tv_k)$, $t \in [0,L_k]$, $|v_k| = 1$. By compactness of the unit sphere a subsequence has $v_k \to v$; let $\gamma(t) = \exp_p(tv)$, defined for all t. For fixed $s,t \geq 0$ and k large, $d(\gamma_k(s),\gamma_k(t)) = |s-t|$, since a subarc of a minimising curve is minimising; by continuity of exp and of d, $d(\gamma(s),\gamma(t)) = |s-t|$.

(2) Let σ_k be a minimising unit speed geodesic from p_k to q_k. It meets C; shift its parameter so that $\sigma_k(0) \in C$, and σ_k is defined on $[-a_k, b_k]$ with $a_k = d(p_k, \sigma_k(0)) \geq d(p_k, C)$ and $b_k \geq d(q_k, C)$, both tending to ∞. The unit tangent vectors at points of the compact C form a compact set, so a subsequence has $\dot\sigma_k(0) \to v \in T_xM$, $x \in C$. As in (1), $\gamma(t) = \exp_x(tv)$ satisfies $d(\gamma(s),\gamma(t)) = |s-t|$ for all $s,t \in \mathbb{R}$. □

For example, on the flat cylinder $S^{n-1} \times \mathbb{R}$ the curves $\{x\} \times \mathbb{R}$ are lines, and (2) applies with $C = S^{n-1} \times \{0\}$. On the hyperbolic plane every geodesic is a line: it is complete (see the discussion after Definition 40.16.9) with $K = -1$ (Example 40.12.8(1)), so by Theorem 40.8.10 each $\exp_p$ is a diffeomorphism, so the geodesic from $\gamma(s)$ through $\gamma(t)$ is the only one joining them, and the minimising geodesic given by Theorem 40.6.8 must be it.

Theorem 40.10.15 (Cheeger–Gromoll splitting theorem)**.** *Let M be a complete connected Riemannian manifold with* $\mathrm{Ric} \geq 0$ *that contains a line. Then M is isometric to a Riemannian product $N \times \mathbb{R}$, where N is a complete manifold with* $\mathrm{Ric} \geq 0$*, and the line corresponds to a curve $\{x\} \times \mathbb{R}$.*

This is [CG71]. The proof uses the *Busemann functions* $b^{\pm}(x) = \lim_{t\to\infty}\big(t - d(x,\gamma(\pm t))\big)$ of the line γ. The triangle inequality gives $b^+ + b^- \leq 0$, with equality on γ. The condition $\mathrm{Ric} \geq 0$ gives, through the comparison of the Laplacian of distance functions with the Euclidean one, that b^+ and b^- are subharmonic in a weak sense; so $b^+ + b^-$ is subharmonic, and by the maximum principle it vanishes identically. Hence $b^+ = -b^-$ is harmonic and smooth, with $|\nabla b^+| = 1$, and the Bochner formula $0 = \frac{1}{2}\Delta|\nabla b^+|^2 = |\mathrm{Hess}\, b^+|^2 + \mathrm{Ric}(\nabla b^+, \nabla b^+)$ (with the analysts' sign of the Laplacian) forces $\mathrm{Hess}\, b^+ = 0$: the gradient of b^+ is a parallel unit vector field, and its flow splits off the factor $\mathbb{R}$. The same paper shows that the universal covering of a compact manifold with $\mathrm{Ric} \geq 0$ is isometric to $N \times \mathbb{R}^k$ with N compact.

Corollary 40.10.16. *A complete connected manifold with positive definite Ricci curvature, for instance with $K > 0$, contains no line. In particular it satisfies the hypothesis of Proposition 40.10.14(2) for no compact C: outside a large compact set it cannot be separated into two unbounded pieces.*

Proof. If it contained a line, it would be isometric to $N \times \mathbb{R}$ by Theorem 40.10.15. In coordinates (x,t) adapted to the product, the metric is $g_N + dt^2$, and the Christoffel symbols with an index t vanish (Theorem 40.2.8); so $\nabla\partial_t = 0$, $R(X,\partial_t)\partial_t = 0$ for all X, and $\mathrm{Ric}(\partial_t,\partial_t) = 0$, a contradiction. The second statement follows from Proposition 40.10.14(2). □

Both hypotheses of the splitting theorem are needed: the hyperbolic plane contains lines but is not a product, since a product $N \times \mathbb{R}$ of dimension 2 is flat, and the flat cylinder shows that the factor N need not be $\mathbb{R}^{n-1}$. The paraboloid $z = x^2 + y^2$, which has $K > 0$, contains rays from every point but no line.

40.11 Submanifolds and the second fundamental form

Classical differential geometry studies curves and surfaces in $\mathbb{R}^3$ through two quadratic forms: the *first fundamental form*, the metric induced from $\mathbb{R}^3$, and the *second fundamental form*, which measures how the surface bends in the ambient space. This section develops both for a submanifold of any Riemannian manifold, proves the equations of Gauss and Codazzi that relate them to the curvature, and then specialises to surfaces in $\mathbb{R}^3$, where Gauss's equation becomes the Theorema Egregium.

Throughout, $(\overline{M}, \bar{g})$ is a Riemannian manifold with Levi-Civita connection $\overline{\nabla}$ and curvature $\overline{R}$, and $M \subseteq \overline{M}$ is a submanifold with the induced metric $g = \bar{g}|_{TM}$, the *first fundamental form*; we write $\langle -, - \rangle$ for both. For $p \in M$, $T_p\overline{M} = T_pM \oplus N_pM$ with $N_pM = (T_pM)^\perp$, the *normal space*, and $v = v^\top + v^\perp$ is the corresponding decomposition. The normal spaces form the *normal bundle* NM, a vector bundle over M: near each point it has a smooth frame, obtained by applying Gram–Schmidt to a frame of $T\overline{M}$ extending one of TM. For $\overline{M} = \mathbb{R}^N$ with the Euclidean metric, $\overline{\nabla}_X Y$ is the directional derivative of the components of Y, and $\overline{R} = 0$.

Lemma 40.11.1. *Every vector field on an open subset of M extends, near each point, to a vector field on an open subset of $\overline{M}$. For vector fields X, Y on M and extensions $\overline{X}, \overline{Y}$, the value of $\overline{\nabla}_{\overline{X}}\overline{Y}$ at $p \in M$ depends only on X and Y, and we denote it by $\overline{\nabla}_X Y$. Moreover $[\overline{X}, \overline{Y}]$ restricts on M to $[X, Y]$.*

Proof. In a slice chart $(x^1, \dots, x^k, y^1, \dots, y^m)$ of $\overline{M}$, in which M is $\{y = 0\}$, a vector field $\sum_i Y^i(x)\partial/\partial x^i$ on M extends by the same formula, with coefficients independent of y. The value of $\overline{\nabla}_{\overline{X}}\overline{Y}$ at p depends only on $\overline{X}_p = X_p$, since a connection is C^∞-linear in its first argument, and it equals $D_t(\overline{Y} \circ \gamma)(0)$ for any curve γ with $\dot\gamma(0) = X_p$ (Proposition 40.4.1); taking γ in M, $\overline{Y} \circ \gamma = Y \circ \gamma$. Finally X and $\overline{X}$ are related by the inclusion ι, and so are Y and $\overline{Y}$, so $[X, Y]$ and $[\overline{X}, \overline{Y}]$ are ι-related (Lemma 34.2.7). □

Theorem 40.11.2 (Gauss formula)**.** *For vector fields X, Y tangent to M,*

$$\overline{\nabla}_X Y = \nabla_X Y + \mathrm{II}(X, Y),$$

where $\nabla_X Y = (\overline{\nabla}_X Y)^\top$ is the Levi-Civita connection of (M, g), and $\mathrm{II}(X, Y) = (\overline{\nabla}_X Y)^\perp$, the second fundamental form*, is a symmetric $C^\infty(M)$-bilinear map with values in the normal bundle.*

Proof. The operator $\nabla^\top_X Y = (\overline{\nabla}_X Y)^\top$ is a connection on TM: it is C^∞-linear in X, and $\overline{\nabla}_X(fY) = (Xf)Y + f\overline{\nabla}_X Y$ with Y tangent gives the Leibniz rule. It is metric: for tangent Y and Z, $X\langle Y, Z\rangle = \langle \overline{\nabla}_X Y, Z\rangle + \langle Y, \overline{\nabla}_X Z\rangle = \langle \nabla^\top_X Y, Z\rangle + \langle Y, \nabla^\top_X Z\rangle$, as the normal parts are orthogonal to Y and Z. By Lemma 40.11.1 and the torsion-freeness of $\overline{\nabla}$, $\overline{\nabla}_X Y - \overline{\nabla}_Y X = [X, Y]$, which is tangent. Its tangent part says that $\nabla^\top$ is torsion-free, so $\nabla^\top = \nabla$ by the uniqueness in Theorem 40.2.8; its normal part says $\mathrm{II}(X, Y) = \mathrm{II}(Y, X)$. Finally II is C^∞-linear in X, and in Y because $\mathrm{II}(X, fY) = ((Xf)Y + f\overline{\nabla}_X Y)^\perp = f\,\mathrm{II}(X, Y)$. □

So the intrinsic connection of M is the tangential part of the ambient one. For a submanifold of Euclidean space this is the familiar description of the covariant derivative as "differentiate in $\mathbb{R}^N$ and project back onto the tangent space", which classical surface theory takes as the definition [dC76, Chapter 4].

Corollary 40.11.3 (Covariant derivative as tangential projection)**.** *Let $M \subseteq \mathbb{R}^N$ be a submanifold with the induced metric, and for $p \in M$ let $P_p \colon \mathbb{R}^N \to T_pM$ be the orthogonal projection. Regard vector fields on M, and vector fields along curves in M, as $\mathbb{R}^N$-valued functions.*

1. *For vector fields X, Y on M, $\nabla_X Y = P(X(Y))$, where $X(Y) = dY(X)$ is the derivative of the components of Y in the direction X. So the Levi-Civita connection of M is the projection connection of Example 38.3.3(4).*

2. *For a curve $\gamma\colon I \to M$ and a vector field V along γ, $D_t V = P(V'(t))$, the projection of the ordinary derivative of $V\colon I \to \mathbb{R}^N$ onto $T_{\gamma(t)}M$.*

3. *V is parallel along γ if and only if $V'(t) \perp T_{\gamma(t)}M$ for all t, and γ is a geodesic if and only if its acceleration $\gamma''(t)$ is normal to M for all t.*

Proof. (1) In $\mathbb{R}^N$ with the Euclidean metric all Christoffel symbols vanish, so $\overline{\nabla}_{\overline{X}}\overline{Y}$ is the derivative of the components of $\overline{Y}$ in the direction $\overline{X}$. At a point $p \in M$ the direction $\overline{X}_p = X_p$ is tangent to M, so this derivative only involves the values of $\overline{Y}$ along a curve in M with velocity X_p, where $\overline{Y} = Y$; hence $\overline{\nabla}_X Y = X(Y)$, and the Gauss formula (Theorem 40.11.2) gives $\nabla_X Y = (X(Y))^{\top} = P(X(Y))$. The projection connection of the subbundle $TM \subseteq M \times \mathbb{R}^N$ is $\nabla_X s = P(X(s))$ by definition.

(2) The operator $V \mapsto P(V')$ sends vector fields along γ to vector fields along γ and is $\mathbb{R}$-linear. It satisfies the Leibniz rule: $(fV)' = f'V + fV'$, and $P(V) = V$ because $V(t)$ is tangent, so $P((fV)') = f'V + fP(V')$. For $V = Y \circ \gamma$, with Y a vector field near $\gamma(I)$, the chain rule gives $(Y \circ \gamma)'(t) = dY(\dot\gamma(t))$, the derivative of Y in the direction $\dot\gamma(t)$, so $P((Y \circ \gamma)') = \nabla_{\dot\gamma} Y$ by (1), whose value at a point depends only on the direction there. By the uniqueness in Proposition 40.4.1, $P(V') = D_t V$.

(3) $D_t V = 0$ means $P(V') = 0$, that is, V' is normal; for $V = \dot\gamma$ this is the statement about geodesics. □

The formula $P(X(Y))$ looks extrinsic, since it uses the embedding; the point of the Levi-Civita theorem is that the result only depends on the metric of M. In classical surface theory one defines the covariant derivative by the projection and then discovers, by writing it with the Christoffel symbols of the first fundamental form, that it is intrinsic; here the logic runs the other way, and the corollary is a computation. The part of the acceleration that is thrown away is the second fundamental form: for a curve γ in M,

$$\gamma'' = D_t\dot\gamma + \mathrm{II}(\dot\gamma, \dot\gamma),$$

the acceleration measured inside M plus the normal acceleration forced on the curve by the bending of M (Theorem 40.11.2, applied along γ as in (2)). A geodesic is a curve with no acceleration inside M; it is accelerated only as much as is needed to stay on M. The same argument, with the covariant derivative $\overline{D}_t$ of $\overline{M}$ in place of the ordinary derivative, shows for a submanifold of any Riemannian manifold $\overline{M}$ that $D_t V = (\overline{D}_t V)^{\top}$, and that a curve in M is a geodesic of M exactly when its acceleration in $\overline{M}$ is normal to M.

Example 40.11.4. 1. *Circles of latitude.* On the unit sphere S^2, with normal $\nu(x) = x$, let $\gamma(t) = (s\cos t, s\sin t, c)$ with $s = \sin\vartheta_0 > 0$ and $c = \cos\vartheta_0$. Then $\gamma'' = (-s\cos t, -s\sin t, 0)$, $\langle\gamma'', \gamma\rangle = -s^2$, and

$$D_t\dot\gamma = \gamma'' - \langle\gamma'', \gamma\rangle\gamma = (-sc^2\cos t, -sc^2\sin t, s^2c) = sc\,(-c\cos t, -c\sin t, s).$$

The vector $(-c\cos t, -c\sin t, s)$ is the unit tangent to the meridian pointing north. So $D_t\dot\gamma = 0$ only for $c = 0$: among the circles of latitude only the equator is a geodesic. The others curve towards the nearer pole inside the sphere, with geodesic curvature $|D_t\dot\gamma|/|\dot\gamma|^2 = |c|/s = |\cot\vartheta_0|$, in accordance with Definition 40.12.1 and the value given after it.

2. *Great circles.* For $\gamma(t) = \cos(t)p + \sin(t)v$ with $p \perp v$ unit vectors, $\gamma'' = -\gamma$ is normal to the sphere at $\gamma(t)$, so great circles are geodesics; by uniqueness of geodesics with given initial velocity they are all of them (Exercise 40.17.7).

3. *The cylinder* $C = \{x^2 + y^2 = 1\} \subseteq \mathbb{R}^3$, with normal $\nu = (x, y, 0)$. The helix $\gamma(t) = (\cos(at + t_0), \sin(at + t_0), bt + z_0)$ has $\gamma'' = -a^2(\cos(at + t_0), \sin(at + t_0), 0)$, a multiple of ν; so helices are geodesics, including the horizontal circles ($b = 0$) and the vertical lines ($a = 0$). Every geodesic is of this form (Exercise 40.17.23). Unrolling the cylinder onto the plane, which is a local isometry, the helices become straight lines.

4. *Parallel transport* along a circle of latitude, which Example 38.5.6 computed with connection forms, is the solution of the linear equation $V' \perp T_\gamma S^2$, that is $V' = \langle V', \gamma\rangle\gamma = -\langle V, \gamma'\rangle\gamma$ (differentiate $\langle V, \gamma\rangle = 0$): the vector turns only as much as needed to remain tangent to the sphere.

The Gauss formula does not need the ambient extensions of Lemma 40.11.1, and it holds for immersions that are not injective, such as a curve with self-intersections. Let $f : M \to \overline{M}$ be an immersion with the induced metric $g = f^*\bar{g}$, and let $\overline{\nabla}^f = f^*\overline{\nabla}$ be the pullback connection on $f^*T\overline{M}$ (Proposition 38.4.5(1)): it differentiates sections of $f^*T\overline{M}$, such as $df(Y)$ for a vector field Y on M, in the direction of tangent vectors of M, and $\overline{\nabla}^f_X(s \circ f) = \overline{\nabla}_{df(X)}s$ for a vector field s on $\overline{M}$. The differential identifies TM with the subbundle $df(TM)$ of $f^*T\overline{M}$, whose orthogonal complement N_f is the normal bundle of f.

One often writes $\overline{\nabla}_{df(X)}(df(Y))$ for $\overline{\nabla}^f_X(df(Y))$. This is justified by Remark 38.5.1: if near p the section $df(Y)$ is of the form $\overline{Y} \circ f$ for a vector field $\overline{Y}$ on an open set of $\overline{M}$, then $\overline{\nabla}^f_X(df(Y)) = \overline{\nabla}_{df(X)}\overline{Y}$, whatever the extension $\overline{Y}$; for an immersion such extensions exist near every point, since f is an embedding near each point. But $df(Y)$ itself is not a vector field on $\overline{M}$: it is defined only along f, and at a self-intersection $f(p) = f(q)$ it has two values. For maps that are not immersions the abbreviation is even misleading: for a constant curve c and a nonconstant field V along it, $\dot{c} = 0$ but $(c^*\overline{\nabla})_{d/dt}V = \dot{V} \neq 0$.

Proposition 40.11.5 (Gauss formula for immersions)**.** *For vector fields X, Y on M,*

$$\overline{\nabla}^f_X\big(df(Y)\big) = df(\nabla_X Y) + \mathrm{II}(X, Y),$$

where ∇ is the Levi-Civita connection of g and $\mathrm{II}(X, Y) = (\overline{\nabla}^f_X(df(Y)))^\perp$ is a symmetric $C^\infty(M)$-bilinear map with values in N_f. In particular $df(\nabla_X Y) = (\overline{\nabla}^f_X(df(Y)))^\top$. For an embedding this is Theorem 40.11.2, and for $\overline{M} = \mathbb{R}^N$, $\overline{\nabla}^f_X(df(Y))$ is the derivative $X(df(Y))$ of the $\mathbb{R}^N$-valued function $df(Y)$.

Proof. $\overline{\nabla}^f$ is metric: $X\langle\sigma, \tau\rangle = \langle\overline{\nabla}^f_X\sigma, \tau\rangle + \langle\sigma, \overline{\nabla}^f_X\tau\rangle$ for sections σ, τ of $f^*T\overline{M}$. Both sides satisfy the same Leibniz rule when σ or τ is multiplied by a function, and locally $f^*T\overline{M}$ has frames $s_a \circ f$ pulled back from frames of $T\overline{M}$; for $\sigma = s \circ f$ and $\tau = t \circ f$ the identity is $df(X)\langle s, t\rangle = \langle\overline{\nabla}_{df(X)}s, t\rangle + \langle s, \overline{\nabla}_{df(X)}t\rangle$ at $f(p)$.

Symmetry: $\overline{\nabla}^f_X(df(Y)) - \overline{\nabla}^f_Y(df(X)) = df([X, Y])$. The difference of the two sides is $C^\infty(M)$-bilinear in X and Y (the terms with derivatives of a function cancel, as for the torsion), so it suffices to take coordinate fields $X = \partial_i$, $Y = \partial_j$ of a chart u of M. With a chart y of $\overline{M}$ and $f^a = y^a \circ f$, $df(\partial_j) = \sum_a \partial_j f^a\,(\partial_a \circ f)$ and

$$\overline{\nabla}^f_{\partial_i}\big(df(\partial_j)\big) = \sum_a \Big(\partial_i\partial_j f^a + \sum_{b,c} \overline{\Gamma}^a_{bc}(f)\,\partial_i f^b\,\partial_j f^c\Big)(\partial_a \circ f),$$

which is symmetric in i and j since $\overline{\Gamma}^a_{bc} = \overline{\Gamma}^a_{cb}$; and $[\partial_i, \partial_j] = 0$.

Conclusion: as in the proof of Theorem 40.11.2, $\nabla'_X Y = df^{-1}\big((\overline{\nabla}^f_X(df(Y)))^\top\big)$ is a connection on TM; it is metric for g because $\overline{\nabla}^f$ is metric and $N_f \perp df(TM)$, and torsion-free by the tangential part of the symmetry. So $\nabla' = \nabla$ (Theorem 40.2.8); the normal part of the symmetry gives $\mathrm{II}(X,Y) = \mathrm{II}(Y,X)$, and II is C^∞-linear in X and in Y as before. For an embedding and an extension $\overline{Y}$ of Y, $df(Y) = \overline{Y} \circ f$, so $\overline{\nabla}^f_X(df(Y)) = \overline{\nabla}_X \overline{Y}$. For $\overline{M} = \mathbb{R}^N$ the Christoffel symbols vanish. □

Definition 40.11.6. For a normal vector field ν along M, the *shape operator* (or *Weingarten map*) is $S_\nu(X) = -(\overline{\nabla}_X \nu)^\top$, and the *normal connection* is $\nabla^\perp_X \nu = (\overline{\nabla}_X \nu)^\perp$.

Proposition 40.11.7 (Weingarten equation)**.** *For tangent X, Y and normal ν, $\langle S_\nu X, Y\rangle = \langle \mathrm{II}(X,Y), \nu\rangle$. In particular S_ν is self-adjoint on each tangent space, and $\overline{\nabla}_X \nu = -S_\nu X + \nabla^\perp_X \nu$.*

Proof. Differentiate $\langle Y, \nu\rangle = 0$: $0 = \langle \overline{\nabla}_X Y, \nu\rangle + \langle Y, \overline{\nabla}_X \nu\rangle = \langle \mathrm{II}(X,Y), \nu\rangle - \langle Y, S_\nu X\rangle$. Self-adjointness follows from the symmetry of II, and the last formula is the decomposition of $\overline{\nabla}_X \nu$ into its two parts. □

Theorem 40.11.8 (Gauss and Codazzi equations)**.** *For vector fields X, Y, Z, W tangent to M:*

1. *(Gauss) $R(X,Y,Z,W) = \overline{R}(X,Y,Z,W) + \langle \mathrm{II}(Y,Z), \mathrm{II}(X,W)\rangle - \langle \mathrm{II}(X,Z), \mathrm{II}(Y,W)\rangle$;*
2. *(Codazzi) $(\overline{R}(X,Y)Z)^\perp = (\nabla_X \mathrm{II})(Y,Z) - (\nabla_Y \mathrm{II})(X,Z)$, where $(\nabla_X \mathrm{II})(Y,Z) = \nabla^\perp_X(\mathrm{II}(Y,Z)) - \mathrm{II}(\nabla_X Y, Z) - \mathrm{II}(Y, \nabla_X Z)$.*

In particular, for orthonormal $u, v \in T_pM$ the sectional curvatures of the plane they span satisfy $K(u,v) = \overline{K}(u,v) + \langle \mathrm{II}(u,u), \mathrm{II}(v,v)\rangle - |\mathrm{II}(u,v)|^2$.

Proof. By the Gauss formula, $\overline{\nabla}_X \overline{\nabla}_Y Z = \overline{\nabla}_X(\nabla_Y Z) + \overline{\nabla}_X(\mathrm{II}(Y,Z))$, and applying the Gauss formula again to the tangent field $\nabla_Y Z$ and the Weingarten equation to the normal field $\mathrm{II}(Y,Z)$,

$$\overline{\nabla}_X \overline{\nabla}_Y Z = \nabla_X \nabla_Y Z + \mathrm{II}(X, \nabla_Y Z) - S_{\mathrm{II}(Y,Z)} X + \nabla^\perp_X(\mathrm{II}(Y,Z)).$$

Also $\overline{\nabla}_{[X,Y]} Z = \nabla_{[X,Y]} Z + \mathrm{II}([X,Y], Z)$. Subtract the same expressions with X and Y exchanged, and use $[X,Y] = \nabla_X Y - \nabla_Y X$. The tangential part gives $(\overline{R}(X,Y)Z)^\top = R(X,Y)Z - S_{\mathrm{II}(Y,Z)}X + S_{\mathrm{II}(X,Z)}Y$; pairing with W and using the Weingarten equation gives (1). The normal part gives

$$(\overline{R}(X,Y)Z)^\perp = \nabla^\perp_X(\mathrm{II}(Y,Z)) + \mathrm{II}(X, \nabla_Y Z) - \nabla^\perp_Y(\mathrm{II}(X,Z)) - \mathrm{II}(Y, \nabla_X Z) - \mathrm{II}(\nabla_X Y - \nabla_Y X, Z),$$

which is (2) after regrouping. For the last statement take $X = W = u$ and $Y = Z = v$ in (1), recalling $K(u,v) = R(u,v,v,u)$. □

Definition 40.11.9. Let M be a hypersurface ($\dim \overline{M} = \dim M + 1$) with a unit normal field ν, which exists locally, and globally if M and $\overline{M}$ are orientable. The *scalar second fundamental form* is $h(X,Y) = \langle \mathrm{II}(X,Y), \nu\rangle$, so $\mathrm{II} = h\,\nu$, and the shape operator $S = S_\nu$ satisfies $\langle SX, Y\rangle = h(X,Y)$. The eigenvalues $\kappa_1, \dots, \kappa_n$ of the self-adjoint map S on T_pM are the *principal curvatures*, its eigenvectors the *principal directions*, $H = \frac{1}{n}\operatorname{tr} S$ is the *mean curvature*, and $\det S = \kappa_1 \cdots \kappa_n$ the *Gauss–Kronecker curvature*. Changing ν to $-\nu$ changes the signs of h, S, the κ_i and H.

Remark 40.11.10. Notations for the second fundamental form vary, and it may help to collect them. Many texts call the scalar form h of a hypersurface itself the second fundamental form, write it II, and write n for the unit normal; the choice of the unit normal is also called a *co-orientation*. For tangent vectors v, w at p, a tangent vector field W extending w and a unit normal field ν,

$$h(v,w) = \langle \mathrm{II}(v,w), \nu\rangle = \langle \overline{\nabla}_v W, \nu\rangle = -\langle \overline{\nabla}_v \nu, w\rangle = \langle S_\nu v, w\rangle.$$

The first equality is the definition of h; the second is the Gauss formula (Theorem 40.11.2), as $\mathrm{II}(v,w)$ is the normal part of $\overline{\nabla}_v W$; the third comes from differentiating $\langle W, \nu\rangle = 0$; and the last is the Weingarten equation (Proposition 40.11.7). The symmetry $h(v,w) = h(w,v)$ comes from the torsion-freeness of $\overline{\nabla}$, through the Gauss formula. Changing the co-orientation ν to $-\nu$ changes the signs of h and S_ν; for a hypersurface of an oriented manifold, a co-orientation is the same as an orientation (Proposition 36.3.4).

For $M \subseteq \mathbb{R}^{n+1}$, $\overline{\nabla}_v \nu = d\nu(v)$ is the derivative of the *Gauss map* $\nu\colon M \to S^n$, regarded as an $\mathbb{R}^{n+1}$-valued function on M, so that $d\nu$ is an $\mathbb{R}^{n+1}$-valued 1-form (a 1-form with values in the trivial bundle $M \times \mathbb{R}^{n+1}$, Definition 38.2.1). It is tangent to M, since $\langle d\nu(v), \nu\rangle = \frac{1}{2} v(|\nu|^2) = 0$; hence $h(v,w) = -\langle d\nu(v), w\rangle$ and $S = -d\nu$, the form in which classical surface theory states it (Remark 40.11.16).

In higher codimension there is no preferred normal, and one keeps the vector-valued form $\mathrm{II}(v,w) = (\overline{\nabla}_v W)^\perp$, with values in the normal space. With $R(u,v,w,z) = \langle R(u,v)w, z\rangle$, the Gauss equation of Theorem 40.11.8(1) reads

$$\langle R(u,v)w,z\rangle = \langle \overline{R}(u,v)w,z\rangle + \langle \mathrm{II}(u,z), \mathrm{II}(v,w)\rangle - \langle \mathrm{II}(u,w), \mathrm{II}(v,z)\rangle,$$

and for a hypersurface of $\mathbb{R}^{n+1}$, where $\overline{R} = 0$ and $\mathrm{II} = h\,\nu$, it becomes $\langle R(u,v)w,z\rangle = h(u,z)h(v,w) - h(u,w)h(v,z)$ (Corollary 40.11.13 below), the generalisation of Gauss's Theorema Egregium. These are the formulas found in most references, with II in place of h in the last one.

Definition 40.11.11. A submanifold $M \subseteq \overline{M}$ is *totally geodesic* if $\mathrm{II} = 0$, and *totally umbilical* if $\mathrm{II}(X,Y) = g(X,Y)\,\eta$ for a normal vector field η; for a hypersurface the second condition says $S = \lambda\,\mathrm{id}$, all principal curvatures being equal at each point. A point where S is a multiple of the identity is an *umbilic*.

By the Gauss formula (Theorem 40.11.2), M is totally geodesic if and only if every geodesic of M is a geodesic of $\overline{M}$: a curve in M has $\overline{\nabla}_{\dot\gamma}\dot\gamma = \nabla_{\dot\gamma}\dot\gamma + \mathrm{II}(\dot\gamma,\dot\gamma)$, and II is determined by its values $\mathrm{II}(v,v)$ by polarisation.

Proposition 40.11.12. *1. Affine hyperplanes of $\mathbb{R}^{n+1}$ are totally geodesic, and spheres of radius r are totally umbilical with $S = -\frac{1}{r}\mathrm{id}$ for the outward normal.*

2. A connected totally umbilical hypersurface $M \subseteq \mathbb{R}^{n+1}$, $n \geq 2$, is an open subset of a hyperplane or of a sphere.

3. In the half-space model of hyperbolic space, $|dx|^2/x_n^2$, the horospheres *$\{x_n = c\}$ are totally umbilical, with $S = \mathrm{id}$ for the normal $\nu = x_n\partial_n$; their induced metric $|dx'|^2/c^2$ is flat. The vertical hyperplanes $\{x_1 = c\}$ are totally geodesic.*

Proof. (1) For a hyperplane the unit normal is constant, so $S = -d\nu = 0$. For the sphere $|x| = r$, $\nu(x) = x/r$, so $d\nu = \frac{1}{r}\mathrm{id}$ and $S = -\frac{1}{r}\mathrm{id}$ (Remark 40.11.10).

(2) Let $S = \lambda\mathrm{id}$ with λ a smooth function, so $d\nu(X) = -\lambda X$ for tangent X. For coordinate fields $X = \partial_i$, $Y = \partial_j$ of a parametrisation x, $\partial_i\partial_j\nu = -\partial_i(\lambda\,\partial_j x) = -(\partial_i\lambda)\partial_j x - \lambda\partial_i\partial_j x$;

exchanging i and j and subtracting gives $(\partial_i\lambda)\partial_j x = (\partial_j\lambda)\partial_i x$. For $i \neq j$ the vectors $\partial_i x$, $\partial_j x$ are independent ($n \geq 2$), so $\partial_i\lambda = 0$: λ is locally constant, hence constant on the connected M. If $\lambda = 0$, v is constant and $\langle x, v\rangle$ is constant (its differential is $\langle dx, v\rangle = 0$): M lies in a hyperplane. If $\lambda \neq 0$, $d(v + \lambda x) = -\lambda\, dx + \lambda\, dx = 0$, so $v + \lambda x = \lambda c$ is constant and $|x - c| = |v|/|\lambda| = 1/|\lambda|$: M lies in a sphere. In both cases M is contained in a hypersurface N, and the inclusion $M \to N$ is smooth (Proposition 33.6.5) and an injective immersion between manifolds of the same dimension, hence open by the inverse function theorem.

(3) With $g_{ij} = \delta_{ij}x_n^{-2}$, the formula of Theorem 40.2.8 gives, for $i, j < n$, $\Gamma^n_{ij} = \frac{1}{2}x_n^2(-\partial_n g_{ij}) = \delta_{ij}/x_n$ and $\Gamma^k_{ij} = 0$ for $k < n$. So $\nabla_{\partial_i}\partial_j = \delta_{ij}x_n^{-1}\partial_n$ is normal to the horosphere: $\mathrm{II}(\partial_i, \partial_j) = \delta_{ij}x_n^{-1}\partial_n$, and with the unit normal $\nu = x_n\partial_n$, $h(\partial_i, \partial_j) = \delta_{ij}x_n^{-1}x_n g(\partial_n, \partial_n) = \delta_{ij}x_n^{-2} = g(\partial_i, \partial_j)$. So $h = g$ and $S = \mathrm{id}$. The induced metric on $x_n = c$ is $|dx'|^2/c^2$, a constant multiple of the Euclidean one. For $\{x_1 = c\}$, the reflection $x_1 \mapsto 2c - x_1$ is an isometry fixing it pointwise; it maps a geodesic of hyperbolic space tangent to the hyperplane to a geodesic with the same initial data, hence to itself, so such geodesics stay in the hyperplane, which is therefore totally geodesic. □

The horospheres illustrate the Gauss equation: an umbilical surface with principal curvatures 1 in a space of curvature -1 has $K = -1 + 1 \cdot 1 = 0$ (Theorem 40.11.8), and indeed it is flat. In $\mathbb{R}^3$ an umbilical surface has $K = \lambda^2 \geq 0$, by the Theorema Egregium below.

Corollary 40.11.13 (Theorema Egregium)**.** *Let $M^n \subseteq \mathbb{R}^{n+1}$ be a hypersurface with unit normal ν. Then $R(X, Y, Z, W) = h(Y, Z)h(X, W) - h(X, Z)h(Y, W)$, and for orthonormal principal directions e_i, e_j the sectional curvature is $K(e_i, e_j) = \kappa_i\kappa_j$. In particular, for a surface in $\mathbb{R}^3$ the Gaussian curvature is $K = \kappa_1\kappa_2 = \det S$: a quantity defined through the embedding is determined by the metric alone, and so is preserved by isometries between surfaces.*

Proof. Apply Theorem 40.11.8(1) with $\overline{R} = 0$ and $\mathrm{II} = h\nu$, $|\nu| = 1$. For principal directions, $h(e_i, e_i) = \kappa_i$ and $h(e_i, e_j) = 0$ for $i \neq j$. For a surface the only plane is the tangent plane, and $\kappa_1\kappa_2$ is the determinant of S. □

Codazzi's equation for a hypersurface of $\mathbb{R}^{n+1}$ says that $(\nabla_X h)(Y, Z)$ is symmetric in all three arguments. Gauss's and Codazzi's equations are the integrability conditions for the embedding: by Bonnet's theorem, a metric and a symmetric form on a simply connected surface satisfying them come from a surface in $\mathbb{R}^3$, unique up to rigid motions [dC76, Chapter 4].

Proposition 40.11.14 (Surfaces in coordinates)**.** *Let $x\colon \Omega \to \mathbb{R}^3$ be a parametrisation of a surface S (Remark 33.6.19), with coordinates $(u^1, u^2) = (u, v)$, $x_i = \partial x/\partial u^i$, and unit normal $\nu = x_1 \times x_2/|x_1 \times x_2|$. Let*

$$g_{ij} = \langle x_i, x_j\rangle = \begin{pmatrix} E & F \\ F & G \end{pmatrix}, \qquad h_{ij} = \langle \partial_i\partial_j x, \nu\rangle = -\langle x_i, \partial_j\nu\rangle.$$

Then the first and second fundamental forms are $E\,du^2 + 2F\,du\,dv + G\,dv^2$ and $h_{11}\,du^2 + 2h_{12}\,du\,dv + h_{22}\,dv^2$, the matrix of S in the basis (x_1, x_2) is $(g_{ij})^{-1}(h_{ij})$, and

$$K = \frac{h_{11}h_{22} - h_{12}^2}{EG - F^2}, \qquad H = \frac{Eh_{22} - 2Fh_{12} + Gh_{11}}{2(EG - F^2)}.$$

More generally, for a submanifold of $\mathbb{R}^N$ parametrised by $x(u^1, \dots, u^n)$, the decomposition $\partial_i\partial_j x = \Gamma^k_{ij}x_k + \mathrm{II}(x_i, x_j)$ of the second derivatives into tangent and normal parts gives the Christoffel symbols and the second fundamental form, and $R_{ijkl} = R(x_i, x_j, x_k, x_l) = \langle \mathrm{II}_{jk}, \mathrm{II}_{il}\rangle - \langle \mathrm{II}_{ik}, \mathrm{II}_{jl}\rangle$ with $\mathrm{II}_{ij} = \mathrm{II}(x_i, x_j)$.

Proof. In $\mathbb{R}^N$, $\overline{\nabla}$ is the directional derivative, and the coordinate field $\partial/\partial u^j$ is x_j along S, so $\overline{\nabla}_{x_i} x_j = \partial_i \partial_j x$. The Gauss formula splits it as $\nabla_{x_i} x_j + \mathrm{II}(x_i, x_j)$, which gives the general statement, the tangent part being $\Gamma^k_{ij} x_k$, and for surfaces $h_{ij} = \langle \partial_i \partial_j x, \nu \rangle$; differentiating $\langle x_i, \nu \rangle = 0$ gives the second expression. From $\langle S x_i, x_j \rangle = h_{ij}$, the matrix A of S satisfies $(g_{ij})A = (h_{ij})$. So $K = \det S = \det(h_{ij})/\det(g_{ij})$ and $2H = \operatorname{tr}((g_{ij})^{-1}(h_{ij}))$, which are the formulas stated, using $(g_{ij})^{-1} = \frac{1}{EG-F^2}\left(\begin{smallmatrix} G & -F \\ -F & E \end{smallmatrix}\right)$. The formula for R_{ijkl} is Theorem 40.11.8(1) with $\overline{R} = 0$. □

For a hypersurface of $\mathbb{R}^{n+1}$ the unit normal is a map $\nu \colon M \to S^n$, the *Gauss map*, and since $T_{\nu(p)} S^n = \nu(p)^\perp = T_p M$ its differential is an endomorphism of $T_p M$. The shape operator is *minus* this differential, and the next proposition writes the relation between the matrices of $d\nu$, of the first fundamental form and of the second in coordinates.

Proposition 40.11.15 (Weingarten equations in coordinates)**.** *Let $M \subseteq \mathbb{R}^{n+1}$ be a hypersurface with unit normal ν and shape operator $S = S_\nu$, parametrised by $x(u^1, \dots, u^n)$, with $x_i = \partial x/\partial u^i$, $g_{ij} = \langle x_i, x_j \rangle$, $h_{ij} = h(x_i, x_j) = \langle \partial_i \partial_j x, \nu \rangle$, and (g^{ij}) the inverse matrix of (g_{ij}).*

1. *$d\nu = -S$: the derivative $\partial_i \nu = \partial(\nu \circ x)/\partial u^i$ is tangent to M and equals $-S(x_i)$.*
2. *Write $\partial_i \nu = \sum_k a_{ki}\, x_k$, so that $A = (a_{ki})$ is the matrix of $d\nu$ in the basis $(x_1, \dots, x_n)$. Then*

$$h_{ij} = -\sum_{k=1}^{n} a_{ki}\, g_{kj}, \qquad \text{that is,} \qquad (h_{ij}) = -A^T (g_{ij}), \qquad A = -(g_{ij})^{-1}(h_{ij}), \qquad a_{ki} = -\sum_l g^{kl} h_{li},$$

   and the matrix of S in the same basis is $-A = (g_{ij})^{-1}(h_{ij})$.

3. *The Gauss–Kronecker curvature is $\det S = \det(h_{ij})/\det(g_{ij})$, and $\det(d\nu) = (-1)^n \det(h_{ij})/\det(g_{ij})$.*
4. *For a surface in $\mathbb{R}^3$, with E, F, G as in Proposition 40.11.14 and $\Delta = EG - F^2$,*

$$a_{11} = \frac{F h_{12} - G h_{11}}{\Delta}, \quad a_{12} = \frac{F h_{22} - G h_{12}}{\Delta}, \quad a_{21} = \frac{F h_{11} - E h_{12}}{\Delta}, \quad a_{22} = \frac{F h_{12} - E h_{22}}{\Delta},$$

   and $K = \det(d\nu) = \det S = (h_{11} h_{22} - h_{12}^2)/\Delta$.

Proof. (1) Differentiating $\langle \nu, \nu \rangle = 1$ gives $\langle \partial_i \nu, \nu \rangle = 0$, so $\partial_i \nu$ is tangent. In $\mathbb{R}^{n+1}$, $\overline{\nabla}_{x_i} \nu$ is the derivative $\partial_i \nu$ of the components of ν, and by Definition 40.11.6, $S(x_i) = -(\overline{\nabla}_{x_i} \nu)^\top = -\partial_i \nu$.

(2) Differentiating $\langle x_j, \nu \rangle = 0$ in u^i gives $h_{ij} = \langle \partial_i \partial_j x, \nu \rangle = -\langle x_j, \partial_i \nu \rangle$, and inserting $\partial_i \nu = \sum_k a_{ki} x_k$ gives $h_{ij} = -\sum_k a_{ki} g_{kj} = -(A^T(g_{ij}))_{ij}$. Transposing, and using that (g_{ij}) and (h_{ij}) are symmetric, $(h_{ij}) = -(g_{ij})A$, so $A = -(g_{ij})^{-1}(h_{ij})$. The matrix of $S = -d\nu$ is $-A$.

(3) $\det S = \det\big((g_{ij})^{-1}(h_{ij})\big) = \det(h_{ij})/\det(g_{ij})$, and $\det(d\nu) = \det(-S) = (-1)^n \det S$.

(4) Here $(g_{ij})^{-1} = \Delta^{-1}\left(\begin{smallmatrix} G & -F \\ -F & E \end{smallmatrix}\right)$, and multiplying out $-(g_{ij})^{-1}(h_{ij})$ gives the four entries. For $n = 2$ the sign $(-1)^n$ is $+1$, and $K = \det S$ by Corollary 40.11.13. □

Remark 40.11.16. With $e = h_{11}$, $f = h_{12}$, $g = h_{22}$ (the letter g here is not the metric), part (4) is the form in which the Weingarten equations appear in [dC76, Chapter 3], where one works with $d\nu$ itself and $\mathrm{II}_p(v) = -\langle d\nu_p(v), v \rangle$. The two conventions differ by the sign $S = -d\nu$, which is the most common source of sign errors: if a_{ki} denotes the matrix of S instead of $d\nu$, the formula of (2) reads $h_{ij} = +\sum_k a_{ki} g_{kj}$. For surfaces the sign does not affect K, because $\det(-S) = \det S$ for 2×2 matrices, but it changes the sign of the mean curvature, and in odd dimension n it changes the sign of the Gauss–Kronecker curvature. For example, for the unit sphere with the outward normal $\nu(p) = p$, the Gauss map is the identity, $d\nu = \mathrm{id}$, $S = -\mathrm{id}$, and $(h_{ij}) = -(g_{ij})$, as in Example 40.11.17(1) below.

Example 40.11.17. 1. *Spheres.* For the sphere of radius r in $\mathbb{R}^{n+1}$ with outward normal $\nu(p) = p/r$, $\overline{\nabla}_X \nu = X/r$, so $S = -\frac{1}{r}\mathrm{id}$, $h = -\frac{1}{r}g$, every direction is principal with $\kappa = -1/r$, and all sectional curvatures are $\kappa_i\kappa_j = 1/r^2$, as in Example 40.8.3.

2. *Cylinders.* For $\{x^2 + y^2 = 1\} \subseteq \mathbb{R}^3$ with $\nu = (x, y, 0)$, $\overline{\nabla}_X \nu$ is the horizontal part of X, so the principal curvatures are -1 (horizontal circles) and 0 (vertical lines), $K = 0$ and $H = -\frac{1}{2}$. The cylinder is locally isometric to the plane (unroll it), in accordance with the Theorema Egregium, although it bends in $\mathbb{R}^3$.

3. *Graphs.* Let S be the graph of $z = \phi(u, v)$ with $\phi(0) = 0$ and $d\phi(0) = 0$, parametrised by $x(u, v) = (u, v, \phi(u, v))$. At the origin $x_1 = e_1$, $x_2 = e_2$, $\nu = e_3$, so $g_{ij} = \delta_{ij}$ and $h_{ij} = \partial_i\partial_j\phi(0)$: the second fundamental form is the Hessian of ϕ, the principal curvatures are its eigenvalues, and $K(0) = \det \mathrm{Hess}\,\phi(0)$. The paraboloid $z = u^2 + v^2$ has $K(0) = 4$, the saddle $z = u^2 - v^2$ has $K(0) = -4$: positive curvature means that the surface lies locally on one side of its tangent plane, negative curvature that it crosses it.

40.12 Global theorems on surfaces

This section collects the classical global theorems on surfaces, in the form in which they appear in [dC76, Chapters 4 and 5]: the Gauss–Bonnet theorem for regions with boundary, Minding's theorem on surfaces of constant curvature, and the theorems of Hadamard, Liebmann and Hilbert on surfaces in $\mathbb{R}^3$. Throughout, M is an oriented Riemannian surface and J is the rotation by $+\pi/2$ in its tangent planes, as in the proof of Theorem 40.8.11; for a surface in $\mathbb{R}^3$ oriented by a unit normal ν, $Jv = \nu \times v$ (Proposition 36.3.4).

Definition 40.12.1. Let α be a curve in M parametrised by arc length, $|\alpha'| = 1$. Since $\langle D_s\alpha', \alpha'\rangle = \frac{1}{2}\frac{d}{ds}|\alpha'|^2 = 0$, the covariant acceleration is $D_s\alpha' = k_g\, J\alpha'$ for a function k_g, the *geodesic curvature* of α. So α is a geodesic if and only if $k_g = 0$. For a surface in $\mathbb{R}^3$, $D_s\alpha'$ is the tangential part of the acceleration α'' (Theorem 40.11.2), and $k_g = \langle \alpha'', \nu \times \alpha'\rangle$.

For example, a circle of latitude at height $z_0 \in (-1, 1)$ on the unit sphere has geodesic curvature $\pm z_0/(1 - z_0^2)^{1/2}$, and the equator, $z_0 = 0$, is a geodesic.

A *simple region* is a compact set $R \subseteq M$ contained in the domain of a positively oriented chart $x\colon \Omega \to M$, whose preimage is homeomorphic to a closed disc and bounded by a simple closed curve that is piecewise smooth with nonvanishing velocity. Its boundary ∂R is parametrised by arc length so that R lies on the left (that is, $J\alpha'$ points into R), and at a vertex where the smooth arcs meet, the *exterior angle* $\theta_i \in (-\pi, \pi)$ is the oriented angle from the incoming to the outgoing velocity; the *interior angle* is $\pi - \theta_i$.

Theorem 40.12.2 (Theorem of turning tangents)**.** *Let R be a simple region, (e_1, e_2) a positively oriented orthonormal frame on a neighbourhood of R, and write the unit tangent of ∂R on each smooth arc as $\alpha' = \cos\varphi\, e_1 + \sin\varphi\, e_2$ with φ continuous on the arc. Then the sum over the arcs of the changes of φ, plus the sum of the exterior angles, equals 2π.*

This is Hopf's Umlaufsatz [Hop35]; see [dC76, Chapters 4 and 5]. For a smooth convex curve in the Euclidean plane it says that the tangent turns exactly once.

Theorem 40.12.3 (Local Gauss–Bonnet theorem)**.** *Let R be a simple region with smooth boundary arcs α_i and exterior angles $\theta_1, \dots, \theta_k$. Then*

$$\sum_i \int_{\alpha_i} k_g\, ds + \iint_R K\, dA + \sum_{j=1}^{k} \theta_j = 2\pi.$$

Proof. Applying Gram–Schmidt to the coordinate fields of the chart gives a positively oriented orthonormal frame (e_1, e_2) near R. Define the 1-form ω by $\nabla_X e_1 = \omega(X)e_2$; then $\nabla_X e_2 = -\omega(X)e_1$, and the computation in the proof of Theorem 40.8.11 gives $d\omega = -K\,dA$. On a smooth arc, $\alpha' = \cos\varphi\, e_1 + \sin\varphi\, e_2$ with φ smooth (lift the map to the circle through the covering $\mathbb{R} \to S^1$, Proposition 19.4.3), and $J\alpha' = -\sin\varphi\, e_1 + \cos\varphi\, e_2$. Differentiating,

$$D_s\alpha' = \varphi' J\alpha' + \cos\varphi\, \nabla_{\alpha'} e_1 + \sin\varphi\, \nabla_{\alpha'} e_2 = \varphi' J\alpha' + \omega(\alpha')J\alpha',$$

so $k_g = \varphi' + \omega(\alpha')$. Summing over the arcs, $\sum_i \int_{\alpha_i} k_g\, ds = \sum_i(\text{change of } \varphi \text{ on } \alpha_i) + \int_{\partial R} \omega$. The first sum is $2\pi - \sum_j \theta_j$ by Theorem 40.12.2, and $\int_{\partial R} \omega = \iint_R d\omega = -\iint_R K\,dA$ by Stokes' theorem in the chart, that is, Green's theorem for a plane region bounded by a piecewise smooth simple closed curve [dC76, Chapter 4] (Theorem 36.4.8 in the smooth case). □

Corollary 40.12.4. *Let T be a* geodesic triangle*: a simple region bounded by three geodesic arcs, with interior angles ϕ_1, ϕ_2, ϕ_3. Then*

$$\phi_1 + \phi_2 + \phi_3 - \pi = \iint_T K\,dA.$$

So the angles of a geodesic triangle add up to more than π where $K > 0$, to π where $K = 0$, and to less than π where $K < 0$. On the unit sphere the area of a geodesic triangle is its angle excess; for instance the octant bounded by the three coordinate planes has three right angles and area $\frac{3\pi}{2} - \pi = \frac{\pi}{2}$, one eighth of 4π.

Proof. Here $k_g = 0$ on the sides and $\theta_j = \pi - \phi_j$, so Theorem 40.12.3 gives $\iint_T K\,dA + 3\pi - (\phi_1 + \phi_2 + \phi_3) = 2\pi$. □

Theorem 40.12.5 (Global Gauss–Bonnet theorem)**.** *Let $R \subseteq M$ be a compact region whose boundary consists of finitely many simple closed piecewise smooth curves $C_1, \dots, C_m$, oriented with R on the left, with exterior angles $\theta_1, \dots, \theta_k$ at their vertices. Then*

$$\iint_R K\,dA + \sum_i \int_{C_i} k_g\, ds + \sum_j \theta_j = 2\pi\chi(R).$$

Sketch of proof. Choose a triangulation of R fine enough that every triangle is a simple region, with F triangles, E edges and V vertices, so that $\chi(R) = V - E + F$. Apply Theorem 40.12.3 to each triangle and add. Every interior edge is traversed twice in opposite directions, so the integrals of k_g over interior edges cancel, and those over boundary edges give the boundary term. The exterior angles of a triangle are π minus its interior angles, and the interior angles of all triangles add up to 2π at each of the V_{int} interior vertices, to $\pi - \theta_j$ at a corner of ∂R, and to π at the other boundary vertices. So the sum of all exterior angles of the triangles is $3\pi F - 2\pi V_{\text{int}} - \pi V_\partial + \sum_j \theta_j$, where V_∂ is the number of boundary vertices, and adding the local formulas gives $\iint_R K\,dA + \sum_i \int_{C_i} k_g\, ds + \sum_j \theta_j = -\pi F + 2\pi V_{\text{int}} + \pi V_\partial$. Each triangle has three edges, each interior edge lies in two triangles and each boundary edge in one, so $3F = 2E_{\text{int}} + E_\partial$, and $E_\partial = V_\partial$ because the boundary is a union of closed curves; with these relations the right side equals $2\pi(V - E + F)$. The omitted verification is the existence of such triangulations [dC76, Chapter 4]. □

For $R = M$ a closed surface there is no boundary, and the formula is Theorem 40.8.11, $\iint_M K\,dA = 2\pi\chi(M)$. For a disc in the plane bounded by a smooth convex curve, $K = 0$ and $\chi = 1$, and the formula says that the total geodesic curvature of the boundary is 2π: the tangent turns once. For the northern cap of the unit sphere above the latitude z_0, the

area $2\pi(1-z_0)$ plus the total geodesic curvature $2\pi(1-z_0^2)^{1/2}\cdot z_0/(1-z_0^2)^{1/2} = 2\pi z_0$ of the boundary circle gives $2\pi = 2\pi\chi(\text{disc})$.

Combining the global Gauss–Bonnet theorem, proved with a triangulation, with Chern's theorem, proved with a vector field, gives the Poincaré–Hopf theorem for surfaces without using Theorem 39.5.1(3).

Corollary 40.12.6 (Poincaré–Hopf for surfaces)**.** *Let M be a compact oriented surface and X a vector field on M with isolated zeros. Then the sum of the indices of X (Definition 39.6.5) is $\chi(M)$.*

Proof. Choose a metric (Proposition 40.2.2). By Theorem 40.12.5 with $R = M$, $\iint_M K\,dA = 2\pi\chi(M)$. By Theorem 39.6.12 for the Levi-Civita connection on TM, $\int_M e(TM,\nabla) = \sum_z \operatorname{ind}_z(X)$, and $e(TM,\nabla) = \frac{1}{2\pi}K\,dA$ (Definition 39.6.2 and Proposition 40.9.3(2)). □

For example, on the torus of revolution the vector field ∂_v that rotates the torus about its axis has no zeros, and indeed $\chi = 0$; on the round sphere the rotation field has two zeros, at the poles, each of index 1 (in a chart at a pole it is $-y\partial_x + x\partial_y$, a rotation of the source $x\partial_x + y\partial_y$, by Lemma 39.6.6(2)).

Many metrics on surfaces are given as a function times the Euclidean metric, and for them the Gaussian curvature has a simple formula.

Proposition 40.12.7 (Conformal metrics on surfaces)**.** *Let $\Omega \subseteq \mathbb{R}^2$ be open, $u\colon \Omega \to \mathbb{R}$ smooth, and $g = e^{2u}(dx^2+dy^2)$. Then the Gaussian curvature of g is*

$$K = -e^{-2u}\Delta u, \qquad \Delta = \partial_x^2 + \partial_y^2.$$

Proof. Here $g_{ij} = e^{2u}\delta_{ij}$, and the formula of Theorem 40.2.8 gives $\Gamma^k_{ij} = u_i\delta_{jk} + u_j\delta_{ik} - u_k\delta_{ij}$, where $u_1 = u_x$ and $u_2 = u_y$. So $\nabla_{\partial_x}\partial_x = u_x\partial_x - u_y\partial_y$ and $\nabla_{\partial_y}\partial_x = u_y\partial_x + u_x\partial_y$. The fields $e_1 = e^{-u}\partial_x$, $e_2 = e^{-u}\partial_y$ form a positively oriented orthonormal frame, and

$$\nabla_{\partial_x}e_1 = -u_x e^{-u}\partial_x + e^{-u}(u_x\partial_x - u_y\partial_y) = -u_y\,e_2, \qquad \nabla_{\partial_y}e_1 = -u_y e^{-u}\partial_x + e^{-u}(u_y\partial_x + u_x\partial_y) = u_x\,e_2.$$

So the 1-form ω with $\nabla_X e_1 = \omega(X)e_2$ is $\omega = -u_y\,dx + u_x\,dy$, and $d\omega = \Delta u\,dx\wedge dy$. The computation in the proof of Theorem 40.8.11 gives $d\omega = -K\,dA = -Ke^{2u}\,dx\wedge dy$, whence the formula. □

This is also the case $n = 2$ of Proposition 40.9.5, proved there with moving frames; the proof above is the same computation written with the Christoffel symbols.

Example 40.12.8.

1. *The hyperbolic plane.* On $H = \{(x,y) : y > 0\}$ with $(dx^2+dy^2)/y^2$, $u = -\log y$ and $\Delta u = 1/y^2$, so $K = -1$.

2. *The Poincaré disc.* On the unit disc $\mathbb{D}$ with $4(dx^2+dy^2)/(1-x^2-y^2)^2$, $u = \log 2 - \log(1-r^2)$ with $r^2 = x^2+y^2$. For a radial function $\Delta = \partial_r^2 + r^{-1}\partial_r$, and $\Delta\log(1-r^2) = -4/(1-r^2)^2$, so $K = -\frac{(1-r^2)^2}{4}\cdot\frac{4}{(1-r^2)^2} = -1$. The *Cayley map* $w = i(1+z)/(1-z)$ is an isometry from the disc onto H: with $z = x+iy$ and w written as a complex number, $|dw/dz| = 2/|1-z|^2$ and $\operatorname{Im} w = (1-|z|^2)/|1-z|^2$, so $|dw|/\operatorname{Im} w = 2|dz|/(1-|z|^2)$.

3. *The sphere.* In the chart of stereographic projection (Example 18.9.4) the round metric of S^2 is $4(dx^2+dy^2)/(1+x^2+y^2)^2$, by a direct computation; here $\Delta\log(1+r^2) = 4/(1+r^2)^2$, and $K = 1$, as it should.

Theorem 40.12.9 (Minding; Cartan)**.** *Let (M, g) and (N, h) be Riemannian manifolds of the same dimension n with the same constant sectional curvature κ, and let $p \in M$, $q \in N$ and a linear isometry $I\colon T_pM \to T_qN$ be given. Then there are neighbourhoods $U \ni p$ and $V \ni q$ and an isometry $F\colon U \to V$ with $F(p) = q$ and $d_pF = I$. In particular two surfaces with the same constant Gaussian curvature are locally isometric, and every surface of constant curvature 0, 1 or -1 is locally isometric to the plane, the unit sphere or the hyperbolic plane.*

Proof. Let s_κ be the solution of $s''+\kappa s = 0$ with $s(0) = 0$ and $s'(0) = 1$ (so $s_\kappa(t) = t, \sin(\sqrt{\kappa}t)/\sqrt{\kappa}$ or $\sinh(\sqrt{-\kappa}t)/\sqrt{-\kappa}$). We first compute the differential of $\exp_p$. Let $v \in T_pM$ be in the domain of $\exp_p$, $v \neq 0$, $\ell = |v|$, $u = v/\ell$, and $\gamma(t) = \exp_p(tv)$; for $w \in T_pM$ write $w = w_\parallel + w_\perp$ with $w_\parallel$ a multiple of u and $w_\perp \perp u$, and let P_t be parallel transport along γ. By Proposition 40.8.8(2), $J(t) = d_{tv}\exp_p(tw)$ is the Jacobi field along γ with $J(0) = 0$ and $D_tJ(0) = w$. The fields $J_1(t) = tP_t(w_\parallel)$ and $J_2(t) = \ell^{-1}s_\kappa(\ell t)P_t(w_\perp)$ are Jacobi fields: $P_t(u) = \dot\gamma(t)/\ell$, so J_1 is a multiple of $t\dot\gamma$, with $D_t^2J_1 = 0$ and $R(J_1, \dot\gamma)\dot\gamma = 0$; and $P_t(w_\perp) \perp \dot\gamma$, so by the curvature formula of Example 40.8.3, $R(J_2, \dot\gamma)\dot\gamma = \kappa|\dot\gamma|^2J_2 = \kappa\ell^2J_2$, while $D_t^2J_2 = \ell s_\kappa''(\ell t)P_t(w_\perp) = -\kappa\ell^2J_2$. They have the initial values $J_1(0) = J_2(0) = 0$, $D_tJ_1(0) = w_\parallel$ and $D_tJ_2(0) = w_\perp$, so $J = J_1 + J_2$ by uniqueness, and since P_1 is an isometry and $P_1(w_\parallel) \perp P_1(w_\perp)$,

$$|d_v\exp_p(w)|^2 = |J(1)|^2 = |w_\parallel|^2 + \left(\frac{s_\kappa(\ell)}{\ell}\right)^2|w_\perp|^2.$$

The same formula holds on N. Now choose $\varepsilon > 0$ such that $\exp_p$ and $\exp_q$ map the balls of radius ε diffeomorphically onto open sets U and V (Proposition 40.6.2), and let $F = \exp_q \circ I \circ (\exp_p|_{B_\varepsilon})^{-1}\colon U \to V$, a diffeomorphism with $F(p) = q$ and $d_pF = I$. For $x = \exp_p(v)$ and $w \in T_pM$, the chain rule gives $d_xF(d_v\exp_p(w)) = d_{Iv}\exp_q(Iw)$, and since I preserves lengths and orthogonality, the formula above gives the same length for $d_v\exp_p(w)$ and $d_{Iv}\exp_q(Iw)$ (for $v = 0$ both differentials are the identity). As the $d_v\exp_p(w)$ fill T_xM, d_xF preserves lengths, hence inner products, so F is an isometry. □

For instance the cylinder and the cone (without its vertex) are locally isometric to the plane, and the pseudosphere, the surface of revolution of the tractrix, to the hyperbolic plane. Minding's theorem is local: the flat torus $\mathbb{R}^2/\mathbb{Z}^2$ is locally but not globally isometric to the plane.

Lemma 40.12.10. *A compact surface $S \subseteq \mathbb{R}^3$ has a point where $K > 0$. More precisely, if $p_0 \in S$ maximises the distance R to the origin, then $K(p_0) \geq 1/R^2$.*

Proof. The function $|x|^2$ on S has a maximum $R^2 > 0$ at some p_0, since S is compact (and not a point). Its differential $2\langle p_0, -\rangle$ vanishes on $T_{p_0}S$, so $\nu = p_0/R$ is a unit normal at p_0. For a unit-speed curve α in S with $\alpha(0) = p_0$, the function $|\alpha|^2$ has a maximum at 0, so $0 \geq (|\alpha|^2)''(0) = 2\langle\alpha''(0), p_0\rangle + 2|\alpha'(0)|^2$, that is, $\langle\alpha''(0), \nu\rangle \leq -1/R$. The normal part of $\alpha''(0)$ is $\mathrm{II}(\alpha'(0), \alpha'(0))$ (Theorem 40.11.2, applied to the covariant acceleration of α in $\mathbb{R}^3$), so $h(w, w) \leq -1/R$ for every unit tangent vector w at p_0. Both principal curvatures are therefore $\leq -1/R$, and $K(p_0) = \kappa_1\kappa_2 \geq 1/R^2$ (Corollary 40.11.13). □

So there is no compact surface in $\mathbb{R}^3$ with $K \leq 0$ everywhere; in particular the flat torus cannot be embedded isometrically in $\mathbb{R}^3$ as a smooth surface.

Lemma 40.12.11. *Let $f\colon X \to Y$ be a local homeomorphism from a compact Hausdorff space to a connected Hausdorff space. Then f is surjective and a covering map with finite fibres.*

Proof. A local homeomorphism is open, and a continuous map from a compact space to a Hausdorff space is closed (Proposition 18.6.3); so $f(X)$ is open, closed and nonempty, hence

equal to Y. A fibre $f^{-1}(y)$ is discrete, because f is injective near each point, and closed in the compact X, hence finite, say $\{x_1, \dots, x_k\}$. Choose pairwise disjoint open sets $U_i \ni x_i$ mapped homeomorphically onto open sets $V_i \ni y$. The set $f(X \setminus \bigcup_i U_i)$ is compact, hence closed, and does not contain y, so $V = \bigcap_i V_i \setminus f(X \setminus \bigcup_i U_i)$ is an open neighbourhood of y with $f^{-1}(V) = \bigsqcup_i (U_i \cap f^{-1}(V))$, each piece mapped homeomorphically onto V. □

Theorem 40.12.12 (Hadamard). *Let $S \subseteq \mathbb{R}^3$ be a compact connected surface with Gaussian curvature $K > 0$ everywhere. Then S is orientable, its Gauss map $\nu \colon S \to S^2$ (for the unit normal making the second fundamental form negative definite) is a diffeomorphism, S is diffeomorphic to S^2, and $\iint_S K \, dA = 4\pi$.*

Proof. At each point $K = \det S_\nu > 0$, so the self-adjoint shape operator is definite, and exactly one of the two unit normals makes it negative definite; near each point this normal is ± a smooth local normal with a locally constant sign, so it is a unit normal field, and S is orientable (Proposition 36.3.4). For tangent X, $\overline{\nabla}_X \nu$ is tangent, because $\langle \nu, \nu \rangle = 1$, so $d_p\nu(X) = \overline{\nabla}_X \nu = -S_\nu X$ (Proposition 40.11.7); as $T_{\nu(p)}S^2 = \nu(p)^\perp = T_pS$ and $\det(-S_\nu) = K > 0$, $d_p\nu$ is an isomorphism, and ν is a local diffeomorphism (Theorem 33.4.14). By Lemma 40.12.11 it is a covering map, and since S^2 is simply connected (Corollary 19.5.2) and S connected, it has one sheet (Corollary 19.4.4(3)): ν is a bijective local diffeomorphism, hence a diffeomorphism. Orient S by ν and S^2 by its outward normal; then T_pS and $T_{\nu(p)}S^2$ are the same oriented plane, $\nu^*(dA_{S^2}) = \det(d\nu)\, dA_S = K \, dA_S$, and $\iint_S K \, dA = \int_{S^2} dA = 4\pi$ (Proposition 36.4.2). □

The value 4π agrees with the Gauss–Bonnet theorem, $2\pi\chi(S^2)$. Hadamard also proved that such a surface is the boundary of a convex body: it lies on one side of each of its tangent planes [dC76, Chapter 5].

Theorem 40.12.13 (Liebmann). *A compact connected surface in $\mathbb{R}^3$ with constant Gaussian curvature K is a round sphere of radius $1/\sqrt{K}$.*

Sketch of proof. By Lemma 40.12.10 the constant K is positive. With the unit normal ν of Theorem 40.12.12 the principal curvatures are negative; write them as $-\lambda_1 \le -\lambda_2$, so that $\lambda_1 \ge \lambda_2 > 0$ and $\lambda_1\lambda_2 = K$. The continuous function λ_1 attains its maximum at some p, where $\lambda_2 = K/\lambda_1$ attains its minimum. *Hilbert's lemma*, a computation with the Codazzi equation in principal coordinates, says that at a non-umbilic point ($\lambda_1 > \lambda_2$) where λ_1 has a local maximum and λ_2 a local minimum, $K \le 0$. So p is umbilic and $\lambda_1(p) = \sqrt{K}$. Hence $\lambda_1 \le \sqrt{K}$ everywhere, while $\lambda_1^2 \ge \lambda_1\lambda_2 = K$; so $\lambda_1 = \lambda_2 = \sqrt{K}$ everywhere, every point is umbilic, and $S_\nu = -\sqrt{K}\,\mathrm{id}$. Then $d\nu = -S_\nu = \sqrt{K}\,\mathrm{id}$, so $x - \nu/\sqrt{K}$ is constant on the connected surface S: every point lies at distance $1/\sqrt{K}$ from this fixed point. The omitted verification is Hilbert's lemma [dC76, Chapter 5]. □

Theorem 40.12.14 (Hilbert). *There is no complete surface of constant negative Gaussian curvature isometrically immersed in $\mathbb{R}^3$. In particular the hyperbolic plane cannot be realised as a complete surface in $\mathbb{R}^3$.*

This is [Hil01]; see [dC76, Chapter 5] for a complete proof. In outline: on such a surface the asymptotic directions (where $h(w, w) = 0$) give coordinates in which the metric has the form $du^2 + 2\cos\theta \, du\, dv + dv^2$, and the Gauss equation becomes the sine-Gordon equation $\theta_{uv} = \sin\theta$; these coordinates cover the whole hyperbolic plane, whose area is infinite, while the Gauss–Bonnet theorem bounds the area of every coordinate quadrilateral by 2π, a contradiction. Incomplete surfaces of constant negative curvature do exist, such as the pseudosphere; and by Theorem 40.12.9 each is locally isometric to the hyperbolic plane.

40.13 Minimal surfaces

A soap film spanning a wire takes the shape that minimises its area. Its surface is therefore a critical point of the area, and the first variation formula below identifies these critical points: they are the hypersurfaces with mean curvature $H = 0$, the *minimal* ones. We give the classical examples, show that a minimal graph has the least area among all surfaces with the same boundary, compute the second variation, and prove Bernstein's theorem: a minimal graph over the whole plane is a plane. Throughout, $M \subseteq \mathbb{R}^{n+1}$ is a hypersurface with unit normal ν, shape operator $S = -d\nu$ and mean curvature $H = \frac{1}{n}\operatorname{tr} S$ (Definition 40.11.9 and Proposition 40.11.15), and we write $|S|^2 = \operatorname{tr}(S^2) = \sum_i \kappa_i^2$.

Proposition 40.13.1 (First and second variation of area)**.** *Let $x\colon \Omega \to \mathbb{R}^{n+1}$ be a parametrisation of M, ϕ a smooth function with compact support $K \subseteq \Omega$, and $x_s = x + s\phi\nu$ (with $\nu = \nu \circ x$). For small $|s|$ the matrix $g(s)_{ij} = \langle \partial_i x_s, \partial_j x_s\rangle$ is positive definite on K; let $A(s) = \int_K \sqrt{\det g(s)}\,du$, the area of the piece $x_s(K)$. Then, with $dA = \sqrt{\det g}\,du$ and $|\nabla\phi|^2 = \sum_{i,j} g^{ij}\partial_i\phi\,\partial_j\phi$,*

$$A'(0) = -n\int_K H\phi\,dA, \qquad \text{and, if } H = 0, \qquad A''(0) = \int_K \big(|\nabla\phi|^2 - |S|^2\phi^2\big)\,dA.$$

Proof. Write $x_i = \partial_i x$ and $\nu_i = \partial_i \nu$. Then $\partial_i x_s = x_i + s(\partial_i\phi\,\nu + \phi\nu_i)$. Since $\langle x_i, \nu\rangle = 0$, $\langle \nu_i, \nu\rangle = 0$ and $|\nu| = 1$,

$$g(s)_{ij} = g_{ij} + s\phi\big(\langle x_i, \nu_j\rangle + \langle \nu_i, x_j\rangle\big) + s^2\big(\partial_i\phi\,\partial_j\phi + \phi^2\langle \nu_i, \nu_j\rangle\big).$$

By Proposition 40.11.15, $\langle x_i, \nu_j\rangle = -h_{ij}$, and $\nu_i = -\sum_k S_i^k x_k$ with $(S_i^k) = g^{-1}h$ the matrix of S, so $\langle \nu_i, \nu_j\rangle = (S^T g\, S)_{ij} = (hg^{-1}h)_{ij}$. Hence $g(s) = g(I + sY + s^2Z)$ with $Y = -2\phi\, g^{-1}h$ and $Z = g^{-1}(\nabla\phi\,\nabla\phi^T) + \phi^2(g^{-1}h)^2$, where $\nabla\phi$ is the column $(\partial_i\phi)$; in particular $g(s)$ is positive definite on K for small $|s|$. For a matrix X, $\det(I + X) = 1 + \operatorname{tr} X + \frac{1}{2}\big((\operatorname{tr} X)^2 - \operatorname{tr}(X^2)\big) + O(|X|^3)$ (expand the characteristic polynomial, or use the eigenvalues), and $\sqrt{1+\epsilon} = 1 + \frac{\epsilon}{2} - \frac{\epsilon^2}{8} + O(\epsilon^3)$. With $X = sY + s^2Z$ this gives

$$\sqrt{\det g(s)} = \sqrt{\det g}\,\Big(1 + \frac{s}{2}\operatorname{tr} Y + s^2\Big(\frac{1}{2}\operatorname{tr} Z + \frac{1}{8}(\operatorname{tr} Y)^2 - \frac{1}{4}\operatorname{tr}(Y^2)\Big) + O(s^3)\Big),$$

uniformly on K. Now $\operatorname{tr} Y = -2\phi\operatorname{tr} S = -2nH\phi$, $\operatorname{tr}(Y^2) = 4\phi^2|S|^2$ and $\operatorname{tr} Z = |\nabla\phi|^2 + \phi^2|S|^2$. So the coefficient of s is $-nH\phi$, and if $H = 0$ the coefficient of s^2 is $\frac{1}{2}(|\nabla\phi|^2 + \phi^2|S|^2) - \phi^2|S|^2 = \frac{1}{2}(|\nabla\phi|^2 - |S|^2\phi^2)$. Integrating over K gives $A'(0)$ and $A''(0)$. □

Remark 40.13.2. *The second variation without $H = 0$, and the Jacobi operator.* The computation in the proof gives, without assuming $H = 0$, the coefficient of s^2 as $\frac{1}{2}\operatorname{tr} Z + \frac{1}{8}(\operatorname{tr} Y)^2 - \frac{1}{4}\operatorname{tr}(Y^2) = \frac{1}{2}(|\nabla\phi|^2 - |S|^2\phi^2 + n^2H^2\phi^2)$, so

$$A''(0) = \int_K \big(|\nabla\phi|^2 - |S|^2\phi^2 + n^2H^2\phi^2\big)\,dA.$$

For the sphere of radius r in $\mathbb{R}^3$ and $\phi = 1$ (the computation is local, so it applies to the whole sphere), $|S|^2 = 2/r^2$ and $n^2H^2 = 4/r^2$, so $A''(0) = \frac{2}{r^2}\cdot 4\pi r^2 = 8\pi$, in agreement with $A(s) = 4\pi(r+s)^2$. When $H = 0$, Green's formula (Proposition 40.5.4, applied on a compact domain with smooth boundary containing the support of ϕ) gives $\int|\nabla\phi|^2 = -\int \phi\,\Delta\phi$, hence

$$A''(0) = -\int_K \phi\, L\phi\,dA, \qquad L = \Delta + |S|^2,$$

the *Jacobi operator*, or stability operator, of the minimal hypersurface: M is *stable* (the area does not decrease to second order under compactly supported normal variations) if and only if $-L$ is nonnegative on such ϕ. For instance a minimal graph is stable, since it minimises area (Proposition 40.13.6 below).

Definition 40.13.3. A hypersurface $M \subseteq \mathbb{R}^{n+1}$ is *minimal* if $H = 0$ everywhere. A minimal hypersurface is *stable* if $A''(0) \geq 0$ for all variations as in Proposition 40.13.1.

By the proposition, M is minimal if and only if $A'(0) = 0$ for every such variation: if $H(u_0) \neq 0$, a function $\phi \geq 0$ supported where H has the sign of $H(u_0)$, with $\phi(u_0) > 0$, gives $A'(0) \neq 0$. The word "minimal" refers to this critical point property; a minimal surface need not minimise area. For a minimal surface in $\mathbb{R}^3$ the principal curvatures are $\pm\kappa$, so $|S|^2 = 2\kappa^2 = -2K$, and the second variation is $\int_K (|\nabla\phi|^2 + 2K\phi^2)\, dA$.

Example 40.13.4. We use Proposition 40.11.14.

1. *Planes* are minimal, with $S = 0$.

2. *The catenoid*, the surface of revolution of the catenary, $x(u, v) = (\cosh v \cos u, \cosh v \sin u, v)$. Here $E = G = \cosh^2 v$, $F = 0$, the unit normal is $\nu = (\cos u, \sin u, -\sinh v)/\cosh v$, and $h_{11} = -1$, $h_{12} = 0$, $h_{22} = 1$. So $H = (Eh_{22} + Gh_{11})/(2EG) = 0$, and $K = -1/\cosh^4 v$.

3. *The helicoid*, swept out by a line rotating about an axis while moving along it, $x(u, v) = (v \cos u, v \sin u, u)$. Here $E = 1 + v^2$, $F = 0$, $G = 1$, $\nu = (-\sin u, \cos u, -v)/\sqrt{1 + v^2}$, and $h_{11} = h_{22} = 0$, $h_{12} = 1/\sqrt{1 + v^2}$. So $H = 0$ and $K = -1/(1 + v^2)^2$.

Proposition 40.13.5 (Minimal surface equation)**.** *Let $u : \Omega \to \mathbb{R}$ be smooth on an open $\Omega \subseteq \mathbb{R}^2$, write $p = u_x$, $q = u_y$ and $W = \sqrt{1 + p^2 + q^2}$, and give the graph of u the upward unit normal $\nu = (-p, -q, 1)/W$. Then its mean curvature is*

$$2H = \frac{(1 + q^2)u_{xx} - 2pq\, u_{xy} + (1 + p^2)u_{yy}}{W^3} = \frac{\partial}{\partial x}\Big(\frac{p}{W}\Big) + \frac{\partial}{\partial y}\Big(\frac{q}{W}\Big).$$

So the graph is minimal if and only if u solves the minimal surface equation $(1 + q^2)u_{xx} - 2pq\, u_{xy} + (1 + p^2)u_{yy} = 0$.

Proof. For $x(x, y) = (x, y, u(x, y))$: $E = 1 + p^2$, $F = pq$, $G = 1 + q^2$, $EG - F^2 = W^2$, and $h_{11} = u_{xx}/W$, $h_{12} = u_{xy}/W$, $h_{22} = u_{yy}/W$, since the second derivatives of x are $(0, 0, u_{xx})$, $(0, 0, u_{xy})$, $(0, 0, u_{yy})$. The formula for H in Proposition 40.11.14 gives the first expression. For the second, $\partial_x(p/W) = (u_{xx}W^2 - p(pu_{xx} + qu_{xy}))/W^3$ and $\partial_y(q/W) = (u_{yy}W^2 - q(pu_{xy} + qu_{yy}))/W^3$, whose sum is the first expression, as $W^2 - p^2 = 1 + q^2$ and $W^2 - q^2 = 1 + p^2$. □

For example *Scherk's surface*, the graph of $u = \log\cos y - \log\cos x$ over the square $|x|, |y| < \pi/2$, is minimal: $p = \tan x$, $q = -\tan y$, $u_{xy} = 0$, and $(1 + q^2)u_{xx} + (1 + p^2)u_{yy} = \sec^2 y \sec^2 x - \sec^2 x \sec^2 y = 0$. It tends to $\pm\infty$ on the sides of the square, and cannot be extended to a larger domain.

Proposition 40.13.6 (Minimal graphs minimise area)**.** *Let $\Omega \subseteq \mathbb{R}^2$ be open and convex, $u : \Omega \to \mathbb{R}$ a solution of the minimal surface equation, and $G \subseteq \Omega \times \mathbb{R}$ a compact piece of its graph that is a surface with boundary, oriented by the upward normal. Let $\Sigma \subseteq \Omega \times \mathbb{R}$ be a compact oriented surface with boundary (Definition 36.4.5) with $\partial\Sigma = \partial G$ as oriented curves. Then* $\mathrm{Area}(\Sigma) \geq \mathrm{Area}(G)$.

Proof. Extend the unit normal of the graph to the unit vector field $N(x,y,z) = (-p,-q,1)/W$ on $\Omega\times\mathbb{R}$, independent of z, and let $\omega = \iota_N(dx\wedge dy\wedge dz)$. As in the proof of Corollary 40.3.13, $d\omega = (\operatorname{div} N)\,dx\wedge dy\wedge dz$, and $\operatorname{div} N = -\partial_x(p/W) - \partial_y(q/W) = -2H = 0$ by Proposition 40.13.5. The open set $\Omega\times\mathbb{R}$ is convex, so $\omega = d\eta$ for a 1-form η by the Poincaré lemma (Theorem 37.2.4). By Stokes' theorem (Theorem 36.4.8), $\int_\Sigma \omega = \int_{\partial\Sigma}\eta = \int_{\partial G}\eta = \int_G \omega$. By Proposition 40.3.7, on a surface with unit normal n defining its orientation, ω restricts to $\langle N, n\rangle\,dA$. On G, $n = N$, so $\int_G \omega = \operatorname{Area}(G)$; on Σ, $\langle N, n\rangle \le 1$, so $\int_\Sigma \omega \le \operatorname{Area}(\Sigma)$. □

The form ω is a *calibration*: a closed form whose restriction to every surface is at most the area form, with equality on the graph. In particular minimal graphs are stable, since the surfaces $x_s(K)$ of Proposition 40.13.1 stay in $\Omega\times\mathbb{R}$ and have the same boundary as the graph for small $|s|$. The comparison only involves surfaces inside the cylinder $\Omega\times\mathbb{R}$ over a convex domain, where the calibration is defined; a minimal surface that is not a graph, such as the catenoid, need not have this property.

The minimal surface equation is a nonlinear elliptic equation, and its entire solutions are much more rigid than harmonic functions: $x^2 - y^2$ is harmonic on all of $\mathbb{R}^2$ without being affine, but the only minimal graphs over the whole plane are planes. The proof below is Nitsche's [Nit57]; it reduces the statement to Liouville's theorem through a theorem of Jörgens on the Monge–Ampère equation.

Lemma 40.13.7 (Jörgens)**.** *Let $\varphi\colon\mathbb{R}^2\to\mathbb{R}$ be smooth with $\varphi_{xx} > 0$ and $\varphi_{xx}\varphi_{yy} - \varphi_{xy}^2 = 1$ everywhere. Then the Hessian of φ is constant.*

Proof. Write $a = \varphi_{xx}$, $b = \varphi_{xy}$, $c = \varphi_{yy}$, so $a > 0$ and $ac - b^2 = 1$; then $c > 0$, the Hessian $\mathcal{H}$ is positive definite, and $a + c \ge 2\sqrt{ac} \ge 2$.

Step 1: the Lewy map. Let $L\colon\mathbb{R}^2\to\mathbb{R}^2$, $L(z) = z + \nabla\varphi(z)$. For $z_1, z_2\in\mathbb{R}^2$, $\langle\nabla\varphi(z_2) - \nabla\varphi(z_1), z_2 - z_1\rangle = \int_0^1 \langle\mathcal{H}(z_1 + t(z_2 - z_1))(z_2 - z_1), z_2 - z_1\rangle\,dt \ge 0$, so

$$|L(z_2) - L(z_1)|^2 = |z_2 - z_1|^2 + 2\langle\nabla\varphi(z_2) - \nabla\varphi(z_1), z_2 - z_1\rangle + |\nabla\varphi(z_2) - \nabla\varphi(z_1)|^2 \ge |z_2 - z_1|^2.$$

So L is injective, and its image is closed: if $L(z_k)\to w$, then (z_k) is a Cauchy sequence, $z_k\to z$ and $L(z) = w$. Its differential $I + \mathcal{H}$ is positive definite, hence invertible, so L is a local diffeomorphism (Theorem 27.11.2) and its image is open. As $\mathbb{R}^2$ is connected, L is a diffeomorphism of $\mathbb{R}^2$ onto itself.

Step 2: a holomorphic function. Write $(\xi,\eta) = L(x,y)$ and $\zeta = \xi + i\eta$, and define $F(\zeta) = w + iv$ with $w = x - \varphi_x$ and $v = \varphi_y - y$, where $(x,y) = L^{-1}(\xi,\eta)$; F is smooth. From $d(\xi,\eta) = (I + \mathcal{H})\,d(x,y)$ and $d(w,v) = N\,d(x,y)$ with $N = \left(\begin{smallmatrix} 1-a & -b \\ b & c-1\end{smallmatrix}\right)$, the Jacobian matrix of (w,v) with respect to (ξ,η) is

$$N(I+\mathcal{H})^{-1} = \frac{1}{2+a+c}\begin{pmatrix}1-a & -b\\ b & c-1\end{pmatrix}\begin{pmatrix}1+c & -b\\ -b & 1+a\end{pmatrix} = \frac{1}{2+a+c}\begin{pmatrix}c-a & -2b\\ 2b & c-a\end{pmatrix},$$

using $\det(I+\mathcal{H}) = 1 + a + c + ac - b^2 = 2 + a + c$ and $ac - b^2 = 1$ to simplify the entries. This matrix satisfies the Cauchy–Riemann equations, so F is an entire function (Proposition 29.2.3), with

$$F' = \frac{(c-a) + 2ib}{2+a+c}, \qquad |F'|^2 = \frac{(a+c)^2 - 4(ac-b^2)}{(a+c+2)^2} = \frac{a+c-2}{a+c+2} < 1.$$

Step 3: Liouville. The derivative F' is entire (Theorem 29.4.2) and bounded, hence constant (Corollary 29.4.7). So $\rho = |F'|^2 < 1$ is constant, hence so is $a + c = 2(1+\rho)/(1-\rho)$, and then the real and imaginary parts of F' show that $c - a$ and b are constant. So a, b and c are constant. □

Theorem 40.13.8 (Bernstein)**.** *Let $u\colon \mathbb{R}^2 \to \mathbb{R}$ be a smooth solution of the minimal surface equation on the whole plane. Then u is affine, $u(x,y) = \alpha x + \beta y + \gamma$: an entire minimal graph is a plane.*

Proof. With the notation of Proposition 40.13.5, write $\mathcal{M} = (1+q^2)u_{xx} - 2pq\,u_{xy} + (1+p^2)u_{yy}$, and consider the smooth functions

$$A = \frac{1+p^2}{W}, \qquad B = \frac{pq}{W}, \qquad C = \frac{1+q^2}{W}.$$

A direct computation, using $p_x = u_{xx}$, $p_y = q_x = u_{xy}$ and $q_y = u_{yy}$, gives

$$\frac{\partial A}{\partial y} - \frac{\partial B}{\partial x} = -\frac{q\mathcal{M}}{W^3}, \qquad \frac{\partial C}{\partial x} - \frac{\partial B}{\partial y} = -\frac{p\mathcal{M}}{W^3}$$

(for the first, both $W^3\partial_y A$ and $W^3\partial_x B$ are polynomials in $p, q, u_{xx}, u_{xy}, u_{yy}$, and their difference is $-q\mathcal{M}$; the second follows by exchanging x and y). As $\mathcal{M} = 0$, the 1-forms $A\,dx + B\,dy$ and $B\,dx + C\,dy$ are closed on $\mathbb{R}^2$, hence exact by the Poincaré lemma (Theorem 37.2.4): there are smooth P, Q with $dP = A\,dx + B\,dy$ and $dQ = B\,dx + C\,dy$. Then $P\,dx + Q\,dy$ is closed, since $P_y = B = Q_x$, so it is $d\varphi$ for a smooth φ. Its Hessian is

$$\varphi_{xx} = A = \frac{1+p^2}{W} > 0, \qquad \varphi_{xy} = B, \qquad \varphi_{yy} = C, \qquad \varphi_{xx}\varphi_{yy} - \varphi_{xy}^2 = \frac{(1+p^2)(1+q^2) - p^2q^2}{W^2} = 1.$$

By Lemma 40.13.7, A, B and C are constant. Then $A + C = (2 + p^2 + q^2)/W = W + 1/W$ is constant, and so is $W \geq 1$, since $t \mapsto t + 1/t$ is injective on $[1, \infty)$. Hence $p^2 = AW - 1$ and $q^2 = CW - 1$ are constant, and so are p and q, being continuous on the connected plane with constant square. So u has constant gradient and is affine. □

Remark 40.13.9. The *Bernstein problem* asks whether an entire minimal graph over $\mathbb{R}^n$, a solution of $\operatorname{div}(\nabla u/\sqrt{1+|\nabla u|^2}) = 0$ on $\mathbb{R}^n$, must be affine. The answer is yes for $n \leq 7$ (De Giorgi for $n = 3$, Almgren for $n = 4$, Simons [Sim68] for $n \leq 7$) and no for $n \geq 8$: Bombieri, De Giorgi and Giusti [BDGG69] constructed a non-affine entire minimal graph over $\mathbb{R}^8$, related to the area-minimising cone over $S^3 \times S^3 \subseteq \mathbb{R}^8$. The existence of minimal surfaces spanning a given curve, Plateau's problem, is a separate and deep theory.

Minimal hypersurfaces in a Riemannian manifold. So far the ambient space was $\mathbb{R}^{n+1}$. In a curved ambient manifold $\overline{M}$ a new phenomenon appears: the normal geodesics leaving a hypersurface converge or diverge according to the curvature of $\overline{M}$, exactly as nearby geodesics do in the Jacobi equation (Definition 40.8.7), and this changes the area of the pushed hypersurface to second order. The relevant curvature is the average over the tangent directions of the sectional curvatures of the planes containing the normal: for an orthonormal basis $e_1, \dots, e_n$ of T_pM and the unit normal ν,

$$\overline{\operatorname{Ric}}(\nu,\nu) = \sum_{a=1}^{n} \overline{R}(e_a,\nu,\nu,e_a) = \sum_{a=1}^{n} \overline{K}(e_a,\nu),$$

by Definition 40.8.1 applied to the orthonormal basis $e_1, \dots, e_n, \nu$ of $T_p\overline{M}$ (the term with ν vanishes). The second variation formula below is the form in which the area variation is usually quoted.

Proposition 40.13.10 (Second variation of area in a Riemannian manifold)**.** *Let $\overline{M}$ be a Riemannian manifold of dimension $n+1$, with Levi-Civita connection $\overline{\nabla}$ and curvature $\overline{R}$, and let $M \subseteq \overline{M}$ be a hypersurface with unit normal field ν, shape operator S and mean curvature H*

(Definition 40.11.9). Let $x\colon \Omega \to M$ be a parametrisation, $\Omega \subseteq \mathbb{R}^n$ open, and let $F\colon \Omega\times(-\varepsilon,\varepsilon) \to \overline{M}$ be smooth with $F(u,0) = x(u)$, with $F(u,s) = x(u)$ for u outside a compact set $K \subseteq \Omega$, and with $\partial_s F(u,0) = \phi(u)\nu(x(u))$ for a smooth function ϕ. Let $g(s)_{ij} = \langle \partial_i F, \partial_j F\rangle$, which is positive definite on K for small $|s|$, and $A(s) = \int_K \sqrt{\det g(s)}\,du$. Then $A'(0) = -n\int_K H\phi\,dA$, and if $H = 0$,

$$A''(0) = \int_K \Big(|\nabla\phi|^2 - \big(|S|^2 + \overline{\mathrm{Ric}}(\nu,\nu)\big)\phi^2 \Big)\,dA.$$

Proof. Step 1: the setting. Write $F_i = \partial_i F$ and $T = \partial_s F$, vector fields along F, and D_s, D_i for the covariant derivatives along the curves $s \mapsto F(u,s)$ and $u^i \mapsto F(u,s)$ (Proposition 40.4.1). Two facts will be used: $D_s F_i = D_i T$ by the symmetry lemma (Lemma 40.4.3), and $D_s D_i T - D_i D_s T = \overline{R}(T, F_i)T$ by Lemma 40.8.6. At $s = 0$ we write $x_i = \partial_i x$, $h_{ij} = h(x_i, x_j)$, and $Z = D_s T|_{s=0}$, the acceleration of the curves $s \mapsto F(u,s)$; for $u \notin K$ these curves are constant, so T and Z vanish there.

Step 2: derivatives of the metric. By the metric property of D_s and the symmetry lemma,

$$g'_{ij} = \langle D_i T, F_j\rangle + \langle F_i, D_j T\rangle, \qquad g''_{ij} = \langle D_s D_i T, F_j\rangle + \langle F_i, D_s D_j T\rangle + 2\langle D_i T, D_j T\rangle.$$

Step 3: derivatives of the area density. For a smooth family of invertible matrices, $(\det g)' = \det g\ \mathrm{tr}(g^{-1}g')$: indeed $\det g(s+t) = \det g(s)\det\big(I + g(s)^{-1}(g(s+t) - g(s))\big)$ and $\det(I+X) = 1 + \mathrm{tr}\,X + O(|X|^2)$. Also $(g^{-1})' = -g^{-1}g'g^{-1}$, obtained by differentiating $g^{-1}g = I$. Hence

$$\frac{(\sqrt{\det g})'}{\sqrt{\det g}} = \frac{1}{2}\,\mathrm{tr}(g^{-1}g'), \qquad \frac{(\sqrt{\det g})''}{\sqrt{\det g}} = \frac{1}{4}\big(\mathrm{tr}(g^{-1}g')\big)^2 + \frac{1}{2}\,\mathrm{tr}(g^{-1}g'') - \frac{1}{2}\,\mathrm{tr}\big((g^{-1}g')^2\big).$$

For a symmetric bilinear form B on T_pM, $\sum_{i,j} g^{ij}B(x_i,x_j) = \sum_a B(e_a,e_a)$ for any orthonormal basis (e_a), the trace of B with respect to the metric; we use this repeatedly.

Step 4: the values at $s = 0$. Since $|\nu| = 1$, $D_i\nu = \overline{\nabla}_{x_i}\nu$ is orthogonal to ν, so it is tangent, and $D_i\nu = -Sx_i$ by Definition 40.11.6. So $D_i T = (\partial_i\phi)\nu - \phi\,Sx_i$, and

$$\langle D_i T, x_j\rangle = -\phi h_{ij}, \qquad \langle D_i T, D_j T\rangle = \partial_i\phi\,\partial_j\phi + \phi^2\langle Sx_i, Sx_j\rangle.$$

Hence $g^{-1}g'$ is the matrix of $-2\phi S$, with trace $-2nH\phi$, and $\mathrm{tr}((g^{-1}g')^2) = 4\phi^2|S|^2$. This already gives $A'(0) = \int_K \frac{1}{2}(-2nH\phi)\,dA = -n\int_K H\phi\,dA$, differentiating under the integral sign (the integrand is smooth on $K\times(-\varepsilon',\varepsilon')$). Further, $\sum g^{ij}\langle D_i T, D_j T\rangle = |\nabla\phi|^2 + \phi^2\sum_a |Se_a|^2 = |\nabla\phi|^2 + \phi^2|S|^2$, as S is self-adjoint. Finally $D_s D_i T = D_i Z + \phi^2\overline{R}(\nu, x_i)\nu$ at $s = 0$, and

$$\sum_{i,j} g^{ij}\langle \overline{R}(\nu,x_i)\nu, x_j\rangle = \sum_a \overline{R}(\nu,e_a,\nu,e_a) = -\sum_a \overline{R}(e_a,\nu,\nu,e_a) = -\overline{\mathrm{Ric}}(\nu,\nu).$$

So $\frac{1}{2}\,\mathrm{tr}(g^{-1}g'') = \sum g^{ij}\langle D_i Z, x_j\rangle - \phi^2\,\overline{\mathrm{Ric}}(\nu,\nu) + |\nabla\phi|^2 + \phi^2|S|^2$.

Step 5: the term with Z. Write $Z = Z^\top + f\nu$ with $Z^\top$ tangent to M. Then $\langle D_i(f\nu), x_j\rangle = f\langle D_i\nu, x_j\rangle = -f h_{ij}$, whose trace is $-nHf$. By the Gauss formula (Theorem 40.11.2), the tangential part of $D_i Z^\top = \overline{\nabla}_{x_i} Z^\top$ is $\nabla_{x_i} Z^\top$, so the trace of $\langle D_i Z^\top, x_j\rangle$ is $\mathrm{div}_M Z^\top$ (Definition 40.5.1(3)). So $\sum g^{ij}\langle D_i Z, x_j\rangle = \mathrm{div}_M Z^\top - nHf$.

Step 6: conclusion. Now let $H = 0$. Collecting Steps 3–5,

$$\frac{(\sqrt{\det g})''(0)}{\sqrt{\det g}} = 0 + \mathrm{div}_M Z^\top - \phi^2\,\overline{\mathrm{Ric}}(\nu,\nu) + |\nabla\phi|^2 + \phi^2|S|^2 - 2\phi^2|S|^2.$$

The vector field $Z^\top$ vanishes outside K, so $\int_K \mathrm{div}_M Z^\top\,dA = 0$ by the divergence theorem (Proposition 40.5.4, applied on a compact domain with smooth boundary in $x(\Omega)$ containing

$x(K)$ in its interior). Differentiating twice under the integral sign gives the formula. (For the variation $F(u,s) = \exp_{x(u)}(s\phi(u)\nu(x(u)))$, which exists for small $|s|$ as at the beginning of Section 40.10, the curves $s \mapsto F(u,s)$ are geodesics, $Z = 0$, and Step 5 is not needed.) □

Remark 40.13.11. *Reading the formula: what each term means.* Write $Q(\phi) = \int_K (|\nabla\phi|^2 - (|S|^2 + \overline{\mathrm{Ric}}(\nu,\nu))\phi^2)\,dA$ for the second variation. Its three terms describe three different effects of pushing M by the amount ϕ in the normal direction.

1. $|\nabla\phi|^2$*: tilting.* Pushing different points by different amounts tilts the hypersurface, and a tilted piece has more area. For a flat piece of a hyperplane in $\mathbb{R}^{n+1}$ this is exact: the pushed hypersurface is the graph of $s\phi$, whose area element is $\sqrt{1+s^2|\nabla\phi|^2} = 1 + \frac{s^2}{2}|\nabla\phi|^2 + O(s^4)$, with second derivative $|\nabla\phi|^2$ at $s = 0$. This term is always ≥ 0, and it is the only one for a hyperplane in $\mathbb{R}^{n+1}$, which is therefore stable.

2. $-|S|^2\phi^2$*: the shape of* M. Push a piece of a hypersurface in $\mathbb{R}^{n+1}$ by the constant amount s: then $\partial_i(x+s\nu) = x_i - sSx_i$, so the area element is multiplied by $\det(I - sS) = \prod_i(1-s\kappa_i) = 1 - s\sum_i \kappa_i + s^2\sum_{i<j}\kappa_i\kappa_j + O(s^3)$. If $H = 0$, then $\sum_{i<j}\kappa_i\kappa_j = \frac{1}{2}((\sum_i\kappa_i)^2 - \sum_i\kappa_i^2) = -\frac{1}{2}|S|^2$, and the second derivative is $-|S|^2$. Geometrically, a minimal surface curves equally in opposite directions ($\kappa_2 = -\kappa_1$ in $\mathbb{R}^3$, like a saddle); pushing it expands it in one principal direction and contracts it in the other, the first-order effects cancel, and the second-order effect is a loss of area.

3. $-\overline{\mathrm{Ric}}(\nu,\nu)\phi^2$*: the curvature of the ambient space.* The pushed hypersurface consists of points on the normal geodesics. If the ambient sectional curvatures $\overline{K}(e_a,\nu)$ are positive, these geodesics converge, as the meridians do when one pushes the equator of a sphere towards a pole, and the area decreases; the amount is the sum $\sum_a \overline{K}(e_a,\nu) = \overline{\mathrm{Ric}}(\nu,\nu)$ over the n tangent directions (Example 40.13.12 below computes it exactly). In a space of constant curvature κ, $\overline{\mathrm{Ric}}(\nu,\nu) = n\kappa$ (Proposition 40.8.4), so $Q(\phi) = \int_K (|\nabla\phi|^2 - (|S|^2 + n\kappa)\phi^2)\,dA$.

The Jacobi operator and the notations of the literature. Green's formula (Proposition 40.5.4, on a compact domain with smooth boundary containing $x(K)$) gives $\int |\nabla\phi|^2\,dA = -\int \phi\,\Delta\phi\,dA$ for compactly supported ϕ, where $\Delta = \mathrm{div}\,\mathrm{grad}$ is the Laplacian of M with its induced metric (the *Laplace–Beltrami operator*). Hence

$$Q(\phi) = -\int_K \big(\Delta\phi + |S|^2\phi + \overline{\mathrm{Ric}}(\nu,\nu)\phi\big)\phi\,dA = -\int_K \phi\,L\phi\,dA, \qquad L = \Delta + |S|^2 + \overline{\mathrm{Ric}}(\nu,\nu),$$

the *Jacobi operator*, or stability operator, of M. A minimal hypersurface is *stable* if $Q(\phi) \geq 0$ for all compactly supported ϕ. Both curvature terms make stability harder; positive Ricci curvature of $\overline{M}$ pushes towards instability, negative towards stability. The literature uses several equivalent conventions, and it helps to have a dictionary:

1. The second fundamental form is often written A, and $|A|^2 = \sum_{i,j} h_{ij}h^{ij} = \sum_i \kappa_i^2$ is our $|S|^2$ (the norm of h and of S agree, as S is h with an index raised). The variation function is often written u, the Laplacian Δ_M, and the area element $d\mu$ or $d\omega$.

2. Many authors define the Jacobi operator with the opposite sign, $L = -\Delta_M - |A|^2 - \overline{\mathrm{Ric}}(\nu,\nu)$, so that $Q(u) = \int_M u\,Lu$ and stability means that L is a nonnegative operator (its eigenvalues on compactly supported functions are ≥ 0). This is the same quadratic form as ours.

3. Some authors use the geometer's Laplacian $\Delta = -\operatorname{div}\operatorname{grad}$, which is a nonnegative operator; then L is written $\Delta - |A|^2 - \overline{\operatorname{Ric}}(\nu,\nu)$.

To translate a formula, check it on the hyperplane in $\mathbb{R}^{n+1}$: there $Q(u) = \int |\nabla u|^2 \geq 0$.

Curves. For $n = 1$, M is a curve in a surface $\overline{M}$, A is its length, and S is its geodesic curvature, so $H = 0$ means that M is a geodesic and then $S = 0$. Since $\overline{\operatorname{Ric}}(\nu,\nu) = \overline{K}$ for a surface (Proposition 40.8.4), the formula becomes $\int (\phi'^2 - K\phi^2)\,dt$ for a unit speed geodesic γ. This is the index form $I(V,V)$ of Proposition 40.10.4 for $V = \phi\nu$: the unit normal ν is parallel along γ (its derivative is orthogonal to ν and, as $\langle D_t\nu, \dot\gamma\rangle = -\langle \nu, D_t\dot\gamma\rangle = 0$, to $\dot\gamma$), so $D_t V = \phi'\nu$ and $R(V,\dot\gamma,\dot\gamma,V) = K\phi^2$. So the second variation of area generalises the second variation of geodesics, and the three terms above correspond to those of Remark 40.10.5.

Example 40.13.12. *The equator of a sphere.* Let $M = S^n = \{x_{n+2} = 0\}$ be the equator of the unit sphere $\overline{M} = S^{n+1} \subseteq \mathbb{R}^{n+2}$, with the constant unit normal $\nu = e_{n+2}$. The derivative of the constant field ν in $\mathbb{R}^{n+2}$ is zero, so its tangential projection $\overline{\nabla}_X \nu$ vanishes (Corollary 40.11.3): $S = 0$, the equator is totally geodesic and in particular minimal. As $\overline{\operatorname{Ric}} = n\overline{g}$ (Proposition 40.8.4), the proposition with $\phi = 1$ (the computation is pointwise, so it applies on all of S^n) predicts $A''(0) = -n\operatorname{vol}(S^n)$ for the variation by normal geodesics. Directly: the normal geodesic from p is the great circle $s \mapsto \cos s\, p + \sin s\, e_{n+2}$, so $F_s(p) = \cos s\, p + \sin s\, e_{n+2}$ maps S^n onto the parallel sphere $\{x_{n+2} = \sin s\}$, with $dF_s = \cos s \cdot \mathrm{id}$ on tangent vectors. So $g(s) = \cos^2 s\, g$, $A(s) = \cos^n s \operatorname{vol}(S^n)$, and $A''(0) = -n\operatorname{vol}(S^n)$, as predicted. The equator is a minimal hypersurface, but it is a maximum of the area among the parallel spheres, not a minimum.

Corollary 40.13.13. *Let $\overline{M}$ be a Riemannian manifold with $\overline{\operatorname{Ric}}(v,v) > 0$ for all $v \neq 0$, and let $M \subseteq \overline{M}$ be a compact minimal hypersurface without boundary with a unit normal field ν. Then pushing M along its normal geodesics, $F_s(p) = \exp_p(s\nu(p))$, decreases its area to second order: M is not stable.*

Proof. Cover M by finitely many parametrisations and take a subordinate partition of unity (ρ_k) (Proposition 33.8.2). The area of $F_s(M)$, computed with the metric $F_s^*\overline{g}$, is $A(s) = \sum_k \int \rho_k \sqrt{\det g(s)}\,du$, each term in the chart of ρ_k. Steps 3 to 6 of the proof of Proposition 40.13.10 compute $(\sqrt{\det g})''(0)$ pointwise, with $\phi = 1$ and $Z = 0$ (the curves are geodesics), as $-(|S|^2 + \overline{\operatorname{Ric}}(\nu,\nu))\sqrt{\det g}$. Since $\sum_k \rho_k = 1$,

$$A''(0) = -\int_M \left(|S|^2 + \overline{\operatorname{Ric}}(\nu,\nu)\right) dA < 0.$$

□

For example, in the round sphere S^{n+1} every compact minimal hypersurface with a unit normal is unstable, as the equator of Example 40.13.12 illustrates. The proof uses a global unit normal field (that is, M is *two-sided*), since it pushes every point of M to the same side. For a one-sided hypersurface, such as the image of the equator $S^2 \subseteq S^3$ in $\mathbb{RP}^3$, which is a totally geodesic copy of $\mathbb{RP}^2$, the variation with $\phi = 1$ is not defined and the argument does not apply.

40.14 The Laplace–Beltrami operator

The Laplacian $\Delta_g = \operatorname{div}\operatorname{grad}$ of Definition 40.5.1, called the *Laplace–Beltrami operator* when one wants to stress that it lives on a Riemannian manifold, has already appeared several

times: in coordinates (Proposition 40.5.2(1)),

$$\Delta_g f = \frac{1}{\sqrt{\det g}} \sum_{i,j} \partial_i \big(\sqrt{\det g}\, g^{ij} \partial_j f \big),$$

as the trace of the Hessian (Proposition 40.5.2(3)), and in the Jacobi operator of a minimal hypersurface (Remark 40.13.11). This section collects its basic properties: it is symmetric and nonpositive, it measures how f differs from its averages, it detects minimal submanifolds, and on spheres its eigenfunctions are the spherical harmonics. We use the analysts' sign, so that $\Delta_g = \sum_i \partial_i^2$ on $\mathbb{R}^n$; the Hodge Laplacian of Section 40.15 below is $-\Delta_g$ on functions. Throughout, (M, g) is a Riemannian manifold of dimension n and we write Δ for Δ_g.

Proposition 40.14.1 (Green's identities)**.** *Let M be oriented and without boundary, and let f, h be smooth functions, one of which has compact support. Then*

$$\int_M f\, \Delta h \operatorname{vol}_g = -\int_M \langle \operatorname{grad} f, \operatorname{grad} h \rangle \operatorname{vol}_g = \int_M h\, \Delta f \operatorname{vol}_g.$$

In particular $\int_M \Delta f \operatorname{vol}_g = 0$ and $\int_M f\, \Delta f \operatorname{vol}_g = -\int_M |\operatorname{grad} f|^2 \operatorname{vol}_g \le 0$ for compactly supported f: the operator Δ is symmetric *and* nonpositive*. Moreover Δ is the only operator $P\colon C^\infty(M) \to C^\infty(M)$ with $\int_M f P h \operatorname{vol}_g = -\int_M \langle \operatorname{grad} f, \operatorname{grad} h \rangle \operatorname{vol}_g$ for all h and all compactly supported f.*

Proof. The vector field $X = f \operatorname{grad} h$ has compact support and $\operatorname{div} X = \langle \operatorname{grad} f, \operatorname{grad} h \rangle + f \Delta h$, by the Leibniz rule $\nabla_v(fY) = v(f)Y + f \nabla_v Y$ inside the trace. By Proposition 40.5.2(5), $(\operatorname{div} X) \operatorname{vol}_g = d(\iota_X \operatorname{vol}_g)$ is the exterior derivative of a compactly supported form, so its integral vanishes by Stokes' theorem (Theorem 36.4.8) with $\partial M = \emptyset$. This is the first equality; the second follows by exchanging f and h, as the middle term is symmetric. For the special cases take $h = 1$, so that $\int \Delta f = \int f\, \Delta 1 = 0$, and $h = f$. For uniqueness, if P has the property, then $u = Ph - \Delta h$ is smooth and $\int_M f u \operatorname{vol}_g = 0$ for all compactly supported f. Taking $f = \rho u$ for a bump function $\rho \ge 0$ gives $\int_M \rho u^2 \operatorname{vol}_g = 0$, so $u = 0$ where $\rho > 0$; as ρ is arbitrary, $u = 0$. □

The identity $\int f \Delta h = \int h \Delta f$ is the analogue for Δ of the symmetry of a symmetric matrix, with the L^2 inner product $(f, h) = \int_M f h \operatorname{vol}_g$ in place of the dot product; $\int f \Delta f \le 0$ says that Δ is "negative semidefinite". As for symmetric matrices, this has consequences for eigenvalues.

Corollary 40.14.2. *Let M be compact, oriented and without boundary.*

1. *A function with $\Delta f \ge 0$ everywhere (a* subharmonic *function), in particular a* harmonic *function, $\Delta f = 0$, is locally constant.*
2. *If $\Delta f = -\lambda f$ with $f \ne 0$, then $\lambda = \int |\operatorname{grad} f|^2 / \int f^2 \ge 0$, and $\lambda = 0$ only for locally constant f. So the eigenvalues of $-\Delta$ are nonnegative.*
3. *Eigenfunctions $\Delta f_1 = -\lambda_1 f_1$, $\Delta f_2 = -\lambda_2 f_2$ with $\lambda_1 \ne \lambda_2$ are orthogonal: $\int_M f_1 f_2 \operatorname{vol}_g = 0$.*
4. *If the equation $\Delta u = h$ has a solution, then $\int_M h \operatorname{vol}_g = 0$.*

Proof. All functions have compact support. (1) $\int \Delta f = 0$ and $\Delta f \ge 0$ force $\Delta f = 0$; then $\int |\operatorname{grad} f|^2 = -\int f \Delta f = 0$, so $df = 0$ and f is constant on each connected component. (2) $\lambda \int f^2 = -\int f \Delta f = \int |\operatorname{grad} f|^2$; if $\lambda = 0$ then $df = 0$. (3) $-\lambda_1 \int f_1 f_2 = \int f_2 \Delta f_1 = \int f_1 \Delta f_2 = -\lambda_2 \int f_1 f_2$. (4) $\int h = \int \Delta u = 0$. □

Example 40.14.3. 1. On the circle $\mathbb{R}/2\pi\mathbb{Z}$ with $g = d\theta^2$, $\Delta = d^2/d\theta^2$; the functions $\cos k\theta$ and $\sin k\theta$ are eigenfunctions of $-\Delta$ with eigenvalue k^2, and part (3) of the corollary is the orthogonality of the Fourier basis. On the flat torus $\mathbb{R}^n/2\pi\mathbb{Z}^n$ the functions $\cos\langle k, x\rangle$ and $\sin\langle k, x\rangle$, $k \in \mathbb{Z}^n$, have eigenvalue $|k|^2$.

2. On the round sphere S^n, the restriction of a linear function $\langle a, x\rangle$ is an eigenfunction of $-\Delta$ with eigenvalue n (Exercise 40.17.25); more eigenfunctions are found in Corollary 40.14.9 below.

3. On a noncompact manifold part (1) fails: x is a nonconstant harmonic function on $\mathbb{R}^n$. Compactness enters through the absence of boundary terms in Green's identities.

What the Laplacian measures. On $\mathbb{R}$, $f''(p) > 0$ means that f bends upwards, so that $\frac{1}{2}(f(p+r) + f(p-r)) > f(p)$ for small r: the average of f over the two points at distance r exceeds $f(p)$, and $f''(p) = \lim_{r\to 0} \frac{2}{r^2}\big(\frac{1}{2}(f(p+r) + f(p-r)) - f(p)\big)$. The Laplacian is the same quantity in higher dimension, with the two points replaced by a small sphere.

Proposition 40.14.4 (The Laplacian as a mean value)**.** *Let $p \in M$, identify T_pM with $\mathbb{R}^n$ by an orthonormal basis, let $S \subseteq T_pM$ be the unit sphere with its Riemannian measure σ, and for small $r > 0$ let $m_f(r) = \frac{1}{\sigma(S)}\int_S f(\exp_p(rv))\, d\sigma(v)$, the average of f over the points at distance r from p in all directions. Then*

$$m_f(r) = f(p) + \frac{r^2}{2n}\Delta f(p) + O(r^3), \qquad \text{so} \qquad \Delta f(p) = \lim_{r\to 0} \frac{2n}{r^2}\big(m_f(r) - f(p)\big).$$

Proof. The map $(r, v) \mapsto f(\exp_p(rv))$ is smooth on $[0, r_0] \times S$ for small r_0, as exp is smooth on an open set containing $0 \in T_pM$. For fixed v, the curve $\gamma(r) = \exp_p(rv)$ is the geodesic with $\gamma(0) = p$ and $\dot\gamma(0) = v$, so by Proposition 40.5.2(4) and Taylor's formula, $f(\exp_p(rv)) = f(p) + r\, df_p(v) + \frac{r^2}{2}\operatorname{Hess} f_p(v, v) + R(r, v)$ with $|R(r, v)| \le Cr^3$, where C bounds the third derivative in r on the compact set $[0, r_0] \times S$. Now average over $v \in S$. The orthogonal maps of $\mathbb{R}^n$ preserve S and are isometries of it, so they preserve σ. The reflection $v_i \mapsto -v_i$ shows $\int_S v_i\, d\sigma = 0$ and $\int_S v_i v_j\, d\sigma = 0$ for $i \ne j$; the coordinate permutations show that $\int_S v_i^2\, d\sigma$ does not depend on i, and as $\sum_i v_i^2 = 1$ on S it equals $\sigma(S)/n$. Writing $df_p(v) = \sum_i a_i v_i$ and $\operatorname{Hess} f_p(v, v) = \sum_{i,j} b_{ij} v_i v_j$, the averages are 0 and $\frac{1}{n}\sum_i b_{ii} = \frac{1}{n}\Delta f(p)$ (Proposition 40.5.2(3)). □

So $\Delta f(p) > 0$ means that near p the values of f are on average larger than $f(p)$: p lies in a "valley" of f. A harmonic function equals its averages up to order r^3; on $\mathbb{R}^2$ it equals them exactly (the mean value property of harmonic functions, Chapter 29). At a local maximum, $\operatorname{Hess} f_p \le 0$ (Proposition 40.5.2(4)), so $\Delta f(p) \le 0$; hence a function with $\Delta f > 0$ everywhere has no local maximum. This explains the heat equation $\partial_t u = \Delta u$: the temperature rises at points whose surroundings are warmer on average.

The Laplacian of the coordinates of a submanifold. For a submanifold $M \subseteq \mathbb{R}^N$ with the induced metric, the coordinates $x^1, \dots, x^N$ restrict to functions on M, and we write $\Delta x = (\Delta x^1, \dots, \Delta x^N)$, a map $M \to \mathbb{R}^N$. Let II be the second fundamental form (Theorem 40.11.2) and $\vec{H} = \frac{1}{n}\sum_a \mathrm{II}(e_a, e_a)$ the *mean curvature vector*, a normal vector; for a hypersurface with unit normal ν, $\vec{H} = H\nu$ (Definition 40.11.9).

Proposition 40.14.5 (Beltrami's formula)**.** *For a submanifold $M^n \subseteq \mathbb{R}^N$, $\operatorname{Hess} x^k(X, Y) = \langle \mathrm{II}(X, Y), e_k\rangle$ for each coordinate x^k, and*

$$\Delta x = n\vec{H}.$$

In particular M is minimal ($\vec{H} = 0$) if and only if its coordinate functions are harmonic on M.

Proof. For a tangent vector field Y, regarded as a map $M \to \mathbb{R}^N$, $Yx^k = \langle Y, e_k\rangle$ is the k-th component of Y. Differentiating along a tangent field X, $X(Yx^k) = \langle D_X Y, e_k\rangle$, where $D_X Y$ is the ordinary derivative of the vector-valued function Y along X. By the Gauss formula and Corollary 40.11.3, $D_X Y = \nabla_X Y + \mathrm{II}(X,Y)$, so $\operatorname{Hess} x^k(X,Y) = X(Yx^k) - (\nabla_X Y)x^k = \langle D_X Y - \nabla_X Y, e_k\rangle = \langle \mathrm{II}(X,Y), e_k\rangle$. Taking the trace, $\Delta x^k = \sum_a \langle \mathrm{II}(e_a, e_a), e_k\rangle = n\langle \vec{H}, e_k\rangle$. □

Example 40.14.6. 1. *The sphere.* For the unit sphere $S^n \subseteq \mathbb{R}^{n+1}$ and tangent fields X, Y, differentiating $\langle Y, x\rangle = 0$ gives $\langle D_X Y, x\rangle = -\langle Y, X\rangle$, so $\mathrm{II}(X,Y) = -\langle X, Y\rangle x$, $\vec{H} = -x$ and $\Delta x = -nx$: each coordinate function is an eigenfunction of $-\Delta$ with eigenvalue n, as in Example 40.14.3(2).

2. *The catenoid* $x(u,v) = (\cosh v \cos u, \cosh v \sin u, v)$ has $E = G = \cosh^2 v$, $F = 0$ (Example 40.13.4), so in these conformal coordinates $\Delta = \cosh^{-2} v\,(\partial_u^2 + \partial_v^2)$ (Example 40.5.3(3)). The coordinates $\cosh v \cos u$, $\cosh v \sin u$ and v are harmonic for $\partial_u^2 + \partial_v^2$ (for the first, ∂_u^2 gives $-\cosh v \cos u$ and ∂_v^2 gives $+\cosh v \cos u$), confirming that the catenoid is minimal.

Corollary 40.14.7. *There is no compact oriented minimal submanifold $M \subseteq \mathbb{R}^N$ without boundary and of positive dimension.*

Proof. By Proposition 40.14.5 the coordinate functions are harmonic on M, hence constant on each component of M (Corollary 40.14.2(1)). So each component of M is a point, which contradicts $\dim M > 0$. □

So a soap film always needs a wire: a closed soap bubble is not a minimal surface, and indeed it has constant mean curvature $H \neq 0$, balancing the pressure difference between inside and outside.

Spherical coordinates and spherical harmonics. On $\mathbb{R}^n \setminus \{0\}$ write $x = r\theta$ with $r = |x| > 0$ and $\theta \in S^{n-1}$, and let $\theta^1, \dots, \theta^{n-1}$ be local coordinates on S^{n-1}. The map $(r, \theta) \mapsto r\theta$ sends ∂_r to θ, a unit vector, and ∂_{θ^a} to $r\,\partial_a\theta$, which is orthogonal to θ (differentiate $|\theta|^2 = 1$). So the Euclidean metric is

$$g = dr^2 + r^2 g_{S^{n-1}},$$

with $\sqrt{\det g} = r^{n-1}\sqrt{\det g_{S^{n-1}}}$, $g^{rr} = 1$, $g^{ra} = 0$ and $g^{ab} = r^{-2} g^{ab}_{S^{n-1}}$.

Proposition 40.14.8. *For a smooth function f on $\mathbb{R}^n \setminus \{0\}$,*

$$\Delta f = \frac{\partial^2 f}{\partial r^2} + \frac{n-1}{r}\frac{\partial f}{\partial r} + \frac{1}{r^2}\Delta_{S^{n-1}} f,$$

where $\Delta_{S^{n-1}} f$ is the Laplacian of the round unit sphere applied to $\theta \mapsto f(r\theta)$ for fixed r.

Proof. Insert the metric above into the coordinate formula for Δ. The terms with $i = j = r$ give $r^{1-n}\partial_r(r^{n-1}\partial_r f) = \partial_r^2 f + \frac{n-1}{r}\partial_r f$, and the terms with i, j among the θ^a give r^{-2} times the coordinate formula for $\Delta_{S^{n-1}}$, since the factor r^{n-1} in $\sqrt{\det g}$ does not depend on θ and cancels. □

For $n = 2$ this is the formula $f_{rr} + \frac{1}{r} f_r + \frac{1}{r^2} f_{\theta\theta}$ of Example 40.5.3(2), and for radial functions it is Exercise 40.17.24.

Corollary 40.14.9 (Spherical harmonics)**.** *Let P be a homogeneous polynomial of degree k on $\mathbb{R}^n$ with $\Delta P = 0$, and $Y = P|_{S^{n-1}}$. Then*

$$\Delta_{S^{n-1}} Y = -k(k+n-2)\,Y.$$

Proof. By homogeneity $P(r\theta) = r^k Y(\theta)$. By Proposition 40.14.8, $0 = \Delta P = \big(k(k-1) + (n-1)k\big) r^{k-2} Y + r^{k-2} \Delta_{S^{n-1}} Y$, and $k(k-1) + (n-1)k = k(k+n-2)$. □

For $k = 1$ the eigenvalue of $-\Delta_{S^{n-1}}$ is $n-1 = \dim S^{n-1}$, as in Example 40.14.6(1). For $k = 2$ and $n = 3$, the harmonic polynomials xy, yz, zx, $x^2 - y^2$ and $x^2 + y^2 - 2z^2$ give eigenfunctions on S^2 with eigenvalue 6; in physics these are the orbitals with angular momentum $\ell = 2$, and the eigenvalue $\ell(\ell+1)$ is the square of the angular momentum.

Remark 40.14.10. *Other Laplacians.* The formula $\Delta f = \operatorname{tr}_g \operatorname{Hess} f = \sum_a (\nabla^2 f)(e_a, e_a)$ makes sense for any tensor field T in place of f: $\nabla^*\nabla T = -\sum_a (\nabla_{e_a}\nabla_{e_a} T - \nabla_{\nabla_{e_a} e_a} T)$ is the *rough* or *connection Laplacian* (with the geometers' sign). On differential forms there is also the Hodge Laplacian $d\delta + \delta d$ of the next section, which equals $-\Delta$ on functions; on forms of positive degree the two differ by a curvature term (the Weitzenböck formula), which is the source of the Bochner vanishing theorems mentioned there.

40.15 The Hodge star and the Laplacian

Definition 40.15.1. Let (M, g) be an oriented Riemannian n-manifold. The metric induces inner products on $\Lambda^k T_p^* M$ (Proposition 12.5.2), the volume form vol_g (Proposition 40.3.3), and the *Hodge star* $*\colon \Omega^k(M) \to \Omega^{n-k}(M)$, characterised pointwise by $\alpha \wedge *\beta = \langle \alpha, \beta \rangle \operatorname{vol}_g$ (Theorem 12.5.3); it satisfies $** = (-1)^{k(n-k)}$ on k-forms. In coordinates $\operatorname{vol}_g = \sqrt{\det(g_{ij})}\, dx^1 \wedge \cdots \wedge dx^n$ in a positively oriented chart.

Definition 40.15.2. On a compact oriented Riemannian manifold define the L^2 inner product on $\Omega^k(M)$ by

$$(\alpha, \beta) = \int_M \alpha \wedge *\beta = \int_M \langle \alpha, \beta \rangle \operatorname{vol}_g,$$

the *codifferential* $\delta = (-1)^{n(k+1)+1} * d *\colon \Omega^k(M) \to \Omega^{k-1}(M)$, and the *Hodge Laplacian* $\Delta = d\delta + \delta d$. A form is *harmonic* if $\Delta\alpha = 0$; the space of harmonic k-forms is $\mathcal{H}^k(M)$.

Proposition 40.15.3. *On a compact oriented Riemannian manifold:*

1. $(d\alpha, \beta) = (\alpha, \delta\beta)$ *for* $\alpha \in \Omega^{k-1}$, $\beta \in \Omega^k$*: the codifferential is the formal adjoint of* d*;*
2. Δ *is self-adjoint and non-negative:* $(\Delta\alpha, \alpha) = \|d\alpha\|^2 + \|\delta\alpha\|^2$*;*
3. $\Delta\alpha = 0$ *if and only if* $d\alpha = 0$ *and* $\delta\alpha = 0$*;*
4. $*\Delta = \Delta *$*, so* $*$ *maps harmonic* k*-forms isomorphically to harmonic* $(n-k)$*-forms;*
5. Δ *on functions is* $-\sum_i \partial_i^2$ *in normal coordinates at a point, and in general* $\Delta f = -\frac{1}{\sqrt{\det g}} \sum_{i,j} \partial_i(\sqrt{\det g}\, g^{ij} \partial_j f)$*.*

Proof. (1) By Stokes' theorem (Theorem 36.4.8), $0 = \int_M d(\alpha \wedge *\beta) = \int_M d\alpha \wedge *\beta + (-1)^{k-1} \int_M \alpha \wedge d*\beta$. Using $** = (-1)^{k(n-k)}$ one rewrites $d * \beta = \pm * * d * \beta$, and the sign in the definition of δ is chosen exactly so that the identity becomes $(d\alpha, \beta) = (\alpha, \delta\beta)$.

(2) $(\Delta\alpha, \beta) = (d\delta\alpha, \beta) + (\delta d\alpha, \beta) = (\delta\alpha, \delta\beta) + (d\alpha, d\beta)$, which is symmetric; taking $\beta = \alpha$ gives the formula. (3) If $\Delta\alpha = 0$ then $\|d\alpha\|^2 + \|\delta\alpha\|^2 = 0$, so both vanish; the converse is clear. (4) follows from the definition of δ and $** = \pm 1$: $*d\delta = \delta d *$ and $*\delta d = d\delta *$ up to the signs, which cancel. (5) is a computation with the coordinate formula for $*$ and d, using $\operatorname{vol}_g = \sqrt{\det g}\, dx^1 \wedge \cdots \wedge dx^n$; in normal coordinates at p the first derivatives of g vanish, so $\Delta f(p) = -\sum_i \partial_i^2 f(p)$. □

Remark 40.15.4. The Levi-Civita connection gives a second Laplacian, the *connection Laplacian* $\nabla^*\nabla$, and the two differ by curvature: the *Weitzenböck formula* states $\Delta = \nabla^*\nabla + \mathcal{R}$, where $\mathcal{R}$ is an endomorphism built from the curvature tensor; on 1-forms $\mathcal{R} = \mathrm{Ric}$. Since $(\nabla^*\nabla\alpha, \alpha) = \|\nabla\alpha\|^2 \geq 0$, this gives vanishing theorems: if $\mathrm{Ric} > 0$ on a compact oriented manifold, every harmonic 1-form vanishes, hence $b_1(M) = 0$ by the Hodge theorem (Bochner's theorem, Chapter 43, where the formula on 1-forms is proved). For proofs see [War83, Chapter 6] and [Lee18, Chapter 4].

Example 40.15.5. On a compact connected oriented manifold the harmonic 0-forms are the constants, since $\Delta f = 0$ implies $df = 0$; by (4) the harmonic n-forms are the multiples of vol_g. On the flat torus $\mathbb{R}^n/\mathbb{Z}^n$, the harmonic forms are exactly the forms with constant coefficients in the coordinates dx^i, a space of dimension $\binom{n}{k}$, which matches $b_k(T^n)$ (Exercise 37.5.1).

40.16 Isometries, Killing fields and space forms

The symmetries of a Riemannian manifold are its isometries, and the infinitesimal symmetries are the vector fields whose flows consist of isometries, the Killing fields. This section shows that both are rigid: an isometry is determined by its value and its differential at one point, a Killing field by its value and its covariant derivative at one point, and so the Killing fields form a Lie algebra of dimension at most $n(n+1)/2$. The spaces with the most symmetry are those of constant curvature, and the theorem of Killing and Hopf classifies them: a complete simply connected manifold of constant curvature is Euclidean space, a round sphere or hyperbolic space, and every complete manifold of constant curvature is a quotient of one of these. At the end we state two rigidity theorems of a more difficult kind, Cheng's maximal diameter theorem and the sphere theorem. Throughout, M has dimension n.

A smooth map $F\colon (N, h) \to (M, g)$ between Riemannian manifolds of the same dimension is a *local isometry* if $F^*g = h$, that is, if every d_xF is a linear isometry. By the inverse function theorem (Theorem 33.4.14) it is then a local diffeomorphism, and it restricts to isometries between small open sets.

Lemma 40.16.1. *Let $F\colon N \to M$ be a local isometry.*

1. *F maps geodesics to geodesics: if $v \in T_xN$ is in the domain of $\exp_x$, then $d_xF(v)$ is in the domain of $\exp_{F(x)}$ and $F(\exp_x(v)) = \exp_{F(x)}(d_xF(v))$.*
2. *If N is connected and $F'\colon N \to M$ is another local isometry with $F'(x_0) = F(x_0)$ and $d_{x_0}F' = d_{x_0}F$ at one point x_0, then $F' = F$.*

Proof. (1) Let $x \in N$, and choose an open $U \ni x$ that F maps diffeomorphically onto an open V, and a chart y on V. Then $y \circ F$ is a chart on U, and since $F^*g = h$, the coefficients of h in the chart $y \circ F$ are the same functions as those of g in the chart y. So the Christoffel symbols (Theorem 40.2.8) and the geodesic equations (Definition 40.2.9) are the same, and a curve c in U is a geodesic if and only if $F \circ c$ is one. Being a geodesic is a local condition, so $F \circ \gamma$ is a geodesic whenever γ is. For v in the domain of $\exp_x$, the curve $F \circ \gamma_v$ is a geodesic defined on $[0, 1]$ with initial velocity $d_xF(v)$; by uniqueness it is the geodesic $\gamma_{d_xF(v)}$, which is therefore defined on $[0, 1]$, and evaluating at $t = 1$ gives the formula.

(2) Let $A = \{x \in N : F'(x) = F(x),\ d_xF' = d_xF\}$, which contains x_0. It is closed, as F, F' and their differentials are continuous (in charts, the Jacobian matrices are continuous). It is open: if $x \in A$ and $\exp_x$ maps a ball $B(0, \varepsilon)$ diffeomorphically onto a neighbourhood W of x (Proposition 40.6.2), then by (1), $F(\exp_x v) = \exp_{F(x)}(d_xFv) = F'(\exp_x v)$ for $|v| < \varepsilon$, so $F = F'$ on W, and hence also $dF = dF'$ on W. As N is connected, $A = N$. □

For example, for a geodesic γ and an isometry F, $F \circ \gamma$ is the geodesic with initial velocity $dF(\dot\gamma(0))$; this is the argument of Exercise 40.17.7, which finds the geodesics of the sphere as fixed sets of reflections.

Definition 40.16.2. A vector field X on M with local flow Φ (Theorem 33.9.2) is a *Killing field* if every Φ_t is an isometry from its domain onto its image: $\langle d\Phi_t(v), d\Phi_t(w)\rangle = \langle v, w\rangle$ whenever Φ_t is defined at the base point of v and w. For vector fields X, Y, Z we put

$$(L_X g)(Y,Z) = X\langle Y,Z\rangle - \langle [X,Y],Z\rangle - \langle Y,[X,Z]\rangle.$$

Proposition 40.16.3. *The expression $L_X g$ is a symmetric tensor, and $(L_X g)_p(v,w) = \frac{d}{dt}\big|_{t=0}\langle d_p\Phi_t(v), d_p\Phi_t(w)\rangle$ for $v, w \in T_pM$ (the* Lie derivative *of g). Moreover $(L_X g)(Y,Z) = \langle \nabla_Y X, Z\rangle + \langle Y, \nabla_Z X\rangle$. Consequently the following are equivalent:*

1. *X is a Killing field;*
2. *$X\langle Y,Z\rangle = \langle [X,Y],Z\rangle + \langle Y,[X,Z]\rangle$ for all vector fields Y, Z;*
3. *for every p, the endomorphism $v \mapsto \nabla_v X$ of T_pM is skew-adjoint: $\langle \nabla_v X, w\rangle + \langle v, \nabla_w X\rangle = 0$.*

*In coordinates, (3) reads $\nabla_i X_j + \nabla_j X_i = 0$, where $\nabla_i X_j = \langle \nabla_{\partial_i} X, \partial_j\rangle$ (*Killing's equation*).*

Proof. Since ∇ is metric and torsion-free, $X\langle Y,Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle$ and $[X,Y] = \nabla_X Y - \nabla_Y X$; substituting gives $(L_X g)(Y,Z) = \langle \nabla_Y X, Z\rangle + \langle Y, \nabla_Z X\rangle$. The right side depends only on the values of Y and Z at each point and is symmetric, so $L_X g$ is a symmetric tensor.

Fix p and vector fields Y, Z near p. Let $V(t) = d\Phi_{-t}(Y_{\Phi_t(p)})$ and $W(t) = d\Phi_{-t}(Z_{\Phi_t(p)})$, curves in T_pM with $V(0) = Y_p$, $W(0) = Z_p$, and by Proposition 33.9.7, $V'(0) = [X,Y]_p$ and $W'(0) = [X,Z]_p$. For $a, b \in T_pM$ put $G(t,a,b) = \langle d_p\Phi_t(a), d_p\Phi_t(b)\rangle$, smooth in t and bilinear in (a,b). Since $d\Phi_t(V(t)) = Y_{\Phi_t(p)}$,

$$\langle Y,Z\rangle(\Phi_t(p)) = G(t, V(t), W(t)),$$

and differentiating at $t = 0$ (the derivative of $t \mapsto \Phi_t(p)$ is X_p), $X_p\langle Y,Z\rangle = \partial_t G(0, Y_p, Z_p) + \langle [X,Y]_p, Z_p\rangle + \langle Y_p, [X,Z]_p\rangle$. So $(L_X g)_p(Y_p, Z_p) = \partial_t G(0, Y_p, Z_p)$, and every pair of tangent vectors at p is of the form (Y_p, Z_p).

(1) $\Rightarrow$ (2): if X is Killing, $G(t,a,b) = \langle a, b\rangle$ does not depend on t, so $L_X g = 0$. (2) $\Rightarrow$ (1): let $v, w \in T_pM$ and $f(t) = G(t, v, w)$, on the interval of t for which $\Phi_t(p)$ is defined. For t_0 in it, $\Phi_{t_0+s} = \Phi_s \circ \Phi_{t_0}$ near p, so $f(t_0 + s) = \langle d\Phi_s(v'), d\Phi_s(w')\rangle$ with $v' = d_p\Phi_{t_0}(v)$ and $w' = d_p\Phi_{t_0}(w)$, and $f'(t_0) = (L_X g)(v', w') = 0$. So f is constant, equal to $f(0) = \langle v, w\rangle$. (2) $\Leftrightarrow$ (3) is the formula $(L_X g)(Y,Z) = \langle \nabla_Y X, Z\rangle + \langle Y, \nabla_Z X\rangle$. □

Example 40.16.4.

1. *Euclidean space.* On $\mathbb{R}^n$, $\nabla_v X = DX(x)v$, so X is a Killing field if and only if the matrix $DX(x)$ is skew-symmetric for every x. This holds for $X(x) = Ax + b$ with $A^T = -A$, whose flow $\Phi_t(x) = e^{tA}x + \int_0^t e^{sA}b\,ds$ consists of rigid motions (e^{tA} is orthogonal, since $(e^{tA})^T e^{tA} = e^{-tA}e^{tA} = I$). Conversely, if $\partial_i X^j + \partial_j X^i = 0$ everywhere, then $T_{kij} = \partial_k\partial_i X^j$ is symmetric in k, i (Theorem 27.10.5) and antisymmetric in i, j, and a tensor with these two properties vanishes: $T_{kij} = -T_{kji} = -T_{jki} = T_{jik} = T_{ijk} = -T_{ikj} = -T_{kij}$. So DX is constant and $X(x) = Ax + b$.

2. *Spheres.* For $A \in \mathfrak{so}(n+1)$ the field $X(x) = Ax$ is tangent to S^n, as $\langle Ax, x\rangle = 0$, and its flow $e^{tA} \in \mathrm{SO}(n+1)$ restricts to isometries of S^n; so X is a Killing field on S^n. For $n = 2$ and $A = L_a$ (Example 34.2.10) it is the field $x \mapsto a \times x$ of the rotations about the axis a.

3. *Lie groups.* On a Lie group with a bi-invariant metric (Example 40.2.5) the left-invariant fields are Killing fields: the flow of X is $\Phi_t(g) = g\exp(tX)$ (Theorem 34.3.3, translated by L_g), which is right multiplication by $\exp(tX)$, an isometry.

Proposition 40.16.5. *Let X be a Killing field and $\gamma\colon I \to M$ a geodesic. Then $X \circ \gamma$ is a Jacobi field along γ, with $D_t(X\circ\gamma) = \nabla_{\dot\gamma} X$.*

Proof. Let $[a,b] \subseteq I$. The domain of the flow is open and contains $\{0\}\times M$ (Theorem 33.9.2), so by compactness of $\gamma([a,b])$ there is $\varepsilon > 0$ such that Φ_s is defined on a neighbourhood of $\gamma([a,b])$ for $|s| < \varepsilon$. Each Φ_s is an isometry onto its image, so by Lemma 40.16.1(1), $f(s,t) = \Phi_s(\gamma(t))$ is a family of geodesics $t \mapsto f(s,t)$. By Proposition 40.8.8(1), $\partial_s f(0,t) = X(\gamma(t))$ is a Jacobi field on $[a,b]$; as $[a,b]$ is arbitrary, it is one on I. The formula for D_t is Proposition 40.4.1. □

Theorem 40.16.6. *Let M be connected, and $p \in M$.*

1. *A Killing field X with $X_p = 0$ and $\nabla_v X = 0$ for all $v \in T_pM$ vanishes identically.*
2. *The Killing fields form a Lie subalgebra $\mathfrak{iso}(M,g)$ of the Lie algebra $\mathfrak{X}(M)$ of vector fields (Proposition 33.9.7), and the linear map $X \mapsto (X_p, \nabla X_p)$, where ∇X_p is the endomorphism $v \mapsto \nabla_v X$ of T_pM, is injective with values in $T_pM \oplus \mathfrak{so}(T_pM)$. Hence $\dim \mathfrak{iso}(M,g) \le n + \frac{n(n-1)}{2} = \frac{n(n+1)}{2}$.*
3. *The bound is attained by $\mathbb{R}^n$ and by S^n, whose Killing fields are exactly those of Example 40.16.4.*

Proof. (1) Let Z be the set of points q with $X_q = 0$ and $\nabla X_q = 0$; it is closed and contains p. If $q \in Z$ and $\exp_q$ maps $B(0,\varepsilon)$ diffeomorphically onto W, then for $|v| < \varepsilon$, $J(t) = X(\exp_q(tv))$ is a Jacobi field (Proposition 40.16.5) with $J(0) = X_q = 0$ and $D_tJ(0) = \nabla_v X = 0$, hence $J = 0$ and $X(\exp_q v) = J(1) = 0$. So $X = 0$ on W, hence also $\nabla X = 0$ on W, and Z is open. As M is connected, $Z = M$.

(2) The map is linear, and it takes values in $T_pM \oplus \mathfrak{so}(T_pM)$ by Proposition 40.16.3(3); by (1) it is injective, which gives the dimension bound. Killing fields form a vector space, since $L_X g$ is linear in X. For the bracket, let X and W satisfy Proposition 40.16.3(2). Then for all Y, Z,

$$\begin{aligned} XW\langle Y,Z\rangle &= X\langle [W,Y],Z\rangle + X\langle Y,[W,Z]\rangle \\ &= \langle [X,[W,Y]],Z\rangle + \langle [W,Y],[X,Z]\rangle + \langle [X,Y],[W,Z]\rangle + \langle Y,[X,[W,Z]]\rangle. \end{aligned}$$

Exchanging X and W and subtracting, the two middle terms cancel, and by the Jacobi identity $[X,[W,Y]] - [W,[X,Y]] = [[X,W],Y]$, so $[X,W]\langle Y,Z\rangle = \langle [[X,W],Y],Z\rangle + \langle Y,[[X,W],Z]\rangle$: $[X,W]$ is a Killing field.

(3) On $\mathbb{R}^n$ the fields $Ax + b$ with A skew form a space of dimension $n(n-1)/2 + n$, and by Example 40.16.4(1) they are all the Killing fields. On S^n the fields $x \mapsto Ax$, $A \in \mathfrak{so}(n+1)$, form a space of dimension $\dim \mathfrak{so}(n+1) = n(n+1)/2$: the map $A \mapsto (x \mapsto Ax)$ is injective because S^n spans $\mathbb{R}^{n+1}$. By (2) there are no other Killing fields on S^n. □

Remark 40.16.7. The isometries of M form a group $\mathrm{Isom}(M,g)$. By the theorem of Myers and Steenrod it is a Lie group, compact if M is compact, and its Lie algebra is identified with the Lie algebra of complete Killing fields (on a compact manifold every Killing field is complete); so $\dim \mathrm{Isom}(M,g) \le n(n+1)/2$ [KN63, Chapter VI]. For instance $\mathrm{Isom}(\mathbb{R}^n)$ is the group of rigid motions and $\mathrm{Isom}(S^n) = \mathrm{O}(n+1)$ (Corollary 40.16.11 below), both of the maximal dimension. A manifold whose isometry group has maximal dimension has constant curvature [KN63, Chapter VI].

Curvature restricts the Killing fields of a compact manifold, by an integration argument of Bochner.

Theorem 40.16.8 (Bochner)**.** *Let M be compact and oriented, and X a Killing field. Then, with $|\nabla X|^2 = \sum_i |\nabla_{e_i} X|^2$ for an orthonormal basis (e_i),*

$$\int_M \big(|\nabla X|^2 - \mathrm{Ric}(X,X)\big)\,\mathrm{vol}_g = 0.$$

Hence if Ric ≤ 0 *everywhere, every Killing field is parallel ($\nabla X = 0$); and if* Ric *is negative definite everywhere, every Killing field is zero, and* $\mathrm{Isom}(M,g)$ *is finite.*

Proof. Let $f = \frac{1}{2}|X|^2$ and $p \in M$, and take normal coordinates at p (Proposition 40.6.2), with $\partial_i = e_i$ orthonormal at p. The i-th coordinate line $\gamma_i(t) = \exp_p(te_i)$ is a unit speed geodesic, so $\partial_i^2 f(p) = (f\circ\gamma_i)''(0)$, and $J = X\circ\gamma_i$ is a Jacobi field with $D_tJ(0) = \nabla_{e_i}X$ (Proposition 40.16.5). Hence

$$(f\circ\gamma_i)'' = \frac{d}{dt}\langle D_tJ, J\rangle = |D_tJ|^2 + \langle D_t^2 J, J\rangle = |D_tJ|^2 - \langle R(J,\dot\gamma_i)\dot\gamma_i, J\rangle,$$

which at $t = 0$ is $|\nabla_{e_i}X|^2 - R(X,e_i,e_i,X)$. Summing over i, and using $R(X,e_i,e_i,X) = R(e_i,X,X,e_i)$ (Proposition 40.8.2) and Definition 40.8.1, Proposition 40.15.3(5) gives

$$\Delta f(p) = -\sum_i \partial_i^2 f(p) = \mathrm{Ric}(X,X) - |\nabla X|^2.$$

By Proposition 40.15.3(2), $\int_M \Delta f\,\mathrm{vol}_g = (\Delta f, 1) = (f, \Delta 1) = 0$, which is the integral formula. If Ric ≤ 0, the integrand $|\nabla X|^2 - \mathrm{Ric}(X,X)$ is a non-negative continuous function with integral 0, so it vanishes: $\nabla X = 0$ and $\mathrm{Ric}(X,X) = 0$; if Ric is negative definite, the latter forces $X = 0$. Then the compact Lie group $\mathrm{Isom}(M,g)$ has Lie algebra 0 (Remark 40.16.7), so it is discrete and compact, hence finite. □

For the flat torus $\mathbb{R}^n/\mathbb{Z}^n$ the Killing fields are therefore the parallel fields, the constant fields $\sum_i c^i\partial_i$, whose flows are the translations; and a compact surface of constant curvature -1 has only finitely many isometries.

We now turn to spaces of constant curvature. Recall that $s_\kappa(t)$ is t, $\sin(\sqrt{\kappa}t)/\sqrt{\kappa}$ or $\sinh(\sqrt{-\kappa}t)/\sqrt{-\kappa}$ for $\kappa = 0$, $\kappa > 0$ or $\kappa < 0$, and that the proof of Theorem 40.12.9 shows: if M has constant sectional curvature κ, $v \ne 0$ lies in the domain of $\exp_p$ and $w = w_\parallel + w_\perp$ is the decomposition of $w \in T_pM$ into a multiple of v and a vector orthogonal to v, then

$$|d_v\exp_p(w)|^2 = |w_\parallel|^2 + \left(\frac{s_\kappa(|v|)}{|v|}\right)^2 |w_\perp|^2. \tag{40.4}$$

Multiplying a metric by a constant $c > 0$ does not change the Levi-Civita connection (the Koszul formula is homogeneous) or $R(X,Y)Z$, and it multiplies the sectional curvatures by $1/c$, since an orthonormal basis for cg is $(u,v)/\sqrt{c}$ for an orthonormal basis (u,v) for g.

Definition 40.16.9. For $n \ge 2$ and $\kappa \in \mathbb{R}$ the *model space* M_κ^n is: Euclidean space $\mathbb{R}^n$ if $\kappa = 0$; the sphere $S^n(r) = \{x \in \mathbb{R}^{n+1} : |x| = r\}$ of radius $r = 1/\sqrt{\kappa}$ with the induced metric if $\kappa > 0$; and the *hyperbolic space* $\{x \in \mathbb{R}^n : x^n > 0\}$ with the metric $|\kappa|^{-1}(dx_1^2 + \cdots + dx_n^2)/(x^n)^2$ if $\kappa < 0$.

All three are complete, connected and simply connected with constant sectional curvature κ. For $\mathbb{R}^n$ this is clear. The sphere $S^n(r)$ has curvature $1/r^2 = \kappa$ (Example 40.11.17(1)), is complete, being compact (Theorem 40.6.8), and is simply connected for $n \ge 2$ (Corollary 19.5.2). Hyperbolic space is simply connected, being a convex subset of $\mathbb{R}^n$, and it has constant

curvature κ by Corollary 40.9.6(1) and the remark on scaling that follows it. It is complete because its isometries act transitively. Indeed the maps $T(x) = \lambda x + b$ with $\lambda > 0$ and $b = (b_1, \dots, b_{n-1}, 0)$ satisfy $T^*\big(|dx|^2/x_n^2\big) = \lambda^2|dx|^2/(\lambda x_n)^2 = |dx|^2/x_n^2$, so they are isometries, and they map $p = (0, \dots, 0, 1)$ to any point. The domain of $\exp_p$ is open, so it contains the ball of radius 2δ in T_pM for some $\delta > 0$; by Lemma 40.16.1(1), applied to these isometries, the same holds at every point. So every unit speed geodesic, starting anywhere, is defined at least on $(-2\delta, 2\delta)$. If a unit speed geodesic γ were defined on $[0, a)$ with $a < \infty$ and could not be extended, the geodesic with initial velocity $\dot\gamma(a - \delta)$ would be defined on $(-2\delta, 2\delta)$ and would agree with $t \mapsto \gamma(a - \delta + t)$ where both are defined, by uniqueness of geodesics; so it would extend γ to $[0, a + \delta)$. Hence geodesics are defined for all time, and hyperbolic space is complete by Theorem 40.6.8.

Theorem 40.16.10 (Killing–Hopf)**.** *Let M and N be complete, connected and simply connected Riemannian manifolds of the same dimension $n \geq 2$, with the same constant sectional curvature κ. For $p \in M$, $q \in N$ and a linear isometry $I\colon T_pM \to T_qN$ there is a unique isometry $F\colon M \to N$ with $F(p) = q$ and $d_pF = I$. In particular M is isometric to the model space M_κ^n.*

Proof. Uniqueness is Lemma 40.16.1(2). By Theorem 40.6.8, $\exp_p$ and $\exp_q$ are defined on the whole tangent spaces. The key observation is the following. Let $U \subseteq T_pM$ be an open set on which $\exp_p$ is a local diffeomorphism, and $F = \exp_q \circ I \circ \psi$ for a smooth local inverse ψ of $\exp_p|_U$. For $x = \exp_p(v)$, $v \in U$, and $w \in T_pM$ the chain rule gives $d_xF(d_v \exp_p(w)) = d_{Iv} \exp_q(Iw)$. The linear isometry I preserves lengths and maps $w_\parallel$ and $w_\perp$ to the components of Iw along and orthogonal to Iv, so by (40.4) on M and on N the vectors $d_v \exp_p(w)$ and $d_{Iv} \exp_q(Iw)$ have the same length (for $v = 0$ both differentials are the identity). As $d_v \exp_p$ is onto T_xM, d_xF preserves lengths, hence inner products: F is a local isometry.

The case $\kappa \leq 0$. By Theorem 40.8.10, $\exp_p$ and $\exp_q$ are diffeomorphisms, so $F = \exp_q \circ I \circ \exp_p^{-1}\colon M \to N$ is a diffeomorphism, a local isometry by the observation, hence an isometry; and $F(p) = q$, $d_pF = I$ as $d_0 \exp = \mathrm{id}$.

The case $\kappa > 0$. Multiplying both metrics by κ we may assume $\kappa = 1$. Let $S = S^n$ be the unit sphere and $o \in S$. We show: for every linear isometry $J\colon T_oS \to T_pM$ there is an isometry $G\colon S \to M$ with $G(o) = p$ and $d_oG = J$. Applying this to M with J and to N with $I \circ J$ gives isometries G_M and G_N, and $F = G_N \circ G_M^{-1}$ has the required properties.

For a unit vector $u \perp o$, the great circle $c(t) = \cos t\, o + \sin t\, u$ has acceleration $c'' = -c$, which is normal to S, so $D_tc' = (c'')^\top = 0$ (Theorem 40.11.2): c is the geodesic with initial velocity u. Hence $\exp_o(v) = \cos|v|\, o + \sin|v|\, v/|v|$ for $v \neq 0$. It maps $B = B(0, \pi)$ bijectively onto $S \setminus \{-o\}$: every $y \neq -o$ is uniquely $\cos\theta\, o + \sin\theta\, u$ with $\theta = \arccos\langle y, o\rangle \in [0, \pi)$ and, if $\theta > 0$, a unit vector $u \perp o$. By (40.4), with $s_1(\ell) = \sin\ell > 0$ for $0 < \ell < \pi$, $d_v \exp_o$ is injective for $v \in B$; so $\exp_o\colon B \to S \setminus \{-o\}$ is a bijective local diffeomorphism, hence a diffeomorphism. By the observation (with the roles of M and N played by S and M), $G_1 = \exp_p \circ J \circ (\exp_o|_B)^{-1}\colon S \setminus \{-o\} \to M$ is a local isometry with $G_1(o) = p$ and $d_oG_1 = J$. Choose $o' \in S$ with $o' \neq \pm o$, and let $G_2 = \exp_{p'} \circ J' \circ (\exp_{o'}|_B)^{-1}\colon S \setminus \{-o'\} \to M$ with $p' = G_1(o')$ and $J' = d_{o'}G_1$, again a local isometry. The open set $W = S \setminus \{-o, -o'\}$ contains o', and it is connected: stereographic projection from $-o$ (Example 18.9.4) maps it homeomorphically onto $\mathbb{R}^n$ minus a point, which is connected for $n \geq 2$ (Lemma 20.5.4). On W the local isometries G_1 and G_2 agree at o' together with their differentials, so $G_1 = G_2$ on W by Lemma 40.16.1(2). Since $-o \neq -o'$, they glue to a local isometry $G\colon S \to M$. By Lemma 40.12.11, G is a covering map (S is compact and M connected and Hausdorff). As M is simply connected and S is path-connected, the monodromy action on a fibre is transitive and $\pi_1(M)$ is trivial, so each fibre is a single point (Corollary 19.4.4(3)). So G is a bijective

local isometry, that is, an isometry, with $G(o) = p$ and $d_oG = J$.

Finally, the model space M^n_κ satisfies the hypotheses, so M is isometric to it. □

Corollary 40.16.11. *1. The model spaces are* frame-homogeneous*: for any two points $p, q \in M^n_\kappa$ and any linear isometry $I\colon T_pM^n_\kappa \to T_qM^n_\kappa$ there is a unique isometry F of M^n_κ with $F(p) = q$ and $d_pF = I$.*

2. The isometries of $\mathbb{R}^n$ are the maps $x \mapsto Ax + b$ with $A \in \mathrm{O}(n)$ and $b \in \mathbb{R}^n$.

3. The isometries of $S^n(r)$ are the restrictions of the maps $A \in \mathrm{O}(n+1)$.

Proof. (1) is Theorem 40.16.10 with $M = N = M^n_\kappa$. (2) These maps are isometries, since $d(Ax + b) = A$ preserves lengths. Given an isometry F, $A = d_0F$ is an orthogonal matrix and $x \mapsto Ax + F(0)$ agrees with F at 0 together with its differential, so it equals F by Lemma 40.16.1(2). (3) Restrictions of orthogonal maps are isometries of $S^n(r)$. Given an isometry F and $o \in S^n(r)$, let A be the linear map with $A(o) = F(o)$ and $A = d_oF$ on $T_oS^n(r) = o^\perp$; it maps the orthonormal basis formed by o/r and an orthonormal basis of $o^\perp$ to an orthonormal basis, since d_oF is an isometry onto $F(o)^\perp$, so $A \in \mathrm{O}(n+1)$. Then $A|_{S^n(r)}$ and F agree at o together with their differentials, so they are equal. □

Corollary 40.16.12 (Space forms)**.** *Let M be a complete connected Riemannian manifold of dimension $n \geq 2$ with constant sectional curvature κ. Then M is isometric to M^n_κ/Γ for a group $\Gamma \cong \pi_1(M)$ of isometries of M^n_κ acting freely, such that the projection $M^n_\kappa \to M^n_\kappa/\Gamma$ is a covering map, with the metric for which this projection is a local isometry.*

Proof. Let $\pi\colon \widetilde{M} \to M$ be a universal covering (Theorem 19.6.5; manifolds are locally path-connected and locally simply connected). $\widetilde{M}$ is a connected manifold (Proposition 20.6.1(4)) with a unique smooth structure making π a local diffeomorphism (Example 33.7.2), and $\tilde{g} = \pi^*g$ makes π a local isometry, so $\widetilde{M}$ has constant curvature κ. It is complete: let $\tilde{\gamma}$ be a maximal geodesic of $\widetilde{M}$ with $\tilde{\gamma}(0) = \tilde{x}$. Then $\pi \circ \tilde{\gamma}$ is a geodesic (Lemma 40.16.1(1)), which extends to a geodesic γ defined on $\mathbb{R}$ by completeness of M. Let $\hat{\gamma}$ be the lift of γ starting at $\tilde{x}$ (Proposition 19.4.3). Near each time, $\hat{\gamma}$ is γ followed by a local inverse of π, which is a local isometry, so $\hat{\gamma}$ is a geodesic; it has the initial velocity of $\tilde{\gamma}$, so it extends $\tilde{\gamma}$ to $\mathbb{R}$, and $\tilde{\gamma}$ was defined on all of $\mathbb{R}$. By Theorem 40.16.10 there is an isometry $\Theta\colon M^n_\kappa \to \widetilde{M}$.

The deck transformations τ of π satisfy $\pi \circ \tau = \pi$, so $\tau^*\tilde{g} = \tau^*\pi^*g = \pi^*g = \tilde{g}$: they are isometries. The group $\mathrm{Deck}(\pi)$ is isomorphic to $\pi_1(M)$ and acts transitively on each fibre of π (Proposition 19.6.3), and freely, since a deck transformation with a fixed point is a lift of π agreeing with the identity at one point, hence the identity by the uniqueness in Proposition 19.6.1. So the fibres of π are the orbits, and π induces a bijection $\widetilde{M}/\mathrm{Deck}(\pi) \to M$, a homeomorphism because π is open. The group $\Gamma = \Theta^{-1}\,\mathrm{Deck}(\pi)\,\Theta$ acts on M^n_κ by isometries, freely, and $\pi \circ \Theta$ identifies M^n_κ/Γ with M; the projection is the covering map $\pi \circ \Theta$, a local isometry. □

Example 40.16.13. 1. *Flat manifolds.* The flat tori $\mathbb{R}^n/\mathbb{Z}^n$ and the flat Klein bottle (Example 40.2.7) are complete flat manifolds, quotients of $\mathbb{R}^n$.

2. *Spherical space forms.* $\mathbb{RP}^n = S^n/\{\pm 1\}$ and the lens spaces $L(p, q) = S^3/(\mathbb{Z}/p)$ (Example 20.4.5) have constant curvature 1, because the maps $(z_1, z_2) \mapsto (\zeta^k z_1, \zeta^{qk} z_2)$ are unitary, hence orthogonal on $\mathbb{R}^4 = \mathbb{C}^2$, and so isometries of S^3. In particular, in dimension 3 there are infinitely many spherical space forms, with fundamental groups $\mathbb{Z}/p$ for all p.

3. *Hyperbolic surfaces.* By the Gauss–Bonnet theorem (Corollary 40.8.12) a compact orientable surface with a metric of constant curvature -1 has genus $g \geq 2$, and its area is $4\pi(g-1)$; by Corollary 40.16.12 it is a quotient of the hyperbolic plane. That every genus $g \geq 2$ actually occurs is shown below (Proposition 40.16.15).

A closed surface with a hyperbolic metric. The flat torus is obtained from a Euclidean square by gluing opposite sides; its four corners come together at one point, where the four right angles add up to 2π, and this is why the flat metric extends across that point. A surface of genus 2 is obtained from an octagon by gluing its sides in pairs, and again all eight corners come together at one point. A Euclidean octagon will not do, as its angles add up to 6π; but in the hyperbolic plane polygons are "thinner", the angle sum of a geodesic polygon being smaller than in the Euclidean plane by its area, and there is a regular octagon with all angles equal to $\pi/4$, whose angles add up to exactly 2π. We construct it in the disc model of the hyperbolic plane, where geodesics and angles are easy to see.

Lemma 40.16.14 (Regular polygons in the hyperbolic plane)**.** *Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ with the metric $g_{\mathbb{D}} = 4|dz|^2/(1-|z|^2)^2$, of constant curvature -1 (Corollary 40.9.6(2)).*

1. *The rotations $z \mapsto e^{i\theta}z$ and the conjugation $z \mapsto \bar{z}$ are isometries of $\mathbb{D}$. Let C be a circle with centre c and radius s such that $|c|^2 = 1 + s^2$, that is, a circle meeting the unit circle at right angles. Then the inversion $J_C(z) = c + s^2(z-c)/|z-c|^2$ maps $\mathbb{D}$ onto itself and is an isometry of $\mathbb{D}$; its fixed points in $\mathbb{D}$ are the points of the arc $C \cap \mathbb{D}$; and this arc, parametrised by arc length, is a geodesic.*

2. *The angle between two tangent vectors at a point of $\mathbb{D}$ for the metric $g_{\mathbb{D}}$ is their Euclidean angle, and $\mathbb{D}$ is complete.*

3. *Let $n \geq 3$, $0 < \rho < 1$, and $v_k = \rho e^{2\pi i k/n}$ for $k \in \mathbb{Z}/n\mathbb{Z}$. For each k there is exactly one circle C_k as in (1) through v_k and v_{k+1}. The arcs of the C_k from v_k to v_{k+1} are geodesic segments of the same length, and they bound a* regular geodesic n-gon *P_ρ, invariant under the rotation by $2\pi/n$ and under conjugation, all of whose interior angles are equal to the angle $\alpha \in (0,\pi)$ with*

$$\tan\frac{\alpha}{2} = \frac{1-\rho^2}{1+\rho^2}\cot\frac{\pi}{n}.$$

As ρ increases from 0 to 1, α decreases strictly and continuously from $\frac{(n-2)\pi}{n}$, the angle of a Euclidean regular n-gon, to 0. In particular the interior angles equal $\frac{2\pi}{n}$ if and only if $n > 4$ and $\rho^2 = \cos\frac{2\pi}{n}$.

Proof. (1) Rotations and conjugation preserve $|z|$ and $|dz|$. For the inversion write $w = z - c$; note $|c| > 1$, so $c \notin \overline{\mathbb{D}}$. The map $w \mapsto w/|w|^2$ has differential $\frac{1}{|w|^2}\big(I - 2\frac{ww^T}{|w|^2}\big)$, a reflection times $1/|w|^2$, so $J_C^*|dz|^2 = s^4|dz|^2/|w|^4$. Next, expanding $|J_C(z)|^2|w|^2 = |c|^2|w|^2 + 2s^2\operatorname{Re}(\bar{c}w) + s^4$ and using $|w|^2 = |z|^2 - 2\operatorname{Re}(\bar{c}z) + |c|^2$, $\operatorname{Re}(\bar{c}w) = \operatorname{Re}(\bar{c}z) - |c|^2$ and $1 - |c|^2 = -s^2$,

$$\big(1-|J_C(z)|^2\big)|w|^2 = -s^2|w|^2 - 2s^2\operatorname{Re}(\bar{c}z) + 2s^2|c|^2 - s^4 = -s^2|z|^2 + s^2|c|^2 - s^4 = s^2\big(1-|z|^2\big).$$

Hence J_C maps $\mathbb{D}$ into $\mathbb{D}$; it is an involution of $\mathbb{C}\setminus\{c\}$, so it maps $\mathbb{D}$ onto $\mathbb{D}$; and

$$J_C^* g_{\mathbb{D}} = \frac{4s^4|dz|^2}{|w|^4}\cdot\frac{|w|^4}{s^4(1-|z|^2)^2} = g_{\mathbb{D}}.$$

$J_C(z) = z$ means $s^2/|w|^2 = 1$, that is, $|z-c| = s$. Let $p \in C \cap \mathbb{D}$ and let u be a unit tangent vector of C at p. Since J_C fixes $C \cap \mathbb{D}$ pointwise, $d_pJ_C(u) = u$, and by Lemma 40.16.1(1) the

geodesic γ with $\dot\gamma(0) = u$ satisfies $J_C \circ \gamma = \gamma$: it stays in the fixed set $C \cap \mathbb{D}$. As $\dot\gamma \neq 0$, on a small interval γ is a parametrisation with constant speed of an open sub-arc of $C \cap \mathbb{D}$ around p; so near p the parametrisation of $C \cap \mathbb{D}$ by arc length differs from γ by a change of parameter $t \mapsto \pm t + \text{const}$, and is a geodesic near p. As p was arbitrary, it is a geodesic.

(2) $g_{\mathbb{D}}$ is the Euclidean inner product multiplied at each point by the positive number $4/(1-|z|^2)^2$, which does not change angles. The isometries of (1) act transitively on $\mathbb{D}$: for $c > 1$ real and $s^2 = c^2 - 1$, $J_C(0) = c - s^2/c = 1/c$, which takes every value in $(0,1)$, and rotations then reach every point. So the argument given above for the half-space shows that $\mathbb{D}$ is complete.

(3) A circle through v_0 and v_1 has its centre on the perpendicular bisector of the segment v_0v_1, which is the line through 0 in the direction $e^{i\pi/n}$, as $|v_0| = |v_1|$. So $c = de^{i\pi/n}$ with $d \in \mathbb{R}$, and the conditions $|c|^2 = 1 + s^2$ and $|v_0 - c|^2 = s^2$ give $\rho^2 - 2\rho d\cos\frac{\pi}{n} + d^2 = d^2 - 1$, that is,

$$d = \frac{1+\rho^2}{2\rho\cos(\pi/n)},$$

which determines C_0 uniquely ($\cos\frac{\pi}{n} > 0$ as $n \geq 3$). The rotation R by $2\pi/n$ maps v_k to v_{k+1} and circles orthogonal to the unit circle to such circles, so by uniqueness $R(C_k) = C_{k+1}$; likewise conjugation maps C_0 to C_{-1}, since it fixes v_0 and maps v_1 to v_{-1}. By (1) the arcs from v_k to v_{k+1} are geodesic segments, and R, an isometry, maps each to the next, so they have the same length; and the polygon P_ρ they bound is invariant under R and conjugation. Hence all interior angles are equal, and the one at $v_0 = \rho$ is bisected by the real axis, so $\frac{\alpha}{2}$ is the angle at v_0 between the real axis and the tangent of C_0, measured on the side of 0; by (2) we may compute it in Euclidean geometry. The tangent of C_0 at v_0 is perpendicular to the radius $r = c - v_0$, so $\frac{\alpha}{2}$ equals the angle between r and the imaginary axis, whose tangent is $\operatorname{Re} r/\operatorname{Im} r$. Here

$$\operatorname{Re} r = d\cos\frac{\pi}{n} - \rho = \frac{1-\rho^2}{2\rho} > 0, \qquad \operatorname{Im} r = d\sin\frac{\pi}{n} = \frac{1+\rho^2}{2\rho}\tan\frac{\pi}{n} > 0,$$

which gives the formula. As ρ runs from 0 to 1, $\frac{1-\rho^2}{1+\rho^2}$ decreases strictly and continuously from 1 to 0, so $\tan\frac{\alpha}{2}$ decreases from $\cot\frac{\pi}{n} = \tan(\frac{\pi}{2} - \frac{\pi}{n})$ to 0, and α from $\pi - \frac{2\pi}{n}$ to 0. Finally $\alpha = \frac{2\pi}{n}$ means $\frac{1-\rho^2}{1+\rho^2} = \tan^2\frac{\pi}{n}$, that is, $\rho^2 = \frac{1-\tan^2(\pi/n)}{1+\tan^2(\pi/n)} = \cos\frac{2\pi}{n}$, which lies in $(0,1)$ exactly when $n > 4$. □

For $n = 8$ the octagon with angles $\frac{\pi}{4}$ has $\rho^2 = \cos\frac{\pi}{4} = 2^{-1/2}$: its vertices lie at Euclidean distance $\rho = 2^{-1/4} \approx 0.84$ from the centre. For $n = 4$ the required angle $\frac{\pi}{2}$ is the Euclidean one, which no hyperbolic square attains: the torus is flat, not hyperbolic.

Proposition 40.16.15 (Closed hyperbolic surfaces)**.** *For every $g \geq 2$ there is a compact surface homeomorphic to the closed orientable surface Σ_g of genus g with a Riemannian metric of constant curvature -1. Its area is $4\pi(g-1)$.*

Sketch of proof. Let $n = 4g > 4$. By Lemma 40.16.14(3) there is a regular geodesic n-gon $P \subseteq \mathbb{D}$ with interior angles $\frac{2\pi}{n}$ and all sides of the same length. Number its sides $\sigma_1, \dots, \sigma_n$ counterclockwise and label them, in blocks of four, by the word

$$a_1b_1a_1^{-1}b_1^{-1}\,a_2b_2a_2^{-1}b_2^{-1}\cdots a_gb_ga_g^{-1}b_g^{-1}:$$

in the j-th block, the sides σ_{4j-3} (labelled a_j) and σ_{4j-1} (labelled a_j^{-1}) are to be glued, and so are σ_{4j-2} and σ_{4j}, each with the orientation that the exponent indicates relative to the counterclockwise direction.

The side pairings are isometries. $\mathbb{D}$ is complete (Lemma 40.16.14(2)), simply connected (it is convex) and of constant curvature -1, so by Theorem 40.16.10 with $M = N = \mathbb{D}$ it is frame-homogeneous. For a pair of glued sides σ, σ' with midpoints m, m', let T be the isometry of $\mathbb{D}$ with $T(m) = m'$ whose differential maps the unit tangent of σ at m to the unit tangent of σ' at m' prescribed by the gluing, and the inward unit normal of P at m to the outward unit normal at m'; such a linear isometry exists, as it maps an orthonormal basis to an orthonormal basis. By Lemma 40.16.1(1), T maps geodesics to geodesics, and σ, σ' are geodesic segments of the same length centred at m, m'; so $T(\sigma) = \sigma'$, and $T(P)$ lies on the other side of σ'. Let X be the quotient of P by the relation identifying each $x \in \sigma$ with $T(x) \in \sigma'$, for the $2g$ pairs of sides.

Omitted verifications. (a) X is homeomorphic to Σ_g: this is the standard description of Σ_g as a $4g$-gon with the above edge word [Hat02], and in particular all n vertices of P become a single point of X. (b) X has a smooth structure and a metric for which every point has a neighbourhood isometric to an open subset of $\mathbb{D}$, compatibly with the interior of P. At interior points of P this is clear. At a point x of a side σ, a half-disc of P around x and the image under T^{-1} of a half-disc of P around $T(x)$ lie on opposite sides of σ and together form a disc in $\mathbb{D}$. At the vertex, the n corner sectors of P, each of angle $\frac{2\pi}{n}$, moved by suitable composites of the pairings T, fit together cyclically around one vertex of P without overlapping and fill a disc, because their angles add up to $n \cdot \frac{2\pi}{n} = 2\pi$; we omit the bookkeeping of which composites occur. This is the case of a single vertex cycle of Poincaré's polygon theorem. Since curvature is computed locally, X has constant curvature -1.

Area. P is a simple region in the chart $\mathbb{D}$ (it is star-shaped with respect to 0), its sides are geodesics, so their geodesic curvature vanishes, $K = -1$, and its exterior angles are $\pi - \frac{2\pi}{n}$. The local Gauss–Bonnet theorem (Theorem 40.12.3) gives $-\operatorname{Area}(P) + n(\pi - \frac{2\pi}{n}) = 2\pi$, so $\operatorname{Area}(P) = (n-4)\pi = 4\pi(g-1)$. The boundary of P has measure zero, so X has the same area, in accordance with Example 40.16.13(3). □

Remark 40.16.16.

1. The construction explains the dichotomy of Corollary 40.8.12. The angle sum of a geodesic n-gon of area A in a surface of constant curvature κ is $(n-2)\pi + \kappa A$ by the local Gauss–Bonnet theorem; gluing a $4g$-gon whose vertices all become one point requires the angle sum 2π, so $\kappa A = (4-4g)\pi$: the curvature must be zero for $g = 1$ (the square of the flat torus) and negative for $g \geq 2$.

2. By Corollary 40.16.12, X is isometric to $\mathbb{D}/\Gamma$ for a group $\Gamma \cong \pi_1(\Sigma_g)$ of isometries of $\mathbb{D}$ acting freely; the side pairings T are the natural candidates for generators of Γ, and the copies $\gamma(P)$ for a picture of the hyperbolic plane tiled by octagons, eight of them meeting at each vertex for $g = 2$, as the plane is tiled by the squares of the flat torus, with $\Gamma = \mathbb{Z}^2$, for $g = 1$. Proving this is the rest of Poincaré's polygon theorem, which we do not develop.

In even dimension the positive curvature case is very restricted.

Corollary 40.16.17. *Let n be even and M complete and connected with constant curvature $\kappa > 0$. Then M is isometric to $S^n(r)$ or to $\mathbb{RP}^n(r) = S^n(r)/\{\pm 1\}$, with $r = 1/\sqrt{\kappa}$.*

Proof. By Corollaries 40.16.12 and 40.16.11(3), $M = S^n(r)/\Gamma$ for a subgroup $\Gamma \subseteq \mathrm{O}(n+1)$ acting freely on the sphere. Every $A \in \mathrm{SO}(n+1)$ has the eigenvalue 1, because $n+1$ is odd: $\det(A-I) = \det(A-AA^T) = \det A \det(I-A^T) = \det(I-A) = (-1)^{n+1}\det(A-I) = -\det(A-I)$. A unit eigenvector, scaled to length r, is a fixed point of A on $S^n(r)$, so $\Gamma \cap \mathrm{SO}(n+1) = \{I\}$, and $\det\colon \Gamma \to \{\pm 1\}$ is injective. If $\Gamma \neq \{I\}$ it is $\{I, A\}$ with $\det A = -1$ and $A^2 = I$; then A is diagonalisable with eigenvalues ± 1 (its minimal polynomial divides $x^2 - 1$), and the eigenvalue 1 is excluded since the action is free, so $A = -I$. □

Finally we state two rigidity theorems for positive curvature, whose proofs use comparison geometry beyond this chapter. The first says that the diameter bound of the Bonnet–Myers theorem (Theorem 40.10.7) is attained only by the round sphere.

Theorem 40.16.18 (Cheng's maximal diameter theorem)**.** *Let M be complete and connected of dimension n, with* $\mathrm{Ric} \geq (n-1)\kappa g$ *for some $\kappa > 0$, and with diameter $\pi/\sqrt{\kappa}$. Then M is isometric to the sphere $S^n(1/\sqrt{\kappa})$.*

This is [Che75]. For example $\mathbb{RP}^n$ with the metric of constant curvature 1 has $\mathrm{Ric} = (n-1)g$ but diameter $\pi/2$, and a product $S^2 \times S^2$ of unit spheres has $\mathrm{Ric} = g$ and diameter $\sqrt{2}\,\pi$, less than the bound $\pi/\sqrt{1/3} = \sqrt{3}\,\pi$ given by Theorem 40.10.7 with $n = 4$ and $3\kappa = 1$.

Theorem 40.16.19 (Sphere theorem)**.** *Let M be complete and simply connected with sectional curvatures in $(\frac{1}{4}, 1]$. Then M is homeomorphic to S^n (Rauch, Berger, Klingenberg), and in fact diffeomorphic to S^n (Brendle and Schoen).*

The topological statement is [dC92, Chapter 13]; the differentiable one is [BS09], which proves more generally that a compact manifold with pointwise $\frac{1}{4}$-pinched curvature is diffeomorphic to a spherical space form. The constant $\frac{1}{4}$ is optimal: the complex projective space $\mathbb{CP}^m$ with the Fubini–Study metric (Chapter 47) has sectional curvatures filling $[\frac{1}{4}, 1]$ after scaling, it is compact and simply connected, and for $m \geq 2$ it is not homeomorphic to a sphere, since $H^2(\mathbb{CP}^m; \mathbb{Z}) \cong \mathbb{Z}$. By Theorem 40.10.7, the hypothesis of the sphere theorem already forces M to be compact.

40.17 Exercises

Exercise 40.17.1. Verify that $\delta = - * d *$ on k-forms when n is even, and compute δ on 1-forms in $\mathbb{R}^3$, identifying it with minus the divergence.

Exercise 40.17.2. Show that in dimension $n = 2k$ the Hodge star on k-forms depends only on the conformal class of the metric, and that $*^2 = (-1)^k$ on Ω^k.

Solution. Under $g \mapsto \lambda g$ with $\lambda > 0$, the inner product on k-forms scales by λ^{-k} and the volume form by $\lambda^{n/2} = \lambda^k$, so $*$ is unchanged by the characterisation $\alpha \wedge *\beta = \langle \alpha, \beta \rangle \mathrm{vol}$. The sign is $(-1)^{k(n-k)} = (-1)^{k^2} = (-1)^k$. □

Exercise 40.17.3. Compute the Laplacian on functions on the round sphere S^2 in spherical coordinates, and verify that the restrictions of linear functions of $\mathbb{R}^3$ are eigenfunctions with eigenvalue 2.

Solution. With $x = (\sin\vartheta\cos\varphi, \sin\vartheta\sin\varphi, \cos\vartheta)$ the metric is $d\vartheta^2 + \sin^2\vartheta\, d\varphi^2$, and Example 40.5.3(2) gives $\Delta f = \frac{1}{\sin\vartheta}\partial_\vartheta(\sin\vartheta\, f_\vartheta) + \frac{1}{\sin^2\vartheta} f_{\varphi\varphi}$. For $f = \cos\vartheta$: $\frac{1}{\sin\vartheta}\partial_\vartheta(-\sin^2\vartheta) = -2\cos\vartheta$. For $f = \sin\vartheta\cos\varphi$: the first term is $\frac{\cos\varphi}{\sin\vartheta}(\cos^2\vartheta - \sin^2\vartheta)$ and the second $-\frac{\cos\varphi}{\sin\vartheta}$, with sum $\frac{\cos\varphi}{\sin\vartheta}(-2\sin^2\vartheta) = -2\sin\vartheta\cos\varphi$; the same computation applies to $\sin\vartheta\sin\varphi$. So $\Delta x_i = -2x_i$, eigenvalue 2 for $-\Delta$, in accordance with Example 40.14.6(1). □

Exercise 40.17.4. Show that $\Delta(\phi \circ f) = \phi'(f)\,\Delta f + \phi''(f)\,|\mathrm{grad}\, f|^2$ for smooth $f : M \to \mathbb{R}$ and $\phi : \mathbb{R} \to \mathbb{R}$. Deduce that if f is harmonic and ϕ is convex, then $\phi \circ f$ is subharmonic, and that on a compact oriented manifold without boundary a harmonic function f has $\int_M |\mathrm{grad}\, f|^2 e^f \,\mathrm{vol}_g = 0$.

Solution. $\operatorname{grad}(\phi\circ f)=\phi'(f)\operatorname{grad} f$, since $d(\phi\circ f)=\phi'(f)\,df$. By the Leibniz rule for the divergence (as in the proof of Proposition 40.14.1), $\operatorname{div}(\phi'(f)\operatorname{grad} f)=\phi'(f)\Delta f+\langle\operatorname{grad}\phi'(f),\operatorname{grad} f\rangle$, and $\operatorname{grad}\phi'(f)=\phi''(f)\operatorname{grad} f$. If $\Delta f=0$ and $\phi''\geq 0$, then $\Delta(\phi\circ f)=\phi''(f)|\operatorname{grad} f|^2\geq 0$. For $\phi=\exp$, $\int_M\Delta(e^f)\operatorname{vol}_g=0$ by Proposition 40.14.1 gives the last statement, and hence $\operatorname{grad} f=0$: another proof of Corollary 40.14.2(1) for harmonic functions. □

Exercise 40.17.5. Show that $x_n^2-\frac{1}{n}|x|^2$ is a harmonic polynomial on $\mathbb{R}^n$, and deduce that $Y=x_n^2-\frac{1}{n}$ satisfies $\Delta_{S^{n-1}}Y=-2nY$ on the unit sphere S^{n-1}. Check that $\int_{S^{n-1}}Y\,d\sigma=0$, as Corollary 40.14.2(4) requires.

Solution. $\Delta(x_n^2)=2$ and $\Delta|x|^2=\sum_i\Delta(x_i^2)=2n$, so $\Delta(x_n^2-\frac{1}{n}|x|^2)=2-2=0$; the polynomial is homogeneous of degree 2 and restricts to Y on the sphere. By Corollary 40.14.9 with $k=2$, $\Delta_{S^{n-1}}Y=-2n\,Y$. By the symmetry argument in the proof of Proposition 40.14.4, $\int x_n^2\,d\sigma=\sigma(S^{n-1})/n$, so $\int Y\,d\sigma=0$; and indeed $Y=-\frac{1}{2n}\Delta_{S^{n-1}}Y$ has integral 0. □

Exercise 40.17.6. Let G be a compact Lie group with a bi-invariant metric. Show that the geodesics through e are the one-parameter subgroups, and that $\nabla_XY=\frac{1}{2}[X,Y]$ for left-invariant fields.

Exercise 40.17.7. Show that the geodesics of the round sphere $S^n\subseteq\mathbb{R}^{n+1}$ are the great circles $\gamma(t)=\cos(t)p+\sin(t)v$ with $|p|=|v|=1$ and $v\perp p$ (up to affine reparametrisation).

Solution. The reflection ρ of $\mathbb{R}^{n+1}$ fixing the plane spanned by p and v pointwise preserves S^n and its metric, so it is an isometry, and it maps geodesics to geodesics. The geodesic γ_v with $\gamma_v(0)=p$, $\dot\gamma_v(0)=v$ is mapped to a geodesic with the same initial data, hence to itself by uniqueness; so γ_v lies in the fixed set of ρ on S^n, the great circle through p and v. Having constant speed 1 (Proposition 40.4.4) and starting at p with velocity v, it is $\cos(t)p+\sin(t)v$. □

Exercise 40.17.8. Show that the holonomy group of a Riemannian product $M_1\times M_2$ at (p_1,p_2) is $\operatorname{Hol}_{p_1}(M_1)\times\operatorname{Hol}_{p_2}(M_2)$, acting on $T_{p_1}M_1\oplus T_{p_2}M_2$.

Solution. The Levi-Civita connection of the product metric is the direct sum of those of the factors, since the direct sum connection is metric and torsion-free. So a vector (v_1,v_2) is transported along a curve (c_1,c_2) to $(P_{c_1}v_1,P_{c_2}v_2)$. Loops at (p_1,p_2) are exactly the pairs of loops at p_1 and p_2, which gives the claim. □

Exercise 40.17.9. Let (M,g) be connected with $\operatorname{Hol}_p(g)=\{1\}$. Show that M is flat and that TM is trivial.

Solution. By Theorem 40.7.2, each vector of an orthonormal basis of T_pM extends to a parallel vector field E_i; the E_i are orthonormal everywhere (parallel transport is an isometry), so they trivialise TM. For parallel fields $R(X,Y)E_i=\nabla_X\nabla_YE_i-\nabla_Y\nabla_XE_i-\nabla_{[X,Y]}E_i=0$, so $R=0$. □

Exercise 40.17.10. Let $f,h\colon(a,b)\to\mathbb{R}$ be smooth with $f>0$ and (f',h') nowhere zero, and let S be the surface of revolution $x(t,\theta)=(f(t)\cos\theta,f(t)\sin\theta,h(t))$, assuming that the profile curve $t\mapsto(f(t),h(t))$ is injective and a homeomorphism onto its image. Show that the area of S is $2\pi\int_a^b f(t)\sqrt{f'(t)^2+h'(t)^2}\,dt$.

Solution. Here $x_t=(f'\cos\theta,f'\sin\theta,h')$ and $x_\theta=(-f\sin\theta,f\cos\theta,0)$, so $E=f'^2+h'^2$, $F=0$ and $G=f^2$. By Proposition 40.3.4(2), $dA=f\sqrt{f'^2+h'^2}\,dt\,d\theta$ on the image of $(a,b)\times(0,2\pi)$, and the missing meridian $\theta=0$ has area 0 (it is a segment in the chart given by $\theta\in(-\pi,\pi)$). Integrating over θ gives the formula. □

Exercise 40.17.11. Show that the area of the part of the unit sphere S^2 between the planes $z = c$ and $z = d$, for $-1 \le c < d \le 1$, is $2\pi(d - c)$: it depends only on the distance between the planes (Archimedes).

Solution. Apart from the poles and a meridian, which have area 0, the sphere is the surface of revolution of Exercise 40.17.10 with $t = z \in (-1, 1)$, $f(z) = \sqrt{1 - z^2}$ and $h(z) = z$. Then $f' = -z/\sqrt{1 - z^2}$ and $f\sqrt{f'^2 + 1} = \sqrt{1 - z^2} \cdot (1 - z^2)^{-1/2} = 1$, so the area of the zone is $2\pi \int_c^d 1\, dz = 2\pi(d - c)$. □

Exercise 40.17.12. On the upper half plane $H = \{z = x + iy : y > 0\}$ with the metric $(dx^2 + dy^2)/y^2$, show that the Killing fields form a Lie algebra of dimension 3, spanned by ∂_x, $x\partial_x + y\partial_y$ and $(x^2 - y^2)\partial_x + 2xy\,\partial_y$. (Show first that the maps $z \mapsto (az + b)/(cz + d)$ with a, b, c, d real and $ad - bc = 1$ are isometries of H.)

Solution. For $\phi(z) = (az + b)/(cz + d)$ one computes $\operatorname{Im}\phi(z) = y/|cz + d|^2$ and $\phi'(z) = 1/(cz + d)^2$, so ϕ maps H to itself. As a map of $\mathbb{R}^2 = \mathbb{C}$, its differential at z is multiplication by the complex number $\phi'(z)$, which multiplies Euclidean lengths by $|\phi'(z)| = 1/|cz + d|^2$. The hyperbolic length of $v \in T_zH$ is $|v|/y$, and that of its image is $|\phi'(z)||v|/\operatorname{Im}\phi(z) = |v|/y$: ϕ is an isometry. The three fields generate the one-parameter groups $z \mapsto z + t$, $z \mapsto e^t z$ and $z \mapsto z/(1 - tz)$ (for $z/(1 - tz)$, the derivative in t at $t = 0$ is $z^2 = (x^2 - y^2) + 2ixy$), which are of the form above (for the last one, with $a = d = 1$, $b = 0$, $c = -t$, defined on all of H since $1 - tz \neq 0$ for $y > 0$); so they are Killing fields. They are linearly independent, and by Theorem 40.16.6 the Killing algebra has dimension at most $2 \cdot 3/2 = 3$. □

Exercise 40.17.13. Show that the helicoid and the catenoid of Example 40.13.4 are locally isometric: in the parameters (u, w), with $v = \sinh w$ on the helicoid, both metrics are $\cosh^2 w\,(du^2 + dw^2)$.

Solution. The helicoid has the metric $(1 + v^2)\,du^2 + dv^2$. Substituting $v = \sinh w$, $dv = \cosh w\, dw$ and $1 + v^2 = \cosh^2 w$ gives $\cosh^2 w\,(du^2 + dw^2)$. The catenoid, with v renamed w, has $E = G = \cosh^2 w$ and $F = 0$. So the map sending the point with parameters $(u, \sinh w)$ on the helicoid to the point with parameters (u, w) on the catenoid preserves the metric: it is a local isometry from the helicoid onto the catenoid, which winds the helicoid infinitely often around the catenoid, as the catenoid is 2π-periodic in u. □

Exercise 40.17.14. Show that a minimal surface in $\mathbb{R}^3$ has Gaussian curvature $K \le 0$, and deduce that there is no compact minimal surface in $\mathbb{R}^3$.

Solution. If $H = 0$, the principal curvatures are κ and $-\kappa$, so $K = -\kappa^2 \le 0$ (Corollary 40.11.13). A compact surface in $\mathbb{R}^3$ has a point with $K > 0$ by Lemma 40.12.10. □

Exercise 40.17.15. Let $F\colon \mathbb{R}^n \to S^n \subseteq \mathbb{R}^{n+1}$, $F(x) = \big(2x, |x|^2 - 1\big)/(|x|^2 + 1)$, be the inverse of the stereographic projection from the north pole. Show that the pullback of the round metric is $4(dx_1^2 + \cdots + dx_n^2)/(1 + |x|^2)^2$, and deduce once more from Corollary 40.9.6(3) that S^n has constant curvature 1.

Solution. Let $s = 1 + |x|^2$, so $F = (2x/s, 1 - 2/s)$ and $ds = 2\sum_k x_k\, dx_k = 2\,x \cdot dx$. Then $dF = \big(2\,dx/s - 2x\,ds/s^2,\ 2\,ds/s^2\big)$, and the pullback metric is

$$|dF|^2 = \frac{4|dx|^2}{s^2} - \frac{8(x \cdot dx)\,ds}{s^3} + \frac{4|x|^2 ds^2}{s^4} + \frac{4ds^2}{s^4} = \frac{4|dx|^2}{s^2} - \frac{16(x \cdot dx)^2}{s^3} + \frac{4s \cdot 4(x \cdot dx)^2}{s^4} = \frac{4|dx|^2}{s^2},$$

where $|dx|^2 = \sum_k dx_k^2$ and products of 1-forms mean symmetric products. So F is an isometry from $\mathbb{R}^n$ with this metric onto S^n minus the north pole, and the curvature of S^n is that of Corollary 40.9.6(3). □

Exercise 40.17.16. Let f be a positive function on an interval I and $g = f(x_n)^2(dx_1^2 + \cdots + dx_n^2)$ on $\{x \in \mathbb{R}^n : x_n \in I\}$, $n \geq 3$. Show that in the frame $e_i = f^{-1}\partial_i$ the sectional curvatures are

$$K(e_i, e_j) = -\frac{f'^2}{f^4} \quad (i, j < n), \qquad K(e_i, e_n) = \frac{f'^2 - f f''}{f^4} \quad (i < n),$$

and that g has constant curvature if and only if $f(t) = 1/(a + bt)$ for constants a, b, the curvature being $-b^2$. Which metric is obtained for $a = 0$, $b = 1$?

Solution. Apply Proposition 40.9.5 with $u = \log f(x_n)$: $u_i = \delta_{in} f'/f$, $u_{ik} = \delta_{in}\delta_{kn}(f'/f)'$, so $\alpha_i = \delta_{in}\mu\, dx_n$ with $\mu = (f'/f)' - (f'/f)^2$, and $|\nabla u|^2 = (f'/f)^2$. For $i, j < n$, $\alpha_i = \alpha_j = 0$ and $\Omega^i_j = -(f'/f)^2 dx_i \wedge dx_j = -(f'^2/f^4)\,\omega^i \wedge \omega^j$. For $i < n$, $\Omega^i_n = -\mu\, dx_i \wedge dx_n - (f'/f)^2 dx_i \wedge dx_n = -(f'/f)' dx_i \wedge dx_n$, and $-(f'/f)'/f^2 = (f'^2 - f f'')/f^4$. Since every Ω^i_j is a multiple of $\omega^i \wedge \omega^j$, the curvature is constant c if and only if both expressions equal c (Theorem 40.9.2(4)). The first forces $c = -b^2 \leq 0$ and $f'/f^2 = \pm b$, that is $(1/f)' = \mp b$, so $f = 1/(a + bt)$ after renaming b; and for this f, $f' = -b/(a + bt)^2$, $f'' = 2b^2/(a + bt)^3$, $f'^2 - f f'' = -b^2/(a + bt)^4$, so the second expression is $-b^2$ as well. For $a = 0$, $b = 1$ we get $f = 1/x_n$: the upper half-space model of hyperbolic space. □

Exercise 40.17.17. On $\mathbb{R}^3$ consider the metric $g = dx^2 + dy^2 + (dz - x\, dy)^2$ (a left-invariant metric on the Heisenberg group). With the coframe $\omega^1 = dx$, $\omega^2 = dy$, $\omega^3 = dz - x\, dy$, find the connection forms, and show that the sectional curvatures of the planes spanned by pairs of frame vectors are $K(e_1, e_2) = -\frac{3}{4}$ and $K(e_1, e_3) = K(e_2, e_3) = \frac{1}{4}$. Compute the Ricci and scalar curvatures.

Solution. $d\omega^1 = d\omega^2 = 0$ and $d\omega^3 = -dx \wedge dy = -\omega^1 \wedge \omega^2$. Try $\omega^1_2 = a\,\omega^3$, $\omega^1_3 = b\,\omega^2$, $\omega^2_3 = c\,\omega^1$. The first structure equation for $i = 1, 2, 3$ gives $(a - b)\,\omega^2 \wedge \omega^3 = 0$, $-(a + c)\,\omega^1 \wedge \omega^3 = 0$ and $-(b - c)\,\omega^1 \wedge \omega^2 = -\omega^1 \wedge \omega^2$, so $a = b = \frac{1}{2}$, $c = -\frac{1}{2}$; by uniqueness these are the connection forms. Then

$$\Omega^1_2 = d\omega^1_2 + \omega^1_3 \wedge \omega^3_2 = -\tfrac{1}{2}\omega^1 \wedge \omega^2 + \tfrac{1}{2}\omega^2 \wedge \tfrac{1}{2}\omega^1 = -\tfrac{3}{4}\omega^1 \wedge \omega^2,$$

$\Omega^1_3 = d\omega^1_3 + \omega^1_2 \wedge \omega^2_3 = \frac{1}{2}\omega^3 \wedge (-\frac{1}{2}\omega^1) = \frac{1}{4}\omega^1 \wedge \omega^3$ and $\Omega^2_3 = d\omega^2_3 + \omega^2_1 \wedge \omega^1_3 = -\frac{1}{2}\omega^3 \wedge \frac{1}{2}\omega^2 = \frac{1}{4}\omega^2 \wedge \omega^3$. Each Ω^i_j is a multiple of $\omega^i \wedge \omega^j$, so $R(e_k, e_l, e_j, e_i) = 0$ unless $\{k, l\} = \{i, j\}$, and $\mathrm{Ric}(e_i, e_j) = \sum_k R(e_k, e_i, e_j, e_k)$ is diagonal, with $\mathrm{Ric}(e_1, e_1) = K_{12} + K_{13} = -\frac{1}{2}$, $\mathrm{Ric}(e_2, e_2) = -\frac{1}{2}$, $\mathrm{Ric}(e_3, e_3) = \frac{1}{2}$, and scalar curvature $-\frac{1}{2}$. So this homogeneous metric has sectional curvatures of both signs. □

Exercise 40.17.18. Let $(\theta^a)_{a<n}$ be an orthonormal coframe on an open subset of the round S^{n-1}, $n \geq 3$, with connection forms θ^a_b, so that $d\theta^a_b + \sum_c \theta^a_c \wedge \theta^c_b = \theta^a \wedge \theta^b$. On $(0, L) \times S^{n-1}$ consider $g = dr^2 + f(r)^2 g_{S^{n-1}}$ with $f > 0$, and the coframe $\omega^a = f\theta^a$, $\omega^n = dr$. Show that $\omega^a_b = \theta^a_b$ and $\omega^a_n = f'\theta^a$, and that

$$K(e_a, e_n) = -\frac{f''}{f}, \qquad K(e_a, e_b) = \frac{1 - f'^2}{f^2}.$$

Check that $f(r) = r$, $\sin r$ and $\sinh r$ give constant curvature 0, 1 and -1.

Solution. $d\omega^a = f'dr \wedge \theta^a + f\, d\theta^a = -\frac{f'}{f}\omega^a \wedge \omega^n - f \sum_b \theta^a_b \wedge \theta^b$, and with the proposed forms $-\sum_b \omega^a_b \wedge \omega^b - \omega^a_n \wedge \omega^n = -f \sum_b \theta^a_b \wedge \theta^b - \frac{f'}{f}\omega^a \wedge \omega^n$, which agrees; also $d\omega^n = 0$ and $-\sum_a \omega^n_a \wedge \omega^a = \sum_a f'\theta^a \wedge f\theta^a = 0$. So these are the connection forms. Then

$\Omega^a_b = d\theta^a_b + \sum_c \theta^a_c \wedge \theta^c_b + \omega^a_n \wedge \omega^n_b = \theta^a \wedge \theta^b - f'^2\theta^a \wedge \theta^b = \frac{1-f'^2}{f^2}\omega^a \wedge \omega^b$, and $\Omega^a_n = d(f'\theta^a) + \sum_c \theta^a_c \wedge f'\theta^c = f''dr \wedge \theta^a + f'\big(d\theta^a + \sum_c \theta^a_c \wedge \theta^c\big) = -\frac{f''}{f}\omega^a \wedge \omega^n$, using the first structure equation of the sphere. For $f = r$: 0 and $(1-1)/r^2 = 0$; for $f = \sin r$: 1 and $(1-\cos^2 r)/\sin^2 r = 1$; for $f = \sinh r$: -1 and $(1-\cosh^2 r)/\sinh^2 r = -1$. □

Exercise 40.17.19. Let M be a compact oriented 4-manifold with a metric of constant curvature κ. Show that $\chi(M) = 3\kappa^2 \operatorname{vol}(M)/(4\pi^2)$, and deduce that T^4 and $S^1 \times S^3$ carry no metric of constant curvature $\kappa \neq 0$. Check the formula directly for the product $S^2 \times S^2$ of two unit spheres, which does not have constant curvature: compute $\operatorname{Pf}(\Omega)$ in a product frame and show that $\frac{1}{4\pi^2}\int \operatorname{Pf}(\Omega) = 4 = \chi(S^2 \times S^2)$.

Solution. The first statement is Corollary 40.9.9(2) with $m = 2$, $(2m-1)!! = 3$. By the Künneth formula $\chi(T^4) = \chi(S^1)^4 = 0$ and $\chi(S^1 \times S^3) = 0$, while $3\kappa^2 \operatorname{vol}/(4\pi^2) > 0$ for $\kappa \neq 0$. For $S^2 \times S^2$ take orthonormal coframes ω^1, ω^2 of the first factor and ω^3, ω^4 of the second, pulled back. The connection forms of the factors, placed in two diagonal blocks, are skew-symmetric and satisfy the first structure equation, so they are the connection forms of the product. The curvature is block diagonal, $\Omega^1_2 = \omega^1 \wedge \omega^2$, $\Omega^3_4 = \omega^3 \wedge \omega^4$ and $\Omega^1_3 = \Omega^1_4 = \Omega^2_3 = \Omega^2_4 = 0$, so $\operatorname{Pf}(\Omega) = \Omega^1_2 \wedge \Omega^3_4 = dA_1 \wedge dA_2$, with integral $(4\pi)^2$, and $16\pi^2/(4\pi^2) = 4 = \chi(S^2)^2$. (The plane spanned by e_1 and e_3 has curvature 0, the one spanned by e_1 and e_2 curvature 1.) □

Exercise 40.17.20. This exercise relates three descriptions of hyperbolic space, each with its natural chart: the upper half-space $H^n = \{x \in \mathbb{R}^n : x_n > 0\}$ with the metric $|dx|^2/x_n^2$, where $|dx|^2 = dx_1^2 + \cdots + dx_n^2$ and the chart is the inclusion $H^n \subseteq \mathbb{R}^n$; the ball $B^n = \{|y| < 1\}$ with the metric $4|dy|^2/(1-|y|^2)^2$, where the chart is the identity of B^n; and the hyperboloid $L = \{x \in \mathbb{R}^{n+1} : x_1^2 + \cdots + x_n^2 - x_{n+1}^2 = -1,\ x_{n+1} > 0\}$, where the chart is the projection $p(x) = (x_1, \ldots, x_n)$.

1. ($n = 2$, starting from the half-plane.) Show that the Cayley map $C(z) = i\frac{1+z}{1-z}$ is a diffeomorphism from the unit disc onto the upper half-plane, with inverse $w \mapsto \frac{w-i}{w+i}$, and compute $C^*\big(|dw|^2/(\operatorname{Im} w)^2\big)$, finding the metric $4|dz|^2/(1-|z|^2)^2$ of the disc. So the disc model is the half-plane model transported by C.

2. (Any n.) Let $\sigma(x) = -e_n + 2(x+e_n)/|x+e_n|^2$, the inversion in the sphere of radius $\sqrt{2}$ centred at $-e_n$. Show that σ maps H^n diffeomorphically onto B^n and that $\sigma^*\big(4|dy|^2/(1-|y|^2)^2\big) = |dx|^2/x_n^2$.

3. Show that p maps L diffeomorphically onto $\mathbb{R}^n$, that the restriction to L of the Lorentz form $dx_1^2 + \cdots + dx_n^2 - dx_{n+1}^2$ is positive definite, and that in the chart p, with coordinates $u = p(x)$, it is $|du|^2 - (u \cdot du)^2/(1+|u|^2)$, where $u \cdot du = \sum_i u_i\, du_i$.

4. Show that the map $y \mapsto x = \big(2y, 1+|y|^2\big)/(1-|y|^2)$ is a diffeomorphism from B^n onto L, inverse to $x \mapsto (x_1, \ldots, x_n)/(1+x_{n+1})$ (the projection from the point $(0, \ldots, 0, -1)$), which pulls the Lorentz metric of L back to $4|dy|^2/(1-|y|^2)^2$. Conclude that the three models are isometric, and that all have constant curvature -1.

Solution. (1) C is holomorphic on $\{z \neq 1\}$ with $C'(z) = 2i/(1-z)^2$. Writing $\frac{1+z}{1-z} = \frac{(1+z)(1-\bar z)}{|1-z|^2}$ and $(1+z)(1-\bar z) = 1 - |z|^2 + (z - \bar z)$, we get $\operatorname{Im} C(z) = \operatorname{Re}\frac{1+z}{1-z} = \frac{1-|z|^2}{|1-z|^2}$, which is positive exactly for $|z| < 1$. Solving $w = i(1+z)/(1-z)$ for z gives $z = (w-i)/(w+i)$, defined on the half-plane, and $|(w-i)/(w+i)| < 1$ there since w is closer to i than to $-i$; so C is a bijection onto the half-plane with holomorphic, hence smooth, inverse. The differential of a holomorphic map is multiplication by the complex derivative, which by the Cauchy–Riemann

equations (Proposition 29.2.3) multiplies lengths by its modulus; so $C^*|dw|^2 = |C'(z)|^2|dz|^2 = 4|dz|^2/|1-z|^4$, and

$$C^*\frac{|dw|^2}{(\operatorname{Im} w)^2} = \frac{4|dz|^2}{|1-z|^4}\cdot\frac{|1-z|^4}{(1-|z|^2)^2} = \frac{4|dz|^2}{(1-|z|^2)^2}.$$

(2) Write $v = x + e_n$. The inversion $J(x) = c + r^2(x-c)/|x-c|^2$, here with $c = -e_n$ and $r^2 = 2$, is an involution of $\mathbb{R}^n \setminus \{c\}$, and its differential is $dJ = \frac{r^2}{|v|^2}\big(I - 2\frac{vv^T}{|v|^2}\big)$, a reflection times $r^2/|v|^2$ (compare Example 35.2.7); so $J^*|dy|^2 = r^4|dx|^2/|v|^4 = 4|dx|^2/|x+e_n|^4$. Moreover

$$|\sigma(x)|^2 = 1 - \frac{4\langle v, e_n\rangle}{|v|^2} + \frac{4}{|v|^2} = 1 - \frac{4x_n}{|x+e_n|^2}, \qquad \text{so} \qquad 1 - |\sigma(x)|^2 = \frac{4x_n}{|x+e_n|^2}.$$

Hence σ maps H^n into B^n, and, being an involution of $\mathbb{R}^n \setminus \{-e_n\}$ mapping $\{x_n > 0\}$ into $\{|y| < 1\}$ and (by the same formula read backwards) $\{|y| < 1\}$ into $\{x_n > 0\}$, it is a diffeomorphism $H^n \to B^n$. Finally $\sigma^*\big(4|dy|^2/(1-|y|^2)^2\big) = 4\cdot\frac{4|dx|^2}{|x+e_n|^4}\cdot\frac{|x+e_n|^4}{16x_n^2} = \frac{|dx|^2}{x_n^2}$.

(3) The inverse of $p|_L$ is $u \mapsto (u, (1+|u|^2)^{1/2})$, smooth on $\mathbb{R}^n$, so p is a diffeomorphism and p is a chart of the submanifold L. On L, $x_{n+1} = (1+|u|^2)^{1/2}$ and $dx_{n+1} = (u\cdot du)/(1+|u|^2)^{1/2}$, so the Lorentz form restricts to $|du|^2 - (u\cdot du)^2/(1+|u|^2)$. By the Cauchy–Schwarz inequality, $(u\cdot\xi)^2 \le |u|^2|\xi|^2 < (1+|u|^2)|\xi|^2$ for $\xi \neq 0$, so this is positive definite.

(4) Let $t = 1-|y|^2$, so $x' = (x_1,\dots,x_n) = 2y/t$ and $x_{n+1} = (1+|y|^2)/t = 2/t - 1$. Then $|x'|^2 - x_{n+1}^2 = (4|y|^2 - (1+|y|^2)^2)/t^2 = -1$ and $x_{n+1} > 0$, so the point is on L; and $x'/(1+x_{n+1}) = (2y/t)/(2/t) = y$, which shows that the two maps are inverse to each other (the second one maps L into B^n, as $|x'|^2 = x_{n+1}^2 - 1 < (1+x_{n+1})^2$). With $dt = -2\,y\cdot dy$,

$$\begin{aligned}|dx'|^2 - dx_{n+1}^2 &= \Big|\frac{2\,dy}{t} - \frac{2y\,dt}{t^2}\Big|^2 - \frac{4\,dt^2}{t^4} = \frac{4|dy|^2}{t^2} - \frac{8(y\cdot dy)\,dt}{t^3} + \frac{4(|y|^2-1)\,dt^2}{t^4}\\ &= \frac{4|dy|^2}{t^2} + \frac{16(y\cdot dy)^2}{t^3} - \frac{16(y\cdot dy)^2}{t^3} = \frac{4|dy|^2}{(1-|y|^2)^2}.\end{aligned}$$

One can also do this computation in the chart p of (3), with the metric $|du|^2 - (u\cdot du)^2/(1+|u|^2)$ and the map $u = 2y/t$. Written with indices, the pulled back metric is $\sum_{k,l} J_{ki}J_{lj}\big(\delta_{kl} - u_ku_l/(1+|u|^2)\big)$, where $J_{ki} = \partial u_k/\partial y_i$, and expanding it term by term is long; organised as products of matrices it is short. With $r = |y|$,

$$J = \frac{2}{t}\Big(I + \frac{2yy^T}{t}\Big), \qquad J^TJ = \frac{4}{t^2}\Big(I + \frac{4yy^T}{t} + \frac{4r^2yy^T}{t^2}\Big) = \frac{4}{t^2}\Big(I + \frac{4yy^T}{t^2}\Big),$$

using $t + r^2 = 1$; and $J^Tu = \frac{4(1+r^2)}{t^3}y$ and $1 + |u|^2 = \frac{(1+r^2)^2}{t^2}$. So the matrix of the pulled back metric is

$$J^T\Big(I - \frac{uu^T}{1+|u|^2}\Big)J = J^TJ - \frac{(J^Tu)(J^Tu)^T}{1+|u|^2} = \frac{4}{t^2}I + \frac{16yy^T}{t^4} - \frac{16yy^T}{t^4} = \frac{4}{(1-|y|^2)^2}I,$$

as before. So the ball, the half-space and the hyperboloid are isometric (by (2) and (4)), and they have constant curvature -1 by Corollary 40.9.6. □

Exercise 40.17.21. Let D be as in Corollary 40.3.13, and u a smooth function near D with $\Delta_0 u = 0$ on D and $u = 0$ on ∂D. Show that $u = 0$ on D. Deduce that a solution of the Dirichlet problem $\Delta_0 u = f$ on D, $u = g$ on ∂D, is unique.

Solution. By Corollary 40.3.14(2) with $f = g = u$, $\int_D |\nabla u|^2\,d\lambda = \int_{\partial D} u\,\partial_\nu u\,dA - \int_D u\,\Delta_0 u\,d\lambda = 0$. So $\nabla u = 0$ on D and u is constant on each connected component of D. Each component C

meets ∂D: otherwise C would be a compact manifold without boundary, open in $\mathbb{R}^{n+1}$ (its points are interior points of D, and C is open in D) and closed, hence empty or all of $\mathbb{R}^{n+1}$, which is not compact. As $u = 0$ on ∂D, $u = 0$ on C. For the Dirichlet problem, the difference of two solutions satisfies the hypotheses. □

Exercise 40.17.22. Let $S = \{z = 1 - x^2 - y^2,\ z \geq 0\}$ with the upward normal, and $F = (-y, x, z)$. Verify Stokes' theorem (Corollary 40.3.16) by computing both sides, using Example 40.3.9 for the flux of curl F.

Solution. $\operatorname{curl} F = (\partial_y F_3 - \partial_z F_2, \partial_z F_1 - \partial_x F_3, \partial_x F_2 - \partial_y F_1) = (0, 0, 2)$. By Example 40.3.9, with $h = 1 - u^2 - v^2$ on the unit disc, the flux of $(0, 0, 2)$ is $\iint 2\, du\, dv = 2\pi$. The boundary is the unit circle in the plane $z = 0$; the upward normal and the rule of Corollary 40.3.16 orient it counterclockwise seen from above, $\gamma(t) = (\cos t, \sin t, 0)$, and $\oint F^1\, dx + F^2\, dy + F^3\, dz = \int_0^{2\pi} (\sin^2 t + \cos^2 t + 0)\, dt = 2\pi$. □

Exercise 40.17.23. Show that every geodesic of the cylinder $C = \{x^2 + y^2 = 1\} \subseteq \mathbb{R}^3$ is one of the helices of Example 40.11.4(3), and that the points $(1, 0, 0)$ and $(1, 0, 1)$ are joined by infinitely many geodesics. Which of them is the shortest?

Solution. At the point $(\cos t_0, \sin t_0, z_0)$ the tangent plane of C is spanned by $(-\sin t_0, \cos t_0, 0)$ and $(0, 0, 1)$, so every initial velocity is $v = a(-\sin t_0, \cos t_0, 0) + b(0, 0, 1)$, which is the initial velocity of the helix with parameters a, b, t_0, z_0. That helix is a geodesic (Example 40.11.4(3)), and a geodesic is determined by its initial position and velocity (Definition 40.2.9), so every geodesic is a helix. For $k \in \mathbb{Z}$ the helix $\gamma_k(t) = (\cos(2\pi k t), \sin(2\pi k t), t)$, $0 \leq t \leq 1$, goes from $(1, 0, 0)$ to $(1, 0, 1)$ winding k times around the cylinder, with length $(4\pi^2 k^2 + 1)^{1/2}$. The shortest is the vertical segment, $k = 0$, of length 1. On the plane obtained by unrolling the cylinder, the γ_k are the segments from $(0, 0)$ to the different preimages $(2\pi k, 1)$ of the end point. □

Exercise 40.17.24. On a Riemannian manifold, prove the Leibniz rules $\operatorname{grad}(fh) = f \operatorname{grad} h + h \operatorname{grad} f$, $\operatorname{div}(fX) = f \operatorname{div} X + \langle \operatorname{grad} f, X \rangle$ and $\Delta_g(fh) = f\Delta_g h + h\Delta_g f + 2\langle \operatorname{grad} f, \operatorname{grad} h \rangle$. On $\mathbb{R}^n \setminus \{0\}$ show that a radial function $u = \phi(r)$ has $\Delta_g u = \phi''(r) + \frac{n-1}{r}\phi'(r)$, and deduce that r^{2-n} ($n \geq 3$) and $\log r$ ($n = 2$) are harmonic.

Solution. The first identity follows from $d(fh) = f\, dh + h\, df$. For the second, $\nabla_v(fX) = v(f)X + f\nabla_v X$, and taking traces gives $\langle \operatorname{grad} f, X \rangle + f \operatorname{div} X$. The third is the second with $X = \operatorname{grad} h$ and f, plus the same with the roles exchanged, using the first. For $u = \phi(r)$, $\operatorname{grad} u = \phi'(r) \operatorname{grad} r$, and by the second identity and Example 40.5.3(4), $\Delta_g u = \langle \operatorname{grad} \phi'(r), \operatorname{grad} r \rangle + \phi'(r)\Delta_g r = \phi''(r) |\operatorname{grad} r|^2 + \phi'(r)\frac{n-1}{r}$, with $|\operatorname{grad} r| = 1$. For $\phi = r^{2-n}$, $\phi'' + \frac{n-1}{r}\phi' = (2-n)r^{-n}\big((1-n) + (n-1)\big) = 0$; for $\phi = \log r$ and $n = 2$, $-r^{-2} + r^{-1} \cdot r^{-1} = 0$. □

Exercise 40.17.25. Let $a \in \mathbb{R}^{n+1}$ and $f(x) = \langle a, x \rangle$ on the round unit sphere S^n. Show that $\operatorname{grad} f = a - f(x)x$, that $\operatorname{Hess} f = -f\, g$, and hence $\Delta_g f = -n f$. Where are the critical points of f, and what does the Hessian say about them?

Solution. For X tangent to S^n at x, $Xf = \langle a, X \rangle = \langle a - \langle a, x \rangle x, X \rangle$, and $a - \langle a, x \rangle x$ is tangent, as it is orthogonal to x; this is the gradient. For a unit speed great circle $\gamma(t) = \cos(t)x + \sin(t)v$, a geodesic (Exercise 40.17.7), $f(\gamma(t)) = \cos(t) f(x) + \sin(t)\langle a, v \rangle$, so $\operatorname{Hess} f(v, v) = (f \circ \gamma)''(0) = -f(x)$ for every unit $v \in T_x S^n$ (Proposition 40.5.2(4)). A symmetric bilinear form is determined by its quadratic form, so $\operatorname{Hess} f = -f\, g$, and its trace is $-n f$. The gradient vanishes where a is a multiple of x, that is at $x = \pm a/|a|$ (for $a \neq 0$), where $f = \pm|a|$ and $\operatorname{Hess} f = \mp|a| g$: the maximum and the minimum of f. So the restrictions of linear functions are eigenfunctions of the Laplacian, with eigenvalue n for the Hodge Laplacian $-\Delta_g$. □

Chapter 41

Symplectic Geometry

41.1 Introduction

A Riemannian metric is a symmetric nondegenerate bilinear form on each tangent space; a *symplectic form* is an *antisymmetric* one, which moreover is closed as a differential form. The change of symmetry changes everything. A Riemannian metric has local invariants, the curvature, while by Darboux's theorem all symplectic forms of the same dimension look alike near a point; on the other hand symplectic forms are rigid globally, and live only on manifolds with nonzero even-dimensional cohomology. Symplectic geometry is the geometry of classical mechanics: the phase space of a mechanical system is the cotangent bundle T^*Q of its configuration space, which carries a canonical symplectic form, and the evolution of the system is the flow of a Hamiltonian vector field. It is also the geometry behind Kähler manifolds (Chapter 47), whose Kähler form is a symplectic form compatible with the complex structure, and behind the Hodge-theoretic notion of polarisation.

The chapter goes as follows.

1. Symplectic linear algebra: $\omega^n \neq 0$, isotropic, coisotropic and Lagrangian subspaces, Lagrangian complements and compatible complex structures (Section 41.2).
2. Symplectic manifolds: the Liouville volume form, the cohomological obstruction $[\omega]^n \neq 0$, examples (Section 41.3).
3. The cotangent bundle: the tautological 1-form, the canonical symplectic form, Lagrangian graphs of closed 1-forms, conormal bundles, and the splitting of $T(T^*M)$ into two Lagrangian subbundles given by a torsion-free connection; the derivative of a 1-form regarded as a map $M \to T^*M$, whose antisymmetric part is the exterior derivative, and the graph of df, whose tangent spaces are given by the Hessian of f (Section 41.4).
4. Hamiltonian vector fields, Poisson brackets and Liouville's theorem; area on the sphere and Archimedes' theorem (Section 41.5).
5. Darboux's theorem and Moser's theorem on area forms, by Moser's method (Section 41.6).
6. Lagrangian submanifolds and Weinstein's neighbourhood theorem (Section 41.7).

References: [CdS01], [MS17], [Lee13, Chapter 22].
Prerequisites. Chapter 11 (Linear Algebra), Chapter 36 (Differential Forms and Integration), Chapter 37 (de Rham Cohomology), Chapter 38 (Connections and Curvature) and Chapter 40 (Riemannian Geometry).

41.2 Symplectic linear algebra

Definition 41.2.1. A *symplectic vector space* is a finite-dimensional real vector space V with a nondegenerate alternating bilinear form ω. For a subspace W, its *symplectic complement* is $W^\omega = \{v \in V : \omega(v, w) = 0 \text{ for all } w \in W\}$. The subspace W is *isotropic* if $W \subseteq W^\omega$ (that is, $\omega|_W = 0$), *coisotropic* if $W^\omega \subseteq W$, *Lagrangian* if $W = W^\omega$, and *symplectic* if $W \cap W^\omega = 0$. A linear map preserving ω is *symplectic*, and the symplectic automorphisms of V form the *symplectic group* $\mathrm{Sp}(V, \omega)$.

By Theorem 11.9.6, $\dim V = 2n$ is even and V has a symplectic basis $e_1, \dots, e_n, f_1, \dots, f_n$; with the dual basis, $\omega = \sum_i e_i^* \wedge f_i^*$. The model is $\mathbb{R}^{2n}$ with coordinates (x, y) and $\omega_0 = \sum_i dx^i \wedge dy^i$, that is, $\omega_0(u, v) = u^T J_0 v$ with $J_0 = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$; its symplectic group is $\mathrm{Sp}_{2n}(\mathbb{R}) = \{A : A^T J_0 A = J_0\}$, whose Lie algebra is $\mathfrak{sp}_{2n}(\mathbb{R})$ (Example 13.7.14). In contrast to inner products, there is no notion of length: $\omega(v, v) = 0$ for every v.

Proposition 41.2.2. *Let (V, ω) be a symplectic vector space of dimension $2n$ and $W \subseteq V$ a subspace.*

1. *An alternating form σ on a $2n$-dimensional space is nondegenerate if and only if $\sigma^n \neq 0$. Symplectic maps have determinant 1.*
2. *$\dim W + \dim W^\omega = 2n$ and $(W^\omega)^\omega = W$.*
3. *Isotropic subspaces have dimension at most n, and a subspace is Lagrangian if and only if it is isotropic of dimension n. Every isotropic subspace is contained in a Lagrangian one.*
4. *Every Lagrangian subspace L has a Lagrangian complement L', and then $v \mapsto \omega(\cdot, v)|_L$ is an isomorphism $L' \to L^*$; a basis $e_1, \dots, e_n$ of L and the dual basis $f_1, \dots, f_n$ of L' ($\omega(e_i, f_j) = \delta_{ij}$) form a symplectic basis. So $(V, \omega) \cong (L \oplus L^*, \omega_L)$ with $\omega_L((u, \alpha), (v, \beta)) = \beta(u) - \alpha(v)$.*
5. *There is a complex structure J with $\omega(Ju, Jv) = \omega(u, v)$ and $g(u, v) = \omega(u, Jv)$ an inner product (a* compatible *complex structure). For such J and a Lagrangian L, JL is a Lagrangian complement of L, equal to the orthogonal complement $L^\perp$ for g.*

Proof. (1) If σ is nondegenerate, $\sigma^n \neq 0$ by Proposition 12.4.7, which also gives the statement on determinants. If σ is degenerate, choose $v \neq 0$ with $\iota_v \sigma = 0$; as ι_v is an antiderivation, $\iota_v(\sigma^n) = n(\iota_v \sigma) \wedge \sigma^{n-1} = 0$, so the top form σ^n vanishes on every basis containing v, hence $\sigma^n = 0$.

(2) The map $V \to W^*$, $v \mapsto \omega(v, \cdot)|_W$, is the composite of the isomorphism $V \to V^*$ given by ω and the surjective restriction $V^* \to W^*$; its kernel is W^ω. So $\dim W^\omega = 2n - \dim W$. Clearly $W \subseteq (W^\omega)^\omega$, and both have the same dimension.

(3) If $W \subseteq W^\omega$, then $2 \dim W \leq \dim W + \dim W^\omega = 2n$. An isotropic W of dimension n has $\dim W^\omega = n$, so $W = W^\omega$; conversely a Lagrangian is isotropic of dimension n by (2). If W is isotropic with $\dim W < n$, then $\dim W^\omega > n > \dim W$, so there is $v \in W^\omega \setminus W$, and $W + \mathbb{R}v$ is isotropic since $\omega(v, v) = 0$; repeat.

(5) Let e_i, f_i be a symplectic basis and $Je_i = f_i$, $Jf_i = -e_i$. Then $J^2 = -1$, J preserves ω (check on basis vectors), and $\omega(u, Jv)$ is the standard inner product in this basis: $\omega(e_i, Je_j) = \omega(e_i, f_j) = \delta_{ij}$, $\omega(f_i, Jf_j) = -\omega(f_i, e_j) = \delta_{ij}$, and the mixed terms vanish. Given compatible J and Lagrangian L, JL is isotropic since $\omega(Ju, Jv) = \omega(u, v) = 0$, hence Lagrangian; and for $u, v \in L$, $g(u, Jv) = \omega(u, J^2 v) = -\omega(u, v) = 0$, so $JL \subseteq L^\perp$, with equality by dimension. In particular $L \cap JL = 0$. (For the relation between ω, J and g in general, any two of which determine the third, see Proposition 11.11.5.)

(4) A Lagrangian complement exists by (5). For $v \in L'$, if $\omega(u, v) = 0$ for all $u \in L$, then $v \in L^\omega \cap L' = L \cap L' = 0$; so $L' \to L^*$ is injective, hence an isomorphism. The dual

basis f_j of L' satisfies $\omega(e_i, f_j) = \delta_{ij}$, and $\omega(e_i, e_j) = 0 = \omega(f_i, f_j)$ as L and L' are isotropic. Identifying L' with L^* by $f \mapsto \omega(\cdot, f)|_L$, for $u, v \in L$ and $f, f' \in L'$ corresponding to α, β: $\omega(u + f, v + f') = \omega(u, f') - \omega(v, f) = \beta(u) - \alpha(v)$. □

Example 41.2.3.
1. In $(\mathbb{R}^2, dx \wedge dy)$ every line is Lagrangian, and $\omega(u, v) = \det(u, v)$ is the signed area of the parallelogram spanned by u and v. Symplectic maps of $\mathbb{R}^2$ are the area- and orientation-preserving linear maps: $\mathrm{Sp}_2(\mathbb{R}) = \mathrm{SL}_2(\mathbb{R})$. In higher dimensions $\mathrm{Sp}_{2n}(\mathbb{R}) \subsetneq \mathrm{SL}_{2n}(\mathbb{R})$: preserving ω is much stronger than preserving the volume $\omega^n/n!$.
2. In $(\mathbb{R}^4, dx^1 \wedge dy^1 + dx^2 \wedge dy^2)$ the planes $\{y = 0\}$, $\{x = 0\}$ and the graph $\{y = Sx\}$ of a symmetric matrix S are Lagrangian; the graph of $A\colon \mathbb{R}^2 \to \mathbb{R}^2$ is Lagrangian if and only if A is symmetric, since $\omega((x, Ax), (x', Ax')) = x^T Ax' - x'^T Ax$. The plane $\{x^2 = y^2 = 0\}$ is symplectic, and the hyperplane $\{y^2 = 0\}$ is coisotropic, with $\{y^2 = 0\}^\omega = \mathbb{R}\partial_{x^2}$.
3. On $\mathbb{C}^n = \mathbb{R}^{2n}$ with $z = x + iy$, $\omega_0 = \sum dx^i \wedge dy^i = \frac{i}{2} \sum dz^i \wedge d\bar{z}^i$, and J = multiplication by i is compatible: $\omega_0(u, iv) = \mathrm{Re}\langle u, v\rangle$ for the standard Hermitian product (Proposition 11.11.3). The real subspace $\mathbb{R}^n$ is Lagrangian, and $i\mathbb{R}^n$ is its orthogonal Lagrangian complement.

41.3 Symplectic manifolds

Definition 41.3.1. A *symplectic form* on a manifold M is a closed 2-form ω that is nondegenerate at every point; (M, ω) is a *symplectic manifold*. A *symplectomorphism* is a diffeomorphism φ with $\varphi^*\omega = \omega$. A submanifold $S \subseteq M$ is isotropic, coisotropic, Lagrangian or symplectic if each $T_pS \subseteq T_pM$ is.

Each tangent space is a symplectic vector space, so $\dim M = 2n$ is even. By Proposition 41.2.2(1), ω^n vanishes nowhere: $\omega^n/n!$ *is a volume form*, the *Liouville form*, and every symplectic manifold is oriented by it. Nondegeneracy means that $X \mapsto \iota_X\omega$ is an isomorphism of vector bundles $TM \to T^*M$, an analogue of the musical isomorphism of a metric (Remark 40.2.4), which identifies vector fields with 1-forms.

Proposition 41.3.2. *If (M, ω) is a compact symplectic manifold without boundary of dimension $2n$, then $[\omega]^k \neq 0$ in $H^{2k}_{\mathrm{dR}}(M)$ for $0 \leq k \leq n$. In particular $H^{2k}_{\mathrm{dR}}(M) \neq 0$, and the spheres S^{2n} with $n \geq 2$ carry no symplectic form.*

Proof. With the orientation given by ω^n, $\int_M \omega^n > 0$ (Corollary 36.4.13), so ω^n is not exact by Stokes' theorem (Theorem 36.4.8), and $[\omega]^n = [\omega^n] \neq 0$. Since $[\omega]^n = [\omega]^k[\omega]^{n-k}$, also $[\omega]^k \neq 0$. For $n \geq 2$, $H^2_{\mathrm{dR}}(S^{2n}) = 0$ (Example 37.2.9). □

Example 41.3.3.
1. $(\mathbb{R}^{2n}, \omega_0)$, and every open subset of it.
2. *Surfaces.* On an oriented surface, an area form is a symplectic form: it is closed for degree reasons, and a 2-form on a 2-dimensional space is nondegenerate if and only if it is nonzero. So every orientable surface is symplectic, and symplectic forms on surfaces are just area forms. Symplectomorphisms of surfaces are the area-preserving diffeomorphisms.
3. *Products.* If (M_1, ω_1), (M_2, ω_2) are symplectic, so is $M_1 \times M_2$ with $\mathrm{pr}_1^*\omega_1 + \mathrm{pr}_2^*\omega_2$; for instance the torus T^{2n} with the form induced from ω_0, and $S^2 \times S^2$. On the other hand $S^2 \times S^4$ is not symplectic, although it is orientable with $H^2 \neq 0$: by the Künneth formula (Theorem 37.3.8) a symplectic form ω would have $[\omega] = c\,a$ with a the pullback of a generator of $H^2(S^2)$, and $a^2 = 0$, as it is pulled back from $H^4(S^2) = 0$; so $[\omega]^3 = 0$, contradicting Proposition 41.3.2.

4. *Kähler manifolds*, such as $\mathbb{CP}^n$ with the Fubini–Study form and all complex projective manifolds, are symplectic (Chapter 47).

5. *Cotangent bundles* carry a canonical symplectic form (Section 41.4).

41.4 The cotangent bundle

Let M be a manifold of dimension n and $\pi\colon T^*M \to M$ its cotangent bundle. A chart with coordinates $q^1, \dots, q^n$ on $U \subseteq M$ gives coordinates (q, p) on T^*U, where a covector $\xi = \sum_i p_i\, dq^i|_q$ has coordinates $(q^1, \dots, q^n, p_1, \dots, p_n)$.

Definition 41.4.1. The *tautological* (or *Liouville*) 1-form λ on T^*M is defined at a point $\xi \in T^*_qM$ by

$$\lambda_\xi(Y) = \xi\big(d\pi(Y)\big), \qquad Y \in T_\xi(T^*M);$$

a point of T^*M is a covector on M, and λ applies it to the projection of Y to M. The *canonical symplectic form* is $\omega = -d\lambda$.

Proposition 41.4.2. *1. In coordinates $\lambda = \sum_i p_i\, dq^i$ and $\omega = \sum_i dq^i \wedge dp_i$; so ω is a symplectic form.*

2. λ is characterised by the property $\alpha^*\lambda = \alpha$ for every 1-form α on M, regarded as a section $\alpha\colon M \to T^*M$. Consequently $\alpha^*\omega = -d\alpha$.

3. For a diffeomorphism $f\colon M \to N$, the cotangent lift *$F\colon T^*M \to T^*N$, $F(\xi) = \xi \circ (d_q f)^{-1}$ for $\xi \in T^*_qM$, satisfies $F^*\lambda_N = \lambda_M$; so it is a symplectomorphism.*

Proof. (1) A tangent vector at $\xi = \sum p_i\, dq^i$ is $Y = \sum_i (a^i \partial_{q^i} + b_i \partial_{p_i})$, and $d\pi(Y) = \sum_i a^i \partial_{q^i}$, so $\lambda(Y) = \sum_i p_i a^i$: $\lambda = \sum_i p_i\, dq^i$. Then $\omega = -\sum_i dp_i \wedge dq^i = \sum_i dq^i \wedge dp_i$, which is closed and, at each point, the standard form in the basis $\partial_{q^i}, \partial_{p_i}$.

(2) Since $\pi \circ \alpha = \mathrm{id}$, $(\alpha^*\lambda)_q(v) = \lambda_{\alpha_q}(d\alpha(v)) = \alpha_q(d\pi\, d\alpha(v)) = \alpha_q(v)$. Conversely, let $\mu = \sum_i (A_i\, dq^i + B^i\, dp_i)$ in coordinates satisfy $\alpha^*\mu = \alpha$ for all $\alpha = \sum_i a_i\, dq^i$; then $\sum_i A_i(q, a(q))\, dq^i + \sum_k B^k(q, a(q))\, da_k = \sum_i a_i\, dq^i$. Constant $a_i = c_i$ give $A_i(q, c) = c_i$; then at a point q_0, the functions $a_k = c_k + \sum_j d_{kj}(q^j - q_0^j)$ give $\sum_k B^k(q_0, c) d_{ki} = 0$ for every matrix (d_{kj}), so $B^k = 0$. Hence $\mu = \lambda$. Finally $\alpha^*\omega = -d\alpha^*\lambda = -d\alpha$.

(3) $\pi_N \circ F = f \circ \pi_M$, so $(F^*\lambda_N)_\xi(Y) = F(\xi)(d\pi_N\, dF(Y)) = F(\xi)(df\, d\pi_M(Y)) = \xi(d\pi_M(Y)) = \lambda_M(Y)$. □

Corollary 41.4.3. *1. The graph $\alpha(M)$ of a 1-form α is a Lagrangian submanifold of T^*M if and only if $d\alpha = 0$. In particular the zero section and the graphs $df(M)$ of differentials of functions are Lagrangian.*

2. The fibres T^*_qM are Lagrangian.

3. For a submanifold $S \subseteq M$, the conormal bundle *$N^*S = \{\xi \in T^*_qM : q \in S,\ \xi|_{T_qS} = 0\}$ is a Lagrangian submanifold of T^*M.*

Proof. (1) $\alpha\colon M \to T^*M$ is an embedding onto its graph, of dimension n, and the restriction of ω to it corresponds to $\alpha^*\omega = -d\alpha$. (2) In coordinates the fibre is $\{q = \text{const}\}$, on which every dq^i vanishes. (3) In a slice chart with $S = \{q^{k+1} = \cdots = q^n = 0\}$, $N^*S = \{q^{k+1} = \cdots = q^n = 0,\ p_1 = \cdots = p_k = 0\}$, a submanifold of dimension n with coordinates $q^1, \dots, q^k, p_{k+1}, \dots, p_n$, on which each term $dq^i \wedge dp_i$ vanishes. □

Along the zero section 0_M the tangent spaces split canonically: $T_{0_q}(T^*M) = T_q(0_M) \oplus T_{0_q}(T^*_qM) \cong T_qM \oplus T^*_qM$, the second summand being the vertical subspace (Definition 38.7.1). Both summands are Lagrangian, and in coordinates $\omega(\sum a^i \partial_{q^i}, \sum b_i \partial_{p_i}) = \sum a^i b_i$: the symplectic form pairs them by the duality $T_qM \times T^*_qM \to \mathbb{R}$, as in Proposition 41.2.2(4). Away from the zero section the vertical subspace is still canonical, but a horizontal complement requires a choice, a connection (Section 38.7).

Proposition 41.4.4. *Let ∇ be a connection on TM, with the induced connection on T^*M and its horizontal distribution $H \subseteq T(T^*M)$ (Proposition 38.7.3). The vertical distribution V is Lagrangian, and H is Lagrangian if and only if ∇ is torsion-free. In that case $T(T^*M) = H \oplus V$ with $H_\xi \cong T_qM$, $V_\xi \cong T^*_qM$ and $\omega(h, v) = \langle v, d\pi(h)\rangle$ for $h \in H_\xi$ and $v \in V_\xi$ (identified with a covector by the vertical lift).*

Proof. V is the tangent bundle of the fibres, Lagrangian by Corollary 41.4.3(2). With $\nabla_{\partial_i}\partial_k = \sum_j \Gamma^j_{ik}\partial_j$, the dual connection satisfies $\nabla_{\partial_i} dq^j = -\sum_k \Gamma^j_{ik}\, dq^k$, so for $\alpha = \sum_j p_j\, dq^j$, $\nabla_{\partial_i}\alpha = \sum_k (\partial_i p_k - \sum_j \Gamma^j_{ik} p_j)\, dq^k$. By Proposition 38.7.3, H is spanned by the horizontal lifts of the coordinate fields,

$$X_i = \partial_{q^i} + \sum_{j,k} \Gamma^j_{ik} p_j\, \partial_{p_k}.$$

Then $\omega(X_i, X_l) = dp_i(X_l) - dp_l(X_i) = \sum_j p_j(\Gamma^j_{li} - \Gamma^j_{il}) = -\sum_j p_j T^j_{il}$, where $T(\partial_i, \partial_l) = \sum_j T^j_{il}\partial_j$ is the torsion (Exercise 38.9.7). This vanishes for all p if and only if $T = 0$; and H has dimension n. Finally $\omega(X_i, \partial_{p_k}) = \delta_{ik}$, which is the stated pairing. □

So a torsion-free connection, for instance the Levi-Civita connection of a metric, splits $T(T^*M)$ into two transverse Lagrangian subbundles paired by duality, extending the canonical splitting along the zero section, where every horizontal distribution is tangent to the zero section.

The derivative of a 1-form regarded as a map. A 1-form α on M can be differentiated in two quite different ways, and it is worth seeing how the two are related. Its exterior derivative $d\alpha$ is a 2-form on M. But α is also a smooth map $\alpha\colon M \to T^*M$, a section of π, and as such it has at each point q a differential, a linear map

$$\alpha_*\colon T_qM \to T_{\alpha_q}(T^*M),$$

which we write α_* rather than $d_q\alpha$ so as not to confuse it with the exterior derivative. Its image is the tangent space of the graph $\alpha(M)$ at the point α_q. For $\alpha = df$ one may ask whether the identity $d(df) = 0$ (Remark 36.2.9) means that the map $df\colon M \to T^*M$ has zero differential. It does not, and it could not: $\pi \circ df = \mathrm{id}_M$, so $d\pi \circ (df)_* = \mathrm{id}$ and $(df)_*$ is injective. We shall see that $(df)_*$ is, apart from a part that is the same for every 1-form, the Hessian of f. What $d(df) = 0$ says is subtler: the exterior derivative only sees the *antisymmetric part* of the derivative of a 1-form, and the Hessian is symmetric. Geometrically, it says that the tangent spaces of the graph of df are Lagrangian, which is Corollary 41.4.3(1) seen pointwise.

Proposition 41.4.5. *Let α be a 1-form on M, $q \in M$, and $\alpha_*\colon T_qM \to T_{\alpha_q}(T^*M)$ the differential of α regarded as a map $M \to T^*M$.*

1. *In coordinates, if $\alpha = \sum_i a_i\, dq^i$ and $v = \sum_j v^j \partial_{q^j}$, then*

$$\alpha_*(v) = \sum_j v^j\, \partial_{q^j} + \sum_{i,j} \frac{\partial a_i}{\partial q^j}(q)\, v^j\, \partial_{p_i}.$$

So in the bases ∂_{q^j} of T_qM and $\partial_{q^j}, \partial_{p_i}$ of $T_{\alpha_q}(T^*M)$, the matrix of α_* is $\begin{pmatrix} I \\ A \end{pmatrix}$, where $A = (\partial a_i/\partial q^j)$ is the Jacobian matrix of the components of α, with rows indexed by i and columns by j.

2. *For $u, v \in T_qM$,*
$$\omega(\alpha_* u, \alpha_* v) = -d\alpha(u, v) = u^T(A - A^T)v,$$
where in the last expression u and v are the columns of their components. So $d\alpha$ is, up to sign, the antisymmetric part of the Jacobian matrix of α.

3. *Let ∇ be a connection on TM, with the induced connection on T^*M and its horizontal distribution (Proposition 38.7.3). Then*
$$\alpha_*(v) = v^h_{\alpha_q} + \mathrm{vl}_{\alpha_q}(\nabla_v \alpha) :$$
the horizontal part of $\alpha_*(v)$ is the horizontal lift of v, the same for all 1-forms, and its vertical part is the covariant derivative $\nabla_v \alpha \in T^*_q M$. If ∇ is torsion-free, then
$$d\alpha(u, v) = (\nabla_u \alpha)(v) - (\nabla_v \alpha)(u), \qquad u, v \in T_qM.$$

Proof. (1) In the coordinates (q, p) of T^*U, the map α is $q \mapsto (q, a_1(q), \dots, a_n(q))$. Its Jacobian matrix has the identity matrix in the rows of the coordinates q^j and the matrix $(\partial a_i/\partial q^j)$ in the rows of the coordinates p_i; this is the formula.

(2) By the definition of the pullback, $(\alpha^*\omega)_q(u, v) = \omega(\alpha_* u, \alpha_* v)$, and $\alpha^*\omega = -d\alpha$ by Proposition 41.4.2(2). For the matrix expression we compute both sides in coordinates. By (1), $\alpha_* u$ has components (u, Au) and $\alpha_* v$ has components (v, Av), and $\omega = \sum_i dq^i \wedge dp_i$ gives
$$\omega(\alpha_* u, \alpha_* v) = \sum_i \big(u^i (Av)_i - v^i (Au)_i\big) = u^T Av - v^T Au = u^T(A - A^T)v.$$
On the other side $d\alpha = \sum_{i,j} \partial_j a_i \, dq^j \wedge dq^i$, so $d\alpha(u, v) = \sum_{i,j} \partial_j a_i \, (u^j v^i - v^j u^i) = v^T Au - u^T Av$, which is the same number with the opposite sign.

(3) The first formula is Proposition 38.7.3(2), applied to the bundle T^*M with the induced connection and to its section α. For the second, extend u and v to vector fields U and V near q. By the Leibniz rule $(\nabla_U \alpha)(V) = U(\alpha(V)) - \alpha(\nabla_U V)$ (Proposition 38.3.6(1)),
$$(\nabla_U \alpha)(V) - (\nabla_V \alpha)(U) = U(\alpha(V)) - V(\alpha(U)) - \alpha(\nabla_U V - \nabla_V U) = U(\alpha(V)) - V(\alpha(U)) - \alpha([U, V]),$$
since $\nabla_U V - \nabla_V U = [U, V]$ for a torsion-free connection. The right-hand side is $d\alpha(U, V)$ by the invariant formula for the exterior derivative (Theorem 36.2.8); evaluate at q. □

So the derivative of α as a map has two parts. The horizontal part carries no information about α. The vertical part is a bilinear form $(u, v) \mapsto (\nabla_u \alpha)(v)$ on T_qM, which depends on the choice of the connection; but its antisymmetric part is $d\alpha$, which does not. For the flat connection of a chart, whose Christoffel symbols vanish, the horizontal lift of ∂_{q^j} is ∂_{q^j} and $(\nabla_{\partial_{q^j}} \alpha)(\partial_{q^i}) = \partial a_i/\partial q^j$, and (3) reduces to (1). The symmetric part of $\nabla\alpha$ has no meaning independent of the connection, except where α vanishes, as we now see for $\alpha = df$.

Corollary 41.4.6. *Let f be a smooth function on M, and give M a Riemannian metric with its Levi-Civita connection.*

1. *The differential of the map $df : M \to T^*M$ at q is*
$$(df)_*(v) = v^h + \mathrm{vl}\big(\mathrm{Hess}\, f(v, \cdot)\big), \qquad v \in T_qM,$$
so the tangent space of the graph $df(M)$ at $d_q f$ is the graph of the linear map $T_qM \to T^*_qM$, $v \mapsto \mathrm{Hess}\, f(v, \cdot)$, in the splitting $H \oplus V$.

2. *For $u, v \in T_qM$,*

$$\omega\big((df)_*u, (df)_*v\big) = -d(df)(u,v) = \operatorname{Hess} f(v,u) - \operatorname{Hess} f(u,v) = 0.$$

Thus the identity $d(df) = 0$, the symmetry of the Hessian, and the fact that the graph of df is Lagrangian are one and the same statement.

3. *Let q be a critical point of f, so that $d_qf = 0_q$ lies on the zero section. In the canonical splitting $T_{0_q}(T^*M) = T_qM \oplus T_q^*M$,*

$$(df)_*(v) = \big(v, \operatorname{Hess} f_q(v,\cdot)\big),$$

where $\operatorname{Hess} f_q(v,w) = \sum_{i,j} v^i w^j\, \partial_i\partial_j f(q)$ in any chart does not depend on the metric. The tangent spaces at 0_q of the graph $df(M)$ and of the zero section 0_M intersect in $\{(v,0) : \operatorname{Hess} f_q(v,\cdot) = 0\}$, and they span $T_{0_q}(T^*M)$ if and only if $\operatorname{Hess} f_q$ is nondegenerate.

Proof. (1) By Proposition 41.4.5(3) with $\alpha = df$, the vertical part of $(df)_*(v)$ is $\nabla_v df = \operatorname{Hess} f(v,\cdot)$ (Definition 40.5.1(2)).

(2) The first equality is Proposition 41.4.5(2). By Proposition 41.4.5(3), since the Levi-Civita connection is torsion-free, $d(df)(u,v) = (\nabla_u df)(v) - (\nabla_v df)(u) = \operatorname{Hess} f(u,v) - \operatorname{Hess} f(v,u)$, and this vanishes, either because $d(df) = 0$ or because the Hessian is symmetric (Proposition 40.5.2(3)).

(3) In the coordinates of the proof of Proposition 41.4.4, the horizontal lifts of the coordinate fields are $X_i = \partial_{q^i} + \sum_{j,k} \Gamma^j_{ik} p_j\, \partial_{p_k}$, which at a point with $p = 0$ equal ∂_{q^i}; so at 0_q the horizontal lift of v is the vector $(v,0)$, tangent to the zero section $\{p = 0\}$. By (1), $(df)_*(v) = (v, \operatorname{Hess} f_q(v,\cdot))$, and the Hessian at a critical point has the stated form and does not depend on the metric by Proposition 40.5.2(4). Write $H_q\colon T_qM \to T_q^*M$, $v \mapsto \operatorname{Hess} f_q(v,\cdot)$. The two tangent spaces are $\{(v, H_qv)\}$ and $\{(v,0)\}$. A vector lies in both when $(v, H_qv) = (w,0)$, that is, $w = v$ and $H_qv = 0$. Their sum contains every $(v,0)$ and every $(0, H_qv) = (v, H_qv) - (v,0)$, so it is $T_qM \oplus H_q(T_qM)$; it is everything if and only if H_q is surjective, that is (as $\dim T_qM = \dim T_q^*M$), an isomorphism, which means that $\operatorname{Hess} f_q$ is nondegenerate. □

Remark 41.4.7. A critical point q with $\operatorname{Hess} f_q$ nondegenerate is called *nondegenerate*, and a function all of whose critical points are nondegenerate is a *Morse function*. Two submanifolds whose tangent spaces at a common point span the whole tangent space meet *transversally* there. So (3) says: the critical points of f are the points where the graph of df meets the zero section, and the nondegenerate ones are those where it meets the zero section transversally. Both are Lagrangian submanifolds of dimension n in the $2n$-dimensional manifold T^*M, so a transverse intersection point is isolated (Exercise 41.8.6). This picture, critical points of functions as intersections of Lagrangian submanifolds, is the starting point of Morse theory and of its symplectic generalisations, which we do not develop.

Example 41.4.8. 1. On $\mathbb{R}^2$ with coordinates $(x,y) = (q^1, q^2)$, let $f = \frac{1}{2}(x^2 - y^2)$. Then $df = x\,dx - y\,dy$, and the map $df\colon \mathbb{R}^2 \to T^*\mathbb{R}^2 = \mathbb{R}^4$ is $(x,y) \mapsto (x,y,x,-y)$. Its differential has matrix $\begin{pmatrix} I \\ A \end{pmatrix}$ with $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, the Hessian matrix of f, which is symmetric. The graph is the plane $\{p_1 = q^1,\ p_2 = -q^2\}$, on which $\omega = dq^1 \wedge dp_1 + dq^2 \wedge dp_2$ restricts to $dq^1 \wedge dq^1 - dq^2 \wedge dq^2 = 0$: it is Lagrangian. It meets the zero section $\{p = 0\}$ only at the origin, the critical point of f, and transversally, as A is invertible: the origin is a nondegenerate critical point (a saddle).

2. On $\mathbb{R}$, let $f = x^3$. Then $df = 3x^2\,dx$ and the graph of df is the parabola $\{p = 3q^2\}$ in the plane $T^*\mathbb{R} = \mathbb{R}^2$. It touches the zero section $\{p = 0\}$ at the origin, tangentially: the

critical point 0 is degenerate, as $f''(0) = 0$. (On a 1-dimensional manifold every curve in T^*M is Lagrangian, so here the Lagrangian condition says nothing.)

3. On $\mathbb{R}^2$, let $\alpha = x\,dy$, so $a_1 = 0$ and $a_2 = x$. The Jacobian matrix is $A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, which is not symmetric: $\alpha_*(\partial_x) = \partial_{q^1} + \partial_{p_2}$ and $\alpha_*(\partial_y) = \partial_{q^2}$, and
$$\omega\big(\alpha_*\partial_x, \alpha_*\partial_y\big) = (dq^2 \wedge dp_2)(\partial_{p_2}, \partial_{q^2}) = -1 = -d\alpha(\partial_x, \partial_y),$$
as $d\alpha = dx \wedge dy$. So the graph of α is not Lagrangian, and α is not the differential of a function. Splitting A into its symmetric and antisymmetric parts corresponds to writing $\alpha = \frac{1}{2}\,d(xy) + \frac{1}{2}(x\,dy - y\,dx)$: the exact part $\frac{1}{2}\,d(xy)$ has the symmetric Jacobian $\frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, the Hessian of $\frac{1}{2}xy$, and the part $\frac{1}{2}(x\,dy - y\,dx)$ has the antisymmetric Jacobian $\frac{1}{2}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, which carries all of $d\alpha$.

41.5 Hamiltonian vector fields

Definition 41.5.1. Let (M, ω) be symplectic. The *Hamiltonian vector field* of a function H is the unique vector field X_H with $\iota_{X_H}\omega = dH$. The *Poisson bracket* of two functions is $\{F, G\} = \omega(X_F, X_G) = dF(X_G) = X_G F$.

On $T^*\mathbb{R}^n$ with $\omega = \sum dq^i \wedge dp_i$, $X_H = \sum_i \big(\frac{\partial H}{\partial p_i}\partial_{q^i} - \frac{\partial H}{\partial q^i}\partial_{p_i}\big)$, since then $\iota_{X_H}\omega = \sum_i(\partial_{p_i}H\,dp_i + \partial_{q^i}H\,dq^i) = dH$. So the integral curves of X_H solve *Hamilton's equations*
$$\dot{q}^i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q^i},$$
and $\{F, G\} = \sum_i(\partial_{q^i}F\,\partial_{p_i}G - \partial_{p_i}F\,\partial_{q^i}G)$.

Lemma 41.5.2. *For vector fields X, Y on a manifold, $\iota_{[X,Y]} = L_X\iota_Y - \iota_Y L_X$ on differential forms.*

Proof. Both sides are local antiderivations of degree -1: the right side is the commutator of the derivation L_X of degree 0 (Proposition 36.2.13) with the antiderivation ι_Y, and the antiderivation rules combine as in Step 2 of that proof. Both vanish on functions, and on an exact 1-form dg the left side is $[X, Y]g$ and the right side is $L_X(Yg) - \iota_Y d L_X g = XYg - YXg$. As in Step 3 of the proof of Proposition 36.2.13, they agree everywhere. □

Proposition 41.5.3. *Let H, F, G be functions on a symplectic manifold.*

1. *H is constant along the integral curves of X_H (conservation of energy), and the flow of X_H preserves ω and the Liouville volume (Liouville's theorem).*
2. *$\{F, G\} = -\{G, F\}$, the bracket satisfies the Jacobi identity and the Leibniz rule $\{F, GK\} = \{F, G\}K + G\{F, K\}$, and $[X_F, X_G] = -X_{\{F,G\}}$.*
3. *F is constant along the flow of X_G if and only if $\{F, G\} = 0$, if and only if G is constant along the flow of X_F.*

Proof. (1) $X_H H = dH(X_H) = \omega(X_H, X_H) = 0$. By Cartan's formula, $L_{X_H}\omega = d\iota_{X_H}\omega + \iota_{X_H}d\omega = d\,dH = 0$, so the flow preserves ω (Proposition 36.2.13) and hence ω^n.

(2) Antisymmetry is that of ω, and the Leibniz rule holds since X_G is linear in G and $\{F, G\} = -X_F G$ is a derivation in G. By Lemma 41.5.2, $\iota_{[X_F,X_G]}\omega = L_{X_F}\iota_{X_G}\omega -$

$\iota_{X_G} L_{X_F}\omega = L_{X_F} dG = d(X_F G) = -d\{F,G\}$, so $[X_F, X_G] = -X_{\{F,G\}}$. For the Jacobi identity, use $\{A,B\} = X_B A$: $\{\{F,G\},K\} = X_K X_G F$, $\{\{K,F\},G\} = X_G\{K,F\} = -X_G X_K F$, and $\{\{G,K\},F\} = -\{F,\{G,K\}\} = -X_{\{G,K\}}F = [X_G, X_K]F$. The sum is $[X_K, X_G]F + [X_G, X_K]F = 0$.

(3) $\frac{d}{dt}F(\varphi_t(p)) = (X_G F)(\varphi_t(p)) = \{F,G\}(\varphi_t(p))$, and antisymmetry. □

Example 41.5.4.

1. *The harmonic oscillator* $H = \frac{1}{2}(q^2 + p^2)$ on $\mathbb{R}^2$: Hamilton's equations $\dot q = p$, $\dot p = -q$ give the rotations $(q,p) \mapsto (q\cos t + p\sin t, -q\sin t + p\cos t)$, which preserve H (circles) and area.

2. *The pendulum* $H = \frac{1}{2}p^2 - \cos q$ on T^*S^1: $\dot q = p$, $\ddot q = \dot p = -\sin q$. The level sets of H are the phase curves: closed curves around $(0,0)$ for $H < 1$ (oscillations), curves going around the cylinder for $H > 1$ (rotations), and the separatrix $H = 1$ through the unstable equilibrium $(\pi, 0)$.

3. *Geodesic flow.* On T^*M for a Riemannian manifold, $H(\xi) = \frac{1}{2}|\xi|^2$ (the metric on covectors, Remark 40.2.4): in coordinates $H = \frac{1}{2}\sum g^{ij}p_i p_j$, and Hamilton's equations are $\dot q^k = \sum_i g^{ki}p_i$, $\dot p_k = -\frac{1}{2}\sum_{i,j}\partial_k g^{ij} p_i p_j$. Eliminating p by $p_i = \sum_j g_{ij}\dot q^j$ and using $\partial_k g^{ij} = -\sum_{a,b} g^{ia}\partial_k g_{ab} g^{bj}$, they become $\frac{d}{dt}\sum_j g_{kj}\dot q^j = \frac{1}{2}\sum_{a,b}\partial_k g_{ab}\dot q^a \dot q^b$, which, expanding the left side and multiplying by (g^{lk}), is the geodesic equation $\ddot q^l + \sum \Gamma^l_{ab}\dot q^a \dot q^b = 0$ (Theorem 40.2.8). So geodesics are the projections of the integral curves of a Hamiltonian vector field, and the geodesic flow preserves the Liouville measure of T^*M.

Example 41.5.5. *Area on the sphere and Archimedes' theorem.* On $S^2 \subseteq \mathbb{R}^3$ use the cylindrical coordinates $(\theta, z) \in (0, 2\pi) \times (-1, 1)$, $x = (r\cos\theta, r\sin\theta, z)$ with $r = \sqrt{1 - z^2}$. Then $x_\theta = (-r\sin\theta, r\cos\theta, 0)$ and $x_z = (-\frac{z}{r}\cos\theta, -\frac{z}{r}\sin\theta, 1)$ are orthogonal, with lengths r and $1/r$, and $x_\theta \times x_z = x$ is the outer normal. So the area form of the outward orientation is

$$dA = |x_\theta \times x_z|\, d\theta \wedge dz = d\theta \wedge dz :$$

the projection from the sphere to the cylinder $\{r = 1\}$ along horizontal rays from the axis preserves area. This is *Archimedes' theorem*; in particular the zone $\{a < z < b\}$ has area $2\pi(b-a)$, independently of where it is, and the whole sphere has area 4π. For the symplectic form dA, $\iota_{\partial_\theta}(d\theta \wedge dz) = dz$, so the Hamiltonian vector field of the height function z is ∂_θ, the rotation about the vertical axis: the height is the conserved quantity of the rotation, its *moment map*.

41.6 Darboux's theorem and Moser's method

Let X_t, $t \in [0,1]$, be a time-dependent vector field, smooth in (t,x), and φ_t its flow, $\frac{d}{dt}\varphi_t(x) = X_t(\varphi_t(x))$ with $\varphi_0 = \mathrm{id}$, defined where the solutions exist (Theorem 27.13.3). For a smooth family of forms ω_t,

$$\frac{d}{dt}\varphi_t^*\omega_t = \varphi_t^*\Big(L_{X_t}\omega_t + \frac{d\omega_t}{dt}\Big); \tag{41.1}$$

this is the time-dependent version of the formula $\frac{d}{dt}\Phi_t^*\omega = \Phi_t^* L_X\omega$ of Proposition 36.2.13, with the same proof [Lee13, Chapter 22]. *Moser's method* constructs a diffeomorphism φ_1 with $\varphi_1^*\omega_1 = \omega_0$ by choosing X_t so that the right side vanishes.

Theorem 41.6.1 (Darboux)**.** *Every point of a symplectic manifold (M,ω) of dimension $2n$ has a chart with coordinates $(x^1,\dots,x^n,y^1,\dots,y^n)$ in which $\omega = \sum_i dx^i \wedge dy^i$.*

Sketch of proof. Choose a chart centred at the given point; after a linear change of coordinates (Theorem 11.9.6) the form $\omega_1 = \omega$ agrees at 0 with the constant form $\omega_0 = \sum dx^i \wedge dy^i$. Let $\omega_t = \omega_0 + t(\omega_1 - \omega_0)$. At 0 all ω_t equal ω_0, so by compactness of $[0,1]$ and continuity of the determinant they are nondegenerate on a ball B around 0 for all $t \in [0,1]$. The closed form $\omega_1 - \omega_0$ is $d\beta$ on B (Theorem 37.2.4), and subtracting a constant 1-form we may assume $\beta(0) = 0$. By nondegeneracy there is a unique X_t with $\iota_{X_t}\omega_t = -\beta$; it is smooth in (t,x) and vanishes at 0, so its flow fixes 0 and, by continuous dependence on initial conditions, is defined for $t \in [0,1]$ on a smaller ball. By (41.1) and Cartan's formula,

$$\frac{d}{dt}\varphi_t^*\omega_t = \varphi_t^*\big(d\iota_{X_t}\omega_t + \iota_{X_t}d\omega_t + \omega_1 - \omega_0\big) = \varphi_t^*(-d\beta + 0 + d\beta) = 0.$$

So $\varphi_1^*\omega_1 = \omega_0$, and composing the chart with φ_1 gives the required coordinates. The omitted verification is formula (41.1). □

So symplectic manifolds have no local invariants: there is no symplectic analogue of curvature. Compare with Riemannian geometry, where a metric is locally Euclidean only if it is flat (Theorem 38.5.10 applied to the Levi-Civita connection).

Theorem 41.6.2 (Moser)**.** *Let Σ be a compact connected oriented surface and ω_0, ω_1 area forms (positive for the orientation) with $\int_\Sigma \omega_0 = \int_\Sigma \omega_1$. Then there is a diffeomorphism φ of Σ with $\varphi^*\omega_1 = \omega_0$. In particular two compact connected orientable surfaces with area forms are symplectomorphic if and only if they are diffeomorphic and have the same area.*

Sketch of proof. The forms $\omega_t = (1-t)\omega_0 + t\omega_1$ are positive, as convex combinations of positive top forms, hence area forms. Since $\int_\Sigma(\omega_1 - \omega_0) = 0$, the form $\omega_1 - \omega_0$ is exact (Corollary 37.3.2), say $d\beta$. Define X_t by $\iota_{X_t}\omega_t = -\beta$; as Σ is compact its flow is defined for all $t \in [0,1]$, and the computation in the proof of Theorem 41.6.1 gives $\varphi_1^*\omega_1 = \omega_0$. The same argument, with "area form" replaced by "volume form", applies in every dimension; it again uses formula (41.1). □

41.7 Lagrangian submanifolds

Lagrangian submanifolds are the most important submanifolds of symplectic manifolds: graphs of closed 1-forms (Corollary 41.4.3), and, as the next example shows, graphs of symplectomorphisms.

Example 41.7.1.

1. Every curve in a surface is Lagrangian: a line in a 2-dimensional symplectic vector space is Lagrangian.

2. *Graphs of symplectomorphisms.* For a diffeomorphism φ of (M,ω), the graph $\Gamma_\varphi = \{(x,\varphi(x))\} \subseteq M \times M$ is Lagrangian for the form $\mathrm{pr}_2^*\omega - \mathrm{pr}_1^*\omega$ if and only if φ is a symplectomorphism: the parametrisation $x \mapsto (x,\varphi(x))$ pulls the form back to $\varphi^*\omega - \omega$.

3. The torus $\{|z_1| = \cdots = |z_n| = 1\} \subseteq (\mathbb{C}^n, \omega_0)$ is Lagrangian. Its tangent space at z is spanned by the vectors iz_je_j, with (e_j) the standard basis, and since $\omega_0(u,v) = \mathrm{Re}\langle u, -iv\rangle$ (Example 41.2.3(3)), $\omega_0(iz_je_j, iz_ke_k) = \mathrm{Re}\langle iz_je_j, z_ke_k\rangle = \delta_{jk}\,\mathrm{Re}(i|z_j|^2) = 0$.

Theorem 41.7.2 (Weinstein's Lagrangian neighbourhood theorem)**.** *Let L be a compact Lagrangian submanifold of a symplectic manifold (M,ω). Then a neighbourhood of L in M is symplectomorphic to a neighbourhood of the zero section in $(T^*L, -d\lambda)$, by a symplectomorphism that is the identity on L (identified with the zero section).*

This is [CdS01] (Lagrangian neighbourhood theorem) or [MS17]. The proof chooses a compatible almost complex structure, uses J to identify the normal bundle of L with TL, and the metric with T^*L (Proposition 41.2.2(4),(5), pointwise), so that a tubular neighbourhood gives a diffeomorphism which is symplectic along L; Moser's method, relative to L, then corrects it to a symplectomorphism. Consequences:

1. *Nearby Lagrangians.* A Lagrangian submanifold C^1-close to L corresponds, under the theorem, to a submanifold of T^*L close to the zero section and transverse to the fibres, hence to the graph of a 1-form, which is closed by Corollary 41.4.3. So the Lagrangians near L are parametrised by the closed 1-forms on L near 0. The graphs of exact forms df are the images of the zero section under the time-one maps of the Hamiltonians $-f \circ \pi$ (Exercise 41.8.4, in a chart).

2. *Symplectomorphisms near the identity.* Applied to the diagonal of $M \times M$, the graph of the identity, which is Lagrangian for $\mathrm{pr}_2^*\omega - \mathrm{pr}_1^*\omega$ by Example 41.7.1(2), the theorem shows that symplectomorphisms C^1-close to the identity correspond to closed 1-forms on M near 0; in particular such a symplectomorphism, corresponding to α, is joined to the identity by the path of symplectomorphisms corresponding to $t\alpha$, $0 \le t \le 1$.

41.8 Exercises

Exercise 41.8.1. For which real numbers a is $\sigma = dx^1 \wedge dy^1 + a\,dx^1 \wedge dx^2 + dx^2 \wedge dy^2$ a symplectic form on $\mathbb{R}^4$? Find a Lagrangian plane for it.

Solution. σ is closed (constant coefficients). $\sigma^2 = 2\,dx^1 \wedge dy^1 \wedge dx^2 \wedge dy^2$, since $dx^1 \wedge dx^2$ wedges to zero with the other two terms and with itself; so σ is symplectic for every a (Proposition 41.2.2(1)). The plane $\{x^1 = x^2 = 0\}$ is Lagrangian, as every term contains dx^1 or dx^2. □

Exercise 41.8.2. Using Example 41.5.5, compute the area of the spherical cap $\{z > \cos\vartheta_0\}$ and compare with the holonomy of the sphere computed in Example 38.5.6(2).

Solution. The cap is $\{\cos\vartheta_0 < z < 1\}$, of area $2\pi(1 - \cos\vartheta_0)$; this is the angle of the rotation produced by parallel transport around its boundary circle, modulo 2π, in accordance with the Gauss–Bonnet theorem for the curvature 1. □

Exercise 41.8.3. Show that S^4, $S^2 \times S^4$ and $S^1 \times S^3$ admit no symplectic form, and that T^4, $S^2 \times S^2$ and $S^2 \times T^2$ do.

Solution. H^2_{dR} vanishes for S^4 and for $S^1 \times S^3$ (Künneth, Theorem 37.3.8), and for $S^2 \times S^4$ the square of every class in H^2 vanishes, as in Example 41.3.3(3); Proposition 41.3.2 applies. Products of symplectic manifolds (tori, area forms on S^2) are symplectic. □

Exercise 41.8.4. Let $f : \mathbb{R}^n \to \mathbb{R}$ be smooth. Show that $\Phi(q,p) = (q, p + df(q))$ is a symplectomorphism of $T^*\mathbb{R}^n$, the time-one map of the Hamiltonian $H(q,p) = -f(q)$, which maps the zero section onto the Lagrangian graph of df.

Solution. $\Phi^* \sum dq^i \wedge dp_i = \sum dq^i \wedge (dp_i + \sum_j \partial_i\partial_j f\, dq^j) = \sum dq^i \wedge dp_i$, since $\sum_{i,j} \partial_i\partial_j f\, dq^i \wedge dq^j = 0$ by symmetry. For $H = -f(q)$, Hamilton's equations are $\dot q = 0$, $\dot p = \nabla f(q)$, with time-one map Φ. □

Exercise 41.8.5. On $T^*\mathbb{R}^n$ compute $\{q^i, q^j\}$, $\{p_i, p_j\}$ and $\{q^i, p_j\}$, and show that the angular momentum $L = q^1p_2 - q^2p_1$ Poisson-commutes with every rotation-invariant Hamiltonian $H = h(|q|^2, |p|^2, \langle q, p\rangle)$ on $T^*\mathbb{R}^2$.

Solution. With $\{F,G\} = \sum_k (\partial_{q^k} F\, \partial_{p_k} G - \partial_{p_k} F\, \partial_{q^k} G)$: $\{q^i, q^j\} = \{p_i, p_j\} = 0$ and $\{q^i, p_j\} = \delta_{ij}$. The flow of X_L is $\dot q = (-q^2, q^1)$, $\dot p = (-p_2, p_1)$, the simultaneous rotation of q and p, which preserves $|q|^2$, $|p|^2$ and $\langle q, p\rangle$; so H is constant along it, and $\{H, L\} = X_L H = 0$ (Proposition 41.5.3(3)). □

Exercise 41.8.6. Let q_0 be a nondegenerate critical point of a smooth function f on M (Remark 41.4.7). Show that q_0 has a neighbourhood containing no other critical point of f. Deduce that a Morse function on a compact manifold has finitely many critical points.

Solution. In a chart around q_0, the critical points of f are the zeros of $F = (\partial_1 f, \dots, \partial_n f)$, a smooth map from an open subset of $\mathbb{R}^n$ to $\mathbb{R}^n$. Its Jacobian matrix at q_0 is $(\partial_j \partial_i f(q_0))$, the matrix of Hess f_{q_0} (Corollary 41.4.6(3)), which is invertible. By the inverse function theorem (Theorem 33.4.14), F maps a neighbourhood W of q_0 bijectively onto an open set; so F vanishes on W only at q_0. If M is compact and f is a Morse function, the set of critical points is closed (the zero set of the continuous map df into T^*M, a preimage of the closed zero section) hence compact, and each of its points has a neighbourhood containing no other; a cover of it by such neighbourhoods has a finite subcover, so it is finite. □

Exercise 41.8.7. Let α be a 1-form on M and ∇ a connection on TM with Christoffel symbols Γ^k_{ij}. Show that in coordinates the vertical part of $\alpha_*(\partial_{q^j})$ (Proposition 41.4.5(3)) is $\sum_k \big(\partial_j a_k - \sum_i \Gamma^i_{jk} a_i\big)\, dq^k$. Deduce that at a point where α vanishes, the vertical part of α_* does not depend on the connection, and that elsewhere it does: for $\alpha = dq^1$ on $\mathbb{R}^2$, find a torsion-free connection for which $\nabla_{\partial_{q^1}} \alpha \neq 0$, although the Jacobian matrix of α vanishes.

Solution. By Proposition 38.3.6(1), $(\nabla_{\partial_j}\alpha)(\partial_k) = \partial_j(a_k) - \alpha(\nabla_{\partial_j}\partial_k) = \partial_j a_k - \sum_i \Gamma^i_{jk} a_i$, which is the formula. Where all a_i vanish it reduces to $\partial_j a_k$, which does not involve Γ. For $\alpha = dq^1$, so $a_1 = 1$ and $a_2 = 0$, take the connection with $\Gamma^1_{11} = 1$ and all other symbols zero; it is torsion-free, since $\Gamma^k_{ij} = \Gamma^k_{ji}$. Then $(\nabla_{\partial_1}\alpha)(\partial_1) = 0 - \Gamma^1_{11} = -1$, while $\partial_j a_k = 0$ for all j, k. □

Chapter 42

Elliptic Operators on Compact Manifolds

42.1 Introduction

This chapter contains the analysis behind the Hodge theorem. The question it answers is the following: given a linear differential operator P on a compact manifold, for instance the Laplacian on forms, when is the equation $Pu = f$ solvable, how large is its space of solutions, and how regular are they? For a general operator nothing good can be said; for *elliptic* operators the answer is as good as in finite dimensions. An elliptic operator on a compact manifold is *Fredholm* between Sobolev spaces (Theorem 42.5.2): its kernel is finite-dimensional and consists of smooth sections, and $Pu = f$ is solvable exactly when f is orthogonal to the finite-dimensional kernel of the adjoint. A formally self-adjoint elliptic operator moreover has a discrete spectrum with smooth eigensections (Theorem 42.5.7). Applied to the Hodge Laplacian, these statements give the decomposition of the space of forms and the existence of harmonic representatives (Chapter 43).

The model case is the torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$ with an operator with constant coefficients, treated in Chapter 32: there the Fourier transform turns P into the multiplication of the Fourier coefficients $\hat{u}(k)$ by a polynomial $p(k)$, ellipticity says that $|p(k)|$ grows like $|k|^m$, and every statement of this chapter can be checked coefficient by coefficient. On a general compact manifold the argument has three steps:

1. the *principal symbol*, the highest-order part of P with derivatives replaced by covectors, makes sense invariantly, and ellipticity means that it is invertible (Section 42.2);

2. the *elliptic estimate* $\|u\|_{s+m} \leq C(\|Pu\|_s + \|u\|_s)$ between Sobolev norms, obtained by comparing P near each point with the constant-coefficient operator given by its symbol at that point (Section 42.4);

3. functional analysis: the estimate, together with the compactness of the inclusions between Sobolev spaces, gives regularity, finite-dimensionality, closed range and the spectral theorem (Section 42.5).

References: [War83, Chapter 6], [Wel08, Chapter 4], [Hör90a].
Prerequisites. Chapter 40 (Riemannian Geometry) and Chapter 32 (Distributions and Sobolev Spaces).
On a first reading. One can read the definitions and examples of Section 42.2, take the elliptic estimate (Theorem 42.4.1) on faith, and go on to its consequences in Section 42.5, which are what Chapter 43 uses.

Throughout, M is a compact smooth manifold, and E, F are complex vector bundles on M with Hermitian metrics; M carries a fixed Riemannian metric and, when needed, an orientation, so that sections can be integrated.

42.2 Differential operators and symbols

Definition 42.2.1. A linear map $P\colon \Gamma(M,E) \to \Gamma(M,F)$ is a *differential operator of order at most* m if in local coordinates and local frames it is given by $Pu = \sum_{|\alpha|\le m} A_\alpha(x)\,\partial^\alpha u$ with smooth matrix-valued coefficients A_α. Its *principal symbol* is

$$\sigma_m(P)(x,\xi) = \sum_{|\alpha|=m} A_\alpha(x)\,(i\xi)^\alpha \in \operatorname{Hom}(E_x,F_x), \qquad \xi \in T^*_xM,$$

and P is *elliptic* if $\sigma_m(P)(x,\xi)$ is invertible for all x and all $\xi \neq 0$.

Here $\xi = \sum_j \xi_j\,dx^j$ and $(i\xi)^\alpha = i^{|\alpha|}\xi_1^{\alpha_1}\cdots\xi_n^{\alpha_n}$. The rule is: *keep only the derivatives of highest order, and replace* ∂_j *by* $i\xi_j$. The factor i is chosen so that $\partial_j(e^{ix\cdot\xi}) = i\xi_j e^{ix\cdot\xi}$: on an exponential $e^{i\langle x,\xi\rangle}$ with large frequency ξ, the operator P acts, to leading order, as multiplication by the symbol. In Chapter 32 the Fourier transform is normalised with $e^{-2\pi i x\cdot\xi}$, and correspondingly the symbol there is $\sum_{|\alpha|=m} a_\alpha(2\pi i\xi)^\alpha$; the two conventions differ by the harmless factor $(2\pi)^m$. Ellipticity means that P *controls all derivatives of order* m *in all directions*; the lower-order terms play no role in it.

Example 42.2.2.

1. On the circle, $P = d/d\theta$ has $\sigma_1(P)(\theta,\xi) = i\xi$, which is nonzero for $\xi \neq 0$: P is elliptic. So is $d/d\theta + a(\theta)$ for any smooth a, with the same symbol.

2. On a Riemannian manifold, the Laplacian on functions is $\Delta f = -\frac{1}{\sqrt{\det g}}\sum_{i,j}\partial_i(\sqrt{\det g}\,g^{ij}\partial_j f)$ (Proposition 40.15.3(5)). Its terms of order 2 are $-\sum_{i,j} g^{ij}\partial_i\partial_j f$, so

$$\sigma_2(\Delta)(x,\xi) = -\sum_{i,j} g^{ij}(i\xi_i)(i\xi_j) = \sum_{i,j} g^{ij}\xi_i\xi_j = |\xi|^2,$$

   the squared length of the covector ξ for the metric. It is nonzero for $\xi \neq 0$, so Δ is elliptic. The first-order terms, which involve the derivatives of the metric, do not enter.

3. On $\mathbb{C}$ with $z = x + iy$, the Cauchy–Riemann operator $\partial_{\bar z} = \frac{1}{2}(\partial_x + i\partial_y)$ has symbol $\frac{i}{2}(\xi_1 + i\xi_2)$, nonzero for $(\xi_1,\xi_2) \neq 0$; it is elliptic. Its solutions are the holomorphic functions, and the local form of elliptic regularity (Theorem 42.4.1) is, in this case, the fact that a weak solution of the Cauchy–Riemann equations is smooth, indeed holomorphic.

4. On $\mathbb{R}^2$ the operator ∂_x is not elliptic: its symbol $i\xi_1$ vanishes for $\xi = dy$. It sees no derivatives in the y direction, and correspondingly $\partial_x u = 0$ has the non-smooth solutions $u(x,y) = h(y)$ for every function h.

5. The exterior derivative $d\colon \Omega^k \to \Omega^{k+1}$ has symbol $\sigma_1(d)(x,\xi)\alpha = i\xi\wedge\alpha$ (Exercise 42.6.1); it maps between bundles whose ranks $\binom{n}{k}$ and $\binom{n}{k+1}$ differ in general, and $\xi\wedge -$ is never injective for $k \ge 1$, since it kills $\xi\wedge\beta$; for $k = 0$ and $n \ge 2$ it is not surjective. So d is not elliptic. Ellipticity is recovered by combining d with its adjoint, as in the next example.

The rule "replace ∂_j by $i\xi_j$" uses coordinates, and the next proposition shows that the result does not depend on them. The idea is to test P on functions that oscillate in the direction of ξ, or, algebraically, on powers of a function vanishing at x with differential ξ.

Proposition 42.2.3. *The principal symbol is well defined: it does not depend on the coordinates and frames, and it is characterised by*

$$\sigma_m(P)(x, d_x f)\, u(x) = \frac{i^m}{m!}\, P\big((f - f(x))^m u\big)(x)$$

for $f \in C^\infty(M)$ and $u \in \Gamma(M, E)$. Symbols are multiplicative, $\sigma_{m+m'}(P'P) = \sigma_{m'}(P')\sigma_m(P)$, so a composite of elliptic operators is elliptic, and the symbol of the formal adjoint is the adjoint of the symbol.

Proof. Expanding $P((f - f(x))^m u)$ by the Leibniz rule, all terms in which fewer than m derivatives fall on $(f - f(x))^m$ vanish at x, and the remaining term is $\sum_{|\alpha|=m} A_\alpha(x)\, \partial^\alpha\big((f - f(x))^m\big)(x) u(x) = m! \sum_{|\alpha|=m} A_\alpha(x)(d_x f)^\alpha u(x)$, which gives the formula. The right-hand side is defined without reference to coordinates, so the symbol is invariantly defined, and multiplicativity follows from the same formula applied twice. □

For instance, for $P = \Delta$ on functions and f with $d_x f = \xi$, the formula reads $|\xi|^2 u(x) = -\frac{1}{2}\Delta((f - f(x))^2 u)(x)$; on $\mathbb{R}^n$ with $f(y) = \langle \xi, y\rangle$ and $u = 1$, the right side is $-\frac{1}{2}\Delta(\langle \xi, y - x\rangle^2)$ at $y = x$, that is $\frac{1}{2} \cdot 2|\xi|^2$.

Example 42.2.4.

1. The Hodge Laplacian $\Delta = d\delta + \delta d$ on $\Omega^k(M)$ has principal symbol $|\xi|^2 \,\mathrm{id}$ (compute in normal coordinates, where the first-order terms of the metric vanish); it is elliptic.
2. The operator $d + \delta \colon \Omega^{\text{even}} \to \Omega^{\text{odd}}$ has symbol $i(\xi \wedge - \iota_\xi)$, whose square is $-|\xi|^2$, so it is invertible for $\xi \neq 0$ and $d + \delta$ is elliptic. Its index is the Euler characteristic (Chapter 43).
3. On a complex manifold, $\bar\partial + \bar\partial^*$ is elliptic (Chapter 46); the Cauchy–Riemann operator $\bar\partial$ itself is elliptic in one variable.
4. Not every operator is elliptic: the wave operator $\partial_t^2 - \Delta$ and the heat operator $\partial_t - \Delta$ are not.

In (4), the heat operator $\partial_t - \sum_j \partial_{x_j}^2$ on $\mathbb{R} \times \mathbb{R}^n$ has principal symbol (of order 2) $\sum_j \xi_j^2$, which vanishes for the covector dt: the operator contains only one derivative in time. The wave operator has symbol $-\tau^2 + |\xi|^2$, which vanishes on the light cone $|\tau| = |\xi|$; Counterexample 42.4.2 shows the consequence for regularity.

The simplest elliptic operator with constant coefficients, the Laplacian of $\mathbb{R}^3$, can be inverted by an explicit kernel. This is the prototype of the parametrices used below.

Proposition 42.2.5 (Fundamental solution of the Laplacian)**.** *On $\mathbb{R}^3$, the locally integrable function $E(x) = -\frac{1}{4\pi|x|}$ satisfies $\sum_i \partial_i^2 E = \delta_0$ in $\mathcal{D}'(\mathbb{R}^3)$. Consequently, for $f \in \mathcal{D}(\mathbb{R}^3)$ the convolution $u = E * f$ solves $\sum_i \partial_i^2 u = f$.*

Proof. E is locally integrable, as $\int_{|x|\le 1} |x|^{-1}\, dx = 4\pi \int_0^1 r\, dr$ in spherical coordinates (Theorem 28.11.2). Away from 0, a direct computation gives $\Delta E = 0$, where $\Delta = \sum_i \partial_i^2$ here. Let $\varphi \in \mathcal{D}(\mathbb{R}^3)$ and $A_\varepsilon = \{\varepsilon \le |x| \le R\}$ with R large. By the divergence theorem (Example 36.4.11) applied to $E\nabla\varphi - \varphi\nabla E$ on A_ε (Green's identity), and since φ vanishes near $|x| = R$,

$$\int_{A_\varepsilon} E\, \Delta\varphi = \int_{A_\varepsilon} (E\, \Delta\varphi - \varphi\, \Delta E) = -\int_{|x|=\varepsilon} \big(E\, \partial_r\varphi - \varphi\, \partial_r E\big)\, dA,$$

where ∂_r is the radial derivative, the sign coming from the inner boundary sphere having outward normal $-x/|x|$. On $|x| = \varepsilon$, $E = -1/(4\pi\varepsilon)$ and $\partial_r E = 1/(4\pi\varepsilon^2)$; the area is $4\pi\varepsilon^2$.

So the first term is at most $\varepsilon \sup|\nabla\varphi| \to 0$, and the second is $\frac{1}{4\pi\varepsilon^2}\int_{|x|=\varepsilon}\varphi\,dA \to \varphi(0)$ by continuity. Letting $\varepsilon \to 0$ (dominated convergence on the left), $(\Delta E)(\varphi) = \int E\,\Delta\varphi = \varphi(0)$. For the last statement, $\Delta(E * f) = E * \Delta f$ by differentiation under the integral sign, and $(E * \Delta f)(x) = \int E(y)\,\Delta_y f(x-y)\,dy = f(x)$ by the first part applied to $\varphi(y) = f(x-y)$. □

The formula $u = E * f$ shows the gain of regularity concretely: convolution with E, which behaves like $|x|^{-1}$ near 0, smooths by two derivatives. On a compact connected oriented manifold the Laplacian has no fundamental solution at all: $\Delta E = \delta_p$ would give $1 = \delta_p(1) = (\Delta E)(1) = E(\Delta 1) = 0$. Correspondingly $\Delta u = f$ is solvable only when $\int_M f\,\mathrm{vol} = 0$ (Exercise 42.6.4), and the role of the fundamental solution is played by the inverse of Δ on the orthogonal complement of the constants, the Green operator of Chapter 43.

42.3 Sobolev spaces of sections

To measure regularity on a manifold we glue the Sobolev spaces of $\mathbb{R}^n$ (Definition 32.4.1) with a partition of unity. For integer $s = m \geq 0$, a section is in H^m if its derivatives up to order m in charts are square integrable; negative and non-integer s are needed because elliptic operators shift the order by m in both directions, and because the dual of H^s is H^{-s}.

Definition 42.3.1. Fix a finite atlas of M with trivialisations of E and a subordinate partition of unity (ψ_i). For $s \in \mathbb{R}$ and $u \in \Gamma(M,E)$ set $\|u\|_s^2 = \sum_i \|\psi_i u\|^2_{H^s(\mathbb{R}^n)}$, using the coordinates and frames to view $\psi_i u$ as a compactly supported $\mathbb{C}^r$-valued function. The completion is the Sobolev space $H^s(E)$ of sections.

Proposition 42.3.2. *The space $H^s(E)$ and the equivalence class of its norm do not depend on the choices. Moreover $H^0(E) = L^2(E)$ with the inner product $\int_M\langle u,v\rangle\,\mathrm{vol}$; the inclusions $H^s(E) \to H^t(E)$ for $s > t$ are compact; $\bigcap_s H^s(E) = \Gamma(M,E)$; and a differential operator of order m extends to a bounded map $H^{s+m}(E) \to H^s(F)$ for every s.*

Proof. Changing charts and frames changes $\psi_i u$ by composition with a diffeomorphism and multiplication by a smooth matrix, which are bounded operations on H^s of compactly supported functions [War83, Chapter 6]; hence the norms are equivalent. For $s = 0$ the statement is Plancherel's theorem plus the fact that the two ways of integrating differ by a bounded density. Compactness follows from the Rellich theorem (Theorem 32.4.5, in the form of Counterexample 32.4.6 for compactly supported distributions) applied to each $\psi_i u$. The identification of $\bigcap_s H^s$ with smooth sections is the Sobolev embedding theorem (Theorem 32.4.3), and boundedness of differential operators follows because differentiation lowers s by the order and multiplication by smooth functions is bounded. □

Example 42.3.3. On the circle $T^1 = \mathbb{R}/\mathbb{Z}$ the Sobolev norm is $\|u\|_s^2 = \sum_k \langle k\rangle^{2s}|\hat u(k)|^2$ with $\langle k\rangle = (1+k^2)^{1/2}$ (Definition 32.4.1), and the membership of a function in H^s is read off from the decay of its Fourier coefficients.

1. The Dirac distribution δ_0 has $\hat\delta_0(k) = 1$ for all k, so $\delta_0 \in H^s$ exactly for $s < -\frac12$; on T^n, exactly for $s < -\frac{n}{2}$, since $\sum_k \langle k\rangle^{2s}$ converges if and only if $2s < -n$.

2. The indicator function u of $[0,\frac12)$ has $\hat u(0) = \frac12$, $\hat u(k) = \frac{1-(-1)^k}{2\pi i k}$ for $k \neq 0$, so $|\hat u(k)|^2 = \frac{1}{\pi^2 k^2}$ for odd k and 0 for even $k \neq 0$. Hence $u \in H^s$ exactly for $s < \frac12$. This is sharp in the Sobolev embedding theorem: in dimension 1, H^s consists of continuous functions for $s > \frac12$, and a function with a jump is in every H^s with $s < \frac12$ but in none with $s > \frac12$.

42.4 Elliptic estimates and regularity

The heart of the theory is an inequality: an elliptic operator controls all derivatives of order m, up to a lower-order error. On the torus with constant coefficients this is a computation with Fourier coefficients (Theorem 32.5.3). For example, for $P = \Delta + 1 = 1 - \sum_j \partial_j^2$ on $T^n = \mathbb{R}^n/\mathbb{Z}^n$, $\widehat{Pu}(k) = (1 + 4\pi^2|k|^2)\hat{u}(k)$, and $1 + |k|^2 \le 1 + 4\pi^2|k|^2 \le 4\pi^2(1 + |k|^2)$, so

$$\|u\|_{s+2} \le \|Pu\|_s \le 4\pi^2\|u\|_{s+2}\,:$$

the operator $\Delta + 1$ is an isomorphism $H^{s+2} \to H^s$ for every s, with inverse $\hat{f}(k) \mapsto \hat{f}(k)/(1 + 4\pi^2|k|^2)$. For $P = \Delta$ the coefficient $k = 0$ is lost, and the estimate needs the extra term $\|u\|_s$; this term is what makes the kernel (here the constants) possible.

Theorem 42.4.1. *Let P be an elliptic differential operator of order m from E to F over the compact manifold M. For every s there is C_s with*

$$\|u\|_{s+m} \le C_s\,(\|Pu\|_s + \|u\|_s) \qquad \text{for all } u \in H^{s+m}(E),$$

*and (*elliptic regularity*) if $u \in H^t(E)$ for some t and $Pu \in H^s(F)$, then $u \in H^{s+m}(E)$; in particular Pu smooth implies u smooth.*

Sketch of proof. Locally one freezes the coefficients: near a point x_0, P differs from the constant-coefficient elliptic operator $P_0 = \sum_{|\alpha|=m} A_\alpha(x_0)\partial^\alpha$ by an operator whose top-order coefficients are small on a small neighbourhood and whose remaining terms have order $< m$. The constant-coefficient estimate (Theorem 32.5.3, transported to $\mathbb{R}^n$ by a cut-off and to the torus) then gives $\|u\|_{s+m} \le C(\|Pu\|_s + \|u\|_{s+m-1})$ for u supported in that neighbourhood, and the error term is absorbed using the interpolation inequality $\|u\|_{s+m-1} \le \varepsilon\|u\|_{s+m} + C_\varepsilon\|u\|_s$, which follows from $\langle\xi\rangle^{s+m-1} \le \varepsilon\langle\xi\rangle^{s+m} + C_\varepsilon\langle\xi\rangle^s$. Summing over a finite cover with a partition of unity gives the global estimate. Regularity is proved by the same estimates applied to difference quotients, or by constructing a *parametrix*: an operator Q with $QP = I + S$ and $PQ = I + S'$ where S, S' are smoothing operators; then $u = Q(Pu) - Su$ is as regular as Pu allows. The omitted verifications are the construction of the parametrix and the coordinate invariance of the estimates; see [War83, Chapter 6], [Wel08, Chapter 4] or [Hör90a, Chapters 17–18]. □

The interpolation inequality used in the sketch deserves a word, as it is the reason why lower-order terms never matter. Let $0 < \varepsilon \le 1$ and $m \ge 1$. Where $\langle\xi\rangle \ge 1/\varepsilon$, $\langle\xi\rangle^{s+m-1} = \langle\xi\rangle^{s+m}/\langle\xi\rangle \le \varepsilon\langle\xi\rangle^{s+m}$; where $\langle\xi\rangle \le 1/\varepsilon$, $\langle\xi\rangle^{s+m-1} = \langle\xi\rangle^{m-1}\langle\xi\rangle^s \le \varepsilon^{1-m}\langle\xi\rangle^s$. So $\langle\xi\rangle^{s+m-1} \le \varepsilon\langle\xi\rangle^{s+m} + \varepsilon^{1-m}\langle\xi\rangle^s$ everywhere, and squaring (with $(a+b)^2 \le 2a^2 + 2b^2$) and integrating against $|\hat{u}|^2$ gives $\|u\|_{s+m-1} \le \sqrt{2}\big(\varepsilon\|u\|_{s+m} + \varepsilon^{1-m}\|u\|_s\big)$: high frequencies are controlled by the strong norm with a small constant, low frequencies by the weak norm with a large one.

Counterexample 42.4.2. Regularity fails for non-elliptic operators. On $\mathbb{R}^2$ with coordinates (t, x) let $P = \partial_t^2 - \partial_x^2$, whose symbol $-\tau^2 + \xi^2$ vanishes on the lines $\tau = \pm\xi$. For every locally integrable g on $\mathbb{R}$, the function $u(t, x) = g(x - t)$ satisfies $Pu = 0$ in $\mathcal{D}'(\mathbb{R}^2)$. Indeed, in the coordinates $s = x - t$, $r = x + t$ one has $\partial_t = \partial_r - \partial_s$, $\partial_x = \partial_r + \partial_s$, so $\partial_t^2 - \partial_x^2 = -4\partial_r\partial_s$ and $dt\,dx = \frac{1}{2}ds\,dr$. For a test function φ, written $\tilde{\varphi}(s, r)$ in the new coordinates,

$$\langle Pu, \varphi\rangle = \int u\,(\partial_t^2 - \partial_x^2)\varphi\,dt\,dx = -2\int g(s)\Big(\int \partial_r(\partial_s\tilde{\varphi})\,dr\Big)ds = 0,$$

since $\partial_s\tilde{\varphi}$ has compact support in r. Taking g the indicator function of $[0, \infty)$ gives a discontinuous solution of $Pu = 0$, although 0 is smooth. For elliptic operators this cannot happen: a distribution u with $\sum_j \partial_j^2 u$ smooth is smooth, by the local form of Theorem 42.4.1.

The discontinuity of the solution travels along the line $x = t$, a direction in which the symbol vanishes: singularities of solutions of $Pu = 0$ can only sit in the directions where $\sigma(P)$ is not invertible. For an elliptic operator there are no such directions, and no singularities.

42.5 Fredholm theory and the spectral decomposition

Definition 42.5.1. The *formal adjoint* P^* of a differential operator P is the differential operator of the same order characterised by $\int_M \langle Pu, v\rangle = \int_M \langle u, P^*v\rangle$ for smooth sections; it exists, is unique, and has symbol $\sigma_m(P^*)(x,\xi) = \sigma_m(P)(x,\xi)^*$ (Proposition 42.2.3), so P^* is elliptic if P is. The operator P is *formally self-adjoint* if $P = P^*$; the Hodge Laplacian is (Proposition 40.15.3).

For instance on the circle, $\int_0^{2\pi} u'\bar v\,d\theta = -\int_0^{2\pi} u\overline{v'}\,d\theta$ for periodic functions, so $(d/d\theta)^* = -d/d\theta$, with symbol $-i\xi = \overline{i\xi}$, the adjoint of the symbol $i\xi$ of $d/d\theta$. This is the reason for the factor i in the definition of the symbol: integration by parts produces a sign $(-1)^m$ for m derivatives, and complex conjugation of i^m produces the same sign.

The following theorem is the infinite-dimensional analogue of the statement that, for a linear map $A\colon V \to W$ between finite-dimensional inner product spaces, $W = \operatorname{im} A \oplus \ker A^*$, so that $Av = w$ is solvable exactly when $w \perp \ker A^*$.

Theorem 42.5.2. *Let P be elliptic of order m on a compact manifold. Then:*

1. $\ker P = \{u \in \Gamma(M,E) : Pu = 0\}$ *is finite-dimensional and consists of smooth sections, and the same holds for* $\ker P^*$*;*
2. $P\colon H^{s+m}(E) \to H^s(F)$ *is Fredholm for every s, with kernel and cokernel independent of s;*
3. *there is an orthogonal decomposition $L^2(F) = \operatorname{im}(P) \oplus \ker P^*$, and $Pu = f$ is solvable with u smooth if and only if f is smooth and orthogonal to* $\ker P^*$.

Proof. (1) Elements of the kernel are smooth by elliptic regularity. On $\ker P$ the elliptic estimate gives $\|u\|_{s+m} \le C_s\|u\|_s$, so the H^s and H^{s+m} norms are equivalent there; since the inclusion $H^{s+m} \to H^s$ is compact (Proposition 42.3.2), the unit ball of $\ker P$ is compact, and $\ker P$ is finite-dimensional (Proposition 30.2.2(3)).

(2) Closed range: let $K \subseteq H^{s+m}$ be the orthogonal complement of $\ker P$. We claim $\|u\|_{s+m} \le C\|Pu\|_s$ on K. If not, there are $u_j \in K$ with $\|u_j\|_{s+m} = 1$ and $Pu_j \to 0$; by compactness of the inclusion, after passing to a subsequence $u_j \to u$ in H^s, and the elliptic estimate gives $\|u_j - u_l\|_{s+m} \le C(\|P(u_j - u_l)\|_s + \|u_j - u_l\|_s) \to 0$, so $u_j \to u$ in H^{s+m} with $\|u\|_{s+m} = 1$, $Pu = 0$ and $u \in K$, a contradiction. Hence P has closed range, and with (1) applied to P and P^* it is Fredholm; the kernel is independent of s by regularity, and so is the cokernel by (3).

(3) The operator P with domain $H^m(E)$ and its formal adjoint satisfy: $f \in L^2(F)$ is orthogonal to $\operatorname{im}(P)$ if and only if $\int\langle Pu, f\rangle = 0$ for all smooth u, that is, $P^*f = 0$ in the sense of distributions, which by elliptic regularity means $f \in \ker P^*$ and f smooth. Since the range is closed, $L^2 = \overline{\operatorname{im} P} \oplus \ker P^* = \operatorname{im} P \oplus \ker P^*$, where $\operatorname{im} P$ is taken on H^m; and if f is smooth and orthogonal to $\ker P^*$, a solution $u \in H^m$ exists and is smooth by regularity. □

The *index* of P is $\operatorname{ind} P = \dim\ker P - \dim\ker P^*$, the number of free parameters in the solutions minus the number of conditions on the right side. For square matrices it is 0; for elliptic operators it can be any integer, and it is a deep invariant (Remark 42.5.10).

Example 42.5.3. On the circle $S^1 = \mathbb{R}/2\pi\mathbb{Z}$ let $P = d/d\theta$ on complex functions. Its symbol $i\xi$ is invertible for $\xi \neq 0$, so P is elliptic of order 1, and integration by parts gives $P^* = -d/d\theta$.

Both kernels consist of the constants. Theorem 42.5.2 predicts that $u' = f$ has a smooth periodic solution exactly when f is orthogonal to $\ker P^*$, that is, when $\int_0^{2\pi} f\,d\theta = 0$; this is visible directly, since $u(\theta) = \int_0^\theta f$ is periodic exactly in that case. The index is $\dim\ker P - \dim\ker P^* = 0$. For the perturbation $P_c = d/d\theta + c$ with a constant $c \in \mathbb{C}$, the kernel is spanned by $e^{-c\theta}$ when this function is 2π-periodic, that is, when $c \in i\mathbb{Z}$, and is 0 otherwise; so the dimension of the kernel jumps. The adjoint $P_c^* = -d/d\theta + \bar{c}$ has kernel spanned by $e^{\bar{c}\theta}$ for exactly the same values of c, so the index stays 0, as predicted by Remark 42.5.10, since the principal symbol does not depend on c.

Example 42.5.4. On $S^1 = \mathbb{R}/2\pi\mathbb{Z}$ consider $P = -\frac{d^2}{d\theta^2} - c$ with a real constant c, which is elliptic and formally self-adjoint. On the Fourier coefficients of $u = \sum_k \hat{u}(k)e^{ik\theta}$ it acts by $\widehat{Pu}(k) = (k^2 - c)\hat{u}(k)$. If c is not the square of an integer, P is bijective on smooth functions, with inverse $\hat{f}(k) \mapsto \hat{f}(k)/(k^2 - c)$, which gains two derivatives. If $c = k_0^2$, the kernel is spanned by $e^{\pm ik_0\theta}$ (by 1 if $k_0 = 0$), and $-u'' - k_0^2 u = f$ is solvable exactly when $\hat{f}(\pm k_0) = 0$, that is, when f is orthogonal to the kernel. This is the classical phenomenon of *resonance*: forcing a string at one of its own frequencies has no periodic response. Theorem 42.5.2 is the statement that this alternative, "either unique solvability or a finite number of conditions", holds for every elliptic operator on every compact manifold.

Example 42.5.5. *The Laplacian on the torus.* On $T^n = \mathbb{R}^n/\mathbb{Z}^n$ with the flat metric, $\Delta = -\sum_j \partial_j^2$ acts on Fourier coefficients by $\widehat{\Delta u}(k) = 4\pi^2|k|^2\hat{u}(k)$, $k \in \mathbb{Z}^n$. Every statement of Theorem 42.5.2 can be seen directly.

1. The kernel consists of the u with $\hat{u}(k) = 0$ for $k \neq 0$: the constants. As Δ is formally self-adjoint, $\ker\Delta^* = \ker\Delta$ is also the constants.

2. The equation $\Delta u = f$ forces $\hat{f}(0) = 0$, that is, $\int_{T^n} f = 0$, which is orthogonality to $\ker\Delta^*$. Conversely, if $\hat{f}(0) = 0$, then $\hat{u}(k) = \hat{f}(k)/(4\pi^2|k|^2)$ for $k \neq 0$ (and $\hat{u}(0)$ arbitrary) defines a solution, unique up to the kernel.

3. The solution gains two derivatives: for $k \neq 0$ we have $|k| \geq 1$, so $2|k|^2 \geq 1 + |k|^2 = \langle k\rangle^2$ and $4\pi^2|k|^2 \geq 2\pi^2\langle k\rangle^2$; hence $\langle k\rangle^{2(s+2)}|\hat{u}(k)|^2 \leq \frac{1}{4\pi^4}\langle k\rangle^{2s}|\hat{f}(k)|^2$, and $f \in H^s$ gives $u \in H^{s+2}$, and f smooth gives u smooth. This is elliptic regularity in its simplest form.

4. The image of $\Delta\colon H^{s+2} \to H^s$ is the closed hyperplane $\{\hat{f}(0) = 0\}$, of codimension 1, and the index is $1 - 1 = 0$.

By contrast, the non-elliptic operator ∂_1 on T^2 multiplies $\hat{u}(k)$ by $2\pi i k_1$, which vanishes for all $k = (0, k_2)$: its kernel, the functions of x_2 alone, is infinite-dimensional, and a solution of $\partial_1 u = f$ gains a derivative only in the x_1-direction.

The operator P^*P is always formally self-adjoint and nonnegative, and it is elliptic when P is, since its symbol is $\sigma_m(P)^*\sigma_m(P)$ (Proposition 42.2.3). It is the mechanism by which the Hodge Laplacian is built from d and its adjoint, and the following corollary is the abstract form of the statement that harmonic forms are closed and coclosed.

Corollary 42.5.6. *Let P be elliptic. Then* $\ker(P^*P) = \ker P$*, and the smooth sections decompose orthogonally as*

$$\Gamma(M,F) = P\big(\Gamma(M,E)\big) \oplus \ker P^*, \qquad \Gamma(M,E) = P^*\big(\Gamma(M,F)\big) \oplus \ker P.$$

Proof. If $P^*Pu = 0$, then $0 = \int \langle P^*Pu, u\rangle = \int |Pu|^2$, so $Pu = 0$; the converse is clear. For the first decomposition, $P(\Gamma(M,E)) \perp \ker P^*$ by the definition of the formal adjoint, and a smooth f is the sum of its orthogonal projection h to the finite-dimensional space $\ker P^*$, which is smooth, and $f - h$, which is smooth and orthogonal to $\ker P^*$, hence of the form Pu with u smooth by Theorem 42.5.2(3). The second decomposition is the first one for the elliptic operator P^*, whose adjoint is P. □

Theorem 42.5.7. *Let P be a formally self-adjoint elliptic operator of order $m > 0$ on a compact manifold, with $P \geq 0$ in the sense that $\int \langle Pu, u\rangle \geq 0$. Then $L^2(E)$ has a complete orthonormal system of smooth eigensections of P, the eigenvalues $0 \leq \lambda_1 \leq \lambda_2 \leq \cdots$ are real, have finite multiplicity, and tend to $+\infty$. Every $u \in L^2(E)$ is the L^2-limit of its eigenfunction expansion, which converges in $\Gamma(M,E)$ when u is smooth.*

Proof. By Theorem 42.5.2 applied to $P + I$, which is elliptic and injective on smooth sections (as $\int \langle (P+I)u, u\rangle \geq \|u\|^2_{L^2}$), the map $P + I \colon H^m(E) \to L^2(E)$ is bijective, with bounded inverse G by the open mapping theorem (Theorem 30.4.2). Composing with the compact inclusion $H^m \to L^2$ shows that $G \colon L^2 \to L^2$ is compact, and it is self-adjoint and positive because $P + I$ is. By the spectral theorem for compact self-adjoint operators (Theorem 30.6.3) there is an orthonormal basis of $L^2(E)$ of eigenvectors of G, with eigenvalues $\mu_k > 0$ tending to 0 (they are nonzero since G is injective). Then $Pu_k = (\mu_k^{-1} - 1)u_k$, so the u_k are eigensections of P with eigenvalues $\lambda_k = \mu_k^{-1} - 1 \to \infty$, smooth by elliptic regularity, and $\lambda_k \geq 0$ by positivity. Finite multiplicity is finite-dimensionality of the eigenspaces, which are kernels of the elliptic operators $P - \lambda_k$. For smooth u, the coefficients $\langle u, u_k\rangle$ decay faster than any power of λ_k, because $\lambda_k^N \langle u, u_k\rangle = \langle P^N u, u_k\rangle$ is bounded; with the Sobolev embedding this gives convergence in every H^s, hence in $\Gamma(M,E)$. □

The proof reduces the spectral theory of the unbounded operator P to that of the compact operator $G = (P + I)^{-1}$; the two have the same eigenvectors, and the eigenvalues correspond by $\lambda = \mu^{-1} - 1$. Eigenvalues of P tend to infinity because those of G tend to zero, which in turn reflects the compactness of $H^m \to L^2$. For the circle the theorem gives back the Fourier basis $e^{ik\theta}$ of $-d^2/d\theta^2$, with eigenvalues k^2; for the round sphere S^2 some eigenfunctions are computed in Exercise 42.6.6. The first nonzero eigenvalue of the Laplacian on functions is the best constant in the Poincaré inequality (Chapter 43), and the eigenvalue expansion solves the heat equation (Exercise 42.6.8).

The positivity assumption is a convenience: it makes $P + I$ injective, so that its inverse exists. Without it the conclusion survives in a slightly weaker form, with eigenvalues that may go to $-\infty$ as well.

Corollary 42.5.8. *Let P be a formally self-adjoint elliptic operator of order $m > 0$ on a compact manifold, not necessarily nonnegative. Then $L^2(E)$ has a complete orthonormal system of smooth eigensections of P; the eigenvalues are real, have finite multiplicity, and $|\lambda| \to \infty$: for every c only finitely many of them satisfy $|\lambda| \leq c$. If moreover P is* bounded below, *$\int \langle Pu, u\rangle \geq -c_0\|u\|^2_{L^2}$ for some c_0 and all smooth u, then only finitely many eigenvalues are negative, and they can be ordered as $\lambda_1 \leq \lambda_2 \leq \cdots \to +\infty$.*

Proof. The operator $P^2 = P^*P$ is elliptic of order $2m$, with principal symbol $\sigma_m(P)^2$ (Proposition 42.2.3), formally self-adjoint, and nonnegative: $\int \langle P^2u, u\rangle = \int |Pu|^2 \geq 0$. By Theorem 42.5.7, $L^2(E)$ is the Hilbert space direct sum of the eigenspaces $E_\nu = \ker(P^2 - \nu)$, which are finite-dimensional, consist of smooth sections, and are nonzero only for a sequence of values $\nu \geq 0$ tending to ∞. Since P commutes with P^2, it maps each E_ν into itself, and on the finite-dimensional space E_ν it is symmetric for the L^2 inner product, by the definition of the formal adjoint. So E_ν has an orthonormal basis of eigenvectors of P (Theorem 11.10.6), with

real eigenvalues λ satisfying $\lambda^2 = \nu$, that is, $\lambda = \pm\sqrt{\nu}$. Together these bases form a complete orthonormal system of smooth eigensections of P. The λ-eigenspace of P lies in E_{λ^2}, hence is finite-dimensional, and $|\lambda| = \sqrt{\nu} \to \infty$. If P is bounded below, apply Theorem 42.5.7 to the nonnegative operator $P + c_0$, which has the same eigensections and eigenvalues shifted by c_0; its eigenvalues tend to $+\infty$, so those of P do too, and only finitely many are negative. □

Example 42.5.9.

1. On $S^1 = \mathbb{R}/2\pi\mathbb{Z}$, the operator $P = -i\,d/d\theta$ is elliptic of order 1 and formally self-adjoint, since $(d/d\theta)^* = -d/d\theta$. Its eigenfunctions are $e^{ik\theta}$ with eigenvalues $k \in \mathbb{Z}$, which go to both $+\infty$ and $-\infty$: P is not bounded below, and its square is $-d^2/d\theta^2$. The same happens for every first-order operator of Dirac type, such as $d + \delta$ on forms (Chapter 43), whose square is the Hodge Laplacian: the eigenvalues of $d + \delta$ are $\pm\sqrt{\nu}$ for the eigenvalues ν of Δ.

2. The operator $-d^2/d\theta^2 - c$ with $c > 0$ is bounded below but not nonnegative. Its eigenvalues are $k^2 - c$, of which finitely many, those with $k^2 < c$, are negative. More generally, adding a smooth potential V to a nonnegative operator, $P = \Delta + V$, gives an operator bounded below by $-\sup|V|$.

3. The order must be positive: for $m = 0$, an operator of order 0 is multiplication by a section of $\mathrm{End}(E)$, and multiplication by the function x on $L^2([0,1])$ has no eigenvectors at all (Example 30.7.4(2)). On a compact manifold the function $\cos\theta$ on S^1 gives the same phenomenon.

Remark 42.5.10. The index $\operatorname{ind} P = \dim\ker P - \dim\ker P^*$ depends only on the homotopy class of the principal symbol, and the Atiyah–Singer index theorem computes it as a characteristic number of the symbol; for $d + \delta$ this gives the Gauss–Bonnet theorem, and for the Dolbeault operator the Riemann–Roch–Hirzebruch theorem (Chapter 50). See [LM89].

42.5.1 Variational description of the eigenvalues

For a symmetric matrix the smallest eigenvalue is the minimum of $\langle Ax, x\rangle$ on the unit sphere. The same is true for elliptic operators, and it gives a way to estimate eigenvalues without computing them. Let P be as in Theorem 42.5.7, with orthonormal eigensections $u_1, u_2, \dots$ and eigenvalues $0 \le \lambda_1 \le \lambda_2 \le \cdots$, listed with multiplicity. The *Rayleigh quotient* of a smooth section $u \neq 0$ is

$$\mathcal{R}(u) = \frac{\int_M \langle Pu, u\rangle}{\int_M |u|^2}.$$

Proposition 42.5.11 (Min-max principle)**.**

1. *For smooth u with coefficients $c_j = \langle u, u_j\rangle$, $\int\langle Pu, u\rangle = \sum_j \lambda_j |c_j|^2$.*

2. *$\lambda_1 = \min_{u\neq 0}\mathcal{R}(u)$, and more generally $\lambda_k = \min\{\mathcal{R}(u) : u \neq 0,\ u \perp u_1, \dots, u_{k-1}\}$; the minima are attained exactly at the eigensections for λ_1, resp. at the combinations of eigensections with eigenvalue λ_k orthogonal to $u_1, \dots, u_{k-1}$.*

3. *(*Courant–Fischer*) $\lambda_k = \min_V \max_{0\neq u\in V}\mathcal{R}(u)$, the minimum over all k-dimensional subspaces V of $\Gamma(M, E)$.*

Proof. (1) The coefficients of the smooth section Pu are $\langle Pu, u_j\rangle = \langle u, Pu_j\rangle = \lambda_j c_j$, as P is formally self-adjoint and λ_j is real. By Parseval's identity (Theorem 30.3.6), $\langle Pu, u\rangle_{L^2} = \sum_j \lambda_j c_j \bar{c}_j$.

(2) If $u \perp u_1, \dots, u_{k-1}$, then $c_j = 0$ for $j < k$, and $\int\langle Pu, u\rangle = \sum_{j\ge k}\lambda_j|c_j|^2 \ge \lambda_k \sum_j |c_j|^2 = \lambda_k\|u\|^2$, with equality if and only if $c_j = 0$ whenever $\lambda_j > \lambda_k$. For $u = u_k$ equality holds.

(3) A k-dimensional V contains a nonzero u orthogonal to the $k-1$ vectors $u_1,\dots,u_{k-1}$ (the map $V\to\mathbb{C}^{k-1}$, $u\mapsto(\langle u,u_j\rangle)_j$, has a nonzero kernel), and $\mathcal{R}(u)\ge\lambda_k$ by (2); so $\max_V\mathcal{R}\ge\lambda_k$. For $V=\operatorname{span}(u_1,\dots,u_k)$, (1) gives $\mathcal{R}(u)=\sum_{j\le k}\lambda_j|c_j|^2/\sum_{j\le k}|c_j|^2\le\lambda_k$. □

The min-max principle compares spectra. If $P'=P+V$ with V a smooth self-adjoint endomorphism with $V\ge c$, then $\mathcal{R}'\ge\mathcal{R}+c$ for every u, so $\lambda_k(P')\ge\lambda_k(P)+c$ for every k: raising the potential raises every eigenvalue. Characterisation (3) does not involve the eigensections, which is what makes such comparisons possible.

Example 42.5.12. *Wirtinger's inequality.* On $S^1=\mathbb{R}/2\pi\mathbb{Z}$, $P=-d^2/d\theta^2$ has eigenvalues $0,1,1,4,4,9,9,\dots$, with eigenfunctions 1, $e^{\pm i\theta}$, $e^{\pm 2i\theta}$, …, and $\int\langle Pu,u\rangle=\int_0^{2\pi}|u'|^2$ by integration by parts. Proposition 42.5.11(2) with $k=2$ says: for a smooth 2π-periodic u with $\int_0^{2\pi}u=0$,

$$\int_0^{2\pi}|u'|^2\,d\theta\ge\int_0^{2\pi}|u|^2\,d\theta,$$

with equality exactly for $u=ae^{i\theta}+be^{-i\theta}$. This sharp *Poincaré inequality* on the circle gives Hurwitz's proof of the isoperimetric inequality (Exercise 42.6.9).

42.5.2 Counting eigenvalues

How fast do the eigenvalues tend to infinity? Let $N(\lambda)$ be the number of eigenvalues $\le\lambda$, counted with multiplicity. For the torus everything is explicit.

Proposition 42.5.13 (Weyl's law on the torus)**.** *Let Δ be the Laplacian on functions on the flat torus $T^n=\mathbb{R}^n/\mathbb{Z}^n$, and ω_n the volume of the unit ball of $\mathbb{R}^n$. Then*

$$N(\lambda)=\#\{k\in\mathbb{Z}^n:4\pi^2|k|^2\le\lambda\},\qquad\lim_{\lambda\to\infty}\frac{N(\lambda)}{\lambda^{n/2}}=\frac{\omega_n}{(2\pi)^n}.$$

Proof. The functions $e^{2\pi i\langle k,x\rangle}$, $k\in\mathbb{Z}^n$, form an orthonormal basis of $L^2(T^n)$ (the n-dimensional version of Example 30.3.7) of eigenfunctions with eigenvalues $4\pi^2|k|^2$, which gives the formula for $N(\lambda)$. Let $L(R)=\#\{k\in\mathbb{Z}^n:|k|\le R\}$, so $N(\lambda)=L(\sqrt{\lambda}/2\pi)$. The unit cubes $k+[0,1)^n$ are disjoint, have volume 1 and diameter $\sqrt{n}$. Those with $|k|\le R$ lie in the ball of radius $R+\sqrt{n}$; and every point y of the ball of radius $R-\sqrt{n}$ lies in the cube of $k=\lfloor y\rfloor$ (componentwise), with $|k|\le|y|+\sqrt{n}\le R$. Comparing volumes,

$$\omega_n(R-\sqrt{n})^n\le L(R)\le\omega_n(R+\sqrt{n})^n\qquad(R\ge\sqrt{n}),$$

so $L(R)/(\omega_nR^n)\to1$, and $N(\lambda)/\lambda^{n/2}\to\omega_n/(2\pi)^n$. □

Equivalently, the k-th eigenvalue grows like $\lambda_k\sim4\pi^2(k/\omega_n)^{2/n}$. The same law holds on every compact Riemannian manifold of dimension n, with the volume entering: $N(\lambda)\sim\omega_n\operatorname{vol}(M)\lambda^{n/2}/(2\pi)^n$ (*Weyl's law*, see [Kac66]). So the spectrum "hears" the dimension and the volume of M. Whether it determines M itself is Kac's question "can one hear the shape of a drum?" [Kac66]; for manifolds the answer is negative, as Milnor found two flat tori of dimension 16 with the same spectrum that are not isometric [Mil64].

42.6 Exercises

Exercise 42.6.1. Compute the principal symbol of d, of δ and of Δ on forms, and verify that Δ is elliptic while d is not (for $n\ge2$).

Solution. $\sigma_1(d)(x,\xi)\alpha = i\xi\wedge\alpha$ and $\sigma_1(\delta)(x,\xi)\alpha = -i\iota_\xi\alpha$, so $\sigma_2(\Delta)(x,\xi) = \xi\wedge\iota_\xi + \iota_\xi(\xi\wedge -) = |\xi|^2\mathrm{id}$ by the algebraic identity of Proposition 12.4.6. The symbol of d is not injective on Λ^k for $k \geq 1$, since $\xi\wedge\xi = 0$. □

Exercise 42.6.2. Compute the spectrum of the Laplacian on functions on the flat torus $\mathbb{R}^n/\mathbb{Z}^n$ and verify Theorem 42.5.7 directly.

Solution. The functions $e^{2\pi i k\cdot x}$, $k\in\mathbb{Z}^n$, are eigenfunctions with eigenvalues $4\pi^2|k|^2$, and they form an orthonormal basis of L^2 (Example 30.3.7); multiplicities are the numbers of lattice points on spheres, and the eigenvalues tend to infinity. □

Exercise 42.6.3. Show that a formally self-adjoint elliptic operator on a compact manifold has index zero, and that the index of an elliptic operator changes sign when P is replaced by P^*.

Solution. The formal adjoint of P^* is P, by the symmetry of the defining identity $\int\langle Pu,v\rangle = \int\langle u,P^*v\rangle$. So $\operatorname{ind}P^* = \dim\ker P^* - \dim\ker P = -\operatorname{ind}P$, and if $P = P^*$ the index equals its own negative, hence is 0. □

Exercise 42.6.4. Give an example of a compact manifold and a smooth function f for which $\Delta u = f$ has no solution, and characterise the f for which it does.

Solution. On a compact connected oriented M, $\ker\Delta$ on functions consists of the constants, so by Theorem 42.5.2 the equation is solvable exactly when $\int_M f\,\mathrm{vol} = 0$; any f with nonzero integral, for instance $f = 1$, gives no solution. □

Exercise 42.6.5. On $\mathbb{C}$ write $\partial_z = \frac12(\partial_x - i\partial_y)$ and $\partial_{\bar z} = \frac12(\partial_x + i\partial_y)$. Compute the principal symbols of ∂_z, $\partial_{\bar z}$ and of the composite $\partial_z\partial_{\bar z}$, and show that $\sum_j \partial_j^2 = 4\partial_z\partial_{\bar z}$. Deduce that a harmonic function on a disc is the real part of a holomorphic function, using that $\partial_z u$ is holomorphic when u is harmonic.

Solution. $\sigma_1(\partial_z) = \frac{i}{2}(\xi_1 - i\xi_2)$ and $\sigma_1(\partial_{\bar z}) = \frac{i}{2}(\xi_1 + i\xi_2)$, both nonzero for $\xi\neq 0$, and by multiplicativity $\sigma_2(\partial_z\partial_{\bar z}) = -\frac14(\xi_1^2+\xi_2^2)$, which is $\frac14$ times the symbol $-|\xi|^2$ of $\partial_x^2+\partial_y^2$. Directly, $4\partial_z\partial_{\bar z} = (\partial_x - i\partial_y)(\partial_x + i\partial_y) = \partial_x^2 + \partial_y^2$. If u is a real harmonic function, then $\partial_{\bar z}(\partial_z u) = \frac14(\partial_x^2+\partial_y^2)u = 0$, so $h = 2\partial_z u = u_x - iu_y$ is holomorphic (it satisfies the Cauchy–Riemann equations). On a disc it has a holomorphic primitive F (Corollary 29.3.2), $F' = u_x - iu_y$. Then $\operatorname{Re}F$ has $\partial_x\operatorname{Re}F = \operatorname{Re}F' = u_x$ and $\partial_y\operatorname{Re}F = -\operatorname{Im}F' = u_y$, using the Cauchy–Riemann equations for F; so $u - \operatorname{Re}F$ is constant, and $u = \operatorname{Re}(F + c)$. □

Exercise 42.6.6. On the round unit sphere S^2 with spherical coordinates (ϑ,φ), in which $g = d\vartheta^2 + \sin^2\vartheta\,d\varphi^2$, show that

$$\Delta f = -\frac{1}{\sin\vartheta}\partial_\vartheta(\sin\vartheta\,\partial_\vartheta f) - \frac{1}{\sin^2\vartheta}\partial_\varphi^2 f.$$

Check that the restrictions of the coordinate functions $z = \cos\vartheta$ and $x = \sin\vartheta\cos\varphi$ are eigenfunctions with eigenvalue 2, and that $3\cos^2\vartheta - 1$ (the restriction of $3z^2 - (x^2+y^2+z^2)$) is an eigenfunction with eigenvalue 6.

Solution. Here $\det g = \sin^2\vartheta$, $g^{\vartheta\vartheta} = 1$ and $g^{\varphi\varphi} = \sin^{-2}\vartheta$, and Proposition 40.15.3(5) gives the formula. For $f = \cos\vartheta$: $\sin\vartheta\,\partial_\vartheta f = -\sin^2\vartheta$, whose ϑ-derivative is $-2\sin\vartheta\cos\vartheta$, so $\Delta f = 2\cos\vartheta$. For $f = \sin\vartheta\cos\varphi$: $\partial_\vartheta(\sin\vartheta\cos\vartheta) = \cos^2\vartheta - \sin^2\vartheta$ and $\partial_\varphi^2 f = -f$, so $\Delta f = \big(-(\cos^2\vartheta - \sin^2\vartheta) + 1\big)\cos\varphi/\sin\vartheta = 2\sin^2\vartheta\cos\varphi/\sin\vartheta = 2f$. For $f = 3\cos^2\vartheta - 1$: $\sin\vartheta\,\partial_\vartheta f = -6\cos\vartheta\sin^2\vartheta$, whose derivative is $6\sin^3\vartheta - 12\cos^2\vartheta\sin\vartheta$, so $\Delta f = -6\sin^2\vartheta + 12\cos^2\vartheta = 18\cos^2\vartheta - 6 = 6f$. (In general the restrictions of the homogeneous harmonic polynomials of degree l are the eigenfunctions with eigenvalue $l(l+1)$, and they span $L^2(S^2)$.) □

Exercise 42.6.7. On $S^1 = \mathbb{R}/2\pi\mathbb{Z}$ let $P = d/d\theta + a(\theta)$ with a smooth and complex valued, and let $A = \int_0^{2\pi} a\, d\theta$. Compute $\ker P$ and $\ker P^*$, and show that both are 0 unless $A \in 2\pi i\mathbb{Z}$, when both are one-dimensional. So the index is always 0, although the kernel jumps.

Solution. $u' = -au$ has the solutions $u = Ce^{-\int_0^\theta a}$ on $\mathbb{R}$, which are 2π-periodic (for $C \neq 0$) if and only if $e^{-A} = 1$, that is $A \in 2\pi i\mathbb{Z}$. The adjoint is $P^* = -d/d\theta + \bar{a}$, with solutions $Ce^{\int_0^\theta \bar{a}}$, periodic if and only if $\bar{A} \in 2\pi i\mathbb{Z}$, which is the same condition, as $2\pi i\mathbb{Z}$ is stable under complex conjugation. For $a = c$ constant, $A = 2\pi c$ and the condition is $c \in i\mathbb{Z}$, as in Example 42.5.3. □

Exercise 42.6.8. Let M be compact, connected and oriented, and u_0 a smooth function on M. Let $\varphi_0, \varphi_1, \dots$ be an orthonormal basis of eigenfunctions of the Laplacian on functions, with eigenvalues $\lambda_0 \leq \lambda_1 \leq \cdots$ (Theorem 42.5.7, numbered from 0). Show that $u(t) = \sum_k e^{-\lambda_k t}\langle u_0, \varphi_k\rangle\varphi_k$ is a smooth solution of the heat equation $\partial_t u + \Delta u = 0$ for $t > 0$ with $u(t) \to u_0$ in L^2 as $t \to 0$, and that $u(t)$ converges, as $t \to \infty$, to the constant function equal to the average of u_0. (For forms, the same argument shows that the heat flow converges to the harmonic part; this is the heat equation proof of the Hodge theorem mentioned in Chapter 43.)

Solution. Write $c_k = \langle u_0, \varphi_k\rangle$; these decay faster than any power of λ_k (proof of Theorem 42.5.7), so the series and all its derivatives in t converge in every Sobolev norm, uniformly for $t \geq 0$, and u is smooth, with $\partial_t u = -\sum_k \lambda_k e^{-\lambda_k t} c_k \varphi_k = -\Delta u$. At $t = 0$ the series is the expansion of u_0. The kernel of Δ on functions consists of the constants ($\|du\|^2 = (\Delta u, u) = 0$ forces $du = 0$, and M is connected), so $\lambda_0 = 0$ with $\varphi_0 = \mathrm{vol}(M)^{-1/2}$ and $\lambda_k \geq \lambda_1 > 0$ for $k \geq 1$. Then $\|u(t) - c_0\varphi_0\|^2 = \sum_{k\geq 1} e^{-2\lambda_k t}|c_k|^2 \leq e^{-2\lambda_1 t}\|u_0\|^2 \to 0$, and $c_0\varphi_0 = \mathrm{vol}(M)^{-1}\int_M u_0\,\mathrm{vol}$ is the average. □

Exercise 42.6.9. (*Hurwitz's proof of the isoperimetric inequality.*) Let $\gamma\colon \mathbb{R}/2\pi\mathbb{Z} \to \mathbb{C}$ be a smooth closed curve with $|\gamma'| = 1$, so of length 2π, bounding the area $A = \frac{1}{2}\int_0^{2\pi} \mathrm{Im}(\overline{\gamma}\gamma')\, d\theta$ (the integral of $\frac{1}{2}(x\, dy - y\, dx)$, Green's formula). Show that $A \leq \pi$, with equality only for circles of radius 1. Deduce that a closed curve of length L bounds area at most $L^2/4\pi$.

Solution. Expand $\gamma = \sum_k c_k e^{ik\theta}$ (converging with all derivatives, as γ is smooth). By Parseval's identity, with $\frac{1}{2\pi}\int_0^{2\pi}$ as normalised measure, $1 = \frac{1}{2\pi}\int |\gamma'|^2 = \sum_k k^2|c_k|^2$, and $\frac{1}{2\pi}\int \bar{\gamma}\gamma' = \sum_k \bar{c}_k\, ikc_k = i\sum_k k|c_k|^2$, so $A = \pi\sum_k k|c_k|^2$. Since $k \leq k^2$ for all integers k, $A \leq \pi\sum_k k^2|c_k|^2 = \pi$, with equality exactly when $c_k = 0$ for $k \neq 0, 1$, that is, $\gamma(\theta) = c_0 + c_1 e^{i\theta}$ with $|c_1| = 1$, a unit circle. (This is Wirtinger's inequality, Example 42.5.12, applied to $\gamma - c_0$.) For a curve of length L, rescaling by $2\pi/L$ gives one of length 2π, and areas scale by $(2\pi/L)^2$, so $A(2\pi/L)^2 \leq \pi$. □

Exercise 42.6.10. Using Exercise 42.6.6 and Proposition 42.5.11, show that every smooth function u on the round sphere S^2 with $\int_{S^2} u = 0$ satisfies $\int_{S^2} |\nabla u|^2 \geq 2\int_{S^2} u^2$, assuming that the eigenvalues of Δ on S^2 are $\ell(\ell+1)$, $\ell \geq 0$.

Solution. The constants are the eigenfunctions for $\lambda_1 = 0$, and the next eigenvalue is $\ell(\ell+1) = 2$ for $\ell = 1$. Since $\int\langle \Delta u, u\rangle = \int |\nabla u|^2$ for functions, and $u \perp 1$, the min-max principle with $k = 2$ gives $\int |\nabla u|^2 \geq 2\int u^2$; equality holds for the restrictions of linear functions, the eigenfunctions with $\ell = 1$. □

Chapter 43

Harmonic Forms and the Hodge Theorem

43.1 Introduction

On a compact oriented Riemannian manifold every de Rham cohomology class contains exactly one harmonic form (Theorem 43.2.5). This is the theorem that makes Hodge theory possible: it turns a topological object, the cohomology of a manifold, into the solution space of an elliptic equation, on which the metric and, in the complex case, the complex structure act. The idea is already visible in finite dimensions: for a complex of finite-dimensional inner product spaces, each cohomology class has a unique representative of smallest norm, and these representatives are the kernel of a "Laplacian" $dd^* + d^*d$ (Lemma 43.2.1). On a compact manifold the same statement holds for the de Rham complex, and the only difficulty is analytic: one needs to know that the Laplacian has closed range and finite-dimensional kernel, which is what Chapter 42 provides.

The consequences drawn here are the Hodge decomposition of the space of forms (with the Helmholtz decomposition of vector fields on a torus as a concrete instance), Poincaré duality via the Hodge star, the conformal invariance of harmonic 1-forms on surfaces, the Poincaré inequality, and Bochner's vanishing theorem, with a complete proof of the Bochner–Weitzenböck formula on 1-forms; the complex analogue is in Chapter 48. References: [War83, Chapter 6], [Wel08, Chapter 4], [GH78, Chapter 0], [Voi02b, Chapter 5].

Prerequisites. Chapter 42 (Elliptic Operators on Compact Manifolds) and Chapter 37 (de Rham Cohomology).

On a first reading. The essential results are Theorem 43.2.2, Theorem 43.2.5 and Corollary 43.2.6. The proof of the Bochner–Weitzenböck formula can be skipped.

Throughout, (M, g) is a compact oriented Riemannian n-manifold, with the L^2 inner product, the codifferential δ and the Laplacian $\Delta = d\delta + \delta d$ of Definition 40.15.2, and $\mathcal{H}^k = \mathcal{H}^k(M, g) = \{\alpha \in \Omega^k(M) : \Delta\alpha = 0\}$.

43.2 The Hodge decomposition

We begin with the finite-dimensional model, where everything is linear algebra.

Lemma 43.2.1 (Hodge theory in finite dimensions)**.** *Let $V^{k-1} \xrightarrow{d} V^k \xrightarrow{d} V^{k+1}$ be linear maps of finite-dimensional real inner product spaces with $d \circ d = 0$, let δ denote the adjoints of the two maps d, and $\Delta = d\delta + \delta d \colon V^k \to V^k$. Then $\ker \Delta = \ker d \cap \ker \delta$, there is an orthogonal decomposition*

$$V^k = \ker \Delta \oplus dV^{k-1} \oplus \delta V^{k+1},$$

and $\ker d = \ker\Delta \oplus dV^{k-1}$. *Hence every class in* $\ker d/dV^{k-1}$ *has a unique representative in* $\ker\Delta$, *which is its unique element of smallest norm.*

Proof. For $v \in V^k$, $\langle \Delta v, v\rangle = \langle \delta v, \delta v\rangle + \langle dv, dv\rangle$, so $\Delta v = 0$ implies $dv = 0$ and $\delta v = 0$; the converse is clear. The images of d and δ are orthogonal: $\langle du, \delta w\rangle = \langle ddu, w\rangle = 0$. In finite dimensions the orthogonal complement of an image is the kernel of the adjoint, so the orthogonal complement of $dV^{k-1} \oplus \delta V^{k+1}$ is $\ker\delta \cap \ker d = \ker\Delta$, which gives the decomposition. If $dv = 0$ and $v = h + du + \delta w$, then $0 = dv = d\delta w$, so $\|\delta w\|^2 = \langle w, d\delta w\rangle = 0$, and $v = h + du$. Finally $\|h + du\|^2 = \|h\|^2 + \|du\|^2$, which is smallest exactly for $du = 0$. □

For $\Omega^\bullet(M)$ the spaces are infinite-dimensional and two things could go wrong: the image of d need not be closed (in L^2), so that "orthogonal complement of the image = kernel of the adjoint" only holds up to closure, and the kernel of Δ could be infinite-dimensional. Ellipticity of Δ rules out both.

Theorem 43.2.2. *For each* k *the space* $\mathcal{H}^k$ *is finite-dimensional and consists of the forms with* $d\alpha = 0$ *and* $\delta\alpha = 0$, *and there is an orthogonal direct sum decomposition*

$$\Omega^k(M) = \mathcal{H}^k \oplus d\Omega^{k-1}(M) \oplus \delta\Omega^{k+1}(M),$$

with $\ker d = \mathcal{H}^k \oplus d\Omega^{k-1}$ *and* $\ker\delta = \mathcal{H}^k \oplus \delta\Omega^{k+1}$. *Moreover there is a unique operator* $G\colon \Omega^k(M) \to \Omega^k(M)$, *the* Green operator, *that vanishes on* $\mathcal{H}^k$, *maps* $(\mathcal{H}^k)^\perp$ *into itself, and satisfies* $\alpha = H\alpha + \Delta G\alpha$, *where* H *is the orthogonal projection onto* $\mathcal{H}^k$; *it commutes with* d, δ *and* Δ.

Proof. Δ is elliptic, formally self-adjoint and non-negative (Example 42.2.4 and Proposition 40.15.3), so by Theorem 42.5.2 its kernel $\mathcal{H}^k$ is finite-dimensional and consists of smooth forms, and

$$\Omega^k(M) = \mathcal{H}^k \oplus \Delta\Omega^k(M)$$

orthogonally: indeed $L^2 = \operatorname{im}\Delta \oplus \ker\Delta$ by Theorem 42.5.2(3), and a smooth form orthogonal to $\mathcal{H}^k$ is $\Delta\beta$ for a smooth β by elliptic regularity. The description of $\mathcal{H}^k$ as $\ker d \cap \ker\delta$ is Proposition 40.15.3(3).

Since $\Delta\beta = d(\delta\beta) + \delta(d\beta)$, the second summand is contained in $d\Omega^{k-1} + \delta\Omega^{k+1}$, so the three spaces span Ω^k. They are pairwise orthogonal: harmonic forms are orthogonal to $d\Omega^{k-1}$ because $(\alpha, d\beta) = (\delta\alpha, \beta) = 0$, and similarly to $\delta\Omega^{k+1}$; and $(d\beta, \delta\gamma) = (dd\beta, \gamma) = 0$. Hence the sum is direct and orthogonal. If $d\alpha = 0$ and $\alpha = \eta + d\beta + \delta\gamma$, then $0 = (d\alpha, \gamma) = (d\delta\gamma, \gamma) = \|\delta\gamma\|^2$, so $\alpha \in \mathcal{H}^k \oplus d\Omega^{k-1}$; the statement for $\ker\delta$ is symmetric.

For the Green operator, define $G\alpha = 0$ on $\mathcal{H}^k$ and, on the orthogonal complement, $G\alpha = \beta$ where β is the unique solution of $\Delta\beta = \alpha$ orthogonal to $\mathcal{H}^k$ (Theorem 42.5.2(3) gives existence and smoothness, and uniqueness holds because two solutions differ by a harmonic form). Then $\alpha = H\alpha + \Delta G\alpha$ by construction. Since Δ commutes with d, δ and Δ, and these operators preserve $\mathcal{H}^\bullet$ and its orthogonal complement (for instance $d\mathcal{H}^k = 0$ and $(d\alpha, \eta) = (\alpha, \delta\eta) = 0$ for η harmonic), the uniqueness of G forces $Gd = dG$, $G\delta = \delta G$ and $G\Delta = \Delta G$. □

Explicitly, the decomposition of a form α is

$$\alpha = H\alpha + d(\delta G\alpha) + \delta(dG\alpha),$$

since $\alpha - H\alpha = \Delta G\alpha = d\delta G\alpha + \delta dG\alpha$. So the Green operator, an inverse of Δ on the orthogonal complement of the harmonic forms, produces the primitives: if α is exact, then $\alpha = d(\delta G\alpha)$, and $\delta G\alpha$ is the primitive of α orthogonal to the closed forms, the primitive of smallest norm.

Example 43.2.3. On the circle $M = \mathbb{R}/2\pi\mathbb{Z}$ with the coordinate θ, write functions as Fourier series $f = \sum_k c_k e^{ik\theta}$ (complex coefficients, $c_{-k} = \bar{c}_k$ for real f). Here $\Delta f = -f''$, and on 1-forms $\delta(g\,d\theta) = -g'$ (by Definition 40.15.2, $\delta = -{*}d{*}$ on 1-forms when $n = 1$, and ${*}(g\,d\theta) = g$). The harmonic functions are the constants and the harmonic 1-forms the constant multiples of $d\theta$, so the decompositions of Theorem 43.2.2 are

$$f = c_0 + \delta\Big(-\sum_{k\neq 0} \frac{c_k}{ik} e^{ik\theta}\, d\theta\Big), \qquad g\,d\theta = c_0\,d\theta + d\Big(\sum_{k\neq 0} \frac{c_k}{ik} e^{ik\theta}\Big) \quad \text{for } g = \sum_k c_k e^{ik\theta},$$

and the Green operator on functions is $G(\sum_k c_k e^{ik\theta}) = \sum_{k\neq 0} c_k e^{ik\theta}/k^2$. The harmonic part of $g\,d\theta$ is $c_0\,d\theta = \big(\frac{1}{2\pi}\int_0^{2\pi} g\big)d\theta$, and the theorem says that $g\,d\theta$ is exact exactly when its average vanishes, which is the familiar condition for $\int_0^\theta g$ to be periodic.

Example 43.2.4. On the flat torus $T^3 = \mathbb{R}^3/\mathbb{Z}^3$ identify 1-forms with vector fields by the metric, $X = (X_1, X_2, X_3) \leftrightarrow X_1\,dx_1 + X_2\,dx_2 + X_3\,dx_3$, and 2-forms with vector fields by $Y \leftrightarrow {*}Y^\flat = Y_1\,dx_2\wedge dx_3 + Y_2\,dx_3\wedge dx_1 + Y_3\,dx_1\wedge dx_2$. Then $df \leftrightarrow \nabla f$, the codifferential of a 2-form is $\delta({*}Y^\flat) = {*}d{*}{*}Y^\flat = {*}dY^\flat$ (here $n = 3$, $k = 2$, so the sign in $\delta = (-1)^{n(k+1)+1} * d*$ is $+1$, and ${*}{*} = 1$), and ${*}dY^\flat \leftrightarrow \operatorname{curl} Y$ (Example 36.2.14). The harmonic 1-forms are the constant ones (Example 40.15.5). So Theorem 43.2.2 in degree 1 is the *Helmholtz decomposition*: every smooth vector field X on T^3 can be written as

$$X = c + \nabla f + \operatorname{curl} Y$$

with c a constant vector, the three summands are uniquely determined by X (although f and Y are not), and they are orthogonal in L^2; moreover $\operatorname{curl} X = 0$ if and only if $X = c + \nabla f$ (the closed 1-forms), and $\operatorname{div} X = 0$ if and only if $X = c + \operatorname{curl} Y$ (the coclosed ones, as $\delta X^\flat = -\operatorname{div} X$). In Fourier coefficients, $X = \sum_k \hat{X}(k) e^{2\pi i k\cdot x}$, the three parts of the term with $k \neq 0$ are the components of $\hat{X}(k)$ along k and orthogonal to k, and $c = \hat{X}(0)$.

Theorem 43.2.5 (Hodge)**.** *The map $\mathcal{H}^k \to H^k_{\mathrm{dR}}(M)$, $\alpha \mapsto [\alpha]$, is an isomorphism: every de Rham class contains a unique harmonic representative, namely the one of smallest L^2 norm. In particular $H^k_{\mathrm{dR}}(M)$ is finite-dimensional, and $\dim \mathcal{H}^k = b_k(M)$.*

Proof. Harmonic forms are closed, so the map is defined. It is injective: if $\alpha \in \mathcal{H}^k$ is exact, $\alpha = d\beta$, then $\|\alpha\|^2 = (\alpha, d\beta) = (\delta\alpha, \beta) = 0$. It is surjective: a closed α decomposes as $\alpha = \eta + d\beta + \delta\gamma$ with η harmonic, and $\delta\gamma = 0$ as in the proof of Theorem 43.2.2, so $\alpha = \eta + d\beta$ is cohomologous to η. If $\alpha = \eta + d\beta$ is any representative, then $\|\alpha\|^2 = \|\eta\|^2 + \|d\beta\|^2 \geq \|\eta\|^2$ with equality only for $\alpha = \eta$, which is the statement about the norm. Finite-dimensionality and the identification with the Betti numbers follow from Theorem 37.3.1. □

The minimisation property has a variational reading: among all closed forms in a class, the harmonic one minimises the energy $\|\alpha\|^2$, and the condition $\delta\alpha = 0$ is the Euler–Lagrange equation, since $\frac{d}{dt}\|\alpha + t\,d\beta\|^2\big|_{t=0} = 2(\alpha, d\beta) = 2(\delta\alpha, \beta)$ must vanish for all β.

Corollary 43.2.6. *The Hodge star induces an isomorphism ${*}\colon \mathcal{H}^k \to \mathcal{H}^{n-k}$, hence $b_k(M) = b_{n-k}(M)$, and the pairing*

$$H^k_{\mathrm{dR}}(M) \times H^{n-k}_{\mathrm{dR}}(M) \to \mathbb{R}, \qquad ([\alpha], [\beta]) \mapsto \int_M \alpha\wedge\beta,$$

is nondegenerate; for $[\beta] = [{*}\alpha]$ with α harmonic and nonzero it takes the value $\|\alpha\|^2 > 0$.

Proof. $*$ commutes with Δ (Proposition 40.15.3(4)) and is an isomorphism of Ω^k onto Ω^{n-k}, hence restricts to an isomorphism on harmonic forms; combine with Theorem 43.2.5. For a nonzero harmonic α, $\int_M \alpha\wedge{*}\alpha = \|\alpha\|^2 > 0$, so the pairing is nondegenerate on the harmonic representatives, which represent all classes. □

Remark 43.2.7. The space $\mathcal{H}^k$ depends on the metric, but its dimension does not, and the isomorphism of Theorem 43.2.5 identifies all of them with $H^k_{\mathrm{dR}}(M)$. This is the mechanism that produces the Hodge structures of Chapter 48: on a compact Kähler manifold the harmonic forms decompose by type, and the resulting decomposition of cohomology is independent of the metric, although the harmonic representatives are not (Exercise 43.4.3).

On a surface part of this dependence disappears: harmonic 1-forms only see the conformal class of the metric. This is the first sign of the relation between harmonic forms on a surface and its structure of Riemann surface.

Proposition 43.2.8. *Let M be an oriented surface and $g' = e^{2u}g$ a conformal metric. Then the Hodge stars of g and g' agree on 1-forms, so a 1-form is harmonic for g if and only if it is harmonic for g' (when M is compact).*

Proof. If (ω^1, ω^2) is an oriented orthonormal coframe for g, then $(e^u\omega^1, e^u\omega^2)$ is one for g'. The Hodge star of g maps $\omega^1 \mapsto \omega^2$ and $\omega^2 \mapsto -\omega^1$ (Theorem 12.5.3), and that of g' maps $e^u\omega^1 \mapsto e^u\omega^2$ and $e^u\omega^2 \mapsto -e^u\omega^1$; by linearity over functions the two stars agree on ω^1 and ω^2, hence on all 1-forms. On a compact surface a 1-form α is harmonic if and only if $d\alpha = 0$ and $d{*}\alpha = 0$ (Proposition 40.15.3(3), as $\delta = -{*}d{*}$ on 1-forms in dimension 2), and these conditions only involve $*$ on 1-forms. □

For a compact Riemann surface, with any metric compatible with its complex structure, $*$ on 1-forms is determined by the complex structure ($*\,dx = dy$ and $*\,dy = -dx$ in a holomorphic coordinate $z = x + iy$), and a real 1-form $\alpha = a\,dx + b\,dy$ is harmonic exactly when $\alpha + i{*}\alpha = (a - ib)\,dz$ is a holomorphic 1-form (the Cauchy–Riemann equations for $a - ib$ are $a_y = b_x$ and $a_x + b_y = 0$, that is, $d\alpha = 0$ and $d{*}\alpha = 0$); this identifies $\mathcal{H}^1$ with the real parts of holomorphic 1-forms, and is the case of curves of the Hodge decomposition $H^1 = H^{1,0} \oplus H^{0,1}$ (Chapter 48).

Proposition 43.2.9. *Let G be a compact connected Lie group with a bi-invariant Riemannian metric (Example 40.2.5) and an orientation. Then the harmonic forms on G are exactly the bi-invariant forms.*

Proof. Left and right translations are isometries, and they preserve the orientation, being connected to the identity through diffeomorphisms; so they commute with the Hodge star, and $*$ maps bi-invariant forms to bi-invariant forms. A bi-invariant form ω is closed (Theorem 37.3.11), and so is ${*}\omega$; hence $\delta\omega = \pm{*}d{*}\omega = 0$ and ω is harmonic (Proposition 40.15.3(3)). The bi-invariant k-forms and the harmonic k-forms both map isomorphically onto $H^k_{\mathrm{dR}}(G)$ (Theorems 37.3.11 and 43.2.5), so the inclusion of the first space into the second is an equality. □

Counterexample 43.2.10. Compactness is essential in the Hodge theorem. On $\mathbb{R}$ with the Euclidean metric, the harmonic functions are the affine functions $ax + b$, a two-dimensional space, while $H^0_{\mathrm{dR}}(\mathbb{R}) = \mathbb{R}$; and the harmonic 1-forms $(ax + b)\,dx$ form a two-dimensional space although $H^1_{\mathrm{dR}}(\mathbb{R}) = 0$. Without compactness harmonic forms need not be closed or coclosed: the function x is harmonic with $dx \neq 0$, and $x\,dx$ is harmonic with $\delta(x\,dx) = -1 \neq 0$, since $\delta = -{*}d{*}$ on 1-forms on $\mathbb{R}$. The integration by parts behind Proposition 40.15.3(3) fails for forms without compact support.

43.3 Applications

Corollary 43.3.1. *For a compact oriented manifold, $\sum_k (-1)^k \dim \mathcal{H}^k = \chi(M)$, which is the index of the elliptic operator $d + \delta\colon \Omega^{\mathrm{even}}(M) \to \Omega^{\mathrm{odd}}(M)$.*

Proof. The first equality is Theorem 43.2.5 together with the definition of χ as the alternating sum of Betti numbers (Corollary 23.6.4). For the second, $(d+\delta)^* = d+\delta$ exchanges even and odd forms, its kernel on even forms is $\bigoplus_{k\,\text{even}} \mathcal{H}^k$ and on odd forms $\bigoplus_{k\,\text{odd}} \mathcal{H}^k$, because $(d+\delta)\alpha = 0$ implies $\Delta\alpha = (d+\delta)^2\alpha = 0$; so the index is $\sum_k (-1)^k \dim \mathcal{H}^k$. □

Corollary 43.3.2 (Poincaré inequality)**.** *Let M be compact and connected, and let $\lambda_1 > 0$ be the smallest nonzero eigenvalue of the Laplacian $\Delta = \delta d$ on functions. Then for every smooth function f with $\int_M f \,\mathrm{vol} = 0$,*

$$\int_M f^2 \,\mathrm{vol} \le \frac{1}{\lambda_1} \int_M |df|^2 \,\mathrm{vol},$$

with equality exactly for the eigenfunctions of eigenvalue λ_1.

Proof. A function u with $\Delta u = 0$ satisfies $\|du\|^2 = (\delta du, u) = 0$, so it is locally constant, hence constant as M is connected: $\mathcal{H}^0 = \mathbb{R}$. By Theorem 42.5.7 there is an orthonormal basis $u_0, u_1, \ldots$ of $L^2(M)$ of smooth eigenfunctions with eigenvalues $0 = \lambda^{(0)} < \lambda_1 = \lambda^{(1)} \le \lambda^{(2)} \le \cdots$, where u_0 is constant. Write $f = \sum_k c_k u_k$; then $c_0 = (f, u_0) = 0$ by hypothesis. Since $\Delta f = \sum_k \lambda^{(k)} c_k u_k$ in L^2,

$$\|df\|^2 = (\Delta f, f) = \sum_{k\ge 1} \lambda^{(k)} c_k^2 \ge \lambda_1 \sum_{k\ge 1} c_k^2 = \lambda_1 \|f\|^2,$$

with equality if and only if $c_k = 0$ whenever $\lambda^{(k)} > \lambda_1$, that is, f lies in the λ_1-eigenspace. □

Example 43.3.3. On the circle $M = \mathbb{R}/2\pi\mathbb{Z}$ with its standard metric, $\Delta f = -f''$. The real solutions of $f'' = -\lambda f$ with $\lambda > 0$ are $a\cos(\sqrt{\lambda}\,\theta) + b\sin(\sqrt{\lambda}\,\theta)$, which are 2π-periodic only for $\sqrt{\lambda} \in \mathbb{Z}$; so the eigenvalues are the k^2, with eigenfunctions $\cos k\theta$, $\sin k\theta$ (and 1 for $k = 0$). Hence $\lambda_1 = 1$, and Corollary 43.3.2 is *Wirtinger's inequality*: $\int_0^{2\pi} f^2 \le \int_0^{2\pi} f'^2$ for 2π-periodic smooth f with $\int_0^{2\pi} f = 0$, with equality exactly for $f = a\cos\theta + b\sin\theta$. Theorem 42.5.7 also shows that the functions $1, \cos k\theta, \sin k\theta$ span a dense subspace of L^2, which is the completeness of Fourier series.

The last application relates harmonic 1-forms to curvature. It rests on a formula expressing the Hodge Laplacian through the Levi-Civita connection. Below ∇ also denotes the connections induced on 1-forms and 2-forms (Proposition 38.3.6), so that $(\nabla_X\alpha)(Y) = X(\alpha(Y)) - \alpha(\nabla_X Y)$ and $(\nabla_X\beta)(Y,Z) = X(\beta(Y,Z)) - \beta(\nabla_X Y, Z) - \beta(Y, \nabla_X Z)$, and (e_i) is a local orthonormal frame. By Lemma 39.6.1, applied to the metric connection ∇ on TM, near every point p there are orthonormal frames with $\nabla e_i = 0$ at p; then also $[e_i, e_j] = \nabla_{e_i} e_j - \nabla_{e_j} e_i = 0$ at p, and the dual coframe (e^i) satisfies $\nabla e^i = 0$ at p, its connection form being $-\theta^T$.

Lemma 43.3.4. *Let α be a 1-form and β a 2-form on M.*

1. $d\alpha(X,Y) = (\nabla_X\alpha)(Y) - (\nabla_Y\alpha)(X)$.
2. *For a vector field V, $d(\iota_V \mathrm{vol}) = (\mathrm{div}\, V)\,\mathrm{vol}$ with $\mathrm{div}\, V = \sum_i \langle \nabla_{e_i} V, e_i\rangle$; hence $\int_M \mathrm{div}\, V \,\mathrm{vol} = 0$.*
3. $\delta\alpha = -\sum_i (\nabla_{e_i}\alpha)(e_i)$ *and* $(\delta\beta)(X) = -\sum_i (\nabla_{e_i}\beta)(e_i, X)$.

Proof. (1) By the invariant formula (Theorem 36.2.8) and the Leibniz rule for ∇, $d\alpha(X,Y) = X(\alpha(Y)) - Y(\alpha(X)) - \alpha([X,Y]) = (\nabla_X\alpha)(Y) + \alpha(\nabla_X Y) - (\nabla_Y\alpha)(X) - \alpha(\nabla_Y X) - \alpha([X,Y])$, and the terms without derivatives of α cancel because $\nabla_X Y - \nabla_Y X = [X,Y]$.

(2) The trace $\sum_i \langle \nabla_{e_i} V, e_i\rangle$ does not depend on the orthonormal frame, so we may compute at a point p in a frame with $\nabla e_i(p) = 0$, oriented, with dual coframe (e^i). Then

$\mathrm{vol} = e^1 \wedge \cdots \wedge e^n$ and, with $V = \sum_i V^i e_i$, $\iota_V \mathrm{vol} = \sum_i (-1)^{i-1} V^i\, e^1 \wedge \cdots \widehat{e^i} \cdots \wedge e^n$. By (1), $de^j(X,Y) = (\nabla_X e^j)(Y) - (\nabla_Y e^j)(X)$ vanishes at p, so at p only the derivatives of the V^i contribute: $d(\iota_V \mathrm{vol}) = \sum_i (-1)^{i-1} dV^i \wedge e^1 \wedge \cdots \widehat{e^i} \cdots \wedge e^n = \sum_i e_i(V^i)\,\mathrm{vol}$. And at p, $e_i(V^i) = e_i\langle V, e_i\rangle = \langle \nabla_{e_i} V, e_i\rangle$. The integral vanishes by Stokes' theorem.

(3) Let δ' be defined by the right-hand sides; again they do not depend on the frame. We show that δ' is a formal adjoint of d; then $(\gamma, \delta\beta - \delta'\beta) = 0$ for all 1-forms γ, so $\delta\beta = \delta'\beta$, and similarly in degree 1. Let γ be a 1-form, and V the vector field with $\langle V, X\rangle = \sum_j \gamma(e_j)\beta(X, e_j)$. At a point p, in a frame with $\nabla e_i(p) = 0$, the derivative of a function $\gamma(e_j)$ along e_i is $(\nabla_{e_i}\gamma)(e_j)$, and similarly for β, so

$$\operatorname{div} V = \sum_{i,j} e_i\big(\gamma(e_j)\beta(e_i,e_j)\big) = \sum_{i,j} (\nabla_{e_i}\gamma)(e_j)\beta(e_i,e_j) + \sum_{i,j} \gamma(e_j)(\nabla_{e_i}\beta)(e_i,e_j) = \langle d\gamma, \beta\rangle - \langle \gamma, \delta'\beta\rangle,$$

where we used $\langle d\gamma, \beta\rangle = \frac{1}{2}\sum_{i,j} d\gamma(e_i,e_j)\beta(e_i,e_j) = \sum_{i,j}(\nabla_{e_i}\gamma)(e_j)\beta(e_i,e_j)$, by (1) and the antisymmetry of β. Integrating with (2), $(d\gamma, \beta) = (\gamma, \delta'\beta)$. For a function f and a 1-form α the same computation with $V = f\alpha^\sharp$, where $\langle \alpha^\sharp, X\rangle = \alpha(X)$, gives $\operatorname{div} V = \sum_i e_i(f)\alpha(e_i) + f\sum_i (\nabla_{e_i}\alpha)(e_i) = \langle df, \alpha\rangle - f\,\delta'\alpha$. □

Proposition 43.3.5 (Bochner–Weitzenböck formula)**.** *For a* 1*-form* α *on* M*,*

$$\Delta\alpha = \nabla^*\nabla\alpha + \mathrm{Ric}(\alpha^\sharp, -), \qquad \nabla^*\nabla\alpha = -\sum_i \big(\nabla_{e_i}\nabla_{e_i}\alpha - \nabla_{\nabla_{e_i} e_i}\alpha\big),$$

and $(\Delta\alpha, \alpha) = \|\nabla\alpha\|^2 + \int_M \mathrm{Ric}(\alpha^\sharp, \alpha^\sharp)\,\mathrm{vol}$*, where* $\|\nabla\alpha\|^2 = \int_M \sum_i |\nabla_{e_i}\alpha|^2\,\mathrm{vol}$*.*

Proof. All terms are independent of the frame, so we compute at a point p in a frame with $\nabla e_i = 0$ and $[e_i, e_j] = 0$ at p; then at p the derivative along e_i of a function $\phi(e_j)$ is $(\nabla_{e_i}\phi)(e_j)$ for any form ϕ. By Lemma 43.3.4(3), $\delta\alpha = -\sum_i (\nabla_{e_i}\alpha)(e_i)$ near p, so at p

$$(d\delta\alpha)(e_j) = e_j(\delta\alpha) = -\sum_i (\nabla_{e_j}\nabla_{e_i}\alpha)(e_i).$$

By the same lemma and (1), $(\delta d\alpha)(e_j) = -\sum_i e_i\big(d\alpha(e_i,e_j)\big) = -\sum_i (\nabla_{e_i}\nabla_{e_i}\alpha)(e_j) + \sum_i (\nabla_{e_i}\nabla_{e_j}\alpha)(e_i)$ at p. Adding,

$$(\Delta\alpha)(e_j) = -\sum_i (\nabla_{e_i}\nabla_{e_i}\alpha)(e_j) + \sum_i \big((\nabla_{e_i}\nabla_{e_j} - \nabla_{e_j}\nabla_{e_i})\alpha\big)(e_i).$$

The first sum is $(\nabla^*\nabla\alpha)(e_j)$ at p. Since $[e_i, e_j] = 0$ at p, the operator in the second sum is the curvature $R^*(e_i, e_j)$ of the connection on 1-forms, and $(R^*(X,Y)\alpha)(Z) = -\alpha(R(X,Y)Z)$: applying $XY - YX - [X,Y]$, which is zero on functions, to $\alpha(Z)$ and expanding with the Leibniz rule, the terms $(\nabla_X\alpha)(\nabla_Y Z) + (\nabla_Y\alpha)(\nabla_X Z)$ cancel and what remains is $(R^*(X,Y)\alpha)(Z) + \alpha(R(X,Y)Z) = 0$. With $A = \alpha^\sharp$ the second sum is

$$-\sum_i \langle A, R(e_i,e_j)e_i\rangle = -\sum_i R(e_i,e_j,e_i,A) = \sum_i R(e_i,e_j,A,e_i) = \mathrm{Ric}(e_j, A),$$

by the antisymmetry of R in its last two arguments (Proposition 40.8.2) and Definition 40.8.1. For the integral formula, at p

$$\langle \nabla^*\nabla\alpha, \alpha\rangle = -\sum_i \langle \nabla_{e_i}\nabla_{e_i}\alpha, \alpha\rangle = \sum_i |\nabla_{e_i}\alpha|^2 - \sum_i e_i\langle \nabla_{e_i}\alpha, \alpha\rangle = \sum_i |\nabla_{e_i}\alpha|^2 - \operatorname{div} W$$

with $W = \sum_i \langle \nabla_{e_i}\alpha, \alpha\rangle e_i$, the vector field with $\langle W, X\rangle = \langle \nabla_X\alpha, \alpha\rangle = \frac{1}{2}X|\alpha|^2$. Integrating with Lemma 43.3.4(2) gives $(\nabla^*\nabla\alpha, \alpha) = \|\nabla\alpha\|^2$, and $\langle \mathrm{Ric}(A, -), \alpha\rangle = \mathrm{Ric}(A, A)$. □

Theorem 43.3.6 (Bochner)**.** *Let (M, g) be a compact oriented Riemannian manifold with* $\mathrm{Ric} \geq 0$*. Then every harmonic* 1*-form is parallel, and $b_1(M) \leq n$. If moreover* $\mathrm{Ric} > 0$ *at some point, then $b_1(M) = 0$.*

Proof. By Proposition 43.3.5, for α harmonic,

$$0 = (\Delta\alpha, \alpha) = \|\nabla\alpha\|^2 + \int_M \mathrm{Ric}(\alpha^\sharp, \alpha^\sharp)\,\mathrm{vol},$$

where $\alpha^\sharp$ is the vector field dual to α. Both terms are non-negative, hence both vanish: $\nabla\alpha = 0$, so α is parallel and determined by its value at one point, giving $\dim \mathcal{H}^1 \leq n$; and $\mathrm{Ric}(\alpha^\sharp, \alpha^\sharp) = 0$ everywhere. If $\mathrm{Ric} > 0$ at p and $\alpha \neq 0$, then α is nowhere zero (being parallel and nonzero somewhere), so $\mathrm{Ric}(\alpha^\sharp, \alpha^\sharp)(p) > 0$, a contradiction; hence $\mathcal{H}^1 = 0$ and $b_1(M) = 0$ by Theorem 43.2.5. □

Example 43.3.7.

1. On the round sphere S^n with $n \geq 2$, $\mathrm{Ric} > 0$, so $b_1 = 0$, which agrees with Example 37.2.9.
2. On a flat torus the harmonic forms are those with constant coefficients (Example 40.15.5), so Theorem 43.2.5 recovers $b_k(T^n) = \binom{n}{k}$.
3. A compact oriented manifold with a metric of positive Ricci curvature has $b_1 = 0$; by Bochner's argument applied to the other Laplacians, positivity assumptions on curvature produce vanishing theorems in all degrees. In the complex setting the same mechanism gives the Kodaira vanishing theorem (Chapter 50).

Both bounds of Theorem 43.3.6 are sharp. The flat torus T^n has $\mathrm{Ric} = 0$ and $b_1 = n$, its harmonic 1-forms dx_i being parallel. The product $S^1 \times S^{n-1}$ ($n \geq 3$) of a circle and a round sphere has $\mathrm{Ric} \geq 0$, with $\mathrm{Ric}(\partial_\theta, \partial_\theta) = 0$ along the circle factor, and $b_1 = 1$: the harmonic form $d\theta$ is parallel, and $\mathrm{Ric}(\partial_\theta, \partial_\theta) = 0$ is exactly what the proof requires.

Remark 43.3.8. The proof of Theorem 43.2.2 used only that $(\Omega^\bullet, d)$ is a complex of sections of vector bundles whose Laplacian is elliptic: an *elliptic complex*. The same argument applies verbatim to any elliptic complex, in particular to the Dolbeault complex $(\Omega^{p,\bullet}, \bar\partial)$ on a compact complex manifold, giving the isomorphism between Dolbeault cohomology and $\bar\partial$-harmonic forms (Chapter 46), which is how the Hodge decomposition of Chapter 48 is proved. There is also a proof of the Hodge theorem by the heat equation, which constructs the harmonic projection as $\lim_{t\to\infty} e^{-t\Delta}$ and yields the local index formula [LM89]; for functions this limit is computed in Exercise 42.6.8.

43.4 Exercises

Exercise 43.4.1. Show that on a compact connected oriented Riemannian n-manifold, $\mathcal{H}^0 = \mathbb{R}$ and $\mathcal{H}^n = \mathbb{R}\,\mathrm{vol}_g$, and deduce $b_0 = b_n = 1$.

Solution. A harmonic function has $df = 0$, hence is constant; apply $*$ for the top degree (Corollary 43.2.6). □

Exercise 43.4.2. Show that the Green operator is self-adjoint, that it is compact as an operator on L^2, and that $\alpha \mapsto H\alpha$ is the orthogonal projection onto $\mathcal{H}^k$ with $H = I - \Delta G$.

Solution. $H = I - \Delta G$ is the identity $\alpha = H\alpha + \Delta G\alpha$ of Theorem 43.2.2. For self-adjointness write $\alpha = H\alpha + \Delta a$ and $\beta = H\beta + \Delta b$ with $a = G\alpha$, $b = G\beta$ orthogonal to $\mathcal{H}^k$. Then $(G\alpha, \beta) = (a, H\beta + \Delta b) = (a, \Delta b)$, since $a \perp \mathcal{H}^k$, and likewise $(\alpha, G\beta) = (\Delta a, b)$; these are equal

because Δ is formally self-adjoint. For compactness, use an orthonormal basis of eigenforms u_k of Δ (Theorem 42.5.7): G kills the u_k with eigenvalue 0 and maps u_k to $\lambda_k^{-1}u_k$ otherwise. As $\lambda_k \to \infty$, G is the norm limit of the finite rank operators obtained by truncating this diagonal operator, so it is compact (Proposition 30.6.2), and it extends from smooth forms to L^2. □

Exercise 43.4.3. Give an example of a compact manifold and two metrics for which the harmonic representatives of a fixed cohomology class differ. Why can the two metrics not be conformal to each other when the manifold is a surface and the class has degree 1?

Solution. On the torus $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ take the flat metric $dx^2 + dy^2$, for which the harmonic representative of $[dx]$ is dx, and the metric $g' = e^{2\phi(x)}dx^2 + dy^2$ with ϕ smooth, 1-periodic and non-constant. An oriented orthonormal coframe for g' is $(e^{\phi}dx, dy)$, so its Hodge star maps $e^{\phi}dx \mapsto dy$ and $dy \mapsto -e^{\phi}dx$. The form $\alpha = c\, e^{\phi(x)}dx$ with $c = (\int_0^1 e^{\phi}\, dx)^{-1}$ is closed (its coefficient depends only on x) and $*'\alpha = c\, dy$ is closed, so α is g'-harmonic; its integrals over the two circles $y = 0$ and $x = 0$ are 1 and 0, as for dx, so it lies in the class $[dx]$ (on the torus a class in H^1 is determined by these two integrals, Exercise 37.5.1). It differs from dx because ϕ is not constant; equivalently, dx is not g'-harmonic, since $*'dx = e^{-\phi}dy$ has $d(e^{-\phi}dy) = -\phi' e^{-\phi}dx \wedge dy \neq 0$. For conformal metrics on a surface this cannot happen, by Proposition 43.2.8; and g' is indeed not conformal to the flat metric, as dx and dy have different g'-lengths where $\phi \neq 0$, while they remain orthogonal. □

Exercise 43.4.4. Let M be a compact orientable surface of genus g. Using Theorem 43.2.5 and Corollary 43.2.6, show that $\dim \mathcal{H}^1 = 2g$ and that the intersection pairing on $\mathcal{H}^1$ is a nondegenerate alternating form.

Solution. $\dim \mathcal{H}^1 = b_1(M) = 2g$ by Theorem 43.2.5 and the computation of the Betti numbers of Σ_g (Example 21.7.5). The pairing $(\alpha, \beta) \mapsto \int_M \alpha \wedge \beta$ is bilinear and alternating, since $\alpha \wedge \beta = -\beta \wedge \alpha$ for 1-forms, and it is nondegenerate on $\mathcal{H}^1$: for $\alpha \neq 0$ harmonic, $*\alpha$ is harmonic (Corollary 43.2.6) and $\int_M \alpha \wedge *\alpha = \|\alpha\|^2 > 0$. (A nondegenerate alternating form forces the dimension to be even, in accordance with $2g$.) □

Exercise 43.4.5. Let $V^0 = \mathbb{R}$, $V^1 = \mathbb{R}^2$, $V^2 = \mathbb{R}$ with the standard inner products, $d\colon V^0 \to V^1$, $t \mapsto (t, t)$, and $d = 0\colon V^1 \to V^2$. Compute the cohomology, the Laplacians and their kernels in each degree, and verify Lemma 43.2.1. Which element of the cohomology class of $(1, 0)$ has the smallest norm?

Solution. $H^0 = \ker(t \mapsto (t, t)) = 0$, $H^1 = \mathbb{R}^2/\mathbb{R}(1, 1) \cong \mathbb{R}$ and $H^2 = V^2 = \mathbb{R}$. The adjoint of $d\colon V^0 \to V^1$ is $\delta(a, b) = a + b$, and the adjoint of the zero map is zero. So $\Delta = \delta d = 2$ on V^0, with kernel 0; on V^1, $\Delta(a, b) = d\delta(a, b) = (a + b, a + b)$, with kernel the line $a + b = 0$; and $\Delta = 0$ on V^2, with kernel V^2. The dimensions 0, 1, 1 of the kernels are those of the cohomology. The class of $(1, 0)$ is $(1, 0) + \mathbb{R}(1, 1)$, and $(1, 0) = (\frac{1}{2}, -\frac{1}{2}) + d(\frac{1}{2})$ with $(\frac{1}{2}, -\frac{1}{2})$ harmonic: it is the element of the class of smallest norm, $\frac{1}{\sqrt{2}} < 1$, the orthogonal projection of $(1, 0)$ to the line $a + b = 0$. □

Exercise 43.4.6. Verify the Bochner–Weitzenböck formula for $\alpha = dz|_{S^2}$ on the round unit sphere $S^2 \subseteq \mathbb{R}^3$. Use that $\Delta z = 2z$ (Exercise 42.6.6), so $\Delta\alpha = d\Delta z = 2\alpha$, and compute $\nabla^*\nabla\alpha$ from the Hessian $\nabla dz = -z\, g$ of the restriction of a linear function to the sphere.

Solution. For a function f, ∇df is the Hessian $(X, Y) \mapsto (\nabla_X df)(Y) = X(Y(f)) - (\nabla_X Y)(f)$. For the restriction of z and tangent fields X, Y on S^2, $Y(z) = \langle Y, e_3\rangle$ because z is linear, so $X(Y(z)) = \langle \overline{\nabla}_X Y, e_3\rangle$, where $\overline{\nabla}$ is the derivative in $\mathbb{R}^3$. By the Gauss formula (Theorem 40.11.2), $\overline{\nabla}_X Y = \nabla_X Y + \mathrm{II}(X, Y)$, and on the unit sphere the normal part is $\mathrm{II}(X, Y) = \langle \overline{\nabla}_X Y, x\rangle x = -\langle Y, \overline{\nabla}_X x\rangle x =$

$-\langle X, Y\rangle x$ at the point x. Hence $(\nabla_X dz)(Y) = \langle \overline{\nabla}_X Y - \nabla_X Y, e_3\rangle = -\langle X, Y\rangle\langle x, e_3\rangle = -\langle X, Y\rangle z$. Hence $\nabla_{e_i}\alpha = -z\, e^i$ in an orthonormal frame, and $\nabla^*\nabla\alpha = -\sum_i \nabla_{e_i}(-z\, e^i)$, which at a point with $\nabla e_i = 0$ is $\sum_i e_i(z)\, e^i = dz = \alpha$. Since Ric $= g$ on the unit sphere, the formula reads $\Delta\alpha = \alpha + \alpha = 2\alpha$, as it should. □

Part VII

Complex Geometry

Chapter 44

Complex Manifolds

44.1 Introduction

A complex manifold is a manifold with holomorphic transition functions. The rigidity of holomorphic functions makes these spaces very different from smooth manifolds: compact complex manifolds have only constant holomorphic functions, and their geometry is recorded instead by the cohomology of holomorphic bundles and by the decomposition of differential forms into types, which is the subject of Hodge theory. This chapter sets up complex manifolds, meromorphic forms on Riemann surfaces and the residue theorem on a compact Riemann surface (the sum of the residues of a meromorphic 1-form is zero), the uniformisation of Riemann surfaces with its consequences (the hyperbolic metric, Picard's theorem), the operators ∂ and $\bar{\partial}$, analytic subvarieties and blow-ups. References: [GH78, Chapter 0], [Huy05, Chapter 2], [Voi02b, Chapter 3].

Prerequisites. Chapter 33 (Smooth Manifolds), Chapter 31 (Several Complex Variables), Chapter 12 (Multilinear Algebra) and Chapter 35 (Vector Bundles).

On a first reading. Section 44.4, on the uniformisation of Riemann surfaces, is not used later and can be skipped.

44.2 Complex manifolds

Definition 44.2.1. A *complex manifold* of dimension n is a topological $2n$-manifold with an atlas of charts $\varphi_i \colon U_i \to \mathbb{C}^n$ whose transition maps are holomorphic; a map of complex manifolds is *holomorphic* if it is so in charts, and a *biholomorphism* is a bijective holomorphic map with holomorphic inverse. The sheaf of holomorphic functions is $\mathcal{O}_X$, making $(X, \mathcal{O}_X)$ a locally ringed space. Every complex manifold is in particular a smooth $2n$-manifold, canonically oriented by the orientation of $\mathbb{C}^n$ (transition maps have Jacobian determinant $|\det_{\mathbb{C}}|^2 > 0$).

Example 44.2.2.

1. Open subsets of $\mathbb{C}^n$; $\mathbb{CP}^n$ with the charts of Example 33.2.9, whose transition maps are rational and nowhere singular.

2. A *complex torus* $X = \mathbb{C}^n/\Lambda$ for a lattice $\Lambda \cong \mathbb{Z}^{2n}$; these are compact, and for $n = 1$ they are the elliptic curves. Two complex tori are rarely biholomorphic even when they are diffeomorphic, which is the first appearance of moduli (Chapter 66).

3. Grassmannians $\mathrm{Gr}(k, n)$ of k-planes in $\mathbb{C}^n$, with charts given by graphs of linear maps; they are compact complex manifolds of dimension $k(n-k)$.

4. A smooth analytic hypersurface $\{f = 0\} \subseteq \mathbb{CP}^n$, for f a homogeneous polynomial with $df \neq 0$ on the zero set, is a compact complex manifold of dimension $n-1$, by the implicit function theorem (Theorem 27.11.3, valid for holomorphic maps by Remark 27.11.5).

5. Not every even-dimensional smooth manifold admits a complex structure: S^4 does not (Exercise 39.8.4), and whether S^6 does is open.

Proposition 44.2.3. *A holomorphic function on a compact connected complex manifold is constant. Consequently a compact complex manifold has no nonconstant holomorphic map to $\mathbb{C}^n$, and $\mathcal{O}_X(X) = \mathbb{C}$.*

Proof. $|f|$ attains a maximum at some p; in a chart around p, the maximum principle (Proposition 31.2.3) makes f constant near p, and the set where f equals $f(p)$ is open and closed by the identity theorem. □

Corollary 44.2.4. *Every compact complex submanifold of $\mathbb{C}^n$ is a finite set of points. In particular no compact complex manifold of positive dimension, such as $\mathbb{CP}^m$ or a complex torus, embeds holomorphically in $\mathbb{C}^n$, in contrast with the Whitney embedding theorem for smooth manifolds (Theorem 33.9.9).*

Proof. The coordinate functions restrict to holomorphic functions on each connected component, which are constant by Proposition 44.2.3; so each component is a point, and a compact discrete set is finite. □

Theorem 44.2.5. *Let $X = \mathbb{C}^n/\Lambda$ and $Y = \mathbb{C}^m/\Lambda'$ be complex tori. Every holomorphic map $f\colon X \to Y$ is induced by an affine map $\mathbb{C}^n \to \mathbb{C}^m$, $z \mapsto Az + b$, with A a complex $m \times n$ matrix such that $A\Lambda \subseteq \Lambda'$. In particular f is a group homomorphism followed by a translation, and $X \cong Y$ as complex manifolds if and only if $n = m$ and $\Lambda' = A\Lambda$ for some $A \in \mathrm{GL}_n(\mathbb{C})$.*

Proof. The projection $\mathbb{C}^n \to X$ is a universal covering, as $\mathbb{C}^n$ is simply connected and Λ acts freely and properly (Theorem 20.4.2). By the lifting criterion (Proposition 19.6.1), $f \circ \pi$ lifts to a continuous map $F\colon \mathbb{C}^n \to \mathbb{C}^m$, holomorphic because π' is a local biholomorphism. For $\lambda \in \Lambda$, $F(z+\lambda) - F(z) \in \Lambda'$ for all z, and by continuity and discreteness of Λ' it is a constant $c(\lambda)$. Hence each partial derivative $\partial F_j/\partial z_k$ is Λ-periodic, so it is a holomorphic function on the compact manifold X, hence constant (Proposition 44.2.3). So $F(z) = Az + b$, and $A\lambda = c(\lambda) \in \Lambda'$. If f is a biholomorphism, apply this to f and f^{-1}: their lifts compose to a lift of the identity, which is a translation by an element of Λ, so A is invertible with $A\Lambda = \Lambda'$ and $n = m$; the converse is clear. □

Corollary 44.2.6. *Every one-dimensional complex torus is biholomorphic to $E_\tau = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$ with $\operatorname{Im}\tau > 0$, and $E_\tau \cong E_{\tau'}$ if and only if $\tau' = \frac{a\tau+b}{c\tau+d}$ for some $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$.*

Proof. A lattice $\mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ is carried by $z \mapsto z/\omega_1$ to $\mathbb{Z} + \mathbb{Z}\tau$ with $\tau = \omega_2/\omega_1 \notin \mathbb{R}$, and replacing ω_2 by $-\omega_2$ if necessary, $\operatorname{Im}\tau > 0$. By Theorem 44.2.5, $E_\tau \cong E_{\tau'}$ if and only if $\alpha(\mathbb{Z} + \mathbb{Z}\tau) = \mathbb{Z} + \mathbb{Z}\tau'$ for some $\alpha \in \mathbb{C}^\times$. Then $\alpha\tau = a\tau' + b$ and $\alpha = c\tau' + d$ for an integer matrix, invertible over $\mathbb{Z}$ because the inverse change of basis is also integral; so $\tau = \frac{a\tau'+b}{c\tau'+d}$, and the determinant is $+1$ because both τ and τ' lie in the upper half plane, as $\operatorname{Im}\frac{a\tau'+b}{c\tau'+d} = \frac{(ad-bc)\operatorname{Im}\tau'}{|c\tau'+d|^2}$. Conversely such a matrix gives the lattice isomorphism. The relation is symmetric, so τ' is also of this form in terms of τ. □

Counterexample 44.2.7. Diffeomorphic complex manifolds need not be biholomorphic: all E_τ are diffeomorphic to $S^1 \times S^1$, but by Corollary 44.2.6 $E_i \ncong E_{2i}$, since $\frac{ai+b}{ci+d} = 2i$ with integers of determinant 1 would give, comparing imaginary parts, $|ci + d|^2 = \frac{1}{2}$, impossible for integers c, d.

Proposition 44.2.8. *Let $f\colon X \to Y$ be a nonconstant holomorphic map of connected Riemann surfaces (complex manifolds of dimension 1), and $p \in X$. There are holomorphic coordinates z centred at p and w centred at $f(p)$ in which f is $z \mapsto z^k$ for a unique integer $k = e_p \geq 1$, the* ramification index. *Consequently f is open, its fibres are discrete, and if X is compact and Y connected, then f is surjective and has finite fibres.*

Proof. Choose coordinates z at p and w at $f(p)$ with $z(p) = 0$, $w(f(p)) = 0$. In them f is a holomorphic function of one variable with $f(0) = 0$, not identically zero near 0 by the identity theorem (Theorem 29.4.8) and connectedness of X. So $f(z) = z^k g(z)$ with $k \geq 1$ and $g(0) \neq 0$. On a small disc where g has no zeros, g'/g has a primitive L (Corollary 29.3.2), and $(ge^{-L})' = 0$, so $g = ce^{L}$ with $c \neq 0$; writing $c = e^{c'}$, the function $h = e^{(L+c')/k}$ satisfies $h^k = g$. The new coordinate $\tilde{z} = zh(z)$ has derivative $h(0) \neq 0$ at 0, so it is a local biholomorphism (Theorem 29.5.12, or Proposition 31.2.5), and $f = \tilde{z}^k$. The integer k is determined by f, as the number of preimages near p of points $w \neq 0$ near 0. The map $z \mapsto z^k$ is open with discrete fibres, so f is. If X is compact, $f(X)$ is compact, hence closed, and open, so $f(X) = Y$ when Y is connected; the fibres are discrete and closed in a compact space, hence finite. □

Proposition 44.2.9. *Let $f\colon X \to Y$ be a nonconstant holomorphic map of compact connected Riemann surfaces. The number $\deg f = \sum_{p \in f^{-1}(q)} e_p$ does not depend on $q \in Y$. It equals the number of preimages of every point q outside the finite set $f(\{p : e_p > 1\})$ of* branch points.

Proof. The ramification points, where f' vanishes in coordinates, are the zeros of a holomorphic function locally, hence discrete, hence finitely many in the compact X. Fix q_0 with preimages $p_1, \dots, p_r$ (Proposition 44.2.8), and choose disjoint coordinate discs D_i around p_i on which f is $z \mapsto z^{e_i}$ onto a disc D around q_0. The set $f(X \setminus \bigcup_i D_i)$ is compact and does not contain q_0, so there is a neighbourhood $D' \subseteq D$ of q_0 avoiding it. For $q \in D' \setminus \{q_0\}$, all preimages of q lie in $\bigcup_i D_i$, and in D_i the equation $z^{e_i} = w(q) \neq 0$ has exactly e_i solutions, each with ramification index 1. So the weighted count equals $\sum_i e_i$ on D'. Hence $q \mapsto \deg_q f$ is locally constant, hence constant on the connected Y. □

Corollary 44.2.10. *Every holomorphic map from a simply connected compact connected complex manifold, such as $\mathbb{CP}^1$, to a complex torus is constant. In particular $\mathbb{CP}^1$ is not biholomorphic, and admits no nonconstant holomorphic map, to an elliptic curve.*

Proof. Let $f\colon X \to T = \mathbb{C}^m/\Lambda$. The projection $\mathbb{C}^m \to T$ is a covering (Theorem 20.4.2), so by the lifting criterion (Proposition 19.6.1) f lifts to a continuous $F\colon X \to \mathbb{C}^m$, which is holomorphic because the projection is a local biholomorphism. By Proposition 44.2.3 each component of F is constant, so f is constant. The projective line is covered by the two charts U_0, U_1, each homeomorphic to $\mathbb{C}$, with path-connected intersection $\mathbb{C}^\times$; by the Seifert–van Kampen theorem (Theorem 19.5.1) $\pi_1(\mathbb{CP}^1)$ is a quotient of the free product of two trivial groups, hence trivial. □

44.3 Meromorphic forms and the residue theorem

On a compact Riemann surface, Stokes' theorem turns the residue theorem of complex analysis into a global constraint: the residues of a meromorphic 1-form add up to zero. This is the simplest case of the *global residue theorem* [GH78, Chapter 5], and it has quick consequences: a meromorphic function has as many zeros as poles, a complex torus carries no meromorphic function with a single simple pole, and the roots of a polynomial satisfy identities that are not obvious from the polynomial.

Definition 44.3.1. Let X be a Riemann surface. A *meromorphic* 1*-form* ω on X is given in every holomorphic chart z by an expression $f(z)\,dz$ with f meromorphic (Definition 29.5.2), such that for two charts z and w the expressions $f(z)\,dz$ and $g(w)\,dw$ agree: $g(w) = f(z(w))\,z'(w)$. It is *holomorphic* if all the f are. The *order* $\operatorname{ord}_p \omega$ is $\operatorname{ord}_0 f$ in a chart centred at p, and the *residue* $\operatorname{Res}_p \omega$ is the residue at 0 of f, that is, the coefficient of z^{-1} in its Laurent expansion.

Since $z'(w)$ does not vanish, the order does not depend on the chart; the residue does not either, by the lemma below. For a meromorphic function h, $dh = h'(z)\,dz$ is a meromorphic 1-form, and so is dh/h if $h \not\equiv 0$. On $\mathbb{CP}^1$, with the charts z and $w = 1/z$, the form dz becomes $dz = -dw/w^2$: it has a pole of order 2 at ∞ and residue 0 there, while $dz/z = -dw/w$ has residue 1 at 0 and -1 at ∞.

Away from its poles, a meromorphic 1-form is a smooth complex-valued 1-form, $\omega = \alpha + i\beta$ with real 1-forms α and β, and line integrals and integrals over manifolds of such forms are taken componentwise. It is *closed*: in a chart, with f holomorphic, the Cauchy–Riemann equations (Proposition 29.2.3) give $\partial_x f = f'$ and $\partial_y f = i f'$, so $df = f'\,(dx + i\,dy) = f'\,dz$ and $d(f\,dz) = df \wedge dz = f'\,dz \wedge dz = 0$. Holomorphic charts preserve the orientation (the Jacobian determinant of a holomorphic map is $|f'|^2 \geq 0$), so a Riemann surface is canonically oriented, and the boundary of a coordinate disc is traversed counterclockwise in the chart.

Lemma 44.3.2. *Let ω be a meromorphic* 1*-form on a Riemann surface X and $p \in X$. For every closed coordinate disc D around p (the preimage of a closed disc under a holomorphic chart) whose only possible pole is p,*

$$\operatorname{Res}_p \omega = \frac{1}{2\pi i} \int_{\partial D} \omega.$$

In particular $\operatorname{Res}_p \omega$ does not depend on the chart.

Proof. If $D = \{|z| \leq r\}$ in a chart z centred at p with $\omega = f\,dz$, the line integral of ω over ∂D is the complex path integral $\int_{|z|=r} f(z)\,dz$, which is $2\pi i \operatorname{Res}_0 f$ by the residue theorem (Theorem 29.5.8). Now let D' be a closed coordinate disc around p for another chart, with no pole other than p. For r small the disc $D_r = \{|z| \leq r\}$ lies in the interior of D', and D' minus the interior of D_r is a compact manifold with boundary $\partial D' \cup \partial D_r$, the second circle with the opposite orientation. As ω is closed there, Stokes' theorem (Theorem 36.4.8) gives $\int_{\partial D'} \omega = \int_{\partial D_r} \omega$. □

Theorem 44.3.3 (Residue theorem on a compact Riemann surface)**.** *Let X be a compact Riemann surface and ω a meromorphic* 1*-form on X. Then ω has finitely many poles, and*

$$\sum_{p \in X} \operatorname{Res}_p \omega = 0.$$

Proof. In a chart the poles of a meromorphic function are isolated, so the poles of ω form a discrete closed subset of the compact X, hence a finite set $\{p_1, \dots, p_k\}$. Choose pairwise disjoint closed coordinate discs D_j around the p_j. Then $X' = X \setminus \bigcup_j \operatorname{int} D_j$ is a compact manifold with boundary $\bigcup_j \partial D_j$, each circle with the orientation opposite to its orientation as the boundary of D_j, and ω is a closed form on X'. By Stokes' theorem and Lemma 44.3.2,

$$0 = \int_{X'} d\omega = \int_{\partial X'} \omega = -\sum_{j=1}^{k} \int_{\partial D_j} \omega = -2\pi i \sum_{j=1}^{k} \operatorname{Res}_{p_j} \omega,$$

which is the claim. □

The proof is the same as that of the residue theorem in the plane, with one difference: on a compact surface there is no "outside", so the integral over the boundary of the complement of the discs is not the integral over a large circle but zero. For $X = \mathbb{CP}^1$ the point ∞ plays the role of the outside: the theorem says that the sum of the residues of a rational form in $\mathbb{C}$ equals minus its residue at ∞.

Corollary 44.3.4. *Let f be a nonconstant meromorphic function on a compact connected Riemann surface X. Then $\sum_p \operatorname{ord}_p f = 0$: f has as many zeros as poles, counted with multiplicity. Hence f takes every value $c \in \mathbb{C}$ the same number of times, counted with multiplicity, namely the number of its poles.*

Proof. At a point p, in a chart centred at p, write $f = z^k g$ with $k = \operatorname{ord}_p f$ and g holomorphic and nonvanishing near 0. Then $df/f = k\,dz/z + dg/g$, and dg/g is holomorphic near 0, so $\operatorname{Res}_p(df/f) = \operatorname{ord}_p f$. Since f is not identically zero (it is nonconstant on the connected X, so not zero on any open set by the identity theorem), df/f is a meromorphic 1-form, and Theorem 44.3.3 gives $\sum_p \operatorname{ord}_p f = 0$. Apply this to $f - c$, which has the same poles as f. □

This is also Proposition 44.2.9 applied to f regarded as a holomorphic map to $\mathbb{CP}^1$; the residue theorem gives it without the local analysis of that proof.

Corollary 44.3.5. *Let X be a compact Riemann surface with a holomorphic 1-form η that does not vanish at $p \in X$. Then no meromorphic function on X is holomorphic on $X \setminus \{p\}$ with a simple pole at p. In particular a meromorphic function on a complex torus $\mathbb{C}/\Lambda$ that is not constant has at least two poles, counted with multiplicity.*

Proof. Suppose f were such a function. In a chart centred at p, $f = c/z +$ (holomorphic) with $c \neq 0$ and $\eta = h(z)\,dz$ with $h(0) \neq 0$. Then $f\eta$ is holomorphic except at p, where $\operatorname{Res}_p(f\eta) = c\,h(0) \neq 0$, contradicting Theorem 44.3.3. On $\mathbb{C}/\Lambda$ the form dz is invariant under the translations by Λ, so it descends to a holomorphic 1-form without zeros. A nonconstant meromorphic function has a pole, since otherwise it would be a holomorphic function on a compact connected manifold, hence constant (Proposition 44.2.3); and a single simple pole is excluded by the first part, applied at that pole. □

On $\mathbb{CP}^1$ the function z has a single simple pole, at ∞, so by the corollary $\mathbb{CP}^1$ has no holomorphic 1-form that is nonzero at ∞. In fact it has no nonzero holomorphic 1-form at all: such a form is $f(z)\,dz$ with f entire, and $f(z)\,dz = -f(1/w)w^{-2}\,dw$ is holomorphic at $w = 0$ only if $f(z) = O(|z|^{-2})$ as $z \to \infty$; then f is bounded, hence constant by Liouville's theorem, hence 0.

Corollary 44.3.6 (Euler–Jacobi formula in one variable)**.** *Let $f \in \mathbb{C}[z]$ have degree $d \geq 2$ and distinct roots $r_1, \dots, r_d$, and let $g \in \mathbb{C}[z]$ have degree at most $d - 2$. Then*

$$\sum_{i=1}^{d} \frac{g(r_i)}{f'(r_i)} = 0.$$

Proof. Apply Theorem 44.3.3 to $\omega = g(z)\,dz/f(z)$ on $\mathbb{CP}^1$. In $\mathbb{C}$ its poles are the roots of f, which are simple since $f'(r_i) \neq 0$, and there $\operatorname{Res}_{r_i} \omega = \lim_{z \to r_i}(z - r_i)g(z)/f(z) = g(r_i)/f'(r_i)$. At ∞, with $w = 1/z$, write $g(1/w) = w^{-e}G(w)$ and $f(1/w) = w^{-d}F(w)$ with polynomials F, G, where $e = \deg g$ and $F(0) \neq 0$ is the leading coefficient of f. Then $\omega = -w^{d-e-2}\frac{G(w)}{F(w)}\,dw$, which is holomorphic at $w = 0$ because $d - e - 2 \geq 0$ (if $g = 0$ there is nothing to prove). So the residues at the roots add up to zero. □

For example, for $f = z^3 - 1$ the roots are the cube roots of unity ζ, with $f'(\zeta) = 3\zeta^2$, and the formula with $g = 1$ and $g = z$ says $\sum_\zeta \zeta^{-2} = 0$ and $\sum_\zeta \zeta^{-1} = 0$; for $g = z^2$, which has degree $d - 1$, the residue at ∞ is -1 and indeed $\sum_\zeta \frac{\zeta^2}{3\zeta^2} = 1$. For $f = z^2 - 1$ and $g = 1$: $\frac{1}{2} + \frac{1}{-2} = 0$.

In higher dimension the residue of a meromorphic n-form is attached to a point where n divisors meet: if near P the divisors are $\{f_1 = 0\}, \dots, \{f_n = 0\}$, meeting only at P, and $\omega = g\, dz_1 \wedge \cdots \wedge dz_n/(f_1 \cdots f_n)$ with g holomorphic, the residue is $\operatorname{Res}_P \omega = \frac{1}{(2\pi i)^n} \int_\Gamma \omega$ over the real n-cycle $\Gamma = \{|f_1| = \cdots = |f_n| = \varepsilon\}$, suitably oriented.

Theorem 44.3.7 (Global residue theorem)**.** *Let M be a compact complex manifold of dimension n, $D_1, \dots, D_n$ effective divisors whose intersection is a finite set of points, and ω a meromorphic n-form on M whose poles are contained in $D_1 \cup \cdots \cup D_n$, of order at most one along each D_i (that is, ω is a holomorphic section of $\Omega^n_M(D_1 + \cdots + D_n)$). Then $\sum_{P \in D_1 \cap \cdots \cap D_n} \operatorname{Res}_P \omega = 0$.*

This is [GH78, Chapter 5]. For $n = 1$ and D_1 the polar divisor it is Theorem 44.3.3 for forms with simple poles. As there, the proof rests on Stokes' theorem, now applied on the complement of tubes around the divisors, together with a local analysis of the residue. Griffiths and Harris use it to prove the Cayley–Bacharach theorem: if two plane cubic curves meet in nine distinct points, every cubic through eight of them passes through the ninth.

44.4 Uniformisation of Riemann surfaces

A *Riemann surface* is a connected complex manifold of dimension 1. The uniformisation theorem reduces the study of all Riemann surfaces to three simply connected ones, the sphere, the plane and the disc, and to groups of automorphisms acting on them. Combined with the hyperbolic metric of the disc it shows that most Riemann surfaces, among them all compact ones of genus at least 2, carry a natural metric of constant curvature -1; and it gives a short proof of Picard's theorem. Throughout, $\mathbb{D} = \{|z| < 1\}$ and $\operatorname{Aut}(X)$ is the group of biholomorphisms of X to itself.

Lemma 44.4.1. *Let X be a Riemann surface.*

1. *If $p : Y \to X$ is a covering map with Y connected, then Y has a unique structure of Riemann surface for which p is a local biholomorphism, and the deck transformations are biholomorphisms.*
2. *If a group Γ acts on X freely and properly by biholomorphisms, then X/Γ has a unique structure of Riemann surface for which $X \to X/\Gamma$ is a local biholomorphism, and a map $f : X/\Gamma \to Z$ to a Riemann surface is holomorphic if and only if its composite with the projection is.*

Proof. These are the proofs of Theorem 33.7.1 and Example 33.7.2, with "holomorphic" in place of "smooth": the charts of Y (of X/Γ) are the charts of X composed with p on sheets over evenly covered chart domains (with local inverses of the projection), and their transition maps are transition maps of X, composed with the biholomorphisms of Γ in the case of X/Γ. Topologically, Y is a surface by Proposition 20.6.1(4), and X/Γ one by (3). A deck transformation τ satisfies $p \circ \tau = p$, so in these charts it is the identity map of coordinates. □

In particular the universal covering $\widetilde{X}$ of a Riemann surface (Theorem 19.6.5) is a simply connected Riemann surface, $\pi_1(X)$ acts on it freely and properly by biholomorphisms as the deck group (Proposition 19.6.3), and $X \cong \widetilde{X}/\pi_1(X)$.

Theorem 44.4.2 (Uniformisation theorem)**.** *Every simply connected Riemann surface is biholomorphic to exactly one of $\mathbb{CP}^1$, $\mathbb{C}$ and $\mathbb{D}$.*

This is due to Koebe and Poincaré (1907); see [For81]. The three surfaces are pairwise non-biholomorphic: $\mathbb{CP}^1$ is compact and the others are not, and a biholomorphism $\mathbb{C} \to \mathbb{D}$ would be a bounded nonconstant entire function (Exercise 29.8.1). For open subsets of $\mathbb{C}$ the theorem is the Riemann mapping theorem (Theorem 29.6.5); the general case constructs the biholomorphism from Green's functions, using Perron's method to solve the Dirichlet problem on the surface.

Lemma 44.4.3. *1.* $\mathrm{Aut}(\mathbb{CP}^1)$ *consists of the Möbius maps* $z \mapsto (az+b)/(cz+d)$, $ad-bc \neq 0$, *and each of them has a fixed point.*

2. $\mathrm{Aut}(\mathbb{C})$ *consists of the maps* $z \mapsto az+b$, $a \neq 0$; *those without fixed points are the translations* $z \mapsto z+b$, $b \neq 0$.

3. $\mathrm{Aut}(\mathbb{D})$ *consists of the maps* $z \mapsto \lambda(z-a)/(1-\bar{a}z)$ *with* $|\lambda| = 1$ *and* $a \in \mathbb{D}$, *and acts transitively on* $\mathbb{D}$.

Proof. (2) is Exercise 29.8.2, and $az+b = z$ has the solution $b/(1-a)$ when $a \neq 1$. (3) is Proposition 29.6.2; the map ϕ_{-a} sends 0 to a. (1) A Möbius map is holomorphic in the charts of $\mathbb{CP}^1$, with the Möbius inverse $z \mapsto (dz-b)/(-cz+a)$, so it is an automorphism, and the Möbius maps act transitively ($z \mapsto z+b$ and $z \mapsto 1/z$ move any point to any other). Given $f \in \mathrm{Aut}(\mathbb{CP}^1)$, choose a Möbius m with $m(f(\infty)) = \infty$; then $m \circ f$ restricts to an automorphism of $\mathbb{C} = \mathbb{CP}^1 \setminus \{\infty\}$, hence is $z \mapsto az+b$ by (2), and $f = m^{-1} \circ (m \circ f)$ is Möbius. If $c = 0$, ∞ is a fixed point; if $c \neq 0$, the equation $cz^2 + (d-a)z - b = 0$ has a root $z_0 \in \mathbb{C}$ (Corollary 29.4.7(2)), and $cz_0 + d \neq 0$, since at $z_0 = -d/c$ the left side is $(ad-bc)/c \neq 0$; so z_0 is a fixed point. □

Proposition 44.4.4 (The Poincaré metric)**.** *Let* $g_{\mathbb{D}} = 4|dz|^2/(1-|z|^2)^2$ *on* $\mathbb{D}$, *that is* $4(dx^2+dy^2)/(1-x^2-y^2)^2$.

1. (Schwarz–Pick) For every holomorphic $f : \mathbb{D} \to \mathbb{D}$,

$$\frac{|f'(z)|}{1-|f(z)|^2} \leq \frac{1}{1-|z|^2},$$

that is, $f^* g_{\mathbb{D}} \leq g_{\mathbb{D}}$: *holomorphic self-maps of the disc do not increase lengths of curves, hence distances.*

2. Every automorphism of $\mathbb{D}$ *is an isometry of* $g_{\mathbb{D}}$.

3. $g_{\mathbb{D}}$ *is complete, with constant curvature* -1.

Proof. (1) Fix $a \in \mathbb{D}$, let $b = f(a)$ and $\phi_a(z) = (z-a)/(1-\bar{a}z)$. A computation gives $\phi_a'(z) = (1-|a|^2)/(1-\bar{a}z)^2$, so $\phi_{-a}'(0) = 1-|a|^2$ and $\phi_b'(b) = 1/(1-|b|^2)$. The map $h = \phi_b \circ f \circ \phi_{-a}$ sends $\mathbb{D}$ to $\mathbb{D}$ and 0 to 0, so $|h'(0)| \leq 1$ by Schwarz's lemma (Lemma 29.6.1), and by the chain rule $h'(0) = f'(a)(1-|a|^2)/(1-|b|^2)$. At a point z, $f^* g_{\mathbb{D}}$ is $4|f'(z)|^2|dz|^2/(1-|f(z)|^2)^2$, so the inequality says $f^* g_{\mathbb{D}} \leq g_{\mathbb{D}}$, and lengths of curves decrease.

(2) Apply (1) to an automorphism and to its inverse: $f^* g_{\mathbb{D}} \leq g_{\mathbb{D}}$ and $g_{\mathbb{D}} = (f^{-1})^* f^* g_{\mathbb{D}}$ force equality.

(3) The curvature is -1 by Proposition 40.12.7 (Example 40.12.8(2)). For completeness, choose $\varepsilon > 0$ such that $\exp_0$ is defined on the ball of radius ε in $T_0\mathbb{D}$. Since $\mathrm{Aut}(\mathbb{D})$ acts transitively by isometries, and isometries map geodesics to geodesics, the same ε works at every point: every geodesic of unit speed is defined on an interval of length ε around each of its times. So a maximal unit speed geodesic defined on $[0, T)$ with $T < \infty$ could be continued beyond T from its point at time $T - \varepsilon/2$, a contradiction. Geodesics are defined for all time, and $g_{\mathbb{D}}$ is complete by Theorem 40.6.8. □

Theorem 44.4.5. *Every Riemann surface X is biholomorphic to $\widetilde{X}/\Gamma$, where $\widetilde{X} \in \{\mathbb{CP}^1, \mathbb{C}, \mathbb{D}\}$ is its universal covering and $\Gamma \cong \pi_1(X)$ is a group of automorphisms of $\widetilde{X}$ acting freely and properly. Moreover:*

1. *$\widetilde{X} = \mathbb{CP}^1$ if and only if $X \cong \mathbb{CP}^1$;*
2. *$\widetilde{X} = \mathbb{C}$ if and only if X is biholomorphic to $\mathbb{C}$, to $\mathbb{C}^\times$ or to a complex torus $\mathbb{C}/\Lambda$;*
3. *in all other cases $\widetilde{X} = \mathbb{D}$, and X carries a unique complete metric of constant curvature -1 for which the projection $\mathbb{D} \to X$ is a local isometry; in holomorphic coordinates it has the form $\lambda(z)|dz|^2$.*

Proof. The first statement is Theorem 44.4.2 and the remark after Lemma 44.4.1. (1) By Lemma 44.4.3(1) every automorphism of $\mathbb{CP}^1$ has a fixed point, so a group acting freely on it is trivial. (2) By Lemma 44.4.3(2), Γ consists of translations $z \mapsto z + \lambda$, and $\Lambda = \{\lambda\}$ is a subgroup of $\mathbb{C} = \mathbb{R}^2$. It is discrete: it is the orbit of 0, which is a fibre of the covering $\mathbb{C} \to X$. By Lemma 34.3.10, Λ is 0, $\mathbb{Z}\omega$ with $\omega \neq 0$, or a lattice. In the first case $X \cong \mathbb{C}$; in the second, $z \mapsto e^{2\pi i z/\omega}$ is a covering $\mathbb{C} \to \mathbb{C}^\times$ whose fibres are the cosets of $\mathbb{Z}\omega$, so it induces a bijective local biholomorphism $\mathbb{C}/\mathbb{Z}\omega \to \mathbb{C}^\times$; in the third X is a complex torus. Conversely these surfaces have universal covering $\mathbb{C}$. (3) The group $\Gamma \subseteq \mathrm{Aut}(\mathbb{D})$ acts by isometries of $g_{\mathbb{D}}$ (Proposition 44.4.4), so $g_{\mathbb{D}}$ descends to a metric on X making the projection a local isometry (Proposition 40.2.6), of curvature -1; it is complete because geodesics of X lift to geodesics of the complete $\mathbb{D}$ (as in the proof of Theorem 40.10.7), and in the holomorphic coordinates given by local inverses of the projection it is $4|dz|^2/(1-|z|^2)^2$. It is unique because the projection is a local diffeomorphism, so the metric on X is determined by its pullback $g_{\mathbb{D}}$. □

Corollary 44.4.6. *Let X be a compact Riemann surface of genus g (Theorem 20.8.6; X is orientable). If $g = 0$ then $X \cong \mathbb{CP}^1$; if $g = 1$ then X is a complex torus $\mathbb{C}/\Lambda$; and if $g \geq 2$ then $X \cong \mathbb{D}/\Gamma$, and X carries a metric of constant curvature -1, compatible with its complex structure, with area $4\pi(g-1)$.*

Proof. If $g = 0$, X is simply connected (Corollary 19.5.2), so X is its own universal covering, compact, hence $\mathbb{CP}^1$. If $g \geq 1$, $\pi_1(X)$ is nontrivial, as its abelianisation $H_1(X;\mathbb{Z})$ (Theorem 21.9.1) is $\mathbb{Z}^{2g}$ (Example 21.7.5), so $\widetilde{X} \neq \mathbb{CP}^1$ by Theorem 44.4.5(1). If $\widetilde{X} = \mathbb{D}$, the metric of curvature -1 has area A with $-A = 2\pi(2-2g)$ by the Gauss–Bonnet theorem (Theorem 40.8.11), so $A = 4\pi(g-1)$ and $g \geq 2$. If $\widetilde{X} = \mathbb{C}$, then X is compact among $\mathbb{C}$, $\mathbb{C}^\times$ and the tori, so it is a torus, of genus 1. Hence $g = 1$ forces $\widetilde{X} = \mathbb{C}$ and $g \geq 2$ forces $\widetilde{X} = \mathbb{D}$. □

Corollary 44.4.7 (Little Picard theorem)**.** *A nonconstant entire function takes every complex value with at most one exception.*

Proof. Suppose the entire function f omits two values; composing with an affine map we may assume they are 0 and 1, so $f : \mathbb{C} \to X = \mathbb{C} \setminus \{0,1\}$. The fundamental group of X is not abelian: $X = U \cup V$ with $U = \{\mathrm{Re}\, z < \frac{3}{4}\} \setminus \{0\}$ and $V = \{\mathrm{Re}\, z > \frac{1}{4}\} \setminus \{1\}$, where $U \cap V$ is a convex strip, hence simply connected, and U, V deformation retract onto the circles of radius $\frac{1}{4}$ around 0 and 1 (radially, along segments that stay in the convex half planes), so $\pi_1(U) \cong \pi_1(V) \cong \mathbb{Z}$ (Lemma 19.2.6 and Theorem 19.4.6) and $\pi_1(X) \cong \mathbb{Z} * \mathbb{Z}$ by Corollary 19.5.2, a non-abelian group (Example 19.5.4). Since X is not compact and $\pi_1(X)$ is not abelian, while $\mathbb{C}$, $\mathbb{C}^\times$ and the tori have abelian fundamental groups 0, $\mathbb{Z}$ and $\mathbb{Z}^2$, Theorem 44.4.5 gives a covering map $p : \mathbb{D} \to X$ that is a local biholomorphism. As $\mathbb{C}$ is simply connected, f lifts to a continuous $\tilde{f} : \mathbb{C} \to \mathbb{D}$ with $p \circ \tilde{f} = f$ (Proposition 19.6.1), which is holomorphic, being locally a local inverse of p composed with f. Then $\tilde{f}$ is a bounded entire function, hence constant (Corollary 29.4.7), and so is f. □

Remark 44.4.8. For compact surfaces the three cases match the sign of the Euler characteristic: $\mathbb{CP}^1$ carries the round metric of curvature 1 (Example 40.12.8(3)), complex tori the flat metric $|dz|^2$, and surfaces of genus $g \geq 2$ the hyperbolic metric, all compatible with the complex structure; the sign of the curvature is forced by the Gauss–Bonnet theorem. Schwarz–Pick also shows that the hyperbolic metric is intrinsic: a holomorphic map $f : X \to Y$ between Riemann surfaces covered by the disc lifts to a holomorphic map $\mathbb{D} \to \mathbb{D}$ of the universal coverings (Proposition 19.6.1), so it does not increase the lengths of curves for the hyperbolic metrics of Theorem 44.4.5(3).

44.5 Almost complex structures and forms of type (p, q)

Definition 44.5.1. An *almost complex structure* on a smooth manifold M is an endomorphism J of TM with $J^2 = -\mathrm{id}$. It makes each T_pM a complex vector space, and gives the decomposition $T_pM \otimes \mathbb{C} = T_p^{1,0} \oplus T_p^{0,1}$ into the $\pm i$ eigenspaces, with $T^{0,1} = \overline{T^{1,0}}$, and dually $\Lambda^k T_p^*M \otimes \mathbb{C} = \bigoplus_{p+q=k} \Lambda^{p,q}$ (Proposition 12.6.2). A complex manifold has a canonical almost complex structure, given in holomorphic coordinates $z^j = x^j + iy^j$ by $J\partial_{x^j} = \partial_{y^j}$, $J\partial_{y^j} = -\partial_{x^j}$; then $T^{1,0}$ has frame $\partial/\partial z^j$ and $\Lambda^{p,q}$ has frame $dz^I \wedge d\bar{z}^J$ with $|I| = p$, $|J| = q$. An almost complex structure is *integrable* if it arises from a complex structure.

Definition 44.5.2. On a complex manifold, $\Omega^{p,q}(X)$ denotes the smooth (p, q)-forms, so $\Omega^k(X, \mathbb{C}) = \bigoplus_{p+q=k} \Omega^{p,q}(X)$. Writing $d = \partial + \bar{\partial}$ with $\partial \colon \Omega^{p,q} \to \Omega^{p+1,q}$ and $\bar{\partial} \colon \Omega^{p,q} \to \Omega^{p,q+1}$ the components of d, one has in coordinates

$$\partial \left(\sum f_{IJ}\, dz^I \wedge d\bar{z}^J \right) = \sum \frac{\partial f_{IJ}}{\partial z^k}\, dz^k \wedge dz^I \wedge d\bar{z}^J, \qquad \bar{\partial} \left(\sum f_{IJ}\, dz^I \wedge d\bar{z}^J \right) = \sum \frac{\partial f_{IJ}}{\partial \bar{z}^k}\, d\bar{z}^k \wedge dz^I \wedge d\bar{z}^J.$$

Proposition 44.5.3. *On a complex manifold $d = \partial + \bar{\partial}$, and $\partial^2 = \bar{\partial}^2 = \partial\bar{\partial} + \bar{\partial}\partial = 0$. A function is holomorphic if and only if $\bar{\partial} f = 0$, and a $(p, 0)$-form α is* holomorphic *(has holomorphic coefficients) if and only if $\bar{\partial}\alpha = 0$; the sheaf of holomorphic p-forms is denoted Ω_X^p, with $\Omega_X^0 = \mathcal{O}_X$ and $\Omega_X^n = K_X$ the* canonical bundle.

Proof. In holomorphic coordinates d of a (p, q)-form has components of types $(p + 1, q)$ and $(p, q + 1)$ only, by the coordinate formula for d; these are ∂ and $\bar{\partial}$. Then $0 = d^2$ decomposes into its components of types $(p + 2, q)$, $(p + 1, q + 1)$ and $(p, q + 2)$, which are ∂^2, $\partial\bar{\partial} + \bar{\partial}\partial$ and $\bar{\partial}^2$. The statement about functions is the Cauchy–Riemann equations (Definition 31.2.1), and for forms it is the same condition on the coefficients. □

Theorem 44.5.4 (Newlander–Nirenberg)**.** *An almost complex structure J on a smooth manifold is integrable if and only if the* Nijenhuis tensor *vanishes,*

$$N_J(X, Y) = [JX, JY] - J[JX, Y] - J[X, JY] - [X, Y] = 0,$$

equivalently if and only if $T^{0,1}$ is closed under the Lie bracket, equivalently if and only if $d = \partial + \bar{\partial}$ on 1-forms with respect to the induced type decomposition.

The implication "integrable $\Rightarrow N_J = 0$" is a computation in holomorphic coordinates, where the brackets of the coordinate fields vanish; the converse is a hard analytic theorem, proved in [Huy05, Chapter 2] and, in the real-analytic case, by the Frobenius theorem (Theorem 33.9.8) applied to $T^{0,1}$ after complexification.

Proposition 44.5.5. *Every almost complex structure on a manifold of real dimension 2 has vanishing Nijenhuis tensor. Consequently, by Theorem 44.5.4, every oriented surface with a Riemannian metric is a Riemann surface, with J the rotation by $\pi/2$ in each tangent plane.*

Proof. N_J is antisymmetric and C^∞-bilinear, so it suffices to evaluate it on X and JX for a local nowhere vanishing field X, which together with JX form a frame. Using $J^2 = -1$, $N_J(X, JX) = [JX, -X] - J[JX, JX] - J[X, -X] - [X, JX] = [X, JX] - [X, JX] = 0$. On an oriented surface with a metric, rotation by $\pi/2$ in the positive sense is a smooth endomorphism with $J^2 = -1$. □

44.6 Analytic subvarieties and blow-ups

Definition 44.6.1. An *analytic subvariety* of a complex manifold X is a closed subset that is locally the common zero set of finitely many holomorphic functions (Definition 31.5.1); it is *smooth*, that is, a complex submanifold, at its regular points. By Theorem 31.5.3 the singular locus is a proper analytic subvariety, the regular locus is dense, and Z has a well-defined dimension on each irreducible component.

Proposition 44.6.2. *Let $Z \subseteq X$ be an analytic subvariety of pure complex dimension k in a compact complex manifold of dimension n. Then integration over the regular locus, $\varphi \mapsto \int_{Z_{\mathrm{reg}}} \varphi$, converges for every smooth $2k$-form φ and defines a closed current of degree $2n - 2k$; its class in $H^{2n-2k}(X; \mathbb{R})$ is the* cycle class $[Z]$. *Moreover $\int_Z \varphi \geq 0$ when φ is a positive (k, k)-form.*

Sketch of proof. Convergence follows from the local structure of analytic sets: near a regular point Z is a submanifold, and near a singular point it is a finite branched cover of a polydisc (Theorem 31.5.3), so the volume of $Z \cap K$ is finite for compact K (in a chart, Wirtinger's inequality bounds the volume by the integral of a power of the standard form $\frac{i}{2}\sum_j dz^j \wedge d\bar{z}^j$ of $\mathbb{C}^n$). Closedness, that is, Stokes' theorem across the singular locus, is proved by cutting out a neighbourhood of the singular set, whose boundary has small area because the singular locus has real codimension at least 2. The positivity statement is local linear algebra. Details are in [GH78, Chapter 0, Section 2] and [Voi02b, Chapter 11]. □

Remark 44.6.3. If X carries a Kähler form ω (Chapter 47) and $Z \neq \emptyset$, then $[Z] \neq 0$: pairing with the closed form ω^k gives $\int_Z \omega^k > 0$, the volume of Z up to a constant. On non-Kähler manifolds cycle classes can vanish; for instance on the Hopf surface of Exercise 44.7.3 the elliptic curves $\{z_1 = 0\}/\langle 2\rangle$ have zero class, since H^2 of the Hopf surface vanishes.

Definition 44.6.4. The *blow-up* of $\mathbb{C}^n$ at the origin is

$$\mathrm{Bl}_0\mathbb{C}^n = \{(z, \ell) \in \mathbb{C}^n \times \mathbb{CP}^{n-1} : z \in \ell\},$$

with the projection π to $\mathbb{C}^n$. It is a complex manifold, π is a biholomorphism over $\mathbb{C}^n \setminus \{0\}$, and $E = \pi^{-1}(0) \cong \mathbb{CP}^{n-1}$ is the *exceptional divisor*. For a point p of a complex manifold X one defines $\mathrm{Bl}_p X$ by gluing $\mathrm{Bl}_0\mathbb{C}^n$ in a chart, and similarly along a complex submanifold $Z \subseteq X$, with $E = \mathbb{P}(N_{Z/X})$.

Proposition 44.6.5. *$\mathrm{Bl}_p X$ is a compact complex manifold if X is, π is proper and holomorphic, and*

$$H^k(\mathrm{Bl}_p X; \mathbb{Z}) \cong H^k(X; \mathbb{Z}) \oplus \bigoplus_{0<j<n} H^{k-2j}(\mathrm{pt}; \mathbb{Z}),$$

so blowing up a point adds one generator in each even degree $2, 4, \ldots, 2n-2$. The exceptional divisor has normal bundle $\mathcal{O}(-1)$ in $\mathrm{Bl}_0\mathbb{C}^n$.

Proof. In the chart $\ell = [1 : w_2 : \cdots : w_n]$ the map $(u, w_2, \ldots, w_n) \mapsto (u, uw_2, \ldots, uw_n)$ identifies $\mathbb{C}^n$ with an open subset of $\mathrm{Bl}_0\mathbb{C}^n$, which shows that the blow-up is a complex manifold and that π is holomorphic and biholomorphic away from E; properness holds

because $\mathbb{CP}^{n-1}$ is compact. In these coordinates E is $\{u = 0\}$, and the transition functions of its normal bundle are those of the tautological bundle, so $N_E \cong O(-1)$. For the cohomology, use the Mayer–Vietoris sequence for $\mathrm{Bl}_p X = (\mathrm{Bl}_p X \setminus E) \cup \pi^{-1}(B)$ with B a small ball, together with $\pi^{-1}(B) \simeq \mathbb{CP}^{n-1}$ and $\mathrm{Bl}_p X \setminus E \cong X \setminus \{p\}$, and Example 22.2.5. □

Remark 44.6.6. Blow-ups are the basic birational modifications: Hironaka's resolution of singularities states that every analytic (or algebraic, in characteristic zero) variety becomes smooth after finitely many blow-ups along smooth centres, and the weak factorisation theorem says that any two birational smooth projective varieties are connected by blow-ups and blow-downs. Both are used to reduce statements about the Hodge conjecture to the smooth projective case (Chapter 74).

44.7 Exercises

Exercise 44.7.1. Show that every complex torus of dimension 1 is biholomorphic to $\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ with $\operatorname{Im}\tau > 0$, and that two such are biholomorphic if and only if the parameters are related by an element of $\mathrm{SL}_2(\mathbb{Z})$ acting by Möbius transformations.

Exercise 44.7.2. Verify that the transition maps of $\mathbb{CP}^n$ are holomorphic and that $\mathbb{CP}^1$ is biholomorphic to the Riemann sphere $\mathbb{C} \cup \{\infty\}$, with S^2 as underlying smooth manifold.

Exercise 44.7.3. The *Hopf surface* is $X = (\mathbb{C}^2 \setminus \{0\})/\langle z \mapsto 2z \rangle$. Show that it is a compact complex manifold diffeomorphic to $S^1 \times S^3$, and deduce that $b_1(X) = 1$; conclude that X carries no Kähler metric (Chapter 48), since odd Betti numbers of compact Kähler manifolds are even.

Solution. The action is free and properly discontinuous, so the quotient is a complex manifold, and it is compact because the annulus $1 \le |z| \le 2$ maps onto it. The diffeomorphism with $S^1 \times S^3$ comes from the polar decomposition $\mathbb{C}^2 \setminus \{0\} \cong S^3 \times \mathbb{R}_{>0}$, with the action becoming translation on the second factor. Then $b_1(S^1 \times S^3) = 1$ by the Künneth formula. □

Exercise 44.7.4. Show that S^2 and S^6 admit almost complex structures (for S^6 use the octonionic cross product), and that S^2's is integrable.

Exercise 44.7.5. Compute $\int_{|z|=2} \frac{z^5}{z^6-1}\,dz$ and $\int_{|z|=2} \frac{dz}{(z-1)^3(z^2+9)}$ (the circle traversed once counterclockwise) by computing a single residue at ∞ on $\mathbb{CP}^1$ instead of the residues inside the circle.

Solution. By Theorem 44.3.3 on $\mathbb{CP}^1$, the integral over the circle, which is $2\pi i$ times the sum of the residues inside it, equals $-2\pi i$ times the sum of the residues outside it, including ∞. For $\omega = z^5\,dz/(z^6-1)$ all six poles lie inside. With $w = 1/z$, $\frac{z^5}{z^6-1} = \frac{w}{1-w^6}$ and $dz = -dw/w^2$, so $\omega = -\frac{dw}{w(1-w^6)}$ has residue -1 at ∞, and the integral is $2\pi i$. (Directly: each pole ζ is simple with residue $\zeta^5/(6\zeta^5) = \frac{1}{6}$.) For the second form the poles $\pm 3i$ lie outside the circle and 1 inside. At ∞ the form is $O(w^3)\,dw$, holomorphic, so the integral is $-2\pi i(\mathrm{Res}_{3i} + \mathrm{Res}_{-3i})$, with $\mathrm{Res}_{\pm 3i} = \frac{1}{(\pm 3i-1)^3(\pm 6i)}$. Now $(3i-1)^3 = 26 - 18i$ and $(3i-1)^3 \cdot 6i = 108 + 156i$, so $\mathrm{Res}_{3i} = \frac{1}{108+156i}$ and $\mathrm{Res}_{-3i} = \frac{1}{108-156i}$ is its conjugate. Their sum is $\frac{216}{108^2+156^2} = \frac{216}{36000} = \frac{3}{500}$, and the integral is $-\frac{3\pi i}{250}$. (Directly: the residue at the triple pole 1 is $\frac{1}{2}\frac{d^2}{dz^2}(z^2+9)^{-1}$ at $z = 1$, which is $\frac{1}{2}\left(-\frac{2}{100} + \frac{8}{1000}\right) = -\frac{3}{500}$.) □

Exercise 44.7.6. Let f be a nonconstant meromorphic function on $X = \mathbb{C}/\Lambda$, $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$, viewed as a Λ-periodic meromorphic function on $\mathbb{C}$. Let $a_1, \dots, a_k$ and $b_1, \dots, b_k$ be its zeros

and poles in a fundamental parallelogram P with vertices a, $a+\omega_1$, $a+\omega_1+\omega_2$, $a+\omega_2$, listed with multiplicity, with a chosen so that no zero or pole lies on ∂P. Show that $\sum_i a_i - \sum_i b_i \in \Lambda$, by computing $\frac{1}{2\pi i}\int_{\partial P} z\frac{f'(z)}{f(z)}\,dz$ in two ways. (This is one direction of Abel's theorem for complex tori.)

Solution. At a zero or pole c of order m (negative for a pole), $f'/f = m/(z-c)+(\text{holomorphic})$, so zf'/f has residue mc. The residue theorem in the plane (Theorem 29.5.8) gives $\frac{1}{2\pi i}\int_{\partial P} zf'/f\,dz = \sum_i a_i - \sum_i b_i$. On the other hand, pair opposite sides: the side from $a+\omega_2$ to $a+\omega_1+\omega_2$ is traversed backwards, and on it f'/f takes the same values as on the side from a to $a+\omega_1$, so the two sides contribute

$$\int_a^{a+\omega_1} \big(z-(z+\omega_2)\big)\frac{f'(z)}{f(z)}\,dz = -\omega_2\int_a^{a+\omega_1}\frac{f'(z)}{f(z)}\,dz = -2\pi i\, n_1\omega_2,$$

where $\frac{1}{2\pi i}\int_a^{a+\omega_1} f'/f\,dz = n_1$ is the winding number around 0 of the closed curve $t\mapsto f(a+t\omega_1)$, $t\in[0,1]$ (closed because f is periodic), an integer. In the same way the other two sides contribute $2\pi i\, n_2\omega_1$ with $n_2\in\mathbb{Z}$. So $\sum_i a_i - \sum_i b_i = n_2\omega_1 - n_1\omega_2 \in \Lambda$. Applying the same computation to f'/f instead of zf'/f gives k zeros and k poles again, in accordance with Corollary 44.3.4. □

Chapter 45

Holomorphic Vector Bundles

45.1 Introduction

Holomorphic line bundles are the objects that encode divisors, projective embeddings and, through their first Chern classes, the $(1,1)$-part of cohomology. The exponential sequence (Theorem 45.2.4) computes the Picard group in terms of cohomology and already contains the case $p = 1$ of the Hodge conjecture: the classes of divisors are exactly the integral classes of type $(1,1)$ (Chapter 69). References: [GH78, Chapter 1], [Huy05, Chapters 2–4], [Voi02b, Chapter 4].

Prerequisites. Chapter 44 (Complex Manifolds), Chapter 26 (Sheaf Cohomology) and Chapter 39 (Characteristic Classes).

X denotes a complex manifold of dimension n.

45.2 Holomorphic bundles and sheaves

Definition 45.2.1. A *holomorphic vector bundle* of rank r on X is a complex manifold E with a holomorphic map $\pi\colon E \to X$ and $\mathbb{C}$-vector space structures on the fibres, locally biholomorphic to $U \times \mathbb{C}^r$ over U and linear on fibres; equivalently, it is given by holomorphic transition functions $g_{ij}\colon U_i \cap U_j \to \mathrm{GL}_r(\mathbb{C})$ satisfying the cocycle condition (Proposition 35.2.6). Its sheaf of holomorphic sections is a locally free $\mathcal{O}_X$-module of rank r, and this is an equivalence between holomorphic bundles and locally free sheaves, as in Proposition 35.2.10.

Example 45.2.2. The holomorphic tangent bundle $T^{1,0}X$, with transition functions the holomorphic Jacobians; the bundles $\Omega^p_X = \Lambda^p(T^{1,0}X)^\vee$ of holomorphic p-forms and the canonical bundle $K_X = \Omega^n_X$; the tautological bundle $\mathcal{O}(-1)$ on $\mathbb{CP}^n$ (Exercise 35.6.3), its dual $\mathcal{O}(1)$ and the powers $\mathcal{O}(k) = \mathcal{O}(1)^{\otimes k}$. The holomorphic sections of $\mathcal{O}(k)$ on $\mathbb{CP}^n$ are the homogeneous polynomials of degree k for $k \geq 0$, and 0 for $k < 0$ (Exercise 45.6.1).

Proposition 45.2.3. *The isomorphism classes of holomorphic line bundles on X form a group* $\mathrm{Pic}(X)$ *under $\otimes$, and*

$$\mathrm{Pic}(X) \cong H^1(X, \mathcal{O}_X^*),$$

where $\mathcal{O}_X^*$ is the sheaf of nowhere vanishing holomorphic functions.

Proof. As in Proposition 35.2.6 and Theorem 35.4.2, line bundles correspond to Čech cocycles with values in $\mathcal{O}_X^*$ modulo coboundaries, and $\check{H}^1 = H^1$ (Remark 26.5.4); tensor product corresponds to multiplication of cocycles. □

Theorem 45.2.4 (Exponential sequence)**.** *The sequence of sheaves* $0 \to \mathbb{Z}_X \to \mathcal{O}_X \xrightarrow{\exp(2\pi i\,\cdot)} \mathcal{O}_X^* \to 0$ *is exact, and its long exact sequence reads*

$$\cdots \to H^1(X,\mathbb{Z}) \to H^1(X,\mathcal{O}_X) \to \operatorname{Pic}(X) \xrightarrow{c_1} H^2(X,\mathbb{Z}) \to H^2(X,\mathcal{O}_X) \to \cdots$$

The map c_1 *is the first Chern class of the underlying smooth line bundle (Theorem 35.4.2). In particular, for compact* X*:*

1. c_1 *is surjective onto the kernel of* $H^2(X,\mathbb{Z}) \to H^2(X,\mathcal{O}_X)$*;*
2. *its kernel* $\operatorname{Pic}^0(X)$ *is* $H^1(X,\mathcal{O}_X)/\operatorname{im} H^1(X,\mathbb{Z})$*;*
3. *if* $H^1(X,\mathcal{O}_X) = H^2(X,\mathcal{O}_X) = 0$*, then* $c_1\colon \operatorname{Pic}(X) \to H^2(X,\mathbb{Z})$ *is an isomorphism.*

Proof. Exactness: the kernel of $\exp(2\pi i\cdot)$ is the constant sheaf $\mathbb{Z}$, and surjectivity holds on stalks because a nonvanishing holomorphic function on a small ball has a holomorphic logarithm (integrate df/f, which is possible on a simply connected domain by Theorem 29.3.3 applied in one variable at a time, or note that the ball is contractible and use the same argument as in Theorem 35.4.2). The long exact sequence is Theorem 15.4.3, using $\operatorname{Pic}(X) \cong H^1(X,\mathcal{O}_X^*)$. That the connecting map is the topological first Chern class follows by comparing with the corresponding sequence for smooth functions, into which this one maps, and using that the sheaf of smooth functions is acyclic. The three consequences are exactness of the long sequence. □

Remark 45.2.5. When X is compact Kähler, the Hodge decomposition (Chapter 48) shows that $H^1(X,\mathbb{Z})$ maps onto a lattice in the complex vector space $H^1(X,\mathcal{O}_X)$ of dimension $b_1/2$, so $\operatorname{Pic}^0(X)$ is a compact complex torus, the *Picard variety*, and the image of c_1, the *Néron–Severi group*, is the group of integral classes of type $(1,1)$ (Chapter 69).

Lemma 45.2.6. *Let* L *be a holomorphic line bundle on* X*.*

1. L *is trivial if and only if it has a nowhere vanishing holomorphic section.*
2. *If* X *is compact and connected and both* L *and its dual* $L^\vee$ *have nonzero holomorphic sections, then* L *is trivial. In particular, if* L *is nontrivial and* $H^0(X,L) \neq 0$*, then* $H^0(X,L^\vee) = 0$*.*

Proof. (1) A nowhere vanishing section s gives the map $\mathcal{O}_X \to L$, $f \mapsto fs$, which is an isomorphism on every fibre; conversely the trivial bundle has the section 1.

(2) First, if U is a connected open set and a, b are holomorphic functions on U, neither identically zero, then ab is not identically zero: otherwise $U \setminus a^{-1}(0)$ is open, nonempty by the identity theorem (Proposition 31.2.3), and b vanishes on it, so $b = 0$ by the identity theorem again. Next, a nonzero section s of L does not vanish identically on any nonempty open set: the set of points near which s vanishes identically is open, and it is closed by the identity theorem applied in local frames, so it is empty because X is connected and $s \neq 0$.

Now let $s \in H^0(X,L)$ and $t \in H^0(X,L^\vee)$ be nonzero. The pairing $t(s)$ is a holomorphic function on X, hence a constant c (Proposition 44.2.3). In a connected coordinate neighbourhood U with a holomorphic frame e and dual frame $e^\vee$, write $s = ae$ and $t = be^\vee$; then $t(s) = ab$ with a, b not identically zero on U, so $c \neq 0$. Therefore $a(x)b(x) = c \neq 0$ everywhere, s vanishes nowhere, and L is trivial by (1). □

Definition 45.2.7. Let L be a smooth complex line bundle on X, and let $\mathcal{A}^{0,q}(L)$ denote the smooth L-valued $(0,q)$-forms. A *$\bar\partial$-operator* on L is a $\mathbb{C}$-linear map $D\colon \mathcal{A}^0(L) \to \mathcal{A}^{0,1}(L)$ with $D(fs) = \bar\partial f \otimes s + fDs$ for smooth functions f. In a local smooth frame e it has the form $D(fe) = (\bar\partial f + f a_e)e$ for a $(0,1)$-form $a_e = De/e$. The operator is *integrable* if $\bar\partial a_e = 0$ for every local frame e.

Proposition 45.2.8. *1. If L is a holomorphic line bundle, the operator $\bar\partial_L(fe) = \bar\partial f \otimes e$, for e a holomorphic frame, is a well-defined integrable $\bar\partial$-operator, and a smooth section s is holomorphic if and only if $\bar\partial_L s = 0$.*

2. Integrability does not depend on the frame: if $e' = ge$, then $a_{e'} = a_e + g^{-1}\bar\partial g$ and $\bar\partial a_{e'} = \bar\partial a_e$.

3. Conversely, for every integrable $\bar\partial$-operator D on a smooth complex line bundle L there is a unique structure of holomorphic line bundle on L with $D = \bar\partial_L$.

In particular, on a Riemann surface, where there are no $(0,2)$-forms, every $\bar\partial$-operator is integrable.

Proof. (2) $D(e') = D(ge) = (\bar\partial g + g a_e)e = (g^{-1}\bar\partial g + a_e)e'$. Moreover $\bar\partial(g^{-1}\bar\partial g) = -g^{-2}\bar\partial g \wedge \bar\partial g + g^{-1}\bar\partial^2 g = 0$, since a 1-form wedged with itself vanishes and $\bar\partial^2 = 0$ (Proposition 44.5.3).

(1) Two holomorphic frames differ by a holomorphic g, with $\bar\partial g = 0$, so by the computation in (2) the operators defined by the two frames agree; the Leibniz rule is clear, and $a_e = 0$ for a holomorphic frame, so $\bar\partial_L$ is integrable. A section fe is holomorphic if and only if f is, that is, $\bar\partial f = 0$.

(3) Let $x \in X$, choose a coordinate polydisc Δ around x on which L has a smooth frame e, and write $a_e = \sum_j f_j\, d\bar z_j$. The condition $\bar\partial a_e = 0$ says $\partial f_j/\partial \bar z_l = \partial f_l/\partial \bar z_j$, so by the $\bar\partial$-Poincaré lemma (Theorem 31.4.2) there is a smooth u on a smaller polydisc Δ' with $\bar\partial u = a_e$. The frame $e' = e^{-u}e$ satisfies $De' = (-\bar\partial u + a_e)e' = 0$. Cover X by open sets U_i with frames e_i such that $De_i = 0$. On overlaps $e_i = g_{ij}e_j$ with g_{ij} smooth and nowhere zero, and $0 = De_i = \bar\partial g_{ij} \otimes e_j$, so the g_{ij} are holomorphic and define a holomorphic structure on L for which the e_i are holomorphic frames. For a section $s = fe_i$ we get $Ds = \bar\partial f \otimes e_i = \bar\partial_L s$, so $D = \bar\partial_L$. For uniqueness, the holomorphic frames of any structure with $\bar\partial_L = D$ are the local frames killed by D, so the holomorphic atlas is determined by D. □

Example 45.2.9. On the smooth trivial line bundle $X \times \mathbb{C}$, the $\bar\partial$-operators are the $D_a = \bar\partial + a$ with a a $(0,1)$-form, integrable exactly when $\bar\partial a = 0$. A holomorphic isomorphism between the structures defined by D_a and D_b is multiplication by a smooth nowhere vanishing function g taking D_a-holomorphic sections to D_b-holomorphic ones; since $D_b(gs) = g(D_a s + (g^{-1}\bar\partial g + b - a)s)$, this happens if and only if $a - b = g^{-1}\bar\partial g$. In particular, if $a - b = \bar\partial u$ for a smooth function u, then $g = e^u$ is an isomorphism. So $a \mapsto (X \times \mathbb{C}, D_a)$ defines a map from the $\bar\partial$-closed $(0,1)$-forms modulo $\bar\partial$ of functions to $\mathrm{Pic}(X)$, whose image consists of line bundles that are trivial as smooth bundles; compare the map $H^1(X, \mathcal{O}_X) \to \mathrm{Pic}(X)$ of Theorem 45.2.4, and Chapter 46, where the quotient above is the Dolbeault group $H^{0,1}_{\bar\partial}(X)$.

45.3 Divisors

Definition 45.3.1. A *divisor* on X is a locally finite formal sum $D = \sum_i a_i Z_i$ with $a_i \in \mathbb{Z}$ and Z_i irreducible analytic hypersurfaces; they form the group $\mathrm{Div}(X)$. A *meromorphic function* is a section of the sheaf $\mathcal{M}_X$ of quotients f/g of holomorphic functions with $g \not\equiv 0$; the order $\mathrm{ord}_Z(f) \in \mathbb{Z}$ of a meromorphic function along an irreducible hypersurface Z is defined by writing $f = u\, h^m$ near a regular point of Z, where h is a local defining function of Z and u is a unit, using that $\mathcal{O}_{X,x}$ is a unique factorisation domain (Theorem 31.3.4). The divisor of f is $(f) = \sum_Z \mathrm{ord}_Z(f)\, Z$, and divisors of this form are *principal*.

Proposition 45.3.2. *There is a homomorphism $\mathrm{Div}(X) \to \mathrm{Pic}(X)$, $D \mapsto \mathcal{O}(D)$, defined by taking local defining functions f_i for D on a cover and the cocycle $g_{ij} = f_i/f_j \in \mathcal{O}^*$. Its kernel is the group of principal divisors, and $\mathcal{O}(D)$ has a canonical meromorphic section with divisor D; for D effective (all $a_i \geq 0$) this section is holomorphic and vanishes exactly on D. Conversely a nonzero meromorphic section of a line bundle L has a divisor D with $L \cong \mathcal{O}(D)$.*

Proof. The functions f_i/f_j are nowhere vanishing and holomorphic on overlaps, because two local defining functions of the same divisor differ by a unit (Theorem 31.3.4), and they clearly form a cocycle; changing the f_i changes the cocycle by a coboundary. The local functions f_i glue to a meromorphic section of $\mathcal{O}(D)$, whose divisor is D. If $\mathcal{O}(D)$ is trivial, the cocycle is a coboundary $f_i/f_j = u_i/u_j$, so the f_i/u_i glue to a global meromorphic function with divisor D; conversely, principal divisors give trivial cocycles. The last statement follows by taking the divisor of the section and comparing cocycles. □

Example 45.3.3. On $\mathbb{CP}^n$ every irreducible hypersurface is the zero set of an irreducible homogeneous polynomial, and $\mathcal{O}(D) \cong \mathcal{O}(\deg D)$; the map $\deg\colon \operatorname{Div}(\mathbb{CP}^n) \to \mathbb{Z}$ induces the isomorphism $\operatorname{Pic}(\mathbb{CP}^n) \cong \mathbb{Z}$, which also follows from Theorem 45.2.4 since $H^1(\mathbb{CP}^n, \mathcal{O}) = H^2(\mathbb{CP}^n, \mathcal{O}) = 0$ (Chapter 46). On a complex torus $X = \mathbb{C}^n/\Lambda$, by contrast, $H^1(X, \mathcal{O}_X) \neq 0$ and $\operatorname{Pic}^0(X)$ is a nontrivial complex torus, the dual torus.

Counterexample 45.3.4. Holomorphic line bundles are not determined by their topology. Let $X = \mathbb{C}/\Lambda$ with $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ a lattice, numbered so that $\operatorname{Im}(\omega_2/\omega_1) > 0$, $\pi\colon \mathbb{C} \to X$ the projection, and $p \neq q$ points of X. The line bundle $L = \mathcal{O}(p-q) = \mathcal{O}(p) \otimes \mathcal{O}(q)^\vee$ is trivial as a smooth bundle but not as a holomorphic bundle.

Smooth triviality. Lift $q - p$ to $c \in \mathbb{C}$ and let $\tau(x) = x + c$, a biholomorphism of X with $\tau(p) = q$. A local defining function of q composed with τ is a local defining function of p, so $\tau^*\mathcal{O}(q) \cong \mathcal{O}(p)$. The translations $x \mapsto x + tc$, $0 \le t \le 1$, form a homotopy from the identity to τ, so by Theorem 35.4.2 and homotopy invariance (Theorem 22.2.2), $c_1(\mathcal{O}(p)) = \tau^* c_1(\mathcal{O}(q)) = c_1(\mathcal{O}(q))$. Hence $c_1(L) = 0$ and L is smoothly trivial, by the same theorem.

Holomorphic nontriviality. If L were holomorphically trivial, the divisor $p - q$ would be principal (Proposition 45.3.2): there would be a meromorphic function f on X whose only pole is a simple pole at q. Then $F = f \circ \pi$ is meromorphic on $\mathbb{C}$, invariant under translation by Λ, with simple poles exactly at the points of $\pi^{-1}(q)$. Choose a lift $\tilde{q}$ of q and let P be the parallelogram with vertices a, $a + \omega_1$, $a + \omega_1 + \omega_2$, $a + \omega_2$, where $a = \tilde{q} - \frac{1}{2}(\omega_1 + \omega_2)$. A boundary point $\tilde{q} + (s - \frac{1}{2})\omega_1 + (t - \frac{1}{2})\omega_2$ has s or t in $\{0, 1\}$, so it is not in $\tilde{q} + \Lambda$; and $\tilde{q}$ is the only point of $\tilde{q} + \Lambda$ inside P. The boundary ∂P, traversed through the vertices in the order listed, is positively oriented, with winding number 1 about $\tilde{q}$ and 0 about the points outside P. By the residue theorem (Theorem 29.5.8), $\int_{\partial P} F\,dz = 2\pi i \operatorname{Res}_{\tilde{q}} F \neq 0$, the residue at a simple pole being nonzero. On the other hand the integrals over opposite sides of P cancel: the side from $a + \omega_2$ to $a + \omega_1 + \omega_2$ is the translate by ω_2 of the side from a to $a + \omega_1$, traversed in the opposite direction, and $F(z + \omega_2) = F(z)$; similarly for the other pair. So $\int_{\partial P} F\,dz = 0$, a contradiction.

Thus L is a nontrivial element of $\operatorname{Pic}^0(X)$, the kernel of c_1 in Theorem 45.2.4. The same argument shows that no meromorphic function on X has a single pole which is simple.

45.4 Hermitian metrics and the Chern connection

Theorem 45.4.1. *Let E be a holomorphic vector bundle with a Hermitian metric h. There is a unique connection ∇ on E, the* Chern connection, *that is metric and satisfies $\nabla^{0,1} = \bar\partial$. In a holomorphic frame with $H_{ij} = h(e_i, e_j)$ one has $\theta = \overline{H}^{-1}\partial\overline{H}$ and curvature $F = \bar\partial\theta$, a $(1,1)$-form with values in* End E. *For a line bundle with local weight $h = e^{-\varphi}$ in a holomorphic frame, $F = \partial\bar\partial\varphi$ and*

$$c_1(L) = \left[\frac{i}{2\pi}F\right] = \left[\frac{i}{2\pi}\partial\bar\partial\varphi\right] \in H^2(X;\mathbb{R}),$$

a class of type $(1,1)$.

Proof. Write $\nabla = \nabla^{1,0} + \nabla^{0,1}$. The condition $\nabla^{0,1} = \bar\partial$ fixes the $(0,1)$-part, and the metric condition $d\,h(s,t) = h(\nabla s, t) + h(s, \nabla t)$ splits into its $(1,0)$- and $(0,1)$-parts, the first of which reads $\partial h(s,t) = h(\nabla^{1,0}s, t) + h(s, \bar\partial t)$ and determines $\nabla^{1,0}$; this proves uniqueness, and the resulting formula in a holomorphic frame, $\theta = \overline{H}^{-1}\partial\overline{H}$, defines a connection with the two properties, which proves existence. Then $F = d\theta + \theta \wedge \theta$ has no $(0,2)$-part because $\bar\partial^2 = 0$ and $\nabla^{0,1} = \bar\partial$, and no $(2,0)$-part by the metric condition; the formula $F = \bar\partial\theta$ follows. For a line bundle $H = e^{-\varphi}$ is real, so $\theta = H^{-1}\partial H = \partial \log H = -\partial\varphi$ and $F = \bar\partial\theta = -\bar\partial\partial\varphi = \partial\bar\partial\varphi$; the Chern class is computed by Chern–Weil theory (Proposition 39.2.7). □

Definition 45.4.2. A Hermitian line bundle (L, h) is *positive* if the real $(1,1)$-form $\frac{i}{2\pi}F$ is positive definite, that is, if it is the fundamental form of a Hermitian metric on X (Chapter 47); L is positive if it admits such a metric. A line bundle is *ample* if a power of it gives an embedding of X in projective space (Chapter 50).

Example 45.4.3. On $\mathbb{CP}^n$, give $\mathcal{O}(-1)$ the metric induced from the standard Hermitian metric of $\mathbb{C}^{n+1}$. On $U_0 = \{x_0 \neq 0\}$, with coordinates $w_j = x_j/x_0$, the section $\sigma(w) = (1, w_1, \dots, w_n)$ is a holomorphic frame with $h(\sigma,\sigma) = 1 + |w|^2 = e^{-\varphi}$, so by Theorem 45.4.1 $\frac{i}{2\pi}F = -\frac{i}{2\pi}\partial\bar\partial \log(1 + |w|^2) = -\omega_{FS}$, where ω_{FS}, the *Fubini–Study form*, is the global $(1,1)$-form given on U_0 by $\frac{i}{2\pi}\partial\bar\partial\log(1+|w|^2)$ (on U_j one uses $\log(|x|^2/|x_j|^2)$ in place of $\log(1+|w|^2) = \log(|x|^2/|x_0|^2)$; on overlaps these differ by $\log|x_k/x_j|^2$, the logarithm of the squared modulus of a nowhere vanishing holomorphic function, which $\partial\bar\partial$ kills); hence $\mathcal{O}(1)$ is positive, with $\frac{i}{2\pi}F = \omega_{FS}$ and $c_1(\mathcal{O}(1)) = [\omega_{FS}]$, the positive generator of $H^2(\mathbb{CP}^n;\mathbb{Z})$ (Example 39.2.8).

45.5 The canonical bundle and adjunction

Proposition 45.5.1 (Adjunction)**.** *Let $Z \subseteq X$ be a smooth hypersurface with associated line bundle $\mathcal{O}(Z)$. Then the normal bundle is $N_{Z/X} \cong \mathcal{O}(Z)|_Z$, and*

$$K_Z \cong (K_X \otimes \mathcal{O}(Z))\big|_Z.$$

For a smooth hypersurface of degree d in $\mathbb{CP}^n$ one gets $K_Z \cong \mathcal{O}(d-n-1)|_Z$.

Proof. A local defining function f_i of Z gives, by df_i, an isomorphism of the conormal bundle $N_{Z/X}^\vee$ with $\mathcal{O}(-Z)|_Z$: the transition functions of the conormal bundle are the f_i/f_j restricted to Z, computed from $df_i = (f_i/f_j)\,df_j$ modulo T^*Z. Taking determinants in the exact sequence $0 \to N_{Z/X}^\vee \to \Omega^1_X|_Z \to \Omega^1_Z \to 0$ gives $K_X|_Z \cong K_Z \otimes N_{Z/X}^\vee \cong K_Z \otimes \mathcal{O}(-Z)|_Z$, which is the formula. For $\mathbb{CP}^n$ use $K_{\mathbb{CP}^n} \cong \mathcal{O}(-n-1)$, which follows from the Euler sequence (Example 35.3.10) by taking determinants, and $\mathcal{O}(Z) = \mathcal{O}(d)$. □

Example 45.5.2. A smooth plane curve of degree d in $\mathbb{CP}^2$ has $K_Z \cong \mathcal{O}(d-3)|_Z$, whence genus $g = (d-1)(d-2)/2$ by Riemann–Roch (Chapter 59); a smooth quadric surface in $\mathbb{CP}^3$ has $K \cong \mathcal{O}(-2)$ and is $\mathbb{CP}^1 \times \mathbb{CP}^1$; a smooth quartic surface in $\mathbb{CP}^3$ has trivial canonical bundle and is a K3 surface, one of the basic examples in Hodge theory.

45.6 Exercises

Exercise 45.6.1. Show that the holomorphic sections of $\mathcal{O}(k)$ on $\mathbb{CP}^n$ are the homogeneous polynomials of degree k in the homogeneous coordinates, and that there are none for $k < 0$.

Solution. A section is given by functions s_i on U_i with $s_i = (x_i/x_j)^{-k} s_j$; the function $F(x) = x_i^k s_i(x/x_i)$ is independent of i, holomorphic on $\mathbb{C}^{n+1} \setminus \{0\}$ and homogeneous of degree k. By Hartogs' theorem (Theorem 31.2.9) it extends to $\mathbb{C}^{n+1}$, and a homogeneous holomorphic function of degree k is a polynomial for $k \geq 0$ and zero for $k < 0$, by looking at its power series. □

Exercise 45.6.2. Let X be a compact Riemann surface. Show that $\deg L = \int_X c_1(L)$ defines a homomorphism $\mathrm{Pic}(X) \to \mathbb{Z}$, that $\deg \mathcal{O}(D) = \sum_p a_p$ for $D = \sum a_p p$, and that a line bundle of negative degree has no nonzero holomorphic sections.

Exercise 45.6.3. Compute the Chern connection and its curvature for the trivial line bundle with metric $e^{-|z|^2}$ on $\mathbb{C}$, and verify that the curvature is a positive $(1,1)$-form.

Exercise 45.6.4. For a complex torus $X = \mathbb{C}/\Lambda$ of dimension one, show that $H^1(X, \mathcal{O}_X) \cong \mathbb{C}$ and deduce that $\mathrm{Pic}^0(X)$ is a complex torus of dimension one.

Chapter 46

Dolbeault Cohomology

46.1 Introduction

Dolbeault cohomology is the cohomology of the $\bar\partial$-operator on forms of type (p,q). By the Dolbeault theorem (Theorem 46.2.3) it computes the sheaf cohomology of the holomorphic p-forms, and on a compact complex manifold harmonic theory for $\bar\partial$ makes it finite-dimensional and gives Serre duality (Theorem 46.3.4). The numbers $h^{p,q}=\dim H^q(X,\Omega^p_X)$ are the *Hodge numbers*; for compact Kähler manifolds they are the dimensions of the pieces of the Hodge decomposition (Chapter 48). References: [GH78, Chapter 0], [Voi02b, Chapter 4], [Huy05, Chapters 2–4], [Wel08].

Prerequisites. Chapter 45 (Holomorphic Vector Bundles) and Chapter 42 (Elliptic Operators on Compact Manifolds).

X is a complex manifold of dimension n and E a holomorphic vector bundle on it.

46.2 The Dolbeault theorem

Definition 46.2.1. Let $\mathcal{A}^{p,q}(E)$ be the sheaf of smooth E-valued (p,q)-forms and $\mathcal{A}^{p,q}(X,E)$ its global sections. In a holomorphic frame of E set $\bar\partial_E(\sum_j \alpha_j e_j)=\sum_j \bar\partial\alpha_j\, e_j$; this is independent of the frame because the transition functions are holomorphic, and $\bar\partial_E^2=0$. The *Dolbeault cohomology* is $H^{p,q}_{\bar\partial}(X,E)=H^q(\mathcal{A}^{p,\bullet}(X,E),\bar\partial_E)$; for E trivial we write $H^{p,q}_{\bar\partial}(X)$.

Lemma 46.2.2. *Let $\Delta'\subseteq\Delta$ be polydiscs with $\overline{\Delta'}\subseteq\Delta$ and α a smooth (p,q)-form on Δ with $q\geq 1$ and $\bar\partial\alpha=0$. Then $\alpha=\bar\partial\beta$ on Δ' for some smooth $(p,q-1)$-form β.*

Proof. Writing $\alpha=\sum_I dz^I\wedge\alpha_I$ with α_I of type $(0,q)$, the condition $\bar\partial\alpha=0$ holds componentwise, since $\bar\partial(dz^I)=0$; so we may assume $p=0$. For $q=1$ this is Theorem 31.4.2. For general q the same induction applies: if α involves only $d\bar z^1,\dots,d\bar z^k$, write $\alpha=d\bar z^k\wedge\alpha_1+\alpha_2$ with α_1,α_2 not involving $d\bar z^k,\dots,d\bar z^n$; the equation $\bar\partial\alpha=0$ implies that the coefficients of α_1 are holomorphic in $z_{k+1},\dots,z_n$. Solving $\partial g/\partial\bar z_k=$ (coefficient of α_1) with the Cauchy transform of Lemma 31.4.1, after a cut-off in z_k alone, gives a form γ with $\alpha-\bar\partial\gamma$ involving only $d\bar z^1,\dots,d\bar z^{k-1}$ on a slightly smaller polydisc; by induction on k we conclude. The details are exactly as in the proof of Theorem 31.4.2; see [GH78, Chapter 0, Section 2]. □

Theorem 46.2.3 (Dolbeault)**.** *The sequence of sheaves $0\to\Omega^p_X(E)\to\mathcal{A}^{p,0}(E)\xrightarrow{\bar\partial_E}\mathcal{A}^{p,1}(E)\xrightarrow{\bar\partial_E}\cdots$ is a fine resolution of the sheaf $\Omega^p_X(E)=\Omega^p_X\otimes E$ of holomorphic E-valued p-forms. Consequently*

$$H^{p,q}_{\bar\partial}(X,E)\cong H^q(X,\Omega^p_X\otimes E).$$

Proof. Exactness at $\mathcal{A}^{p,0}$ says that the E-valued $(p,0)$-forms killed by $\bar\partial_E$ are the holomorphic ones (Proposition 44.5.3), and exactness at $\mathcal{A}^{p,q}$ for $q \geq 1$ holds on stalks by Lemma 46.2.2, applied in a holomorphic frame of E on small polydiscs. The sheaves $\mathcal{A}^{p,q}(E)$ are modules over the sheaf of smooth functions, hence fine and acyclic (Theorem 26.4.5); apply Proposition 26.7.1(3). □

Theorem 46.2.4 (Holomorphic Poincaré lemma)**.** *The holomorphic de Rham complex* $0 \to \mathbb{C}_X \to \Omega^0_X \xrightarrow{\partial} \Omega^1_X \xrightarrow{\partial} \cdots \to \Omega^n_X \to 0$ *is exact. Hence* $H^k(X;\mathbb{C}) \cong \mathbb{H}^k(X, \Omega^\bullet_X)$*, and there is a spectral sequence (the* Frölicher *or* Hodge-to-de Rham *spectral sequence)*

$$E_1^{p,q} = H^q(X, \Omega^p_X) \Rightarrow H^{p+q}(X;\mathbb{C}).$$

Proof. On holomorphic forms $d = \partial$. Consider the double complex of sheaves $\mathcal{A}^{p,q}$ with differentials ∂ and $\bar\partial$, whose total complex is the complexified de Rham complex. By Theorem 46.2.3 each column $(\mathcal{A}^{p,\bullet}, \bar\partial)$ is a resolution of Ω^p_X; by Lemma 15.7.2, applied to stalks, the inclusion $\Omega^\bullet_X \to \mathrm{Tot}(\mathcal{A}^{\bullet,\bullet}) = \mathcal{A}^\bullet_{\mathbb{C}}$ is a quasi-isomorphism. The de Rham complex is a resolution of $\mathbb{C}_X$ (Theorem 37.3.1), so $\Omega^\bullet_X$ has cohomology sheaf $\mathbb{C}_X$ in degree 0 and none elsewhere, which is the exactness. The isomorphism with $H^k(X;\mathbb{C})$ follows from Proposition 26.7.1(1) and Theorem 37.3.1, and the spectral sequence is the first one of Proposition 26.7.1(2). □

Proposition 46.2.5. *Let* $f : X \to Y$ *be a holomorphic map of complex manifolds. Pullback of forms maps* $\mathcal{A}^{p,q}(Y)$ *into* $\mathcal{A}^{p,q}(X)$ *and commutes with* ∂ *and* $\bar\partial$*, so it induces maps* $f^* \colon H^{p,q}_{\bar\partial}(Y) \to H^{p,q}_{\bar\partial}(X)$*, with* $(g \circ f)^* = f^* \circ g^*$*. The wedge product induces products* $H^{p,q}_{\bar\partial}(X) \times H^{p',q'}_{\bar\partial}(X) \to H^{p+p',q+q'}_{\bar\partial}(X)$*, making* $\bigoplus_{p,q} H^{p,q}_{\bar\partial}(X)$ *a graded-commutative algebra (for the total degree* $p+q$*), and* f^* *is a homomorphism of algebras.*

Proof. In local holomorphic coordinates w_j on Y, $f^* dw_j = d(w_j \circ f) = \partial(w_j \circ f)$ because $w_j \circ f$ is holomorphic, so $f^* dw_j$ has type $(1,0)$, and by conjugation $f^* d\bar w_j$ has type $(0,1)$; hence f^* preserves types. Since $f^* d = d f^*$ and $d = \partial + \bar\partial$ with ∂ of type $(1,0)$ and $\bar\partial$ of type $(0,1)$ (Proposition 44.5.3), comparing the components of type $(p+1,q)$ and $(p,q+1)$ of $f^* d\alpha = d f^* \alpha$ for α of type (p,q) gives $f^*\partial\alpha = \partial f^*\alpha$ and $f^*\bar\partial\alpha = \bar\partial f^*\alpha$. So f^* maps $\bar\partial$-closed forms to $\bar\partial$-closed forms and $\bar\partial$-exact forms to $\bar\partial$-exact forms. Functoriality holds already for forms.

Similarly, taking the component of type $(p+p', q+q'+1)$ of the Leibniz rule $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^{p+q}\alpha \wedge d\beta$ gives $\bar\partial(\alpha \wedge \beta) = \bar\partial\alpha \wedge \beta + (-1)^{p+q}\alpha \wedge \bar\partial\beta$. Hence the wedge product of $\bar\partial$-closed forms is $\bar\partial$-closed, and if $\bar\partial\beta = 0$ then $\bar\partial\gamma \wedge \beta = \bar\partial(\gamma \wedge \beta)$ is exact; so the product is well defined on classes. Graded commutativity and compatibility with f^* hold for forms. □

Definition 46.2.6. The *Bott–Chern* and *Aeppli cohomology* groups of X are

$$H^{p,q}_{BC}(X) = \frac{\{\alpha \in \mathcal{A}^{p,q}(X) : \partial\alpha = \bar\partial\alpha = 0\}}{\partial\bar\partial\mathcal{A}^{p-1,q-1}(X)}, \qquad H^{p,q}_{A}(X) = \frac{\{\alpha \in \mathcal{A}^{p,q}(X) : \partial\bar\partial\alpha = 0\}}{\partial\mathcal{A}^{p-1,q}(X) + \bar\partial\mathcal{A}^{p,q-1}(X)}.$$

Lemma 46.2.7. *The identity on forms induces natural linear maps*

$$H^{p,q}_{BC}(X) \to H^{p,q}_{\bar\partial}(X), \qquad H^{p,q}_{BC}(X) \to H^{p+q}_{\mathrm{dR}}(X;\mathbb{C}), \qquad H^{p,q}_{\bar\partial}(X) \to H^{p,q}_{A}(X).$$

Complex conjugation induces isomorphisms $H^{p,q}_{BC}(X) \cong H^{q,p}_{BC}(X)$ *of real vector spaces, conjugate-linear over* $\mathbb{C}$*.*

Proof. The quotients in Definition 46.2.6 make sense because $\partial\bar\partial\beta$ is ∂- and $\bar\partial$-closed and $\partial\alpha = 0$, $\bar\partial\alpha = 0$ imply $\partial\bar\partial\alpha = 0$ (using $\partial^2 = \bar\partial^2 = 0$ and $\partial\bar\partial = -\bar\partial\partial$). If $\partial\alpha = \bar\partial\alpha = 0$, then α is $\bar\partial$-closed and $d\alpha = 0$; and $\partial\bar\partial\beta = -\partial(\partial\beta)$ is $\bar\partial$-exact, while $\partial\bar\partial\beta = d(\bar\partial\beta)$ is d-exact because $\bar\partial^2\beta = 0$. This gives the first two maps. For the third, a $\bar\partial$-closed form is $\partial\bar\partial$-closed, and $\bar\partial$-exact forms lie in the denominator of H_A. Finally $\overline{\partial\alpha} = \bar\partial\bar\alpha$, so conjugation preserves the conditions defining H_{BC} and maps $\partial\bar\partial\mathcal{A}^{p-1,q-1}$ onto $\bar\partial\partial\mathcal{A}^{q-1,p-1} = \partial\bar\partial\mathcal{A}^{q-1,p-1}$. □

Remark 46.2.8. Unlike $H^{p,q}_{\bar\partial}$, the Bott–Chern groups have a definition symmetric in p and q, and they map to de Rham cohomology. On a compact Kähler manifold the $\partial\bar\partial$-lemma (Chapter 48) says that a d-closed form of pure type which is d-exact or $\bar\partial$-exact is of the form $\partial\bar\partial\beta$; this is exactly the injectivity of the first two maps of Lemma 46.2.7. On general compact complex manifolds these maps need not be injective, and Bott–Chern cohomology is a finer invariant than Dolbeault or de Rham cohomology.

46.3 Harmonic theory for $\bar\partial$

Definition 46.3.1. Let X be compact with a Hermitian metric and E a Hermitian holomorphic bundle. With the L^2 inner products on $\mathcal{A}^{p,q}(X,E)$, let $\bar\partial^*_E$ be the formal adjoint of $\bar\partial_E$ (Definition 42.5.1) and $\square_E = \bar\partial_E\bar\partial^*_E + \bar\partial^*_E\bar\partial_E$, the *$\bar\partial$-Laplacian*. The space of *$\bar\partial$-harmonic forms* is $\mathcal{H}^{p,q}(X,E) = \ker\square_E$.

Lemma 46.3.2. *In the situation of Definition 46.3.1, a form $\alpha \in \mathcal{A}^{p,q}(X,E)$ is $\square_E$-harmonic if and only if $\bar\partial_E\alpha = 0$ and $\bar\partial^*_E\alpha = 0$. In particular, for $q = 0$ the harmonic forms are exactly the holomorphic sections of $\Omega^p_X \otimes E$, for every choice of the metrics, and*

$$\mathcal{H}^{p,0}(X,E) = H^{p,0}_{\bar\partial}(X,E) = H^0(X,\Omega^p_X \otimes E).$$

Proof. Since X is compact, the defining property of formal adjoints gives $\langle\square_E\alpha,\alpha\rangle = \|\bar\partial_E\alpha\|^2 + \|\bar\partial^*_E\alpha\|^2$ in L^2, so $\square_E\alpha = 0$ forces $\bar\partial_E\alpha = \bar\partial^*_E\alpha = 0$; the converse is clear from the definition of $\square_E$. For $q = 0$, $\bar\partial^*_E$ maps $\mathcal{A}^{p,0}$ to $\mathcal{A}^{p,-1} = 0$, so the harmonic $(p,0)$-forms are the $\bar\partial_E$-closed ones, which are the holomorphic sections of $\Omega^p_X \otimes E$ (Proposition 44.5.3, in a holomorphic frame of E); and $H^{p,0}_{\bar\partial}(X,E)$ is the kernel of $\bar\partial_E$ on $\mathcal{A}^{p,0}(X,E)$, as there are no forms of type $(p,-1)$. □

Theorem 46.3.3. *$\square_E$ is elliptic, with principal symbol $\frac{1}{2}|\xi|^2$. Consequently $\mathcal{H}^{p,q}(X,E)$ is finite-dimensional, there is an orthogonal decomposition $\mathcal{A}^{p,q}(X,E) = \mathcal{H}^{p,q}(X,E) \oplus \bar\partial_E\mathcal{A}^{p,q-1}(X,E) \oplus \bar\partial^*_E\mathcal{A}^{p,q+1}(X,E)$, and every Dolbeault class has a unique harmonic representative:*

$$H^q(X,\Omega^p_X \otimes E) \cong H^{p,q}_{\bar\partial}(X,E) \cong \mathcal{H}^{p,q}(X,E).$$

In particular the Hodge numbers $h^{p,q}(X) = \dim H^q(X,\Omega^p_X)$ of a compact complex manifold are finite.

Proof. The symbol of $\bar\partial$ is $i\xi^{0,1}\wedge-$ and that of $\bar\partial^*$ is $-i\iota_{\xi^{0,1}}$, so the symbol of $\square_E$ is $\xi^{0,1}\wedge\iota + \iota\,\xi^{0,1}\wedge = |\xi^{0,1}|^2 = \frac{1}{2}|\xi|^2$, by the same computation as in Exercise 42.6.1. The rest is the proof of Theorems 43.2.2 and 43.2.5 with d, δ replaced by $\bar\partial_E$, $\bar\partial^*_E$ (Remark 43.3.8): the only inputs were ellipticity, self-adjointness and $\bar\partial^2_E = 0$. Combine with Theorem 46.2.3. □

Theorem 46.3.4 (Serre duality)**.** *Let X be a compact complex manifold of dimension n and E a holomorphic vector bundle. The pairing*

$$H^q(X,E) \times H^{n-q}(X,K_X \otimes E^\vee) \to H^n(X,K_X) \xrightarrow{\int} \mathbb{C}, \qquad ([\alpha],[\beta]) \mapsto \int_X \alpha\wedge\beta,$$

computed with Dolbeault representatives of types $(0,q)$ *and* $(n,n-q)$ *and the evaluation* $E\otimes E^\vee\to\mathbb{C}$*, is perfect. In particular* $h^{p,q}=h^{n-p,n-q}$ *for the Hodge numbers.*

Proof. The pairing is well defined by Stokes' theorem, since $\int_X\bar\partial(\alpha\wedge\gamma)=\int_X d(\alpha\wedge\gamma)=0$ for forms of total type $(n,n-1)$. Choose Hermitian metrics and define the conjugate-linear operator $\bar*_E\colon\mathcal{A}^{0,q}(E)\to\mathcal{A}^{n,n-q}(E^\vee)$ by $\alpha\wedge\bar*_E\beta=\langle\alpha,\beta\rangle\,\mathrm{vol}$, using the metric to identify $\overline{E}$ with $E^\vee$. As for the real Hodge star, $\bar\partial^*=-\bar*\bar\partial\bar*$, so $\bar*_E$ maps $\Box_E$-harmonic forms bijectively onto $\Box_{K\otimes E^\vee}$-harmonic forms. For a nonzero harmonic α, $\int_X\alpha\wedge\bar*_E\alpha=\|\alpha\|^2>0$, so the pairing is nondegenerate on harmonic representatives, which by Theorem 46.3.3 represent all classes. For the Hodge numbers take $E=\Omega^p_X$, so $K_X\otimes E^\vee\cong\Omega^{n-p}_X$ by the perfect pairing $\Omega^p\otimes\Omega^{n-p}\to K_X$. □

46.4 Examples and the Frölicher inequality

Proposition 46.4.1. *For a compact complex manifold,* $b_k(X)\le\sum_{p+q=k}h^{p,q}(X)$*, and* $\chi(X)=\sum_{p,q}(-1)^{p+q}h^{p,q}(X)$*. Equality holds for all* k *if and only if the Frölicher spectral sequence degenerates at* E_1*.*

Proof. In a spectral sequence of finite-dimensional vector spaces, $\dim E^{p,q}_{r+1}\le\dim E^{p,q}_r$ and $\sum_{p+q=k}\dim E^{p,q}_\infty=b_k$ (Theorem 16.3.4), so $b_k\le\sum_{p+q=k}h^{p,q}$, with equality for all k if and only if all differentials d_r, $r\ge1$, vanish. The Euler characteristic is unchanged when passing to cohomology of a differential of total degree one, so it is the same on every page. □

Example 46.4.2.

1. For a complex torus $X=\mathbb{C}^n/\Lambda$, the forms $dz^I\wedge d\bar z^J$ with constant coefficients are harmonic for the flat metric and span each $\mathcal{H}^{p,q}$, so $h^{p,q}=\binom{n}{p}\binom{n}{q}$, and $\sum_{p+q=k}h^{p,q}=\binom{2n}{k}=b_k$: the Frölicher spectral sequence degenerates.
2. For $\mathbb{CP}^n$, $h^{p,q}=1$ if $p=q\le n$ and 0 otherwise. Since $b_{2k}=1$ and $b_{2k+1}=0$, this follows from the Hodge decomposition (Chapter 48) once one knows $h^{k,k}\ge1$, which is witnessed by the class of ω_{FS}^k; see [GH78, Chapter 0, Section 7].
3. For the Hopf surface of Exercise 44.7.3, $b_1=1$, $h^{0,1}=1$ and $h^{1,0}=0$. So $b_1=h^{1,0}+h^{0,1}$ (the Frölicher spectral sequence of a compact complex surface always degenerates), but $h^{1,0}\ne h^{0,1}$: the Hodge symmetry $h^{p,q}=h^{q,p}$ of compact Kähler manifolds fails [BHPVdV04, Chapter IV].

Example 46.4.3. The Hodge numbers of $\mathbb{CP}^1$ can be computed directly. First $h^{0,0}=1$ by Proposition 44.2.3. A holomorphic 1-form is $f(z)\,dz$ on the chart $\mathbb{C}$, with f entire; in the chart $w=1/z$ at infinity, $dz=-w^{-2}dw$, so $-f(1/w)w^{-2}$ must extend holomorphically to $w=0$. Hence $|f(z)|\le C|z|^{-2}$ for $|z|\ge1$, so f is bounded, hence constant by Liouville's theorem (Corollary 29.4.7), and the constant is 0. So $h^{1,0}=0$ by Lemma 46.3.2. Serre duality (Theorem 46.3.4, with $n=1$ and $E=\mathcal{O}$ or $E=\Omega^1$) gives $h^{0,1}=h^{1,0}=0$ and $h^{1,1}=h^{0,0}=1$. Since $b_0=b_2=1$ and $b_1=0$, equality holds in Proposition 46.4.1. Moreover $H^1(\mathbb{CP}^1,\mathcal{O})=0$, and $H^2(\mathbb{CP}^1,\mathcal{O})=0$ because there are no forms of type $(0,2)$ on a curve; by Theorem 45.2.4, $c_1\colon\mathrm{Pic}(\mathbb{CP}^1)\to H^2(\mathbb{CP}^1;\mathbb{Z})\cong\mathbb{Z}$ is an isomorphism.

Counterexample 46.4.4. Dolbeault cohomology is not a homotopy invariant, unlike de Rham cohomology. The plane $\mathbb{C}$ is contractible, so $H^k_{\mathrm{dR}}(\mathbb{C})=0$ for $k>0$ and $H^0_{\mathrm{dR}}(\mathbb{C})=\mathbb{C}$. But $H^{0,0}_{\bar\partial}(\mathbb{C})$ and $H^{1,0}_{\bar\partial}(\mathbb{C})$ are the spaces of entire functions and of holomorphic 1-forms $f(z)\,dz$ (there is nothing to divide by in type $(p,0)$), both infinite-dimensional: they contain the z^k and the $z^k\,dz$. The two computations are reconciled by $\partial\colon H^{0,0}_{\bar\partial}(\mathbb{C})\to H^{1,0}_{\bar\partial}(\mathbb{C})$, $f\mapsto f'\,dz$,

which is surjective, because every entire function has an entire primitive (Corollary 29.3.2), and has kernel the constants. In the Frölicher spectral sequence of Theorem 46.2.4 the first differential is induced by ∂, and here it cancels the two infinite-dimensional groups against each other, leaving $H^0_{\mathrm{dR}}(\mathbb{C}) = \mathbb{C}$.

Remark 46.4.5. Cohomology of line bundles on projective space is computed by Bott's formula: $H^q(\mathbb{CP}^n, \mathcal{O}(k))$ vanishes for $0 < q < n$ and all k, H^0 = homogeneous polynomials of degree k, and $H^n(\mathbb{CP}^n, \mathcal{O}(k)) \cong H^0(\mathbb{CP}^n, \mathcal{O}(-k-n-1))^\vee$ by Serre duality, since $K_{\mathbb{CP}^n} = \mathcal{O}(-n-1)$ (Proposition 45.5.1). Vanishing theorems for positive bundles are the subject of Chapter 50; the algebraic computation is in Chapter 56.

46.5 Exercises

Exercise 46.5.1. Let X be a compact Riemann surface of genus g. Using Serre duality and the Frölicher inequality, show that $h^{0,0} = h^{1,1} = 1$ and $h^{1,0} = h^{0,1}$, and that $h^{1,0} + h^{0,1} \geq 2g$ with equality if and only if the Frölicher spectral sequence degenerates (it always does in dimension one, by Chapter 48).

Solution. $h^{0,0} = 1$ since holomorphic functions are constant (Proposition 44.2.3); Serre duality gives $h^{1,1} = h^{0,0}$ and $h^{1,0} = h^{0,1}$. The inequality is Proposition 46.4.1 for $k = 1$, with $b_1 = 2g$. □

Exercise 46.5.2. Show that $H^{0,1}_{\bar\partial}(\mathbb{C}) = 0$, but that $H^{0,1}_{\bar\partial}(\mathbb{C}^2 \setminus \{0\})$ is infinite-dimensional.

Exercise 46.5.3. For a compact Riemann surface X and a line bundle L, show that $H^1(X, L) \cong H^0(X, K_X \otimes L^\vee)^\vee$, and deduce that $H^1(X, L) = 0$ if $\deg L > 2g - 2$.

Exercise 46.5.4. Show that the holomorphic Euler characteristic $\chi(X, E) = \sum_q (-1)^q \dim H^q(X, E)$ of a compact complex manifold is the index of the elliptic operator $\bar\partial_E + \bar\partial_E^* \colon \mathcal{A}^{0,\mathrm{even}}(E) \to \mathcal{A}^{0,\mathrm{odd}}(E)$.

Chapter 47

Kähler Manifolds

47.1 Introduction

A Kähler metric is a Hermitian metric whose fundamental 2-form is closed. This single condition makes the Riemannian and the complex structure compatible to second order (Proposition 47.3.2), it holds on every projective manifold (Corollary 47.4.3), and it is exactly what the Hodge decomposition of Chapter 48 needs. This chapter develops Kähler metrics and the pointwise linear algebra of the Lefschetz operator (Theorem 47.5.2), which becomes the hard Lefschetz theorem in Chapter 49. References: [GH78, Chapter 0], [Huy05, Chapters 1 and 3], [Voi02b, Chapters 3 and 6].

Prerequisites. Chapter 46 (Dolbeault Cohomology), Chapter 40 (Riemannian Geometry) and Chapter 13 (Representations of Groups and Lie Algebras).

Throughout, X is a complex manifold of complex dimension n and J is its almost complex structure (multiplication by i on tangent vectors). Forms are complex valued unless said otherwise, and $\Lambda^{p,q}$ denotes forms of type (p,q).

47.2 Hermitian metrics and the fundamental form

Definition 47.2.1. A *Hermitian metric* on X is a Riemannian metric g with $g(JU,JV)=g(U,V)$ for all tangent vectors U,V. Its *fundamental form* is the real 2-form $\omega(U,V)=g(JU,V)$.

Proposition 47.2.2. *Let g be a Hermitian metric and extend it complex bilinearly to complexified tangent vectors. In holomorphic coordinates $z^j=x^j+iy^j$ put $h_{j\bar k}=2g(\partial/\partial z^j,\partial/\partial\bar z^k)$. Then $(h_{j\bar k})$ is a positive definite Hermitian matrix at every point and*

$$\omega=\frac{i}{2}\sum_{j,k}h_{j\bar k}\,dz^j\wedge d\bar z^k,$$

a real form of type $(1,1)$. Conversely, every real $(1,1)$-form $\frac{i}{2}\sum h_{j\bar k}\,dz^j\wedge d\bar z^k$ whose matrix $(h_{j\bar k})$ is positive definite (such a form is called positive*) is the fundamental form of exactly one Hermitian metric. For the standard metric of $\mathbb{C}^n$ one gets $\omega_0=\frac{i}{2}\sum_j dz^j\wedge d\bar z^j=\sum_j dx^j\wedge dy^j$.*

Proof. The J-invariance of g says that g pairs the $+i$- and $-i$-eigenspaces of J with each other and each of them with itself trivially: $g(\partial_j,\partial_k)=g(J\partial_j,J\partial_k)=i^2g(\partial_j,\partial_k)$ forces $g(\partial_j,\partial_k)=0$, where $\partial_j=\partial/\partial z^j$, and similarly for $\bar\partial_j=\partial/\partial\bar z^j$. So $g=\frac12\sum h_{j\bar k}(dz^j\otimes d\bar z^k+d\bar z^k\otimes dz^j)$. Since g is real, $\overline{h_{j\bar k}}=h_{k\bar j}$, and for $U=\sum u^j\partial_j+\overline{u^j}\bar\partial_j$ real, $g(U,U)=\sum h_{j\bar k}u^j\overline{u^k}$, so $(h_{j\bar k})$ is positive definite. Since $J\partial_j=i\partial_j$ and $J\bar\partial_k=-i\bar\partial_k$, $\omega(\partial_j,\bar\partial_k)=g(i\partial_j,\bar\partial_k)=\frac{i}{2}h_{j\bar k}$, $\omega(\bar\partial_k,\partial_j)=-\frac{i}{2}h_{j\bar k}$ and ω vanishes on pairs of the same type; this is the displayed formula. Conversely, a

positive $(1,1)$-form determines $h_{j\bar{k}}$ and hence g by the same formulas, and g is then a J-invariant Riemannian metric. For $\mathbb{C}^n$, $g(\partial_j, \bar{\partial}_k) = \frac{1}{4}(g(\partial_{x^j}, \partial_{x^k}) + g(\partial_{y^j}, \partial_{y^k})) = \frac{1}{2}\delta_{jk}$, and $\frac{i}{2} dz \wedge d\bar{z} = dx \wedge dy$. □

Remark 47.2.3. At every point x, the metric g_x, the fundamental form ω_x and the complex structure J_x form a compatible triple on $T_x X$ in the sense of Proposition 11.11.5, so any two of them determine the third.

1. g and J give $\omega(U,V) = g(JU,V)$: this is the definition.
2. ω and J give $g(U,V) = \omega(U,JV)$, provided $\omega(JU,JV) = \omega(U,V)$ and $\omega(U,JU) > 0$ for $U \neq 0$. These two conditions say, in coordinates, that ω is a real $(1,1)$-form with positive definite matrix $(h_{j\bar{k}})$, which is the converse part of Proposition 47.2.2. So a Hermitian manifold can equally be described by (X,ω) with ω a positive $(1,1)$-form, which is how Kähler metrics are usually given.
3. g and ω give J through $\omega(U,V) = g(JU,V)$. Here the pair is restricted twice: the endomorphism A defined by $\omega(U,V) = g(AU,V)$ must satisfy $A^2 = -1$ for A to be an almost complex structure, and the almost complex structure must moreover be integrable (Theorem 44.5.4) to come from a complex manifold.

Correspondingly, a linear map of $T_x X$ preserving two of the three structures preserves the third, and the maps preserving all three form the unitary group $\mathrm{U}(n)$. The analogous "two out of three" statement one level up concerns the Kähler condition: for a Hermitian metric, $d\omega = 0$ holds if and only if J is parallel for the Levi-Civita connection of g (Remark 47.3.3), and then g, ω and J are all parallel. Dropping the integrability of J and keeping $d\omega = 0$ gives the *almost Kähler* manifolds, which are the symplectic manifolds with a compatible almost complex structure (Chapter 41).

Proposition 47.2.4. *For a Hermitian metric, $\omega^n/n! = \mathrm{vol}_g$ is the Riemannian volume form for the canonical orientation. The restriction of ω to a complex submanifold Y is the fundamental form of the restricted metric. Hence for a nonempty compact complex submanifold, or analytic subvariety, Z of pure dimension k,*

$$\int_Z \frac{\omega^k}{k!} = \mathrm{vol}(Z_{\mathrm{reg}}) > 0.$$

Proof. At a point choose, by Gram–Schmidt for the Hermitian form $(h_{j\bar{k}})$, a linear change of coordinates making $h_{j\bar{k}} = \delta_{jk}$ there. Then $\omega = \sum_j dx^j \wedge dy^j$ at that point, so $\omega^n = n!\, dx^1 \wedge dy^1 \wedge \cdots \wedge dx^n \wedge dy^n$, which is the volume form of the orthonormal basis $\partial_{x^1}, \partial_{y^1}, \ldots$ in the canonical orientation. The tangent spaces of Y are J-stable, so the restricted metric is Hermitian, and $\omega(U,V) = g(JU,V)$ for U, V tangent to Y is its fundamental form. The formula for Z follows by applying the first statement to Z_{reg}, which is a nonempty open complex manifold of dimension k; the integral converges by Proposition 44.6.2. □

47.3 The Kähler condition

Definition 47.3.1. A Hermitian metric is *Kähler* if its fundamental form is closed, $d\omega = 0$; then ω is called a *Kähler form*. A complex manifold is *Kähler* if it admits a Kähler metric, and the class $[\omega] \in H^2(X;\mathbb{R})$ is then a *Kähler class*.

Proposition 47.3.2. *For a Hermitian metric on X the following are equivalent:*

1. $d\omega = 0$;

2. *around every point p there are holomorphic coordinates centred at p with $h_{j\bar{k}} = \delta_{jk} + O(|z|^2)$ (the metric* osculates to order two *to the flat metric);*

3. *locally $\omega = i\partial\bar{\partial}\varphi$ for a smooth real function φ, a* local Kähler potential.

Proof. (2)⇒(1): in the coordinates of (2), the coefficients of $d\omega$ at p are first derivatives of the $h_{j\bar{k}}$ at 0, which vanish; since p is arbitrary, $d\omega = 0$. (3)⇒(1): $d(i\partial\bar{\partial}\varphi) = i(\partial + \bar{\partial})\partial\bar{\partial}\varphi = 0$ by Proposition 44.5.3.

(1)⇒(2): after a linear change of coordinates, $h_{j\bar{k}}(0) = \delta_{jk}$. By Taylor's theorem

$$h_{j\bar{k}} = \delta_{jk} + \sum_l \left(a_{jkl} z^l + b_{jkl}\bar{z}^l\right) + O(|z|^2),$$

and $\overline{h_{j\bar{k}}} = h_{k\bar{j}}$ gives $b_{jkl} = \overline{a_{kjl}}$. The $(2,1)$-component of $d\omega$ is $\partial\omega = \frac{i}{2}\sum \partial_l h_{j\bar{k}}\, dz^l \wedge dz^j \wedge d\bar{z}^k$, and its vanishing at 0 says that $a_{jkl} = a_{lkj}$. Define $w^k = z^k + \frac{1}{2}\sum_{j,l} a_{jkl} z^j z^l$. Since $dw^k = dz^k$ at 0, the map $z \mapsto w$ is a diffeomorphism near 0 (Theorem 27.11.2), and its inverse is holomorphic because its derivative, the inverse of a complex linear map, is complex linear; so the w^k are holomorphic coordinates near 0. By the symmetry $a_{jkl} = a_{lkj}$, $dw^k = dz^k + \sum_{j,l} a_{jkl} z^l\, dz^j$. Hence

$$\sum_k dw^k \wedge d\bar{w}^k = \sum_{j,k}\left(\delta_{jk} + \sum_l a_{jkl} z^l + \sum_l \overline{a_{kjl}}\,\bar{z}^l\right) dz^j \wedge d\bar{z}^k + O(|z|^2),$$

so $\omega - \frac{i}{2}\sum_k dw^k \wedge d\bar{w}^k$ has coefficients $O(|z|^2)$ in the basis $dz^j \wedge d\bar{z}^k$, hence also in the basis $dw^j \wedge d\bar{w}^k$, which differs from it by a smooth change of basis, and $|z|$ and $|w|$ are comparable near 0. This is (2).

(1)⇒(3): on a polydisc Δ around a point, $\omega = d\eta$ for a real 1-form η by the Poincaré lemma (Theorem 37.2.4). Write $\eta = \eta^{1,0} + \eta^{0,1}$ with $\eta^{1,0} = \overline{\eta^{0,1}}$. The $(0,2)$-component of $d\eta = \omega$ is $\bar{\partial}\eta^{0,1} = 0$, so on a smaller polydisc $\eta^{0,1} = \bar{\partial} f$ for a smooth complex function f by Lemma 46.2.2, and then $\eta^{1,0} = \partial\bar{f}$. The $(1,1)$-component gives

$$\omega = \partial\eta^{0,1} + \bar{\partial}\eta^{1,0} = \partial\bar{\partial} f + \bar{\partial}\partial\bar{f} = \partial\bar{\partial}(f - \bar{f}),$$

so $\varphi = -i(f - \bar{f}) = 2\operatorname{Im} f$ is real and $i\partial\bar{\partial}\varphi = \omega$. □

Remark 47.3.3. The Kähler condition has several other equivalent forms: J is parallel for the Levi-Civita connection; the Levi-Civita connection of g, transported to $T^{1,0}X$, is the Chern connection of the Hermitian holomorphic bundle $T^{1,0}X$ (Theorem 45.4.1); the holonomy group of g is contained in $U(n)$ [Huy05, Chapter 4]. Condition (2) is the useful one in practice: any first-order identity between operators built from g and J that holds on $\mathbb{C}^n$ with the flat metric holds on every Kähler manifold, since it can be checked at the centre of such coordinates. The Kähler identities of Chapter 48 are proved this way.

47.4 Examples

Proposition 47.4.1 (Fubini–Study metric)**.** *Let σ be a holomorphic section of the projection $\mathbb{C}^{n+1} \setminus \{0\} \to \mathbb{CP}^n$ over an open set U. The form*

$$\omega_{FS} = \frac{i}{2\pi}\partial\bar{\partial}\log|\sigma|^2$$

does not depend on σ, and so defines a real $(1,1)$-form on $\mathbb{CP}^n$. It is a Kähler form, invariant under the action of $U(n+1)$, and $[\omega_{FS}] = c_1(\mathcal{O}(1))$, the positive generator of $H^2(\mathbb{CP}^n;\mathbb{Z})$.

Proof. Another section is $\lambda\sigma$ with λ holomorphic and nowhere zero on U (shrinking U so that $\log\lambda$ exists), and $\log|\lambda\sigma|^2 = \log|\sigma|^2 + \log\lambda + \overline{\log\lambda}$. Since $\bar\partial\log\lambda = 0$ and $\partial\overline{\log\lambda} = 0$, $\partial\bar\partial$ kills the last two terms, so ω_{FS} is well defined; it is real because $\overline{i\partial\bar\partial u} = -i\bar\partial\partial u = i\partial\bar\partial u$ for real u, and closed by Proposition 47.3.2. For $A \in U(n+1)$, $A\circ\sigma$ is a section over $A(U)$ with $|A\sigma| = |\sigma|$, so ω_{FS} is $U(n+1)$-invariant. As $U(n+1)$ acts transitively on $\mathbb{CP}^n$, positivity need only be checked at $[1:0:\cdots:0]$. With the section $\sigma(w) = (1, w_1, \dots, w_n)$ in affine coordinates, $\log|\sigma|^2 = \log(1+|w|^2) = |w|^2 + O(|w|^4)$, so at $w = 0$

$$\omega_{FS} = \frac{i}{2\pi}\sum_j dw^j\wedge d\bar w^j = \frac{1}{\pi}\sum_j du^j\wedge dv^j \qquad (w^j = u^j + iv^j),$$

which is positive. Finally, σ is a nowhere-vanishing section of $\mathcal{O}(-1) \subseteq \mathbb{CP}^n\times\mathbb{C}^{n+1}$, and the dual frame σ^* of $\mathcal{O}(1)$ has norm $|\sigma^*|^2 = |\sigma|^{-2} = e^{-\varphi}$ for the dual of the standard metric, with $\varphi = \log|\sigma|^2$. By Theorem 45.4.1, $c_1(\mathcal{O}(1)) = [\frac{i}{2\pi}\partial\bar\partial\varphi] = [\omega_{FS}]$. This is the positive generator of $H^2(\mathbb{CP}^n;\mathbb{Z})$ by Example 45.4.3. □

Example 47.4.2. *The Fubini–Study form in coordinates.* On $U_0 = \{[1:w_1:\cdots:w_n]\} \cong \mathbb{C}^n$, with $\sigma(w) = (1,w)$ and $|w|^2 = \sum_j |w_j|^2$, $\bar\partial\log(1+|w|^2) = \sum_k w_k\,d\bar w_k/(1+|w|^2)$, and applying ∂,

$$\omega_{FS} = \frac{i}{2\pi}\Big(\frac{\sum_j dw_j\wedge d\bar w_j}{1+|w|^2} - \frac{(\sum_j \bar w_j\,dw_j)\wedge(\sum_k w_k\,d\bar w_k)}{(1+|w|^2)^2}\Big) = \frac{i}{2}\sum_{j,k}\frac{h_{j\bar k}}{\pi}\,dw_j\wedge d\bar w_k, \quad h_{j\bar k} = \frac{(1+|w|^2)\delta_{jk} - \bar w_j w_k}{(1+|w|^2)^2}.$$

This shows positivity at every point directly, without the homogeneity argument: for $\xi\in\mathbb{C}^n$, by the Cauchy–Schwarz inequality $|\sum_k w_k\xi_k|^2 \le |w|^2|\xi|^2$,

$$\sum_{j,k}\bar\xi_j h_{j\bar k}\xi_k = \frac{(1+|w|^2)|\xi|^2 - |\sum_k w_k\xi_k|^2}{(1+|w|^2)^2} \ge \frac{|\xi|^2}{(1+|w|^2)^2} > 0.$$

For $n=1$, with $w = u+iv$ and $dw\wedge d\bar w = -2i\,du\wedge dv$, $\omega_{FS} = \frac{1}{\pi}\frac{du\wedge dv}{(1+|w|^2)^2}$, and the area of $\mathbb{CP}^1$ is $\int_{U_0}\omega_{FS} = \frac{1}{\pi}\int_0^{2\pi}\int_0^\infty \frac{r\,dr\,d\theta}{(1+r^2)^2} = 1$, the complement of U_0 being a point. This agrees with $[\omega_{FS}] = c_1(\mathcal{O}(1))$, whose degree is 1; in general $\int_{\mathbb{CP}^n}\omega_{FS}^n = 1$, so the Riemannian volume of $\mathbb{CP}^n$ for this normalisation is $\frac{1}{n!}$, as $\omega^n/n!$ is the volume form. On $\mathbb{CP}^1$ the metric of $\pi\omega_{FS}$ is invariant under the transitive isometric action of $U(2)$, so its Gaussian curvature K is constant; its area is π, so $K\pi = 4\pi$ by the Gauss–Bonnet theorem (Theorem 40.12.5), $K = 4$, and $\mathbb{CP}^1$ with this metric is isometric to the round sphere of radius $\frac{1}{2}$ (Theorem 40.16.10).

Corollary 47.4.3. *A complex submanifold of a Kähler manifold is Kähler with the restricted metric. In particular every* projective manifold*, that is, every closed complex submanifold $X\subseteq\mathbb{CP}^N$, is Kähler, and its Kähler class $[\omega_{FS}|_X]$ is the restriction of the hyperplane class $c_1(\mathcal{O}(1))$.*

Proof. By Proposition 47.2.4, the fundamental form of the restricted metric is the restriction of ω, and restriction of forms commutes with d. □

Example 47.4.4.

1. $\mathbb{C}^n$ with ω_0, and complex tori $\mathbb{C}^n/\Lambda$ with the form induced by ω_0, which is invariant under translations. Products of Kähler manifolds are Kähler, with $\omega = \mathrm{pr}_1^*\omega_1 + \mathrm{pr}_2^*\omega_2$.
2. Every Hermitian metric on a Riemann surface is Kähler, since $d\omega$ is a 3-form on a real surface.

3. Every smooth projective variety is Kähler (Corollary 47.4.3). The blow-up of a compact Kähler manifold along a compact complex submanifold is again Kähler [Voi02b, Chapter 3].

4. Not every compact Kähler manifold is projective: a complex torus $\mathbb{C}^n/\Lambda$ with $n \geq 2$ and Λ general has no nonconstant meromorphic functions, hence no embedding into projective space [BL04]; see also Chapter 61. Voisin even showed that there are compact Kähler manifolds that are not homotopy equivalent to any projective manifold [Voi04].

Proposition 47.4.5. *Let X be a compact Kähler manifold with Kähler form ω. Then $[\omega]^k \neq 0$ in $H^{2k}(X;\mathbb{R})$ for $0 \leq k \leq n$; in particular $b_{2k}(X) \geq 1$. The cycle class (Proposition 44.6.2) of a nonempty analytic subvariety Z of pure dimension k is nonzero in $H^{2n-2k}(X;\mathbb{R})$.*

Proof. By Proposition 47.2.4, $\int_X \omega^k \wedge \omega^{n-k} = n!\,\mathrm{vol}(X) > 0$. If $\omega^k = d\eta$ were exact, then $\omega^n = d(\eta \wedge \omega^{n-k})$ would be exact and its integral would vanish by Stokes' theorem. For Z, the class $[Z]$ is represented by the current of integration over Z_{reg}, whose value on the closed form ω^k is $\int_Z \omega^k = k!\,\mathrm{vol}(Z_{\mathrm{reg}}) > 0$; since a closed current that is exact vanishes on closed forms, $[Z] \neq 0$ (Theorem 37.4.3). □

Lemma 47.4.6 (Wirtinger inequality for 2-planes)**.** *Let g be a Hermitian metric with fundamental form ω, and let u, v be orthonormal tangent vectors at a point. Then $|\omega(u,v)| \leq 1$, with equality if and only if $v = \pm Ju$, that is, if and only if the real 2-plane spanned by u and v is a complex line.*

Proof. $\omega(u,v) = g(Ju,v)$, and $|Ju| = |u| = 1$ because g is J-invariant. By the Cauchy–Schwarz inequality $|g(Ju,v)| \leq |Ju||v| = 1$, with equality if and only if $v = \pm Ju$. A real plane containing u and Ju is J-stable, hence a complex line. □

Proposition 47.4.7. *Let (X,ω) be a Kähler manifold and $C \subseteq X$ a compact complex curve (a compact complex submanifold of dimension 1). If Σ is a compact oriented surface and $f : \Sigma \to X$ an immersion with $f_*[\Sigma] = [C]$ in $H_2(X;\mathbb{R})$, then $\mathrm{area}(f) \geq \mathrm{area}(C) = \int_C \omega$, with equality if $f(\Sigma)$ is a complex curve with its complex orientation. So holomorphic curves minimise area in their homology class.*

Proof. At a point of Σ choose an oriented orthonormal basis u, v of $T\Sigma$ for the pulled-back metric; the area form of f^*g takes the value 1 on (u,v), while $f^*\omega$ takes the value $\omega(df\,u, df\,v) \leq 1$ by Lemma 47.4.6. So $f^*\omega \leq dA$ pointwise, and

$$\mathrm{area}(f) = \int_\Sigma dA \geq \int_\Sigma f^*\omega = \int_C \omega = \mathrm{area}(C),$$

where the middle equality holds because ω is closed and $f_*[\Sigma] = [C]$ (Theorem 37.3.1), and the last is Proposition 47.2.4 with $k = 1$. If $f(\Sigma)$ is a complex curve with the complex orientation, equality holds pointwise by the equality case of the lemma. □

Remark 47.4.8. The same argument, with the Wirtinger inequality $\frac{\omega^k}{k!} \leq \mathrm{vol}$ on real $2k$-planes, shows that compact complex submanifolds of any dimension minimise volume in their homology class [GH78, Chapter 0, Section 2]. A closed form that is bounded by the volume on every plane, with equality on a class of submanifolds, is called a *calibration*; the Kähler form is the first example, and the holomorphic volume form of a Calabi–Yau manifold calibrates the *special Lagrangian* submanifolds [Joy00].

Counterexample 47.4.9. The Hopf surface $X = (\mathbb{C}^2 \setminus \{0\})/\langle z \mapsto 2z \rangle$ is a compact complex manifold diffeomorphic to $S^1 \times S^3$ (Exercise 44.7.3), and it is not Kähler: $H^2(X;\mathbb{R}) = 0$, whereas a compact Kähler manifold of positive dimension has $H^2 \neq 0$ by Proposition 47.4.5. Another obstruction, that $b_1(X) = 1$ is odd, is explained in Chapter 48.

47.5 Linear algebra of the Lefschetz operator

Definition 47.5.1. Let V be a real vector space of dimension $2n$ with a complex structure J and a J-invariant inner product g, with fundamental form ω. Let $\Lambda^\bullet = \Lambda^\bullet V^* \otimes_{\mathbb{R}} \mathbb{C}$ with the Hermitian inner product induced by g, for which the wedge products of an orthonormal basis of V^* form an orthonormal basis. Define:

- the *Lefschetz operator* $L(\alpha) = \omega \wedge \alpha$, of degree +2;
- its adjoint $\Lambda = L^*$, of degree −2;
- the *counting operator* $H = \sum_k (k-n)\pi_k$, where π_k is the projection onto Λ^k.

On a Hermitian manifold these operators act pointwise on forms. A form α with $\Lambda\alpha = 0$ is called *primitive*, and P^k denotes the primitive forms of degree k.

Theorem 47.5.2. *The operators satisfy*

$$[L, \Lambda] = H, \qquad [H, L] = 2L, \qquad [H, \Lambda] = -2\Lambda,$$

so $e = L$, $f = \Lambda$, $h = H$ define a representation of $\mathfrak{sl}_2$ on $\Lambda^\bullet$ in which Λ^k is the weight space of weight $k - n$. Consequently:

1. *for $0 \le k \le n$, $L^{n-k} \colon \Lambda^k \to \Lambda^{2n-k}$ is an isomorphism;*
2. *$P^k = 0$ for $k > n$, and every $\alpha \in \Lambda^k$ can be written uniquely as $\alpha = \sum_{j \ge 0} L^j \alpha_j$ with $\alpha_j \in P^{k-2j}$ (the* Lefschetz *or* primitive decomposition*);*
3. *for $k \le n$, a form $\alpha \in \Lambda^k$ is primitive if and only if $L^{n-k+1}\alpha = 0$;*
4. *L, Λ map $\Lambda^{p,q}$ to $\Lambda^{p+1,q+1}$ and $\Lambda^{p-1,q-1}$, and if α has type (p, q) then each α_j has type $(p - j, q - j)$.*

Proof. The last two commutators hold because L raises the degree by 2 and Λ lowers it by 2: for $\alpha \in \Lambda^k$, $HL\alpha - LH\alpha = (k + 2 - n)L\alpha - (k - n)L\alpha = 2L\alpha$, and similarly for Λ.

For $[L, \Lambda] = H$, choose an orthonormal basis $e_1, Je_1, \dots, e_n, Je_n$ of V (possible by induction: $e_1^\perp$ is J-stable when e_1 is a unit vector and $e_1 \perp Je_1$, as $g(e_1, Je_1) = g(Je_1, J^2 e_1) = -g(Je_1, e_1)$). Let $x^1, y^1, \dots, x^n, y^n$ be the dual basis, so that $\omega = \sum_j \omega_j$ with $\omega_j = x^j \wedge y^j$. The algebra $\Lambda^\bullet$ is the graded tensor product of the four-dimensional algebras $A_j = \Lambda^\bullet\langle x^j, y^j\rangle$, with the product inner product; $L = \sum_j L_j$ where L_j is multiplication by ω_j in the j-th factor, and so $\Lambda = \sum_j \Lambda_j$ with Λ_j the adjoint of L_j acting in the j-th factor. Since L_j and Λ_j have even degree, operators in different factors commute, and

$$[L, \Lambda] = \sum_{j,k} [L_j, \Lambda_k] = \sum_j [L_j, \Lambda_j].$$

In A_j, with orthonormal basis $1, x^j, y^j, x^j \wedge y^j$, the operator L_j maps $1 \mapsto x^j \wedge y^j$ and kills the other basis vectors, and Λ_j maps $x^j \wedge y^j \mapsto 1$ and kills the others. So $[L_j, \Lambda_j]$ is -1 on 1, 0 on x^j, y^j and $+1$ on $x^j \wedge y^j$: it is multiplication by $(\deg_j - 1)$, where $\deg_j$ is the degree in the j-th factor. On a product of basis elements of total degree $k = \sum_j \deg_j$, $[L, \Lambda]$ is therefore multiplication by $\sum_j (\deg_j - 1) = k - n$, which is H.

The consequences are statements about finite-dimensional $\mathfrak{sl}_2$-representations. The map $(e, f, h) \mapsto (f, e, -h)$ is an automorphism of $\mathfrak{sl}_2$, so $e' = \Lambda$, $f' = L$, $h' = -H$ is again a representation, in which Λ^k has weight $n - k$. Corollary 13.6.4 for (e', f', h') says: $f'^j = L^j$ is

an isomorphism from weight j to weight $-j$, which for $j = n-k$ is (1); the primitive vectors $\ker e' = \ker \Lambda$ occur only in weights $j \geq 0$, that is, in degrees $k \leq n$; and

$$\Lambda^\bullet = \bigoplus_{j\geq 0} \bigoplus_{i=0}^{j} L^i P^{n-j},$$

which, restricted to degree k, is (2). For (3), write $\alpha = \sum_i L^i \alpha_i$ with $\alpha_i \in P^{k-2i}$. A primitive vector of degree $m \leq n$ generates an irreducible representation with basis $\alpha, L\alpha, \ldots, L^{n-m}\alpha$ and $L^{n-m+1}\alpha = 0$ (Theorem 13.6.2). So $L^{n-k+1}L^i\alpha_i = 0$ for $i = 0$, while for $i \geq 1$ the exponent $n-k+1+i \leq n-(k-2i)$ and $L^{n-k+1+i}\alpha_i \neq 0$ if $\alpha_i \neq 0$. These terms lie in different summands of the direct sum above, so $L^{n-k+1}\alpha = 0$ if and only if $\alpha_i = 0$ for all $i \geq 1$, that is, if and only if $\alpha = \alpha_0$ is primitive.

For (4), L has type $(1,1)$ because ω does (Proposition 47.2.2); since the spaces $\Lambda^{p,q}$ are mutually orthogonal (the basis $dz^I \wedge d\bar{z}^K$ of an orthonormal complex frame is orthogonal), the adjoint Λ has type $(-1,-1)$. For $\theta \in \mathbb{R}$ let T_θ act on $\Lambda^{p,q}$ by $e^{i(p-q)\theta}$. It commutes with L and Λ, hence maps $L^j P^{k-2j}$ to itself, and by uniqueness in (2) $(T_\theta\alpha)_j = T_\theta(\alpha_j)$. If α has type (p,q) then $T_\theta\alpha = e^{i(p-q)\theta}\alpha$, so $T_\theta\alpha_j = e^{i(p-q)\theta}\alpha_j$ for all θ, and α_j has only components of types (p',q') with $p'-q' = p-q$ and $p'+q' = k-2j$, that is, $(p',q') = (p-j, q-j)$. □

Remark 47.5.3. For a primitive form $\alpha \in P^k$ of type (p,q), the Hodge star operator is given by *Weil's formula*

$$*\alpha = (-1)^{k(k+1)/2}\, i^{p-q}\, \frac{L^{n-k}\alpha}{(n-k)!},$$

and it follows that $i^{p-q}(-1)^{k(k-1)/2}\,\alpha \wedge \bar{\alpha} \wedge \omega^{n-k}$ is a positive multiple of the volume form when $\alpha \neq 0$ [Huy05, Section 1.2]. For instance, when $n = 1$ and $\alpha = dz$, one has $*dz = -i\,dz$ and $i\,dz \wedge d\bar{z} = 2\,dx \wedge dy$. Integrated over a compact Kähler manifold, this positivity becomes the Hodge–Riemann bilinear relations of Chapter 49, the source of the polarisations of the Hodge structures of Part IX.

47.6 Exercises

Exercise 47.6.1. Compute $\int_{\mathbb{CP}^1} \omega_{FS}$ directly in the affine coordinate w, and deduce that $\int_{\mathbb{CP}^n} \omega_{FS}^n = 1$.

Solution. On $\mathbb{CP}^1$, $\omega_{FS} = \frac{i}{2\pi}\partial\bar{\partial}\log(1+|w|^2) = \frac{i}{2\pi}\frac{dw\wedge d\bar{w}}{(1+|w|^2)^2}$, and the point at infinity has measure zero. In polar coordinates $\frac{i}{2}dw \wedge d\bar{w} = r\,dr\,d\theta$, so

$$\int_{\mathbb{CP}^1} \omega_{FS} = \frac{1}{\pi}\int_0^{2\pi}\int_0^{\infty} \frac{r\,dr}{(1+r^2)^2}\,d\theta = \frac{1}{\pi}\cdot 2\pi\cdot\frac{1}{2} = 1.$$

A line generates $H_2(\mathbb{CP}^n;\mathbb{Z})$ (it is the 2-skeleton of the standard cell structure), so $[\omega_{FS}]$ is the generator α of $H^2(\mathbb{CP}^n;\mathbb{Z})$ that is positive on lines, and α^n generates $H^{2n}(\mathbb{CP}^n;\mathbb{Z})$ (Theorem 22.5.1); it is positive on the fundamental class since ω_{FS}^n is a positive multiple of the volume form (Proposition 47.2.4). So $\int_{\mathbb{CP}^n} \omega_{FS}^n = 1$. □

Exercise 47.6.2. Show that $\dim P^k = \binom{2n}{k} - \binom{2n}{k-2}$ for $k \leq n$, and list the dimensions of P^0, P^1, P^2 when $n = 2$.

Solution. By Theorem 47.5.2(2), $\Lambda^k = P^k \oplus L\Lambda^{k-2}$, and $L\colon \Lambda^{k-2} \to \Lambda^k$ is injective for $k \leq n$ by (1) (as L^{n-k+2} is injective on Λ^{k-2}). For $n = 2$ the dimensions are 1, 4 and $6-1 = 5$. □

Exercise 47.6.3. Show that the Kähler form of a compact Kähler manifold of positive dimension is never of the form $i\partial\bar\partial\varphi$ for a smooth function φ on all of X, while on $\mathbb{C}^n$ one has $\omega_0 = i\partial\bar\partial(\frac{1}{2}|z|^2)$.

Solution. $i\partial\bar\partial\varphi = d(i\bar\partial\varphi)$, because $d = \partial + \bar\partial$ and $\bar\partial\bar\partial\varphi = 0$. So $[\omega] = 0$, contradicting Proposition 47.4.5 for $k = 1$. On $\mathbb{C}^n$, $i\partial\bar\partial\frac{1}{2}\sum_j z^j\bar z^j = \frac{i}{2}\sum_j dz^j \wedge d\bar z^j = \omega_0$. □

Exercise 47.6.4. Let $C \subseteq X$ be the image of the line $\{z_1 = 0\}$ in the Hopf surface of Counterexample 47.4.9. Show that C is an elliptic curve and give a second proof that X is not Kähler using $\int_C \omega$.

Solution. $C = (\mathbb{C}^\times)/\langle 2\rangle \cong \mathbb{C}/(2\pi i\mathbb{Z}+\log 2\,\mathbb{Z})$ via the exponential, a complex torus of dimension one. If ω were a Kähler form, then $\omega = d\eta$ because $H^2(X;\mathbb{R}) = 0$, so $\int_C \omega = \int_C d\eta = 0$ by Stokes' theorem; but $\int_C \omega = \mathrm{vol}(C) > 0$ by Proposition 47.2.4. □

Exercise 47.6.5. Show that the set of Kähler classes of a compact Kähler manifold is an open convex cone in the real vector space $H^{1,1}(X) \cap H^2(X;\mathbb{R})$ of classes represented by closed real $(1,1)$-forms.

Solution. If ω_1, ω_2 are Kähler forms and $s, t > 0$, then $s\omega_1 + t\omega_2$ is a closed positive real $(1,1)$-form, hence Kähler; this gives convexity and the cone property. For openness, choose closed real $(1,1)$-forms $\beta_1, \dots, \beta_m$ whose classes form a basis of the space. Positivity is an open condition on the coefficient matrix and X is compact, so $\omega + \sum_i c_i\beta_i$ is positive for all small $|c_i|$. □

Exercise 47.6.6. Let $V = V_1 \oplus V_2$ be an orthogonal sum of Hermitian vector spaces of complex dimensions n_1, n_2. Show that $L = L_1 \otimes 1 + 1 \otimes L_2$, $\Lambda = \Lambda_1 \otimes 1 + 1 \otimes \Lambda_2$ on $\Lambda^\bullet V^* = \Lambda^\bullet V_1^* \otimes \Lambda^\bullet V_2^*$, and that the $\mathfrak{sl}_2$-representation of Theorem 47.5.2 is the tensor product of those for V_1 and V_2. Deduce the decomposition of $\Lambda^\bullet(\mathbb{C}^n)^*$ into irreducibles using the Clebsch–Gordan formula.

Solution. $\omega = \omega_1 + \omega_2$, and the tensor product inner product is the induced one, which gives the formulas for L and Λ (the operators have even degree, so there are no signs). The counting operators satisfy $H = H_1 \otimes 1 + 1 \otimes H_2$ since $n = n_1 + n_2$; this is the tensor product representation. For $n = 1$ the representation on $\Lambda^\bullet(\mathbb{C})^* \cong \mathbb{C}^4$ is $V(1) \oplus 2V(0)$ (from the proof of Theorem 47.5.2: $1, x \wedge y$ span $V(1)$, and x, y are primitive of weight 0). Hence $\Lambda^\bullet(\mathbb{C}^n)^* \cong (V(1) \oplus 2V(0))^{\otimes n}$, which decomposes by the Clebsch–Gordan formula $V(a) \otimes V(b) \cong V(a+b) \oplus V(a+b-2) \oplus \cdots \oplus V(|a-b|)$ (Exercise 13.12.2). □

Chapter 48

The Hodge Decomposition

48.1 Introduction

On a compact Kähler manifold the complex cohomology splits as

$$H^k(X;\mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X), \qquad \overline{H^{p,q}(X)} = H^{q,p}(X), \qquad H^{p,q}(X) \cong H^q(X,\Omega_X^p)$$

(Theorem 48.5.2). This *Hodge decomposition* is the basic structure of the whole subject: Part IX axiomatises it as a Hodge structure, and the Hodge conjecture asks which rational classes of type (p,p) come from algebraic cycles. The proof combines the Hodge theorem of Chapter 43 with the *Kähler identities* (Theorem 48.3.3), which give $\Delta_d = 2\Delta_{\bar\partial}$, so that the Laplacian preserves types. We also prove the $\partial\bar\partial$-lemma, derive the topological restrictions on Kähler manifolds, and compute examples. References: [GH78, Chapter 0, Sections 6–7], [Huy05, Chapter 3], [Voi02b, Chapter 6], [Wel08, Chapter V].

Prerequisites. Chapter 47 (Kähler Manifolds) and Chapter 43 (Harmonic Forms and the Hodge Theorem).

48.2 Conventions

Notation 48.2.1. Let X be a complex manifold of complex dimension n with a Hermitian metric g and fundamental form ω. Complex forms carry the Hermitian inner product $\langle\alpha,\beta\rangle$ extending the real one of Definition 40.15.1 (linear in α, conjugate linear in β). The Hodge star $*$ is extended complex linearly, so that $\alpha\wedge *\bar\beta = \langle\alpha,\beta\rangle\,\mathrm{vol}_g$, and when X is compact the L^2 product is $(\alpha,\beta) = \int_X \alpha\wedge *\bar\beta$. The operators d, δ and $\Delta = \Delta_d = d\delta + \delta d$ are extended complex linearly; since they are real, δ is still the formal adjoint of d. We write ∂^*, $\bar\partial^*$ for the formal adjoints of ∂, $\bar\partial$, and

$$\Delta_\partial = \partial\partial^* + \partial^*\partial, \qquad \Delta_{\bar\partial} = \bar\partial\bar\partial^* + \bar\partial^*\bar\partial,$$

so that $\Delta_{\bar\partial}$ is the operator $\Box$ of Definition 46.3.1. The operators L, Λ, H are those of Definition 47.5.1, applied pointwise, and $\Pi^{p,q}$ is the projection onto forms of type (p,q). Finally $c(\alpha) = \bar\alpha$ is complex conjugation of forms. It is conjugate linear and satisfies $(c\alpha, c\beta) = \overline{(\alpha,\beta)}$, $c\partial c = \bar\partial$, and $cLc = L$, $c\Lambda c = \Lambda$, since ω is real.

Lemma 48.2.2. *The complex linear Hodge star maps $\Lambda^{p,q}$ to $\Lambda^{n-q,n-p}$, forms of different types are orthogonal, and*

$$\partial^* = -*\bar\partial*, \qquad \bar\partial^* = -*\partial*, \qquad \delta = \partial^* + \bar\partial^*,$$

where ∂^* has type $(-1,0)$ and $\bar\partial^*$ has type $(0,-1)$. Moreover $c\,\partial^* c = \bar\partial^*$.

Proof. At a point choose holomorphic coordinates with $h_{j\bar{k}} = \delta_{jk}$ there (as in the proof of Proposition 47.2.4). Then $dz^j = dx^j + i\,dy^j$ with dx^j, dy^j orthonormal, so $\langle dz^j, dz^k\rangle = 2\delta_{jk}$ and $\langle dz^j, d\bar{z}^k\rangle = \langle dx^j, dx^k\rangle - \langle dy^j, dy^k\rangle = 0$ (the term $i \cdot \overline{(-i)}$ is -1). Hence the products $dz^I \wedge d\bar{z}^J$ are pairwise orthogonal, and forms of different types are orthogonal.

If $\beta \in \Lambda^{p,q}$, then $\alpha \wedge *\beta = \langle \alpha, \bar{\beta}\rangle \operatorname{vol} = 0$ for every α of degree $p+q$ and type other than (q,p), since $\bar{\beta}$ has type (q,p). The wedge pairing $\Lambda^{r,s} \times \Lambda^{n-r,n-s} \to \Lambda^{n,n}$ is perfect and pairs $\Lambda^{r,s}$ trivially with all other types of complementary degree, so the component of $*\beta$ of type $(n-r, n-s)$ vanishes for every $(r,s) \neq (q,p)$. Thus $*\beta$ has type $(n-q, n-p)$.

The real dimension of X is even, so $\delta = -*d*$ by Definition 40.15.2. On a form of type (p,q), the operator $-*\partial*$ has type $(0,-1)$, since $(p,q) \mapsto (n-q, n-p) \mapsto (n-q+1, n-p) \mapsto (p, q-1)$, and likewise $-*\bar{\partial}*$ has type $(-1,0)$. So $\delta = D' + D''$ with $D' = -*\bar{\partial}*$ of type $(-1,0)$ and $D'' = -*\partial*$ of type $(0,-1)$. For α of type $(p-1,q)$ and β of type (p,q), orthogonality of types gives $(\partial\alpha, \beta) = (d\alpha, \beta) = (\alpha, \delta\beta) = (\alpha, D'\beta)$, so $D' = \partial^*$; similarly $D'' = \bar{\partial}^*$. Finally $(c\,\partial^* c\,\alpha, \beta) = \overline{(\partial^* c\alpha, c\beta)} = \overline{(c\alpha, \partial c\beta)} = (\alpha, c\,\partial c\,\beta) = (\alpha, \bar{\partial}\beta)$. □

48.3 The Kähler identities

Lemma 48.3.1. *Let P be a differential operator on forms, defined on every Hermitian manifold of dimension n, whose coefficients at each point, in any holomorphic coordinates, are given by universal expressions in the values at that point of the $h_{j\bar{k}}$ and their first derivatives. If $P = 0$ for the flat metric on $\mathbb{C}^n$, then $P = 0$ on every Kähler manifold of dimension n.*

Proof. Let $p \in X$ and choose holomorphic coordinates centred at p with $h_{j\bar{k}} = \delta_{jk} + O(|z|^2)$ (Proposition 47.3.2). In this chart, the metric of X and the flat metric $h_{j\bar{k}} = \delta_{jk}$ have the same values and the same first derivatives at p. So the coefficients of P at p are the same for both metrics, and those of the flat metric vanish. □

Lemma 48.3.2. *Let W be a finite-dimensional complex vector space with a Hermitian inner product, and extend the inner product to $\Lambda^\bullet W$ as above. For $\theta \in W$ let $\varepsilon(\theta) = \theta \wedge -$ and let $\varepsilon(\theta)^*$ be its adjoint. Then*

$$\varepsilon(\theta)\varepsilon(\theta') = -\varepsilon(\theta')\varepsilon(\theta), \qquad \varepsilon(\varphi)^*\varepsilon(\theta) + \varepsilon(\theta)\varepsilon(\varphi)^* = \langle \theta, \varphi\rangle\,\mathrm{id}.$$

Proof. The first identity holds because $\theta \wedge \theta' = -\theta' \wedge \theta$. Both sides of the second are linear in θ and conjugate linear in φ, so it suffices to take $\theta = e_a$, $\varphi = e_b$ from an orthonormal basis $e_1, \dots, e_m$ and to evaluate on $e_I = e_{i_1} \wedge \cdots \wedge e_{i_r}$, $i_1 < \cdots < i_r$. The adjoint $\varepsilon(e_b)^*$ sends e_I to $(-1)^{s-1} e_{I\setminus b}$ if $b = i_s \in I$, and to 0 if $b \notin I$. If $a = b$, exactly one of the two terms is nonzero on e_I, depending on whether $a \in I$, and it equals e_I. If $a \neq b$, both terms vanish unless $b \in I$ and $a \notin I$. In that case moving e_b into place in $e_a \wedge e_I$ costs one extra sign compared with e_I, so the two terms cancel. □

Theorem 48.3.3 (Kähler identities)**.** *On a Kähler manifold,*

$$[\Lambda, \bar{\partial}] = -i\partial^*, \qquad [\Lambda, \partial] = i\bar{\partial}^*, \qquad [\bar{\partial}^*, L] = i\partial, \qquad [\partial^*, L] = -i\bar{\partial},$$

and $[L, d] = 0$, $[\Lambda, \delta] = 0$.

Proof. The operator $P = [\Lambda, \bar{\partial}] + i\partial^*$ satisfies the hypothesis of Lemma 48.3.1. Indeed Λ and $*$ are algebraic operators whose coefficients are universal functions of the $h_{j\bar{k}}$, and P is obtained from them and from $\bar{\partial}$ by composition, using $\partial^* = -*\bar{\partial}*$ (Lemma 48.2.2). The derivatives in $\bar{\partial}$ fall at most once on these coefficients. So it suffices to prove $P = 0$ on $\mathbb{C}^n$ with the flat metric.

There $\omega = \frac{i}{2}\sum_k dz^k \wedge d\bar z^k$. Write $e_k = \varepsilon(dz^k)$ and $\bar e_k = \varepsilon(d\bar z^k)$, which are constant operators. Then

$$L = \frac{i}{2}\sum_k e_k \bar e_k, \qquad \Lambda = L^* = -\frac{i}{2}\sum_k \bar e_k^* e_k^*, \qquad \bar\partial = \sum_l \bar e_l\, \partial_{\bar z^l}, \qquad \partial = \sum_k e_k\, \partial_{z^k},$$

where ∂_{z^k}, $\partial_{\bar z^l}$ act on the coefficients and commute with the constant operators. By Lemma 48.3.2 and the inner products $\langle d\bar z^l, d\bar z^k\rangle = 2\delta_{kl}$, $\langle d\bar z^l, dz^k\rangle = 0$ computed in Lemma 48.2.2, we have $e_k^* \bar e_l = -\bar e_l e_k^*$ and $\bar e_k^* \bar e_l = 2\delta_{kl} - \bar e_l \bar e_k^*$. Hence

$$\bar e_k^* e_k^* \bar e_l = -\bar e_k^* \bar e_l e_k^* = -2\delta_{kl} e_k^* + \bar e_l \bar e_k^* e_k^*, \qquad \text{that is,} \qquad [\bar e_k^* e_k^*, \bar e_l] = -2\delta_{kl} e_k^*,$$

and therefore $[\Lambda, \bar\partial] = -\frac{i}{2}\sum_k (-2) e_k^* \partial_{\bar z^k} = i\sum_k e_k^* \partial_{\bar z^k}$. On the other hand, integration by parts against compactly supported forms shows that the formal adjoint of ∂_{z^k} acting on coefficients is $-\partial_{\bar z^k}$, because $\int \partial_z f \cdot \bar u = -\int f\, \overline{\partial_{\bar z} u}$. So $\partial^* = \sum_k (\partial_{z^k})^* e_k^* = -\sum_k e_k^* \partial_{\bar z^k}$, and $[\Lambda, \bar\partial] = -i\partial^*$.

Conjugating with c and using $c\Lambda c = \Lambda$, $c\bar\partial c = \partial$, $c\,\partial^* c = \bar\partial^*$ and $c(-i)c = i$ gives $[\Lambda, \partial] = i\bar\partial^*$. Taking formal adjoints of the first two identities gives $[\bar\partial^*, L] = (-i\partial^*)^* = i\partial$ and $[\partial^*, L] = (i\bar\partial^*)^* = -i\bar\partial$. Finally, $d(\omega \wedge \alpha) = \omega \wedge d\alpha$ because $d\omega = 0$ and ω has even degree, so $[L, d] = 0$, and its adjoint is $[\Lambda, \delta] = 0$. □

Remark 48.3.4. The signs in the Kähler identities depend on conventions: the normalisation $\omega = \frac{i}{2}\sum h_{j\bar k} dz^j \wedge d\bar z^k$ used here, the sign of δ, and whether inner products are linear in the first or in the second variable. They agree with [Huy05, Proposition 3.1.12]. The proof via Lemma 48.3.1 is the typical use of the osculation property of Kähler metrics. The identities fail for non-Kähler Hermitian metrics, where torsion terms involving $d\omega$ appear.

48.4 Comparison of Laplacians

Theorem 48.4.1. *On a Kähler manifold, $\partial\bar\partial^* + \bar\partial^*\partial = 0$, $\bar\partial\partial^* + \partial^*\bar\partial = 0$, and*

$$\Delta_d = 2\Delta_{\bar\partial} = 2\Delta_\partial.$$

Consequently Δ_d preserves types: it commutes with every $\Pi^{p,q}$, with ∂, $\bar\partial$, ∂^*, $\bar\partial^*$, and with c. Moreover Δ_d commutes with $*$, L and Λ.

Proof. By Theorem 48.3.3, $i\bar\partial^* = [\Lambda, \partial]$, so

$$i(\partial\bar\partial^* + \bar\partial^*\partial) = \partial\Lambda\partial - \partial\partial\Lambda + \Lambda\partial\partial - \partial\Lambda\partial = 0,$$

using $\partial^2 = 0$. The second identity is its conjugate. Expanding,

$$\Delta_d = (\partial + \bar\partial)(\partial^* + \bar\partial^*) + (\partial^* + \bar\partial^*)(\partial + \bar\partial) = \Delta_\partial + \Delta_{\bar\partial} + (\partial\bar\partial^* + \bar\partial^*\partial) + (\bar\partial\partial^* + \partial^*\bar\partial) = \Delta_\partial + \Delta_{\bar\partial}.$$

Using $\partial^* = i[\Lambda, \bar\partial]$ and $\bar\partial^* = -i[\Lambda, \partial]$,

$$\Delta_\partial = i(\partial\Lambda\bar\partial - \partial\bar\partial\Lambda + \Lambda\bar\partial\partial - \bar\partial\Lambda\partial), \qquad \Delta_{\bar\partial} = -i(\bar\partial\Lambda\partial - \bar\partial\partial\Lambda + \Lambda\partial\bar\partial - \partial\Lambda\bar\partial),$$

so $\Delta_\partial - \Delta_{\bar\partial} = i\left(-(\partial\bar\partial + \bar\partial\partial)\Lambda + \Lambda(\bar\partial\partial + \partial\bar\partial)\right) = 0$ by Proposition 44.5.3. Hence $\Delta_d = 2\Delta_{\bar\partial} = 2\Delta_\partial$.

$\Delta_{\bar\partial}$ has type $(0,0)$, so Δ_d commutes with the $\Pi^{p,q}$. Δ_d commutes with d and δ, hence with their components of each type, which are $\partial, \bar\partial, \partial^*, \bar\partial^*$. It is real, so it commutes with c, and it commutes with $*$ by Proposition 40.15.3. For L: by the Kähler identities $[L, \delta] = -[\partial^*, L] - [\bar\partial^*, L] = i(\bar\partial - \partial)$ and $[L, d] = 0$, so

$$[L, \Delta_d] = d[L, \delta] + [L, \delta]d = i\left((\partial + \bar\partial)(\bar\partial - \partial) + (\bar\partial - \partial)(\partial + \bar\partial)\right) = 0,$$

again because $\partial^2 = \bar\partial^2 = \partial\bar\partial + \bar\partial\partial = 0$. Taking adjoints, $[\Lambda, \Delta_d] = 0$. □

48.5 The Hodge decomposition

Definition 48.5.1. For a complex manifold X, let $H^{p,q}(X) \subseteq H^{p+q}(X;\mathbb{C})$ be the subspace of classes that can be represented by a closed form of type (p,q). For a Hermitian metric, let $\mathcal{H}^k_{\mathbb{C}}$ be the complex Δ_d-harmonic k-forms and $\mathcal{H}^{p,q} = \mathcal{H}^{p+q}_{\mathbb{C}} \cap \mathcal{A}^{p,q}(X)$.

Theorem 48.5.2 (Hodge decomposition)**.** *Let X be a compact Kähler manifold. Then*

1. $\mathcal{H}^k_{\mathbb{C}} = \bigoplus_{p+q=k} \mathcal{H}^{p,q}$*, and $\mathcal{H}^{p,q}$ is also the space of $\Delta_{\bar\partial}$-harmonic, and of Δ_∂-harmonic, forms of type (p,q);*
2. $H^k(X;\mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X)$*, and $\mathcal{H}^{p,q} \to H^{p,q}(X)$, $\alpha \mapsto [\alpha]$, is an isomorphism;*
3. $\overline{H^{p,q}(X)} = H^{q,p}(X)$ *(*Hodge symmetry*);*
4. $H^{p,q}(X) \cong H^{p,q}_{\bar\partial}(X) \cong H^q(X,\Omega^p_X)$.

The subspaces $H^{p,q}(X)$ depend only on the complex structure of X, not on the Kähler metric.

Proof. (1) If $\Delta_d\alpha = 0$ then $\Delta_d\Pi^{p,q}\alpha = \Pi^{p,q}\Delta_d\alpha = 0$ by Theorem 48.4.1, so the components of a harmonic form are harmonic. The second statement holds because $\Delta_d = 2\Delta_{\bar\partial} = 2\Delta_\partial$.

(2) Applying Theorem 43.2.5 to real and imaginary parts, and Theorem 37.3.1, the map $\mathcal{H}^k_{\mathbb{C}} \to H^k(X;\mathbb{C})$ is an isomorphism. By (1), $H^k(X;\mathbb{C})$ is the direct sum of the images of the $\mathcal{H}^{p,q}$, and each image lies in $H^{p,q}(X)$ because harmonic forms are closed. Conversely, let α be a closed form of type (p,q) and let h be its harmonic projection. By Theorem 43.2.2 (applied to real and imaginary parts), $\alpha = h + d\beta$ for some β, and h is the orthogonal projection of α onto $\mathcal{H}^k_{\mathbb{C}}$. Since forms of different types are orthogonal (Lemma 48.2.2), α is orthogonal to $\mathcal{H}^{r,s}$ for $(r,s) \neq (p,q)$, so $h \in \mathcal{H}^{p,q}$ by (1). So $H^{p,q}(X)$ is exactly the image of $\mathcal{H}^{p,q}$, and (2) follows.

(3) Δ_d is a real operator, so c maps $\mathcal{H}^{p,q}$ onto $\mathcal{H}^{q,p}$. Alternatively, the conjugate of a closed (p,q)-form is a closed (q,p)-form.

(4) By (1), $\mathcal{H}^{p,q}$ is the space of $\Delta_{\bar\partial}$-harmonic (p,q)-forms, which is isomorphic to $H^{p,q}_{\bar\partial}(X) \cong H^q(X,\Omega^p_X)$ by Theorem 46.3.3.

The definition of $H^{p,q}(X)$ does not involve a metric. □

Corollary 48.5.3. *Let X be a compact Kähler manifold of dimension n, with Hodge numbers $h^{p,q} = \dim H^q(X,\Omega^p_X)$. Then:*

1. $b_k = \sum_{p+q=k} h^{p,q}$*, so the Frölicher spectral sequence degenerates at E_1;*
2. $h^{p,q} = h^{q,p} = h^{n-p,n-q} = h^{n-q,n-p}$*;*
3. b_{2k+1} *is even for every k;*
4. $h^{p,p} \geq 1$ *for $0 \leq p \leq n$.*

Proof. (1) is Theorem 48.5.2(2) and (4) together with Proposition 46.4.1. In (2), the first equality is Hodge symmetry and the second is Serre duality (Theorem 46.3.4). For (3), $b_{2k+1} = \sum_{p+q=2k+1} h^{p,q} = 2\sum_{p<q,\ p+q=2k+1} h^{p,q}$, since $p \neq q$ when $p+q$ is odd. For (4), ω^p is a closed form of type (p,p) and $[\omega]^p \neq 0$ by Proposition 47.4.5. □

Remark 48.5.4. The Hodge numbers are displayed in the *Hodge diamond*, with $h^{n,n}$ at the top, $h^{0,0}$ at the bottom, and $h^{p,q}$ in row $p+q$. Row k adds up to b_k, and the diamond is symmetric

under the reflections $h^{p,q} \leftrightarrow h^{q,p}$ (Hodge symmetry) and $h^{p,q} \leftrightarrow h^{n-p,n-q}$ (Serre duality). For a compact Kähler surface it reads

$$\begin{array}{ccccc} & & h^{2,2} & & \\ & h^{2,1} & & h^{1,2} & \\ h^{2,0} & & h^{1,1} & & h^{0,2} \\ & h^{1,0} & & h^{0,1} & \\ & & h^{0,0} & & \end{array} \quad = \quad \begin{array}{ccccc} & & 1 & & \\ & q & & q & \\ p_g & & h^{1,1} & & p_g \\ & q & & q & \\ & & 1 & & \end{array}$$

where $q = h^{0,1}$ is the *irregularity* and $p_g = h^{2,0}$ the *geometric genus*.

Remark 48.5.5. The *Hodge filtration* $F^pH^k(X;\mathbb{C}) = \bigoplus_{r\geq p} H^{r,k-r}(X)$ determines the decomposition, since $H^{p,q} = F^p \cap \overline{F^q}$ when $p+q=k$. Because the Frölicher spectral sequence degenerates, F^p is also the image of the hypercohomology of the truncated complex $\Omega_X^{\geq p}$ in $H^k(X;\mathbb{C})$ [Voi02b, Chapter 8]. The filtration varies holomorphically in families, whereas the decomposition does not. This is why the theory of Hodge structures in Part IX is phrased with filtrations.

Proposition 48.5.6. *On a compact Kähler manifold every holomorphic p-form is closed and harmonic, and $H^0(X,\Omega_X^p) = \mathcal{H}^{p,0} \cong H^{p,0}(X) \subseteq H^p(X;\mathbb{C})$. In particular a holomorphic form that is exact, or merely cohomologous to zero, vanishes, and $b_1(X) = 2\dim H^0(X,\Omega_X^1)$.*

Proof. If α is a holomorphic p-form, then $\bar\partial\alpha = 0$ and $\bar\partial^*\alpha = 0$, the latter because $\bar\partial^*\alpha$ would have type $(p,-1)$. So $\Delta_{\bar\partial}\alpha = 0$, hence $\Delta_d\alpha = 0$ and α is closed and harmonic. Conversely, if $\alpha \in \mathcal{H}^{p,0}$ then $0 = (\Delta_{\bar\partial}\alpha,\alpha) = \|\bar\partial\alpha\|^2 + \|\bar\partial^*\alpha\|^2$, so $\bar\partial\alpha = 0$ and α is holomorphic. The harmonic representative of a class is unique, so a harmonic form cohomologous to zero vanishes. Finally $b_1 = h^{1,0} + h^{0,1} = 2h^{1,0}$. □

Proposition 48.5.7. *Let $f\colon X \to Y$ be a holomorphic map of compact Kähler manifolds. Then $f^*H^{p,q}(Y) \subseteq H^{p,q}(X)$. The cup product maps $H^{p,q}\times H^{r,s}$ into $H^{p+r,q+s}$. If $\dim X = n$, the Poincaré duality pairing $H^k(X;\mathbb{C})\times H^{2n-k}(X;\mathbb{C}) \to \mathbb{C}$ restricts to perfect pairings $H^{p,q}(X)\times H^{n-p,n-q}(X) \to \mathbb{C}$, and it is zero on $H^{p,q}\times H^{r,s}$ for all other pairs of types.*

Proof. Pullback by a holomorphic map and wedge product preserve types of forms, and the wedge product corresponds to the cup product (Theorem 37.3.1). For $(r,s) \neq (n-p,n-q)$ with $r+s = 2n-p-q$, the wedge product of forms of types (p,q) and (r,s) has type $(p+r,q+s) \neq (n,n)$ and so is zero, being a $2n$-form. The pairing on $H^k \times H^{2n-k}$ is perfect (Theorem 37.3.10), and by Theorem 48.5.2 it is block diagonal with blocks $H^{p,q}\times H^{n-p,n-q}$. Each block is therefore perfect. □

Proposition 48.5.8. *Let X be a compact Kähler manifold of dimension n and $Z \subseteq X$ an analytic subvariety of pure codimension p. Then its cycle class $[Z] \in H^{2p}(X;\mathbb{C})$ (Proposition 44.6.2) lies in $H^{p,p}(X)$.*

Proof. Let $k = n-p = \dim Z$. The restriction of a form of type (r,s) to the complex manifold Z_{reg} of dimension k has type (r,s) there, so it vanishes if $r+s = 2k$ and $(r,s) \neq (k,k)$. Hence $\int_Z \beta = 0$ for every closed form β of type $(r,s) \neq (k,k)$ with $r+s = 2k$. By definition of the cycle class, $\int_X \eta\wedge\beta = \int_Z \beta$ for a closed form η representing $[Z]$ (Theorem 37.4.3). Write $[Z] = \sum_{a+b=2p} z^{a,b}$ with $z^{a,b} \in H^{a,b}(X)$. By Proposition 48.5.7, $z^{a,b}$ pairs only with $H^{n-a,n-b}$, and it pairs to zero with it unless $(n-a,n-b) = (k,k)$. So $z^{a,b}$ pairs to zero with everything unless $(a,b) = (p,p)$, and by perfectness $z^{a,b} = 0$ for $(a,b) \neq (p,p)$. □

Remark 48.5.9. The cycle class of Z is the image of an integral class in $H^{2p}(X;\mathbb{Z})$ (Chapter 63). So algebraic cycles give *Hodge classes*, the elements of $H^{2p}(X;\mathbb{Q}) \cap H^{p,p}(X)$. The Hodge conjecture (Chapter 68) predicts that on a projective manifold every Hodge class is a rational linear combination of cycle classes.

48.6 The $\partial\bar\partial$-lemma

Lemma 48.6.1. *Let X be compact Kähler. Then $\mathcal{A}^{p,q}(X) = \mathcal{H}^{p,q} \oplus \partial\mathcal{A}^{p-1,q}(X) \oplus \partial^*\mathcal{A}^{p+1,q}(X)$ orthogonally, and $\ker\partial = \mathcal{H}^{p,q} \oplus \partial\mathcal{A}^{p-1,q}(X)$. The same holds with $\bar\partial$, $\bar\partial^*$ in place of ∂, ∂^*.*

Proof. For $\bar\partial$ the decomposition is Theorem 46.3.3, with $\mathcal{H}^{p,q}$ the $\Delta_{\bar\partial}$-harmonic forms (Theorem 48.5.2(1)). Applying the conjugate linear isometry c to the decomposition of $\mathcal{A}^{q,p}(X)$, and using $c\bar\partial c = \partial$, $c\bar\partial^* c = \partial^*$ and $c\mathcal{H}^{q,p} = \mathcal{H}^{p,q}$, gives the decomposition for ∂. If $\partial\alpha = 0$ and $\alpha = h + \partial\beta + \partial^*\gamma$, then $0 = (\partial\alpha, \gamma) = (\partial\partial^*\gamma, \gamma) = \|\partial^*\gamma\|^2$, because $\partial h = 0$ and $\partial^2 = 0$. So $\alpha \in \mathcal{H}^{p,q} \oplus \operatorname{im}\partial$; the argument for $\bar\partial$ is the same. □

Theorem 48.6.2 ($\partial\bar\partial$-lemma)**.** *Let X be a compact Kähler manifold and α a d-closed form of type (p,q). The following are equivalent:*

1. *α is d-exact;*
2. *α is ∂-exact;*
3. *α is $\bar\partial$-exact;*
4. *$\alpha = \partial\bar\partial\gamma$ for a form γ of type $(p-1, q-1)$.*

Proof. (4) implies the others: $\partial\bar\partial\gamma = d(\bar\partial\gamma) = \partial(\bar\partial\gamma) = \bar\partial(-\partial\gamma)$.

Assume (1), (2) or (3). Then α is orthogonal to $\mathcal{H}^{p,q}$: for h harmonic, $(d\beta, h) = (\beta, \delta h) = 0$, and likewise with $\partial^* h = 0$ or $\bar\partial^* h = 0$ (harmonic forms are killed by all of these, by Theorem 48.5.2(1)). Since $\partial\alpha = 0$ (the $(p+1,q)$-part of $d\alpha = 0$), Lemma 48.6.1 gives $\alpha = \partial\gamma$ with γ of type $(p-1,q)$. Decompose $\gamma = h' + \bar\partial\mu + \bar\partial^*\nu$ by the same lemma. Then $\partial h' = 0$, and $\partial\bar\partial^*\nu = -\bar\partial^*\partial\nu$ by Theorem 48.4.1, so

$$\alpha = \partial\bar\partial\mu - \bar\partial^*\partial\nu.$$

Now $\bar\partial\alpha = 0$ and $\bar\partial\partial\bar\partial\mu = -\partial\bar\partial\bar\partial\mu = 0$, hence $\bar\partial\bar\partial^*\partial\nu = 0$ and $\|\bar\partial^*\partial\nu\|^2 = (\bar\partial\bar\partial^*\partial\nu, \partial\nu) = 0$. Therefore $\alpha = \partial\bar\partial\mu$, with μ of type $(p-1, q-1)$. □

Corollary 48.6.3. *Let X be compact Kähler. Every Dolbeault class in $H^{p,q}_{\bar\partial}(X)$ has a d-closed representative, and $[\alpha] \mapsto [\alpha]_{\bar\partial}$, for α a closed (p,q)-form, is a well-defined isomorphism $H^{p,q}(X) \to H^{p,q}_{\bar\partial}(X)$ that does not depend on any metric.*

Proof. If α and α' are closed (p,q)-forms in the same de Rham class, then $\alpha - \alpha'$ is a d-exact closed form of type (p,q), hence $\bar\partial$-exact by Theorem 48.6.2. So the map is well defined. For a Kähler metric, restricting it to harmonic representatives gives the isomorphism $\mathcal{H}^{p,q} \cong H^{p,q}_{\bar\partial}(X)$ of Theorem 46.3.3. The harmonic representative of a Dolbeault class is d-closed. □

Corollary 48.6.4. *Let X be compact Kähler and let α, α' be closed real $(1,1)$-forms with $[\alpha] = [\alpha']$ in $H^2(X;\mathbb{R})$. Then $\alpha' = \alpha + i\partial\bar\partial\varphi$ for a smooth real function φ. In particular any two Kähler forms with the same Kähler class differ by $i\partial\bar\partial\varphi$.*

Proof. By Theorem 48.6.2, $\alpha' - \alpha = \partial\bar\partial\mu$ for a complex function μ. Since $\alpha' - \alpha$ is real, $\partial\bar\partial\mu = \overline{\partial\bar\partial\mu} = \bar\partial\partial\bar\mu = -\partial\bar\partial\bar\mu$. So $\alpha' - \alpha = \frac{1}{2}\partial\bar\partial(\mu - \bar\mu) = i\partial\bar\partial \operatorname{Im}\mu$, and $\varphi = \operatorname{Im}\mu$ works. □

Remark 48.6.5. Deligne, Griffiths, Morgan and Sullivan deduced from the $\partial\bar\partial$-lemma that the real homotopy type of a compact Kähler manifold is determined by its real cohomology ring (the manifold is *formal*). In particular all Massey products vanish, which gives obstructions to Kählerness beyond the Hodge numbers [DGMS75]. The $\partial\bar\partial$-lemma, and hence the Hodge decomposition, also hold for compact complex manifolds bimeromorphic to Kähler ones [DGMS75].

48.7 Examples

Example 48.7.1. Let X be a compact Riemann surface. It is Kähler (Example 47.4.4), so b_1 is even, and its Hodge diamond has $h^{0,0} = h^{1,1} = 1$ and $h^{1,0} = h^{0,1} = g$, where $g = b_1/2$. So the *genus* g, a topological invariant, equals both the dimension of the space of holomorphic 1-forms $H^0(X, K_X)$ and $\dim H^1(X, \mathcal{O}_X)$. For the Riemann sphere $g = 0$; for an elliptic curve $\mathbb{C}/\Lambda$, $g = 1$ and $H^0(X, K_X)$ is spanned by dz.

Example 48.7.2. For $\mathbb{CP}^n$, $b_{2k} = 1$ for $0 \le k \le n$ and all other Betti numbers vanish (Theorem 22.5.1). By Corollary 48.5.3(4), $h^{k,k} \ge 1$, so $H^{2k}(\mathbb{CP}^n;\mathbb{C}) = H^{k,k}$ is spanned by $[\omega_{FS}]^k$, and $h^{p,q} = 0$ for $p \ne q$. In particular $\mathbb{CP}^n$ has no nonzero holomorphic forms of positive degree, and $H^q(\mathbb{CP}^n, \mathcal{O}) = 0$ for $q > 0$.

Example 48.7.3. Let $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ with $\operatorname{Im}\tau > 0$, and let a, b be the classes in $H_1(E_\tau;\mathbb{Z})$ of the images of the segments $[0,1]$ and $[0,\tau]$. Integration identifies $H^1(E_\tau;\mathbb{C})$ with $\mathbb{C}^2$ through $[\alpha] \mapsto (\int_a \alpha, \int_b \alpha)$. Then $H^{1,0}$ is spanned by $[dz] \mapsto (1,\tau)$ and $H^{0,1}$ by $[d\bar z] \mapsto (1,\bar\tau)$. The lattice $H^1(E_\tau;\mathbb{Z})$ and the vector space $H^1(E_\tau;\mathbb{C})$ do not depend on τ, but the line $H^{1,0} = [1:\tau] \in \mathbb{P}^1$ does, and it recovers τ. So the Hodge decomposition captures the complex structure, which is not a topological invariant. This is the first example of a *period map* (Chapter 70).

Example 48.7.4. For a smooth quartic surface $X \subseteq \mathbb{CP}^3$ (a K3 surface) the Hodge diamond is

$$\begin{array}{ccccc} & & 1 & & \\ & 0 & & 0 & \\ 1 & & 20 & & 1 \\ & 0 & & 0 & \\ & & 1 & & \end{array}$$

Here $b_1 = 0$ by the Lefschetz hyperplane theorem (Chapter 50), $K_X \cong \mathcal{O}_X$ by adjunction (Proposition 45.5.1) so that $p_g = 1$, and $b_2 = 22$ comes from the Euler characteristic $\chi = c_2(X) = 24$ [BHPVdV04, Chapter VIII]. The subspace $H^{1,1} \cap H^2(X;\mathbb{Q})$ is where the Hodge conjecture for divisors, the Lefschetz $(1,1)$-theorem, lives.

Example 48.7.5. Let C and C' be compact Riemann surfaces of genera g and g', so $h^{0,0} = h^{1,1} = 1$ and $h^{1,0} = h^{0,1} = g$ for C (Example 48.7.1). By Exercise 48.8.1, the surface $S = C \times C'$ has

$$h^{1,0}(S) = g + g', \qquad h^{2,0}(S) = gg', \qquad h^{1,1}(S) = 2 + 2gg',$$

since $H^{1,1}(S)$ receives $H^{1,1}(C)\otimes H^{0,0}(C')$, $H^{0,0}(C)\otimes H^{1,1}(C')$, $H^{1,0}(C)\otimes H^{0,1}(C')$ and $H^{0,1}(C)\otimes H^{1,0}(C')$. As a check, $b_2(S) = 2h^{2,0} + h^{1,1} = 2 + 4gg'$, which is $1 + (2g)(2g') + 1$ by the Künneth

formula. For two elliptic curves, $g = g' = 1$, the Hodge diamond is

$$\begin{array}{ccccc} & & 1 & & \\ & 2 & & 2 & \\ 1 & & 4 & & 1 \\ & 2 & & 2 & \\ & & 1 & & \end{array}$$

which is also that of every two-dimensional complex torus (Example 46.4.2). The $H^{1,1}$ of S always contains the classes of the two fibres, and the remaining $2gg'$ dimensions come from $H^1(C) \otimes H^1(C')$; which rational classes lie there is governed by the homomorphisms between the Jacobians of C and C' (for elliptic curves this is worked out in Chapter 68).

Counterexample 48.7.6. The Hopf surface (Exercise 44.7.3) has $b_1 = 1$, which is odd, so by Corollary 48.5.3(3) it is not Kähler, which gives a second proof of Counterexample 47.4.9. It also shows how the theorem fails: $h^{1,0} = 0 \neq 1 = h^{0,1}$ (Example 46.4.2).

Counterexample 48.7.7. Let G be the complex Heisenberg group of matrices $\left(\begin{smallmatrix} 1 & z_1 & z_3 \\ 0 & 1 & z_2 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, $z \in \mathbb{C}^3$, and $\Gamma \subseteq G$ the subgroup with entries in $\mathbb{Z}[i]$. The *Iwasawa manifold* $X = \Gamma\backslash G$ is a compact complex manifold of dimension 3 carrying a holomorphic 1-form that is not closed. So X is not Kähler, by Proposition 48.5.6.

Proof. Left multiplication by an element with entries (a_1, a_2, a_3) is $(z_1, z_2, z_3) \mapsto (z_1 + a_1, z_2 + a_2, z_3 + a_1 z_2 + a_3)$. This map is holomorphic, and Γ acts freely and properly discontinuously, being a discrete subgroup acting by translations of the group. So X is a complex manifold. It is compact: using $a_1, a_2 \in \mathbb{Z}[i]$ one moves z_1, z_2 into the unit square, and then using a_3 one moves z_3 there, so the image of a compact set surjects onto X. The form $\theta = dz_3 - z_1\, dz_2$ is invariant under left multiplication: $d(z_3 + a_1 z_2 + a_3) - (z_1 + a_1)\, d(z_2 + a_2) = dz_3 - z_1\, dz_2$. So it descends to a holomorphic 1-form on X, and $d\theta = -dz_1 \wedge dz_2 \neq 0$. □

48.8 Exercises

Exercise 48.8.1. Let X and Y be compact Kähler manifolds. Show that $h^{p,q}(X \times Y) = \sum h^{p_1,q_1}(X)\, h^{p_2,q_2}(Y)$, the sum over $p_1 + p_2 = p$, $q_1 + q_2 = q$.

Solution. $X \times Y$ is Kähler (Example 47.4.4). The Künneth isomorphism $H^\bullet(X;\mathbb{C}) \otimes H^\bullet(Y;\mathbb{C}) \to H^\bullet(X \times Y;\mathbb{C})$, $a \otimes b \mapsto \mathrm{pr}_X^* a \cup \mathrm{pr}_Y^* b$ (Theorem 22.4.2), maps $H^{p_1,q_1}(X) \otimes H^{p_2,q_2}(Y)$ into $H^{p_1+p_2,q_1+q_2}(X \times Y)$ by Proposition 48.5.7. The source is the direct sum of these pieces and the target is the direct sum of its $H^{p,q}$. So the isomorphism maps $\bigoplus_{p_1+p_2=p,\ q_1+q_2=q} H^{p_1,q_1} \otimes H^{p_2,q_2}$ onto $H^{p,q}(X \times Y)$. □

Exercise 48.8.2. Calabi and Eckmann constructed complex structures on $S^{2a+1} \times S^{2b+1}$ for all $a, b \geq 0$ [CE53]. Show that none of them is Kähler, except on $S^1 \times S^1$.

Solution. The complex dimension is $a + b + 1 \geq 1$. By the Künneth formula, the cohomology of $S^{2a+1} \times S^{2b+1}$ lives in degrees $0, 2a+1, 2b+1$ and $2a+2b+2$. If $a, b \geq 1$, then $H^2 = 0$, contradicting Proposition 47.4.5. If $a = 0 < b$, then $b_1 = 1$ is odd, contradicting Corollary 48.5.3(3), and the case $b = 0 < a$ is symmetric. The remaining case $a = b = 0$ is a torus of real dimension 2, and every complex structure on it is Kähler (Example 47.4.4). □

Exercise 48.8.3. Show that ω_{FS} is the only $U(n+1)$-invariant Kähler form on $\mathbb{CP}^n$ in its cohomology class.

Solution. Let ω' be another one. By Corollary 48.6.4, $\omega' = \omega_{FS} + i\partial\bar\partial\varphi$ with φ real. For $g \in U(n+1)$, $g^*\omega' = \omega'$ and $g^*\omega_{FS} = \omega_{FS}$, so $i\partial\bar\partial(\varphi \circ g) = i\partial\bar\partial\varphi$, since g is holomorphic. Averaging over $U(n+1)$ with its normalised Haar measure gives an invariant function $\tilde\varphi = \int \varphi \circ g \, dg$ with $i\partial\bar\partial\tilde\varphi = i\partial\bar\partial\varphi$ (differentiate under the integral). As $U(n+1)$ acts transitively, $\tilde\varphi$ is constant. Hence $i\partial\bar\partial\varphi = 0$ and $\omega' = \omega_{FS}$. □

Exercise 48.8.4. Let X be compact Kähler. Show that the map $H^1(X;\mathbb{R}) \to H^1(X, \mathcal{O}_X)$ induced by the inclusion $\mathbb{R}_X \to \mathcal{O}_X$ is an isomorphism of real vector spaces. (This is the starting point for the Picard variety $H^1(X, \mathcal{O}_X)/H^1(X;\mathbb{Z})$.)

Solution. Projecting forms to their $(0,q)$-components defines a morphism from the de Rham resolution of $\mathbb{C}_X$ to the Dolbeault resolution of $\mathcal{O}_X$ over the inclusion $\mathbb{C}_X \to \mathcal{O}_X$, because the $(0,q+1)$-part of $d\alpha$ is $\bar\partial$ of the $(0,q)$-part of α. So the map sends $[\alpha]$ to $[\alpha^{0,1}]_{\bar\partial}$. Both sides have real dimension $b_1 = 2h^{0,1}$. If α is a real harmonic 1-form with $\alpha^{0,1}$ $\bar\partial$-exact, then $\alpha^{0,1}$ is harmonic (by Theorem 48.5.2(1)) and $\bar\partial$-exact, hence zero. Then $\alpha^{1,0} = \overline{\alpha^{0,1}} = 0$, so $\alpha = 0$ and the map is injective. □

Exercise 48.8.5. Let X be compact Kähler. Show that L and Λ map harmonic forms to harmonic forms, and that the primitive components (Theorem 47.5.2) of a harmonic form are harmonic.

Solution. L and Λ commute with Δ_d (Theorem 48.4.1), so they preserve $\ker\Delta_d$. Let $\alpha = \sum_j L^j\alpha_j$ be the primitive decomposition of a harmonic form. Then $\Lambda\Delta_d\alpha_j = \Delta_d\Lambda\alpha_j = 0$, so $\Delta_d\alpha_j$ is primitive, and $0 = \Delta_d\alpha = \sum_j L^j(\Delta_d\alpha_j)$ is a primitive decomposition of 0. By uniqueness, every $\Delta_d\alpha_j$ vanishes. □

Exercise 48.8.6. Show that on a compact complex surface every holomorphic 1-form is closed, even without assuming the surface is Kähler. (Hint: for a holomorphic 1-form θ, compute $\int_X d\theta \wedge \overline{d\theta}$ in two ways.)

Solution. $d\theta = \partial\theta$ is a holomorphic 2-form, so $d\theta \wedge \overline{d\theta}$ is a form of type $(2,2)$. In local coordinates, $d\theta = f\, dz_1 \wedge dz_2$, and $d\theta \wedge \overline{d\theta} = |f|^2\, dz_1 \wedge dz_2 \wedge d\bar z_1 \wedge d\bar z_2$, which is $4|f|^2\, dx_1 \wedge dy_1 \wedge dx_2 \wedge dy_2$, a non-negative multiple of the volume form. On the other hand $d\theta \wedge \overline{d\theta} = d(\theta \wedge \overline{d\theta})$, because $\overline{d\theta}$ is closed. So its integral vanishes by Stokes' theorem, which forces $f = 0$. The Iwasawa manifold (Counterexample 48.7.7) shows that this fails in dimension 3. □

Chapter 49

Hard Lefschetz and the Hodge–Riemann Relations

49.1 Introduction

Let X be a compact Kähler manifold of dimension n with Kähler form ω. Cup product with the Kähler class is an operator L on $H^\bullet(X;\mathbb{R})$ of degree 2. The *hard Lefschetz theorem* (Theorem 49.2.2) says that

$$L^{n-k}\colon H^k(X;\mathbb{R}) \to H^{2n-k}(X;\mathbb{R})$$

is an isomorphism for $k \le n$. The proof takes the pointwise $\mathfrak{sl}_2$-action of Theorem 47.5.2 and passes it to harmonic forms, using that L and Λ commute with the Laplacian. The result is a canonical decomposition of cohomology into primitive pieces. On them the intersection form, twisted by powers of ω, has definite signs: these are the *Hodge–Riemann bilinear relations* (Theorem 49.4.2). Their consequences include the *Hodge index theorem* (Theorem 49.5.1) and, in Part IX, the polarisations of the Hodge structures of projective manifolds. References: [GH78, Chapter 0, Section 7], [Huy05, Section 3.3], [Voi02b, Chapter 6].

Prerequisites. Chapter 48 (The Hodge Decomposition).

Throughout, X is a compact Kähler manifold of dimension n with Kähler form ω, and we use the notation of Notation 48.2.1. We identify $H^k(X;\mathbb{C})$ with the space $\mathcal{H}^k_{\mathbb{C}}$ of complex harmonic forms and $H^k(X;\mathbb{R})$ with the real harmonic forms (Theorem 43.2.5).

49.2 The $\mathfrak{sl}_2$-action on cohomology

Definition 49.2.1. The *Lefschetz operator* on cohomology is $L\colon H^k(X;\mathbb{R}) \to H^{k+2}(X;\mathbb{R})$, $L(a) = [\omega] \cup a$, which is represented on forms by $\alpha \mapsto \omega \wedge \alpha$. A class $a \in H^k(X;\mathbb{R})$ with $k \le n$ is *primitive* if $L^{n-k+1}a = 0$. We write $P^k(X) = P^k(X;\mathbb{R})$ for the primitive classes of degree k, $P^k(X;\mathbb{C})$ for their complexification, and $P^{p,q}(X) = P^{p+q}(X;\mathbb{C}) \cap H^{p,q}(X)$.

Theorem 49.2.2 (Hard Lefschetz)**.** *Let X be a compact Kähler manifold of dimension n with Kähler class $[\omega]$. Then:*

1. *the pointwise operators L, Λ and H preserve the space $\mathcal{H}^\bullet$ of harmonic forms, which becomes a finite-dimensional representation of $\mathfrak{sl}_2$ in which $\mathcal{H}^k$ has weight $k - n$;*

2. *for $0 \le k \le n$, $L^{n-k}\colon H^k(X;\mathbb{R}) \to H^{2n-k}(X;\mathbb{R})$ is an isomorphism;*

3. (Lefschetz decomposition) *for every* k,
$$H^k(X;\mathbb{R}) = \bigoplus_{j\geq 0} L^j P^{k-2j}(X),$$
and L^j *is injective on* $P^{k-2j}(X)$;

4. *a class of degree* $k \leq n$ *is primitive if and only if its harmonic representative* α *satisfies* $\Lambda\alpha = 0$, *that is,* α *is primitive at every point;*

5. *the decompositions are compatible with the Hodge decomposition:* $P^k(X;\mathbb{C}) = \bigoplus_{p+q=k} P^{p,q}(X)$ *and* $H^{p,q}(X) = \bigoplus_j L^j P^{p-j,q-j}(X)$.

Proof. (1) By Theorem 48.4.1, L and Λ commute with Δ_d, so they preserve $\mathcal{H}^\bullet = \ker\Delta_d$; so does H, which acts on $\mathcal{H}^k$ as multiplication by $k-n$. The commutation relations $[L,\Lambda] = H$, $[H,L] = 2L$, $[H,\Lambda] = -2\Lambda$ hold pointwise by Theorem 47.5.2, hence on $\mathcal{H}^\bullet$. The space $\mathcal{H}^\bullet$ is finite-dimensional by Theorem 43.2.2, and it is real since all three operators are real.

(2) and (3) now follow exactly as statements (1) and (2) of Theorem 47.5.2 were derived from Corollary 13.6.4: that argument used only a finite-dimensional $\mathfrak{sl}_2$-representation in which the degree k part has weight $k-n$. Under the Hodge isomorphism, L on harmonic forms is the cup product with $[\omega]$, since $\omega\wedge\alpha$ is harmonic and represents $[\omega]\cup[\alpha]$ (Theorem 37.3.1). This gives (2), and it gives (3) with the primitive subspaces defined as $\ker\Lambda\cap\mathcal{H}^{k-2j}$.

(4) Part (3) of Theorem 47.5.2 was also derived from the representation theory alone. Applied to $\mathcal{H}^\bullet$, it says that for $\alpha\in\mathcal{H}^k$ with $k\leq n$, $\Lambda\alpha = 0$ if and only if $L^{n-k+1}\alpha = 0$. So the subspaces $\ker\Lambda\cap\mathcal{H}^k$ used in (3) are the $P^k(X)$ of Definition 49.2.1.

(5) L maps $H^{p,q}$ to $H^{p+1,q+1}$. So the kernel of L^{n-k+1} on $H^k(X;\mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}$ is the direct sum of its kernels on the summands, which gives the first statement. The second follows by comparing types in (3). □

Remark 49.2.3. Primitivity is defined by L alone, so $P^k(X)$ and the Lefschetz decomposition depend only on the Kähler class $[\omega]$, not on the metric. The operator Λ on cohomology was defined using harmonic forms, but it too depends only on $[\omega]$, by Proposition 49.2.4.

Proposition 49.2.4. *Let* V *be a finite-dimensional vector space over a field* K *of characteristic zero, graded by degree. Let* e *and* h *be endomorphisms such that* h *acts on the degree* k *part by* $k-n$ *and* $[h,e] = 2e$. *Then there is at most one endomorphism* f *with* $[e,f] = h$ *and* $[h,f] = -2f$. *If* $K\subseteq K'$ *is a field extension and such an* f *exists over* K', *then it is defined over* K. *In particular, if* X *is projective and* $[\omega]$ *is the restriction of the hyperplane class, then* L *and* Λ *preserve* $H^\bullet(X;\mathbb{Q})$, *and the Lefschetz decomposition is defined over* $\mathbb{Q}$.

Proof. It suffices to treat $K = \mathbb{C}$ for uniqueness, after extending scalars. If f and f' both work, then $g = f - f'$ satisfies $[e,g] = 0$ and $[h,g] = -2g$. The Lie algebra $\mathfrak{sl}_2$ acts on $\mathrm{End}(V)$ through the triple (e,f,h) by $x\cdot\phi = [x,\phi]$. In this representation g is killed by e and has weight -2. By Theorem 13.6.2, a nonzero vector killed by e has a non-negative weight, so $g = 0$. The conditions on f are linear equations with coefficients in K. A system of linear equations over K that has a solution over K' has one over K, since the rank of a matrix does not change under field extension, and by uniqueness the two solutions coincide. For the last statement, $[\omega] = c_1(\mathcal{O}(1))|_X$ lies in the image of $H^2(X;\mathbb{Z})$ (Proposition 47.4.1), so L preserves $H^\bullet(X;\mathbb{Q})$, and the rest follows. □

Corollary 49.2.5. *Let* X *be a compact Kähler manifold of dimension* n. *Then* $L\colon H^k\to H^{k+2}$ *is injective for* $k<n$ *and surjective for* $k\geq n-1$. *Consequently*
$$b_k\leq b_{k+2}\quad(k<n),\qquad h^{p,q}\leq h^{p+1,q+1}\quad(p+q<n),\qquad \dim P^k = b_k - b_{k-2}\quad(k\leq n).$$

Proof. For $k < n$, L^{n-k} is injective on H^k and equals $L^{n-k-1} \circ L$, so L is injective on H^k. For $k \geq n-1$, put $j = 2n-k-2 \leq n-1$. Then $L^{n-j} = L \circ L^{n-j-1}$ maps H^j onto $H^{2n-j} = H^{k+2}$, and L^{n-j-1} lands in H^k, so L is onto H^{k+2}. The inequality for Hodge numbers follows since L has type $(1,1)$, and the last formula from $H^k = P^k \oplus LH^{k-2}$ with L injective on H^{k-2}. □

Example 49.2.6. Let $X = \mathbb{C}^n/\Lambda$ be a complex torus with a flat Kähler metric, so that ω has constant coefficients. The harmonic forms are the forms with constant coefficients (Example 46.4.2), so $H^\bullet(X;\mathbb{R}) \cong \Lambda^\bullet V^*$ with $V = \mathbb{R}^{2n}$, and L is the wedge product with the constant form ω. In this case Theorem 49.2.2 is exactly the linear algebra of Theorem 47.5.2. For instance, for $n = 2$, $H^2(X;\mathbb{R})$ has dimension 6 and is the sum of $\mathbb{R}[\omega]$ and the 5-dimensional primitive part (Exercise 47.6.2).

Example 49.2.7. The manifold $S^2 \times S^3 \times S^3$ has Poincaré polynomial $(1+t^2)(1+t^3)^2 = 1 + t^2 + 2t^3 + 2t^5 + t^6 + t^8$. Its real dimension is 8, so a Kähler structure would have $n = 4$, and Corollary 49.2.5 would give $b_2 \leq b_4$. But $b_2 = 1 > 0 = b_4$, so no complex structure on this manifold is Kähler.

49.3 Pointwise Hodge–Riemann relations

Proposition 49.3.1. *Let V be a complex vector space of dimension n with a Hermitian inner product and fundamental form ω, as in Definition 47.5.1. Let α be a primitive form of type (p,q) and degree $k = p+q \leq n$. Then*

$$i^{p-q}(-1)^{k(k-1)/2}\,\alpha \wedge \bar{\alpha} \wedge \omega^{n-k} = (n-k)!\,|\alpha|^2 \,\mathrm{vol}.$$

More generally, if α and β are primitive of type (p,q), then $i^{p-q}(-1)^{k(k-1)/2}\alpha \wedge \bar{\beta} \wedge \omega^{n-k} = (n-k)!\,\langle \alpha, \beta \rangle \,\mathrm{vol}$.

Proof. The second statement follows from the first by polarisation, since both sides are Hermitian forms in (α, β). Choose a basis with $h_{j\bar{k}} = \delta_{jk}$, so that $dz_j = dx_j + i\,dy_j$ with dx_j, dy_j orthonormal, and put $\nu_j = dx_j \wedge dy_j = \frac{i}{2} dz_j \wedge d\bar{z}_j$. Then $\omega = \sum_j \nu_j$, and $\mathrm{vol} = \nu_1 \wedge \cdots \wedge \nu_n$. For a set R of indices write $\nu_R = \bigwedge_{j \in R} \nu_j$, and for disjoint A, B write $D = A \cup B$ and $dz_A \wedge d\bar{z}_B$ for the wedge products in increasing order.

Step 1: splitting. Every monomial $dz_I \wedge d\bar{z}_J$ equals $\pm(2/i)^{|I \cap J|} dz_A \wedge d\bar{z}_B \wedge \nu_C$ with $C = I \cap J$, $A = I \setminus C$, $B = J \setminus C$. So $\Lambda^{p,q}$ is the orthogonal direct sum of the subspaces $dz_A \wedge d\bar{z}_B \wedge S_D$, where S_D is spanned by the ν_R with $R \cap D = \emptyset$. Write $\alpha = \sum_{A,B} dz_A \wedge d\bar{z}_B \wedge f_{AB}$ with $f_{AB} \in S_D$ homogeneous of degree $r = p - |A| = q - |B|$ in the ν_j. In the proof of Theorem 47.5.2, $\Lambda = \sum_j \Lambda_j$, where Λ_j acts on the j-th factor, kills 1, dz_j and $d\bar{z}_j$, and sends ν_j to 1. Hence $\Lambda(dz_A \wedge d\bar{z}_B \wedge f) = dz_A \wedge d\bar{z}_B \wedge \Lambda f$, where $\Lambda \nu_R = \sum_{j \in R} \nu_{R \setminus j}$, and α is primitive if and only if every f_{AB} is.

Step 2: the summands are orthogonal for the left-hand side. Let α and β lie in the summands of (A,B) and (A',B'). For $j \in A$, the form α contains dz_j but not $d\bar{z}_j$. A top-degree monomial needs exactly one $d\bar{z}_j$, and it cannot come from a power of ω or from a ν_j in $\bar{\beta}$, both of which would repeat dz_j. So it must come from $\bar{\beta} = d\bar{z}_{A'} \wedge dz_{B'} \wedge \bar{f}'$, which forces $j \in A'$. Thus $\alpha \wedge \bar{\beta} \wedge \omega^{n-k} = 0$ unless $A \subseteq A'$, and by symmetry $A = A'$ and $B = B'$. The summands are also orthogonal for $\langle\ ,\ \rangle$, so it suffices to treat $\alpha = dz_A \wedge d\bar{z}_B \wedge f$ with $|A| = a$, $|B| = b$, $f = \sum_{|R|=r} f_R \nu_R \in S_D$ primitive, and $m = n - a - b$ indices outside D. Then $n - k = m - 2r$.

Step 3: the ν-part. Only the ν_j with $j \notin D$ survive in ω^{n-k} after wedging with $dz_A \wedge d\bar{z}_B \wedge d\bar{z}_A \wedge dz_B$. Moreover $\big(\sum_{j \notin D} \nu_j\big)^{m-2r} = (m-2r)! \sum_{|T|=m-2r} \nu_T$, and $\nu_R \nu_{R'} \nu_T$ is the product of

all v_j, $j \notin D$, exactly when R, R', T partition the complement of D. So

$$f \wedge \bar{f} \wedge \Big(\sum_{j\notin D} v_j\Big)^{m-2r} = (m-2r)! \sum_{R\cap R'=\emptyset} f_R \overline{f_{R'}}\, v_{\{1,\dots,n\}\setminus D}.$$

We claim that $\sum_{R\cap R'=\emptyset} f_R\overline{f_{R'}} = (-1)^r \sum_R |f_R|^2$ for primitive f. Primitivity says that $\sum_{R'\supseteq Q} f_{R'} = 0$ for every set Q of size $r-1$. For $|S| = s < r$, each $R' \supseteq S$ of size r contains exactly $r-s$ sets Q of size $r-1$ with $S \subseteq Q$, so $\sum_{R'\supseteq S} f_{R'} = \frac{1}{r-s}\sum_Q \sum_{R'\supseteq Q} f_{R'} = 0$. By inclusion–exclusion, $[R\cap R' = \emptyset] = \sum_{S\subseteq R\cap R'} (-1)^{|S|}$. Hence

$$\sum_{R'} \overline{f_{R'}}\,[R\cap R'=\emptyset] = \sum_{S\subseteq R}(-1)^{|S|} \sum_{R'\supseteq S} \overline{f_{R'}} = (-1)^r\,\overline{f_R},$$

because only $S = R$ survives. Multiplying by f_R and summing over R proves the claim.

Step 4: the dz-part and the signs. Reordering gives

$$dz_A\wedge d\bar z_B\wedge d\bar z_A\wedge dz_B = (-1)^{ab+b}(dz_A\wedge d\bar z_A)\wedge(dz_B\wedge d\bar z_B) = (-1)^{ab+b+a(a-1)/2+b(b-1)/2}(-2i)^{a+b}\,v_D,$$

using $dz_A \wedge d\bar z_A = (-1)^{a(a-1)/2}\bigwedge_{j\in A}(dz_j\wedge d\bar z_j)$ and $dz_j\wedge d\bar z_j = -2i\,v_j$. The forms f and $\bar f$ have even degree, so they can be moved freely. Therefore $i^{p-q}(-1)^{k(k-1)/2}\alpha\wedge\bar\alpha\wedge\omega^{n-k}$ equals $(m-2r)!\sum_R |f_R|^2$ vol times

$$i^{a-b}(-2i)^{a+b}(-1)^{k(k-1)/2+ab+b+a(a-1)/2+b(b-1)/2+r}.$$

Now $i^{a-b}(-i)^{a+b} = (-1)^{a+b}i^{2a} = (-1)^b$. With $s = a+b$ and $k = s+2r$, we have $k(k-1)/2 \equiv s(s-1)/2 + r \equiv a(a-1)/2 + b(b-1)/2 + ab + r \pmod 2$. So the total sign is $+1$ and the factor is 2^{a+b}. Finally $|\alpha|^2 = |dz_A\wedge d\bar z_B|^2|f|^2 = 2^{a+b}\sum_R |f_R|^2$, since the dz_j, $d\bar z_j$ are orthogonal of squared length 2 (Lemma 48.2.2) and the v_R are orthonormal. As $m-2r = n-k$, this is the assertion. □

Remark 49.3.2. Proposition 49.3.1 also follows from Weil's formula $*\alpha = (-1)^{k(k+1)/2} i^{p-q} L^{n-k}\alpha/(n-k)!$ for primitive α of type (p,q) (Remark 47.5.3), using $\beta\wedge *\bar\alpha = \langle\beta,\alpha\rangle$ vol. The direct computation above avoids proving Weil's formula in general.

49.4 The Hodge–Riemann bilinear relations

Definition 49.4.1. For $k \le n$ define the bilinear form Q on $H^k(X;\mathbb{R})$, and its complex bilinear extension to $H^k(X;\mathbb{C})$, by

$$Q(a,b) = (-1)^{k(k-1)/2}\int_X a\cup b\cup[\omega]^{n-k}.$$

Theorem 49.4.2 (Hodge–Riemann bilinear relations)**.** *Let X be a compact Kähler manifold of dimension n and $k \le n$. Then:*

1. *$Q(b,a) = (-1)^k Q(a,b)$, and Q is nondegenerate on $H^k(X;\mathbb{R})$;*
2. *$Q(H^{p,q}, H^{p',q'}) = 0$ unless $(p',q') = (q,p)$;*
3. *the Lefschetz decomposition is Q-orthogonal: $Q(L^jP^{k-2j}, L^{j'}P^{k-2j'}) = 0$ for $j \ne j'$;*
4. *for $0 \ne a \in P^{p,q}(X)$, $p+q=k$, we have $i^{p-q}Q(a,\bar a) > 0$.*

Proof. (1) The symmetry is graded commutativity of the cup product. For nondegeneracy, if $Q(a,b)=0$ for all b, then $a\cup[\omega]^{n-k}$ pairs trivially with H^k, so $L^{n-k}a=0$ by Poincaré duality (Theorem 37.3.10), and $a=0$ by Theorem 49.2.2.

(2) is Proposition 48.5.7, as L^{n-k} has type $(n-k,n-k)$.

(3) Let $j<j'$, $a\in P^{k-2j}$ and $b\in P^{k-2j'}$. Then $Q(L^ja,L^{j'}b)=\pm\int_X b\cup L^{n-k+j+j'}a$, and $n-k+j+j'\geq n-(k-2j)+1$, so $L^{n-k+j+j'}a=0$ by primitivity. The case $j>j'$ follows by (1).

(4) Let α be the harmonic representative of a. It has type (p,q) (Theorem 48.5.2) and is primitive at every point (Theorem 49.2.2(4)), and $\bar\alpha$ represents $\bar a$. By Proposition 49.3.1,

$$i^{p-q}Q(a,\bar a)=\int_X i^{p-q}(-1)^{k(k-1)/2}\alpha\wedge\bar\alpha\wedge\omega^{n-k}=(n-k)!\,\|\alpha\|^2>0.$$

□

Remark 49.4.3. In the language of Chapter 66: if $[\omega]$ is integral, then $P^k(X;\mathbb{Q})$ with the decomposition $P^k(X;\mathbb{C})=\bigoplus P^{p,q}$ and the form Q is a *polarised Hodge structure* of weight k. By Theorem 49.4.2(3), $H^k(X;\mathbb{Q})$ is an orthogonal direct sum of such structures, with Q twisted by signs on the summands. This is why the Hodge structures of projective manifolds are polarisable, a fact that underlies the semisimplicity arguments of Part IX.

Example 49.4.4. For a compact Riemann surface ($n=1$) and $k=1$, every class is primitive, and part (4) says that $i\int_X\alpha\wedge\bar\alpha>0$ for every nonzero holomorphic 1-form α. These are *Riemann's bilinear relations*. For $E_\tau=\mathbb{C}/(\mathbb{Z}+\tau\mathbb{Z})$ and $\alpha=dz$,

$$i\int_{E_\tau}dz\wedge d\bar z=i\cdot(-2i)\operatorname{area}(E_\tau)=2\operatorname{Im}\tau>0,$$

so the positivity condition $\operatorname{Im}\tau>0$ on the period τ is an instance of the Hodge–Riemann relations. It is the condition cutting out the period domain, here the upper half plane (Chapter 70).

49.5 The Hodge index theorem

Theorem 49.5.1 (Hodge index theorem)**.** *Let X be a compact Kähler manifold of even dimension n. The signature of the intersection form $(a,b)\mapsto\int_X a\cup b$ on $H^n(X;\mathbb{R})$ is*

$$\sigma(X)=\sum_{p,q=0}^{n}(-1)^p h^{p,q}(X).$$

Proof. Let $I(a,b)=\int_X a\cup b$ on H^n. By Theorem 49.4.2(3) and the Hodge decomposition, $H^n(X;\mathbb{R})$ is the I-orthogonal direct sum of the real subspaces $W_{j,p,q}=L^j\left(P^{p,q}+P^{q,p}\right)\cap H^n(X;\mathbb{R})$ over $j\geq0$ and $p\leq q$ with $p+q=n-2j=:k$. Here $P^{p,q}$ is paired by I only with $P^{q,p}$ (Theorem 49.4.2(2)). We compute the sign of I on each piece. Note that k is even. For $a\in P^{p,q}$ we have $I(L^ja,L^j\bar a)=\int a\cup\bar a\cup[\omega]^{2j}=(-1)^{k(k-1)/2}Q(a,\bar a)$, since $n-k=2j$. So $I(L^ja,L^j\bar a)=(-1)^{k(k-1)/2}i^{q-p}\cdot i^{p-q}Q(a,\bar a)$ has the sign $\varepsilon_{p,q}=(-1)^{k(k-1)/2+(q-p)/2}$. Writing $k=2m$, $k(k-1)/2\equiv m$ and $(q-p)/2=m-p$, so $\varepsilon_{p,q}=(-1)^p$.

- If $p<q$, the real points of $L^j(P^{p,q}\oplus P^{q,p})$ are the $L^j(a+\bar a)$, $a\in P^{p,q}$, and $I(L^j(a+\bar a),L^j(a+\bar a))=2I(L^ja,L^j\bar a)$, because $I(L^ja,L^ja)=0$ by type. So I is definite of sign $(-1)^p$ on a real space of dimension $2\dim P^{p,q}$.
- If $p=q$, then I is definite of sign $(-1)^p$ on the real points of $L^jP^{p,p}$, which has real dimension $\dim P^{p,p}$.

Hence $\sigma(X) = \sum_{p+q\le n,\ p+q \text{ even}} (-1)^p \dim P^{p,q}$. By Theorem 49.2.2(5) and Corollary 49.2.5, $\dim P^{p,q} = h^{p,q} - h^{p-1,q-1}$ for $p+q \le n$. Substituting $p' = p-1$, $q' = q-1$ in the second sum gives

$$\sigma(X) = \sum_{p+q=n} (-1)^p h^{p,q} + 2 \sum_{p+q<n,\ p+q \text{ even}} (-1)^p h^{p,q}.$$

On the other hand, in $\sum_{p,q}(-1)^p h^{p,q}$ the terms with $p+q$ odd cancel in pairs by Hodge symmetry $h^{p,q} = h^{q,p}$, since then $(-1)^p = -(-1)^q$. The terms with $p+q > n$ even equal those with $p+q < n$ by Serre duality $h^{p,q} = h^{n-p,n-q}$, since n is even. So the two expressions agree. □

Corollary 49.5.2. *Let X be a compact Kähler surface with irregularity $q = h^{0,1}$ and geometric genus $p_g = h^{2,0}$. The intersection form on $H^2(X;\mathbb{R})$ has signature $(2p_g+1, h^{1,1}-1)$, and its restriction to $H^{1,1}_{\mathbb{R}} = H^{1,1}(X) \cap H^2(X;\mathbb{R})$ has signature $(1, h^{1,1}-1)$. So $b^+(X) = 2p_g + 1$, and a real $(1,1)$-class a with $\int_X a \cup [\omega] = 0$ satisfies $\int_X a^2 \le 0$, with equality only for $a = 0$.*

Proof. For $n = 2$ the pieces in the proof of Theorem 49.5.1 are: $L(P^{0,0}) = \mathbb{R}[\omega]$ with sign $+$; $P^{0,2} + P^{2,0} = H^{0,2} \oplus H^{2,0}$ with sign $+$ and real dimension $2p_g$ (every class of type $(2,0)$ is primitive since $H^{3,1} = 0$); and $P^{1,1}$ with sign $-$ and dimension $h^{1,1} - 1$. The real $(1,1)$-classes are $\mathbb{R}[\omega] \oplus P^{1,1}_{\mathbb{R}}$. A class $a \in H^{1,1}_{\mathbb{R}}$ with $\int a \cup [\omega] = 0$ is orthogonal to $[\omega]$, so it lies in $P^{1,1}_{\mathbb{R}}$, where the form is negative definite. □

Remark 49.5.3. For a projective surface, applied to classes of divisors, Corollary 49.5.2 is the classical Hodge index theorem: the intersection form on the Néron–Severi group has exactly one positive eigenvalue (Chapter 62). Grothendieck's standard conjecture of Hodge type asks for the analogue of Theorem 49.4.2(4) for algebraic cycles in arbitrary characteristic (Chapter 77).

49.6 Further remarks and counterexamples

Remark 49.6.1. Hard Lefschetz is a special property of Kähler manifolds. Thurston gave a compact symplectic 4-manifold, a nilmanifold with $b_1 = 3$, that admits no Kähler structure [Thu76]. Its first Betti number is odd, so Corollary 48.5.3 already rules out a Kähler structure. Moreover, for every symplectic form ω the Lefschetz map $[\omega] \cup -\colon H^1 \to H^3$ fails to be an isomorphism: otherwise $(a,b) \mapsto \int a \cup b \cup [\omega]$ would be a nondegenerate skew-symmetric form on the odd-dimensional space H^1. For singular projective varieties, hard Lefschetz fails for ordinary cohomology but holds for intersection cohomology [BBD82]. In characteristic p, Deligne proved hard Lefschetz for ℓ-adic cohomology as part of his second proof of the Weil conjectures [Del80].

Counterexample 49.6.2. The sign in Theorem 49.4.2(4) genuinely requires primitivity. On $X = \mathbb{CP}^1 \times \mathbb{CP}^1$ with $\omega = \mathrm{pr}_1^*\omega_{FS} + \mathrm{pr}_2^*\omega_{FS}$, let h_1, h_2 be the pullbacks of the generators, so $[\omega] = h_1 + h_2$, $h_1^2 = h_2^2 = 0$ and $\int h_1 h_2 = 1$. For $k = 2$, $Q(a,a) = -\int a^2$. Then $Q(h_1 - h_2, h_1 - h_2) = 2 > 0$ for the primitive class $h_1 - h_2$, as predicted for type $(1,1)$, while $Q([\omega],[\omega]) = -2 < 0$ for the non-primitive class $[\omega]$.

49.7 Exercises

Exercise 49.7.1. Compute the signature of $\mathbb{CP}^2$, of $\mathbb{CP}^1 \times \mathbb{CP}^1$, and of a smooth quartic surface in $\mathbb{CP}^3$ from Theorem 49.5.1, using the Hodge numbers of Chapter 48.

Solution. For a surface $\sigma = 2 + 2p_g - h^{1,1}$ (the q-terms cancel). For $\mathbb{CP}^2$: $p_g = 0$, $h^{1,1} = 1$, so $\sigma = 1$. For $\mathbb{CP}^1 \times \mathbb{CP}^1$: $p_g = 0$ and $h^{1,1} = 2$ by Exercise 48.8.1, so $\sigma = 0$, matching the hyperbolic form of Counterexample 49.6.2. For the quartic (Example 48.7.4): $p_g = 1$ and $h^{1,1} = 20$, so $\sigma = -16$. □

Exercise 49.7.2. Let X be compact Kähler of dimension $n \geq 1$ and $0 \neq a \in H^1(X;\mathbb{R})$. Show that $a \cup [\omega]^{n-1} \neq 0$, and that $(a, b) \mapsto \int_X a \cup b \cup [\omega]^{n-1}$ is a symplectic form on $H^1(X;\mathbb{R})$.

Solution. The first statement is hard Lefschetz for $k = 1$. The form is Q for $k = 1$, so it is skew-symmetric and nondegenerate by Theorem 49.4.2(1). This gives another proof that b_1 is even. □

Exercise 49.7.3. Let X be a compact Kähler surface with $b_2(X) = 1$. Show that $p_g = 0$ and $h^{1,1} = 1$, and that a generator e of $H^2(X;\mathbb{Z})/\text{torsion} \cong \mathbb{Z}$ satisfies $\int_X e^2 = 1$.

Solution. By Theorem 48.5.2, $b_2 = 2p_g + h^{1,1}$, and $h^{1,1} \geq 1$ (Corollary 48.5.3), so $p_g = 0$ and $h^{1,1} = 1$. By Corollary 49.5.2 the intersection form has signature $(1, 0)$, so $\int e^2 > 0$. By Poincaré duality the cup product pairing on $H^2(X;\mathbb{Z})/\text{torsion}$ is unimodular (Corollary 23.6.3), so $\int e^2 = \pm 1$, hence 1. □

Exercise 49.7.4. Determine the Lefschetz decomposition of $H^\bullet(\mathbb{CP}^1 \times \mathbb{CP}^1;\mathbb{R})$ for $[\omega] = h_1 + h_2$, and for $[\omega] = h_1 + 2h_2$.

Solution. In both cases $P^0 = \mathbb{R}$ and $H^4 = L^2P^0$. For $[\omega] = h_1 + h_2$, P^2 is the kernel of $L\colon H^2 \to H^4$, $a_1h_1 + a_2h_2 \mapsto (a_1 + a_2)h_1h_2$, which is spanned by $h_1 - h_2$, and $H^2 = LP^0 \oplus P^2$. For $[\omega] = h_1 + 2h_2$, L sends $a_1h_1 + a_2h_2$ to $(2a_1 + a_2)h_1h_2$, so P^2 is spanned by $h_1 - 2h_2$. The primitive subspace depends on the Kähler class. □

Exercise 49.7.5. Show that the character of the $\mathfrak{sl}_2$-representation $H^\bullet(X;\mathbb{R})$ determines the Betti numbers and vice versa, and write $H^\bullet(\mathbb{CP}^n;\mathbb{R})$ and $H^\bullet$ of a compact Riemann surface of genus g as sums of irreducibles $V(m)$.

Solution. The weight $k - n$ has multiplicity b_k, so the character is $\sum_k b_k t^{k-n}$. For $\mathbb{CP}^n$, $H^\bullet = V(n)$, spanned by $1, [\omega], \ldots, [\omega]^n$. For a Riemann surface, $H^\bullet = V(1) \oplus 2g\,V(0)$: the classes 1 and $[\omega]$ span $V(1)$, and H^1 consists of primitive classes of weight 0. □

Chapter 50

Vanishing and Embedding Theorems

50.1 Introduction

This chapter answers the question: which compact Kähler manifolds are projective? The answer is Kodaira's embedding theorem (Theorem 50.6.5). A compact complex manifold embeds in projective space if and only if it carries a positive line bundle, if and only if it has a Kähler form with integral cohomology class. The analytic input is the Kodaira–Nakano vanishing theorem (Theorem 50.4.1). It comes from the Bochner–Kodaira–Nakano identity, a version of the Kähler identities with coefficients in a Hermitian bundle, in which the curvature appears. Vanishing also gives a Hodge-theoretic proof of the Lefschetz hyperplane theorem (Theorem 50.5.1). Together these results form the bridge from Kähler geometry to algebraic geometry: by Chow's theorem (Chapter 58) a projective manifold is algebraic, so the Hodge theory of the previous chapters applies to smooth projective varieties. References: [GH78, Chapter 1, Sections 2–4], [Huy05, Chapter 5], [Voi02b, Chapter 7], [Kod54].

Prerequisites. Chapter 49 (Hard Lefschetz and the Hodge–Riemann Relations) and Chapter 45 (Holomorphic Vector Bundles).

Throughout, X is a compact complex manifold of dimension n. For a line bundle L with a Hermitian metric h, we write F_h for the curvature of its Chern connection and $\omega_h = \frac{i}{2\pi} F_h$ for its *curvature form*, a closed real $(1,1)$-form representing $c_1(L)$ (Theorem 45.4.1). We write $L^k = L^{\otimes k}$ and $L(-D) = L \otimes \mathcal{O}(-D)$.

50.2 Positive line bundles

Proposition 50.2.1. *Positivity (Definition 45.4.2) has the following properties.*

1. *$\mathcal{O}(1)$ on $\mathbb{CP}^N$ is positive, and if $f : Y \to X$ is a holomorphic immersion and L is positive, then f^*L is positive.*
2. *Metrics multiply under tensor products and curvature forms add: $\omega_{h_1 \otimes h_2} = \omega_{h_1} + \omega_{h_2}$ and $\omega_{h^{-1}} = -\omega_h$ for the dual metric. Hence tensor products of positive bundles are positive.*
3. *If L is positive and M is any line bundle, then $L^k \otimes M$ is positive for all $k \geq k_0$.*
4. *If (L, h) is positive, then ω_h is a Kähler form. So X is Kähler and $c_1(L)_{\mathbb{R}}$ is a Kähler class.*
5. *Conversely, if a Kähler form ω represents $c_1(L)_{\mathbb{R}}$, then L has a Hermitian metric with $\omega_h = \omega$. In particular L is positive.*

Proof. (1) The first statement is Example 45.4.3. The Chern connection of f^*h is the pullback, so $\omega_{f^*h} = f^*\omega_h$, and the pullback of a positive $(1,1)$-form by a holomorphic immersion

is positive, since the restriction of a positive definite Hermitian form to a subspace is positive definite. (2) In local holomorphic frames the weights add: $h_1 \otimes h_2 = e^{-(\varphi_1+\varphi_2)}$, and $h^{-1} = e^{\varphi}$. Apply the formula $F = \partial\bar\partial\varphi$. (3) Choose metrics. Since X is compact and ω_{h_L} is positive, there is C with $\omega_{h_M} \geq -C\omega_{h_L}$. Then $\omega_{h_L^k\otimes h_M} = k\omega_{h_L} + \omega_{h_M} \geq (k-C)\omega_{h_L}$, which is positive for $k > C$. (4) ω_h is a closed positive real $(1,1)$-form, hence a Kähler form (Proposition 47.2.2). (5) X is compact Kähler with Kähler form ω. Let h_0 be any metric on L. Then ω_{h_0} and ω are cohomologous closed real $(1,1)$-forms, so $\omega = \omega_{h_0} + i\partial\bar\partial\psi$ for a smooth real function ψ (Corollary 48.6.4). The metric $h = h_0 e^{-2\pi\psi}$ has local weights $\varphi = \varphi_0 + 2\pi\psi$, so $\omega_h = \omega_{h_0} + i\partial\bar\partial\psi = \omega$. □

Lemma 50.2.2. *Let X be a compact Kähler manifold. The map $H^2(X;\mathbb{Z}) \to H^2(X,\mathcal{O}_X)$ induced by $\mathbb{Z}_X \subseteq \mathcal{O}_X$ is the composite of $H^2(X;\mathbb{Z}) \to H^2(X;\mathbb{C})$ with the map $[\alpha] \mapsto [\alpha^{0,2}]_{\bar\partial} \in H^{0,2}_{\bar\partial}(X) \cong H^2(X,\mathcal{O}_X)$. Consequently, every class $a \in H^2(X;\mathbb{Z})$ whose image in $H^2(X;\mathbb{C})$ lies in $H^{1,1}(X)$ is $c_1(L)$ for a holomorphic line bundle L.*

Proof. The inclusion $\mathbb{Z}_X \to \mathcal{O}_X$ factors through $\mathbb{C}_X$. The de Rham complex $\mathcal{A}^\bullet_{\mathbb{C}}$ resolves $\mathbb{C}_X$ and the Dolbeault complex $\mathcal{A}^{0,\bullet}$ resolves $\mathcal{O}_X$ (Theorems 37.3.1 and 46.2.3), by fine, hence acyclic, sheaves. Projection onto the $(0,q)$-component commutes with the differentials: the $(0,q+1)$-part of $d\alpha$ is $\bar\partial$ applied to the $(0,q)$-part of α, because ∂ raises the first degree and $\bar\partial$ of a form of type (r,s) with $r \geq 1$ has type $(r,s+1)$. So it is a morphism of resolutions over $\mathbb{C}_X \to \mathcal{O}_X$, and it induces the map on cohomology (Proposition 15.6.6, which is natural). If the image of a lies in $H^{1,1}(X)$, it is represented by a closed form of type $(1,1)$, whose $(0,2)$-part is zero. So a maps to 0 in $H^2(X,\mathcal{O}_X)$, and $a = c_1(L)$ for some L by Theorem 45.2.4(1). □

Definition 50.2.3. A *Hodge manifold* is a compact complex manifold with a Kähler form ω whose class $[\omega] \in H^2(X;\mathbb{R})$ lies in the image of $H^2(X;\mathbb{Z})$; such an ω is a *Hodge metric*, and $[\omega]$ an *integral Kähler class*.

Corollary 50.2.4. *A compact complex manifold carries a positive line bundle if and only if it is a Hodge manifold.*

Proof. If L is positive, then ω_h is a Kähler form representing the integral class $c_1(L)$ (Proposition 50.2.1(4)). Conversely, let ω be a Hodge metric and $a \in H^2(X;\mathbb{Z})$ a class mapping to $[\omega]$. As ω has type $(1,1)$, Lemma 50.2.2 gives L with $c_1(L) = a$, so $c_1(L)_{\mathbb{R}} = [\omega]$, and L is positive by Proposition 50.2.1(5). □

Proposition 50.2.5. *On a compact Riemann surface X, a line bundle L is positive if and only if $\deg L = \int_X c_1(L) > 0$. Moreover $\deg K_X = 2g-2$, where $g = b_1(X)/2$ is the genus.*

Proof. If L is positive, $\deg L = \int_X \omega_h > 0$. Conversely, if $\deg L > 0$, choose any Hermitian metric on X and rescale its fundamental form to a Kähler form ω with $\int_X \omega = \deg L$ (every Hermitian metric on a curve is Kähler, Example 47.4.4). Since $H^2(X;\mathbb{R}) \cong \mathbb{R}$ via integration, $[\omega] = c_1(L)_{\mathbb{R}}$, and Proposition 50.2.1(5) applies. For the last statement, $K_X = T^*X$, so $\deg K_X = -\int_X c_1(TX) = -\int_X e(TX) = -\chi(X) = -(2-2g)$ by Theorem 39.5.1 and Proposition 39.2.7(4). □

50.3 The Bochner–Kodaira–Nakano identity

Notation 50.3.1. Let X be a Hermitian manifold and (E,h) a Hermitian holomorphic vector bundle, with Chern connection $D = D' + \bar\partial_E$, where D' has type $(1,0)$. On $\mathcal{A}^{p,q}(X,E)$ we use

the pointwise inner product induced by g and h and, for X compact, the L^2 product. The operators L and Λ act on the form part. Let D'^* and $\bar\partial_E^*$ be the formal adjoints, and put

$$\Delta' = D'D'^* + D'^*D', \qquad \Delta'' = \bar\partial_E\bar\partial_E^* + \bar\partial_E^*\bar\partial_E.$$

The curvature $F = D^2$ is a $(1,1)$-form with values in $\operatorname{End} E$, and as operators $F = D'\bar\partial_E + \bar\partial_E D'$, since $D'^2 = 0$ and $\bar\partial_E^2 = 0$ (Theorem 45.4.1).

Lemma 50.3.2. *Let (E,h) be a Hermitian holomorphic vector bundle and $p \in X$. There is a holomorphic frame $e_1,\dots,e_r$ of E near p with $h(e_a,e_b) = \delta_{ab} + O(|z|^2)$ in holomorphic coordinates z centred at p. In such a frame the connection form of D vanishes at p.*

Proof. Start with a holomorphic frame that is orthonormal at p, and write $H_{ab} = h(e_a,e_b) = \delta_{ab} + \sum_l(\partial_l H_{ab}(0)z^l + \bar\partial_l H_{ab}(0)\bar z^l) + O(|z|^2)$. Put $e'_a = e_a - \sum_{b,l}\partial_l H_{ab}(0)\,z^l e_b$, again a holomorphic frame near p. To first order,

$$h(e'_a,e'_b) = H_{ab} - \sum_l \partial_l H_{ab}(0)z^l - \sum_l \overline{\partial_l H_{ba}(0)}\,\bar z^l + O(|z|^2),$$

and $\overline{\partial_l H_{ba}} = \bar\partial_l\overline{H_{ba}} = \bar\partial_l H_{ab}$ because H is Hermitian. So all first-order terms cancel. The connection form $\overline{H}^{-1}\partial\overline{H}$ of Theorem 45.4.1 involves only first derivatives of H, so it vanishes at p. □

Theorem 50.3.3 (Kähler identities with coefficients)**.** *Let X be Kähler and (E,h) a Hermitian holomorphic vector bundle. Then*

$$[\Lambda,\bar\partial_E] = -iD'^*, \qquad [\Lambda,D'] = i\bar\partial_E^*.$$

Proof. Both identities are between first-order differential operators. In holomorphic coordinates and a holomorphic frame of E, the coefficients of these operators at a point are universal expressions in the metric coefficients $h_{j\bar k}$ of X, the matrix H of h, and their first derivatives at that point. Indeed $\bar\partial_E$ acts componentwise, $D' = \partial + \theta$ where the connection form θ involves first derivatives of H, and the formal adjoints are obtained by integration by parts, which differentiates the pointwise inner products and the volume form once. Given p, choose coordinates osculating to order two (Proposition 47.3.2) and a frame as in Lemma 50.3.2. The data then agree to first order at p with the flat metric on $\mathbb{C}^n$ and the trivial bundle $\mathbb{C}^r$ with the standard metric. For these, an E-valued form is an r-tuple of scalar forms, $\bar\partial_E = \bar\partial$ and $D' = \partial$ act componentwise, and the identities are Theorem 48.3.3. So both sides agree at p, as in Lemma 48.3.1. □

Theorem 50.3.4 (Bochner–Kodaira–Nakano identity)**.** *Let X be Kähler and (E,h) a Hermitian holomorphic vector bundle with curvature F. Then on $\mathcal{A}^{p,q}(X,E)$*

$$\Delta'' = \Delta' + [iF,\Lambda],$$

where iF acts by wedge product combined with the action of $\operatorname{End} E$.

Proof. By Theorem 50.3.3, $\bar\partial_E^* = -i[\Lambda,D']$ and $D'^* = i[\Lambda,\bar\partial_E]$. Writing $\bar\partial$ for $\bar\partial_E$,

$$\Delta'' = -i\left(\bar\partial\Lambda D' - \bar\partial D'\Lambda + \Lambda D'\bar\partial - D'\Lambda\bar\partial\right),$$
$$\Delta' = i\left(D'\Lambda\bar\partial - D'\bar\partial\Lambda + \Lambda\bar\partial D' - \bar\partial\Lambda D'\right).$$

Subtracting, the terms $\bar\partial\Lambda D'$ and $D'\Lambda\bar\partial$ cancel, and

$$\Delta'' - \Delta' = i(\bar\partial D' + D'\bar\partial)\Lambda - i\Lambda(D'\bar\partial + \bar\partial D') = iF\Lambda - \Lambda iF = [iF,\Lambda].$$

□

50.4 The Kodaira–Nakano vanishing theorem

Theorem 50.4.1 (Kodaira–Nakano vanishing)**.** *Let X be a compact complex manifold of dimension n and L a positive line bundle. Then*

$$H^q(X, \Omega^p_X \otimes L) = 0 \quad \text{for } p+q > n, \qquad H^q(X, \Omega^p_X \otimes L^{-1}) = 0 \quad \text{for } p+q < n.$$

Proof. Choose a metric h on L with $\omega = \omega_h$ positive, and give X the Kähler metric with fundamental form ω. Then $iF_h = 2\pi\omega$, so $[iF_h, \Lambda] = 2\pi[L, \Lambda] = 2\pi H$, which is multiplication by $2\pi(p+q-n)$ on $\mathcal{A}^{p,q}(X, L)$ (Theorem 47.5.2). By Theorem 46.3.3, every class in $H^q(X, \Omega^p_X \otimes L)$ is represented by some α with $\Delta''\alpha = 0$. By Theorem 50.3.4,

$$0 = (\Delta''\alpha, \alpha) = \|D'\alpha\|^2 + \|D'^*\alpha\|^2 + 2\pi(p+q-n)\|\alpha\|^2.$$

If $p+q > n$, all three terms are non-negative, so $\alpha = 0$. The second statement follows by Serre duality (Theorem 46.3.4): $H^q(X, \Omega^p \otimes L^{-1})$ is dual to $H^{n-q}(X, K_X \otimes (\Omega^p)^\vee \otimes L)$, and $K_X \otimes (\Omega^p)^\vee \cong \Omega^{n-p}$ by the perfect pairing $\Omega^p \otimes \Omega^{n-p} \to K_X$. Here $(n-p) + (n-q) > n$. □

Corollary 50.4.2 (Kodaira vanishing)**.** *If L is positive, then $H^q(X, K_X \otimes L) = 0$ and $H^{n-q}(X, L^{-1}) = 0$ for $q > 0$. More generally, if $N \otimes K_X^{-1}$ is positive, then $H^q(X, N) = 0$ for $q > 0$.*

Proof. Take $p = n$ in Theorem 50.4.1, and $p = 0$ in the second statement. For the last claim write $N = K_X \otimes (N \otimes K_X^{-1})$. □

Example 50.4.3. On $\mathbb{CP}^n$, $K = \mathcal{O}(-n-1)$ (Proposition 45.5.1). Since $\mathcal{O}(k) \otimes K^{-1} = \mathcal{O}(k+n+1)$ is positive for $k > -n-1$, we get $H^q(\mathbb{CP}^n, \mathcal{O}(k)) = 0$ for $q > 0$ and $k > -n-1$. By the second statement of Theorem 50.4.1 with $p = 0$, also $H^q(\mathbb{CP}^n, \mathcal{O}(k)) = 0$ for $q < n$ and $k < 0$. Together, $H^q(\mathbb{CP}^n, \mathcal{O}(k)) = 0$ for $0 < q < n$ and all k, which is part of Bott's formula (Remark 46.4.5).

Corollary 50.4.4. *Let X be a compact Riemann surface of genus g and L a line bundle with $\deg L > 2g - 2$. Then $H^1(X, L) = 0$.*

Proof. $\deg(L \otimes K_X^{-1}) = \deg L - (2g-2) > 0$, so $L \otimes K_X^{-1}$ is positive (Proposition 50.2.5), and Corollary 50.4.2 applies. □

Counterexample 50.4.5. Positivity cannot be dropped. For $L = \mathcal{O}_X$, $H^n(X, K_X) \cong H^0(X, \mathcal{O}_X)^\vee \cong \mathbb{C}$ by Serre duality. On an elliptic curve E, where $K_E \cong \mathcal{O}_E$ (because dz is a nowhere vanishing holomorphic 1-form), every line bundle of degree 0 has degree $\leq 2g - 2 = 0$, and $H^1(E, \mathcal{O}_E) \cong \mathbb{C}$ is nonzero, so Corollary 50.4.4 is sharp.

Remark 50.4.6. The Kodaira vanishing theorem has been generalised to nef and big line bundles (Kawamata–Viehweg) [Laz04, Chapter 4], and it has an algebraic proof by reduction to positive characteristic [DI87]. However, it fails in characteristic p in general [Ray78]. Its dual form $H^q(X, L^{-1}) = 0$ for $q < n$ will be used for coherent cohomology in Chapter 56.

50.5 The Lefschetz hyperplane theorem

Theorem 50.5.1 (Lefschetz hyperplane theorem)**.** *Let X be a compact Kähler manifold of dimension n and $Y \subseteq X$ a smooth hypersurface such that $\mathcal{O}(Y)$ is positive, for instance a smooth hyperplane section of a projective manifold. Then the restriction map*

$$H^k(X; \mathbb{Q}) \to H^k(Y; \mathbb{Q})$$

is an isomorphism for $k < n-1$ and injective for $k = n-1$. The same holds for $H^q(X, \Omega^p_X) \to H^q(Y, \Omega^p_Y)$ with $k = p+q$.

Proof. Write $\mathcal{O}(-Y)$ for the ideal sheaf of Y. It is locally generated by a defining function, since in local coordinates with $Y = \{z_1 = 0\}$ a holomorphic function g vanishing on Y satisfies $g(z) = z_1 \int_0^1 \partial_1 g(tz_1, z_2, \dots, z_n)\, dt$. Holomorphic functions on Y extend locally (constantly in z_1), so $\mathcal{O}_X \to i_*\mathcal{O}_Y$ is onto. Tensoring $0 \to \mathcal{O}_X(-Y) \to \mathcal{O}_X \to i_*\mathcal{O}_Y \to 0$ with Ω^p_X gives

$$0 \to \Omega^p_X \otimes \mathcal{O}(-Y) \to \Omega^p_X \to i_*(\Omega^p_X|_Y) \to 0.$$

By Theorem 50.4.1, $H^q(X, \Omega^p_X \otimes \mathcal{O}(-Y)) = 0$ for $p + q < n$. Since $H^q(X, i_*\mathcal{G}) = H^q(Y, \mathcal{G})$ (Example 26.7.4), the long exact sequence shows that $H^q(X, \Omega^p_X) \to H^q(Y, \Omega^p_X|_Y)$ is an isomorphism for $p + q < n - 1$ and injective for $p + q = n - 1$.

Next, the conormal sequence $0 \to \mathcal{O}(-Y)|_Y \to \Omega^1_X|_Y \to \Omega^1_Y \to 0$ (Proposition 45.5.1) gives, by taking exterior powers of a sequence of bundles whose first term is a line bundle, the exact sequence

$$0 \to \Omega^{p-1}_Y \otimes \mathcal{O}(-Y)|_Y \to \Omega^p_X|_Y \to \Omega^p_Y \to 0.$$

Locally this is the decomposition $\Lambda^p(A \oplus C) = \Lambda^p C \oplus (A \otimes \Lambda^{p-1} C)$ for a line A. Now $\mathcal{O}(Y)|_Y$ is positive on Y (Proposition 50.2.1(1)) and $\dim Y = n - 1$. So $H^q(Y, \Omega^{p-1}_Y \otimes \mathcal{O}(-Y)|_Y) = 0$ for $(p-1) + q < n - 1$, and hence $H^q(Y, \Omega^p_X|_Y) \to H^q(Y, \Omega^p_Y)$ is an isomorphism for $p + q < n - 1$ and injective for $p + q = n - 1$. Composing proves the statement for $H^q(\Omega^p)$.

The composite $H^q(X, \Omega^p_X) \to H^q(Y, \Omega^p_Y)$ is induced by restriction of forms. Under the canonical isomorphisms $H^q(\Omega^p) \cong H^{p,q}$ of Corollary 48.6.3 on X and on Y (which is Kähler by Corollary 47.4.3), it corresponds to the restriction $H^{p,q}(X) \to H^{p,q}(Y)$, since restriction of forms preserves types and commutes with d and $\bar\partial$. By the Hodge decomposition, $H^k(X; \mathbb{C}) \to H^k(Y; \mathbb{C})$ is therefore an isomorphism for $k < n - 1$ and injective for $k = n - 1$. This map is the complexification of the restriction on rational cohomology, and rank is unchanged by extension of scalars. □

Remark 50.5.2. Lefschetz's theorem holds with integer coefficients and for homotopy groups: $\pi_k(X, Y) = 0$ for $k \le n - 1$, so that $H_k(Y; \mathbb{Z}) \to H_k(X; \mathbb{Z})$ is an isomorphism for $k < n - 1$ and surjective for $k = n - 1$. The proof uses Morse theory on the affine variety $X \setminus Y$, which has the homotopy type of a CW complex of dimension $\le n$ (Andreotti–Frankel) [Mil63, Section 7].

Example 50.5.3. Let $Y \subseteq \mathbb{CP}^{n+1}$ be a smooth hypersurface of degree d, so $\dim Y = n$ and $\mathcal{O}(Y) = \mathcal{O}(d)$ is positive. Then $H^k(Y; \mathbb{Q}) \cong H^k(\mathbb{CP}^{n+1}; \mathbb{Q})$ for $k < n$, so $b_k(Y) = b_k(\mathbb{CP}^n)$ for $k < n$, and by Poincaré duality also for $k > n$. Only the middle cohomology $H^n(Y)$ is interesting. For a surface in $\mathbb{CP}^3$, $b_1 = 0$, which was used for the quartic surface in Example 48.7.4. Also $h^{0,q}(Y) = 0$ for $0 < q < n$, that is, $H^q(Y, \mathcal{O}_Y) = 0$ in that range.

50.6 The Kodaira embedding theorem

Definition 50.6.1. Let L be a holomorphic line bundle on X. A point x is a *base point* of L if every $s \in H^0(X, L)$ vanishes at x. If L has no base points, the map $\varphi_L \colon X \to \mathbb{P}(H^0(X, L)^\vee)$ sends x to the hyperplane of sections vanishing at x. After choosing a basis $s_0, \dots, s_N$ of $H^0(X, L)$, it is $x \mapsto [s_0(x) : \cdots : s_N(x)]$, computed in any local frame, so it is holomorphic. L is *very ample* if φ_L is an embedding, and *ample* if some power L^k is very ample. A compact complex manifold is *projective* if it admits a holomorphic embedding into some $\mathbb{CP}^N$.

Lemma 50.6.2. *Let L have no base points. Then φ_L is an embedding if and only if*

1. *for all $x \neq y$ there is $s \in H^0(X, L)$ with $s(x) = 0 \neq s(y)$* (sections separate points)*, and*

2. *for all x, the differentials at x of the sections vanishing at x span $L_x \otimes T^*_x X$* (sections separate tangents)*, where $T^*_x X$ is the holomorphic cotangent space.*

Proof. $\varphi_L(x) = \varphi_L(y)$ exactly when the hyperplanes of sections vanishing at x and at y coincide, so (1) is injectivity. Near x, choose the basis with $s_0(x) \neq 0$. Then φ_L is given in affine coordinates by the functions s_j/s_0. A section s vanishing at x satisfies $d_x s = d_x(s/s_0) \otimes s_0(x)$. Every s is congruent modulo multiples of s_0 to a section vanishing at x, and $d_x(s_0/s_0) = 0$, so the differentials $d_x(s/s_0)$ of sections vanishing at x span the same space as the $d_x(s_j/s_0)$. Thus (2) says that the differentials of the affine coordinates of φ_L at x span $T^*_x X$, that is, φ_L is an immersion at x. An injective immersion of a compact manifold is an embedding (Proposition 33.6.7), and the image of a holomorphic embedding is a complex submanifold by the holomorphic implicit function theorem. □

Lemma 50.6.3. *Let $n \geq 2$, let $x_1, \dots, x_s \in X$ be distinct points, and let $\sigma\colon \tilde{X} \to X$ be the blow-up at these points, with exceptional divisors $E_i = \sigma^{-1}(x_i) \cong \mathbb{P}(T_{x_i} X)$ and $E = \sum_i E_i$ (Definition 44.6.4). Then:*

1. *$\mathcal{O}(E_i)|_{E_i} \cong \mathcal{O}_{E_i}(-1)$;*
2. *$K_{\tilde{X}} \cong \sigma^* K_X \otimes \mathcal{O}((n-1)E)$;*
3. *for every line bundle M on X, $\sigma^*\colon H^0(X, M) \to H^0(\tilde{X}, \sigma^* M)$ is bijective, and it maps the sections vanishing at all x_i onto $H^0(\tilde{X}, \sigma^* M(-E))$;*
4. *if $s \in H^0(X, M)$ vanishes at x_i, the restriction of $\sigma^* s \in H^0(\tilde{X}, \sigma^* M(-E))$ to E_i is the section of $M_{x_i} \otimes \mathcal{O}_{E_i}(1)$ given by the linear form $d_{x_i} s \in M_{x_i} \otimes T^*_{x_i} X$ on $T_{x_i} X$;*
5. *over a coordinate ball B around x_i, $\sigma^{-1}(B) = \{(z, \ell) \in B \times \mathbb{CP}^{n-1} : z \in \ell\}$ (with x_i at the origin), $\mathcal{O}(E_i)$ restricts to the tautological bundle $\mathrm{pr}_2^* \mathcal{O}(-1)$, and the canonical section s_{E_i} of $\mathcal{O}(E_i)$ corresponds to the tautological section $(z, \ell) \mapsto z$.*

Proof. (5) The tautological section vanishes exactly on E_i, to first order: in the chart $(u, w_2, \dots, w_n) \mapsto (u, uw_2, \dots, uw_n)$ it is u times the frame $(1, w_2, \dots, w_n)$. So it is a section with divisor E_i, and the bundle is $\mathcal{O}(E_i)$ by Proposition 45.3.2. (1) Restrict (5) to $E_i = \{0\} \times \mathbb{CP}^{n-1}$; alternatively combine Proposition 44.6.5 with adjunction. (2) In the chart above, $\sigma^*(dz_1 \wedge \cdots \wedge dz_n) = u^{n-1} du \wedge dw_2 \wedge \cdots \wedge dw_n$. So the pullback of a local frame of K_X is a section of $K_{\tilde{X}}$ vanishing to order $n-1$ along E, which gives $\sigma^* K_X \cong K_{\tilde{X}} \otimes \mathcal{O}(-(n-1)E)$. (3) σ^* is injective since σ is surjective. Given $\tilde{s} \in H^0(\tilde{X}, \sigma^* M)$, its restriction to $\tilde{X} \setminus E \cong X \setminus \{x_i\}$ is a section of M there. In a local frame near x_i it is a holomorphic function on a punctured ball, which extends by Hartogs' theorem (Theorem 31.2.9), since $n \geq 2$. The extension s satisfies $\sigma^* s = \tilde{s}$ off E, hence everywhere. A section $\sigma^* s$ vanishes on E if and only if $s(x_i) = 0$ for all i, and a section vanishing on the smooth hypersurface E is divisible by s_E. (4) Write $s = f\, e$ in a local frame with $f(0) = 0$, $f = \sum_j a_j z_j + O(|z|^2)$. In the chart, $f \circ \sigma = u(a_1 + \sum_{j \geq 2} a_j w_j) + O(u^2)$, and dividing by the local equation u of E and setting $u = 0$ gives $a_1 + \sum_{j \geq 2} a_j w_j$. This is the linear form $\sum_j a_j z_j$ written in the trivialisation of $\mathcal{O}_{\mathbb{CP}^{n-1}}(1)$ over $\{z_1 \neq 0\}$ by z_1, and $\sum a_j z_j = d_0 f$. □

Lemma 50.6.4. *Let L be positive, M any line bundle and $l \geq 1$ an integer. Let $s \in \{1, 2\}$ and $(x_1, \dots, x_s)$ a tuple of distinct points. There are a neighbourhood W of $(x_1, \dots, x_s)$ in X^s and k_0 such that, for every tuple $(x'_1, \dots, x'_s) \in W$ of distinct points and every $k \geq k_0$, the line bundle $\sigma^*(L^k \otimes M) \otimes \mathcal{O}(-lE)$ on the blow-up $\tilde{X}$ at $x'_1, \dots, x'_s$ is positive. For $n = 1$ the same holds with $\tilde{X} = X$ and $E = \sum_i x'_i$.*

Proof. Choose disjoint coordinate charts around the x_i containing closed coordinate balls $\bar{B}(x_i, 3\varepsilon)$, and let W be the set of tuples with $x'_i \in B(x_i, \varepsilon)$. Let ω_0 be the Euclidean Kähler form of the chart. Fix metrics on L and M. By compactness there are constants $c, C > 0$ with $\omega_{h_M} \geq -C\omega_{h_L}$ on X and $\omega_{h_L} \geq c\,\omega_0$ on each $\bar{B}(x_i, 3\varepsilon)$.

Fix x'_i and write $w = z - z(x'_i)$. Put a metric on $\mathcal{O}(E_i)$ as follows. On $\sigma^{-1}(B(x'_i, \varepsilon))$ it is the metric h_1 induced by (5) of Lemma 50.6.3 from the tautological bundle, so that $|s_{E_i}|_{h_1} = |w|$. On $\tilde{X} \setminus E_i$ it is the metric h_2 with $|s_{E_i}|_{h_2} = 1$. Glue them by $h = \rho(|w|)h_1 + (1 - \rho(|w|))h_2$, where ρ is a smooth cut-off equal to 1 on $[0, \varepsilon/2]$ and to 0 on $[3\varepsilon/4, \infty)$. Let θ_i be the curvature form of the dual metric on $\mathcal{O}(-E_i)$. Then:

- on $\sigma^{-1}(B(x'_i, \varepsilon/2))$, $\theta_i = \mathrm{pr}_2^*\omega_{FS} \geq 0$, since $\mathcal{O}(-E_i) \cong \mathrm{pr}_2^*\mathcal{O}(1)$ with the Fubini–Study metric (Proposition 47.4.1);
- outside $\sigma^{-1}(B(x'_i, 3\varepsilon/4))$, $\theta_i = 0$, since s_{E_i} is a holomorphic frame of constant norm;
- on the region $\varepsilon/2 \leq |w| \leq 3\varepsilon/4$, $\theta_i = \frac{i}{2\pi}\partial\bar{\partial}\log\big(\rho(|w|)|w|^2 + 1 - \rho(|w|)\big)$, a smooth form whose coefficients depend continuously on (z, x'_i). So $\theta_i \geq -C_1\omega_0$ there, with C_1 independent of $x'_i \in B(x_i, \varepsilon)$.

The curvature form of $N_k = \sigma^*(L^k \otimes M) \otimes \mathcal{O}(-lE)$ with the product metric is $\Theta_k = \sigma^*(k\omega_{h_L} + \omega_{h_M}) + l\sum_i \theta_i \geq (k - C)\sigma^*\omega_{h_L} + l\sum_i \theta_i$. Let $k > C + lC_1/c$. Away from the balls $B(x'_i, 3\varepsilon/4)$, σ is biholomorphic and $\Theta_k \geq (k - C)\sigma^*\omega_{h_L} > 0$. On the annuli, $\Theta_k \geq ((k - C)c - lC_1)\sigma^*\omega_0 > 0$. On $\sigma^{-1}(B(x'_i, \varepsilon/2))$,

$$\Theta_k \geq (k - C)c\,\sigma^*\omega_0 + l\,\mathrm{pr}_2^*\omega_{FS} \geq \min((k - C)c, l)\,\big(\mathrm{pr}_1^*\omega_0 + \mathrm{pr}_2^*\omega_{FS}\big)\big|_{\sigma^{-1}(B)},$$

which is positive, because $\sigma^{-1}(B)$ is a complex submanifold of $B \times \mathbb{CP}^{n-1}$, on which $\mathrm{pr}_1^*\omega_0 + \mathrm{pr}_2^*\omega_{FS}$ is a Kähler form (Corollary 47.4.3). For $n = 1$ the same construction works with $\mathbb{CP}^0$ a point: h_1 is the flat metric, $\mathrm{pr}_2^*\omega_{FS} = 0$, and $\sigma^*\omega_0 = \omega_0$ is positive. So $k_0 = \lfloor C + lC_1/c \rfloor + 1$ works for all tuples in W. □

Theorem 50.6.5 (Kodaira embedding theorem)**.** *Let X be a compact complex manifold and L a positive line bundle. Then there is k_0 such that L^k is very ample for all $k \geq k_0$. In particular X is projective.*

Proof. Step 1: vanishing on blow-ups. Let $\sigma\colon \tilde{X} \to X$ be the blow-up at distinct points $x'_1, \ldots, x'_s$ ($s \in \{1, 2\}$) and $m \in \{1, 2\}$. By Lemma 50.6.3(2),

$$\sigma^*L^k \otimes \mathcal{O}(-mE) = K_{\tilde{X}} \otimes N_k, \qquad N_k = \sigma^*(L^k \otimes K_X^{-1}) \otimes \mathcal{O}(-(m + n - 1)E),$$

also for $n = 1$. By Lemma 50.6.4 with $M = K_X^{-1}$ and $l = m + n - 1 \geq 1$, every tuple has a neighbourhood W and a k_0 such that N_k is positive for $k \geq k_0$ and tuples in W. Then $H^1(\tilde{X}, \sigma^*L^k \otimes \mathcal{O}(-mE)) = 0$ by Corollary 50.4.2, applied on the compact complex manifold $\tilde{X}$.

Step 2: consequences. For $s = 1$, $m = 1$, the exact sequence $0 \to \sigma^*L^k(-E) \to \sigma^*L^k \to \sigma^*L^k|_E \to 0$ and Step 1 show that $H^0(\tilde{X}, \sigma^*L^k) \to H^0(E, \sigma^*L^k|_E)$ is onto. The bundle $\sigma^*L^k|_E = E \times L^k_{x'}$ is trivial and E is compact and connected, so the target is $L^k_{x'}$. By Lemma 50.6.3(3), $H^0(X, L^k) \to L^k_{x'}$ is onto, so x' is not a base point. For $s = 1$, $m = 2$, the sequence $0 \to \sigma^*L^k(-2E) \to \sigma^*L^k(-E) \to \sigma^*L^k(-E)|_E \to 0$ shows that $H^0(\tilde{X}, \sigma^*L^k(-E))$ maps onto $H^0(E, L^k_{x'} \otimes \mathcal{O}_E(1))$, using Lemma 50.6.3(1). This space contains every $v \otimes \lambda$ with λ a linear form on $T_{x'}X$. By parts (3) and (4) of that lemma, the differentials of the sections of L^k vanishing at x' span $L^k_{x'} \otimes T^*_{x'}X$. For $s = 2$, $m = 1$, similarly $H^0(X, L^k) \to L^k_{x'_1} \oplus L^k_{x'_2}$ is onto, so some section vanishes at x'_1 but not at x'_2. For $n = 1$ the same arguments work with E a sum of points and $\mathcal{O}(-x')|_{x'} \cong T^*_{x'}X$.

Step 3: uniformity. Cover the compact space X by finitely many neighbourhoods W from Step 1 for $s = 1$, $m \in \{1, 2\}$. We get k_1 such that for all $k \geq k_1$, L^k has no base points and its sections separate tangents at every point. So φ_{L^k} is an immersion for all $k \geq k_1$. An immersion is locally injective (Proposition 33.6.7), so every x has a neighbourhood V_x on which $\varphi_{L^{k_1}}$ is injective. Then $U = \bigcup_x V_x \times V_x$ is a neighbourhood of the diagonal in $X \times X$ such that $\varphi_{L^{k_1}}(x) \neq \varphi_{L^{k_1}}(y)$ for $(x, y) \in U$ with $x \neq y$. The complement $K = (X \times X) \setminus U$ is compact and consists of pairs of distinct points, so finitely many neighbourhoods from Step 1 with $s = 2$ cover it. This gives k_2 such that sections of L^k separate the points of every pair in K for $k \geq k_2$.

Step 4: conclusion. Let $k \geq \max(k_2, 2k_1)$. For $(x, y) \in U$ with $x \neq y$, pick $s \in H^0(X, L^{k_1})$ with $s(x) = 0 \neq s(y)$ and $t \in H^0(X, L^{k-k_1})$ with $t(y) \neq 0$, which exists because $k - k_1 \geq k_1$. Then $st \in H^0(X, L^k)$ vanishes at x but not at y. Together with Step 3, sections of L^k separate all pairs of points and all tangents, so L^k is very ample by Lemma 50.6.2. □

Corollary 50.6.6. *For a compact complex manifold X the following are equivalent:*

1. *X is projective;*
2. *X carries a positive line bundle;*
3. *X is a Hodge manifold;*
4. *X is Kähler and some Kähler class lies in the image of $H^2(X; \mathbb{Q}) \to H^2(X; \mathbb{R})$.*

Proof. (1) $\Rightarrow$ (2): restrict $\mathcal{O}(1)$ (Proposition 50.2.1(1)). (2) $\Leftrightarrow$ (3) is Corollary 50.2.4, and (2) $\Rightarrow$ (1) is Theorem 50.6.5. (3) $\Rightarrow$ (4) is clear. For (4) $\Rightarrow$ (3), if $[\omega]$ is rational, then $N[\omega]$ is integral for a suitable integer $N > 0$, and $N\omega$ is a Hodge metric. □

Corollary 50.6.7. *A compact Kähler manifold with $H^0(X, \Omega^2_X) = 0$ is projective. In particular every compact Riemann surface is projective, and so is every compact Kähler surface with $p_g = 0$.*

Proof. By the Hodge decomposition, $H^{2,0} = 0 = H^{0,2}$, so $H^2(X; \mathbb{R}) = H^{1,1}(X) \cap H^2(X; \mathbb{R})$ consists of classes of closed real $(1,1)$-forms. The Kähler classes form an open subset of this space: if ω is Kähler and $\beta_1, \ldots, \beta_m$ are closed real $(1,1)$-forms whose classes form a basis, then $\omega + \sum_i c_i \beta_i$ is positive for all small $|c_i|$, by compactness. The image of $H^2(X; \mathbb{Q})$ is dense in $H^2(X; \mathbb{R}) = H^2(X; \mathbb{Q}) \otimes \mathbb{R}$. So there is a rational Kähler class, and Corollary 50.6.6 applies. For a Riemann surface $\Omega^2 = 0$. □

Remark 50.6.8. By Chow's theorem (Chapter 58), a closed complex submanifold of $\mathbb{CP}^N$ is the zero set of homogeneous polynomials, so projective manifolds are smooth projective varieties. Not every compact Kähler manifold is projective: a complex torus $\mathbb{C}^n/\Lambda$ is projective if and only if Λ satisfies the Riemann bilinear relations, that is, some integral class of type $(1,1)$ is positive, which fails for a general lattice when $n \geq 2$ (Chapter 61, [BL04]).

50.7 Exercises

Exercise 50.7.1. Let X be a compact Riemann surface of genus g and L a line bundle of degree $\geq 2g + 1$. Show directly, without blowing up, that L is very ample.

Solution. For points x, y, possibly equal, $\deg L(-x-y) = \deg L - 2 > 2g - 2$, so $H^1(X, L(-x-y)) = 0$ by Corollary 50.4.4. The exact sequence $0 \to L(-x-y) \to L \to L \otimes (\mathcal{O}_X/\mathcal{O}(-x-y)) \to 0$ shows that $H^0(X, L)$ maps onto the two-dimensional space $L_x \oplus L_y$ if $x \neq y$, and onto the 1-jets $L_x \otimes \mathcal{O}_{X,x}/\mathfrak{m}_x^2$ if $x = y$. This is separation of points and tangents. Base-point freeness follows similarly from $H^1(X, L(-x)) = 0$. □

Exercise 50.7.2. Let L be a positive line bundle on X and E a holomorphic vector bundle of the form $M_1 \oplus \cdots \oplus M_r$ with M_i line bundles. Show that $H^q(X, E \otimes L^k) = 0$ for all $q > 0$ and $k \gg 0$. (The statement holds for all holomorphic vector bundles, and even for coherent sheaves; this is Serre's vanishing theorem, see Chapter 56.)

Solution. Cohomology commutes with finite direct sums. For each i, $M_i \otimes L^k \otimes K_X^{-1}$ is positive for $k \gg 0$ by Proposition 50.2.1(3), so $H^q(X, M_i \otimes L^k) = 0$ for $q > 0$ by Corollary 50.4.2. □

Exercise 50.7.3. Let $Y \subseteq \mathbb{CP}^3$ be a smooth surface of degree d. Show that $q(Y) = 0$ and $p_g(Y) = h^0(\mathbb{CP}^3, \mathcal{O}(d-4))$, using Theorem 50.5.1, adjunction and Kodaira vanishing.

Solution. $q = h^{0,1}(Y) = h^{0,1}(\mathbb{CP}^3) = 0$ by the Lefschetz theorem for $H^1(\mathcal{O})$ ($p + q = 1 < 2$). By adjunction $K_Y = \mathcal{O}(d-4)|_Y$, so $p_g = h^0(Y, \mathcal{O}_Y(d-4))$. The sequence $0 \to \mathcal{O}(-4) \to \mathcal{O}(d-4) \to \mathcal{O}_Y(d-4) \to 0$ on $\mathbb{CP}^3$ and $H^1(\mathbb{CP}^3, \mathcal{O}(-4)) = 0$ (Example 50.4.3) give $H^0(\mathbb{CP}^3, \mathcal{O}(d-4)) \to H^0(Y, \mathcal{O}_Y(d-4))$ onto. It is injective because $H^0(\mathbb{CP}^3, \mathcal{O}(-4)) = 0$. For $d = 4$ this gives $p_g = 1$. □

Exercise 50.7.4. Show that a compact complex manifold with $H^2(X; \mathbb{Q}) = 0$ and $\dim X \geq 1$ is never projective, and deduce again that the Hopf surface is not projective.

Solution. A projective manifold has an integral Kähler class $[\omega]$, and $[\omega] \neq 0$ (Proposition 47.4.5). The Hopf surface has $H^2 = 0$ (Counterexample 47.4.9). □

Exercise 50.7.5. Show that products of projective manifolds and blow-ups of projective manifolds at finitely many points are projective.

Solution. If ω_1, ω_2 are Hodge metrics on X_1, X_2, then $\mathrm{pr}_1^*\omega_1 + \mathrm{pr}_2^*\omega_2$ is a Hodge metric on $X_1 \times X_2$, so Corollary 50.6.6 applies. For a blow-up at one or two points, Lemma 50.6.4 with $M = \mathcal{O}_X$ and $l = 1$ shows that $\sigma^*L^k \otimes \mathcal{O}(-E)$ is positive for $k \gg 0$ when L is positive. The same proof works for any finite number of points, with pairwise disjoint balls. □

Exercise 50.7.6. Let X be a compact Riemann surface and L a line bundle with $\deg L < 0$. Deduce $H^0(X, L) = 0$ from Theorem 50.4.1, and show directly from Theorem 50.3.4 that a holomorphic section s of a Hermitian line bundle whose curvature satisfies $iF < 0$ everywhere vanishes.

Solution. L^{-1} is positive (Proposition 50.2.5), and the second statement of Theorem 50.4.1 with $p = q = 0 < 1 = n$ gives $H^0(X, L) = 0$. Directly: on sections ($p = q = 0$), $\Lambda s = 0$, so $[iF, \Lambda]s = -\Lambda(iF\,s) = -f\,s$, where $iF = f\,\omega$ with $f < 0$ (for $n = 1$, $\Lambda\omega = 1$). Since $\Delta'' s = \bar\partial^* \bar\partial s = 0$ for holomorphic s, $0 = \|D's\|^2 + \|D'^* s\|^2 - \int_X f|s|^2\,\omega$, and all terms are non-negative, so $s = 0$. □

Chapter 51

Deformations and Moduli

51.1 Introduction

A compact Riemann surface of genus 1 is a torus $\mathbb{C}/\Lambda$, and different lattices can give non-isomorphic Riemann surfaces, although all of them are diffeomorphic to $S^1 \times S^1$. So a fixed smooth manifold can carry many complex structures, and they vary continuously. Riemann already counted that the compact Riemann surfaces of genus $g \geq 2$ depend on $3g-3$ complex parameters, his "moduli". Two questions arise: how do complex structures vary in families (*deformation theory*), and is there a space whose points are the isomorphism classes, a *moduli space*? This chapter gives a first introduction to both.

1. Families of compact complex manifolds, and Ehresmann's theorem: in a family over a connected base all fibres are diffeomorphic (Section 51.2).

2. The moduli space of elliptic curves: isomorphism classes of complex tori of dimension one correspond to points of $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$, which has an explicit fundamental domain (Section 51.3).

3. The Kodaira–Spencer map, which sends a tangent vector to the base of a family to a class in $H^1(X, T_X)$, the space of *first-order deformations*; we compute it for the family of elliptic curves and show that $\mathbb{P}^1$ is rigid (Section 51.4).

4. An overview of the main theorems: Kuranishi's versal family, obstructions in $H^2(X, T_X)$, the Bogomolov–Tian–Todorov theorem, stability of Kähler manifolds, and moduli spaces of curves (Section 51.5).

Deformation theory is where Hodge theory becomes variational: the Hodge decomposition of the fibres of a family varies holomorphically, and the derivative of this variation is computed from the Kodaira–Spencer class (Chapter 70). References: [Huy05, Chapter 6], [Voi02b, Chapter 9].

Prerequisites. Chapter 33 (Smooth Manifolds), Chapter 29 (Complex Analysis), Chapter 19 (Homotopy and the Fundamental Group), Chapter 44 (Complex Manifolds), Chapter 45 (Holomorphic Vector Bundles) and Chapter 46 (Dolbeault Cohomology).

Throughout, $T_X = T^{1,0}X$ denotes the holomorphic tangent bundle of a complex manifold X, and $H^1(X, T_X)$ its first Dolbeault cohomology $H^{0,1}_{\bar\partial}(X, T_X)$ (Definition 46.2.1); its elements are represented by $\bar\partial$-closed $(0,1)$-forms with values in T_X, that is, by expressions $\sum_{j,k} a_{jk}\, d\bar z^j \otimes \partial/\partial z^k$ in local coordinates.

51.2 Families of complex manifolds

Definition 51.2.1. A *family of compact complex manifolds* over a connected complex manifold B is a holomorphic map $\pi\colon \mathcal{X} \to B$ of complex manifolds that is proper, surjective and a submersion (its differential is surjective at every point). The *fibres* $X_t = \pi^{-1}(t)$ are compact complex submanifolds of $\mathcal{X}$. The family is *trivial* if there is a biholomorphism $\mathcal{X} \cong B \times X$ over B, and *locally trivial* if every point of B has a neighbourhood over which it is trivial. A *deformation* of a compact complex manifold X is a family $\pi\colon \mathcal{X} \to B$ with a point $0 \in B$ and an isomorphism $X \cong X_0$.

The fibres are complex submanifolds by the holomorphic version of the implicit function theorem: near a point of $\mathcal{X}$ there are holomorphic coordinates $(z^1, \dots, z^n, t^1, \dots, t^m)$ in which $\pi(z,t) = t$. Indeed, if $t^1, \dots, t^m$ are coordinates on B near $\pi(x)$, the functions $t^k \circ \pi$ have independent differentials at x, and can be completed to a system of coordinates by n further linear functions z^j of some chart; the map (z,t) has invertible complex linear derivative, so by the inverse function theorem (Theorem 27.11.2) it is a diffeomorphism near x, and its inverse is holomorphic, since the derivative of the inverse is the inverse of a complex linear map. We call such coordinates *adapted* to π.

Example 51.2.2.

1. *The family of elliptic curves.* Let $\mathbb{H} = \{\tau \in \mathbb{C} : \operatorname{Im}\tau > 0\}$. The group $\mathbb{Z}^2$ acts on $\mathbb{C} \times \mathbb{H}$ by $(m,n)\cdot(z,\tau) = (z + m + n\tau, \tau)$, freely and properly discontinuously, by holomorphic maps. The quotient $\mathcal{E} = (\mathbb{C} \times \mathbb{H})/\mathbb{Z}^2$ is a complex manifold, the projection to $\mathbb{H}$ is a family, and its fibre over τ is the complex torus $E_\tau = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$. Every fibre is diffeomorphic to $\mathbb{R}^2/\mathbb{Z}^2$, by $(x,y) \mapsto x + \tau y$.

2. *Plane curves.* The homogeneous polynomials of degree d in three variables form a vector space of dimension $\binom{d+2}{2}$; those defining a smooth curve form an open subset U of it, and $\{(F,[x]) \in U \times \mathbb{P}^2 : F(x) = 0\} \to U$ is a family of compact Riemann surfaces of genus $\frac{(d-1)(d-2)}{2}$. Since F and λF, and F and $F \circ A$ for $A \in \mathrm{GL}_3(\mathbb{C})$, give isomorphic curves, one expects about $\binom{d+2}{2} - 9$ effective parameters, which is 1 for cubics and $6 = 3 \cdot 3 - 3$ for quartics, in accordance with Riemann's count for genus 1 and 3.

3. *Hopf surfaces.* For $\alpha \in \mathbb{C}$ with $|\alpha| > 1$, the group generated by $z \mapsto \alpha z$ acts freely and properly discontinuously on $\mathbb{C}^2 \setminus \{0\}$, and the quotient X_α is a compact complex surface. These surfaces form a family over $\{|\alpha| > 1\}$ (the quotient of $(\mathbb{C}^2 \setminus \{0\}) \times \{|\alpha| > 1\}$ by $(z,\alpha) \mapsto (\alpha z, \alpha)$). Every X_α is diffeomorphic to $S^1 \times S^3$ (Exercise 51.6.1); as $b_1(S^1 \times S^3) = 1$ is odd, no Hopf surface is Kähler (Chapter 48).

4. *A trivial and a non-trivial family with isomorphic fibres.* The projection $B \times X \to B$ is trivial. On the other hand, let $E = E_i$ and let $\mathbb{Z}$ act on $\mathbb{C} \times E$ by $k \cdot (t,z) = (t + k, i^k z)$; the quotient is a family over $\mathbb{C}/\mathbb{Z} \cong \mathbb{C}^*$ all of whose fibres are isomorphic to E. It is not trivial: going once around the base induces the automorphism $z \mapsto iz$ of E, which acts on $H_1(E;\mathbb{Z}) = \mathbb{Z}[i]$ as multiplication by i, whereas in a product the transport along a loop acts trivially on homology. Such families are *isotrivial*. They are locally trivial; in contrast, Remark 51.5.4 describes a family whose fibres are all isomorphic except one.

The fibres of a family over a connected base are all diffeomorphic, so deformation theory studies the complex structures on a fixed smooth manifold. This is a theorem about smooth maps.

Theorem 51.2.3 (Ehresmann)**.** *A proper surjective submersion $f : M \to B$ of smooth manifolds is a locally trivial fibre bundle: every $b \in B$ has a neighbourhood U with a diffeomorphism $f^{-1}(U) \cong U \times f^{-1}(b)$ over U.*

Proof. Take U a coordinate ball centred at b with coordinates $x^1, \dots, x^m$. Since df is surjective, the coordinate vector fields $\partial/\partial x^i$ lift locally to vector fields on $f^{-1}(U)$, and a partition of unity (Theorem 33.8.2) glues local lifts to vector fields v_i on $f^{-1}(U)$ with $df(v_i) = \partial/\partial x^i$, since the condition is affine. As f is proper, the flows of the v_i exist as long as the flows of $\partial/\partial x^i$ stay in U (Theorem 33.9.2: an integral curve cannot leave a compact set in finite time, and f^{-1} of a compact subset of U is compact). The map $U \times f^{-1}(b) \to f^{-1}(U)$, $(x, y) \mapsto \phi^1_{x^1} \circ \cdots \circ \phi^m_{x^m}(y)$, composing the flows for the times x^i, is then a diffeomorphism over U, with inverse obtained by flowing back. □

Corollary 51.2.4. *All fibres of a family of compact complex manifolds over a connected base are diffeomorphic. In particular their Betti numbers agree.*

Proof. By Theorem 51.2.3, the set of $t \in B$ whose fibre is diffeomorphic to a given fibre X_{t_0} is open, and so is its complement. As B is connected, it is all of B. □

The trivialisation is only smooth: Ehresmann's theorem says nothing about the complex structures, which genuinely vary, as the next section shows for elliptic curves.

51.3 The moduli space of elliptic curves

A *lattice* in $\mathbb{C}$ is a subgroup $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ generated by two $\mathbb{R}$-linearly independent numbers; the quotient $\mathbb{C}/\Lambda$ is a compact Riemann surface, an *elliptic curve*, diffeomorphic to $S^1 \times S^1$. Every compact Riemann surface of genus 1 is of this form, by the uniformisation theorem (Theorem 44.4.2), but we shall not need this. We determine when two such tori are isomorphic.

Proposition 51.3.1. *Let $\Lambda, \Lambda' \subseteq \mathbb{C}$ be lattices. Every holomorphic map $f : \mathbb{C}/\Lambda \to \mathbb{C}/\Lambda'$ is induced by an affine map $z \mapsto \alpha z + \beta$ with $\alpha\Lambda \subseteq \Lambda'$. Consequently $\mathbb{C}/\Lambda \cong \mathbb{C}/\Lambda'$ if and only if $\Lambda' = \alpha\Lambda$ for some $\alpha \in \mathbb{C}^*$.*

Proof. The projection $\mathbb{C} \to \mathbb{C}/\Lambda'$ is a covering map and $\mathbb{C}$ is simply connected, so $f \circ p : \mathbb{C} \to \mathbb{C}/\Lambda'$, with $p : \mathbb{C} \to \mathbb{C}/\Lambda$, lifts to a continuous map $F : \mathbb{C} \to \mathbb{C}$ (Proposition 19.6.1), which is holomorphic because the projections are local biholomorphisms. For $\lambda \in \Lambda$, $F(z + \lambda) - F(z)$ lies in the discrete set Λ' and depends continuously on z, so it is a constant $c(\lambda)$. Hence $F'(z + \lambda) = F'(z)$: the entire function F' is Λ-periodic, hence bounded (its values are those on a compact fundamental parallelogram), hence constant by Liouville's theorem (Corollary 29.4.7). So $F(z) = \alpha z + \beta$, and $\alpha\lambda = c(\lambda) \in \Lambda'$. If f is an isomorphism, the same applied to f^{-1} gives $\alpha^{-1}\Lambda' \subseteq \Lambda$, so $\alpha\Lambda = \Lambda'$ with $\alpha \neq 0$. Conversely, $z \mapsto \alpha z$ induces an isomorphism $\mathbb{C}/\Lambda \to \mathbb{C}/\alpha\Lambda$. □

Every lattice has a basis ω_1, ω_2 with $\operatorname{Im}(\omega_2/\omega_1) > 0$ (exchange the two vectors if necessary), and then $\Lambda = \omega_1(\mathbb{Z} + \mathbb{Z}\tau)$ with $\tau = \omega_2/\omega_1 \in \mathbb{H}$. So every elliptic curve is isomorphic to some $E_\tau = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$. The group $\mathrm{SL}_2(\mathbb{Z})$ acts on $\mathbb{H}$ by Möbius transformations,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \tau = \frac{a\tau + b}{c\tau + d}, \qquad \operatorname{Im} \frac{a\tau + b}{c\tau + d} = \frac{\operatorname{Im} \tau}{|c\tau + d|^2},$$

the formula for the imaginary part following from $(a\tau + b)(c\bar\tau + d) = ac|\tau|^2 + ad\tau + bc\bar\tau + bd$ and $ad - bc = 1$.

Theorem 51.3.2. *$E_\tau \cong E_{\tau'}$ if and only if $\tau' = \gamma \cdot \tau$ for some $\gamma \in \mathrm{SL}_2(\mathbb{Z})$. So the isomorphism classes of elliptic curves correspond bijectively to the points of $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$. Every orbit meets the set*

$$\mathcal{F} = \{\tau \in \mathbb{H} : |\operatorname{Re} \tau| \leq \tfrac{1}{2},\ |\tau| \geq 1\},$$

and two distinct points of $\mathcal{F}$ are in the same orbit only if they lie on the boundary and are exchanged by $\tau \mapsto \tau + 1$ or by $\tau \mapsto -1/\tau$.

Proof. By Proposition 51.3.1, $E_\tau \cong E_{\tau'}$ if and only if $\mathbb{Z} + \mathbb{Z}\tau' = \alpha(\mathbb{Z} + \mathbb{Z}\tau)$ for some α. Then $\alpha\tau$ and α form a basis of $\mathbb{Z} + \mathbb{Z}\tau'$, as do τ' and 1; so $\tau' = \alpha(a\tau + b)$ and $1 = \alpha(c\tau + d)$ with $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2(\mathbb{Z})$, and dividing, $\tau' = (a\tau + b)/(c\tau + d)$. Since both τ and τ' lie in $\mathbb{H}$, the formula for the imaginary part, valid for det $= \pm 1$ with the factor det, forces $ad - bc = 1$. Conversely, if $\tau' = \gamma \cdot \tau$, then $\alpha = (c\tau + d)^{-1}$ satisfies $\alpha(\mathbb{Z} + \mathbb{Z}\tau) = \mathbb{Z} + \mathbb{Z}\tau'$, as γ is invertible over $\mathbb{Z}$.

Every orbit meets $\mathcal{F}$. Recall that $\mathrm{SL}_2(\mathbb{Z})$ contains $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, acting by $\tau \mapsto \tau + 1$, and $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, acting by $\tau \mapsto -1/\tau$. Let $\tau \in \mathbb{H}$. The numbers $|c\tau + d|$ for $(c, d) \in \mathbb{Z}^2 \setminus \{0\}$ are bounded below by a positive constant and take only finitely many values below any bound (the lattice $\mathbb{Z} + \mathbb{Z}\tau$ is discrete), so $\mathrm{Im}(\gamma \cdot \tau) = \mathrm{Im}\,\tau/|c\tau + d|^2$ attains a maximum over $\gamma \in \mathrm{SL}_2(\mathbb{Z})$; choose γ attaining it and a power $T^k\colon \tau \mapsto \tau + k$ with $|\mathrm{Re}(T^k\gamma \cdot \tau)| \le \frac{1}{2}$. The point $\sigma = T^k\gamma \cdot \tau$ still has maximal imaginary part. If $|\sigma| < 1$, then $S\colon \sigma \mapsto -1/\sigma$ would give $\mathrm{Im}(-1/\sigma) = \mathrm{Im}\,\sigma/|\sigma|^2 > \mathrm{Im}\,\sigma$, a contradiction. So $\sigma \in \mathcal{F}$.

Points of $\mathcal{F}$ in the same orbit. Let $\tau = x + iy$ and $\gamma \cdot \tau$ both lie in $\mathcal{F}$, with $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \neq \pm 1$. Exchanging the roles of τ and $\gamma \cdot \tau$ (and replacing γ by γ^{-1}) we may assume $\mathrm{Im}(\gamma \cdot \tau) \ge \mathrm{Im}\,\tau$, that is, $|c\tau + d| \le 1$. Write $\rho = e^{2\pi i/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ and $\rho + 1 = -\bar{\rho} = e^{\pi i/3}$ for the two corners of $\mathcal{F}$. For $\tau \in \mathcal{F}$ we have $y^2 = |\tau|^2 - x^2 \ge 1 - \frac{1}{4}$, so $y \ge \sqrt{3}/2$, and $|c|\sqrt{3}/2 \le |c|y = |\mathrm{Im}(c\tau + d)| \le 1$ forces $c \in \{0, \pm 1\}$.

- If $c = 0$, then $ad = 1$, so $a = d = \pm 1$ and $\gamma \cdot \tau = \tau \pm b$ with $b \neq 0$. Both τ and $\tau \pm b$ have real part in $[-\frac{1}{2}, \frac{1}{2}]$, so $|b| = 1$ and $x = \pm\frac{1}{2}$: the two points lie on the vertical sides and are exchanged by $\tau \mapsto \tau \pm 1$.

- If $c = \pm 1$, replacing γ by $-\gamma$ (which acts in the same way) we may take $c = 1$, so $|\tau + d| \le 1$. Now $|\tau + d|^2 = (x + d)^2 + y^2$. If $|d| \ge 2$ this is at least $(\frac{3}{2})^2 > 1$. If $d = \pm 1$, it is $|\tau|^2 \pm 2x + 1 \ge 1 \pm 2x + 1 \ge 1$, with equality only if $|\tau| = 1$ and $x = \mp\frac{1}{2}$, that is, $\tau = \rho$ for $d = 1$ and $\tau = \rho + 1$ for $d = -1$. If $d = 0$, it is $|\tau|^2 \ge 1$, with equality only if $|\tau| = 1$.

  Case $d = 0$, $|\tau| = 1$. Then $-b = ad - bc = 1$ and $\gamma \cdot \tau = (a\tau - 1)/\tau = a - 1/\tau = a - \bar{\tau}$, of real part $a - x$. This lies in $[-\frac{1}{2}, \frac{1}{2}]$ only if $a = 0$, or $a = \pm 1$ and $x = \pm\frac{1}{2}$. If $a = 0$, then $\gamma \cdot \tau = -1/\tau$, and $\tau, -1/\tau = -\bar{\tau}$ are points of the arc $|\tau| = 1$ exchanged by $\tau \mapsto -1/\tau$. If $a = \pm 1$, then $x = a/2$, so τ is a corner, and $\gamma \cdot \tau = a - \bar{\tau} = (a - x) + iy = \tau$.

  Case $d = \pm 1$, τ a corner. Then $b = ad - 1$ and $\gamma \cdot \tau = a - (\tau + d)^{-1}$. For $\tau = \rho$, $d = 1$: $\rho + 1 = -\bar{\rho}$ and $(\rho + 1)^{-1} = -\rho$, so $\gamma \cdot \rho = a + \rho$, which is in $\mathcal{F}$ only for $a \in \{0, 1\}$, giving ρ itself or $\rho + 1$. For $\tau = \rho + 1$, $d = -1$, similarly $\gamma \cdot \tau = a - \rho^{-1} = a - \bar{\rho} = a + \rho + 1$, giving $\rho + 1$ or ρ.

In all cases either $\gamma \cdot \tau = \tau$ or the two points lie on the boundary of $\mathcal{F}$ and are exchanged by $\tau \mapsto \tau \pm 1$ or by $\tau \mapsto -1/\tau$. □

So the *moduli space of elliptic curves* is $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$, which is $\mathcal{F}$ with its boundary glued: the vertical sides are identified by $\tau \mapsto \tau + 1$ and the two halves of the arc $|\tau| = 1$ by $\tau \mapsto -1/\tau$. Topologically it is a plane. The case analysis in the proof also shows that the stabiliser of a point of $\mathcal{F}$ in $\mathrm{SL}_2(\mathbb{Z})$ is $\{\pm 1\}$, except at $\tau = i$ (stabiliser of order 4, generated by S) and at the corners ρ, $\rho + 1$ (order 6; for ρ it is generated by ST, as $ST \cdot \tau = -1/(\tau + 1)$); correspondingly every elliptic curve has the automorphism $z \mapsto -z$, E_i has in addition $z \mapsto iz$, and E_ρ has $z \mapsto -\rho z$, a sixth root of unity. These two curves have *complex multiplication* by $\mathbb{Z}[i]$ and $\mathbb{Z}[\rho]$, the Gaussian and Eisenstein integers.

Remark 51.3.3. 1. *The j-invariant.* There is a holomorphic function $j\colon \mathbb{H} \to \mathbb{C}$, invariant under $\mathrm{SL}_2(\mathbb{Z})$, that induces a bijection $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z}) \to \mathbb{C}$; with the Eisenstein series

$g_2(\tau) = 60\sum'(m+n\tau)^{-4}$ and $g_3(\tau) = 140\sum'(m+n\tau)^{-6}$ it is $j = 1728 g_2^3/(g_2^3 - 27g_3^2)$, with $j(i) = 1728$ and $j(\rho) = 0$ [Ser73, Chapter VII]. So the moduli space of elliptic curves is the affine line, the *j-line*.

2. *Quadratic forms.* The reduction of τ into $\mathcal{F}$ is the reduction theory of positive definite binary quadratic forms (Theorem 10.8.3) in another guise: the form $|x+y\tau|^2 = x^2 + 2\operatorname{Re}(\tau)xy + |\tau|^2y^2$ satisfies $|2\operatorname{Re}\tau| \le 1 \le |\tau|^2$, the reduction inequalities $|b| \le a \le c$, exactly when $\tau \in \mathcal{F}$. For a positive definite form $ax^2+bxy+cy^2$ of discriminant $D<0$, the root $\tau = \frac{-b+i\sqrt{|D|}}{2a}$ of $a\tau^2+b\tau+c$ in $\mathbb{H}$ gives an elliptic curve with complex multiplication, and $j(\tau)$ is an algebraic integer of degree the class number $h(D)$; for class number one it is a rational integer, for instance $j\big(\frac{1+i\sqrt{163}}{2}\big) = -640320^3$, which explains why $e^{\pi\sqrt{163}}$ is so close to an integer (Remark 10.8.9) [Cox89].

3. *Not a fine moduli space.* There is no family over $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$ whose fibre over each point is the corresponding curve and which induces all families: the isotrivial family of Example 51.2.2(4) has constant "moduli point" $[i]$ without being trivial. The automorphisms of the curves are the obstruction. Such a space, which classifies the objects but does not carry a universal family, is a *coarse moduli space*.

51.4 The Kodaira–Spencer map

Let $\pi\colon \mathcal{X} \to B$ be a family, $0 \in B$ and $X = X_0$. We want to measure, to first order, how the complex structure of X_t changes as t moves away from 0 in a direction $v \in T_0B$. Locally over B nothing happens: the family looks like a product $U \times \Delta$ near each point of X. The information is in how these local products are glued.

Choose holomorphic coordinates $t = (t^1, \dots, t^m)$ on B centred at 0. By compactness of X and the existence of adapted coordinates, after shrinking B to a small polydisc Δ there are open sets $\mathcal{W}_i$ covering $\pi^{-1}(\Delta)$, $i = 1, \dots, N$, with adapted coordinates (z_i, t), $z_i = (z_i^1, \dots, z_i^n)$, identifying $\mathcal{W}_i$ with $U_i \times \Delta$ for open $U_i \subseteq \mathbb{C}^n$; the sets $X \cap \mathcal{W}_i$ cover X. On overlaps the coordinates are related by holomorphic *transition maps*

$$z_j = f_{ij}(z_i, t), \qquad f_{ik}(z_i, t) = f_{jk}(f_{ij}(z_i, t), t).$$

For a tangent vector $v = \sum_l v^l \partial/\partial t^l \in T_0B$ define, on $X \cap \mathcal{W}_i \cap \mathcal{W}_j$, the holomorphic vector field

$$\theta_{ij}(v) = \sum_{k=1}^{n} \frac{\partial f_{ij}^k}{\partial v}(z_i, 0)\, \frac{\partial}{\partial z_j^k}, \qquad \frac{\partial}{\partial v} = \sum_l v^l \frac{\partial}{\partial t^l},$$

the velocity, at $t = 0$, of the point with fixed coordinate z_i, as seen in the coordinates z_j.

Lemma 51.4.1. *The vector fields $\theta_{ij} = \theta_{ij}(v)$ satisfy $\theta_{ik} = \theta_{ij} + \theta_{jk}$ on triple overlaps; in particular $\theta_{ii} = 0$ and $\theta_{ji} = -\theta_{ij}$. If the adapted coordinates are changed to $z_i' = g_i(z_i, t)$, the new vector fields are $\theta_{ij}' = \theta_{ij} + \eta_j - \eta_i$, where η_i is the holomorphic vector field $\sum_k \frac{\partial g_i^k}{\partial v}(z_i, 0)\, \partial/\partial z_i'^k$ on $X \cap \mathcal{W}_i$.*

Proof. Differentiate $f_{ik}(z_i, t) = f_{jk}(f_{ij}(z_i, t), t)$ with respect to v at $t = 0$, by the chain rule:

$$\frac{\partial f_{ik}}{\partial v} = D_z f_{jk} \cdot \frac{\partial f_{ij}}{\partial v} + \frac{\partial f_{jk}}{\partial v}.$$

The first term is the vector θ_{ij}, written in the coordinates z_j, transformed into the coordinates z_k by the Jacobian matrix $D_z f_{jk}$ of the change of coordinates; that is, it is the same vector

field θ_{ij} expressed in the coordinates z_k. So $\theta_{ik} = \theta_{ij} + \theta_{jk}$. Taking $i = j = k$ gives $\theta_{ii} = 2\theta_{ii}$, and $k = i$ gives $\theta_{ji} = -\theta_{ij}$. For the change of coordinates, the new transition maps are $f'_{ij} = g_j \circ f_{ij} \circ g_i^{-1}$, where g_i^{-1} inverts $z_i \mapsto g_i(z_i, t)$ for fixed t; the same differentiation, applied to $f'_{ij}(g_i(z_i,t),t) = g_j(f_{ij}(z_i,t),t)$, gives $\eta_i + \theta'_{ij} = \theta_{ij} + \eta_j$ as vector fields. □

A family of holomorphic vector fields (θ_{ij}) with $\theta_{ik} = \theta_{ij} + \theta_{jk}$ is a *Čech* 1*-cocycle* of the sheaf of holomorphic vector fields (Definition 26.5.1), and those of the form $\eta_j - \eta_i$ are the coboundaries. We convert it into a Dolbeault class, which does not depend on the cover.

Lemma 51.4.2. *Let E be a holomorphic vector bundle on a complex manifold X, (U_i) an open cover, and θ_{ij} holomorphic sections of E over $U_i \cap U_j$ with $\theta_{ik} = \theta_{ij} + \theta_{jk}$.*

1. *There are smooth sections σ_i of E over U_i with $\sigma_j - \sigma_i = \theta_{ij}$ on $U_i \cap U_j$; for instance $\sigma_i = \sum_k \rho_k \theta_{ki}$ for a partition of unity (ρ_k) subordinate to the cover.*
2. *The forms $\bar\partial\sigma_i$ agree on overlaps, and define a $\bar\partial$-closed form $\delta(\theta) \in \mathcal{A}^{0,1}(X, E)$, whose class in $H^{0,1}_{\bar\partial}(X, E)$ does not depend on the choice of the σ_i.*
3. *If $\theta_{ij} = \eta_j - \eta_i$ with η_i holomorphic on U_i, then $[\delta(\theta)] = 0$. If (V_a) is a refinement of the cover, with $V_a \subseteq U_{\lambda(a)}$, then the restricted cocycle $\theta_{\lambda(a)\lambda(b)}$ gives the same class.*

Proof. (1) Each $\rho_k\theta_{ki}$, extended by 0 outside $\operatorname{supp}\rho_k$, is a smooth section on U_i, and the sum is locally finite. Then $\sigma_j - \sigma_i = \sum_k \rho_k(\theta_{kj} - \theta_{ki}) = \sum_k \rho_k\theta_{ij} = \theta_{ij}$, using $\theta_{kj} = \theta_{ki} + \theta_{ij}$.

(2) On $U_i \cap U_j$, $\bar\partial\sigma_j - \bar\partial\sigma_i = \bar\partial\theta_{ij} = 0$ since θ_{ij} is holomorphic. The glued form is locally $\bar\partial$-exact, hence $\bar\partial$-closed. If σ'_i is another choice, then $\sigma'_i - \sigma_i = \sigma'_j - \sigma_j$ on overlaps, so these differences glue to a global smooth section s, and the two forms differ by $\bar\partial s$.

(3) If $\theta_{ij} = \eta_j - \eta_i$, we may take $\sigma_i = \eta_i$, which are holomorphic, so $\delta(\theta) = 0$. For a refinement, the sections $\sigma_{\lambda(a)}|_{V_a}$ satisfy the required relation and have the same $\bar\partial$. □

Definition 51.4.3. The *Kodaira–Spencer map* of the family π at 0 is

$$\kappa\colon T_0B \to H^1(X, T_X), \qquad \kappa(v) = [\delta(\theta(v))].$$

By Lemmas 51.4.1 and 51.4.2 it does not depend on the choice of adapted coordinates (two systems of adapted coordinates can be compared on a common refinement), and it is clearly $\mathbb{C}$-linear in v. The space $H^1(X, T_X)$ is the space of *first-order deformations* of X.

Proposition 51.4.4. 1. *For a trivial family $B \times X \to B$, $\kappa = 0$.*

2. *(Base change) If $\varphi\colon B' \to B$ is holomorphic with $\varphi(0') = 0$, the pulled-back family $\mathcal{X}' = B' \times_B \mathcal{X} \to B'$ has Kodaira–Spencer map $\kappa' = \kappa \circ d\varphi_{0'}$.*

Proof. (1) With adapted coordinates (z_i, t) coming from charts z_i of X, the transition maps $f_{ij}(z_i, t) = f_{ij}(z_i)$ do not depend on t, so $\theta = 0$. (2) The fibre product $\mathcal{X}' = \{(s, x) \in B' \times \mathcal{X} : \varphi(s) = \pi(x)\}$ is, in adapted coordinates (z_i, t) of $\mathcal{X}$, the set of points (s, z_i, t) with $t = \varphi(s)$; so it is a complex submanifold of $B' \times \mathcal{X}$ with coordinates (z_i, s), adapted to the projection to B', and its transition maps are $z_j = f_{ij}(z_i, \varphi(s))$. Its fibre over $0'$ is X_0. By the chain rule, $\partial f_{ij}(z_i, \varphi(s))/\partial w = \partial f_{ij}/\partial(d\varphi_{0'}(w))$ at $s = 0'$, so $\theta'_{ij}(w) = \theta_{ij}(d\varphi_{0'}(w))$. □

So if $\kappa(v) \neq 0$, the family is not trivial near 0, and more: its restriction to any holomorphic disc $\varphi\colon \Delta \to B$ through 0 with tangent vector v is not trivial, since by (2) its Kodaira–Spencer map sends $\partial/\partial s$ to $\kappa(v) \neq 0$, while by (1) a trivial family has $\kappa = 0$.

Example 51.4.5. *The family of elliptic curves.* Let $\mathcal{E} \to \mathbb{H}$ be the family of Example 51.2.2(1), and $\tau_0 \in \mathbb{H}$. Adapted coordinates are given by (z, τ) on small open sets of $\mathbb{C} \times \mathbb{H}$ that map injectively to $\mathcal{E}$. Two such charts differ by an element of $\mathbb{Z}^2$, so the transition maps are $z_j = z_i + m_{ij} + n_{ij}\tau$ with integers m_{ij}, n_{ij}, and

$$\theta_{ij}\Big(\frac{\partial}{\partial \tau}\Big) = n_{ij}\frac{\partial}{\partial z}.$$

Here $\partial/\partial z$ is a global holomorphic vector field on $E = E_{\tau_0}$, invariant under translations. To find σ_i as in Lemma 51.4.2, write $z = x + \tau_0 y$ with x, y real; the function $y = \operatorname{Im} z / \operatorname{Im} \tau_0$ satisfies $y(z + m + n\tau_0) = y(z) + n$. So $\sigma_i = y(z_i)\,\partial/\partial z$ satisfies $\sigma_j - \sigma_i = n_{ij}\,\partial/\partial z = \theta_{ij}$, and

$$\kappa\Big(\frac{\partial}{\partial \tau}\Big) = \Big[\frac{\partial y}{\partial \bar z}\, d\bar z \otimes \frac{\partial}{\partial z}\Big] = \frac{i}{2 \operatorname{Im} \tau_0}\Big[d\bar z \otimes \frac{\partial}{\partial z}\Big],$$

since $\operatorname{Im} z = (z - \bar z)/2i$ has $\partial/\partial \bar z$ equal to $-1/2i = i/2$. This class is nonzero. As T_E is trivialised by $\partial/\partial z$, it suffices to see that $d\bar z$ is not $\bar\partial g$ for a smooth function g on E. In the real coordinates (x, y) of $\mathbb{R}^2/\mathbb{Z}^2$, $\partial/\partial \bar z = \alpha\,\partial/\partial x + \beta\,\partial/\partial y$ with constants α, β (as x and y are real linear functions of z, $\bar z$), so if $\partial g/\partial \bar z = 1$, integrating over $[0,1]^2$ and using periodicity, $1 = \int_{[0,1]^2} (\alpha g_x + \beta g_y)\, dx\, dy = 0$ by Fubini's theorem and the fundamental theorem of calculus, a contradiction. So κ is injective at every point: the family of elliptic curves is nowhere locally trivial. Here $H^1(E, T_E)$ is one-dimensional (Proposition 51.4.6), so κ is an isomorphism.

The dimension of $H^1(X, T_X)$ can often be computed with Serre duality. For a compact Riemann surface X, $T_X^\vee = K_X$, and Serre duality (Theorem 46.3.4) gives

$$H^1(X, T_X) \cong H^0(X, K_X \otimes K_X)^\vee,$$

the dual of the space of holomorphic *quadratic differentials* $q = f(z)\, dz^2$.

Proposition 51.4.6. *1. $H^1(\mathbb{P}^1, T_{\mathbb{P}^1}) = 0$: the Riemann sphere has no first-order deformations.*

2. For an elliptic curve E, $H^1(E, T_E)$ is one-dimensional.

Proof. (1) A holomorphic quadratic differential on $\mathbb{P}^1$ is $f(z)\, dz^2$ with f entire on the chart $\mathbb{C}$; in the coordinate $w = 1/z$ near ∞, $dz = -w^{-2} dw$, so it is $f(1/w) w^{-4}\, dw^2$, which is holomorphic at $w = 0$ only if $f(z) z^4$ stays bounded as $z \to \infty$. Then f is bounded, hence constant by Liouville's theorem, and the constant is 0. So $H^0(K^2) = 0$, and $H^1(T) = 0$ by Serre duality.

(2) K_E is trivialised by the global form dz, so $H^0(E, K_E^2) \cong H^0(E, \mathcal{O}_E)$, the holomorphic functions on the compact connected E. Such a function h attains its maximum modulus at some point, so it is constant near that point by the maximum modulus principle (Theorem 29.4.9), and the set where h equals this constant is open and closed, hence all of E. So $H^0(E, K_E^2)$ is one-dimensional. □

For a curve of genus $g \geq 2$, the Riemann–Roch theorem gives $\dim H^0(X, K_X^2) = 3g - 3$ (Chapter 59), so $\dim H^1(X, T_X) = 3g - 3$, Riemann's count of moduli. For a complex torus $\mathbb{C}^n/\Lambda$, T_X is trivial and $\dim H^1(X, T_X) = n \dim H^{0,1}_{\bar\partial}(X) = n^2$; this matches the count of linear complex structures on $\mathbb{R}^{2n}$: $\mathrm{GL}_{2n}(\mathbb{R})/\mathrm{GL}_n(\mathbb{C})$ has real dimension $4n^2 - 2n^2 = 2n^2$.

Remark 51.4.7. *Deformations as a differential equation.* By Ehresmann's theorem a deformation of X over a disc can be viewed as a family of complex structures on the fixed manifold X. A complex structure J_t close to J_0 is determined by its space of $(0,1)$-vectors, which is the graph of a map $T^{0,1} \to T^{1,0}$, that is, by a form $\varphi(t) \in \mathcal{A}^{0,1}(X, T_X)$ with $\varphi(0) = 0$. The new

$\bar\partial$-operator is $\bar\partial + \varphi$ acting through the contraction of φ with ∂f, and the integrability of J_t (Theorem 44.5.4) becomes the *Maurer–Cartan equation*

$$\bar\partial\varphi + \tfrac{1}{2}[\varphi,\varphi] = 0.$$

Writing $\varphi(t) = t\varphi_1 + t^2\varphi_2 + \cdots$, the first-order part satisfies $\bar\partial\varphi_1 = 0$, and $[\varphi_1]$ is the Kodaira–Spencer class of $\partial/\partial t$. The second-order part requires $\bar\partial\varphi_2 = -\frac{1}{2}[\varphi_1,\varphi_1]$: the class of $[\varphi_1,\varphi_1]$ in $H^2(X,T_X)$ is the first *obstruction* to extending φ_1 to an actual deformation [Huy05, Chapter 6].

51.5 Kuranishi theory and moduli spaces

We state the main theorems of the theory; their proofs combine the Maurer–Cartan equation with the harmonic theory of the $\bar\partial$-Laplacian and power series estimates.

Theorem 51.5.1 (Kodaira–Spencer, Kuranishi)**.** *Let X be a compact complex manifold.*

1. *There is a deformation $\mathcal{X} \to (K,0)$ of X over a complex analytic space K, the* Kuranishi space*, which is* complete*: every deformation of X over a germ $(B,0)$ is obtained from it by pull-back along a holomorphic map $(B,0)\to(K,0)$, whose differential at 0 is determined by the Kodaira–Spencer map. The Zariski tangent space of K at 0 is $H^1(X,T_X)$.*
2. *K is the zero set, near 0, of a holomorphic map $H^1(X,T_X)\to H^2(X,T_X)$ vanishing to second order at 0. In particular, if $H^2(X,T_X)=0$, then K is a neighbourhood of 0 in $H^1(X,T_X)$, and X is* unobstructed*.*
3. *If $H^0(X,T_X)=0$, that is, X has no holomorphic vector fields, the map in (1) is unique, and the family is* universal*.*
4. *If $H^1(X,T_X)=0$, then X is* rigid*: every deformation of X is trivial near 0.*

This is [Huy05, Chapter 6] and [Voi02b, Chapter 9]. For curves, $H^2(X,T_X)=0$ for dimension reasons, so curves are unobstructed; with Proposition 51.4.6, $\mathbb{P}^1$ is rigid, an elliptic curve has a one-dimensional smooth Kuranishi space (the family over $\mathbb{H}$ near τ_0), and a curve of genus $g\geq 2$ has a smooth $(3g-3)$-dimensional Kuranishi space, universal because such curves have no holomorphic vector fields (T_X is a line bundle of negative degree $2-2g$, Chapter 59).

Theorem 51.5.2 (Bogomolov–Tian–Todorov)**.** *A compact Kähler manifold X with trivial canonical bundle is unobstructed: its Kuranishi space is smooth, of dimension $h^1(X,T_X)=h^{n-1,1}(X)$.*

This is [Huy05, Chapter 6]. The equality $h^1(T_X)=h^{n-1,1}$ comes from the isomorphism $T_X\cong\Omega_X^{n-1}$ given by contraction with a nowhere vanishing holomorphic n-form. It applies to Calabi–Yau manifolds (Chapter 52) and to K3 surfaces, whose Kuranishi space has dimension $h^{1,1}=20$.

Theorem 51.5.3 (Kodaira–Spencer)**.** *Let $\pi\colon\mathcal{X}\to B$ be a family. If X_0 is Kähler, then X_t is Kähler for t in a neighbourhood of 0. The Hodge numbers $h^{p,q}(X_t)$ are upper semicontinuous functions of t for any family, and constant near 0 if X_0 is Kähler.*

This is [Voi02b, Chapter 9]. Constancy for Kähler fibres follows from semicontinuity and the Hodge decomposition: $\sum_{p+q=k}h^{p,q}(X_t)=b_k$ is constant by Corollary 51.2.4, and a sum of upper semicontinuous functions with constant sum consists of locally constant functions. Being projective, in contrast, is not an open condition: a general small deformation of a projective K3 surface is not projective.

Remark 51.5.4. *Jumping of the complex structure.* The Hirzebruch surfaces $\Sigma_n = \mathbb{P}(\mathcal{O} \oplus \mathcal{O}(n))$ over $\mathbb{P}^1$ are diffeomorphic for all even n, and there is a family over $\mathbb{C}$ whose fibre over $t \neq 0$ is $\Sigma_0 = \mathbb{P}^1 \times \mathbb{P}^1$ and whose fibre over 0 is Σ_2 [Kod86]. So isomorphism classes of fibres can jump at special points, and the set of isomorphism classes of complex structures on a fixed manifold is in general not a Hausdorff space: the point $[\Sigma_2]$ lies in the closure of the point $[\Sigma_0]$. Correspondingly $h^0(T_{\Sigma_2}) = 7 > 6 = h^0(T_{\Sigma_0})$: the special fibre has more symmetry.

Remark 51.5.5. *Moduli of curves.* The compact Riemann surfaces of genus $g \geq 2$ form a coarse moduli space $\mathcal{M}_g$, a normal quasi-projective variety of dimension $3g - 3$, as Riemann predicted; it is the quotient of the *Teichmüller space* $\mathcal{T}_g$, which classifies complex structures together with a marking of the fundamental group and is biholomorphic to a bounded domain in $\mathbb{C}^{3g-3}$, by the action of the *mapping class group*; its singularities come from curves with nontrivial automorphisms [HM98]. The *period map* sends a curve to its Jacobian (Chapter 61); by Torelli's theorem it is injective, and its derivative is computed from the Kodaira–Spencer map by cup product, a basic instance of the variations of Hodge structure of Chapter 70.

51.6 Exercises

Exercise 51.6.1. Show that every Hopf surface X_α (Example 51.2.2(3)) is diffeomorphic to $S^1 \times S^3$.

Solution. Write $\alpha = re^{i\theta}$ with $r > 1$. The map $\mathbb{C}^2 \setminus \{0\} \to \mathbb{R} \times S^3$, $z \mapsto (\log|z|/\log r, z/|z|)$, is a diffeomorphism transforming $z \mapsto \alpha z$ into $(s, w) \mapsto (s + 1, e^{i\theta}w)$. So X_α is the quotient of $\mathbb{R} \times S^3$ by the group generated by $(s, w) \mapsto (s + 1, e^{i\theta}w)$, the *mapping torus* of the rotation $w \mapsto e^{i\theta}w$ of $S^3 \subseteq \mathbb{C}^2$. The map $[s, w] \mapsto ([s], e^{-i\theta s}w)$, with $[s] \in \mathbb{R}/\mathbb{Z}$, is well defined from this quotient to $S^1 \times S^3$: replacing (s, w) by $(s + 1, e^{i\theta}w)$ gives $e^{-i\theta(s+1)}e^{i\theta}w = e^{-i\theta s}w$. In the same way $([s], u) \mapsto [s, e^{i\theta s}u]$ is well defined, and the two maps are smooth and inverse to each other. □

Exercise 51.6.2. Show that the group of automorphisms of E_τ fixing 0 is $\{\pm 1\}$ unless τ is equivalent to i or ρ, where it is cyclic of order 4 or 6.

Solution. By Proposition 51.3.1, such an automorphism is $z \mapsto \alpha z$ with $\alpha\Lambda = \Lambda$, $\Lambda = \mathbb{Z} + \mathbb{Z}\tau$. Then $|\alpha| = 1$ (the shortest nonzero vectors of Λ are permuted) and α is a root of the characteristic polynomial of an integer matrix of size 2, so $\alpha + \bar{\alpha} \in \mathbb{Z}$, and $\alpha \in \{\pm 1, \pm i, \pm\rho, \pm\rho^2\}$. If $\alpha = \pm i$, then $i\Lambda = \Lambda$, so Λ is closed under multiplication by $\mathbb{Z}[i]$. Let $\lambda \in \Lambda$ be a nonzero element of smallest absolute value. Then $\lambda\mathbb{Z}[i] \subseteq \Lambda$; conversely, for $\mu \in \Lambda$ there is $q \in \mathbb{Z}[i]$ with $|\mu/\lambda - q| < 1$ (round the real and imaginary parts, as in the division with remainder of Example 8.5.5), so $\mu - q\lambda \in \Lambda$ is shorter than λ, hence 0. So $\Lambda = \lambda\mathbb{Z}[i]$, $E_\tau \cong E_i$, and τ is equivalent to i by Theorem 51.3.2. If α is a primitive third or sixth root of unity, then ρ is a power of α, so $\rho\Lambda = \Lambda$, and the same argument with the Eisenstein integers $\mathbb{Z}[\rho]$, which have a division with remainder of the same kind (Proposition 10.7.5, where ρ is written ω), gives $\tau \sim \rho$. Conversely $i\mathbb{Z}[i] = \mathbb{Z}[i]$ and $\rho\mathbb{Z}[\rho] = \mathbb{Z}[\rho]$. □

Exercise 51.6.3. For $\lambda \in \mathbb{C} \setminus \{0, 1\}$ let C_λ be the cubic curve $y^2 = x(x - 1)(x - \lambda)$, completed in $\mathbb{P}^2$. Show that the substitutions $x \mapsto 1 - x$ and $x \mapsto x/\lambda$ (with suitable changes of y) give isomorphisms $C_\lambda \cong C_{1-\lambda}$ and $C_\lambda \cong C_{1/\lambda}$, so that the six values λ, $1 - \lambda$, $1/\lambda$, $1/(1 - \lambda)$, $\lambda/(\lambda - 1)$, $(\lambda - 1)/\lambda$ give isomorphic curves, and check that

$$J(\lambda) = \frac{(\lambda^2 - \lambda + 1)^3}{\lambda^2(\lambda - 1)^2}$$

takes the same value at these six points. (In fact $j = 256J$ is the j-invariant, and $C_\lambda \cong C_\mu$ if and only if $J(\lambda) = J(\mu)$.)

Solution. With $x = 1-x'$ and $y = iy'$: $-y'^2 = (1-x')(-x')(1-x'-\lambda)$, that is, $y'^2 = x'(x'-1)(x'-(1-\lambda))$. With $x = \lambda x'$ and $y = \lambda^{3/2}y'$: $\lambda^3 y'^2 = \lambda x'(\lambda x'-1)(\lambda x'-\lambda) = \lambda^3 x'(x'-1/\lambda)(x'-1)$. The two maps $\lambda \mapsto 1-\lambda$ and $\lambda \mapsto 1/\lambda$ generate a group of order 6 acting on $\mathbb{C}\setminus\{0,1\}$, with the six listed values as orbit. For invariance it suffices to check the generators: $\lambda^2-\lambda+1$ and $\lambda(\lambda-1)$ are invariant under $\lambda \mapsto 1-\lambda$; under $\lambda \mapsto 1/\lambda$, the numerator becomes $\lambda^{-6}(1-\lambda+\lambda^2)^3$ and the denominator $\lambda^{-2}\lambda^{-2}(1-\lambda)^2 = \lambda^{-4}(\lambda-1)^2$, so $J(1/\lambda) = \lambda^{-6}\lambda^4(\lambda^2-\lambda+1)^3/(\lambda-1)^2 = J(\lambda)$. □

Exercise 51.6.4. Count parameters, as in Example 51.2.2(2), for the following families of curves, and compare with Riemann's count $3g-3$:

1. smooth plane quartics, of genus 3, modulo projective transformations of $\mathbb{P}^2$;
2. the curves $y^2 = f(x)$ with f a polynomial with $2g+2$ distinct roots (the *hyperelliptic curves* of genus g), which are determined by the set of roots of f on $\mathbb{P}^1$ modulo the Möbius transformations of $\mathbb{P}^1$.

Solution. (1) Quartic forms in three variables have $\binom{6}{2} = 15$ coefficients; subtracting 1 for scaling and $\dim \mathrm{PGL}_3(\mathbb{C}) = 8$ gives $6 = 3\cdot 3-3$. (2) $2g+2$ points of $\mathbb{P}^1$, modulo the group $\mathrm{PGL}_2(\mathbb{C})$ of dimension 3, give $2g+2-3 = 2g-1$ parameters. For $g = 2$ this is $3 = 3g-3$, and in fact every curve of genus 2 is hyperelliptic. For $g \geq 3$, $2g-1 < 3g-3$, so the hyperelliptic curves form a proper subfamily: a general curve of genus $g \geq 3$ is not hyperelliptic. □

Chapter 52

Calabi–Yau Manifolds

52.1 Introduction

A compact Kähler manifold with vanishing first Chern class carries, in every Kähler class, a unique Ricci-flat Kähler metric. This is Yau's solution of the Calabi conjecture (Theorem 52.3.1). Such manifolds, and in particular those with trivial canonical bundle, are called Calabi–Yau manifolds. They are among the richest examples in Hodge theory. Their Hodge numbers are constrained but far from trivial, their deformations are unobstructed and governed by $H^{n-1,1}$, and they are the setting of mirror symmetry. Elliptic curves, K3 surfaces and the quintic threefold are the basic examples. We compute the Hodge numbers of the quintic threefold completely (Proposition 52.4.4). References: [GHJ03], [Joy00, Chapter 6], [Yau78], [Bea83].
Prerequisites. Chapter 50 (Vanishing and Embedding Theorems), Chapter 49 (Hard Lefschetz and the Hodge–Riemann Relations) and Chapter 39 (Characteristic Classes).

Throughout, X is a compact Kähler manifold of dimension n with Kähler form $\omega = \frac{i}{2}\sum h_{j\bar{k}}\,dz^j \wedge d\bar{z}^k$ in local holomorphic coordinates (Proposition 47.2.2). We write $c_1(X) = c_1(T^{1,0}X) = -c_1(K_X)$.

52.2 The Ricci form

Definition 52.2.1. The *Ricci form* of the Kähler metric ω is the real $(1,1)$-form given in local holomorphic coordinates by

$$\mathrm{Ric}(\omega) = -i\partial\bar{\partial}\log\det(h_{j\bar{k}}).$$

The metric is *Ricci-flat* if $\mathrm{Ric}(\omega) = 0$ and *Kähler–Einstein* if $\mathrm{Ric}(\omega) = \lambda\omega$ for a constant λ.

Proposition 52.2.2. *$\mathrm{Ric}(\omega)$ is a well-defined closed real $(1,1)$-form, equal to 2π times the curvature form $\frac{i}{2\pi}F$ of the anticanonical bundle $K_X^{-1} = \Lambda^n T^{1,0}X$ with the metric induced by ω. Hence*

$$\left[\frac{1}{2\pi}\mathrm{Ric}(\omega)\right] = c_1(X) \in H^2(X;\mathbb{R}).$$

Proof. In a holomorphic chart, $e = \partial_1 \wedge \cdots \wedge \partial_n$ is a holomorphic frame of K_X^{-1} whose norm for the induced metric is $|e|^2 = \det(g(\partial_j, \bar{\partial}_k)) = 2^{-n}\det(h_{j\bar{k}})$. So the local weight is $\varphi = -\log\det(h_{j\bar{k}}) + n\log 2$, and by Theorem 45.4.1 the curvature is $F = \partial\bar{\partial}\varphi = -\partial\bar{\partial}\log\det(h_{j\bar{k}})$, with $c_1(K_X^{-1}) = [\frac{i}{2\pi}F] = [\frac{1}{2\pi}\mathrm{Ric}(\omega)]$. This also shows that $\mathrm{Ric}(\omega)$ is independent of the chart: under a change of coordinates, $\det(h_{j\bar{k}})$ is multiplied by $|J|^2$ with J the holomorphic Jacobian determinant, and $\partial\bar{\partial}\log|J|^2 = 0$. It is real and closed as a curvature form of a line bundle. □

Remark 52.2.3. $\mathrm{Ric}(\omega)(U,V) = \mathrm{Ric}_g(JU,V)$, where Ric_g is the Ricci tensor of the underlying Riemannian metric [Huy05, Chapter 4]. So Ricci-flat Kähler metrics are Ricci-flat Riemannian metrics, and Proposition 52.2.2 shows that $c_1(X)_{\mathbb{R}} = 0$ is a necessary condition for a Kähler metric to be Ricci-flat.

52.3 The Calabi conjecture

Theorem 52.3.1 (Calabi–Yau)**.** *Let X be a compact Kähler manifold with Kähler form ω, and let ρ be a closed real $(1,1)$-form with $[\rho] = 2\pi c_1(X)$. Then there is a unique Kähler form ω' with $[\omega'] = [\omega]$ and $\mathrm{Ric}(\omega') = \rho$. In particular, if $c_1(X)_{\mathbb{R}} = 0$, every Kähler class contains a unique Ricci-flat Kähler form.*

Uniqueness is due to Calabi and existence to Yau [Yau78]. Proposition 52.3.2 reduces the theorem to a nonlinear elliptic equation and proves uniqueness; existence is proved by the continuity method with a priori estimates up to order 2 and 3 [Joy00, Chapter 5].

Proposition 52.3.2. *In the situation of Theorem 52.3.1, there is a smooth real function F with $\rho = \mathrm{Ric}(\omega) - i\partial\bar\partial F$ and $\int_X (e^F - 1)\omega^n = 0$. A Kähler form $\omega' = \omega + i\partial\bar\partial\varphi$ in the class of ω satisfies $\mathrm{Ric}(\omega') = \rho$ if and only if*

$$(\omega + i\partial\bar\partial\varphi)^n = e^F\omega^n$$

(the complex Monge–Ampère equation*). Solutions of this equation are unique up to additive constants, so ω' is unique.*

Proof. ρ and $\mathrm{Ric}(\omega)$ are cohomologous closed real $(1,1)$-forms, so $\rho = \mathrm{Ric}(\omega) - i\partial\bar\partial F$ by Corollary 48.6.4, and adding a constant to F achieves the normalisation. Every Kähler form in $[\omega]$ is $\omega + i\partial\bar\partial\varphi$ for a real φ, by the same corollary. The ratio of volume forms $\omega'^n/\omega^n = \det(h'_{j\bar k})/\det(h_{j\bar k})$ is a positive function, so $\mathrm{Ric}(\omega') - \mathrm{Ric}(\omega) = -i\partial\bar\partial\log(\omega'^n/\omega^n)$. Hence $\mathrm{Ric}(\omega') = \rho$ if and only if $i\partial\bar\partial(\log(\omega'^n/\omega^n) - F) = 0$, if and only if $\log(\omega'^n/\omega^n) = F + c$ with c constant (a real function with $\partial\bar\partial u = 0$ on a compact manifold is pluriharmonic, hence harmonic for the Kähler Laplacian, hence constant). Integrating, $\int\omega'^n = \int\omega^n$ because the forms are cohomologous, and the normalisation of F forces $c = 0$.

For uniqueness, let $\omega_1 = \omega + i\partial\bar\partial\varphi_1$ and $\omega_2 = \omega + i\partial\bar\partial\varphi_2$ with $\omega_1^n = \omega_2^n$, and let $\psi = \varphi_1 - \varphi_2$. Then

$$0 = \omega_1^n - \omega_2^n = i\partial\bar\partial\psi \wedge T, \qquad T = \sum_{k=0}^{n-1} \omega_1^k \wedge \omega_2^{n-1-k}.$$

Multiply by ψ and integrate. Since T is closed, Stokes' theorem gives $0 = \int_X \psi\, i\partial\bar\partial\psi \wedge T = -\int_X i\,\partial\psi \wedge \bar\partial\psi \wedge T$. At each point, ω_1 and ω_2 are positive, so T is a sum of products of positive $(1,1)$-forms, and $i\,\partial\psi \wedge \bar\partial\psi \wedge T$ is a non-negative multiple of the volume form, positive where $\partial\psi \neq 0$ (diagonalise ω_1 and ω_2 simultaneously at the point). Hence $\partial\psi = 0$, so $d\psi = 0$ as ψ is real, and ψ is constant. □

52.4 Calabi–Yau manifolds and their Hodge numbers

Definition 52.4.1. A *Calabi–Yau manifold* is a compact Kähler manifold X with trivial canonical bundle $K_X \cong \mathcal{O}_X$. Equivalently, there is a nowhere vanishing holomorphic n-form Ω, the *holomorphic volume form*. It is *strict* if moreover $h^{p,0}(X) = 0$ for $0 < p < n$.

Proposition 52.4.2. *Let X be a Calabi–Yau manifold of dimension n.*

1. *$c_1(X) = 0$, so X carries Ricci-flat Kähler metrics (Theorem 52.3.1). Moreover $h^{0,q} = h^{n,q}$ for all q; in particular $h^{n,0} = h^{0,n} = 1$.*

2. *Contraction with Ω is an isomorphism of holomorphic bundles $T^{1,0}X \to \Omega_X^{n-1}$, $v \mapsto \iota_v\Omega$, and more generally $\Lambda^p T^{1,0}X \cong \Omega_X^{n-p}$. Hence $H^1(X, T_X) \cong H^{n-1,1}(X)$.*

Proof. (1) $c_1(X) = -c_1(K_X) = 0$, and $h^{n,q} = h^q(X, K_X) = h^q(X, \mathcal{O}_X) = h^{0,q}$ (Theorem 48.5.2(4)). For $q = 0$ this is 1, and $h^{0,n} = h^{n,0}$ by Hodge symmetry. (2) At a point choose coordinates with $\Omega = f\, dz^1 \wedge \cdots \wedge dz^n$, $f \neq 0$. Then $\iota_{\partial_j}\Omega = (-1)^{j-1} f\, dz^1 \wedge \cdots \widehat{dz^j} \cdots \wedge dz^n$, and these form a basis of Ω^{n-1}, so contraction is a holomorphic bundle isomorphism; the same computation with $\iota_{\partial_{j_1}} \cdots \iota_{\partial_{j_p}}$ gives $\Lambda^p T^{1,0}X \cong \Omega_X^{n-p}$. By Dolbeault's theorem, $H^1(X, T_X) \cong H^1(X, \Omega^{n-1}) \cong H^{n-1,1}(X)$. □

Example 52.4.3.

1. Complex tori $\mathbb{C}^n/\Lambda$, with $\Omega = dz^1 \wedge \cdots \wedge dz^n$ and the flat metric, which is Ricci-flat. They are not strict for $n \geq 2$.

2. Elliptic curves are the Calabi–Yau manifolds of dimension 1, since K is trivial exactly in genus 1.

3. A *K3 surface* is a strict Calabi–Yau surface, that is, $K_X \cong \mathcal{O}$ and $h^{1,0} = 0$. Smooth quartic surfaces in $\mathbb{CP}^3$ are K3 surfaces (Example 48.7.4).

4. A smooth hypersurface of degree $n+2$ in $\mathbb{CP}^{n+1}$ has trivial canonical bundle by adjunction, $K = \mathcal{O}(n+2-n-2) = \mathcal{O}$ (Proposition 45.5.1), and for $n \geq 2$ it is strict by the Lefschetz hyperplane theorem (Theorem 50.5.1), since $h^{p,0}(X) = h^{p,0}(\mathbb{CP}^{n+1}) = 0$ for $0 < p < n$.

5. Products of Calabi–Yau manifolds are Calabi–Yau, with $\Omega = \mathrm{pr}_1^*\Omega_1 \wedge \mathrm{pr}_2^*\Omega_2$; they are not strict.

Proposition 52.4.4 (The quintic threefold)**.** *Let $X \subseteq \mathbb{CP}^4$ be a smooth quintic hypersurface. Then X is a strict Calabi–Yau threefold with Hodge diamond determined by*

$$h^{0,0} = h^{3,0} = 1, \qquad h^{1,0} = h^{2,0} = 0, \qquad h^{1,1} = 1, \qquad h^{2,1} = 101,$$

and Euler characteristic $\chi(X) = -200$.

Proof. K_X is trivial and $h^{1,0} = h^{2,0} = 0$ by Example 52.4.3(4), with $n = 3$. By the Lefschetz hyperplane theorem, $b_1(X) = 0$ and $b_2(X) = b_2(\mathbb{CP}^4) = 1$, so $h^{1,1} = b_2 - 2h^{2,0} = 1$. By Poincaré duality $b_4 = 1$ and $b_5 = 0$, and by the Hodge decomposition $b_3 = h^{3,0} + h^{2,1} + h^{1,2} + h^{0,3} = 2 + 2h^{2,1}$. So $\chi(X) = 1 + 1 - b_3 + 1 + 1 = 2 - 2h^{2,1}$.

We compute $\chi(X) = \int_X c_3(TX)$ (Theorem 39.5.1). Let h be the hyperplane class. The tangent bundle of $\mathbb{CP}^4$ has total Chern class $(1+h)^5$ (Example 39.2.8). The normal bundle of X is $\mathcal{O}(X)|_X = \mathcal{O}(5)|_X$ (Proposition 45.5.1), with $c = 1 + 5h$. From $0 \to TX \to T\mathbb{CP}^4|_X \to N \to 0$ and the Whitney formula,

$$c(TX) = \frac{(1+h)^5}{1+5h} = (1 + 5h + 10h^2 + 10h^3)(1 - 5h + 25h^2 - 125h^3) = 1 + 10h^2 - 40h^3$$

restricted to X, modulo h^4. So $c_1(X) = 0$, as it must be, and $\chi(X) = -40\int_X h^3$. Finally $\int_X h^3 = 5$: the class of X is Poincaré dual to $c_1(\mathcal{O}(X)) = 5h$ [GH78, Chapter 1, Section 1], and $\int_{\mathbb{CP}^4} h^4 = 1$. Hence $\chi(X) = -200$ and $h^{2,1} = 101$. □

Remark 52.4.5. The number 101 is the number of moduli of quintic threefolds: the quintic polynomials in five variables form a space of dimension $\binom{9}{4} = 126$, and removing scalars and the 24-dimensional group PGL_5 leaves $126 - 1 - 24 = 101$ parameters. This matches $h^1(X, T_X) = h^{2,1}$ (Proposition 52.4.2), which is the dimension of the space of first-order deformations of the complex structure. By the Bogomolov–Tian–Todorov theorem the deformations of a Calabi–Yau manifold are unobstructed, so its Kuranishi space is smooth of dimension $h^{n-1,1}$ [Tia87], [Tod89] (Theorem 51.5.2).

Proposition 52.4.6 (Complete intersection Calabi–Yau threefolds)**.** *Let $X = V(f_1, \dots, f_r) \subseteq \mathbb{CP}^{3+r}$ with f_i homogeneous of degree $d_i \geq 2$ and $\sum_i d_i = 4 + r$, and assume that every partial intersection $V(f_1, \dots, f_j)$, $1 \leq j \leq r$, is smooth of codimension j. Then X is a strict Calabi–Yau threefold with $h^{1,1} = 1$, and*

$(d_1, \dots, d_r)$	(5)	$(2,4)$	$(3,3)$	$(2,2,3)$	$(2,2,2,2)$
$c_2(X)$	$10h^2$	$7h^2$	$6h^2$	$5h^2$	$4h^2$
$c_3(X)$	$-40h^3$	$-22h^3$	$-16h^3$	$-12h^3$	$-8h^3$
$\int_X h^3 = \prod_i d_i$	5	8	9	12	16
$\chi(X)$	-200	-176	-144	-144	-128
$h^{2,1}(X)$	101	89	73	73	65

These five are all the possibilities with every $d_i \geq 2$.

Proof. Write $X_j = V(f_1, \dots, f_j)$, so that $X_j \subseteq X_{j-1}$ is a smooth hypersurface cut out by a section of $\mathcal{O}(d_j)$. By adjunction (Proposition 45.5.1) applied r times, $K_X = \mathcal{O}(\sum_i d_i - 4 - r)|_X = \mathcal{O}_X$. By the Lefschetz hyperplane theorem (Theorem 50.5.1) applied to each $X_j \subseteq X_{j-1}$, $H^k(X) \cong H^k(\mathbb{CP}^{3+r})$ for $k < 3$ (with $\mathbb{Q}$-coefficients); so $b_1 = 0$ and $b_2 = 1 = 2h^{2,0} + h^{1,1}$, and since $h^{1,1} \geq 1$ (the Kähler class is nonzero, Proposition 47.4.5), $h^{1,0} = h^{2,0} = 0$ and $h^{1,1} = 1$, which makes X strict. As in the proof of Proposition 52.4.4, $b_4 = 1$, $b_3 = 2 + 2h^{2,1}$ and $\chi(X) = 2 - 2h^{2,1}$.

The normal bundle of X is $\bigoplus_i \mathcal{O}(d_i)|_X$ (adjunction at each step), so by the Whitney formula $c(TX) = (1+h)^{4+r} \prod_i (1 + d_i h)^{-1}$ restricted to X, modulo h^4. For instance for $(2,4)$,

$$(1 + 6h + 15h^2 + 20h^3)(1 - 6h + 28h^2 - 120h^3) = 1 + 7h^2 - 22h^3,$$

using $(1+2h)^{-1}(1+4h)^{-1} = (1 - 2h + 4h^2 - 8h^3)(1 - 4h + 16h^2 - 64h^3)$ modulo h^4; the other columns are the same computation. The degree $\int_X h^3 = \prod_i d_i$ holds because the class of X is $\prod_i d_i\, h^r$, each hypersurface having class $d_i h$ (as in the proof of Proposition 52.4.4). Then $\chi(X) = \int_X c_3(TX)$ (Theorem 39.5.1) gives the fourth row, and $h^{2,1} = 1 - \chi/2$ the last. Finally, $\sum_i (d_i - 1) = 4$ with all $d_i \geq 2$ has exactly the five solutions listed, up to order. □

52.5 Structure and symmetries

Theorem 52.5.1 (Beauville–Bogomolov decomposition)**.** *Let X be a compact Kähler manifold with $c_1(X)_{\mathbb{R}} = 0$. Then there is a finite étale cover $\tilde{X} \to X$ with*

$$\tilde{X} \cong T \times \prod_i Y_i \times \prod_j Z_j,$$

where T is a complex torus, the Y_i are simply connected strict Calabi–Yau manifolds with holonomy $\mathrm{SU}(\dim Y_i)$*, and the Z_j are simply connected* hyperkähler *manifolds, with holonomy* $\mathrm{Sp}(\dim Z_j/2)$ *and $H^0(Z_j, \Omega^2)$ spanned by a symplectic form.*

This is [Bea83]. Its proof uses the Ricci-flat metric of Theorem 52.3.1 and the de Rham and Berger classifications of holonomy groups. In dimension 2 the only simply connected factors are K3 surfaces, which are both strict Calabi–Yau and hyperkähler.

Remark 52.5.2. The first step is Bochner's argument. For a Ricci-flat metric, harmonic 1-forms are parallel (Theorem 43.3.6), and more generally every holomorphic p-form is parallel. So the holomorphic forms are determined by the holonomy representation, and $h^{p,0}$ is the dimension of the space of holonomy-invariant forms of type $(p,0)$ at a point [GHJ03]. For holonomy exactly $\mathrm{SU}(n)$ the only invariants are 1 and the volume form, which gives $h^{p,0} = 0$ for $0 < p < n$.

Remark 52.5.3. *Mirror symmetry* predicts that strict Calabi–Yau threefolds come in pairs $(X, X^\vee)$ with $h^{1,1}(X) = h^{2,1}(X^\vee)$ and $h^{2,1}(X) = h^{1,1}(X^\vee)$, the complex geometry of X being equivalent to the symplectic geometry of $X^\vee$. The mirror of the quintic has $h^{1,1} = 101$ and $h^{2,1} = 1$, and the resulting prediction of the numbers of rational curves on the quintic [CdlOGP91] was a starting point of enumerative mirror symmetry.

Remark 52.5.4. For projective Calabi–Yau manifolds the Hodge conjecture is known in some degrees. On any projective threefold it holds in all degrees, by the Lefschetz $(1,1)$-theorem in degree 2 and hard Lefschetz in degree 4 (Chapter 69); the interesting group H^3 carries no Hodge classes, being of odd degree. For K3 surfaces it holds in all degrees (Chapter 72). For higher-dimensional hyperkähler varieties it is largely open.

52.6 Exercises

Exercise 52.6.1. Show that a smooth quartic surface $X \subseteq \mathbb{CP}^3$ has $\chi(X) = 24$ by computing $c_2(TX)$ as in Proposition 52.4.4, and deduce $h^{1,1} = 20$.

Solution. $c(TX) = (1+h)^4/(1+4h) = (1+4h+6h^2)(1-4h+16h^2) = 1+6h^2$ modulo h^3, so $c_1 = 0$ and $\chi(X) = 6\int_X h^2 = 6 \cdot 4 = 24$. With $b_1 = b_3 = 0$ and $b_0 = b_4 = 1$ we get $b_2 = 22$, and $h^{1,1} = b_2 - 2h^{2,0} = 22 - 2 = 20$. □

Exercise 52.6.2. Show that the Fubini–Study metric on $\mathbb{CP}^n$ is Kähler–Einstein with $\mathrm{Ric}(\omega_{FS}) = 2\pi(n+1)\omega_{FS}$.

Solution. By $U(n+1)$-invariance (Proposition 47.4.1), $\mathrm{Ric}(\omega_{FS})$ is an invariant closed real $(1,1)$-form. At the point $w = 0$ of the affine chart, $\omega_{FS} = \frac{i}{2\pi}\partial\bar\partial \log(1+|w|^2)$ has $h_{j\bar k} = \frac{1}{\pi}\frac{(1+|w|^2)\delta_{jk} - \bar w_j w_k}{(1+|w|^2)^2}$, whose determinant is $\pi^{-n}(1+|w|^2)^{-n-1}$. So $\mathrm{Ric}(\omega_{FS}) = (n+1)i\partial\bar\partial\log(1+|w|^2) = 2\pi(n+1)\omega_{FS}$ on the chart, hence everywhere. □

Exercise 52.6.3. Show that a smooth hypersurface $X \subseteq \mathbb{CP}^{n+1}$ of degree d has $c_1(X) = (n+2-d)h|_X$, so that it is Fano ($c_1 > 0$), Calabi–Yau or of general type ($c_1 < 0$) according as $d < n+2$, $d = n+2$ or $d > n+2$.

Solution. $c(TX) = (1+h)^{n+2}/(1+dh)$ has degree-one part $(n+2)h - dh$. Alternatively, $K_X = \mathcal{O}(d-n-2)|_X$ by adjunction. □

Part VIII

Algebraic Geometry

Chapter 53

Algebraic Varieties

53.1 Introduction

This chapter develops classical algebraic geometry over an algebraically closed field: affine and projective varieties, their regular functions and morphisms, dimension, tangent spaces and smoothness, rational maps, and blow-ups. The Nullstellensatz of Chapter 14 translates geometry into commutative algebra. Closed subsets correspond to radical ideals, varieties to prime ideals, points to maximal ideals, and affine varieties to finitely generated domains. Chapter 54 generalises everything to schemes, which are needed for non-closed fields, for families, and for nilpotents. The classical language is enough for most statements about complex projective manifolds, and it is the one in which the Hodge conjecture is usually stated. References: [Har77, Chapter I], [GH78, Chapter 1].

Prerequisites. Chapter 14 (Commutative Algebra), Chapter 18 (General Topology) and Chapter 25 (Sheaves on Topological Spaces).

Throughout, k is an algebraically closed field, of any characteristic. Rings are commutative with 1, and $k[x_1, \dots, x_n]$ is the polynomial ring, with its usual addition and multiplication.

53.2 Affine algebraic sets and the Zariski topology

Definition 53.2.1. *Affine n-space* is $\mathbb{A}^n = k^n$. For a subset $T \subseteq A = k[x_1, \dots, x_n]$ put $Z(T) = \{a \in \mathbb{A}^n : f(a) = 0 \text{ for all } f \in T\}$, and for $Y \subseteq \mathbb{A}^n$ put $I(Y) = \{f \in A : f(a) = 0 \text{ for all } a \in Y\}$. The sets $Z(T)$ are the *algebraic sets*.

Proposition 53.2.2. *1. $Z(0) = \mathbb{A}^n$, $Z(1) = \emptyset$, $Z(T) = Z((T))$, and $Z(I) \cup Z(J) = Z(IJ) = Z(I \cap J)$ and $\bigcap_\alpha Z(I_\alpha) = Z(\sum_\alpha I_\alpha)$ for ideals. So the algebraic sets are the closed sets of a topology on $\mathbb{A}^n$, the* Zariski topology.

2. *$I(Y)$ is a radical ideal, $Z(I(Y)) = \overline{Y}$, and $I(Z(J)) = \sqrt{J}$ for every ideal J.*

3. *$Y \mapsto I(Y)$ and $J \mapsto Z(J)$ are mutually inverse inclusion-reversing bijections between closed subsets of $\mathbb{A}^n$ and radical ideals of A. Under them, points correspond to maximal ideals.*

Proof. (1) If $a \notin Z(I) \cup Z(J)$, pick $f \in I$, $g \in J$ with $f(a) \neq 0 \neq g(a)$; then $fg \in IJ$ does not vanish at a. So $Z(IJ) \subseteq Z(I) \cup Z(J) \subseteq Z(I \cap J) \subseteq Z(IJ)$. The rest is immediate. (2) If f^m vanishes on Y, so does f, so $I(Y)$ is radical. $Z(I(Y))$ is a closed set containing Y, and if $Z(J) \supseteq Y$ then $J \subseteq I(Y)$, so $Z(J) \supseteq Z(I(Y))$; hence $Z(I(Y)) = \overline{Y}$. The equality $I(Z(J)) = \sqrt{J}$ is the Nullstellensatz (Theorem 14.7.2(3)), the inclusion $\supseteq$ being clear. (3) follows from (2), and points correspond to maximal ideals by Theorem 14.7.2(1). □

Definition 53.2.3. A topological space Y is *irreducible* if it is nonempty and not the union of two proper closed subsets; equivalently, every nonempty open subset is dense. It is *Noetherian* if every descending chain of closed subsets is eventually constant.

Proposition 53.2.4. *1. A closed set $Y \subseteq \mathbb{A}^n$ is irreducible if and only if $I(Y)$ is prime.*

2. *$\mathbb{A}^n$, and every subspace of it, is Noetherian.*

3. *Every Noetherian space Y is a finite union $Y = Y_1 \cup \cdots \cup Y_r$ of irreducible closed subsets, unique up to order if $Y_i \not\subseteq Y_j$ for $i \neq j$; the Y_i are the* irreducible components.

Proof. (1) If $Y = Y_1 \cup Y_2$ with Y_i proper closed, pick $f_i \in I(Y_i) \setminus I(Y)$ by Proposition 53.2.2(3). Then $f_1 f_2 \in I(Y)$, so $I(Y)$ is not prime. Conversely, if $fg \in I(Y)$, then $Y = (Y \cap Z(f)) \cup (Y \cap Z(g))$, so $Y \subseteq Z(f)$ or $Y \subseteq Z(g)$ by irreducibility. (Also $Y \neq \emptyset$ if and only if $I(Y) \neq A$.) (2) A descending chain of closed sets gives an ascending chain of ideals, which stabilises because A is Noetherian (Theorem 14.5.3). For a subspace, its closed sets are traces of closed sets of $\mathbb{A}^n$. (3) If some closed subset is not such a finite union, choose a minimal one Y', which exists by the Noetherian property. It is not irreducible, so $Y' = Y'_1 \cup Y'_2$ with smaller closed sets, which are finite unions, a contradiction. For uniqueness, if $Y_1 \cup \cdots \cup Y_r = Y'_1 \cup \cdots \cup Y'_s$ are irredundant, then $Y'_1 = \bigcup_i (Y'_1 \cap Y_i)$ forces $Y'_1 \subseteq Y_i$ for some i, and similarly $Y_i \subseteq Y'_j$, so $j = 1$ and $Y'_1 = Y_i$; induct. □

Definition 53.2.5. An *affine variety* is an irreducible closed subset $Y \subseteq \mathbb{A}^n$ with the induced topology. Its *coordinate ring* is $A(Y) = k[x_1, \ldots, x_n]/I(Y)$, a finitely generated k-algebra that is a domain. A *quasi-affine variety* is a nonempty open subset of an affine variety.

Example 53.2.6. 1. $\mathbb{A}^n$ is an affine variety, since $I(\mathbb{A}^n) = 0$ is prime (k is infinite, so a polynomial vanishing everywhere is 0).

2. If f is irreducible, then (f) is prime because $k[x_1, \ldots, x_n]$ is a unique factorisation domain (Theorem 8.6.11), so the *hypersurface* $Z(f)$ is an affine variety with $I(Z(f)) = (f)$.

3. The closed subsets of $\mathbb{A}^1$ are $\mathbb{A}^1$ and the finite sets, since $k[x]$ is a principal ideal domain. So the Zariski topology is not Hausdorff: any two nonempty open sets meet.

4. The Zariski topology on $\mathbb{A}^2$ is not the product topology on $\mathbb{A}^1 \times \mathbb{A}^1$. The closed sets of the product topology are the finite unions of points, horizontal and vertical lines and $\mathbb{A}^2$, whereas the parabola $Z(y - x^2)$ is also Zariski closed.

5. The *twisted cubic* $Y = \{(t, t^2, t^3)\} = Z(y - x^2, z - x^3) \subseteq \mathbb{A}^3$ is an affine variety with $A(Y) \cong k[x, y, z]/(y - x^2, z - x^3) \cong k[t]$, a domain.

53.3 Regular functions and morphisms

Definition 53.3.1. Let Y be a quasi-affine variety in $\mathbb{A}^n$. A function $f : Y \to k$ is *regular at* $P \in Y$ if there are an open neighbourhood $U \ni P$ and polynomials g, h with h nowhere zero on U and $f = g/h$ on U. It is *regular* on an open set if it is regular at each of its points. The regular functions on open subsets form a sheaf of k-algebras $\mathcal{O}_Y$. The *local ring* at P is the stalk $\mathcal{O}_{Y,P}$, and the *function field* $K(Y)$ is the set of regular functions on nonempty open subsets modulo agreement on a nonempty open subset.

Lemma 53.3.2. *A regular function $f : Y \to k = \mathbb{A}^1$ is continuous for the Zariski topologies. If two regular functions on an irreducible Y agree on a nonempty open subset, they are equal.*

Proof. The closed subsets of $\mathbb{A}^1$ are finite sets, so it suffices that $f^{-1}(a)$ is closed. Locally $f = g/h$, and $f^{-1}(a) \cap U = Z(g - ah) \cap U$ is closed in U; being closed is a local property. For the second statement, $\{f = f'\} = (f - f')^{-1}(0)$ is closed and contains a dense open subset. □

Theorem 53.3.3. *Let Y be an affine variety with coordinate ring $A = A(Y)$. For $P \in Y$ let $\mathfrak{m}_P \subseteq A$ be the maximal ideal of functions vanishing at P. Then:*

1. $\mathcal{O}_{Y,P} \cong A_{\mathfrak{m}_P}$;
2. *for $f \in A$, the* basic open set $D(f) = Y \setminus Z(f)$ *satisfies $\mathcal{O}(D(f)) \cong A_f$; in particular $\mathcal{O}(Y) \cong A$;*
3. $K(Y) \cong \mathrm{Frac}(A)$, *and the basic open sets form a basis of the topology.*

Proof. The basic open sets form a basis because $Y \setminus Z(J) = \bigcup_{f \in J} D(f)$. A polynomial vanishing on Y lies in $I(Y)$, so A is a ring of functions on Y. (1) Send $g/h \in A_{\mathfrak{m}_P}$, $h(P) \neq 0$, to the germ of the function g/h on $D(h)$. This is surjective by definition. It is injective: if g/h vanishes near P, then g vanishes on a nonempty open subset of Y, hence on Y by Lemma 53.3.2, so $g = 0$. So all local rings embed compatibly in $\mathrm{Frac}(A)$, and a regular function on an open set U is determined by its germs, so $\mathcal{O}(U) \subseteq \bigcap_{P \in U} A_{\mathfrak{m}_P} \subseteq \mathrm{Frac}(A)$, with equality by the definition of regularity.

(2) The points of $D(f)$ correspond to the maximal ideals of A not containing f, which are the maximal ideals of A_f (Proposition 14.3.2). So we must show $\bigcap_{\mathfrak{m} \not\ni f} A_{\mathfrak{m}} = A_f$ inside $\mathrm{Frac}(A)$. The inclusion $\supseteq$ is clear. Let φ lie in the intersection, and let $J = \{a \in A_f : a\varphi \in A_f\}$, an ideal of A_f. For each maximal ideal $\mathfrak{n}$ of A_f we have $\varphi = g/h$ with $h \notin \mathfrak{n}$, so $h \in J \setminus \mathfrak{n}$. Thus J lies in no maximal ideal, $1 \in J$, and $\varphi \in A_f$. Taking $f = 1$ gives $\mathcal{O}(Y) = A$.

(3) Every nonempty open set contains a nonempty basic open set, and $\mathcal{O}(D(f)) = A_f$ with restriction maps the inclusions of subrings of $\mathrm{Frac}(A)$. So $K(Y) = \bigcup_{f \neq 0} A_f = \mathrm{Frac}(A)$. □

Definition 53.3.4. Let X, Y be varieties, meaning for now quasi-affine varieties (Definition 53.4.4 below extends this to quasi-projective ones). A *morphism* $\varphi \colon X \to Y$ is a continuous map such that for every open $V \subseteq Y$ and every regular f on V, the function $f \circ \varphi$ is regular on $\varphi^{-1}(V)$. Varieties and morphisms form a category, and an isomorphism is a morphism with an inverse morphism. Equivalently, a morphism is a morphism of ringed spaces $(X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ whose map of sheaves is composition of functions (Definition 25.6.1).

Proposition 53.3.5. *Let X be a variety and $Y \subseteq \mathbb{A}^n$ an affine variety. Then $\varphi \mapsto \varphi^*$, $\varphi^*(f) = f \circ \varphi$, is a bijection*

$$\mathrm{Hom}(X, Y) \to \mathrm{Hom}_{k\text{-}alg}(A(Y), \mathcal{O}(X)).$$

Consequently $Y \mapsto A(Y)$ is an anti-equivalence between the category of affine varieties and that of finitely generated k-algebras that are domains.

Proof. Given a k-algebra map $\theta \colon A(Y) \to \mathcal{O}(X)$, let $\xi_i = \theta(\bar{x}_i)$ and define $\psi(P) = (\xi_1(P), \dots, \xi_n(P))$. For $f \in I(Y)$, $f(\psi(P)) = \theta(\bar{f})(P) = 0$, so ψ maps into Y, and $f \circ \psi = \theta(\bar{f})$ for all polynomials f. Then $\psi^{-1}(Y \cap Z(f)) = \theta(\bar{f})^{-1}(0)$ is closed (Lemma 53.3.2), so ψ is continuous. A function that is locally g/h pulls back to $(g \circ \psi)/(h \circ \psi)$, a quotient of regular functions, hence regular, so ψ is a morphism with $\psi^* = \theta$. Conversely, a morphism φ is determined by the functions $x_i \circ \varphi = \varphi^*(\bar{x}_i)$. For the anti-equivalence, use $\mathcal{O}(X) = A(X)$ for affine X (Theorem 53.3.3). Every finitely generated domain $B = k[x_1, \dots, x_n]/\mathfrak{p}$ is $A(Z(\mathfrak{p}))$, since $I(Z(\mathfrak{p})) = \sqrt{\mathfrak{p}} = \mathfrak{p}$. □

Example 53.3.6. 1. $\mathbb{A}^1 \setminus \{0\}$ is isomorphic to the affine variety $Z(xy - 1) \subseteq \mathbb{A}^2$, with coordinate ring $k[x, x^{-1}]$. More generally $D(f) \subseteq Y$ is isomorphic to the affine variety with coordinate ring $A(Y)_f \cong A(Y)[t]/(tf - 1)$.

2. $\mathbb{A}^2 \setminus \{0\}$ is not isomorphic to an affine variety. Its regular functions are $\mathcal{O}(D(x)) \cap \mathcal{O}(D(y)) = k[x,y]_x \cap k[x,y]_y = k[x,y]$ by Theorem 53.3.3. So if it were affine, the inclusion into $\mathbb{A}^2$ would induce an isomorphism of coordinate rings and would therefore be an isomorphism by Proposition 53.3.5, which it is not.

3. The morphism $\mathbb{A}^1 \to Z(y^2 - x^3)$, $t \mapsto (t^2, t^3)$, is a bijective morphism that is not an isomorphism: it corresponds to the inclusion $k[t^2, t^3] \subseteq k[t]$, which is not surjective.

53.4 Projective varieties

Definition 53.4.1. *Projective n-space* is $\mathbb{P}^n = (k^{n+1} \setminus \{0\})/k^\times$, with points written $[a_0 : \cdots : a_n]$. Let $S = k[x_0, \ldots, x_n]$ with its grading by degree, and $S_+ = (x_0, \ldots, x_n)$. For a set T of homogeneous polynomials, $Z(T) \subseteq \mathbb{P}^n$ is the set of points at which all $f \in T$ vanish (this is well defined since $f(\lambda a) = \lambda^{\deg f} f(a)$). For $Y \subseteq \mathbb{P}^n$, $I(Y)$ is the ideal generated by the homogeneous polynomials vanishing on Y, a homogeneous ideal. The sets $Z(\mathfrak{a})$, $\mathfrak{a}$ a homogeneous ideal, are the closed sets of the *Zariski topology* on $\mathbb{P}^n$, as in Proposition 53.2.2(1).

Proposition 53.4.2 (Projective Nullstellensatz)**.** *Let $\mathfrak{a} \subseteq S$ be a homogeneous ideal. Then $Z(\mathfrak{a}) = \emptyset$ if and only if $\sqrt{\mathfrak{a}} \supseteq S_+$, and otherwise $I(Z(\mathfrak{a})) = \sqrt{\mathfrak{a}}$. Hence closed subsets of $\mathbb{P}^n$ correspond bijectively to homogeneous radical ideals not containing S_+, and irreducible closed subsets to homogeneous primes not containing S_+.*

Proof. Let $C \subseteq \mathbb{A}^{n+1}$ be the affine zero set of $\mathfrak{a}$. Since $\mathfrak{a}$ is homogeneous, C is a union of lines through 0 (or empty), and $Z(\mathfrak{a}) = (C \setminus \{0\})/k^\times$. So $Z(\mathfrak{a}) = \emptyset$ if and only if $C \subseteq \{0\}$, if and only if $\sqrt{\mathfrak{a}} \supseteq I(\{0\}) = S_+$ by the affine Nullstellensatz. Otherwise $C \neq \{0\}$ contains 0 if $\mathfrak{a} \neq S$, and a homogeneous f of positive degree vanishes on $Z(\mathfrak{a})$ if and only if it vanishes on C, if and only if $f \in \sqrt{\mathfrak{a}}$. The radical of a homogeneous ideal is homogeneous: if $f^m \in \mathfrak{a}$ and f_d is the top component of f, then f_d^m is the top component of f^m, so it lies in $\mathfrak{a}$ (Lemma 8.10.4), and $f_d \in \sqrt{\mathfrak{a}}$; now induct on the number of components of $f - f_d$. Hence $I(Z(\mathfrak{a})) = \sqrt{\mathfrak{a}}$.

For irreducibility, argue as in Proposition 53.2.4(1) with homogeneous f, g. This suffices because a homogeneous ideal $\mathfrak{p}$ such that $fg \in \mathfrak{p}$ implies $f \in \mathfrak{p}$ or $g \in \mathfrak{p}$ for homogeneous f, g is prime. Indeed, let $fg \in \mathfrak{p}$ with $f, g \notin \mathfrak{p}$. Subtracting from f and g their homogeneous components of high degree that lie in $\mathfrak{p}$ does not change the hypotheses, so we may assume that the top components f_d and g_e are not in $\mathfrak{p}$. But $f_d g_e$ is the top component of fg, so it lies in $\mathfrak{p}$, a contradiction. □

Proposition 53.4.3. *Let $U_i = \mathbb{P}^n \setminus Z(x_i)$. The map $\varphi_i \colon U_i \to \mathbb{A}^n$, $[a] \mapsto (a_j/a_i)_{j \neq i}$, is a homeomorphism for the Zariski topologies. Under it, closed subsets of U_i correspond to closed subsets of $\mathbb{A}^n$ through dehomogenisation $f \mapsto f(x_0, \ldots, 1, \ldots, x_n)$ and homogenisation.*

Proof. φ_i is bijective with inverse $(b_j) \mapsto [b_0 : \cdots : 1 : \cdots : b_n]$. For homogeneous f of degree d, $\varphi_i(Z(f) \cap U_i)$ is the zero set of the dehomogenisation $f(x_0, \ldots, 1, \ldots, x_n)$. Conversely, for $g \in k[y_j : j \neq i]$ of degree d, the homogenisation $x_i^d g(x_0/x_i, \ldots)$ is a homogeneous polynomial whose zero set in U_i is $\varphi_i^{-1}(Z(g))$. So φ_i and its inverse are closed maps. □

Definition 53.4.4. A *projective variety* is an irreducible closed subset $Y \subseteq \mathbb{P}^n$, and its *homogeneous coordinate ring* is $S(Y) = S/I(Y)$, a graded domain. A *quasi-projective variety* is a nonempty open subset of a projective variety. A function $f \colon Y \to k$ on a quasi-projective variety is *regular at P* if near P it is g/h with g, h homogeneous of the same degree and h nowhere zero. This gives the sheaf $\mathcal{O}_Y$, the local rings $\mathcal{O}_{Y,P}$ and the function field $K(Y)$ as before. From now on a *variety* is a quasi-projective variety, and morphisms are defined as in Definition 53.3.4.

Proposition 53.4.5. *Let $Y \subseteq \mathbb{P}^n$ be a quasi-projective variety. Then φ_i restricts to an isomorphism of $Y \cap U_i$ (if nonempty) onto a quasi-affine variety, and onto an affine variety if Y is projective, with coordinate ring the degree-zero part $S(Y)_{(x_i)}$ of $S(Y)_{x_i}$. In particular, every quasi-affine variety is quasi-projective (via $\mathbb{A}^n \cong U_0$), every variety has an open cover by affine varieties, and $K(Y) \cong K(Y \cap U_i)$.*

Proof. By Proposition 53.4.3, φ_i is a homeomorphism onto its image, which is locally closed, and irreducible since $Y \cap U_i$ is a nonempty open subset of an irreducible space. A quotient g/h of homogeneous polynomials of degree d equals the quotient of the dehomogenisations of g and h on U_i, and conversely a quotient of polynomials in the affine coordinates can be homogenised to a quotient of homogeneous polynomials of the same degree. So regular functions correspond. If Y is projective, $\varphi_i(Y \cap U_i)$ is closed, and dehomogenisation identifies its ideal with the degree-zero part of the localised ideal. A quasi-affine variety is covered by basic open sets $D(f)$ of its closure, which are affine (Example 53.3.6(1)). Finally, every nonempty open subset of Y meets U_i in a dense open subset, so the function fields agree. □

Theorem 53.4.6. *If Y is a projective variety, then $\mathcal{O}(Y) = k$. Consequently every morphism from a projective variety to an affine variety is constant.*

Proof. Let $f \in \mathcal{O}(Y)$ and $L = \operatorname{Frac}(S(Y))$. We view $K(Y)$ as the degree-zero part of L, by $g/h \mapsto g/h$. Let Σ be the set of i with $x_i \notin I(Y)$; the other x_i are zero in $S(Y)$. For $i \in \Sigma$, f restricted to $Y \cap U_i$ lies in $S(Y)_{(x_i)}$ by Proposition 53.4.5 and Theorem 53.3.3, so $x_i^{N_i} f \in S(Y)_{N_i}$ for some N_i, as elements of L. Let $N = \sum_{i \in \Sigma} N_i$. The space $S(Y)_N$ is spanned by the monomials of degree N in the x_i, $i \in \Sigma$, and each of them is divisible by $x_i^{N_i}$ for some $i \in \Sigma$. So $S(Y)_N \cdot f \subseteq S(Y)_N$, and by induction $S(Y)_N \cdot f^q \subseteq S(Y)_N$ for all $q \geq 1$. Choose $i \in \Sigma$. Then $x_i^N f^q \in S(Y)$ for all q, so $S(Y)[f] \subseteq x_i^{-N} S(Y)$, a finitely generated $S(Y)$-module. As $S(Y)$ is Noetherian, $S(Y)[f]$ is a finitely generated $S(Y)$-module, and f is integral over $S(Y)$ (Proposition 14.6.2(1)). Taking the degree-zero parts of the coefficients of an integral equation shows that f is integral over $S(Y)_0 = k$. As k is algebraically closed and f lies in a field, $f \in k$. For the second statement, compose with the coordinate functions of the target. □

Example 53.4.7.

1. A *projective hypersurface* is $Z(f) \subseteq \mathbb{P}^n$ with f irreducible and homogeneous. Plane curves are the case $n = 2$.

2. The *d-uple (Veronese) embedding* $v_d \colon \mathbb{P}^n \to \mathbb{P}^N$, $N = \binom{n+d}{n} - 1$, sends a to the vector of all monomials of degree d in a. Its image is closed, cut out by the quadratic relations $z_M z_{M'} = z_{M''} z_{M'''}$ whenever $MM' = M''M'''$, and v_d is an isomorphism onto it [Har77, Chapter I, Exercises 2.12 and 3.4]. On the image, the inverse is given on $\{z_{x_i^d} \neq 0\}$ by $z \mapsto [z_{x_i^{d-1} x_0} : \cdots : z_{x_i^{d-1} x_n}]$. For $n = 1$, $d = 3$ the image is the *twisted cubic* in $\mathbb{P}^3$, the closure of the affine twisted cubic.

3. The *Segre embedding* $\mathbb{P}^m \times \mathbb{P}^n \to \mathbb{P}^{mn+m+n}$, $(a, b) \mapsto (a_i b_j)$, has closed image $Z(z_{ij} z_{kl} - z_{il} z_{kj})$. This image is used to give $\mathbb{P}^m \times \mathbb{P}^n$ the structure of a projective variety. For $m = n = 1$ it is the smooth quadric surface $Z(z_{00} z_{11} - z_{01} z_{10})$ in $\mathbb{P}^3$.

Proposition 53.4.8. *Let $X \subseteq \mathbb{A}^m$ and $Y \subseteq \mathbb{A}^n$ be affine varieties. Then $X \times Y \subseteq \mathbb{A}^{m+n}$ is an affine variety with coordinate ring $A(X) \otimes_k A(Y)$, and it is the product of X and Y in the category of varieties.*

Proof. $X \times Y = Z(I(X) + I(Y))$ is closed. It is irreducible: if $X \times Y = Z_1 \cup Z_2$ with Z_i closed, each fibre $\{x\} \times Y \cong Y$ is irreducible, so it lies in Z_1 or in Z_2. The set $X_i = \{x : \{x\} \times Y \subseteq Z_i\} = \bigcap_{y \in Y} \{x : (x, y) \in Z_i\}$ is closed, and $X = X_1 \cup X_2$, so $X = X_1$ or $X = X_2$. The natural map $A(X) \otimes_k A(Y) \to \{\text{functions on } X \times Y\}$ is injective: write an element as $\sum_i a_i \otimes b_i$ with the b_i linearly independent over k; if it vanishes, then for each x the function $\sum_i a_i(x) b_i$ vanishes

on Y, so all $a_i(x) = 0$, so all $a_i = 0$. Its image is the ring of polynomial functions on $X \times Y$, which is $A(X \times Y)$. The universal property follows from Proposition 53.3.5 and the universal property of $\otimes_k$ for k-algebras (Proposition 8.10.2). □

53.5 Dimension

Definition 53.5.1. The *dimension* of a topological space is the supremum of the lengths r of chains $Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_r$ of irreducible closed subsets. For an irreducible closed subset $Z \subseteq Y$, the *codimension* of Z in Y is the supremum of the lengths of such chains with $Z_0 = Z$. Varieties of dimension 1 and 2 are *curves* and *surfaces*.

Lemma 53.5.2. *Let A be a finitely generated k-algebra that is a domain, of dimension d. Then every maximal chain of primes of A has length d. Consequently* $\operatorname{ht}\mathfrak{p} + \dim A/\mathfrak{p} = d$ *for every prime $\mathfrak{p}$, and every maximal ideal has height d.*

Proof. Induction on $d = \operatorname{trdeg}_k \operatorname{Frac}(A)$ (Theorem 14.9.3). If $d = 0$, A is a domain algebraic over k, hence a field (Proposition 14.6.2(3)), and the only chain is 0. Let $d > 0$ and let $0 = \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_m$ be a maximal chain. Then $m \geq 1$, and no prime lies strictly between 0 and $\mathfrak{p}_1$. By Noether normalisation (Theorem 14.6.5), A is finite over a polynomial ring $B = k[y_1, \ldots, y_d]$. Let $\mathfrak{q} = \mathfrak{p}_1 \cap B$.

The prime $\mathfrak{q}$ is nonzero. Indeed, take $a \in \mathfrak{p}_1 \setminus 0$ and an equation $a^r + b_1 a^{r-1} + \cdots + b_r = 0$ over B with r minimal. Then $b_r \neq 0$, since otherwise we could divide by a, and $b_r \in aA \cap B \subseteq \mathfrak{q}$. If there were a prime $0 \subsetneq \mathfrak{q}' \subsetneq \mathfrak{q}$, then going down for the integral extension $B \subseteq A$ of domains with B normal (Remark 14.6.4; B is a unique factorisation domain, hence normal by Proposition 14.6.2(4)) would give a prime $\mathfrak{p}' \subsetneq \mathfrak{p}_1$ lying over $\mathfrak{q}'$, so $\mathfrak{p}' \neq 0$, which is a contradiction. So $\mathfrak{q}$ is a minimal nonzero prime of the unique factorisation domain B. It contains an irreducible factor f of any of its nonzero elements, and (f) is a nonzero prime, so $\mathfrak{q} = (f)$. If f involves y_i, the images of the other y_j in $B/(f)$ are algebraically independent, since a relation $g(y_j : j \neq i) \in (f)$ is impossible for $g \neq 0$, and y_i is algebraic over them. So $\operatorname{trdeg} \operatorname{Frac}(B/\mathfrak{q}) = d - 1$. As $A/\mathfrak{p}_1$ is integral over $B/\mathfrak{q}$, it also has transcendence degree $d - 1$, and by induction the maximal chain $\mathfrak{p}_1/\mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_m/\mathfrak{p}_1$ of $A/\mathfrak{p}_1$ has length $d - 1$. Hence $m = d$.

For a prime $\mathfrak{p}$, a chain of length $\operatorname{ht}\mathfrak{p}$ ending at $\mathfrak{p}$ and a chain of length $\dim A/\mathfrak{p}$ starting at $\mathfrak{p}$ cannot be refined, since they have maximal length. Their concatenation is therefore a maximal chain of A, of length d. □

Theorem 53.5.3. *Let Y be a variety.*

1. $\dim Y = \operatorname{trdeg}_k K(Y)$*; if Y is affine,* $\dim Y = \dim A(Y)$.
2. $\dim U = \dim Y$ *for every nonempty open $U \subseteq Y$, and* $\dim \mathcal{O}_{Y,P} = \dim Y$ *for every $P \in Y$.*
3. *If $Z \subsetneq Y$ is closed and irreducible, then* $\dim Z < \dim Y$*, and* $\dim Z + \operatorname{codim}(Z, Y) = \dim Y$.

Proof. First let Y be affine with $A = A(Y)$. Irreducible closed subsets of Y correspond to primes of A, reversing inclusions (Propositions 53.2.2 and 53.2.4), so $\dim Y = \dim A = \operatorname{trdeg} \operatorname{Frac}(A) = \operatorname{trdeg} K(Y)$ by Theorem 14.9.3 and Theorem 53.3.3. Also $\dim \mathcal{O}_{Y,P} = \dim A_{\mathfrak{m}_P} = \operatorname{ht}\mathfrak{m}_P = \dim Y$ by Lemma 53.5.2. For U open, $Z \mapsto \overline{Z}$ maps chains of irreducible closed subsets of U injectively to chains in Y, so $\dim U \leq \dim Y$. Conversely, let $P \in U$ and take a chain of primes of length $\dim Y$ contained in $\mathfrak{m}_P$. It gives a chain of irreducible closed subsets of Y all containing P, and their intersections with U form a chain of the same length in U, since a nonempty open subset of an irreducible space is dense.

Statement (3) for affine Y is Lemma 53.5.2, since $\operatorname{codim}(Z, Y) = \operatorname{ht} I(Z)/I(Y)$ and $\operatorname{ht} \geq 1$ for a nonzero prime.

In general, cover Y by affine open subsets Y_i (Proposition 53.4.5); they have $K(Y_i) = K(Y)$, so $\dim Y_i = \operatorname{trdeg} K(Y)$. Given a chain $Z_0 \subsetneq \cdots \subsetneq Z_r$ in Y, choose Y_i meeting Z_0; the sets $Z_j \cap Y_i$ form a chain in Y_i (their closures are the Z_j). So $\dim Y = \max_i \dim Y_i = \operatorname{trdeg} K(Y)$, and the other statements reduce to the affine case in the same way: a proper closed irreducible Z meets some Y_i in a proper subset, since otherwise Z would contain the dense set Y_i. □

Theorem 53.5.4 (Krull's theorem for varieties)**.** *Let Y be an affine variety of dimension r and $f \in A(Y)$ neither zero nor a unit. Then every irreducible component of $Y \cap Z(f)$ has dimension $r - 1$. A variety $Y \subseteq \mathbb{A}^n$ has dimension $n - 1$ if and only if $Y = Z(f)$ for an irreducible polynomial f.*

Proof. The components of $Z(f)$ correspond to the minimal primes over (f) in $A(Y)$. They have height ≤ 1 by Krull's principal ideal theorem (Theorem 14.9.5), hence height 1 as they are nonzero, hence dimension $r - 1$ by Lemma 53.5.2. For the second statement, $Z(f)$ has dimension $n - 1$ by the first. Conversely, if $\dim Y = n - 1$, then $I(Y)$ has height 1. It contains an irreducible factor f of any of its nonzero elements, and $0 \subsetneq (f) \subseteq I(Y)$ with (f) prime, so $I(Y) = (f)$. □

Theorem 53.5.5 (Projective dimension theorem)**.** *Let $Y \subseteq \mathbb{P}^n$ be a projective variety of dimension $r \geq 1$ and f a homogeneous polynomial of positive degree with $f \notin I(Y)$. Then $Y \cap Z(f)$ is nonempty and all its irreducible components have dimension $r - 1$.*

Proof. Let $C(Y) \subseteq \mathbb{A}^{n+1}$ be the affine cone, the affine variety with coordinate ring $S(Y)$. The fraction field of $S(Y)$ is $K(Y)(x_i)$ for any $x_i \notin I(Y)$, with x_i transcendental over $K(Y)$ (elements of $K(Y)$ have degree zero), so $\dim C(Y) = r + 1$ by Theorem 53.5.3. The same holds for every projective variety, in particular for the components of $Y \cap Z(f)$, whose cones are the components of $C(Y) \cap Z(f)$ other than $\{0\}$. The set $C(Y) \cap Z(f)$ contains 0, and f is not a unit in $S(Y)$, so by Theorem 53.5.4 all its components have dimension $r \geq 1$. In particular they are not $\{0\}$, so they are cones over components of $Y \cap Z(f)$ of dimension $r - 1$. □

Corollary 53.5.6. *Any $r \leq \dim Y$ hypersurfaces in $\mathbb{P}^n$ meet a projective variety Y. In particular two curves in $\mathbb{P}^2$ always meet, and $\mathbb{P}^2$ is not isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$, which contains disjoint curves.*

Proof. Apply Theorem 53.5.5 repeatedly to a component of dimension $\geq \dim Y - j$ after j steps. (If some f vanishes on the current component, the dimension does not drop.) The curves $\mathbb{P}^1 \times \{0\}$ and $\mathbb{P}^1 \times \{\infty\}$ are disjoint. □

53.6 Tangent spaces and smoothness

Definition 53.6.1. Let P be a point of a variety Y, with local ring $(\mathcal{O}_{Y,P}, \mathfrak{m}_P)$ and residue field k. The *Zariski tangent space* is $T_P Y = (\mathfrak{m}_P/\mathfrak{m}_P^2)^\vee$. The variety is *nonsingular* (or *smooth*) at P if $\mathcal{O}_{Y,P}$ is a regular local ring (Definition 14.11.1), that is, if $\dim_k T_P Y = \dim Y$, and *singular* at P otherwise. Y is nonsingular if it is nonsingular at every point.

Lemma 53.6.2. *For every Noetherian local ring $(R, \mathfrak{m}, k)$, $\dim R \leq \dim_k \mathfrak{m}/\mathfrak{m}^2$. Hence $\dim T_P Y \geq \dim Y$.*

Proof. By Nakayama's lemma (Corollary 14.4.3), $\mathfrak{m}$ is generated by $e = \dim \mathfrak{m}/\mathfrak{m}^2$ elements, and $\mathfrak{m}$ is minimal over them, so $\dim R = \operatorname{ht} \mathfrak{m} \leq e$ by Theorem 14.9.5. The second statement uses $\dim \mathcal{O}_{Y,P} = \dim Y$. □

Theorem 53.6.3 (Jacobian criterion)**.** *Let $Y \subseteq \mathbb{A}^n$ be an affine variety of dimension r with $I(Y) = (f_1, \dots, f_t)$, and $P \in Y$. Then* $\dim T_P Y = n - \operatorname{rank} J(P)$*, where $J = (\partial f_i/\partial x_j)$ is the Jacobian matrix. So Y is nonsingular at P if and only if* $\operatorname{rank} J(P) = n - r$.

Proof. Let $M = (x_1 - P_1, \dots, x_n - P_n) \subseteq k[x]$. The linear map $\theta\colon M \to k^n$, $g \mapsto (\partial g/\partial x_j(P))_j$, is onto with kernel M^2 (expand g in powers of $x - P$), so $M/M^2 \cong k^n$. The maximal ideal $\mathfrak{m}$ of $A(Y)$ at P satisfies $\mathfrak{m}/\mathfrak{m}^2 \cong M/(M^2 + I(Y))$, and localising at $\mathfrak{m}$ does not change $\mathfrak{m}/\mathfrak{m}^2$. The image of $I(Y)$ in $M/M^2 \cong k^n$ is spanned by the $\theta(f_i)$, the rows of $J(P)$, because $\theta(gf_i) = g(P)\theta(f_i)$ as $f_i(P) = 0$. So $\dim \mathfrak{m}/\mathfrak{m}^2 = n - \operatorname{rank} J(P)$. □

Example 53.6.4.

1. The cusp $Z(y^2 - x^3)$ and the node $Z(y^2 - x^2(x+1))$ (characteristic $\neq 2$) are singular exactly at the origin, where the Jacobian $(\partial f/\partial x, \partial f/\partial y)$ vanishes.
2. A hypersurface $Z(f) \subseteq \mathbb{A}^n$ is singular exactly at the common zeros of f and all $\partial f/\partial x_j$.
3. $\mathbb{A}^n$, $\mathbb{P}^n$ and the smooth quadric $Z(xw - yz) \subseteq \mathbb{P}^3$ are nonsingular.

53.7 Rational maps and birational equivalence

Lemma 53.7.1. *Two morphisms $\varphi, \psi\colon X \to Y$ of varieties that agree on a nonempty open subset are equal.*

Proof. Embed Y in some $\mathbb{P}^n$. The set $E = \{x : \varphi(x) = \psi(x)\}$ is closed. Indeed, near a point of X, φ and ψ map into standard open sets U_a and U_b, so on a neighbourhood V they are given by regular functions $\varphi = [f_0 : \cdots : f_n]$ with $f_a = 1$ and $\psi = [g_0 : \cdots : g_n]$ with $g_b = 1$. On V, E is the common zero set of the regular functions $f_i g_j - f_j g_i$. The closed set E contains a dense open subset of the irreducible X, so $E = X$. □

Definition 53.7.2. A *rational map* $\varphi\colon X \dashrightarrow Y$ is an equivalence class of pairs (U, φ_U), where $U \subseteq X$ is nonempty open and $\varphi_U\colon U \to Y$ is a morphism, two pairs being equivalent if the morphisms agree on the intersection (which is possible by Lemma 53.7.1). It is *dominant* if $\varphi_U(U)$ is dense. A *birational map* is a dominant rational map with a dominant rational inverse, and X, Y are *birational* if there is one. A variety birational to $\mathbb{P}^n$ is *rational*.

Theorem 53.7.3. *The functor $X \mapsto K(X)$ is an anti-equivalence between the category of varieties with dominant rational maps and the category of finitely generated field extensions of k. In particular X and Y are birational if and only if $K(X) \cong K(Y)$ over k, if and only if they have isomorphic nonempty open subsets.*

Proof. A dominant φ defined on U pulls back regular functions on nonempty open subsets of Y to regular functions on nonempty open subsets of U, since preimages of dense open sets under a dominant map are nonempty. This gives $\varphi^*\colon K(Y) \to K(X)$. Conversely, let $\theta\colon K(Y) \to K(X)$ be a k-embedding. Choose an affine open $V \subseteq Y$ with $A(V) = k[y_1, \dots, y_n]$. The elements $\theta(y_i)$ are regular on some nonempty open $U \subseteq X$, and θ restricts to $A(V) \to \mathcal{O}(U)$. By Proposition 53.3.5 this is φ^* for a morphism $\varphi\colon U \to V$. It is dominant: if its image lay in a proper closed subset $Z(g)$ with $0 \neq g \in A(V)$, then $\theta(g) = 0$, contradicting injectivity. These constructions are inverse to each other by Lemma 53.7.1 and Theorem 53.3.3. Every finitely generated extension $K = k(y_1, \dots, y_n)$ is $K(V)$ for the affine variety with coordinate ring $k[y_1, \dots, y_n] \subseteq K$ (Proposition 53.3.5).

If $\varphi\colon X \dashrightarrow Y$ and $\psi\colon Y \dashrightarrow X$ are inverse birational maps, defined on U and V, then $\psi \circ \varphi = \mathrm{id}$ on $\varphi^{-1}(V)$ and $\varphi \circ \psi = \mathrm{id}$ on $\psi^{-1}(U)$. So φ restricts to an isomorphism $\varphi^{-1}(\psi^{-1}(U)) \to \psi^{-1}(U)$. The converse is clear. □

Proposition 53.7.4. *Every variety of dimension r is birational to a hypersurface in $\mathbb{A}^{r+1}$.*

Proof. The finitely generated extension $K(Y)/k$ of transcendence degree r is separably generated: there are $x_1, \dots, x_r$, algebraically independent, such that $K(Y)$ is a finite separable extension of $k(x_1, \dots, x_r)$. In characteristic 0 this holds for every transcendence basis. In characteristic p it holds because k is perfect [Mat86, Section 26]. By the primitive element theorem (Theorem 9.6.3), $K(Y) = k(x_1, \dots, x_r, y)$ with y separable algebraic over $k(x)$. Clearing denominators in the minimal polynomial of y gives an irreducible $f \in k[x_1, \dots, x_r, t]$ with $f(x, y) = 0$, and $K(Z(f)) \cong K(Y)$. Apply Theorem 53.7.3. □

Theorem 53.7.5. *The singular points of a variety Y form a proper closed subset.*

Proof. Closedness is local, so let $Y \subseteq \mathbb{A}^n$ be affine of dimension r with $I(Y) = (f_1, \dots, f_t)$. By Lemma 53.6.2 and Theorem 53.6.3, $\operatorname{rank} J(P) \le n - r$ everywhere, and the singular set is the common zero set of the $(n-r)$-minors of J, which is closed. It is proper: by Proposition 53.7.4 and Theorem 53.7.3, Y has an open subset isomorphic to an open subset of a hypersurface $Z(f) \subseteq \mathbb{A}^{r+1}$ with f irreducible, so it suffices to find a nonsingular point on $Z(f)$. The singular points are $Z(f, \partial f/\partial x_1, \dots)$ (Example 53.6.4). If they were all of $Z(f)$, then each $\partial f/\partial x_j$ would lie in $I(Z(f)) = (f)$, hence vanish, by degree. In characteristic 0 this makes f constant. In characteristic p it makes f a polynomial in $x_1^p, \dots, x_{r+1}^p$, hence a p-th power since k is perfect, contradicting irreducibility. So the nonsingular points of $Z(f)$ form a nonempty open set, which meets the open subset isomorphic to an open subset of Y, because $Z(f)$ is irreducible. Isomorphic open subsets have isomorphic local rings. □

Example 53.7.6.

1. $\mathbb{P}^n$ and $\mathbb{A}^n$ are birational, and so are $\mathbb{P}^1 \times \mathbb{P}^1$ and $\mathbb{P}^2$: all three have function field $k(x_1, x_2)$ for $n = 2$. By Corollary 53.5.6, $\mathbb{P}^1 \times \mathbb{P}^1$ and $\mathbb{P}^2$ are not isomorphic.
2. The cuspidal cubic $Z(y^2 - x^3)$ is rational: $t = y/x$ gives $K = k(t)$. It is singular, so it is not isomorphic to $\mathbb{P}^1$ or $\mathbb{A}^1$.
3. The *Cremona transformation* $\mathbb{P}^2 \dashrightarrow \mathbb{P}^2$, $[x : y : z] \mapsto [yz : xz : xy]$, is birational and equal to its own inverse, but it is not defined at the three coordinate points.
4. A nonsingular plane cubic is not rational. This follows from the genus (Chapter 59): rational nonsingular projective curves are isomorphic to $\mathbb{P}^1$, which has genus 0, while plane cubics have genus 1.

53.8 Blow-ups

Definition 53.8.1. The *blow-up* of $\mathbb{A}^n$ at 0 is the closed subset $\mathrm{Bl}_0\mathbb{A}^n = \{(x, y) \in \mathbb{A}^n \times \mathbb{P}^{n-1} : x_i y_j = x_j y_i \text{ for all } i, j\}$ with the projection $\pi\colon \mathrm{Bl}_0\mathbb{A}^n \to \mathbb{A}^n$. For a closed subvariety $Y \subseteq \mathbb{A}^n$ through 0, the *blow-up* (or *strict transform*) $\tilde{Y}$ is the closure of $\pi^{-1}(Y \setminus \{0\})$ in $\mathrm{Bl}_0\mathbb{A}^n$.

Proposition 53.8.2. *$\mathrm{Bl}_0\mathbb{A}^n$ is a nonsingular variety of dimension n. The map π is an isomorphism over $\mathbb{A}^n \setminus \{0\}$, and $E = \pi^{-1}(0) = \{0\} \times \mathbb{P}^{n-1}$ parametrises the lines through 0. For Y as above, $\tilde{Y} \to Y$ is birational, and $\tilde{Y} \cap E$ consists of limits of secant directions of Y at 0.*

Proof. On $\{y_i \neq 0\}$ put $y_i = 1$. The equations become $x_j = x_i y_j$, so this open set is the graph of a morphism $\mathbb{A}^n \to \mathbb{A}^{n-1}$, which is isomorphic to $\mathbb{A}^n$ with coordinates $(x_i, y_j)_{j \neq i}$. These charts are irreducible of dimension n and nonsingular, and pairwise they meet in nonempty open sets, so $\mathrm{Bl}_0\mathbb{A}^n$ is irreducible, of dimension n and nonsingular. Over $x \neq 0$, the equations force $y = [x]$, so $x \mapsto (x, [x])$ is an inverse of π there. Over 0 the equations are empty, so $E = \{0\} \times \mathbb{P}^{n-1}$. For Y, the map $\tilde{Y} \to Y$ is an isomorphism over $Y \setminus \{0\}$, which is dense in $\tilde{Y}$ by definition, and a point $(0, [v]) \in \tilde{Y}$ is a limit of points $(x, [x])$ with $x \in Y \setminus \{0\}$. □

Example 53.8.3. Let $\operatorname{char} k \neq 2$. For the node $Y = Z(y^2 - x^2(x+1))$, in the chart $y = xt$ the preimage is $Z(x^2(t^2 - x - 1))$, the union of $E = Z(x)$ (with multiplicity 2) and the strict transform $\tilde{Y} = Z(t^2 - x - 1)$. The latter is nonsingular and meets E at the two points $t = \pm 1$, the two tangent directions of the branches: blowing up separates the branches. For the cusp $Z(y^2 - x^3)$, the strict transform is $Z(t^2 - x)$, nonsingular and meeting E at one point. In both cases one blow-up *resolves* the singularity.

Remark 53.8.4. Blow-ups at points glue to blow-ups of any variety at a nonsingular point, and more generally along closed subvarieties. For $k = \mathbb{C}$ this is the construction of Definition 44.6.4. Hironaka proved that in characteristic 0 every variety is birational to a nonsingular projective variety, and in fact that singularities can be resolved by a sequence of blow-ups along nonsingular centres [Hir64]. In positive characteristic this is known only in low dimension.

53.9 Exercises

Exercise 53.9.1. Show that $\mathbb{A}^1$ minus finitely many points is affine, and determine its coordinate ring.

Solution. If the points are $a_1, \dots, a_m$, the set is $D(f)$ for $f = \prod_i (x - a_i)$, which is affine with coordinate ring $k[x]_f$ by Example 53.3.6(1). □

Exercise 53.9.2. Let $\operatorname{char} k \neq 2$. Show that every irreducible conic $Z(q) \subseteq \mathbb{P}^2$ (q a quadratic form) is isomorphic to $\mathbb{P}^1$.

Solution. Diagonalise q by a linear change of coordinates. It has rank 3, since a quadratic form of rank ≤ 2 over an algebraically closed field factors into linear forms. Over k it is then equivalent to $xz - y^2$. The morphism $\mathbb{P}^1 \to Z(xz - y^2)$, $[s : t] \mapsto [s^2 : st : t^2]$, is the Veronese map v_2, an isomorphism onto its image (Example 53.4.7). Its inverse is given by $[x : y]$ on $\{x \neq 0\}$ and by $[y : z]$ on $\{z \neq 0\}$. □

Exercise 53.9.3. Let $\operatorname{char} k \neq 2$. Show that the cone $Z(x^2 + y^2 - z^2) \subseteq \mathbb{A}^3$ is singular only at the origin, and compute its tangent space there.

Solution. The Jacobian is $(2x, 2y, -2z)$, which vanishes only at 0, and $\dim = 2$, so the cone is nonsingular away from 0 (Theorem 53.6.3). At 0 the Jacobian is zero, so $T_0 Y = k^3$ has dimension $3 > 2$. □

Exercise 53.9.4. Show that n hypersurfaces in $\mathbb{P}^n$ have a common point, and that the Segre quadric $Z(xw - yz) \subseteq \mathbb{P}^3$ contains two families of lines such that two lines of the same family are disjoint and two lines of different families meet in one point.

Solution. The first statement is Corollary 53.5.6 with $Y = \mathbb{P}^n$. The quadric is the image of the Segre map $\mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^3$, $([a_0 : a_1], [b_0 : b_1]) \mapsto [a_0 b_0 : a_0 b_1 : a_1 b_0 : a_1 b_1]$, which is a bijection onto it. The images of $\{a\} \times \mathbb{P}^1$ and $\mathbb{P}^1 \times \{b\}$ are lines. Two lines of the same family are images of disjoint sets, and $\{a\} \times \mathbb{P}^1$ meets $\mathbb{P}^1 \times \{b\}$ exactly in (a, b). □

Exercise 53.9.5. Show that the projective twisted cubic $C = v_3(\mathbb{P}^1) \subseteq \mathbb{P}^3$ is cut out by the three quadrics $xz - y^2$, $yw - z^2$, $xw - yz$, but not by any two of them.

Solution. The three quadrics say that the matrix $\left(\begin{smallmatrix} x & y & z \\ y & z & w \end{smallmatrix}\right)$ has rank ≤ 1. Such a point with $x \neq 0$ is $[1 : t : t^2 : t^3]$ with $t = y/x$, and with $x = 0$ it is $[0 : 0 : 0 : 1]$, so the common zeros are exactly C. The quadrics $xz - y^2$ and $yw - z^2$ also vanish on the line $y = z = 0$, which is not contained in C. For the other pairs, $\{xz = y^2, xw = yz\}$ contains the line $x = y = 0$, and $\{yw = z^2, xw = yz\}$ contains the line $z = w = 0$. □

Chapter 54

Schemes

54.1 Introduction

A scheme is a locally ringed space that is locally the spectrum of a ring. Schemes extend the varieties of Chapter 53 in three directions that Hodge theory needs. They work over arbitrary base rings, so that a variety defined over $\mathbb{Q}$, or a family over a base, is a single object; this is how the conjugates X^{σ} of the theory of absolute Hodge classes are defined (Chapter 75). They allow nilpotent functions, which carry infinitesimal information such as tangent vectors and deformations. And they have non-closed points, which make the generic behaviour of families visible. References: [Har77, Chapter II], [The].

Prerequisites. Chapter 53 (Algebraic Varieties) and Chapter 14 (Commutative Algebra).

Rings are commutative with 1. For a ring A we use the notation $V(I)$, $D(f)$ of Definition 14.2.1. For a prime $\mathfrak{p}$, $k(\mathfrak{p}) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$ is its residue field.

54.2 The spectrum of a ring

Proposition 54.2.1. *Let A be a ring and $X = \operatorname{Spec} A$.*

1. *The sets $V(I)$ are the closed sets of a topology, the* Zariski topology*: $V(IJ) = V(I \cap J) = V(I) \cup V(J)$ and $V(\sum I_\alpha) = \bigcap V(I_\alpha)$. Moreover $V(I) \subseteq V(J)$ if and only if $\sqrt{J} \subseteq \sqrt{I}$.*
2. *The sets $D(f)$ form a basis of the topology, $D(f) \cap D(g) = D(fg)$, and $D(f) \subseteq D(g)$ if and only if $f \in \sqrt{(g)}$.*
3. *The closure of $\{\mathfrak{p}\}$ is $V(\mathfrak{p})$. So the closed points are the maximal ideals, and $\mathfrak{p} \mapsto V(\mathfrak{p})$ is a bijection from X to the set of irreducible closed subsets; $\mathfrak{p}$ is the unique* generic point *of $V(\mathfrak{p})$.*
4. *X and every $D(f)$ are quasi-compact.*
5. *A ring map $\varphi\colon A \to B$ induces a continuous map $\operatorname{Spec} B \to \operatorname{Spec} A$, $\mathfrak{q} \mapsto \varphi^{-1}(\mathfrak{q})$, and the preimage of $D(f)$ is $D(\varphi(f))$.*

Proof. (1) A prime contains IJ if and only if it contains I or J, and it contains $\sum I_\alpha$ if and only if it contains every I_α. The last statement holds because $\sqrt{I}$ is the intersection of the primes containing I (Proposition 14.2.2). (2) $X \setminus V(I) = \bigcup_{f \in I} D(f)$, and $\mathfrak{p}$ omits fg if and only if it omits both. $D(f) \subseteq D(g)$ means $V(g) \subseteq V(f)$, that is, $f \in \sqrt{(g)}$ by (1). (3) The smallest $V(I)$ containing $\mathfrak{p}$ is $V(\mathfrak{p})$. If $V(I)$ is irreducible, then $\sqrt{I}$ is prime: if $fg \in \sqrt{I}$, then $V(I) = (V(I) \cap V(f)) \cup (V(I) \cap V(g))$, so $V(I) \subseteq V(f)$, say, and $f \in \sqrt{I}$. Then $V(I) = V(\sqrt{I})$ is the closure of the point $\sqrt{I}$, which is unique because $V(\mathfrak{p}) = V(\mathfrak{q})$ forces $\mathfrak{p} = \mathfrak{q}$. (4) If $X = \bigcup_\alpha D(f_\alpha)$,

then no prime contains all f_α, so they generate the unit ideal, and $1 = \sum_{i=1}^{r} a_i f_{\alpha_i}$ involves finitely many. For $D(f) \cong \operatorname{Spec} A_f$ (by Proposition 14.3.2(3)) the same argument applies. (5) $\varphi^{-1}(\mathfrak{q})$ omits f exactly when $\mathfrak{q}$ omits $\varphi(f)$. □

Definition 54.2.2. The *structure sheaf* $\mathcal{O} = \mathcal{O}_{\operatorname{Spec} A}$ assigns to an open $U \subseteq \operatorname{Spec} A$ the ring of functions $s : U \to \bigsqcup_{\mathfrak{p} \in U} A_\mathfrak{p}$ with $s(\mathfrak{p}) \in A_\mathfrak{p}$ that are *locally quotients*: every point of U has a neighbourhood $V \subseteq U$ and $a, f \in A$ with $f \notin \mathfrak{p}$ and $s(\mathfrak{p}) = a/f$ in $A_\mathfrak{p}$ for all $\mathfrak{p} \in V$. The locally ringed space $(\operatorname{Spec} A, \mathcal{O})$ is the *affine scheme* of A.

Theorem 54.2.3. *Let A be a ring.*

1. *For every $\mathfrak{p}$, the stalk $\mathcal{O}_\mathfrak{p}$ is isomorphic to $A_\mathfrak{p}$; in particular* Spec A *is a locally ringed space.*
2. *For every $f \in A$, $\mathcal{O}(D(f)) \cong A_f$; in particular $\Gamma(\operatorname{Spec} A, \mathcal{O}) \cong A$.*

Proof. (1) Evaluation at $\mathfrak{p}$ gives $\mathcal{O}_\mathfrak{p} \to A_\mathfrak{p}$. It is surjective, since a/f with $f \notin \mathfrak{p}$ defines a section on $D(f)$. It is injective: if s, t agree at $\mathfrak{p}$ and are $a/f, b/g$ near $\mathfrak{p}$, then $h(ag - bf) = 0$ for some $h \notin \mathfrak{p}$, so $a/f = b/g$ in every $A_\mathfrak{q}$ with $\mathfrak{q} \in D(fgh)$, a neighbourhood of $\mathfrak{p}$.

(2) Define $\psi : A_f \to \mathcal{O}(D(f))$ by $a/f^n \mapsto (\mathfrak{p} \mapsto a/f^n)$. *Injective:* if $a/f^n = 0$ in every $A_\mathfrak{p}$, $\mathfrak{p} \in D(f)$, then the annihilator of a is contained in no prime of $D(f)$. So $V(\operatorname{Ann}(a)) \subseteq V(f)$, and $f \in \sqrt{\operatorname{Ann}(a)}$, that is, $f^m a = 0$ and $a/f^n = 0$ in A_f. *Surjective:* let $s \in \mathcal{O}(D(f))$. Cover $D(f)$ by open sets on which $s = a_i/g_i$. Shrinking them to basic open sets $D(h_i) \subseteq D(g_i)$, and writing $h_i^m = c g_i$, so that $a_i/g_i = c a_i/h_i^m$, we may assume $s = a_i/h_i$ on $D(h_i)$. By (4) of Proposition 54.2.1, finitely many $D(h_1), \dots, D(h_r)$ suffice. On $D(h_i h_j)$ both a_i/h_i and a_j/h_j represent s. By the injectivity just proved, applied to $D(h_i h_j)$, we get $(h_i h_j)^N (a_i h_j - a_j h_i) = 0$, and we may take one N for all pairs. Replacing a_i by $a_i h_i^N$ and h_i by h_i^{N+1} we get $a_i h_j = a_j h_i$. Since $D(f) \subseteq \bigcup D(h_i)$, $f^n = \sum_i b_i h_i$ for some n and b_i. Let $a = \sum_i b_i a_i$. Then $h_j a = \sum_i b_i a_i h_j = \sum_i b_i h_i a_j = f^n a_j$, so $a/f^n = a_j/h_j$ on $D(h_j)$ for every j, and $s = \psi(a/f^n)$. □

Theorem 54.2.4. *For every locally ringed space X and ring A, the map*

$$\operatorname{Hom}_{\mathrm{LRS}}(X, \operatorname{Spec} A) \to \operatorname{Hom}_{\mathrm{Ring}}(A, \Gamma(X, \mathcal{O}_X)), \qquad (f, f^\#) \mapsto f^\#_{\operatorname{Spec} A},$$

is bijective. In particular $A \mapsto \operatorname{Spec} A$ is an anti-equivalence between rings and affine schemes, and a ring map φ induces the morphism Spec φ *whose map on stalks is the localisation $A_{\varphi^{-1}(\mathfrak{q})} \to B_\mathfrak{q}$.*

Proof. Given $\psi : A \to \Gamma(X, \mathcal{O}_X)$, define $f(x) = \psi_x^{-1}(\mathfrak{m}_x)$, where $\psi_x : A \to \mathcal{O}_{X,x}$. This is a prime. Then $f^{-1}(D(a)) = X_{\psi(a)} = \{x : \psi(a)_x \notin \mathfrak{m}_x\}$, which is open, since a germ that is a unit is a unit nearby. So f is continuous. On $X_{\psi(a)}$ the section $\psi(a)$ is invertible, so ψ induces $A_a \to \mathcal{O}_X(X_{\psi(a)})$. By Theorem 54.2.3 these maps define a morphism of sheaves $\mathcal{O}_{\operatorname{Spec} A} \to f_* \mathcal{O}_X$ on the basis of basic open sets, hence a morphism of sheaves. On the stalk at x it is the map $A_{f(x)} \to \mathcal{O}_{X,x}$ induced by ψ_x, which is local by the definition of $f(x)$. Conversely, let $(f, f^\#)$ be a morphism with global map ψ. Since $f^\#_x : A_{f(x)} \to \mathcal{O}_{X,x}$ is local and compatible with ψ_x, we get $f(x) A_{f(x)} = (f^\#_x)^{-1}(\mathfrak{m}_x)$, so $f(x) = \psi_x^{-1}(\mathfrak{m}_x)$. On $D(a)$ the map $f^\#$ is determined by ψ through the universal property of A_a. So the two constructions are inverse. For $X = \operatorname{Spec} B$, $\Gamma(X, \mathcal{O}_X) = B$ by Theorem 54.2.3. □

54.3 Schemes and their properties

Definition 54.3.1. A *scheme* is a locally ringed space $(X, \mathcal{O}_X)$ in which every point has an open neighbourhood U such that $(U, \mathcal{O}_X|_U)$ is isomorphic to an affine scheme. A *morphism* of schemes is a morphism of locally ringed spaces. For a scheme S, a scheme *over* S is a

morphism $X \to S$, and morphisms over S commute with the structure maps. We write $X(T) = \mathrm{Hom}_S(T, X)$ for the *T-valued points*, and $X(R) = X(\operatorname{Spec} R)$. The residue field at x is $k(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$.

Proposition 54.3.2. *Let X_i be schemes, $U_{ij} \subseteq X_i$ open subschemes with $U_{ii} = X_i$, and $\varphi_{ij} \colon U_{ij} \to U_{ji}$ isomorphisms such that $\varphi_{ij}(U_{ij} \cap U_{ik}) = U_{ji} \cap U_{jk}$ and $\varphi_{ik} = \varphi_{jk} \circ \varphi_{ij}$ on $U_{ij} \cap U_{ik}$. Then there is a scheme X, unique up to unique isomorphism, with open subschemes X_i glued along the φ_{ij}. Morphisms defined on the X_i that agree on overlaps glue to a morphism $X \to Y$.*

Proof. Glue the topological spaces along the homeomorphisms φ_{ij}, then glue the sheaves, which is possible by the cocycle condition. The result is locally ringed and locally affine because each X_i is. Morphisms of locally ringed spaces glue because continuous maps and sheaf maps do (Proposition 25.6.4). □

Example 54.3.3.

1. $\operatorname{Spec} k$ for a field k is a point. $\operatorname{Spec} \mathbb{Z}$ has closed points (p) with residue fields $\mathbb{F}_p$, and a generic point (0) with residue field $\mathbb{Q}$, dense in $\operatorname{Spec} \mathbb{Z}$.

2. *Affine space* over A is $\mathbb{A}^n_A = \operatorname{Spec} A[x_1, \dots, x_n]$. For $A = k$ algebraically closed, its closed points are the points of k^n (Theorem 14.7.2), and it has one further point for each irreducible closed subset of dimension > 0, namely its generic point.

3. *Projective space* $\mathbb{P}^n_A$ is glued from $U_i = \operatorname{Spec} A[x_0/x_i, \dots, x_n/x_i]$ along $U_i \supseteq D(x_j/x_i) \cong D(x_i/x_j) \subseteq U_j$. It is also $\operatorname{Proj} A[x_0, \dots, x_n]$ (Section 54.6).

4. The *affine line with doubled origin* is glued from two copies of $\mathbb{A}^1_k$ along the identity of $\mathbb{A}^1_k \setminus \{0\}$. It is a scheme that is not isomorphic to any variety (Example 54.5.3).

5. $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ is a single point with a non-reduced local ring. A morphism $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2) \to X$ over k is a k-point x of X together with a tangent vector in $(\mathfrak{m}_x/\mathfrak{m}_x^2)^\vee$ (Exercise 54.10.2).

Definition 54.3.4. A scheme X is *reduced* if all local rings are reduced, *irreducible* if its space is, *integral* if every $\mathcal{O}_X(U)$, $U \neq \emptyset$, is a domain, *locally Noetherian* if it has an affine open cover by spectra of Noetherian rings, and *Noetherian* if moreover it is quasi-compact. A morphism $f \colon X \to Y$ is *locally of finite type* if there are affine open covers $Y = \bigcup \operatorname{Spec} B_i$ and $f^{-1}(\operatorname{Spec} B_i) = \bigcup_j \operatorname{Spec} A_{ij}$ with each A_{ij} a finitely generated B_i-algebra, *of finite type* if moreover finitely many A_{ij} suffice for each i, and *finite* if $f^{-1}(\operatorname{Spec} B_i) = \operatorname{Spec} A_i$ with A_i a finite B_i-module. An *open immersion* is an isomorphism onto an open subscheme. A *closed immersion* is a morphism that is a homeomorphism onto a closed subset and such that $f^\# \colon \mathcal{O}_Y \to f_*\mathcal{O}_X$ is surjective. A *closed subscheme* is an equivalence class of closed immersions.

Proposition 54.3.5. *A scheme is integral if and only if it is reduced and irreducible. An integral scheme X has a unique generic point η, and its* function field *is $K(X) = \mathcal{O}_{X,\eta}$, which is the fraction field of A for every nonempty affine open $\operatorname{Spec} A \subseteq X$.*

Proof. If X is integral, it is reduced. If it were reducible, there would be disjoint nonempty open sets U_1, U_2, and $\mathcal{O}(U_1 \sqcup U_2) = \mathcal{O}(U_1) \times \mathcal{O}(U_2)$ would not be a domain. Conversely, let X be reduced and irreducible, and let $U = \operatorname{Spec} A$ be a nonempty affine open. Then A is reduced with irreducible spectrum, so its nilradical 0 is prime (Proposition 54.2.1(3)), and A is a domain. For general U and $fg = 0$ in $\mathcal{O}(U)$, U is the union of the closed sets where the germs of f and of g lie in the maximal ideal. On each affine open these are $V(f)$ and $V(g)$. By irreducibility one of them, say $V(f)$, is all of U, so f is nilpotent on each affine open, hence 0. The generic point of X is the generic point of each nonempty open subset, which for $\operatorname{Spec} A$ is the prime (0), with local ring $A_{(0)} = \mathrm{Frac}(A)$. □

Proposition 54.3.6. *Let X be a scheme and K a field. Morphisms $\operatorname{Spec} K \to X$ correspond bijectively to pairs of a point $x \in X$ and a field homomorphism $k(x) \to K$. If X is of finite type over an algebraically closed field k, then the closed points of X are exactly the points with $k(x) = k$, and $X(k)$ is the set of closed points.*

Proof. A morphism from the one-point space $\operatorname{Spec} K$ has image a point x, and the stalk map $\mathcal{O}_{X,x} \to K$ is local, so it kills $\mathfrak{m}_x$ and factors through $k(x)$. Conversely such data define a morphism, by Theorem 54.2.4 applied on an affine neighbourhood. For X of finite type over k, a point x in an affine open $\operatorname{Spec} A$ is closed in $\operatorname{Spec} A$ if and only if its prime is maximal, if and only if $k(x) = k$, by Zariski's lemma (Theorem 14.7.1) and algebraic closedness. If $k(x) = k$, then x is closed in every affine open containing it, and its closure does not meet any open set not containing x, since open sets are stable under generisation. So x is closed in X. Conversely, a closed point of X is closed in an affine open containing it. □

54.4 Fibre products

Theorem 54.4.1. *Fibre products exist in the category of schemes. For affine schemes,* $\operatorname{Spec} A \times_{\operatorname{Spec} R} \operatorname{Spec} B = \operatorname{Spec}(A \otimes_R B)$.

Sketch of proof. For affine schemes this follows from Theorem 54.2.4 and the universal property of $A \otimes_R B$ (Proposition 8.10.2): a morphism $T \to \operatorname{Spec}(A \otimes_R B)$ is a ring map $A \otimes_R B \to \Gamma(T, \mathcal{O}_T)$, that is, a compatible pair of ring maps from A and B. In general one glues. If $X \times_S Y$ exists and $U \subseteq X$ is open, then the preimage of U in $X \times_S Y$ is $U \times_S Y$. Hence, covering S, then X and Y by affine opens, the affine fibre products glue along these canonical open subschemes by Proposition 54.3.2. The omitted verifications are the cocycle conditions, which follow from the uniqueness of fibre products, and the universal property of the glued scheme, which is local [Har77, Theorem II.3.3]. □

Definition 54.4.2. For a morphism $f : X \to Y$ and $y \in Y$, the *fibre* is $X_y = X \times_Y \operatorname{Spec} k(y)$, a scheme over the residue field $k(y)$ whose underlying space is homeomorphic to $f^{-1}(y)$ [Har77, Exercise II.3.10]. For $Y' \to Y$, the *base change* of X is $X_{Y'} = X \times_Y Y'$. A property of morphisms is *stable under base change* if $X_{Y'} \to Y'$ has it whenever $X \to Y$ does.

Example 54.4.3.

1. $\mathbb{A}^m_R \times_R \mathbb{A}^n_R = \mathbb{A}^{m+n}_R$, since $R[x] \otimes_R R[y] = R[x, y]$.
2. $\operatorname{Spec} \mathbb{C} \times_{\operatorname{Spec} \mathbb{R}} \operatorname{Spec} \mathbb{C} = \operatorname{Spec}(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}) = \operatorname{Spec}(\mathbb{C} \times \mathbb{C})$, two points. So a fibre product of one-point schemes can have several points, and the underlying set of a fibre product is not the fibre product of the underlying sets.
3. For $f : \operatorname{Spec} \mathbb{Z}[i] \to \operatorname{Spec} \mathbb{Z}$, the fibre over (p) is $\operatorname{Spec} \mathbb{F}_p[x]/(x^2 + 1)$. It is two reduced points if $p \equiv 1 \bmod 4$, one point with residue field $\mathbb{F}_{p^2}$ if $p \equiv 3 \bmod 4$, and the non-reduced point $\operatorname{Spec} \mathbb{F}_2[x]/(x + 1)^2$ if $p = 2$. The generic fibre is $\operatorname{Spec} \mathbb{Q}(i)$.

54.5 Separated and proper morphisms

Definition 54.5.1. A morphism $f : X \to Y$ is *separated* if the diagonal $\Delta : X \to X \times_Y X$ is a closed immersion. It is *proper* if it is separated, of finite type, and *universally closed*: every base change $X_{Y'} \to Y'$ is a closed map. A scheme over a field k is *complete* if it is proper over $\operatorname{Spec} k$.

Proposition 54.5.2. *Every morphism of affine schemes is separated. Open and closed immersions are separated, and so are compositions and base changes of separated morphisms. A morphism $X \to Y$ is separated if and only if the diagonal has closed image.*

Proof. For $\operatorname{Spec} A \to \operatorname{Spec} R$ the diagonal corresponds to the multiplication $A \otimes_R A \to A$, which is surjective, so Δ is a closed immersion (a surjection of rings $B \to B/I$ gives a homeomorphism onto $V(I)$ and a surjection of sheaves, by Theorem 54.2.3). For the remaining statements, see [Har77, Corollary II.4.2 and Corollary II.4.6]. The last one holds because Δ is always a locally closed immersion: over an affine open of Y and affine opens U of X, the diagonal maps U by a closed immersion into the open subscheme $U \times_Y U$. □

Example 54.5.3. The affine line with doubled origin X is not separated over k. Let $i_1, i_2 \colon \mathbb{A}^1_k \to X$ be the two open immersions. The morphism $(i_1, i_2) \colon \mathbb{A}^1_k \to X \times_k X$ maps $t \neq 0$ into the diagonal but maps 0 to the pair of the two origins, which is not in the diagonal. So the preimage of $\Delta(X)$ is $\mathbb{A}^1_k \setminus \{0\}$, which is not closed. Hence $\Delta(X)$ is not closed.

Theorem 54.5.4 (Valuative criteria)**.** *Let $f : X \to Y$ be a morphism of finite type with X Noetherian. For every valuation ring R with fraction field K and every commutative square formed by* $\operatorname{Spec} K \to X$ *and* $\operatorname{Spec} R \to Y$*, consider lifts* $\operatorname{Spec} R \to X$ *making both triangles commute. Then f is separated if and only if there is at most one lift for every such square, and proper if and only if there is exactly one. It suffices to consider discrete valuation rings.*

This is [Har77, Theorems II.4.3 and II.4.7 and Exercise II.4.11]. Geometrically: a punctured disc $\operatorname{Spec} K$ in X whose image extends over the centre of $\operatorname{Spec} R$ in Y extends in at most one way (separated: limits are unique) or in exactly one way (proper: limits exist).

Theorem 54.5.5. *For every ring A, $\mathbb{P}^n_A \to \operatorname{Spec} A$ is proper. Consequently every* projective *morphism, that is, a closed immersion into some $\mathbb{P}^n_Y$ followed by the projection, is proper, and every projective variety is complete.*

Proof. Since $\mathbb{P}^n_A = \mathbb{P}^n_{\mathbb{Z}} \times \operatorname{Spec} A$ and properness is stable under base change [Har77, Corollary II.4.8], we may take $A = \mathbb{Z}$. The morphism is of finite type, being covered by the $n+1$ affine spaces U_i. It is separated: on $U_i \times U_j$ the diagonal is $U_i \cap U_j \to U_i \times U_j$, given by the ring map $\mathbb{Z}[x/x_i] \otimes \mathbb{Z}[x/x_j] \to \mathbb{Z}[x/x_i]_{x_j/x_i}$, which is surjective because x_i/x_j is the image of an element of the second factor. By Proposition 54.5.2 this suffices. For properness, use Theorem 54.5.4. A morphism $\operatorname{Spec} K \to \mathbb{P}^n_{\mathbb{Z}}$ lands in some U_i, so by Theorem 54.2.4 it is given by elements $a_j = x_j/x_i \in K$ with $a_i = 1$. Let v be the valuation of R (with $v(0) = \infty$) and choose m with $v(a_m)$ minimal; then $a_m \neq 0$. All $b_j = a_j/a_m$ lie in R, and the ring map $\mathbb{Z}[x_0/x_m, \dots, x_n/x_m] \to R$, $x_j/x_m \mapsto b_j$, defines $\operatorname{Spec} R \to U_m$. On $\operatorname{Spec} K$ it agrees with the given morphism, because on $U_i \cap U_m$ the coordinates $x_j/x_i = (x_j/x_m)(x_m/x_i)$ are sent to $b_j/b_i = a_j$. Uniqueness follows from separatedness. Closed immersions are proper, and compositions of proper morphisms are proper [Har77, Corollary II.4.8]. □

54.6 Projective schemes

Definition 54.6.1. Let $S = \bigoplus_{d \geq 0} S_d$ be a graded ring and $S_+ = \bigoplus_{d>0} S_d$. The set $\operatorname{Proj} S$ consists of the homogeneous primes not containing S_+, with the topology whose closed sets are the $V_+(\mathfrak{a})$, $\mathfrak{a}$ homogeneous. For homogeneous $f \in S_+$ let $D_+(f) = \{\mathfrak{p} : f \notin \mathfrak{p}\}$, and let $S_{(f)}$ be the degree-zero part of S_f, and $S_{(\mathfrak{p})}$ the degree-zero part of the localisation at the homogeneous elements not in $\mathfrak{p}$. The structure sheaf consists of the functions $s(\mathfrak{p}) \in S_{(\mathfrak{p})}$ that are locally quotients a/f with a, f homogeneous of the same degree.

Proposition 54.6.2. $\operatorname{Proj} S$ *is a scheme. The stalk at $\mathfrak{p}$ is $S_{(\mathfrak{p})}$, the sets $D_+(f)$ form an open cover, and $D_+(f) \cong \operatorname{Spec} S_{(f)}$. There is a morphism* $\operatorname{Proj} S \to \operatorname{Spec} S_0$*. If $S = A[x_0, \dots, x_n]$, then* $\operatorname{Proj} S = \mathbb{P}^n_A$*, and if $S = A[x_0, \dots, x_n]/I$ with I homogeneous, then* $\operatorname{Proj} S$ *is a closed subscheme of $\mathbb{P}^n_A$.*

Proof. The stalks are computed as in Theorem 54.2.3(1). Every $\mathfrak{p} \in \operatorname{Proj} S$ omits some homogeneous element of S_+, so the $D_+(f)$ cover. Let f have degree $e > 0$. The map $\mathfrak{p} \mapsto (\mathfrak{p}S_f) \cap S_{(f)}$ is a homeomorphism $D_+(f) \to \operatorname{Spec} S_{(f)}$. Its inverse sends $\mathfrak{q}$ to the homogeneous ideal whose degree d part is $\{a \in S_d : a^e/f^d \in \mathfrak{q}\}$; checking that this is a homogeneous prime and that the two maps are inverse homeomorphisms is routine [Har77, Proposition II.2.5]. Under it, $S_{(\mathfrak{p})}$ corresponds to the localisation of $S_{(f)}$ at $\mathfrak{q}$, and the locally-quotient descriptions of the two structure sheaves match, so the ringed spaces are isomorphic. The inclusions $S_0 \to S_{(f)}$ glue to the morphism to $\operatorname{Spec} S_0$. For $S = A[x_0, \dots, x_n]$, $S_{(x_i)} = A[x_0/x_i, \dots, x_n/x_i]$, and the gluing agrees with Example 54.3.3(3). A surjection $S \to S/I$ induces surjections $S_{(x_i)} \to (S/I)_{(x_i)}$, hence closed immersions $\operatorname{Spec}(S/I)_{(x_i)} \to \operatorname{Spec} S_{(x_i)}$, which glue to a closed immersion $\operatorname{Proj} S/I \to \mathbb{P}^n_A$. □

54.7 Varieties as schemes

Theorem 54.7.1. *Let k be algebraically closed. There is a fully faithful functor t from the category of varieties over k (Chapter 53) to the category of schemes over k. The space $t(V)$ is the set of irreducible closed subsets of V, and V is homeomorphic to the subspace of closed points of $t(V)$. The image consists of the integral separated schemes of finite type over k that admit an immersion into some $\mathbb{P}^n_k$, and $t(V)$ is proper over k if and only if V is projective.*

Sketch of proof. For an affine variety V with coordinate ring A, $t(V) = \operatorname{Spec} A$. Irreducible closed subsets of V correspond to primes of A (Proposition 53.2.4), and points to maximal ideals. By Theorem 53.3.3 and Theorem 54.2.3, the regular functions on an open subset $U \subseteq V$ are the sections of $\mathcal{O}_{\operatorname{Spec} A}$ over the corresponding open subset. In general $t(V)$ is glued from affine pieces. Full faithfulness reduces, by gluing, to the affine case, where both categories are anti-equivalent to the category of finitely generated domains (Proposition 53.3.5 and Theorem 54.2.4). The omitted verifications are that the construction is compatible with gluing and that quasi-projective varieties give separated schemes, which is Proposition 54.5.2 and Theorem 54.5.5; see [Har77, Proposition II.2.6 and Proposition II.4.10]. □

Remark 54.7.2. From now on, a *variety over a field k* is an integral separated scheme of finite type over k. For k algebraically closed and quasi-projective varieties this agrees with Chapter 53 by Theorem 54.7.1. There are complete varieties that are not projective (Nagata, Hironaka), but every complete curve is projective, and so is every smooth complete surface; see [Har77, Appendix B].

54.8 Flat and smooth morphisms

Definition 54.8.1. A morphism $f : X \to Y$ is *flat* at x if $\mathcal{O}_{X,x}$ is a flat $\mathcal{O}_{Y,f(x)}$-module (Definition 8.9.7), and *flat* if it is flat at every point. It is *faithfully flat* if moreover it is surjective.

Proposition 54.8.2.

1. *Open immersions are flat, and $\mathbb{A}^n_Y \to Y$ and $\mathbb{P}^n_Y \to Y$ are flat. Compositions and base changes of flat morphisms are flat.*
2. *Let R be a principal ideal domain, for instance a discrete valuation ring. An R-module is flat if and only if it is torsion-free. Hence $X \to \operatorname{Spec} R$ is flat if and only if every local ring $\mathcal{O}_{X,x}$ is torsion-free over R. If X is integral, this holds if and only if X dominates $\operatorname{Spec} R$.*

Proof. (1) Localisations are flat (Proposition 14.3.2(2)), and so are free modules such as $A[x_1, \dots, x_n]$ over A; local rings of $\mathbb{A}^n_A$ are localisations of $A[x]$. The statements on composition and base change are [Har77, Proposition III.9.2]. (2) If M is flat and $0 \neq t \in R$, tensoring

the injection $t\colon R \to R$ with M shows that t is injective on M. Conversely, a torsion-free M is the union of its finitely generated submodules, which are torsion-free, hence free (Theorem 8.11.3). A union of flat submodules is flat, since an element of the kernel of $N' \otimes M \to N \otimes M$ comes from some finitely generated submodule of M. For integral X, torsion-freeness fails exactly when some nonzero $t \in R$ vanishes on X, that is, when X maps into a closed point. □

Example 54.8.3.

1. The family $X = \operatorname{Spec} k[t,x,y]/(xy-t) \to \mathbb{A}^1_k = \operatorname{Spec} k[t]$ is flat, since X is integral and dominates $\mathbb{A}^1$. Its fibres are the smooth conics $xy = t$ for $t \neq 0$, which degenerate to the pair of lines $xy = 0$.

2. The blow-up $\mathrm{Bl}_0\mathbb{A}^2 \to \mathbb{A}^2$ is not flat. Its fibres have dimension 0 except over the origin, where the fibre is $\mathbb{P}^1$, while flat morphisms of finite type between varieties have equidimensional fibres [Har77, Proposition III.9.5].

3. A closed immersion $\operatorname{Spec} A/I \to \operatorname{Spec} A$ with $0 \neq I$ and A a domain is not flat, since A/I is torsion.

Definition 54.8.4. A morphism $f\colon X \to Y$ is *smooth of relative dimension n* if every point of X has affine open neighbourhoods $\operatorname{Spec} B$ of x and $\operatorname{Spec} A$ of $f(x)$, with $f(\operatorname{Spec} B) \subseteq \operatorname{Spec} A$, such that

$$B \cong A[x_1,\dots,x_{n+r}]/(g_1,\dots,g_r) \qquad \text{with } \det\left(\partial g_i/\partial x_{n+j}\right)_{1\le i,j\le r} \text{ invertible in } B.$$

It is *étale* if it is smooth of relative dimension 0.

Proposition 54.8.5. *Let k be algebraically closed and X a variety over k of dimension n. Then $X \to \operatorname{Spec} k$ is smooth (of relative dimension n) if and only if X is nonsingular in the sense of Definition 53.6.1.*

Proof. Both conditions are local, so let X be affine. Suppose $B = k[x_1,\dots,x_{n+r}]/(g_1,\dots,g_r)$ with the minor invertible, and let P be a closed point. Every irreducible component of $Z(g_1,\dots,g_r) \subseteq \mathbb{A}^{n+r}$ has dimension $\geq n$ by Theorem 53.5.4 applied r times, so $\dim \mathcal{O}_{X,P} \geq n$. The Jacobian of (g_i) has rank r at P, so $\dim T_P X \leq n$ by the computation in Theorem 53.6.3, which only used generators of the ideal defining B. By Lemma 53.6.2, $\mathcal{O}_{X,P}$ is regular of dimension n. Conversely, let $X \subseteq \mathbb{A}^N$ be nonsingular at P with $I(X) = (f_1,\dots,f_t)$. The Jacobian has rank $r = N - n$ at P, so after renumbering, $\det(\partial f_i/\partial x_{n+j})_{i,j\le r}$ is nonzero at P, hence invertible on a neighbourhood $D(h)$. The scheme $X' = Z(f_1,\dots,f_r) \cap D(h)$ is smooth of relative dimension n by construction. So by the first part its local ring at P is regular of dimension n, in particular a domain (Remark 14.11.3). Hence X' has a unique irreducible component through P, of dimension n, which contains $X \cap D(h)$ and so equals it near P. As X' is reduced near P, $X = X'$ near P. □

Counterexample 54.8.6. Over a non-perfect field, regular does not imply smooth. Let p be an odd prime, $k = \mathbb{F}_p(t)$ and $X = \operatorname{Spec} k[x,y]/(y^2 - x^p + t)$. At the closed point $\mathfrak{m} = (y, x^p - t)$, whose residue field is $k(t^{1/p})$, the relation $x^p - t = y^2 \in \mathfrak{m}^2$ shows that $\mathfrak{m}/\mathfrak{m}^2$ is spanned by y. So X is regular there, of dimension 1. But X is not smooth over k. Smoothness is stable under base change, since a presentation as in Definition 54.8.4 base changes to one. After base change to $\bar{k}$, with $s = t^{1/p}$, the curve becomes $y^2 = (x-s)^p$, whose Jacobian $(-p(x-s)^{p-1}, 2y) = (0, 2y)$ vanishes at the point $(s, 0)$. That point is singular by Theorem 53.6.3, so $X_{\bar{k}}$ is not smooth by Proposition 54.8.5.

54.9 Base change and varieties over non-closed fields

Definition 54.9.1. Let X be a scheme over a field k and $K \supseteq k$ a field extension. The *base change* is $X_K = X \times_{\operatorname{Spec} k} \operatorname{Spec} K$. The scheme X is *geometrically reduced*, *irreducible* or *integral* if $X_{\bar{k}}$ is reduced, irreducible or integral for an algebraic closure $\bar{k}$. A *geometric point* is a morphism $\operatorname{Spec} \Omega \to X$ with Ω algebraically closed.

Example 54.9.2.

1. $X = \operatorname{Spec} \mathbb{R}[x]/(x^2+1) = \operatorname{Spec} \mathbb{C}$ is integral over $\mathbb{R}$, but $X_{\mathbb{C}} = \operatorname{Spec} \mathbb{C} \sqcup \operatorname{Spec} \mathbb{C}$ is not irreducible.
2. $X = \operatorname{Spec} \mathbb{F}_p(t)[x]/(x^p - t)$ is a point with residue field $\mathbb{F}_p(t^{1/p})$, integral, but $X_{\bar{k}}$ is non-reduced, since $x^p - t = (x - t^{1/p})^p$.
3. The conic $X = \operatorname{Spec} \mathbb{R}[x,y]/(x^2+y^2+1)$ has no $\mathbb{R}$-points, but it is a nonempty integral scheme whose closed points have residue field $\mathbb{C}$. $X(\mathbb{C})$ is an affine conic with an action of complex conjugation without fixed points.
4. For a variety X over k and a Galois extension K/k, $\operatorname{Gal}(K/k)$ acts on $X(K) = \operatorname{Hom}_k(\operatorname{Spec} K, X)$ through its action on $\operatorname{Spec} K$, and $X(k) \subseteq X(K)^{\operatorname{Gal}(K/k)}$. For a variety over $\mathbb{Q}$, the set $X(\bar{\mathbb{Q}})$ with its Galois action is the starting point of the ℓ-adic cohomology of Chapter 65.

Remark 54.9.3. A variety X over $\mathbb{C}$ is defined by finitely many polynomials, so it is defined over a finitely generated subfield $k_0 \subseteq \mathbb{C}$: $X = X_0 \times_{k_0} \mathbb{C}$. Moreover, X_0 extends to a scheme of finite type over a finitely generated $\mathbb{Z}$-algebra, a technique called *spreading out*. For an automorphism σ of $\mathbb{C}$, the *conjugate variety* is $X^\sigma = X \times_{\mathbb{C},\sigma} \mathbb{C}$. It is a scheme isomorphic to X as an abstract scheme but not over $\mathbb{C}$, and $X^\sigma(\mathbb{C})$ is in general not homeomorphic to $X(\mathbb{C})$ [Ser64]. These constructions are used in the theory of absolute Hodge classes (Chapter 75).

54.10 Exercises

Exercise 54.10.1. Describe the points of $\operatorname{Spec} \mathbb{Z}[x]$: show that they are (0), the primes (f) with f irreducible, and the maximal ideals (p, f) with p prime and f irreducible modulo p.

Solution. Let $\mathfrak{p}$ be a prime. If $\mathfrak{p} \cap \mathbb{Z} = (p)$ with p prime, then $\mathfrak{p}$ corresponds to a prime of $\mathbb{F}_p[x]$, a principal ideal domain, namely 0 or $(\bar{f})$ with $\bar{f}$ irreducible. This gives (p) and (p, f), and the latter is maximal. If $\mathfrak{p} \cap \mathbb{Z} = 0$, then $\mathfrak{p}$ corresponds to a prime of $\mathbb{Q}[x]$ (localisation at $\mathbb{Z} \setminus 0$), which is 0 or (g) with g irreducible. Then $\mathfrak{p} = g\mathbb{Q}[x] \cap \mathbb{Z}[x] = (f)$, where f is g made primitive (Gauss's lemma, Lemma 8.6.10). The primes (p) are the primes (f) with $f = p$ constant. □

Exercise 54.10.2. Let X be a scheme over an algebraically closed field k and $D = \operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$. Show that $X(D)$ is in bijection with pairs of a closed point $x \in X(k)$ and a vector of $(\mathfrak{m}_x/\mathfrak{m}_x^2)^\vee$.

Solution. The image of D is a point x, and the morphism is a local k-algebra map $\varphi \colon \mathcal{O}_{X,x} \to k[\varepsilon]$. Composing with $k[\varepsilon] \to k$ gives $k(x) = k$. Write $\varphi(a) = a(x) + \delta(a)\varepsilon$. Then δ is a k-linear derivation at x that kills k and $\mathfrak{m}_x^2$, since $\varepsilon^2 = 0$. So δ is a linear form on $\mathfrak{m}_x/\mathfrak{m}_x^2$, and conversely every such form defines a local homomorphism. □

Exercise 54.10.3. Let R be a discrete valuation ring with fraction field K and residue field κ. Describe $\operatorname{Spec} R$ and the fibres of $X \to \operatorname{Spec} R$ for $X = \operatorname{Spec} R[x,y]/(xy - \pi)$, where π is a uniformiser.

Solution. Spec R has two points: the closed point $\mathfrak{m}$ and the generic point (0), which is open. The generic fibre is the smooth conic $xy = \pi$ over K, isomorphic to $\mathbb{A}^1_K \setminus \{0\}$. The special fibre is $xy = 0$ over κ, two lines meeting in a point. X is a domain dominating Spec R, so the morphism is flat (Proposition 54.8.2). □

Exercise 54.10.4. Show that $\operatorname{Proj} k[x, y, z]/(xz - y^2) \cong \mathbb{P}^1_k$ for every field k.

Solution. The ring $S = k[x, y, z]/(xz - y^2)$ is isomorphic to the subring $k[s^2, st, t^2] \subseteq k[s, t]$ of polynomials of even degree, with degrees halved. For a graded ring T and $T^{(2)} = \bigoplus_d T_{2d}$ one has $\operatorname{Proj} T^{(2)} \cong \operatorname{Proj} T$, since $D_+(f^2)$ and $D_+(f)$ have the same ring $T_{(f)} = T^{(2)}_{(f^2)}$ [Har77, Exercise II.5.13]. Hence $\operatorname{Proj} S \cong \operatorname{Proj} k[s, t] = \mathbb{P}^1_k$. □

Chapter 55

Quasi-coherent and Coherent Sheaves

55.1 Introduction

Quasi-coherent sheaves are to schemes what modules are to rings. On an affine scheme $\operatorname{Spec} A$ they are exactly the sheaves $\widetilde{M}$ attached to A-modules, and in general they are glued from such. Coherent sheaves, the finitely generated ones, include vector bundles, ideal sheaves of subvarieties, structure sheaves of closed subschemes and sheaves of differentials. Their cohomology is the subject of Chapter 56. This chapter develops the basic theory: the equivalence between modules and quasi-coherent sheaves on affine schemes, closed subschemes and ideal sheaves, the twisting sheaves $\mathcal{O}(n)$ and Serre's theorem that $\mathcal{F}(n)$ is globally generated for $n \gg 0$, and sheaves of differentials, with the Euler sequence and the canonical sheaf of $\mathbb{P}^n$. References: [Har77, Sections II.5 and II.8], [The].

Prerequisites. Chapter 54 (Schemes).

For a ringed space $(X, \mathcal{O}_X)$, $\mathcal{O}_X$-modules, their tensor products, $\mathcal{H}om$, direct and inverse images are as in Definition 25.6.3. For $f \in \Gamma(X, \mathcal{O}_X)$ we write $X_f = \{x \in X : f_x \notin \mathfrak{m}_x\}$, an open set on which f is invertible.

55.2 The sheaf associated to a module

Definition 55.2.1. Let A be a ring and M an A-module. The sheaf $\widetilde{M}$ on $\operatorname{Spec} A$ assigns to an open U the set of functions $s : U \to \bigsqcup_{\mathfrak{p} \in U} M_{\mathfrak{p}}$ with $s(\mathfrak{p}) \in M_{\mathfrak{p}}$ that are locally of the form m/f with $m \in M$ and $f \in A$ nonvanishing on the neighbourhood. It is an $\mathcal{O}_{\operatorname{Spec} A}$-module.

Proposition 55.2.2. *Let A be a ring, M, N modules and $X = \operatorname{Spec} A$.*

1. *$(\widetilde{M})_{\mathfrak{p}} \cong M_{\mathfrak{p}}$ and $\widetilde{M}(D(f)) \cong M_f$; in particular $\Gamma(X, \widetilde{M}) = M$.*
2. *$M \mapsto \widetilde{M}$ is an exact, fully faithful functor from A-modules to $\mathcal{O}_X$-modules, commuting with direct sums, and $(M \otimes_A N)^{\sim} \cong \widetilde{M} \otimes_{\mathcal{O}_X} \widetilde{N}$.*
3. *For a ring map $A \to B$ and $f : \operatorname{Spec} B \to \operatorname{Spec} A$: $f_*(\widetilde{N}) \cong ({}_A N)^{\sim}$ for a B-module N, and $f^*(\widetilde{M}) \cong (B \otimes_A M)^{\sim}$.*

Proof. (1) The proof of Theorem 54.2.3 applies word for word with A replaced by M in the numerators. (2) Exactness can be checked on stalks, where it is exactness of localisation (Proposition 14.3.2(2)). Full faithfulness: a morphism $\widetilde{M} \to \widetilde{N}$ is determined by its effect on global sections, by (1) on basic open sets, and every A-linear map induces one. Direct sums and tensor products commute with localisation, and the sheaf tensor product has stalks $M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} N_{\mathfrak{p}}$; the natural map is therefore an isomorphism on stalks. (3) For $g \in A$ with

image $\bar{g} \in B$, $f_*(\widetilde{N})(D(g)) = \widetilde{N}(D(\bar{g})) = N_{\bar{g}} = (N_A)_g$, compatibly with restrictions. For the pullback, both sides represent the functor $\mathcal{G} \mapsto \mathrm{Hom}_{\mathcal{O}_{\mathrm{Spec}\, A}}(\widetilde{M}, f_*\mathcal{G})$ on quasi-coherent $\mathcal{G}$, by adjunction and (2); see [Har77, Proposition II.5.2]. □

55.3 Quasi-coherent sheaves

Definition 55.3.1. An $\mathcal{O}_X$-module $\mathcal{F}$ on a scheme X is *quasi-coherent* if X has an affine open cover $X = \bigcup \mathrm{Spec}\, A_i$ with $\mathcal{F}|_{\mathrm{Spec}\, A_i} \cong \widetilde{M_i}$ for A_i-modules M_i. If X is Noetherian, $\mathcal{F}$ is *coherent* if moreover the M_i can be taken finitely generated.

Lemma 55.3.2. *Let X be a scheme that is Noetherian, or quasi-compact and separated. Let $\mathcal{L}$ be an invertible sheaf, that is, locally isomorphic to $\mathcal{O}_X$, $f \in \Gamma(X, \mathcal{L})$, and $X_f = \{x : f_x \notin \mathfrak{m}_x \mathcal{L}_x\}$. Let $\mathcal{F}$ be quasi-coherent.*

1. *If $s \in \Gamma(X, \mathcal{F})$ restricts to 0 on X_f, then $s \otimes f^n = 0$ in $\Gamma(X, \mathcal{F} \otimes \mathcal{L}^n)$ for some n.*
2. *If $t \in \mathcal{F}(X_f)$, then $t \otimes f^n$ extends to a global section of $\mathcal{F} \otimes \mathcal{L}^n$ for some n.*

Proof. Cover X by finitely many affine opens $U_i = \mathrm{Spec}\, A_i$ with $\mathcal{L}|_{U_i} \cong \mathcal{O}_{U_i}$ and $\mathcal{F}|_{U_i} = \widetilde{M_i}$. Every $U_i \cap U_j$ is quasi-compact: every open subset of a Noetherian scheme is, and in the separated case $U_i \cap U_j$ is affine, being the preimage of the closed subscheme $\Delta(X)$ under $U_i \times U_j \to X \times X$. Under the trivialisation f becomes $f_i \in A_i$ and $X_f \cap U_i = D(f_i)$. (1) $s|_{U_i} \in M_i$ vanishes in $(M_i)_{f_i}$ by Proposition 55.2.2(1), so $f_i^n s|_{U_i} = 0$ for large n, which we take uniform over the finitely many i. (2) $t|_{D(f_i)} \in (M_i)_{f_i}$, so $f_i^n t$ extends to a section t_i over U_i. On $U_i \cap U_j$, which is quasi-compact and so covered by finitely many affine opens, $t_i - t_j$ vanishes on X_f. By (1) applied to $U_i \cap U_j$, $f^m(t_i - t_j) = 0$ there. So the sections $f^m t_i$ glue to a global section of $\mathcal{F} \otimes \mathcal{L}^{n+m}$ extending $t \otimes f^{n+m}$. □

Theorem 55.3.3. *Let $X = \mathrm{Spec}\, A$. An $\mathcal{O}_X$-module $\mathcal{F}$ is quasi-coherent if and only if the natural map $\Gamma(X, \mathcal{F})^\sim \to \mathcal{F}$ is an isomorphism. Hence $M \mapsto \widetilde{M}$ is an equivalence between A-modules and quasi-coherent $\mathcal{O}_X$-modules, with inverse $\Gamma(X, -)$. A sequence of quasi-coherent sheaves on X is exact if and only if the sequence of global sections is exact.*

Proof. The natural map is defined on $D(g)$ by $M_g \to \mathcal{F}(D(g))$, $m/g^n \mapsto g^{-n} m|_{D(g)}$, where $M = \Gamma(X, \mathcal{F})$. If $\mathcal{F}$ is quasi-coherent, apply Lemma 55.3.2 with $\mathcal{L} = \mathcal{O}_X$ and $f = g$ to the affine, hence quasi-compact and separated, scheme X: part (2) gives surjectivity and part (1) injectivity of $M_g \to \mathcal{F}(D(g))$. So the map is an isomorphism on a basis. The converse is clear. The equivalence follows with Proposition 55.2.2(2). For exactness, kernels and cokernels of maps $\widetilde{M} \to \widetilde{N}$ are the sheaves associated to the kernel and cokernel, since $\sim$ is exact. So a sequence of quasi-coherent sheaves is isomorphic to $\sim$ applied to its sequence of global sections, and $\sim$ is exact and faithful: $\widetilde{M} = 0$ implies $M = 0$. □

Corollary 55.3.4. *On any scheme, kernels, cokernels, images, direct sums and tensor products of quasi-coherent sheaves are quasi-coherent, and pullbacks of quasi-coherent sheaves are quasi-coherent. $\mathcal{F}$ is quasi-coherent if and only if $\mathcal{F}|_U \cong \mathcal{F}(U)^\sim$ for every affine open U. On a Noetherian scheme the same statements hold for coherent sheaves (with pullbacks along morphisms of Noetherian schemes), and on $\mathrm{Spec}\, A$ with A Noetherian, $\widetilde{M}$ is coherent if and only if M is finitely generated.*

Proof. All statements are local, and for affine schemes they follow from Theorem 55.3.3 and Proposition 55.2.2. For the characterisation of quasi-coherence on every affine open, restrict a cover with $\mathcal{F}|_{U_i} = \widetilde{M_i}$ to the affine open U: then $\mathcal{F}|_U$ is quasi-coherent in the sense of the definition on U, and Theorem 55.3.3 applies. For coherence, submodules and quotients of

finitely generated modules over a Noetherian ring are finitely generated (Proposition 14.5.2). If $\widetilde{M}$ is coherent on $\operatorname{Spec} A$, then M is locally, on a finite cover by $D(f_i)$, generated by finitely many elements m_{ij}/f_i^n; the m_{ij} generate M, since a module is zero when all its localisations at the f_i vanish. □

Proposition 55.3.5. *Let $f : X \to Y$ be a morphism with X Noetherian, or with X quasi-compact and separated. If $\mathcal{F}$ is quasi-coherent on X, then $f_*\mathcal{F}$ is quasi-coherent on Y.*

Proof. The question is local on Y, so let Y be affine. Cover X by finitely many affine opens U_i. The intersections $U_{ij} = U_i \cap U_j$ are covered by finitely many affine opens U_{ijk} (in the separated case U_{ij} itself is affine). By the sheaf axiom,

$$0 \to f_*\mathcal{F} \to \bigoplus_i f_*(\mathcal{F}|_{U_i}) \to \bigoplus_{i,j,k} f_*(\mathcal{F}|_{U_{ijk}})$$

is exact. Each term on the right is the direct image of a quasi-coherent sheaf under a morphism of affine schemes, which is quasi-coherent by Proposition 55.2.2(3). So $f_*\mathcal{F}$ is a kernel of quasi-coherent sheaves. □

Proposition 55.3.6. *Let X be a scheme. For a closed immersion $i : Y \to X$ the* ideal sheaf *$\mathcal{I}_Y = \ker(\mathcal{O}_X \to i_*\mathcal{O}_Y)$ is quasi-coherent, and $Y \mapsto \mathcal{I}_Y$ is a bijection from closed subschemes of X to quasi-coherent ideal sheaves of $\mathcal{O}_X$. For $X = \operatorname{Spec} A$, every closed subscheme is $\operatorname{Spec} A/I$ for a unique ideal I. If X is Noetherian, $\mathcal{I}_Y$ is coherent.*

Sketch of proof. The question is local, so let $X = \operatorname{Spec} A$. The underlying space of Y is closed in X, so Y is quasi-compact. It is separated, since a closed immersion is separated (Proposition 54.5.2). By Proposition 55.3.5, $i_*\mathcal{O}_Y$ is quasi-coherent, so the kernel $\mathcal{I}_Y = \widetilde{I}$ is quasi-coherent with $I = \Gamma(X, \mathcal{I}_Y)$. As $\mathcal{O}_X \to i_*\mathcal{O}_Y$ is onto, $i_*\mathcal{O}_Y \cong (A/I)^{\sim}$, and its support $V(I)$ is the image of Y. Then $Y \to \operatorname{Spec} A/I$ is a homeomorphism with an isomorphism of structure sheaves, so $Y \cong \operatorname{Spec} A/I$. Conversely, a quasi-coherent ideal $\widetilde{I}$ defines $\operatorname{Spec} A/I$, and on a general X these glue. The omitted verifications are the compatibilities of these constructions with restriction to smaller affine opens [Har77, Corollary II.5.10]. □

Definition 55.3.7. The *support* of a sheaf $\mathcal{F}$ is $\{x : \mathcal{F}_x \neq 0\}$. For a coherent sheaf it is closed: on $\operatorname{Spec} A$ with M finitely generated, $\operatorname{Supp} \widetilde{M} = V(\operatorname{Ann} M)$. The *fibre* of $\mathcal{F}$ at x is $\mathcal{F}(x) = \mathcal{F}_x \otimes k(x)$.

Proposition 55.3.8. *Let $\mathcal{F}$ be coherent on a Noetherian scheme X.*

1. *(Nakayama) If sections $s_1, \dots, s_r$ of $\mathcal{F}$ near x span $\mathcal{F}(x)$, they generate $\mathcal{F}$ on a neighbourhood of x.*
2. *$x \mapsto \dim_{k(x)} \mathcal{F}(x)$ is upper semicontinuous.*
3. *If X is reduced, $\mathcal{F}$ is locally free (locally isomorphic to $\mathcal{O}_X^r$) if and only if this function is locally constant.*

Proof. (1) By Nakayama's lemma (Corollary 14.4.3) the s_i generate the stalk $\mathcal{F}_x$. The cokernel of $\mathcal{O}^r \to \mathcal{F}$ is coherent with x outside its closed support. (2) follows from (1): generators at x generate nearby, so the fibre dimension cannot jump up near x. (3) If $X = \operatorname{Spec} A$ is reduced and the fibre dimension is r near x, choose $s_1, \dots, s_r$ as in (1) and let $\mathcal{K}$ be the kernel of the resulting surjection $\mathcal{O}^r \to \mathcal{F}$. For each y nearby, $\mathcal{O}^r(y) \to \mathcal{F}(y)$ is a surjection of r-dimensional spaces, hence injective. So an element $(a_1, \dots, a_r)$ of $\Gamma(\mathcal{K})$ has all a_i vanishing at every point, hence nilpotent, hence 0. So $\mathcal{K} = 0$. □

55.4 Locally free sheaves and the Picard group

Definition 55.4.1. An $\mathcal{O}_X$-module $\mathcal{E}$ is *locally free of rank r* if it is locally isomorphic to $\mathcal{O}_X^r$, and *invertible* if $r = 1$. Its dual is $\mathcal{E}^\vee = \mathcal{H}om(\mathcal{E}, \mathcal{O}_X)$, its determinant $\det \mathcal{E} = \Lambda^r \mathcal{E}$. The *Picard group* $\operatorname{Pic}(X)$ is the group of isomorphism classes of invertible sheaves under $\otimes$, with inverse $\mathcal{L}^{-1} = \mathcal{L}^\vee$.

Proposition 55.4.2. $\operatorname{Pic}(X) \cong H^1(X, \mathcal{O}_X^*)$. *Locally free sheaves of rank r on X correspond to isomorphism classes of* algebraic vector bundles*: schemes $E \to X$ locally isomorphic to $\mathbb{A}^r_U \to U$ with linear transition functions in $\mathrm{GL}_r(\mathcal{O}_X(U \cap V))$. The correspondence sends E to its sheaf of sections.*

Proof. An invertible sheaf trivialised on an open cover (U_i) by generators e_i gives units $g_{ij} \in \mathcal{O}^*(U_i \cap U_j)$ with $e_j = g_{ij} e_i$. They form a Čech cocycle, changing the e_i changes it by a coboundary, and tensor product multiplies cocycles. Conversely a cocycle glues copies of $\mathcal{O}_{U_i}$. So invertible sheaves trivialised on (U_i) correspond to $\check{H}^1((U_i), \mathcal{O}^*)$, and passing to the limit over covers gives $\check{H}^1(X, \mathcal{O}^*) = H^1(X, \mathcal{O}^*)$ (Remark 26.5.4). The same cocycles with values in GL_r glue the $\mathbb{A}^r_{U_i}$ to a vector bundle, as for manifolds (Proposition 35.2.6). □

55.5 Twisting sheaves on projective schemes

Definition 55.5.1. Let S be a graded ring and M a graded S-module. The sheaf $\widetilde{M}$ on $\operatorname{Proj} S$ has sections over $D_+(f)$ equal to $M_{(f)}$, the degree-zero part of M_f, and stalks $M_{(\mathfrak{p})}$. For $n \in \mathbb{Z}$, $M(n)$ is M with degrees shifted, $M(n)_d = M_{n+d}$, and $\mathcal{O}(n) = S(n)^\sim$. For an $\mathcal{O}$-module $\mathcal{F}$ put $\mathcal{F}(n) = \mathcal{F} \otimes \mathcal{O}(n)$ and $\Gamma_*(\mathcal{F}) = \bigoplus_{n \in \mathbb{Z}} \Gamma(X, \mathcal{F}(n))$. For a scheme X with a closed immersion into $\mathbb{P}^r_A$, $\mathcal{O}_X(1)$ is the restriction of $\mathcal{O}(1)$.

Proposition 55.5.2. *Let S be generated by S_1 as an S_0-algebra and $X = \operatorname{Proj} S$. Then $\mathcal{O}(n)$ is invertible, $\mathcal{O}(n) \otimes \mathcal{O}(m) \cong \mathcal{O}(n+m)$, and $\widetilde{M}(n) \cong (M(n))^\sim$. For $S = A[x_0, \dots, x_r]$ and $X = \mathbb{P}^r_A$,*

$$\Gamma(X, \mathcal{O}(n)) \cong S_n \quad \text{for all } n \in \mathbb{Z},$$

the homogeneous polynomials of degree n (zero for $n < 0$); so $\Gamma_*(\mathcal{O}_X) = S$.

Proof. Since S is generated by S_1, the $D_+(f)$ with $f \in S_1$ cover X, and on $D_+(f)$ the module $S(n)_{(f)}$ is free over $S_{(f)}$ with basis f^n. This proves invertibility, and the other isomorphisms follow similarly from $M(n)_{(f)} = M_{(f)} \cdot f^n$. For $\mathbb{P}^r_A$, a global section of $\mathcal{O}(n)$ is a compatible family of degree-n elements $t_i \in S_{x_i}$. The x_i are non-zero-divisors, so all S_{x_i} embed in $S' = S_{x_0 \cdots x_r}$ and compatibility means that the t_i are one element t of $\bigcap_i S_{x_i}$, whose homogeneous components lie in the intersection. An element of S' is a sum of Laurent monomials x^a, $a \in \mathbb{Z}^{r+1}$, with coefficients in A. It lies in S_{x_i} exactly when only monomials with $a_j \geq 0$ for $j \neq i$ occur. Hence $\bigcap_i S_{x_i} = S$, and the degree-n part is S_n. □

Theorem 55.5.3 (Serre)**.** *Let X be a closed subscheme of $\mathbb{P}^r_A$ with A Noetherian, and $\mathcal{F}$ a coherent sheaf on X. Then there is n_0 such that for all $n \geq n_0$ the sheaf $\mathcal{F}(n)$ is generated by finitely many global sections. Consequently $\mathcal{F}$ is a quotient of a finite direct sum of sheaves $\mathcal{O}_X(-n)$.*

Proof. Let $i \colon X \to \mathbb{P}^r_A$ be the inclusion. Then $i_*\mathcal{F}$ is coherent (Proposition 55.3.5, with finiteness checked on affines), and $i_*(\mathcal{F}(n)) = (i_*\mathcal{F})(n)$ since $\mathcal{O}_X(1) = i^*\mathcal{O}(1)$. So we may assume $X = \mathbb{P}^r_A$. Cover X by $U_i = D_+(x_i) = X_{x_i}$, where $x_i \in \Gamma(X, \mathcal{O}(1))$, and write $\mathcal{F}|_{U_i} = \widetilde{M_i}$ with M_i generated by finitely many sections s_{ij}. By Lemma 55.3.2(2) with $\mathcal{L} = \mathcal{O}(1)$ and $f = x_i$, for n large each $s_{ij} \otimes x_i^n$ extends to a global section of $\mathcal{F}(n)$, and we can take one n for

the finitely many sections. On U_i, x_i^n generates $\mathcal{O}(n)$, so the sections $s_{ij} \otimes x_i^n$ generate $\mathcal{F}(n)|_{U_i}$. The same holds for all larger n (multiply by x_i). Finally, N global sections generating $\mathcal{F}(n)$ give a surjection $\mathcal{O}^N \to \mathcal{F}(n)$, that is, $\mathcal{O}(-n)^N \to \mathcal{F}$. □

Definition 55.5.4. An invertible sheaf $\mathcal{L}$ on a scheme X of finite type over A is *very ample* (relative to A) if $\mathcal{L} \cong i^*\mathcal{O}(1)$ for an immersion $i\colon X \to \mathbb{P}^r_A$, and *ample* if for every coherent $\mathcal{F}$ the sheaf $\mathcal{F} \otimes \mathcal{L}^n$ is globally generated for $n \gg 0$. By Theorem 55.5.3, very ample sheaves on projective schemes over Noetherian rings are ample. The converse relation, that some power of an ample sheaf is very ample, is in Chapter 57.

55.6 Sheaves of differentials

Definition 55.6.1. Let $f : X \to Y$ be a morphism of schemes. The *sheaf of relative differentials* $\Omega_{X/Y}$ is the quasi-coherent $\mathcal{O}_X$-module obtained by gluing the sheaves $(\Omega_{B/A})^\sim$ on affine opens $\operatorname{Spec} B \subseteq f^{-1}(\operatorname{Spec} A)$, together with the derivation $d\colon \mathcal{O}_X \to \Omega_{X/Y}$. The gluing is possible because Ω commutes with localisation (Proposition 14.12.2(5)). For X smooth of relative dimension n over Y, the *canonical sheaf* is $\omega_{X/Y} = \Lambda^n \Omega_{X/Y}$, and the *tangent sheaf* is $\mathcal{T}_{X/Y} = \Omega^\vee_{X/Y}$.

Proposition 55.6.2. *1. For morphisms $X \to Y \to Z$ there is an exact sequence $f^*\Omega_{Y/Z} \to \Omega_{X/Z} \to \Omega_{X/Y} \to 0$.*

2. For a closed subscheme $Y \subseteq X$ over Z with ideal sheaf $\mathcal{I}$ there is an exact sequence $\mathcal{I}/\mathcal{I}^2 \to \Omega_{X/Z}|_Y \to \Omega_{Y/Z} \to 0$ (the conormal sequence*).*

3. Ω commutes with base change: $\Omega_{X_{Y'}/Y'} \cong g'^*\Omega_{X/Y}$.

4. If $X \to Y$ is smooth of relative dimension n, then $\Omega_{X/Y}$ is locally free of rank n. In particular, a nonsingular variety of dimension n over an algebraically closed field has $\Omega_{X/k}$ locally free of rank n.

Proof. (1), (2) are Proposition 14.12.2(3), (4) on affine opens. (3) holds because $\Omega_{(B\otimes_A A')/A'} \cong \Omega_{B/A} \otimes_A A'$, which follows from the universal property. (4) Locally $B = A[x_1, \dots, x_{n+r}]/(g_1, \dots, g_r)$ with $\det(\partial g_i/\partial x_{n+j})$ invertible. By Proposition 14.12.2(2), (4), $\Omega_{B/A}$ is the quotient of the free module on $dx_1, \dots, dx_{n+r}$ by the relations $dg_i = \sum_j (\partial g_i/\partial x_j) dx_j = 0$. Since the $r \times r$ block for $dx_{n+1}, \dots, dx_{n+r}$ is invertible, these relations express $dx_{n+1}, \dots, dx_{n+r}$ uniquely in terms of $dx_1, \dots, dx_n$, and $\Omega_{B/A}$ is free on $dx_1, \dots, dx_n$. The last statement uses Proposition 54.8.5. □

Theorem 55.6.3 (Euler sequence)**.** *On $X = \mathbb{P}^r_A$ there is an exact sequence of locally free sheaves*

$$0 \to \Omega_{X/A} \to \mathcal{O}(-1)^{r+1} \to \mathcal{O}_X \to 0,$$

where the second map sends the basis section e_j to x_j. Consequently $\omega_{\mathbb{P}^r_A/A} \cong \mathcal{O}(-r-1)$.

Proof. Write $\mathcal{O}(-1)^{r+1} = (S(-1)^{r+1})^\sim$ with basis $e_0, \dots, e_r$ of degree 1, and let π be the map $e_j \mapsto x_j$. It is surjective: on U_i, the section e_i/x_i maps to 1. On $U_i = \operatorname{Spec} A[x_0/x_i, \dots, x_r/x_i]$, Ω is free on the $d(x_j/x_i)$, $j \neq i$. Define

$$\varphi_i\colon \Omega|_{U_i} \to \mathcal{O}(-1)^{r+1}|_{U_i}, \qquad d(x_j/x_i) \mapsto \frac{1}{x_i^2}\left(x_i e_j - x_j e_i\right).$$

The composite with π is 0. The kernel of π on U_i consists of the $\sum_j a_j e_j$ (a_j of degree -1) with $\sum_j a_j x_j = 0$, that is, $a_i = -\sum_{j\neq i} a_j x_j/x_i$, and then $\sum_j a_j e_j = \sum_{j\neq i}(a_j x_i)\,\varphi_i(d(x_j/x_i))$ with $a_j x_i$

of degree 0. The images $\varphi_i(d(x_j/x_i))$ are independent, since the coefficient of e_j is $1/x_i$. So φ_i is an isomorphism onto $\ker \pi|_{U_i}$. On $U_i \cap U_k$, writing $x_j/x_i = (x_j/x_k)(x_k/x_i)$ and $d(x_k/x_i) = -(x_k/x_i)^2 d(x_i/x_k)$ shows $\varphi_k(d(x_j/x_i)) = \varphi_i(d(x_j/x_i))$ by a direct computation. So the φ_i glue. Taking determinants in the exact sequence gives $\omega \otimes \mathcal{O} \cong \det(\mathcal{O}(-1)^{r+1}) = \mathcal{O}(-r-1)$. □

Proposition 55.6.4 (Adjunction). *Let $Y \subseteq X$ be nonsingular varieties over an algebraically closed field, of dimensions $n-1$ and n, with ideal sheaf $\mathcal{I}$. Then $\mathcal{I}$ is invertible, $\mathcal{I}/\mathcal{I}^2 \cong \mathcal{I}|_Y$ is the conormal sheaf, the conormal sequence $0 \to \mathcal{I}/\mathcal{I}^2 \to \Omega_X|_Y \to \Omega_Y \to 0$ is exact on the left, and*

$$\omega_Y \cong \omega_X \otimes \mathcal{I}^{-1}|_Y.$$

For a nonsingular hypersurface Y of degree d in $\mathbb{P}^n$, $\omega_Y \cong \mathcal{O}_Y(d-n-1)$.

Proof. Near a point of Y, $\mathcal{O}_X$ is a regular local ring, hence a unique factorisation domain (Remark 14.11.3), and the height-one prime of Y is principal, generated by g. So $\mathcal{I}$ is invertible and $\mathcal{I}/\mathcal{I}^2 = \mathcal{I} \otimes \mathcal{O}_Y$. The map $\mathcal{I}/\mathcal{I}^2 \to \Omega_X|_Y$ sends g to dg. Both sides of the right-exact conormal sequence are locally free, of ranks 1, n and $n-1$ (Proposition 55.6.2(4)). A surjection of locally free sheaves of ranks n and $n-1$ has locally free kernel of rank 1, which is the image of $\mathcal{I}/\mathcal{I}^2$. A surjection between invertible sheaves is an isomorphism, since locally it is multiplication by a function that generates the unit ideal. So the sequence is exact. Taking determinants gives $\omega_X|_Y \cong \omega_Y \otimes \mathcal{I}/\mathcal{I}^2$. For a hypersurface of degree d, $\mathcal{I} = \mathcal{O}(-d)$ and $\omega_{\mathbb{P}^n} = \mathcal{O}(-n-1)$. □

55.7 Exercises

Exercise 55.7.1. Let $X = \operatorname{Spec} R$ for a discrete valuation ring R with fraction field K, and let $\mathcal{F}$ be the sheaf with $\mathcal{F}(X) = 0$ and $\mathcal{F}(\{\eta\}) = K$ on the open generic point. Show that $\mathcal{F}$ is an $\mathcal{O}_X$-module that is not quasi-coherent, while j_*K for $j\colon \{\eta\} \to X$ is quasi-coherent.

Solution. If $\mathcal{F}$ were quasi-coherent, it would be $\widetilde{\Gamma(X,\mathcal{F})} = 0$ (Theorem 55.3.3), but $\mathcal{F}_\eta = K \neq 0$. On the other hand $j_*K = \widetilde{K}$, since $\widetilde{K}(\{\eta\}) = K_{(0)} = K$ and $\widetilde{K}(X) = K$. □

Exercise 55.7.2. Show that $\mathcal{O}(-1)$ on $\mathbb{P}^r_k$ is the sheaf of sections of the tautological line bundle $\{(x,\ell) \in \mathbb{A}^{r+1} \times \mathbb{P}^r : x \in \ell\}$, and that $\mathcal{O}(1)$ has no nonzero section vanishing on a hyperplane twice.

Solution. On U_i the tautological bundle is trivialised by the section $\ell \mapsto (x_0/x_i, \dots, x_r/x_i)$, and on $U_i \cap U_j$ these trivialisations differ by x_i/x_j. This is the cocycle of $\mathcal{O}(-1)$, whose generator on U_i is $1/x_i$: $1/x_j = (x_i/x_j)(1/x_i)$. A section of $\mathcal{O}(1)$ is a linear form by Proposition 55.5.2, and a nonzero linear form vanishes on its hyperplane to order exactly 1. □

Exercise 55.7.3. Using Proposition 55.6.4, show that a nonsingular plane cubic has trivial canonical sheaf and that a nonsingular plane quartic has $\omega \cong \mathcal{O}(1)$, so that $\Gamma(Y,\omega_Y)$ contains the three linear forms. (In fact it is spanned by them, and the genus is 3; see Chapter 59.)

Solution. $d-n-1 = 3-3 = 0$ and $4-3 = 1$. The restriction $\Gamma(\mathbb{P}^2,\mathcal{O}(1)) \to \Gamma(Y,\mathcal{O}_Y(1))$ is injective because a linear form vanishing on Y would be divisible by the quartic equation. So $\Gamma(Y,\omega_Y)$ contains a 3-dimensional space. □

Exercise 55.7.4. Compute $\Omega_{X/k}$ for the cusp $X = \operatorname{Spec} k[x,y]/(y^2 - x^3)$ and show that it is not locally free.

Solution. By Example 14.12.3, $\Omega = (B\,dx \oplus B\,dy)/(2y\,dy - 3x^2dx)$. Its fibre at the origin has dimension 2, while at other points it has dimension 1 (where $x \neq 0$ or $y \neq 0$, the relation is nonzero modulo the maximal ideal, in characteristic $\neq 2, 3$). As X is reduced, Proposition 55.3.8(3) shows that Ω is not locally free. □

Chapter 56

Cohomology of Coherent Sheaves

56.1 Introduction

The sheaf cohomology of Chapter 26 applied to quasi-coherent sheaves on schemes is computable. On affine schemes it vanishes in positive degrees (Theorem 56.2.3), so on separated schemes it is computed by Čech complexes of affine covers. On projective space everything can be computed explicitly (Theorem 56.3.1). For a projective scheme over a field the cohomology groups of coherent sheaves are finite-dimensional, those of $\mathcal{F}(n)$ vanish for $n \gg 0$, and the Euler characteristic $\chi(\mathcal{F}(n))$ is a polynomial in n, the Hilbert polynomial. Serre duality relates H^i and H^{n-i}. For complex projective manifolds these groups coincide with the Dolbeault cohomology of Chapter 46 by GAGA (Chapter 58), and so compute the Hodge numbers. References: [Har77, Chapter III], [The].

Prerequisites. Chapter 55 (Quasi-coherent and Coherent Sheaves) and Chapter 26 (Sheaf Cohomology).

For an A-module M and $f_1, \dots, f_r \in A$, the *Čech complex* $\check{C}^\bullet(f; M)$ is $\prod_{i_0} M_{f_{i_0}} \to \prod_{i_0<i_1} M_{f_{i_0} f_{i_1}} \to \cdots$ with the alternating differential of Definition 26.5.1. It is the Čech complex of $\widetilde{M}$ for the cover of $\bigcup_i D(f_i)$ by the $D(f_i)$.

56.2 Vanishing on affine schemes

Lemma 56.2.1. *If $f_1, \dots, f_r$ generate the unit ideal of A, then the augmented Čech complex $0 \to M \to \check{C}^\bullet(f; M)$ is exact for every A-module M.*

Proof. A module that becomes zero after inverting each f_j is zero, because every prime omits some f_j (Proposition 14.3.3). Localisation is exact, so it suffices to show that the complex becomes exact after inverting f_j. After inverting f_j it is the augmented Čech complex of M_{f_j} for the elements $f_1, \dots, f_r$ of A_{f_j}, one of which is a unit. The operator $(k\alpha)_{i_0 \cdots i_{p-1}} = \alpha_{j i_0 \cdots i_{p-1}}$ is then defined, since $M_{f_j f_{i_0} \dots} = M_{f_{i_0} \dots}$, and $dk + kd = \mathrm{id}$ as in the proof of Lemma 26.5.2. So the localised complex is contractible. □

Lemma 56.2.2 (Cartan)**.** *Let X be a topological space and $\mathcal{B}$ a basis of quasi-compact open sets closed under finite intersections. Let $\mathcal{F}$ be a sheaf of abelian groups such that $\check{H}^p(\mathcal{V}, \mathcal{F}) = 0$ for all $p > 0$, all $U \in \mathcal{B}$ and all finite covers $\mathcal{V}$ of U by members of $\mathcal{B}$. Then $H^p(U, \mathcal{F}) = 0$ for all $U \in \mathcal{B}$ and $p > 0$.*

Proof. Embed $\mathcal{F}$ into an injective sheaf $\mathcal{I}$ (Theorem 26.2.2) and let $\mathcal{Q} = \mathcal{I}/\mathcal{F}$.

Step 1: $\mathcal{I}(U) \to \mathcal{Q}(U)$ is onto for $U \in \mathcal{B}$. A section $s \in \mathcal{Q}(U)$ lifts locally, and by quasi-compactness we find a finite cover of U by $U_1, \dots, U_m \in \mathcal{B}$ and lifts $t_i \in \mathcal{I}(U_i)$. Then

$t_i - t_j \in \mathcal{F}(U_i \cap U_j)$ is a Čech 1-cocycle, so by hypothesis $t_i - t_j = c_j - c_i$ with $c_i \in \mathcal{F}(U_i)$. The sections $t_i + c_i$ agree on overlaps and glue to a lift of s.

Step 2: $\mathcal{Q}$ satisfies the hypothesis. For a finite cover $\mathcal{V}$ of $U \in \mathcal{B}$ by members of $\mathcal{B}$, all intersections lie in $\mathcal{B}$, so by Step 1 the sequence $0 \to \check{C}^\bullet(\mathcal{V}, \mathcal{F}) \to \check{C}^\bullet(\mathcal{V}, \mathcal{I}) \to \check{C}^\bullet(\mathcal{V}, \mathcal{Q}) \to 0$ is exact. Injective sheaves are flasque (Lemma 26.3.2), so $\check{H}^p(\mathcal{V}, \mathcal{I}) = 0$ for $p > 0$ (Lemma 26.5.2). The long exact sequence gives $\check{H}^p(\mathcal{V}, \mathcal{Q}) \cong \check{H}^{p+1}(\mathcal{V}, \mathcal{F}) = 0$.

Step 3. By the long exact sequence of $0 \to \mathcal{F} \to \mathcal{I} \to \mathcal{Q} \to 0$ on U, $H^1(U, \mathcal{F}) = \operatorname{coker}(\mathcal{I}(U) \to \mathcal{Q}(U)) = 0$ by Step 1, and $H^{p+1}(U, \mathcal{F}) \cong H^p(U, \mathcal{Q})$ for $p \geq 1$. By Step 2 and induction on p, applied to all sheaves satisfying the hypothesis at once, $H^p(U, \mathcal{F}) = 0$ for all $p \geq 1$. □

Theorem 56.2.3 (Serre)**.** *Let $X = \operatorname{Spec} A$ and $\mathcal{F}$ a quasi-coherent sheaf on X. Then $H^p(X, \mathcal{F}) = 0$ for all $p > 0$.*

Proof. Take $\mathcal{B}$ to be the basic open sets $D(g)$. They are quasi-compact (Proposition 54.2.1) and $D(g) \cap D(h) = D(gh)$. A finite cover of $D(g) \cong \operatorname{Spec} A_g$ by basic open sets $D(f_i)$ with $D(f_i) \subseteq D(g)$ is a cover of $\operatorname{Spec} A_g$ by the $D(f_i)$, and the f_i generate the unit ideal of A_g. By Theorem 55.3.3, $\mathcal{F} = \widetilde{M}$ and the Čech complex of the cover is $\check{C}^\bullet(f; M_g)$, which is exact in positive degrees by Lemma 56.2.1. Apply Lemma 56.2.2. □

Counterexample 56.2.4. Theorem 56.2.3 needs quasi-coherence. Let $X = \mathbb{A}^1_k = \operatorname{Spec} k[t]$ with its Zariski topology, and let 0, 1 be the corresponding closed points. Consider the morphism of sheaves of abelian groups

$$\mathbb{Z}_X \to (i_0)_*\mathbb{Z} \oplus (i_1)_*\mathbb{Z}$$

to the skyscraper sheaves at 0 and 1, taking the values of a locally constant function at the two points, and let $\mathcal{K}$ be its kernel. It is surjective on stalks: the stalk of $(i_0)_*\mathbb{Z}$ at a point y is $\mathbb{Z}$ if every open neighbourhood of y contains 0, that is, if $y = 0$, since $\{0\}$ is closed, and 0 otherwise; and at $y = 0$ the map on stalks is the identity of $\mathbb{Z}$. So $0 \to \mathcal{K} \to \mathbb{Z}_X \to (i_0)_*\mathbb{Z} \oplus (i_1)_*\mathbb{Z} \to 0$ is exact. Since X is irreducible, every nonempty open subset is irreducible, hence connected, so $\mathbb{Z}_X(U) = \mathbb{Z}$ for all nonempty U, with the identity as restriction maps; thus $\mathbb{Z}_X$ is flasque and $H^1(X, \mathbb{Z}_X) = 0$ (Proposition 26.3.3), and $\Gamma(X, \mathbb{Z}_X) = \mathbb{Z}$ maps to $\mathbb{Z} \oplus \mathbb{Z}$ diagonally. The long exact sequence gives

$$H^1(X, \mathcal{K}) \cong (\mathbb{Z} \oplus \mathbb{Z})/\mathbb{Z} \cong \mathbb{Z} \neq 0,$$

although X is affine. The sheaf $\mathcal{K}$, the extension by zero of $\mathbb{Z}$ from $X \setminus \{0, 1\}$ (Definition 25.4.3), is not a sheaf of $\mathcal{O}_X$-modules.

Corollary 56.2.5. *Let X be a quasi-compact separated scheme, $\mathcal{U}$ a finite cover by affine open subsets and $\mathcal{F}$ quasi-coherent. Then $H^p(X, \mathcal{F}) \cong \check{H}^p(\mathcal{U}, \mathcal{F})$. In particular $H^p(X, \mathcal{F}) = 0$ for $p \geq |\mathcal{U}|$.*

Proof. Finite intersections of affine opens in a separated scheme are affine (as in the proof of Lemma 55.3.2), so Theorem 56.2.3 and Leray's theorem (Theorem 26.5.3) apply. The alternating Čech complex of a cover with m members vanishes in degrees $\geq m$. □

Remark 56.2.6. Conversely, Serre's criterion says that a Noetherian scheme on which H^1 of every quasi-coherent (even every coherent) ideal sheaf vanishes is affine [Har77, Theorem III.3.7]. Grothendieck's vanishing theorem says that $H^p(X, \mathcal{F}) = 0$ for $p > \dim X$ for every sheaf of abelian groups on a Noetherian space X [Har77, Theorem III.2.7].

56.3 The cohomology of projective space

Theorem 56.3.1. *Let A be a Noetherian ring, $S = A[x_0, \dots, x_r]$ and $X = \mathbb{P}^r_A$ with $r \geq 1$.*

1. $\bigoplus_n H^0(X, \mathcal{O}(n)) \cong S$.
2. $H^i(X, \mathcal{O}(n)) = 0$ *for* $0 < i < r$ *and all* n.
3. $H^r(X, \mathcal{O}(n))$ *is the free A-module with basis the Laurent monomials $x_0^{a_0} \cdots x_r^{a_r}$ with all $a_i \leq -1$ and $\sum a_i = n$. In particular $H^r(X, \mathcal{O}(-r-1)) \cong A$, spanned by $(x_0 \cdots x_r)^{-1}$, and $H^r(X, \mathcal{O}(n)) = 0$ for $n > -r-1$.*
4. *The multiplication pairing $H^0(X, \mathcal{O}(n)) \times H^r(X, \mathcal{O}(-n-r-1)) \to H^r(X, \mathcal{O}(-r-1)) \cong A$ is perfect.*

Proof. Let $\mathcal{F} = \bigoplus_n \mathcal{O}(n)$, a quasi-coherent sheaf. By Corollary 56.2.5 with the standard cover $U_i = D_+(x_i)$, $H^i(X, \mathcal{O}(n))$ is the degree-n part of the cohomology of the graded complex

$$\prod_{i_0} S_{x_{i_0}} \to \prod_{i_0 < i_1} S_{x_{i_0} x_{i_1}} \to \cdots \to S_{x_0 \cdots x_r},$$

which is the Čech complex of $\mathcal{F}$. (1) is Proposition 55.5.2. (3) H^r is the cokernel of $\prod_k S_{x_0 \cdots \widehat{x_k} \cdots x_r} \to S_{x_0 \cdots x_r}$. The target is free on all Laurent monomials, and the image is spanned by those in which some exponent a_k is ≥ 0. The monomials with all $a_i \leq -1$ give a basis of the cokernel. (4) In the basis of (3), $x^b \cdot x^a$ with $b_i \geq 0$, $\sum b_i = n$, and $a_i \leq -1$, $\sum a_i = -n-r-1$ is zero in H^r unless $b_i + a_i = -1$ for all i. So the monomial bases $\{x^b\}$ and $\{x^{-b-\mathbf{1}}\}$ are dual.

(2) By induction on r. The case $r = 1$ is empty. Let $r \geq 2$ and $M^i = \bigoplus_n H^i(X, \mathcal{O}(n))$. *Every element of M^i, $i > 0$, is killed by a power of x_r.* Indeed, localising the Čech complex at x_r gives the Čech complex of S_{x_r} for the cover of U_r by the affine sets $U_i \cap U_r$. Its cohomology is $H^i(U_r, \mathcal{F}|_{U_r}) = 0$ for $i > 0$, as U_r is affine (Theorem 56.2.3), and localisation is exact. *Multiplication by x_r is injective on M^i for $0 < i < r$.* Let $H = V_+(x_r) \cong \mathbb{P}^{r-1}_A$. The exact sequence $0 \to S(-1) \xrightarrow{x_r} S \to S/(x_r) \to 0$ gives $0 \to \mathcal{O}(n-1) \to \mathcal{O}(n) \to \mathcal{O}_H(n) \to 0$, with $H^i(X, \mathcal{O}_H(n)) = H^i(H, \mathcal{O}_H(n))$ (Example 26.7.4). In the long exact sequence

$$H^{i-1}(H, \mathcal{O}_H(n)) \to H^i(X, \mathcal{O}(n-1)) \xrightarrow{x_r} H^i(X, \mathcal{O}(n)),$$

the left term vanishes for $2 \leq i \leq r-1$ by induction ($0 < i-1 < r-1$). For $i = 1$, the map $H^0(X, \mathcal{O}(n)) = S_n \to H^0(H, \mathcal{O}_H(n)) = (S/x_r)_n$ is onto, so again x_r is injective on H^1. An element killed by a power of x_r, on which x_r is injective, is zero. So $M^i = 0$ for $0 < i < r$. □

Corollary 56.3.2. *On $\mathbb{P}^1_k$, $h^0(\mathcal{O}(n)) = \max(n+1, 0)$ and $h^1(\mathcal{O}(n)) = \max(-n-1, 0)$. So $\chi(\mathcal{O}(n)) = n+1$ for all n, and $H^1(\mathbb{P}^1, \mathcal{O}(-2)) \cong k$.*

Proof. Count the monomials $x_0^{a_0} x_1^{a_1}$ with $a_0 + a_1 = n$ and $a_i \geq 0$, respectively $a_i \leq -1$. □

56.4 Finiteness and Serre vanishing

Theorem 56.4.1 (Serre)**.** *Let X be a projective scheme over a Noetherian ring A with a very ample sheaf $\mathcal{O}_X(1)$, and $\mathcal{F}$ a coherent sheaf on X. Then:*

1. *$H^i(X, \mathcal{F})$ is a finitely generated A-module for every i;*
2. *there is n_0 with $H^i(X, \mathcal{F}(n)) = 0$ for all $i > 0$ and $n \geq n_0$.*

Proof. Pushing forward along the closed immersion $X \to \mathbb{P}^r_A$ changes neither the cohomology nor coherence (Example 26.7.4), so let $X = \mathbb{P}^r_A$. Both statements hold for $i > r$ by Corollary 56.2.5. We use descending induction on i, for all coherent sheaves simultaneously. By Theorem 55.5.3 there is an exact sequence $0 \to \mathcal{R} \to \mathcal{E} \to \mathcal{F} \to 0$ with $\mathcal{E}$ a finite direct sum of sheaves $\mathcal{O}(q_j)$ and $\mathcal{R}$ coherent (Corollary 55.3.4). By Theorem 56.3.1, $H^i(\mathcal{E})$ is finitely generated and $H^i(\mathcal{E}(n)) = 0$ for $i > 0$ and n large. The exact sequence $H^i(\mathcal{E}(n)) \to H^i(\mathcal{F}(n)) \to H^{i+1}(\mathcal{R}(n))$ and the induction hypothesis for $\mathcal{R}$ give both statements for $\mathcal{F}$ in degree i. For (1), A is Noetherian, so an extension of a submodule of a finitely generated module by a quotient of a finitely generated module is finitely generated. □

Corollary 56.4.2. *For a projective scheme X over a field k and $\mathcal{F}$ coherent, all $H^i(X, \mathcal{F})$ are finite-dimensional, and they vanish for $i > \dim X$. In particular $\Gamma(X, \mathcal{O}_X)$ is a finite-dimensional k-algebra, equal to k if X is geometrically integral.*

Proof. Finiteness is Theorem 56.4.1(1), and vanishing above the dimension is Grothendieck's theorem (Remark 56.2.6). If X is integral, $\Gamma(X, \mathcal{O}_X)$ is a domain finite over k, hence a field finite over k. If X is geometrically integral, its base change to $\bar{k}$ is $\Gamma(X_{\bar{k}}, \mathcal{O}) = \bar{k}$ by Theorem 53.4.6 (cohomology commutes with flat base change of fields, by the Čech description), so $\Gamma(X, \mathcal{O}_X) = k$. □

56.5 Euler characteristic and Hilbert polynomial

Definition 56.5.1. For a coherent sheaf $\mathcal{F}$ on a projective scheme X over a field k, the *Euler characteristic* is $\chi(\mathcal{F}) = \sum_i (-1)^i h^i(\mathcal{F})$, where $h^i = \dim_k H^i$. It is additive in short exact sequences, by the long exact sequence.

Theorem 56.5.2. *Let $X \subseteq \mathbb{P}^r_k$ be a projective scheme over an infinite field k and $\mathcal{F}$ a coherent sheaf on X. There is a polynomial $P_{\mathcal{F}} \in \mathbb{Q}[t]$, the* Hilbert polynomial, *with $\chi(\mathcal{F}(n)) = P_{\mathcal{F}}(n)$ for all $n \in \mathbb{Z}$. Its degree is $\dim \operatorname{Supp} \mathcal{F}$ ($-\infty$ if $\mathcal{F} = 0$), and $h^0(\mathcal{F}(n)) = P_{\mathcal{F}}(n)$ for $n \gg 0$.*

Proof. Induction on $d = \dim \operatorname{Supp} \mathcal{F}$. If $d \leq 0$, the support is finite, $\mathcal{O}(1)$ is trivial on a neighbourhood of it, so $\mathcal{F}(n) \cong \mathcal{F}$ and χ is constant, equal to $h^0(\mathcal{F})$, which is nonzero if $\mathcal{F} \neq 0$ (a sheaf with finite support is a sum of skyscrapers). Let $d \geq 1$. The *associated points* of $\mathcal{F}$, the points x such that $\mathfrak{m}_x$ is an associated prime of $\mathcal{F}_x$, are finite in number (Proposition 14.8.2, on a finite affine cover). Since k is infinite, there is a linear form h vanishing at none of them: S_1 is not a finite union of proper subspaces. Then multiplication by h is injective on $\mathcal{F}(-1) \to \mathcal{F}$, because its kernel would have an associated point among those of $\mathcal{F}$. So there is an exact sequence

$$0 \to \mathcal{F}(n-1) \xrightarrow{h} \mathcal{F}(n) \to \mathcal{F}_H(n) \to 0, \qquad \mathcal{F}_H = \mathcal{F}/h\mathcal{F}.$$

$\operatorname{Supp} \mathcal{F}_H = \operatorname{Supp} \mathcal{F} \cap V_+(h)$ has dimension $d-1$ by Theorem 53.5.5, since h vanishes at no generic point of a component of the support (these are associated points). By induction $\chi(\mathcal{F}(n)) - \chi(\mathcal{F}(n-1)) = P_{\mathcal{F}_H}(n)$ is a polynomial of degree $d-1$, so $\chi(\mathcal{F}(n))$ is a polynomial of degree d. The last statement follows from Theorem 56.4.1(2). □

Definition 56.5.3. For a closed subscheme $X \subseteq \mathbb{P}^r_k$ of dimension d with Hilbert polynomial $P_X = P_{\mathcal{O}_X}$, the *degree* of X is $d!$ times the leading coefficient of P_X, and the *arithmetic genus* is $p_a(X) = (-1)^d(P_X(0) - 1)$.

Example 56.5.4.

1. $P_{\mathbb{P}^r}(n) = \binom{n+r}{r}$ for all n, by Theorem 56.3.1. So $\mathbb{P}^r$ has degree 1 and $p_a = 0$.

2. For a hypersurface $X = V_+(f)$ of degree d, the sequence $0 \to \mathcal{O}(n-d) \to \mathcal{O}(n) \to \mathcal{O}_X(n) \to 0$ gives $P_X(n) = \binom{n+r}{r} - \binom{n-d+r}{r}$, of degree $r-1$ with leading coefficient $d/(r-1)!$. So X has degree d.

3. For a plane curve of degree d, $P_X(n) = dn + 1 - \frac{(d-1)(d-2)}{2}$, so $p_a = \frac{(d-1)(d-2)}{2}$: 0 for lines and conics, 1 for cubics, 3 for quartics.

4. The twisted cubic $C = v_3(\mathbb{P}^1) \subseteq \mathbb{P}^3$ has $\mathcal{O}_C(n) = \mathcal{O}_{\mathbb{P}^1}(3n)$, so $P_C(n) = 3n + 1$: degree 3, genus 0.

56.6 Serre duality

Theorem 56.6.1 (Serre duality on projective space)**.** *Let $X = \mathbb{P}^r_k$ and $\omega = \mathcal{O}(-r-1)$, with the isomorphism $H^r(X, \omega) \cong k$ of Theorem 56.3.1. For every coherent $\mathcal{F}$ the pairing $\operatorname{Hom}(\mathcal{F}, \omega) \times H^r(X, \mathcal{F}) \to H^r(X, \omega) \cong k$ is perfect.*

Proof. Both $\mathcal{F} \mapsto \operatorname{Hom}(\mathcal{F}, \omega)$ and $\mathcal{F} \mapsto H^r(X, \mathcal{F})^\vee$ are contravariant and left exact: the second because $H^{r+1} = 0$, so H^r is right exact. The pairing gives a natural transformation between them, which is an isomorphism for $\mathcal{F} = \mathcal{O}(q)$ by Theorem 56.3.1(4), since $\operatorname{Hom}(\mathcal{O}(q), \omega) = H^0(\mathcal{O}(-q-r-1))$, and hence for finite sums of such. Every coherent $\mathcal{F}$ has a presentation $\mathcal{E}_1 \to \mathcal{E}_0 \to \mathcal{F} \to 0$ by finite sums of twisting sheaves (Theorem 55.5.3, applied twice). By the five lemma the transformation is an isomorphism for $\mathcal{F}$. □

Theorem 56.6.2 (Serre duality)**.** *Let X be a nonsingular projective variety of dimension n over an algebraically closed field k, with canonical sheaf $\omega_X = \Lambda^n \Omega_{X/k}$. There is an isomorphism $H^n(X, \omega_X) \cong k$, and for every locally free sheaf $\mathcal{E}$ and every i the pairing*

$$H^i(X, \mathcal{E}) \times H^{n-i}(X, \mathcal{E}^\vee \otimes \omega_X) \to H^n(X, \omega_X) \cong k$$

is perfect. More generally $\operatorname{Ext}^i(\mathcal{F}, \omega_X) \cong H^{n-i}(X, \mathcal{F})^\vee$ for coherent $\mathcal{F}$.

This is [Har77, Theorem III.7.6 and Corollary III.7.7], where it is deduced from Theorem 56.6.1 and a computation of Ext sheaves for the embedding $X \subseteq \mathbb{P}^r$, and [Har77, Corollary III.7.12] for the identification of the dualising sheaf with ω_X. For $k = \mathbb{C}$, via GAGA (Chapter 58), it agrees with the analytic Serre duality of Theorem 46.3.4.

Corollary 56.6.3. *For a nonsingular projective variety over k put $h^{p,q} = \dim H^q(X, \Omega^p_X)$. Then $h^{p,q} = h^{n-p,n-q}$, and $\chi(\omega_X) = (-1)^n \chi(\mathcal{O}_X)$.*

Proof. Ω^p is locally free, and $(\Omega^p)^\vee \otimes \omega \cong \Omega^{n-p}$ by the perfect pairing $\Omega^p \otimes \Omega^{n-p} \to \omega$. Apply Theorem 56.6.2. For the second, take $\mathcal{E} = \mathcal{O}_X$. □

56.7 Semicontinuity and base change

Theorem 56.7.1 (Semicontinuity)**.** *Let $f : X \to Y$ be a projective morphism of Noetherian schemes and $\mathcal{F}$ a coherent sheaf on X, flat over Y. For $y \in Y$ let $\mathcal{F}_y$ be the restriction to the fibre X_y. Then:*

1. *$y \mapsto h^i(X_y, \mathcal{F}_y)$ is upper semicontinuous for every i;*
2. *$y \mapsto \chi(\mathcal{F}_y)$ is locally constant, and so is the Hilbert polynomial of $\mathcal{F}_y$;*
3. *(cohomology and base change) if Y is integral and $h^i(X_y, \mathcal{F}_y)$ is constant, then $R^i f_* \mathcal{F}$ is locally free and $R^i f_* \mathcal{F} \otimes k(y) \cong H^i(X_y, \mathcal{F}_y)$ for all y.*

This is [Har77, Theorem III.12.8, Corollary III.12.9 and Theorem III.9.9]. The proof represents the cohomology of all fibres by one bounded complex of finitely generated locally free sheaves on Y. The theorem implies that Hodge numbers are upper semicontinuous in algebraic families; in the Kähler case they are in fact constant, by the Hodge decomposition and the constancy of Betti numbers in smooth families (Chapter 70).

Counterexample 56.7.2. Individual h^i can jump, although χ cannot. For the family of line bundles $\mathcal{L}_t$ of degree 0 on a fixed elliptic curve E, parametrised by $t \in E$ (the Poincaré bundle on $E \times E$), one has $h^0(\mathcal{L}_t) = 1$ if $\mathcal{L}_t$ is trivial and 0 otherwise, while $\chi(\mathcal{L}_t) = 0$ is constant (Chapter 59).

56.8 Exercises

Exercise 56.8.1. Compute $H^1(\mathbb{A}^2_k \setminus \{0\}, \mathcal{O})$ with the Čech complex of the cover $D(x) \cup D(y)$, and deduce again that $\mathbb{A}^2_k \setminus \{0\}$ is not affine.

Solution. The complex is $k[x,y]_x \times k[x,y]_y \to k[x,y]_{xy}$, $(a,b) \mapsto b - a$. Its cokernel has basis the monomials $x^{-i}y^{-j}$ with $i, j \geq 1$, so H^1 is infinite-dimensional. The scheme is quasi-compact and separated, so the Čech complex computes cohomology (Corollary 56.2.5), and it is not affine by Theorem 56.2.3. □

Exercise 56.8.2. Let $C \subseteq \mathbb{P}^2_k$ be a plane curve of degree $d \geq 1$. Show that $h^0(\mathcal{O}_C) = 1$ and $h^1(\mathcal{O}_C) = \frac{(d-1)(d-2)}{2}$.

Solution. The sequence $0 \to \mathcal{O}(-d) \to \mathcal{O} \to \mathcal{O}_C \to 0$ and Theorem 56.3.1 give $0 \to k \to H^0(\mathcal{O}_C) \to H^1(\mathcal{O}(-d)) = 0 \to 0$ and $0 \to H^1(\mathcal{O}_C) \to H^2(\mathcal{O}(-d)) \to H^2(\mathcal{O}) = 0$. The space $H^2(\mathbb{P}^2, \mathcal{O}(-d))$ has basis the monomials $x^a y^b z^c$ with $a, b, c \leq -1$ and $a + b + c = -d$, and there are $\binom{d-1}{2}$ of them. □

Exercise 56.8.3. Show that $\chi(\mathbb{P}^r, \mathcal{O}(n)) = \binom{n+r}{r}$ for all $n \in \mathbb{Z}$, interpreting the binomial coefficient as the polynomial $\frac{(n+1)\cdots(n+r)}{r!}$, and check the value at $n = -r-1$ against Serre duality.

Solution. For $n \geq 0$ only H^0 contributes and has dimension $\binom{n+r}{r}$. For $-r-1 < n < 0$ all cohomology vanishes and the polynomial vanishes too. For $n \leq -r-1$, only H^r contributes, with dimension equal to the number of monomials with exponents ≤ -1 summing to n, which is $\binom{-n-1}{r}$. This equals $(-1)^r\binom{n+r}{r}$ as polynomials in n. At $n = -r-1$ we get $(-1)^r$, matching $h^r(\omega) = h^0(\mathcal{O}) = 1$. □

Exercise 56.8.4. Let $X \subseteq \mathbb{P}^r_k$ be a projective variety of dimension $d \geq 1$ over an infinite field, and H a hyperplane not containing X. Show that $X \cap H$ has the same degree as X.

Solution. From $0 \to \mathcal{O}_X(n-1) \to \mathcal{O}_X(n) \to \mathcal{O}_{X\cap H}(n) \to 0$ we get $P_{X\cap H}(n) = P_X(n) - P_X(n-1)$. If P_X has leading term $e\,n^d/d!$, the difference has leading term $e\,n^{d-1}/(d-1)!$. So both have degree e. □

Chapter 57

Divisors and Line Bundles

57.1 Introduction

Divisors are the codimension-one cycles. They are the first case of the algebraic cycles whose cohomology classes the Hodge conjecture is about, and the case in which the conjecture is a theorem (the Lefschetz $(1,1)$-theorem, Chapter 69). This chapter treats Weil divisors and the class group, Cartier divisors and their identification with line bundles, linear systems and morphisms to projective space, ample line bundles, intersection numbers on surfaces, and the Néron–Severi group. References: [Har77, Sections II.6, II.7 and V.1], [Laz04, Chapter 1], [Bea96, Chapter I].

Prerequisites. Chapter 56 (Cohomology of Coherent Sheaves).

Throughout, schemes are Noetherian and separated. $K(X)$ is the function field of an integral scheme and $\mathcal{K}_X$ the constant sheaf $K(X)$.

57.2 Weil divisors

Definition 57.2.1. Let X be an integral scheme that is *regular in codimension one*: every local ring $\mathcal{O}_{X,x}$ of dimension one is regular, hence a discrete valuation ring (Theorem 14.11.2). A *prime divisor* is a closed integral subscheme Y of codimension one; its generic point η_Y has local ring a DVR with valuation v_Y on $K(X)$. A *Weil divisor* is an element $D = \sum n_Y Y$ of the free abelian group $\operatorname{Div}(X)$ on the prime divisors; it is *effective*, $D \geq 0$, if all $n_Y \geq 0$. For $f \in K(X)^*$ put $(f) = \sum_Y v_Y(f) Y$, a *principal divisor*. Divisors D, D' are *linearly equivalent*, $D \sim D'$, if $D - D'$ is principal, and the *class group* is $\operatorname{Cl}(X) = \operatorname{Div}(X)/\{\text{principal divisors}\}$.

Lemma 57.2.2. *For $f \in K(X)^*$, $v_Y(f) = 0$ for all but finitely many Y, so (f) is a divisor, and $f \mapsto (f)$ is a homomorphism $K(X)^* \to \operatorname{Div}(X)$.*

Proof. f is a regular function with regular inverse on some nonempty open U. The complement $X \setminus U$ is a proper closed subset of the Noetherian space X, so it contains only finitely many prime divisors (they are among its finitely many irreducible components). For Y meeting U, f is a unit at η_Y, so $v_Y(f) = 0$. Additivity is $v_Y(fg) = v_Y(f) + v_Y(g)$. □

Proposition 57.2.3.

1. *Let A be a Noetherian domain. If A is a unique factorisation domain, then $\operatorname{Cl}(\operatorname{Spec} A) = 0$. In particular $\operatorname{Cl}(\mathbb{A}^n_k) = 0$.*
2. *If $Z \subseteq X$ is closed and $U = X \setminus Z$, there is an exact sequence $\bigoplus \mathbb{Z} Z_i \to \operatorname{Cl}(X) \to \operatorname{Cl}(U) \to 0$, the sum running over the irreducible components Z_i of Z of codimension one.*
3. *For $X = \mathbb{P}^n_k$, the degree $\deg \sum n_Y Y = \sum n_Y \deg Y$, where $\deg Y$ is the degree of a homogeneous generator of the ideal of Y, induces $\operatorname{Cl}(\mathbb{P}^n_k) \cong \mathbb{Z}$, generated by a hyperplane H.*

Proof. (1) A prime divisor of $\operatorname{Spec} A$ corresponds to a prime $\mathfrak{p}$ of height one. It contains an irreducible factor f of any of its nonzero elements, and (f) is a nonzero prime in a UFD, so $\mathfrak{p} = (f)$. Then $Y = (f)$ is principal, because $v_{(f)}(f) = 1$ and $v_{(g)}(f) = 0$ for irreducibles g not associate to f. (2) $Y \mapsto Y \cap U$ gives a surjection $\operatorname{Div}(X) \to \operatorname{Div}(U)$, since a prime divisor of U is the trace of its closure. Its kernel is generated by the Z_i, and principal divisors map to principal divisors, $(f)|_U = (f|_U)$. (3) A prime divisor of $\mathbb{P}^n$ is $V_+(g)$ with g irreducible homogeneous, by Theorem 53.5.4 applied to the cone. For $f = g/h$ with g, h homogeneous of the same degree, (f) has degree $\deg g - \deg h = 0$, so deg is defined on Cl. It is onto, as $\deg H = 1$. If $D = \sum n_i Y_i$ with $Y_i = V_+(g_i)$ has degree 0, then $f = \prod_i g_i^{n_i}$ is a quotient of homogeneous polynomials of the same degree, hence an element of $K(\mathbb{P}^n)^*$, and $(f) = D$ because $v_{Y_j}(g_i) = \delta_{ij}$. So deg is injective. □

Example 57.2.4. Let k be a field of characteristic $\neq 2$ and $X = \operatorname{Spec} k[x, y, z]/(xy - z^2)$, the quadric cone. It is normal, and the line $Y = \{y = z = 0\}$ is a prime divisor with $2Y = (y)$, since y vanishes to order 2 along Y (on Y, x is a unit near the generic point and $y = z^2/x$). One shows $\operatorname{Cl}(X) \cong \mathbb{Z}/2\mathbb{Z}$, generated by Y [Har77, Example II.6.5.2]. So Y is not locally principal at the vertex, although $2Y$ is.

57.3 Cartier divisors and invertible sheaves

Definition 57.3.1. Let X be an integral scheme. A *Cartier divisor* is a global section of $\mathcal{K}_X^*/\mathcal{O}_X^*$. It is given by an open cover (U_i) and $f_i \in K(X)^*$ with $f_i/f_j \in \mathcal{O}^*(U_i \cap U_j)$; the f_i are *local equations*. It is *principal* if it is the image of some $f \in K(X)^*$, and *effective* if all $f_i \in \mathcal{O}(U_i)$. The group of Cartier divisors modulo principal ones is $\operatorname{CaCl}(X)$. For a Cartier divisor D, $\mathcal{O}(D) \subseteq \mathcal{K}_X$ is the subsheaf generated by f_i^{-1} on U_i.

Proposition 57.3.2. *Let X be an integral scheme.*

1. *$\mathcal{O}(D)$ is invertible, $\mathcal{O}(D - D') \cong \mathcal{O}(D) \otimes \mathcal{O}(D')^{-1}$, and $D \mapsto \mathcal{O}(D)$ induces an isomorphism $\operatorname{CaCl}(X) \cong \operatorname{Pic}(X)$.*

2. *If X is moreover regular in codimension one and* locally factorial, *meaning that all local rings are unique factorisation domains, for instance if X is regular (Remark 14.11.3), then Weil and Cartier divisors coincide, and $\operatorname{Cl}(X) \cong \operatorname{CaCl}(X) \cong \operatorname{Pic}(X)$.*

Proof. (1) On U_i, $\mathcal{O}(D)$ is free with generator f_i^{-1}, so it is invertible, and multiplication of local equations corresponds to tensor product of sub-$\mathcal{O}_X$-modules of $\mathcal{K}_X$. $\mathcal{O}(D) \cong \mathcal{O}_X$ if and only if there is a global generator f^{-1}, if and only if D is principal, so the map is injective on classes. It is onto: an invertible sheaf $\mathcal{L}$ on an integral scheme embeds in $\mathcal{L} \otimes \mathcal{K}_X \cong \mathcal{K}_X$ (a constant sheaf, as X is irreducible), and a local generator e_i of $\mathcal{L}$ becomes some $f_i^{-1} \in K(X)^*$, so $\mathcal{L} \cong \mathcal{O}(D)$ with local equations f_i. (2) A Cartier divisor gives the Weil divisor $\sum_Y v_Y(f_Y)Y$, where f_Y is a local equation near η_Y. Conversely a Weil divisor is locally principal: near x, each prime divisor through x corresponds to a height-one prime of the UFD $\mathcal{O}_{X,x}$, which is principal, generated by some g. So D has the local equation $\prod g^{n_Y}$ on a neighbourhood of x. These constructions are inverse, and principal divisors correspond. □

Example 57.3.3. $\operatorname{Pic}(\mathbb{P}_k^n) \cong \mathbb{Z}$, generated by $\mathcal{O}(1) = \mathcal{O}(H)$, by Propositions 57.2.3(3) and 57.3.2. For the quadric cone of Example 57.2.4, $\operatorname{Pic}(X) = 0$ while $\operatorname{Cl}(X) = \mathbb{Z}/2$: the line Y is a Weil divisor that is not Cartier.

57.4 Linear systems and morphisms to projective space

Proposition 57.4.1. *Let X be a scheme over a ring A. Morphisms $\varphi\colon X \to \mathbb{P}^n_A$ over A correspond bijectively to invertible sheaves $\mathcal{L}$ with sections $s_0, \dots, s_n \in \Gamma(X, \mathcal{L})$ generating $\mathcal{L}$, up to isomorphism of such data. The correspondence sends φ to $\varphi^*\mathcal{O}(1)$ and $s_i = \varphi^* x_i$.*

Proof. Given $(\mathcal{L}, s_i)$, the open sets X_{s_i} cover X because the s_i generate. On X_{s_i} the quotients s_j/s_i are regular functions, and the ring map $A[x_0/x_i, \dots, x_n/x_i] \to \Gamma(X_{s_i}, \mathcal{O})$, $x_j/x_i \mapsto s_j/s_i$, defines $X_{s_i} \to U_i$ (Theorem 54.2.4). These glue, since the quotients satisfy the relations of the gluing of $\mathbb{P}^n$. Then $\varphi^*\mathcal{O}(1) \cong \mathcal{L}$, the generator x_i of $\mathcal{O}(1)$ on U_i being sent to s_i. Conversely $\varphi^* x_i$ generate $\varphi^*\mathcal{O}(1)$, and φ is recovered on each $\varphi^{-1}(U_i) = X_{\varphi^* x_i}$ from the quotients. □

Definition 57.4.2. Let X be a projective variety over an algebraically closed field k and D a Cartier divisor. The *complete linear system* $|D|$ is the set of effective divisors linearly equivalent to D. A *linear system* is a subset corresponding to a linear subspace of $\Gamma(X, \mathcal{O}(D))$ as below, and its *base points* are the common points of its members.

Proposition 57.4.3. *In the situation of Definition 57.4.2, sending a nonzero $s \in \Gamma(X, \mathcal{O}(D))$ to its divisor of zeros $(s)_0$ gives a bijection $(\Gamma(X, \mathcal{O}(D)) \setminus 0)/k^* \cong |D|$. So $|D|$ is a projective space of dimension $h^0(\mathcal{O}(D)) - 1$, finite by Corollary 56.4.2.*

Proof. A section s of $\mathcal{O}(D) \subseteq \mathcal{K}$ is an element $f \in K(X)$ with $f_i f$ regular on each U_i, and $(s)_0$ is the effective divisor with local equations $f_i f$, which equals $D + (f)$. Conversely, if $D' = D + (f) \geq 0$, then $f \in \Gamma(X, \mathcal{O}(D))$. Two sections give the same divisor if and only if their quotient is a unit on X, that is, an element of $\Gamma(X, \mathcal{O}_X)^* = k^*$ (Corollary 56.4.2). □

Proposition 57.4.4. *Let X be a projective scheme over an algebraically closed field k and $\varphi\colon X \to \mathbb{P}^n_k$ the morphism given by sections $s_0, \dots, s_n$ generating $\mathcal{L}$, spanning $V \subseteq \Gamma(X, \mathcal{L})$. Then φ is a closed immersion if and only if V separates points (for closed $P \neq Q$ there is $s \in V$ vanishing at P but not at Q) and separates tangent vectors (for every closed P, the sections vanishing at P span $\mathfrak{m}_P \mathcal{L}_P / \mathfrak{m}_P^2 \mathcal{L}_P$).*

This is [Har77, Proposition II.7.3]; it is the algebraic version of Lemma 50.6.2. The proof uses that a proper morphism that is injective on closed points and on tangent spaces is a closed immersion, by Nakayama's lemma applied to $\varphi_* \mathcal{O}_X$.

57.5 Ample line bundles

Theorem 57.5.1. *Let X be of finite type over a Noetherian ring A and $\mathcal{L}$ invertible. Then $\mathcal{L}$ is ample (Definition 55.5.4) if and only if $\mathcal{L}^m$ is very ample over A for some $m > 0$.*

Proof. If $\mathcal{L}^m$ is very ample, X is an open subscheme of a projective scheme $\bar{X}$ with $\mathcal{L}^m = \mathcal{O}(1)|_X$. A coherent $\mathcal{F}$ on X extends to a coherent sheaf on $\bar{X}$ [Har77, Exercise II.5.15]. Applying Theorem 55.5.3 to the extensions of $\mathcal{F}, \mathcal{F} \otimes \mathcal{L}, \dots, \mathcal{F} \otimes \mathcal{L}^{m-1}$ shows that $\mathcal{F} \otimes \mathcal{L}^n$ is globally generated for $n \gg 0$.

Conversely, let $\mathcal{L}$ be ample. *Step 1: affine cover by sets X_s.* Let $P \in X$, U an affine neighbourhood on which $\mathcal{L}$ is trivial, and $\mathcal{I}$ the ideal sheaf of the reduced closed subscheme $X \setminus U$. For some $n > 0$, $\mathcal{I} \otimes \mathcal{L}^n$ is globally generated, so there is $s \in \Gamma(X, \mathcal{I} \otimes \mathcal{L}^n) \subseteq \Gamma(X, \mathcal{L}^n)$ with $s_P \notin \mathfrak{m}_P \mathcal{L}_P^n$. Then $P \in X_s \subseteq U$, and X_s is a basic open subset of the affine U (where s becomes a function), hence affine. By quasi-compactness finitely many $X_{s_1}, \dots, X_{s_k}$ cover X, and replacing the s_i by powers we may assume they are all sections of the same $\mathcal{L}^n$.

Step 2: coordinates. Let $B_i = \Gamma(X_{s_i}, \mathcal{O})$, a finitely generated A-algebra with generators b_{ij}. By Lemma 55.3.2(2) there is N such that $s_i^N b_{ij}$ extends to a global section c_{ij} of $\mathcal{L}^{nN}$ for all i, j. The sections s_i^N and c_{ij} of $\mathcal{L}^{nN}$ generate it, because the s_i^N have no common

zero. Let $\varphi\colon X \to \mathbb{P}^M_A$ be the morphism they define (Proposition 57.4.1), with homogeneous coordinates y_i, y_{ij} corresponding to s_i^N, c_{ij}.

Step 3: φ is an immersion. We have $\varphi^{-1}(D_+(y_i)) = X_{s_i}$, and the map $X_{s_i} \to D_+(y_i)$ corresponds to the ring map $A[y/y_i] \to B_i$ sending y_{ij}/y_i to $c_{ij}/s_i^N = b_{ij}$. It is surjective, so $X_{s_i} \to D_+(y_i)$ is a closed immersion. Hence φ is a closed immersion of X into the open subset $\bigcup_i D_+(y_i)$ of $\mathbb{P}^M_A$, that is, an immersion, and $\mathcal{L}^{nN} = \varphi^*\mathcal{O}(1)$ is very ample. □

Theorem 57.5.2 (Cohomological criterion)**.** *Let X be proper over a Noetherian ring A and $\mathcal{L}$ invertible. Then $\mathcal{L}$ is ample if and only if for every coherent $\mathcal{F}$ there is n_0 with $H^i(X, \mathcal{F} \otimes \mathcal{L}^n) = 0$ for all $i > 0$ and $n \geq n_0$.*

Proof. If $\mathcal{L}$ is ample, some $\mathcal{L}^m$ is very ample (Theorem 57.5.1). As X is proper, the immersion is closed, so X is projective. Apply Theorem 56.4.1(2) to $\mathcal{F} \otimes \mathcal{L}^r$, $0 \leq r < m$.

Conversely, let P be a closed point with ideal sheaf $\mathcal{I}_P$. From $0 \to \mathcal{I}_P\mathcal{F} \to \mathcal{F} \to \mathcal{F} \otimes k(P) \to 0$ and $H^1(\mathcal{I}_P\mathcal{F} \otimes \mathcal{L}^n) = 0$ for n large, $\Gamma(\mathcal{F} \otimes \mathcal{L}^n) \to \mathcal{F} \otimes \mathcal{L}^n \otimes k(P)$ is onto. By Nakayama's lemma (Proposition 55.3.8(1)), $\mathcal{F} \otimes \mathcal{L}^n$ is then generated by global sections on a neighbourhood of P, a neighbourhood that depends on n. Apply this to $\mathcal{F} = \mathcal{O}_X$ to find n_0 and a neighbourhood V of P on which $\mathcal{L}^{n_0}$ is globally generated. Apply it to $\mathcal{F} \otimes \mathcal{L}^r$, $0 \leq r < n_0$, to find n_1 and neighbourhoods V_r on which $\mathcal{F} \otimes \mathcal{L}^{r+n_1}$ is globally generated. On $W_P = V \cap \bigcap_r V_r$, every $\mathcal{F} \otimes \mathcal{L}^n$ with $n \geq n_1$ is globally generated, being a tensor product $(\mathcal{F} \otimes \mathcal{L}^{r+n_1}) \otimes (\mathcal{L}^{n_0})^q$. The W_P for closed points P cover the quasi-compact X, since every nonempty closed subset contains a closed point, so finitely many suffice. □

Example 57.5.3.

1. $\mathcal{O}(d)$ on $\mathbb{P}^n_k$, $n \geq 1$, is ample if and only if $d > 0$: for $d > 0$ it is very ample (Example 53.4.7(2)), and for $d \leq 0$ and $\mathcal{F} = \mathcal{O}(-1)$ the sheaf $\mathcal{F} \otimes \mathcal{O}(d)^m = \mathcal{O}(dm - 1)$ is nonzero without nonzero global sections (Proposition 55.5.2).

2. The restriction of an ample sheaf to a closed subscheme is ample, and a tensor product of ample sheaves is ample.

3. For a smooth projective variety over $\mathbb{C}$, a line bundle is ample if and only if the associated holomorphic line bundle is positive (Definition 45.4.2). This combines Theorem 50.6.5 with GAGA (Chapter 58).

57.6 Intersection numbers on surfaces

Definition 57.6.1. Let X be a nonsingular projective surface over an algebraically closed field k. For invertible sheaves $\mathcal{L}$, $\mathcal{M}$ put

$$(\mathcal{L} \cdot \mathcal{M}) = \chi(\mathcal{O}_X) - \chi(\mathcal{L}^{-1}) - \chi(\mathcal{M}^{-1}) + \chi(\mathcal{L}^{-1} \otimes \mathcal{M}^{-1}),$$

and for divisors $D \cdot E = (\mathcal{O}(D) \cdot \mathcal{O}(E))$, $D^2 = D \cdot D$.

Proposition 57.6.2. *The pairing $(D, E) \mapsto D \cdot E$ is symmetric, and for an effective divisor C (for instance a curve) and any divisor D,*

$$D \cdot C = \chi(\mathcal{O}_C) - \chi(\mathcal{O}_X(-D)|_C).$$

Proof. Symmetry is clear. Tensor $0 \to \mathcal{O}(-C) \to \mathcal{O} \to \mathcal{O}_C \to 0$ with $\mathcal{O}$ and with $\mathcal{O}(-D)$ and use additivity of χ: $\chi(\mathcal{O}) - \chi(\mathcal{O}(-C)) = \chi(\mathcal{O}_C)$ and $\chi(\mathcal{O}(-D)) - \chi(\mathcal{O}(-D-C)) = \chi(\mathcal{O}(-D)|_C)$. Subtracting the second equation from the first gives the formula. □

Theorem 57.6.3. *The pairing $(D, E) \mapsto D \cdot E$ is bilinear on* $\mathrm{Pic}(X)$. *For an irreducible curve C, $D \cdot C = \deg(\mathcal{O}(D)|_{\tilde{C}})$, the degree of the pullback to the normalisation. If C and D are curves meeting transversally in finitely many points, then $C \cdot D = \#(C \cap D)$, and if C, D are distinct irreducible curves, $C \cdot D \geq 0$.*

This is [Har77, Theorem V.1.1, Proposition V.1.4 and Exercise V.1.1]; see also [Bea96, Chapter I], where Definition 57.6.1 is taken as the definition. By Proposition 57.6.2 and the Riemann–Roch theorem on curves (Chapter 59), $D \cdot C$ is the degree of $\mathcal{O}(D)|_C$, which is additive in D. Linearity in the other variable follows by writing any divisor as a difference of very ample ones, which are linearly equivalent to nonsingular curves by Bertini's theorem.

Example 57.6.4.

1. On $\mathbb{P}^2$, $H \cdot H = \chi(\mathcal{O}) - 2\chi(\mathcal{O}(-1)) + \chi(\mathcal{O}(-2)) = 1 - 0 + 0 = 1$ by Theorem 56.3.1. By bilinearity, curves of degrees d and e satisfy $C \cdot D = de$ (*Bézout's theorem*).

2. On $\mathbb{P}^1 \times \mathbb{P}^1$, with fibres $F_1 = \{pt\} \times \mathbb{P}^1$ and $F_2 = \mathbb{P}^1 \times \{pt\}$, $\chi(\mathcal{O}(a, b)) = (a + 1)(b + 1)$ by the Künneth formula for the Čech complexes, and the definition gives $F_1^2 = F_2^2 = 0$ and $F_1 \cdot F_2 = 1$.

3. If $\pi\colon \tilde{X} \to X$ is the blow-up of a point with exceptional curve $E \cong \mathbb{P}^1$, then $E^2 = -1$, since $\mathcal{O}(E)|_E \cong \mathcal{O}_{\mathbb{P}^1}(-1)$ (Lemma 50.6.3(1) in the complex case; the algebraic proof is the same).

Theorem 57.6.5 (Hodge index theorem)**.** *Let H be an ample divisor on a nonsingular projective surface X. If $D \cdot H = 0$ and D is not numerically trivial (Definition 57.7.1), then $D^2 < 0$. Equivalently, the intersection form on* $\mathrm{Num}(X) \otimes \mathbb{R}$ *has signature* $(1, \rho - 1)$.

This is [Har77, Theorem V.1.9], proved there with Riemann–Roch on surfaces. Over $\mathbb{C}$ it is the part of Corollary 49.5.2 concerning classes of divisors, once intersection numbers are identified with cup products of cycle classes (Chapter 63).

Theorem 57.6.6 (Nakai–Moishezon criterion)**.** *A divisor D on a nonsingular projective surface is ample if and only if $D^2 > 0$ and $D \cdot C > 0$ for every irreducible curve C.*

This is [Har77, Theorem V.1.10]. In higher dimensions the criterion reads $D^{\dim V} \cdot V > 0$ for every closed subvariety V [Laz04, Theorem 1.2.23].

57.7 The Néron–Severi group

Definition 57.7.1. Let X be a nonsingular projective variety over an algebraically closed field. A divisor D is *numerically trivial*, $D \equiv 0$, if $\deg(\mathcal{O}(D)|_C) = 0$ for every irreducible curve $C \subseteq X$. $\mathrm{Num}(X) = \mathrm{Pic}(X)/{\equiv}$. Two divisors are *algebraically equivalent* if they are members of a family of divisors parametrised by a connected base, and $\mathrm{NS}(X) = \mathrm{Pic}(X)/\{\text{algebraic equivalence}\}$ is the *Néron–Severi group*. The rank of $\mathrm{Num}(X)$ is the *Picard number* $\rho(X)$.

Theorem 57.7.2 (Theorem of the base)**.** $\mathrm{NS}(X)$ *is a finitely generated abelian group, algebraic equivalence implies numerical equivalence, and* $\mathrm{Num}(X)$ *is a free abelian group of finite rank $\rho(X)$, the quotient of* $\mathrm{NS}(X)$ *by its torsion.*

This is due to Severi and Néron; see [Laz04, Section 1.1]. Over $\mathbb{C}$ it follows from Hodge theory. The kernel $\mathrm{Pic}^0(X)$ of the first Chern class is the Picard torus (Remark 45.2.5), and $\mathrm{NS}(X)$ embeds into $H^2(X; \mathbb{Z})$ with image the integral classes of type $(1, 1)$ (Chapter 69). So $\rho(X) \leq h^{1,1}(X)$.

Example 57.7.3. 1. $\rho(\mathbb{P}^n) = 1$ and $\rho(\mathbb{P}^1 \times \mathbb{P}^1) = 2$, since $\mathrm{Pic}(\mathbb{P}^1 \times \mathbb{P}^1) \cong \mathbb{Z}^2$, generated by the two rulings.

2. For a quartic surface $X \subseteq \mathbb{P}^3$, $1 \le \rho(X) \le h^{1,1} = 20$ (Example 48.7.4). The very general quartic has $\rho = 1$ (Noether–Lefschetz), and the Fermat quartic $x^4 + y^4 + z^4 + w^4 = 0$ has $\rho = 20$.

3. For $E \times E'$ a product of elliptic curves, $\rho = 2 + \operatorname{rank} \mathrm{Hom}(E, E')$, which is 2, 3 or 4 (Chapter 61).

57.8 Exercises

Exercise 57.8.1. Let $Y \subseteq \mathbb{P}^n_k$ be an irreducible hypersurface of degree d. Show that $\mathrm{Cl}(\mathbb{P}^n \setminus Y) \cong \mathbb{Z}/d\mathbb{Z}$.

Solution. By Proposition 57.2.3(2), $\mathrm{Cl}(\mathbb{P}^n \setminus Y) = \mathrm{Cl}(\mathbb{P}^n)/\mathbb{Z}[Y] = \mathbb{Z}/d\mathbb{Z}$, using (3) and $\deg Y = d$. □

Exercise 57.8.2. Show that the conics in $\mathbb{P}^2$ form a linear system $|2H|$ of dimension 5 without base points, and that the morphism it defines is the Veronese embedding $v_2 \colon \mathbb{P}^2 \to \mathbb{P}^5$.

Solution. $\Gamma(\mathbb{P}^2, \mathcal{O}(2))$ is the 6-dimensional space of quadratic forms (Proposition 55.5.2), so $\dim |2H| = 5$ by Proposition 57.4.3. The monomials x_i^2 have no common zero, so there are no base points, and the morphism of Proposition 57.4.1 given by the six quadratic monomials is v_2. □

Exercise 57.8.3. Let C be an irreducible curve on a nonsingular projective surface with $C^2 < 0$. Show that C is the only member of $|C|$, that is, $h^0(\mathcal{O}(C)) = 1$.

Solution. Let $D \in |C|$ and write $D = aC + D'$ with $a \ge 0$ and D' effective not containing C. If $a \ge 1$, then $(a-1)C + D'$ is effective and linearly equivalent to 0. By Proposition 57.4.3 with $h^0(\mathcal{O}) = 1$, the only effective divisor in $|0|$ is 0, so $a = 1$, $D' = 0$ and $D = C$. If $a = 0$, then $C^2 = C \cdot D' \ge 0$ by Theorem 57.6.3, since D' is a sum of curves different from C. This contradicts $C^2 < 0$. □

Exercise 57.8.4. Show that if $\mathcal{L}$ is ample on X and $Y \subseteq X$ is a closed subscheme, then $\mathcal{L}|_Y$ is ample, using Theorem 57.5.2 (for X proper over a Noetherian ring).

Solution. For coherent $\mathcal{G}$ on Y, $i_*\mathcal{G}$ is coherent on X and $H^j(Y, \mathcal{G} \otimes \mathcal{L}|_Y^n) = H^j(X, i_*\mathcal{G} \otimes \mathcal{L}^n)$ by the projection formula and Example 26.7.4. These vanish for $j > 0$ and $n \gg 0$. □

Chapter 58

Analytic Spaces and GAGA

58.1 Introduction

A complex algebraic variety X has an underlying complex analytic space X^{an}: the set $X(\mathbb{C})$ with the Euclidean topology and the sheaf of holomorphic functions. For projective varieties the two worlds agree. Chow's theorem says that closed analytic subsets of $\mathbb{P}^n$ are algebraic, and Serre's GAGA theorems say that coherent algebraic and analytic sheaves, and their cohomology, correspond. Consequently the Hodge theory of compact Kähler manifolds (Part VII) applies to smooth projective varieties, and the Hodge filtration can be defined algebraically through the algebraic de Rham complex, whose hypercohomology computes $H^\bullet(X^{\mathrm{an}};\mathbb{C})$ by Grothendieck's comparison theorem. This algebraicity is the starting point of the theory of absolute Hodge classes (Chapter 75). References: [Ser56], [Har77, Appendix B], [GH78, Chapter 1], [Gro66].

Prerequisites. Chapter 57 (Divisors and Line Bundles), Chapter 50 (Vanishing and Embedding Theorems) and Chapter 44 (Complex Manifolds).

58.2 Complex analytic spaces

Definition 58.2.1. A *local model* is a closed subset $V = \{f_1 = \cdots = f_k = 0\}$ of an open set $U \subseteq \mathbb{C}^n$, with f_i holomorphic on U, together with the sheaf of rings $\mathcal{O}_V = (\mathcal{O}_U/(f_1,\dots,f_k))|_V$. A *complex analytic space* is a Hausdorff locally ringed space over $\mathbb{C}$ that is locally isomorphic to a local model. Morphisms are morphisms of locally ringed spaces over $\mathbb{C}$. A sheaf of $\mathcal{O}$-modules is *coherent* if it is locally the cokernel of a map $\mathcal{O}^p \to \mathcal{O}^q$.

Theorem 58.2.2 (Oka and Cartan)**.** *The structure sheaf of a complex analytic space is coherent, and the ideal sheaf of holomorphic functions vanishing on a closed analytic subset is coherent. Kernels, cokernels and images of morphisms of coherent sheaves are coherent.*

This is [GR84, Chapters 2 and 4]. The proof is based on the Weierstrass division theorem (Theorem 31.3.3). Complex manifolds are complex analytic spaces, and conversely an analytic space whose local rings are regular is a complex manifold.

58.3 The analytification functor

Proposition 58.3.1. *Let X be a scheme locally of finite type over $\mathbb{C}$. There is a complex analytic space X^{an} with a morphism of locally ringed spaces $\varphi\colon X^{\mathrm{an}} \to X$, such that every morphism from a complex analytic space to X (as locally ringed spaces over $\mathbb{C}$) factors uniquely through φ. The*

underlying set of X^{an} is $X(\mathbb{C})$, the set of closed points. $X \mapsto X^{\mathrm{an}}$ is a functor, and for X affine, $X = \operatorname{Spec} \mathbb{C}[x_1, \dots, x_n]/(f_1, \dots, f_k)$, X^{an} is the local model $\{f = 0\} \subseteq \mathbb{C}^n$.

Sketch of proof. For affine X take the local model. Its universal property follows from that of the polynomial ring: a morphism from an analytic space Z to $\mathbb{A}^n_{\mathbb{C}}$ is an n-tuple of holomorphic functions on Z (Theorem 54.2.4), and it factors through X exactly when the f_i vanish on them. By the universal property the affine constructions glue (Proposition 54.3.2). The underlying set is $X(\mathbb{C})$ because points of an analytic space have residue field $\mathbb{C}$, and these are the closed points (Proposition 54.3.6). The omitted verification is the gluing, which is formal; see [Har77, Appendix B] and [Gro71, Exposé XII]. □

Proposition 58.3.2. *Let X be of finite type over $\mathbb{C}$.*

1. *X is smooth over $\mathbb{C}$ if and only if X^{an} is a complex manifold, and then $\dim_{\mathbb{C}} X^{\mathrm{an}} = \dim X$ at each point.*

2. *X is separated if and only if X^{an} is Hausdorff, and proper if and only if X^{an} is compact.*

3. *X is connected if and only if X^{an} is connected.*

4. *For a coherent sheaf $\mathcal{F}$ on X, $\mathcal{F}^{\mathrm{an}} = \varphi^*\mathcal{F}$ is coherent, and $\mathcal{F} \mapsto \mathcal{F}^{\mathrm{an}}$ is exact, because $\mathcal{O}_{X^{\mathrm{an}},x}$ is flat over $\mathcal{O}_{X,x}$. Moreover $(\Omega_{X/\mathbb{C}})^{\mathrm{an}} = \Omega_{X^{\mathrm{an}}}$, the sheaf of holomorphic 1-forms.*

Proof. (1) If X is smooth, locally $X = \operatorname{Spec} \mathbb{C}[x_1, \dots, x_{n+r}]/(g_1, \dots, g_r)$ with the Jacobian minor with respect to $x_{n+1}, \dots, x_{n+r}$ invertible (Definition 54.8.4). By the holomorphic implicit function theorem (a consequence of Theorem 27.11.2 and the Cauchy–Riemann equations), X^{an} is locally the graph of a holomorphic map from an open set of $\mathbb{C}^n$, so it is a complex manifold of dimension n. Conversely, if X^{an} is a manifold of dimension n at x, the completions of $\mathcal{O}_{X,x}$ and $\mathcal{O}_{X^{\mathrm{an}},x}$ agree (both are $\mathbb{C}[[x_1, \dots, x_N]]/(f)$), so $\mathcal{O}_{X,x}$ is regular, and X is smooth at x by Proposition 54.8.5. Statements (2)–(4) are in [Gro71, Exposé XII] and [Ser56, Section 2]. □

58.4 Chow's theorem and GAGA

Theorem 58.4.1 (Serre's GAGA)**.** *Let X be a projective scheme over $\mathbb{C}$.*

1. *For every coherent sheaf $\mathcal{F}$ on X the natural maps $H^i(X, \mathcal{F}) \to H^i(X^{\mathrm{an}}, \mathcal{F}^{\mathrm{an}})$ are isomorphisms.*

2. *$\mathcal{F} \mapsto \mathcal{F}^{\mathrm{an}}$ is an equivalence between coherent sheaves on X and coherent analytic sheaves on X^{an}.*

This is [Ser56, Théorèmes 1–3]. The proof reduces to $X = \mathbb{P}^n$ by pushing forward, verifies (1) for the sheaves $\mathcal{O}(q)$ by comparing the two computations of their cohomology, and deduces (1) in general by induction on n and exact sequences. For (2), the analytic version of Serre's generation theorem (Theorem 55.5.3) presents every coherent analytic sheaf as a cokernel of a map between sums of sheaves $\mathcal{O}(q)^{\mathrm{an}}$, and (1) applied to $\mathcal{H}om$ sheaves shows that such maps are algebraic.

Corollary 58.4.2 (Chow's theorem)**.** *Every closed analytic subset of $\mathbb{P}^n_{\mathbb{C}}$ is the zero set of finitely many homogeneous polynomials.*

Proof. Let $Z \subseteq \mathbb{P}^{n,\mathrm{an}}$ be closed analytic. Its ideal sheaf $\mathcal{I} \subseteq \mathcal{O}^{\mathrm{an}}$ is coherent (Theorem 58.2.2). By Theorem 58.4.1(2), the inclusion $\mathcal{I} \to \mathcal{O}^{\mathrm{an}}$ is the analytification of a morphism $\mathcal{J} \to \mathcal{O}_{\mathbb{P}^n}$ of coherent sheaves, which is injective because analytification is exact and faithful. So $\mathcal{J}$ is an ideal sheaf, defining a closed subscheme Y (Proposition 55.3.6), and Y^{an} has ideal sheaf $\mathcal{J}^{\mathrm{an}} = \mathcal{I}$, so its underlying set is Z. The closed subscheme Y is cut out by homogeneous polynomials (Proposition 54.6.2 and Theorem 55.5.3). □

Corollary 58.4.3. *Let X and Y be projective varieties over $\mathbb{C}$.*

1. *Every holomorphic map $X^{\mathrm{an}} \to Y^{\mathrm{an}}$ is the analytification of a unique morphism $X \to Y$.*
2. *$\mathrm{Pic}(X) \to \mathrm{Pic}(X^{\mathrm{an}})$ is an isomorphism, and every meromorphic function on X^{an} is rational.*
3. *A compact complex manifold that admits a holomorphic embedding into some $\mathbb{CP}^N$ is X^{an} for a nonsingular projective variety X, unique up to isomorphism.*

Proof. (1) The graph of f is a closed analytic subset of $(X \times Y)^{\mathrm{an}} \subseteq \mathbb{P}^{N,\mathrm{an}}$ (Segre), hence algebraic by Chow's theorem, and it is a closed subvariety Γ whose projection to X is bijective and an analytic isomorphism. By Theorem 58.4.1(2) applied to $\mathcal{O}_X \to p_*\mathcal{O}_\Gamma$, whose analytification is an isomorphism, the projection is an algebraic isomorphism, and f is the composite with the second projection. Uniqueness holds because $X(\mathbb{C})$ is dense in X. (2) Invertible sheaves correspond under Theorem 58.4.1(2), since the equivalence is compatible with $\otimes$. A meromorphic function f with polar divisor D is the quotient fs_D/s_D of two holomorphic sections of the analytic line bundle $\mathcal{O}(D)$. This bundle and its sections are algebraic by Theorem 58.4.1, so f is rational. (3) The image is a closed analytic subset, hence algebraic by Chow's theorem, and it is nonsingular by Proposition 58.3.2(1). Uniqueness is (1). □

Corollary 58.4.4. *Every holomorphic map $f : \mathbb{CP}^n \to \mathbb{CP}^m$ is of the form $[x] \mapsto [F_0(x) : \cdots : F_m(x)]$ for homogeneous polynomials $F_0, \ldots, F_m$ of a common degree $d \geq 0$ with no common zero in $\mathbb{C}^{n+1} \setminus \{0\}$. If $m < n$, every such map is constant.*

Proof. By Corollary 58.4.3(1), f is a morphism $\mathbb{P}^n \to \mathbb{P}^m$. By Proposition 57.4.1 it is given by the invertible sheaf $\mathcal{L} = f^*\mathcal{O}(1)$ and the generating sections $s_i = f^*x_i$. By Example 57.3.3, $\mathcal{L} \cong \mathcal{O}(d)$ for some $d \in \mathbb{Z}$, and by Theorem 56.3.1 its global sections are the homogeneous polynomials of degree d if $d \geq 0$ and 0 if $d < 0$; since the s_i generate, $d \geq 0$, and generating means having no common zero. If $d = 0$ the F_i are constants and f is constant. Suppose $d \geq 1$ and $m < n$. Put $Y_{-1} = \mathbb{P}^n$ and $Y_j = Y_{j-1} \cap Z(F_j)$. By induction every irreducible component of Y_j has dimension at least $n - j - 1$ and $Y_j \neq \emptyset$ for $j \leq m$: a component Y of Y_{j-1} has dimension $\geq n - j \geq n - m \geq 1$, and $Y \cap Z(F_j)$ is either Y (if $F_j \in I(Y)$) or nonempty with components of dimension $\dim Y - 1$ (Theorem 53.5.5). So $Y_m \neq \emptyset$, and the F_i have a common zero, a contradiction. □

Counterexample 58.4.5. GAGA fails without properness. The exponential is a holomorphic function on $\mathbb{A}^{1,\mathrm{an}} = \mathbb{C}$ that is not a polynomial, so $\Gamma(\mathbb{A}^1, \mathcal{O}) \neq \Gamma(\mathbb{C}, \mathcal{O}^{\mathrm{an}})$. The holomorphic map $\exp : \mathbb{C} \to \mathbb{C}^*$ is not algebraic, since an algebraic map $\mathbb{A}^1 \to \mathbb{A}^1 \setminus 0$ is constant: it corresponds to a unit of $\mathbb{C}[t]$. Chow's theorem also fails in $\mathbb{C}^n$: the graph of exp is a closed analytic subset of $\mathbb{C}^2$ that is not algebraic, since it meets the line $y = 1$ in infinitely many points.

Counterexample 58.4.6. Not every compact complex manifold is algebraic, even when it is Kähler: a general complex torus of dimension ≥ 2 is not of the form X^{an} for any variety (Example 47.4.4), and the Hopf surface is not even Kähler. There are also nonsingular complete varieties that are not projective, such as Hironaka's example [Har77, Appendix B,

Example 3.4.1]; their analytifications are compact complex manifolds that are algebraic but not projective.

58.5 Algebraic de Rham cohomology

Definition 58.5.1. Let X be a smooth scheme over a field k. The *algebraic de Rham complex* is $\Omega^\bullet_{X/k} = (\mathcal{O}_X \xrightarrow{d} \Omega^1_{X/k} \xrightarrow{d} \Omega^2_{X/k} \to \cdots)$, with $\Omega^p = \Lambda^p \Omega^1$ and d the k-linear exterior derivative. The *algebraic de Rham cohomology* is the hypercohomology $H^n_{\mathrm{dR}}(X/k) = \mathbb{H}^n(X, \Omega^\bullet_{X/k})$ in the Zariski topology, and the *Hodge filtration* is $F^p H^n_{\mathrm{dR}}(X/k) = \operatorname{im}\left(\mathbb{H}^n(X, \Omega^{\geq p}_{X/k}) \to H^n_{\mathrm{dR}}(X/k)\right)$. The spectral sequence of the filtration by the truncations $\Omega^{\geq p}$ is the *Hodge-to-de Rham spectral sequence* $E_1^{p,q} = H^q(X, \Omega^p_{X/k}) \Rightarrow H^{p+q}_{\mathrm{dR}}(X/k)$ (Proposition 26.7.1).

Theorem 58.5.2 (Grothendieck)**.** *For a smooth scheme X of finite type over $\mathbb{C}$ there are natural isomorphisms $H^n_{\mathrm{dR}}(X/\mathbb{C}) \cong H^n(X^{\mathrm{an}}; \mathbb{C})$.*

Proof. We prove it for X projective; the general case is in [Gro66], proved with resolution of singularities. The morphism $\varphi\colon X^{\mathrm{an}} \to X$ gives maps $\Omega^p_{X/\mathbb{C}} \to \varphi_* \Omega^p_{X^{\mathrm{an}}}$ compatible with d, hence a map of hypercohomologies $\mathbb{H}^n(X, \Omega^\bullet_{X/\mathbb{C}}) \to \mathbb{H}^n(X^{\mathrm{an}}, \Omega^\bullet_{X^{\mathrm{an}}})$ compatible with the filtrations by the truncations. On E_1 it is $H^q(X, \Omega^p_{X/\mathbb{C}}) \to H^q(X^{\mathrm{an}}, \Omega^p_{X^{\mathrm{an}}})$, an isomorphism by Theorem 58.4.1(1) and Proposition 58.3.2(4). A morphism of first-quadrant spectral sequences that is an isomorphism on E_1 is an isomorphism on all pages. As the filtrations are finite, the five lemma gives an isomorphism on the abutments. Finally $\mathbb{H}^n(X^{\mathrm{an}}, \Omega^\bullet_{X^{\mathrm{an}}}) \cong H^n(X^{\mathrm{an}}; \mathbb{C})$ by the holomorphic Poincaré lemma (Theorem 46.2.4). □

Corollary 58.5.3. *Let X be a smooth projective variety over $\mathbb{C}$. Then X^{an} is a compact Kähler manifold, the Hodge-to-de Rham spectral sequence of X degenerates at E_1, and under Theorem 58.5.2*

$$F^p H^n(X^{\mathrm{an}}; \mathbb{C}) = \bigoplus_{r \geq p} H^{r,n-r}(X^{\mathrm{an}}) \quad \text{corresponds to} \quad F^p H^n_{\mathrm{dR}}(X/\mathbb{C}), \qquad H^{p,q}(X^{\mathrm{an}}) \cong H^q(X, \Omega^p_{X/\mathbb{C}}).$$

If $X = X_0 \times_k \mathbb{C}$ for a subfield $k \subseteq \mathbb{C}$ and a smooth projective X_0 over k, then $H^n_{\mathrm{dR}}(X/\mathbb{C}) = H^n_{\mathrm{dR}}(X_0/k) \otimes_k \mathbb{C}$, compatibly with the Hodge filtrations. So the Hodge filtration is defined over k.

Proof. X^{an} is a compact complex submanifold of $\mathbb{CP}^N$ (Proposition 58.3.2), hence Kähler (Corollary 47.4.3). The E_1 terms have total dimension $\sum_{p+q=n} h^{p,q} = b_n$ by Corollary 48.5.3, so all differentials vanish. The statement on filtrations is Remark 48.5.5 transported by the comparison, which respects the filtrations by construction. For base change, $\Omega^p_{X/\mathbb{C}} = \Omega^p_{X_0/k} \otimes_k \mathbb{C}$ (Proposition 55.6.2(3)), and hypercohomology can be computed by Čech complexes of a finite affine cover (Corollary 56.2.5), which commute with the flat base change $- \otimes_k \mathbb{C}$. □

Corollary 58.5.4. *Let X be a smooth projective variety over $\mathbb{C}$ and $\sigma \in \operatorname{Aut}(\mathbb{C})$. The conjugate variety X^σ (Remark 54.9.3) has the same Betti numbers and Hodge numbers as X. The map σ induces a σ-linear isomorphism $H^n_{\mathrm{dR}}(X/\mathbb{C}) \to H^n_{\mathrm{dR}}(X^\sigma/\mathbb{C})$ that preserves the Hodge filtrations.*

Proof. The scheme X^σ is obtained from X by the base change $\sigma\colon \operatorname{Spec}\mathbb{C} \to \operatorname{Spec}\mathbb{C}$. It is again smooth and projective, and $\Omega^p_{X^\sigma} = \sigma^* \Omega^p_X$. So $H^q(X^\sigma, \Omega^p) = H^q(X, \Omega^p) \otimes_{\mathbb{C},\sigma} \mathbb{C}$ by flat base change of Čech complexes, and the same holds for de Rham cohomology and its Hodge filtration. Now apply Theorem 58.5.2 and Corollary 58.5.3. □

Remark 58.5.5. The spaces X^{an} and $(X^{\sigma})^{\mathrm{an}}$ need not be homeomorphic, and can have different fundamental groups [Ser64]. The isomorphism of Corollary 58.5.4 does not preserve the rational structures $H^n(X^{\mathrm{an}};\mathbb{Q})$. Whether it maps Hodge classes to Hodge classes is the question of *absolute Hodge classes* (Chapter 75); Deligne proved it for abelian varieties [Del82]. Deligne and Illusie gave an algebraic proof of the degeneration of the Hodge-to-de Rham spectral sequence in characteristic 0 [DI87].

58.6 Smooth projective varieties as Kähler manifolds

Theorem 58.6.1. *The functor $X \mapsto X^{\mathrm{an}}$ is an equivalence between the category of nonsingular projective varieties over $\mathbb{C}$ and the category of compact complex manifolds that admit a positive line bundle (equivalently, a Hodge metric, Corollary 50.2.4), with holomorphic maps. A line bundle $\mathcal{L}$ on a nonsingular projective variety is ample if and only if $\mathcal{L}^{\mathrm{an}}$ is positive.*

Proof. Full faithfulness is Corollary 58.4.3(1). A compact complex manifold with a positive line bundle embeds into some $\mathbb{CP}^N$ (Theorem 50.6.5), so it is X^{an} by Corollary 58.4.3(3). Conversely $O(1)$ restricted to $X^{\mathrm{an}} \subseteq \mathbb{CP}^N$ is positive (Proposition 50.2.1(1)). If $\mathcal{L}$ is ample, $\mathcal{L}^m$ is very ample (Theorem 57.5.1), so $\mathcal{L}^{m,\mathrm{an}}$ is positive, and so is $\mathcal{L}^{\mathrm{an}}$, by dividing the curvature by m. If $\mathcal{L}^{\mathrm{an}}$ is positive, then $\mathcal{L}^{k,\mathrm{an}}$ is very ample for large k by Theorem 50.6.5. The sections involved are algebraic by Theorem 58.4.1(1), and the embedding is algebraic by Corollary 58.4.3(1). So $\mathcal{L}^k$ is very ample and $\mathcal{L}$ is ample. □

Remark 58.6.2. Consequently the Hodge numbers $h^{p,q} = \dim H^q(X, \Omega^p)$ of a smooth projective variety are algebraic invariants, computable by coherent cohomology. The cohomology classes of algebraic cycles (Chapter 63) are Hodge classes (Proposition 48.5.8), and by Chow's theorem the analytic subvarieties of X^{an} are exactly the algebraic ones. So on a projective manifold the Hodge conjecture may equivalently be stated with analytic cycles. For compact Kähler manifolds the analogous statement with analytic cycles is false (Chapter 73).

58.7 Exercises

Exercise 58.7.1. Show that every compact Riemann surface is C^{an} for a unique nonsingular projective curve C, and that its field of meromorphic functions is a finitely generated extension of $\mathbb{C}$ of transcendence degree 1.

Solution. A compact Riemann surface is Kähler with $H^{2,0} = 0$, hence projective by Corollary 50.6.7, that is, it embeds into some $\mathbb{CP}^N$. Apply Corollary 58.4.3(3). Meromorphic functions are rational by Corollary 58.4.3(2), so the field is $K(C)$, of transcendence degree $\dim C = 1$ (Theorem 53.5.3). □

Exercise 58.7.2. Let $C \subseteq \mathbb{P}^2_{\mathbb{C}}$ be a nonsingular curve of degree d. Show that C^{an} is a Riemann surface of topological genus $(d-1)(d-2)/2$.

Solution. By Exercise 56.8.2, $h^1(C, O_C) = (d-1)(d-2)/2$. By GAGA this is $h^{0,1}(C^{\mathrm{an}})$, and by the Hodge decomposition $b_1 = 2h^{0,1}$ (Example 48.7.1), so the genus $b_1/2$ equals $(d-1)(d-2)/2$. □

Exercise 58.7.3. Let $X \subseteq \mathbb{P}^3_{\mathbb{C}}$ be a nonsingular quartic surface. Compute $h^{0,1}$, $h^{0,2}$ and $h^{2,0}$ algebraically.

Solution. $h^{0,q} = h^q(X, O_X)$ by Corollary 58.5.3. The sequence $0 \to O(-4) \to O \to O_X \to 0$ on $\mathbb{P}^3$ and Theorem 56.3.1 give $h^0(O_X) = 1$, $h^1(O_X) = 0$ and $h^2(O_X) = h^3(\mathbb{P}^3, O(-4)) = 1$. By Serre duality, or directly from $\omega_X \cong O_X$ (Proposition 55.6.4), $h^{2,0} = h^0(\omega_X) = 1$. □

Chapter 59

Algebraic Curves and Riemann Surfaces

59.1 Introduction

Curves are the one-dimensional case, in which almost everything can be computed. Over $\mathbb{C}$, nonsingular projective curves are the same as compact Riemann surfaces (Theorem 59.2.5). The Riemann–Roch theorem (Theorem 59.3.2) computes dimensions of spaces of sections, and from it follow the classification of curves of genus 0 and 1, the Riemann–Hurwitz formula, and the theory of the canonical embedding. The Jacobian of a curve is the first nontrivial example of a Hodge structure that is not of Tate type, and its theory (Abel–Jacobi) leads to the abelian varieties of Chapter 61 and the intermediate Jacobians of Chapter 71. References: [Har77, Chapter IV], [GH78, Chapter 2].

Prerequisites. Chapter 58 (Analytic Spaces and GAGA).

Throughout, k is algebraically closed, and a *curve* is a nonsingular projective variety of dimension one over k, unless stated otherwise. Points are closed points. For a curve X, divisors are finite formal sums $D = \sum n_P P$ (Weil and Cartier divisors agree by Proposition 57.3.2), $\deg D = \sum n_P$, $\mathcal{O}(D)$ is the associated invertible sheaf, $l(D) = h^0(\mathcal{O}(D))$, and K is a *canonical divisor*, $\mathcal{O}(K) \cong \omega_X = \Omega_{X/k}$.

59.2 Curves and their function fields

Proposition 59.2.1. *Let X be a nonsingular curve (not necessarily projective). Every local ring $\mathcal{O}_{X,P}$ is a discrete valuation ring of $K(X)$ with residue field k. A rational map from X to a projective variety Y extends to a morphism $X \to Y$.*

Proof. $\mathcal{O}_{X,P}$ is a regular local ring of dimension one, hence a DVR (Theorem 14.11.2), with residue field k by the Nullstellensatz. For the extension, let $Y \subseteq \mathbb{P}^n$ and let the map be defined near P except at P, given by rational functions $[f_0 : \cdots : f_n]$. Let v be the valuation at P and choose m with $v(f_m)$ minimal among the nonzero f_i. Then $f_i/f_m \in \mathcal{O}_{X,P}$ for all i, and $[f_0/f_m : \cdots : f_n/f_m]$ is a morphism on a neighbourhood of P that agrees with the map where both are defined. It lands in Y because Y is closed. (This is the valuative argument of Theorem 54.5.5.) $\square$

Theorem 59.2.2. *The functor $X \mapsto K(X)$ is an anti-equivalence between the category of curves with nonconstant morphisms and the category of function fields of one variable over k (finitely generated extensions of transcendence degree 1). Every nonconstant morphism of curves $f : X \to Y$ is finite*

and surjective, and its degree $\deg f = [K(X) : K(Y)]$ *satisfies*

$$\sum_{P \mapsto Q} e_P = \deg f \qquad \text{for every } Q \in Y,$$

where the ramification index $e_P = v_P(t)$ *for a uniformiser* t *at* Q.

Proof. By Theorem 53.7.3, nonconstant morphisms $X \to Y$ give k-embeddings $K(Y) \to K(X)$, and every such embedding comes from a dominant rational map $X \dashrightarrow Y$, which is a morphism by Proposition 59.2.1. This gives full faithfulness. That every function field K is $K(X)$ for a curve X is [Har77, Theorem I.6.9]: the points of X are the discrete valuation rings of K containing k, and X is covered by the affine curves $\operatorname{Spec} A'$, where A' is the integral closure of $k[t]$ in K for a nonconstant $t \in K$.

Let $f : X \to Y$ be nonconstant. Its image is closed (Theorem 54.5.5) and irreducible, not a point, hence Y. For an affine $V = \operatorname{Spec} B \subseteq Y$, $f^{-1}(V)$ consists of the points whose valuation rings contain B. These are the points of $\operatorname{Spec} A'$, where A' is the integral closure of B in $K(X)$, because a DVR is integrally closed and every local ring of A' at a maximal ideal is a DVR. By the finiteness of integral closure [Har77, Theorem I.3.9A], A' is a finite B-module, so f is finite. Localising at Q, $A'_Q = A' \otimes_B \mathcal{O}_{Y,Q}$ is a finitely generated torsion-free module over the DVR $\mathcal{O}_{Y,Q}$, hence free (Theorem 8.11.3), of rank $[K(X) : K(Y)]$ since $A'_Q \otimes K(Y) = K(X)$. So $\dim_k A'_Q/tA'_Q = \deg f$. The semilocal ring A'_Q has maximal ideals the points P over Q, and by the Chinese remainder theorem $A'_Q/tA'_Q \cong \prod_{P \mapsto Q} \mathcal{O}_{X,P}/t\mathcal{O}_{X,P}$, where the factor at P has dimension $v_P(t) = e_P$. □

Corollary 59.2.3. *For nonconstant* $f \in K(X)$, $\deg(f)_0 = \deg(f)_\infty = [K(X) : k(f)]$, *where* $(f)_0$ *and* $(f)_\infty$ *are the divisors of zeros and poles. So principal divisors have degree* 0, *and* deg *induces a surjection* $\operatorname{Pic}(X) \to \mathbb{Z}$ *with kernel* $\operatorname{Pic}^0(X)$. *For a nonconstant morphism* $f : X \to Y$ *and a divisor* D *on* Y, $\deg f^*D = \deg f \cdot \deg D$.

Proof. f defines a morphism $X \to \mathbb{P}^1$ (Proposition 59.2.1) of degree $[K(X) : k(f)]$, and $(f)_0$, $(f)_\infty$ are the pullbacks of the points 0 and ∞, with multiplicities e_P. Apply Theorem 59.2.2. The last statement is the same formula applied to each point of D. □

Lemma 59.2.4. *If* $\deg D < 0$ *then* $l(D) = 0$. *If* $\deg D = 0$, *then* $l(D) \leq 1$, *with equality if and only if* $D \sim 0$.

Proof. A nonzero section of $\mathcal{O}(D)$ gives an effective divisor $D' \sim D$ (Proposition 57.4.3), and $\deg D' \geq 0$, with equality only for $D' = 0$, by Corollary 59.2.3. □

Theorem 59.2.5. *The functor* $X \mapsto X^{\mathrm{an}}$ *is an equivalence between complex curves and compact Riemann surfaces. The topological genus of* X^{an} *equals* $g(X) = h^1(X, \mathcal{O}_X) = h^0(X, \omega_X)$.

Proof. A compact Riemann surface is Kähler with $H^{2,0} = 0$, hence projective (Corollary 50.6.7), hence of the form X^{an} (Corollary 58.4.3); full faithfulness is also part of that corollary. By GAGA and the Hodge decomposition (Example 48.7.1), $b_1(X^{\mathrm{an}})/2 = h^{0,1} = h^1(X, \mathcal{O}_X) = h^{1,0} = h^0(X, \omega_X)$. □

59.3 The Riemann–Roch theorem

Definition 59.3.1. The *genus* of a curve is $g = h^1(X, \mathcal{O}_X)$. By Serre duality (Theorem 56.6.2) also $g = h^0(X, \omega_X) = l(K)$.

Theorem 59.3.2 (Riemann–Roch). *For every divisor D on a curve X of genus g,*

$$l(D) - l(K - D) = \deg D + 1 - g.$$

Proof. First, $\chi(\mathcal{O}(D)) = \deg D + 1 - g$. This holds for $D = 0$, since $h^0(\mathcal{O}_X) = 1$ (Corollary 56.4.2) and $h^1 = g$. For a point P, the sequence $0 \to \mathcal{O}(D) \to \mathcal{O}(D + P) \to k(P) \to 0$, where $k(P)$ is the skyscraper sheaf at P, gives $\chi(\mathcal{O}(D + P)) = \chi(\mathcal{O}(D)) + 1$. Both sides of the formula change by 1 when a point is added or removed, and every divisor is reached from 0 this way. Second, $h^1(\mathcal{O}(D)) = h^0(\omega \otimes \mathcal{O}(-D)) = l(K - D)$ by Serre duality. □

Corollary 59.3.3. *Let X be a curve of genus g.*

1. $\deg K = 2g - 2$.
2. *If* $\deg D > 2g - 2$, *then* $l(D) = \deg D + 1 - g$.
3. *If* $\deg D \geq 2g$, *then* $|D|$ *has no base points; if* $\deg D \geq 2g + 1$, *then D is very ample.*
4. $X \cong \mathbb{P}^1$ *if and only if* $g = 0$, *if and only if there are points* $P \neq Q$ *with* $P \sim Q$.

Proof. (1) Put $D = K$: $g - 1 = \deg K + 1 - g$. (2) $\deg(K - D) < 0$, so $l(K - D) = 0$ by Lemma 59.2.4. (3) By (2), $l(D - P) = l(D) - 1$ if $\deg D - 1 > 2g - 2$, so P is not a base point. Similarly $l(D - P - Q) = l(D) - 2$ for all P, Q (possibly equal) if $\deg D - 2 > 2g - 2$, which means that $|D|$ separates points and tangent vectors, and D is very ample by Proposition 57.4.4. (4) If $g = 0$, then $l(P) = 2$ by (2), and a nonconstant $f \in \Gamma(\mathcal{O}(P))$ has a single simple pole, so $[K(X) : k(f)] = 1$ by Corollary 59.2.3, and $X \cong \mathbb{P}^1$ by Theorem 59.2.2. On $\mathbb{P}^1$ all points are linearly equivalent. Conversely, if $P \sim Q$ with $P \neq Q$, then $l(P) \geq 2$, and as before $X \cong \mathbb{P}^1$, which has $g = h^1(\mathcal{O}_{\mathbb{P}^1}) = 0$ (Corollary 56.3.2). □

Example 59.3.4. Let $g = 1$ and fix a point O. Then $K \sim 0$, since $\deg K = 0$ and $l(K) = 1$ (Lemma 59.2.4). The divisor $3O$ is very ample with $l(3O) = 3$, so it embeds X as a plane cubic. Choosing coordinates $1, x, y$ with poles of order 0, 2, 3 at O ($l(2O) = 2$, $l(3O) = 3$), the seven functions $1, x, y, x^2, xy, x^3, y^2$ in the 6-dimensional space $\Gamma(\mathcal{O}(6O))$ are linearly dependent, which gives a Weierstrass equation $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$. The map $P \mapsto \mathcal{O}(P - O)$ is a bijection $X \to \mathrm{Pic}^0(X)$ (Exercise 59.7.1), which gives X the structure of an abelian group with identity O: the *elliptic curve*.

59.4 The Riemann–Hurwitz formula

Theorem 59.4.1 (Riemann–Hurwitz). *Let $f : X \to Y$ be a nonconstant morphism of curves of degree n such that $K(X)/K(Y)$ is separable and every ramification index e_P is prime to the characteristic (always true in characteristic 0). Then*

$$2g(X) - 2 = n\,(2g(Y) - 2) + \sum_{P \in X} (e_P - 1).$$

Proof. Let ω be a nonzero rational differential on Y, so that $\deg(\omega) = 2g(Y) - 2$ by Corollary 59.3.3(1). Its pullback $f^*\omega$ is a nonzero rational differential on X, by separability. At $P \mapsto Q$ let t be a uniformiser at Q and s at P, so $t = us^e$ with u a unit and $e = e_P$. Then $dt = (eus^{e-1} + s^e\, du/ds)\, ds$, which has order exactly $e - 1$ at P because $e \neq 0$ in k. Writing $\omega = h\, dt$ near Q gives $v_P(f^*\omega) = e_P v_Q(\omega) + e_P - 1$. Summing over P and using $\sum_{P \mapsto Q} e_P = n$ (Theorem 59.2.2) gives $\deg(f^*\omega) = n \deg(\omega) + \sum_P (e_P - 1)$. □

Example 59.4.2.

1. A *hyperelliptic* curve $y^2 = h(x)$, with h a polynomial of degree $2g+1$ or $2g+2$ without repeated roots (characteristic $\neq 2$), has a degree-2 map to $\mathbb{P}^1$ branched at the $2g+2$ roots of h (and ∞ if $\deg h$ is odd). Riemann–Hurwitz gives $2g(X) - 2 = 2(-2) + 2g + 2$, so $g(X) = g$.
2. For the Fermat curve $x^d + y^d = z^d$ and the projection $[x : y : z] \mapsto [x : z]$ of degree d, the ramification points are the d points with $y = 0$, each with $e = d$. So $2g-2 = -2d+d(d-1)$, and $g = (d-1)(d-2)/2$, as for every plane curve of degree d (Example 56.5.4).
3. Over $\mathbb{C}$, Riemann–Hurwitz is the Euler characteristic computation $\chi(X) = n\chi(Y) - \sum(e_P - 1)$ for a branched covering of surfaces.

Corollary 59.4.3 (Lüroth)**.** *Let k be algebraically closed of characteristic 0. Every field L with $k \subsetneq L \subseteq k(t)$ is of the form $L = k(s)$ for some $s \in L$. Equivalently, a curve that receives a nonconstant morphism from $\mathbb{P}^1$ is isomorphic to $\mathbb{P}^1$.*

Proof. Choose $u \in L \setminus k$. It is transcendental over k, since k is algebraically closed, so $k(t)$ is algebraic and finitely generated, hence finite, over $k(u)$. As a $k(u)$-subspace of $k(t)$, L is finite over $k(u)$, so it is a function field of one variable over k. By Theorem 59.2.2, $L = K(Y)$ for a curve Y, and the inclusion $L \subseteq k(t) = K(\mathbb{P}^1)$ comes from a nonconstant morphism $f : \mathbb{P}^1 \to Y$, separable because the characteristic is 0. By Theorem 59.4.1, $-2 = \deg f\,(2g(Y)-2) + \sum_P (e_P - 1) \geq \deg f\,(2g(Y)-2)$, so $g(Y) = 0$ and $Y \cong \mathbb{P}^1$ by Corollary 59.3.3(4). Hence $L \cong K(\mathbb{P}^1)$ is generated over k by one element. The second statement is the same argument. □

Counterexample 59.4.4. The separability hypothesis in Theorem 59.4.1 cannot be dropped. Let $\operatorname{char} k = p > 0$ and $F : \mathbb{P}^1 \to \mathbb{P}^1$, $[x : y] \mapsto [x^p : y^p]$. It is bijective on points, since every element of the algebraically closed field k has exactly one p-th root ($x^p - a^p = (x-a)^p$). On function fields it is the inclusion $k(t^p) \subseteq k(t)$, of degree p and purely inseparable, since t is the only root of $X^p - t^p = (X-t)^p$. At a point $a \in k$, the uniformiser $x - a^p$ at $F(a) = a^p$ pulls back to $x^p - a^p = (x-a)^p$, so $e_a = p$: every point is ramified, in accordance with $\sum_{P \mapsto Q} e_P = \deg F = p$ and a single preimage. The Riemann–Hurwitz formula would give $-2 = -2p + \sum_P (p-1)$, a sum with infinitely many nonzero terms. The proof breaks down because pullbacks of differentials vanish: $d(x^p) = px^{p-1}\,dx = 0$.

59.5 Hyperelliptic curves and the canonical embedding

Definition 59.5.1. A curve of genus $g \geq 2$ is *hyperelliptic* if it has a morphism of degree 2 to $\mathbb{P}^1$, equivalently a divisor D with $\deg D = 2$ and $l(D) = 2$ (a g^1_2).

Theorem 59.5.2. *Let X be a curve of genus $g \geq 1$. Then $|K|$ has no base points. If $g \geq 2$, then K is very ample if and only if X is not hyperelliptic. Every curve of genus 2 is hyperelliptic.*

Proof. By Riemann–Roch, $l(K-P) = g-2+l(P)$, and $l(P) = 1$ since $X \not\cong \mathbb{P}^1$ (Corollary 59.3.3(4)). So $l(K-P) = g-1 = l(K)-1$ and P is not a base point. Similarly $l(K-P-Q) = g-3+l(P+Q)$, so $l(K-P-Q) = l(K)-2$ for all P, Q if and only if $l(P+Q) = 1$ for all P, Q, that is, if and only if there is no g^1_2 among the divisors $P+Q$. Every g^1_2 is of this form, since it contains an effective divisor of degree 2. By Proposition 57.4.4, K is very ample exactly when X is not hyperelliptic. For $g = 2$, $\deg K = 2$ and $l(K) = 2$, so K is a g^1_2. □

Example 59.5.3. A nonsingular plane quartic has genus 3 and $\omega \cong \mathcal{O}(1)$ (Exercise 55.7.3), so the plane embedding is the canonical embedding, and plane quartics are not hyperelliptic. Conversely every non-hyperelliptic curve of genus 3 is canonically a plane quartic, since its canonical image in $\mathbb{P}^2$ is a curve of degree $\deg K = 4$.

59.6 The Jacobian and the Abel–Jacobi theorem

Definition 59.6.1. Let X be a compact Riemann surface of genus g. The *Jacobian* is the complex torus $J(X) = H^0(X,\Omega^1)^\vee / H_1(X;\mathbb{Z})$, where $\gamma \in H_1(X;\mathbb{Z})$ acts by $\omega \mapsto \int_\gamma \omega$. Fixing a base point P_0, the *Abel–Jacobi map* is $u\colon X \to J(X)$, $u(P) = (\omega \mapsto \int_{P_0}^{P} \omega)$ modulo periods; it extends additively to divisors, and on degree-zero divisors it does not depend on P_0.

Proposition 59.6.2. *$H_1(X;\mathbb{Z}) \to H^0(X,\Omega^1)^\vee$ is injective with image a lattice, so $J(X)$ is a compact complex torus of dimension g. Moreover $\mathrm{Pic}^0(X^{\mathrm{an}}) \cong H^1(X,\mathcal{O})/H^1(X;\mathbb{Z}) \cong J(X)$ as complex tori.*

Proof. Holomorphic 1-forms are closed (Proposition 48.5.6), so the integrals depend only on homology classes. By the Hodge decomposition, $H^1(X;\mathbb{R}) \to H^{0,1} \cong H^1(X,\mathcal{O})$ is an isomorphism of real vector spaces (Exercise 48.8.4), so the image of $H^1(X;\mathbb{Z})$ is a lattice in the g-dimensional complex space $H^1(X,\mathcal{O})$, and by Theorem 45.2.4(2), $\mathrm{Pic}^0(X^{\mathrm{an}}) \cong H^1(X,\mathcal{O})/H^1(X;\mathbb{Z})$. Serre duality identifies $H^1(X,\mathcal{O})$ with $H^0(X,\Omega^1)^\vee$ via $(\alpha,\omega) \mapsto \int_X \alpha\wedge\omega$ for a Dolbeault representative α, and Poincaré duality identifies $H^1(X;\mathbb{Z})$ with $H_1(X;\mathbb{Z})$. Under these identifications, the class of a closed real 1-form η Poincaré dual to γ is sent to $\omega \mapsto \int_X \eta^{0,1}\wedge\omega = \int_X \eta\wedge\omega = \int_\gamma \omega$, since $\eta^{1,0}\wedge\omega = 0$ for type reasons. So the lattices correspond. □

Theorem 59.6.3 (Abel and Jacobi)**.** *Let X be a compact Riemann surface of genus g.*

1. *(Abel) A divisor D of degree 0 is principal if and only if $u(D) = 0$ in $J(X)$. Equivalently, under Proposition 59.6.2 the class of $\mathcal{O}(D)$ in $\mathrm{Pic}^0(X)$ corresponds to $u(D)$.*

2. *(Jacobi inversion) Every divisor class of degree g contains an effective divisor. So the map $X^{(g)} = X^g/S_g \to \mathrm{Pic}^g(X)$, and hence $u\colon X^{(g)} \to J(X)$, is surjective. It is injective over a dense open subset.*

Sketch of proof. (2) By Riemann–Roch, $l(D) \geq \deg D + 1 - g = 1$ for $\deg D = g$, so D is linearly equivalent to an effective divisor. Choosing points $P_1,\dots,P_g$ successively, each outside the base locus of $|K - P_1 - \cdots - P_{i-1}|$, gives $l(K - P_1 - \cdots - P_g) = 0$, hence $l(P_1 + \cdots + P_g) = 1$ by Riemann–Roch; for such divisors the fibre of $X^{(g)} \to \mathrm{Pic}^g(X)$ is the single point $|D|$. (1) One direction: if $D = (f)$, let $D_t = f^*(t)$ for $t \in \mathbb{P}^1$. Then $t \mapsto u(D_t - D_\infty)$ is a holomorphic map $\mathbb{P}^1 \to J(X)$, which lifts to the universal cover $\mathbb{C}^g$ because $\mathbb{P}^1$ is simply connected, and is therefore constant. It vanishes at $t = \infty$, so $u(D) = u(D_0 - D_\infty) = 0$. The converse, which we omit, constructs a meromorphic function with divisor D as the exponential of an abelian integral of the third kind whose periods are integral [GH78, Chapter 2, Section 2]. □

Remark 59.6.4. The Jacobian is determined by the weight-one Hodge structure $H^1(X;\mathbb{Z})$ with $H^1(X;\mathbb{C}) = H^{1,0} \oplus H^{0,1}$, and it is polarised by the intersection form, which satisfies the Riemann bilinear relations (Example 49.4.4). So $J(X)$ is an abelian variety (Chapter 61). By Torelli's theorem the polarised Jacobian determines X [GH78, Chapter 2, Section 7]. Griffiths' intermediate Jacobians (Chapter 71) generalise this construction to higher-dimensional varieties.

59.7 Exercises

Exercise 59.7.1. Let X be a curve of genus 1 with a point O. Show that $P \mapsto \mathcal{O}(P - O)$ is a bijection $X \to \mathrm{Pic}^0(X)$.

Solution. Injective: if $P-O \sim Q-O$ with $P \neq Q$, then $P \sim Q$ and $X \cong \mathbb{P}^1$ by Corollary 59.3.3(4), contradicting $g = 1$. Surjective: for D of degree 0, $l(D+O) = 1$ by Riemann–Roch ($\deg(D+O) = 1 > 0 = 2g-2$), so $D + O \sim P$ for an effective divisor P of degree 1, that is, a point. □

Exercise 59.7.2. Show that $g(X) \geq g(Y)$ for every nonconstant morphism of curves $X \to Y$ in characteristic 0. Deduce that there is no nonconstant morphism $\mathbb{P}^1 \to Y$ if $g(Y) \geq 1$ (*Lüroth's theorem* for curves).

Solution. Riemann–Hurwitz gives $2g(X) - 2 \geq n(2g(Y) - 2)$. If $g(Y) \geq 1$ the right-hand side is $\geq 2g(Y) - 2$, so $g(X) \geq g(Y)$, and if $g(Y) = 0$ there is nothing to prove. □

Exercise 59.7.3. Show that a curve of genus 2 over a field of characteristic $\neq 2$ is of the form $y^2 = h(x)$ with h of degree 5 or 6 without repeated roots.

Solution. By Theorem 59.5.2, $|K|$ gives a degree-2 morphism $X \to \mathbb{P}^1$. So $K(X)$ is a quadratic extension of $k(x)$, of the form $k(x)(\sqrt{h})$ with $h \in k[x]$ squarefree (characteristic $\neq 2$). By Riemann–Hurwitz there are $2g + 2 = 6$ branch points, which are the roots of h together with ∞ if $\deg h$ is odd. So $\deg h \in \{5, 6\}$. □

Exercise 59.7.4. Let $C \subseteq \mathbb{P}^2_{\mathbb{C}}$ be a nonsingular curve of degree d and $p \notin C$ a general point. By projecting from p and applying Riemann–Hurwitz, recover $g(C) = (d-1)(d-2)/2$.

Solution. Projection from p is a morphism $C \to \mathbb{P}^1$ of degree d. Its ramification points are the points where the line through p is tangent to C. For general p they are simple ($e = 2$), and their number is the number of tangent lines to C through p, which is the degree $d(d-1)$ of the polar curve $\sum_i p_i \partial F/\partial x_i = 0$ meeting C (Bézout, Example 57.6.4). So $2g - 2 = -2d + d(d-1)$ and $g = (d-1)(d-2)/2$. □

Chapter 60

Linear Algebraic Groups

60.1 Introduction

A Hodge structure on a real vector space V is the same as an action of the real algebraic group $\mathbb{S} = \mathrm{Res}_{\mathbb{C}/\mathbb{R}}\mathbb{G}_m$, the Deligne torus (Proposition 60.4.5). The symmetry group of a rational Hodge structure, its Mumford–Tate group (Chapter 67), is the smallest algebraic subgroup of $\mathrm{GL}(V)$ defined over $\mathbb{Q}$ through which this action factors. This chapter provides the language: affine group schemes and their Hopf algebras, the embedding of every algebraic group into some GL_n, representations, tori and characters, Lie algebras, reductive groups and Weil restriction. The ground field is a field k of characteristic 0 unless stated otherwise. References: [Wat79], [Mil17], [Bor91], [DM82].

Prerequisites. Chapter 54 (Schemes), Chapter 34 (Lie Groups) and Chapter 13 (Representations of Groups and Lie Algebras).

60.2 Affine group schemes

Definition 60.2.1. An *affine group scheme* over k is an affine k-scheme $G = \operatorname{Spec} A$ whose functor of points $R \mapsto G(R) = \mathrm{Hom}_{k\text{-alg}}(A, R)$ takes values in groups, functorially in the k-algebra R. Equivalently (Yoneda), $A = \boldsymbol{O}(G)$ is a *commutative Hopf algebra*: it has k-algebra maps $\Delta\colon A \to A \otimes A$ (comultiplication), $\varepsilon\colon A \to k$ (counit) and $S\colon A \to A$ (antipode), dual to multiplication, identity and inverse, satisfying the dual group axioms. A *linear algebraic group* is an affine group scheme of finite type. Homomorphisms are morphisms of schemes compatible with the group laws.

Example 60.2.2.

1. The *additive group* $\mathbb{G}_a = \operatorname{Spec} k[t]$, $\Delta(t) = t \otimes 1 + 1 \otimes t$, with $\mathbb{G}_a(R) = (R, +)$.
2. The *multiplicative group* $\mathbb{G}_m = \operatorname{Spec} k[t, t^{-1}]$, $\Delta(t) = t \otimes t$, with $\mathbb{G}_m(R) = (R^*, \cdot)$. Its subgroup $\mu_n = \operatorname{Spec} k[t]/(t^n - 1)$ of n-th roots of unity.
3. $\mathrm{GL}_n = \operatorname{Spec} k[x_{ij}, 1/\det]$ with $\Delta(x_{ij}) = \sum_l x_{il} \otimes x_{lj}$, and its closed subgroups SL_n ($\det = 1$), the orthogonal group $\mathrm{O}(q) = \{g : q(gv) = q(v)\}$ of a nondegenerate quadratic form, the symplectic group Sp_{2g} of a nondegenerate alternating form, and the diagonal torus $\mathbb{G}_m^n$.
4. For a finite-dimensional vector space V, $\mathrm{GL}(V)(R) = \mathrm{Aut}_R(V \otimes_k R)$.

Definition 60.2.3. A *representation* of G on a k-vector space V is a natural action of $G(R)$ on $V \otimes_k R$ by R-linear maps. Equivalently it is a *comodule*: a linear map $\rho\colon V \to V \otimes A$ with

$(\rho \otimes \mathrm{id})\rho = (\mathrm{id} \otimes \Delta)\rho$ and $(\mathrm{id} \otimes \varepsilon)\rho = \mathrm{id}$. The element $g \in G(R)$, that is $g\colon A \to R$, acts by $(\mathrm{id} \otimes g) \circ \rho$ extended R-linearly. The *regular representation* is A itself with $\rho = \Delta$.

Lemma 60.2.4. *Every comodule is the union of its finite-dimensional subcomodules.*

Proof. Let $v \in V$ and choose a basis (a_i) of A. Write $\rho(v) = \sum_i v_i \otimes a_i$ (finite sum) and $\Delta(a_i) = \sum_{j,l} c_{ijl} a_j \otimes a_l$. Applying the two sides of the coassociativity identity to v gives $\sum_i \rho(v_i) \otimes a_i = \sum_{i,j,l} c_{ijl} v_i \otimes a_j \otimes a_l$, and comparing the coefficients of a_l in the last factor gives $\rho(v_l) = \sum_{i,j} c_{ijl} v_i \otimes a_j$. So $W = \mathrm{span}(v_i)$ is a subcomodule, and $v = \sum_i \varepsilon(a_i) v_i \in W$ by the counit identity. □

Theorem 60.2.5. *Every linear algebraic group G is isomorphic to a closed subgroup of some GL_n.*

Proof. Let $A = \mathcal{O}(G)$ with algebra generators $f_1, \dots, f_m$. By Lemma 60.2.4 for the regular representation, there is a finite-dimensional subcomodule $V \subseteq A$ containing all f_i. Choose a basis $v_1, \dots, v_n$ of V and write $\Delta(v_j) = \sum_i v_i \otimes a_{ij}$ with $a_{ij} \in A$. Coassociativity shows that $x_{ij} \mapsto a_{ij}$ defines a Hopf algebra map $k[x_{ij}, 1/\det] \to A$, that is, a homomorphism $\rho\colon G \to \mathrm{GL}_n$ (the determinant maps to a unit, whose inverse is the image of the determinant of $S(a_{ij})$). It is surjective on rings, because $v_j = (\varepsilon \otimes \mathrm{id})\Delta(v_j) = \sum_i \varepsilon(v_i) a_{ij}$ lies in the image, and the v_j include the generators f_i. So ρ is a closed immersion. □

Remark 60.2.6. In characteristic 0 every linear algebraic group is smooth (Cartier's theorem) [Mil17], so its reduced structure causes no trouble and $\dim G = \dim \mathrm{Lie}(G)$. In characteristic p, $\mu_p = \mathrm{Spec}\, k[t]/(t^p - 1) = \mathrm{Spec}\, k[t]/(t-1)^p$ is a non-reduced group scheme with one point.

60.3 Representations and Tannakian categories

Definition 60.3.1. The finite-dimensional representations of G form an abelian category $\mathrm{Rep}_k(G)$ with tensor products, duals and the unit object k (trivial representation), and a *fibre functor* $\omega\colon \mathrm{Rep}_k(G) \to \mathrm{Vec}_k$ forgetting the action. For a representation V, the tensor spaces are $T^{r,s}(V) = V^{\otimes r} \otimes (V^\vee)^{\otimes s}$, and a *tensor* $t \in T^{r,s}(V)$ is *fixed* by G if $g(t \otimes 1) = t \otimes 1$ for all $g \in G(R)$ and all R.

Theorem 60.3.2.

1. *(Tannaka duality) G is recovered from $(\mathrm{Rep}_k(G), \omega)$: $G(R)$ is the group of families (λ_V) of R-linear automorphisms of $V \otimes R$, natural in V and compatible with tensor products and the unit.*
2. *(Chevalley) Every closed subgroup $H \subseteq G$ is the stabiliser of a line L in some finite-dimensional representation W of G.*
3. *If $H \subseteq \mathrm{GL}(V)$ is reductive (Definition 60.6.1), then H is the subgroup of $\mathrm{GL}(V)$ fixing all tensors in $\bigoplus_{r,s} T^{r,s}(V)$ that are fixed by H.*

Statement (1) is [DM82, Theorem 2.11], (2) is Chevalley's theorem [Mil17], and (3) is [Del82, Proposition 3.1(c)]. Statement (3) is the reason why Mumford–Tate groups, defined as stabilisers of Hodge tensors, capture all the Hodge classes in tensor spaces (Chapter 67). Without reductivity (3) fails. Every tensor fixed by the group B of invertible upper triangular 2×2 matrices is fixed by GL_2: a B-fixed vector in a representation of GL_2 is a highest weight vector of weight 0, so it spans a trivial subrepresentation. But $B \neq \mathrm{GL}_2$.

Proposition 60.3.3. *Let G be a linear algebraic group and V a representation. For $t \in T^{r,s}(V)$ the stabiliser $G_t = \{g : gt = t\}$ is a closed subgroup of G, defined over the field of definition of t. For $v \in V$, the stabiliser of the line kv is a closed subgroup.*

Proof. Write $\rho(t) = \sum_i t_i \otimes a_i$ with the t_i a basis of $T^{r,s}(V)$ containing $t = t_1$. Then $g \in G(R)$ fixes t if and only if $g(a_1) = 1$ and $g(a_i) = 0$ for $i \geq 2$, which are polynomial equations. For the line, g stabilises kv if and only if $gv \wedge v = 0$ in $\Lambda^2 V$, again polynomial equations. □

60.4 Tori, characters and the Deligne torus

Definition 60.4.1. A *character* of G is a homomorphism $G \to \mathbb{G}_m$; they form an abelian group $X^*(G)$ under pointwise multiplication (written additively). A *cocharacter* is a homomorphism $\mathbb{G}_m \to G$. A *split torus* is a group isomorphic to $\mathbb{G}_m^r$, and a *torus* is a group T such that $T_{\bar{k}}$ is a split torus. T is *split* over a field extension L if T_L is.

Proposition 60.4.2. *Let $T = \mathbb{G}_m^r$. Then $X^*(T) \cong \mathbb{Z}^r$, the characters being the monomials $t^m = t_1^{m_1} \cdots t_r^{m_r}$, and $O(T) = k[X^*(T)]$ is the group algebra. A representation of T is the same as a $X^*(T)$-graded vector space $V = \bigoplus_\chi V_\chi$, where T acts on V_χ through χ. In particular every representation of a split torus is a direct sum of one-dimensional ones.*

Proof. $O(T) = k[t_i^{\pm 1}]$ has basis the monomials t^m, which are group-like: $\Delta(t^m) = t^m \otimes t^m$. A character is a group-like unit, and the group-like elements of a group algebra are the group elements (a group-like element $\sum c_m t^m$ satisfies $\sum c_m t^m \otimes t^m = \sum c_m c_{m'} t^m \otimes t^{m'}$, forcing a single term with $c_m = 1$). For a comodule V put $V_m = \{v : \rho(v) = v \otimes t^m\}$. Writing $\rho(v) = \sum_m v_m \otimes t^m$, coassociativity gives $\rho(v_m) = v_m \otimes t^m$, so $v_m \in V_m$, and the counit gives $v = \sum_m v_m$. The sum of the V_m is direct because the t^m are linearly independent. □

Remark 60.4.3. For a torus T over k, split over a finite Galois extension L, $\mathrm{Gal}(L/k)$ acts on $X^*(T_L) \cong \mathbb{Z}^r$, and $T \mapsto X^*(T_L)$ is an anti-equivalence between tori over k split by L and free abelian groups of finite rank with a $\mathrm{Gal}(L/k)$-action [Mil17]. Representations of T are $X^*(T_L)$-graded L-vector spaces V_L with a semilinear Galois action compatible with the grading.

Definition 60.4.4. The *Deligne torus* is $\mathbb{S} = \mathrm{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m$ (Section 60.7), the real algebraic group with $\mathbb{S}(R) = (R \otimes_{\mathbb{R}} \mathbb{C})^*$. Concretely $\mathbb{S} = \mathrm{Spec}\, \mathbb{R}[a, b, 1/(a^2 + b^2)]$, $\mathbb{S}(\mathbb{R}) = \mathbb{C}^*$ via $z = a + ib$, and $\mathbb{S}(\mathbb{C}) = (\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C})^* \cong \mathbb{C}^* \times \mathbb{C}^*$, where $z \in \mathbb{C}^* = \mathbb{S}(\mathbb{R})$ maps to $(z, \bar{z})$. The characters of $\mathbb{S}_{\mathbb{C}}$ are $z^p \bar{z}^q$, $(p, q) \in \mathbb{Z}^2$. The *weight cocharacter* $w \colon \mathbb{G}_m \to \mathbb{S}$ is the inclusion $\mathbb{R}^* \subseteq \mathbb{C}^*$, and the *norm* $\mathrm{Nm} \colon \mathbb{S} \to \mathbb{G}_m$ is $z \mapsto z\bar{z}$.

Proposition 60.4.5. *Let V be a finite-dimensional real vector space. Representations $h \colon \mathbb{S} \to \mathrm{GL}(V)$ correspond bijectively to decompositions $V_{\mathbb{C}} = \bigoplus_{p,q} V^{p,q}$ with $\overline{V^{p,q}} = V^{q,p}$, via*

$$V^{p,q} = \{v \in V_{\mathbb{C}} : h(z)v = z^{-p}\bar{z}^{-q}v \text{ for all } z \in \mathbb{C}^*\}.$$

Under this correspondence $w(r)$ acts on $V^{p,q}$ by $r^{-(p+q)}$, so $V = \bigoplus_n V_n$ with $V_{n,\mathbb{C}} = \bigoplus_{p+q=n} V^{p,q}$ is the weight decomposition.

Proof. $\mathbb{S}_{\mathbb{C}}$ is a split torus with character group $\mathbb{Z}^2$ generated by z and $\bar{z}$. By Proposition 60.4.2, a representation of $\mathbb{S}_{\mathbb{C}}$ on $V_{\mathbb{C}}$ is a $\mathbb{Z}^2$-grading, which we index by $(p, q) \mapsto$ the character $z^{-p}\bar{z}^{-q}$ (Deligne's sign convention). The subgroup $\mathbb{S}(\mathbb{R}) = \mathbb{C}^*$ is Zariski dense in $\mathbb{S}_{\mathbb{C}}$, since a Laurent polynomial in $z, \bar{z}$ vanishing on $\{(z, \bar{z})\}$ is zero, so the grading is determined by the action of $\mathbb{C}^*$ as displayed. The representation is defined over $\mathbb{R}$ if and only if $V_{\mathbb{C}}$ is stable under the action of $\mathbb{C}^*$ and $h(z)$ commutes with complex conjugation on $V_{\mathbb{C}}$. If $v \in V^{p,q}$, then $h(z)\bar{v} = \overline{h(z)v} = \overline{z^{-p}\bar{z}^{-q}v} = z^{-q}\bar{z}^{-p}\bar{v}$, so $\bar{v} \in V^{q,p}$. Conversely, a grading with $\overline{V^{p,q}} = V^{q,p}$ defines $h(z)$ by the formula, and $h(z)$ commutes with conjugation by the same computation, hence is real. The statement about w is the case $z = r \in \mathbb{R}^*$. □

60.5 The Lie algebra

Definition 60.5.1. Let $k[\varepsilon] = k[t]/(t^2)$. The *Lie algebra* of G is $\mathrm{Lie}(G) = \ker(G(k[\varepsilon]) \to G(k))$, a k-vector space identified with the tangent space $(\mathfrak{m}_e/\mathfrak{m}_e^2)^\vee$ at the identity (Exercise 54.10.2). The bracket is induced by the commutator in $G(k[\varepsilon_1, \varepsilon_2])$, and a representation ρ of G differentiates to a representation $d\rho$ of $\mathrm{Lie}(G)$.

Example 60.5.2. $\mathrm{Lie}(\mathrm{GL}_n) = \{1 + \varepsilon X\} \cong \mathfrak{gl}_n$, with the commutator bracket. Since $\det(1+\varepsilon X) = 1 + \varepsilon \operatorname{tr} X$, $\mathrm{Lie}(\mathrm{SL}_n) = \{X : \operatorname{tr} X = 0\}$. Similarly $\mathrm{Lie}(\mathrm{O}(q)) = \{X : B(Xv, w) + B(v, Xw) = 0\}$, where B is the bilinear form of q, and $\mathrm{Lie}(\mathbb{G}_m^r) = k^r$ with zero bracket. For $k = \mathbb{R}$ or $\mathbb{C}$, $\mathrm{Lie}(G)$ is the Lie algebra of the Lie group $G(k)$ (Definition 34.2.6).

Proposition 60.5.3. *In characteristic 0, for connected G, $H \mapsto \mathrm{Lie}(H)$ is an injective, inclusion-preserving map from connected closed subgroups of G to Lie subalgebras of $\mathrm{Lie}(G)$. A subspace $W \subseteq V$ of a representation is G-stable if and only if it is $\mathrm{Lie}(G)$-stable, and a vector is fixed by G if and only if it is killed by $\mathrm{Lie}(G)$.*

See [Bor91] and [Mil17]. It fails in characteristic p: μ_p and $\{1\}$ have different Lie algebras over a field of characteristic p but the same points, and $\mathbb{G}_a$ acting on k^2 by $t \mapsto \left(\begin{smallmatrix} 1 & t^p \\ 0 & 1 \end{smallmatrix}\right)$ has Lie acting trivially without fixing everything.

60.6 Reductive groups

Definition 60.6.1. A linear algebraic group G over k is *unipotent* if it is isomorphic to a closed subgroup of the upper triangular unipotent matrices. The *unipotent radical* $R_u(G)$ is the largest connected normal unipotent subgroup of $G_{\bar{k}}$. A connected G is *reductive* if $R_u(G) = 1$, and *semisimple* if moreover its centre is finite.

Theorem 60.6.2. *Let k have characteristic 0 and G be connected.*

1. *G is reductive if and only if every finite-dimensional representation of G is semisimple (a direct sum of irreducible representations).*
2. *If G is reductive, then $G = Z(G)^\circ \cdot G^{\mathrm{der}}$, where the connected centre $Z(G)^\circ$ is a torus and the derived group G^{der} is semisimple, and $Z(G)^\circ \cap G^{\mathrm{der}}$ is finite.*
3. *Tori, GL_n, SL_n, Sp_{2g} and $\mathrm{SO}(q)$ are reductive; $\mathbb{G}_a$ and the group of invertible upper triangular matrices are not.*

See [Mil17] and [Bor91]. For example, the representation of $\mathbb{G}_a$ on k^2 by $t \mapsto \left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right)$ has the invariant line ke_1 without an invariant complement, so $\mathbb{G}_a$ is not reductive.

Proposition 60.6.3 (Compact real forms)**.** *Let G be a connected linear algebraic group over $\mathbb{R}$ and θ an involution of $G_{\mathbb{C}}$ such that $G^{(\theta)} = \{g \in G(\mathbb{C}) : \theta(\bar{g}) = g\}$ is compact. Then G is reductive. In particular, if $G \subseteq \mathrm{GL}(V)$ and there is a positive definite inner product on V such that G is stable under $g \mapsto g^T$ (the adjoint), then G is reductive.*

Proof. The group $K = G^{(\theta)}$ is a compact real form of $G(\mathbb{C})$: it is the fixed group of the antiholomorphic involution $g \mapsto \theta(\bar{g})$, so its Lie algebra $\mathfrak{k}$ is a real form of $\mathrm{Lie}(G_{\mathbb{C}})$, that is, $\mathrm{Lie}(G_{\mathbb{C}}) = \mathfrak{k} \oplus i\mathfrak{k}$. Let W be a complex representation of $G_{\mathbb{C}}$ and $W' \subseteq W$ a subrepresentation. By Corollary 34.5.3, W has a K-invariant Hermitian inner product, and the orthogonal complement W'' of W' is K-stable, hence $\mathfrak{k}$-stable. As $\mathfrak{k}$ spans $\mathrm{Lie}(G_{\mathbb{C}})$ over $\mathbb{C}$, W'' is $\mathrm{Lie}(G_{\mathbb{C}})$-stable, hence $G_{\mathbb{C}}$-stable by Proposition 60.5.3. So every representation of $G_{\mathbb{C}}$, and therefore of G, is semisimple, and G is reductive by Theorem 60.6.2(1). For the second statement, extend

the inner product to a Hermitian one on $V_{\mathbb{C}}$ and take $\theta(g) = (g^*)^{-1}$, where g^* is the Hermitian adjoint of $g \in G(\mathbb{C})$. Then $\theta(\bar{g}) = (g^T)^{-1}$ in a real orthonormal basis, so $G^{(\theta)} = G(\mathbb{C}) \cap U(V_{\mathbb{C}})$, which is compact. □

Remark 60.6.4. A variant of this argument, with the Weil operator of a polarisation in place of the transpose, shows that Mumford–Tate groups of polarisable Hodge structures are reductive (Chapter 67).

60.7 Weil restriction of scalars

Definition 60.7.1. Let L/k be a finite field extension and Y an affine scheme over L. The *Weil restriction* $\mathrm{Res}_{L/k}Y$ is the functor on k-algebras $R \mapsto Y(R \otimes_k L)$.

Proposition 60.7.2. *$\mathrm{Res}_{L/k}Y$ is represented by an affine k-scheme. If Y is a linear algebraic group of dimension d, then $\mathrm{Res}_{L/k}Y$ is a linear algebraic group of dimension $d[L:k]$. If L/k is Galois, then $(\mathrm{Res}_{L/k}Y)_L \cong \prod_{\sigma \in \mathrm{Gal}(L/k)} Y^{\sigma}$.*

Proof. Choose a basis $e_1, \dots, e_m$ of L over k. For $Y = \mathbb{A}^n_L$, a point of $Y(R \otimes_k L)$ is an n-tuple $y_j = \sum_i x_{ij} e_i$ with $x_{ij} \in R$, so $\mathrm{Res}_{L/k}\mathbb{A}^n_L = \mathbb{A}^{nm}_k$. If $Y = V(f_1, \dots, f_s) \subseteq \mathbb{A}^n_L$, substitute $y_j = \sum_i x_{ij} e_i$ into f_l and expand in the basis: $f_l = \sum_i F_{li}(x) e_i$ with $F_{li} \in k[x]$. Then $\mathrm{Res}_{L/k}Y = V(F_{li}) \subseteq \mathbb{A}^{nm}_k$. The group structure is inherited functorially. For the dimension and the Galois statement, use $(R \otimes_k L) \otimes_k L \cong \prod_{\sigma} R \otimes_k L$ for R an L-algebra, which gives $\mathrm{Res}_{L/k}Y(R) = \prod_{\sigma} Y^{\sigma}(R)$ for L-algebras R. □

Example 60.7.3.

1. $\mathbb{S} = \mathrm{Res}_{\mathbb{C}/\mathbb{R}}\mathbb{G}_m$ is a two-dimensional torus over $\mathbb{R}$ that is not split, with $\mathbb{S}_{\mathbb{C}} \cong \mathbb{G}_m^2$; complex conjugation swaps the two characters z and $\bar{z}$.
2. For a number field E, $T_E = \mathrm{Res}_{E/\mathbb{Q}}\mathbb{G}_m$ has $T_E(\mathbb{Q}) = E^*$. If E is a CM field, the Mumford–Tate group of an abelian variety with complex multiplication by E is a subtorus of T_E (Chapter 61).
3. The kernel $\mathbb{U}(1) = \ker(\mathrm{Nm}\colon \mathbb{S} \to \mathbb{G}_m)$ is the circle group, with $\mathbb{U}(1)(\mathbb{R}) = S^1$, a torus that is compact and not split over $\mathbb{R}$.

60.8 Exercises

Exercise 60.8.1. Show that $X^*(\mathrm{GL}_n) = \mathbb{Z}$, generated by det, and that $X^*(\mathrm{SL}_n) = 0$.

Solution. A character is a unit $\chi \in k[x_{ij}, 1/\det]$ with $\Delta(\chi) = \chi \otimes \chi$. The units of this ring are $c \det^m$, since det is irreducible, and group-likeness forces $c = 1$. For SL_n, the group $\mathrm{SL}_n(\bar{k})$ is generated by the elementary matrices $1 + tE_{ij}$, $i \neq j$, and each family $t \mapsto 1 + tE_{ij}$ is a homomorphism $\mathbb{G}_a \to \mathrm{SL}_n$. A character of $\mathbb{G}_a$ is a group-like unit of $k[t]$, hence 1. So every character of SL_n is trivial on a generating set of $\mathrm{SL}_n(\bar{k})$, and as SL_n is reduced, it is trivial. □

Exercise 60.8.2. Describe the representations of $\mathbb{S}$ on $\mathbb{R}^2$ given by $h(a+ib) = \left(\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right)$ and by $h(z) = z\bar{z} \cdot \mathrm{id}$, in terms of the decomposition of Proposition 60.4.5.

Solution. The first is multiplication by z on $\mathbb{C} = \mathbb{R}^2$. On $V_{\mathbb{C}} = \mathbb{C}^2$ its eigenvectors are $(1, -i)$ with eigenvalue z and $(1, i)$ with eigenvalue $\bar{z}$. With $h(z)v = z^{-p}\bar{z}^{-q}v$, these are $V^{-1,0}$ and $V^{0,-1}$, a Hodge structure of weight -1 (the Hodge structure of H_1 of an elliptic curve). The second acts by $z\bar{z} = z^1\bar{z}^1$ on all of $V_{\mathbb{C}}$, so $V = V^{-1,-1}$, of weight -2. □

Exercise 60.8.3. Show directly that the group B of invertible upper triangular 2×2 matrices has a representation that is not semisimple, and compute its unipotent radical.

Solution. The standard representation on k^2 has the invariant line ke_1, and an invariant complement would be spanned by a common eigenvector outside ke_1, which does not exist because $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ has only the eigenline ke_1. The unipotent radical is the normal subgroup U of unipotent upper triangular matrices, and $B/U \cong \mathbb{G}_m^2$ is a torus. □

Chapter 61

Abelian Varieties

61.1 Introduction

A complex abelian variety is a complex torus $\mathbb{C}^g/\Lambda$ that embeds in projective space. Tori are the simplest compact Kähler manifolds, and their cohomology is completely explicit: exterior powers of H^1. For this reason abelian varieties are the main testing ground of the Hodge conjecture. They include the known cases (divisors, Lefschetz classes), the first instances of exceptional Hodge classes (Weil classes), and Deligne's theorem that Hodge classes on abelian varieties are absolutely Hodge (Part IX). This chapter proves Riemann's criterion for a torus to be algebraic, describes line bundles, polarisations, the dual and the Jacobian, and discusses endomorphisms and complex multiplication. References: [BL04], [Mum70], [GH78, Chapter 2, Section 6].

Prerequisites. Chapter 59 (Algebraic Curves and Riemann Surfaces), Chapter 57 (Divisors and Line Bundles), Chapter 48 (The Hodge Decomposition) and Chapter 60 (Linear Algebraic Groups).

Throughout, V is a complex vector space of dimension g, $\Lambda \subseteq V$ a lattice (a discrete subgroup of rank $2g$ spanning V over $\mathbb{R}$), and $T = V/\Lambda$ the complex torus, with its structure of compact complex Lie group (group law written additively). $\bar{V}^\vee$ denotes the space of conjugate-linear forms $V \to \mathbb{C}$.

61.2 Complex tori and their cohomology

Proposition 61.2.1. *1. $H_1(T;\mathbb{Z}) = \Lambda$, and cup product gives ring isomorphisms $H^k(T;\mathbb{Z}) \cong \Lambda^k \operatorname{Hom}(\Lambda,\mathbb{Z})$, the alternating k-forms on Λ.*

2. The Hodge decomposition is $H^1(T;\mathbb{C}) = \operatorname{Hom}_{\mathbb{R}}(\Lambda\otimes\mathbb{R},\mathbb{C}) = V^\vee \oplus \bar{V}^\vee$, with $H^{1,0} = V^\vee$ (spanned by the dz_j) and $H^{0,1} = \bar{V}^\vee$ (spanned by the $d\bar{z}_j$), and $H^{p,q}(T) = \Lambda^p V^\vee \otimes \Lambda^q \bar{V}^\vee$.

Proof. (1) T is diffeomorphic to $(S^1)^{2g}$ via a $\mathbb{Z}$-basis of Λ, and $\mathbb{R}^{2g} \to T$ is the universal covering, so $\pi_1(T) = H_1(T;\mathbb{Z}) = \Lambda$. The ring structure follows from the Künneth formula (Theorem 22.4.2) and $H^\bullet(S^1) = \Lambda^\bullet\mathbb{Z}$. (2) With a flat metric the harmonic forms are the constant-coefficient forms (Example 46.4.2); a constant 1-form is a real-linear form on V, and it splits uniquely into complex-linear and conjugate-linear parts. □

Definition 61.2.2. A complex torus is an *abelian variety* if it is projective. A homomorphism of tori is a holomorphic group homomorphism, and an *isogeny* is a surjective homomorphism with finite kernel. More generally, over any field an *abelian variety* is a complete connected group variety; such a group is automatically commutative and projective [Mum70, Chapter II].

Lemma 61.2.3. *Every holomorphic map $f : T \to T'$ of complex tori with $f(0) = 0$ is a homomorphism, induced by a complex-linear map $F : V \to V'$ with $F(\Lambda) \subseteq \Lambda'$.*

Proof. Lift f to $F : V \to V'$ with $F(0) = 0$. For $\lambda \in \Lambda$, $F(v+\lambda)-F(v) \in \Lambda'$ depends continuously on v, hence is constant. So the partial derivatives of F are Λ-periodic holomorphic functions, so they descend to holomorphic functions on the compact connected manifold T, which are constant by the maximum principle (Proposition 31.2.3). So F is linear, and $F(\Lambda) \subseteq \Lambda'$ because $F(\lambda) = F(0 + \lambda) - F(0) \in \Lambda'$. □

Proposition 61.2.4. *Let $T = V/\Lambda$ be a complex torus of dimension g and $n \geq 1$.*

1. *Multiplication by n, $[n] : T \to T$, is a surjective homomorphism with kernel $T[n] = \frac{1}{n}\Lambda/\Lambda \cong (\mathbb{Z}/n\mathbb{Z})^{2g}$. The torsion subgroup of T is $(\Lambda \otimes \mathbb{Q})/\Lambda \cong (\mathbb{Q}/\mathbb{Z})^{2g}$, and it is dense in T.*
2. *A homomorphism $f : T \to T'$ of tori of the same dimension is an isogeny if and only if its linear lift $F : V \to V'$ is an isomorphism; then $|\ker f| = [\Lambda' : F(\Lambda)]$.*
3. *If $f : T \to T'$ is an isogeny with $|\ker f| = d$, there is an isogeny $h : T' \to T$ with $h \circ f = [d]_T$ and $f \circ h = [d]_{T'}$. In particular being isogenous is an equivalence relation.*

Proof. (1) $[n]$ is induced by $v \mapsto nv$, which is bijective on V; so $[n]$ is surjective, and $x = v + \Lambda$ lies in the kernel if and only if $nv \in \Lambda$. A $\mathbb{Z}$-basis of Λ identifies $\frac{1}{n}\Lambda/\Lambda$ with $(\frac{1}{n}\mathbb{Z}/\mathbb{Z})^{2g} \cong (\mathbb{Z}/n\mathbb{Z})^{2g}$, and the union over n is $(\Lambda\otimes\mathbb{Q})/\Lambda$. The rational points $\Lambda\otimes\mathbb{Q}$ are dense in $\Lambda \otimes \mathbb{R} = V$, so their image is dense in T.

(2) By Lemma 61.2.3, f is induced by a $\mathbb{C}$-linear F with $F(\Lambda) \subseteq \Lambda'$. If F is an isomorphism, $F(\Lambda)$ is a lattice in V' contained in Λ', hence of finite index, f is surjective, and $\ker f = F^{-1}(\Lambda')/\Lambda \cong \Lambda'/F(\Lambda)$ is finite. Conversely, if f has finite kernel, then $\ker F$ is a complex subspace whose image in T is contained in the finite group $\ker f$; a connected subspace mapping into a finite set maps to a point, so $\ker F \subseteq \Lambda$, which forces $\ker F = 0$; equal dimensions make F an isomorphism.

(3) The group $\Lambda'/F(\Lambda)$ has order d, so $d\Lambda' \subseteq F(\Lambda)$. Hence $H = dF^{-1} : V' \to V$ maps Λ' into Λ and induces a homomorphism $h : T' \to T$ with $HF = d$ and $FH = d$, which is an isogeny by (2). Reflexivity and transitivity of isogeny are clear, and (3) gives symmetry. □

61.3 Line bundles and the Riemann conditions

Proposition 61.3.1. *Identify $H^2(T;\mathbb{Z})$ with the alternating forms $E : \Lambda\times\Lambda \to \mathbb{Z}$ (Proposition 61.2.1). A class E lies in $H^{1,1}(T)$ if and only if $E(iu, iv) = E(u, v)$ for all $u, v \in V$ (extending E $\mathbb{R}$-bilinearly), if and only if $E = \operatorname{Im} H$ for a (unique) Hermitian form H on V, namely $H(u, v) = E(iu, v) + iE(u, v)$. Every such E is $c_1(L)$ for a holomorphic line bundle L, so*

$$\mathrm{NS}(T) := \operatorname{im}(c_1) = \{H \text{ Hermitian on } V : \operatorname{Im} H(\Lambda, \Lambda) \subseteq \mathbb{Z}\}.$$

Proof. A real alternating form E on V is a constant real 2-form. Its $(2, 0)$ and $(0, 2)$ components vanish exactly when E is of type $(1, 1)$ for the complex structure i, that is, $E(iu, iv) = E(u, v)$: the $(2, 0) + (0, 2)$ part is $\frac{1}{2}(E - E(i\cdot, i\cdot))$. Given this, $H(u, v) = E(iu, v) + iE(u, v)$ is Hermitian: $H(iu, v) = E(-u, v) + iE(iu, v) = i(E(iu, v) + iE(u, v)) = iH(u, v)$, and $\overline{H(v, u)} = E(iv, u) - iE(v, u) = E(iu, v) + iE(u, v)$, using $E(iv, u) = -E(u, iv) = -E(iu, i^2v) = E(iu, v)$. Conversely Im of a Hermitian form satisfies the condition. The last statement is Lemma 50.2.2. □

Theorem 61.3.2 (Riemann)**.** *A complex torus $T = V/\Lambda$ is an abelian variety if and only if there is a positive definite Hermitian form H on V with $\operatorname{Im} H(\Lambda, \Lambda) \subseteq \mathbb{Z}$. Such an H (or its alternating form $E = \operatorname{Im} H$) is a* polarisation*; it is the first Chern class of an ample line bundle.*

Proof. For a constant real 2-form ω put $E_\omega = -\omega$ and $H_\omega(u,v) = E_\omega(iu,v) + iE_\omega(u,v)$. If ω is of type $(1,1)$, then H_ω is Hermitian (Proposition 61.3.1) with $\operatorname{Re} H_\omega(u,u) = -\omega(iu,u) = \omega(u,iu)$. So ω is a positive $(1,1)$-form, that is, the fundamental form of a Hermitian metric (Proposition 47.2.2), if and only if H_ω is positive definite. Conversely every Hermitian H is H_ω for $\omega = -\operatorname{Im} H$.

Suppose H is positive definite with $\operatorname{Im} H$ integral on Λ. Then $\omega = -\operatorname{Im} H$ is a positive constant $(1,1)$-form, closed because its coefficients are constant, and its class is integral. So T is a Hodge manifold (Definition 50.2.3) and projective by Corollary 50.6.6.

Conversely, let T be projective, with an ample line bundle L and a metric of positive curvature form ω_h (Theorem 58.6.1). Average over translations: $\bar\omega = \int_T t_x^*\omega_h\,dx$ for the normalised Haar measure (Theorem 34.5.1). Each $t_x^*\omega_h$ is positive and cohomologous to ω_h, because translations are homotopic to the identity, so $\bar\omega$ is a positive translation-invariant closed $(1,1)$-form in the integral class $c_1(L)$. A translation-invariant form has constant coefficients. So $H_{\bar\omega}$ is positive definite, and $\operatorname{Im} H_{\bar\omega} = -\bar\omega$ is integral on Λ, being (up to sign) the integral class $c_1(L)$ in $H^2(T;\mathbb{Z}) = \Lambda^2 \operatorname{Hom}(\Lambda,\mathbb{Z})$. □

Remark 61.3.3. The proof shows that ample line bundles correspond to positive definite H once a constant form ω is matched with the alternating form $-\omega$; this is the convention of [BL04], where $c_1(L) = \operatorname{Im} H$. With the opposite matching one gets negative definite forms instead; Riemann's condition is insensitive to this, since $-H$ is positive definite if H is negative definite.

Proposition 61.3.4. *Let E be a polarisation. There is a basis $\lambda_1,\dots,\lambda_g,\mu_1,\dots,\mu_g$ of Λ with $E(\lambda_j,\mu_k) = d_j\delta_{jk}$ and $E(\lambda_j,\lambda_k) = E(\mu_j,\mu_k) = 0$, where $d_1 \mid d_2 \mid \cdots \mid d_g$ are positive integers, the* type *of the polarisation. It is* principal *if all $d_j = 1$. Then $\frac{1}{d_1}\mu_1,\dots,\frac{1}{d_g}\mu_g$ is a $\mathbb{C}$-basis of V, and in it $\Lambda = Z\mathbb{Z}^g + D\mathbb{Z}^g$ with $D = \operatorname{diag}(d_j)$ and Z a symmetric complex $g\times g$ matrix with $\operatorname{Im} Z$ positive definite. Conversely every such Z in the* Siegel upper half space $\mathfrak{H}_g$ *defines a polarised abelian variety of type D.*

Proof. The symplectic basis is the elementary divisor theorem for alternating forms over $\mathbb{Z}$ (Frobenius), proved like the Smith normal form (Theorem 8.11.2); see [BL04]. Since H is positive definite, the real span of the μ_j is E-isotropic, and H is real and positive on it, so the μ_j are $\mathbb{C}$-linearly independent: a complex relation $\sum c_j\mu_j = 0$ with $c = a + ib$ gives $\sum a_j\mu_j = -i\sum b_j\mu_j$, and pairing with H forces $a = b = 0$. Writing $\lambda_j = \sum_k Z_{kj}\mu_k/d_k$, the conditions that the λ's span an isotropic subspace and that H is positive definite translate into $Z = Z^T$ and $\operatorname{Im} Z > 0$ by a direct computation [BL04]. The converse is the same computation read backwards together with Theorem 61.3.2. □

Example 61.3.5. For $g = 1$, $\mathfrak{H}_1$ is the upper half plane, and $\Lambda = \mathbb{Z}\tau + \mathbb{Z}$ with $\operatorname{Im}\tau > 0$. Every one-dimensional torus is an abelian variety, as Riemann's condition only asks $\operatorname{Im}\tau > 0$ (compare Example 49.4.4). The alternating form $E(\tau,1) = 1$ gives $H(u,v) = u\bar v/\operatorname{Im}\tau$.

Counterexample 61.3.6. For $g \geq 2$ the general complex torus is not algebraic. Fix a real basis identifying V with $\mathbb{R}^{2g}$ and Λ with $\mathbb{Z}^{2g}$, so that complex tori correspond to complex structures J on $\mathbb{R}^{2g}$. For each of the countably many nonzero integral alternating forms E, the condition $E(Ju,Jv) = E(u,v)$ defines a proper closed real-analytic subset of the space of complex structures when $g \geq 2$ [BL04]. Outside the countable union of these subsets, $\mathrm{NS}(T) = 0$ by Proposition 61.3.1, so there is no ample line bundle and T is not projective. For such tori the Hodge conjecture in degree 2 has no analytic cycles to work with; more generally the analytic version of the conjecture fails for some complex tori (Chapter 73).

Theorem 61.3.7 (Appell–Humbert)**.** *Holomorphic line bundles on T correspond bijectively to pairs (H,α) with H Hermitian, $E = \operatorname{Im} H$ integral on Λ, and $\alpha\colon \Lambda \to U(1)$ a* semicharacter*:*

$\alpha(\lambda+\mu) = \alpha(\lambda)\alpha(\mu)e^{i\pi E(\lambda,\mu)}$. *The bundle* $L(H,\alpha)$ *is the quotient of* $V\times\mathbb{C}$ *by* $\lambda\cdot(v,t) = (v+\lambda, \alpha(\lambda)e^{\pi H(v,\lambda)+\frac{\pi}{2}H(\lambda,\lambda)}t)$, *and* $c_1(L(H,\alpha)) = E$. *Its sections are the* theta functions, *and if* H *is positive definite of type* $(d_1,\dots,d_g)$, *then* $h^0(L) = d_1\cdots d_g$. *Moreover* L^2 *is globally generated and* L^3 *is very ample (Lefschetz).*

This is [BL04]; see also [Mum70].

61.4 The dual abelian variety

Proposition 61.4.1. $\mathrm{Pic}^0(T)\cong\hat{T} = \bar{V}^\vee/\hat{\Lambda}$, *where* $\hat{\Lambda} = \{l\in\bar{V}^\vee : \operatorname{Im} l(\Lambda)\subseteq\mathbb{Z}\}$, *a complex torus of dimension* g, *the* dual torus. *A polarisation* H *defines a homomorphism* $\varphi_H\colon T\to\hat{T}$, $v\mapsto H(v,\cdot)$, *which is an isogeny of degree* $(d_1\cdots d_g)^2$, *and an isomorphism if the polarisation is principal. If* T *is an abelian variety, so is* $\hat{T}$.

Proof. By the exponential sequence, $\mathrm{Pic}^0(T) = H^1(T,\mathcal{O})/H^1(T;\mathbb{Z})$ (Theorem 45.2.4), and $H^1(T,\mathcal{O}) = H^{0,1} = \bar{V}^\vee$ (Proposition 61.2.1). The map $H^1(T;\mathbb{R}) = \mathrm{Hom}_{\mathbb{R}}(V,\mathbb{R})\to\bar{V}^\vee$ sends a real linear form a to its conjugate-linear part l_a, and $a = 2\operatorname{Re} l_a$. So the image of $H^1(T;\mathbb{Z}) = \mathrm{Hom}(\Lambda,\mathbb{Z})$ is $\{l : 2\operatorname{Re} l(\Lambda)\subseteq\mathbb{Z}\}$, which the $\mathbb{C}$-linear automorphism $l\mapsto 2il$ of $\bar{V}^\vee$ carries onto $\hat{\Lambda}$, since $\operatorname{Im}(2il) = 2\operatorname{Re} l$. The map $v\mapsto H(v,\cdot)$ is complex-linear from V to $\bar{V}^\vee$ and maps Λ into $\hat{\Lambda}$ because $\operatorname{Im} H(\Lambda,\Lambda)\subseteq\mathbb{Z}$. It is injective since H is nondegenerate, so its kernel on T is $\hat{\Lambda}'/\Lambda$ with $\hat{\Lambda}' = \{v : \operatorname{Im} H(v,\Lambda)\subseteq\mathbb{Z}\}$, of order $\det E = (d_1\cdots d_g)^2$ in a symplectic basis. The inverse of H gives a polarisation on $\hat{T}$ [BL04]. □

61.5 Jacobians

Theorem 61.5.1. *Let* C *be a compact Riemann surface of genus* g. *The Jacobian* $J(C) = H^0(C,\Omega^1)^\vee/H_1(C;\mathbb{Z})$ *(Definition 59.6.1) is a principally polarised abelian variety, the polarisation being the intersection form on* $H_1(C;\mathbb{Z})$.

Proof. $J(C)$ is a complex torus of dimension g (Proposition 59.6.2). The intersection form E on $H_1(C;\mathbb{Z})$ is unimodular (Poincaré duality, Corollary 23.6.3), so a symplectic basis $a_1,\dots,a_g,b_1,\dots,b_g$ exists with $E(a_j,b_k) = \delta_{jk}$. Extend E $\mathbb{R}$-linearly to $V = H^0(\Omega^1)^\vee\cong H_1(C;\mathbb{R})$. Under Poincaré duality $H_1(C;\mathbb{R})\cong H^1(C;\mathbb{R})$ and the real isomorphism $H^1(C;\mathbb{R})\to H^{1,0}$, $\alpha\mapsto\alpha^{1,0}$, the form E becomes $\pm\int_C\alpha\wedge\beta = \pm 2\operatorname{Re}\int_C\alpha^{1,0}\wedge\overline{\beta^{1,0}}$, and the complex structure of V becomes that of $H^{1,0}$ (up to conjugation). So $E(iu,iv) = E(u,v)$, and the Hermitian form $H(u,v) = E(iu,v) + iE(u,v)$ is, up to sign and a positive factor, $i\int_C\alpha^{1,0}\wedge\overline{\beta^{1,0}}$, which is definite by the Hodge–Riemann relations (Theorem 49.4.2, Example 49.4.4). Replacing E by $-E$ if necessary, H is positive definite, and $J(C)$ is an abelian variety by Theorem 61.3.2. The polarisation is principal as E is unimodular. □

Remark 61.5.2. By Torelli's theorem the principally polarised Jacobian determines C. The Jacobians form a subvariety of dimension $3g-3$ of the moduli space $\mathfrak{H}_g/\mathrm{Sp}_{2g}(\mathbb{Z})$ of principally polarised abelian varieties, which has dimension $g(g+1)/2$. For $g\le 3$ the dimensions agree and the general principally polarised abelian variety is a Jacobian; for $g\ge 4$ it is not, and characterising Jacobians is the *Schottky problem* [BL04].

61.6 Endomorphisms and complex multiplication

Proposition 61.6.1. *Let* $E = \mathbb{C}/(\mathbb{Z}\tau+\mathbb{Z})$. *Then* $\mathrm{End}(E)$ *is* $\mathbb{Z}$, *or an order in the imaginary quadratic field* $\mathbb{Q}(\tau)$; *the latter happens if and only if* τ *is quadratic over* $\mathbb{Q}$.

Proof. By Lemma 61.2.3 an endomorphism is multiplication by some $\alpha \in \mathbb{C}$ with $\alpha\Lambda \subseteq \Lambda$. Then $\alpha = a + b\tau$ and $\alpha\tau = c + d\tau$ with integers a, b, c, d. If $b \neq 0$, eliminating α gives $b\tau^2 + (a-d)\tau - c = 0$, so τ is quadratic, imaginary as $\operatorname{Im}\tau > 0$, and $\alpha \in \mathbb{Q}(\tau)$ is integral (it is an eigenvalue of an integer matrix). If $b = 0$ then $\alpha = a \in \mathbb{Z}$. Conversely, if $a\tau^2 + b\tau + c = 0$ with integers $a \neq 0$, b, c, then $\alpha = a\tau$ satisfies $\alpha \cdot 1 = a\tau \in \Lambda$ and $\alpha\tau = -b\tau - c \in \Lambda$, so multiplication by $\alpha \notin \mathbb{Z}$ is an endomorphism. □

Definition 61.6.2. An abelian variety A of dimension g has *complex multiplication* (CM) if $\operatorname{End}(A) \otimes \mathbb{Q}$ contains a commutative semisimple subalgebra of dimension $2g$ over $\mathbb{Q}$, for instance a CM field F (a totally imaginary quadratic extension of a totally real field) with $[F : \mathbb{Q}] = 2g$.

Theorem 61.6.3. *Let A be an abelian variety over $\mathbb{C}$.*

1. *(Poincaré reducibility) A is isogenous to a product $A_1^{n_1} \times \cdots \times A_r^{n_r}$ with the A_i simple and pairwise non-isogenous, and $\operatorname{End}(A) \otimes \mathbb{Q} \cong \prod_i M_{n_i}(D_i)$ with $D_i = \operatorname{End}(A_i) \otimes \mathbb{Q}$ division algebras.*

2. *A polarisation defines the* Rosati involution $f \mapsto f^\dagger$ *on $\operatorname{End}(A) \otimes \mathbb{Q}$, and $\operatorname{tr}(f f^\dagger) > 0$ for $f \neq 0$.*

3. *(Shimura–Taniyama) If A has CM by a CM field F of degree $2g$, then $A \cong \mathbb{C}^g/\Phi(\mathfrak{a})$ for a fractional ideal $\mathfrak{a}$ of F and a* CM type $\Phi = \{\phi_1, \dots, \phi_g\}$*, a set of embeddings $F \to \mathbb{C}$ containing exactly one of each pair of complex conjugate embeddings.*

This is [BL04] and [Mum70]. The endomorphism algebras D_i are classified by Albert into four types. For the Hodge conjecture, CM abelian varieties are the case where the Mumford–Tate group is a torus (Chapter 67), and where exceptional Hodge classes first appear (Chapter 72).

Example 61.6.4.
1. $E_i = \mathbb{C}/\mathbb{Z}[i]$ has $\operatorname{End}(E_i) = \mathbb{Z}[i]$ and $E_\rho = \mathbb{C}/\mathbb{Z}[\rho]$, $\rho = e^{2\pi i/3}$, has $\operatorname{End}(E_\rho) = \mathbb{Z}[\rho]$.

2. For E without CM, $\operatorname{End}(E \times E) \otimes \mathbb{Q} = M_2(\mathbb{Q})$, and $\operatorname{NS}(E \times E) \otimes \mathbb{Q}$ has rank 3; for E with CM it has rank 4. This is Example 57.7.3(3).

3. The Jacobian of the curve $y^2 = x^5 - 1$ has CM by $\mathbb{Q}(\zeta_5)$, a CM field of degree $4 = 2g$.

61.7 Exercises

Exercise 61.7.1. Let E_1, E_2 be elliptic curves. Show that $\operatorname{NS}(E_1 \times E_2) \cong \mathbb{Z}^2 \oplus \operatorname{Hom}(E_1, E_2)$, and deduce that its rank is 2, 3 or 4.

Solution. Write $\Lambda = \Lambda_1 \oplus \Lambda_2$. An integral alternating form of type $(1,1)$ on Λ restricts to such forms on Λ_1 and Λ_2, each a multiple of the generator ($\mathbb{Z}^2$), and its mixed part $E(\lambda_1, \lambda_2)$ is a pairing of type $(1,1)$ between Λ_1 and Λ_2, which via E_1's principal polarisation is the same as a $\mathbb{Z}$-linear map $\Lambda_1 \to \Lambda_2$ commuting with i after identification, that is, a homomorphism $E_1 \to E_2$ (Lemma 61.2.3). By Proposition 61.6.1 and its analogue for homomorphisms, $\operatorname{Hom}(E_1, E_2)$ has rank 0, 1 or 2. □

Exercise 61.7.2. Show that the Hodge classes in $H^2(T; \mathbb{Q})$ of a complex torus form the space $\operatorname{NS}(T) \otimes \mathbb{Q}$, and that for $T = E^g$, E without CM, this space has dimension $g(g+1)/2$.

Solution. The first statement is Proposition 61.3.1. For E^g, $\mathrm{NS} \otimes \mathbb{Q}$ is the space of $g \times g$ Hermitian matrices with entries in $\mathrm{End}(E) \otimes \mathbb{Q} = \mathbb{Q}$ that are Hermitian for the Rosati involution, that is, symmetric rational matrices, of dimension $g(g+1)/2$ [BL04]. □

Exercise 61.7.3. Show that multiplication by n on T is an isogeny of degree n^{2g} with kernel $\frac{1}{n}\Lambda/\Lambda \cong (\mathbb{Z}/n)^{2g}$.

Solution. The map is induced by $v \mapsto nv$ on V, surjective, with kernel $\{v : nv \in \Lambda\}/\Lambda = \frac{1}{n}\Lambda/\Lambda$, which has n^{2g} elements since $\Lambda \cong \mathbb{Z}^{2g}$. □

Chapter 62

Algebraic Cycles and Chow Groups

62.1 Introduction

An algebraic cycle is a formal integer combination of subvarieties. The Hodge conjecture asks which cohomology classes come from cycles, so cycles and the relations between them are the objects on the algebraic side of the conjecture. This chapter introduces the Chow groups, cycles modulo rational equivalence, with their functoriality (proper pushforward, flat pullback) and the intersection product on smooth varieties, and compares the adequate equivalence relations (rational, algebraic, homological, numerical) that appear in the standard conjectures. We follow Fulton [Ful98]. Many constructions are quoted from there, but the computations for affine and projective spaces and the elementary exact sequences are proved. Chapter 63 attaches cohomology classes to cycles.

Prerequisites. Chapter 57 (Divisors and Line Bundles) and Chapter 54 (Schemes).

Throughout, schemes are separated and of finite type over a field k, a *variety* is an integral such scheme, and *subvariety* means closed subvariety.

62.2 Cycles and rational equivalence

Definition 62.2.1. The group $Z_k(X)$ of *k-cycles* is the free abelian group on the k-dimensional subvarieties of X; write $Z_\bullet(X) = \bigoplus_k Z_k(X)$. For a closed subscheme $Y \subseteq X$ with irreducible components Y_i of dimension k and generic points η_i, its *fundamental cycle* is $[Y] = \sum_i \ell(\mathcal{O}_{Y,\eta_i})[Y_i]$, where ℓ is the length of the Artinian local ring. For a $(k+1)$-dimensional subvariety W and $f \in K(W)^*$, the divisor $\operatorname{div}(f) = \sum_V \operatorname{ord}_V(f)[V] \in Z_k(X)$, the sum over codimension-one subvarieties V of W, is defined with the order function $\operatorname{ord}_V(f) = \ell(\mathcal{O}_{W,V}/(a)) - \ell(\mathcal{O}_{W,V}/(b))$ for $f = a/b$ [Ful98, Section 1.2]. The subgroup of *cycles rationally equivalent to zero* is generated by these divisors, and the *Chow group* is

$$\mathrm{CH}_k(X) = Z_k(X)/\{\text{cycles rationally equivalent to } 0\}.$$

For X equidimensional of dimension n put $\mathrm{CH}^p(X) = \mathrm{CH}_{n-p}(X)$.

Example 62.2.2.

1. $\mathrm{CH}_n(X) = Z_n(X)$ is free on the n-dimensional irreducible components.
2. If X is a nonsingular variety, ord_V is the valuation of the DVR $\mathcal{O}_{X,V}$ (Definition 57.2.1), so $\mathrm{CH}^1(X) = \mathrm{Cl}(X) \cong \mathrm{Pic}(X)$ (Proposition 57.3.2).
3. For a nonsingular projective curve C, $\mathrm{CH}_0(C) = \mathrm{CH}^1(C) = \mathrm{Pic}(C)$, and the degree gives $\mathrm{CH}_0(C) \to \mathbb{Z}$ with kernel $\mathrm{Pic}^0(C)$ (Corollary 59.2.3); over $\mathbb{C}$ this kernel is the Jacobian (Theorem 59.6.3).

Proposition 62.2.3 (Localisation sequence)**.** *Let $Y \subseteq X$ be closed and $U = X \setminus Y$. Then the sequence*

$$\mathrm{CH}_k(Y) \xrightarrow{i_*} \mathrm{CH}_k(X) \xrightarrow{j^*} \mathrm{CH}_k(U) \to 0$$

is exact, where i_* is induced by the inclusion of cycles and j^* by restriction $[V] \mapsto [V \cap U]$.

Proof. On cycles, $0 \to Z_k(Y) \to Z_k(X) \to Z_k(U) \to 0$ is exact: $V \mapsto V \cap U$ is a bijection from k-dimensional subvarieties of X not contained in Y to those of U (with inverse the closure), and those contained in Y are the subvarieties of Y. Both maps preserve rational equivalence, since the divisor of $f \in K(W)^*$ restricts to the divisor of f on $W \cap U$, and a divisor on $W \subseteq Y$ is a divisor on W in X. So j^* is surjective and $j^* \circ i_* = 0$. If $\alpha \in Z_k(X)$ restricts to a cycle rationally equivalent to zero on U, say $\alpha|_U = \sum \operatorname{div}(f_l)$ with $f_l \in K(W_l)^*$, $W_l \subseteq U$, then $\alpha - \sum \operatorname{div}(f_l)$, computed on the closures $\overline{W_l}$ (which have the same function fields), restricts to 0 on U, so it is supported on Y. □

Proposition 62.2.4 (Homotopy invariance)**.** *Let $p\colon X \times \mathbb{A}^1 \to X$ be the projection. Then $p^*\colon \mathrm{CH}_k(X) \to \mathrm{CH}_{k+1}(X \times \mathbb{A}^1)$, $[V] \mapsto [V \times \mathbb{A}^1]$, is surjective. More generally flat pullback along the projection of a vector bundle of rank r is surjective (in fact an isomorphism) $\mathrm{CH}_k(X) \to \mathrm{CH}_{k+r}(E)$.*

Sketch of proof. By the localisation sequence and Noetherian induction on X, it suffices to treat X affine and irreducible, and to show that every $(k+1)$-dimensional subvariety V of $X \times \mathbb{A}^1$ is, modulo cycles pulled back from X, rationally equivalent to cycles supported over a proper closed subset of X. If V dominates a subvariety W of dimension $k+1$ of X, then V is a divisor in $W \times \mathbb{A}^1$, defined over the generic point of W by a polynomial $f \in K(W)[t]$. The divisor of f on $W \times \mathbb{A}^1$ is $[V]$ plus cycles lying over a proper closed subset of W, which are handled by induction. If V dominates a subvariety of dimension k, then $V = W \times \mathbb{A}^1$ is a pullback. The omitted verification is the bookkeeping in this induction [Ful98, Proposition 1.9 and Theorem 3.3]. □

Theorem 62.2.5. *$\mathrm{CH}_k(\mathbb{A}^n) = 0$ for $k < n$ and $\mathrm{CH}_n(\mathbb{A}^n) = \mathbb{Z}$. For $\mathbb{P}^n$, $\mathrm{CH}_k(\mathbb{P}^n) \cong \mathbb{Z}$ for $0 \le k \le n$, generated by the class of a linear subspace L_k of dimension k.*

Proof. For $\mathbb{A}^n$ use induction on n and Proposition 62.2.4: $\mathrm{CH}_k(\mathbb{A}^n) = \mathrm{CH}_k(\mathbb{A}^{n-1} \times \mathbb{A}^1)$ is a quotient of $\mathrm{CH}_{k-1}(\mathbb{A}^{n-1})$, which vanishes for $k - 1 < n - 1$; and $\mathrm{CH}_0(\mathbb{A}^0) = \mathbb{Z}$, $\mathrm{CH}_k(\mathbb{A}^0) = 0$ for $k > 0$. For $\mathbb{P}^n$, the localisation sequence for $\mathbb{P}^{n-1} \subseteq \mathbb{P}^n$ with complement $\mathbb{A}^n$ gives $\mathrm{CH}_k(\mathbb{P}^{n-1}) \to \mathrm{CH}_k(\mathbb{P}^n) \to \mathrm{CH}_k(\mathbb{A}^n) \to 0$. For $k < n$ the right term vanishes, so $\mathrm{CH}_k(\mathbb{P}^n)$ is generated by $[L_k]$ by induction; for $k = n$ it is $\mathbb{Z}[\mathbb{P}^n]$. To see that $[L_k]$ has infinite order, use the degree: a k-cycle α and a general linear subspace M of dimension $n - k$ meet in finitely many points, and $\alpha \mapsto \#(\alpha \cap M)$ (with multiplicities) is invariant under rational equivalence and equals 1 on $[L_k]$ [Ful98, Chapter 1]. □

Example 62.2.6. Let $Y = V(f) \subseteq \mathbb{P}^n$ be an irreducible hypersurface of degree d and $U = \mathbb{P}^n \setminus Y$. Then $\mathrm{CH}^1(U) \cong \mathbb{Z}/d$. Indeed $\operatorname{div}(f/x_0^d) = [Y] - d[H_0]$ on $\mathbb{P}^n$, with $H_0 = V(x_0)$, so $[Y] = d[H]$ in $\mathrm{CH}^1(\mathbb{P}^n) = \mathbb{Z}[H]$ (Theorem 62.2.5). The localisation sequence (Proposition 62.2.3) reads $\mathrm{CH}_{n-1}(Y) = \mathbb{Z}[Y] \to \mathrm{CH}_{n-1}(\mathbb{P}^n) = \mathbb{Z}[H] \to \mathrm{CH}_{n-1}(U) \to 0$, with image $d\mathbb{Z}[H]$. For instance the complement of a smooth conic in $\mathbb{P}^2$ has $\mathrm{Pic}(U) = \mathrm{CH}^1(U) \cong \mathbb{Z}/2$ (Example 62.2.2(2)), a nonzero torsion Picard group on a smooth affine surface: the restriction of a line L is not principal on U, but $2L$ restricts to the principal divisor of ℓ^2/q, with ℓ and q equations of L and of the conic.

Counterexample 62.2.7. The degree of Theorem 62.3.1(3) needs completeness. On $\mathbb{A}^1 = \operatorname{Spec} k[t]$ every point a is the divisor of the rational function $t - a$, so $[a] = 0$ in $\mathrm{CH}_0(\mathbb{A}^1)$, and $\mathrm{CH}_0(\mathbb{A}^1) = 0$, in accordance with Theorem 62.2.5. On $\mathbb{P}^1$ the same function has divisor

$[a]-[\infty]$: the pole at infinity, invisible on $\mathbb{A}^1$, is what makes the degree of a principal divisor zero rather than one.

62.3 Functoriality

Theorem 62.3.1. *1. (Proper pushforward) For a proper morphism $f\colon X\to Y$, the map $f_*[V]=\deg(V/f(V))[f(V)]$ if $\dim f(V)=\dim V$ and 0 otherwise, where $\deg(V/f(V))=[K(V):K(f(V))]$, induces homomorphisms $f_*\colon \mathrm{CH}_k(X)\to\mathrm{CH}_k(Y)$, functorial in f.*

2. (Flat pullback) For a flat morphism $f\colon X\to Y$ of relative dimension r, $f^*[V]=[f^{-1}(V)]$ induces $f^*\colon \mathrm{CH}_k(Y)\to\mathrm{CH}_{k+r}(X)$.

3. For X complete, the degree $\deg\colon \mathrm{CH}_0(X)\to\mathbb{Z}$, $\sum n_P[P]\mapsto\sum n_P[k(P):k]$, *is well defined; it is the pushforward to $\mathrm{CH}_0(\operatorname{Spec}k)=\mathbb{Z}$.*

This is [Ful98, Theorem 1.4 and Theorem 1.7]. For (1), the key case is a proper surjective morphism of curves, where the pushforward of a principal divisor is principal by the norm: $f_*\mathrm{div}(g)=\mathrm{div}(N_{K(X)/K(Y)}(g))$. For curves this is Corollary 59.2.3 applied to g and its norm.

62.4 Chern classes and the intersection product

Definition 62.4.1. For a line bundle L on X and a k-dimensional subvariety V, write $L|_V\cong\mathcal{O}_V(D)$ for a Cartier divisor D on V and put $c_1(L)\cap[V]=[D]\in\mathrm{CH}_{k-1}(X)$. This is well defined and extends to $c_1(L)\cap-\colon \mathrm{CH}_k(X)\to\mathrm{CH}_{k-1}(X)$ [Ful98, Chapter 2]. For a vector bundle E of rank r, with projective bundle $p\colon\mathbb{P}(E)\to X$ and tautological line bundle $\mathcal{O}(1)$, the *Chern classes* $c_i(E)\cap-$ are the unique operations with $\sum_{i=0}^{r}p^*c_i(E)\cap c_1(\mathcal{O}(1))^{r-i}\cap p^*\alpha=0$ and $c_0=1$ (Grothendieck's definition) [Ful98, Chapter 3].

Theorem 62.4.2. *The Chern classes of vector bundles on X, acting on $\mathrm{CH}_\bullet(X)$, commute with each other and satisfy naturality under flat pullback and proper pushforward (projection formula), the Whitney formula $c(E)=c(E')c(E'')$ for $0\to E'\to E\to E''\to 0$, and the splitting principle. For X smooth over $\mathbb{C}$ they map to the topological Chern classes of Chapter 39 under the cycle class map.*

This is [Ful98, Theorem 3.2]; the comparison with topology is in [Ful98, Chapter 19].

Theorem 62.4.3. *Let X be a nonsingular variety of dimension n. There is a product $\mathrm{CH}^p(X)\otimes\mathrm{CH}^q(X)\to\mathrm{CH}^{p+q}(X)$, $\alpha\otimes\beta\mapsto\alpha\cdot\beta$, making $\mathrm{CH}^\bullet(X)=\bigoplus_p\mathrm{CH}^p(X)$ a commutative, associative graded ring with unit $[X]$, such that:*

1. *if subvarieties V, W intersect* properly, *that is, every component Z of $V\cap W$ has codimension* $\operatorname{codim}V+\operatorname{codim}W$, *then $[V]\cdot[W]=\sum_Z i(Z;V\cdot W)[Z]$ with positive intersection multiplicities, given by Serre's formula $i(Z;V\cdot W)=\sum_j(-1)^j\ell_{\mathcal{O}_{X,Z}}\operatorname{Tor}_j^{\mathcal{O}_{X,Z}}(\mathcal{O}_V,\mathcal{O}_W)$; if the intersection is also transverse at the generic point of Z, then $i(Z;V\cdot W)=1$;*
2. *for a morphism $f\colon X\to Y$ of nonsingular varieties there is a ring homomorphism $f^*\colon\mathrm{CH}^\bullet(Y)\to\mathrm{CH}^\bullet(X)$, and for f proper the* projection formula *$f_*(f^*\alpha\cdot\beta)=\alpha\cdot f_*\beta$ holds;*
3. *on $\mathrm{CH}^1(X)=\mathrm{Pic}(X)$ the product with CH^p is $c_1(L)\cap-$, and for $n=2$ the product $\mathrm{CH}^1\times\mathrm{CH}^1\to\mathrm{CH}^2\xrightarrow{\deg}\mathbb{Z}$ is the intersection number of Definition 57.6.1.*

This is [Ful98, Chapters 6–8]; the construction uses deformation to the normal cone. Serre's formula is discussed in [Ful98, Chapter 7]. Chow's moving lemma gives an older construction for quasi-projective X: cycles can be moved within their rational equivalence class to intersect properly.

Example 62.4.4. $\mathrm{CH}^\bullet(\mathbb{P}^n) = \mathbb{Z}[h]/(h^{n+1})$, with $h = [H]$ the class of a hyperplane: $h^k = [L_{n-k}]$, since k general hyperplanes meet transversally in a linear subspace. So a subvariety of codimension p and degree d has class dh^p, and *Bézout's theorem* says that properly intersecting subvarieties of degrees d and e intersect in a cycle of degree de.

Example 62.4.5. CH_0 can be enormous. For a complex projective surface S with $h^{2,0}(S) \neq 0$, Mumford showed that the kernel of $\deg\colon \mathrm{CH}_0(S) \to \mathbb{Z}$ is not "finite-dimensional": it is not parametrised by any variety [Mum68]. By contrast, Bloch's conjecture predicts that $\mathrm{CH}_0(S) = \mathbb{Z}$ when $p_g(S) = q(S) = 0$; it is known for surfaces not of general type. These phenomena are the starting point of the Bloch–Beilinson conjectures relating Chow groups to Hodge theory.

62.5 Correspondences

Definition 62.5.1. Let X, Y be nonsingular projective varieties, X of pure dimension d. A *correspondence of degree r* from X to Y is an element of $\mathrm{CH}^{d+r}(X \times Y)$. It acts on Chow groups by $\Gamma_*(\alpha) = \mathrm{pr}_{2*}(\Gamma \cdot \mathrm{pr}_1^*\alpha)$, and correspondences compose by $\Gamma' \circ \Gamma = \mathrm{pr}_{13*}(\mathrm{pr}_{12}^*\Gamma \cdot \mathrm{pr}_{23}^*\Gamma')$.

Proposition 62.5.2. *Composition of correspondences is associative, $(\Gamma' \circ \Gamma)_* = \Gamma'_* \circ \Gamma_*$, the diagonal Δ_X acts as the identity, and the graph of a morphism $f\colon X \to Y$ acts as f_*, its transpose as f^*.*

This is [Ful98, Proposition 16.1.1]; it follows from the projection formula and the compatibility of pullback and pushforward in fibre squares. Correspondences are the morphisms in the category of Chow motives, and the idempotent correspondences, projectors, cut motives out of varieties (Chapter 77).

62.6 Adequate equivalence relations

Definition 62.6.1. Let X be a nonsingular projective variety and $\alpha \in Z^p(X)$.

1. α is *algebraically equivalent to zero* if there are a nonsingular projective curve C, points $c_0, c_1 \in C$ and a cycle Γ on $C \times X$ with $\alpha = \Gamma(c_1) - \Gamma(c_0)$, where $\Gamma(c) = \mathrm{pr}_{2*}(\Gamma \cdot [\{c\} \times X])$.
2. For a Weil cohomology theory $H^\bullet$ with a cycle class map $\mathrm{cl}\colon Z^p(X) \to H^{2p}(X)$ compatible with pullbacks, proper pushforwards, products and degrees, for instance singular cohomology when $k = \mathbb{C}$ (Chapter 63), α is *homologically equivalent to zero* if $\mathrm{cl}(\alpha) = 0$.
3. α is *numerically equivalent to zero* if $\deg(\alpha \cdot \beta) = 0$ for all $\beta \in Z^{n-p}(X)$.

We write $\mathrm{CH}^p(X) \to \mathrm{Alg}^p(X) \to \mathrm{Hom}^p(X) \to \mathrm{Num}^p(X)$ for the quotients of $Z^p(X)$ by these relations.

Proposition 62.6.2. *Rational equivalence implies algebraic equivalence, which implies homological equivalence (for any cohomology theory as in Definition 62.6.1(2)), which implies numerical equivalence.*

Proof. Rational equivalence is the special case $C = \mathbb{P}^1$ [Ful98, Proposition 1.6]: the divisor of f on W is $\Gamma(0) - \Gamma(\infty)$ for the closure Γ of the graph of $f\colon W \dashrightarrow \mathbb{P}^1$. If $\alpha = \Gamma(c_1) - \Gamma(c_0)$, then $\mathrm{cl}(\alpha) = \Gamma_*(\mathrm{cl}(c_1) - \mathrm{cl}(c_0))$, and $\mathrm{cl}(c_1) = \mathrm{cl}(c_0)$ in $H^2(C)$ since both are the point class, which is determined by the degree. Homological equivalence implies numerical equivalence because the cycle class map is assumed compatible with products and degrees: $\deg(\alpha \cdot \beta) = \int \mathrm{cl}(\alpha) \cup \mathrm{cl}(\beta)$. The previous step used the compatibility with pullback and pushforward. $\square$

Remark 62.6.3. All three inclusions can be strict. Rational and algebraic equivalence differ already for points on an elliptic curve, since $\mathrm{Pic}^0(E) \neq 0$. Algebraic and homological equivalence differ in codimension ≥ 2: the *Griffiths group* $\mathrm{Hom}^p(X)/\mathrm{Alg}^p(X)$ can be nonzero, and even not finitely generated, for 1-cycles on a general quintic threefold (Griffiths, Clemens; Chapter 71). Whether homological and numerical equivalence coincide is Grothendieck's standard conjecture D (Chapter 77); it is known for divisors (Matsusaka) and in characteristic 0 it follows from the Hodge conjecture.

62.7 Riemann–Roch

Definition 62.7.1. For a vector bundle E with Chern roots $x_1, \dots, x_r$ (splitting principle), the *Chern character* and *Todd class* are

$$\mathrm{ch}(E) = \sum_i e^{x_i} = r + c_1 + \tfrac{1}{2}(c_1^2 - 2c_2) + \cdots, \qquad \mathrm{td}(E) = \prod_i \frac{x_i}{1 - e^{-x_i}} = 1 + \tfrac{1}{2}c_1 + \tfrac{1}{12}(c_1^2 + c_2) + \cdots$$

in $\mathrm{CH}^\bullet(X) \otimes \mathbb{Q}$. For a coherent sheaf $\mathcal{F}$ on a nonsingular variety, $\mathrm{ch}(\mathcal{F})$ is defined through a finite locally free resolution.

Theorem 62.7.2 (Grothendieck–Riemann–Roch)**.** *Let $f\colon X \to Y$ be a proper morphism of nonsingular quasi-projective varieties and $\mathcal{F}$ a coherent sheaf on X. Then in* $\mathrm{CH}^\bullet(Y) \otimes \mathbb{Q}$

$$\mathrm{ch}\Big(\sum_i (-1)^i R^i f_* \mathcal{F}\Big) \cdot \mathrm{td}(T_Y) = f_*\big(\mathrm{ch}(\mathcal{F}) \cdot \mathrm{td}(T_X)\big).$$

*In particular, for X projective and $Y = \operatorname{Spec} k$ (*Hirzebruch–Riemann–Roch*),*

$$\chi(X, \mathcal{F}) = \int_X \mathrm{ch}(\mathcal{F})\,\mathrm{td}(T_X).$$

This is [Ful98, Theorem 15.2 and Corollary 15.2.1]. Hirzebruch proved the second formula over $\mathbb{C}$ [Hir66]. Over $\mathbb{C}$ it is also a special case of the Atiyah–Singer index theorem for the Dolbeault complex (Chapter 42).

Example 62.7.3.

1. For a curve of genus g and a line bundle L: $\mathrm{td}(T_C) = 1 + \frac{1}{2}c_1(T_C)$ and $\int c_1(T_C) = 2 - 2g$, so HRR gives $\chi(L) = \deg L + 1 - g$, which is Theorem 59.3.2.
2. For a surface S with canonical class K and $c_2 = c_2(T_S)$: $\mathrm{td}(T_S) = 1 - \frac{1}{2}K + \frac{1}{12}(K^2 + c_2)$. For $\mathcal{F} = \mathcal{O}_S$ this gives *Noether's formula* $\chi(\mathcal{O}_S) = \frac{1}{12}(K^2 + c_2)$, and for a divisor D, $\chi(\mathcal{O}(D)) = \chi(\mathcal{O}_S) + \frac{1}{2}D \cdot (D - K)$.
3. For $S = \mathbb{P}^2$: $K^2 = 9$ and $c_2 = 3$, so $\chi(\mathcal{O}) = 1$. For a K3 surface: $K = 0$, $c_2 = 24$ (Exercise 52.6.1), so $\chi(\mathcal{O}) = 2$, matching $h^0 - h^1 + h^2 = 1 - 0 + 1$.

Example 62.7.4. Theorem 62.7.2 has content beyond the case $Y = \operatorname{Spec} k$; here is a case where both sides can be computed. Let X be a nonsingular projective variety, $i\colon D \to X$ the inclusion of a nonsingular prime divisor, $d = [D] \in \mathrm{CH}^1(X)$, and $\mathcal{F} = \mathcal{O}_D$. Since i is a closed immersion, $R^q i_* \mathcal{O}_D = 0$ for $q > 0$, and the exact sequence $0 \to \mathcal{O}_X(-D) \to \mathcal{O}_X \to i_*\mathcal{O}_D \to 0$ gives $\mathrm{ch}(i_*\mathcal{O}_D) = 1 - e^{-d}$ (the Chern character is additive in exact sequences, by the Whitney formula of Theorem 62.4.2 and the splitting principle). So the left side is $(1 - e^{-d})\,\mathrm{td}(T_X)$. On the right, the exact sequence $0 \to T_D \to i^*T_X \to N_{D/X} \to 0$ with $N_{D/X} = i^*\mathcal{O}_X(D)$ and the multiplicativity of the Todd class give

$$\mathrm{td}(T_D) = i^*\Big(\mathrm{td}(T_X) \cdot \frac{1 - e^{-d}}{d}\Big),$$

where $(1-e^{-d})/d = 1 - \tfrac{1}{2}d + \tfrac{1}{6}d^2 - \cdots$ is the inverse of $\operatorname{td}(\mathcal{O}_X(D))$. By the projection formula (Theorem 62.4.3(2)), $i_*i^*\beta = \beta \cdot i_*[D] = \beta \cdot d$, so the right side is $\operatorname{td}(T_X)\,(1-e^{-d})$ as well. The denominators of the Todd class cancel against the normal bundle: this is the mechanism of the general proof, which factors a projective morphism into a closed immersion and a projection $\mathbb{P}^n_Y \to Y$ and treats the two cases separately [Ful98, Chapter 15].

Example 62.7.5. Let $f\colon X \to Y$ be a finite étale morphism of degree n between nonsingular projective varieties (Definition 54.8.4). Then $f^*\Omega_Y \cong \Omega_X$, so $T_X \cong f^*T_Y$, and $R^q f_*\mathcal{O}_X = 0$ for $q > 0$ because f is finite, hence affine. Theorem 62.7.2 and the projection formula give

$$\operatorname{ch}(f_*\mathcal{O}_X)\operatorname{td}(T_Y) = f_*\big(f^*\operatorname{td}(T_Y)\big) = \operatorname{td}(T_Y)\cdot f_*[X] = n\operatorname{td}(T_Y),$$

and since $\operatorname{td}(T_Y)$ is invertible, $\operatorname{ch}(f_*\mathcal{O}_X) = n$: the rank-$n$ vector bundle $f_*\mathcal{O}_X$ has all its Chern classes zero in $\mathrm{CH}^\bullet(Y)\otimes\mathbb{Q}$. Taking degrees, $\chi(X,\mathcal{O}_X) = \chi(Y, f_*\mathcal{O}_X) = n\,\chi(Y,\mathcal{O}_Y)$. For curves this is $1-g(X) = n(1-g(Y))$, the Riemann–Hurwitz formula for unramified coverings (Theorem 59.4.1).

62.8 Exercises

Exercise 62.8.1. Compute $\mathrm{CH}^\bullet(\mathbb{P}^1\times\mathbb{P}^1)$ using the localisation sequence, and show that it is $\mathbb{Z}[a,b]/(a^2,b^2)$ with a, b the classes of the two rulings.

Solution. Remove the two lines $\{\infty\}\times\mathbb{P}^1$ and $\mathbb{P}^1\times\{\infty\}$; the complement is $\mathbb{A}^2$. Localisation and Theorem 62.2.5 show that CH_1 is generated by the two lines and CH_0 by a point, and $\mathrm{CH}_2 = \mathbb{Z}$. The two line classes are independent, as their intersection numbers with the lines are $a\cdot a = 0$, $a\cdot b = 1$, $b\cdot b = 0$ (Example 57.6.4). The ring structure follows: $a^2 = b^2 = 0$ and $ab = [\mathrm{pt}]$. □

Exercise 62.8.2. Use Example 62.4.4 to show that two plane conics without common components meet in at most 4 points, and in exactly 4 counted with multiplicity.

Solution. $[C]\cdot[C'] = (2h)(2h) = 4h^2 = 4[\mathrm{pt}]$ in $\mathrm{CH}^2(\mathbb{P}^2)$, and since the intersection is proper, it equals $\sum_P i(P)[P]$ with $i(P)\geq 1$. Taking degrees, $\sum i(P) = 4$. □

Exercise 62.8.3. Check Noether's formula for a smooth quartic surface and for $\mathbb{P}^1\times\mathbb{P}^1$.

Solution. For the quartic, $K = 0$, $c_2 = 24$, and $\chi(\mathcal{O}) = 2$ as computed in Example 62.7.3. For $\mathbb{P}^1\times\mathbb{P}^1$, $K = -2a-2b$ has $K^2 = 8$, $c_2 = \chi_{\mathrm{top}} = 4$, and $\frac{8+4}{12} = 1 = \chi(\mathcal{O})$. □

Chapter 63

The Cycle Class Map

63.1 Introduction

A subvariety Z of codimension p in a complex projective manifold X has a cohomology class $[Z] \in H^{2p}(X;\mathbb{Z})$. Topologically it is the Poincaré dual of the fundamental class of Z, which exists although Z may be singular. Analytically it is represented by the current of integration over Z. The resulting *cycle class map* $\mathrm{cl}\colon \mathrm{CH}^p(X) \to H^{2p}(X;\mathbb{Z})$ is a ring homomorphism compatible with pullbacks, pushforwards and Chern classes. Its image consists of Hodge classes (Corollary 63.4.2), and the Hodge conjecture asks whether, after tensoring with $\mathbb{Q}$, every Hodge class is in the image. This chapter constructs the map, states its properties, and shows that the conjecture can equivalently be phrased with Chern classes of vector bundles. References: [Ful98, Chapter 19], [Voi02b, Chapter 11], [GH78, Chapter 0, Section 4].

Prerequisites. Chapter 62 (Algebraic Cycles and Chow Groups), Chapter 58 (Analytic Spaces and GAGA), Chapter 37 (de Rham Cohomology), Chapter 23 (Topological Manifolds and Poincaré Duality) and Chapter 48 (The Hodge Decomposition).

Throughout, X is a nonsingular complex variety of dimension n, identified with the complex manifold X^{an} (Chapter 58), and cohomology is singular cohomology of X^{an}.

63.2 Fundamental classes

Lemma 63.2.1. *Let S be a closed complex analytic subset of complex dimension $\le d$ of a complex manifold. Then $H_i^{BM}(S;\mathbb{Z}) = 0$ for $i > 2d$, and $H_{2d}^{BM}(S;\mathbb{Z})$ is free on the irreducible components of dimension d.*

Sketch of proof. Analytic sets admit triangulations compatible with a stratification by complex submanifolds, the strata of S having real dimension $\le 2d$ [Ful98, Chapter 19]. So the locally finite chain complex of S vanishes above degree $2d$, which gives the first statement. For the second, let $\Sigma = \mathrm{Sing}(S) \cup S_{<d}$, where $S_{<d}$ is the union of the components of dimension $< d$. It has complex dimension $\le d-1$, and the long exact sequence $H_{2d}^{BM}(\Sigma) \to H_{2d}^{BM}(S) \to H_{2d}^{BM}(S \setminus \Sigma) \to H_{2d-1}^{BM}(\Sigma)$ has outer terms zero by the first statement. The complex manifold $S \setminus \Sigma$ is canonically oriented, of real dimension $2d$, with one connected component for each d-dimensional irreducible component of S (the regular locus of an irreducible analytic set is connected), and H_{2d}^{BM} of an oriented $2d$-manifold is free on its connected components (Remark 23.8.1). □

Definition 63.2.2. Let $Z \subseteq X$ be a subvariety of codimension p. Its *fundamental class* $[Z] \in H_{2n-2p}^{BM}(Z;\mathbb{Z})$ is the class corresponding to Z in Lemma 63.2.1. Its *cycle class* $\mathrm{cl}(Z) \in H^{2p}(X;\mathbb{Z})$ is the image of $[Z]$ under $H_{2n-2p}^{BM}(Z) \cong H^{2p}(X, X \setminus Z) \to H^{2p}(X)$ (Remark 23.8.2).

It extends linearly to $\mathrm{cl}\colon Z^p(X) \to H^{2p}(X;\mathbb{Z})$. For X compact, $\mathrm{cl}(Z)$ is the Poincaré dual of $i_*[Z] \in H_{2n-2p}(X;\mathbb{Z})$.

Proposition 63.2.3. *Let X be compact. The image of $\mathrm{cl}(Z)$ in $H^{2p}(X;\mathbb{R}) \cong H^{2p}_{\mathrm{dR}}(X)$ is the class of the current of integration over Z_{reg} of Proposition 44.6.2: for every closed $(2n-2p)$-form φ,*

$$\int_X \mathrm{cl}(Z) \wedge \varphi = \int_{Z_{\mathrm{reg}}} \varphi.$$

Sketch of proof. For Z nonsingular this is Theorem 37.4.3. In general both sides are determined by their restriction to the complement of $\Sigma = \mathrm{Sing}(Z)$. The topological class is determined there by Lemma 63.2.1, and the current by the fact that integration over Z_{reg} is a closed current whose restriction to $X \setminus \Sigma$ is the integration current of a closed submanifold. Two closed currents of degree $2p$ that agree off an analytic set of real codimension $> 2p$ in X are equal (the difference would be a closed current supported on Σ of degree below its codimension, which vanishes by the support theorem) [GH78, Chapter 3], [Voi02b, Section 11.1]. □

63.3 The cycle class map and its properties

Theorem 63.3.1. *Let X be a nonsingular complex variety.*

1. cl *factors through rational equivalence and gives homomorphisms* $\mathrm{cl}\colon \mathrm{CH}^p(X) \to H^{2p}(X;\mathbb{Z})$, *which form a ring homomorphism* $\mathrm{CH}^\bullet(X) \to H^{2\bullet}(X;\mathbb{Z})$ *for the intersection product and the cup product.*
2. *It is compatible with pullback along morphisms of nonsingular varieties and with proper pushforward (Gysin maps in cohomology), and with exterior products.*
3. $\mathrm{cl}(c_i(E) \cap [X]) = c_i(E)$, *the topological Chern class of the underlying complex vector bundle.*
4. *For X projective,* $\deg \alpha = \int_X \mathrm{cl}(\alpha)$ *for* $\alpha \in \mathrm{CH}^n(X)$.

This is [Ful98, Chapter 19]. For (1), rational equivalence is generated by the fibres $\Gamma(0) - \Gamma(\infty)$ of a family $\Gamma \subseteq \mathbb{P}^1 \times X$ (Proposition 62.6.2), and $\mathrm{cl}(\Gamma(0)) = \mathrm{cl}(\Gamma(\infty))$ because the two points of $\mathbb{P}^1$ have the same class in $H^2(\mathbb{P}^1;\mathbb{Z})$ and $\mathrm{cl}(\Gamma(t)) = \mathrm{pr}_{2*}(\mathrm{cl}(\Gamma) \cup \mathrm{pr}_1^*\mathrm{cl}(t))$. Multiplicativity reduces, by the moving lemma or deformation to the normal cone, to transverse intersections, where it is the statement that cup product is Poincaré dual to transverse intersection (Remark 23.8.2).

Example 63.3.2.

1. On $\mathbb{P}^n$, $\mathrm{cl}(h^k)$ is the k-th power of the positive generator of $H^2(\mathbb{CP}^n;\mathbb{Z})$, and $\mathrm{cl}\colon \mathrm{CH}^\bullet(\mathbb{P}^n) \to H^{2\bullet}(\mathbb{CP}^n;\mathbb{Z})$ is an isomorphism of rings (Example 62.4.4 and Theorem 22.5.1).
2. For a divisor D, $\mathrm{cl}(D) = c_1(\mathcal{O}(D))$, by (3) applied to $E = \mathcal{O}(D)$ and $c_1(\mathcal{O}(D)) \cap [X] = [D]$. So on $\mathrm{CH}^1(X) = \mathrm{Pic}(X)$ the cycle class map is the first Chern class of Theorem 45.2.4.
3. For a point x of a projective X, $\mathrm{cl}(x)$ is the generator of $H^{2n}(X;\mathbb{Z}) \cong \mathbb{Z}$ dual to the fundamental class. So cl on CH_0 is the degree, and it loses all the information in the huge kernel of Example 62.4.5.

Example 63.3.3.

1. Let C be a nonsingular projective curve of genus g. On $\mathrm{CH}^1(C) = \mathrm{Pic}(C)$ the cycle class map is the degree $\mathrm{CH}^1(C) \to H^2(C;\mathbb{Z}) \cong \mathbb{Z}$ (Example 63.3.2(2) and (3)), which is surjective, and its kernel $\mathrm{CH}^1(C)_{\mathrm{hom}}$ is the group of divisor classes of

degree 0, isomorphic to the Jacobian $J(C)$ by the Abel–Jacobi theorem (Theorem 59.6.3). For $g \geq 1$ this kernel is an uncountable group; for an elliptic curve and points $p \neq q$ the class of $p - q$ is nonzero (Counterexample 45.3.4 and GAGA). For $g = 0$, cl is an isomorphism.

2. Let $Z = V(f_1, \dots, f_p) \subseteq \mathbb{P}^n$ be a complete intersection of hypersurfaces of degrees $d_1, \dots, d_p$, meeting properly and generically transversally. Then $[Z] = [V(f_1)] \cdots [V(f_p)]$ in $\mathrm{CH}^p(\mathbb{P}^n)$, so by Theorem 63.3.1(1) and Exercise 63.6.1, $\mathrm{cl}(Z) = d_1 \cdots d_p\, h^p$, and $\int \mathrm{cl}(Z) \cup h^{n-p} = d_1 \cdots d_p$ is the degree of Z (Bézout).

63.4 Cycle classes are Hodge classes

Definition 63.4.1. Let X be a compact Kähler manifold. The group of *rational Hodge classes* of degree $2p$ is $\mathrm{Hdg}^p(X) = H^{2p}(X;\mathbb{Q}) \cap H^{p,p}(X)$, the intersection taken inside $H^{2p}(X;\mathbb{C})$. The group of *integral Hodge classes* $\mathrm{Hdg}^p(X;\mathbb{Z})$ is the preimage of $H^{p,p}(X)$ under $H^{2p}(X;\mathbb{Z}) \to H^{2p}(X;\mathbb{C})$; it contains the torsion subgroup.

Corollary 63.4.2. *For a nonsingular projective complex variety X, $\mathrm{cl}(\mathrm{CH}^p(X)) \subseteq \mathrm{Hdg}^p(X;\mathbb{Z})$, and $\mathrm{cl}(\mathrm{CH}^p(X)) \otimes \mathbb{Q} \subseteq \mathrm{Hdg}^p(X)$.*

Proof. By Proposition 63.2.3, the image of $\mathrm{cl}(Z)$ in $H^{2p}(X;\mathbb{C})$ is the class of the integration current over Z, which lies in $H^{p,p}(X)$ by Proposition 48.5.8. Extend linearly. □

Corollary 63.4.3. *Let $X \subseteq \mathbb{P}^N$ be a nonsingular projective variety of dimension n, let $h \in H^2(X;\mathbb{Z})$ be the class of a hyperplane section, and let $Z = \sum_i a_i Z_i \in Z^p(X)$ be a nonzero effective cycle ($a_i \geq 0$). Then $\int_X \mathrm{cl}(Z) \cup h^{n-p} > 0$. In particular $\mathrm{cl}(Z)$ is not a torsion class, and a nonzero effective cycle is not homologically, hence not rationally, equivalent to zero.*

Proof. By Example 63.3.2(2), $h = c_1(\mathcal{O}(1)|_X)$, which is represented by the restriction ω of the Fubini–Study form (Example 45.4.3 and naturality of Chern–Weil forms), a Kähler form on X. By Proposition 63.2.3 and Proposition 47.2.4,

$$\int_X \mathrm{cl}(Z_i) \cup h^{n-p} = \int_{Z_{i,\mathrm{reg}}} \omega^{n-p} = (n-p)!\,\mathrm{vol}(Z_{i,\mathrm{reg}}) > 0,$$

and summing with the coefficients $a_i \geq 0$, not all zero, gives a positive number. A torsion class, or the class 0, would give 0. □

Counterexample 63.4.4. Corollary 63.4.3 is a constraint on which Hodge classes can be classes of *effective* cycles, and the Hodge conjecture cannot be strengthened to effective cycles. On $X = \mathbb{P}^1 \times \mathbb{P}^1$ with the classes a, b of the two rulings, $\int (a-b) \cup (a+b) = 0$ for the hyperplane class $a + b$ of the Segre embedding (as $a^2 = b^2 = 0$), so $a - b$ is an integral Hodge class, the class of the divisor $\{p\} \times \mathbb{P}^1 - \mathbb{P}^1 \times \{q\}$, but not the class of any effective cycle. In general the classes of effective cycles of codimension p form a monoid, closed under addition, which by the corollary meets the hyperplane $\{\alpha : \int_X \alpha \cup h^{n-p} = 0\}$ only in 0.

Remark 63.4.5. The *Hodge conjecture* (Chapter 68) asserts that $\mathrm{cl}(\mathrm{CH}^p(X)) \otimes \mathbb{Q} = \mathrm{Hdg}^p(X)$ for every nonsingular projective complex variety. The integral version, $\mathrm{cl}(\mathrm{CH}^p(X)) = \mathrm{Hdg}^p(X;\mathbb{Z})$, is false: Atiyah and Hirzebruch found torsion classes that are not algebraic [AH62], and Kollár found non-torsion counterexamples (Chapter 73). For $p = 1$ the integral statement is true: this is the Lefschetz $(1,1)$-theorem (Chapter 69), which for projective X follows from Lemma 50.2.2 together with Example 63.3.2(2) and GAGA.

63.5 Cycles and coherent sheaves

Proposition 63.5.1. *Let X be a nonsingular projective variety and $Z \subseteq X$ a subvariety of codimension p. Then* $\operatorname{ch}(\mathcal{O}_Z) = [Z] + (\text{terms in } \mathrm{CH}^{>p}(X) \otimes \mathbb{Q})$, *where* ch *is computed with a finite locally free resolution of $\mathcal{O}_Z$. Consequently the $\mathbb{Q}$-subspace of $H^{2\bullet}(X;\mathbb{Q})$ spanned by the classes of algebraic cycles equals the $\mathbb{Q}$-subalgebra generated by the Chern classes of algebraic vector bundles on X.*

Proof. The first statement is in [Ful98, Chapter 15]: it follows from Grothendieck–Riemann–Roch (Theorem 62.7.2) applied to a resolution of singularities of Z, or directly from the fact that the lowest nonzero term of ch of a sheaf supported in codimension p is its fundamental cycle. On a nonsingular projective variety every coherent sheaf has a finite resolution by locally free sheaves [Har77, Exercises III.6.8 and III.6.9], so $\operatorname{ch}(\mathcal{O}_Z)$ is a polynomial with rational coefficients in Chern classes of vector bundles. By descending induction on p, every $[Z]$ lies in the $\mathbb{Q}$-algebra generated by Chern classes of vector bundles. Conversely, Chern classes of vector bundles are classes of algebraic cycles, $c_i(E) \cap [X] \in \mathrm{CH}^i(X)$ (Theorem 63.3.1(3)), and products of cycle classes are cycle classes (Theorem 63.3.1(1)). □

Remark 63.5.2. Proposition 63.5.1 suggests versions of the Hodge conjecture for compact Kähler manifolds: with analytic cycles, with Chern classes of holomorphic vector bundles, or with Chern classes of coherent analytic sheaves. All of these fail for suitable complex tori: Voisin constructed a four-dimensional torus with a nonzero Hodge class in H^4 that is not a combination of Chern classes of coherent sheaves [Voi02a]. So projectivity is essential (Chapter 73).

Remark 63.5.3. The cycle class map refines to the *Deligne cycle class* with values in Deligne cohomology $H^{2p}_{\mathcal{D}}(X;\mathbb{Z}(p))$. Its restriction to homologically trivial cycles is the Abel–Jacobi map to the intermediate Jacobian (Chapter 71), which sees information about $\mathrm{CH}^p(X)$ invisible to cohomology, such as Pic^0 for $p = 1$.

63.6 Exercises

Exercise 63.6.1. Let $X \subseteq \mathbb{P}^n$ be a hypersurface of degree d. Show that $\operatorname{cl}(X) = dh$ in $H^2(\mathbb{CP}^n;\mathbb{Z})$.

Solution. $[X] = \operatorname{div}(f/x_0^d) + d[H_0]$ in $Z^1(\mathbb{P}^n)$ for a defining polynomial f, so $[X] = dh$ in $\mathrm{CH}^1(\mathbb{P}^n)$, and $\operatorname{cl}(h)$ is the generator. Alternatively, $\operatorname{cl}(X) = c_1(\mathcal{O}(X)) = c_1(\mathcal{O}(d)) = dh$ by Example 63.3.2(2). □

Exercise 63.6.2. Let X be a nonsingular projective variety of dimension n and $\Delta \subseteq X \times X$ the diagonal. Show that $\deg(\Delta \cdot \Delta) = \chi_{\mathrm{top}}(X)$, and more generally the *Lefschetz fixed point formula*: for a morphism $f\colon X \to X$ with graph Γ_f, $\deg(\Gamma_f \cdot \Delta) = \sum_i (-1)^i \operatorname{tr}(f^* \mid H^i(X;\mathbb{Q}))$.

Solution. Choose a basis (e_a) of $H^\bullet(X;\mathbb{Q})$ with dual basis (e^a) for the Poincaré pairing, $\int_X e_a \cup e^b = \delta_a^b$. By the Künneth formula, $\operatorname{cl}(\Delta) = \sum_a (-1)^{\deg e_a} e_a \otimes e^a$ (the sign makes $\int_{X\times X} \operatorname{cl}(\Delta) \cup (\alpha \otimes \beta) = \int_X \alpha \cup \beta$). Then $\operatorname{cl}(\Gamma_f) = (1 \times f)^* \operatorname{cl}(\Delta)$ up to the ordering of factors, and $\int \operatorname{cl}(\Gamma_f) \cup \operatorname{cl}(\Delta) = \sum_a (-1)^{\deg e_a} \int_X f^* e_a \cup e^a$, which is the alternating trace. For $f = \mathrm{id}$ the traces are the Betti numbers. Degrees of intersections are integrals of cup products by Theorem 63.3.1(4). □

Exercise 63.6.3. Show that every torsion class in $H^{2p}(X;\mathbb{Z})$ is an integral Hodge class, and that for X with $H^{2p}(X;\mathbb{Z})$ torsion-free and $H^{2p}(X;\mathbb{C}) = H^{p,p}(X)$ every integral class is a Hodge class.

Solution. A torsion class maps to $0 \in H^{2p}(X;\mathbb{C})$, which lies in $H^{p,p}$. The second statement is immediate from the definition. □

Chapter 64

Index Theorems and Riemann–Roch

64.1 Introduction

An elliptic operator on a compact manifold has a finite-dimensional kernel and cokernel (Theorem 42.5.2), and the difference of their dimensions, the *index*, is remarkably stable: it does not change under deformations, and the Atiyah–Singer index theorem computes it from characteristic classes. For the three classical elliptic complexes the index is a familiar invariant: the Euler characteristic, the signature, and the holomorphic Euler characteristic $\chi(X,E)=\sum_q(-1)^q\dim H^q(X,E)$. The corresponding cases of the index theorem are the Gauss–Bonnet–Chern theorem, Hirzebruch's signature theorem and the Hirzebruch–Riemann–Roch theorem. This chapter identifies the three indices with complete proofs, states the index theorem and its three cases, proves Hirzebruch–Riemann–Roch directly for line bundles on curves and for all line bundles on projective spaces, and draws the integrality consequences (Noether's formula, divisibility of Chern numbers) that make these theorems useful in algebraic geometry. References: [Hir66], [AS68a], [AS68b], [LM89], [GH78].

Prerequisites. Chapter 43 (Harmonic Forms and the Hodge Theorem), Chapter 46 (Dolbeault Cohomology), Chapter 39 (Characteristic Classes), Chapter 59 (Algebraic Curves), Chapter 56 (Cohomology of Coherent Sheaves) and Chapter 63 (The Cycle Class Map).

Conventions. Manifolds are compact, and characteristic classes are taken in de Rham cohomology (or rational cohomology), identified by the de Rham theorem. For a real vector bundle V we write $V_{\mathbb{C}}=V\otimes_{\mathbb{R}}\mathbb{C}$. For a class a in the cohomology of a compact oriented manifold, $\int_M a$ is the integral of its top-degree component.

64.2 The index of an elliptic operator

Definition 64.2.1. The *index* of an elliptic differential operator $P\colon\Gamma(M,E)\to\Gamma(M,F)$ on a compact manifold is $\operatorname{ind}P=\dim\ker P-\dim\ker P^{*}$, where P^{*} is the formal adjoint (Definition 42.5.1).

By Theorem 42.5.2, $\ker P^{*}$ is isomorphic to the cokernel of P on smooth sections, so $\operatorname{ind}P=\dim\ker P-\dim\operatorname{coker}P$, and it is the Fredholm index of $P\colon H^{s+m}(E)\to H^{s}(F)$ for every s. As the Fredholm index is locally constant on the space of Fredholm operators (Chapter 30), $\operatorname{ind}P$ is unchanged when P is deformed through a continuous family of elliptic operators of the same order; in particular it depends only on the principal symbol up to homotopy through invertible symbols.

Proposition 64.2.2 (Euler characteristic operator)**.** *For a compact oriented Riemannian manifold, $d+\delta\colon\Omega^{\mathrm{even}}(M)\to\Omega^{\mathrm{odd}}(M)$ is elliptic with index $\chi(M)$.*

Proof. This is Corollary 43.3.1; ellipticity holds because $(d+\delta)^2 = \Delta$ has principal symbol $|\xi|^2$, so the symbol of $d+\delta$ is injective for $\xi \neq 0$, and it maps between bundles of the same rank (2^{n-1}). □

Proposition 64.2.3 (Dolbeault operator)**.** *Let X be a compact complex manifold with a Hermitian metric and E a holomorphic vector bundle with a Hermitian metric. The operator $\bar\partial_E + \bar\partial_E^*\colon \mathcal{A}^{0,\mathrm{even}}(X,E) \to \mathcal{A}^{0,\mathrm{odd}}(X,E)$ is elliptic, and its index is $\chi(X,E) = \sum_q (-1)^q \dim H^q(X,E)$.*

Proof. Let $D = \bar\partial_E + \bar\partial_E^*$ on $\mathcal{A}^{0,\bullet}(X,E)$. As $\bar\partial_E^2 = 0$, $D^2 = \bar\partial_E\bar\partial_E^* + \bar\partial_E^*\bar\partial_E = \square_E$, which is elliptic with principal symbol $\frac12|\xi|^2$ (Theorem 46.3.3); so D is elliptic, and it is formally self-adjoint, so the formal adjoint of its even part is its odd part. If $D\alpha = 0$, then $\square_E\alpha = 0$; conversely $(\square_E\alpha,\alpha) = \|D\alpha\|^2$, so $\ker D = \ker\square_E = \bigoplus_q \mathcal{H}^{0,q}(X,E)$, split into even and odd q. By Theorem 46.3.3, $\mathcal{H}^{0,q}(X,E) \cong H^q(X,E)$, and the index is $\sum_q(-1)^q \dim H^q(X,E)$. □

Proposition 64.2.4 (Signature operator)**.** *Let M be a compact oriented Riemannian manifold of dimension $4m$, and τ the operator on $\Omega^\bullet(M)\otimes\mathbb{C}$ equal to $i^{k(k-1)+2m}*$ on k-forms. Then $\tau^2 = 1$, τ anticommutes with $d+\delta$, and $d+\delta\colon \Omega^+ \to \Omega^-$, between the ±1-eigenspaces of τ, is elliptic with index the signature $\sigma(M)$ (Definition 23.6.5).*

Sketch of proof. We omit the verification of $\tau^2 = 1$ and $\tau(d+\delta) = -(d+\delta)\tau$, which follow from $**= (-1)^{k(n-k)}$ and $\delta = -*d*$ in even dimension (Proposition 40.15.3) by tracking signs [LM89, Chapter 3]. Granting these, $d+\delta$ maps Ω^+ to Ω^-, the kernel of $d+\delta$ on $\Omega^\pm$ is the space $\mathcal{H}^\pm$ of harmonic forms in $\Omega^\pm$, and the index is $\dim\mathcal{H}^+ - \dim\mathcal{H}^-$, exactly as in Proposition 64.2.2. As τ commutes with Δ and maps harmonic k-forms to harmonic $(4m-k)$-forms, for $k < 2m$ the space $\mathcal{H}^k \oplus \mathcal{H}^{4m-k}$ is τ-stable, and $\alpha \mapsto \alpha \pm \tau\alpha$ identifies $\mathcal{H}^k$ with its ±1-eigenspaces, which therefore have equal dimensions and cancel in the index. On $\mathcal{H}^{2m}$, $\tau = i^{2m(2m-1)+2m}* = *$, and for a real harmonic $2m$-form α with $*\alpha = \pm\alpha$, $\int_M \alpha\wedge\alpha = \pm\int_M \alpha\wedge*\alpha = \pm\|\alpha\|^2$. So the intersection form is positive definite on the self-dual and negative definite on the anti-self-dual harmonic forms, which are orthogonal for it, and by the Hodge theorem the index $\dim\mathcal{H}^{2m}_+ - \dim\mathcal{H}^{2m}_-$ is the signature. □

64.3 The Atiyah–Singer index theorem

Definition 64.3.1. For a complex vector bundle with Chern roots $x_1,\dots,x_r$ and a real vector bundle V whose complexification has Chern roots $\pm y_1,\dots,\pm y_s$ (and 0 if the rank is odd), define

$$\mathrm{td}(E) = \prod_i \frac{x_i}{1-e^{-x_i}}, \qquad L(V) = \prod_j \frac{y_j}{\tanh y_j}, \qquad \hat A(V) = \prod_j \frac{y_j/2}{\sinh(y_j/2)}.$$

L and $\hat A$ are even in each y_j, hence polynomials in the Pontryagin classes: $L = 1 + \frac13 p_1 + \frac{1}{45}(7p_2 - p_1^2) + \cdots$ and $\hat A = 1 - \frac{1}{24}p_1 + \cdots$. For a compact oriented $4m$-manifold, the *L-genus* is $\int_M L(TM)$; for a compact complex manifold, the *Todd genus* is $\int_X \mathrm{td}(T_X)$.

Theorem 64.3.2 (Atiyah–Singer)**.** *The index of an elliptic operator on a compact manifold depends only on the class of its principal symbol in the compactly supported K-theory of the tangent bundle, and it is given by a universal cohomological formula in the Chern character of that class and the Todd class of $TM_{\mathbb{C}}$. For the three classical operators the formula gives:*

1. *(Gauss–Bonnet–Chern) $\chi(M) = \int_M e(TM)$ for a compact oriented manifold;*
2. *(Hirzebruch signature theorem) $\sigma(M) = \int_M L(TM)$ for a compact oriented $4m$-manifold;*

3. *(Hirzebruch–Riemann–Roch)* $\chi(X,E)=\int_X \operatorname{ch}(E)\operatorname{td}(T_X)$ *for a holomorphic vector bundle* E *on a compact complex manifold* X*, where* T_X *is the holomorphic tangent bundle.*

This is [AS68a] and [AS68b]; for the formulation and the heat equation proof see [LM89]. The signature theorem and Riemann–Roch for projective manifolds were proved earlier by Hirzebruch, using Thom's cobordism theory [Hir66]. Case (1) is also Theorem 39.5.1(3). For the Dirac operator of a spin manifold the index is the $\hat{A}$-genus, which is therefore an integer on spin manifolds.

Remark 64.3.3. For a projective manifold X and an algebraic vector bundle E, case (3) also follows from the Grothendieck–Riemann–Roch theorem (Theorem 62.7.2): it gives $\chi(X,E)=\deg(\operatorname{ch}(E)\operatorname{td}(T_X))$ in the Chow ring, and the cycle class map is a ring homomorphism compatible with Chern classes and degrees (Theorem 63.3.1), so the right-hand side is $\int_X \operatorname{ch}(E)\operatorname{td}(T_X)$; by GAGA the algebraic and analytic Euler characteristics agree (Chapter 58).

64.4 Riemann–Roch: direct verifications

Proposition 64.4.1. *Hirzebruch–Riemann–Roch holds for line bundles on a nonsingular projective curve* X *of genus* g*:* $\chi(X,\mathcal{O}(D))=\int_X \operatorname{ch}(\mathcal{O}(D))\operatorname{td}(T_X)=\deg D+1-g$.

Proof. The left side is $l(D)-h^1(\mathcal{O}(D))=l(D)-l(K-D)$ by Serre duality (Theorem 56.6.2), which is $\deg D+1-g$ by the Riemann–Roch theorem (Theorem 59.3.2). On the right, $\operatorname{ch}(\mathcal{O}(D))=1+c_1(\mathcal{O}(D))$ and $\operatorname{td}(T_X)=1+\frac{1}{2}c_1(T_X)$ in $H^\bullet(X)$, as $H^{>2}(X)=0$; so the integrand of degree 2 is $c_1(\mathcal{O}(D))+\frac{1}{2}c_1(T_X)$. Now $\int_X c_1(\mathcal{O}(D))=\deg D$, because $c_1(\mathcal{O}(D))$ is the cycle class of D and the cycle class map is compatible with degrees (Theorem 63.3.1); and $c_1(T_X)=-c_1(K_X)$ with $\deg K=2g-2$ (Corollary 59.3.3). So the right side is $\deg D-(g-1)$. □

Corollary 64.4.2. *Let* E *be a holomorphic vector bundle on a compact connected Riemann surface* C *of genus* g*, and* $\deg E=\int_C c_1(E)$*. Then*

$$\chi(C,E)=\deg E+\operatorname{rk}(E)\,(1-g).$$

For a line bundle this reads $\chi(C,L)=\int_C\big(c_1(L)+\frac{1}{2}c_1(C)\big)=\deg L+\frac{1}{2}\deg K_C^\vee$*, where* $c_1(C)=c_1(T_C)$ *and* $T_C=K_C^\vee$*, the dual of the canonical bundle.*

Proof. On a curve only classes of degree 0 and 2 survive, so $\operatorname{ch}(E)=\operatorname{rk}(E)+c_1(E)$ and $\operatorname{td}(T_C)=1+\frac{1}{2}c_1(T_C)$ (Definition 39.3.3), and the degree-2 part of their product is $c_1(E)+\frac{1}{2}\operatorname{rk}(E)c_1(T_C)$. Since $T_C=K_C^\vee$ (both have the local frame $\partial/\partial z$, dual to dz), $\int_C c_1(T_C)=-\deg K_C=2-2g$ (Proposition 39.2.7(4) and Proposition 50.2.5). Theorem 64.3.2(3) gives $\chi(C,E)=\deg E+\operatorname{rk}(E)(1-g)$. □

Remark 64.4.3. The same theorem appears in several guises, and it helps to translate between them. For a divisor D on a curve, $\chi(C,\mathcal{O}(D))=h^0-h^1$, and Serre duality $h^1(\mathcal{O}(D))=h^0(K_C-D)$ turns Corollary 64.4.2 into the classical form $l(D)-l(K-D)=\deg D+1-g$ of Theorem 59.3.2. The Hirzebruch form $\int\operatorname{ch}(E)\operatorname{td}(X)$ writes the correction term through the Todd class of the tangent bundle, $c_1(X):=c_1(T_X)=-c_1(K_X)$; this is why $\frac{1}{2}c_1(C)$, $\frac{1}{2}\deg K_C^\vee=1-g$ and $-\frac{1}{2}K$ appear, for instance in the surface formula $\chi(S,L)=\chi(S,\mathcal{O}_S)+\frac{1}{2}L\cdot(L-K_S)$ of Corollary 64.4.6. In every form, the Euler characteristic depends only on topological data (degrees and Chern classes), while h^0 alone does not; vanishing theorems (Chapter 50) are what turn the formula into a computation of h^0.

Lemma 64.4.4. *On* $\mathbb{P}^n$ *with hyperplane class* h*,* $\operatorname{td}(T_{\mathbb{P}^n})=\big(h/(1-e^{-h})\big)^{n+1}$ *and* $\operatorname{ch}(\mathcal{O}(k))=e^{kh}$.

Proof. The Euler sequence $0 \to \mathcal{O} \to \mathcal{O}(1)^{\oplus(n+1)} \to T_{\mathbb{P}^n} \to 0$ (Example 35.3.10) and the multiplicativity of the Todd class in exact sequences (Proposition 39.3.4) give $\operatorname{td}(T)\operatorname{td}(\mathcal{O}) = \operatorname{td}(\mathcal{O}(1))^{n+1}$, with $\operatorname{td}(\mathcal{O}) = 1$ and $\operatorname{td}(\mathcal{O}(1)) = h/(1-e^{-h})$, as $c_1(\mathcal{O}(1)) = h$ (Example 39.2.8). Also $\operatorname{ch}(\mathcal{O}(k)) = e^{c_1(\mathcal{O}(k))} = e^{kh}$. □

Proposition 64.4.5. *For all $n \geq 1$ and $k \in \mathbb{Z}$,*

$$\chi(\mathbb{P}^n, \mathcal{O}(k)) = \int_{\mathbb{P}^n} \operatorname{ch}(\mathcal{O}(k))\operatorname{td}(T_{\mathbb{P}^n}) = \binom{n+k}{n} = \frac{(k+1)(k+2)\cdots(k+n)}{n!}.$$

Proof. The left side. By Theorem 56.3.1, $H^q(\mathbb{P}^n, \mathcal{O}(k)) = 0$ for $0 < q < n$; $h^0(\mathcal{O}(k))$ is the number of monomials of degree k in $n+1$ variables, which is $\binom{n+k}{n}$ for $k \geq 0$ and 0 for $k < 0$; and $h^n(\mathcal{O}(k))$ is the number of Laurent monomials $x_0^{a_0}\cdots x_n^{a_n}$ with all $a_i \leq -1$ and $\sum a_i = k$, which is the number of monomials of degree $-k-n-1$, that is $\binom{-k-1}{n}$ for $k \leq -n-1$ and 0 otherwise. For $k \geq 0$ the polynomial $P(k) = (k+1)\cdots(k+n)/n!$ equals $\binom{n+k}{n}$; for $-n \leq k \leq -1$ it vanishes, as does χ; and for $k \leq -n-1$, $P(k) = (-1)^n(-k-1)(-k-2)\cdots(-k-n)/n! = (-1)^n\binom{-k-1}{n} = (-1)^n h^n(\mathcal{O}(k))$. So $\chi(\mathbb{P}^n, \mathcal{O}(k)) = P(k)$ in all cases.

The right side. By Lemma 64.4.4 and $\int_{\mathbb{P}^n} h^n = 1$, the integral is the coefficient of h^n in $e^{kh}(h/(1-e^{-h}))^{n+1}$, a power series convergent near 0, that is, the residue at 0 of the meromorphic function $e^{kh}(1-e^{-h})^{-n-1}$:

$$\frac{1}{2\pi i}\oint_{|h|=\varepsilon} \frac{e^{kh}\,dh}{(1-e^{-h})^{n+1}}.$$

The substitution $u = 1 - e^{-h}$ is biholomorphic near 0 with $u(0) = 0$, so it maps the small circle to a loop winding once around 0; since $e^{-h} = 1-u$, $e^{kh} = (1-u)^{-k}$ and $dh = du/(1-u)$, the integral becomes $\frac{1}{2\pi i}\oint (1-u)^{-k-1}u^{-n-1}\,du$, the coefficient of u^n in $(1-u)^{-k-1} = \sum_j \binom{k+j}{j}u^j$ (Chapter 29). This coefficient is $\binom{k+n}{n} = P(k)$, read as a polynomial in k. □

Corollary 64.4.6 (Noether's formula)**.** *For a compact complex surface S, $\chi(S, \mathcal{O}_S) = \frac{1}{12}\int_S (c_1^2 + c_2)$, where $c_i = c_i(T_S)$. Consequently $\int_S (c_1^2 + c_2)$ is divisible by 12, and for a line bundle L with $l = c_1(L)$,*

$$\chi(S, L) = \chi(S, \mathcal{O}_S) + \frac{1}{2}\int_S l(l + c_1) = \chi(S, \mathcal{O}_S) + \frac{1}{2}L \cdot (L - K_S),$$

where $L \cdot M = \int_S c_1(L)c_1(M)$ is the intersection number and K_S is the canonical bundle.

Proof. By Definition 64.3.1, the degree-4 part of the Todd class is $\frac{1}{12}(c_1^2 + c_2)$ (Definition 39.3.3), and Theorem 64.3.2(3) with $E = \mathcal{O}_S$ gives the formula; the left side is an integer. For L, the degree-4 part of $(1 + l + \frac{1}{2}l^2)(1 + \frac{1}{2}c_1 + \frac{1}{12}(c_1^2 + c_2))$ is $\frac{1}{2}l^2 + \frac{1}{2}lc_1 + \frac{1}{12}(c_1^2 + c_2)$. Finally $K_S = \Lambda^2 T_S^\vee$ and, with Chern roots x_1, x_2 of T_S (Theorem 39.3.2), $T_S^\vee$ has roots $-x_1, -x_2$ (Proposition 39.2.7(4)) and $\Lambda^2 T_S^\vee$ has the single root $-x_1 - x_2$. So $c_1(K_S) = -c_1$, and $\int_S l(l + c_1) = L \cdot L - L \cdot K_S$. □

Lemma 64.4.7. *For a complex vector bundle E, $(E_{\mathbb{R}})_{\mathbb{C}} \cong E \oplus \bar{E}$, and hence $p_1(E_{\mathbb{R}}) = c_1(E)^2 - 2c_2(E)$.*

Proof. The complex structure J of E extends $\mathbb{C}$-linearly to $(E_{\mathbb{R}})_{\mathbb{C}}$, where it has eigenvalues $\pm i$; the $+i$-eigenbundle $\{v - iJv\}$ is isomorphic to E by $v \mapsto \frac{1}{2}(v - iJv)$, and the $-i$-eigenbundle to $\bar{E}$. By Definition 39.5.2 and the Whitney formula, $p_1(E_{\mathbb{R}}) = -c_2(E \oplus \bar{E}) = -(c_2(E) + c_1(E)c_1(\bar{E}) + c_2(\bar{E})) = c_1^2 - 2c_2$, using $c_1(\bar{E}) = -c_1(E)$ and $c_2(\bar{E}) = c_2(E)$ (Proposition 39.2.7(4)). □

Corollary 64.4.8 (Gauss–Bonnet for complex manifolds)**.** *Let X be a compact complex manifold of dimension n and $c_i(X) = c_i(T_X)$ the Chern classes of its holomorphic tangent bundle. Then*

$$\chi(X) = \int_X c_n(X) = \sum_{p,q} (-1)^{p+q} h^{p,q}(X) = \sum_p (-1)^p \chi(X, \Omega_X^p).$$

For a curve of genus g this is $\int_C c_1(C) = 2 - 2g$, and for a surface $\chi(S) = \int_S c_2(S)$, so Noether's formula reads $\chi(S, \mathcal{O}_S) = \frac{1}{12}(K_S^2 + \chi(S))$.

Proof. The real tangent bundle TX with the complex structure J is isomorphic, as a complex vector bundle, to $T_X = T^{1,0}X$, by $v \mapsto \frac{1}{2}(v - iJv)$ (as in Lemma 64.4.7), and the orientation of X is the one induced by J. So $e(TX) = c_n(T_X)$ and $\int_X c_n(X) = \chi(X)$ by Theorem 39.5.1(2),(3). The Frölicher spectral sequence gives $\chi(X) = \sum_{p,q}(-1)^{p+q} h^{p,q}(X)$ (Proposition 46.4.1), and grouping the terms with the same p gives $\sum_p (-1)^p \chi(X, \Omega_X^p)$.

The equality $\sum_p (-1)^p \chi(X, \Omega_X^p) = \int_X c_n(X)$ also follows from Hirzebruch–Riemann–Roch (Theorem 64.3.2(3)), which is how it is often presented. If $x_1, \dots, x_n$ are the Chern roots of T_X, those of $\Omega_X^1 = T_X^\vee$ are $-x_1, \dots, -x_n$, so $\operatorname{ch}(\Omega_X^p) = \operatorname{ch}(\Lambda^p \Omega_X^1)$ is the p-th elementary symmetric function of the e^{-x_i}, and

$$\sum_p (-1)^p \operatorname{ch}(\Omega_X^p) \operatorname{td}(T_X) = \prod_{i=1}^n (1 - e^{-x_i}) \cdot \prod_{i=1}^n \frac{x_i}{1 - e^{-x_i}} = x_1 \cdots x_n = c_n(X).$$

Integrating and summing Theorem 64.3.2(3) over p gives the claim. For a curve, $c_1(C) = c_1(T_C)$ has degree $2 - 2g$ (Corollary 64.4.2), and for a surface $c_1^2 = K_S^2$ since $c_1(S) = -c_1(K_S)$ (Corollary 64.4.6). □

Example 64.4.9.

1. $\mathbb{P}^2$: $c(T) = (1 + h)^3$, so $c_1^2 = 9$, $c_2 = 3$ and Noether's formula gives $\chi(\mathcal{O}) = 12/12 = 1$, as it must. By Lemma 64.4.7, $p_1 = 9 - 6 = 3$, and the signature theorem $\sigma = \frac{1}{3}\int p_1 = 1$ agrees with the intersection form (1) on $H^2(\mathbb{P}^2) = \mathbb{Z}h$.
2. $\mathbb{P}^1 \times \mathbb{P}^1$: $c_1 = 2a + 2b$ and $c_2 = 4ab$, where a, b are the pullbacks of the point classes, with $a^2 = b^2 = 0$ and $\int ab = 1$. So $c_1^2 = 8$, $c_2 = 4$, $\chi(\mathcal{O}) = 1$, and $p_1 = 8 - 8 = 0$, in agreement with the hyperbolic intersection form on H^2, of signature 0.
3. A K3 surface has $c_1 = 0$ and $\chi(\mathcal{O}) = h^0 - h^1 + h^2 = 1 - 0 + 1 = 2$ (Chapter 52), so Noether's formula gives $\int c_2 = 24$, the topological Euler characteristic; then $p_1 = -48$ and $\sigma = -16$. As $b_2 = 22$, the intersection form has 3 positive and 19 negative eigenvalues.

Counterexample 64.4.10. The characteristic numbers appearing in the index theorems are a priori rational, and their integrality is a real constraint. There is no compact complex surface with $\int c_1^2 = \int c_2 = 1$, since $12 \nmid 2$ would contradict Noether's formula. By the signature theorem, $\int_M p_1(TM) = 3\sigma(M)$ is divisible by 3 for every compact oriented 4-manifold M; so no such manifold has first Pontryagin number 1, although nothing in the formal properties of characteristic classes excludes it.

Proposition 64.4.11 (Riemann–Roch on complex tori)**.** *Let $T = \mathbb{C}^g/\Lambda$ be a complex torus and L a holomorphic line bundle on T. Then*

$$\chi(T, L) = \frac{1}{g!} \int_T c_1(L)^g.$$

If L is positive, then $H^q(T, L) = 0$ for $q > 0$, and $h^0(T, L) = \frac{1}{g!}\int_T c_1(L)^g > 0$.

Proof. The vector fields $\partial/\partial z_1, \dots, \partial/\partial z_g$ descend to T and form a global holomorphic frame of T_T, which is therefore trivial, so all its Chern classes vanish and all its Chern roots may be taken to be 0; hence $\operatorname{td}(T_T) = 1$. With one Chern root, $\operatorname{ch}(L) = e^{c_1(L)}$ (Definition 39.3.3). By Theorem 64.3.2(3), $\chi(T, L) = \int_T e^{c_1(L)}$, and only the term of degree $2g$ contributes. If L is positive, then since $K_T = \Omega^g_T$ is trivial (it has the frame $dz_1 \wedge \cdots \wedge dz_g$), $H^q(T, L) = H^q(T, \Omega^g_T \otimes L) = 0$ for $q > 0$ by Kodaira–Nakano vanishing (Theorem 50.4.1), so $h^0 = \chi$. Finally $c_1(L)$ is represented by a positive $(1,1)$-form ω_L, the fundamental form of a Hermitian metric (Definition 45.4.2), and $\int_T \omega_L^g = g!\,\operatorname{vol}(T) > 0$ (Proposition 47.2.4). □

Example 64.4.12.

1. For $g = 1$ the proposition says $\chi(E, L) = \deg L$, in accordance with Proposition 64.4.1 for genus 1.
2. On $T = E \times E'$ with $L = \operatorname{pr}_1^* \mathcal{O}(p) \otimes \operatorname{pr}_2^* \mathcal{O}(q)$, $c_1(L) = a + b$ with a, b the classes of the fibres $\{p\} \times E'$ and $E \times \{q\}$, so $c_1(L)^2 = 2ab$. The bundles $\mathcal{O}(p)$ and $\mathcal{O}(q)$ have degree 1, hence are positive (Proposition 50.2.5), and the product of positive metrics has curvature form $\operatorname{pr}_1^* \omega_1 + \operatorname{pr}_2^* \omega_2 > 0$, so L is positive and $h^0(T, L) = \frac{1}{2} \cdot 2 = 1$: the only sections are the multiples of the product of the canonical sections. A line bundle with $h^0 = 1$ of this kind is a *principal polarisation*; its section is, in the analytic description of Chapter 61, a theta function.

64.5 Exercises

Exercise 64.5.1. Using Hirzebruch–Riemann–Roch on $\mathbb{P}^1 \times \mathbb{P}^1$, show that $\chi(\mathcal{O}(p, q)) = (p + 1)(q + 1)$ for the line bundle of bidegree (p, q), and check it for $p, q \geq 0$ by counting bihomogeneous polynomials.

Solution. With $l = pa + qb$ and $c_1 = 2a + 2b$, Corollary 64.4.6 gives $\chi = 1 + \frac{1}{2}\int (pa + qb)((p + 2)a + (q + 2)b) = 1 + \frac{1}{2}(p(q + 2) + q(p + 2)) = pq + p + q + 1$. For $p, q \geq 0$ the higher cohomology vanishes (by the Künneth formula and Theorem 56.3.1) and the sections are the polynomials of bidegree (p, q), of which there are $(p + 1)(q + 1)$. □

Exercise 64.5.2. Show that the Todd genus of $\mathbb{P}^n$ is 1, and that the Todd genus of a compact complex manifold equals $\chi(X, \mathcal{O}_X) = \sum_q (-1)^q h^{0,q}(X)$.

Solution. The second statement is Theorem 64.3.2(3) with $E = \mathcal{O}_X$, since $\operatorname{ch}(\mathcal{O}_X) = 1$. For $\mathbb{P}^n$, Proposition 64.4.5 with $k = 0$ gives $\binom{n}{n} = 1$. □

Exercise 64.5.3. Let $S \subseteq \mathbb{P}^3$ be a smooth surface of degree d. Using the adjunction formula $c_1(T_S) = (4 - d)h$ and Noether's formula, show that $\int_S c_2(T_S) = d^3 - 4d^2 + 6d$, and check that this gives 24 for a quartic.

Solution. From the exact sequence $0 \to \mathcal{O}_{\mathbb{P}^3}(-d) \to \mathcal{O}_{\mathbb{P}^3} \to \mathcal{O}_S \to 0$, additivity of χ and Proposition 64.4.5, $\chi(\mathcal{O}_S) = 1 - \frac{(1-d)(2-d)(3-d)}{6} = 1 + \frac{(d-1)(d-2)(d-3)}{6}$. As $\int_S h^2 = d$, $\int c_1^2 = (4-d)^2 d$. Noether's formula gives $\int c_2 = 12\chi(\mathcal{O}_S) - (4 - d)^2 d = 12 + 2(d - 1)(d - 2)(d - 3) - d(4 - d)^2 = d^3 - 4d^2 + 6d$. For $d = 4$: $64 - 64 + 24 = 24$. □

Chapter 65

Étale Cohomology

65.1 Introduction

Singular cohomology needs the complex topology. For varieties over other fields, Grothendieck replaced open subsets by étale maps and built étale cohomology. With coefficients $\mathbb{Z}/\ell^n\mathbb{Z}$, and in the limit $\mathbb{Z}_\ell$ and $\mathbb{Q}_\ell$, for a prime ℓ different from the characteristic, it has all the formal properties of singular cohomology. It agrees with singular cohomology over $\mathbb{C}$ (Artin's comparison theorem, Theorem 65.4.3) and carries an action of the Galois group of the field of definition. This Galois action is the arithmetic counterpart of the Hodge structure: the Tate conjecture (Chapter 76) is the analogue of the Hodge conjecture with Galois invariants in place of Hodge classes. Most of the foundational results are quoted from SGA 4 and Milne's book. References: [Mil80], [AGV72], [AGV73], [Del77].

Prerequisites. Chapter 54 (Schemes), Chapter 26 (Sheaf Cohomology) and Chapter 9 (Fields and Galois Theory).

Throughout, ℓ is a prime and schemes are Noetherian. For a field k, k^s is a separable closure and $G_k = \mathrm{Gal}(k^s/k)$ the absolute Galois group, a profinite group (Remark 9.10.7).

65.2 Étale morphisms and the étale site

Proposition 65.2.1. *Let $f : X \to Y$ be a morphism locally of finite type. The following are equivalent: (a) f is étale (Definition 54.8.4); (b) f is flat and* unramified*, meaning $\Omega_{X/Y} = 0$; (c) f is flat, and every fibre X_y is a disjoint union of spectra of finite separable field extensions of $k(y)$. Étale morphisms are stable under composition and base change, and a morphism between étale Y-schemes is étale. For varieties over $\mathbb{C}$, f is étale if and only if f^{an} is a local biholomorphism.*

Proof. For the equivalences see [Mil80, Chapter I]. If f is étale, locally $X = \operatorname{Spec} A[x_1,\dots,x_r]/(g_1,\dots,g_r)$ with invertible Jacobian determinant. Over $\mathbb{C}$ the holomorphic inverse function theorem then shows that f^{an} is a local biholomorphism, as in Proposition 58.3.2(1). Conversely, if f^{an} is a local biholomorphism, the completed local rings of X and Y at corresponding points agree. So f is flat and unramified at closed points, hence étale near them, and étaleness is an open condition. $\square$

Definition 65.2.2. The *small étale site* X_{et} has objects the étale morphisms $U \to X$, and a family $(U_i \to U)$ is a *covering* if the union of the images is U. An *étale sheaf* is a presheaf $\mathcal{F}$ of abelian groups on X_{et} such that for every covering the sequence $0 \to \mathcal{F}(U) \to \prod_i \mathcal{F}(U_i) \to \prod_{i,j} \mathcal{F}(U_i \times_U U_j)$ is exact. Étale sheaves form an abelian category with enough injectives, and *étale cohomology* $H^i(X_{\mathrm{et}}, \mathcal{F})$ is the right derived functor of $\mathcal{F} \mapsto \mathcal{F}(X)$.

Example 65.2.3. 1. Every quasi-coherent sheaf $\mathcal{F}$ defines an étale sheaf $U \mapsto \Gamma(U, f^*\mathcal{F})$ (faithfully flat descent), and $H^i(X_{\text{et}}, \mathcal{F}) = H^i(X, \mathcal{F})$ [Mil80, Chapter III].

2. $\mathbb{G}_m \colon U \mapsto \Gamma(U, \mathcal{O}_U)^*$ and $\mu_n \colon U \mapsto \{f \in \Gamma(U, \mathcal{O}_U) : f^n = 1\}$ are étale sheaves.

3. For an abelian group A, the constant sheaf A_X assigns to U the locally constant functions $U \to A$.

Proposition 65.2.4. *For a field k, the functor $\mathcal{F} \mapsto \varinjlim_L \mathcal{F}(\operatorname{Spec} L)$, the colimit over finite separable extensions $L \subseteq k^s$, is an equivalence between étale sheaves on* $\operatorname{Spec} k$ *and discrete G_k-modules (abelian groups with a continuous action). Under it, $H^i((\operatorname{Spec} k)_{\text{et}}, \mathcal{F}) \cong H^i(G_k, M)$, the continuous Galois cohomology.*

Sketch of proof. An étale k-scheme is a disjoint union of spectra of finite separable extensions (Proposition 65.2.1), so a sheaf is determined by its values on the $\operatorname{Spec} L$, and the sheaf condition for the Galois covering $\operatorname{Spec} L' \to \operatorname{Spec} L$ (L'/L Galois) says $\mathcal{F}(L) = \mathcal{F}(L')^{\operatorname{Gal}(L'/L)}$. This identifies sheaves with discrete G_k-modules M, with $\mathcal{F}(L) = M^{G_L}$. Global sections become M^{G_k}, and taking derived functors gives Galois cohomology [Mil80, Chapters II and III]. □

Counterexample 65.2.5. The Zariski topology is too coarse to see topology. Let X be an irreducible variety and A an abelian group. Every nonempty open subset of X is irreducible, hence connected, so the constant sheaf A_X has $A_X(U) = A$ for every nonempty open U, with the identity maps as restrictions. Thus A_X is flasque, and $H^i(X_{\text{Zar}}, A_X) = 0$ for $i > 0$ (Proposition 26.3.3). For an elliptic curve E over $\mathbb{C}$ this gives $H^1(E_{\text{Zar}}, \mathbb{Z}/n) = 0$, whereas $H^1(E^{\text{an}}; \mathbb{Z}/n) \cong (\mathbb{Z}/n)^2$. The étale topology repairs this for torsion coefficients (Example 65.3.2 and Theorem 65.4.3), but not for $\mathbb{Z}$: for a normal connected variety, $H^1(X_{\text{et}}, \mathbb{Z})$ is the group of continuous homomorphisms from the profinite étale fundamental group to $\mathbb{Z}$, which is 0 [Mil80, Chapter III]. This is why ℓ-adic cohomology is defined as an inverse limit of cohomology with finite coefficients (Definition 65.4.1).

65.3 The Kummer sequence

Proposition 65.3.1 (Kummer sequence)**.** *Let n be invertible on X. Then $0 \to \mu_n \to \mathbb{G}_m \xrightarrow{f \mapsto f^n} \mathbb{G}_m \to 0$ is an exact sequence of étale sheaves. If X is proper and connected over an algebraically closed field k with n invertible in k, there is an exact sequence*

$$0 \to \operatorname{Pic}(X)/n \to H^2(X_{\text{et}}, \mu_n) \to \operatorname{Br}(X)[n] \to 0, \qquad H^1(X_{\text{et}}, \mu_n) \cong \operatorname{Pic}(X)[n],$$

where $\operatorname{Br}(X) = H^2(X_{\text{et}}, \mathbb{G}_m)$ is the cohomological Brauer group and $[n]$ denotes n-torsion.

Proof. Exactness on the left is clear. For surjectivity, let $f \in \Gamma(U, \mathcal{O}_U)^*$. The morphism $U' = \operatorname{Spec}_U \mathcal{O}_U[t]/(t^n - f) \to U$ is finite, flat and surjective, and étale because $\partial(t^n - f)/\partial t = nt^{n-1}$ is invertible on U' (n and t are units). On U', $f = t^n$ is an n-th power. So f is locally an n-th power in the étale topology. The long exact cohomology sequence uses $H^0(X, \mathbb{G}_m) = \Gamma(X, \mathcal{O}_X)^* = k^*$, which is n-divisible as k is algebraically closed, and $H^1(X_{\text{et}}, \mathbb{G}_m) = \operatorname{Pic}(X)$ (Hilbert's theorem 90 in Grothendieck's form, [Mil80, Chapter III]). It reads $0 \to H^1(\mu_n) \to \operatorname{Pic}(X) \xrightarrow{n} \operatorname{Pic}(X) \to H^2(\mu_n) \to \operatorname{Br}(X) \xrightarrow{n} \operatorname{Br}(X)$, which gives both statements. □

Example 65.3.2. For an elliptic curve E over an algebraically closed field of characteristic prime to n, $H^1(E_{\text{et}}, \mu_n) \cong \operatorname{Pic}^0(E)[n] \cong E[n] \cong (\mathbb{Z}/n)^2$, using $\operatorname{Pic}^0(E) \cong E$ (Exercise 59.7.1) and the structure of n-torsion [Mum70]. So H^1 has rank $2 = b_1$, as over $\mathbb{C}$.

65.4 ℓ-adic cohomology

Definition 65.4.1. Let X be a variety over a field k of characteristic $\neq \ell$, and $\bar{X} = X \times_k \bar{k}$. The *$\ell$-adic cohomology* is

$$H^i(\bar{X}, \mathbb{Z}_\ell) = \varprojlim_m H^i(\bar{X}_{\text{et}}, \mathbb{Z}/\ell^m\mathbb{Z}), \qquad H^i(\bar{X}, \mathbb{Q}_\ell) = H^i(\bar{X}, \mathbb{Z}_\ell) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell.$$

The *Tate twist* is $\mathbb{Z}_\ell(1) = \varprojlim_m \mu_{\ell^m}(\bar{k})$, a free $\mathbb{Z}_\ell$-module of rank one on which G_k acts through the cyclotomic character, and $\mathbb{Z}_\ell(r) = \mathbb{Z}_\ell(1)^{\otimes r}$, $H^i(\bar{X}, \mathbb{Q}_\ell(r)) = H^i(\bar{X}, \mathbb{Q}_\ell) \otimes \mathbb{Q}_\ell(r)$. The group G_k acts on $H^i(\bar{X}, \mathbb{Q}_\ell(r))$ through its action on $\bar{X}$.

Theorem 65.4.2. *Let X be a nonsingular projective variety of dimension n over k, and $\ell \neq \operatorname{char} k$.*

1. *$H^i(\bar{X}, \mathbb{Q}_\ell)$ is finite-dimensional, and zero outside $0 \leq i \leq 2n$.*
2. *(Poincaré duality) $H^{2n}(\bar{X}, \mathbb{Q}_\ell(n)) \cong \mathbb{Q}_\ell$ via a trace map, and cup product $H^i(\bar{X}, \mathbb{Q}_\ell) \times H^{2n-i}(\bar{X}, \mathbb{Q}_\ell(n)) \to \mathbb{Q}_\ell$ is perfect and G_k-equivariant.*
3. *(Künneth) $H^\bullet(\bar{X} \times \bar{Y}, \mathbb{Q}_\ell) \cong H^\bullet(\bar{X}, \mathbb{Q}_\ell) \otimes H^\bullet(\bar{Y}, \mathbb{Q}_\ell)$.*
4. *(Cycle classes) There is a cycle class map $\mathrm{cl}_\ell \colon \mathrm{CH}^p(\bar{X}) \to H^{2p}(\bar{X}, \mathbb{Q}_\ell(p))$, a ring homomorphism compatible with pullback, pushforward and degrees, and G_k-equivariant; the class of a cycle defined over a finite extension k' is fixed by $G_{k'}$.*
5. *(Lefschetz trace formula) For an endomorphism f of $\bar{X}$, $\deg(\Gamma_f \cdot \Delta) = \sum_i (-1)^i \operatorname{tr}(f^* \mid H^i(\bar{X}, \mathbb{Q}_\ell))$.*
6. *(Hard Lefschetz) Cup product with the class of a hyperplane section induces isomorphisms $H^{n-i}(\bar{X}, \mathbb{Q}_\ell) \to H^{n+i}(\bar{X}, \mathbb{Q}_\ell(i))$.*

Statements (1)–(5) are [Mil80, Chapter VI] and [Del77]; (6) is due to Deligne [Del80].

Theorem 65.4.3 (Artin's comparison theorem)**.** *Let X be a variety over $\mathbb{C}$ and A a finite abelian group. Then $H^i(X_{\text{et}}, A) \cong H^i(X^{\text{an}}; A)$, compatibly with cup products. Hence $H^i(X, \mathbb{Z}_\ell) \cong H^i(X^{\text{an}}; \mathbb{Z}) \otimes \mathbb{Z}_\ell$ for X nonsingular and projective (or more generally when the singular cohomology groups are finitely generated), and cl_ℓ corresponds to $\mathrm{cl} \otimes \mathbb{Z}_\ell$ of Chapter 63.*

This is [AGV73, Exposé XI, Théorème 4.4] and [Mil80, Chapter III]. The proof compares both theories on a basis of neighbourhoods given by "elementary fibrations", which are $K(\pi, 1)$ spaces in both topologies. The isomorphism with $\mathbb{Z}_\ell$-coefficients follows by the universal coefficient theorem and passing to the limit.

Definition 65.4.4. A *Weil cohomology theory* for nonsingular projective varieties over k, with coefficients in a field K of characteristic 0, is a contravariant functor $X \mapsto H^\bullet(X)$ to graded-commutative finite-dimensional K-algebras, satisfying Poincaré duality, the Künneth formula, and having a cycle class map compatible with pullback, pushforward and products, with the class of a point generating H^{2n} and satisfying the weak and hard Lefschetz theorems. Examples: singular cohomology with $\mathbb{Q}$-coefficients for $k \subseteq \mathbb{C}$; ℓ-adic cohomology for $\ell \neq \operatorname{char} k$ (Theorem 65.4.2); algebraic de Rham cohomology in characteristic 0 (Chapter 58); crystalline cohomology in characteristic p.

Remark 65.4.5. For X over a finite field $\mathbb{F}_q$, the Frobenius F acts on $\bar{X}$, its fixed points are $X(\mathbb{F}_{q^m})$ for F^m, and the trace formula (Theorem 65.4.2(5)) gives $\#X(\mathbb{F}_{q^m}) = \sum_i (-1)^i \operatorname{tr}(F^{m*} \mid H^i)$, so the zeta function is rational. Deligne proved that the eigenvalues of Frobenius on H^i are algebraic integers of absolute value $q^{i/2}$ [Del74a], the analogue of the Hodge decomposition's weight: this is the Riemann hypothesis part of the Weil conjectures.

65.5 Galois actions and the Tate conjecture

Example 65.5.1. Let E be an elliptic curve over k and $T_\ell(E) = \varprojlim E[\ell^m](\bar{k})$ its *Tate module*, free of rank 2 over $\mathbb{Z}_\ell$ with an action of G_k. Then $H^1(\bar{E}, \mathbb{Z}_\ell) \cong \mathrm{Hom}(T_\ell(E), \mathbb{Z}_\ell)$ equivariantly, by Example 65.3.2 and duality. For k a number field and E without complex multiplication, the image of G_k in $\mathrm{GL}_2(\mathbb{Z}_\ell)$ is open (Serre's open image theorem), the ℓ-adic analogue of the fact that the Mumford–Tate group of E is GL_2 (Chapter 67).

Remark 65.5.2. By Theorem 65.4.2(4), the classes of cycles defined over k lie in the invariants $H^{2p}(\bar{X}, \mathbb{Q}_\ell(p))^{G_k}$, and those of cycles defined over finite extensions lie in the classes fixed by an open subgroup. The *Tate conjecture* asserts that for k finitely generated over its prime field the $\mathbb{Q}_\ell$-span of the cycle classes is exactly $H^{2p}(\bar{X}, \mathbb{Q}_\ell(p))^{G_k}$ (Chapter 76).

65.6 Exercises

Exercise 65.6.1. Let k be a field. Show that $H^1((\operatorname{Spec} k)_{\mathrm{et}}, \mathbb{Z}/n) \cong \mathrm{Hom}_{\mathrm{cont}}(G_k, \mathbb{Z}/n)$, and describe it for $k = \mathbb{F}_q$.

Solution. By Proposition 65.2.4, this is $H^1(G_k, \mathbb{Z}/n)$ with trivial action, which is the group of continuous homomorphisms. For $\mathbb{F}_q$, $G_k = \hat{\mathbb{Z}}$ generated by Frobenius, so the group is $\mathbb{Z}/n$, corresponding to the unique cyclic extension of degree d for each $d \mid n$. □

Exercise 65.6.2. Let k be a field and n invertible in k. Show that $H^1(G_k, \mu_n) \cong k^*/k^{*n}$ (Kummer theory).

Solution. Apply the Kummer sequence on $\operatorname{Spec} k$ and Proposition 65.2.4: $k^* \xrightarrow{n} k^* \to H^1(G_k, \mu_n) \to H^1(G_k, (k^s)^*)$, and the last group is 0 by Hilbert's theorem 90. □

Exercise 65.6.3. Show that $H^2(\mathbb{P}^1_{\bar{k}}, \mu_n) \cong \mathbb{Z}/n$ for n prime to the characteristic, and that $H^2(\mathbb{P}^1_{\bar{k}}, \mathbb{Z}_\ell(1)) \cong \mathbb{Z}_\ell$.

Solution. In Proposition 65.3.1, $\mathrm{Pic}(\mathbb{P}^1) = \mathbb{Z}$ and the Brauer group of a curve over an algebraically closed field vanishes (Tsen's theorem) [Mil80, Chapter III], so $H^2(\mu_n) = \mathbb{Z}/n$. Passing to the limit over $n = \ell^m$ gives $\mathbb{Z}_\ell$. □

Part IX

Hodge Theory

Chapter 66

Hodge Structures

66.1 Introduction

The cohomology $H^k(X;\mathbb{Z})$ of a compact Kähler manifold comes with the Hodge decomposition of $H^k(X;\mathbb{C})$. The abstract structure that remembers exactly this, a lattice together with a decomposition of its complexification compatible with complex conjugation, is a *Hodge structure*. Hodge structures form an abelian category with tensor products (Theorem 66.3.3). Polarisable ones form a semisimple category (Theorem 66.5.3), and the cohomology of projective manifolds is polarisable. In this language the Hodge conjecture is a statement about the Hodge classes of the Hodge structures $H^{2p}(X;\mathbb{Q})$ (Chapter 68). References: [Voi02b, Chapter 7], [Del71], [PS08, Chapter 2].

Prerequisites. Chapter 49 (Hard Lefschetz and the Hodge–Riemann Relations) and Chapter 60 (Linear Algebraic Groups).

Throughout, A denotes one of the rings $\mathbb{Z}$, $\mathbb{Q}$ or $\mathbb{R}$, and for an A-module H_A we write $H_{\mathbb{C}} = H_A \otimes_A \mathbb{C}$, with complex conjugation acting on the second factor.

66.2 Pure Hodge structures

Definition 66.2.1. An *A-Hodge structure of weight $k \in \mathbb{Z}$* is a finitely generated A-module H_A with a decomposition

$$H_{\mathbb{C}} = \bigoplus_{p+q=k} H^{p,q}, \qquad \overline{H^{p,q}} = H^{q,p}.$$

The *Hodge numbers* are $h^{p,q} = \dim_{\mathbb{C}} H^{p,q}$, and the *Hodge filtration* is the decreasing filtration $F^p H_{\mathbb{C}} = \bigoplus_{r \geq p} H^{r,k-r}$. For $A = \mathbb{Z}$ only the torsion-free quotient matters for the decomposition, since $H_{\mathbb{C}} = H_{\mathbb{Z}} \otimes \mathbb{C}$ kills torsion.

Proposition 66.2.2. *Decompositions as in Definition 66.2.1 correspond bijectively to finite decreasing filtrations $F^\bullet$ of $H_{\mathbb{C}}$ such that $H_{\mathbb{C}} = F^p \oplus \overline{F^{k-p+1}}$ for all p. The inverse bijection is $H^{p,q} = F^p \cap \overline{F^q}$.*

Proof. Given the decomposition, $\overline{F^{k-p+1}} = \bigoplus_{r \geq k-p+1} H^{k-r,r} = \bigoplus_{s \leq p-1} H^{s,k-s}$, which is a complement of F^p; and $F^p \cap \overline{F^q}$ with $p+q=k$ is $\bigoplus_{r \geq p} H^{r,k-r} \cap \bigoplus_{s \leq p} H^{s,k-s} = H^{p,q}$. Conversely, given the filtration, put $H^{p,q} = F^p \cap \overline{F^q}$. Then $\overline{H^{p,q}} = H^{q,p}$. The sum of the $H^{p,q}$ is direct: $H^{p,q} \subseteq F^p$ and $H^{p',q'} \subseteq \overline{F^{q'}} \subseteq \overline{F^{k-p+1}}$ for $p' < p$. To see that it is all of $H_{\mathbb{C}}$, show $F^p = H^{p,k-p} \oplus F^{p+1}$ by descending induction. Intersecting $H_{\mathbb{C}} = F^{p+1} \oplus \overline{F^{k-p}}$ with $F^p \supseteq F^{p+1}$ gives $F^p = F^{p+1} \oplus (F^p \cap \overline{F^{k-p}}) = F^{p+1} \oplus H^{p,k-p}$. □

Example 66.2.3. 1. The *Tate structure* $\mathbb{Z}(n)$ is the rank-one lattice $(2\pi i)^n\mathbb{Z} \subseteq \mathbb{C}$, of weight $-2n$ and type $(-n,-n)$. We write $H(n) = H \otimes \mathbb{Z}(n)$ for the *Tate twist*, of weight $k-2n$.

2. For a compact Kähler manifold X, $H^k(X;\mathbb{Z})$ is a $\mathbb{Z}$-Hodge structure of weight k by Theorem 48.5.2.

3. For a compact Riemann surface of genus g, H^1 has $h^{1,0} = h^{0,1} = g$. For a K3 surface, H^2 has Hodge numbers $(1,20,1)$.

4. A Hodge structure of weight k with $H^{p,q} = 0$ unless $p = q$ is of *Tate type*; for instance $H^{2p}(\mathbb{CP}^n;\mathbb{Z}) \cong \mathbb{Z}(-p)$.

Remark 66.2.4. The factor $(2\pi i)^n$ in $\mathbb{Z}(n)$ is not decoration. Integration of a closed 1-form over the circle $S^1 \subseteq \mathbb{C}^*$, or the connecting map of the exponential sequence $0 \to \mathbb{Z} \to \mathcal{O} \xrightarrow{\exp(2\pi i\cdot)} \mathcal{O}^* \to 0$, produce periods in $2\pi i\mathbb{Z}$, and the cycle class of a codimension-p cycle naturally lies in $H^{2p}(X;\mathbb{Z}(p))$. Over $\mathbb{Q}$ the twist only changes the weight, but it matters for the comparison with de Rham and ℓ-adic cohomology (Chapter 75).

66.3 Morphisms and the category of Hodge structures

Definition 66.3.1. A *morphism* of A-Hodge structures $H \to H'$ of weights k and $k+2r$ (a morphism *of type* (r,r)) is an A-linear map φ with $\varphi_{\mathbb{C}}(H^{p,q}) \subseteq H'^{p+r,q+r}$. Morphisms of type $(0,0)$ between structures of the same weight are simply called morphisms.

Proposition 66.3.2. *A morphism $\varphi\colon H \to H'$ of Hodge structures of the same weight is* strict *for the Hodge filtrations: $\varphi(F^pH_{\mathbb{C}}) = F^pH'_{\mathbb{C}} \cap \varphi(H_{\mathbb{C}})$.*

Proof. φ preserves the decompositions. So for $x \in H_{\mathbb{C}}$ with $\varphi(x) \in F^p$, decompose $x = \sum_r x^{r,k-r}$. Then $\varphi(x) = \sum_r \varphi(x^{r,k-r})$ with $\varphi(x^{r,k-r}) \in H'^{r,k-r}$, and $\varphi(x) \in F^p$ forces $\varphi(x^{r,k-r}) = 0$ for $r < p$. So $\varphi(x) = \varphi(\sum_{r\geq p} x^{r,k-r}) \in \varphi(F^p)$. □

Theorem 66.3.3. *For $A = \mathbb{Q}$ or $\mathbb{R}$, the Hodge structures of weight k form an abelian category: kernels, cokernels and images of morphisms carry induced Hodge structures of weight k. The category of all A-Hodge structures (direct sums of pure ones of various weights) has tensor products, duals and internal* Hom*:*

$$(H\otimes H')^{p,q} = \bigoplus_{a+a'=p,\,b+b'=q} H^{a,b}\otimes H'^{a',b'}, \qquad (H^\vee)^{p,q} = (H^{-p,-q})^\vee, \qquad \mathrm{Hom}(H,H') = H^\vee\otimes H',$$

of weights $k+k'$, $-k$ and $k'-k$. Morphisms of type $(0,0)$ between H and H' of the same weight are exactly the elements of $\mathrm{Hom}(H,H')_A \cap \mathrm{Hom}(H,H')^{0,0}$.

Proof. Let $\varphi\colon H \to H'$ be a morphism. Then $\ker\varphi_{\mathbb{C}} = \bigoplus_{p,q}(\ker\varphi_{\mathbb{C}} \cap H^{p,q})$, because φ preserves the decomposition, and this decomposition is stable under conjugation since φ is defined over A. So $\ker\varphi_A$ is a Hodge structure. Similarly $\varphi_{\mathbb{C}}(H_{\mathbb{C}}) = \bigoplus \varphi(H^{p,q})$ and $H'_{\mathbb{C}}/\varphi(H_{\mathbb{C}}) = \bigoplus H'^{p,q}/\varphi(H^{p,q})$. The coimage and image coincide because the underlying category of vector spaces is abelian and the Hodge structures on both are induced from $H^{p,q}$. The formulas for tensor products and duals define Hodge structures because conjugation acts factorwise, and a linear map is of type $(0,0)$ exactly when it maps each $H^{p,q}$ into $H'^{p,q}$, that is, when it lies in $\mathrm{Hom}(H,H')^{0,0}$. □

Counterexample 66.3.4. Strictness is special. The category of filtered vector spaces is not abelian: the identity of a vector space V, from the trivial filtration $F^0 = V$, $F^1 = 0$ to the filtration $F^0 = F^1 = V$, $F^2 = 0$, is a bijective morphism of filtered spaces that is not an isomorphism, since its kernel and cokernel vanish but it is not strict. Proposition 66.3.2 excludes this for Hodge structures, because the complex conjugate filtration is also respected.

66.4 Hodge structures as representations

Proposition 66.4.1. *A real Hodge structure on $H_{\mathbb{R}}$, that is, a direct sum of real Hodge structures of various weights, is the same as a representation $h\colon \mathbb{S} \to \mathrm{GL}(H_{\mathbb{R}})$ of the Deligne torus, via $h(z)v = z^{-p}\bar{z}^{-q}v$ for $v \in H^{p,q}$. It is pure of weight k if and only if $h \circ w\colon \mathbb{G}_m \to \mathrm{GL}(H_{\mathbb{R}})$ is $r \mapsto r^{-k}$. A $\mathbb{Q}$-structure $H_{\mathbb{Q}}$ gives a $\mathbb{Q}$-Hodge structure if and only if the weight cocharacter $h \circ w$ is defined over $\mathbb{Q}$. The* Weil operator *$C = h(i)^{-1}$ acts on $H^{p,q}$ as i^{p-q}; it is a real operator with $C^2 = (-1)^k$ on a structure of weight k.*

Proof. The first statement is Proposition 60.4.5, which also gives $h(w(r)) = r^{-(p+q)}$. The weight decomposition $H_{\mathbb{R}} = \bigoplus_k H_k$ is the eigenspace decomposition of $h \circ w$, and it is defined over $\mathbb{Q}$ exactly when $h \circ w$ is. Finally $h(i)$ acts on $H^{p,q}$ by $i^{-p}\bar{i}^{-q} = i^{-p}(-i)^{-q} = i^{q-p}$, so $C = h(i)^{-1}$ acts as i^{p-q}, and $C^2 = (-1)^{p-q} = (-1)^{p+q}$. □

66.5 Polarisations

Definition 66.5.1. A *polarisation* of a $\mathbb{Q}$-Hodge structure H of weight k is a bilinear form $Q\colon H_{\mathbb{Q}} \times H_{\mathbb{Q}} \to \mathbb{Q}$ such that

1. $Q(y,x) = (-1)^k Q(x,y)$;
2. $Q(H^{p,q}, H^{p',q'}) = 0$ unless $p' = q$ and $q' = p$ (for the $\mathbb{C}$-bilinear extension);
3. $i^{p-q}Q(x,\bar{x}) > 0$ for every nonzero $x \in H^{p,q}$; equivalently $Q(Cx,\bar{x}) > 0$ for all nonzero $x \in H_{\mathbb{C}}$.

Equivalently, $Q\colon H \otimes H \to \mathbb{Q}(-k)$ is a morphism of Hodge structures satisfying (3). H is *polarisable* if it admits a polarisation.

Theorem 66.5.2. *Let X be a compact Kähler manifold of dimension n with a Kähler class $[\omega]$ in the image of $H^2(X;\mathbb{Q})$ (for instance X projective). Then for $k \le n$ the primitive cohomology $P^k(X;\mathbb{Q})$ is a $\mathbb{Q}$-Hodge structure polarised by $Q_k(a,b) = (-1)^{k(k-1)/2}\int_X a \cup b \cup [\omega]^{n-k}$. The whole $H^k(X;\mathbb{Q})$ is polarisable: in terms of the Lefschetz decomposition, a polarisation is*

$$\tilde{Q}\Big(\sum_j L^j a_j, \sum_j L^j b_j\Big) = \sum_j Q_{k-2j}(a_j, b_j), \qquad a_j, b_j \in P^{k-2j}(X;\mathbb{Q}).$$

Proof. P^k is a sub-Hodge structure, as the kernel of the morphism L^{n-k+1} of type $(n-k+1, n-k+1)$ (Theorem 66.3.3), and it is defined over $\mathbb{Q}$ because $[\omega]$ is rational. Conditions (1)–(3) for Q_k on P^k are Theorem 49.4.2. For H^k, the summands $L^j P^{k-2j}$ of the Lefschetz decomposition (Theorem 49.2.2) are sub-Hodge structures defined over $\mathbb{Q}$, since L^j is a morphism of type (j,j), and $L^j\colon P^{k-2j} \to H^k$ is injective. So $\tilde{Q}$ is well defined, $(-1)^k$-symmetric, and pairs $H^{p,q}$ only with $H^{q,p}$. For $a \in P^{p',q'}$ with $p'+q' = k-2j$, the element $x = L^j a$ lies in $H^{p'+j,q'+j}$, and $i^{(p'+j)-(q'+j)}\tilde{Q}(x,\bar{x}) = i^{p'-q'}Q_{k-2j}(a,\bar{a}) > 0$ by the primitive case. As the summands are $\tilde{Q}$-orthogonal and the $H^{p,q}$-components of different summands pair to zero, (3) holds on all of H^k. For comparison, $Q_k(L^j a, L^j b) = (-1)^j Q_{k-2j}(a,b)$, because both are $\pm\int a \cup b \cup [\omega]^{n-k+2j}$ and $\frac{k(k-1)}{2} - \frac{(k-2j)(k-2j-1)}{2} = 2jk - 2j^2 - j \equiv j \pmod 2$. So $\tilde{Q}$ is Q_k with the sign changed on the summands with j odd; Q_k itself is not a polarisation of H^k in general (Counterexample 49.6.2). □

Theorem 66.5.3. *The category of polarisable $\mathbb{Q}$-Hodge structures of weight k is semisimple: every sub-Hodge structure of a polarisable Hodge structure has a complement, and every polarisable Hodge structure is a finite direct sum of irreducible ones. Sub-Hodge structures and quotients of polarisable Hodge structures are polarisable.*

Proof. Let Q polarise H and let $W \subseteq H$ be a sub-Hodge structure. Then $W_{\mathbb{C}} = \bigoplus (W_{\mathbb{C}} \cap H^{p,q})$, and the restriction of Q to W satisfies (1)–(3), since (3) is a positivity condition that restricts to subspaces. So $Q|_W$ is a polarisation, in particular nondegenerate: if $Q(x, W) = 0$ for $x \in W^{p,q}$, then $Q(Cx, \bar{x}) = 0$ forces $x = 0$. The orthogonal $W^{\perp} = \{y \in H_{\mathbb{Q}} : Q(y, W) = 0\}$ is a sub-Hodge structure: if $y \in W^{\perp}_{\mathbb{C}}$ has components $y^{p,q}$, then for $w \in W^{p',q'}$, $Q(y^{p,q}, w)$ is the only possibly nonzero term of $Q(y, w) = 0$ when $(p, q) = (q', p')$, so each $y^{p,q} \in W^{\perp}_{\mathbb{C}}$. As $W \cap W^{\perp} = 0$ by nondegeneracy and dimensions add up, $H = W \oplus W^{\perp}$. The quotient $H/W \cong W^{\perp}$ is polarised by $Q|_{W^{\perp}}$. Semisimplicity follows by induction on the dimension. □

Counterexample 66.5.4. Not every Hodge structure is polarisable. In dimension 2 every weight-one $\mathbb{Q}$-Hodge structure is polarisable: $V_{\mathbb{C}} = V^{1,0} \oplus V^{0,1}$ with $V^{1,0}$ spanned by $(1, \tau)$, $\tau \notin \mathbb{R}$, is polarised by $\pm \det$, and it is the Hodge structure of an elliptic curve. But a weight-one Hodge structure of dimension 4 whose $H^{1,0}$ is a general 2-dimensional subspace of $\mathbb{C}^4$ with $H^{1,0} \cap \overline{H^{1,0}} = 0$ admits no polarisation: it is the Hodge structure of a non-algebraic complex torus (Counterexample 61.3.6 and Theorem 66.7.1).

Proposition 66.5.5. *Let H be a torsion-free $\mathbb{Z}$-Hodge structure of weight k and Q a polarisation of $H_{\mathbb{Q}}$. The group of automorphisms of H (automorphisms of $H_{\mathbb{Z}}$ whose complexification preserves every $H^{p,q}$) that preserve Q is finite.*

Proof. Put $h(x, y) = Q(Cx, \bar{y})$ on $H_{\mathbb{C}}$, with C the Weil operator (Proposition 66.4.1). By conditions (1)–(3) of Definition 66.5.1, h is a positive definite Hermitian form: condition (3) is $h(x, x) > 0$ for $x \neq 0$, and Hermitian symmetry follows from (1) and $Q(Cx, Cy) = Q(x, y)$, which holds by (2) because C acts on $H^{p,q} \times H^{q,p}$ by $i^{p-q} i^{q-p} = 1$. An automorphism g preserving the decomposition commutes with C, which is a scalar on each $H^{p,q}$; if it also preserves Q it preserves h. So the matrices of such g in a $\mathbb{Z}$-basis of $H_{\mathbb{Z}}$ lie in the unitary group of h, which is a bounded subset of the real matrices (all g in it have operator norm 1 for the norm of h), and they have integer entries. A bounded set of integer matrices is finite. □

66.6 Hodge classes

Definition 66.6.1. A *Hodge class* in a $\mathbb{Q}$-Hodge structure H of even weight $2p$ is an element of $\mathrm{Hdg}(H) = H_{\mathbb{Q}} \cap H^{p,p}$. Equivalently, $\mathrm{Hdg}(H) = \mathrm{Hom}_{\mathrm{HS}}(\mathbb{Q}(-p), H)$.

Proposition 66.6.2. *Morphisms of Hodge structures $H \to H'$ of the same weight are exactly the Hodge classes in $\mathrm{Hom}(H, H')$, and $\mathrm{End}_{\mathrm{HS}}(H) = \mathrm{Hdg}(\mathrm{End}\, H)$. Morphisms of Hodge structures map Hodge classes to Hodge classes. For a compact Kähler manifold X, $\mathrm{Hdg}^p(X) = \mathrm{Hdg}(H^{2p}(X; \mathbb{Q}))$ (Definition 63.4.1) is a $\mathbb{Q}$-subalgebra of $\bigoplus_p H^{2p}(X; \mathbb{Q})$ under cup product.*

Proof. The first statement is the last part of Theorem 66.3.3. A morphism of type (r, r) maps $H_{\mathbb{Q}} \cap H^{p,p}$ into $H'_{\mathbb{Q}} \cap H'^{p+r,p+r}$. Cup product $H^{2p} \otimes H^{2q} \to H^{2p+2q}$ is a morphism of Hodge structures, as the wedge product of forms of types (p, p) and (q, q) has type $(p + q, p + q)$, and it preserves rational classes. □

Example 66.6.3.

1. In $H^2(X; \mathbb{Q})$ of a compact Kähler manifold, the Hodge classes are the rational classes of type $(1, 1)$; for a complex torus they form $\mathrm{NS}(T) \otimes \mathbb{Q}$ (Exercise 61.7.2).
2. Hodge classes can be scarce even when $H^{p,p}$ is large: for a very general projective K3 surface of given degree, the rational classes of type $(1, 1)$ in H^2 are only the multiples of the polarisation, although $h^{1,1} = 20$.
3. The polarisation Q of a Hodge structure H of weight k is a Hodge class in $(H^{\vee})^{\otimes 2}(-k)$, a Hodge structure of weight 0.

66.7 Weight one and complex tori

Theorem 66.7.1. *The functor $T = V/\Lambda \mapsto H_1(T;\mathbb{Z}) = \Lambda$, with the decomposition $\Lambda_{\mathbb{C}} = V \oplus \bar{V}$ into the $(-1,0)$- and $(0,-1)$-parts, is an equivalence between the category of complex tori and the category of torsion-free $\mathbb{Z}$-Hodge structures of type $\{(-1,0),(0,-1)\}$. Its inverse is $H \mapsto H_{\mathbb{C}}/(F^0 H_{\mathbb{C}} + H_{\mathbb{Z}})$. Under it, abelian varieties correspond exactly to the polarisable Hodge structures, and a polarisation of the Hodge structure is (up to sign) a Riemann form.*

Proof. $\Lambda \otimes_{\mathbb{Z}} \mathbb{R} = V$ as real vector spaces, so $\Lambda_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C} = V^{1,0} \oplus V^{0,1}$, the $\pm i$-eigenspaces of the complex structure, and the projection to $V^{1,0} \cong V$ restricts to the identity of V on $\Lambda_{\mathbb{R}} = V$. So $V \cong \Lambda_{\mathbb{C}}/V^{0,1} = \Lambda_{\mathbb{C}}/F^0$, and $T = \Lambda_{\mathbb{C}}/(F^0 + \Lambda)$. Conversely, for a Hodge structure H of this type, $H_{\mathbb{R}} \to H_{\mathbb{C}}/F^0 = H^{-1,0}$ is an isomorphism of real vector spaces, since $H_{\mathbb{R}} \cap F^0 = H_{\mathbb{R}} \cap H^{0,-1} = 0$ (a real vector in $H^{0,-1}$ lies in $H^{-1,0} \cap H^{0,-1} = 0$). So $H_{\mathbb{Z}}$ is a lattice in the complex vector space $H^{-1,0}$. Homomorphisms of tori are $\mathbb{C}$-linear maps preserving lattices (Lemma 61.2.3), which are exactly the morphisms of Hodge structures. A polarisation Q of weight -1 is an alternating integral (after scaling) form on Λ with $Q(V,V) = 0$ and $iQ(x,\bar{x}) > 0$ on $V^{1,0}$, up to the sign conventions; this is Riemann's condition in the form of Theorem 61.3.2. □

Remark 66.7.2. Dually, $H^1(T;\mathbb{Z}) = \mathrm{Hom}(\Lambda,\mathbb{Z})$ is of weight one with $H^{1,0} = V^\vee$. For a curve C, $H^1(C;\mathbb{Z})$ is polarised by the cup product, and the corresponding torus is the Jacobian (Theorem 61.5.1). In higher weight there is no such dictionary: a Hodge structure of weight 3 does not in general come from a complex torus in a way compatible with polarisations, which is the origin of the difference between Griffiths' intermediate Jacobians and Weil's (Chapter 71).

Corollary 66.7.3. *A complex torus with a polarisation has only finitely many group automorphisms preserving the polarisation.*

Proof. By Theorem 66.7.1, group automorphisms of the torus are the automorphisms of the Hodge structure H_1, and a polarisation of the torus is a polarisation of this Hodge structure. Apply Proposition 66.5.5. □

Counterexample 66.7.4. The polarisation is needed in Proposition 66.5.5. Let $E = \mathbb{C}/\Lambda$ be an elliptic curve and $T = E \times E = \mathbb{C}^2/\Lambda^2$. Every matrix in $\mathrm{GL}_2(\mathbb{Z})$ acts $\mathbb{C}$-linearly on $\mathbb{C}^2$ and preserves Λ^2, so it is an automorphism of the Hodge structure $H_1(T;\mathbb{Z})$ (Theorem 66.7.1), and $\mathrm{GL}_2(\mathbb{Z})$ is infinite. Consequently no single polarisation of $H_1(T)$ is preserved by all of these automorphisms; each polarisation is preserved only by a finite subgroup.

Example 66.7.5. Let $E_\tau = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$ with $\mathrm{Im}\,\tau > 0$, and $H = H_1(E_\tau;\mathbb{Q}) = \mathbb{Q} + \mathbb{Q}\tau \subseteq \mathbb{C}$, of type $\{(-1,0),(0,-1)\}$. An $\mathbb{R}$-linear endomorphism of $H_{\mathbb{R}} = \mathbb{C}$ preserves the decomposition $H_{\mathbb{C}} = H^{-1,0} \oplus H^{0,-1}$ exactly when it commutes with the complex structure, that is, when it is multiplication by some $\alpha \in \mathbb{C}$ (as in the proof of Theorem 66.7.1). So $\mathrm{End}_{\mathrm{HS}}(H) = \{\alpha \in \mathbb{C} : \alpha(\mathbb{Q} + \mathbb{Q}\tau) \subseteq \mathbb{Q} + \mathbb{Q}\tau\}$, which is $\mathrm{End}(E_\tau) \otimes \mathbb{Q}$: clearing denominators, α preserves $\mathbb{Q} + \mathbb{Q}\tau$ if and only if $n\alpha$ preserves $\mathbb{Z} + \mathbb{Z}\tau$ for some integer $n \geq 1$. By Proposition 61.6.1,

$$\mathrm{End}_{\mathrm{HS}}(H) = \begin{cases} \mathbb{Q}(\tau) & \text{if } [\mathbb{Q}(\tau):\mathbb{Q}] = 2 \quad (E_\tau \text{ has } \textit{complex multiplication}), \\ \mathbb{Q} & \text{otherwise.} \end{cases}$$

For instance $\tau = i$ gives $\mathbb{Q}(i)$. Elliptic curves with complex multiplication are thus exactly those whose Hodge structure has extra symmetry; they are countably many up to isomorphism.

66.8 Exercises

Exercise 66.8.1. Show that $\mathbb{Z}(m) \otimes \mathbb{Z}(n) \cong \mathbb{Z}(m+n)$ and $\mathbb{Z}(n)^\vee \cong \mathbb{Z}(-n)$, and that a rank-one $\mathbb{Q}$-Hodge structure is isomorphic to $\mathbb{Q}(n)$ for a unique n.

Solution. The lattices multiply, $(2\pi i)^m\mathbb{Z} \otimes (2\pi i)^n\mathbb{Z} \to (2\pi i)^{m+n}\mathbb{Z}$, and the types add. A rank-one structure H has $H_{\mathbb{C}}$ one-dimensional, so it is a single $H^{p,q}$ with $\overline{H^{p,q}} = H^{q,p}$ forcing $p = q$; the weight is $2p$, and any nonzero rational vector gives an isomorphism with $\mathbb{Q}(-p)$. □

Exercise 66.8.2. Compute the Hodge numbers of $\Lambda^2 H$ for the weight-one Hodge structure $H = H^1(E)$ of an elliptic curve, and of $H^1(E)^{\otimes 2}$.

Solution. $H^{1,0}$ and $H^{0,1}$ are one-dimensional. $\Lambda^2 H$ is one-dimensional of type $(1,1)$, isomorphic to $\mathbb{Q}(-1)$ (it is $H^2(E)$). $H^{\otimes 2}$ has Hodge numbers $h^{2,0} = 1$, $h^{1,1} = 2$, $h^{0,2} = 1$, and $\mathrm{Sym}^2 H$ has $(1,1,1)$. □

Exercise 66.8.3. Let $H = H^1(E_\tau; \mathbb{Q})$ with basis dual to $1, \tau$ and Q the cup product pairing. Verify condition (3) of Definition 66.5.1 directly.

Solution. $H^{1,0}$ is spanned by $[dz]$, and $iQ([dz], \overline{[dz]}) = i\int_{E_\tau} dz \wedge d\bar{z} = 2\operatorname{Im}\tau > 0$, as in Example 49.4.4. □

Chapter 67

Mumford–Tate Groups

67.1 Introduction

The Mumford–Tate group of a rational Hodge structure H is the smallest algebraic group over $\mathbb{Q}$ whose real points contain the image of the Deligne torus. It is the symmetry group of H: a rational subspace of any tensor space built from H is a sub-Hodge structure exactly when it is stable under the Mumford–Tate group, and Hodge classes are exactly the invariants of the special Mumford–Tate group (Theorem 67.3.2). So determining all Hodge classes on a variety, the first step in any attack on the Hodge conjecture, amounts to computing a group and its invariants. For abelian varieties this is a problem in the representation theory of reductive groups, and it is how the known cases, and the exceptional Weil classes, are found (Chapter 72). References: [Del82], [BL04], [GGK12].

Prerequisites. Chapter 66 (Hodge Structures) and Chapter 61 (Abelian Varieties).

Throughout, H is a $\mathbb{Q}$-Hodge structure with representation $h\colon \mathbb{S} \to \mathrm{GL}(H_{\mathbb{R}})$ (Proposition 66.4.1), and $T^{m,n}(H) = H^{\otimes m} \otimes (H^\vee)^{\otimes n}$ are the tensor spaces, which are Hodge structures (Theorem 66.3.3).

67.2 Definition and first properties

Definition 67.2.1. The *Mumford–Tate group* $\mathrm{MT}(H)$ is the smallest algebraic subgroup $G \subseteq \mathrm{GL}(H_{\mathbb{Q}})$, defined over $\mathbb{Q}$, such that $h(\mathbb{S}(\mathbb{R})) \subseteq G(\mathbb{R})$. The *Hodge group*, or special Mumford–Tate group, $\mathrm{Hg}(H)$ is the smallest algebraic subgroup defined over $\mathbb{Q}$ with $h(\mathbb{U}(1)(\mathbb{R})) \subseteq \mathrm{Hg}(H)(\mathbb{R})$, where $\mathbb{U}(1) \subseteq \mathbb{S}$ is the circle group (Example 60.7.3).

Proposition 67.2.2. *1.* $\mathrm{MT}(H)$ *and* $\mathrm{Hg}(H)$ *exist and are connected, and* $\mathrm{Hg}(H) \subseteq \mathrm{MT}(H)$.

2. If H is pure of weight $k \neq 0$, then $\mathrm{MT}(H) = \mathbb{G}_m \cdot \mathrm{Hg}(H)$, where $\mathbb{G}_m$ is the group of scalars; if $k = 0$, $\mathrm{MT}(H) = \mathrm{Hg}(H)$.

3. $\mathrm{MT}(H)$ *acts on each* $T^{m,n}(H)$, *and* $\mathrm{MT}(H \oplus H') \subseteq \mathrm{MT}(H) \times \mathrm{MT}(H')$ *projects onto each factor.*

Proof. (1) Intersections of algebraic subgroups defined over $\mathbb{Q}$ are algebraic subgroups defined over $\mathbb{Q}$, and the Noetherian property of $\mathrm{GL}(H_{\mathbb{Q}})$ makes the intersection of all subgroups with the property a finite intersection, which still has it. The identity component of $\mathrm{MT}(H)$ is defined over $\mathbb{Q}$ and contains the image of the connected group $\mathbb{S}(\mathbb{R}) = \mathbb{C}^*$, so it equals $\mathrm{MT}(H)$; likewise for $\mathrm{Hg}(H)$. (2) $\mathbb{S}(\mathbb{R}) = \mathbb{R}_{>0} \cdot \mathbb{U}(1)(\mathbb{R})$, and $h(r) = r^{-k}$ for $r \in \mathbb{R}_{>0}$ by the weight condition. So $h(\mathbb{S}(\mathbb{R})) \subseteq (\mathbb{G}_m \cdot \mathrm{Hg}(H))(\mathbb{R})$, which gives $\subseteq$. Conversely $\mathrm{MT}(H)$ contains the scalars $h(\mathbb{R}_{>0})$ if $k \neq 0$, hence the $\mathbb{Q}$-group $\mathbb{G}_m$, and it contains $\mathrm{Hg}(H)$. If $k = 0$, h is trivial on $\mathbb{R}_{>0}$. (3) The action on tensors is functorial. For the direct sum, the image of $\mathrm{MT}(H \oplus H')$

in $\mathrm{GL}(H_{\mathbb{Q}})$ is a $\mathbb{Q}$-subgroup containing $h(\mathbb{S})$, so it contains $\mathrm{MT}(H)$, and it is contained in it because $\mathrm{MT}(H) \times \mathrm{MT}(H')$ contains the image of $h \oplus h'$. □

Theorem 67.2.3. *A $\mathbb{Q}$-subspace $W \subseteq T^{m,n}(H)$ is a sub-Hodge structure if and only if it is stable under* $\mathrm{MT}(H)$. *A vector $t \in T^{m,n}(H)_{\mathbb{Q}}$ spans a sub-Hodge structure if and only if* $\mathrm{MT}(H)$ *stabilises the line $\mathbb{Q}t$.*

Proof. By Proposition 66.4.1, $W_{\mathbb{C}} = \bigoplus (W_{\mathbb{C}} \cap T^{p,q})$ if and only if $W_{\mathbb{R}}$ is stable under $h(\mathbb{S}(\mathbb{R}))$, since the $T^{p,q}$ are the eigenspaces of the torus. The stabiliser of W in $\mathrm{GL}(T^{m,n}(H)_{\mathbb{Q}})$, pulled back to $\mathrm{GL}(H_{\mathbb{Q}})$, is an algebraic subgroup defined over $\mathbb{Q}$ (Proposition 60.3.3). It contains $h(\mathbb{S}(\mathbb{R}))$ if and only if it contains $\mathrm{MT}(H)$, by minimality. The second statement is the case $W = \mathbb{Q}t$. □

Corollary 67.2.4. $\mathrm{End}_{\mathrm{HS}}(H) = \mathrm{End}(H_{\mathbb{Q}})^{\mathrm{MT}(H)}$, *the endomorphisms commuting with the Mumford–Tate group. In particular H is irreducible as a Hodge structure if and only if $H_{\mathbb{Q}}$ is irreducible as a representation of* $\mathrm{MT}(H)$.

Proof. An endomorphism is a morphism of Hodge structures if and only if its graph, a subspace of $H \oplus H$, is a sub-Hodge structure. By Theorem 67.2.3 applied to $H \oplus H$ (whose Mumford–Tate group is the diagonal copy of $\mathrm{MT}(H)$), this means that the graph is $\mathrm{MT}(H)$-stable, that is, the endomorphism commutes with $\mathrm{MT}(H)$. Sub-Hodge structures are the $\mathrm{MT}(H)$-stable subspaces. □

67.3 Reductivity and Hodge classes

Theorem 67.3.1. *If H is polarisable, then* $\mathrm{MT}(H)$ *and* $\mathrm{Hg}(H)$ *are reductive.*

Proof. By Theorem 67.2.3, the $\mathrm{MT}(H)$-stable subspaces of $H_{\mathbb{Q}}$ are the sub-Hodge structures. Since H is polarisable, every sub-Hodge structure has a complement (Theorem 66.5.3), so $H_{\mathbb{Q}}$ is a semisimple representation of $\mathrm{MT}(H)$. It remains semisimple after extension of scalars to $\bar{\mathbb{Q}}$, as we are in characteristic 0 [Mil17]. Let $U = R_u(\mathrm{MT}(H)_{\bar{\mathbb{Q}}})$. On each irreducible summand W of $H_{\bar{\mathbb{Q}}}$, the unipotent normal subgroup U has a nonzero fixed vector (Kolchin's theorem [Bor91]), and the fixed space W^U is stable under $\mathrm{MT}(H)$ because U is normal, so $W^U = W$. Hence U acts trivially on the faithful representation $H_{\bar{\mathbb{Q}}}$, so $U = 1$ and $\mathrm{MT}(H)$ is reductive. The same argument applies to $\mathrm{Hg}(H)$, a normal subgroup of the reductive group $\mathrm{MT}(H)$ (it is normalised by $\mathrm{MT}(H)$ because $h(\mathbb{S})$ is commutative), whose representation $H_{\bar{\mathbb{Q}}}$ remains semisimple. □

Theorem 67.3.2. *Let H be a $\mathbb{Q}$-Hodge structure, pure of weight k.*

1. *A rational tensor $t \in T^{m,n}(H)_{\mathbb{Q}}$ is a Hodge class, that is, of type (p,p) with $2p = k(m-n)$, if and only if it is fixed by* $\mathrm{Hg}(H)$.
2. *If H is polarisable,* $\mathrm{Hg}(H)$ *is the subgroup of* $\mathrm{GL}(H_{\mathbb{Q}})$ *fixing all Hodge classes in all tensor spaces $T^{m,n}(H)$.*

Proof. (1) The circle $z \in \mathbb{U}(1)(\mathbb{R})$, $|z| = 1$, acts on $T^{p,q}$ by $z^{-p}\bar{z}^{-q} = z^{q-p}$. So a rational t is fixed by $h(\mathbb{U}(1)(\mathbb{R}))$ if and only if all its components have $p = q$, that is, t is a Hodge class. The stabiliser of t is an algebraic subgroup defined over $\mathbb{Q}$ (Proposition 60.3.3), and it contains $h(\mathbb{U}(1)(\mathbb{R}))$ if and only if it contains $\mathrm{Hg}(H)$. (2) By (1), $\mathrm{Hg}(H)$ is contained in the subgroup fixing all Hodge classes. By Theorem 67.3.1 $\mathrm{Hg}(H)$ is reductive, so by Theorem 60.3.2(3) it is the subgroup fixing all tensors fixed by $\mathrm{Hg}(H)$, which by (1) are the Hodge classes. □

Corollary 67.3.3. *Let A be a complex abelian variety and $H = H^1(A;\mathbb{Q})$. Then $H^\bullet(A;\mathbb{Q}) \cong \Lambda^\bullet H$ as Hodge structures, and the Hodge classes on A are $\bigoplus_p \mathrm{Hdg}^p(A) = (\Lambda^\bullet H)^{\mathrm{Hg}(H)}$. The same holds for all powers A^r, with $H^1(A^r) = H^{\oplus r}$ and the same Hodge group.*

Proof. $H^\bullet(A;\mathbb{Q}) = \Lambda^\bullet H^1(A;\mathbb{Q})$ by Proposition 61.2.1, compatibly with Hodge decompositions (cup product is a morphism of Hodge structures), and $\Lambda^\bullet H$ is a direct summand of the tensor algebra. Apply Theorem 67.3.2(1). For A^r, the Hodge group of $H^{\oplus r}$ is the diagonal image of $\mathrm{Hg}(H)$ by Proposition 67.2.2(3). □

Proposition 67.3.4. *Let H be pure of weight k with a polarisation Q. Then $\mathrm{Hg}(H)$ is contained in the automorphism group $\mathrm{Aut}(H_\mathbb{Q}, Q)$ of the form, which is the symplectic group $\mathrm{Sp}(H_\mathbb{Q}, Q)$ for k odd and the orthogonal group $\mathrm{O}(H_\mathbb{Q}, Q)$ for k even. Consequently $\mathrm{MT}(H)$ is contained in the group of similitudes $\{g : Q(gx, gy) = \lambda(g)Q(x, y)\}$.*

Proof. Viewed as an element of $T^{0,2}(H) = (H^\vee)^{\otimes 2}$, a Hodge structure of weight $-2k$, the form Q is rational and lies in $\bigoplus_{p+q=k}(H^{p,q})^\vee \otimes (H^{q,p})^\vee$ by condition (2) of Definition 66.5.1. By Theorem 66.3.3 the dual $(H^{p,q})^\vee$ has type $(-p,-q)$, so Q has type $(-k,-k)$ and is a Hodge class. By Theorem 67.3.2(1) it is fixed by $\mathrm{Hg}(H)$, that is, $Q(gx, gy) = Q(x, y)$ for $g \in \mathrm{Hg}(H)$. The form is alternating for k odd and symmetric for k even. The last statement follows from Proposition 67.2.2(2), since scalars are similitudes. □

Proposition 67.3.5. *For a $\mathbb{Q}$-Hodge structure H pure of weight k, the following are equivalent: $\mathrm{Hg}(H) = 1$; every element of $H_\mathbb{Q}$ is a Hodge class; H is of Tate type, that is, $k = 2p$ is even and $H \cong \mathbb{Q}(-p)^{\oplus r}$. In that case $\mathrm{MT}(H) = \mathbb{G}_m$ (scalars) if $p \neq 0$, and $\mathrm{MT}(H) = 1$ if $p = 0$.*

Proof. If $\mathrm{Hg}(H) = 1$, every rational vector is fixed, hence a Hodge class by Theorem 67.3.2(1). If every rational vector is a Hodge class, then $H_\mathbb{C}$, spanned by $H_\mathbb{Q}$, equals $H^{p,p}$, and a basis of $H_\mathbb{Q}$ gives $H \cong \mathbb{Q}(-p)^{\oplus r}$. If H is of Tate type, $\mathbb{U}(1)(\mathbb{R})$ acts on $H^{p,p}$ by $z^{-p}\bar z^{-p} = |z|^{-2p} = 1$, so the smallest $\mathbb{Q}$-group containing its image is trivial. The last statement follows from Proposition 67.2.2(2): $\mathrm{MT}(H) = \mathbb{G}_m \cdot \mathrm{Hg}(H) = \mathbb{G}_m$ if the weight $2p$ is nonzero, and $\mathrm{MT}(H) = \mathrm{Hg}(H) = 1$ if $p = 0$. □

Example 67.3.6. Let S be a nonsingular projective surface and $H = H^2(S;\mathbb{Q})$. By Proposition 67.3.5, $\mathrm{Hg}(H)$ is trivial exactly when $h^{2,0}(S) = 0$, and then every class in $H^2(S;\mathbb{Q})$ is a Hodge class; by the Lefschetz $(1,1)$-theorem (Chapter 69) all of them are then classes of divisors. When $h^{2,0} \neq 0$, the Hodge group fixes every Hodge class, in particular the class of a hyperplane section and all of $\mathrm{NS}(S) \otimes \mathbb{Q}$, and it preserves the polarisation of Theorem 66.5.2 (Proposition 67.3.4); so it acts on the orthogonal complement of $\mathrm{NS}(S) \otimes \mathbb{Q}$ for the intersection form, the *transcendental part*, preserving the intersection form, and it is nontrivial there.

67.4 Examples

Proposition 67.4.1. *Let E be a complex elliptic curve and $H = H^1(E;\mathbb{Q})$. If E has complex multiplication by the imaginary quadratic field K, then $\mathrm{MT}(H) = \mathrm{Res}_{K/\mathbb{Q}}\mathbb{G}_m$, a torus of dimension 2, and $\mathrm{Hg}(H)$ is the norm-one subtorus. If E has no complex multiplication, then $\mathrm{MT}(H) = \mathrm{GL}_2$ and $\mathrm{Hg}(H) = \mathrm{SL}_2$.*

Proof. By Corollary 67.2.4 and Theorem 66.7.1, the commutant of $\mathrm{MT}(H)$ in $\mathrm{End}(H_\mathbb{Q})$ is $\mathrm{End}(E)\otimes\mathbb{Q}$. In the CM case, K acts on the 2-dimensional space $H_\mathbb{Q}$, making it a 1-dimensional K-vector space, so the commutant of K in $\mathrm{GL}(H_\mathbb{Q})$ is the torus $T_K = \mathrm{Res}_{K/\mathbb{Q}}\mathbb{G}_m$, and $\mathrm{MT}(H) \subseteq T_K$. It contains the scalars (weight $1 \neq 0$) and the Weil operator $C = h(i)^{-1}$, which acts as i on $H^{1,0}$ and $-i$ on $H^{0,1}$ and is not a scalar. So $\dim \mathrm{MT}(H) = 2$ and $\mathrm{MT}(H) = T_K$. Without CM, the

commutant is $\mathbb{Q}$. The group $\mathrm{MT}(H)$ is connected and reductive (Theorem 67.3.1). A connected reductive subgroup of GL_2 is a torus or contains SL_2. A torus of GL_2 containing a non-scalar element has as commutant a 2-dimensional commutative algebra, not $\mathbb{Q}$. So $\mathrm{MT}(H) \supseteq \mathrm{SL}_2$, and with the scalars, $\mathrm{MT}(H) = \mathrm{GL}_2$. The Hodge groups follow from Proposition 67.2.2(2) and the fact that Hg preserves the polarisation (it fixes the Hodge class Q), so lies in $\mathrm{SL}_2 = \mathrm{Sp}_2$. □

Proposition 67.4.2. *If A is an abelian variety with complex multiplication by a CM field F of degree $2g$, then $\mathrm{MT}(H^1(A;\mathbb{Q}))$ is a torus, contained in $\mathrm{Res}_{F/\mathbb{Q}}\mathbb{G}_m$. Conversely, a polarisable weight-one Hodge structure whose Mumford–Tate group is a torus is of CM type.*

Proof. F acts on $H_{\mathbb{Q}}$ by Hodge endomorphisms, and $\dim_{\mathbb{Q}} H_{\mathbb{Q}} = 2g = [F:\mathbb{Q}]$, so $H_{\mathbb{Q}}$ is a 1-dimensional F-vector space. Its commutant in $\mathrm{GL}(H_{\mathbb{Q}})$ is $\mathrm{Res}_{F/\mathbb{Q}}\mathbb{G}_m$, a torus, which contains MT by Corollary 67.2.4; subgroups of tori that are connected are tori. Conversely, if MT is a torus T, its commutant contains the $\mathbb{Q}$-algebra generated by $T(\mathbb{Q})$, which is commutative semisimple; one shows that it contains a maximal commutative subalgebra of dimension $2g$ [BL04]. □

Theorem 67.4.3. *For a very general principally polarised abelian variety A of dimension g (outside a countable union of proper analytic subsets of the moduli space), $\mathrm{MT}(H^1(A)) = \mathrm{GSp}_{2g}$ and $\mathrm{Hg}(H^1(A)) = \mathrm{Sp}_{2g}$. Consequently $\mathrm{Hdg}^\bullet(A^r)$ is generated by divisor classes for every r, and the Hodge conjecture holds for all powers of A.*

Sketch of proof. For each of the countably many $\mathbb{Q}$-subgroups $G \subsetneq \mathrm{GSp}_{2g}$, the set of points of the Siegel space $\mathfrak{H}_g$ (Proposition 61.3.4) where h factors through $G(\mathbb{R})$ is a proper analytic subset, since $\mathrm{GSp}_{2g}(\mathbb{R})$ acts transitively on $\mathfrak{H}_g$ and the h of a general point generates. Outside these subsets $\mathrm{MT} = \mathrm{GSp}_{2g}$. By classical invariant theory for Sp_{2g}, the invariants in $\Lambda^\bullet(H^{\oplus r})$ are generated by the symplectic forms pairing the various copies, which are the classes of divisors [BL04]. By Corollary 67.3.3 these are all the Hodge classes. Rational Hodge classes of degree 2 are classes of divisors: a multiple lifts to an integral class of type $(1,1)$, which is c_1 of a line bundle (Lemma 50.2.2), algebraic by GAGA (Theorem 58.4.1), whose class is that of a divisor (Example 63.3.2(2)). □

Remark 67.4.4. The Mumford–Tate group is not determined by the endomorphism algebra. Mumford constructed abelian fourfolds with $\mathrm{End}(A) = \mathbb{Z}$ whose Mumford–Tate group is much smaller than GSp_8: a form of $\mathbb{G}_m \cdot (\mathrm{SL}_2)^3$ acting through the tensor product of three standard representations [Mum69]. For such A the ring of Hodge classes is not detected by endomorphisms.

Remark 67.4.5. For A defined over a number field k, the *Mumford–Tate conjecture* predicts that the identity component of the Zariski closure of the image of G_k in $\mathrm{GL}(H^1(\bar{A},\mathbb{Q}_\ell))$ equals $\mathrm{MT}(H^1(A;\mathbb{Q})) \otimes \mathbb{Q}_\ell$, under Artin's comparison isomorphism (Theorem 65.4.3). It relates the Hodge conjecture to the Tate conjecture (Chapter 76), and it is known for elliptic curves (Serre) and in many cases for abelian varieties.

67.5 Exercises

Exercise 67.5.1. Show that $\mathrm{MT}(\mathbb{Q}(n)) = \mathbb{G}_m$ for $n \neq 0$ and $\mathrm{MT}(\mathbb{Q}(0)) = 1$.

Solution. $h(z)$ acts on $\mathbb{Q}(n)$, of type $(-n,-n)$, by $z^n\bar{z}^n = |z|^{2n}$. For $n \neq 0$ the image of $\mathbb{S}(\mathbb{R})$ is $\mathbb{R}_{>0}$, whose $\mathbb{Q}$-Zariski closure in GL_1 is $\mathbb{G}_m$. For $n = 0$ the action is trivial. □

Exercise 67.5.2. Let E_1, E_2 be non-isogenous elliptic curves without complex multiplication. Show that $\mathrm{Hg}(H^1(E_1) \oplus H^1(E_2)) = \mathrm{SL}_2 \times \mathrm{SL}_2$, and deduce that $\mathrm{Hdg}^1(E_1 \times E_2)$ has dimension 2.

Solution. By Proposition 67.2.2(3) the Hodge group is a connected reductive subgroup of $\mathrm{SL}_2 \times \mathrm{SL}_2$ projecting onto both factors. By Goursat's lemma for the simple groups SL_2, it is either the product or the graph of an isomorphism $\mathrm{SL}_2 \to \mathrm{SL}_2$. In the second case the two representations would be isomorphic as Hodge group representations, giving a nonzero Hodge class in $\mathrm{Hom}(H^1(E_1), H^1(E_2))$, hence an isogeny (Theorem 66.7.1), which is excluded. The invariants of $\mathrm{SL}_2 \times \mathrm{SL}_2$ in $\Lambda^2(H_1 \oplus H_2) = \Lambda^2 H_1 \oplus (H_1 \otimes H_2) \oplus \Lambda^2 H_2$ are $\Lambda^2 H_1 \oplus \Lambda^2 H_2$, of dimension 2. $\square$

Chapter 68

The Hodge Conjecture

68.1 Introduction

On a nonsingular complex projective variety there are two ways of producing rational cohomology classes of type (p,p). The algebro-geometric way takes the class of a subvariety of codimension p (Chapter 63). The Hodge-theoretic way takes all rational classes whose type is (p,p). The first set is contained in the second (Proposition 68.2.2). The Hodge conjecture asserts that, after taking $\mathbb{Q}$-linear combinations, they are equal. The inclusion is easy; the conjecture is a *construction problem*: from a class satisfying two linear conditions one must produce subvarieties, and there is no general method for producing subvarieties of codimension ≥ 2. This chapter states the conjecture and its equivalent forms, proves the elementary reductions, tells the history of its two corrections, and records the reasons for believing it and the reasons it is hard. We follow the exposition of the source *The Hodge Conjecture* (2026) and Deligne's official problem description [Del06], checking each claim against the literature.

Prerequisites. Chapter 66 (Hodge Structures) and Chapter 63 (The Cycle Class Map).

Throughout, X is a nonsingular complex projective variety of dimension n, identified with the compact Kähler manifold X^{an} (Theorem 58.6.1), and $0 \leq p \leq n$.

68.2 Algebraic classes and Hodge classes

Definition 68.2.1. The group of *rational Hodge classes* of codimension p is $\mathrm{Hdg}^p(X) = H^{2p}(X;\mathbb{Q}) \cap H^{p,p}(X)$ (Definition 63.4.1). The space of *algebraic classes* is the image of the cycle class map,

$$Z^p_{\mathrm{alg}}(X) = \mathrm{cl}(\mathrm{CH}^p(X))\otimes\mathbb{Q} = \Big\{ \sum_i a_i \,\mathrm{cl}(Z_i) : a_i \in \mathbb{Q},\ Z_i \subseteq X \text{ subvarieties of codimension } p \Big\} \subseteq H^{2p}(X;\mathbb{Q}).$$

Proposition 68.2.2 (The easy inclusion)**.** $Z^p_{\mathrm{alg}}(X) \subseteq \mathrm{Hdg}^p(X)$.

Proof. This is Corollary 63.4.2. The content is that a form of type (a,b) with $a > \dim Z$ or $b > \dim Z$ restricts to zero on Z_{reg} (Proposition 48.5.8). □

Conjecture 68.2.3 (Hodge conjecture)**.** *For every nonsingular complex projective variety X and every p,*

$$Z^p_{\mathrm{alg}}(X) = \mathrm{Hdg}^p(X).$$

That is, every rational cohomology class of type (p,p) on X is a rational linear combination of classes of algebraic subvarieties of codimension p. We write $\mathrm{HC}(X,p)$ for the statement for given X and p, and $\mathrm{HC}(X)$ for all p.

Remark 68.2.4. Every hypothesis is necessary. The coefficients must be rational: the integral version is false, both for torsion classes (Atiyah–Hirzebruch) and for non-torsion classes (Kollár). X must be projective: the analogous statement for compact Kähler manifolds is false (Voisin). Hodge's original, more general conjecture on the whole cohomology was false as stated (Grothendieck). These failures are the subject of Chapter 73 and Section 68.4.

Proposition 68.2.5 (Equivalent forms)**.** *For nonsingular projective X the following are equivalent to* $\mathrm{HC}(X,p)$*:*

1. *(homological form) every class in $H_{2n-2p}(X;\mathbb{Q})$ whose Poincaré dual is of type (p,p) is a rational combination of fundamental classes of subvarieties of dimension $n-p$;*
2. *(Hodge-theoretic form) the map $\mathrm{CH}^p(X)_{\mathbb{Q}} \to \mathrm{Hom}_{\mathrm{HS}}(\mathbb{Q}(0), H^{2p}(X;\mathbb{Q})(p))$ induced by* cl *is surjective;*
3. *(vector bundle form) $\mathrm{Hdg}^p(X)$ is contained in the $\mathbb{Q}$-algebra generated by the Chern classes of algebraic vector bundles on X.*

Proof. (1) is Poincaré duality, which maps fundamental classes to cycle classes (Definition 63.2.2) and Hodge classes to the corresponding classes in homology. (2) holds because $\mathrm{Hom}_{\mathrm{HS}}(\mathbb{Q}(0), H^{2p}(p))$ is the space of rational classes of type $(0,0)$ in $H^{2p}(p)$, which is $\mathrm{Hdg}^p(X)$ twisted (Definition 66.6.1). (3) follows from Proposition 63.5.1: the $\mathbb{Q}$-span of algebraic classes of all codimensions equals the $\mathbb{Q}$-algebra generated by Chern classes of vector bundles, and its degree-$2p$ part is $Z^p_{\mathrm{alg}}(X)$. □

Remark 68.2.6. Applied to a product $X \times Y$, the conjecture predicts that every morphism of Hodge structures $H^k(X;\mathbb{Q}) \to H^{k+2r-2n}(Y;\mathbb{Q})(r-n)$ that is a Hodge class in $H^{2r}(X\times Y;\mathbb{Q})$ (via Künneth and Poincaré duality) is induced by an algebraic correspondence (Definition 62.5.1). This is the form in which the conjecture is used: for instance, it would make the Künneth projectors and the Lefschetz inverse algebraic (Chapter 77).

68.3 First reductions and elementary cases

Proposition 68.3.1. $\mathrm{HC}(X,0)$ *and* $\mathrm{HC}(X,n)$ *hold:* $\mathrm{Hdg}^0(X) = H^0(X;\mathbb{Q})$ *is spanned by the classes of the connected components, and* $\mathrm{Hdg}^n(X) = H^{2n}(X;\mathbb{Q})$ *is spanned by the classes of points, one in each connected component.*

Proof. H^0 and H^{2n} are entirely of type $(0,0)$ and (n,n). The class of a connected component is its fundamental class, and the class of a point generates H^{2n} of its component (Example 63.3.2(3)). □

Proposition 68.3.2 (Hard Lefschetz reduction)**.** *Let $2p \le n$. Then* $\mathrm{HC}(X,p)$ *implies* $\mathrm{HC}(X,n-p)$*.*

Proof. Let $h = \mathrm{cl}(H)$ be the class of a hyperplane section, an algebraic class. By Theorem 49.2.2, $L^{n-2p}\colon H^{2p}(X;\mathbb{Q}) \to H^{2n-2p}(X;\mathbb{Q})$, $\alpha \mapsto h^{n-2p}\cup\alpha$, is an isomorphism; it is defined over $\mathbb{Q}$ and of type $(n-2p, n-2p)$, so it maps $\mathrm{Hdg}^p(X)$ isomorphically onto $\mathrm{Hdg}^{n-p}(X)$ (its inverse is also of type $(-(n-2p),-(n-2p))$ and defined over $\mathbb{Q}$). If $\beta \in \mathrm{Hdg}^{n-p}(X)$, then $\beta = h^{n-2p}\alpha$ with $\alpha \in \mathrm{Hdg}^p(X)$, which is algebraic by hypothesis, and β is a product of algebraic classes, hence algebraic (Theorem 63.3.1(1)). □

Remark 68.3.3. The case $p = 1$ is the Lefschetz $(1,1)$-theorem (Chapter 69). With Proposition 68.3.2 it gives $\mathrm{HC}(X,n-1)$, and therefore the Hodge conjecture for all varieties of

dimension at most 3. The first open case is $p = 2$ on fourfolds. The converse of Proposition 68.3.2, that $\mathrm{HC}(X, n-p)$ implies $\mathrm{HC}(X, p)$, is not known in general: it would require the inverse of L^{n-2p} to be induced by an algebraic correspondence, which is the Lefschetz standard conjecture (Chapter 77).

Proposition 68.3.4. *Let $f : X \to Y$ be a surjective morphism of nonsingular projective varieties of the same dimension. If $\mathrm{HC}(X, p)$ holds, then $\mathrm{HC}(Y, p)$ holds.*

Proof. f is generically finite of some degree $d \geq 1$, so $f_*[X] = d[Y]$ in $\mathrm{CH}^0(Y)$ (Theorem 62.3.1). The pullback f^* and the Gysin pushforward f_* in cohomology are morphisms of Hodge structures (of type $(0,0)$ between the relevant degrees, as $\dim X = \dim Y$), and both map algebraic classes to algebraic classes (Theorem 63.3.1(2)). By the projection formula, $f_* f^* \beta = f_*(1) \cup \beta = d\beta$. So for $\beta \in \mathrm{Hdg}^p(Y)$, $f^*\beta \in \mathrm{Hdg}^p(X)$ is algebraic, and $\beta = \frac{1}{d} f_*(f^*\beta)$ is algebraic. □

Example 68.3.5. If X has a filtration by closed subvarieties $X = X_0 \supseteq X_1 \supseteq \cdots \supseteq X_m = \emptyset$ with each $X_i \setminus X_{i+1}$ a disjoint union of affine spaces, then $H^{\mathrm{odd}}(X;\mathbb{Z}) = 0$ and $H^{2p}(X;\mathbb{Z})$ is free on the classes of the closures of the cells of codimension p [Ful98, Example 19.1.11]. So $\mathrm{HC}(X)$ holds, even integrally. This covers projective spaces, Grassmannians, flag varieties and smooth projective toric varieties; here the Hodge structures are of Tate type and there is no room for a non-algebraic Hodge class. For $\mathbb{P}^n$ it is Example 63.3.2(1).

Example 68.3.6. Let E, E' be complex elliptic curves. By Künneth, $H^2(E \times E';\mathbb{Q}) = H^2(E) \oplus (H^1(E) \otimes H^1(E')) \oplus H^2(E')$, and the outer summands are spanned by the classes of $\{\mathrm{pt}\} \times E'$ and $E \times \{\mathrm{pt}\}$. The Hodge classes in the middle summand correspond to morphisms of Hodge structures $H_1(E) \to H_1(E')$ (after Poincaré duality on E), that is, by Theorem 66.7.1, to elements of $\mathrm{Hom}(E, E') \otimes \mathbb{Q}$. The Hodge classes are nonzero exactly when E and E' are isogenous. Each homomorphism φ has a graph $\Gamma_\varphi \subseteq E \times E'$, whose class has Künneth component in $H^1 \otimes H^1$ equal to the class corresponding to φ, so $\mathrm{HC}(E \times E', 1)$ holds. (This is also a consequence of the Lefschetz $(1,1)$-theorem.)

Remark 68.3.7. $\mathrm{HC}(X)$ and $\mathrm{HC}(Y)$ do not formally imply $\mathrm{HC}(X \times Y)$. The Künneth summands $H^i(X) \otimes H^j(Y)$ with i and j odd contain Hodge classes that are not products of Hodge classes on the factors. Example 68.3.6 is the simplest case, and the Weil classes on abelian fourfolds (Chapter 72) are the serious ones. For blow-ups, on the other hand, the blow-up formula $H^\bullet(\tilde{X}) = H^\bullet(X) \oplus \bigoplus_{j=1}^{c-1} H^{\bullet-2j}(W)(-j)$ for the blow-up along a nonsingular centre W of codimension c is induced by algebraic correspondences. So HC for X and W in the relevant degrees gives it for $\tilde{X}$ [Voi02b, Theorem 7.31].

68.4 History and two corrections

Remark 68.4.1 (Lefschetz). In 1924 Lefschetz proved that on a nonsingular projective surface an integral class of type $(1,1)$ in H^2 is the class of a divisor, using normal functions. The modern proof via the exponential sequence is due to Kodaira and Spencer [KS53]. Lefschetz also proved the hyperplane theorem and, with a gap, the hard Lefschetz theorem, whose first complete proof is Hodge's, by harmonic forms. The framework of the conjecture (hyperplane sections, primitive cohomology, divisors versus cohomology) is Lefschetz's.

Remark 68.4.2 (Hodge). At the International Congress of 1950 Hodge proposed a generalisation of Lefschetz's theorem to all codimensions [Hod52]. He stated it with integral coefficients, and he also proposed a stronger conjecture about the whole cohomology. Both versions were false, for different reasons.

Remark 68.4.3 (First correction: integral to rational). Atiyah and Hirzebruch showed in 1962 that there are nonsingular projective varieties with torsion classes in $H^{2p}(X;\mathbb{Z})$ (automatically Hodge classes) that are not algebraic [AH62]. The obstruction is topological: algebraic classes come from complex vector bundles, so the differentials of the Atiyah–Hirzebruch spectral sequence, starting with $d_3 = \mathrm{Sq}^3_{\mathbb{Z}}$, must vanish on them. Kollár later gave non-torsion counterexamples: on a very general hypersurface of suitable degree in $\mathbb{P}^4$ the generator of $H^4(X;\mathbb{Z}) \cong \mathbb{Z}$ is not algebraic, although a multiple of it is (Chapter 73). Rational coefficients remove both phenomena.

Definition 68.4.4. The *coniveau filtration* on $H^k(X;\mathbb{Q})$ is

$$N^c H^k(X;\mathbb{Q}) = \sum_{Z} \ker\left(H^k(X;\mathbb{Q}) \to H^k(X \setminus Z;\mathbb{Q})\right),$$

the sum over closed algebraic subsets $Z \subseteq X$ of codimension $\geq c$: the classes *supported in codimension* c.

Proposition 68.4.5. *$N^c H^k(X;\mathbb{Q})$ is a sub-Hodge structure of $H^k(X;\mathbb{Q})$ contained in $F^c H^k(X;\mathbb{C})$. Moreover $N^p H^{2p}(X;\mathbb{Q}) = Z^p_{\mathrm{alg}}(X)$.*

Sketch of proof. A class supported on Z lies in the image of the Gysin map $H^{k-2c'}(\tilde{Z};\mathbb{Q})(-c') \to H^k(X;\mathbb{Q})$ from a resolution of singularities $\tilde{Z}$ of a component of Z of codimension $c' \geq c$, by Deligne's theory of weights; the Gysin map is a morphism of Hodge structures of type (c',c'), so its image is a sub-Hodge structure contained in $F^{c'} \subseteq F^c$ [Del71]. For $k = 2p$ and $c = p$, the Gysin image of $H^0(\tilde{Z})$ is spanned by $\mathrm{cl}(Z)$. □

Remark 68.4.6 (Second correction: Grothendieck's amendment). Hodge's general conjecture asserted $N^c H^k(X;\mathbb{Q}) = H^k(X;\mathbb{Q}) \cap F^c H^k(X;\mathbb{C})$. In 1969 Grothendieck observed that this is false for trivial reasons [Gro69a]: the left side is a sub-Hodge structure, while the right side, a rational subspace contained in $H^{k-c,c} \oplus \cdots \oplus H^{c,k-c}$, need not be one, and he gave an abelian variety where they differ. He proposed the *generalised Hodge conjecture*: $N^c H^k(X;\mathbb{Q})$ is the largest sub-Hodge structure of $H^k(X;\mathbb{Q})$ contained in $F^c H^k(X;\mathbb{C})$. For $k = 2p$ and $c = p$ the largest sub-Hodge structure contained in $F^p \cap \overline{F^p} = H^{p,p}$ is spanned by $\mathrm{Hdg}^p(X)$, so the generalised conjecture contains Conjecture 68.2.3.

Remark 68.4.7 (The Clay problem). In 2000 the Hodge conjecture became one of the seven Millennium Prize Problems of the Clay Mathematics Institute, with an official description by Deligne [Del06], which emphasises the reformulation through absolute Hodge classes (Chapter 75).

68.5 Why one might believe it, and why it is hard

Remark 68.5.1 (Evidence).

1. *It holds where it can be checked*: for $p \in \{0, 1, n-1, n\}$, hence in dimension ≤ 3; for cellular varieties; for cubic fourfolds [Zuc77]; for many abelian varieties (Chapter 72). No verified case has failed.

2. *Hodge loci are algebraic*. In a family of projective manifolds, the locus where a given integral class stays of type (p,p) is a countable union of algebraic subvarieties, by the theorem of Cattani, Deligne and Kaplan [CDK95]. This is what the conjecture predicts, since loci where cycles exist are algebraic (Hilbert schemes), and the theorem is proved without it (Chapter 70).

3. *Hodge classes on abelian varieties are absolute* [Del82]: they are compatible with all automorphisms of $\mathbb{C}$, as algebraic classes are and a random Hodge class would not be (Chapter 75).

4. *Coherence*: the conjecture follows from, and is required by, the expected properties of pure motives and the standard conjectures (Chapter 77).

Remark 68.5.2 (Difficulty). 1. *No construction machine in codimension* ≥ 2. For divisors, an integral class of type $(1,1)$ lifts to a line bundle by the exponential sequence, and a high twist of a line bundle has sections whose zero loci are divisors (Chapter 69). This uses that $\mathrm{Pic}(X) = H^1(X, \mathcal{O}_X^*)$ is the cohomology of a sheaf of abelian groups. In codimension $p \geq 2$, Bloch's formula $\mathrm{CH}^p(X) \cong H^p(X, \mathcal{K}_p)$ expresses the Chow group through a sheaf of K-groups, but this transfers the difficulty rather than removing it.

2. *Deformation theory obstructs.* Hilbert schemes of subvarieties of codimension ≥ 2 are typically singular of excess dimension, and deformations of a cycle do not see the target Hodge class.

3. *The transcendental input is weak.* Hodge theory says nothing about where in X a cycle should lie. Griffiths' normal functions (Chapter 71) convert the problem into the existence of singularities of certain sections, which is as hard in general.

4. *The conjecture is close to false.* Its integral, Kähler and general versions all fail (Chapter 73). A proof must distinguish it from these neighbours.

68.6 Exercises

Exercise 68.6.1. Show that a $\mathbb{Q}$-Hodge structure of odd weight has no Hodge classes in any Tate twist, so that Hodge classes live only in even degrees. Explain why the Künneth summands $H^{\mathrm{odd}}(X) \otimes H^{\mathrm{odd}}(Y)$ of a product are where new Hodge classes appear.

Solution. $V(m)$ has weight $k-2m$, odd, so it has no component of type $(0,0)$, and $\mathrm{Hom}_{\mathrm{HS}}(\mathbb{Q}(0), V(m)) = 0$. The tensor product of two odd-weight structures has even weight, and it can contain Hodge classes not coming from Hodge classes on the factors, which have none in odd degree (Example 68.3.6). □

Exercise 68.6.2. Verify $\mathrm{HC}(\mathbb{P}^m \times \mathbb{P}^k)$ directly.

Solution. By Künneth, $H^\bullet = \mathbb{Q}[a,b]/(a^{m+1}, b^{k+1})$ with a, b the pullbacks of hyperplane classes, all classes are of Tate type and spanned by the monomials $a^i b^j$, which are the classes of the products of linear subspaces $L \times L'$ (Example 63.3.2(1) and Theorem 63.3.1(2)). □

Exercise 68.6.3. Let a finite group G act on X with nonsingular projective quotient $Y = X/G$. Show that $\mathrm{HC}(X,p)$ implies $\mathrm{HC}(Y,p)$.

Solution. The quotient map $X \to Y$ is surjective and finite, of degree $|G|$, so Proposition 68.3.4 applies. □

Chapter 69

The Lefschetz (1,1) Theorem

69.1 Introduction

The Lefschetz $(1,1)$-theorem is the one case of the Hodge conjecture that is completely understood, and it holds even with integral coefficients: on a nonsingular projective variety every integral class of type $(1,1)$ in H^2 is the class of a divisor (Theorem 69.3.1). Its proof uses the exponential sequence, which exists because the Picard group is the cohomology of a sheaf of abelian groups. With hard Lefschetz it gives the Hodge conjecture in codimension $n-1$, hence in dimension at most three. This chapter proves the theorem, draws the consequences, and explains precisely which step has no analogue in higher codimension. References: [GH78, Chapter 1, Section 1], [Voi02b, Section 11.3], [KS53].

Prerequisites. Chapter 68 (The Hodge Conjecture) and Chapter 58 (Analytic Spaces and GAGA).

Throughout, X is a compact Kähler manifold of dimension n; from Section 69.3 on it is projective.

69.2 The theorem for compact Kähler manifolds

Theorem 69.2.1 (Lefschetz $(1,1)$, Kähler form)**.** *Let X be a compact Kähler manifold. The image of the first Chern class $c_1\colon \operatorname{Pic}(X) \to H^2(X;\mathbb{Z})$ is exactly the group of integral Hodge classes $\operatorname{Hdg}^1(X;\mathbb{Z})$, the classes whose image in $H^2(X;\mathbb{C})$ is of type $(1,1)$.*

Proof. The Chern class of a holomorphic line bundle is represented by $\frac{i}{2\pi}F$, a closed real $(1,1)$-form (Theorem 45.4.1), so $c_1(\operatorname{Pic}(X)) \subseteq \operatorname{Hdg}^1(X;\mathbb{Z})$. Conversely, let $a \in H^2(X;\mathbb{Z})$ have image of type $(1,1)$. By the *key lemma* (Lemma 50.2.2), the map $H^2(X;\mathbb{Z}) \to H^2(X,\mathcal{O}_X)$ induced by $\mathbb{Z}_X \subseteq \mathcal{O}_X$ is the composite of $H^2(X;\mathbb{Z}) \to H^2(X;\mathbb{C})$ with the projection to $H^{0,2}(X) \cong H^2(X,\mathcal{O}_X)$. So a maps to 0 in $H^2(X,\mathcal{O}_X)$. In the long exact sequence of the exponential sequence $0 \to \mathbb{Z} \to \mathcal{O}_X \to \mathcal{O}_X^* \to 0$ (Theorem 45.2.4),

$$H^1(X,\mathcal{O}_X^*) = \operatorname{Pic}(X) \xrightarrow{c_1} H^2(X;\mathbb{Z}) \to H^2(X,\mathcal{O}_X),$$

so a is in the image of c_1. □

Corollary 69.2.2. $\operatorname{NS}(X) := c_1(\operatorname{Pic}(X)) = \operatorname{Hdg}^1(X;\mathbb{Z})$, *and* $\operatorname{Pic}(X)$ *is an extension* $0 \to \operatorname{Pic}^0(X) \to \operatorname{Pic}(X) \to \operatorname{NS}(X) \to 0$ *of a finitely generated abelian group by the complex torus* $\operatorname{Pic}^0(X) = H^1(X,\mathcal{O}_X)/H^1(X;\mathbb{Z})$ *of dimension* $h^{0,1} = b_1/2$. *The* Picard number $\rho(X) = \operatorname{rank}\operatorname{NS}(X)$ *satisfies* $\rho(X) \le h^{1,1}(X)$.

Proof. The first statement is Theorem 69.2.1, and the extension is the exponential sequence. $\mathrm{NS}(X) \subseteq H^2(X;\mathbb{Z})$ is finitely generated, and $\mathrm{Pic}^0(X)$ is a torus by Remark 45.2.5 and Exercise 48.8.4. Finally $\mathrm{NS}(X) \otimes \mathbb{R}$ injects into the real classes of type $(1,1)$, which form a real vector space of dimension $h^{1,1}$ (a real structure on $H^{1,1}$, since $\overline{H^{1,1}} = H^{1,1}$). □

Proposition 69.2.3. *Let X be a compact Kähler manifold, and let $U(1)_X$ be the sheaf of locally constant functions with values in $U(1) = \{z \in \mathbb{C} : |z| = 1\}$. The inclusion $U(1)_X \subseteq \mathcal{O}_X^*$ induces a homomorphism $b\colon H^1(X, U(1)_X) \to \mathrm{Pic}(X)$, whose image consists of the* flat unitary *line bundles, those given by locally constant transition functions of modulus one.*

1. *b maps the kernel of the connecting map $\delta\colon H^1(X, U(1)_X) \to H^2(X;\mathbb{Z})$ isomorphically onto $\mathrm{Pic}^0(X)$, and this kernel is $H^1(X;\mathbb{R})/H^1(X;\mathbb{Z})$, a real torus of dimension $b_1(X)$.*

2. *A holomorphic line bundle is flat unitary if and only if its first Chern class is a torsion element of $H^2(X;\mathbb{Z})$. In particular every torsion class in $H^2(X;\mathbb{Z})$ lies in* $\mathrm{NS}(X)$.

Proof. The map $t \mapsto e^{2\pi i t}$ gives a short exact sequence of sheaves $0 \to \mathbb{Z}_X \to \mathbb{R}_X \to U(1)_X \to 0$: on a connected neighbourhood a locally constant function is constant, and $\mathbb{R} \to U(1)$ is surjective with kernel $\mathbb{Z}$. The inclusions $\mathbb{R}_X \subseteq \mathcal{O}_X$ and $U(1)_X \subseteq \mathcal{O}_X^*$ map it to the exponential sequence of Theorem 45.2.4, and by naturality of the long exact sequence (Theorem 15.4.3) we get a commutative diagram with exact rows

$$\begin{array}{ccccccccc}
H^1(X;\mathbb{Z}) & \longrightarrow & H^1(X;\mathbb{R}) & \xrightarrow{\beta} & H^1(X,U(1)_X) & \xrightarrow{\delta} & H^2(X;\mathbb{Z}) & \longrightarrow & H^2(X;\mathbb{R}) \\
\| & & \downarrow a & & \downarrow b & & \| & & \\
H^1(X;\mathbb{Z}) & \longrightarrow & H^1(X,\mathcal{O}_X) & \xrightarrow{\beta'} & \mathrm{Pic}(X) & \xrightarrow{c_1} & H^2(X;\mathbb{Z}) & &
\end{array}$$

where a is induced by $\mathbb{R}_X \subseteq \mathcal{O}_X$. By Exercise 48.8.4, a is an isomorphism of real vector spaces. By exactness, $\ker\delta = \beta(H^1(X;\mathbb{R})) \cong H^1(X;\mathbb{R})/H^1(X;\mathbb{Z})$ and $\mathrm{Pic}^0(X) = \ker c_1 = \beta'(H^1(X,\mathcal{O}_X)) \cong H^1(X,\mathcal{O}_X)/H^1(X;\mathbb{Z})$, and a induces an isomorphism between these quotients, since the left square commutes. As $b\beta = \beta' a$, this isomorphism is the restriction of b, proving (1); the image of $H^1(X;\mathbb{Z})$ in $H^1(X;\mathbb{R})$ is a lattice by the universal coefficient theorem (Theorem 22.2.3).

(2) If $L = b(y)$, then $c_1(L) = \delta(y)$ maps to 0 in $H^2(X;\mathbb{R})$ by exactness of the top row, so it is torsion, because the kernel of $H^2(X;\mathbb{Z}) \to H^2(X;\mathbb{R})$ is the torsion subgroup (again by the universal coefficient theorem). Conversely, let $c_1(L)$ be torsion. Then it maps to 0 in $H^2(X;\mathbb{R})$, so $c_1(L) = \delta(y)$ for some y, and $L \otimes b(y)^{-1}$ has $c_1 = 0$. By (1), $L \otimes b(y)^{-1} = b(y')$ is flat unitary, and so is $L = b(yy')$. The same argument, starting from a torsion class $t = \delta(y)$, shows $t = c_1(b(y)) \in \mathrm{NS}(X)$. □

Example 69.2.4. Let $T = \mathbb{C}^2/\Lambda$ be a complex torus of dimension 2 and $\sigma = dz_1 \wedge dz_2$. By Example 46.4.2, $h^{2,0} = h^{0,2} = 1$ and $h^{1,1} = 4$, with $H^{2,0}(T)$ spanned by $[\sigma]$ and $H^{0,2}(T)$ by $[\bar\sigma]$; and $H^2(T;\mathbb{Z}) \cong \mathbb{Z}^6$ (Example 22.4.3). A real class $a \in H^2(T;\mathbb{R})$ is of type $(1,1)$ if and only if $\int_T a \wedge \sigma = 0$. Indeed, write $a = a^{2,0} + a^{1,1} + a^{0,2}$ (Theorem 48.5.2), with $a^{2,0} = \overline{a^{0,2}}$ because a is real, and $a^{0,2} = c[\bar\sigma]$. For type reasons only $a^{0,2}$ contributes to $\int_T a \wedge \sigma$, so the integral is $c\int_T \bar\sigma \wedge \sigma$, and $\bar\sigma \wedge \sigma$ is a nonzero constant multiple of $dx_1 \wedge dy_1 \wedge dx_2 \wedge dy_2$ ($z_j = x_j + iy_j$); hence it vanishes if and only if $c = 0$, that is, $a^{0,2} = a^{2,0} = 0$.

By Theorem 69.2.1, $\mathrm{NS}(T)$ is therefore the kernel of $\varphi\colon H^2(T;\mathbb{Z}) \to \mathbb{C}$, $\varphi(a) = \int_T a \wedge \sigma$. If $e_1, \dots, e_6$ is a $\mathbb{Z}$-basis of $H^2(T;\mathbb{Z})$ with *periods* $\pi_k = \varphi(e_k)$, then

$$\rho(T) = 6 - \dim_{\mathbb{Q}}(\mathbb{Q}\pi_1 + \cdots + \mathbb{Q}\pi_6),$$

since the kernel of $\mathbb{Z}^6 \to \mathbb{C}$, $m \mapsto \sum_k m_k \pi_k$, has the same rank as the kernel of its extension to $\mathbb{Q}^6$. The real extension $H^2(T;\mathbb{R}) \to \mathbb{C}$ has as kernel the real classes of type $(1,1)$, of real dimension $h^{1,1} = 4$, so it is surjective and the π_k span $\mathbb{C}$ over $\mathbb{R}$. Hence the $\mathbb{Q}$-dimension above is at least 2 and $\rho(T) \le 4$, as predicted by Corollary 69.2.2. The Picard number of T is thus governed by the linear relations over $\mathbb{Q}$ among six complex numbers: $\rho(T) = 4$ when the periods span only a two-dimensional $\mathbb{Q}$-vector space, as for $E \times E$ with $E = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}i)$ (Example 69.3.4(2), since $\mathrm{Hom}(E,E) \supseteq \mathbb{Z}[i]$ has rank 2), and $\rho(T) = 0$ when they are linearly independent over $\mathbb{Q}$, which is the situation of Exercise 69.5.1.

69.3 The projective case: classes of divisors

Theorem 69.3.1 (Lefschetz $(1,1)$)**.** *Let X be a nonsingular complex projective variety. Every class in $\mathrm{Hdg}^1(X;\mathbb{Z})$ is the class of a divisor: the cycle class map $\mathrm{CH}^1(X) \to \mathrm{Hdg}^1(X;\mathbb{Z})$ is surjective. In particular $\mathrm{HC}(X,1)$ holds, even integrally.*

Proof. By Theorem 69.2.1, a class $a \in \mathrm{Hdg}^1(X;\mathbb{Z})$ is $c_1(L)$ for a holomorphic line bundle L on X^{an}. By GAGA (Corollary 58.4.3(2)), L is the analytification of an algebraic line bundle, still denoted L. Let H be a hyperplane section. By Serre's theorem (Theorem 55.5.3), $L(m) = L \otimes \mathcal{O}(mH)$ is globally generated for $m \gg 0$, so it has a nonzero section, whose divisor D_1 is effective with $\mathcal{O}(D_1) \cong L(m)$ (Proposition 57.3.2). Likewise $\mathcal{O}(mH)$ has the divisor $D_2 = mH$. So $L \cong \mathcal{O}(D_1 - D_2)$, and $a = c_1(L) = \mathrm{cl}(D_1) - \mathrm{cl}(D_2)$ by Example 63.3.2(2). □

Corollary 69.3.2. *For a nonsingular projective variety X of dimension n, $\mathrm{HC}(X, n-1)$ holds. Consequently the Hodge conjecture holds for all nonsingular projective varieties of dimension at most 3.*

Proof. $\mathrm{HC}(X,1)$ holds by Theorem 69.3.1, and implies $\mathrm{HC}(X, n-1)$ for $n \ge 2$ by Proposition 68.3.2. Together with the trivial cases $p = 0$ and $p = n$ (Proposition 68.3.1), this covers all p when $n \le 3$. □

Remark 69.3.3. For curves on a threefold, the integral version fails in general: Proposition 68.3.2 only gives that some multiple of an integral Hodge class in H^4 is algebraic, since L^{n-2} need not be an isomorphism over $\mathbb{Z}$. Kollár's counterexamples to the integral Hodge conjecture are integral classes in H^4 of threefolds (Chapter 73). On the other hand, Voisin proved the integral Hodge conjecture for curve classes on uniruled threefolds and on Calabi–Yau threefolds [Voi06].

Example 69.3.4.

1. For $X \subseteq \mathbb{P}^3$ a nonsingular quartic surface, $\mathrm{NS}(X)$ is a sublattice of $H^2(X;\mathbb{Z}) \cong \mathbb{Z}^{22}$ of rank $\rho \in [1,20]$, and $\rho = 20$ is attained (for instance by the Fermat quartic).
2. For $E \times E'$, $\mathrm{NS} \cong \mathbb{Z}^2 \oplus \mathrm{Hom}(E,E')$ (Exercise 61.7.1), which by Theorem 69.3.1 is also the lattice of integral classes of type $(1,1)$, a statement about Hodge structures (Example 68.3.6).

Example 69.3.5. Let $X = \mathbb{P}^1 \times \mathbb{P}^1$, which is isomorphic to a nonsingular quadric surface in $\mathbb{P}^3$ (Example 45.5.2), with projections pr_1, pr_2, and let h be the positive generator of $H^2(\mathbb{P}^1;\mathbb{Z})$. By the Künneth formula (Theorem 22.4.2), $H^2(X;\mathbb{Z}) = \mathbb{Z}a \oplus \mathbb{Z}b$ with $a = \mathrm{pr}_1^* h$ and $b = \mathrm{pr}_2^* h$. By naturality of c_1 (Theorem 35.4.2), $a = c_1(\mathrm{pr}_1^* \mathcal{O}(1))$ and $b = c_1(\mathrm{pr}_2^* \mathcal{O}(1))$; these are the classes of the divisors $\{p\} \times \mathbb{P}^1$ and $\mathbb{P}^1 \times \{p\}$, the two rulings of the quadric. So every integral class is algebraic and $\rho(X) = b_2(X) = 2$, without any Hodge theory. Conversely this determines the Hodge numbers: $\rho(X) \le h^{1,1}(X)$ (Corollary 69.2.2) and $h^{2,0} + h^{1,1} + h^{0,2} = b_2 = 2$ with

$h^{2,0} = h^{0,2}$ force $h^{1,1} = 2$ and $h^{2,0} = 0$. The intersection numbers are $a^2 = b^2 = 0$ and $ab = 1$: two lines of the same ruling are disjoint, and two lines of different rulings meet transversally in one point.

Theorem 69.3.6 (Noether–Lefschetz)**.** *For $d \geq 4$, a very general nonsingular surface of degree d in $\mathbb{P}^3$ has* $\mathrm{Pic}(X) = \mathbb{Z} \cdot \mathcal{O}_X(1)$, *so* $\rho(X) = 1$.

This is proved in [Voi03] and [GH85]. The proof shows, with the infinitesimal variation of Hodge structure (Chapter 70), that the locus of surfaces on which a given integral class stays of type $(1,1)$ is a proper subvariety of the parameter space, unless the class is a multiple of the hyperplane class; a very general surface lies outside all these countably many loci. For $d \leq 3$ the statement is false: quadrics contain two families of lines, and cubic surfaces contain 27 lines.

69.4 Why the proof does not generalise

Remark 69.4.1. The proof of Theorem 69.3.1 has three steps, and none generalises to codimension $p \geq 2$.

1. *A cohomological object parametrising cycles.* Divisors modulo linear equivalence are line bundles, $\mathrm{Pic}(X) = H^1(X, \mathcal{O}_X^*)$, the cohomology of a sheaf of abelian groups, and the exponential sequence relates it to $H^2(X;\mathbb{Z})$. For codimension p the Chow group $\mathrm{CH}^p(X) \cong H^p(X, \mathcal{K}_p)$ (Bloch's formula) involves a sheaf of Quillen K-groups, which has no comparison with singular cohomology that is exact in the needed way.

2. *The key lemma.* The obstruction to lifting an integral class is its image in $H^2(X, \mathcal{O}_X) = H^{0,2}$, a coherent cohomology group, and it vanishes exactly for type $(1,1)$. For $p \geq 2$, the analogous vanishing of the components in $H^{p-1,p+1}, \ldots, H^{0,2p}$ has no sheaf-theoretic interpretation producing an object like a line bundle. Deligne cohomology $H^{2p}_{\mathcal{D}}(X;\mathbb{Z}(p))$ fits into an exact sequence $0 \to J^p(X) \to H^{2p}_{\mathcal{D}}(X;\mathbb{Z}(p)) \to \mathrm{Hdg}^p(X;\mathbb{Z}) \to 0$, so every integral Hodge class lifts to Deligne cohomology (Chapter 71); the missing step is to lift further to cycles.

3. *Sections of line bundles.* A line bundle twisted by a high multiple of an ample bundle has sections, whose zero loci are divisors. In codimension ≥ 2 there is no analogue: a vector bundle of rank p with enough sections gives cycles as zero loci of sections, but the Chern classes of such bundles need not span the Hodge classes, and producing a bundle with a prescribed Chern class is the vector bundle form of the conjecture itself (Proposition 68.2.5(3)).

69.5 Exercises

Exercise 69.5.1. Let T be a complex torus of dimension 2 with $\rho(T) = 0$. Show that T carries no nonconstant meromorphic functions and no curves.

Solution. A curve C would give a line bundle $\mathcal{O}(C)$ with $c_1(\mathcal{O}(C)) = \mathrm{cl}(C) \neq 0$, since $\int_T \mathrm{cl}(C) \wedge \omega = \int_C \omega > 0$ for a Kähler form ω (Proposition 47.4.5). So $\mathrm{NS}(T) \neq 0$, a contradiction. A nonconstant meromorphic function f has a nonzero polar divisor, which is a union of curves. □

Exercise 69.5.2. Let S be a nonsingular projective surface with $p_g(S) = h^{2,0}(S) = 0$. Show that every class in $H^2(S;\mathbb{Z})$ is the class of a divisor, and that $\rho(S) = b_2(S)$.

Solution. If $h^{2,0} = h^{0,2} = 0$, then $H^2(S;\mathbb{C}) = H^{1,1}$, so every integral class is an integral Hodge class, hence algebraic by Theorem 69.3.1. So $\mathrm{NS}(S) = H^2(S;\mathbb{Z})$ and $\rho = b_2$. □

Exercise 69.5.3. Show that on a nonsingular cubic threefold $X \subseteq \mathbb{P}^4$ the Hodge conjecture holds, and compute Hdg^1 and Hdg^2.

Solution. By Corollary 69.3.2 the conjecture holds in dimension 3. By the Lefschetz hyperplane theorem and Poincaré duality, $H^2 = \mathbb{Q}h$ and $H^4 = \mathbb{Q}h^2/3$ (the class of a line, since $\int_X h^3 = 3$), both algebraic, so $\mathrm{Hdg}^1 = H^2(X;\mathbb{Q})$ and $\mathrm{Hdg}^2 = H^4(X;\mathbb{Q})$. The interesting group H^3, of dimension 10 with $h^{2,1} = 5$, has no Hodge classes, being of odd degree. □

Chapter 70

Variations of Hodge Structure

70.1 Introduction

In a smooth projective family $f : \mathcal{X} \to S$, the fibres are diffeomorphic (Ehresmann's theorem, Chapter 51), so their cohomology groups form a local system with a flat connection, the Gauss–Manin connection. The Hodge filtrations of the fibres vary holomorphically, and their derivatives move each F^p at most into F^{p-1} (Griffiths transversality). The resulting object, a *variation of Hodge structure*, is recorded by the *period map* to a period domain. Variations are the natural setting for asking how Hodge classes behave in families. The locus where a class stays of type (p,p) is an analytic subvariety, and by the theorem of Cattani, Deligne and Kaplan it is algebraic, as the Hodge conjecture predicts. References: [Voi02b, Chapters 9 and 10], [Voi03], [CMSP03], [Gri68].

Prerequisites. Chapter 66 (Hodge Structures), Chapter 38 (Connections and Curvature) and Chapter 58 (Analytic Spaces and GAGA).

Throughout, $f : \mathcal{X} \to S$ is a *smooth projective family*: a proper holomorphic submersion of complex manifolds with connected base S, whose fibres $X_t = f^{-1}(t)$ are projective (for instance the analytification of a smooth projective morphism of varieties).

70.2 The Gauss–Manin connection

Definition 70.2.1. By Ehresmann's theorem (Theorem 51.2.3), $\mathcal{H}^k_{\mathbb{Z}} = R^k f_* \mathbb{Z}$ is a local system on S with fibres $H^k(X_t;\mathbb{Z})$. The holomorphic vector bundle $\mathcal{H}^k = \mathcal{H}^k_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathcal{O}_S$ carries the flat connection ∇ whose flat sections are the local sections of $\mathcal{H}^k_{\mathbb{C}}$ (Theorem 38.5.10), the *Gauss–Manin connection*. Transport along a loop gives the *monodromy representation* $\pi_1(S,t) \to \mathrm{GL}(H^k(X_t;\mathbb{Z}))$.

Theorem 70.2.2. *In a smooth projective family:*

1. *the Hodge numbers $h^{p,q}(X_t)$ are independent of t;*
2. *the Hodge filtrations $F^pH^k(X_t;\mathbb{C})$ form holomorphic subbundles $\mathcal{F}^p \subseteq \mathcal{H}^k$ (the* Hodge bundles*);*
3. *(*Griffiths transversality*) $\nabla \mathcal{F}^p \subseteq \mathcal{F}^{p-1} \otimes \Omega^1_S$.*

Sketch of proof. (1) The numbers $h^{p,q}(X_t) = \dim H^q(X_t, \Omega^p_{X_t})$ are upper semicontinuous in t: this is Theorem 56.7.1 for an algebraic family, and Kodaira–Spencer's semicontinuity theorem in the analytic case [Voi02b, Chapter 9]. Their sum over $p+q=k$ is the Betti number b_k, constant by Ehresmann's theorem. A sum of upper semicontinuous integer functions

that is constant forces each to be locally constant, hence constant as S is connected. (2) With constant Hodge numbers, the Hodge filtration of X_t can be computed from harmonic forms for a family of Kähler metrics, and depends smoothly on t. Holomorphicity follows from (3) applied in the $(0,1)$-direction: differentiating a family of classes of type $\geq p$ in an antiholomorphic direction does not leave F^p [Voi02b, Chapter 10]. (3) Trivialise the family smoothly (Theorem 51.2.3) and represent a local section of $\mathcal{F}^p$ by closed forms α_t on the fixed manifold X_0 that are of type $\geq p$ for the complex structure J_t. Differentiating in a holomorphic direction v at $t=0$, $\nabla_v[\alpha] = [\partial_v \alpha_t|_{t=0}]$. A form of type $\geq p$ for J_t is, in terms of J_0, a form of type $\geq p$ plus terms of order t of type $\geq p-1$, because the $(1,0)$-forms for J_t differ from those for J_0 by first-order terms of type $(0,1)$. So $\partial_v \alpha_t|_{t=0}$ has type $\geq p-1$. □

Definition 70.2.3. A *polarised variation of Hodge structure* of weight k on a complex manifold S is a local system $\mathcal{H}_\mathbb{Z}$ of finitely generated abelian groups, a flat bilinear form Q, and a filtration of $\mathcal{H} = \mathcal{H}_\mathbb{Z} \otimes \mathcal{O}_S$ by holomorphic subbundles $\mathcal{F}^p$ such that each fibre is a Hodge structure of weight k polarised by Q, and $\nabla \mathcal{F}^p \subseteq \mathcal{F}^{p-1} \otimes \Omega^1_S$. By Theorem 70.2.2 and Theorem 66.5.2, the primitive cohomology of a smooth projective family, with the polarisation given by a relatively ample class, is one.

70.3 Period domains and period maps

Definition 70.3.1. Fix a lattice $H_\mathbb{Z}$, a bilinear form Q with $Q(y,x) = (-1)^k Q(x,y)$, and Hodge numbers $h^{p,q}$. The *period domain* D is the set of filtrations $F^\bullet$ of $H_\mathbb{C}$ with $\dim F^p = \sum_{r \geq p} h^{r,k-r}$ that define Hodge structures of weight k polarised by Q. It is an open subset, in the analytic topology, of the *compact dual* $\check{D} = \{F^\bullet : Q(F^p, F^{k-p+1}) = 0\}$, a closed subvariety of a flag variety. The real group $G = \mathrm{Aut}(H_\mathbb{R}, Q)$ acts transitively on D with compact isotropy groups.

Example 70.3.2.

1. Weight 1, $h^{1,0} = h^{0,1} = g$, Q symplectic: D is the Siegel upper half space $\mathfrak{H}_g$ (Proposition 61.3.4), with $G = \mathrm{Sp}_{2g}(\mathbb{R})$, a Hermitian symmetric domain. For $g = 1$ it is the upper half plane.
2. Weight 2 with $h^{2,0} = 1$, the case of K3 surfaces ($h^{1,1} = 20$): a Hodge structure is determined by the line $F^2 = H^{2,0} = \mathbb{C}\omega$, and $D = \{[\omega] \in \mathbb{P}(H_\mathbb{C}) : Q(\omega,\omega) = 0,\ Q(\omega,\bar{\omega}) > 0\}$, an open subset of a quadric, with $G = \mathrm{SO}(3,19)$. This is a Hermitian symmetric domain of type IV.
3. For weight ≥ 2 with $h^{2,0} \geq 2$, or weight 3 with $h^{3,0} \neq 0$ and $h^{2,1} \neq 0$ (the quintic threefold), D is not Hermitian symmetric, and Griffiths transversality is a genuine restriction: period maps are tangent to a distribution that is not integrable.

Example 70.3.3. Over the upper half plane $\mathfrak{H}_1 = \{\tau : \mathrm{Im}\,\tau > 0\}$ consider the family of elliptic curves $E_\tau = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$, the quotient of $\mathbb{C} \times \mathfrak{H}_1$ by the action of $\mathbb{Z}^2$ given by $(m,n) \cdot (z,\tau) = (z + m + n\tau, \tau)$. The loops a, b given by the segments from 0 to 1 and from 0 to τ form a basis of $H_1(E_\tau;\mathbb{Z})$ that is flat, since the family is trivialised smoothly by $(x + y\tau, \tau) \mapsto (x,y)$, $x,y \in \mathbb{R}/\mathbb{Z}$. The holomorphic form dz has periods $\int_a dz = 1$ and $\int_b dz = \tau$. So in the flat coordinates given by the dual basis of H^1, the line $F^1 = H^{1,0}(E_\tau)$ is spanned by $(1,\tau)$, and the period map $\mathfrak{H}_1 \to \mathfrak{H}_1$ of Example 70.3.2(1) is the identity. This is the *tautological* variation: every polarised variation of weight one and rank two is locally the pullback of it by its period map.

Definition 70.3.4. Given a polarised variation on S and a base point, the *period map* $\mathcal{P} : \tilde{S} \to D$ from the universal cover sends $\tilde{t}$ to the Hodge filtration of the fibre, transported to $H_\mathbb{C} = \mathcal{H}_{t_0}$ by the flat connection. It descends to $S \to \Gamma\backslash D$, where Γ is the image of the monodromy representation.

Proposition 70.3.5. *The period map is holomorphic and horizontal: its differential maps T_tS into $\bigoplus_p \operatorname{Hom}(F^p/F^{p+1}, F^{p-1}/F^p) \subseteq T_{\mathcal{P}(t)}\check{D}$. For a family of projective manifolds its differential at t is the composite of the Kodaira–Spencer map $T_tS \to H^1(X_t, T_{X_t})$ with the contraction and cup product $H^1(X_t, T) \to \bigoplus_p \operatorname{Hom}(H^q(\Omega^p), H^{q+1}(\Omega^{p-1}))$.*

This is Theorem 70.2.2(2), (3) restated, together with Griffiths' computation of the differential [Voi02b, Chapter 10]. The second description is the *infinitesimal variation of Hodge structure*, the main tool in the proof of the Noether–Lefschetz theorem (Theorem 69.3.6).

Example 70.3.6. The *Legendre family* $E_\lambda\colon y^2 = x(x-1)(x-\lambda)$, $\lambda \in \mathbb{C}\setminus\{0,1\}$, is a family of elliptic curves. With the holomorphic form $\omega_\lambda = dx/y$ and a symplectic basis a, b of $H_1(E_\lambda;\mathbb{Z})$ transported along paths, the period map is $\lambda \mapsto \tau(\lambda) = \int_b \omega_\lambda / \int_a \omega_\lambda \in \mathfrak{H}_1$. The periods $\int_\gamma \omega_\lambda$ satisfy the Picard–Fuchs equation $\lambda(1-\lambda)\varpi'' + (1-2\lambda)\varpi' - \frac{1}{4}\varpi = 0$, a differential equation expressing the Gauss–Manin connection. The monodromy is a subgroup of finite index of $\mathrm{SL}_2(\mathbb{Z})$, the congruence subgroup $\Gamma(2)$.

Proposition 70.3.7. *Let $H_\mathbb{Z}$, Q be as in Definition 70.3.1, of weight 2 with $h^{2,0} = 1$. Then for a polarised Hodge structure the filtration is determined by F^2, namely $F^1 = (F^2)^\perp$, and every holomorphic family of such polarised Hodge structures over a complex manifold, with $\mathcal{F}^p$ holomorphic subbundles of a flat bundle $\mathcal{H}$ preserving Q and $H_\mathbb{Z}$, satisfies Griffiths transversality automatically.*

Proof. By condition (2) of Definition 66.5.1, $Q(H^{2,0}, H^{2,0}\oplus H^{1,1}) = 0$, so $F^1 = H^{2,0}\oplus H^{1,1} \subseteq (F^2)^\perp$; both have dimension $\dim H - 1$ (Q is nondegenerate), so they are equal. Transversality asks $\nabla\mathcal{F}^2 \subseteq \mathcal{F}^1 \otimes \Omega^1$ and $\nabla\mathcal{F}^1 \subseteq \mathcal{F}^0 \otimes \Omega^1$; the second is empty since $\mathcal{F}^0 = \mathcal{H}$. For the first, let ω be a local holomorphic generator of $\mathcal{F}^2$. Then $Q(\omega,\omega) = 0$, and since Q is flat, differentiating along a vector field v gives $2Q(\nabla_v\omega, \omega) = 0$, so $\nabla_v\omega \in \omega^\perp = \mathcal{F}^1$. □

Counterexample 70.3.8. Without such a special shape, transversality is a genuine condition, and holomorphic families of Hodge structures need not be variations of Hodge structure. Let $H_\mathbb{Z} = \mathbb{Z}^2$ and consider Hodge structures of weight 2 with $h^{2,0} = h^{0,2} = 1$ and $h^{1,1} = 0$ (not polarised). Such a structure is a line $F^2 = H^{2,0} = \mathbb{C}\omega \subseteq \mathbb{C}^2$ with ω and $\bar\omega$ independent, so the Hodge structures form $D' = \mathbb{P}^1(\mathbb{C})\setminus\mathbb{P}^1(\mathbb{R})$, two copies of the upper half plane. The tautological family over D', with $\mathcal{F}^2 = \mathcal{F}^1$ the tautological line in the trivial flat bundle $D'\times\mathbb{C}^2$, is a holomorphic family of Hodge structures. Transversality would require $\nabla\mathcal{F}^2 \subseteq \mathcal{F}^1 = \mathcal{F}^2$, so that $\mathcal{F}^2$ would be a flat, hence locally constant, subbundle; but the tautological line varies. So this family satisfies all the axioms except polarisation and transversality, and by Theorem 70.2.2(3) it is not the variation of Hodge structure of any smooth projective family.

Corollary 70.3.9. *If the period map of a polarised variation of Hodge structure is constant, its monodromy group Γ is finite. For instance, in the Legendre family (Example 70.3.6) the monodromy is infinite, so the period map is not constant and the curves E_λ are not all isomorphic.*

Proof. Transport by the flat connection identifies the fibres over the universal cover with a fixed $(H_\mathbb{Z}, Q)$, and the period map is equivariant: for $\gamma \in \pi_1(S,t)$ with monodromy $\rho(\gamma)$, $\mathcal{P}(\gamma\tilde t) = \rho(\gamma)\mathcal{P}(\tilde t)$ (Definition 70.3.4). If $\mathcal{P}$ is constant with value $F^\bullet$, then every $\rho(\gamma)$ preserves $F^\bullet$, $H_\mathbb{Z}$ and Q, so Γ lies in the finite group of Proposition 66.5.5. For the Legendre family $\Gamma = \Gamma(2)$ is infinite (it contains $\begin{pmatrix}1&2\\0&1\end{pmatrix}$, of infinite order). If all E_λ were isomorphic, the $\tau(\lambda)$ would all lie in one orbit of $\mathrm{SL}_2(\mathbb{Z})$ (Chapter 61), a countable set; a continuous map from the connected space $\tilde S$ to $\mathfrak{H}_1$ with countable image is constant. □

70.4 Global invariants and the theorem of the fixed part

Theorem 70.4.1 (Theorem of the fixed part)**.** *Let $f : \mathcal{X} \to S$ be a smooth projective morphism of complex varieties, $\bar{\mathcal{X}}$ a smooth compactification of $\mathcal{X}$, and $t \in S$. Then $H^0(S, R^k f_*\mathbb{Q}) = H^k(X_t;\mathbb{Q})^{\pi_1(S,t)}$ is a sub-Hodge structure of $H^k(X_t;\mathbb{Q})$, and it is the image of the restriction map $H^k(\bar{\mathcal{X}};\mathbb{Q}) \to H^k(X_t;\mathbb{Q})$ (the* global invariant cycle theorem*).*

This is [Del71, Théorème 4.1.1 and Corollaire 4.1.2]; the proof uses mixed Hodge theory (Chapter 74). A consequence: a Hodge class on one fibre that is invariant under monodromy is the restriction of a Hodge class on the total space, and so remains a Hodge class on every fibre. Deligne also proved that the monodromy representation on $H^k(X_t;\mathbb{Q})$ is semisimple.

70.5 Hodge loci

Definition 70.5.1. Let $\mathcal{H}$ be a variation of Hodge structure of weight $2p$ on S. For a flat section λ of $\mathcal{H}_{\mathbb{Z}}$ over a simply connected open $U \subseteq S$, the *Hodge locus* of λ is $\{t \in U : \lambda_t \in F^p\mathcal{H}_t\}$, the set where λ is a Hodge class. The *Hodge locus* of the variation is the union, over all local flat sections, of these sets, a subset of S.

Proposition 70.5.2. *The Hodge locus of a flat integral section λ over U is a closed analytic subset of U, locally defined by* $\operatorname{rank}\mathcal{H} - \operatorname{rank}\mathcal{F}^p$ *holomorphic equations, so each component has codimension at most* $\dim \mathcal{H}/\mathcal{F}^p = h^{p-1,p+1} + h^{p-2,p+2} + \cdots + h^{0,2p}$*. For a real class of weight $2p$, lying in F^p is the same as being of type (p,p).*

Proof. The composite $U \to \mathcal{H}|_U \to \mathcal{H}/\mathcal{F}^p$, $t \mapsto$ the image of λ_t, is a holomorphic section of the holomorphic bundle $\mathcal{H}/\mathcal{F}^p$ of rank $\sum_{r<p} h^{r,2p-r} = \sum_{j\geq 1} h^{p-j,p+j}$. Its zero set is the Hodge locus, cut out locally by as many holomorphic functions as the rank. For real λ, $\lambda \in F^p$ implies $\lambda = \bar{\lambda} \in \overline{F^p}$, so $\lambda \in F^p \cap \overline{F^p} = H^{p,p}$. □

Theorem 70.5.3 (Cattani–Deligne–Kaplan)**.** *Let $\mathcal{H}$ be a polarised variation of Hodge structure of weight $2p$ over a nonsingular complex algebraic variety S (for instance from a smooth projective family). Then the Hodge locus of $\mathcal{H}$, restricted to integral classes λ with $Q(\lambda,\lambda)$ bounded, is a finite union of algebraic subvarieties of S, and the whole Hodge locus is a countable union of algebraic subvarieties.*

This is [CDK95]. The Hodge conjecture would imply it: on the Hodge locus of λ, the class is algebraic on each fibre, and the locus where a given family of cycles exists is constructible by the theory of Hilbert schemes, with countably many Hilbert schemes involved. The theorem is proved unconditionally with the several-variable SL_2-orbit theorem describing the asymptotics of the period map near the boundary of S. Recent work of Bakker, Klingler and Tsimerman uses o-minimal geometry to reprove and extend it [BKT20].

Remark 70.5.4. Grothendieck's *variational Hodge conjecture* predicts that if a Hodge class λ stays of type (p,p) on all fibres of a family, and is algebraic on one fibre X_0, then it is algebraic on all fibres. By Theorem 70.4.1 it would follow from the Hodge conjecture applied to the total space. It is known for $p = 1$ and in some other cases, and it remains open in general (Chapter 73).

70.6 Exercises

Exercise 70.6.1. Show that in a smooth projective family of curves the genus is constant, and in a smooth family of K3 surfaces the Hodge numbers of H^2 are $(1, 20, 1)$ on every fibre.

Solution. Both follow from Theorem 70.2.2(1): the genus is $h^{1,0}$. For K3 surfaces it suffices that one fibre is a K3 surface with these numbers, for instance a quartic (Exercise 52.6.1). □

Exercise 70.6.2. Show that Griffiths transversality is automatic for variations of weight one: every holomorphic family of Hodge filtrations $F^1 \subseteq H_{\mathbb{C}}$ is horizontal.

Solution. For weight one the filtration is $0 = F^2 \subseteq F^1 \subseteq F^0 = H_{\mathbb{C}}$, and transversality asks $\nabla F^1 \subseteq F^0 \otimes \Omega^1$, which holds trivially. □

Exercise 70.6.3. In a family of K3 surfaces with a nonconstant period map, show that the Hodge locus of an integral class $\lambda \in H^2$ that is not generically of type $(1,1)$ has codimension at most 1.

Solution. By Proposition 70.5.2, the codimension is at most $\dim \mathcal{H}/\mathcal{F}^1 = h^{0,2} = 1$. Concretely, the locus is where $Q(\lambda, \omega_t) = 0$ for a local holomorphic generator ω_t of F^2, one holomorphic equation. □

Chapter 71

Intermediate Jacobians and Normal Functions

71.1 Introduction

Cycle classes forget much of the structure of Chow groups: a cycle whose class vanishes can still be very far from zero. Griffiths' intermediate Jacobian $J^p(X)$, a complex torus built from the odd-degree Hodge structure $H^{2p-1}(X)$, receives the *Abel–Jacobi map* from homologically trivial cycles of codimension p. For $p = 1$ this is the Picard variety and the classical Abel–Jacobi map. Deligne cohomology packages the cycle class and the Abel–Jacobi invariant into one group. Families of intermediate Jacobians and their *normal functions* give the only serious analytic method for constructing cycles; it reproves the Lefschetz $(1,1)$-theorem and proves the Hodge conjecture for cubic fourfolds. References: [Voi02b, Chapter 12], [Voi03], [Gri68], [CMSP03].

Prerequisites. Chapter 70 (Variations of Hodge Structure), Chapter 61 (Abelian Varieties) and Chapter 63 (The Cycle Class Map).

Throughout, X is a compact Kähler manifold of dimension n, usually projective, and $1 \le p \le n$.

71.2 Intermediate Jacobians

Definition 71.2.1. The *p-th intermediate Jacobian* of X is

$$J^p(X) = \frac{H^{2p-1}(X;\mathbb{C})}{F^p H^{2p-1}(X;\mathbb{C}) + H^{2p-1}(X;\mathbb{Z})},$$

where $H^{2p-1}(X;\mathbb{Z})$ means its image in $H^{2p-1}(X;\mathbb{C})$.

Proposition 71.2.2. *$J^p(X)$ is a compact complex torus of dimension $\tfrac{1}{2}b_{2p-1}(X)$. Via Poincaré duality,*

$$J^p(X) \cong \frac{F^{n-p+1}H^{2n-2p+1}(X;\mathbb{C})^\vee}{H_{2n-2p+1}(X;\mathbb{Z})},$$

where a homology class acts by integration. $J^1(X) = \mathrm{Pic}^0(X)$, and $J^n(X)$ is the Albanese torus $\mathrm{Alb}(X) = H^0(X,\Omega^1)^\vee / H_1(X;\mathbb{Z})$.

Proof. H^{2p-1} has odd weight $2p-1$, so $H_{\mathbb{C}} = F^p \oplus \overline{F^{(2p-1)-p+1}} = F^p \oplus \overline{F^p}$ (Proposition 66.2.2). Hence the real subspace $H^{2p-1}(X;\mathbb{R})$ meets F^p trivially and maps isomorphically onto $H_{\mathbb{C}}/F^p$ (dimensions: $b_{2p-1} = 2\dim_{\mathbb{C}} H_{\mathbb{C}}/F^p$). So the image of $H^{2p-1}(X;\mathbb{Z})$ modulo torsion is a lattice

in the complex vector space $H_{\mathbb{C}}/F^p$, of complex dimension $\frac{1}{2}b_{2p-1}$. For the second description, the Poincaré pairing $H^{2p-1}\times H^{2n-2p+1}\to\mathbb{C}$ pairs F^p with F^{n-p+1} to zero (types add up to more than n in one index) and is perfect, so $H_{\mathbb{C}}/F^p\cong(F^{n-p+1}H^{2n-2p+1})^\vee$, and $H^{2p-1}(X;\mathbb{Z})\cong H_{2n-2p+1}(X;\mathbb{Z})$ modulo torsion. For $p=1$, $H^1(X;\mathbb{C})/F^1=H^{0,1}=H^1(X,\mathcal{O})$, giving $\mathrm{Pic}^0(X)$ (Corollary 69.2.2); for $p=n$ the second description gives $H^0(X,\Omega^1)^\vee/H_1(X;\mathbb{Z})$. □

Proposition 71.2.3. *A holomorphic map $f\colon X\to Y$ of compact Kähler manifolds induces holomorphic group homomorphisms $f^*\colon J^p(Y)\to J^p(X)$, with $(g\circ f)^*=f^*\circ g^*$ and $\mathrm{id}^*=\mathrm{id}$. If $H^{2p-1}(X;\mathbb{Q})=0$, then $J^p(X)=0$.*

Proof. The pullback $f^*\colon H^{2p-1}(Y;\mathbb{C})\to H^{2p-1}(X;\mathbb{C})$ maps $H^{r,s}(Y)$ into $H^{r,s}(X)$ (Proposition 48.5.7), hence $F^pH^{2p-1}(Y)$ into $F^pH^{2p-1}(X)$, and it maps the image of $H^{2p-1}(Y;\mathbb{Z})$ into the image of $H^{2p-1}(X;\mathbb{Z})$, being induced by the pullback of integral cohomology. So it induces a $\mathbb{C}$-linear map $H^{2p-1}(Y;\mathbb{C})/F^p\to H^{2p-1}(X;\mathbb{C})/F^p$ carrying lattice into lattice, hence a holomorphic homomorphism of the quotient tori; functoriality holds already on cohomology. The last statement is the dimension formula of Proposition 71.2.2, since $b_{2p-1}(X)=\dim_{\mathbb{Q}}H^{2p-1}(X;\mathbb{Q})$. □

Example 71.2.4. Let $T=\mathbb{C}^g/\Lambda$ be a complex torus and $x_0=0$. The holomorphic 1-forms on T are the constant forms $\sum_j a_j\,dz_j$ (Example 46.4.2 and Lemma 46.3.2), so $H^0(T,\Omega^1)^\vee\cong\mathbb{C}^g$, the functional $\omega\mapsto\int\omega$ being determined by its values on $dz_1,\dots,dz_g$. For a path γ in T from 0 to x, lifted to a path in $\mathbb{C}^g$ from 0 to $\tilde{x}$, $\int_\gamma dz_j=\tilde{x}_j$; for a closed loop the lift ends at a point of Λ, and every $\lambda\in\Lambda$ arises from the loop $t\mapsto t\lambda$. So the period lattice is Λ, $\mathrm{Alb}(T)=\mathbb{C}^g/\Lambda=T$, and the Albanese map $x\mapsto\mathrm{AJ}(x-0)$ of Exercise 71.7.2 is the identity of T. Likewise $J^1(T)=\mathrm{Pic}^0(T)=H^{0,1}(T)/H^1(T;\mathbb{Z})$, where $H^{0,1}(T)$ is spanned by the forms $d\bar{z}_j$ and $H^1(T;\mathbb{Z})=\mathrm{Hom}(\Lambda,\mathbb{Z})$ embeds by sending an $\mathbb{R}$-linear form on $\mathbb{C}^g$ with integer values on Λ to its $(0,1)$-part; this is the *dual torus* of T. For $g=1$ it is again isomorphic to T (Chapter 59).

Remark 71.2.5. The complex structure of $J^p(X)$ is the one for which F^p is killed; it varies holomorphically with X by Griffiths transversality. But $J^p(X)$ is in general not an abelian variety: the Hodge–Riemann form on H^{2p-1} is definite on each $H^{r,2p-1-r}$ with alternating signs, so it gives a positive definite form on $H_{\mathbb{C}}/F^p$ only when the Hodge structure has *level one*, $H^{2p-1}=H^{p,p-1}\oplus H^{p-1,p}$. Weil's intermediate Jacobian uses a different complex structure that makes it polarised, but it does not vary holomorphically. When H^{2p-1} has level one, both agree and $J^p(X)$ is a principally polarised abelian variety, for instance for curves, and for $p=2$ on the cubic threefold (Example 71.6.4).

71.3 The Abel–Jacobi map

Definition 71.3.1. Let X be projective and $Z\in Z^p(X)$ a cycle with $\mathrm{cl}(Z)=0$ in $H^{2p}(X;\mathbb{Z})$. Then $Z=\partial\Gamma$ for a real $(2n-2p+1)$-chain Γ (the support of Z has real dimension $2n-2p$, and its fundamental class vanishes in $H_{2n-2p}(X;\mathbb{Z})$). The *Abel–Jacobi invariant* is

$$\mathrm{AJ}(Z)=\left(\omega\mapsto\int_\Gamma\omega\right)\in\frac{F^{n-p+1}H^{2n-2p+1}(X;\mathbb{C})^\vee}{H_{2n-2p+1}(X;\mathbb{Z})}=J^p(X),$$

for ω closed forms of type $\geq n-p+1$ representing classes in F^{n-p+1}.

Proposition 71.3.2. $\mathrm{AJ}(Z)$ *is well defined, and* $\mathrm{AJ}\colon Z^p(X)_{\mathrm{hom}}\to J^p(X)$ *is a homomorphism. It factors through rational equivalence, giving* $\mathrm{AJ}\colon \mathrm{CH}^p(X)_{\mathrm{hom}}\to J^p(X)$. *For $p=1$, and for curves, it is the Abel–Jacobi map of Theorem 59.6.3.*

Sketch of proof. If Γ and Γ' both bound Z, then $\Gamma - \Gamma'$ is a cycle and $\int_{\Gamma-\Gamma'} \omega$ is a period, an element of the image of $H_{2n-2p+1}(X;\mathbb{Z})$. The integral does not depend on the representative ω of the class: if $\omega = d\eta$ with ω of type $\geq n-p+1$, one can choose η of type $\geq n-p+1$, because the degeneration of the Hodge-to-de Rham spectral sequence (Corollary 48.5.3) makes d strict for the filtration by type [Voi02b, Chapter 8]; and then $\int_\Gamma d\eta = \int_Z \eta = 0$ because Z has complex dimension $n-p$ and η has more than $n-p$ holomorphic indices. For rational equivalence one shows that $t \mapsto \mathrm{AJ}(Z_t - Z_0)$ is a holomorphic map from $\mathbb{P}^1$ to the torus $J^p(X)$, hence constant, as in the proof of Theorem 59.6.3 [Voi02b, Chapter 12]. □

Remark 71.3.3. For curves, the Abel–Jacobi map $\mathrm{CH}^1(C)_{\mathrm{hom}} \to J^1(C)$ is an isomorphism (Theorem 59.6.3). In higher dimension its kernel can be enormous. For 0-cycles on a surface S with $h^{2,0}(S) > 0$, Mumford proved that the maps $\mathrm{Sym}^m S \times \mathrm{Sym}^m S \to \mathrm{CH}_0(S)_{\mathrm{hom}}$, $(a,b) \mapsto a-b$, are not surjective for any m [Mum68]: the group $\mathrm{CH}_0(S)_{\mathrm{hom}}$ is "infinite-dimensional", while its image under the Albanese map $\mathrm{CH}_0(S)_{\mathrm{hom}} \to \mathrm{Alb}(S)$ is a torus of finite dimension. On the other side, Griffiths' theorem (Section 71.5) shows that AJ can detect cycles which are homologically, but not algebraically, trivial.

71.4 Deligne cohomology

Definition 71.4.1. The *Deligne complex* is $\mathbb{Z}(p)_{\mathcal{D}} = \left(\mathbb{Z}(p) \to \mathcal{O}_X \to \Omega^1_X \to \cdots \to \Omega^{p-1}_X\right)$, with $\mathbb{Z}(p) = (2\pi i)^p\mathbb{Z}$ in degree 0, and *Deligne cohomology* is its hypercohomology $H^k_{\mathcal{D}}(X;\mathbb{Z}(p))$. For $p=1$, $\mathbb{Z}(1)_{\mathcal{D}} \simeq \mathcal{O}^*_X[-1]$ by the exponential sequence, so $H^2_{\mathcal{D}}(X;\mathbb{Z}(1)) = \mathrm{Pic}(X)$.

Proposition 71.4.2. *For X compact Kähler there is an exact sequence*

$$0 \to J^p(X) \to H^{2p}_{\mathcal{D}}(X;\mathbb{Z}(p)) \to \mathrm{Hdg}^p(X;\mathbb{Z}(p)) \to 0.$$

For $p=1$ it is $0 \to \mathrm{Pic}^0(X) \to \mathrm{Pic}(X) \to \mathrm{NS}(X) \to 0$.

Proof. The Deligne complex is the cone, shifted, of $\mathbb{Z}(p) \to \Omega^{<p}_X$. So there is a long exact sequence

$$H^{2p-1}(X;\mathbb{Z}(p)) \to \mathbb{H}^{2p-1}(X,\Omega^{<p}) \to H^{2p}_{\mathcal{D}}(X;\mathbb{Z}(p)) \to H^{2p}(X;\mathbb{Z}(p)) \to \mathbb{H}^{2p}(X,\Omega^{<p}).$$

By the holomorphic Poincaré lemma and the degeneration of the Hodge-to-de Rham spectral sequence (Corollary 48.5.3), $\mathbb{H}^k(X,\Omega^{<p}) = H^k(X;\mathbb{C})/F^pH^k(X;\mathbb{C})$. So the kernel of the last map consists of the integral classes whose image in $H^{2p}(X;\mathbb{C})$ lies in F^p; a real class in F^p of weight $2p$ lies in $F^p \cap \overline{F^p} = H^{p,p}$, so these are the integral Hodge classes. The cokernel of the first map is $J^p(X)$ (the twist by $(2\pi i)^p$ only rescales the lattice). □

Example 71.4.3.

1. For $p=0$ the Deligne complex is $\mathbb{Z}$ in degree 0, so $H^k_{\mathcal{D}}(X;\mathbb{Z}(0)) = H^k(X;\mathbb{Z})$.
2. For $p=1$, $\mathbb{Z}(1)_{\mathcal{D}} \simeq \mathcal{O}^*_X[-1]$ gives $H^k_{\mathcal{D}}(X;\mathbb{Z}(1)) = H^{k-1}(X,\mathcal{O}^*_X)$: for compact connected X, $H^1_{\mathcal{D}}(X;\mathbb{Z}(1)) = \mathbb{C}^*$ (the nonzero constants, by Proposition 44.2.3) and $H^2_{\mathcal{D}}(X;\mathbb{Z}(1)) = \mathrm{Pic}(X)$.
3. For X a point and $p \geq 1$, the complex is $\mathbb{Z}(p) \to \mathbb{C}$ in degrees 0, 1, so $H^1_{\mathcal{D}}(\mathrm{pt};\mathbb{Z}(p)) = \mathbb{C}/(2\pi i)^p\mathbb{Z}$ and the other groups vanish. For $p=1$ this is $\mathbb{C}/2\pi i\mathbb{Z} \cong \mathbb{C}^*$ via exp, in accordance with (2).

Deligne cohomology is thus a mixture of integral cohomology and of the holomorphic data $\mathbb{C}/F^p$; it is not a topological invariant, and for $p \geq 1$ its groups are in general neither finitely generated nor vector spaces.

Remark 71.4.4. There is a *Deligne cycle class* $\mathrm{cl}_{\mathcal{D}}\colon \mathrm{CH}^p(X) \to H^{2p}_{\mathcal{D}}(X;\mathbb{Z}(p))$ lifting cl and restricting to AJ on homologically trivial cycles [Voi02b, Chapter 12]. In these terms the integral Hodge conjecture asks whether $\mathrm{cl}_{\mathcal{D}}$ followed by the projection to $\mathrm{Hdg}^p(X;\mathbb{Z}(p))$ is surjective. For $p = 1$ both maps are isomorphisms, $\mathrm{CH}^1(X) = \mathrm{Pic}(X) = H^2_{\mathcal{D}}(X;\mathbb{Z}(1))$; for $p \geq 2$, $\mathrm{cl}_{\mathcal{D}}$ is neither injective nor surjective in general.

71.5 Griffiths' theorem

Theorem 71.5.1 (Griffiths)**.** *Let $X \subseteq \mathbb{P}^4$ be a general quintic threefold. There are homologically trivial 1-cycles on X, differences of lines, that are not algebraically equivalent to zero, even after multiplication by any nonzero integer. So the Griffiths group $\mathrm{Griff}^2(X) = Z^2(X)_{\mathrm{hom}}/Z^2(X)_{\mathrm{alg}}$ is not torsion.*

This is [Gri69]. The argument: algebraically trivial cycles map under AJ to an abelian subvariety of $J^2(X)$ whose tangent space lies in $H^{1,2}$, which for a quintic threefold is forced to be zero by the variation of Hodge structure. But the Abel–Jacobi images of differences of lines, deformed in the family of quintics, have a nonzero infinitesimal invariant, so they are not torsion. Clemens later showed that $\mathrm{Griff}^2(X) \otimes \mathbb{Q}$ is infinite-dimensional for the general quintic [Cle83].

71.6 Normal functions

Definition 71.6.1. Let $f : \mathcal{X} \to S$ be a smooth projective family. The intermediate Jacobians of the fibres form a family of complex tori $\mathcal{J}^p \to S$, the quotient of the holomorphic bundle $\mathcal{H}^{2p-1}/\mathcal{F}^p$ by the local system $R^{2p-1}f_*\mathbb{Z}$. A *normal function* is a holomorphic section ν of $\mathcal{J}^p$ that is *horizontal*: its local lifts $\tilde{\nu}$ to $\mathcal{H}^{2p-1}$ satisfy $\nabla\tilde{\nu} \in \mathcal{F}^{p-1} \otimes \Omega^1_S$. A family of homologically trivial cycles Z_t gives the normal function $t \mapsto \mathrm{AJ}(Z_t)$.

Remark 71.6.2 (Poincaré and Lefschetz). Lefschetz's original proof of the $(1,1)$-theorem for a surface S used normal functions. Take a Lefschetz pencil of curves $C_t \subseteq S$. An integral class λ of type $(1,1)$ restricts to classes on the C_t that, after subtracting a multiple of the hyperplane class, have degree 0, and a topological construction produces from λ a normal function ν with $\nu(t) \in J(C_t) = \mathrm{Pic}^0(C_t)$. By Jacobi inversion (Theorem 59.6.3), $\nu(t)$ is the class of a divisor of degree 0 on C_t, and these divisors sweep out a divisor on S whose class is λ up to multiples of the hyperplane class (*Poincaré's existence theorem*). The same strategy for $p = 2$ fails because the intermediate Jacobians of the hyperplane sections are not generated by cycles. For cubic fourfolds, where the relevant J^2 of the cubic threefold hyperplane sections is covered by Abel–Jacobi images of lines, Zucker carried it through [Zuc77].

Remark 71.6.3. Green and Griffiths showed that the Hodge conjecture for a primitive Hodge class λ of degree $2p$ on X is equivalent to the existence of singularities, over the discriminant locus of a sufficiently ample linear system, of the normal function attached to λ [GG07]. This converts the conjecture into a problem about the boundary behaviour of normal functions, studied by Brosnan, Pearlstein, Schnell and others.

Example 71.6.4. For a nonsingular cubic threefold $X \subseteq \mathbb{P}^4$, $H^3(X)$ has Hodge numbers $h^{3,0} = 0$, $h^{2,1} = 5$, so it has level one and $J^2(X)$ is a principally polarised abelian variety of dimension 5. The Abel–Jacobi map from the Fano surface of lines on X to $J^2(X)$ identifies $J^2(X)$ with the Albanese variety of the Fano surface. Clemens and Griffiths showed that $J^2(X)$ is not a product of Jacobians of curves, and deduced that X is not rational, although it is unirational [CG72b].

71.7 Exercises

Exercise 71.7.1. Compute $\dim J^p(X)$ for $X = \mathbb{P}^n$, for an abelian variety of dimension g, and for a quintic threefold with $p = 2$.

Solution. $\mathbb{P}^n$ has no odd cohomology, so all $J^p = 0$. For an abelian variety, $b_{2p-1} = \binom{2g}{2p-1}$, so $\dim J^p = \frac{1}{2}\binom{2g}{2p-1}$. For the quintic, $b_3 = 204$ (Proposition 52.4.4), so $\dim J^2 = 102$. □

Exercise 71.7.2. Show that for $p = n$ the Abel–Jacobi map on 0-cycles of degree zero is $\sum n_i x_i \mapsto (\omega \mapsto \sum_i n_i \int_{x_0}^{x_i} \omega)$, and that it gives the *Albanese map* $X \to \mathrm{Alb}(X)$, $x \mapsto \mathrm{AJ}(x - x_0)$.

Solution. For $Z = \sum n_i(x_i - x_0)$ with $\sum n_i = 0$, a chain Γ with boundary Z is a sum of paths from x_0 to x_i with multiplicities n_i, and $F^1H^1 = H^{1,0}$ is spanned by holomorphic 1-forms (Proposition 48.5.6). The formula follows from Definition 71.3.1 with $p = n$. □

Chapter 72

Known Cases and Evidence

72.1 Introduction

This chapter collects the cases in which the Hodge conjecture is known, and explains the mechanism in each. They fall into three groups. First, cases where every Hodge class is forced to be a combination of known algebraic classes: small dimension, cellular varieties, varieties whose Hodge classes are generated by divisors, and hypersurfaces outside the middle degree. Second, abelian varieties, where the Mumford–Tate group (Chapter 67) makes the Hodge classes computable; there the difficulty concentrates in the *Weil classes*. Third, geometric methods: the decomposition of the diagonal for varieties with small Chow groups, normal functions for cubic fourfolds, and the Kuga–Satake construction for K3 surfaces. We state precisely what is proved, with proofs of the elementary parts and references for the rest. References: the source *The Hodge Conjecture* (Chapter 6), [Lew99], [BL04], [Voi03].
Prerequisites. Chapter 69 (The Lefschetz (1,1) Theorem), Chapter 67 (Mumford–Tate Groups), Chapter 70 (Variations of Hodge Structure) and Chapter 71 (Intermediate Jacobians and Normal Functions).

Throughout, X is a nonsingular complex projective variety of dimension n.

72.2 Cases from general theory

Definition 72.2.1. The *Lefschetz ring* of X is the $\mathbb{Q}$-subalgebra $\mathrm{Lef}^\bullet(X) \subseteq \bigoplus_p H^{2p}(X;\mathbb{Q})$ generated by $\mathrm{Hdg}^1(X)$, the rational classes of divisors. Its elements are *Lefschetz classes*.

Proposition 72.2.2. *Lefschetz classes are algebraic. Hence* $\mathrm{HC}(X)$ *holds whenever* $\mathrm{Hdg}^\bullet(X) = \mathrm{Lef}^\bullet(X)$.

Proof. Divisor classes are algebraic by the Lefschetz $(1,1)$-theorem (Theorem 69.3.1), and products of algebraic classes are algebraic (Theorem 63.3.1(1)). □

Theorem 72.2.3. *The Hodge conjecture holds in the following cases:*

1. $p \in \{0, 1, n-1, n\}$, *and hence for* $\dim X \le 3$ *(Corollary 69.3.2);*
2. *cellular varieties, such as flag varieties and toric varieties (Example 68.3.5);*
3. *very general abelian varieties and all their powers (Theorem 67.4.3);*
4. *smooth complete intersections, in all degrees except the middle one (Proposition 72.4.1);*
5. *uniruled fourfolds [CM78] and, more generally, varieties whose group of* 0*-cycles is supported on a subvariety of dimension* ≤ 3*, in codimension* 2 *[BS83];*

6. *cubic fourfolds [Zuc77];*

7. *Fermat hypersurfaces of many degrees and dimensions [Shi79].*

Items (1)–(4) are proved in this book. For (5)–(7) see the references and Sections 72.5 and 72.4.

Proposition 72.2.4. *Let X and Y be nonsingular projective varieties, and suppose that $H^\bullet(Y;\mathbb{Q})$ is spanned by classes of algebraic cycles, for instance Y cellular (Example 68.3.5). Then* $\mathrm{HC}(X\times Y)$ *holds if and only if* $\mathrm{HC}(X)$ *holds.*

Proof. Choose a basis $y_j = \mathrm{cl}(W_j) \in H^{2k_j}(Y;\mathbb{Q})$ of algebraic classes; they are of type (k_j,k_j) (Corollary 63.4.2), and $H^\bullet(Y;\mathbb{Q})$ has no odd part. By the Künneth formula (Theorem 22.4.2), every $\beta \in H^{2p}(X\times Y;\mathbb{Q})$ is uniquely $\beta = \sum_j \mathrm{pr}_1^*\alpha_j \cup \mathrm{pr}_2^* y_j$ with $\alpha_j \in H^{2p-2k_j}(X;\mathbb{Q})$, and the same holds over $\mathbb{C}$. Since pullbacks and cup products preserve types (Proposition 48.5.7), the component of type (r,s) of α_j contributes a class of type $(r+k_j, s+k_j)$. If β is of type (p,p), comparing components of type (p,p) and using the uniqueness of the decomposition shows that each α_j is of type $(p-k_j,p-k_j)$, hence a Hodge class. If $\mathrm{HC}(X)$ holds, $\alpha_j = \mathrm{cl}(Z_j)$ for a rational cycle Z_j, and $\beta = \sum_j \mathrm{cl}(Z_j\times W_j)$ is algebraic (Theorem 63.3.1(1) and (2)).

Conversely, let $\alpha \in \mathrm{Hdg}^p(X)$ and let $y_0 = \mathrm{cl}(\mathrm{pt}) \in H^{2\dim Y}(Y;\mathbb{Q})$. The class $\mathrm{pr}_1^*\alpha \cup \mathrm{pr}_2^* y_0$ is a Hodge class on $X\times Y$; it is the pullback of α to $X\times\{\mathrm{pt}\}$ pushed forward to $X\times Y$, and if it is algebraic, so is its pushforward $\mathrm{pr}_{1*}(\mathrm{pr}_1^*\alpha \cup \mathrm{pr}_2^* y_0) = \alpha \cup \mathrm{pr}_{1*}\mathrm{pr}_2^* y_0 = \alpha$, by the projection formula and $\mathrm{pr}_{1*}\mathrm{pr}_2^* y_0 = 1$. □

Example 72.2.5. The Hodge conjecture holds for $S\times\mathbb{P}^m$, $S\times\mathrm{Gr}(k,n)$ and $S\times F$ with F a flag variety, for every nonsingular projective surface or threefold S, although these products can have arbitrary dimension: by Proposition 72.2.4 and Theorem 72.2.3(1),(2). The same argument shows that the Hodge classes of $X\times Y$ are all products of Hodge classes of X with classes of Y in this situation; the interesting Hodge classes on products come from factors with odd cohomology, as for $E\times E'$ (Example 68.3.6).

72.3 Abelian varieties

Remark 72.3.1. For an abelian variety A, the Hodge classes on all powers A^r are the invariants of the Hodge group $\mathrm{Hg} = \mathrm{Hg}(H^1(A))$ (Corollary 67.3.3), and the divisor classes on A^r are the invariants of the *Lefschetz group* $\mathrm{Lef}(A)$, the connected centraliser of $\mathrm{End}(A)\otimes\mathbb{Q}$ in the symplectic group of a polarisation [Mur84]. Always $\mathrm{Hg}\subseteq\mathrm{Lef}(A)$. If equality holds, the Hodge ring of every power of A is generated by divisors, and the Hodge conjecture holds for all powers of A.

Theorem 72.3.2. $\mathrm{Hg}(A) = \mathrm{Lef}(A)$*, and hence the Hodge conjecture holds for all powers of A, in each of the following cases: A is a product of elliptic curves [Ima76], [Mur84]; A is simple of prime dimension (Tankeev) [Tan82]; A is of dimension at most* 5 *and has no factor of Weil type (Moonen–Zarhin) [MZ99]. More generally the Hodge conjecture holds for A itself (not necessarily its powers) whenever* $\mathrm{Hdg}^\bullet(A) = \mathrm{Lef}^\bullet(A)$*, which holds for A of dimension* ≤ 3 *by Theorem 72.2.3(1).*

Proposition 72.3.3 (Weil classes)**.** *Let A be a complex abelian fourfold and $K\subseteq\mathrm{End}(A)\otimes\mathbb{Q}$ an imaginary quadratic field such that, for the two embeddings $\sigma,\bar\sigma\colon K\to\mathbb{C}$, the eigenspaces of K on $H^{1,0}(A)$ both have dimension* 2 (Weil type)*. Then $H^1(A;\mathbb{Q})$ is a* 4*-dimensional K-vector space. Let $V_\sigma, V_{\bar\sigma}\subseteq H^1(A;\mathbb{C})$ be the eigenspaces of K. There is a* 2*-dimensional $\mathbb{Q}$-subspace $W_K\subseteq\Lambda^4H^1(A;\mathbb{Q}) = H^4(A;\mathbb{Q})$ with $W_K\otimes\mathbb{C} = \Lambda^4V_\sigma\oplus\Lambda^4V_{\bar\sigma}$, and it consists of Hodge classes, the* Weil classes*. For a general abelian fourfold of Weil type,* $\mathrm{Hdg}^2(A) = \mathrm{Lef}^2(A)\oplus W_K$ *with* $W_K\cap\mathrm{Lef}^2(A) = 0$*, so the Weil classes are not combinations of intersections of divisors.*

Proof. K acts on $H_{\mathbb{C}} = H^1(A;\mathbb{C})$ with eigenspaces V_σ, $V_{\bar\sigma}$, each of dimension 4, and K acts by Hodge endomorphisms, so $V_\sigma = V_\sigma^{1,0} \oplus V_\sigma^{0,1}$. By hypothesis $\dim V_\sigma^{1,0} = 2$, and then $\dim V_\sigma^{0,1} = 2$. The space $\Lambda^4_K H^1(A;\mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{C}$ decomposes as $\Lambda^4 V_\sigma \oplus \Lambda^4 V_{\bar\sigma}$, and $\Lambda^4 V_\sigma = \Lambda^2 V_\sigma^{1,0} \otimes \Lambda^2 V_\sigma^{0,1}$ is of type $(2,2)$, and likewise for $\bar\sigma$. The subspace $\Lambda^4 V_\sigma \oplus \Lambda^4 V_{\bar\sigma}$ of $\Lambda^4 H^1(A;\mathbb{C})$ is defined over K, since the eigenspace decomposition is, and it is stable under the nontrivial automorphism of K, which exchanges V_σ and $V_{\bar\sigma}$. By Galois descent it is the complexification of a $\mathbb{Q}$-subspace W_K, of dimension 2, whose elements are rational classes of type $(2,2)$. That $W_K \cap \mathrm{Lef}^2 = 0$ for general A of Weil type, where $\mathrm{End}(A) \otimes \mathbb{Q} = K$ and the Lefschetz group is a unitary group $\mathrm{U}(2,2)$ while $\mathrm{Hg} = \mathrm{SU}(2,2)$, is [Wei79]; see also [BL04]. □

Remark 72.3.4. Weil classes are the essential difficulty for abelian varieties: they are the first Hodge classes not generated by divisors, and Moonen and Zarhin showed that in dimension ≤ 5 all exceptional Hodge classes come from Weil classes. The Hodge conjecture for Weil classes was proved by Schoen for $K = \mathbb{Q}(\sqrt{-3})$ and $K = \mathbb{Q}(i)$ in special cases [Sch88], by van Geemen in further cases, and by Markman for all abelian fourfolds of Weil type of discriminant 1 [Mar23]. Weil classes are absolute Hodge classes by Deligne's theorem (Chapter 75).

72.4 Hypersurfaces and complete intersections

Proposition 72.4.1. *Let $X \subseteq \mathbb{P}^{n+1}$ be a nonsingular hypersurface, or more generally a nonsingular complete intersection of dimension n in some $\mathbb{P}^N$, and h the hyperplane class. Then for $2p \neq n$, $H^{2p}(X;\mathbb{Q}) = \mathbb{Q}h^p$. Hence* $\mathrm{HC}(X,p)$ *holds for $2p \neq n$.*

Proof. By the Lefschetz hyperplane theorem (Theorem 50.5.1), applied repeatedly for complete intersections, $H^k(X;\mathbb{Q}) \cong H^k(\mathbb{P}^N;\mathbb{Q})$ for $k < n$, which is $\mathbb{Q}h^{k/2}$ for k even. For $k > n$, Poincaré duality gives $b_k = b_{2n-k}$, which is 1 for k even, and $h^{k/2} \neq 0$ by Proposition 47.4.5, so $H^k = \mathbb{Q}h^{k/2}$. Powers of h are algebraic, being classes of complete intersections of hyperplane sections. □

Remark 72.4.2. In the middle degree $n = 2p$, a very general hypersurface of degree d has $\mathrm{Hdg}^p(X) = \mathbb{Q}h^p$ as soon as H^n is not of Tate type up to h^p, for instance for surfaces with $d \geq 4$ (Theorem 69.3.6) and for fourfolds with $d \geq 3$, so the conjecture holds there trivially. Special hypersurfaces are the interesting cases. For cubic fourfolds, Zucker proved the conjecture with normal functions (Remark 71.6.2), and Voisin proved the integral version [Voi06]. For Fermat hypersurfaces $x_0^d + \cdots + x_{n+1}^d = 0$, the Hodge classes are computable via the action of $(\mu_d)^{n+2}$ and characters (Jacobi sums), and Shioda proved the conjecture for many (d,n), including d prime and $d = 4$ [Shi79].

72.5 Uniruled varieties and the decomposition of the diagonal

Theorem 72.5.1 (Bloch–Srinivas)**.** *Suppose that $\mathrm{CH}_0(X)_{\mathbb{Q}}$ is supported on a closed subvariety of dimension ≤ 3, that is, there is $W \subseteq X$ of dimension ≤ 3 with $\mathrm{CH}_0(W)_{\mathbb{Q}} \to \mathrm{CH}_0(X)_{\mathbb{Q}}$ surjective. Then* $\mathrm{HC}(X,2)$ *holds. This applies to rationally connected varieties, for which $\mathrm{CH}_0(X)_{\mathbb{Q}} = \mathbb{Q}$.*

This is [BS83]. The method is the *decomposition of the diagonal*: the hypothesis gives an equality $N\Delta_X = Z_1 + Z_2$ in $\mathrm{CH}^n(X \times X)$ with Z_1 supported on $D \times X$ for a divisor D and Z_2 supported on $X \times W$. Letting both sides act on $H^4(X;\mathbb{Q})$ as correspondences, the identity (times N) factors through the cohomology of a resolution of D, where Hodge classes of degree 2 on a variety of lower dimension are handled by induction and the $(1,1)$-theorem,

plus a contribution through W, of dimension ≤ 3, where the conjecture is known. The same argument gives the conjecture for uniruled fourfolds, first proved by Conte and Murre [CM78].

72.6 K3 surfaces, hyper-Kähler varieties and Kuga–Satake

Remark 72.6.1. For a single K3 surface the conjecture holds, by dimension. The interesting questions are on products. *Mukai's conjecture*, proved by Buskin [Bus19], says that every rational Hodge isometry $H^2(S;\mathbb{Q}) \to H^2(S';\mathbb{Q})$ between K3 surfaces, a Hodge class in $H^4(S \times S';\mathbb{Q})$, is algebraic. Markman extended this to hyper-Kähler varieties of $K3^{[n]}$-type.

Remark 72.6.2 (Kuga–Satake). Kuga and Satake attached to a polarised K3 surface S an abelian variety $\mathrm{KS}(S)$, of dimension 2^{19}, built from the even Clifford algebra of the lattice of primitive classes, together with an embedding of Hodge structures $H^2_{\mathrm{prim}}(S;\mathbb{Q}) \hookrightarrow H^1(\mathrm{KS}(S)) \otimes H^1(\mathrm{KS}(S))$ [KS67]. The Hodge conjecture predicts that this embedding is induced by an algebraic correspondence between S and $\mathrm{KS}(S)^2$ (the *Kuga–Satake Hodge conjecture*). It is known for some families, for instance by Paranjape for double planes branched along six lines [Par88]. Deligne used the Kuga–Satake construction, which works in families and in characteristic p, to prove the Weil conjectures for K3 surfaces [Del72], and to show that Hodge classes on K3 surfaces and their products with abelian varieties behave like absolute Hodge classes.

72.7 The evidence assessed

Remark 72.7.1. The known cases share a feature: the Hodge classes are either generated by divisors, or produced by an explicit geometric construction adapted to the variety (normal functions for cubic fourfolds, the decomposition of the diagonal, families of cycles on special abelian fourfolds, derived equivalences of K3 surfaces). There is no general mechanism. At the same time no case has failed, the Hodge loci behave as the conjecture predicts (Theorem 70.5.3), and Hodge classes on abelian varieties satisfy the arithmetic consequence of the conjecture (Deligne, Chapter 75). The first genuinely open cases include Weil classes on general abelian fourfolds of Weil type of arbitrary discriminant, abelian varieties of dimension ≥ 6 with exceptional classes, and products of two K3 surfaces in degree 4 away from Hodge isometries.

72.8 Exercises

Exercise 72.8.1. For a nonsingular cubic fourfold $X \subseteq \mathbb{P}^5$, show that $H^4(X;\mathbb{Q})$ has Hodge numbers $(0,1,21,1,0)$ and that $\mathrm{HC}(X,p)$ holds for $p \neq 2$ by Proposition 72.4.1.

Solution. As in Proposition 52.4.4, $c(TX) = (1+h)^6/(1+3h)$, whose degree-4 part is $(15 - 3\cdot 20 + 9\cdot 15 - 27\cdot 6 + 81)h^4 = 9h^4$, and $\int_X h^4 = 3$, so $\chi(X) = 27$. The odd Betti numbers vanish and $b_0 = b_2 = b_6 = b_8 = 1$ (Proposition 72.4.1), so $b_4 = 27 - 4 = 23$. The Hodge numbers $h^{4,0} = 0$, $h^{3,1} = 1$, $h^{2,2} = 21$ follow from Griffiths' description of the primitive cohomology of hypersurfaces by residues [Voi03]. The second statement is Proposition 72.4.1 with $n = 4$. □

Exercise 72.8.2. Let E be an elliptic curve without complex multiplication. Show that $\mathrm{Hdg}^\bullet(E^r) = \mathrm{Lef}^\bullet(E^r)$ for all r.

Solution. By Proposition 67.4.1, $\mathrm{Hg}(E) = \mathrm{SL}_2$, and the invariants of SL_2 acting diagonally on $\Lambda^\bullet(H^{\oplus r})$ are generated, by classical invariant theory, by the invariants in $\Lambda^2(H^{\oplus r})$, which are the divisor classes (Exercise 61.7.2). Apply Corollary 67.3.3. □

Chapter 73

Variants That Fail

73.1 Introduction

The Hodge conjecture is surrounded by statements that are equally natural and demonstrably false. Each failure identifies a mechanism that Hodge theory does not control, and so a constraint on any proof. We treat four variants: the integral Hodge conjecture, false for torsion classes (Atiyah–Hirzebruch) and for non-torsion classes (Kollár); the Kähler version, false even with Chern classes of coherent sheaves (Voisin); Hodge's general conjecture, false for formal reasons and repaired by Grothendieck; and the variational Hodge conjecture, which is open but belongs here as the statement one would need to propagate algebraicity in families. The chapter ends with a diagnostic list. References: the source *The Hodge Conjecture* (Chapter 7), [AH62], [Kol92], [Voi02a], [Gro69a].

Prerequisites. Chapter 72 (Known Cases and Evidence).

Throughout, X is a nonsingular complex projective variety of dimension n unless stated otherwise.

73.2 The integral Hodge conjecture

Definition 73.2.1. The *integral Hodge conjecture* $\mathrm{IHC}(X,p)$ asserts that the group $\mathrm{Hdg}^p(X;\mathbb{Z})$ of integral Hodge classes (Definition 63.4.1) is generated by classes of subvarieties. Its *defect* is the group $Z^{2p}(X) = \mathrm{Hdg}^p(X;\mathbb{Z})/\mathrm{cl}(\mathrm{CH}^p(X))$.

Proposition 73.2.2. *$Z^{2p}(X)$ is finitely generated, and it is finite if and only if $\mathrm{HC}(X,p)$ holds. $Z^2(X)=0$, and $Z^{2n}(X)=0$.*

Proof. $\mathrm{Hdg}^p(X;\mathbb{Z})$ is a subgroup of the finitely generated group $H^{2p}(X;\mathbb{Z})$. The group $Z^{2p}(X)\otimes\mathbb{Q}$ is $\mathrm{Hdg}^p(X)/Z^p_{\mathrm{alg}}(X)$, which vanishes exactly when $\mathrm{HC}(X,p)$ holds, and a finitely generated group is finite if and only if its rationalisation vanishes. $Z^2=0$ is the Lefschetz $(1,1)$-theorem (Theorem 69.3.1), and $H^{2n}(X;\mathbb{Z})$ is generated by the classes of points, one for each component. □

Theorem 73.2.3 (Atiyah–Hirzebruch)**.** *There are nonsingular projective varieties X with torsion classes in $H^4(X;\mathbb{Z})$, automatically integral Hodge classes, that are not algebraic. In particular* IHC *fails.*

This is [AH62]. The obstruction is topological. A subvariety Z of codimension p defines an element $[\mathcal{O}_Z]$ of topological K-theory $K^0(X)$, through a finite resolution by vector bundles, and its image in the graded piece of the topological filtration is $\mathrm{cl}(Z)$ up to the factor $(p-1)!$, since $c_p(\mathcal{O}_Z) = (-1)^{p-1}(p-1)!\,\mathrm{cl}(Z)$ while $\mathrm{ch}_p(\mathcal{O}_Z) = \mathrm{cl}(Z)$ (Proposition 63.5.1). So algebraic

classes survive to E_∞ in the Atiyah–Hirzebruch spectral sequence $H^s(X;K^t(\mathrm{pt}))\Rightarrow K^{s+t}(X)$, whose first possibly nonzero differential is $d_3=\mathrm{Sq}^3_{\mathbb{Z}}=\beta\circ\mathrm{Sq}^2\circ\rho$, the integral lift of the Steenrod square (Chapter 24). Hence an algebraic class α satisfies $\mathrm{Sq}^3_{\mathbb{Z}}(\alpha)=0$. Atiyah and Hirzebruch then produce, by the Godeaux–Serre construction of a quotient $X=Y/G$ of a complete intersection by a free action of a finite group G such as $(\mathbb{Z}/p)^3$, whose cohomology agrees with that of the classifying space BG in low degrees, a torsion class with $\mathrm{Sq}^3_{\mathbb{Z}}\neq 0$.

Remark 73.2.4. Totaro observed that the cycle class map factors through complex cobordism, $\mathrm{CH}^p(X)\to MU^{2p}(X)\otimes_{MU^*}\mathbb{Z}\to H^{2p}(X;\mathbb{Z})$, which gives a finer obstruction [Tot97]. Soulé and Voisin produced further torsion counterexamples by other methods [SV05]. All these classes vanish in $H^{2p}(X;\mathbb{Q})$, so they do not affect the rational conjecture: Hodge theory is blind to torsion.

Theorem 73.2.5 (Kollár)**.** *Let $\ell\geq 5$ be a prime and d a multiple of ℓ^3. For a very general hypersurface $X\subseteq\mathbb{P}^4$ of degree d, every curve $C\subseteq X$ has degree divisible by ℓ. Consequently the generator α of $H^4(X;\mathbb{Z})\cong\mathbb{Z}$ is an integral Hodge class that is not algebraic, and $Z^4(X)\neq 0$ contains an element of order ℓ.*

Proof of the consequence. The divisibility statement is [Kol92]. Granting it: $\dim X=3$, and by the Lefschetz hyperplane theorem (Theorem 50.5.1) and Poincaré duality, $H^2(X;\mathbb{Z})\cong\mathbb{Z}h$ and $H^4(X;\mathbb{Z})\cong\mathbb{Z}$, generated by a class α with $\int_X\alpha\cup h=1$ (the pairing $H^2\times H^4\to\mathbb{Z}$ is perfect on the torsion-free quotients; integral Lefschetz for H^2 uses the integral version of the hyperplane theorem, Remark 50.5.2). Since $H^4(X;\mathbb{C})=\mathbb{C}h^2$ is of type $(2,2)$, α is an integral Hodge class. A 1-cycle $\sum a_iC_i$ has class $(\sum a_i\deg C_i)\alpha$, because $\int\mathrm{cl}(C)\cup h=\deg C$ (Theorem 63.3.1(4)). If all degrees are divisible by ℓ, the algebraic classes lie in $\ell\mathbb{Z}\alpha$, so α is not algebraic and $\mathbb{Z}/\ell$ is a quotient of $Z^4(X)$. □

Corollary 73.2.6. *In the situation of Theorem 73.2.5, the defect $Z^4(X)$ is a finite cyclic group whose order is divisible by ℓ and divides d. In particular* $\mathrm{HC}(X,2)$ *holds although* $\mathrm{IHC}(X,2)$ *fails.*

Proof. With α the generator of $H^4(X;\mathbb{Z})=\mathbb{Z}\alpha$ normalised by $\int_X\alpha\cup h=1$, the class h^2 satisfies $\int_X h^2\cup h=\deg X=d$, so $h^2=d\alpha$. It is algebraic, the class of the intersection of X with two hyperplanes, a curve of degree d. So $d\mathbb{Z}\alpha\subseteq\mathrm{cl}(\mathrm{CH}^2(X))\subseteq\ell\mathbb{Z}\alpha$, the second inclusion by Theorem 73.2.5. Hence $\mathrm{cl}(\mathrm{CH}^2(X))=m\mathbb{Z}\alpha$ with $\ell\mid m\mid d$, and $Z^4(X)=\mathrm{Hdg}^2(X;\mathbb{Z})/\mathrm{cl}(\mathrm{CH}^2(X))=\mathbb{Z}\alpha/m\mathbb{Z}\alpha\cong\mathbb{Z}/m$. Rationally $\alpha=\frac{1}{d}h^2$ is algebraic. □

Remark 73.2.7. The divisibility is proved by specialisation and countability. The relative Hilbert scheme of curves in the universal family of hypersurfaces has countably many components, each proper over the parameter space. For a very general member X, every curve on X therefore lies in a component that dominates the parameter space, so it specialises to a 1-cycle of the same degree on every member, in particular on a special singular member X_0. Kollár builds X_0 from a cyclic covering of degree ℓ so that every 1-cycle on it has degree divisible by ℓ; this is where $\ell^3\mid d$ is used. (Kollár's statement allows any integer ℓ prime to 6.) The Hodge structure here is as simple as possible; the obstruction is purely about degrees of curves, on which Hodge theory has nothing to say.

Theorem 73.2.8.

1. *(Voisin)* $\mathrm{IHC}(X,2)$ *holds for uniruled threefolds and for Calabi–Yau threefolds [Voi06].*
2. *(Colliot-Thélène–Voisin) The group $Z^4(X)$ is a birational invariant of nonsingular projective varieties. Its torsion subgroup is a quotient of the unramified cohomology group $H^3_{\mathrm{nr}}(X,\mathbb{Q}/\mathbb{Z}(2))$, and is isomorphic to it when $\mathrm{CH}_0(X)$ is supported on a surface, for instance when X is rationally connected. Since unramified cohomology vanishes for rational varieties, a nonzero torsion element of $Z^4(X)$ obstructs rationality [CTV12].*

73.3 The Kähler counterexample

Theorem 73.3.1 (Voisin). *There is a complex torus T of dimension 4 with a nonzero class in $\mathrm{Hdg}^2(T)$ that is not in the $\mathbb{Q}$-span of the classes $\mathrm{ch}_2(\mathcal{F})$ of coherent analytic sheaves on T, nor of classes of analytic subvarieties. Hence the Hodge conjecture fails for compact Kähler manifolds in both the cycle form and the sheaf form.*

This is [Voi02a]. The torus is of Weil type (Proposition 72.3.3): the construction of Weil classes uses only Hodge theory, not projectivity. In outline, Voisin chooses the lattice so that $\mathrm{NS}(T) = 0$ and the Hodge classes in H^4 are primitive for a Kähler class ω, that is, $\int_T \alpha \wedge \omega^2 = 0$. For a coherent sheaf $\mathcal{F}$ on T, $c_1(\mathcal{F}) = 0$ because $\mathrm{NS}(T) = 0$, and a positivity property of c_2 coming from Hermite–Einstein metrics (a Bogomolov-type inequality) gives $\int_T c_2(\mathcal{F}) \wedge \omega^2 \geq 0$ with equality only if $c_2(\mathcal{F}) = 0$. Since $c_2(\mathcal{F})$ is a Hodge class, it is primitive, so $c_2(\mathcal{F}) = 0$. Analytic subvarieties are covered because their structure sheaves are coherent sheaves.

Remark 73.3.2. All the Hodge-theoretic structure (Hodge decomposition, hard Lefschetz, Hodge–Riemann relations, the $\partial\bar{\partial}$-lemma) holds on Voisin's torus. So no proof of the Hodge conjecture can use only the Kähler package: it must use projectivity, through sections of ample line bundles, algebraic Hilbert schemes, GAGA, or arithmetic models. For $p = 1$ the Kähler version holds in the form of Theorem 69.2.1: integral $(1,1)$-classes are Chern classes of holomorphic line bundles.

73.4 Hodge's general conjecture

Proposition 73.4.1. *Hodge's original general conjecture $N^c H^k(X;\mathbb{Q}) = H^k(X;\mathbb{Q}) \cap F^c H^k(X;\mathbb{C})$ fails in general: the left side is a sub-Hodge structure (Proposition 68.4.5), while the right side need not be.*

Proof. Let $V = H^k(X;\mathbb{Q})$ and $W = V \cap F^c$. Since V is stable under conjugation, $W \subseteq F^c \cap \overline{F^c}$, which is $\bigoplus_{c \leq p \leq k-c} V^{p,k-p}$. A rational subspace of this sum need not be the rational form of a sum of Hodge summands, that is, need not be a sub-Hodge structure. Grothendieck exhibits an abelian variety whose H^3 contains a rational line in F^1 whose smallest containing sub-Hodge structure is not contained in F^1 [Gro69a]. For such a line, the right side contains it, and the left side, a sub-Hodge structure contained in F^1 (Proposition 68.4.5), cannot. □

Remark 73.4.2. The lesson is formal: the objects of Hodge theory are sub-Hodge structures, not rational subspaces satisfying a filtration condition. Grothendieck's repaired *generalised Hodge conjecture* (Remark 68.4.6) is open in essentially every case where the Hodge conjecture is, and in many where it is known.

73.5 The variational Hodge conjecture

Conjecture 73.5.1 (Variational Hodge conjecture). *Let $f : \mathcal{X} \to S$ be a smooth projective morphism over a connected nonsingular base, and λ a flat section of $R^{2p} f_* \mathbb{Q}$ with $\lambda_s \in \mathrm{Hdg}^p(X_s)$ for all s. If λ_{s_0} is algebraic for one s_0, then λ_s is algebraic for all s.*

Remark 73.5.2. The variational conjecture follows from the Hodge conjecture applied to a compactification of the total space, by the theorem of the fixed part (Theorem 70.4.1). Bloch's semiregularity theorem proves the infinitesimal version: a local complete intersection cycle whose semiregularity map is injective deforms together with its cohomology class [Blo72].

The Cattani–Deligne–Kaplan theorem (Theorem 70.5.3) says that the locus where λ stays Hodge is algebraic; the variational conjecture says that on this locus λ stays algebraic. The first is a theorem and the second is open, which summarises the state of the subject.

73.6 A diagnostic list

Remark 73.6.1. A proof of the Hodge conjecture must be able to distinguish it from its false neighbours:

1. *It must lose denominators.* Since the integral version is false (Kollár), cycles can only be produced after clearing denominators. In known cases the denominators enter through push-pull, $f_* f^* = \deg f$, or through factorials such as $c_p(\mathcal{O}_Z) = (-1)^{p-1}(p-1)!\,\mathrm{cl}(Z)$.
2. *It must use projectivity*, not just the Kähler package (Voisin).
3. *It must be insensitive to torsion* (Atiyah–Hirzebruch): it cannot produce integral cycles from integral Hodge classes.
4. *It must respect sub-Hodge structures* (Grothendieck): it must work with Hodge classes, not with rational classes in a filtration step.

73.7 Exercises

Exercise 73.7.1. Show that every torsion class in $H^{2p}(X;\mathbb{Z})$ is an integral Hodge class, and that $Z^{2p}(X)$ is torsion if $\mathrm{HC}(X,p)$ holds.

Solution. A torsion class maps to $0 \in H^{2p}(X;\mathbb{C})$, which lies in $H^{p,p}$. The second statement is Proposition 73.2.2. □

Exercise 73.7.2. Why does Theorem 73.2.5 not contradict the fact that a quintic threefold contains lines, which have degree 1?

Solution. The theorem needs d divisible by $\ell^3 \geq 125$, and the hypersurface very general. A quintic has $d = 5$, which is not divisible by 125. □

Chapter 74

Mixed Hodge Structures

74.1 Introduction

The cohomology of a smooth but non-compact variety, or of a singular projective one, is not a pure Hodge structure. For example, $H^1(\mathbb{C}^*;\mathbb{Q})$ is one-dimensional, so it cannot carry a Hodge structure of odd weight. Deligne showed that the cohomology of every complex algebraic variety carries a *mixed Hodge structure*: an increasing *weight filtration* whose graded pieces are pure Hodge structures of various weights. Morphisms of mixed Hodge structures are strict for both filtrations (Theorem 74.3.2), and this rigidity has consequences for compact varieties too. Three of them are proved in Section 74.6: the global invariant cycle theorem, used in Chapter 70; the fact that algebraicity of a Hodge class can be tested on a Zariski open subset, modulo lower dimensions; and the birational invariance of the Hodge conjecture, modulo lower dimensions. References: [Del71], [Del74b], [PS08], [Voi02b, Chapter 8].

Prerequisites. Chapter 66 (Hodge Structures), Chapter 17 (Derived Categories) and Chapter 58 (Analytic Spaces and GAGA).

Conventions. Filtrations are finite and exhaustive. $W_\bullet$ always denotes an increasing filtration ($W_{k-1} \subseteq W_k$) and $F^\bullet$ a decreasing one ($F^{p+1} \subseteq F^p$). For an increasing filtration, $\mathrm{Gr}^W_k = W_k/W_{k-1}$. The Tate structure $\mathbb{Q}(m)$ is $(2\pi i)^m\mathbb{Q} \subseteq \mathbb{C}$, pure of weight $-2m$ and type $(-m,-m)$ (Example 66.2.3), and $H(m) = H \otimes \mathbb{Q}(m)$. Varieties are complex algebraic varieties, and their cohomology is the singular cohomology of the associated analytic space.

74.2 Mixed Hodge structures

Definition 74.2.1. A *mixed Hodge structure* (H, W, F) consists of

1. a finite-dimensional $\mathbb{Q}$-vector space H;
2. a finite increasing filtration $W_\bullet$ of H by $\mathbb{Q}$-subspaces, the *weight filtration*;
3. a finite decreasing filtration $F^\bullet$ of $H_\mathbb{C} = H \otimes_\mathbb{Q} \mathbb{C}$ by $\mathbb{C}$-subspaces, the *Hodge filtration*;

such that for every k the filtration induced by F on $\mathrm{Gr}^W_k H_\mathbb{C} = W_kH_\mathbb{C}/W_{k-1}H_\mathbb{C}$, namely

$$F^p\mathrm{Gr}^W_k H_\mathbb{C} = (F^p \cap W_kH_\mathbb{C} + W_{k-1}H_\mathbb{C})/W_{k-1}H_\mathbb{C},$$

is the Hodge filtration of a pure $\mathbb{Q}$-Hodge structure of weight k on $\mathrm{Gr}^W_k H$ (Proposition 66.2.2). A *morphism* $f : (H, W, F) \to (H', W', F')$ is a $\mathbb{Q}$-linear map with $f(W_k) \subseteq W'_k$ and $f_\mathbb{C}(F^p) \subseteq F'^p$ for all k, p. The *Hodge numbers* are $h^{p,q}(H) = \dim_\mathbb{C}(\mathrm{Gr}^W_{p+q}H_\mathbb{C})^{p,q}$, and the *weights* of H are the k with $\mathrm{Gr}^W_k H \neq 0$.

Example 74.2.2. 1. A pure Hodge structure of weight k is a mixed Hodge structure with $W_{k-1} = 0$ and $W_k = H$, and morphisms of pure Hodge structures of the same weight are the same as morphisms of the associated mixed Hodge structures. A direct sum of pure Hodge structures of different weights is a mixed Hodge structure, with W_k the sum of the summands of weight $\leq k$.

2. $H^1(\mathbb{C}^*;\mathbb{Q})$ is one-dimensional, spanned by the class of $\frac{1}{2\pi i}\frac{dz}{z}$, which integrates to 1 over the unit circle. With $W_1 = 0$, $W_2 = H^1$, $F^1 = H^1(\mathbb{C}^*;\mathbb{C})$ and $F^2 = 0$, it is the pure Tate structure $\mathbb{Q}(-1)$ of weight 2 and type $(1,1)$. This is the structure given by Theorem 74.4.3 below, with $\bar{X} = \mathbb{P}^1$ and $D = \{0,\infty\}$.

3. Let C be a nodal plane cubic curve, obtained from $\mathbb{P}^1$ by identifying two points. Then $H^1(C;\mathbb{Q}) \cong \mathbb{Q}$ is pure of weight 0 and type $(0,0)$ (Example 74.5.2 below): the loop through the node is not detected by holomorphic 1-forms.

4. *Kummer structures.* Let $q \in \mathbb{C}^*$, choose $\log q$, and put $\lambda = \log q/2\pi i$. On $H = \mathbb{Q}e_0 \oplus \mathbb{Q}e_1$ let $W_{-3} = 0$, $W_{-2} = W_{-1} = \mathbb{Q}e_1$, $W_0 = H$, and $F^{-1} = H_{\mathbb{C}}$, $F^0 = \mathbb{C}(e_0 - \lambda e_1)$, $F^1 = 0$. This is a mixed Hodge structure K_q, an extension of $\mathbb{Q}(0)$ by $\mathbb{Q}(1)$ (Exercise 74.7.2). Another choice of $\log q$ changes λ by an integer n, and $e_0 \mapsto e_0 + ne_1$ is an isomorphism between the two structures, so K_q depends only on q.

Counterexample 74.2.3. A mixed Hodge structure need not be the direct sum of its graded pieces, not even after extending scalars to $\mathbb{R}$. In K_q, a complement of W_{-2} that is a sub-mixed Hodge structure is a line $L = \mathbb{Q}(e_0 + te_1)$ with $t \in \mathbb{Q}$ whose induced Hodge filtration is that of $\mathbb{Q}(0)$, that is, with $L_{\mathbb{C}} \subseteq F^0 = \mathbb{C}(e_0 - \lambda e_1)$. This forces $t = -\lambda$. So K_q splits if and only if $\lambda \in \mathbb{Q}$, that is, q is a root of unity; and the same argument with $t \in \mathbb{R}$ shows that $K_q \otimes \mathbb{R}$ does not split when $|q| \neq 1$, since then $\lambda \notin \mathbb{R}$. Carlson showed that the analogous extensions of $\mathbb{Z}(0)$ by $\mathbb{Z}(1)$ over $\mathbb{Z}$ are classified exactly by $q \in \mathbb{C}/2\pi i\mathbb{Z} \cong \mathbb{C}^*$ [Car80]: Kummer structures are the Hodge-theoretic incarnation of the exponential map. Contrast this with pure polarisable Hodge structures, which form a semisimple category (Theorem 66.5.3).

74.3 The Deligne splitting and strictness

Theorem 74.3.1 (Deligne)**.** *Let (H, W, F) be a mixed Hodge structure. For $p, q \in \mathbb{Z}$ put*

$$I^{p,q} = F^p \cap W_{p+q,\mathbb{C}} \cap \Big(\overline{F^q} \cap W_{p+q,\mathbb{C}} + \sum_{j\geq 1} \overline{F^{q-j}} \cap W_{p+q-j-1,\mathbb{C}}\Big).$$

Then $H_{\mathbb{C}} = \bigoplus_{p,q} I^{p,q}$, and

$$W_{k,\mathbb{C}} = \bigoplus_{p+q\leq k} I^{p,q}, \qquad F^p = \bigoplus_{p'\geq p} I^{p',q}, \qquad \overline{I^{p,q}} \subseteq I^{q,p} \oplus \bigoplus_{p'<q,\, q'<p} I^{p',q'}.$$

The projection $W_{p+q,\mathbb{C}} \to \mathrm{Gr}^W_{p+q} H_{\mathbb{C}}$ maps $I^{p,q}$ isomorphically onto $(\mathrm{Gr}^W_{p+q})^{p,q}$. Conversely, if W is a filtration by $\mathbb{Q}$-subspaces and F a filtration of $H_{\mathbb{C}}$ admitting a bigrading $H_{\mathbb{C}} = \bigoplus I^{p,q}$ with these three properties, then (H, W, F) is a mixed Hodge structure.

The first part is [Del71]; see [PS08, Chapter 3] for the explicit formula. The converse is immediate: on Gr^W_k the images of the $I^{p,q}$ with $p+q = k$ give a decomposition with $\overline{H^{p,q}} = H^{q,p}$, since the error terms have $p' + q' < k$, and the filtration induced by F is $\bigoplus_{p'\geq p} H^{p',k-p'}$. The subspaces $I^{p,q}$ form the *Deligne splitting*. It splits W and F simultaneously, but it is not compatible with complex conjugation in general (Counterexample 74.2.3). Every morphism

f of mixed Hodge structures maps $I^{p,q}(H)$ into $I^{p,q}(H')$, as is clear from the formula, since f preserves W, F and, being defined over $\mathbb{Q}$, also $\overline{F}$.

Theorem 74.3.2 (Deligne). *1. Every morphism $f : H \to H'$ of mixed Hodge structures is* strict *for W and for F: $f(W_k) = W'_k \cap f(H)$ and $f_{\mathbb{C}}(F^p) = F'^p \cap f_{\mathbb{C}}(H_{\mathbb{C}})$.*

2. Mixed Hodge structures form an abelian category, in which kernels, images and cokernels carry the induced filtrations.

3. For every k, the functor $H \mapsto \mathrm{Gr}^W_k H$ from mixed Hodge structures to pure Hodge structures of weight k is exact.

4. $H \otimes H'$, $H^\vee$ and $\mathrm{Hom}(H, H') = H^\vee \otimes H'$ carry mixed Hodge structures, with $W_k(H \otimes H') = \sum_{a+b=k} W_a H \otimes W_b H'$, $F^p(H \otimes H') = \sum_{a+b=p} F^a H \otimes F^b H'$, $W_k(H^\vee) = (W_{-k-1}H)^\perp$ and $F^p(H^\vee) = (F^{1-p}H)^\perp$.

Proof. (1) By Theorem 74.3.1, $f_{\mathbb{C}}$ maps $I^{p,q}(H)$ into $I^{p,q}(H')$. Let $y = f(x) \in W'_{k,\mathbb{C}}$ and decompose $x = \sum x^{p,q}$ with $x^{p,q} \in I^{p,q}(H)$. Then $y = \sum f(x^{p,q})$ is the decomposition of y, so $f(x^{p,q}) = 0$ whenever $p + q > k$, and $y = f(\sum_{p+q\le k} x^{p,q}) \in f(W_{k,\mathbb{C}})$. The same argument, with $p' \ge p$ in place of $p + q \le k$, proves strictness for F. Strictness for W over $\mathbb{Q}$ follows from strictness over $\mathbb{C}$, because $f(W_k)$ and $W'_k \cap f(H)$ are $\mathbb{Q}$-subspaces with the same complexification.

(2) Since $f_{\mathbb{C}}$ preserves the bigradings, its kernel $K_{\mathbb{C}}$, image $f(H_{\mathbb{C}})$ and cokernel $C_{\mathbb{C}}$ are bigraded: $K_{\mathbb{C}} = \bigoplus (K_{\mathbb{C}} \cap I^{p,q}(H))$, $f(H_{\mathbb{C}}) = \bigoplus f(I^{p,q}(H))$ and $C_{\mathbb{C}} = \bigoplus I^{p,q}(H')/f(I^{p,q}(H))$. Give K, $f(H)$ and C the filtrations induced as subspaces or quotients. In each case W_k and F^p are the sums of the pieces with $p + q \le k$, respectively with first index $\ge p$. For $f(H)$ the filtrations induced from H (quotient) and from H' (subspace) agree, which is the strictness proved in (1). The bigradings satisfy the conjugation condition of Theorem 74.3.1, because K, $f(H)$ and C are defined over $\mathbb{Q}$ and the components of a vector in a bigraded subspace lie in it. By the converse part of Theorem 74.3.1, K, $f(H)$ and C are mixed Hodge structures, and the canonical map $H/K \to f(H)$ is an isomorphism of mixed Hodge structures.

(3) If $0 \to H' \to H \to H'' \to 0$ is an exact sequence of mixed Hodge structures, then its complexification is exact and each map preserves the bigradings, so $0 \to I^{p,q}(H') \to I^{p,q}(H) \to I^{p,q}(H'') \to 0$ is exact for each (p, q). Since $\mathrm{Gr}^W_k H_{\mathbb{C}} \cong \bigoplus_{p+q=k} I^{p,q}(H)$, the sequence of graded pieces is exact.

(4) Put $I^{p,q}(H \otimes H') = \bigoplus I^{a,b}(H) \otimes I^{a',b'}(H')$, the sum over $a + a' = p$, $b + b' = q$. This bigrading splits the filtrations W and F of the statement, and it satisfies the conjugation condition because the error terms of the factors multiply to error terms. As $W_k(H \otimes H')$ is defined over $\mathbb{Q}$, the converse part of Theorem 74.3.1 applies. For the dual, use $I^{p,q}(H^\vee) = (\bigoplus_{(p',q')\neq(-p,-q)} I^{p',q'}(H))^\perp$ in the same way. □

Lemma 74.3.3 (Principle of weights). *Let $f : H \to H'$ be a morphism of mixed Hodge structures.*

1. If the weights of H are all $< m$ and those of H' are all $\ge m$, or the other way round, then $f = 0$.

2. If H' is pure of weight k, then $f(H) = f(W_k H)$ and $f(W_{k-1}H) = 0$.

Proof. (1) If the weights of H are $< m$ and those of H' are $\ge m$, then $H = W_{m-1}H$ and $f(H) \subseteq W'_{m-1}H' = 0$. In the other case $W_{m-1}H = 0$ and $H' = W'_{m-1}H'$, so strictness (Theorem 74.3.2(1)) gives $f(H) = f(H) \cap W'_{m-1}H' = f(W_{m-1}H) = 0$.

(2) $W'_{k-1}H' = 0$ and $W'_k H' = H'$, so $f(W_{k-1}H) \subseteq W'_{k-1}H' = 0$, and strictness gives $f(W_k H) = W'_k H' \cap f(H) = f(H)$. □

Remark 74.3.4. Lemma 74.3.3 is used over and over below: a map compatible with mixed Hodge structures cannot move classes of one weight to classes of another. Its power comes from the fact that the maps it is applied to (restrictions, Gysin maps, differentials of spectral sequences) are defined topologically, while the weights are defined algebro-geometrically. It has no analogue for filtered vector spaces, where strictness fails (Counterexample 66.3.4).

74.4 The logarithmic de Rham complex

Definition 74.4.1. Let $\bar{X}$ be a complex manifold of dimension n and $D \subseteq \bar{X}$ a divisor with *simple normal crossings*: $D = D_1 \cup \cdots \cup D_N$ with each D_i a nonsingular hypersurface, and near every point there are holomorphic coordinates $z_1, \ldots, z_n$ with $D = \{z_1 \cdots z_r = 0\}$. Put $X = \bar{X} \setminus D$ and let $j\colon X \to \bar{X}$ be the inclusion.

1. The sheaf $\Omega^p_{\bar{X}}(\log D)$ of *logarithmic p-forms* is the $\mathcal{O}_{\bar{X}}$-submodule of $j_*\Omega^p_X$ generated locally by the wedge products of p of the forms $dz_1/z_1, \ldots, dz_r/z_r, dz_{r+1}, \ldots, dz_n$. It is locally free, and stable under d, which gives the *logarithmic de Rham complex* $\Omega^\bullet_{\bar{X}}(\log D)$.

2. The *weight filtration* $W_m\Omega^p_{\bar{X}}(\log D)$ is the subsheaf spanned by the forms $\alpha \wedge dz_{i_1}/z_{i_1} \wedge \cdots \wedge dz_{i_m}/z_{i_m}$ with α holomorphic: forms with at most m logarithmic factors.

3. $D^{(m)}$ is the disjoint union of the m-fold intersections $D_{i_1} \cap \cdots \cap D_{i_m}$, $i_1 < \cdots < i_m$, with $D^{(0)} = \bar{X}$. It is a complex manifold of dimension $n - m$, with the map $a_m\colon D^{(m)} \to \bar{X}$.

Lemma 74.4.2. *The residue map $\alpha \wedge dz_{i_1}/z_{i_1} \wedge \cdots \wedge dz_{i_m}/z_{i_m} \mapsto \alpha|_{D_{i_1}\cap\cdots\cap D_{i_m}}$ induces an isomorphism of complexes*

$$\mathrm{Res}_m\colon \mathrm{Gr}^W_m\Omega^\bullet_{\bar{X}}(\log D) \longrightarrow (a_m)_*\Omega^{\bullet-m}_{D^{(m)}}.$$

Sketch of proof. It is easiest to construct the inverse. For $I = \{i_1 < \cdots < i_m\}$ write $D_I = D_{i_1} \cap \cdots \cap D_{i_m}$ and $dz_I/z_I = dz_{i_1}/z_{i_1} \wedge \cdots \wedge dz_{i_m}/z_{i_m}$. Given a holomorphic form β on D_I, extend it locally to a holomorphic form $\tilde{\beta}$ on $\bar{X}$ and send it to the class of $\tilde{\beta} \wedge dz_I/z_I$ in Gr^W_m. The class does not depend on the extension: two extensions differ by a combination of forms $z_i\gamma$ and $dz_i \wedge \gamma$ with $i \in I$, and $z_i\gamma \wedge dz_I/z_I$ and $dz_i \wedge \gamma \wedge dz_I/z_I$ have at most $m-1$ logarithmic factors, since $z_i \cdot dz_i/z_i = dz_i$ and $dz_i \wedge dz_i/z_i = 0$. It does not depend on the coordinates: replacing z_i by uz_i with u a unit changes dz_i/z_i by the holomorphic form du/u, which lowers the weight. The map is surjective by the definition of W_m, it commutes with d because $d(dz_i/z_i) = 0$, and Res_m is its inverse. That it is injective, that is, that a sum $\sum_I \tilde{\beta}_I \wedge dz_I/z_I$ lies in W_{m-1} only if every β_I vanishes, is a computation in the local basis $dz_I/z_I \wedge dz_J$, $I \subseteq \{1, \ldots, r\}$, $J \subseteq \{r+1, \ldots, n\}$, which we omit [Del71]. □

The residue depends on the ordering $i_1 < \cdots < i_m$ through a sign, and in the comparison with rational cohomology it comes with the factor $(2\pi i)^{-m}$, because $\frac{1}{2\pi i}\frac{dz}{z}$ is the rational class of Example 74.2.2(2). This is why the Tate twist $(-m)$ appears below.

Theorem 74.4.3 (Deligne)**.** *Let X be a nonsingular variety, and $\bar{X}$ a nonsingular compactification with $D = \bar{X} \setminus X$ a simple normal crossings divisor; such a compactification exists by Nagata's compactification theorem and Hironaka's resolution of singularities [Hir64].*

1. *The inclusion $\Omega^\bullet_{\bar{X}}(\log D) \to j_*\Omega^\bullet_X$ is a quasi-isomorphism, and $j_*\Omega^\bullet_X$ computes $Rj_*\mathbb{C}_X$; hence $\mathbb{H}^k(\bar{X}, \Omega^\bullet_{\bar{X}}(\log D)) \cong H^k(X;\mathbb{C})$.*

2. *Let $W_mH^k(X;\mathbb{C})$ be the image of $\mathbb{H}^k(\bar{X}, W_{m-k}\Omega^\bullet_{\bar{X}}(\log D))$ and $F^pH^k(X;\mathbb{C})$ the image of $\mathbb{H}^k(\bar{X}, \Omega^{\geq p}_{\bar{X}}(\log D))$. Then W is defined over $\mathbb{Q}$, and $(H^k(X;\mathbb{Q}), W, F)$ is a mixed Hodge structure, independent of the choice of $\bar{X}$ and functorial for morphisms of nonsingular varieties.*

3. *The weights of $H^k(X;\mathbb{Q})$ lie in $[k,2k]$, and $W_kH^k(X;\mathbb{Q}) = \operatorname{im}\left(H^k(\bar{X};\mathbb{Q}) \to H^k(X;\mathbb{Q})\right)$.*

4. *There is a spectral sequence, the* weight spectral sequence,

$$E_1^{-m,k+m} = H^{k-m}(D^{(m)};\mathbb{Q})(-m) \Longrightarrow H^k(X;\mathbb{Q}),$$

whose terms are pure Hodge structures of weight $k+m$, whose differential d_1 is an alternating sum of Gysin maps, and which degenerates at E_2, with $E_2^{-m,k+m} = \operatorname{Gr}^W_{k+m}H^k(X;\mathbb{Q})$. The Hodge-to-de Rham spectral sequence $E_1^{p,q} = H^q(\bar{X}, \Omega^p_{\bar{X}}(\log D)) \Rightarrow H^{p+q}(X;\mathbb{C})$ degenerates at E_1.

This is [Del71, Section 3]; see also [PS08, Chapter 4]. The weight spectral sequence is the spectral sequence of the filtered complex $(\Omega^\bullet_{\bar{X}}(\log D), W)$, with E_1 computed by Lemma 74.4.2. Once one knows that its differentials are morphisms of Hodge structures, degeneration at E_2 is the principle of weights: for $r \geq 2$, $d_r\colon E_r^{-m,k+m} \to E_r^{-m+r,k+m-r+1}$ goes between subquotients of pure Hodge structures of weights $k+m$ and $k+m+1-r \neq k+m$, so it vanishes by Lemma 74.3.3(1).

Example 74.4.4. Let $\bar{C}$ be a compact Riemann surface of genus g, $s \geq 1$, and $C = \bar{C}\backslash\{p_1,\dots,p_s\}$. Then $D^{(1)} = \{p_1,\dots,p_s\}$ and $D^{(2)} = \emptyset$, and the weight spectral sequence reduces to the exact sequence of mixed Hodge structures

$$0 \to H^1(\bar{C};\mathbb{Q}) \to H^1(C;\mathbb{Q}) \xrightarrow{\text{Res}} \mathbb{Q}(-1)^s \xrightarrow{\Sigma} \mathbb{Q}(-1) \to 0,$$

where Res takes the residues at the punctures and Σ, the Gysin map $H^0(D^{(1)})(-1) \to H^2(\bar{C})$, is the sum. So $W_1H^1(C) = H^1(\bar{C})$ is pure of weight 1 and $\operatorname{Gr}^W_2H^1(C) \cong \mathbb{Q}(-1)^{s-1}$; exactness at $\mathbb{Q}(-1)^s$ is the residue theorem. For $\mathbb{P}^1$ minus the three points $0,1,\infty$, $H^1 \cong \mathbb{Q}(-1)^2$ is spanned by the classes of $\frac{1}{2\pi i}\frac{dz}{z}$ and $\frac{1}{2\pi i}\frac{dz}{z-1}$.

Counterexample 74.4.5. The weights of $H^k(X)$ for X nonsingular need not include k or $2k$, and cohomology of even degree can be pure of odd weight. Let $E \subseteq \mathbb{P}^2$ be a smooth cubic curve and $X = \mathbb{P}^2 \setminus E$, an affine surface. The weight spectral sequence ($D^{(2)} = \emptyset$) gives the Gysin sequence

$$H^0(E)(-1) \to H^2(\mathbb{P}^2) \to H^2(X) \to H^1(E)(-1) \to H^3(\mathbb{P}^2) = 0.$$

The first map sends 1 to $\operatorname{cl}(E) = 3h \neq 0$, so it is an isomorphism over $\mathbb{Q}$, and $H^2(X;\mathbb{Q}) \cong H^1(E;\mathbb{Q})(-1)$ is pure of weight 3.

74.5 Singular varieties

Theorem 74.5.1 (Deligne)**.** *For every variety X and every k, $H^k(X;\mathbb{Q})$ and $H^k_c(X;\mathbb{Q})$ carry functorial mixed Hodge structures, extending those of Theorem 74.4.3. The long exact sequence of a pair (X,Z) with Z closed, and the sequence $\cdots \to H^k_c(X\setminus Z) \to H^k_c(X) \to H^k_c(Z) \to \cdots$, are exact sequences of mixed Hodge structures, and cup products and Künneth isomorphisms are morphisms of mixed Hodge structures. The weights of $H^k(X)$ lie in $[0,2k]$; if X is proper they lie in $[0,k]$, and if X is nonsingular in $[k,2k]$. If X is proper and $\pi\colon \tilde{X} \to X$ is a resolution of singularities, then $W_{k-1}H^k(X;\mathbb{Q}) = \ker\left(\pi^*\colon H^k(X;\mathbb{Q}) \to H^k(\tilde{X};\mathbb{Q})\right)$.*

This is [Del74b], via simplicial resolutions of X by nonsingular varieties; see also [PS08, Chapter 5]. For proper X with a resolution $\pi\colon \tilde{X} \to X$ that is an isomorphism outside a

closed subset $Z \subseteq X$ with preimage $E = \pi^{-1}(Z)$, there is an exact sequence of mixed Hodge structures

$$\cdots \to H^{k-1}(E) \to H^k(X) \to H^k(\tilde{X}) \oplus H^k(Z) \to H^k(E) \to H^{k+1}(X) \to \cdots,$$

the Mayer–Vietoris sequence of the proper modification, which computes the mixed Hodge structure of X inductively on the dimension.

Example 74.5.2. Let C be the nodal cubic, $\pi\colon \mathbb{P}^1 \to C$ its normalisation, Z the node and $E = \pi^{-1}(Z) = \{0, \infty\}$. The sequence above reads

$$0 \to H^0(C) \to H^0(\mathbb{P}^1) \oplus H^0(Z) \to H^0(E) \to H^1(C) \to H^1(\mathbb{P}^1) \oplus H^1(Z) = 0.$$

The map $\mathbb{Q}^2 \to \mathbb{Q}^2$, $(a, b) \mapsto (a - b, a - b)$, has a one-dimensional cokernel, so $H^1(C) \cong \mathbb{Q}(0)$, a quotient of the pure structure $H^0(E)$ of weight 0. Accordingly $W_0H^1(C) = \ker(H^1(C) \to H^1(\mathbb{P}^1)) = H^1(C)$, as Theorem 74.5.1 predicts, and $H^1(C)$ has no Hodge decomposition of weight 1.

74.6 Applications of weights

Corollary 74.6.1. *Let $\bar{X}$ be a nonsingular projective variety of dimension n, $U \subseteq \bar{X}$ a nonempty open subset, and $Z = \bar{X} \setminus U$ with irreducible components $Z_1, \dots, Z_N$ of codimensions $c_1, \dots, c_N$. Let $g_i\colon \tilde{Z}_i \to \bar{X}$ be resolutions of singularities of the Z_i composed with the inclusions.*

1. *The image of the restriction $H^k(\bar{X}; \mathbb{Q}) \to H^k(U; \mathbb{Q})$ is $W_kH^k(U; \mathbb{Q})$.*
2. *The kernel of the restriction is $\sum_i g_{i*}H^{k-2c_i}(\tilde{Z}_i; \mathbb{Q})(-c_i)$, the sum of the images of the Gysin maps.*

Sketch of proof. (1) By resolution of singularities there is a proper birational morphism $\pi\colon \tilde{X} \to \bar{X}$ from a nonsingular projective variety that is an isomorphism over U and such that $\tilde{X} \setminus \pi^{-1}(U)$ is a simple normal crossings divisor. By Theorem 74.4.3(3), the image of $H^k(\tilde{X}) \to H^k(U)$ is $W_kH^k(U)$. The image of $H^k(\bar{X})$ is contained in it, because restriction from $\bar{X}$ factors through π^*. Conversely, for $\alpha \in H^k(\tilde{X})$ the class $\pi_*\alpha \in H^k(\bar{X})$, defined by Poincaré duality, restricts on U to $\alpha|_U$, since π is an isomorphism over U. So the two images agree.

(2) The Gysin images restrict to zero on U, since the classes are supported on Z. Conversely, the kernel is the image of $H^k_Z(\bar{X}; \mathbb{Q}) \cong H^{BM}_{2n-k}(Z; \mathbb{Q})(-n)$ (Poincaré–Alexander duality), by a morphism of mixed Hodge structures into the pure structure $H^k(\bar{X})$ of weight k. By Lemma 74.3.3(2) the kernel is the image of the part of weight $\leq k$, and Deligne shows that the lowest weight part of the Borel–Moore homology of Z is the image of the homology of the resolutions $\tilde{Z}_i$ [Del74b]. This last identification is the verification we omit. □

Theorem 74.6.2 (Global invariant cycle theorem)**.** *Let $f : \mathcal{X} \to S$ be a smooth projective morphism of nonsingular varieties with S connected, $\bar{\mathcal{X}}$ a nonsingular projective compactification of $\mathcal{X}$, and $t \in S$ with fibre X_t. Then*

$$\operatorname{im}\left(H^k(\bar{\mathcal{X}}; \mathbb{Q}) \to H^k(X_t; \mathbb{Q})\right) = \operatorname{im}\left(H^k(\mathcal{X}; \mathbb{Q}) \to H^k(X_t; \mathbb{Q})\right) = H^k(X_t; \mathbb{Q})^{\pi_1(S,t)},$$

and this subspace is a sub-Hodge structure of $H^k(X_t; \mathbb{Q})$.

Proof. The Leray spectral sequence $E_2^{a,b} = H^a(S, R^b f_*\mathbb{Q}) \Rightarrow H^{a+b}(\mathcal{X}; \mathbb{Q})$ of a smooth projective morphism degenerates at E_2 [Del68]. Its edge map $H^k(\mathcal{X}) \to E_\infty^{0,k} \subseteq E_2^{0,k} = H^0(S, R^k f_*\mathbb{Q}) \subseteq H^k(X_t)$ is the restriction to the fibre, and by degeneration $E_\infty^{0,k} = E_2^{0,k}$, so the restriction

is surjective onto $H^0(S, R^k f_*\mathbb{Q})$. Since $R^k f_*\mathbb{Q}$ is a local system (Theorem 51.2.3) and S is connected, $H^0(S, R^k f_*\mathbb{Q})$ is the space of $\pi_1(S,t)$-invariants in the stalk $H^k(X_t;\mathbb{Q})$. This proves the second equality.

The restriction $r\colon H^k(\mathcal{X};\mathbb{Q}) \to H^k(X_t;\mathbb{Q})$ is induced by the inclusion of nonsingular varieties $X_t \to \mathcal{X}$, so it is a morphism of mixed Hodge structures (Theorem 74.4.3(2)), and its target is pure of weight k because X_t is nonsingular and projective. By Lemma 74.3.3(2), $r(H^k(\mathcal{X})) = r(W_kH^k(\mathcal{X}))$, and $W_kH^k(\mathcal{X})$ is the image of $H^k(\bar{\mathcal{X}})$ by Corollary 74.6.1(1). Since restriction from $\bar{\mathcal{X}}$ to X_t factors through $\mathcal{X}$, this proves the first equality. Finally, the image of $H^k(\bar{\mathcal{X}}) \to H^k(X_t)$ is the image of a morphism of pure Hodge structures of weight k, hence a sub-Hodge structure (Theorem 66.3.3). □

Remark 74.6.3. Theorem 74.6.2 is the proof of Theorem 70.4.1 promised in Chapter 70. The only input not proved in this book is the degeneration of the Leray spectral sequence, which Deligne deduces from the hard Lefschetz theorem (Chapter 49) applied to the fibres [Del68]. The coniveau filtration is handled in the same way: $N^cH^k(X)$ is a sum of kernels of restrictions $H^k(X) \to H^k(X \setminus Z)$, which by Corollary 74.6.1(2) are images of Gysin maps and so sub-Hodge structures (Proposition 68.4.5).

Proposition 74.6.4. *Keep the notation of Corollary 74.6.1, and let $\alpha \in \mathrm{Hdg}^p(\bar{X})$. Suppose that*

1. *there is an algebraic cycle V on $\bar{X}$ with $\mathbb{Q}$-coefficients such that $\alpha|_U = \mathrm{cl}(V)|_U$ in $H^{2p}(U;\mathbb{Q})$, and*
2. *for each i with $c_i \le p$, the Hodge conjecture holds for $\tilde{Z}_i$ in codimension $p - c_i$.*

Then α is algebraic.

Proof. Put $\beta = \alpha - \mathrm{cl}(V)$. It is a Hodge class, by Corollary 63.4.2, and $\beta|_U = 0$. By Corollary 74.6.1(2), β lies in the image of

$$g_* = \sum_i g_{i*}\colon A = \bigoplus_i H^{2p-2c_i}(\tilde{Z}_i;\mathbb{Q})(-c_i) \longrightarrow H^{2p}(\bar{X};\mathbb{Q}),$$

where the summands with $c_i > p$ are zero. Each Gysin map g_{i*} is a morphism of Hodge structures of weight $2p$: it is Poincaré dual to pushforward in homology, and it maps forms of type (a,b) to currents of type $(a+c_i, b+c_i)$. The source A is a direct sum of polarisable Hodge structures of weight $2p$ (Theorem 66.5.2). By Theorem 66.5.3, $\ker g_*$ has a complement A' in A that is a sub-Hodge structure, and $g_*|_{A'}\colon A' \to g_*(A)$ is an isomorphism of Hodge structures onto the sub-Hodge structure $g_*(A)$. The preimage $\gamma = (\gamma_i) \in A'$ of β is then a Hodge class, since isomorphisms of Hodge structures preserve Hodge classes (Proposition 66.6.2); that is, $\gamma_i \in \mathrm{Hdg}^{p-c_i}(\tilde{Z}_i)$. By hypothesis (2), $\gamma_i = \mathrm{cl}(V_i)$ for cycles V_i on $\tilde{Z}_i$ with $\mathbb{Q}$-coefficients. By compatibility of the cycle class with proper pushforward (Theorem 63.3.1(2)), $\beta = \sum_i g_{i*}\mathrm{cl}(V_i) = \mathrm{cl}(\sum_i g_{i*}V_i)$. Hence $\alpha = \mathrm{cl}(V + \sum_i g_{i*}V_i)$ is algebraic. □

Corollary 74.6.5. *Assume that the Hodge conjecture holds for all nonsingular projective varieties of dimension $< n$, and let $\bar{X}$ be an irreducible nonsingular projective variety of dimension n and $U \subseteq \bar{X}$ a nonempty open subset. Then a Hodge class $\alpha \in \mathrm{Hdg}^p(\bar{X})$ is algebraic if and only if $\alpha|_U$ is the restriction of the class of an algebraic cycle on $\bar{X}$.*

Proof. One direction is trivial. For the other, since $\bar{X}$ is irreducible and U is nonempty, every component of $\bar{X} \setminus U$ has codimension $c_i \ge 1$, so $\dim \tilde{Z}_i < n$ and hypothesis (2) of Proposition 74.6.4 holds. □

Corollary 74.6.6. *Assume that the Hodge conjecture holds for all nonsingular projective varieties of dimension $< n$, and let $\bar{X}$ and $\bar{X}'$ be birational irreducible nonsingular projective varieties of dimension n. Then the Hodge conjecture holds for $\bar{X}$ if and only if it holds for $\bar{X}'$.*

Proof. Choose nonempty open subsets $U \subseteq \bar{X}$ and $U' \subseteq \bar{X}'$ and an isomorphism $\phi\colon U' \to U$; we identify U and U' through ϕ. Suppose the Hodge conjecture holds for $\bar{X}'$, and let $\alpha \in \mathrm{Hdg}^p(\bar{X})$. The restriction $\alpha|_U$ lies in $W_{2p}H^{2p}(U;\mathbb{Q})$ by Corollary 74.6.1(1), which is a pure Hodge structure of weight $2p$, since the weights of $H^{2p}(U)$ are $\geq 2p$ (Theorem 74.4.3(3)). As the restriction is a morphism of mixed Hodge structures, $\alpha|_U$ is rational and lies in F^p, so it is a Hodge class of this pure structure (Proposition 66.2.2: in weight $2p$, $F^p \cap \overline{F^p}$ is the (p,p)-part). The restriction $r'\colon H^{2p}(\bar{X}';\mathbb{Q}) \to W_{2p}H^{2p}(U;\mathbb{Q})$ is a surjective morphism of pure Hodge structures of weight $2p$, by Corollary 74.6.1(1) applied to $\bar{X}'$. As in the proof of Proposition 74.6.4, semisimplicity (Theorem 66.5.3) gives a Hodge class $\alpha' \in \mathrm{Hdg}^p(\bar{X}')$ with $r'(\alpha') = \alpha|_U$. By hypothesis $\alpha' = \mathrm{cl}(V')$ for a cycle V' on $\bar{X}'$ with $\mathbb{Q}$-coefficients. Let V be the closure in $\bar{X}$ of the cycle $V'|_U$. Since the cycle class commutes with restriction to open subsets (Theorem 63.3.1(2)), $\mathrm{cl}(V)|_U = \mathrm{cl}(V'|_U) = \alpha'|_U = \alpha|_U$. By Corollary 74.6.5, α is algebraic. The converse follows by symmetry. □

Remark 74.6.7. The arguments of Proposition 74.6.4 and Corollary 74.6.6 use semisimplicity to lift Hodge classes along surjections, and this is a statement about $\mathbb{Q}$-structures. Over $\mathbb{Z}$ an integral Hodge class in the image of a morphism need not be the image of an integral Hodge class, and the integral analogues must be proved by other means: for classes of degree 4, the birational invariance of the defect $Z^4(X)$ is a theorem of Colliot-Thélène and Voisin (Theorem 73.2.8).

74.7 Exercises

Exercise 74.7.1. Compute the mixed Hodge structure on $H^\bullet((\mathbb{C}^*)^n;\mathbb{Q})$.

Solution. By the Künneth formula, which is compatible with mixed Hodge structures (Theorem 74.5.1), and Example 74.2.2(2), $H^k((\mathbb{C}^*)^n) \cong \Lambda^k(\mathbb{Q}(-1)^n) \cong \mathbb{Q}(-k)^{\binom{n}{k}}$, pure of weight $2k$ and type (k,k), spanned by the products of the classes $\frac{1}{2\pi i}\frac{dz_j}{z_j}$. The weights are the largest allowed by Theorem 74.4.3(3). □

Exercise 74.7.2. Show that the Kummer structure K_q of Example 74.2.2(4) is a mixed Hodge structure with $\mathrm{Gr}^W_{-2}K_q \cong \mathbb{Q}(1)$ and $\mathrm{Gr}^W_0 K_q \cong \mathbb{Q}(0)$, and that $K_q \cong \mathbb{Q}(1) \oplus \mathbb{Q}(0)$ if and only if q is a root of unity.

Solution. The only nonzero graded pieces are $\mathrm{Gr}^W_{-2} = \mathbb{Q}e_1$ and $\mathrm{Gr}^W_0 = H/\mathbb{Q}e_1$. On Gr^W_{-2} the induced filtration is $F^{-1} = \mathbb{C}e_1$ and $F^0 = F^0 \cap \mathbb{C}e_1 = 0$, since $e_0 - \lambda e_1 \notin \mathbb{C}e_1$; this is the Hodge filtration of type $(-1,-1)$, that of $\mathbb{Q}(1)$ (identify e_1 with $2\pi i$). On Gr^W_0 the image of F^0 is everything and $F^1 = 0$, which is type $(0,0)$. So K_q is a mixed Hodge structure. An isomorphism $K_q \cong \mathbb{Q}(1) \oplus \mathbb{Q}(0)$ maps $W_{-2}K_q$ onto $\mathbb{Q}(1)$, so the preimage of $\mathbb{Q}(0)$ is a complement of W_{-2} that is a sub-mixed Hodge structure; by Counterexample 74.2.3 one exists if and only if $\lambda = \log q/2\pi i \in \mathbb{Q}$. Writing $\lambda = a/b$ with $b \geq 1$, this says $q^b = e^{2\pi i a} = 1$; conversely if $q^b = 1$ then $\log q \in 2\pi i \frac{1}{b}\mathbb{Z}$. □

Exercise 74.7.3. Let X be a nonsingular variety. Show that $h^{p,q}(H^k(X)) = 0$ unless $0 \leq p, q \leq k$ and $k \leq p+q \leq 2k$.

Solution. The condition on $p+q$ is Theorem 74.4.3(3). The complex $\Omega_{\bar{X}}^{\geq k+1}(\log D)$ is zero in degrees $\leq k$, so its hypercohomology vanishes in degree k, and $F^{k+1}H^k(X;\mathbb{C}) = 0$. Hence F^{k+1} vanishes on every Gr_m^W and $h^{p,q} = 0$ for $p > k$; by the symmetry $h^{p,q} = h^{q,p}$ of pure Hodge structures, also for $q > k$. Finally $F^0 = H_{\mathbb{C}}$ because $\Omega^{\geq 0}(\log D)$ is the whole complex, so $h^{p,q} = 0$ for $p < 0$, and for $q < 0$ by symmetry. □

Exercise 74.7.4. Let E be an elliptic curve with origin 0 and $a \in E$, $a \neq 0$. Show that $H^1(E \setminus \{0\})$ is pure of weight 1, that $H^1(E \setminus \{0, a\})$ has $\mathrm{Gr}_1^W \cong H^1(E)$ and $\mathrm{Gr}_2^W \cong \mathbb{Q}(-1)$, and compute the Hodge numbers of both.

Solution. By Example 74.4.4 with $g = 1$: for $s = 1$, $\mathrm{Gr}_2^W \cong \mathbb{Q}(-1)^0 = 0$, so $H^1(E \setminus \{0\}) = H^1(E)$, with $h^{1,0} = h^{0,1} = 1$. For $s = 2$, $\mathrm{Gr}_1^W = H^1(E)$ and $\mathrm{Gr}_2^W \cong \mathbb{Q}(-1)$, so $h^{1,0} = h^{0,1} = h^{1,1} = 1$ and $\dim H^1 = 3$. (Whether this extension splits depends on a: it splits if and only if a is a torsion point of E, by Carlson's description of extensions [Car80]; we do not prove this.) □

Chapter 75

Absolute Hodge Classes

75.1 Introduction

The cohomology of a nonsingular projective variety over $\mathbb{C}$ comes in several flavours: Betti cohomology with its rational structure, algebraic de Rham cohomology with its Hodge filtration, and ℓ-adic cohomology with, for varieties defined over smaller fields, a Galois action. The Hodge conjecture concerns classes that are special in the first two at once. An automorphism σ of the field $\mathbb{C}$, in general wildly discontinuous, transforms a variety X into a *conjugate variety* X^σ and acts on de Rham and ℓ-adic cohomology, but not on Betti cohomology. Classes of algebraic cycles are carried to classes of algebraic cycles, so they remain rational and of type (p,p) after conjugation. Deligne called a Hodge class *absolute* if it shares this property. Absolute Hodge classes sit between algebraic classes and Hodge classes, and, unlike algebraic classes, one can prove things about them: Deligne's theorem that all Hodge classes on abelian varieties are absolute (Theorem 75.7.1) is the strongest general result in the direction of the Hodge conjecture. References: [Del82], [CS14], [Voi07], and the source *The Hodge Conjecture* (Chapter 8).

Prerequisites. Chapter 68 (The Hodge Conjecture), Chapter 65 (Étale Cohomology), Chapter 58 (Analytic Spaces and GAGA), Chapter 67 (Mumford–Tate Groups), Chapter 72 (Known Cases and Evidence) and Chapter 74 (Mixed Hodge Structures).

Conventions. In this chapter X is a nonsingular projective variety over $\mathbb{C}$ of dimension n, and ℓ runs over all primes. We write $\mathbb{Q}(p) = (2\pi i)^p\mathbb{Q} \subseteq \mathbb{C}$ and $H^k_B(X)(p) = H^k(X^{\mathrm{an}};\mathbb{Q}) \otimes \mathbb{Q}(p) \subseteq H^k(X^{\mathrm{an}};\mathbb{C})$ for twisted Betti cohomology, $H^k_{\mathrm{dR}}(X) = H^k_{\mathrm{dR}}(X/\mathbb{C})$ (Definition 58.5.1), and $H^k_\ell(X)(p) = H^k_{\mathrm{et}}(X, \mathbb{Q}_\ell(p))$ (Definition 65.4.1; here X is already over an algebraically closed field).

75.2 Conjugate varieties and comparison isomorphisms

Definition 75.2.1. Let $\sigma \in \mathrm{Aut}(\mathbb{C})$. The *conjugate variety* is $X^\sigma = X \times_{\operatorname{Spec}\mathbb{C},\sigma} \operatorname{Spec}\mathbb{C}$, the base change of X along the morphism $\operatorname{Spec}\mathbb{C} \to \operatorname{Spec}\mathbb{C}$ induced by σ. The projection $\sigma_X\colon X^\sigma \to X$ is an isomorphism of schemes, but not of $\mathbb{C}$-schemes. If $X \subseteq \mathbb{P}^N$ is cut out by homogeneous polynomials f_j, then X^σ is cut out by the polynomials f_j^σ obtained by applying σ to the coefficients, and on closed points σ_X^{-1} is $(x_0 : \cdots : x_N) \mapsto (\sigma(x_0) : \cdots : \sigma(x_N))$.

Example 75.2.2. Let E be the elliptic curve $y^2 = x^3 + ax + b$. Then E^σ is $y^2 = x^3 + \sigma(a)x + \sigma(b)$, so $j(E^\sigma) = \sigma(j(E))$, and $E^\sigma \cong E$ if and only if σ fixes $j(E)$. If $j(E)$ is transcendental, then $\sigma(j(E))$ can be any transcendental number, and E has uncountably many pairwise non-isomorphic conjugates. If E is defined over $\mathbb{Q}$, then $E^\sigma = E$ for every σ, but σ_E is a discontinuous bijection of $E(\mathbb{C})$ for $\sigma \neq \mathrm{id}$, complex conjugation.

Definition 75.2.3. For $\sigma \in \mathrm{Aut}(\mathbb{C})$ define

1. $\sigma_*\colon H^k_{\mathrm{dR}}(X) \to H^k_{\mathrm{dR}}(X^\sigma)$, the σ-linear isomorphism induced by $\sigma_X^*\Omega^\bullet_{X/\mathbb{C}} = \Omega^\bullet_{X^\sigma/\mathbb{C}}$ (Corollary 58.5.4); it maps $F^pH^k_{\mathrm{dR}}(X)$ onto $F^pH^k_{\mathrm{dR}}(X^\sigma)$;
2. $\sigma_* = \sigma_X^*\colon H^k_\ell(X)(p) \to H^k_\ell(X^\sigma)(p)$, the pullback in étale cohomology along the isomorphism of schemes σ_X, which identifies the sheaves μ_{ℓ^m}; it is $\mathbb{Q}_\ell$-linear.

Remark 75.2.4. There is no map $H^k_B(X) \to H^k_B(X^\sigma)$ for general σ: σ_X is not continuous for the classical topology unless σ is the identity or complex conjugation, and X^{an} and $(X^\sigma)^{\mathrm{an}}$ need not even be homeomorphic (Remark 58.5.5). The *comparison isomorphisms*

$$c_{\mathrm{dR}}\colon H^k_B(X)(p) \otimes_{\mathbb{Q}} \mathbb{C} \xrightarrow{\sim} H^k_{\mathrm{dR}}(X), \qquad c_\ell\colon H^k_B(X)(p) \otimes_{\mathbb{Q}} \mathbb{Q}_\ell \xrightarrow{\sim} H^k_\ell(X)(p)$$

of Theorem 58.5.2 and Theorem 65.4.3 (the second uses $\mathbb{Z}(1) \otimes \mathbb{Z}/\ell^m \cong \mu_{\ell^m}$, $2\pi i/\ell^m \mapsto e^{2\pi i/\ell^m}$) give two further structures on $H^k_{\mathrm{dR}}(X)$ and $H^k_\ell(X)$: a $\mathbb{Q}$-structure and the complex conjugation it defines. The map σ_* preserves the algebraic structure (the Hodge filtration, cup products, algebraic cycle classes) but has no reason to preserve these transcendental ones. In particular it preserves F^p but not $\overline{F^q}$, so it need not map $H^{p,q}(X)$ to $H^{p,q}(X^\sigma)$.

Proposition 75.2.5. *The Betti numbers, the Hodge numbers, the dimension of* $\mathrm{Hdg}^1(X)$ *and the Picard number of* X *and* X^σ *agree.*

Proof. Betti and Hodge numbers: Corollary 58.5.4. For divisors, σ_X^* gives a bijection between divisors on X and divisors on X^σ that preserves linear and algebraic equivalence, since these are defined algebraically; so the Néron–Severi groups are isomorphic and the Picard numbers agree. By the Lefschetz $(1,1)$-theorem (Theorem 69.3.1), $\dim \mathrm{Hdg}^1(X)$ is the Picard number. □

75.3 Absolute Hodge classes

Definition 75.3.1 (Deligne)**.** Write $H^{2p}_{\mathbb{A}}(X)(p) = H^{2p}_{\mathrm{dR}}(X) \times \prod_\ell H^{2p}_\ell(X)(p)$ and $t = (t_{\mathrm{dR}}, (t_\ell)_\ell)$ for its elements.

1. t is a *Hodge cycle* on X if there is $t_B \in H^{2p}_B(X)(p)$ with $c_{\mathrm{dR}}(t_B) = t_{\mathrm{dR}}$ and $c_\ell(t_B) = t_\ell$ for all ℓ, and $t_{\mathrm{dR}} \in F^pH^{2p}_{\mathrm{dR}}(X)$.
2. t is an *absolute Hodge cycle* if $\sigma_* t = (\sigma_* t_{\mathrm{dR}}, (\sigma_* t_\ell)_\ell)$ is a Hodge cycle on X^σ for every $\sigma \in \mathrm{Aut}(\mathbb{C})$.

The absolute Hodge cycles form a $\mathbb{Q}$-vector space $C^p_{\mathrm{AH}}(X)$: σ_* is additive and fixes rational scalars.

Lemma 75.3.2. *The map* $t \mapsto (2\pi i)^{-p} t_B$ *is an isomorphism from the* $\mathbb{Q}$*-vector space of Hodge cycles on* X *onto* $\mathrm{Hdg}^p(X)$*. Its inverse sends* α *to* $(c_{\mathrm{dR}}((2\pi i)^p\alpha), (c_\ell((2\pi i)^p\alpha))_\ell)$*.*

Proof. t_B is determined by t, because c_ℓ is injective. Put $\alpha = (2\pi i)^{-p} t_B \in H^{2p}(X^{\mathrm{an}}; \mathbb{Q})$. Since F^p is a complex subspace, $t_{\mathrm{dR}} \in F^p$ if and only if $\alpha \in F^p$. A rational class is fixed by complex conjugation, so $\alpha \in F^p$ if and only if $\alpha \in F^p \cap \overline{F^p} = H^{p,p}$ (Proposition 66.2.2), that is, $\alpha \in \mathrm{Hdg}^p(X)$. The inverse map is clear. □

We write $\mathrm{Hdg}^p_{\mathrm{AH}}(X) \subseteq \mathrm{Hdg}^p(X)$ for the image of $C^p_{\mathrm{AH}}(X)$, and call its elements *absolute Hodge classes*.

Proposition 75.3.3. *Classes of algebraic cycles are absolute Hodge:* $Z^p_{\mathrm{alg}}(X) \subseteq \mathrm{Hdg}^p_{\mathrm{AH}}(X) \subseteq \mathrm{Hdg}^p(X)$.

Proof. Let Z be a cycle of codimension p with $\mathbb{Q}$-coefficients. There are cycle classes $\mathrm{cl}_{\mathrm{dR}}(Z) \in F^p H^{2p}_{\mathrm{dR}}(X)$ and $\mathrm{cl}_\ell(Z) \in H^{2p}_\ell(X)(p)$ (Theorem 65.4.2(4)), both defined algebraically, and with suitable sign conventions they are the images of $(2\pi i)^p \mathrm{cl}(Z)$ under c_{dR} and c_ℓ: for ℓ-adic cohomology this is Theorem 65.4.3, and for de Rham cohomology it is [Del82]. So $\mathrm{cl}_{\mathbb{A}}(Z) = (\mathrm{cl}_{\mathrm{dR}}(Z), (\mathrm{cl}_\ell(Z))_\ell)$ is a Hodge cycle. Since the cycle classes are defined algebraically, they are compatible with transport of structure along the isomorphism of schemes σ_X: $\sigma_* \mathrm{cl}_{\mathbb{A}}(Z) = \mathrm{cl}_{\mathbb{A}}(Z^\sigma)$, where $Z^\sigma = \sigma_X^{-1}(Z)$ is a cycle on X^σ. By the same argument on X^σ, this is a Hodge cycle. The second inclusion is the case $\sigma = \mathrm{id}$. □

Remark 75.3.4. The factor $(2\pi i)^p$ is forced: already for a point on $\mathbb{P}^1$, the de Rham class is $\pm 2\pi i$ times the Betti class (Exercise 75.9.2). This is why the rational structure relevant for cycle classes of codimension p is $H^{2p}_B(X)(p)$ and not $H^{2p}(X^{\mathrm{an}}; \mathbb{Q})$.

Conjecture 75.3.5 (Absolute Hodge conjecture)**.** *Every Hodge class is absolute Hodge:* $\mathrm{Hdg}^p_{\mathrm{AH}}(X) = \mathrm{Hdg}^p(X)$.

By Proposition 75.3.3, the Hodge conjecture for X and X^σ in all codimensions implies Conjecture 75.3.5. The converse is not expected to be provable directly: being absolute Hodge is a condition on cohomology and its comparison isomorphisms, and it never produces a subvariety.

Remark 75.3.6. Several variants are in use. Omitting the ℓ-adic components gives *de Rham–Betti absolute* classes, which is the version in the source *The Hodge Conjecture*; Deligne's theorem below holds for the full definition, and all known results hold for both. André's *motivated cycles* [And96b] are a smaller class, generated by algebraic cycles and the inverse Lefschetz operators (Corollary 75.4.3); they are absolute Hodge.

75.4 Operations on absolute Hodge classes

Proposition 75.4.1. *Let $f : X \to Y$ be a morphism of nonsingular projective varieties, of relative dimension $d = \dim X - \dim Y$. Then absolute Hodge classes are preserved by*

1. *pullback $f^* : \mathrm{Hdg}^p(Y) \to \mathrm{Hdg}^p(X)$ and pushforward $f_* : \mathrm{Hdg}^p(X) \to \mathrm{Hdg}^{p-d}(Y)$;*
2. *cup product, so $\bigoplus_p \mathrm{Hdg}^p_{\mathrm{AH}}(X)$ is a $\mathbb{Q}$-subalgebra of $\bigoplus_p \mathrm{Hdg}^p(X)$;*
3. *the Künneth projections $H^{2p}(X \times Y) \to H^i(X) \otimes H^{2p-i}(Y)$: the Künneth components of an absolute Hodge class on $X \times Y$ are absolute Hodge;*
4. *correspondences: if $\gamma \in \mathrm{Hdg}^q_{\mathrm{AH}}(X \times Y)$ then $\gamma_* = \mathrm{pr}_{2*}(\mathrm{pr}_1^*(-) \cup \gamma)$ maps absolute Hodge classes on X to absolute Hodge classes on Y.*

Proof. Each operation is defined in Betti, de Rham and ℓ-adic cohomology. In de Rham and ℓ-adic cohomology it is defined algebraically, from pullbacks, cup products and trace maps, so it commutes with σ_*: for instance $\sigma_*(f^*t) = (f^\sigma)^*(\sigma_* t)$, where $f^\sigma : X^\sigma \to Y^\sigma$ is the conjugate morphism. The operations are compatible with the comparison isomorphisms c_{dR} and c_ℓ (with the Tate twists indicated: f_* goes from $H^{2p}(X)(p)$ to $H^{2p-2d}(Y)(p-d)$), and on Betti cohomology they are morphisms of Hodge structures, so they map Hodge cycles to Hodge cycles [Del82]. Given an absolute Hodge cycle t on Y and $\sigma \in \mathrm{Aut}(\mathbb{C})$, the cycle $\sigma_*(f^*t) = (f^\sigma)^*(\sigma_* t)$ is the image of the Hodge cycle $\sigma_* t$ on Y^σ under an operation that

preserves Hodge cycles, hence a Hodge cycle. The other cases are identical: for (3), the Künneth decomposition is built from pr_1^*, pr_2^* and cup product, so the projections commute with σ_* and preserve Hodge cycles, because the Künneth decomposition of $H^{2p}(X \times Y)$ is a decomposition into sub-Hodge structures; and (4) is a composite of (1) and (2). □

Corollary 75.4.2. *The Künneth components $\pi_i \in H^{2n-i}(X) \otimes H^i(X) \subseteq H^{2n}(X \times X)$ of the class of the diagonal are absolute Hodge classes.*

Proof. The class of the diagonal is algebraic, hence absolute Hodge by Proposition 75.3.3; apply Proposition 75.4.1(3). □

Corollary 75.4.3. *Let $h \in H^2(X)$ be the class of an ample divisor and Λ the inverse Lefschetz operator of $L = h \cup -$ (Proposition 49.2.4). The class $\lambda \in H^{2n-2}(X \times X)(n-1)$ of Λ, obtained through the Künneth decomposition and Poincaré duality, is an absolute Hodge class.*

Proof. Let $\sigma \in \mathrm{Aut}(\mathbb{C})$ and let h^σ be the class of the conjugate divisor on X^σ, with operators L^σ and Λ^σ. In de Rham cohomology, σ_* intertwines L with L^σ (it preserves cup products and cycle classes) and preserves degrees. Hence $\sigma_*\Lambda\sigma_*^{-1}$ satisfies the equations $[L^\sigma, f] = h_{\deg}$, $[h_{\deg}, f] = -2f$, where $h_{\deg}$ acts by $k - n$ in degree k, and by the uniqueness in Proposition 49.2.4 it equals $\Lambda^\sigma_{\mathrm{dR}}$. The same argument applies in ℓ-adic cohomology, using hard Lefschetz there (Theorem 65.4.2(6)). On the other hand, the Betti operator Λ^σ_B is defined over $\mathbb{Q}$ (Proposition 49.2.4), and its images under c_{dR} and c_ℓ satisfy the defining equations, so they are $\Lambda^\sigma_{\mathrm{dR}}$ and Λ^σ_ℓ by uniqueness again. Finally, Λ^σ_B is a morphism of Hodge structures of type $(-1,-1)$. Indeed, let $h(z)$, $z \in \mathbb{S}(\mathbb{R}) = \mathbb{C}^*$, be the action of the Deligne torus on $H^\bullet(X^\sigma;\mathbb{R})$ (Proposition 66.4.1). Since L^σ maps $H^{p,q}$ to $H^{p+1,q+1}$, $h(z)L^\sigma h(z)^{-1} = |z|^{-2}L^\sigma$, and $h(z)$ commutes with the grading operator. So $|z|^{-2}h(z)\Lambda^\sigma_B h(z)^{-1}$ solves the defining equations for L^σ, and by uniqueness $h(z)\Lambda^\sigma_B h(z)^{-1} = |z|^2\Lambda^\sigma_B$ for all z, which says exactly that Λ^σ_B maps $H^{p,q}$ to $H^{p-1,q-1}$. So the class of Λ^σ is a Hodge class on $X^\sigma \times X^\sigma$ (Proposition 66.6.2), and $\sigma_*\lambda$ is a Hodge cycle. □

Remark 75.4.4. Grothendieck's Lefschetz standard conjecture asks for λ to be algebraic (Chapter 77). Corollary 75.4.3 proves the weaker statement that λ is absolute Hodge, and the Hodge conjecture would bridge the gap. This is typical: statements whose algebraic form is open are often theorems in the absolute Hodge setting.

75.5 Galois action and fields of definition

Every X is defined over a subfield of $\mathbb{C}$ finitely generated over $\mathbb{Q}$, since it is cut out by finitely many polynomials: $X = X_0 \times_k \mathbb{C}$ with $k \subseteq \mathbb{C}$ finitely generated over $\mathbb{Q}$ and X_0 a nonsingular projective k-variety. Let $\bar{k}$ be the algebraic closure of k in $\mathbb{C}$ and $G_k = \mathrm{Gal}(\bar{k}/k)$. For $\tau \in \mathrm{Aut}(\mathbb{C}/k)$ we have $X^\tau = X$ canonically, so τ_* acts on $H^{2p}_{\mathbb{A}}(X)(p)$. On $H^{2p}_\ell(X)(p) \cong H^{2p}_{\mathrm{et}}(X_0 \times_k \bar{k}, \mathbb{Q}_\ell(p))$, where the isomorphism is invariance of étale cohomology of proper varieties under extension of algebraically closed fields [Mil80, Chapter VI], the action of τ is the Galois action of $\tau|_{\bar{k}} \in G_k$.

Proposition 75.5.1. *In this situation, $\mathrm{Aut}(\mathbb{C}/k)$ maps $C^p_{\mathrm{AH}}(X)$ to itself, and it acts on it through a finite quotient of G_k. Consequently there is a finite extension $k' \subseteq \bar{k}$ of k such that for every absolute Hodge cycle t, the ℓ-adic components $t_\ell \in H^{2p}_{\mathrm{et}}(X_0 \times_k \bar{k}, \mathbb{Q}_\ell(p))$ are fixed by $G_{k'}$.*

Proof. If t is absolute Hodge and $\tau \in \mathrm{Aut}(\mathbb{C}/k)$, then $\tau_* t$ is a Hodge cycle on $X^\tau = X$, and $\sigma_*(\tau_* t) = (\sigma\tau)_* t$ is a Hodge cycle on $X^{\sigma\tau}$ for every σ; so $\tau_* t$ is absolute Hodge. The space

$V = C^p_{\mathrm{AH}}(X)$ is finite-dimensional over $\mathbb{Q}$, since $t \mapsto t_B$ embeds it in $H^{2p}_B(X)(p)$. Fix a prime ℓ. The map $V \otimes \mathbb{Q}_\ell \to H^{2p}_\ell(X)(p)$, $t \otimes a \mapsto a t_\ell$, is injective, because it is the restriction of the comparison isomorphism c_ℓ to $V_B \otimes \mathbb{Q}_\ell$, where $V_B = \{t_B : t \in V\}$. It is equivariant, and the action of τ on the right depends only on $\tau|_{\bar{k}}$. So the action on V is given by a homomorphism $\rho\colon G_k \to \mathrm{GL}(V)$, and ρ is continuous for the ℓ-adic topology on $\mathrm{GL}(V \otimes \mathbb{Q}_\ell)$, because the Galois action on ℓ-adic cohomology is continuous. The image $\rho(G_k)$ is therefore a compact subgroup of $\mathrm{GL}(V \otimes \mathbb{Q}_\ell)$, and it is contained in the countable group $\mathrm{GL}(V)$. A countable compact Hausdorff group is finite: by the Baire category theorem one of its points is not nowhere dense, hence is open, so by homogeneity every point is open, and a discrete compact space is finite. Hence $\ker \rho$ is an open subgroup of G_k, of the form $G_{k'}$ with k'/k finite, and it fixes every t, hence every t_ℓ. □

Corollary 75.5.2. *If $X = X_0 \times_k \mathbb{C}$ as above, the ℓ-adic components of absolute Hodge classes are* Tate classes*: classes in $H^{2p}_{\mathrm{et}}(X_0 \times_k \bar{k}, \mathbb{Q}_\ell(p))$ fixed by an open subgroup of G_k. If the absolute Hodge conjecture holds for X, then every Hodge class of X gives, through c_ℓ, a Tate class.*

Proof. The first statement is Proposition 75.5.1. For the second, a Hodge class α determines the Hodge cycle of Lemma 75.3.2, which is absolute Hodge by hypothesis, and its ℓ-adic component is $c_\ell((2\pi i)^p \alpha)$. □

Remark 75.5.3. Corollary 75.5.2 is the bridge from Hodge theory to arithmetic. Algebraic classes of cycles defined over k' are Tate classes for $G_{k'}$ (Theorem 65.4.2(4)), and the Tate conjecture predicts the converse (Chapter 76). Absolute Hodge classes give Tate classes without producing cycles.

75.6 Principle B

Theorem 75.6.1 (Deligne's Principle B)**.** *Let $f\colon \mathcal{X} \to S$ be a smooth projective morphism of nonsingular complex varieties with S connected, and let t be a global section of the local system $R^{2p} f_* \mathbb{Q}(p)$ such that $t_s \in H^{2p}_B(X_s)(p)$ is (the Betti component of) a Hodge cycle for every $s \in S$. If t_{s_0} is absolute Hodge for one point $s_0 \in S$, then t_s is absolute Hodge for every $s \in S$.*

Sketch of proof. Fix $\sigma \in \mathrm{Aut}(\mathbb{C})$. The relative de Rham cohomology $\mathcal{H}^{2p}_{\mathrm{dR}}(\mathcal{X}/S)$ is an algebraic vector bundle on S with an algebraic flat connection, the Gauss–Manin connection (Definition 70.2.1), which is defined algebraically by Katz and Oda [KO68]; hence conjugation by σ carries it to the Gauss–Manin connection of $f^\sigma\colon \mathcal{X}^\sigma \to S^\sigma$. We use three facts. First, the de Rham components $t_{\mathrm{dR},s}$ form a flat section of $\mathcal{H}^{2p}_{\mathrm{dR}}$, and flat sections of the Gauss–Manin connection are algebraic, by its regularity; so $\sigma_* t_{\mathrm{dR}}$ is a flat algebraic section over S^σ. Second, the ℓ-adic components form a global section of the étale local system $R^{2p} f_* \mathbb{Q}_\ell(p)$, and σ_* carries it to a global section over S^σ. We omit the proofs of these two facts [Del82]. Third, by hypothesis, $u = \sigma_* t$ is a Hodge cycle at the point $\sigma(s_0) \in S^\sigma$, with Betti component u_B.

Since u_ℓ is a global section, it is invariant under the monodromy of S^σ; as c_ℓ is compatible with monodromy, so is $u_B \otimes 1$, and hence u_B is the value at $\sigma(s_0)$ of a global section of $R^{2p} f^\sigma_* \mathbb{Q}(p)$. Its de Rham and ℓ-adic incarnations are flat and agree with u_{dR} and u_ℓ at $\sigma(s_0)$, hence everywhere, because flat sections of a local system on a connected space are determined by one value. So u is rational at every point of S^σ. By the global invariant cycle theorem (Theorem 74.6.2), the global sections of $R^{2p} f^\sigma_* \mathbb{Q}(p)$ form a Hodge structure V that maps to each fibre by an injective morphism of Hodge structures: V is the quotient of H^{2p} of a nonsingular projective compactification of $\mathcal{X}^\sigma$ by the kernel of the restriction map to global sections, which does not depend on the fibre. The element $u_B \in V$ is of type $(0,0)$ in the fibre

at $\sigma(s_0)$, hence in V, hence in every fibre. Thus $u = \sigma_* t$ is a Hodge cycle at every point of S^σ. As σ was arbitrary, t_s is absolute Hodge for every s. □

Remark 75.6.2. Principle B is the absolute Hodge analogue of the variational Hodge conjecture (Conjecture 73.5.1), which is open for algebraic classes. The difference is that absolute Hodge cycles are defined by conditions on cohomology, which propagate along flat sections, while algebraic cycles are geometric objects whose existence on one fibre does not obviously imply existence on the others.

75.7 Deligne's theorem for abelian varieties

Theorem 75.7.1 (Deligne)**.** *Let A be a complex abelian variety. Every Hodge class on A is absolute Hodge:* $\mathrm{Hdg}^p_{\mathrm{AH}}(A) = \mathrm{Hdg}^p(A)$ *for all p. The same holds for Hodge classes in all tensor spaces $T^{m,m'}(H^1(A;\mathbb{Q}))(r)$ of weight 0, defined analogously.*

Outline of proof. The proof is [Del82]; see also [CS14]. It combines four ingredients.

1. *Principle B* (Theorem 75.6.1).

2. *Principle A.* If $t_1, \dots, t_r$ are absolute Hodge tensors on A, and $G \subseteq \mathrm{GL}(H^1(A;\mathbb{Q}))$ is the subgroup fixing them, then every tensor fixed by G is absolute Hodge. Applied when G is the Hodge group, this reduces the theorem to finding absolute Hodge tensors whose stabiliser is $\mathrm{Hg}(A)$ (Theorem 67.3.2).

3. *Weil classes are absolute Hodge.* Let K be an imaginary quadratic field and consider abelian varieties of Weil type for K (Proposition 72.3.3 treats dimension 4; the construction is the same in every even dimension), with a fixed type of polarisation. They form a connected family over which the Weil classes form a flat family of Hodge classes. For the *split* type, the family contains abelian varieties of the form $B \otimes_{\mathbb{Q}} K$, on which the Weil classes are algebraic. By Principle B the Weil classes are absolute Hodge on every member.

4. *Reduction to CM type, and the CM case.* A Hodge class t on A is fixed by $\mathrm{Hg}(A)$, hence remains Hodge on every abelian variety in the connected family parametrised by the Mumford–Tate domain of A, and this family contains abelian varieties of CM type. By Principle B it suffices to treat A of CM type. For those, the Hodge group is a torus (Proposition 67.4.2), and one shows that it is the stabiliser of the pullbacks of split Weil classes along homomorphisms from A to abelian varieties of Weil type, together with divisor classes. Principle A and (3) conclude.

□

Remark 75.7.2. Theorem 75.7.1 says that a Hodge class on an abelian variety, however transcendentally it was produced, behaves under all automorphisms of $\mathbb{C}$ as if it were algebraic. Its limit is equally clear: the Weil classes on a general abelian fourfold of Weil type are absolute Hodge but, apart from the cases listed in Chapter 72, not known to be algebraic. Deligne used the theorem, with Milne, to build an unconditional Tannakian category of motives for absolute Hodge cycles containing the motives of abelian varieties [DM82]. A deformation argument of the same kind, combined with the Kuga–Satake construction, had earlier given Deligne's proof of the Weil conjectures for K3 surfaces [Del72].

75.8 Fields of definition of Hodge loci

Lemma 75.8.1. *Let $Z \subseteq \mathbb{P}^N_{\mathbb{C}}$ be a closed subvariety. If the set $\{\sigma(Z) : \sigma \in \mathrm{Aut}(\mathbb{C}/\bar{\mathbb{Q}})\}$ is countable, then Z is defined over $\bar{\mathbb{Q}}$.*

Proof. Fix a monomial order. The homogeneous ideal of Z has a unique reduced Gröbner basis, and the field $k_0 \subseteq \mathbb{C}$ generated by its coefficients is the smallest field of definition of Z; for $\sigma \in \mathrm{Aut}(\mathbb{C})$ the reduced Gröbner basis of $\sigma(Z)$ is obtained by applying σ to the coefficients, so $\sigma(Z) = \tau(Z)$ if and only if $\sigma|_{k_0} = \tau|_{k_0}$. If $k_0 \not\subseteq \bar{\mathbb{Q}}$, it contains a transcendental element c, and for each transcendental $c' \in \mathbb{C}$ there is $\sigma \in \mathrm{Aut}(\mathbb{C}/\bar{\mathbb{Q}})$ with $\sigma(c) = c'$ (extend $c \mapsto c'$ from $\bar{\mathbb{Q}}(c)$ to $\mathbb{C}$, using transcendence bases and algebraic closures). The values $\sigma(c)$, and hence the restrictions $\sigma|_{k_0}$ and the conjugates $\sigma(Z)$, are then uncountable, a contradiction. So $k_0 \subseteq \bar{\mathbb{Q}}$. □

Proposition 75.8.2 (Voisin)**.** *Let $f : \mathcal{X} \to S$ be a smooth projective morphism of nonsingular quasi-projective varieties defined over $\bar{\mathbb{Q}}$. Let $Z \subseteq S$ be an irreducible closed subvariety, $s \in Z$ and $\alpha \in \mathrm{Hdg}^p(X_s)$ such that the parallel transport of α along every path in Z is a Hodge class, and suppose Z is maximal with this property, that is, an irreducible component of the Hodge locus (Definition 70.5.1). If α is absolute Hodge, then Z is defined over $\bar{\mathbb{Q}}$, and its conjugates are components of Hodge loci.*

Sketch of proof. Z is algebraic by the Cattani–Deligne–Kaplan theorem (Theorem 70.5.3). By Principle B, applied over the nonsingular locus of Z, the class is absolute Hodge at every point of Z. Let $\sigma \in \mathrm{Aut}(\mathbb{C}/\bar{\mathbb{Q}})$; then $\mathcal{X}^\sigma = \mathcal{X}$ and $S^\sigma = S$. As in the proof of Theorem 75.6.1, σ transports the flat family of absolute Hodge cycles over Z to a flat family of Hodge cycles over $\sigma(Z)$. So $\sigma(Z)$ is contained in a component Z' of the Hodge locus of a class, and applying the same argument to σ^{-1} and Z' shows $\sigma^{-1}(Z') \subseteq Z$, so $Z' = \sigma(Z)$. The Hodge loci of all rational classes have only countably many components in total, since there are countably many rational classes at a base point and each Hodge locus is an analytic subset with countably many components. So Z has countably many conjugates, and Lemma 75.8.1 (after embedding S in a projective space over $\bar{\mathbb{Q}}$ and taking closures) shows that Z is defined over $\bar{\mathbb{Q}}$ [Voi07]. □

Corollary 75.8.3. *In a family of abelian varieties defined over $\bar{\mathbb{Q}}$, every component of the Hodge locus is defined over $\bar{\mathbb{Q}}$.*

Proof. Combine Theorem 75.7.1 with Proposition 75.8.2. □

Remark 75.8.4. Voisin also showed a partial converse: if a component of a Hodge locus is defined over $\bar{\mathbb{Q}}$ and its conjugates are components of Hodge loci, then the Hodge classes over it are absolute Hodge, under a mild condition on the variation of Hodge structure [Voi07]. So the absolute Hodge conjecture is, to a large extent, a statement about the arithmetic of Hodge loci. For families of abelian varieties Corollary 75.8.3 reflects the fact that special subvarieties of Shimura varieties are defined over $\bar{\mathbb{Q}}$.

75.9 Exercises

Exercise 75.9.1. Let X be a nonsingular projective surface and $\sigma \in \mathrm{Aut}(\mathbb{C})$. Show that σ_* maps $\mathrm{Hdg}^1(X)$, viewed through Lemma 75.3.2, isomorphically onto $\mathrm{Hdg}^1(X^\sigma)$, and that $\mathrm{Hdg}^2(X) = H^4(X;\mathbb{Q})$ also consists of absolute Hodge classes. Conclude that the absolute Hodge conjecture holds for surfaces.

Solution. By the Lefschetz $(1,1)$-theorem, $\mathrm{Hdg}^1(X)$ is spanned by divisor classes, which are absolute Hodge (Proposition 75.3.3), and $\sigma_*\mathrm{cl}(D) = \mathrm{cl}(D^\sigma)$. The same applies to σ^{-1} on X^σ, so the map is bijective. $H^4(X;\mathbb{Q})$ is spanned by the class of a point, which is algebraic. $\mathrm{Hdg}^0 = H^0$ is spanned by the class of X. So all Hodge classes on X are algebraic, hence absolute Hodge. □

Exercise 75.9.2. On $\mathbb{P}^1$ with coordinate z, the line bundle $O(1)$ has transition function z on $U_0 \cap U_1$, and its de Rham Chern class is represented in Čech–de Rham cohomology by the closed form dz/z on $U_0 \cap U_1$. Using a partition of unity, show that the corresponding smooth 2-form η satisfies $\int_{\mathbb{P}^1} \eta = \pm 2\pi i$. Deduce that the de Rham class of a point is $\pm 2\pi i$ times the Betti class, so that it lies in $H^2_B(\mathbb{P}^1)(1)$.

Solution. Let ρ_0 be a smooth function on $\mathbb{P}^1$ equal to 1 for $|z| \le 1$ and to 0 for $|z| \ge 2$, and $\rho_1 = 1 - \rho_0$. The Čech cocycle dz/z is the difference of the 1-forms $\rho_0\, dz/z$ on U_1 and $-\rho_1\, dz/z$ on U_0 (each extended by zero), and the corresponding global 2-form is $\eta = d(\rho_0\, dz/z)$, which equals $-d(\rho_1\, dz/z)$ where both are defined, since $d(dz/z) = 0$. It vanishes for $|z| \le 1$ and for $|z| \ge 2$. By Stokes' theorem on the annulus $\frac{1}{2} \le |z| \le 3$, $\int \eta = \int_{|z|=3} \rho_0\, dz/z - \int_{|z|=1/2} \rho_0\, dz/z = 0 - 2\pi i$. The sign depends on the conventions for the Čech–de Rham isomorphism. As $\int_{\mathbb{P}^1}$ of the Betti class of a point is 1, the de Rham class is $\pm 2\pi i$ times the Betti class. □

Exercise 75.9.3. Let X and Y be nonsingular projective varieties and $\gamma \in \mathrm{Hdg}^q(X \times Y)$ an absolute Hodge class such that $\gamma_*\colon H^{2p}(X;\mathbb{Q}) \to H^{2p+2q-2\dim X}(Y;\mathbb{Q})$ is surjective. Show that if the absolute Hodge conjecture holds for X in codimension p, then it holds for Y in codimension $p + q - \dim X$.

Solution. γ_* is a surjective morphism of polarisable Hodge structures (up to a Tate twist), so by semisimplicity (Theorem 66.5.3) every Hodge class on Y in that degree is γ_* of a Hodge class on X, as in the proof of Proposition 74.6.4. By hypothesis that class is absolute Hodge, and γ_* preserves absolute Hodge classes (Proposition 75.4.1(4)). □

Exercise 75.9.4. Let C be a nonsingular projective curve and J its Jacobian. Using the Abel–Jacobi map $C^g \to J$ and Theorem 75.7.1, show that every Hodge class on C^m, for every $m \ge 1$, is absolute Hodge.

Solution. Fix a base point and let $u\colon C \to J$ be the Abel–Jacobi map; $u^*\colon H^1(J;\mathbb{Q}) \to H^1(C;\mathbb{Q})$ is an isomorphism of Hodge structures. By the Künneth formula, $H^\bullet(C^m)$ is the direct sum of the sub-Hodge structures $H^{a_1}(C) \otimes \cdots \otimes H^{a_m}(C)$, $a_i \in \{0,1,2\}$, and a Hodge class on C^m is a sum of Hodge classes in these summands, which are absolute Hodge if the summands are, by Proposition 75.4.1(3). Fix a summand, let $S = \{i : a_i = 1\}$ and $T = \{i : a_i = 2\}$. Since $H^0(C)$ is spanned by 1 and $H^2(C)$ by the class of a point, a Hodge class in the summand has the form $\mathrm{pr}_S^*(\gamma) \cup \prod_{i\in T} \mathrm{pr}_i^*[\mathrm{pt}]$ with γ a Hodge class in the summand $H^1(C)^{\otimes S}$ of $H^{|S|}(C^S)$. The pullback $(u^S)^*$ maps the corresponding Künneth summand $H^1(J)^{\otimes S}$ of $H^{|S|}(J^S)$ isomorphically onto $H^1(C)^{\otimes S}$, as a morphism of Hodge structures, so $\gamma = (u^S)^*\delta$ with δ a Hodge class on the abelian variety J^S. By Theorem 75.7.1 δ is absolute Hodge, and so is γ by Proposition 75.4.1(1). The classes $\mathrm{pr}_i^*[\mathrm{pt}]$ are algebraic, and cup products and pullbacks preserve absolute Hodge classes. □

Chapter 76

The Tate Conjecture

76.1 Introduction

The Tate conjecture is the arithmetic analogue of the Hodge conjecture. The Hodge decomposition is replaced by the action of a Galois group on ℓ-adic cohomology, and Hodge classes by *Tate classes*, the classes fixed by an open subgroup of the Galois group. Classes of algebraic cycles defined over the base field are Tate classes, and Tate conjectured the converse for varieties over finitely generated fields [Tat65]. Over finite fields there is no Hodge theory and the Tate conjecture is the only statement available; over $\mathbb{C}$ there is no interesting Galois group and the Hodge conjecture is the only statement available; over number fields both apply, and the Mumford–Tate conjecture connects them. The best-known evidence is the case of divisors on abelian varieties, which is Tate's isogeny theorem over finite fields [Tat66a] and Faltings' theorem over number fields [Fal83]. References: [Tat65], [Tot17], [Mil80], and the source *The Hodge Conjecture* (Chapter 8).

Prerequisites. Chapter 65 (Étale Cohomology), Chapter 61 (Abelian Varieties) and Chapter 75 (Absolute Hodge Classes).

Conventions. In this chapter k is a field finitely generated over its prime field, $\bar{k}$ an algebraic closure, $G_k = \mathrm{Aut}(\bar{k}/k)$, which is the Galois group of the separable closure of k in $\bar{k}$, and ℓ a prime different from the characteristic of k. X is a nonsingular projective geometrically irreducible variety over k, $\bar{X} = X \times_k \bar{k}$, and $H^i(\bar{X}, \mathbb{Q}_\ell(j))$ is ℓ-adic cohomology with the continuous action of G_k (Definition 65.4.1). For a finite extension k'/k inside $\bar{k}$ we write $G_{k'} \subseteq G_k$ for the corresponding open subgroup.

76.2 The statement

Definition 76.2.1. The space of *Tate classes of codimension p over k* is $T^p_\ell(X/k) = H^{2p}(\bar{X}, \mathbb{Q}_\ell(p))^{G_k}$. The space of *geometric Tate classes* is $T^p_\ell(\bar{X}) = \bigcup_{k'} T^p_\ell(X_{k'}/k')$, the union over finite extensions k'/k, that is, the classes fixed by some open subgroup of G_k.

Lemma 76.2.2. *Let Z be an algebraic cycle of codimension p on $\bar{X}$ with $\mathbb{Q}$-coefficients. Then Z is defined over a finite extension k' of k, and $\mathrm{cl}_\ell(Z) \in T^p_\ell(X_{k'}/k')$. If Z is defined over k, then $\mathrm{cl}_\ell(Z) \in T^p_\ell(X/k)$.*

Proof. Each component of Z is cut out by finitely many polynomials with coefficients in $\bar{k}$, which lie in a finite extension k'. For $g \in G_{k'}$, g fixes Z, and the cycle class is G_k-equivariant (Theorem 65.4.2(4)), so $g\,\mathrm{cl}_\ell(Z) = \mathrm{cl}_\ell(gZ) = \mathrm{cl}_\ell(Z)$. The same argument with G_k gives the last statement. □

Conjecture 76.2.3 (Tate)**.** *The cycle class map induces a surjection*

$$\mathrm{CH}^p(X) \otimes_{\mathbb{Z}} \mathbb{Q}_\ell \longrightarrow H^{2p}(\bar{X}, \mathbb{Q}_\ell(p))^{G_k}.$$

We write $T^p(X/k)$ for this statement, and $T^p(\bar{X})$ for the statement that the geometric Tate classes are spanned over $\mathbb{Q}_\ell$ by classes of cycles on $\bar{X}$.

Lemma 76.2.4. *If k'/k is a finite extension and $T^p(X_{k'}/k')$ holds, then $T^p(X/k)$ holds. In particular $T^p(\bar{X})$ is equivalent to $T^p(X_{k'}/k')$ for all finite k'/k.*

Proof. Let k_s be the separable closure of k in k', so that k'/k_s is purely inseparable and $G_{k'} = G_{k_s}$. First we show $T^p(X_{k_s}/k_s)$. The projection $\pi\colon X_{k'} \to X_{k_s}$ is finite and a homeomorphism, so for every subvariety $W \subseteq X_{k'}$ the preimage of $\pi(W)$ is W as a set, and $\pi^*[\pi(W)] = m[W]$ for some integer $m \geq 1$. As $\bar{X}$ is the same for k' and k_s, this gives $\mathrm{cl}_\ell(W) = \frac{1}{m}\mathrm{cl}_\ell(\pi(W))$ in $H^{2p}(\bar{X}, \mathbb{Q}_\ell(p))$. So classes of cycles on $X_{k'}$ are combinations of classes of cycles on X_{k_s}, and $T^p(X_{k_s}/k_s)$ holds.

Now let $v \in H^{2p}(\bar{X}, \mathbb{Q}_\ell(p))^{G_k}$. It is G_{k_s}-invariant, so $v = \sum_i a_i \mathrm{cl}_\ell(Z_i)$ with $a_i \in \mathbb{Q}_\ell$ and Z_i cycles on X_{k_s}. Let $g_1, \dots, g_d$ be representatives of the cosets G_k/G_{k_s}, where $d = [k_s : k]$. On G_{k_s}-invariant vectors the operator $\pi_\Gamma = \frac{1}{d}\sum_j g_j$ does not depend on the representatives, it fixes v, and $\pi_\Gamma \mathrm{cl}_\ell(Z_i) = \frac{1}{d}\mathrm{cl}_\ell(\sum_j g_j Z_i)$. The cycle $\sum_j g_j Z_i$ on $\bar{X}$ is the pullback of the pushforward of Z_i along the finite separable morphism $X_{k_s} \to X$, hence the class of a cycle defined over k. So $v = \sum_i a_i \pi_\Gamma \mathrm{cl}_\ell(Z_i)$ is a combination of classes of cycles defined over k.

For the last statement, a geometric Tate class is fixed by some $G_{k'}$, so $T^p(X_{k'}/k')$ for all k' implies $T^p(\bar{X})$. Conversely, if $T^p(\bar{X})$ holds and v is a Tate class over k', then $v = \sum a_i \mathrm{cl}_\ell(Z_i)$ with cycles Z_i on $\bar{X}$, all defined over some finite extension k'' of k' (Lemma 76.2.2). The argument above, with k'/k replaced by k''/k', used only that v is invariant under the smaller Galois group and is a combination of classes of cycles defined over the larger field; so it shows that v is a combination of classes of cycles defined over k'. □

Counterexample 76.2.5. The hypothesis that k is finitely generated cannot be dropped. Let X be a nonsingular projective surface over $\mathbb{C}$ with $h^{2,0}(X) > 0$, for instance a quartic surface in $\mathbb{P}^3$. For $k = \mathbb{C}$ the Galois group is trivial, so every class in $H^2(X, \mathbb{Q}_\ell(1))$ is invariant. But the $\mathbb{Q}_\ell$-span of divisor classes has dimension $\rho(X) \leq h^{1,1} < b_2(X)$, by Artin's comparison theorem (Theorem 65.4.3) and the Lefschetz $(1,1)$-theorem. So $T^1(X/\mathbb{C})$ fails. The same happens over $\bar{\mathbb{Q}}$, and over $\bar{\mathbb{F}}_p$ for a surface with $\rho(X) < b_2(X)$.

Remark 76.2.6. Over a finite field $k = \mathbb{F}_q$, Tate formulated a companion conjecture: the rank of the group of codimension-p cycles on X modulo numerical equivalence equals the order of the pole of the zeta function $Z(X, q^{-s})$ at $s = p$ [Tat65]. Since the eigenvalues of Frobenius on $H^{2p}(\bar{X}, \mathbb{Q}_\ell)$ have absolute value q^p [Del74a], that order is the multiplicity of the eigenvalue q^p of the geometric Frobenius on H^{2p}. This refinement is tied to $T^p(X/k)$, to the equality of homological and numerical equivalence (Chapter 77), and to the semisimplicity of the Frobenius action; see [Tot17].

76.3 Divisors on abelian varieties

For an abelian variety A over k, the *Tate module* is $T_\ell A = \varprojlim_m A[\ell^m](\bar{k})$, a free $\mathbb{Z}_\ell$-module of rank $2\dim A$ with a continuous action of G_k, and $V_\ell A = T_\ell A \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$. As in Example 65.5.1, $H^1(\bar{A}, \mathbb{Q}_\ell) \cong \mathrm{Hom}(V_\ell A, \mathbb{Q}_\ell)$ and $H^i(\bar{A}, \mathbb{Q}_\ell) \cong \Lambda^i H^1(\bar{A}, \mathbb{Q}_\ell)$, compatibly with G_k [Mil80].

Theorem 76.3.1 (Tate, Zarhin, Faltings)**.** *Let A and B be abelian varieties over k, where k is finite or of characteristic $\neq 2$. Then:*

1. *the natural map* $\mathrm{Hom}_k(A,B)\otimes_{\mathbb{Z}}\mathbb{Z}_\ell \to \mathrm{Hom}_{G_k}(T_\ell A, T_\ell B)$ *is an isomorphism;*
2. *the representation of* G_k *on* $V_\ell A$ *is semisimple.*

This is due to Tate for finite fields [Tat66a], Zarhin for finitely generated fields of characteristic $p>2$ [Zar75], and Faltings for finitely generated fields of characteristic 0, where it is the heart of his proof of the Mordell conjecture [Fal83]. Injectivity in (1), and the fact that the cokernel is torsion-free, hold over any field [Mum70]; the content is surjectivity. Over a finite field the key finiteness input is that there are only finitely many isomorphism classes of polarised abelian varieties of given dimension and degree over k, and over number fields Faltings proves this finiteness with his height.

Corollary 76.3.2. *For an abelian variety A over k (as in Theorem 76.3.1), $T^1(A/k)$ holds: the Tate classes in $H^2(\bar{A},\mathbb{Q}_\ell(1))$ are $\mathbb{Q}_\ell$-linear combinations of classes of divisors defined over k.*

Sketch of proof. We use two facts from the theory of abelian varieties [Mum70]. First, $H^2(\bar{A},\mathbb{Q}_\ell(1)) = \Lambda^2 H^1(\bar{A},\mathbb{Q}_\ell)(1)$ is the space of alternating $\mathbb{Q}_\ell$-bilinear forms $V_\ell A\times V_\ell A\to \mathbb{Q}_\ell(1)$, and through the Weil pairing $e_\ell\colon V_\ell A\times V_\ell A^\vee\to\mathbb{Q}_\ell(1)$ such a form is $E_\phi(x,y)=e_\ell(x,\phi y)$ for a unique $\phi\in\mathrm{Hom}(V_\ell A, V_\ell A^\vee)$ that is symmetric, $\phi^\vee=\phi$ (with the sign conventions of [Mum70]); a form is G_k-invariant if and only if ϕ is G_k-equivariant, because e_ℓ is. Second, the class $c_1(L)\in H^2(\bar{A},\mathbb{Q}_\ell(1))$ of a line bundle corresponds to $\phi=\lambda_L$, the homomorphism $\lambda_L\colon A\to A^\vee$, $a\mapsto t_a^*L\otimes L^{-1}$, and $L\mapsto\lambda_L$ identifies $\mathrm{NS}(A)\otimes\mathbb{Q}$ with the symmetric elements of $\mathrm{Hom}_k(A,A^\vee)\otimes\mathbb{Q}$.

Now let E be a G_k-invariant class. Then $E=E_\phi$ with ϕ symmetric and equivariant, so by Theorem 76.3.1(1), applied to $B=A^\vee$, $\phi=\sum_j a_j\psi_j$ with $a_j\in\mathbb{Q}_\ell$ and $\psi_j\in\mathrm{Hom}_k(A,A^\vee)$. Replacing ψ_j by $\frac12(\psi_j+\psi_j^\vee)$ we may take the ψ_j symmetric, since ϕ is. Each symmetric ψ_j is λ_{L_j} for some $L_j\in\mathrm{NS}(A)\otimes\mathbb{Q}$, with L_j defined over k. Hence $E=\sum_j a_j c_1(L_j)$. □

Example 76.3.3. Let C_1 and C_2 be nonsingular projective curves over k with Jacobians J_1, J_2. By the Künneth formula, $H^2(\bar{C}_1\times\bar{C}_2)=H^2(\bar{C}_1)\oplus\big(H^1(\bar{C}_1)\otimes H^1(\bar{C}_2)\big)\oplus H^2(\bar{C}_2)$, and the Tate classes in the middle summand, twisted by (1), are the G_k-equivariant maps $V_\ell J_1\to V_\ell J_2$. By Theorem 76.3.1 these are combinations of homomorphisms $J_1\to J_2$, which are induced by divisors on $C_1\times C_2$ (divisorial correspondences). So $T^1(C_1\times C_2/k)$ holds. This was Tate's motivating application [Tat66a].

76.4 K3 surfaces and other known cases

Theorem 76.4.1. *The Tate conjecture T^1 holds in the following cases.*

1. *Abelian varieties over finite fields and over finitely generated fields of characteristic $\neq 2$ (Corollary 76.3.2), and more generally varieties whose H^2 is controlled by an abelian variety, such as products of curves (Example 76.3.3).*
2. *K3 surfaces over finitely generated fields of characteristic 0 [And96a].*
3. *K3 surfaces over finite fields of characteristic $p\geq 3$: for ordinary K3 surfaces by Nygaard [Nyg83], for those of finite height by Nygaard and Ogus [NO85], and in general by work of Charles [Cha13], Maulik [Mau14] and Madapusi Pera [MP15].*

The proofs for K3 surfaces use the Kuga–Satake construction (Remark 72.6.2), which embeds H^2 of a K3 surface into H^2 of an abelian variety, and transport Theorem 76.3.1 through it. The difficulty is arithmetic: the Kuga–Satake abelian variety must be defined over a finitely generated field, and in positive characteristic one needs integral models of orthogonal Shimura varieties.

Remark 76.4.2. For a surface X over a finite field, Tate showed that $T^1(X/k)$ is equivalent to the finiteness of the ℓ-primary part of the Brauer group $\mathrm{Br}(X)$, and, with Artin, conjectured a formula for the special value of the zeta function in terms of $|\mathrm{Br}(X)|$ and the discriminant of the Néron–Severi lattice, an analogue of the Birch and Swinnerton-Dyer conjecture [Tat66b]. The Tate conjecture for divisors thus has a cohomological content (surjectivity of a cycle map) and a finiteness content (Brauer groups) that are equivalent.

76.5 The Mumford–Tate conjecture

Let $k \subseteq \mathbb{C}$ be a number field and X nonsingular projective over k. For each i, Artin's comparison theorem identifies $H^i(\bar{X}, \mathbb{Q}_\ell)$ with $H^i(X_{\mathbb{C}}; \mathbb{Q}) \otimes \mathbb{Q}_\ell$ (Theorem 65.4.3, using invariance of étale cohomology under extension of algebraically closed fields [Mil80]). So G_k acts on $H \otimes \mathbb{Q}_\ell$, where $H = H^i(X_{\mathbb{C}}; \mathbb{Q})$ is a Hodge structure with Mumford–Tate group $\mathrm{MT}(H)$ (Definition 67.2.1).

Definition 76.5.1. Let V be a finite-dimensional $\mathbb{Q}_\ell$-vector space with a continuous representation $\rho\colon G_k \to \mathrm{GL}(V)$. Its *$\ell$-adic algebraic monodromy group* $G_\ell(V)$ is the Zariski closure of $\rho(G_k)$ in $\mathrm{GL}(V)$, an algebraic group over $\mathbb{Q}_\ell$, and $G_\ell^0(V)$ is its identity component. For $V = H \otimes \mathbb{Q}_\ell$ with $H = H^i(X_{\mathbb{C}}; \mathbb{Q})$ we write G_ℓ and G_ℓ^0.

Lemma 76.5.2. *In the situation of Definition 76.5.1, $G_\ell^0(V)$ does not change if k is replaced by a finite extension, and the vectors of V fixed by some open subgroup of G_k are exactly the invariants of $G_\ell^0(V)$.*

Proof. If $\Gamma_0 \subseteq \Gamma$ are subgroups of $\mathrm{GL}(V)(\mathbb{Q}_\ell)$ with $\Gamma = \bigcup_{j=1}^{r} g_j \Gamma_0$, then the Zariski closures satisfy $\bar{\Gamma} = \bigcup_j g_j \bar{\Gamma}_0$, so $\bar{\Gamma}_0$ is a closed subgroup of finite index in $\bar{\Gamma}$; it is then a union of connected components of $\bar{\Gamma}$ and contains $\bar{\Gamma}^0$. The subgroup $\rho^{-1}(G_\ell^0(V)(\mathbb{Q}_\ell))$ has finite index in G_k and is closed, hence open, say G_{k_0}; the Zariski closure of $\rho(G_{k_0})$ lies in $G_\ell^0(V)$ and has finite index in $G_\ell(V)$, so it equals $G_\ell^0(V)$. For any finite k'/k, the Zariski closure of $\rho(G_{k'})$ has finite index in $G_\ell(V)$, so its identity component is $G_\ell^0(V)$. This proves the first statement. For the second, the stabiliser of a vector is Zariski closed, so a vector fixed by $\rho(G_{k'})$ is fixed by its Zariski closure, which contains $G_\ell^0(V)$; conversely a $G_\ell^0(V)$-invariant vector is fixed by $\rho(G_{k_0})$. □

Conjecture 76.5.3 (Mumford–Tate)**.** $G_\ell^0 = \mathrm{MT}(H) \otimes_{\mathbb{Q}} \mathbb{Q}_\ell$ *inside* $\mathrm{GL}(H \otimes \mathbb{Q}_\ell)$.

Theorem 76.5.4 (Deligne)**.** *If $X = A$ is an abelian variety and $H = H^1(A_{\mathbb{C}}; \mathbb{Q})$, then $G_\ell^0 \subseteq \mathrm{MT}(H) \otimes_{\mathbb{Q}} \mathbb{Q}_\ell$.*

Sketch of proof. Let $\mathrm{MT}' \subseteq \mathrm{GL}(H) \times \mathbb{G}_m$ be the subgroup fixing every Hodge class in every tensor space $T^{m,m'}(H)(r)$ of weight 0, where $\mathbb{G}_m$ acts on $\mathbb{Q}(1)$ through the identity character. Its image under the first projection is $\mathrm{MT}(H)$; this is a reformulation of Theorem 67.3.2 which we do not prove [Del82]. By Noetherianity, MT' is the stabiliser of finitely many Hodge classes $t_1, \dots, t_N$. The spaces $T^{m,m'}(H)(r)$ are Tate twists of Künneth summands of $H^\bullet(A_{\mathbb{C}}^M)$ for suitable M (using a polarisation to identify $H^\vee$ with $H(1)$), so by Theorem 75.7.1 the t_j are absolute Hodge, and by Proposition 75.5.1 their ℓ-adic incarnations are fixed by an open subgroup $G_{k'}$. The group G_k acts on $T^{m,m'}(H \otimes \mathbb{Q}_\ell)(r)$ through $(\rho_\ell, \chi_\ell)\colon G_k \to \mathrm{GL}(H \otimes \mathbb{Q}_\ell) \times \mathbb{G}_m(\mathbb{Q}_\ell)$, χ_ℓ the cyclotomic character. Hence $(\rho_\ell, \chi_\ell)(G_{k'}) \subseteq \mathrm{MT}'(\mathbb{Q}_\ell)$, so $\rho_\ell(G_{k'}) \subseteq \mathrm{MT}(H)(\mathbb{Q}_\ell)$, and taking Zariski closures, $G_\ell^0 \subseteq \mathrm{MT}(H)_{\mathbb{Q}_\ell}$. □

Proposition 76.5.5. *Let A be an abelian variety over a number field $k \subseteq \mathbb{C}$ such that all endomorphisms of $A_{\mathbb{C}}$ are defined over k. Then G_ℓ and $\mathrm{MT}(H)_{\mathbb{Q}_\ell}$ have the same commutant in $\mathrm{End}(H \otimes \mathbb{Q}_\ell)$, namely $\mathrm{End}(A) \otimes \mathbb{Q}_\ell$.*

Proof. By Theorem 76.3.1(1) with $B = A$, the commutant of $\rho_\ell(G_k)$, which equals the commutant of its Zariski closure G_ℓ, is $\mathrm{End}_k(A) \otimes \mathbb{Q}_\ell$, acting on $V_\ell A$ and dually on $H \otimes \mathbb{Q}_\ell$. By Corollary 67.2.4, the commutant of $\mathrm{MT}(H)$ in $\mathrm{End}(H)$ is $\mathrm{End}_{\mathrm{HS}}(H) = \mathrm{End}(A_{\mathbb{C}}) \otimes \mathbb{Q}$, and taking commutants commutes with the flat base change $\mathbb{Q} \to \mathbb{Q}_\ell$ (it is the solution space of linear equations). As $\mathrm{End}(A_{\mathbb{C}}) = \mathrm{End}_k(A)$ by hypothesis, the two commutants agree. □

Remark 76.5.6. Proposition 76.5.5 and Theorem 76.5.4 show that $G_\ell^0 \subseteq \mathrm{MT}_{\mathbb{Q}_\ell}$ are reductive groups with the same commutant (G_ℓ^0 is reductive because $V_\ell A$ is semisimple, Theorem 76.3.1(2)). This is consistent with the Mumford–Tate conjecture but does not imply it: reductive subgroups with the same commutant, such as $\mathrm{Sp}_{2g} \subseteq \mathrm{SL}_{2g}$ acting on $\mathbb{Q}_\ell^{2g}$, can differ. The conjecture is known for elliptic curves: for E without complex multiplication, Serre's open image theorem [Ser72] gives $G_\ell = \mathrm{GL}_2$, which is $\mathrm{MT}(H)_{\mathbb{Q}_\ell}$ by Proposition 67.4.1. It is known for abelian varieties of CM type, and for many abelian varieties with $\mathrm{End}(A_{\mathbb{C}}) = \mathbb{Z}$, depending on the dimension [Pin98]; it is open in general.

Proposition 76.5.7. *Let X be a nonsingular projective variety over a number field $k \subseteq \mathbb{C}$, and $p \geq 0$. Suppose that the geometric Tate classes in $H^{2p}(\bar{X}, \mathbb{Q}_\ell(p))$ are exactly the images under c_ℓ of the classes $(2\pi i)^p \alpha \otimes a$, $\alpha \in \mathrm{Hdg}^p(X_{\mathbb{C}})$, $a \in \mathbb{Q}_\ell$; this is what the Mumford–Tate conjecture for $H^{2p}(X_{\mathbb{C}})$ predicts. Then the Hodge conjecture for $X_{\mathbb{C}}$ in codimension p is equivalent to $T^p(\bar{X})$.*

Proof. We use that every algebraic class on $X_{\mathbb{C}}$ is the class of a cycle defined over $\bar{k}$: a cycle on $X_{\mathbb{C}}$ is defined over a finitely generated extension of $\bar{k}$, so it is a member of a family of cycles parametrised by a variety over $\bar{k}$, and specialising to a $\bar{k}$-point does not change its cohomology class, which is locally constant in the family. So the $\mathbb{Q}$-space Z^p_{alg} of algebraic classes is the same for $X_{\mathbb{C}}$ and $\bar{X}$, and c_ℓ identifies $Z^p_{\mathrm{alg}} \otimes \mathbb{Q}_\ell$ with the $\mathbb{Q}_\ell$-span of the ℓ-adic cycle classes, by Theorem 65.4.3. By hypothesis c_ℓ identifies $\mathrm{Hdg}^p(X_{\mathbb{C}}) \otimes \mathbb{Q}_\ell$ with $T^p_\ell(\bar{X})$. Hence $T^p(\bar{X})$ says $Z^p_{\mathrm{alg}} \otimes \mathbb{Q}_\ell = \mathrm{Hdg}^p \otimes \mathbb{Q}_\ell$, which, since $Z^p_{\mathrm{alg}} \subseteq \mathrm{Hdg}^p$ are $\mathbb{Q}$-subspaces of the same finite-dimensional space, is equivalent to $Z^p_{\mathrm{alg}} = \mathrm{Hdg}^p$. □

Remark 76.5.8. Neither the Hodge nor the Tate conjecture is known to imply the other. The Hodge conjecture says nothing over finite fields, and the Tate conjecture over number fields is linked to it only through the Mumford–Tate conjecture, which is itself open. The two known inclusions, algebraic $\subseteq$ Hodge and algebraic $\subseteq$ Tate, are joined by Deligne's theorem, which gives Hodge $\subseteq$ Tate for abelian varieties (Corollary 75.5.2 and Theorem 75.7.1), and Theorem 76.5.4 is the group-theoretic form of the same statement. For abelian varieties Faltings' theorem proves the Tate conjecture for divisors, and the Lefschetz $(1,1)$-theorem the Hodge conjecture for divisors; in codimension ≥ 2 both are open.

76.6 Exercises

Exercise 76.6.1. Let $k = \mathbb{F}_q$ and let F be the geometric Frobenius, acting on $H^{2p}(\bar{X}, \mathbb{Q}_\ell)$. Assuming that F acts semisimply, show that $\dim T^p_\ell(\bar{X})$ is the number of eigenvalues of F on $H^{2p}(\bar{X}, \mathbb{Q}_\ell)$, counted with multiplicity, of the form ζq^p with ζ a root of unity.

Solution. $G_{\mathbb{F}_{q^m}}$ is topologically generated by F^m, and the geometric Frobenius acts on $\mathbb{Q}_\ell(1)$ by q^{-1}. So a class in $H^{2p}(\bar{X}, \mathbb{Q}_\ell(p)) = H^{2p}(\bar{X}, \mathbb{Q}_\ell) \otimes \mathbb{Q}_\ell(p)$ is fixed by $G_{\mathbb{F}_{q^m}}$ if and only if, as a class in $H^{2p}(\bar{X}, \mathbb{Q}_\ell)$, it satisfies $F^m v = q^{pm} v$. Since F is semisimple, the union over m of the eigenspaces of F^m for q^{pm} is the sum of the eigenspaces of F for the eigenvalues λ with $\lambda^m = q^{pm}$ for some m, that is, $\lambda = \zeta q^p$. The union is an increasing union of subspaces, so it is the largest of them. □

Exercise 76.6.2. Let E be an elliptic curve over a number field $k \subseteq \mathbb{C}$ with complex multiplication by an imaginary quadratic field K, all of whose endomorphisms are defined over k. Show that G_ℓ^0 is contained in the torus $(\mathrm{Res}_{K/\mathbb{Q}}\mathbb{G}_m)_{\mathbb{Q}_\ell}$, and deduce that $G_\ell^0 \subseteq \mathrm{MT}(H^1(E_{\mathbb{C}}))_{\mathbb{Q}_\ell}$ directly, without Theorem 76.5.4.

Solution. The action of $K = \mathrm{End}(E) \otimes \mathbb{Q}$ on $V_\ell E$ commutes with G_k, since the endomorphisms are defined over k. $V_\ell E$ is free of rank one over $K \otimes \mathbb{Q}_\ell$, so the commutant of $K \otimes \mathbb{Q}_\ell$ in $\mathrm{End}(V_\ell E)$ is $K \otimes \mathbb{Q}_\ell$ itself. Hence $\rho_\ell(G_k) \subseteq (K \otimes \mathbb{Q}_\ell)^\times = (\mathrm{Res}_{K/\mathbb{Q}}\mathbb{G}_m)(\mathbb{Q}_\ell)$, and the same holds for the Zariski closure. By Proposition 67.4.1, this torus is $\mathrm{MT}(H^1(E_{\mathbb{C}}))$ (the same argument applies to the dual H^1). □

Exercise 76.6.3. Let E be a supersingular elliptic curve over $\mathbb{F}_p$, and $k = \mathbb{F}_{p^2}$ a field over which all endomorphisms of E are defined, so that $\mathrm{End}(E_k) \otimes \mathbb{Q}$ is a quaternion algebra of rank 4. Using Theorem 76.3.1, show that $\rho(E \times E) = 6$ over k, and that all of $H^2(\overline{E \times E}, \mathbb{Q}_\ell(1))$ consists of Tate classes over k. Why is this impossible over $\mathbb{C}$?

Solution. By Example 76.3.3, the Néron–Severi group of $E \times E$ has rank $2 + \mathrm{rank}\,\mathrm{Hom}(E, E) = 2 + 4 = 6 = b_2(E \times E)$. By Corollary 76.3.2 the Tate classes are spanned by divisor classes, and conversely divisor classes over k are Tate; since the divisor classes span a 6-dimensional space, every class in H^2 is a Tate class. Over $\mathbb{C}$ the rank of $\mathrm{End}(E)$ is at most 2 (Proposition 61.6.1), so $\rho(E \times E) \leq 4 < 6$; equivalently $h^{2,0}(E \times E) = 1$ prevents all of H^2 from being of type $(1, 1)$. □

Chapter 77

The Standard Conjectures and Motives

77.1 Introduction

In the 1960s Grothendieck proposed a small set of conjectures about algebraic cycles, the *standard conjectures*, designed to prove the Weil conjectures and to build a category of *motives*: a universal cohomology theory through which every Weil cohomology factors. Deligne proved the Weil conjectures by other means, and the standard conjectures remain open. They are much weaker than the Hodge conjecture, which implies them in characteristic 0 (Theorem 77.3.9), yet even the Lefschetz standard conjecture, which asks for one explicit class to be algebraic, is open in general. This chapter states the conjectures, proves the implications among them, constructs pure motives, proves Jannsen's semisimplicity theorem, reformulates the Hodge conjecture motivically, and describes the conjectures of Bloch and Beilinson on the kernel of the cycle class map, the other half of the theory of algebraic cycles. References: [Gro69b], [Kle68], [Kle94], [Sch94], [And04], [Jan94], and the source *The Hodge Conjecture* (Chapters 8 and 9).

Prerequisites. Chapter 62 (Algebraic Cycles and Chow Groups), Chapter 49 (Hard Lefschetz and the Hodge–Riemann Relations) and Chapter 76 (The Tate Conjecture).

Conventions. k is a field, and varieties are nonsingular projective over k; X has pure dimension n unless stated otherwise. We fix a Weil cohomology theory $H^\bullet$ with coefficients in a field K of characteristic 0 (Definition 65.4.4): ℓ-adic cohomology of $X \times_k \bar{k}$ in general, or Betti cohomology of $X_{\mathbb{C}}$ with $K = \mathbb{Q}$ when $k \subseteq \mathbb{C}$. Tate twists are suppressed from the notation. For $\sim$ one of rational, homological (with respect to $H^\bullet$) or numerical equivalence (Definition 62.6.1) we write

$$A^p_{\sim}(X) = (Z^p(X) \otimes_{\mathbb{Z}} \mathbb{Q})/\!\sim, \qquad A^\bullet_{\sim}(X) = \bigoplus_p A^p_{\sim}(X),$$

so that $A^p_{\mathrm{rat}}(X) = \mathrm{CH}^p(X)_{\mathbb{Q}}$ and $A^p_{\mathrm{hom}}(X)$ is the $\mathbb{Q}$-span of the image of the cycle class map in $H^{2p}(X)$, whose elements we call *algebraic classes*. We fix an ample divisor on X with class $h \in A^1_{\mathrm{hom}}(X)$ and write $L = h \cup -\colon H^i(X) \to H^{i+2}(X)$.

77.2 Correspondences in cohomology

Proposition 77.2.1. *Let X, Y, W be varieties, X of dimension n.*

1. *A class $\gamma \in H^{2n+2r}(X \times Y)$ acts by $\gamma_*(x) = \mathrm{pr}_{2*}(\mathrm{pr}_1^* x \cup \gamma)$, which maps $H^i(X)$ to $H^{i+2r}(Y)$, and the Künneth formula together with Poincaré duality give an isomorphism $H^{2n+2r}(X \times Y) \cong \bigoplus_i \mathrm{Hom}_K(H^i(X), H^{i+2r}(Y))$, $\gamma \mapsto (\gamma_*|_{H^i(X)})_i$.*

2. *For $\gamma \in H^\bullet(X \times Y)$ and $\delta \in H^\bullet(Y \times W)$, the class $\delta \circ \gamma = \mathrm{pr}_{13*}(\mathrm{pr}_{12}^*\gamma \cup \mathrm{pr}_{23}^*\delta)$ satisfies $(\delta \circ \gamma)_* = \delta_* \circ \gamma_*$, the class of the diagonal acts as the identity, and $\mathrm{cl}(\Gamma' \circ \Gamma) = \mathrm{cl}(\Gamma') \circ \mathrm{cl}(\Gamma)$ for algebraic correspondences (Definition 62.5.1).*

3. *(Lefschetz trace formula) For $\alpha, \beta \in H^{2n}(X \times X)$, $\int_{X\times X} \alpha \cup {}^t\beta = \sum_i (-1)^i \operatorname{tr}\big((\beta \circ \alpha)_* \mid H^i(X)\big)$, where ${}^t\beta$ is the pullback of β under the exchange of the factors. The right side is the* supertrace $\mathrm{str}((\beta \circ \alpha)_*)$.

These are formal consequences of the axioms of a Weil cohomology [Kle68]; (2) is the cohomological version of Proposition 62.5.2, and (3) for $\beta = \Delta_X$ is the Lefschetz trace formula of Theorem 65.4.2(5). By (2), an algebraic correspondence maps algebraic classes to algebraic classes: $\mathrm{cl}(\Gamma)_*\mathrm{cl}(Z) = \mathrm{cl}(\Gamma_* Z)$. We call a K-linear endomorphism of $H^\bullet(X)$ *algebraic* if it is γ_* for γ in the $\mathbb{Q}$-span of classes of algebraic cycles on $X \times X$. Sums and composites of algebraic endomorphisms are algebraic, and $L = (\Delta_* h)_*$ is algebraic.

Lemma 77.2.2. *$A^p_{\mathrm{num}}(X)$ is a finite-dimensional $\mathbb{Q}$-vector space of dimension at most $\dim_K H^{2p}(X)$, and the pairing $A^p_{\mathrm{num}}(X) \times A^{n-p}_{\mathrm{num}}(X) \to \mathbb{Q}$, $(\alpha, \beta) \mapsto \deg(\alpha \cdot \beta)$, is perfect. If $K = \mathbb{Q}$, then $A^p_{\mathrm{hom}}(X)$ is finite-dimensional too.*

Proof. Homological equivalence implies numerical equivalence (Proposition 62.6.2), and $\deg(\alpha \cdot \beta) = \int \mathrm{cl}(\alpha) \cup \mathrm{cl}(\beta)$. Let $\alpha_1, \dots, \alpha_N \in A^p_{\mathrm{num}}(X)$ be linearly independent. The map $A^{n-p}_{\mathrm{num}}(X) \to \mathbb{Q}^N$, $\beta \mapsto (\deg(\alpha_i \cdot \beta))_i$, is surjective: otherwise a nonzero linear form on $\mathbb{Q}^N$ vanishes on its image, which gives a nontrivial combination of the α_i that pairs to zero with every cycle, that is, is zero in A^p_{num}. Choose β_j with $\deg(\alpha_i \cdot \beta_j) = \delta_{ij}$. If $\sum_i c_i \mathrm{cl}(\alpha_i) = 0$ with $c_i \in K$, cupping with $\mathrm{cl}(\beta_j)$ and integrating gives $c_j = 0$. So the $\mathrm{cl}(\alpha_i)$ are K-linearly independent in $H^{2p}(X)$ and $N \le \dim_K H^{2p}(X)$. The pairing is nondegenerate by definition, hence perfect between finite-dimensional spaces. If $K = \mathbb{Q}$, $A^p_{\mathrm{hom}}(X)$ is a subspace of $H^{2p}(X)$. □

Remark 77.2.3. For ℓ-adic cohomology, $A^p_{\mathrm{hom}}(X)$ is a $\mathbb{Q}$-subspace of a $\mathbb{Q}_\ell$-vector space, and it is not known in general that it is finite-dimensional over $\mathbb{Q}$, or that $A^p_{\mathrm{hom}}(X) \otimes_{\mathbb{Q}} \mathbb{Q}_\ell \to H^{2p}(X)$ is injective. Both follow from the equality of homological and numerical equivalence (Conjecture 77.3.3 below), by the proof of Lemma 77.2.2.

77.3 The standard conjectures

By hard Lefschetz (an axiom of Weil cohomologies; for ℓ-adic cohomology a theorem of Deligne, Theorem 65.4.2(6)), $L^{n-i}\colon H^i(X) \to H^{2n-i}(X)$ is an isomorphism for $i \le n$. As in Chapter 49, this gives the Lefschetz decomposition $H^i = \bigoplus_j L^j P^{i-2j}$, with $P^j = \ker(L^{n-j+1}\colon H^j \to H^{2n-j+2})$ for $j \le n$, and a unique operator Λ with $[L, \Lambda] = h_{\deg}$ and $[h_{\deg}, \Lambda] = -2\Lambda$, where $h_{\deg}$ acts on H^i by $i - n$ (Proposition 49.2.4); explicitly $\Lambda(L^r x) = r(n - j - r + 1)L^{r-1}x$ for $x \in P^j$. So $H^\bullet(X)$ is a representation of $\mathfrak{sl}_2$ with $e = L$, $f = \Lambda$, $h = h_{\deg}$, in which H^i is the weight space of weight $i - n$ and P^j consists of lowest weight vectors of weight $j - n$.

Conjecture 77.3.1 (Lefschetz standard conjecture $B(X)$)**.** *The operator Λ is algebraic.*

Conjecture 77.3.2 (Künneth standard conjecture $C(X)$)**.** *The Künneth components $\pi_i \in H^{2n-i}(X) \otimes H^i(X) \subseteq H^{2n}(X \times X)$ of the class of the diagonal are algebraic; equivalently, the projectors of $H^\bullet(X)$ onto the summands $H^i(X)$ are algebraic.*

Conjecture 77.3.3 (Standard conjecture $D(X)$)**.** *Homological and numerical equivalence coincide on $Z^\bullet(X) \otimes \mathbb{Q}$: $A^\bullet_{\mathrm{hom}}(X) = A^\bullet_{\mathrm{num}}(X)$.*

Conjecture 77.3.4 (Hodge standard conjecture Hdg(X))**.** *For $p \le n/2$, the symmetric bilinear form $(x,y) \mapsto (-1)^p \int_X L^{n-2p}(x \cup y)$ is positive definite on the $\mathbb{Q}$-space $A^p_{\mathrm{hom}}(X) \cap P^{2p}(X)$ of primitive algebraic classes.*

The equivalence in $C(X)$ is Proposition 77.2.1(1): $(\pi_i)_*$ is the projector onto $H^i(X)$.

Lemma 77.3.5. *Let V be a finite-dimensional representation of $\mathfrak{sl}_2$ over a field of characteristic 0, with weight spaces V_w for h.*

1. *The projector of V onto V_w, along the other weight spaces, is a polynomial in h.*
2. *The projector onto the isotypic component of $V(m)$ is a polynomial in the Casimir element $c = ef + fe + \frac{1}{2}h^2$, which acts on $V(m)$ by $\frac{1}{2}m(m+2)$.*
3. *For $j \ge 0$, $e^j : V_{-j} \to V_j$ is bijective, and its inverse is the restriction to V_j of $P(f^je^j)f^j$ for a polynomial P.*
4. *If v is a lowest weight vector of weight $-m$ and $0 \le j \le m$, then $f^je^jv = c_{m,j}v$ with $c_{m,j}$ a nonzero integer.*

Proof. We use the basis $v_0, \dots, v_m$ of $V(m)$ from Theorem 13.6.2, with $hv_i = (m-2i)v_i$, $fv_i = (i+1)v_{i+1}$, $ev_i = (m-i+1)v_{i-1}$, and the decomposition of V into irreducibles. (1) h is diagonalisable with finitely many eigenvalues $w_1, \dots, w_s$, and the projector onto V_{w_a} is $\prod_{b \ne a}(h - w_b)/(w_a - w_b)$. (2) On v_0, $ev_0 = 0$ gives $efv_0 = [e,f]v_0 = hv_0 = mv_0$ and $fev_0 = 0$, so $cv_0 = (m + \frac{1}{2}m^2)v_0$; as c commutes with e, f, h, it acts on $V(m)$ by this scalar. The scalars $\frac{1}{2}m(m+2)$ are distinct for distinct $m \ge 0$, so the projectors are Lagrange polynomials in c. (4) The lowest weight vector of $V(m)$ is v_m. Applying e^j multiplies it by $\prod_{s=0}^{j-1}(s+1) = j!$ and gives v_{m-j}; applying f^j multiplies by $\prod_{t=0}^{j-1}(m-j+t+1)$ and returns to v_m. All factors are positive. (3) The weight $-j$ vector of $V(m)$, when it exists ($m \ge j$, $m \equiv j \bmod 2$), is v_i with $i = (m+j)/2$; applying e^j and then f^j multiplies it by $\prod_{s=0}^{j-1}(m-i+s+1)\prod_{t=0}^{j-1}(i-j+t+1)$, a product of integers $\ge (m-j)/2 + 1 \ge 1$. So f^je^j acts on V_{-j} diagonalisably with finitely many nonzero eigenvalues $c_1, \dots, c_r$, and $P(f^je^j)$ inverts it for the Lagrange polynomial with $P(c_a) = c_a^{-1}$. Then $P(f^je^j)f^j \circ e^j$ is the identity of V_{-j}, so e^j is injective there; as $\dim V_{-j} = \dim V_j$, it is bijective, with inverse $P(f^je^j)f^j|_{V_j}$. □

Proposition 77.3.6. *Assume $B(X)$. Then:*

1. *$C(X)$ holds;*
2. *for $p \le n/2$, $L^{n-2p} : A^p_{\mathrm{hom}}(X) \to A^{n-p}_{\mathrm{hom}}(X)$ is bijective* (Grothendieck's conjecture $A(X,L)$)*;*
3. *if $x = \sum_j L^j x_j$ is the Lefschetz decomposition of an algebraic class, with x_j primitive, then every x_j is algebraic.*

Proof. By $B(X)$, L and Λ are algebraic, hence so is every noncommutative polynomial in them, in particular $h_{\deg} = [L, \Lambda]$ and the Casimir element. (1) By Lemma 77.3.5(1), the projector onto H^i is a polynomial in $h_{\deg}$. (2) L^{n-2p} maps algebraic classes to algebraic classes and is injective on H^{2p} by hard Lefschetz. By Lemma 77.3.5(3), its inverse $H^{2n-2p} \to H^{2p}$ is the restriction of an algebraic endomorphism, so it maps algebraic classes to algebraic classes. (3) The summand L^jP^{i-2j} of H^i is the intersection of H^i with the isotypic component of $V(n-i+2j)$, since a primitive class in P^{i-2j} is a lowest weight vector of weight $i-2j-n$. By Lemma 77.3.5(1),(2) the projector onto L^jP^{i-2j} is algebraic, so L^jx_j is algebraic. By Lemma 77.3.5(4), $x_j = c^{-1}\Lambda^jL^jx_j$ for a nonzero integer c, so x_j is algebraic. □

Proposition 77.3.7. *If $k \subseteq \mathbb{C}$ and $H^\bullet$ is Betti cohomology, then* $\mathrm{Hdg}(X)$ *holds.*

Proof. Algebraic classes in $H^{2p}(X_{\mathbb{C}};\mathbb{Q})$ are rational classes of type (p,p) (Corollary 63.4.2). The class h is represented by a Kähler form ω, the restriction of a multiple of the Fubini–Study form. For a nonzero primitive rational class x of type (p,p), the Hodge–Riemann relation (Theorem 49.4.2(4)) with $k = 2p$ and $\bar{x} = x$ reads $Q(x,x) > 0$, where $Q(x,x) = (-1)^{2p(2p-1)/2}\int x \cup x \cup [\omega]^{n-2p} = (-1)^p \int L^{n-2p}(x \cup x)$, since $p(2p-1) \equiv p \pmod 2$. □

Proposition 77.3.8. *$B(X)$ and* $\mathrm{Hdg}(X)$ *together imply $D(X)$.*

Proof. First let $p \le n/2$, and put $\langle x, y\rangle = \int L^{n-2p}(x \cup y)$ on H^{2p}. The Lefschetz decomposition is orthogonal for $\langle\ ,\ \rangle$: for $a \in P^{2p-2j}$, $b \in P^{2p-2j'}$ and $j < j'$, $\langle L^j a, L^{j'} b\rangle = \int b \cup L^{n-2p+j+j'} a$ vanishes because $n - 2p + j + j' \ge n - (2p - 2j) + 1$, so $L^{n-2p+j+j'}a = 0$ by primitivity. Let $x \in A^p_{\mathrm{hom}}(X)$ with $\langle x, y\rangle = 0$ for every $y \in A^p_{\mathrm{hom}}(X)$, and write $x = \sum_j L^j x_j$. By Proposition 77.3.6(3) the x_j are algebraic, so $y = L^j x_j$ is allowed, and $0 = \langle x, L^j x_j\rangle = \langle L^j x_j, L^j x_j\rangle = \int L^{n-2(p-j)}(x_j \cup x_j)$. By $\mathrm{Hdg}(X)$ in codimension $p - j$, $x_j = 0$. Hence $\langle\ ,\ \rangle$ is nondegenerate on $A^p_{\mathrm{hom}}(X)$. Now if $x \in A^p_{\mathrm{hom}}(X)$ is numerically trivial, then $\langle x, y\rangle = \deg(x \cdot h^{n-2p} \cdot y) = 0$ for all algebraic y, so $x = 0$.

Now let $p > n/2$ and let $x \in A^p_{\mathrm{hom}}(X)$ be numerically trivial. By Proposition 77.3.6(2) applied to $n - p < n/2$, $L^{2p-n}\colon A^{n-p}_{\mathrm{hom}} \to A^p_{\mathrm{hom}}$ is bijective, so $x = L^{2p-n}y$ with y algebraic, and every algebraic class of codimension p is $L^{2p-n}u$ with $u \in A^{n-p}_{\mathrm{hom}}$. Then $\deg(y \cdot L^{2p-n}u) = \deg(x \cdot u) = 0$ for all such u, so y is numerically trivial, hence $y = 0$ by the first case, and $x = 0$. □

Theorem 77.3.9. *Let $k \subseteq \mathbb{C}$ and let $H^\bullet$ be Betti cohomology. If the Hodge conjecture holds for $X \times X$, then $B(X)$, $C(X)$ and $D(X)$ hold.*

Proof. The operator Λ is defined over $\mathbb{Q}$ and is a morphism of Hodge structures of type $(-1,-1)$ (see the proof of Corollary 75.4.3), so its class in $H^{2n-2}(X \times X;\mathbb{Q})$ is a Hodge class (Proposition 66.6.2). By the Hodge conjecture it is algebraic, which is $B(X)$. Then $C(X)$ follows from Proposition 77.3.6, and $D(X)$ from Propositions 77.3.7 and 77.3.8. □

Remark 77.3.10. Grothendieck showed that the standard conjectures for X and $X \times X$ over a finite field imply the Riemann hypothesis part of the Weil conjectures, by an argument modelled on Serre's Kähler analogue: the graph of Frobenius is an algebraic correspondence, and the positivity of $\mathrm{Hdg}(X \times X)$ bounds its eigenvalues [Gro69b], [Kle68]. Deligne's proof [Del74a] avoided the standard conjectures. Katz and Messing then deduced $C(X)$ over finite fields from Deligne's theorem [KM74]. In positive characteristic, $\mathrm{Hdg}(X)$ is known in general only for surfaces, where it is the Hodge index theorem [Har77, Theorem V.1.9].

Theorem 77.3.11 (Known cases)**.**

1. *$B(X)$ holds if $\dim X \le 2$, if X is an abelian variety [Lie68], for products of varieties for which it holds [Kle68], for varieties whose cohomology is spanned by algebraic classes (projective spaces, Grassmannians, flag varieties), for smooth hypersurfaces and complete intersections in projective space (Exercise 77.6.3), and for hyper-Kähler varieties deformation equivalent to Hilbert schemes of points on K3 surfaces [CM13].*
2. *$C(X)$ holds over finite fields [KM74], and whenever $B(X)$ holds.*
3. *$D(X)$ holds for divisors in any characteristic, and in characteristic 0 for abelian varieties [Lie68].*
4. *$\mathrm{Hdg}(X)$ holds in characteristic 0 (Proposition 77.3.7) and for surfaces in any characteristic.*

See [Kle94] for a survey and for the classical cases. For a surface, the components of Λ on $H^2 \to H^0$ and $H^4 \to H^2$ are expressed through h and the class of a point, and the component $H^3 \to H^1$, the inverse of $L\colon H^1 \to H^3$, is induced by a divisorial correspondence built from the Picard and Albanese varieties. Milne showed that Hdg for abelian varieties in positive characteristic would follow from the Hodge conjecture for complex abelian varieties of CM type [Mil02].

Remark 77.3.12. $B(X)$ asks for the algebraicity of one specific Hodge class on $X \times X$, while the Hodge conjecture asks for all of them, and still $B(X)$ is open for most varieties of dimension ≥ 3. For a threefold, $B(X)$ essentially amounts to the algebraicity of the inverse of $L\colon H^2(X) \to H^4(X)$, a class in $H^2(X) \otimes H^2(X)$ whose transcendental part is not accessible by divisors. The class of Λ is absolute Hodge (Corollary 75.4.3), but no algebraic cycle representing it is known in general.

77.4 Pure motives

Definition 77.4.1. Let $\sim$ be rational, homological or numerical equivalence. The category $\mathcal{M}_\sim(k)$ of *pure motives* over k (with $\mathbb{Q}$-coefficients) has as objects triples (X, e, m) with X a nonsingular projective variety of pure dimension, $e \in A^{\dim X}_\sim(X \times X)$ an idempotent, $e \circ e = e$, and $m \in \mathbb{Z}$, together with their finite formal direct sums. Morphisms between triples are

$$\operatorname{Hom}\big((X, e, m), (Y, f, m')\big) = f \circ A^{\dim X + m' - m}_\sim(X \times Y) \circ e,$$

with composition of correspondences, and are extended to direct sums by matrices. The *motive* of X is $h(X) = (X, \Delta_X, 0)$, and a morphism $\phi\colon X \to Y$ gives $h(\phi) = {}^t\Gamma_\phi\colon h(Y) \to h(X)$. The *unit motive* is $\mathbf{1} = h(\operatorname{Spec} k)$, the *Lefschetz motive* is $\mathbb{L} = (\operatorname{Spec} k, \mathrm{id}, -1)$, and the *Tate twist* is $(X, e, m)(r) = (X, e, m + r)$.

Composition is well defined because $\sim$ is compatible with pushforward, pullback and intersection product (Proposition 62.5.2). The category $\mathcal{M}_\sim(k)$ is $\mathbb{Q}$-linear and additive, and *pseudo-abelian*: an idempotent endomorphism $\phi = f \circ \phi \circ f$ of (X, f, m) has image (X, ϕ, m). The product $(X, e, m) \otimes (Y, f, m') = (X \times Y, e \times f, m + m')$ makes it a tensor category, with $(X, e, m) \otimes \mathbb{L} = (X, e, m - 1)$ [Sch94].

Example 77.4.2. In $\mathcal{M}_{\mathrm{rat}}(k)$, $h(\mathbb{P}^1) \cong \mathbf{1} \oplus \mathbb{L}$. Let x be a k-point of $\mathbb{P}^1$, and put $e_0 = [x \times \mathbb{P}^1]$ and $e_2 = [\mathbb{P}^1 \times x]$ in $\mathrm{CH}^1(\mathbb{P}^1 \times \mathbb{P}^1)$. The diagonal has bidegree $(1, 1)$ and $\mathrm{CH}^1(\mathbb{P}^1 \times \mathbb{P}^1)$ is free on e_0 and e_2, so $\Delta = e_0 + e_2$. By the composition formula of Definition 62.5.1, on $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$:

1. $e_0 \circ e_0 = \mathrm{pr}_{13*}([x \times \mathbb{P}^1 \times \mathbb{P}^1] \cdot [\mathbb{P}^1 \times x \times \mathbb{P}^1]) = \mathrm{pr}_{13*}[x \times x \times \mathbb{P}^1] = e_0$;
2. $e_2 \circ e_2 = \mathrm{pr}_{13*}([\mathbb{P}^1 \times x \times \mathbb{P}^1] \cdot [\mathbb{P}^1 \times \mathbb{P}^1 \times x]) = \mathrm{pr}_{13*}[\mathbb{P}^1 \times x \times x] = e_2$;
3. $e_2 \circ e_0 = \mathrm{pr}_{13*}([x \times \mathbb{P}^1 \times \mathbb{P}^1] \cdot [\mathbb{P}^1 \times \mathbb{P}^1 \times x]) = \mathrm{pr}_{13*}[x \times \mathbb{P}^1 \times x] = 0$, since pr_{13} contracts the middle factor;
4. $e_0 \circ e_2 = \mathrm{pr}_{13*}([\mathbb{P}^1 \times x \times \mathbb{P}^1]^2) = 0$, since $[x]^2 = 0$ in $\mathrm{CH}^\bullet(\mathbb{P}^1)$.

So e_0, e_2 are orthogonal idempotents with sum Δ, and $h(\mathbb{P}^1) = (\mathbb{P}^1, e_0, 0) \oplus (\mathbb{P}^1, e_2, 0)$. The correspondences $a = [\mathbb{P}^1] \in \mathrm{CH}^0(\operatorname{Spec} k \times \mathbb{P}^1)$ and $b = [x] \in \mathrm{CH}^1(\mathbb{P}^1 \times \operatorname{Spec} k)$ satisfy $b \circ a = \mathrm{id}_{\mathbf{1}}$ and $a \circ b = e_0$, so $(\mathbb{P}^1, e_0, 0) \cong \mathbf{1}$. Similarly $a' = [x] \in \mathrm{CH}^1(\operatorname{Spec} k \times \mathbb{P}^1)$, a morphism $\mathbb{L} \to h(\mathbb{P}^1)$, and $b' = [\mathbb{P}^1] \in \mathrm{CH}^0(\mathbb{P}^1 \times \operatorname{Spec} k)$, a morphism $h(\mathbb{P}^1) \to \mathbb{L}$, satisfy $b' \circ a' = \mathrm{id}_{\mathbb{L}}$ and $a' \circ b' = e_2$, so $(\mathbb{P}^1, e_2, 0) \cong \mathbb{L}$. In cohomology, e_0 and e_2 are the Künneth components π_0 and π_2.

Theorem 77.4.3 (Jannsen). *$\mathcal{M}_{\mathrm{num}}(k)$ is a semisimple abelian category. Conversely, if $\sim$ is an adequate equivalence relation such that $\mathcal{M}_\sim(k)$ is semisimple abelian, then $\sim$ is numerical equivalence.*

Proof of the first statement. Step 1: for every X of dimension n, the algebra $E = A^n_{\mathrm{num}}(X \times X)$, with composition of correspondences, is semisimple. It is finite-dimensional over $\mathbb{Q}$ by Lemma 77.2.2, so its Jacobson radical R is a nilpotent two-sided ideal. Let $\alpha \in Z^n(X \times X) \otimes \mathbb{Q}$ have image in R, and let β be any cycle of dimension n on $X \times X$. Put $\gamma = \beta \circ \alpha$. Its image lies in R, so the image of $\gamma^{\circ s}$ in E is 0 for some $s \geq 1$, and then so is the image of $\gamma^{\circ t}$ for all $t \geq s$. By Proposition 77.2.1(3) with ${}^t\Delta = \Delta$,

$$\mathrm{str}((\gamma_*)^t) = \int \mathrm{cl}(\gamma^{\circ t}) \cup \mathrm{cl}(\Delta) = \deg(\gamma^{\circ t} \cdot \Delta) = 0 \qquad (t \geq s),$$

because $\gamma^{\circ t}$ is numerically trivial. Let $\lambda_1, \dots, \lambda_a$ and $\mu_1, \dots, \mu_b$ be the eigenvalues, with multiplicity, of γ_* on H^{even} and on H^{odd}, in an algebraic closure of K. Then $\sum_i \lambda_i^t - \sum_j \mu_j^t = 0$ for $t \geq s$, so the rational function

$$R(z) = \sum_i \frac{\lambda_i z}{1 - \lambda_i z} - \sum_j \frac{\mu_j z}{1 - \mu_j z} = \sum_{t \geq 1} \Big(\sum_i \lambda_i^t - \sum_j \mu_j^t \Big) z^t$$

is a polynomial. Its poles at the points $1/\lambda$, for nonzero eigenvalues λ, must therefore cancel, which means that every nonzero value occurs equally often among the λ_i and the μ_j. Hence $\sum_i \lambda_i^t = \sum_j \mu_j^t$ for all $t \geq 1$, and in particular $\mathrm{str}(\gamma_*) = 0$. By the trace formula again, $\deg(\alpha \cdot {}^t\beta) = \mathrm{str}((\beta \circ \alpha)_*) = 0$. As β was arbitrary and every cycle of dimension n on $X \times X$ is of the form ${}^t\beta$, α is numerically trivial. So $R = 0$, and E is semisimple.

Step 2: every object is a direct summand of some $h(Y)(m)$. The triple (X, e, m) is the image of the idempotent e of $h(X)(m)$. Given finitely many triples (X_i, e_i, m_i), choose $m \geq \max m_i$. Then $(X_i, e_i, m_i) = (X_i, e_i, m) \otimes \mathbb{L}^{\otimes(m - m_i)}$, and $\mathbb{L}^{\otimes r}$ is a summand of $h(\mathbb{P}^1)^{\otimes r} = h((\mathbb{P}^1)^r)$ by Example 77.4.2 (which holds modulo numerical equivalence as well). So each (X_i, e_i, m_i) is a summand of $h(X_i \times (\mathbb{P}^1)^{m - m_i})(m)$, and their direct sum is a summand of $h(Y)(m)$ with Y the disjoint union of the $X_i \times (\mathbb{P}^1)^{m - m_i}$.

Step 3: conclusion. By Step 2, the endomorphism algebra of an object M is $\varepsilon E \varepsilon$ for an idempotent ε of $E = \mathrm{End}(h(Y)(m)) = A^{\dim Y}_{\mathrm{num}}(Y \times Y)$ (for disconnected Y, a product of such algebras over pairs of components, with the same argument). A corner $\varepsilon E \varepsilon$ of a finite-dimensional semisimple algebra is semisimple. Finally, a $\mathbb{Q}$-linear additive pseudo-abelian category in which every object has a finite-dimensional semisimple endomorphism algebra is abelian and semisimple [Jan92]. □

The converse statement is also in [Jan92]. Jannsen's theorem was a surprise: Grothendieck expected semisimplicity to require the standard conjectures.

Remark 77.4.4. What the standard conjectures are still needed for is the *Tannakian* structure. The natural commutativity constraint on $\mathcal{M}_{\mathrm{num}}(k)$ gives $h(X)$ the wrong dimension (the supertrace $\sum(-1)^i b_i$, which can be negative); to correct it one changes the sign of the constraint on the odd part, which requires the Künneth projectors to be algebraic modulo numerical equivalence. With C and D, $\mathcal{M}_{\mathrm{num}}(k) = \mathcal{M}_{\mathrm{hom}}(k)$ is a semisimple Tannakian category, the realisation $H^\bullet$ is a fibre functor, and $\mathcal{M}_{\mathrm{num}}(k)$ is the category of representations of a pro-reductive group, the *motivic Galois group* [Jan92], [And04], [DM82]. Unconditional substitutes exist: Deligne's motives for absolute Hodge cycles [DM82] and André's motives for motivated cycles [And96b] are Tannakian and contain the motives of abelian varieties.

Proposition 77.4.5 (The Hodge conjecture, motivically)**.** *Let $k = \mathbb{C}$ and let $\sim$ be homological equivalence for Betti cohomology. The Hodge realisation, which sends (X, e, m) to the graded polarisable $\mathbb{Q}$-Hodge structure $e_* H^\bullet(X; \mathbb{Q})(m)$, with $H^i(X)(m)$ placed in degree $i - 2m$, is a faithful functor from $\mathcal{M}_{\mathrm{hom}}(\mathbb{C})$ to graded polarisable $\mathbb{Q}$-Hodge structures. It is full if and only if the Hodge conjecture holds for all nonsingular complex projective varieties.*

Proof. A morphism $h(X)(m) \to h(Y)(m')$ is an element $\gamma \in A^{n+r}_{\mathrm{hom}}(X \times Y)$ with $n = \dim X$, $r = m' - m$, and it acts by $\gamma_* \colon H^i(X)(m) \to H^{i+2r}(Y)(m')$, which preserves the grading. Faithfulness is Proposition 77.2.1(1): an element of A_{hom} is determined by its class, which is determined by its action. Under the isomorphism of Proposition 77.2.1(1), which is compatible with Hodge structures up to the Tate twist, the grading-preserving morphisms of Hodge structures $\bigoplus_i H^i(X)(m) \to \bigoplus_i H^{i+2r}(Y)(m')$ correspond to the Hodge classes in $H^{2n+2r}(X \times Y; \mathbb{Q})$ (Proposition 66.6.2), and the morphisms of motives to the algebraic classes. So the realisation is full on the objects $h(X)(m)$ if and only if $\mathrm{Hdg}^{n+r}(X \times Y) = Z^{n+r}_{\mathrm{alg}}(X \times Y)$ for all X, Y, r. Taking $X = \operatorname{Spec} \mathbb{C}$ gives the Hodge conjecture for Y; conversely the Hodge conjecture for all varieties includes $X \times Y$. Fullness for general objects follows, since they are summands of the $h(X)(m)$ and morphisms between summands are cut out by the idempotents. □

Remark 77.4.6. Proposition 77.4.5 is the cleanest statement of what the Hodge conjecture is *for*: the Hodge structure of a variety would determine its motive, and every Hodge-theoretic relation between varieties would be realised by correspondences. Under the Hodge conjecture, the standard conjectures hold (Theorem 77.3.9), $\mathcal{M}_{\mathrm{hom}}(\mathbb{C})$ is Tannakian, and the motivic Galois group of the Tannakian subcategory generated by $h(X)$ is the Mumford–Tate group of $H^\bullet(X; \mathbb{Q})$ (Chapter 67), an identification of a geometric object with a Hodge-theoretic one. If the Hodge conjecture were false, the motives for absolute Hodge cycles (Remark 77.4.4) would survive, but they would no longer be built out of subvarieties.

77.5 The Bloch–Beilinson conjectures

The Hodge conjecture concerns the image of the cycle class map. Its kernel, $\mathrm{CH}^p(X)_{\mathrm{hom}} = \ker(\mathrm{cl})$, is governed by a separate set of conjectures.

Theorem 77.5.1 (Mumford)**.** *Let S be a nonsingular complex projective surface with $h^{2,0}(S) > 0$. Then the group $\mathrm{CH}_0(S)_{\deg 0}$ of 0-cycles of degree 0 modulo rational equivalence is* infinite-dimensional*: for no variety T and cycle Z on $T \times S$ is the map $T \to \mathrm{CH}_0(S)_{\deg 0}$, $t \mapsto Z_t - Z_{t_0}$, surjective.*

This is [Mum68]; compare Example 62.4.5. The proof shows that a surjective family would force every holomorphic 2-form to vanish, by pulling it back along the family. Chow groups thus see Hodge-theoretic information that cohomology classes of cycles do not.

Conjecture 77.5.2 (Bloch)**.** *If S is a nonsingular complex projective surface with $h^{2,0}(S) = 0$, then the Albanese map $\mathrm{CH}_0(S)_{\deg 0} \to \mathrm{Alb}(S)$ is an isomorphism.*

Bloch's conjecture is known for surfaces not of general type [BKL76] and in scattered cases of general type, and open in general [Blo10]. By the Bloch–Srinivas method (Theorem 72.5.1), its conclusion implies a decomposition of the diagonal, and conversely the conclusion of Mumford's theorem is the Hodge-theoretic obstruction to one.

Conjecture 77.5.3 (Bloch–Beilinson filtration)**.** *For every nonsingular projective variety X over a field of characteristic 0 and every p, there is a finite decreasing filtration $F^\bullet$ on $\mathrm{CH}^p(X)_{\mathbb{Q}}$ such that:*

1. *$F^0 = \mathrm{CH}^p(X)_{\mathbb{Q}}$, $F^1 = \mathrm{CH}^p(X)_{\mathrm{hom},\mathbb{Q}}$ and $F^{p+1} = 0$;*
2. *the filtrations are compatible with intersection products, $F^i \cdot F^j \subseteq F^{i+j}$, and with the action of correspondences;*
3. *a correspondence $\Gamma \in \mathrm{CH}^{\dim X + r}(X \times Y)_{\mathbb{Q}}$ acts on $\mathrm{Gr}^\nu_F \mathrm{CH}^p(X)_{\mathbb{Q}}$ through its cohomology class, and it acts by 0 if $\Gamma_* \colon H^{2p-\nu}(X) \to H^{2p-\nu+2r}(Y)$ vanishes.*

See [Jan94] and [Blo10]; the formulation is due to Beilinson and Bloch, and Murre gave an equivalent one in terms of a decomposition of the diagonal into orthogonal idempotents lifting the Künneth components to Chow groups [Mur93]. On Gr^1_F the filtration is expected to be detected by the Abel–Jacobi map (Chapter 71), and Gr^ν_F is expected to be an Ext^ν in a category of mixed motives. Beilinson and Bloch conjectured moreover that for X defined over a number field, $F^2\,\mathrm{CH}^p(X_{\bar{\mathbb{Q}}})_{\mathbb{Q}} = 0$, so that the Abel–Jacobi map is injective on $\mathrm{CH}^p(X_{\bar{\mathbb{Q}}})_{\mathrm{hom},\mathbb{Q}}$ [Bei87].

Remark 77.5.4. Conjecture 77.5.3 gives Bloch's conjecture up to torsion. Let S be a surface with $h^{2,0}(S) = 0$. By the Lefschetz $(1,1)$-theorem, $H^2(S;\mathbb{Q})$ is spanned by classes of divisors, so there is a correspondence $\Gamma = \sum a_{ij} D_i \times D_j$, with divisors D_i and $a_{ij} \in \mathbb{Q}$, that acts on $H^2(S)$ as the identity. On 0-cycles $\Gamma_* = 0$, since $(D_i \times D_j)\cdot(z \times S) = 0$ for a 0-cycle z in general position. The diagonal also acts on H^2 as the identity, so by (3) with $p = \nu = 2$, the identity of $\mathrm{Gr}^2_F\,\mathrm{CH}_0(S)_{\mathbb{Q}}$ equals $\Gamma_* = 0$, and $F^2\,\mathrm{CH}_0(S)_{\mathbb{Q}} = 0$. The Albanese kernel is expected to lie in F^2, which is the case when the Künneth component π_3 is realised, as Murre showed it can be, by a correspondence acting through the Albanese variety [Mur93]; then the Albanese kernel is torsion. Kimura's notion of *finite-dimensional* motives gives another route: motives of curves and abelian varieties are finite-dimensional, and finite-dimensionality of the motive of S would imply Bloch's conjecture for S [Kim05].

Remark 77.5.5. The Hodge conjecture describes Gr^0_F: the image of the cycle class map is as large as Hodge theory permits. The Bloch–Beilinson conjectures describe F^1: the kernel is as structured as motivic expectations require. Neither is known to imply the other, but a complete theory of algebraic cycles needs both, and both are expected to come from a category of mixed motives. The absolute Hodge classes of Chapter 75, the motivated cycles of André and the unconditional results of this chapter all live on the cohomological side; on the side of cycles, the Lefschetz $(1,1)$-theorem (Chapter 69) remains the only general method for producing an algebraic cycle from a cohomology class.

77.6 Exercises

Exercise 77.6.1. Let X be a nonsingular complex projective variety. Show that the Künneth components π_i of the class of the diagonal are Hodge classes on $X \times X$, and deduce $C(X)$ from the Hodge conjecture for $X \times X$ without using $B(X)$.

Solution. The class of the diagonal is algebraic, hence a Hodge class. The Künneth decomposition $H^{2n}(X \times X) = \bigoplus_i H^{2n-i}(X) \otimes H^i(X)$ is a decomposition into sub-Hodge structures, and the components of a Hodge class along such a decomposition are Hodge classes (the components of a rational class of type (n,n) are rational and of type (n,n)). The Hodge conjecture for $X \times X$ makes each π_i algebraic. $\square$

Exercise 77.6.2. Show that $h(\mathbb{P}^n) \cong \bigoplus_{i=0}^{n} \mathbb{L}^{\otimes i}$ in $\mathcal{M}_{\mathrm{rat}}(k)$, and that $B(\mathbb{P}^n)$ holds, by writing Λ explicitly as a correspondence.

Solution. Let h_1, h_2 be the pullbacks of the hyperplane class to $\mathbb{P}^n \times \mathbb{P}^n$ and $p_i = h_1^{n-i} h_2^i \in \mathrm{CH}^n(\mathbb{P}^n \times \mathbb{P}^n)$. Then $(p_i)_*(h^j) = \mathrm{pr}_{2*}(h_1^{j+n-i} h_2^i) = \deg(h^{j+n-i}) h^i = \delta_{ij} h^i$, and $p_a \circ p_b = \mathrm{pr}_{13*}(h_1^{n-b} h_2^b h_2^{n-a} h_3^a) = \deg(h^{n+b-a}) h_1^{n-b} h_3^a = \delta_{ab} p_a$. The group $\mathrm{CH}^n(\mathbb{P}^n \times \mathbb{P}^n)$ is free on the classes $h_1^{n-i} h_2^i$, and $\gamma = \sum_i a_i h_1^{n-i} h_2^i$ acts by $\gamma_*(h^j) = a_j h^j$; as Δ acts as the identity, $\Delta = \sum_i p_i$. The correspondences $a = h^i \in \mathrm{CH}^i(\mathrm{Spec}\,k \times \mathbb{P}^n)$, a morphism $\mathbb{L}^{\otimes i} \to h(\mathbb{P}^n)$, and $b = h^{n-i} \in \mathrm{CH}^{n-i}(\mathbb{P}^n \times \mathrm{Spec}\,k)$, a morphism $h(\mathbb{P}^n) \to \mathbb{L}^{\otimes i}$, satisfy $b \circ a = \deg(h^n) = 1$ and $a \circ b = h_1^{n-i} h_2^i = p_i$, so $(\mathbb{P}^n, p_i, 0) \cong \mathbb{L}^{\otimes i}$. For B: the only primitive class is $1 \in P^0$,

so $\Lambda(h^j) = j(n-j+1)h^{j-1}$. The correspondence $\lambda = \sum_{i=1}^{n} i(n-i+1)h_1^{n-i}h_2^{i-1}$ satisfies $\lambda_*(h^j) = \sum_i i(n-i+1)\deg(h^{j+n-i})h^{i-1} = j(n-j+1)h^{j-1}$, so $\lambda_* = \Lambda$ and Λ is algebraic. □

Exercise 77.6.3. Let $X \subseteq \mathbb{P}^{n+1}$ be a nonsingular hypersurface over $\mathbb{C}$. Show that $B(X)$ holds.

Solution. By the Lefschetz hyperplane theorem and Poincaré duality, $H^i(X;\mathbb{Q})$ is $\mathbb{Q}h^{i/2}$ for $i \neq n$ even, 0 for $i \neq n$ odd, and $H^n = \mathbb{Q}h^{n/2} \oplus H^n_{\text{prim}}$ (with the first summand only for n even), where $H^n_{\text{prim}} = P^n$. The operator Λ kills P^n and acts on the powers of h by $\Lambda(h^i) = i(n-i+1)h^{i-1}$. Let $d = \deg X = \int h^n$, and consider the algebraic class $\lambda = \frac{1}{d}\sum_{j=1}^{n} j(n-j+1)h_1^{n-j}h_2^{j-1} \in H^{2n-2}(X \times X)$, where h_1, h_2 are the pullbacks of h. It acts by $\lambda_*(x) = \frac{1}{d}\sum_j j(n-j+1)\big(\int_X x \cup h^{n-j}\big)h^{j-1}$, and for $x \in H^i$ only the term with $i = 2j$ can contribute. On h^j it gives $j(n-j+1)h^{j-1} = \Lambda(h^j)$, because $\int h^n = d$. For $x \in P^n$, the only possible term has $j = n/2$ (so n is even), and $\int x \cup h^{n/2} = \int L^{n/2}x = 0$ because $Lx = 0$ by primitivity (if $n \geq 2$; the case $n = 0$ is trivial). So λ_* and Λ agree on a basis, and Λ is algebraic. The same argument applies to smooth complete intersections, whose cohomology has the same shape. □

Exercise 77.6.4. Explain why Jannsen's theorem does not imply the standard conjecture D, although both concern numerical equivalence. Show, using the converse part of Theorem 77.4.3, that $D(X)$ for all X is equivalent to the semisimplicity of $\mathcal{M}_{\text{hom}}(k)$.

Solution. Jannsen's theorem is a statement about the quotient A_{num} of A_{hom}; D says that the kernel of $A_{\text{hom}} \to A_{\text{num}}$ vanishes. A quotient algebra can be semisimple whatever the kernel is, so the first says nothing about the second. If D holds for all varieties, then $\mathcal{M}_{\text{hom}}(k) = \mathcal{M}_{\text{num}}(k)$, which is semisimple abelian by Jannsen. Conversely, if $\mathcal{M}_{\text{hom}}(k)$ is semisimple abelian, the converse part of the theorem says that homological equivalence is numerical equivalence, which is D for all varieties. □

Exercise 77.6.5. Show that $\mathrm{CH}_0(\mathbb{P}^2) = \mathbb{Z}$, via the degree, and $\mathrm{CH}_0(\mathbb{P}^1 \times \mathbb{P}^1) = \mathbb{Z}$, in accordance with Bloch's conjecture.

Solution. Two points of $\mathbb{P}^2$ lie on a line $\ell \cong \mathbb{P}^1$, and any two points of $\mathbb{P}^1$ are rationally equivalent, being the zero and the pole of a rational function; pushing forward along $\ell \to \mathbb{P}^2$, the two points are rationally equivalent on $\mathbb{P}^2$. So $\mathrm{CH}_0(\mathbb{P}^2)$ is generated by the class of one point, and the degree shows it is $\mathbb{Z}$. On $\mathbb{P}^1 \times \mathbb{P}^1$, the points (a,b) and (a',b') are joined by the chain $(a,b),(a',b),(a',b')$, whose consecutive points lie on a fibre $\cong \mathbb{P}^1$. Both surfaces have $h^{2,0} = 0$ and trivial Albanese, so this is Bloch's conjecture for them. □

Part X

Miscellany

Bibliography

[Ada60] J. F. Adams, *On the non-existence of elements of Hopf invariant one*, Annals of Mathematics. Second Series **72** (1960), 20–104.

[AGV72] M. Artin, A. Grothendieck, and J.-L. Verdier, *Théorie des topos et cohomologie étale des schémas. Tome 1: Théorie des topos (SGA 4)*, Lecture Notes in Mathematics, vol. 269, Springer-Verlag, Berlin, 1972.

[AGV73] _____, *Théorie des topos et cohomologie étale des schémas. Tome 3 (SGA 4)*, Lecture Notes in Mathematics, vol. 305, Springer-Verlag, Berlin, 1973.

[AH62] M. F. Atiyah and F. Hirzebruch, *Analytic cycles on complex manifolds*, Topology **1** (1962), 25–45.

[Ahl78] Lars V. Ahlfors, *Complex analysis*, third ed., McGraw-Hill Book Co., New York, 1978.

[AM69] M. F. Atiyah and I. G. Macdonald, *Introduction to commutative algebra*, Addison-Wesley, Reading, MA, 1969.

[And96a] Yves André, *On the Shafarevich and Tate conjectures for hyper-Kähler varieties*, Mathematische Annalen **305** (1996), no. 2, 205–248.

[And96b] _____, *Pour une théorie inconditionnelle des motifs*, Publications Mathématiques de l'IHÉS **83** (1996), 5–49.

[And04] _____, *Une introduction aux motifs (motifs purs, motifs mixtes, périodes)*, Panoramas et Synthèses, vol. 17, Société Mathématique de France, Paris, 2004.

[Art42] Emil Artin, *Galois theory*, Notre Dame Mathematical Lectures, vol. 2, University of Notre Dame, Notre Dame, IN, 1942.

[AS68a] Michael F. Atiyah and Isadore M. Singer, *The index of elliptic operators. I*, Annals of Mathematics. Second Series **87** (1968), 484–530.

[AS68b] _____, *The index of elliptic operators. III*, Annals of Mathematics. Second Series **87** (1968), 546–604.

[Axl15] Sheldon Axler, *Linear algebra done right*, third ed., Undergraduate Texts in Mathematics, Springer, Cham, 2015.

[BBD82] A. A. Beĭlinson, J. Bernstein, and P. Deligne, *Faisceaux pervers*, Astérisque **100** (1982), 5–171.

[BDGG69] Enrico Bombieri, Ennio De Giorgi, and Enrico Giusti, *Minimal cones and the Bernstein problem*, Inventiones Mathematicae **7** (1969), 243–268.

[Bea83] Arnaud Beauville, *Variétés Kähleriennes dont la première classe de Chern est nulle*, Journal of Differential Geometry **18** (1983), no. 4, 755–782.

[Bea96] _____, *Complex algebraic surfaces*, second ed., London Mathematical Society Student Texts, vol. 34, Cambridge University Press, Cambridge, 1996.

[Bei87] Alexander A. Beilinson, *Height pairing between algebraic cycles*, K-theory, Arithmetic and Geometry (Moscow, 1984–1986), Lecture Notes in Mathematics, vol. 1289, Springer-Verlag, Berlin, 1987, pp. 1–25.

[Ber55] Marcel Berger, *Sur les groupes d'holonomie homogène des variétés à connexion affine et des variétés riemanniennes*, Bulletin de la Société Mathématique de France **83** (1955), 279–330.

[BHPVdV04] Wolf P. Barth, Klaus Hulek, Chris A. M. Peters, and Antonius Van de Ven, *Compact complex surfaces*, second ed., Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge, vol. 4, Springer-Verlag, Berlin, 2004.

[BKL76] Spencer Bloch, Arnold Kas, and David Lieberman, *Zero cycles on surfaces with $p_g = 0$*, Compositio Mathematica **33** (1976), no. 2, 135–145.

[BKT20] Benjamin Bakker, Bruno Klingler, and Jacob Tsimerman, *Tame topology of arithmetic quotients and algebraicity of Hodge loci*, Journal of the American Mathematical Society **33** (2020), no. 4, 917–939.

[BL04] Christina Birkenhake and Herbert Lange, *Complex abelian varieties*, second ed., Grundlehren der Mathematischen Wissenschaften, vol. 302, Springer-Verlag, Berlin, 2004.

[Blo72] Spencer Bloch, *Semi-regularity and de Rham cohomology*, Inventiones Mathematicae **17** (1972), 51–66.

[Blo10] _____, *Lectures on algebraic cycles*, second ed., New Mathematical Monographs, vol. 16, Cambridge University Press, Cambridge, 2010.

[BM58] R. Bott and J. Milnor, *On the parallelizability of the spheres*, Bulletin of the American Mathematical Society **64** (1958), 87–89.

[Bor09] Émile Borel, *Les probabilités dénombrables et leurs applications arithmétiques*, Rendiconti del Circolo Matematico di Palermo **27** (1909), 247–271.

[Bor91] Armand Borel, *Linear algebraic groups*, second ed., Graduate Texts in Mathematics, vol. 126, Springer-Verlag, New York, 1991.

[Bre97] Glen E. Bredon, *Sheaf theory*, second ed., Graduate Texts in Mathematics, vol. 170, Springer-Verlag, New York, 1997.

[Bre11] Haim Brezis, *Functional analysis, sobolev spaces and partial differential equations*, Universitext, Springer, New York, 2011.

[BS83] Spencer Bloch and V. Srinivas, *Remarks on correspondences and algebraic cycles*, American Journal of Mathematics **105** (1983), no. 5, 1235–1253.

[BS09] Simon Brendle and Richard Schoen, *Manifolds with 1/4-pinched curvature are space forms*, Journal of the American Mathematical Society **22** (2009), no. 1, 287–307.

[BT24] Stefan Banach and Alfred Tarski, *Sur la décomposition des ensembles de points en parties respectivement congruentes*, Fundamenta Mathematicae **6** (1924), 244–277.

[BT82] Raoul Bott and Loring W. Tu, *Differential forms in algebraic topology*, Graduate Texts in Mathematics, vol. 82, Springer-Verlag, New York-Berlin, 1982.

[BtD85] Theodor Bröcker and Tammo tom Dieck, *Representations of compact lie groups*, Graduate Texts in Mathematics, vol. 98, Springer-Verlag, New York, 1985.

[Bus19] Nikolay Buskin, *Every rational Hodge isometry between two K3 surfaces is algebraic*, Journal für die reine und angewandte Mathematik **755** (2019), 127–150.

[BV04] Stephen Boyd and Lieven Vandenberghe, *Convex optimization*, Cambridge University Press, Cambridge, 2004.

[Car66] Lennart Carleson, *On convergence and growth of partial sums of Fourier series*, Acta Mathematica **116** (1966), 135–157.

[Car80] James A. Carlson, *Extensions of mixed Hodge structures*, Journées de Géométrie Algébrique d'Angers, Juillet 1979, Sijthoff & Noordhoff, Alphen aan den Rijn, 1980.

[CDK95] Eduardo Cattani, Pierre Deligne, and Aroldo Kaplan, *On the locus of Hodge classes*, Journal of the American Mathematical Society **8** (1995), no. 2, 483–506.

[CdlOGP91] Philip Candelas, Xenia C. de la Ossa, Paul S. Green, and Linda Parkes, *A pair of Calabi–Yau manifolds as an exactly soluble superconformal theory*, Nuclear Physics B **359** (1991), no. 1, 21–74.

[CdS01] Ana Cannas da Silva, *Lectures on symplectic geometry*, Lecture Notes in Mathematics, vol. 1764, Springer-Verlag, Berlin, 2001.

[CE53] Eugenio Calabi and Beno Eckmann, *A class of compact, complex manifolds which are not algebraic*, Annals of Mathematics. Second Series **58** (1953), 494–500.

[CE56] Henri Cartan and Samuel Eilenberg, *Homological algebra*, Princeton University Press, Princeton, NJ, 1956.

[CG71] Jeff Cheeger and Detlef Gromoll, *The splitting theorem for manifolds of nonnegative Ricci curvature*, Journal of Differential Geometry **6** (1971), 119–128.

[CG72a] _____, *On the structure of complete manifolds of nonnegative curvature*, Annals of Mathematics. Second Series **96** (1972), 413–443.

[CG72b] C. Herbert Clemens and Phillip A. Griffiths, *The intermediate Jacobian of the cubic threefold*, Annals of Mathematics. Second Series **95** (1972), 281–356.

[Cha33] D. G. Champernowne, *The construction of decimals normal in the scale of ten*, Journal of the London Mathematical Society **8** (1933), 254–260.

[Cha13] François Charles, *The Tate conjecture for K3 surfaces over finite fields*, Inventiones Mathematicae **194** (2013), no. 1, 119–145.

[Che44] Shiing-Shen Chern, *A simple intrinsic proof of the Gauss–Bonnet formula for closed Riemannian manifolds*, Annals of Mathematics. Second Series **45** (1944), no. 4, 747–752.

[Che75] Shiu Yuen Cheng, *Eigenvalue comparison theorems and its geometric applications*, Mathematische Zeitschrift **143** (1975), 289–297.

[CK90] C. C. Chang and H. J. Keisler, *Model theory*, third ed., Studies in Logic and the Foundations of Mathematics, vol. 73, North-Holland, Amsterdam, 1990.

[Cle83] Herbert Clemens, *Homological equivalence, modulo algebraic equivalence, is not finitely generated*, Publications Mathématiques de l'IHÉS **58** (1983), 19–38.

[CM78] Alberto Conte and Jacob P. Murre, *The Hodge conjecture for fourfolds admitting a covering by rational curves*, Mathematische Annalen **238** (1978), no. 1, 79–88.

[CM13] François Charles and Eyal Markman, *The standard conjectures for holomorphic symplectic varieties deformation equivalent to Hilbert schemes of K3 surfaces*, Compositio Mathematica **149** (2013), no. 3, 481–494.

[CMSP03] James Carlson, Stefan Müller-Stach, and Chris Peters, *Period mappings and period domains*, Cambridge Studies in Advanced Mathematics, vol. 85, Cambridge University Press, Cambridge, 2003.

[Coh63] Paul J. Cohen, *The independence of the continuum hypothesis*, Proceedings of the National Academy of Sciences of the U.S.A. **50** (1963), 1143–1148.

[Con78] John B. Conway, *Functions of one complex variable I*, second ed., Graduate Texts in Mathematics, vol. 11, Springer-Verlag, New York, 1978.

[Cox89] David A. Cox, *Primes of the form $x^2 + ny^2$: Fermat, class field theory, and complex multiplication*, John Wiley & Sons, New York, 1989.

[CS14] François Charles and Christian Schnell, *Notes on absolute Hodge classes*, Hodge Theory, Mathematical Notes, vol. 49, Princeton University Press, Princeton, NJ, 2014.

[CTV12] Jean-Louis Colliot-Thélène and Claire Voisin, *Cohomologie non ramifiée et conjecture de Hodge entière*, Duke Mathematical Journal **161** (2012), no. 5, 735–801.

[dC76] Manfredo Perdigão do Carmo, *Differential geometry of curves and surfaces*, Prentice-Hall, Englewood Cliffs, NJ, 1976.

[dC92] ———, *Riemannian geometry*, Mathematics: Theory & Applications, Birkhäuser Boston, Boston, MA, 1992.

[Ded72] Richard Dedekind, *Stetigkeit und irrationale Zahlen*, Vieweg, Braunschweig, 1872.

[Del68] Pierre Deligne, *Théorème de Lefschetz et critères de dégénérescence de suites spectrales*, Publications Mathématiques de l'IHÉS **35** (1968).

[Del71] ———, *Théorie de Hodge. II*, Institut des Hautes Études Scientifiques. Publications Mathématiques **40** (1971), 5–57.

[Del72] ———, *La conjecture de Weil pour les surfaces K3*, Inventiones Mathematicae **15** (1972), 206–226.

[Del74a] ———, *La conjecture de Weil. I*, Publications Mathématiques de l'IHÉS **43** (1974), 273–307.

[Del74b] ———, *Théorie de Hodge. III*, Institut des Hautes Études Scientifiques. Publications Mathématiques **44** (1974), 5–77.

[Del77] ———, *Cohomologie étale (SGA $4\frac{1}{2}$)*, Lecture Notes in Mathematics, vol. 569, Springer-Verlag, Berlin, 1977.

[Del80] ———, *La conjecture de Weil. II*, Publications Mathématiques de l'IHÉS **52** (1980), 137–252.

[Del82] ———, *Hodge cycles on abelian varieties (notes by J. S. Milne)*, Hodge Cycles, Motives, and Shimura Varieties, Lecture Notes in Mathematics, vol. 900, Springer-Verlag, Berlin, 1982, pp. 9–100.

[Del06] ———, *The Hodge conjecture*, The Millennium Prize Problems, Clay Math. Inst., Cambridge, MA, 2006, pp. 45–53.

[DF04] David S. Dummit and Richard M. Foote, *Abstract algebra*, third ed., John Wiley & Sons, Hoboken, NJ, 2004.

[DGMS75] Pierre Deligne, Phillip Griffiths, John Morgan, and Dennis Sullivan, *Real homotopy theory of Kähler manifolds*, Inventiones Mathematicae **29** (1975), no. 3, 245–274.

[DI87] Pierre Deligne and Luc Illusie, *Relèvements modulo p^2 et décomposition du complexe de de Rham*, Inventiones Mathematicae **89** (1987), no. 2, 247–270.

[Die60] J. Dieudonné, *Foundations of modern analysis*, Pure and Applied Mathematics, vol. 10, Academic Press, New York-London, 1960.

[DM82] Pierre Deligne and James S. Milne, *Tannakian categories*, Hodge Cycles, Motives, and Shimura Varieties, Lecture Notes in Mathematics, vol. 900, Springer-Verlag, Berlin, 1982, pp. 101–228.

[dR84] Georges de Rham, *Differentiable manifolds*, Grundlehren der mathematischen Wissenschaften, vol. 266, Springer-Verlag, Berlin, 1984.

[EFT94] Heinz-Dieter Ebbinghaus, Jörg Flum, and Wolfgang Thomas, *Mathematical logic*, second ed., Undergraduate Texts in Mathematics, Springer-Verlag, New York, 1994.

[EG92] Lawrence C. Evans and Ronald F. Gariepy, *Measure theory and fine properties of functions*, Studies in Advanced Mathematics, CRC Press, Boca Raton, FL, 1992.

[Eis95] David Eisenbud, *Commutative algebra with a view toward algebraic geometry*, Graduate Texts in Mathematics, vol. 150, Springer-Verlag, New York, 1995.

[EML45] Samuel Eilenberg and Saunders Mac Lane, *General theory of natural equivalences*, Transactions of the American Mathematical Society **58** (1945), 231–294.

[End01] Herbert B. Enderton, *A mathematical introduction to logic*, second ed., Harcourt/Academic Press, Burlington, MA, 2001.

[Eng89] Ryszard Engelking, *General topology*, revised and completed ed., Sigma Series in Pure Mathematics, vol. 6, Heldermann Verlag, Berlin, 1989.

[Fal83] Gerd Faltings, *Endlichkeitssätze für abelsche Varietäten über Zahlkörpern*, Inventiones Mathematicae **73** (1983), no. 3, 349–366.

[Fal85] K. J. Falconer, *The geometry of fractal sets*, Cambridge Tracts in Mathematics, vol. 85, Cambridge University Press, Cambridge, 1985.

[Fal90] Kenneth Falconer, *Fractal geometry: Mathematical foundations and applications*, John Wiley & Sons, Chichester, 1990.

[Fed69] Herbert Federer, *Geometric measure theory*, Grundlehren der mathematischen Wissenschaften, vol. 153, Springer-Verlag New York Inc., New York, 1969.

[FF60] Herbert Federer and Wendell H. Fleming, *Normal and integral currents*, Annals of Mathematics **72** (1960), 458–520.

[FH91] William Fulton and Joe Harris, *Representation theory. A first course*, Graduate Texts in Mathematics, vol. 129, Springer-Verlag, New York, 1991.

[Fol99] Gerald B. Folland, *Real analysis: Modern techniques and their applications*, second ed., Pure and Applied Mathematics, John Wiley & Sons, New York, 1999.

[For81] Otto Forster, *Lectures on Riemann surfaces*, Graduate Texts in Mathematics, vol. 81, Springer-Verlag, New York, 1981.

[Fre82] Michael Hartley Freedman, *The topology of four-dimensional manifolds*, Journal of Differential Geometry **17** (1982), no. 3, 357–453.

[Ful98] William Fulton, *Intersection theory*, second ed., Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge, vol. 2, Springer-Verlag, Berlin, 1998.

[Gö30] Kurt Gödel, *Die Vollständigkeit der Axiome des logischen Funktionenkalküls*, Monatshefte für Mathematik und Physik **37** (1930), 349–360.

[Gö31] ———, *Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I*, Monatshefte für Mathematik und Physik **38** (1931), 173–198.

[Gö38] ———, *The consistency of the axiom of choice and of the generalized continuum-hypothesis*, Proceedings of the National Academy of Sciences of the U.S.A. **24** (1938), 556–557.

[GG07] Mark Green and Phillip Griffiths, *Algebraic cycles and singularities of normal functions*, Algebraic Cycles and Motives. Vol. 1, London Mathematical Society Lecture Note Series, vol. 343, Cambridge University Press, Cambridge, 2007, pp. 206–263.

[GGK12] Mark Green, Phillip Griffiths, and Matt Kerr, *Mumford–Tate groups and domains: Their geometry and arithmetic*, Annals of Mathematics Studies, vol. 183, Princeton University Press, Princeton, NJ, 2012.

[GH78] Phillip Griffiths and Joseph Harris, *Principles of algebraic geometry*, Wiley-Interscience, New York, 1978.

[GH85] ———, *On the Noether–Lefschetz theorem and some remarks on codimension-two cycles*, Mathematische Annalen **271** (1985), no. 1, 31–51.

[GHJ03] Mark Gross, Daniel Huybrechts, and Dominic Joyce, *Calabi–Yau manifolds and related geometries*, Universitext, Springer-Verlag, Berlin, 2003.

[GM69] Detlef Gromoll and Wolfgang Meyer, *On complete open manifolds of positive curvature*, Annals of Mathematics. Second Series **90** (1969), 75–90.

[GM03] Sergei I. Gelfand and Yuri I. Manin, *Methods of homological algebra*, second ed., Springer Monographs in Mathematics, Springer-Verlag, Berlin, 2003.

[God58] Roger Godement, *Topologie algébrique et théorie des faisceaux*, Hermann, Paris, 1958.

[Goo44] R. L. Goodstein, *On the restricted ordinal theorem*, The Journal of Symbolic Logic **9** (1944), 33–41.

[GR65] Robert C. Gunning and Hugo Rossi, *Analytic functions of several complex variables*, Prentice-Hall, Englewood Cliffs, NJ, 1965.

[GR84] Hans Grauert and Reinhold Remmert, *Coherent analytic sheaves*, Grundlehren der mathematischen Wissenschaften, vol. 265, Springer-Verlag, Berlin, 1984.

[Gre78] Werner Greub, *Multilinear algebra*, second ed., Universitext, Springer-Verlag, New York, 1978.

[Gri68] Phillip A. Griffiths, *Periods of integrals on algebraic manifolds. I*, American Journal of Mathematics **90** (1968), 568–626.

[Gri69] _____, *On the periods of certain rational integrals. I, II*, Annals of Mathematics. Second Series **90** (1969), 460–495, 496–541.

[Gro66] Alexander Grothendieck, *On the de Rham cohomology of algebraic varieties*, Publications Mathématiques de l'IHÉS **29** (1966), 95–103.

[Gro69a] _____, *Hodge's general conjecture is false for trivial reasons*, Topology **8** (1969), 299–303.

[Gro69b] _____, *Standard conjectures on algebraic cycles*, Algebraic Geometry (Internat. Colloq., Tata Inst. Fund. Res., Bombay, 1968), Oxford University Press, London, 1969, pp. 193–199.

[Gro71] _____, *Revêtements étales et groupe fondamental (SGA 1)*, Lecture Notes in Mathematics, vol. 224, Springer-Verlag, Berlin, 1971.

[Hal60] Paul R. Halmos, *Naive set theory*, D. Van Nostrand, Princeton, NJ, 1960.

[Hal74] _____, *Finite-dimensional vector spaces*, second ed., Undergraduate Texts in Mathematics, Springer-Verlag, New York, 1974.

[Har77] Robin Hartshorne, *Algebraic geometry*, Graduate Texts in Mathematics, vol. 52, Springer-Verlag, New York-Heidelberg, 1977.

[Har02] Philip Hartman, *Ordinary differential equations*, Classics in Applied Mathematics, vol. 38, Society for Industrial and Applied Mathematics (SIAM), Philadelphia, PA, 2002, Corrected reprint of the second (1982) edition.

[Hat02] Allen Hatcher, *Algebraic topology*, Cambridge University Press, Cambridge, 2002.

[Hen49] Leon Henkin, *The completeness of the first-order functional calculus*, Journal of Symbolic Logic **14** (1949), 159–166.

[Hil01] David Hilbert, *Ueber Flächen von constanter Gaussscher Krümmung*, Transactions of the American Mathematical Society **2** (1901), 87–99.

[Hir64] Heisuke Hironaka, *Resolution of singularities of an algebraic variety over a field of characteristic zero. I, II*, Annals of Mathematics. Second Series **79** (1964), 109–203, 205–326.

[Hir66] Friedrich Hirzebruch, *Topological methods in algebraic geometry*, third ed., Grundlehren der mathematischen Wissenschaften, vol. 131, Springer-Verlag, New York, 1966.

[Hir76] Morris W. Hirsch, *Differential topology*, Graduate Texts in Mathematics, vol. 33, Springer-Verlag, New York-Heidelberg, 1976.

[HM98] Joe Harris and Ian Morrison, *Moduli of curves*, Graduate Texts in Mathematics, vol. 187, Springer-Verlag, New York, 1998.

[Hod52] W. V. D. Hodge, *The topological invariants of algebraic varieties*, Proceedings of the International Congress of Mathematicians, Cambridge, Mass., 1950, vol. 1, Amer. Math. Soc., Providence, R. I., 1952, pp. 182–192.

[Hop35] Heinz Hopf, *Über die Drehung der Tangenten und Sehnen ebener Kurven*, Compositio Mathematica **2** (1935), 50–62.

[Hör90a] Lars Hörmander, *The analysis of linear partial differential operators. I*, second ed., Grundlehren der mathematischen Wissenschaften, vol. 256, Springer-Verlag, Berlin, 1990.

[Hör90b] ———, *An introduction to complex analysis in several variables*, third ed., North-Holland Mathematical Library, vol. 7, North-Holland Publishing Co., Amsterdam, 1990.

[Hum72] James E. Humphreys, *Introduction to lie algebras and representation theory*, Graduate Texts in Mathematics, vol. 9, Springer-Verlag, New York, 1972.

[Hut81] John E. Hutchinson, *Fractals and self-similarity*, Indiana University Mathematics Journal **30** (1981), no. 5, 713–747.

[Huy05] Daniel Huybrechts, *Complex geometry: An introduction*, Universitext, Springer-Verlag, Berlin, 2005.

[HW08] G. H. Hardy and E. M. Wright, *An introduction to the theory of numbers*, sixth ed., Oxford University Press, Oxford, 2008, Revised by D. R. Heath-Brown and J. H. Silverman.

[Ima76] Hideo Imai, *On the Hodge groups of some abelian varieties*, Kodai Mathematical Seminar Reports **27** (1976), no. 3, 367–372.

[IR90] Kenneth Ireland and Michael Rosen, *A classical introduction to modern number theory*, second ed., Graduate Texts in Mathematics, vol. 84, Springer-Verlag, New York, 1990.

[Ive86] Birger Iversen, *Cohomology of sheaves*, Universitext, Springer-Verlag, Berlin, 1986.

[Jan92] Uwe Jannsen, *Motives, numerical equivalence, and semi-simplicity*, Inventiones Mathematicae **107** (1992), no. 3, 447–452.

[Jan94] ———, *Motivic sheaves and filtrations on Chow groups*, Motives (Seattle, WA, 1991), Proceedings of Symposia in Pure Mathematics, vol. 55, Part 1, American Mathematical Society, Providence, RI, 1994, pp. 245–302.

[Jec03] Thomas Jech, *Set theory*, the third millennium ed., Springer Monographs in Mathematics, Springer-Verlag, Berlin, 2003.

[Joy00] Dominic D. Joyce, *Compact manifolds with special holonomy*, Oxford Mathematical Monographs, Oxford University Press, Oxford, 2000.

[Kac66] Mark Kac, *Can one hear the shape of a drum?*, American Mathematical Monthly **73** (1966), no. 4, part II, 1–23.

[Kan58] Daniel M. Kan, *Adjoint functors*, Transactions of the American Mathematical Society **87** (1958), 294–329.

[Kel50] John L. Kelley, *The Tychonoff product theorem implies the axiom of choice*, Fundamenta Mathematicae **37** (1950), 75–76.

[Ker58] Michel A. Kervaire, *Non-parallelizability of the n-sphere for $n > 7$*, Proceedings of the National Academy of Sciences of the United States of America **44** (1958), 280–283.

[Kim05] Shun-Ichi Kimura, *Chow groups are finite dimensional, in some sense*, Mathematische Annalen **331** (2005), no. 1, 173–201.

[Kle68] Steven L. Kleiman, *Algebraic cycles and the Weil conjectures*, Dix exposés sur la cohomologie des schémas, North-Holland, Amsterdam, 1968, pp. 359–386.

[Kle94] ———, *The standard conjectures*, Motives (Seattle, WA, 1991), Proceedings of Symposia in Pure Mathematics, vol. 55, Part 1, American Mathematical Society, Providence, RI, 1994, pp. 3–20.

[KM74] Nicholas M. Katz and William Messing, *Some consequences of the Riemann hypothesis for varieties over finite fields*, Inventiones Mathematicae **23** (1974), 73–77.

[KN63] Shoshichi Kobayashi and Katsumi Nomizu, *Foundations of differential geometry. Vol. I*, Interscience Publishers, New York-London, 1963.

[Kna02] Anthony W. Knapp, *Lie groups beyond an introduction*, second ed., Progress in Mathematics, vol. 140, Birkhäuser Boston, Boston, MA, 2002.

[KO68] Nicholas M. Katz and Tadao Oda, *On the differentiation of de Rham cohomology classes with respect to parameters*, Journal of Mathematics of Kyoto University **8** (1968), 199–213.

[Kod54] Kunihiko Kodaira, *On Kähler varieties of restricted type (an intrinsic characterization of algebraic varieties)*, Annals of Mathematics. Second Series **60** (1954), 28–48.

[Kod86] ———, *Complex manifolds and deformation of complex structures*, Grundlehren der Mathematischen Wissenschaften, vol. 283, Springer-Verlag, New York, 1986.

[Kol23] André Kolmogoroff, *Une série de Fourier–Lebesgue divergente presque partout*, Fundamenta Mathematicae **4** (1923), 324–328.

[Kol92] János Kollár, *Trento examples*, Classification of Irregular Varieties (Trento, 1990), Lecture Notes in Mathematics, vol. 1515, Springer-Verlag, Berlin, 1992.

[KP82] Laurie Kirby and Jeff Paris, *Accessible independence results for Peano arithmetic*, Bulletin of the London Mathematical Society **14** (1982), 285–293.

[KS53] Kunihiko Kodaira and Donald C. Spencer, *Divisor class groups on algebraic varieties*, Proceedings of the National Academy of Sciences of the USA **39** (1953), 872–877.

[KS67] Michio Kuga and Ichirô Satake, *Abelian varieties attached to polarized K3-surfaces*, Mathematische Annalen **169** (1967), 239–242.

[KS90] Masaki Kashiwara and Pierre Schapira, *Sheaves on manifolds*, Grundlehren der mathematischen Wissenschaften, vol. 292, Springer-Verlag, Berlin, 1990.

[KS06] ———, *Categories and sheaves*, Grundlehren der mathematischen Wissenschaften, vol. 332, Springer-Verlag, Berlin, 2006.

[Kun11] Kenneth Kunen, *Set theory*, Studies in Logic, vol. 34, College Publications, London, 2011.

[Lan30] Edmund Landau, *Grundlagen der Analysis*, Akademische Verlagsgesellschaft, Leipzig, 1930.

[Lan93] Serge Lang, *Real and functional analysis*, third ed., Graduate Texts in Mathematics, vol. 142, Springer-Verlag, New York, 1993.

[Lan02] ———, *Algebra*, revised third ed., Graduate Texts in Mathematics, vol. 211, Springer-Verlag, New York, 2002.

[Laz04] Robert Lazarsfeld, *Positivity in algebraic geometry. I*, Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge, vol. 48, Springer-Verlag, Berlin, 2004.

[Lee13] John M. Lee, *Introduction to smooth manifolds*, second ed., Graduate Texts in Mathematics, vol. 218, Springer, New York, 2013.

[Lee18] ———, *Introduction to riemannian manifolds*, second ed., Graduate Texts in Mathematics, vol. 176, Springer, Cham, 2018.

[Lew99] James D. Lewis, *A survey of the Hodge conjecture*, second ed., CRM Monograph Series, vol. 10, American Mathematical Society, Providence, RI, 1999.

[Lie68] David I. Lieberman, *Numerical and homological equivalence of algebraic cycles on Hodge manifolds*, American Journal of Mathematics **90** (1968), 366–374.

[Lin69] Per Lindström, *On extensions of elementary logic*, Theoria **35** (1969), 1–11.

[LM89] H. Blaine Lawson and Marie-Louise Michelsohn, *Spin geometry*, Princeton Mathematical Series, vol. 38, Princeton University Press, Princeton, NJ, 1989.

[LS77] Roger C. Lyndon and Paul E. Schupp, *Combinatorial group theory*, Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 89, Springer-Verlag, Berlin-New York, 1977.

[Mar02] David Marker, *Model theory: An introduction*, Graduate Texts in Mathematics, vol. 217, Springer-Verlag, New York, 2002.

[Mar23] Eyal Markman, *The monodromy of generalized Kummer varieties and algebraic cycles on their intermediate Jacobians*, Journal of the European Mathematical Society **25** (2023), no. 1, 231–321.

[Mat86] Hideyuki Matsumura, *Commutative ring theory*, Cambridge Studies in Advanced Mathematics, vol. 8, Cambridge University Press, Cambridge, 1986.

[Mat95] Pertti Mattila, *Geometry of sets and measures in Euclidean spaces: Fractals and rectifiability*, Cambridge Studies in Advanced Mathematics, vol. 44, Cambridge University Press, Cambridge, 1995.

[Mau14] Davesh Maulik, *Supersingular K3 surfaces for large primes*, Duke Mathematical Journal **163** (2014), no. 13, 2357–2425.

[McC01] John McCleary, *A user's guide to spectral sequences*, second ed., Cambridge Studies in Advanced Mathematics, vol. 58, Cambridge University Press, Cambridge, 2001.

[Mil58] John Milnor, *On simply connected* 4*-manifolds*, Symposium internacional de topología algebraica, Universidad Nacional Autónoma de México and UNESCO, Mexico City, 1958, pp. 122–128.

[Mil63] ———, *Morse theory*, Annals of Mathematics Studies, vol. 51, Princeton University Press, Princeton, NJ, 1963.

[Mil64] ———, *Eigenvalues of the Laplace operator on certain manifolds*, Proceedings of the National Academy of Sciences of the USA **51** (1964), 542.

[Mil80] James S. Milne, *étale cohomology*, Princeton Mathematical Series, vol. 33, Princeton University Press, Princeton, NJ, 1980.

[Mil97] John W. Milnor, *Topology from the differentiable viewpoint*, Princeton Landmarks in Mathematics, Princeton University Press, Princeton, NJ, 1997.

[Mil02] James S. Milne, *Polarizations and Grothendieck's standard conjectures*, Annals of Mathematics. Second Series **155** (2002), no. 2, 599–610.

[Mil17] ———, *Algebraic groups: The theory of group schemes of finite type over a field*, Cambridge Studies in Advanced Mathematics, vol. 170, Cambridge University Press, Cambridge, 2017.

[ML98] Saunders Mac Lane, *Categories for the working mathematician*, second ed., Graduate Texts in Mathematics, vol. 5, Springer-Verlag, New York, 1998.

[MP15] Keerthi Madapusi Pera, *The Tate conjecture for K3 surfaces in odd characteristic*, Inventiones Mathematicae **201** (2015), no. 2, 625–668.

[MS74] John W. Milnor and James D. Stasheff, *Characteristic classes*, Annals of Mathematics Studies, vol. 76, Princeton University Press, Princeton, NJ, 1974.

[MS17] Dusa McDuff and Dietmar Salamon, *Introduction to symplectic topology*, third ed., Oxford Graduate Texts in Mathematics, vol. 27, Oxford University Press, Oxford, 2017.

[MT68] Robert E. Mosher and Martin C. Tangora, *Cohomology operations and applications in homotopy theory*, Harper & Row, New York, 1968.

[Mum68] David Mumford, *Rational equivalence of* 0*-cycles on surfaces*, Journal of Mathematics of Kyoto University **9** (1968), 195–204.

[Mum69] ———, *A note of Shimura's paper "Discontinuous groups and abelian varieties"*, Mathematische Annalen **181** (1969), 345–351.

[Mum70] ———, *Abelian varieties*, Tata Institute of Fundamental Research Studies in Mathematics, vol. 5, Oxford University Press, London, 1970.

[Mun00] James R. Munkres, *Topology*, second ed., Prentice Hall, Upper Saddle River, NJ, 2000.

[Mur84] V. Kumar Murty, *Exceptional Hodge classes on certain abelian varieties*, Mathematische Annalen **268** (1984), no. 2, 197–206.

[Mur93] Jacob P. Murre, *On a conjectural filtration on the Chow groups of an algebraic variety. I. The general conjectures and some examples*, Indagationes Mathematicae. New Series **4** (1993), no. 2, 177–188.

[MZ99] Ben Moonen and Yuri G. Zarhin, *Hodge classes on abelian varieties of low dimension*, Mathematische Annalen **315** (1999), no. 4, 711–733.

[Neu79] Peter M. Neumann, *A lemma that is not Burnside's*, The Mathematical Scientist **4** (1979), no. 2, 133–141.

[Nit57] Johannes C. C. Nitsche, *Elementary proof of Bernstein's theorem on minimal surfaces*, Annals of Mathematics. Second Series **66** (1957), 543–544.

[NO85] Niels Nygaard and Arthur Ogus, *Tate's conjecture for K3 surfaces of finite height*, Annals of Mathematics. Second Series **122** (1985), no. 3, 461–507.

[Nyg83] Niels O. Nygaard, *The Tate conjecture for ordinary K3 surfaces over finite fields*, Inventiones Mathematicae **74** (1983), no. 2, 213–237.

[Par88] Kapil Paranjape, *Abelian varieties associated to certain K3 surfaces*, Compositio Mathematica **68** (1988), no. 1, 11–22.

[Per94] Grigori Perelman, *Proof of the soul conjecture of Cheeger and Gromoll*, Journal of Differential Geometry **40** (1994), no. 1, 209–212.

[PH77] Jeff Paris and Leo Harrington, *A mathematical incompleteness in Peano arithmetic*, Handbook of Mathematical Logic (Jon Barwise, ed.), North-Holland, Amsterdam, 1977, pp. 1133–1142.

[Pin98] Richard Pink, *l-adic algebraic monodromy groups, cocharacters, and the Mumford–Tate conjecture*, Journal für die reine und angewandte Mathematik **495** (1998), 187–237.

[Pre87] David Preiss, *Geometry of measures in* $\mathbf{R}^n$*: distribution, rectifiability, and densities*, Annals of Mathematics **125** (1987), no. 3, 537–643.

[PS08] Chris A. M. Peters and Joseph H. M. Steenbrink, *Mixed hodge structures*, Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge, vol. 52, Springer-Verlag, Berlin, 2008.

[Qui69] Daniel Quillen, *On the formal group laws of unoriented and complex cobordism theory*, Bulletin of the American Mathematical Society **75** (1969), 1293–1298.

[Rav86] Douglas C. Ravenel, *Complex cobordism and stable homotopy groups of spheres*, Pure and Applied Mathematics, vol. 121, Academic Press, Orlando, FL, 1986.

[Ray78] Michel Raynaud, *Contre-exemple au "vanishing theorem" en caractéristique $p > 0$*, C. P. Ramanujam—a tribute, Tata Institute of Fundamental Research Studies in Mathematics, vol. 8, Springer, Berlin-New York, 1978, pp. 273–278.

[Rie16] Emily Riehl, *Category theory in context*, Dover Publications, Mineola, NY, 2016.

[Ros36] Barkley Rosser, *Extensions of some theorems of Gödel and Church*, Journal of Symbolic Logic **1** (1936), 87–91.

[Rot95] Joseph J. Rotman, *An introduction to the theory of groups*, fourth ed., Graduate Texts in Mathematics, vol. 148, Springer-Verlag, New York, 1995.

[RS80] Michael Reed and Barry Simon, *Methods of modern mathematical physics. I. functional analysis*, revised and enlarged ed., Academic Press, New York, 1980.

[Rud69] Mary Ellen Rudin, *A new proof that metric spaces are paracompact*, Proceedings of the American Mathematical Society **20** (1969), 603.

[Rud76] Walter Rudin, *Principles of mathematical analysis*, third ed., McGraw-Hill, New York, 1976.

[Rud87] ———, *Real and complex analysis*, third ed., McGraw-Hill Book Co., New York, 1987.

[Rud91] ———, *Functional analysis*, second ed., International Series in Pure and Applied Mathematics, McGraw-Hill, New York, 1991.

[Sch88] Chad Schoen, *Hodge classes on self-products of a variety with an automorphism*, Compositio Mathematica **65** (1988), no. 1, 3–32.

[Sch94] Anthony J. Scholl, *Classical motives*, Motives (Seattle, WA, 1991), Proceedings of Symposia in Pure Mathematics, vol. 55, Part 1, American Mathematical Society, Providence, RI, 1994, pp. 163–187.

[SE62] N. E. Steenrod and D. B. A. Epstein, *Cohomology operations*, Annals of Mathematics Studies, vol. 50, Princeton University Press, Princeton, NJ, 1962.

[Ser53] Jean-Pierre Serre, *Cohomologie modulo 2 des complexes d'Eilenberg-MacLane*, Commentarii Mathematici Helvetici **27** (1953), 198–232.

[Ser56] ———, *Géométrie algébrique et géométrie analytique*, Annales de l'Institut Fourier **6** (1956), 1–42.

[Ser64] ———, *Exemples de variétés projectives conjuguées non homéomorphes*, Comptes Rendus de l'Académie des Sciences de Paris **258** (1964), 4194–4196.

[Ser72] ———, *Propriétés galoisiennes des points d'ordre fini des courbes elliptiques*, Inventiones Mathematicae **15** (1972), no. 4, 259–331.

[Ser73] ———, *A course in arithmetic*, Graduate Texts in Mathematics, vol. 7, Springer-Verlag, New York, 1973.

[Ser77] ———, *Linear representations of finite groups*, Graduate Texts in Mathematics, vol. 42, Springer-Verlag, New York, 1977.

[Ser80] ———, *Trees*, Springer-Verlag, Berlin-New York, 1980.

[Shi79] Tetsuji Shioda, *The Hodge conjecture for Fermat varieties*, Mathematische Annalen **245** (1979), no. 2, 175–184.

[Sho67] Joseph R. Shoenfield, *Mathematical logic*, Addison-Wesley, Reading, MA, 1967.

[Sim68] James Simons, *Minimal varieties in Riemannian manifolds*, Annals of Mathematics. Second Series **88** (1968), 62–105.

[Smi13] Peter Smith, *An introduction to Gödel's theorems*, second ed., Cambridge University Press, Cambridge, 2013.

[Spa66] Edwin H. Spanier, *Algebraic topology*, McGraw-Hill Book Co., New York, 1966.

[Spi65] Michael Spivak, *Calculus on manifolds*, W. A. Benjamin, New York-Amsterdam, 1965.

[SS03] Elias M. Stein and Rami Shakarchi, *Complex analysis*, Princeton Lectures in Analysis, vol. 2, Princeton University Press, Princeton, NJ, 2003.

[Ste15] Ian Stewart, *Galois theory*, fourth ed., CRC Press, Boca Raton, FL, 2015.

[Sto48] A. H. Stone, *Paracompactness and product spaces*, Bulletin of the American Mathematical Society **54** (1948), 977–982.

[Sto68] Robert E. Stong, *Notes on cobordism theory*, Mathematical Notes, Princeton University Press, Princeton, NJ; University of Tokyo Press, Tokyo, 1968.

[SV05] Christophe Soulé and Claire Voisin, *Torsion cohomology classes and algebraic cycles on complex projective manifolds*, Advances in Mathematics **198** (2005), no. 1, 107–127.

[Sza09] Tamás Szamuely, *Galois groups and fundamental groups*, Cambridge Studies in Advanced Mathematics, vol. 117, Cambridge University Press, Cambridge, 2009.

[Tan82] Sergei G. Tankeev, *Cycles on simple abelian varieties of prime dimension*, Izvestiya Akademii Nauk SSSR. Seriya Matematicheskaya **46** (1982), no. 1, 155–170.

[Tat65] John T. Tate, *Algebraic cycles and poles of zeta functions*, Arithmetical Algebraic Geometry (Proc. Conf. Purdue Univ., 1963), Harper & Row, New York, 1965, pp. 93–110.

[Tat66a] John Tate, *Endomorphisms of abelian varieties over finite fields*, Inventiones Mathematicae **2** (1966), 134–144.

[Tat66b] ———, *On the conjectures of Birch and Swinnerton-Dyer and a geometric analog*, Séminaire Bourbaki, 18e année (1965/66), 1966, Exposé 306.

[The] The Stacks project authors, *The Stacks project*, `https://stacks.math.columbia.edu`.

[Tho54] René Thom, *Quelques propriétés globales des variétés différentiables*, Commentarii Mathematici Helvetici **28** (1954), 17–86.

[Thu76] William P. Thurston, *Some simple examples of symplectic manifolds*, Proceedings of the American Mathematical Society **55** (1976), no. 2, 467–468.

[Tia87] Gang Tian, *Smoothness of the universal deformation space of compact Calabi–Yau manifolds and its Petersson–Weil metric*, Mathematical Aspects of String Theory (San Diego, 1986), Advanced Series in Mathematical Physics, vol. 1, World Scientific, Singapore, 1987, pp. 629–646.

[TMR53] Alfred Tarski, Andrzej Mostowski, and Raphael M. Robinson, *Undecidable theories*, Studies in Logic and the Foundations of Mathematics, North-Holland, Amsterdam, 1953.

[Tod89] Andrey N. Todorov, *The Weil–Petersson geometry of the moduli space of $SU(n \geq 3)$ (Calabi–Yau) manifolds. I*, Communications in Mathematical Physics **126** (1989), no. 2, 325–346.

[Tot97] Burt Totaro, *Torsion algebraic cycles and complex cobordism*, Journal of the American Mathematical Society **10** (1997), 467–493.

[Tot17] ———, *Recent progress on the Tate conjecture*, Bulletin of the American Mathematical Society. New Series **54** (2017), no. 4, 575–590.

[Tur37] Alan M. Turing, *On computable numbers, with an application to the Entscheidungsproblem*, Proceedings of the London Mathematical Society (2) **42** (1937), 230–265.

[vD13] Dirk van Dalen, *Logic and structure*, fifth ed., Universitext, Springer, London, 2013.

[Voi02a] Claire Voisin, *A counterexample to the Hodge conjecture extended to Kähler varieties*, International Mathematics Research Notices (2002), no. 20, 1057–1075.

[Voi02b] ———, *Hodge theory and complex algebraic geometry. I*, Cambridge Studies in Advanced Mathematics, vol. 76, Cambridge University Press, Cambridge, 2002.

[Voi03] ———, *Hodge theory and complex algebraic geometry. II*, Cambridge Studies in Advanced Mathematics, vol. 77, Cambridge University Press, Cambridge, 2003.

[Voi04] ———, *On the homotopy types of compact Kähler and complex projective manifolds*, Inventiones Mathematicae **157** (2004), no. 2, 329–343.

[Voi06] ———, *On integral Hodge classes on uniruled or Calabi–Yau threefolds*, Moduli Spaces and Arithmetic Geometry, Advanced Studies in Pure Mathematics, vol. 45, Mathematical Society of Japan, Tokyo, 2006, pp. 43–73.

[Voi07] ———, *Hodge loci and absolute Hodge classes*, Compositio Mathematica **143** (2007), no. 4, 945–958.

[War83] Frank W. Warner, *Foundations of differentiable manifolds and lie groups*, Graduate Texts in Mathematics, vol. 94, Springer-Verlag, New York-Berlin, 1983.

[Wat79] William C. Waterhouse, *Introduction to affine group schemes*, Graduate Texts in Mathematics, vol. 66, Springer-Verlag, New York, 1979.

[Wei79] André Weil, *Abelian varieties and the Hodge ring*, Œuvres Scientifiques. Collected Papers. Vol. III, Springer-Verlag, New York, 1979, pp. 421–429.

[Wei94] Charles A. Weibel, *An introduction to homological algebra*, Cambridge Studies in Advanced Mathematics, vol. 38, Cambridge University Press, Cambridge, 1994.

[Wel08] Raymond O. Wells, *Differential analysis on complex manifolds*, third ed., Graduate Texts in Mathematics, vol. 65, Springer, New York, 2008.

[Whi35] Hassler Whitney, *A function not constant on a connected set of critical points*, Duke Mathematical Journal **1** (1935), no. 4, 514–517.

[Whi49] J. H. C. Whitehead, *On simply connected, 4-dimensional polyhedra*, Commentarii Mathematici Helvetici **22** (1949), 48–92.

[Yau78] Shing-Tung Yau, *On the Ricci curvature of a compact Kähler manifold and the complex Monge–Ampère equation. I*, Communications on Pure and Applied Mathematics **31** (1978), no. 3, 339–411.

[Zar75] Yuri G. Zarhin, *Endomorphisms of abelian varieties over fields of finite characteristic*, Izvestiya Akademii Nauk SSSR. Seriya Matematicheskaya **39** (1975), no. 2, 272–277.

[Zer08] Ernst Zermelo, *Untersuchungen über die Grundlagen der Mengenlehre. I*, Mathematische Annalen **65** (1908), 261–281.

[Zor35] Max Zorn, *A remark on method in transfinite algebra*, Bulletin of the American Mathematical Society **41** (1935), 667–670.

[Zuc77] Steven Zucker, *The Hodge conjecture for cubic fourfolds*, Compositio Mathematica **34** (1977), no. 2, 199–209.